SKETCHES FOR THREE-DIMENSIONAL YANG–MILLS THEORY

Abstract. We give detailed sketches for extensions of the scalar construction of three-dimensional lattice Yang–Mills theory. Each section develops one main result, including curvature tensors, short-distance laws, compact gauge groups, Wilson loops, local nets, dynamics and a confinement criterion. We explain the constructions and estimates, retain their technical arguments, and specify the estimates and limit passages required by the sketches.

Contents

1. Introduction 1
2. Tensor correlations on a fixed torus 2
3. Real-time tensor fields 56
4. Point splitting at two scales 79
5. Plane fields and anisotropic variations 154
6. The two-term short-distance law 255
7. The lattice two-loop coefficient 305
8. Natural scaling and failure of dilation covariance 419
9. Compact gauge groups 453
10. Renormalized Wilson loops 543
11. Centre flux 590
12. Vanishing latent heat 637
13. Infrared bounds 672
14. The Haag net of the action density 715
15. A continuum Langevin process 784
16. Positive-time gradient flow 821
17. A finite-size criterion for confinement 862
Appendix A. Index of notation and conventions 950
References 950

1. Introduction

The curvature-bilinear manuscript [8] constructs the Euclidean correlations of three-dimensional SU(2) Yang–Mills theory and proves their Osterwalder–Schrader axioms on Schwartz tests flat at coincidences. We develop further consequences and extensions of that construction. The first concerns local fields in real time.

Theorem 0 (Real-time tensor fields, sketched). Every clustering component of the curvature-bilinear correlations admits a reconstruction by local tempered tensor fields in $1 + 2$ dimensions. Their energy-momentum spectrum lies in the closed forward cone, and each component has a cyclic vacuum unique among translation-invariant vectors. ■□

The precise setting and reconstruction are given in §3. The Euclidean construction, stress conservation and rotation argument are used from [8]; their proofs are not repeated here.

The remaining sections each develop one principal result. Point splitting relates a close pair of scalar insertions to a curvature tensor. Weighted backgrounds describe anisotropic variations. The two-term short-distance law and its lattice coefficient determine a correction to natural scaling.

Other sketches concern compact gauge groups, Wilson loops, centre flux, latent heat, infrared bounds, a Haag net, and stochastic and deterministic dynamics. The final section gives a finite-size confinement criterion.

The arguments are drawn from [7]. We explain their reductions and estimates, and retain the technical reasoning for the additional results. Each statement headed *Sketch* gives the intended conclusion and the route to it. The general-group construction and continuum gradient flow require the uniform estimates and limiting identifications specified in their sketches; these steps are not proved here. The confinement criterion assumes its stated box inequality along the continuum trajectory.

We use the scalar construction of [1] and the curvature-bilinear theory of [8]. Lattice spacing is $a_K = L^{-K}$, with odd $L \geq 13$, and the physical inverse-coupling label is $b = a_K \beta_K$. Thus a fixed-label sequence has $\beta_K = L^K b$. The reference coupling that controls the renormalization steps is distinct from $b^{-1/2}$. We recall the necessary conventions where each sketch uses them. Full convergence on a fixed torus, subsequential infinite-volume convergence, and selection of a clustering component have different meanings; their order is specified in each argument.

2. Tensor correlations on a fixed torus

The curvature main manuscript constructs tensor correlations by taking subsequences and then constructs their Osterwalder–Schrader spaces. Here the volume is fixed. Comparing two consecutive lattice cutoffs gives a summable error, and hence a limit along the full sequence. The same comparison identifies several lattice representatives of a tensor insertion and controls changes in the physical coupling label. The detailed comparisons below retain the supplied proof sketches; the following statement collects the conclusions they are intended to establish.

Write $T_\ell = (\mathbb{R}/\ell\mathbb{Z})^3$, $a_K = L^{-K}$, and $\Lambda_{K,\ell} = a_K \mathbb{Z}^3/\ell\mathbb{Z}^3$, where the blocking factor L is odd. The physical label is $\mathsf{b} = a_K \beta$; thus a fixed label means Wilson measures at $\beta = L^K \mathsf{b}$. Initially use the reference regime $0 < g \leq \gamma_W^{\mathrm{ref}}$ and

$$\mathsf{b} \in I(g) = (g^{-2} + \bar{B}, X_g - \bar{B}], \qquad \ell \in \mathfrak{L}(g) = \{2L^m : m \geq m_{\mathfrak{L}}(g)\}.$$

Here $B_\sharp$ is the fixed error bound in the main manuscript's first-passage construction, $\bar{B} = B_\sharp L/(L-1)$, $X_g = L(g^{-2} + B_\sharp)$, and $m_{\mathfrak{L}}(g)$ is its admitted-side threshold. In this regime $(K, L^K \mathsf{b})$ is a first-passage datum for every $K \geq 1$. An exact dilation, described below, removes this initial restriction on the positive label and on the side $2L^m$.

Let $V = \mathrm{Sym}_3(\mathbb{R})$ and let $O_K^T(H)$ be the main manuscript's lattice representative of $\frac{1}{2}\int \mathrm{tr}(HT)\,\mathrm{d}x$. Its diagonal terms are the three plaquette-plane forms, and its off-diagonal terms are the three bond forms. For a matrix weight set

$$|H| = \sup_x \max_{i,j} |H_{ij}(x)|, \qquad \|H\|_{\mathrm{Lip}} = |H| + \mathrm{Lip}_\infty H,$$

$$\mathcal{S}_{n;K}(H_1, \dots, H_n) = \langle O_K^T(H_1); \dots; O_K^T(H_n) \rangle_{K,\ell,L^K\mathsf{b}}.$$

The Lipschitz seminorm uses the sup distance on the torus, entry by entry. Support separations in this overview use that distance. For bounded Borel weights, $\bar{O}_K^T(H)$ uses the cell means

$$H^{(K)}(x) = a_K^{-3} \int_{x + a_K[-1/2,1/2)^3} H(y)\,\mathrm{d}y$$

at the least corner x of the plaquette or bond. This specifies the sampling convention even when point values have no meaning.

We also need to compare insertions built from different plaquette corners. For an ordered pair of signed directions $l = (\alpha, \alpha')$, $\alpha' \neq \pm\alpha$, let $W_{x,l}$ be its plaquette loop, $\lambda(l)$ its normal direction, and

$s(l) = \operatorname{sgn}(\alpha)\operatorname{sgn}(\alpha')\varepsilon_{\lambda(l)|\alpha||\alpha'|}$ its orientation. Put

$$w(U) = \frac{U - U^{-1}}{2i}, \qquad Q_x(l,l') = \frac{1}{2}\operatorname{tr}[w(W_{x,l})w(W_{x,l'})],$$
$$\mathcal{Q}_{l,l';\xi}(\phi) = a_K^{-1}\sum_x \phi(x + a_K\xi)Q_x(l,l'), \qquad |\xi|_\infty \le \frac{1}{2}.$$

A class insertion $\mathcal{E}$ is a finite sum of these corner forms and of plane forms $\mathcal{O}_\lambda(\phi) = O_K^T(\phi E_{\lambda\lambda})$, with real Lipschitz weights. Here E_{ij} are the matrix units. Its quadratic symbol $\operatorname{Sym}\mathcal{E}$ is the sum of

$$\phi E_{\lambda\lambda}, \qquad \phi s(l)s(l')(E_{\lambda(l)\lambda(l')} + E_{\lambda(l')\lambda(l)}),$$

respectively; its norm $\|\mathcal{E}\|$ is the sum of $\sup|\phi| + \operatorname{Lip}_\infty\phi$ over its terms. In particular $\operatorname{Sym} O_K^T(H) = H$.

Sketch A (Full fixed-torus limit). Fix $n \ge 2$ and $d > 0$. Suppose the weights H_a vanish off closed sets $\Sigma_a \subset T_\ell$ with pairwise sup distances at least d. For Lipschitz weights, there are C, p, K_1 such that

$$|\mathcal{S}_{n;K+1}(H_1,\dots,H_n) - \mathcal{S}_{n;K}(H_1,\dots,H_n)| \le C\ell^{3(n+1)}K^pL^{-\theta_nK}\prod_a \|H_a\|_{\mathrm{Lip}},$$
$$\theta_n = \frac{1}{1260n(8n+3)}, \qquad K \ge K_1.$$

Consequently $\mathcal{S}_n^{\ell,\mathsf{b}} = \lim_K \mathcal{S}_{n;K}$ exists along the full sequence, with the tail bounded by the sum of the displayed right-hand side over $K' \ge K$. For $\bar{O}_K^T$ the same assertions hold for bounded Borel weights, with $\prod_a |H_a|$ and a possibly different constant. The two limits agree for Lipschitz weights.

Class insertions $\mathcal{E}_a, \mathcal{E}'_a$ over these sets with equal symbols have cumulants differing by at most

$$C^{\mathrm{u}}K^pL^{-\theta_nK}\prod_a(\|\mathcal{E}_a\| + \|\mathcal{E}'_a\|).$$

Their common limit is $\mathcal{S}_n^{\ell,\mathsf{b}}(\operatorname{Sym}\mathcal{E}_1,\dots,\operatorname{Sym}\mathcal{E}_n)$. The Lipschitz seminorms in this comparison can be omitted when the representatives carry the same weights at the same sites.

The cutoff estimates are uniform for b in any compact interval $J \subset I(g)$. The limits are continuous in the label; if b is an interior point of $I(g)$ and $a_K\beta_K \to \mathsf{b}$, the full cutoff sequence at β_K has the same limit. Exact dilation extends these conclusions to every $\mathsf{b} > 0$ and every side $2L^m$, $m \ge 0$. ■□

2.1. **Comparing two cutoffs through one block field.** The new estimate concerns the relation between the measures at K and $K+1$. Bounds on each measure separately would only provide compactness. The comparison instead evaluates their insertions over one block field at a common depth. This introduces three errors: replacing each insertion by its value at a background, comparing the two background values, and comparing the two laws of the block field. Prop. 2.1 gives this argument for plane and bond forms; Lem. 2.11 includes corner forms. Expanding a tensor insertion into its six entries then gives 6^n terms.

The two depths have different purposes. Set

$$\sigma = \left\lfloor \frac{K}{1260n} \right\rfloor, \qquad s = \left\lfloor \frac{\sigma}{8n+3} \right\rfloor.$$

At depth s the supports are far enough apart for the separated-source calculus. Each insertion has size at most a constant times ℓ^3L^{4s} times its weight norm. Replacing it by the first source member evaluated at the background of the depth-σ block field leaves a large-field error, a loss-of-memory error decaying like a polynomial times $L^{-(\sigma-s)/2}$, and a connecting-polymer error decaying exponentially in L^sd. The inequality $\sigma - s \ge (8n+2)s$ pays for the growth L^{4ns} of a product of n insertions. After that cost, the slowest decay is L^{-s}.

The background comparison is the substantive local step. In a suitable gauge the fine plaquette variables are curls of a small Lie-algebra-valued field to second order; the coarse variables are their block averages. The variance about an average is controlled by the defect in Jensen's inequality. Summing that defect relates it to the action lost in averaging. The comparison of background actions controls this loss, and the minimality of the coarse background controls the difference between that background and the averaged fine background. This gives, for one insertion, a defect bounded by a constant times

$$\ell^3 L^{4\sigma} K^{1/2} L^{-(K-\sigma)/2} (\sup|\phi| + L^{-\sigma} \operatorname{Lip}_\infty \phi).$$

Finally, the two block-field laws differ by at most a constant times $\ell^3 K^{\mathsf{p}_0} L^{-K/10}$ on bounded gauge-invariant functions. Here p_0 denotes the fixed polynomial exponent of that comparison. The small choice of σ pays for the $L^{4n\sigma}$ growth in both of these errors: their remaining decays are at least $L^{-K/3}$ and $L^{-K/11}$, respectively. Thus $s \geq \theta_n K - 2$ gives the stated summable rate.

2.2. **Sampling, symbols, and changes of label.** Cell means remove the regularity cost at its source. The cell of a coarse site y is the union of the cells of its L^3 fine sites $B(y)$, so

$$\phi^{(K)}(y) = L^{-3} \sum_{x \in B(y)} \phi^{(K+1)}(x).$$

The weights therefore match exactly across one averaging step. Point sampling introduces a mismatch of order $a_K \operatorname{Lip}_\infty \phi$ instead. The uniform sup-norm bound of Lem. 2.10(b) makes this mismatch vanish in the limit. Uniform convergence on sup-norm balls also transfers bounded pointwise continuity from the finite-cutoff cell means to the limit. In each argument separately, the resulting functional is a countably additive measure, vanishes on Lebesgue null sets, and has an L^1 density. This is the precise measure-theoretic conclusion for separated weights.

Equality of quadratic symbols is stronger than agreement of a formal leading expression. To prove it, the comparison replaces each loop by a canonical loop in the same plane, with its orientation sign. Moving its anchor costs $O(a_K \operatorname{Lip}_\infty \phi)$. The plane insertion is reduced to corner products by the exact identity

$$a_p = \frac{1}{2} Q_x(l, l) + \frac{1}{2} a_p^2.$$

The last term is fourth order at the backgrounds and lies within the comparison error. The resulting canonical form depends only on the symbol. Thus bond averages, clovers, and individual corners produce the same separated continuum field, with a quantified error before taking the limit.

Uniformity in the label requires tracking where it enters the comparison. Let $\Theta_\star$ be the fixed threshold for comparing the two backgrounds. On $J \subset I(g)$ the block-law constants and their threshold are common; the background-radius bounds are common; and $L^\sigma \mathsf{b} \geq \Theta_\star$ holds throughout J once it holds at its least label. At fixed K the compact-group integral is continuous in $\beta = L^K \mathsf{b}$. Uniform cutoff convergence therefore gives a continuous limit in b. For $\mathsf{b}_K = a_K \beta_K \to \mathsf{b}$, split the error into the uniform cutoff tail at b_K and continuity of the limiting function at b.

For an initially inadmissible side or label, dilation replaces (ℓ, b, K) by $(L^z \ell, L^z \mathsf{b}, K - z)$, keeps the underlying graph and bare coupling unchanged, and multiplies an n-fold cumulant by L^{zn}. For sufficiently large z, the reference coupling $g' = (L^z \mathsf{b}/2)^{-1/2}$ admits this new side and places the new label in the interior of $I(g')$. This is the exact transfer used in Thm. A.

The same theorem retains translations, signed-permutation covariance, coordinate-plane reflection positivity, and the separated fixed-torus stress identity. Derivatives in that identity act on periodic continuum test functions. Its two-point refinements are stated with their own scale restrictions. Extensions through coinciding supports, their contact ambiguity, and stronger rates under weaker separation are separate tasks below; they are not consequences of the separated-support convergence estimate.

Sources. Thm. YM3_I:D follows an insertion through the levels as a source under a majorant. A *majorant* on $\Lambda_{K,\ell}$ is a pair (s, S) of a number $\mathsf{s} \geq 0$, its *size*, and a set S of plaquettes; its weight

is the plaquette source $\mathsf{s}\mathbf{1}_S$, and $\mathsf{m}_j = 12M^3\alpha_0^2 L^{K-j}\mathsf{s}$, $j \geq 0$, are the numbers that the calculus of Thm. YM3_I:D attaches to that weight at the level j, $\alpha_0 \in (0,1]$ a basic constant of the calculus. A *source under* (s, S) is a triple of: a *first member* F, a continuous function of the configurations of $\Lambda_{K,\ell}$, invariant under the gauge transformations, with $|F(U)| \leq a_K^{-1}\mathsf{s}\sum_{p\in S} a_p(U)$; a local family at the level 0 of norm at most $\mathsf{c}^\flat\mathsf{m}_0$, $\mathsf{c}^\flat$ a basic number of the calculus; and local families at the levels $j \geq 1$ of norms at most $\frac{1}{2}\mathsf{m}_j$; each family vanishes at the regions of its level that hold no plaquette of S, and each sums at its level to F, up to the value at the trivial field.

A *source system under majorants* is a tuple of sources under majorants (s_a, S_a); all the numbers of the calculus are then formed at the weights $\mathsf{s}_a\mathbf{1}_{S_a}$. A source under (s, S) is a source under every (s', S') with $\mathsf{s}' \geq \mathsf{s}$ and $S' \supset S$, the majorant entering only the right members of three inequalities and the two clauses of vanishing; the sum of two sources, member by member, is a source under the sum of the sizes and the union of the sets, and c times a source is one under $|c|$ times the size. For the value at short distance the calculus reads besides, of a real source, the kernel $\mathsf{t}(p,p') = \mathsf{t}(p',p)$ of its first member to the second order at a small field, vanishing unless $p, p' \in S$, with $2\sum_{p'} |\mathsf{t}(p,p')| \leq \mathsf{s}$.

[7, Lem. C.4] says which insertions are sources.

Proposition 2.1 (Two consecutive cutoffs on one torus for the plaquette forms of one plane and the bond forms)**.** Let $0 < g \leq \gamma_W^{\mathrm{ref}}$, $b \in I(g)$, $\ell \in \mathfrak{L}(g)$, $n \geq 2$ and $\mathsf{d} > 0$; by Cor. YM3_I:118.4 the pairs $(K, L^K b)$ are first-passage data at g with label b. For a bounded Borel function ψ on T_ℓ and $x \in \Lambda_{K,\ell}$ put $\psi^{(K)}(x) = a_K^{-3}\int_{x+a_K[-\frac{1}{2},\frac{1}{2})^3}\psi$. A *cell insertion* at the cutoff K is one of the following: the plaquette form $\mathcal{O}(w)$ of one plane $\{\kappa,\kappa'\}$, with $w(p) = \psi^{(K)}(x)$ at the plaquette p spanned at x by $a_K e_\kappa$, $a_K e_{\kappa'}$ and $w = 0$ at the plaquettes of the two other planes; the sum of the three, the action density with the weight $\psi^{(K)}$; and the bond form $Y[\mathsf{w}]$ of one direction μ, with $\mathsf{w}(\langle x, x + a_K e_\mu\rangle) = \psi^{(K)}(x)$ and $\mathsf{w} = 0$ at the bonds of the two other directions. Let $\mathcal{E}_1, \dots, \mathcal{E}_n$ be cell insertions, of any kinds, whose functions ψ_a are real and vanish off closed sets at pairwise sup distances $\geq \mathsf{d}$, and $\kappa_K = \langle \mathcal{E}_1; \dots; \mathcal{E}_n\rangle_{K,\ell,L^K b}$.

(a) There are numbers C, p, formed from the constants, n and b, and a cutoff K_1, formed besides from g, ℓ and d, such that for $K \geq K_1$

$$|\kappa_{K+1} - \kappa_K| \leq C\,\ell^{3(n+1)}K^p L^{-\theta_n K}\prod_a \sup|\psi_a|, \qquad \theta_n = \frac{1}{1260\,n(8n+3)}.$$

So κ_K converges along the full sequence $K \to \infty$, and the limit is n-linear and symmetric. The same holds, with $\sup|\psi_a| + \mathrm{Lip}_\infty\psi_a$ in the place of $\sup|\psi_a|$, for the forms of the three kinds whose weights are real functions ψ_a with $\mathrm{Lip}_\infty\psi_a < \infty$ read at the barycentres of the plaquettes and at the midpoints of the bonds, $w(p) = \psi_a(x(p))$, $\mathsf{w}(\mathfrak{b}) = \psi_a(x_\mathfrak{b})$, under the same separation of the supports; and for such weights the two limits agree.

(b) In this clause the separation of the head is replaced by a weaker one. Let $\psi_1, \dots, \psi_n$ be real Lipschitz functions on T_ℓ, none of them 0, let $\mathcal{E}_1 = Y[\mathsf{w}]$ be the bond form of one direction μ with $\mathsf{w}(\mathfrak{b}) = \psi_1(x_\mathfrak{b})$ at the bonds $\mathfrak{b}$ of the direction μ and $\mathsf{w} = 0$ at the others, let $\mathcal{E}_a = \mathcal{O}(w_a)$, $w_a(p) = \psi_a(x(p))$, $2 \leq a \leq n$, be action densities, and let κ_K be the joint cumulant of these n forms. If the n supports have no common point and their spread is at least $\mathsf{d}_0 > 0$ (the definition preceding Lem. YM3_I:114.6), then from a cutoff on $|\kappa_{K+1} - \kappa_K| \leq C\,K^p L^{-K/120}\prod_a \|\psi_a\|_\sharp$, $\|\psi\|_\sharp = \max\{\sup|\psi|, 40M\mathrm{Lip}\,\psi\}$, with C, p and the cutoff formed from the constants, n, ℓ, b and d_0.

(c) If all the $\mathcal{E}_a$ are action densities with real Lipschitz weights read at the barycentres, $\mathcal{E}_a = \mathcal{O}(w_a)$, $w_a(p) = \psi_a(x(p))$, and κ_K is their joint cumulant, the convergence is given by Thm. YM3_I:AR, equation YM3_I:(2203), Cor. YM3_I:117.1 and Cor. YM3_I:118.4: from a cutoff on, $|\kappa_{K+1} - \kappa_K| \leq C\,\ell^{3(n+1)}K^p L^{-K/10}\prod_a \|\psi_a\|_\sharp$.

Clauses (a) and (b) carry the convergence of Thm. YM3_I:AR over to the plaquette form of one plane, to the bond form and to weights that are means over the lattice cubes; clause (c) is that theorem for the action density. ■

Proof sketch. (a) At the two cutoffs K and $K+1$, at the one label b on the one torus, put

$$\sigma = \lfloor K/(1260\,n)\rfloor, \qquad s = \lfloor \sigma/(8n+3)\rfloor;$$

then $\sigma \le \lceil K/1260\rceil$, $\sigma - s \ge (8n+2)s$, $s \ge \theta_n K - 2$ and $4n\sigma \le K/315$. Every insertion is the first member of a source under a majorant at both cutoffs ([7, Lem. 2.11], which holds on every side with (Sd.1), so on every $\ell \in \mathfrak{L}(g)$ by Prop. YM3_I:74.1): under (s_a, S_a) with $\mathsf{s}_a = \sup|\psi_a|$ for a plaquette form or an action density and $\mathsf{s}_a = 59\sup|\psi_a|$ for a bond form, S_a the plaquettes whose corners lie within $\frac{3}{2}a_K$ of the closed set off which ψ_a vanishes; a source under a majorant is one under every larger majorant, so that these majorants serve, and corners of plaquettes of different S_a are at sup distance $\ge \mathsf{d} - 3a_K \ge \mathsf{d}/2$ once $6a_K \le \mathsf{d}$. The two cumulants are compared through three errors, each bounded by a number formed from the constants and n times the factors written.

(i) *One cutoff.* By Thm. YM3_I:D, on the small fields, the chain with the sources is run up to the level $K - s$, at which the n insertions have sizes at most $4\ell^3 L^{4s}\mathsf{s}_a$; each insertion is replaced by its *read-out*, its first member read at the background of the block field of the depth σ, a function of that block field alone. The cumulant of the n insertions and the cumulant of their n read-outs differ by at most

$$\Big[2L^{4n}V_\ell\, L^{-(4n+2)s} + n\,\epsilon(\sigma - s) + 2e^{3-L^s\mathsf{d}/(2M)}\Big]\prod_a 4\ell^3 L^{4s}\mathsf{s}_a.$$

The first term is the large-field term (Prop. YM3_I:115.2), $V_\ell = 4M^{-3}\ell^3 e^{\mathsf{B}(X_g)\ell^3}$ times a number formed from the constants and n. The second is the loss of memory of the progression with sources between the depths s and σ: the sources switched on at the level $K - \sigma$ in the place of the level 0 change the effective terms of the level $K - s$ by at most $\epsilon(\sigma - s)$ times their sizes, $\epsilon(\nu) = C_{\mathrm{m}}(\nu+1)^{2\mathsf{p}_0+2}L^{-\nu/2}$ with a number C_{m} of the constants, because the linearised steps contract on the small fields, as the flows of Props. YM3_I:114.2, YM3_I:114.3 forget their start. The third bounds the polymers that join two supports, whose linear size is at least $L^s\mathsf{d}/(2M) - 2$ in the units of the level $K - s$.

(ii) *The tie at the depth σ.* The read-outs of the two cutoffs are functions of one block field V of the depth σ, vanishing where V has large fields, of sizes at most $\frac{3}{2}B_3^2\ell^3 L^{4\sigma}\mathsf{s}_a$; over one such V they differ by at most

$$\tau_a = C_{\mathrm{t}}\,\ell^3 L^{4\sigma}K^{1/2}L^{-(K-\sigma)/2}\sup|\psi_a|$$

for weights that are means over the lattice cubes, and by the same with $\sup|\psi_a| + L^{-\sigma}\mathrm{Lip}_\infty\psi_a$ in the place of $\sup|\psi_a|$ for a weight read at the points of the two lattices, C_{t} a number of the constants; so, under one law of V, the cumulants of the read-outs of the two cutoffs differ by at most $\sum_a \tau_a \prod_{b\ne a}\frac{3}{2}B_3^2\ell^3L^{4\sigma}\mathsf{s}_b$. The tie is a statement of its own, the counterpart of equation YM3_I:(2156) for the products of two plaquette loops at a site. Let $(t, \varepsilon_{\mathrm{r}}, V)$ be data of the comparison (the setting preceding Lem. YM3_I:113.8), $t = K - \sigma$, U^{A}, U^{B} the backgrounds over V of the bare lattices $\Lambda_{K,\ell}$, $\Lambda_{K+1,\ell}$, and $N_t = \ell^3L^{3\sigma}$ the number of the sites of the level t; for two loops l, l' at a site x let $Q_x(l,l') = \langle w(W_{x,l}), w(W_{x,l'})\rangle$, $\langle X, Y\rangle = \frac{1}{2}\operatorname{tr}XY$, $W_{x,l}$ the plaquette loop of l at x. Then for $\xi \in [-1,1]^3$ and a complex function ϕ on T_ℓ with $\mathrm{Lip}_\infty\phi < \infty$,

$$\Big|L^{t+1}\sum_{x\in\Lambda_{K+1,\ell}}\phi(x+a_{K+1}\xi)\,Q_x(l,l')(U^{\mathrm{B}}) - L^t\sum_{y\in\Lambda_{K,\ell}}\phi(y+a_K\xi)\,Q_y(l,l')(U^{\mathrm{A}})\Big| \le C_{\mathrm{t}}\,\|\phi\|_{(t)}\,\varepsilon_{\mathrm{r}}^2 N_t\, t^{1/2}L^{-t/2},$$

$\|\phi\|_{(t)}$ the norm in equation YM3_I:(2154); and for a weight v_{f} on the sites of $\Lambda_{K+1,\ell}$ and its block mean $\mathsf{v}_{\mathrm{c}}(y) = L^{-3}\sum_{x\in B(y)}\mathsf{v}_{\mathrm{f}}(x)$ on $\Lambda_{K,\ell}$, $B(y)$ the L^3 sites of the block of y, the same holds with v_{f}, v_{c} in the place of the two readings of ϕ and $\sup|\mathsf{v}_{\mathrm{f}}|$ in the place of $\|\phi\|_{(t)}$. The proof follows that of Prop. YM3_I:113.9: in a gauge on a box about a site of the coarse lattice the background is e^{ia} with

a small 1-form a with values in the Lie algebra, the plaquette variables $w(W_{x,l})$ of the fine lattice are the curls of a up to the second order, and those of the coarse lattice their averages (one averaging step with values in the Lie algebra); the variance of the curls about their averages, plaquette by plaquette, is bounded by the defect of Jensen's inequality, whose sum over the plaquettes is $2L^{-2}$ times the difference $L\sum_p a_p - \sum_{p'} a_{p'}$ of the actions of the fine background and of its block average, a difference bounded through the comparison of the two background actions, equation YM3_I:(2155), and the minimality of the background of the coarse lattice; and the background of the coarse lattice is compared with the block average of the fine background, a nearly minimal competitor, through its excess of action.

The bond density is a sum of four such products, $q_{\mathfrak{b}} = \frac{1}{4}\sum_{\varepsilon,\varepsilon'=\pm}\varepsilon\varepsilon' Q_x((\mu,\varepsilon\lambda),(\mu,\varepsilon'\lambda'))$ at $\xi = \frac{1}{2}e_\mu$, and the plaquette density of one plane is one such product up to a_p^2, by $a_p = \frac{1}{2}Q_x(l,l) + \frac{1}{2}a_p^2$ at the loop l of the plane, a_p^2 being of the fourth order in the field at the backgrounds; the action density is equation YM3_I:(2156) itself. With $a_K^{-1} = L^\sigma L^t$ and $\varepsilon_{\mathrm{r}} \le 1$, $t \le K$, this gives τ_a; for means over the lattice cubes the weights of the two cutoffs are tied by the block mean, $\psi^{(K)}(y) = L^{-3}\sum_{x\in B(y)}\psi^{(K+1)}(x)$, since the cell of y is the union of the cells of the $x \in B(y)$, and the tie asks no regularity.

(iii) *The laws.* The laws of the block field of the depth $\sigma \le \lceil K/1260\rceil$ at the two cutoffs differ, on the bounded gauge-invariant functions, by at most $\mathsf{C}_{\mathrm{bf}}\ell^3 K^{\mathsf{p}_0}L^{-K/10}$ times their sup norm for $K \ge K_{\mathrm{bf}}$ (Cor. 2.8), with the numbers C_{bf}, K_{bf} of Def. 2.7 (5) at the label b and $\mathsf{p}_0 = 2\mathfrak{p}_0 + \frac{1}{2}$; for the cumulant of the read-outs this costs $\mathsf{C}_{\mathrm{bf}}\ell^3 K^{\mathsf{p}_0}L^{-K/10}\prod_a \frac{3}{2}B_3^2\ell^3 L^{4\sigma}\mathsf{s}_a$.

The balance. In (i), $\sigma - s \ge (8n+2)s$ gives $L^{4ns}\epsilon(\sigma-s) \le C_{\mathrm{m}}(\sigma+1)^{2\mathfrak{p}_0+2}L^{-s}$; and from a cutoff on $L^{4n}V_\ell L^{-s} \le 1$ and $2e^{3-L^s\mathsf{d}/(2M)}L^{(4n+1)s} \le 1$; so (i) is at most $(4\ell^3)^n[3 + nC_{\mathrm{m}}(\sigma+1)^{2\mathfrak{p}_0+2}]L^{-s}\prod_a \mathsf{s}_a$. In (ii) and (iii), $4n\sigma \le K/315$ gives $L^{4n\sigma}L^{-(K-\sigma)/2} \le L^{-K/3}$ and $L^{4n\sigma}L^{-K/10} \le L^{-K/11}$. As $L^{-s} \le L^2 L^{-\theta_n K}$, $\theta_n < \frac{1}{11}$, $\sigma + 1 \le K$ and $\mathsf{s}_a \le 59\sup|\psi_a|$, the three errors give (a) with $p = 2\mathfrak{p}_0 + 2$; the factor $\ell^{3(n+1)}$ comes from the n sizes and the law, and C is formed from the constants, n and C_{bf}, so from the constants, n and b.

The cutoff K_1 is the least from which $K \ge K_{\mathrm{bf}}$, $s \ge 1$, $L^s\mathsf{d} \ge 6M$, $6a_K \le \mathsf{d}$, $L^\sigma b \ge \Theta_\star$ (the number of equation YM3_I:(2148)), the two inequalities above and the thresholds of the chain of n sources at the depth s hold; through V_ℓ, the distance and K_{bf} it reads besides g, ℓ and d, while C and p read neither. The Lipschitz numbers enter at one place, the tie, with the factor $L^{-\sigma}$. The two readings of a Lipschitz weight, at the points and through the means over the lattice cubes, differ by at most $a_K\mathrm{Lip}_\infty\psi_a$, the size of the weight of a source; the bound of the moments of first members of sources at one fixed depth then makes the two cumulants at the cutoff K differ by at most a number formed from the constants, n, g, ℓ and d times $a_K\sum_a \mathrm{Lip}_\infty\psi_a\prod_{b\ne a}\sup|\psi_b|$, which tends to 0; so the two limits agree.

Since $\sum_K K^p L^{-\theta_n K} < \infty$, κ_K converges along the full sequence; multilinearity and symmetry pass to the limit.

(b) Only the bond form needs a tie: the cumulant is the mixed derivative at 0, in the parameters of the action densities, of the mean of the bond form under the measure with these densities as sources; along the chain the derivative of the bond form obeys a linear recursion, whose sizes are those of a member of the level j and of what a step creates, so that no disc in a parameter of the bond form is needed; the polymers that meet all the n supports, which make the contact terms, have at a depth u a linear size of at least $\mathsf{d}_0 L^u/M - 2$ in the units of their level, as in the polymer estimate in Thm. YM3_I:AR, so that the spread, and no pairwise distance, enters; and the comparison of the bond forms of the two backgrounds over one block field, the tie of (ii), gives the rate $L^{-K/120}$. (c) is Thm. YM3_I:AR, equation YM3_I:(2203) with Cor. YM3_I:117.1 and Cor. YM3_I:118.4. □

2.3. **The pointwise source identity.** We use the actual sourced history of [8, Thm. D, Prop. 5.59, §7.2], with its admitted torus, coupling range, source ball and fixed analytic radii. The coefficient

sequence is $x_j = L^{-1}x_{j-1} - e_j$; the counterterms e_j are fixed independently of the sources. For two plaquette weights put $w_\lambda = \lambda_1 w_1 + \lambda_2 w_2$, and let $E_\lambda^{(j)}$ be the effective action at height j for these same source parameters. The new point here is an identity at every admitted real chart point, with the whole source parameter left in its exponential, rather than only an almost-everywhere identity for the pushed-forward measure.

We recall the size conventions used in that identity. Set $\omega_n = (n+1)L^{-(n-1)}$. At height j the source-size recursion and the remaining analytic room are

$$s_j = (1+\beta_{\mathrm{s}})\delta(x_j)/\alpha_0, \qquad \theta_j = K_{\mathbf{M}} s_j, \qquad N_{\star,j} = w_\star/(4K_\Lambda s_j),$$
$$m_{j,a} = 12M^3\alpha_0^2 L^{K-j}|w_a|_\infty, \qquad c^\flat = 81 C_P^2 C_{\mathcal{A}}^2/K_\Lambda,$$
$$d_{1,a} = c^\flat\theta_0 m_{0,a}, \qquad n_{j,a} = m_{j,a} + \sum_{i=1}^{j}\omega_{j+1-i}d_{i,a}, \qquad d_{j+1,a} = \theta_j n_{j,a},$$
$$N_j^{\mathrm{rm}} = N_{\star,j} - \left(\sum_{n=1}^{j}\omega_n\right)(E_0 + \tfrac{3}{2}\alpha_0^2 B_\sharp).$$

All constants and radii in these formulas are those of the fixed weighted source construction just cited. The admitted parameter domains are the successive intersections

$$\begin{aligned}\Lambda_0^{\mathrm{w}} &= \{\lambda : c^\flat\sum_a |\lambda_a| m_{0,a} < N_{\star,0}\},\\ \Lambda_j^{\mathrm{w}} &= \{\lambda \in \Lambda_{j-1}^{\mathrm{w}} : \sum_a |\lambda_a| n_{j,a} < N_j^{\mathrm{rm}}\}.\end{aligned} \tag{2.i}$$

We write $\Lambda^{\mathrm{w}} = \Lambda_{K-1}^{\mathrm{w}}$ for the common admitted domain through the required heights. The norm $N_i(g)$ below is the bare-array norm at height i; the map $\mathbb{V}'[g]^{\mathrm{tot}}(P;B)$ is the sum of its local evaluations in the fourth source slot of the actual step, over its finite domain set $\mathcal{D}_i$.

For clarity, write $w_i(P,B)$ for the fluctuation density of this step, including the interaction and the chart Jacobian, on its real cutoff domain Rg_{x_i}. Its normalizer and normalized measure are

$$\Xi(x_i;P) = \int_{\mathsf{Rg}_{x_i}} w_i(P,B)\,\mathrm{d}B, \qquad \nu_i[P](\mathrm{d}B) = \Xi(x_i;P)^{-1} w_i(P,B)\,\mathrm{d}B.$$

For complex P this measure need not be positive. The normalization and insertion identities below first control its total variation; positivity is then used only on the specified real chart class. The regular backgrounds and enlarged cubes are those of [8, §7.2, eq. 7.xxvi], including the stated range $d_* \geq 3$ of the background estimate.

Evaluation radii in the remaining sketches. Write $\ell = 2L^m$ and $\eta_T = L^{-T}$. Here K is the bare cutoff, T is a regularity level of a real bare configuration U, and $1 \leq j \leq T$ is the level of the pointed local family being evaluated. Regularity at radius ε means $|\partial U - 1| < \varepsilon\eta_T^2$ and $|J^{(T)}(U)| < \varepsilon$, with the lattice current of the main construction. Applying the detailed calculations with the radius of the main manuscript requires an admission check. The calculations use

$$d_*^{\mathrm{old}}(j;T,\varepsilon) = \min\left\{\frac{c_A\alpha_0 L^{2(T-j)}}{M\varepsilon}, \frac{L^{m+K-j}-4}{2M}\right\} - 1,$$

whereas the regular-family estimate in the main manuscript uses

$$d_*^{\mathrm{main}}(j;T,\varepsilon) = \min\left\{\frac{c_A\alpha_0 L^{2(T-j)}}{M\varepsilon}, \frac{L^{m+K-j}-5}{2M}\right\} - 2.$$

The source's automatic admission calculation gives a lower bound of 4 for the minimum before the final subtraction, and thereby gives $d_*^{\mathrm{old}} \geq 3$. That calculation does not give $d_*^{\mathrm{main}} \geq 3$. One must verify admission of the working and upper levels with d_*^{main}, then use this radius in the flat-top and

regular-background estimates. The calculations below describe these comparisons, but the displayed bound alone does not supply that admission check.

At the canonical source choice we retain the following numerical part of the choice explicitly. Put

$$N_1 = \frac{L}{L-1}(E_0 + \tfrac{3}{2}\alpha_0^2 B_\sharp), \qquad N_\star \geq \frac{2L-1}{L-1}N_1, \qquad K_\Lambda^\omega = (1+\beta_{\mathrm{s}})^4 K_\Lambda,$$
$$K_{\mathbf{M}}(K_\Lambda^\omega)\left(\delta(\gamma^{-2}) + s_\gamma N_1 + e^{-\gamma_2 \mathfrak{b}(\gamma^{-2})^2/2}\right) \leq E_0/4, \tag{2.ii}$$
$$s_\gamma \leq \frac{w_\star}{4K_\Lambda^\omega N_\star}, \qquad 8K_P s_\gamma \leq w_\star.$$

The allowance $N_\star$ is chosen after $E_0, B_\sharp$ and before γ. The last three requirements are upper bounds tending to zero with γ, with all other quantities already fixed. They can therefore be retained in the ordered common choice of Prop. YM3_I:69.1. They are the source-room conditions also imposed in §2.3.1 and Lem. 14.14; a half-size bound alone is not used in the source induction.

Lemma 2.2 (Normalized step integrals)**.** At an actual step of height i in [7, Def. B.16], let $P \in \mathcal{U}_{i+1}^\circ(T_\ell; \alpha_0, \alpha_1)$. The integral defining $\Xi(x_i; P)$ is absolutely convergent and nonzero. Its normalized measure $\nu_i[P]$ has finite total variation and mass one. For every bare array g through level i, the function $B \mapsto \mathbb{V}'[g]^{\mathrm{tot}}(P; B)$ is bounded and continuous on Rg_{x_i}, with

$$|\mathbb{V}'[g]^{\mathrm{tot}}(P; B)| \leq \#\mathcal{D}_i \, K_\Lambda s_i N_i(g). \tag{2.iii}$$

The first and mixed insertion identities are equation YM3_I:(2298). For a real chart point of norm less than r_α, the normalized measure is a probability measure by Prop. YM3_I:124.5. ■

Proof. The rule sending an array to the fourth potential bounds each of its local members by $K_\Lambda s_i N_i(g)$ on the step domain. Summing over the finitely many domains gives eq. 2.iii. Each member is holomorphic in the fluctuation field there, hence continuous on the real box. To evaluate the normalization, insert μg in the source slot and restrict μ so that its array norm is smaller than the remaining room. The whole-torus identity of Thm. YM3_I:V, equation YM3_I:(902), expresses the integral as the exponential of the sum of the localized output, so it is nonzero and absolutely convergent. Here $\tau = 1$ removes the additional quadratic potential. At $\mu = 0$, division by this nonzero integral gives mass one and finite total variation. The bound eq. 2.iii justifies differentiation of the normalized integral; the first derivative of its logarithm is the mean of the inserted total, and the mixed derivative is its covariance. This is precisely the argument for equation YM3_I:(2298). Positivity is asserted only at the real chart points covered by Prop. YM3_I:124.5. □

For the tower identity we use the real two-source frame of Lem. 4.5(a), retaining its actual canonical places, flat-top hypotheses and weighted progression. The chart, lowering map and chain classes are those of Section YM3_I:124. In particular, a member of $\mathsf{Cc}_n(r)$ is a real bare potential A satisfying

$$\|v_A^{[n]}\| \leq r, \qquad v_A^{[n]} = (A, J^{(n)}(e^{i\eta_n A})), \qquad e^{i\eta_n A} = U^u, \quad U \in \mathcal{C}(\mathbf{B}_n, W), \quad \varpi(W) < \varepsilon_{\mathcal{W}}\alpha_0. \tag{2.iv}$$

Thus the norm condition is accompanied by an actual background condition. The two radii in equation YM3_I:(2285) are denoted here by ϱ_{ch} and ϱ_{fl}.

Lemma 2.3 (The domains of a chain background)**.** In this frame, let $1 \leq n \leq K$, $r \geq 0$ and $A \in \mathsf{Cc}_n(r)$. The array actions through level n are defined at $\widetilde{U} = e^{i\eta_n A}$. If, in addition, $r < r' \leq \varrho_{\mathrm{ch}}$, then $\widetilde{U}$ is regular of level n at radius $3r'$, and

$$P[v_A^{[n]}] \in \mathcal{U}_n^\circ(T_\ell; \alpha_0, \alpha_1), \qquad 0 \leq a_p(\widetilde{U}) < \tfrac{9}{2}(r')^2 \eta_n^4.$$

If $r = 0$, then $A = 0$ and $\widetilde{U} = \mathbf{1}$. ■

Proof. Under the additional radius condition the chart norm is at most $r < r'$. Its image lies in the simple domain with radii $(3r', r', r')$, and in the core because $r' \le r_\alpha$. This is the usable-chart construction in equation YM3_I:(933) and the domain verification preceding equation YM3_I:(2287). Hence

$$|\partial\widetilde{U} - \mathbf{1}| < 3r'\eta_n^2, \qquad |J^{(n)}(\widetilde{U})| < r',$$

which gives regularity. To check the earlier array actions, use the background U and block field W in eq. 2.iv. The small-plaquette clause of Prop. YM3_I:52.3 puts W in the actual step domain. At height $n-1$ the canonical sum rule has its array actions defined at every representative of this orbit. This gives all levels through $n-1$; at level n the plain-core clause in the proof of the same proposition supplies the remaining pairs. At $n = 1$ there are no earlier levels to check. The trivial pairs lie in their cores as well. These are exactly the domain conditions for the absolute array actions. Their domains are gauge invariant, so they hold also at $U^u = \widetilde{U}$.

For an $SU(2)$ plaquette, $a_p(\widetilde{U}) = \frac{1}{2}|\partial\widetilde{U}(p) - \mathbf{1}|^2$; the regularity bound gives the asserted estimate. Finally the chart norm dominates $|A|$, so $r = 0$ forces $A = 0$. □

Write $\mathsf{Sum}[f](U)$ for the finite sum over the domains and base points of the family f, evaluated at the pairs of U. For $w_\lambda = \lambda_1 w_1 + \lambda_2 w_2$ and $\lambda \in \Lambda^{\mathrm{w}} = \Lambda^{\mathrm{w}}_{K-1}$, set

$$D_i^\lambda(U) = \mathcal{O}(w_\lambda)(U) + \sum_{j=1}^{i} \mathsf{Sum}[E_\lambda^{(j)} - E^{(j)}](U), \qquad \mathcal{O}(w)(U) = L^K \sum_p w(p) a_p(U),$$
$$\Phi_{a,i} = \partial_a D_i^\lambda|_0, \qquad \mathsf{F}_i = \partial_1\partial_2 D_i^\lambda|_0, \tag{2.v}$$
$$\Phi^{\mathrm{ir}}_{a,i}(U) = \sum_{j=1}^{i} (\mathsf{Sum}[\dot{E}_a^{(j)}](U) - \mathsf{Sum}[\dot{E}_a^{(j)}](\mathbf{1})).$$

These functions are defined when the array actions through level i are defined at U. Then the relative exponent is holomorphic in λ and $D_i^0 = 0$, by [7, Prop. B.8].

Proposition 2.4 (The two-source tower)**.** In the frame just specified, let $1 \le i < K$ and $r \ge 0$ satisfy

$$r < \varrho_{\mathrm{ch}}, \qquad \delta_i < \varrho_{\mathrm{fl}}, \qquad 6L^{-2}r + 8C_{\mathrm{db}}\delta_i < \varepsilon_{\mathcal{W}}\alpha_0. \tag{2.vi}$$

For $A \in \mathsf{Cc}_{i+1}(r)$ put $\widetilde{U} = e^{i\eta_{i+1}A}$, $v = v_A^{[i+1]}$, $P = P[v]$ and $\nu = \nu_i[P]$. For $B \in \mathsf{Rg}_{x_i}$ put

$$\widetilde{U}^{\mathrm{lw}}(B) = e^{i\eta_i A^{\mathrm{lw}}(v, g_i B)}.$$

Then ν is a probability measure. The relative exponents and their derivatives in eq. 2.v are defined at $\widetilde{U}$ through level $i+1$ and at $\widetilde{U}^{\mathrm{lw}}(B)$ through level i. Their latter values, as well as $\mathcal{O}(w_a)(\widetilde{U}^{\mathrm{lw}}(B))$, are bounded continuous functions of B. For every $\lambda \in \Lambda^{\mathrm{w}}$,

$$e^{D^\lambda_{i+1}(\widetilde{U})} = \int e^{D_i^\lambda(\widetilde{U}^{\mathrm{lw}}(B))}\, \mathrm{d}\nu(B). \tag{2.vii}$$

Consequently,

$$\Phi_{a,i+1}(\widetilde{U}) = \int \Phi_{a,i}(\widetilde{U}^{\mathrm{lw}})\, \mathrm{d}\nu,$$
$$\mathsf{F}_{i+1}(\widetilde{U}) = \int \mathsf{F}_i(\widetilde{U}^{\mathrm{lw}})\, \mathrm{d}\nu + \big\langle \Phi_{1,i}(\widetilde{U}^{\mathrm{lw}}); \Phi_{2,i}(\widetilde{U}^{\mathrm{lw}}) \big\rangle_\nu. \tag{2.viii}$$

■

Proof. Prop. YM3_I:124.2 applies under eq. 2.vi. It gives

$$A^{\mathrm{lw}}(v, g_i B) \in \mathsf{Cc}_i(L^{-1}r + \tfrac{11}{10}c_{\mathfrak{h}}\delta_i),$$

and Lem. 2.3 puts P in the level-$(i+1)$ core and verifies the array actions at both backgrounds. Restriction to earlier levels preserves this definedness: it retains a subset of the required core

conditions. The chart point is real and has norm less than r_α, so Prop. YM3_I:124.5 gives the asserted positivity and normalization of ν.

The source enters only through the bare array. At height i, the difference $g_\lambda = f_{\lambda,i} - f_{0,i}$ has zero counterterms and equals

$$g_\lambda = ((E_\lambda^{(j)} - E^{(j)}, 0)_{j\le i}, \emptyset) \sqcup \mathsf{W}_{L^{K-i}w_\lambda}.$$

Indeed $E_0^{(j)} = E^{(j)}$ and the counterterms were kept fixed. The action of a zero-counterterm array is the sum of its pointed totals minus their values at $\mathbf{1}$. Its weight member adds $\eta_i^{-1}\sum_p L^{K-i}w_\lambda(p)a_p(U) = \mathcal{O}(w_\lambda)(U)$. Since the plaquette action vanishes at $\mathbf{1}$, this gives

$$S[g_\lambda](U) = D_i^\lambda(U) - D_i^\lambda(\mathbf{1}).$$

Apply the array sum rule, equation YM3_I:(2289), to g_λ. The constant at $\mathbf{1}$ cancels, yielding

$$\mathbb{V}'[g_\lambda]^{\mathrm{tot}}(P;B) = D_i^\lambda(\widetilde{U}^{\mathrm{lw}}(B)) - D_i^\lambda(\widetilde{U}). \tag{2.ix}$$

The same calculation for the first and mixed derivatives of g_λ, and for its weight member alone, gives the analogous identities for $\Phi_{a,i}$, F_i and $\mathcal{O}(w_a)$. Lem. 2.2 makes these potential totals bounded and continuous on the fluctuation box; hence so are the listed functions. Finally $\Phi_{a,i}^{\mathrm{ir}} = \Phi_{a,i} - \mathcal{O}(w_a) - \sum_{j\le i}\mathsf{Sum}[\dot{E}_a^{(j)}](\mathbf{1})$, which gives the remaining continuity assertion.

Let $\Xi_\lambda(P)$ be the step integral with array $f_{\lambda,i}$ and with all other data unchanged. The finite-source estimate puts this array in the admitted norm ball. The actual one-step theorem therefore gives an absolutely convergent integral and the identity

$$\Xi_\lambda(P) = \exp\mathsf{Sum}[\mathsf{St}_i(f_{\lambda,i})](P) \neq 0. \tag{2.x}$$

Here the calibration equals the running coefficient, so $\tau = 1$ and the additional quadratic potential, bounded by a constant times $|1-\tau|$, is zero. The whole torus is an admitted localization domain, and the output radii are (α_0, α_1). These are the hypotheses for the whole-torus identity of Thm. YM3_I:V; the same specialization is used in [7, Prop. B.8].

At zero source this integral is precisely the normalization of $\nu_i[P]$, equation YM3_I:(2293). The provision is linear in its bare array, so division of the two integrals and eq. 2.ix give

$$\frac{\Xi_\lambda(P)}{\Xi_0(P)} = e^{-D_i^\lambda(\widetilde{U})}\int e^{D_i^\lambda(\widetilde{U}^{\mathrm{lw}}(B))}\,\mathrm{d}\nu(B).$$

On the other hand the recursion [7, eq. 200] and eq. 2.x give

$$\frac{\Xi_\lambda(P)}{\Xi_0(P)} = \exp\mathsf{Sum}[E_\lambda^{(i+1)} - E^{(i+1)}](P) = \exp(D_{i+1}^\lambda(\widetilde{U}) - D_i^\lambda(\widetilde{U})).$$

The sum at P equals the sum at its pair, which is the pair of $\widetilde{U}$. Comparing the two quotients proves eq. 2.vii.

To differentiate this identity, the pointwise holomorphy in the sources needs a bound uniform in the fluctuation field. Define the finite numbers

$$K_{i,a} = 2L^K\sum_p |w_a(p)| + \sum_{j\le i}\sum_{X\in\mathcal{D}_j}\#(X\cap\mathbb{T}^{(j)})d_{j,a},$$

where $d_{j,a}$ are the Lipschitz bounds in [7, eq. 196]. All pairs entering the lowered array actions lie in their cores. The family norm bounds each of their values, and $0 \le a_p(U) \le 2$ on real configurations. Consequently

$$|D_i^\lambda(\widetilde{U}^{\mathrm{lw}}(B))| \le \sum_{a=1}^{2}|\lambda_a|K_{i,a} \quad (\lambda\in\Lambda^{\mathrm{w}},\ B\in\mathsf{Rg}_{x_i}).$$

Thus the exponential is uniformly bounded on each compact source set, independently of B. Cauchy's formula and integration against the probability measure justify differentiation. Since $D_i^0 = 0$, the first derivatives give the first identity in eq. 2.viii. The mixed derivative gives the integral of

$\mathsf{F}_i + \Phi_{1,i}\Phi_{2,i}$. Subtracting the product of the two first-derivative identities leaves the covariance in its second identity. □

2.3.1. *The calibrated cutoff comparison.* We record the choices used in the comparison of two successive cutoffs. They must be made before fixing the torus or its physical coupling. The setting has three parts, denoted by (Th1)–(Th3).

In (Th1), choose the real construction constants and canonical auxiliary data as in Lem. YM3_I:118.1. Impose the global background, response, localization and actual-domain hypotheses of Section YM3_I:113, and the source and recursion hypotheses of Section YM3_I:114. In particular the fixed pair radii are (α_0, α_1), $\alpha_0 \le a_{\mathrm{H}}$, the lag weights are $L^{-(j-i)}$, and

$$E_0^{\mathrm{s}} = E_0, \qquad N_\star^{\mathrm{s}} = \frac{L}{L-1}(E_0 + \tfrac{3}{2}\alpha_0^2 B_\sharp).$$

The constants $K_{\mathrm{Q}}^\vee, K_{\mathrm{L}}^\vee, \mu_\star, \sigma_{\mathrm{src}}$ and the three smallness inequalities are those in equations YM3_I:(2170) and the surrounding source construction. Take

$$\varrho = \min\left\{\frac{20\mu_\star}{\sqrt{3}K_{\mathrm{L}}^\vee\gamma}, \frac{\sigma_{\mathrm{src}}}{2}(\gamma^{-2} - \bar{B})\right\} > 0, \qquad \mathsf{Src} = \mathsf{Src}_{\mathsf{Bc}}(2\varrho, 2\varrho).$$

Thus the coupling sequence, Gaussian structures, auxiliary data and pair cores are independent of the source.

In (Th2), fix $0 < \eth < 1$ before decreasing γ, and use the tail calibration of Section YM3_I:115:

$$c' = (1-\eth)c_\star, \qquad C_{\mathrm{maj}}^\Sigma = C_{\mathrm{maj}} + (6L^3+1)\log 2, \qquad C_{\mathrm{low}}(x) = c_{\mathrm{low}} + \tfrac{9}{2}\log x.$$

The ceiling γ is chosen so that $\log\gamma^{-2}$ exceeds both the case-II history threshold and the summability threshold specified there. These thresholds, the source smallness conditions, and the background and response conditions are part of the calibration. The proof of Lem. YM3_I:118.1 verifies that they can be imposed simultaneously. None depends on the finite lattice.

For later use put $c_{\log} = \log L/\log\gamma^{-2}$ and

$$c_\rho = \max\left\{2B_3, \frac{C_H A_1}{(1-L^{-3/2})A_0}\right\}.$$

We write $\mathsf{t}_\natural$ for the fixed calibration denoted by $\tau_\natural$ in equation YM3_I:(2134). It is the least $\tau \ge \gamma^{-2}$ at which

$$2c_\rho\varepsilon(\tau) \le \min\{\varepsilon_{\mathcal{W}}\alpha_0, c_2^{\mathrm{SO}}, \varepsilon_{\mathrm{v}}/4\},$$
$$\delta(\tau) \le \varepsilon_{\mathcal{W}}, \qquad 4C_{\mathrm{db}}\delta(\tau) \le \min\{1, r_{\mathrm{B}}, r_{\mathrm{B}}^{\mathrm{s}}/2\}.$$

The thresholds are continuous, decreasing and tend to zero; their positive fixed ceilings therefore make this minimum finite. All three inequalities persist at larger τ.

In (Th3), fix

$$K \ge 2, \quad \mathsf{b}' \ge \gamma^{-2}, \quad m \in \mathbb{N}, \quad MR(\mathsf{b}') \le L^m, \quad \ell = 2L^m, \quad x_j = L^{K-j}\mathsf{b}' \quad (0 \le j \le K).$$

Choose an integer top level $1 \le t_* \le K-1$ such that $x_{t_*} \ge \mathsf{t}_\natural$. These are precisely the finite-lattice data in Section YM3_I:115. The setting (E) retains (Th1), (Th2), the label and the side, allowing K to vary. A compact label interval $J = [b_-, b_+]$ is admissible on this side when $b_- \ge \gamma^{-2}$ and $MR(b_+) \le L^m$, as in equation YM3_I:(2201).

For a comparison involving n insertions, the depths are $u = \lceil K/[180(3n+4)]\rceil$ and $s = \lceil K/9\rceil$. The errors and smallness conditions are those in Section YM3_I:116; the additional support condition is $d_\infty(\underline{f})L^u \ge 6M$ when $n \ge 2$. For bounded functions of block fields one takes $n = 1$ and omits this support condition. §2.3.2 displays all the resulting errors and conditions (k0)–(k4), (k6).

Lemma 2.5. Let $0 < g \le \gamma$ and $\ell = 2L^m \in \mathfrak{L}(g)$. Then

$$J_g = [g^{-2} + B_\sharp - L^{-1}\bar{B},\ X_g + \bar{B}]$$

is an admissible label interval on T_ℓ, and it contains $I(g)$ and every first-passage label at g. The larger interval $\mathsf{J}_g = [\gamma^{-2}, L(X_g + \bar{B})]$ is also admissible on T_ℓ. ■

Proof. The lower endpoint is at least γ^{-2}, since $B_\sharp - L^{-1}\bar{B} = B_\sharp(L-2)/(L-1) \ge 0$. The side floor in equation YM3_I:(1528) gives $L^m \ge LMR(X_g + \bar{B})$, and hence the required room $MR(\max J_g) \le L^m$. The containment of $I(g)$ follows from its endpoints. The first-passage label bound is equation YM3_I:(1531). For the larger interval, the radius scaling inequality in the proof of Prop. YM3_I:74.1 gives $R(LX) \le LR(X)$ for $X \ge \gamma^{-2}$. Apply it at $X = X_g + \bar{B}$ and use the same side floor to obtain $MR(\max \mathsf{J}_g) \le LMR(X_g + \bar{B}) \le L^m$. □

2.3.2. *The laws of the block fields.* The cutoff comparison also controls bounded functions of a fixed block field. We prove the resulting total-variation convergence here. We use the calibrated small-field setting of Section YM3_I:115: all constants, auxiliary data, source radii and tail thresholds are fixed before the cutoff, with $E_0^{\mathrm{s}} = E_0$, $N_\star^{\mathrm{s}} = L(E_0 + 3\alpha_0^2 B_\sharp/2)/(L-1)$, $\mathsf{b} \ge \gamma^{-2}$, $MR(\mathsf{b}) \le L^m$ and $\ell = 2L^m$. This is the setting denoted by (E). The precise calibration, including the source and tail smallness conditions, is that of Prop. YM3_I:115.2, not merely the displayed lower bound on b.

At cutoffs K and $K+1$ use the same physical depth, and put

$$u = \lceil K/1260 \rceil, \qquad s = \lceil K/9 \rceil, \qquad t_1 = K - u, \qquad t_0 = K - s.$$

$$x_{[d]} = L^d \mathsf{b}, \qquad N_{[d]} = \ell^3 L^{3d}, \qquad y_{[d]} = (\log x_{[d]})^{2\mathfrak{p}_0}.$$

The errors in Prop. YM3_I:116.1 and the discarded mass in Prop. YM3_I:115.2 are

$$\begin{aligned}
\epsilon_{\mathrm{fg}} &= C_{\mathrm{old}} C_{\mathrm{fg}} \frac{L}{L-1} E_0 N_{[u]} L^{-(s-u)}, \\
\epsilon_{\mathrm{cl}} &= 4\mathsf{K}_{\mathrm{w}} c_\rho^2 A_0^2 (1 + 2\varrho/\mathsf{b}) y_{[K]} N_{[s]} \sqrt{K} L^{-(K-s)/2}, \\
\epsilon_{\mathrm{sh}} &= 3(s-u) N_{[s]} \exp[C_{\mathrm{sh}} - \kappa_{\mathrm{b}} A_1^2 y_{[u]}/4], \\
\mathsf{R}_{t_1} &= 2M^{-3} N_{[u]} \exp[-c' A_0^2 y_{[u]} + (\mathsf{C}_{\mathrm{maj}}^{\Sigma} + 2 + C_{\mathrm{low}}(\mathsf{b}))\ell^3].
\end{aligned}$$

Here $c' = (1 - \eth) c_\star > 0$, with the fixed $0 < \eth < 1$ of that calibration. We retain the following names for the comparison conditions:

$$\text{(k0)}\quad 1 \le u < s < K, \qquad \text{(k1)}\quad L^u \mathsf{b} \ge \Theta_\star,$$

$$\text{(k2)}\quad \epsilon_{\mathrm{fg}} \le 1/8, \qquad \text{(k3)}\quad \epsilon_{\mathrm{cl}}, \epsilon_{\mathrm{sh}} \le 1/20,$$

$$\text{(k4)}\quad 256\mathsf{K}_u c_\rho^2 A_0^2 N_{[s]} L^{-(K-s)} \le \gamma_{\mathrm{r}} A_1^2, \qquad \text{(k6)}\quad \mathsf{R}_{t_1} \le 1/2.$$

There is no support-separation condition in this measure statement. The constants $\Theta_\star$, K_u, K_{w} and the other comparison constants are those in Sections YM3_I:116 and YM3_I:115, with identical level units. In particular the block lattice at depth d is $\Lambda_{d,\ell}$ on both cutoffs.

For uniformity in b, let $\mathsf{J} = [\mathsf{b}_-, \mathsf{b}_+]$ satisfy $\mathsf{b}_- \ge \gamma^{-2}$ and $MR(\mathsf{b}_+) \le L^m$. Define $\bar{\epsilon}_{\mathrm{sh}}$ by replacing b by b_- in $y_{[u]}$, and define $\bar{\mathsf{R}}$ by that same replacement in its negative exponent and by replacing $C_{\mathrm{low}}(\mathsf{b})$ with $C_{\mathrm{low}}(\mathsf{b}_+)$. They dominate their unbarred counterparts. The uniform threshold K_0 in Thm. YM3_I:AR, taken at $n = 1$, makes (k0)–(k4) and (k6) valid for all $K \ge K_0$. Its proof shows that $K^a e^{a'K}(\bar{\mathsf{R}} + \bar{\epsilon}_{\mathrm{sh}}) \to 0$ for every fixed real a, a'. The reason is $u \ge K/1260$ and $2\mathfrak{p}_0 \ge 2$ in the negative exponents.

We first record the measure-theoretic passage to the limit.

Lemma 2.6 (A summable sequence of probability measures)**.** Let $(\Omega, \mathfrak{B})$ be a measurable space, let $K_* \in \mathbb{N}$, and let μ^K, $K \geq K_*$, be probability measures on $\mathfrak{B}$. Let $q_K \geq 0$ be numbers with $\sum_{K \geq K_*} q_K < \infty$ and

$$\sup_{E \in \mathfrak{B}} |\mu^{K+1}(E) - \mu^K(E)| \leq q_K \qquad (K \geq K_*).$$

Then $\mu(E) := \lim_{K\to\infty} \mu^K(E)$ exists for every $E \in \mathfrak{B}$,

$$\sup_{E \in \mathfrak{B}} |\mu^K(E) - \mu(E)| \leq \sum_{k \geq K} q_k \qquad (K \geq K_*),$$

and μ is a probability measure on $\mathfrak{B}$. ■

Proof. Step 1: the limit and the bound. For $K' > K \geq K_*$ and $E \in \mathfrak{B}$,

$$|\mu^{K'}(E) - \mu^K(E)| \leq \sum_{k=K}^{K'-1} q_k \leq \sum_{k \geq K} q_k,$$

and the right-hand side tends to 0 as $K \to \infty$. Hence $(\mu^K(E))_K$ is a Cauchy sequence, and its limit $\mu(E)$ exists. Letting $K' \to \infty$ in the display gives the asserted bound, uniformly in E.

Step 2: elementary properties. As limits of the corresponding properties of the μ^K, we have $\mu(E) \geq 0$ for all $E \in \mathfrak{B}$, $\mu(\emptyset) = 0$, $\mu(\Omega) = 1$, and μ is finitely additive.

Step 3: σ-additivity. Let $E_1, E_2, \cdots \in \mathfrak{B}$ be pairwise disjoint and put $R_N := \bigcup_{i>N} E_i$. By finite additivity,

$$\mu\Big(\bigcup_i E_i\Big) = \sum_{i \leq N} \mu(E_i) + \mu(R_N).$$

Let $\eta > 0$, and choose $K \geq K_*$ with $\sum_{k \geq K} q_k \leq \eta$. By Step 1, $\mu(R_N) \leq \mu^K(R_N) + \eta$ for every N. Since μ^K is a measure and R_N decreases to $\emptyset$, we have $\mu^K(R_N) \to 0$ as $N \to \infty$; hence $\limsup_N \mu(R_N) \leq \eta$. As $\eta > 0$ is arbitrary and $\mu(R_N) \geq 0$, $\mu(R_N) \to 0$, and $\mu(\bigcup_i E_i) = \sum_i \mu(E_i)$. □

Definition 2.7 (The laws of the block fields)**.** Assume (E), with label b and side $\ell = 2L^m$.

(1) At cutoff K set $\beta = L^K \mathsf{b}$ and

$$\rho_0(U) = e^{-A^{a_K}(\mathsf{b},U)}, \qquad Z_K = \int \rho_0 \, \mathrm{d}U > 0, \qquad \langle H \rangle_K = Z_K^{-1} \int H \rho_0 \, \mathrm{d}U.$$

Here dU is product Haar probability measure and $A^{a_K}(\mathsf{b}, U) = \beta \sum_p a_p(U)$ is the Wilson action.

(2) For $K \geq \max\{d, 1\}$ the law at depth d is

$$\mu_d^K(E) = \langle \mathbf{1}_E(\mathcal{M}^{K-d}U) \rangle_K \qquad (E \subset \mathcal{V}(\Lambda_{d,\ell}) \text{ Borel}). \tag{2.xi}$$

The averaging map is Borel and gauge covariant. Its map from $\Lambda_{d+1,\ell}$ to $\Lambda_{d,\ell}$ is independent of the bare cutoff: its definition uses precisely these two lattices. These properties are proved in Prop. YM3_I:4.5. Thus μ_d^K is a Borel probability measure.

(3) Write $\mathfrak{B}_d$ for the sigma-algebra of invariant Borel sets in $\mathcal{V}(\Lambda_{d,\ell})$. Gauge transformations are bijections, so complements and countable unions of invariant sets remain invariant. The map $\mathcal{M}$ in the consistency identity below is the fixed map from depth $d+1$ to depth d.

(4) For $K \geq 2$, use the errors above and set

$$\mathfrak{o}_K = 8\mathsf{R}_{t_1} + 12\epsilon_{\mathrm{fg}} + 8\epsilon_{\mathrm{cl}} + 12\epsilon_{\mathrm{sh}}. \tag{2.xii}$$

(5) For an interval $\mathsf{J} = [\mathsf{b}_-, \mathsf{b}_+]$ as above, put

$$\mathsf{C}_{\mathrm{bf}} := 2\Big[1 + 12\, C_{\mathrm{old}} C_{\mathrm{fg}} \tfrac{L}{L-1} E_0 \, L^4 + 32\, \mathsf{K}_{\mathrm{w}} c_\rho^2 A_0^2 (1 + \tfrac{2\varrho}{\mathsf{b}_-}) \big(\log(L\mathsf{b}_+)\big)^{2\mathfrak{p}_0} L^{7/2}\Big] \tag{2.xiii}$$

and

$$K_{\mathrm{bf}} := \min\Big\{k \in \mathbb{Z} :\ k \geq K_0,\ \text{and}\ \ 8\,\bar{\mathsf{R}} + 12\,\bar{\epsilon}_{\mathrm{sh}} \leq \tfrac{1}{2}\,\mathsf{C}_{\mathrm{bf}}\,\ell^3 K^{\mathsf{p}_0} L^{-K/10}\ \ \text{at all integers } K \geq k\Big\},$$

where $\mathsf{p}_0 = 2\mathfrak{p}_0 + 1/2$ and K_0 is the uniform threshold specified above. This minimum exists: the bracket defining C_{bf} is at least 1, and the barred errors decay faster than every exponential in K. Its dependencies are only ℓ, the endpoints of J and the fixed constants.

■

The comparison of the two lattices made above also controls the laws of Def. 2.7, and Lem. 2.6 turns this control into their convergence as the cutoff tends to infinity.

Corollary 2.8 (Convergence of the laws of the block fields)**.** Assume the setting (E), and take u, s, t_1, t_0 and the comparison errors defined above.

(a) Let $K \geq 2$ be an integer at which the conditions (k0), (k1), (k2), (k3), (k4) and (k6) hold. Then, for every integer $0 \leq d \leq u$ and every bounded Borel function $F : \mathcal{V}(\Lambda_{d,\ell}) \to \mathbb{C}$ invariant under the gauge transformations of $\Lambda_{d,\ell}$,

$$\Big|\big\langle F(\mathcal{M}^{K+1-d}U)\big\rangle_{K+1} - \big\langle F(\mathcal{M}^{K-d}U)\big\rangle_{K}\Big| \ \leq\ \mathfrak{o}_K\ \sup|F|. \tag{2.xiv}$$

If moreover J is an interval of labels under (Tor) on the side ℓ with $\mathsf{b} \in \mathsf{J}$, then at every integer $K \geq K_{\mathrm{bf}}$ the conditions (k0)–(k4) and (k6) hold, and

$$\mathfrak{o}_K\ \leq\ \mathsf{C}_{\mathrm{bf}}\,\ell^3\,K^{\mathsf{p}_0}\,L^{-K/10}; \tag{2.xv}$$

here C_{bf} and K_{bf} are the numbers of Def. 2.7(5), which depend on the label b only through J.

(b) Let K_{bf} be taken at $\mathsf{J} = \{\mathsf{b}\}$. For every integer $d \geq 0$ and every $E \in \mathfrak{B}_d$ the limit $\mu_d(E) := \lim_{K\to\infty} \mu_d^K(E)$ exists, and

$$\sup_{E\in\mathfrak{B}_d} |\mu_d^K(E) - \mu_d(E)| \ \leq\ \sum_{K'\geq K} \mathfrak{o}_{K'} < \infty \qquad \text{for every integer } K \geq K_{\mathrm{bf}} \text{ with } \lceil K/1260\rceil \geq d. \tag{2.xvi}$$

The set function μ_d is a probability measure on $\mathfrak{B}_d$. The family $(\mu_d)_{d\geq 0}$ is consistent: if $E \in \mathfrak{B}_d$, then $\mathcal{M}^{-1}(E) \in \mathfrak{B}_{d+1}$ and

$$\mu_{d+1}(\mathcal{M}^{-1}E) = \mu_d(E), \tag{2.xvii}$$

where $\mathcal{M} : \mathcal{V}(\Lambda_{d+1,\ell}) \to \mathcal{V}(\Lambda_{d,\ell})$ is the averaging map of Def. 2.7(3).

(c) For every integer $d \geq 0$, every integer $K \geq \max\{d, 1\}$ and every Borel set $E \subset \mathcal{V}(\Lambda_{d,\ell})$,

$$\mu_d^K(E) = \int \bar{\mathbf{1}}_E \,\mathrm{d}\mu_d^K, \qquad \bar{\mathbf{1}}_E(V) := \int \mathbf{1}_E(V^v)\,\mathrm{d}v, \tag{2.xviii}$$

where dv is the product of one Haar probability measure of G for each site of $\Lambda_{d,\ell}$, a measure on the group of the gauge transformations v of $\Lambda_{d,\ell}$. The function $\bar{\mathbf{1}}_E$ is Borel, takes its values in $[0, 1]$ and is gauge invariant. Consequently the limit in (b), the bound eq. 2.xvi and the identity eq. 2.xvii hold for all Borel sets: $\mu_d^K \to \mu_d$ in total variation, that is, uniformly on the Borel subsets of $\mathcal{V}(\Lambda_{d,\ell})$; μ_d is a Borel probability measure on $\mathcal{V}(\Lambda_{d,\ell})$ that is gauge invariant; and the image of μ_{d+1} under $\mathcal{M}$ is μ_d.

■□

Proof. We first compare the normalized means at depth u. On either lattice $\bullet \in \{\mathrm{A}, \mathrm{B}\}$, let $\Lambda_0^\bullet$ be the full unnormalized law at that depth, $\Lambda_1^\bullet$ its small-field chain restricted to $\mathsf{In}_{[u]}$, and $\Lambda_2^\bullet$ the chain restarted from the classical action at depth s, again restricted to $\mathsf{In}_{[u]}$. We regard them as functionals on bounded gauge-invariant Borel functions F of the depth-u field. These are precisely the chain and classical-start measures used in Props. YM3_I:115.2 and YM3_I:116.1, with source

zero. All three functionals are positive and have strictly positive mass. For the full law this is immediate. The chain is the image of the nonnegative small-field weight times the bare density, by Prop. YM3_I:115.2. For the classical start, the initial density is positive, the chart kernels preserve positivity, and the final exponential density does not vanish on the open set $\mathsf{In}_{[u]}$ containing the trivial configuration; these are the positivity statements in the proof of Prop. YM3_I:116.1.

We use the following elementary normalization bound. If positive functionals P, Q satisfy $P(1) > 0$ and

$$|Q(F) - P(F)| \le \epsilon \|F\|_\infty P(1), \qquad 0 \le \epsilon \le 1/2,$$

then $Q(1) \ge (1-\epsilon)P(1) > 0$, and subtraction of the two normalized fractions gives

$$\left| \frac{Q(F)}{Q(1)} - \frac{P(F)}{P(1)} \right| \le \frac{2\epsilon}{1-\epsilon} \|F\|_\infty \le 4\epsilon \|F\|_\infty.$$

The discarded-mass estimate in Prop. YM3_I:115.2 gives $0 \le \Lambda_0^\bullet - \Lambda_1^\bullet$ on this algebra and

$$(\Lambda_0^\bullet - \Lambda_1^\bullet)(1) \le \mathsf{R}_{t_1} \Lambda_0^\bullet(1).$$

Consequently their normalized means differ by at most $4\mathsf{R}_{t_1}\|F\|_\infty$.

Forgetting the early array changes the final exponent by at most ϵ_{fg}, by equation YM3_I:2180 and the definition of ϵ_{fg} above. Write the two final densities as $p_1 = e^{c^\bullet + \Phi^\bullet}$ and $p_2 = e^{\hat{c}^\bullet + \hat{\Phi}^\bullet}$ on $\mathsf{In}_{[u]}$. Positivity gives $p_1 = |p_1|$ almost everywhere and $p_2 = |p_2| > 0$ there. Thus, with $\varkappa = |e^{c^\bullet - \hat{c}^\bullet}|$,

$$\frac{p_1}{p_2} \in \varkappa[e^{-\epsilon_{\mathrm{fg}}}, e^{\epsilon_{\mathrm{fg}}}] \quad \text{almost everywhere.}$$

Since $\epsilon_{\mathrm{fg}} \le 1/8$, $e^{\epsilon_{\mathrm{fg}}} - 1 \le (5/4)\epsilon_{\mathrm{fg}} \le 1/2$. Apply the normalization bound to $P = \varkappa\Lambda_2^\bullet$, $Q = \Lambda_1^\bullet$. The factor $\varkappa$ cancels, and these two normalized means differ by at most $5\epsilon_{\mathrm{fg}}\|F\|_\infty$.

The two classical starts can be written on the common depth-s configuration space as $k^\bullet \hat{F}_0^\bullet \, \mathrm{d}W$, followed by the same block averaging down to depth u. The final total-variation estimate in the proof of Prop. YM3_I:116.1 is

$$\int |k^{\mathrm{A}} \hat{F}_0^{\mathrm{A}} - k^{\mathrm{B}} \hat{F}_0^{\mathrm{B}}| \, \mathrm{d}W \le (2\epsilon_{\mathrm{cl}} + 3\epsilon_{\mathrm{sh}}) \Lambda_2^{\mathrm{A}}(1).$$

We will also use the relative version of the mismatch bound. For every bounded nonnegative gauge-invariant function H of the final block field, the proof of Prop. YM3_I:116.1 gives

$$\int H(\mathcal{M}^{s-u} W) \, k^\bullet \mathbf{1}_{\{k^\circ = 0\}} \hat{F}_0^\bullet \, \mathrm{d}W \le \epsilon_{\mathrm{sh}} \Lambda_2^\bullet(H).$$

Indeed, at a mismatching test some fluctuation of lattice $\bullet$ exceeds half its threshold. Prop. YM3_I:114.4 bounds the marked next measure by

$$\exp(C_{\mathrm{sh}} - \kappa_{\mathrm{b}} \mathfrak{b}(x_{(k)})^2/4)$$

times the ordinary next measure. Each later positive cut and image operation preserves this order. Sum over at most $s-u$ steps and $3N_{[s]}$ bonds at each step, and use $\mathfrak{b}(x_{(k)})^2 \ge A_1^2 y_{[u]}$ to obtain the displayed bound. This proves an inequality of measures before testing against H.

The total-variation proof bounds separately the mass where a fluctuation test differs and the difference of the classical action on the common cut. Inserting F in this inequality costs only $\|F\|_\infty$. By (k3), $2\epsilon_{\mathrm{cl}} + 3\epsilon_{\mathrm{sh}} \le 1/4$, so normalization gives an error $(8\epsilon_{\mathrm{cl}} + 12\epsilon_{\mathrm{sh}})\|F\|_\infty$. Adding the five comparisons

$$\Lambda_0^{\mathrm{A}}, \ \Lambda_1^{\mathrm{A}}, \ \Lambda_2^{\mathrm{A}}, \ \Lambda_2^{\mathrm{B}}, \ \Lambda_1^{\mathrm{B}}, \ \Lambda_0^{\mathrm{B}}$$

bounds the difference of the normalized full laws by

$$[8\mathsf{R}_{t_1} + 10\epsilon_{\mathrm{fg}} + 8\epsilon_{\mathrm{cl}} + 12\epsilon_{\mathrm{sh}}]\|F\|_\infty \le \mathfrak{o}_K \|F\|_\infty.$$

For $d \le u$, apply this to $F \circ \mathcal{M}^{u-d}$. Block averaging is Borel and gauge covariant, and its maps at equal physical depth coincide. Hence this composition remains bounded and gauge invariant, with no increase of its supremum norm. This proves eq. 2.xiv.

For the rate, the ceiling inequalities in Thm. YM3_I:AR give $4u-s \le -K/10+4$ and $(7s-K)/2 \le -K/9+7/2$. Therefore

$$12\epsilon_{\mathrm{fg}} \le 12 C_{\mathrm{old}} C_{\mathrm{fg}} \frac{L}{L-1} E_0 L^4 \ell^3 L^{-K/10},$$

and, since $\log(L^K \mathsf{b}_+) \le K \log(L\mathsf{b}_+)$,

$$8\epsilon_{\mathrm{cl}} \le 32 \mathsf{K}_{\mathrm{w}} c_\rho^2 A_0^2 (1 + 2\varrho/\mathsf{b}_-)(\log(L\mathsf{b}_+))^{2\mathfrak{p}_0} L^{7/2} \ell^3 K^{2\mathfrak{p}_0+1/2} L^{-K/9}.$$

Their sum is at most half the right side of eq. 2.xv; the other half bounds $8\mathsf{R}_{t_1} + 12\epsilon_{\mathrm{sh}}$ by the choice of K_{bf}. This proves the uniform rate.

For (b), fix d and let K_* be the least $K \ge K_{\mathrm{bf}}$ with $\lceil K/1260 \rceil \ge d$. For every invariant Borel set E, apply (a) to $F = \mathbf{1}_E$. The resulting differences of the probability measures are bounded by the summable sequence $\mathfrak{o}_K$, since a fixed power of K times $L^{-K/10}$ is summable. Lem. 2.6 gives the limit, the rate and countable additivity. Gauge covariance makes $\mathcal{M}^{-1}E$ invariant, and the exact finite-cutoff identity

$$\mu_{d+1}^K(\mathcal{M}^{-1}E) = \mu_d^K(E)$$

passes to the limit. This proves consistency.

For (c), invariance of the Wilson density and Haar measure makes the bare law gauge invariant. Gauge covariance of block averaging and extension of a coarse site gauge to the fine sites give the same invariance for μ_d^K. The action $(V, v) \mapsto V^v$ is continuous, so Fubini's theorem shows that $\bar{\mathbf{1}}_E$ is Borel. Right invariance of Haar measure gives

$$\bar{\mathbf{1}}_E(V^{v'}) = \int \mathbf{1}_E(V^{vv'})\, \mathrm{d}v = \bar{\mathbf{1}}_E(V).$$

Integrating the invariance of μ_d^K over v proves eq. 2.xviii. Now apply (a) to this function, whose values lie in $[0,1]$. The same summable bound holds for every Borel set E, so Lem. 2.6 applies on the full Borel sigma-algebra. The finite-cutoff consistency and gauge invariance pass to these limits. For invariant E, $\bar{\mathbf{1}}_E = \mathbf{1}_E$, so this extension agrees with the measure in (b). □

2.3.3. *Notation and hypotheses. Notation.* All constants depend only on the basic constants (a common real choice as in Prop. YM3_I:69.1, among them $\mathfrak{p}_0$, A_0, $M = L^{m'}$, C_{lat} and the last, γ) and on the arguments shown; a constant is called basic if it depends on the basic constants alone. The blocking factor L is odd; $a_K = L^{-K}$, $\mathbb{L}_K = a_K\mathbb{Z}^3$; T_ℓ is the torus of side $\ell = 2L^m$ and $\Lambda_{K,\ell}$ its lattice of spacing a_K; $\langle\cdot\rangle_{K,\ell,\beta}$ is Wilson's measure on $\Lambda_{K,\ell}$, $\langle \mathcal{A}_1; \dots; \mathcal{A}_n \rangle$ a cumulant, $\mathcal{G}_{K,\beta}$ the set of the periodic Gibbs states on $\mathbb{L}_K$ at β, that is, of the Gibbs states of Wilson's weight $e^{-\beta\sum_p a_p}$ that are invariant under the translations of a subgroup of finite index of $a_K\mathbb{Z}^3$ ([7, §K.1]); by Prop. YM3_I:103.4 every $\nu \in \mathcal{G}_{K,\beta}$ is invariant under $a_K\mathbb{Z}^3 \rtimes W_3$ and reflection positive at the lattice mirrors $\{n \cdot x = c\}$, $n \in \{e_\mu, e_\mu \pm e_\lambda\}$, of the nine families.

On $\mathbb{R}^3$ and on T_ℓ, $|x|$ is the Euclidean norm and $|x|_\infty = \max_\mu |x_\mu|$ the sup norm; the balls $B(x,r)$ (open) and $\bar{B}(x,r)$ (closed), the distances $|x-y|$ of points and the distance dist of sets are Euclidean. Where sets or supports are said to be pairwise $\ge \delta$ apart, to have a diameter $\le \varrho$, or where a weight vanishes within δ of a set, the sup distance dist_∞ is meant, unless the Euclidean distance is named, as in Thm. B(a), (c). In Lems. [7, Lem. C.15] and [7, Lem. C.16], in [7, Thm. BV](e) and in Thm. B(c), (d) the separations of sets are read with the Euclidean distance $\mathrm{dist}(A,B) = \inf_{x\in A, y\in B}|x-y|$, which these statements name; the factor $1/\sqrt{3}$ between it and the sup distance is paid inside their proofs.

The depth at the level k of the renormalisation is $s = K-k$ ([7, result G.2]); the depth of a distance d is the least $s = s(d) \ge 0$ with $L^s d \ge C_W(s)$, where $C_W(s) = c_W(\log \bar{x}_s)^{2\mathfrak{p}_0}$, $\bar{x}_s = L^{s+1}(g^{-2} + \bar{B})$

and c_W is basic; it depends on g, it does not decrease as d falls or as g^{-2} grows, and $s(d) \to \infty$ as $d \to 0$; if $s(d) \geq 1$, then $C_W(s) \leq L^s d < LC_W(s)$ at $s = s(d)$, so that d is about $C_W(s)$ units of the level of that depth, and, for $0 < d < 1$, $Y_{s(d)} := \log \bar{x}_{s(d)} \leq 2 \log \frac{1}{d} + C(g)$. The reference coupling $g > 0$ is that of YM3_I:74, $\mathfrak{d}(g)$ is the set of its first-passage data (K, β) (YM3_I:74, YM3_I:74), g_t is the running coupling at the level t, and $\mathsf{b} = a_K \beta$ is the label ([7, result G.2]). $B_\sharp$ bounds $|L^{-1} x_{j-1} - x_j|$ as in [7, §G.2], $\bar{B} = B_\sharp L/(L-1)$, $X_g = L(g^{-2} + B_\sharp)$ as in YM3_I:(1526), $I(g) = (g^{-2} + \bar{B}, X_g - \bar{B}]$ as in YM3_I:(1527), and $\mathfrak{L}(g) = \{2L^m : m \geq m_{\mathfrak{L}}(g)\}$ (YM3_I:(1528), YM3_I:(1528)) is the set of the sides at g, ℓ_g its least member.

Every $(K, \beta) \in \mathfrak{d}(g)$ has $g^{-2} < a_K \beta \leq L(g^{-2} + \bar{B})$, and $(K, L^K \mathsf{b}) \in \mathfrak{d}(g)$ for every integer $K \geq 1$ if $\mathsf{b} \in I(g)$. (H) is the condition $0 < g \leq \gamma_W^{\mathrm{ref}}$, where $\gamma_W^{\mathrm{sd}} \leq \gamma_W^{\mathrm{ref}} \leq \gamma$ are the thresholds of the choices in equations YM3_I:(2326)–(2327); (T) is (H) together with $\ell \in \mathfrak{L}(g)$ and $\mathsf{b} \in I(g)$, and then $\langle \cdot \rangle_K := \langle \cdot \rangle_{K,\ell,L^K \mathsf{b}}$.

The field. $V = \mathrm{Sym}_3(\mathbb{R}) = V_A \oplus V_E \oplus V_T$, the multiples of $\mathbf{1}$, the traceless diagonal matrices and the matrices with zero diagonal, is the splitting under the group W_3 of the 48 signed permutation matrices; $V_{\mathbb{C}} = \mathrm{Sym}_3(\mathbb{C})$; $E_{\lambda\lambda'}$ are the matrix units and $e_{ij} = E_{ij} + E_{ji}$. As $F_{\mu\nu} = \varepsilon_{\mu\nu\lambda} B_\lambda$ in three dimensions, the six fields are the entries of $T_{\lambda\lambda'} = \langle B_\lambda, B_{\lambda'} \rangle$. A weight is a bounded Borel function H on $\mathbb{R}^3$ (or on T_ℓ) with values in V or $V_{\mathbb{C}}$ and compact support, $|H| = \sup_x \max_{\lambda\lambda'} |H_{\lambda\lambda'}(x)|$; an isometry $rx = Rx + v$ acts by $(rH)(x) = RH(r^{-1}x)R^t$, so that a reflection acts on points and on indices.

At a site x the 24 plaquette loops are $W_{x,l}$, $l = (\alpha, \alpha')$ an ordered pair of signed directions with $\alpha' \neq \pm\alpha$; $\lambda(l)$ is the normal of the plane of l and $s(l) = \operatorname{sgn} \alpha \operatorname{sgn} \alpha' \, \varepsilon_{\lambda(l)|\alpha||\alpha'|}$ its orientation. With $w(U) = \frac{U - U^{-1}}{2i}$, $Q_x(l, l') = \frac{1}{2} \mathrm{tr}\, w(W_{x,l}) w(W_{x,l'})$, $a_p = 1 - \frac{1}{2} \mathrm{tr}\, U(\partial p)$, and, for a bond $b = \langle x, x + a_K e_\mu \rangle$ with $\lambda_b < \lambda_b'$ the two other directions, $q_b = \frac{1}{4} \sum_{\epsilon,\epsilon' = \pm} \epsilon\epsilon' Q_x((\mu, \epsilon\lambda_b), (\mu, \epsilon'\lambda_b'))$, the lattice form of $\frac{1}{2} \int \mathrm{tr}(HT)$ is [7, eq. 1] ($x(p)$, x_b the midpoints, $\lambda(p)$ the normal of p). $O_K^T(f\mathbf{1}) = \mathcal{O}(f)$ is the action density, and $\mathcal{O}_\lambda(\phi) = O_K^T(\phi E_{\lambda\lambda}) = a_K^{-1} \sum_{\lambda(p) = \lambda} \phi(x(p)) a_p$ are the three plane fields. $\bar{O}_K^T(H)$ is [7, eq. 1] with $H(x(p))$, $H(x_b)$ replaced by the mean of H over the cell $x + a_K[-\frac{1}{2}, \frac{1}{2})^3$, x the least corner of p, resp. of b. A corner field is $\mathcal{Q}_{l,l';\xi}(\phi) = a_K^{-1} \sum_x \phi(x + a_K \xi) Q_x(l, l')$, $|\xi|_\infty \leq \frac{1}{2}$; a site form is $a_K^{-1} \sum_x \sum_{l,l'} v(x; l, l') Q_x(l, l')$, of norm $\|v\| = \sup_x \sum_{l,l'} |v(x; l, l')|$; a mixed source is a pair $\mathfrak{s} = (w, \mathsf{w})$ of real weights of the plaquettes and of the bonds, with the lattice form $\mathcal{O}^T(\mathfrak{s}) = a_K^{-1}[\sum_p w(p) a_p - \sum_b \mathsf{w}(b) q_b]$ and $|\mathfrak{s}| = \sup |w| + 8 \sup |\mathsf{w}|$. $\mathfrak{S}(O)$ is the set of the tuples $X = (H_1, \dots, H_n)$, $n \geq 0$, of continuous weights with pairwise disjoint supports in an open set O, and $\mathfrak{F}(O)$ its complex span; rX, θF, $\tau_v F$ and $\bar{F}$ act entry by entry, $F' \cdot F$ concatenates, and $A_K^{\nu'}(X) = \prod_a [O_K^T(H_a) - \nu'(O_K^T(H_a))]$, $A^\nu[\mathcal{A}] = \mathcal{A} - \nu(\mathcal{A})$. A pair at scale d is a pair of real weights vanishing off closed balls $\bar{B}(x_0, d_-)$, $\bar{B}(y_0, d_-)$, $0 < d_- < d$.

Free values and the stress tensor. Let

$$\mathcal{W}(H_1, H_2) = \tfrac{3}{2} \iint \mathrm{tr}[H_1(x) K_3(x - y) H_2(y) K_3(x - y)] \, \mathrm{d}x \, \mathrm{d}y, \qquad K_3(u) = \frac{3\hat{u}\hat{u}^t - \mathbf{1}}{4\pi |u|^3}. \tag{2.xix}$$

The free values are the separated connected Wick contractions, with no contraction within one insertion, as in [7, Thm. V](f)(i). Use three independent colors with off-diagonal covariance $\langle B_\lambda^c(x) B_\mu^d(y) \rangle = \delta^{cd} \mathsf{b}^{-1} K_3(x - y)_{\lambda\mu}$ for $x \neq y$. The cumulants of the smeared bilinears formally denoted by $\frac{1}{2} \int \mathrm{tr}(H_a T^{\mathrm{free}})$, $T^{\mathrm{free}} = \sum_c B^c (B^c)^t$, at separated supports are: $\mathsf{b}^{-2} \mathcal{W}(H_1, H_2)$ at two points; the kernel $3\mathsf{b}^{-3} k_3$, $k_3(x, y, z) = \mathrm{tr}[K_3(x - y) K_3(y - z) K_3(z - x)]$, for three action densities and $3\mathsf{b}^{-4} k_4$, k_4 the sum over the three rings $(x_{\varsigma(1)}, \dots, x_{\varsigma(4)})$ of $\mathrm{tr} \prod_l K_3(x_{\varsigma(l)} - x_{\varsigma(l+1)})$, for four; and $3\mathsf{b}^{-n} K_{\underline{\lambda}}^{(n)}(\phi)$ for n plane fields $\mathcal{O}_{\lambda_a}(\phi_a)$.

On the lattice, $3\mathsf{b}^{-n} K_K^{(n)}(w_1, \dots, w_n)$ is the cumulant of the plaquette forms $a_K^{-1} \sum_p w_a(p) a_p$ for three free lattice photon fields, that is with a_p replaced by $\frac{1}{2} \sum_c (F_p^c)^2$, the F^c independent Gaussian

plaquette variables of covariance $\beta^{-1}C^{\mathrm{free}}$, C^{free} the free lattice covariance; likewise, for mixed sources the free lattice value of the cumulant of $\mathcal{O}^T(\mathfrak{s}_1)$, $\mathcal{O}^T(\mathfrak{s}_2)$ is the cumulant obtained when every a_p and every $Q_x(l,l')$ is replaced by its quadratic part in three such independent fields; it is $\frac{3}{2}\mathsf{b}^{-2}\mathrm{Tr}(\mathsf{W}_1 C^{\mathrm{free}}\mathsf{W}_2 C^{\mathrm{free}})$, W_a the symmetric matrix with $\mathcal{O}^T(\mathfrak{s}_a) = (2a_K)^{-1}\sum_c F^{c\,t}\mathsf{W}_a F^c$ after this replacement. $\mathsf{w}_0 = \frac{1}{12}388^{-6}$.

For $\eta \in C^2_c(\mathbb{R}^3;\mathbb{R}^3)$ put $H^{\mathrm{cons}}[\eta] = (\nabla\cdot\eta)\mathbf{1} - \nabla\eta - (\nabla\eta)^t$, $(\nabla\eta)_{ij} = \partial_i\eta_j$, and $\|\eta\|_{C^2} = \sup|\eta| + \sup|\nabla\eta| + \sup|\nabla^2\eta|$; then $\frac{1}{2}\mathrm{tr}(H^{\mathrm{cons}}[\eta]T) = \sum_{ij}\partial_i\eta_j\Theta_{ij}$ with the stress tensor $\Theta_{ij} = \frac{1}{2}\delta_{ij}\mathrm{tr}\,T - T_{ij}$: insertions of $H^{\mathrm{cons}}[\eta]$ express the conservation of Θ, and $\frac{1}{2}\mathrm{tr}(s_{ij}T) = \Theta_{ij}$ for $s_{ij} = \delta_{ij}\mathbf{1} - e_{ij}$. For a unit vector n, $\Theta_n = \mathbf{1} - 2nn^t$.

Chessboards and limits. A board of side $\mathsf{l} = L^{-i}$, $i\in\mathbb{Z}$, divides space into cubes, its cells, by bond mirrors, its walls; for $\ell' \in \mathfrak{L}(g)$, $\ell' \geq 2\mathsf{l}$, and D a function of the bonds of one cell, $\|D\|_{\mathrm{ch},\ell'}$ is the N-th root of $|\langle\prod_\kappa D_\kappa\rangle_{K,\ell',\beta}|$, the D_κ being the $N = (\ell'/\mathsf{l})^3$ mirror images of D in the cells of $T_{\ell'}$, conjugated after an odd number of reflections as in the main chessboard convention. A Gibbs sequence at g is (K_j,β_j,ν_j) with $(K_j,\beta_j)\in\mathfrak{d}(g)$, $K_j\to\infty$, $\nu_j\in\mathcal{G}_{K_j,\beta_j}$; a limit at g is $\sigma(X) = \lim_j \nu_j(A^{\nu_j}_{K_j}(X))$, $X\in\mathfrak{S}(\mathbb{R}^3)$, along a Gibbs sequence with $a_{K_j}\beta_j\to\mathsf{b}$, its label.

Its trace part $\sigma_0(f_1,\dots,f_n) = \sigma(f_1\mathbf{1},\dots,f_n\mathbf{1})$ is a scalar limit, to which Thm. YM3_I:AP (the scalar action density on $\mathbb{R}^3$), with Thm. YM3_I:AL applies, with a canonical decomposition $(\mathsf{X}_0,\lambda_0,(\sigma_{0,\zeta})_{\zeta\in\mathsf{X}_0})$. With θ the reflection in $\{x_3=0\}$, $\mathcal{H}(\sigma)$ is the Hilbert space of the form $\langle[F'],[F]\rangle = \sigma(\theta\bar F'\cdot F)$ on $\mathfrak{F}(\{x_3>0\})$, and $\Omega = [\emptyset]$. More generally, let σ have the properties of [7, Thm. BV](c), as every limit at g has, and let u be a unit vector at which σ is reflection positive: $\sigma(\theta_u\bar F\cdot F)\geq 0$ for $F\in\mathfrak{F}(\{u\cdot x>0\})$, θ_u the reflection in $\{u\cdot x=0\}$, acting on points and on indices by $E\mapsto\Theta_u E\Theta_u$.

The *picture of* u is formed in the same way: the Hilbert space $\mathcal{H}_u$ of the form $\langle[F'],[F]\rangle_u = \sigma(\theta_u\bar F'\cdot F)$ on $\mathfrak{F}(\{u\cdot x>0\})$, $\Omega_u = [\emptyset]_u$, the self-adjoint $H_u\geq 0$ with $e^{-tH_u}[F]_u = [\tau_{tu}F]_u$, $t\geq 0$, the unitary group $\mathcal{U}^u(v)[F]_u = [\tau_v F]_u$, $v\in u^\perp$, which commutes with it, and their joint spectral measure $\mathcal{P}_u$ on $[0,\infty)\times u^\perp$, with $e^{-tH_u}\mathcal{U}^u(v) = \int e^{-tE+iv\cdot q}\,\mathrm{d}\mathcal{P}_u(E,q)$; the picture of e_3 is $(\mathcal{H}(\sigma),H,\mathcal{U},\mathcal{P})$ of [7, Thm. BW](a). On a torus, $\mathcal{S}_{n;K}(X) = \langle O^T_K(H_1);\dots;O^T_K(H_n)\rangle_K$.

Theorem A (The tori)**.** Assume (T), $n\geq 2$, $d>0$, $\theta_n = 1/(1260n(8n+3))$, and let the supports of $X = (H_1,\dots,H_n)$ be pairwise $\geq d$ apart.

(a) There are an exponent p depending only on the basic constants (one may take $\mathsf{p} = 2\mathfrak{p}_0+2$), a number C^{od} depending only on n and on the constant C_{bf} of the comparison of two cutoffs at the label b, and a threshold K_1 depending besides on g, b, ℓ, d, on the threshold K_{bf} of that comparison and on the number $\Theta_\star$ (C_{bf}, K_{bf} of Def. 2.7(5), formed at a compact interval $J\subset I(g)$ that holds b, so that they read the label only through J (Cor. 2.8(a)), $\Theta_\star$ of Section YM3_I:113), such that for $K\geq K_1$ and Lipschitz weights

$$|\mathcal{S}_{n;K+1}(X) - \mathcal{S}_{n;K}(X)| \leq 6^n C^{\mathrm{od}}\ell^{3(n+1)}K^{\mathsf{p}}L^{-\theta_n K}\prod_a(|H_a| + \mathrm{Lip}\,H_a). \tag{2.xx}$$

So $\mathcal{S}_n = \mathcal{S}_n^{\ell,\mathsf{b}} = \lim_K \mathcal{S}_{n;K}$ exists along the full sequence of cutoffs, with $|\mathcal{S}_{n;K}(X) - \mathcal{S}_n(X)| \leq 6^n C^{\mathrm{od}}\ell^{3(n+1)}\sum_{K'\geq K}K'^{\mathsf{p}}L^{-\theta_n K'}\prod_a(|H_a|+\mathrm{Lip}\,H_a)$; it is n-linear and symmetric, $|\mathcal{S}_n(X)|\leq 60^n K_n^{\mathrm{ub}}(g,\ell,\frac{d}{2})\prod_a|H_a|$ with the number of Lem. 2.10(b), and the centred moments converge to $\sigma^{\ell,\mathsf{b}}(X) = \sum_\pi\prod_{B\in\pi}\mathcal{S}_{|B|}(X_B)$, the sum over the partitions π of the entries without singletons; $\sigma^{\ell,\mathsf{b}}(\emptyset) = 1$, $\sigma^{\ell,\mathsf{b}}((H)) = 0$. With $\bar O^T_K$, eq. 2.xx holds for bounded Borel weights vanishing off closed sets Σ_a pairwise $\geq d$ apart, with $\prod_a|H_a|$ on the right and a number C^{av} of the same kind as C^{od}; the limit extends $\mathcal{S}_n$, and in each argument $A\mapsto\mathcal{S}_n(\dots,1_A E,\dots)$, $E\in V$, is a countably additive measure on Σ_a vanishing on Lebesgue null sets, so it is given by a density in $L^1(\Sigma_a;V)$.

(b) $\mathcal{S}_n(rX) = \mathcal{S}_n(X)$ for $rx = Rx + v$, $R \in W_3$, $v \in \mathbb{R}^3$; $\sigma^{\ell,\mathsf{b}}(\theta\bar{F} \cdot F) \geq 0$ for the reflection θ in a plane $\{x_\mu = c\}$, acting on points and indices, and $F \in \mathfrak{F}(\{c < x_\mu < c + \frac{\ell}{2}\})$; at $H_a = f_a\mathbf{1}$, $\mathcal{S}_n$ is the limit of the truncated functions of the action density.

(c) A class insertion over a closed set Σ is a finite sum $\mathcal{E}$ of terms $\mathcal{O}(\phi 1_\lambda) = a_K^{-1} \sum_{\lambda(p)=\lambda} \phi(x(p))a_p$ and $\mathcal{Q}_{l,l';\xi}(\phi)$ with real Lipschitz ϕ supported in Σ; its symbol $\operatorname{Sym}\mathcal{E}$ is the sum of the weights $\phi E_{\lambda\lambda}$, resp. $\phi\, s(l)s(l')(E_{\lambda(l)\lambda(l')} + E_{\lambda(l')\lambda(l)})$, of its terms, and $\|\mathcal{E}\|$ is the sum of their numbers $\sup|\phi| + \operatorname{Lip}\phi$. There is C^{u} depending only on n, g, ℓ, d such that for $K \geq K_1 + 1$ and class insertions $\mathcal{E}_a$, $\mathcal{E}'_a$ over closed sets pairwise $\geq d$ apart with $\operatorname{Sym}\mathcal{E}_a = \operatorname{Sym}\mathcal{E}'_a$,

$$|\langle \mathcal{E}_1; \dots; \mathcal{E}_n\rangle_K - \langle \mathcal{E}'_1; \dots; \mathcal{E}'_n\rangle_K| \leq C^{\mathrm{u}} K^{\mathsf{p}} L^{-\theta_n K} \prod_a (\|\mathcal{E}_a\| + \|\mathcal{E}'_a\|),$$

and the Lipschitz numbers drop out of the bound if $\mathcal{E}_a$, $\mathcal{E}'_a$ carry the same weights at the same sites. So $\lim_K \langle \mathcal{E}_1; \dots; \mathcal{E}_n\rangle_K = \mathcal{S}_n(\operatorname{Sym}\mathcal{E}_1, \dots, \operatorname{Sym}\mathcal{E}_n)$: the bond form, the clover form and the single corners give one field, and $\operatorname{Sym} O_K^T(H) = H$.

(d) Let (H_1, H_2) be a pair at scale $d < 1$ with $3d \leq |x_0 - y_0| \leq \frac{\ell}{4}$. There are a threshold $s_1(g, \ell)$ on the depth and a number $\mathsf{e}^T_{g,\ell}(d)$, tending to 0 like $(\log\frac{1}{d})^{-2\mathsf{p}_0}$, such that if $s(d) \geq s_1(g, \ell)$, then for Lipschitz pairs, and for bounded ones with $\bar{O}_K^T$,

$$|\mathcal{S}_2(H_1, H_2) - \mathsf{b}^{-2}\mathcal{W}(H_1, H_2)| \leq \mathsf{e}^T_{g,\ell}(d)\, \mathsf{b}^{-2}|H_1||H_2|, \tag{2.xxi}$$

and the same holds for class insertions over the two balls, with $\operatorname{Sym}\mathcal{E}_a$ for H_a. At finite cutoff, for $(K, \beta) \in \mathfrak{d}(g)$ with $K \geq s(d) + 2$ and $4a_K \leq d - d_-$, and for continuous pairs, eq. 2.xxi holds for $\langle O_K^T(H_1); O_K^T(H_2)\rangle_{K,\ell,\beta}$ with $a_K\beta$ for b and the further error $\mathsf{b}^{-2}[c_K|H_1||H_2| + \frac{9}{8}\mathrm{rm}_K(H_1, H_2)]$, where $c_K = 28000\, C_{\mathrm{lat}}(1 + C_{\mathsf{k}})(a_K/d)^2$ with C_{k} basic, and $\mathrm{rm}_K \geq 0$ depends only on K, d, the two weights and the choice of O_K^T or $\bar{O}_K^T$, and tends to 0 as $K \to \infty$. One may take $s_1(g, \ell) \leq \hat{s}(g, \ell) := \max\{s_\flat(g), \lceil 2\Lambda_\ell/\log L\rceil\}$, with a depth $s_\flat(g)$ depending only on g and $\Lambda_\ell = (12\log\ell/(c'A_0^2))^{1/(2\mathsf{p}_0)}$, c' basic; from this depth on, the error of the finite-cutoff law against the lattice Wick value of [7, Lem. C.10] is independent of ℓ. Further, under the same conditions on d: (i) $|\mathcal{S}_2(rH_1, rH_2) - \mathcal{S}_2(H_1, H_2)| \leq 10\, \mathsf{e}^T_{g,\ell}(d)\mathsf{b}^{-2}|H_1||H_2|$ for $rx = x_0 + R(x - x_0)$, $R \in O(3)$; (ii) for the reflection θ in a plane $\{x_\mu = c\}$, a point z at distance $\frac{3}{2}d$ from it, a Lipschitz function ϕ with $1_{B(z,d/128)} \leq \phi \leq 1_{B(z,d/64)}$ and $E \in V$: $\mathcal{S}_2(\theta(\phi E), \phi E) \geq (\mathsf{w}_0 - \mathsf{e}^T_{g,\ell}(d))\mathsf{b}^{-2}\mathrm{tr}\, E^2$; so no non-zero real combination of the six fields vanishes. And at every d: (iii) for $0 < d_- < d < 1$, $\frac{3}{2}d \leq c - x_{0,\mu} \leq \frac{\ell}{2} - \frac{3}{2}d$ and real Lipschitz H vanishing off $\bar{B}(x_0, d_-)$, $0 \leq \mathcal{S}_2(\theta H, H) \leq 3600\, C_3^*(g, \ell)\mathsf{b}^{-2}|H|^2$, with $C_3^*(g, \ell) \leq c_3(g)(1 + \log\ell)^8 e^{16\Lambda_\ell}$ independent of d.

(e) For $q \geq 2$, $\eta \in C^2(T_\ell; \mathbb{R}^3)$ and Lipschitz $H_2, \dots, H_q$, the q supports pairwise $\geq d$ apart: $\mathcal{S}_q(H^{\mathrm{cons}}[\eta], H_2, \dots, H_q) = 0$, and $\sigma^{\ell,\mathsf{b}}((H^{\mathrm{cons}}[\eta]) \cdot X') = 0$ for $X' \in \mathfrak{S}(T_\ell)$ off $\operatorname{supp}\eta$. There are C, p, K_0 depending only on g, ℓ, q, d such that for all $(K, \beta) \in \mathfrak{d}(g)$ with $K \geq K_0$ and all bounded H_a,

$$|\langle O_K^T(H^{\mathrm{cons}}[\eta]); O_K^T(H_2); \dots; O_K^T(H_q)\rangle_{K,\ell,\beta}| \leq C K^{\mathsf{p}} L^{-K/56} \|\eta\|_{C^2} \prod_{a \geq 2} |H_a|,$$

also with $\bar{O}_K^T$; the bound is uniform in the label, and it stays true, with another C, if $O_K^T(H^{\mathrm{cons}}[\eta])$ is changed by a site form plus a plaquette form, carried within $2a_K$ of $\operatorname{supp}\eta$, whose weights are at most $36a_K\|\eta\|_{C^2}$. At $H_a = h_a\mathbf{1}$ this is the conservation of the stress tensor among action densities.

(f) The numbers p, C^{od}, C^{av} and K_1 of (a) may be taken the same at all the labels of a compact interval $J \subset I(g)$ (Lem. 2.11(b)), and $\mathsf{b} \mapsto \mathcal{S}_n^{\ell,\mathsf{b}}(X)$ is continuous on $I(g)$. For Lipschitz weights, and for bounded ones with $\bar{O}_K^T$: if b is an inner point of $I(g)$ and $a_K\beta_K \to \mathsf{b}$, then $\langle O_K^T(H_1); \dots; O_K^T(H_n)\rangle_{K,\ell,\beta_K} \to \mathcal{S}_n^{\ell,\mathsf{b}}(X)$ along the full sequence of cutoffs.

By [7, Lem. C.8], (a)–(f) hold on every torus of side $2L^m$, $m \geq 0$, and at every label $\mathsf{b} > 0$, with the constants of a triple $(g', L^z\mathsf{b}, L^z\ell)$ with (T), taken at the distance $L^z d$, and powers of L^z; for large z one may take $g' = (L^z\mathsf{b}/2)^{-1/2}$, and then $L^z\mathsf{b}$ is an inner point of $I(g')$. ■□

The remaining estimate for overlapping sources. In the setting (T), consider the source derivatives of the step maps on their admitted analytic domains, with the weighted local-family norms and source sizes of the construction. For two bond forms Y_a, Y_b whose weights overlap, one still needs to show that the derivative of order two created at depth u decomposes into a local part $\mathfrak{l}$ of norm at most cL^u and a remainder of norm at most $cK^{\mathsf{p}}L^{u/2}$, in the units of the two source sizes. The local part vanishes at the trivial background. At the real background $U_{u'}$ of a small block field of a coarser level, of depth $u' \leq u$, one also needs the evaluated sum estimate

$$\left| \sum_{X,z} \mathfrak{l}(X, U_{u'}, z) \right| \leq K^{\mathsf{p}}\mathsf{b}^{-2} N L^{-2(u-u')},$$

where the sum runs over every polymer and base point and N is the number of sites of that coarser level. The family-norm bounds on the decomposition hold on the admitted analytic domains; the last estimate is required at these real backgrounds. The constants may depend on the number of derivatives, the fixed torus and a compact label interval, but not on the cutoff or the depths. The sources in the overlapping cases are the off-diagonal bond forms in clause (b) below.

The cancellation behind the evaluated sum estimate concerns the linear parts of the two insertions in the fluctuation field. If these parts have coefficient vectors r_a, r_b, their pairing through the fluctuation covariance C_u satisfies

$$|\langle r_a, C_u r_b \rangle| \leq \|C_u\|_{2\to 2} \|r_a\|_2 \|r_b\|_2.$$

Thus a uniform covariance bound and the estimates

$$\|r_a\|_2^2 \leq C K^{\mathsf{p}}\mathsf{b}^{-2} N L^{-2(u-u')} \|\phi_a\|_\sharp^2$$

give the required estimate, after absorbing fixed constants and the two source sizes. The proposed gain comes from writing the linear curvature response as a discrete curl and summing by parts: a difference then falls on the product of the weight and the background curvature. Over one lattice spacing at depth u, the local gauge bounds at depth u' give the factor $L^{-(u-u')}$ in the L^2 norm, whose square gives the displayed gain. To complete this argument for the actual off-diagonal step insertions, one still needs the quantitative decomposition above and the identification and bounds of its linear coefficient vectors, including the remaining nonlinear terms. These are the additional input here; the local gauge estimate and the step-map Cauchy estimate alone do not supply that identification.

Theorem B (Coinciding points, overlapping insertions, more points)**.**

(a) Assume (H), $\ell \in \mathfrak{L}(g)$, $(K, \beta) \in \mathfrak{d}(g)$, $0 < d < 1$ and $s_{\mathrm{th}}(n, g) \leq s(d) \leq K$, s_{th} a threshold depth depending on n and g. For complex weights whose supports lie within the Euclidean distance $0.87d$ of points x_a and are at pairwise Euclidean distances $\geq 1.26d$, with no bound on their mutual distances,

$$|\langle O_K^T(H_1); \dots; O_K^T(H_n) \rangle_{K,\ell,\beta}| \leq 60^n C_n(g,\ell) \big[2 \log \tfrac{1}{d} + C(g)\big]^{8n\mathfrak{p}_0} \mathsf{b}^{-n} \prod_a |H_a| :$$

the power counting d^{-3} of an entry, the dimension of $\operatorname{tr} FF$, up to logarithms. On a side that is small over the canonical sequence of (K, β) (equation YM3_I:(2312)), for $g \leq \gamma_W^{\mathrm{sd}}$ the least side ℓ_g (Lem. YM3_I:126.3), the number $C_n(g, \ell)$ is free of ℓ; on every other $\ell \in \mathfrak{L}(g)$ it carries the factor $\max\{1, \mathsf{c}(\ell)\}^n$, $\mathsf{c}(\ell) = 2\ell^3 e^{\mathsf{B}(\bar{x}_0)\ell^3}/(c' A_0^2)$, with the function B and the number c' of equation YM3_I:(2311). Read at the reference coupling γ_W^{sd} after a dilation, the bound takes a scale-covariant form: let $\delta_\star = [L((\gamma_W^{\mathrm{sd}})^{-2} + \bar{B})]^{-1}$ and $\beta > X_{\gamma_W^{\mathrm{sd}}}$, let $z \in \mathbb{Z}$

be fixed by the first-passage window of β at γ_W^{sd}, so that $\delta_\star \mathsf{b} \le L^{-z} < (\gamma_W^{\mathrm{sd}})^2 \mathsf{b}$, and let $L^z\ell$ be the least side at γ_W^{sd}; then for every $d < \delta_\star \mathsf{b}$ with $s_{\mathrm{th}}(n, \gamma_W^{\mathrm{sd}}) \le s(L^z d) \le K - z$, the depth and the threshold being formed at γ_W^{sd}, the bound holds with $[2\log \frac{\mathsf{b}}{d} + C]^{8n\mathfrak{p}_0}$ in the place of $[2\log \frac{1}{d} + C(g)]^{8n\mathfrak{p}_0}$ and numbers depending only on n and the basic constants, and this statement is invariant under $(K, \ell, \mathsf{b}, d) \mapsto (K-1, L\ell, L\mathsf{b}, Ld)$.

For supports only at pairwise Euclidean distances $\ge d$ the bound holds at $d'' = \min\{\frac{3}{4}d, L^{-s_{\mathrm{th}}(n,g)}\}$, for $K \ge s(d'')$, with $\sum_Q \sup_Q |H_a|$ in the place of $|H_a|$ and b^{-n} kept, Q running through the boxes of a partition of T_ℓ into at most $(2\ell/d'')^3$ boxes of sides $\le \frac{99}{100}d''$: each box lies within the Euclidean distance $0.87d''$ of its centre and pieces of two weights are at Euclidean distances $\ge d \ge 1.26d''$, so every product of the pieces $H_a \mathbf{1}_Q$ has the geometry of the first display at d''. In the amalgam norm $d''^3 \sum_Q \sup_Q |H_a|$ this is the bound with $\mathsf{b} d''^3$ in the place of b: the power d^{-3} of an entry.

At $n = 2$ and for $g \le \gamma_W^{\mathrm{sd}}$ the off-diagonal components need no logarithm and no number of the side: for every $\ell \in \mathfrak{L}(g)$, $(K, \beta) \in \mathfrak{d}(g)$, $h > 0$ with $1.31h < 1$ and $s(1.31h) \le K$, and complex weights $\mathsf{w}_1, \mathsf{w}_2$ of the bonds with x_b in closed cubes Q_1, Q_2 of side $\frac{3}{2}h$ at a sup distance $\ge \frac{5}{2}h$ wherever $\mathsf{w}_i(b) \ne 0$, no bound being asked on the distance of the cubes, the bond forms $Y_i = a_K^{-1} \sum_b \mathsf{w}_i(b) q_b$ have $|\langle Y_1; Y_2\rangle_{K,\ell,\beta}| \le 36\, C_\infty(g)\, \mathsf{b}^{-2} \sup|\mathsf{w}_1| \sup|\mathsf{w}_2|$, with the number $C_\infty(g)$ of [7, Lem. C.7], which depends on g alone; so the off-diagonal part of $\mathcal{S}_2^{\ell,\mathsf{b}}$ has the power counting of [7, Prop. C.22] with $q = 0$ and a number A free of ℓ.

The bound passes to $\mathcal{S}_n^{\ell,\mathsf{b}}$ for weights vanishing off closed sets in balls of radius $0.86d$ at pairwise Euclidean distances $\ge 1.27d$, with a number depending on ℓ. So $\mathcal{S}_n^{\ell,\mathsf{b}}$ has the power counting of [7, Prop. C.22] with $q = 8n\mathfrak{p}_0$, a number $A = A_n(g,\ell)\mathsf{b}^{-n}$ and $h_0 \in \ell 2^{-\mathbb{N}}$ with $h_0 \le \min\{\frac{\ell}{64n}, \frac{1}{2}L^{-s_{\mathrm{th}}(n,g)}\}$: it extends over all the diagonals to a distribution $\tilde{\mathcal{S}}_n$ of order $\le 3n-2$ with the same power counting and one more logarithm. The lattice functionals $\tilde{u}_K$ formed from the cumulants at the cutoff K with the subtractions of [7, Prop. C.22], the sum running over the products of Whitney cubes of the sides h with $s(1.52h) \le K$ alone, obey $|\tilde{u}_K(F)| \le C_E A\, \mathfrak{l}(\lambda)^{8n\mathfrak{p}_0+1} \|F\|_{(\lambda),3n-2}$, one number for all K, for $8h_K \le \lambda \le h_0$, $h_K \le 2C_W(K)a_K$ the least side admitted at the cutoff K, and every F supported in a product of closed cubes of side 2λ, wherever their centres lie; and $\tilde{u}_K(F) \to \tilde{\mathcal{S}}_n(F)$ for every smooth F.

Within this power counting the extension is unique exactly up to the sums $\sum_{\pi,\alpha} \partial^\alpha \delta_\pi \otimes T_{\pi,\alpha}$ over the partitions π into at most $n-1$ blocks and $|\alpha| \le 3|\pi|$. At two points the real, exchange-symmetric, translation-invariant contact terms in $\mathfrak{E}_2(0)$ form a twelve-dimensional space under W_3 on $\mathbb{R}^3$ or T_ℓ, and a six-dimensional space under $O(3)$ on $\mathbb{R}^3$, by Lem. 3.6(f). $\mathcal{S}_2$ is no measure across the diagonal.

(b) The convergence rates in this clause remain targets until the actual step insertions satisfy the overlapping-source estimates described in §2.3.3. Under (T) let $r \ge 2$ tensor entries $O_K^T(\phi_a E_a)$, $E_a \in V$, and n action densities $\mathcal{O}(f_i)$ be given, all with Lipschitz weights, and let $d > 0$. Their joint cumulants κ_K converge as $K \to \infty$ at a rate: $|\kappa_{K+1} - \kappa_K| \le CK^{\mathsf{p}} L^{-cK} \prod_a \|\phi_a\|_\sharp |E_a| \prod_i \|f_i\|_\sharp$ in the norms $\|\phi\|_\sharp = \max\{\sup|\phi|, 40M \operatorname{Lip}\phi\}$, $|E|$ the largest modulus of the entries of E, for K beyond a threshold, with C, p depending on r, n, ℓ and a compact interval of labels about b and the threshold depending besides on the geometry of the supports (on d in the cases where d is named), in the following cases:

(i) $c = \frac{1}{120r}$ if the tensor supports are pairwise $\ge d$ apart, nothing being asked of the supports of the f_i, which may meet one another and the tensor supports

(ii) $c = \frac{1}{120r}$ also if $\phi_a \phi_b = 0$ for $a \ne b$, the tensor weights not overlapping (their supports may touch), provided that of any $r+n$ points, one from each support, the supports of the f_i included, two are $\ge d$ apart, so that no point is common to all the supports (at $n = 0$ the numbers C, p depend on d as well)

(iii) $c = \frac{1}{360r}$ for off-diagonal E_a, $r \geq 3$, if no three tensor supports share a point, nothing being asked of the f_i

(iv) $c = \frac{1}{864}$ for $r = 2$, off-diagonal E_a and $n \geq 1$, the two tensor supports overlapping at will, if of any $n+2$ points, one from each support, two are $\geq d$ apart.

(c) The arbitrary-order interacting conclusion remains a target. It requires the remainder estimate described in §5.1.2, with its real plaquette weights, first-creation bound, numerical ranges and error e_n. Keep $36d < \ell$. The cyclic formula is exact for three free lattice photon fields; the corresponding interacting conclusion is the following. For the plane fields, with real Lipschitz ϕ_a in the open balls $B(x_a, d)$, these in a ball of radius $\leq \frac{\ell}{4}$, $\operatorname{Lip}\phi_a \leq |\phi_a|/(\vartheta_0 d)$ and $|x' - y'| \geq d$ for $x' \in \operatorname{supp}\phi_a$, $y' \in \operatorname{supp}\phi_b$, $a \neq b$ (Euclidean distances, as for the w_a): $|\langle \mathcal{O}_{\lambda_1}(\phi_1); \dots; \mathcal{O}_{\lambda_n}(\phi_n)\rangle_{K,\ell,\beta} - 3\mathsf{b}^{-n} K^{(n)}_{\underline{\lambda}}(\phi)| \leq [\mathsf{e}_n + C_n(\frac{a_K}{\vartheta_0 d} + \frac{d^3}{\ell^3})]\mathsf{b}^{-n}\prod_a |\phi_a|$, $C_n = \frac{3}{2}n![6^n C_{\mathrm{lat}}(1 + 3C_{\mathrm{lat}})^{n-1} + 43 \cdot 11^n]$; the same with $3^n C_n$ for real combinations $\sum_\lambda k_{a,\lambda}\mathcal{O}_\lambda(\phi_a)$, $k_a \in [-1,1]^3$, among them the diagonal entries $\mathcal{O}(\phi) - 2\mathcal{O}_m(\phi) = O^T_K(\phi s_{mm})$ of the stress tensor.

For $2 \leq n \leq 7$ and $g \leq \min\{\gamma_n, \gamma_W^{\mathrm{sd}}\}$, with d below a threshold depending on n and g and the balls $B(x_a, d)$ in one ball of radius $9d$, the lattice inequality in §5.1.2, with $\mathcal{K}^{(n)}_K$ formed on T_{ℓ_g} and with $\varepsilon^{\mathrm{rt}}_n(g,d)$ of [7, Lem. C.16](a) added to e_n, holds also in every periodic Gibbs state and on every torus of $\mathfrak{L}(g)$; $\varepsilon^{\mathrm{rt}}_n(g,d)$ tends to 0 with d like a positive power of d times a power of $\log\frac{1}{d}$.

(d) Let g be below a basic threshold. On the least torus, for $n = 2, 3, 4$ and a label $\mathsf{b} \in I(g)$, the limit $\mathcal{S}^{\ell_g,\mathsf{b}}_n$ of Thm. A(a) has on each set of points named below a Borel kernel: $\mathcal{S}^{\ell_g,\mathsf{b}}_n(f_1\mathbf{1}, \dots, f_n\mathbf{1}) = \int s_n \prod_a f_a$ for continuous f_a whose supports form a product inside that set, s_n the kernel of the truncated n-point function of the action density, and $\mathcal{S}^{\ell_g,\mathsf{b}}_2(f_1 E_{\lambda\lambda}, f_2 E_{\mu\mu}) = \iint \tau_{\lambda\mu} f_1 f_2$ at $0 < |x-y| < d_\sigma(g)$. For a limit σ at g the kernel of three action densities is $s_3(\cdot; \mathbf{1}, \mathbf{1}, \mathbf{1})$ of [7, Thm. BW](d), the third centred moment being the third cumulant, and $\tau_{\lambda\mu} = s_2(\cdot; E_{\lambda\lambda}, E_{\mu\mu})$; for a component σ_ξ they are the kernels of its re-centred σ^c_ξ. On T_{ℓ_g} almost everywhere, and for every limit at g and every re-centred component at every such triple of points, the kernel s_3 of three action densities obeys

$$|s_3 - 3\mathsf{b}^{-3}k_3| \leq \varepsilon_A(g,d)\,\mathsf{b}^{-3}r^{-3}d^{-6} \quad \text{for } 0 < |x-y| = r \leq d = \operatorname{dist}(z,\{x,y\}) \leq d_A(g),$$
$$\varepsilon_A \leq C(g)\Big[\big(\log\tfrac{1}{d}\big)^{39\mathfrak{p}_0+2} d^{1/2} + \Big(\frac{d}{\ell_g}\Big)^3\Big],$$

uniformly in $\frac{r}{d} \in (0,1]$, $d_A(g) > 0$ depending on g. At a cutoff the same law holds for narrow weights, in every periodic Gibbs state and on T_{ℓ_g}: for real plaquette weights w_1, w_2, w_3 carried by sets $S_a \subset B(x_a, \rho)$ with $\operatorname{dist}(S_1, S_2) \geq r$, $|x_1 - x_2| \leq 8r$, $\operatorname{dist}(S_a, S_3) \geq d$ and $|x_a - x_3| \leq 18d$ for $a = 1, 2$, where $0 < \rho \leq r \leq d$, d below a threshold depending on g, and $s(\rho) \leq K$ (the geometry of [7, Lem. C.15](c)), the third cumulant of the forms $a_K^{-1}\sum_p w_a(p)a_p$ is $3\mathsf{b}^{-3}K^{(3)}_K(w_1, w_2, w_3)$ up to $[C(g)Y^{39\mathfrak{p}_0+2}d^{1/2} + C(g)d^{3/2}]\rho^9 r^{-3}d^{-6}\mathsf{b}^{-3}\prod_a |w_a|$, $Y = \log \bar{x}_{s(d)}$; for Lipschitz bumps $w_a = f_a \circ x$ with $\operatorname{Lip} f_a \leq |f_a|/(\vartheta\rho)$, $0 < \vartheta \leq 1$, the same holds against $3\mathsf{b}^{-3}\int k_3 \prod_a f_a$ with $C(g)(\frac{a_K}{\vartheta\rho} + (\frac{d}{\ell_g})^3)$ more in the bracket. For $d = |c - z| \leq \frac{1}{2}d_A(g)$ and $0 < r \leq \frac{d}{2}$, almost everywhere on the torus and at every (c, z) where the kernels are continuous, in particular for limits and re-centred components,

$$\Big|\tfrac{2\pi}{3}\mathsf{b}r^3 s_3(c + \tfrac{r}{2}e_\lambda, c - \tfrac{r}{2}e_\lambda, z) - t_\lambda(c,z)\Big| \leq \Big[\varepsilon'(g,d) + 2.7\frac{r^2}{d^2}\Big]\mathsf{b}^{-2}d^{-6}, \qquad \varepsilon'(g,d)$$
$$\leq C(g)\big(\log\tfrac{1}{d}\big)^{39\mathfrak{p}_0+2} d^{1/2},$$

$t_\lambda = \sum_\mu(\tau_{\lambda\mu} - \frac{1}{3}\sum_{\lambda'}\tau_{\lambda'\mu})$, the kernel of the traceless diagonal component $E_{\lambda\lambda} - \frac{1}{3}\mathbf{1}$ against the action density, formed with the kernels $\tau_{\lambda\mu}$ of the same torus, limit or component: the

tensor is recovered, to leading order, from two action densities that approach one another along e_λ, the right member being small against the natural size $\mathsf{b}^{-2}d^{-6}$ as $d \to 0$ and $r/d \to 0$. Four action densities at two scales are free in the same sense, on the torus and at a cutoff: on T_{ℓ_g}, for $r \le |x_1 - x_2|, |x_3 - x_4| \le 2r$ and $r \le d = \operatorname{dist}(\{x_1, x_2\}, \{x_3, x_4\}) \le d'_A(g)$, the kernel s_4 of the truncated four-point function of the action density has $|s_4 - 3\mathsf{b}^{-4}k_4| \le \varepsilon_4(g,d)\mathsf{b}^{-4}r^{-6}d^{-6}$ almost everywhere, $\varepsilon_4 \le C(g)[(\log \frac{1}{d})^{47\mathfrak{p}_0+2}d^{1/2} + (\frac{d}{\ell_g})^3]$, uniformly in $\frac{r}{d} \in (0,1]$; and at a cutoff, in every periodic Gibbs state and on T_{ℓ_g}, for real plaquette weights $w_1, \dots, w_4$ in two such narrow pairs $\{1,2\}$, $\{3,4\}$, with $\operatorname{dist}(S_3, S_4) \ge r$, $|x_3 - x_4| \le 8r$, $\operatorname{dist}(S_a, S_b) \ge d$ and $|x_a - x_b| \le 18d$ for $a \le 2 < b$, d below a threshold depending on g, the fourth cumulant of the plaquette forms is $3\mathsf{b}^{-4}K_K^{(4)}(w_1, \dots, w_4)$ up to $[C(g)Y^{47\mathfrak{p}_0+2}d^{1/2} + C(g)d^{3/2}]\rho^{12}r^{-6}d^{-6}\mathsf{b}^{-4}\prod_a |w_a|$.

■□

Corollary 2.9 (A tensor against the action density)**.** Let σ be a limit at g under (H), or a re-centred component σ^c_ξ. There are real-analytic functions A, B on $(0,\infty)$ with $s_2(x,y;e,\mathbf{1}) = \operatorname{tr}(e\,[A(r)\mathbf{1} + B(r)\hat{u}\hat{u}^t])$, $u = x - y$, $r = |u|$, for all $e \in V$; $F = 3A + B$ is the kernel of two action densities, and

$$B'(r) + \frac{3}{r}B(r) = \frac{1}{4}F'(r), \qquad \text{so} \qquad B(r) = r^{-3}\Big(c + \frac{1}{4}\int_1^r t^3F'(t)\,\mathrm{d}t\Big)$$

for one real number c. ■□

Lemma 2.10 (Exact symmetries; separated insertions)**.** (a) Let $rx = Rx + v$, with $R \in W_3$, $Re_\mu = s_\mu e_{\pi(\mu)}$, and $v \in a_K\mathbb{Z}^3$, act on configurations by moving the bonds. Then $rO_K^T(H) = O_K^T(rH)$ for every complex weight, on $\Lambda_{K,\ell}$ and on $\mathbb{L}_K$; in components $ra_p = a_{rp}$ and $rq_b = s_{\lambda_b}s_{\lambda'_b}q_{rb}$. Under a law ν invariant under these maps, as the torus measures and the periodic Gibbs states are (Prop. YM3_I: 103.4), $\nu(q_b) = 0$ and $\nu(a_p) = e_\nu$ does not depend on p, so that $\nu(O_K^T(H)) = e_\nu a_K^{-1}\sum_p H_{\lambda(p)\lambda(p)}(x(p))$: the off-diagonal components need no subtraction; moments and cumulants are invariant. $O_K^T(H)$ depends only on the bonds within a_K of a point where $H \ne 0$.

(b) Assume (H), $\ell \in \mathfrak{L}(g)$, $q \ge 2$, $\delta > 0$. In the calculus of sources of Thm. YM3_I:D let $C_{(m)}$, $m \ge 1$, be the numbers, depending only on m, with which the mixed derivative families of order m of a source system under majorants are bounded at the level t by $C_{(m)}L^{4m(T-t)}$ times the product of the sizes of the sources, T the working level; put $K_r = \sum_\pi \prod_{B\in\pi} C_{(|B|)}$, the sum over the partitions of $\{1, \dots, r\}$; and let $h_q(s)$ be the largest left member, at the depth s, of the smallness conditions that this calculus asks of the families of the orders $2, \dots, q$ for majorants at least $3M$ units of the working level apart, a function of the depth depending only on q and g and tending to 0 as $s \to \infty$.

Let $s_q(\delta)$ be the least $s \ge 0$ with $L^s\delta \ge 3M$ and $h_q(s) \le 1$; at $s = s_q(\delta)$ put $M_r = K_r[1 + L^{4r}V_\ell(4q)L^{-(4q+1)s}]$, where $V_\ell(b) = 4M^{-3}\ell^3 e^{\mathsf{B}(X_g)\ell^3}\mathsf{C}_{\mathrm{ag}}(b+1, \eth)$, with the function B and the number C_{ag} of equation eq. 4.xxxii at a fixed $\eth \in (0,1)$, so that $K_rL^{4r}V_\ell(b)L^{-(b+1)s}$, $b \ge 4r$, bounds, in units of the sizes, the part of a centred mean of r first members that is carried by the histories in which a large field is created below the working level (Prop. 4.54); and

$$K_q^{\mathrm{ub}}(g,\ell,\delta) = (4\ell^3L^{4s})^q \sum_\pi (|\pi| - 1)! \prod_{B\in\pi} M_{|B|},$$

the sum over all partitions of $\{1, \dots, q\}$. Let $(K,\beta) \in \mathfrak{d}(g)$ with $K \ge s_q(\delta) + 2$ and $3a_K \le \delta$, and let each $\mathcal{F}_a$ be one of: $O_K^T(H_a)$ or $\bar{O}_K^T(H_a)$ with H_a vanishing off a closed set Σ_a (size $\mathsf{s}_a = 60|H_a|$); a site form with weights at the sites within $\frac{1}{2}a_K$ of Σ_a (size $23\|v_a\|$); $\mathcal{O}(w_a)$ with $w_a(p) = 0$ unless $x(p)$ is within a_K of Σ_a (size $|w_a|$); or a sum of two of these (the sum of the sizes); the Σ_a being

pairwise $\geq 2\delta$ apart. Then

$$|\langle \mathcal{F}_1; \dots; \mathcal{F}_q \rangle_{K,\ell,\beta}| \leq K_q^{\mathrm{ub}}(g,\ell,\delta) \prod_a \mathsf{s}_a.$$

■

Proof sketch. (a) $\operatorname{tr} U^{-1} = \operatorname{tr} U$ on $SU(2)$ gives $ra_p = a_{rp}$, and $w(U^{-1}) = -w(U)$ gives the signs in rq_b; inserted into [7, eq. 1] they give $rO_K^T(H) = O_K^T(rH)$, as $(rH)_{\pi\lambda,\pi\lambda'}(rx) = s_\lambda s_{\lambda'} H_{\lambda\lambda'}(x)$. The reflection through b normal to λ_b lies in the group, fixes b and reverses q_b; the group is transitive on the plaquettes. (b) By [7, Lem. C.4] each $\mathcal{F}_a$ is the first member of a source, in the calculus of Thm. YM3_I:D, under the majorant made of its size and of the plaquettes of its form (for $O_K^T(H)$ the size is $1 + 59$, for its plaquette and its bond part), and so, the notion being monotone in the majorant, under the majorant made of its size and of the plaquettes whose corners lie within $\frac{3}{2}a_K$ of Σ_a; as $3a_K \leq \delta$, corners of plaquettes of different majorants are $\geq \delta$ apart. At the depth $s = s_q(\delta)$, $L^s\delta$ exceeds the range of the block terms, and the mean of a centred product of r first members is at most $M_r \prod_a 4\ell^3 L^{4s} \mathsf{s}_a$, up to constants common to all the subproducts, which cancel in cumulants; the Möbius formula for cumulants gives K_q^{ub}; the number grows with ℓ and as $\delta \to 0$, since only the sizes of the sources enter. □

Lemma 2.11 (Two successive cutoffs)**.** Assume (T). An entry is one of $\mathcal{O}(w)$, $\mathcal{O}(w1_\lambda)$,

$$a_K^{-1} \sum_{b \| \mu} w(x_b) q_b$$

and $\mathcal{Q}_{l,l';\xi}(w)$, with a real function w read at every cutoff at the barycentres of the plaquettes, at the midpoints of the bonds, resp. at the shifted sites. (a) eq. 2.xx holds for n entries with supports pairwise $\geq d$ apart, without the factor 6^n and with $|w_a|$ for $|H_a|$; and for entries whose weights are the cell means of bounded Borel functions, with $\prod_a |w_a|$ on the right. (b) Every compact interval $J \subset I(g)$ is an interval of labels under (Tor) on every side $\ell \in \mathfrak{L}(g)$ (§2.3.1): its labels exceed $g^{-2} \geq \gamma^{-2}$, and $MR(x) \leq L^{m_{\mathfrak{L}}(g)-1}$ for $x \leq L(g^{-2} + \bar{B})$ by YM3_I:(1528). With C_{bf} and K_{bf} formed at J (Def. 2.7(5)), which serve the comparison of Cor. 2.8(a) at every label of J, the numbers p, C^{od} and C^{av} of Thm. A(a) serve all labels of J, and so does K_1, formed with K_{bf} of J and at the least label of J. (c) Let (t, ε_r, V) be data of the comparison of the cutoffs K and $K+1$ (the setting preceding Lem. YM3_I:113.8): a level $1 \leq t \leq K$ of the cutoff K with $L^{K-t}\mathsf{b} \geq \Theta_\star$, the radius

$$\varepsilon_r = 2c_\rho \varepsilon(L^{K-t}\mathsf{b})$$

of that Definition, and a block field V of the depth $K - t$ with $\varpi(V) < \varepsilon_r$; let U^A and U^B be its backgrounds at the level t of the cutoff K and at the level $t+1$ of the cutoff $K+1$, and $\eta_t = L^{-t}$, $N_t = \ell^3 L^{3(K-t)}$. The *corner form* of a weight v on the sites and of loops l, l' at a configuration U of a lattice is $\eta^{-1} \sum_x v(x) Q_x(l,l')(U)$, with $\eta = \eta_{t+1}$ on $\Lambda_{K+1,\ell}$ at U^B and $\eta = \eta_t$ on $\Lambda_{K,\ell}$ at $\mathcal{M}U^B$ and at U^A; as $a_K^{-1} = L^{K-t}\eta_t^{-1}$ and

$$a_{K+1}^{-1} = L^{K-t}\eta_{t+1}^{-1},$$

an entry is L^{K-t} times a corner form, and $L^{K-t}N_t = \ell^3 L^{4(K-t)}$ gives the factor $L^{4\sigma}$ below, $\sigma = K - t$. B_3 is the fixed background regularity constant of Thm. YM3_I:E; its bound at these data is $B_3\varepsilon_r$, as used in the comparison preceding Lem. YM3_I:113.8. For loops l, l' and l_1, l_1' in the same planes and complex v on the sites,

$$\left| \eta_{t+1}^{-1} \sum_x v(x) \left[s(l)s(l')Q_x(l,l') - s(l_1)s(l_1')Q_x(l_1,l_1') \right](U^B) \right| \leq 2K_0^{\mathrm{cf}} \sup|v| \, \varepsilon_r^2 N_t L^{-t/2},$$

K_0^{cf} basic: at a background the corner forms depend, in the mean, on their planes and signs alone. (d) Let U^A be the background of the coarser lattice and $\mathcal{M}U^B$ the block average of U^B. For weights

v_f, v_c on the sites of the fine and of the coarse lattice, with $v_{\max} = \max\{\sup|v_f|, \sup|v_c|\}$ and the mismatch $m = \max_y |L^{-3}\sum_{x\in B(y)} v_f(x) - v_c(y)|$, $B(y)$ the block of y:

$$\Big|\eta_{t+1}^{-1}\sum_x v_f(x)Q_x(l,l')(U^B) - \eta_t^{-1}\sum_y v_c(y)Q_y(l,l')(\mathcal{M}U^B)\Big| \le [4B_3^2 m + K_0^{\mathrm{cf}} v_{\max} L^{-t/2}]\varepsilon_r^2 N_t;$$

and $\mathcal{M}U^B$, a nearly minimal competitor at the level t, has $|\eta_t^{-1}\sum_x v(x)[Q_x(l,l')(\mathcal{M}U^B) - Q_x(l,l')(U^A)]| \le 12\sup|v|\,(C\,t\,A\,\Delta A)^{1/2}$, A its action and ΔA its excess. So an entry with a Lipschitz weight, read at the two cutoffs, changes over one block field of depth σ by at most $C\ell^3 L^{4\sigma} K^{1/2} L^{-(K-\sigma)/2}(\sup|w| + L^{-\sigma}\mathrm{Lip}\,w)$, and an entry whose weights at the two cutoffs are tied by the block mean by the same with $\sup|w|$ alone. ■

Proof sketch. (a) The entries are first members of sources under majorants ([7, Lem. C.4]), of the sizes $\sup|w|$, $59\sup|w|$, resp. $23\sup|w|$, and the corners of the plaquettes of two majorants are $\ge d - 3a_K \ge d' = \frac{d}{2}$ apart. *First link, at one cutoff.* By Thm. YM3_I:D a source system of n members is carried by its first members read at the backgrounds of the block field of a depth σ up to the error

$$2\mathfrak{K}_n'\big[2L^{4n}V_\ell(b)L^{-(b+1)s} + n\,\epsilon_{\mathrm{fg}}(\sigma - s) + 2e^{3-L^s d'/M}\big](4\ell^3 L^{4s})^n \prod_a \mathsf{s}_a$$

at a second depth $s < \sigma$, for every real $b \ge 4n$, with V_ℓ and M_r of Lem. 2.10(b) formed at the depth s and $\mathfrak{K}_n' = \sum_\pi |\pi|! \prod_{B\in\pi} M_{|B|}$: the first term is carried by the histories in which a large field is created below the working level of the depth s (Prop. 4.54); the second is the forgetting of what the members of the sources make below the level $K - \sigma$, $\epsilon_{\mathrm{fg}}(\nu) = C_{\mathrm{fg}}(\nu+1)^{2\mathfrak{p}_0+2}L^{-\nu/2}$ with C_{fg} basic; the third is the decay over the distance d' in the units of the working level; no number of it reads K or β. *Second link, between the cutoffs.* The laws of the block field of the depth σ at the cutoffs K and $K+1$ differ in total variation by at most $C_{\mathrm{bf}}\ell^3 K^{\mathfrak{p}_0} L^{-K/10}$, $\mathfrak{p}_0 = 2\mathfrak{p}_0 + \frac{1}{2}$ the exponent of Def. 2.7(5) (Cor. 2.8(a)); over one block field an entry read at the two backgrounds changes by the defect of (d); and a first member read at a background is at most $\frac{3}{2}B_3^2\ell^3 L^{4\sigma}$ times its size.

With $\sigma = \lfloor K/(1260n)\rfloor$, $s = \lfloor \sigma/(8n+3)\rfloor$ and $b = 4n+1$ the three errors are at most $L^2 L^{-\theta_n K}$, $L^{-K/3}$ and $L^{-K/11}$ times the sizes, up to powers of K and of ℓ: the first as $\sigma - s \ge (8n+2)s$ makes each term of the bracket at most a power of K times $L^{-(4n+1)s}$, and $L^{-s} \le L^2 L^{-\theta_n K}$; the second as $L^{4n\sigma}L^{-(K-\sigma)/2} \le L^{-K/3}$; the third as $L^{4n\sigma}L^{-K/10} \le L^{-K/11}$: this is θ_n. A Lipschitz weight read at the points of two lattices has the mismatch $m \le \frac{3}{2}a_K \mathrm{Lip}\,w$, whence the term $L^{-\sigma}\mathrm{Lip}\,w$; cell means have $m = 0$, as the cell of a coarse site is the union of the cells of the sites of its block, whence sup norms. (b) The first link reads the label only through the data of the depth s of $(K, L^K\mathsf{b})$, whose numbers are uniform on the compact J, the numbers of its error being free of the label.

Otherwise the label enters the proof of (a) only through C_{bf} and K_{bf}, which depend on it only through J (Cor. 2.8(a)), through the condition $L^\sigma\mathsf{b} \ge \Theta_\star$ of the data of the comparison at the depth σ, which holds at every label of J once it holds at the least one, and through the radius ε_r of the backgrounds, which the calibration conditions in equation YM3_I:(2148) bound at the levels where $L^\sigma\mathsf{b} \ge \Theta_\star$, and of which the defects of (c), (d) read only $\varepsilon_r \le 1$. (c) Over a block, in a suitable gauge, each loop at the background is $\pm$ one plaquette variable of the block, depending only on the plane of the loop, up to $O(\varpi^2)$, ϖ the size of the background; the variance of the plaquette variables of a block is bounded, plaquette by plaquette, by the defect of Jensen's inequality, whose sum is the action that one averaging step loses, and this is controlled by the weighted actions of the backgrounds (Prop. YM3_I:113.9). (d) The first display is (c) carried across one averaging step, the block mean tying the weights; for the second, $\mathcal{M}U^B$ is nearly minimal at the level t (Props. YM3_I:113.5, YM3_I:113.7), and a nearly minimal competitor has nearly the corner forms of the background, by its energy; this gives the factor $t^{1/2}$ and, with $t \le K$, the defect. □

Step 1: the tori, (a)–(c) *and* (f). By [7, eq. 1], $O_K^T(H) = \sum_\lambda \mathcal{O}(H_{\lambda\lambda}1_\lambda) + \sum_{\lambda<\lambda'} O^{\mathrm{od}}_{\lambda\lambda'}(H_{\lambda\lambda'})$, $O^{\mathrm{od}}_{\lambda\lambda'}(\phi) = -a_K^{-1}\sum_{b\|\mu}\phi(x_b)q_b$ for μ the third direction, with weights supported in $\operatorname{supp} H$, of sup norm $\le |H|$ and Lipschitz number $\le \operatorname{Lip} H$; so a cumulant of the $O_K^T(H_a)$ is a sum of 6^n cumulants of entries. Lem. 2.11(a) gives eq. 2.xx, and summing over $K' \ge K$ the limit with its rate; the bound of $\mathcal{S}_n$ is Lem. 2.10(b) at $\delta = \frac{d}{2}$; the formula for the centred moments holds at every K.

With $\bar O_K^T$ the cumulant at fixed K is a polynomial in finitely many cell means of the H_a, each continuous under bounded pointwise convergence, and the rate is uniform on balls of the sup norm: so the limit is continuous under bounded pointwise convergence, which makes it countably additive in each argument, and it vanishes when all the cell means vanish, in particular on null sets; the theorem of Radon–Nikodym gives the density. For (b), Lem. 2.10(a) gives the invariance under $a_K\mathbb{Z}^3 \rtimes W_3$ at every cutoff; a real and a lattice translation of a weight differ by a weight of sup norm $\le a_K \operatorname{Lip} H$, and the bounds are in sup norms. As L is odd, $\{x_\mu = c\}$ with $c \in a_{K_0}(\mathbb{Z} + \frac12)$ is a bond mirror of all the lattices with $K \ge K_0$, at which Wilson's measure is reflection positive (Cor. YM3_I:85.3, Prop. YM3_I:91.4) and $\theta O_K^T(H) = O_K^T(\theta H)$; such c are dense, and the reflection at another c is conjugate to one of these by a translation.

At $H_a = f_a\mathbf{1}$, $O_K^T(f_a\mathbf{1}) = \mathcal{O}(f_a)$. For (c): every class insertion is, at the rate, its standard form, built from the corner fields with the loops $l_\lambda = (\kappa,\kappa')$, $\kappa<\kappa'$ the two directions other than λ, at the sites. The shift ξ costs a site form of norm $\le \frac12 a_K \operatorname{Lip}\phi$, bounded by Lem. 2.10(b); loops in one plane are exchanged, with the sign $s(l)s(l')$, inside the rate of Lem. 2.11(a), the difference being tied at the depth σ with the defect of Lem. 2.11(c); and $a_p = \frac12 Q_x(l_\lambda, l_\lambda) + \frac12 a_p^2$, whose second term is of fourth order at the backgrounds. The standard form depends on the symbol alone; $O_K^T(\operatorname{Sym}\mathcal{E})$ is a class insertion with the same symbol, and (a) gives the limit; with the same weights at the same sites no shift and no Lipschitz number occur.

For (f), take the numbers C_{bf}, K_{bf} of Def. 2.7(5) at a compact interval $J \subset I(g)$ that holds b in its interior: by Cor. 2.8(a) they read the label only through J, and the comparison of the block fields of two successive cutoffs holds with them at every label of J. The rate of (a) reads the label at these places alone: C_{bf} and K_{bf}; the condition $L^\sigma\mathsf{b} \ge \Theta_\star$, which is monotone in the label; the radius of the backgrounds, which the calibration conditions in equation YM3_I:(2148) bound at the levels where $L^\sigma\mathsf{b} \ge \Theta_\star$; and the data of the depth s of $(K, L^K\mathsf{b})$, whose numbers are uniform on the compact J.

So, by Lem. 2.11(b), one exponent p, one C^{od}, one C^{av} and one threshold K_1 serve all labels of J, for the entries and for the cell means. At a fixed K, $\mathcal{S}_{n;K}$ is continuous in the label, an integral over a compact group against a density continuous in $\beta = L^K\mathsf{b}$; so $\mathcal{S}_n^{\ell,\cdot}$ is a uniform limit on J of functions continuous in the label, hence continuous on J; with $\mathsf{b}_K = a_K\beta_K \in J$ for large K,

$$|\mathcal{S}_{n;K}(\mathsf{b}_K) - \mathcal{S}_n(\mathsf{b})| \le |\mathcal{S}_{n;K}(\mathsf{b}_K) - \mathcal{S}_n(\mathsf{b}_K)| + |\mathcal{S}_n(\mathsf{b}_K) - \mathcal{S}_n(\mathsf{b})| \to 0,$$

as in Cor. YM3_I:117.1; with $\bar O_K^T$ the same.

Step 2: every label and side; all of (H). [7, Lem. C.8](a) gives

$$\langle O_K^T(H_1);\dots;O_K^T(H_n)\rangle_{K,\ell_*,L^K\mathsf{b}_*} = L^{zn}\langle O^T_{K-z}(H_1^{(z)});\dots;O^T_{K-z}(H_n^{(z)})\rangle_{K-z,L^z\ell_*,L^{K-z}(L^z\mathsf{b}_*)},$$

which carries Thm. A to every label and side, with $L^z d$ for d; with $g' = (L^z\mathsf{b}/2)^{-1/2}$ the label $L^z\mathsf{b}$ is an inner point of $I(g')$, which carries (f). Lems. [7, Lem. C.12] and [7, Lem. C.13], [7, Prop. C.21] for its limits and its growth, and Steps 6, 7 and 10–13 of the proof of [7, Thms. BU–BX] are first carried out at couplings $g \le \gamma_W^{\mathrm{sd}}$; [7, Lem. C.8](b) carries the data, the states, the limits, the Hilbert spaces, the semigroups, [7, eq. 263] and the decomposition to all of (H), and $\sigma = L^{nz}\sigma_z(X^{(z)})$ carries the bounds of [7, Prop. C.21](b), (c) and its growth, the powers of L^z entering C_G and C; the count of [7, Prop. C.21](a) at a cutoff holds on all of (H) with the numbers set in its statement. In particular one may take $d_0(g) = L^{-z}\min_i d_0(g_i')$, with the error $\max_i \mathsf{e}^\infty_{g_i'}(L^z d)$ at d, for z large enough at g.

Step 3: two points on one torus. By [7, Lem. C.10](a) and $|K_3(u)_{\lambda\lambda'}| \le (2\pi|u|^3)^{-1}$, $|u| \ge d$ on the supports, the Wick value is a Riemann sum of $\mathcal{W}$: with $a_K^3 \sum_{p,p'} |\mathfrak{t}_H(p,p')| \le 45 d^3 |H|$, by the row sums $2\sum_{p'} |\mathfrak{t}_H(p,p')| \le 3|H|$ of the kernel of $O_K^T(H)$ (Step 14(a)) and the counts of the plaquettes and the bonds in a ball,

$$\Big| 6 \sum \mathfrak{t}_1 \mathfrak{t}_2 CC - \mathcal{W}(H_1, H_2) \Big| \le 12150\, E_1 (\tfrac{1}{\pi} + E_1) |H_1||H_2| + \mathrm{rm}_K(H_1, H_2),$$

where rm_K is the difference between a Riemann sum over the anchors, that is the barycentres, the midpoints or the sites of the cells, and the integral $\mathcal{W}$; it depends only on K, d and the weights, and it tends to 0, by the uniform continuity of K_3 on the two balls, for continuous weights and for cell means. With $|\mathcal{W}| \le 54 |H_1||H_2|$ ([7, Lem. C.9](a)) and the bounds on v^2, eq. 2.xxi follows at finite cutoff with

$$\mathsf{e}_{g,\ell}^T(d) = \tfrac{9}{8} \big[3600\, \varepsilon_{g,\ell}^e(s) + 12150\, E^\infty (\tfrac{1}{\pi} + E^\infty) \big] + 162 \bar{B} g^2 L^{2-s}, \qquad E^\infty = E_1 - C_{\mathrm{lat}} (a_K/d)^2,$$

and $\frac{9}{8} \cdot 12150 \, [E_1(\frac{1}{\pi} + E_1) - E^\infty(\frac{1}{\pi} + E^\infty)] \le c_K$; the term $162 \bar{B} g^2 L^{2-s}$ is 54 times $3\bar{B} g^2 L^{2-s}$. As $K \to \infty$, (a) gives eq. 2.xxi; $\mathsf{e}^T \to 0$ like $(\log \frac{1}{d})^{-2\mathfrak{p}_0}$ because $Y_s^{2\mathfrak{p}_0} \varepsilon_{g,\ell}^e(s)$ is bounded and $t = L^s d \ge C_W(s) \to \infty$. The class: (c) at $n = 2$. The bound on $s_1(g, \ell)$ and the independence of ℓ: [7, Lem. C.10](b). (i): $\mathcal{W}$ is invariant ([7, Lem. C.9](a)) and $|rH| \le 3|H|$, so the law at (H_1, H_2) and at (rH_1, rH_2) gives $(1 + 9)\mathsf{e}^T$. (ii): [7, Lem. C.9](b) and the law at the pair $(\theta(\phi E), \phi E)$, whose centres are $3d$ apart, with $|\phi E|^2 \le \mathrm{tr}\, E^2$. (iii): [7, Lem. C.10](c) at $c \in a_{K_0}(\mathbb{Z} + \frac{1}{2})$ and the limit; a general c by (b).

Step 4: conservation on the torus. Cut η by a partition of unity into pieces of diameter < 1; by multilinearity it suffices to treat one piece. In its slot, by [7, eq. 264], $\mathsf{b} O_K^T(H^{\mathrm{cons}}[\eta]) = \mathrm{St}_K[\eta] - \Delta_K[\eta]$, and $\mathrm{St}_K[\eta]$ may be replaced by $-\mathrm{Rm}_K[\eta]$, the null terms $\mathcal{N}_\eta$, $\mathcal{N}'_\eta$ vanishing against the far entries. This leaves the remainders of [7, Prop. C.17](a): the cubic one $\mathcal{S}[\mathsf{b}\eta]$, the five local forms of J_η, the form $\frac{3}{2} a_K \mathcal{O}(w_\eta)$ and the forms of $\Delta_K[\eta]$, each with a conditional mean, given the block field of depth ϕK, that is constant on T_ℓ up to $K^{\mathfrak{p}} L^{-(\frac{1}{28} - 3\phi)K} \|\eta\|_{C^2}$: by [7, Prop. C.17](c) for the cubic one; by power counting at the backgrounds of the level $K - \phi K$, as in the proof of [7, Prop. C.17](b), for the local forms, those of the order 2 (the three of J_η, with weights at most $(3 + 5\mathsf{b})\|\eta\|_{C^2}$, the form $\frac{3}{2} a_K \mathcal{O}(w_\eta)$ and the two forms of $\Delta_K[\eta]$ with weights at most $60 \mathsf{b} a_K \sup |\nabla^2 \eta|$) having oscillations at most $C K^{2\mathfrak{p}_0} L^{4\phi K - K} \|\eta\|_{C^2}$, and the two of the orders 3 and 4 of J_η, with weights at most $L^K \mathsf{b} \sup |\nabla \eta|$, oscillations at most $C K^{3\mathfrak{p}_0} L^{4\phi K - \frac{K}{2}} \|\eta\|_{C^2}$; and by the cut at the level $K - \lceil K/14 \rceil$ for the form of vanishing symbol of $\Delta_K[\eta]$ alone.

A joint cumulant of entries far apart is at most the product of the oscillations of their conditional means at that level, a spectator costing $L^{4\phi K}$; at $\phi = 1/(56(4q-1))$ the exponent is $-\frac{1}{28} + (4q-1)\phi = -\frac{1}{56}$. (a) gives the limit, and the moments vanish, being sums of products of cumulants of which one holds the slot of $H^{\mathrm{cons}}[\eta]$. The estimate of [7, Prop. C.17](c) is uniform on the interval of labels, which gives the uniformity in the label; $\bar{O}_K^T$ as in Step 1; a change by forms with weights $\le 36 a_K \|\eta\|_{C^2}$ costs, by Lem. 2.10(b), a_K times a number independent of K. At $H_a = h_a \mathbf{1}$ the spectators are action densities.

The rate $L^{-K/36}$ also holds by the argument of [7, proof of Thms. BU–BX, Step 8] on T_ℓ.

Step 14: coinciding points, overlaps and more points. (a): $O_K^T(H)$ is the first member of a source of Thm. YM3_I:D under a majorant of size $60|H|$, carried by the plaquettes p with $H_{\lambda(p)\lambda(p)}(x(p)) \ne 0$ and by the plaquettes through the bonds b with $H_{\lambda_b \lambda'_b}(x_b) \ne 0$ (Lem. 2.10(b)): its plaquette part has the size $|H|$, and its bond part $-a_K^{-1} \sum_b H_{\lambda_b \lambda'_b}(x_b) q_b$ the size $59|H|$. Indeed q_b is a bilinear form in the vector parts $w(W_{x,(\mu_b,\alpha)})$, $\alpha \in \{\pm\lambda_b, \pm\lambda'_b\}$, of the four plaquette loops at the bond $b = \langle x, x + a_K e_{\mu_b} \rangle$; for a form $-a_K^{-1} \sum_b \mathsf{w}(b) q_b$ the members at the height 0 and at every level are carried by the at most six cubes with a common corner that hold the four plaquettes through b, and they are bounded there as those of a plaquette source of the weight $59 \sup |\mathsf{w}|$ on these plaquettes;

to the second order at a small field the form is a_K^{-1} times the quadratic form $-\sum_b \mathsf{w}(b)\langle \mathsf{u}_{b,\lambda_b}, \mathsf{u}_{b,\lambda'_b}\rangle$ in the plaquette variables, $\mathsf{u}_{b,\lambda}$ linear forms with the coefficients $0, \pm\frac12$, whose symmetric kernel has the row sums $2\sum_{p'} |\mathfrak{t}(p,p')| \le 2\sup|\mathsf{w}|$, so that the kernel $\mathfrak{t}_H$ of $O_K^T(H)$ has $2\sum_{p'}|\mathfrak{t}_H(p,p')| \le 3|H|$.

So at the depth $s(d)$ an insertion of size d is a source of size $v|H|$ in the units of the working level, and Thm. YM3_I:D bounds the cumulant of n such sources, wherever they lie, by $60^n k'_n(g,s)\mathsf{b}^{-n}\prod_a |H_a|$ on a side that is small over the canonical sequence of (K,β) (equation YM3_I:(2312)), for $g \le \gamma_W^{\mathrm{sd}}$ the least side ℓ_g (Lem. YM3_I:126.3), the bound reading only the sizes of the sources; $k'_n(g,s) \le C_n(g) Y_s^{8n\mathfrak{p}_0}$, of the $8\mathfrak{p}_0$ of one entry $2\mathfrak{p}_0$ coming from the supremum of the plaquette action, in units of the working level, on the small fields, and $6\mathfrak{p}_0$ from the number $(LC_W(s))^3$ of the unit cubes of the working level in a ball of radius d; and $Y_{s(d)} \le 2\log\frac{1}{d} + C(g)$.

On every other side $\ell \in \mathfrak{L}(g)$ the histories with a large field on T_ℓ, summed against their weights (Prop. 4.54), add the factor $\max\{1, \mathsf{c}(\ell)\}^n$, $\mathsf{c}(\ell) = 2\ell^3 e^{\mathsf{B}(\bar{x}_0)\ell^3}/(c' A_0^2)$, with the function B and the number c' of equation YM3_I:(2311), as V_ℓ does in Lem. 2.10(b): this is the ℓ of $C_n(g,\ell)$. The scale-covariant form is the same bound read at the coupling γ_W^{sd} after the dilation of [7, Lem. C.8] by L^z: $(K-z,\beta)$ is a first-passage datum at γ_W^{sd} with the label $L^z\mathsf{b} \in ((\gamma_W^{\mathrm{sd}})^{-2}, L((\gamma_W^{\mathrm{sd}})^{-2} + \bar{B})]$, which gives $\delta_\star \mathsf{b} \le L^{-z} < (\gamma_W^{\mathrm{sd}})^2\mathsf{b}$, so that $\log\frac{1}{L^z d}$ differs from $\log\frac{\mathsf{b}}{d}$ by at most a basic number, and the factor L^{zn} of the dilation turns $(L^z\mathsf{b})^{-n}$ into b^{-n}; the depth and the threshold are those of the dilated distance at γ_W^{sd}, and the shift $(K,\ell,\mathsf{b},d) \mapsto (K-1, L\ell, L\mathsf{b}, Ld)$ is one more step of the dilation.

Supports only at pairwise Euclidean distances $\ge d$ are cut into the boxes Q of sides $\le \frac{99}{100}d''$ of a partition of T_ℓ into at most $(2\ell/d'')^3$ boxes, $d'' = \min\{\frac34 d, L^{-s_{\mathrm{th}}(n,g)}\}$: a piece $H_a 1_Q$ lies within the Euclidean distance $\frac{\sqrt3}{2}\cdot\frac{99}{100}d'' < 0.8574\,d'' \le 0.87\,d''$ of the centre of Q, and pieces of different entries are at Euclidean distances $\ge d \ge \frac43 d'' \ge 1.26\,d''$; the bound at the distance d'' for each product of pieces, summed by the n-linearity, the sum of the products being the product of the sums, gives $\prod_a\big(\sum_Q \sup_Q |H_a|\big)$ in the place of $\prod_a |H_a|$, at the same b^{-n}: that is $(\mathsf{b}d''^3)^{-n}$ times the product of the norms $d''^3\sum_Q \sup_Q |H_a|$, a factor d''^{-3} for each entry.

At $n=2$, for $g \le \gamma_W^{\mathrm{sd}}$, every $\ell \in \mathfrak{L}(g)$, $1.31h < 1$ and $s(1.31h) \le K$, two off-diagonal bond forms $Y[\mathsf{w}_\iota] = -a_K^{-1}\sum_b \mathsf{w}_\iota(b) q_b$ whose bonds have their midpoints in cubes of side $\frac32 h$ at the sup distance $\ge \frac52 h$ have $|\langle Y[\mathsf{w}_1]; Y[\mathsf{w}_2]\rangle_{K,\ell,\beta}| \le 36\, C_\infty(g)\mathsf{b}^{-2}\sup|\mathsf{w}_1|\sup|\mathsf{w}_2|$, with the number $C_\infty(g)$ of [7, Lem. C.7], of g alone. This rests on [7, Lem. C.7], applied at $d = 1.31h$: at every $0 < d < 1$ with $s(d) \le K$, two real bond forms of one direction each, with $|x_b - x_\iota| \le d - 2a_K$ on $\operatorname{supp}\mathsf{w}_\iota$ and $|x_1 - x_2|_\infty \ge 3d + a_K$, with no upper bound on that distance, have $|\langle Y[\mathsf{w}_1]; Y[\mathsf{w}_2]\rangle_{K,\ell,\beta}| \le C_\infty(g)\mathsf{b}^{-2}\sup|\mathsf{w}_1|\sup|\mathsf{w}_2|$.

The means vanish, by Lem. 2.10(a); a bond mirror parts the two forms at the distance $\ge \frac32 d$ from both points, and reflection positivity there (Cor. YM3_I:85.3, Prop. YM3_I:91.4), that is Schwarz's inequality for the positive form, bounds the cumulant by the square roots of the two mirror pairs $\langle \theta Y[\mathsf{w}_\iota]; Y[\mathsf{w}_\iota]\rangle$. By the transfer operator of the normal direction, positive by reflection positivity, a mirror pair is, as a function of the distance t of the form from the mirror, a sum of exponentials with positive coefficients, symmetric under $t \mapsto \frac{\ell}{2} - t$, as the reflection fixes a second plane of the torus at the distance $\frac{\ell}{2}$; so on $[\frac32 d, \frac{\ell}{2} - \frac32 d]$ it is largest at $t = \frac32 d$, where the form and its image are at the distance $3d$.

The proof of [7, Lem. C.7] bounds such a mirror pair on every torus of $\mathfrak{L}(g)$ and at every $K \ge s(d)$ by a number of g alone times b^{-2} and the sizes, the cutoffs $K = s(d)$ and $K = s(d)+1$ among them; for $d < d_0(g)$ and $K \ge s(d)+2$ this is also the range of [7, Lem. C.12]: on every torus of $\mathfrak{L}(g)$ the pair has its Wick value of [7, Lem. C.10](a) on T_{ℓ_g} up to a number of g alone times v^2 and the sizes, and the Wick value of two entries of size d at the distance $3d$ is at most a basic multiple of $v^2 \le \frac98\mathsf{b}^{-2}$ times the sizes, by $|C(p,q)| \le C a_K^3 d^{-3}$ and $a_K^3\sum_{p,p'}|\mathfrak{t}_H(p,p')| \le 45 d^3|H|$, as in [7, Lem. C.10](c); for $d \ge d_0(g)$ the forms are cut into pieces of a size below $d_0(g)$, whose pairs are

bounded so, in a number that depends on g alone; and where K lies below the depth that these pieces ask, the crude bound $c\,a_K^{-4}$ of a form serves, as in [7, Lem. C.10](c).

At $d = 1.31h$, x_ι the centres of the cubes: $|x_b - x_\iota| \le 1.2991h \le d - 2a_K$ and $|x_1 - x_2|_\infty \ge 4h \ge 3d + a_K$; and a complex weight of the bonds is the sum of six weights, each i^ϵ times a real weight carried by the bonds of one direction, of sup norm at most that of the weight: thirty-six pairs. The bound passes to $\mathcal{S}_n$ by Thm. A(a) for $\bar{O}_K^T$ and continuity, which gives the power counting of [7, Prop. C.22] with $q = 8n\mathfrak{p}_0$ and $h_0 \in \ell 2^{-\mathbb{N}}$ with $h_0 \le \min\{\frac{\ell}{64n}, \frac{1}{2}L^{-s_{\mathrm{th}}(n,g)}\}$; [7, Prop. C.22] gives the extension and the contact terms; the same subtractions made at each cutoff, with the products of cubes restricted to those of a side h with $s(1.52h) \le K$, give lattice functionals that are bounded uniformly in K and converge to the extension.

The counts at two points are those of Lem. 3.6(f): take real, exchange-symmetric, translation-invariant contact terms in $\mathfrak{E}_2(0)$. They are $\sum_{|\alpha|\le 3} \partial^\alpha \delta(x-y)\, c_\alpha(e, e')$ with bilinear c_α, symmetric under the exchange of the two entries; under $O(3)$ on $\mathbb{R}^3$, which contains $-\mathbf{1}$, the odd orders drop, and there remain $\mathrm{tr}\, e\, \mathrm{tr}\, e'$ and $\mathrm{tr}(ee')$ at the order 0 and the contractions of $\partial_i\partial_j$ with $\delta_{ij}\mathrm{tr}\, e\, \mathrm{tr}\, e'$, $\delta_{ij}\mathrm{tr}(ee')$, $e_{ij}\mathrm{tr}\, e' + e'_{ij}\mathrm{tr}\, e$ and $(ee' + e'e)_{ij}$ at the order 2: six; under W_3, on $\mathbb{R}^3$ or T_ℓ, there are twelve. $\mathcal{S}_2$ is no measure across the diagonal: at the scale d a unit cube holds $\gtrsim d^{-3}$ disjoint mirror pairs of balls, each of mass $\ge (\mathsf{w}_0 - \mathsf{e}^T)\mathsf{b}^{-2}\mathrm{tr}\, E^2$ by Thm. A(d)(ii). (b): the action densities stay in the exponent: their weights form the source $w_z = \sum_i z_i f_i$ of the sourced bare density $\rho_0^{w_z} = \exp[-\beta A_K + \sum_i z_i \mathcal{O}(f_i)]$ of equation YM3_I:(10), which Thm. YM3_I:D carries through all the levels as a source at the plaquettes; the tensor entries, in the third and fourth cases the bond forms $Y_a = -a_K^{-1}\sum_{b\|\mu_a} \phi_a(x_b) q_b$ of the off-diagonal labels E_a, enter through their sources of Lem. 2.10(b) and are derivatives at the strength 0: $\kappa_K = \partial_{z_1}\cdots\partial_{z_n}\partial_{\tau_1}\cdots\partial_{\tau_r} \log \int \exp[\sum_a \tau_a Y_a]\, \rho_0^{w_z}\, \mathrm{d}U$ at $z = 0$, $\tau = 0$; and likewise, in the first two cases, where the tensor weights do not overlap, for diagonal labels with the plaquette forms $Y_a = a_K^{-1}\sum_p \phi_a(x(p))(E_a)_{\lambda(p)\lambda(p)} a_p$, and for every $E_a \in V$ with $Y_a = O_K^T(\phi_a E_a)$, by [7, eq. 1] the plaquette form of the diagonal part of E_a plus the bond forms of its off-diagonal part.

Differentiated at the strength 0 along the tensor entries, the step maps of Thm. YM3_I:D give a recursion for the members of each order, each derivative of a step being bounded by Cauchy's estimate in the strengths. The input of §2.3.3 is used at order two: for two bond forms Y_a, Y_b whose weights overlap, one renormalisation step at the depth u creates a part $\mathfrak{l}$ bounded by cL^u and a remainder bounded by $cK^{\mathsf{p}}L^{u/2}$ in the units of the sizes, and $\mathfrak{l}$, read at a background of a lower depth u', is at most $K^{\mathsf{p}}\mathsf{b}^{-2}NL^{-2(u-u')}$, N the number of the sites of the level of that depth: the arrays of order two born above a depth are forgotten below it, and this carries the case of two overlapping entries.

The remaining step is to establish this second-derivative estimate, including its stated forgetting bound, for these bond forms; and it is here that the third and fourth cases use that the labels E_a are off-diagonal. Where the tensor supports are $\ge d$ apart, a member of order ≥ 2 vanishes unless its polymer reaches across the distance d; where only $\phi_a\phi_b = 0$, the arrays of order $m \ge 2$ are at most $K^{\mathsf{p}}L^{-mu/2}$ in the units of the sizes, the weights not overlapping, and with the condition on the $r + n$ supports this gives the same rate. For $r \ge 3$ off-diagonal entries of which any two may overlap and no three supports share a point, each overlapping pair is carried by its arrays of order two, which are forgotten below their depth as for two entries; an array of order $m \ge 3$ joins m entries whose supports have no common point, so, the supports being compact, its polymer must cross a positive distance fixed by their geometry, and it is bounded as where the supports are apart; the depths proportional to K then give $c = \frac{1}{360r}$.

Two cutoffs are then compared as in Lem. 2.11, at depths proportional to K, which gives the rates. (c), with Euclidean distances between sets, here and in (d): at the depth $s(d)$ the n-th cumulant of the quadratic insertions in the Gaussian part is the sum over the cyclic orders, $3\mathsf{b}^{-n}K_K^{(n)}$, the value for three free lattice photon fields, for which the left member vanishes identically; the uniform

interacting remainder is the estimate still to be established in §5.1.2, with its explicit first-creation term bounded by the weights ϖ_t; for the plane fields the free lattice cumulants differ from $3\mathsf{b}^{-n}K_{\underline{\lambda}}^{(n)}$ by at most $\frac{1}{3}C_n(\frac{a_K}{\vartheta_0 d}+\frac{d^3}{\ell^3})\prod_a|\phi_a|$, by the comparison of the lattice kernel with the free kernel (equations YM3_I:(2245)–(2246)).

Every periodic Gibbs state and every torus of $\mathfrak{L}(g)$, for $2\le n\le 7$: establishing the remainder estimate on T_{ℓ_g} and [7, Lem. C.16](a), the weights being bounded functions of the plaquettes in balls of radius d within one ball of radius $9d$, give the lattice inequality in §5.1.2 with $\varepsilon_n^{\mathrm{rt}}(g,d)$ added to e_n, and $\varepsilon_n^{\mathrm{rt}}(g,d)\to 0$ with d since some $\sigma\in(0,1)$ has $4-\frac{n}{2}-3\sigma>0$ exactly when $n\le 7$. (d): [7, Lem. C.15](c) gives the laws at a cutoff on T_{ℓ_g}, the large fields being bounded as in [7, Lem. C.15](a), and the comparison of the free lattice cycles with k_3, k_4, by equations YM3_I:(2245)–(2246), gives the terms $(d/\ell_g)^3$; [7, Lem. C.16](c) carries the laws to every periodic Gibbs state, at three points with the further $C(g)d^{3/2}\rho^9 r^{-3}d^{-6}\prod_a|w_a|$ and at four with $C(g)d^{3/2}\rho^{12}r^{-6}d^{-6}\prod_a|w_a|$, $d^{3/2}$ times the natural sizes; first passing to a continuum subsequential limit at fixed positive widths, then letting the widths tend to 0 and using Lebesgue's differentiation theorem gives the kernels, with estimates uniform in the states, those of the truncated functions; the law of four points is that on the torus and at a cutoff; for limits and re-centred components σ_ξ^c, [7, Lem. C.14] and [7, Thm. BW](d), (f), the kernels being continuous, and for a limit the third centred moment being the third cumulant; the point-split law follows from the three-point law, [7, Lem. C.9](d) and the kernels of [7, Thm. BV](e), the constant 2.7 covering 2.66 and the passage from $\mathrm{dist}(z,\{x,y\})$ to $|c-z|$.

Proof of Cor. 2.9. By [7, Thm. BW](c), (d) the kernel $s_2(x,y;e,\mathbf{1})$ is real-analytic, a function of $u=x-y$, and linear in e, so $s_2(x,y;e,\mathbf{1})=\mathrm{tr}(e\,\tau(u))$ with $\tau(u)\in V$; invariance under $O(3)$ gives $\tau(Ru)=R\tau(u)R^t$, so $\tau(u)$ commutes with the stabiliser of $\hat u$ and is $A(r)\mathbf{1}+B(r)\hat u\hat u^t$, and $\mathrm{tr}\,\tau=3A+B=F$. Conservation against one action density, $\sigma((H^{\mathrm{cons}}[\eta]),(f\mathbf{1}))=0$ for $\mathrm{supp}\,\eta$ off $\mathrm{supp}\,f$, reads $\iint[(\nabla\cdot\eta)(x)F(x-y)-2\sum_{ij}\partial_i\eta_j(x)\tau_{ij}(x-y)]f(y)\,\mathrm{d}x\,\mathrm{d}y=0$; integration by parts in x, for all such η and f, gives $2\sum_i\partial_i\tau_{ij}=\partial_jF$ off $u=0$, that is $2(A'+B'+\frac{2}{r}B)=F'$, as $\sum_i\partial_i(B\hat u_i\hat u_j)=(B'+\frac{2}{r}B)\hat u_j$. With $A=\frac{1}{3}(F-B)$ this is $B'+\frac{3}{r}B=\frac{1}{4}F'$, and $(r^3B)'=\frac{1}{4}r^3F'$. □

Remark 2.12. The limits on $\mathbb{R}^3$ are taken along subsequences of Gibbs sequences, and each theorem speaks of the limit of the sequence chosen; the label, the bounds and the laws at short distance are the same for all of them. The uniqueness of the vacuum in [7, Thm. BW](f) is that of each component σ_ξ, and the vacuum of $\mathcal{H}(\sigma)$ is unique exactly when the index space X is a point. [7, Thm. BW] concerns limits through periodic Gibbs states; of the limits along growing tori [7, Thm. BV](c) is asserted, with positivity at the coordinate mirrors alone. The laws at short distance are laws to leading order, with errors that tend to 0 with the distance, and Thm. A and Thm. B(a), (b) concern one torus at a time, with constants that depend on its side.

2.3.4. *Notation and hypotheses. Data.* The radius, the constants and the data are those of the main theorems (Prop. YM3_I:69.1 and Section YM3_I:82; the constants chosen in the order of that proof; the uniform free-response geometry, radius and decay conditions in Section YM3_I:52.6, retained in Section YM3_I:120.3; $c_\star$ by equation YM3_I:(1429)): L odd, M a power of L, A_0, A_1, $P=2\mathfrak{p}_0\ge 18$, $\eth\in(0,1)$, $c'=(1-\eth)c_\star$, the last constant γ, and the ceilings $\gamma_1=\gamma_W^{\mathrm{sd}}\le\gamma_2=\gamma_W^{\mathrm{ref}}\le\gamma$ of the choices in equations YM3_I:(2326)–(2327). $B_\sharp\ge 0$ is the flow bound: $|L^{-1}x_{j-1}-x_j|\le B_\sharp$ at every step ([7, §G.2]), and $\bar B=B_\sharp L/(L-1)$ (Section YM3_I:74); both are numbers of the constants alone. T_ℓ, $\ell=2L^m$, is the torus of side ℓ, $\Lambda_{K,\ell}$ its lattice of spacing $a_K=L^{-K}$, $\mathbb{L}_K=a_K\mathbb{Z}^3$; $|\cdot|$ and $|\cdot|_\infty$ are the Euclidean and the maximum norm, on a torus minimised over lifts; $\hat u=u/|u|$.

A first-passage datum (K,β) at the reference coupling g (YM3_I:74.2, YM3_I:74.1) has the running couplings $x_0=\beta,\dots,x_K$, with $|L^{-1}x_{t-1}-x_t|\le B_\sharp$ for $1\le t\le K$ (Prop. YM3_I:74.1), so that, the differences telescoping, $|L^{-z}x_{K-z}-x_K|\le B_\sharp\sum_{0\le i<z}L^{-i}\le\bar B$ for $0\le z\le K$; $y_t=\log x_t$, $g_t=x_t^{-1/2}$, the large-field threshold $p_0(g_t)^2=A_0^2y_t^P$; the first passage $x_K\le X_g<x_{K-1}$ YM3_I:74 at

$X_g = L(g^{-2} + B_\sharp)$, so that $X_g + \bar B = L(g^{-2} + \bar B)$ YM3_I:(1526); and the label $b = a_K\beta \in I^{\mathrm{lab}}(g) := [g^{-2}, \bar x_0]$, $\bar x_0 = L(g^{-2} + \bar B) = X_g + \bar B$; for $\mathsf b$ in the interval $I(g) = (g^{-2} + \bar B, X_g - \bar B] \subset I^{\mathrm{lab}}(g)$ of Section YM3_I:74, $X_g - \bar B = L(g^{-2} + \bar B) - 2\bar B$, $(K, L^K\mathsf b)$ is a first-passage datum for every integer $K \ge 1$. $\mathfrak L(g) = \{2L^m : m \ge m_{\mathfrak L}(g)\}$ is the family of the sides at g (equation YM3_I:(1528)), $\ell' = 2\mathsf l$, $\mathsf l = L^{m_{\mathfrak L}(g)}$, its least member, $\mathsf l \ge LM$.

A side $\ell = 2L^m$ is *admissible* if $MR(x_K) \le L^m$, $R(x)$ the least power of L that is at least $(\log x)^r$, r a constant; it is *small* if it is small over the running couplings in the sense of equation YM3_I:(2312): admissible, $LM \le L^m$, $\ell \ge N_{\mathsf H}$, the lower bound of condition (H), and $\mathsf B(x_K)\ell^3 \le \frac12 c' p_0(g_K)^2$, $\mathsf B$ the function equation YM3_I:(2311) of the volume factor $e^{\mathsf B\ell^3}$ (equation eq. 4.xxxii); for $g \le \gamma_1$ the side ℓ' is small for every datum (Lem. YM3_I:126.3). As R does not decrease, $x_K \le X_g < \bar x_0$ and $L^{m_{\mathfrak L}(g)} \ge LMR(\bar x_0)$ YM3_I:(1528), every side $2L^m \in \mathfrak L(g)$ has $LMR(x_K) \le L^m$ for every datum at g: the sides of $\mathfrak L(g)$ are admissible.

Depth. $Y_s = (s+1)\log L + \log(g^{-2} + \bar B) = \log \bar x_s$, $\bar x_s = L^{s+1}(g^{-2} + \bar B)$ and $C_W(s) = C_{W,0}A_0^2 Y_s^P$ as in equation YM3_I:(2321), with the room constant $C_{W,0} = \max\{1, 4L^2M/A_0^2\}$ (equation YM3_I:(2320)), so that $C_W(s) \ge 4L^2MY_s^P$ and $L^{-2}C_W(s) \ge 4MY_s^P \ge M$; $\mathsf{Lm}_s = L^2C_W(s)Y_s^{3\mathfrak p_0}$, $x_s^- = 4L^{s-2}g^{-2}/(5\mathsf{Lm}_s)$. For $0 < d < 1$, $s(d)$ is the least s with $L^s d \ge C_W(s)$; so $Y_{s(d)} \le 2\log\frac1d + C(g)$. At the working level $\mathsf k = \max\{K - s(d), 2\}$: $y = y_{\mathsf k}$, $\hat d = d/a_{K-\mathsf k}$, $\upsilon^{-1} = a_{K-\mathsf k}x_{\mathsf k}$, and $\upsilon^2 \le \frac98 b^{-2}$; if $s(d) \le K$, then $\hat d \ge L^{s(d)-2}d \ge L^{-2}C_W(s(d)) \ge M$ and $d \ge C_W(s(d))a_{s(d)} \ge 4L^2Ma_K \ge 676a_K$.

The free template of Section YM3_I:120 has, for $0 \le n \le K$, the orthogonal projection $\Pi_n = \Pi_n^{\mathrm{tp}}$ onto the background space of the level n of the exact curvatures, with its kernel $\Pi_n(p,q)$ on the plaquettes (Section YM3_I:120); the free kernel is Π_0, the background kernel of the depth s' is $C^{\mathrm{bg}}_{K,s'} = \Pi_{K-s'}$, $0 \le s' \le K$, and the slice of the step j is $\mathcal C_{(j)} = \Pi_{j-1} - \Pi_j$, $1 \le j \le K$ YM3_I:(2227). So $\Pi_0 = \Pi_n + \sum_{1\le j\le n}\mathcal C_{(j)}$ for every n, and $\Pi_k - \Pi_T = \sum_{k<j\le T}\mathcal C_{(j)}$ for $k \le T$ YM3_I:(2227); in particular the free kernel is the sum of the background kernel Π_K of the top level and of all the slices.

With $\eta_n = L^{-n}$ and $\mathsf r = |x(p) - x(q)|_\infty$, $|\Pi_n(p,q)| \le C^{\mathrm{bg}}_{\mathrm{tp}}\eta_n^3$ and $|\mathcal C_{(j)}(p,q)| \le C_{\mathrm{tp}}\eta_{j-1}^3 e^{-\delta_{\mathrm{tp}}\mathsf r/a_{K-j+1}}$ (Props. YM3_I:120.5 and YM3_I:120.6). The background kernel of the working level is $C^{\mathrm{bg}} = C^{\mathrm{bg}}_{K,K-\mathsf k} = \Pi_{\mathsf k}$, and the free kernel is C^{bg} plus the slices $\mathcal C_{(j)}$, $j \le \mathsf k$, the slice $\mathcal C_{(j)}$ decaying exponentially beyond the scale a_{K-j+1}. $K_3(u) = (3\hat u\hat u^{\mathrm T} - \mathbf 1)/(4\pi|u|^3)$.

Large fields; the chain. The chain has the levels $0 \le j \le K$, $\eta_j = L^{-j}$, local families with the sizes of YM3_I:(759), the geometry of Prop. YM3_I:45.1 and [7, Lem. O.1], and the contraction numbers $\theta^{[j]}$. A history is a sequence of large-field regions up to a step; $Z_t(h)$ is its region at the level t, $\tilde\nu_K(h) = Z^{-1}\int \mathfrak w_h\rho_0\,\mathrm dU$ the law of the histories; $\mathsf w_t$ is the probability that the first large field is created at the step t, so that $\sum_{t'\le t}\mathsf w_{t'} = \tilde\nu_K(\{Z_t \ne \emptyset\})$; $\mathsf W_T = \sum_{t\le T}\mathsf w_t$. $\Phi^{(t)}[X]$ is the conditional mean of X given the block field of the level t on the history without large fields. For sources with majorants (Lem. 2.15) $\mathsf D_j$ is the least sup distance, in units of a_{K-j}, of corners of plaquettes of different sets S_a. For two sources at the working level, $\mathsf k < j_1 \le T$ are the middle and the upper level that go with $(\mathsf k, \hat d, y)$ (equation YM3_I:(2316)), $n_T = T - \mathsf k$, $L^{n_T} \ge \hat d y^{3\mathfrak p_0}$; for $\mathsf L \ge 0$, $c > 0$,

$$\mathrm{sm}_T^{\mathsf L,c} = \max\Big\{x_T^{-1/2},\ x_{\mathsf k}e^{-d_*(T)},\ x_{\mathsf k}e^{-(\mathsf D_{\mathsf k}/M-2)},\ \max_{\mathsf k\le i<T}(\delta_i + \varepsilon_i^{\mathrm{co}}),\ x_{\mathsf k}^2\mathsf L L^{3(K-\mathsf k)}e^{-cp_0(g_T)^2}\Big\},$$

and sm_T is its value at $c = \frac12 c'$ and $\mathsf L = \mathsf L_\ell := 2M^{-3}\ell^3$, the number of the weights of a small side: by the rarity of large fields on one torus (Prop. 4.54), $\mathsf w_t \le \tilde\nu_K(\{Z_t \ne \emptyset\}) \le 2M^{-3}\ell^3L^{3(K-t)}e^{-c'p_0(g_t)^2}e^{\mathsf B(x_K)\ell^3}$ for $1 \le t \le K$, for $g \le \gamma_2$, $K \ge 3$ and every admissible side, and on a small side $\mathsf B(x_K)\ell^3 \le \frac12 c'p_0(g_K)^2 \le \frac12 c'p_0(g_t)^2$, as $x_t \ge x_K$; so $\mathsf w_t \le \mathsf L_\ell L^{3(K-t)}e^{-\frac12 c'p_0(g_t)^2}$ and $\mathsf W_T \le \mathsf L_\ell L^{3(K-T)}e^{-\frac12 c'p_0(g_T)^2}$ there. The symbols $\alpha_0, \beta_s, \kappa, K_M, K_\Lambda, C_{\mathcal A}$ retain their meanings in the common choice of Prop. YM3_I:69.1 and Section YM3_I:82 and C_P is the number of

Def. YM3_I:51.2; $\delta(t) = A_1 t^{-1/2}(\log t)^{\mathfrak{p}_0}$ and $\delta_i = \delta(x_i)$; $\varepsilon_i^{\mathrm{co}}$ is the error e_i^{co} defined before equation YM3_I:(2302), d_* that of equation YM3_I:(2262), M_{pl} that of equation YM3_I:(2285); c_X, c_I, C_{M}, C_i, c_{sb} and c_{tl} are fixed in eq. 2.xxii–eq. 2.xxiii and the preceding paragraph; p_0 is the exponent of Cor. 2.8; C_{tp}, $C_{\mathrm{tp}}^{\mathrm{bg}}$, δ_{tp} are the numbers of Props. YM3_I:120.5 and YM3_I:120.6, C_{lat} that of equation YM3_I:(2245), and $C_\star^{\mathrm{fl}}$, $C_\flat^{\mathrm{tm}}$ those of equation eq. 6.xxi; c_θ and S_3 are numbers of the constants alone.

The short-distance estimates use the constants of Lems. YM3_I:125.2 and YM3_I:125.5. We write these constants with the font used here: $\mathsf{C}_{\mathrm{sz}} = C_{\mathrm{sz}}$, $\mathsf{C}_{\mathsf{M}} = C_{\mathsf{M}}$, $\mathsf{c}_{\mathrm{up}} = c_{\mathrm{up}}$, $\mathsf{c}_I = c_I$ and $\mathsf{C}_i = C_i$, with the same convention for C_3', C_3''. In particular $\mathsf{c}_X = 405 B_3^2 A_0^2/2$. Fix these choices before the source, lattice and levels, and put

$$\mathsf{c}_{\mathfrak{p}} = \sup_{a\in\mathbb{Z}_{\ge 0}} (a+2)^{4\mathfrak{p}_0} L^{-a}, \qquad \mathsf{c}_{\mathrm{sb}} = \mathsf{C}_{\mathrm{sz}} + \mathsf{c}_{\mathrm{up}}, \qquad \mathsf{c}_{\mathrm{tl}} = 2L^8 \mathsf{c}_{\mathfrak{p}} + 1. \tag{2.xxii}$$

The supremum is finite because exponential decay dominates the fixed power of a. For $\vartheta \in \{1,2\}$ define

$$\begin{aligned} \mathsf{C}_B^{(\vartheta)} &= \mathsf{C}_{\mathsf{M}} + 4\mathsf{c}_X\mathsf{c}_I + 6\mathsf{c}_I^2 \\ &\quad + \mathsf{c}_{\mathrm{tl}}\big[\mathsf{C}_{\mathsf{M}} + (2+\vartheta^2+\vartheta^2\mathsf{c}_{\mathrm{tl}})\mathsf{c}_{\mathrm{sb}}^2 + 2\vartheta\mathsf{c}_{\mathrm{sb}}(\mathsf{c}_X+\mathsf{c}_I)\big], \\ \mathsf{C}_{\mathrm{B}} &= \mathsf{C}_7 + 2(1+A_1)^4\mathsf{C}_5 + \mathsf{C}_3' + \mathsf{C}_3'' + \mathsf{C}_6 + \mathsf{C}_B^{(1)}, \\ \mathsf{C}_B' &= \mathsf{C}_B^{(2)}, \qquad \mathsf{C}_B^e = \mathsf{C}_7 + 2(1+A_1)^4\mathsf{C}_5 + \mathsf{C}_3' + \mathsf{C}_3'' + \mathsf{C}_6 + \mathsf{C}_B'. \end{aligned} \tag{2.xxiii}$$

These are fixed numbers, and $\mathsf{C}_{\mathrm{B}} \le \mathsf{C}_B^e$.

Lemma 2.13 (The coefficient of the collected error)**.** Under the hypotheses of Cor. YM3_I:125.6, the coefficient C_B in its estimate may be chosen to be C_{B} in eq. 2.xxiii. ■

Proof. Use the notation of that corollary: $p = \mathfrak{p}_0$, $q = \widehat{d}_{\mathsf{k}}$, $y = y_{\mathsf{k}}$, $n = T - \mathsf{k} \ge 1$, and $s_T \le 1$. Divide the errors by $\widehat{m}_1\widehat{m}_2$, and write $\mathfrak{m} = n^4 q^6 y^{15p}$. The first four terms in equation YM3_I:(2319) are the direct terms of the seven-error estimate. Its remaining terms are bounded by

$$[\mathsf{C}_7 + 2(1+A_1)^4\mathsf{C}_5 + \mathsf{C}_3' + \mathsf{C}_3'' + \mathsf{C}_6]\mathfrak{m} s_T.$$

Indeed $x_T^{-1} \le s_T^2 \le s_T$, $W_T \le t_T \le s_T$, and $1 + A_1 y^p \le (1+A_1)y^p$.

In the remainder of equation YM3_I:(2315), use $i_T \le \mathsf{c}_I q^3 y^{3p} s_T$ and $z_T \le \mathsf{c}_X y^{-7p} + i_T$. The separated-support term costs $\mathsf{C}_{\mathsf{M}}\mathfrak{m} s_T$. The cross terms cost $(4\mathsf{c}_X\mathsf{c}_I + 2\mathsf{c}_I^2)\mathfrak{m} s_T$, and the top term is at most

$$2z_T^2 \le 4\mathsf{c}_X^2 y^{-14p} + 4\mathsf{c}_I^2 \mathfrak{m} s_T.$$

For its last line, the size and tail bounds give

$$2L^8\mathsf{c}_{\mathfrak{p}} + n \le \mathsf{c}_{\mathrm{tl}} n, \qquad b_0 \le \mathsf{c}_{\mathrm{sb}} n q^3 x_{\mathsf{k}} y^{2p}, \qquad x_{\mathsf{k}}^2 t_T \le s_T.$$

Consequently the bracket in that line is at most

$$[\mathsf{C}_{\mathsf{M}} + (3+\mathsf{c}_{\mathrm{tl}})\mathsf{c}_{\mathrm{sb}}^2 + 2\mathsf{c}_{\mathrm{sb}}(\mathsf{c}_X+\mathsf{c}_I)] n^3 q^6 y^{5p} x_{\mathsf{k}}^2.$$

Multiplication by its remaining factor, at most $\mathsf{c}_{\mathrm{tl}} n t_T$, gives the last part of $\mathsf{C}_B^{(1)}\mathfrak{m} s_T$. Adding these coefficients proves the assertion. □

We use this admissible value of the collected coefficient in the depth error of equation YM3_I:(2330). We use the following names for its auxiliary functions:

$$\mathsf{Lm}_s = \mathsf{L}_s = L^2 C_W(s) Y_s^{3\mathfrak{p}_0}, \qquad \mathsf{sm}_g(s) = \sigma_g(s). \tag{2.xxiv}$$

Here L_s is defined in equation YM3_I:(2322), and σ_g, ϵ_g, Y_s^-, x_s^- and $s_1(g)$ are the functions and threshold of equations YM3_I:(2328)–(2332). The proof there gives $s^a \mathsf{sm}_g(s) \to 0$ for every $a \ge 0$

and boundedness of $Y_s^{2\mathfrak{p}_0}\epsilon_g(s)$. Changing the fixed collected coefficient to the admissible value above preserves that proof: its only additional occurrence is a fixed multiple of a polynomial times $\mathsf{sm}_g(s)$.

Laws. $\langle\cdot\rangle_\ell = \langle\cdot\rangle_{K,\ell,\beta}$ is the Wilson measure on T_ℓ, $\tilde\mu_\ell$ its periodic extension to $\mathbb{L}_K$, $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ the set of the periodic Gibbs states on $\mathbb{L}_K$; a *law* at (K,β) is a member of $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ or a $\tilde\mu_\ell$, $\ell \in \mathfrak{L}(g)$, and for the latter distances are those of T_ℓ. $\langle X_1;\dots;X_n\rangle_\nu$ is the joint cumulant (complex multilinear), $\nu(X;Y) = \langle X;Y\rangle_\nu$, $A[F] = A^\nu[F] = F - \nu(F)$; for a tuple $X = (H_1,\dots,H_n)$, $A^\nu(X) = \prod_a A^\nu[O_K^T(H_a)]$, extended linearly to combinations of tuples; along a sequence of laws ν_j, $A_j = A^{\nu_j}$, and $A(X) = A^\rho(X)$ for the law ρ at hand. $p_\ell = \ell^{-3}\log Z_{K,\ell,\beta}$ is the pressure, and $G_g = C_S + C_g + 2C_{\mathrm{tt}}C'_{\mathrm{tt}}E_0$ is formed with the constants of [7, §N], with which the canonical source energies form a system of localised source energies ([7, Lem. N.1]).

A *mirror* θ is the reflection in a coordinate plane through midpoints of bonds; it acts on points, on functions of the field by $(\theta F)(U) = F(\theta U)$, and on tensor labels. The *nine families* are the planes whose normals are e_μ or $(e_\mu \pm e_\lambda)/\sqrt{2}$, $\mu \neq \lambda$ (eighteen unit normals); every periodic Gibbs state is invariant under the lattice isometries and reflection positive, in the sense of the reflection form below, at the lattice planes of the nine families (Prop. YM3_I:103.4). For a mirror θ_c in $\{x_{\mu_0} = c\}$ and a law ν, $\|F\|_{\nu,c} = \nu(\overline{\theta_c F}F)^{1/2}$ on the functions of the field on $\{x_{\mu_0} > c\}$.

Boards. A board of $\mathbb{L}_K$ has a side $\mathsf{l}_0 = a_i$, $i \le K-1$ an integer (negative i admitted, so that $\mathsf{l} = L^{m_{\mathfrak{L}}(g)}$ is a side of a board), and an offset $o \in a_K\mathbb{Z}^3$; it is the partition of $\mathbb{R}^3$ into the cubes $o + \mathsf{l}_0\mathbf{n} + [-\frac12\mathsf{l}_0, \frac12\mathsf{l}_0)^3$, $\mathbf{n} \in \mathbb{Z}^3$, its *cells*, so that the offset is the centre of a cell, and its *walls* are the planes $\{x_\mu = o_\mu + \mathsf{l}_0(t+\frac12)\}$, $t \in \mathbb{Z}$, $\mu = 1,2,3$. As $\mathsf{l}_0/a_K = L^{K-i}$ is odd, the walls lie in $a_K(\mathbb{Z}+\frac12)$ and pass through midpoints of bonds: they are planes of mirrors. A *function of a cell* is a function of the variables of the bonds with both ends in the open cell; its images under successive reflections in the walls are its *copies*, with complex conjugation after an odd number of reflections. For a side $\ell_0 \in 2\mathsf{l}_0\mathbb{N}$ the chessboard seminorm $\|\Phi\|_{\mathrm{ch},\ell_0} \ge 0$ of a bounded function Φ of a cell has as its N_0-th power, $N_0 = (\ell_0/\mathsf{l}_0)^3$, the $\tilde\mu_{\ell_0}$-mean of the product of the N_0 copies of Φ in a period box.

Field and label conventions. The field is O_K^T of [7, eq. 1], with the same $w(G)$, a_p, q_b and midpoint convention. We use $\langle X,Y\rangle = \frac12\operatorname{tr}(XY)$ and the sup norm $|H|_\infty$. In this section $E_{\lambda\lambda'}$ denotes the symmetric label with entries 1 at (λ,λ') and (λ',λ), rather than a matrix unit. A loop is a pair $l = (\alpha,\alpha')$, $\alpha,\alpha' \in \{\pm1,\pm2,\pm3\}$, $|\alpha| \neq |\alpha'|$; $W_x(l)$ is the holonomy from x along e_α, then $e_{\alpha'}$, around the plaquette $p(x,l)$, and $\sigma(l) = \operatorname{sgn}\alpha\operatorname{sgn}\alpha'\operatorname{sgn}(|\alpha'|-|\alpha|)$. We write $S_n^T[\nu](H_1,\dots,H_n)$ for the joint cumulant of the $O_K^T(H_a)$ under ν. The free covariance is $\mathcal{W}$ in eq. 2.xix. The test norm used here is

$$N(H) = \max_{\lambda\lambda'}(|H_{\lambda\lambda'}|_\infty + d\ \operatorname{Lip} H_{\lambda\lambda'}).$$

Forms. A *cell* is a plaquette, a positively oriented bond with the four loops that start along it, or a site with all its 24 loops. A *form* is $F = a_K^{-1}\sum_p h(p)a_p$ or $F = a_K^{-1}\sum_{\mathsf{c}}\sum_{l,l'} v(\mathsf{c},l,l')\langle w(W_l), w(W_{l'})\rangle$, the cells c being bonds, resp. sites, with real weights of finite support (complex weights where this is said); its *size* $|F|_\star$ is $\sup|h|$, $59\|v\|$, resp. $23\|v\|$, $\|v\| = \sup_{\mathsf{c}}\sum_{l,l'}|v(\mathsf{c},l,l')|$; its *carrier* is the set of the centres of the cells of its support, $S[F]$ the set of the plaquettes of these cells; its *kernel* is $\mathsf{t}_F(p,p') = \frac12 h(p)\delta_{pp'}$, resp. $\mathsf{t}_F(p',p'') = \frac12\sum v(\mathsf{c},l,l')\sigma(l)\sigma(l')[\delta_{p'p(l)}\delta_{p''p(l')} + \delta_{p'p(l')}\delta_{p''p(l)}]$. For a weight w of the bonds, $Y[\mathsf{w}] = -a_K^{-1}\sum \mathsf{w}(\mathsf{e})q_{\mathsf{e}}$ is a bond form of size $59\sup|\mathsf{w}|$; $O_K^T(H)$ is a sum of six forms, each on the plaquettes of one normal or on the bonds of one direction, of total size at most $180|H|_\infty$. The *Gaussian value* of two forms is

$$G(F_1,F_2) = 6\upsilon^2\sum \mathsf{t}_{F_1}(p,p')\,\mathsf{t}_{F_2}(q,q')\,C^{\mathrm{bg}}(p,q)\,C^{\mathrm{bg}}(p',q'),$$

Wick's value for three free photon fields, $2(N^2-1) = 6$ at $N = 2$.

Limits. S denotes a volume-first limit, or a continuum limit along growing tori, of the action density, with its defining sequence ([7, §K.1]). A tuple is *separated* if its supports are pairwise disjoint. Along a sequence of laws at which all $S_n^T[\nu_j](X)$ converge, $\sigma^T(X) = \sum_\pi\prod_{I\in\pi} S^T_{\#I}(X_I)$,

the sum over the partitions into blocks of at least two, is the limit of the means of the products of the insertions centred in the states, and $\sigma^T(\emptyset)=1$. A *limit through periodic Gibbs states* is a limit along $\nu_j\in\mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$, $K_j\to\infty$, that converges at all separated tuples in $C^1_c(\mathbb{R}^3;V)$; H,P_1,P_2,Ω are the Hamiltonian, the momenta and the vacuum reconstructed from σ^T at $\{x_3=0\}$ (Prop. YM3_I:94.1; classes $[F]$ of combinations F of tuples in $\{x_3>0\}$, $\langle[F'],[F]\rangle=\sigma^T(\theta\bar F'\cdot F)$), with $e^{-tH}[F]=[\tau_{te_3}F]$, the unitary group $\mathcal{U}(v)[F]=[\tau_{(v,0)}F]$, $v\in\mathbb{R}^2$, of the horizontal translations, and the joint spectral measure $\mathcal{P}$ of (H,P_1,P_2); and alike at a plane of each of the nine families: the *nine pictures*. τ_v is the translation by v.

Numbers. Each symbol $C.$, $c.$, $k.$, $s.(g)$, and each error function $\varepsilon.,\omega.,o.,e.,$ Df. of s, is a function of the constants, of g and of the arguments shown alone; it does not depend on K, β, ℓ, the law, a weight or a position unless these are shown, and error functions tend to 0 as $s\to\infty$. $s_1(g)$ is the first depth of the envelopes of equations YM3_I:(2328)–(2336), with the notation in eq. 2.xxiv, $s_p(g)$ (p even) that of Lem. 2.21.

Theorem C. Let (K,β) be a first-passage datum at g and $b=a_K\beta$.

(A) Let $0<g\le\gamma_2$ and let ℓ be small. Then the following hold.

(A1) For $n\ge2$, $0<d<1$, $K\ge K_0'(n,d)$ and forms $F_1,\dots,F_n$ on $\Lambda_{K,\ell}$, with complex weights allowed at bonds and sites, whose carriers are pairwise at distance $\ge d$,

$$|\langle F_1;\dots;F_n\rangle_\ell|\le k^\circ(n,d)\,b^{-n}\prod_a|F_a|_\star,$$

where $d''=\min\{\frac34 d,L^{-s_{\mathrm{th}}(n)}\}$, $K_0'(n,d)=s(d'')$, $k^\circ(n,d)=(2\ell_{\max}/d'')^{3n}k_n'(g,s(d''))$, $\ell^3_{\max}=\frac12 c'A_0^2Y^P_{s(d'')}$, with s_{th}, k_n' of Lem. 2.17: the power d^{-3} for each entry, times a power of $\log\frac1d$. No upper bound on the distances is asked.

(A2) Let $s_1(g)\le s(d)\le K$, $3d\le|x_1-x_2|\le4d$, and let the carrier of F_i lie within $d-2a_K$ of x_i. Then $|\langle F_1;F_2\rangle_\ell-G(F_1,F_2)|\le\varepsilon_g^\natural(s(d))\,v^2|F_1|_\star|F_2|_\star$, $\varepsilon_g^\natural$ of Lem. 2.19. For complex tensor weights H_i supported within $d-2a_K$ of x_i the same holds for $O_K^T(H_1)$, $O_K^T(H_2)$, the Gaussian value being formed with their kernels (Lem. 2.15) and the error being $4\cdot60^2\varepsilon_g^\natural(s(d))v^2|H_1|_\infty|H_2|_\infty$. If $H_i=\varphi_iE_{\lambda_i\lambda_i'}$, φ_i real and Lipschitz with support in $B(x_i,d-2a_K)$, and $\lambda_1\ne\lambda_1'$, then

$$|\langle O_K^T(H_1);O_K^T(H_2)\rangle_\ell-b^{-2}\mathcal{W}(H_1,H_2)|\le e^{\mathrm{od}}\,b^{-2}N(H_1)N(H_2),$$

with the number e^{od} of Step 2, at most $\bar e_g^{\mathrm{od}}(s(d))$, $\bar e_g^{\mathrm{od}}$ an error function.

(A3) Let $s_1(g)\le s(d)\le K$, and let the carrier of $Y=Y[\mathsf{w}]$, w real on the bonds of one direction, lie within $d-2a_K$ of a point at a distance between $\frac32 d$ and $2d$ from the plane of a mirror θ. Then

$$0\le\langle\theta Y;Y\rangle_\ell\le C^\circ_{\mathrm{od}}(g)\,b^{-2}\sup|\mathsf{w}|^2,\qquad C^\circ_{\mathrm{od}}(g)=\tfrac98[600C_{\mathsf{k}}^2+59^2\bar\varepsilon_g^\natural],\quad \bar\varepsilon_g^\natural=\sup_{s\ge s_1(g)}\varepsilon_g^\natural(s),$$

C_{k} of Lem. 2.16. For a real form F at bonds, resp. sites, with the weight v, placed in the same way, $0\le\langle\theta F;F\rangle_\ell\le\frac98[6N^2C_{\mathsf{k}}^2+c^2\bar\varepsilon_g^\natural]\,b^{-2}\|v\|^2$, $(N,c)=(30,59)$, resp. $(10,23)$. For real tensor weights H_i as in (A2), $|\langle O_K^T(H_1);O_K^T(H_2)\rangle_\ell|\le\frac98[12150C_{\mathsf{k}}^2+60^2\bar\varepsilon_g^\natural]\,b^{-2}|H_1|_\infty|H_2|_\infty$.

(B) Let $g\le\gamma_1$, $s_8(g)\le s(d)\le K$, let $\theta_1',\theta_2',\theta_3'$ be mirrors of $T_{\ell'}$, θ'_μ normal to e_μ, and let Y be as in (A3), carried within $d-2a_K$ of a point at a distance between $\frac32 d$ and $2d$ from a plane of θ'_{μ_0} and, up to $a_K/2$, midway between the planes of the two other mirrors. With $\theta'^\eta=\theta_1'^{\eta_1}\theta_2'^{\eta_2}\theta_3'^{\eta_3}$,

$$0\le M_8=\Big\langle\prod_{\eta\in\{0,1\}^3}\theta'^\eta Y\Big\rangle_{\ell'}\le C_8\,b^{-8}\sup|\mathsf{w}|^8,\qquad 0\le M_8-\langle\theta'_{\mu_0}Y;Y\rangle^4_{\ell'}$$
$$\le(\tfrac98)^4 59^8\omega^\sharp_{g,8}(s(d))\,b^{-8}\sup|\mathsf{w}|^8,$$

$C_8 = (\frac{9}{8})^4[(106C_{\mathsf{k}}'^2)^4 + 59^8\bar\omega_{g,8}]$, $\bar\omega_{g,8} = \sup_{s\ge s_8(g)}\omega_{g,8}(s)$, with C_{k}' of Lem. 2.16 and $\omega_{g,8}$, $\omega_{g,8}^\sharp$ of Lem. 2.21.

(C) Let $g \le \gamma_1$ and let ν, ν' be laws at (K,β).

(C1) For $n \ge 2$, $d, R > 0$, every K, and forms $F_1, \dots, F_n$ on $\mathbb{L}_K$ whose carriers lie in balls of radius R and are pairwise at distance $\ge d$: $|\langle F_1; \dots; F_n\rangle_\nu| \le C_n^T(d;R)\, b^{-n} \prod_a |F_a|_\star$, with C_n^T of Step 4; complex weights cost a factor 2^n. So

$$|S_n^T[\nu](H_1,\dots,H_n)| \le 60^n C_n^T(d;R)\, b^{-n} \prod_a |H_a|_\infty$$

for supports in such balls at mutual distance $\ge d$, and $|\nu(F) - \nu'(F)| \le 2C_1^T(R)\, b^{-1}|F|_\star$.

(C2) If $s(d) \le K$, $\operatorname{supp} H_i \subset B(x_i, d - 3a_K)$ and $|x_1 - x_2|_\infty \ge 3d + a_K$ (on a torus for all lifts):

$$|S_2^T[\nu](H_1,H_2)| \le 180^2 C_\infty^\diamond(g)\, b^{-2} |H_1|_\infty |H_2|_\infty,$$

$C_\infty^\diamond$ of Step 5. For $Y[\mathsf{w}_i]$, w_i real on the bonds of one direction, carried within $d - 2a_K$ of x_i:

$$|\nu(Y[\mathsf{w}_1]; Y[\mathsf{w}_2])| \le C_\infty^{\mathrm{per}}(g)\, b^{-2} \sup|\mathsf{w}_1| \sup|\mathsf{w}_2|,$$

with a number $C_\infty^{\mathrm{per}} \ge e^{2G_g\mathsf{l}^3} C_8^{1/4}$.

(C3) Let $s_{16}(g) \le s = s(d) \le K$, $3d \le |x_1 - x_2| \le 4d$, let the carriers of real forms F_i lie within $d - 2a_K$ of x_i, $\hat m = \upsilon^2 |F_1|_\star |F_2|_\star$, and let F_i' be the forms on $T_{\ell'}$ with the weights of F_i on the board of side l whose cell holds the pair in its middle. Then

$$|\nu(F_i) - \tilde\mu_{\ell'}(F_i)| \le e^{G_g\mathsf{l}^3} o_{g,16}(s)^{1/8} \upsilon |F_i|_\star, \quad |\nu(F_1;F_2) - \tilde\mu_{\ell'}(F_1;F_2)| \le \mathrm{Df}_g(s)\, \hat m,$$
$$|\nu(F_1;F_2) - G(F_1',F_2')| \le [\omega_{g,16}' + \mathrm{Df}_g](s)\, \hat m,$$

the Gaussian value being formed on $T_{\ell'}$ and Df_g given in Step 5; the error function and the first depth are the same for all the laws. The same holds for $F_i = O_K^T(H_i)$, H_i real tensor weights supported within $d - 2a_K$ of x_i, with $60|H_i|_\infty$ in the place of $|F_i|_\star$.

(D) Let $g \le \gamma_1$, let (K_j, β_j) be first-passage data at g and ν_j laws at (K_j,β_j): all periodic Gibbs states, or all $\tilde\mu_{\ell_j}$ with $\ell_j \to \infty$. Tuples $X = (H_1,\dots,H_n)$ are in $C_c(\mathbb{R}^3; V)$ with pairwise disjoint supports.

(D1) Along a subsequence ($K_j \to \infty$ is not asked) $a_{K_j}\beta_j \to \mathsf{b} \in I^{\mathrm{lab}}(g)$ and $S_n^T[\nu_j](X) \to S_n^T(X)$ for all $n \ge 2$ and all X; S_n^T is symmetric, multilinear, bounded as in (C1) with b for b, and in each entry a V^*-valued Radon measure off the other supports. For laws ν_j' at the same data, $\lim(\nu_j - \nu_j')(O_{K_j}^T(H))$ exists along a further subsequence, of modulus $\le 120 C_1^T(R)\mathsf{b}^{-1}|H|_\infty$ if $\operatorname{supp} H$ lies in a ball of radius R. If each ν_j is a weak limit of measures $\tilde\mu_\ell$ at (K_j,β_j), then for any numbers Λ_j there are sides $\ell_j \ge \max\{\Lambda_j, j\}$ in $\mathfrak{L}(g)$ with $S_n^T[\tilde\mu_{\ell_j}](X) \to S_n^T(X)$ along the full sequence: a limit through limit states of the tori is a limit along growing tori.

(D2) Over every volume-first limit S and every continuum limit along growing tori there is such an S^T, with $S_n^T(f_1\mathbf{1},\dots,f_n\mathbf{1}) = S_n(f_1,\dots,f_n)$, $f_i \in C_c^1$.

(D3) If $K_j \to \infty$: $S_n^T(rX) = S_n^T(X)$ for $rx = Ax + y$, $A \in W_3$; and for every coordinate plane $\{x_\mu = c\}$, with its reflection θ acting on points and on indices, all finitely many tuples X_i (the empty one included) with supports in $\{x_\mu > c\}$ and all complex c_i: $\sum_{i,k} \bar c_i c_k \sigma^T(\theta X_i \cdot X_k) \ge 0$. For limits through periodic Gibbs states the same holds at the planes with the normals $e_\mu \pm e_\lambda$, the Hilbert space is separable, and $H \ge |P_1|$, $H \ge |P_2|$. The numbers $\mathfrak{c}_j = a_{K_j}^{-4}(\nu_j(a_p) - \tilde\mu_{\ell'}(a_p))$, the same at all plaquettes, obey $|\mathfrak{c}_j| \le C(g)\mathsf{b}^{-1}$ for all large j; along a further subsequence $\mathfrak{c}_j \to \mathfrak{c}$, and along it the moments centred at the means of $\tilde\mu_{\ell'}$ in the place of those of the ν_j converge too, to $\sum_B \prod_{i\in B}(\mathfrak{c}\int \operatorname{tr} H_i)\, \sigma^T(X_{B^c})$, the sum over the subsets B of the entries, a functional with the same properties whose one-point function is $\mathfrak{c}\int \operatorname{tr} H$.

(D4) For $\mathsf{b} \in I(g)$, $\ell \in \mathfrak{L}(g)$ and continuous $H_a : T_\ell \to V$ with disjoint supports, $S_n^{T,(\ell,\mathsf{b})}(H) = \lim_K \langle O_K^T(H_1); \dots; O_K^T(H_n)\rangle_{K,\ell,L^K\mathsf{b}}$ exists along the full sequence of cutoffs (for $g \le \gamma_2$) and obeys the bound of (D1). Along a subsequence of every $\ell_i \to \infty$ these limits converge, and the limit is also the limit of $S_n^T[\tilde\mu_{\ell_i}]$ at suitable cutoffs $K_i \to \infty$: a limit along growing tori, with the properties of (D3) for these limits.

(D5) Let $K_j \to \infty$, $s(d) \ge s_{16}(g)$, $3d \le |x_1 - x_2| \le 4d$, and H_i Lipschitz with compact support in the open ball $B(x_i, d)$. Then

$$|S_2^T(H_1, H_2) - \mathsf{b}^{-2}\mathcal{W}(H_1, H_2)| \le 36\, \bar e_g(s(d))\, \mathsf{b}^{-2} N(H_1) N(H_2),$$

the function $\bar e_g$ being the same for all sequences; the same holds for the limits of (D4), on every ℓ. At $H_i = f_i\mathbf{1}$ the leading term is $\frac{9}{16\pi^2}\mathsf{b}^{-2}\iint f_1(x) f_2(y)|x-y|^{-6}\,\mathrm{d}x\,\mathrm{d}y$, the scalar leading coefficient in Thm. YM3_I:AW.

(E) Let $g \le \gamma_2$, $K \ge 3$, and let $\ell = 2L^m$ be admissible.

(E1) With $\tilde\nu_K$ the law of the histories and Z_t the large-field region of the level t, for $1 \le t \le K$:

$$0 \le \mathsf{w}_t \le \tilde\nu_K(\{Z_t \ne \emptyset\}) \le C_\flat M^{-3}\ell^3 L^{3(K-t)} \exp[-\tfrac34 c_\star p_0(g_t)^2];$$

$C_\flat$ is a function of the constants at two admissible sets of values alone, and does not depend on g, K, β, ℓ, t; $\frac34 c_\star \ge \frac12 c'$, and $\frac34 c_\star \ge c'$ if $\eth \ge \frac14$. The same holds with $(c_2^\natural, C_\natural)$ for $(\frac34 c_\star, C_\flat)$, $c_2^\natural$ the constant of Def. 4.47 with which Prop. 4.52 bounds the rarity of a large field at a point, and $C_\natural$ a function of the constants.

(E2) Let (c, C) be a pair for which (E1) holds with (c, C) for $(\frac34 c_\star, C_\flat)$, as $(\frac12 c', C_\flat)$, $(\frac34 c_\star, C_\flat)$ and $(c_2^\natural, C_\natural)$ do; $\sigma_c = \frac{2}{\log L}(\frac{6}{cA_0^2})^{1/P}$, $\hat s(g,\ell) = \max\{s_\flat(g), \lceil \sigma_c (\log\ell)^{1/P}\rceil\}$, the numbers below being formed at (c, C) and $s_\flat(g)$ being the first depth of the envelopes at (c, C), chosen so that also $2c\mathfrak{p}_0 A_0^2 y^{P-1} \ge 13$ at the working level. Let $\hat s \le s(d) \le K$, $|x_1 - x_2| \ge 3d$ (no upper bound), and let F_i be sums of forms within $d - 2a_K$ of x_i, of size the sum of the sizes. Then (i) $|\langle F_1; F_2\rangle_\ell - G(F_1, F_2)| \le \varepsilon_g^\circ(s(d)) \upsilon^2 |F_1|_\star |F_2|_\star$, with $Y_s^P \varepsilon_g^\circ(s)$ bounded; (ii) $|\langle F_1; F_2\rangle_\ell| \le C_\star^\circ(g) b^{-2} |F_1|_\star |F_2|_\star$; (iii) if x_1 is at a distance $\ge \frac32 d$ from both planes of θ, $0 \le \langle \theta F_1; F_1\rangle_\ell \le C_\star^\circ(g) b^{-2} |F_1|_\star^2$; (iv) for all d with $s(d) \le K$ and F placed as in (iii), $0 \le \langle \theta F; F\rangle_\ell \le C_3^\circ(g,\ell) b^{-2} (|F|_\star/59)^2$, which is $C_3^\circ b^{-2} \sup|\mathsf{w}|^2$ at $F = Y[\mathsf{w}]$, w real on the bonds of one direction, and $C_3^\circ(g,\ell) \le c_3(g)(1+\log\ell)^8 \exp[16(\frac{6\log\ell}{cA_0^2})^{1/P}]$, so that $\ell^{-\epsilon} C_3^\circ(g,\ell)$ is bounded for every $\epsilon > 0$; (v) if all points of the carriers of F_1 and F_2 are $\ge r$ apart and $K \ge s(\min\{\frac{r}{8}, L^{-\hat s}\})$: $|\langle F_1; F_2\rangle_\ell| \le (\ell \max\{\frac{8}{r}, L^{\hat s}\} + 1)^6 C_\star^\circ(g) b^{-2} |F_1|_\star |F_2|_\star$. On the sides with $\log\ell \le (s_\flat(g)/\sigma_c)^P$ the first depth is $\hat s = s_\flat(g)$, free of ℓ.

(E3) Let $\mathsf{L} < \infty$ and $c'' \ge \frac12 c'$ be numbers with $\mathsf{w}_t \le \mathsf{L} L^{3(K-t)} e^{-c'' p_0(g_t)^2}$ for $1 \le t \le K$. Then: (i) Lem. 2.17, for sources under majorants, holds on T_ℓ with $\max\{1, M^3\mathsf{L}/(c'A_0^2)\}^n k_n'$ for k_n', under its hypotheses $n \ge 2$ and $s_{\mathrm{th}}(n) \le s(d) \le K$ and in its geometry: $|\langle F_1; \dots; F_n\rangle_\ell| \le \max\{1, M^3\mathsf{L}/(c'A_0^2)\}^n k_n'(g, s(d))\, b^{-n} \prod_a \mathsf{s}_a$, with no factor of a covering; (ii) from (i), by the covering of Step 1, (A1) holds on T_ℓ with 2ℓ for $2\ell_{\max}$ and $\max\{1, M^3\mathsf{L}/(c'A_0^2)\}^n k_n'$ for k_n'. Two such pairs are had on every admissible side: $\mathsf{L} = 2M^{-3}\ell^3 e^{\mathsf{B}(\bar x_0)\ell^3}$, $c'' = c'$, by the rarity of large fields on one torus (Prop. 4.54), as B increases and $x_K \le X_g < \bar x_0$; and the pair of (E1), $\mathsf{L} = C_\flat M^{-3}\ell^3$, $c'' = \frac34 c_\star \ge \frac12 c'$. So (i) and (ii) hold on every admissible side with $\max\{1, c(\ell)\}^n k_n'$ for k_n', (ii) with 2ℓ for $2\ell_{\max}$, where $c(\ell) = 2\ell^3 e^{\mathsf{B}(\bar x_0)\ell^3}/(c'A_0^2)$, the number $M^3\mathsf{L}/(c'A_0^2)$ at the first pair; with the weights of (E1), $c(\ell)$ may be replaced by $C_\flat \ell^3/(c'A_0^2)$, a power of ℓ.

■□

Theorem D (Contact extensions and mirror bounds)**.** Let $g \le \gamma_1$ unless another range is given, and let $\sigma = \sigma^T$, with cumulants S_n^T, be a limit through periodic Gibbs states.

(F8) (*Coinciding points.*) Let $g \le \gamma_2$, $\mathsf{b} \in I(g)$, $\ell \in \mathfrak{L}(g)$, $n \ge 2$. $S_n^{T,(\ell,\mathsf{b})}$ extends to a symmetric, translation invariant distribution $\tilde S_n$ on T_ℓ^n of order $\le 3n-2$ with

$$|\tilde S_n(F)| \le C\mathsf{b}^{-n}(1+\log\tfrac{h_0}{\lambda})^{8n\mathfrak{p}_0+1} \max_{|\gamma|\le 3n-2} \lambda^{|\gamma|} \sup|\partial^\gamma F|$$

for F supported in a product of closed sup-balls of radius $\lambda \le h_0$ about any points, coinciding or not (C, h_0 of g, ℓ, n); the lattice cumulants with their Taylor parts at the diagonals subtracted (Lem. 2.25) obey this bound for all K and all $8h_K \le \lambda \le h_0$, $h_K \le 2C_W(K)a_K$ the least scale at the cutoff K, with one number for all cutoffs, and they converge to $\tilde S_n$ at every smooth F; two extensions with this bound differ by a sum $\sum_{|\pi|\le n-1}\sum_{|\alpha|\le 3|\pi|}\partial_\xi^\alpha\delta_\pi \otimes T_{\pi,\alpha}$ of contact terms, and every such sum may be added; at $n=2$ the translation invariant, real, symmetric contact terms covariant under the cubic group form a space of dimension twelve, under $O(3)$ on $\mathbb{R}^3$ one of dimension six.

The reconstruction (the Hilbert space, the semigroup, the momenta and the kernels of [7, Lem. D.13]) uses the values of σ at separated tuples alone, so no choice of the extension and no contact term enters it; beyond the separated tuples it rests on the insertion bound of [7, Thm. BZ(F2)(vi)]. For $g \le \gamma_1$ every limit point of $\langle Y[\varphi_1]; Y[\varphi_2]\rangle_{K,\ell,\beta}$ as $K, \ell \to \infty$, $\ell \in \mathfrak{L}(g)$, in any order, is bounded at separated cubes by $36C_\infty^{\mathrm{od}}(g)(\inf b)^{-2}\sup|\varphi_1|\sup|\varphi_2|$, with no power of the distance and no logarithm, and so extends to $\mathbb{R}^6$ as a distribution of order ≤ 4.

(F9) (*Mirror bound in every state; the power* 6.) Let $s(d) \le K$. If $X = A[O_K^T(H)]$ is carried in $B(x_0, d-2a_K)$ and x_0 is at a distance $\ge \frac{3}{2}d$ from the plane of θ, then $0 \le \nu(\theta X \cdot X) \le C_\infty(g)\, b^{-2}|H|_\infty^2$ in every periodic Gibbs state ν, $C_\infty(g)$ defined in eq. 2.xxv; for a bond form $Y = Y[\mathsf{w}]$ with a complex weight carried there, $\nu(Y) = 0$ and $0 \le \nu(\overline{\theta Y} Y) \le 36C_\infty^{\mathrm{od}}(g)b^{-2}|\mathsf{w}|_\infty^2$, with $C_\infty^{\mathrm{od}}(g)$ alone for real weights of one direction. For $H_1, H_2 \in C_c(\mathbb{R}^3; V)$ with disjoint supports and off-diagonal values,

$$|\sigma(H_1,H_2)| \le A_6 C_\infty^{\mathrm{od}}(g)\,\mathsf{b}^{-2} \iint \|H_1(x)\|_1\|H_2(y)\|_1 \min\{|x-y|_\infty, 3\}^{-6}\,\mathrm{d}x\,\mathrm{d}y, \qquad \|E\|_1 = \sum_{\lambda<\lambda'}|E_{\lambda\lambda'}|,$$

$A_6 = 64\cdot 36\cdot((8/\pi^3)32^3)^2$; for the whole tensor the same holds with a pure number A in the place of A_6, with $C_\infty(g)$ in the place of $C_\infty^{\mathrm{od}}(g)$ and with $\|E\|_1 = \sum_{\lambda\le\lambda'}|E_{\lambda\lambda'}|$; by (D5) the power 6 is the true one.

■□

Lemma 2.14 (Matrices and corner forms)**.** (a) For complex 2×2 matrices, with the operator norm: $|\langle X, Y\rangle| \le |X||Y|$; if $\det V = \det V' = 1$ and T is invertible, then $w(TVT^{-1}) = Tw(V)T^{-1}$, $w(V^{-1}) = -w(V)$, $|V^{-1} - \mathbf{1}| = |V - \mathbf{1}|$ and $|V^{-1} - V'^{-1}| = |V - V'|$; so $|w(V)| \le |V - \mathbf{1}|$ and $|w(V) - w(V')| \le |V - V'|$.

(b) For loops l, l' of one cell the corner form $Q = \langle w(W_l), w(W_{l'})\rangle$ is a polynomial in the bond variables, 0 at the trivial configuration, gauge invariant, and real at SU(2) configurations, where $|Q| \le a_{p(l)} + a_{p(l')}$. Let $\eta, b_0, b_1 > 0$ be arbitrary. At complex configurations with $|U(\mathsf{e})| \le e^{\eta b_1}$ on the bonds of the cell and $|\partial U(p) - \mathbf{1}| < b_0\eta^2$ at its two plaquettes, $|Q| \le 4e^{4t\eta b_1}b_0^2\eta^4$, $t = 1$ at bonds and $t = 2$ at sites. If $|U(\mathsf{e})| \le e^{1/13}$ at the bonds of $p = p(l)$, $p' = p(l')$ and at their reverses, $|\partial U(p) - \mathbf{1}| \le \zeta_0$ and $|\partial U(p') - \mathbf{1}| \le \zeta_0$ with $e^{4/13}\zeta_0 \le 1$, and B' is a 1-form with $\rho = \max|B'| \le \frac{1}{128}$, then $|Q(e^{iB'}U) - Q(U)| \le 784\rho^2 + 112e^{2t/13}\rho\zeta_0$. At the small points of a chart of height i, $|Q - \sigma(l)\sigma(l')\langle \mathrm{ang}_{p(l)}, \mathrm{ang}_{p(l')}\rangle| \le 12M_{\mathrm{pl}}^2\eta_i^4\mathsf{rt}^{-3}$, where ang_p is the angle of p and $\mathsf{rt} \ge 2$ is the quotient of the radii of the chart over the point (Lem. YM3_I:124.4). ■

Proof sketch. (a) The trace of a 2×2 matrix is the sum of two entries, each at most the norm. For $\det V = 1$, $V^{-1} = (\operatorname{tr} V)\mathbf{1} - V$; so $V^{-1} - \mathbf{1} = (\operatorname{tr} E)\mathbf{1} - E$ with $E = V - \mathbf{1}$, and $V^{-1} - V'^{-1} = (\operatorname{tr} E)\mathbf{1} - E$ with $E = V - V'$; for 2×2 matrices $(\operatorname{tr} E)\mathbf{1} - E = JE^{\mathrm{T}}J^{-1}$ with the unitary $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, and $|E^{\mathrm{T}}| = |E|$; finally $w(V) = [(V - \mathbf{1}) - (V^{-1} - \mathbf{1})]/2i$. (b) For $V = \exp(i\phi\, n\cdot\sigma) \in \mathrm{SU}(2)$, $w(V) = \sin\phi\; n\cdot\sigma$ and $\langle w(V), w(V)\rangle = \sin^2\phi \le 2(1-\cos\phi)$, which with the Schwarz inequality for $\langle\cdot,\cdot\rangle$ gives $|Q| \le a_{p(l)} + a_{p(l')}$. The complex bounds follow from (a) applied to the holonomies of the two plaquettes; the bound on a shift and the second order follow from the expansion of the holonomy in the 1-form and in the chart, $w(W_l) = \sigma(l)\operatorname{ang}_{p(l)}$ to the first order; every bound of a vector part at a complex configuration goes through (a). □

Lemma 2.15 (Forms are sources)**.** (a) Let $\bar{k} \ge 1$ be an integer, the *top index*. Call F a *source of top index $\bar{k}$ under the majorant* (s, S), $\mathsf{s} \ge 0$ a number and S a set of plaquettes, if F is continuous, gauge invariant, $|F| \le L^K \mathsf{s} \sum_{p\in S} a_p$, and F is the sum, at height 0, of a family in the fourth place over a point (§15.3.1) of size $\le c^\flat \mathsf{m}_0$, $c^\flat = 81 C_P^2 C_{\mathcal{A}}^2 / K_\Lambda$, and, at each height $1 \le j \le \bar{k}$, of a family without base point (Def. YM3_I:49.5) of size $\le \frac{1}{2}\mathsf{m}_j$, $\mathsf{m}_j = 12 M^3 \alpha_0^2 L^{K-j}\mathsf{s}$, both vanishing on the regions that miss S. The top index is that of the data of the chain at which the source is used: the working level k for n sources (Lems. 2.17 and 2.18, where the chain runs up to the level $\mathsf{k}+1$), and $T-1$, T the upper level, for two sources and for far pairs (Lems. 2.19 and 2.21), where the one-step covariance of the level j, $\mathsf{k} < j \le T$, is formed with the members of the height $j-1$; since $n_T \ge 2$, there $T - 1 > \mathsf{k}$.

A source is a source of the top index of the data at hand. Call it *quadratic*, with the kernel t, if besides t is a real symmetric function on $S\times S$ with $2\sum_{p'} |\mathsf{t}(p,p')| \le \mathsf{s}$ such that $|a_K F - \sum \mathsf{t}(p,p')\langle \operatorname{ang}_p, \operatorname{ang}_{p'}\rangle| \le M_{\mathrm{pl}}^2 \eta_i^4 \mathsf{r}\mathsf{t}^{-3}(\#S)\mathsf{s}$ at the small points of the chart of height i. Then every form F is a source under $(|F|_\star, S[F])$ of every top index, since it has its member at every height $j \ge 1$ of the chain, linearly in its weights, and it is quadratic with the kernel t_F if it is real; a plaquette form with the weight h is one under $(\sup|h|, \operatorname{supp} h)$, with $\mathsf{t} = \frac{1}{2}h\delta$; for real weights at bonds or sites $2\sum_{p'} |\mathsf{t}_F(p,p')| \le 8\|v\|$; $O_K^T(H)$ is one under $(60|H|_\infty, S_H)$, S_H the plaquettes of its six forms, with $2\sum_{p'} |\mathsf{t}_H(p,p')| \le 3|H|_\infty$.

Under sums the sizes add and the sets unite, and a source under (s, S) is one under (s', S) for every $\mathsf{s}' \ge \mathsf{s}$.

(b) The proof of the two-point law of the action density on a torus of small side (Thm. YM3_I:AU), and the proofs of the statements on which it rests (Thm. YM3_I:AT, Prop. YM3_I:125.1, Lems. YM3_I:125.2, YM3_I:125.4 and YM3_I:125.5; the weighted progression of Prop. YM3_I:122.1; the first step and the term of the history without large-field regions, the first-step part of [7, Lem. B.19] and Prop. YM3_I:123.1; the quotients of the sources and of a step, their tower and the reduction of the covariance to one-step covariances, Prop. YM3_I:123.1 and equation YM3_I:(2268), the quotient definitions in [7, §B.9], the quotient identities preceding equation YM3_I:(2273), equation YM3_I:(2273), [7, Lem. B.20] and Prop. YM3_I:123.2; the carried family and the flat top, Prop. YM3_I:123.3, equation YM3_I:(2283); and the sum over the level of the first creation, equation eq. 4.xxxi), use the source only through the properties of (a) and its boundedness; they therefore apply to every source in the sense of (a) whose top index is that of the data of the statement applied, and to every quadratic source where the kernel is used.

(c) The step map and its Lipschitz bound hold for forms: for weights v, v' at the cells the outputs of one step differ by at most $\theta^{[j]} 8 n_{\mathfrak{c}} e^{4t_{\mathfrak{c}}/13} M^3 \alpha_0^2 \|v - v'\|$ in the norms of the families of Def. YM3_I:49.5, a pure number times the number for plaquette weights ($n_{\mathfrak{c}} = 18$, $t_{\mathfrak{c}} = 1$ at bonds; $n_{\mathfrak{c}} = 7$, $t_{\mathfrak{c}} = 2$ at sites); the step map depends only on its input array. ■

Proof sketch. (a) The cubes of the partition of height k that hold a corner of a plaquette of a cell form a connected region of at most 6 (bond) or 7 (site) cubes with a common corner, and

a region of height k serves at most $n_{\mathfrak{c}} M^3 \eta_k^{-3}$ cells. At height 0, with $\mathbf{U}$ the configuration of the bond variables and B' the 1-form in the fourth place, of which the family is a function, the family $\sum_{X_0(\mathsf{c})=Y}[f_{\mathsf{c}}(e^{iB'}\mathbf{U}) - f_{\mathsf{c}}(\mathbf{U})]$, $f_{\mathsf{c}} = \sum_{l,l'} v(\mathsf{c},l,l')\langle w(W_l), w(W_{l'})\rangle$, $X_0(\mathsf{c})$ the region of the cell, is holomorphic and gauge covariant, and Lem. 2.14(b) bounds it at complex configurations; with the count, $n_{\mathfrak{c}} \cdot 3138/(81\cdot 12) \le 59$, resp. 23, it gives the size $c^\flat \mathsf{m}_0$ with the factor 59, resp. 23, of $|F|_\star$.

At height $j \ge 1$ the member over a region X is $\eta_j^{-1} \sum_{X_j(\mathsf{c})=X} f_{\mathsf{c}}$, of size $\le \frac{1}{2}\mathsf{m}_j$ by the same count. To second order $w(W_l) = \sigma(l)\,\mathrm{ang}_{p(l)}$, whence $\sum \mathsf{t}_F(p,p')\langle \Phi_p, \Phi_{p'}\rangle = \sum v(\mathsf{c},l,l')\sigma(l)\sigma(l')\langle \Phi_{p(l)}, \Phi_{p(l')}\rangle$, with the error of Lem. 2.14(b). Finally $O_K^T(H) = a_K^{-1}\sum_p w_H(p) a_p + Y[\mathsf{w}_H]$ with the plaquette weight $w_H(p) = H_{\lambda(p)\lambda(p)}(x(p))$ and the bond weight $\mathsf{w}_H(\mathsf{e}) = H_{\lambda\lambda'}(x_{\mathsf{e}})$: a plaquette form of size $\le |H|_\infty$ and a bond form of size $\le 59|H|_\infty$, whose sizes add to $60|H|_\infty$. (b) The source enters these proofs at four places.

First, through the clause on F itself: its continuity and its gauge invariance in the first step, where the Wilson density is multiplied by e^{F_λ}, $F_\lambda = \sum_a \lambda_a F_a$ (the first-step part of [7, Lem. B.19]), and in the sum over the level of the first creation (equation eq. 4.xxxi); its pointwise majorant, which bounds F at the backgrounds of the small fields of the level t by $L^K \mathsf{s} \sum_{p\in S} \sup a_p$, the supremum over these backgrounds, that is by the size of the source at the level t, and by $2L^K \mathsf{s} \# S$ at the height 0, in the quotients, their tower, the reduction and the identity of the quotients (Prop. YM3_I:123.1 and equation YM3_I:(2268), the quotient identities preceding equation YM3_I:(2273), equation YM3_I:(2273) and Prop. YM3_I:123.2); it is the source of the power $2\mathfrak{p}_0$ of Lem. 2.17.

Second, at the height 0, through its family in the fourth place over a point and the size of that family (the height-zero part of [7, Prop. B.8]; here the clause of the height 0 of (a)). Third, at the heights $1 \le j \le \bar{k}$, through its families without base point, their sizes and the action formula ([7, Lem. P.6]), the one-step covariance of the level $j+1$ being formed with the member of the height j ([7, Lem. B.20]); here the clause of the heights of (a), up to the top index. Fourth, at the second order at the small points of a chart, through its kernel (Lem. YM3_I:124.4), in the carried family and the flat top (Props. YM3_I:125.1 and YM3_I:123.3, equation YM3_I:(2283)); here the kernel t of (a).

Every other step of these proofs is stated there for arbitrary bare arrays and local families, for bounded Borel functions of the field, or for numbers. (c) The step map is formed from its input array alone; its Lipschitz constant for the arrays of plaquette weights, with the count of (a) and Lem. 2.14(b), gives the factor $8n_{\mathfrak{c}} e^{4t_{\mathfrak{c}}/13}$. □

Lemma 2.16 (The kernel)**.** Let $s' \le K$, $d \ge a_{s'}$, $\hat{d} = d/a_{s'}$, and $|u| = |x(p) - x(q)| \ge d$. Then

$$|C^{\mathrm{bg}}_{K,s'}(p,q) - \epsilon_p\epsilon_q a_K^3 K_3(u)_{\lambda(p)\lambda(q)}| \le (a_K/d)^3 E_1, \qquad E_1 = C_{\mathrm{lat}}[(a_K/d)^2 + (d/\ell)^3] + C^{\mathrm{fl}}_\star \hat{d}^3 e^{-\delta_{\mathrm{tp}} L\hat{d}/\sqrt{3}},$$

$\epsilon_p = \pm 1$ the orientation of p; so $|C^{\mathrm{bg}}(p,q)| \le C_{\mathsf{k}}(a_K/d)^3$, $C_{\mathsf{k}} = 1 + 3C_{\mathrm{lat}} + C^{\mathrm{fl}}_\star C^{\mathrm{tm}}_\flat$, and in the form in which far pairs use it the same holds with $C'_{\mathsf{k}} = 3^{3/2} C_{\mathrm{tp}} \mathsf{S}_3 + C^{\mathrm{bg}}_{\mathrm{tp}}$. If $8a_K \le d$ and $4d < \ell$, a ball of radius d holds at most $10(d/a_K)^3$ sites, at most as many centres of bonds of one direction, and as many centres of plaquettes of one normal. ■

Proof sketch. The free kernel is the background kernel plus the fluctuation kernel $C^{\mathrm{fl}}_{K,s'} = \Pi_0 - \Pi_{K-s'}$, the sum of the slices of the scales below $a_{s'}$ YM3_I:(2227). The comparison of the lattice free kernel with $a_K^3 K_3$, with the number C_{lat} of equation YM3_I:(2245), gives the first term of E_1, and the exponential decay of the fluctuation kernel beyond the scale $a_{s'}$, $|C^{\mathrm{fl}}_{K,s'}(p,q)| \le C^{\mathrm{fl}}_\star (a_K/a_{s'})^3 e^{-\delta_{\mathrm{tp}} L\mathsf{r}/a_{s'}}$ for $\mathsf{r} = |x(p) - x(q)|_\infty \ge a_{s'}/\sqrt{3}$ (Prop. YM3_I:120.6 at $t_0 = 3^{-1/2}$), with $\mathsf{r} \ge d/\sqrt{3}$, gives the second; condition (H) fixes the constants, and $|K_3(u)_{\lambda\lambda'}| \le (2\pi)^{-1}|u|^{-3}$. The count is a count of lattice points in a ball. □

Lemma 2.17 (n sources on a small torus)**.** Let $0 < g \le \gamma_2$, $n \ge 2$, let ℓ be small, and let $s_{\mathrm{th}}(n) \le s(d) \le K$, where $s_{\mathrm{th}}(n)$ is the least integer $s \ge 3$ with $2T_n(3)\theta(L^{s-2}g^{-2}) \le 1$, $\theta(t) =$

$K_M(1+\beta_s)\delta(t)/\alpha_0$, which does not increase on $[e^P,\infty)$, $\delta(t)=A_1 t^{-1/2}(\log t)^{\mathfrak{p}_0}$, and T_n is the series [7, eq. 3]. Let (s_a, S_a) be majorants with points x_a, $|x(p)-x_a|\le\frac{7}{8}d$ on S_a, $|x(p)-x(q)|\ge\frac{5}{4}d$ for $p\in S_a$, $q\in S_{a'}$, $a\ne a'$, and let F_a be (complex) sources of top index k, the working level, under them. Then

$$|\langle F_1;\dots;F_n\rangle_\ell|\le k'_n(g,s(d))\,b^{-n}\prod_a \mathsf{s}_a,\qquad k'_n(g,s)\le C_n(g)Y_s^{8n\mathfrak{p}_0},$$

so $k'_n(g,s(d))\le C_n(g)[2\log\frac{1}{d}+C(g)]^{8n\mathfrak{p}_0}$. No upper bound on the distances is asked. ■

Proof sketch. Insert $F_\lambda=\sum_a\lambda_a F_a$ as the source into the weighted progression of [7, eq. 200] (Prop. YM3_I:122.1). A chain rule for the step map in any number of parameters (Cauchy estimates on polydiscs) and locality give, for every set β of $r=\#\beta\ge 2$ sources: the mixed derivative $E^{(j)}_\beta=\partial_\beta E^{(j)}_\lambda|_0$ of the local family of level j vanishes on the polymers that miss some S_a, $a\in\beta$, so that its norm is at most $e^{-4\beta_s\kappa[\mathsf{D}_j/M-2]_+}$ times its size $\mathsf{d}_{j,\beta}$; splitting into the top block and the rest, by induction on r and on j, $\mathsf{d}_{j+1,\beta}\le c_\theta\hat G_r\prod_{a\in\beta}\hat g_{j,a}$, $\hat G_r$ given by a recursion in r and $\hat g_{j,a}$ the sizes of the sources at the level j, as long as $\theta^{[j]}T_n(\mathsf{D}_j/M)\le\frac12$; the threshold $s_{\mathrm{th}}(n)$ gives this for $1\le j\le\mathsf{k}$, since $\mathsf{D}_j/M\ge 3$ and $x_j>L^{s-2}g^{-2}$ there.

The chain runs up to the level $\mathsf{k}+1$, and its top index is the working level k, so that the sources need their families without base point at the heights $1\le j\le\mathsf{k}$ alone. On the history without large fields, with e^Ψ the density of the level $t\le\mathsf{k}$, Ψ formed with the source F_λ, the identity $\partial_B e^\Psi=e^\Psi\sum_\pi\prod_{\beta\in\pi}\partial_\beta\Psi$ gives, almost everywhere and for any complex constants c_a,

$$\Phi^{(t)}\Big[\prod_{a\in B}(F_a-c_a)\Big]=\sum_{\pi\in\mathrm{Part}(B)}\prod_{\beta\in\pi}\varkappa^{(t)}_\beta,\qquad \varkappa^{(t)}_{\{a\}}=\partial_a\Psi|_0-c_a,\qquad \varkappa^{(t)}_\beta=\partial_\beta\Psi|_0\quad(\#\beta\ge2),$$

since the constants multiply the density by $e^{-\sum_a\lambda_a c_a}$; the blocks of two or more elements do not depend on the c_a, and all the blocks are bounded by the sizes YM3_I:(759). The sum over the level t of the first creation (equation eq. 4.xxxi) adds the weights $\mathsf{w}_t\le 2M^{-3}\ell^3L^{3(K-t)}e^{-\frac12 c'p_0(g_t)^2}$ (Lem. YM3_I:125.3), and on a small side $2M^{-3}\ell^3\le M^{-3}c'A_0^2Y_s^P$; the Gaussian decay in $K-t$ makes the sum converge. Möbius inversion gives the cumulant from the moments of the F_a-c_a. These steps alone give a power of $\frac1d$: the sizes of the blocks hold the factor $x_{\mathsf{k}}^{r/2}\le\bar x_s^{r/2}$, and the sum over t the factor L^{3s}.

Three estimates remove both factors. (i) $\hat d\ge 4MY_s^P$ (Lem. YM3_I:126.1, equation YM3_I:(2323)), and corners of plaquettes of different sets S_a are at the sup distance $\ge\frac{5}{4\sqrt3}d-a_K$; so $\mathsf{D}_{\mathsf{k}}/M\ge 2Y_s^P$, as in Lem. YM3_I:126.1, and a block of an order $r\ge2$ at the working level is at most a number times $\hat d^3(x_{\mathsf{k}}^{1/2}y^{\mathfrak{p}_0})^r e^{-(\mathsf{D}_{\mathsf{k}}/M-2)}\prod_{a\in\beta}\upsilon\mathsf{s}_a$, in which $e^{2-2Y_s^P}\bar x_s^{r/2}=e^{2-2Y_s^P+rY_s/2}$ is bounded for $r\le n$ (at the levels $t<\mathsf{k}$ the blocks are larger by $L^{(3+r/2)(\mathsf{k}-t)}(\mathsf{k}-t+2)^{r\mathfrak{p}_0}$, and they enter the sum over t alone). (ii) In the sum over t, $y_{\mathsf{k}-m}\ge y+m\log\frac{3L}{4}$ for $m\ge0$, $(y+u)^P\ge y^P+u^P$, and $u^P\ge u^2$ at $u=m\log\frac{3L}{4}$ give $e^{-\frac12c'p_0(g_{\mathsf{k}-m})^2}\le e^{-\frac12c'A_0^2y^P}e^{-\frac12c'A_0^2(m\log\frac{3L}{4})^2}$; the first factor is kept, and as $y>Y_s^-=Y_s-\log(2L^3)$ (Lem. YM3_I:126.1, equation YM3_I:(2324), equations YM3_I:(2328)–(2336), with the notation in eq. 2.xxiv) it is at most $e^{-\frac12c'A_0^2(Y_s^-)^P}$, against which $L^{3s}\bar x_s^r$ is bounded. (iii) Take for c_a the constant part of the first order of F_a at the working level, $c_a=\sum_{i\le\mathsf{k}}\sum_{(X,z)}\dot E_a^{(i)}(X,\mathbf{1},z)$, $\dot E_a^{(i)}$ the derivative in λ_a at 0 of the local family of the level i.

Then the singleton $\varkappa^{(\mathsf{k})}_{\{a\}}$ is the first-order part of F_a at the block field of that level, at most its supremum over the small fields, the size $\mathsf{X}_{\mathsf{k},a}\le\mathsf{c}_X y^P\hat d^3\upsilon\mathsf{s}_a$ of the size definitions preceding and in Prop. YM3_I:123.2 (Lem. YM3_I:125.2, equation YM3_I:(2310), at each pair of the majorants), plus an irrelevant part, smaller by a factor that tends to 0 as $s\to\infty$; neither holds a factor x_{k}. So the leading term of a moment is the product of the singletons, at most $\prod_a C\,Y_s^P(LC_W(s))^3\upsilon\mathsf{s}_a$; the

partitions with a block of an order ≥ 2 and the sum over t add at most a number of n and g times this, and with $\upsilon^2 \leq \frac{9}{8} b^{-2}$ Möbius inversion gives $k'_n(g,s) \leq C_n(g) Y_s^{8n\mathfrak{p}_0}$.

Of the power $8\mathfrak{p}_0$ for each entry, $2\mathfrak{p}_0$ come from the supremum of $x_{\mathsf{k}} a_p$ over the small fields, of order $A_0^2 y^{2\mathfrak{p}_0}$, and $6\mathfrak{p}_0$ from the number $\hat{d}^3 < (LC_W(s))^3$ of the unit cubes of the working level in a ball of radius d. Of the side the proof reads the settings of the chain at the n majorants, the conditions of the sum over the level of the first creation and the bound on the weights w_t, which serves only to bound that sum. The weights enter only the tail term of each block of the Möbius inversion, the sum over the level of the first creation in the moment of that block: linearly in their prefactor, and through the Gaussian sum of the estimate (ii) above, in which $e^{-c'' p_0(g_t)^2}$ with $c'' \geq \frac{1}{2} c'$ may stand for $e^{-\frac{1}{2} c' p_0(g_t)^2}$.

As the settings and the conditions hold on every admissible side for $K \geq 3$ (Step 14), the lemma holds on every admissible side, for $K \geq 3$ and weights $\mathsf{w}_t \leq \mathsf{L} L^{3(K-t)} e^{-c'' p_0(g_t)^2}$, $1 \leq t \leq K$, $c'' \geq \frac{1}{2} c'$, with the factor $\max\{1, M^3 \mathsf{L}/(c' A_0^2)\}^n$ on k'_n: as $Y_s \geq 1$, each block gains at most the factor $\max\{1, M^3 \mathsf{L}/(c' A_0^2)\}$ over the tail of a small side, and there are at most n blocks (Thm. C(E3)). □

Lemma 2.18 (As many sources as unit cubes)**.** Let $0 < g' \leq \gamma_2$, (K', β) a first-passage datum at g', ℓ a side small over its running couplings, $0 < d < 1$ with $s = s(d) \leq K'$, the depth, the working level and υ being formed at g', and let n be an integer with $2 \leq n \leq \ell^3$. Let (s_a, S_a), $1 \leq a \leq n$, be majorants placed as in Lem. 2.17 and F_a sources of top index k, the working level, under them, and suppose

$$2 Y_s^P \geq u_*(n), \qquad y_{K'} \geq 2 \log L + \mathfrak{p}_0 \log 3, \qquad L^{3s} \leq x_{K'}^{20},$$

where $u_*(n) = \max\{4, (2/A_\kappa)(n[c_1 + \log(c^* n)] + \log 3)\} \leq Cn \log(en)$, $A_\kappa = 4\beta_s \kappa$, $c^* = \max\{4, \frac{1}{2} c^\flat\}$, c_1 a number of the constants. Then there are complex numbers c_a such that for every nonempty $B \subset \{1, \dots, n\}$

$$\Big| \Big\langle \prod_{a \in B} (F_a - c_a) \Big\rangle_{K', \ell, \beta} \Big| \leq \prod_{a \in B} \mathsf{A}_n(s) \upsilon \mathsf{s}_a, \qquad \mathsf{A}_n(s) \leq C n^{4 + \mathfrak{p}_0} (LC_W(s))^3 \bar{x}_s Y_s^P.$$

■

Proof sketch. It is the proof of Lem. 2.17 with two changes. First, the mixed derivative families of every order $r \leq n$ are bounded with the numbers $(c^* r)^r$ in the place of $\hat{G}_r$, by Cauchy estimates on polydiscs of radii proportional to $1/r$; for this it suffices that $T_m(u)(c^* m)^m \leq 1$ for $2 \leq m \leq n$ at $u = \mathsf{D}_j / M$, which holds as soon as $u \geq 4$ and $\frac{1}{2} A_\kappa u \geq m[c_1 + \log(c^* m)] + \log 3$, so for $u \geq u_*(n)$, and the first hypothesis gives $\mathsf{D}_j / M \geq u_*(n)$. Second, the moments are bounded directly, from the conditional cumulants at the upper level and the sum over the partitions, and not through the cumulants of the sources; the sum over the level of the first creation (equation eq. 4.xxxi) is bounded by the count of the cubes of a small side, through the last two hypotheses. Hence a bound in product form, with no factorial in n beyond the power in $\mathsf{A}_n(s)$. □

Lemma 2.19 (Two sources at one scale)**.** (a) Let $g \leq \gamma_2$, ℓ small, $s_1(g) \leq s(d) \leq K$, $3d \leq |x_1 - x_2| \leq 4d$, and let (F_a, t_a) be real quadratic sources of top index $T - 1$ under (s_a, S_a) in the sense of Lem. 2.15(a), with their families without base point at the heights $1 \leq j \leq T - 1$, T the upper level (forms have a member at every height $j \geq 1$), S_a within $d - a_K$ of x_a. Then

$$\Big| \langle F_1; F_2 \rangle_\ell - 6\upsilon^2 \sum \mathsf{t}_1(p,p') \mathsf{t}_2(q,q') C^{\mathrm{bg}}(p,q) C^{\mathrm{bg}}(p',q') \Big| \leq \varepsilon_g^\natural(s(d)) \upsilon^2 \mathsf{s}_1 \mathsf{s}_2, \qquad |\langle F_1; F_2 \rangle_\ell| \leq C_g^\sharp(s(d)) \upsilon^2 \mathsf{s}_1 \mathsf{s}_2,$$

where $\varepsilon_g^\natural(s) = \varepsilon_g^\flat(s) + 2(1+A_1)^4\mathsf{C}_5 s^2 (LC_W(s))^6 Y_s^{2P}\mathrm{sm}_g(s)$,

$$\begin{aligned}\varepsilon_g^\flat(s) = {}& (\mathsf{C}_2+\mathsf{C}_3)(Y_s^-)^{-2\mathfrak{p}_0} + \mathsf{C}_3\, s\,(Y_s^-)^{-5\mathfrak{p}_0}\\ &+ \mathsf{C}_6(Y_s^-)^{-6\mathfrak{p}_0} + \mathsf{C}_8\big((LC_W(s))^3+1\big)(Y_s^-)^{-9\mathfrak{p}_0}\\ &+ 4\mathsf{c}_X^2(Y_s^-)^{-14\mathfrak{p}_0}\\ &+ (\mathsf{C}_B' + \mathsf{C}_7 + \mathsf{C}_3' + \mathsf{C}_3'' + \mathsf{C}_6)\, s^4 (LC_W(s))^6 Y_s^{11\mathfrak{p}_0}\mathrm{sm}_g(s),\end{aligned}$$

$Y_s^- = Y_s - \log(2L^3)$ and sm_g the functions of equations YM3_I:(2328)–(2336), with the notation in eq. 2.xxiv, C_i the constants of Lem. YM3_I:125.5, with the convention in eq. 2.xxii–eq. 2.xxiii: $\varepsilon_g^\flat$ bounds all the errors of the proof but those of the Gaussian values at the flat top, which the second summand bounds, and both tend to 0 as $s\to\infty$ (the decay argument preceding equation YM3_I:(2332)); and

$$C_g^\sharp(s) = \mathsf{C}_2(LC_W(s))^6 Y_s^{2P} + 4\mathsf{c}_X^2 + \mathsf{C}_B' s^4 (LC_W(s))^6 Y_s^{6\mathfrak{p}_0}\mathrm{sm}_g(s) \le C(g) Y_s^{16\mathfrak{p}_0}.$$

(b) Let ℓ be admissible, $K\ge 3$, $s = s(d)\le K$, $\mathsf{Lm}_s \le L^{s-3}$, $x_s^- \ge e^P$; then the settings of the chain hold at the levels $\mathsf{k} < j_1 \le T \le K-2$, with $n_T \le s$, $676a_K \le d$, $20d < \ell$, $M \le \hat d < LC_W(s)$. Let $|x_1 - x_2| \ge 3d$, $|L^{-1}x_{t-1} - x_t| \le B_\sharp$ for all t, $B_\sharp$ the flow bound of the running couplings and $\bar B = LB_\sharp/(L-1)$ (Section YM3_I:74), $y_K \ge 2\log(1+\bar B)$; let $\mathsf{L}\ge 0$, $c>0$ with $\mathsf{w}_t \le \mathsf{L}L^{3(K-t)}e^{-cp_0(g_t)^2}$ $(t\le T)$, $\mathsf{W}_T \le \mathsf{L}L^{3(K-T)}e^{-cp_0(g_T)^2}$, $2c\mathfrak{p}_0 A_0^2 y^{P-1} \ge 13$, and $\mathrm{sm}_T^{\mathsf{L},c} \le 1$. Then on T_ℓ the error in (a) is at most

$$\{(\mathsf{C}_2+\mathsf{C}_3)y^{-2\mathfrak{p}_0} + \mathsf{C}_3 n_T y^{-5\mathfrak{p}_0} + \mathsf{C}_6 y^{-6\mathfrak{p}_0} + \mathsf{C}_8(\hat d^3+1)y^{-9\mathfrak{p}_0} + 4\mathsf{c}_X^2 y^{-14\mathfrak{p}_0} + \mathsf{C}_B^e n_T^4 \hat d^6 y^{15\mathfrak{p}_0}\mathrm{sm}_T^{\mathsf{L},c}\}\upsilon^2\mathsf{s}_1\mathsf{s}_2,$$

and the remainder Rest_T' of the proof of (a) obeys $|\mathrm{Rest}_T'| \le [4\mathsf{c}_X^2 y^{-14\mathfrak{p}_0} + \mathsf{C}_B' n_T^4 \hat d^6 y^{6\mathfrak{p}_0}\mathrm{sm}_T^{\mathsf{L},c}]\upsilon^2\mathsf{s}_1\mathsf{s}_2$; for two plaquette weights $\mathsf{C}_B^{(1)}$ may replace C_B'.

(c) In the setting of (a), almost everywhere at the upper level T and for all the quotients,

$$\big|\Phi^{(T)}[F_1F_2] - \Phi^{(T)}[F_1]\,\Phi^{(T)}[F_2] - G\big| \le \varepsilon_g(s(d))\,\upsilon^2\mathsf{s}_1\mathsf{s}_2,$$

$G = 6\upsilon^2\sum \mathsf{t}_1(p,p')\mathsf{t}_2(q,q')C^{\mathrm{bg}}(p,q)C^{\mathrm{bg}}(p',q')$ the Gaussian value of (a) and $\varepsilon_g = \epsilon_g$ the error function in equation YM3_I:(2330), with the coefficient fixed in Lem. 2.13; and almost everywhere at every level $j\le T$ the conditional covariance $\Phi^{(j)}[F_1F_2] - \Phi^{(j)}[F_1]\Phi^{(j)}[F_2]$ is at most the crude size M_j' of the size definitions preceding and in Prop. YM3_I:123.2, formed at the majorants, which Lem. YM3_I:125.2 bounds by $[\mathsf{m}_\star L^{4n}(n+1)^{2\mathfrak{p}_0} + 2n_T\mathsf{sb}_0^2]\upsilon^2\mathsf{s}_1\mathsf{s}_2$, $n = [\mathsf{k}-j]_+$. ∎

Proof sketch. (a) The covariance is a sum of one-step covariances plus a rest, $\langle F_1; F_2\rangle = \sum_{\mathsf{k}<j\le T}\mathrm{qm}_j + \mathrm{Rest}_T'$, $\mathrm{qm}_j = \int \mathrm{Qt}_j[X_{j-1,1}; X_{j-1,2}]\,\mathrm{d}\mathsf{m}_{j|T}$, with the step quotients Qt_j of the quotient definitions in [7, §B.9], their measures $\mathsf{m}_{j|T}$ and the first-order parts $X_{j-1,a}$ of the sources (Prop. YM3_I:123.2, which holds for these sources by Lem. 2.15(b)). The steps $j_1 < j \le T$ are bounded crudely, by $2\mathsf{X}_{j-1,1}\mathsf{X}_{j-1,2}$ each: the error Er_2 of equation YM3_I:(2307). For $\mathsf{k} < j \le j_1$ the one-step covariance is almost everywhere the total of the mixed derivative family $\mathsf{K}^{(j)}$ of the step j ([7, Lem. B.20]); carried to the upper level T (Prop. YM3_I:123.3) and evaluated at the trivial background it is the number $\mathsf{sum}_T^{(j)}$, up to the error $\mathsf{Ft}_T^{(j)}$ of the flat top, which holds at every point of the small-field set of the level T (Prop. YM3_I:123.3, equation YM3_I:(2283), in its sup-norm form); the mass $1-\mathsf{W}_T$ of $\mathsf{m}_{T|T}$ costs $\mathsf{W}_T(\bar{\mathsf{n}}_j^{\mathrm{G}} + \mathsf{e}_{j,T})$; and $\mathsf{sum}_T^{(j)}$ is the Gaussian number $\mathsf{n}_{j,T}^{\mathrm{G}}$ up to the error $\mathsf{e}_{j,T}$ of equations YM3_I:(2303)–(2305) (Prop. YM3_I:125.1).

With kernels t_a in the place of $\frac12 w_a\delta$ this proof reads the source at three places alone: the covariance of one step at a triple is the covariance of the first-order variations of the two sources under the fluctuation measure of that step; the second order of a source at a small point of the chart is the quadratic form of its kernel; and the forms of the slices are built with the kernels. So it gives the Gaussian numbers $2(N^2-1)\upsilon_j^2\sum \mathsf{t}_1(p,p')\mathsf{t}_2(q,q')\mathcal{C}_{(j)}(p,q)\mathcal{C}_{(j)}(p',q') +$

$4(N^2-1)\sum_{j<i'\le T} v_j v_{i'} \sum \mathsf{t}_1(p,p')\mathsf{t}_2(q,q')\mathcal{C}_{(j)}(p',q')\mathcal{C}_{(i')}(p,q)$, with the same errors $\mathsf{e}_{j,T}$ and the same constants, for every real quadratic source, the bond forms under $(59\sup|\mathsf{w}|, S[\mathsf{w}])$ among them (Lem. 2.15(a)).

The Gaussian numbers sum to $6v^2\sum \mathsf{t}_1\mathsf{t}_2 DD$, $D=\sum_{\mathsf{k}<j\le T}\mathcal{C}_{(j)} = \Pi_{\mathsf{k}} - \Pi_T$, since the slices telescope and the terms (j,i'), (i',j) agree under the symmetry of the kernels, and $C^{\mathrm{bg}} = \Pi_{\mathsf{k}}$; the Gaussian numbers above j_1, the replacing of the couplings v_j of the steps by v and the dropping of Π_T, with $|\Pi_T(p,q)| \le C_{\mathrm{tp}}^{\mathrm{bg}}\eta_T^3$ (Prop. YM3_I:120.5), cost the errors Er_6, Er_7, Er_8 of equation YM3_I:(2307). With the deposit inequality $L^{3a}\mathsf{e}_{\mathsf{k}-a} \le L^{-10a}\mathsf{e}_{\mathsf{k}}$, $\mathsf{e}_k = e^{-\frac{1}{2}c' p_0(g_k)^2}$ (the proof of Lem. YM3_I:125.4; its hypotheses hold by equation YM3_I:(2326) and the logarithmic floors immediately following that equation),

$$|\mathrm{Rest}'_T| \le [4\mathsf{c}_X^2 y^{-14\mathfrak{p}_0} + \mathsf{C}'_B n_T^4 \hat{d}^6 y^{6\mathfrak{p}_0}\mathrm{sm}_T]v^2\mathsf{s}_1\mathsf{s}_2.$$

Collected, the left member of (a) is at most $|\mathrm{Rest}'_T| + \mathsf{Er}_2 + \sum_{\mathsf{k}<j\le j_1}\mathsf{Ft}_T^{(j)} + \mathsf{W}_T\sum_{\mathsf{k}<j\le j_1}\bar{\mathsf{n}}_j^{\mathrm{G}} + 2\sum_{\mathsf{k}<j\le j_1}\mathsf{e}_{j,T} + \mathsf{Er}_6 + \mathsf{Er}_7 + \mathsf{Er}_8$. By the bounds of the errors at the attached levels (Lem. YM3_I:125.5), with $Y_s^- < y \le Y_s$ (Lem. YM3_I:126.1, equation YM3_I:(2324)), $1 \le n_T \le s$, $\hat{d} < LC_W(s)$ and $\mathrm{sm}_T \le \mathrm{sm}_g(s)$, all these terms but the sum of the $\mathsf{e}_{j,T}$ are together at most $\varepsilon_g^\flat(s)v^2\mathsf{s}_1\mathsf{s}_2$, and $2\sum_{\mathsf{k}<j\le j_1}\mathsf{e}_{j,T} \le 2(1+A_1)^4\mathsf{C}_5 s^2(LC_W(s))^6 Y_s^{2P}\mathrm{sm}_g(s)v^2\mathsf{s}_1\mathsf{s}_2$.

The crude bound is the same sum with the Gaussian numbers bounded by their sizes. (b) In the proof of (a) the side enters only through the weights w_t and the sum over the level of the first creation (equation eq. 4.xxxi), and the distance only through a lower bound; the factor $e^{-cp_0(g_t)^2}$ is used at three places alone: in the deposit inequality at c, in its monotonicity in t, and in the bound on w_t. For the first, the proof of Lem. YM3_I:125.4 applies with $c'/2$ replaced by c. To make this substitution explicit, put $P = 2\mathfrak{p}_0$, $y = y_{\mathsf{k}}$ and $\Delta = y_{\mathsf{k}-a} - y$ for $1 \le a \le \mathsf{k}$. The flow bound and $y_K \ge 2\log(1+\bar{B})$ give

$$\Delta \ge a\log L + \log(1-\bar{B}/x_{\mathsf{k}}) > 2a, \qquad y(a\log L - \Delta) \le \frac{8}{3e}\frac{\bar{B}}{1+\bar{B}} < 1.$$

Keeping the linear and quadratic terms of the power gives

$$(y+\Delta)^P - y^P \ge Py^{P-1}a\log L + Py^{P-2}\left[\frac{P-1}{2}\Delta^2 - y(a\log L - \Delta)\right] \ge Py^{P-1}a\log L.$$

Thus, for $\mathsf{e}_j = e^{-cp_0(g_j)^2}$,

$$L^{3a}\mathsf{e}_{\mathsf{k}-a} \le L^{-(2c\mathfrak{p}_0 A_0^2 y^{P-1}-3)a}\mathsf{e}_{\mathsf{k}} \le L^{-10a}\mathsf{e}_{\mathsf{k}}.$$

The case $a=0$ is immediate. This uses exactly the rate condition $2c\mathfrak{p}_0 A_0^2 y^{P-1} \ge 13$ in (b), at the working level k; no stronger rate condition at K is needed. The settings follow from the arithmetic of the working level under the two depth conditions, the flow bound and the lower bound on y_K. (c) A block of two is the conditional covariance $\Phi_{12}^{(n)} = \Phi^{(n)}[F_1F_2] - \Phi^{(n)}[F_1]\Phi^{(n)}[F_2]$.

Almost everywhere, by the tower of the quotients (equation YM3_I:(2273)), $\Phi_{12}^{(n)} = \mathrm{Qt}_n[\Phi_{12}^{(n-1)}] + \mathrm{Qt}_n[\Phi^{(n-1)}[F_1];\Phi^{(n-1)}[F_2]]$, and $\Phi^{(n-1)}[F_a]$ is a constant plus the first-order part $X_{n-1,a}$ plus an irrelevant part (equation YM3_I:(2267) and Prop. YM3_I:123.1); this gives the crude bound at every level (the proof of Prop. YM3_I:123.2, through Lem. 2.15(b)). For $\mathsf{k} < j \le j_1$ the one-step covariance $\mathrm{Qt}_j[X_{j-1,1};X_{j-1,2}]$ is almost everywhere the total $\mathsf{F}_j^{(j)}$ of the family $\mathsf{K}^{(j)}$ ([7, Lem. B.20]), and $\mathrm{Qt}_{i+1}[\mathsf{F}_i^{(j)}] = \mathsf{F}_{i+1}^{(j)}$ almost everywhere (the proof of Prop. YM3_I:123.3).

So, by induction on n from k to T, the remainder $R_n = \Phi_{12}^{(n)} - \sum_{\mathsf{k}<j\le\min\{n,j_1\}}\mathsf{F}_n^{(j)}$ is almost everywhere at most M_{k}, plus twice the products of the sizes of the first-order and of the irrelevant parts at the levels $\mathsf{k} \le j < n$, plus twice the products of the sizes of the first-order parts at the steps above j_1. At $n = T$ the measure m_T is carried by the small-field set of the level T, at every point of

which $\mathsf{F}_T^{(j)}$ is within $\mathsf{Ft}_T^{(j)}$ of the number $\mathsf{sum}_T^{(j)}$ (Prop. YM3_I:123.3, equation YM3_I:(2283), before any integration); $\mathsf{sum}_T^{(j)}$ is $\mathsf{n}_{j,T}^{\mathrm{G}}$ up to $\mathsf{e}_{j,T}$, and the Gaussian numbers give G up to $\mathsf{Er}_6 + \mathsf{Er}_7 + \mathsf{Er}_8$, as in (a).

No sum over the levels of the first creation enters, so that there are no tails and no weight W_T: the error is at most the bracket of Cor. YM3_I:125.6 times $\upsilon^2 \mathsf{s}_1 \mathsf{s}_2$, and with $Y_s^- < y \le Y_s$, $n_T \le s$, $\hat{d} < LC_W(s)$ and $\mathrm{sm}_T \le \mathrm{sm}_g(s)$ this bracket is at most $\varepsilon_g(s)$, term by term. □

Lemma 2.20 (Symmetries and copies)**.** (a) For a lattice isometry r with $re_\mu = s_\mu e_{\pi(\mu)}$: $q_{(r\mathsf{e})^+}(rU) = s_\lambda s_{\lambda'} q_{\mathsf{e}}(U)$, $\lambda < \lambda'$ the directions other than that of e; so $rO_K^T(H) = O_K^T(rH)$, and there is a linear bijection $r^\sharp$ of the weights, with $\|r^\sharp v\| = \|v\|$, real weights kept real and carriers and sets $S[\cdot]$ carried by r, such that the image of a form is the form of the same kind with the weight $r^\sharp v$; in particular $Y[r^\sharp \mathsf{w}](rU) = Y[\mathsf{w}](U)$ and $\theta Y[\mathsf{w}] = Y[\theta^\sharp \mathsf{w}]$.

(b) A law that is invariant under the reflections in the site planes has $\nu(q_{\mathsf{e}}) = 0$; all laws are such, so $\nu(Y[\mathsf{w}]) = 0$.

(c) For a board of side l_0 and $\ell_0 = 2\mathsf{l}_0$, a form whose carrier lies in a cell at a distance $\ge \rho > a_K$ from the walls is a function of the cell; its copies on T_{ℓ_0} are forms of the same kind, size and mean, with the mirror images as carriers; copies in different cells of a period box are $\ge 2\rho$ apart, two in one cell at the original distance.

(d) If $3d \le |x_0 - y_0| \le 4d$, the midpoint is within $\frac{a_K}{2}$ of the offset of the board of side l in each coordinate, and the weights are carried within $d - 2a_K$ of x_0, y_0, then the corners of their plaquettes lie within the sup distance $3d$ of the centre of the cell, and the sixteen copies on $T_{\ell'}$ form eight pairs, partners $3d$ to $4d$ apart, different pairs at sup distance $\ge \mathsf{l} - 6d \ge 11M$, the suprema of the weights kept; the same holds for tensor weights and arbitrary majorants.

(e) For majorants with plaquettes within $d - a_K$ of x_0, y_0, $3d \le |x_0 - y_0| \le 4d$, the tents $\tilde{f}_a = \mathsf{s}_a \sum_{p \in S_a} \max\{0, 1 - 2a_K^{-1}|u - x(p)|_\infty\}$ are Lipschitz, equal $\mathsf{s}_a \mathbf{1}_{S_a}$ at the barycentres, have their supports in $B(x_a, d)$, and $d \le \mathrm{dist}(\mathrm{supp}\, \tilde{f}_1, \mathrm{supp}\, \tilde{f}_2) \le 6d$. Let besides $0 < g \le \gamma_2$, (K, β) a first-passage datum at g, ℓ small, $0 < d < 1$ and $s_1(g) \le s = s(d) \le K$, so that $\mathsf{Lm}_s \le L^{s-3}$ and $x_s^- \ge e^P$ (equation YM3_I:(2331)). Then $s \ge 3$, $2 \le \mathsf{k} \le K - 1$, $M \le L^{-2} C_W(s) \le \hat{d} < LC_W(s)$, $\mathsf{D}_\mathsf{k} \ge 3M$ for the pair, $20d \le \ell$, and the middle and the upper level have $\mathsf{k} < j_1 \le T \le K - 2$, $1 \le n_T \le s$, $L^{n_T} < \mathsf{Lm}_s$, $x_T > x_s^-$.

Let $n \ge 1$ pairs of such majorants be given, $p = 2n$, the corners of plaquettes of different pairs at the sup distance $\ge 11M$. Then D_k, the least over all $a \ne a'$, is $\ge 3M$: partners give $3M$, members of different pairs $11ML^{K-\mathsf{k}}$. The settings of the chain for the p sources hold with the top index $T - 1$, with the smallness conditions at the heights $0, \dots, T - 1$ and $d_*(t) \ge 3$ for $1 \le t \le T$; the levels, the lattices, the histories, the measures and the quotients of the chain are the same for all pairs, and for the chain of each pair alone, and do not depend on the sources; and $\theta^{[j]} T_p(\mathsf{D}_j/M) \le \frac{1}{2}$ for $1 \le j < T$ as soon as

$$\mathrm{th}_{g,p}(s) = 2s^{2+p\mathfrak{p}_0} \mathsf{Lm}_s^{p/2-1} T_p^+ \, \theta(x_s^-) \le 1, \qquad T_p^+ = T_p(3) + (\tfrac{4}{3})^{p/2} L,$$

θ the function of Lem. 2.17; $\mathrm{th}_{g,p}(s)$ tends to 0 as $s \to \infty$, being a power of s times $L^{-s/2}$. ■

Proof sketch. (a) r carries the loops at e to those at $r\mathsf{e}$; reversal sends $w(V)$ to $w(V^{-1}) = -w(V)$, conjugation keeps $\langle \cdot, \cdot \rangle$, and the sign is the product of the signs of the two vectors u_λ, $u_{\lambda'}$. (b) The reflection in the site plane through x_{e} normal to λ flips s_λ alone, so that it changes the sign of q_{e}; torus measures are invariant under it, periodic Gibbs states by Prop. YM3_I:103.4. (c) Each coordinate of a copy lies between two consecutive walls, at a distance $\ge \rho$ from both, and the mean is kept by the invariance of $\tilde{\mu}_{\ell_0}$ under the lattice isometries. (d) The corners lie within the sup distance $3d$ of the centre of the cell, and $\mathsf{l} \ge LM$. (e) A tent is carried within $\frac{\sqrt{3}}{2} a_K$ of its barycentre and vanishes at the other barycentres; the supports need not come near the centres, which is why only $d \le \mathrm{dist} \le 6d$ holds, and nothing uses an upper bound on the distance.

The settings and the numbers come from the arithmetic of the working level (Lem. YM3_I:126.1) and from Lem. YM3_I:126.2, by which the objects of the construction satisfy the settings of the chain; neither involves a source, and of the majorants the settings for the p sources read the weights $\mathsf{s}_a\mathbf{1}_{S_a}$ and the bound $\mathsf{D}_\mathsf{k} \ge 3M$ alone. *The threshold.* $T_p(u) = \sum_{n'\ge1}(\frac43)^{p/2}(n'+1)^{1+p\mathfrak{p}_0}L^{1+(p/2-1)n'}e^{-4\beta_s\kappa[L^{n'-1}u-2]_+}$ does not increase. If $L^qu \ge 3$, $q \ge 1$, the q terms with $n' \le q$ are each at most $(\frac43)^{p/2}(q+1)^{1+p\mathfrak{p}_0}L^{1+(p/2-1)q}$, and the term of the index $n' = q+n''$ is at most $(q+1)^{1+p\mathfrak{p}_0}L^{(p/2-1)q}$ times the term of the index n'' of $T_p(3)$, as $n'+1 \le (q+1)(n''+1)$ and $L^{n'-1}u \ge 3L^{n''-1}$; so $T_p(u) \le (q+1)^{2+p\mathfrak{p}_0}L^{(p/2-1)q}T_p^+$.

For $1 \le j \le \mathsf{k}$, $\mathsf{D}_j/M \ge 3$ and $T_p(\mathsf{D}_j/M) \le T_p(3) \le T_p^+$; for $\mathsf{k} < j < T$, $q = j-\mathsf{k} \le n_T-1$ has $L^q\mathsf{D}_j/M = \mathsf{D}_\mathsf{k}/M \ge 3$, $q+1 \le n_T \le s$ and $L^{(p/2-1)q} < \mathsf{Lm}_s^{p/2-1}$. And $\theta^{[j]} = \theta(x_j) \le \theta(x_s^-)$, since $x_j \ge x_T > x_s^- \ge e^P$ and θ does not increase on $[e^P,\infty)$. So $\theta^{[j]}T_p(\mathsf{D}_j/M) \le \frac12\mathrm{th}_{g,p}(s) \le \frac12$ for $1 \le j < T$. □

Lemma 2.21 (Far pairs; centred sources)**.** Let $0 < g \le \gamma_2$, ℓ small, $n \ge 1$, $p = 2n$, $s = s(d) \le K$. For pairs $\tau \le n$ let $3d \le |z_{(\tau,1)} - z_{(\tau,2)}| \le 4d$, let the majorants have their plaquettes within $d - a_K$ of z_a, and let corners of plaquettes of different pairs be at sup distance $\ge 11M$; let (F_a, t_a) be real quadratic sources of top index $T-1$ under the majorants, with their families without base point at the heights $1 \le j \le T-1$ (as forms have them), $\tilde F_a = F_a - \langle F_a\rangle_\ell$, $\tilde P_J = \prod_{\tau\in J}\tilde F_{(\tau,1)}\tilde F_{(\tau,2)}$, G_τ the Gaussian value of the pair, $\hat m_a = \upsilon\mathsf{s}_a$, $\hat m_\tau = \hat m_{(\tau,1)}\hat m_{(\tau,2)}$, $\hat m_J = \prod_{\tau\in J}\hat m_\tau$. For $s \ge s_p(g)$, a first depth $\ge s_1(g)$, and all nonempty J and B:

$$|\langle\tilde P_J\rangle_\ell - \textstyle\prod_J G_\tau| \le \omega'_{g,p}(s)\hat m_J, \qquad |\langle\tilde P_J\rangle_\ell - \prod_J\langle\tilde P_\tau\rangle_\ell| \le \omega'^\sharp_{g,p}(s)\hat m_J, \qquad |G_\tau| \le g_0\hat m_\tau;$$

if N_2 pairs lie in B and N_1 meet it in one element,

$$\Big|\Big\langle\prod_B\tilde F_a\Big\rangle_\ell\Big| \le [e_g(s)^{N_1}(g_0+\delta_g(s))^{N_2} + \mathrm{Bl}(p)\mathrm{far}_{g,p}(s)(1+g_0+\varepsilon_g(s))^n + \mathrm{tl}_{g,p}(s)]\prod_B\hat m_a,$$

so that this mean is $\le o_{g,p}(s)\prod_B\hat m_a$, $o_{g,p} = e_g + \mathrm{Bl}(p)\mathrm{far}_{g,p}(1+g_0+\varepsilon_g)^n + \mathrm{tl}_{g,p}$, if B holds no pair, and $\le \mathrm{Mo}_{g,p}(s)\prod_B\hat m_a$ in every case, $\sup_{s\ge s_p(g)}\mathrm{Mo}_{g,p}(s) < \infty$; $Y_s^P\omega'^\sharp_{g,p}(s)$ is bounded. Here

$$\omega'_{g,p} = n\delta_g(1+g_0+\delta_g)^{n-1} + \mathrm{Bl}(p)\mathrm{far}_{g,p}(1+g_0+\varepsilon_g)^n + \mathrm{sm}_g(1+g_0)^n + \mathrm{tl}_{g,p}, \qquad \omega'^\sharp_{g,p} = \omega'_{g,p}[1+n(1+g_0+\omega'_{g,p})^{n-1}],$$

with $g_0 = \frac32 81^2C_\mathsf{k}'^2$ for arbitrary majorants and $g_0 = 422C_\mathsf{k}'^2$ for majorants of bond weights of one direction. For bond forms $Y[\mathsf{w}_a]$ with real weights on the bonds of one direction, whose means vanish, $|G_\tau| \le 106C_\mathsf{k}'^2\upsilon^2\sup|\mathsf{w}_{(\tau,1)}|\sup|\mathsf{w}_{(\tau,2)}|$, and the same functions at $g_0 = 106C_\mathsf{k}'^2/59^2$, written $\omega_{g,p}$, $\omega^\sharp_{g,p}$, serve in units of $\hat m_a = 59\upsilon\sup|\mathsf{w}_a|$. Here $\mathrm{far}_{g,p}(s) = C_p^{\mathrm{far}}(LC_W(s))^3\bar x_s^{p/2}Y_s^{p\mathfrak{p}_0}e^{-11L^{s-2}/\mathsf{Lm}_s}$, ε_g is the error function of equations YM3_I:(2328)–(2336), with the notation in eq. 2.xxiv for a close pair, $\delta_g = \varepsilon_g + e_g^2$, e_g and $\mathrm{tl}_{g,p}$ are the error functions of the singletons and of the tails,

$$e_g(s) = 2\mathsf{c}_X(Y_s^-)^{-7\mathfrak{p}_0} + 2(\mathsf{c}_I + \mathsf{c}_{\mathrm{tl}}\mathsf{c}_{\mathrm{sb}})\,s^2(LC_W(s))^3Y_s^{3\mathfrak{p}_0}\mathrm{sm}_g(s),$$

$$\mathrm{tl}_{g,p}(s) = \mathrm{Bl}(p)\,\mathsf{c}_\sigma^p\mathsf{c}_{\mathrm{sb}}^p(1+S_p^{\mathrm{tl}})\,s^{1+3p/2}(LC_W(s))^{3p}Y_s^{2p\mathfrak{p}_0}\,M^{-3}c'A_0^2Y_s^PL^{3s}\bar x_s^p\,e^{-\frac12c'A_0^2(\log x_s^-)^P},$$

$\mathsf{c}_\sigma = 4 + \mathsf{C}_\mathsf{M}^{1/2}/\mathsf{C}_{\mathrm{sz}} + \sqrt2$, C_{sz} of Lem. YM3_I:125.2 and the constant convention preceding eq. 2.xxii, $S_p^{\mathrm{tl}} = \sum_{a\ge0}L^{(3+4p)a+4p}(a+2)^{2p\mathfrak{p}_0}e^{-\frac12c'A_0^2(a\log\frac{3L}{4})^2}$, $\mathrm{Mo}_{g,p} = (1+g_0+\delta_g)^n + \mathrm{Bl}(p)\mathrm{far}_{g,p}(1+g_0+\varepsilon_g)^n + \mathrm{tl}_{g,p}$, $\mathrm{Bl}(p)$ is the number of partitions of p elements, and $s_p(g)$ is the least integer $\ge s_1(g)$ from which $\mathrm{th}_{g,p}$ of Lem. 2.20(e), $\mathrm{far}_{g,p}$ and e_g are at most 1 and $2\mathsf{c}_{\mathrm{tl}}s\,\mathrm{sm}_g(s) \le 1$; it exists, since these four functions tend to 0. ■

Proof sketch. By Lem. 2.20(e) all pairs share one working level, one upper level T and one tower of quotients, and, as $\mathrm{th}_{g,p}(s) \le 1$ for $s \ge s_p(g)$, $\theta^{[j]}T_p(\mathsf{D}_j/M) \le \frac12$ for $1 \le j < T$, so that the mixed

derivative families of every order $\le p$ are bounded up to the level T (the bound of the proof of Lem. 2.17). At T, almost everywhere, $\Phi^{(T)}[\prod_B \tilde F_a] = \sum_\pi \prod_{\beta\in\pi} \varkappa_\beta^{(T)}$ (the identity of the proof of Lem. 2.17, which holds at every level of the chain in use, here up to T, at $c_a = \langle F_a\rangle_\ell$); the conditional cumulants of order ≥ 2 do not depend on the centring constants.

A block that links two pairs is at most $\mathrm{far}_{g,p}(s)\prod \hat m_a$: by the vanishing of mixed derivatives on polymers that miss a support it is exponentially small in the distance $11M$ measured in units of the level T, that is in $L^{K-T} > L^{s-2}/\mathsf{Lm}_s$. A block of one pair is the conditional covariance $\Phi^{(T)}[F_{(\tau,1)}F_{(\tau,2)}] - \Phi^{(T)}[F_{(\tau,1)}]\Phi^{(T)}[F_{(\tau,2)}]$, the quotients being those of the chain of that pair alone (Lem. 2.20(e)); by Lem. 2.19(c) it is almost everywhere within $\varepsilon_g(s)\hat m_\tau$ of G_τ. A singleton obeys, by the sum over the level of the first creation (equation eq. 4.xxxi) applied to F_a minus its flat-top constant,

$$|\Phi^{(T)}[F_a] - \langle F_a\rangle_\ell| \le 2\mathsf{Y}_{T,a} + \rho'_a \le e_g(s)\hat m_a,$$

$\mathsf{Y}_{T,a}$ the size of F_a at the level T and ρ'_a its first-creation sum; for bond forms the mean is 0, and the bound is used only in its last line, so that centring replaces the vanishing of the mean. So the partitions of B without a linking block add up to the product, over the pairs that meet B, of $\varkappa^{(T)}_{\{(\tau,1),(\tau,2)\}} + \varkappa^{(T)}_{\{(\tau,1)\}}\varkappa^{(T)}_{\{(\tau,2)\}} = G_\tau + O(\delta_g(s))\hat m_\tau$ if both members lie in B, and of a singleton if one does. Crude bounds of the conditional moments at all the levels $t \le T$ (for a block of one pair, the crude clause of Lem. 2.19(c)), multiplied by w_t and summed, give the tails $\mathrm{tl}_{g,p}$.

Summing over the at most $\mathrm{Bl}(p)$ partitions gives the bounds of the lemma, and $\langle\tilde P_\tau\rangle = G_\tau + O(\delta_g)\hat m_\tau$ gives the second from the first. *The size of G_τ.* For plaquettes $p \in S_{(\tau,1)}$, $q \in S_{(\tau,2)}$, $|x(p) - x(q)| \ge 3d - 2(d - a_K) > d$; so $|C^{\mathrm{bg}}(p,q)| \le C'_{\mathsf{k}}(a_K/d)^3$ (Lem. 2.16), and $|G_\tau| \le 6C'^2_{\mathsf{k}}(a_K/d)^6(\sum|\mathsf{t}_{(\tau,1)}|)(\sum|\mathsf{t}_{(\tau,2)}|)v^2$. The two sums are counted by volume. The midpoints of the bonds of one direction form a translate of $a_K\mathbb{Z}^3$. Let w be a weight on the bonds of one direction with $S[\mathsf{w}]$ within $d - a_K$ of z_a.

A bond of its support has its four plaquettes in $S[\mathsf{w}]$, and its midpoint is the midpoint of the centres of two of them, opposite in one plane, so that it lies within $d - a_K$ of z_a; the open cubes of side a_K about these midpoints are disjoint and lie within $d - a_K + \frac{\sqrt3}{2}a_K < d$ of z_a, in $B(z_a, d)$, a Euclidean ball as $20d \le \ell$; so the support of w has at most $\frac{4\pi}{3}(d/a_K)^3$ bonds. For a bond form $Y[\mathsf{w}]$ on the bonds of one direction, q_b being a quarter of a signed sum of four corner forms, $\sum_{p,p'}|\mathsf{t}(p,p')| \le \sum_b|\mathsf{w}(b)| \le \frac{4\pi}{3}(d/a_K)^3\sup|\mathsf{w}|$, and $6(\frac{4\pi}{3})^2 < 106$. For a majorant of such a form, $\#S \le 4\cdot\frac{4\pi}{3}(d/a_K)^3$, each bond lying in four plaquettes, and $\sum|\mathsf{t}| \le \frac12(\#S)\mathsf{s}$, as $2\sum_{p'}|\mathsf{t}(p,p')| \le \mathsf{s}$; so $6(\frac{8\pi}{3})^2 < 422$.

For an arbitrary majorant, $\#S \le 81(d/a_K)^3$ (the plaquette count following equation YM3_I: (2300)) and $\sum|\mathsf{t}| \le \frac12(\#S)\mathsf{s}$ give $6(\frac{81}{2})^2 = \frac32 81^2$. □

Proposition 2.22 (Pressure, domination, boards, extraction)**.** (a) For $0 < g \le \gamma$, $\ell_1 \le \ell_2$ in $\mathfrak{L}(g)$ and every datum, $0 \le p_{\ell_1} - p_{\ell_2} \le G_g$; from Thm. YM3_I:D alone, $0 \le p_{\ell_1} - p_{\ell_2} \le G^F_g = C_g + C_{\mathrm{maj}} + 2 + 2C_{\mathrm{tt}}C'_{\mathrm{tt}}E_0 \le 2C_g + 2C_{\mathrm{tt}}C'_{\mathrm{tt}}E_0$.

(b) Let K, $\beta > 0$, a board of side l_0, a side $\ell_0 \ge 2\mathsf{l}_0$ and G with $p_{\ell_0} - p_\ell \le G$ on an unbounded set of sides be given. For bounded functions $\Phi_{\mathbf{k}}$ of distinct cells and every $\nu \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$ (and for $\tilde\mu_\ell$ if the cells are distinct modulo the period): $|\nu(\prod\Phi_{\mathbf{k}})| \le \prod e^{G\mathsf{l}_0^3}\|\Phi_{\mathbf{k}}\|_{\mathrm{ch},\ell_0}$. No reflection invariance of ν is used.

(c) Let $h > 0$ with $\mathsf{l}_0 \in h\mathbb{N}$, as $h = a_K$, or $h = a_{K-t}$ for a level t. If $n(r + 2\rho + h) < \mathsf{l}_0$, any n cubes of side r admit an offset in $h\mathbb{Z}^3$ with all their points at a distance $\ge\rho$ from the walls, also after translations by $\mathsf{l}_0\mathbb{Z}^3$; at $h = a_{K-t}$ the walls lie along faces of the blocks of the level t.

(d) Let functionals $\Lambda^{(j)}_n$ on tuples in $C_c(\mathbb{R}^3;V)$ with disjoint supports be linear in each entry on the tuples in $B(0,R')$ from an index $j_0(n,d,R,R')$ on, and bounded there by $c_n(d;R)\prod|H_a|_\infty$ at the tuples of type $(d;R)$ (supports in balls of radius R, at mutual distance $\ge d$). Then every

subsequence has a subsequence that converges at all tuples, to multilinear limits with the same bounds. ■

Proof sketch. (a) At the source 0 the localised source energies of [7, Lem. N.1] give $E/\ell^3 - C_g \le p_\ell \le E/\ell^3 + C_S$ and $|E_\ell/\ell^3 - E_{\ell'}/\ell'^3| \le 2C_{\mathrm{tt}}C'_{\mathrm{tt}}E_0$, and p_ℓ does not increase with ℓ by the chessboard estimate (Cor. YM3_I:86.6). Without Thms. YM3_I:AJ and YM3_I:AD, $\tilde{Z}(0) = e^{-E}Z$ with $e^{-C_g\ell^3} \le \tilde{Z}(0) \le e^{(C_{\mathrm{maj}}+2)\ell^3}$ gives G_g^F. (b) The chessboard estimate on T_ℓ (Prop. YM3_I:86.3) gives $|\langle\prod \Phi_{\mathbf{k}}\rangle_\ell| \le \prod \|\Phi_{\mathbf{k}}\|_{\mathrm{ch},\ell}$, and comparing Z_ℓ with Z_{ℓ_0} cell by cell gives $\|\Phi\|_{\mathrm{ch},\ell} \le e^{\mathsf{l}_0^3(p_{\ell_0}-p_\ell)}\|\Phi\|_{\mathrm{ch},\ell_0}$ (Cor. YM3_I:86.6).

For every periodic Gibbs state ν and every unbounded set of sides, the domination theorem (Prop. YM3_I:91.9) gives $|\nu(\prod \Phi_{\mathbf{k}})| \le \liminf_\ell \prod \|\Phi_{\mathbf{k}}\|_{\mathrm{ch},\ell}$, the lower limit taken along the sides of the set; along the sides with $p_{\ell_0} - p_\ell \le G$ the comparison bounds the right member by $\prod e^{G\mathsf{l}_0^3}\|\Phi_{\mathbf{k}}\|_{\mathrm{ch},\ell_0}$. (c) In each direction there are l_0/h offsets modulo l_0, and fewer than $n((r+2\rho)/h+1) < \mathsf{l}_0/h$ of them are bad. (d) The bound gives the equicontinuity $|\Lambda(X) - \Lambda(X')| \le c_n \sum_a |H_a - H'_a|_\infty \prod_{a'\ne a} \max\{|H_{a'}|_\infty, |H'_{a'}|_\infty\}$; a countable set of Lipschitz functions is dense with control of the supports; a diagonal sequence converges on it, hence everywhere. □

Lemma 2.23 (The rate on a fixed torus)**.** Let $g \le \gamma_2$, $\mathsf{b} \in I(g)$, $\ell \in \mathfrak{L}(g)$, $q \ge 2$, $d > 0$, $\theta_q = 1/(1260q(8q+3))$, $\mathsf{p} = \max\{2\mathsf{p}_0+2, \mathsf{p}_0, 1\}$, where $\mathsf{C}_{\mathrm{bf}}\ell^3 K^{\mathsf{p}_0} L^{-K/10}$, $K \ge K_{\mathrm{bf}}$, is the comparison of the laws of the block fields at two cutoffs of Cor. 2.8. There are C, of the constants, q and C_{bf}, and K_1, besides of g, b, ℓ, d, such that for $K \ge K_1$ and $G_K = \langle \mathcal{E}_1; \dots; \mathcal{E}_q\rangle_{K,\ell,L^K\mathsf{b}}$, the $\mathcal{E}_a$ any of $\mathcal{O}_K(\varphi) = O_K^T(\varphi\mathbf{1})$, $O_K^T(\varphi E_{\lambda\lambda})$ (the plaquettes of the plane normal to λ alone) and $O_K^T(\varphi E_{\lambda\lambda'})$, $\lambda \ne \lambda'$, read at the centres of the plaquettes and the bonds, with real φ_a, Lipschitz in $|\cdot|_\infty$, whose supports are pairwise at sup distance $\ge d$:

$$|G_{K+1} - G_K| \le C\ell^{3(q+1)}K^{\mathsf{p}}L^{-\theta_q K}\prod_a(\sup|\varphi_a| + \operatorname{Lip}\varphi_a).$$

So $\lim_K G_K$ exists, with the tail of the series as the error; any number of the entries may be off-diagonal, and one pair (C, K_1) serves a compact interval of labels. The same holds with $\prod_a \sup|\psi_a|$ for bounded Borel functions ψ_a that vanish off closed sets at pairwise sup distances $\ge d$, entered through their cell means: at a site x of $\Lambda_{K,\ell}$ the mean $\psi_a^{(K)}(x) = a_K^{-3}\int_{x+a_K[-\frac12,\frac12)^3}\psi_a$ is the weight of the plaquette with the corner x and the edges $a_K e_\kappa$, $a_K e_{\kappa'}$, for the component $\lambda\lambda$, $\{\kappa,\kappa',\lambda\} = \{1,2,3\}$, and of the bond $\langle x, x + a_K e_\mu\rangle$, for the component $\lambda\lambda'$, $\lambda \ne \lambda'$, μ the third direction, for all six components.

As L is odd, the cells of $\Lambda_{K+1,\ell}$ at the sites of the block $B(y)$ of a site y of $\Lambda_{K,\ell}$ make up the cell of y, so that $\psi^{(K)}(y) = L^{-3}\sum_{x\in B(y)}\psi^{(K+1)}(x)$: the weights of two cutoffs are tied exactly. So the limit exists for test functions of no regularity and agrees with that for Lipschitz functions. For one function evaluated at the points of the lattices of two cutoffs no bound of this kind by the sup norms holds. For one off-diagonal entry among action densities the rate is free of their number: if $n \ge 1$, $\lambda \ne \lambda'$, and $\varphi, f_1, \dots, f_n$ are real Lipschitz functions on T_ℓ, none of them 0, whose $n+1$ supports have no common point, then $\kappa_K = \langle O_K^T(\varphi E_{\lambda\lambda'}); \mathcal{O}_K(f_1); \dots; \mathcal{O}_K(f_n)\rangle_{K,\ell,L^K\mathsf{b}}$ obeys

$$|\kappa_{K+1} - \kappa_K| \le CK^pL^{-K/120}(\sup|\varphi| + \operatorname{Lip}\varphi)\prod_{i=1}^n(\sup|f_i| + \operatorname{Lip} f_i) \qquad (K \ge K_2),$$

with C, p, K_2 of the constants, n, ℓ, the end points of a compact interval of labels that holds b and the spread d_∞ of the $n+1$ supports (the definition preceding Lem. YM3_I:114.6). ■

Proof sketch. Let $s < \sigma' \le \lceil K/1260\rceil$ be depths, the second at most the depth up to which Cor. 2.8 compares the laws of the block fields, and $\sigma' \ge (8q+3)s$. At each of the cutoffs K, $K+1$ the cumulant is, up to three errors, the cumulant of the first members of the sources, that is of the functions $\mathcal{E}_a$ themselves and not of their families at the heights, evaluated at the background of the

block field of the depth σ': the progression with the sources switched on at the level $K-\sigma'$ forgets what lies below, with the error $\epsilon(\sigma'-s)$, $\epsilon(\nu)=C(\nu+1)^{2\mathfrak{p}_0+2}L^{-\nu/2}$; the large fields cost $L^{-(b'+1)s}$, for any $b'\ge 4q$; and the distance of the supports costs $e^{3-L^s d/M}$.

Each of the three multiplies $(4\ell^3L^{4s})^q\prod_a(\sup|\varphi_a|+\operatorname{Lip}\varphi_a)$, the bound of the entries at the depth s; as $L^{4qs}L^{-(\sigma'-s)/2}\le L^{-s}$ for $\sigma'-s\ge(8q+2)s$ and $L^{4qs}L^{-(b'+1)s}\le L^{-s}$ for $b'\ge 4q$, together they are of the order $K^{2\mathfrak{p}_0+2}L^{-s}$. Over one block field V of the depth σ', with $t=K-\sigma'$, the background U^{A} on $\Lambda_{K,\ell}$ and the background U^{B} on $\Lambda_{K+1,\ell}$ are compared in three links. (1) $U'=\mathcal{M}(U^{\mathrm{B}})$, U^{B} averaged once, is a configuration of the coarse lattice over the same block field, a competitor whose excess energy over U^{A} is at most $C\varepsilon_{\mathrm{r}}^2\eta_t N_t$, ε_{r} the radius of the small fields of the depth σ' and N_t the number of the cubes of the level t: its energy is at least that of U^{A}, by the minimality of U^{A}, and at most that of U^{B} up to a number of the same order, by the inequality of Jensen of (2); and the energy of U^{B} exceeds that of U^{A} by at most a number of the same order, since the prolongation of U^{A} to the fine lattice is a competitor of U^{B} whose energy exceeds that of U^{A} by at most such a number. (2) The fine background against its average: for a bilinear form of the loops at one site or at one bond, the defect of Jensen's inequality $\sum_p \mathsf{W}(p',p)|g_p|^2\ge L^{-2}|G'_{p'}|^2$, with the averaging kernel W, g_p the fine and $G'_{p'}$ the averaged curvature, is the variance of the fine curvature on the block of p', and its sum over the blocks is the difference of the two energies; so the curvature is nearly constant in mean square under the shifts of a plaquette in its own plane within a block, with the error $L^{-t/2}$, and the weights enter only through the mismatch of their block means, which is 0 for cell means and at most $\frac{3}{2}a_K\operatorname{Lip}\varphi$ for one function read at the centres on the two lattices. (3) The competitor against U^{A}: along the t consecutive levels each pair of backgrounds differs by one gauge transformation and one small 1-form A_k, with $\sum_k L^{t-k}\|A_k\|^2$ at most a number times the excess, which costs $(t\eta_t)^{1/2}$.

So the corner fields $a_K^{-1}\sum_x\varphi(x+a_K\xi)Q_x(l,l')$, $Q_x(l,l')$ the corner forms of Lem. 2.14(b), of which $O_K^T(\varphi E_{\lambda\lambda'})$ is a sum of four with the coefficients $\pm\frac14$, agree at the two backgrounds up to $C\ell^3L^{4\sigma'}K^{1/2}L^{-(K-\sigma')/2}(\sup|\varphi|+L^{-\sigma'}\operatorname{Lip}\varphi)$ for Lipschitz weights, and up to the same number with $\sup|\psi|$ for weights tied by the cell mean; for one function evaluated at the points of two lattices the mismatch of the block means is not small in the sup norm, and there the Lipschitz number pays. The block-field laws at K and $K+1$ differ by $\mathsf{C}_{\mathrm{bf}}\ell^3K^{\mathfrak{p}_0}L^{-K/10}$ (Cor. 2.8(a), whose numbers read the label only through an interval that holds it).

With $\sigma'=\lfloor K/(1260q)\rfloor$ and $s=\lfloor\sigma'/(8q+3)\rfloor$ the four errors give the rate θ_q. For one off-diagonal entry among action densities whose supports have no common point another road keeps the action densities as sources in the exponent of the progression, as in the comparison of two cutoffs with sources (Thm. YM3_I:AR), and passes the one off-diagonal entry alone to the block field as a bounded function: it is put into the chain as a member whose tower is split at the depth $\lceil K/36\rceil$; what it creates below the level $K-\lceil K/36\rceil$ is born at a small coupling, and above it the member is carried along the run and forgotten.

No factor $L^{4q\sigma'}$ enters, the rate is free of n, and it is $L^{-K/120}$. □

Proposition 2.24 (What the properties of the limits do not decide)**.** Let σ be a limit through periodic Gibbs states, $g\le\gamma_1$, and for real ζ let Λ_ζ be the linear map of V that is 1 on the diagonal and ζ on the off-diagonal matrices, $\sigma^\zeta(X)=\sigma(\Lambda_\zeta X)$.

(a) σ^ζ is a real symmetric functional, bounded as in (D1) with the factor $\max\{1,|\zeta|\}^n$, invariant under translations and W_3, reflection positive at the nine families; $[F]\mapsto[\Lambda_\zeta F]$ is an isometry of its Hilbert space into that of σ that intertwines the semigroups, the translations and the vacua; so σ^ζ has the cone of [7, Thm. BZ(F4)], the insertion bound of [7, Thm. BZ(F2)] with the factor $\max\{1,|\zeta|\}$ for each entry, and the plateau of [7, Thm. BZ(F6)].

(b) For $\zeta\ne1$, σ^ζ is not invariant under the rotations about e_3; it is reflection positive at no plane outside the nine families; and it does not obey the free law of [7, Thm. BZ(F1)] at any label. ■

Proof sketch. (a) Λ_ζ commutes with $\rho(w)$ for $w \in W_3$, since a signed permutation carries diagonal matrices to diagonal and off-diagonal to off-diagonal ones, and the mirrors of the nine families act on the labels by elements of W_3; the rest follows from the isometry. (b) Let $D = E_{11} - E_{22}$ and $O = E_{12}$, the symmetric matrix with the entries 1 at $(1,2)$ and $(2,1)$, so that $\rho(R_t)D = \cos 2t\, D + \sin 2t\, O$, $\rho(R_t)O = -\sin 2t\, D + \cos 2t\, O$ for the rotations R_t about e_3, and let $C = E_{33}$. If σ and σ^ζ were both invariant under the R_t, the mixed kernel would obey $(\zeta - 1)\sin 2t\ \mathcal{G}(R_t u)(O, C) = 0$, so $\mathcal{G}(\cdot)(O, C) \equiv 0$; but by [7, Thm. BZ(F1)] at short distance $\mathcal{G}(u)(O, C)$ is close to b^{-2} times the mixed free kernel $3K_3(u)_{13}K_3(u)_{23}$, which does not vanish identically.

By [7, Prop. D.16](b) σ^ζ is then positive at no further plane. If σ^ζ obeyed the free law at a label b', the law at two diagonal entries, where $\sigma^\zeta = \sigma$, would give $\mathsf{b}' = \mathsf{b}$; and the mixed law depends on ζ with its sign, $\sigma^\zeta(H_1, H_2) = \zeta\sigma(H_1, H_2)$ for H_1 off-diagonal and H_2 diagonal, while the mixed free covariance does not vanish identically; so $\zeta = 1$. In the proof of [7, Prop. D.15] every input but conservation holds for σ^ζ: it is conservation that excludes it at all distances, and the mixed free law at short distance. □

Lemma 2.25 (Extension over coinciding points)**.** Let $\mathsf{M} = \mathbb{R}^3$ or T_ℓ, $n \geq 2$, $A, h_0 > 0$, $q \geq 0$, and let u be a functional of n entries, n-linear on $\prod_a C_c^\infty(O_a; V_\mathbb{C})$ for open O_a with pairwise disjoint closures, with the bound

$$|u(f_1, \dots, f_n)| \leq A\,(1 + \log \tfrac{h_0}{h})^q \prod_a \sup |f_a|$$

for $0 < h \leq h_0$ and all f_a supported in closed cubes of side $\frac{3}{2}h$, wherever they lie, at pairwise sup distances $\geq \frac{5}{2}h$; on T_ℓ the number h_0 lies in $\ell 2^{-\mathbb{N}}$ and $h_0 \leq \ell/(64n)$.

(a) There is a distribution $\tilde{u}$ on M^n of order $\leq 3n - 2$ that agrees with u on separated tensor products and has

$$|\tilde{u}(F)| \leq C_{\mathrm{ex}} A (1 + \log \tfrac{h_0}{\lambda})^{q+1} \max_{|\gamma| \leq 3n-2} \lambda^{|\gamma|} \sup |\partial^\gamma F|$$

for every F supported in a product of closed sup-balls of radius $\lambda \leq h_0$ about any points, C_{ex} depending only on n and q, with the label norm and auxiliary smooth cutoffs fixed; if u is invariant under permutations of the entries, translations and a closed subgroup of $O(3)$ that maps M to itself, $\tilde{u}$ can be chosen invariant.

(b) Two extensions with this bound differ by $\sum_{|\pi| \leq n-1} \sum_{|\alpha| \leq 3|\pi|} \partial_\xi^\alpha \delta_\pi \otimes T_{\pi,\alpha}$, π the partitions of the entries, δ_π the diagonal of π, ξ the differences of the points in the blocks, and $T_{\pi,\alpha}$ distributions on $\mathsf{M}^{|\pi|}$ with the bound of (a) at the power $\lambda^{|\alpha|}$; and every such sum can be added. A distribution with the bound of (a) at a power $\lambda^{\kappa'}$, $\kappa' > 3k$, on M^k is 0.

(c) At $n = 2$ the translation invariant terms of (b) are $\sum_{|\alpha| \leq 3} \int \mathsf{c}_\alpha(\partial^\alpha f_1(x), f_2(x))\, \mathrm{d}x$ with bilinear forms c_α; the real ones that are symmetric in the two entries and covariant under $O(3)$ on $\mathbb{R}^3$ form a space of dimension six, spanned in Fourier space by $\operatorname{tr} H \operatorname{tr} H'$, $\operatorname{tr} HH'$, $p^2 \operatorname{tr} H \operatorname{tr} H'$, $p^2 \operatorname{tr} HH'$, $(p \cdot Hp)\operatorname{tr} H' + (p \cdot H'p)\operatorname{tr} H$, $(Hp) \cdot (H'p)$; covariant under the cubic group on T_ℓ, they form a space of dimension twelve. ■

Proof sketch. (a) Apply Lem. 3.6(a)–(d) at $\kappa = r = 0$, with the stated torus convention. Its construction reads as follows. M^n minus the big diagonal is the union of the admissible products of dyadic cubes of sides $h_j = 2^{-j}h_0$, pairwise at sup distances $\geq 3h_j$, that lie in no larger admissible product (on T_ℓ the dyadic cubes tile the torus, as $h_0 \in \ell 2^{-\mathbb{N}}$, and each cluster lies in a sup ball of radius $13nh_0 < \frac{\ell}{4}$, which lifts to $\mathbb{R}^3$ and defines the differences within that cluster); there is a partition of unity $\varphi_{\mathbf{Q}}$ with $|\partial^\gamma \varphi_{\mathbf{Q}}| \leq c_\gamma h(\mathbf{Q})^{-|\gamma|}$, and on each product u extends to the smooth functions with the bound, by an absolutely convergent expansion in products.

The cubes of a product at mutual distances $\leq 12h$ form its clusters, a partition $\pi(\mathbf{Q})$ with k blocks, where $k \leq n - 1$ below the coarsest side h_0; put $\tilde{u}(F) = \sum_{\mathbf{Q}} u_{\mathbf{Q}}(\varphi_{\mathbf{Q}}(F - P_{\mathbf{Q}}F))$, $P_{\mathbf{Q}}F$ the Taylor polynomial of order $3k$ averaged over the piece of size h of the stratum $\Delta_{\pi(\mathbf{Q})}$ next

to $\mathbf{Q}$ when $k \le n-1$, and $P_{\mathbf{Q}} = 0$ otherwise. On the products with $k \le n-1$ the remainder is of order $(h/\lambda)^{3k+1}$, the number of products of side h near a given point is of order $(\lambda/h)^{3k}$, and the sum over the scales converges with one more logarithm; the order of the extension is $3(n-1)+1 = 3n-2$. Choose the grid partition and the Taylor averages covariantly under $h_0\mathbb{Z}^3$. Average the resulting extension over the parameters $t \in [0,h_0)^3$ with normalized Lebesgue measure; periodicity in t makes this mean invariant under every translation. Then average over permutations and the compact orthogonal subgroup, as in Lem. 3.6(d). Its support-radius comparison and the fixed-scale localization in part (c) preserve the stated bound, with a constant depending on n, q. (b) A distribution supported on the big diagonal with the bound of (a) is, stratum by stratum, a sum of transversal derivatives of order $\le 3|\pi|$ of δ_π; the vanishing for $\kappa' > 3k$ is the scaling of the bound. (c) A translation invariant contact term at $n = 2$ is a polynomial of order ≤ 3 in the relative momentum p; $-\mathbf{1} \in O(3)$ acts trivially on V and changes the sign of p, so only the orders 0 and 2 remain, and the invariant theory of $O(3)$ on symmetric matrices gives two forms at the order 0 and four at the order 2; under the cubic group $3+9$. □

Proof of Thms. C and D. Step 1: (A1). Cut T_ℓ into at most $(2\ell/d'')^3$ boxes of radius $\le 0.86\,d''$, and each form into its pieces on these boxes. A choice of nonzero pieces, one of each form, has the geometry of Lem. 2.17 at d'', with the majorants of Lem. 2.15: the pieces lie within $\frac{7}{8}d''$ of points, and the plaquettes of pieces of different forms are at least $d - 2a_K \ge \frac{5}{4}d''$ apart, as $d \ge \frac{4}{3}d''$; $s(d'') \ge s_{\mathrm{th}}(n)$ since $L^{s(d'')} \ge 1/d''$, and $\ell \le \ell_{\max}$ on a small side. Multilinearity gives

$$|\langle F_1;\dots;F_n\rangle_\ell| \le (2\ell_{\max}/d'')^{3n} k'_n(g,s(d''))\, b^{-n} \prod_a |F_a|_\star .$$

Step 2: (A2), (A3). (A2) is Lem. 2.19(a) with Lem. 2.15(a): a real form is a real quadratic source under $(|F|_\star, S[F])$ with the kernel t_F, and its plaquettes lie within $d - a_K$ of x_i when its cells lie within $d - 2a_K$; for complex tensor weights the real and imaginary parts give four real pairs, and $O_K^T(H)$ is a source under $(60|H|_\infty, S_H)$. For the continuum form, Lem. 2.16 gives $C^{\mathrm{bg}}(p,q) = \epsilon_p\epsilon_q a_K^3 K_3(x(p)-x(q))_{\lambda(p)\lambda(q)} + O((a_K/d)^3 E_1)$; the sums over plaquettes and bonds are Riemann sums of the integral, with an error $(6/\pi)(a_K/d)$ for each factor at Lipschitz weights; v^2 against b^{-2} costs $8\bar{B}g^2L^{2-s}$; and

$$\tfrac{3}{2}\mathrm{tr}[E_{\lambda_1\lambda_1'}KE_{\lambda_2\lambda_2'}K] = 3[K_{\lambda_1\lambda_2}K_{\lambda_1'\lambda_2'} + K_{\lambda_1\lambda_2'}K_{\lambda_1'\lambda_2}], \quad (\lambda_1 \ne \lambda_1',\ \lambda_2 \ne \lambda_2'),$$
$$\tfrac{3}{2}\mathrm{tr}[E_{\lambda\lambda'}KE_{\gamma\gamma}K] = 3K_{\lambda\gamma}K_{\lambda'\gamma} \quad (\lambda \ne \lambda')$$

identify the Gaussian value with $b^{-2}\mathcal{W}$. So, with $s = s(d)$,

$$e^{\mathrm{od}} = \tfrac{9}{8}[59^2\varepsilon_g^\natural(s) + 600E_2(\tfrac{1}{\pi}+E_2) + 636/C_W(s)] + 8\bar{B}g^2L^{2-s}, \qquad E_2 = E_1 + \tfrac{6}{\pi}\tfrac{a_K}{d},$$

which is at most an error function of s, since $a_K/d \le C_W(s)^{-1}$, $d < LC_W(s)L^{-s}$ and $M \le L^{-2}C_W(s) \le \hat{d} < LC_W(s)$ (Lem. 2.20(e)), so that also the term $C_\star^{\mathrm{fl}}\hat{d}^3 e^{-\delta_{\mathrm{tp}}L\hat{d}/\sqrt{3}}$ of E_1 is at most an error function of s. For (A3) apply (A2) to $(\theta Y, Y)$, whose points are $3d$ to $4d$ apart. By Lem. 2.16 and its count, at most $10(d/a_K)^3$ centres of bonds of one direction in a ball of radius d,

$$|G(\theta Y,Y)| \le 6C_{\mathsf{k}}^2(a_K/d)^6\Big(\sum|\mathsf{t}|\Big)^2 v^2 \le 600C_{\mathsf{k}}^2 v^2 \sup|\mathsf{w}|^2,$$

and with $v^2 \le \frac{9}{8}b^{-2}$, $|Y|_\star = 59\sup|\mathsf{w}|$ the bound follows. Since $\langle Y\rangle_\ell = 0$ (Lem. 2.20(b)), $\langle \theta Y; Y\rangle_\ell = \langle \theta Y \cdot Y\rangle_\ell \ge 0$ by reflection positivity at a plane through midpoints of bonds (Cor. YM3_I:85.3); for other forms the centred form $F - \langle F\rangle_\ell$ gives the sign. For forms at bonds and sites the count gives $\sum|\mathsf{t}| \le N(d/a_K)^3\|v\|$, $N = 30$, resp. 10, and for $O_K^T(H)$ the same count gives the number $12150 = 6\cdot 45^2$, with $60^2\bar{\varepsilon}_g^\natural$ from the size $60|H|_\infty$.

Step 3: (B). By Lem. 2.20(a) the copies $\theta'^\eta Y$ are bond forms of the same size; by Lem. 2.20(d) partners across the plane of θ'_{μ_0} are $3d$ to $4d$ apart and other pairs $\ge 11M$; their means vanish

by Lem. 2.20(b). Lem. 2.21 with $n = 4$, for bond forms, gives $M_8 \le (g_0^4 + \bar\omega_{g,8})\hat m_J$ with $\hat m_J = 59^8 \upsilon^8 \sup|\mathsf{w}|^8$ and $59^2 g_0 = 106 C_{\mathsf{k}}'^2$, so $M_8 \le C_8 b^{-8} \sup|\mathsf{w}|^8$; and $|M_8 - \prod_\tau \langle P_\tau\rangle| \le \omega_{g,8}^\sharp \hat m_J$ with $\langle P_\tau\rangle = \langle \theta'_{\mu_0} Y; Y\rangle_{\ell'}$ for all four pairs. The signs come from reflection positivity: M_8 is a mean of mirror copies, and the chessboard estimate gives $\langle \theta'_{\mu_0} Y \cdot Y\rangle_{\ell'} \le M_8^{1/8} M_8^{1/8}$ times the seminorms 1 of the six other cells.

Step 4: (C1). With $\delta = \min\{d, \frac12, \mathsf{l}/(4n)\}$ cut each form into its pieces on the cubes of side δ (at most $(2R/\delta + 2)^{3n}$ tuples of pieces); only the boundedness of the weights is used. For each tuple of pieces Prop. 2.22(c) gives a board of side l with all pieces at distance $\frac{\delta}{2}$ from the walls; on a torus, pieces are first moved by periods into one period box. Centre by $c_a = \tilde\mu_{\ell'}(F_a)$, write the cumulant through moments, and dominate each moment by Prop. 2.22(b). The eighth power of the seminorm of a product of centred pieces in one cell is a centred moment of at most $8m$ forms on $T_{\ell'}$ at mutual distance $\ge \delta$ (Lem. 2.20(c)), bounded by (A1):

$$\Big\|\prod_a (F_a - c_a)\Big\|_{\mathrm{ch},\ell'} \le \mathrm{Bl}_{8m}^{1/8} \prod_a b^{-1} |F_a|_\star, \qquad \mathrm{Bl}_N = \sum_{\pi \in \mathrm{Part}_2(N)} \prod_{I\in\pi} k^\circ(\#I, \delta),$$

$\mathrm{Part}_2(N)$ the partitions into blocks of at least two. So

$$C_n^T(d;R) = \max\{(2R/\delta+2)^{3n} \mathsf{f}_n e^{nG_g \mathsf{l}^3} \mathsf{Bm}_n^n,\ C_n^{\mathrm{cr}}(d;R)\},$$

$\mathsf{f}_n = \sum_\pi (\#\pi - 1)!$, $\mathsf{Bm}_n = \max_{m\le n} \max\{1, \mathrm{Bl}_{8m}^{1/8}\}$; below the cutoff $K_n = \max_{r\le 8n} K_0'(r,\delta)$ bounds by suprema give $C_n^{\mathrm{cr}}(d;R) = \mathsf{f}_n[6\bar x_0 L^{K_n}(2RL^{K_n}+1)^3]^n$. The one-point number $C_1^T(R)$ is the case $n = 1$. Complex weights: real and imaginary parts.

Step 5: (C2), (C3). For (C2) let $D = |x_1 - x_2|_\infty$, on a torus minimised over the lifts, be attained in the coordinate μ_0. If $D > 3$, the supports lie in balls of radius 1 more than 1 apart, and (C1) at $(\frac12; 1)$ bounds the covariance. Let $D \le 3$. One board of side l serves both insertions: it has a wall normal to e_{μ_0} at the midpoint of x_{1,μ_0} and x_{2,μ_0}, up to $\frac{a_K}{2}$, and its offset o at $\frac12(x_{1,\nu}+x_{2,\nu})$, up to $\frac{a_K}{2}$, in the two coordinates $\nu \ne \mu_0$. The distance t_i of x_i to the wall lies in $[\frac{D}{2} - \frac{a_K}{2}, \frac{D}{2} + \frac{a_K}{2}] \subset [\frac34 d, 2)$, and $|x_{i,\nu} - o_\nu| < 2$, so that the corners of the plaquettes of either insertion lie within 3 of the middle of its cell in the coordinates $\nu \ne \mu_0$.

Of the eight copies of one insertion on $T_{\ell'}$, the partners across the wall are $2t_i \in [3d', 4d']$ apart, $d' = \max\{d, \frac{t_i}{2}\}$, and copies that differ in a reflection normal to an e_ν, $\nu \ne \mu_0$, are at least $\mathsf{l} - 6 \ge 11M$ apart ($\mathsf{l} \ge LM$, $M \ge 3$); the carriers lie within $d' - 2a_K$ of their points, and $s(d') \le s(d) \le K$. So Lem. 2.21 with $n = 4$ at the distance d', for bond forms, whose means vanish, bounds the eighth power of the seminorm of a bond insertion by $(g_0^4 + \bar\omega_{g,8})\hat m_J = C_8 b^{-8} \sup|\mathsf{w}|^8$, as in Step 3, on this board, which the two insertions fix together, and its centred one-direction case bounds the seminorm of each of the forms of a tensor insertion; domination bounds the covariance by the product of the two seminorms; if $s(d') < s_8(g)$, (A1) is used at the distance $d_\flat = L^{-s_8(g)}$.

This gives

$$C_\infty^\diamond = \max\{2(c_\infty^\diamond)^2, C_2^T(\tfrac12;1)\}, \qquad c_\infty^\diamond = e^{G_g \mathsf{l}^3} \max\{(\tfrac98)^{1/2}((422 C_{\mathsf{k}}'^2)^4 + \bar\omega_{g,8}^{\mathrm{o}})^{1/8},\ \mathrm{Bl}_8(d_\flat)^{1/8},\ 12\bar x_0 L^{K_\flat}(2L^{K_\flat}+1)^3\},$$

$\bar\omega_{g,8}^{\mathrm{o}}$ the supremum of the error of Lem. 2.21 for majorants of one direction, $\mathrm{Bl}_8(d_\flat)$ the number of Step 4 at the distance $d_\flat$ and $K_\flat$ its cutoff. For bond forms the mean vanishes in every law, and C_∞^{per} is formed in the same way with $C_8^{1/4}$; complex weights of all three directions cost the factor 36. For (C3) place the pair in the middle of a cell (Lem. 2.20(d)) and use Lem. 2.21 with $n = 8$ for the sixteen copies (in its proof the blocks of one pair rest on the clause of Lem. 2.19(c), almost everywhere at the upper level): for $X_i = F_i - c_i$, $c_i = \tilde\mu_{\ell'}(F_i)$, and $m = \tilde\mu_{\ell'}(X_1X_2) = \tilde\mu_{\ell'}(F_1; F_2)$,

$$\|X_1X_2 - m\|_{\mathrm{ch},\ell'}^8 = \sum_J (-m)^{8-\#J} \langle \tilde P_J\rangle_{\ell'} \le \omega_{g,16}'^\sharp(s)(|m| + \hat m)^8,$$

since $\langle \tilde{P}_J \rangle$ replaced by $m^{\#J}$ gives $(m-m)^8 = 0$ and $|\langle \tilde{P}_J \rangle - m^{\#J}| \le \omega'^{\sharp}_{g,16} \hat{m}^{\#J}$. For one point, $\|X_i\|^8_{\mathrm{ch},\ell'}$ is a centred moment of eight of the sixteen forms, one from each pair, a set that holds no pair, so it is at most $o_{g,16}(s)(v|F_i|_\star)^8$. Domination now gives the three bounds of (C3), with

$$\mathrm{Df}_g(s) = e^{G_g|^3}[\omega'^{\sharp}_{g,16}(s)]^{1/8}(1+g_0+\omega'_{g,16}(s)) + e^{2G_g|^3} o_{g,16}(s)^{1/4},$$

since $|m| \le (g_0+\omega'_{g,16})\hat{m}$, and $|m - G(F_1', F_2')| \le \omega'_{g,16}\hat{m}$ by Lem. 2.21 at one pair. For $O_K^T(H_i)$ the same, with the majorants $(60|H_i|_\infty, S_{H_i})$.

Step 6: (D1)–(D3). (C1), Prop. 2.22(d), the compactness of $I^{\mathrm{lab}}(g)$ and the theorem of F. Riesz give (D1); the second clause is (C1) at $n=1$; for the third, the local functions $O_K^T(H)$ and the weak convergence give some $\ell_j \ge \max\{\Lambda_j, j\}$ in $\mathfrak{L}(g)$ at which the first j tuples of a dense countable family are matched within $\frac{1}{j}$. $O_K^T(f\mathbf{1}) = \mathcal{O}_K(f)$ gives (D2), along a subsequence of the defining sequence of S. Lem. 2.20(a) gives $rO_K^T(H) = O_K^T(rH)$; the invariance and the reflection positivity of the laws at the lattice mirrors pass to the limits, since lattice isometries approximate the real ones and the bounds are equicontinuous; periodic Gibbs states are positive at the nine families.

The semigroup of symmetric contractions $[F] \mapsto [\tau_{te_3}F]$ is e^{-tH}, Stone's theorem gives the momenta and the joint spectral measure, and the lattice energy–momentum inequality (Prop. YM3_I: 99.3) gives $H \ge |P_k|$. Centred at the means of $\tilde{\mu}_{\ell'}$, the moments in ν_j are $\sum_B \prod_{i\in B}(\nu_j - \tilde{\mu}_{\ell'})(O^T_{K_j}(H_i))\,\nu_j(A_j(X_{B^c}))$. The bond parts have mean 0 in both laws, and all plaquettes have one mean in each law (Prop. YM3_I:103.4), so that $(\nu_j - \tilde{\mu}_{\ell'})(O^T_{K_j}(H)) = \mathfrak{c}_j\, a^3_{K_j} \sum_p H_{\lambda(p)\lambda(p)}(x(p))$, a Riemann sum of $\mathfrak{c}_j \int \mathrm{tr}\, H$, with $\mathfrak{c}_j = a^{-4}_{K_j}(\nu_j(a_p) - \tilde{\mu}_{\ell'}(a_p))$; (C1) at $n=1$, applied to a plaquette form, bounds $|\mathfrak{c}_j|$ by $C(g)\mathsf{b}^{-1}$ for large j.

So the $\mathfrak{c}_j$ converge along a further subsequence, to $\mathfrak{c}$, and along it the recentred moments converge to the functional of (D3), whose one-point function is $\mathfrak{c}\int \mathrm{tr}\, H$; it is reflection positive, since the means of $\tilde{\mu}_{\ell'}$ are invariant under the lattice mirrors.

Step 7: (D4). For Lipschitz entries Lem. 2.23 makes the cumulants Cauchy in K, and its form for cell means gives the same for bounded Borel weights; continuous entries are approximated by Lipschitz ones uniformly in K, by the torus form of (C1); Prop. 2.22(d) applies in ℓ to periodised tuples, which are defined and linear once ℓ is large against the supports; a diagonal choice of the cutoffs gives the limits along growing tori.

Step 8: (D5) and [7, Thm. BZ(F1)]. On $T_{\ell'}$, (C3) and the continuum form of Step 2, with $\frac{3}{2}\mathrm{tr}[E_{\lambda\lambda}KE_{\gamma\gamma}K] = \frac{3}{2}K^2_{\lambda\gamma}$, give the law for all 36 pairs of components; then $j\to\infty$. For two off-diagonal components the error is $\bar{e}_g^{\mathrm{od}} + \frac{9}{8}59^2\mathrm{Df}_g$, for an off-diagonal and a diagonal one $\bar{e}_g^{\mathrm{od}} + \frac{9}{8}59\,\mathrm{Df}_g$, for two diagonal ones $\frac{9}{8}[\omega'_{g,16} + \mathrm{Df}_g + 150\bar{E}_1(\frac{1}{\pi}+\bar{E}_1) + 159/C_W(s)] + 2\bar{B}g^2L^{2-s}$, $\bar{E}_1$ the bound of E_1 by a function of s ((C3) being used with the majorants of the respective kinds); $\bar{e}_g$ is their maximum. At $H_i = f_i\mathbf{1}$, $\frac{3}{2}\mathrm{tr}\, K_3(u)^2 = \frac{9}{16\pi^2}|u|^{-6}$; the leading term is covariant under $O(3)$. For (F1), Lem. 2.21 with arbitrary majorants, centred, at $n=8$ bounds the means, $|\nu(O_K^T(H)) - \tilde{\mu}_{\ell'}(O_K^T(H))| \le 60e^{G_g|^3}\mathrm{mn}_g(s)v|H|_\infty$, where $\mathrm{mn}_g = [e_g^8 + \mathrm{Bl}(16)\mathrm{far}_{g,16}(1+g_0+\varepsilon_g)^8 + \mathrm{tl}_{g,16}]^{1/8} \le o^{1/8}_{g,16}$ comes from its moment bound at $N_1 = 8$, so sup norms suffice, and the law holds at a finite cutoff for continuous weights, the Riemann sums of continuous weights giving $r_K \to 0$ and the lattice term of E_1 the term $C(g)(a_K/d)^2$; letting $j\to\infty$,

$$e_g^\infty(d) = \tfrac{9}{8}\Big\{3600[\omega'_{g,16} + e^{G_g|^3}(\omega'^{\sharp}_{g,16})^{1/8}(1+g_0+\omega'_{g,16}) + e^{2G_g|^3}\mathrm{mn}_g^2](s) + 12150E_1^\infty(\tfrac{1}{\pi} + E_1^\infty)\Big\} + 162\bar{B}g^2L^{2-s}, \qquad s = s(d),$$

$E_1^\infty = E_1 - C_{\mathrm{lat}}(a_K/d)^2$. The errors are powers of $Y^{-1}_{s(d)}$, the worst the root $[\omega'^{\sharp}_{g,16}]^{1/8}$, and $Y_{s(d)} \sim \log\frac{1}{d}$. This root can be avoided: in Step 5 the eighth power of the seminorm of $X_1X_2 - m$ is an alternating sum of means of products of the sixteen copies, and at the upper level the parts of these means

without a block that links two pairs recombine to $\prod_\tau(\Psi_\tau - m)$, Ψ_τ the conditional means of the pairs, with $|\Psi_\tau - m| \le (\delta_g + \omega'_{g,16})\hat m$ almost everywhere at the upper level (the blocks of one pair by Lem. 2.19(c), the singletons as in the proof of Lem. 2.21, and $|G(F'_1, F'_2) - m| \le \omega'_{g,16}\hat m$ as in Step 5); so the seminorm itself is at most a multiple of $(\delta_g + \omega'_{g,16})\hat m$ plus the eighth roots of the terms far and tl, which decay faster than every power of Y_s^{-1}, and the means are bounded with mn_g, so that the second term of Df_g may be taken $e^{2G_g\mathsf{l}^3}\mathrm{mn}_g^2$; as ε_g, $\omega'_{g,16}$ and $1/C_W(s)$ are of order $Y_s^{-2\mathfrak{p}_0}$ and mn_g^2 is of smaller order, the error is at most $C(g)(\log\frac{1}{d})^{-2\mathfrak{p}_0}$; by [7, Lem. D.19] the same holds for $g \le \gamma_2$, as $(\log\frac{1}{L^z d})^{-2\mathfrak{p}_0} \le 2^{2\mathfrak{p}_0}(\log\frac{1}{d})^{-2\mathfrak{p}_0}$ for $d \le L^{-2z}$.

On the tori of $\mathfrak{L}(g)$ the torus limits of (D4) are limits along cutoffs at the labels $L^K\mathsf{b}$, and the same bound passes to them. The law fixes b, since $\mathcal{W} \ne 0$. Since $\mathcal{W}$ is invariant under $O(3)$ and $|rH|_\infty \le 3|H|_\infty$, the law gives $|\sigma(rH_1, rH_2) - \sigma(H_1, H_2)| \le (9+1)e_g^\infty \mathsf{b}^{-2}|H_1|_\infty|H_2|_\infty$. Non-degeneracy: $\mathcal{W}(\theta(\varphi E), \varphi E) \ge w_0 \operatorname{tr} E^2$ for this bump. □

The mirror bound and the power-six kernel estimate. (F9). In a periodic Gibbs state ν, $\langle F', F\rangle_{\nu,c} = \nu(\overline{\theta_c F'}F)$ is a positive semi-definite Hermitian form on the local functions on $\{x_{\mu_0} > c\}$, the translation by $a_K e_{\mu_0}$ is a contraction for it, and so the mirror may be moved away: $\|F\|_{\nu,c} \le \|F\|_{\nu,c'}$ for $c \le c'$, $c' - c \in a_K\mathbb{Z}$. It therefore suffices that x_0 lies at a distance between $\frac{3}{2}d$ and $2d$ from the plane. There the plane is a wall of a board of side l, θX is the copy of X in the neighbouring cell, and, as $\nu(\theta X \cdot X) = \nu(\theta(F-c)\cdot(F-c)) - (\nu(F)-c)^2$ for $F = O_K^T(H)$ and c its mean on $T_{\ell'}$, domination by $T_{\ell'}$ gives $\nu(\theta X \cdot X) \le e^{2G_g\mathsf{l}^3}\|F - c\|^2_{\mathrm{ch},\ell'}$, the eighth power of the seminorm being the mean of eight copies. For bond forms (B) bounds it, and for the whole tensor Lem. 2.21 with arbitrary majorants, centred, at $n = 4$; if $s(d) < s_8(g)$, (C1) at the distance $L^{-s_8(g)}$ bounds the covariance directly. So

$$C_\infty^{\mathrm{od}}(g) = \max\{e^{2G_g\mathsf{l}^3}C_8^{1/4},\, 59^2 C_2^T(L^{-s_8(g)};1)\}, \qquad C_\infty(g) = \max\{e^{2G_g\mathsf{l}^3}(C'_8)^{1/4},\, 60^2 C_2^T(L^{-s_8(g)};1)\}, \tag{2.xxv}$$

$C'_8 = (\frac{9}{8})^4 60^8[g_0^4 + \bar\omega'_{g,8}]$, $g_0 = \frac{3}{2}81^2 C'^2_{\mathsf{k}}$; complex weights of all directions cost 36. For the kernel bound at off-diagonal entries, [7, Prop. D.15](a) bounds the two-point kernel at u by $\Gamma(|u|/2\sqrt{3})$, and the mirror bound for bond forms applied to bumps at the height t bounds $\Gamma(t)$ by a multiple of $C_\infty^{\mathrm{od}}(g)\mathsf{b}^{-2}\max\{1, t^{-1}\}^6$; the holomorphy of the two-point function in the relative position and the mean value inequality on a polydisc of radius proportional to $|x-y|$ turn this bound for bumps at the scale $|x-y|$ into the pointwise bound $A_6 C_\infty^{\mathrm{od}}(g)\mathsf{b}^{-2}\min\{|x-y|_\infty, 3\}^{-6}$ of the kernel. For the whole tensor the same steps, with the mirror bound for $X = A[O_K^T(H)]$ and its number $C_\infty(g)$, give the bound with a pure number A in the place of A_6. By (D5) the kernel is of order $|x-y|^{-6}$ at short distance, so the power is the true one. □

Contact extensions on tori and the mirror bound. (F8). Every side $\ell = 2L^m \in \mathfrak{L}(g)$ is admissible over the run of every datum at g: $LM\,R(x_K) \le L^m$ (equation YM3_I:(1533), as in the paragraph *Data* of the Setting).

On T_ℓ the clause (i) of (E3), on Lem. 2.17 (the bound of that Lemma on an admissible side with $\max\{1, c(\ell)\}^n k'_n$ for k'_n, without the factor of the covering), at the geometry of Lem. 2.17 at $d = 1.52h$ (a cube of side $\frac{3}{2}h$ lies within $1.3h \le \frac{7}{8}d$ of its centre, and cubes at sup distance $\ge \frac{5}{2}h$ are more than $\frac{5}{4}d$ apart), bounds the cumulants of n entries at the cubes of side $\frac{3}{2}h$ at sup distances $\ge \frac{5}{2}h$, for $0 < h \le h_0$, where $h_0 \in \ell 2^{-\mathbb{N}}$ and $h_0 \le \min\{\ell/(64n), \frac{1}{2}L^{-s_{\mathrm{th}}(n)}\}$, so that $1.52h \le 1.52h_0 < L^{-s_{\mathrm{th}}(n)}$, by $60^n \max\{1, c(\ell)\}^n C_n(g)[2\log\frac{1}{1.52h} + C(g)]^{8n\mathfrak{p}_0} b^{-n}\prod|H_a|_\infty$ uniformly in the cutoff, which is at most $C(1+\log\frac{h_0}{h})^{8n\mathfrak{p}_0} b^{-n}\prod|H_a|_\infty$; through the cell means of Lem. 2.23 the bound passes to the limits of (D4), and Lem. 2.25 gives the extension, its bound, the contact terms and the counts.

At a cutoff the same construction, with the products of the cubes restricted to the scales h with $s(1.52h) \le K$, gives subtracted lattice cumulants with the bound for all $\lambda \ge 8h_K$, $h_K \le 2C_W(K)a_K$ the least scale, uniformly in K; by Lem. 2.23 they converge to $\tilde{S}_n$; their difference to the lattice cumulants depends on F only through the derivatives of the orders $\le 3k$ on the strata of k blocks and the values at distances $< 6h_K$. On $\mathbb{R}^3$: let two cubes of side $\frac{3}{2}h$ lie at sup distance $\ge \frac{5}{2}h$, $d = 1.31h$, $s(d) \le K$. Their centres are at sup distance $\ge 4h \ge 3d + a_K$, the cubes lie within $d - 2a_K$ of their centres, and a mirror plane $\{x_{\mu_0} = c\}$ between them, normal to a coordinate in which the centres are $\ge 4h$ apart, has each centre at a distance $\ge 2h - \frac{a_K}{2} \ge \frac{3}{2}d$.

On a torus T_ℓ, $\ell \in \mathfrak{L}(g)$, the mirror θ has the second plane $\{x_{\mu_0} = c + \frac{\ell}{2}\}$, at a distance $\ge \frac{\ell}{4} - \frac{a_K}{2} \ge 2h - \frac{a_K}{2} \ge \frac{3}{2}d$ from each centre, as the sup distance $\ge 4h$ of the centres on T_ℓ is at most $\frac{\ell}{2}$. For a bond form Y with a complex weight of all directions, carried within $d - 2a_K$ of a point at a distance $\ge \frac{3}{2}d$ from both planes, and τ the translation by $a_K e_{\mu_0}$, the invariance of the torus measure gives $\psi(k) := \langle \overline{\theta \tau^k Y}\, \tau^k Y\rangle_\ell = \langle \theta \bar{Y}\, \tau^{2k} Y\rangle_\ell$, and the Schwarz inequality for the positive form of θ on the half torus between the two planes (Cor. YM3_I:85.3) gives $\psi(k)^2 \le \psi(k-1)\psi(k+1)$ as long as the translates keep the distance $\ge \frac{3}{2}d$ from both planes.

A non-negative sequence with this property takes its largest value at an end of its range, and there the point lies at a distance between $\frac{3}{2}d$ and $2d$ from one of the two planes. That plane is a wall of a board of side l, $\frac{\ell}{2}$ being a multiple of l, the cells are distinct modulo the period, and domination by $T_{\ell'}$ with (B), or (C1) at the distance $L^{-s_8(g)}$ if $s(d) < s_8(g)$, bounds ψ there as in (F9); so the mirror bound of (F9) for bond forms with complex weights of all directions, $0 \le \langle \overline{\theta Y} Y\rangle_\ell \le 36 C_\infty^{\mathrm{od}}(g) b^{-2} |\mathsf{w}|_\infty^2$, holds on the tori of $\mathfrak{L}(g)$. With the Schwarz inequality for the positive form of that mirror, the means of bond forms being 0, this mirror bound gives $|\langle Y[\mathsf{w}_1]; Y[\mathsf{w}_2]\rangle_{K,\ell,\beta}| \le 36 C_\infty^{\mathrm{od}}(g) b^{-2} |\mathsf{w}_1|_\infty |\mathsf{w}_2|_\infty$ on every $\ell \in \mathfrak{L}(g)$: the bound of Lem. 2.25 with $q = 0$ and a number free of the side, which every limit point inherits. □

First-creation and tail refinements. Step 14: (E1)–(E3). (E1): $\tilde{\nu}_K(\{Z_t \ne \emptyset\}) = \sum_{t' \le t} \mathsf{w}_{t'}$, the weights of the first creations, and the run may be stopped at the level t without changing this number. A first creation forces the failure of a local test in some test cube of the step; averaging the test over the gauge variables of its cube with the density of the gauge fixing of the step gives Borel functions $0 \le F \le 1$ that are local, gauge invariant and covariant under the lattice isometries, and the average is undone exactly inside the sum over the histories of any admissible set of values with the same A_0; so w_t is at most a sum of means of such functions, two for each test cube, its plaquette test and its averaged witness test, and at $t = 1$ one for each density cube too: at most $2N_\tau + N_\delta$ for each cell of the level t, N_τ the number of the test cubes and N_δ that of the density cubes of a cell (N_δ entering at $t = 1$ alone), and at most $(2N_\tau + N_\delta) M^{-3} \ell^3 L^{3(K-t)}$ in all.

These means pass to a second admissible set of values with $\frac{93}{100}A_1$ for A_1 and the other constants that enter $c_\star$ kept, whose deposit constant $c_\star''$ has $(\frac{93}{100})^2 c_\star \le c_\star'' \le c_\star$: a product of test functions with disjoint supports is dominated by the probability, at the second set of values, that all the centres lie in the large-field region. Let s_t be the side of a cell of the level t. If the torus has at least $2L^2$ cells of the level t along a side, take the board of side $L^2 s_t$ whose offset is the centre of the test function F, its *witness*; the chessboard estimate with the mirrors in the walls of this board bounds $\langle F\rangle$ by the N-th root of the mean of the product of the copies of F in its N cells on the torus, and this product asks a large field in every cell of the board.

A creation reaches less than $49 s_t < L^2 s_t$ and so serves one cell alone: such a history has at least N creations. Thm. YM3_I:D, which bounds the density of these histories from above by $\exp[-\frac{9}{10} c_\star'' p_0(g_t)^2 N]$, at the fraction $\frac{9}{10}$ of the deposit, and that of the empty history from below, with the volume term $C y_t (L^2 M R(x_t))^3 N$, which $\frac{1}{40} c_\star p_0(g_t)^2 N$ pays, gives $\langle F\rangle \le \exp[-\frac{3}{4} c_\star p_0(g_t)^2]$, as $\frac{9}{10}(\frac{93}{100})^2 - \frac{1}{40} \ge \frac{3}{4}$; with fewer cells of the level t Thm. YM3_I:D is used without mirrors, at the first set of values; a threshold on y_K is removed by stopping the couplings at the last level above it.

The union bound over the cells gives $C_\flat$. The pair $(c_2^\natural, C_\natural)$: if $Z_t \neq \emptyset$, some cell of the level t has its centre in Z_t, and the rarity at a point of Prop. 4.52 with the union bound over the at most $M^{-3}\ell^3 L^{3(K-t)}$ cells gives the bound. (E2): Lem. 2.19(b) with $\mathsf{L} = CM^{-3}\ell^3$; half of the exponent $cp_0(g_t)^2$ pays ℓ^3 once $\frac{1}{2}cA_0^2(\frac{1}{2}s\log L)^P \geq 3\log\ell$, which is $s \geq \sigma_c(\log\ell)^{1/P}$; the other half pays $\log C + 3s\log L + 2Y_s$ and the envelopes from $s_\flat(g)$ on; the depth condition $2c\mathfrak{p}_0 A_0^2 y^{P-1} \geq 13$ is the rate condition of the deposit lemma at c. For (iv) there are three cases.

If $s(d) \geq \hat{s}$, (iv) is (iii). If $s(d) < \hat{s}$, then $d \geq L^{-s(d)} \geq L^2 d'$, $d' = L^{-\hat{s}-1}$; let $s' = s(d') \geq \hat{s}$. If besides $K \geq s'$, cover by balls of radius d', at which (ii) and (iii) hold; this costs the factor $(4/d'+2)^6 \leq 5^6 L^{6\hat{s}+6}$. If $K < s'$, the suprema of the forms give the number $100L^{8s'}\bar{x}_0^2$. So C_3° is the larger of $(4/d'+2)^6$ times the number of (ii) and (iii), in the units of (iv), and of $100L^{8s'}\bar{x}_0^2$; as $s' \leq \hat{s} + 1 + \lceil \log_L C_W(2\hat{s}+2)\rceil$, $L^{8s'} \leq L^{8\hat{s}+16} C_W (2\hat{s}+2)^8$, and with $L^{\hat{s}} \leq L^{s_\flat(g)+1}\exp[2(\frac{6\log\ell}{cA_0^2})^{1/P}]$ this gives the bound of (iv), whose factor $(1+\log\ell)^8\exp[16(\frac{6\log\ell}{cA_0^2})^{1/P}]$ comes from the third case, the covering costing less; for (v) cover by balls of radius $\min\{\frac{r}{8}, L^{-\hat{s}}\}$. (E3): let $\ell = 2L^m$ be admissible, $K \geq 3$, and $s_{\mathrm{th}}(n) \leq s(d) \leq K$.

The proof of Lem. 2.17 reads the side at the following places, and each of them holds on an admissible side. The arithmetic of the working level (Lem. YM3_I:126.1) asks nothing of the side. Lem. YM3_I:126.2 asks (Sd.1) alone, $MR(x_K) \leq L^m$, and gives the frame, the order $x_0 > x_1 > \cdots > x_K$ of the run (so $y_K \leq y$), (Sd.2) and (Sd.3) ($LM \leq L^m$, $\ell \geq N_{\mathsf{H}}$), the data of the construction at every level $\leq K$, and the settings of the chain at the levels $\mathsf{k} < j_1 = T = \mathsf{k}+1$ at every pair of the majorants; the setting of the chain for n sources ([7, Prop. B.21] at $p = n$) asks these data and the frame alone, and the threshold $s_{\mathrm{th}}(n)$ reads no side.

The sum over the level of the first creation (equation eq. 4.xxxi) holds at all the levels $\leq K$ (Thm. YM3_I:D and Cor. YM3_I:69.4, as in the proof of Lem. YM3_I:126.2), and the rarity on one torus, Prop. 4.54 at the level K, whose hypotheses hold by that canonical construction, bounds the weights before any smallness of the side is used:

$$\mathsf{w}_t \leq \tilde{\nu}_K(\{Z_t \neq \emptyset\}) \leq 2M^{-3}\ell^3 e^{\mathsf{B}(x_K)\ell^3} L^{3(K-t)} e^{-c' p_0(g_t)^2}, \qquad \mathsf{B}(x_K) \leq \mathsf{B}(\bar{x}_0).$$

From there on the side enters the proof only through this number $2M^{-3}\ell^3 e^{\mathsf{B}(\bar{x}_0)\ell^3}$, in the tail of the sum over the level of the first creation, where a small side has $2M^{-3}\ell^3 \leq M^{-3}c'A_0^2 Y_s^P$, and through the number of the boxes of the covering. As $Y_s \geq 1$ the quotient of the two numbers is at most $c(\ell)$, each block of a partition takes at most the factor $\max\{1, c(\ell)\}$, and there are at most n blocks: the bound of Lem. 2.17 holds on every admissible side with $\max\{1, c(\ell)\}^n k'_n(g, s(d))$ for $k'_n(g, s(d))$, which is (E3)(i) at this pair; the covering of Step 1, with 2ℓ for $2\ell_{\max}$, gives (E3)(ii). The same argument serves every pair (L, c'') of (E3): the weights enter only the tail term of each block, linearly in L and through a Gaussian sum in which $c'' \geq \frac{1}{2}c'$ may stand for $\frac{1}{2}c'$, and no smallness of w_t is used; so, as $Y_s \geq 1$, each block takes at most the factor $\max\{1, M^3\mathsf{L}/(c'A_0^2)\}$, which is $\max\{1, c(\ell)\}$ at the pair above, and (E3)(i) and (ii) hold with it. The weights of (E1), $\mathsf{w}_t \leq C_\flat M^{-3}\ell^3 L^{3(K-t)} e^{-\frac{1}{2}c' p_0(g_t)^2}$, give $C_\flat \ell^3/(c'A_0^2)$ in the place of $c(\ell)$. □

3. Real-time tensor fields

Let $V = \mathrm{Sym}_3(\mathbb{R})$. The gauge-invariant curvature tensor $T = (T_{ij})$ is normalized by

$$\mathcal{T}(H) = \frac{1}{2}\int \mathrm{tr}(H(x)T(x))\,\mathrm{d}x, \qquad H \in C_c^\infty(\mathbb{R}^3; V_{\mathbb{C}}).$$

This notation denotes a smeared bilinear, rather than a pointwise product of separately constructed gauge-dependent curvatures. Its trace gives the action density $\mathcal{O}(f) = \mathcal{T}(f\mathbf{1})$. An orthogonal matrix R acts on tensor labels by $\rho(R)E = RER^{\mathsf{T}}$. Reflections act on these labels as well as on the points.

Fix a clustering component σ of the Euclidean correlations. Reflection in $x_3 = 0$ gives a Hilbert space $\mathcal{H}$, the vacuum Ω, a positive generator H of Euclidean time translations, and spatial momenta

P_1, P_2. These objects and the analytic insertion bounds are supplied by [8, §§6,8]. The purpose here is to obtain operator-valued tempered distributions on Minkowski space, including their tensor covariance and locality. The Euclidean theorem on flat tests alone is not being quoted as a theorem about distributions on all ordinary Euclidean tests.

Sketch B (Reconstruction of the tensor fields). The component σ reconstructs a covariant local tensor field on $\mathbb{R}^{1+2}$, defined on a common invariant domain and tempered in its real-time test functions. Its translation spectrum lies in the closed forward cone. The vacuum is cyclic and unique among translation-invariant vectors. Multiplying a component with r Euclidean-time indices by i^r gives its Hermitian real-time component. Their separated Euclidean correlations agree with σ. ■□

3.1. **What the continuation must accomplish.** The input consists of separated Euclidean insertions. Their analytic continuation must do more than assign numbers to imaginary times: it must produce distributions with a common domain, identify adjoints in a real tensor frame, and make products commute at spacelike separation. The bounded insertion operators give a route to these tasks. They keep positive Euclidean time gaps while the spatial and temporal arguments are continued, and their power bounds control removal of those gaps.

The main manuscript has already constructed bounded slab insertions and analytic point kernels. We use those objects without repeating their lattice estimates or the proof of rotation covariance. A slab insertion between two positive time gaps is a bounded operator. Concentrating its test near a point gives the regularized point operators; the uniform L^1 insertion bound controls their dependence on the regularization. Their products, with positive gaps left between consecutive factors, give holomorphic vector-valued kernels.

The spectrum controls the permitted complex translations. For $z \in \mathbb{C}^3$, the tube

$$\operatorname{Re} z_3 > |(\operatorname{Im} z_1, \operatorname{Im} z_2)|$$

keeps the translation operator bounded. Consecutive differences of insertion points must lie in this tube; the absolute positions alone do not specify a product's analytic domain. Rotation covariance identifies the continuations obtained from different choices of the Euclidean time direction. Reflection supplies the adjoint relation. The tensor action in that relation is essential for selecting the Hermitian real-time components.

At the boundary of the tube, the point kernels need not have pointwise limits. Their polynomial bounds give distributional boundary values. Smearing those boundary values defines the fields and their products on the vacuum domain. The same bounds give temperateness and the energy-domain extensions in the detailed argument. Locality follows by comparing the continuations in the two orders on a region of spacelike separation and taking their boundary values there. The conclusion concerns the common invariant domain; passing to a net of bounded local operators is the separate task in §14.

There is also a Euclidean extension issue. The theory on tests flat at coincidences does not select contact terms supported on the diagonals. The additional extension discussion below describes the freedom left by the separated correlations. It does not assert that an arbitrary choice of such terms inherits reflection positivity on ordinary Euclidean tests.

For an open set O, let $\mathfrak{S}_V(O)$ denote the tuples of $C_c^1(O;V)$ entries with pairwise disjoint supports, including the empty tuple, and let $\mathbb{C}\mathfrak{S}_V(O)$ be their complex linear span. We also write $\mathfrak{S}_V^1(O)$ for this space of tuples when distinguishing it from tuples of continuous entries. Put $\mathbb{E}_+ = \{x_3 > 0\}$ and $V_0 = \mathbb{R}\mathbf{1}$. The stress tensor is $\Theta_{ij} = \frac{1}{2}\delta_{ij} \operatorname{tr} T - T_{ij}$; its labels are $\mathsf{m}_{ij} = \delta_{ij}\mathbf{1} - e_i e_j^{\mathsf{T}} - e_j e_i^{\mathsf{T}}$, so $\frac{1}{2}\operatorname{tr}(\mathsf{m}_{ij}T) = \Theta_{ij}$. The notation $\mathcal{T}_K(H)$ below is another name for $O_K^T(H)$. Isometries and linear maps of V act entry by entry. Put $\mathcal{D}_{n,c} = \mathbb{C}\mathfrak{S}_V(\{n \cdot x > c\})$, $\Theta_n = 1 - 2nn^{\mathsf{T}}$, and $(\theta_{n,c}H)(x) = \Theta_n H(\theta_{n,c}x)\Theta_n$, where $\theta_{n,c}x = x - 2(n \cdot x - c)n$. Positivity at (n, c) means $\sigma(\theta_{n,c}\bar{F} \cdot F) \geq 0$ for $F \in \mathcal{D}_{n,c}$.

Definition 3.1 (Data with labels; real time; the frame)**.**

(1) A *symmetric functional on labelled tuples* is a map σ on $\mathfrak{S}^1_V(\mathbb{R}^3)$ with $\sigma(\varnothing)=1$, invariant under the permutations of the entries and n-linear on $\prod_a C^1_c(O_a;V)$ for open sets O_a with pairwise disjoint closures; it is continued linearly to $\mathbb{C}\mathfrak{S}^1_V$. It is *reconstruction data with labels* with the numbers $C_V, p_V \geq 0$, $c_0 \in (0,\frac12]$, if: $(\mathrm{Rec1})_V$ $\sigma(\mathsf{r}X)=\sigma(X)$ for every isometry r; $(\mathrm{Rec2})_V$ $\sigma(\theta\bar F\cdot F)\geq 0$ on $\mathbb{C}\mathfrak{S}^1_V(\mathbb{E}_+)$, the space $\mathcal{H}(\sigma)$ is separable, and H, $\mathcal{U}$, $\mathcal{P}$ are formed as in [7, Def. 2.2] (3); $(\mathrm{Rec3})_V$ $\mathcal{P}(\{E<|q|\})=0$; $(\mathrm{Rec4})_V$ for $\eta\in(0,1]$, $F,F'\in\mathbb{C}\mathfrak{S}^1_V(\{x_3>\eta\})$ and $h\in C^1_c(\{|x_3|<c_0\eta\};V)$,

$$|\sigma(\theta\bar F\cdot(h)\cdot F')|\leq C_V\eta^{-p_V}\|h\|_{L^1}\,\|[\tau_{-\eta e_3}F]\|\,\|[\tau_{-\eta e_3}F']\|.$$

(Rec5) is the condition $\ker H=\mathbb{C}\Omega$; it is not part of the data.

(2) A point of $\mathbb{R}^{1+2}$ is $x=(t,\vec x)$; $U(x)=e^{itH}\mathcal{U}(\vec x)$; $\mathcal{D}_\infty=\bigcap_j \operatorname{dom} H^j$; $\mathcal{S}_V=\mathcal{S}(\mathbb{R}^{1+2};V_{\mathbb{C}})$ and $f^*=\rho(\theta)\bar f$. $\iota(t,\vec x)=(\vec x,-it)$; $\mathcal{L}$ is the restricted Lorentz group of $\mathbb{R}^{1+2}$, and $\hat\Lambda=\iota\Lambda\iota^{-1}\in \mathrm{SO}(3,\mathbb{C})$. For $\mathfrak{z}\in\mathbb{C}^3$, $d_\square(\mathfrak{z})=\operatorname{Re}\mathfrak{z}_3-|\operatorname{Im}\mathfrak{z}_1|-|\operatorname{Im}\mathfrak{z}_2|$ and $d_\circ(\mathfrak{z})=\operatorname{Re}\mathfrak{z}_3-|(\operatorname{Im}\mathfrak{z}_1,\operatorname{Im}\mathfrak{z}_2)|$.

(3) The *frame* (M1)–(M5) concerns a tensor carrying one label on a space with a scalar field $\phi(f)$, $f\in\mathcal{S}(\mathbb{R}^{1+2})$, a unique vacuum and a representation U of $\mathbb{R}^{1+2}\rtimes\mathcal{L}$; with $\vartheta_1=\vartheta_2=1$, $\vartheta_3=-1$: (M1) sesquilinear forms $\mathfrak{q}_{ij}=\mathfrak{q}_{ji}$ on $\mathcal{D}_\infty\times\mathcal{D}_\infty$ with $|\mathfrak{q}_{ij}(\psi',\psi)|\leq c_T\|(1+H)^l\psi'\|\,\|(1+H)^l\psi\|$, $\overline{\mathfrak{q}_{ij}(\psi',\psi)}=\vartheta_i\vartheta_j\mathfrak{q}_{ij}(\psi,\psi')$, $\mathfrak{q}_{ij}(\Omega,\Omega)=0$, and the operators $T^E_{ij}(F)$ they define, with the real-time components $T_{00}=-T^E_{33}$, $T_{0j}=-iT^E_{3j}$, $T_{jk}=T^E_{jk}$; (M2) continuous kernels $\mathsf{s}_{ij}(x;y_1,\dots,y_m)$ of one tensor entry among m scalar entries, covariant under $\mathrm{O}(3)$ and the translations, $\mathsf{s}_{ij}(\mathsf{R}x+b;\mathsf{R}y+b)=\mathsf{R}_{ii'}\mathsf{R}_{jj'}\mathsf{s}_{i'j'}(x;y)$, related to the forms by the product formula of two point operators at different heights, $\mathsf{s}^{F',G}_{ij}(x,y)=\mathfrak{q}_{ij}(\psi'_x,\psi_x)$ as written out in the proof of Lem. 3.5 (c); (M3) $T^E_{ij}(F_1)\phi(f)\psi=\phi(f)T^E_{ij}(F_1)\psi$ on the span $\mathcal{D}_0$ of Ω and the vectors $\phi(f_1)\cdots\phi(f_n)\Omega$, for spacelike separated supports; (M4) (a) $T_{0\nu}(\partial_0F)\psi=\sum_{k=1,2}T_{k\nu}(\partial_kF)\psi$ on $\mathcal{D}_0$; (M5) $U(\Lambda)T[a,b](F)\Omega=T[\Lambda a,\Lambda b](F\circ\Lambda^{-1})\Omega$, $T[a,b]=a^\mu b^\nu T_{\mu\nu}$.

■

Theorem E (Covariance, reconstruction, locality)**.** Let $0<g\leq\gamma^{\mathrm{sd}}_W$ and let σ be a tensor limit at g with label b_*.

(a) $\sigma(\mathsf{R}X)=\sigma(X)$ for every $\mathsf{R}\in\mathrm{O}(3)$ and every $X\in\mathfrak{S}_V(\mathbb{R}^3)$.

(b) $\sigma(\theta_{n,c}\bar F\cdot F)\geq 0$ for every unit vector n, every $c\in\mathbb{R}$ and every $F\in\mathbb{C}\mathfrak{S}_V(\{n\cdot x>c\})$, where $(\theta_{n,c}H)(x)=\Theta_nH(x-2(n\cdot x-c)n)\Theta_n$ and $\Theta_n=1-2nn^{\mathrm{T}}$.

(c) σ^1 is reconstruction data with labels at $V=\mathrm{Sym}_3(\mathbb{R})$ with the invariant norm, with $c_0=\frac12$, every $p_V>\frac{17}{2}$ and $C_V=2^{p_V}C_{p_V}b_*^{-1}\leq 2^{p_V}C_{p_V}g^2$; its invariance holds at all tuples of continuous entries. The map $[F]_{\sigma^1}\mapsto[F]_\sigma$ extends to a unitary map of $\mathcal{H}(\sigma^1)$ onto $\mathcal{H}(\sigma)$ that carries Ω, H, $\mathcal{U}$, $\mathcal{P}$ to Ω, H, $\mathcal{U}$, $\mathcal{P}$.

(d) (i) For every n there is exactly one continuous function $\mathsf{s}_n(x;e)$ on the non-coinciding configurations, n-linear in $e\in V^n$, with $\sigma(H_1,\dots,H_n)=\int\mathsf{s}_n(x;H_1(x_1),\dots,H_n(x_n))\,\mathrm{d}x$ for all $X\in\mathfrak{S}_V(\mathbb{R}^3)$; it is real-analytic, symmetric when points and labels are permuted together, $\mathsf{s}_n(\mathsf{r}x_1,\dots,\mathsf{r}x_n;\rho(\mathsf{R})e_1,\dots,\rho(\mathsf{R})e_n)=\mathsf{s}_n(x;e)$ for every isometry $\mathsf{r}x=\mathsf{R}x+v$, and $|\mathsf{s}_n(x;e)|\leq\big(C_V(2n^2/\min\{1,\mathsf{d}(x)\})^{p_V}\big)^n\prod_a|e_a|$ at every point, $\mathsf{d}(x)=\min_{a\neq b}|x_a-x_b|$; so $|\sigma(X)|\leq(C_V(2n^2/d)^{p_V})^n\prod_a\|H_a\|_{L^1}$ at pairwise distances $\geq d$, $0<d\leq1$, and $|\mathsf{S}_n(\varphi)|\leq(8\pi^2 3^{\lceil p_V\rceil+2}(2e^2)^{p_V}C_V)^n(n!)^{2p_V}|\varphi|_{(\lceil p_V\rceil+4)n}$. (ii) There is a strongly continuous unitary representation U of $\mathbb{R}^{1+2}\rtimes\mathcal{L}$ on $\mathcal{H}(\sigma)$ with $U(x,1)=e^{itH}\mathcal{U}(\vec x)$, $U(a,\Lambda)\Omega=\Omega$, $U(a,\Lambda)\mathcal{D}_0=\mathcal{D}_0$, the joint spectrum in the closed forward cone, and

$$U(a,\Lambda)T(f)U(a,\Lambda)^{-1}=T(\hat\Lambda f(\Lambda^{-1}(\cdot-a))\hat\Lambda^{\mathrm{T}})\quad\text{on }\mathcal{D}_0,$$

with the transpose, not the adjoint. (iii) $T(f)T(f')\psi = T(f')T(f)\psi$ for $\psi \in \mathcal{D}_0$ whenever $f, f' \in \mathcal{S}_V$ have spacelike separated supports: all six components are local with respect to all six.

(e) Let $0 < g \le \gamma$. Every sequence of first-passage data at g with $K_j \to \infty$ and periodic Gibbs states has a subsequence along which the tensor limit exists. Every such limit σ satisfies (a), (b) and the conclusion of [7, Thm. E] (c), and σ^1 is reconstruction data with labels with $c_0 = \frac{1}{2}$, every $p_V > \frac{17}{2}$ and

$$C_V = 3^{p_V} L^{4z} \max_{k=1,2} 2^{p_V} C_{p_V}[g'_k]\, g'^2_k,$$

where z is the least integer ≥ 1 with $L^z g^{-2} \ge 2(\gamma_W^{\mathrm{sd}})^{-2}$, $g_1'^{-2} = L^z g^{-2}/2$, $g_2'^{-2} = L^{z+1} g^{-2}/4$, and $C_p[g']$ is the number C_p of [7, Thm. C] (c) at the coupling g': a number formed from the constants, g and p_V, one triple for all the tensor limits at g; so (c) and (d) hold for it.

■□

Theorem F (Components and Wightman tensor fields)**.** Let σ be the tensor limit of a sequence (K_j, β_j, ν_j) of first-passage data and periodic Gibbs states at g, and $A^\nu_K(X) = \prod_a [\mathcal{T}_K(H_a) - \nu(\mathcal{T}_K(H_a))]$.

(a) *(Decomposition; $0 < g \le \gamma_W^{\mathrm{ref}}$.)* There are a compact metrisable space X, a Borel probability measure λ on X with support X, and real symmetric functionals σ_ξ, $\xi \in \mathsf{X}$, on labelled tuples, with $\xi \mapsto \sigma_\xi(X)$ continuous and $\xi \mapsto \sigma_\xi$ injective, such that σ has iterated plateaux in the sense of [7, Def. 2.2] (5) at every $(X^1, \dots, X^q)$, $q \ge 1$, $X^i \in \mathfrak{S}_V(\mathbb{R}^3)$, and its iterated plateau there is $\int \prod_i \sigma_\xi(X^i)\, \mathrm{d}\lambda(\xi)$.

(b) *(Each component; $0 < g \le \gamma_W^{\mathrm{ref}}$.)* For every $\xi \in \mathsf{X}$: (i) there are translation-invariant events E_j with $\nu_j(E_j) > 0$ such that $\rho_j = \nu_j(\cdot \,|\, E_j) \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$ and $\sigma_\xi(X) = \lim_j \rho_j(A^{\nu_j}_{K_j}(X))$ along the full sequence, indeed $\|E_{\rho_j}[A^{\nu_j}_{K_j}(X) \,|\, \mathcal{I}] - \sigma_\xi(X)\|_{L^\infty(\rho_j)} \to 0$, $\mathcal{I}$ the invariant σ-field; (ii) $\sigma_\xi((H)) = \frac{c_{\pi(\xi)}}{3} \int \mathrm{tr}\, H$ with a real number $c_{\pi(\xi)}$, and the recentred functional $\sigma^c_\xi(X) = \sum_{J \subset \{1..n\}} \prod_{a \notin J} (-\sigma_\xi((H_a)))\, \sigma_\xi(X_J)$ is the tensor limit of (K_j, β_j, ρ_j); so it has all the properties of a tensor limit, and $[F] \mapsto [\mathsf{S}_\xi F]$, $\mathsf{S}_\xi X = \sum_J \prod_{a \notin J} \sigma_\xi((H_a)) X_J$, is a unitary map of $\mathcal{H}(\sigma_\xi)$ onto $\mathcal{H}(\sigma^c_\xi)$ that carries Ω_ξ to Ω and intertwines H and $\mathcal{P}$; (iii) $\ker H_\xi = \mathbb{C}\Omega_\xi$; (iv) there is exactly one map π of X onto the space X_0 of the canonical decomposition $(\mathsf{X}_0, \lambda_0, (\sigma_{0,\zeta}))$ of the scalar part $\sigma|_{V_0}$ with $\sigma_\xi(f_1\mathbf{1}, \dots, f_n\mathbf{1}) = \sigma_{0,\pi(\xi)}(f_1, \dots, f_n)$; it is continuous and onto, and $\pi_* \lambda = \lambda_0$.

(c) *(Wightman fields; $0 < g \le \gamma_W^{\mathrm{sd}}$.)* For every ξ, the restriction of σ^c_ξ to C^1_c entries is reconstruction data with labels with $c_0 = \frac{1}{2}$, every $p_V > \frac{17}{2}$ and the numbers of Thm. E (c), and it has (Rec5). Its reconstruction is a Wightman tensor field of six components in $1 + 2$ dimensions: the fields $T(f)$, $f \in \mathcal{S}_V$, on $\mathcal{D}_0$ are tempered; $\langle \psi, T(f)\psi' \rangle = \langle T(f^*)\psi, \psi' \rangle$, so that T_{33} and T_{jk}, $j, k \le 2$, are Hermitian and T_{3j} is Hermitian after the factor i; $U(a,\Lambda) T(f) U(a,\Lambda)^{-1} = T(\hat{\Lambda} f(\Lambda^{-1}(\cdot - a))\hat{\Lambda}^{\mathrm{T}})$; the joint spectrum lies in the closed forward cone; Ω_ξ is the only invariant vector and is cyclic; all components are local with respect to all. The growth condition of [7, Thm. C] (h) holds with the same numbers for all components. The given limit σ is the integral over λ of its components in the sense of (a).

(d) *(The frame over a component; $0 < g \le \gamma_W^{\mathrm{sd}}$.)* On the scalar subspace of $\mathcal{H}(\sigma^c_\xi)$ the frame (M1)–(M5) holds for the tensor formed at the label matrices m_{ij} of the stress tensor, with the forms $q_0(\cdot, \cdot; \mathsf{m}_{ij})$, the kernels $\mathsf{s}_{ij}(x; y) = \mathsf{s}_{1+m}(x, y; \mathsf{m}_{ij}, \mathbf{1}, \dots, \mathbf{1})$, the scalar field $\phi(f) = T(f\mathbf{1})$ and the restriction to the scalar subspace of the representation U of Thm. E (d) (ii) for σ^c_ξ. The scalar subspace reduces $U(a,\Lambda)$: $\phi(f)$ keeps it; $U(a,\Lambda)\phi(f)U(a,\Lambda)^{-1} = \phi(f(\Lambda^{-1}(\cdot - a)))$ on $\mathcal{D}_0$, since $\hat{\Lambda}\mathbf{1}\hat{\Lambda}^{\mathrm{T}} = \mathbf{1}$; $U(a,\Lambda)\Omega = \Omega$; and Ω with the vectors $\phi(f_1) \cdots \phi(f_n)\Omega$ spans a dense subspace of it, by Prop. 3.3 (a) (v) for the restriction of σ^c_ξ to the tuples with entries in V_0,

whose space is the scalar subspace and whose fields are the restrictions of the $T(f\mathbf{1})$. In particular the real-time stress tensor is conserved, (M4) (a).

(e) *(The frame on $\mathcal{H}_0$; $0 < g \le \gamma_W^{\mathrm{sd}}$.)* For σ itself, (M1) holds, the kernels $\mathsf{s}_{ij}(x;y)$ of one tensor entry among scalar entries are covariant as in (M2) and continuous at all non-coinciding points, $[T^E_{ij}(F), \phi(f)] = 0$ between scalar vectors at spacelike separation as in (M3), and $\langle \Omega, \phi(f_1)\cdots\phi(f_n)T[t](F)\Omega\rangle$ is invariant under $(f,t,F) \mapsto (f\circ\Lambda^{-1}, \Lambda t, F\circ\Lambda^{-1})$ as in (M5). ■□

Proposition 3.2 (Covariance at one label suffices)**.** Let $0 < g \le \gamma_W^{\mathrm{sd}}$ and let σ be a tensor limit at g. If $\sigma(\mathsf{R}X) = \sigma(X)$ for all $\mathsf{R} \in \mathrm{O}(3)$ at the tuples with exactly one entry outside V_0, then the kernels of (M2) are covariant at all non-coinciding points and (M3), (M5) hold on $\mathcal{H}_0$. ■

By Thm. E (a) every tensor limit at g satisfies the condition of Prop. 3.2, and its conclusion is contained in Thm. F (e). Its proof uses rotational covariance only for tuples with exactly one entry outside V_0. It follows the proofs of Props. YM3_I:108.3, YM3_I:110.2 and YM3_I:110.1, with one labelled insertion.

Proposition 3.3 (Reconstruction for data with labels)**.**

(a) Let σ be reconstruction data with labels with the numbers C_V, $p = p_V$, c_0; (Rec5) is not supposed. Then $\mathcal{H}(\sigma)$ is separable, and there are a strongly continuous unitary representation U of $\mathbb{R}^{1+2} \rtimes \mathcal{L}$ with $U(x,1) = e^{itH}\mathcal{U}(\vec{x})$, operators $\phi(f)$, $f \in \mathcal{S}_V$, on the span $\mathcal{D}_0$ of Ω and of the vectors $\phi(f_1)\cdots\phi(f_n)\Omega$, functions $\mathbf{W}_n(\mathfrak{z};e)$ holomorphic on $\{d_\circ > 0\}^{n-1}$ and n-linear in e, and vectors $\Psi_n(z;e)$ with $[X] = \int \Psi_n(x; h_1(x_1), \dots, h_n(x_n))\,\mathrm{d}x$ for the tuples in $\mathbb{E}_+$ ordered in the third coordinate, such that: (i) the spectral measure of $U(\cdot,1)$ is carried by the closed forward cone and $U(a,\Lambda)\Omega = \Omega$; with (Rec5), the invariant vectors are the multiples of Ω; (ii) $\phi(f)\mathcal{D}_0 \subset \mathcal{D}_0$, $U(a,\Lambda)\mathcal{D}_0 = \mathcal{D}_0$, $f \mapsto \langle\psi, \phi(f)\psi'\rangle$ is tempered, and $\langle\psi, \phi(f)\psi'\rangle = \langle\phi(f^*)\psi, \psi'\rangle$; (iii) $U(a,\Lambda)\phi(f)U(a,\Lambda)^{-1} = \phi(D(\Lambda)f(\Lambda^{-1}(\cdot - a)))$ on $\mathcal{D}_0$, where $D(\Lambda) = \rho_{\mathbb{C}}(\hat{\Lambda})$ and $\rho_{\mathbb{C}}$ is the holomorphic continuation of $\rho|_{\mathrm{SO}(3)}$ to $\mathrm{SO}(3,\mathbb{C})$; at $V = \mathrm{Sym}_3(\mathbb{R})$, $D(\Lambda)A = \hat{\Lambda}A\hat{\Lambda}^{\mathrm{T}}$; (iv) $\phi(f)\phi(f')\psi = \phi(f')\phi(f)\psi$ on $\mathcal{D}_0$ at spacelike separated supports, for all components; (v) $\mathcal{D}_0$ is dense; (vi) $|\mathbf{W}_n(\mathfrak{z};e)| \le (C_V\eta^{-p})^n \prod_k |e_k|$ for $e \in V^n$, $d_\circ(\mathfrak{z}_k) \ge 2\eta$, $0 < \eta \le 1$; $\langle\Omega, \phi(f_1)\cdots\phi(f_n)\Omega\rangle = \lim_{\epsilon\downarrow 0}\int \mathbf{W}_n(\iota(x_2 - x_1 + i\epsilon\alpha_1), \dots; f_1(x_1), \dots, f_n(x_n))\,\mathrm{d}x$ for all α_k in the open forward cone; $\sigma(X) = \int \mathbf{W}_n(x_2 - x_1, \dots; h_1(x_1), \dots)\,\mathrm{d}x$ at the tuples ordered in the third coordinate; (vii) there is exactly one function $\mathbf{S}_n$, holomorphic on the union of the tubes $\{d_\circ(\mathsf{R}^{-1}(z_{\varsigma(k+1)} - z_{\varsigma(k)})) > 0,\ 1 \le k < n\}$ over $\mathsf{R} \in \mathrm{SO}(3)$ and the permutations ς, which on each of them is $\mathbf{W}_n$ written in the frame R at the labels $\rho(\mathsf{R})^{-1}e_{\varsigma(k)}$; its restriction s_n to the real non-coinciding configurations is real-analytic, is the only continuous kernel of σ, is symmetric in points and labels together, and is invariant under every isometry; (viii) Ω and the vectors $\Psi_n(x;e)$ at real x with $0 < x_{13} < \cdots < x_{n3}$ are total, and so are Ω and the classes of the tuples whose entries lie in disjoint slabs.

Clauses (i)–(vi) hold as well for complex symmetric functionals invariant under the orientation-preserving isometries, with $\overline{\sigma(W)} = \sigma(\theta\bar{W})$ and with $(\mathrm{Rec2})_V$–$(\mathrm{Rec4})_V$.

(b) For data as in (a) the two-fold plateau $m_2(X,Y) = \lim_{|v|\to\infty}\sigma(X\cdot\tau_v Y)$ exists, and σ has a Hermitian canonical decomposition: a compact metrisable X, a Borel probability measure λ with support X, and complex symmetric functionals σ_ξ with $\overline{\sigma_\xi(X)} = \sigma_\xi(\theta X)$, continuous and injective in ξ, such that the iterated plateaux of σ ([7, Def. 2.2] (5)) are $\int\prod_i \sigma_\xi(X^i)\,\mathrm{d}\lambda$; it is unique up to exactly one homeomorphism, and its components are real, so that it is a canonical decomposition, if and only if $m_2(F,F) \ge 0$ for the real combinations F of tuples.

Every component has $(\mathrm{Rec2})_V$–$(\mathrm{Rec4})_V$ with the numbers of σ and (Rec5), is reflection positive at every mirror, and $|\sigma_\xi(X\cdot\tau_v Y) - \sigma_\xi(X)\sigma_\xi(Y)| \le K(X,Y)|v|^{-1}$ for $|v| \ge r(X,Y)$, with K, r formed from C_V, p, c_0 and the two tuples, the same for all ξ. Let V_0 be a subrepresentation such that the classes of the tuples with entries in V_0 are total, and let

$(\mathsf{X}_0, \lambda_0, (\sigma_{0,\zeta}))$ be the Hermitian canonical decomposition of $\sigma|_{V_0}$, which is reconstruction data with labels in V_0 with the numbers of σ. There is exactly one map $\pi : \mathsf{X} \to \mathsf{X}_0$ with $\sigma_\xi|_{V_0} = \sigma_{0,\pi(\xi)}$ for every ξ; it is continuous and onto, $\pi_*\lambda = \lambda_0$, and $g \mapsto g \circ \pi$ is a unitary map of $L^2(\mathsf{X}_0, \lambda_0)$ onto $L^2(\mathsf{X}, \lambda)$.

So for every tuple X there is a bounded Borel function g_X on X_0 with $\sigma_\xi(X) = g_X(\pi(\xi))$ for λ-almost every ξ, and the iterated plateaux of σ are $\int \prod_i g_{X^i} \,\mathrm{d}\lambda_0$: the decomposition of $\sigma|_{V_0}$ bears that of σ as measure spaces. The map π is a homeomorphism if and only if every g_X agrees λ_0-almost everywhere with a continuous function, in particular if $\dim \ker H < \infty$. ■

Proof sketch. (a) On the circular cone $|e^{-\mathfrak{z}_3 E + i(\mathfrak{z}_1 q_1 + \mathfrak{z}_2 q_2)}| \le e^{-d_\circ(\mathfrak{z})E}$; this gives the contractions $\mathcal{T}(\mathfrak{z})$, holomorphic inside the tube. By $(\mathrm{Rec4})_V$ the forms $(F, F') \mapsto \sigma(\theta\bar{F} \cdot (h) \cdot F')$ are bounded, whence insertion operators $B_\eta(h)$ with $\|B_\eta(h)\| \le C_V \eta^{-p} \|h\|_{L^1}$, covariant under the translations, $B_\eta(\tau_y h') = \mathcal{T}(y + (\eta - \eta')e_3) B_{\eta'}(h') \mathcal{T}(-y + (\eta - \eta')e_3)$. Since translations act on them Lipschitz-continuously (on the cone each derivative of $e^{-(\eta/2 + y_3)E + i\vec{y}\cdot q}$ is at most $4/(e\eta)$), the limits $\mathsf{B}(\eta, \eta; e) = \lim_{\epsilon \downarrow 0} B_\eta(\delta_\epsilon e)$, $\delta_\epsilon = \epsilon^{-3}\chi(\cdot/\epsilon)$ with $\chi \ge 0$ radial of integral 1, exist in norm, and $\mathsf{b}(w|e|w') = \mathcal{T}(w - se_3)\mathsf{B}(s, s; e)\mathcal{T}(w' - se_3)$ are holomorphic point operators with $B_\eta(h) = \int \mathsf{b}(y + \eta e_3 | h(y) | -y + \eta e_3) \,\mathrm{d}y$.

Ordered tuples are products of insertion operators and semigroup factors; this gives $\mathbf{W}_n$, Ψ_n and the integral formulas. Any n distinct points are ordered in some frame $\mathsf{R} \in \mathrm{SO}(3)$ whose cap $\{\mathsf{R}e_3 \cdot \hat{v} > \frac{1}{2}\}$ contains a given direction (the zones $\{|u \cdot (x_a - x_b)| < \gamma_N |x_a - x_b|\}$, $\gamma_N = 1/(2N^2)$ with $N \ge n$, cover less than a quarter of the sphere); the pieces in two frames agree at the real points of the intersection of two convex tubes, hence on it, and they form one function $\mathbf{S}_n$. Totality: the map $y \mapsto \Phi(y)$, the limit of the classes of tuples of bumps at y, has the real-analytic Gram kernel $\mathsf{s}_{2n}(\theta y, y')$, so its matrix elements are real-analytic; for a vector orthogonal to the Ψ_n they vanish on the ordered configurations, hence on the connected set of the non-coinciding ones.

The fields are defined by the forms q_0 with $\mathsf{A}_l(e) = \Gamma(l)^{-2} \iint (ss')^{l-1} e^{-s-s'} \mathsf{B}(s, s'; e) \,\mathrm{d}s \,\mathrm{d}s'$, $\mathsf{B}(s, s'; e) = \mathsf{b}(se_3 | e | s'e_3)$, convergent for $l > p$; then $\mathcal{T}(w)\phi(f)\mathcal{T}(w') = \int \mathsf{b}(w + \iota x | f(x) | w' - \iota x) \,\mathrm{d}x$ for w, w' in the open tube, and $\|(1 + H)^j \phi(f) \psi\| \le c_j(f) \|(1 + H)^{2l + j} \psi\|$. *Boundary values* (vi): with $c_k = \iota(\frac{i}{2}\epsilon\alpha_k)$, α_k in the open forward cone, the contractions $\mathcal{T}(2c_k)$ commute with H and tend strongly to 1 as $\epsilon \downarrow 0$; inserted between the fields, they turn $\langle \Omega, \phi(f_1) \cdots \phi(f_n) \Omega \rangle$ into $\int \mathbf{W}_n(\iota(x_2 - x_1 + i\epsilon\alpha_1), \ldots; f_1(x_1), \ldots, f_n(x_n)) \,\mathrm{d}x$, up to an error that the energy bounds, applied from the right to the left, send to 0.

The density of $\mathcal{D}_0$ (v): let $\psi \perp \mathcal{D}_0$. For a word w in the operators $\phi(f)$ and $\mathcal{T}(w)$, $\psi \perp \mathsf{w}\Omega$, by induction on the number of the letters $\mathcal{T}$: the matrix element is continuous on the closed tube and holomorphic inside in the argument w of the leftmost such letter; it vanishes at $w = \iota a$, $a \in \mathbb{R}^{1+2}$, because $\mathcal{T}(\iota a) = U(a)$ moves the rest of the word into a word with translated test functions and one such letter fewer; so it vanishes on the half planes in time by the reflection principle, and on the tube by the identity theorem in the space variables. As $\mathcal{T}(w)\phi(g_k e)\mathcal{T}(w') \to \mathsf{b}(w|e|w')$ in norm for $g_k \ge 0$ of integral 1 whose supports shrink to a point, $\psi \perp \Psi_n(z; e)$, and $\psi = 0$ by (viii).

The representation D: with the covering $\mathsf{p} : \mathrm{SL}(2, \mathbb{C}) \to \mathrm{SO}(3, \mathbb{C})$, the differential of $\rho \circ \mathsf{p}$ on $\mathrm{SU}(2)$ continues complex-linearly to $\mathfrak{sl}(2, \mathbb{C})$ and integrates to a holomorphic representation of the simply connected $\mathrm{SL}(2, \mathbb{C})$, trivial on ± 1, hence to a holomorphic representation $\rho_\mathbb{C}$ of $\mathrm{SO}(3, \mathbb{C})$ that continues ρ; $\Lambda \mapsto \hat{\Lambda}$ is a homomorphism of $\mathcal{L}$ into $\mathrm{SO}(3, \mathbb{C})$, so $D(\Lambda) = \rho_\mathbb{C}(\hat{\Lambda})$ is a representation, and $D(\Lambda)e^* = (D(\Lambda)e)^*$. Every $\Lambda \in \mathcal{L}$ is $(1 \oplus \mathsf{R}_1) B(s) (1 \oplus \mathsf{R}_2)$ with rotations $\mathsf{R}_1, \mathsf{R}_2$ of the space plane and the boost $B(s)$: a spatial rotation and a boost send the future unit vector e_0 to Λe_0, and their inverse followed by Λ fixes e_0 and acts as a rotation of its orthogonal spatial plane; $\widehat{1 \oplus \mathsf{R}} = \mathsf{R} \oplus 1$, and $\widehat{B(s)}$ is the rotation in the plane (e_1, e_3) by the imaginary angle is.

Lorentz covariance: $\mathbf{W}_n$ is invariant under the real rotations about e_3 and under those in the plane (e_1, e_3), and this last invariance continues holomorphically in the angle to imaginary angles,

the boosts, the labels being moved by $e^{is\,\mathrm{d}\rho(\omega_1)}$, $\omega_1 = E_{13} - E_{31}$, $d\rho$ the differential of ρ; no bound on this factor is needed. So $\mathbf{W}_n(\hat{\Lambda}_{\mathfrak{z}}; D(\Lambda)e) = \mathbf{W}_n(\mathfrak{z}; e)$, and the boundary values at the cone vectors $\Lambda^{-1}\alpha_k$ give $\langle \Omega, \phi(\Lambda f_1)\cdots\phi(\Lambda f_n)\Omega\rangle = \langle \Omega, \phi(f_1)\cdots\phi(f_n)\Omega\rangle$ with $(\Lambda f)(x) = D(\Lambda) f(\Lambda^{-1}x)$.

As $(\Lambda f)^* = \Lambda(f^*)$, the map $\phi(f_1)\cdots\phi(f_n)\Omega \mapsto \phi(\Lambda f_1)\cdots\phi(\Lambda f_n)\Omega$ is isometric on the dense $\mathcal{D}_0$ and extends to a unitary operator $U(\Lambda)$; these form a representation with $U(\Lambda)U(a)U(\Lambda)^{-1} = U(\Lambda a)$, whose strong continuity reads of D only that it is locally bounded. With the spectrum of $U(\cdot, 1)$, the image of $\mathcal{P}$, carried by the closed forward cone by $(\mathrm{Rec3})_V$, this gives (i)–(iii). *Locality*: the componentwise argument is that of Prop. YM3_I:110.2. Fix the labels and compactly supported scalar tests with two adjacent supports spacelike separated. The height deformation used there stays in tilted tubes with every consecutive margin at least 3η. The Cauchy–Riemann equations and integration by parts move derivatives onto the tests, and Lem. YM3_I:109.4 gives a finite smeared bound $Q_e(g)$ uniform along the deformation. For g the product test, the integrals for the two adjacent orders differ by at most $2\eta Q_e(D'g)$, where $D' = \frac{1}{2}(\partial_{t_{k+1}} - \partial_{t_k})$ and $k, k+1$ are the exchanged entries. Their endpoint boundary values therefore agree as $\eta \to 0$. Graph-norm continuity extends the equality to Schwartz tests, as in the cited proof.

These are the proofs for scalar data of Props. YM3_I:108.3, YM3_I:109.1, YM3_I:110.1, YM3_I:110.2 and Thm. YM3_I:AO among the construction results, with a label at each entry and with D in the place of the trivial representation; besides them they read the spectral theorem, the identity theorem, the reflection principle in one variable and the covering of $\mathrm{SO}(3,\mathbb{C})$ by $\mathrm{SL}(2,\mathbb{C})$. They read the reality of σ and its invariance under an orientation-reversing isometry at one place only, the adjoint $B_\eta(h)^* = B_\eta(\theta\bar{h})$, where $\overline{\sigma(W)} = \sigma(\theta\bar{W})$ is all that is used; this gives (i)–(vi) for the complex functionals of the last sentence of (a). (b) For a combination Y of tuples put $\mathsf{e}(Y) = P_0[\tau_{se_3}Y]$, P_0 the projection onto $\ker H$ and s so large that $\tau_{se_3}Y$ lies over $\mathbb{E}_+$: it does not depend on s, it is invariant under the translations, and these vectors are dense in $\ker H$.

In a frame R, by the cone, $\sigma(A\cdot\tau_w Y) \to \langle \mathsf{e}(\theta\bar{A}'), \mathsf{e}(Y')\rangle$, $A' = \mathsf{R}^{-1}A$, $Y' = \mathsf{R}^{-1}Y$, uniformly as $|w| \to \infty$ with $\mathsf{R}e_3\cdot w \geq \frac{1}{2}|w|$; two frames whose caps meet give the same limit, and the caps cover the sphere; so the two-fold plateau exists, $m_2(A, Y) = \langle \mathsf{e}(\theta\bar{A}), \mathsf{e}(Y)\rangle$. *The rotations are trivial at infinity*: for a unit vector u, $\beta_u(F, G) = m_2(\theta_u\bar{F}, G)$ is a positive Hermitian form; for $u' \perp u$ the involution $\theta_u\theta_{u'}$ acts on its space as a self-adjoint unitary operator ≥ 0, which is 1; hence $m_2(\theta_u A, Y)$ does not depend on u, $m_2(\mathsf{r}A, Y) = m_2(A, Y)$ and $\mathsf{e}(\mathsf{r}Y) = \mathsf{e}(Y)$ for the orientation-preserving isometries r, and $J\mathsf{e}(Y) = \mathsf{e}(\theta Y)$ defines a unitary involution J of $\ker H$ with $m_2(\bar{A}, Y) = \langle J\mathsf{e}(A), \mathsf{e}(Y)\rangle$.

The operators at infinity: $M(X)$ is the bounded operator of $\ker H$ with $\langle \mathsf{e}(Z), M(X)\mathsf{e}(Y)\rangle = \lim_{|v|\to\infty} m_2(\theta\bar{Z}, X\cdot\tau_v Y)$, the limit of $\sigma(\tau_w\theta\bar{Z}\cdot X\cdot\tau_v Y)$ as w and v go to infinity on opposite sides of X; for X ordered in a frame, $M(X) = P_0\mathsf{b}_X|_{\ker H}$, b_X the product of the insertion operators and semigroup factors of the ordered tuple, and a partition of unity reduces every tuple to such ones. Then $M(F)^* = M(\theta\bar{F})$, $M(\mathsf{r}X) = M(X)$ for the orientation-preserving and $M(\mathsf{r}X) = JM(X)J$ for the other isometries, $M(X\cdot\tau_v Y) \to M(X)M(Y)$ strongly as $|v| \to \infty$, and, σ being symmetric, the $M(X)$ commute.

They generate a separable commutative C^*-algebra; its space of characters is X, the vacuum gives λ, whose support is X because Ω is cyclic, hence separating, for the algebra; $\ker H$ is carried unitarily onto $L^2(\mathsf{X}, \lambda)$, $\sigma_\xi(X)$ is the Gelfand transform of $M(X)$, so that $\overline{\sigma_\xi(X)} = \sigma_\xi(\theta X)$, and the iterated plateaux of σ are $\langle \Omega, \prod_i M(X^i)\Omega\rangle$. Uniqueness follows from the density of the polynomials in the $\sigma_\xi(X)$ and their conjugates. *Reality*: $m_2(\bar{F}, F) = \langle J\mathsf{e}(F), \mathsf{e}(F)\rangle$, so $m_2(F, F) \geq 0$ for the real combinations F if and only if $J \geq 0$, that is $J = 1$, that is $M(X)^* = M(X)$ for every tuple, that is every σ_ξ is real.

The components: an inequality of σ at the tuples $F\cdot\tau_{tu}Y$, the spectator Y sent to infinity along u, becomes an inequality between elements of the algebra on a dense set of $\ker H$, and, the Gelfand transforms being continuous and $\operatorname{supp}\lambda = \mathsf{X}$, an inequality at every ξ (test it with $\lambda(O)^{-1/2}\mathbf{1}_O$ for small neighbourhoods O of ξ); so every σ_ξ is reflection positive at every mirror, has $(\mathrm{Rec4})_V$ with

the numbers of σ, and the translations act on its space by contractions and unitary operators, which gives $(\mathrm{Rec2})_V$, one countable family of tuples making all these spaces separable.

The spectral measures disintegrate, $\mu_F = \int \mu_F^\xi \,\mathrm{d}\lambda(\xi)$ for F over $\mathbb{E}_+$; so $\mu_F^\xi(\{E < |q|\}) = 0$ for λ-almost every ξ, and, this number being lower semicontinuous in ξ, for every ξ: $(\mathrm{Rec3})_V$. At $\{(0,0)\}$ the disintegration gives $\int \|P_0^\xi[F]_\xi\|^2 \,\mathrm{d}\lambda = \|P_0[F]\|^2 = \int |\sigma_\xi(F)|^2 \,\mathrm{d}\lambda$, while $\|P_0^\xi[F]_\xi\| \geq |\sigma_\xi(F)|$, P_0^ξ the projection onto $\ker H_\xi$: so $\ker H_\xi = \mathbb{C}\Omega_\xi$ for almost every ξ. *The rate*: for data, and for Hermitian data, $|\sigma(X \cdot \tau_v Y) - m_2(X,Y)| \leq K(X,Y)|v|^{-1}$ for $|v| \geq r(X,Y)$, with K, r formed from C_V, p, c_0 and the two tuples alone.

In each frame of a finite family whose caps cover the sphere and in which X and Y are ordered, the difference $f(v)$ is the integral of $e^{-(v_3'-c)E+i\vec{v}'\cdot q}$, $v' = \mathsf{R}^{-1}v$, against a complex measure off $(0,0)$ whose variation is at most the product of the numbers $C_V\eta^{-p}\|h\|_{L^1}$ of the entries of X and Y. The image κ_v of this measure, with the exponential, under $(E,q) \mapsto s = E^2 - |q|^2$ has the moments $\Delta^k f(v)$, which are independent of the frame. Its damping gives $\int e^{a\sqrt{s}}\, d|\kappa_v|(s) < \infty$ for every $a > 0$ below the positive time margin, since $0 \leq \sqrt{s} \leq E$. This square-root exponential condition determines complex measures within this class by their moments. Indeed, the real and imaginary parts of the difference of two such measures have zero moments, including order zero. Their positive and negative parts have equal masses and moments and inherit the same integrability condition; after normalization when the common mass is nonzero, Hardy's criterion applies to them [2, Thm. 1]. Thus κ_v is independent of the frame, and its masses $f_{\mathsf{B}}(v) = \kappa_v(\mathsf{B})$ on windows B are functions outside a ball $\{|v| < r_2\}$, with $\Delta f_{\mathsf{B}}(v) = \int_{\mathsf{B}} s\, \kappa_v(ds)$.

For $\mathsf{B} = [m_1^2, m_2^2)$, $|(\Delta - m_1^2)\,\mathrm{Re}(e^{i\varphi} f_{\mathsf{B}})| \leq (m_2^2 - m_1^2)\mathsf{V}_{\mathsf{B}}$, V_{B} the variation of the part, and the comparison principle for $\Delta - m_1^2$ outside the ball, with the radial solution $\frac{r_2}{r}e^{-m_1(r-r_2)}$ of the dimension three, gives $|f_{\mathsf{B}}(v)| \leq \mathsf{V}_{\mathsf{B}}(\frac{r_2}{|v|} + \frac{m_2^2-m_1^2}{m_1^2})$; the harmonic part at the mass 0, which tends to 0 at infinity, is at most $\mathsf{V}_{\{0\}}\frac{r_2}{|v|}$; the sum over the windows $[(1+\epsilon)^j, (1+\epsilon)^{j+1})$, $j \in \mathbb{Z}$, with $\epsilon \downarrow 0$, gives the rate. One rate serves all the components, which are Hermitian data with the numbers of σ; with the continuity in ξ and $\mathrm{supp}\,\lambda = \mathsf{X}$ the bound by $\sigma_\xi(X)\sigma_\xi(Y)$ holds at every ξ, so that $\langle [F]_\xi, e^{-tH_\xi}[F]_\xi\rangle \to |\sigma_\xi(F)|^2$ and $\ker H_\xi = \mathbb{C}\Omega_\xi$ for every ξ.

The subrepresentation: the defining limits of the operators at infinity of $\sigma|_{V_0}$ and of σ agree at the tuples with entries in V_0; when their classes are total, $\sigma|_{V_0}$ and σ have one space and one $\ker H$, the operators of these tuples generate a C^*-subalgebra with the same unit, for which Ω is cyclic too, π is the restriction of the characters, the only map with $\sigma_\xi|_{V_0} = \sigma_{0,\pi(\xi)}$ because $\zeta \mapsto \sigma_{0,\zeta}$ is injective, $\pi_*\lambda = \lambda_0$ by the vacuum state, π is onto because its image is compact of full measure, and $g \mapsto g \circ \pi$ is unitary because both decompositions carry the one space $\ker H$ unitarily onto their L^2; π is injective if and only if the subalgebra is the whole algebra, by the theorem of Stone and Weierstrass, that is if and only if every g_X agrees almost everywhere with a continuous function; and if $\dim\ker H = N < \infty$, then X and X_0 have N points each. □

Lemma 3.4 (The circular cone from five mirrors; for data)**.** Let σ be a real symmetric multilinear functional on labelled tuples, with $\sigma(\varnothing) = 1$ and bounds independent of the position, invariant under the translations. For a unit vector u at whose mirror $\{u \cdot x = 0\}$ σ is reflection positive, let $\mathcal{H}_u$, H_u, $\mathcal{P}_u$ be formed as in [7, Def. 2.2] (3) with u in the place of e_3.

(a) Let u' be a unit vector orthogonal to u. If σ is reflection positive at the mirrors with the normals u and $(u+u')/\sqrt{2}$, then $H_u \geq |u' \cdot P_u|$.

(b) Let (u, u', u'') be an orthonormal basis and $n_\beta = \cos\beta\, u + \sin\beta\, u'$. If σ is reflection positive at the four planes through the axis $\mathbb{R}u''$ with the normals $n_0, n_{\pi/4}, n_{\pi/2}, n_{3\pi/4}$ and at one plane with the normal $(n_{\beta_1} + u'')/\sqrt{2}$, $\beta_1 \in \frac{\pi}{4}\mathbb{Z}$, then $\mathcal{P}_{n_\beta}(\{E < |q|\}) = 0$ for every $\beta \in \frac{\pi}{4}\mathbb{Z}$.

(c) *(Eight measures in one plane; analysis alone.)* Let $R > 0$, $\kappa > 0$, $C < \infty$, $\mathsf{V} = \{(E,q) \in \mathbb{R}^2 : E \geq |q|\}$, and for each of the eight angles $\beta \in \frac{\pi}{4}\{0, 1, \ldots, 7\}$ let m_β be a complex Borel

measure on V with $|m_\beta|(\mathsf{V}) \le C$; for $\rho > 0$ and complex c put

$$f_\beta(c;\rho) = \int_{\mathsf{V}} \exp(2RE - \rho[E\cos(c-\beta) - iq\sin(c-\beta)])\,\mathrm{d}m_\beta(E,q)$$

where the integrand is bounded. Suppose that $f_\beta(t;\rho) = f_{\beta'}(t;\rho)$ for any two of the eight angles β, β', all $\rho > 0$ and all real t with $\rho\cos(t-\beta) > 2R$ and $\rho\cos(t-\beta') > 2R$; that $\mu = e^{2RE}m_0$ is a finite positive measure; and that for one of the eight angles, β_1, the measure m_{β_1} is carried by $\{E \ge \kappa\}$. Then $\mu(\{(E,q) \in \mathsf{V} : E^2 - q^2 < \kappa^2\}) = 0$.

No invariance under W_3, under a reflection or under a rotation is supposed. ■

Proof sketch. (a) is the argument of Prop. YM3_I:94.4, applied at the mirrors with the normals u and $(u+u')/\sqrt{2}$; the labels ride along, the one number $\sigma(\theta_u\bar{F}\cdot\tau_w F)$ being read in the two pictures. (b) By the symmetry of σ (the entries commute) one and the same number $\sigma(A\cdot\tau_x B)$ is a matrix element in every picture whose mirror separates A from $\tau_x B$, also when x has gone around A. After a cut $|q\cdot u''| \ge \kappa$ in the momentum normal to the plane of the eight normals n_β, which is common to the eight pictures, this gives eight complex measures m_β, $\beta \in \frac{\pi}{4}\mathbb{Z}$, carried by V by (a) in each picture, whose Laplace–Fourier transforms $\int \exp(2RE - \rho[E\cos(c-\beta) - iq\sin(c-\beta)])\,\mathrm{d}m_\beta$ agree wherever two of them are defined, by the uniqueness of the Fourier transform in the common momentum; the one of the picture u is positive after the weight e^{2RE}, and by (a) at the fifth mirror the one of the picture n_{β_1} is carried by $\{E \ge |q\cdot u''|\} \subset \{E \ge \kappa\}$.

Clause (c) gives $\mu(\{E^2 - q^2 < \kappa^2\}) = 0$ for every $\kappa > 0$, and so the circular cone in each picture. (c) Write $c = t + i\gamma$. On V, $E\cosh\gamma + q\sinh\gamma \ge Ee^{-|\gamma|}$; so on $S_\beta(\rho) = \{\rho\cos(t-\beta)e^{-|\gamma|} > 2R\}$ the integrand of f_β is bounded by 1, and $f_\beta(\cdot;\rho)$ is holomorphic there (Morera), bounded by C and of the period 2π. The sets $S_\beta(\rho)$ cover the band $\{|\gamma| < \Gamma(\rho)\}$, $\Gamma(\rho) = \log(\rho\cos\frac{\pi}{8}/2R)$, and two of the functions agree on real intervals of the overlaps, hence on the overlaps (identity theorem): they glue to one holomorphic function $f(\cdot;\rho)$ on the band, $|f| \le C$.

Near β_1, $|f(c;\rho)| \le Ce^{-\kappa\rho_\delta}$ for $|c-\beta_1| \le \delta$, $\rho_\delta = \rho e^{-\delta}\cos\delta - 2R$, by the carrier of m_{β_1}. With $N(c) = \log|2\sin((c-\beta_1)/2)|$, harmonic off $\beta_1 + 2\pi\mathbb{Z}$, the function $\log|f| - \log C + \kappa\rho_\delta h$, $h = (\mathsf{M} - N)/(\mathsf{M} + \log(2/\delta))$, $\mathsf{M} = \log(2\sinh(\Gamma'/2))$, $\Gamma' = \Gamma(\rho) - 1$, is subharmonic on the band $\{|\gamma| < \Gamma'\}$ without the discs $|c - \beta_1 - 2\pi k| \le \delta$ and ≤ 0 on its boundary; by the maximum principle $|f| \le Ce^{-\kappa\rho_\delta h}$ there. At the points $i\gamma_0$, $f(i\gamma_0;\rho) = \int \exp(-\rho(E\cosh\gamma_0 + q\sinh\gamma_0))\,\mathrm{d}\mu$ is the Laplace transform of the positive measure μ, and $h(i\gamma_0) \to 1$ as $\rho \to \infty$. For fixed $\gamma_0 \ne 0$, choose $\delta > 0$ small enough that $i\gamma_0$ lies outside the excluded discs. First letting $\rho \to \infty$ and then $\delta \downarrow 0$ gives $E\cosh\gamma_0 + q\sinh\gamma_0 \ge \kappa$ almost everywhere. Apply this first on a countable dense subset of $\mathbb{R}\setminus\{0\}$ and use continuity in γ_0; the inequality then holds for every real γ_0 on one set of full measure, and the infimum over γ_0 of the left member is 0 if $E = |q|$ and $(E^2 - q^2)^{1/2}$ if $E > |q|$. □

Lemma 3.5 (Conservation of kernels and of operators; for data)**.** Let σ' be scalar reconstruction data with a unique vacuum on $\mathcal{H}$, with a tensor of one label satisfying (M1), (M2), forms $\mathfrak{q}_{ij}$ and kernels s_{ij}.

(a) A subspace $\mathcal{V} \subset \mathcal{D}_\infty$, dense in $\mathcal{H}$ and invariant under $e^{-\tau H}$, $\tau > 0$, is a core of $(1+H)^N$ for every N.

(b) $\mathcal{V}_1$, the span of the $e^{-\tau H}[F']$, and, if $\mathcal{H} \ne \mathbb{C}\Omega$, $\mathcal{V}_2$, the span of the $\mathcal{U}(v)\mathsf{B}_0(s,s')[F]$, B_0 the scalar point operators, are such subspaces.

(c) If $\sum_i \int \partial_i\zeta(x)\,\mathsf{s}_{ij}(x;y)\,\mathrm{d}x = 0$ for all $m \ge 1$, $y \in (\mathbb{R}^3)^m_{\ne}$ and $\zeta \in C^\infty_c$ off the y_k, then the form $CL_j(\psi',\psi) = \mathfrak{q}_{3j}(H\psi',\psi) - \mathfrak{q}_{3j}(\psi',H\psi) - i\sum_k[\mathfrak{q}_{kj}(P_k\psi',\psi) - \mathfrak{q}_{kj}(\psi',P_k\psi)]$, $P_k = \int q_k\,\mathrm{d}\mathcal{P}$, vanishes on $\mathcal{D}_\infty \times \mathcal{D}_\infty$, and $T_{0\nu}(\partial_0 F)\psi = \sum_{k=1,2} T_{k\nu}(\partial_k F)\psi$ on $\mathcal{D}_\infty$. Conversely (M4) (a) gives the identity of the kernels.

■

Proof sketch. (a) A vector orthogonal to $\mathcal{V}$ in the graph norm of $(1+H)^N$ gives a finite measure $(1+E)^N\mu$ that annihilates all the functions $e^{-\tau E}$, hence vanishes by the Stone–Weierstrass theorem. (b) The closure of $\mathcal{V}_2$ reduces H and $\mathcal{U}$ and contains $\phi(f)\mathcal{D}_\infty$, hence all the $\phi(f_1)\cdots\phi(f_n)\Omega$; if it were proper, its orthogonal complement would hold an invariant vector, hence Ω by the unique vacuum, and $\phi(f)\Omega=0$ would follow. (c) Let $a>0$, let F', G be scalar tuples over $\mathbb{E}_+$ with smooth entries, $F'(z')$, $G(z)$ the products of their entries at the points z', z, and $P_k=\int q_k\,\mathrm{d}\mathcal{P}$, so that $\mathcal{U}(v)=e^{iv\cdot P}$. For $x\neq y$ in $\{|x_3|<a\}$ with $x_3<y_3$ put

$$\mathsf{s}_{ij}^{F',G}(x,y)=\int \overline{F'(z')}\,G(z)\,\mathsf{s}_{ij}(x;y,\theta(z'+ae_3),z+ae_3)\,\mathrm{d}z'\,\mathrm{d}z;$$

the operator formula of (M2), the product of the point operator of $\mathfrak{q}_{ij}$ at x and of the scalar point operator at y between $[F']$ and $[G]$, reads $\mathsf{s}_{ij}^{F',G}(x,y)=\mathfrak{q}_{ij}(\psi'_x,\psi_x)$ with $\psi'_x=\mathcal{U}(-\vec{x})e^{-(a+x_3)H}[F']$ and $\psi_x=\mathcal{U}(-\vec{x})e^{-(h_*-x_3)H}\mathcal{U}(\vec{y})\mathsf{B}_0(y_3-h_*,a-y_3)\mathcal{U}(-\vec{y})[G]$, for any $h_*\in(x_3,y_3)$. As $\partial_{x_3}\psi'_x=-H\psi'_x$, $\partial_{x_3}\psi_x=H\psi_x$, $\partial_{x_k}\psi'_x=-iP_k\psi'_x$ and $\partial_{x_k}\psi_x=-iP_k\psi_x$ for $k=1,2$, one has $\sum_i\partial_{x_i}\mathsf{s}_{ij}^{F',G}(x,y)=-CL_j(\psi'_x,\psi_x)$, and $|CL_j(\psi',\psi)|\le 6c_T\|(1+H)^{l+1}\psi'\|\,\|(1+H)^{l+1}\psi\|$.

Given $\tau,s,s'>0$, $v\in\mathbb{R}^2$ and F, the choice $a=\frac12(\tau+s+s')$, $x=(0,\tau-a)$, $y=(v,a-s')$, $G=\tau_{(v,0)}F$ gives $\psi'_x=e^{-\tau H}[F']$ and $\psi_x=\mathcal{U}(v)\mathsf{B}_0(s,s')[F]$. The identity of the kernels supposed in (c), integrated against $\overline{F'}G$, kills the divergence near x; so $CL_j=0$ on $\mathcal{V}_1\times\mathcal{V}_2$, and on $\mathcal{D}_\infty\times\mathcal{D}_\infty$ by (a), (b) and the bound. Read at $U(b)^{-1}\psi'$, $U(b)^{-1}\psi$, $b\in\mathbb{R}^{1+2}$, multiplied by the test function of the operator identity and integrated by parts in b, this is that identity. Conversely, $\mathcal{D}_0$ is invariant under the operators $U(b)$, so it is a core of every $(1+H)^N$, as in (a) with the group in the place of the semigroup, the finite measure $(1+E)^N\mu$ having a vanishing Fourier transform; the bounds $\|T_{\mu\nu}(F_1)\psi\|\le C\|(1+H)^{2l}\psi\|$ of the operators of (M1) carry (M4) (a) from $\mathcal{D}_0$ to $\mathcal{D}_\infty$, and, the test function being arbitrary, $CL_j=0$ on $\mathcal{D}_\infty\times\mathcal{D}_\infty$.

At a configuration $(x,y_1,\dots,y_m)$ with pairwise different heights take a slab that holds x and the point next to it in height and no other point; the other points are smeared by bumps into G above and, through their mirror images, into F' below; the divergence of $\mathsf{s}_{ij}^{F',G}$ is $-CL_j(\psi'_x,\psi_x)=0$, and approximate units for the bumps give $\sum_i\int\partial_i\zeta(x)\,\mathsf{s}_{ij}(x;y)\,\mathrm{d}x=0$ for ζ carried near x. A rotation that makes the heights pairwise different reaches every configuration, by the covariance of the kernels in (M2), and a partition of unity every ζ. □

Lemma 3.6 (Extension over the diagonals; contact terms)**.** Let $\mathbb{M}$ be $\mathbb{R}^3$ or a torus T_ℓ, $n\ge2$, and V a real vector space of finite dimension with a norm. A *functional of n entries* is a map u on the tuples $(f_1,\dots,f_n)$ of functions $f_a\in C_c^\infty(\mathbb{M};V_\mathbb{C})$ with pairwise disjoint supports that is n-linear on $\prod_a C_c^\infty(O_a;V_\mathbb{C})$ for all open sets O_a with pairwise disjoint closures; the entries may as well take their values in n different spaces $V_1,\dots,V_n$, with $\bigotimes_a V_{a,\mathbb{C}}$ in the place of $V_\mathbb{C}^{\otimes n}$ below, and nothing else changes (this is used in (e)). We write $\|G\|_{(h),r}=\max_{|\gamma|\le r}h^{|\gamma|}\sup|\partial^\gamma G|$, on $\mathbb{M}$ and on $\mathbb{M}^n$; $\mathsf{d}_\infty(x)=\min_{a\neq b}|x_a-x_b|_\infty$ for $x\in\mathbb{M}^n$; and $\Delta=\{\mathsf{d}_\infty=0\}$, the union of the diagonals.

(1) *Strata.* For a partition π of $\{1,\dots,n\}$ with $k=|\pi|$ blocks, $B(a)$ the block of a and $a_B=\min B$: $\Delta_\pi=\{x: x_a=x_b \text{ whenever } B(a)=B(b)\}$, and $\Sigma_k=\bigcup_{|\pi|\le k}\Delta_\pi$, so that $\Sigma_{n-1}=\Delta$. The coordinates at π are $y_B=x_{a_B}\in\mathbb{M}$ and $\xi_a=x_a-x_{a_{B(a)}}$ for $a\neq a_{B(a)}$; $\iota_\pi(y)_a=y_{B(a)}$, so that $\Delta_\pi=\{\xi=0\}=\iota_\pi(\mathbb{M}^k)$. For a distribution T' on $\mathbb{M}^k$ and a multi-index α of ξ, $(\partial_\xi^\alpha\delta_\pi\otimes T')(F)=(-1)^{|\alpha|}\,T'((\partial_\xi^\alpha F)\circ\iota_\pi)$.

(2) *The bound* $(\mathrm{W};\kappa,q,r)$, $\kappa\in\mathbb{R}$, $q\ge0$, $r\in\mathbb{N}_0$, with numbers $A,h_0>0$ (on T_ℓ: $h_0\in\ell2^{-\mathbb{N}}$, $h_0\le\ell/(64n)$): for every $0<h\le h_0$, all closed cubes $Q_1^*,\dots,Q_n^*\subset\mathbb{M}$ of the side $\frac32 h$, wherever they lie, at pairwise sup distances $\ge\frac52 h$, this lower bound being the only condition on their distances, and all $f_a\in C_c^\infty(\operatorname{int}Q_a^*;V_\mathbb{C})$,

$$|u(f_1,\dots,f_n)|\le A\,h^\kappa\,l(h)^q\prod_a\|f_a\|_{(h),r},\qquad l(h)=1+\log(h_0/h).$$

(3) *The class.* A function F on $\mathbb{M}^n$ is λ-*supported* if $\operatorname{supp} F \subset \prod_a \{|x_a - z_a|_\infty \le \lambda\}$ for some $z \in \mathbb{M}^n$, the z_a coinciding or not. $\mathfrak{E}_n(\kappa)$ is the set of the $(V_{\mathbb{C}}^{\otimes n})^*$-valued distributions T on $\mathbb{M}^n$ for which there are C, q, m, h_1 with $|T(F)| \le C\lambda^\kappa(1+\log\frac{h_1}{\lambda})^q \|F\|_{(\lambda),m}$ for all $0<\lambda\le h_1$ and all λ-supported F: the *power counting* κ, up to logarithms, also where the points meet. On $\mathbb{M} = T_\ell$ the norms $\|\cdot\|_{(\lambda),m}$, the λ-supported functions, the classes $\mathfrak{E}_n(\kappa)$, the coordinates at π and the distributions $\partial_\xi^\alpha \delta_\pi \otimes T'$ of (1) are those of [7, Def. 2.4]. We put $\omega_k = \lfloor 3k - \kappa \rfloor$.

(4) *Whitney products.* A product $Q = \prod_a Q_a$ of half-open cubes $Q_a \subset \mathbb{M}$ of one side $h(Q) \le h_0$ is *admissible* if $\operatorname{dist}_\infty(Q_a, Q_b) \ge 3h(Q)$ for $a \ne b$; $c(Q)$ is its centre, and Q^+, Q^* are the closed concentric products of the sides $\frac54 h(Q)$ and $\frac32 h(Q)$, so that the cubes Q_a^* are at pairwise sup distances $\ge \frac52 h(Q)$. Let $h_j = 2^{-j}h_0$, $j \ge 0$, and let $\mathcal{Q}$ be the set of the admissible products of dyadic cubes of the sides h_j that lie in no larger admissible product of dyadic cubes. For $Q \in \mathcal{Q}$, $\pi(Q)$ is the partition of $\{1,\dots,n\}$ into the connected components of the graph $a \sim b \iff |c(Q)_a - c(Q)_b|_\infty \le 12h(Q)$, and $k(Q) = |\pi(Q)|$.

Then $\mathcal{Q}$ is a partition of $\mathbb{M}^n \setminus \Delta$; on Q, $\mathtt{d}_\infty \ge 3h(Q)$, and if $h(Q) < h_0$, $\mathtt{d}_\infty < 10h(Q)$ and $k(Q) \le n-1$; there are $\varphi_Q \in C_c^\infty(\operatorname{int} Q^+)$, $\varphi_Q \ge 0$, with $\sum_{Q\in\mathcal{Q}} \varphi_Q = 1$ on $\mathbb{M}^n \setminus \Delta$ and $|\partial^\gamma \varphi_Q| \le c_\gamma h(Q)^{-|\gamma|}$. Fix a smooth μ on $\mathbb{R}^3$ carried by $\{|y|_\infty < 1\}$ with $\int \mu = 1$. For $Q \in \mathcal{Q}$ with $h = h(Q)$, $\pi = \pi(Q)$, $k = k(Q) \le n-1$ and $\omega_k \ge 0$ put $\mu_Q(y') = \prod_{B\in\pi} h^{-3}\mu((y'_B - c(Q)_{a_B})/h)$ for $y' \in \mathbb{M}^k$ and

$$P_Q F(x) = \int \mu_Q(y') \sum_{|\gamma|\le\omega_k} \frac{(x - \iota_\pi(y'))^\gamma}{\gamma!}\, \partial^\gamma F(\iota_\pi(y'))\, \mathrm{d}y',$$

the Taylor polynomial of F of total order ω_k averaged over the piece of the size h of the stratum Δ_π next to Q; $P_Q = 0$ for the other Q.

Let u be a functional of n entries with $(\mathrm{W};\kappa,q,r)$, and put $N_0 = \lfloor\frac{3n}{2}\rfloor + 1 + nr$ and $m = \max\{\omega_{n-1}+1, N_0\}$.

(a) For every admissible product Q, dyadic or not, in particular for every $Q \in \mathcal{Q}$, there is exactly one linear functional u_Q on $C_c^\infty(\operatorname{int} Q^+; V_{\mathbb{C}}^{\otimes n})$ with $u_Q(g_1 \otimes \cdots \otimes g_n) = u(g_1,\dots,g_n)$ and

$$|u_Q(G)| \le C_{\mathrm{F}}\, A\, h(Q)^\kappa\, l(h(Q))^q\, \|G\|_{(h(Q)),N_0},$$

C_{F} formed from n, r and the norm of V; it reads (W) at the side $h(Q)$ alone, and two of these functionals agree where both are defined. There is exactly one distribution u° on $\mathbb{M}^n \setminus \Delta$ with $u^\circ(f_1 \otimes \cdots \otimes f_n) = u(f_1,\dots,f_n)$ for the tuples of the definition, and $u^\circ = u_Q$ on $C_c^\infty(\operatorname{int} Q^+)$: off the diagonals u is given by exactly one distribution.

(b) The sum

$$\tilde u(F) = \sum_{Q\in\mathcal{Q}} u_Q(\varphi_Q(F - P_Q F))$$

converges absolutely for $F \in C_c^m(\mathbb{M}^n; V_{\mathbb{C}}^{\otimes n})$, and $\tilde u$ is a distribution of order at most m that extends u°. At $\kappa = 0$, $r = 0$: $\omega_k = 3k$ and $m = 3n-2$.

(c) For $0 < \lambda \le h_0$ and every λ-supported F,

$$|\tilde u(F)| \le C_{\mathrm{ext}}\, A\, \lambda^\kappa\, l(\lambda)^{q+\epsilon}\, \|F\|_{(\lambda),m},$$

where $\epsilon = 1$ if $3k - \kappa$ is an integer ≥ 0 for some $k \in \{1,\dots,n-1\}$, $\epsilon = 0$ otherwise, and C_{ext} is formed from n, κ, q, r, the norm of V and μ, and does not depend on u; so $\tilde u \in \mathfrak{E}_n(\kappa)$. *(A range of sides.)* If u has the inequality of $(\mathrm{W};\kappa,q,r)$ at the sides $h_\flat \le h \le h_0$ alone, for some $h_\flat \in (0, h_0]$, then the sum $\tilde u^\flat(F)$ over the $Q \in \mathcal{Q}$ with $h(Q) \ge h_\flat$ has the same bound, with the same C_{ext}, for $8h_\flat \le \lambda \le h_0$ and every λ-supported F.

(d) If u is invariant under a group generated by permutations of the entries, translations of $\mathbb{M}$ acting on all the entries at once, and a closed subgroup of $\mathrm{O}(3)$ that maps $\mathbb{M}$ onto itself, acting by $f \mapsto \rho(\mathsf{R}) f(\mathsf{R}^{-1}\cdot)$, then u° has an extension in $\mathfrak{E}_n(\kappa)$ of order at most m that is invariant under this group.

(e) *(Contact terms.)* A distribution T on $\mathbb{M}^n$ with support in Δ lies in $\mathfrak{E}_n(\kappa)$ if and only if

$$T = \sum_{|\pi| \le n-1} \sum_{|\alpha| \le \omega_{|\pi|}} \partial_\xi^\alpha \delta_\pi \otimes T_{\pi,\alpha} \qquad \text{with} \quad T_{\pi,\alpha} \in \mathfrak{E}_{|\pi|}(\kappa + |\alpha|),$$

the sum over the partitions π with at most $n-1$ blocks and the multi-indices α of ξ at π. So two extensions of u° in $\mathfrak{E}_n(\kappa)$ differ by such a sum, and every such sum may be added to an extension. A term with $|\alpha| > \omega_{|\pi|}$ and a coefficient in $\mathfrak{E}_{|\pi|}(\kappa + |\alpha|)$ vanishes, and for every π and every α with $|\alpha| \le \omega_{|\pi|}$ there are nonzero examples when $V \neq \{0\}$. The coefficients in this representation need not be unique. If $\kappa > 3(n-1)$, the extension in $\mathfrak{E}_n(\kappa)$ is unique.

(f) *(Two points.)* Let $n = 2$, $\kappa = 0$, $V = \mathrm{Sym}_3(\mathbb{R})$ with $\rho(\mathsf{R})H = \mathsf{R}H\mathsf{R}^{\mathrm{T}}$, and let $T \in \mathfrak{E}_2(0)$ have its support in $\{x_1 = x_2\}$, be invariant under the translations, real, and symmetric under the exchange of the two entries. Then $T(f_1 \otimes f_2) = \sum_{|\alpha| \le 3} \int \mathsf{c}_\alpha(\partial^\alpha f_1(x), f_2(x))\,\mathrm{d}x$ with bilinear forms c_α on $V_{\mathbb{C}}$. If $\mathbb{M} = \mathbb{R}^3$ and T is covariant under $\mathrm{O}(3)$, the symbol $\hat{T}(p)(H, H') = \sum_\alpha \mathsf{c}_\alpha(H, H')(ip)^\alpha$ is a real combination of the six forms

$$\operatorname{tr} H \operatorname{tr} H', \quad \operatorname{tr} HH', \quad p^2 \operatorname{tr} H \operatorname{tr} H', \quad p^2 \operatorname{tr} HH', \quad (p \cdot Hp) \operatorname{tr} H' + (p \cdot H'p) \operatorname{tr} H, \quad (Hp) \cdot (H'p);$$

if T is covariant under the cubic group W_3, on $\mathbb{R}^3$ or on T_ℓ, then $\hat{T}$ is a real combination of twelve forms.

■

Proof sketch. (4) The facts on $\mathcal{Q}$: if Q is admissible, so is the product of the next generation that holds a given point; $|x_a - x_b|_\infty \ge \mathsf{d}_\infty(x) - 2h_j \ge 3h_j$ once $5h_j \le \mathsf{d}_\infty(x)$, so every $x \notin \Delta$ lies in exactly one member of $\mathcal{Q}$; if $h(Q) < h_0$ the parent of Q is not admissible, whence $\mathsf{d}_\infty < 10h(Q)$ on Q and two points in one block of $\pi(Q)$ are linked by a chain of centres at most $12h(Q)$ apart. Members of $\mathcal{Q}$ whose sets Q^+ meet have sides in the ratio $\frac12$, 1 or 2, and a point lies in at most $3 \cdot 2^{3n}$ sets Q^+; φ_Q is a normalised product of cutoffs of the scale $h(Q)$. A count used below: at most $\mathrm{Bl}(n)(24n)^{3n}(3R/h_j)^{3k}$ members of $\mathcal{Q}$ have the side h_j, k clusters and all the coordinates of their centres within $R \ge h_j$ of those of a given point of $\mathbb{M}^n$, a partition, one cube for each block and the cubes of the other indices within $12(n-1)h_j$ of it being chosen.

(a) Let $h = h(Q)$, $\mathsf{s} = \frac32 h$, $c = c(Q)$. A $G \in C_c^\infty(\operatorname{int} Q^+; V_{\mathbb{C}}^{\otimes n})$, continued periodically with the cell Q^*, is the sum of its Fourier series in the modes $\mathsf{E}_{\mathbf{k}}(x) = \prod_a e^{2\pi i \mathbf{k}_a \cdot (x_a - c_a)/\mathsf{s}}$, $\mathbf{k} \in \mathbb{Z}^{3n}$, with all derivatives, uniformly; with cutoffs $\chi_a \in C_c^\infty(\operatorname{int} Q_a^*)$, equal to 1 on Q_a^+, fixed once at the scale h, $\|\chi_a e^{2\pi i \mathbf{k}_a \cdot (\cdot - c_a)/\mathsf{s}}\|_{(h),r} \le C_r(1 + |\mathbf{k}_a|)^r$, and every mode times $\prod_a \chi_a$ is a product of entries carried by the cubes Q_a^*, of the side $\frac32 h$ and at pairwise sup distances $\ge \frac52 h$.

Put $u_Q(G) = \sum_{\mathbf{k}} \hat{G}_{\mathbf{k}}\, u(\chi_1 e^{2\pi i \mathbf{k}_1 \cdot (\cdot - c_1)/\mathsf{s}}, \dots, \chi_n e^{2\pi i \mathbf{k}_n \cdot (\cdot - c_n)/\mathsf{s}})$, componentwise in a basis of V. By (W) the terms are at most $Ah^\kappa l(h)^q C_r^n (1 + |\mathbf{k}|)^{nr} |\hat{G}_{\mathbf{k}}|$; by the Cauchy–Schwarz inequality, with $\sum_{\mathbf{k}} (1 + |\mathbf{k}|)^{-2s} < \infty$ for $2s > 3n$, and by Parseval, $\sum_{\mathbf{k}} (1 + |\mathbf{k}|)^{nr} |\hat{G}_{\mathbf{k}}| \le C \|G\|_{(h),N_0}$: this is the origin of $N_0 = \lfloor \frac{3n}{2} \rfloor + 1 + nr$. At a product $G = g_1 \otimes \cdots \otimes g_n$ the coefficients are products, the series of the entries converge in $\|\cdot\|_{(h),r}$, and $u_Q(G) = u(g_1, \dots, g_n)$.

A function carried by a product of open sets O_a is a limit in every C^N, within one compact subset of $\prod_a O_a$, of finite sums of products of functions carried by the O_a (truncated Fourier series times cutoffs); so two functionals that are bounded by a C^N norm and equal to u on products agree, which gives the uniqueness of u_Q and the agreement of u_Q, $u_{Q'}$ on $\operatorname{int} Q^+ \cap \operatorname{int} Q'^+$. Then $u^\circ(G) = \sum_Q u_Q(\varphi_Q G)$ is a finite sum on every compact subset of $\mathbb{M}^n \setminus \Delta$, a distribution; on a tuple with disjoint supports, a partition of unity of $\mathbb{M}$ by dyadic cubes of a side h_j with $6h_j$ below the

least distance of the supports and the multilinearity of u give $u^\circ = u$, the products of the cubes that meet the supports being admissible; and u° is unique by the same density.

(b) Four facts on P_Q, at $Q \in \mathcal{Q}$ with $h = h(Q)$, $\pi = \pi(Q)$, $k = k(Q) \le n-1$, $\omega = \omega_k \ge 0$. (P1) For $x \in Q^+$ and y' in the support of μ_Q, $|x_a - y'_{B(a)}|_\infty \le \frac{5}{8}h + 12(n-1)h + h \le 12nh$. (P2) $P_Q F = 0$ if F vanishes near Δ_π: it reads the derivatives of F on Δ_π alone. (P3) For $h \le \lambda$ and $|\gamma| \le N_0$, on Q^+, $h^{|\gamma|}|\partial^\gamma(F - P_Q F)| \le C(h/\lambda)^{\omega+1}\|F\|_{(\lambda),m}$: since $\int \mu_Q = 1$ and ∂^γ of the Taylor polynomial of order ω is the Taylor polynomial of order $\omega - |\gamma|$ of $\partial^\gamma F$, Taylor's formula with (P1) gives $(36n^2h)^{\omega+1-|\gamma|}\lambda^{-\omega-1}\|F\|_{(\lambda),\omega+1}$ for $|\gamma| \le \omega$, and for $|\gamma| > \omega$, $\partial^\gamma P_Q F = 0$. (P4) If F is λ-supported at z and $\lambda \le h$, then, in the coordinates at π and after an integration by parts in y', $h^{|\gamma|}|\partial^\gamma P_Q F| \le C\sum_{j\le\omega}(h/\lambda)^{j-3k}\|F\|_{(\lambda),\omega}$ on Q^+, because μ_Q and its derivatives are of the size h^{-3k} and $(\partial_\xi^\alpha F)\circ\iota_\pi$, of the size $\lambda^{-|\alpha|}$, is carried by a set of volume $\le (2\lambda)^{3k}$; and $P_Q F = 0$ unless $|c(Q)_a - z_a|_\infty \le 12nh$ for all a.

Now let $F \in C_c^m$ be carried by a ball of radius $R_F \ge h_0$. By (P3) at $\lambda = h_0$, (a), the count of (4) and $\|\varphi_Q G\|_{(h),N_0} \le C\|G\|_{(h),N_0}$ on Q^+, the terms of the members of the side h_j with k clusters add up to at most $C\,A\,\|F\|_{(h_0),m}((R_F + 12nh_0)/h_j)^{3k} h_j^\kappa l(h_j)^q (h_j/h_0)^{(\omega_k+1)_+}$; the exponent of h_j is $\varepsilon_k = \kappa - 3k + (\omega_k+1)_+ > 0$, equal to $\lfloor 3k-\kappa\rfloor + 1 - (3k-\kappa) \in (0,1]$ if $\omega_k \ge 0$ and to $\kappa - 3k$ if $\omega_k < 0$; for $k = n$ only $h_j = h_0$ occurs. So the series converges absolutely, $\tilde u$ is bounded by a C^m norm on every compact set, and by (P2) $\tilde u = u^\circ$ off Δ. At $\kappa = 0$, $r = 0$: $\omega_{n-1} + 1 = 3n - 2 \ge \lfloor \frac{3n}{2}\rfloor + 1$.

(c) Let F be λ-supported at z. Only finitely many members with $h(Q) > \lambda$ give a term, and

$$\tilde u(F) = \underbrace{\sum_{h(Q)\le\lambda} u_Q\big(\varphi_Q(F - P_Q F)\big)}_{\text{I}} + \underbrace{u^\circ\Big(\sum_{h(Q)>\lambda}\varphi_Q F\Big)}_{\text{II}} - \underbrace{\sum_{h(Q)>\lambda} u_Q(\varphi_Q P_Q F)}_{\text{III}}.$$

In I the members that give a term have their centres within $13n\lambda$ of z, and (P3) at the scale λ with the count gives $C\,A\,\lambda^\kappa\|F\|_{(\lambda),m}\sum_k\sum_{h_j\le\lambda}(h_j/\lambda)^{\varepsilon_k} l(h_j)^q$, which is at most $C\,A\,\lambda^\kappa l(\lambda)^q\|F\|_{(\lambda),m}$, as $l(2^{-i}\lambda) \le 2(1+i)l(\lambda)$. In II the function is smooth, carried where $\mathsf{d}_\infty \ge \frac{11}{4}\lambda$, with $\|\cdot\|_{(\lambda),N_0} \le C\|F\|_{(\lambda),N_0}$; a partition of unity by all the products of dyadic cubes of one side $h' \in (\frac{\lambda}{8}, \frac{\lambda}{4}]$ meets its support only in admissible products, at most 19^{3n} of them, and (a) bounds each term by $C\,A\,\lambda^\kappa l(\lambda)^q\|F\|_{(\lambda),N_0}$.

In III, by (P4), at most $(24n+1)^{3n}$ members of each side $h_j > \lambda$ give a term, each at most $C\,A\,\lambda^\kappa l(\lambda)^q (h_j/\lambda)^{\kappa+\omega_k-3k}\|F\|_{(\lambda),\omega_k}$; the exponent $\kappa + \omega_k - 3k$ is ≤ 0, and it is 0 exactly when $3k - \kappa$ is an integer; so the sum over $\lambda < h_j \le h_0$ is a convergent geometric series or has at most $1 + \log_2(h_0/\lambda) \le 2\,l(\lambda)$ equal terms: this is the one logarithm more. *A range of sides:* if (W) holds only at the sides $\ge h_\flat$ and $8h_\flat \le \lambda$, I loses its members below $h_\flat$ and was bounded term by term; III is unchanged; and II, now $\sum_{h(Q)>\lambda} u_Q(\varphi_Q F)$, is the sum over the products of the side $h' > \frac{\lambda}{8} \ge h_\flat$ of the partition of II, by the agreement in (a) at sides $\ge h_\flat$, and is bounded in the same way.

(d) The grids, so $\mathcal{Q}$ and $\pi(Q)$, and with covariant choices φ_Q, μ_Q, are invariant under the translations by $h_0\mathbb{Z}^3$, so $\tilde u$ is; the mean of $\tilde u(\tau_t\,\cdot)$ over $t \in [0,h_0)^3$ is invariant under all the translations, and the mean over the permutations and over the Haar measure of the closed subgroup of O(3) keeps this invariance. Each of these maps carries λ-supported functions to $\sqrt{3}\lambda$-supported ones and changes $\|\cdot\|_{(\lambda),m}$ by a factor formed from n and m; so it maps $\mathfrak{E}_n(\kappa)$ to itself with uniform numbers and extensions of the invariant u° to extensions, and means of extensions with one bound are such extensions.

(e) *If:* for λ-supported F the function $g = (\partial_\xi^\alpha F)\circ\iota_\pi$ on $\mathbb{M}^k$ is λ-supported, and $\|g\|_{(\lambda),m'} \le n^{m'}\lambda^{-|\alpha|}\|F\|_{(\lambda),m'+|\alpha|}$, as $\partial_{y_B} = \sum_{a\in B}\partial_{x_a}$; so $|T_{\pi,\alpha}(g)| \le C\lambda^{\kappa+|\alpha|}(1+\log\frac{h_1}{\lambda})^q\, n^{m'}\lambda^{-|\alpha|}\|F\|_{(\lambda),m'+|\alpha|}$, the power counting κ. A distribution $T' \in \mathfrak{E}_k(\kappa')$ with $\kappa' > 3k$ is 0: a test function on $\mathbb{M}^k$ is the sum of at most $C\lambda^{-3k}$ pieces carried by balls of radius λ, each of value $\le C\lambda^{\kappa'}$ up to logarithms, and $\lambda \to 0$; this is the case $|\alpha| > \omega_k$, where $\kappa + |\alpha| > 3k$.

The Lebesgue measure of $\mathbb{M}^k$ times a fixed nonzero element of the dual lies in $\mathfrak{E}_k(\kappa')$ exactly when $\kappa' \le 3k$, that is, at $\kappa' = \kappa + |\alpha|$, exactly when $|\alpha| \le \omega_k$; this supplies a nonzero example for every allowed pair (π, α) when $V \ne \{0\}$. *Only if:* a descent over $k = n-1, \dots, 1$ for $T \in \mathfrak{E}_n(\kappa)$, with its numbers C, q, m, h_1, carried by Σ_k. Off Σ_{k-1} the strata Δ_π with $|\pi| = k$ are disjoint; in the tube $U_\pi = \{|\xi|_\infty < \frac{1}{4}\min\{\mathsf{d}_\infty(y), h_1\}\}$ about the open stratum, which meets no other of them, and with a cutoff χ carried by $\{|\xi|_\infty < 1\}$ and equal to 1 on $\{|\xi|_\infty \le \frac{1}{2}\}$, the coefficients

$$t_{\pi,\alpha}(g) = (-1)^{|\alpha|}\, T\Big(\chi(\xi/\varepsilon)\,\frac{\xi^\alpha}{\alpha!}\,g(y)\Big), \qquad |\alpha| \le \omega_k, \quad 0 < \varepsilon \le \tfrac{1}{4}\min\Big\{\min_{\operatorname{supp} g} \mathsf{d}_\infty, h_1\Big\},$$

do not depend on ε and are distributions on $\mathbb{M}^k \setminus \Delta^{(k)}$ alone, $\Delta^{(k)}$ the diagonals of $\mathbb{M}^k$. *Claim A:* if G is carried by U_π and $\partial_\xi^\alpha G(y,0) = 0$ for $|\alpha| \le \omega_k$, then $T(G) = 0$; for $(1 - \chi(\xi/\varepsilon))G$ vanishes near Δ_π and is carried by U_π, which meets the support of T in the open stratum alone, so that $T(G) = T(\chi(\xi/\varepsilon)G)$ for small ε, and a partition of unity at the scale ε in y cuts $\chi(\xi/\varepsilon)G$ into at most $C\varepsilon^{-3k}$ pieces, each 2ε-supported with $\|\cdot\|_{(2\varepsilon),m} \le C\varepsilon^{(\omega_k+1)_+}$, so that $|T(G)| \le C\varepsilon^{\kappa-3k+(\omega_k+1)_+}$ times logarithms, which tends to 0.

Claim B: on $C_c^\infty(\mathbb{M}^n \setminus \Sigma_{k-1})$, by Taylor's formula in ξ and Claim A, $T = \sum_{|\pi|=k}\sum_{|\alpha|\le\omega_k} \partial_\xi^\alpha \delta_\pi \otimes t_{\pi,\alpha}$. The coefficients have a power counting of their own: for cubes R_B^* of the side $\frac{3}{2}h$ at pairwise sup distances $\ge \frac{5}{2}h$, $h \le \frac{1}{4}h_1$, and g carried by $\prod_B R_B^*$, the function $\chi(2\xi/h)\frac{\xi^\alpha}{\alpha!}g(y)$ is $2h$-supported with $\|\cdot\|_{(2h),m} \le Ch^{|\alpha|}\|g\|_{(h),m}$; so $t_{\pi,\alpha}$ is a functional of k entries with $(\mathrm{W}; \kappa + |\alpha|, q, m)$, with derivatives of the order m in the norms, which is why (a)–(c) are stated for every r.

For $k \ge 2$, (a)–(c) at $r = m$ extend $t_{\pi,\alpha}$ to $T_{\pi,\alpha} \in \mathfrak{E}_k(\kappa + |\alpha|)$; for $k = 1$ there are no diagonals in $\mathbb{M}^1$ and $T_{\pi,\alpha} = t_{\pi,\alpha}$ lies in $\mathfrak{E}_1(\kappa + |\alpha|)$ by the same inequality. By Claim B, $T - \sum_{|\pi|=k}\sum_{|\alpha|\le\omega_k} \partial_\xi^\alpha \delta_\pi \otimes T_{\pi,\alpha}$ vanishes off Σ_{k-1}, and it lies in $\mathfrak{E}_n(\kappa)$ by the first part; the descent ends at $\Sigma_0 = \varnothing$. If $\kappa > 3(n-1)$, all ω_k with $k \le n-1$ are negative and the sum is empty.

(f) At $n = 2$ the one stratum has $k = 1$, $y = x_1$, $\xi = x_2 - x_1$ and $\omega_1 = 3$; the coefficients T_α are translation-invariant distributions on $\mathbb{M}$, hence constants $\mathsf{c}_\alpha\, dy$, and an integration by parts gives the form; conversely a constant lies in $\mathfrak{E}_1(\kappa')$ for $\kappa' \le 3$. Covariance reads $\hat{T}(\mathsf{R}p)(\rho(\mathsf{R})H, \rho(\mathsf{R})H') = \hat{T}(p)(H, H')$, and the exchange $\hat{T}(-p)(H', H) = \hat{T}(p)(H, H')$. Since -1 lies in both groups and $\rho(-1) = 1$, $\hat{T}$ is even in p; so only the terms of the orders 0 and 2 in p occur, with real coefficients, $\hat{T}(p)$ is symmetric in (H, H'), and the number of the independent terms is that of the invariants of $\mathrm{Sym}^2 V^* \otimes \mathrm{Sym}^j(\mathbb{R}^3)^*$, $j = 0, 2$. Under $\mathrm{SO}(3)$, $V = [0] \oplus [2]$, $\mathrm{Sym}^2 V = 2[0] \oplus 2[2] \oplus [4]$ and $\mathrm{Sym}^2\mathbb{R}^3 = [0] \oplus [2]$: $2 + 4 = 6$; the six forms are invariant. Separating the homogeneous terms of degrees 0 and 2, their independence follows by evaluating at $H = H' \in \{\mathrm{diag}(1,-1,0), \mathrm{diag}(1,0,0)\}$ and $p \in \{e_1, e_2, e_3\}$ within each degree. Under the cubic group $V = A_1 \oplus E \oplus T_2$, $\mathrm{Sym}^2 V = 3A_1 \oplus 3E \oplus 3T_2 \oplus T_1$ and $\mathrm{Sym}^2\mathbb{R}^3 \cong V$: $3 + 9 = 12$; the characters give the same numbers, and 0 for the odd orders. □

Remark 3.7 (The point bound and the bounds of kernels, for data). For reconstruction data with labels the point bound gives pointwise bounds of the kernels by powers of the least distance of the points (Prop. 3.3 and Thm. E (d) (i)). The converse fails: at $V = \mathbb{R}$, $\sigma(f_1, \dots, f_n) = n! \prod_k \int f_k$ is symmetric and has constant kernels, hence every bound of the kind of (E1), but no bound of one insertion between two vectors. The reconstruction uses nothing at coinciding points.

Detailed sketch of tensor reconstruction and locality. Step 3: data (c). $(\mathrm{Rec1})_V$ is the composition of the translations ([7, Thm. C] (b)) with (a); $(\mathrm{Rec2})_V$ is [7, Thm. C] (b), the space being separable by the bounds and a countable dense family of tuples; $(\mathrm{Rec3})_V$ is [7, Thm. C] (e), or follows from (a) and $H \ge |P_1|$ by the unitary rotations about e_3, which turn $\mathcal{P}$; $(\mathrm{Rec4})_V$ is [7, Thm. C] (c). The classes of the C_c^1 tuples are dense, because a continuous entry is the uniform limit of its mollifications and σ is continuous under such limits; so the map of the two spaces is unitary. Since $g^{-2} \le b_*$, the

numbers do not depend on σ:

$$C_V = 2^{p_V} C_{p_V} b_*^{-1} \le 2^{p_V} C_{p_V} g^2.$$

Step 4: kernels, representation, locality (d). Prop. 3.3 (a) at $V = \mathrm{Sym}_3(\mathbb{R})$ applies to σ^1. Its clause (vii) gives the kernels, extended to continuous entries by the continuity of Step 3; for the bound at every point, a direction u with $|u \cdot (x_a - x_b)| \ge |x_a - x_b|/n^2$ for all pairs exists, and the bound of $\mathbf{S}_n$ in the tube of the frame of u at $\eta = \frac{1}{2}\min\{1, \mathsf{d}(x)\}/n^2$ gives

$$|\mathsf{s}_n(x;e)| \le \big(C_V(2n^2/\min\{1,\mathsf{d}(x)\})^{p_V}\big)^n \prod_a |e_a|;$$

integrating gives the bound of σ, and $\mathsf{S}_n(\varphi) = \int \mathsf{s}_n \varphi$ with the Taylor bound of φ near the diagonals gives the growth with the norm $|\cdot|_{(\lceil p_V\rceil+4)n}$. Its clauses (i)–(iii) and (v) give the representation, with $D(\Lambda)A = \hat\Lambda A \hat\Lambda^{\mathrm{T}}$, and (iv) the locality. The fields of Prop. 3.3 are the $T(f)$ of [7, Thm. C] (d): both are determined on $\mathcal{D}_\infty$ by $\langle \psi, T(f)\psi'\rangle = \int \lim_{s\downarrow 0} \langle U(x)^{-1}\psi, \mathsf{B}(s,s;f(x))U(x)^{-1}\psi'\rangle \,\mathrm{d}x$, which involves σ alone, and the totality used in Prop. 3.3 (a) (viii) is $\mathcal{H}^{\mathrm{ord}} = \mathcal{H}(\sigma)$, [7, Thm. C] (f).

Step 5: all couplings (e). Let $0 < g \le \gamma$. Extract a subsequence with converging labels, $b_j \to b_*$, and let z, g_1', g_2' be as in (e); $g_2' < g_1' \le \gamma_W^{\mathrm{sd}}$, and by [7, Lem. 2.14] (b) the label $L^z b_*$ lies in the interior of $I(g')$ for one $g' \in \{g_1', g_2'\}$. By the same Lemma the dilated data are, from an index on, first-passage data at g', [7, Thm. C] (b) at g' gives a further subsequence with a tensor limit σ', and

$$\sigma(X) = L^{zn}\sigma'(X^{(z)})$$

exists. The dilation commutes with O(3), carries mirrors to mirrors and $H^{\mathrm{cons}}[\eta]$ to $H^{\mathrm{cons}}[\eta^{[z]}]$; so (a), (b) and the conservation pass from σ' to σ, and by [7, Lem. 2.14] (b) $(\mathrm{Rec4})_V$ holds with $(3^{p_V} L^{4z} C_V[\sigma'], p_V, \frac{1}{2})$, where $C_V[\sigma'] \le 2^{p_V} C_{p_V}[g']\, g'^2$ by [7, Thm. C] (c) at g'; this is at most the number C_V of (e), which does not depend on the sequence. Then (c), (d) follow as in Steps 3, 4. □

Real-time conservation. The real-time conservation conclusion of [7, Thm. E(d)] follows from Lem. 3.5: the identity of the kernels gives, and is given by,

$$T_{0\nu}(\partial_0 F)\psi = \sum_{k=1,2} T_{k\nu}(\partial_k F)\psi \qquad (\psi \in \mathcal{D}_\infty).$$

□

Fields over a clustering component and their frame. Step 6: Wightman fields (c). By (b) (ii), σ_ξ^c is a tensor limit at g, so Thm. E (c), (d) apply to it with the same numbers; it has (Rec5) by (b) (iii) and the unitary map of (b) (ii). Prop. 3.3 (a) (i) with (Rec5) gives the uniqueness of the invariant vector; the Hermiticity is

$$\langle \psi, T(f)\psi'\rangle = \langle T(f^*)\psi, \psi'\rangle, \qquad f^* = \rho(\theta)\bar f,$$

under which a component with r indices equal to 3 is Hermitian after the factor i^r; the representation, the spectrum, the locality and the cyclicity are Thm. E (d) and [7, Thm. C] (f) for σ_ξ^c. The growth is [7, Thm. C] (h), whose numbers do not depend on the limit.

Step 7: the frame over a component (d). The frame is formed at the label matrices m_{ij} of the stress tensor, the labels of $\frac{1}{2}\delta_{ij}\operatorname{tr} T - T_{ij}$; (M1), (M2), (M3), (M5) are linear in the label. (M1) and the operator formula of (M2) are [7, Thm. C] (d) (vi) for σ_ξ^c, and the covariance of the kernels is Thm. E (d) (i) with $\rho(\mathsf{R})\mathsf{m}_{ij} = \sum_{i'j'} \mathsf{R}_{i'i}\mathsf{R}_{j'j}\mathsf{m}_{i'j'}$, at $\mathsf{s}_{ij}(x;y) = \mathsf{s}_{1+m}(x,y;\mathsf{m}_{ij},\mathbf{1},\dots,\mathbf{1})$. The scalar subspace reduces H, $\mathcal{U}$, $\mathcal{P}$, and it is invariant under the families $\mathcal{T}(\mathfrak{z})$ and $\mathsf{b}(w|\mathbf{1}|w')$, which are closed under the adjoint; so its projection Π_0 commutes with $\phi(f) = T(f\mathbf{1})$ on $\mathcal{D}_\infty$, and with $T_{ij}^E(F_1) = \Pi_0 T(F_1\mathsf{m}_{ij})\Pi_0$,

$$T_{ij}^E(F_1)\phi(f)\psi = \Pi_0 T(F_1\mathsf{m}_{ij})\phi(f)\psi = \Pi_0\phi(f)T(F_1\mathsf{m}_{ij})\psi = \phi(f)T_{ij}^E(F_1)\psi$$

at spacelike separation by the locality: (M3). The scalar subspace reduces $U(a,\Lambda)$ as well: the restriction of $(\sigma^c_\xi)^1$ to the tuples with entries in V_0 is reconstruction data with the label space V_0, whose space is the scalar subspace, and by Prop. 3.3 (a) (v) the span of Ω and of the vectors $\phi(f_1)\cdots\phi(f_n)\Omega$ is dense in it; as $\hat\Lambda\mathbf{1}\hat\Lambda^{\mathrm{T}} = \hat\Lambda\hat\Lambda^{\mathrm{T}} = \mathbf{1}$, Thm. E (d) (ii) gives $U(a,\Lambda)\phi(f)U(a,\Lambda)^{-1} = \phi(f(\Lambda^{-1}(\cdot - a)))$ on $\mathcal{D}_0$, and with $U(a,\Lambda)\Omega = \Omega$ every $U(a,\Lambda)$ maps this span onto itself; U being a unitary representation, the scalar subspace and its orthogonal complement are invariant, and Π_0 commutes with every $U(a,\Lambda)$.

So U restricts to a representation on the scalar subspace, and (M5) is Π_0 applied to the Lorentz covariance on Ω of Thm. E (d) (ii), with the passage from the Euclidean labels to $T_{\mu\nu}$ of Def. 3.1 (3). (M4) (a): conservation passes to σ^c_ξ by [7, Lem. 2.24] (b), so [7, Thm. E] (c), in the second form of [7, eq. 5] with ζ vanishing near the scalar points, gives $\sum_i \int \partial_i\zeta(x)\,\mathsf{s}_{ij}(x;y)\,\mathrm{d}x = \sigma^c_\xi(H^{\mathrm{cons}}[\zeta e_j], \text{the scalar entries}) = 0$; the scalar subspace has a unique vacuum, by (b) (iii); and Lem. 3.5 gives (M4) (a). At the labels $e_ie_j^{\mathrm{T}} + e_je_i^{\mathrm{T}} = \delta_{ij}\mathbf{1} - \mathsf{m}_{ij}$ the identity of the kernels fails: among one scalar entry their divergence is $\partial_{x_j}\mathsf{s}_2(x,y;\mathbf{1},\mathbf{1})$, which does not vanish identically by the free law (E6) of [7, Thm. G]; the components T_{ij} themselves are not conserved.

Step 8: the frame on $\mathcal{H}_0$ (e). For σ itself the same compression applies without the unique vacuum: (M1) is [7, Thm. C] (d) (vi), the covariant kernels come from Thm. E (d) (i), and (M3), (M5) are the compressions of Thm. E (d) (iii), (ii), as identities between scalar vectors, for instance

$$\langle\Omega, \phi(f_1)\cdots\phi(f_n)T[\Lambda t](F\circ\Lambda^{-1})\Omega\rangle = \langle\Omega, \phi(f_1\circ\Lambda)\cdots\phi(f_n\circ\Lambda)T[t](F)\Omega\rangle.$$

□

Proof of Prop. 3.2. The argument follows the proofs of Props. YM3_I:108.3, YM3_I:110.2 and YM3_I:110.1 for scalar data, with one entry carrying a label; its steps are these.

Step 1: the scalar part and one label. Let $e' \in V \setminus V_0$. For a tuple whose entries all lie in V_0, $f_1\mathbf{1} = f_1(\mathbf{1} - e') + f_1e'$ writes it as the sum of two tuples with exactly one entry outside V_0, at which $\sigma(\mathsf{R}X) = \sigma(X)$ is supposed; so the scalar part of σ is invariant under $\mathrm{O}(3)$, and with the translations it is invariant under all the isometries. With [7, Thm. C] (b), (c), (e), which give the positivity, the point bound and the circular cone, the space being separable by the bounds and a countable dense family of tuples, its restriction to C^1_c entries is scalar reconstruction data (Rec); Thm. YM3_I:AP gives this as well, as in Step 6 of the proof of [7, Thm. C]. The insertion and point operators with one tensor label and their energy bounds are those of [7, Thm. C] (d), and [7, Prop. 2.18] (c) gives the kernel $\mathsf{s}_{1+m}(x,y;e,\mathbf{1},\dots,\mathbf{1})$ of one tensor entry among m scalar entries, continuous and real-analytic at all non-coinciding points; by continuity, the supposed covariance at one label is $\mathsf{s}_{1+m}(\mathsf{R}x+v,\mathsf{R}y+v;\rho(\mathsf{R})e,\mathbf{1},\dots,\mathbf{1}) = \mathsf{s}_{1+m}(x,y;e,\mathbf{1},\dots,\mathbf{1})$ for $\mathsf{R}\in\mathrm{O}(3)$, $v\in\mathbb{R}^3$.

Step 2: one holomorphic function at one label; (M2). For a frame $\mathsf{R}_u \in \mathrm{SO}(3)$ with $\mathsf{R}_ue_3 = u$ and an order ς of the $1+m$ points along u, the piece on the tube of the configurations ordered along u in the order ς is the function $\mathbf{W}_{1+m}$ of [7, Thm. C] (d) (iii) written in the frame R_u at the label $\rho(\mathsf{R}_u)^{-1}e$; in the components of the label it is

$$\sum_{i',j'} (\mathsf{R}_u)_{ii'}(\mathsf{R}_u)_{jj'}\,\mathbf{W}_{1+m}(\mathsf{R}_u^{-1}\cdot; E_{i'j'}, \mathbf{1},\dots,\mathbf{1}),$$

$E_{i'j'}$ the symmetrised matrix units, that is $\rho(\mathsf{R}_u)\mathbf{W}_{1+m}(\mathsf{R}_u^{-1}\cdot)$ at the one label. At the real points of the intersection of two tubes, the configurations ordered in both frames, both pieces are equal to the kernel of Step 1, by the covariance at one label; the intersection is convex and these real points form an open subset of its real section, so the two pieces agree on it, and they form one holomorphic function on the union of the tubes. As $|(\mathsf{R}_u)_{ii'}| \le 1$, its bound is that of $\mathbf{W}_{1+m}$ times at most 9, the number of the terms (i',j'). Its restriction to the real non-coinciding configurations is the kernel; so the kernels $\mathsf{s}_{ij}(x;y) = \mathsf{s}_{1+m}(x,y;\mathsf{m}_{ij},\mathbf{1},\dots,\mathbf{1})$ are covariant at all non-coinciding points, and with the product formula of [7, Thm. C] (d) (vi) this is (M2).

Step 3: locality; (M3). The matrix elements of products of the fields between vectors of $\mathcal{D}_\infty$ are the boundary values of the one function of Step 2 from the tubes of the orders of the real times. For a pair of spacelike separated points, one of them carrying the label, the tubes of the two orders are joined by the tubes of the frames tilted by an angle of the order η towards the direction of the separation, on which the function is one and the same; apply the uniform smeared-bound argument in the proof of Prop. 3.3 (a) to the holomorphic kernel and polynomial bound of Step 2, with its single tensor label fixed and the remaining labels scalar. The Cauchy–Riemann identities and integration by parts give the same bound $2\eta Q_e(D'g)$ for the difference of the two orders, with $Q_e(D'g)$ independent of η; hence the boundary values agree as $\eta \to 0$. So $\langle \psi', [T(F_1 \mathsf{m}_{ij}), \phi(f)]\psi\rangle = 0$ at spacelike separation for ψ', ψ in the span of the vectors $e^{-\tau H}[F]$, $\tau > 0$, F scalar, which lie in $\mathcal{D}_\infty$, are dense in $\mathcal{H}_0$ and form a core of every power of $1 + H$ there (Lem. 3.5 (a), (b), at the span $\mathcal{V}_1$); the bound of (M1) and the commutation of Π_0 with $\phi(f)$ on $\mathcal{D}_\infty$, as in Step 7 of the proof of Thm. F, carry this to $T^E_{ij}(F_1)\phi(f)\psi = \phi(f)T^E_{ij}(F_1)\psi$ on $\mathcal{D}_0$: (M3).

Step 4: Lorentz covariance; (M5). The invariance of $\mathbf{W}_{1+m}$ at one label under the real rotations R_ζ in the plane of e_1, e_3, the label being turned by $\rho(\mathsf{R}_\zeta)$, continues holomorphically in the angle ζ to the imaginary angles $\zeta = is$, which act on the arguments, through ι, as the boosts of rapidity s in the plane of (t, x_1); the label is then moved by $\rho_{\mathbb{C}}(\mathsf{R}_{is})$, $\rho_{\mathbb{C}}$ the holomorphic continuation of $\rho|_{\mathrm{SO}(3)}$, that is by $e \mapsto \hat{\Lambda} e \hat{\Lambda}^{\mathrm{T}}$, and no bound on this factor is needed, the identity being one of holomorphic functions at fixed arguments. With the real rotations in the plane of e_1, e_2, which fix e_3, this gives the invariance of $\langle \Omega, \phi(f_1) \cdots \phi(f_m) T(F)\Omega\rangle$, the tensor entry at the last place, under $(f_k, F) \mapsto (f_k \circ \Lambda^{-1}, \hat{\Lambda} F(\Lambda^{-1}\cdot)\hat{\Lambda}^{\mathrm{T}})$ for all $\Lambda \in \mathcal{L}$.

The scalar data of Step 1 carry the representation U of $\mathbb{R}^{1+2} \rtimes \mathcal{L}$ on $\mathcal{H}_0$ with

$$U(\Lambda)\phi(f_1) \cdots \phi(f_m)\Omega = \phi(f_1 \circ \Lambda^{-1}) \cdots \phi(f_m \circ \Lambda^{-1})\Omega$$

(Prop. 3.3 (a) at V_0); these vectors are dense in $\mathcal{H}_0$, and the invariance reads $U(\Lambda)\Pi_0 T(F)\Omega = \Pi_0 T(\hat{\Lambda} F(\Lambda^{-1}\cdot)\hat{\Lambda}^{\mathrm{T}})\Omega$. At the label matrices m_{ij}, with the passage from the Euclidean labels to $T_{\mu\nu}$ of Def. 3.1 (3), this is (M5). □

The next two propositions use the generic conditions (D0)–(D4) of [7, Def. 6.1, p. 185], distinct from the clauses (D1)–(D5) of theorem C. Condition (D0) is $\sigma(\emptyset) = 1$, symmetry and n-linearity on $\prod_a C^1_c(O_a; V)$ when the open sets O_a have pairwise disjoint closures, with linear extension to combinations. Condition (D1) is the bound

$$|\sigma(H_1, \ldots, H_n)| \leq C_n(\delta, \varrho_0) \prod_a |H_a|_\infty$$

for $n \geq 1$, supports of sup diameter at most ϱ_0 and pairwise sup distance at least 2δ, uniformly in their positions; complex entries cost a factor 2^n. Conditions (D2) and (D3) are invariance under all translations and under W_3, respectively. Condition (D4) is positivity at (n, c) for every $c \in \mathbb{R}$ and every

$$n \in \mathfrak{N}_{18} = \{\pm e_\mu : 1 \leq \mu \leq 3\} \cup \{(\pm e_\mu \pm e_\lambda)/\sqrt{2} : \mu < \lambda\}.$$

Here o_{12} has entries 1 at $(1, 2)$ and $(2, 1)$ and zero elsewhere. If P_A, P_E, P_T are the projections onto scalar, traceless diagonal and off-diagonal matrices, then $Z_z = P_A + P_E + zP_T$; it multiplies just the off-diagonal labels by z.

Proposition 3.8 (Wightman fields from labelled functionals)**.** Let σ be a real functional on $\mathfrak{S}_V(\mathbb{R}^3)$ with (D0), (D1) and $\sigma((H)) = 0$, and suppose:

(R1) $\sigma \circ \iota = \sigma$ for every isometry ι of $\mathbb{R}^3$;
(R2) σ is positive at $(e_3, 0)$, and $\mathcal{H}(\sigma)$ is separable;
(R3) $\mathcal{P}(\{E < |q|\}) = 0$;
(R4) there are C_V and p with $|\sigma(\theta \bar{F} \cdot (h) \cdot F')| \leq C_V \eta^{-p} \|h\|_{L^1} \|[\tau_{-\eta e_3} F]\| \|[\tau_{-\eta e_3} F']\|$ for $\eta \in (0, 1]$, $F, F' \in \mathcal{D}_{e_3,\eta}$ and $h \in C^1_c(\{|x_3| < \eta/2\}; V)$.

Then:

(i) for $n \geq 2$ there is exactly one continuous function s_n on $\{x \in (\mathbb{R}^3)^n : x_i \neq x_j \ (i \neq j)\} \times V^n$, n-linear in e, with $\sigma(H_1, \dots, H_n) = \int s_n(x; H_1(x_1), \dots, H_n(x_n))\,\mathrm{d}x$ for all tuples, and for the tuples of continuous weights if σ is given there; it is real analytic, symmetric, invariant under the isometries acting on points and labels, and

$$|s_n(x;e)| \leq (cC_V)^n \max\{1, 2n^2/\delta(x)\}^{pn} \prod_k |e_k|_\infty, \qquad \delta(x) = \min_{i \neq j} |x_i - x_j|,$$

c a number depending on (V, ρ) and its norm; so $|\sigma(X)| \leq (cC_V)^n (2n^2/d)^{pn} \prod_a \|H_a\|_{L^1}$ if the supports are at mutual distances $\geq d$, $0 < d \leq 1$. For $\varphi : \mathbb{R}^{3n} \to V_{\mathbb{C}}^{\otimes n}$ whose 6^n components are Schwartz functions vanishing with all their derivatives where two of the n points coincide, $S_n(\varphi) = \int s_n \varphi$ converges absolutely and

$$|S_n(\varphi)| \leq (c' C_V)^n (n!)^{2p} |\varphi|_{k_E n}, \qquad k_E = \max\{\lceil p \rceil, 4\}, \quad c' = 6\pi^2 2^p e^{2p} 3^{k_E} c,$$

$|\varphi|_k$ the greatest of the norms $\sup(1 + |x|^2)^{k/2} |\partial^\alpha \varphi_\iota|$, $|\alpha| \leq k$, over the components; $S_n(H_1 \otimes \cdots \otimes H_n) = \sigma(H_1, \dots, H_n)$ at smooth entries with compact disjoint supports, and S_n is the only linear functional bounded for one of the norms with these values. The truncated functions have the kernels that the formula of the cumulants writes in the s_m, $m \leq n$.

(ii) Ω and the classes of the tuples whose entries lie in pairwise disjoint slabs $\{t < x_3 < t'\}$ are total in $\mathcal{H}(\sigma)$.

(iii) On $\mathcal{D}_\infty = \bigcap_k \operatorname{dom} H^k$ there are operators $T(f)$, $f \in \mathcal{S}(\mathbb{R}^{1+2}; V_{\mathbb{C}})$, linear in f, with $\langle \psi, T(f)\psi' \rangle = \int \lim_{s \downarrow 0} \langle U(x)^{-1}\psi, B(s, s; f(x)) U(x)^{-1}\psi' \rangle\,\mathrm{d}x$, B the point operators that (R4) provides, and $\langle \psi, T(f)\psi' \rangle = \langle T(\rho(\Theta_{e_3})\bar{f})\psi, \psi' \rangle$, so that a component with r indices equal to 3 is Hermitian after the factor i^r; and there is a strongly continuous unitary representation U of the Poincaré group of $\mathbb{R}^{1+2}$ extending $x \mapsto U(x)$, with $U\Omega = \Omega$, the spectrum of the translations in the closed forward cone, and $U(a, \Lambda) T(f) U(a, \Lambda)^{-1} = T(\rho_{\mathbb{C}}(\hat{\Lambda}) f(\Lambda^{-1}(\cdot - a)))$ on the span $\mathcal{D}_0$ of Ω and of the vectors $T(f_1) \cdots T(f_n)\Omega$.

Here $\hat{\Lambda} = \iota \Lambda \iota^{-1} \in \mathrm{SO}(3, \mathbb{C})$ for $\Lambda \in \mathrm{SO}_0(1, 2)$, with $\iota(t, \vec{x}) = (\vec{x}, -it)$: a time t sits at the Euclidean height $x_3 = -it$, as $U(t, \vec{x}) = \mathsf{E}((\vec{x}, -it))$; the boost of (t, x_1) by s goes to R_{is}, and a rotation $1 \oplus R$, $R \in \mathrm{SO}(2)$ acting on $\vec{x}$, to $R \oplus 1$; and $\rho_{\mathbb{C}}(\hat{\Lambda}) E = \hat{\Lambda} E \hat{\Lambda}^{\mathrm{t}}$, so that $\rho_{\mathbb{C}}(\mathsf{R}_{is}) = \exp(is\,\mathrm{d}\rho(E_{13} - E_{31}))$, $d\rho(\omega) E = \omega E + E\omega^{\mathrm{t}}$.

(iv) $T(f)T(f') = T(f')T(f)$ on $\mathcal{D}_0$ when the supports of f and f' are spacelike separated, all six components with all six; $\mathcal{D}_0$ is dense; the Wightman functions $\langle \Omega, T(f_1) \cdots T(f_n)\Omega \rangle$ are tempered and equal $\lim_{\epsilon \downarrow 0} \int \mathbf{W}_n(\iota(x_2 - x_1 + i\epsilon\alpha_1), \dots; f_1(x_1), \dots)\,\mathrm{d}x$ for all α_k in the forward cone, with the ι of (iii).

(v) The kernels $\mathbf{W}_n$ of the product formula are holomorphic on Tb^{n-1}, with $|\mathbf{W}_n(\zeta; e)| \leq (C_V \eta^{-p})^n \prod_k |e_k|_\infty$ for $e \in V^n$ where $\operatorname{Re} \zeta_3^k - |(\operatorname{Im} \zeta_1^k, \operatorname{Im} \zeta_2^k)| \geq 2\eta$ for all k, $0 < \eta \leq 1$; $\hat{\Lambda}$ of (iii) maps $\mathsf{Tb} = \iota(\mathbb{R}^{1+2} + iV_+)$, $V_+ = \{\alpha : \alpha^0 > |\vec{\alpha}|\}$, onto itself, and $\mathbf{W}_n(\hat{\Lambda}\zeta^1, \dots, \hat{\Lambda}\zeta^{n-1}; \rho_{\mathbb{C}}(\hat{\Lambda}) e_1, \dots, \rho_{\mathbb{C}}(\hat{\Lambda}) e_n) = \mathbf{W}_n(\zeta; e)$ for every $\Lambda \in \mathrm{SO}_0(1, 2)$. The s_n continue to one holomorphic function $\mathbf{S}_n$ on the union of the tubes $\{z \in \mathbb{C}^{3n} : R^{-1}(z_{\varsigma(k+1)} - z_{\varsigma(k)}) \in \mathsf{Tb},\ 1 \leq k < n\}$ of the real frames $R \in \mathrm{SO}(3)$ and the permutations ς, on the tube of (R, ς) equal to $\mathbf{W}_n(R^{-1}(z_{\varsigma(2)} - z_{\varsigma(1)}), \dots, R^{-1}(z_{\varsigma(n)} - z_{\varsigma(n-1)}); \rho(R)^{-1} e_{\varsigma(1)}, \dots, \rho(R)^{-1} e_{\varsigma(n)})$; $\mathbf{S}_n$ is symmetric when points and labels are permuted together, $\mathbf{S}_n(Rz_1 + a, \dots, Rz_n + a; \rho(R)e) = \mathbf{S}_n(z; e)$ for the real isometries $x \mapsto Rx + a$, $R \in \mathrm{O}(3)$, and $|\mathbf{S}_n(z; e)| \leq (cC_V \eta^{-p})^n \prod_k |e_k|_\infty$ for $e \in V^n$, c the number of (i), where for some (R, ς) the differences $R^{-1}(z_{\varsigma(k+1)} - z_{\varsigma(k)})$ have the margin 2η of the first bound.

If besides $\ker H = \mathbb{C}\Omega$, every vector invariant under the translations is a multiple of Ω, and all the Wightman axioms hold. ∎

Proof. Point operators and kernels. By (R4) the form $h \mapsto \sigma(\theta\bar F \cdot (h) \cdot F')$ is bounded in $\|h\|_{L^1}$; with the covariance under translations this gives point operators with $\|B(s,s;e)\| \le C_V s^{-p}|e|_\infty$ and the product formula of [7, Prop. 6.14](f), whose kernels are holomorphic on Tb^{n-1} by (R3), $\mathsf{E}(\zeta)$ being a contraction on $\overline{\mathsf{Tb}}$; the first bound of (v) is $\prod \|B(\eta,\eta;e_k)\|$. By (R1) the same holds in every frame $R \in \mathrm{SO}(3)$: there the heights are taken along Re_3, the kernel is $\mathbf{W}_n(R^{-1}\cdot;\rho(R)^{-1}e)$, the labels being turned back by $\rho(R)^{-1}$, and the bound holds with $\prod_k |\rho(R)^{-1}e_k|_\infty \le c^n \prod_k |e_k|_\infty$; $c = 3$ serves, as each row of R has $\sum_l |R_{kl}| \le \sqrt{3}$. The kernels of two frames, and of two orders of the points, agree at the common real points of their tubes, as the proof of (i) shows; the intersection of two tubes is convex and holds the real parts of its points, so they agree on it by the identity theorem, and they form the one function $\mathbf{S}_n$ of (v), symmetric as σ is. Its covariance under the real isometries holds at the real points by (i), and on each tube, which an isometry carries onto a tube, by the identity theorem. (i) For a unit vector v the map $u \mapsto u \cdot v$ carries the surface measure of S^2 to the uniform law on $[-1,1]$, so the band $\{|u \cdot v| < n^{-2}\}$ has the fraction n^{-2}; for pairwise different $x_1, \dots, x_n$ the u with $|u \cdot (x_i - x_j)| \ge n^{-2}|x_i - x_j|$ for all $i \ne j$ fill at least the fraction $1 - (n-1)/(2n) > \frac{1}{2}$ of the sphere. In a frame whose third axis is such a u the points are ordered in height with gaps at least $n^{-2}\delta(x)$, and the kernel of that frame is real analytic near x with the bound, at $\eta = \min\{1, \delta(x)/(2n^2)\}$. For two frames that both order x, both kernels are continuous on a product of small balls about the x_i and have the same integrals against all products $f_1(x_1)\cdots f_n(x_n)$; so they are equal.

The same reason gives uniqueness, symmetry and covariance. For a general tuple, cover the product of the supports by finitely many such products of balls, cut each entry by a partition of unity, and sum by the n-linearity on neighbourhoods with disjoint closures. For S_n: at a point x with $\delta(x) < 1$, φ is flat at the point where the two nearest points coincide, and Taylor's formula of order $m = \lceil pn \rceil$ along the segment gives $\max\{1, 2n^2/\delta(x)\}^{pn}|\varphi(x)| \le (2n^2)^{pn} 3^k |\varphi|_k (1+|x|^2)^{-k/2}$, $k = k_E n \ge 3n+1$; $\int (1+|x|^2)^{-k/2}\,\mathrm{d}x \le \pi^{2n}$ and $n^n \le e^n n!$. (ii) The vectors $\Phi(y)$ of n points of the half space have the Gram kernel s_{2n} at $(\theta y, y')$, real analytic by (i); so for ψ orthogonal to the classes of (ii), $y \mapsto \langle \psi, \Phi(y)\rangle$ is real analytic on the connected set of the non-coinciding n-tuples of the half space, and vanishes where the heights are ordered. (iii), (iv), (v) The fields are built from the point operators and $U(x)$, and the Wightman functions are the boundary values of the $\mathbf{W}_n$ from every direction α of the open cone V_+, tempered by the power p; this follows the reconstruction of Thm. YM3_I:AO, with labels; (ii) gives cyclicity, and $B(s,s';e)^* = B(s',s;\rho(\Theta_{e_3})e)$ gives the adjoints. *The invariance of the* $\mathbf{W}_n$ *in (v).* $\hat\Lambda\iota(x+i\alpha) = \iota(\Lambda x + i\Lambda\alpha)$, and $\Lambda V_+ = V_+$: so $\hat\Lambda\mathsf{Tb} = \mathsf{Tb}$. For a rotation $\Lambda = 1 \oplus R$ of $\vec x$, $\hat\Lambda = R \oplus 1$ is a real rotation that keeps the heights; at real $\zeta \in \mathsf{Tb}^{n-1}$, that is at configurations ordered in height, $\mathbf{W}_n = s_n$, and the covariance of s_n gives the identity, which the identity theorem carries to Tb^{n-1}.

For the boost $B(s)$ of (t, x_1), $\hat\Lambda = \mathsf{R}_{is}$: fix a real $\zeta \in \mathsf{Tb}^{n-1}$ and e; there is $\delta > 0$ such that $\mathsf{R}_\tau \zeta^k \in \mathsf{Tb}$ for all k and all complex τ at a distance $< \delta$ from the segment $\{is' : |s'| \le |s|\}$, a convex open set on which $\gamma(\tau) = \mathbf{W}_n(\mathsf{R}_\tau\zeta^1, \dots, \mathsf{R}_\tau\zeta^{n-1}; \rho_{\mathbb{C}}(\mathsf{R}_\tau)e_1, \dots, \rho_{\mathbb{C}}(\mathsf{R}_\tau)e_n)$ is holomorphic; at real τ, $|\tau| < \delta$, the turned points are still ordered in height, and the covariance of s_n under the real rotation R_τ gives $\gamma(\tau) = \gamma(0)$; so γ is constant, and at $\tau = is$ this is the identity at real ζ, which the identity theorem carries to Tb^{n-1}.

The boosts $B(s)$ and the rotations $1 \oplus R$ generate $\mathrm{SO}_0(1,2)$, and $\Lambda \mapsto \hat\Lambda$ is a homomorphism: the identity holds for every Λ. *(iii).* By the substitution $x_k = \Lambda y_k$ and $\iota(\Lambda y + i\epsilon\alpha) = \hat\Lambda\iota(y + i\epsilon\Lambda^{-1}\alpha)$, with $\Lambda^{-1}\alpha \in V_+$, the invariance of the $\mathbf{W}_n$ makes the Wightman functions invariant under $f_k \mapsto \Lambda_a f_k = \rho_{\mathbb{C}}(\hat\Lambda) f_k(\Lambda^{-1}(\cdot - a))$, the translations being had already; as $\bar\iota = \Theta_{e_3}\iota$ and Λ is real, $\rho_{\mathbb{C}}(\hat\Lambda)$ commutes with $e \mapsto \rho(\Theta_{e_3})\bar e$, so the adjoints of (iii) go over too, and $U(a,\Lambda)T(f_1)\cdots T(f_n)\Omega := T(\Lambda_a f_1)\cdots T(\Lambda_a f_n)\Omega$ is well defined and isometric on $\mathcal{D}_0$, which is dense; it extends to a unitary representation, strongly continuous as the fields are continuous in f, which fixes Ω and has the law of (iii) on $\mathcal{D}_0$.

(iv). As $\mathcal{D}_0$ is dense it suffices that a Wightman function does not change when two neighbours f_k, f_{k+1} with spacelike separated supports are exchanged, first for f_j of compact support, to which cut-offs that do not enlarge the supports reduce the general case. Let x run through the product of the supports. At each x let $\hat\xi$ be the unit vector of $\vec x_{k+1}-\vec x_k$, and tilt e_3 towards it: $u_\vartheta = \cos\vartheta\, e_3 + \sin\vartheta(\hat\xi,0)$, $\sin\vartheta = A\eta$, a real unit vector. The points $z_j = \iota(x_j) + \eta h_j^\lambda e_3$, $|\lambda|\le 1$, $0<\eta\le\eta_0$, with heights h_j^λ formed from the supports alone, increasing with j at $\lambda=1$ and in the order with $k,k+1$ exchanged at $\lambda=-1$, lie in the tube of the real frame with third axis u_ϑ, its differences having the margin 3η: across the pair $k,k+1$ the spacelike separation, $\hat\xi\cdot(\vec x_{k+1}-\vec x_k) > |t_{k+1}-t_k|$, pays for the exchanged heights, and $|\operatorname{Im}\mathfrak{z} - (u_\vartheta\cdot\operatorname{Im}\mathfrak{z})u_\vartheta| = |t_{j+1}-t_j|\sin\vartheta$ for $\mathfrak{z} = z_{j+1}-z_j$.

There $\mathbf{S}_n$, symmetric in the points with their labels, is bounded by $(cC_V\eta^{-p})^n\prod_k|e_k|_\infty$ by (v). The integral $I(\lambda,\eta)$ of $\mathbf{S}_n(z;f_1(x_1),\dots,f_n(x_n))$ over x has $\partial_\eta I$ and $\partial_\lambda I/\eta$ equal to integrals of the same kind with derivatives of the f_j in the times, after an integration by parts. The one-label smoothing and integration-by-parts argument in the proof of Prop. 3.3, applied to each tensor component, gives Taylor's formula with the integral remainder and bounds all of them uniformly in $\eta\in(0,\eta_0]$. So $|I(1,\eta)-I(-1,\eta)|\le C\eta\to 0$; and $I(1,\eta)$, $I(-1,\eta)$ tend to the Wightman functions of the two orders, the labels being permuted with the points. *The last sentence.* A vector ψ invariant under the translations has $e^{itH}\psi = U(t,\vec 0)\psi = \psi$ for every real t, by (iii), so $H\psi=0$ by Stone's theorem; here $\ker H = \mathbb{C}\Omega$ enters and gives $\psi\in\mathbb{C}\Omega$.

With this uniqueness of the vacuum, (iii), (iv) and the separability of $\mathcal{H}(\sigma)$ in (R2) give all the Wightman axioms. □

Proposition 3.9 (Rescaling the off-diagonal labels)**.** Let σ satisfy (D0)–(D4), $z\ne 0$ real, and $\sigma^z = \sigma\circ Z_z$. Then σ^z satisfies (D0)–(D4), with $\max\{1,|z|\}^n C_n(\delta,\varrho_0)$ in (D1), and has the trace part of σ; $[F]_{\sigma^z}\mapsto[Z_zF]_\sigma$ is a unitary map of $\mathcal{H}(\sigma^z)$ onto $\mathcal{H}(\sigma)$ that commutes with the translations and the semigroup, under which the slab and point operators and the kernels of σ^z are those of σ with Z_z applied to the labels. If $\sigma(\varphi_1E,\varphi_2E)\ne 0$ for $E=o_{12}$ and some φ_1,φ_2 with disjoint supports, and $z^2\ne 1$, then σ and σ^z are not both covariant at order 2. For a tensor limit at $g\le\gamma_W^{\mathrm{sd}}$ the two-point law of [7, Prop. 6.15], at all pairs of weights, holds for σ^z with some label $b'>0$ in the place of b if and only if $z=1$; at the pairs of off-diagonal weights alone it holds with the label b of σ if and only if $z^2=1$. ■

Proof. Z_z is the identity on V_A and commutes with $\rho(W_3)$, in particular with $\rho(\Theta_n)$ for $n\in\mathfrak{N}_{18}$; this gives the first part. Let R be the rotation by $\pi/4$ about e_3, so that $\rho(R)E = E_{22}-E_{11}\in V_E$. If both were covariant at order 2, then

$$z^2\sigma(\varphi_1E,\varphi_2E) = \sigma^z(\varphi_1E,\varphi_2E) = \sigma^z(R(\varphi_1E),R(\varphi_2E)) = \sigma(R(\varphi_1E),R(\varphi_2E)) = \sigma(\varphi_1E,\varphi_2E).$$

For the law: at $z=1$, $\sigma^z=\sigma$. Let σ^z obey the law with an error $\epsilon'(d)\to 0$, at all pairs with a label b', or at the pairs of off-diagonal weights alone with the label b. In the first case, at the pairs of diagonal weights, which Z_z leaves alone, its label is that of σ: there $\sigma^z=\sigma$, and at $(\varphi_1^d\mathbf{1},\varphi_2^d\mathbf{1})$, with the bumps of [7, Prop. 6.15](a), the kernel of $\mathcal{W}$ is $\frac{3}{2}\operatorname{tr}K_3(u)^2 = 9(4\pi|u|^3)^{-2}>0$, $u=x-y$, so that $\mathcal{W}(\varphi_1^d\mathbf{1},\varphi_2^d\mathbf{1})$ is a positive number independent of d, and b'^{-2} and b^{-2} are both the limit of $\sigma(\varphi_1^d\mathbf{1},\varphi_2^d\mathbf{1})/\mathcal{W}(\varphi_1^d\mathbf{1},\varphi_2^d\mathbf{1})$ as $d\downarrow 0$: $b'=b$.

With the label b, in both cases, at off-diagonal entries $\sigma^z(H_1,H_2) = z^2\sigma(H_1,H_2)$, and $|z^2-1|\,b^{-2}|\mathcal{W}(H_1,H_2)|\le(z^2\epsilon_g(d)+\epsilon'(d))b^{-2}|H_1|_\infty|H_2|_\infty$ for all small d forces $z^2=1$ by the lower bound of $\mathcal{W}$ ([7, Prop. 6.15](a)); conversely at $z=\pm1$ these pairs are unchanged. At the mixed pair $H_1=\varphi_1^d\mathbf{1}$, $H_2=\varphi_2^d o_{12}$, with the bumps of [7, Prop. 6.15](a) placed with $x_2-x_1=\frac{7}{2}d(e_1+e_2)/\sqrt{2}$, one has $\sigma^z(H_1,H_2) = z\,\sigma(H_1,H_2)$, and the kernel of $\mathcal{W}$ is $\frac{3}{2}\operatorname{tr}[K_3(u)o_{12}K_3(u)] = 3(K_3(u)^2)_{12} = 9\hat u_1\hat u_2(4\pi|u|^3)^{-2}$, with $K_3(u) = (3\hat u\hat u^{\mathrm t}-\mathbf{1})(4\pi|u|^3)^{-1}$, $\hat u = u/|u|$, u the difference of the two points; on the supports $\hat u$ lies within the angle $\arcsin\frac{2}{7}<\frac{\pi}{4}$ of $\pm(e_1+e_2)/\sqrt{2}$, so $\hat u_1\hat u_2>0$, and $\mathcal{W}(H_1,H_2)=w_\times>0$ does not depend on d, $\mathcal{W}$ being invariant under dilations.

So $|z-1|\, b^{-2} w_\times \le (|z|\epsilon_g(d) + \epsilon'(d)) b^{-2} |\varphi|^2_\infty$ for all small d, and $z = 1$. This is [7, Prop. 6.15](d) at $a = e = 1$. □

Proposition 3.10 (Reconstruction with values in V)**.** Let σ be a real symmetric functional on $\mathfrak{S}(\mathbb{R}^3)$, multilinear at disjoint supports, with $\sigma(\emptyset) = 1$, invariant under all isometries acting on points and indices, positive at $\{x_3 = 0\}$, with separable $\mathcal{H}(\sigma)$, with $H, \mathcal{U}, \mathcal{P}$ as in [7, Thm. BW](a), the circular cone $\mathcal{P}(\{E < |q|\}) = 0$, and [7, eq. 263] for some C_V and p_V. Then: (i) there is exactly one continuous s_n at non-coinciding points, n-linear in the labels, with $\sigma(X) = \int s_n(x; H_1(x_1), \dots, H_n(x_n))\, \mathrm{d}x$ for all X; it is real-analytic, symmetric and invariant, and $|s_n(x; e)| \le (cC_V)^n \max\{1, 2n^2/\delta(x)\}^{p_V n} \prod_k |e_k|$; (ii) the fields $T(f)$ of [7, Thm. BW](e) are defined on $\mathcal{D}_\infty$, with the bound and the adjoint stated there, covariant under the translations, and their vacuum expectations are tempered, the boundary values of the kernels continued to the forward tube; (iii) Ω and the classes of the tuples in disjoint slabs are total in $\mathcal{H}(\sigma)$, and $\mathcal{D}_0$ is dense; (iv) there is a strongly continuous unitary representation of $\mathbb{R}^{1+2} \rtimes \mathcal{L}$ on $\mathcal{H}(\sigma)$ as in [7, Thm. BW](e), acting on the labels by $D(\Lambda)e = \hat\Lambda e \hat\Lambda^t$, with spectrum in the closed forward cone; (v) the fields commute at spacelike separation.

Nothing is supposed or said of $\ker H$; if $\ker H = \mathbb{C}\Omega$, all Wightman axioms hold. ■

Proof sketch. It is the proof of Thm. YM3_I:AO with values in V. The ordered kernels $\mathbf{W}_n$ of [7, Prop. C.18](c) continue to the circular tube, and [7, eq. 263] gives $|\mathbf{W}_n| \le (C_V \eta^{-p_V})^n \prod |e_k|$ at distance 2η from its boundary, with no factorial. For n distinct points more than half of the directions u have $|u \cdot (x_a - x_b)| \ge n^{-2} |x_a - x_b|$ for all $a \ne b$ (zones of Archimedes); rotating such a u to e_3, the frames in $SO(3)$ glue the ordered kernels into one continuous, real-analytic function s_n, and the bound at $\eta = \min\{1, \delta(x)/(2n^2)\}$ is (i). The limits $\lim_{s\downarrow 0} \langle \psi, \mathsf{B}(s, s; e)\psi' \rangle$ exist for $\psi, \psi' \in \operatorname{dom} H^l$, $l > p_V$, the power being paid by $(1 + H)^l$; this defines $T(f)$, tempered, with $\langle \Omega, T(f_1) \cdots T(f_n)\Omega \rangle$ the boundary values of the $\mathbf{W}_n$ at ι of the real points, from the directions of the forward cone.

The totality (iii) follows from (i): for $X \in \mathfrak{S}(\{x_3 > 0\})$ cut every entry by a smooth partition of unity in x_3 of mesh ϵ; the terms in which two entries lie in one slab or in neighbouring slabs form a remainder R_ϵ, and $\|[R_\epsilon]\|^2 = \sigma(\theta \bar R_\epsilon \cdot R_\epsilon)$ is, by (i), an integral of s_{2n} against the weights of $\theta \bar R_\epsilon$ and R_ϵ, where s_{2n} is continuous, so bounded, on the compact product of the supports of the entries of θX and of X, which are pairwise disjoint, while the weights of R_ϵ are carried by a set of measure $O(\epsilon)$; so $[R_\epsilon] \to 0$, and $[X]$ is a limit of sums of classes of tuples in disjoint slabs.

A vector orthogonal to $\mathcal{D}_0$ gives functions holomorphic in the forward tube whose boundary values vanish; their values at imaginary times are its scalar products with the classes of the tuples in disjoint slabs, so by the totality it is 0, and $\mathcal{D}_0$ is dense. The complex rotations R_{is} of the plane (x_3, x_1) are the boosts: invariance gives exact norms along them, so they are unitary on the closed span of the point vectors, which is $\mathcal{H}(\sigma)$ by (iii), and with the translations and the rotations about e_3 they generate $U(a, \Lambda)$ on $\mathcal{H}(\sigma)$, with $U(a, \Lambda)\mathcal{D}_0 = \mathcal{D}_0$; D is a continuous homomorphism of $\mathcal{L}$ into $GL(V_\mathbb{C})$, $\hat\Lambda^t \hat\Lambda = \mathbf{1}$, with $D(\Lambda)e^* = (D(\Lambda)e)^*$ for $e^* = \Theta_{e_3} \bar e \Theta_{e_3}$, as $\overline{\hat\Lambda} = \Theta_{e_3} \hat\Lambda \Theta_{e_3}$.

Locality: the kernels continue to tubes tilted by an angle of the order of η, and the symmetry of s_n under permutations of points and labels gives the commutation at spacelike separation, all components with all. □

Proposition 3.11 (Fields and kernels from the square cone)**.** Let $0 < g \le \gamma_W^{\mathrm{sd}}$ and let σ be a limit at g. [7, Thm. BV](a)–(c) and [7, eq. 263], with the square cone $\mathcal{P}(\{E < \max_k |q_k|\}) = 0$ in the place of the circular one, give the following, independently of the invariance under $O(3)$: the part of [7, Thm. BW](d), (e) that rests on the square cone, the one step at which the rotations enter Lorentz covariance, and bounded slab operators for entries side by side. (a) There are bounded operators $\mathsf{b}(w|e|w')$, for w, w' in the open square tube and $e \in V_\mathbb{C}$, linear in e and holomorphic in norm, with $\|\mathsf{b}(w|e|w')\| \le \sqrt{2} C_V s^{-p_V} |e|$ for $0 < s \le \min\{1, d_\square(w), d_\square(w')\}$, $d_\square(z) = \operatorname{Re} z_3 - |\operatorname{Im} z_1| - |\operatorname{Im} z_2|$, and $\mathsf{B}_{2\eta}((\tau_{\eta e_3} h)) = \int \mathsf{b}(y + \eta e_3 | h(y)| - y + \eta e_3)\, \mathrm{d}y$; and $|\mathbf{W}_n(z; e)| \le (C_V \eta^{-p_V})^n \prod_k |e_k|$ where

$d_\square(z^k) \geq 2\eta$, $0 < \eta \leq 1$. (b) The fields $T(f)$ of [7, Thm. BW](e) are defined on $\mathcal{D}_\infty$, with the same bound and adjoint, covariant under the translations and the dihedral group of order eight that fixes e_3, with joint spectrum in $\{E \geq \max_k |q_k|\}$ and Ω cyclic in $\mathcal{H}(\sigma)$; their vacuum expectations are tempered and are the boundary values of the $\mathbf{W}_n$ from every direction of $\{\alpha^0 > |\alpha^1| + |\alpha^2|\}$; and they commute in the square form: $T(f)T(f') = T(f')T(f)$ on $\mathcal{D}_0$ whenever $\max_k |x^k - x'^k| > |t - t'|$ on the supports. Lorentz covariance needs the rotations at one step only: for one one-parameter group r_t of rotations, the identity $\frac{d}{dt}\mathbf{W}_n(r_t\xi; r_t e_1 r_t^t, \dots, r_t e_n r_t^t)\big|_{t=0} = 0$ at all points ξ and labels e is equivalent to invariance under $O(3)$. (c) For $m \geq 0$ the kernel $s(x; y_1, \dots, y_m; e) = \lim_{\epsilon \downarrow 0} \sigma(\delta_\epsilon^x e, \delta_\epsilon^{y_1}\mathbf{1}, \dots, \delta_\epsilon^{y_m}\mathbf{1})$ of one tensor entry among m action densities, δ_ϵ^z a bump of width ϵ at z, exists at all non-coinciding points, is real-analytic there and represents σ at such tuples; where for one direction μ the coordinate x_μ differs from all $y_{k,\mu}$, it is the matrix element of a point operator between two point vectors of the scalar theory, the points below and above x in that direction. The trace part is data of Thm. YM3_I:AO, and its point vectors exist and are norm continuous at all non-coinciding points of the half space. (d) For $l > 0$ and a tuple $Y = (h_1, \dots, h_n) \in \mathfrak{S}(\{0 < x_3 < l\})$, its entries side by side or not, let $d(Y) = \min\{1, \operatorname{dist}(\operatorname{supp} Y, \{x_3 = 0\} \cup \{x_3 = l\}), \min_{a \neq b} \operatorname{dist}(\operatorname{supp} h_a, \operatorname{supp} h_b)\}$. The slab operator $\mathsf{B}_l(Y)[F] = [Y \cdot \tau_{l e_3} F]$ is bounded,

$$\|\mathsf{B}_l(Y)\| \leq \mathsf{K}_n \big(\sqrt{2} C_V (3/d(Y))^{p_V}\big)^n \prod_a \|h_a\|_{L^1}, \qquad \mathsf{K}_n = (10n)^{\lceil p_V n \rceil n(n-1)/2};$$

and $\sigma(X) = \int s_n(x; h_1(x_1), \dots, h_n(x_n))\, \mathrm{d}x$ for every $X = (h_1, \dots, h_n) \in \mathfrak{S}(\mathbb{R}^3)$, with one function s_n continuous at all non-coinciding points, given at points of pairwise different heights by $\mathbf{W}_n$ at the points ordered in x_3, and $|s_n(x; e)| \leq \mathsf{K}_n (\sqrt{2} C_V (3/d_\infty(x))^{p_V})^n \prod_a |e_a|$, $d_\infty(x) = \min\{1, \min_{a \neq b} |x_a - x_b|_\infty\}$. ■

Proof sketch. (a) As in [7, Prop. C.18](c), with [7, eq. 263] in the place of the finite function C_{pt}: covariance moves the point of the plate into the exponentials of the spectral measure, which are contractions on the square tube by the square cone, and the bound is [7, eq. 263] at the distance $d_\square$. (b) As in the proof of Prop. 3.10, with the square tube for the circular one: boundary values as tempered distributions from the directions of the square cone, the translations and the dihedral group of W_3 fixing e_3 acting, and [7, Prop. C.18](b) for the cyclicity. The commutation in the square form: by a partition of unity of f and f' into compact pieces it suffices to treat two pieces, and for them to fix a horizontal direction k with $|x^k - x'^k| - |t - t'| \geq c > 0$ on their supports.

First, at a configuration of n points ordered in x^k with $|\Delta x^k| > |\Delta t|$ for every pair, $\iota(t, \vec{x}) = (\vec{x}, -it)$, the points $\iota(x) + \eta h e_3$, $h = (h_1, \dots, h_n)$ increasing along an order of the points, lie in the e_3-tube of that order for $\eta > 0$, for either order of the two entries that are exchanged, and inside the e_k-tube with a margin free of η, as $\operatorname{Re} \Delta z_k = \Delta x^k > |\Delta t| = |\operatorname{Im} \Delta z_3|$; the tubes are convex and have a common real open set, on which σ has one kernel, so the two boundary values as $\eta \downarrow 0$ are the one function of the picture e_k at $\iota(x)$. Second, $F(a, b) = \langle T(g_1(\cdot - b_1)) \cdots T(g_m(\cdot - b_m))\Omega, [T(f), T(f')] T(f_1(\cdot - a_1)) \cdots T(f_{m'}(\cdot - a_{m'}))\Omega \rangle$ is continuous where $\operatorname{Im} a_1$, the $\operatorname{Im}(a_{i+1} - a_i)$, $\operatorname{Im} b_1$ and the $\operatorname{Im}(b_{i+1} - b_i)$ lie in the closed cone $\{\alpha^0 \geq |\alpha^1| + |\alpha^2|\}$, dual to the square cone, and holomorphic in $(a, \bar{b})$ inside, by the spectrum and the bounds of T on $\mathcal{D}_\infty$; it vanishes on the open set of the real shifts far along e_k, by the first step; the edge-of-the-wedge theorem against 0 gives $F = 0$, and Schwartz f, f' follow by density.

The boosts are the complex rotations, and their covariance is the analytic continuation of the rotation covariance from one infinitesimal generator. (c) The trace part has the bounds, the invariance and the positivity required by Thm. YM3_I:AO, whose reconstruction gives the point vectors $\Phi_n(y)$ of the scalar theory. For (x, y) with $x_\mu \neq y_{k,\mu}$ for all k cut by two planes $x_\mu = c_\pm$ next to x_μ, and write the kernel as $\langle \Phi(\text{reflected points below}), \mathcal{U}\, \mathsf{B}_e(x_\mu - c_-, c_+ - x_\mu)\, \mathcal{U}^{-1} \Phi(\text{points above}) \rangle$ in the picture e_μ; the three pictures agree where they overlap, by the uniqueness of continuous kernels.

The remaining points form a null set of codimension three; there holomorphy in each coordinate separately with one bound gives joint holomorphy, by Bernstein's estimate of the Chebyshev coefficients, so the kernel is real-analytic at all non-coinciding points. (d) First the continuation and the bound, then the integral. By the symmetry of σ the class $[Y \cdot \tau_{le_3} F]$ does not depend on the order of the entries. At points x of pairwise different heights let $\mathsf{G}_l(x;e)$ be the product of the operators $\mathsf{b}(w|e|w')$ of (a), taken in the order of the heights, whose arguments are the half gaps of the heights and the distances to the faces of the slab; the spectators enter as two vectors $\mathcal{U}(v)[F']$, $\mathcal{U}(w)[F]$, and $f(x;v,w) = \langle \mathcal{U}(v)[F'], \mathsf{G}_l(x;e)\mathcal{U}(w)[F]\rangle$ is analytic in (v,w).

Take a pair a,b of entries with $|x_a^k - x_b^k| \geq d = d(Y)$ in a horizontal direction k, and let ζ be their relative height, the mean height and the other points being fixed: the two orders of the pair give functions $f_\pm(\zeta;v,w)$, holomorphic in $\pm \mathrm{Re}\,\zeta > 0$, with $|f_\pm| \leq \mathsf{M}(|\mathrm{Re}\,\zeta|/2)^{-\mathsf{P}}$ by (a), P the power of the bound of (a) in the half gap of the pair ($\mathsf{P} = 2p_V$ at $n = 2$). For spectators shifted far along e_k, planes $x_k = \text{const}$ part them from the entries; in the picture e_k all is then ordered in x_k, ζ is a translation along the mirror, and the kernel is holomorphic in $|\mathrm{Im}\,\zeta| < |x_a^k - x_b^k|$, equal to f_+ and to f_- at the real ζ of either sign, by the symmetry of σ and the uniqueness of continuous kernels.

For such spectators the Cauchy integral over $|\zeta| = r < d$ of $(\zeta^2 + r^2)^{\mathsf{N}} f(\zeta;v,w)$, $\mathsf{N} = \lceil \mathsf{P} \rceil$, reproduces $(\zeta_0^2 + r^2)^{\mathsf{N}} f(\zeta_0;v,w)$ inside the circle; it is analytic in (v,w), the factor $(\zeta^2 + r^2)^{\mathsf{N}}$ taming f on the circle, and the identity theorem in (v,w) carries the continuation across $\mathrm{Re}\,\zeta = 0$ to all spectators. On the circle $|\zeta^2 + r^2| = 2r|\mathrm{Re}\,\zeta|$, and at real ζ_0 inside $|\zeta_0^2 + r^2| \geq r^2$, so the maximum principle gives $|f| \leq 2^{\mathsf{N}}(2/r)^{\mathsf{P}}\mathsf{M}$ there, while where $|\mathrm{Re}\,\zeta| \geq r$ the bound (a) serves directly; at $n = 2$, $r = \frac{2}{3}d$ gives the factor $2^{\mathsf{N}}(3/d)^{2p_V}\mathsf{M}$.

Made once for each of the $\frac{1}{2}n(n-1)$ pairs, with the total power at most $p_V n$ and radii of the order $d(Y)/n$, this gives $\|\mathsf{G}_l(x;e)\| \leq \mathsf{K}_n(\sqrt{2}C_V(3/d(Y))^{p_V})^n \prod_a |e_a|$ at all points of pairwise different heights, and the continuation of G_l with this bound across the sets of equal heights, where the entries are apart in a horizontal direction. Then the integrand is bounded, and σ charges no set of points with two equal heights: by [7, Prop. C.18](b) the unordered remainder of a cut in x_3 tends to 0, and at ordered tuples the class is the product of the operators of (a) applied to $[F]$, by an induction on the number of the entries; so $[Y \cdot \tau_{le_3} F]$ is the integral, over the points of pairwise different heights, a set of full measure, of $\mathsf{G}_l(x;h_1(x_1),\dots,h_n(x_n))[F]$, which gives the L^1 bound.

So $s_n(x;e) = \langle \Omega, \mathsf{G}_l(x;e)\Omega\rangle$ is continuous at all non-coinciding points, and $\sigma(X) = \langle \Omega, [X]\rangle = \int s_n$ after a translation of X into a slab. □

Step 12: the field; real time. By [7, proof of Thms. BU–BX, Steps 10–11] and [7, Prop. C.18](b), the restriction of σ to C^1 entries has the hypotheses of Prop. 3.10, with p_V and C_V, and the Hilbert spaces of σ and of this restriction are one, the classes of C^1 tuples being dense. Prop. 3.10 gives [7, Thm. BW](e) and the kernels of (d), which represent σ at all continuous entries by the convergence on $\mathfrak{S}(\mathbb{R}^3)$; at supports ordered in x_3 the kernel is the boundary value of $\mathbf{W}_n$ ([7, Prop. C.18](c)). The growth: for φ vanishing with all its derivatives at coinciding points, Taylor's formula at the nearest point with two equal entries, to the order $\lceil p_V n \rceil \leq 9n$, absorbs $\delta(x)^{-p_V n}$; then $(1+|x|^2)^{-k/2}$, $k = 9n > 3n$ the index of the norm below, is integrable, there are 6^n components, and $(2n^2)^{p_V n} \leq (2e^2)^{p_V n}(n!)^{2p_V}$, which at $p_V = 9$ gives $(n!)^{18}|\varphi|_{9n}$.

For [7, Thm. BW(g)], let $\eta \in C_c^2(\mathbb{R}^3;\mathbb{R}^3)$ be carried by the slab $\{|y_3| < a\}$, $a > 0$, and let the spectators be $\tau_{ae_3}F$ and $\theta\tau_{ae_3}F'$, $F, F' \in \mathfrak{F}(\{x_3 > 0\})$, on either side of it. As $H^{\mathrm{cons}}[\eta] = \sum_{ij} \partial_i \eta_j \, s_{ij}$, [7, Thm. BV](f), with the point operators of [7, Thm. BW](b) and an integration by parts in y, gives $\sum_i \partial_{y_i} \langle [F'], \mathcal{U}(\vec{y})\mathsf{B}(a+y_3, a-y_3; s_{ij})\mathcal{U}(-\vec{y})[F]\rangle = 0$ for $|y_3| < a$ and every j; both members are holomorphic in y, and at $y = \iota(x)$, where $y_3 = -it$ and $\partial_{y_3} = i\partial_t$, this is, smeared with f, $-i\,\Theta_{3j}^E(\partial_0 f) = \sum_{k\leq 2} \Theta_{kj}^E(\partial_k f)$ between the vectors $e^{-aH}[F']$ and $e^{-aH}[F]$: the identity of [7, Thm. BW(g)] at $\nu = j \leq 2$, and, multiplied by $-i$, at $\nu = 0$, $j = 3$.

These vectors span a core of every power of $1 + H$, as $(1+H)^m e^{-aH}$ is bounded for every m, the $[F]$ are dense and the spectral truncations of a vector lie in the range of e^{-aH}; read on the core of

$(1+H)^{2l}$, $l > p_V$, the bound $\|(1+H)^j T(f)\psi\| \le c_j(f)\|(1+H)^{2l+j}\psi\|$ of [7, Thm. BW(e)] at $j=0$ extends the identity to $\mathcal{D}_0$.

4. Point splitting at two scales

Two nearby scalar insertions retain directional information. To read it, keep the pair a distance r apart and place a spectator at a larger distance d. The limit $r \to 0$ is then taken while d stays positive. There is also a third scale: the support radius of each bump used to approximate a point.

For the coordinate direction λ, write

$$\mathcal{O}_\lambda(\varphi) = a_K^{-1} \sum_{\lambda(p)=\lambda} \varphi(x(p)) a_p, \qquad \mathcal{O} = \sum_\lambda \mathcal{O}_\lambda, \qquad T_\lambda = \mathcal{O}_\lambda - \tfrac{1}{3}\mathcal{O}.$$

Thus T_λ is a diagonal traceless tensor component, not the stress tensor. Let χ_c^ρ be the radial tent of mass one, radius ρ and centre c, and put $f_\pm^c = \chi^{r/8}_{c \pm r e_\lambda/2}$. Denote joint cumulants by $S_{n;K}$. At label b on the least admissible torus, define

$$A_{r,K} = \frac{2\pi}{3} b r^3 \int \varphi_1(c) S_{3;K}(f_+^c, f_-^c, \varphi_2)\, \mathrm{d}c.$$

Sketch C (The close pair and the tensor insertion). Fix $0 < \beta_H < 1$, a sufficiently small reference coupling, and a label $b \in I(g)$. Take $6d \le \min\{1, b\}$. Suppose that the Lipschitz weights φ_1, φ_2 lie in balls of radius d, with supports at distance at least d and centres at distance at most $3d$. For sufficiently small r, with $128 a_K \le r$ and the point-splitting working level below the cutoff,

$$\left| A_{r,K} - \langle T_\lambda(\varphi_1); \mathcal{O}(\varphi_2)\rangle_K \right| \le C(d) b^{-2} \left[r^{\beta_H/72} + (a_K/r)^{1/18} \right] |\varphi_1|_\infty |\varphi_2|_\infty.$$

Here the constant and the small-radius threshold also depend on the construction, β_H, g, b, and on $d\,\mathrm{Lip}(\varphi_1)/|\varphi_1|_\infty$ when $\varphi_1 \neq 0$; a zero weight gives the identity trivially. At fixed r both expressions have limits as $K \to \infty$. Their difference then tends to zero as $r \to 0$ with d fixed. The construction also gives directional tensor comparisons and three- and four-point laws with their separate scale restrictions. ■□

Sketch of the argument. The Gaussian calculation explains the normalization. The three-point function is a cyclic contraction of three free curvature kernels. When two points approach one another, its singular factor has order r^{-3}. Multiplication by $\frac{2\pi}{3} b r^3$ leaves precisely the free harmonic of the traceless tensor seen by the distant insertion. Plane fields give the diagonal components; the bond bilinears supply the off-diagonal components. Their joint comparison identifies the tensor in every direction.

Agreement of free contractions is insufficient for the limiting identity. Place the close pair and the tensor insertion in one renormalization calculation with sources. Their normalized difference is carried through the same backgrounds and fluctuation laws. Above the scale where the pair is no longer resolved, that difference is nearly constant. Centring removes the constant, while the remaining response and mixed-source errors give the positive power of r in the display. The histories with a first large-field creation are summed with their probability weights. This controls the part of the correlation not accounted for by the explicit Gaussian contractions.

The lattice error has a different role. At finite cutoff the two bumps must remain much farther apart than one mesh spacing. The term $(a_K/r)^{1/18}$ records this restriction. Taking $r \to 0$ first at fixed K would lose the estimate. Removing the cutoff first, or retaining the stated relation between the two scales, makes both errors small.

One must also distinguish kernels from their averaged values. The fixed-torus construction initially gives Borel kernels defined almost everywhere. Narrow-bump comparisons give bounds on their means and hence at their Lebesgue points. They do not select values on an exceptional set. For periodic-Gibbs limits and their components, the independently established continuous kernels turn the corresponding bounds into pointwise assertions.

This point-splitting result identifies a particular tensor insertion beside separated spectators. It does not define every coincident product or supply an unrestricted operator-product expansion. Such a conclusion would require control of additional products, their subtractions and their contact terms.

4.1. **Two short scales and point splitting.** A pair of nearby action-density insertions has a tensor-valued leading term. We compare the close pair with this tensor at a second, larger distance, retaining the dependence on both scales. The three-point comparison, its extension to plane fields, the fixed-distance point-splitting limit and the four-point law have different hypotheses and are stated separately.

The proof begins with the local kernels and their Gaussian contractions, then controls their remainders through the renormalisation steps. The treatment of narrow test functions is needed to distinguish continuous continuum kernels from the almost-everywhere classes obtained on a fixed torus.

4.1.1. *Notation and hypotheses. Constants and conventions.* Fix one common choice of the real and sourced construction constants, as in Prop. YM3_I:69.1 and Section YM3_I:82, at a fixed source radius $2\varrho > 0$. The real construction uses the restriction of this choice, with no change in the value of any retained constant. In particular, the hypotheses of Thms. YM3_I:AJ and YM3_I:AD are imposed at this same choice: the blocking factor $L \geq 13$ is odd, the cube size is $M = L^{m'}$, $a_j = \eta_j = L^{-j}$, and $A_0, A_1, \mathfrak{p}_0 \geq 1$, $\bar{B}$, $B_\sharp$ (with $\bar{B} = B_\sharp L/(L-1)$), α_0, β_{s}, κ, together with the remaining constants of the real and sourced steps, have these fixed values. By C, c, c', with or without an index, we denote positive numbers made from these constants only, and by $C(g)$, $C(g, \mathsf{b})$, $C(g, \ell)$ numbers that may also depend on the displayed arguments; such a number depends on nothing that is not displayed, in particular not on K, β, ℓ, a level, a length, a position, a direction, a sign or a weight. We put $\delta(t) = A_1(\log t)^{\mathfrak{p}_0} t^{-1/2}$, $\mathfrak{b}(t) = A_1(\log t)^{\mathfrak{p}_0}$, $\varepsilon(t) = A_0(\log t)^{\mathfrak{p}_0} t^{-1/2}$; δ and ε decrease on $[e^{2\mathfrak{p}_0}, \infty)$.

Couplings, data, sides. The reference coupling satisfies $0 < g \leq \gamma$, with the ceiling fixed by Prop. YM3_I:82.2. A first-passage datum (K, β) at g consists of a bare inverse coupling β and the number K of renormalisation steps after which, for the first time, the running inverse coupling that starts at β is at most $L(g^{-2} + B_\sharp)$, as in Prop. YM3_I:74.1. Its run is $x_0 = \beta > x_1 > \cdots > x_K > g^{-2}$; we write $y_j = \log x_j$, $g_j = x_j^{-1/2}$, $\delta_j = \delta(x_j)$, $\mathfrak{b}_j = \mathfrak{b}(x_j)$, $\varepsilon_j = \varepsilon(x_j)$. The label $\mathsf{b} = a_K \beta$ is a length, $g^{-2} < \mathsf{b} \leq \bar{\mathsf{b}} := L(g^{-2} + \bar{B})$. We put $I(g) = (g^{-2} + \bar{B},\, L(g^{-2} + B_\sharp) - \bar{B}]$; for $\mathsf{b} \in I(g)$ the pair $(K, L^K \mathsf{b})$ is a first-passage datum at g with the label b for every $K \geq 1$. $\mathfrak{L}(g)$ is the set of the sides at g, each of the form $\ell = 2L^m$, and $\ell_g = \min \mathfrak{L}(g) \geq 26$ is the least side; T_ℓ is the torus of side ℓ with the lattice $a_K \mathbb{Z}^3$, and the lattice (ℓ, K) is this torus with the coarser lattices of its levels.

States and limits. $\mu_{K,\ell,\beta}$ is the measure of the torus T_ℓ at the cutoff K and the bare coupling β, $\langle \cdot \rangle_{K,\ell,\beta}$ its expectation, $\tilde{\mu}_{K,\ell,\beta}$ its periodic extension to $\mathbb{L}_K = a_K \mathbb{Z}^3$, and $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ is the set of periodic Gibbs states (equation YM3_I:(1819) and the definition of periodic states following it). By a state at (K, β) we mean either a periodic extension $\tilde{\mu}_{K,\ell,\beta}$ with $\ell \in \mathfrak{L}(g)$ or a periodic Gibbs state in $\mathcal{G}^{\mathrm{per}}_{K,\beta}$. The test pairs $\mathfrak{T}_{T_\ell}(r; \vartheta)$ of scale r and shape ϑ are those of [7, Lem. I.1].

Along first-passage data (K_j, β_j) at g with $K_j \to \infty$ and $a_{K_j} \beta_j \to \mathsf{b} > 0$, the torus connected functions converge to $S_n^{(\ell, \mathsf{b})}$ on real Lipschitz tuples with pairwise disjoint supports, independently of the defining sequence, by Thm. YM3_I:A. The first-passage labels lie in $(g^{-2}, \bar{\mathsf{b}}]$ by Prop. YM3_I:74.1; in particular their limiting labels are positive. A law on $\mathbb{R}^3$ is a functional on separated tuples of real C^1_c functions with the properties of the growing-torus or volume-first limits in Props. YM3_I:92.1, YM3_I:92.3 and YM3_I:92.4. Limits of periodic-Gibbs sequences and the components of their canonical decompositions are examples, by Thms. YM3_I:AP and YM3_I:AL.

For limits of periodic-Gibbs sequences and their canonical components, Thm. YM3_I:AP and Prop. YM3_I:108.3 give unique continuous moment kernels, real analytic and invariant under Euclidean isometries. The connected kernels have the same properties by the finite moment–cumulant formula. For an arbitrary growing-torus law, the inherited bounds initially give distributions on separated configurations. Full rotation invariance has the additional Ward criterion of Thm. YM3_I:AQ; the analytic-kernel conclusion is available when the reconstruction hypotheses (Rec) of Prop. YM3_I:108.3 hold. The pointwise assertions below for such a law explicitly assume continuity of the kernels they evaluate. The finite-cutoff, integrated and almost-everywhere assertions retain their stated hypotheses.

Fields. For a plaquette p, $x(p)$ is its barycentre, $\lambda(p) \in \{1,2,3\}$ its dual direction (the direction orthogonal to its plane), and $a_p = 1 - \frac{1}{2}\operatorname{tr}\partial U(p)$. For a real weight w on the plaquettes with finite support, $\mathcal{O}(w) = a_K^{-1}\sum_p w(p)a_p$, $|w|_\infty = \max_p |w(p)|$ and $\Sigma(w) = \{x(p) : w(p) \neq 0\}$; for a bounded function f, $\mathcal{O}(f) = \mathcal{O}(f \circ x)$. The plane fields are $\mathcal{O}_\lambda(\varphi) = \mathcal{O}(\mathbf{1}_{\{\lambda(p)=\lambda\}}\,\varphi \circ x)$, with plane weights, so that $\mathcal{O} = \sum_\lambda \mathcal{O}_\lambda$; $T_\lambda = \mathcal{O}_\lambda - \frac{1}{3}\mathcal{O}$ are the diagonal elements of the traceless tensor.

Off the diagonal stand the bond forms. For $V \in SU(2)$ put $\mathrm{w}(V) = (V - V^{-1})/2i$, and $\langle X, Y\rangle = \frac{1}{2}\operatorname{tr} XY$. A bond $b = \langle x, x + a_K e_\mu\rangle$ has the base point x, the direction $\mu_b = \mu$ and the midpoint $x_b = x + \frac{1}{2}a_K e_\mu$. For signed directions α, α' with $\alpha \neq \pm\alpha'$ let $W_x(\alpha, \alpha')$ be the product of U along the closed path $x \to x + a_K e_\alpha \to x + a_K e_\alpha + a_K e_{\alpha'} \to x + a_K e_{\alpha'} \to x$, where $e_{-\alpha} = -e_\alpha$. For a direction $\alpha \neq \mu_b$ the loops $W_x(\mu_b, \alpha)$ and $W_x(\mu_b, -\alpha)$ run around the two plaquettes of the plane spanned by μ_b and α that contain b, both from the base point x along b first, and we put $u_{b,\alpha} = -\frac{1}{2}[\mathrm{w}(W_x(\mu_b, \alpha)) - \mathrm{w}(W_x(\mu_b, -\alpha))]$; with $\lambda_b < \lambda_b'$ the two directions other than μ_b, $q_b = \langle u_{b,\lambda_b}, u_{b,\lambda_b'}\rangle$.

For a real weight w on the bonds, $Y[\mathsf{w}] = -a_K^{-1}\sum_b \mathsf{w}(b)q_b$, and $\mathcal{O}^{\mathrm{od}}_{\lambda\lambda'}(\varphi) = Y[\mathsf{w}]$ at $\mathsf{w}(b) = \varphi(x_b)\mathbf{1}_{\{\mu_b=\mu\}}$, where $\{\lambda, \lambda', \mu\} = \{1,2,3\}$: the bond weight is taken at the midpoint. Written in the angles F_p of the plaquette variables, the linear part of $u_{b,\lambda}$ is $\sum_p u_{b,\lambda}(p)F_p$, with $u_{b,\lambda}(p) = \frac{1}{2}\epsilon_{\lambda\mu_b\lambda'}\epsilon_p$ at the two plaquettes with the barycentres $x_b \pm \frac{1}{2}a_K e_\lambda$ and $u_{b,\lambda}(p) = 0$ at all other plaquettes; here $\{\lambda, \mu_b, \lambda'\} = \{1,2,3\}$, $\epsilon_{\lambda\mu_b\lambda'}$ is the sign of this permutation and $\epsilon_p = \pm 1$ is the orientation sign of p.

For a unit vector n,

$$T_n(\varphi) = \sum_\lambda n_\lambda^2\, T_\lambda(\varphi) + \sum_{\lambda<\lambda'} n_\lambda n_{\lambda'}\, \mathcal{O}^{\mathrm{od}}_{\lambda\lambda'}(\varphi).$$

A mixed source is a pair $\mathfrak{s} = (w, \mathsf{w})$ of real weights on the plaquettes and on the bonds, $\mathcal{O}^T(\mathfrak{s}) = \mathcal{O}(w) + Y[\mathsf{w}]$, $|\mathfrak{s}|_\infty = |w|_\infty + 8|\mathsf{w}|_\infty$.

Connected functions and natural sizes. For weights $w_1, \dots, w_n$ and $w_\lambda = \sum_a \lambda_a w_a$, $S_n^T = S^T_{n;K,\ell,\beta}(w_1, \dots, w_n) = \partial_{\lambda_1}\cdots\partial_{\lambda_n}\log\langle e^{\mathcal{O}(w_\lambda)}\rangle_{K,\ell,\beta}|_{\lambda=0}$; for functions f_a it is taken at $w_a = f_a \circ x$; $\langle X; Y\rangle$ denotes S_2^T, and $S_n^T[\mathfrak{s}]$ the same in a state $\mathfrak{s}$. For a function f on $\mathbb{R}^3$, $f^{\langle\ell\rangle}$ is its periodisation. We write $\mathsf{u}_a = \mathsf{b}^{-1}|w_a|_\infty$. Three bumps of width ρ, two of them at the near distance r and the third at the far distance d, are measured against $\mathrm{nat}_3 = \rho^9 r^{-3}\mathsf{d}^{-6}\mathsf{b}^{-3}\prod_{a\le 3}|w_a|_\infty$, and two such pairs against $\mathrm{nat}_4 = \rho^{12}r^{-6}\mathsf{d}^{-6}\mathsf{b}^{-4}\prod_{a\le 4}|w_a|_\infty$ (at functions $\|f_a\|_\infty$ replaces $|w_a|_\infty$); a kernel of three points is measured against $\mathsf{b}^{-3}r_*^{-3}\mathsf{d}_*^{-6}$ and a kernel of two points against $\mathsf{b}^{-2}|x-y|^{-6}$. These are the natural sizes.

Placed weights. Real weights w_1, w_2 are placed at (ρ, r) if there are centres x_a with $\Sigma(w_a) \subset \mathbb{B}(x_a, \rho)$ (closed balls), $\operatorname{dist}(\Sigma(w_1), \Sigma(w_2)) \ge r$ and $|x_1 - x_2| \le 8r$. Weights w_1, w_2, w_3 are placed at (ρ, r, d) if besides $\Sigma(w_3) \subset \mathbb{B}(x_3, \rho)$, $\operatorname{dist}(\Sigma(w_a), \Sigma(w_3)) \ge \mathsf{d}$ and $|x_a - x_3| \le 18\mathsf{d}$ for $a = 1, 2$. Weights $w_1, \dots, w_4$ form two pairs at (ρ, r, d) if (w_1, w_2) and (w_3, w_4) are placed at (ρ, r) and $\operatorname{dist}(\Sigma(w_a), \Sigma(w_b)) \ge \mathsf{d}$, $|x_a - x_b| \le 18\mathsf{d}$ for $a \le 2 < b$. A set β of sources crosses if it meets both clusters, $\{1,2\}$ and $\{3\}$ (or $\{3,4\}$), and $c(\beta)$ is the number of whole pairs in β.

Free objects. $\mathbb{K}_3(u) = (4\pi|u|^3)^{-1}(3\hat u\hat u^{\mathsf{T}} - 1) = (4\pi)^{-1}\nabla\nabla|u|^{-1}$, $\hat u = u/|u|$; $k_3(x,y,z) = \operatorname{tr}\mathbb{K}_3(x-y)\mathbb{K}_3(y-z)\mathbb{K}_3(z-x)$; $k_4(x_1, \dots, x_4) = \frac{1}{2}\sum_\varsigma \operatorname{tr}\prod_l \mathbb{K}_3(x_{\varsigma(l)} - x_{\varsigma(l+1)})$, the sum over the six cyclic

orders of $\{1,2,3,4\}$ (three rings, each once); $\mathcal{K}_{\lambda\mu}(\varphi_1,\varphi_2)=\iint \varphi_1(x)\varphi_2(y)[\mathbb{K}_3(x-y)_{\lambda\mu}]^2\,\mathrm{d}x\,\mathrm{d}y$; and for a unit vector n and $X\neq 0$

$$\mathsf{t}^\circ(n,X)=\tfrac{3}{2}\mathsf{b}^{-2}\big[\langle n,\mathbb{K}_3(X)^2 n\rangle-\tfrac{1}{3}\operatorname{tr}\mathbb{K}_3(X)^2\big]=\frac{3}{32\pi^2}\,\mathsf{b}^{-2}|X|^{-6}(3(n\cdot\hat{X})^2-1).$$

C_K^{free} is the free curvature kernel on the bare plaquettes of the lattice (ℓ,K), $\mathcal{C}_{(j)}$, $1\le j\le K$, are its slices by level, and $\upsilon_j=(a_{K-j+1}x_{j-1})^{-1}$. Over a base level $k_\flat$ we put $\mathcal{S}_i=\sum_{k_\flat<j\le i}\upsilon_j\mathcal{C}_{(j)}$; for a nonempty set β of sources with $m=|\beta|$, $R_{i;\beta}=\sum w_{a_1}\mathcal{S}_i w_{a_2}\cdots\mathcal{S}_i w_{a_m}$, the sum over the orderings $(a_1,\dots,a_m)$ of β, the weights acting as diagonal matrices; $q_{i;\beta}(F)=\frac{1}{2a_K}\sum_{p,q}R_{i;\beta}(p,q)\langle F_p,F_q\rangle$ for $\mathfrak{su}(2)$-valued functions F of the plaquettes; and $g_{i;\beta}=\frac{3}{2}\sum_\varsigma\operatorname{Tr}\prod_l(w_{\varsigma(l)}\mathcal{S}_i)$, the sum over the cyclic orders ς of β. At $\beta=\{1,2\}$ we write $q_n=q_{n;12}$, so $q_n(F)=a_K^{-1}\sum_{p,q}w_1(p)\mathcal{S}_n(p,q)w_2(q)\langle F_p,F_q\rangle$, and $\mathcal{Q}_i=q_{i+1}-q_i$. The lattice ring of three weights is $3\operatorname{Tr}\prod_a(w_a\mathsf{b}^{-1}C_K^{\mathrm{free}})$. Finally $\chi_x^\rho=\frac{3}{\pi\rho^3}(1-|\cdot-x|/\rho)_+$ is the radial tent of radius ρ and mass 1.

The renormalisation group. We use the following signs. The level j is the lattice of spacing a_{K-j}, level 0 being the bare lattice; for a length ϱ, $\hat{\varrho}_j=\varrho/a_{K-j}$. The chain classes $\mathsf{Cc}_n(r_n)$ have the radii $r_K=0$, $r_i=L^{-1}r_{i+1}+\frac{11}{10}\mathsf{c}_\mathfrak{h}\delta_i$ ($\mathsf{c}_\mathfrak{h}\ge 1$); for $A\in\mathsf{Cc}_n(r_n)$ we put $\tilde{U}=e^{i\eta_n A}$ and $\mathsf{ang}_p^{[n]}(A)=\eta_n(\mathrm{d}A)(p)$, and $\mathbf{1}$ is the trivial configuration. The witness depth is $s(\varrho)=\min\{s\ge 0: L^s\varrho\ge C_W(s)\}$, where $C_W(s)=C_{W,0}A_0^2(\log[L^{s+1}(g^{-2}+\bar{B})])^{2\mathfrak{p}_0}$, with the room constant $C_{W,0}=\max\{1,4L^2M/A_0^2\}$, so that $C_{W,0}A_0^2\ge 4L^2M$ (equations YM3_I:(2320) and YM3_I:(2322)); the working level is $\mathsf{k}(\varrho)=\max\{K-s(\varrho),2\}$; $Y_s=(s+1)\log L+\log(g^{-2}+\bar{B})$, so that $C_W(s)=C_{W,0}A_0^2Y_s^{2\mathfrak{p}_0}$, $Y_\varrho=Y_{s(\varrho)}$, and Lm_s is a constant times $Y_s^{5\mathfrak{p}_0}$.

For sources $w_1,\dots,w_p$ the weighted progression $E_\lambda^{(j)}$, $1\le j\le K$, is the sequence of families of local terms of level j that the renormalisation steps make with the source w_λ, for λ in a neighbourhood of 0; $E^{(j)}=E_0^{(j)}$ and $E_\beta^{(j)}=\partial_\beta E_\lambda^{(j)}|_0$. The relative exponent is

$$D_i^\lambda(U)=\mathcal{O}(w_\lambda)(U)+\sum_{i'\le i}\mathsf{Sum}[E_\lambda^{(i')}-E^{(i')}](U),\qquad \mathsf{Sum}[\mathsf{f}](U)=\sum_{(X,z)}\mathsf{f}(X,U,z),$$

the sum over the domains X of the level and their base points z; $G_{i;\beta}(A)=\partial_\beta D_i^\lambda(e^{i\eta_i A})|_0$, the pair functional on configurations is defined by $\mathsf{F}_n(e^{i\eta_n A})=G_{n;12}(A)$, the jets are $\mathsf{J}_n=G_{K;[n]}(0)$, and $\Delta[\mathsf{f}]=\mathsf{Sum}[\mathsf{f}](\tilde{U})-\mathsf{Sum}[\mathsf{f}](\mathbf{1})$. A family f of level i has the norms $\|\mathsf{f}\|_{i,\kappa'}=\sup|\mathsf{f}(X,P,z)|e^{\kappa' d_i(X)}$, d_i the linear size of a domain, and $\lambda_{\mathrm{out}}=(1+4\beta_{\mathrm{s}})\kappa$; π_i is the partition into cubes of side Ma_{K-i}; for a weight w_a, Cr_a is the set of the corners of the plaquettes of $\operatorname{supp}w_a$, $\mathsf{N}_i(w_a)$ the number of the cubes of π_i that hold a site of Cr_a, and D_i^{ab} the least sup distance of Cr_a from Cr_b in level-i units.

The step map $\mathsf{St}^{[j]}$ is the renormalisation step of the height j, viewed as a function from bare arrays $\mathfrak{f}=((G^{(i)},e_i)_{i\le j},(H))$ of level j to families of level $j+1$; $N(\mathfrak{f})$ is the size of an array and $\mathfrak{n}[\mathfrak{f}]\le N(\mathfrak{f})$ the size of the family that it puts in the fourth place of the one-step theorem; $N_\star^{[j]}$ and $\theta^{[j]}=K_{\mathbf{M}}(1+\beta_{\mathrm{s}})\delta_j/\alpha_0$ are numbers of the step j, and W_w is the weight member of a weight w. Further construction objects are defined where they enter.

Definition 4.1 (Ceilings, thresholds, numbers)**.**

(1) The ceiling (Γ1): $0<g\le\gamma_{\mathrm{jt}}$; $g^{-2}\ge\max\{x_{\mathrm{J}},e^{2\mathfrak{p}_0},e^{y_{\mathfrak{p}}}\}$; $\delta(g^{-2})\le\min\{1/2c_{\mathrm{rt}},1/c_\epsilon,1/2C_{\mathrm{cu}}\}$; $\theta(g^{-2}):=K_{\mathbf{M}}(1+\beta_{\mathrm{s}})\delta(g^{-2})/\alpha_0\le\tau^\natural$. The ceiling ($\Gamma$2): $0<g\le\gamma_W^{\mathrm{sd}}$, $g^{-2}\ge x_{\mathrm{J}}$, $\log g^{-2}\ge\mathsf{Y}_{\mathrm{lt}}(\frac{1}{2})$; data under ($\Gamma$2) have $K\ge 2$. (Γ) is (Γ1) together with (Γ2). Here γ_{jt} is defined in eq. 4.iv, γ_W^{sd} in equation YM3_I:(2327), and x_{J} in [7, eq. 228]; [7, Prop. B.21] verifies the required actual-step hypotheses for every finite number of sources. The least-torus threshold Y_{lt} is specified below; $y_{\mathfrak{p}},c_{\mathrm{rt}},c_\epsilon,C_{\mathrm{cu}},\tau^\natural$ are numbers formed from the constants in Lems. 4.3, 4.6 and 4.14.

For $0<\eth<1$, put $c'=(1-\eth)c_\star$ and $C_{\rm sd}=8L^6(\mathsf{C}^\Sigma_{\rm maj}+2+c_{\rm low}+9/2)$. Choose $\mathsf{Y}_{\rm lt}(\eth)\ge 1$ so that all the history thresholds in Prop. 4.54 hold above it and

$$2C_{\rm sd}y[MR(e^y)]^3\le \tfrac12 c'A_0^2y^{2\mathfrak{p}_0}\qquad (y\ge \mathsf{Y}_{\rm lt}(\eth)). \tag{4.i}$$

Such a finite threshold exists: $R(e^y)<Ly^r$ and $2\mathfrak{p}_0>3r+2$, as in the proof of Lem. YM3_I: 126.3. On the least admissible side that proof gives

$$\mathsf{B}(x_K)\ell_g^3\le C_{\rm sd}y_K[MR(e^{y_K})]^3.$$

Thus $\log g^{-2}\ge \mathsf{Y}_{\rm lt}(\eth)$ implies $2\mathsf{B}(x_K)\ell_g^3\le c'p_K^2/2$, the stronger small-side condition in Prop. 4.54.

Since δ decreases on $[e^{2\mathfrak{p}_0},\infty)$, each member is a condition "$g\le$ a number formed from the constants". $(\Gamma^{(4)})$ is (Γ) with $c_\epsilon^{(4)}$ of Lem. 4.14 in the place of c_ϵ. As $c_\epsilon\le c_\epsilon^{(4)}$ (Lem. 4.14(a), where $C_{\rm cu}$, $c_{\rm cu}$ and the numbers $c_\epsilon^{(n)}$ are formed), $(\Gamma1)$ with $c_\epsilon^{(4)}$ in the place of c_ϵ contains $(\Gamma1)$, and $(\Gamma^{(4)})$ contains (Γ): every g under $(\Gamma^{(4)})$ satisfies (Γ). For the bond forms we use the ceiling $\gamma_{\rm N}=x_{\rm N}^{-1/2}$, where $x_{\rm N}$ is the least real number $\ge x_{\rm jt}=\gamma_{\rm jt}^{-2}$ such that every $t\ge x_{\rm N}$ satisfies $t\ge\max\{x_{\rm J},e^{2\mathfrak{p}_0},e^{y_{\mathfrak{p}}}\}$, $\delta(t)\le\min\{1/2c_{\rm rt},1/c_\epsilon\}$, $\theta(t)=K_{\mathbf{M}}(1+\beta_{\rm s})\delta(t)/\alpha_0\le\tau^\natural$, $\mathfrak{b}(t)\ge1$ and $c_{\rm A}\alpha_0\ge 5M\mathsf{c}_\natural\delta(t)(1+\log t)$, with $c_{\rm A}$ of Lem. 4.8.

So $\gamma_{\rm N}\le\gamma_{\rm jt}$, and every $0<g\le\gamma_{\rm N}$ satisfies $(\Gamma1)$ but for its member $\delta(g^{-2})\le 1/2C_{\rm cu}$.

(2) $r_1(\mathsf{b})>0$ is the number of Lem. 4.6, formed from the constants and b; it does not increase in b. $\mathsf{d}_4(g)>0$ is the number fixed in Prop. 4.19, formed from the constants and g, with $\mathsf{d}_4(g)\le\min\{e^{-2(15\mathfrak{p}_0+2)},r_1(\bar{\mathsf{b}})\}$; $\mathsf{d}_5(g)=\min\{2\mathsf{d}_4(g),\min\{1,16r_1(\bar{\mathsf{b}})\},\frac{1}{32}\ell_g\}$. For two pairs, $\mathsf{d}_4^{(4)}(g)$ is fixed in the same Proposition, and $\mathsf{d}_5^{(4)}(g)=\min\{2\mathsf{d}_4^{(4)}(g),\frac{1}{48}\ell_g\}$.

(3) $\bar{\mathsf{M}}^{(24)}=1+2L^{24}\mathsf{C}_{\rm ag}(25,\frac12)$, where $\mathsf{C}_{\rm ag}$ is defined by eq. 4.xxxii; $\Lambda_3\ge1$ is the number of Lem. 4.21; $C_{\rm lat}$ is the absolute constant in equations YM3_I:(2245)–(2246), comparing $C_K^{\rm free}$ with $\mathbb{K}_3$ on every side; $C^{(3)}=3\cdot81^3(1+3C_{\rm lat})^2C_{\rm lat}+2.1\cdot10^7$, $C^{(4)}=3[4\cdot81^4C_{\rm lat}(1+3C_{\rm lat})^3+1.49\cdot10^8]$, $C_{\rm Rs}=7400+729C_{\rm lat}(2+3C_{\rm lat})$, the constants of the lattice sums of Lem. 4.22; $C_{\rm PS}=62+1089C_{\rm lat}(2+3C_{\rm lat})$ and $C_{\rm PS}^{\rm od}=62+729(2+5C_{\rm lat})(1+C_{\rm lat})$, the constants of the lattice sums at two balls on every torus (Lem. 4.33); $C_3^{\rm tr3}(g)$, $C_4^{\rm tr3}(g)$, $C_2^{\rm tr}(g)$ are the numbers of Prop. 4.28, formed from the constants and g.

(4) With the number $C(g)$ of Prop. 4.19: $\epsilon^{\rm nar}(g,t)=C(g)(\log\frac1t)^{39\mathfrak{p}_0+2}t^{1/2}$, $\epsilon_2^{\rm nar}(g,t)=C(g)(\log\frac1t)^{19\mathfrak{p}_0}t^{1/2}$, $\epsilon_4^{\rm nar}(g,t)=C(g)(\log\frac1t)^{47\mathfrak{p}_0+2}t^{1/2}$; and

$$\epsilon_{\rm A}(g,t)=512\big[\epsilon^{\rm nar}(g,\tfrac t2)+3C^{(3)}(5t/\ell_g)^3+13\Lambda_3^3(\bar{\mathsf{M}}^{(24)})^3\bar{\mathsf{b}}^3t^{3/2}\big],$$
$$\mathsf{E}_3(g,t)=512\big[\epsilon^{\rm nar}(g,\tfrac t2)+3C^{(3)}(5t/\ell_g)^3+(13\Lambda_3^3(\bar{\mathsf{M}}^{(24)})^3+C_3^{\rm tr3}(g))\bar{\mathsf{b}}^3t^{3/2}\big],$$
$$\epsilon_{\rm P}(g,t)=64\big[\epsilon_2^{\rm nar}(g,\tfrac t2)+3\Lambda_3^2(\bar{\mathsf{M}}^{(24)})^2\bar{\mathsf{b}}^2(\tfrac t2)^{3/2}+\tfrac32C_{\rm Rs}(t/2\ell_g)^3\big],$$
$$\mathsf{E}_2(g,t)=\epsilon_2^{\rm nar}(g,t)+[3\Lambda_3^2(\bar{\mathsf{M}}^{(24)})^2+C_2^{\rm tr}(g)]\bar{\mathsf{b}}^2t^{3/2}+\tfrac32C_{\rm Rs}(t/\ell_g)^3,\qquad \epsilon_\sigma(g,t)=64\sup_{0<t'\le t}\mathsf{E}_2(g,\tfrac{t'}2).$$

So $\epsilon_{\rm A},\mathsf{E}_3\le C(g)[(\log\frac1t)^{39\mathfrak{p}_0+2}t^{1/2}+(t/\ell_g)^3]$ and $\epsilon_{\rm P},\epsilon_\sigma\le C(g)[(\log\frac1t)^{19\mathfrak{p}_0}t^{1/2}+(t/\ell_g)^3]$ for small t; all of them are functions of (g,t) alone.

(5) *The threshold $r_\alpha(g,\mathsf{b})$ of Thm. K.* For a label b and $0<r\le L^{-4}$ consider every first-passage datum (K,β) at g with $a_K\beta=\mathsf{b}$, $128a_K\le r$ and $s(\frac r2)\le K$, and the near pair $w_{1,2}=\chi^{r/8}_{c\pm\frac r2e_\lambda}\circ x$ of that Theorem, a test pair of the scale $\frac r2$; n_1 and n_0 are the least levels with $a_{K-n_1}\ge(\frac r2)^{3/4}$ and $a_{K-n_0}\ge r^{1/2}$. The conditions are:

($\alpha1$) $\mathsf{k}(\frac r2)<n_1<n_0<K$, $2(r+a_K)\le a_{K-n_0}$ and $\frac{27}{8M^3}(r/2a_{K-n_1})^3\le1$;

($\alpha2$) for $\mathsf{k}(\frac r2)\le i<n_1$, $n_0\le n<K$ and $m=n-i-1$: $(4\mathsf{h}_i+4)a_{K-i-1}\le\ell_g$ and $7\mathsf{h}_iL^{-2m}r_n\le\delta_i/2\Lambda_i$, where h_i and Λ_i are the numbers of Prop. 4.13 for the near pair at (i,n), formed with $R_w=\frac54\hat r_{i+1}+1$ and $\bar{\mathsf{e}}=2\bar{\mathsf{w}}_1\bar{\mathsf{w}}_2\mathsf{E}_i$, $\bar{\mathsf{w}}_a=81\hat r_i^3|w_a|_\infty/a_{K-i}$;

(α3) $\frac{r}{2} \le \min\{r_0(\mathsf{b}), r_1(\mathsf{b})\}$;
(α4) $s(\frac{r}{2}) \ge 3$ and $\mathsf{Lm}_{s(r/2)} \le L^{s(r/2)-3}$;
(α5) $2\sqrt{3}\,\delta_{\mathrm{tp}} L(L-1)M(\log\frac{2}{r})^{2\mathfrak{p}_0} \ge 3\log L + \log 2$, so that Lem. 4.17(b) applies at the level $\mathsf{k}(\frac{r}{2})$ to two plaquettes of the two supports of the near pair.

$r_\alpha(g,\mathsf{b})$ is the largest number $r_\star \in (0, L^{-4}]$ such that (α1)–(α5) hold at every $0 < r \le r_\star$ and every such datum. It is positive. (α1) holds for $r \le L^{-4}$ at every datum. (α2), (α4) and (α5) hold for r below a number formed from the constants and g: by the working level (Lem. 4.3(c)) and step 6 of the proof of Prop. 4.13, $\mathsf{h}_i \le C(1+Y)^{2\mathfrak{p}_0}$ and $\Lambda_i \le C(1+Y)$ with $Y = Y_{r/2} \le 2\log\frac{2}{r} + C(g)$; $r_n \le \frac{8}{5}\mathsf{c}_{\mathfrak{h}}(\frac{5}{4})^{1/2} L^{(n-i)/2}\delta_i$; and $L^{-(n-i)} = a_{K-i}/a_{K-n} < (\frac{r}{2})^{3/4} r^{-1/2} \le r^{1/4}$; none of these bounds depends on b or on K. (α3) holds for $\frac{r}{2} \le \min\{\inf_{\mathsf{b}'} r_0(\mathsf{b}'), r_1(\bar{\mathsf{b}})\}$, the infimum over $g^{-2} < \mathsf{b}' \le \bar{\mathsf{b}}$ being positive by (B.1) and r_1 not increasing. So $r_\alpha(g,\mathsf{b}) \ge r_\alpha^-(g)$ for all $\mathsf{b} \in (g^{-2}, \bar{\mathsf{b}}]$, with a positive number $r_\alpha^-(g)$ formed from the constants and g.

(6) *The ceiling* (Γ3) *and the threshold* $r_\alpha^{\mathcal{W}}(g,\mathsf{b})$. ($\Gamma$3) asks: $c_{\mathrm{A}}\alpha_0 \ge 8MB_3\varepsilon(t)(1+\log t)$ for all $t \ge g^{-2}$; $g \le 2C_{\mathrm{A}}C_{\mathrm{E},3}M^2B_3^2A_0^2/C_{\mathrm{E},1}\alpha_0^2$; and $7488B_3\varepsilon(g^{-2}) \le \epsilon_{\mathrm{G}}$, $C_{\mathcal{W}}\varepsilon(g^{-2}) \le \frac{1}{120}$ with $C_{\mathcal{W}} = \max\{720C_{\mathrm{G}}^\sharp, 2\}B_3$. Here c_{A}, C_{A} and $C_{\mathrm{E},1}$ are the numbers of Lem. 4.8 and of its proof, $C_{\mathrm{E},3}$ is the number in [7, Lem. B.10] at $q = 2$, $2B_3\varepsilon_n$ is the radius at which the small-field backgrounds are regular (Lem. 4.20(d)), and ϵ_{G}, $C_{\mathrm{G}}^\sharp$ are the numbers of the lemma on a regular gauge on a cube, which precedes Lem. 4.36.

As $\varepsilon(t)(1+\log t) \to 0$ and ε decreases on $[e^{2\mathfrak{p}_0}, \infty)$, each member is a condition "$g \le$ a number formed from the constants and β_{H}". $r_\alpha^{\mathcal{W}}(g,\mathsf{b})$ is the largest number $r_\star \in (0, r_\alpha(g,\mathsf{b})]$ such that, at every $0 < r \le r_\star$ and every datum as in (5), the numbers h_i and Λ_i of (α2), formed with $C_{\mathcal{W}}\varepsilon_n$ in the place of r_n, satisfy for $\mathsf{k}(\frac{r}{2}) \le i < n_1$, $n_0 \le n \le K$ and $m = n - i - 1$: $(4\mathsf{h}_i + 4)a_{K-i-1} \le \ell_g$, $7\mathsf{h}_i L^{-2m} C_{\mathcal{W}}\varepsilon_n \le \delta_i/2\Lambda_i$ and $(\mathsf{h}_i + 1)L^{-m} \le 3$, and besides $Lr^{1/2} \le (2\bar{\mathsf{M}}^{(24)})^{-1/24}$.

The bounds of (5), with $\varepsilon_n = (A_0/A_1)\delta_n$, give $r_\alpha^{\mathcal{W}}(g,\mathsf{b}) \ge r_\alpha^{\mathcal{W}-}(g)$, a positive number formed from the constants, β_{H} and g.

■

4.1.2. *The three lengths and the comparison.* Point splitting asks whether two nearby scalar insertions determine a tensor insertion when tested from a positive distance away. The pair separation is r, the distance to the remaining insertion is d, and the radius of each approximating bump is ρ. These lengths perform different jobs. The free curvature kernel is singular at distance r; the distant insertion sees its direction through a tensor coefficient; and ρ measures how accurately a bump samples a point. The estimates must keep all three lengths visible. An estimate with ρ fixed proportional to d would lose the regime $r/d \to 0$ that defines the point-splitting limit.

For three weights in the geometry of theorem G, the normalization is

$$\mathrm{nat}_3 = \rho^9 r^{-3} d^{-6} b^{-3} \prod_{a=1}^{3} |w_a|_\infty, \qquad \mathrm{nat}_4 = \rho^{12} r^{-6} d^{-6} b^{-4} \prod_{a=1}^{4} |w_a|_\infty \tag{4.ii}$$

for a pair with one spectator and for two separated pairs, respectively. Each bump contributes its volume. Each close pair contributes one curvature line of degree -3, and passage between the two clusters contributes two such lines. These powers explain which errors have to be uniform as ρ/r and r/d decrease. The three-point and four-point conclusions retain different coupling thresholds and different numbers of source derivatives.

The free calculation identifies the tensor and fixes its normalization. Put

$$\mathbb{K}_3(u) = \frac{3\hat{u}\hat{u}^{\mathsf{T}} - 1}{4\pi|u|^3}, \qquad D(n) = 3n \otimes n - 1.$$

The connected three-point kernel is a cyclic product of three $\mathbb{K}_3$ kernels. At the close pair its singular factor is $(4\pi r^3)^{-1}D(n)$. The normalized pair therefore carries the same quadratic form as the traceless tensor in direction n. For $n = e_\lambda$, the tensor weight is $\delta_{\lambda(p)\lambda} - 1/3 = D(e_\lambda)_{\lambda(p)\lambda(p)}/3$. The factor $2\pi br^3/3$ in theorem K makes these two quadratic forms agree. This equality concerns the explicit quadratic terms; the rest of the argument controls their difference in the interacting measure.

4.1.3. *The source derivatives through the levels.* Introduce the close-pair sources and the tensor source in one weighted progression. Let $G_{n;\beta}$ denote its derivative at level n in the source set β. Using one progression is essential: the difference is then formed on the same configurations and with the same conditional measures. For the second source derivative, the chain rule gives the decomposition of lemma 4.7,

$$E_{12}^{(j+1)} = \mathsf{K}^{(j+1)} + \mathsf{R}^{(j+1)}.$$

Here K is obtained by differentiating the step twice in the two original weight members. It supplies the free quadratic kernel. The remaining terms contain an earlier source derivative or the carried mixed derivative. Their norms gain the small factors in lemma 4.9. Both families vanish on a localization domain unless that domain meets both source supports. At levels finer than their separation, this forces a large domain and gives exponential decay.

Above that separation scale, locality alone no longer makes the derivative small. The induction instead compares it with an explicit quadratic form assembled from the curvature slices. In the notation of lemma 4.17, the slice sum obeys

$$\left| \sum_{j \in \mathcal{J}} \upsilon_j \mathcal{C}_{(j)}(p,q) \right| \le C b^{-1} a_K^3 |x(p) - x(q)|_\infty^{-3}.$$

The flow estimate compares υ_j with b^{-1}, while the exponential decay of each slice makes the lacunary sum finite. The error in replacing the retained slices by the full free kernel has three parts: the change of the running coupling, the coarsest background covariance, and the omitted fine slices. Their separate bounds are retained in lemma 4.17.

For several sources, differentiating the exact conditional integral produces the partitions of the source set. The induction weights in lemma 4.18 are chosen so that a product over such a partition is bounded by the weight of the whole set. A whole close pair needs the factor $q^6 u^3 \max\{1, u^3\}$, where $q = \rho/r$ and u is r in the working-level units. Cutting that pair into two singletons costs $\hat{\rho}^6 = q^6 u^6$. This explains the extra maximum: u may grow with a power of $\log(1/d)$, so treating it as bounded would invalidate the partition estimate. The resulting jet estimates are uniform in the two ratios in eq. 4.ii.

The jets evaluated at the identity are still not the full connected functions. Lemma 4.20 makes this distinction explicit. Stop each history either at its first large-field creation or at the terminal level if no such creation occurs. The resulting probability measure $\mathbb{P}$ is independent of the sources. If Y_β is the corresponding remainder derivative, then

$$S_n^T - \mathsf{J}_n = \sum_{\pi \in \mathrm{Part}([n])} \kappa_{\mathbb{P}}(\mathsf{Y}_\beta : \beta \in \pi). \tag{4.iii}$$

The bounds on moments of the stopping scale control every term of this finite sum. On the least torus, the first-creation estimates bound these moments uniformly in the cutoff. The dependence on the side is retained when this argument is used on a larger torus.

4.1.4. *Cancellation of the pair against the tensor.* The decisive estimate concerns

$$\Delta_n = (2\pi br^3/3) G_{n;12} - G_{n;T}$$

. The supplied proof first treats chain classes, then actual small-field backgrounds. On a regular cube, a real gauge makes the potential, its first lattice differences and its current small. The lattice Hodge identity applied to a cutoff of the curvature then gives its C^{β_H} estimate. These are estimates for lattice forms at their stated spacing. Their role is to replace the curvature at the two nearby supports by its value at their common centre, with an error tending to zero as r shrinks.

Separate the explicit kernel families created before the pair has been averaged out from all remaining families. The first part is the frozen part in lemma 4.36; its leading quadratic form cancels the tensor weight by the identity involving $D(e_\lambda)$. The remainder has small norm. The exact one-step averaging identity does not increase oscillation modulo constants, so this cancellation can be carried to coarser levels. The source proves first an error of size $r^{\beta_H/36} + (a_K/r)^{1/9}$ for this oscillation.

To attach the distant source, the tower identity of lemma 4.40 produces a covariance of Δ_n with the spectator and terms crossing the two clusters. Bound the covariance in two ways. Its oscillation estimate contains the small pair-minus-tensor error. Its family norm contains the exponential cost of reaching the distant support. Taking the smaller estimate, and then its geometric-mean bound, retains both advantages. Summing over levels now gives the square root of the former error. This is the source of $\beta_H/72$ and $1/18$ in theorem K; those exponents do not follow from the free contraction alone.

Finally integrate over the pair centre and sum the spectator pieces. Replacing the smeared tensor weight by φ_1 uses its Lipschitz continuity. Consequently the constant depends on $d\,\mathrm{Lip}(\varphi_1)/|\varphi_1|_\infty$ for a nonzero weight. At fixed r, the scalar three-point function has its cutoff limit; the comparison then determines the tensor limit. Only after that limit is taken does $r \to 0$ at fixed d give the point-splitting identity. The lattice error records why the reverse order is unavailable. The kernel statements likewise retain their order: bounds on arbitrarily narrow bumps give a Borel kernel and its almost-everywhere bounds; continuity supplied by reconstruction is the additional fact that permits evaluation at every separated configuration.

4.1.5. *Principal results.*

Theorem G (Three points at two scales)**.** Let g satisfy (Γ).

(A.1) *At a cutoff, on the least torus.* Let (K,β) be a first-passage datum at g, $\mathsf{b} = a_K\beta$, $\ell = \ell_g$; let $0 < \rho \le r \le \mathsf{d} \le \mathsf{d}_4(g)$ and $s(\rho) \le K$, and let w_1, w_2, w_3 be real weights placed at (ρ, r, d), for instance $w_a = f_a \circ x$ with bounded real Borel functions f_a. Then

$$\Big|S_3^T(w_1,w_2,w_3) - 3\,\mathrm{Tr}\prod_a (w_a \mathsf{b}^{-1} C_K^{\mathrm{free}})\Big| \le [\epsilon^{\mathrm{nar}}(g,\mathsf{d}) + 13\Lambda_3^3 \mathsf{M}'^3 \mathsf{b}^3 \mathsf{d}^{3/2}]\,\mathrm{nat}_3,$$

where $\mathsf{M}' = \int \alpha^{-24} d\mathbb{P} \le \bar{\mathsf{M}}^{(24)}$ (Lem. 4.20). The bound holds at every ratio $r/\mathsf{d} \in (0,1]$.

(A.2) *The kernel on the least torus.* Let b be a limit of labels of first-passage data at g (every $\mathsf{b} \in I(g)$ is one), and $\mathcal{D} = \{(x,y,z) \in T^3_{\ell_g} : 0 < |x-y| = r_* \le \mathsf{d}_* = \mathrm{dist}(z,\{x,y\}) \le \mathsf{d}_5(g)\}$. At every triple of $\mathcal{D}$, for every $0 < \rho < \min\{\frac{1}{4}r_*, r_1(\bar{\mathsf{b}})\}$ and all real Lipschitz f_1, f_2, f_3 with supports in balls of radius ρ inside the open balls of radius $\frac{1}{4}r_*$ about x, y, z,

$$\Big|S_3^{(\ell_g,\mathsf{b})}(f_1,f_2,f_3) - 3\mathsf{b}^{-3}\int k_3 \prod_a f_a\Big| \le \epsilon_{\mathrm{A}}(g,\mathsf{d}_*)\,\mathsf{b}^{-3} r_*^{-3}\mathsf{d}_*^{-6}\,\rho^9 \prod_a \|f_a\|_\infty.$$

Consequently $S_3^{(\ell_g,\mathsf{b})}$ has on $\mathcal{D}$ a Borel kernel s_3, and at every Lebesgue point of s_3 in $\mathcal{D}$, so almost everywhere on $\mathcal{D}$,

$$|\mathsf{s}_3 - 3\mathsf{b}^{-3}k_3| \le \epsilon_{\mathrm{A}}(g,\mathsf{d}_*)\,\mathsf{b}^{-3} r_*^{-3}\mathsf{d}_*^{-6},$$

uniformly in $r_*/\mathsf{d}_* \in (0,1]$. Choose the closest pair first and call it (x,y). The triple lies in $\mathcal{D}$ whenever the resulting far distance $\mathsf{d}_* = \mathrm{dist}(z,\{x,y\})$ is at most $\mathsf{d}_5(g)$.

(A.3) *Means at every triple.* At every triple of $\mathcal{D}$ and for every $0 < \rho < \min\{\frac{1}{4}r_*, r_1(\bar{\mathsf{b}})\}$ the mean of $\mathsf{s}_3 - 3\mathsf{b}^{-3}k_3$ over the product of the three balls of radius ρ about its points is at most $(\frac{3}{4\pi})^3\epsilon_{\mathrm{A}}(g,\mathsf{d}_*)\mathsf{b}^{-3}r_*^{-3}\mathsf{d}_*^{-6}$ in modulus; with the radial tents χ^ρ about the three points in place of the normalised indicators of the balls the same holds with $(\frac{3}{\pi})^3$ in place of $(\frac{3}{4\pi})^3$.

(A.4) *Every state; limits and components.* (i) Let (K,β) be a first-passage datum at g, $\mathfrak{s}$ a state at (K,β), $0 < \rho \le r \le \mathsf{d} \le 1$, $\rho \le \frac{1}{2}$, $s(\rho) \le K$, $\rho \le r_1(\mathsf{b})$, and let f_1, f_2, f_3 be bounded real Borel functions on $\mathbb{R}^3$ with $\operatorname{supp} f_a \subset \mathbb{B}(x_a,\rho)$, $\operatorname{dist}(\operatorname{supp} f_1, \operatorname{supp} f_2) \ge r$, $\operatorname{dist}(\operatorname{supp} f_a, \operatorname{supp} f_3) \ge \mathsf{d}$ for $a = 1,2$, all supports lying in one ball of radius at most $\frac{1}{8}\ell_g - 1$. Then

$$\big|S_3^T[\mathfrak{s}](f_1,f_2,f_3) - S_{3;K,\ell_g,\beta}^T(f_1^{\langle\ell_g\rangle}, f_2^{\langle\ell_g\rangle}, f_3^{\langle\ell_g\rangle})\big| \le C_3^{\mathrm{tr}3}(g)\,\mathsf{b}^3\mathsf{d}^{3/2}\,\mathrm{nat}_3;$$

so (A.1) holds for $S_3^T[\mathfrak{s}]$ with $C_3^{\mathrm{tr}3}(g)\mathsf{b}^3\mathsf{d}^{3/2}$ more in the bracket. (ii) Let (K_j,β_j) be first-passage data at g with $K_j \to \infty$ and $a_{K_j}\beta_j \to \mathsf{b}$, ν_j a state at (K_j,β_j), and σ' a law on $\mathbb{R}^3$ with $S_3^T[\nu_j] \to S_3^{\sigma'}$ on separated triples. Assume that $S_3^{\sigma'}$ has a continuous kernel on distinct triples, as it does for a periodic-Gibbs limit or one of its canonical components. Then that kernel s_3 satisfies, at every $(x,y,z) \in (\mathbb{R}^3)^3$ with $0 < |x-y| = r_* \le \mathsf{d}_* = \operatorname{dist}(z,\{x,y\}) \le \mathsf{d}_5(g)$,

$$|\mathsf{s}_3(x,y,z) - 3\mathsf{b}^{-3}k_3(x,y,z)| \le \mathsf{E}_3(g,\mathsf{d}_*)\,\mathsf{b}^{-3}r_*^{-3}\mathsf{d}_*^{-6},$$

uniformly in r_*/d_*. (iii) Every Gibbs sequence at g that converges, with its limit, and the approximants of every component of a canonical decomposition of that limit, with the component, are such data along every subsequence on which the labels converge.

(A.5) *Gibbs sequences at couplings above the ceilings.* Let $0 < g_0 \le \gamma$ be a reference coupling with $g \le g_0$. There is an integer $z_+ \ge 0$ formed from the constants, g_0 and g such that for every Gibbs sequence at g_0 that converges, and every subsequence along which its labels converge, to b_0 say, the continuous kernel s_3 of the limit, and that of every component of a canonical decomposition of it, satisfy at every triple with $0 < r_* \le \mathsf{d}_* \le L^{-z_+}\mathsf{d}_5(g)$

$$|\mathsf{s}_3 - 3\mathsf{b}_0^{-3}k_3| \le \max_{0\le z'\le z_+} \mathsf{E}_3(g, L^{z'}\mathsf{d}_*)\;\mathsf{b}_0^{-3}r_*^{-3}\mathsf{d}_*^{-6}.$$

■□

Theorem H (A close pair seen from the coarser levels)**.** Let $0 < g \le \gamma$ be a reference coupling, (K,β) a first-passage datum at g with its run, $\mathsf{b} = a_K\beta$, and $\ell \in \mathfrak{L}(g)$. For a level n and $A \in \mathsf{Cc}_n(r_n)$ put $\tilde{U} = e^{i\eta_n A}$.

(B.1) *A pair as wide as its distance.* There are numbers $\gamma_{\mathrm{O}}, C, c, c' > 0$ formed from the constants, and a number $r_0(\mathsf{b}) > 0$ formed from these and b, with $\inf\{r_0(\mathsf{b}') : g^{-2} < \mathsf{b}' \le \bar{\mathsf{b}}\} > 0$, such that the following holds. Let $g \le \min\{\gamma_{\mathrm{jt}}, \gamma_{\mathrm{O}}\}$; $0 < r \le r_0(\mathsf{b})$; $s = s(r)$ with $3 \le s \le K$ and $\mathsf{Lm}_s \le L^{s-3}$; $\vartheta \in (0,1)$; $(f_1,f_2) \in \mathfrak{T}_{T_\ell}(r;\vartheta)$ with $w_a = f_a \circ x \not\equiv 0$; $k_0 = \max\{K-s, 2\}$; and let q_n be formed over the base level k_0. Then for every $k_0 < n \le K$ and every $A \in \mathsf{Cc}_n(r_n)$

$$|\mathsf{F}_n(\tilde{U}) - \mathsf{F}_n(\mathbf{1}) - q_n(\mathsf{ang}^{[n]}(A))| \le \Big\{C\mathfrak{b}_n^2\delta_n + C(\log x_{k_0})^c\big[(r/\mathsf{b})^{1/2} + e^{-c'Y_s^{2\mathfrak{p}_0}}\big]\Big\}\,\hat{r}_n^3\,\mathsf{u}_1\mathsf{u}_2.$$

None of $\gamma_{\mathrm{O}}, C, c, c'$ depends on ϑ, K, ℓ or the level.

(B.2) *A narrow pair.* Let g satisfy (Γ1); let $0 < \rho \le r \le e^{-2\sqrt{3}}$, $s(\rho) \le K$, $\rho \le r_1(\mathsf{b})$; let $w_1, w_2 \not\equiv 0$ be real weights placed at (ρ, r), and let q_n be formed over the base level $\mathsf{k}(r)$. Then for every $\mathsf{k}(r) \le n \le K$ and every $A \in \mathsf{Cc}_n(r_n)$

$$|\mathsf{F}_n(\tilde{U}) - \mathsf{F}_n(\mathbf{1}) - q_n(\mathsf{ang}^{[n]}(A))| \le \big[C_{\mathrm{N}}\mathfrak{b}_n^2\delta_n + \epsilon'_{\mathrm{N}}(r)\big]\Big(\frac{\rho}{r}\Big)^6\hat{r}_n^3\,\mathsf{u}_1\mathsf{u}_2, \qquad \epsilon'_{\mathrm{N}}(r) = C'_{\mathrm{N}}Y_r^{15\mathfrak{p}_0+2}\big[g\,r^{1/2} + e^{-Y_r^{2\mathfrak{p}_0}/\sqrt{3}}\big].$$

$C_{\mathrm{N}}, C'_{\mathrm{N}}$ are formed from the constants, ϵ'_{N} depends on g and r alone; nothing in the bracket depends on ρ, K, ℓ, the position, the direction, the sign or the shape of the weights, and $\epsilon'_{\mathrm{N}}(r)(\log\frac{1}{r})^N \to 0$ as $r \to 0$ for every N.

(B.3) *The explicit form.* In both cases $\sup_{A\in\mathsf{Cc}_n(r_n)} |q_n(\mathsf{ang}^{[n]}(A))| \le C\,\mathfrak{b}_n^2(\rho/r)^6\hat{r}_n^3\mathsf{u}_1\mathsf{u}_2$, with $\rho = r$ in (B.1).

■□

Theorem I (The pair against the plane fields)**.** Let g satisfy (Γ), let b be as in (A.2), and let s_3 be the kernel of (A.2). For points c, z put $\mathsf{d} = |c - z|$, let $0 < r \le \frac{1}{2}\mathsf{d}$, $\mathsf{d} \le \frac{16}{17}\mathsf{d}_5(g)$, $\bar{\epsilon}_{\mathrm{A}} = \sup\{\epsilon_{\mathrm{A}}(g,t) : \frac{3}{4}\mathsf{d} \le t \le \frac{17}{16}\mathsf{d}\}$, and write $(c \pm \frac{r}{2}n, z)$ for the triple $(c + \frac{r}{2}n, c - \frac{r}{2}n, z)$; e_λ is the unit vector of the direction λ, and M_ρ denotes the mean over the balls of radius ρ about the arguments.

(C.1) *Against the free harmonic.* (i) For almost every $(c, h, z) \in (T_{\ell_g})^3$, with $h = rn$, $r = |h|$ in the range above,

$$\big|\tfrac{2\pi}{3}\mathsf{b}r^3\mathsf{s}_3(c \pm \tfrac{r}{2}n, z) - \mathsf{t}^\circ(n, c - z)\big| \le \big[12\bar{\epsilon}_{\mathrm{A}} + 2.7(r/\mathsf{d})^2\big]\mathsf{b}^{-2}\mathsf{d}^{-6}.$$

(ii) For every (c, n, r, z) in the range above the same holds with $\mathsf{s}_3(c \pm \frac{r}{2}n, z)$ replaced by $\limsup_{\rho\to 0}$ of its mean $\mathsf{M}_\rho\mathsf{s}_3(c \pm \frac{r}{2}n, z)$; and, for every $0 < \rho < \min\{\frac{1}{4}r, r_1(\bar{\mathsf{b}})\}$, with $\mathsf{s}_3(c \pm \frac{r}{2}n, z)$ replaced by $\int \mathsf{s}_3\,\chi^\rho_{c+\frac{r}{2}n} \otimes \chi^\rho_{c-\frac{r}{2}n} \otimes \chi^\rho_z$ and $7900\rho/r$ added in the bracket. (iii) For a law σ' as in (A.4)(ii), in particular for every periodic-Gibbs limit at g and every component of its canonical decomposition, the bound of (i) holds for its continuous kernel at every (c, r, z) and every unit vector n, with $12\sup\{\mathsf{E}_3(g,t) : \frac{3}{4}\mathsf{d} \le t \le \frac{17}{16}\mathsf{d}\}$ in place of $12\bar{\epsilon}_{\mathrm{A}}$.

(C.2) *The plane laws.* (i) At a cutoff: let (K, β) be a first-passage datum at g, $\mathsf{b} = a_K\beta$, $0 < \rho \le \mathsf{d} \le \mathsf{d}_4(g)$, $s(\rho) \le K$, $\vartheta \in (0, 1]$, and let φ_1, φ_2 be real Lipschitz functions on T_{ℓ_g} with $\operatorname{supp}\varphi_a \subset \mathbb{B}(x_a, \rho)$, $\operatorname{Lip}\varphi_a \le \|\varphi_a\|_\infty/\vartheta\rho$, the balls $\mathbb{B}(x_a, \rho)$ at a distance $\ge \mathsf{d}$ and $|x_1 - x_2| \le 18\mathsf{d}$; no sign condition. Then for all λ, μ

$$\big|\langle\mathcal{O}_\lambda(\varphi_1);\mathcal{O}_\mu(\varphi_2)\rangle_{K,\ell_g,\beta} - \tfrac{3}{2}\mathsf{b}^{-2}\mathcal{K}_{\lambda\mu}(\varphi_1,\varphi_2)\big| \le \Big[\epsilon_2^{\mathrm{nar}}(g,\mathsf{d}) + 3\Lambda_3^2\mathsf{M}'^2\mathsf{b}^2\mathsf{d}^{3/2} + \tfrac{3}{2}C_{\mathrm{Rs}}\Big(\frac{a_K}{\vartheta\rho} + \Big(\frac{\mathsf{d}}{\ell_g}\Big)^3\Big)\Big]\Big(\frac{\rho}{\mathsf{d}}\Big)^6\mathsf{b}^{-2}\|\varphi_1\|_\infty\|\varphi_2\|_\infty.$$

(ii) Let (K_j, β_j) be first-passage data at g with $K_j \to \infty$, $a_{K_j}\beta_j \to \mathsf{b}$, and $\Omega_{\mathrm{P}} = \{(x, y) \in T^2_{\ell_g} : 0 < |x - y| < 2\mathsf{d}_4(g)\}$. There is a subsequence along which the nine forms $\langle\mathcal{O}_\lambda(\varphi_1);\mathcal{O}_\mu(\varphi_2)\rangle_{K_j,\ell_g,\beta_j}$ converge for all real Lipschitz φ_1, φ_2 with $\operatorname{supp}\varphi_1 \times \operatorname{supp}\varphi_2 \subset \Omega_{\mathrm{P}}$; the nine limits $T_{\lambda\mu}$ along one such subsequence are a limit point of the plane laws. Every limit point has Borel kernels $\tau_{\lambda\mu}$ on Ω_{P} with

$$\big|\tau_{\lambda\mu}(x, y) - \tfrac{3}{2}\mathsf{b}^{-2}[\mathbb{K}_3(x - y)_{\lambda\mu}]^2\big| \le \epsilon_{\mathrm{P}}(g, |x - y|)\,\mathsf{b}^{-2}|x - y|^{-6} \quad \text{almost everywhere on } \Omega_{\mathrm{P}};$$

before a kernel is named, at every $(x_0, y_0) \in \Omega_{\mathrm{P}}$ with $\mathsf{d}_0 = |x_0 - y_0|$ and for Lipschitz φ_1, φ_2 supported in balls of radius ρ inside the balls of radius $\frac{1}{8}\mathsf{d}_0$ about x_0, y_0: $|T_{\lambda\mu}(\varphi_1, \varphi_2) - \frac{3}{2}\mathsf{b}^{-2}\mathcal{K}_{\lambda\mu}(\varphi_1, \varphi_2)| \le \epsilon_{\mathrm{P}}(g, \mathsf{d}_0)\mathsf{b}^{-2}\mathsf{d}_0^{-6}\rho^6\|\varphi_1\|_\infty\|\varphi_2\|_\infty$. The kernel $t_\lambda = \sum_\mu(\tau_{\lambda\mu} - \frac{1}{3}\sum_{\lambda'}\tau_{\lambda'\mu})$ of the limit of $\langle T_\lambda;\mathcal{O}\rangle$ satisfies $|t_\lambda(x, y) - \mathsf{t}^\circ(e_\lambda, x - y)| \le 6\epsilon_{\mathrm{P}}(g, |x - y|)\mathsf{b}^{-2}|x - y|^{-6}$ almost everywhere.

(C.3) *The pair against the kernel of $\langle T_\lambda;\mathcal{O}\rangle$.* Take one limit point as in (C.2)(ii), along a subsequence of (K_j, β_j); along it $S^T_{3;K_j,\ell_g,\beta_j} \to S_3^{(\ell_g,\mathsf{b})}$. For every c, z, r, λ in the range above

$$\limsup_{\rho\to 0}\big|\tfrac{2\pi}{3}\mathsf{b}r^3\mathsf{M}_\rho\mathsf{s}_3(c \pm \tfrac{r}{2}e_\lambda, z) - \mathsf{M}_\rho t_\lambda(c, z)\big| \le \big[12\bar{\epsilon}_{\mathrm{A}} + 6\epsilon_{\mathrm{P}}(g,\mathsf{d}) + 2.7(r/\mathsf{d})^2\big]\mathsf{b}^{-2}\mathsf{d}^{-6},$$

and the same with the radial tents of mass 1 in place of the means. For almost every (c,h,z), $h=rn$, in the range above,

$$\big|\tfrac{2\pi}{3}\mathsf{b}r^3\mathsf{s}_3(c\pm\tfrac{r}{2}n,z)-t_\lambda(c,z)\big|\le\big[12\bar\epsilon_{\rm A}+6\epsilon_{\rm P}(g,\mathsf{d})+2.7(r/\mathsf{d})^2+0.057|n-e_\lambda|\big]\mathsf{b}^{-2}\mathsf{d}^{-6}.$$

(C.4) *Limits and components on* $\mathbb{R}^3$. Let (K_j,β_j), ν_j, σ' be as in (A.4)(ii), now with $S_n^T[\nu_j]\to S_n^{\sigma'}$ for $n=2,3$, assume that $S_3^{\sigma'}$ and $S_2^{\sigma'}$ have continuous kernels s_3 and s_2^T on distinct configurations, and $\Omega_\sigma=\{(x,y)\in(\mathbb{R}^3)^2:0<|x-y|<\mathsf{d}_5(g)\}$. (a) There is a subsequence along which $\langle\mathcal{O}_\lambda(\varphi_1);\mathcal{O}_\mu(\varphi_2)\rangle_{\nu_j}$ converges for all λ,μ and all real Lipschitz φ_1,φ_2 with compact supports, $\operatorname{supp}\varphi_1\times\operatorname{supp}\varphi_2\subset\Omega_\sigma$. (b) For every such subsequence the limits have Borel kernels $\tau^\sigma_{\lambda\mu}$ on Ω_σ with $|\tau^\sigma_{\lambda\mu}(x,y)-\frac{3}{2}\mathsf{b}^{-2}[\mathbb{K}_3(x-y)_{\lambda\mu}]^2|\le(\frac{3}{\pi})^2\epsilon_\sigma(g,|x-y|)\mathsf{b}^{-2}|x-y|^{-6}$ almost everywhere; $\tau^\sigma_{\lambda\mu}(x,y)$ depends on $x-y$ alone (almost everywhere); and $\sum_{\lambda,\mu}\tau^\sigma_{\lambda\mu}=\mathsf{s}_2^T$ almost everywhere, so that $|\mathsf{s}_2^T(x,y)-\frac{9}{16\pi^2}\mathsf{b}^{-2}|x-y|^{-6}|\le 9\epsilon_\sigma(g,|x-y|)\mathsf{b}^{-2}|x-y|^{-6}$ at every $(x,y)\in\Omega_\sigma$. (c) With $t^\sigma_\lambda=\sum_\mu(\tau^\sigma_{\lambda\mu}-\frac{1}{3}\sum_{\lambda'}\tau^\sigma_{\lambda'\mu})$ there is one null set $\mathcal{N}\subset\Omega_\sigma$, fixed before r is named, such that for every $(c,z)\notin\mathcal{N}$ with $0<\mathsf{d}=|c-z|\le\frac{1}{2}\mathsf{d}_5(g)$, every $0<r\le\frac{1}{2}\mathsf{d}$ and every λ

$$\big|\tfrac{2\pi}{3}\mathsf{b}r^3\mathsf{s}_3(c\pm\tfrac{r}{2}e_\lambda,z)-t^\sigma_\lambda(c,z)\big|\le\big[\epsilon'_\sigma(g,\mathsf{d})+2.7(r/\mathsf{d})^2\big]\mathsf{b}^{-2}\mathsf{d}^{-6},\qquad \epsilon'_\sigma(g,\mathsf{d})=12\sup_{\frac{3}{4}\mathsf{d}\le t\le\frac{17}{16}\mathsf{d}}\mathsf{E}_3(g,t)+6\epsilon_\sigma(g,\mathsf{d}),$$

and $\epsilon'_\sigma(g,\mathsf{d})\le C(g)(\log\frac{1}{\mathsf{d}})^{39\mathfrak{p}_0+2}\mathsf{d}^{1/2}$; at every (c,z) in this range the same holds with $\sum_\mu(\mathsf{t}_{\lambda\mu}-\frac{1}{3}\sum_{\lambda'}\mathsf{t}_{\lambda'\mu})(\chi^\rho_c,\chi^\rho_z)$ in place of $t^\sigma_\lambda(c,z)$, $\mathsf{t}_{\lambda\mu}$ the limit forms, for every $0<\rho<\frac{1}{8}\mathsf{d}$, with $21\rho/\mathsf{d}$ added in the bracket. (d) At a cutoff, for every first-passage datum (K,β) at g, every state $\mathfrak{s}$ at (K,β), points x_0,y_0 with $0<\mathsf{d}_0=|x_0-y_0|\le 2\mathsf{d}_4(g)$, $\mathsf{d}=\frac{1}{2}\mathsf{d}_0$, $s(\rho)\le K$, and bounded real Borel f_1,f_2 on $\mathbb{R}^3$ supported in balls of radius ρ inside the closed balls of radius $\frac{1}{8}\mathsf{d}_0$ about x_0,y_0:

$$\Big|\langle\mathcal{O}_\lambda(f_1);\mathcal{O}_\mu(f_2)\rangle_{\mathfrak{s}}-\tfrac{3}{2}\mathsf{b}^{-2}\sum_{\lambda(p)=\lambda,\,\lambda(q)=\mu}f_1(x(p))f_2(x(q))\,C_K^{\rm free}(\operatorname{pr}p,\operatorname{pr}q)^2\Big|\le\Big[\mathsf{E}_2(g,\mathsf{d})-\tfrac{3}{2}C_{\rm Rs}\Big(\frac{\mathsf{d}}{\ell_g}\Big)^3\Big]\Big(\frac{\rho}{\mathsf{d}}\Big)^6\mathsf{b}^{-2}\|f_1\|_\infty\|f_2\|_\infty,$$

with $C_K^{\rm free}$ of the lattice (ℓ_g,K) and pr the projection onto T_{ℓ_g}.

■□

Theorem J (The traceless tensor in every direction)**.** Let $0<g\le\gamma$ be a reference coupling and $\ell\in\mathfrak{L}(g)$. There are an integer $K_\circ(g)$ and a number $\mathsf{d}_{\rm cl}(g)>0$ formed from the constants and g, and a finite function $\epsilon^{\rm cl}(g,\ell,\cdot)$ on $(0,\frac{1}{8}\ell]$ formed from the constants, g and ℓ, with $\epsilon^{\rm cl}(g,\ell,\mathsf{d}')\le C(g,\ell)(\log\frac{1}{\mathsf{d}'})^{19\mathfrak{p}_0}\mathsf{d}'^{1/2}$ for $\mathsf{d}'\le\mathsf{d}_{\rm cl}(g)$, such that the following hold.

(D.1) *The bond forms against the plane fields, on every torus.* Let (K,β) be a first-passage datum at g with $K\ge K_\circ(g)$, $\mathsf{b}=a_K\beta$; $0<16r\le\mathsf{d}'\le\frac{1}{8}\ell$ with $s(r)\le K$; $x_1,x_2\in T_\ell$ with $|x_1-x_2|=\mathsf{d}'$; $\vartheta\in(0,1]$; φ_1,φ_2 real Lipschitz functions on T_ℓ with $\operatorname{supp}\varphi_a\subset\mathbb{B}(x_a,r)$ and

$\operatorname{Lip}\varphi_a \le \|\varphi_a\|_\infty/\vartheta r$. Then for all $\lambda \neq \lambda'$ and all γ, μ

$$\begin{aligned}
&\Big|\langle \mathcal{O}^{\mathrm{od}}_{\lambda\lambda'}(\varphi_1); \mathcal{O}_\gamma(\varphi_2)\rangle_{K,\ell,\beta} \\
&\qquad - 3\mathsf{b}^{-2}\iint \varphi_1(x)\varphi_2(y)\mathbb{K}_3(x-y)_{\lambda\gamma}\mathbb{K}_3(x-y)_{\lambda'\gamma}\,\mathrm{d}x\,\mathrm{d}y\Big| \\
&\le \Big[8\epsilon^{\mathrm{cl}}(g,\ell,\mathsf{d}') + 3C^{\mathrm{od}}_{\mathrm{PS}}\frac{a_K}{\vartheta r} + 3C^{\mathrm{od}}_{\mathrm{PS}}\Big(\frac{\mathsf{d}'}{\ell}\Big)^3\Big] \\
&\quad \Big(\frac{r}{\mathsf{d}'}\Big)^6 \mathsf{b}^{-2}\|\varphi_1\|_\infty\|\varphi_2\|_\infty, \\
&\big|\langle \mathcal{O}_\lambda(\varphi_1); \mathcal{O}_\mu(\varphi_2)\rangle_{K,\ell,\beta} - \tfrac{3}{2}\mathsf{b}^{-2}\mathcal{K}_{\lambda\mu}(\varphi_1,\varphi_2)\big| \\
&\le \Big[\epsilon^{\mathrm{cl}}(g,\ell,\mathsf{d}') + \tfrac{3}{2}C_{\mathrm{PS}}\frac{a_K}{\vartheta r} + \tfrac{3}{2}C_{\mathrm{PS}}\Big(\frac{\mathsf{d}'}{\ell}\Big)^3\Big] \\
&\quad \Big(\frac{r}{\mathsf{d}'}\Big)^6 \mathsf{b}^{-2}\|\varphi_1\|_\infty\|\varphi_2\|_\infty.
\end{aligned}$$

(D.2) *The kernels of $\langle T_n;\mathcal{O}\rangle$ for all directions at once.* Let (K_j,β_j) be first-passage data at g with $K_j\to\infty$ and $a_{K_j}\beta_j\to\mathsf{b}$; $\Omega^\circ = \{(x,y)\in T^2_\ell : 0<|x-y|<\frac{7}{9}\mathsf{d}_{\mathrm{cl}}(g)\}$;

$$\bar\epsilon^{\mathrm{cl}}(t) = \sup\{8\epsilon^{\mathrm{cl}}(g,\ell,\mathsf{d}'') + 3\max\{C_{\mathrm{PS}},C^{\mathrm{od}}_{\mathrm{PS}}\}(\mathsf{d}''/\ell)^3 : \frac{7}{9}t\le \mathsf{d}''\le\frac{9}{7}t\},$$

so that $\bar\epsilon^{\mathrm{cl}}(t) \le C(g,\ell)[(\log\frac{1}{t})^{19\mathfrak{p}_0}t^{1/2} + (t/\ell)^3]$. Along a subsequence all the forms $\langle\mathcal{O}_\lambda(f);\mathcal{O}_\mu(h)\rangle_{K_j,\ell,\beta_j}$ and $\langle\mathcal{O}^{\mathrm{od}}_{\lambda\lambda'}(f);\mathcal{O}_\gamma(h)\rangle_{K_j,\ell,\beta_j}$ converge, at all real Lipschitz f,h with $\operatorname{supp}f\times\operatorname{supp}h\subset\Omega^\circ$.

For every such subsequence the limits have Borel kernels on Ω° that lie, almost everywhere, within $4.2\,\bar\epsilon^{\mathrm{cl}}(|x-y|)\mathsf{b}^{-2}|x-y|^{-6}$ of the free kernels $\frac{3}{2}\mathsf{b}^{-2}[\mathbb{K}_3(x-y)_{\lambda\mu}]^2$ and $3\mathsf{b}^{-2}\mathbb{K}_3(x-y)_{\lambda\gamma}\mathbb{K}_3(x-y)_{\lambda'\gamma}$; and outside one null set of Ω°, for all unit vectors n at once, the kernel t_n of the limit of $\langle T_n(f);\mathcal{O}(h)\rangle$ satisfies

$$|t_n(x,y) - \mathsf{t}^\circ(n,x-y)| \le 38\,\bar\epsilon^{\mathrm{cl}}(|x-y|)\,\mathsf{b}^{-2}|x-y|^{-6}.$$

(D.3) *The pair against t_n.* Let moreover g satisfy (Γ) and $\ell=\ell_g$, and take s_3 of (A.2) and a limit point of (D.2). For every c,z,r,n with $0<r\le\frac{1}{2}\mathsf{d}$, $\mathsf{d}=|c-z|\le\frac{16}{17}\mathsf{d}_5(g)$ and $\mathsf{d}<\frac{7}{9}\mathsf{d}_{\mathrm{cl}}(g)$,

$$\limsup_{\rho\to0}|\tfrac{2\pi}{3}\mathsf{b}r^3\mathsf{M}_\rho\mathsf{s}_3(c\pm\tfrac{r}{2}n,z) - \mathsf{M}_\rho t_n(c,z)| \le [12\bar\epsilon_{\mathrm{A}} + 38\bar\epsilon^{\mathrm{cl}}(\mathsf{d}) + 2.7(r/\mathsf{d})^2]\mathsf{b}^{-2}\mathsf{d}^{-6},$$

the same with the radial tents of mass 1 in place of the means, and the same bound holds for $|\frac{2\pi}{3}\mathsf{b}r^3\mathsf{s}_3(c\pm\frac{r}{2}n,z) - t_n(c,z)|$ at almost every (c,rn,z) in this range.

■□

Theorem K (Pair minus tensor field; the limit $r\to0$)**.** Fix $\beta_{\mathrm{H}}\in(0,1)$. There is a number $\gamma_{\mathrm{R}}\in(0,\gamma_{\mathrm{jt}}]$ formed from the constants and β_{H} such that every $0<g\le\gamma_{\mathrm{R}}$ satisfies (Γ) and $g\le\gamma_{\mathrm{O}}$ of (B.1), and besides the two conditions $\frac{3}{2}C_\natural c_\gamma g\le\log2$, where $C_\natural=(L^8+8c_3)L^4$ with c_3 of Lem. 4.6(c) and c_γ of Lem. 4.6(f), and $c_{\mathrm{A}}\alpha_0\ge 32M\mathsf{c}_\natural\delta(t)(1+\log t)$ for all $t\ge g^{-2}$, which contains the condition $c_{\mathrm{A}}\alpha_0\ge 5M\mathsf{c}_\natural\delta(t)(1+\log t)$ for all $t\ge g^{-2}$, the last member of the list of γ_{N} in Def. 4.1(1), together with $g\le 19C_{\mathrm{A}}C_{\mathrm{E},3}M^2\mathsf{c}_\natural^2A_1^2/2C_{\mathrm{E},1}\alpha_0^2$, with the numbers of Def. 4.1(6); each of them is a condition "$g\le$ a number formed from the constants", and together they contain the ceilings of all the lemmas used in the proof.

The following holds for $0<g\le\gamma_{\mathrm{R}}$ on the least torus $\ell=\ell_g$. Let (K,β) be a first-passage datum at g, $\mathsf{b}=a_K\beta$, λ a direction, $\mathsf{d}>0$ with $6\mathsf{d}\le\min\{1,\mathsf{b}\}$, and φ_1,φ_2 bounded real functions on T_{ℓ_g} with $\operatorname{supp}\varphi_a\subset\mathbb{B}(y_a,\mathsf{d})$ and $\operatorname{dist}(\operatorname{supp}\varphi_1,\operatorname{supp}\varphi_2)\ge\mathsf{d}$. Let $r_\alpha(g,\mathsf{b})>0$ be the number of

Def. 4.1(5), formed from the constants, g and b. For every r with

$$128a_K \le r \le \min\{r_\alpha(g,\mathsf{b}),(\mathsf{d}/5LM)^2\}, \qquad 3 \le s(\tfrac{r}{2}) \le K, \qquad \tfrac{r}{2} \le \min\{r_0(\mathsf{b}), r_1(\mathsf{b})\}, \quad \mathsf{Lm}_{s(r/2)} \le L^{s(r/2)-3},$$

the following holds. For $c \in \operatorname{supp}\varphi_1$ let $w_{1,2} = \chi^{r/8}_{c\pm\frac{r}{2}e_\lambda} \circ x$ and $w_T(p) = (\delta_{\lambda(p)\lambda} - \frac{1}{3})\chi^{r/8}_c(x(p))$, so that $\mathcal{O}(w_T) = T_\lambda(\chi^{r/8}_c)$; let $\mathsf{c} = \frac{2\pi}{3}\mathsf{b}r^3$; let w_3 be $\varphi_2 \circ x$ restricted to the plaquettes with barycentres in a cube of side $\frac{r}{4}$ whose centre z has $|c-z| \le 6\mathsf{d}$, and $\mu_3 = r^3\|\varphi_2\|_\infty$. Let Y_β be the remainders of Lem. 4.20 for the lists (w_1,w_2,w_3) and (w_T,w_3), $\mathsf{Y}_\Delta = \mathsf{c}\mathsf{Y}_{12} - \mathsf{Y}_T$ and $\mathsf{Y}_{\Delta 3} = \mathsf{c}\mathsf{Y}_{123} - \mathsf{Y}_{T3}$.

(E.1) *The part that is not explicit.* There is a finite function $\mathsf{C}_\mathrm{R}(g,\mathsf{b};\cdot)$ on $(0,1]$, formed from the constants, β_H, g and b, such that

$$\Big|\int \mathsf{Y}_{\Delta 3}\,\mathrm{d}\mathbb{P} + \kappa_\mathbb{P}(\mathsf{Y}_\Delta,\mathsf{Y}_3)\Big| \le \mathsf{C}_\mathrm{R}(g,\mathsf{b};\mathsf{d})\big[r^{\beta_\mathrm{H}/72} + (a_K/r)^{1/18}\big]\mathsf{b}^{-2}\mathsf{d}^{-6}\mu_3.$$

If g satisfies besides (Γ3) of Def. 4.1(6), one may take $\mathsf{C}_\mathrm{R}(g,\mathsf{b};\mathsf{d}) = C(g,\mathsf{b})\,\mathsf{d}^{-5/2}$, and the pair minus the tensor field is nearly constant up to the last level: with $\Delta_n = \mathsf{c}G_{n;12} - G_{n;T}$ and $\Gamma_n = \mathsf{c}G_{n;123} - G_{n;T3}$, formed with one progression of the four sources, with $\varepsilon^\mathcal{W} = C(g,\mathsf{b})[r^{\beta_\mathrm{H}/72} + (a_K/r)^{1/18}]$, and for $r \le r^\mathcal{W}_\alpha(g,\mathsf{b})$, there are numbers c_n such that at every level $n \le K$ with $a_{K-n} \ge r^{1/2}$, for m_n-almost every small-field background W of the level n (Lem. 4.20),

$$|\Delta_n(U_n(W)) - \mathsf{c}_n| \le \varepsilon^\mathcal{W}\varepsilon_n^2 a_{K-n}^{-4}, \qquad |\mathsf{c}_n - \mathsf{c}_K| \le \varepsilon^\mathcal{W}\varepsilon_n^2 a_{K-n}^{-4}, \qquad |\Gamma_n(U_n(W)) - \Gamma_K(\mathbf{1})| \le C(\mathsf{b})\,\mathsf{d}^{-5/2}\varepsilon^\mathcal{W}\mathsf{b}^{-2}\mathsf{d}^{-6}\mu_3.$$

(E.2) *The true functions.* There are numbers $C_\mathrm{A}(g,\mathsf{b})$, formed from the constants, β_H, g and b, and $r_\mathcal{W}(g,\mathsf{b},\mathsf{d}) > 0$, formed besides with d, such that for $r \le r_\mathcal{W}(g,\mathsf{b},\mathsf{d})$

$$|\mathsf{c}\,S_3^T(w_1,w_2,w_3) - S_2^T(w_T,w_3)| \le C_\mathrm{A}(g,\mathsf{b})\,\mathsf{d}^{-18}\big[r^{\beta_\mathrm{H}/72} + (a_K/r)^{1/18}\big]\mu_3.$$

(E.3) *At a fixed label: the limit in the cutoff, and $r \to 0$.* Let φ_1,φ_2 be as above, Lipschitz, with $|y_1 - y_2| \le 3\mathsf{d}$, let $f^c_\pm = \chi^{r/8}_{c\pm\frac{r}{2}e_\lambda}$, and for a first-passage datum (K,β) at g with the label $\mathsf{b}' = a_K\beta$ put

$$\mathsf{A}_{r,K} = \tfrac{2\pi}{3}\mathsf{b}'r^3\int \mathrm{d}c\,\varphi_1(c)\,S^T_{3;K,\ell_g,\beta}(f^c_+,f^c_-,\varphi_2).$$

If either φ_1 or φ_2 is zero, the assertions are immediate by multilinearity; assume both are nonzero in the normalized estimates that follow. At every label $\mathsf{b}' \in (g^{-2},\bar{\mathsf{b}}]$ there are numbers $r_\sharp > 0$ and $\mathsf{C}(\mathsf{d})$, formed from the constants, β_H, g, b', d and $\mathsf{d}\operatorname{Lip}\varphi_1/\|\varphi_1\|_\infty$, such that: (i) for $0 < r \le r_\sharp$ and every first-passage datum (K,β) at g with the label b', $128a_K \le r$ and $s(\frac{r}{2}) \le K$, $|\mathsf{A}_{r,K} - \langle T_\lambda(\varphi_1);\mathcal{O}(\varphi_2)\rangle_{K,\ell_g,\beta}| \le \mathsf{C}(\mathsf{d})[r^{\beta_\mathrm{H}/72} + (a_K/r)^{1/18}]\mathsf{b}'^{-2}\|\varphi_1\|_\infty\|\varphi_2\|_\infty$. Let now $\mathsf{b} \in I(g)$ and $\beta_K = L^K\mathsf{b}$, so that (K,β_K) is a first-passage datum at g with the label b for every $K \ge 1$; let $\mathsf{A}_{r,K}$ be formed at (K,β_K), and $r_\sharp$, $\mathsf{C}(\mathsf{d})$ at the label b.

Then (ii) the limit $t_\lambda(\varphi_1,\varphi_2) = \lim_{K\to\infty}\langle T_\lambda(\varphi_1);\mathcal{O}(\varphi_2)\rangle_{K,\ell_g,\beta_K}$ exists; (iii)

$$\mathsf{A}_r = \lim_K \mathsf{A}_{r,K} = \frac{2\pi}{3}\mathsf{b}r^3\int \mathrm{d}c\,\varphi_1(c)S_3^{(\ell_g,\mathsf{b})}(f^c_+,f^c_-,\varphi_2)$$

exists and

$$|\mathsf{A}_r - t_\lambda(\varphi_1,\varphi_2)| \le \mathsf{C}(\mathsf{d})r^{\beta_\mathrm{H}/72}\mathsf{b}^{-2}\|\varphi_1\|_\infty\|\varphi_2\|_\infty$$

for $0 < r \le r_\sharp$; so $\mathsf{A}_r \to t_\lambda(\varphi_1,\varphi_2)$ as $r \to 0$ at a fixed d.

(E.4) *Labels that tend to* b. For $\mathsf{b} \in I(g)$, the φ_a of (E.3) and every sequence of first-passage data (K_j,β_j) at g with $K_j \to \infty$ and $a_{K_j}\beta_j \to \mathsf{b}$: $\langle T_\lambda(\varphi_1);\mathcal{O}(\varphi_2)\rangle_{K_j,\ell_g,\beta_j} \to t_\lambda(\varphi_1,\varphi_2)$.

■□

Theorem L (Four points at two scales)**.** Let g satisfy $(\Gamma^{(4)})$, let (K,β) be a first-passage datum at g, $\mathsf{b}=a_K\beta$, $0<\rho\le r\le \mathsf{d}\le \mathsf{d}_4^{(4)}(g)$, $s(\rho)\le K$, and let $w_1,\dots,w_4$ form two pairs at (ρ,r,d).

(a) On every side $\ell\in\mathfrak{L}(g)$ with $64\mathsf{d}\le\ell$,

$$\Big|\mathsf{J}_4-\tfrac{3}{2}\sum_{\varsigma}\operatorname{Tr}\prod_{l}(w_{\varsigma(l)}\mathsf{b}^{-1}C_K^{\mathrm{free}})\Big|\le\epsilon_4^{\mathrm{nar}}(g,\mathsf{d})\,\mathrm{nat}_4,$$

the sum over the six cyclic orders of $\{1,2,3,4\}$; on the least torus

$$|S_4^T-\mathsf{J}_4|\le 77\Lambda_3^4\mathsf{M}'^4\mathsf{b}^4\mathsf{d}^{3/2}\mathrm{nat}_4;$$

and in every state at (K,β), for bounded real Borel functions on $\mathbb{R}^3$ in the geometry of (A.4)(i) for two pairs, the connected function differs from that of the least torus at the periodised functions by at most

$$C_4^{\mathrm{tr}3}(g)\mathsf{b}^4\mathsf{d}^{3/2}\mathrm{nat}_4.$$

(b) If $w_a=f_a\circ x$ with real Lipschitz f_a on T_{ℓ_g} such that $\operatorname{supp}f_a\subset\mathbb{B}(x_a,\rho)$, x_a the centres with which the w_a form the two pairs (so that $|x_1-x_2|,|x_3-x_4|\le 8r$ and $|x_a-x_b|\le 18\mathsf{d}$ for $a\le 2<b$), the balls $\mathbb{B}(x_a,\rho)$ are at least r apart in $\{1,2\}$ and in $\{3,4\}$ and at least d apart across, $\operatorname{Lip}f_a\le\|f_a\|_\infty/\vartheta\rho$, $\vartheta\in(0,1]$, and all supports lie in one ball of radius $\frac12\mathsf{d}^+$ with $4\mathsf{d}^+<\ell_g$, then

$$\Big|S_4^T(f)-3\mathsf{b}^{-4}\int k_4\prod_a f_a\Big|\le\Big[\epsilon_4^{\mathrm{nar}}(g,\mathsf{d})+77\Lambda_3^4\mathsf{M}'^4\mathsf{b}^4\mathsf{d}^{3/2}+3C^{(4)}\Big(\frac{a_K}{\vartheta\rho}+\Big(\frac{\mathsf{d}^+}{\ell_g}\Big)^3\Big)\Big]\mathrm{nat}_4.$$

Here $8a_K\le\rho$, by the working level at $\varrho=\rho$, and $\mathsf{d}\le\mathsf{d}^+$, the supports of f_1 and f_3 being at least d apart in a ball of diameter d^+; if an f_a vanishes, both members are 0.

(c) Let b be a limit of labels of first-passage data at g and $\mathcal{D}_4=\{(x_1,\dots,x_4):r_*\le|x_1-x_2|,|x_3-x_4|\le 2r_*,\ r_*\le\mathsf{d}_*=\operatorname{dist}(\{x_1,x_2\},\{x_3,x_4\})\le\mathsf{d}_5^{(4)}(g)\}$. Then $S_4^{(\ell_g,\mathsf{b})}$ has on $\mathcal{D}_4$ a Borel kernel s_4 with, almost everywhere on $\mathcal{D}_4$ and uniformly in $r_*/\mathsf{d}_*\in(0,1]$,

$$|\mathsf{s}_4-3\mathsf{b}^{-4}k_4|\le\epsilon_4(g,\mathsf{d}_*)\,\mathsf{b}^{-4}r_*^{-6}\mathsf{d}_*^{-6},\qquad \epsilon_4(g,t)=4096\big[\epsilon_4^{\mathrm{nar}}(g,\tfrac{t}{2})+77\Lambda_3^4(\bar{\mathsf{M}}^{(24)})^4\bar{\mathsf{b}}^4t^{3/2}+3C^{(4)}(12t/\ell_g)^3\big].$$

■□

Remark 4.2 (The ceilings). Let x_W be the threshold of Section YM3_I:126, equation YM3_I:(2326), and put

$$x_{\mathrm{jt}}=\max\left\{x_W,\Big(\frac{20M\mathsf{c}_{\mathfrak{h}}A_1(4\mathfrak{p}_0)^{\mathfrak{p}_0}}{c_A\alpha_0}\Big)^4\right\},\qquad\gamma_{\mathrm{jt}}=x_{\mathrm{jt}}^{-1/2}.\tag{4.iv}$$

The notation x_{jt}, γ_{jt} denotes these same numbers. The defining inequalities for x_W include those for x_{J}, so $x_{\mathrm{jt}}\ge x_W\ge\max\{\gamma^{-2},x_{\mathrm{J}}\}$. For $t>1$, the inequality $\log u\le u$, applied to $u=t^{1/(4\mathfrak{p}_0)}$, gives

$$\delta(t)=A_1t^{-1/2}(\log t)^{\mathfrak{p}_0}\le A_1(4\mathfrak{p}_0)^{\mathfrak{p}_0}t^{-1/4}.$$

Consequently $t\ge x_{\mathrm{jt}}$ implies $5\mathsf{c}_{\mathfrak{h}}\delta(t)\le c_A\alpha_0/(4M)$. Also $\gamma_W^{\mathrm{ref}}=x_W^{-1/2}$ and $\gamma_W^{\mathrm{sd}}\le\gamma_W^{\mathrm{ref}}$ by Section YM3_I:126, equation YM3_I:(2327). Hence $\gamma_{\mathrm{jt}}\le\gamma_W^{\mathrm{ref}}\le\gamma$ and $\gamma_W^{\mathrm{sd}}\le\gamma$. Every coupling under (Γ1), (Γ2), (Γ) or $(\Gamma^{(4)})$, or below γ_{N} or γ_{R}, therefore satisfies $0<g\le\gamma$. For $g\le\gamma_{\mathrm{jt}}$ one also has $g^{-2}\ge e^{2\mathfrak{p}_0}$. All these ceilings are positive numbers depending only on the fixed constants.

4.1.6. *Construction estimates.* We use the construction results at their stated hypotheses, including Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD. The finite construction of Thm. YM3_I:D and its sourced form, Thm. YM3_I:AJ, enters wherever the objects of the theory (the run of a first-passage datum, the effective actions of all levels with their bounds, and the weighted progression with any number of sources) satisfy the actual-step hypotheses of [7, Prop. B.21]; this holds for $0 < g \le \gamma$ on every side of $\mathfrak{L}(g)$, with $g^{-2} \ge x_{\mathrm{J}}$ where sources are present. [7, Prop. B.8] constructs the weighted progression for sources on the bare plaquettes; mixed sources enter through clauses (5) and (6) of the lemma on bond sources, which follows Lem. 4.31: at $0 < g \le \gamma_{\mathrm{N}}$, on every side of $\mathfrak{L}(g)$, the progression and the representation of the torus expectations are formed by the same construction, with the weight member of the mixed source in the source slot of the bare arrays; the frame of the pair functional is read with the hulls of the sources in the place of their supports; and the tower identity holds for a pair of mixed sources in that frame.

Thms. YM3_I:AJ and YM3_I:AD enter through the two-sided stability bound of the torus measures, which together with their reflection positivity gives the chessboard bound of Lem. 4.25. Through this bound they are used in (A.4), (A.5), (C.1)(iii), (C.4) and for the states in Thm. L(a); they are not used in Thm. H, in (A.1)–(A.3), (C.1)(i), (ii), (C.2), (C.3), in Thm. J, or in Thm. L on the tori. Thm. K, like every statement of this section, assumes the construction inputs just listed.

We use the fixed construction setting just specified, the torus limits of Thm. YM3_I:A, and the periodic-Gibbs limits and decompositions of Thms. YM3_I:AP and YM3_I:AL. On one renormalisation step (whose provision is that of Section YM3_I:52.4) and on the progression: Prop. 6.31, with the fourth-family norm eq. 6.xvii; the bracket localisation of the families with its count; the locality and linear-weight dependence of [7, Prop. B.8]; the chart and normalized real measures of Lems. [7, Lem. B.3] and 2.2, with the cumulant bounds of Lem. 4.14; the expansion $|a_p(e^{i\eta_i A}) - \frac{1}{2}|\mathsf{ang}_p^{[i]}(A)|^2| \le \mathsf{M}_{\mathrm{pl}}^2 \eta_{i-1}^4 (r_i/\varrho_{\mathrm{ch}})^3$ on the chain classes, and $|\mathsf{ang}_p^{[i]}(A)| \le 2\eta_i^2 r_i$; the regular-field evaluation and domain sums of Lems. [7, Lem. B.22] and [7, Lem. B.10]; the lowering maps of the chain classes, by which $A' \in \mathsf{Cc}_{i+1}(r_{i+1})$ and a fluctuation field B give $A^{\mathrm{lw}}(B) \in \mathsf{Cc}_i(r_i)$, and the pair tower identity of Prop. 2.4, extended to several sources in the lemma preceding Prop. 4.16; the history decomposition eq. 4.xxxi, its sourced quotients in [7, Prop. B.15], and the tail bounds of Prop. 4.54.

On the free kernel, Prop. YM3_I:120.6 and equations YM3_I:(2245)–(2246) give the decay of the slices $|\mathcal{C}_{(j)}(p,q)| \le C_{\mathrm{tp}} a_K^3 \mathsf{r}^{-3} t_j^3 e^{-\delta_{\mathrm{tp}} t_j}$, $\mathsf{r} = |x(p) - x(q)|_\infty$, $t_j = \mathsf{r}/a_{K-j+1}$; and the comparison $|\epsilon_p \epsilon_q C_K^{\mathrm{free}}(p,q) - a_K^3 \mathbb{K}_3(u)_{\lambda(p)\lambda(q)}| \le C_{\mathrm{lat}} a_K^3 [a_K^2 |u|^{-5} + \ell^{-3}]$, $|C_K^{\mathrm{free}}(p,q)| \le (1 + 3C_{\mathrm{lat}}) a_K^3 |u|^{-3}$, for $p \ne q$, u the least lift of $x(p) - x(q)$, $|u| \ge 2^{-1/2} a_K$, $\epsilon_p = \pm 1$ the orientation signs. On the working level (Lem. YM3_I:126.1): for $0 < \varrho < 1$ with $s = s(\varrho) \le K$ and $k = \mathsf{k}(\varrho)$, $Y = Y_s$: $s \ge 3$, $2 \le k \le K - 1$, $4MY^{2\mathfrak{p}_0} \le \hat{\varrho}_k < LC_{W,0} A_0^2 Y^{2\mathfrak{p}_0}$, $Y \ge \log \frac{1}{\varrho}$, $y_k \le Y$, $K - k \le s \le Y/\log L$; in particular $a_K \le a_{K-k} = \varrho/\hat{\varrho}_k \le \varrho/4M$, and $Y \ge 2\sqrt{3}$ when $\varrho \le e^{-2\sqrt{3}}$.

The lower bounds come from the definition of $s(\varrho)$ and from $C_{W,0} A_0^2 \ge 4L^2 M$: $L^s \varrho \ge C_W(s) \ge 4L^2 M Y^{2\mathfrak{p}_0} > L^2$ and $\varrho < 1$ give $s \ge 3$; $K - k = \min\{s, K-2\} \ge s - 2$ gives $\hat{\varrho}_k = L^{K-k}\varrho \ge L^{-2} C_W(s) \ge 4MY^{2\mathfrak{p}_0}$; and $L^s \varrho \ge C_W(s) \ge 1$ gives $Y \ge s \log L \ge \log \frac{1}{\varrho}$. Besides, $L^m \ge LMy_K$ for the sides $\ell = 2L^m \in \mathfrak{L}(g)$. Finally the dilation of lattices: $x \mapsto L^z x$ carries the lattice (ℓ, K) to the lattice $(L^z \ell, K - z)$ and $\mu_{K,\ell,\beta}$ to $\mu_{K-z, L^z\ell, \beta}$.

4.1.7. *Estimates and proofs.* The argument has eight parts: the calculus of the derivative families (Lems. 4.3–4.10); the kernel families (Lem. 4.11 to Prop. 4.13); the induction over the levels and the jets (Lem. 4.14 to Prop. 4.19); the connected functions on the least torus (Lems. 4.20–4.24); every state (Lem. 4.25 to Lem. 4.30); the bond forms (Lem. 4.31 to Prop. 4.34); pair minus tensor field (the lemma on a regular gauge on a cube, and Lems. 4.36 to 4.41); and the proofs of the theorems. The calculus of the derivative families.

Lemma 4.3 (The run)**.** Let $0 < g \le \gamma$ with $g^{-2} \ge x_{\mathrm{J}}$, which holds for every $0 < g \le \gamma_{\mathrm{jt}}$ (Remark 4.2), and (K, β) a first-passage datum at g with its run and label b.

(a) $x_K > g^{-2} \ge x_{\mathrm{J}} \ge \max\{e^{2\mathfrak{p}_0}, eL, 4\bar{B}\}$, so $y_K \ge \max\{2\mathfrak{p}_0, 1 + \log L\}$. For $0 \le i \le j \le K$: $\frac{4}{5}\mathsf{b} \le a_{K-j}x_j \le \frac{4}{3}\mathsf{b}$; $\frac{3}{4}L^{j-i}x_j \le x_i \le \frac{5}{4}L^{j-i}x_j$, in particular $x_j \ge \frac{4}{5}L^{-(j-i)}x_i$; $y_j \le y_i$; $\delta_j \le (\frac{5}{4})^{1/2}L^{(j-i)/2}\delta_i$ and $\delta_i \le \delta_j$; $g^{-2} < \mathsf{b} \le \bar{\mathsf{b}}$; and $\frac{11}{10}\mathsf{c}_{\mathfrak{h}}\delta_i \le r_i \le \frac{8}{5}\mathsf{c}_{\mathfrak{h}}\delta_i$ for $1 \le i < K$, so $x_n r_n^2 \le \frac{64}{25}\mathsf{c}_{\mathfrak{h}}^2\mathfrak{b}_n^2$. Of these, $g^{-2} < \mathsf{b} \le \bar{\mathsf{b}}$ holds, together with the flow bound $|x_j - L^{-(j-i)}x_i| \le \bar{B}$ $(0 \le i \le j \le K)$, for the label and the run of every first-passage datum at every $0 < g \le \gamma$.

(b) Let $g^{-2} \ge \max\{e^{2\mathfrak{p}_0}, e^{y_{\mathfrak{p}}}\}$ and $\delta(g^{-2}) \le \min\{1/2c_{\mathrm{rt}}, 1/c_\epsilon\}$, where $c_\delta = (\frac{5}{4}L)^{1/2}$, $c_{\mathrm{rt}} = \max\{8\mathsf{c}_{\mathfrak{h}}c_\delta/5\varrho_{\mathrm{ch}}, 1/\varrho_{\mathfrak{fl}}\}$, $c_\epsilon = c_{\mathrm{cu}} + 16\mathsf{c}_{\mathbf{C}}\mathsf{c}_{\mathfrak{h}}c_\delta/5r_\alpha$, $y_{\mathfrak{p}} = \max\{1, (LM)^3A_1^{-2}(1+2\log^+(c_{\mathfrak{p}}/A_1))\}$. Here ϱ_{ch} and $\varrho_{\mathfrak{fl}}$ are the radii denoted by ρ_{ch} and $\rho_{\mathfrak{fl}}$ in equation YM3_I:(2285), and $r_\alpha = a^{\mathrm{us}}(\alpha_0, \alpha_1)$ is the usable chart radius of equation YM3_I:(933). The covariance and fluctuation-tail constants are

$$\mathsf{c}_{\mathbf{C}} = 2/\gamma_r, \qquad c_{\mathfrak{p}} = C_{\mathrm{ex}}((L+2)M)^3 e^{4\lambda_1},$$

as in equations YM3_I:(2294) and YM3_I:(2296). For this regular one-step expansion, $\gamma_2 = 2(LM)^{-3}$ by equation YM3_I:(889); hence its tail factor $e^{-\gamma_2\mathfrak{b}_i^2/2}$ is $e^{-(LM)^{-3}\mathfrak{b}_i^2}$. Finally, c_{cu} is the constant of Lem. 4.14.

Then at all heights $1 \le i < K$: $2c_{\mathrm{rt}}\delta_i \le 1$, $c_\epsilon\delta_i \le 1$, $y_i \ge y_{\mathfrak{p}}$; hence $2r_i \le \varrho_{\mathrm{ch}}$, $r_{i+1} \le \frac{8}{5}\mathsf{c}_{\mathfrak{h}}c_\delta\delta_i$, $\mathfrak{p}_{\star,i} \le \delta_i$ and $\epsilon_i' \le c_\epsilon\delta_i \le 1$, where $\mathfrak{p}_{\star,i} = c_{\mathfrak{p}}\exp[-(LM)^{-3}\mathfrak{b}_i^2]$ and ϵ_i' is the number of Lem. 4.14; if besides $\delta(g^{-2}) \le 1/2C_{\mathrm{cu}}$, then $\epsilon_i^{\mathrm{cu}} \le 2C_{\mathrm{cu}}\delta_i \le 1$. By the conditions on the constants $\mathfrak{b}_j \ge (LM)^{3/2} \ge 1$, so $g_j \le \delta_j$ and $x_j^{-1} \le \delta_j^2 \le r_j^2$ $(j < K)$; and $\theta^{[j]} \le \theta^{[K-1]} \le \theta(g^{-2})$.

(c) For $0 < \varrho < 1$ with $s(\varrho) \le K$: $\log\frac{1}{\varrho} \le Y_\varrho \le 2\log\frac{1}{\varrho} + C(g)$, $a_{K-\mathsf{k}(\varrho)} \le \varrho/4M$, and $\delta_{\mathsf{k}(\varrho)} \le A_1 Y_\varrho^{\mathfrak{p}_0}(\varrho/\mathsf{b})^{1/2}$.

■

Proof. (a) At every $0 < g \le \gamma$ the run of a first-passage datum has $x_0 > x_1 > \cdots > x_K > g^{-2}$ and $|x_j - L^{-(j-i)}x_i| \le \bar{B}$ for $i \le j$, and its label has $g^{-2} < \mathsf{b} \le \bar{\mathsf{b}}$ (Prop. YM3_I:74.1). The threshold x_{J} is at least $\max\{e^{2\mathfrak{p}_0}, eL, 4\bar{B}\}$ by its definition (the definitions preceding [7, eq. 228]); so $g^{-2} \ge x_{\mathrm{J}}$ gives $x_j \ge x_K > g^{-2} \ge 4\bar{B}$, hence $\bar{B} \le \frac{1}{4}x_j$ and $\frac{3}{4}x_j \le L^{-(j-i)}x_i \le \frac{5}{4}x_j$, which at $i = 0$ gives the bounds of $a_{K-j}x_j$ against $\mathsf{b} = a_K x_0$. As y decreases along the run and $x_i \le \frac{5}{4}L^{j-i}x_j$, $\delta_j/\delta_i = (y_j/y_i)^{\mathfrak{p}_0}(x_i/x_j)^{1/2} \le (\frac{5}{4})^{1/2}L^{(j-i)/2}$.

As $x_i \ge x_j > g^{-2} \ge e^{2\mathfrak{p}_0}$ lie in $[e^{2\mathfrak{p}_0}, \infty)$, where δ decreases, $\delta_i \le \delta_j$. The radii follow from their recursion and these bounds. (b) $x_i > g^{-2} \ge e^{2\mathfrak{p}_0}$ and δ decreases there, so every condition stated at g^{-2} holds at every height; $2r_i \le \varrho_{\mathrm{ch}}$ and $r_{i+1} \le \frac{8}{5}\mathsf{c}_{\mathfrak{h}}c_\delta\delta_i$ by (a). For $y \ge y_{\mathfrak{p}}$: $(LM)^{-3}A_1^2 y^{2\mathfrak{p}_0} \ge \frac{y}{2} + \log^+(c_{\mathfrak{p}}/A_1)$, so $\mathfrak{p}_{\star,i} \le A_1 e^{-y_i/2} \le A_1 y_i^{\mathfrak{p}_0}x_i^{-1/2} = \delta_i$; then $\epsilon_i' \le c_{\mathrm{cu}}\delta_i + 2\mathsf{c}_{\mathbf{C}}r_{i+1}/r_\alpha \le c_\epsilon\delta_i$ and $\epsilon_i^{\mathrm{cu}} \le C_{\mathrm{cu}}(\delta_i + \mathfrak{p}_{\star,i}) \le 2C_{\mathrm{cu}}\delta_i$. (c) As $K - s \le \mathsf{k}(\varrho) \le K - s + 2$, the working level gives $\varrho L^s \le L^2\hat{\varrho}_k < L^3 C_{W,0}A_0^2 Y^{2\mathfrak{p}_0}$, whence

$$Y = s\log L + \log L(g^{-2} + \bar{B}) < \log\tfrac{1}{\varrho} + 2\mathfrak{p}_0\log Y + \log[L^4 C_{W,0}A_0^2(g^{-2} + \bar{B})],$$

and $2\mathfrak{p}_0\log Y \le \frac{1}{2}Y + 2\mathfrak{p}_0\log 4\mathfrak{p}_0$; further $a_{K-k} = \varrho/\hat{\varrho}_k \le \varrho/4M$ and $\delta_k = A_1 y_k^{\mathfrak{p}_0}x_k^{-1/2} \le A_1 Y^{\mathfrak{p}_0}(5a_{K-k}/4\mathsf{b})^{1/2}$. □

The frame used below includes the identity of the step data with the canonical construction. It also records the numerical room needed to lower a field through the chain. Keeping both requirements explicit allows us to apply the estimates at an arbitrary lower level.

Definition 4.4. Let $\mathfrak{w} = (\ell, K)$, $\ell = 2L^m$, and let $x_0, \dots, x_K$ be a finite running sequence. The pair setting consists of the following conditions.

(Jt.1) Fix $2 \le k_0 < K$, $d \ge a_K$ and real Lipschitz functions f_1, f_2, each supported in a closed ball of radius d. Put $w_a(p) = f_a(x(p))$ and $s_0 = K - k_0$.

(Jt.2) The setting (J) of [7, Def. B.16] holds through the upper level K, with $p = 2$ and the calibration equal to this running sequence. Thus its index k is $K-1$, and its actual step data, including (Pv), are required. Both sampled sources are nonzero, and

$$\widehat{d}_{k_0} := d/a_{K-k_0} \ge M, \qquad D_{k_0} \ge 3M.$$

The running and weighted bounds are

$$\begin{aligned} &|x_j - L^{-(j-i)}x_i| \le \bar{B} \quad (0 \le i \le j \le K), \qquad x_K \ge 4\bar{B}, \qquad y_K \ge \max\{2\mathfrak{p}_0, 1 + \log L\}, \\ &C_0^\flat \mathsf{S}_A \theta^{[K-1]} \le 1, \qquad \varepsilon_j \le c_A\alpha_0/(8MB_3) \quad (1 \le j \le K), \qquad \varepsilon_j \le \varepsilon_\flat \quad (0 \le j < K). \end{aligned} \tag{4.v}$$

The weighted data restrict to every earlier height. At heights $1 \le j < k_0$ they satisfy

$$s^{[j]} \le s^\star/2, \qquad \tfrac{4}{3}C_1^\flat L\theta^{[j]}\mathsf{T}(D_j/M) \le 1.$$

These are the hypotheses for the mixed derivative bound at the lower cut, in addition to the running bounds just stated.

The flat-top radii are $r_K = 0$ and $r_i = L^{-1}r_{i+1} + \frac{11}{10}\mathsf{c}_\mathfrak{h}\delta_i$. Set

$$c_R = \max\left\{\tfrac{8}{5}\sqrt{5L/4}\,\mathsf{c}_\mathfrak{h}/\varrho_{\mathrm{ch}}, \varrho_{\mathrm{fl}}^{-1}\right\}, \qquad e_i^{\mathrm{KF}} = \epsilon_i^{\mathrm{cu}} + \tfrac{16}{5}\sqrt{5L/4}\,\mathsf{c}_{\mathbf{C}}\mathsf{c}_\mathfrak{h}\delta_i/r_\alpha.$$

Require $\ell \ge N_{\mathsf{H}}$ and, for $k_0 \le i < K$,

$$\begin{aligned} c_R\delta_i \le \tfrac{1}{2}, \qquad e_i^{\mathrm{KF}} \le 1, \\ [\tfrac{48}{5}\sqrt{5/4}L^{-3/2}\mathsf{c}_\mathfrak{h} + 8C_{\mathrm{db}}]\delta_i < \varepsilon_{\mathcal{W}}\alpha_0. \end{aligned} \tag{4.vi}$$

(Jt.3) The two-source weighted progression is the one of [7, Def. B.7], with $k = K-1$ and $t_j = x_j$. It uses the same canonical data as (Jt.2), when that condition is imposed. Its source is $w_\lambda = \lambda_1 w_1 + \lambda_2 w_2$, $w_\lambda^{[j]} = L^{K-j}w_\lambda$, and $C_0^\flat \mathsf{S}_A\theta^{[K-1]} \le 1$.

(Jt.4) Both sources are nonzero, $s^{[j]} \le s^\star/2$ for $0 \le j < k_0$, and

$$\tfrac{4}{3}C_1^\flat L\theta^{[j]}\mathsf{T}(D_j/M) \le 1 \quad (1 \le j < k_0).$$

(Jt.5) An intermediate integer T is fixed with $k_0 < T \le K$.

(Jt.6) For $k_0 \le i < K$,

$$5\mathsf{c}_\mathfrak{h}\delta_i \le c_A\alpha_0/(4M). \tag{4.vii}$$

The *frame at* k_0 means (Jt.1)–(Jt.3) and (Jt.6). The full pair setting includes (Jt.4) and (Jt.5) as well. ■

Lemma 4.5 (The frame of the pair functional; tents)**.** Let $0 < g \le \gamma_{\mathrm{jt}}$, $g^{-2} \ge x_{\mathrm{J}}$, (K, β) a first-passage datum at g and $\ell \in \mathfrak{L}(g)$.

(a) Let $2 \le k < K$ and let f, h be real Lipschitz functions on T_ℓ, neither $f \circ x$ nor $h \circ x$ identically 0, each with its support in a closed ball of a radius $\bar{\mathsf{d}} \ge a_K$ with $\bar{\mathsf{d}}/a_{K-k} \ge M$, such that the corners of the plaquettes of the two supports are at least $3M$ level-k spacings apart in the sup distance. Then $(f \circ x, h \circ x)$ satisfies, with the lower level k, the four conditions (Jt.1)–(Jt.3) and (Jt.6) of Def. 4.4: the data, the flat top at K, the weighted progression of the two sources, and the smallness of δ_i at the heights $k \le i < K$; we call these four conditions the frame at k.

The two further conditions of that setting, (Jt.4) at the levels below the lower level and (Jt.5) on an intermediate level, hold, together with the other four, for a test pair $(f_1, f_2) \in \mathfrak{T}_{T_\ell}(r; \vartheta)$, $0 < r < 1$, $\vartheta \in (0,1)$, with $f_a \circ x \not\equiv 0$, at the lower level $\max\{K - s, 2\}$ when $3 \le s = s(r) \le K$ and $\mathsf{Lm}_s \le L^{s-3}$ with the intermediate level specified in the proof. The frame at k holds in the same way for the two sampled linear combinations of n functions

with pairwise disjoint supports falling into two clusters, provided both combinations are nonzero, the two unions of supports lie in such balls, and the corners of the two clusters are $3M$ level-k spacings apart.

(b) Let real weights w_1, w_2 be placed at (ρ, r), $0 < \rho \le r \le e^{-2\sqrt{3}}$, $s(\rho) \le K$. Then $a_K \le \rho/4M$; with the tent τ of height 1 and radius $\frac{1}{3}a_K$, the functions $f_a = \sum_p w_a(p)\tau(\cdot - x(p))$ are Lipschitz, $f_a \circ x = w_a$, $\operatorname{supp} f_a \subset \mathbb{B}(x_a, 2r)$, and (a) applies at $k = \mathsf{k}(r)$ with $\bar{\mathsf{d}} = 2r$. For $0 \le i \le K$: (P1) $\sum_p |w_a(p)| \le 81(\rho/a_K)^3 |w_a|_\infty$; (P2) $\mathsf{N}_i(w_a) \le (2\hat{\rho}_i/M + 3)^3 \le 108(1 + t_i)$, $t_i = \frac{27}{8M^3}\hat{\rho}_i^3$; (P3) $\mathsf{D}_i^{12} \ge \hat{r}_i/\sqrt{3} - 1$; and every corner of a plaquette of $\operatorname{supp} w_1 \cup \operatorname{supp} w_2$ lies within the sup distance $R_w \le 11\hat{r}_{i+1}$, in level-$(i+1)$ units, of a bare site next to x_1. For weights placed at (ρ, r, d), or forming two pairs at (ρ, r, d), with $\mathsf{d} \le e^{-2\sqrt{3}}$, the same holds, (P3) holds across the clusters with $\hat{\mathsf{d}}_i$ for $\hat{r}_i$, and (a) applies to the two clusters at $k = \mathsf{k}(\mathsf{d})$, each lying in a ball of radius $10\mathsf{d}$; for two weights with $\Sigma(w_a) \subset \mathbb{B}(x_a, \rho)$ at a distance $\ge \mathsf{d} \ge \rho$, (a) applies at $k = \mathsf{k}(\mathsf{d})$.

■

Proof. (a) First consider a pair satisfying the four geometric hypotheses in the statement. The first-passage construction gives $x_0 > \cdots > x_K > g^{-2}$ and the running comparison in eq. 4.v; see Prop. YM3_I:74.1. Since $\ell \in \mathfrak{L}(g)$, it also gives $MR(x_K) \le L^m$. The lattice is therefore admissible over every initial segment of the run. In particular $L^m \ge LM$ and $\ell \ge N_{\mathsf{H}}$, by the proof of Lem. YM3_I:126.2.

[7, Prop. B.21], applied with $k = K - 1$, supplies the actual setting (J), its weighted progression and the identification (Pv). This is the same canonical datum at every height, with the counterterms of this run. The inequalities defining x_{J} apply at every x_j: they give the regular-background radius, the half-ceiling and $C_0^\flat \mathsf{S}_A \theta^{[K-1]} \le 1$. The cutoff bound $\varepsilon_j \le \varepsilon_\flat$ is part of the actual history verification in that proposition.

The supports satisfy $\widehat{d}_k \ge M$ and $D_k \ge 3M$ by hypothesis. Their separation at earlier levels is $D_j = L^{k-j} D_k$, so the decreasing function T obeys $\mathsf{T}(D_j/M) \le \mathsf{T}(3)$ for $j < k$. Equation [7, eq. 228] therefore gives all the mixed derivative hypotheses at the lower cut. The threshold inequalities in equation YM3_I:(2326) give eq. 4.vi. The factor 2 in the corresponding lowering-domain inequality makes its conclusion strict. They also give $y_K \ge \max\{2\mathfrak{p}_0, 1 + \log L\}$ and $x_K \ge 4\bar{B}$. Thus (Jt.1)–(Jt.3) hold. Finally $x_i > g^{-2} \ge x_{\mathrm{jt}}$ and the threshold computation in Remark 4.2 give (Jt.6).

This verification uses only the four geometric properties in the statement. It uses neither the filling nor the sign condition of a test pair, nor the formula for the working level. For n functions in two clusters, apply it to the two sums over the clusters, with real coefficients for which both sampled sums are nonzero. Shrinking a support preserves the radius and separation conditions.

For a test pair at $k_0 = \max\{K - s, 2\}$, Lem. YM3_I:126.1 gives $d \ge a_K$, $\widehat{d}_{k_0} \ge M$ and $D_{k_0} \ge 3M$. The preceding argument gives its frame. At all heights below k_0, [7, eq. 228] gives $s^{[j]} \le s^\star/2$, including $j = 0$, and the same separation argument proves (Jt.4). Choose

$$n_T = \left\lceil \log_L(\widehat{d}_{k_0} y_{k_0}^{3\mathfrak{p}_0}) \right\rceil, \qquad T = k_0 + n_T.$$

The room bound $\mathsf{Lm}_s \le L^{s-3}$ and the final assertion of Lem. YM3_I:126.1 give $k_0 < T \le K - 2$. This proves (Jt.5) and the claimed full pair setting.

(b) The working level at $\varrho = \rho$ gives $a_K \le a_{K-\mathsf{k}(\rho)} \le \rho/4M$. Two distinct barycentres are at least $2^{-1/2} a_K$ apart, so the tents are disjoint and $f_a(x(p)) = w_a(p)$.

A corner is at the distance $2^{-1/2} a_K$ from its barycentre: corners of the two supports are $\ge r - \sqrt{2}a_K$ apart, $\ge (r - \sqrt{2}a_K)/\sqrt{3}$ in the sup distance, that is $\ge \hat{r}_i/\sqrt{3} - 1$ level-i units. At $\varrho = r$ the working level gives $2 \le \mathsf{k}(r) \le K - 1$, $2\hat{r}_{\mathsf{k}(r)} \ge M$ and $\mathsf{D}^{12}_{\mathsf{k}(r)} \ge \frac{4}{\sqrt{3}} M Y_r^{2\mathfrak{p}_0} - 1 \ge 3M$, as $Y_r \ge 2\sqrt{3}$. (P1): the barycentres of one orientation form a translate of $a_K\mathbb{Z}^3$, with at most $(2\rho/a_K + 1)^3 \le 27(\rho/a_K)^3$ points in a ball of radius $\rho \ge a_K$, and there are three orientations. (P2): an interval of length

$2\hat\rho_i+\eta_i$ meets at most $2\hat\rho_i/M+3$ intervals of length M, and $(2y+3)^3\le 4(8y^3+27)$. R_w: all points of $\Sigma(w_1)\cup\Sigma(w_2)$ are within $9r$ of x_1.

The three-point and four-point cases are the same with d for r across the clusters, and $9r+a_K\le 10\mathsf{d}$. □

Lemma 4.6 (The calculus of the derivative families)**.** Let $0<g\le\gamma$ with $g^{-2}\ge x_{\mathrm J}$, (K,β) a first-passage datum at g with $K\ge 2$, $\ell\in\mathfrak{L}(g)$, and $w_1,\dots,w_p$ real weights, none identically 0, $p\ge 1$ arbitrary. For $\emptyset\ne\beta\subset[p]$, $m=|\beta|$, $0\le j\le K-1$, put

$$\mathsf{m}_{j,a}=12M^3\alpha_0^2L^{K-j}|w_a|_\infty,\quad \varkappa_j=\frac{M^{3/2}}{2\mathsf{C}_{\mathcal A}\mathfrak{b}_j},\quad \gamma_j=\theta^{[j]}\varkappa_j,\quad \mathsf{U}_{j;\beta}=\varkappa_j^{m-1}(N_\star^{[j]})^{1-m}\prod_{a\in\beta}2\mathsf{m}_{j,a},\quad \bar\nu_{j;\beta}=\gamma_j\mathsf{U}_{j;\beta},$$

with the constant $\mathsf{C}_{\mathcal A}$ of the one-step theorem; the bare arrays of the weighted progression, $\mathfrak{f}_{\lambda,j}=((E_\lambda^{(i)},e_i)_{i\le j},(\mathsf{W}_{L^{K-j}w_\lambda}))$ for λ near 0, where the counterterms e_i of the regular progression of the run are real numbers that depend neither on λ nor on the sources, and the weight member W_w is linear in w, so that $E_\lambda^{(j+1)}-E^{(j+1)}=\mathsf{St}^{[j]}(\mathfrak{f}_{\lambda,j})-\mathsf{St}^{[j]}(\mathfrak{f}_{0,j})$ (equation [7, eq. 200]); their value at $\lambda=0$, $\mathfrak{f}_{0,j}=((E^{(i)},e_i)_{i\le j},(0))$; and their derivatives $\mathfrak{f}_{\beta,j}=\partial_\beta\mathfrak{f}_{\lambda,j}|_{\lambda=0}=((E_\beta^{(i)},0)_{i\le j},(H_\beta))$ with $H_{\{a\}}=\mathsf{W}_{L^{K-j}w_a}$ and $H_\beta=0$ for $m\ge 2$; and $\bar{\mathsf{h}}_{j;\beta}=\mathfrak{n}[\mathfrak{f}_{\beta,j}]/\mathsf{U}_{j;\beta}$, $\bar\Psi_{j;\beta}=\|E_\beta^{(j+1)}\|_{j+1,\lambda_{\mathrm{out}}}/\bar\nu_{j;\beta}$. Then for $1\le j\le K-1$:

(a) *Chain rule.* $E_\beta^{(j+1)}=\sum_{\pi\in\mathrm{Part}(\beta)}(\prod_{\beta'\in\pi}\partial_{s_{\beta'}})\mathsf{St}^{[j]}(\mathfrak{f}_{0,j}+\sum_{\beta'\in\pi}s_{\beta'}\mathfrak{f}_{\beta',j})\big|_{s=0}$; $E_\beta^{(j)}$ is m-linear in $(w_a)_{a\in\beta}$, depends on no other source, and is the same in every progression that holds the sources of β.

(b) *Derivatives of the step.* $N(\mathfrak{f}_{0,j})\le\frac12 N_\star^{[j]}$, and for bare arrays $\mathfrak{f},\mathfrak{g}_1,\dots,\mathfrak{g}_q$ with $N(\mathfrak{f})\le\frac12N_\star^{[j]}$

$$\big\|\partial_{s_1}\cdots\partial_{s_q}\mathsf{St}^{[j]}(\mathfrak{f}+\textstyle\sum_l s_l\mathfrak{g}_l)\big|_0\big\|_{j+1,\lambda_{\mathrm{out}}}\le(2(q-1))^{q-1}\,\theta^{[j]}\varkappa_j^q\,(N_\star^{[j]})^{1-q}\prod_l\mathfrak{n}[\mathfrak{g}_l].$$

(c) *Lags.* If $\mathfrak{g}=((G^{(i)},0)_{i\le j},(H))$ and each $G^{(i)}(X,\cdot,z)$ vanishes unless X meets a set Cr of bare sites, then $\mathfrak{n}[\mathfrak{g}]\le\|H\|_{j,\kappa}+\sum_{i\le j}\omega^{\mathrm{sp}}_{j-i}(i)\|G^{(i)}\|_{i,\kappa}$, where $\omega_0^{\mathrm{sp}}=1$, $\omega_l^{\mathrm{sp}}(i)=\varpi_l+L^{-4l}\min\{L^{3l},c_3\mathsf{n}_i^{\mathrm{cb}}(\mathsf{Cr})\}$, $0\le\varpi_l\le L^{8-5l}$, $\mathsf{n}_i^{\mathrm{cb}}(\mathsf{Cr})$ is the number of the cubes of π_i that hold a site of Cr, and c_3 is a constant.

(d) *Support.* $E_\beta^{(i)}(X,\cdot,z)=0$ unless every $\operatorname{supp}w_a$, $a\in\beta$, has a plaquette with its corners in X; and then $d_i(X)\ge[\max_{a,b\in\beta}\mathsf{D}_i^{ab}/M-2]_+$.

(e) *Start.* $\bar\Psi_{0;\beta}\le m^m(\mathsf{c}^\flat/2)^m$ with a constant $\mathsf{c}^\flat$; and if $\sum_p|w_a(p)|\le 81(\rho/a_K)^3|w_a|_\infty$ for a width $\rho\ge a_K$, then $\|\mathsf{W}_{L^{K-j}w_a}\|_{j,\kappa}\le\varsigma_j\mathsf{m}_{j,a}$, $\varsigma_j=\min\{1,t_j\}$.

(f) *Units.* $\gamma_j=c_\gamma x_j^{-1/2}$, $\bar\nu_{j;\beta}=\bar c_m x_j^{-m/2}\prod_{a\in\beta}(|w_a|_\infty/a_{K-j})$; for $i\le j$: $(\frac45)^{m/2}L^{(j-i)m/2}\bar\nu_{j;\beta}\le\bar\nu_{i;\beta}\le(\frac43)^{m/2}L^{(j-i)m/2}\bar\nu_{j;\beta}$, $\gamma_i\le(\frac43)^{1/2}L^{-(j-i)/2}\gamma_j$, $\sum_{i<j}\gamma_i\le 0.45\gamma_j$; and $\varkappa_j\le\frac16L^{-3/2}$.

(g) *Recursions.* $\bar\Psi_{j;\beta}\le\sum_{\pi\in\mathrm{Part}(\beta)}(2(|\pi|-1))^{|\pi|-1}\prod_{\beta'\in\pi}\bar{\mathsf{h}}_{j;\beta'}$; and for $m\ge2$ and every $a_*\in\beta$, with ω^{sp} formed with $\mathsf{n}_i^{\mathrm{cb}}=\mathsf{N}_i(w_{a_*})$,

$$\bar{\mathsf{h}}_{j;\beta}\le\gamma_j(\tfrac43)^{m/2}\sum_{i\le j}\omega^{\mathrm{sp}}_{j-i}(i)\,\sigma_{i;\beta}\,L^{(j-i+1)m/2}\,\bar\Psi_{i-1;\beta},\qquad \sigma_{i;\beta}=e^{-c_\sigma[\max_{a,b\in\beta}\mathsf{D}_i^{ab}/M-2]_+},$$

$$c_\sigma=\min\{\tfrac12,4\beta_{\mathrm s}\kappa\}.$$

No assumption of (a)–(g) depends on p; the order m enters through the displayed numbers alone. There are numbers $\tau^\natural,\gamma^\natural>0$ formed from the constants such that $\theta^{[K-1]}\le\tau^\natural$ gives $\gamma_j\le\gamma^\natural$ for all $j<K$ and $2C_\omega\mathsf{c}^\flat\theta^{[0]}\le1$, C_ω the constant of Lem. 4.9; and a number $r_1(\mathsf{b})>0$, formed from the constants and b and not increasing in b, such that every width $0<\rho\le r_1(\mathsf{b})$ with $s(\rho)\le K$ has $\rho\ge4Ma_K$ and $E_\sigma(\hat\rho_1/2)\le\varkappa_0^4$, where $E_\sigma(u)=\exp(-c_\sigma[u/\sqrt3M-3]_+)$. ■

Proof. (a) The progression is $E_\lambda^{(j+1)} - E^{(j+1)} = \mathsf{St}^{[j]}(\mathfrak{f}_{\lambda,j}) - \mathsf{St}^{[j]}(\mathfrak{f}_{0,j})$ (equation [7, eq. 200]), and the multilinear derivatives at $\lambda = 0$ of a composition are sums over partitions; the dependence through w_λ alone gives the multilinearity. (b) Within the holomorphic, gauge-invariant input class of Prop. 6.31, the fourth family must vanish at the block field 0 and have finite norm eq. 6.xvii on the polydisc $|B(b)| < \mathsf{C}_{\mathcal{A}}\mathfrak{b}_j$. A family that vanishes at $B = 0$ and is bounded on that polydisc is at most its supremum times $|B|/\mathsf{C}_{\mathcal{A}}\mathfrak{b}_j$ (Schwarz's lemma); so $\mathsf{St}^{[j]}(\mathfrak{f})$ is formed for $\varkappa_j\mathfrak{n}[\mathfrak{f}] \le N_\star^{[j]}$, with

$$|\mathsf{St}^{[j]}(\mathfrak{f}) - \mathsf{St}^{[j]}(\mathfrak{f}')|(X,P,z) \le \theta^{[j]}\varkappa_j\,\mathfrak{n}[\mathfrak{f}-\mathfrak{f}']\,e^{-\lambda_{\mathrm{out}}d_{j+1}(X)},$$

and it is holomorphic in the parameters of $\mathfrak{f} + \sum s_l\mathfrak{g}_l$ on $\{\varkappa_j\mathfrak{n}[\cdot] < N_\star^{[j]}\}$. Dividing the Lipschitz bound by $|s|$ and letting $s \to 0$ gives $\|\mathsf{DSt}^{[j]}(\mathfrak{h})[\mathfrak{g}_1]\|_{j+1,\lambda_{\mathrm{out}}} \le \theta^{[j]}\varkappa_j\mathfrak{n}[\mathfrak{g}_1]$ wherever $\varkappa_j\mathfrak{n}[\mathfrak{h}] < N_\star^{[j]}$; this is (b) at $q = 1$. For $q \ge 2$, as $\varkappa_j\mathfrak{n}[\mathfrak{f}] \le N(\mathfrak{f}) \le \frac12 N_\star^{[j]}$, the point $\mathfrak{f} + \sum_{l\ge2} s_l\mathfrak{g}_l$ stays in that set on the polydisc of the radii $N_\star^{[j]}/2(q-1)\varkappa_j\mathfrak{n}[\mathfrak{g}_l]$ in $s_2,\dots,s_q$; Cauchy's inequality in these $q-1$ variables, applied to the first derivative in the direction $\mathfrak{g}_1$, gives the factor $\prod_{l\ge2} 2(q-1)\varkappa_j\mathfrak{n}[\mathfrak{g}_l]/N_\star^{[j]}$, that is (b), with $(2(q-1))^{q-1} = 1, 2, 16, 216$ for $q = 1,\dots,4$.

This factor $1/\mathfrak{b}_j$ for every order in the sources is the whole of the loss $\mathfrak{b}_j^2$ that a bound through the size N would cost. $N(\mathfrak{f}_{0,j}) \le \frac12 N_\star^{[j]}$ is the half-ball estimate in the proof of Prop. YM3_I:122.2: the admitted bound $s_j = \theta^{[j]}/K_{\mathbf{M}} \le s_\omega^\star/2$ gives $N_j(f_{0,j}) \le N_\star \le w_\star/(4K_\Lambda s_\omega^\star) \le w_\star/(8K_\Lambda s_j) = N_{\star,j}/2$. [7, Prop. B.21] verifies this ceiling at the actual run. (c) Use the wide-domain and octet decomposition in the proof of Prop. YM3_I:51.16, with zero counterterms and all relative localization radii equal to 1. Put $l = j - i \ge 1$ and $D_l = \max\{2L^{l-1} - 2, 0\}$. The wide domains have size at least D_l. Their rooted sum gives the factor $\varpi_l = L^{3l}e^{-D_l}$ of equation YM3_I:(884). For $l = 1$ this is L^3; for $l \ge 2$,

$$D_l \ge 2(l-1)(L-1) \ge 8(l-1)\log L, \qquad \varpi_l \le L^{8-5l},$$

where the second inequality uses $L \ge 13$. For a narrow domain, equation YM3_I:(864) bounds the octet term minus its constant by

$$C_{\mathrm{oc}}\|G^{(i)}\|_{i,\kappa}(d_i(X)+2)^2L^{4-4l}e^{-\kappa d_i(X)}.$$

Sum first over domains meeting Cr and then over their base points. The rooted count in [7, Lem. B.10] applies with the marked sites of Cr in place of plaquette corners: each relevant domain contains one of the $\mathfrak{n}_i^{\mathrm{cb}}(\mathsf{Cr})$ marked cubes, and $\#(X \cap \mathbb{T}^{(i)}) \le 8M^3(d_i(X)+1)$. The remaining sum is bounded by the same convergent series of a polynomial in $d_i(X)$ times its exponential decay. This replaces the L^{3l} possible positions in the unrestricted count by $c_3\mathfrak{n}_i^{\mathrm{cb}}(\mathsf{Cr})$. Taking the better of the two counts, with the normalization of equation YM3_I:(885), gives the narrow contribution $L^{-4l}\min\{L^{3l}, c_3\mathfrak{n}_i^{\mathrm{cb}}(\mathsf{Cr})\}$. The fixed constants, including L^4, are those in the bracket normalization and in c_3. There is no growing-radius factor here: the relative radii are 1. The level-j family and H have no lag and retain their ordinary norm bounds. This proves (c). (d) is the locality of the progression with the linear size of a domain. (e) The one-step theorem applied at the height 0, where the source stands in the fourth place itself, with Schwarz's lemma as in (b); the weight member: $|a_p| < \frac12\alpha_0^2\eta_j^4$ on the cores, and the count of plaquettes. (f) is arithmetic along the run (Lem. 4.3). (g) follows from (a), (b), (e) in the units, and from (c) with $\mathsf{Cr} = \mathsf{Cr}_{a_*}$, (d) and (f).

The number $\gamma^\natural$, formed from the constants, is the smallness that Lem. 4.9 reads: $\gamma^\natural \le (3.6\,C_\omega)^{-1}$, so that $\epsilon_j^{(1)} = C_\omega(0.9\gamma_j + \mathsf{c}^\flat\theta^{[0]}) \le \frac34$ in Lem. 4.9(a) once $2C_\omega\mathsf{c}^\flat\theta^{[0]} \le 1$; $\gamma^\natural$ is at most the number under which the feedback $\sum_{i'<j}\gamma_{i'} \le 0.45\gamma_j$ is absorbed in the recursion of the proof of Lem. 4.9(b); and $\gamma^\natural \le (5C_1L^{7/2}\hat{C})^{-1}$, which the proof of Lem. 4.9(e) reads, where $\hat{C} = \sup_{v>0}(1+v^3)\max\{1, v^{12}\}E_\sigma(v/2)$ is finite, E_σ decaying exponentially, and C_1 is the constant of the bound $\omega_l^{\mathrm{sp}}(i) \le C_1L^{-4l}(1+t_i)$ in that proof.

Then $\tau^\natural = \min\{6L^{3/2}\gamma^\natural, (2C_\omega \mathsf{c}^\flat)^{-1}\}$: for $j < K$, $\delta_j \le \delta_{K-1}$ (Lem. 4.3(a)) gives $\theta^{[j]} \le \theta^{[K-1]}$, in particular $\theta^{[0]} \le \theta^{[K-1]}$, and with $\varkappa_j \le \frac{1}{6}L^{-3/2}$ of (f), $\theta^{[K-1]} \le \tau^\natural$ gives $\gamma_j = \theta^{[j]}\varkappa_j \le \gamma^\natural$ and $2C_\omega \mathsf{c}^\flat \theta^{[0]} \le 1$. For r_1, let $C_7 = \sup_{u\ge 1}[4\mathfrak{p}_0 \log(1+\log u) - c_\sigma u/4\sqrt{3}LM]$, finite, let $\Lambda_1(\mathsf{b}) \ge 1$ be the least number such that for all $\Lambda \ge \Lambda_1(\mathsf{b})$

$$\frac{c_\sigma}{\sqrt{3}L}\Lambda^{2\mathfrak{p}_0} \ge 3c_\sigma + C_7 + 4\log^+\!\big(2\mathsf{C}_{\mathcal{A}}A_1M^{-3/2}\big) + 4\mathfrak{p}_0\big[\log(1+\log^+\mathsf{b}) + \log(1+\Lambda)\big],$$

and $r_1(\mathsf{b}) = e^{-\Lambda_1(\mathsf{b})}$. The left member grows like $\Lambda^{2\mathfrak{p}_0}$, $\mathfrak{p}_0 \ge 1$, the right one like $\log\Lambda$, so $\Lambda_1(\mathsf{b})$ is finite; it is formed from the constants and b, it holds no K, and since the right member increases with b, r_1 does not increase in b. Let $0 < \rho \le r_1(\mathsf{b})$ with $s(\rho) \le K$, $\Lambda = \log\frac{1}{\rho} \ge \Lambda_1(\mathsf{b}) \ge 1$ and $u = \rho/a_K = \hat\rho_0$. The working level at $\varrho = \rho < 1$ gives $u \ge \hat\rho_{\mathsf{k}(\rho)} \ge 4MY_\rho^{2\mathfrak{p}_0} \ge 4M\Lambda^{2\mathfrak{p}_0} \ge 4M$, so $\rho \ge 4Ma_K$. Next, $1/\varkappa_0 = 2\mathsf{C}_{\mathcal{A}}A_1M^{-3/2}y_0^{\mathfrak{p}_0}$, and $x_0 = \beta = \mathsf{b}/a_K = \mathsf{b}u/\rho$ gives $y_0 = \log\mathsf{b} + \log u + \Lambda$, whence $\log y_0 \le \log(1+\log^+\mathsf{b}) + \log(1+\log u) + \log(1+\Lambda)$, as $x+y+z \le (1+x)(1+y)(1+z)$ for $x,y,z \ge 0$. As $\hat\rho_1 = u/L$, the bound $E_\sigma(\hat\rho_1/2) \le \varkappa_0^4$ follows from

$$c_\sigma\Big(\frac{u}{2\sqrt{3}LM} - 3\Big) \ge 4\log^+\!\big(2\mathsf{C}_{\mathcal{A}}A_1M^{-3/2}\big) + 4\mathfrak{p}_0\log y_0.$$

Halve the first term: by $u \ge 4M\Lambda^{2\mathfrak{p}_0}$ one half is at least $\frac{c_\sigma}{\sqrt{3}L}\Lambda^{2\mathfrak{p}_0}$, and the other half is at least $4\mathfrak{p}_0\log(1+\log u) - C_7$; with the bound of $\log y_0$, what is left is the display that defines $\Lambda_1(\mathsf{b})$, at $\Lambda = \log\frac{1}{\rho}$. The term $\log(1+\log u)$, through which y_0 grows with K, is beaten by the half of the exponent, so no number in the argument depends on K. □

Lemma 4.7 (The kernel family and the rest family)**.** In the data of Lem. 4.6 with $p \ge 2$, let $\dot{\mathfrak{e}}_{a,j} = ((E_a^{(i)},0)_{i\le j},(0))$ and $\mathfrak{u}_{a,j} = ((0,0)_{i\le j},(\mathsf{W}_{L^{K-j}w_a}))$, so that $\mathfrak{f}_{\{a\},j} = \dot{\mathfrak{e}}_{a,j} + \mathfrak{u}_{a,j}$, and $\mathsf{B}_j(\mathfrak{g}_1,\mathfrak{g}_2) = \partial_{s_1}\partial_{s_2}\mathsf{St}^{[j]}(\mathfrak{f}_{0,j} + s_1\mathfrak{g}_1 + s_2\mathfrak{g}_2)|_0$. With the kernel family $\mathsf{K}^{(j+1)} = \mathsf{B}_j(\mathfrak{u}_{1,j},\mathfrak{u}_{2,j})$, for $1 \le j \le K-1$

$$E_{12}^{(j+1)} = \mathsf{K}^{(j+1)} + \mathsf{R}^{(j+1)}, \qquad \mathsf{R}^{(j+1)} = \mathsf{B}_j(\dot{\mathfrak{e}}_{1,j},\mathfrak{u}_{2,j}) + \mathsf{B}_j(\mathfrak{u}_{1,j},\dot{\mathfrak{e}}_{2,j}) + \mathsf{B}_j(\dot{\mathfrak{e}}_{1,j},\dot{\mathfrak{e}}_{2,j}) + \mathsf{DSt}^{[j]}(\mathfrak{f}_{0,j})[\mathfrak{f}_{12,j}],$$

and $\mathsf{R}^{(j+1)}$ is a pointed family of level $j+1$ that vanishes off Cr_1 and off Cr_2; $\mathsf{K}^{(j+1)}(X,\cdot,z) = 0$ unless a plaquette of $\operatorname{supp} w_1$ and one of $\operatorname{supp} w_2$ have their corners in X. ■

Proof. Lem. 4.6(a) at $\beta = \{1,2\}$ has two partitions. B_j is additive in each entry: for bare arrays $\mathfrak{a},\mathfrak{b},\mathfrak{g}$ the value $h(t) = \mathsf{St}^{[j]}(\mathfrak{f}_{0,j} + t_1\mathfrak{a} + t_2\mathfrak{b} + t_3\mathfrak{g})(X,P,z)$ is holomorphic on a polydisc about 0, the size being subadditive, and $\partial_{s_1}\partial_{s_2}h(s_1,s_1,s_2)|_0 = (\partial_{t_1}+\partial_{t_2})\partial_{t_3}h(0)$. The first and second derivatives of the step are pointed families with the locality of the step, and Lem. 4.6(d) gives where they vanish. □

Lemma 4.8 (One level at a regular configuration)**.** Let g satisfy (Γ1), (K,β) be a first-passage datum at g and $\ell \in \mathfrak{L}(g)$.

(a) Let the frame at a level k_0 hold for a pair of real Lipschitz functions (Lem. 4.5(a)), and let $k_0 \le n < K$, $1 \le i' \le n$, $A \in \mathsf{Cc}_n(r_n)$, Cr a set of bare sites, $D' \ge 0$, and f a pointed family of level i' that vanishes off Cr and on the domains X with $d_{i'}(X) < D'$. Then

$$|\Delta[\mathsf{f}]| \le C_\Delta\, \mathsf{n}_{i'}^{\mathrm{cb}}(\mathsf{Cr})\, \|\mathsf{f}\|_{i',\lambda_{\mathrm{out}}}\, r_n^2\, L^{-4(n-i')}e^{-D'/2}.$$

(b) Let G be a family of local terms of level i that vanishes unless $\mathcal{P}(X)$, the set of the plaquettes with their corners in X, meets a set $\mathcal{S}$ of plaquettes and $d_i(X) \ge D_i$, and let N_i be the number of the cubes of π_i that hold a corner of a plaquette of $\mathcal{S}$. At every configuration U whose pairs of level i lie in the cores, $\sum_{(X,z)}|G(X,U,z)| \le 8M^3C_{\mathrm{E},1}\mathsf{N}_ie^{-D_i}\|G\|_{i,\lambda_{\mathrm{out}}}$; if U is besides regular of a level $t' \ge i$ at a radius ε with $4M\varepsilon \le c_{\mathrm{A}}\alpha_0$, then $\sum_{(X,z)}|G(X,U,z) - G(X,\mathbf{1},z)| \le 2C_{\mathrm{f}}L^{-4(t'-i)}\mathsf{N}_ie^{-D_i/2}\|G\|_{i,\lambda_{\mathrm{out}}}$.

The numbers $C_\Delta, C_\mathrm{f}, C_{\mathrm{E},1}, c_\mathrm{A}$ are formed from the constants. Of (Γ1) the proof of (a) uses only, besides the frame at k_0, the two bounds $2r_n \le \varrho_\mathrm{ch}$ and $x_n^{-1} \le r_n^2$ at the heights concerned, and the proof of (b) only $g \le \gamma$ and $g^{-2} \ge x_\mathrm{J}$; both use besides the bound $L^m \ge LMy_K$ of the sides $\ell = 2L^m \in \mathfrak{L}(g)$. So (a) holds whenever the frame at k_0 and these two bounds hold, and (b) holds whenever $g \le \gamma$ and $g^{-2} \ge x_\mathrm{J}$, with no frame. ■

Proof. (a) Let $k_0 \le n < K$. Then $r_n > 0$ and $2r_n \le \varrho_\mathrm{ch}$ (Lem. 4.3(a), (b)); in the frame at k_0, Lem. 2.3, at the radii $r_n < r' = 2r_n \le \varrho_\mathrm{ch}$, gives that $\tilde U$ is regular of level n at the radius $\varepsilon := 6r_n$ on T_ℓ and that the actions of the bare arrays over the absolute system of height n are formed at it; this is where the frame is read. For one domain X of size d and $\mathsf{E} = \mathsf{f}(X, \cdot, z)$, the evaluation of a local term at a regular configuration gives either $|\mathsf{E}(\tilde U) - \mathsf{E}(\mathbf{1})| \le C_\mathrm{A}\|\mathsf{f}\| e^{-\lambda_\mathrm{out} d}(\varepsilon/\alpha_0)^2 L^{-4(n-i')} M^2(1+d)^2$, or $d \ge [d_*]_+$ and $|\mathsf{E}(\tilde U) - \mathsf{E}(\mathbf{1})| \le 2\|\mathsf{f}\| e^{-\lambda_\mathrm{out} d}$, with the depth $d_* = \min\{c_\mathrm{A}\alpha_0\tau/6M, (L^{m+K-i'}-4)/2M\} - 1$, $\tau = L^{2(n-i')}/r_n$.

A contributing X holds one of the $\mathsf{n}_{i'}^\mathrm{cb}$ cubes, has $d \ge D'$ and at most $8M^3(d+1)$ base points; the sum over domains through a cube is bounded by [7, eq. 320]. In the tail, $e^{-[d_*]_+/2} \le e^{1/2}[e^{-c_\mathrm{A}\alpha_0\tau/12M} + e^{-(L^{m+K-i'}-4)/4M}]$; the first part is $\le C\tau^{-2}$; for the second, $L^m \ge LMy_K$ on the sides of $\mathfrak{L}(g)$, $x_n \le \frac{5}{3}L^{K-n}x_K$ and $x_n^{-1} \le r_n^2$ give $e^{-(L^{m+K-i'}-4)/4M} \le \frac{5}{3}e^{1/M}r_n^2 L^{-4(n-i')}$. So the bound holds on every side $\ell \in \mathfrak{L}(g)$. (b) is the same count at a configuration whose pairs lie in the cores, and, for the difference, at a configuration regular of level t', with $\vartheta = L^{-(t'-i)}$. □

Lemma 4.9 (Sizes at three lengths)**.** Let g satisfy (Γ1), (K, β) be a first-passage datum at g and $\ell \in \mathfrak{L}(g)$, and let real weights be placed at (ρ, r), or at (ρ, r, d), or form two pairs at (ρ, r, d), with $0 < \rho \le r \le \mathsf{d} \le e^{-2\sqrt3}$, $s(\rho) \le K$, $\rho \le r_1(\mathsf{b})$. Put $q = \rho/r$, $u_i = \hat r_i$, $v_i = \hat{\mathsf{d}}_i$, $\Xi_j^r = q^6 \min\{1, u_j\}^3$, $\mathsf{z}(\beta) = q^{3c(\beta)}(\rho/\mathsf{d})^6$ for crossing β, $\mathsf{D}(u) = [u/\sqrt3 M - 3]_+$, $C_\sigma = \sup_u \max\{1, u^6\}E_\sigma(u)$ and $\bar\nu_j = \bar\nu_{j;12}$. There are numbers $C_\omega, \mathsf{A}_\mathrm{N}, C_\mathrm{B}, \mathsf{A}_m^\times, C_m^\Psi, C_\times$ formed from the constants such that for $1 \le j \le K-1$:

(a) $\mathfrak{n}[\mathsf{u}_{a,j}] \le \varsigma_j \mathsf{m}_{j,a}$, $\mathfrak{n}[\dot{\mathfrak{e}}_{a,j}] \le \epsilon_j^{(1)}\varsigma_j \mathsf{m}_{j,a}$ with $\epsilon_j^{(1)} = C_\omega(0.9\gamma_j + \mathsf{c}^\flat\theta^{[0]}) \le \frac34$, and $\bar{\mathsf{h}}_{j;a}, \bar\Psi_{j;a} \le \varsigma_j$;

(b) for a pair, $\bar{\mathsf{h}}_{j;12} \le \mathsf{A}_\mathrm{N}\gamma_j\Xi_j^r$ and $\bar\Psi_{j;12} \le 2\varsigma_j^2 + \mathsf{A}_\mathrm{N}\gamma_j\Xi_j^r$;

(c) $\|\mathsf{K}^{(j+1)}\|_{j+1,\lambda_\mathrm{out}} \le \frac12\varsigma_j^2\bar\nu_j$ and $\|\mathsf{R}^{(j+1)}\|_{j+1,\lambda_\mathrm{out}} \le [\frac32\epsilon_j^{(1)}\varsigma_j^2 + \mathsf{A}_\mathrm{N}\gamma_j\Xi_j^r]\bar\nu_j$;

(d) below the level of the distance: if $1 \le n \le K-1$, $u_n \ge \sqrt3 M$, and the pairs of all levels $\le n$ of U lie in the cores, then $|\sum_{i\le n}\mathsf{Sum}[E_{12}^{(i)}](U)| \le C_\mathrm{B}q^6 e^{-u_n/2\sqrt3 M}\bar\nu_n$;

(e) for crossing β, $m = |\beta| \le 4$, with $\Xi_j^3(\beta) = \mathsf{z}(\beta)\min\{1, v_j\}^{3/2}$: $\bar{\mathsf{h}}_{j;\beta} \le \mathsf{A}_m^\times\gamma_j\Xi_j^3(\beta)$, and $\mathsf{N}_i E_\sigma(v_i)\bar\Psi_{i-1;\beta} \le 108\, C_m^\Psi \mathsf{z}(\beta)\min\{1, v_i\}^{3/2}E_\sigma(v_i/6)$ for $1 \le i \le K$; and at $n = \mathsf{k}(\mathsf{d})$, with $P = v_n$ and $A \in \mathsf{Cc}_n(r_n)$, $|G_{n;\beta}(A)| \le C_\times e^{-P/2\sqrt3 M}\mathsf{z}(\beta)\bar\nu_{n;\beta}$;

(f) *without the placing:* the bounds (a)–(d), and those of (e) but the last one, at the level $\mathsf{k}(\mathsf{d})$, hold with the same numbers for real weights, none identically 0, that form one pair, or a pair and a third source, or two pairs, with numbers $0 < \rho \le r \le \mathsf{d} \le 1$ such that for $0 \le i \le K$: $a_K \le \rho$; (P1) and (P2); the corners of the plaquettes of the supports are at least $u_i/\sqrt3 - 1$ level-i spacings apart in a cluster and at least $v_i/\sqrt3 - 1$ across; $\rho \ge \frac23 M a_K$; $E_\sigma(\hat\rho_1/2) \le \varkappa_0^4$; and $\gamma_{K-1} \le \gamma^\natural$. No bound on the distances of the centres and no bound $\mathsf{d} \le e^{-2\sqrt3}$ is asked in (f); these serve only the frame, the working level, the number R_w and the rings.

■

Proof. All of it is Lem. 4.6 with (P1)–(P3) of Lem. 4.5, $\rho \ge \frac23 M a_K$ (so $\varsigma_0 = 1$), $E_\sigma(\hat\rho_1/2) \le \varkappa_0^4$, $\gamma_j \le \gamma^\natural$, and three inequalities of arithmetic, $\omega_l^\mathrm{sp}(i)L^{l+1}\varsigma_{i-1} \le C_\omega\varsigma_{i+l}$, $\omega_l^\mathrm{sp}(i) \le C_1 L^{-4l}(1+t_i)$ and $(1+t_i)\varsigma_{i-1}^2 \le 2L^6 t_i^2$, which use only $a_K \le \rho$ and (P2). (a) By induction on j, with (c) of Lem. 4.6 at $\mathsf{Cr} = \mathsf{Cr}_a$, (e) and the first of these inequalities; then (b) of Lem. 4.6 at $q = 1$. (b) In the recursion (g) of Lem. 4.6 the support rule and (P3) give $\sigma_{i;12} \le E_\sigma(u_i)$; the kernel parts are

$(1+t_i)2\varsigma_{i-1}^2 \le 4L^6q^6u_i^6$, and

$$\sum_{i\le j} L^{-3(j-i)}\min\{u_i^6, C_\sigma\} \le 2.002\, C_\sigma \min\{1,u_j\}^3$$

($u_i = L^{j-i}u_j$: two geometric sums); the term $i=1$ uses $E_\sigma(u_1) \le E_\sigma(u_1/2)E_\sigma(\hat\rho_1/2) \le E_\sigma(u_1/2)\varkappa_0^4$; the feedback $\sum_{i'<j}\gamma_{i'} \le 0.45\gamma_j$ is absorbed under $\gamma_j \le \gamma^\natural$. (c) $\mathsf{K}^{(j+1)} = \mathsf{B}_j(\mathfrak{u}_1,\mathfrak{u}_2)$ and (b) of Lem. 4.6 at $q=2$; for $\mathsf{R}^{(j+1)}$, Lem. 4.7 with (a), (b) and $\frac12(2+\epsilon)\epsilon \le \frac32\epsilon$. (d) By the support rule $E_{12}^{(i)}$ vanishes on the domains with $d_i(X) < \mathsf{D}(u_i)$; Lem. 4.8(b) with (b), $\bar\nu_{i-1} \le \frac43 L^{n-i+1}\bar\nu_n$, and $\sum_l L^{l+1}e^{-\frac58\mathsf{D}(L^l u_n)} \le Ce^{-u_n/2\sqrt3 M}$. (e) Induction on m and, for one β, on the level, through one inequality on the natural sizes $\mu_i(\beta') = \hat\rho_i^{3|\beta'|}u_i^{-3c(\beta')}v_i^{-6[\beta' \text{ crosses}]}$, which is $\hat\rho_i^3$ at a point, $q^6u_i^3$ at a whole pair and $\mathsf{z}(\beta')$ at a crossing set, at every level (there $3|\beta'| = 3c(\beta')+6$): for crossing β and a partition π of it with $|\pi|\ge 2$,

$$\prod_{\beta'\in\pi}\mu_i(\beta') \le \max\{1,v_i^{12}\}\,\mathsf{z}(\beta);$$

the quotient is $u_i^{3s}v_i^{6-6k_\times}$, s the number of whole pairs of β that π cuts and $k_\times$ the number of its crossing blocks (with k blocks that do not cross, $3k+6k_\times = 6+3s$), and $u_i \le v_i$, $k_\times \le 2$, and $k_\times = 2$ forces $s=2$: at $k_\times = 0$ the quotient is $\le v_i^{3s+6}$, at $k_\times = 1$ (then $s = k \ge 1$) it is $u_i^{3s} \le \max\{1,v_i^6\}$, at $k_\times = 2$ it is $(u_i/v_i)^6 \le 1$. At $v_i < 1$ the display gives the bound $\mathsf{z}(\beta)$ alone (at $k_\times = 2$ the quotient is 1 when $r = \mathsf{d}$); the factor $\min\{1,v_i\}^{3/2}$ comes, at $k_\times \ge 1$, from the crossing blocks, whose factor of the induction is kept, and at $k_\times = 0$ from the quotient.

The blocks. At a level $i-1 \ge 1$: a point by (a), $\bar{\mathsf{h}}_{i-1;a} \le \varsigma_{i-1} \le t_{i-1} = L^3t_i \le L^3\mu_i(\{a\})$; a whole pair c by (b), $\bar{\mathsf{h}}_{i-1;c} \le \mathsf{A}_{\mathrm N}\gamma^\natural q^6\min\{1,Lu_i\}^3 \le \mathsf{A}_{\mathrm N}\gamma^\natural L^3\mu_i(c)$; a crossing proper subset β' by the first bound of (e) at $|\beta'| < m$, $\bar{\mathsf{h}}_{i-1;\beta'} \le \mathsf{A}^\times_{|\beta'|}\gamma^\natural\mathsf{z}(\beta')\min\{1,Lv_i\}^{3/2} \le \mathsf{A}^\times_{|\beta'|}\gamma^\natural L^{3/2}\mu_i(\beta')\min\{1,v_i\}^{3/2}$. With $\mathsf{a}_{(2)} = \max\{1,\mathsf{A}_{\mathrm N}\gamma^\natural\}$ and $\mathsf{a}_{(m+1)} = \max\{\mathsf{a}_{(m)},\mathsf{A}^\times_m\gamma^\natural\}$, every partition π of β with $|\pi|\ge 2$ has

$$\prod_{\beta'\in\pi}\bar{\mathsf{h}}_{i-1;\beta'} \le (L^3\mathsf{a}_{(m)})^m\min\{1,v_i\}^{3/2}\max\{1,v_i^{12}\}\,\mathsf{z}(\beta):$$

if π has a crossing block, the factor $\min\{1,v_i\}^{3/2}$ of one of them is kept and the display bounds the rest; if it has none, $k_\times = 0$ and the quotient is $\le v_i^{3s+6} \le \min\{1,v_i\}^{3/2}\max\{1,v_i^{12}\}$, as $s\le 2$. The block $\pi = \{\beta\}$ is the first bound of (e) for β itself at the level $i-1$: $\bar{\mathsf{h}}_{i-1;\beta} \le \mathsf{A}^\times_m\gamma^\natural L^{3/2}\mathsf{z}(\beta)\min\{1,v_i\}^{3/2}$.
The second bound. Put $T_i = (1+t_i)E_\sigma(v_i)\bar\Psi_{i-1;\beta}$ and $\hat C = \sup_v(1+v^3)\max\{1,v^{12}\}E_\sigma(v/2)$. As $t_i \le \hat\rho_i^3 \le v_i^3$, $E_\sigma(v) \le E_\sigma(v/2)^2$ and $E_\sigma(v/2) \le E_\sigma(v/6)^3 \le 1$, we have $(1+t_i)\max\{1,v_i^{12}\}E_\sigma(v_i) \le \hat C\,E_\sigma(v_i/6)$; so the first recursion of (g) of Lem. 4.6 gives for $2\le i\le K$

$$T_i \le \hat C\big[L^{3/2}\mathsf{A}^\times_m\gamma^\natural + \mathsf{b}_m(L^3\mathsf{a}_{(m)})^m\big]\mathsf{z}(\beta)\min\{1,v_i\}^{3/2}E_\sigma(v_i/6), \qquad \mathsf{b}_m = \sum_{\pi\in\mathrm{Part}(\beta),\,|\pi|\ge2}(2(|\pi|-1))^{|\pi|-1} = 2,22,326.$$

At $i=1$, (e) of Lem. 4.6 gives $\bar\Psi_{0;\beta} \le \mathsf{a}^\flat_m := m^m(\mathsf{c}^\flat/2)^m$; as $\rho \ge \frac23 Ma_K$ and $M\ge L$, $v_1 \ge u_1 \ge \hat\rho_1 \ge \frac23$, so that $\mathsf{z}(\beta) = \hat\rho_1^{3m}u_1^{-3c(\beta)}v_1^{-6} \ge (\hat\rho_1/v_1)^{3m} \ge (\frac23)^{12}\max\{1,v_1^{12}\}^{-1}$ and $\min\{1,v_1\}^{3/2} \ge (\frac23)^{3/2}$; hence $T_1 \le \mathsf{a}^\flat_m(\frac32)^{27/2}\hat C\,\mathsf{z}(\beta)\min\{1,v_1\}^{3/2}E_\sigma(v_1/6)$. With $C_m^\Psi = 2\hat C[\mathsf{b}_m(L^3\mathsf{a}_{(m)})^m + \mathsf{a}^\flat_m(\frac32)^{27/2}]$, $\mathsf{A}^\times_m = 2.5\,C_1L^2C_m^\Psi$ and $\gamma^\natural \le (5C_1L^{7/2}\hat C)^{-1}$, a smallness of the same kind as the one that absorbs the feedback in (b) and part of the choice of $\gamma^\natural$, the feedback $\hat CL^{3/2}\mathsf{A}^\times_m\gamma^\natural$ is $\le \frac12 C_m^\Psi$, and $T_i \le C_m^\Psi\mathsf{z}(\beta)\min\{1,v_i\}^{3/2}E_\sigma(v_i/6)$ for $1\le i\le K$; with $\mathsf{N}_i \le 108(1+t_i)$ (P2) this is the second bound of (e). *The first bound.* By the second recursion of (g), with $\sigma_{i;\beta} \le E_\sigma(v_i)$ (the corners across are $v_i/\sqrt3 - 1$ apart, and $[(v_i/\sqrt3-1)/M-2]_+ \ge \mathsf{D}(v_i)$), $\omega^{\mathrm{sp}}_{j-i}(i) \le C_1L^{-4(j-i)}(1+t_i)$, $v_{j-l} = L^lv_j$ and

$\min\{1, L^l v_j\}^{3/2} \le L^{3l/2} \min\{1, v_j\}^{3/2}$:

$$\bar{\mathsf{h}}_{j;\beta} \le (\tfrac{4}{3})^{\frac{m}{2}} C_1 L^{\frac{m}{2}} \gamma_j \sum_{l=0}^{j-1} L^{(-4+\frac{m}{2})l} T_{j-l} \le (\tfrac{4}{3})^{\frac{m}{2}} C_1 L^{\frac{m}{2}} C_m^{\Psi} \gamma_j \, \mathsf{z}(\beta) \min\{1, v_j\}^{3/2} \sum_{l\ge 0} L^{(-4+\frac{m}{2}+\frac{3}{2})l};$$

the exponent $-4 + \frac{m}{2} + \frac{3}{2} \le -\frac{1}{2}$ ($m \le 4$) gives $\sum_l \le 1.39$, and $\bar{\mathsf{h}}_{j;\beta} \le \mathsf{A}_m^{\times} \gamma_j \Xi_j^3(\beta)$. The induction runs over $m = 2, 3, 4$ and, for one β, over the level: T_i reads the first bounds at the levels $\le i-1$ only. The bound of $G_{n;\beta}$ at the level of d is Lem. 4.8(b) with the factors $e^{-\frac{1}{2}\mathsf{D}(v_i)}$, summed over $i \le n$. (f): the proofs of (a)–(d) and of the first two bounds of (e) use of the weights only the properties listed in (f), the separation of the corners standing for (P3); the last bound of (e) uses besides the working level $\mathsf{k}(\mathsf{d})$, which asks $\mathsf{d} \le e^{-2\sqrt{3}}$. □

Lemma 4.10 (One source of any width)**.** Let g satisfy (Γ1), let the frame at k_0 hold for a pair that holds the source w_a, and let w_a satisfy (P1), (P2) at a width ρ with $\frac{2}{3} M a_K \le \rho$. Then for $k_0 \le n \le K$ and $A \in \mathsf{Cc}_n(r_n)$, with $q_{n;a}(F) = \frac{1}{2a_K} \sum_p w_a(p) |F_p|^2$,

$$|G_{n;a}(A) - G_{n;a}(0) - q_{n;a}(\mathsf{ang}^{[n]}(A))| \le C_{\mathrm{S}} \, \mathfrak{b}_n^2 \delta_n \, \hat{\rho}_n^3 \, \mathsf{u}_a.$$

■

Proof. At $n = K$, $A = 0$ and both members vanish; let $k_0 \le n < K$. $G_{n;a}(A) = \mathcal{O}(w_a)(\tilde{U}) + \sum_{i\le n} \mathsf{Sum}[E_a^{(i)}](\tilde{U})$ and $a_p(\mathbf{1}) = 0$. The weight member: the expansion of a_p on the chain class and (P1) give $|\mathcal{O}(w_a)(\tilde{U}) - q_{n;a}(\mathsf{ang}^{[n]}(A))| \le C r_n^3 \hat{\rho}_n^3 x_n \mathsf{u}_a \le C \mathfrak{b}_n^2 \delta_n \hat{\rho}_n^3 \mathsf{u}_a$ (Lem. 4.3). The families: Lem. 4.8(a), in the frame at k_0, at Cr_a, $D' = 0$, with $\|E_a^{(i)}\| \le \varsigma_{i-1} \mathsf{m}_{i-1,a} \mathsf{g}_i$, $\mathsf{g}_i = 2\gamma_{i-1}$ ($i \ge 2$), $\mathsf{g}_1 = \mathsf{c}^{\flat} \theta^{[0]}$ (Lem. 4.9(a)), and $\mathsf{N}_i \varsigma_{i-1} \le 216 L^3 t_i$; then every level gives the same natural size

$$t_i \, \mathsf{m}_{i-1,a} L^{-4(n-i)} = \tfrac{81}{2} \alpha_0^2 L \, \hat{\rho}_n^3 |w_a|_\infty / a_{K-n}$$

times g_i, and $\sum_{i\le n} \mathsf{g}_i \le C\delta_n$, $|w_a|_\infty r_n^2 / a_{K-n} \le C \mathfrak{b}_n^2 \mathsf{u}_a$. □

The kernel families.

Lemma 4.11 (From a real segment to the second order)**.** Let $R > 0$, $0 < \mathsf{e} \le \mathsf{M}$, and let ψ be holomorphic on $\{|\zeta| < R\}$ with $|\psi| \le \mathsf{M}$ there, $|\psi(t)| \le \mathsf{e}$ for real $t \in (-R, R)$, and $\psi(0) = \psi'(0) = 0$. With $\Lambda = \max\{1, \log(\mathsf{M}/\mathsf{e})\}$, $|\psi(\zeta)| \le 4e \, \mathsf{e} \, \Lambda^2 |\zeta|^2 / R^2$ for $|\zeta| \le R/2\Lambda$. ■

Proof. On the upper half disc $h(\zeta) = \frac{2}{\pi} \arg \frac{R+\zeta}{R-\zeta}$ is harmonic, 0 on the diameter and 1 on the arc; $\log|\psi| - \log \mathsf{e} - h \log(\mathsf{M}/\mathsf{e})$ is subharmonic and bounded above, with nonpositive boundary values but at two points, so it is ≤ 0 (the two-constants theorem); the same on the lower half disc. As $h(\zeta) \le 2|\zeta|/R$, $|\psi| \le e \, \mathsf{e}$ on $|\zeta| \le R/2\Lambda$; $\psi(\zeta)/\zeta^2$ is holomorphic, and the maximum principle on that disc gives the claim. □

Lemma 4.12 (A local gauge)**.** In units of a level, let the bare lattice have the spacing $\xi \le \frac{1}{13}$, let x_0 be a bare site, $N_s = \{x : |x - x_0|_\infty \le s\}$, $\rho \ge 2$ with $4\rho + 4$ at most the side of the torus, and let $v = (A, \mathbf{J})$ be a real pair of $\mathfrak{su}(2)$-valued 1-forms on the bare bonds with $|A| \le \alpha \le \frac{1}{60}$, $|\nabla^\xi A| \le \gamma_\circ$, $|\mathbf{J}| \le \iota$. With $c_\mu = A_\mu(x_0)$, $X(x) = \sum_\mu (x - x_0)_\mu c_\mu$ and $u = e^{iX}$ on $N_{2\rho+1}$, there is a real pair $v' = (A', \mathbf{J}')$ which at the bonds with base point in N_ρ is the gauge transform of v under u, with $\|v'\| \le \max\{\rho(6\gamma_\circ + 19\alpha^2), \iota\}$ in the norm of the chart, $|A'(b)| \le 3s\gamma_\circ + (9s + 8\xi)\alpha^2$ at the bonds with base point in N_s, $s \le \rho$, and $|\nabla^\xi A'| \le \frac{100}{99}(\gamma_\circ + 3\alpha^2)$ on N_ρ. In the sharp first-difference bound, the whole difference stencil must lie in N_ρ; the chart-norm bound is global.

The same holds in a local form, with data on $N_{\rho+1}$ alone: let U be a configuration, $\mathsf{g}\colon N_{\rho+1} \to SU(2)$ and A a real $\mathfrak{su}(2)$-valued 1-form on the bonds $b = \langle b_-, b_+ \rangle$ with both end points in $N_{\rho+1}$ such that $\mathsf{g}(b_-) U(b) \mathsf{g}(b_+)^{-1} = e^{i\xi A(b)}$, $|A| \le \alpha \le \frac{1}{60}$ and $|\nabla^\xi A| \le \gamma_\circ$ there, the current $\mathbf{J}(U)$ of U being at most ι there; with c_μ and X as above and $h = e^{iX} \mathsf{g}$ on $N_{\rho+1}$, there is a real pair v' on all

the bare bonds which, at the bonds with base point in N_ρ, is the image of U and of $\mathbf{J}(U)$ under the gauge transformation h, and which obeys the three bounds above; and if $\|v'\|$ is less than the radius r_α of the ball of the chart, every local term of the level whose domain lies in $N_{\rho-1}$, formed at U, has the same value at $P[v']$ restricted to that domain. ■

Proof. For hermitian Y, a, c with $|a| + |c| \le \frac14$ let $\Psi(Y,a,c) = -i\log(e^{iY}e^{ia}e^{-i(Y+c)})$; matrix calculus gives $|\Psi| \le |a-c| + 3|Y||c| + 2(|a|+|c|)^2$ and a Lipschitz bound in (Y,a). Put $A^u_\mu(x) = \xi^{-1}\Psi(X(x), \xi A_\mu(x), \xi c_\mu)$, so that $e^{i\xi A^u_\mu(x)} = u(x)e^{i\xi A_\mu(x)}u(x+\xi e_\mu)^{-1}$, a real form; $|X| \le 3s\alpha$ and $|A_\mu - c_\mu| \le 3s\gamma_\circ$ on N_s along three segments. Cut off with $\chi(x) = \min\{1, \max\{0, 2-|x-x_0|_\infty/\rho\}\}$: $A' = \chi A^u$, $\mathbf{J}' = \chi u \mathbf{J} u^{-1}$. The point is that all first difference quotients of A are bounded; lowered by m levels, $|A|$ falls like L^{-m} and $|\nabla A|$ like L^{-2m}, and only the constant part is too large, which a gauge linear in x removes at the price $|c|^2$.

In the local form A^u is formed by the same formula from the potential A of the transform of U under g, so that $e^{i\xi A^u(b)} = h(b_-)U(b)h(b_+)^{-1}$ at the bonds with both end points in $N_{\rho+1}$, and the cut-off $\chi(x) = \min\{1, \max\{0, (\rho+1-2\xi-|x-x_0|_\infty)/(1-2\xi)\}\}$, of the width $1-2\xi$ and equal to 1 on N_ρ, takes the place of the cut-off of the width ρ: $A' = \chi A^u$, $\mathbf{J}' = \chi h\mathbf{J}(U)h^{-1}$. Both vanish at the bonds whose base point lies at a distance $\ge \rho+1-2\xi$ from x_0, and the other bonds have both end points in $N_{\rho+1}$, where A, h and the bound ι are given. As $\nabla^\xi_\nu(\chi A^u)(x) = \chi(x+\xi e_\nu)\nabla^\xi_\nu A^u(x) + A^u(x)(\chi(x+\xi e_\nu) - \chi(x))/\xi$, $|\chi(x+\xi e_\nu) - \chi(x)| \le \frac{13}{11}\xi$, and, on $N_{\rho+1}$, $|\nabla^\xi A^u| \le \frac{100}{99}(\gamma_\circ + 3\alpha^2)$ and $|A^u| \le 3(\rho+1)\gamma_\circ + (9\rho+9.62)\alpha^2$ (the third and the second bound, at $s = \rho+1$, by the Lipschitz bound of Ψ and the three segments in $N_{\rho+1}$, with $8\xi \le 0.62$), for $\rho \ge 2$

$$|\nabla^\xi A'| \le \tfrac{100}{99}(\gamma_\circ + 3\alpha^2) + \tfrac{13}{11}[3(\rho+1)\gamma_\circ + (9\rho+9.62)\alpha^2] \le \rho(6\gamma_\circ + 19\alpha^2), \qquad |A'| \le \rho(6\gamma_\circ + 19\alpha^2),$$

which with $|\mathbf{J}'| \le \iota$ gives the global chart bound; on stencils contained in N_ρ, the cutoff equals 1 and the sharper local bounds are unchanged. The bonds that a local term with its domain in $N_{\rho-1}$ reads have their end points in N_ρ, where $P[v']$ is the image of U under h, and a local term is a gauge invariant function of the pair on its domain. □

Proposition 4.13 (The kernel families)**.** Let g satisfy (Γ1), let the frame at k_0 hold for (w_1, w_2), let the forms q_n be formed over a base level $k_\flat \le k_0$, and let $k_0 \le i < n \le K$, $m = n-i-1$, $A \in \mathsf{C}\mathsf{c}_n(r_n)$. As $k_\flat \le i$, $\mathcal{S}_{i+1} - \mathcal{S}_i = \upsilon_{i+1}\mathcal{C}_{(i+1)}$, so that $\mathcal{Q}_i = q_{i+1} - q_i$ is the slice $\mathcal{Q}_i(F) = a_K^{-1}\upsilon_{i+1}\sum_{p,q} w_1(p)\mathcal{C}_{(i+1)}(p,q)w_2(q)\langle F_p, F_q\rangle$. At $n = K$, $A = 0$ and the member LM below is 0; the numbers below are formed for $n < K$. Put $\mathsf{w}_a = a_K^{-1}\eta_i^4\sum_p |w_a(p)|$; let R_w be such that every corner of a plaquette of $\operatorname{supp} w_1 \cup \operatorname{supp} w_2$ lies within the sup distance R_w, in level-$(i+1)$ units, of one bare site; $\alpha = L^{-m}r_n$, $\gamma_\circ = L^{-2m}r_n$, $\mathsf{c}_i = C_{\mathrm{tp}}e^{-\delta_{\mathrm{tp}}[\mathsf{D}_i^{12}-1]_+}$, $\mathsf{M}_i^{\mathsf{K}} = 8M^3C_{\mathrm{E},1}\mathsf{N}_{i+1}(w_1)\|\mathsf{K}^{(i+1)}\|_{i+1,\lambda_{\mathrm{out}}}$, $\mathsf{M}_i = 2\mathsf{M}_i^{\mathsf{K}} + 4L^{-4}\mathsf{c}_i g_i^2\delta_i^2\mathsf{w}_1\mathsf{w}_2$, and, with $\varrho_\wedge = \min\{\varrho_{\mathrm{ch}}, \varrho_{\mathrm{fl}}\}$, $\epsilon = \epsilon_i^{\mathrm{cu}}$ and $\mathsf{c}_{\mathbf{C}}$ of Lem. 4.14(a), $\epsilon' = \epsilon + 2\mathsf{c}_{\mathbf{C}}\delta_i/r_\alpha$, the number $C_{\mathrm{tp}}^{\mathrm{row}}$ of Prop. YM3_I:120.6 that bounds the row sums $\sum_b |\mathsf{T}_i^{\mathrm{cv}}(p,b)| \le C_{\mathrm{tp}}^{\mathrm{row}}\eta_i^2$ of the matrix $\mathsf{T}_i^{\mathrm{cv}}$ of step 3 (Prop. YM3_I:120.6, equation YM3_I:(2239)), $\bar X = (4L^{-2}\mathsf{c}_{\mathfrak{h}} + 2\mathsf{c}_{\mathfrak{h}}^2)\delta_i^2$ and $\bar\rho = \mathsf{M}_{\mathrm{pl}}^2\delta_i^3/\varrho_\wedge^3$,

$$\mathsf{E}_i = 12L^{-4}(C_{\mathrm{tp}}^{\mathrm{row}})^2 g_i^2\delta_i^2\epsilon' + 6\sqrt{3}L^{-2}(C_{\mathrm{tp}}^{\mathrm{row}})^3 g_i^3\delta_i\epsilon(1 + 2\mathsf{c}_{\mathbf{C}} + 2\epsilon) + \tfrac94(C_{\mathrm{tp}}^{\mathrm{row}})^4 g_i^4[\epsilon + 4\epsilon^2 + 4(\mathsf{c}_{\mathbf{C}} + \epsilon)\epsilon^2 + 2\epsilon'(2\mathsf{c}_{\mathbf{C}} + \epsilon)] + 4\bar X\bar\rho + 2\bar\rho^2,$$

and for $\delta_i \le \mathsf{r} \le \frac12\varrho_{\mathrm{ch}}$ let $\mathsf{E}_i(\mathsf{r})$ be the same number with $g_i^2\mathsf{r}^2$, $g_i^3\mathsf{r}$ in the places of $g_i^2\delta_i^2$, $g_i^3\delta_i$, with $\epsilon + 2\mathsf{c}_{\mathbf{C}}\mathsf{r}/r_\alpha$ in the place of ϵ', $4L^{-2}\mathsf{c}_{\mathfrak{h}}\mathsf{r}\delta_i + 2\mathsf{c}_{\mathfrak{h}}^2\delta_i^2$ in that of $\bar X$ and $\mathsf{M}_{\mathrm{pl}}^2\max\{\mathsf{r}/\varrho_{\mathrm{ch}}, \delta_i/\varrho_{\mathrm{fl}}\}^3$ in that of $\bar\rho$. All summands are ≥ 0 and the last one is $2\bar\rho^2$, so $2\mathsf{M}_{\mathrm{pl}}^4\varrho_\wedge^{-6}\delta_i^6 \le \mathsf{E}_i$. As $g_i \le \delta_i$ (Lem. 4.3(b)) and $\epsilon_i^{\mathrm{cu}} \le 1$, E_i and, for $\mathsf{r} \le \delta_i + r_{i+1}$, $\mathsf{E}_i(\mathsf{r})$ are at most $C_{\mathsf{E}}\delta_i^4(\delta_i + \epsilon_i^{\mathrm{cu}})$, with a number C_{E} formed from the constants; and as $\epsilon_i^{\mathrm{cu}} \le C_{\mathrm{cu}}(\delta_i + \mathsf{p}_{\star,i})$ (Lem. 4.14(a)) and $\mathsf{p}_{\star,i} \le \max\{1, (c_{\mathsf{p}}/A_1)e^{y_{\mathsf{p}}/2}\}\delta_i$ at every height (the quotient $\mathsf{p}_{\star,i}/\delta_i = (c_{\mathsf{p}}/A_1)y_i^{-\mathfrak{p}_0}\exp[\frac{y_i}{2} - (LM)^{-3}A_1^2y_i^{2\mathfrak{p}_0}]$ is ≤ 1 for $y_i \ge y_{\mathsf{p}}$ by the proof

of Lem. 4.3(b), and $\le (c_{\mathsf{p}}/A_1)e^{y_{\mathsf{p}}/2}$ for $1 \le y_i < y_{\mathsf{p}}$; $y_i > \log g^{-2} \ge \log \gamma^{-2} > 1$ since $\gamma \le \frac{1}{2}$), they are at most $C\delta_i^5$.

Let $\bar{\mathsf{e}}$ be any number $\ge 2\mathsf{w}_1\mathsf{w}_2\mathsf{E}_i$; $\Lambda_i = \max\{1, \log(\mathsf{M}_i/\bar{\mathsf{e}})\}$, $\mathsf{h}_i = \max\{2, R_w + 1 + M(2 + \Lambda_i + 2\log^+(\delta_i/\gamma_\circ))\}$, $\mathsf{s} = \mathsf{h}_i\gamma_\circ(6 + 19r_n)$. Then $\mathsf{LM} = |\Delta[\mathsf{K}^{(i+1)}] - \mathcal{Q}_i(\mathsf{ang}^{[n]}(A))|$ satisfies:

(i) always $\mathsf{LM} \le 2\mathsf{w}_1\mathsf{w}_2\mathsf{E}_i(\max\{\delta_i, \alpha\})$, where $\max\{\delta_i, \alpha\} \le \min\{\frac{1}{2}\varrho_{\mathrm{ch}}, \delta_i + r_{i+1}\}$ ($\alpha \le r_{i+1}$, as $r_j \ge L^{-1}r_{j+1}$; $2\delta_i \le \varrho_\wedge$ and $2r_{i+1} \le \varrho_{\mathrm{ch}}$ in the frame, and under (Γ1) by Lem. 4.3(b)), so that $\mathsf{E}_i(\max\{\delta_i, \alpha\}) \le C\delta_i^5$;

(ii) if $\mathsf{s} \le \delta_i/2\Lambda_i$: $\mathsf{LM} \le 8e\,\bar{\mathsf{e}}\Lambda_i^2\mathsf{s}^2/\delta_i^2 + L^{-4}\mathsf{c}_i g_i^2\mathsf{w}_1\mathsf{w}_2[18(\alpha^2 + \alpha'^2)(\gamma_\circ + \gamma') + 48R_w\alpha\gamma_\circ^2]$, with $\alpha' = (R_w + 1)(3\gamma_\circ + 10\alpha^2)$, $\gamma' = \frac{100}{99}(\gamma_\circ + 3\alpha^2)$.

The weights enter only through w_a, R_w, D_i^{12}, $\mathsf{N}_{i+1}(w_1)$ and $\|\mathsf{K}^{(i+1)}\|$; no test pair, filling or sign is asked. For weights placed at (ρ, r) as in (B.2), with $k_0 = \mathsf{k}(r)$, this gives for $\mathsf{k}(r) \le i < n \le K$

$$\mathsf{LM} \le \{C_{\mathrm{K}}\mathfrak{b}_i^4\delta_i\Lambda_\sharp^2\mathsf{h}_\sharp^2u_i^3 + C'_{\mathrm{K}}\mathsf{c}_i(1 + u_i)^2u_i^3\,\mathfrak{b}_n^2r_n\}(\rho/r)^6\hat{r}_n^3\mathsf{u}_1\mathsf{u}_2, \qquad \Lambda_\sharp \le CY_r, \quad \mathsf{h}_\sharp \le CY_r^{2\mathfrak{p}_0},$$

and neither $\Lambda_\sharp$ nor $\mathsf{h}_\sharp$ depends on ρ. Of (Γ1) the proofs of (i) and (ii) use only the frame at k_0 and, at the heights concerned, $2r_n \le \varrho_{\mathrm{ch}}$ (so $r_n \le \frac{1}{120}$, as $\varrho_{\mathrm{ch}} \le \frac{1}{60}$ by equation YM3_I:(2285)), $\epsilon_i^{\mathrm{cu}} \le 1$ and $x_n^{-1} \le r_n^2$, besides the bound $\ell \ge 2LMy_K$ of the sides $\ell \in \mathfrak{L}(g)$; (i) and (ii), with the bounds of E_i and $\mathsf{E}_i(\cdot)$ stated above, hold whenever these hold. The narrow display holds under the data of (B.2), where step 6 reads besides Lem. 4.9(c) and $\delta_i \le \delta_n$. ∎

Proof. 1. *Lowering.* $L^{-m}A$, viewed at the level $i + 1$, is a real point v of the chart with $|A_m| \le \alpha$, $|\nabla A_m| \le \gamma_\circ$, current $\le \gamma_\circ$, and $\mathsf{ang}^{[i+1]}(A_m) = \mathsf{ang}^{[n]}(A)$; $\mathsf{Sum}[\mathsf{K}^{(i+1)}](\tilde{U}) = \mathcal{K}_i(v) := \mathsf{Sum}[\mathsf{K}^{(i+1)}](P[v])$, and $\mathcal{Q}_i(v)$ is the third term.

2. *Holomorphy and no first order.* $\mathcal{K}_i$ is holomorphic on the ball of radius r_α of the chart, $|\mathcal{K}_i| \le \mathsf{M}_i^{\mathsf{K}}$, and it is invariant under constant gauge rotations; as $\sum_{k=1}^3 \mathrm{Ad}(i\sigma_k) = -\mathrm{id}$, its first derivative at 0 vanishes.

3. *The real ball.* At the real points $\|v\| \le \delta_i$, $\mathcal{K}_i(v)$ is a covariance under the probability measure $\nu_{i,v}$ of one step of the plaquette terms at the lowered configuration; this rests on a sum rule at every point of the chart, which (a) and (b) give.

(a) *The sum rule at every point.* Prop. YM3_I:51.15 proves the sum rule of one local term at the two admitted background points, in equation YM3_I:(872), and from it the sum rule of the totals at the chart points v_A of the chain classes (Prop. YM3_I:124.2, equation YM3_I:(2289)). The finite-vertex argument in that proof uses only membership in the localization domain $\mathfrak{D}(T_\ell)$ until it identifies the two endpoint values with background values; so for a term f on a domain X whose radius exceeds 1, as do those of the localisation of bare arrays, and every point $\mathfrak{u} = (P, \mathcal{A}) \in \mathfrak{D}(T_\ell)$, critical or not,

$$\sum_{Y' \in \mathcal{D}_i\colon Y' \supset X^\uparrow} \mathbb{V}[f](Y'; \mathfrak{u}|_{Y'}) = f(\rho_X P \dot{+} \mathfrak{h}_{T_\ell}(\mathbf{1}; \mathfrak{u})) - f(\rho_X P).$$

Indeed, with the interpolating function $F(s) = \Phi_{T_\ell}[f](s, 1; \mathfrak{u})$ of equation YM3_I:(869), including its distinct rescaling factors for the potential and the current, the resummation of brackets (eq. 5.xxvi) writes $F(\mathbf{1}) - F(0)$ as the sum of the brackets $F^*[\mathcal{K}|\mathsf{N}_0]$ over the sets $\mathcal{K}$ of fine cubes off the cubes N_0 of $X^\uparrow$; a bracket for which $\mathcal{K} \cup \mathsf{N}_0$ has more than one wall-component vanishes, F being local for the roots N_0 (Prop. YM3_I:51.15 and Lem. 5.25); if it has one, its union is a region $Y' \supset X^\uparrow$ and the bracket is $\mathbb{V}[f](Y'; \mathfrak{u}|_{Y'})$ by the zero-extension consistency in Prop. YM3_I:51.15, and $\mathcal{K} \mapsto Y'$ is a bijection onto these regions; finally $F(\mathbf{1}) = f(\rho_X P \dot{+} \mathfrak{h}_{T_\ell}(\mathbf{1}; \mathfrak{u}))$ by the definition of F, and $F(0) = f(\rho_X P)$ by the zero-value statement in Prop. YM3_I:51.15. Identifying the endpoint values with the two background values gives equation YM3_I:(872); the finite-bracket identity itself requires only the stated domain membership.

(b) *The totals at every point of the chart.* The weight member of a weight w is, on a domain X of level i, $\eta_i^{-1}\sum_{p\colon X(p) = X} w(p)a_p$, every bare plaquette p having one domain $X(p)$; let $w = w_a^{[i]} =$

$L^{K-i}w_a$, so that $\eta_i^{-1}L^{K-i}=L^K$, and let $\mathfrak{f}_i[\mathsf{W}_{w_a^{[i]}}]$ be the bare array whose only member is this weight member. Applied to its members at the point $\mathfrak{u}=(P[v],g_iB)$, with the pair identity in equation YM3_I:(2286) for the configuration of the substituted pair, (a) gives at every point $v=(A,\mathbf{J})$ of the chart of level $i+1$ with $\|v\|<\varrho_{\mathrm{ch}}$, real or not, in a chain class or not, and every $B\in\mathsf{Rg}_{x_i}$ (as $\delta_i<\varrho_{\mathfrak{fl}}$)

$$\mathbb{V}'\big[\mathfrak{f}_i[\mathsf{W}_{w_a^{[i]}}]\big]^{\mathrm{tot}}(P[v];B)=L^K\sum_p w_a(p)\Big[a_p\big(e^{i\eta_iA^{\mathrm{lw}}(v,g_iB)}\big)-a_p\big(e^{i\eta_{i+1}A}\big)\Big],$$

This proves at every such chart point the algebraic identity whose real-background form is equation YM3_I:(2290).

(c) *The covariance.* By Lem. 2.2 and equation YM3_I:(2298), $\mathsf{Sum}[\mathsf{K}^{(i+1)}](P)$ is, at every triple P of the core, the covariance under $\nu_i[P]$ of the two totals of (b); the terms $a_p(e^{i\eta_{i+1}A})$ are constants. So at every real v with $\|v\|\le\delta_i$

$$\mathcal{K}_i(v)=L^{2K}\sum_{p,q}w_1(p)w_2(q)\big\langle a_p(\tilde U^{\mathrm{lw}});a_q(\tilde U^{\mathrm{lw}})\big\rangle_{\nu_{i,v}},\qquad \tilde U^{\mathrm{lw}}(B)=e^{i\eta_iA^{\mathrm{lw}}(v,g_iB)},$$

where $\nu_{i,v}$ is a probability measure on Rg_{x_i} and $|\langle F;F'\rangle_{\nu_{i,v}}|\le 2\sup|F|\sup|F'|$ (Prop. YM3_I:124.5).

(d) *The plaquette terms.* (v,g_iB) is a real point with $\|v\|\le\delta_i$ and $\sup|g_iB|<g_i\mathfrak{b}_i=\delta_i$. The linear part of the lowered angle in the fluctuation is $\mathsf{sg}_i\sum_b\mathsf{T}_i^{\mathrm{cv}}(p,b)\mathcal{A}(b)$, where $\mathsf{T}_i^{\mathrm{cv}}=(\mathsf{T}_i^{\mathrm{cv}}(p,b))$ is the real matrix, indexed by the bare plaquettes and the positively oriented fluctuation bonds, of the curvature response of one step composed with its elimination map (the definitions preceding Lem. YM3_I:124.3), $\mathsf{sg}_i\in\{1,-1\}$ is the sign of Lem. YM3_I:124.3, and $\sum_b|\mathsf{T}_i^{\mathrm{cv}}(p,b)|\le C_{\mathrm{tp}}^{\mathrm{row}}\eta_i^2$ (Prop. YM3_I:120.6, equation YM3_I:(2239)). Lem. YM3_I:124.4 at $r=a=\delta_i$, $\mathsf{rt}=\varrho_\wedge/\delta_i\ge2$, gives

$$a_p(\tilde U^{\mathrm{lw}}(B))=\tfrac12|F_p|^2+\langle F_p,\xi_p\rangle+\tfrac12\langle\xi_p,\xi_p\rangle+\mathsf{o}_p(B),\qquad \xi_p=\mathsf{sg}_ig_i\sum_b\mathsf{T}_i^{\mathrm{cv}}(p,b)B(b),$$

with $F_p=\mathsf{ang}_p^{[i+1]}(A)$, $|\mathsf{o}_p|\le\mathsf{M}_{\mathrm{pl}}^2\eta_i^4\mathsf{rt}^{-3}=\eta_i^4\bar\rho$, $|F_p|\le2L^{-2}\eta_i^2\delta_i$ and $|\xi_p|\le2\mathsf{c}_{\mathfrak{h}}\eta_i^2\delta_i$. So $\mathsf{X}_p=\langle F_p,\xi_p\rangle+\frac12\langle\xi_p,\xi_p\rangle$ has $\sup|\mathsf{X}_p|\le\eta_i^4\bar X$, and $\langle a_p;a_q\rangle_{\nu_{i,v}}$ differs from $\langle\mathsf{X}_p;\mathsf{X}_q\rangle_{\nu_{i,v}}$ by at most $\eta_i^8(4\bar X\bar\rho+2\bar\rho^2)$.

(e) *The Gaussian part.* Under $\nu_{i,v}$ the joint cumulants of the orders 1, 3, 4 of the components of B are at most $\epsilon=\epsilon_i^{\mathrm{cu}}$, and those of the order 2 lie within $\epsilon'=\epsilon+2\mathsf{c}_{\mathbf{C}}\delta_i/r_\alpha$ of $\mathbf{C}_0$, the covariance of the step at $v=0$ (Prop. YM3_I:124.5, equations YM3_I:(2296)–YM3_I:(2297), at $\|v\|\le\delta_i$); and $\mathbf{C}_0=(\mathsf{A}_i^{\mathrm{cv}})^{-1}\otimes1$ with $\mathsf{T}_i^{\mathrm{cv}}(\mathsf{A}_i^{\mathrm{cv}})^{-1}(\mathsf{T}_i^{\mathrm{cv}})^{\mathsf{T}}=\eta_i\mathcal{C}_{(i+1)}$, $\mathsf{A}_i^{\mathrm{cv}}$ the real symmetric matrix of one step at the trivial configuration on the fluctuation bonds (equation YM3_I:(2297)).

A covariance of products is the sum, over the partitions that are no refinements of the two factors, of the products of the cumulants of the blocks. With $\mathbf{C}_0$ in the place of the second cumulants, the part of $\langle\langle F_p,\xi_p\rangle;\langle F_q,\xi_q\rangle\rangle$ is $g_i^2\eta_i\mathcal{C}_{(i+1)}(p,q)\langle F_p,F_q\rangle$, that of $\frac14\langle\langle\xi_p,\xi_p\rangle;\langle\xi_q,\xi_q\rangle\rangle$ is $\frac32g_i^4\eta_i^2\mathcal{C}_{(i+1)}(p,q)^2$ (three colours, two pairings), the mixed covariances $\langle\langle F_p,\xi_p\rangle;\frac12\langle\xi_q,\xi_q\rangle\rangle$ have none, and the rest is at most η_i^8 times the first three summands of E_i, by the row sums.

Multiplied by $L^{2K}w_1(p)w_2(q)$ and summed ($L^{2K}\eta_i^8\sum_p|w_1(p)|\sum_q|w_2(q)|=\mathsf{w}_1\mathsf{w}_2$, $L^{2K}g_i^2\eta_i=a_K^{-1}\upsilon_{i+1}$), this gives, with $\mathsf{k}_i=\frac32g_i^4\eta_i^2L^{2K}\sum_{p,q}w_1(p)w_2(q)\mathcal{C}_{(i+1)}(p,q)^2$ and $\mathcal{Q}_i(v)=\mathcal{Q}_i(\mathsf{ang}^{[i+1]}(A))$,

$$|\mathcal{K}_i(v)-\mathsf{k}_i-\mathcal{Q}_i(v)|\le\mathsf{w}_1\mathsf{w}_2\mathsf{E}_i.$$

(f) *Why a ball.* A ball of the chart is closed under $v\mapsto\zeta v$, under cut-offs and under small gauge transformations, which steps 4 and 5 use; a chain class is closed under none of the three, and this is why the totals are taken at every point of the chart. The radius δ_i enters only the expansion, the bound of $|F_p|$ and the ratio $\|v\|/r_\alpha$; read at a radius $\delta_i\le\mathsf{r}\le\frac12\varrho_{\mathrm{ch}}$ ($a=\delta_i$,

$\mathsf{rt} = \min\{\varrho_{\mathrm{ch}}/\mathsf{r}, \varrho_{\mathrm{fl}}/\delta_i\} \geq 2$), the same argument gives $|\mathcal{K}_i(v) - \mathsf{k}_i - \mathcal{Q}_i(v)| \leq \mathsf{w}_1\mathsf{w}_2\mathsf{E}_i(\mathsf{r})$ at the real points with $\|v\| \leq \mathsf{r}$; at $\mathsf{r} = \max\{\delta_i, \alpha\}$, at the point v of step 1 and at 0, this is (i).

4. *Two constants.* For real v' with $\|v'\| \leq \delta_i/2\Lambda_i$ put $\psi(\zeta) = \mathcal{K}_i(\zeta v') - \mathcal{K}_i(0) - \zeta^2\mathcal{Q}_i(v')$: holomorphic for $|\zeta| < \delta_i/\|v'\|$, $\psi(0) = \psi'(0) = 0$, $|\psi| \leq \mathsf{M}_i$, $|\psi| \leq \bar{\mathsf{e}}$ on the real segment (step 3 at tv' and at 0); Lem. 4.11 gives $|\psi(1)| \leq 4e\bar{\mathsf{e}}\Lambda_i^2\|v'\|^2/\delta_i^2$ (if $\bar{\mathsf{e}} > \mathsf{M}_i$, then $|\psi| \leq \mathsf{M}_i < \bar{\mathsf{e}}$ on the disc and Schwarz's lemma of the order two gives more). Of the complex ball only the bound M_i is used, through $\log \mathsf{M}_i$; the smallness comes from the real ball, where the step measure is a probability measure, and any $\bar{\mathsf{e}} \geq 2\mathsf{w}_1\mathsf{w}_2\mathsf{E}_i$ serves.

5. *The gauge.* Let first $4\mathsf{h}_i + 4$ be at most the side ℓL^{K-i-1} of the torus in level-$(i+1)$ units. Lem. 4.12 on the box N_{h_i} about the pair gives v' with $\|v'\| \leq \mathsf{s}$; the domains in the box take the same values at v and at v', those outside have $d_{i+1}(X) \geq (\mathsf{h}_i - 1 - R_w)/M - 2$ and cost $2\mathsf{M}_i^{\mathsf{K}} e^{-[\cdot]_+} \leq \bar{\mathsf{e}}\gamma_\circ^2/\delta_i^2$ by the choice of h_i; replacing $\mathcal{Q}_i(v')$ by $\mathcal{Q}_i(v)$ costs the last term of (ii), since $|\mathcal{C}_{(i+1)}| \leq \mathsf{c}_i\eta_i^3$ on the supports. If $4\mathsf{h}_i + 4 > \ell L^{K-i-1}$, then $L^m \leq L^{K-i-1} < (4\mathsf{h}_i + 4)/\ell \leq \mathsf{h}_i$, as $\ell \geq 8$ and $\mathsf{h}_i \geq 2$; so $\|v\| \leq \max\{\alpha, \gamma_\circ\} = \alpha = L^m\gamma_\circ \leq \mathsf{h}_i\gamma_\circ \leq \mathsf{s}$, and step 4 applies at $v' = v$ itself, with no gauge and no domains outside a box: (ii) holds without its last term.

6. *The narrow pair.* Take $\bar{\mathsf{e}} = 2\bar{\mathsf{w}}_1\bar{\mathsf{w}}_2\mathsf{E}_i$ with $\bar{\mathsf{w}}_a = 81\hat{\rho}_i^3|w_a|_\infty/a_{K-i} \geq \mathsf{w}_a$. By Lem. 4.9(c), $\|\mathsf{K}^{(i+1)}\| \leq \frac{1}{2}t_i^2\bar{\nu}_i = Cx_i^{-1}\bar{\mathsf{w}}_1\bar{\mathsf{w}}_2$: the factor $\hat{\rho}_i^6$ cancels in $\mathsf{M}_i/\bar{\mathsf{e}} \leq C(1 + \mathsf{N}_{i+1})x_i^{-1}\delta_i^{-6} \leq C(1 + u_{\mathsf{k}(r)}^3)x_{\mathsf{k}(r)}^2$ (by the lower bound $2\mathsf{M}_{\mathrm{pl}}^4\varrho_\wedge^{-6}\delta_i^6 \leq \mathsf{E}_i$), so $\Lambda_i \leq CY_r$ by the working level; $R_w \leq 11u_{i+1}$ and $\log^+(\delta_i/\gamma_\circ) \leq 2m\log L \leq 2Y_r$ give $\mathsf{h}_i \leq CY_r^{2\mathfrak{p}_0}$. If $\mathsf{s} \leq \delta_i/2\Lambda_i$, (ii) with $\bar{\mathsf{w}}_1\bar{\mathsf{w}}_2x_i^{-1}L^{-4m}r_n^2 \leq C\mathfrak{b}_n^2u_i^3(\rho/r)^6\hat{r}_n^3\mathsf{u}_1\mathsf{u}_2$; else $L^{3(n-i)} \leq C\mathsf{h}_i^2\Lambda_i^2$, and (i) gives the first term. □

The induction over the levels; the jets.

Lemma 4.14 (One step: cumulants, quadratic polynomials, the Gaussian step)**.**

(a) In the data of one renormalisation step at a height i, for a real point v of the chart of the level $i+1$ with $\|v\| < r_\alpha$, every $n \geq 1$ and components $B^{\alpha_1}(b_1), \dots, B^{\alpha_n}(b_n)$ of the fluctuation field, repetitions allowed, the joint cumulants under the measure $\nu_{i,v}$ of the step are formed and satisfy $|\kappa(B^{\alpha_1}(b_1), \dots)| \leq \epsilon_i^{\mathrm{cu},n}$ for $n \neq 2$ and $|\kappa(B^{\alpha_1}(b_1), B^{\alpha_2}(b_2)) - \mathbf{C}_v((b_1,\alpha_1),(b_2,\alpha_2))| \leq \epsilon_i^{\mathrm{cu},2}$, where

$$\epsilon_i^{\mathrm{cu},n} = C_{\mathrm{E},0}(K_\Pi\mathsf{n}_i^{\mathbb{V}} + K_P^\Pi\mathsf{p}_{\star,i})(n\mathsf{n}_{\mathrm{f}}/w_\star)^n$$

with construction constants, $\mathsf{n}_i^{\mathbb{V}} \leq c_{\mathbb{V}}\delta_i$; $\mathbf{C}_v$ is the covariance of the step, $|\mathbf{C}_v| \leq \mathsf{c}_{\mathbf{C}}$ and $|\mathbf{C}_v - \mathbf{C}_0| \leq 2\mathsf{c}_{\mathbf{C}}\|v\|/r_\alpha$ entry by entry, $\mathbf{C}_0 = (\mathsf{A}_i^{\mathrm{cv}})^{-1} \otimes 1$, and $\mathsf{T}_i^{\mathrm{cv}}(\mathsf{A}_i^{\mathrm{cv}})^{-1}(\mathsf{T}_i^{\mathrm{cv}})^{\mathsf{T}} = \eta_i\mathcal{C}_{(i+1)}$ (at $n \leq 4$ the bounds on the cumulants are those of the proof of Prop. YM3_I:124.5, and the statements on $\mathbf{C}_v$ and $\mathbf{C}_0$ are in Prop. YM3_I:124.5).

Here $\mathsf{A}_i^{\mathrm{cv}}$ is the real symmetric matrix of one step at the trivial configuration on the positively oriented fluctuation bonds $b \in \mathfrak{X}^+$, positive definite as $\mathbf{C}_0$ is (Prop. YM3_I:124.5 at $v = 0$); $\mathsf{T}_i^{\mathrm{cv}} = (\mathsf{T}_i^{\mathrm{cv}}(p,b))$ is the real matrix, indexed by the bare plaquettes and the bonds of $\mathfrak{X}^+$, of the curvature response of one step composed with its elimination map (the definitions preceding Lem. YM3_I:124.3); and there are a sign $\mathsf{sg}_i \in \{1,-1\}$ and the number $C_{\mathrm{tp}}^{\mathrm{row}}$ of Prop. YM3_I:120.6 such that the linear part of the lowered angle in a fluctuation $\mathcal{A}$ is $\mathsf{sg}_i\sum_b \mathsf{T}_i^{\mathrm{cv}}(p,b)\mathcal{A}(b)$ and $\sum_b|\mathsf{T}_i^{\mathrm{cv}}(p,b)| \leq C_{\mathrm{tp}}^{\mathrm{row}}\eta_i^2$ (Lem. YM3_I:124.3 and Prop. YM3_I:120.6, equation YM3_I:(2239), on a side ℓ at least the constant N_{H} fixed in Section YM3_I:120.3, as in the frame).

At a real point (v, g_iB), $v = (A, \mathbf{J})$, $B \in \mathsf{Rg}_{x_i}$, with $r \geq \|v\|$ and $\mathsf{rt} = \min\{\varrho_{\mathrm{ch}}/r, \varrho_{\mathrm{fl}}/\delta_i\} \geq 2$, Lem. YM3_I:124.4 (at $a = \delta_i$) splits the angle of the lowered configuration:

$$\mathsf{ang}_p^{[i]}(A^{\mathrm{lw}}(v, g_iB)) = \mathsf{ang}_p^{[i+1]}(A) + \psi_p(B) + \zeta_p(B), \qquad \psi_p(B) = \mathsf{sg}_ig_i\sum_b \mathsf{T}_i^{\mathrm{cv}}(p,b)B(b),$$

$|\psi_p| \leq 2\mathsf{c}_{\mathfrak{h}}\eta_i^2\delta_i$, $|\zeta_p| \leq 4\mathsf{M}_{\mathrm{gr}}\eta_i^2\mathsf{rt}^{-2}$, with the number M_{gr} of equation YM3_I:(2285); under the centred Gaussian with covariance $\mathbf{C}_0$, ψ has the covariance $g_i^2\eta_i\mathcal{C}_{(i+1)} = a_Kv_{i+1}\mathcal{C}_{(i+1)}$,

as $\mathsf{sg}_i^2 = 1$ and $g_i^2\eta_i = a_K \upsilon_{i+1}$. At $v = v_{A'}$, $A' \in \mathsf{Cc}_{i+1}(r_{i+1})$, one takes $r = r_{i+1}$; under (Γ1) Lem. 4.3(b) gives $\mathsf{rt} \ge 2$ and $|\zeta_p| \le C\eta_i^2\delta_i^2$. We put $\epsilon_i^{\mathrm{cu}} = \epsilon_i^{\mathrm{cu},4}$; with $C_\star = C_{\mathrm{E},0}\max\{K_\Pi c_{\mathbb{V}}, K_P^\Pi\}$, so that $\epsilon_i^{\mathrm{cu},n} \le C_\star(\delta_i + \mathsf{p}_{\star,i})(n\mathsf{n_f}/w_\star)^n$, we take $C_{\mathrm{cu}} = C_\star(4\mathsf{n_f}/w_\star)^4$, $c_{\mathrm{cu}}^{(n)} = 2C_\star(2n\mathsf{n_f}/w_\star)^{2n}$ and $c_{\mathrm{cu}} = c_{\mathrm{cu}}^{(3)}$, so that $\epsilon_i^{\mathrm{cu}} \le C_{\mathrm{cu}}(\delta_i + \mathsf{p}_{\star,i})$ and $\epsilon_i^{\mathrm{cu},2n} \le c_{\mathrm{cu}}^{(n)}(\delta_i + \mathsf{p}_{\star,i})/2$; we put $\epsilon_i'^{(n)} = \epsilon_i^{\mathrm{cu},2n} + 2\mathsf{c_C} r_{i+1}/r_\alpha$ and $\epsilon_i' = \epsilon_i'^{(3)}$, and $c_\epsilon^{(n)}$ is the number c_ϵ of Lem. 4.3(b) with $c_{\mathrm{cu}}^{(n)}$ in place of c_{cu}.

We have $4\mathsf{n_f}/w_\star > 1$ ($\mathsf{n_f} = \frac{1}{2}M^{3/2}$ and $w_\star < 1$: equation YM3_I:(2294) and the proof of Prop. YM3_I:124.5); so with $x = \mathsf{n_f}/w_\star$, $(nx)^n \le (n'x)^n \le (n'x)^{n'}$ for $n \le n'$, $n' \ge 4$, and $\epsilon_i^{\mathrm{cu},n} \le \epsilon_i^{\mathrm{cu},2m}$ for $n \le 2m$, $m = 2, 3, 4$: at $\|v\| \le r_{i+1}$ the joint cumulants of all the orders $\le 2m$ lie within $\epsilon_i'^{(m)}$ of those of the centred Gaussian with covariance $\mathbf{C}_0$ (at the order two through $|\mathbf{C}_v - \mathbf{C}_0| \le 2\mathsf{c_C} r_{i+1}/r_\alpha$). Besides $\epsilon_i'^{(2)} \le \epsilon_i' \le \epsilon_i'^{(4)}$ and $c_\epsilon^{(2)} \le c_\epsilon \le c_\epsilon^{(4)}$; so (Γ1) with $c_\epsilon^{(4)}$ in place of c_ϵ contains (Γ1), and ($\Gamma^{(4)}$) contains (Γ).

(b) Let I be finite, ν a probability measure with bounded support on $\mathbb{R}^I$, $\mathbf{C}$ a real symmetric positive semidefinite matrix on I with $|\mathbf{C}(\mathsf{i},\mathsf{j})| \le \mathsf{c}$, $\mathsf{c} \ge 1$, $0 \le \epsilon \le 1$, and let the cumulants of ν of orders $n \le 6$ be within ϵ of those of the centred Gaussian G with covariance $\mathbf{C}$. For quadratic polynomials P_1, P_2, P_3 in the coordinates, with $\|P\|_{\mathrm{pol}}$ the sum of the moduli of their coefficients of orders one and two, and $1 \le m \le 3$: $|\kappa_\nu(P_1,\dots,P_m) - \kappa_{\mathsf{G}}(P_1,\dots,P_m)| \le c_6\epsilon\prod_l \|P_l\|_{\mathrm{pol}}$, $c_6 = 1218(\mathsf{c}+1)^5$. If only the cumulants of the orders $n \le 4$ are within ϵ of those of G, the same holds for $m \le 2$ with $c_4 = 60(\mathsf{c}+1)^3$; if those of the orders $n \le 8$ are, it holds for $m \le 4$ (four sources) with $c_8 = 33120(\mathsf{c}+1)^7$; and $c_4 \le c_6 \le c_8$.

(c) Let $k_\flat \le i < K$, $\mathsf{s} = \upsilon_{i+1}\mathcal{C}_{(i+1)}$, and let ψ be a centred Gaussian vector of $\mathfrak{su}(2)$-valued functions of the plaquettes with $\mathbb{E}\psi_p^\alpha\psi_q^{\alpha'} = a_K\delta_{\alpha\alpha'}\mathsf{s}(p,q)$. For every F' and every nonempty set β of sources,

$$\sum_{\pi\in\mathrm{Part}(\beta)} \kappa\big(q_{i;\beta'}(F'+\psi) : \beta' \in \pi\big) = (g_{i+1;\beta} - g_{i;\beta}) + q_{i+1;\beta}(F').$$

■

Proof. (a) The bound on the cumulants of one step (the proof of Prop. YM3_I:124.5) is proved with n insertions of one sort, and depends on the number n only through the factor $(n\mathsf{n_f}/w_\star)^n$; so it holds at every order. The remaining statements in (a) follow from the results cited there, and the comparison of the orders is the arithmetic stated in (a). (b) The cumulant of products $X_{S_1},\dots,X_{S_m}$ is the sum, over the partitions σ of $S = \bigsqcup S_l$ with $\sigma \vee \{S_1,..,S_m\} = \{S\}$, of the products of the cumulants of the blocks (from the expansions of exp and log in $\mathbb{C}[z]/(z_l^2)$). Replacing the factors one by one costs $|\sigma|\epsilon(\mathsf{c}+1)^{|\sigma|-1} \le 2m\epsilon(\mathsf{c}+1)^{2m-1}$, as $|S| \le 2m$, and a set of $2m$ elements has B_{2m} partitions, $B_4 = 15$, $B_6 = 203$, $B_8 = 4140$: so $c_{2m} = 2m\,B_{2m}(\mathsf{c}+1)^{2m-1}$ serves, which is $c_6 = 6\cdot 203(\mathsf{c}+1)^5$ for $m \le 3$, c_4 for $m \le 2$ and c_8 for $m \le 4$. (c) Put $w_\lambda = \sum_a \lambda_a w_a$, $R_i^\lambda = w_\lambda(1 - \mathcal{S}_i w_\lambda)^{-1}$, $g_i^\lambda = -\frac{3}{2}\operatorname{Tr}\log(1 - \mathcal{S}_i w_\lambda)$ and $q_i^\lambda(F) = \frac{1}{2a_K}\langle F, R_i^\lambda F\rangle$; their coefficients of $\lambda^\beta = \prod_{a\in\beta}\lambda_a$ are $R_{i;\beta}$, $g_{i;\beta}$ and $q_{i;\beta}$.

The Gaussian integral, with the factor 3 from the dimension of $\mathfrak{su}(2)$, is

$$\log\mathbb{E}\,e^{q_i^\lambda(F'+\psi)} = -\tfrac{3}{2}\operatorname{Tr}\log(1 - \mathsf{s}R_i^\lambda) + \tfrac{1}{2a_K}\langle F', R_i^\lambda(1-\mathsf{s}R_i^\lambda)^{-1}F'\rangle,$$

and $1 - \mathsf{s}R_i^\lambda = (1 - \mathcal{S}_{i+1}w_\lambda)(1 - \mathcal{S}_i w_\lambda)^{-1}$, $R_i^\lambda(1-\mathsf{s}R_i^\lambda)^{-1} = R_{i+1}^\lambda$, as $\mathcal{S}_{i+1} = \mathcal{S}_i + \mathsf{s}$. The coefficient of λ^β of the left member is the sum over the partitions of the joint cumulants; that of the right member is $(g_{i+1;\beta} - g_{i;\beta}) + q_{i+1;\beta}(F')$. This is Wick's rule for the photon fields of one step. □

Lemma 4.15 (The tower identity at several sources)**.** Let $0 < g \le \gamma_{\mathrm{jt}}$, $g^{-2} \ge x_{\mathrm{J}}$, (K,β) a first-passage datum at g, $\ell \in \mathfrak{L}(g)$, $n \ge 2$, and let $w_a = f_a \circ x \not\equiv 0$, $a \in [n]$, with real Lipschitz functions f_a on T_ℓ with pairwise disjoint supports (for instance the tents of Lem. 4.5(b)) that fall into two

nonempty clusters $[n] = I_1 \sqcup I_2$ lying in the frame at a level k. Let $k \le i < K$, $A' \in \mathsf{Cc}_{i+1}(r_{i+1})$, $\tilde U = e^{i\eta_{i+1}A'}$, $v_{A'} = (A', J^{(i+1)}(\tilde U))$ the real point of the chart of level $i+1$ that A' gives, with $\|v_{A'}\| \le r_{i+1}$, $\nu = \nu_{i,A'} := \nu_{i,v_{A'}}$ the measure of the step, and $B \mapsto A^{\mathrm{lw}}(B) = A^{\mathrm{lw}}(v_{A'}, g_i B)$ the lowering map (Prop. YM3_I:124.2). Then for every nonempty $\beta \subset [n]$

$$G_{i+1;\beta}(A') = \sum_{\pi \in \mathrm{Part}(\beta)} \kappa_\nu(G_{i;\beta'} \circ A^{\mathrm{lw}} : \beta' \in \pi).$$

The same holds for real mixed sources $\mathfrak{s}_a \ne 0$, $a \in [n]$, at $0 < g \le \gamma_{\mathrm{N}}$ with $K \ge 2$, the progression being formed with their weight members in the source slot (the lemma on bond sources, which follows Lem. 4.31), when the two unions of the hulls of the clusters I_1, I_2 satisfy the conditions of the frame at k of clause (6) of that lemma. For mixed sources the frame is that of clause (6), read with the hulls in the place of the supports, and the tower identity for a pair is the one stated in that clause. ■

Proof. For λ near 0 in $\mathbb{C}^n$ put $\Phi(\lambda) = \exp D^\lambda_{i+1}(\tilde U) - \int \exp D^\lambda_i(e^{i\eta_i A^{\mathrm{lw}}(B)})\,\mathrm{d}\nu(B)$, formed with the progression of $(w_1, \dots, w_n)$. Each relative exponent is a finite sum of functions that are holomorphic in λ, and the integrand is bounded uniformly in B for λ near 0, as the progression with any number of sources lies in the setting of one step (Thm. YM3_I:D); so Φ is holomorphic near 0. Let $t \in \mathbb{R}^n$ be nonzero on I_1 and on I_2, and put $f^t = \sum_{a\in I_1} t_a f_a$, $h^t = \sum_{a\in I_2} t_a f_a$. These are real Lipschitz functions; $f^t \circ x$ and $h^t \circ x$ do not vanish identically, the supports of the w_a being disjoint; and their supports lie in the two unions of supports.

So (f^t, h^t) lies in the frame at k, and the tower identity among the construction results for a pair (equation eq. 2.vii) gives $\exp D^\mu_{i+1}[f^t, h^t](\tilde U) = \int \exp D^\mu_i[f^t, h^t](e^{i\eta_i A^{\mathrm{lw}}})\,\mathrm{d}\nu$ for μ near 0 in $\mathbb{C}^2$, with the same measure ν and the same lowering map, neither of which depends on the sources. The progression depends on the parameters and the sources only through w_λ; at $\lambda(\mu) = (\mu_1 t|_{I_1}, \mu_2 t|_{I_2})$ one has $w_{\lambda(\mu)} = \mu_1 f^t \circ x + \mu_2 h^t \circ x$, so $D^{\lambda(\mu)} = D^\mu[f^t, h^t]$, and at $\mu = (s, s)$: $\Phi(st) = 0$ for all small real s. With the expansion $\Phi = \sum_N \Phi_N$ in homogeneous polynomials, $\sum_N s^N \Phi_N(t) = 0$ for small s; so $\Phi_N(t) = 0$ for every N and every t of a dense subset of $\mathbb{R}^n$, hence $\Phi_N = 0$ and $\Phi = 0$ near 0.

As $D^0_{i+1} = 0$ and $D^0_i = 0$, the multilinear jets of the two members of $\exp D^\lambda_{i+1}(\tilde U) = \int \exp D^\lambda_i\,\mathrm{d}\nu$ agree in $\mathbb{C}[\lambda_1, \dots, \lambda_n]/(\lambda_a^2)$, and the logarithm of a moment series is the cumulant series; differentiation under the integral is allowed by Cauchy's formula in λ. The coefficient of λ^β is the claim. At $n = 2$ the lemma is the identity among the construction results. For mixed sources the argument is the same, with the weight member of $\mathfrak{s}_\lambda = \sum_a \lambda_a \mathfrak{s}_a$, which is linear in $\mathfrak{s}_\lambda$ (clause (1) of the lemma on bond sources), in the source slot. The $t \in \mathbb{R}^n$ at which $\sum_{a\in I_1} t_a \mathfrak{s}_a$ or $\sum_{a\in I_2} t_a \mathfrak{s}_a$ is 0 form two proper linear subspaces, every $\mathfrak{s}_a$ being nonzero; off them, a dense set, the two sums are nonzero mixed sources whose hulls lie in the two unions of the hulls of the clusters, so they are in the frame of clause (6), and the identity for a pair stated there, with the same measure ν and the same lowering map, stands in the place of equation eq. 2.vii. □

Proposition 4.16 (The induction over the levels)**.** Let $n \in \{2, 3, 4\}$ and let g satisfy (Γ1), with $c_\epsilon^{(4)}$ in place of c_ϵ if $n = 4$; let (K, β) be a first-passage datum at g, $\ell \in \mathfrak{L}(g)$, and let real weights $w_1, \dots, w_n$ with the two clusters $\{1\}, \{2\}$, or $\{1, 2\}, \{3\}$, or $\{1, 2\}, \{3, 4\}$ lie in the frame at a level k; let $k \le k_1 < K$, $0 \le k_\flat \le k_1$, and form the free objects over the base level $k_\flat$. Let numbers $\mathsf{v}_\beta > 0$, c°_β and $\epsilon_\circ \ge 0$, $\emptyset \ne \beta \subset [n]$, satisfy

(I1) $|G_{k_1;\beta}(A) - c^\circ_\beta - q_{k_1;\beta}(\mathsf{ang}^{[k_1]}(A))| \le \epsilon_\circ \mathsf{v}_\beta$ for $A \in \mathsf{Cc}_{k_1}(r_{k_1})$;
(I2) $\varrho_{i;\beta} := \eta_i^4 \sum_{p,q} |R_{i;\beta}(p, q)|/(a_K x_i) \le L^{-3(i-k_1)} \mathsf{v}_\beta$ for $k_1 \le i < K$;
(I3) $\prod_{\beta' \in \pi} \mathsf{v}_{\beta'} \le \mathsf{v}_\beta$ for every partition π of β.

Put $Z = Z_0(1 + \mathfrak{b}_{k_1})^2$ and $\varpi = \epsilon_\circ + \sum_{i=k_1}^{K-1} L^{-3(i-k_1)}(\delta_i + c_{2n}\epsilon_i'^{(n)})$, with a number Z_0 formed from the constants and the numbers c_{2n} of Lem. 4.14(b), and let $(K - k_1)\varpi \le 1$. Then $\varpi \le$

$\epsilon_\circ + \frac{6}{5}(1 + c_{2n}c_\epsilon^{(n)})\delta_{k_1}$, and for $k_1 \le i \le K$, $A \in \mathsf{Cc}_i(r_i)$ and every β, $m = |\beta|$,

$$|G_{i;\beta}(A) - c_\beta^\circ - (g_{i;\beta} - g_{k_1;\beta}) - q_{i;\beta}(\mathsf{ang}^{[i]}(A))| \le \mathsf{a}_m Z^m \varpi\, \mathsf{v}_\beta, \qquad (\mathsf{a}_1, \mathsf{a}_2, \mathsf{a}_3) = (1, 14, 285),$$

a_4 a number; in particular $|\mathsf{J}_n - c^\circ_{[n]} - g_{K;[n]} + g_{k_1;[n]}| \le \mathsf{a}_n Z^n \varpi \mathsf{v}_{[n]}$. Of the base level only the relation $\mathcal{S}_{i+1} - \mathcal{S}_i = \upsilon_{i+1}\mathcal{C}_{(i+1)}$, $i \ge k_1$, is used. ■

Proof. Let $\mathsf{e}_{i;\beta}$ be the supremum over $\mathsf{Cc}_i(r_i)$ of the left member. 1. *The tower.* For $A' \in \mathsf{Cc}_{i+1}(r_{i+1})$ and the measure $\nu = \nu_{i,A'}$ of the step, $G_{i+1;\beta}(A') = \sum_\pi \kappa_\nu(G_{i;\beta'} \circ A^{\mathrm{lw}} : \beta' \in \pi)$ (the lemma on the tower identity at several sources, which precedes this proposition, for the two clusters of the frame). 2. *The lowered angle.* The point $v_{A'}$ of the chart is real, with $\|v_{A'}\| \le r_{i+1} < \varrho_{\mathrm{ch}} \le r_\alpha$ by the definition of the chain classes (equation eq. 2.iv); $\nu = \nu_{i,v_{A'}}$ is a probability measure carried by the real B with $|B(b)| < \mathfrak{b}_i$ (Prop. YM3_I:124.5), and $A^{\mathrm{lw}}(B) \in \mathsf{Cc}_i(r_i)$ (Prop. YM3_I:124.2 with the radius recurrence YM3_I:(2291)).

By the splitting of Lem. 4.14(a) at $r = r_{i+1}$, $\mathsf{ang}_p^{[i]}(A^{\mathrm{lw}}(B)) = F'_p + \psi_p(B) + \zeta_p(B)$, with $F' = \mathsf{ang}^{[i+1]}(A')$, $\psi_p(B) = \mathsf{sg}_i g_i \sum_b \mathsf{T}_i^{\mathrm{cv}}(p,b)B(b)$ linear in the fluctuation field B, $|\psi_p| \le 2\mathsf{c}_\mathfrak{h}\eta_i^2\delta_i$ and $|\zeta_p| \le C\eta_i^2\delta_i^2$ by the smallness of Lem. 4.3(b); besides $|F'_p| \le 2\eta_{i+1}^2 r_{i+1} \le \mathsf{c}_\mathfrak{h}\eta_i^2\delta_i$. So $Y_\beta = G_{i;\beta} \circ A^{\mathrm{lw}}$ is within $\mathsf{e}_{i;\beta} + Z\delta_i L^{-3(i-k_1)}\mathsf{v}_\beta$ of the quadratic polynomial $P_\beta = c_\beta^\circ + g_{i;\beta} - g_{k_1;\beta} + q_{i;\beta}(F' + \psi)$ in the coordinates $B^\alpha(b)$: the term with ζ is at most $Ca_K^{-1}\eta_i^4\delta_i^3 \sum_{p,q} |R_{i;\beta}(p,q)| = C\mathfrak{b}_i^2\delta_i\varrho_{i;\beta}$.

The linear part of P_β has the coefficients $\mathsf{sg}_i g_i a_K^{-1} \sum_{p,q} R_{i;\beta}(p,q)\mathsf{T}_i^{\mathrm{cv}}(q,b)F'^\alpha_p$, its quadratic part the coefficients $\frac{1}{2}g_i^2 a_K^{-1} \sum_{p,q} R_{i;\beta}(p,q)\mathsf{T}_i^{\mathrm{cv}}(p,b)\mathsf{T}_i^{\mathrm{cv}}(q,b')$ of $B^\alpha(b)B^\alpha(b')$; with the row sums $\sum_b |\mathsf{T}_i^{\mathrm{cv}}(p,b)| \le C_{\mathrm{tp}}^{\mathrm{row}}\eta_i^2$, $x_i g_i \delta_i = \mathfrak{b}_i$ and $x_i g_i^2 = 1$,

$$\|P_\beta\|_{\mathrm{pol}} \le \big[2\mathsf{c}_\mathfrak{h} C_{\mathrm{tp}}^{\mathrm{row}}\mathfrak{b}_i + \tfrac{3}{2}(C_{\mathrm{tp}}^{\mathrm{row}})^2\big]\varrho_{i;\beta};$$

so, by (I2), $\mathfrak{b}_i \le \mathfrak{b}_{k_1}$ and $|B(b)| < \mathfrak{b}_i$, $\|P_\beta\|_{\mathrm{pol}}$ and $\sup|P_\beta - \mathrm{const}|$ are $\le ZL^{-3(i-k_1)}\mathsf{v}_\beta$, Z_0 being at least $2\mathsf{c}_\mathfrak{h}C_{\mathrm{tp}}^{\mathrm{row}} + \frac{3}{2}(C_{\mathrm{tp}}^{\mathrm{row}})^2$ and the constant of the term with ζ. 3. *The Gaussian value.* Under the Gaussian with covariance $\mathbf{C}_0$, $\psi = \mathsf{sg}_i g_i \mathsf{T}_i^{\mathrm{cv}} B$ has the covariance $g_i^2\eta_i\mathcal{C}_{(i+1)} = a_K\upsilon_{i+1}\mathcal{C}_{(i+1)}$ (Lem. 4.14(a): $\mathsf{sg}_i^2 = 1$, $\mathsf{T}_i^{\mathrm{cv}}(\mathsf{A}_i^{\mathrm{cv}})^{-1}(\mathsf{T}_i^{\mathrm{cv}})^{\mathsf{T}} = \eta_i\mathcal{C}_{(i+1)}$, $g_i^2\eta_i = a_K\upsilon_{i+1}$), and Lem. 4.14(c) gives $\sum_\pi \kappa_{\mathsf{G}}(P_{\beta'}) = c_\beta^\circ + g_{i+1;\beta} - g_{k_1;\beta} + q_{i+1;\beta}(F')$. 4.

From G *to* ν by Lem. 4.14(a), (b), with $\mathsf{c} = \max\{1, \mathsf{c}_{\mathbf{C}}\}$: as $\|v_{A'}\| \le r_{i+1}$, all the cumulants of the orders $\le 2n$ lie within $\epsilon = \epsilon_i'^{(n)}$ of those of G, and $\epsilon_i'^{(n)} \le c_\epsilon^{(n)}\delta_i \le 1$ (under (Γ1), as $c_\epsilon^{(2)} \le c_\epsilon$, and with $c_\epsilon^{(4)}$ at $n = 4$); the number is c_{2n}; *from* P *to* Y by $|\kappa(X,Y)| \le \|X\|^\sim\|Y\|^\sim$, $|\kappa(X,Y,W)| \le 2\|X\|^\sim\|Y\|^\sim\|W\|^\sim$, $\|X\|^\sim = \inf_c \sup|X - c|$. 5. *The recursion.* With (I3) every product of sizes over a partition is at most $Z^m\mathsf{v}_\beta$ times pure numbers, and $\varepsilon_{i;\beta} = \mathsf{e}_{i;\beta}/Z^m\mathsf{v}_\beta$ obeys $\varepsilon_{i+1;a} \le \varepsilon_{i;a} + \tau_i$, $\varepsilon_{i+1;ab} \le \varepsilon_{i;ab} + 2\tau_i + 4\varpi^2 + 4\theta_i\varpi$ and the like at three sources, $\tau_i = L^{-3(i-k_1)}(\delta_i + c_{2n}\epsilon_i'^{(n)})$, $\theta_i = L^{-3(i-k_1)}$; summed over at most $K - k_1$ heights with $(K - k_1)\varpi \le 1$ this gives 1, 14 and 285.

At $i = K$, $\mathsf{Cc}_K = \{0\}$ and $q_{K;\beta}(0) = 0$. The bound of ϖ: $\epsilon_i'^{(n)} \le c_\epsilon^{(n)}\delta_i$ (as $\epsilon'_i \le c_\epsilon\delta_i$ in Lem. 4.3(b)), $\delta_i \le (\frac{5}{4})^{1/2}L^{(i-k_1)/2}\delta_{k_1}$, $(\frac{5}{4})^{1/2}\sum_k L^{-5k/2} \le \frac{6}{5}$. For two sources Steps 1–5 use only eq. 2.viii and the cumulants up to order four. □

Lemma 4.17 (The slices)**.** Let (K, β) be a first-passage datum at $g \le \gamma$, $p \ne q$ plaquettes, $\mathsf{r} = |x(p) - x(q)|_\infty$, $c_L^{(N)} = \sup_\tau \sum_{j\in\mathbb{Z}} (L^j\tau)^N e^{-\delta_{\mathrm{tp}}L^j\tau}$ and $\mathsf{m}_K = \frac{5}{4}C_{\mathrm{tp}}c_L^{(3)}$. (a) For every set $\mathcal{J} \subset \{1, \dots, K\}$: $|\sum_{j\in\mathcal{J}} \upsilon_j \mathcal{C}_{(j)}(p,q)| \le \mathsf{m}_K \mathsf{b}^{-1} a_K^3 \mathsf{r}^{-3}$. (b) If $t = \mathsf{r}/a_{K-k}$ satisfies $\delta_{\mathrm{tp}}L(L-1)t \ge 3\log L + \log 2$, then

$$\Big|\sum_{k<j\le K} \upsilon_j \mathcal{C}_{(j)}(p,q) - \mathsf{b}^{-1}C_K^{\mathrm{free}}(p,q)\Big| \le \mathsf{b}^{-1}a_K^3\big[\tfrac{5}{4}C_{\mathrm{tp}}c_L^{(2)}\bar{B}\,\mathsf{r}^{-2}\mathsf{b}^{-1} + C_{\mathrm{tp}}^{\mathrm{bg}} + 2L^3 C_{\mathrm{tp}}\mathsf{r}^{-3}t^3 e^{-\delta_{\mathrm{tp}}Lt}\big],$$

with $C_{\mathrm{tp}}^{\mathrm{bg}}$ from Prop. YM3_I:120.5, evaluated at $s = 0$ in equation YM3_I:(2236). ■

Proof. (a) $\upsilon_j = (a_{K-j+1}x_{j-1})^{-1} \le \frac{24}{23}\mathsf{b}^{-1} \le \frac{5}{4}\mathsf{b}^{-1}$ at every $0 < g \le \gamma$: the flow bound and $g^{-2} < \mathsf{b}$, which Lem. 4.3(a) states for every first-passage datum at every $0 < g \le \gamma$, give $|a_{K-j+1}x_{j-1} - \mathsf{b}| \le a_{K-j+1}\bar{B} \le \bar{B}/L$, and $\bar{B} \le \frac{13}{24}g^{-2} < \frac{13}{24}\mathsf{b}$ at $g \le \gamma$ ([7, Lem. G.3] gives this bound), so $a_{K-j+1}x_{j-1} \ge \frac{23}{24}\mathsf{b}$ as $L \ge 13$; and the decay of the slices: $\sum_j (\mathsf{r}/a_{K-j+1})^3 e^{-\delta_{\mathrm{tp}}\mathsf{r}/a_{K-j+1}} \le c_L^{(3)}$, a lacunary sum. (b) Equation YM3_I:(2227) gives $\sum_{k<j\le K}\mathcal{C}_{(j)} = C_K^{\mathrm{free}} - C_{K,0}^{\mathrm{bg}} - \sum_{j\le k}\mathcal{C}_{(j)}$, and Prop. YM3_I:120.5 bounds $|C_{K,0}^{\mathrm{bg}}(p,q)|$ by $C_{\mathrm{tp}}^{\mathrm{bg}}a_K^3$; $|\upsilon_j - \mathsf{b}^{-1}| = |\mathsf{b} - a_{K-j+1}x_{j-1}|\upsilon_j\mathsf{b}^{-1} \le \frac{5}{4}\bar{B}a_{K-j+1}\mathsf{b}^{-2}$ by the same bounds of the run, which with the decay gives the first term; the slices of the levels $j \le k$ are at $t_j \ge Lt$ and give the last term. □

Lemma 4.18 (The explicit sizes; the weights of the induction)**.** Let weights be placed at (ρ, r, d), or form two pairs at (ρ, r, d), with $\mathsf{d} \le e^{-2\sqrt{3}}$, $s(\rho) \le K$; let $k_\flat = \mathsf{k}(r)$, $k_1 = \mathsf{k}(\mathsf{d})$, $u = \hat{r}_{k_1}$, $\hat{\rho} = \hat{\rho}_{k_1} = qu$, $P = \hat{\mathsf{d}}_{k_1}$, and I_1, I_2 the clusters. Put $\mathsf{w}(\emptyset) = 1$, $\mathsf{w}(\{a\}) = \hat{\rho}^3$, w(a whole pair) $= q^6u^3\max\{1, u^3\}$, $c_* = \max\{102, 4\cdot 81\cdot 3^{3/2}(1+\mathsf{m}_K)\}$, and

$$\mathsf{v}_\beta = c_*^{|\beta|}\,\mathsf{w}(\beta\cap I_1)\,\mathsf{w}(\beta\cap I_2)\,\mathsf{u}_\beta, \qquad \mathsf{u}_\beta = \prod_{a\in\beta}\mathsf{u}_a.$$

Then (I2) and (I3) of Prop. 4.16 hold; $\mathsf{v}_{[3]} \le c_*^3P^9\mathrm{nat}_3$ and $\mathsf{v}_{[4]} \le c_*^4P^{12}\mathrm{nat}_4$; for crossing β, $\mathsf{z}(\beta)\mathsf{u}_\beta \le P^{-6}\mathsf{v}_\beta$, and if $\mathsf{e}_{\mathrm{far}} = 8e^{-\delta_{\mathrm{tp}}LP/2\sqrt{3}} \le 1$, then $\varrho_{k_1;\beta} \le \mathsf{e}_{\mathrm{far}}\mathsf{v}_\beta$ and $|g_{k_1;[n]}| \le \frac{3}{2}(n-1)!(81\cdot 3^{3/2}\mathsf{m}_K)^n\mathsf{e}_{\mathrm{far}}^2\mathrm{nat}_n$. The factor $\max\{1, u^3\}$ cannot be dropped: u may be a power of $\log\frac{1}{\mathsf{d}}$. ■

Proof. By Lem. 4.17(a), $|\mathcal{S}_i(p,q)| \le \mathsf{m}_K\mathsf{b}^{-1}a_K^3\mathsf{r}^{-3}$ with $\mathsf{r} \ge r/\sqrt{3}$ in a cluster and $\ge \mathsf{d}/\sqrt{3}$ across. An ordering of β is a path with $m-1$ edges, at most $c(\beta)$ of them inside a whole pair; with (P1),

$$\varrho_{i;\beta} \le \tfrac{5}{4}m!\,81^m(3^{3/2}\mathsf{m}_K)^{m-1}\rho^{3m}r^{-3c}\mathsf{d}^{-3(m-1-c)}a_{K-i}^{-3}\mathsf{u}_\beta,$$

and with $a = a_{K-k_1}$ the powers of a cancel: $\rho^{3m}r^{-3c}\mathsf{d}^{-3(m-1-c)}a^{-3} = \hat{\rho}^{3m}u^{-3c}P^{-3(m-1-c)} \le \mathsf{w}(\beta\cap I_1)\mathsf{w}(\beta\cap I_2)$, $a_{K-i}^{-3} = L^{-3(i-k_1)}a^{-3}$: (I2). (I3): either $\beta\cap I_\iota$ lies in one block, or it is a whole pair cut into two points, and $\hat{\rho}^6 = q^6u^6 \le q^6u^3\max\{1,u^3\}$. The top sets: $q^6u^3\hat{\rho}^3 = P^6\rho^9r^{-3}\mathsf{d}^{-6}$. At the level k_1 the slices of the levels $\le k_1$ at distances $\ge \mathsf{d}/\sqrt{3}$ are exponentially small, $|\mathcal{S}_{k_1}(p,q)| \le \mathsf{e}_{\mathrm{far}}\mathsf{m}_K\mathsf{b}^{-1}a_K^3\mathsf{r}^{-3}$; a path through a crossing set has a far edge, a ring through all sources two. □

Proposition 4.19 (The jets of narrow bumps)**.** There are numbers $C(g)$ and $\mathsf{d}_4(g), \mathsf{d}_4^{(4)}(g) \in (0, \min\{e^{-2(15\mathfrak{p}_0+2)}, r_1(\bar{\mathsf{b}})\}]$, formed from the constants and g, such that the following holds for g under (Γ1), every first-passage datum (K,β) at g, every side $\ell \in \mathfrak{L}(g)$ with $64\mathsf{d} \le \ell$, and $s(\rho) \le K$.

(a) For real weights placed at (ρ, r, d) with $0 < \rho \le r \le \mathsf{d} \le \mathsf{d}_4(g)$: $|\mathsf{J}_3 - 3\,\mathrm{Tr}\prod_a(w_a\mathsf{b}^{-1}C_K^{\mathrm{free}})| \le \epsilon^{\mathrm{nar}}(g,\mathsf{d})\,\mathrm{nat}_3$.

(b) For real weights w_1, w_2 with $\Sigma(w_a) \subset \mathbb{B}(x_a, \rho)$, $\mathrm{dist}(\Sigma(w_1), \Sigma(w_2)) \ge \mathsf{d}$, $|x_1 - x_2| \le 18\mathsf{d}$, $0 < \rho \le \mathsf{d} \le \mathsf{d}_4(g)$: $|\mathsf{J}_2 - \frac{3}{2}\mathsf{b}^{-2}\sum_{p,q} w_1(p)w_2(q)C_K^{\mathrm{free}}(p,q)^2| \le \epsilon_2^{\mathrm{nar}}(g,\mathsf{d})(\rho/\mathsf{d})^6\mathsf{u}_1\mathsf{u}_2$.

(c) Under (Γ1) with $c_\epsilon^{(4)}$ in place of c_ϵ, for two pairs at (ρ, r, d) with $\mathsf{d} \le \mathsf{d}_4^{(4)}(g)$: $|\mathsf{J}_4 - \frac{3}{2}\sum_\varsigma \mathrm{Tr}\prod_l(w_{\varsigma(l)}\mathsf{b}^{-1}C_K^{\mathrm{free}})| \le \epsilon_4^{\mathrm{nar}}(g,\mathsf{d})\,\mathrm{nat}_4$.

The normalized error factors and constants are uniform in ρ/r, r/d, K, ℓ, position and direction; in particular they hold for $w_a = f_a \circ x$ with bounded Borel f_a, and for plane weights. ■

Proof. 1. *Levels and data.* $k_1 = \mathsf{k}(\mathsf{d})$, $k_\flat = \mathsf{k}(r) \le k_1$, $P = \hat{\mathsf{d}}_{k_1}$; by the working level $4MY_{\mathsf{d}}^{2\mathfrak{p}_0} \le P < CY_{\mathsf{d}}^{2\mathfrak{p}_0}$, $K - k_1 \le Y_{\mathsf{d}}/\log L$, $\delta_{k_1} \le A_1Y_{\mathsf{d}}^{\mathfrak{p}_0}(\mathsf{d}/\mathsf{b})^{1/2}$. Lem. 4.5(b) puts the two clusters in the frame at k_1; the smallness holds at all heights (Lem. 4.3); (I2), (I3) hold with the weights of Lem. 4.18.

2. *(I1).* At a single source, $c_a^\circ = G_{k_1;a}(0)$ and Lem. 4.10 give the tolerance $C\mathfrak{b}_{k_1}^2\delta_{k_1}\mathsf{v}_a$. At the whole pair, $c^\circ = G_{k_1;12}(0)$: by the multilinearity $G_{k_1;12}$ is the pair functional of the pair alone, and (B.2) at $n = k_1 \ge \mathsf{k}(r)$ (its proof, given below, rests on Lems. 4.3–4.13 alone) gives

$[C_{\mathrm{N}}\mathfrak{b}_{k_1}^2\delta_{k_1}+\epsilon'_{\mathrm{N}}(r)]q^6u^3\mathsf{u}_{12}\le[\dots]\mathsf{v}_{12}$. At a crossing set, $c^\circ_\beta=0$: Lem. 4.9(e) and Lem. 4.18 give $|G_{k_1;\beta}|\le Ce^{4Y_{\mathsf{d}}-\frac{2}{\sqrt3}Y_{\mathsf{d}}^{2\mathfrak{p}_0}}\mathsf{v}_\beta$, and $|q_{k_1;\beta}|\le C\mathfrak{b}_{k_1}^2\mathsf{e}_{\mathrm{far}}\mathsf{v}_\beta$. So

$$\epsilon_\circ=C_\circ\{\mathfrak{b}_{k_1}^2\delta_{k_1}+\epsilon'_{\mathrm{N}}(r)+e^{4Y_{\mathsf{d}}-\frac{2}{\sqrt3}Y_{\mathsf{d}}^{2\mathfrak{p}_0}}+\mathfrak{b}_{k_1}^2\mathsf{e}_{\mathrm{far}}\}.$$

Here $C_\circ$ is a number formed from the constants that dominates the constants of these four bounds (of Lem. 4.10, of (B.2), of Lem. 4.9(e) with Lem. 4.18, and of the bound of $q_{k_1;\beta}$), and those of the two bounds of (I1) in step 6. The tolerance of (B.2) falls together with its yardstick; for this reason no relation between r and d is needed, and the bounds hold at every ratio $r/\mathsf{d}\in(0,1]$.

3. *The induction* (Prop. 4.16) at $i=K$: $|\mathsf{J}_n-g_{K;[n]}+g_{k_1;[n]}|\le\mathsf{a}_nZ^n\varpi c_*^nP^{3n}\mathrm{nat}_n$.

4. *The rings.* Lem. 4.17(b) at $k=\mathsf{k}(r)$ for p,q in different supports ($t\ge\hat r_{\mathsf{k}(r)}/\sqrt3$) gives $|\mathcal{S}_K-\mathsf{b}^{-1}C_K^{\mathrm{free}}|\le\mathrm{br}\,\mathsf{b}^{-1}a_K^3\mathsf{r}^{-3}$ with

$$\mathrm{br}=25C_{\mathrm{tp}}c_L^{(2)}\bar B\,\mathsf{d}/\mathsf{b}+8000C_{\mathrm{tp}}^{\mathrm{bg}}\mathsf{d}^3+2L^3C_{\mathrm{tp}}\sup_{t\ge\frac{4}{\sqrt3}MY_r^{2\mathfrak{p}_0}}t^3e^{-\delta_{\mathrm{tp}}Lt};$$

telescoping a product of n kernels, $|g_{K;[n]}-\frac32\sum_\varsigma\mathrm{Tr}\prod_l(w_{\varsigma(l)}\mathsf{b}^{-1}C_K^{\mathrm{free}})|\le C\,\mathrm{br}\,\mathrm{nat}_n$; at $n=3$ the subtrahend is $3\,\mathrm{Tr}\prod_a(w_a\mathsf{b}^{-1}C_K^{\mathrm{free}})$, two cyclic orders with one trace. With Lem. 4.18 for $g_{k_1;[n]}$:

$$\epsilon^{\mathrm{nar}}_{(n)}=\mathsf{a}_nZ^nc_*^nP^{3n}\varpi+C\,[\mathsf{e}_{\mathrm{far}}^2+\mathrm{br}].$$

5. *The threshold* $\mathsf{d}_4(g)$. It is fixed by six conditions: (i) $\mathsf{d}\le e^{-2(15\mathfrak{p}_0+2)}$, so that by Lem. 4.3(c) $\epsilon'_{\mathrm{N}}(r)\le C(g)(\log\frac1{\mathsf{d}})^{15\mathfrak{p}_0+2}\mathsf{d}^{1/2}$ for all $r\le\mathsf{d}$; (ii) $\frac4{\sqrt3}M\delta_{\mathrm{tp}}L(L-1)(\log\frac1{\mathsf{d}})^2\ge3\log L+\log2$; (iii) $\log\frac1{\mathsf{d}}\ge\max\{5,\sqrt3/\delta_{\mathrm{tp}}LM\}$, so that $e^{4Y_{\mathsf{d}}-\frac{2}{\sqrt3}Y_{\mathsf{d}}^2}\le\mathsf{d}$, $\mathsf{e}_{\mathrm{far}}\le8\mathsf{d}^2$ and the last member of br is $\le C\mathsf{d}$; (iv) $8\mathsf{d}^2\le1$, $\overline{\mathrm{br}}\le1$; (v) $(Y_{\mathsf{d}}/\log L)[\bar\epsilon_\circ+\frac65(1+c_6c_\epsilon)A_1Y_{\mathsf{d}}^{\mathfrak{p}_0}g\mathsf{d}^{1/2}]\le1$; (vi) $\mathsf{d}\le r_1(\bar{\mathsf{b}})$, so that $\rho\le r_1(\mathsf{b})$ holds of itself. In (iv) and (v) the numbers br and $\epsilon_\circ$, which read r and the datum, stand replaced by majorants that read g and d alone. As $r\le\mathsf{d}$, $s(r)\ge s(\mathsf{d})$ and $Y_r\ge Y_{\mathsf{d}}$; with $\mathsf{d}/\mathsf{b}<g^2\mathsf{d}$,

$$\mathrm{br}\le\overline{\mathrm{br}}=25C_{\mathrm{tp}}c_L^{(2)}\bar B\,g^2\mathsf{d}+8000C_{\mathrm{tp}}^{\mathrm{bg}}\mathsf{d}^3+2L^3C_{\mathrm{tp}}\sup_{t\ge\frac{4}{\sqrt3}MY_{\mathsf{d}}^{2\mathfrak{p}_0}}t^3e^{-\delta_{\mathrm{tp}}Lt}.$$

In $\epsilon_\circ$, $\epsilon'_{\mathrm{N}}(r)$ is at most the bound of (i); $\mathfrak{b}_{k_1}\le A_1Y_{\mathsf{d}}^{\mathfrak{p}_0}$ and $\delta_{k_1}\le A_1Y_{\mathsf{d}}^{\mathfrak{p}_0}g\mathsf{d}^{1/2}$ (step 1, $\mathsf{b}>g^{-2}$); and $\mathsf{e}_{\mathrm{far}}\le8\mathsf{d}^2$ by (iii), as $P\ge4MY_{\mathsf{d}}^{2\mathfrak{p}_0}$. So

$$\epsilon_\circ\le\bar\epsilon_\circ=C_\circ\{A_1^3Y_{\mathsf{d}}^{3\mathfrak{p}_0}g\mathsf{d}^{1/2}+C(g)(\log\tfrac1{\mathsf{d}})^{15\mathfrak{p}_0+2}\mathsf{d}^{1/2}+e^{4Y_{\mathsf{d}}-\frac{2}{\sqrt3}Y_{\mathsf{d}}^{2\mathfrak{p}_0}}+8A_1^2Y_{\mathsf{d}}^{2\mathfrak{p}_0}\mathsf{d}^2\}.$$

Y_{d} is a function of g and d, with $\log\frac1{\mathsf{d}}\le Y_{\mathsf{d}}\le2\log\frac1{\mathsf{d}}+C(g)$ (Lem. 4.3(c)). So (i)–(vi) are conditions on (g,d) alone, and they hold for all small d, every majorised member tending to 0 with d. $\mathsf{d}_4(g)$ is half the supremum of the numbers d_* such that (i)–(vi) hold at every $0<\mathsf{d}\le\mathsf{d}_*$: a positive number formed from the constants and g, which reads no K, β, ℓ, ρ, r; at every $\mathsf{d}\le\mathsf{d}_4(g)$, every $r\le\mathsf{d}$, every datum at g and every label $\mathsf{b}\in(g^{-2},\bar{\mathsf{b}}]$ the conditions hold with br and $\epsilon_\circ$ themselves.

As $\varpi\le\epsilon_\circ+\frac65(1+c_6c_\epsilon)\delta_{k_1}$ at $n=3$ (Prop. 4.16) and $K-k_1\le Y_{\mathsf{d}}/\log L$, (v) secures $(K-k_1)\varpi\le1$; at $n=2$ (step 6) the same (v) secures it, as $c_4c_\epsilon^{(2)}\le c_6c_\epsilon$ (Lem. 4.14) and $C_\circ$ dominates the constants of step 6. $\mathsf{d}_4^{(4)}(g)$ is formed in the same way with $n=4$, that is with $c_8c_\epsilon^{(4)}$ in (v) and the tolerance $\epsilon_\circ$ of two pairs, under (Γ1) with $c_\epsilon^{(4)}$. Then $Z^n\le CY_{\mathsf{d}}^{2n\mathfrak{p}_0}$, $P^{3n}\le CY_{\mathsf{d}}^{6n\mathfrak{p}_0}$, $\mathfrak{b}_{k_1}^2\delta_{k_1}\le CY_{\mathsf{d}}^{3\mathfrak{p}_0}g\mathsf{d}^{1/2}$, and the exponents are $8n\mathfrak{p}_0+15\mathfrak{p}_0+2$: $39\mathfrak{p}_0+2$ at $n=3$, $47\mathfrak{p}_0+2$ at $n=4$.

6. *Two sources (b).* The same with the base level equal to the start level $k_1=\mathsf{k}(\mathsf{d})$, $\mathsf{v}_a=c_*\hat\rho^3\mathsf{u}_a$, $\mathsf{v}_{12}=c_*^2\hat\rho^6\mathsf{u}_{12}$; at the pair $\mathcal{S}_{k_1}=0$, so $q_{k_1;12}=g_{k_1;12}=0$, and Lem. 4.9(d) at $n=k_1$ gives $|G_{k_1;12}|\le Ce^{2Y_{\mathsf{d}}-\frac{2}{\sqrt3}Y_{\mathsf{d}}^{2\mathfrak{p}_0}}\mathsf{v}_{12}$; then $g_{K;12}=\frac32\sum w_1(p)w_2(q)\mathcal{S}_K(p,q)^2$ and Lem. 4.17(b) at $k=k_1$; the logarithms count $4\mathfrak{p}_0+12\mathfrak{p}_0+3\mathfrak{p}_0=19\mathfrak{p}_0$. □

The connected functions on the least torus.

Lemma 4.20 (The mixture; the conditioning stopped at any level)**.** Let $0 < g \le \gamma$ with $g^{-2} \ge x_{\mathrm{J}}$, (K, β) a first-passage datum at g with $K \ge 2$, $\ell \in \mathfrak{L}(g)$, and $w_1, \dots, w_p$ real weights, $p \ge 1$ arbitrary. For $0 \le k \le K$ let $\Omega_k = \mathcal{V}_k \sqcup \bigsqcup_{t=1}^{k} \mathcal{V}_{t-1}$ and $\mathbb{P}_k = Z_k^{-1}[\mathsf{m}_k \oplus \bigoplus_{t \le k} \mathsf{g}_t^{\mathrm{w}} \mathsf{m}_{t-1}]$, where $\mathcal{V}_t$ is the space of the block fields of the level t, m_t the measure of the never-large history up to the level t, and $\mathsf{g}_t^{\mathrm{w}} \ge 0$ the weight of a first large field at the step t in eq. 4.xxxi; $t' = k$ on the first part and $t' = t - 1$ on the part t; $\alpha = a_{K-t'}$; $U_0(W) = W$, and for $t' \ge 1$ let $U_{t'}(W)$ be the background configuration of W in the small-field region $\mathcal{V}_{t'}^{\chi}$. Write $\Omega = \Omega_K$, $\mathbb{P} = \mathbb{P}_K$, and $\mathsf{Y}_\beta(\omega) = \partial_\beta[D_{t'}^{\lambda}(U_{t'}(W)) - D_K^{\lambda}(\mathbf{1})]|_{\lambda=0}$ on Ω. Then:

(a) $\mathbb{P}_k$ is a Borel probability measure formed with no source; its parts with $t \le k$ are parts of $\mathbb{P}$; the part t of $\mathbb{P}$ has the mass w_t, and $\int \alpha^{-b} d\mathbb{P} = \mathbb{P}(\mathcal{V}_K) + \sum_t L^{b(K-t+1)} \mathsf{w}_t$ for every real b.

(b) $\langle e^{\mathcal{O}(w_\lambda)} \rangle_{K,\ell,\beta} = \int \exp[D_{t'}^{\lambda}(U_{t'}(W))] \, \mathrm{d}\mathbb{P}_k$ for λ near 0; coefficientwise, $\langle \prod_{\iota \in \gamma} \mathcal{O}(w_\iota) \rangle_{K,\ell,\beta} = \int [\exp \mathbb{W}]_\gamma \, \mathrm{d}\mathbb{P}$ in $\mathbb{C}[\lambda_\iota]/(\lambda_\iota^2)$, with $\mathbb{W}$ the multilinear jet of $\lambda \mapsto D_{t'}^{\lambda}(U_{t'}(W))$; each $[\mathbb{W}]_\gamma$ is a bounded Borel function that depends on the sources of γ alone.

(c) $S_n^T - \mathsf{J}_n = \sum_{\pi \in \mathrm{Part}([n])} \kappa_{\mathbb{P}}(\mathsf{Y}_{\beta'} : \beta' \in \pi)$; and with $G_\beta(\omega) = G_{t';\beta}(U_{t'}(W))$ on Ω_k, $S_n^T = \sum_\pi \kappa_{\mathbb{P}_k}(G_{\beta'} : \beta' \in \pi)$.

(d) Where $t' \ge 1$, $U_{t'}(W)$ has its pairs of all levels $\le t'$ in the cores and is regular of the level t' at the radius $2B_3\varepsilon_{t'}$, with $8MB_3\varepsilon_{t'} \le c_{\mathrm{A}}\alpha_0$.

(e) On the least torus, under (Γ2): $\int \alpha^{-b} d\mathbb{P} \le 1 + 2L^b \mathsf{C}_{\mathrm{ag}}(b+1, \frac{1}{2})$ for every $b \ge 0$, so $\mathsf{M}' = \int \alpha^{-24} d\mathbb{P} \le \bar{\mathsf{M}}^{(24)}$; and $\int \alpha^{-b} d\mathbb{P}_k \le a_{K-k}^{-b} + \mathsf{M}'$, $\int_{\{\alpha < \varrho\}} \alpha^{-b} d\mathbb{P}_k \le \varrho^{24-b} \mathsf{M}'$ for $0 \le b \le 24$, $\varrho \le a_{K-k}$. On a side $\ell \in \mathfrak{L}(g)$: $\int \alpha^{-b} d\mathbb{P} \le 1 + 4L^b M^{-3} \ell^3 e^{\mathsf{B}(X_g)\ell^3} \mathsf{C}_{\mathrm{ag}}(b+1, \eth)$, finite and growing with ℓ, with the function B and the letter $\eth$ and $X_g = L(g^{-2} + B_\sharp)$.

■

Proof. Equation eq. 4.xxxi writes the torus expectation of a bounded gauge invariant function F as Z^{-1} times the integral against m_k of its quotient at the level k plus the integrals against $\mathsf{g}_t^{\mathrm{w}} \mathsf{m}_{t-1}$ of its quotients at the levels $t - 1$, $t \le k$, at every level $k \le K$. At $F = 1$ this makes $\mathbb{P}_k$ a probability measure; at $F = e^{\mathcal{O}(w_\lambda)}$ the quotient at a level t is, on $\mathcal{V}_t^{\chi}$, $\exp D_t^{\lambda}(U_t(W))$, by [7, Prop. B.15], applied to the finite progression of [7, Prop. B.8]. Coefficients of the multilinear jets give (b); the logarithm of the moment series is the cumulant series, and constants drop out of the cumulants of orders ≥ 2, which gives (c), at every level k. (d) is [7, Lem. B.17], in the setting verified by [7, Prop. B.21] at $x_K \ge x_{\mathrm{J}}$. (e) The mass w_t is at most the probability that the first large field arises at the step t; on the least torus Prop. 4.54, under $\log g^{-2} \ge \mathsf{Y}_{\mathrm{lt}}(\frac{1}{2})$, gives $\sum_t L^{b(K-t+1)} \mathsf{w}_t \le 2L^b \mathsf{C}_{\mathrm{ag}}(b+1, \frac{1}{2})$ for every real $b \ge 0$; on a larger torus the number of places costs $\ell^3 e^{\mathsf{B}(X_g)\ell^3}$. The parts $t \le k$ of $\mathbb{P}_k$ are those of $\mathbb{P}$, and on the first part $\alpha = a_{K-k}$. □

Lemma 4.21 (The remainder)**.** Let g satisfy (Γ), (K, β) be a first-passage datum at g, $\ell = \ell_g$, $n \in \{3, 4\}$, and let real weights satisfy the conditions of weights placed at (ρ, r, d) (if $n = 3$), or of two pairs at (ρ, r, d) (if $n = 4$), but for the upper bounds on the distances of the centres, with $0 < \rho \le r \le \mathsf{d} \le 1$, $\rho \le \frac{1}{2}$, $s(\rho) \le K$, $\rho \le r_1(\mathsf{b})$, all supports in one ball of radius $\le \frac{1}{8}\ell_g - 1$. There are numbers $\Lambda_3 \ge 1$ and C_{J} formed from the constants such that at $\mathbb{P}$-almost every ω, for every nonempty β,

$$|\mathsf{Y}_\beta(\omega)| \le \Lambda_3 \, \alpha^{-6} \, \mathsf{s}(\beta) \prod_{a \in \beta} |w_a|_\infty, \qquad \mathsf{s}(\beta) = \rho^3 \text{ (a point)}, \quad \rho^6 r^{-3} \text{ (a whole pair)},$$
$$(\rho/r)^{3c(\beta)} \rho^6 \mathsf{d}^{-9/2} \ (\beta \text{ crossing});$$

hence $|S_n^T - \mathsf{J}_n| \le \mathsf{k}_n \Lambda_3^n \mathsf{M}'^n \mathsf{b}^n \mathsf{d}^{3/2} \mathrm{nat}_n$ with $\mathsf{k}_3 = 13$, $\mathsf{k}_4 = 77$, and $|\mathsf{J}_n| \le C_{\mathrm{J}} (\mathsf{b}/\mathsf{d})^{n/2} \mathrm{nat}_n$, uniformly in $\rho \le r \le \mathsf{d}$ and in K. For two weights w_1, w_2 with $\Sigma(w_a) \subset \mathbb{B}(x_a, \rho)$ whose supports are at a

distance $\geq \mathsf{d}$, with $0 < \rho \leq \mathsf{d} \leq 1$, $\rho \leq \frac{1}{2}$, $s(\rho) \leq K$, $\rho \leq r_1(\mathsf{b})$ and both supports in one ball of radius $\leq \frac{1}{8}\ell_g - 1$, at $\mathbb{P}$-almost every ω

$$|\mathsf{Y}_a(\omega)| \leq \Lambda_3\, \alpha^{-6}\rho^3 |w_a|_\infty, \qquad |\mathsf{Y}_{12}(\omega)| \leq \Lambda_3\, \alpha^{-6}\rho^6 \mathsf{d}^{-3} |w_1|_\infty |w_2|_\infty \leq \Lambda_3\, \alpha^{-6}\rho^6 \mathsf{d}^{-9/2} |w_1|_\infty |w_2|_\infty,$$

and

$$|S_2^T - \mathsf{J}_2| \leq 3\Lambda_3^2 \mathsf{M}'^2 \rho^6 \mathsf{d}^{-9/2} |w_1|_\infty |w_2|_\infty;$$

if besides the balls are $\geq \mathsf{d}$ apart,

$$|\mathsf{J}_2| \leq C_{\mathsf{J}}(\mathsf{b}/\mathsf{d})\rho^6 \mathsf{d}^{-6}\mathsf{b}^{-2} |w_1|_\infty |w_2|_\infty.$$

■

Proof. For $|\beta| \geq 2$, $\mathsf{Y}_\beta = \sum_{i \leq t'} (\mathsf{Sum}[E_\beta^{(i)}](U) - \mathsf{Sum}[E_\beta^{(i)}](\mathbf{1})) - \sum_{t' < i \leq K} \mathsf{Sum}[E_\beta^{(i)}](\mathbf{1})$, $U = U_{t'}(W)$; at a point $\mathcal{O}(w_a)(U) - \mathcal{O}(w_a)(\mathbf{1})$ comes besides, and for $t' \geq 1$ it is $\leq 162 B_3^2 \varepsilon_{t'}^2 \rho^3 \alpha^{-4} |w_a|_\infty$ on the regular configuration of Lem. 4.20(d). The first sum is bounded by Lem. 4.8(b) at the regular background of Lem. 4.20(d), the second at $\mathbf{1}$, both with the sizes of Lem. 4.9(f) at the width ρ, whose conditions hold: (P1), (P2) and the separation of the corners follow, as in the proof of Lem. 4.5(b), from the balls that hold the supports and from the distances of the supports alone, and $\rho \geq \frac{2}{3} M a_K$, $E_\sigma(\hat{\rho}_1/2) \leq \varkappa_0^4$ from $\rho \leq r_1(\mathsf{b})$, $s(\rho) \leq K$ (Lem. 4.6); the sums over the levels on both sides of t' are geometric.

For $t' \geq 1$, with $j = t' - 1$, $a_{K-j} = \alpha/L$ and $x_j > 1$: $\bar{\nu}_{j;\beta} \leq \bar{c}_m (L/\alpha)^m \prod |w_a|_\infty$, $\Xi_j^r \leq \rho^6 r^{-3} L^3 \alpha^{-3}$, and $\mathsf{z}(\beta) \min\{1, v_j\}^{3/2} \leq \mathsf{s}(\beta) L^{3/2} \alpha^{-3/2}$; so $|\mathsf{Y}_\beta| \leq C \alpha^{-6} \mathsf{s}(\beta) \prod |w_a|_\infty$ ($\alpha \leq 1$, $m \leq 4$). The part $t = 1$ of Ω has $t' = 0$, $\alpha = a_K$, and there $W \in \mathcal{V}_0$ is any configuration and $U_0(W) = W$: $D_0^\lambda = \mathcal{O}(w_\lambda)$, so the first sum is empty; at a point $0 \leq a_p \leq 2$ and (P1) give $|\mathcal{O}(w_a)(W)| \leq 2 a_K^{-1} \sum_p |w_a(p)| \leq 162 \rho^3 a_K^{-4} |w_a|_\infty = 162 \rho^3 \alpha^{-4} |w_a|_\infty$, and $\mathcal{O}(w_a)(\mathbf{1}) = 0$; the second sum runs over $1 \leq i \leq K$ at $\mathbf{1}$ and is led by $i = 1$, with the sizes above at $j = 0$: $\bar{\nu}_{0;\beta} \leq \bar{c}_m \alpha^{-m} \prod |w_a|_\infty$ ($x_0 > 1$), $\Xi_0^r \leq \rho^6 r^{-3} \alpha^{-3}$, $\mathsf{z}(\beta) \min\{1, v_0\}^{3/2} \leq \mathsf{s}(\beta) \alpha^{-3/2}$.

So the bound holds at every ω of this part, with $\Lambda_3 \geq 162$. The weights $\mathsf{v}_1 = \mathsf{v}_2 = \rho^3 r^{-3/2}$, $\mathsf{v}_3 = \rho^3 \mathsf{d}^{-9/2}$ (and $\mathsf{v}_a = \rho^3 r^{-3/2} \mathsf{d}^{-9/8}$ at $n = 4$) have $\mathsf{s}(\beta) \leq \mathsf{v}^\beta$ and $\mathsf{v}^{[n]} = \mathsf{s}([n])$, so every product over a partition is $\leq \mathsf{s}([n])$; Lem. 4.20(c), the bounds $|\kappa_2| \leq 2$, $|\kappa_3| \leq 6$ times the mean of the product of the bounds, and the moments of α^{-1} give $13 = 1 + 3 \cdot 2 + 6$ at $n = 3$; finally $\mathsf{s}([3]) \prod |w_a|_\infty = \mathsf{b}^3 \mathsf{d}^{3/2} \mathrm{nat}_3$. The bound of J_n is Lem. 4.8(b) at $\mathbf{1}$ with the bounds of Lem. 4.9(e) at all the levels, in the form (f).

At two sources whose supports are at a distance $\geq \mathsf{d}$ the two weights form one pair at the widths (ρ, d) in the sense of Lem. 4.9(f): (P1), (P2) and the separation of the corners follow from the balls $\mathbb{B}(x_a, \rho)$ and from the distance $\geq \mathsf{d}$ of the supports, and $\rho \geq \frac{2}{3} M a_K$, $E_\sigma(\hat{\rho}_1/2) \leq \varkappa_0^4$ from $\rho \leq r_1(\mathsf{b})$, $s(\rho) \leq K$, as above. The proof of the bound above, run for the list of the two weights, gives at a point and at the whole pair, read at $r = \mathsf{d}$, the display at two sources, as $\rho^6 \mathsf{d}^{-3} \leq \rho^6 \mathsf{d}^{-9/2}$ for $\mathsf{d} \leq 1$; with $S_2^T - \mathsf{J}_2 = \int \mathsf{Y}_{12} d\mathbb{P} + \kappa_{\mathbb{P}}(\mathsf{Y}_1, \mathsf{Y}_2)$ (Lem. 4.20(c)), $|\kappa_2| \leq 2$ and the moments of α^{-1} it gives $3 = 1 + 2$.

If besides the balls are $\geq \mathsf{d}$ apart, put a dummy bump w_2', the indicator of the plaquettes with barycentres in $\mathbb{B}(x_1 + 3\rho e, \rho)$, e the unit vector from x_2 to x_1; the three are at the lengths (ρ, ρ, d), and by multilinearity the objects of the pair are those of the list with the dummy: $S_2^T - \mathsf{J}_2 = \int \mathsf{Y}_{12} d\mathbb{P} + \kappa_{\mathbb{P}}(\mathsf{Y}_1, \mathsf{Y}_2)$, with $|\mathsf{Y}_a| \leq \Lambda_3 \alpha^{-6} \rho^3 |w_a|_\infty$ and $|\mathsf{Y}_{12}| \leq \Lambda_3 \alpha^{-6} \rho^6 \mathsf{d}^{-9/2} |w_1|_\infty |w_2|_\infty$; and the bound of J_n above, at the crossing set that the two weights form in this list, is the bound of J_2. □

Lemma 4.22 (Lattice sums and continuum integrals)**.** For every side ℓ and every cutoff K, that is on the torus T_ℓ with the lattice $a_K \mathbb{Z}^3$, and with the least lifts of the differences, the following hold.

(a) For $4a_K \le \rho \le r \le \mathsf{d} \le \mathsf{d}^+$, $4\mathsf{d}^+ < \ell$, $\vartheta \in (0,1]$ and Lipschitz f_1, f_2, f_3 with $\operatorname{supp} f_a \subset \mathbb{B}(x_a, \rho)$, $\operatorname{Lip} f_a \le \|f_a\|_\infty/\vartheta\rho$, $\operatorname{dist}(\operatorname{supp} f_1, \operatorname{supp} f_2) \ge r$, $\operatorname{dist}(\operatorname{supp} f_a, \operatorname{supp} f_3) \ge \mathsf{d}$, all supports in one ball of radius $\frac12\mathsf{d}^+$:

$$\Big|\operatorname{Tr}\prod_a((f_a\circ x)C_K^{\text{free}}) - \int k_3\prod_a f_a\Big| \le C^{(3)}\Big[\frac{a_K}{\vartheta\rho} + \Big(\frac{\mathsf{d}^+}{\ell}\Big)^3\Big]\rho^9 r^{-3}\mathsf{d}^{-6}\prod_a\|f_a\|_\infty.$$

(b) For two pairs with the same data ($8a_K \le \rho$, the balls $\ge r$ apart in a pair and $\ge \mathsf{d}$ apart across): $|\frac12\sum_\varsigma \operatorname{Tr}\prod_l((f_{\varsigma(l)}\circ x)C_K^{\text{free}}) - \int k_4\prod f_a| \le C^{(4)}[\frac{a_K}{\vartheta\rho} + (\frac{\mathsf{d}^+}{\ell})^3]\rho^{12}r^{-6}\mathsf{d}^{-6}\prod\|f_a\|_\infty$.

(c) For $8a_K \le \rho \le \mathsf{d}$, $64\mathsf{d} \le \ell$, Lipschitz φ_a in $\mathbb{B}(x_a,\rho)$ with $\operatorname{Lip}\varphi_a \le \|\varphi_a\|_\infty/\vartheta\rho$, the balls $\ge \mathsf{d}$ apart and $|x_1 - x_2| \le 18\mathsf{d}$:

$$\Big|\sum_{\lambda(p)=\lambda,\,\lambda(q)=\mu}\varphi_1(x(p))\varphi_2(x(q))C_K^{\text{free}}(p,q)^2 - \mathcal{K}_{\lambda\mu}(\varphi_1,\varphi_2)\Big| \le C_{\text{Rs}}\Big[\frac{a_K}{\vartheta\rho} + \Big(\frac{\mathsf{d}}{\ell}\Big)^3\Big]\Big(\frac{\rho}{\mathsf{d}}\Big)^6\|\varphi_1\|_\infty\|\varphi_2\|_\infty.$$

■

Proof. Step A. In a ring, and in a square, every sign ϵ_p stands twice, so $C_K^{\text{free}}(p_l, p_{l+1})$ may be replaced by $\epsilon\epsilon C_K^{\text{free}}$ and then, by equations YM3_I:(2245)–(2246), by $a_K^3\mathbb{K}_3(u_l)_{\lambda(p_l)\lambda(p_{l+1})}$; both are $\le (1+3C_{\text{lat}})a_K^3|u|^{-3}$, and telescoping costs $C_{\text{lat}}a_K^3[a_K^2|u_l|^{-5} + \ell^{-3}]$ times the other factors, with at most $81(\rho/a_K)^3$ plaquettes for each source. *Step B.* For each choice of the directions the remaining sum is a Riemann sum over the cubes of side a_K about the barycentres, which for one orientation form a translate of $a_K\mathbb{Z}^3$; summed over the directions, $\sum\prod_l\mathbb{K}_3(\cdot)_{\lambda_l\lambda_{l+1}} = \operatorname{tr}\prod_l\mathbb{K}_3$. On the cubes that count the differences are at least 0.56 times their least values, $\|\mathbb{K}_3(u)\| \le \frac{1}{2\pi}|u|^{-3}$, $\|\partial\mathbb{K}_3(u)\| \le \frac{6}{4\pi}|u|^{-4}$ (Lem. 4.24(b)), and the Lipschitz bounds of the f_a give the error $a_K/\vartheta\rho$ times the natural size; the numbers $C^{(3)}, C^{(4)}, C_{\text{Rs}}$ are these counts. □

Lemma 4.23 (Kernels of forms bounded on small bumps)**.** Let $U_1, \dots, U_p$ be open subsets of $\mathbb{R}^3$ or of a torus, $\rho_0 > 0$, $\mathsf{A} \ge 0$, and T a p-linear form on $\prod_a \operatorname{Lip}_c(U_a)$ with $|T(f_1,\dots,f_p)| \le \mathsf{A}\rho^{3p}\prod_a\|f_a\|_\infty$ whenever $\rho \le \rho_0$ and each f_a has its support in a ball of radius ρ in U_a. Then: (a) $T(f) = \int k\prod_a f_a$ with a Borel function k, $|k| \le (\frac{3}{\pi})^p\mathsf{A}$ almost everywhere, and at every point where T has a continuous kernel; (b) T extends to bounded Borel functions with compact supports in the U_a, and for balls $\mathbb{B}(z_a,\rho) \subset U_a$, $\rho \le \rho_0$, the mean of k over $\prod_a\mathbb{B}(z_a,\rho)$ is at most $(\frac{3}{4\pi})^p\mathsf{A}$ in modulus, at every choice of the centres; with the radial tents, $|\int k\prod_a\chi^\rho_{z_a}| \le (\frac{3}{\pi})^p\mathsf{A}$. ■

Proof. Let $k_\rho(x_1,\dots,x_p) = T(\chi^\rho_{x_1},\dots,\chi^\rho_{x_p})$; $\|\chi^\rho\|_\infty = 3/\pi\rho^3$ gives $|k_\rho| \le (\frac{3}{\pi})^p\mathsf{A}$. On functions with supports in fixed compact sets T is bounded in the sup norms (a Lipschitz partition of unity of width ρ_0), and the Riemann sums of $f_a * \chi^\rho$ converge uniformly, so $\int k_\rho\prod f_a = T(f_1*\chi^\rho,\dots) \to T(f)$; a weak-$*$ limit point of k_ρ in L^∞ is a kernel. (b) By dominated convergence T extends with the same bound; at $f_a = \mathbf{1}_{\mathbb{B}(z_a,\rho)}$, $|\int_{\prod\mathbb{B}(z_a,\rho)}k| \le \mathsf{A}\rho^{3p}$, which is the bound of the mean. □

Lemma 4.24 (The free template)**.** Let $X \ne 0$, $\mathsf{d} = |X|$, $\mathsf{b} > 0$, n, n' unit vectors.

(a) $\mathbb{K}_3(X)^2 = (16\pi^2\mathsf{d}^6)^{-1}(3\hat X\hat X^{\mathsf{T}} + 1)$; $\frac32\mathsf{b}^{-2}\sum_\mu\{[\mathbb{K}_3(X)_{\lambda\mu}]^2 - \frac13\sum_{\lambda'}[\mathbb{K}_3(X)_{\lambda'\mu}]^2\} = \mathsf{t}^\circ(e_\lambda, X)$; $\frac32\mathsf{b}^{-2}\sum_{\lambda,\mu}[\mathbb{K}_3(X)_{\lambda\mu}]^2 = \frac{9}{16\pi^2}\mathsf{b}^{-2}\mathsf{d}^{-6}$; $\sum_\lambda n_\lambda^2\mathsf{t}^\circ(e_\lambda,X) + \sum_{\lambda<\lambda'}n_\lambda n_{\lambda'}3\mathsf{b}^{-2}[\mathbb{K}_3(X)^2]_{\lambda\lambda'} = \mathsf{t}^\circ(n,X)$; and $|\mathsf{t}^\circ(n,X) - \mathsf{t}^\circ(n',X)| \le 0.057|n-n'|\mathsf{b}^{-2}\mathsf{d}^{-6}$.

(b) In the operator norm, $\|\mathbb{K}_3(y)\| = \frac{2}{4\pi}|y|^{-3}$, $\|(h\cdot\nabla)\mathbb{K}_3(y)\| \le \frac{6}{4\pi}|h||y|^{-4}$, $\|(h\cdot\nabla)^2\mathbb{K}_3(y)\| \le \frac{24}{4\pi}|h|^2|y|^{-5}$.

(c) For $0 < r \le \frac12\mathsf{d}$ and $c - z = X$: $|\frac{2\pi}{3}\mathsf{b}r^3\cdot 3\mathsf{b}^{-3}k_3(c+\frac{r}{2}n, c-\frac{r}{2}n, z) - \mathsf{t}^\circ(n,X)| \le 2.66\,(r/\mathsf{d})^2\mathsf{b}^{-2}\mathsf{d}^{-6}$; and for $0 < \rho < \frac14 r$ the same function averaged against the radial tents χ^ρ about the three points differs from its value by at most $7900\,(\rho/r)\mathsf{b}^{-2}\mathsf{d}^{-6}$.

(d) For $0 < \rho \le \frac18\mathsf{d}$: $|\mathcal{K}_{\lambda\mu}(\chi^\rho_c,\chi^\rho_z) - [\mathbb{K}_3(X)_{\lambda\mu}]^2| \le 2.28\,(\rho/\mathsf{d})\mathsf{d}^{-6}$.

■

Proof. (a) With $E = \hat{X}\hat{X}^{\mathsf{T}}$, $E^2 = E$: $(3E-1)^2 = 3E+1$; $\sum_\mu [\mathbb{K}_3(X)_{\lambda\mu}]^2 = [\mathbb{K}_3(X)^2]_{\lambda\lambda}$ and $\operatorname{tr}\mathbb{K}_3(X)^2 = 6(16\pi^2\mathsf{d}^6)^{-1}$; the identity in n is the expansion of $\langle n, \mathbb{K}_3(X)^2 n\rangle$; the last bound is $\frac{18}{32\pi^2} \le 0.057$ times $|n-n'|$. (b) $\mathbb{K}_3 = (4\pi)^{-1}\nabla\nabla|y|^{-1}$; the third and fourth derivatives of $|y|^{-1}$ contracted with h, in a frame with $\hat{y} = e_1$, have eigenvalues bounded by 6 and 24 times $|h|^k|y|^{-3-k}$. (c) With $D(n) = 3nn^{\mathsf{T}} - 1$, $h = \frac{r}{2}n$ and $\phi(t) = \operatorname{tr}[D(n)\mathbb{K}_3(X-th)\mathbb{K}_3(X+th)]$: $\mathbb{K}_3(rn) = (4\pi r^3)^{-1}D(n)$, so the first member is $\frac{1}{2}\mathsf{b}^{-2}\phi(1)$, while $\frac{1}{2}\mathsf{b}^{-2}\phi(0) = \mathsf{t}^\circ(n,X)$ by (a). ϕ is even, since $\operatorname{tr}[DAB] = \operatorname{tr}[DBA]$ for symmetric matrices, so $|\phi(1)-\phi(0)| \le \frac{1}{2}\sup|\phi''|$; with $|X \pm th| \ge \frac{3}{4}\mathsf{d}$, the trace norm 4 of $D(n)$ and (b),

$$|\phi''| \le \frac{4}{(4\pi)^2}(24\cdot 2 + 2\cdot 6\cdot 6 + 2\cdot 24)\frac{r^2}{4}\Big(\frac{4}{3}\Big)^8 \mathsf{d}^{-8} = 10.63\,(r/\mathsf{d})^2\mathsf{d}^{-6},$$

and $\frac{1}{4}\cdot 10.63 \le 2.66$. The tents: the mean value theorem with (b) for the three factors of k_3. (d) $|\nabla(\mathbb{K}^2_{3,\lambda\mu})| \le \frac{3}{2\pi^2}|u|^{-7}$ on a segment in $\{|u| \ge \frac{3}{4}\mathsf{d}\}$, and $\frac{3}{2\pi^2}(\frac{4}{3})^7\cdot 2 \le 2.28$. □

Every state.

Lemma 4.25 (The chessboard bound)**.** Let $0 < g \le \gamma$, (K,β) a first-passage datum at g, $\ell' \in \mathfrak{L}(g)$, and consider the board of $\mathbb{L}_K$ into cubes of side $\frac{1}{2}\ell'$, the cells, with any offset in $a_K\mathbb{Z}^3$; let $\mathsf{Q}_0 = \exp[\frac{1}{8}\ell'^3(2C_{\mathrm{S}} + C_g)]$ with the constants C_{S}, C_g of the two-sided stability bound. For every $\ell \in \mathfrak{L}(g)$ with $\ell \ge \ell'$, every $\nu \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$, every cell and every bounded complex Borel function F of the variables of that cell,

$$|\tilde{\mu}_{K,\ell,\beta}(F)|,\ |\nu(F)| \le \mathsf{Q}_0\|F\|_{\mathrm{ch}}, \qquad \|F\|^8_{\mathrm{ch}} = \tilde{\mu}_{K,\ell',\beta}\Big(\prod_{\text{the eight cells of } T_{\ell'}} F^{[\cdot]}\Big) \ge 0,$$

$F^{[\cdot]}$ are the reflected copies of F, conjugated when an odd number of coordinate reflections is used. ■

Proof. The bound is a chain of four links. The cells are cubes of side $\mathsf{l} = \frac{1}{2}\ell'$; the torus T_ℓ holds $\mathsf{N}_\ell = (2\ell/\ell')^3$ of them and $T_{\ell'}$ eight. For F as in the statement let F^{all}_ℓ be the product of the reflected copies of F, conjugated with the parity convention in the statement, over all the cells of T_ℓ, and $Z_{K,\ell,\beta}[F^{\mathrm{all}}_\ell]$ the integral of the density of T_ℓ with this insertion. (1) *Reflection positivity on T_ℓ.* The torus measures are reflection positive in the planes of the board, and the chessboard estimate ([7, Cor. B.1](b), (c)) gives $|\tilde{\mu}_{K,\ell,\beta}(F)| \le \|F\|_{\mathrm{ch},\ell}$, where $\|F\|^{\mathsf{N}_\ell}_{\mathrm{ch},\ell} = Z_{K,\ell,\beta}[F^{\mathrm{all}}_\ell]/Z_{K,\ell,\beta} \ge 0$; on $T_{\ell'}$ this is the number of the statement, $\|F\|_{\mathrm{ch},\ell'} = \|F\|_{\mathrm{ch}}$. (2) *Two sides, with the insertions.* As ℓ/ℓ' is a power of L, the subadditivity in the volume of the integrals with the reflected copies inserted (Prop. YM3_I:86.5, Cor. YM3_I:86.6) gives $Z_{K,\ell,\beta}[F^{\mathrm{all}}_\ell]^{1/\mathsf{N}_\ell} \le Z_{K,\ell',\beta}[F^{\mathrm{all}}_{\ell'}]^{1/8}$; with $p_\ell = \ell^{-3}\log Z_{K,\ell,\beta}$, $Z^{1/\mathsf{N}_\ell}_{K,\ell,\beta} = e^{\mathsf{l}^3 p_\ell}$ and $Z^{1/8}_{K,\ell',\beta} = e^{\mathsf{l}^3 p_{\ell'}}$ this is clause (e) of the same corollary:

$$\|F\|_{\mathrm{ch},\ell} \le \|F\|_{\mathrm{ch},\ell'}\, e^{\mathsf{l}^3(p_{\ell'} - p_\ell)}.$$

(3) *Stability, with no insertion.* The two-sided stability bound among the construction results, which rests on Thms. YM3_I:AJ and YM3_I:AD ([7, Lem. N.1]), concerns the partition functions without insertions: $Z_{K,\ell,\beta} = e^{E_{K,\ell,\beta}}\widetilde{Z}_\ell$ with the vacuum constant $E_{K,\ell,\beta}$ and a number $e^{-C_g\ell^3} \le \widetilde{Z}_\ell \le e^{C_{\mathrm{S}}\ell^3}$, and $|E_{K,\ell,\beta}\ell^{-3} - E_{K,\ell',\beta}\ell'^{-3}| \le C_{\mathrm{S}}$ for two sides $\ell, \ell' \in \mathfrak{L}(g)$. So

$$p_{\ell'} - p_\ell \le (E_{K,\ell',\beta}\ell'^{-3} + C_{\mathrm{S}}) - (E_{K,\ell,\beta}\ell^{-3} - C_g) \le 2C_{\mathrm{S}} + C_g,$$

and (1)–(3), with $\mathsf{l}^3 = \frac{1}{8}\ell'^3$, give $|\tilde{\mu}_{K,\ell,\beta}(F)| \le \mathsf{Q}_0\|F\|_{\mathrm{ch}}$. (4) *Periodic Gibbs states.* For $\nu \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$ the domination theorem (Prop. YM3_I:91.9, for every board of $\mathbb{L}_K$ and every offset) gives $|\nu(F)| \le \liminf_k \|F\|_{\mathrm{ch},\ell_k}$ along every sequence $\ell_k \to \infty$ in $\mathfrak{L}(g)$, and (2), (3) at $\ell = \ell_k$ bound the right member by $\mathsf{Q}_0\|F\|_{\mathrm{ch}}$. No link reads what F is made of, and Q_0 depends neither on $\ell \ge \ell'$ nor on K. □

Lemma 4.26 (The fold)**.** Let $\mathfrak{N}_I = \mathbb{C}[\lambda_\iota : \iota \in I]/(\lambda_\iota^2)$, $[x]_\gamma$ the coefficient of λ^γ, $\mathfrak{m} = \{[x]_\emptyset = 0\}$, and $\|x\|_u = \sum_\gamma |[x]_\gamma|/u^\gamma$ for numbers $u_\iota > 0$.

(a) $\|\cdot\|_u$ is submultiplicative; for $y \in \mathfrak{m}$, $\|e^y - 1\|_u \le (1 + \|y\|_u)^{|I|} - 1$ and $\|\log(1+y)\|_u \le \|y\|_u(1+\|y\|_u)^{|I|-1}$; the logarithm of the moment series is the cumulant series.

(b) In the data of Lem. 4.25 let $f_1, \dots, f_n$ be bounded real functions that vanish off the interior of one cell, w_ι, $\iota \in I = [n] \times \{\text{the eight cells}\}$, the eight mirror images of the $f_a \circ x$ on $T_{\ell'}$, and $(\Omega, \mathbb{P})$ a probability space with a measurable $\mathbb{W}\colon \Omega \to \mathfrak{m}$ such that $\langle \prod_{\iota\in\gamma} \mathcal{O}(w_\iota)\rangle_{K,\ell',\beta} = \int [\exp \mathbb{W}]_\gamma d\mathbb{P}$ for all $\gamma \subset I$. Let $u_{(a,\cdot)} = u_a > 0$, $\mathsf{c} = \sum \mathsf{c}^{(\cdot)}$ a sum of constants of $\mathfrak{m}$ each in the variables of one cell, of any size, and $\mathsf{Z} \ge 0$ measurable with $\|\mathbb{W} - \mathsf{c}\|_u \le \mathsf{Z}$ almost everywhere. Put $\mathsf{y} = \int[(1+\mathsf{Z})^n - 1]d\mathbb{P}$, $\mathsf{h} = \mathsf{y}(1+\mathsf{y})^{n-1}$, $\eta_0 = [\int(1 + \mathsf{Z} + 8\mathsf{h})^{8n} d\mathbb{P}]^{1/8}$, $\eta = (2^n - 1)\mathsf{Q}_0\eta_0$. Then for every nonempty $\beta \subset [n]$, every $\ell \in \mathfrak{L}(g)$ with $\ell \ge \ell'$, and $\mathfrak{s} = \tilde{\mu}_{K,\ell,\beta}$ or $\mathfrak{s} \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$:

$$|S^T_{|\beta|}[\mathfrak{s}]((f_a)_{a\in\beta}) - S^T_{|\beta|}[\tilde{\mu}_{K,\ell',\beta}]((f_a)_{a\in\beta})| \le \eta(1+\eta)^{n-1}u^\beta.$$

The bound holds at every size of Z, of the constants c and of η. ∎

Proof. (a) is the algebra of exp and log in an algebra with $|I|$ nilpotent generators. (b) Let $\mathsf{k} = \log \sum_\beta \tilde{\mu}_{K,\ell',\beta}(\prod_{a\in\beta}\mathcal{O}(f_a))\lambda^\beta$ and $\Xi_\beta = [\exp(\sum_a \lambda_a \mathcal{O}(f_a) - \mathsf{k})]_\beta$, a function of the variables of the cell. The mirror images of Ξ_β are the same expressions in the mirrored sources, so by the representation $\|\Xi_\beta\|^8_{\mathrm{ch}} = \int[\exp(\mathbb{W} - \mathsf{k}^{\mathrm{all}})]_{\beta\times\{\text{cells}\}} d\mathbb{P}$, $\mathsf{k}^{\mathrm{all}}$ the sum of the eight copies of k. Each copy has $e^{\mathsf{k}^{(\cdot)}} = \int \exp(\pi_{\mathrm{cell}}\mathbb{W})d\mathbb{P}$, so $\mathsf{k}^{(\cdot)} - \mathsf{c}^{(\cdot)} = \log(1 + \mathsf{y}^{(\cdot)})$ with $\|\mathsf{y}^{(\cdot)}\|_u \le \mathsf{y}$ and $\|\mathsf{k}^{(\cdot)} - \mathsf{c}^{(\cdot)}\|_u \le \mathsf{h}$; the constants of any size cancel, $\|\mathbb{W} - \mathsf{k}^{\mathrm{all}}\|_u \le \mathsf{Z} + 8\mathsf{h}$, and $|[\exp(\cdot)]_{\beta\times\{\text{cells}\}}| \le (1 + \mathsf{Z} + 8\mathsf{h})^{8n}(u^\beta)^8$.

So $\|\Xi_\beta\|_{\mathrm{ch}} \le \eta_0 u^\beta$, and Lem. 4.25 gives $|\mathfrak{s}(\Xi_\beta)| \le \mathsf{Q}_0\eta_0 u^\beta$. With $\mathsf{b}' = \sum_\beta \mathfrak{s}(\Xi_\beta)\lambda^\beta = \mathsf{M}_\mathfrak{s} e^{-\mathsf{k}}$, $\mathsf{M}_\mathfrak{s}$ the moment series of $\mathfrak{s}$: $\|\mathsf{b}' - 1\|_u \le \eta$ and $\|\log \mathsf{M}_\mathfrak{s} - \mathsf{k}\|_u = \|\log \mathsf{b}'\|_u \le \eta(1+\eta)^{n-1}$. □

Lemma 4.27 (Mirror copies)**.** Zero sources are omitted before forming the weighted jet norm; coefficients involving their directions vanish by multilinearity. Assume the remaining source weights are nonzero. Let g satisfy (Γ), $\ell' = \ell_g$, and let $f_1, \dots, f_n$, $n \in \{2, 3, 4\}$, be as in (A.4)(i) (for $n = 4$, two pairs; for $n = 2$, two bounded real Borel functions with $\operatorname{supp} f_a \subset \mathbb{B}(x_a, \rho)$, the two supports at a distance $\ge \mathsf{d}$, $0 < \rho \le \mathsf{d} \le 1$, $\rho \le \frac12$, $s(\rho) \le K$, $\rho \le r_1(\mathsf{b})$, both supports in one ball of radius $\le \frac18\ell_g - 1$), placed by the offset in the interior of one cell, so that two mirror copies are at least $\frac14\ell_g \ge 6.5$ apart. Let $\mathbb{W}$ be the multilinear jet of $\lambda \mapsto D^\lambda_{t'}(U_{t'}(W))$ for the $8n$ sources w_ι on the space of Lem. 4.20, $[\mathsf{c}]_\gamma = \partial^\gamma D^\lambda_K(\mathbf{1})|_0$ for γ in one cell and $[\mathsf{c}]_\gamma = 0$ for γ that link two cells, and $u_a = \mathsf{v}_a|w_a|_\infty$ with $\mathsf{v}_1 = \mathsf{v}_2 = \rho^3 r^{-3/2}$, $\mathsf{v}_3 = \rho^3\mathsf{d}^{-9/2}$ $(n = 3)$, $\mathsf{v}_a = \rho^3 r^{-3/2}\mathsf{d}^{-9/8}$ $(n = 4)$, $\mathsf{v}_a = \rho^3\mathsf{d}^{-9/4}$ $(n = 2)$.

Then $\|\mathbb{W}(\omega) - \mathsf{c}\|_u \le \mathsf{Z}_n\alpha(\omega)^{-6}$ almost everywhere, $\mathsf{Z}_n = 8(2^n - 1)\Lambda_3 + 2^{8n}\max_m \Lambda^{\mathrm{L}}_m$, where the numbers Λ^{L}_m, formed from the constants, bound the linking jets: for γ linking two copies, $|\sum_{i\le t} \mathsf{Sum}[E^{(i)}_\gamma](U)| \le \Lambda^{\mathrm{L}}_m \mathsf{b}^{-m/2} \prod_{\text{copies}} \mathsf{s}(\gamma \cap \text{copy}) \prod_{\iota\in\gamma} |w_\iota|_\infty$ at every U whose pairs of the levels $\le t$ lie in the cores. ∎

Proof. Within one cell the coefficients of $\mathbb{W} - \mathsf{c}$ are the Y_β of the copy (by multilinearity the mixture names no source), bounded by Lem. 4.21 and $\mathsf{s}(\beta) \le \mathsf{v}^\beta$; there are $8(2^n - 1)$ such sets. At $n = 2$ the two bumps are taken as one pair at the widths (ρ, d) in the sense of Lem. 4.9(f), since (P1), (P2) and the separation of the corners follow from the balls $\mathbb{B}(x_a, \rho)$ and from the distance $\ge \mathsf{d}$ of the supports, as in the proof of Lem. 4.21; that proof, with the size $\rho^6\mathsf{d}^{-3}$ of a whole pair at the distance d in the place of $\rho^6 r^{-3}$, gives at $\mathbb{P}$-almost every ω

$$|\mathsf{Y}_a| \le \Lambda_3\alpha^{-6}\rho^3|w_a|_\infty, \qquad |\mathsf{Y}_{12}| \le \Lambda_3\alpha^{-6}\rho^6\mathsf{d}^{-3}|w_1|_\infty|w_2|_\infty \le \Lambda_3\alpha^{-6}\rho^6\mathsf{d}^{-9/2}|w_1|_\infty|w_2|_\infty \quad (\mathsf{d} \le 1),$$

and $\rho^3 \le \mathsf{v}_a$, $\rho^6\mathsf{d}^{-9/2} = \mathsf{v}_1\mathsf{v}_2$. For a set γ that links two copies, the calculus of Lem. 4.6 at $8n$ sources, with the copies as cells and $\mathsf{D}^{\iota\iota'}_i \ge \hat{\mathsf{D}}^\times_i/\sqrt{3} - 1$ across, where $\mathsf{D}^\times \ge \frac14\ell_g$ is the least distance

on T_{ℓ_g} of the supports of two different mirror copies and $\hat{\mathsf{D}}_i^{\times} = \mathsf{D}^{\times}/a_{K-i}$, gives by the recursion (g), with $\sigma_{i;\gamma} \le E_\sigma(\hat{\mathsf{D}}_i^{\times})$, bounds on $\bar{\mathsf{h}}_{j;\gamma}$ and $\bar{\Psi}_{j;\gamma}$ by products of the one-cell sizes; with Lem. 4.8(b) and the geometric sums this is the bound of the linking jets. The products of natural sizes over a partition do not exceed that of the whole set. The numbers Λ_m^{L} are finite and formed from the constants, two copies being at least $\frac{1}{4}\ell_g \ge 6.5$ apart; no condition on g beyond (Γ) is needed. □

Proposition 4.28 (Transfer to every state)**.** Let g satisfy (Γ) and (K,β) be a first-passage datum at g.

(a) For $n \in \{3,4\}$ and the data of (A.4)(i) (for $n=4$ two pairs, all supports in one ball of radius $\le \frac{1}{8}\ell_g - 1$), every $\ell \in \mathfrak{L}(g)$ and every state $\mathfrak{s}$ at (K,β): $|S_n^T[\mathfrak{s}](f) - S_{n;K,\ell_g,\beta}^T(f^{\langle \ell_g\rangle})| \le C_n^{\mathrm{tr}3}(g)\mathsf{b}^n\mathsf{d}^{3/2}\mathrm{nat}_n$, where $C_n^{\mathrm{tr}3}(g) = \eta(1+\eta)^{n-1}$, $\eta = (2^n-1)\mathsf{Q}_0(g)\eta_0$, $\eta_0 = (1+\mathsf{Z}_n + 8\mathsf{h})^n(\bar{\mathsf{M}}^{(48n)})^{1/8}$, $\mathsf{h} = \mathsf{y}(1+\mathsf{y})^{n-1}$, $\mathsf{y} = (1+\mathsf{Z}_n)^n\bar{\mathsf{M}}^{(6n)}$, $\bar{\mathsf{M}}^{(b)} = 1 + 2L^b\mathsf{C}_{\mathrm{ag}}(b+1,\frac{1}{2})$ and $\mathsf{Q}_0(g) = \exp[\frac{1}{8}\ell_g^3(2C_{\mathrm{S}} + C_g)]$.

(b) For two bumps: let $0 < \rho \le \mathsf{d} \le 1$, $\rho \le \frac{1}{2}$, $s(\rho) \le K$, $\rho \le r_1(\mathsf{b})$, f_1, f_2 bounded real Borel functions on $\mathbb{R}^3$ with $\operatorname{supp} f_a \subset \mathbb{B}(x_a,\rho)$, the two supports at a distance $\ge \mathsf{d}$, both in one ball of radius $\le \frac{1}{8}\ell_g - 1$, $t_a\colon \{1,2,3\} \to [-1,1]$, $w_a(p) = t_a(\lambda(p))f_a(x(p))$, and $\bar{w}_a$ the same with $f_a^{\langle \ell_g\rangle}$ on T_{ℓ_g}. Then for every state $\mathfrak{s}$ at (K,β), $|\langle \mathcal{O}(w_1);\mathcal{O}(w_2)\rangle_{\mathfrak{s}} - \langle \mathcal{O}(\bar{w}_1);\mathcal{O}(\bar{w}_2)\rangle_{K,\ell_g,\beta}| \le C_2^{\mathrm{tr}}(g)(\rho/\mathsf{d})^6\mathsf{d}^{3/2}\|f_1\|_\infty\|f_2\|_\infty$, where $C_2^{\mathrm{tr}}(g) = \eta(1+\eta)$ is the number of (a) formed at $n=2$, with Z_2 of Lem. 4.27.

■

Proof. Every periodic Gibbs state is invariant under lattice translations by Prop. YM3_I:103.4; choose the offset of the board so that the tuple lies in the interior of one cell. Lem. 4.26 with $\ell' = \ell_g$, the representation of Lem. 4.20(b) at the $8n$ mirrored sources, $\mathsf{Z} = \mathsf{Z}_n\alpha^{-6}$ by Lem. 4.27, and the moments of Lem. 4.20(e): $\mathsf{y} \le (1+\mathsf{Z}_n)^n\bar{\mathsf{M}}^{(6n)}$ and $\eta_0 \le (1+\mathsf{Z}_n+8\mathsf{h})^n(\bar{\mathsf{M}}^{(48n)})^{1/8}$, so η is formed from the constants and g. With the weights of Lem. 4.27, $u^{[n]} = \mathsf{s}([n])\prod|w_a|_\infty = \mathsf{b}^n\mathsf{d}^{3/2}\mathrm{nat}_n$. For (b) the same argument runs at $n=2$, with Lem. 4.27 at $n=2$, whose bounds read only the balls of radius ρ and the distance $\ge \mathsf{d}$ of the supports; a mirror map carries a_p to a_{rp} with $x(rp) = rx(p)$ and $\lambda(rp) = \lambda(p)$, so plane weights go to plane weights, and $u_1u_2 = (\rho/\mathsf{d})^6\mathsf{d}^{3/2}|w_1|_\infty|w_2|_\infty$. □

Lemma 4.29 (Limits and components)**.**

(a) Every Gibbs sequence at g that converges has subsequences on which the labels converge (they lie in $(g^{-2},\bar{\mathsf{b}}]$); along every such subsequence the members are states at their data, the limit σ is a law on $\mathbb{R}^3$ with $S_n^T[\nu_j] \to S_n^\sigma$ on separated tuples, and for every component σ_ξ of a canonical decomposition of σ its approximants ρ_j are periodic Gibbs states at the same data with $S_n^T[\rho_j] \to S_n^{\sigma_\xi}$.

(b) For a law with continuous moment kernels through order three, the connected kernel is $\mathsf{s}_3 = \mathsf{s}_3^{\mathrm{m}} - \mathsf{s}_2\otimes\mathsf{s}_1 - \cdots + 2\mathsf{s}_1\otimes\mathsf{s}_1\otimes\mathsf{s}_1$, and $\mathsf{s}_2^T = \mathsf{s}_2 - \mathsf{s}_1\otimes\mathsf{s}_1$. These kernels are continuous. For the periodic-Gibbs limits and components in (a), they are also real analytic and Euclidean invariant, by Prop. YM3_I:108.3.

(c) Let $\eta(t) = 1$ $(t \le 0)$, $1 - 3t^2 + 2t^3$ $(0 \le t \le 1)$, 0 $(t \ge 1)$, $\psi_\rho(v) = \eta(5|v|/\rho - 4)$ and $f_a^\rho = \psi_\rho(\cdot - z_a)/\int\psi_\rho$. These are C^1 probability densities supported in $\mathbb{B}(z_a,\rho)$ with $\rho^9\prod_{a\le 3}\|f_a^\rho\|_\infty \le 0.102$; if s_3 is continuous at (z_1,z_2,z_3) and $|\int(\mathsf{s}_3 - 3\mathsf{b}^{-3}k_3)\prod f_a^\rho| \le \mathsf{A}$ for all small ρ, then $|\mathsf{s}_3 - 3\mathsf{b}^{-3}k_3|(z_1,z_2,z_3) \le \mathsf{A}$.

■

Proof. (a) follows from the component approximation in Thm. YM3_I:AP and the canonical decomposition of Thm. YM3_I:AL; a joint cumulant of order ≥ 2 does not change if constants are added to its entries, so the centring of the approximants does not enter. (b) Apply the finite moment–cumulant formula to the assumed continuous moment kernels. For the laws in (a), their

analytic and invariance properties follow from Thm. YM3_I:AP and Prop. YM3_I:108.3. For a self-centred limit, $S_1^\sigma = 0$. (c) ψ_ρ is constant on $|v| \le \frac{4}{5}\rho$, and $\mathsf{s}_3 - 3\mathsf{b}^{-3}k_3$ is continuous at the point. □

Lemma 4.30 (One subsequence for all pairs of bumps)**.** Let Ω be an open subset of $\{(x,y) : 0 < |x-y| < \mathsf{d}_\Omega\}$ in T_ℓ^2 or in $(\mathbb{R}^3)^2$, and let bilinear forms $\mathsf{t}^{(j)}_{\lambda\mu}$, $\lambda,\mu \in \{1,2,3\}$, $j \in \mathbb{N}$, on bounded real Borel functions be such that for every $(x_0,y_0) \in \Omega$, with closed balls $\bar U, \bar V$ about x_0, y_0 of a radius $\varrho(x_0,y_0) > 0$ for which $\bar U \times \bar V \subset \Omega$, there is $\mathsf{B} < \infty$ with $|\mathsf{t}^{(j)}_{\lambda\mu}(f,h)| \le \mathsf{B}\|f\|_\infty \|h\|_\infty$ for all large j and all f,h supported in $\bar U, \bar V$. Then there is a subsequence along which $\mathsf{t}^{(j)}_{\lambda\mu}(\varphi_1,\varphi_2)$ converges for all λ,μ and all real Lipschitz φ_1,φ_2 with compact supports, $\operatorname{supp}\varphi_1 \times \operatorname{supp}\varphi_2 \subset \Omega$; the limits are bilinear. If on every such product of balls the limit forms satisfy the assumption of Lem. 4.23, they have Borel kernels on Ω, unique almost everywhere. ■

Proof. $\mathrm{Lip}_c(U)$ is a subset of the separable space $C_0(U)$, so it has a countable dense subset in the sup norm. Take a countable dense set of centres in Ω and a diagonal subsequence along which the countably many bounded sequences $\mathsf{t}^{(j)}_{\lambda\mu}(f,h)$, f,h in these dense sets, converge; by the uniform bound and bilinearity the sequences converge at all Lipschitz f,h in the balls. A pair φ_1,φ_2 as in the claim is decomposed by a Lipschitz partition of unity into cubes of a side small against the distances of $\operatorname{supp}\varphi_1 \times \operatorname{supp}\varphi_2$ to the diagonal and to the boundary of Ω: finitely many pairs, each in one product of balls. Two kernels that represent one form on an open product agree almost everywhere (mollification), and a numbering of the countable cover glues the local kernels into one Borel function. □

The bond forms.

Lemma 4.31 (Dilation to two couplings below the ceiling)**.** Let $0 < g \le \gamma$, let $z = z(g) \ge 1$ be the least integer with $L^{z-1}(g^{-2} + \bar B) - \bar B \ge x_{\mathrm N} = \gamma_{\mathrm N}^{-2}$ (Def. 4.1(1)), and define g_1', g_2' by $g_1'^{-2} = L^z(g^{-2} + \bar B) - \bar B$, $g_2'^{-2} = L^{-1}(g_1'^{-2} + 2B_\sharp) - B_\sharp$.

(a) $0 < g_1' < g_2' \le \gamma_{\mathrm N}$, and both depend on g and the constants alone.

(b) For a first-passage datum (K,β) at g with $K \ge z+1$ there is $\iota \in \{1,2\}$ such that $(K-z,\beta)$ is a first-passage datum at g_ι'; its run is $(x_0,\dots,x_{K-z})$, and its label is $L^z\mathsf{b}$.

(c) $L^z\ell \in \mathfrak{L}(g_\iota')$ for $\ell \in \mathfrak{L}(g)$; if $s(t) \ge z$, then $s_\iota'(L^z t) \le s(t) - z$ for the witness depth s_ι' at g_ι', and if $s(t) < z$, then $L^z t > 1$.

(d) $U \mapsto U^{(z)}$, $U^{(z)}(L^z b) = U(b)$, carries $\mu_{K,\ell,\beta}$ to $\mu_{K-z,L^z\ell,\beta}$, with $a_{L^z p}(U^{(z)}) = a_p(U)$ and $q_{L^z b}(U^{(z)}) = q_b(U)$. So $\mathcal{O}^T(\mathfrak{s})(U) = L^z\mathcal{O}'^T(\mathfrak{s}^{(z)})(U^{(z)})$ and $\langle \mathcal{O}^T(\mathfrak{s}_1); \mathcal{O}^T(\mathfrak{s}_2)\rangle_{K,\ell,\beta} = L^{2z}\langle \mathcal{O}'^T(\mathfrak{s}_1^{(z)}); \mathcal{O}'^T(\mathfrak{s}_2^{(z)})\rangle_{K-z,L^z\ell,\beta}$, primes marking the objects of the torus $T_{L^z\ell}$ at the cut-off $K-z$ and $\mathfrak{s}^{(z)}(L^z\cdot) = \mathfrak{s}(\cdot)$; and the numbers $\|\varphi\|_\infty$, $r\,\mathrm{Lip}\,\varphi$, a_K/r, d'/ℓ, r/d' and the integrals $\iint \varphi_1\varphi_2 \mathbb{K}_3(x-y)_{\lambda\gamma}\mathbb{K}_3(x-y)_{\lambda'\gamma'}$ are the same for $(\varphi_a; r, \mathsf{d}', \ell, K)$ and for $(\varphi_a^{(z)}; L^z r, L^z\mathsf{d}', L^z\ell, K-z)$.

Hence each display of (D.1) at (K,ℓ,β) is the same display at $(K-z, L^z\ell, \beta)$, with the same tolerance, both members being multiplied by L^{-2z}. The identity $\mathcal{O}^T(\mathfrak{s})(U) = L^z\mathcal{O}'^T(\mathfrak{s}^{(z)})(U^{(z)})$ holds at every configuration U of $\mathbb{L}_K$ as well, the primes now marking the fields of $\mathbb{L}_{K-z}$; in particular $\mathcal{O}(f)(U) = L^z\mathcal{O}'(f^{(z)})(U^{(z)})$ with $f^{(z)} = f(L^{-z}\cdot)$. So for every law ν on the configurations of $\mathbb{L}_K$ and its image $\nu^{(z)}$ under the dilation, $S_n^T[\nu](f_1,\dots,f_n) = L^{nz}S_n^T[\nu^{(z)}](f_1^{(z)},\dots,f_n^{(z)})$ for bounded real functions f_a with compact supports, and the joint cumulants of the fields $\mathcal{O}^T(\mathfrak{s}_a)$ of mixed sources scale in the same way; and $\nu \mapsto \nu^{(z)}$ is a bijection of $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ onto $\mathcal{G}^{\mathrm{per}}_{K-z,\beta}$.

These statements of (d) hold with every integer $0 \le z' < K$ in the place of z.

■

Proof. (a), (b) are arithmetic of the inverse couplings. Put $G = g^{-2} + \bar{B}$ and $X_{g'} = L(g'^{-2} + B_\sharp)$ for a coupling g'. One step of the run is $x_{j-1} = L(x_j + b_j)$ with $|b_j| \le B_\sharp$ (the flow bound in [7, §G.2]); as $\bar{B} = B_\sharp L/(L-1)$ sums the geometric series, z steps give $L^z(x_K - \bar{B}) + \bar{B} \le x_{K-z} \le L^z(x_K + \bar{B}) - \bar{B}$, and with $g^{-2} < x_K \le X_g$ and $X_g + \bar{B} = LG$:

$$L^z(g^{-2} - \bar{B}) + \bar{B} < x_{K-z} \le L^{z+1}G - \bar{B}.$$

(a) $g_2'^{-2} = L^{z-1}G - \bar{B} + 2B_\sharp/L \ge L^{z-1}G - \bar{B} \ge x_{\mathrm{N}}$ by the choice of z, and $g_1'^{-2} - g_2'^{-2} = (L-1)L^{z-1}G - 2B_\sharp/L > 0$; both read g and the constants alone. (b) $X_{g_1'} = L^{z+1}G - \bar{B}$ is the upper end above, and $X_{g_2'} = g_1'^{-2} + 2B_\sharp$; so the half-open intervals $I_\iota = (g_\iota'^{-2} + 2B_\sharp, X_{g_\iota'}]$, $\iota = 1, 2$, abut, are disjoint, and $I_2 \cup I_1 = (g_2'^{-2} + 2B_\sharp, X_{g_1'}]$. For the lower end, $g_2'^{-2} + 2B_\sharp \le L^{z-1}G + 2\bar{B}$ and

$$\big[L^z(g^{-2} - \bar{B}) + \bar{B}\big] - \big[L^{z-1}G + 2\bar{B}\big] = L^{z-1}\big[(L-1)g^{-2} - (L+1)\bar{B}\big] - \bar{B} \ge (L^{z-1} - 1)\bar{B} \ge 0,$$

since $\bar{B} \le \frac{13}{24}g^{-2}$ at $g \le \gamma$ ([7, Lem. G.3] gives this bound) gives $(L-1)g^{-2} \ge \frac{24}{13}(L-1)\bar{B} \ge (L+2)\bar{B}$, as $11L \ge 50$. So x_{K-z} lies in exactly one I_ι. There $x_{K-z} \le X_{g_\iota'}$ and $x_{K-z-1} = L(x_{K-z} + b_{K-z}) \ge L(x_{K-z} - B_\sharp) > L(g_\iota'^{-2} + B_\sharp) = X_{g_\iota'}$, and the run decreases: $K - z$ is the first index at which the run that starts at β is at most $X_{g_\iota'}$, which is the first passage at the coupling $g_\iota' \le \gamma_{\mathrm{N}} \le \gamma$. The run of $(K - z, \beta)$ is the same sequence cut at $K - z$, and its label is $a_{K-z}\beta = L^z\mathsf{b}$.

The margin $2B_\sharp$ in the lower ends of the I_ι is what gives $x_{K-z-1} > X_{g_\iota'}$, the second half of the first passage. (c) Put $G = g^{-2} + \bar{B}$. Then $g_1'^{-2} + \bar{B} = L^z G$ and $g_2'^{-2} + \bar{B} \le L^z G$. The radius-scaling inequality YM3_I:(1534), applied to $X = LG$ and $X' = L(g_\iota'^{-2} + \bar{B})$, gives $R(X') \le L^z R(X)$. The admissibility formula YM3_I:(1528) therefore holds with $m + z$ at g_ι' whenever it holds with m at g; its fixed C_{s} threshold is also preserved. Moreover $C_W^{g_\iota'}(s - z) \le C_W^g(s)$ for $s \ge z$, directly from the displayed definition of C_W. Thus a witness at depth $s(t)$ transfers to depth $s(t) - z$. If $s(t) < z$, then $L^{s(t)}t \ge C_W^g(s(t)) \ge 1$, whence $L^z t > 1$. (d) is the dilation; $\mathsf{b}' = L^z\mathsf{b}$, so $\mathsf{b}^{-2} = L^{2z}\mathsf{b}'^{-2}$, and $\mathbb{K}_3(L^z u) = L^{-3z}\mathbb{K}_3(u)$ makes the integrals invariant. The identity of the fields reads of the configuration only $a_{L^z p}(U^{(z)}) = a_p(U)$ and $q_{L^z b}(U^{(z)}) = q_b(U)$, and of the lattice only $x(L^z p) = L^z x(p)$, $x_{L^z b} = L^z x_b$ and $a_{K-z} = L^z a_K$; so it holds at every configuration of $\mathbb{L}_K$, and the joint cumulants under a law are multilinear in the fields.

The dilation maps the bonds, the plaquettes and the finite sets of bonds of $\mathbb{L}_K$ onto those of $\mathbb{L}_{K-z}$. The Gibbs kernel of a finite set of bonds has the exponent $-\beta\sum_p a_p$ over the plaquettes that contain a bond of the set, which reads β and the incidences of the bonds and the plaquettes alone; so the image of a Gibbs state at (a_K, β) satisfies the equations of the kernels at (a_{K-z}, β), and invariance under a subgroup of finite index of $a_K\mathbb{Z}^3$ becomes invariance under its image under $x \mapsto L^z x$, a subgroup of finite index of $a_{K-z}\mathbb{Z}^3$. The inverse dilation carries the periodic Gibbs states at $(K - z, \beta)$ back in the same way, so $\nu \mapsto \nu^{(z)}$ is a bijection of $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ onto $\mathcal{G}^{\mathrm{per}}_{K-z,\beta}$ (equation YM3_I:(1819) and the definition of periodic states following it).

The same arguments hold at every integer $0 \le z' < K$. □

Lemma 4.32 (Bond sources)**.** Let $0 < g \le \gamma_{\mathrm{N}}$, (K, β) a first-passage datum at g with $K \ge 2$, $\ell \in \mathfrak{L}(g)$, and let $\mathfrak{s} = (w, \mathsf{w})$ be a real mixed source whose barycentres $x(p)$, $w(p) \ne 0$, and midpoints x_b, $\mathsf{w}(b) \ne 0$, lie in a ball of a radius $\rho \ge a_K$. Put $\mathsf{m}_j = 12M^3\alpha_0^2 L^{K-j}|\mathfrak{s}|_\infty$ and $\mathsf{u} = \mathsf{b}^{-1}|\mathfrak{s}|_\infty$, let ς_j be as in Lem. 4.6(e) at the width ρ, and let

$$\mathsf{W}_{\mathfrak{s}}(p, q) = w(p)\delta_{pq} - \sum_b \mathsf{w}(b)\big[u_{b,\lambda_b}(p)u_{b,\lambda_b'}(q) + u_{b,\lambda_b'}(p)u_{b,\lambda_b}(q)\big],$$

a symmetric matrix that joins two plaquettes only through a bond of the support, with the coefficients $u_{b,\lambda}(p)$ of the paragraph on the fields. There are numbers C formed from the constants such that:

(1) the weight member of the source $L^{K-j}\mathfrak{s}$ at the level j is linear in $\mathfrak{s}$, vanishes on the domains that hold no corner of a plaquette p with $w(p) \neq 0$ and no site of a bond b with $\mathsf{w}(b) \neq 0$, and its norm in $\|\cdot\|_{j,\kappa}$ is at most $C\varsigma_j \mathsf{m}_j$;
(2) at the first level, $\|E^{(1)}_{\mathfrak{s}}\|_{1,\lambda_{\mathrm{out}}} \leq C\theta^{[0]}\mathsf{m}_0$;
(3) for $1 \leq n \leq K$, $A \in \mathsf{Cc}_n(r_n)$, $\tilde{U} = e^{i\eta_n A}$ and $F = \mathsf{ang}^{[n]}(A)$:

$$|\mathcal{O}^T(\mathfrak{s})(\tilde{U}) - \frac{1}{2a_K}\sum_{p,q} \mathsf{W}_{\mathfrak{s}}(p,q)\langle F_p, F_q\rangle| \leq C\mathfrak{b}_n^2\delta_n\hat{\rho}_n^3\mathsf{u},$$

and $\eta_n^4 \sum_{p,q} |\mathsf{W}_{\mathfrak{s}}(p,q)|/(a_K x_n) \leq C\hat{\rho}_n^3\mathsf{u}$;
(4) Lem. 4.14(c) and Prop. 4.16 hold with the matrices $\mathsf{W}_{\mathfrak{s}_a}$ in the place of the diagonal matrices w_a in $R_{i;\beta}$, $q_{i;\beta}$ and $g_{i;\beta}$, for two mixed sources in the frame of (6), the tower identity of (6) standing in Step 1 of the proof of Prop. 4.16;
(5) the weighted progression and the representation of the torus expectations through the never-large histories, which name no source, are formed for mixed sources at $g \leq \gamma_{\mathrm{N}}$ on every side $\ell \in \mathfrak{L}(g)$, with the weight member of (1) in the source slot of the bare arrays; the lemma on the tower identity at several sources (it precedes Prop. 4.16), for clusters in the frame of (6), the representation of Lem. 4.20 and the almost everywhere bound of Lem. 4.21 on the remainders hold for mixed sources, the moments of α^{-1} being those of Lem. 4.20(e) on that side;
(6) *the frame and the tower identity.* The hull of a mixed source $\mathfrak{s} = (w, \mathsf{w})$ is the set of the plaquettes p with $w(p) \neq 0$ and of the four plaquettes through each bond b with $\mathsf{w}(b) \neq 0$. For $2 \leq k < K$, two mixed sources, neither 0, are in the frame at k, that is they satisfy the four conditions (Jt.1)–(Jt.3) and (Jt.6) of the frame at k of Lem. 4.5(a), read with the hulls in the place of the supports and with the members of (1) in the slot, if each hull has its barycentres in a closed ball of a radius $\bar{\mathsf{d}} \geq a_K$ with $\bar{\mathsf{d}}/a_{K-k} \geq M$ and the corners of the plaquettes of the two hulls are at least $3M$ level-k spacings apart in the sup distance; this holds at $k = \mathsf{k}(\mathsf{d})$ for hulls in balls of radius $\leq \mathsf{d}$ at a distance $\geq \mathsf{d}$, $0 < \mathsf{d} \leq e^{-2\sqrt{3}}$, $s(\mathsf{d}) \leq K$, as in Lem. 4.5(b). In the frame, for $k \leq i < K$, $A' \in \mathsf{Cc}_{i+1}(r_{i+1})$, $\tilde{U} = e^{i\eta_{i+1}A'}$ and λ near 0,

$$\exp D^{\lambda}_{i+1}(\tilde{U}) = \int \exp D^{\lambda}_i(e^{i\eta_i A^{\mathrm{lw}}(B)})\, \mathrm{d}\nu_{i,A'}(B),$$

with the measure $\nu_{i,A'}$ of the step and the lowering map; and the one-source Lem. 4.10 holds for each source $\mathfrak{s}$ of such a pair, with $\frac{1}{2a_K}\sum_{p,q} \mathsf{W}_{\mathfrak{s}}(p,q)\langle F_p, F_q\rangle$ in the place of $q_{n;a}(F)$, its frame read with the hulls and its count (P1) replaced by the second bound of (3).

■

Proof. (1) $\mathcal{O}^T(\mathfrak{s}) = a_K^{-1}\sum_p w(p)a_p - a_K^{-1}\sum_b \mathsf{w}(b)q_b$, and q_b is a gauge invariant function of the variables of the four plaquettes through b; so the source is a local term of the bare level, linear in $\mathfrak{s}$, which lives at the sites of its support. On the cores the plaquette variables are within $C\alpha_0\eta_j^2$ of $\mathbf{1}$, so that $|q_b| \leq |u_{b,\lambda_b}||u_{b,\lambda'_b}| \leq C\alpha_0^2\eta_j^4$ there, as $|a_p| < \frac{1}{2}\alpha_0^2\eta_j^4$; at most $81(\rho/a_K)^3$ bonds have their midpoints in a ball of the radius $\rho \geq a_K$; with $|\mathfrak{s}|_\infty = |w|_\infty + 8|\mathsf{w}|_\infty$ the count of Lem. 4.6(e) gives the bound. (2) The one-step theorem at the height 0, with this member in the fourth place, and Schwarz's lemma, as in Lem. 4.6(e). (3) On the chain class the angle of a plaquette variable of $\tilde{U}$ is $\pm F_p$ up to $C\eta_n^2 r_n^2$, and $\mathrm{w}(e^{i\theta}) = \theta + O(|\theta|^3)$; so $u_{b,\lambda} = \sum_p u_{b,\lambda}(p)F_p$ up to $C\eta_n^2 r_n^2$, and, as $|F_p| \leq 2\eta_n^2 r_n$, $|q_b - \langle\sum_p u_{b,\lambda_b}(p)F_p, \sum_q u_{b,\lambda'_b}(q)F_q\rangle| \leq C\eta_n^4 r_n^3$, the bound that the expansion of a_p on the chain class has.

Summed against w, this gives the bond part of $\frac{1}{2a_K}\langle F, \mathsf{W}_{\mathfrak{s}}F\rangle$ with the error $Ca_K^{-1}(\rho/a_K)^3|\mathsf{w}|_\infty\eta_n^4 r_n^3$; with the plaquette part as in the proof of Lem. 4.10, $a_{K-n}x_n \geq \frac{4}{5}\mathsf{b}$ and $x_n r_n^2 \leq C\mathfrak{b}_n^2$, this is the first bound. Each $u_{b,\lambda}$ has two coefficients of modulus $\frac{1}{2}$, so $\sum_{p,q}|\mathsf{W}_{\mathfrak{s}}(p,q)| \leq \sum_p |w(p)| + 2\sum_b |\mathsf{w}(b)| \leq$

$81(\rho/a_K)^3|\mathfrak{s}|_\infty$, and $\eta_n^4 a_K^{-4} = a_{K-n}^{-4}$ gives the second. (4) In the proof of Lem. 4.14(c) the sources enter only through $R_i^\lambda = \mathsf{W}_\lambda(1-\mathcal{S}_i\mathsf{W}_\lambda)^{-1}$, $\mathsf{W}_\lambda = \sum_a \lambda_a \mathsf{W}_{\mathfrak{s}_a}$, which is symmetric as W_λ and $\mathcal{S}_i$ are, and through the Gaussian integral of a quadratic form with a symmetric matrix; the identity $1-\mathfrak{s}R_i^\lambda = (1-\mathcal{S}_{i+1}\mathsf{W}_\lambda)(1-\mathcal{S}_i\mathsf{W}_\lambda)^{-1}$ does not use that W_λ is diagonal.

The proof of Prop. 4.16 uses the sources only through the tower identity of its Step 1, which (6) gives for two mixed sources in the frame, through (I1)–(I3), through the sums $\sum_{p,q}|R_{i;\beta}(p,q)|$, which the second bound of (3) controls as (P1) does for plaquette weights, and through the lowered angles at the plaquettes of the hulls. (5) Props. [7, Prop. B.8] and [7, Prop. B.15], together with eq. 4.xxxi, give the weighted progression and the history representation for sources on the bare plaquettes, on every side of $\mathfrak{L}(g)$ with $g^{-2} \geq x_{\mathrm{J}}$ (Thm. YM3_I:D). For mixed sources the same construction is made with the weight member of (1) in the source slot of the bare arrays; the other data of the progression name no source.

The proofs among the construction results read a source only through that member, and of the member they read these properties: the entry of the objects into the settings of one step, for any number of sources, $K \geq 2$ and every side, reads that it is a local term of the bare level with the bound of (1); the never-large histories and the measures of the mixture are formed with no source; the representation reads that $\mathcal{O}^T(\mathfrak{s})$ is a bounded gauge invariant function, linear in $\mathfrak{s}$ and 0 at $\mathbf{1}$, whose member is that of (1), so that the quotient at a level t of $e^{\mathcal{O}^T(\mathfrak{s}_\lambda)}$ is $\exp D_t^\lambda$; and the tower identity reads the member in the slot of the step map, which takes any member there with the bound of (1).

By (1) a mixed source has these properties. The proofs of the lemma on the tower identity at several sources, which precedes Prop. 4.16, and of Lem. 4.20 use the sources only through the progression and the tower identity for a pair, which (6) gives for mixed sources; the almost everywhere bound of Lem. 4.21 uses them only through (1), (2), the support rule of Lem. 4.6(d) and the sizes of Lem. 4.9(f), which hold for mixed sources by (1)–(3) with the hulls in the place of the supports; on a side ℓ the moments of α^{-1} are bounded by Lem. 4.20(e). (6) The frame at k is formed by the four conditions (Jt.1)–(Jt.3) and (Jt.6) of the setting of the pair functional among the construction results, read with the hulls in the place of the supports: the proof of Lem. 4.5(a), using Def. 4.4, reads of the two sources only the four geometric properties listed there, now of the hulls, which hold all the plaquettes that the members of the sources read, and the members in the slot.

The tower identity for a pair is that among the construction results, read with the member of (1) in the slot: its proof forms the measure $\nu_{i,A'}$ and the lowering map with no source and reads the sources only through the relative exponents. Hulls at a distance $\geq \mathsf{d}$ have corners at least $(\mathsf{d}-\sqrt{2}a_K)/\sqrt{3}$ apart in the sup distance, which is at least $3M$ level-$\mathsf{k}(\mathsf{d})$ spacings by the working level, as in the proof of Lem. 4.5(b); a ball of radius d has $\mathsf{d}/a_{K-\mathsf{k}(\mathsf{d})} \geq 4M$. In Lem. 4.10 the member of $\mathfrak{s}$ is expanded on the chain class by the first bound of (3), and the families by the count of its proof with the second bound of (3). □

Lemma 4.33 (Lattice sums at two balls on every torus)**.** On every torus T_ℓ with the lattice $a_K\mathbb{Z}^3$, with the least lifts of the differences, let $4a_K \leq r$, $16r \leq \mathsf{d}'$, $8\mathsf{d}' \leq \ell$, $x_1, x_2 \in T_\ell$ with $|x_1 - x_2| = \mathsf{d}'$, $\vartheta \in (0,1]$, and let φ_1, φ_2 be real Lipschitz functions with $\operatorname{supp}\varphi_a \subset \mathbb{B}(x_a, r)$ and $\operatorname{Lip}\varphi_a \leq \|\varphi_a\|_\infty/\vartheta r$. Put $\mathsf{e}_{\mathrm{PS}} = [a_K/\vartheta r + (\mathsf{d}'/\ell)^3](r/\mathsf{d}')^6\|\varphi_1\|_\infty\|\varphi_2\|_\infty$.

(a) For all λ, μ: $\big|\sum_{\lambda(p)=\lambda,\,\lambda(q)=\mu}\varphi_1(x(p))\varphi_2(x(q))\,C_K^{\mathrm{free}}(p,q)^2 - \mathcal{K}_{\lambda\mu}(\varphi_1,\varphi_2)\big| \leq C_{\mathrm{PS}}\,\mathsf{e}_{\mathrm{PS}}$.

(b) Let $\{\lambda, \lambda', \mu\} = \{1,2,3\}$, γ a direction, $\mathsf{w}(b) = \varphi_1(x_b)\mathbf{1}_{\{\mu_b=\mu\}}$, $w_2(q) = \varphi_2(x(q))\mathbf{1}_{\{\lambda(q)=\gamma\}}$, and $C_\alpha(b,q) = \sum_p u_{b,\alpha}(p)C_K^{\mathrm{free}}(p,q)$ for a direction $\alpha \neq \mu$. Then

$$\Big|-\sum_{b,q}\mathsf{w}(b)w_2(q)\,C_\lambda(b,q)C_{\lambda'}(b,q) - \iint \varphi_1(x)\varphi_2(y)\,\mathbb{K}_3(x-y)_{\lambda\gamma}\mathbb{K}_3(x-y)_{\lambda'\gamma}\,\mathrm{d}x\,\mathrm{d}y\Big| \leq C_{\mathrm{PS}}^{\mathrm{od}}\,\mathsf{e}_{\mathrm{PS}}.$$

■

Proof. Points of the two balls are at least $\mathsf{d}' - 2r \geq \frac{7}{8}\mathsf{d}'$ and at most $\mathsf{d}' + 2r \leq \frac{9}{8}\mathsf{d}' < \frac{1}{4}\ell$ apart, so all the least lifts are taken in one ball of radius $\frac{1}{4}\ell$. (a) Steps A and B of the proof of Lem. 4.22 at these distances. In a square every sign ϵ_p stands twice, so equations YM3_I:(2245)–(2246) replace $C_K^{\mathrm{free}}(p,q)^2$ by $a_K^6[\mathbb{K}_3(u)_{\lambda\mu}]^2$ at the cost $C_{\mathrm{lat}}(2+3C_{\mathrm{lat}})a_K^6|u|^{-3}[a_K^2|u|^{-5}+\ell^{-3}]$ for each pair, and a ball of radius $r \geq a_K$ holds at most $27(r/a_K)^3$ barycentres of one orientation; with $|u| \geq \frac{7}{8}\mathsf{d}'$ and $a_K r \leq \frac{1}{4}r^2 \leq \frac{1}{1024}\mathsf{d}'^2$ this costs at most $729(\frac{8}{7})^3 C_{\mathrm{lat}}(2+3C_{\mathrm{lat}})\,\mathsf{e}_{\mathrm{PS}} \leq 1089\,C_{\mathrm{lat}}(2+3C_{\mathrm{lat}})\,\mathsf{e}_{\mathrm{PS}}$.

The Riemann sums over the cubes of side a_K about the barycentres, which for one orientation form a translate of $a_K\mathbb{Z}^3$, with $\|\mathbb{K}_3(u)\| \leq \frac{1}{2\pi}|u|^{-3}$, $\|\partial\mathbb{K}_3(u)\| \leq \frac{6}{4\pi}|u|^{-4}$ (Lem. 4.24(b)) and the Lipschitz bounds of the φ_a, give the rest of $C_{\mathrm{PS}}\mathsf{e}_{\mathrm{PS}}$. (b) $u_{b,\alpha}(p) \neq 0$ only at the two plaquettes $p_\pm$ with the barycentres $x_b \pm \frac{1}{2}a_K e_\alpha$, which lie in the plane of μ and α, and there $u_{b,\alpha}(p_\pm) = \frac{1}{2}\epsilon_{\alpha\mu\alpha'}\epsilon_{p_\pm}$, $\{\alpha,\mu,\alpha'\} = \{1,2,3\}$. By equation YM3_I:(2245), $C_\lambda(b,q)$ is $\epsilon_{\lambda\mu\lambda'}\epsilon_q a_K^3$ times the mean of $\mathbb{K}_3(\cdot)_{\lambda'\gamma}$ at the two differences $x(p_\pm) - x(q)$, and $C_{\lambda'}(b,q)$ is $\epsilon_{\lambda'\mu\lambda}\epsilon_q a_K^3$ times the like mean of $\mathbb{K}_3(\cdot)_{\lambda\gamma}$, up to the terms of the comparison.

As $\epsilon_{\lambda\mu\lambda'}\epsilon_{\lambda'\mu\lambda} = -1$ and $\epsilon_q^2 = 1$, $-C_\lambda(b,q)C_{\lambda'}(b,q)$ is $a_K^6\,\mathbb{K}_3(x_b - x(q))_{\lambda\gamma}\mathbb{K}_3(x_b - x(q))_{\lambda'\gamma}$ up to these terms and to the shifts by $\pm\frac{1}{2}a_K e_\lambda$, $\pm\frac{1}{2}a_K e_{\lambda'}$, which cost $a_K\|\partial\mathbb{K}_3\|$. The midpoints of the bonds of one direction form a translate of $a_K\mathbb{Z}^3$, as the barycentres of the plaquettes of one orientation do, and steps A and B as in (a) give $C_{\mathrm{PS}}^{\mathrm{od}}\mathsf{e}_{\mathrm{PS}}$. The numbers C_{PS} and $C_{\mathrm{PS}}^{\mathrm{od}}$ of Def. 4.1(3) are these counts. □

Proposition 4.34 (Two mixed sources on every side)**.** For $0 < g \leq \gamma_{\mathrm{N}}$ let $t_{\mathrm{N}}(g) \in (0, \frac{1}{2}e^{-2\sqrt{3}}]$ be the threshold, formed from the constants and g, below which the conditions of the induction at two sources with the base level equal to the start level hold at g and at every label in $(g^{-2}, \bar{\mathsf{b}}]$ (the conditions of step 5 of the proof of Prop. 4.19 that its step 6 reads, with the members that $g \leq \gamma_{\mathrm{N}}$ gives), and put $r_g = \frac{1}{2}\min\{r_1(\bar{\mathsf{b}}), t_{\mathrm{N}}(g)\}$. For $\ell \in \mathfrak{L}(g)$ there is a finite function $\epsilon_\sharp^{\mathrm{ev}}(g,\ell,\cdot)$ on $(0, \frac{1}{8}\ell]$, formed from the constants, g and ℓ (it is written out at the end of the proof), with $\epsilon_\sharp^{\mathrm{ev}}(g,\ell,\mathsf{d}') \leq C(g,\ell)(\log\frac{1}{\mathsf{d}'})^{19\mathfrak{p}_0}\mathsf{d}'^{1/2}$ for $\mathsf{d}' \leq \frac{7}{6}t_{\mathrm{N}}(g)$, such that the following holds. Let (K,β) be a first-passage datum at g, $\mathsf{b} = a_K\beta$, $x_1, x_2 \in T_\ell$ with $|x_1 - x_2| = \mathsf{d}'$, $0 < 16r \leq \mathsf{d}' \leq \frac{1}{8}\ell$, $s(\min\{r, r_g\}) \leq K$. For real mixed sources $\mathfrak{s}_a = (w_a, \mathsf{w}_a)$ with $\{x(p) : w_a(p) \neq 0\} \cup \{x_b : \mathsf{w}_a(b) \neq 0\} \subset \mathbb{B}(x_a, r)$,

$$\Big|\langle \mathcal{O}^T(\mathfrak{s}_1); \mathcal{O}^T(\mathfrak{s}_2)\rangle_{K,\ell,\beta} - \tfrac{3}{2}\mathsf{b}^{-2}\,\mathrm{Tr}(\mathsf{W}_{\mathfrak{s}_1}C_K^{\mathrm{free}}\mathsf{W}_{\mathfrak{s}_2}C_K^{\mathrm{free}})\Big| \leq \epsilon_\sharp^{\mathrm{ev}}(g,\ell,\mathsf{d}')\Big(\frac{r}{\mathsf{d}'}\Big)^6\mathsf{b}^{-2}|\mathfrak{s}_1|_\infty|\mathfrak{s}_2|_\infty,$$

where $\mathsf{W}_{\mathfrak{s}}(p,q) = w(p)\delta_{pq} - \sum_b \mathsf{w}(b)[u_{b,\lambda_b}(p)u_{b,\lambda_b'}(q) + u_{b,\lambda_b'}(p)u_{b,\lambda_b}(q)]$ and $u_{b,\alpha}(p)$ are the coefficients of the plaquette angles in the linear part of $u_{b,\alpha}$, given in the paragraph on the fields (it is the matrix of the lemma on bond sources, which follows Lem. 4.31). Consequently both displays of (D.1) hold at $g \leq \gamma_{\mathrm{N}}$, with $\epsilon_\sharp^{\mathrm{ev}}$ in place of ϵ^{cl}, for every K with $s(\min\{r, r_g\}) \leq K$. ■

Proof. A bond weight enters the progression through its weight member, a local term on the plaquettes of the two planes that contain the bond. Put $\rho = \min\{r, r_g\}$; the working level at ρ ($\rho \leq \frac{1}{4}e^{-2\sqrt{3}}$, $s(\rho) \leq K$) gives $a_K \leq \rho/4M \leq \frac{1}{52}\rho$. The hull of $\mathfrak{s}_a$ (clause (6) of the lemma on bond sources, which follows Lem. 4.31) has its barycentres in $\mathbb{B}(x_a, r + \frac{1}{2}a_K)$, so that for p, q in the two hulls

$$\tfrac{6}{7}\mathsf{d}' \leq |x(p) - x(q)| \leq \tfrac{8}{7}\mathsf{d}' =: \mathsf{d}^+,$$

all these points lying in a ball of radius $< \frac{1}{4}\ell$, where the least lifts are taken. Put $\mathsf{d} = \min\{\frac{6}{7}\mathsf{d}', t_{\mathrm{N}}(g)\}$, the floor of the distances that is read.

The case $r < r_g$. The width $r_T = r + \frac{1}{2}a_K \leq \frac{105}{104}r$ has $r_T \leq \mathsf{d}$, $s(r_T) \leq K$, and $r_T \leq \frac{1}{2}\cdot 1.01\,r_1(\bar{\mathsf{b}}) \leq r_1(\mathsf{b})$, as r_1 does not increase and $\mathsf{b} \leq \bar{\mathsf{b}}$; so the two sources are in the frame at $\mathsf{k}(\mathsf{d})$ (clause (6) of the lemma on bond sources). By the lemma on bond sources the calculus of Lem. 4.6, the sizes of Lem. 4.9(f), the one-source Lem. 4.10 and the induction at two sources (Prop. 4.16, base level equal to start level $\mathsf{k}(\mathsf{d})$, as in step 6 of the proof of Prop. 4.19) apply to the mixed sources, with

the explicit form $\frac{1}{2a_K}\langle F, \mathsf{W}_{\mathfrak{s}} F\rangle$ of a source, the orientation signs entering through $u_{b,\lambda}(p)$. Of (Γ1) these use only the members that $g \le \gamma_{\mathrm{N}}$ gives: the member $\delta(g^{-2}) \le 1/2C_{\mathrm{cu}}$ is used only for the kernel families (Prop. 4.13), which do not occur at two sources whose distance fixes the base level. The difference of the two members of the display is at most the sum of three members.

(i) *The remainder.* $S_2^T - \mathsf{J}_2 = \int \mathsf{Y}_{12}\,\mathrm{d}\mathbb{P} + \kappa_{\mathbb{P}}(\mathsf{Y}_1, \mathsf{Y}_2)$ (Lem. 4.20(c) for mixed sources, by clause (5) of the lemma on bond sources). The two sources form one pair at the widths (r_T, d) in the sense of Lem. 4.9(f), so the proof of Lem. 4.21, with the sizes of clauses (1)–(3) of the lemma on bond sources, gives at $\mathbb{P}$-almost every ω the bounds $|\mathsf{Y}_a| \le \Lambda_T \alpha^{-6} r_T^3 |\mathfrak{s}_a|_\infty$ and $|\mathsf{Y}_{12}| \le \Lambda_T \alpha^{-6} r_T^6 \mathsf{d}^{-9/2} |\mathfrak{s}_1|_\infty |\mathfrak{s}_2|_\infty$, with a number $\Lambda_T \ge 1$ formed from the constants. With the moments of α^{-1} on the side ℓ, $\int \alpha^{-24} d\mathbb{P} \le \bar{\mathsf{M}}_\ell := 1 + 4L^{24} M^{-3} \ell^3 e^{\mathsf{B}(X_g)\ell^3} \mathsf{C}_{\mathrm{ag}}(25, \eth)$ (Lem. 4.20(e)), which grows with ℓ, this member is at most $3\Lambda_T^2 \bar{\mathsf{M}}_\ell^2 \bar{\mathsf{b}}^2 \mathsf{d}^{3/2} (r_T/\mathsf{d})^6 \mathsf{b}^{-2} |\mathfrak{s}_1|_\infty |\mathfrak{s}_2|_\infty$.

(ii) *The induction.* The Gaussian step of Lem. 4.14(c) with $\mathsf{W}_{\mathfrak{s}}$ in place of w gives at the top the value $g_{K;12} = \frac{3}{2}\operatorname{Tr}(\mathsf{W}_{\mathfrak{s}_1} \mathcal{S}_K \mathsf{W}_{\mathfrak{s}_2} \mathcal{S}_K)$, $\mathcal{S}_K = \sum_{\mathsf{k}(\mathsf{d}) < j \le K} \upsilon_j \mathcal{C}_{(j)}$, Wick's rule for two quadratic forms of the photon field. Prop. 4.16 and Lem. 4.9(d), as in step 6 of the proof of Prop. 4.19, bound $|\mathsf{J}_2 - g_{K;12}|$ by $\mathsf{E}_{\mathrm{J}}(g, \mathsf{d})(r_T/\mathsf{d})^6 \mathsf{b}^{-2} |\mathfrak{s}_1|_\infty |\mathfrak{s}_2|_\infty$, with a function E_{J} of (g, d) alone that is at most $C(g)(\log \frac{1}{\mathsf{d}})^{19\mathfrak{p}_0} \mathsf{d}^{1/2}$ for $\mathsf{d} \le t_{\mathrm{N}}(g)$.

(iii) *The free kernel.* For p, q in the two hulls, $\mathsf{r} = |x(p) - x(q)|_\infty \in [\mathsf{d}/\sqrt{3}, \mathsf{d}^+]$ and $t = \mathsf{r}/a_{K-\mathsf{k}(\mathsf{d})} \ge \hat{\mathsf{d}}_{\mathsf{k}(\mathsf{d})}/\sqrt{3}$, the floor of the working level, at which the condition of Lem. 4.17(b) holds by the conditions that fix $t_{\mathrm{N}}(g)$ (as condition (ii) of step 5 of the proof of Prop. 4.19). So Lem. 4.17(b) at $k = \mathsf{k}(\mathsf{d})$, multiplied by r^3, gives $|\mathcal{S}_K - \mathsf{b}^{-1} C_K^{\mathrm{free}}|(p, q) \le \mathsf{b}^{-1} a_K^3 \mathsf{r}^{-3}\, \mathsf{e}_{\mathrm{fr}}(g; \mathsf{d}, \mathsf{d}^+)$, with $\mathsf{b}^{-1} < g^2$ and

$$\mathsf{e}_{\mathrm{fr}}(g; \mathsf{d}, \mathsf{d}^+) = \tfrac{5}{4} C_{\mathrm{tp}} c_L^{(2)} \bar{B} g^2\, \mathsf{d}^+ + C_{\mathrm{tp}}^{\mathrm{bg}} (\mathsf{d}^+)^3 + 2L^3 C_{\mathrm{tp}} \sup_{t \ge \hat{\mathsf{d}}_{\mathsf{k}(\mathsf{d})}/\sqrt{3}} t^3 e^{-\delta_{\mathrm{tp}} L t}.$$

Telescoping the two kernels in $\operatorname{Tr}(\mathsf{W}_{\mathfrak{s}_1} \cdot \mathsf{W}_{\mathfrak{s}_2} \cdot)$, with Lem. 4.17(a), $|C_K^{\mathrm{free}}(p, q)| \le (1 + 3C_{\mathrm{lat}}) a_K^3 |u|^{-3}$, $\mathsf{r}^{-3} \le 3^{3/2} \mathsf{d}^{-3}$ and the second bound of clause (3) of the lemma on bond sources for $\sum_{p,q} |\mathsf{W}_{\mathfrak{s}_a}(p, q)|$, this member is at most $C\, \mathsf{e}_{\mathrm{fr}}(g; \mathsf{d}, \mathsf{d}^+)(r_T/\mathsf{d})^6 \mathsf{b}^{-2} |\mathfrak{s}_1|_\infty |\mathfrak{s}_2|_\infty$: a member polynomial in d^+, besides the far term at the floor.

The members (i) and (ii), which carry the renormalisation group, read of the distances the floor d alone; the largest distance d^+ enters the one explicit member (iii), the comparison of the sum of the slices with the free kernel, through a bound polynomial in d^+.

The case $r \ge r_g$. Then $\rho = r_g$ and $s(r_g) \le K$. Tile the lift by the half-open cubes Q of side r_g and cut $\mathfrak{s}_a = \sum_Q \mathfrak{s}_a^Q$, $\mathfrak{s}_a^Q$ the restriction of $\mathfrak{s}_a$ to the plaquettes and the bonds whose barycentres or midpoints lie in Q; $|\mathfrak{s}_a^Q|_\infty \le |\mathfrak{s}_a|_\infty$, and at most $(2r/r_g + 2)^3 \le 64(r/r_g)^3$ pieces of each source are not 0. The hull of a piece is a part of the hull of $\mathfrak{s}_a$ and has its barycentres within $\frac{\sqrt{3}}{2} r_g + \frac{1}{2} a_K \le r_g$ of the centre of its cube; so every pair of pieces, neither 0, is in the data of the first case at the widths (r_g, d), with $r_g \le \mathsf{d}$, $s(r_g) \le K$, $r_g \le r_1(\mathsf{b})$, the centres of the cubes in the place of x_1, x_2, and the same bounds $\frac{6}{7}\mathsf{d}' \le |x(p) - x(q)| \le \mathsf{d}^+$ on the hulls. Both members of the display are bilinear in $(\mathfrak{s}_1, \mathfrak{s}_2)$, as $\mathcal{O}^T$ and $\mathsf{W}_{\mathfrak{s}}$ are linear in $\mathfrak{s}$, and the sum over the pairs of pieces costs $64^2 (r/r_g)^6 (r_g/\mathsf{d})^6 = 4096 (r/\mathsf{d})^6$.

Together. $(\mathsf{d}'/\mathsf{d})^6 = (\frac{7}{6})^6 \max\{1, 6\mathsf{d}'/7t_{\mathrm{N}}(g)\}^6$ and $4096(\frac{7}{6})^6(\frac{105}{104})^6 \le 11060$ give the display with

$$\epsilon_\sharp^{\mathrm{ev}}(g, \ell, \mathsf{d}') = 11060 \max\Big\{1, \frac{6\mathsf{d}'}{7t_{\mathrm{N}}(g)}\Big\}^6 \Big[\mathsf{E}_{\mathrm{J}}(g, \mathsf{d}) + C\, \mathsf{e}_{\mathrm{fr}}(g; \mathsf{d}, \tfrac{8}{7}\mathsf{d}') + 3\Lambda_T^2 \bar{\mathsf{M}}_\ell^2 \bar{\mathsf{b}}^2 \mathsf{d}^{3/2}\Big], \qquad \mathsf{d} = \min\{\tfrac{6}{7}\mathsf{d}', t_{\mathrm{N}}(g)\},$$

a finite function on $(0, \frac{1}{8}\ell]$ formed from the constants, g and ℓ. For $\mathsf{d}' \le \frac{7}{6} t_{\mathrm{N}}(g)$ one has $\mathsf{d} = \frac{6}{7}\mathsf{d}'$ and $\mathsf{d}^+ = \frac{4}{3}\mathsf{d}$, and each of the three members is at most $C(g, \ell)(\log \frac{1}{\mathsf{d}'})^{19\mathfrak{p}_0} \mathsf{d}'^{1/2}$: the first by step 6 of the proof of Prop. 4.19, the second as the member br of its step 4 under the conditions of its step 5, the third as $\mathsf{d}^{3/2} \le \mathsf{d}^{1/2}$.

The plane and the bond forms. For Lipschitz bumps the lattice form is turned into the integrals of (D.1) by Lem. 4.33, whose data hold: $4a_K \le r$, $16r \le \mathsf{d}'$, $8\mathsf{d}' \le \ell$. For the plane fields take $\mathfrak{s}_a = (\mathbf{1}_{\{\lambda(\cdot)=\lambda_a\}}\varphi_a \circ x, 0)$, $\lambda_1 = \lambda$, $\lambda_2 = \mu$, so that $|\mathfrak{s}_a|_\infty \le \|\varphi_a\|_\infty$ and the main term is $\frac{3}{2}\mathsf{b}^{-2}\sum_{p,q} w_1(p)w_2(q)C_K^{\mathrm{free}}(p,q)^2$; Lem. 4.33(a) times $\frac{3}{2}\mathsf{b}^{-2}$ gives the second display of (D.1). For the bond form take $\mathfrak{s}_1 = (0, \mathsf{w})$, $\mathsf{w}(b) = \varphi_1(x_b)\mathbf{1}_{\{\mu_b=\mu\}}$, so that $Y[\mathsf{w}] = \mathcal{O}^{\mathrm{od}}_{\lambda\lambda'}(\varphi_1)$ and $|\mathfrak{s}_1|_\infty \le 8\|\varphi_1\|_\infty$, and $\mathfrak{s}_2 = (\mathbf{1}_{\{\lambda(\cdot)=\gamma\}}\varphi_2 \circ x, 0)$.

For the symmetric matrix $C = C_K^{\mathrm{free}}$, $\mathrm{Tr}(\mathsf{W}_{\mathfrak{s}_1}C\mathsf{W}_{\mathfrak{s}_2}C) = -2\sum_{b,q}\mathsf{w}(b)w_2(q)C_\lambda(b,q)C_{\lambda'}(b,q)$ with $C_\alpha(b,q) = \sum_p u_{b,\alpha}(p)C(p,q)$, so the main term is $-3\mathsf{b}^{-2}\sum_{b,q}\mathsf{w}(b)w_2(q)C_\lambda(b,q)C_{\lambda'}(b,q)$, and Lem. 4.33(b) times $3\mathsf{b}^{-2}$ gives the first display, with $8\epsilon_\sharp^{\mathrm{ev}}$ in the place of $8\epsilon^{\mathrm{cl}}$. □

Pair minus tensor field.

Lemma 4.35 (A regular gauge on a cube; the curvature is Hölder continuous)**.** Fix $\beta_{\mathrm{H}} \in (0,1)$. There are numbers $\epsilon_{\mathrm{G}} \in (0,1]$ and $C_{\mathrm{G}}^\sharp, C_{\mathrm{H}} \ge 1$, formed from the constants and β_{H}, such that the following holds for $1 \le n \le K$; lengths are in level-n units, $\xi = \eta_n$, and $Q_s(x_0)$ is the set of the bare sites at the sup distance at most s from a bare site x_0.

(a) Let $12a_{K-n} \le \ell$ and let A be a real $\mathfrak{su}(2)$-valued 1-form on the bonds of $Q_2(x_0)$ such that $|A|$, $|\nabla^\xi A|$ and the current of $e^{i\xi A}$ are at most $\mathsf{r}_\circ \le \frac{1}{120}$ there. Then the linear angles $F_p = \xi(\mathrm{d}A)(p)$ satisfy $|F_p| \le 2\xi^2\mathsf{r}_\circ$ and $|F_{p_{\mu\nu}(x)} - F_{p_{\mu\nu}(x')}| \le C_{\mathrm{H}}\,\xi^2\mathsf{r}_\circ|x-x'|^{\beta_{\mathrm{H}}}$ for the plaquettes $p_{\mu\nu}(x)$, $p_{\mu\nu}(x')$ of one orientation with $x, x' \in Q_{1/2}(x_0)$.

(b) Let $W \in \mathcal{V}_n^\chi$, $U = U_n(W)$, and let $\varrho \ge 1$ with $(8\varrho+2)a_{K-n} < \ell$ and $(4\varrho)^2(1+4\varrho)\,4B_3\varepsilon_n \le \epsilon_{\mathrm{G}}$. Then there are $\mathsf{g}\colon Q_{4\varrho}(x_0) \to SU(2)$ and a real 1-form A on the bonds of $Q_{2\varrho}(x_0)$ with $\mathsf{g}(b_-)U(b)\mathsf{g}(b_+)^{-1} = e^{i\xi A(b)}$ there and, on $Q_\varrho(x_0)$, $|A| \le 80C_{\mathrm{G}}^\sharp B_3\varrho^2\varepsilon_n$ and $|\nabla^\xi A| \le 80C_{\mathrm{G}}^\sharp B_3\varrho\,\varepsilon_n$; the plaquette variables and the current of $e^{i\xi A}$ are those of U conjugated by g, and the current has a modulus $< 2B_3\varepsilon_n$.

None of these numbers depends on n, K, ℓ, the spacing, x_0 or the configuration. ■

Proof. (a) The 2-form $f_{\mu\nu} = \xi^{-2}F_{p_{\mu\nu}} = \nabla^\xi_\mu A_\nu - \nabla^\xi_\nu A_\mu$ is antisymmetric and closed, $\mathrm{d}f = 0$, and its lattice codifferential differs from the current of $e^{i\xi A}$ by at most $40|A||\nabla^\xi A| + 32|A|^3$ in the sup bounds, by the expansion of the lattice current; so $|f| \le 2\mathsf{r}_\circ$ and $|\mathrm{d}^*f| \le 2\mathsf{r}_\circ$. For an antisymmetric lattice 2-form f on a cube $Q_R(x_0)$, $R \ge 8\xi$, with $|f| \le m_0$ and $|\mathrm{d}f|, |\mathrm{d}^*f| \le m_1$, the lattice Hodge identity $\Delta(\chi f) = \mathrm{d}^*\mathrm{d}(\chi f) + \mathrm{d}\mathrm{d}^*(\chi f)$, for a cut-off χ equal to 1 on $Q_{R/2}(x_0)$ and written as $\chi f = G_\xi * \Delta(\chi f)$ with the lattice Green function G_ξ of $\xi\mathbb{Z}^3$, and the bounds $|\nabla G_\xi(z)| \le C(|z|+\xi)^{-2}$, $|\nabla^2 G_\xi(z)| \le C(|z|+\xi)^{-3}$, which are uniform in ξ, give $|f(x) - f(x')| \le C(\beta_{\mathrm{H}})(m_0 + Rm_1)(|x-x'|/R)^{\beta_{\mathrm{H}}}$ for $x, x' \in Q_{R/2}(x_0)$. At $R = 1$, $m_0 = m_1 = 2\mathsf{r}_\circ$ this is (a). (b) Put $\alpha = 4B_3\varepsilon_n$ and lift the sites of $Q_{4\varrho}(x_0)$ to the box $\Omega = x_0 + \xi\{-M, \dots, M\}^3$, $M = \lfloor 4\varrho/\xi \rfloor$, of side $s = (2M+1)\xi \le 9\varrho$. The room hypothesis makes the lift injective. By Lem. 4.20(d), U is regular of level n at radius $\alpha_0 = 2B_3\varepsilon_n$. The current-to-matrix-divergence calculation in Lem. YM3_I:9.4 gives $|D_U^{\mathsf{T}}\partial U| \le \xi^{-1}(\alpha_0\xi^2)^2 + \xi^2\alpha_0 < \alpha\xi^2$ at every full box bond; its plaquette bound is also $< \alpha\xi^2$. Choose $\epsilon_{\mathrm{G}} \le \min\{1, c_\square\}$ and $C_{\mathrm{G}}^\sharp \ge \max\{1, B_\square\}$, where $c_\square, B_\square$ are the box-gauge constants in the proof of Thm. YM3_I:F. The stated smallness implies $s\alpha \le c_\square$.

That box estimate gives a real gauge and potential on Ω with $|A| < B_\square s\alpha \le 36B_\square B_3\varrho\varepsilon_n$, and the same bound for first differences at base sites of depth at least $3/2$. Every positive bond with an endpoint in Q_ϱ has base depth at least $3\varrho - 2\xi > 3/2$. Restricting the potential to $Q_{2\varrho}$ gives both displayed bounds, since $\varrho \ge 1$. Plaquette and current moduli are preserved by the real gauge. □

In Lems. 4.36–4.41 the data are those of Thm. K; one progression holds the sources w_1, w_2, w_T, w_3, so that by multilinearity the objects of the two lists (w_1, w_2, w_3) and (w_T, w_3) are obtained from it. We write $\Delta_n = \mathsf{c}G_{n;12} - G_{n;T}$ and $\Gamma_n = \mathsf{c}G_{n;123} - G_{n;T3}$, functions of the configurations of the level n whose pairs lie in the cores, and $\mathrm{nat}_n^{\mathcal{W}} = \varepsilon_n^2 a_{K-n}^{-4}$.

Lemma 4.36 (Pair minus tensor field on all small-field backgrounds)**.** Let $r \le r_\alpha(g,\mathsf{b})$; let n_1 and n_0 be the least levels with $a_{K-n_1} \ge (\frac{r}{2})^{3/4}$ and $a_{K-n_0} \ge r^{1/2}$, $N_0 = n_0 + m_1$, where $m_1 \ge 1$ is the least integer with $L^{m_1/2} \ge C_7 A_0/(0.85\,\mathsf{c}_\mathfrak{h} A_1)$, $C_7 = 2B_3 + 69\cdot 1280\, C_{\mathrm{G}}^\sharp B_3$, an integer formed from the constants and β_{H} (through $C_{\mathrm{G}}^\sharp$ of the lemma on a regular gauge on a cube, which precedes this lemma), $\epsilon = r^{\beta_{\mathrm{H}}/36} + (a_K/r)^{1/9}$, and let $x_\sharp \ge e^{2\mathfrak{p}_0}$ be the least number such that every $x \ge x_\sharp$ has $17408 B_3 \varepsilon(x) \le \epsilon_{\mathrm{G}}$ and $1280 C_{\mathrm{G}}^\sharp B_3 \varepsilon(x) \le \frac{1}{100}$, with the numbers ϵ_{G}, $C_{\mathrm{G}}^\sharp$ of the lemma on a regular gauge on a cube.

Let $L^{N_0-n_1} \ge 2M$, and let $\epsilon_i^{\mathrm{e}} = C_\omega[1.45\gamma_{i-1} + \frac{1}{2}\mathsf{c}^\flat\theta^{[0]}] \le 1$ for $\mathsf{k}(\frac{r}{2}) \le i < n_1$; this number bounds the single-source arrays of the near pair at the level i, $\mathfrak{n}[\dot{\mathfrak{e}}_{a,i}] \le 2\epsilon_i^{\mathrm{e}}\varsigma_i \mathsf{m}_{i,a}$ ($a = 1, 2$, ς_i formed at the width $\frac{r}{8}$), as in Lem. 4.9(a). For $N_0 \le n^\sharp < K$ with $x_{n^\sharp} \ge x_\sharp$ there are real numbers c_n, $N_0 \le n \le n^\sharp$, such that for m_n-almost every $W \in \mathcal{V}_n^\chi$

$$|\Delta_n(U_n(W)) - \mathsf{c}_n| \le \varepsilon^{\mathcal{W}}\,\mathrm{nat}_n^{\mathcal{W}}, \qquad |\mathsf{c}_n - \mathsf{c}_{n'}| \le 2.01\,\tilde{\varepsilon}\,\mathrm{nat}_n^{\mathcal{W}} \quad (n \le n' \le n^\sharp),$$

with $\varepsilon^{\mathcal{W}} \le \tilde{\varepsilon} \le C(g,\mathsf{b})\,\epsilon$. ■

Proof. (i) We use the regular-field estimate of [7, Lem. B.22] and the insertion identity of Lem. 2.2 on the small-field backgrounds. The first is the evaluation of a local term at a configuration that is regular of a level, with the factor ε_n^2: for $1 \le i \le n \le K$, $W \in \mathcal{V}_n^\chi$ and a pointed family f of level i that vanishes unless one of N given cubes of π_i is a cube of the domain, and on the domains with $d_i(X) < D$,

$$\sum_{(X,z)} |\mathsf{f}(X, U_n(W), z) - \mathsf{f}(X, \mathbf{1}, z)| \le C\,\mathsf{N}\,\|\mathsf{f}\|_{i,\lambda_{\mathrm{out}}}\,\varepsilon_n^2\,L^{-4(n-i)}e^{-D/2}.$$

It is [7, eq. 230] at the radius $\varepsilon = 2B_3\varepsilon_n$ at which $U_n(W)$ is regular of the level n (Lem. 4.20(d), $4M\varepsilon \le c_{\mathrm{A}}\alpha_0$), which gives for one domain of size d the factor $(\varepsilon/\alpha_0)^2 L^{-4(n-i)} M^2(1+d)^2$, together with the tails of the proof of Lem. 4.8(a): the domains beyond the depth $d_* = \min\{c_{\mathrm{A}}\alpha_0 L^{2(n-i)}/M\varepsilon, (L^{m+K-i}-4)/2M\} - 1$ have $e^{-[d_*]_+/2} \le C\varepsilon_n^2 L^{-4(n-i)}$, by $e^{-s/2} \le 8s^{-2}$, by $L^{m+K-n} \ge 13My_n$ on the sides $\ell = 2L^m$ of $\mathfrak{L}(g)$ (from $L^m \ge LMy_K$ and Lem. 4.3(a)) and by $x_n^{-1} \le \varepsilon_n^2/A_0^2$; no ceiling beyond those of the data is read. (Lem. 4.8(b) is the same count without the factor ε_n^2.) The second is the averaging identity of one step: the quotient Qt_{j+1} of one step, a positive normalised mean that takes bounded Borel functions of the block fields of the level j to functions of those of the level $j+1$, takes the action $S[\mathfrak{g}](U_j(\cdot))$ of a bare array $\mathfrak{g}$ of the level j to $(S[\mathfrak{g}] + \mathsf{Sum}[\mathsf{DSt}^{[j]}(\mathfrak{f}_{0,j})[\mathfrak{g}]])(U_{j+1}(\cdot))$, m_{j+1}-almost everywhere; so the seminorm $\inf_c \operatorname{ess\,sup}_{\mathsf{m}_n} |\Phi(U_n(W)) - c|$ of such actions does not grow from one level to the next. (ii) *The split.* For $n_0 \le j < K$ let $\mathfrak{x}_j$ be the bare array whose families are c times the kernel families $\mathsf{K}^{(i)}$ of the near pair at the levels $\mathsf{k}(\frac{r}{2}) < i \le n_1$ and whose slot holds minus the weight member of w_T; its action $X^{\mathrm{fz}} = S[\mathfrak{x}_j]$, the frozen part, does not depend on j.

Let $\mathfrak{r}_j$ hold all the other families of Δ of the levels $\le n_0$. Then $\Delta_n = \Delta_n(\mathbf{1}) + X_n + S_n$, where X_n adds to X^{fz} the families that the first derivatives of the steps above n_0 make from $\mathfrak{x}$ alone, and S_n is the action of $\mathfrak{r}_n$ together with the families $\mathsf{s}^{(i)}$, $n_0 < i \le n$, that the first derivatives of the steps make from $\mathfrak{r}$ and from the $\mathsf{s}^{(i')}$ of the lower levels, and with the second derivatives $\mathsf{c}\,\mathsf{B}_{i-1}(\mathfrak{f}_{\{1\},i-1}, \mathfrak{f}_{\{2\},i-1})$ of the steps (Lem. 4.7). The evaluation of (i), the lags of Lem. 4.6(c) and the sizes of Lem. 4.9 give $|S_n| \le C\varepsilon_S\,\mathrm{nat}_n^{\mathcal{W}}$ at every small-field background of the level n, with $\varepsilon_S = 2\varepsilon_{\mathfrak{r}} + C(r/a_{K-n_0})^3$ and the number $\varepsilon_{\mathfrak{r}}$ of step (i) of the proof of the lemma on the chain classes, which follows this lemma; that step uses the same split on the chain classes. (iii) The explicit parts cancel.

With $D(n) = 3n \otimes n - 1$, the pair, multiplied by c, carries the explicit form with the kernel $\mathsf{b}^{-1}C_K^{\mathrm{free}} \approx \mathsf{b}^{-1}a_K^3\mathbb{K}_3(re_\lambda) = \mathsf{b}^{-1}a_K^3(4\pi r^3)^{-1}D(e_\lambda)$, while the tensor source carries $\frac{1}{2a_K}\sum_p w_T(p)|F_p|^2$ with $\delta_{\lambda(p)\lambda} - \frac{1}{3} = \frac{1}{3}D(e_\lambda)_{\lambda(p)\lambda(p)}$; the moment condition $\sum_k \mathsf{a}_k D(n_k) = \mathsf{a}_T D(e_\lambda)$, here with one pair of direction e_λ and $\mathsf{a}_1 = \mathsf{a}_T = 1$, makes them equal. (iv) *The frozen part on the*

backgrounds. Let $N_0 \le n \le n^\sharp$, $W \in \mathcal{V}_n^\chi$, $U = U_n(W)$, x_0 a bare site with $|x_0 - c|_\infty \le a_K$, and let lengths be in level-n units, $\xi = \eta_n$.

The gauge. As $x_n \ge x_\sharp$ and $34 a_{K-n} < \ell$, clause (b) of the lemma on a regular gauge on a cube, which precedes this lemma, applies at $\varrho = 4$ and gives g on $Q_{16}(x_0)$ and a potential A on the bonds of $Q_8(x_0)$ with $\mathsf{g}(b_-)U(b)\mathsf{g}(b_+)^{-1} = e^{i\xi A(b)}$, and on $Q_4(x_0)$ the bounds $|A|, |\nabla^\xi A| \le C_6\varepsilon_n \le \frac{1}{100}$, $C_6 = 1280 C_{\mathrm{G}}^\sharp B_3$, and a current of modulus $< 2B_3\varepsilon_n$. *The cut-off.* Let $h = \prod_\mu \chi(|x_\mu - x_{0,\mu}|)$ with $\chi = 1$ on $[0,3]$, $\chi = 0$ on $[\frac{15}{4}, \infty)$, $|\chi'| \le 2$ and $\operatorname{Lip}\chi' \le 11$; put $\hat A = hA$ on the bonds of $Q_4(x_0)$, $\hat A = 0$ on all other bonds of T_ℓ, and $\hat U = e^{i\xi\hat A}$.

Then $\hat U$ is the transform of U by g on every bond with an end in $Q_3(x_0)$, and the product rules for hA with the expansion of the lattice current give

$$\max\{\sup|\hat A|,\ \sup|\nabla^\xi \hat A|,\ \sup|J(\hat U)|\} \le (2B_3 + 69\,C_6)\,\varepsilon_n = C_7\varepsilon_n \qquad \text{on all of } T_\ell.$$

The shift of the level. Seen from the level $n^\flat = n - m_1 \ge n_0$, the same configuration has the potential $\hat A^\flat = L^{-m_1}\hat A$, with $|\hat A^\flat| \le L^{-m_1}C_7\varepsilon_n$, differences $\le L^{-2m_1}C_7\varepsilon_n$ and current $\le L^{-3m_1}C_7\varepsilon_n$ in level-$n^\flat$ units; with $\varepsilon_n = (A_0/A_1)\delta_n$, $\delta_n \le (\frac{5}{4})^{1/2}L^{m_1/2}\delta_{n^\flat}$ (Lem. 4.3(a)) and the choice of m_1,

$$L^{-m_1}C_7\varepsilon_n \le (\tfrac{5}{4})^{1/2}\cdot 0.85\,\mathsf{c}_\mathfrak{h}\delta_{n^\flat} < \tfrac{11}{10}\mathsf{c}_\mathfrak{h}\delta_{n^\flat} \le r_{n^\flat}.$$

This is the norm clause of the chain class of radius $r_{n^\flat}$, and it is all that the steps below read of the field. *The pair.* Step (ii) of the proof of the lemma on the chain classes is read at the level $n^\flat$ with U in the place of $\tilde U$, with two changes. (1) The gauge $u = e^{iX}$ of Lem. 4.12 at the pair is formed for $\hat A^\flat$ on the bonds within two level-$n^\flat$ units of x_0, where $\hat U$ is the transform of U; it gives the angles F', Hölder continuous as in that step, through the estimate of the proof of clause (a) of the lemma on a regular gauge on a cube on $Q_1(x_0)$ at $m_0 = 2.08\,r_{n^\flat}$ and $m_1 = 3.7\,r_{n^\flat}$, and the plaquette term of the weight member of w_T in this gauge, as $a_p(U) = a_p(\hat U)$ at the plaquettes of $\operatorname{supp} w_T$, which lie in $Q_{1/2}(x_0)$. (2) For the kernel families of the levels $\mathsf{k}(\frac{r}{2}) \le i < n_1$, $m = n^\flat - i - 1$: Lem. 4.12 for the real pair of the level $i+1$ that $\hat A^\flat$ and its current give on all of T_ℓ (the potential multiplied by L^{-m}, the current by L^{-3m}), on the box $N_{\rho'}$ about x_0, $\rho' = \min\{\mathsf{h}_i, 3L^{n-i-1} - 1\}$, with the same u, gives a real point v' of the chart with $\|v'\| \le \mathsf{h}_i L^{-2m} r_{n^\flat}(6 + 19 r_{n^\flat}) \le \delta_i/2\Lambda_i$, by $(\alpha 2)$ at $n^\flat \ge n_0$, the pair having $|L^{-m}\hat A^\flat| \le L^{-m} r_{n^\flat}$ and differences and current $\le L^{-2m} r_{n^\flat}$.

A domain X of the level $i+1$ with all its sites within $\rho' - 1$ of x_0 lies in $Q_3(x_0)$, where $\hat U$ is the transform of U, so that $\mathsf{K}^{(i+1)}(X, U, z) = \mathsf{K}^{(i+1)}(X, P[v']|_X, z)$ by the gauge invariance and the locality of a local term; the other domains have $d_{i+1}(X) \ge 2L^{n-i-1}/M - 2$, or the depth of step 5 of the proof of Prop. 4.13 if $\rho' = \mathsf{h}_i$, and are paid by the norm (Lem. 4.8(b) at U and at $P[v']$). Steps 2–4 of the proof of Prop. 4.13 (holomorphy, the real ball, two constants), which read of the field only the real point v' and its norm, give at v' the kernel step with the angles F'. As $L^{n-i-1} \ge L^{N_0 - n_1} \ge 2M$ and $\epsilon_i^{\mathrm{e}} \le 1$, the domains that leave the box cost at most $C\mathsf{b}\,r\,\mathrm{nat}_n^{\mathcal{W}}$ together.

With $r_{n^\flat}^2 a_{K-n^\flat}^{-4} \le C\,\mathrm{nat}_n^{\mathcal{W}}$ (Lem. 4.3(a), $a_{K-n^\flat} = L^{-m_1}a_{K-n}$) this bounds the frozen part: $|X^{\mathrm{fz}}(U)| \le \varepsilon_X^{\mathcal{W}}\mathrm{nat}_n^{\mathcal{W}}$, with $\varepsilon_X^{\mathcal{W}} \le C[\varepsilon_X + \mathsf{b}\,r]$ and the number ε_X of step (ii) of the proof of the lemma on the chain classes. (v) Carried upward with (i) and the lags of Lem. 4.6(c), $\|X_n\| \le [\varepsilon_X^{\mathcal{W}} + C\gamma_{N_0-1} + C\gamma_n^{8/9}(\varepsilon_X^{\mathcal{W}})^{1/9}]\mathrm{nat}_n^{\mathcal{W}}$; the exponents $\frac{8}{9}$, $\frac{1}{9}$, and $\frac{1}{2}$ below, come from interpolations between the oscillation bound and the bound of the sizes, and the choice of the levels, $\mathsf{k}(\frac{r}{2}) < n_1 < n_0$ with $a_{K-n_1} \ge (\frac{r}{2})^{3/4}$, costs the exponent $\frac{1}{4}$.

Together $\varepsilon^{\mathcal{W}} \le \tilde\varepsilon \le C(g, \mathsf{b})[r^{\beta_{\mathrm{H}}/36} + (a_K/r)^{1/9}]$ for $r \le r_\alpha(g, \mathsf{b})$, the levels being ordered by $(\alpha 1)$–$(\alpha 5)$ of Def. 4.1(5); the constants c_n of two levels differ by at most these oscillations plus the part of the steps between that the sizes of the sources at those levels bound, which is of the same order. □

Lemma 4.37 (Pair minus tensor field on the chain classes)**.** Let $r \le r_\alpha(g,\mathsf{b})$, let n_1 and n_0 be the levels of Lem. 4.36, and $\epsilon = r^{\beta_H/36} + (a_K/r)^{1/9}$. There are numbers c'_n, $n_0 \le n \le K$, and a number $\varepsilon_\alpha \le C(g,\mathsf{b})\,\epsilon$ such that

$$\sup_{A\in \mathsf{Cc}_n(r_n)} |\Delta_n(e^{i\eta_n A}) - \mathsf{c}'_n| \le \varepsilon_\alpha\, r_n^2 a_{K-n}^{-4} \qquad (n_0 \le n \le K).$$

■

Proof. (i) *The split* is that of step (ii) of the proof of Lem. 4.36. For $n_0 \le j < K$, $\mathfrak{x}_j$ is the array whose families are c times the kernel families $\mathsf{K}^{(i)}$ of the near pair at the levels $\mathsf{k}(\frac{r}{2}) < i \le n_1$ and whose slot holds minus the weight member of w_T; its action X^{fz}, the frozen part, does not depend on j. $\mathfrak{r}_j$ holds all the other families of Δ of the levels $\le n_0$. Lem. 4.6 and the sizes of Lem. 4.9 give $\mathfrak{n}[\mathfrak{x}_j] \le C a_{K-j}^{-4}$ and $\mathfrak{n}[\mathfrak{r}_j] \le \varepsilon_{\mathfrak{r}} a_{K-j}^{-4}$, where $\varepsilon_{\mathfrak{r}}$ is a constant times $\gamma_{n_0} + s(\frac{r}{2})\theta^{[0]} + (r/a_{K-n_1})^3$ and terms exponentially small in $\hat r_{\mathsf{k}(r/2)}$ and in r/a_{K-1}, so that $\varepsilon_{\mathfrak{r}} \le C(g,\mathsf{b})[r^{1/4} + (a_K/r)^{1/4}]$.

On the chain classes $\Delta_n = \Delta_n(\mathbf{1}) + X_n + S_n$, where X_n adds to X^{fz} the families that the first derivatives of the steps above n_0 make from $\mathfrak{x}$ alone, and S_n collects the rest: the action of $\mathfrak{r}_n$, the families that the first derivatives make from $\mathfrak{r}$ and from them, and the second derivatives of the steps; Lem. 4.8(a), in the frame at $\mathsf{k}(\frac{r}{2})$ in which Lem. 4.5(a) puts the near pair, a test pair of the scale $\frac{r}{2}$, under the conditions $3 \le s(\frac{r}{2}) \le K$ and $\mathsf{Lm}_{s(r/2)} \le L^{s(r/2)-3}$ of the data (and $\mathsf{k}(\frac{r}{2}) < n_1 < n_0 \le n < K$ by (α1)), and the lags of Lem. 4.6(c) give $|S_n| \le C\varepsilon_S r_n^2 a_{K-n}^{-4}$ with $\varepsilon_S = 2\varepsilon_{\mathfrak{r}} + C(r/a_{K-n_0})^3$. (ii) *The frozen part.* Let $n_0 \le n \le K$, $A \in \mathsf{Cc}_n(r_n)$, x_0 a bare site with $|x_0 - c|_\infty \le a_K$, let $u = e^{iX}$, X linear, be the gauge of Lem. 4.12 at the pair, about x_0, A' the potential of $\tilde U = e^{i\eta_n A}$ in this gauge and $F' = \eta_n \mathrm{d}A'$ its angles. Near the pair $|A'|$, $|\nabla A'|$ and the current are at most a constant times r_n (Lem. 4.12, at $\alpha = \gamma_\circ = r_n$, in the units of the level n): on N_2, $|A'| \le 6r_n + 18.7 r_n^2 \le 6.2\, r_n$ and $|\nabla^{\eta_n} A'| \le \frac{100}{99}(r_n + 3r_n^2) \le 1.04\, r_n$, and the current, conjugated by the gauge, is at most r_n. So the 2-form $f = \eta_n^{-2} F'$ is closed, $|f| \le 2.08\, r_n$, and, as its codifferential differs from the current by at most $40|A'||\nabla^{\eta_n} A'| + 32|A'|^3$, $|\mathrm{d}^* f| \le 3.7\, r_n$ on $Q_1(x_0)$, since $2r_n \le \varrho_{\mathrm{ch}}$ (Lem. 4.3(b)) and $\varrho_{\mathrm{ch}} \le \frac{1}{60}$ by equation YM3_I:(2285).

The lemma on a regular gauge on a cube, which precedes Lem. 4.36, is read through the estimate of the proof of its clause (a) on $Q_1(x_0)$, at $m_0 = 2.08\, r_n$ and $m_1 = 3.7\, r_n$: it makes F' Hölder continuous with the exponent β_H on $Q_{1/2}(x_0)$, on the scale of the pair. Prop. 4.13, in the frame of step (i), at the levels $\mathsf{k}(\frac{r}{2}) \le i < n_1$, where ($\alpha$2) of Def. 4.1(5) gives its condition in (ii), taken before the last replacement of step 5 of its proof, so that $\mathcal{Q}_i$ stands at F' and the last term of (ii) does not occur, replaces $\mathsf{c}\sum_i(\mathsf{Sum}[\mathsf{K}^{(i+1)}](\tilde U) - \mathsf{Sum}[\mathsf{K}^{(i+1)}](\mathbf{1}))$ by $\mathsf{c}\, q_{n_1}(F')$ with an error at most $C\sum_i \hat r_i^3 \mathsf{b}_i^2(\delta_i + \epsilon_i^{\mathrm{cu}})\Lambda_i^2 \mathsf{h}_i^2\, r_n^2 a_{K-n}^{-4} \le C(g,\mathsf{b})\, r^{1/4} r_n^2 a_{K-n}^{-4}$; the weight member of w_T is expanded in the gauge u at the pair.

With $\xi = \eta_n$, the plaquette angles of $\tilde U$ in this gauge are $F'_p + \Psi_p$ with the commutator part $|\Psi_p| \le 24\xi^2|A'|^2$; Lem. 4.12 at $\alpha = \gamma_\circ = r_n$ gives $|A'(b)| \le 3sr_n + (9s + 8\xi)r_n^2 \le 4sr_n + 8\xi r_n^2$ at the bonds with base point within s of x_0, and $|F'_p| \le 2.08\,\xi^2 r_n$; as $a_p = \frac{1}{2}|\theta|^2 + O(|\theta|^4)$ in the plaquette angle θ,

$$|a_p(\tilde U) - \tfrac{1}{2}|F'_p|^2| \le |F'_p||\Psi_p| + \tfrac{1}{2}|\Psi_p|^2 + C|F'_p + \Psi_p|^4 \le C\xi^4 r_n^3 (s+\xi)^2$$

at the plaquettes within $s \le \frac{1}{2}$ of x_0. The plaquettes of $\operatorname{supp} w_T$ lie within $s \le r/a_{K-n}$ of x_0, so $s + \xi \le 2r/a_{K-n_0}$; summed against w_T, with $\sum_p |w_T(p)| \le C a_K^{-3}$, this replaces the tensor source by $\frac{1}{2a_K}\sum_p w_T(p)|F'_p|^2$ up to $C(r/a_{K-n_0})^2 r_n^2 a_{K-n}^{-4} \le C\, r\, r_n^2 a_{K-n}^{-4}$, as $a_{K-n_0} \ge r^{1/2}$: in the gauge at the pair the angles are small at the scale $s + \xi$ of the tensor source. By the Hölder continuity and the comparison of $\mathcal{S}_{n_1}$ with $\mathsf{b}^{-1} C_K^{\mathrm{free}}$ and of C_K^{free} with $a_K^3 \mathbb{K}_3$ (Lem. 4.17(b), with (α5)), both main terms equal $\frac{1}{6a_K^4}\sum_{\mu,\nu} D(e_\lambda)_{\mu\nu}\langle F^\circ_\mu, F^\circ_\nu\rangle$, F°_μ the angle of the orientation μ at the centre, up to $C[(r/a_{K-n_1})^{\beta_H} + a_K/r]\, r_n^2 a_{K-n}^{-4}$ and terms of the same order; the moment condition

of step (iii) of the proof of Lem. 4.36 cancels the main terms. So $\sup_{\mathsf{Cc}_n(r_n)} |X^{\mathrm{fz}}| \le \varepsilon_X r_n^2 a_{K-n}^{-4}$ with $\varepsilon_X \le C(g,\mathsf{b})[r^{\beta_{\mathrm{H}}/4} + a_K/r]$. (iii) *Carrying upward.* On the chain classes the averaging identity of one step holds at every point: for $n_0 \le m < K$, $A' \in \mathsf{Cc}_{m+1}(r_{m+1})$, the measure $\nu = \nu_{m,A'}$ of the step and every bare array $\mathfrak{g}$ of the level m of size below $N_\star^{[m]}$,

$$\int S[\mathfrak{g}](e^{i\eta_m A^{\mathrm{lw}}(B)})\,\mathrm{d}\nu(B) = (S[\mathfrak{g}] + \mathsf{Sum}[\mathsf{DSt}^{[m]}(\mathfrak{f}_{0,m})[\mathfrak{g}]])\,(e^{i\eta_{m+1}A'}) + c_{\mathfrak{g}},$$

with a number $c_{\mathfrak{g}}$ that does not depend on A'. It is the first derivative, in the direction of $\mathfrak{g}$, of the exact step integral in Lem. 2.2 at the real chart point A', which writes the action of the level $m+1$ through the mean under $\nu_{m,A'}$ of the exponential of the action of the level m at the lowered configurations; the measure $\nu_{m,A'}$ and the lowering map name no array, and the step takes any bare array of that size in its fourth place (Lem. 4.6(b)). As $A^{\mathrm{lw}}(B) \in \mathsf{Cc}_m(r_m)$ and ν is a probability measure, the oscillation of the right member over $\mathsf{Cc}_{m+1}(r_{m+1})$ is at most the oscillation of $S[\mathfrak{g}]$ over $\mathsf{Cc}_m(r_m)$.

A family that the steps above n_0 make from $\mathfrak{x}$, first created at the level $m+1$, is bounded in two ways: by the norm of the calculus, with the factor γ_m (Lem. 4.6(b), (c)), and, through this identity at $\mathfrak{g} = \mathfrak{x}_m$, by the oscillation of X^{fz} at the levels m and $m+1$, at most $2\varepsilon_X L^{4(n-m)} r_n^2 a_{K-n}^{-4}$. The smaller of the two, summed over m with $\sum_{l\ge 0} \min\{aL^{-l/2}, bL^{4l}\} \le 2.4 a^{8/9} b^{1/9}$ ($L \ge 13$), bounds the oscillation of X_n by $[\varepsilon_X + C\gamma_{K-1}^{8/9}\varepsilon_X^{1/9}] r_n^2 a_{K-n}^{-4}$. Together the oscillation of Δ_n on $\mathsf{Cc}_n(r_n)$ is at most $\varepsilon_\alpha r_n^2 a_{K-n}^{-4}$ with $\varepsilon_\alpha = \varepsilon_X + C\gamma_{K-1}^{8/9}\varepsilon_X^{1/9} + C\varepsilon_S \le C(g,\mathsf{b})\,\epsilon$, as $\gamma_{K-1} \le 1$; c'_n is the centre of this oscillation.

At $n = K$ the chain class is $\{0\}$ and $r_K = 0$. □

Lemma 4.38 (Pair minus tensor field up to the last level)**.** Let g satisfy besides (Γ3) of Def. 4.1(6), let $r \le r_\alpha^{\mathcal{W}}(g,\mathsf{b})$, let n_0 be the level of Lem. 4.36, and $\varepsilon^{\mathcal{W}} = C(g,\mathsf{b})[r^{\beta_{\mathrm{H}}/72} + (a_K/r)^{1/18}]$ with a suitable number $C(g,\mathsf{b}) \ge 1$. There are numbers c_n, $n_0 \le n \le K$, such that for m_n-almost every $W \in \mathcal{V}_n^{\chi}$

$$|\Delta_n(U_n(W)) - \mathsf{c}_n| \le \varepsilon^{\mathcal{W}}\,\mathrm{nat}_n^{\mathcal{W}}, \qquad |\mathsf{c}_n - \mathsf{c}_K| \le \varepsilon^{\mathcal{W}}\,\mathrm{nat}_n^{\mathcal{W}},$$

and, for $w_3 \not\equiv 0$ and $\mathsf{d}' = |c - z|$, $|\Gamma_n(U_n(W))| + |\Gamma_K(\mathbf{1})| \le C(\mathsf{b})\,\mathsf{d}'^{-5/2}\varepsilon^{\mathcal{W}}\mathsf{b}^{-2}\mathsf{d}'^{-6}\mu_3$. ■

Proof. The proof of Lem. 4.36 is made at every level $n_0 \le n \le K$, with the radius $C_{\mathcal{W}}\varepsilon_n$ of the backgrounds in the place of the radius of a chain class and without the shift of the level. (i) By the first two members of (Γ3) the domains that leave the cores at the radius $2B_3\varepsilon_n$ have a depth that makes the tails of Lem. 4.8(b) at most $C\varepsilon_n^2 L^{-4(n-i)}$; so a pointed family of level i that vanishes off a set of sites changes on the backgrounds of the level n by at most a constant times its norm, the number of the cubes that hold a site of the set, $L^{-4(n-i)}$ and $\mathrm{nat}_n^{\mathcal{W}}$. (ii) Lem. 4.35(b) at $\varrho = 3$, which applies as $7488B_3\varepsilon_n \le \epsilon_{\mathrm{G}}$ by (Γ3) and $26a_{K-n} < \ell_g$ (as $\ell_g \ge 2LMy_K \ge 338$), gives about a bare site x_0 next to c a gauge g on $Q_{12}(x_0)$ and a potential A on the bonds of $Q_6(x_0)$ with $\mathsf{g}(b_-)U(b)\mathsf{g}(b_+)^{-1} = e^{i\eta_n A(b)}$ there, $U = U_n(W)$; on $Q_3(x_0)$ the potential, its differences and the current are at most $\mathsf{r}_\circ := C_{\mathcal{W}}\varepsilon_n \le \frac{1}{120}$.

Lem. 4.12 is used in a local form, with data on $N_{\rho+1}$: *in the units of a level, with the bare spacing $\xi \le \frac{1}{13}$, let $\rho \ge 2$ with $4\rho+4$ at most the side of the torus, and let $\mathsf{g}\colon N_{\rho+1} \to SU(2)$ and a real 1-form A on the bonds with both end points in $N_{\rho+1}$ satisfy $\mathsf{g}(b_-)U(b)\mathsf{g}(b_+)^{-1} = e^{i\xi A(b)}$, $|A| \le \alpha \le \frac{1}{60}$ and $|\nabla^\xi A| \le \gamma_\circ$ there, the current of U being at most ι there; with $c_\mu = A_\mu(x_0)$, $X(x) = \sum_\mu (x - x_0)_\mu c_\mu$ and $h = e^{iX}\mathsf{g}$ on $N_{\rho+1}$ there is a real pair v' on all the bare bonds which, at the bonds with base point in N_ρ, is the image of U and of its current under the gauge transformation h, and which obeys the three bounds of Lem. 4.12; if $\|v'\|$ is less than the radius of the ball of the chart, every local term of the level on a domain $X \subset N_{\rho-1}$ that is formed at U has the same value at $P[v']|_X$.* Its proof is that of Lem. 4.12, with the transformed potential formed on $N_{\rho+1}$

and the cut-off $\chi(x) = \min\{1, \max\{0, (\rho + 1 - 2\xi - |x - x_0|_\infty)/(1 - 2\xi)\}\}$, of the width $1 - 2\xi$ and equal to 1 on N_ρ, in the place of the cut-off of the width ρ: as $|\chi(x + \xi e_\nu) - \chi(x)| \le \frac{13}{11}\xi$, $|\nabla^\xi A'| \le \frac{100}{99}(\gamma_\circ + 3\alpha^2) + \frac{13}{11}[3(\rho+1)\gamma_\circ + (9\rho + 9.62)\alpha^2] \le \rho(6\gamma_\circ + 19\alpha^2)$ for $\rho \ge 2$; and the bonds that a local term on $X \subset N_{\rho-1}$ reads have their end points in N_ρ, where $P[v']$ is the image of U under h, while a local term is a gauge invariant function of the pair on its domain.

Viewed from the level $i+1$, $\mathsf{k}(\frac{r}{2}) \le i < n_1$, $m = n-i-1$, the data on $Q_3(x_0)$ have $|A| \le \alpha = L^{-m}\mathsf{r}_\circ$, $|\nabla A| \le \gamma_\circ = L^{-2m}\mathsf{r}_\circ$ and a current at most $L^{-3m}\mathsf{r}_\circ \le \gamma_\circ$, and the cube N_{h_i+1} of the level $i+1$ lies in $Q_3(x_0)$, as $(\mathsf{h}_i + 1)L^{-m} \le 3$ (Def. 4.1(6)). The local form at $\rho = \mathsf{h}_i$, where $(4\mathsf{h}_i + 4)a_{K-i-1} \le \ell_g$, gives a real point v' of the chart with $\|v'\| \le \mathsf{h}_i\gamma_\circ(6 + 19\mathsf{r}_\circ) \le 7\mathsf{h}_i L^{-2m} C_{\mathcal{W}}\varepsilon_n \le \delta_i/2\Lambda_i$ (Def. 4.1(6)), inside the ball of the chart; the linear gauge X is the same function at all the levels i, so that one h serves them all.

Steps 2–4 of the proof of Prop. 4.13 at the real point v', the domains in N_{h_i-1} taking the same values at U and at $P[v']$, and the others, of a linear size at least $D_* = (\mathsf{h}_i - 1 - R_w)/M - 2$, costing $2\mathsf{M}_i^{\mathsf{K}} e^{-[D_*]_+} \le \bar{\mathsf{e}}\gamma_\circ^2/\delta_i^2$ by the choice of h_i, replace $\mathsf{c}\sum_i(\mathsf{Sum}[\mathsf{K}^{(i+1)}](U) - \mathsf{Sum}[\mathsf{K}^{(i+1)}](\mathbf{1}))$ by $\mathsf{c}\, q_{n_1}(F')$, F' the linear angles in the gauge h, up to $C\sum_i \hat{r}_i^3 \mathfrak{b}_i^2(\delta_i + \epsilon_i^{\mathrm{cu}})\Lambda_i^2 \mathsf{h}_i^2\, C_{\mathcal{W}}^2 \mathrm{nat}_n^{\mathcal{W}}$. In the gauge h, in the units of the level n, $|A'| \le 4s\mathsf{r}_\circ + 8\eta_n \mathsf{r}_\circ^2$ at the bonds with base point within $s \le 2$ of x_0, $|\nabla A'| \le 1.04\,\mathsf{r}_\circ$ and $|F_p'| \le 2.08\,\eta_n^2 \mathsf{r}_\circ$ there, and $|a_p(U) - \frac{1}{2}|F_p'|^2| \le C\eta_n^4 \mathsf{r}_\circ^3 (s + \eta_n)^2$ at the plaquettes within $s \le \frac{1}{2}$ of x_0; this replaces the tensor source by $\frac{1}{2a_K}\sum_p w_T(p)|F_p'|^2$ up to $C(r/a_{K-n_0})^2 C_{\mathcal{W}}^2 \mathrm{nat}_n^{\mathcal{W}}$.

Lem. 4.35(a) is read through the estimate of its proof on $Q_1(x_0)$, at $m_0 = 2.08\,\mathsf{r}_\circ$ and $m_1 = 3.7\,\mathsf{r}_\circ$ (on N_2, $|A'| \le 6.2\,\mathsf{r}_\circ$, and the codifferential of $\eta_n^{-2}F'$ differs from the current by at most $40|A'||\nabla A'| + 32|A'|^3$): it makes F' Hölder continuous with the exponent β_{H} on the scale of the pair. The comparison of $\mathcal{S}_{n_1}$ with $\mathfrak{b}^{-1}C_K^{\mathrm{free}}$ and of C_K^{free} with $a_K^3\mathbb{K}_3$ (Lem. 4.17(b), with (α5)) brings both main terms to $\frac{1}{6a_K^4}\sum_{\mu\nu} D(e_\lambda)_{\mu\nu}\langle F_\mu^\circ, F_\nu^\circ\rangle$, F° the angles at the centre, up to $C[(r/a_{K-n_1})^{\beta_{\mathrm{H}}} + a_K/r]C_{\mathcal{W}}^2\mathrm{nat}_n^{\mathcal{W}}$; the moment condition cancels them, and the frozen part on the backgrounds is at most $\varepsilon_X^{\mathcal{W}}\mathrm{nat}_n^{\mathcal{W}}$. (iii) From one level to the next the averaging identity of one step, written for arbitrary bare arrays, gives m_{n+1}-almost everywhere $\Delta_{n+1} = \mathsf{Qt}_{n+1}[\Delta_n] + \mathsf{b}_n$, where Qt_{n+1} is the quotient of the step, a positive normalised mean, and b_n is c times the covariance, under the step, of the quotients of the two single sources of the near pair.

The one-source bound $|G_{n;a} - G_{n;a}(\mathbf{1})| \le Cr^3|w_a|_\infty \mathrm{nat}_n^{\mathcal{W}}$ on the backgrounds gives $|\mathsf{b}_n| \le CA_0^2 y_{n_0}^{2\mathfrak{p}_0}(r/a_{K-n_0})^3\mathrm{nat}_n^{\mathcal{W}} \le C(\mathsf{b})\, r\,\mathrm{nat}_n^{\mathcal{W}}$. A quotient does not increase the oscillation, so the centres c_n of the oscillations of two consecutive levels differ by at most twice the oscillation bound plus $C(\mathsf{b})\, r\,\mathrm{nat}_n^{\mathcal{W}}$; as $\mathrm{nat}^{\mathcal{W}}$ falls by the factor $\frac{5}{4}L^{-3}$ from one level to the next, the sum over the levels gives the second bound. (iv) For Γ_n the averaging identity of one step, in its form for two and for three arrays, gives m_{i+1}-almost everywhere, for $n_0 \le i < K$, $\Gamma_{i+1} = \mathsf{Qt}_{i+1}[\Gamma_i] + \mathsf{Qt}_{i+1}[\Delta_i; G_{i;3}] + \mathsf{c}\,\mathsf{Sum}[\mathsf{C}^{(i+1)}]$; here $\mathsf{Qt}_{i+1}[\,\cdot\,;\,\cdot\,]$ is the covariance of the quotients of the step, and the family $\mathsf{C}^{(i+1)}$ is made of the mixed second derivatives of the step in the arrays of the sets $\{1,3\}$ and $\{2\}$ and in those of $\{2,3\}$ and $\{1\}$, and of its third derivative in the arrays of the three single sources.

A quotient does not raise the essential supremum, so the bound of Γ_n is that of Γ_{n_0} plus the sum over $n_0 \le i < n$ of the bounds of the two other terms. The covariance is at most the smaller of the bound by the oscillations of Δ_i and of the far source on the backgrounds and the bound by the norms, $\mathsf{C}^{(i+1)}$ is bounded by the norms, and Γ_{n_0} by the first display of Lem. 4.39 at $U_{n_0}(W)$, whose pairs of all levels $\le n_0$ lie in the cores (Lem. 4.20(d)); the sum over the levels is that of the proof of Lem. 4.40, with the bound of the oscillation of Δ on the backgrounds in the place of ε_α.

All the numbers so formed are at most $C(g, \mathsf{b})[r^{\beta_{\mathrm{H}}/36} + (a_K/r)^{1/9}]$ for $r \le r_\alpha^{\mathcal{W}}$; with $(\frac{\mathsf{b}}{\mathsf{d}'})^{3/2}(\frac{r}{\mathsf{d}'})^3 \le \mathsf{b}^{3/2}\mathsf{d}'^{-2}r^{1/2}$ and $(\frac{\mathsf{b}}{\rho_0})^{3/2}e^{-\mathsf{d}'/8M\rho_0} \le C\mathsf{b}^{3/2}\mathsf{d}'^{-5/2}r^{1/2}$, $\rho_0 = a_{K-n_0}$, the claims follow. □

Lemma 4.39 (Crossing functionals at levels below the distance)**.** For $\beta \in \{13, 23, 123\}$ of the list (w_1, w_2, w_3) and $\beta = T3$ of the list (w_T, w_3), $m = |\beta|$, $1 \le i \le K$, and every configuration U whose

pairs of the level i lie in the cores,

$$\sum_{(X,z)} |E^{(i)}_\beta(X,U,z)| \le C_\times (\tfrac{8r}{3\mathsf{d}})^6 \bar\nu_{i-1;\beta}\, e^{-D_i^\sharp/2}, \qquad D_i^\sharp = [\mathsf{d}/2Ma_{K-i} - 2]_+.$$

So if $8Ma_{K-n} \le \mathsf{d}$, then for every such U of the level n:

$$|G_{n;\beta}(U)| \le C'_\times \mathsf{b}^{-m/2} a_{K-n}^{-m/2} e^{-\mathsf{d}/4Ma_{K-n}} (\frac{8r}{3\mathsf{d}})^6 \prod_{a\in\beta} |w_a|_\infty.$$

■

Proof. By the support rule (Lem. 4.6(d)) a crossing family lives on domains that reach from the pair to the piece, of linear size at least $D_i^\sharp$; the near supports lie in $\mathbb{B}(c, \frac58 r)$, the piece in $\mathbb{B}(z, \frac14 r)$, so the corners of the two kinds are at least $\frac12\mathsf{d}$ apart in the sup distance; its norm is bounded by the sizes of Lem. 4.9(e), in the form (f), at the width r and a crossing distance $\mathsf{d}_\times \ge \frac38\mathsf{d}$, with $(r/\mathsf{d}_\times)^6 \le (\frac{8r}{3\mathsf{d}})^6$; Lem. 4.8(b) at U gives the sum with the factor $e^{-D_i^\sharp/2}$, and $\bar\nu_{i-1;\beta} \le \bar c_m (5L/4\mathsf{b})^{m/2} a_{K-i}^{-m/2} \prod |w_a|_\infty$; the sum over $i \le n$ is led by $i = n$. □

Lemma 4.40 (The difference of the jets)**.** Let $w_3 \not\equiv 0$. Then the centre z of its cube has $(1 - \frac{1}{64})\mathsf{d} \le |c - z| \le 6\mathsf{d}$, and

$$\big|\mathsf{c}\,\mathsf{J}_3(w_1,w_2,w_3) - \mathsf{J}_2(w_T,w_3)\big| \le C(g,\mathsf{b})\,\mathsf{d}^{-12}\big[r^{\beta_{\mathrm{H}}/72} + (a_K/r)^{1/18}\big]\mathsf{b}^{-2}\mathsf{d}^{-6}\mu_3.$$

■

Proof. A barycentre in the cube lies in $\operatorname{supp}\varphi_2$, at a distance $\ge \mathsf{d}$ from c and $\le \frac{\sqrt3}{8} r \le \frac{1}{64}\mathsf{d}$ from z, as $r \le (\mathsf{d}/5LM)^2$. Put $\mathsf{d}' = |c - z|$ and $\rho_0 = a_{K-n_0}$; the data give $16r \le \mathsf{d}' \le \min\{1, \mathsf{b}\}$, $4M\rho_0 \le \mathsf{d}'$ and $64\mathsf{d}' \le \ell_g$ ($\mathsf{d}' \le 6\mathsf{d}$, $r \le (\mathsf{d}/5LM)^2$, $\rho_0 < Lr^{1/2}$, $\ell_g \ge 2LMy_K \ge 338$). The corners of the plaquettes of the near supports and of the piece are at least $\frac12\mathsf{d}'$ apart in the sup distance, so Lem. 4.39 holds with d' in the place of d. Lem. 4.9 is used in its form (f), which bounds the crossing distance by 1 in the place of the bound $e^{-2\sqrt3}$ of the placing: at the width r and the crossing distance $\mathsf{d}_\times = \frac38\mathsf{d}' \in [6r, 1]$, with the corners of the two kinds at least $\frac12\mathsf{d}' \ge \mathsf{d}_\times/\sqrt3$ apart, its conditions hold as in the proof of Lem. 4.39. By the support rule (Lem. 4.6(d)) a domain of the level $i+1$ that holds corners of both kinds has a linear size at least $D_{i+1} = [\mathsf{d}'/2Ma_{K-i-1} - 2]_+$.

The tower. Let $n_0 \le i < K$, $A' \in \mathsf{Cc}_{i+1}(r_{i+1})$ and $\nu = \nu_{i,A'}$. The tower identity at several sources (step 1 of the proof of Prop. 4.16; the two clusters, the near sources and the piece, lie in the frame at the level $n_0 - 1$, as $4M\rho_0 \le \mathsf{d}'$), at the five partitions of $\{1,2,3\}$ and the two of $\{T,3\}$, with every $G_{i;\beta}$ taken at $A^{\mathrm{lw}}(B)$, gives

$$\Gamma_{i+1}(A') = \int \Gamma_i \,\mathrm{d}\nu + \kappa_\nu(\Delta_i, G_{i;3}) + \mathsf{c}\big[\kappa_\nu(G_{i;13}, G_{i;2}) + \kappa_\nu(G_{i;23}, G_{i;1}) + \kappa_\nu(G_{i;1}, G_{i;2}, G_{i;3})\big].$$

As $A^{\mathrm{lw}}(B) \in \mathsf{Cc}_i(r_i)$ and ν is a probability measure, the mean is at most $\sup_{\mathsf{Cc}_i(r_i)} |\Gamma_i|$; as $\mathsf{Cc}_K(r_K) = \{0\}$, $|\Gamma_K(\mathbf{1})| = |\mathsf{c}\mathsf{J}_3 - \mathsf{J}_2|$ is at most $\sup_{\mathsf{Cc}_{n_0}(r_{n_0})} |\Gamma_{n_0}|$ plus the sum over $n_0 \le i < K$ of the suprema of the four cumulants.

The covariance of Δ and of the far source. By the oscillations: the explicit parts of Δ_i cancel by the moment condition of step (iii) of the proof of Lem. 4.36, and Lem. 4.37 bounds the oscillation of Δ_i on $\mathsf{Cc}_i(r_i)$ by $2\varepsilon_\alpha r_i^2 a_{K-i}^{-4}$; Lem. 4.10 and $|\mathsf{ang}_p^{[i]}| \le 2\eta_i^2 r_i$ bound that of $G_{i;3}$ by $C\mu_3 r_i^2 a_{K-i}^{-4}$; with $x_i r_i^2 \le C\mathsf{b}_i^2$ and $a_{K-i}x_i \ge \frac45\mathsf{b}$, $|\kappa_\nu(\Delta_i, G_{i;3})| \le C\varepsilon_\alpha \mathsf{b}_i^4 \mathsf{b}^{-2} a_{K-i}^{-6}\mu_3$. By the norms: $\kappa_\nu(\Delta_i, G_{i;3})$ is the value of Sum of the mixed second derivative of the step in the array of Δ_i, of the size at most Ca_{K-i}^{-4} (Lem. 4.6, and Lem. 4.9 in the form (f)), and in the array of the far source; this family vanishes on the domains with $d_{i+1}(X) < D_{i+1}$, and Lem. 4.6(b), the sizes of Lem. 4.9(a), in the form (f), and Lem. 4.8(b) give $|\kappa_\nu(\Delta_i, G_{i;3})| \le C\mathsf{b}^{-1} a_{K-i}^{-7} e^{-D_{i+1}}\mu_3$.

The smaller of the two bounds is at most their geometric mean, $C\varepsilon_\alpha^{1/2}\mathfrak{b}_i^2\mathsf{b}^{-3/2}a_{K-i}^{-13/2}e^{-D_{i+1}/2}\mu_3$; with $\mathfrak{b}_i = A_1 y_i^{\mathfrak{p}_0}$ and $y_i \le \log(4\mathsf{b}/3a_{K-i})$ the sum over i is led by the levels with a_{K-i} about d' and gives the first term below.

The crossing terms. $\kappa_\nu(G_{i;1}, G_{i;2}, G_{i;3})$ is the value of Sum of the third derivative of the step in the arrays of the three single sources, and $\kappa_\nu(G_{i;13}, G_{i;2})$, $\kappa_\nu(G_{i;23}, G_{i;1})$ are the values of Sum of its mixed second derivatives in the arrays of a crossing set and of a single source; these families vanish on the domains with $d_{i+1}(X) < D_{i+1}$. Lem. 4.6(b) at $q = 3$, the sizes of Lem. 4.9(a), in the form (f), and Lem. 4.8(b) give $\mathsf{c}|\kappa_\nu(G_{i;1}, G_{i;2}, G_{i;3})| \le C\mathsf{c}\,\bar\nu_{i;123}\varsigma_i^3 e^{-D_{i+1}} \le C\mathsf{b}^{-1/2}r^3 a_{K-i}^{-21/2}e^{-D_{i+1}}\mu_3$, the widths of the three sources being of the order of r; the two others are smaller, by a factor γ_i (Lem. 4.9(e), in the form (f)). The sum over i, led by the levels with a_{K-i} about d', gives the second term below.

The start. At the configurations of $\mathsf{Cc}_{n_0}(r_{n_0})$, whose pairs lie in the cores,

$$\Gamma_{n_0} = \sum_{i\le n_0} \mathsf{Sum}[\mathsf{c}E_{123}^{(i)} - E_{T3}^{(i)}];$$

the first display of Lem. 4.39, with

$$\bar\nu_{i-1;\beta} \le \bar c_m (5L/4\mathsf{b})^{m/2} a_{K-i}^{-m/2} \prod_{a\in\beta} |w_a|_\infty$$

and

$$e^{-D_i^\sharp/2} \le e\cdot e^{-\mathsf{d}'/4Ma_{K-i}},$$

summed over $i \le n_0$, where the quotient of two consecutive terms is at most $L^{3/2}e^{-(L-1)} < 1$ as $4M\rho_0 \le \mathsf{d}'$, gives

$$|\Gamma_{n_0}| \le C(\mathsf{b}/\rho_0)^{3/2} e^{-\mathsf{d}'/4M\rho_0}\mathsf{b}^{-2}\mathsf{d}'^{-6}\mu_3,$$

at most the third term below.

The result is

$$|\Gamma_K(\mathbf{1})| \le C\Big[\Big(\frac{\mathsf{b}}{\mathsf{d}'}\Big)^{1/2}\mathsf{L}_{\mathsf{d}'}\varepsilon_\alpha^{1/2} + \Big(\frac{\mathsf{b}}{\mathsf{d}'}\Big)^{3/2}\Big(\frac{r}{\mathsf{d}'}\Big)^3 + \Big(\frac{\mathsf{b}}{\rho_0}\Big)^{3/2} e^{-\mathsf{d}'/8M\rho_0}\Big]\mathsf{b}^{-2}\mathsf{d}'^{-6}\mu_3, \qquad \mathsf{L}_{\mathsf{d}'} = \Big(1 + \log^+\frac{\mathsf{b}}{\mathsf{d}'}\Big)^{2\mathfrak{p}_0},$$

where $\varepsilon_\alpha \le C(g,\mathsf{b})\epsilon$ is the number of Lem. 4.37. Then $\varepsilon_\alpha^{1/2} \le C(g,\mathsf{b})[r^{\beta_{\mathrm{H}}/72} + (a_K/r)^{1/18}]$, $(r/\mathsf{d}')^3 \le \mathsf{d}'^{-3}r^{\beta_{\mathrm{H}}/72}$, $e^{-s} \le 2s^{-2}$ at $s = \mathsf{d}'/8M\rho_0$ with $\rho_0 < Lr^{1/2}$, and $(1-\frac{1}{64})\mathsf{d} \le \mathsf{d}' \le 6\mathsf{d} \le 1$ give the claim. □

Lemma 4.41 (Uniformity in the label)**.** Let $0 < g \le \gamma_{\mathrm{R}}$ and let $\mathcal{K} \subset (g^{-2}, \bar{\mathsf{b}}]$ be compact. Then $C_{\mathrm{A}}(g,\mathsf{b})$, $\mathsf{C}_{\mathrm{R}}(g,\mathsf{b};\mathsf{d})$ and $\mathsf{C}(\mathsf{d})$ of Thm. K are bounded, and $r_\alpha(g,\mathsf{b})$, $r_{\mathcal{W}}(g,\mathsf{b},\mathsf{d})$, $r_\sharp$, $r_0(\mathsf{b})$, $r_1(\mathsf{b})$ are bounded away from 0, uniformly in $\mathsf{b} \in \mathcal{K}$; here $r_\sharp$ and $\mathsf{C}(\mathsf{d})$ are the numbers that steps 2 and 3 of the proof of (E.3) form at a first-passage datum at g with the label b, for every $\mathsf{b} \in (g^{-2}, \bar{\mathsf{b}}]$. ■

Proof. The thresholds. r_1 does not increase in b, so $r_1(\mathsf{b}) \ge r_1(\bar{\mathsf{b}}) > 0$, and the infimum of r_0 over the labels at g is positive by (B.1). The threshold $r_\alpha(g,\mathsf{b})$ is the largest number below which the inequalities (α1)–(α5) of Def. 4.1(5) hold, and the bound $r \le r_\alpha(g,\mathsf{b})$ enters Lem. 4.36 and the lemma on the chain classes through these inequalities; by the bounds given there each of them holds as soon as r is below a number formed from the constants and g, or, for (α3), below $2\min\{\inf_{\mathsf{b}'} r_0(\mathsf{b}'), r_1(\bar{\mathsf{b}})\}$; so $r_\alpha(g,\mathsf{b}) \ge r_\alpha^-(g) > 0$ for every label at g. $r_{\mathcal{W}}(g,\mathsf{b},\mathsf{d})$ adds to r_α the inequalities of step 0 of the proof of (E.2), which depend on b only through $a_\sharp = 4\mathsf{b}/5x_\sharp \ge 4g^{-2}/5x_\sharp$; and $r_\sharp$ is the least of these thresholds and of those of the data. So all the thresholds are bounded below by positive numbers formed from the constants, β_{H}, g and d, for all labels in $(g^{-2}, \bar{\mathsf{b}}]$.

The constants. The label enters the proofs of Lems. 4.36–4.40 and of Thm. K through five kinds of quantities. (1) The run: by Lem. 4.3(a), $\frac{4}{5}\mathsf{b} \le a_{K-j}x_j \le \frac{4}{3}\mathsf{b}$, so that x_j, y_j, δ_j, ε_j, $\mathfrak{b}_j$, $\gamma_j = c_\gamma x_j^{-1/2}$,

$\theta^{[j]}$ and r_j lie between two functions of b/a_{K-j}, each continuous in b, and $\frac{3}{4\mathsf{b}} \le \upsilon_j \le \frac{5}{4\mathsf{b}}$. (2) The prefactor $\mathsf{c} = \frac{2\pi}{3}\mathsf{b}r^3$ and the units $\mathsf{u}_a = \mathsf{b}^{-1}|w_a|_\infty$. (3) The slices, through b^{-1} and $\bar{B}/\mathsf{b}$ (Lem. 4.17). (4) The number $a_\sharp = 4\mathsf{b}/5x_\sharp$, through $a_*^{-1} \le L\max\{8M, 5x_\sharp/4\mathsf{b}\}\ell_\epsilon/\mathsf{d}$. (5) The thresholds, bounded below as above.

Member by member: $\gamma_{n_0} \le c_\gamma(5Lr^{1/2}/4\mathsf{b})^{1/2}$; $s(\frac{r}{2})\theta^{[0]}$, with $x_0 = \mathsf{b}/a_K$; $\sum_i \hat{r}_i^3\mathfrak{b}_i^2(\delta_i + \epsilon_i^{\mathrm{cu}})\Lambda_i^2\mathsf{h}_i^2$, with $x_i \ge \frac{4}{5}\mathsf{b}/a_{K-i}$, while $\Lambda_i \le C(1+Y_{r/2})$ and $\mathsf{h}_i \le C(1+Y_{r/2})^{2\mathfrak{p}_0}$ are formed with g and r; the cost, at most $C\mathsf{b}\,r\,\mathrm{nat}_n^{\mathcal{W}}$, of the domains that leave the cube of the gauge, and the part of the steps between two levels, of the same order (proof of Lem. 4.36); $y_{n_0} \le \log(4\mathsf{b}/3\rho_0)$; the bracket of the proof of Lem. 4.40, through $(\mathsf{b}/\mathsf{d}')^{1/2}$, $(\mathsf{b}/\mathsf{d}')^{3/2}$, $(\mathsf{b}/\rho_0)^{3/2}$ and $\mathsf{L}_{\mathsf{d}'}$; the size $b_\Delta = (C\mathsf{b}+C)\Lambda_3$ of Y_Δ, the terms T_1–T_4 and the moments M_b of the proof of (E.2); the number $\epsilon_0(g,\mathsf{b})$, the reciprocal of the constant in $\sum_i \epsilon_i^{\mathrm{e}} \le C(g,\mathsf{b})\,\epsilon$; the quotients that form C_{R}; and the constant $C(\mathsf{b},\mathsf{d})$ of step 3 of the proof of (E.3).

Each majorant so formed is a sum of products of a factor that does not contain b and of a function of b made by finitely many products, powers, logarithms and maxima from b, b^{-1}, the constants, β_{H}, g, d and the lower bounds of the thresholds; such a function is continuous on the compact interval $[g^{-2}, \bar{\mathsf{b}}]$, where $g^{-2} \ge e^{2\mathfrak{p}_0} > 1$, and so bounded there. Hence $C_{\mathrm{A}}(g,\mathsf{b})$, $\mathsf{C}_{\mathrm{R}}(g,\mathsf{b};\mathsf{d})$ and $\mathsf{C}(\mathsf{d})$ are bounded on all of $(g^{-2}, \bar{\mathsf{b}}]$ by numbers formed from the constants, β_{H}, g, d and $\mathsf{d}\,\mathrm{Lip}\,\varphi_1/\|\varphi_1\|_\infty$, in particular on $\mathcal{K}$. □

Proofs of the theorems.

Proof of Thm. H. (B.2). 1. *The data.* By Lem. 4.5(b) the tents put the pair in the frame at $k_0 = \mathsf{k}(r)$; (Γ1) gives $g \le \gamma_{\mathrm{jt}}$ and $g^{-2} \ge x_{\mathrm{J}}$, so Lem. 4.6 applies, with (P1)–(P3), and $\rho \le r_1(\mathsf{b})$ gives $\rho \ge 4Ma_K$ and $E_\sigma(\hat{\rho}_1/2) \le \varkappa_0^4$; the smallness holds at all heights (Lem. 4.3(b)). At $n = K$, $A = 0$ and both members vanish; let $\mathsf{k}(r) \le n < K$.

2. *The decomposition.* Differentiate the defining formula for D_n^λ twice: the source term is linear in λ, so $\mathsf{F}_n = \sum_{i\le n}\mathsf{Sum}[E_{12}^{(i)}]$; with Lem. 4.7 and $q_n = \sum_{i=\mathsf{k}(r)}^{n-1}\mathcal{Q}_i$, $q_{\mathsf{k}(r)} = 0$,

$$\mathsf{F}_n(\tilde{U}) - \mathsf{F}_n(\mathbf{1}) - q_n(\mathsf{ang}^{[n]}(A)) = \underbrace{\sum_{i=1}^{\mathsf{k}(r)}\Delta[E_{12}^{(i)}]}_{\mathrm{I}} + \underbrace{\sum_{j=\mathsf{k}(r)}^{n-1}\Delta[\mathsf{R}^{(j+1)}]}_{\mathrm{II}} + \underbrace{\sum_{i=\mathsf{k}(r)}^{n-1}\left(\Delta[\mathsf{K}^{(i+1)}] - \mathcal{Q}_i(\mathsf{ang}^{[n]}(A))\right)}_{\mathrm{III}}.$$

3. *The passage to the yardstick* $\mathsf{Y}_n = (\rho/r)^6\hat{r}_n^3\mathsf{u}_1\mathsf{u}_2$. For a pointed family f of level $j+1$ that vanishes off Cr_1 and on the domains with $d_{j+1}(X) < D'$, with $\|\mathsf{f}\| \le \Phi\bar{\nu}_j$, Lem. 4.8(a) in the frame at $k_0 = \mathsf{k}(r)$ of step 1, $\bar{\nu}_j \le \frac{25}{16}\bar{c}_2 x_j\mathsf{u}_1\mathsf{u}_2$, $x_j \le \frac{5}{3}L^{n-j}x_n$ and $x_n r_n^2 \le C\mathfrak{b}_n^2$ give

$$|\Delta[\mathsf{f}]| \le C_{\mathrm{U}}\,\mathfrak{b}_n^2\,\mathsf{N}_{j+1}(w_1)\,\Phi\,u_j^{-3}e^{-D'/2}\,u_n^3\mathsf{u}_1\mathsf{u}_2.$$

4. *I, the levels at and below the working level, are cut by the distance.* By the support rule $D' = \mathsf{D}(u_i)$; Lem. 4.9(b) and (P2) give $\mathsf{N}_i\bar{\Psi}_{i-1;12} \le Cq^6(1+u_i^6)$; with $u_i = L^l u_{\mathsf{k}(r)}$ and $u_{\mathsf{k}(r)}/\sqrt{3}M \ge \frac{4}{\sqrt{3}}Y_r^{2\mathfrak{p}_0}$ the sum is led by $i = \mathsf{k}(r)$: $|\mathrm{I}| \le C\mathfrak{b}_n^2 e^{-Y_r^{2\mathfrak{p}_0}/\sqrt{3}}\mathsf{Y}_n$.

5. *II, the rest families.* Step 3 with $D' = 0$ and Lem. 4.9(c); $\mathsf{N}_{j+1}\varsigma_j^2 u_j^{-3} \le 216L^3q^6u_{j+1}^3$, $\mathsf{N}_{j+1}\Xi_j^r u_j^{-3} \le 216q^6$, $\sum_{j<n}\gamma_j \le 0.45\gamma_n \le C\delta_n/\mathfrak{b}_n$, and $u_{\mathsf{k}(r)}^3[\gamma_{\mathsf{k}(r)} + \theta^{[0]}] \le CY_r^{6\mathfrak{p}_0}\delta_{\mathsf{k}(r)}$, $\delta_{\mathsf{k}(r)} \le A_1Y_r^{\mathfrak{p}_0}(r/\mathsf{b})^{1/2}$ (Lem. 4.3(c)):

$$|\mathrm{II}| \le \{C\mathfrak{b}_n\delta_n + C\mathfrak{b}_n^2Y_r^{7\mathfrak{p}_0}(r/\mathsf{b})^{1/2}\}\mathsf{Y}_n.$$

6. *III, the kernel families.* Prop. 4.13, narrow form; $\mathsf{c}_i \le Ce^{-\delta_{\mathrm{tp}}u_i/\sqrt{3}}$, so $\sum_i \mathsf{c}_i(1+u_i)^2u_i^3 \le C$, and $\sum_{i\ge\mathsf{k}(r)}\mathfrak{b}_i^4\delta_i u_i^3 \le 1.12\,\mathfrak{b}_{\mathsf{k}(r)}^4u_{\mathsf{k}(r)}^3\delta_{\mathsf{k}(r)}$:

$$|\mathrm{III}| \le \{CY_r^{15\mathfrak{p}_0+2}(r/\mathsf{b})^{1/2} + C\mathfrak{b}_n^2r_n\}\mathsf{Y}_n.$$

7. *The sum.* $\mathfrak{b}_n \le \mathfrak{b}_{\mathsf{k}(r)} \le A_1 Y_r^{\mathfrak{p}_0}$, $r_n \le \frac{8}{5}\mathsf{c}_\mathfrak{h}\delta_n$ and $(r/\mathsf{b})^{1/2} \le g\, r^{1/2}$ give the bound. The factor $(\rho/r)^6$ costs nothing: F_n is bilinear in (w_1, w_2), every estimate depends on the weights only through $\sum_p |w_a(p)|$ or through a norm that (P1) bounds, and the one place where a sup norm could enter, Λ_i, sees a quotient (Prop. 4.13, step 6). By Lem. 4.3(c), $\log\frac{1}{r} \le Y_r \le 2\log\frac{1}{r} + C(g)$, whence $\epsilon'_{\mathrm{N}}(r)(\log\frac{1}{r})^N \to 0$.

(B.1). 1. Lem. 4.5(a) and its proof put the test pair in all six conditions of Def. 4.4 at $k_0 = \max\{K-s, 2\}$, under $3 \le s \le K$ and $\mathsf{Lm}_s \le L^{s-3}$, as required in (B.1); there the number $\theta^{[j]}/K_{\mathbf{M}}$ of every step is at most half of the largest value that the one-step theorem admits, $\hat r_{k_0} \ge M$, so $\varsigma_i = 1$ for $i \le k_0$, and $\mathsf{D}_i^{12} \ge \hat r_i/\sqrt{3} - 1$. 2. Below k_0, Prop. YM3_I:122.2 gives $\|E_{12}^{(j)}\| \le C\nu_{j-1}$, with ν_j defined in [7, eq. 199]; its running, half-ceiling and separation hypotheses are those verified in the pair frame of Lem. 4.5(a); from k_0 on, Lem. 4.6(b), (c) and the start at k_0 give the rest families with $\|\mathsf{R}^{(j+1)}\|_{j+1,\lambda_{\mathrm{out}}} \le \epsilon_j^{(2)}\varsigma_j\bar\nu_j$,

$$\epsilon_j^{(2)} \le C\big[x_j^{-1/2}(1+\mathsf{e}_\downarrow) + \theta^{[k_0]}\big], \qquad \mathsf{e}_\downarrow = \sum_{i\le k_0} e^{-4\beta_{\mathrm{s}}\kappa[\mathsf{D}_i^{12}/M-2]_+}\varkappa_{i-1}^{-2}\,:$$

no number in it grows with $\log\frac{1}{r}$ or with $n-k_0$, as the derivatives of the step gain the factor $1/\mathfrak{b}_j$ per order in the sources (Lem. 4.6(b)) and the lags of Lem. 4.6(c) fall like L^{-4l} and sum geometrically over the levels. 3. The decomposition of step 2 above, over the base level k_0, with Lem. 4.8(a), in the frame at k_0, for I and II and Prop. 4.13 at the test pair for III, both in the form stated after them, whose conditions are the frame, the smallness at the heights concerned ($2r_n \le \varrho_{\mathrm{ch}}$, so that $r_n \le \frac{1}{120}$ as $\varrho_{\mathrm{ch}} \le \frac{1}{60}$; $\epsilon_i^{\mathrm{cu}} \le 1$; $x_n^{-1} \le r_n^2$) and the side bound $\ell \ge 2LMy_K$: for a test pair at $k_0 = \max\{K-s, 2\}$ these hold for $g \le \min\{\gamma_{\mathrm{jt}}, \gamma_{\mathrm{O}}\}$ and $r \le r_0(\mathsf{b})$, the frame by Lem. 4.5(a), the side bound as $\ell \in \mathfrak{L}(g)$, and the smallness at the heights $\ge k_0$ being among the conditions that fix γ_{O} and $r_0(\mathsf{b})$.

The shape ϑ does not enter, since any $\bar{\mathsf{e}} \ge 2\mathsf{w}_1\mathsf{w}_2\mathsf{E}_i$ is admissible. 4. In the yardstick $\hat r_n^3\mathsf{u}_1\mathsf{u}_2$ the main part of II is $\le C\mathfrak{b}_n^2[\gamma_n(1+\mathsf{e}_\downarrow) + (n-k_0)\theta^{[k_0]}]$, III is $\le C\hat r_{k_0}^3\mathfrak{b}_{k_0}^4(\delta_{k_0} + \epsilon_{k_0}^{\mathrm{cu}})\Lambda^2\mathsf{h}^2 + C\mathfrak{b}_n^2 r_n$, by the bound $\mathsf{E}_i \le C_{\mathsf{E}}\delta_i^4(\delta_i + \epsilon_i^{\mathrm{cu}})$ of Prop. 4.13, and I is $\le C\mathfrak{b}_n^2 e^{-c'Y_s^{2\mathfrak{p}_0}}$. With $\theta^{[k_0]} \le C(r/\mathsf{b})^{1/2}y_{k_0}^{\mathfrak{p}_0}$, $K - k_0 \le s \le Cy_{k_0}$ and $\mathsf{e}_\downarrow \le Cy_{k_0}^{2\mathfrak{p}_0}e^{-c'Y_s^{2\mathfrak{p}_0}}$, and the smallness that the recursions need holding for $g \le \gamma_{\mathrm{O}}$ and $r \le r_0(\mathsf{b})$, this is the bound of (B.1); its last term $C\mathfrak{b}_n^2 r_n \le C\mathfrak{b}_n^2\delta_n$ is the price of writing the form with the linear angles of one chart. 5.

The conditions that fix $r_0(\mathsf{b})$ depend on b through $(r/\mathsf{b})^{1/2}$, through $y_{k_0} \le \log\frac{\mathsf{b}}{r} + C\log\log\frac{\mathsf{b}}{r}$ and through $x_{k_0} \le \frac{24}{23}\mathsf{b}\hat r_{k_0}/r$; on $(g^{-2}, \bar{\mathsf{b}}]$ these are bounded by their values at the two ends, so r_0 has a positive infimum over the labels at g.

(B.3). $|\mathsf{ang}_p^{[n]}(A)| \le 2\eta_n^2 r_n$, Lem. 4.17(a) with $\mathsf{r} \ge r/\sqrt{3}$, and (P1):

$$|q_n(\mathsf{ang}^{[n]}(A))| \le C\, r_n^2\, \mathsf{b}\, a_{K-n}^{-4}\rho^6 r^{-3}\mathsf{u}_1\mathsf{u}_2 = C\, r_n^2\, \mathsf{b}\, a_{K-n}^{-1}(\rho/r)^6\hat r_n^3\mathsf{u}_1\mathsf{u}_2 \le C\mathfrak{b}_n^2(\rho/r)^6\hat r_n^3\mathsf{u}_1\mathsf{u}_2,$$

as $\mathsf{b}\, a_{K-n}^{-1} \le \frac{5}{4}x_n$ and $x_n r_n^2 \le C\mathfrak{b}_n^2$. □

Proof of Thm. G. 1. *(A.1).* Prop. 4.19(a) bounds J_3 against the lattice ring ($64\mathsf{d} \le \ell_g$ as $\ell_g \ge 26$ and $\mathsf{d} \le e^{-34}$; $\rho \le r_1(\mathsf{b})$ by $\mathsf{d} \le \mathsf{d}_4(g) \le r_1(\bar{\mathsf{b}}) \le r_1(\mathsf{b})$), and Lem. 4.21 bounds $S_3^T - \mathsf{J}_3$ by $13\Lambda_3^3\mathsf{M}'^3\mathsf{b}^3\mathsf{d}^{3/2}\mathrm{nat}_3$ (all supports lie within $20\mathsf{d} \le \frac{1}{8}\ell_g - 1$ of x_1).

2. *The geometry of (A.2).* For a triple $(z_1, z_2, z_3) = (x, y, z)$ of $\mathcal{D}$ put $r = \frac{1}{2}r_*$, $\mathsf{d} = \frac{1}{2}\mathsf{d}_*$; bumps f_a with supports in balls of radius $\rho < \min\{\frac{1}{4}r_*, r_1(\bar{\mathsf{b}})\}$ inside the open balls of radius $\frac{1}{4}r_*$ about the z_a are placed at (ρ, r, d): their supports are $\ge r_* - \frac{1}{2}r_* = r$ apart in the pair and $\ge \mathsf{d}_* - \frac{1}{2}r_* \ge \mathsf{d}$ across, with $|x_1 - x_2| \le 3r$, $|x_a - x_3| \le 5\mathsf{d}$, all in $\mathbb{B}(z_1, \frac{9}{4}\mathsf{d}_*)$; this holds for every ratio $r_*/\mathsf{d}_* \in (0, 1]$, and $\mathsf{d}_* \le \mathsf{d}_5(g)$ gives $\mathsf{d} \le \mathsf{d}_4(g)$.

3. *Lattice to continuum.* Lem. 4.22(a) with $\mathsf{d}^+ = 5\mathsf{d}_*$ and a Lipschitz number ϑ of the f_a:

$$|3\,\mathrm{Tr}\prod_a((f_a\circ x)\mathsf{b}^{-1}C_K^{\mathrm{free}}) - 3\mathsf{b}^{-3}\int k_3\prod f_a| \le 3C^{(3)}[a_K/\vartheta\rho + (5\mathsf{d}_*/\ell_g)^3]\,\mathrm{nat}_3.$$

4. *The limit.* Along first-passage data with $K_j\to\infty$ and labels $\to\mathsf{b}$, $S_3^T(f)\to S_3^{(\ell_g,\mathsf{b})}(f)$ (Thm. YM3_I:A), $a_{K_j}/\vartheta\rho\to 0$, and the brackets are bounded with $\bar{\mathsf{b}}$ and $\bar{\mathsf{M}}^{(24)}$ (Lem. 4.20(e)); with $r^{-3}\mathsf{d}^{-6} = 512\,r_*^{-3}\mathsf{d}_*^{-6}$ this is the bound of the form in (A.2).

5. *Kernels.* Lem. 4.23(a) on $\prod_a U_a$, with $(\frac{3}{\pi})^3\le 1$: a Borel kernel, bounded at its Lebesgue points by the bound of the form at the centre; the cover of $\mathcal{D}$ by such products and the glueing of Lem. 4.30 give one kernel on $\mathcal{D}$; the closest pair of a triple is (x,y) after a relabelling. *(A.3)* is Lem. 4.23(b).

6. *(A.4).* (i) is Prop. 4.28(a) at $n=3$. (ii) For a triple with $\mathsf{d}_*\le\mathsf{d}_5(g)$, the bumps of Lem. 4.29(c) about it are in the data of (i) at every large j (the cube of side $\frac{9}{2}\mathsf{d}_*\le\frac{1}{2}\ell_g$ about x holds them, so the projection keeps all distances and the integrals of k_3); (i), step 4 at the label b, and $S_3^T[\nu_j]\to S_3^{\sigma'}$ give $|\int(\mathsf{s}_3 - 3\mathsf{b}^{-3}k_3)\prod f_a^\rho| \le [\epsilon_{\mathrm{A}} + 512C_3^{\mathrm{tr}3}\mathsf{b}^3\mathsf{d}_*^{3/2}]\mathsf{b}^{-3}r_*^{-3}\mathsf{d}_*^{-6}\cdot 0.102 \le \mathsf{E}_3(g,\mathsf{d}_*)\mathsf{b}^{-3}r_*^{-3}\mathsf{d}_*^{-6}$, and Lem. 4.29(c) gives the bound at the point. (iii) is Lem. 4.29(a).

7. *(A.5).* Let (K_j,β_j,ν_j) be a Gibbs sequence at g_0; its members are periodic Gibbs states, $\nu_j\in\mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$. For large j, β_j exceeds $L(g^{-2}+B_\sharp)$; let K_j' be the first-passage index of β_j at g and $z_j = K_j - K_j'$; as $g\le g_0$, $0\le z_j\le z_+$ with z_+ formed from g_0, g and the constants. (1) *The states.* For $0\le z<K$ the dilation $U\mapsto U^{(z)}$, $U^{(z)}(L^z b) = U(b)$, maps the bonds, the plaquettes and the finite sets of bonds of $\mathbb{L}_K$ onto those of $\mathbb{L}_{K-z}$, with $a_{L^z p}(U^{(z)}) = a_p(U)$. The Gibbs kernel of a finite set of bonds has the exponent $-\beta\sum_p a_p$ over the plaquettes that contain a bond of the set, which reads β and the lattice and not its spacing; so the image $\nu^{(z)}$ of a Gibbs state at (a_K,β) satisfies the equations of the kernels at (a_{K-z},β), and invariance under a subgroup Γ' of finite index of $a_K\mathbb{Z}^3$ becomes invariance under $L^z\Gamma'$, of finite index in $a_{K-z}\mathbb{Z}^3$.

So the dilation carries $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ onto $\mathcal{G}^{\mathrm{per}}_{K-z,\beta}$ (equation YM3_I:(1819) and the definition of periodic states following it), and $\nu_j^{(z_j)}$ is a periodic Gibbs state at the first-passage datum (K_j',β_j) at g, with the label $L^{z_j}a_{K_j}\beta_j$. (2) *The fields.* At every configuration U of $\mathbb{L}_K$, $x(L^z p) = L^z x(p)$ and $a_{K-z} = L^z a_K$ give $\mathcal{O}(f)(U) = L^z\mathcal{O}'(f^{(z)})(U^{(z)})$, $f^{(z)} = f(L^{-z}\cdot)$, the prime marking the field at the cutoff $K-z$, as in Lem. 4.31(d), which is the dilation; so in every law ν on $\mathbb{L}_K$, $S_3^T[\nu](f) = L^{3z}S_3^T[\nu^{(z)}](f^{(z)})$. (3) *The approximants* of a component of a canonical decomposition of the limit are periodic Gibbs states at the same data, by the definition of the components and their approximants in Lem. 4.29(a); (1) and (2) apply to them. (4) *The limit.* On a subsequence $z_j = z$ is constant and the labels converge, to b_0 say; along it $S_3^T[\nu_j^{(z)}](f) = L^{-3z}S_3^T[\nu_j](f(L^z\cdot))\to L^{-3z}S_3^\sigma(f(L^z\cdot))$ on separated triples, σ the limit, and this functional has the continuous kernel $\mathsf{s}_3^{(z)} = L^{-12z}\mathsf{s}_3(L^{-z}\cdot)$; likewise for a component and its approximants.

The proof of (A.4)(ii) uses of the law σ' the convergence $S_3^T[\nu_j]\to S_3^{\sigma'}$ on separated triples and the continuity of the kernel of $S_3^{\sigma'}$ at the triple (Lem. 4.29(c)); so (A.4)(ii) at g applies to the dilated sequence, whose labels tend to $L^z\mathsf{b}_0$, at the triples with $L^z\mathsf{d}_*\le\mathsf{d}_5(g)$. In kernels $\mathsf{s}_3(x) = L^{12z}\mathsf{s}_3^{(z)}(L^z x)$, and the natural size is invariant: $L^{12z}(L^z\mathsf{b}_0)^{-3}(L^z r_*)^{-3}(L^z\mathsf{d}_*)^{-6} = \mathsf{b}_0^{-3}r_*^{-3}\mathsf{d}_*^{-6}$, $k_3(L^z\cdot) = L^{-9z}k_3$. The approximants of a component are periodic Gibbs states at the same couplings and are dilated alike. □

Proof of Thm. I. 1. *(C.1).* At $x,y = c\pm\frac{r}{2}n$: $r_* = r$; the two distances $|c-z\pm\frac{r}{2}n|$ lie in $[\mathsf{d}-\frac{r}{2},\mathsf{d}+\frac{r}{2}]$ and the sum of their squares is $2\mathsf{d}^2+\frac{1}{2}r^2$, so $\mathsf{d}_*\in[\frac{3}{4}\mathsf{d},\frac{17}{16}\mathsf{d}]$, $r\le\mathsf{d}_*\le\mathsf{d}_5(g)$. (A.2) multiplied by $\frac{2\pi}{3}\mathsf{b}r^3$ gives $\frac{2\pi}{3}(\frac{4}{3})^6\le 12$ times $\bar\epsilon_{\mathrm{A}}\mathsf{b}^{-2}\mathsf{d}^{-6}$, and Lem. 4.24(c) gives $2.66\le 2.7$; "almost everywhere" is in (c,h,z), a linear change of the variables (x,y,z). (ii) (A.3) at every triple; k_3 is continuous off

the diagonals, so its means tend to its values; with tents, (A.3) in the tent form and the second part of Lem. 4.24(c). (iii) (A.4)(ii) at every triple and Lem. 4.24(c).

2. *(C.2)(i).* Prop. 4.19(b) bounds J_2; Lem. 4.21 at two sources gives

$$|S_2^T - \mathsf{J}_2| \le 3\Lambda_3^2 \mathsf{M}'^2 \mathsf{b}^2 \mathsf{d}^{3/2} (\rho/\mathsf{d})^6 \mathsf{b}^{-2} |w_1|_\infty |w_2|_\infty$$

; at the plane weights $w_1 = \mathbf{1}_{\{\lambda(p)=\lambda\}}\varphi_1 \circ x$, $w_2 = \mathbf{1}_{\{\lambda(p)=\mu\}}\varphi_2 \circ x$, Lem. 4.22(c) turns the lattice square into $\mathcal{K}_{\lambda\mu}$ ($8a_K \le \rho$ by $s(\rho) \le K$; the assumption $|x_1 - x_2| \le 18\mathsf{d}$ is the one that this step and the rings need).

3. *(C.2)(ii).* For $(x_0, y_0) \in \Omega_{\mathrm{P}}$ and bounded Borel f, h supported in the closed balls of radius $\frac{1}{8}\mathsf{d}_0$ about x_0, y_0, the jets, the remainder, $|C_K^{\mathrm{free}}(p,q)| \le (1 + 3C_{\mathrm{lat}}) a_K^3 |u|^{-3}$ and the count of plaquettes give

$$|\langle \mathcal{O}_\lambda(f); \mathcal{O}_\mu(h)\rangle_{K_j,\ell_g,\beta_j}| \le \mathsf{B}(g, \mathsf{d}_0) \|f\|_\infty \|h\|_\infty, \qquad \mathsf{B} = 4^{-6} g^4 [\tfrac{3}{2} \cdot 729 (1 + 3C_{\mathrm{lat}})^2 + \mathsf{E}_2(g, \tfrac{1}{2}\mathsf{d}_0)],$$

with no Lipschitz number (the Riemann sums of step 2 involve $a_K/\vartheta\rho$ and give no bound in the sup norms). Lem. 4.30 gives one subsequence for all pairs of balls; along it step 2, with $a_{K_j}/\vartheta\rho \to 0$, gives the bump form at every centre: the balls of radius $\frac{1}{8}\mathsf{d}_0$ are $\ge \frac{1}{2}\mathsf{d}_0 =: \mathsf{d}$ apart, $|x_1 - x_2| \le \frac{5}{2}\mathsf{d}$, and $(\rho/\mathsf{d})^6 = 64(\rho/\mathsf{d}_0)^6$. Lem. 4.23 at $p = 2$ gives the kernels and the almost everywhere bound, and Lem. 4.24(a) gives $t_\lambda - \mathsf{t}^\circ(e_\lambda, \cdot) = \sum_\mu (\delta_{\lambda\mu} - \frac{1}{3}\sum_{\lambda'} \delta_{\lambda'\mu})$ with $\delta_{\lambda\mu} = \tau_{\lambda\mu} - \frac{3}{2}\mathsf{b}^{-2}[\mathbb{K}_{3,\lambda\mu}]^2$: three terms and a third of nine, so the factor 6.

4. *(C.3).* Along the subsequence of the limit point $S_3^T \to S_3^{(\ell_g, \mathsf{b})}$, since that limit is taken along all data with labels $\to \mathsf{b}$. By (A.3) and by Lem. 4.23(b) applied to the bump form of (C.2)(ii) at the centre (c, z), the means over balls of radius ρ of $\mathsf{s}_3 - 3\mathsf{b}^{-3}k_3$ and of $\tau_{\lambda\mu} - \frac{3}{2}\mathsf{b}^{-2}[\mathbb{K}_{3,\lambda\mu}]^2$ are bounded at every centre, with the factors $(\frac{3}{4\pi})^3, (\frac{3}{4\pi})^2 \le 1$; k_3 and $[\mathbb{K}_3]^2$ are continuous off the diagonals; adding with Lem. 4.24(a), (c) gives the display, and the tents likewise. The cone form: (C.1)(i) at n, (C.2)(ii) at (c, z) almost everywhere (a null set of (c, z) is a null set of (c, h, z)), and $|\mathsf{t}^\circ(n, X) - \mathsf{t}^\circ(e_\lambda, X)| \le 0.057 |n - e_\lambda| \mathsf{b}^{-2} |X|^{-6}$.

5. *(C.4).* (d) is Prop. 4.28(b) with step 2 on the least torus (the projection is injective and keeps distances on $\mathbb{B}(x_0, \frac{3}{2}\mathsf{d}_0)$, so the integrals and the norms are those of the periodised functions); it gives the bound $\mathsf{B}(g, \mathsf{d}_0)$ of step 3 along ν_j for j large. (a) Lem. 4.30 on Ω_σ. (b) The bump form at every centre of Ω_σ with $64\mathsf{E}_2(g, \frac{1}{2}\mathsf{d}_0)\mathsf{b}^{-2}\mathsf{d}_0^{-6}\rho^6$; Lem. 4.23; the bound at a point by centres $\zeta_m \to (x, y)$ of distances $\mathsf{d}_{\zeta_m} < |x - y|$, whence the supremum in ϵ_σ. Every ν_j is invariant under the translations of $\mathbb{L}_{K_j}$ by Prop. YM3_I:103.4, so the limit forms are invariant under all translations, and by Fubini's theorem $\tau_{\lambda\mu}^\sigma(x, y)$ is a function of $x - y$ almost everywhere. $\sum_{\lambda,\mu} \langle \mathcal{O}_\lambda; \mathcal{O}_\mu\rangle = S_2^T$, which converges along the whole sequence to $S_2^{\sigma'}$ with the continuous kernel s_2^T; with Lem. 4.24(a) this gives the bound of s_2^T almost everywhere, and at every point by continuity, $\epsilon_\sigma(g, \cdot)$ being nondecreasing. (c) Let $\mathcal{N}$ be the union of the null sets of (b); it is fixed before r is named.

Off $\mathcal{N}$, (b) and Lem. 4.24(a) give $|t_\lambda^\sigma - \mathsf{t}^\circ(e_\lambda, \cdot)| \le 6(\frac{3}{\pi})^2 \epsilon_\sigma \mathsf{b}^{-2}\mathsf{d}^{-6} \le 6\epsilon_\sigma \mathsf{b}^{-2}\mathsf{d}^{-6}$, and (C.1)(iii) at $n = e_\lambda$ holds at every point, since s_3 is continuous; $12 \sup \mathsf{E}_3 + 6\epsilon_\sigma = \epsilon'_\sigma$. With tents of radius $\rho < \frac{1}{8}\mathsf{d}$ at (c, z): the bump form of (b) at the centre (c, z), and Lem. 4.24(d) times $\frac{3}{2}$, six times, give $21\rho/\mathsf{d}$. □

Proof of Thm. J. 1. *(D.1) below* γ_{N} is Prop. 4.34. 2. *(D.1) at every* $g \le \gamma$. Let z, g'_1, g'_2 be as in Lem. 4.31 and $K_\circ(g) = z + \max_\iota s'_\iota(r_{g'_\iota})$. For $K \ge K_\circ(g)$, $(K - z, \beta)$ is a first-passage datum at some $g'_\iota \le \gamma_{\mathrm{N}}$, on the side $L^z\ell \in \mathfrak{L}(g'_\iota)$, and the conditions of Prop. 4.34 hold at $(L^z r, L^z \mathsf{d}')$; by Lem. 4.31(d) each display at (K, ℓ, β) is that display at $(K - z, L^z\ell, \beta)$. So (D.1) holds with

$$\epsilon^{\mathrm{cl}}(g, \ell, \mathsf{d}') = \max_{\iota=1,2} \epsilon_\sharp^{\mathrm{ev}}(g'_\iota, L^z\ell, L^z\mathsf{d}'), \qquad \mathsf{d}_{\mathrm{cl}}(g) = \tfrac{7}{6} L^{-z} \min_\iota t_{\mathrm{N}}(g'_\iota),$$

a finite function of (g, ℓ, d'), with the rate for $\mathsf{d}' \le \mathsf{d}_{\mathrm{cl}}(g)$. On the dilated torus, at the distance $L^z\mathsf{d}'$, the members of $\epsilon_\sharp^{\mathrm{ev}}$ that carry the renormalisation group, the remainder and the induction,

read the distances through the floor $\min\{\frac{6}{7}L^z\mathsf{d}', t_{\mathrm{N}}(g'_\iota)\}$ alone; the largest distance $\frac{8}{7}L^z\mathsf{d}'$ enters the one explicit member, the comparison of the sum of the slices with the free kernel (Lem. 4.17(b)), which is polynomial in that distance; and the passage from the floor to $L^z\mathsf{d}'$ costs the factor $\max\{1, 6L^z\mathsf{d}'/7t_{\mathrm{N}}(g'_\iota)\}^6$ of $\epsilon^{\mathrm{ev}}_\sharp$ (proof of Prop. 4.34). 3.

(D.2). The forms are bounded in the sup norms at every pair of small balls in Ω°, as in step 3 of the proof of Thm. I; Lem. 4.30 gives one subsequence for the plane and the bond forms; (D.1) with $a_{K_j}/\vartheta r \to 0$ gives the bump forms at every centre; Lem. 4.23, with centres in the window $[\frac{7}{9}t, \frac{9}{7}t]$ and $(\frac{3}{\pi})^2(\frac{9}{7})^6 \le 4.2$, gives the kernels. For T_n: the kernel of $\langle T_\lambda; \mathcal{O}\rangle$ is within $3(4.2+4.2)\bar\epsilon^{\mathrm{cl}} = 25.2\,\bar\epsilon^{\mathrm{cl}}$ of $\mathsf{t}^\circ(e_\lambda, \cdot)$, that of $\langle \mathcal{O}^{\mathrm{od}}_{\lambda\lambda'}; \mathcal{O}\rangle = \sum_\gamma \langle \mathcal{O}^{\mathrm{od}}_{\lambda\lambda'}; \mathcal{O}_\gamma\rangle$ within $12.6\,\bar\epsilon^{\mathrm{cl}}$ of $3\mathsf{b}^{-2}[\mathbb{K}^2_3]_{\lambda\lambda'}$; with $\sum_\lambda n_\lambda^2 = 1$, $\sum_{\lambda<\lambda'} |n_\lambda n_{\lambda'}| \le 1$ and Lem. 4.24(a), $25.2 + 12.6 \le 38$.

The kernel t_n is a quadratic polynomial in n whose coefficients are the nine kernels, so one null set serves all n. 4. *(D.3)* is (C.1) and (D.2) by the triangle inequality: almost everywhere in (c, h, z) by Fubini's theorem; for the means and the tents, the bump forms of (D.1) at the centre (c, z) itself, extended to indicators of balls by Lem. 4.23(b), bound the means of the kernels with factors below those of the almost everywhere bound. □

Proof of Thm. K. (E.2). 0. *The stopping level.* Let $\epsilon = r^{\beta_{\mathrm{H}}/36} + (a_K/r)^{1/9}$, $\ell_\epsilon = \max\{1, \log\frac{1}{\epsilon}\}$, $a^\flat = \min\{\mathsf{d}/8M\ell_\epsilon, a_\sharp\}$ with $a_\sharp = 4\mathsf{b}/5x_\sharp$, so that $a_{K-n} \le a_\sharp$ gives $x_n \ge x_\sharp$ (Lem. 4.3(a)), n_* the largest level with $a_{K-n_*} \le a^\flat$, and $a_* = a_{K-n_*} \in (a^\flat/L, a^\flat]$. The conditions that define $r_{\mathcal{W}}(g, \mathsf{b}, \mathsf{d})$, namely $r \le \min\{r_\alpha, (\mathsf{d}/5LM)^2, e^{-1}\}$, $8ML^{m_1+2}r^{1/2}(1+\log\frac{1}{r}) \le \mathsf{d}$, $L^{m_1+2}r^{1/2} \le a_\sharp$ and $r^{-1/4} \ge 2M$, secure $N_0 < n_*$, $x_{n_*} \ge x_\sharp$ and $L^{N_0-n_1} \ge 2M$, as $L^{N_0-n_1} = L^{m_1}a_{K-n_0}/a_{K-n_1} \ge r^{-1/4}$ (so Lem. 4.36 holds with $n^\sharp = n_*$ when $\epsilon^{\mathrm{e}}_i \le 1$), $8Ma_* \le \mathsf{d}$, $e^{-\mathsf{d}/4Ma_*} \le \epsilon^2$ and $a_*^{-1} \le C\ell_\epsilon/\mathsf{d}$.

1. *The identity.* Lem. 4.20(c) at $k = n_*$, $\mathbb{P}_* = \mathbb{P}_{n_*}$, for the two lists, G_3 being the same function in both:

$$\mathsf{c}S_3^T(w_1, w_2, w_3) - S_2^T(w_T, w_3) = \int \Gamma\, d\mathbb{P}_* + \kappa_{\mathbb{P}_*}(\Delta, G_3) + \mathsf{c}[\kappa_{\mathbb{P}_*}(G_{13}, G_2) + \kappa_{\mathbb{P}_*}(G_{23}, G_1)] + \mathsf{c}\,\kappa_{\mathbb{P}_*}(G_1, G_2, G_3) =: T_1 + T_2 + T_3 + T_4,$$

with Δ, Γ, G_β taken at $U_{t'}(W)$.

2. *Sizes.* $r^3|w_{1,2}|_\infty \le 489$, $|w_T|_\infty \le \frac{2}{3}|w_1|_\infty$, $\mathsf{c}|w_1|_\infty|w_2|_\infty r^3 \le C\mathsf{b}$, $|w_3|_\infty \le \mu_3 r^{-3}$; the bounds of Lem. 4.21 hold at every point of the small-field region, as their proof uses of W only that the background is regular.

3. *T_1, T_3: the crossings.* At $t' \le n_*$, $8Ma_{K-t'} \le \mathsf{d}$, and Lem. 4.39 with $e^{-\mathsf{d}/4Ma_*} \le \epsilon^2$ gives $|T_1| \le C(\mathsf{b})a_*^{-3/2}\mathsf{d}^{-6}\epsilon^2\mu_3$ and $|T_3| \le Ca_*^{-1}r^3\mathsf{d}^{-6}\epsilon^2\mu_3\mathsf{M}_6$, $\mathsf{M}_b = \int \alpha^{-b} d\mathbb{P}_*$: no negative power of r is left.

4. T_4. $|T_4| \le C\mathsf{b}r^3\Lambda_3^3\mathsf{M}_{18}\mu_3$.

5. *T_2: pair minus tensor field.* $T_2 = \int(\Delta - \mathsf{c}_{n_*})(G_3 - \int G_3)d\mathbb{P}_*$, and $|G_3 - \int G_3| \le \Lambda_3\mu_3(\alpha^{-6} + \mathsf{M}_6)$ by step 2. On the parts with $t' \ge N_0$ each measure is absolutely continuous with respect to $\mathsf{m}_{t'}$, and Lem. 4.36 gives $|\Delta - \mathsf{c}_{n_*}| \le 3.01\,\tilde\varepsilon\,\alpha^{-4}$ almost everywhere. The number c_{n_*}: the first part of $\mathbb{P}_*$, on which $t' = n_*$ and $\alpha = a_*$, has positive mass, as the density of m_{n_*} on $\mathcal{V}^\chi_{n_*}$ is a positive exponential by [7, eq. 224] and [7, eq. 225]. This region is open, its defining tests are strict, and they hold at the trivial field, where all curvature and fluctuation defects vanish. It therefore has positive Haar measure; at almost every point of it $|\Delta - \mathsf{c}_{n_*}| \le \varepsilon^{\mathcal{W}}a_*^{-4}$ (Lem. 4.36) and $|\mathsf{Y}_\Delta| = |\Delta - \Delta_K(\mathbf{1})| \le b_\Delta a_*^{-6}$, $b_\Delta = (C\mathsf{b} + C)\Lambda_3$ (Lem. 4.21 at every point, step 2), so that $|\mathsf{c}_{n_*} - \Delta_K(\mathbf{1})| \le (\varepsilon^{\mathcal{W}} + b_\Delta)a_*^{-6}$.

On the parts with $t' < N_0$, $\alpha < a_{K-N_0} = L^{m_1}a_{K-n_0} < L^{m_1+1}r^{1/2} < a_*$, so that there $|\Delta - \mathsf{c}_{n_*}| \le |\mathsf{Y}_\Delta| + |\Delta_K(\mathbf{1}) - \mathsf{c}_{n_*}| \le (2b_\Delta + \varepsilon^{\mathcal{W}})\alpha^{-6}$ and $\mathsf{M}_6 \le (1 + \mathsf{M}')\alpha^{-6}$; and $\int_{\{\alpha<\varrho\}} \alpha^{-12} d\mathbb{P}_* \le \varrho^{12}\mathsf{M}'$ (Lem. 4.20(e)) at $\varrho = a_{K-N_0}$ gives a factor Cr^6.

6. *Together.* $\mathsf{M}_b \le a_*^{-b} + \mathsf{M}' \le C(g, \mathsf{b})\ell_\epsilon^b\mathsf{d}^{-b}$, $\epsilon^{1/2}\ell_\epsilon^{10} \le C$ and $r^{3-\beta_{\mathrm{H}}/72}(1 + \log\frac{1}{r})^{18} \le C$ give $|\mathsf{c}S_3^T - S_2^T| \le C_{\mathrm{A}}(g, \mathsf{b})\mathsf{d}^{-18}[r^{\beta_{\mathrm{H}}/72} + (a_K/r)^{1/18}]\mu_3$. If the smallness $\epsilon^{\mathrm{e}}_i \le 1$ of the single-source arrays

of the near pair, which Lem. 4.36 asks at the levels $\mathsf{k}(\frac{r}{2}) \le i < n_1$, fails, then, as $\sum_{\mathsf{k}(r/2)\le i<n_0} \epsilon_i^{\mathrm{e}} \le C_\omega[2.2\gamma_{n_0} + \frac12 s(\frac{r}{2})\mathsf{c}^\flat\theta^{[0]}] \le C(g,\mathsf{b})\,\epsilon$ (with $\gamma_{n_0} \le C(\mathsf{b})r^{1/4}$ and $s(\frac{r}{2})\theta^{[0]} \le C(\mathsf{b})(a_K/r)^{1/4}$), the number ϵ exceeds a number $\epsilon_0(g,\mathsf{b}) > 0$, the bracket is $\ge (\frac12\epsilon_0)^{1/2}$, and the bounds of Lem. 4.21 give the claim with a larger C_{A}.

(E.1). By Lem. 4.20(c) for the two lists,

$$[\mathsf{c}S_3^T - S_2^T] - [\mathsf{c}\mathsf{J}_3 - \mathsf{J}_2] = \int \mathsf{Y}_{\Delta 3}\,d\mathbb{P} + \kappa_{\mathbb{P}}(\mathsf{Y}_\Delta, \mathsf{Y}_3) + \mathsf{E}, \qquad \mathsf{E} = \mathsf{c}[\kappa_{\mathbb{P}}(\mathsf{Y}_{13}, \mathsf{Y}_2) + \kappa_{\mathbb{P}}(\mathsf{Y}_{23}, \mathsf{Y}_1) + \kappa_{\mathbb{P}}(\mathsf{Y}_1, \mathsf{Y}_2, \mathsf{Y}_3)],$$

and Lem. 4.21 gives $|\mathsf{E}| \le C\Lambda_3^3\mathsf{M}'\mathsf{b}^3\mathsf{d}^{3/2}r^3 \cdot \mathsf{b}^{-2}\mathsf{d}^{-6}\mu_3$. For $r \le r_{\mathcal{W}}$ the first bracket is bounded by (E.2) and the second by Lem. 4.40 (if $w_3 \equiv 0$ all members vanish): together $C(g,\mathsf{b})\mathsf{d}^{-12}[r^{\beta_{\mathrm{H}}/72} + (a_K/r)^{1/18}]\mathsf{b}^{-2}\mathsf{d}^{-6}\mu_3$, and $|\mathsf{E}|$ is of this form as $r^3 \le r^{\beta_{\mathrm{H}}/72}$. For $r > r_{\mathcal{W}}$, Lem. 4.21 bounds the left member of (E.1) by $C(\mathsf{b})\bar{\mathsf{M}}^{(24)}[\mathsf{d}^{-9/2}+1]\mu_3$, while the bracket is $\ge r_{\mathcal{W}}^{\beta_{\mathrm{H}}/72}$; the larger of the two quotients so formed is $\mathsf{C}_{\mathrm{R}}(g,\mathsf{b};\mathsf{d})$.

(E.1) under (Γ3). The stronger bounds stated in (E.1) are Lem. 4.38, at the levels $n \ge n_0$, that is at the levels with $a_{K-n} \ge r^{1/2}$, with $\mathsf{d}' = |c-z| \in [(1-\frac{1}{64})\mathsf{d}, 6\mathsf{d}]$. Let $r \le r_\alpha^{\mathcal{W}}(g,\mathsf{b})$. On the part of Ω with $t' = n \ge n_0$ the measure $\mathbb{P}$ is absolutely continuous with respect to m_n, $\mathsf{Y}_{\Delta 3} = \Gamma_n(U_n(W)) - \Gamma_K(\mathbf{1})$ and, with $\mathsf{c}_* = \mathsf{c}_K - \Delta_K(\mathbf{1})$, $\mathsf{Y}_\Delta - \mathsf{c}_* = \Delta_n(U_n(W)) - \mathsf{c}_K$, so that Lem. 4.38 bounds both by $\varepsilon^{\mathcal{W}}$ times their natural sizes, $\mathrm{nat}_n^{\mathcal{W}} \le \alpha^{-4}$; $\kappa_{\mathbb{P}}(\mathsf{Y}_\Delta, \mathsf{Y}_3) = \int(\mathsf{Y}_\Delta - \mathsf{c}_*)(\mathsf{Y}_3 - \int \mathsf{Y}_3 d\mathbb{P})\,d\mathbb{P}$, and $|\mathsf{c}_*| \le C(g,\mathsf{b})$, since $\mathbb{P}(\alpha < \alpha_1) \le \frac12$ for $\alpha_1 = (2\bar{\mathsf{M}}^{(24)})^{-1/24}$ (Lem. 4.20(e)), while $a_{K-n_0} < Lr^{1/2} \le \alpha_1$, so that on the set $\{\alpha \ge \alpha_1\}$, of positive measure, both $|\mathsf{Y}_\Delta - \mathsf{c}_*| \le 2\varepsilon^{\mathcal{W}}\alpha_1^{-4}$ and the bound of Y_Δ by Lem. 4.21 hold.

The parts with $t' < n_0$ have $\alpha < a_{K-n_0} < Lr^{1/2}$, where nothing cancels and the moments of Lem. 4.20(e) give the factors r^6 and r^9; for $r > r_\alpha^{\mathcal{W}}$ Lem. 4.21 bounds the left member while the bracket is $\ge (r_\alpha^{\mathcal{W}})^{\beta_{\mathrm{H}}/72}$. Together the left member of (E.1) is at most $C(g,\mathsf{b})\mathsf{d}^{-5/2}[r^{\beta_{\mathrm{H}}/72} + (a_K/r)^{1/18}]\mathsf{b}^{-2}\mathsf{d}^{-6}\mu_3$.

(E.3). 1. For $\mathsf{b} \in I(g)$ every $(K, L^K\mathsf{b})$ is a first-passage datum at g with the label b. 2. *The pieces.* Cut φ_2 by the indicators of the cubes of side $\frac{r}{4}$; at most $9^3(\mathsf{d}/r)^3$ of them meet its support, and as $|y_1 - y_2| \le 3\mathsf{d}$ the centre z of each has $|c-z| \le 5\mathsf{d} + \frac{\sqrt3}{8}r \le 6\mathsf{d}$ for $c \in \operatorname{supp}\varphi_1$, so every piece is a w_3 of the data. By (E.2), summed over the pieces and integrated against $\varphi_1(c)\,\mathrm{d}c$, $\int|\varphi_1| \le \frac{4\pi}{3}\mathsf{d}^3\|\varphi_1\|_\infty$:

$$\Big|\mathsf{A}_{r,K} - \int \mathrm{d}c\,\varphi_1(c)\,S_2^T(w_T^c, \varphi_2)\Big| \le C(g,\mathsf{b})\mathsf{d}^{-12}[r^{\beta_{\mathrm{H}}/72} + (a_K/r)^{1/18}]\|\varphi_1\|_\infty\|\varphi_2\|_\infty.$$

3. *The smeared tensor.* $\int \mathrm{d}c\,\varphi_1(c)\mathcal{O}(w_T^c) = T_\lambda(\varphi_1 * \chi^{r/8})$ exactly, $\|\varphi_1 * \chi^{r/8} - \varphi_1\|_\infty \le \frac{r}{8}\operatorname{Lip}\varphi_1$, and the two-point bound of Lem. 4.21 in the sup norms (supports at a distance $\ge \frac12\mathsf{d}$, after cutting $\varphi_1 * \chi^{r/8} - \varphi_1$ into pieces) gives $|\langle T_\lambda(\varphi_1 * \chi^{r/8} - \varphi_1); \mathcal{O}(\varphi_2)\rangle| \le C(\mathsf{b},\mathsf{d})\,r\operatorname{Lip}\varphi_1\|\varphi_2\|_\infty$; this is (i), with $r_\sharp$ the least of the thresholds of (E.2) and of the data. Steps 2 and 3 read the datum through (E.2) and Lem. 4.21 at it; so (i) holds, with the same proof, at every first-passage datum (K,β) at g whose label $\mathsf{b}' = a_K\beta$ lies in $(g^{-2}, \bar{\mathsf{b}}]$, with $\mathsf{A}_{r,K}$ formed at (K,β) with the prefactor $\frac{2\pi}{3}\mathsf{b}'r^3$ and with the numbers $r_\sharp$ and $\mathsf{C}(\mathsf{d})$ formed at b'. 4.

(ii). At a fixed r, $\mathsf{A}_{r,K} \to \mathsf{A}_r$ as $K \to \infty$: for each c, $S^T_{3;K,\ell_g,\beta_K}(f_+^c, f_-^c, \varphi_2) \to S_3^{(\ell_g,\mathsf{b})}(f_+^c, f_-^c, \varphi_2)$ by Thm. YM3_I:A (Lipschitz functions with disjoint supports), with a bound uniform in c and K (Lem. 4.21 and Prop. 4.19), and dominated convergence in c. Hence, by (i),

$$\limsup_{K,K'\to\infty} \big|\langle T_\lambda(\varphi_1); \mathcal{O}(\varphi_2)\rangle_{K,\ell_g,\beta_K} - \langle T_\lambda(\varphi_1); \mathcal{O}(\varphi_2)\rangle_{K',\ell_g,\beta_{K'}}\big| \le 2\mathsf{C}(\mathsf{d})r^{\beta_{\mathrm{H}}/72}\mathsf{b}^{-2}\|\varphi_1\|_\infty\|\varphi_2\|_\infty,$$

and $r \to 0$ shows that the sequence is a Cauchy sequence; $s(\frac{r}{2}) \le K$ and $128a_K \le r$ hold for all large K, so the conditions of (i) on the cutoff hold along the sequence. 5. *(iii):* $K \to \infty$ in (i). As g satisfies (Γ), the limit points of (C.2) along data with $\beta_j = L^{K_j}\mathsf{b}$ all give the form t_λ on these pairs.

(E.4). Let $\mathsf{b}_j = a_{K_j}\beta_j \to \mathsf{b} \in I(g)$, and let $\mathsf{A}_r^{(j)}$ be $\mathsf{A}_{r,K}$ formed at (K_j,β_j) with $\frac{2\pi}{3}\mathsf{b}_j r^3$. As $g^{-2} < \mathsf{b} < \bar{\mathsf{b}}$, for j large the labels lie in a compact subset of $(g^{-2},\bar{\mathsf{b}}]$. The bound (i), in its form at every label of $(g^{-2},\bar{\mathsf{b}}]$ (step 3 above; a label b_j may lie outside $I(g)$), holds at (K_j,β_j) with the prefactor $\frac{2\pi}{3}\mathsf{b}_j r^3$, and by Lem. 4.41 its constants and thresholds are uniform in j. At a fixed r, $\mathsf{A}_r^{(j)} \to \mathsf{A}_r$: the prefactor $\frac{2\pi}{3}\mathsf{b}_j r^3$ tends to $\frac{2\pi}{3}\mathsf{b} r^3$; for each c, $S^T_{3;K_j,\ell_g,\beta_j}(f_+^c,f_-^c,\varphi_2) \to S_3^{(\ell_g,\mathsf{b})}(f_+^c,f_-^c,\varphi_2)$ by Thm. YM3_I:A, one limit along all data whose labels tend to b; and the bounds of Lem. 4.21 on $|S_3^T - \mathsf{J}_3|$ and on $|\mathsf{J}_3|$, summed over the pieces of φ_2, are uniform in c and in j, as $\mathsf{b}_j \le \bar{\mathsf{b}}$, so that dominated convergence applies in c. So

$$\limsup_j \big|\langle T_\lambda(\varphi_1);\mathcal{O}(\varphi_2)\rangle_{K_j,\ell_g,\beta_j} - t_\lambda(\varphi_1,\varphi_2)\big| \le \limsup_j \big(|\langle T_\lambda;\mathcal{O}\rangle_j - \mathsf{A}_r^{(j)}| + |\mathsf{A}_r^{(j)} - \mathsf{A}_r|\big) + |\mathsf{A}_r - t_\lambda| \le C\, r^{\beta_{\mathrm{H}}/72},$$

and $r \to 0$. □

Proof of Thm. L. (a) Prop. 4.19(c), Lem. 4.21 at $n=4$, and Prop. 4.28(a) at $n=4$. (b) The assumptions of (b) give $\operatorname{supp} f_a \subset \mathbb{B}(x_a,\rho)$, with the balls $\mathbb{B}(x_a,\rho)$ at least r apart in each pair and at least d apart across, as Lem. 4.22(b) asks; the $w_a = f_a \circ x$ form two pairs at (ρ,r,d) with the centres x_a, with the bounds $|x_1-x_2|, |x_3-x_4| \le 8r$ and $|x_a - x_b| \le 18\mathsf{d}$, $a \le 2 < b$, of the data; and $8a_K \le \rho$ by $s(\rho) \le K$ (Lem. 4.5(b)) and $\mathsf{d} \le \mathsf{d}^+$, as (b) notes. The bounds of (a) on $|S_4^T - \mathsf{J}_4|$ and on $|\mathsf{J}_4 - \frac{3}{2}\sum_\varsigma \operatorname{Tr}\prod_l (w_{\varsigma(l)}\mathsf{b}^{-1}C_K^{\mathrm{free}})|$, the identity $\frac{3}{2}\sum_\varsigma \operatorname{Tr}\prod_l (w_{\varsigma(l)}\mathsf{b}^{-1}C_K^{\mathrm{free}}) = 3\mathsf{b}^{-4}\cdot\frac{1}{2}\sum_\varsigma \operatorname{Tr}\prod_l ((f_{\varsigma(l)}\circ x)C_K^{\mathrm{free}})$ and Lem. 4.22(b) give (b). (c) For a point of $\mathcal{D}_4$ take the open balls of radius $\frac{1}{4}r_*$ about the x_a, $r = \frac{1}{2}r_*$, $\mathsf{d} = \frac{1}{2}\mathsf{d}_*$, and bumps f_a supported in balls $\mathbb{B}(x'_a,\rho)$ of radius ρ that lie inside them: these balls are $\ge r$ apart in a pair and, since $r_* \le \mathsf{d}_*$, $\ge \mathsf{d}$ apart across, as (b) asks, and the $f_a \circ x$ form two pairs at (ρ,r,d) with the centres x'_a (in a pair the centres are $\le \frac{5}{2}r_* = 5r$ apart, across $\le 11\mathsf{d}$), all inside a ball of radius $6\mathsf{d}_* = \frac{1}{2}\mathsf{d}^+$; then (b), the limit $K \to \infty$ at the label b (Thm. YM3_I:A), Lem. 4.23 at $p=4$ with $(\frac{3}{\pi})^4 < 1$, and $r^{-6}\mathsf{d}^{-6} = 4096\, r_*^{-6}\mathsf{d}_*^{-6}$.

The rings (1234), (1243) of k_4 have the size $r_*^{-6}\mathsf{d}_*^{-6}$, the ring (1324) the size d_*^{-12}. □

4.1.8. *Scope.* Every statement of this section is made at short distances, on one torus or in states at a cutoff and in their limits, and the constants that depend on the side grow with it. The side of the torus enters through conditions that ask it to be large against the lengths in play, and through constants that grow with it, among them: the conditions that the supports lie in a ball small against it; the conditions that the boxes and the cubes of the local gauges do not wrap around the torus, namely $4\rho+4$ at most the side in Lem. 4.12, in the units of the level, the case distinction of step 5 of the proof of Prop. 4.13 on whether its box fits in the torus, which reads $\ell \ge 8$, $(4\mathsf{h}_i+4)a_{K-i-1} \le \ell_g$ in (α2) and in Def. 4.1(6), and $12a_{K-n} \le \ell$, $(8\varrho+2)a_{K-n} < \ell$ for the regular gauge on a cube, read at $\varrho = 4$ in Lem. 4.36 and at $\varrho = 3$ up to $n = K$ in the proof of the bounds of (E.1) under (Γ3), where $26 < \ell_g$; the bound $L^m \ge LMy_K$ of the sides of $\mathfrak{L}(g)$, which gives $\ell_g \ge 338$, in the tails of Lem. 4.8, in Prop. 4.13, in Lem. 4.40 and for $26 < \ell_g$; the distance $\frac{1}{4}\ell_g \ge 6.5$ of two mirror copies in Lem. 4.27; the membership $\ell \in \mathfrak{L}(g)$ wherever the objects of the theory are entered, as in the frame of Lem. 4.5; the terms in d/ℓ of the comparison of C_K^{free} with $\mathbb{K}_3$; the moments of Lem. 4.20(e); and the constant Q_0 of Lem. 4.25.

In the bond forms the members that carry the renormalisation group, the remainder and the induction of the proof of Prop. 4.34, read the distances through the floor $\min\{\frac{6}{7}\mathsf{d}', t_{\mathrm{N}}(g)\}$; the largest distance $\frac{8}{7}\mathsf{d}'$ enters the one explicit member, the comparison of the sum of the slices with C_K^{free} (Lem. 4.17(b)), which is polynomial in that distance and a part of $\epsilon_\sharp^{\mathrm{ev}}$. Thms. YM3_I:AJ and YM3_I:AD enter through the two-sided stability bound of the torus measures and the chessboard bound of Lem. 4.25, in (A.4), (A.5), (C.1)(iii), (C.4) and for the states of Thm. L(a); Thm. K, like every statement of this section, uses Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD.

The crossing functionals of Lem. 4.39 are joint cumulants of the steps given the block fields of a scale below d. The comparisons with t° and with the kernels t_λ and t_n are additive, in the unit $\mathsf{b}^{-2}\mathsf{d}^{-6}$; $\mathsf{t}^\circ(n,X)$ vanishes on the cone $3(n\cdot\hat{X})^2=1$. On a torus a kernel is a class modulo null sets, while at fixed (c,z) the triples $(c\pm\frac{r}{2}n,z)$ form a null set; there (C.1)–(C.3) and (D.3) are statements almost everywhere, or through means, tents and cones of directions. For the laws on $\mathbb{R}^3$ satisfying the continuity hypotheses in (A.4)(ii) and (C.4), including the periodic-Gibbs limits and their canonical components, s_3 and s_2^T are continuous, and the kernels $\tau^\sigma_{\lambda\mu}$ and t^σ_λ are classes modulo null sets: (C.1)(iii) holds at every point and for every unit vector n; (C.4)(c) holds for s_3 itself, without means, for every $0<r\le\frac12\mathsf{d}$ and every λ, at every (c,z) off one null set fixed before r is named, and at every (c,z) with the limit forms at the tents χ^ρ_c,χ^ρ_z, $0<\rho<\frac18\mathsf{d}$, in the place of $t^\sigma_\lambda(c,z)$ and $21\rho/\mathsf{d}$ added in the bracket.

The kernels $\tau_{\lambda\mu}$, t_λ, t_n of (C.2)–(C.4) and (D.2) belong to limit points, each chosen once for all pairs of bumps; at a label of $I(g)$, for $g\le\gamma_{\mathrm{R}}$ and on the least torus, the forms $\langle T_\lambda(\varphi_1);\mathcal{O}(\varphi_2)\rangle$ of (E.3) have one limit along all cutoffs.

Step 9: narrow bumps; the kernels of the plane fields. In this step, as in Lems. [7, Lem. C.15] and [7, Lem. C.16], in [7, Thm. BV](e), in Thm. B(c), (d) and in Step 14(c), (d), the distance of two sets is the Euclidean one, $\operatorname{dist}(A,B)=\inf_{x\in A,y\in B}|x-y|$, and the separations of sets are read with it; the factor $1/\sqrt{3}$ between it and the sup distance is paid inside the proofs of these lemmas, and the numbers $\frac78d'$, $\frac67d'$, $\frac32t$, $\frac52t$, $3t$, $(2r/d')^6$ and $E_2(g,\frac12d')$ below are those of this reading. Let first g be below the threshold of [7, Lem. C.15], $\ell\in\mathfrak{L}(g)$, and let the two bumps be read as mixed sources carried by $B(x_a,r)$, $|x_1-x_2|=d'$, $16r\le d'\le\frac{\ell}{8}$; their balls are $\ge\frac78d'$ apart.

For d' below a threshold depending on g, [7, Lem. C.15](a), (b) at the separation $d=\frac67d'$, the distance $d'=\frac76d$ of the centres lying between d and $18d$, as (a) asks, and below $8d$, as (b) asks: the cumulant of the two bumps is its free lattice value up to $\varepsilon_2(g,d)(r/d)^6\mathsf{b}^{-2}|\mathfrak{s}_1||\mathfrak{s}_2|$ plus the part of the large fields, which on T_ℓ is at most $C(g,\ell)(r/d)^6d^{3/2}\mathsf{b}^{-2}|\mathfrak{s}_1||\mathfrak{s}_2|$; and $(r/d)^6=(\frac76)^6(r/d')^6$. Distances d' above that threshold, up to $\frac{\ell}{8}$: the argument of [7, Lem. C.15](a), (b) at a fixed small separation $d=t(g)$, a threshold depending on g, asks no ceiling on the actual distance of the two balls but that they lie in a ball of radius $<\frac14\ell$, and gives a tolerance that is a finite function of that distance; where the bumps are wider than the widths that the law admits at the separation d, both members being bilinear, the sources are cut into pieces carried by the cubes of a grid of a mesh depending on g, the number of pairs of pieces being at most a basic multiple of the sixth power of the width over the mesh, and the sum of the laws of the pairs of pieces costs, against $(r/d')^6$, the finite factor $(d'/d)^6$ times a basic number.

So $\varepsilon^{\mathrm{cl}}(g,\ell,\cdot)$ is a finite function on $(0,\frac{\ell}{8}]$, and $\varepsilon^{\mathrm{cl}}(g,\ell,d')\le C(g,\ell)(\log\frac{1}{d'})^{19\mathfrak{p}_0}d'^{1/2}$ for small d', the large fields adding $C(g,\ell)d'^{3/2}$. For Lipschitz bumps the lattice sums against the free kernel, $\sum_{\lambda(p)=\lambda,\lambda(q)=\mu}\phi_1(x(p))\phi_2(x(q))C^{\mathrm{free}}(p,q)^2$, differ from $\iint\phi_1\phi_2[K_3]^2_{\lambda\mu}$ by at most $C_\flat(\frac{a_K}{\vartheta r}+\frac{d'^3}{\ell^3})(\frac{r}{d'})^6|\phi_1||\phi_2|$, by the comparison of the lattice kernel with the free kernel (equations YM3_I: (2245)–(2246)), the term $a_K/(\vartheta r)$ being the Lipschitz defect of a Riemann sum of a bump of width r and $(d'/\ell)^3$ the images on the torus; the off-diagonal bond forms likewise with $C'_\flat$, the factor 8 coming from the four corner forms of a bond and the two directions.

Couplings $0<g\le\gamma$ above the threshold of [7, Lem. C.15] are reached by the dilation of Step 8, whose arithmetic reads $g^{-2}>\bar{B}$ alone and whose two couplings lie below any basic threshold once L^zg^{-2} is large, at L^zd' and $L^z\ell$, both members of the law being multiplied by L^{-2z}; this defines $\varepsilon^{\mathrm{cl}}$ and $K_\circ(g)$. The tolerance depends on d' alone: there is no gain as $r\to0$ at fixed d'. In every periodic Gibbs state, for g below a basic threshold: put $t=\frac12d'$; bumps of radius r in the balls of radius $\frac18d'$ about two points at the distance d' have supports $\ge\frac32t$ apart, at centres $\le\frac52t$ apart, in one ball of radius $3t\le\frac18\ell_g-1$ when d' is below a threshold depending on g.

On T_{ℓ_g}, [7, Lem. C.15](a) at $(\rho, d') = (r, t)$ gives the law of the plaquette forms with $\varepsilon_2(g,t) + C(g)t^{3/2}$ in units of $(r/t)^6 \mathsf{b}^{-2}|\phi_1||\phi_2|$; [7, Lem. C.16](b) at $n = 2$, $d_\times = t$, carries it to every periodic Gibbs state with the further $C_2^{\mathrm{tr}}(g)(r/t)^6 t^{3/2}|\phi_1||\phi_2|$, at most $C(g)t^{3/2}$ in these units, as $\mathsf{b} \le L(g^{-2} + \bar{B})$; and the Riemann sums on T_{ℓ_g} add $\frac{3}{2}C_\flat(\frac{a_K}{\vartheta r} + (\frac{t}{\ell_g})^3)$ in the same units. This is the law in every state with $E_2(g,t) = \varepsilon_2(g,t) + C(g)t^{3/2} + \frac{3}{2}C_\flat(t/\ell_g)^3$ at $t = \frac{1}{2}d'$ for $\varepsilon^{\mathrm{cl}}$, and $(r/t)^6 = (2r/d')^6$; $E_2(g,t) \le C(g)(\log \frac{1}{t})^{19\mathfrak{p}_0} t^{1/2}$.

First take the chosen continuum subsequential limit at each fixed positive width; then let the widths tend to 0. The estimates are uniform in the underlying states. The limits have bounded Borel kernels within $(\frac{3}{\pi})^2 \varepsilon_\sigma(g, |x-y|)\mathsf{b}^{-2}|x-y|^{-6}$ of the free ones almost everywhere, by Lebesgue's differentiation theorem, functions of $x - y$ alone by the translations; for a limit at g the kernels are continuous ([7, Thm. BW](d)), so the bound holds at every pair; the sum over λ, μ is the scalar kernel, the trace part being the scalar limit.

4.2. **Sources and the first large-field region.** The comparison in Prop. 6.6 needs a source domain that does not shrink when an early large-field region is created. We construct the source dependence with the real Gaussian structure at every step. Its total constant term agrees near zero with that of the weighted progression; the individual localized families need not agree.

Fix the real canonical run $x_0, \dots, x_K$, its counterterms $e_i = L^{-1}x_{i-1} - x_i$, and its all-small-field histories. We retain the global core, actual-sum-rule, source-radius and normalizer hypotheses of Lem. YM3_I:114.1 and Prop. YM3_I:114.2, together with the canonical history tests of Prop. YM3_I:123.1. In particular $|e_i| \le B_\sharp$, the source potential $\mathbb{W}^{\mathrm{src}}[v]$ has its actual, rather than merely formal, sum rule, and all steps use their prescribed gauge factors and density tests. Write $E_{\mathrm{s}} = E_0^{\mathrm{s}}$ and $N_{\mathrm{s}} = N_\star^{\mathrm{s}}$. The numerical conditions used below are

$$E_{\mathrm{s}} \ge 2E_{\mathsf{N}}, \qquad N_{\mathrm{s}} \ge \frac{L}{L-1}(E_{\mathrm{s}} + \tfrac{3}{2}\alpha_0^2 B_\sharp),$$
$$K_{\mathbf{M}}\big(\delta(\gamma^{-2}) + \mathsf{s}_\gamma N_{\mathrm{s}} + e^{-\gamma_2 \mathfrak{b}(\gamma^{-2})^2/2}\big) \le E_{\mathrm{s}}/4, \qquad 8K_\Lambda \mathsf{s}_\gamma N_{\mathrm{s}} \le w_\star, \qquad 8K_P \mathsf{s}_\gamma \le w_\star,$$

where $\mathsf{s}_\gamma = (1+\beta_{\mathrm{s}})\delta(\gamma^{-2})/\alpha_0$. We also retain $4\beta_{\mathrm{s}}\kappa \ge 1$ and the normalizer decay bound $\delta_{\mathsf{N}} M \ge \kappa + 1$. Put

$$K^{\mathrm{d}} = \tfrac{351}{2}L^3 K_\Pi \le K_{\mathbf{M}}^{\mathrm{f}}, \qquad \mu_\star = \min\{w_\star/32, E_{\mathrm{s}}/(16K_{\mathbf{M}}^{\mathrm{f}})\}, \qquad \sigma_{\mathrm{src}} = \mu_\star/K_{\mathrm{Q}}^\vee.$$

Here $K_{\mathbf{M}}^{\mathrm{f}}$ and $K_{\mathrm{Q}}^\vee, K_{\mathrm{L}}^\vee$ are the constants in that source construction. The level-wise source domain is the open unit ball of

$$\|w\|_{\mathrm{lw}} = \max_{0 \le j < K} \max\left\{ \frac{|v_j^w|_\infty}{\sigma_{\mathrm{src}}}, \frac{\mathrm{sw}_j(v_j^w)}{x_j^{-1/2}\mu_\star/K_{\mathrm{L}}^\vee} \right\}, \qquad v_j^w = -L^{K-j} w/x_j. \tag{4.viii}$$

The swing compares bare plaquettes whose barycentres have sup distance at most M in level-j units. Since the amplitude is included, this is a norm on the finite-dimensional complex space of bare plaquette weights.

Proposition 4.42 (The sourced history density)**.** On $\|w\|_{\mathrm{lw}} < 1$ there are local pointed families $F_j[w]$, $1 \le j \le K$, jointly holomorphic in the source and in every open core slice, with

$$\|F_j[w]\|_{j,\kappa+1} \le E_{\mathrm{s}}, \qquad \|\Delta F_j[w]\|_{j,\kappa+1} \le 2E_{\mathrm{s}}\|w\|_{\mathrm{lw}}, \qquad \Delta F_j[w] = F_j[w] - F_j[0]. \tag{4.ix}$$

A member indexed by X depends only on the source plaquettes in X. At zero source these are the ordinary progression's families. Define the finite sums

$$\mathcal{E}_t(w) = \sum_{j \le t} \sum_{X,z} \Delta F_j[w](X, \mathbf{1}, z), \qquad \mathcal{E} = \mathcal{E}_K, \tag{4.x}$$

$$\mathsf{R}_t(w, W) = \mathcal{O}(w)(U_t(W)) + \sum_{j \le t} \sum_{X,z} [\Delta F_j[w](X, U_t(W), z) - \Delta F_j[w](X, \mathbf{1}, z)]. \tag{4.xi}$$

At $t = 0$ set $\mathcal{E}_0 = 0$ and $\mathsf{R}_0(w, U) = \mathcal{O}(w)(U)$. If m_t^w is the all-small-field history measure with bare input $e^{\mathcal{O}(w)}\rho_0\,\mathrm{d}U$, and $m_t = m_t^0$, then

$$m_t^w = e^{\mathcal{E}_t(w)+\mathsf{R}_t(w,\cdot)}m_t. \tag{4.xii}$$

For $t \geq 1$ the right side is evaluated on the passing density set and is zero elsewhere. Moreover, on the common neighborhood of zero in any finite set of source parameters, $\mathcal{E}_t$ equals the total source energy of the weighted progression of Prop. YM3_I:122.1. In particular its mixed second derivative at $t = K$ is J_2. ∎

Proof. Starting with the empty array, use the actual old-potential rule and the canonical localized step at x_j, with $\tau = 1$:

$$\mathfrak{f}_j[w] = ((F_i[w], e_i)_{1\leq i\leq j}, \emptyset), \qquad F_{j+1}[w] = \mathbf{N}^{(j+1)} + \mathbf{F}[\mathbb{V}'[\mathfrak{f}_j[w]] - \mathbb{W}^{\mathrm{src}}[v_j^w]].$$

This is the real-Gaussian source recursion of Prop. YM3_I:114.2, with the counterterms of the running coupling retained. Their contribution to the array size is explicit:

$$N(\mathfrak{f}_j[w]) = \sum_{i=1}^{j} L^{-(j-i)}(\|F_i[w]\|_{i,\kappa} + \tfrac{3}{2}\alpha_0^2|e_i|) \leq \frac{L}{L-1}(E_{\mathrm{s}} + \tfrac{3}{2}\alpha_0^2 B_\sharp) \leq N_{\mathrm{s}}.$$

The induction therefore has the same ceilings as that proposition. The old potential has norm at most $w_\star/8$. Lem. YM3_I:114.1 and eq. 4.viii give

$$\mathsf{n}^{\mathrm{f}}(\mathbb{W}^{\mathrm{src}}[v_j^w]) \leq 2K_{\mathrm{Q}}^\vee|v_j^w|_\infty + K_{\mathrm{L}}^\vee x_j^{1/2}\,\mathrm{sw}_j(v_j^w) \leq 3\mu_\star.$$

Thus the fourth potential has norm at most $7w_\star/32$; the three internal potentials together have norm at most $3w_\star/8$. Lem. YM3_I:77.1 applies. The unperturbed output is bounded by $E_{\mathrm{s}}/4$, the normalizer by $E_{\mathrm{s}}/2$, and the perturbation estimate (1601), p. 631, bounds the source contribution by $3K^{\mathrm{d}}\mu_\star \leq 3E_{\mathrm{s}}/16$. The new norm is at most $15E_{\mathrm{s}}/16 < E_{\mathrm{s}}$. The inequalities on the decay rates give the norm with rate $\kappa + 1$ used here.

The same step is jointly holomorphic in its potentials and open core variables. Since v_j^w is linear in w, induction proves the asserted holomorphy. The local rule for a member on X uses only earlier members and source plaquettes inside X, proving locality. At $w = 0$ the source potential vanishes, so the recursion is exactly the ordinary one. Finally, on the complex line through w the holomorphic difference $F_j[\zeta w/\|w\|_{\mathrm{lw}}] - F_j[0]$ vanishes at zero and has norm at most $2E_{\mathrm{s}}$ for $|\zeta| < 1$. Schwarz's lemma proves the second bound in eq. 4.ix; the assertion at $w = 0$ is immediate.

We next verify that this recursion gives the same history operator used for the weighted progression. With $j = t - 1$, its all-small-field step factor is

$$g_t = (\chi_t \circ \mathcal{M})\,\mathfrak{q}_j^0\mathfrak{q}_t\mathfrak{G}_j[x_j]\chi_j^{\mathrm{sf}}, \qquad m_t^w = \mathcal{M}_*(g_t m_j^w).$$

The factors are bounded, nonnegative and Borel, and independent of w. Here $\mathfrak{q}_j^0, \mathfrak{q}_t$ are the compatibility tests of this history, and χ_t is its full density test. For a passing output W, every point $V_W(B)$ of the actual small-field chart satisfies

$$\mathfrak{q}_j^0(V_W(B)) = \mathfrak{q}_t(V_W(B)) = \chi_j(V_W(B)) = 1.$$

These are the chart and test properties used in the proof of Prop. YM3_I:123.1. Thus, if $F' = e^c F$ on the passing input set, the image of $g_t F'\,\mathrm{d}V$ has the pointwise chart density $\chi_t(W)e^c\mathsf{I}_j[F](W)$. Indeed, write the image integral against a bounded Borel test function, insert $\chi_t(\mathcal{M}V)$, and apply the physical chart formula to $\mathfrak{q}_j^0\mathfrak{q}_t F'$. On the chart the displayed identities replace this integrand by $e^c F$. This proves the claim for every passing W.

The source-potential sum rule in Lem. YM3_I:114.1 identifies the sum of the fourth potential with the change along that fibre of

$$S[\mathfrak{f}_j[w]](U) - x_j A^{\eta_j}(v_j^w, U).$$

Its subtracted linear terms sum to zero by criticality. Thm. YM3_I:W supplies the other three sum rules, the same Gaussian covariance and the same positive flat normalizing factor $\exp(\mathsf{cs}_j)$, independently of w. The exact cluster identity then gives the next chart integral, exactly as in Prop. YM3_I:114.2. Keeping the counterterms changes only the following action identity:

$$L^t x_t + \sum_{i=1}^{t} L^i e_i = x_0.$$

Consequently, after t steps the pointwise exponent is

$$A_t^w(U) = -x_0 A^{\eta_0}(U) + \mathcal{O}(w)(U) + \sum_{i\le t}\sum_{X,z} F_i[w](X,U,z) + \sum_{j<t} \mathsf{cs}_j.$$

To see this directly, the counterterms combine with $-x_t A^{\eta_t}$ by the displayed identity, and the flat values subtracted in the bare-array action cancel the flat values in the vacuum factor. Induction using the preceding chart identity gives density $\exp A_t^w(U_t(W))$ on the passing set. The backgrounds are in all required cores, and the local functions are invariant on their critical gauge orbits. The chart formula proves Borel measurability; the finite number of bounded core terms proves boundedness. At $w = 0$ this density is strictly positive. Dividing it out and adding and subtracting the flat values proves eq. 4.xii.

For completeness, the comparison of source energies uses this pointwise chart identity, rather than a choice of a version of a measure density. Both constructions start with the identical bare density $e^{\mathcal{O}(w)}\rho_0$ and use the identical factors g_t just displayed. The exact weighted sum rule, on its admitted parameter neighborhood, therefore evaluates the same successive chart integrals. At $W = \mathbf{1}$ all density and compatibility tests pass: their tested curvatures and their deviations from the trivial datum are zero. The canonical background is $U_t(\mathbf{1}) = \mathbf{1}$; this also follows from the trivial-triple assertion of Prop. YM3_I:51.9 and the uniqueness of the canonical orbit. Hence $\mathcal{O}(w)(\mathbf{1}) = 0$ and all centered terms vanish. The common, source-independent flat factors cancel, giving

$$e^{\mathcal{E}_t(w)} = e^{\mathcal{E}_t^{\text{weighted}}(w)}.$$

On a connected common neighborhood of zero their difference is holomorphic, takes values in $2\pi i\mathbb{Z}$, and is zero at zero. It is therefore zero throughout that neighborhood. This proves the claimed equality and the identification of its mixed derivative. □

Proposition 4.43 (Envelopes and first creation)**.** Retain the preceding hypotheses and the calibrated real history bounds of Thm. YM3_I:D. Suppose that the source is supported in one physical ball of radius 1. For $1 \le t \le K$ put

$$d_*(t) = \min\left\{ \frac{c_A \alpha_0}{2M B_3 \varepsilon_t}, \frac{L^{m+K-t} - 4}{2M} \right\} - 1. \tag{4.xiii}$$

Assume $M\alpha_0 \le \alpha_1$, $\alpha_0 + \alpha_1 \le a^\sharp$, and $d_*(t) \ge 3$. There are numbers $\mathsf{G}_t, \mathsf{H}_t$ such that

$$|\mathsf{R}_t(w,\cdot)| \le \|w\|_{\mathrm{lw}} \mathsf{G}_t, \qquad |\mathcal{E}(w) - \mathcal{E}_t(w)| \le \|w\|_{\mathrm{lw}} \mathsf{H}_t, \tag{4.xiv}$$

on the passing set, with

$$\begin{aligned} &\mathsf{G}_0 \le C b_+ L^{4K}, \\ &\mathsf{G}_t \le C\left[b_+ \varepsilon_t^2 L^{4(K-t)} + (\varepsilon_t^2/\alpha_0^2 + e^{-d_*(t)}) L^{3(K-t)} \right] && (1 \le t \le K), \\ &\mathsf{H}_t \le C L^{3(K-t)} && (0 \le t \le K). \end{aligned}$$

Here C depends only on the fixed construction constants, and b_+ is the upper coupling bound used in Prop. 6.6. On the least torus and for fixed g these imply

$$\mathsf{G}_K \le C(g), \qquad \mathsf{G}_{t-1} + \mathsf{H}_{t-1} \le C_{\mathrm{cnt}}(g) L^{4(K-t+1)}, \qquad C_{\mathrm{cnt}}(g) \ge 1. \tag{4.xv}$$

Let $\mathsf{Cr}_t(w)$ be the contribution whose first large-field region is created at step t, and set $\mathsf{w}_t = \mathsf{Cr}_t(0)/Z[0]$. Then

$$Z[w] = m_K^w(\mathcal{V}_K) + \sum_{t=1}^{K} \mathsf{Cr}_t(w), \tag{4.xvi}$$

$$\left| e^{-\mathcal{E}(w)} m_K^w(\mathcal{V}_K)/Z[0] \right| \le e^{\|w\|_{\mathrm{lw}} \mathsf{G}_K},$$

$$\left| e^{-\mathcal{E}(w)} \mathsf{Cr}_t(w)/Z[0] \right| \le e^{\|w\|_{\mathrm{lw}} (\mathsf{G}_{t-1} + \mathsf{H}_{t-1})} \mathsf{w}_t, \tag{4.xvii}$$

$$\sum_{t=1}^{K} L^{8(K-t)} \mathsf{w}_t \le C_8(g) < \infty, \tag{4.xviii}$$

uniformly in the datum at fixed g. All the sourced quantities in these formulas are holomorphic on the level-wise unit ball. ∎

Proof. Put $r = \|w\|_{\mathrm{lw}}$. The amplitude part of the norm and the coupling bounds give $|w|_\infty \le C b_+ r$. There are at most CL^{3K} bare plaquettes in the support ball. At a passing output of level t, the background is regular of that level at radius $2B_3\varepsilon_t$. Consequently

$$0 \le a_p(U_t) \le 2B_3^2 \varepsilon_t^2 L^{-4t}, \qquad |\mathcal{O}(w)(U_t)| \le C b_+ r \varepsilon_t^2 L^{4(K-t)}.$$

At $t = 0$ use $a_p \le 2$, giving $C b_+ r L^{4K}$.

We spell out the sum of centered local terms. The regular-background estimate (2261), p. 827, applies to any local pointed member E at level $j \le t$:

$$|E(U_t) - E(\mathbf{1})| \le \|E\|_X \left[C(\varepsilon_t/\alpha_0)^2 L^{-4(t-j)} M^2 (1 + d_j(X))^2 + 2\mathbf{1}_{\{d_j(X) \ge d_*(j)\}} \right],$$

where

$$d_*(j) = \min \left\{ \frac{c_A \alpha_0 L^{2(t-j)}}{2MB_3\varepsilon_t}, \frac{L^{m+K-j} - 4}{2M} \right\} - 1.$$

Apply it to $E = \Delta F_j[w]$ and use eq. 4.ix. Locality makes this member zero unless X meets a source plaquette. Anchor such a domain at a level-j cube meeting the support ball; there are at most $CL^{3(K-j)}$ choices. The rooted-domain sum with rate $\kappa + 1$ absorbs the base-point count and the polynomial $(1 + d)^2$, and its tail beyond $d_*(j)$ is at most $Ce^{-d_*(j)}$. These are the same rooted sums used in (2262), p. 827. Since

$$d_*(j) \ge L^{t-j}(d_*(t) + 1) - 1, \qquad L^{3(t-j)} e^{-d_*(j)} \le L^{-(t-j)} e^{-d_*(t)}$$

when $d_*(t) \ge 3$, the two level sums are bounded by

$$C r L^{3(K-t)} (\varepsilon_t^2/\alpha_0^2 + e^{-d_*(t)}) \sum_{j \le t} L^{-(t-j)}.$$

This geometric sum is bounded independently of t and K, proving the stated G_t bound. For the energy tail, the same anchoring and the uncentered family bound give

$$|\mathcal{E} - \mathcal{E}_t| \le C r \sum_{j > t} L^{3(K-j)} \le C r L^{3(K-t)}.$$

This includes $t = 0$; at $t = K$ the tail is zero.

The radius condition is available at every datum used in Prop. 6.6. Decrease γ_{C} so that

$$\varepsilon(x) \le c_A \alpha_0/(8MB_3) \quad (x \ge \gamma_{\mathrm{C}}^{-2}).$$

The least-torus choice has $L^m \ge LM$, and $L \ge 13$, $M \ge 1$ imply $L^{m+K-t} \ge LM \ge 8M + 4$. Both entries of the minimum in eq. 4.xiii are therefore at least 4, so $d_*(t) \ge 3$. The first-passage window bounds x_K and b_+ in terms of g, and the regularity radii are bounded by fixed constants throughout

the run. This proves eq. 4.xv. The dependence of C_{cnt} on g is allowed here; in particular it absorbs the factor b_+ in G_0.

Let g_t^{w} be the sum of the nonnegative step factors for the other outcomes, after the all-small-field prefix of length $t-1$. These factors too are source independent. Partition the histories according to their first such outcome. Summing all later extensions leaves the prefix mass, by the normalized test partition. This gives eq. 4.xvi and the explicit formula

$$\mathsf{Cr}_t(w) = \int g_t^{\mathrm{w}}\,\mathrm{d}m_{t-1}^{w}, \qquad \mathsf{w}_t = Z[0]^{-1}\int g_t^{\mathrm{w}}\,\mathrm{d}m_{t-1} \geq 0.$$

This is the decomposition (2268), p. 829. Substitution of eq. 4.xii, followed by eq. 4.xiv, proves eq. 4.xvii. The all-small-field zero-source mass is at most $Z[0]$, since it is one nonnegative summand of the partition. On each finite lattice the factors are bounded and the input source exponential is entire; integration and the preceding finite holomorphic sums prove the holomorphy assertions.

Finally the event of first creation at t is contained in $\{Z_t \neq \emptyset\}$. Lem. YM3_I:115.1 and the partition-function lower bound used immediately after it give, exactly as in (2185), p. 805,

$$\mathsf{w}_t \leq 2M^{-3}N_t \exp\{-c' A_0^2 (\log x_t)^{2\mathfrak{p}_0} + (C_{\mathrm{maj}}^{\Sigma} + 2 + C_{\mathrm{low}}(b_+))\ell_g^3\}.$$

The canonical least torus satisfies the side assumptions of that history estimate. Here $N_t = \ell_g^3 L^{3(K-t)}$, $x_t \geq L^{K-t}b^{\vee}$, and $b^{\vee} = g^{-2} - \bar{B} > 1$ after decreasing γ_{C}. With $n = K - t$, the sum in eq. 4.xviii is consequently bounded by a fixed multiple, depending on g, of

$$\sum_{n\geq 0} \exp\{11n\log L - c' A_0^2 (\log b^{\vee} + n\log L)^{2\mathfrak{p}_0}\}.$$

It converges because $2\mathfrak{p}_0 > 1$: from some index depending only on g and the fixed constants the negative term exceeds $12n\log L$, and the remaining tail is bounded by $\sum_n L^{-n}$. This gives $C_8(g)$ uniformly in K. No uniform bound as g tends to zero is required. □

Lemma 4.44 (Packed source amplitudes)**.** Let u_ι be real Lipschitz functions with disjoint nonzero sets, $\|u_\iota\|_\infty \leq A_\iota$, and $\operatorname{Lip} u_\iota \leq A_\iota/(\vartheta d)$. For $w = (\sum_\iota \zeta_\iota u_\iota)\circ x$ and $B = \max_\iota |\zeta_\iota| A_\iota$, one has

$$\|w\|_{\mathrm{lw}} \leq B/\mathsf{A}_\star(d), \qquad \mathsf{A}_\star(d) = \min\left\{\sigma_{\mathrm{src}} b^{\vee}, \frac{\mu_\star}{K_{\mathrm{L}}^{\vee}} \frac{b^{\vee}}{(b^{\wedge})^{1/2}} \left(\frac{\vartheta d}{4\sqrt{3}M}\right)^{1/2}\right\},$$

where $b^{\vee} = g^{-2} - \bar{B}$ and $b^{\wedge} = b_+$. ■

Proof. Disjointness gives $|w|_\infty \leq B$. The sum of the weighted functions has Lipschitz constant at most $2B/(\vartheta d)$: for points in different nonzero sets, bound each value by its Lipschitz constant times the distance to the complement of that set. Thus, with $n = K - j$,

$$\mathrm{sw}_j(w) \leq 2B\min\{1, \sqrt{3}ML^{-n}/(\vartheta d)\}.$$

The running-coupling bounds $L^n b^{\vee} \leq x_j \leq L^n b^{\wedge}$ now imply

$$|v_j^w|_\infty \leq B/b^{\vee},$$

$$x_j^{1/2}\,\mathrm{sw}_j(v_j^w) \leq \frac{2B(b^{\wedge})^{1/2}}{b^{\vee}} \min\{L^{n/2}, \sqrt{3}ML^{-n/2}/(\vartheta d)\}$$

$$\leq \frac{2B(b^{\wedge})^{1/2}}{b^{\vee}} \left(\frac{\sqrt{3}M}{\vartheta d}\right)^{1/2}.$$

The last inequality uses $\min(a,b) \leq \sqrt{ab}$. Insertion in eq. 4.viii proves the claim. □

Corollary 4.45 (Packed sources in finitely many balls)**.** Retain the numerical, canonical-run and actual-history hypotheses of Props. 4.42 and 4.43, and the calibrated rarity hypotheses of Prop. 4.54. Work on the least torus at a first-passage datum at fixed g. Let $d > 0$, $0 < \vartheta \leq 1$ and $\vartheta d \leq 1$. Let I

have $q \geq 1$ elements. For $i \in I$, let $F_i = \sum_\iota h_{i,\iota}$ be a finite sum of real Lipschitz functions whose nonzero sets, for all pairs (i,ι), are pairwise disjoint. Suppose

$$\|h_{i,\iota}\|_\infty \leq A_i, \qquad \operatorname{Lip} h_{i,\iota} \leq A_i/(\vartheta d), \qquad A_i > 0,$$

and that every bump vanishes off the union of $N \geq 1$ physical balls of radius 1. Use the amplitude $\mathsf{A}_\star(d)$ of Lem. 4.44, and put

$$u_i = A_i/\mathsf{A}_\star(d), \qquad w_z = \Big(\sum_{i\in I} z_i F_i\Big) \circ x, \qquad \mathbb{P} = \{z : |z_i| u_i < 1 \text{ for every } i\}.$$

Then

$$\begin{aligned} &\Omega(z) = Z[w_z] e^{-\mathcal{E}(w_z)}/Z[0] \quad \text{is holomorphic on } \mathbb{P}, \\ &\Omega(0) = 1, \qquad |[z^B]\Omega| \leq Q_{N,q}(g) \prod_{i\in B} u_i \quad (\emptyset \neq B \subset I). \end{aligned} \tag{4.xix}$$

Here $[z^B]\Omega = (\prod_{i\in B} \partial_{z_i})\Omega(0)$. The number $Q_{N,q}(g) \geq 1$ depends only on N, q, g and the fixed construction data, not on the cutoff, label, balls, packing, d, ϑ or functions. ■□

Proof. Lem. 4.44 gives $\|w_z\|_{\mathrm{lw}} \leq \max_i |z_i| u_i$. The bump index may include several functions with the same parameter z_i; the same maximum bounds each of them. Thus w_z is in the level-wise ball on $\mathbb{P}$. The finite lattice partition function is entire in its source, the source energy is holomorphic on this ball, and $Z[0] > 0$. This proves holomorphy and $\Omega(0) = 1$.

In the proof of Prop. 4.43, the support enters only through the number of bare plaquettes and the number of level-j cubes meeting it. For a union of N balls each count is at most N times its one-ball bound. The local-family bound is applied to the whole source; no decomposition of that source in the level-wise norm is needed. Consequently the proof gives envelopes

$$\mathsf{G}_t^{[N]} = N\mathsf{G}_t, \qquad \mathsf{H}_t^{[N]} = N\mathsf{H}_t, \qquad \mathsf{G}_K^{[N]} \leq NC(g), \qquad \mathsf{G}_{t-1}^{[N]} + \mathsf{H}_{t-1}^{[N]} \leq NC_{\mathrm{cnt}}(g) L^{4(K-t+1)}.$$

Split $\Omega = \Omega_\emptyset + \sum_{t=1}^K \Omega_t$ by eq. 4.xvi. The bounds eq. 4.xvii, with these envelopes, give

$$|\Omega_\emptyset| \leq e^{NC(g)} \quad \text{on } \mathbb{P}, \qquad |\Omega_t| \leq e\mathsf{w}_t \quad \text{on } \{|z_i| u_i \mathsf{Cnt}_t < 1\}, \qquad \mathsf{Cnt}_t = NC_{\mathrm{cnt}}(g) L^{4(K-t+1)} \geq 1.$$

Cauchy's formula, separately in the variables of B, yields

$$|[z^B]\Omega_\emptyset| \leq e^{NC(g)} u^B, \qquad |[z^B]\Omega_t| \leq e\mathsf{w}_t \mathsf{Cnt}_t^{|B|} u^B \leq e\mathsf{w}_t [NC_{\mathrm{cnt}}(g) L^4]^q L^{4q(K-t)} u^B, \qquad u^B = \prod_{i\in B} u_i.$$

Prop. 4.54, with exponent $4q$ and $t = K$, bounds the sum of the last weights by

$$\sum_{t=1}^K L^{4q(K-t)} \mathsf{w}_t \leq C_{4q}(g), \qquad C_{4q}(g) = 4M^{-3} \ell_g^3 e^{\mathsf{B}(X_g)\ell_g^3} \mathsf{C}_{\mathrm{ag}}(4q+1).$$

Indeed $x_K \leq X_g$, the function B in eq. 4.xxxii increases, and the remaining factor $e^{-c' p_K^2/2}$ is at most 1. The displayed bound is finite at fixed g. Therefore one may take

$$Q_{N,q}(g) = e^{NC(g)} + e[NC_{\mathrm{cnt}}(g) L^4]^q C_{4q}(g).$$

All factors have the asserted dependence. □

4.2.1. *The lattice spectral inequality at several steps.* The spectral restriction must be used at a fixed physical displacement while the lattice spacing tends to zero. Its one-step form therefore needs the following elementary extension.

Lemma 4.46. Let T be a positive self-adjoint contraction and V a commuting unitary operator. If

$$(1-T)^2 \geq T(2 - V - V^*),$$

then, for every integer $k \geq 1$,

$$(1-T^k)^2 \geq T^k(2 - V^k - V^{*k}).$$

For the lattice transfer operators of Prop. YM3_I:99.3, this implies the five-point inequality

$$u(0) - 4u(te_\mu) + u(2te_\mu) + u(te_\mu + te_\lambda) + u(te_\mu - te_\lambda) \geq 0, \qquad t = ka,$$

where $u(w) = \nu(\overline{\theta F}\, \tau_w F)$ in the corresponding reflection Hilbert space.

Conversely, suppose strongly commuting self-adjoint operators $H \geq 0$ and P have joint spectral measure $\mathcal{P}$. If, on a dense set of vectors, the bounded quadratic form

$$1 - 4e^{-tH} + e^{-2tH} + e^{-tH}(e^{itP} + e^{-itP})$$

is nonnegative for every t in a subset of $(0, \infty)$ with zero as a limit point, then $\mathcal{P}\{(E, q) : E < |q|\} = 0$. ■

Proof. At a point of the joint spectrum of T, V, write $t_0 = e^{-x} > 0$ and $v = e^{iy}$, where $x \geq 0$. The assumed inequality is $\sinh(x/2) \geq |\sin(y/2)|$. The addition formulas, by induction, give

$$\sinh(kx/2) \geq k \sinh(x/2) \geq k|\sin(y/2)| \geq |\sin(ky/2)|.$$

This is the desired inequality at t_0^k, v^k; at $t_0 = 0$ it is immediate. Joint spectral calculus proves the operator inequality. Taking its matrix element at $[F]$ and using the translation identity in Lem. YM3_I:99.1 gives the five displayed correlations.

For the continuum assertion put

$$f_t(E, q) = 4e^{-tE}(\sinh^2(tE/2) - \sin^2(tq/2)).$$

This is the real bounded function $1 - 4e^{-tE} + e^{-2tE} + 2e^{-tE}\cos(tq)$. Density extends the assumed positivity to the whole Hilbert space, so $\mathcal{P}\{f_t < 0\} = 0$. Choose a sequence $t_j \downarrow 0$ from the indicated set. Outside the union of these countably many null sets,

$$\frac{2}{t_j}\sinh(t_j E/2) \geq \frac{2}{t_j}|\sin(t_j q/2)|$$

for every j. Taking the ordinary numerical limit gives $E \geq |q|$. The spectral projection of the complementary set is therefore zero. □

4.3. **Local large-field probabilities.** The history bounds of Section YM3_I:70 include an exponential of the volume. To bound the probability that a given point belongs to a large-field region, we first attach a gauge invariant test to each possible creation. Reflecting that test across a board forces many creations. The root in the chessboard estimate then removes the volume factor.

We use the real construction of Thm. YM3_I:D, with the constants constructed in Section YM3_I:69 and the final source ceilings of Section YM3_I:82. The couplings obey $x_{j-1} = L(x_j + b_j)$, $|b_j| \leq B_\sharp$, and the lattice is admissible through level K. Write $y_j = \log x_j$, $g_j = x_j^{-1/2}$ and

$$p_j = p_0(g_j) = A_0 y_j^{\mathfrak{p}_0}, \quad \varepsilon_j = g_j p_j, \quad \delta_j = (A_1/A_0)\varepsilon_j, \quad s_j = M\check{R}_j a_{K-j}, \quad w_j = M_2\check{R}_j a_{K-j}.$$

Here $\check{R}_j$ is the least nonnegative integral power of L at least y_j^r, $M_2 = L^2 M_1$, $M \geq L^2 M_2$, $L \geq 13$, and $2\mathfrak{p}_0 > 3r + 2$. A cell of level j has side s_j; a test cube has side Lw_j, and its enlargement by a layers has side $(L + 2a)w_j$. Distances in this section are periodic sup distances. The probability of a history is $\widetilde{\nu}_k(h) = \mathfrak{t}(h)/Z$, where $\mathfrak{t}(h) = \int \mathfrak{w}_h \rho_0 \, \mathrm{d}U$, $\mathfrak{w}_h = \mathbf{1}_h \mathfrak{G}_h$, and ρ_0 is the Wilson density. The exact history sum gives $\sum_h \mathfrak{t}(h) = Z$.

Definition 4.47 (The two choices and their threshold)**.** Put

$$\kappa_{\mathrm{w}} = \min\{(4\Theta_1)^{-1}, \theta_c\}, \qquad \lambda_\natural = \frac{\kappa_{\mathrm{w}}}{2\pi}, \qquad \lambda'_\natural = \frac{\lambda_\natural}{2 + 30L}.$$

A lowered choice keeps the constants before A_1 and the value of A_0, replaces A_1 by $A_1' = \lambda_\natural' A_1$, and completes the construction with $B_\sharp' \geq B_\sharp$. Primes denote the constants of this second choice. Set

$$\begin{aligned} c_1^\natural &= \frac{c_\star'}{4 \cdot 49^3}, \qquad C_1^\natural = \mathsf{C}_{\mathrm{maj}}' + c_{\mathrm{low}}' + \frac{13}{2}, \\ c_{\mathrm{low}}' &= C_5' + 3\log 7 + 24(B_3')^2, \\ c_2^\natural &= \frac{1}{2}\min\left\{\frac{c_1^\natural}{8}, \frac{\lambda_\natural^2 A_1^2}{2A_0^2}\right\}, \\ N_\tau &= \left(\frac{M}{LM_2}\right)^3, \qquad N_\delta = \left(\frac{M}{M_2}\right)^3, \\ C_3^\natural &= 27(N_\tau + N_\delta) + 540 N_\tau \big((L+6)L^2M_2\big)^3 (2L)^{3/2}. \end{aligned} \tag{4.xx}$$

The positive constants $c_\star', \mathsf{C}_{\mathrm{maj}}', C_5'$ are those of the deposit and history bounds in Section YM3_I:70 at the second choice. Let $\bar{B} = LB_\sharp/(L-1)$, $Y_0 = \max\{\log\max(e, 2B_\sharp), 2\log(2L)/(L^{1/r}-1)\}$, and let Y_{mark}' be the threshold below equation YM3_I:(1455) with $w = \eth = 1/2$, $c = c_\star'$, $c_{\mathrm{Dep}} = c_\star'$ and $c_a = 0$, at the second choice. Explicitly it is

$$Y_{\mathrm{mark}}' = \max\left\{(Y^{\mathrm{II}})'\left(c_\star'/4, 48^3(D_0' + \lambda_L')\right), (Y^{\mathrm{S}})'(c_\star'/4)\right\}, \qquad D_0' = \mathsf{C}_{\mathrm{maj}}' + 27(2+\log(2L))C_\flat'.$$

Here $C_\flat'$ is the vacuum surface constant used in equation YM3_I:(1487); Y^{II} and Y^{S} are the explicit functions in Section YM3_I:67. Finally put

$$Y_{\mathrm{pay}}(c, C) = \max\left\{1, \log(1+\bar{B}), \left(\frac{4}{cA_0}\right)^{1/\mathfrak{p}_0}, \left(\frac{4CM^3L^3}{cA_0^2}\right)^{1/(2\mathfrak{p}_0 - 3r - 2)}\right\}.$$

The threshold $\mathsf{Y}^\natural$ is the least number at least 1 such that every $y \geq \mathsf{Y}^\natural$ satisfies

(Y.1) $y \geq \log(\gamma'^{-2})$, (Y.2) $y \geq Y_0$,

(Y.3) $y \geq Y_{\mathrm{mark}}'$, (Y.4) $y \geq Y_{\mathrm{pay}}(c_1^\natural, C_1^\natural)$,

(Y.5) $C_{\mathrm{cu}}\sqrt{2L}(A_0/A_1)e^{-\delta_{\mathrm{lc}} M_2 y^r/4} \leq \lambda_\natural$,

(Y.6) $y \geq \dfrac{\log C_3^\natural + 3r + 3/2}{c_2^\natural A_0^2}$,

(Y.7) $y \geq 2\log(1+\bar{B})$, $2c_2^\natural \mathfrak{p}_0 A_0^2 y^{2\mathfrak{p}_0 - 1} \geq 4$. (4.xxi)

The constants C_{cu} and δ_{lc} are those of the local background comparison YM3_I:(1012) at the original choice. All these numbers are independent of the lattice and the coupling sequence. We also write $\mathsf{Y}_i^\natural$ for the lower bound on y in (Y.i), taking the maximum with 1 in (Y.6). Thus $\mathsf{Y}^\natural = \max_i \mathsf{Y}_i^\natural$; in particular the four thresholds used for the witness estimate are exactly those in (Y.1)–(Y.4). ■

Lemma 4.48. For a choice made by the ordered construction of Section YM3_I:69, a lowered choice exists by resuming that construction after A_0. If a lattice is admissible at the original choice and $x_K \geq \gamma'^{-2}$, it is admissible at the lowered choice. The lattices, block averages, local background maps and their stable domains, plaquette tests, gauge densities and Wilson measure are the same for both choices. The fluctuation thresholds satisfy $\delta_j' = \lambda_\natural' \delta_j$. ■

Proof. In the ordered parameter choice, A_1 is free in $(0, \infty)$ and has no attached condition. The next constant is A_0. Its eight conditions are

$$A_0 \geq c_{\mathrm{im}} A_1, \quad A_0 > c_{\mathrm{ct}} A_1, \quad A_0 \geq C_{\mathrm{sf}} L^3 A_1, \quad A_1/A_0 \leq c_{\mathrm{Q}}, \quad A_1/A_0 \leq c_{\mathrm{no}},$$
$$A_0 \geq 4B_3 C_H A_1, \quad A_0/A_1 \geq 64L^3, \quad A_0/A_1 \geq 384\pi L^3 B_5 \mathsf{S}(\delta_{0,\mathrm{tw}}).$$

Every other constant in these conditions precedes A_1. Decreasing A_1 while keeping A_0 preserves all eight inequalities. The admissible partial choice can be completed by the following continuation of Section YM3_I:69. Rechoose C_0, C_1, then the radii $\alpha_1, \alpha_0, \alpha_2, \alpha_3, \alpha_4, a_\flat$ in the stated order, retaining the strict reserves for every difference of radii. The lower bounds on θ_c involve only the earlier fixed Θ_1, M and chart constants, so its original value may be retained. Evaluate the normalizer constants, increase E_0 as required, and choose $B'_\sharp$ above both its new lower bound and $B_\sharp$. All these are finite choices before the coupling ceiling. Finally reduce γ' so that the real numerical inequalities hold on $x \geq \gamma'^{-2}$. The last part of Section YM3_I:69 expresses them as finitely many quantities tending to zero, or finite lower thresholds, on that interval. The source conditions of Section YM3_I:82 impose only further such restrictions on this final ceiling. This constructs the two choices actually used here; no extension assertion for arbitrary partially specified lists is required.

The listed common objects depend only on the constants before A_1, on A_0, and on $(K, m, x_\bullet)$. In particular their definitions do not use the later fluctuation thresholds. The formula for δ_j gives the asserted scaling. The data conditions $\varepsilon_j \leq 1$ and $M\check{R}_j \leq L^{m+K-j}$ are unchanged, $\delta'_j \leq \delta_j$, and the same increments satisfy $|b_j| \leq B'_\sharp$. The remaining coupling condition is precisely $x_K \geq \gamma'^{-2}$. □

4.3.1. *Witnesses for a creation.* Write $V_j = \mathcal{M}^j U$. For a test cube Q of step t, set $Y_Q = (Q^{\sim 3})^{(t)}$ and dress a bond of level $t-1$ by

$$V^{\mathrm{dr}}_{t-1}(b) = V_{t-1}([b_-]_t, b_-) V_{t-1}(b) V_{t-1}([b_+]_t, b_+)^{-1}.$$

Define $F_{t,Q}(U) = 0$ exactly when all the following conditions hold: the plaquette test of Q is one; the local background $V^{\mathrm{bg}}_{t-1,Q}$ is stably defined; all contour averages $V_{t-1}(y, x)$, $y \in Y_Q$, $x \in B(y)$, are defined; and

$$|V^{\mathrm{dr}}_{t-1}(b)(V^{\mathrm{bg}}_{t-1,Q}(b))^{-1} - 1| < \lambda_\natural \delta_{t-1} \tag{4.xxii}$$

for every non-pivot bond incident to $Q^{\sim 2}$. Otherwise set $F_{t,Q} = 1$. At step one include the witnesses $F_{0,Q} = 1 - \chi_0(Q)$ for the bare density cubes. Their support cubes are respectively $Q^{\sim 4}$ and $Q^{\sim 1}$.

Lemma 4.49. Under (Y.2), these witnesses are Borel, gauge invariant, and covariant under the lattice isometries that preserve the partitions. Their support cubes have diameter at most $(L+8)w_t \leq (L+8)L^{-2}s_k$ when $t \leq k$. If also (Y.5) holds, a cell $\mathfrak{c}$ is created at step t and $\mathfrak{w}_h(U) > 0$, then either a witness whose cube meets $\mathfrak{c}$ equals one, or there is a test cube Q meeting $\mathfrak{c}$ and a non-coarse site $x \in (Q^{\sim 3})^{(t-1)}$ such that $[x]_t \in P''_t(h)$ and the defined contour variable satisfies

$$|V_{t-1}([x]_t, x) - 1| > \lambda_\natural \delta_{t-1}. \tag{4.xxiii}$$

Under (Y.1), on every history at the lowered choice with $\mathfrak{w}'_h(U) > 0$, a witness with value one has its centre in Z'_t. ■

Proof. The stable domains of the local background maps are open, and the maps are continuous there. Contour averages and iterated block averages are Borel on their domains. Thus all four tests defining F are Borel. The local background reads only the inner bonds of $Q^{\sim 4}$. The blocks needed for every contour or block average are contained in that cube, since all its faces are block faces. This proves the stated support. The bare density witness reads only the plaquettes of $Q^{\sim 1}$. The side bounds follow from $M \geq L^2 M_2$ and the monotonicity of s_t in Lem. YM3_I:67.1.

Under a gauge transformation u, the dressed bond transforms with the values $u([b_-]_t), u([b_+]_t)$ at the coarse endpoints. The local background has exactly the same transformation law, so their quotient is conjugated. The plaquette test and the domains of the contour and background maps are gauge invariant. This proves gauge invariance of F. An isometry preserving the block partitions carries each contour, pivot bond, local minimizing orbit and axial representative to the corresponding object of the image cube. It also carries the stable domain onto the image stable domain, because it is a homeomorphism intertwining the two partial maps. Thus it preserves all four tests. In particular

these conclusions apply to the isometries with translations in the lattice of spacing s_k/L: every partition used above has side at most $Lw_t \le s_k/L$, an integral power of L.

For the creation alternative, first observe that, with $w_\pm = V_{t-1}([b_\pm]_t, b_\pm)$,

$$|V_{t-1}(b) - V^{\mathrm{bg}}_{t-1,Q}(b)| \le |w_- - 1| + |w_+ - 1| + |V^{\mathrm{dr}}_{t-1}(b) - V^{\mathrm{bg}}_{t-1,Q}(b)|.$$

Consequently a bond discrepancy at least $3\lambda_\natural\delta_{t-1}$ either makes $F = 1$ or forces one endpoint contour to satisfy eq. 4.xxiii. If $Q^{\sim 3} \subset B(P_t'')$, that endpoint lies in the asserted gauge-fixing region. It cannot be a coarse site, where the contour variable is one.

The created cells are the hulls of the failed plaquette cubes, failed Q-test cubes, failed fluctuation cubes, selected intermediate bonds, and failed low-field cells; at step one there are also failed bare density cubes. A failed plaquette or bare density test directly gives a witness. A failed Q-test gives either an undefined local background or a non-pivot discrepancy at least $2\delta_{t-1}$. Its cube satisfies $Q^{\sim 3} \subset B(P_t'')$, so the preceding observation applies.

In each of the remaining three cases a non-pivot bond incident to a test cube $Q \subset \Omega_t^{\sim -1}$ meeting $\mathfrak{c}$ satisfies

$$|\mathcal{A}_{t-1}(b)| \ge \kappa_{\mathrm{w}}\delta_{t-1}.$$

Indeed the respective failed tests have thresholds δ_{t-1}, $\delta_{t-1}/(4\Theta_1)$ and $\theta_c\delta_{t-1}$. The geometry gives $Q^{\sim 4} \subset \Omega_t \subset B(P_t'')$. The regularity and P, Q tests on the support of the history meet the hypotheses of Prop. YM3_I:53.12. Its principal-log estimate gives

$$|V_{t-1}(b) - V^{\mathrm{bg,step}}_{t-1}(b)| \ge (2/\pi)|\mathcal{A}_{t-1}(b)| \ge 4\lambda_\natural\delta_{t-1}.$$

The local comparison YM3_I:(1012) is applicable on this cube: its four-layer enlargement is contained in Ω_t and both endpoints of b belong to $Q^{\sim 3}$. It gives a difference of backgrounds at most $C_{\mathrm{cu}}\varepsilon_t e^{-\delta_{\mathrm{lc}}M_2\check{R}_t/4}$. Since $\varepsilon_t \le \sqrt{2L}(A_0/A_1)\delta_{t-1}$ and $\check{R}_t \ge y_K^r$, condition (Y.5) bounds this by $\lambda_\natural\delta_{t-1}$. The discrepancy from the cube background is therefore at least $3\lambda_\natural\delta_{t-1}$, as required.

Finally suppose that a witness centre belongs to Ω_t'. Its whole test cube lies there. The lowered plaquette and Q tests are one, the local background is stably defined, and its bond discrepancy is less than $2\delta'_{t-1}$. Prop. YM3_I:53.12, now at the lowered choice, gives all contour averages and bounds them by $15L\delta'_{t-1}$. Hence the dressed discrepancy is less than $(2 + 30L)\delta'_{t-1} = \lambda_\natural\delta_{t-1}$. Thus $F = 0$. The bare density test is also one on Ω_1'. A witness with value one consequently has its centre outside Ω_t', hence in $Z_t' = (\Lambda_t')^c$. □

4.3.2. *Gauge integration and the chessboard estimate.* For a class function $0 \le \phi \le 1$, let $\mathsf{gv}_{t,x}$ equal $\phi(V_{t-1}([x]_t, x))$ when the contour is defined and $[x]_t \in P_t''(h)$, and zero otherwise. Only non-coarse sites are used. The density of one gauge variable is

$$f_j^{\mathrm{gv}}(u) = z_j^{-1}\mathbf{1}_{\{|u-1|<\varepsilon_j\}}e^{-x_j(1-\frac{1}{2}\operatorname{tr} u)}. \tag{4.xxiv}$$

Lemma 4.50. For a finite set J of distinct pairs (t, x), class functions $0 \le \phi_{t,x} \le 1$, and a bounded nonnegative gauge invariant Borel function u of the bare field,

$$\sum_h \int \mathfrak{w}_h u \prod_{(t,x)\in J} \mathsf{gv}_{t,x}\,\mathrm{d}U \le \left(\prod_{(t,x)\in J}\int_G \phi_{t,x} f^{\mathrm{gv}}_{t-1}\,\mathrm{d}v\right)\int u\,\mathrm{d}U. \tag{4.xxv}$$

For the indicator of $|v - 1| > \lambda_\natural\delta_j$, its integral q_j satisfies

$$q_j \le 20\exp\left[\tfrac{3}{2}y_j - \tfrac{1}{2}\lambda_\natural^2(A_1/A_0)^2p_j^2\right]. \tag{4.xxvi}$$

■

Proof. Induct on the last step. Fix a prefix history and the plaquette outcome of its final step. Summing the remaining outcome indicators gives at most one. The remaining last-step gauge factor is a product of the densities eq. 4.xxiv over $B(P_t'') \setminus P_t''$. Average over independent gauge

transformations at these sites, extended constantly over each finer block and equal to one at the coarse centres. This preserves the prefix, the plaquette outcome, Haar measure and u. Earlier contour factors are only conjugated, and their class functions are unchanged. A last-step contour becomes $V_{t-1}([x]_t, x)v(x)^{-1}$. If the necessary contour domains fail, its contribution is zero; otherwise Haar translation turns each site factor into $\int \phi_{t,x} f^{\mathrm{gv}}_{t-1}$, or into one at a site not in J. The plaquette indicators sum to the prefix indicator. Applying the induction hypothesis proves the displayed inequality, including $J = \emptyset$.

On SU(2), $1 - \frac{1}{2}\operatorname{tr} v = |v-1|^2/2$. The Haar radial density for $r = |v-1|$ is $(2/\pi)r^2\sqrt{1-r^2/4}\,\mathrm{d}r$. Since $x_j^{-1/2} \le \varepsilon_j \le 1$,

$$z_j \ge \frac{\sqrt{3}}{3\pi\sqrt{e}} x_j^{-3/2} > \frac{1}{20} x_j^{-3/2}.$$

On the tail set the exponential is at most $e^{-x_j \lambda_\natural^2 \delta_j^2/2}$. Now $x_j \delta_j^2 = (A_1/A_0)^2 p_j^2$, which proves the tail bound. □

Lemma 4.51. Under (Y.1)–(Y.4), every witness F of step t satisfies

$$\langle F \rangle \le e^{-c_1^\natural p_t^2}. \tag{4.xxvii}$$

More generally, at a level $k \ge t$, the same bound holds for its chessboard seminorm on any of the eight boards of side s_k with offset $(L-1)s_k\sigma/(2L)$, $\sigma \in \{0,1\}^3$, whose cell contains its support cube. These boards preserve all construction partitions. For a family of witnesses with support cubes at pairwise distance greater than s_k, their joint mean is at most $\prod_F e^{-c_1^\natural p_{t(F)}^2/8}$. ■

Proof. Use the eight boards of cell side s_k whose offsets are $(L-1)s_k\sigma/(2L)$, $\sigma \in \{0,1\}^3$. In one coordinate the walls of the two families have minimum separation $(L-1)s_k/(2L)$. An interval shorter than this separation lies between two consecutive walls of at least one family. The support diameter $(L+8)s_k/L^2$ is shorter, since $L \ge 13$. Choosing the family in each coordinate puts every witness in a cell of one of these boards. All board reflections and translations preserve its construction, by Lem. 4.49.

Let $N = (\ell/s_k)^3$ be the number of cells and let Π be the product of all reflected copies of F. Each copy is a witness of the same step t. On a history at the lowered choice with $\mathfrak{w}'_h(U) > 0$, $\Pi = 1$ forces all N witness centres into Z'_t. By Prop. YM3_I:67.2, such a centre is within $\rho_{q,t} = 7s_q + 11\sum_{j=q+1}^{t} s_j$ of a creation at some $q \le t$. The centre of its board cell is at most $s_k/2 + \rho_{q,t} + s_q/2$ from the centre of that creation. Since $2\rho_{q,t} + s_q < 48 s_t \le 48 s_k$, a creation can be assigned at most 49^3 board centres. There are therefore at least $N/49^3$ creations.

For the mark $\Theta = (c'_\star/2)\sum_q p_q^2 n_q^{\mathrm{cr}}$, this gives $\Theta \ge 2c_1^\natural p_t^2 N$. Equation YM3_I:(1487), at $w = \eth = 1/2$ and $c_a = 0$, and Prop. YM3_I:70.2 yield

$$\langle \Pi \rangle \le \exp\left[-2c_1^\natural p_t^2 N + C_1^\natural y_k (M\check{R}_k)^3 N\right].$$

The normalization is the same Wilson partition function for the two choices. Condition (Y.3) is exactly the marked-mass hypothesis. Condition (Y.4) and $\check{R}_k < L y_k^r$, $y_k \ge 1$, give $C_1^\natural y_k (M\check{R}_k)^3 \le c_1^\natural p_k^2 \le c_1^\natural p_t^2$. Thus $\langle \Pi \rangle^{1/N} \le e^{-c_1^\natural p_t^2}$. The chessboard estimate of Prop. YM3_I:86.3 gives eq. 4.xxvii and the seminorm assertion.

For several witnesses assign each to one of the eight boards and let Φ_σ be the product in that class. Distinct witnesses in a class occupy distinct cells, by the separation assumption. Since $\Phi_\sigma^8 = \Phi_\sigma$, Hölder's inequality and then the chessboard estimate give $\langle \prod \Phi_\sigma \rangle \le \prod \langle \Phi_\sigma \rangle^{1/8} \le \prod_F e^{-c_1^\natural p_{t(F)}^2/8}$. □

4.3.3. *From witnesses to points.*

Proposition 4.52. Suppose $y_K \ge \mathsf{Y}^\natural$. For any prescribed cell $\mathfrak{c}$ of level $t \le k$, the probability that it is created at step t is at most $e^{-c_2^\natural p_t^2}$. For every $x \in T_\ell$ and $1 \le n \le k \le K$,

$$\widetilde{\nu}_k\{x \in Z_n\} \le 2 \cdot 24^3 e^{-c_2^\natural p_n^2}. \tag{4.xxviii}$$

The same bound holds for a prescribed cell contained in Z_n. For $0 \le a < k$ the right side for $n = k - a$ is at most

$$2 \cdot 24^3 e^{-c_2^\natural p_k^2} L^{-2c_2^\natural \mathfrak{p}_0 A_0^2 y_k^{2\mathfrak{p}_0 - 1} a}. \tag{4.xxix}$$

■

Proof. At most $27(N_\tau + N_\delta)$ witness cubes meet $\mathfrak{c}$. Indeed a partition of side $a \le s_t$ has at most $(s_t/a + 2)^3 \le 27(s_t/a)^3$ cubes meeting it. Apply this to the test cubes, and, at $t = 1$, to the bare density cubes; their quotients are bounded respectively by $M/(LM_2)$ and M/M_2. Each test cube has at most $((L+6)LM_2\check{R}_t)^3$ sites in its three-layer enlargement. There are therefore at most $27N_\tau$ times this many possible gauge-tail sites.

Multiply the alternative of Lem. 4.49 by $\mathfrak{w}_h\rho_0$ and sum and integrate over histories. Lem. 4.50, first with $u = F\rho_0$ and then with $u = \rho_0$, and Lem. 4.51 bound the creation probability by

$$e_t = 27(N_\tau + N_\delta)e^{-c_1^\natural p_t^2/8} + 27N_\tau((L+6)LM_2\check{R}_t)^3 q_{t-1}.$$

Here weakening the first exponent to $c_1^\natural/8$ keeps the same constant as the bound for separated creations. The inequalities $p_{t-1} \ge p_t$, $y_{t-1} \le y_t + \log(2L)$ and $\check{R}_t < Ly_t^r$ imply

$$e_t \le e^{-2c_2^\natural p_t^2} C_3^\natural e^{(3r+3/2)y_t} \le e^{-c_2^\natural p_t^2}.$$

The last inequality follows from (Y.6), $y_t \ge y_K \ge 1$ and $2\mathfrak{p}_0 \ge 2$.

If $x \in Z_n$, Prop. YM3_I:67.2 supplies a creation of level $t \le n$ within distance $\rho_{t,n}$. There are at most $24^3 L^{3(n-t)}$ possible cells: their count is at most $(2\rho_{t,n}/s_t + 2)^3$, and

$$2\rho_{t,n}/s_t + 2 \le 16 + 22\sum_{j=1}^{n-t} L^j \le 24L^{n-t}.$$

We record the coupling estimate needed to sum these bounds, including the small defect from an exact geometric progression. Put $P = 2\mathfrak{p}_0$, $\Delta = y_{n-a} - y_n$, and $v = \bar{B}/x_n$. Condition (Y.7) gives $x_n \ge (1+\bar{B})^2 \ge 4\bar{B}$. The affine flow bound gives, for $a \ge 1$,

$$\Delta \ge a\log L + \log(1-v) > 2a, \qquad a\log L - \Delta \le \tfrac{4}{3}\bar{B}e^{-y_n}, \qquad y_n(a\log L - \Delta) \le \frac{8}{3e}\frac{\bar{B}}{1+\bar{B}} < 1.$$

The last inequality uses $ze^{-z/2} \le 2/e$. The first two terms of the power expansion therefore give

$$y_{n-a}^P - y_n^P \ge Py_n^{P-1}a\log L + Py_n^{P-2}\left[\frac{P-1}{2}\Delta^2 - y_n(a\log L - \Delta)\right] \ge Py_n^{P-1}a\log L.$$

By (Y.7), the rate $c_2^\natural P A_0^2 y_n^{P-1}$ is at least four. Thus the union bound over cells and levels is at most

$$24^3\sum_{a=0}^{n-1} L^{3a}e^{-c_2^\natural p_{n-a}^2} \le 24^3 e^{-c_2^\natural p_n^2}\sum_{a\ge 0} L^{-a} \le 2\cdot 24^3 e^{-c_2^\natural p_n^2}.$$

A cell event implies the point event at its centre. The same power estimate with $n = k$ proves eq. 4.xxix. □

4.4. **Gaussian restrictions and history tails.**

4.4.1. *Removing the fluctuation boxes.*

Lemma 4.53. Consider a complete Gaussian structure on a torus satisfying Thm. YM3_I:R, with at most three admitted homogeneous quadratic origins of sort Σ_{q} and no origin of sort Σ_{v}. Let ν^{r} be its normalized measure restricted to the fluctuation box of radius $\mathfrak{b}$, and let ν° be the normalized unrestricted measure with the same quadratic origins. For n insertions W_i of sort Σ_{q}, with nonzero norms $a_i = \|W_i\|_{\star,\lambda_V}$, set

$$r_i = \frac{w_\star}{n a_i}, \qquad \Pi_{\mathrm{ins}} = \prod_i r_i^{-1}, \qquad N_{\mathrm{ins}} = \sum_{X:\, X \text{ meets every insertion}} e^{-\lambda_{\mathrm{out}} d(X)}.$$

Here "meets" means that X contains the enlarged localization domain of at least one nonzero member of that insertion. Then

$$\left| \langle W_1^{\mathrm{tot}}; \dots; W_n^{\mathrm{tot}} \rangle_{\nu^{\mathrm{r}}} - \langle W_1^{\mathrm{tot}}; \dots; W_n^{\mathrm{tot}} \rangle_{\nu^{\circ}} \right| \le K_P^{\Pi} \mathsf{p}_\star \Pi_{\mathrm{ins}} N_{\mathrm{ins}}, \tag{4.xxx}$$

where, in the notation of equation YM3_I:(705),

$$\mathsf{p}_\star = \mathsf{C}_{\mathrm{ex}} ((L+2)M)^3 e^{4\lambda_1} e^{-\gamma_2 \mathfrak{b}^2/2}.$$

If $\lambda_{\mathrm{out}} \ge c_{\mathrm{E}}$, then for $n = 2$ or 4 and single field components at arbitrary bonds, the right side is at most $C_{\mathrm{bx}} e^{-\gamma_2 \mathfrak{b}^2/2}$, where C_{bx} is independent of the side, cutoff and bonds. Both measures are even, so odd joint cumulants of field components vanish. In the unrestricted measure the second cumulant is its covariance and the fourth is zero. ■

Proof. Add the single origin $\sum_i z_i W_i$ on the polydisc $|z_i| < r_i$. Its norm is at most $\sum_i |z_i| a_i < w_\star$, so the system has at most four admitted origins. Prop. YM3_I:47.4 makes its cluster sums holomorphic. The cluster integral identity of Thm. YM3_I:R gives the generating function as

$$\int e^{\sum_i z_i W_i^{\mathrm{tot}}} \, \mathrm{d}\nu^{\mathrm{r}} = \exp\left[\sum_X (E_z(X) - E_0(X)) \right].$$

The same identity holds at threshold infinity. All these integrals are absolutely convergent. To justify differentiation, replace the quadratic origins by their real parts and the inserted origin by $\sum_i r_i |W_i|$. Monotonicity of the potential norm leaves it admitted. Its absolutely integrable density majorizes the integrands on the closed source polydisc; Cauchy's formula on a smaller polydisc and Fubini's theorem justify every derivative.

The mixed derivative of $E_z(X)$ vanishes unless X meets every insertion: otherwise one source coordinate changes no potential in X, and the locality clause of Prop. YM3_I:47.4 makes $E_z(X)$ independent of that coordinate. Thus cumulants are the sums of the mixed cluster derivatives on the set in N_{ins}. Equation YM3_I:(724), applied at every z, bounds $|E_z(X) - E_z^{\circ}(X)|$ by $K_P^{\Pi} \mathsf{p}_\star e^{-\lambda_{\mathrm{out}} d(X)}$; there are no Σ_{v} origins, and the inserted origin remains of sort Σ_{q}. At infinite threshold the same structure and norms apply, while $\mathsf{p}_\star$ becomes zero. Cauchy's estimate with radii increasing to r_i gives the factor Π_{ins}. Summing proves eq. 4.xxx.

A component at one bond may be placed on a fixed elementary localization domain Y_i. Its norm is bounded by $\frac{1}{2} M^{3/2} e^{\lambda_V d(Y_i)}$, since $s \le \frac{1}{2} M^{3/2} (1 + M^{-3} s^2)$ for $s \ge 0$. The elementary domains have uniformly bounded size. The polymer sum anchored at one such domain is bounded independently of the torus, by the exponential domain-counting estimate YM3_I:(684) of Prop. YM3_I:45.1. Thus $\Pi_{\mathrm{ins}} N_{\mathrm{ins}}$ is uniformly bounded for $n = 2, 4$. The symmetry $B \mapsto -B$ proves evenness. Completing the Gaussian square gives logarithmic generating function $\frac{1}{2} \sum_{i,j} z_i z_j (\varphi_i, C \varphi_j)$ in the unrestricted measure, proving its stated cumulants. □

In the applications to the quadratic systems of Prop. 6.4, the Gaussian structure and the quadratic origin are precisely the admitted one-step system at field scale zero in Prop. 6.26. The complete structure is that of Prop. YM3_I:120.3; its covariance after the quadratic origin is included is M_i^{-1}. Taking $\mathfrak{b} = \mathfrak{b}(x_i) = A_1 (\log x_i)^{\mathfrak{p}_0}$ gives both exponential error bounds used in the field-scale

expansions. Here $\lambda_{\mathrm{out}} = (1 + 4\beta_{\mathrm{s}})\kappa > c_{\mathrm{E}}$: the entropy floor in the choice of κ in Section YM3_I:69 already implies $\kappa > c_{\mathrm{E}}$.

4.4.2. *First creation on a fixed torus.* We use the all-small-field history measures m_t, step factors g_t, and conditional quotients Φ_t of Section YM3_I:123. The history law is $\widetilde{\nu}_k(h) = Z^{-1} \int \mathfrak{w}_h \rho_0 \, \mathrm{d}U$; the measure on the all-small-field history is $m_t[F] = (\mathcal{M}^t)_*(F\mathfrak{w}_{h_t}\rho_0 \, \mathrm{d}U)$, and $m_t = m_t[1]$. Its quotient is the bounded density $\Phi_t[F] = dm_t[F]/dm_t$, defined up to an m_t-null set. The exact identities YM3_I:(2269)–(2271) hold for every bounded gauge invariant insertion F, provided the prescribed tests and the ceilings $\varepsilon_j \le \varepsilon_\flat$ hold through the levels used. In particular

$$\langle F \rangle = Z^{-1} \int \Phi_T[F] \, \mathrm{d}m_T + Z^{-1} \sum_{t=1}^{T} \int \Phi_{t-1}[F] g_t^{\mathrm{w}} \, \mathrm{d}m_{t-1}, \qquad \sum_{t \le T} w_t = \widetilde{\nu}_K\{Z_T \neq \emptyset\}. \tag{4.xxxi}$$

These formulas are the partition of histories according to their first non-small-field outcome, followed by the exact sum over all extensions of each prefix. They are valid independently of the special source-dependent representation of Φ_t.

Proposition 4.54. Use the actual real history masses of Thm. YM3_I:D on an admissible lattice. Fix $0 < \eth < 1$, put $c' = (1 - \eth)c_\star$, and assume the case-II absorption and summability thresholds of Prop. YM3_I:67.6 at $(c_\star, \eth)$, as well as $y_K \ge \max\{Y_1, 2\log(1 + \bar{B})\}$, with Y_1 from Lem. YM3_I:67.1. These are the hypotheses used for the calibrated history tail in Lem. YM3_I:115.1; the couplings here may be any admissible running sequence. Define

$$\begin{aligned} \mathsf{B}(x) &= \mathsf{C}^{\Sigma}_{\mathrm{maj}} + 2 + c_{\mathrm{low}} + \tfrac{9}{2}\log x, \qquad \mathsf{C}^{\Sigma}_{\mathrm{maj}} = \mathsf{C}_{\mathrm{maj}} + (6L^3 + 1)\log 2, \\ \mathsf{C}_{\mathrm{ag}}(q) &= \exp\left[\frac{(3+q)^2(\log L)^2}{2c' A_0^2 (\log(L/2))^2}\right] \qquad (q \ge 0). \end{aligned} \tag{4.xxxii}$$

We also write $\mathsf{C}_{\mathrm{ag}}(q, \eth)$ when the dependence on $\eth$ needs to be displayed. For $1 \le t \le k \le K$, with $N_t = \ell^3 L^{3(K-t)}$,

$$0 \le w_t \le \widetilde{\nu}_k\{Z_t \neq \emptyset\} \le 2M^{-3} N_t e^{-c' p_t^2} e^{\mathsf{B}(x_K)\ell^3}. \tag{4.xxxiii}$$

If $\mathsf{B}(x_K)\ell^3 \le c' p_K^2/2$, one may replace its last two exponentials by $e^{-c' p_t^2/2}$. Write $P_t = \widetilde{\nu}_K\{Z_t \neq \emptyset\}$. Then $m_t(\mathcal{V}_t)/Z = 1 - P_t$ and, for every $q \ge 0$,

$$\sum_{t'=1}^{t} L^{q(K-t')} P_{t'} \le 4M^{-3}\ell^3 e^{\mathsf{B}(x_K)\ell^3} \mathsf{C}_{\mathrm{ag}}(q+1) L^{-(K-t)} e^{-c' p_K^2/2}. \tag{4.xxxiv}$$

If the stronger condition $2\mathsf{B}(x_K)\ell^3 \le c' p_K^2/2$ holds, then the probability in eq. 4.xxxiii is also at most $L^{3(K-t)} e^{-c' p_t^2/2}$; its weighted sum is at most $2\mathsf{C}_{\mathrm{ag}}(q+1) L^{-(K-t)} e^{-c' p_K^2/2}$. Both weighted-sum bounds also hold with $w_{t'}$ in place of $P_{t'}$. All history probabilities of an event determined through level t are unchanged when the terminal level $k \ge t$ is changed. ∎

Proof. We give the dependence on the sequence explicitly. Prop. YM3_I:70.1 and equation YM3_I:(1485) bound the real masses by the summation input of Prop. YM3_I:67.6. Its proof and the proof of Prop. YM3_I:67.5 bound the normalized weight for an exact set W_j of creations by

$$e^{\mathsf{C}_{\mathrm{maj}} N_K} \prod_j (2^{-n_j^{\mathrm{bit}}} \zeta_j)^{\#W_j}, \qquad \zeta_j = e^{-c' p_j^2 - 2p_j}.$$

The multiplicity estimate YM3_I:(1447) cancels the binary factors even if a creation cell is prescribed. Since Z_t is nonempty exactly when a creation occurs at some $j \le t$, the union bound in this finite subset product gives

$$\sum_{h : Z_t \neq \emptyset} e^{-E_{(K)}} \mathfrak{t}(h) \le \Pi_K(\mathsf{C}_{\mathrm{maj}}, c') \sum_{j \le t} \#\mathcal{P}_j e^{-c' p_j^2} \le 2M^{-3} N_t e^{-c' p_t^2} \Pi_K(\mathsf{C}^{\Sigma}_{\mathrm{maj}}, c').$$

The last sum is the stopped summability estimate YM3_I:(1451), under the threshold assumed in the statement. Dividing by the partition lower bound in Prop. YM3_I:70.2, and using $\Pi_K(\mathsf{C}^{\Sigma}_{\mathrm{maj}}, c') \le e^{(\mathsf{C}^{\Sigma}_{\mathrm{maj}}+2)N_K}$, proves eq. 4.xxxiii. This proof uses the admissible flow bound, not an exact geometric line. Consistency follows by summing the extensions of a prefix; the small-field ceilings are part of the construction hypotheses. Equation eq. 4.xxxi gives the assertion for w_t.

For $a = K-t$, the inequality $y_t - y_K \ge a\log(L/2)$ and the first three terms of the power expansion, with $y_K \ge 1$ and $P = 2\mathfrak{p}_0 \ge 2$, imply $p_t^2 \ge p_K^2 + A_0^2 a^2 (\log(L/2))^2$. Completing the square gives

$$L^{(3+q)a} e^{-c' p_t^2/2} \le \mathsf{C}_{\mathrm{ag}}(q) e^{-c' p_K^2/2}. \tag{4.xxxv}$$

Use $q+1$ and sum $\sum_{a \ge K-t} L^{-a} \le 2L^{-(K-t)}$ to obtain eq. 4.xxxiv. Monotonicity $p_t \ge p_K$ absorbs the volume exponential under the weaker small-side condition. Under the stronger condition use also $2M^{-3} \le 1$ and $\ell^3 \le e^{\mathsf{B}(x_K)\ell^3}$, since $\mathsf{B}(x_K) \ge 1$. This gives the final assertions. □

For a first-passage datum at g, $x_K \le X_g$, so the preceding bounds remain valid with $\mathsf{B}(X_g)$ in place of $\mathsf{B}(x_K)$. The canonical choice takes its coupling ceiling below the fixed thresholds in this proposition. Its admissibility on all levels is proved in Thm. YM3_I:D, and the first-passage and stopped-data conditions in Prop. YM3_I:74.1 and [7, Lem. G.3]. On a small side the specialization is Lem. YM3_I:125.3; the least admissible side is small below the ceiling of Lem. YM3_I:126.3.

5. Plane fields and anisotropic variations

Plane-dependent changes of the Wilson action probe the diagonal curvature bilinears. Mixed corner terms probe its off-diagonal components. Their derivatives also produce stress insertions. This connects the tensor correlations with anisotropic variations of the model, but the connection must retain the boundary terms of a localized variation.

For the background estimate, fix the axial trees used for lattice gauge fixing. Let Y_0 be the space of block-data variations that vanish on those trees and have zero next linearized block average.

Sketch D (Plane insertions and anisotropic response). At the admitted coupling and finite-geometry scales, separated plane and stress insertions have bounds uniform in the cutoff. The lattice Ward identity passes to the continuum with its stress flux through the cutoff shell. Near a real regular reference background, sufficiently small plane and symmetric corner weights give a unique constrained critical background, holomorphic in the weights and block datum, with localized response. The Hessian in the block datum has a uniform positive lower bound for its real part on $Y_0 + iY_0$. These conclusions retain the regular-reference, region-separation and norm conditions of the construction. ■□

Sketch of the argument. To see the boundary term, let ω be antisymmetric and choose a smooth cutoff χ equal to one near the scalar spectator supports. The displacement is $\chi(x)\omega x$. Write $\mathfrak{B}_{\omega,\chi}$ for the full stress-flux insertion carried by the shell where χ varies, including the lattice correction to its principal term. On a fixed torus, after the cutoff is removed, the identity has the form

$$\sum_i S_n(f_1, \dots, (\omega x)\cdot \nabla f_i, \dots, f_n) = \lim_{K\to\infty} \langle \mathfrak{B}_{\omega,\chi}; \mathcal{O}_K(f_1); \dots; \mathcal{O}_K(f_n)\rangle.$$

The derivatives on the left are continuum derivatives; the identity from which it comes uses lattice differences. Bounds on the shell insertion allow the limit, but do not make the shell flux vanish. In a rotation-invariant volume-first limit it does vanish. For a general limit along growing tori, vanishing of this flux is the remaining condition in this Ward formulation of rotation invariance.

The background calculation takes place on a finite lattice. A real reference background is regular when its local coordinates, constrained inverse and small-field bounds lie within the construction domains. Write a nearby field in exponential coordinates over this reference, fix the lattice gauge slice and prescribe its block average. The variables are finite-lattice bond fields and row data; the continuum derivatives in the Ward identity do not act on these variables.

The critical equation and the constraint are solved together with their multiplier. Their linear part is inverted using the regular-reference propagators. The remaining constraint and action terms have small derivatives in scaled local norms. One multiplier equation has a bounded linear term without a small factor; weighting the multiplier component in the pair norm makes the complete map a contraction. This is why the proof uses a coupled system rather than a scalar implicit equation. Differentiating its fixed point gives the local response bounds. For a perturbation of the block datum at c_0, for example, the response of A decays exponentially with the block distance from c_0.

The norms matter. The background bound contains a scaled first-gradient Hölder seminorm. Uniform control of that response requires the corresponding regularity of the weights; a supremum bound alone cannot replace it. The positive Hessian estimate also retains its real-reference and longer-sequence conditions. Moving between gauges at complex scalar data uses the separate scalar transfer argument.

For the arbitrary-order free short-distance law, the additional task is a uniform interacting n-source estimate at every order. The separated bounds, the specified low-order laws and the regular-reference background construction do not use this extension. The argument below specifies the required estimate but does not prove it at arbitrary order.

5.1. **Plane fields, stress and anisotropic variations.** We compare the curvature bilinears with plane insertions and the stress tensor obtained from anisotropic changes of the lattice action. Uniform estimates permit the removal of the cutoff; the collapse and remainder estimates identify the conserved stress insertion.

The Ward identities retain their boundary terms and quantitative hypotheses. The two-scale comparisons use the results of §4.1; they do not identify arbitrary coincident-point products without a specified extension.

5.1.1. *Notation and hypotheses. Model, constants and data.* We work in the common real and sourced construction setting of Prop. YM3_I:69.1 and Section YM3_I:82: the Wilson action of SU(2) on $T_\ell = \mathbb{R}^3/\ell\mathbb{Z}^3$, $\ell = 2L^m$, at the spacing $a = a_K = L^{-K}$, $L \geq 13$ odd; $a_p = 1 - \frac{1}{2}\mathrm{tr}\,\partial U(p)$ and $A = \sum_p a_p$; $\langle\cdot\rangle_{K,\ell,\beta}$ and $\langle\cdot\,;\ldots;\cdot\rangle_{K,\ell,\beta}$ are the means and the joint cumulants of $Z^{-1}_{K,\ell,\beta} e^{-\beta A} dU$, and $S^T_{n;K,\ell,\beta}$ are the connected functions of $\mathcal{O}$. For a plaquette p, $x(p)$ is its barycentre, $\lambda(p)$ its normal direction and $P(p)$ the set of its two directions; for a bond b, x_b is its midpoint and μ_b its direction.

The constants are those of the list of constants and conditions among the construction results, fixed in its order, γ last. For a reference coupling $0 < g \leq \gamma$, $\mathfrak{L}(g)$ is the set of the sides at g (equation YM3_I:(1528)), ℓ_g the least of them, and $\mathfrak{D}(g)$ the set of the first-passage data (K, β) at g (Cor. YM3_I:74.2), with the label $\mathsf{b} = a_K\beta$; $X_g = L(g^{-2} + B_\sharp)$, $\bar{B} = B_\sharp L/(L-1)$, $I(g) = (g^{-2} + \bar{B}, X_g - \bar{B}]$. Every label of $\mathfrak{D}(g)$ lies in $[\gamma^{-2}, X_g + \bar{B}]$; $MR(X_g + \bar{B}) \leq L^m$ on every side $2L^m$ in $\mathfrak{L}(g)$, $R(x)$ being the least power of L above $(\log x)^r$ with the exponent r among the construction results; and for $\mathsf{b} \in I(g)$ every pair $(K, L^K\mathsf{b})$, $K \in \mathbb{N}$, is a first-passage datum at g.

The bounds $\gamma^{\mathrm{sd}}_W \leq \gamma^{\mathrm{ref}}_W = x_W^{-1/2} \leq \gamma$ on the reference coupling (the choices in equations YM3_I:(2326)–(2327)), the threshold x_{J}, the radii α_0, α_1 of the cores, the numbers $\mathfrak{p}_0$, A_0, A_1, c', κ, M, C_{lat} and the function $B(\cdot)$ are those among the construction results, and so is the depth function: $s(\rho)$ is the least integer $s \geq 0$ with $L^s\rho \geq C_W(s)$, where $C_W(s) = C_{W,0}A_0^2 Y_s^{2\mathfrak{p}_0}$ and $Y_s = (s+1)\log L + \log(g^{-2} + \bar{B})$. The choice preceding equation YM3_I:(2326) satisfies every inequality defining x_{J} in [7, eq. 228]. Since x_{J} is the least such threshold, $x_W \geq \max\{\gamma^{-2}, x_{\mathrm{J}}\}$, and hence $g^{-2} \geq x_{\mathrm{J}}$ for $0 < g \leq \gamma^{\mathrm{ref}}_W$. $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ is the set of the periodic Gibbs states at (a_K, β); a *state* at (K, β) is one of them or the measure of a torus with a side in $\mathfrak{L}(g)$; for a state ν we write $A^\nu[F] = F - \nu(F)$.

We write $p_\ell = \ell^{-3}\log Z_{K,\ell,\beta}$ for the pressure of T_ℓ per unit of physical volume, formed with the plain partition function $Z_{K,\ell,\beta}$ of the measure above (Section YM3_I:86); at zero source Thms. YM3_I:AJ and YM3_I:AD write $Z_{K,\ell,\beta} = e^{E_{(K)}[\ell]}\widetilde{Z}_\ell(0)$, with the vacuum constant $E_{(K)}[\ell]$ of the side ℓ and the normalised partition function $\widetilde{Z}_\ell(0)$ among the construction results. For $(K,\beta)\in\mathfrak{D}(g)$ and $\ell\ge\ell'$ in $\mathfrak{L}(g)$, $0\le p_{\ell'}-p_\ell\le G_g$, with the number $G_g = C_{\mathrm{S}} + C_g + C_{\mathrm{gap}}$ among the construction results.

The lower bound is reflection positivity: $Z_{K,\ell,\beta}\le Z_{K,\ell',\beta}^{(\ell/\ell')^3}$, [7, Cor. B.1](e). The upper bound follows from $|E_{(K)}[\ell]\ell^{-3} - E_{(K)}[\ell']\ell'^{-3}|\le C_{\mathrm{gap}}$, $\widetilde{Z}_{\ell'}(0)\le e^{C_{\mathrm{S}}\ell'^3}$ and $\widetilde{Z}_\ell(0)\ge e^{-C_g\ell^3}$ (Props. YM3_I:74.4 and YM3_I:74.5, through [7, Cor. B.1], Prop. YM3_I:89.2 and [7, Lem. N.1], which compare two sides; C_{S} and $C_{\mathrm{gap}} = 2C_{\mathrm{tt}}C'_{\mathrm{tt}}E_0$ are the numbers of [7, §N], and C_g that of [7, eq. 288]). Distances are Euclidean unless they are called sup distances; Lip_∞ is the Lipschitz constant for the sup distance, $\|f\|_\sharp = \max\{\sup|f|, 40M\,\mathrm{Lip} f\}$, $\|\eta\|_{C^2}$ is the sum of the suprema of $|\eta|$, $|\nabla\eta|$, $|\nabla^2\eta|$, and $\|\varphi\|_{C^{1,1}} = \max\{\sup|\varphi|, \sup|\nabla\varphi|, \mathrm{Lip}\,\nabla\varphi\}$.

Constants in the statements depend only on the constants among the construction results and on the arguments shown, never on K, β, the state or the test functions.

Lemma 5.1 (A bound on separated torus cumulants)**.** Fix the common construction choice and $0<g\le\gamma$. Put $\delta_e = \kappa - \log 9216 > 0$ and

$$S(\kappa) = \frac{2^{12}}{1-e^{-\delta_e}}, \qquad C'_\Sigma = \frac{27}{26}\,5^3 + \left(\frac{6}{e\delta_e}\right)^3\frac{e^{2\delta_e}}{1-e^{-\delta_e}}, \qquad c_V = C_{\mathrm{S}}(2+\log g^{-2})+1.$$

For $n\ge 2$ and $d>0$, define

$$C_n(d) = \left(\frac{n}{2\varrho}\right)^n\left[4^n\log 2 + 81\sqrt{3}\,C_{\mathrm{S}}S(\kappa)C'_\Sigma d^{-3}\right].$$

At every first-passage datum (K,β) at g, on every $\ell\in\mathfrak{L}(g)$, real Lipschitz functions $f_1,\dots,f_n$ with two supports at Euclidean distance at least d satisfy

$$|S^T_{n;K,\ell,\beta}(f_1,\dots,f_n)|\le C_n(d)e^{nc_V\ell^3}\prod_i\|f_i\|_\sharp. \tag{5.i}$$

The constants are independent of K, β, ℓ and the functions; $C_n(d)$ depends only on n,d,ϱ and the fixed construction constants, while c_V may also depend on g. If each support is in a sup-ball of radius $R>0$ and $\ell\ge 16n(R+1)$, Prop. YM3_I:89.4 instead gives a bound independent of ℓ. ■

Proof. Prop. YM3_I:88.3, with $Q_\ell = \exp[C_{\mathrm{S}}(2+\log g^{-2})\ell^3]$, gives

$$|S^T_n(f)|\le\left(\frac{n}{2\varrho}\right)^n\left[\log 2\,(2Q_\ell+2)^n + 81\sqrt{3}C_{\mathrm{S}}S(\kappa)C'_\Sigma\ell^3 d^{-3}\right]\prod_i\|f_i\|_\sharp.$$

Here C'_Σ is the scale-sum constant in equation YM3_I:(1776). Since $Q_\ell\ge 1$, $(2Q_\ell+2)^n\le 4^nQ_\ell^n$; and $\ell^3\le e^{n\ell^3}$. Absorbing these two factors proves eq. 5.i. Symmetry of a cumulant allows the separated pair to occupy any two positions. The per-ball assertion is exactly Prop. YM3_I:89.4, under the additional support and side assumptions stated above. □

In this section c_V denotes the displayed enlarged constant. Every use below is through eq. 5.i; no identity with a pressure or a vacuum-energy coefficient is used.

Plane fields and rings. A *weight* is a real or complex function w of the plaquettes; $\mathcal{O}(w) = a^{-1}\sum_p w(p)a_p$, $\mathcal{O}(f) = \mathcal{O}(f\circ x)$, $\mathcal{O}_\lambda(f) = a^{-1}\sum_{\lambda(p)=\lambda} f(x(p))a_p$, so that $\mathcal{O} = \mathcal{O}_1+\mathcal{O}_2+\mathcal{O}_3$, and $\check{D}_m = \mathcal{O} - 2\mathcal{O}_m$. $C_K^{\mathrm{free}}(p,q)$ is the matrix of the orthogonal projection onto the exact lattice curvatures, and

$$\mathcal{K}_K^{(n)}(w_1,\dots,w_n) = \tfrac{1}{2}\sum_\varsigma\sum_{p_1,\dots,p_n}\prod_a w_a(p_a)\prod_{i\in\mathbb{Z}/n}C_K^{\mathrm{free}}(p_{\varsigma(i)},p_{\varsigma(i+1)}),$$

the sum over the $(n-1)!$ cyclic orders ς; $\mathcal{K}^{(n)}_{K,\ell'}$ is the same sum on $T_{\ell'}$. With $\mathbb{K}_3(u) = (3\hat{u}\hat{u}^{\mathrm{T}} - \mathbf{1})/(4\pi|u|^3)$, the continuum ring $\mathcal{K}^{(n)}_\lambda(\phi)$ is the same expression with $\int \prod_a \phi_a(y_a)\,\mathrm{d}y_a$ for the sum over the plaquettes and $\mathbb{K}_3(y_{\varsigma(i)} - y_{\varsigma(i+1)})_{\lambda_{\varsigma(i)}\lambda_{\varsigma(i+1)}}$ for C^{free}_K. We put $\mathcal{K}_{\lambda\mu}(\phi_1,\phi_2) = \iint \phi_1(x)\phi_2(y)[\mathbb{K}_3(x-y)_{\lambda\mu}]^2\,\mathrm{d}x\,\mathrm{d}y$, $k_3(x,y,z) = \mathrm{tr}[\mathbb{K}_3(x-y)\mathbb{K}_3(y-z)\mathbb{K}_3(z-x)]$ and, for a unit vector n, $\mathsf{t}^\circ(n,X) = \frac{3}{32\pi^2}\mathsf{b}^{-2}|X|^{-6}(3(n\cdot\hat{X})^2 - 1)$.

Since $\mathbb{K}_3(X)^2 = (16\pi^2|X|^6)^{-1}(3\hat{X}\hat{X}^{\mathrm{T}} + \mathbf{1})$, we have $\frac{3}{2}\mathsf{b}^{-2}\sum_{\lambda,\mu}[\mathbb{K}_3(X)_{\lambda\mu}]^2 = \frac{9}{16\pi^2}\mathsf{b}^{-2}|X|^{-6}$, the free two-point kernel of $\mathcal{O}$, and $\frac{3}{2}\mathsf{b}^{-2}\sum_\mu\{[\mathbb{K}_3(X)_{\lambda\mu}]^2 - \frac{1}{3}\sum_{\lambda'}[\mathbb{K}_3(X)_{\lambda'\mu}]^2\} = \mathsf{t}^\circ(e_\lambda, X)$. For plaquettes $p \neq q$ and u a least lift of $x(p) - x(q)$, equations YM3_I:(2245)–(2246) bound the lattice kernel by $|C^{\mathrm{free}}_K(p,q)| \le (1 + 3C_{\mathrm{lat}})a^3|u|^{-3}$ and compare it with $\mathbb{K}_3$, $|\epsilon_p\epsilon_q C^{\mathrm{free}}_K(p,q) - a^3\mathbb{K}_3(u)_{\lambda(p)\lambda(q)}| \le C_{\mathrm{lat}}a^3(a^2|u|^{-5} + \ell^{-3})$, $\epsilon_p = \pm 1$ the orientations (YM3_I:(2246) and YM3_I:(2245) of equations YM3_I:(2245)–(2246)).

Its differences along one plane obey

$$|C^{\mathrm{free}}_K(p,q) - C^{\mathrm{free}}_K(p',q)| \le C_\nabla\, a^3|x(p) - x(p')|\,|u|^{-4} \qquad \text{for parallel, equally oriented } p,\ p' \text{ with } |x(p)-x(p')| \le \tfrac{1}{4}|u| \text{ and } |u| \ge 8a,$$

$C_\nabla = 256\sqrt{3}\,C_{\mathrm{h}}(1 + 3C_{\mathrm{lat}})$, C_{h} the constant of the interior gradient estimate for discrete harmonic functions on $\mathbb{Z}^3$. Indeed the Laplacian $dd^* + d^*d$ of the cubic lattice acts on 2-forms plane by plane as the scalar lattice Laplacian, and it commutes with the projection; so it carries the column $C^{\mathrm{free}}_K(\cdot,q)$, the projection of the unit form δ_q, to $dd^*\delta_q$, which vanishes at every plaquette that shares no bond with q, in particular at every plaquette whose barycentre is more than a away from $x(q)$. On each plane the column is therefore discrete harmonic at the distance $\ge 2a$ from $x(q)$. A lattice path from $x(p)$ to $x(p')$ of length $\le \sqrt{3}|x(p) - x(p')|$ keeps the distance $\frac{1}{2}|u|$ from $x(q)$; on the balls of radius $\frac{1}{4}|u|$ about its points the first bound bounds the column by $4^3(1 + 3C_{\mathrm{lat}})a^3|u|^{-3}$, and the gradient estimate on these balls bounds each step by $4^4C_{\mathrm{h}}(1 + 3C_{\mathrm{lat}})a^4|u|^{-4}$.

Lattice calculus. $\mathbf{w}(V) = (V - V^{-1})/2i$ and $\langle X, Y\rangle = \frac{1}{2}\mathrm{tr}XY$. For a positively oriented bond b and a plaquette $p \ni b$, $W_{b,p}$ is the holonomy of p from the initial point of b, along b; $p(b,\pm\rho)$, $\rho \neq \mu_b$, are the plaquettes through b in the directions $\pm\rho$; $\mathbf{u}_{b,\rho} = -\frac{1}{2}[\mathbf{w}(W_{b,p(b,\rho)}) - \mathbf{w}(W_{b,p(b,-\rho)})]$, $\mathbf{u}_{b,\mu_b} = 0$, the displacement of b, and $\mathfrak{d}_{b,p,\rho} = \langle \mathbf{u}_{b,\rho}, \mathbf{w}(W_{b,p})\rangle$, the elementary densities; $\mathbf{u}_{b,\rho}$ is the half sum of the field strengths of the two plaquettes at b on the sides $\pm e_\rho$, and $G_\rho(b) = \mathbf{w}(W_{b,p(b,\rho)}) + \mathbf{w}(W_{b,p(b,-\rho)})$ is the transported difference of these field strengths. With $p_\pm = p \pm ae_\rho$,

$$\mathfrak{r}_\rho(p) = \sum_{b\subset\partial p} \mathfrak{d}_{b,p,\rho} - \tfrac{1}{2}(a_{p_+} - a_{p_-}), \qquad \mathfrak{s}_{l\rho}(p) = \sum_{b\subset\partial p} \frac{(x_b - x(p))_l}{a}\,\mathfrak{d}_{b,p,\rho}, \qquad \mathfrak{r}^\sharp_\rho(p) = \mathfrak{r}_\rho(p) - \Psi_\rho(p) + \Psi_\rho(p_-),$$

the orbital remainder, the spin densities and the cubic remainder, where the orbital potential is $\Psi_\rho(p) = \frac{1}{2}a_{p\cup p_+} - a_p - a_{p_+} + \frac{1}{2}a_pa_{p_+}$ for $\rho \in P(p)$, $a_{p\cup p_+}$ being the plaquette term of the 1×2 rectangle, and $\Psi_\rho(p) = -\frac{1}{16}\sum_y\langle Y_y, Y_y\rangle$, $Y_y = \mathbf{w}(W^y_p) - \mathrm{Ad}(U(y, y+ae_\rho))\mathbf{w}(W^{y+ae_\rho}_{p_+})$, for $\rho = \lambda(p)$, the sum running over the corners y of p and W^y_p being the holonomy of p from y. For a weight v and a vector field η the stress insertion is

$$\Theta_v[\eta] = \sum_p v(p)\,a^{-1}\sum_{b\subset\partial p}\sum_\rho(\eta^\rho(x_b) - \eta^\rho(x(p)))\mathfrak{d}_{b,p,\rho},$$

and $\Theta_\beta[\eta]$ is its value at $v \equiv \beta$. A *field* is a triple η of real functions on the barycentres; $\mathsf{H}(\eta) = \max_\rho \sup|\eta^\rho| + \max_\rho \mathrm{Lip}_\infty\eta^\rho$, and

$$\mathfrak{K}^{\mathrm{v}}[\eta] = a^{-1}\sum_p\sum_\rho \eta^\rho(x(p))\,\mathfrak{r}_\rho(p), \qquad \mathfrak{K}^\sharp[\eta] = a^{-1}\sum_p \eta^{\lambda(p)}(x(p))\,\mathfrak{r}^\sharp_{\lambda(p)}(p)$$

are the orbital insertion and the cubic insertion. We write $\mathrm{div}^s\eta(y) = \sum_\rho (2a)^{-1}[\eta^\rho(y+ae_\rho) - \eta^\rho(y-ae_\rho)]$, $\nabla^+_\rho\varphi(y) = a^{-1}[\varphi(y+ae_\rho)-\varphi(y)]$, $\nabla^c_\rho\varphi(y) = a^{-1}[\varphi(y+\frac{a}{2}e_\rho)-\varphi(y-\frac{a}{2}e_\rho)]$, $e^\eta(p) = [\nabla^c_\mu\eta^\mu + \nabla^c_\nu\eta^\nu](x(p))$ with $\{\mu,\nu\} = P(p)$, and $\mathrm{St}_K[\eta] = \Theta_\beta[\eta] - \mathsf{b}\,\mathcal{O}(\mathrm{div}^s\eta)$. With v_η the derivation that displaces U along η and $\mathrm{div}\,v_\eta$ its divergence, the identities ([7, Lem. J.1], with the integration by parts [7, Lem. J.1]) are $v_\eta A(\beta,\cdot) = \mathfrak{K}^{\mathrm{v}}[\beta\eta] - \mathsf{b}\,\mathcal{O}(\mathrm{div}^s\eta) + \Theta_\beta[\eta]$ and $\mathrm{div}\,v_\eta = -\frac{3}{2}a\,\mathcal{O}(e^\eta)$. So the null term $N_\beta[\eta] = v_\eta A(\beta,\cdot) - \mathrm{div}\,v_\eta$ of Lem. 5.9(a) is $\mathrm{St}_K[\eta] + \mathfrak{K}^{\mathrm{v}}[\beta\eta] + \frac{3}{2}a\,\mathcal{O}(e^\eta)$; and since, identically in U, $\mathfrak{K}^{\mathrm{v}}[\beta\eta] = -\beta\sum_{p,\rho}(\nabla^+_\rho\eta^\rho)(x(p))\Psi_\rho(p) + \beta\mathfrak{K}^\sharp[\eta]$ (Lem. 5.9(b)), $N_\beta[\eta] = \mathbb{T}_\beta[\eta] + \beta\mathfrak{K}^\sharp[\eta]$ with the *stress functional*

$$\mathbb{T}_\beta[\eta] = \mathrm{St}_K[\eta] - \beta\sum_{p,\rho}(\nabla^+_\rho\eta^\rho)(x(p))\,\Psi_\rho(p) + \tfrac{3}{2}a\,\mathcal{O}(e^\eta),$$

which reads η through its differences alone and vanishes at a constant field; the cubic insertion $\beta\mathfrak{K}^\sharp[\eta]$, which completes it to the null term, is its *cubic part*.

Tensor entries. $V = \mathrm{Sym}_3(\mathbb{R})$ and $|H|_\infty = \sup_x \max_{l,r}|H_{lr}(x)|$. For $H \in C_c(\mathbb{R}^3;V)$ we put $\mathcal{T}_K(H) = \mathcal{O}(w_H) - a^{-1}\sum_b H_{\lambda_b\lambda'_b}(x_b)\langle \mathbf{u}_{b,\lambda_b}, \mathbf{u}_{b,\lambda'_b}\rangle$, $w_H(p) = H_{\lambda(p)\lambda(p)}(x(p))$, $\lambda_b < \lambda'_b$ being the directions normal to b; formally $\mathcal{T}_K(H) = \frac{1}{2}\int \mathrm{Tr}(HT)$ with $T_{lr} = \mathrm{tr}B_lB_r$, and $\mathcal{T}_K(f\mathbf{1}) = \mathcal{O}(f)$, $\mathcal{T}_K(f(\mathbf{1}-2E_{mm})) = \check{D}_m(f)$. We put $H^{\mathrm{cons}}[\eta] = (\mathrm{div}\,\eta)\mathbf{1} - \nabla\eta - (\nabla\eta)^{\mathrm{T}}$ and $\Delta_K[\eta] = \mathrm{St}_K[\eta] - \mathsf{b}\,\mathcal{T}_K(H^{\mathrm{cons}}[\eta])$.

At smooth fields the density of $\mathrm{St}_K[\eta]$ against $\partial_l\eta^r$ is $\mathsf{b}[\frac{1}{2}\delta_{lr}|B|^2 - \langle B_l, B_r\rangle]$, so that of $\mathsf{b}\,\mathcal{T}_K(H^{\mathrm{cons}}[\eta])$ is the same and the classical density of $\Delta_K[\eta]$ vanishes. A *tensor limit* at g along (K_j,β_j,ρ_j), with $(K_j,\beta_j) \in \mathfrak{D}(g)$, $K_j \to \infty$ and $\rho_j \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$, is a functional with $\sigma(H_1,\dots,H_n) = \lim_j \rho_j(\prod_a A^{\rho_j}[\mathcal{T}_{K_j}(H_a)])$ for all $H_a \in C_c(\mathbb{R}^3;V)$ with pairwise disjoint supports; the labels need not converge. The stress entries are $T_{lr}(\varphi) = \beta\sum_p \varphi(x(p))[\mathfrak{s}_{lr}(p) - \delta_{lr}a_p]$.

Rotations. For $\omega \in \mathfrak{so}(3)$, $|\omega| = \max|\omega_{\rho\lambda}|$, and ωy is taken in the chart $(-\frac{\ell}{2},\frac{\ell}{2})^3$; $\nabla^s_\omega\varphi(y) = \sum_\rho(\omega y)_\rho(2a)^{-1}[\varphi(y+ae_\rho) - \varphi(y-ae_\rho)]$, and $\mathfrak{S}_\omega[v] = \sum_p v(p)\sum_{\rho,\lambda}\omega_{\rho\lambda}\mathfrak{s}_{\lambda\rho}(p)$ is the spin insertion. A cut-off function at K is a real function χ on $\mathbb{R}^3$ that vanishes outside $(-\frac{\ell}{2}+2a,\frac{\ell}{2}-2a)^3$, regarded as a function on T_ℓ (Section YM3_I:112); it has the margin $\delta_0 \geq 2a$ if it vanishes outside $(-\frac{\ell}{2}+\delta_0,\frac{\ell}{2}-\delta_0)^3$; it covers $f_1,\dots,f_n$ if $\chi = 1$ at the barycentre and at the bond midpoints of every plaquette at which some $f_i \neq 0$; $\mathrm{Var}(\chi)$ is the closed set of the points near which χ is not constant. The cut-off variation density and the flux insertion of Section YM3_I:112 are

$$h_{\omega,\chi}(p) = a^{-1}\sum_{b\subset\partial p}\sum_\rho [\chi(x_b) - \chi(x(p))](\omega x_b)_\rho\,\mathfrak{d}_{b,p,\rho}, \qquad \mathfrak{B}_{\omega,\chi} = \mathsf{b}\,\mathcal{O}(\nabla^s_\omega\chi) - \beta\sum_p h_{\omega,\chi}(p) - \tfrac{3}{2}a\,\mathcal{O}(v_\chi),$$

with $v_\chi(p) = \sum_{\rho\in P(p)}(\omega x(p))_\rho\nabla^c_\rho\chi(x(p))$; $\mathfrak{B}_{\omega,\chi}$ depends on U near $\mathrm{Var}(\chi)$ only, and $\beta\sum_p h_{\omega,\chi}(p) = \Theta_\beta[\chi\,\omega y] - \beta\,\mathfrak{S}_\omega[\chi]$. With $\mathcal{O}(f.) = \mathcal{O}(f_1);\dots;\mathcal{O}(f_n)$, the bulk term and the cubic remainder are

$$\mathfrak{U}_{\omega,\chi}(f) = -\beta\langle \mathfrak{S}_\omega[\chi] + \mathfrak{K}^{\mathrm{v}}[\chi\,\omega y]; \mathcal{O}(f.)\rangle + a^{-1}\sum_i \langle \mathfrak{S}_\omega[f_i] + \mathfrak{K}^{\mathrm{v}}[f_i\,\omega y]; (\mathcal{O}(f_l))_{l\neq i}\rangle,$$

$$\mathfrak{U}^\sharp[\eta](f) = -\beta\langle \mathfrak{K}^\sharp[\eta]; \mathcal{O}(f.)\rangle + a^{-1}\sum_i \langle \mathfrak{K}^\sharp[f_i\eta]; (\mathcal{O}(f_l))_{l\neq i}\rangle,$$

and, for χ covering the f_i, the rotational Ward identity at finite cutoff is

$$\sum_i \langle \mathcal{O}(f_1);\dots;\mathcal{O}(\nabla^s_\omega f_i);\dots;\mathcal{O}(f_n)\rangle = \mathfrak{U}_{\omega,\chi}(f) + \langle \mathfrak{B}_{\omega,\chi}; \mathcal{O}(f.)\rangle.$$

The spread of sets $S_0,\dots,S_n$ is $\inf\max_{i<i'}|y_i - y_{i'}|_\infty$ over $y_i \in S_i$; it is at least the sup distance of any two of them.

5.1.2. *Principal results.*

The arbitrary-order interacting comparison. The cyclic Gaussian calculation is explicit. Extending it to the interacting Wilson measure at arbitrary order still requires uniform estimates for the small-field remainder and for histories with a first large-field creation. For a fixed integer $n \geq 2$, the quantitative estimate sought would provide a coupling threshold γ_n, depth thresholds $s_n(g)$ and $\mathsf{d}_n(g,\ell)$, and errors $\varepsilon_n(g,s)$ and $\omega_n(g,s;c)$, uniform in the cutoff, label, position and weights, with the properties below. $\varepsilon_n(g,s) \to 0$ as $s \to \infty$ and use the explicit large-field majorant

$$\omega_n(g,s;c) = L^{3s}(\mathsf{Z}_s^\vee)^n\big[L^{4n}c_\Sigma + s\big]\exp\big[-cA_0^2(Y_s^*)^{2\mathfrak{p}_0}\big], \qquad c_\Sigma = \sum_{l\geq 0} L^{-2l}(l+2)^{2n\mathfrak{p}_0},$$

$Y_s^* = \log(\frac{4}{5}L^{-t_1(s)}x_s^-)$, $x_s^- = \frac{4}{5}L^{s-2}g^{-2}(L^2C_W(s)Y_s^{3\mathfrak{p}_0})^{-1}$, $Y_s = \log \bar{x}_s$, and $\mathsf{Z}_s^\vee \geq 1$, $t_1(s) \geq 1$ are numbers formed with the basic constants, n, g and s: $\mathsf{Z}_s^\vee$ bounds, in the units of the levels above the working level of depth s, the sizes of the n sources carried there by Thm. YM3_I:D, and is at most a power of $\bar{x}_s$ times a polynomial in Y_s, and $L^{t_1(s)}$ is at most a polynomial in Y_s; so $\mathsf{L}_w\omega_n(g,s;c)$ bounds the large-field histories summed against their weights w_t, each with the n-th power of the sizes of the sources at its level.

Let $n \geq 2$, $0 < g \leq \gamma_n$, $(K,\beta) \in \mathfrak{D}(g)$, $\ell \in \mathfrak{L}(g)$, $0 < \mathsf{d} < 1$, $s_n(g) \leq s = s(\mathsf{d}) \leq K$, and let $c, \mathsf{L}_w > 0$ satisfy $\mathsf{w}_t \leq \mathsf{L}_w L^{3(K-t)}e^{-cp_0(g_t)^2}$ $(1 \leq t \leq K)$, $\varsigma_K(c) \geq 4n+5$ and $\mathsf{L}_w\omega_n(g,s;c) \leq \frac{1}{8}$; here w_t is the probability of a first creation of a large field at level t, $p_0(g_t) = A_0(\log g_t^{-2})^{\mathfrak{p}_0}$, and $\varsigma_K(c) = 2c\mathfrak{p}_0A_0^2(\log g_K^{-2})^{2\mathfrak{p}_0-1}$. For real weights w_a and points x_a with $|x(p)-x_a| < \mathsf{d}$ where $w_a(p) \neq 0$ and $|x(p)-x(q)| \geq \mathsf{d}$ where $w_a(p) \neq 0 \neq w_b(q)$, $a \neq b$,

$$\big|\langle \mathcal{O}(w_1);\dots;\mathcal{O}(w_n)\rangle_{K,\ell,\beta} - 3\mathsf{b}^{-n}\mathcal{K}_K^{(n)}(w)\big| \leq \mathsf{e}_n\,\mathsf{b}^{-n}\prod_a \sup|w_a|, \qquad \mathsf{e}_n = \varepsilon_n(g,s) + 2(1 + \bar{B}g^2L^{2-s})^n\mathsf{L}_w\omega_n(g,s;c).$$

These target estimates are needed with $c = c'$ and $\mathsf{L}_w = 2M^{-3}\ell^3e^{B(X_g)\ell^3}$ for $\mathsf{d} \leq \mathsf{d}_n(g,\ell)$ and every $K \geq s(\mathsf{d})$, and then $\mathsf{e}_n \to 0$ as $\mathsf{d} \to 0$ (for $n \geq 6$ provided $\log g^{-2} \geq (\frac{4n+5}{2c'\mathfrak{p}_0A_0^2})^{1/(2\mathfrak{p}_0-1)}$).

The uniform interacting remainder has not been established here. The cyclic Gaussian calculation and the displayed large-field majorant do not prove it. The free-law conclusions described below remain aims of the sketch at the orders for which this comparison is required.

5.1.3. *What a variation measures.* A plane weight changes the coefficient of the plaquette action in one coordinate plane. Differentiating in that weight inserts the corresponding diagonal curvature bilinear. Symmetric products of two corner curvatures provide the mixed components. This interpretation organizes the section: first bound separated insertions, then take their cutoff limits, and finally identify the response of the constrained backgrounds used by the construction. The bounds on arbitrary separated cumulants, the specified low-order short-distance laws and the weighted-background construction are different assertions and use different estimates.

For the separated cumulants, localize each insertion into cells of a reflection board and centres it at its mean on a reference torus. Products of mirror copies are expanded by the moment–cumulant formula. The estimates on one torus then give uniform bounds in other periodic states through the chessboard inequality and domination by conditional torus laws. A diagonal subsequence gives the plane-field limits. Full-sequence convergence on a fixed torus requires the separate consecutive-cutoff estimate in theorem M. That estimate contains the Lipschitz norms of the test functions because a change of mesh samples them at different barycentres. The fixed-cutoff bound itself uses only their suprema.

The same separation matters in the Ward identity. A displacement $\chi(x)\omega x$, with ω antisymmetric, agrees with a rotation near the spectator supports but varies in the shell where χ changes. Haar invariance and integration by parts in the bond variables give the exact finite-lattice identity. Its divergence terms are null inside the relevant expectation; its remaining principal term is the stress insertion, with the cubic and lattice remainders displayed in lemma 5.9. The insertion estimates

control those remainders and pass the identity to the continuum. The shell stress flux remains in that identity. Its vanishing requires the stated limiting argument; boundedness of the shell insertion alone does not give vanishing. The stress-conservation and rotation sections explain the subsequent use of this identity.

There is a further proposed short-distance comparison at arbitrary order. For free photons its cyclic Gaussian expression is exact. The large-field estimates also provide the displayed bounds for histories that leave the small-field region. The interacting comparison also requires a uniform interacting small-field remainder at every source order, with the n-dependent constants specified below. The free cyclic calculation and large-field bound give the other parts of the comparison; they do not supply this remainder. The separated cumulant bounds, the stated low-order laws and the regular-reference response just described have their own supplied arguments and do not require that extension.

Theorem M (Plane fields)**.** Let $0 < g \le \gamma_W^{\mathrm{ref}}$ and $\ell' \in \mathfrak{L}(g)$.

(a) *One torus.* For $r \ge 2$ and $0 < \mathsf{d} < 1$ there are $K_0(r,\mathsf{d})$, depending on g, r, d, and $k(r,\mathsf{d})$, depending besides on ℓ', such that

$$|\langle \mathcal{O}(w_1); \dots; \mathcal{O}(w_r)\rangle_{K,\ell',\beta}| \le k(r,\mathsf{d}) \prod_a \sup|w_a|$$

for all $(K,\beta) \in \mathfrak{D}(g)$ with $K \ge K_0(r,\mathsf{d})$ and all real weights whose sets of barycentres are mutually at distance $\ge \mathsf{d}$; the weights need no regularity and no sign, and they may depend on the plane; with $2^r k(r,\mathsf{d})$ the same holds for complex weights. For every $p > 7$, $k(r,\mathsf{d}) \le C_{r,p}\mathsf{d}^{-pr}$. Besides, the growth in the order is controlled: for every $0 < g \le \gamma$ with $g^{-2} \ge x_{\mathrm{J}}$, every side $\ell \in \mathfrak{L}(g)$, every $q \ge 2$, every $\mathsf{d} \ge a_K$, all planes and all bounded complex functions $u_1, \dots, u_q$ on T_ℓ whose supports are at pairwise sup distances $\ge 2\mathsf{d}$, and for K above a threshold depending on g, q, d,

$$|\langle \mathcal{O}_{\lambda_1}(u_1); \dots; \mathcal{O}_{\lambda_q}(u_q)\rangle_{K,\ell,\beta}| \le A_\ell^q\, (q!)^{\frac{21}{2}+\mathfrak{p}_0}\, e^{36 c_L q^2} \max\{1, \mathsf{d}^{-4q}\} \prod_a \sup|u_a|, \qquad c_L = \frac{(\log L)^2}{2c' A_0^2 (\log \frac{L}{2})^2},$$

with A_ℓ depending on g and ℓ.

(b) *Every state.* For $n \ge 2$ and $\mathsf{d}_0, R > 0$ there is $C_n(\mathsf{d}_0; R)$, depending on $g, \ell', n, \mathsf{d}_0, R$, such that for every $(K,\beta) \in \mathfrak{D}(g)$, all planes and all bounded real f_i supported in closed balls of radius R at mutual distances $\ge \mathsf{d}_0$,

$$|\langle \mathcal{O}_{\lambda_1}(f_1); \dots; \mathcal{O}_{\lambda_n}(f_n)\rangle_s| \le C_n(\mathsf{d}_0; R) \prod_i \sup|f_i|,$$

where s is any member of $\mathcal{G}^{\mathrm{per}}_{K,\beta}$, or the measure of T_ℓ with $\ell = 2L^m \ge L\ell'$ in $\mathfrak{L}(g)$ and the supports in $(-\frac{\ell}{4}, \frac{\ell}{4})^3$. The same holds for bounded weights, and $|\nu'(\mathcal{O}_\lambda(f)) - \nu(\mathcal{O}_\lambda(f))| \le C_1(R)\sup|f|$ for any two such states ν, ν'.

(c) *Limit points.* Let $(K_j, \beta_j) \in \mathfrak{D}(g)$, $K_j \to \infty$, and let s_j be either any periodic Gibbs state at (K_j, β_j), or the measure of T_{ℓ_j} with $\ell_j \in \mathfrak{L}(g)$, $\ell_j \ge L\ell'$, $\ell_j \to \infty$, the cumulants being formed once the supports lie in $(-\frac{\ell_j}{4}, \frac{\ell_j}{4})^3$. Every subsequence has a further subsequence along which the cumulants of (b) converge for all $n \ge 2$, all planes and all real $f_i \in C^1_c(\mathbb{R}^3)$ with disjoint supports. The limits S_n are real, multilinear and symmetric; they obey the bound of (b) and $|S_n| \le B^n (n!)^{4+p} \prod_i \sup|f_i|$ for every $p > 7$, with B depending on $g, \ell', \mathsf{d}_0, R, p$ and not on n; they are invariant under $x \mapsto rx + v$, $r \in W_3$, $v \in \mathbb{R}^3$, the planes being permuted as the axes, and their sum over the planes is the limit of S_n^T.

With $\Pi(J) = \sum_\pi \prod_{I\in\pi} S_{\#I}(I)$, π running over the partitions of a finite family J of pairs (plane, function) into blocks of at least two members, $\sum_{\alpha,\alpha'} \bar z_\alpha z_{\alpha'} \Pi(\theta J_\alpha \cup J_{\alpha'}) \ge 0$ for

families J_α on one side of a mirror θ: at every coordinate mirror, the planes being fixed, in both cases; and, for limits of periodic Gibbs states, also at the diagonal mirrors with normal $e_\mu \pm e_\nu$, which exchange the planes μ, ν in the reflected entries.

(d) *The full sequence on a torus.* Let $0 < g \le \gamma$ with $g^{-2} \ge x_{\mathrm{J}}$, $\mathsf{b} \in I(g)$, $\ell \in \mathfrak{L}(g)$, $n \ge 2$, $\mathsf{d} > 0$ and $\theta_n = 1/(1260n(8n+3))$. There are C, depending on n and b, and K_1, depending on $g, \ell, \mathsf{d}, \mathsf{b}, n$, such that for $K \ge K_1$, all planes and all real ϕ_a with $\mathrm{Lip}_\infty \phi_a < \infty$ and supports at pairwise sup distances $\ge \mathsf{d}$, $G_K = \langle \mathcal{O}_{\lambda_1}(\phi_1); \dots; \mathcal{O}_{\lambda_n}(\phi_n) \rangle_{K,\ell,L^K\mathsf{b}}$ obeys

$$|G_{K+1} - G_K| \le C\, \ell^{3(n+1)} K^{2\mathfrak{p}_0+2} L^{-\theta_n K} \prod_a (\sup|\phi_a| + \mathrm{Lip}_\infty \phi_a).$$

So $\lim_K G_K$ exists with the summed rate; for continuous ϕ_a with supports at pairwise sup distances $\ge 2\mathsf{d}$ it exists without a rate; and the same holds with $\mathsf{b}\breve{D}_m$ for an entry. With one plane field among action densities the rate is better: for real Lipschitz $k, f_1, \dots, f_n$ with supports pairwise $\ge \mathsf{d}$ apart, $K \ge K_1$ and a C depending on $g, \ell, n, \mathsf{d}, \mathsf{b}$,

$$\big|\langle \mathcal{O}_\lambda(k); \mathcal{O}(f.) \rangle_{K+1,\ell,L^{K+1}\mathsf{b}} - \langle \mathcal{O}_\lambda(k); \mathcal{O}(f.) \rangle_{K,\ell,L^K\mathsf{b}}\big| \le C K^{2\mathfrak{p}_0(n+2)+2} L^{-K/80} \|k\|_\sharp \prod_i \|f_i\|_\sharp.$$

The Lipschitz constants cannot be dropped from the bound on $|G_{K+1} - G_K|$: it fails with $\prod_a \sup|\phi_a|$ alone for bounded functions sampled at the barycentres, for instance for the indicator of the points of a ball that are barycentres at an even cutoff and at no smaller cutoff, whereas the bound on $|G_K|$ itself needs sup norms only, by (a). One pair (C, K_1) serves a compact interval of labels in $I(g)$; every $\mathsf{b} \ge 2\max\{\gamma^{-2}, x_{\mathrm{J}}\}$ is admitted at $g = (\mathsf{b}/2)^{-1/2}$, and by the dilation $G_K(\phi; \mathsf{b}, \ell) = L^{zn} G_{K-z}(\phi(L^{-z}\cdot); L^z\mathsf{b}, L^z\ell)$ the same holds at every label $\mathsf{b} > 0$ on every side $2L^m$.

(e) *Two points at one scale.* Let $0 < g \le \gamma_W^{\mathrm{sd}}$, $\ell = \ell_g$, $\vartheta \in (0,1)$, $(K, \beta) \in \mathfrak{D}(g)$, $0 < \mathsf{d} \le \mathsf{d}_0^{\mathrm{ref}}(g, \vartheta)$ with $s(\mathsf{d}) \le K$, $\mathsf{d}_0^{\mathrm{ref}}(g, \vartheta)$ being the threshold in equation YM3_I:(2335), and let (f, h) be a test pair of $\mathfrak{T}_{T_\ell}(\mathsf{d}; \vartheta)$ ([7, Lem. I.1]). Then, uniformly in K,

$$\Big|\langle \mathcal{O}_\lambda(f); \mathcal{O}_{\lambda'}(h) \rangle_{K,\ell,\beta} - \tfrac{3}{2}\mathsf{b}^{-2} \mathcal{K}_{\lambda\lambda'}(f, h)\Big| \le C(g, \vartheta) \big(\log \tfrac{1}{\mathsf{d}}\big)^{-2\mathfrak{p}_0} \mathsf{b}^{-2} \|f\|_\infty \|h\|_\infty.$$

Since $\mathcal{K}_{\lambda\lambda'}(f, h) \ge \frac{6}{175}(\frac{\vartheta}{20})^{10} \|f\|_\infty \|h\|_\infty$, the nine two-point functions are positive for small d. So is the two-point function of $\mathcal{O}_1 - \mathcal{O}_2$ at the test pairs with $\hat{u}_1^2 + \hat{u}_2^2 \le \frac{1}{4}$ for all $u \in \operatorname{supp} f - \operatorname{supp} h$, for instance at those carried by balls of radius d whose centres are $4\mathsf{d}e_3$ apart: there

$$\begin{aligned} &[\mathbb{K}_3(u)]_{11}^2 - 2[\mathbb{K}_3(u)]_{12}^2 + [\mathbb{K}_3(u)]_{22}^2 \\ &\quad = \frac{9(\hat{u}_1^2 - \hat{u}_2^2)^2 - 6(\hat{u}_1^2 + \hat{u}_2^2) + 2}{16\pi^2 |u|^6} \ge \frac{1}{32\pi^2 |u|^6}, \end{aligned}$$

and, since $|u| \le 10\mathsf{d}$ on $\operatorname{supp} f - \operatorname{supp} h$ ([7, Lem. I.1](a)) and the masses of f, h are at least $\vartheta\mathsf{d}^3\|f\|_\infty$, $\vartheta\mathsf{d}^3\|h\|_\infty$ ([7, Lem. I.1]), its main term is at least

$$\frac{3}{2}\mathsf{b}^{-2}(32\pi^2)^{-1} 10^{-6} \vartheta^2 \|f\|_\infty \|h\|_\infty,$$

against 4 times the error of the display; so no plane field is a multiple of $\mathcal{O}$. The law holds with 9 times the error for $\mathcal{O} - 2\mathcal{O}_m$. For two entries $\mathsf{b}[\mathcal{O} - 2\mathcal{O}_m]$, both the main term and the error are multiplied by b^2. The two-point law on a torus of small side, Thm. YM3_I:AU in the form of Prop. YM3_I:126.4 against the lattice kernel, holds for all complex weights w_1, w_2 carried by balls of radius $\mathsf{d} - a_K$ about points x_0, y_0 with $3\mathsf{d} \le |x_0 - y_0| \le 4\mathsf{d}$, with 16 times its tolerance (4 times for real weights), whatever the dependence of the weights on the plane.

(f) *Free photons and the conditional continuum comparison.* For three free lattice photon fields the cyclic expression $3\mathsf{b}^{-n}\mathcal{K}_K^{(n)}(w)$ is the exact cumulant. For the interacting plane fields, the aim is to obtain the corresponding comparison from the estimate to be established in §5.1.2, with its stated data and error e_n. For real Lipschitz ϕ_a with $\operatorname{supp}\phi_a \subset B(x_a,\mathsf{d}) \subset B(c,\frac{\ell}{4})$, $\operatorname{Lip}\phi_a \le \|\phi_a\|_\infty/(\vartheta_0\mathsf{d})$, $0<\vartheta_0\le 1$, and supports $\ge \mathsf{d}$ apart, the plane fields $\mathcal{O}_{\lambda_a}(\phi_a)$ obey the law with $\mathcal{K}_\lambda^{(n)}(\phi)$ for $\mathcal{K}_K^{(n)}(w)$ and with the bracket $\mathsf{e}_n + C_n'(\frac{a_K}{\vartheta_0\mathsf{d}} + \frac{\mathsf{d}^3}{\ell^3})$, where $C_n' = \frac{3}{2}n![6^n C_{\mathrm{lat}}(1+3C_{\mathrm{lat}})^{n-1} + 43\cdot 11^n]$; with $3^nC_n'$ the same holds for the combinations $\sum_\lambda k_{a\lambda}\mathcal{O}_\lambda$, $k_a\in[-1,1]^3$, such as $\mathcal{O}-2\mathcal{O}_m$. Multiplying any of these entries by b multiplies both the main term and its error by b for that entry.

(g) *The law in every state.* (i) Let $0<g\le\gamma_W^{\mathrm{sd}}$, $n\ge 1$ and $\sigma\in(0,1)$. There are $\mathsf{d}_2>0$ and C,c, depending on g,n,σ, such that for $(K,\beta)\in\mathfrak{D}(g)$, $0<\mathsf{d}\le\mathsf{d}_2$ with $s(\mathsf{d})\le K$, all bounded real f_a vanishing outside closed balls of radius d inside one ball of radius $9\mathsf{d}$, at mutual distances $\ge\mathsf{d}$, and every state s at (K,β),

$$|S_n^T[s](f) - S_{n;K,\ell_g,\beta}^T(f)| \le \mathsf{e}_n'(g,\mathsf{d})\,\mathsf{b}^{-n}\prod_a \|f_a\|_\infty, \qquad \mathsf{e}_n'(g,\mathsf{d}) \le C\big(\log\tfrac{1}{\mathsf{d}}\big)^{cn}\mathsf{d}^{\,3(1-\sigma)-[\frac{n}{2}-1]_+};$$

The free law in every state remains a target until the interacting estimate in §5.1.2 is established on T_{ℓ_g}. Once it is established, and $g\le\min\{\gamma_n,\gamma_W^{\mathrm{sd}}\}$, the preceding comparison and (f) give the free law in every state for $2\le n\le 7$, under the depth hypotheses of $(\mathrm{Q_n})$. (ii) For the arbitrary-order extension, one must establish that interacting estimate on $T_{\ell'}$ at every order $2\le m\le 8n$. Let $n\ge 2$, $0<g\le\min_{2\le m\le 8n}\gamma_m$ with $\log g^{-2}\ge(\frac{32n+5}{2c'\mathfrak{p}_0A_0^2})^{1/(2\mathfrak{p}_0-1)}$, and $\ell'\in\mathfrak{L}(g)$. For $0<\mathsf{d}\le\min_{2\le m\le 8n}\mathsf{d}_m(g,\ell')$, every $(K,\beta)\in\mathfrak{D}(g)$ with $K\ge s(\mathsf{d})$, every $s\in\mathcal{G}_{K,\beta}^{\mathrm{per}}$ and the measure s of every torus T_ℓ, $\ell\ge\ell'$ in $\mathfrak{L}(g)$, and real plaquette weights with the support and separation conditions of §5.1.2, inside a cube of side $\frac{1}{4}\ell'$,

$$\big|\kappa_s(\mathcal{O}(w_1),\dots,\mathcal{O}(w_n)) - 3\mathsf{b}^{-n}\mathcal{K}_{K,\ell'}^{(n)}(w)\big| \le \mathsf{e}_n^\sharp(g,\ell',\mathsf{d})\,\mathsf{b}^{-n}\prod_a \sup|w_a|,$$

where $\mathsf{e}_n^\sharp$ depends on no state, side or cutoff and tends to 0 with d. The same argument is intended to give the following comparison for every non-empty $B\subset\{1,\dots,n\}$. For $n=1$ this target requires the corresponding interacting estimates for $2\le m\le 8$, in their coupling and depth ranges:

$$\big|\kappa_s((\mathcal{O}(w_a))_{a\in B}) - \kappa_{T_{\ell'}}((\mathcal{O}(w_a))_{a\in B})\big| \le \eta^\sharp(1+\eta^\sharp)^{n-1}\mathsf{b}^{-\#B}\prod_{a\in B}\sup|w_a|,$$

$\kappa_{T_{\ell'}}$ being the joint cumulant under the measure of $T_{\ell'}$, with $\eta^\sharp\to 0$ as $\mathsf{d}\to 0$. The same holds for the plane fields and their combinations $\mathcal{O}-2\mathcal{O}_m$, with the corresponding three-plane coefficient factors. For $\mathsf{b}[\mathcal{O}-2\mathcal{O}_m]$, multiply the comparison bound by b for every such entry.

(h) *Short distance in every state and in every limit.* Let $0<g\le\gamma_\sigma$, where $\gamma_\sigma\le\gamma_W^{\mathrm{sd}}$ is a constant formed with the constants, $\mathsf{d}_\sigma(g)>0$ a threshold and $\Omega_\sigma=\{(x,y):0<|x-y|<\mathsf{d}_\sigma(g)\}$; put $E_2(g,t)$, $E_3(g,t)$ for the tolerances at the distance t of the laws of two and of three points at narrow bumps, which Step 4 proves on the least torus and (i) and (iii) below state, functions of g and t formed with the constants, $E_2(g,t)\le C(g)(\log\frac{1}{t})^{19\mathfrak{p}_0}t^{1/2}$, $E_3(g,t)\le C(g)[(\log\frac{1}{t})^{39\mathfrak{p}_0+2}t^{1/2}+(t/\ell_g)^3]$.

(i) At a cutoff, in every state: for $(K,\beta)\in\mathfrak{D}(g)$, every state s at (K,β), points x_0,y_0 with $\mathsf{d}_0=|x_0-y_0|\le\mathsf{d}_\sigma(g)$, $\rho>0$ with $s(\rho)\le K$, and bounded real f_1,f_2 supported in balls

of radius ρ inside $B(x_0,\frac18\mathsf{d}_0)$, $B(y_0,\frac18\mathsf{d}_0)$,

$$\Big|\langle\mathcal{O}_\lambda(f_1);\mathcal{O}_\mu(f_2)\rangle_s-\tfrac32\mathsf{b}^{-2}\sum_{\lambda(p)=\lambda}\sum_{\lambda(q)=\mu}f_1(x(p))f_2(x(q))C_K^{\mathrm{free}}(p,q)^2\Big|\le E_2(g,\tfrac12\mathsf{d}_0)\Big(\frac{2\rho}{\mathsf{d}_0}\Big)^6\mathsf{b}^{-2}\|f_1\|_\infty\|f_2\|_\infty,$$

C_K^{free} being that of T_{ℓ_g}; for real Lipschitz ϕ_a with $\operatorname{Lip}\phi_a\le\|\phi_a\|_\infty/(\vartheta\rho)$ the same holds with $\frac32\mathsf{b}^{-2}\mathcal{K}_{\lambda\mu}(\phi_1,\phi_2)$ and with the bracket $E_2+C\frac{a_K}{\vartheta\rho}$. By the change of units $x\mapsto L^zx$ this narrow form holds at every $0<g\le\gamma$, on every side of $\mathfrak{L}(g)$, for distances d_0 below a threshold depending on g.

(ii) Let $(K_j,\beta_j)\in\mathfrak{D}(g)$, $K_j\to\infty$, with labels $\mathsf{b}_j\to\mathsf{b}$, and let s_j be states at (K_j,β_j). Along a subsequence $\langle\mathcal{O}_\lambda(\phi_1);\mathcal{O}_\mu(\phi_2)\rangle_{s_j}$ converges for all real Lipschitz ϕ_a with $\operatorname{supp}\phi_1\times\operatorname{supp}\phi_2\subset\Omega_\sigma$, and every such limit has Borel kernels $\tau_{\lambda\mu}(x,y)=\hat\tau_{\lambda\mu}(x-y)$ with

$$\begin{aligned}&|\tau_{\lambda\mu}(x,y)-\tfrac32\mathsf{b}^{-2}[\mathbb{K}_3(x-y)_{\lambda\mu}]^2|\\&\qquad\le(\tfrac3\pi)^2\epsilon_\sigma(g,|x-y|)\,\mathsf{b}^{-2}|x-y|^{-6}\\&\qquad\qquad\text{almost everywhere,}\end{aligned}$$

$$\epsilon_\sigma(g,t)=64\sup_{0<t'\le t}E_2(g,\tfrac12t')\le C(g)(\log\tfrac1t)^{19\mathfrak{p}_0}t^{1/2};$$

before a kernel is named, at every centre and for bumps of radius ρ in $B(x_0,\frac18\mathsf{d}_0)$, $B(y_0,\frac18\mathsf{d}_0)$, the limit differs from $\frac32\mathsf{b}^{-2}\mathcal{K}_{\lambda\mu}(\phi_1,\phi_2)$ by at most

$$64E_2(g,\frac12\mathsf{d}_0)\mathsf{b}^{-2}\mathsf{d}_0^{-6}\rho^6\|\phi_1\|_\infty\|\phi_2\|_\infty.$$

If besides, for all $n\ge2$, the $S_n^T[s_j]$ converge on separated tuples to the connected functions $S_n^{\sigma'}$ of reconstruction data σ' on $\mathbb{R}^3$, let s_n^T be the continuous kernel of $S_n^{\sigma'}$ $(n=2,3)$: with the continuous kernels $\mathsf{s}_1,\mathsf{s}_2,\mathsf{s}_3^{\mathrm m}$ of the moments of σ' (Prop. YM3_I: 108.3), $\mathsf{s}_2^T=\mathsf{s}_2-\mathsf{s}_1\otimes\mathsf{s}_1$ and

$$\begin{aligned}\mathsf{s}_3^T(x,y,z)=\mathsf{s}_3^{\mathrm m}(x,y,z)-\mathsf{s}_2(x,y)\mathsf{s}_1(z)-\mathsf{s}_2(x,z)\mathsf{s}_1(y)-\mathsf{s}_2(y,z)\mathsf{s}_1(x)\\+2\mathsf{s}_1(x)\mathsf{s}_1(y)\mathsf{s}_1(z).\end{aligned}$$

Then $\sum_{\lambda,\mu}\tau_{\lambda\mu}=\mathsf{s}_2^T$ almost everywhere on Ω_σ, so that $|\mathsf{s}_2^T(x,y)-\frac{9}{16\pi^2}\mathsf{b}^{-2}|x-y|^{-6}|\le9\epsilon_\sigma(g,|x-y|)\mathsf{b}^{-2}|x-y|^{-6}$ on Ω_σ.

(iii) Three points at two scales: in the latter case, at every (x,y,z) with $0<|x-y|=r_*\le\mathsf{d}_*=\operatorname{dist}(z,\{x,y\})\le\mathsf{d}_\sigma(g)$,

$$|\mathsf{s}_3^T(x,y,z)-3\mathsf{b}^{-3}k_3(x,y,z)|\le E_3(g,\mathsf{d}_*)\,\mathsf{b}^{-3}r_*^{-3}\mathsf{d}_*^{-6},$$

uniformly in $r_*/\mathsf{d}_*\in(0,1]$.

(iv) Point splitting: off one null set of pairs (c,z) with $0<\mathsf{d}=|c-z|\le\frac12\mathsf{d}_\sigma(g)$, for all $0<r\le\frac12\mathsf{d}$ and all λ,

$$\big|\tfrac{2\pi}{3}\mathsf{b}r^3\mathsf{s}_3^T(c+\tfrac r2e_\lambda,c-\tfrac r2e_\lambda,z)-t_\lambda(c,z)\big|\le[\epsilon'_\sigma(g,\mathsf{d})+2.7(r/\mathsf{d})^2]\mathsf{b}^{-2}\mathsf{d}^{-6},\qquad\epsilon'_\sigma(g,\mathsf{d})\le C(g)(\log\tfrac1{\mathsf{d}})^{39\mathfrak{p}_0+2}\mathsf{d}^{1/2},$$

$t_\lambda=\sum_\mu(\tau_{\lambda\mu}-\frac13\sum_{\lambda'}\tau_{\lambda'\mu})$ being the kernel of the limit of $\langle\mathcal{O}_\lambda-\frac13\mathcal{O};\mathcal{O}\rangle$; and at every such (c,z), for every unit vector n,

$$\begin{aligned}|\frac{2\pi}{3}\mathsf{b}r^3\mathsf{s}_3^T(c+\frac r2n,c-\frac r2n,z)-\mathsf{t}^\circ(n,c-z)|\le[12\sup_{\frac34\mathsf{d}\le t\le\frac{17}{16}\mathsf{d}}E_3(g,t)\\+2.7(r/\mathsf{d})^2]\mathsf{b}^{-2}\mathsf{d}^{-6}.\end{aligned}$$

(v) On the least torus at a fixed label $\mathsf{b} \in I(g)$, for g small enough in terms of the constants. For real Lipschitz ϕ_1, ϕ_2 with disjoint supports, $\lim_K \langle \mathcal{O}_\lambda(\phi_1) - \frac{1}{3}\mathcal{O}(\phi_1); \mathcal{O}(\phi_2)\rangle_{K,\ell_g,L^K\mathsf{b}}$ exists. Let moreover $\mathsf{d} > 0$ with $6\mathsf{d} \le \min\{1, \mathsf{b}\}$ and $0 < \vartheta_0 \le 1$, let ϕ_1, ϕ_2 be real Lipschitz functions each carried by a ball of radius d, with supports $\ge \mathsf{d}$ apart, $|c - z| \le 6\mathsf{d}$ for $c \in \operatorname{supp}\phi_1$ and $z \in \operatorname{supp}\phi_2$, and $\operatorname{Lip}\phi_1 \le \|\phi_1\|_\infty/(\vartheta_0\mathsf{d})$; for $r > 0$ put $\rho = \frac{r}{8}$ and let $f^c_\pm(y) = \frac{3}{\pi\rho^3}(1 - |y - c \mp \frac{r}{2}e_\lambda|/\rho)_+$ be the radial tents of radius ρ and mass 1 about $c \pm \frac{r}{2}e_\lambda$, of sup norm $\frac{3}{\pi\rho^3} = \frac{1536}{\pi r^3}$. There are $r_* > 0$, depending on $g, \mathsf{b}, \mathsf{d}$, and $C(g, \mathsf{b}, \mathsf{d}, \vartheta_0)$ such that for $0 < r \le r_*$ and every K with $128a_K \le r$, that is $16a_K \le \rho$, and $s(\rho) \le K$

$$\Big|\tfrac{2\pi}{3}\mathsf{b}r^3 \int \phi_1(c)\, S^T_{3;K,\ell_g,L^K\mathsf{b}}(f^c_+, f^c_-, \phi_2)\,\mathrm{d}c - \big\langle \mathcal{O}_\lambda(\phi_1) - \tfrac{1}{3}\mathcal{O}(\phi_1); \mathcal{O}(\phi_2)\big\rangle_{K,\ell_g,L^K\mathsf{b}}\Big| \le C(g, \mathsf{b}, \mathsf{d}, \vartheta_0)\big[r^{\beta_H/7^2} + (a_K/r)^{1/18}\big]\mathsf{b}^{-2}\|\phi_1\|_\infty\|\phi_2\|_\infty,$$

β_H being the Hölder exponent. As the depth function is non-increasing, $s(\rho) \le K$ gives $s(\frac{r}{2}) \le K$; and r_* is taken so that for $0 < r \le r_*$ the depth $s = s(\frac{r}{2})$ obeys $s \ge 3$ and $L^2 C_W(s) Y_s^{3\mathfrak{p}_0} \le L^{s-3}$ (both hold for small r, since $s(\frac{r}{2}) \to \infty$ as $r \to 0$ and the second holds at all large s) and $\frac{r}{2} \le \min\{r_0(\mathsf{b}), r_1(\mathsf{b})\}$, the radii, depending on g and b, up to which Step 4 forms the induction over the levels and the frame of the pair. Here $C(g, \mathsf{b}, \mathsf{d}, \vartheta_0) = C(g, \mathsf{b}, \vartheta_0)\mathsf{d}^{-9/2}$, of which the part of the three-point function beyond its explicit Gaussian part takes $C(g, \mathsf{b})\mathsf{d}^{-5/2}$.

The explicit Gaussian parts of the two members, which take the rest, are compared on the lattice of T_{ℓ_g}: the near leg, between the two tents, against $a_K^3\mathbb{K}_3(re_\lambda)$, by the comparison of the lattice kernel with $\mathbb{K}_3$ in the setting, at the cost $(a_K/r)^2 + (r/\ell_g)^3$, the mean of $\mathbb{K}_3$, whose entries are harmonic off 0, over the two radial tents being its value at re_λ, and $\frac{2\pi}{3}r^3 \cdot 3\mathbb{K}_3(re_\lambda) = \frac{3}{2}(E_{\lambda\lambda} - \frac{1}{3}\mathbf{1})$; the far legs by the bound on the differences of C_K^{free} along one plane of the setting; and the shift of the argument of ϕ_1 by at most r costs $r\operatorname{Lip}\phi_1 \le r\|\phi_1\|_\infty/(\vartheta_0\mathsf{d})$.

Hence, at this fixed d, the limit in K is the limit as $r \to 0$ of

$$\frac{2\pi}{3}\mathsf{b}r^3 \int \phi_1(c)\, S_3^{(\ell_g,\mathsf{b})}(f^c_+, f^c_-, \phi_2)\,\mathrm{d}c.$$

The limit in K exists for all real Lipschitz ϕ_1, ϕ_2 with disjoint supports, and it is identified in this way for all ϕ_1, ϕ_2 as above.

■□

Theorem N (One stress insertion; stress entries; conservation)**.**

(a) *Bounds in every geometry.* Let $0 < g \le \gamma_W^{\mathrm{ref}}$, $\ell' \in \mathfrak{L}(g)$, $n \ge 1$, $0 < \rho < 1$, $\omega \in \mathfrak{so}(3)$, and let χ be a cut-off function at K, Lipschitz on $\mathbb{R}^3$. For every $(K, \beta) \in \mathfrak{D}(g)$,

$$\Big|\Big\langle \beta \sum_p h_{\omega,\chi}(p); \mathcal{O}(f.)\Big\rangle_{K,\ell',\beta}\Big| \le \mathsf{b}\, C_n^\flat(\rho)\, |\omega|\operatorname{Lip}_\infty(\chi) \prod_i \|f_i\|,$$

(i) with $\|f_i\| = \sup|f_i|$ for bounded real f_i (or real weights), $\operatorname{Var}(\chi)$ and the $\operatorname{supp} f_i$ being pairwise $\ge \rho$ apart; (ii) with $\|f_i\| = \|f_i\|_\sharp$ for real Lipschitz f_i in any mutual position, $\operatorname{Var}(\chi)$ being at distance $\ge \rho$ from each $\operatorname{supp} f_i$. $C_n^\flat(\rho) = 468\ell' k_n^\flat(\rho)$, where $k_n^\flat(\rho)$ depends on g, ℓ', n, ρ, not on χ, on the f_i or on the place of the shell. In (ii) the number may be taken free of the side and of the state: on every torus T_ℓ, $\ell \ge \ell'$ in $\mathfrak{L}(g)$, and in every $\nu \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$, for K above a threshold depending on g, ℓ', n, ρ, the left member is at most $156\,\mathsf{b}\, C_n^e(\rho)|\omega|\operatorname{Lip}(\chi)R_1(\chi)N_\chi \prod_i N_i \|f_i\|_\sharp$, where $R_1(\chi) \le \frac{1}{2} + \frac{3}{2}\ell$ is the largest $|x_b|_1$ at the bonds where χ varies, N_χ and N_i are the numbers of the cubes of a mesh $h(n, \rho)$ that meet

the shell and the supports, and $C_n^e(\rho)$ depends on g, ℓ', n, ρ; at every label $\mathsf{b} > 0$ on every side $2L^m$ the same holds for all K with a number depending on $\mathsf{b}, \ell, n, \rho$.

Both bounds are cases of one rule for a rough entry F of Lem. 5.7 whose carrier lies, with respect to the f_i, as $\mathrm{Var}(\chi)$ in (i), resp. (ii): the left member with F in the place of $\beta \sum_p h_{\omega,\chi}(p)$ is at most $k_n^\flat(\rho)\|F\| \prod_i \|f_i\|$ on $T_{\ell'}$, and, in (ii), at most $C_n^e(\rho) N_F \|F\| \prod_i N_i \|f_i\|_\sharp$ free of the side and of the state, N_F being the number of the cubes of the mesh that meet the carrier of F; here $-\beta \sum_p h_{\omega,\chi}(p) = \mathsf{b}\,\mathrm{Pf}(k^h)$ with

$$\|\mathrm{Pf}(k^h)\| \le 156|\omega|\,\mathrm{Lip}(\chi) R_1(\chi) \le 468\ell'|\omega|\,\mathrm{Lip}_\infty(\chi)$$

. The same rule gives the bounds for the whole flux insertion $\mathfrak{B}_{\omega,\chi} = \mathsf{b}\,\mathrm{Pf}(k^h) + \mathcal{O}(v^{\mathfrak{B}})$, of norm $\le (158\mathsf{b} + 3a)|\omega|\,\mathrm{Lip}(\chi) R'(\chi)$, $R'(\chi) = \max\{R_1(\chi), R_1'(\chi)\}$, where $R_1'(\chi) \le \frac{3}{2}\ell$ is the number of Lem. 5.7, the largest $|x(p)|_1$ at the plaquettes at which χ is not constant on the points $x(p) \pm a e_\mu$, $x(p) \pm \frac{a}{2} e_\mu$, $\mu = 1, 2, 3$, among which are those where $\nabla_\omega^s \chi$ or v_χ does not vanish, which hold every plaquette at which $v^{\mathfrak{B}} \ne 0$, so that the largest $|x(p)|_1$ at these may stand for $R_1'(\chi)$ throughout: on $T_{\ell'}$ with $\frac{158\mathsf{b}+3a}{156} C_n^\flat(\rho)$ in the place of $\mathsf{b}\, C_n^\flat(\rho)$, and free of the side and of the state with $(158\mathsf{b} + 3a) C_n^e(\rho) |\omega|\,\mathrm{Lip}(\chi) R'(\chi) N_\chi$ in the place of $156\,\mathsf{b}\, C_n^e(\rho) |\omega|\,\mathrm{Lip}(\chi) R_1(\chi) N_\chi$.

For $\mathrm{supp}\,\eta$ at distance $\ge \rho$ from each $\mathrm{supp}\, f_i$ (and, in (i), the $\mathrm{supp}\, f_i$ pairwise $\ge \rho$ apart) it gives them for $\Theta_\beta[\eta] = \mathsf{b}\,\mathrm{Pf}(k^\eta)$, of norm $\le 156\,\mathsf{b} \max_\rho \mathrm{Lip}\,\eta^\rho$, and for $\mathfrak{K}^{\mathrm{v}}[\beta\eta]$, which beside fields that do not read the bonds within $2a$ of $\mathrm{supp}\,\eta$ is the rough entry $-\Theta_\beta[\eta] + \mathsf{b}\,\mathcal{O}(\mathrm{div}^s \eta) - \frac{3}{2} a\, \mathcal{O}(e^\eta)$ of norm $\le (159\mathsf{b} + 3a) \max_\rho \mathrm{Lip}\,\eta^\rho$ (the null term of Lem. 5.9(a) drops): on $T_{\ell'}$ with $156\,\mathsf{b}\, k_n^\flat(\rho) \max_\rho \mathrm{Lip}_\infty \eta^\rho$, resp. $(159\mathsf{b} + 3a) k_n^\flat(\rho) \max_\rho \mathrm{Lip}_\infty \eta^\rho$, in the place of $\mathsf{b}\, C_n^\flat(\rho) |\omega|\,\mathrm{Lip}_\infty(\chi)$, and free of the side and of the state with $156\,\mathsf{b}\, C_n^e(\rho) \max_\rho \mathrm{Lip}\,\eta^\rho N_\eta$, resp. $(159\mathsf{b} + 3a) C_n^e(\rho) \max_\rho \mathrm{Lip}\,\eta^\rho N_\eta$, in the place of $156\,\mathsf{b}\, C_n^e(\rho) |\omega|\,\mathrm{Lip}(\chi) R_1(\chi) N_\chi$, N_η being the number of the cubes of the mesh that meet the $\frac{\rho}{2}$-neighbourhood of $\mathrm{supp}\,\eta$.

Bounded fields, or real weights that depend on the plane, may be added in sup norms if they are $\ge \rho$ apart from the shell, from one another and from all the other fields.

(b) *Stress entries.* $T_{lr}(\varphi)$ is b times a rough entry (Lem. 5.7) of norm at most an absolute constant times $\sup|\varphi|$. So for $0 < g \le \gamma_W^{\mathrm{ref}}$ the joint cumulants on one torus of r entries $T_{l_a r_a}(\varphi_a)$ and $\mathcal{O}(w_a)$ with supports pairwise $\ge \mathsf{d}$ apart are bounded uniformly in $K \ge K_0'(r, \mathsf{d})$, for merely bounded φ_a, w_a; for two entries this holds at every $0 < g \le \gamma$: $|\langle T_{lr}(\varphi); \mathcal{O}(f)\rangle| \le C\mathsf{b} \sup|\varphi| \sup|f|$ and $|\langle T_{lr}(\varphi); T_{l'r'}(\varphi')\rangle| \le C\mathsf{b}^2 \sup|\varphi| \sup|\varphi'|$, C depending on g, the side and the distance. Along subsequences $\langle T; \mathcal{O}\rangle$ and $\langle T; T\rangle$ converge for continuous entries with disjoint supports, and the limits are symmetric in (l, r), translation invariant, covariant under W_3 with the signs of a tensor, and reflection positive.

(c) *The value, Lipschitz norms.* Let $\mathsf{b} \ge 2x_W$, $g = (\mathsf{b}/2)^{-1/2}$, $\ell' \in \mathfrak{L}(g)$, $\beta = L^K \mathsf{b}$, $n \ge 1$, $0 < \mathsf{d} < 1$ and $c < \frac{1}{14}$. There are C, p, K_*, depending on $\mathsf{b}, \ell', n, \mathsf{d}, c$, such that for $K \ge K_*$, every real C^2 vector field η and all real Lipschitz f_i, the $n+1$ supports being pairwise at sup distance $\ge \mathsf{d}$,

$$|\langle \mathfrak{K}^{\mathrm{v}}[\beta\eta]; \mathcal{O}(f.)\rangle| + |\langle \Theta_\beta[\eta] - \mathsf{b}\,\mathcal{O}(\mathrm{div}^s \eta); \mathcal{O}(f.)\rangle| \le C K^p L^{-cK} \|\eta\|_{C^2} \prod_i \|f_i\|_\sharp .$$

By $x \mapsto L^z x$ the same holds at every label $\mathsf{b} > 0$ on every side $2L^m$.

(d) *The value, sup norms.* In the setting of (c): for every field η and all real weights w_a, the $n+1$ carriers being pairwise at sup distance $\ge \mathsf{d}$, $|\beta\langle \mathfrak{K}^\sharp[\eta]; \mathcal{O}(w_1); \dots; \mathcal{O}(w_n)\rangle| \le C K^p L^{-cK} \mathsf{H}(\eta) \prod_a \sup|w_a|$, and the same with rough entries or block forms (Lems. 5.7 and 5.9) in the places of the $\mathcal{O}(w_a)$; for C^2 fields η vanishing within sup distance d of the carriers, the two members of (c) with $\mathcal{O}(w_a)$ for $\mathcal{O}(f_a)$ are $\le C K^p L^{-cK} \|\eta\|_{C^2} \prod_a \sup|w_a|$. At $n = 1$, for every real vector field η, Lipschitz for the sup distance, and every bounded

real h (a function or a weight) with $\eta = 0$ within sup distance d of $\{x(p) : h(p) \neq 0\}$,

$$|\langle \Theta_\beta[\eta]; \mathcal{O}(h)\rangle - \mathsf{b}\langle \mathcal{O}(\mathrm{div}^s\eta); \mathcal{O}(h)\rangle| + |\langle \mathfrak{K}^{\mathrm{v}}[\beta\eta]; \mathcal{O}(h)\rangle| \le CK^pL^{-K/200}(\mathsf{H}(\eta) + \max_\rho \mathrm{Lip}_\infty \eta^\rho) \sup|h|.$$

For two fields with carriers $\ge \mathsf{d}$ apart, $|\beta^2\langle \mathfrak{K}^\sharp[\eta]; \mathfrak{K}^\sharp[\eta']\rangle| \le CK^{2p}L^{-2cK}\mathsf{H}(\eta)\mathsf{H}(\eta')$, and, with $\mathsf{H}^+ = \mathsf{H} + \max_\rho \mathrm{Lip}_\infty$,

$$|\langle \mathfrak{K}^{\mathrm{v}}[\beta\eta]; \mathfrak{K}^{\mathrm{v}}[\beta\eta']\rangle| \le CK^pL^{-K/200}\mathsf{H}^+(\eta)\mathsf{H}^+(\eta')$$

and

$$|\langle \mathrm{St}_K[\eta]; \mathrm{St}_K[\eta']\rangle| \le CK^pL^{-K/200}\mathsf{b}\,\mathsf{H}^+(\eta) \max_\rho \mathrm{Lip}_\infty \eta'^\rho.$$

All this holds at every fixed label $\mathsf{b} > 0$ on every side $2L^m$. At every first-passage datum $(K, \beta) \in \mathfrak{D}(g)$ on every side of $\mathfrak{L}(g)$, with numbers that depend on g and on no label, the clauses on the cubic insertion alone (one $\beta\mathfrak{K}^\sharp[\eta]$ beside n weights, rough entries or block forms, and two of them) hold for $0 < g \le \gamma$ with $g^{-2} \ge x_{\mathrm{J}}$, and the other clauses, which also read the bilinear half of the orbital insertion and the term $a\,\mathcal{O}(e^\eta)$, for $0 < g \le \gamma_W^{\mathrm{ref}}$, where $g^{-2} \ge x_{\mathrm{J}}$ holds of itself.

(e) *Conservation and collapse in every periodic Gibbs state.* Let $0 < g \le \gamma$ and $R, \mathsf{d} > 0$. There are C, p, K_0, depending on g, d, R, such that for all $(K, \beta) \in \mathfrak{D}(g)$ with $K \ge K_0$, all $\nu \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$, all $\eta, \eta' \in C^2_c(\mathbb{R}^3; \mathbb{R}^3)$ and $H' \in C_c(\mathbb{R}^3; V)$ supported in a ball of radius R, with $\mathrm{supp}\,\eta$ at sup distance $\ge \mathsf{d}$ from $\mathrm{supp}\,\eta'$ and from $\mathrm{supp}\,H'$, and with $A = A^\nu$:

$$|\nu(A[\mathrm{St}_K[\eta']]A[\mathrm{St}_K[\eta]])| \le CK^pL^{-K/14}\|\eta'\|_{C^2}\|\eta\|_{C^2}, \qquad |\nu(\mathfrak{K}^{\mathrm{v}}[\beta\eta])| \le CK^pL^{-K/28}\|\eta\|_{C^2},$$

$$|\nu(A[\mathcal{T}_K(H^{\mathrm{cons}}[\eta'])]A[\mathcal{T}_K(H^{\mathrm{cons}}[\eta])])| \le CK^pL^{-K/20}\|\eta'\|_{C^2}\|\eta\|_{C^2},$$

$$|\nu(A[\mathcal{T}_K(H^{\mathrm{cons}}[\eta])]A[\mathcal{T}_K(H')])| \le CK^pL^{-K/60}\|\eta\|_{C^2}|H'|_\infty,$$

and $|\nu(A[\Delta_K[\eta]]A[Z])| \le CK^pL^{-K/60}\|\eta\|_{C^2}\mathsf{z}$ for $Z = \mathcal{T}_K(H')$, $\mathsf{z} = |H'|_\infty$, and for $Z = \mathrm{St}_K[\eta']$ or $\Delta_K[\eta']$, $\mathsf{z} = \|\eta'\|_{C^2}$. The last bound says that the stress insertion collapses onto $\mathsf{b}\,\mathcal{T}_K(H^{\mathrm{cons}}[\eta])$ in every periodic Gibbs state.

(f) *Any number of entries.* Let $0 < g \le \gamma$, $n \ge 2$, $1 \le n_1 \le n$, $R > 0$, $0 < c < \frac{1}{28}$ and $u = \lceil \frac{1/28 - c}{4n-1} K \rceil$. There are C, p, depending on g, n, R, c, and K_0, depending on g, n, c, such that for all $(K, \beta) \in \mathfrak{D}(g)$ with $K \ge K_0$, all $\nu \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$, every d with $\mathsf{d}L^u \ge K^2$, all $\eta_a \in C^2_c(\mathbb{R}^3; \mathbb{R}^3)$ $(a \le n_1)$ and $H_a \in C_c(\mathbb{R}^3; V)$ $(a > n_1)$ with supports in a ball of radius R at pairwise sup distances $\ge \mathsf{d}$, and for X_a any of $\mathcal{T}_K(H^{\mathrm{cons}}[\eta_a])$, $\mathrm{St}_K[\eta_a]$, $\Delta_K[\eta_a]$, $\mathrm{Rm}_K[\eta_a]$ (Lem. 5.9(c)), $\mathfrak{K}^{\mathrm{v}}[\beta\eta_a]$ $(a \le n_1)$ and $X_a = \mathcal{T}_K(H_a)$ $(a > n_1)$,

$$\Big|\nu\Big(\prod_a A^\nu[X_a]\Big)\Big| \le CK^pL^{-cn_1K} \prod_{a \le n_1} \|\eta_a\|_{C^2} \prod_{a > n_1} |H_a|_\infty,$$

and the same holds for the joint cumulant under ν; with $H_a = h_a\mathbf{1}$ the spectators are action densities. At $n_1 = 1$, $c = \frac{1}{56}$ and a fixed $\mathsf{d} > 0$ this holds for $K \ge K_0(\mathsf{d})$.

(g) *Every tensor limit is conserved.* Let $0 < g \le \gamma$ and let σ be a tensor limit at g. Then $\sigma(H^{\mathrm{cons}}[\eta], H_2, \dots, H_n) = 0$ for every $n \ge 2$, $\eta \in C^2_c(\mathbb{R}^3; \mathbb{R}^3)$ and $H_a \in C_c(\mathbb{R}^3; V)$ whose supports are pairwise disjoint and disjoint from $\mathrm{supp}\,\eta$. In particular $\sigma(H^{\mathrm{cons}}[\theta\eta], H^{\mathrm{cons}}[\eta]) = 0$ for $\mathrm{supp}\,\eta \subset \{x_3 > 0\}$, $\theta\eta = \theta \circ \eta \circ \theta$, θ the mirror $x_3 \mapsto -x_3$, so that in the reconstruction of σ the vector of $\mathcal{T}(H^{\mathrm{cons}}[\eta])$ vanishes; and along the sequence $\mathsf{b}_j^2\rho_j(A[\mathcal{T}(H^{\mathrm{cons}}[\theta\eta])]A[\mathcal{T}(H^{\mathrm{cons}}[\eta])]) - \rho_j(A[\mathfrak{K}^{\mathrm{v}}[\beta_j\theta\eta]]A[\mathfrak{K}^{\mathrm{v}}[\beta_j\eta]]) \to 0$.

For $0 < g \le \gamma_W^{\mathrm{sd}}$ tensor limits exist: by [7, Thm. C](a) of [7, §2], the centred products of any number of tensor entries at separated supports are bounded in every periodic Gibbs

state by numbers that depend on no cutoff, side, state or position, so every sequence of first-passage data at g with $K_j \to \infty$ and of periodic Gibbs states has a subsequence along which all of them converge. Tensor limits are covariant as tensors under the translations and W_3, reflection positive at the nine families of mirrors, and their trace part is the scalar limit, $\sigma(f_1\mathbf{1},\dots,f_n\mathbf{1})$ being the limit of the centred products of the $\mathcal{O}(f_a)$.

■□

Remark 5.2 (Normalisation; the role of the collapse). Conservation at disjoint supports does not fix the normalisation of a tensor: $X_{lr} = |x|^{-2}[F_l n_r + F_r n_l - \delta_{lr} F\cdot n + n_l n_r F\cdot n]$, $n = x/|x|$, is symmetric, traceless and conserved off 0, with the flux $\frac{16\pi}{3}F$. And Thm. N(g) rests on the collapse in (e): the decorrelation of the orbital insertion alone (the second bound of (e)) does not make the six tensor fields conserved; it is the identification of the stress insertion with $\mathsf{b}\,\mathcal{T}_K(H^{\mathrm{cons}}[\eta])$, up to $\Delta_K[\eta]$, that does.

Theorem O (Ward identities on tori)**.** Let $0 < g \le \gamma$, $\ell \in \mathfrak{L}(g)$, $n \ge 1$, $\mathsf{d} > 0$, $0 < \delta_0 < \frac{\ell}{2}$ and $c < \frac{1}{14}$.

(a) *Rotations on a fixed torus.* There are C, p, depending on $g, \ell, n, \mathsf{d}, \delta_0, c$, such that for all K, $\mathsf{b}' \in I(g)$, $\beta = L^K\mathsf{b}'$, $\omega \in \mathfrak{so}(3)$, every cut-off function χ at K of class $C^{1,1}$ with the margin δ_0, and all real f_i vanishing outside $(-\frac{\ell}{2}+\delta_0, \frac{\ell}{2}-\delta_0)^3$ and covered by χ:

(i) if $n \ge 2$, $f_i \in C^{1,1}$ and the spread of the supports is $\ge \mathsf{d}$ (the supports may touch if no point is common to all of them), $|\mathfrak{U}_{\omega,\chi}(f)| \le C|\omega|K^pL^{-cK}\|\chi\|_{C^{1,1}}\prod_i \|f_i\|_{C^{1,1}}$;

(ii) if $n = 1$, f_1 is Lipschitz and $\chi = 1$ on the sup d-neighbourhood of $\operatorname{supp} f_1$, $|\mathfrak{U}_{\omega,\chi}(f_1)| \le C|\omega|K^pL^{-cK}\|\chi\|_{C^{1,1}}\|f_1\|_\sharp$;

(iii) whatever the supports, if χ and χ' both equal 1 on the d-neighbourhood of $\bigcup_i \operatorname{supp} f_i$, $|\langle \mathfrak{B}_{\omega,\chi} - \mathfrak{B}_{\omega,\chi'}; \mathcal{O}(f_\cdot)\rangle| \le C|\omega|K^pL^{-cK}(\|\chi\|_{C^{1,1}} + \|\chi'\|_{C^{1,1}})\prod_i \|f_i\|_\sharp$.

If $n \ge 2$, the f_i are of class C^3, $\operatorname{dist}_\infty(\operatorname{supp} f_1, \operatorname{supp} f_2) \ge \mathsf{d}$ and a fixed χ covers the f_i at all large K, then $\mathfrak{U}_{\omega,\chi}(f) + \langle \mathfrak{B}_{\omega,\chi}; \mathcal{O}(f_\cdot)\rangle \to \sum_i S_n^{(\ell,\mathsf{b}')}(f_1,\dots,(\omega y)\cdot\nabla f_i,\dots,f_n)$ at the rate $K^pL^{-K/10} + L^{-K}$, and the flux alone converges to the same limit at the rate K^pL^{-cK}; the limit does not depend on χ. With nothing asked of the bulk term, for $\omega = \omega_{13}$, the generator of the turns in the plane of e_1, e_3, $f^t = f\circ e^{-t\omega}$, $f_i \in C^3$ supported in a ball inside the margin with $\operatorname{dist}(\operatorname{supp} f_1, \operatorname{supp} f_2) \ge \sqrt{3}\mathsf{d}$, and χ covering all the f^t, $0 \le t \le \frac{\pi}{2}$:

$$\Big|\int_0^{\pi/2} [\mathfrak{U}_{\omega,\chi}(f^t) + \langle \mathfrak{B}_{\omega,\chi}; \mathcal{O}(f^t_\cdot)\rangle]\,dt\Big| \le C a_K \prod_i \|f_i\|_{C^3}, \qquad \int_0^{\pi/2} \sum_i S_n^{(\ell,\mathsf{b}')}(f_1^t,\dots,(\omega y)\cdot\nabla f_i^t,\dots,f_n^t)\,\mathrm{d}t = 0.$$

(b) *Rotations along growing tori.* Let $n \ge 2$, $R, \mathsf{d}_0 > 0$, $\chi_R(y) = \chi_1(|y|/R)$ with $\chi_1 \in C^\infty$, $\chi_1 = 1$ on $[0, \frac{4}{3}]$ and $\chi_1 = 0$ on $[\frac{5}{3}, \infty)$. There are C, p, K_0, depending on g, n, R, d_0, c and on no side, such that

$$|\mathfrak{U}_{\omega,\chi_R}(f)_{K,\ell,\beta}| \le C|\omega|K^pL^{-cK}\prod_i \|f_i\|_{C^{1,1}}$$

for all $(K,\beta) \in \mathfrak{D}(g)$ with $K \ge K_0$, every $\ell \in \mathfrak{L}(g)$ with $2R + 2a < \frac{\ell}{2}$ and all smooth f_i supported in $\{|y| < \frac{R}{2}\}$ at mutual distances $\ge \mathsf{d}_0$; without its cubic member $\mathfrak{U}^\sharp[\chi_R\,\omega y](f)$ the bulk term is $\le C|\omega|K^{1+\mathfrak{p}_0}L^{-c_*K}\prod_i \|f_i\|_{C^{1,1}}$ for every $c_* < \frac{1}{2}$. The same bounds hold under the weak limit points of the torus laws. Along every sequence (K_j, ℓ_j, β_j) with $K_j, \ell_j \to \infty$ along which the connected functions converge to S_n, $\langle \mathfrak{B}_{\omega,\chi_R}; \mathcal{O}(f_\cdot)\rangle \to -\frac{d}{dt}S_n(f(e^{-t\omega}\cdot))|_{t=0}$ for every R; so the limit is invariant under rotations if and only if this flux tends to 0.

For the volume-first scalar limits, which are invariant under O(3), the flux through every shell tends to 0. With the limits taken one after the other: at a label $\mathsf{b}' \in I(g)$, on each

torus T_ℓ, $\ell \in \mathfrak{L}(g)$, the limit in K of the flux is $\sum_i S_n^{(\ell,\mathsf{b}')}(\dots,(\omega y)\cdot\nabla f_i,\dots)$ through every enclosing shell, by (a). If along sides $\ell_k \in \mathfrak{L}(g)$, $\ell_k \to \infty$, the limits $S_n^\circ(X) = \lim_k S_n^{(\ell_k,\mathsf{b}')}(X)$ exist for all $n \ge 2$ and all tuples X of real Lipschitz functions with compact supports at positive distances, then for smooth f_i with compact supports at positive mutual distances and all cut-off functions χ_k on T_{ℓ_k} of class $C^{1,1}$ that vanish near the seam and cover the f_i (for instance χ_R at a fixed R, or χ_{R_k} with $R_k \to \infty$ and $2R_k < \frac12\ell_k - 1$, a far shell)

$$\lim_k \lim_K \langle \mathfrak{B}_{\omega,\chi_k}; \mathcal{O}(f.)\rangle_{K,\ell_k,L^K\mathsf{b}'} = \sum_i S_n^\circ(f_1,\dots,(\omega y)\cdot\nabla f_i,\dots,f_n) = -\tfrac{d}{dt} S_n^\circ(f(e^{-t\omega}\cdot))\big|_{t=0},$$

through any shell; and there are integers K_k^* such that S_n° is the limit of the connected functions along $(K_k,\ell_k,L^{K_k}\mathsf{b}')$ for every sequence $K_k \ge K_k^*$. For each fixed order $n \ge 2$, there is $c_\ell(n,g) > 0$ such that the same limit is obtained by taking the spacing limit first on each torus whenever $\ell_j \to \infty$ and $\ell_j \le c_\ell(n,g) K_j^{2\mathfrak{p}_0/3}$. Thus $S_n = \lim_j S_n^{(\ell_j,\mathsf{b}_j)}$ at any first-passage labels, and raising the cutoffs at fixed sides and labels does not change this limit. The constant is specified in the proof. For every $K \ge 1$, the labels of the first-passage data satisfy

$$I(g) \subset \{L^{-K}\beta : (K,\beta) \in \mathfrak{D}(g)\} \subset \big(g^{-2} + B_\sharp - L^{-1}\bar B,\, L(g^{-2}+\bar B)\big].$$

(c) *One stress entry: trace and diagonal.* Let $0 < \mathsf{b}_- \le \mathsf{b} \le \mathsf{b}_+$, $\beta = L^K\mathsf{b}$, $\ell = 2L^m$ any side, and let two of the $n+1$ supports be $\ge \mathsf{d}$ apart, the others may touch; $\lambda_L = \log(L/2)/\log L$ and $c_n = \lambda_L/(4(6+3n))$. For γ small enough in terms of the other constants, and with C, p depending on $\ell, \mathsf{b}_\pm, n, \mathsf{d}$,

$$|\langle T_{mm}(\varphi) - \mathsf{b}\check D_m(\varphi); \mathcal{O}(f.)\rangle| + |\langle (T_{11}+T_{22}+T_{33})(\varphi) - \mathsf{b}\,\mathcal{O}(\varphi); \mathcal{O}(f.)\rangle| \le C K^p L^{-c_n K} \|\varphi\|_\sharp \prod_i \|f_i\|_\sharp,$$

so that $-T_{33} = \mathsf{b}[\mathcal{O}_3 - \mathcal{O}_1 - \mathcal{O}_2]$ inside these functions. For every $\epsilon > 0$, $\langle (T_{lr} - T_{rl})(\varphi); \mathcal{O}(f.)\rangle = O(a^{1-\epsilon})\mathrm{Lip}\,\varphi$ and $|\langle T_{lr}(\varphi); \mathcal{O}(f.)\rangle| \le C_\epsilon a^{-\epsilon} \sup|\varphi| \prod_i \|f_i\|_\sharp$; and $\langle \mathfrak{K}^{\mathrm{v}}[\beta\eta] - \beta\mathfrak{K}^\sharp[\eta]; \mathcal{O}(f.)\rangle = O(K^p L^{-c_n K})\mathrm{Lip}\,\eta$ for Lipschitz vector fields η, in contact with the f_i or not. At pairwise separated supports and a fixed label $\lim_K \langle T_{mm}(\varphi); \mathcal{O}(f.)\rangle$ exists.

(d) *All Ward identities with one stress entry.* Let γ be as in (c), $\gamma^{-2} \le \mathsf{b}_- \le \mathsf{b} \le \mathsf{b}_+$, $MR(\mathsf{b}_+) \le L^m$, $\ell = 2L^m$; let $\xi = e_r$, ωy or y, where for $\xi = \omega y$ and $\xi = y$, taken in $(-\frac{\ell}{2},\frac{\ell}{2})^3$, the functions χ and f_i vanish within a fixed distance of $\partial(-\frac{\ell}{2},\frac{\ell}{2})^3$; and either $n \ge 2$, $f_i \in C^2$ with pairwise disjoint supports, $I \subset \{1,\dots,n\}$ and $\chi \in C^2$ equal to 1 near $\operatorname{supp} f_i$ for $i \in I$ and to 0 near the others, or $n = 1$, $I = \emptyset$ and $\chi = 0$ near $\operatorname{supp} f_1$. Then

$$\lim_K \Big\langle \sum_{l,r} T_{lr}(\xi^r \partial_l \chi); \mathcal{O}(f.)\Big\rangle_{K,\ell,L^K\mathsf{b}} = -\sum_{i\in I} S_n^{(\ell,\mathsf{b})}(f_1,\dots,\xi\cdot\nabla f_i,\dots,f_n) + \delta_{\xi,y}\big(|I| S_n^{(\ell,\mathsf{b})}(f) - \mathsf{b}\, S_{n+1}^{(\ell,\mathsf{b})}(\chi, f)\big),$$

and at the cutoff the two members differ by $\mathfrak{U}^\sharp[\chi\xi](f)$ up to $O(K^pL^{-c_nK})$ and up to the convergence of the right member. Since $|\mathfrak{U}^\sharp[\chi\xi](f)| \le CK^pL^{-cK}$ for every $c < \frac{1}{14}$ (Lem. 5.11, the spread of $\{\chi\xi \ne 0\}$ and of the supports being positive), the right member converges at the rate $K^pL^{-K/10}$ and $c_n < \frac{1}{36}$, the left member converges at the rate $K^pL^{-c_nK}$, the slowest of the three. For $I = \emptyset$ this is the conservation of the stress tensor beside action densities; for $\xi = e_r$ and χ a periodic function of y_3 alone, $\langle T_{3r}(\chi'); \mathcal{O}(f.)\rangle \to -\sum_{i\in I} S_n^{(\ell,\mathsf{b})}(\dots,\partial_r f_i,\dots)$. Thus, inside these cumulants on one torus, T_{3r} integrated over the planes $\{y_3 = \text{const}\}$ acts on the enclosed entries as $-\partial_r$, so that $-T_{33}$, the Euclidean energy density, acts as ∂_3.

(e) *Translations.* Let $(K,\beta)\in\mathfrak{D}(g)$, $n\geq 2$, ρ a direction and $\mathsf{d}'>0$. For real $f_i\in C^2$ with Lipschitz $\partial_\rho^2 f_i$ and $\operatorname{dist}(\operatorname{supp} f_1,\operatorname{supp} f_2)\geq \mathsf{d}'+2a$,

$$\Big|\beta\langle\mathfrak{K}^\sharp[e_\rho];\mathcal{O}(f_\cdot)\rangle - a^{-1}\sum_i\langle\mathfrak{K}^{\mathrm{v}}[f_ie_\rho];(\mathcal{O}(f_l))_{l\neq i}\rangle\Big| = \Big|\sum_i S^T_{n;K,\ell,\beta}(f_1,\dots,\nabla^s_\rho f_i,\dots,f_n)\Big| \\ \leq (n+1)2^n C_n(\mathsf{d}')e^{nc_V\ell^3}a_K^2\prod_i N_i,$$

$N_i=\max\{\|f_i\|_\sharp,\|\partial_\rho f_i\|_\sharp,\frac12\|\partial_\rho^2 f_i\|_\sharp\}$, with $C_n(\mathsf{d}')$ and c_V as in Lem. 5.1; at $n=2$ the left member vanishes identically. For $I\subset\{1,\dots,n\}$, the f_i with $i\in I$ of class C^3 with Lipschitz $\partial_\rho^3 f_i$, the others Lipschitz, $\operatorname{dist}(\operatorname{supp} f_1,\operatorname{supp} f_2)\geq \mathsf{d}'+a$, and χ a real function on T_ℓ with $\chi(x(p))=\chi(x_b)=1$ at every plaquette p at which some f_i, $i\in I$, does not vanish and at its four bonds b, and $\chi(x(p))=\chi(x_b)=0$ at every plaquette p at which some f_i, $i\notin I$, does not vanish and at its four bonds b,

$$\langle N_\beta[\chi e_\rho];\mathcal{O}(f_\cdot)\rangle - a^{-1}\sum_{i\in I}\langle\mathfrak{K}^{\mathrm{v}}[f_ie_\rho];(\mathcal{O}(f_l))_{l\neq i}\rangle = -\sum_{i\in I}S^T_{n;K,\ell,\beta}(f_1,\dots,\partial_\rho f_i,\dots,f_n)+\mathsf{r},$$

$|\mathsf{r}|\leq\frac{a_K^2}{6}C_n(\mathsf{d}')e^{nc_V\ell^3}\sum_{i\in I}\|\partial_\rho^3 f_i\|_\sharp\prod_{l\neq i}\|f_l\|_\sharp$, $N_\beta[\chi e_\rho]=\mathbb{T}_\beta[\chi e_\rho]+\beta\mathfrak{K}^\sharp[\chi e_\rho]$ the null term of Lem. 5.9(a). Thus the stress functional $\mathbb{T}_\beta[\chi e_\rho]$ of a cut-off translation, taken together with its cubic part $\beta\mathfrak{K}^\sharp[\chi e_\rho]$ and with the contact terms $-a^{-1}\sum_{i\in I}\langle\mathfrak{K}^{\mathrm{v}}[f_ie_\rho];(\mathcal{O}(f_l))_{l\neq i}\rangle$, generates the translation of the enclosed entries up to $O(a^2)$, with connected functions bounded uniformly in K. If the closed a-neighbourhoods of the supports lie in balls of radius R and $\ell\geq 16n(R+1)$, the numbers may be taken free of ℓ.

■□

Remark 5.3 (The spread is needed). For $n\geq 3$ the bound (a)(i) fails at supports with a common point. The symmetric difference is no derivation: by the invariance of the measure under the lattice translations, $\sum_i S^T_n(\dots,\nabla^s_\rho f_i,\dots)$ is a^2 times connected functions of difference quotients of the f_i, at $n=3$ exactly

$$\sum_{i=1}^3 S^T_3(\dots,\nabla^s_\rho f_i,\dots)=\tfrac{a^2}{2}\,S^T_3(\nabla^+_\rho f_1,\nabla^+_\rho f_2,\nabla^+_\rho f_3),$$

and connected functions of action densities at coinciding points are of the order a^{-3} (Prop. 5.18(c)). For free photons, let the real f_i have compact supports inside the margin and bounded derivatives through order five, and let $\chi=1$ on a neighbourhood of their supports. Then $a\,\mathfrak{U}_{\omega,\chi}(f)\to 3\mathsf{b}^{-3}\int\sum_\rho(\omega y)_\rho\,\partial_\rho f_1\,\partial_\rho f_2\,\partial_\rho f_3$ at $n=3$, while the identity with the true generator survives at supports with a common point: $\sum_i S^T_n(\dots,(\omega y)\cdot\nabla f_i,\dots)-\langle\mathfrak{B}_{\omega,\chi};\mathcal{O}(f_\cdot)\rangle$ is $O(a)$ at $n=2$ and $O(a\log\frac1a)$ at $n=3$. For SU(2), at every $g\leq\gamma$, on every side $\ell\in\mathfrak{L}(g)$ and for every $n\geq 3$ there are smooth $f_1,\dots,f_n$ with a common point, a rotation ω and a cut-off function χ such that

$$450\,(n-1)!\,\mathsf{b}'^{-n}\leq a_K|\mathfrak{U}_{\omega,\chi}(f)|\leq 2.3\cdot 10^5\,(n-1)!\,\mathsf{b}'^{-n}$$

for all large K, uniformly in the labels $\mathsf{b}'\in I(g)$. At $n=1$ the neighbourhood in (a)(ii) is needed: there are a real Lipschitz f_1, a rotation ω and a cutoff $\chi\in C^{1,1}$ covering f_1, equal to one on its support but on no neighbourhood of it, for which the free-photon model has $\liminf_K\mathfrak{U}_{\omega,\chi}(f_1)>0$.

Theorem P (One plane field between two times)**.** Let $0<g\leq\gamma_W^{\mathrm{ref}}$, $\ell'\in\mathfrak{L}(g)$, $(K_j,\beta_j)\in\mathfrak{D}(g)$ with $K_j\to\infty$, $\nu_j\in\mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$, and $(X)_j=\prod_i[\mathcal{O}_{K_j}(f_i)-\nu_j(\mathcal{O}_{K_j}(f_i))]$. Every subsequence has a further subsequence along which the following hold.

(i) $\sigma(X)=\lim_j\nu_j((X)_j)$ and $\sigma^{(m)}(u;X)=\lim_j\nu_j((X)_j[\check D_{m,j}(u)-\nu_j(\check D_{m,j}(u))])$ exist for all real $u,f_i\in C^1_c(\mathbb{R}^3)$ with disjoint supports; they are real and multilinear, $\sum_m\sigma^{(m)}(u;X)=\sigma(u\cdot X)$, the product with one more scalar factor, they are translation invariant and covariant under W_3: a quarter turn that carries e_3 to e_i carries $\sigma^{(3)}$ to $\sigma^{(i)}$.

(ii) On the Hilbert space $\mathcal{H}(\sigma)$ of the reconstruction data of the limit, with the Hamiltonian $H \geq 0$, the reflection θ, the translations $U(\vec{x})$, whose generators P_1, P_2 commute with H, and the classes $[G]$, the joint spectrum lies in the cones $H \geq |P_k|$, $k = 1, 2$: every periodic Gibbs state obeys the five-point inequality ([7, eq. 241]) of Lem. 4.46 at the mirror normal to e_3 and the direction e_k, which passes to the limit and puts the spectral measure of every class on $\{\cosh(tE) + \cos(tq_k) \geq 2\}$ for a dense set of $t > 0$, hence on $\{E \geq |q_k|\}$. So $H \geq 0$ and $H \geq |P_k|$ are the hypotheses of Lem. 5.13 in the direction x_3 and, in its last sentence, in the directions x_1, x_2; and there is a unique family of bounded operators $B^{(m)}(s, s')$, $s, s' > 0$, with $B^{(m)}(s, s')^* = B^{(m)}(s', s)$, $e^{-tH} B^{(m)}(s, s') e^{-t'H} = B^{(m)}(s + t, s' + t')$ and

$$\sigma^{(m)}(u; \theta\overline{\tau_{se_3} G'} \cdot \tau_{s'e_3} G) = \int u(x) \, \langle [G'], U(\vec{x}) B^{(m)}(s + x_3, s' - x_3) U(-\vec{x}) [G] \rangle \, \mathrm{d}x$$

for u supported in $\{-s < x_3 < s'\}$ and G, G' supported in $\{x_3 > 0\}$.

(iii) $\|B^{(m)}(s, s')\|$ is bounded by a finite function of $\min\{s, s', 1\}$, g and ℓ', and for u supported in $\{\gamma' - s \leq x_3 \leq s' - \gamma'\}$ the left member of (ii) is at most $c(\gamma') \|u\|_{L^1} \|[G']\| \, \|[G]\|$. For $0 < g \leq \gamma_W^{\mathrm{sd}}$ every such limit with converging labels arises, along a further subsequence, from a tensor limit σ_T (Thm. N(g)): $\sigma(X) = \sigma_T(f_1 \mathbf{1}, \dots, f_n \mathbf{1})$ and $\sigma^{(m)}(u; X) = \sigma_T(u(\mathbf{1} - 2E_{mm}), f_1 \mathbf{1}, \dots, f_n \mathbf{1})$. Let $\mathcal{H}(\sigma_T)$ be the reconstruction of σ_T at θ, spanned by the classes $[F]$ of the tuples F of entries in $C_c(\{x_3 > 0\}; V)$ with $\langle [F'], [F] \rangle = \sigma_T(\theta \bar{F}' \cdot F)$, with its Hamiltonian and its translations, whose joint spectrum lies in the same cones by the same five-point inequality; its *scalar subspace* $\mathcal{H}_0$, the closed span of the classes of the tuples of scalar entries $f_a \mathbf{1}$, is $\mathcal{H}(\sigma)$ and reduces the Hamiltonian and the translations, and Π_0 is the projection onto it.

For $e \in V$ the *point operator* of the tensor entry at e is the family $B^e(s, s')$ of bounded operators on $\mathcal{H}(\sigma_T)$ that Lem. 5.13 gives for $u \mapsto \sigma_T(ue, \cdot)$, with $\sigma_T(ue, \theta\overline{\tau_{se_3} F'} \cdot \tau_{s'e_3} F) = \int u(x) \langle [F'], U(\vec{x}) B^e(s + x_3, s' - x_3) U(-\vec{x}) [F] \rangle \, \mathrm{d}x$ as in (ii); it is linear in e, and $\|B^e(s, s')\| \leq C|e| \min\{s, s', 1\}^{-p}$ for every $p > \frac{17}{2}$, $|e|$ the Euclidean norm of e and C depending on g, p and the limit label and not on the side, by the bound on the chessboard seminorm of a centred tensor entry. Then $B^{(m)}(s, s') = \Pi_0 B^{\mathbf{1} - 2E_{mm}}(s, s') \Pi_0$ on $\mathcal{H}_0$, the compression to the scalar subspace of the point operator at $\mathbf{1} - 2E_{mm}$, so that $\|B^{(m)}(s, s')\| \leq C \min\{s, s', 1\}^{-p}$; and for $Z = (u; X)$ and R_α the turn by the angle α in the plane of e_i, e_3,

$$\sigma^{(3)}(Z \circ R_\alpha) + \sigma^{(3)}(Z \circ R_{-\alpha}) = 2 \cos^2 \alpha \, \sigma^{(3)}(Z) + 2 \sin^2 \alpha \, \sigma^{(i)}(Z).$$

Thus the $\check{D}_m$ behave under turns as the diagonal of a symmetric tensor; moreover $\alpha \mapsto \sigma^{(3)}(Z \circ R_\alpha)$ is of the form $c_0 + c_1 \cos 2\alpha + c_2 \sin 2\alpha$ with numbers c_0, c_1, c_2, and its second derivative at $\alpha = 0$ is $2[\sigma^{(i)}(Z) - \sigma^{(3)}(Z)]$.

■□

One step of the renormalisation group, at every level and with its never-large term, runs with three complex plane couplings, about its prescribed background with free centres (Thm. Q); on the labels of histories with large-field regions the moved backgrounds, the centres of the steps and the counterterms are those of Thm. R. For a triple $P = (U, A', J)$ at the level n, $\mathbf{U}_P = e^{i\eta_n A'} U$, $\eta_n = L^{-n}$, is its complex configuration (Def. YM3_I:49.1); on a region X let $\mathrm{hol}_n(P; X)$ be the Hölder-$\frac{1}{2}$ number, transports included, of $L^{2n}(\partial \mathbf{U}_P - 1)$ at distances ≤ 1 in level-n units; the *core with the third radius* $\mathcal{U}_n(X)$ is the set of triples in the hereditary core of Def. YM3_I:49.1 at the radii (α_0, α_1), with all the tower conditions retained as in §5.2, with $\mathrm{hol}_n(P; X) < \alpha_2$; $\rho_Z^{k \to j}(U, A', J) = (U, \lambda A', \lambda^3 J)|_Z$, $\lambda = L^{-(k-j)}$; $\mathcal{D}_n$ and d_n are as in Section YM3_I:45.

A region $Z \in \mathcal{D}_j$ is *short* at k if $d_j(Z) + 4 \leq L^{k-j}$. The *box* Z^{box} of a short Z is the least box of level-j cubes that holds a lift of Z (a wall-connected union of cubes of the covering that the covering map carries one to one, cube by cube, onto Z; such unions exist, Z being wall-connected, and any

two of them differ by a period), enlarged by one level-j cube on each of its six sides and carried back by the covering map: it does not depend on the lift, its sides are at most $d_j(Z)+4 \le L^{k-j}$ level-j cubes, it lies in $\mathcal{D}_j$, holds Z, and holds both end points of every bond with an end point in Z. Then X_Z is the union of the level-k cubes that hold a level-j cube of Z^{box}: one or two along each axis, so at most eight, with $X_Z \in \mathcal{D}_k$ and $d_k(X_Z)=0$.

For every $Z \in \mathcal{D}_j$, Z^{up} is the union of the level-k cubes that hold a level-j cube of Z. A *term* on $\mathcal{U}_j(Z)$ is a bounded function on it that is holomorphic on its slices, a function of the pair $(\mathbf{U}_P, J)$, and gauge invariant.

The extended list. Thms. Q and R are stated at the choices of the following extension of the list $\mathcal{T}^{\mathrm{S}}[2\varrho]$ of Thms. YM3_I:AJ and YM3_I:AD (the common real and sourced choice of Section YM3_I:82, including eq. 2.ii). The base choice also retains [7, eq. 194] and the radius, decay and block-number conditions of Lem. 5.34. These are imposed when choosing the response parameters in the proof of Prop. YM3_I:69.1, before the polymer-window size M: the finitely many local constants here depend only on L, so the response block number $\mathfrak{M}$ can be raised and its radii and decay rate reduced at that stage. For Lem. 5.19 take this same response block number at least M_{ps} and M_H, the floors in Thm. YM3_I:I and Lem. 5.34. Enlarge w_{lin} to at least $\max\{g_{\mathrm{p}}, w_S\}$, where w_S is from Prop. YM3_I:29.1, and reduce α_{lin} so that $2\alpha_{\mathrm{lin}} \le a_Q$ and the four local-system radii are at most a_H. These choices depend only on L. The separated setting has $R_1 \ge L$ and terminal torus side at least $6R_1\mathfrak{M}$, hence at least $6LM_{\mathrm{ps}}$. On a real regular label, $F'(0)=J(U)$ at full bonds; the scalar part of $D_U^{\mathsf{T}}\partial U$ is bounded by $\xi^{-1}\max_{p\ni b}|\partial U(p)-I|^2$. Thus equation YM3_I:(579) gives the full matrix-divergence bound at twice the regularity radius. The ceiling on $C_m\varepsilon$ in part (c) supplies this radius and the four local-system radii, while its width ceiling supplies the enlarged w_{lin}. These are conditions on the common response choice; they are not new analytic assumptions. It adds 108 entries, in this order: four lower bounds on chosen constants among the construction results, each by a function of earlier constants; 85 further determined constants, each a positive explicit formula in earlier constants, placed before γ (four abbreviations are used among the formulas); and 15 further upper bounds on γ. No condition is laid on L, on E_0 or on a window constant.

The following constants retain their meanings in the real construction and its sourced extension (the proof of Prop. YM3_I:69.1 and Section YM3_I:82):

$$\mathsf{C}_{\mathrm{ex}}, \mathsf{C}_{\mathrm{cs}}, \mathsf{C}_{\mathrm{mn}}, \lambda_{\mathrm{s}}, \gamma_{\mathrm{r}}, c_{\mathrm{r}}, \mathsf{u}_{\mathrm{G}}, \lambda_{\mathrm{r,gs}}, \bar{c}_{\mathrm{P}}, \beta_{\mathrm{s}}, \kappa, \kappa'', \bar{r}_H, \bar{\kappa}, \lambda_V, \vartheta_*, \mathsf{c}_{\mathrm{os}}, \mathsf{c}_{\mathfrak{h}}, \varepsilon_{\mathfrak{h}}, r_P^{\circ}, \mathsf{C}_{\mathcal{A}}, \mathsf{s}^{\star}, K_\Lambda, N_\star, K_P,$$
$$w_\star, \varepsilon_{\mathcal{W}}, \mathsf{o}, B^{(\mathrm{cg})}, A_0, A_1, \mathfrak{p}_0, \alpha_0, \alpha_1, B_3, E_0, M, L'$$

the plaquette constant $\mathsf{c}_{\mathrm{pl}}=600$ and the number C_P of equation YM3_I:(799), $K_{\mathrm{c}}=96(1+2\mathsf{c}_{\mathrm{pl}})^2C_{\mathrm{E}}$, $C_{\mathrm{E}}=4096$, the numbers b_{D}, $\mathsf{B}_\sharp(0)$, $\mathsf{B}_\sharp(1/2)$ and $\mathsf{B}_\sharp=\mathsf{B}_\sharp(0)$ of Prop. 5.35, and $a^\sharp$, $K^\sharp$ of the tower hypotheses of §5.2.

Here $\lambda_{\mathrm{r,gs}}$ denotes the canonical history Gaussian gap, written λ_{r} in equation YM3_I:(1137) with the specialization of equation YM3_I:(1161):

$$\lambda_{\mathrm{r,gs}} = \min\left\{\frac{\gamma_0}{2}, \left(1+2(B_{\mathrm{nm}}^2/\gamma_0)^2\right)^{-1}\right\}.$$

The symbols B_{nm} and γ_0 have the values in that specialization. This history gap is distinct in notation from the plain-step gap $\lambda_{\mathrm{r}}^{\mathrm{pv}}$ defined below.

The entries are:

1. the lower bound $\mathsf{C}_{\mathrm{ex}} \ge \max\{2\mathsf{C}_{\mathrm{cs}}, 8\}$;
2. directly behind $\bar{c}_{\mathrm{P}}$: $\lambda_{\mathrm{r}}^{\mathrm{pv}} := \min\{\frac{1}{2}\gamma_{\mathrm{r}}, (1+2c_{\mathrm{r}}^2)^{-1}\}$ and $c_q := \max\{\bar{c}_{\mathrm{P}}+\mathsf{C}_{\mathrm{mn}}^3/\lambda_{\mathrm{r,gs}}+2,\ \mathsf{C}_{\mathrm{mn}}^3(\frac{1}{2}\mathsf{u}_{\mathrm{G}}+c_{\mathrm{r}}+3)/\lambda_{\mathrm{r}}^{\mathrm{pv}}+2\}$;
3. with the abbreviation $E_q := 1+2\lambda_{\mathrm{s}}+\mathsf{C}_{\mathrm{cs}}(c_q+2)$, the lower bounds $(1+4\beta_{\mathrm{s}})\kappa \ge E_q$ and $\frac{1}{2}\kappa \ge 2\lambda_{\mathrm{s}}+8\log 2+\frac{\mathsf{C}_{\mathrm{cs}}}{L}E_q$;

4. directly behind $\bar r_H$: $r_H^q := \kappa'' + \lambda_{\mathrm{s}} + \mathsf{C}_{\mathrm{cs}}(c_q + 2)$, $\kappa' := \kappa'' + 1$, $r_{\mathrm{w}}^q := \max\{\frac{\mathsf{C}_{\mathrm{cs}}}{L} r_H^q + \lambda_{\mathrm{s}}, \frac{8\kappa'}{L}\}$, $r_t := \lambda_{\mathrm{s}}$, $r^q := r_{\mathrm{w}}^q + \lambda_{\mathrm{s}}$, $r_1 := \kappa'' - 8\log 2$, $r_1' := \kappa'' - 16\log 2$;
5. the lower bounds $\bar\kappa \geq 1 + \mathsf{C}_{\mathrm{cs}} r_H^q + \log \mathsf{C}_{\mathrm{cs}}$ and $\bar\kappa \geq 3 + 16\kappa' + 2\lambda_{\mathrm{s}} + 2\lambda_V$;
6. the lower bound $\mathsf{C}_{\mathrm{mn}}(LM)^3 \geq c_q$;
7. directly before γ, in this order: $c_K := 2/\gamma_{\mathrm{r}} + 4(c_{\mathrm{r}} + \frac12)^2/\gamma_{\mathrm{r}}^2$; $B' := 6 \cdot 10^5 c_K (LM)^3 e^{16\kappa'}$; $S' := 4096 e^{8\kappa'}$; $b_A := 6.1201^2 M^3$; $K_T^{\mathrm{pl}} := \max\{12L^4M^3K_{\mathrm{c}}, 2b_A, 112b_A + \frac12 M^3\}$; $a_Z := \mathsf{c}_{\mathrm{os}}\alpha_0$; $r^{\mathrm{pl}} := \min\{r_P^\circ, \varepsilon_{\mathfrak h}/\sqrt{2}, 1/(3\mathsf{c}_{\mathfrak h}), a_Z/(2\mathsf{c}_{\mathfrak h})\}$; $\delta' := \min\{1, \varepsilon_{\mathfrak h}/\mathsf{C}_{\mathcal A}, \mathsf{s}^\star\alpha_0/(1+\beta_{\mathrm{s}}), A_1(2\mathfrak p_0/e)^{\mathfrak p_0}\}$; $r_2^{\mathrm{pl}} := \mathsf{C}_{\mathcal A}\delta'$; $r_3^{\mathrm{pl}} := \min\{r^{\mathrm{pl}}, r_2^{\mathrm{pl}}\}$; $\bar u_f := K_\Lambda \mathsf{s}^\star N_\star$; $\varrho^{\mathrm{pl}} := \min\{\frac14 r^{\mathrm{pl}}, (r^{\mathrm{pl}})^3/(256B'K_T^{\mathrm{pl}}S')\}$; $\sigma_3 := \min\{(r^{\mathrm{pl}})^2/(64B'K_T^{\mathrm{pl}}S'),\ r^{\mathrm{pl}}\varrho^{\mathrm{pl}}/(4B'K_T^{\mathrm{pl}}S')\}$; $\varepsilon_1 := [8\mathsf{C}_{\mathrm{cs}}^2 L^3 e^{\mathsf{C}_{\mathrm{cs}}(\kappa''+c_q+r_H^q)}]^{-1}$; $v_0 := (2^{17}L^3)^{-1}$; $v_d := (2^{17}L^3)^{-1}$; $\hat\rho := \varepsilon_1/(8e)$; $\eta := 1$; $\gamma_2^q := (LM)^{-3}$; $\theta_m := \frac12\vartheta_*$; $\varepsilon_m^{\mathrm{pl}} := \min\{\frac12\vartheta_*, (LM)^{-3/2}\}$; $\theta_1' := \frac12\theta_m$; $\theta_1'' := \frac14(1/\varepsilon_m^{\mathrm{pl}} + 4\varepsilon_m^{\mathrm{pl}}/\theta_m^2)^{-1}$; $v_{\mathrm{ce}} := \mathsf{C}_{\mathcal A}$; $v^+ := \sqrt2\,\mathsf{C}_{\mathcal A} + 1$; $a_{\mathrm{cap}} := \varepsilon_1 v_0/(96e)$; $c_1^+ := \min\{\frac14 r_3^{\mathrm{pl}},\ a_{\mathrm{cap}}(r^{\mathrm{pl}})^3/(128K_T^{\mathrm{pl}}M^3)\}$; $K_Z := 4B'K_T^{\mathrm{pl}}S'/r^{\mathrm{pl}}$; $B_0 := 6\cdot10^5 c_K(LM)^3$; $K_6 := 1296M^3\max\{1, 2C_P\}B_3L^{-2}$; with the abbreviation $\theta_* := \min\{\theta_1', \theta_1''\}$,

$$\sigma_e := \min\{\sigma_3,\ a_{\mathrm{cap}}(r^{\mathrm{pl}})^2/(16K_T^{\mathrm{pl}}M^3),\ c_1^+/(2K_Z),\ a_Z\theta_*/(4\sqrt2 K_Z\mathsf{c}_{\mathfrak h}v^+L),\ \theta_* A_1/(4B_0K_6(2L)^{1/2}A_0)\};$$

$\bar K_{\mathrm{or}} := \mathsf{C}_{\mathrm{cs}}(LM)^3 e^{\mathsf{C}_{\mathrm{cs}}(c_q + r_H^q + \kappa'')}$; $Q^{\mathrm{pl}} := e\bar K_{\mathrm{or}}/v_d$; $Q' := e^{\kappa - r^q}$; $\bar K_{\mathrm{w}} := \frac12(6(LM)^3)^{1/2}$; $c^{\mathrm{pl}} := \mathsf{C}_{\mathrm{mn}}^{-1}\gamma_2^q(1-\theta_m)^2A_1^2/4$;
8. then, in this order: $\theta_{\mathrm{w}} := \min\{1, E_0 e^{-(\kappa''-\kappa)}\}$; $c_{\mathfrak h}^H := \max\{54C_P\mathsf{b}_{\mathrm D}e^{8\bar\kappa}\mathsf{B}_\sharp(\frac12)/\mathsf{B}_\sharp,\ 9\mathsf{c}_{\mathfrak h}\}$; $c_0'' := 5200(c_{\mathfrak h}^H + \mathsf{c}_{\mathfrak h})$; $\alpha_2^{\mathrm{pl}} := 2c_0''\varepsilon_{\mathfrak h}$, the third radius α_2 of the cores above; $c_{\mathfrak h}'' := c_0'' + 102\,\mathsf{c}_{\mathfrak h}L^{-5/2}\alpha_2^{\mathrm{pl}}$; $K_H^0 := 1.02\cdot 746$; $K_H := \max\{2K_H^0B_3, 10\mathsf{c}_{\mathfrak h}M^3\}$; $C_{\mathrm{gs}} := \max\{\alpha_0/\varepsilon_{\mathfrak h}, 2\mathsf{c}_{\mathfrak h}/\mathsf{c}_{\mathrm{os}}, 1200\mathsf{c}_{\mathfrak h}\}$; with the abbreviations $m_0 := \min\{\alpha_2^{\mathrm{pl}}, \alpha_0\}$ and $m_1 := \max\{\alpha_2^{\mathrm{pl}}, \alpha_0\}$: $c'' := \max\{C_{\mathrm{gs}}, c_{\mathfrak h}''\}$; $c_3^a := L^{-5/2} + c''r^{\mathrm{pl}}/m_0$; $c_4^a := \max\{c''/m_0, 1/r^{\mathrm{pl}}\}$; $C_V^{\mathrm{pl}} := 2^{10}LC_{\mathrm E}(1-1/L)^{-1}$;

$$K_J^0 := \max\{1,\ 10\mathsf{c}_{\mathfrak h}M^3L^{-5/2}m_1,\ 8(1+84C_P(M/L)^3)L^{-5/2}m_1\};$$

$K_J := 2K_J^0$; $K_{\mathrm{c}}^a := 2B'S'K_J$;

$$K_{\mathrm{ta}} := 2^8\cdot 24(LM)^3[K_JK_{\mathrm{c}}^a + 2K_T^{\mathrm{pl}}(K_{\mathrm{c}}^a)^2/(r^{\mathrm{pl}})^2];$$

$a_Z' := \min\{a_Z, \mathsf{c}_{\mathfrak h}\varepsilon_{\mathfrak h}\}$; $\sigma_5' := a_Z'\theta_*/(4\sqrt2 K_Z\mathsf{c}_{\mathfrak h}v^+L)$; $\sigma_*^a := \min\{\sigma_3, \sigma_e, \sigma_5', 1\}$; $c_g^{\mathrm{pl}} := 2\max\{1/\varepsilon_{\mathcal W}, K_H\}$;

$$w_5 := \min\{K_J^0/(1.01\,C_V^{\mathrm{pl}}c_4^aK_{\mathrm{ta}}),\ \min\{\varrho^{\mathrm{pl}}, \frac18 r^{\mathrm{pl}}, K_Z\sigma_*^a\}/K_{\mathrm{c}}^a,\ \sigma_*^aK_T^{\mathrm{pl}}/(6144M^3\alpha_0^2),\ (\sigma_*^aK_T^{\mathrm{pl}}/(2C_V^{\mathrm{pl}}(c_3^a)^2K_{\mathrm{ta}}))^{1/2},\ \sigma_*^a,\ m_0\theta_*A_1/(2K_{\mathrm{c}}^ac_g^{\mathrm{pl}}(2L)^{1/2}A_0)\};$$

$\sigma_5 := \frac12 w_5$; $\tau_5 := \frac14 w_5$; $b_r := \mathsf{o}LMB^{(\mathrm{cg})}\alpha_0$; $a_3 := 8.4(\alpha_1(1+2b_r) + b_r)$; $C_2^Q := 29.4a_3^2 + 14.7a_3$; $\bar C_2 := \max\{C_2^Q, 1\}$; $a_4 := 7.4a_3 + C_2^Q$; $C_H^Q := 18$; $c_s^Q := \min\{1, \mathsf{o}L, 1/(12Ma_3)\}$, the constant c_s of Thm. Q(a); $c_3^Q := \min\{\frac14 a^\sharp, \alpha_0/(4K^\sharp), \alpha_1, \frac14, \alpha_2^{\mathrm{pl}}/18\}$;

$$\lambda_Q := \min\{1/(4a_3),\ a_3/C_2^Q,\ \alpha_1/a_4,\ a^\sharp/(2\alpha_0 + a_4),\ [\alpha_0/(K^\sharp(2\alpha_0 + a_4^2))]^{1/2}\};$$

$$C_Q := \max\{2\lambda_Q^{-4},\ 32(c_s^Q)^{-4},\ 72M^2\bar C_2^2(c_3^Q)^{-2}\};$$

$K_{\mathrm{Lf}} := 2C_{\mathrm E}(140C_Q + 1)$; $u^c := 2^9\cdot 129\,K_{\mathrm{Lf}}L/(L-1)$; $\bar u_f^c := \bar u_f + u^c$; $c_J^c := 2^{21}L^3K_{\mathrm{c}}^a(7M^3/r_P^\circ + \bar u_f^c/r_3^{\mathrm{pl}})$; $w_6 := \min\{w_5, 1/c_J^c\}$, the constant c_{w} of Thm. Q(c); $\sigma_6 := \frac12 w_6$; $\tau_6 := \frac14 w_6$;

9. fifteen upper bounds on γ, each of the form $\sup_{x\geq\gamma^{-2}} F(x)\leq\Phi$ with $F(x)\to 0$ as $x\to\infty$ and $\Phi>0$ formed with earlier constants; with the thresholds $\delta(x)=A_1x^{-1/2}(\log x)^{\mathfrak{p}_0}$, $\varepsilon(x)=A_0x^{-1/2}(\log x)^{\mathfrak{p}_0}$ of YM3_I:(634) and
$$\xi(x):=\exp(-\mathsf{C}_{\mathrm{mn}}^{-1}\gamma_2^q(1-\theta_m)^2A_1^2(\log x)^{2\mathfrak{p}_0})$$
they are: $3K_P\delta(x)\leq a_{\mathrm{cap}}$; $\bar{u}_f^c\mathsf{C}_{\mathcal{A}}\delta(x)/r_3^{\mathrm{pl}}\leq a_{\mathrm{cap}}$; $\frac{64}{3}(7M^3+\bar{u}_f^c)c_1^+\mathsf{C}_{\mathcal{A}}\delta(x)/(r_3^{\mathrm{pl}})^2\leq\frac{1}{3}a_{\mathrm{cap}}$; $\mathsf{C}_{\mathcal{A}}\delta(x)\leq\frac{1}{2}r_3^{\mathrm{pl}}$; $(\theta_1'+\theta_1'')\delta(x)\leq\frac{1}{2}c_1^+$; $6(LM)^3\xi(x)\leq 1$; $2(LM)^3\xi(x)^{1/2}\leq\varepsilon_1L^3$; $[(1-\theta_m)A_1(\log x)^{\mathfrak{p}_0}]^{-1}\leq 1$; $2Q^{\mathrm{pl}}Q'L^{-2c^{\mathrm{pl}}\log(x/2)}\leq 1$; $Q'\bar{K}_{\mathrm{w}}e^{-c^{\mathrm{pl}}(\log(x/2))^2}\leq\hat{\rho}v_d$; $\mathsf{c}_{\mathfrak{h}}v^+\delta(x)\leq\frac{1}{2}a_Z'$; $2B_3\varepsilon(x)\leq 1$; $2\varepsilon(x)\leq\varepsilon_{\mathcal{W}}\alpha_0$; $K_H\varepsilon(x)\leq\frac{1}{2}\alpha_2^{\mathrm{pl}}$; $8K_\Lambda N_\star(1+\beta_{\mathrm{s}})\delta(x)/\alpha_0\leq w_\star$.

Each formula names only constants that stand before it, and none reads K, m, a level, a sequence of couplings, a weight or a field. The list so extended is an order table of admissible shape; choices of it exist; the restriction of a choice to the constants of $\mathcal{T}^{\mathrm{S}}[2\varrho]$ is a choice of that list, and its restriction to the constants of the list $\mathcal{T}^{\mathrm{F}+}$ of Thm. YM3_I:D (the real choices in Prop. YM3_I:69.1) is a choice of that list. Thm. R lowers γ further where it says so.

Theorem Q (Three plane couplings)**.** Let the constants be a choice of the extended list stated before this theorem, which adds to the list of Thms. YM3_I:AJ and YM3_I:AD four lower bounds on chosen constants, each by a function of earlier constants, further determined constants placed before γ and further upper bounds on γ, and no new condition on L itself; its choices exist, they restrict to choices of the lists of Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD, and at each of them the standing conditions of the frame of [7, Def. B.14] hold, as do the locality for all pairs of regions and the gauge covariance at all decoupling parameters of the Gaussian structure among the construction results, the tower hypotheses of §5.2 and the inequality $\alpha_0+\alpha_1\leq a^\sharp$ with the constant $a^\sharp$ in the radius conditions YM3_I:(867). At every choice $L>4\cdot 10^{19}$, as a consequence of the condition $L\geq 2\mathsf{C}_{\mathrm{ex}}$ among the construction results and of the lower bound $\mathsf{C}_{\mathrm{ex}}\geq 2\mathsf{C}_{\mathrm{cs}}$ of the extended list, with $\mathsf{C}_{\mathrm{cs}}\geq 24C_{\mathrm{E}}^2e^{24}$.

(a) *Lifted local terms.* For all K,m and $1\leq j<k<K$ with $2L^{m+K-k}\geq C_LM$, that is, for which the lattice (ℓ,K), $\ell=2L^m$, is allowed at the height k (Section YM3_I:50.4), $C_L\geq 8\mathsf{n}_{\mathrm{e}}L$ being the room and $\mathsf{n}_{\mathrm{e}}\geq 1$ the halo width of Section YM3_I:50.4, so that $m+K-k\geq m'+2$, $\mathcal{D}_j$ and $\mathcal{D}_k$ are defined and a box of at most two cubes of the level k along each axis does not wrap around the torus, which has at least $8L$ such cubes along each axis, every $Z\in\mathcal{D}_j$ short at k, every $P'\in\mathcal{U}_k(X_Z)$ and every term E on $\mathcal{U}_j(Z)$,
$$|E(\rho_Z^{k\to j}P')-E(\mathbf{1})|\leq C_Q\sup|E|\,(d_j(Z)+2)^4L^{-4(k-j)},$$
and the same with $(d_j(Z)+2)^2$ in the place of $(d_j(Z)+2)^4$ and with $72M^2\bar{C}_2^2(c_3^Q)^{-2}\leq C_Q$ in the place of C_Q when $L^{-(k-j)}\leq\lambda_Q$ and $(d_j(Z)+4)L^{-(k-j)}\leq c_s$, with the constants λ_Q, $c_s=c_s^Q$, $\bar{C}_2$ and c_3^Q of item 8 of the list; C_Q is a function of constants placed before γ, it lays no condition on any constant, and it depends on no j,k,K,m,Z,P',E. The same holds on every larger base in $\mathcal{D}_k$ that holds the box of Z, for cores at any positive radii, and on the plain cores for its local terms of level j. The power 4 is attained: for every short Z and every plaquette with its bonds in the boundary of Z there are a term E with $\sup|E|\leq\frac{1}{2}$ and a triple P' with $E(\rho_Z^{k\to j}P')-E(\mathbf{1})\geq c_QL^{-4(k-j)}$, $c_Q>0$ a constant. For the abelian group $\mathrm{U}(1)^3$ the inequality fails.

(b) *The lift.* Let K,m and $1\leq k<K$ be as in (a), $2L^{m+K-k}\geq C_LM$, and let $\kappa_{\mathrm{lf}}\geq 4\log 18432$ be a rate. For terms $g^{(j)}(Z,\cdot)$ on $\mathcal{U}_j(Z)$, $1\leq j\leq k$, $Z\in\mathcal{D}_j$, with $B_j=\sup_Z e^{\kappa_{\mathrm{lf}}d_j(Z)}\sup|g^{(j)}(Z,\cdot)|<\infty$, the lift
$$\mathrm{Lf}[g](X;P')=\sum_{j\leq k}\sum_{Z\in\mathcal{D}_j:\,X_Z=X}[g^{(j)}(Z;\rho_Z^{k\to j}P')-g^{(j)}(Z;\mathbf{1})],\qquad X\in\mathcal{D}_k,\ P'\in\mathcal{U}_k(X),$$

($X_Z = Z^{\mathrm{up}}$, the union of the cubes of the level k that hold a cube of Z, for the Z that are not short, among them all $Z \in \mathcal{D}_k$, for which $Z^{\mathrm{up}} = Z$) is a term on $\mathcal{U}_k(X)$, linear in g, depending on g at the Z in X alone, and

$$|\mathrm{Lf}[g](X;P')| \le 2C_E(140C_Q+1)\sum_{j\le k} L^{-(k-j)}B_j\, e^{-\kappa_{\mathrm{lf}} d_k(X)}, \qquad C_E = 4096,$$

a bound that is uniform in the level.

(c) *One step at all levels.* Let K, m with $L^m \ge MR(x_K)$ and let $x_0 > \cdots > x_K \ge \gamma^{-2}$ be an admissible sequence of couplings in the frame of [7, Def. B.14]; by [7, eq. 215] the frame has $C_L \le 2LR(\gamma^{-2}) \le 2LR(x_K)$, so that $2L^{m+K-k} \ge 2LMR(x_K) \ge C_L M$ at every $k < K$, and (a) and (b) apply at every level. Let $\sigma, \tau \ge 0$ with $\sigma + 2\tau \le c_{\mathrm{w}}$, $c_{\mathrm{w}} = w_6 > 0$ the constant of item 8, placed before γ, and let ω be a complex plaquette function whose plane weights w_κ, $\kappa \in \{12, 13, 23\}$, obey $\sup|w_\kappa| \le \sigma(x_K - \bar{B})$ and $M\mathrm{Lip}_\infty w_\kappa \le \tau x_K$, with no tie between the planes.

Put $v_n = -L^{K-n}\omega/x_n$, $g_n = x_n^{-1/2}$, $y_n = \log(x_n/2)$, $\mathsf{b}_n = A_1(\log x_n)^{\mathfrak{p}_0}$ the threshold of the level n (equation YM3_I:(634)), and, for a weight w, $A_n(w;U) = A^{\eta_n}(w,U) = L^n\sum_p w(p)a_p(U)$, the weighted action of [7, eq. 216]. For $1 \le n \le K$, $X \in \mathcal{D}_n$ and $\mathsf{a} = (\mathsf{a}_1, \mathsf{a}_2) \in (0, \alpha_0] \times (0, \alpha_2]$, $\alpha_2 = \alpha_2^{\mathrm{pl}}$ the third radius of item 8, the *core of the size* a, $\mathcal{U}_n(X;\mathsf{a})$, is the set of triples P in the same hereditary core of Def. YM3_I:49.1 at the radii (α_0, α_1) with $|\partial\mathbf{U}_P(p) - \mathbf{1}| < \mathsf{a}_1 L^{-2n}$ at the plaquettes formed on X and $\mathrm{hol}_n(P;X) < \mathsf{a}_2$; $\mathcal{U}_n(X) = \mathcal{U}_n(X;(\alpha_0,\alpha_2))$ is the core with the third radius, and $N_n(\mathsf{a}) = \sigma\mathsf{a}_2/\alpha_2 + 2\tau L^{-(K-n)}\mathsf{a}_1/\alpha_0$.

The following objects are formed by recursion on the step $k = 0, 1, \dots, K-1$, which integrates the fluctuation field of the level k at the coupling x_k over the fibre of a block field W of the level $k+1$, at the triple $Q = P_W$. *The classical action and the trees:* the classical action $\Psi^{(k)}(Q;\cdot)$ of the step is its Gaussian form, with the covariance $C_*(Q)$ of the step, together with its linear term and its higher potentials, the weight v_k and the earlier trees being entered as potentials, so that the action of the step carries no explicit weight; $c_*^{(k)}$ is its stationary point; the tree $\mathrm{ta}_k(Z';\omega,\cdot)$, $Z' \in \mathcal{D}_{k+1}$, is the value $\Psi^{(k)}(Q;c_*^{(k)})$, written as a sum of local terms and re-sorted into terms on the cores $\mathcal{U}_{k+1}(Z')$; and the sum of the trees is $\mathrm{Tr}_n(P) = \sum_{j=1}^n (x_{j-1}/x_n)\sum_{Z\in\mathcal{D}_j}\mathrm{ta}_{j-1}(Z;\omega,\rho_Z P)$, $\mathrm{Tr}_0 = 0$, so that $x_{k+1}\mathrm{Tr}_{k+1}(Q) = x_k[\mathrm{Tr}_k(\rho Q) + \Psi^{(k)}(Q;c_*^{(k)})]$.

The non-classical array: the non-classical term $\delta_{k+1}(Z';\omega,\cdot)$ consists of the potential V^J of the Jacobians of the chart (the localised exponent J_W of the Jacobians in the weight wt_W of the chart integral below, one of the potentials of the step) and of the potential V' of the arrays of the step, both taken at $c_*^{(k)}$, and of the ideal term E_k^{id} below at the centre $g_k^{-1}c_*^{(k)}$ less the same object at $\omega = 0$, all re-sorted in the same way; $\mathrm{Lf}_k = \mathrm{Lf}[g]$ is the lift of (b) at $g^{(j)} = \delta_j$; $S_n^c[\omega](U) = S[\mathfrak{f}_n](U) + \sum_{j\le n}\sum_{Z\in\mathcal{D}_j}[\delta_j(Z;\omega,p_Z^{(j)}(U)) - \delta_j(Z;\omega,\mathbf{1})]$, where $S[\mathfrak{f}_n]$ is the action of the bare arrays of the never-large term at the zero source ([7, Prop. B.15], at $w = 0$) and $p_Z^{(j)}(U)$ the plain pair of U on Z, the configuration and its current of the level j on $\mathrm{Bd}(Z)$; and $E_n[\omega] = E_{(n)} + \sum_{i\le n}\sum_{X\in\mathcal{D}_i}\delta_i(X;\omega,\mathbf{1})$, $E_{(n)}$ the number of that term.

The centres, the ideal term and the tiny terms: the true centres are $\nu_j^{(n)}(V) = g_j^{-1}c_*^{(j)}$, read at the triple ρP_V of the level $j+1$, $0 \le j < n$, and the centre of the step is $\mathsf{m} = \nu_k^{(k+1)}(W)$; for a system V of potentials $V^o(Y;\cdot)$, $Y \in \mathcal{D}_k$, and a centre n, $\Xi_{\mathsf{n}}[V](Q) = \int d\mathsf{G}_{\mathsf{n}}(B)\,\mathbf{1}\{B \in \mathrm{Rg}_{x_k}\}\exp\sum_{Y,o}V^o(Y;B-\mathsf{n})$, G_{n} being the Gaussian measure of the covariance $C_*(Q)$ and the mean n; V_{m} is the system of the three potentials of the step about the centre m (the potentials of the step, the potential of the arrays, and the remainder of the shift of the centre), without the carried tiny terms; $E_k^{\mathrm{id}}(X';Q,\mu)$, $X' \in \mathcal{D}_{k+1}$, is the cluster sum at

the centre 0 of the system V_μ on X', so that $\Xi_0[V_\mu](Q) = \exp\sum_{X'} E_k^{\mathrm{id}}(X'; Q|_{X'}, \mu|_{X'})$; and $\mathrm{Rf}^{(k+1)}(X'; Q, \nu)$ is the cluster sum at the centre ν_k of the system V_{ν_k} enlarged, for $k \geq 1$, by the carried term $\mathrm{Rf}^{(k)}$ as one further potential, less the cluster sum at the centre 0 of V_{ν_k} without it.

The classical factor of the step: $c_k(W) = n_k(W) \exp[\mathcal{E}_k(W) - x_k \Psi^{(k)}(Q; c_*^{(k)}) + \sum_{Y \in \mathcal{D}_k} (V^J + V')(Y; Q|_Y; c_*^{(k)}|_Y)]$, where $\mathcal{E}_k(W)$ is the exponent of $\Phi_k[\omega]$ below at $V_W(0)$ ($\Phi_0[\omega] = \rho_0$), V^J and V' are the potential of the Jacobians of the chart and the potential of the arrays of the step, as in δ_{k+1}, and $n_k(W) = \mathsf{pf}_k(x_k)\, \mathsf{Z}^{(k)}$ is the normalisation of the step, with the prefactor equation YM3_I:(907) and the Gaussian integral in Prop. YM3_I:50.2 at the triple of W; so $\mathrm{wt}_W(B) \Phi_k[\omega](V_W(B))$ is $c_k(W)$ times the density of G_m at B times $\exp\sum_{Y,o} V^o_\mathsf{m}(Y; B - \mathsf{m})$.

Then for $1 \leq n \leq K$ the never-large term of $\rho_0 = \exp[-A(x_0 - L^K\omega, \cdot)]$ at the level n has the density

$$f_n(V) = \Phi_n[\omega](V) \exp \sum_{X \in \mathcal{D}_n} \mathrm{Rf}^{(n)}(X; P_V, \nu^{(n)}(V)) \quad \text{on } \mathcal{V}_n^\chi, \qquad f_n = 0 \quad \text{off it},$$

$\mathcal{V}_n^\chi$ being the set of the configurations of the level n that pass all the small-field tests ([7, Def. B.14]), and

$$\Phi_n[\omega] = \exp[E_n[\omega] - x_n A_n(1 + v_n; U_n(V)) - x_n \mathrm{Tr}_n(P_V) + S_n^c[\omega](U_n(V))], \qquad v_n = -L^{K-n}\omega/x_n,$$

where Tr_n, the sum of the trees (the classical remainders) of the earlier steps, the non-classical array $S_n^c[\omega]$ and the number $E_n[\omega]$ are those above, and P_V is the frozen triple of the background $U_n(V)$ of [7, Prop. B.15] and Thm. YM3_I:E in the setting of Section YM3_I:52.7. Each step $0 \leq k < K$, at every $W \in \mathcal{V}_{k+1}^\chi$ and with $f_0 = \rho_0$, is the identity

$$\int_{\mathrm{Rg}_{x_k}} \mathrm{wt}_W(B) f_k(V_W(B))\, \mathrm{d}B = c_k(W)\, \Xi_0[V_\mathsf{m}](Q) \exp \sum_{X' \in \mathcal{D}_{k+1}} \mathrm{Rf}^{(k+1)}(X'; Q, \nu^{(k+1)}(W)), \qquad c_k(W)\, \Xi_0[V_\mathsf{m}](Q) = \Phi_{k+1}[\omega](W),$$

whose left member is $f_{k+1}(W)$, the chart integral $\mathsf{I}[f_k](W)$ at $\mathsf{t} = x_k$ ([7, Def. B.2] and equation [7, eq. 191], [7, eq. 191]): Rg_{x_k} is the set of the scaled fields below the threshold $\mathfrak{b}(x_k)$ ([7, §B.2]), $V_W(B)$ the chart of the fibre over W ([7, Lem. B.3]), and $\mathrm{wt}_W(B) = \mathsf{pf}_k(x_k)\, z_{V_0}^{(k)}(\Gamma) \exp[-x_k G_W(B) + \mathsf{J}_W(B)]$, $V_0 = V_k^{\mathrm{bf}}(W)$, the weight of the chart integral; $Q = P_W$, $\mathsf{m} = \nu_k^{(k+1)}(W)$ is the centre of the step, and $c_k(W)$, Ξ_0 and V_m are those above; so each step is classical, times ideal, times the exponential of a tiny term.

The ideal term is $\exp\sum_{X'} E_k^{\mathrm{id}}(X'; \cdot)$ with $|E_k^{\mathrm{id}}| \leq \frac{1}{4} e^{-\kappa'' d_{k+1}(X')}$, so it does not vanish. $\mathrm{Rf}^{(n)}(X; P, \nu)$ is holomorphic on $\mathcal{U}_n(X)$ times the polydisc of the centres, one 1-form ν_j for every earlier level $j < n$, with $|\mathrm{Re}\, \nu_j(b)| < \vartheta_{j,n} \theta_1' \mathsf{b}_j$ and $|\mathrm{Im}\, \nu_j(b)| < \vartheta_{j,n} \theta_1'' \mathsf{b}_j$, $\vartheta_{j,n} = \frac{1}{2}(1 + 2^{-(n-j-1)})$, gauge invariant when the centres are carried along, and $|\mathrm{Rf}^{(n)}| \leq \bar{K}_{\mathrm{w}} e^{-c^{\mathrm{pl}} y_n^2} e^{-\kappa d_n(X)}$, $y_n = \log(x_n/2)$, with the constants $\bar{K}_{\mathrm{w}}$ and c^{pl} of item 7; the true centres obey $|\nu_j^{(n)}(V)(b)| \leq \frac{\theta_*}{2} L^{-2(n-j-1)} \mathsf{b}_j$, $\mathsf{b}_j = A_1 (\log x_j)^{\mathfrak{p}_0}$ the threshold of the level j (equation YM3_I:(634)); the trees obey $e^{r_1 d_n(Z)} |\mathrm{ta}_{n-1}(Z; \omega, P)| \leq K_{\mathrm{ta}} N_n(\mathsf{a})^2$ on the cores $\mathcal{U}_n(Z; \mathsf{a})$ of the size a, at every $\mathsf{a} \in (0, \alpha_0] \times (0, \alpha_2]$, with one K_{ta} and one rate r_1 at all levels; the non-classical terms obey $e^{r_1 d_n(Z)} |\delta_n(Z; \omega, P)| \leq 129 \min\{1, (\sigma + 2\tau)/c_{\mathrm{w}}\}$ on $\mathcal{U}_n(Z)$; and the lift of the non-classical array, which is (b) at $g^{(j)} = \delta_j$, $B_j \leq 129$ and the rate $\kappa_{\mathrm{lf}} = r_1$, obeys $e^{r_1 d_k(X)} |\mathrm{Lf}_k(X; \omega, P')| \leq 129 K_{\mathrm{Lf}} L/(L-1)$, $K_{\mathrm{Lf}} = 2C_E(140 C_Q + 1)$, as $\sum_{j \leq k} L^{-(k-j)} \leq L/(L-1)$.

The rate r_1 is admissible in (b): $r_1 = \kappa'' - 8\log 2$ with $\kappa'' = (1+4\beta_{\mathrm{s}})\kappa + 1 + \lambda_{\mathrm{s}}$ and $\lambda_{\mathrm{s}} \geq 1$ (Section YM3_I:82, where $\lambda_{\mathrm{s}} = \log 18432$ and the formula for κ'' is recalled), so that by item 3, $r_1 \geq E_q + 2 - 8\log 2 > \mathsf{C}_{\mathrm{cs}} > 4\log 18432$. Every number is a constant placed before γ or an absolute constant. Everything is holomorphic in ω on the open set of the plaquette functions that lie, for some $\sigma, \tau \geq 0$ with $\sigma + 2\tau \leq c_{\mathrm{w}}$, in the class with strict inequalities $|v_n|_\infty < \sigma$ and $\mathrm{sw}_n(v_n) < 2\tau L^{-(K-n)}$ at every $0 \leq n \leq K$, $\mathrm{sw}_n(v)$ being the largest difference $|v(p_{\mu\nu}(x')) - v(p_{\mu\nu}(x))|$ over the parallel plaquettes with $x' - x$ along e_μ or e_ν and of length at most one spacing of the level n.

At $\omega = 0$ the trees, the δ_n and Tr_n vanish, the true centres are $\nu_j^{(n)}(V) = 0$, and $\mathrm{Rf}^{(n)}(X; P, 0) = 0$ for every $P \in \mathcal{U}_n(X)$, at the zero centres; hence $f_n = \Phi_n[0]$ is the never-large term. The background is $U_n(V)$ from [7, Prop. B.15], and the centres of the Gaussian integrals are free arguments; the cores with the third radius coincide with the plain cores at the levels $n \leq (\frac{L}{2} + 29.39)/\log L$, so at the levels $n \leq 13$ for every odd $L \geq 13$, and are proper subsets of them from some level on; at $k = 0$, where the step is written with bond labels, its centre is at most $K_Z\sigma$ on the whole plain core, K_Z a constant placed before γ.

■□

Theorem R (Weighted backgrounds)**.** In the setting of Thm. Q, part (a) uses the regular-reference and geometric setting of Def. 5.4; its analytic response and Hessian estimates are proved in Prop. 5.5. Parts (c) and (d) use the complex scalar gauge-transfer estimates proved in Lem. 5.21. Evaluation of the algebraic identity in (e) on these moved backgrounds uses the same construction and its stated regularity and geometric conditions; the algebraic identity and its bounds for already given families do not require that construction.

(a) *Critical backgrounds.* Use the real regular reference, natural-pair and separated-region setting of Def. 5.4. For the plane and symmetric corner weights $\mathbf{h} = (h, v)$, Prop. 5.5 gives a unique critical background in the slice $RD^{\mathsf{T}}A = 0$, $\mathcal{Q}(A) = b$, among fields with $\sup_y \|A\|_y^{\beta_H} \leq r_*$, where this norm includes the scaled first-gradient Hölder seminorm defined there. It is holomorphic in the row datum and the weights on $\sup|b| < \rho_*$, $\|\mathbf{h}\| < z_*$, real at real data, and obeys the localized response bounds eq. 5.iv–eq. 5.v. Its Hessian in the block datum satisfies eq. 5.vi on $Y_0 + iY_0$, with the longer-sequence hypotheses in that proposition. The unweighted comparison is at the same reference background. The Hölder norm of the weights cannot be replaced by their supremum in a uniform Hölder estimate for this response. There is a separate finite-lattice assertion at $U_0 = 1$, when all regions are the whole torus. For real plane weights $v_{\mu\nu} = \epsilon k_{\mu\nu}$, $|k| \leq 1$, or for a single real mixed tensor term $\epsilon\varkappa\langle f_{\mu\nu}, f_{\mu\lambda}\rangle$, $|\varkappa| \leq 1$, with $0 \leq \epsilon < 1$, the critical backgrounds near the zero block datum form a real-analytic family and their block Hessian at that datum obeys $(1-\epsilon)Z_0 \leq Z_\epsilon \leq (1+\epsilon)Z_0$. This assertion uses each of these two perturbations separately.

A common constant $c \neq -1$ in the plane weights factors out: the weights $(h, v + c)$ have the same critical fields as $\mathbf{h}' = (h, v)/(1+c)$ and their multipliers and Hessians are multiplied by $1 + c$. The smallness conditions are those on $\mathbf{h}'$. For real $c > -1$ the positive Hessian floor is multiplied by $1 + c$ as well.

After lowering γ by the ceilings specified in the proof, the following holds on every history. At each step $t \geq 2$, the background and row datum of the run satisfy the hypotheses above, and for real weights with $\|\mathbf{h}\| \leq z_N^+$ the block Hessian is at least $\frac{1}{2}\gamma^{\mathrm{pc}}\|B\|^2$ on real $B \in Y_0$. At the first step the real part of the second variation of the action itself is at least that amount for $B \in Y_0 + iY_0$ and complex weights with $\|h\|_\infty + \sup|v| \leq z_N^1$. The latter assertion uses the two-region natural-pair hypothesis and the explicit background and block-field ceilings in the proof.

(b) *Walk terms.* Under the natural-pair and separated-region hypotheses of Prop. 5.37, there are numbers $r_\tau, r'_\tau > 0$ and a power M^τ of L, depending only on L, such that all four

weighted systems are invertible for $|v|_\infty < r_\tau$, radii at most r'_τ, and block number at least M^τ. Their inverses satisfy the mean-square divergence-source estimate eq. 5.xliv, and their local walk terms satisfy eq. 5.xlv, at a common exponential rate. The Hessian changes by the local plaquette form of Lem. 5.36. Grouping a walk by its visited pieces gives zero for a disconnected set S and a bound with the factor $\epsilon_*^{\#S-1}$ for a connected set, after a further increase of M^τ for the prescribed ϵ_*, as proved after Prop. 5.37. The pieces and their adjacency are those specified there.

(c) *The moved background on a label.* On the label of a history with large-field regions, in every frame of Thms. YM3_I:AJ and YM3_I:AD (every such frame satisfies the standing conditions under which these labels are formed), after lowering γ by three further upper bounds of the shape of the list, $\sup_{x\geq\gamma^{-2}} \max\{w_{\mathrm{G}}, w_{\mathrm{lin}}(L)\}(\log x)^{-r} \leq 1$ with $w_{\mathrm{G}} = 4c_U^{-1}(7\log L + \log 2)$, which gives $MR(\gamma^{-2}) \geq \max\{w_{\mathrm{G}}, w_{\mathrm{lin}}(L)\}$, $\sup_{x\geq\gamma^{-2}} C_m\varepsilon(x) \leq \alpha_\star$ and $\sup_{x\geq\gamma^{-2}} 2K'_5 C_m\varepsilon(x) \leq b_2$, where C_m is fixed before γ and depends on L and the previously chosen construction constants, including B_3, and is the constant with which a label supplies its background, regular at the radius $C_m\varepsilon_k$ with the envelope $e(x) = C_m\varepsilon_{j(x)}$, ε_j being the radius of regularity of the layer j; $\alpha_\star$ is a positive number of L with $\alpha_\star \leq \min\{\alpha_{\mathrm{lin}}(L), e_0(L)\}$, the numbers of Lem. 5.19, so that the second ceiling gives the radius of regularity $\alpha \leq \alpha_{\mathrm{lin}}(L)$ and the bound $\sup e \leq e_0(L)$ on the envelope under which Lem. 5.19 applies; K'_5 is a positive number formed with L through the constant K_{lin} of Lem. 5.19 and the constant K_5 below; and $b_2 > 0$ is the radius with which the substitution with three pairs of radii among the construction results (Prop. 5.27) carries the cores at half their radii into the cores at three quarters of them, and for anisotropies an of the plaquettes with $t(\mathrm{an}) < t_0$, $t_0 > 0$ a number of L with $t_0 \leq t_{\mathrm{lin}}(L) \leq \frac{1}{4}$, $t_{\mathrm{lin}}(L)$ the number of Lem. 5.19, t being the supremum plus the scaled Hölder-$\frac{1}{2}$ quotients: the moved background $W = e^{i\eta_k A}U$, $\eta_k = L^{-k}$ the spacing of the level, exists, is critical for $\sum_p(1+\mathrm{an}(p))a_p$ on the surface of the prescribed block averages up to a complex gauge, and is holomorphic in an; if $U = \tilde{U}$ at the bonds with an end point within the weighted distance R of a point x_1, then, with $e'(x) = e(x)\exp[-c_U(\mathsf{R} - d_B(x, x_1))_+]$,

$$l(b)\,|\eta_k A[U](b) - \eta_k A[\tilde{U}](b)| \leq K_U\, t(\mathrm{an})\, e'(b_-) \qquad \text{at every bond } b,$$

b_- the initial point of b, with no gauge transformation between; the same bound holds, at every plaquette p, site x and row c of the block constraints, for the curvatures, $l(p)^2\eta_k^2|D_U A[U](p) - D_{\tilde{U}} A[\tilde{U}](p)| \leq K_U t(\mathrm{an})e'(x_p)$, x_p the base point of p, for the divergences, $l(x)^2\eta_k^2|D^*_U A[U](x) - D^*_{\tilde{U}} A[\tilde{U}](x)| \leq K_U t(\mathrm{an})e'(x)$, and for the multipliers ω of the pair, $|\omega[U](c) - \omega[\tilde{U}](c)| \leq K_U t(\mathrm{an})e'(c_-)$, c_- the base point of c; if $\mathsf{R} \geq w_*$, then at every point x_* of a layer $n \geq 2$ with $d_B(x_*, x_1) \leq \mathsf{R} - w_*$, in one gauge that is a member of gauge systems of both U and $\tilde{U}$, the Hölder-$\frac{1}{2}$ quotients of $\eta_k^2(D_U A[U] - D_{\tilde{U}} A[\tilde{U}])$, over the plaquettes of one orientation in the cube of the bare sites within $\frac{1}{4}l_n$ bare spacings of x_*, and of $\eta_k(A[U] - A[\tilde{U}])$, over the bonds of one direction based within $\frac{1}{8}l_n - 2$ bare spacings of x_*, obey $(\frac{1}{4}l_n)^{1/2}[\cdot]_{1/2} \leq K_U t(\mathrm{an})e'(x_*)l_n^{-2}$, resp. $l_n^{1/2}[\cdot]_{1/2} \leq K_U t(\mathrm{an})e'(x_*)l_n^{-1}$; and the multiplier ω_0 of the plain background obeys $|\omega_0[U](c) - \omega_0[\tilde{U}](c)| \leq K_O e'(c_-)$.

Here $l(b), l(p), l(x)$ are the scales of the layers, $l_n = L^n$, D_U is the covariant lattice exterior derivative and D^*_U its adjoint, and d_B is the weighted distance of Def. YM3_I:12.3; c_U, K_U, K_O depend on L alone, $c_U \leq 1/(6L)$, $K_U = 7K_{\mathrm{lin}}^2(1+2L^3)^2 e^{c_U w_*}$ with the constant K_{lin} of Lem. 5.19 and $w_* = w + 21L$, $w = w(L)$ being the width of the cut-off used in the proof. The map $(\mathrm{an}, U) \mapsto A$ is jointly continuous, and the moved background, its triples and its currents are Borel functions of the configuration; every critical point on the fibre of norm $\leq r_2(L)$ is carried to W by a unique small gauge transformation that induces the identity at every top of the sequence; the full covariant gradient obeys $l(x)^2|\eta_k^2(\nabla_\kappa A_\nu)(x)| \leq K_5\, t(\mathrm{an})\, e(x)$; W lies

in the cores at three quarters of their radii; the construction commutes with the translations by vectors of the coarsest lattice; and $\sum_p |(1+\mathrm{an}(p))a_p(W) - a_p(U)| \le K|\mathrm{an}|_\infty \sum_p a_p(U)$.

(d) *The centres on a label.* Let $0 < \theta_3 \le 1$ be fixed before γ. In the setting of (c), let γ be lowered further by upper bounds of the shape of the list formed with θ_3, A_0, A_1, $\mathfrak{p}_0$, r, ε_v and numbers of L, among them $C_\chi(1+2\theta_3)\delta(x) < \frac{1}{8}r_c$ for $x \ge \gamma^{-2}$, with two numbers C_χ, r_c of L; this keeps the box $|\mathrm{Re}\, B_1(b)| + |\mathrm{Im}\, B_1(b)| < (1+2\theta_3)\mathsf{b}_k$ of the complex block data of the step, of the unscaled radius $(1+2\theta_3)\delta(x_k)$, and with it the polydisc $|\mathsf{n}(b)| < \theta_3\mathsf{b}_k$ below, of the unscaled radius $\theta_3\delta(x_k)$, inside the ball, of a radius of L alone, in which the system of the centre has exactly one solution.

Let the envelope of (c) be taken with C'/L in the place of C_m, $C' = \max\{LC_m, 2LC_\chi(1+2\theta_3)A_1/(\rho_\mathrm{u}A_0)\}$, $\rho_\mathrm{u} = \rho_\mathrm{u}(L)$ being the radius of that ball; C' is the one number that is enlarged. Among the upper bounds on γ above are the ceilings $\sup_{x\ge\gamma^{-2}} C'\varepsilon(x) \le c(L)$, $c(L) > 0$ numbers of L, and the block-number and small-radius conditions of Prop. 5.37, under which Cor. 5.38 gives the global energy bound for the linear system; the nonlinear field-energy bound is supplied by Lem. 5.20. Let the anisotropies obey moreover $t(\mathrm{an}) < t_\theta$, where $t_\theta = \min\{t_0, t_\mathrm{u}(L), c_\mathrm{ce}\theta_3 A_1/A_0\}$, $t_\mathrm{u}(L)$ being the number of L below which the moved backgrounds with a complex block datum are formed and $c_\mathrm{ce} > 0$ a number formed with L and C'; so $t_\theta \in (0, t_0]$ is formed with θ_3 and with constants placed before γ.

At every step $0 \le k < K$ and every regular base of the step (a regular background of the coarser level together with its block datum), the *centre* of the step is defined by two conditions: it is a complex stationary point n of the weighted action on the fibre of the block field, and it is in the block axial gauge. There is exactly one n with $|\mathsf{n}(b)| < \theta_3\mathsf{b}_k$ at all bonds b of the step that obeys the two conditions; it is $\mathsf{n} = g_k^{-1}a''$, a'' the 1-form, in the axial gauge, of the finer block average of $W[\mathrm{an}; b_0]$ read about the finer block average of $W[0; b_0]$. Here both backgrounds are formed over the same regular reference background U_b on the longer sequence, with the same complex block datum b_0 obtained from the base of the step; $|\mathsf{n}(b)| \le C_\theta t(\mathrm{an})\mathsf{b}_k \le \frac{1}{2}\theta_3\mathsf{b}_k$, C_θ depending on L, θ_3, A_1/A_0 and C_m. At $b_0 = 0$, the base being U_b with its own block averages, $\sum_b |\mathsf{n}(b)|^2 \le C^2|\mathrm{an}|_\infty^2 A(\beta_t, U_\mathrm{b})$, C a number of L, the sum running over all the bonds of the step for $1 \le k < K$ and over its external bonds at $k = 0$, $A(\beta_t, U_\mathrm{b})$ being the action of the reference background with the couplings of the step; and n is holomorphic in an, real at real an and real b_0, and 0 when an is constant.

The linearised system of the centre (the second variation of the constrained weighted action with its multipliers, the averaging constraint and the block axial gauge) is invertible, in ℓ^2 and in a weighted maximum norm, with bounds depending on L alone. The centres $\mathsf{n}^{(i)}$ of the earlier steps $0 \le i < k$, formed by the same rule about the moved background of the label, vanish at $\mathrm{an} = 0$, are holomorphic in an and obey $|g_i\mathsf{n}^{(i)}(b)| \le Ct(\mathrm{an})\varepsilon_i$, that is $|\mathsf{n}^{(i)}(b)| \le Ct(\mathrm{an})(A_0/A_1)\mathsf{b}_i \le \frac{1}{2}\theta_3\mathsf{b}_i$, with one number C of L for all levels, $\varepsilon_i = \varepsilon(x_i)$ the radius of regularity of the level i and $g_i = x_i^{-1/2}$; and their squares, summed over the stored bonds, which are the external bonds of these steps, obey $\sum_{i<k}\sum_b |\mathsf{n}^{(i)}(b)|^2 \le C^2|\mathrm{an}|_\infty^2 A(\beta_k, U)$, $A(\beta_k, U)$ being the action of the background with the couplings of the level k.

(e) *The counterterms on a label.* On the label of a history with large-field regions the part of every coupling counterterm among the construction results that sees the anisotropy moves exactly into the local terms and the boundary terms (Lem. 5.22): the anisotropy stands in the main weight and the local terms, and in the moved background whenever (c) is applied, and in no counterterm; the counterterms, the values at the trivial triple and the trivial values of the sums are unchanged; at a fixed triple, with the original families and counterterms held fixed, the changes in the re-filed local and boundary entries are linear in an with explicit Lipschitz constants. The following size bounds require a margin. Let $0 < \theta < 1$; let the

families $E^{(j)}$ before the re-filing obey $\|E^{(j)}\|_{j,\kappa+1} \le (1-\theta)E_0$ at every level and every source; let $\|\mathrm{an}\|$ be the norm of the anisotropies in the format of the families, with $|\mathrm{an}|_\infty \le \sigma_3\|\mathrm{an}\|$, the anisotropies ranging over $\|\mathrm{an}\| < 1$, and, if $B_\sharp > 0$, let $\sigma_3 \le \theta E_0 e^{-7(\kappa+1)}/(\frac{3}{2}B_\sharp\alpha_0^2)$.

Then the re-filed families $\hat{E}^{(j)}$ obey $\|\hat{E}^{(j)}\|_{j,\kappa+1} \le E_0$ and $\|\hat{E}^{(j)} - E^{(j)}\|_{j,\kappa+1} \le \theta E_0\|\mathrm{an}\|$; if $\|E^{(j)} - E_0^{(j)}\|_{j,\kappa+1} \le D$, $E_0^{(j)}$ the family at zero source, then $\|\hat{E}^{(j)} - E_0^{(j)}\|_{j,\kappa+1} \le D + \theta E_0\|\mathrm{an}\|$, and for the re-filed families this bound stands in the place of the quadratic bound among the construction results; in a frame of Thms. YM3_I:AJ and YM3_I:AD the added parts of the boundary entries are at most $\theta M^3 E_0 e^{-7}\|\mathrm{an}\|e^{-\kappa d_j(X)}$; the trivial values are unchanged. ■□

Definition 5.4 (Weighted critical backgrounds)**.** Fix $0 < \beta_H < 1$. Use a separated sequence through level $1 \le k < K$, with bare spacing $\eta = L^{-k}$, the interior and outward determining rows, and no inward or edge rows. Retain the natural-pair, block-number, separation and torus-size hypotheses of the regular response construction in Props. YM3_I:31.1 and YM3_I:34.2. In particular the terminal torus has at least $6R_1\mathfrak{M}$ sites per direction. The common block number and width are chosen large enough for the ordinary Hölder kernels and the positive fluctuation form used below: $\mathfrak{M} \ge M^{\mathrm{pc}}$, $R_1\mathfrak{M} \ge \mathfrak{r}^{\mathrm{pr}}$ and $\delta_* R_1\mathfrak{M} \ge 512\log L$. Here M^{pc} is enlarged to make the cutoff defect in the proof at most $1/8$, and the regularity ceiling a^{pc} is decreased to give the local inverse comparison there.

Let U_0 be real, critical for the unweighted action on the fibre of its block averages. Write $\mathcal{Q}$ for the straightened row map and $a_{\mathcal{Q}}$ for its fixed analytic potential radius, chosen so that this map and its local Taylor bounds hold on $\sup_y \|A\|_y < 2a_{\mathcal{Q}}$. Write $K_A = \mathcal{Q}'(A)$, $K_0 = K_A|_{A=0}$, and

$$F(A) = \eta^{-1}\sum_p a_p(e^{i\eta A}U_0), \qquad F'(0) = K_0^{\mathsf{T}}\omega_0.$$

The fine pairing has weight η^3, and the dimensionless row pairing has weight ξ_c^{-1}. Suppose U_0 is regular at radius α: its plaquette defects are at most $\alpha\eta^2\xi_y^{-2}$ and $\xi_y^3|F'(0)| \le \alpha$. Require the pair $(U_0, F'(0))$ to belong to the natural class of that sequence with radii (α,α,α) and real factor U_0. This retains the local real gauges in that class.

The weights are $\mathbf{h} = (h, v)$. The plane weight v is a function of inner plaquettes. At each bare site x, the symmetric matrix $h_{ww'}(x)$ is indexed by the 24 oriented corners $w = (\iota, \iota')$, $|\iota| \ne |\iota'|$, with $\iota, \iota' \in \{\pm1, \pm2, \pm3\}$. If f_w is the field strength of the corresponding plaquette loop based at x, put

$$T_{\mathbf{h}}(A) = T_h(A) + T_v(A), \quad T_h(A) = \eta^3\sum_{x,w,w'} h_{ww'}(x)\langle f_w, f_{w'}\rangle, \quad T_v(A) = \eta^{-1}\sum_p v(p)a_p.$$

All loops are contained in the ambient region. Set $|h(x)| = \max_w \sum_{w'} 2|h_{ww'}(x)|$. The norm $\|\mathbf{h}\|$ is the supremum over tops y of the sum of the local suprema of $|h|$ and $|v|$ and their C^{β_H} seminorms multiplied by $\xi_y^{\beta_H}$. The pairs in these seminorms have one base point in the cell of y and distance at most $\xi_y/4$; plane weights are compared in the same plane.

For potentials use

$$\|A\|_y = \max\{\xi_y|A|_y, \xi_y^2|\nabla A|_y\}, \qquad \|A\|_y^{\beta_H} = \max\{\|A\|_y, \xi_y^{2+\beta_H}[\nabla A]_{\beta_H,y}\}.$$

The last seminorm uses the transports of U_0 along inner combs, over the same distances. With N_y denoting the appropriate potential, row or weight norm, write

$$\|u\|_{\theta,y_0} = \sup_y e^{\theta d_{\mathbf{B}}(y,y_0)}N_y(u).$$

For the Hessian assertion add the next region Ω_{k+1} and require the same natural-pair bounds on this longer sequence. Let Y_0 be the real top-layer fields about Ω_{k+1} whose next linearized averages vanish and which vanish on the block axial trees. Its norm and pairing are the top-row ones. ■

5.1.4. *The constrained background and its multiplier.* The weighted-background problem is finite dimensional at each lattice cutoff. Its uniform estimates, however, must not deteriorate with the number of regions or their layers. Fix a real regular reference U_0, critical for the unweighted action on the fibre of its block averages. Write a nearby field as $e^{i\eta A}U_0$. Let $\mathcal{Q}(A)$ be the straightened averaging map, put $K_0 = \mathcal{Q}'(0)$, and write

$$F(A) = \eta^{-1}\sum_p a_p(e^{i\eta A}U_0), \qquad F'(0) = K_0^{\mathsf{T}}\omega_0.$$

The multiplier ω_0 is part of the reference critical equation. It need not vanish when the reference is nontrivial. Thus the linearized constrained Hessian contains its correction W_0, defined by

$$\langle X, W_0 Y\rangle = \langle \omega_0, \mathcal{Q}''(0)[X,Y]\rangle.$$

Using $F''(0)$ alone would discard a term in the derivative of the constraint equation.

Let $\Delta = F''(0)$. The propagators invert the gauge-fixed operator built from $\Delta - W_0$, and their elimination gives maps H and $\widetilde{G}$ with

$$K_0 H = 1, \qquad K_0\widetilde{G} = 0, \qquad \widetilde{G}(\Delta - W_0 + DRD^{\mathsf{T}}) = 1 - HK_0. \tag{5.ii}$$

Here $RD^{\mathsf{T}}A = 0$ is the chosen gauge slice. The exponential and Hölder bounds for these maps are inherited from the regular-reference inverse estimates at the specified region separation and block sizes. They provide a uniform inverse for the coupled equation, rather than an inverse for each local coordinate considered independently.

Write $C(A) = \mathcal{Q}(A) - K_0 A$ and $F'(A) = F'(0) + \Delta A + \mathcal{V}(A)$. For a weighted action term $T_{\mathbf{h}}$ and prescribed row datum b, the critical equation and constraint are equivalent to

$$\begin{aligned} A &= H(b - C(A)) + \widetilde{G}(C'(A)^{\mathsf{T}}\omega - F'(0) - \mathcal{V}(A) - W_0 A - T'_{\mathbf{h}}(A)), \\ \omega &= H^{\mathsf{T}}(F'(A) + T'_{\mathbf{h}}(A) + DRD^{\mathsf{T}}A - C'(A)^{\mathsf{T}}\omega). \end{aligned} \tag{5.iii}$$

The symbol b in this system is the block datum. The proof uses eq. 5.ii to recover both original equations from a fixed point. Gauge invariance removes the slice term at real data, and holomorphy carries that identity to the admitted complex parameter domain.

Most derivatives of the right side of eq. 5.iii contain the small potential, the reference regularity radius or the small weights. One field-to-multiplier derivative has only a fixed bound K_8. This is why the pair norm weights the multiplier component:

$$\|(A,\omega)\|_{\theta,y_0} = \max\{\|A\|^{\beta_H}_{\theta,y_0}, \Theta^{-1}\|\omega\|_{\theta,y_0}\}, \qquad \Theta \geq 4K_8.$$

Choosing the remaining radii in order gives contraction constant $1/2$ for $0 \leq \theta \leq \delta_*/4$. The displacement at $(0,\omega_0)$ is bounded by $C(\sup|b| + \alpha\|\mathbf{h}\|)$, so iteration remains in the chosen ball. This supplies existence, uniqueness in the stated Hölder norm, holomorphy and reality at real data.

Differentiating the fixed point in a single row gives $(1 - \partial_X \Phi^{\mathbf{h}})d_b X e_{c_0} = (He_{c_0}, 0)$. The inverse has norm at most 2, and the right side has the exponential decay of H. Hence

$$\|d_b A e_{c_0}\|^{\beta_H}_y \leq 2B_* e^{-\delta_* d_{\mathbf{B}}(y,y_{c_0})/4}|e|.$$

The weight derivative is treated by the same inverse. Its source is $T'_{\mathbf{k}}(A)$, bounded by the product of the weight norm and the reference plus perturbed field strength. The first variation has a covariant-divergence term. Estimating it in the first-gradient Hölder norm uses the Hölder seminorm of the weights, through the product estimate for that divergence source. A supremum bound for the weights cannot supply this part of the response estimate.

5.1.5. *The Hessian bound.* The positive Hessian bound requires the longer region sequence. Its domain is $Y_0 + iY_0$, where real fields in Y_0 vanish on the block axial trees and have zero next linearized average. These conditions remove the gauge and retained block directions. The proof compares each local reduced inverse with the flat inverse in a real gauge. The resolvent estimate makes their difference $O(\alpha)$ with exponential decay. A partition of unity assembles these local inverses; derivatives of its cutoffs give a small commutator defect when the comparison blocks are sufficiently large. The resulting real unweighted form has a uniform positive floor. Holomorphic perturbation in the row datum and weights then gives exactly

$$\operatorname{Re}\langle \overline{B}, ZB\rangle \geq (\gamma^{\mathrm{pc}} - e_0\alpha - \kappa_0 s)\|B\|^2, \qquad B \in Y_0 + iY_0,$$

with $Z = d_b\omega$ and s as defined in proposition 5.5. The reference in this comparison is the unweighted Hessian over the same U_0.

Proposition 5.5 (Weighted critical response)**.** In Def. 5.4, there are positive radii a_*, r_*, ρ_*, z_* and constants B_*, Θ, C_T, depending only on L and β_H, with $a_* \leq \min\{a^{\mathrm{pc}}, a_{\mathcal{Q}}, L^{-2}/4\}$ and the following property. If $\alpha \leq a_*$, $\sup|b| < \rho_*$ and $\|\mathbf{h}\| < z_*$, there is exactly one critical background $e^{i\eta A}U_0$ with

$$RD^{\mathsf{T}}A = 0, \qquad \mathcal{Q}(A) = b, \qquad \sup_y \|A\|_y^{\beta_H} \leq r_*.$$

Its pair (A, ω) is holomorphic in $(b, \mathbf{h})$ and real at real data, and equals $(0, \omega_0)$ at $(0, 0)$. It satisfies $\sup_y \|A\|_y^{\beta_H} < r_*$. For a row c_0 and a datum e_{c_0} supported there,

$$\begin{aligned} \|d_b A\, e_{c_0}\|_y^{\beta_H} &\leq 2B_* e^{-\delta_* d_{\mathbf{B}}(y, y_{c_0})/4}|e|, \\ |d_b\omega\, e_{c_0}|_y &\leq 2\Theta B_* e^{-\delta_* d_{\mathbf{B}}(y, y_{c_0})/4}|e|. \end{aligned} \tag{5.iv}$$

For every weight direction $\mathbf{k}$ and $0 \leq \theta \leq \delta_*/4$,

$$\max\{\|d_{\mathbf{h}}A\,\mathbf{k}\|_{\theta, y_0}^{\beta_H}, \Theta^{-1}\|d_{\mathbf{h}}\omega\,\mathbf{k}\|_{\theta, y_0}\} \leq 2C_T(\alpha + \sup_y \|A\|_y^{\beta_H})\|\mathbf{k}\|_{\theta, y_0}. \tag{5.v}$$

Here δ_* is the fixed propagator rate, with the response kernels taken at rate $3\delta_*/8$ before their weighted summation. With the longer-sequence hypotheses, let $Z = d_b\omega$ be the Hessian of the weighted critical value and put $s = \max\{\sup|b|/\rho_*, \|\mathbf{h}\|/z_*\}$. After decreasing the radii, there are $\gamma^{\mathrm{pc}}, e_0, \kappa_0 > 0$ such that

$$\operatorname{Re}\langle \bar{B}, ZB\rangle \geq (\gamma^{\mathrm{pc}} - e_0\alpha - \kappa_0 s)\|B\|^2, \qquad B \in Y_0 + iY_0. \tag{5.vi}$$

The reference for the perturbation in s is the unweighted Hessian at $b = 0$ over this same U_0. ■

Proof. We keep the exact constrained response at the reference. Its Hessian includes the multiplier correction. Put $\Delta = F''(0)$, and define $W(A, \omega)$ by $\langle X, W(A, \omega)Y\rangle = \langle \omega, \mathcal{Q}''(A)[X, Y]\rangle$. Thus $W_0 = W(0, \omega_0)$. Set

$$G_1 = (\Delta - W_0 + DRD^{\mathsf{T}} + K_0^{\mathsf{T}} a K_0)^{-1}, \quad H = G_1 K_0^{\mathsf{T}}(K_0 G_1 K_0^{\mathsf{T}})^{-1}, \quad \widetilde{G} = G_1 - HK_0G_1.$$

The physical and dimensionless row conversions are those in the proof of Prop. YM3_I:34.2; a is the positive row coefficient of that construction, including its boundary extension. This extension cancels on $N(K_0)$. Elimination of the auxiliary variables in Prop. YM3_I:31.1 gives

$$K_0H = 1, \quad K_0\widetilde{G} = 0, \quad \widetilde{G}(\Delta - W_0 + DRD^{\mathsf{T}}) = 1 - HK_0, \quad RD^{\mathsf{T}}H = 0.$$

The ordinary response kernels and their first differences decay exponentially. Their Hölder bounds follow as in Lem. 5.34, with the fixed exponents β_H and $(1 + \beta_H)/2$ in that proof. The exponent gap in its ordinary-input argument is positive. For a divergence source of exponent β_H, use instead the same-exponent parametrix in the proof of Lem. 5.19, with that exponent in the Green-kernel cancellation estimate of Lem. YM3_I:19.4. The local multiplier term W_0 has size $C\alpha$ and is one of the lower-order terms in that parametrix. Thus all constants are uniform in the region sequence and its depth.

Write

$$C(A) = \mathcal{Q}(A) - K_0 A, \qquad F'(A) = F'(0) + \Delta A + \mathcal{V}(A).$$

The critical equation with the slice term and its constraint is equivalent to the following two equations:

$$A = H(b - C(A)) + \widetilde{G}(C'(A)^{\mathsf{T}}\omega - F'(0) - \mathcal{V}(A) - W_0 A - T'_{\mathbf{h}}(A)),$$
$$\omega = H^{\mathsf{T}}(F'(A) + T'_{\mathbf{h}}(A) + DRD^{\mathsf{T}}A - C'(A)^{\mathsf{T}}\omega).$$

Denote this map by $\Phi^{\mathbf{h}}$. The exact identities above recover both equations from its fixed point. Conversely a solution gives a fixed point by the same elimination. The slice term vanishes at a real fixed point by gauge invariance and the scalar projection identity; holomorphy extends that identity to the complex parameter domain.

Here is the uniform smallness estimate behind the contraction. The Taylor kernels of the row map are confined to the double block beneath a row. The local action kernels are confined to a plaquette star. In the scaled local norms their bounds are

$$\|C'(A)\| + \|C'(A)^{\mathsf{T}}\| \le Cr, \quad \|\mathcal{V}'(A)\| \le Cr, \quad \|W(A,\omega) - W_0\| \le C(|\omega - \omega_0| + r|\omega_0|)$$

on $\sup \|A\|_y \le r$, while $|\omega_0| \le C\alpha$. These are the local estimates used in Prop. YM3_I:33.1; their passage to exponentially weighted norms costs a fixed summation constant. The first component of $\partial_X \Phi^0$ is therefore small. In the second component, the only term without a small factor is $H^{\mathsf{T}}(D^{\mathsf{T}}D + DRD^{\mathsf{T}})\delta A$, which has a fixed bound $K_8\|\delta A\|$. Choose $\Theta \ge 4K_8$ and measure pairs by

$$\|X\|_{\theta,y_0} = \max\{\|A\|^{\beta_H}_{\theta,y_0}, \Theta^{-1}\|\omega\|_{\theta,y_0}\}.$$

First reducing a_*, r_* makes the two remaining derivative bounds at most $1/8$ each. The product inequality $[fg]_{\beta_H} \le |f|[g]_{\beta_H} + [f]_{\beta_H}|g|$ gives the same bounds for the Hölder terms. The added weighted action contributes at most $C\|\mathbf{h}\|$ to the derivative: its leading term is the covariant divergence of the weight times the field strength, and its other terms are the local curvature commutators. The divergence estimate just described uses the C^{β_H} norm of that product. Reduce z_* to absorb this contribution. We obtain

$$\|\partial_X \Phi^{\mathbf{h}}\|_{\theta,y_0} \le \tfrac{1}{2}, \qquad 0 \le \theta \le \delta_*/4. \tag{5.vii}$$

This contraction proves uniqueness in the stated norm, including its first-gradient Hölder term.

At $X_0 = (0, \omega_0)$ the displacement of the map is bounded by $C(\sup|b| + \alpha\|\mathbf{h}\|)$. Reducing ρ_*, z_* puts this displacement within half the chosen ball. Iteration gives the fixed point and its holomorphy, uniformly on smaller parameter balls. Conjugation gives reality. If a critical point in the slice has $\sup \|A\|^{\beta_H}_y \le r_*$, the row equation $K_A H = 1 + C'(A)H$ has an inverse bounded by its Neumann series. It gives the multiplier bound needed to place that point in the same pair ball. Thus it is the fixed point. The actual block field is obtained from the reduced datum by the accumulated averaging gauge, as in Cor. YM3_I:41.3.

Differentiate the fixed-point equation in a single row. It gives

$$(1 - \partial_X \Phi^{\mathbf{h}}) d_b X\, e_{c_0} = (He_{c_0}, 0).$$

The right side has norm at most $B_*|e|$ in the exponential norm centred at y_{c_0}. The inverse on the left has norm at most 2, proving eq. 5.iv. In a weight direction the right side is formed from $T'_{\mathbf{k}}(A)$. This first variation is linear in the field strength and is bounded by $C_T(\alpha + \sup\|A\|^{\beta_H}_y)\|\mathbf{k}\|_{\theta,y_0}$. The same inverse gives eq. 5.v.

We next record the unweighted lower bound used for the Hessian. The longer-sequence hypothesis is needed here. Let T^{ax} be the fluctuation form in nonpivot axial coordinates. In the local inversion proof of Thm. YM3_I:L, the flat axial form T^0 has the gap in equation YM3_I:(593). The finite

support of the pivot-coordinate map and the exponential bound for the row inverse give an upper bound $T^0 \le a_{\mathrm{pc}} I$ by the row and column sum test. Hence

$$a_{\mathrm{pc}}^{-1} I \le (T^0)^{-1} \le \gamma_{\mathrm{c}}^{-1} I.$$

The natural class supplies a real local gauge on each comparison cube in which the potential and its first differences are small. In that gauge, the local system differs from the flat one by the local defect bounded in the proof of Thm. YM3_I:L. The resolvent expansion bounds the difference of the inverse blocks by $C\alpha e^{-\delta d(b,b')}$. Summing its rows and columns, and decreasing the regularity radius, gives

$$\|C_\kappa - (T^0)^{-1}\| \le (2a_{\mathrm{pc}})^{-1}.$$

Here C_κ is the block of the local inverse from the nonpivot-field source to the nonpivot field. Elimination of its other components identifies it with the inverse of the local reduced form. At real data it is real and symmetric. The last bound therefore gives

$$(2a_{\mathrm{pc}})^{-1} I \le C_\kappa \le p_{\mathrm{pc}} I, \qquad p_{\mathrm{pc}} = \gamma_{\mathrm{c}}^{-1} + (2a_{\mathrm{pc}})^{-1}.$$

Returning to the original local gauge is unitary and preserves these inequalities. With the partition from that same local inversion proof, put

$$P = \sum_\kappa h_\kappa C_\kappa h_\kappa, \qquad \sum_\kappa h_\kappa^2 = 1.$$

The local inverse bounds give $(2a_{\mathrm{pc}})^{-1} I \le P \le p_{\mathrm{pc}} I$. To compare PT^{ax} with the identity, lift a nonpivot field B to the auxiliary solution Φ in the reduced system. Its equation has only the last source component $T^{\mathrm{ax}} B$, and the ordinary inverse estimate gives $\|\Phi\|_\infty \le C_{\mathrm{pc}} \|B\|_\infty$. On the enlarged comparison cube the local and global equations agree. Multiply by a cutoff which equals one on the support of h_κ and apply the local inverse. The main term is $h_\kappa B$. The error is the inverse applied to the cutoff commutator. Every difference of the cutoff costs $\mathfrak{M}^{-1}$; the enlarged cubes have bounded overlap. Multiplying by h_κ and summing consequently gives

$$PT^{\mathrm{ax}} = 1 - E, \qquad \|E\|_\infty \le 2 N_G B_{\mathrm{pc}} C'_{\mathrm{cut}} C_{\mathrm{cut}} C_{\mathrm{pc}} \mathfrak{M}^{-1} \le \tfrac{1}{8}.$$

Here N_G is the overlap bound of the enlarged comparison cubes, B_{pc} the local inverse bound, C_{cut} the bound for the scaled cutoff differences, and C'_{cut} the commutator bound per such difference. The constant C_{pc} is the lifting bound for Φ above. All depend only on L. The last inequality is imposed by the common block floor. Although PT^{ax} need not be symmetric, it is similar to the real symmetric matrix $P^{1/2} T^{\mathrm{ax}} P^{1/2}$. Its eigenvalues lie in $[7/8, 9/8]$. Consequently $T^{\mathrm{ax}} \ge 7/(8p_{\mathrm{pc}})$. For $B \in Y_0$, the next-average constraint recovers its pivot entries from the nonpivot entries: $B = \mathsf{C}\check{B}$. Here C solves the linear next-average equations for the pivot bonds in the axial gauge. Its entries are bounded by C_K, and each bond shares a coordinate block with at most $n_{\mathrm{pc}} = 36L^3$ positively oriented bonds. The finite block-coordinate estimate gives $\|B\|^2 \le 2(C_K n_{\mathrm{pc}})^2 \|\check{B}\|^2$. Hence the unweighted fluctuation form has a positive floor γ^{pc} on Y_0. The difference between that Lagrangian form and the Hessian of the critical value is the second derivative of the row map paired with the reference current. Its local kernel bound gives a loss at most $e_0 \alpha \|B\|^2$. Thus at $(b, \mathbf{h}) = (0, 0)$,

$$\langle B, Z(0,0;U_0) B \rangle \ge (\gamma^{\mathrm{pc}} - e_0 \alpha) \|B\|^2.$$

The derivative of the multiplier is the block Hessian, because the critical equation cancels the field variation in the first derivative of the critical value. The analytic response and its localized derivative estimates above bound the rows and columns of this Hessian by a summable exponential kernel on the top layer. Cauchy's estimate on the common parameter balls, followed by that same row and column sum, gives

$$\|Z(b, \mathbf{h}; U_0) - Z(0,0;U_0)\|_{Y_0^{\mathbb{C}} \to (Y_0^{\mathbb{C}})^*} \le \kappa_0 s.$$

Reducing the stated radii once leaves the larger analytic balls needed for Cauchy's estimate. Taking the real part proves eq. 5.vi. The unweighted comparison throughout is at the same reference U_0. □

5.1.6. *Construction estimates.* We use the following results among the construction results. Thm. YM3_I:D, and Thms. YM3_I:AJ and YM3_I:AD. The weighted progression and the never-large term with a source, with the statements on one term and on families: Section YM3_I:51.9, Prop. YM3_I:51.15 and Thm. YM3_I:V; and the sections on sources, equation [7, eq. 218] and [7, Prop. B.15]. The rarity of a first creation of a large field on one torus, Prop. 4.54, and the rarity of the large-field region uniformly in the volume, Prop. 4.52. The law of the field and of the history, Prop. YM3_I:44.1, with the measures in equation YM3_I:(647).

The count of polymers, [7, Lem. O.1], whose extension to any number of sources is Lem. 5.6. The frame, [7, Def. B.14]; its background and the history with no large-field region, [7, Prop. B.15]; and the local step with a source, Prop. 6.31. The hereditary cores, Def. YM3_I:49.1 and §5.2; the Hölder bound for real backgrounds, Thm. [7, Thm. BT]; and the ordinary Hölder response, Prop. 5.35. The boards of the lattices, [7, §B.1]; reflection positivity at the site and bond mirrors, which are the walls of the boards, Prop. YM3_I:91.4; the chessboard estimate, Prop. YM3_I:86.3 and, for the torus measures, [7, Cor. B.1]; the domination of periodic Gibbs states by the torus measures through their conditional laws, Prop. YM3_I:91.9; and the bound $0 \le p_{\ell'} - p_\ell \le G_g$ of the setting on the pressures formed with the plain partition functions of Section YM3_I:86, [7, Cor. B.1] with Prop. YM3_I:89.2 and [7, Lem. N.1].

The bound on connected functions of $\mathcal{O}$ on a torus at separated supports, $|S^T_{n;K,\ell,\beta}(f)| \le C_n(\mathsf{d}) e^{nc_V \ell^3} \prod_i \|f_i\|_\sharp$ for real Lipschitz f_i two of whose supports are $\ge \mathsf{d}$ apart, as proved in Lem. 5.1, and its per-ball form, Prop. YM3_I:89.4, for supports in sup-balls of radius R and $\ell \ge 16n(R+1)$. The two-point law on a torus of small side, Thm. YM3_I:AU with Prop. YM3_I:126.4, and the kernel bounds $|C^{\mathrm{free}}_K(p,q)| \le (1+3C_{\mathrm{lat}})a^3|u|^{-3}$ and $|\epsilon_p\epsilon_q C^{\mathrm{free}}_K(p,q) - a^3 \mathbb{K}_3(u)_{\lambda(p)\lambda(q)}| \le C_{\mathrm{lat}} a^3 (a^2|u|^{-5} + \ell^{-3})$, u the least lift of $x(p) - x(q)$ and $\epsilon_p = \pm 1$ the orientations, of equations YM3_I:(2245)–(2246).

The comparison of two lattices at equal label, Thm. YM3_I:AR; the laws of the block fields, Def. 2.7 and Cor. 2.8; and the limits $S^{(\ell,\mathsf{b})}_n$ of $S^T_{n;K,\ell,L^K\mathsf{b}}$ at a fixed label, with the rate $K^p L^{-K/10}$, Thm. YM3_I:AS, with the all-side extension in Cor. YM3_I:119.2. Reflection positivity of the periodic Gibbs states at all the families of mirrors, Prop. YM3_I:103.4, and the reconstruction data of the limits of Gibbs sequences, Thm. YM3_I:AP. The large-field part: the histories and their labels defined in Section YM3_I:44, with the finite exploration in equations YM3_I:(643)–(644); the separated layers, physical row spaces and unweighted constrained systems of Sections YM3_I:23 and YM3_I:27; their weighted mean-square extension in Prop. 5.37 and Cor. 5.38; and the layer-weighted distance of Lem. YM3_I:12.3. The same-exponent Hölder estimate for the moved background is proved in Lem. 5.19, with its normed spaces and envelope class.

The lattice calculus of [7, §J], [7, result J] and equations YM3_I:(2104)–(2105) and YM3_I:112, the integration by parts [7, Lem. J.1] of [7, Lem. J.1], and the rotational Ward identity at finite cutoff, Prop. YM3_I:112.1. For the step with plane couplings and the backgrounds: the lattices allowed at a height, with the room and the width of the halos, Section YM3_I:50.4; the scaled fields below the threshold, [7, §B.2], the chart of the fibre, [7, Lem. B.3], and the chart integral, [7, Def. B.2] and equation [7, eq. 191] with [7, eq. 191], its prefactor equation YM3_I:(907) and the Gaussian integral in Prop. YM3_I:50.2; the weighted action [7, eq. 216] and the density [7, eq. 217] at zero source; the common real and sourced choices in Prop. YM3_I:69.1 and Section YM3_I:82; and the substitution with three pairs of radii, Prop. 5.27.

The arbitrary-order short-distance comparison remains a sketch at $(\mathrm{Q_n})$ of §5.1.2. Its uniform interacting small-field remainder remains to be established. The free Gaussian contribution is computed by expanding $-\frac{3}{2}\log\det(1 - \mathcal{S}W_\lambda)$: the mixed coefficients are the cyclic contractions of the covariance $\mathcal{S}$. The explicit first-creation weights give the large-field term in ω_n. These two

calculations do not supply the remaining arbitrary-order interacting error. From the other sections we use: for $0 < g \le \gamma_W^{\mathrm{sd}}$, the bound on centred products of any number of tensor entries at separated supports in every periodic Gibbs state, uniformly in the cutoff, the side, the state and the position, [7, Thm. C](a), and the bound $\|\mathcal{T}_K(H) - c\|_{\mathrm{ch}} \le 2\mathsf{C}_T Y_i^{\mathsf{q}_T} a_i^{-11/2} \mathsf{b}^{-1} |H|_\infty$, $Y_i = 2[\mathsf{c}_T + \log \frac{1}{a_i}]$, on the chessboard seminorm of a centred tensor entry in a cell of side a_i, [7, eq. 4] of [7, Thm. B](a), with $\mathsf{C}_T, \mathsf{q}_T, \mathsf{c}_T$ the constants C_E, q_E, c_E of that bound ([7, §2]); and, for $0 < g \le \gamma_W^{\mathrm{sd}}$, the invariance $\sigma(RX) = \sigma(X)$, $R \in \mathrm{O}(3)$, of tensor limits at tuples with one non-scalar entry among scalar ones ([7, §5], [7, Thm. U], which gives [7, Thm. R](c) at these tuples).

5.1.7. *Auxiliary estimates.*

Lemma 5.6 (Several sources)**.** Let $0 < g \le \gamma$ with $g^{-2} \ge x_{\mathrm{J}}$, which holds for every $0 < g \le \gamma_W^{\mathrm{ref}}$; let $(K, \beta) \in \mathfrak{D}(g)$ with $K \ge 3$, $x_0 = \beta > x_1 > \cdots > x_K$ its run, $g^{-2} < x_K \le X_g$, and $\ell = 2L^m$ with $MR(x_K) \le L^m$, which holds on every side $\ell \in \mathfrak{L}(g)$; let $s \ge 1$ be a depth and $k = \max\{K - s, 2\}$ the working level, so that $2 \le k \le K - 1$; and let (F_a, o_a, w_a), $a = 1, \dots, p$, be dominated local sources with the comparison weights w_a. A *dominated local source* (F, o, w) consists of a real function F of the configurations of the bare bonds that is a gauge invariant polynomial in the entries of the bond variables; a local representation o of F, that is, families of local terms at the heights $1 \le j \le k$ whose sums, less their values at the trivial configuration, are F, and a family at the height 0 that sums to the fluctuation variation $F(e^{iB'}\mathbf{U}) - F(\mathbf{U})$ over the background $\mathbf{U}$ of the level-1 block field; and a real weight $w \ge 0$ of the plaquettes with $|F| \le \mathcal{O}(w)$ on all configurations, with sizes of o at most those of a source of the first size $|w|_\infty$, $m_0(o) \le 12\mathsf{c}^\flat M^3 \alpha_0^2 L^K |w|_\infty$ and $m_j(o) \le 12 M^3 \alpha_0^2 L^{K-j} |w|_\infty$ for $1 \le j \le k$, $m_0(o)$ being the size of the family at the height 0, $m_j(o)$ twice the norm $\|\cdot\|_{j,\kappa}$ of the family at the height j and $\mathsf{c}^\flat = 81 C_P^2 \mathsf{C}_{\mathcal{A}}^2 / K_\Lambda$, and such that the corners of the plaquettes of $\operatorname{supp} w$ carry o: its members vanish on the regions that hold none of these corners. Zero sources are omitted before the parameter radii are formed.

(a) Let $\Lambda_0 = \{\lambda \in \mathbb{C}^p : \sum_a |\lambda_a| m_{0,a} < N_\star^{[0]}\}$ and, for $1 \le j \le k$, $\Lambda_j = \{\lambda \in \Lambda_{j-1} : \sum_a |\lambda_a| n_{j,a} < N_\star^{[j]} - (\sum_{n \le j} \omega_n)(E_0 + \frac{3}{2}\alpha_0^2 B_\sharp)\}$, where $m_{j,a} = m_j(o_a)$, $n_{j,a} = m_{j,a} + \sum_{i=1}^{j} \omega_{j+1-i} d_{i,a}$, $d_{1,a} = \theta(x_0) m_{0,a}$, $d_{j+1,a} = \theta(x_j) n_{j,a}$, $\omega_n = (n+1) L^{-(n-1)}$, θ is the size of a source of (b), and $N_\star^{[j]} = w_\star \alpha_0 / (4 K_\Lambda (1 + \beta_{\mathrm{s}}) \delta(x_j))$ is the bound on the size of the bare array that the step of the level j carries (equations [7, eq. 196] and eq. 2.i). These are open convex neighbourhoods of 0, decreasing in j; their thresholds read no source and not p, so the sources share the room of each step, and a polydisc $\{|\lambda_a| < \rho_a\}$ lies in Λ_{t-1} when $\sum_a \rho_a m_{0,a} \le N_\star^{[0]}$ and $\sum_a \rho_a n_{j,a}$ is at most the right member of the condition of Λ_j for $1 \le j < t$; in particular $\Lambda_k = \bigcap_{0 \le j \le k} \Lambda_j$ holds the polydisc of the radii $\rho_a = p^{-1} \min_{0 \le j \le k} N_j / n_{j,a}$, N_j being the right member of the condition of Λ_j and $n_{0,a} = m_{0,a}$: these radii are p^{-1} times numbers that depend on the sizes of the sources alone.

For $1 \le t \le k+1$ and $\lambda \in \Lambda_{t-1}$, $\tau_{h_t}[\rho_0 e^{\sum_a \lambda_a F_a}] = e^{D_t^\lambda(U_t(\cdot))} \mathsf{m}_t$ on the small block fields of the level t, with D_t^λ holomorphic in λ and formed from the members $E_\lambda^{(i)}(X)$ of the weighted progression; $\partial_B E_\lambda^{(i)}(X) = 0$ unless X meets the carrier of every F_a, $a \in B$, and the mixed derivatives at $\lambda = 0$ of the level i carry the factor $\exp(-[L^{t-i}\mathsf{D}_t/M - 2]_+)$, D_t the least sup distance of the carriers in level-t units. The centred moments at the never-large history are $\Phi[\prod_{a \in B}(F_a - c_a)] = \sum_\pi \prod_{\beta \in \pi} \kappa_\beta$, π over the partitions of B into blocks β, with $\kappa_\beta = \partial_\beta D_t^\lambda|_{\lambda=0}$ for $\#\beta \ge 2$ and $\kappa_{\{a\}} = \partial_a D_t^\lambda|_{\lambda=0} - c_a$ at the one-element blocks.

Here $\rho_0 = e^{-\beta A}$ is the bare density, h_t the never-large history of the level t and $\tau_{h_t}[\cdot]$ its term, $\mathsf{m}_t = \tau_{h_t}[\rho_0]$ a finite positive measure on the block fields of the level t, U_t the background, c_a the constant of F_a at the level t, and $\Phi[Y]$ the bounded density of $\tau_{h_t}[Y\rho_0]$ with respect to m_t.

(b) Let $p \ge 2$, $0 < g \le \gamma_W^{\mathrm{ref}}$, $0 < \mathsf{d} < 1$ with $s = s(\mathsf{d}) \le K$ and $2\mathsf{T}_p(3)\theta(L^{s-2}g^{-2}) \le 1$, and let the weights $w_a \ne 0$ be carried by balls $B(x_a, \frac{7}{8}\mathsf{d})$ at mutual distances $\ge \frac{5}{4}\mathsf{d}$. Then $|\langle F_1; \dots; F_p\rangle_{K,\ell,\beta}| \le k_p^{\mathrm{loc}}(g, s; \ell)\, \mathsf{b}^{-p} \prod_a |w_a|_\infty$, with k_p^{loc} depending on g, p, ℓ, s alone. Here $\mathsf{T}_p(u) = \sum_{n\ge1}(n+1)L^{-(n-1)}\big[(\frac{4}{3})^{1/2}L^{n/2}(1+n)^{\mathfrak{p}_0}\big]^p e^{-4\beta_s\kappa[L^{n-1}u-2]_+}$ is the series of the mixed derivatives of the order p and $\theta(x) = K_{\mathbf{M}}(1+\beta_s)A_1(\log x)^{\mathfrak{p}_0}x^{-1/2}/\alpha_0$ the size of a source at the coupling x, with the constants $\beta_s, K_{\mathbf{M}}$.

(c) In (b), let moreover $2\max\{\tilde{\mathsf{T}}_p(3), (\frac{4}{3})^{1/2}L\,\mathsf{S}_{\mathrm{nat}}\}\,\theta(L^{s-2}g^{-2}) \le 1$, where $\tilde{\mathsf{T}}_p$ is the series T_p with the further factor $L^{n/4}$ in its n-th term and $\mathsf{S}_{\mathrm{nat}} = \sum_{n\ge1}(n+1)^{1+\mathfrak{p}_0}L^{-n/4}$; this condition implies that of (b), and $\tilde{s}(p)$ denotes the least integer $s \ge 3$ with it. If the sizes of the local representation of F_1 at the heights $1 \le j \le k$, and its values on the backgrounds of the levels $1 \le t \le k$, are smaller than those of a source of the weight w_1 by the factors $L^{-j/4}$, resp. $L^{-t/4}$, that is, $m_j(o_1) \le 12M^3\alpha_0^2L^{K-j}L^{-j/4}|w_1|_\infty$ and $|F_1(U_t(W))| \le L^{-t/4}\,2B_3^2\varepsilon(x_t)^2L^{K-4t}\sum_p w_1(p)$ at the backgrounds of the configurations $W \in \mathcal{V}_t^\chi$, then

$$|\langle F_1; \dots; F_p\rangle_{K,\ell,\beta}| \le L^{-(k-1)/4}\,\tilde{k}_p(g, s; \ell)\,\mathsf{b}^{-p}\prod_a |w_a|_\infty,$$

with a number $\tilde{k}_p(g, s; \ell) \ge k_p^{\mathrm{loc}}(g, s; \ell)$ that depends on g, p, ℓ, s alone.

(d) Let $c = \sum_i \kappa_i m_i$ be a block form of Lem. 5.9(b): each monomial m is a product of $n_2(m)$ factors $\langle \mathbf{w}, \mathbf{w}'\rangle$, $n_3(m)$ factors $\det(\mathbf{w}, \mathbf{w}', \mathbf{w}'')$ and $n_a(m)$ factors $1 - \frac{1}{2}\mathrm{tr}\,V$, where V runs over the holonomies of the lassos of the block of 27 sites centred at a site x, the holonomies of its plaquettes carried to x along paths of at most three steps in the block, and $\mathbf{w} = \mathbf{w}(V)$; the order of m is $2n_2(m) + 3n_3(m) + 2n_a(m)$, and $\|c\| = \sum_i |\kappa_i| 2^{n_a(m_i)}$ is the norm of c of that lemma. For a real function k of the sites, the source $\mathrm{Fm}^c(k) = a^{-1}\sum_x k(x)c_x(U)$ of Lem. 5.9(b) is a dominated local source with the comparison weight $\varpi\mathbf{1}_{\mathrm{Pl}(k)}$ for every $\varpi \ge 76\|c\||k|_\infty$, $\mathrm{Pl}(k)$ being the set of the plaquettes of the blocks of the sites with $k \ne 0$, and $|\mathrm{Fm}^c(k)| \le \mathcal{O}(2\|c\||k|_\infty\mathbf{1}_{\mathrm{Pl}(k)})$ on all configurations; if all its monomials have an order ≥ 3, the hypothesis of (c) holds for it with the comparison weight $\Gamma_c^\circ\|c\||k|_\infty\mathbf{1}_{\mathrm{Pl}(k)}$, $\Gamma_c^\circ = \max\{76,\ 34\max\{1, 5.04\alpha_0\}^{\mathrm{Ord}(c)-2},\ 2\max\{1, 2B_3\}^{\mathrm{Ord}(c)-2}\}$, $\mathrm{Ord}(c)$ the largest order of its monomials, even with L^{-2j} and L^{-2t} in the places of $L^{-j/4}$ and $L^{-t/4}$.

(e) Let $\ell \in \mathfrak{L}(g)$, let (F_a, o_a, w_a), $a = 1, 2$, be two dominated local sources whose carriers are at the sup distance d', and let $2 \le t \le k$, k the working level, $s' = K - t$ (so that $1 \le s' \le K-2$, as $k \le K-1$), with $L^{s'}\mathsf{d}' \ge 3M$, so that $\mathsf{D}_t \ge 3M$, and $\mathsf{h}_2(s') \le 1$, where $\mathsf{h}_2(s') = \frac{4}{3}\mathsf{k}_2 L\,\theta(\frac{3}{4}L^{s'+1}g^{-2})\,\mathsf{T}_2^\circ(L^{s'}\mathsf{d}'/M)$ with the series $\mathsf{T}_2^\circ(u) = \sum_{n\ge1}(n+1)^{2\mathfrak{p}_0+1}e^{-4\beta_s\kappa[L^{n-1}u-2]_+}$ and the number $\mathsf{k}_2 = \max\{\frac{1}{2}(\mathsf{c}^\flat)^2, 5\}$. Then, with $y_a \le 4\ell^3L^{4(K-t)}|w_a|_\infty$: the real part of the density $D_{t,2}$ of F_2 at the never-large history differs from the real part of its constant c_2 by at most C_1y_2, and the mixed part $D_{t,12}$ of the density of F_1F_2 is at most $C_2e^{3-\mathsf{D}_t/M}y_1y_2$.

■

Proof sketch. $MR(x_K) \le L^m$ on the sides of $\mathfrak{L}(g)$:

$$L^{m_{\mathfrak{L}}(g)} \ge LM\,R(X_g + \bar{B})$$

by equation YM3_I:(1528), and $x_K \le X_g + \bar{B}$. (a) is the derivative calculus for the never-large term with one source (Prop. 6.31 and equation [7, eq. 218]), carried out in p parameters. Each source enters each step as one more local term, and the p sources share the room of each step, the bound of the step on the size of what it carries: with $m_{0,a}$ the size of F_a at the height 0 and $n_{j,a} = m_{j,a} + \sum_{i=1}^j \omega_{j+1-i}d_{i,a}$ the size at the level j of F_a together with all that the steps below the level j have made of it, as in the statement, the parameter λ ranges at the level t, $1 \le t \le k+1$,

over the set $\Lambda_{t-1} = \bigcap_{0\le j<t} \Lambda_j$, the sets Λ_j being those of the statement,

$$\Lambda_0 = \Big\{\lambda : \sum_a |\lambda_a|\, m_{0,a} < N_0\Big\}, \qquad \Lambda_j = \Big\{\lambda \in \Lambda_{j-1} : \sum_a |\lambda_a|\, n_{j,a} < N_j\Big\} \quad (1 \le j \le k),$$

with $N_0 = N_\star^{[0]}$ and $N_j = N_\star^{[j]} - (\sum_{n\le j} \omega_n)(E_0 + \frac{3}{2}\alpha_0^2 B_\sharp)$ for $1 \le j \le k$, $\omega_n = (n+1)L^{-(n-1)}$ as in the statement: $N_\star^{[j]}$ is the bound on the size of what the step of the level j carries, and for $1 \le j \le k$ the array of the level j without the sources has a size at most $(\sum_{n\le j} \omega_n)(E_0 + \frac{3}{2}\alpha_0^2 B_\sharp) \le \frac{1}{2} N_\star^{[j]}$ (the weighted-lag array bound in the proof of [7, Prop. B.8], together with $8K_\Lambda N_\star s_j \le w_\star$ from item 9 before Thm. Q), to which the sources add at most $\sum_a |\lambda_a|\, n_{j,a}$; so $N_j \ge \frac{1}{2} N_\star^{[j]}$ is the room that the step of the level j leaves to the sources, a number that reads no p. The sets Λ_j are open and convex and decrease with j, and Λ_{t-1} holds the polydisc of the radii $\min_{0\le j<t} N_j/(p\, n_{j,a})$, $n_{0,a} = m_{0,a}$, since on it $\sum_a |\lambda_a|\, n_{j,a} < N_j$ for $0 \le j < t$; so Λ_k, which lies in every Λ_{t-1}, holds the polydisc of the radii $\min_{0\le j\le k} N_j/(p\, n_{j,a})$.

These radii are p^{-1} times numbers that depend on the sizes of the sources alone, while the thresholds N_j read no p. On Λ_{t-1} the members of the progression up to the level t are holomorphic in λ; the derivative in λ_a of a member vanishes unless its polymer meets the carrier of F_a, and the count of polymers in [7, Lem. O.1], applied at every level $1 \le j \le k+1$ to the polymers that meet a given set of sites, gives the decay in the distance of the carriers: a polymer of the level i that meets the carriers of all the F_a, $a \in B$, has a linear size at least $L^{t-i}\mathsf{D}_t/M - 2$, whence the factor $\exp(-[L^{t-i}\mathsf{D}_t/M - 2]_+)$ of the statement; the formula for the centred moments is the moment–cumulant formula for the quotient of the never-large term with the sources by the term without them, in which the centring moves the first cumulant alone: $\kappa_{\{a\}} = \partial_a D_t^\lambda|_{\lambda=0} - c_a$, and $\kappa_\beta = \partial_\beta D_t^\lambda|_{\lambda=0}$ for $\#\beta \ge 2$. (b) The cumulant is a mixed derivative at 0 of the logarithm of the normalised partition function with the sources.

The histories with a first creation of a large field at a level t weigh at most w_t. Along the never-large history the mixed derivative families of the order p are bounded in the norm of the rate $\kappa^+ = (1 + 4\beta_s)\kappa$; read at the rate κ they keep the factor $e^{-4\beta_s\kappa[D_j/M-2]_+}$, D_j the least sup distance of the carriers in level-j units. On the way from the finer levels up to the working level k, a family of the level i enters the step of the level $j \ge i$ with the weight $(n+1)L^{-(n-1)}$ of the lag $n = j + 1 - i$, while the size of a source against the bound of the step is at the level $i-1$ at most $(\frac{4}{3})^{1/2} L^{n/2} (1+n)^{\mathfrak{p}_0}$ times that at the level j; so the series T_p of the statement sums the mixed derivatives of the order p over the levels finer than the working level,

$$\mathsf{T}_p(u) = \sum_{n\ge 1} (n+1) L^{-(n-1)} \Big[(\tfrac{4}{3})^{1/2} L^{n/2} (1+n)^{\mathfrak{p}_0} \Big]^p e^{-4\beta_s \kappa [L^{n-1}u - 2]_+},$$

and the induction over the levels up to $k+1$ closes if $\theta(x_j)\, \mathsf{T}_p(D_j/M) \le \frac{1}{2}$ for $1 \le j \le k$. Since $D_j/M \ge 3$, $\theta(x_j) \le \theta(L^{s-2} g^{-2})$ and T_p does not increase, this holds under the condition $2\mathsf{T}_p(3)\theta(L^{s-2}g^{-2}) \le 1$ of the statement. Cauchy's estimates in each parameter then give the factor $\mathsf{b}^{-p} \prod_a |w_a|_\infty$. (c) is a recursion of numbers. If the sizes of F_1 fall like $L^{-j/4}$, the recursion of (a) gives $n_{j,1} \le \Gamma L^{-j/4}$ times the size at the level j of a source of the weight w_1, Γ a number formed with the constants: against the induction hypothesis at the level j, the lag n contributes at most $(\frac{4}{3})^{1/2} L\, \theta(x_j) (n+1)^{1+\mathfrak{p}_0} L^{-n/4}$, and the induction closes with the sum $\mathsf{S}_{\mathrm{nat}} = \sum_{n\ge 1} (n+1)^{1+\mathfrak{p}_0} L^{-n/4}$ of the statement.

In the induction of (b) the falling sizes of the first source compare the level $i-1$ with the level j with the further factor $L^{n/4}$, so that the series $\tilde{\mathsf{T}}_p$ of the statement, which is T_p with the further factor $L^{n/4}$ in its n-th term, takes the place of T_p. Both inductions close under the condition $2\max\{\tilde{\mathsf{T}}_p(3), (\frac{4}{3})^{1/2} L\, \mathsf{S}_{\mathrm{nat}}\}\, \theta(L^{s-2}g^{-2}) \le 1$ of (c), which implies the one of (b) at the same s; then the bound of (b) holds with $L^{-(k-1)/4} \tilde{k}_p(g, s; \ell)$ in the place of $k_p^{\mathrm{loc}}(g, s; \ell)$, $\tilde{k}_p \ge k_p^{\mathrm{loc}}$ a number of

the same kind formed with these series, for the sizes at the levels $1 \le j \le k$ and the values on the backgrounds of the levels $1 \le t \le k$.

So the recursion closes under the condition of (c); the least integer $s \ge 3$ at which it holds with $p = r$ is the number $\tilde s(r)$ of the statement, and we write $\tilde k_r$ for the number $\tilde k_r(g, s; \ell)$ of the bound of (c). (d) Let $c = \sum_i \kappa_i m_i$ be a block form of Lem. 5.9(b), $\|c\| = \sum_i |\kappa_i| 2^{n_a(m_i)}$ its norm as in the statement, with m_i written also for the order of the monomial m_i, and k a real function of the sites; the source is $\mathrm{Fm}^c(k) = a^{-1} \sum_x k(x) c_x(U)$. At a height $j \ge 1$ it is represented by the family $\eta_j^{-1} \sum_x L^{K-j} k(x) c_x$, $\eta_j = L^{-j}$, so that $\eta_j^{-1} L^{K-j} = a^{-1}$, the sum over the sites x whose block lies in a given region of the level j; such a region has at most 8 cubes and the linear size 0, and at most $8M^3\eta_j^{-3}$ sites have one region (each of the at most eight cubes contains $(M/\eta_j)^3$ bare sites).

On the cores of radii (b_0, b_1) in Def. YM3_I:49.1 the vector part of the holonomy of a lasso is at most $5.04\, b_0 \eta_j^2$ in level-j units and its plaquette term at most $(5.04\, b_0 \eta_j^2)^2$, so a monomial of the order m is at most $(5.04\, b_0)^m \eta_j^{2m}$, and the family has the size $8M^3 L^{K-j} |k|_\infty \sum_i |\kappa_i| (5.04\, b_0)^{m_i} \eta_j^{2m_i - 4}$. The sizes are taken at $(b_0, b_1) = (\alpha_0, \alpha_1)$: as $5.04\, \alpha_0 L^{-2} \le 1$, $(5.04\, \alpha_0)^{m_i} \eta_j^{2m_i - 4} \le 25.41\, \alpha_0^2$, and with the factor 2 of the size of a representation at a height $j \ge 1$ this is at most $34 \|c\| |k|_\infty$ in the unit $12 M^3 \alpha_0^2 L^{K-j}$ of the size of a source of the weight 1.

At the height 0 the representation is the family in the fourth place $a^{-1} \sum_x k(x) [c_x(e^{iB'} U) - c_x(U)]$, B' the fluctuation field with its values at the pivot bonds, which sums to the fluctuation variation of the source; a lasso has at most ten factors, and telescoping in them gives the size $911\, \mathsf{c}^\flat M^3 \alpha_0^2 L^K \|c\| |k|_\infty$, that is $\frac{911}{12} \|c\| |k|_\infty \le 76 \|c\| |k|_\infty$ in the unit $12 \mathsf{c}^\flat M^3 \alpha_0^2 L^K$ of the height 0. So $76 \|c\| |k|_\infty$ bounds the sizes at all the heights, being at least $34 \|c\| |k|_\infty$ at the heights $j \ge 1$ and $\frac{911}{12} \|c\| |k|_\infty$ at the height 0. On the real configurations one factor of a monomial is at most $a_p + a_{p'}$, p, p' plaquettes of its lassos: a factor $\langle w, w' \rangle$ or $\det(w, w', w'')$, since $|w|^2 \le 2a_p$ and $|w| \le 1$, or a plaquette term itself; every other factor is at most 1, or 2 if it is a plaquette term; so $|\mathrm{Fm}^c(k)| \le \mathcal{O}(2 \|c\| |k|_\infty \mathbf{1}_{\mathrm{Pl}(k)})$, $\mathrm{Pl}(k)$ the plaquettes of the blocks of the sites with $k \ne 0$, and the set $\{k \ne 0\}$ carries all the representations: $\mathrm{Fm}^c(k)$ is a dominated local source with the comparison weight $\varpi \mathbf{1}_{\mathrm{Pl}(k)}$ for every $\varpi \ge 76 \|c\| |k|_\infty$.

If all the orders are ≥ 3, the factor $(5.04\, \alpha_0 \eta_j^2)^{m_i - 2} \le \max\{1, 5.04\, \alpha_0\}^{\mathrm{Ord}(c) - 2} L^{-2j}$, $\mathrm{Ord}(c)$ the largest order, gives the fall L^{-2j} of the sizes; on the backgrounds of the level t the plaquette terms are below $\bar a = 2 B_3^2 \varepsilon_t^2 L^{-4t}$, ε_t the radius of regularity of the level t, a factor of the order $q \in \{2, 3\}$ is at most $(2\bar a)^{q/2}$, and the values fall like L^{-2t}. With the comparison weight $\Gamma_c^\circ \|c\| |k|_\infty \mathbf{1}_{\mathrm{Pl}(k)}$ of the statement, $\Gamma_c^\circ = \max\{76,\, 34 \max\{1, 5.04\, \alpha_0\}^{\mathrm{Ord}(c) - 2},\, 2 \max\{1, 2B_3\}^{\mathrm{Ord}(c) - 2}\}$, this is the hypothesis of (c), with L^{-2j} and L^{-2t}. (e) The density of a source at the never-large history of the level t is its first derivative in (a); its oscillation is bounded by the size of the source at that level, which is y_2 up to a constant, and the mixed part is the second mixed derivative, which carries the factor of the distance from (a), with $\mathsf{D}_t \ge L^{s'} \mathsf{d}' \ge 3M$.

The second mixed derivatives are summed over the levels finer than t as in (b) at $p = 2$, $\frac{4}{3} L\, \mathsf{T}_2^\circ$ being the series T_2, and the condition $\mathsf{h}_2(s') \le 1$ closes that induction up to the level t. □

Lemma 5.7 (Rough entries)**.** A petal at a site x is a pair $\pi = (\alpha, \alpha')$ of signed directions with $|\alpha| \ne |\alpha'|$, $W_x(\pi)$ the holonomy from x of the plaquette that it spans, and $q_x(\pi, \pi') = \langle \mathbf{w}(W_x(\pi)), \mathbf{w}(W_x(\pi')) \rangle$. For real $k = (k_{\pi\pi'}(x))$ put $\mathrm{Pf}(k) = a^{-1} \sum_x \sum_{\pi, \pi'} k_{\pi\pi'}(x) q_x(\pi, \pi')$ and $|k|_1 = \sup_x \sum_{\pi, \pi'} |k_{\pi\pi'}(x)|$. A *rough entry* is $F = \mathrm{Pf}(k) + \mathcal{O}(v)$ with a real weight v; its norm is $\|F\| = 13 |k|_1 + |v|_\infty$, and it is carried by the barycentres of the plaquettes of $\mathrm{supp}\, v$ and of those with a corner at which $k \ne 0$. Let $0 < g \le \gamma_W^{\mathrm{ref}}$, $\ell' \in \mathfrak{L}(g)$, $r \ge 2$ and $0 < \mathsf{d} < 1$. There are $K_0'(r, \mathsf{d})$ and $k'(r, \mathsf{d}) \le C_{r,p} \mathsf{d}^{-pr}$ $(p > 7)$ such that

$$|\langle F_1; \dots; F_r \rangle_{K, \ell', \beta}| \le k'(r, \mathsf{d}) \prod_a \|F_a\|$$

for all $(K,\beta)\in\mathfrak{D}(g)$ with $K\ge K_0'(r,\mathsf{d})$ and all rough entries whose carriers are pairwise $\ge\mathsf{d}$ apart. Moreover $\Theta_\beta[\eta]=\mathsf{b}\,\mathrm{Pf}(k^\eta)$ with $|k^\eta|_1\le 12\max_\rho \mathrm{Lip}_\infty\eta^\rho$, so $\|\mathrm{Pf}(k^\eta)\|\le 156\max_\rho\mathrm{Lip}_\infty\eta^\rho$; $-\beta\sum_p h_{\omega,\chi}(p)=\mathsf{b}\,\mathrm{Pf}(k^h)$ with $|k^h|_1\le 12|\omega|\,\mathrm{Lip}(\chi)R_1(\chi)\le 36\ell|\omega|\,\mathrm{Lip}_\infty\chi$; $\mathfrak{B}_{\omega,\chi}=\mathsf{b}\,\mathrm{Pf}(k^h)+\mathcal{O}(v^{\mathfrak{B}})$ with $|v^{\mathfrak{B}}|_\infty\le(2\mathsf{b}+3a)|\omega|\,\mathrm{Lip}(\chi)R_1'(\chi)$, where $R_1'(\chi)\le\frac32\ell$ is the number of Thm. N(a), the largest $|x(p)|_1$, taken in the chart $(-\frac{\ell}{2},\frac{\ell}{2})^3$, at the plaquettes where $\nabla^s_\omega\chi$ or v_χ does not vanish (at their barycentres $x(p)$, χ is not constant on the points $x(p)\pm ae_\rho$, $x(p)\pm\frac{a}{2}e_\rho$, $\rho=1,2,3$); the entries $\mathsf{b}\,\mathrm{Pf}(k^\eta)$, $\mathsf{b}\,\mathrm{Pf}(k^h)$ and $\mathfrak{B}_{\omega,\chi}$ are carried within $2a$ of $\operatorname{supp}\eta$, resp. $\mathrm{Var}(\chi)$; and $T_{lr}(\varphi)=\mathsf{b}\,(\mathrm{Pf}(k^{lr})+\mathcal{O}(v^{lr}))$ with $13|k^{lr}|_1+|v^{lr}|_\infty\le C_T\sup|\varphi|$, C_T an absolute constant. ■

Proof sketch. A rough entry is a dominated local source in the sense of Lem. 5.6: a gauge invariant polynomial with a local representation at every height, with $|F|\le\mathcal{O}(w)$ for a weight w of size $\|F\|$ on its carrier. Put $\sum_a\lambda_aF_a$ into the weighted progression (Lem. 5.6(a)). Its families are holomorphic in λ near 0 with sizes that depend on the $\sup|w_a|$ alone, and they are local: $\partial_BE^{(i)}_\lambda(X)=0$ unless X meets every carrier in B, so that $d_i(X)\ge D_i/M-2$, D_i being the least sup distance of these carriers in level-i units, and a factor $e^{-\kappa^+[D_i/M-2]_+}$ appears. We split the mean over the level t of the first creation of a large field. Up to that level the history is the never-large one, and

$$\tau_{h_t}\big[\rho_0e^{\sum_a\lambda_aF_a}\big]=e^{D_t^\lambda(U_t(\cdot))}\,\mathsf{m}_t,\qquad \Phi\Big[\prod_{a\in B}(F_a-c_a)\Big]=\sum_\pi\prod_{\beta\in\pi}\kappa_\beta,\quad \kappa_\beta=\partial_\beta D_t^\lambda|_{\lambda=0}\ (\#\beta\ge 2),\quad \kappa_{\{a\}}=\partial_aD_t^\lambda|_{\lambda=0}-c_a,$$

with the signs of Lem. 5.6(a); each κ_β with $\#\beta\ge2$ is bounded by the mixed derivatives above. Behind the level t only a positive weight enters, of the mass $\mathsf{w}_t\le 2M^{-3}\ell'^3e^{B(X_g)\ell'^3}L^{3(K-t)}e^{-c'p_0(g_t)^2}$, which is smaller than every power of $L^{-(K-t)}$, while the centred insertions carried to the level $t-1$ are at most powers of L^{K-t}. The mixed derivatives of the orders $\le r$ converge at the depth $s(\mathsf{d}'')$, $\mathsf{d}''=\min\{\frac45\mathsf{d},L^{-\bar s(r)}\}$, where $\bar s(r)$ is the least integer $s\ge3$ with $2\mathsf{T}_r(3)\theta(L^{s-2}g^{-2})\le1$, with the series T_r and the size θ of Lem. 5.6(b); at every $s\ge\bar s(r)$ the condition holds, since θ does not increase.

Apply [7, Prop. 2.10](a′), (b) to the dominated local sources just constructed. Its summability condition is the one defining $\bar s(r)$ above, and it permits the fixed side ℓ': the local cumulant constant may be chosen with $k_r(g,s;\ell')\le C_r(g,\ell')Y_s^{8r\mathfrak{p}_0}$. Cutting every weight into at most $(2\ell'/\mathsf{d}'')^3$ pieces and using $\mathsf{b}^{-r}\le g^{2r}$ therefore gives $k'(r,\mathsf{d})\le C_r(g,\ell')g^{2r}(2\ell'/\mathsf{d}'')^{3r}Y^{8r\mathfrak{p}_0}_{s(\mathsf{d}'')}$. The same proposition bounds $Y_{s(\mathsf{d}'')}\le2\log(1/\mathsf{d}'')+C(g)$; since $\mathsf{d}''=\min\{4\mathsf{d}/5,L^{-\bar s(r)}\}$, this proves the asserted $C_{r,p}\mathsf{d}^{-pr}$ bound for every $p>7$. The same decomposition, at the least depth s with $L^s\mathsf{d}\ge3M$ at which the mixed derivatives of the order q are summable, gives the second bound of Thm. M(a): $L^s\le\max\{1,LM\mathsf{u}_q/\mathsf{d}\}$ with $\mathsf{u}_q=O(q\log q)$, the sums over partitions are at most $C^q(q!)^{3/2+\mathfrak{p}_0}$, and the rarity of a first creation, at which all the q insertions are bounded by their suprema, costs $\exp[c_L(4q+4)^2]$. The formulas for k^η, k^h and k^{lr} are the definitions written in petals: for $b=\langle x,x+ae_\mu\rangle$ and $p=p(b,\alpha'')$, $\mathfrak{d}_{b,p,\rho}=-\frac12[q_x((\mu,\rho),(\mu,\alpha''))-q_x((\mu,-\rho),(\mu,\alpha''))]$, and the shell lies in $|x|_1\le\frac12+\frac32\ell$.

For the whole flux insertion, $\mathfrak{B}_{\omega,\chi}-\mathsf{b}\,\mathrm{Pf}(k^h)=\mathcal{O}(v^{\mathfrak{B}})$ with $v^{\mathfrak{B}}(p)=\mathsf{b}\,\nabla^s_\omega\chi(x(p))-\frac32a\,v_\chi(p)$; as $\sum_\rho|(\omega y)_\rho|\le2|\omega||y|_1$ and the difference quotients of χ in $\nabla^s_\omega\chi$ and in v_χ are at most $\mathrm{Lip}(\chi)$, $|v^{\mathfrak{B}}(p)|\le(2\mathsf{b}+3a)|\omega|\,\mathrm{Lip}(\chi)|x(p)|_1$, and $v^{\mathfrak{B}}(p)\ne0$ only at the plaquettes where $\nabla^s_\omega\chi$ or v_χ does not vanish, at whose barycentres χ is not constant on the points $x(p)\pm ae_\rho$, $x(p)\pm\frac{a}{2}e_\rho$: this gives the factor $R_1'(\chi)$ of Thm. N(a). □

Lemma 5.8 (Chessboard bound)**.** Let $0<g\le\gamma$, $\ell'\in\mathfrak{L}(g)$, $(K,\beta)\in\mathfrak{D}(g)$, and let a board of cubic cells of side l' dividing $\frac{\ell'}{2}$ be placed on $T_{\ell'}$, on every T_ℓ with $\ell\ge\ell'$ in $\mathfrak{L}(g)$ and on $a_K\mathbb{Z}^3$; $N'=(\ell'/\mathsf{l}')^3$. For a bounded Borel function F of the bonds inside one cell put $\|F\|_{\mathrm{ch}}=|\langle\prod_\kappa F^{[\kappa]}\rangle_{K,\ell',\beta}|^{1/N'}$, the product running over the mirror copies of F, conjugated on odd reflection parity, in the N' cells of

$T_{\ell'}$. Then

$$|s(F_1 \cdots F_q)| \le \prod_i e^{G_g \mathsf{l}'^3} \|F_i\|_{\mathrm{ch}}$$

for functions $F_1, \dots, F_q$ of distinct cells, s being a periodic Gibbs state at (K, β) or the measure of T_ℓ, $\ell \in \mathfrak{L}(g)$, $\ell \ge \ell'$. In particular, for real F_1, F_2 of two distinct cells and real c_1, c_2: $|s(F_1) - c_1| \le e^{G_g \mathsf{l}'^3} \|F_1 - c_1\|_{\mathrm{ch}}$ and $|s(A^s[F_1]A^s[F_2])| \le 2e^{2G_g \mathsf{l}'^3} \|F_1 - c_1\|_{\mathrm{ch}} \|F_2 - c_2\|_{\mathrm{ch}}$. ■

Proof sketch. On T_ℓ, $\ell \ge \ell'$, reflection positivity at the walls gives the chessboard estimate $|\langle F_1 \cdots F_q \rangle_{K,\ell,\beta}| \le \prod_i \langle \prod_\kappa F_i^{[\kappa]} \rangle_{K,\ell,\beta}^{1/N_\ell}$, $N_\ell = (\ell/\mathsf{l}')^3$. The product of the mirror copies over all the cells of T_ℓ is a periodisation of the product over the cells of $T_{\ell'}$, and the chessboard seminorms on the two sides compare by the inequality $\langle \prod_\kappa F^{[\kappa]} \rangle_{K,\ell,\beta}^{1/N_\ell} \le \langle \prod_\kappa F^{[\kappa]} \rangle_{K,\ell',\beta}^{1/N'} e^{\mathsf{l}'^3 (p_{\ell'} - p_\ell)}$ ([7, Cor. B.1](e)). It is reflection positivity at the walls applied once more, to the periodised product: the pattern of the copies has the period $2\mathsf{l}'$, which divides ℓ', so that a slab of that length is a half slab and its mirror image, the transfer operator across it is positive, and $\operatorname{Tr} T^n \le (\operatorname{Tr} T)^n$ in each of the three directions; this bounds the unnormalised N_ℓ-th root by the unnormalised N'-th root, and the ratio $Z_{K,\ell',\beta}^{1/N'} / Z_{K,\ell,\beta}^{1/N_\ell}$ of the plain partition functions per cell is $e^{\mathsf{l}'^3 (p_{\ell'} - p_\ell)} \le e^{G_g \mathsf{l}'^3}$, the pressures $p_\ell = \ell^{-3} \log Z_{K,\ell,\beta}$ being formed with the plain partition functions (Section YM3_I:86).

A periodic Gibbs state is dominated by the torus measures through its conditional laws (Prop. YM3_I:91.9), and this domination carries the bound to every periodic Gibbs state. The last two bounds are the cases $q = 1$ and $q = 2$, with $s(A^s[F_1]A^s[F_2]) = s((F_1 - c_1)(F_2 - c_2)) - (s(F_1) - c_1)(s(F_2) - c_2)$. □

Lemma 5.9 (Exact identities)**.**

(a) Let $\mathbf{X}$ be a polynomial field of directions that vanishes off a finite set Λ of bonds: at each positively oriented bond b a $\mathfrak{g}$-valued polynomial $\mathbf{X}(b)$ in the plaquette variables, 0 for $b \notin \Lambda$; let $v_{\mathbf{X}} F = \sum_b \sum_{\mathsf{a}} \langle \sigma_{\mathsf{a}}, \mathbf{X}(b) \rangle \partial_b^{\mathsf{a}} F$ and $\operatorname{div} v_{\mathbf{X}} = \sum_b \sum_{\mathsf{a}} \partial_b^{\mathsf{a}} \langle \sigma_{\mathsf{a}}, \mathbf{X}(b) \rangle$, $\sigma_1, \sigma_2, \sigma_3$ the Pauli matrices and ∂_b^{a} the derivative along $e^{i\epsilon\sigma_{\mathsf{a}}} U(b)$ at $\epsilon = 0$. Then $\nu((\beta v_{\mathbf{X}} A - \operatorname{div} v_{\mathbf{X}}) Y) = 0$ for every Gibbs state or torus measure ν at (a_K, β) and every bounded Borel Y that does not depend on the bonds of Λ; so every joint cumulant under ν with the entry $\beta v_{\mathbf{X}} A - \operatorname{div} v_{\mathbf{X}}$ and further entries not depending on Λ vanishes. $N_\beta[\eta] = \Theta_\beta[\eta] + \mathfrak{K}^{\mathrm{v}}[\beta\eta] - \mathfrak{b}\, \mathcal{O}(\operatorname{div}^s \eta) + \frac{3}{2} a\, \mathcal{O}(e^\eta)$ is such a null term.

With the contact terms, for all weights f_i and every vector field η, $\langle N_\beta[\eta]; \mathcal{O}(f_\cdot) \rangle + \sum_i \langle \mathrm{Ct}[f_i; \eta]; (\mathcal{O}(f_l))_{l \ne i} \rangle = 0$, where $\mathrm{Ct}[f; \eta] = \mathcal{O}(\operatorname{div}^s(f\eta)) - a^{-1} \Theta_f[\eta] - a^{-1} \mathfrak{K}^{\mathrm{v}}[f\eta]$.

(b) Identically in U, $\mathfrak{K}^{\mathrm{v}}[\eta] = -\sum_p \sum_\rho (\nabla_\rho^+ \eta^\rho)(x(p)) \Psi_\rho(p) + \mathfrak{K}^\sharp[\eta]$; for $\rho \in P(p)$, $\mathfrak{r}_\rho^\sharp(p) = 0$ and $-(a_p + a_{p+ae_\rho}) \le \Psi_\rho(p) \le 0$; $\mathfrak{r}^\sharp$ transforms as a vector under the coordinate mirrors of the lattice: $\mathfrak{r}_\rho^\sharp$ is odd under the mirror normal to ρ and even under the two mirrors that contain e_ρ, so that it is odd, for every ρ, under the inversion $y \mapsto 2z - y$ through a site z, the product of the three mirrors through z. Hence $\langle \mathfrak{K}^\sharp[\eta] \rangle_{K,\ell,\beta} = 0$, and by translation invariance also $\langle \mathfrak{K}^{\mathrm{v}}[\eta] \rangle_{K,\ell,\beta} = 0$; and if a field ζ has $\zeta^\rho(y + se_\rho) = \zeta^\rho(y)$ for all y, s and ρ, then $\langle \mathfrak{K}^\sharp[\zeta]; \mathcal{O}(f) \rangle_{K,\ell,\beta} = 0$ for every weight f.

A *block form* is a combination $c = \sum_i \kappa_i m_i$ of monomials in the lassos of the block $\mathrm{Bk}(x) = x + a\{-1, 0, 1\}^3$ of 27 sites centred at a site x: a lasso is a path of at most three steps from x inside the block to a corner y of a plaquette of the block, the holonomy of that plaquette from y, and the path back; with $w = \mathbf{w}(W)$ and $a_{\mathrm{lz}} = 1 - \frac{1}{2} \operatorname{tr} W$ the vector part and the plaquette term of the holonomy W of a lasso, a monomial is a product of n_2 factors $\langle w, w' \rangle$, n_3 factors $\det(w, w', w'')$ and n_a factors a_{lz}, and $2n_2 + 3n_3 + 2n_a$ is its order in the field strength. The same lassos are taken at every site, c_x is the form at the block of x, the order of c is the least order of its monomials, and its norm is the number $\|c\| = \sum_i |\kappa_i| 2^{n_a(m_i)}$ of Lem. 5.6(d).

For a real function k of the sites, $\mathrm{Fm}^c(k) = a^{-1}\sum_x k(x)c_x(U)$ is a dominated local source of size $\le 76\|c\||k|_\infty$, carried by $\{k \ne 0\}$, with $|\mathrm{Fm}^c(k)| \le \mathcal{O}(2\|c\||k|_\infty \mathbf{1}_{\mathrm{Pl}(k)})$, $\mathrm{Pl}(k)$ the plaquettes of the blocks of the sites with $k \ne 0$; and if all the monomials of c have order ≥ 3, as for the commutator terms, the defects of the linear Bianchi identity, a_p^2 and $a_p a_{p_+}$, its sizes fall like L^{-2j} with the level j and its values on the backgrounds like L^{-2t} with the level t, with the comparison weight $\Gamma_c^\circ\|c\||k|_\infty\mathbf{1}_{\mathrm{Pl}(k)}$ of Lem. 5.6(d). $\Psi_{\lambda(p)}(p)$ is a block form of order 2 and norm 1, and $\mathfrak{r}_\rho^\sharp(p)$, $\rho = \lambda(p)$, a block form of order 2 and norm ≤ 8, both at the block of the lowest corner of p; so for a real weight k the sums $a^{-1}\sum_p k(p)\Psi_{\lambda(p)}(p)$ and $a^{-1}\sum_p k(p)\mathfrak{r}^\sharp_{\lambda(p)}(p)$ over the plaquettes of one normal are sources $\mathrm{Fm}^c(\tilde k)$, $\tilde k(x)$ the value of k at the plaquette of that normal with the lowest corner x, and they are dominated local sources of the sizes $\le 76|k|_\infty$ and $\le 608|k|_\infty$; as a function $|\mathfrak{r}_\rho^\sharp(p)| \le C\bar a^{3/2}$, $\bar a$ the largest a_q near p, and $\beta\mathfrak{K}^\sharp$ is treated by parity (Lems. 5.10, 5.11).

(c) For a vector field k, $\beta\sum_{p,\rho}\mathsf{k}^\rho(x(p))\Psi_\rho(p) = -\frac14\mathfrak{N}[\mathsf{k}] + \sum_{v=1}^5\mathfrak{J}_v[\mathsf{k}] - \frac14 c_{\mathsf{k}}$, where $\mathfrak{N}[\mathsf{k}] = \beta v_{\mathbf{X}^{\mathsf{k}}}A - \operatorname{div} v_{\mathbf{X}^{\mathsf{k}}}$ is the null term of (a) for the field of directions $\mathbf{X}^{\mathsf{k}}(b) = \sum_{\rho\ne\mu_b}\mathsf{k}^\rho(x_b)G_\rho(b)$, $c_{\mathsf{k}} = 6\sum_b\sum_{\rho\ne\mu_b}\mathsf{k}^\rho(x_b)$ is a number, and $\mathfrak{J}_1,\dots,\mathfrak{J}_5$ are sums of local densities of the orders $2,2,2,3,4$ with weights at most $3|\mathsf{k}|_\infty + \frac32\mathsf{b}\mathrm{Lip}_\infty\mathsf{k}$, $\frac18\mathsf{b}\mathrm{Lip}_\infty\mathsf{k}$, $\frac12\mathsf{b}\mathrm{Lip}_\infty\mathsf{k}$, $\frac14\beta|\mathsf{k}|_\infty$ and $\beta|\mathsf{k}|_\infty$, vanishing beyond the sup distance a of $\operatorname{supp}\mathsf{k}$. Consequently, with $\mathsf{k}_\eta^\rho = \nabla_\rho^+\eta^\rho$, $\mathrm{St}_K[\eta] = N_\beta[\eta] - \frac14\mathfrak{N}[\mathsf{k}_\eta] - \mathrm{Rm}_K[\eta] - \frac14 c_{\mathsf{k}_\eta}$, where $\mathrm{Rm}_K[\eta] = \beta\mathfrak{K}^\sharp[\eta] - \sum_v\mathfrak{J}_v[\mathsf{k}_\eta] + \frac32 a\,\mathcal{O}(e^\eta)$ is the remainder of the stress insertion.

(d) $\Delta_K[\eta] = F^0 + F^1$, rough entries carried within $2a$ of $\operatorname{supp}\eta$, with $\|F^0\| \le 286\mathsf{b}\sup|\nabla\eta|$ and $\|F^1\| \le 780\mathsf{b}\,a\sup|\nabla^2\eta|$; F^0 is null: its classical density, a quadratic form in the field strength with coefficients linear in $\nabla\eta$, vanishes identically.

(e) For a positively oriented bond b and directions λ,ρ: $\sum_{p\ni b}a^{-1}(x_b - x(p))_\lambda\mathfrak{d}_{b,p,\rho} = \langle\mathbf{u}_{b,\lambda},\mathbf{u}_{b,\rho}\rangle$, so that $\sum_p\mathfrak{s}_{\lambda\rho}(p) = \sum_b\langle\mathbf{u}_{b,\lambda},\mathbf{u}_{b,\rho}\rangle$ is a symmetric positive semidefinite matrix; for $\lambda\ne\mu_b$ and $p_\pm = p(b,\pm\lambda)$, $|\mathbf{u}_{b,\lambda}|^2 = (a_{p_+}+a_{p_-}) - \frac14(a_{p_+}^2 + a_{p_-}^2) + \Psi_\lambda(p_-)$ and $\Psi_\lambda(p_-) = -\frac14|G_\lambda(b)|^2 - \frac14(a_{p_+}^2+a_{p_-}^2)$; for $\nu\in P(p)$, $\mathfrak{s}_{\nu\nu}(p) = \frac32 a_p + \frac14(a_{p+ae_\nu}+a_{p-ae_\nu}) - \frac12 a_p^2 + \frac12(\Psi_\nu(p)+\Psi_\nu(p-ae_\nu))$, and $\mathfrak{s}_{\lambda\rho}(p) = 0$ if $\lambda = \lambda(p)$. Consequently, identically in U,

$$T_{mm}(\varphi) - \mathsf{b}\check D_m(\varphi) = \beta\sum_{p:\,m\in P(p)}\varphi(x(p))\Big[\tfrac12\big(\Psi_m(p)+\Psi_m(p-ae_m)\big) - \tfrac12 a_p^2\Big] + \mathsf{b}\,\mathcal{O}(\tilde\varphi_m),$$

$\tilde\varphi_m(p) = \frac14[\varphi(x(p)+ae_m) - 2\varphi(x(p)) + \varphi(x(p)-ae_m)]\mathbf{1}\{m\in P(p)\}$, so $|\tilde\varphi_m| \le \frac12 a\,\mathrm{Lip}\,\varphi$; and $\sum_m T_{mm}(\varphi) - \mathsf{b}\,\mathcal{O}(\varphi) = \sum_m(T_{mm}(\varphi) - \mathsf{b}\check D_m(\varphi))$.

(f) *The null term against sources that meet its carrier.* For a complex vector field k and a complex function f on T_ℓ put $\mathsf{e}^{\mathsf{k}}(p) = \sum_{\rho\in P(p)}[\mathsf{k}^\rho(x(p)+\frac{a}{2}e_\rho) + \mathsf{k}^\rho(x(p)-\frac{a}{2}e_\rho)]$ and

$$\mathfrak{C}'[\mathsf{k};f] = \sum_p\sum_{b\subset\partial p}\sum_{\rho\ne\mu_b}a^{-1}(f(x(p)) - f(x_b))\mathsf{k}^\rho(x_b)\,\langle G_\rho(b),\mathbf{w}(W_{b,p})\rangle, \qquad \mathfrak{R}[\mathsf{k};f] = \mathsf{b}^{-1}\mathrm{div}\,v_{\mathbf{X}^{f\mathsf{k}}} + \mathfrak{C}'[\mathsf{k};f],$$

with $\mathbf{X}^{\mathsf{k}}$ and $\mathfrak{N}[\mathsf{k}]$ as in (c). Then: (N1) $v_{\mathbf{X}^{\mathsf{k}}}\mathcal{O}(f) = \mathsf{b}^{-1}\mathfrak{N}[f\mathsf{k}] + \mathfrak{R}[\mathsf{k};f]$, identically in U; (N2) $\int\mathfrak{N}[\mathsf{k}]\,e^{\mathcal{O}(w)}e^{-\beta A}\,\mathrm{d}U = \int(v_{\mathbf{X}^{\mathsf{k}}}\mathcal{O}(w))e^{\mathcal{O}(w)}e^{-\beta A}\,\mathrm{d}U$ for every complex function w on T_ℓ; (N3) for a finite set A and complex functions f_l, $l\in\mathsf{A}$,

$$\big\langle\mathfrak{N}[\mathsf{k}];(\mathcal{O}(f_l))_{l\in\mathsf{A}}\big\rangle_{K,\ell,\beta} = \sum_{j=1}^{\#\mathsf{A}}\sum_{(q_1,\dots,q_j)}\mathsf{b}^{-(j-1)}\Big\langle\mathfrak{R}\big[f_{q_1}\cdots f_{q_{j-1}}\mathsf{k};f_{q_j}\big];(\mathcal{O}(f_l))_{l\in\mathsf{A}\setminus\{q_1,\dots,q_j\}}\Big\rangle_{K,\ell,\beta},$$

the inner sum over the sequences of j distinct elements of A (for $\mathsf{A}=\emptyset$ the left member is the mean, and it is 0); (N4) $\mathfrak{R}[\mathsf{k};f] = \mathsf{b}^{-1}c_{f\mathsf{k}} - 3\mathsf{b}^{-1}\sum_p a_p\,\mathsf{e}^{f\mathsf{k}}(p) + \mathfrak{C}'[\mathsf{k};f]$ is a number, the density a_p with weights $\le 12\,\mathsf{b}^{-1}\sup|f|\sup|\mathsf{k}|$ and the densities $\langle G_\rho(b),\mathbf{w}(W_{b,p})\rangle$ of

the order 2 with weights $\le \frac{1}{2}\mathrm{Lip}_\infty f \sup|\mathsf{k}|$: no factor β enters. Its weight at a plaquette p vanishes unless a bond b of p has $\mathsf{k}(x_b) \neq 0$ and $f \neq 0$ at x_b or at $x(p)$, so that its densities are carried within the sup distance $2a$ of $\operatorname{supp}\mathsf{k}$ and of $\operatorname{supp} f$, and the number vanishes unless $\mathsf{k}(x_b)f(x_b) \neq 0$ at some bond.

Consequently, if the spread D of $\operatorname{supp}\mathsf{k}$ and the $\operatorname{supp} f_l$, $l \in \mathsf{A}$, is $> 4a$, the terms of (N3) with $j = \#\mathsf{A}$ vanish, the numbers drop from the other terms, and in each of them the entry is carried by a set whose spread with the supports of the remaining f_l is $\ge D - 2a$. This rests on the merging of sets: if $S_0, \dots, S_n$ have the spread D, $Q \subset \{1, \dots, n\}$, $2r < D$, and T lies within the sup distance r of S_0 and of every S_q, $q \in Q$, then $T = \emptyset$ if $Q = \{1, \dots, n\}$, and otherwise the spread of T and the S_l, $l \notin Q$, is $\ge D - r$.

■

Proof sketch. (a) The Haar measure is invariant; with $\partial_b^{\mathsf{a}}\langle\sigma_{\mathsf{a}}, \mathbf{w}(W_{b,p})\rangle = 1 - a_p$, an integration by parts in each bond variable against $e^{-\beta A}$ gives $\nu((\beta v_{\mathbf{X}} A - \operatorname{div} v_{\mathbf{X}})Y) = 0$ in finite volume, and in a Gibbs state through its conditional laws. With sources the same integration by parts against $e^{-A(\beta - a^{-1}w_z, \cdot)}$, $w_z = \sum_i z_i f_i$, followed by the mixed derivative in z at 0, gives the contact terms, since $v_\eta A(v, \cdot)$ is linear in the weight v. (b) For $q_\pm = p(b, \pm\rho)$ in one plane, $\langle \mathbf{w}(W_{b,q_-}), \mathbf{w}(W_{b,q_+})\rangle = -2\Psi_\rho(q_-) - a_{q_-} - a_{q_+}$; so $\mathfrak{r}_\rho(p) = \Psi_\rho(p) - \Psi_\rho(p_-)$ in the plane, and the split is a summation by parts.

Under the mirror r normal to ρ, $\mathfrak{d}_{\mathsf{r}b,\mathsf{r}p,\rho}(\mathsf{r}U) = -\mathfrak{d}_{b,p,\rho}(U)$ and $\Psi_\rho(\mathsf{r}p)(\mathsf{r}U) = \Psi_\rho(p_-)(U)$, and the measure of a torus is invariant under r. Under a coordinate mirror that contains e_ρ these quantities are carried to the same quantities at the image bond and plaquette: the holonomy $W_{b,p}$ is taken from the initial point of b along b, and where the mirror reverses the orientation of b both factors of $\mathfrak{d}_{b,p,\rho}$ change their sign together; so $\mathfrak{r}_\rho^\sharp$ is even under it, and the three coordinate mirrors through a site compose to the inversion through it.

For ζ: $\langle \mathfrak{K}^\sharp[\zeta]; \mathcal{O}(f)\rangle = a^{-1}\sum_{p'} f(p')\langle \mathfrak{K}^\sharp[\zeta]; a_{p'}\rangle$, and for each p' and ρ the mirror normal to ρ through $x(p')$ keeps $a_{p'}$, the measure and ζ^ρ, carries the plaquettes normal to ρ among themselves, and changes the sign of $\mathfrak{r}_\rho^\sharp$ on them. A block form has local representations at every height of size $76\|c\|\|k\|_\infty$ (the proof of Lem. 5.6(d), with the lassos from the centre of the block, which reach its 27 sites in at most three steps); the field strength is $O(L^{-2j})$ in level-j units on the backgrounds, which gives the fall for monomials of order ≥ 3, with the comparison weight $\Gamma_c^\circ\|c\|\|k\|_\infty \mathbf{1}_{\mathrm{Pl}(k)}$ of Lem. 5.6(d). (c) Take in (a) $\mathbf{X}(b) = \sum_\rho \mathsf{k}^\rho(x_b)[\mathbf{w}(W_{b,p(b,\rho)}) + \mathbf{w}(W_{b,p(b,-\rho)})]$; then $v_{\mathbf{X}} a_p = \sum_{b\subset\partial p}\langle \mathbf{X}(b), \mathbf{w}(W_{b,p})\rangle$, $\operatorname{div} v_{\mathbf{X}}$ is the number c_{k} minus $3\sum_p a_p$ times a weight formed from k, and the in-plane remainder is $-2a_p^2$, of order 4. (d) At a constant field strength, with $G = \nabla\eta$ and $T_{lr} = \operatorname{tr} B_l B_r$, the classical density of $\Delta_K[\eta]$ is $\mathsf{b}[\operatorname{tr}G\operatorname{tr}T - \operatorname{Tr}GT] - \frac{\mathsf{b}}{2}\operatorname{tr}G\operatorname{tr}T - \frac{\mathsf{b}}{2}\operatorname{Tr}(H^{\mathrm{cons}}T) = 0$, since $\operatorname{Tr}(H^{\mathrm{cons}}T) = \operatorname{tr}G\operatorname{tr}T - 2\operatorname{Tr}GT$; the Taylor remainder of η gives F^1. (e) $\sum_{p\ni b} a^{-1}(x_b - x(p))_\lambda \mathbf{w}(W_{b,p}) = \mathbf{u}_{b,\lambda}$ and the form is bilinear; with $w_\pm = \mathbf{w}(W_{b,p_\pm})$, $|w_\pm|^2 = a_{p_\pm}(2 - a_{p_\pm})$ and $\Psi_\lambda(p_-) = -\frac{1}{4}|w_+ + w_-|^2 - \frac{1}{4}(a_{p_-}^2 + a_{p_+}^2)$, and the parallelogram law $4|\mathbf{u}_{b,\lambda}|^2 = |w_+ - w_-|^2 = 2|w_+|^2 + 2|w_-|^2 - |w_+ + w_-|^2$ gives the first identity; the formula for $\mathfrak{s}_{\nu\nu}$ is the same computation at the two bonds of p of the direction other than ν, and no vector $x_b - x(p)$ has a component normal to p.

Inserting these in $T_{mm}(\varphi) = \beta\sum_p \varphi(x(p))[\mathfrak{s}_{mm}(p) - a_p]$ and $\mathsf{b}\check{D}_m(\varphi) = \beta\sum_p \varphi(x(p))a_p(1 - 2\cdot \mathbf{1}\{\lambda(p) = m\})$, the plaquettes normal to m cancel, and moving the shifts $a_{p\pm ae_m}$ onto φ gives $\tilde{\varphi}_m$. Finally $\sum_m \check{D}_m = 3\mathcal{O} - 2\mathcal{O} = \mathcal{O}$. (f) $\mathcal{O}(f) = a^{-1}\sum_p f(x(p))a_p$ and $v_{\mathbf{X}} a_p = \sum_{b\subset\partial p}\langle \mathbf{X}(b), \mathbf{w}(W_{b,p})\rangle$; with $f(x(p)) = f(x_b) + (f(x(p)) - f(x_b))$ and $f(x_b)\mathbf{X}^{\mathsf{k}}(b) = \mathbf{X}^{f\mathsf{k}}(b)$, the first part is $a^{-1}v_{\mathbf{X}^{f\mathsf{k}}}A = \mathsf{b}^{-1}\beta v_{\mathbf{X}^{f\mathsf{k}}}A = \mathsf{b}^{-1}(\mathfrak{N}[f\mathsf{k}] + \operatorname{div} v_{\mathbf{X}^{f\mathsf{k}}})$ and the second is $\mathfrak{C}'[\mathsf{k}; f]$: this is (N1). (N2) is the integration by parts of (a) along $\mathbf{X}^{\mathsf{k}}$ against $e^{\mathcal{O}(w)}e^{-\beta A}$.

At the source $w_z = \sum_{l\in\mathsf{A}} z_l f_l$, (N2) says that the mean of $\mathfrak{N}[\mathsf{k}] - \sum_q z_q v_{\mathbf{X}^{\mathsf{k}}}\mathcal{O}(f_q)$ under the measure with the source vanishes, and this function is affine in z; the mixed derivative in z at 0 gives $\langle \mathfrak{N}[\mathsf{k}]; (\mathcal{O}(f_l))_{l\in\mathsf{A}}\rangle = \sum_{q\in\mathsf{A}}\langle v_{\mathbf{X}^{\mathsf{k}}}\mathcal{O}(f_q); (\mathcal{O}(f_l))_{l\neq q}\rangle$, and inserting (N1), whose first part is a null term

of the same kind with $f_q\mathsf{k}$ for k, an induction on $\#\mathsf{A}$ gives (N3). (N4): $\operatorname{div} v_{\mathbf{X}^\mathsf{k}} = c_\mathsf{k} - 3\sum_p a_p\, \mathsf{e}^\mathsf{k}(p)$, by $\partial_b^\mathsf{a}\langle\sigma_\mathsf{a}, \mathbf{w}(W_{b,p})\rangle = 1 - a_p$; $|\mathsf{e}^{f\mathsf{k}}| \le 4 \sup|f\mathsf{k}|$, $|x(p) - x_b|_\infty = \frac{a}{2}$, and the boxes of the densities lie within the sup distance $\frac{3}{2}a$ of $x(p)$.

The merging: if $Q = \{1, \dots, n\}$, points of $S_0, \dots, S_n$ within r of one point of T are pairwise at most $2r < D$ apart, which the spread forbids; otherwise, for $t \in T$, $y_l \in S_l$ ($l \notin Q$) and points $y_0 \in S_0$, $y_q \in S_q$ within r of t, some pair of the family $y_0, (y_q), (y_l)$ is at least D apart, it is no pair among $y_0, (y_q)$, and so two of the y_l, or one y_l and t, are at least $D - r$ apart. □

Lemma 5.10 (The chain with an entry; the cubic insertion)**.** Let $\gamma^{-2} \le \mathsf{b}_- \le \mathsf{b}_+$, $MR(\mathsf{b}_+) \le L^m$, $\ell = 2L^m$, $0 < \tau \le \frac{1}{4}$, $0 < \phi < \frac{1}{42}$, $u = \lceil \phi K \rceil$, $t = K - u$ and $c_0' = \min\{\frac{1}{14} - \frac{\tau}{7}, \frac{1}{14} - 3\phi\}$. Let K_ϕ be the least integer $K \ge 2$ with $(1-\phi)K \ge 2$, $L^{\phi K}\gamma^{-2} \ge \mathsf{t}_\natural$ and $\phi K > 16\mathfrak{p}_0^2 c_{\log}/(\log L)^2$, $\mathsf{t}_\natural$ being the calibration of §2.3.1 and $c_{\log} = \log L/\log\gamma^{-2}$; it depends on the constants and on ϕ alone. For $K \ge K_\phi$ and $\mathsf{b}' \in [\mathsf{b}_-, \mathsf{b}_+]$ let $\rho_0 dU$ be the measure of T_ℓ at $\beta = L^K\mathsf{b}'$; the data (K, b', m) with the top level t form the setting (Th) (§2.3.1), and the small-field chain is formed along the couplings $x_j = L^{K-j}\mathsf{b}'$ up to the level t. Let $\mathcal{M}^k$ be the block map to the level k, Cut_k the field cut and $\mathrm{In}_k \subset \mathrm{Cut}_k$ the inner ball of the level k (the definitions preceding Lem. YM3_I:113.10), $\mathfrak{G}_k[x_k] \ge 0$ the gauge-fixing factor and $\chi^{\mathrm{sf}}_{x_k} \in \{0,1\}$ the fluctuation test of the step from the level k ([7, §B.2]), and

$$\mathsf{p}_k(U) = \mathbf{1}_{\mathrm{Cut}_k}(\mathcal{M}^k U) \prod_{j<k} [\mathbf{1}_{\mathrm{Cut}_j}\, \mathfrak{G}_j[x_j]\, \chi^{\mathrm{sf}}_{x_j}](\mathcal{M}^j U),$$

the weight of the small-field chain up to the level k: a bounded Borel function with values in $[0,\infty)$, with $\mathsf{p}_0 = \mathbf{1}_{\mathrm{Cut}_0}$ and $\mathsf{p}_{k+1} = (\mathbf{1}_{\mathrm{Cut}_{k+1}} \circ \mathcal{M}^{k+1})\, ((\mathfrak{G}_k[x_k]\chi^{\mathrm{sf}}_{x_k}) \circ \mathcal{M}^k)\, \mathsf{p}_k$; and $(\mathbf{1}_{\mathrm{In}_t} \circ \mathcal{M}^t)\, \mathsf{p}_t$ is the weight $\mathfrak{w}^{\mathrm{sf}}_t$ of the small-field chain of the definitions preceding Prop. YM3_I:115.2. Let $\nu_k = (\mathcal{M}^k)_*(\mathsf{p}_k \rho_0 dU)$, the measure of the level k of the chain among the construction results at the source 0 (Prop. YM3_I:114.2 and the proof of Prop. YM3_I:115.2), a finite positive measure with $\nu_{k+1} = \mathbf{1}_{\mathrm{Cut}_{k+1}} \mathbf{T}(\mathfrak{G}_k[x_k]\chi^{\mathrm{sf}}_{x_k}\nu_k)$, $\mathbf{T}$ the image under the block averaging; let $Q_k h$ be the density of $\mathbf{1}_{\mathrm{Cut}_{k+1}}\mathbf{T}(\mathfrak{G}_k[x_k]\chi^{\mathrm{sf}}_{x_k} h\, \nu_k)$ with respect to ν_{k+1}, the mean of h over the range Rg_{x_k} of the chart of the fibre of the step ([7, Lem. B.3]), and $Q_{t\leftarrow 0} = Q_{t-1}\cdots Q_0$, so that $(\mathcal{M}^t)_*(\mathsf{p}_t Y \rho_0 dU) = (Q_{t\leftarrow 0} Y)\, \nu_t$ for bounded Borel Y: $(Q_{t\leftarrow 0}Y) \circ \mathcal{M}^t$ is the conditional mean of Y given $\mathcal{M}^t U$ under $\mathsf{p}_t \rho_0 dU$.

An *entry of the type* $(m_J, c, \hat{c}, N)$ is a finite family $J = (\mathsf{Q}_\iota, j_\iota, \varkappa_\iota)_\iota$ of boxes $\mathsf{Q}_\iota = \{x + a\sum_\mu n_\mu e_\mu : 0 \le n_\mu \le r_\mu\}$, $(r_1, r_2, r_3) \in \{0,1,2\}^3$ with at least two $r_\mu \ge 1$, x the lowest site of the box, of densities j_ι on them and of complex weights $\varkappa_\iota$, at most N indices having one lowest site. A density on a box is a polynomial in the entries of the complex configurations $\mathbf{U}(b) \in \mathrm{SL}(2,\mathbb{C})$ at its bonds, $\mathbf{U}(\bar{b}) = \mathbf{U}(b)^{-1}$, invariant under the gauge transformations of the box; it has the height $\hat{c}$ if $|j| \le \hat{c}$ wherever $|\mathbf{U}(b)| \le 2$ at all the bonds, and the order m_J with the constant c if $|j| \le c\,\epsilon^{m_J}$ wherever, besides, $|\partial\mathbf{U}(q) - \mathbf{1}| \le \epsilon$ at all the plaquettes q of the box, $0 < \epsilon \le 1$.

Zero insertions and zero scalar sources are omitted before inverse norms and source radii are formed; the multilinear estimates are immediate when an insertion vanishes. We write $J(U) = \sum_\iota \varkappa_\iota j_\iota(U|_{\mathsf{Q}_\iota})$ and $|\varkappa|_\infty = \max_\iota |\varkappa_\iota|$; the carrier of J is the set of the sites of the boxes with $\varkappa_\iota \ne 0$.

(a) For $K \ge K_\phi$, an entry of the type $(m_J, c, \hat{c}, N)$ with $m_J \ge 2$, $|s| < r_J$ with $r_J^{-1} \le C|\varkappa|_\infty$, and $1 \le k \le t$:

$$(\mathcal{M}^k)_*(\mathsf{p}_k e^{sJ}\rho_0 dU) = \exp\Big[\sum_{i\le k} c_i(s)\Big] \mathbf{1}_{\mathrm{Cut}_k} e^{\Phi^s_k}\, \mathrm{d}V,$$

the numbers $c_i(s)$ and the action Φ^s_k being holomorphic in s and sums of local families, each of which depends on $\varkappa$ in its domain only. $Q_{t\leftarrow 0}J$ is a number plus $\partial_s\Phi^s_t|_{s=0}$ ν_t-almost everywhere on Cut_t, and its oscillation is at most $CK^{2\mathfrak{p}_0}L^{4u-K}|\varkappa|_\infty$ for $m_J = 2$ and $CK^{3\mathfrak{p}_0}L^{4u-\frac{3}{2}K}|\varkappa|_\infty$ for $m_J \ge 3$. Here C depends on the constants, ℓ, b_+, ϕ and the type of the entry.

(b) There are $K_2 \geq K_\phi$, C_2, p_2, depending on these data alone, such that for $K \geq K_2$ and every $\mathsf{b}' \in [\mathsf{b}_-, \mathsf{b}_+]$ the following holds. For every field Ξ there is a number a with $|Q_{t\leftarrow 0}(\beta\mathfrak{K}^\sharp[\Xi]) - \mathsf{a}| \leq C_2\mathsf{H}(\Xi)K^{p_2}L^{-c_0'K}$ ν_t-almost everywhere on Cut_t. The same holds at real sources: let $n \geq 1$, $f_1, \dots, f_n$ real Lipschitz functions, $\mathsf{M}_t \leq CK^{2\mathfrak{p}_0}\ell^3 L^{3u}$ the inverse radius of the sources of the chain at the level t, and x real in the polydisc $\mathbb{D}_t$ of the radii $r_i = (8n\mathsf{M}_t\|f_i\|_\sharp)^{-1}$; then $w_x = \sum_i x_i f_i$ lies in the class of sources of the setting (Th), and with the measure $\rho_0 e^{\mathcal{O}(w_x)}dU$ in the place of $\rho_0 dU$, its chain measure $\nu_t^{w_x}$ and its conditional mean $Q_{t\leftarrow 0}^{w_x}$, and the entry with the moved weight $\mathfrak{K}^\sharp[(\beta - a^{-1}w_x)\Xi]$ in the place of $\beta\mathfrak{K}^\sharp[\Xi]$, the same bound holds with a number $\mathsf{a}(x)$, the numbers C_2, p_2, K_2 being free of x and of b'.

The number $\mathsf{a}(x)$ is the sum, over the levels $j \leq t$, of the values at the configuration $\mathbf{1}$ of local families of the level j whose member at a polymer X reads w_x and Ξ on the plaquettes of X alone and vanishes if Ξ vanishes at the barycentres of these plaquettes. For free photons the cubic piece vanishes identically.

■

Proof sketch. (a) The entry is typed anew at each level. At a height $j \geq 1$ the sum of the $\varkappa_\iota j_\iota$ over the boxes whose cubes of the level j form a given polymer is a pointed family of the level j: at most $8N(ML^j)^3$ boxes have one polymer, and on the cores each density is at most $c\,\alpha_0^{m_J}L^{-2m_J j}$ by its order, so that the family has the size $8NM^3 c\,\alpha_0^{m_J}|\varkappa|_\infty L^{-(2m_J-3)j}$; this is the power counting, of order $|\varkappa|_\infty L^{(3-2m_J)j}$ in level-j units. At the height 0 the family in the fourth place $\sum_\iota \varkappa_\iota[j_\iota(e^{iB'}\mathbf{U}) - j_\iota(\mathbf{U})]$, B' the fluctuation field with its values at the pivot bonds, sums to $J(e^{iB'}\mathbf{U}) - J(\mathbf{U})$ and has the size of a number times $N\hat{c}\,\delta(x_0)|\varkappa|_\infty$, by the height and Schwarz's lemma in the fluctuation field.

So for $|s| < r_J$, $r_J^{-1} \leq C|\varkappa|_\infty$ with C reading N, c and $\hat{c}$, the entry enters each step as one more local term, inside the radii of the one-step theorem with sources (Prop. 6.31); the members of the families are holomorphic in s and read s only on the polymers that the entry meets. The identity of the measures follows by induction on k from the recursion of $\mathfrak{p}_k$: the images $\nu_k^s = (\mathcal{M}^k)_*(\mathfrak{p}_k e^{sJ}\rho_0 dU)$ obey $\nu_{k+1}^s = \mathbf{1}_{\mathrm{Cut}_{k+1}}\mathbf{T}(\mathfrak{G}_k[x_k]\chi_{x_k}^{\mathrm{sf}}\nu_k^s)$, the small-field pushforward in [7, Lem. B.3], a chart integral over Rg_{x_k} that the one-step theorem evaluates; one function $e^{\Phi_k^s}$ is both the density reached at the level k and the integrand of the step from the level k.

The derivative of the exponent in s at 0 is the conditional mean up to the number $\dot{c}_t = \sum_i \partial_s c_i(0)$. On Cut_t the plaquette variables of the background are within $\mathsf{e}_t^{\mathrm{c}}L^{-2t}$ of $\mathbf{1}$, $\mathsf{e}_t^{\mathrm{c}}$ the radius of the field cut, so that the entry at the background is at most $N(\ell L^K)^3 c\,(\mathsf{e}_t^{\mathrm{c}})^{m_J}L^{-2m_J t}|\varkappa|_\infty$; with what the families of the earlier levels carry, $|\Phi_t^s - \Phi_t^{s'}| \leq |s - s'|\,\mathsf{o}_t$ with o_t of the order of the stated oscillation, which is at most $2\mathsf{o}_t$. The threshold $K \geq K_\phi$ makes t a top level of the setting (Th): $(1-\phi)K \geq 2$ gives $1 \leq t \leq K-1$, $L^{\phi K}\gamma^{-2} \geq \mathsf{t}_\natural$ gives $x_t = L^u\mathsf{b}' \geq \mathsf{t}_\natural$, and the third condition keeps the size of a source at the coupling x_k, $k \leq t$, below the bound of the step; this size θ_k obeys $L\theta_k \leq \Theta_K L^{-(K-k)/2}$, Θ_K a power of K. (b) Let $S = \beta\mathfrak{K}^\sharp[\Xi] = \sum_p \varkappa(p)\mathfrak{r}_{\lambda(p)}^\sharp(p)$, $\varkappa(p) = L^{2K}\mathsf{b}'\,\Xi^{\lambda(p)}(x(p))$, so that $\max|\varkappa| \leq L^{2K}\mathsf{b}_+\mathsf{H}(\Xi)$; the density vanishes to the third order at $\mathbf{1}$ (Lem. 5.9(b)).

At a real source the weight is moved, $\varkappa(p) = L^{2K}(\mathsf{b}' - w_x(x(p)))\Xi^{\lambda(p)}(x(p))$, and the chain carries the source; we write the source 0, the other case being the same with the index w_x. Let $s_0 = S$ and $s_j = S \circ U_j$ on Cut_j, the entry read at the background of the level j. For a pointed family f of a level $\leq n$ let $\Sigma_n[\mathsf{f}](V)$, $V \in \mathrm{Cut}_n$, be the sum over the polymers of the members of f at the background $U_n(V)$, and $\mathrm{Sum}[\mathsf{f}](\mathbf{1})$ the same sum at the configuration $\mathbf{1}$, a number. Let $S_n\mathsf{f}$ be the first-order response of the step from the level n to a family f of a level $i \leq n$, a family of the level $n+1$ with $\|S_n\mathsf{f}\|_{n+1} \leq \theta_n L^{-(n-i)}\|\mathsf{f}\|_i$, so that $Q_n\Sigma_n[\mathsf{f}] = \Sigma_{n+1}[\mathsf{f} + S_n\mathsf{f}]$ on Cut_{n+1}; and let $D^{(j)} = D^{(j)}[w;\varkappa]$ be the family born at the level j from the entry at the source w and the weight $\varkappa$: $Q_{j-1}s_{j-1} - s_j = \Sigma_j[D^{(j)}]$ on Cut_j.

With $T_t = \mathrm{id}$ and $T_n\mathsf{g} = T_{n+1}(\mathsf{g} + S_n\mathsf{g})$, the carriage of a family from the level n to the level t with all that it feeds, $Q_{t-1}\cdots Q_n\Sigma_n[\mathsf{g}] = \Sigma_t[T_n\mathsf{g}]$; so, with $n_0 = K - \lceil \frac{K}{7} \rceil$ and $Q_{t\leftarrow n_0} = Q_{t-1}\cdots Q_{n_0}$,

$$Q_{t\leftarrow 0}S = Q_{t\leftarrow n_0}s_{n_0} + \sum_{j\le n_0} \Sigma_t[T_j D^{(j)}] \qquad \text{on } \mathrm{Cut}_t.$$

The births are small by the parity in the fluctuation field: the part of the variation of the density that is odd in the fluctuation field has the size $g_{j-1} = x_{j-1}^{-1/2}$ and the Gaussian mean 0, so that it meets only the odd part of the interaction, of the size $g_{j-1}(\log x_{j-1})^{\mathfrak{p}_0}$, while the even part has the size g_{j-1}^2; so $\|D^{(j)}\| \le C(\log x_{j-1})^{\mathfrak{p}_0} g_{j-1}^2 L^{-3(j-1)} \max|\varkappa|$ once $K-j$ exceeds a fixed construction threshold, and $L^j\|D^{(j)}\| \le CK^{\mathfrak{p}_0}L^{K-j}\mathsf{H}(\Xi)$. $T_jD^{(j)}$ is $D^{(j)}$ plus the sum of the chains $S_{k_m}\cdots S_{k_1}D^{(j)}$, $j \le k_1 < \cdots < k_m < t$, of the families that it feeds; each chain is stopped at a level n of its own, chosen with the free fraction τ, at which the variation $\mathrm{Sum}[\mathsf{f}](U_n(V)) - \mathrm{Sum}[\mathsf{f}](\mathbf{1})$ of a family f of the level i is at most $\beta_n L^i\|\mathsf{f}\|_i$, $\beta_n = CK^pL^{3(K-n)-K}$, the square of the radius of regularity of the level n entering. $Q_{t\leftarrow n}$ does not enlarge an oscillation, and each feeding costs $L\theta_k \le \Theta_K L^{-(K-k)/2}$, so that what is fed is forgotten.

The variation of $\mathrm{Sum}[D^{(j)}]$ over Cut_{n_0}, $j < n_0$, is where the parity of the cubic density enters. The step is covariant under the isometries of the lattice of its level, and $\mathfrak{r}^\sharp$ transforms as a vector (Lem. 5.9(b)); so under the inversion $r_z : y \mapsto 2z - y$ through a site z of the lattice of the level j, $\mathrm{Sum}[D^{(j)}[w;\varkappa]](r_zU) = -\mathrm{Sum}[D^{(j)}[r_z^*w; r_z^*\varkappa]](U)$, $(r_z^*v)(p) = v(r_zp)$. With a partition of unity $\psi_z \ge 0$ over these sites, $\psi_z \circ r_z = \psi_z$, $\psi_z(p) = 0$ unless $|x(p) - z|_\infty \le \frac{1}{2}L^{j-K}$, and $\Sigma_z[w;\varkappa] = \mathrm{Sum}[D^{(j)}[w;\psi_z\varkappa]]$, this gives the exact split

$$\Sigma_z[w;\varkappa](U) = \tfrac{1}{2}\big(\Sigma_z[w;\varkappa](U) - \Sigma_z[w;\varkappa](r_zU)\big) + \tfrac{1}{2}\Sigma_z[w;\varkappa - r_z^*\varkappa](U) + \tfrac{1}{2}\big(\Sigma_z[w;r_z^*\varkappa](U) - \Sigma_z[r_z^*w;r_z^*\varkappa](U)\big).$$

(I) The first part compares the background with its image under r_z. In the Hölder gauge (Thm. [7, Thm. BT] for real backgrounds) the curvature, a 2-form, is even under the inversion, so that the linear part of the potential is kept and the potential and its image differ by the odd part of the Hölder remainder, at most $R^2\vartheta^{2+b}e_3$ on a cube of the radius R about z, $\vartheta = L^{-(n_0-j)}$, $b = \frac{3}{4}$, e_3 of the order of the radius of regularity of the level n_0: a gain ϑ^b. (II) The second part carries the asymmetry of the weight, $|\varkappa - r_z^*\varkappa| \le L^{2K}\mathrm{Lip}(\Xi)L^{j-K}$ where $\psi_z \ne 0$. (III) The third part is the asymmetry of the source in the step: it vanishes at the source 0, and at a real source it is of the second order at the trivial triple, of the size $\mathrm{Lip}(w_x)\,\delta(x_{j-1})$, $\delta(x_{j-1})$ of the order of $L^{-(K-j)/2}$ being the threshold of the level.

Summed over $j < n_0$, the parts (I), (II) and (III) are $O(L^{-3K/14})$, $O(L^{-4K/7})$ and $O(L^{-K/14})$; the stopped entry contributes $O(L^{-5K/14})$ and the family born at n_0 $O(L^{-3K/7})$; forgetting leaves $O(L^{-(\frac{1}{14}-\frac{\tau}{7})K})$ of what is fed, and against the factor L^{3u} of the levels above t it leaves $O(L^{-(\frac{1}{14}-3\phi)K})$; all this up to powers of K. The least of these exponents is c_0'; at the source 0 the last two meet at $\frac{1}{14}$ as $\tau, \phi \to 0$, and at a real source the part (III) meets them there. The number a, the sum of the values at the configuration $\mathbf{1}$ of the families born and fed up to their levels of stopping, depends on no V, and each of these families reads the source and the weight on its polymer alone and vanishes where the weight vanishes.

The numbers are uniform on $[\mathsf{b}_-, \mathsf{b}_+]$ because the frame of [7, Def. B.14] holds for the chain along $x_j = L^{K-j}\mathsf{b}'$ at every label of the interval and b' enters only through $x_i \in [\gamma^{-2}L^{K-i}, \mathsf{b}_+L^{K-i}]$ and $\max|\varkappa|$, each time in a monotone way; on the polydisc $\mathbb{D}_t$ the source stays in the class of the setting (Th) and the mass of the small-field chain stays away from 0, and at real x the measures of the chain are positive. □

Lemma 5.11 (The cubic remainder)**.** In the setting of Lem. 5.10, let $n \ge 1$, $\mathsf{d} > 0$ and $c < \frac{1}{14}$. There are C, p, free of K and of b', such that for all $K \ge 2$, $\mathsf{b}' \in [\mathsf{b}_-, \mathsf{b}_+]$, $\beta = L^K\mathsf{b}'$, all real Lipschitz

f_i and all fields η, the spread of $\{\eta \neq 0\}, \operatorname{supp} f_1, \dots, \operatorname{supp} f_n$ being $\geq \mathsf{d}$,

$$|\mathfrak{U}^\sharp[\eta](f)| \leq CK^p L^{-cK} \mathsf{H}(\eta) \prod_i \|f_i\|_\sharp.$$

The spread condition holds if two of the sets are at sup distance $\geq \mathsf{d}$, and if $\eta = 0$ within the sup distance d of one support. By $x \mapsto L^z x$ the same holds at every label and on every side. ■

Proof sketch. Moving the weight of the entry with the sources produces the contact terms:

$$\mathfrak{U}^\sharp[\eta](f) = -\partial_s \partial_{z_1} \cdots \partial_{z_n} \log \Big\langle \exp\Big(s\, \mathfrak{K}^\sharp[(\beta - a^{-1} \textstyle\sum_i z_i f_i)\eta] + \sum_i z_i \mathcal{O}(f_i) \Big) \Big\rangle \Big|_{s=z=0}.$$

After discarding the histories with large fields, Lem. 5.10(b), applied to the chain with the real sources x in its polydisc $\mathbb{D}_t$ and to the field η, makes the conditional mean of the entry with the moved weight equal, up to $\varepsilon = K^p L^{-c_0' K}$, to its number $\mathsf{a}(x)$; and $\mathsf{a}(x)$ is a sum $\sum_i c_i(x)$ of functions each of which does not depend on x_i, up to a remainder below every power of L^{-K}. Indeed, let $c_i(x)$ collect the members of the families of $\mathsf{a}(x)$ at the polymers on whose plaquettes f_i is the first of the sources that vanishes at all the barycentres; such a member reads the sources on its polymer alone, so it does not read x_i; the members at the other polymers, which meet $\{\eta \neq 0\}$ and all the $\operatorname{supp} f_i$, are left over; under the spread such a polymer of the level j has the linear size $\geq \mathsf{d} L^{K-j}/M - 3$, so that these members cost $e^{-\mathsf{d} L^{K-j}/M}$ against the sizes.

Such sums have no mixed derivative. At complex sources the mean is crudely at most $B = C\ell^3 L^{3K} \max|\varkappa|$, $\varkappa$ the weight of the entry. Interpolating between the two bounds, the mixed derivative is at most $2^n \varepsilon^{1-\sigma} B^\sigma (\frac{4n}{\pi\sigma})^n \prod_i r_i^{-1}$ with $r_i^{-1} \leq CK^{2\mathfrak{p}_0} \ell^3 L^{3u} \|f_i\|_\sharp$, and σ, ϕ, τ small give every $c < \frac{1}{14}$. The finitely many $K < K_2$ are paid by the crude bound of the cumulant by the suprema of its entries, which goes into C. For $n \geq 2$ the bound holds also where $\{\eta \neq 0\}$ covers the supports, as in the uses below, since the spread of the $n+1$ sets is at least that of the supports among themselves. □

Lemma 5.12 (Chessboard bound in one cell)**.** Let $0 < g \leq \gamma$, $\ell' = 2\mathsf{l} \in \mathfrak{L}(g)$, $0 < \phi \leq \frac{1}{84}$, $u = \lceil \phi K \rceil$, $t = K - u$, $0 < \delta \leq \frac{1}{8}\mathsf{l}$ and $m \geq 1$. There are C, p, K_3, depending on $g, \ell', \phi, m, \delta$, such that for $(K, \beta) \in \mathfrak{D}(g)$ with $K \geq K_3$, every board of side l with offset in $a_{K-t}\mathbb{Z}^3 = L^{-u}\mathbb{Z}^3$, every cell, every d with $\mathsf{d} L^u \geq K^2$, and m slots $X_1, \dots, X_m$ supported in the cell at sup distance $\geq \delta$ from its walls and pairwise $\geq \mathsf{d}$ apart,

$$\Big\| \prod_a (X_a - \langle X_a \rangle_{K,\ell',\beta}) \Big\|_{\mathrm{ch}} \leq C \prod_a \varpi(X_a)\, \mathsf{n}(X_a),$$

where a slot is $\mathcal{T}_K(H)$, with $\varpi = K^p L^{4u}$ and $\mathsf{n} = |H|_\infty$, or one of

$$\mathcal{T}_K(H^{\mathrm{cons}}[\eta]), \mathrm{St}_K[\eta], \Delta_K[\eta], \mathrm{Rm}_K[\eta], \mathfrak{K}^{\mathrm{v}}[\beta\eta],$$

with $\varpi = K^p L^{3u - K/28}$ and $\mathsf{n} = \|\eta\|_{C^2}$. For $m = 1$ and $\phi = \frac{1}{400}$ we have $\|\mathrm{St}_K[\eta] - \langle\cdot\rangle\|_{\mathrm{ch}} \leq CK^p L^{-K/28} \|\eta\|_{C^2}$ and

$$\|\Delta_K[\eta] - \langle\cdot\rangle\|_{\mathrm{ch}} + \|\mathcal{T}_K(H^{\mathrm{cons}}[\eta]) - \langle\cdot\rangle\|_{\mathrm{ch}} \leq CK^p L^{-K/36} \|\eta\|_{C^2};$$

for a plain tensor entry $\|\mathcal{T}_K(H') - \langle\cdot\rangle\|_{\mathrm{ch}} \leq CK^{2\mathfrak{p}_0} L^{K/100} |H'|_\infty$. At $m = 1$ and $\phi = \frac{1}{400}$, moreover, $\|\mathrm{Rm}_K[\eta] - \langle\cdot\rangle\|_{\mathrm{ch}} = \|\mathrm{St}_K[\eta] - \langle\cdot\rangle\|_{\mathrm{ch}}$, and the bound $CK^p L^{-K/28} \|\eta\|_{C^2}$ holds also for $\|F_\eta\|_{\mathrm{ch}}$, where

$$F_\eta = \mathfrak{K}^{\mathrm{v}}[\beta\eta] - \tfrac{1}{4}\mathfrak{N}[\mathsf{k}_\eta] = \beta \mathfrak{K}^\sharp[\eta] - \sum_{v=1}^{5} \mathfrak{J}_v[\mathsf{k}_\eta] + \tfrac{1}{4} c_{\mathsf{k}_\eta}, \qquad \langle F_\eta \rangle_{K,\ell',\beta} = 0,$$

with $\mathsf{k}_\eta^\rho = \nabla_\rho^+ \eta^\rho$ and the objects of Lem. 5.9(c); for every state s of Lem. 5.8, $s(\mathfrak{K}^{\mathrm{v}}[\beta\eta]) = s(F_\eta)$, so that $|s(\mathfrak{K}^{\mathrm{v}}[\beta\eta])| \leq e^{G_g \mathsf{l}^3} \|F_\eta\|_{\mathrm{ch}}$ by Lem. 5.8 at $q = 1$, $c_1 = 0$. ■

Proof sketch. By Lem. 5.9(c), $\mathrm{St}_K[\eta] = N_\beta[\eta] - \frac14\mathfrak{N}[\mathsf{k}_\eta] - \mathrm{Rm}_K[\eta] - \frac14 c_{\mathsf{k}_\eta}$, so that the slots $\mathrm{St}_K[\eta]$ and $-\mathrm{Rm}_K[\eta]$ differ by null terms and a number; inside the mean of the $8m$ mirror copies on $T_{\ell'}$ the null terms drop, since no other copy depends on their bonds. A product of centred slots in one cell expands into joint cumulants over the partitions into blocks of at least two members. Apply [7, Prop. 3.15](a), with its common insertion radius r_1, size $\bar{\mathsf{d}}$ and polymer sum $\mathsf{t}_\times$. It bounds the higher mixed jets of the exponent as well as the separate conditional means and gives

$$|\langle J_0;\dots;J_r\rangle| \le \mathrm{op}_{r+1}\prod_a \omega_t(J_a) + 9\cdot 2^{r+1}(r+2)\bar{\mathsf{d}}\,\mathsf{t}_\times r_1^{-r} + 4\,\mathrm{op}'_{r+1}\mathsf{R}_t\prod_a \sup|J_a|, \qquad \mathrm{op}_n = \sum_\pi (|\pi|-1)!,$$

π over the partitions of n points, with op'_n as in [7, Prop. 3.15]; ω_t is the oscillation of the conditional mean (Lem. 5.10) and R_t the weight of the histories with large fields. After allowing the carrier collars, take $\mathsf{d}_\times \ge c\min\{\mathsf{d},\delta\}$. The hypothesis $\mathsf{d}L^u \ge K^2$ gives $\mathsf{d}_\times L^u \ge cK^2$ for all sufficiently large K, uniformly in d. The prefactor estimates in the proof of [7, Prop. 3.24] bound the second term by a polynomial in K times $L^{AK}e^{-cK^2}$; thus the condition of [7, Prop. 3.15](a) holds and this term is below any prescribed L^{-BK}. The large-field term is controlled there separately. Six of the seven pieces of Rm_K are small by power counting, the seventh is the cubic one, of oscillation $K^pL^{-c'_0K}$. The principal part of Δ_K is null (Lem. 5.9(d)): on the backgrounds of the level $K - \lceil\frac{K}{14}\rceil$ its value is at most $CK^{2\mathfrak{p}_0}L^{\frac72\lceil K/14\rceil-\frac{K}{2}}\mathsf{b}\sup|\nabla\eta|$, what is born below that level is born at a small coupling and is forgotten, and a conditional mean is never larger than a supremum.

Each crude tensor entry costs L^{4u}. At $m=1$: by (a)–(c) of Lem. 5.9, $\mathfrak{K}^{\mathrm{v}}[\beta\eta] = \frac14\mathfrak{N}[\mathsf{k}_\eta] + \beta\mathfrak{K}^\sharp[\eta] - \sum_v \mathfrak{J}_v[\mathsf{k}_\eta] + \frac14 c_{\mathsf{k}_\eta}$, so that F_η consists of six of the seven pieces of $\mathrm{Rm}_K[\eta]$ and a number. The cubic piece has the oscillation $K^pL^{-K/28}\|\eta\|_{C^2}$ by Lem. 5.10(b) at $\tau=\frac14$, where $c'_0 = \frac{1}{28}$ for $\phi \le \frac{1}{84}$; the five pieces $\mathfrak{J}_v$, of the orders $2,2,2$ with weights of the order of $\|\eta\|_{C^2}$ and of the orders $3,4$ with weights at most $\beta\|\eta\|_{C^2}$, have oscillations at most $K^{3\mathfrak{p}_0}L^{4u-K/2}\|\eta\|_{C^2}$ by Lem. 5.10(a); so no factor L^{3u} enters the bound of F_η; in the bounds of the slots of the kind of η it enters through Δ_K alone.

The means of $\mathfrak{K}^{\mathrm{v}}[\beta\eta]$ and of $\mathfrak{N}[\mathsf{k}_\eta]$ on $T_{\ell'}$ are 0 (Lem. 5.9(a), (b)), so is that of F_η, and $s(\mathfrak{N}[\mathsf{k}_\eta]) = 0$ in every state s by Lem. 5.9(a). □

Lemma 5.13 (From sup norms to an integrable density)**.** Let $H \ge 0$ be self-adjoint, $T_t = e^{-tH}$, and for every $\eta > 0$ let $\rho \mapsto S_\eta[\rho]$ be linear from $C_c((-\eta,\eta))$ to the bounded operators, with $\|S_\eta[\rho]\| \le A_J \sup|\rho|$ for $\mathrm{supp}\,\rho$ in a compact $J \subset (-\eta,\eta)$, and $S_\eta[\rho(\cdot - c)] = T_{\delta+c}S_{\eta-\delta}[\rho]T_{\delta-c}$ for $|c| \le \delta < \eta$. Then there is a unique family of bounded operators $B(s,s')$, $s,s' > 0$, with $T_tB(s,s')T_{t'} = B(s+t,s'+t')$ and $S_\eta[\rho] = \int \rho(c)B(\eta+c,\eta-c)\,\mathrm{d}c$; for $0<\gamma'<\eta$ and $\mathrm{supp}\,\rho \subset [-(\eta-\gamma'),\eta-\gamma']$, $\|S_\eta[\rho]\| \le \|S_{\gamma'/2}[\mathrm{tent}_{\gamma'/8}]\|\,\|\rho\|_{L^1}$; and $B(s,s')^* = B(s',s)$ if $S_\eta[\rho]^* = S_\eta[\bar\rho(-\cdot)]$. The same holds in the space directions for commuting momenta P with $H \ge |P_k|$ in the direction k. ■

Proof sketch. For G holomorphic and bounded on the strip $\{|\mathrm{Re}\,z| < h'\}$, $h' > 2h$, integrating $\Gamma(z)\sin^{-2}(\pi z/2h)$, $\Gamma' = G$, over the boundary of the strip gives

$$G(0) = \int \mu_2(\mathrm{d}s)\int \mathrm{tent}_{2h}(c)\,G(c+is)\,\mathrm{d}c, \qquad \mu_2 = \mu_1 * \mu_1, \quad \mu_1(\mathrm{d}s) = \tfrac{\pi}{4h}\mathrm{sech}^2(\tfrac{\pi s}{2h})\,\mathrm{d}s.$$

Thus an average in real time inverts a tent smearing in Euclidean time. Apply it to $z \mapsto \langle T_{\delta+\bar z}\psi_2, S[\rho]T_{\delta-z}\psi_1\rangle$ for vectors ψ_1,ψ_2, holomorphic by the covariance, and use $\rho_0 * \rho_1 = \rho_1 * \rho_0$: the μ_2-average of $e^{-isH}S_{\gamma'/2}[\mathrm{tent}]e^{isH}$ is the operator $\mathbb{B}$ of the density, $S_\eta[\rho] = \int \rho(c)T_{\eta-\gamma'/2+c}\mathbb{B}T_{\eta-\gamma'/2-c}\,\mathrm{d}c$. In space, $G(0) = \int h_{d'}(x)\frac12[G(x+id') + G(x-id')]\,\mathrm{d}x$ with $h_{d'} = (2d')^{-1}\mathrm{sech}(\pi x/2d')$, with the imaginary translations that $H \ge |P|$ allows. □

Lemma 5.14 (Functions of the block field)**.** For a history h let $\mathfrak{w}_h \ge 0$ be its weight in the law of the field and of the history and $\mathsf{t}(h) = \int \mathfrak{w}_h\rho_0\,\mathrm{d}U$ its mass, so that $\sum_h \mathsf{t}(h) = Z$ (equations YM3_I:(646)–(648) and Prop. YM3_I:44.1).

(i) In the setting of Lem. 5.10, let $1 \le t \le K-1$ be a level up to which the chain is formed (a top level of the chain at the label $\mathfrak{b}'$), h_t the history without large-field regions of the level t, $\mathrm{In}_t = \{V : \varpi(V) < 2\varepsilon(x_t)\} \subset \mathrm{Cut}_t$ the inner ball of the level t (the definitions preceding Lem. YM3_I:113.10; $\varpi(V)$ is the largest $|\partial V(p') - \mathbf{1}|$ over the plaquettes p' of the level t), and $\mathfrak{w}_t^{\mathrm{sf}}$ the weight of the level t of the small-field chain with its number R_t (the definitions preceding Prop. YM3_I:115.2, the definition preceding Prop. YM3_I:115.2, YM3_I:(2187)): $\mathfrak{w}_t^{\mathrm{sf}}(U) = \mathbf{1}_{\mathrm{In}_t}(\mathcal{M}^t U) \prod_{i<t} [\mathbf{1}_{\mathrm{Cut}_i} \mathfrak{G}_i[x_i] \chi_{x_i}^{\mathrm{sf}}](\mathcal{M}^i U)$, with the gauge-fixing factors $\mathfrak{G}_i[x_i]$ and the fluctuation tests $\chi_{x_i}^{\mathrm{sf}}$ among the construction results, so that $\mathfrak{w}_t^{\mathrm{sf}} = (\mathbf{1}_{\mathrm{In}_t} \circ \mathcal{M}^t)\, \mathsf{p}_t$ with $\mathsf{p}_t(U) := \mathbf{1}_{\mathrm{Cut}_t}(\mathcal{M}^t U) \prod_{i<t} [\mathbf{1}_{\mathrm{Cut}_i} \mathfrak{G}_i[x_i] \chi_{x_i}^{\mathrm{sf}}](\mathcal{M}^i U)$.

The measure ν_t of the level t of the chain is $(\mathcal{M}^t)_*(\mathsf{p}_t \rho_0 \,\mathrm{d}U)$ (Prop. YM3_I:114.2, the measure recursion in Prop. YM3_I:114.2), and $Q_{t\leftarrow 0}$ is the conditional mean given $\mathcal{M}^t U$ under $\mathsf{p}_t \rho_0 \,\mathrm{d}U$; the tower of Lem. 5.10 is read with this p_t. By Prop. YM3_I:115.2, $\mathfrak{w}_t^{\mathrm{sf}}$ is a bounded Borel function with values in $[0,\infty)$, $(\mathcal{M}^t)_*(\mathfrak{w}_t^{\mathrm{sf}} \rho_0 \,\mathrm{d}U) = \mathbf{1}_{\mathrm{In}_t} \nu_t$ YM3_I:(2188), $0 \le \mathfrak{w}_{h_t} \le \mathfrak{w}_t^{\mathrm{sf}}$ at every point YM3_I:(2189), and $0 \le Z - \int \mathfrak{w}_t^{\mathrm{sf}} \rho_0 \,\mathrm{d}U \le Z - \mathfrak{t}(h_t) \le \mathsf{R}_t Z$ YM3_I:(2190): R_t bounds the weight of the histories with a large field up to the level t.

For every bounded real Borel gauge invariant Y, real a_0 and $\epsilon \ge 0$ with $|Q_{t\leftarrow 0} Y - a_0| \le \epsilon\, \nu_t$-almost everywhere on In_t, and every bounded Borel function ψ of the block fields of the level t, $|\langle Y; \psi \circ \mathcal{M}^t \rangle| \le 2[\epsilon + 2(\sup|Y| + |a_0|)\mathsf{R}_t] \sup|\psi|$; if besides $\mathsf{R}_t \le \frac{1}{2}$, then $|a_0| \le \sup|Y| + \epsilon$.

(ii) In the setting of Lem. 5.6 at $0 < g \le \gamma$ with $g^{-2} \ge x_{\mathrm{J}}$ (a first-passage datum $(K,\beta) \in \mathfrak{D}(g)$, a side $\ell \in \mathfrak{L}(g)$, a depth s and the working level k), let $2 \le T \le k$ be a level, h_T the history without large-field regions of the level T, $\mathsf{m}_T = \tau_{h_T}[\rho_0]$ (Lem. 5.6(a)), and $W_T < 1$ the weight of a first creation of a large field at a level $\le T$, so that $Z - \mathfrak{t}(h_T) = W_T Z$. A dominated local source F_2 acts on a dominated local source F_1 at the sup distance d', with $L^{K-T}\mathsf{d}' \ge 3M$ and $\mathsf{h}_2(K-T) \le 1$ (the conditions of Lem. 5.6(e) at $t = T$), through a bounded function of the block field of the level T: let r_2 be the real part of the bounded density $D_{T,2}$ of $\tau_{h_T}[F_2\rho_0]$ with respect to m_T, c real, E the essential supremum of $|r_2 - c|$ for m_T, and $\psi = (r_2 - c)\mathbf{1}\{|r_2 - c| \le E\}$, a bounded real Borel function with $\sup|\psi| \le E$ and $\psi = r_2 - c$ m_T-almost everywhere; then

$$|\langle F_1; F_2\rangle - \langle F_1; \psi \circ \mathcal{M}^T\rangle| \le C_2 e^{3-\mathsf{D}_T/M} y_1 y_2 + 2\sup|F_1|(\sup|F_2| + \sup|\psi| + |c|)W_T,$$

with $y_a \le 4\ell^3 L^{4(K-T)}$ times the sizes of the sources and $\mathsf{D}_T \ge L^{K-T}\mathsf{d}'$ the least sup distance of the carriers in level-T units; at $c = \mathrm{Re}\, c_2$, c_2 the constant of F_2 at the level T, $E \le C_1 y_2$ and $|c| \le \sup|F_2| + C_1 y_2$. The display is additive in F_1.

■

Proof sketch. The law of the field and of the history, Prop. YM3_I:44.1 with the measures in equation YM3_I:(647), gives

$$\mathbf{T}^k(\Psi\rho_0 \,\mathrm{d}U) = \sum_h \tau_h[\Psi\rho_0],$$
$$\int F \,\mathrm{d}\tau_h[\Psi\rho_0] = \int F(\mathcal{M}^k U)\, \mathfrak{w}_h(U) \Psi(U) \rho_0(U) \,\mathrm{d}U$$

for every bounded gauge invariant Ψ, when $\varepsilon_j \le \varepsilon_\flat = c_2^{\mathrm{SO}}$ for $j < k$; hence $Z\langle \Psi \cdot \psi \circ \mathcal{M}^k\rangle = \sum_h \int \psi \,\mathrm{d}\tau_h[\Psi\rho_0]$, and, as $\mathfrak{w}_h \ge 0$ and $\sum_h \mathfrak{t}(h) = Z$,

$$\left| Z\langle \Psi \cdot \psi \circ \mathcal{M}^k\rangle - \int \psi \,\mathrm{d}\tau_{h_k}[\Psi\rho_0] \right| \le \sup|\Psi| \sup|\psi| \,(Z - \mathfrak{t}(h_k)).$$

(i) At the level t replace $\mathfrak{w}_{h_t}$ by $\mathfrak{w}_t^{\mathrm{sf}}$: the difference $\mathfrak{w}_t^{\mathrm{sf}} - \mathfrak{w}_{h_t}$ is ≥ 0 at every point, of the mass $\int \mathfrak{w}_t^{\mathrm{sf}} \rho_0 \,\mathrm{d}U - \mathfrak{t}(h_t) \le Z - \mathfrak{t}(h_t) \le \mathsf{R}_t Z$; with the histories other than h_t this gives $|Z\langle \Psi \cdot \psi \circ \mathcal{M}^t\rangle - \int \Psi\, \psi(\mathcal{M}^t U)\, \mathfrak{w}_t^{\mathrm{sf}} \rho_0 \,\mathrm{d}U| \le 2\sup|\Psi| \sup|\psi| \mathsf{R}_t Z$, the factor 2 counting the two parts.

By the tower property, $\mathfrak{w}_t^{\mathrm{sf}} = (\mathbf{1}_{\mathrm{In}_t} \circ \mathcal{M}^t)\mathsf{p}_t$ and $\mathrm{In}_t \subset \mathrm{Cut}_t$, $\int \psi(\mathcal{M}^t U)\,(Y - a_0)\,\mathfrak{w}_t^{\mathrm{sf}} \rho_0 \,\mathrm{d}U = \int_{\mathrm{In}_t} \psi\,(Q_{t\leftarrow 0} Y - a_0)\,\mathrm{d}\nu_t$, of a modulus at most $\epsilon \int_{\mathrm{In}_t} |\psi|\,\mathrm{d}\nu_t \le \epsilon \sup|\psi|\, Z$, since $\nu_t \ge 0$ and $\nu_t(\mathrm{In}_t) = \int \mathfrak{w}_t^{\mathrm{sf}} \rho_0 \,\mathrm{d}U \le Z$. Applied at $\Psi = Y - a_0$, for ψ and for 1, with $\langle Y; \psi \circ \mathcal{M}^t\rangle = \langle (Y - a_0)\,\psi \circ \mathcal{M}^t\rangle - \langle Y - a_0\rangle\langle \psi \circ \mathcal{M}^t\rangle$, this is the bound of (i). If $\mathsf{R}_t \le \frac{1}{2}$: $|Q_{t\leftarrow 0} Y| \le \sup|Y|$ ν_t-almost everywhere and $\nu_t(\mathrm{In}_t) \ge (1 - \mathsf{R}_t) Z > 0$, so In_t holds points at which $|a_0| \le |Q_{t\leftarrow 0} Y| + \epsilon \le \sup|Y| + \epsilon$. (ii) By the first display at the level T, where $Z - \mathsf{t}(h_T) = W_T Z$, the means $\langle F_1 F_2\rangle$, $\langle F_1\,\psi \circ \mathcal{M}^T\rangle$, $\langle F_1\rangle$, $\langle F_2\rangle$ and $\langle \psi \circ \mathcal{M}^T\rangle$ are the integrals against $Z^{-1}\mathsf{m}_T$ of the bounded densities of these functions at h_T, up to the weight W_T times their suprema, and $Z^{-1}\mathsf{m}_T(1) = 1 - W_T$.

The densities of F_1, F_2 and $F_1 F_2$ are $D_{T,1}$, $D_{T,2}$ and $D_{T,1} D_{T,2} + D_{T,12}$; the real part r_2 of $D_{T,2}$ is again a density of the real source F_2, two bounded densities of one function agree m_T-almost everywhere, and $r_2 = \psi + c$ there; so the density of $F_1 F_2$ is $D_{T,12} + D_{T,1}(\psi + c)$ almost everywhere, and subtracting, $\langle F_1; F_2\rangle - \langle F_1; \psi \circ \mathcal{M}^T\rangle$ is $Z^{-1}\int D_{T,12}\,\mathrm{d}\mathsf{m}_T$, at most $C_2 e^{3 - \mathsf{D}_T/M} y_1 y_2$ by Lem. 5.6(e), plus two brackets of errors that together are at most $2\sup|F_1|(\sup|F_2| + \sup|\psi| + |c|)W_T$. At $c = \mathrm{Re}\, c_2$: $|r_2 - c| \le |D_{T,2} - c_2| \le C_1 y_2$ on the small block fields of the level T, which carry m_T (Lem. 5.6(e)), and $|r_2| \le \sup|F_2|$ there.

Every member but $D_{T,12}$ is formed with F_1 linearly, and $D_{T,12}$ is additive in F_1. □

Lemma 5.15 (One entry against separated sources)**.** Let $0 < g \le \gamma$, $\ell \in \mathfrak{L}(g)$, $n \ge 1$, $\mathsf{d} > 0$, $0 < c_* < \frac{1}{2}$, and let $J = \sum_\iota \varkappa_\iota j_\iota$ be an entry of the order $m \ge 2$ (Lem. 5.10) of the type $(m, c, \hat{c}, N)$: at complex configurations with $|\mathbf{U}(b)| \le 2$ at all the bonds of its box, $|j_\iota| \le c\, r^m$ where the plaquette variables of the box are within $r \le 1$ of $\mathbf{1}$, and $|j_\iota| \le \hat{c}$ everywhere; and at most N indices have boxes with one lowest site. There are C, K_1, depending on g, ℓ, n, d, c_* and the type, such that for $K \ge K_1$, every label $\mathsf{b}' \in [\gamma^{-2}, X_g + \bar{B}]$ (so at every first-passage datum $(K, \beta) \in \mathfrak{D}(g)$, and at $\beta = L^K \mathsf{b}'$ for every $\mathsf{b}' \in I(g)$) and all real Lipschitz f_q, if the spread of the $n + 1$ sets formed by the carrier of J and the $\mathrm{supp}\, f_q$ is $\ge \mathsf{d}$, as it is when two of them are at sup distance $\ge \mathsf{d}$,

$$|\langle J; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\rangle_{K,\ell,L^K\mathsf{b}'}| \le C|\varkappa|_\infty L^{-\mathsf{g}_m K} K^{(2n+1)\mathfrak{p}_0} L^{(\frac{1}{2} - c_*)K} \prod_q \|f_q\|_\sharp, \qquad \begin{array}{l} \mathsf{g}_2 = 1,\ \mathsf{g}_m \\ \qquad = \frac{3}{2}\ (m \ge 3). \end{array}$$

On all the sides at once: let besides $R > 0$ and $\ell' \in \mathfrak{L}(g)$ with $\ell' \ge 2R + 2\mathsf{d}$, and let the weights $\varkappa_\iota$ be real and the densities real on the G-valued configurations. There are C', K_1', depending on g, ℓ', n, d, R, c_* and the type, and on no side, such that for all $(K, \beta) \in \mathfrak{D}(g)$ with $K \ge K_1'$, every side $\ell \ge \ell'$ in $\mathfrak{L}(g)$, every entry of the type whose boxes lie in one sup ball of radius R, and all real Lipschitz f_q on $\mathbb{R}^3$ that vanish outside that ball, read on T_ℓ, with the carrier of the entry at sup distance $\ge \mathsf{d}$ from every $\mathrm{supp}\, f_q$ (the sources lie anywhere among themselves),

$$|\langle J; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\rangle_{K,\ell,\beta}| \le C'|\varkappa|_\infty L^{-\mathsf{g}_m K} K^{1+\mathfrak{p}_0} L^{(\frac{1}{2} - c_*)K} \prod_q \|f_q\|_\sharp;$$

the same holds under every periodic Gibbs state at (K, β) and under every weak limit point of the torus laws as $\ell \to \infty$ in $\mathfrak{L}(g)$, and, with the further factor 2^{n+1}, for complex weights and complex Lipschitz f_q. ■

Proof sketch. Put $\phi = (\frac{1}{2} - c_*)/(3n + 4)$, $u = \lceil \phi K \rceil$, $t = K - u$. Since $M\, R(X_g + \bar{B}) \le L^m$ and R increases, the frame of [7, Def. B.14] holds for the chain along $x_j = L^{K-j}\mathsf{b}'$ at every label of $[\gamma^{-2}, X_g + \bar{B}]$, and from a K_1 on t is a top level of it with $\mathsf{R}_t \le \frac{1}{2}$ and $8a \le \mathsf{d}$.

The cut. Let $D \ge \mathsf{d}$ be the spread. For every site y_0 of the carrier there is a source q with $\mathrm{dist}_\infty(y_0, \mathrm{supp}\, f_q) \ge \frac{1}{2}D$: otherwise there are points $y_q \in \mathrm{supp}\, f_q$ with $|y_0 - y_q|_\infty < \frac{1}{2}D$, and $y_0, y_1, \dots, y_n$ are pairwise closer than D, against the definition of the spread. Give each index ι of J the least such q for the lowest site of its box; this cuts J into at most n entries J_q of the same type, the carrier of J_q being at sup distance $\ge \frac{1}{2}D - 2a \ge \frac{1}{4}\mathsf{d}$ from $\mathrm{supp}\, f_q$, while the other sources lie

anywhere. The cumulant is linear in its first entry and symmetric in the others; in the cumulant of J_q the far source is f_q.

The chain with the entry and a complex source. The chain of Lem. 5.10(a) is formed with the entry sJ_q and with the source $w = \sum_q \zeta_q f_q$ at once, for $|s| < r_J$ and

$$\|w\|_\flat = \max\{\sup|w|, \frac{40}{\sqrt{3}} M \mathrm{Lip}_\infty w\} < 2\varrho$$

(the polydisc of the sources of Thms. YM3_I:AJ and YM3_I:AD): its families $E^{(i)}[w,s]$ are pointed families of the levels i, holomorphic in (w,s) and local, the member at a domain X reading w on the plaquettes of X alone, and s only if a box of the entry lies in X; with them $(\mathcal{M}^t)_*(\mathfrak{w}_t^{\mathrm{sf}} e^{sJ_q}\rho_0^w \,\mathrm{d}U) = \exp[\sum_{i\le t} c_i(w,s)]\mathbf{1}_{\mathrm{In}_t} e^{\Phi_t^{w,s}} dV$, $\rho_0^w = \rho_0 e^{\mathcal{O}(w)}$, and on Cut_t

$$|\Phi_t^{w,s} - \Phi_t^{w,s'}| \le |s-s'|\,\mathsf{o}_t, \qquad |\Phi_t^{w,0} - \Phi_t^{0,0}| \le N_t \mathsf{C}_t \|w\|_\flat,$$

wherever the sources lie, with $N_t = \ell^3 L^{3u}$ the number of the sites of the level t and $\mathsf{C}_t \le CK^{2\mathfrak{p}_0}$. The entry enters each step as one more local term, as in Lem. 5.10(a); bounding the radius $\mathsf{e}_t^{\mathrm{c}} \le 1$ of the field cut, raised to the order of the entry, by $\mathsf{e}_t^{\mathrm{c}}$ itself, the size of the entry at the level t and what the steps made of it give $\mathsf{o}_t \le CK^{\mathfrak{p}_0} L^{4u-\mathfrak{g}_m K}|\varkappa|_\infty$.

The bound. Let $T(s,\zeta) = \int e^{sJ_q + \sum_{q'}\zeta_{q'}\mathcal{O}(f_{q'})}\, \mathfrak{w}_t^{\mathrm{sf}}\rho_0 \,\mathrm{d}U$. The cumulant under the positive measure $\mathfrak{w}_t^{\mathrm{sf}}\rho_0\,\mathrm{d}U$ is $\partial_s\partial_{\zeta_1}\cdots\partial_{\zeta_n}\log T(0,0)$, and it differs from the cumulant under the measure of the torus by at most $4\,\mathrm{op}_{n+1}\mathsf{R}_t \sup|J_q| \prod_{q'} 6\ell^3 L^{4K} \sup|f_{q'}|$ (Prop. YM3_I:115.2, its cumulant bound), which is smaller than every power of L^{-K}. By the identity of measures, $\partial_s \log T(0,\zeta) = \partial_s \mathsf{C}(0,\zeta) + \langle \partial_s \mathsf{D}\rangle_\zeta$, where $\mathsf{C}(s,\zeta) = \sum_{i\le t}(c_i(w,s) - c_i(w,0))$ collects the values of the families at the trivial triple, $\mathsf{D} = \Phi_t^{w,s} - \Phi_t^{w,0}$, and $\langle\cdot\rangle_\zeta$ is the mean under $e^{\Phi_t^{w,0} - \Phi_t^{0,0}}$ times the image of $\mathfrak{w}_t^{\mathrm{sf}}\rho_0\,\mathrm{d}U$, normalised.

On the polydisc $|\zeta_{q'}| < (2nN_t\mathsf{C}_t\|f_{q'}\|_\sharp)^{-1}$ the exponent $\Phi_t^{w,0} - \Phi_t^{0,0}$ is at most $\frac12$, so $|\langle G\rangle_\zeta| \le 3.1\sup|G|$, and the second member is at most $3.1\,\mathsf{o}_t$. In the first, a member at a domain that does not hold both a box of J_q and a plaquette of the far source f_q does not depend on ζ_q (locality), and has no mixed derivative; the members at the domains X of the levels i that hold both have $d_i(X) \ge (\frac14 \mathsf{d} - a)L^{K-i}/M - 2$, and with the count of the domains their sum is at most a power of L^K times $e^{-\mathsf{d}L^u/8M}$. Cauchy's estimates in the n sources on this polydisc cost $\prod_{q'}(nCK^{2\mathfrak{p}_0}L^{3u}\|f_{q'}\|_\sharp)$.

Together, $K^{(2n+1)\mathfrak{p}_0} L^{(3n+4)u - \mathfrak{g}_m K}$, and $(3n+4)u \le (\frac12 - c_*)K + 3n + 4$.

On all the sides. Let $\mathsf{l} = L^{-i}$ be the largest power of L with $\mathsf{l} \le \frac12 \mathsf{d}$; then $\ell' \ge 2R + 4\mathsf{l}$. On $T_{\ell'}$ and on every T_ℓ, $\ell \ge \ell'$, a board of side l puts the boxes of the entry and the supports of the sources, which lie in one sup ball of radius R inside one period box of side ℓ', in different cells. The cumulant on T_ℓ is a mixed derivative at 0 of a quotient whose coefficients are means on T_ℓ of products of functions of different cells of that period box; the chessboard bound (Lem. 5.8) bounds each by a product of chessboard seminorms on $T_{\ell'}$, at the price $e^{\mathsf{l}^3(p_{\ell'} - p_\ell)} \le e^{G_g \mathsf{l}^3}$ for a cell; this ratio $Z_{K,\ell',\beta}^{1/N'} / Z_{K,\ell,\beta}^{1/N_\ell}$ of the plain partition functions per cell, N' and N_ℓ being the numbers of the cells of $T_{\ell'}$ and of T_ℓ, is the one number that depends on ℓ, and its bound is free of ℓ.

After a subtraction of Wick type in the functions of the cells, which removes the cumulants of the families inside one cell, the chessboard seminorm of the function of the cell of the entry, the mean on $T_{\ell'}$ of the product of its mirror copies, is by the moment–cumulant formula a sum over the partitions of the copies all of whose blocks join different cells; each such cumulant is small by the bound on one torus, read at a free depth u with $C_{\mathrm{u}} K^{1/(2\mathfrak{p}_0)} \le u \le \frac12 K$, $C_{\mathrm{u}} \ge 1$ a number formed with the data and the type: a factor $|\varkappa|_\infty K^{1+\mathfrak{p}_0} L^{4u - \mathfrak{g}_m K}$ for each copy of the entry and a factor of Cauchy's estimate for each copy of a source.

The number of the cells that the ball of radius R meets, at most $(2R/\mathsf{l} + 3)^3$, enters C', and at $u = \lceil \frac14(\frac12 - c_*)K \rceil$ the bound follows. Every periodic Gibbs state is dominated by the torus

measures through its conditional laws (Prop. YM3_I:91.9), and the weak limit points of the torus laws inherit the bound; for complex data the real and imaginary parts are taken apart. □

Lemma 5.16 (The cubic remainder uniformly in the side)**.** Let $0 < g \le \gamma$, $\ell' \in \mathfrak{L}(g)$, $n \ge 2$, $\mathsf{d} > 0$, $c < \frac{1}{14}$, and let Q be a cube of side $\frac{1}{4}\ell'$. There are C, p, K_0, depending on $g, \ell', n, \mathsf{d}, c$ and independent of the larger side ℓ, such that for all $\ell \ge \ell'$ in $\mathfrak{L}(g)$, all $(K,\beta) \in \mathfrak{D}(g)$ with $K \ge K_0$, all Lipschitz fields η and real Lipschitz f_q carried by Q with the supports of the f_q pairwise $\ge \mathsf{d}$ apart,

$$|\mathfrak{U}^\sharp[\eta](f)_{T_\ell}| + |\mathfrak{U}^\sharp[\eta](f)_{T_\ell} - \mathfrak{U}^\sharp[\eta](f)_{T_{\ell'}}| \le C K^p L^{-cK} \mathsf{H}(\eta) \prod_q \|f_q\|_\sharp;$$

the finitely many sides of $\mathfrak{L}(g)$ below ℓ' are covered by Lem. 5.11. ■

Proof sketch. Every label of $\mathfrak{D}(g)$ lies in $[\gamma^{-2}, X_g + \bar{B}]$, and $MR(X_g + \bar{B}) \le L^m$ on every side of $\mathfrak{L}(g)$; so the frame of [7, Def. B.14] holds for the chain along $x_j = L^{K-j}\mathsf{b}$ at the label of every first-passage datum, and Lem. 5.10(b) holds with one triple of numbers for all of them, also with real sources $\sum_i x_i f_i$ in its polydisc. The chain is formed with several entries and complex sources as well, as in the proof of Lem. 5.15 (Lem. 5.10(a), polarisation and locality in each entry).

The reduction. With $\mathcal{S}[\Xi] = a^{-2}\sum_p \Xi^{\lambda(p)}(x(p))\,\mathfrak{r}^\sharp_{\lambda(p)}(p)$ for a field Ξ we have $\beta\mathfrak{K}^\sharp[\eta] = \mathcal{S}[\mathsf{b}\eta]$ and $a^{-1}\mathfrak{K}^\sharp[f_i\eta] = \mathcal{S}[f_i\eta]$, so $\mathfrak{U}^\sharp[\eta](f) = -\langle \mathcal{S}[\mathsf{b}\eta]; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\rangle + \sum_i \langle \mathcal{S}[f_i\eta]; (\mathcal{O}(f_l))_{l\ne i}\rangle$, with $\mathsf{H}(\mathsf{b}\eta) \le (X_g + \bar{B})\mathsf{H}(\eta)$ and $\mathsf{H}(f_i\eta) \le 2\|f_i\|_\sharp \mathsf{H}(\eta)$. It suffices therefore to compare $G_\ell = \langle \mathcal{S}[\Xi]; \mathcal{O}(h_1); \dots; \mathcal{O}(h_{n'})\rangle_{K,\ell,\beta}$ with $G_{\ell'}$, $n' \in \{n-1, n\}$, $n' \ge 1$, for Lipschitz fields Ξ and real Lipschitz h_q carried by Q whose supports are pairwise $\ge \mathsf{d}$ apart, nothing being asked of Ξ relative to them.

The fold with one factor for each copy. Place Q in the middle of one cell of the board of side $\mathsf{l} = \frac{1}{2}\ell'$; it lies at sup distance $\ge \frac{1}{8}\ell' \ge 1$ from the walls. On every T_ℓ, $\ell \ge \ell'$, $\mathcal{S}[\Xi]$ and the $\mathcal{O}(h_q)$ are the realisations of one model function each, and their mirror copies in the eight cells of $T_{\ell'}$ are again a cubic entry $\mathcal{S}[\Xi^\kappa]$, $\mathsf{H}(\Xi^\kappa) \le \mathsf{H}(\Xi)$, and sources $\mathcal{O}(h_{(q,\kappa)})$ with $\|h_{(q,\kappa)}\|_\sharp = \|h_q\|_\sharp$; copies in different cells are ≥ 1 apart. The chessboard bound (Lem. 5.8) carries the cumulant on T_ℓ to the mean on $T_{\ell'}$ of the product of the eight mirror copies, at the price $e^{G_g \mathsf{l}^3}$ for each function, and by the moment–cumulant formula of the copies (an identity of formal power series, as in the proof of Thm. M(g)(ii)) the difference $G_\ell - G_{\ell'}$ reads the joint cumulants on $T_{\ell'}$ of the families B of copies that join different cells alone, the families inside one cell entering nowhere. If each of these is at most $\Gamma\prod_{(i,\kappa)\in B}\omega_i$, where $\Gamma \ge 1$, with one factor ω_0 for each copy of the entry and one factor ω_q for each copy of the source h_q, then

$$|G_\ell - G_{\ell'}| \le \mathsf{c}_{n'}\, e^{(n'+1)G_g\mathsf{l}^3}\, \Gamma^{(n'+1)/2} \prod_{i=0}^{n'} \omega_i,$$

To see the exponent, use the within-cell normalization and logarithm identities of Lem. 4.26. In a fold of q entries, every remaining block joins cells and has at least two members; a partition of the $8q$ copies has at most $4q$ such blocks. Its bound is at most $\Gamma^{4q}\prod_i \omega_i^8$, and the eighth root gives $\Gamma^{q/2}\prod_i \omega_i$. The subsequent logarithm partitions preserve this exponent, since their subset sizes sum to $q = n' + 1$. This proves the display, including the stated pressure factor. Here $\mathsf{c}_{n'}$ is a pure number. With $u = \lceil \phi K \rceil$, $t = K - u$, the factors $\omega_0 = K^p L^{-c_1 K}\mathsf{H}(\Xi)$ and $\omega_q = K^{2\mathfrak{p}_0} L^{3u}\|h_q\|_\sharp$ serve, in three cases. (i) No copy of the entry: the family holds two copies of sources in different cells, ≥ 1 apart, and the bound eq. 5.i on $T_{\ell'}$ at separated supports applies. (ii) One copy $\mathcal{S}[\Xi^\kappa]$: moving the weight of the entry with the sources writes the cumulant as $-\mathfrak{U}^\sharp[\Xi^\kappa/\mathsf{b}](h)$ plus, when two or more sources are present, the cubic remainders $-\mathfrak{U}^\sharp[h_\alpha \Xi^\kappa/\mathsf{b}^2]((h_{\alpha'})_{\alpha'\ne\alpha})$ (the supports of the h's being disjoint), and with one source the contact member is a mean of $\mathfrak{K}^\sharp$, which vanishes; a source of another cell is ≥ 1 from the entry, and $h_\alpha\Xi^\kappa$ is carried by $\operatorname{supp} h_\alpha$, at sup distance $\ge \min\{\mathsf{d}, 1\}$ from the other supports: the spread condition holds, and Lem. 5.11 on $T_{\ell'}$ at the rate c_1 applies. (iii)

Two to eight copies of the entry, in different cells, the sources anywhere: the cumulant is a mixed derivative of the logarithm of the generating function of the chain with these entries and the complex source $w_\zeta = \sum \zeta_\alpha h_\alpha$, at the level t. For case (iii), retain the higher mixed jets as in the proof of [7, Prop. 3.15](b). [7, Prop. 3.13](v) permits the fixed complex source w and the several insertion parameters $\mathbf{s}$ together. Write their small-field generating function as $T_w(\mathbf{s}) = T_w(0) e^{\mathsf{C}_w(\mathbf{s})} \langle e^{\mathsf{D}_{w,\mathbf{s}}} \rangle_w$, with the normalized block-field functional $\langle\cdot\rangle_w$. For q copies of the entry and I with $|I| \geq 2$, locality and the Cauchy estimate in that proof give $\max\{|\partial_I \mathsf{C}_w(0)|, \sup|\partial_I \mathsf{D}_{w,\mathbf{s}}|_{\mathbf{s}=0}\} \leq 2E_0 \mathsf{t}'_\times \prod_{a\in I}(q/r_a)$, where r_a are the insertion radii and $\mathsf{t}'_\times$ is the polymer sum of that proposition. Copies of the entry lie in distinct cells with a fixed positive margin. Their inverse radii are at most $CL^{2K}\mathsf{H}(\Xi^a)$, whereas $\mathsf{t}'_\times \leq CK^p L^{AK} e^{-cL^u}$. Hence the mixed-jet bound is below $\prod_{a\in I} \epsilon_a$ for large K, uniformly on a fixed fraction of the displayed, K-dependent source polydisc. Zero entries are omitted. At real sources the block-field functional is positive; at complex sources it has norm at most 3.1 on the smaller polydisc, by the exponent bound in the proof of Lem. 5.15. The moment and logarithm partition formulas in [7, Prop. 3.15](b) therefore apply in both cases, with finite constants adjusted in the complex case. At a real source x in the polydisc of the radii $(CK^{2\mathfrak{p}_0} L^{3u} \|h_\alpha\|_\sharp)^{-1}$, Lem. 5.10(b), applied to the field $\Xi^\kappa/(\mathsf{b} - w_x)$, makes the conditional mean of each copy given the block field of the level t a number up to $\epsilon_a = C\mathsf{H}(\Xi) K^{p_2} L^{-c'_0 K}$. Subtracting these constants from the first jets, and using the higher mixed-jet bound above, the same partition formulas bound the cumulant by a constant times $\prod_a \epsilon_a$. At a complex source, the norm bound for the normalized functional and the mixed-jet estimate instead give a constant times the product of the crude bounds $\hat{\mathsf{o}}_a = C\mathsf{H}(\Xi) K^{\mathfrak{p}_0} L^{4u+\frac{1}{2}K}$ of the oscillations of the cubic entries, of the order 3 and of the weight a^{-2}. The estimate with two constants for a function holomorphic on a polydisc of the radii r_α, bounded by B there and by ϵ at its real points, $|\partial_{\zeta_1} \cdots \partial_{\zeta_{n''}}(0)| \leq \epsilon^{1-\sigma} B^\sigma (\frac{4n''}{\pi\sigma})^{n''} \prod_\alpha r_\alpha^{-1}$, gives one factor $\epsilon_a^{1-\sigma} \hat{\mathsf{o}}_a^\sigma \leq C\mathsf{H}(\Xi) K^p L^{-c_1 K}$ for each copy of the entry, $c_1 = (1-\sigma)c'_0 - \sigma(\frac{1}{2} + 4\phi)$.

In all three cases the histories with a large field up to the level t cost on $T_{\ell'}$ less than every power of L^{-K}. With $\delta = \frac{1}{14} - c$, $\tau = \delta$, $\phi = \delta/(12(n+1))$ and $\sigma = \frac{1}{4}\delta$ one has $c_1 - 3n\phi > c$, and the product $\omega_0 \prod_q \omega_q$ gives the bound of the difference with the rate c and with numbers independent of the larger side ℓ; on $T_{\ell'}$ itself the bound is Lem. 5.11. □

Lemma 5.17 (The flow with an insertion)**.** Let $\ell = 2L^m$ be any side, $0 < \mathsf{b}_- \leq \mathsf{b} \leq \mathsf{b}_+$, $\beta = L^K \mathsf{b}$, $n \geq 1$, $\mathsf{d} > 0$ and f_i real Lipschitz; let $S_\varkappa = \sum_l \varkappa(l)\, \mathrm{dn}_l$ be an insertion of brick densities dn_l (gauge invariant polynomials on boxes of at most five sites along each axis, each holding a plaquette, vanishing at $\mathbf{1}$) of the order r, and let two of the $n+1$ sets formed by the carrier of $S_\varkappa$ and the $\operatorname{supp} f_i$ be at sup distance $\geq \mathsf{d}$, the others in any position. For γ small enough in terms of the other constants there are C, p, K_0, depending on $\ell, \mathsf{b}_\pm, n, \mathsf{d}$ and the type, such that for $K \geq K_0$:

(a) if $r \geq 3$ and $|\varkappa|_\infty \leq \beta h$: $|\langle S_\varkappa; \mathcal{O}(f.)\rangle| \leq CK^p L^{-\lambda_L K/4} h \prod_i \|f_i\|_\sharp$; no norm of $\varkappa$ but its supremum enters;

(b) if $r = 2$, $0 \leq \sigma \leq 1$, $|\varkappa|_\infty \leq L^{\sigma K} h$ and $0 < \epsilon \leq 1$: $|\langle S_\varkappa; \mathcal{O}(f.)\rangle| \leq C_\epsilon K^p L^{-(1-\sigma-\epsilon)K} h \prod_i \|f_i\|_\sharp$; in particular $|\langle \mathcal{O}(w); \mathcal{O}(f.)\rangle| \leq C_\epsilon a^{-\epsilon} \max|w| \prod_i \|f_i\|_\sharp$ for every real weight w, in contact with the f_i or not;

(c) if $r = 2$, the insertion is real with $|\varkappa|_\infty \leq \beta h$, and its densities are of derivative kind, that is, $|G_\nu(b)|^2$ or $\Psi_{\lambda(p)}(p)$: $|\langle S_\varkappa; \mathcal{O}(f.)\rangle| \leq CK^p L^{-c_n K} h \prod_i \|f_i\|_\sharp$, $c_n = \lambda_L/(4(6+3n))$; for every $0 < \theta_1 < 1$ and every real insertion of the order 2 with $|\varkappa|_\infty \leq \beta h$, $\langle S_\varkappa - J_{t_1}; \mathcal{O}(f.)\rangle$ obeys the same bound with $L^{-\lambda_L \theta_1 K/4}$, the numbers C, p, K_0 depending besides on θ_1, where $t_1 = K - \lceil \theta_1 K \rceil$ and J_{t_1} is the classical value of the insertion at the background of the level t_1, a function of the block field.

■

Proof sketch. Run the chain with the insertion, as in Lem. 5.10(a), down to a level t at which the large fields are rare. A transfer inequality bounds the cumulant by

$$2\delta_t(\varkappa)\prod_i(8nM_t\|f_i\|_\sharp)+\mathrm{Con}_t+4\,\mathrm{op}_{n+1}\mathsf{R}_t\sup|S_\varkappa|\prod_i(6\ell^3L^{4K}\sup|f_i|),$$

with δ_t the oscillation of the conditional mean of the insertion given the block field of the level t, M_t the inverse radius of the sources at that level, and Con_t the mixed derivative of the constants of the chain, which is small by locality when a pair of sets is separated (when the separated pair is the insertion and one source, the constants of the sources that meet the insertion are handled by a slow cut-off at the scale d, at the price $(2+480M/\mathsf{d})^{n-1}$). By power counting an insertion of the order r has the size $|\varkappa|_\infty L^{(3-2r)i}$ at the level i, and what the steps make of it is forgotten at the rate $L^{-\lambda_L/2}$ per level, $(L/2)^{-x}=L^{-\lambda_L x}$; this gives (a) and (b).

For (c) two levels are used. Let $0<\theta_1<1$, $\tau_1=\lceil\theta_1K\rceil$, $t_1=K-\tau_1$, and $t=K-\tau$, $\tau=\lceil\theta K\rceil$, $\theta=\lambda_L\theta_1/(4(7+3n))<\theta_1$. The insertion is put into the chain only up to the level t_1, as early members; there it equals its classical value J_{t_1} on the background of the level t_1 plus what the steps up to t_1 made of it, which in level-t_1 terms has the size $C|\varkappa|_\infty L^{-(t_1-1)}L^{-\lambda_L\tau_1/2}$ (what a step makes of an insertion of the order 2 is forgotten at the rate $L^{-\lambda_L/2}$ per level). From the level t_1 to the level t this part is forgotten by the factor $L^{-(t-t_1)}$ of the run, on the field cut of the level t; at the level t the histories with a large field are discarded, at a cost below every power of L^{-K}, and the volume $N_t=\ell^3L^{3\tau}$ and Cauchy's estimates in the n sources cost $L^{3(n+1)\tau}$ up to powers of K; the mixed derivatives of the constants of the chain in the sources are small by locality, two of the sets being separated.

The derivative in the insertion is taken as a difference quotient at a small step s, its second derivative being bounded crudely. Together the exponent is $(7+3n)\tau+1-\frac12\lambda_L\tau_1\le-\frac14\lambda_L\theta_1K+8+3n$: this is the second clause of (c), for every θ_1. For the first clause the classical part is bounded besides: J_{t_1} is a function of the block field of the level t_1, its cumulant with the sources is bounded by Cauchy's estimates in the sources at that level, at the price $L^{3n\tau_1}$, times its supremum, and for densities of derivative kind the supremum is at most $\beta\ell^3L^{3K}\,C\varepsilon_{t_1}^2(1+K\log L)^2L^{-6t_1}$ by the estimate below; the exponent is $K+3K-6(K-\tau_1)+3n\tau_1=-2K+(6+3n)\tau_1$, which at $\theta_1=1/(6+3n)$ is at most $-K+6+3n$.

With the second clause at this θ_1 the rate is $\min\{1,\frac14\lambda_L\theta_1\}=c_n$. For a lattice field B on a box of radius R with $|B|\le A_0$ and $|\mathrm{div}^+B|,|\mathrm{curl}^-B|\le\epsilon_1$, the div–curl estimate $|\partial_j^\pm B_i|\le C_*[A_0/R+\epsilon_1(1+\log R)]$ on the quarter box, applied with the Bianchi identity on a cube, gives $|G_\nu(b)|^2,|\Psi_{\lambda(p)}(p)|\le C\varepsilon_i^2\eta_i^6(1+i\log L)^2$ at the regular backgrounds of the level i, $\eta_i=L^{-i}$, which is the order that (c) needs. □

Proposition 5.18 (The mean plaquette)**.** Let $0<g\le\gamma$, $\ell\in\mathfrak{L}(g)$, $n\ge1$, and $E(\beta)=\langle a_p\rangle$ on a periodic lattice of N_s sites along each axis (it does not depend on p).

(a) For every $R\ge3$, $N_s\ge R$ and β with $116R^3(\beta/2)^{-1/2}\le\frac15$ and $\bar\eta(R,\beta)\le\frac14$, $|\beta E(\beta)-1|\le2\bar\eta(R,\beta)^{1/2}$, with $\bar\eta(R,\beta)\le\frac{9\log(2\beta)+55}{R^3}+\frac{18\log(1+256R^3)+15}{R}+1.21\cdot10^9R^9\beta^{-1/2}$: at $R=\lceil\log\beta\rceil$ this tends to 0 uniformly in all volumes $N_s\ge\log\beta$.

(b) On the side ℓ, with $N_s=\ell L^K$: there are C,K_0, depending on g,ℓ,n, such that $\sup_{\mathsf{b}'\in I(g)}|\beta^nE^{(n-1)}(\beta)-(-1)^{n-1}(n-1)!|\le CK^{2n+3}L^{-K/20}$ for $\beta=L^K\mathsf{b}'$ and $K\ge K_0$.

(c) For all real Lipschitz $f_1,\dots,f_n$ on T_ℓ, with nothing asked of the supports, $|a_K^3S^T_{n;K,\ell,L^K\mathsf{b}'}(f)-3(n-1)!\,\mathsf{b}'^{-n}\int_{T_\ell}\prod_i f_i|\le CK^{2n+3}L^{-K/20}\prod_i\max\{\sup|f_i|,\frac{40}{\sqrt3}M\mathrm{Lip}_\infty f_i\}$.

■

Proof sketch. (a) Cut the lattice into blocks of R to $2R$ sites along each axis, and in each block take a comb tree. Bounding the partition function with plaquette weights $\beta_p\in[\beta_-,\beta_+]$ from above (dropping a few plaquettes) and from below (freezing a few bonds), both bounds become the same

Gaussian integral, $\sum_B (2|B|-2)\log \mathsf{g}(\beta_B) - \mathsf{D}$ with $\mathsf{g}(\beta) = (2/\pi)^{1/2}\beta^{-3/2}$ and a number D of the geometry, up to an error $N\eta$ that is small per site; this uses weak coupling on blocks of $R \le \beta^{1/20}$ spacings alone. The convexity of $\log Z$ in β then pins down its first derivative. (b) For the higher derivatives analyticity in β is needed.

On a torus of a fixed physical side, Thms. YM3_I:AJ and YM3_I:AD write $Z[\zeta f] = \exp(E + \sum_{j,X} e_j(X;\zeta f))\widetilde{Z}(\zeta f)$ for real f with $\max\{\sup|f|, \frac{40}{\sqrt{3}} M\mathrm{Lip}_\infty f\} \le 1$, with $|e_j(X;w)| \le C_{\mathrm{S}} e^{-\kappa d_j(X)}$, $|\widetilde{Z}| \le e^{C_{\mathrm{S}}\ell^3}$ and $\widetilde{Z}(0) \ge e^{-C_g\ell^3}$; the factor $\widetilde{Z}$ may vanish. Put $N = (\ell L^K)^3$, $h_f(\zeta) = \widetilde{Z}(\zeta f)/\widetilde{Z}(0)$ and $D_f(\zeta) = (3N)^{-1}[\sum_{j,X} e_j(X;\zeta f) + \sum_p \log(1 - \zeta f(x(p))/\mathsf{b}')]$, the per-site exponent; by the count of polymers $|D_f| \le M_0$ on a disc $|\zeta| < \rho_1$, with M_0 and ρ_1 free of K. Then $h_f(0) = 1$, $h_f > 0$ on the real axis and $|h_f| \le e^{\Gamma}$ with $\Gamma = (C_{\mathrm{S}} + C_g)\ell^3$, a number free of K.

First, Cauchy's estimate on circles of radius $\frac{1}{2}\rho_1$ gives $|h_f - 1| \le \frac{1}{2}$ on $|\zeta| \le \frac{1}{4}\rho_1 e^{-\Gamma}$, so that $\log h_f$ is defined there and $|(\log h_f)^{(n)}(0)| \le n!\,(4e^{\Gamma}/\rho_1)^n$: this member does not grow with the cutoff, and it is of the order $1/N$ per site. Second, the Poisson–Jensen formula and Harnack's inequality, applied to the non-negative function $\Gamma - \log|h_f|$, give $\int_{-\rho_1/2}^{\rho_1/2} |\log|h_f(x)||\,\mathrm{d}x \le 9\rho_1\Gamma$; with this, the two-sided real-axis bound of (a), at the plaquette weights $\beta(1 - xf(x(p))/\mathsf{b}')$, controls the per-site exponent in L^1 on the segment: $\int_{-\rho_1/2}^{\rho_1/2} |D_f(x)|\,\mathrm{d}x \le \rho_1\delta$ with $\delta = \frac{1}{3}(2\bar{\eta}(R,\beta) + Ra_K + R^{-3}) + 3(C_{\mathrm{S}} + C_g)L^{-3K}$.

Third, A. Markov's inequality for the derivatives of the Taylor polynomial of D_f of the order $\mathsf{m} = \max\{n, \lceil \log_2(M_0/\delta)\rceil\}$, with Cauchy's estimate for the rest of the series, gives $|D_f^{(n)}(0)| \le 4\,(2/\rho_1)^n \mathsf{m}^{2n}(\mathsf{m}+1)^2\delta$, and at $R = \lfloor \beta^{1/20} \rfloor$ this is the rate. By Jensen's formula h_f has at most $\Gamma/\log 2$ zeros in $|\zeta| < \frac{1}{2}\rho_1$, a number that does not grow with the cutoff; the bound on $(\log h_f)^{(n)}(0)$ above does not need it. At $f \equiv 1$, where $S_n^T(1,\dots,1) = a_K^{-n}(-1)^{n-1}3N\,E^{(n-1)}(\beta)$, the identity of (c) below gives (b). (c) is the same argument at a general f: $S_n^T(f,\dots,f) = (n-1)!\mathsf{b}'^{-n}\sum_p f(x(p))^n + 3N D_f^{(n)}(0) + (\log h_f)^{(n)}(0)$, with the per-site exponent D_f and the factor h_f of Thms. YM3_I:AJ and YM3_I:AD; polarisation gives distinct entries. □

Use the separated regions and the actual averaging constraints of Prop. 5.37, with its block-number and small-triple hypotheses in addition to the width condition below. Require also the geometric and regularity hypotheses of Thm. YM3_I:I and Prop. YM3_I:29.1, with their block-number, torus-size and gap floors. The matrix-valued plaquette-divergence bound in that theorem follows from the regular curvature and current bounds by equation YM3_I:(579), with a factor 2 absorbed in the regularity ceiling. These are numerical floors and small-radius conditions on the data, not assumptions about the inverse estimate to be proved. Also retain the four local-system radii at most a_H, the block floor $\mathfrak{M} \ge M_H$, and the permitted decay rate of Lem. 5.34. Write $\xi = \eta_k$ and $\ell_n = \xi L^n$ for the physical layer scale. For clarity distinguish D_0, the covariant differential from sites to bonds, from D_1, the covariant differential from bonds to plaquettes. Let U be a real background, critical for the unweighted action on the kernel of the averaging constraint, regular at radius $\alpha \le \alpha_{\mathrm{lin}}(L) \le (16L^2)^{-1}$, with curvature bounded by the envelope $e(x) = \varepsilon_{j(x)}$ in the scaled norm below, and $\sup e \le e_0(L)$. The layer radii satisfy $0 < \varepsilon_j \le \varepsilon_{j+1} \le L\varepsilon_j$. Require $R_1\mathfrak{M} \ge w_{\mathrm{lin}}(L)$.

At each bare site x_* of layer $n \ge 2$, fix a real adjoint gauge g_{x_*} on the cube of physical sup radius $\ell_n/4$ such that

$$|g_{x_*}(b_-)\,\mathrm{Ad}\,U(b)g_{x_*}(b_+)^{-1} - I| \le L^2\alpha/L^n.$$

The gauge hypotheses are part of the setting. All Hölder quotients below compare components of the same orientation in these gauges. The cube for a potential is the concentric cube of radius $\ell_n/8 - 2\xi$; those for a curvature and a source two-form have radius $\ell_n/4$. Quotients are taken over distinct admissible base points in the indicated cube, with the usual physical Euclidean distance.

We use the following scaled spaces. For a field of inverse length degree r, put

$$|f|_{r,h} = \sup_i \frac{\ell(i)^r |f(i)|}{h(x_i)}, \qquad [f]_{r,1/2,h} = \sup_{n\ge 2, x_*} \frac{\ell_n^{r+1/2}}{h(x_*)} [g_{x_*} f]_{1/2;C(x_*)}.$$

For an index i, let $\ell(i)$ be the largest layer scale of the sites in the stencil defining that field value or first difference. For a row use its prescribed scale ξ_c. The envelope is read at the indicated base point x_i. This fixes the norm convention also for indices meeting a layer boundary. The adjacent-layer comparison of Lem. YM3_I:21.2 makes it equivalent, with constants depending only on L, to the base-point scale convention. Set $\|f\|_{r,1/2,h} = |f|_{r,h} + [f]_{r,1/2,h}$. For the first two layers only the supremum term is required; their ratios $\ell_n/\xi \le L$ are uniformly bounded. The curvature hypothesis above means $\|F_U\|_{2,1/2,e} \le 1$.

The row units are dimensionless. If Q_{ph} is the physical averaging derivative with row pairing of weight ξ_c^3, put

$$QA(c) = \xi_c Q_{\mathrm{ph}} A(c), \qquad \omega(c) = \xi_c^3 \omega_{\mathrm{ph}}(c).$$

Use the row pairing of weight ξ_c^{-1} for $Q^{\mathsf T}$. Then $Q^{\mathsf T}\omega = Q_{\mathrm{ph}}^{\mathsf T}\omega_{\mathrm{ph}}$, since both pairings with A are $\sum_c \frac{1}{2}\operatorname{tr}(Q_{\mathrm{ph}} A(c)\omega(c))$. This conversion changes units only; it retains the same row set and averaging germ. Row data b and multipliers have degree zero.

Let $h = e\theta$, where $\theta > 0$ and $|\log\theta(x) - \log\theta(x')| \le c_U d(x,x')$. Choose c_U at most one eighth of each decay rate used below and at most $1/(6L)$. Thus, for fixed constants $C_h = C_h(L)$ and $0 < \nu \le \delta_2/2$,

$$h(z)/h(y) \le C_h e^{\nu d(y,z)}. \tag{5.viii}$$

Take the same condition on the local comparison cubes. Define

$$\|(A,\omega)\|_{\mathcal{X}_h} = \|A\|_{1,1/2,h} + \|D_1 A\|_{2,1/2,h} + |D_0^{\mathsf T} A|_{2,h} + |\omega|_{0,h},$$
$$\|(Y,f,b)\|_{\mathcal{F}_h} = \|Y\|_{2,1/2,h} + |f|_{3,h} + |b|_{0,h}.$$

The ordinary and divergence sources are read only through $f + D_1^{\mathsf T} Y$. For a complex anisotropy an, its size $t(\mathrm{an})$ is its supremum plus the scaled Hölder-1/2 seminorm on the same plaquette cubes.

Lemma 5.19 (Linear response at a variable anisotropy)**.** There are positive numbers $w_{\mathrm{lin}}, \alpha_{\mathrm{lin}}, e_0$, $t_{\mathrm{lin}}, K_{\mathrm{lin}}$, depending only on L, such that the following holds in the setting above. If $t(\mathrm{an}) \le t_{\mathrm{lin}} \le 1/4$, the system

$$QA = b, \qquad T_\tau A = f + D_1^{\mathsf T} Y + Q^{\mathsf T}\omega, \qquad \tau = 1 + \mathrm{an}, \tag{5.ix}$$

has a unique solution, linear in the data, with

$$\|(A,\omega)\|_{\mathcal{X}_h} \le K_{\mathrm{lin}} \|(Y,f,b)\|_{\mathcal{F}_h}. \tag{5.x}$$

Here $T_\tau = D_1^{\mathsf T}\tau c D_1 + V_0' + D_0 R D_0^{\mathsf T}$ is the second variation of the weighted action at the reference field, with the slice term. The coefficient c_p is determined by $\partial U(p) = c_p I + i\xi^2 F_U(p)$, and V_0' contains the curvature commutator terms of that second variation. The nonlinear block-constraint remainder and its multiplier are kept in the forcing when the lemma is applied to the nonlinear equation. The row and slice conventions are those of the setting. All constants are uniform in the volume, the number of layers and the permitted envelopes. For the mean-scheme rows of a label, rotated into the comb frame, the same estimate holds with $2K_{\mathrm{lin}}$. ∎

Proof of the linear estimate. We first treat the unweighted operator. The block constraint must be retained in this argument: the inverse of the gauge-fixed operator without the constraint is not the required Green operator. Put

$$T = T_\tau\big|_{\mathrm{an}=0}, \qquad G = (T + Q^{\mathsf T}Q)^{-1}, \qquad M = QGQ^{\mathsf T},$$

and define

$$H = GQ^{\mathsf T}M^{-1}, \qquad \widetilde{G} = G - HQG. \tag{5.xi}$$

Thus $QH = 1$, $Q\widetilde{G} = 0$ and $\widetilde{G}T = 1 - HQ$. The rows are the interior and outward determining bonds; inward bonds are discarded, so there are no edge rows. On a row of level j, the derivative of the iterated comb average is

$$QA(c) = \xi\,(\mathbf{Q}(W_{j-1})\cdots\mathbf{Q}(W_0)A)(c).$$

The factors read only the double block beneath that row. Separation keeps these blocks inside the domains on which the averages are defined. The row conversion above therefore identifies this Q with the physical averaging derivative in the regular constrained system, without changing the set of constraints.

The regular current bound supplies the matrix-divergence hypothesis used for that system. Indeed, at a full bond the quaternion decomposition gives

$$D_U^{\mathsf{T}}\partial U = aI + i\xi^2 J(U), \qquad |a| \le \xi^{-1}\max_{p\ni b}|\partial U(p) - I|^2.$$

Since the adjacent plaquette scale is at least the bond scale and the latter is at least ξ, the regularity bounds give $|D_U^{\mathsf{T}}\partial U| \le 2\alpha\xi^2\ell(b)^{-3}$. This is equation YM3_I:(579); our choice $2\alpha_{\mathrm{lin}} \le a_Q$ permits the application below.

Here are the unweighted bounds that will be used. Write $|f|_{(r)} = \sup_i \ell(i)^r|f(i)|$ and $\|A\|_X = \max\{|A|_{(1)}, |\nabla A|_{(2)}\}$. For sufficiently small regularity radius, there are $K_1, K_7 \ge 1$, depending only on L, such that

$$\begin{gathered}\|\widetilde{G}f\|_X \le K_7|f|_{(3)}, \qquad \|Hb\|_X \le K_1\sup|b|, \qquad \sup|H^{\mathsf{T}}f| \le K_1|f|_{(3)},\\ \sup|(D_1H)^{\mathsf{T}}Y| \le 3K_1|Y|_{(2)}, \qquad \sup|(D_0^{\mathsf{T}}H)^{\mathsf{T}}\lambda| \le 3K_1|\lambda|_{(2)}, \qquad |R\lambda|_{(2)} \le (1+K_1^3)|\lambda|_{(2)}.\end{gathered} \tag{5.xii}$$

These estimates are unchanged by passing to the units of any layer. The averaging rows are dimensionless, their pairing has weight $\ell(c)^{-1}$, and their transposes are taken for that pairing. They are precisely the bounds of Prop. YM3_I:29.1, applied after the row identification just made. Its proof also records their exponential versions. For the transpose estimates, test against a datum supported at one row c. The row pairing contributes a factor $\ell(c)$, while the number of bonds or plaquettes in a cell of scale ℓ_y is at most $3(\ell_y/\xi)^3$. The resulting kernel sum has factor $\ell(c)/\ell_y$, which is absorbed by the exponential decay. The same calculation with the first differences gives the fourth and fifth estimates, including the zero-extension terms at the boundary. These are equations YM3_I:(459) and the following divergence estimate. The scalar estimate follows from $R = 1 - G'Q'^{\mathsf{T}}(Q'G'^2Q'^{\mathsf{T}})^{-1}Q'G'$. At a bond meeting the complement of the ambient region, the zero-extension difference has local scale ξ and is bounded by $|A|_{(1)}$. Thus these estimates also include boundary differences. For an empty ambient region all spaces are zero. The operators preserve real fields, so the estimates extend to complex fields by their real and imaginary parts, with constants enlarged by a fixed factor.

The same operators have exponential decay in the weighted distance. Their first-difference Hölder estimates follow from the ordinary kernel estimates in Lem. 5.34. Multiplying the input at z and output at y by the envelope ratio costs $h(z)/h(y)$. The permitted growth of this ratio consumes only part of the exponential decay. The remaining kernel is summable by Lem. YM3_I:21.2. This gives the envelope versions of eq. 5.xii, with constants independent of the torus and the number of layers.

For $q = f + D_1^{\mathsf{T}}Y$, elimination of the multiplier in $TA = q + Q^{\mathsf{T}}\omega$, $QA = b$, gives the exact formulas

$$A = \widetilde{G}q + Hb, \qquad \omega = H^{\mathsf{T}}(THb - q). \tag{5.xiii}$$

Indeed $(T + Q^{\mathsf{T}}Q)A = q + Q^{\mathsf{T}}(\omega + b)$, and applying QG determines $\omega + b$. The identity $M^{-1} = H^{\mathsf{T}}TH + 1$ gives the displayed formula for ω. The six bounds control the ordinary forcing, the lift, and all multiplier terms. The remaining issue is the first-slot divergence $D_1^{\mathsf{T}}Y$.

Cover each layer of index at least two by cubes of size comparable to its layer scale. In the local gauges, write $U = e^{i\xi a}$, with a and its first differences bounded in the units of that layer. A partition

of unity reduces the divergence source to the flat equation on these cubes. Convolution with the lattice Laplace Green kernel gives a local potential; its first differences involve second differences of that kernel. These preserve the Hölder exponent by Lem. YM3_I:19.4. The cancellation in that lemma is essential: a bound on the absolute kernel alone would give a logarithmic loss.

Cut off these local potentials and carry them back to the original gauge. Their sum Φ satisfies

$$T\Phi = D_1^{\mathsf{T}} Y + J, \qquad |J|_{3,h} \le C\|Y\|_{2,1/2,h}.$$

Here J consists of cutoff commutators, the terms containing the local potential and curvature, the lower-order term $V_0'\Phi$, the term $-D_0(1-R)D_0^{\mathsf{T}}\Phi$, and the untreated sources in layers zero and one. The first differences of the local gauge potential bound the coefficient terms; the cutoff derivatives have the inverse powers of the local scale. In the first two layers the scale-to-spacing ratio is bounded by L, so a bare difference is bounded by the corresponding scaled supremum. These are ordinary sources of degree three. The parametrix identity is

$$\widetilde{G} D_1^{\mathsf{T}} Y = \Phi - HQ\Phi - \widetilde{G} J. \tag{5.xiv}$$

The row datum $Q\Phi$ is bounded in the dimensionless row norm: each average reads a double block on which the scale and envelope are comparable, and the factor $\ell(c)$ in the physical path average cancels the scale ℓ^{-1} of Φ. The remainder J has the degree-three norm displayed above. Thus both correction terms lie in the ordinary-input classes of Lem. 5.34. Its estimates are read at $s = 1$, with the same rows and their row pairing. This proves the required potential and curvature bounds. The first-difference Hölder bounds for the correction terms are the ordinary kernel bounds; those for Φ are the same-exponent Green-kernel bounds just used. Their supremum estimates also bound the divergence. To pass from the comparison cubes of radius $\ell/56$ to the cubes in the norm, cover the latter by comparison cubes and use the local gauge transition bounds. On an overlap those transitions have differences bounded by $C(L)\alpha/\ell$; the Hölder product inequality and the supremum bound therefore control the change of gauge. The bounded overlap gives a constant depending only on L. No Hölder estimate is obtained here by applying an estimate with a positive loss of exponent at zero loss.

For completeness, the potential Hölder norm can also be read directly from its curvature and divergence. On a cube C_R in bare units, let the adjoint transports differ from the identity by at most v/R. If $6v \le \sigma_H$, then

$$\sup_{C_{R/2-2}} |E| + (R/2-2)^{1/2}[E]_{1/2;C_{R/2-2}} \le 2C_H\big((R-1)^{-3/2}\|E\|_2 + R|DE| + R|D_0^* E|\big).$$

To obtain it, apply Prop. YM3_I:18.1 to each component, whose flat Laplacian is the divergence of its flat curvature and divergence. Replacing the covariant quantities by the flat ones adds at most Cv times the supremum on the next larger cube. On radii increasing from $3R/4$ by $R/8, R/16, \dots$, choose v so that this factor is at most $1/12$. Iteration absorbs the supremum; the terminal error is at most $12^{8-\log_2 R}\|E\|_2$. A second application on cubes of radius $R/4$ gives the Hölder bound. For $R < 64$ the supremum is bounded by the finite-volume ℓ^2 norm. At $R = L^n/4$, the condition follows by reducing the regularity radius so that $\alpha \le \sigma_H/(8L^2)$.

We have proved the estimate at zero anisotropy with a constant $K^{(0)}$. At a nonzero anisotropy, the same solution solves the unweighted system with data

$$(Y - \operatorname{an} c D_1 A,\ f - V_0'[\operatorname{an}]A,\ b).$$

The Hölder product inequality bounds the first change by $Ct(\operatorname{an})\|D_1 A\|_{2,1/2,h}$; the second is bounded by $C|\operatorname{an}|_\infty \alpha |A|_{1,h}$. Choose $K^{(0)} C t_{\mathrm{lin}} \le 1/2$ and absorb these terms. The bound holds with $K_{\mathrm{lin}} = 2K^{(0)}$.

The real part of the form of T_τ on $N(Q)$ is coercive at these radii, as proved below. Hence a homogeneous solution has $A = 0$, by pairing its equation with $\bar{A}$, and then $Q^{\mathsf{T}}\omega = 0$ gives $\omega = 0$.

The finite square system is therefore invertible. This proves existence and uniqueness as well as the estimate.

Finally, the mean-scheme rows of a label, rotated into the comb frame, have the form $Q + E$. Their comparison kernel is supported in the double block of the row and obeys

$$|EA|_{0,h} \le c_E \alpha |A|_{1,h}, \qquad |E^{\mathsf{T}}\omega|_{3,h} \le c_E \alpha |\omega|_{0,h}.$$

The system with these rows is the one just solved with data $(Y, f + E^{\mathsf{T}}\omega, b - EA)$. Taking $K_{\mathrm{lin}} c_E \alpha \le 1/2$ proves its bound, with $2K_{\mathrm{lin}}$, by a contraction. This is the row comparison needed when the estimate is applied to a label. □

Lemma 5.20 (The energy of the displacement)**.** In the setting of Lem. 5.19, decrease the regularity radius and the envelope ceiling by constants depending only on L. Put

$$\mathcal{E}(A) = \xi^3 \sum_p |D_1 A(p)|^2 + \xi^3 \sum_b \ell(b)^{-2} |A(b)|^2, \qquad |b|^2_{\mathrm{row}} = \sum_c \ell(c)^{-1} |b(c)|^2.$$

For the linear solution with data (Y, f, b),

$$\mathcal{E}(A) \le K^{(2)} \left(\xi^3 \sum_p |Y(p)|^2 + \xi^3 \sum_b \ell(b)^2 |f(b)|^2 + |b|^2_{\mathrm{row}} \right). \tag{5.xv}$$

For the small fixed point with zero block datum,

$$\mathcal{E}(A[\mathrm{an}; 0]) \le C |\mathrm{an}|^2_\infty \xi^3 \sum_p |F_U(p)|^2. \tag{5.xvi}$$

At a permitted complex block datum b_0, the difference $A' = A[\mathrm{an}; b_0] - A[0; b_0]$ instead satisfies

$$\mathcal{E}(A') \le C |\mathrm{an}|^2_\infty \left(\xi^3 \sum_p |F_U(p)|^2 + |b_0|^2_{\mathrm{row}} \right). \tag{5.xvii}$$

The constants are independent of the number of layers and the volume. The last two statements concern fixed points in the small ball of the linear estimate. ■

Proof. We need an estimate on the field energy, including the forcing from the nonlinear block constraint. A contraction in a stronger energy norm of the field and multiplier is unnecessary.

At zero anisotropy the operator $T + Q^{\mathsf{T}} Q$ is real, symmetric and positive by Thm. YM3_I:I. Its restriction to $N(Q)$ is T, so the constrained Green operator is positive on that subspace. Apply the first estimate of eq. 5.xii to an eigenvector of the largest eigenvalue of $\ell^{-1} \widetilde{G} \ell^{-1}$. If that eigenvalue is λ, it gives

$$\lambda \sup_b \ell(b)^2 |g(b)| \le K_7 \sup_b \ell(b)^2 |g(b)|.$$

Thus the quadratic form of T on $N(Q)$ controls the weighted potential square sum with constant K_7^{-1}. Its expression as a curvature form, a slice term and V_0' gives the second lower bound. The plaquette coefficient satisfies $c_p \ge 1/2$, while the form of V_0' is bounded by $C_V \alpha$ times the weighted potential square sum. Since $|\mathrm{an}|_\infty \le 1/4$, reducing α so that $3 C_V K_7 \alpha \le 1$ gives

$$\mathrm{Re}(\bar{A}, T_\tau A)_\xi \ge c^{(2)} \mathcal{E}(A) + \tfrac{1}{2} \| R D_0^* A \|^2_\xi, \qquad A \in N(Q), \tag{5.xviii}$$

where $c^{(2)} = \min\{1/8, (4K_7)^{-1}\}$. Pairing the equation with $\bar{A}$ removes the multiplier. Cauchy–Schwarz bounds the remaining pairing by $\mathcal{E}(A)^{1/2}$ times the square root of the first two data sums in eq. 5.xv.

For nonzero row data use the lift $A^\circ = Hb$. Then $A - A^\circ \in N(Q)$ and its equation has the additional forcing $-T_\tau A^\circ$. The bounds on H, its first differences and their transposes in eq. 5.xii, applied by the weighted Schur test, give

$$\mathcal{E}(A^\circ) + \| D_0^* A^\circ \|^2_\xi \le C_H |b|^2_{\mathrm{row}}.$$

The row weight is $\ell(c)^{-1}$; using an unweighted row sum would not give a constant uniform in the layers. Apply eq. 5.xviii to $A - A^\circ$ and use this lift estimate. This proves eq. 5.xv.

At the nonlinear fixed point the data are

$$(-\mathrm{an}F_U - Y_3(A),\ -f_3(A) + (dC_2(U,A))^{\mathsf{T}}(\omega_0+\omega),\ -C_2(U,A)).$$

Here Y_3, f_3 are the terms of order at least two in the displacement in the first variation, and C_2 is the nonlinear remainder of the block constraint. The last entry is not zero. The small-ball bound on A and the multiplier implies that the square-sum norm of the nonlinear data is at most $c_{\mathrm{ms}} \sup e\, \mathcal{E}(A)^{1/2}$. To see the row estimate in the same units, write $l(c) = \ell(c)/\xi$. The Taylor kernel of the block average and Cauchy–Schwarz give

$$|C_2(U,A)(c)| \le C l(c)^{-1} \sum_{b \text{ beneath } c} |\xi A(b)|^2.$$

There are at most $Cl(c)^3$ bonds beneath the row, their scales are comparable to $l(c)$, and each bond lies beneath at most a constant number of rows. Hence the row square sum is bounded by $C(\sup_b \ell(b)|A(b)|)^2 \mathcal{E}(A)$. The same bilinear kernel bounds the transposed derivative paired with $\omega_0+\omega$, whose scaled supremum is at most $C \sup e$. The plaquette and bond remainders are treated by their local product bounds, using one small factor and one energy factor.

Choose the envelope ceiling so that $K^{(2)} c_{\mathrm{ms}}^2 (\sup e)^2 \le 1/16$. The nonlinear data can then be absorbed in eq. 5.xv. The remaining forcing is $-\mathrm{an}F_U$, which proves eq. 5.xvi.

At $\mathrm{an} = 0$ and datum b_0 the same argument gives $\mathcal{E}(A[0;b_0]) \le 4K^{(2)}|b_0|_{\mathrm{row}}^2$. Subtract the two fixed-point equations at the same b_0. The terms linear in an are bounded by its supremum times the square root of $\xi^3 \sum |F_U|^2 + \mathcal{E}(A[0;b_0])$. The remaining differences have the factor $C \sup e$; the multiplier difference is bounded by testing against the same row lift. One further reduction of the envelope ceiling absorbs these differences. This proves eq. 5.xvii. □

Lemma 5.21 (Complex scalar gauge transfer)**.** Use the regular-reference setting of Lem. 5.19. After decreasing its envelope ceiling and the admitted radii by constants depending only on L, the following estimates hold. Consider a complex potential a in a sufficiently small ball for $\mathcal{X}_e$ and the small gauge $e^{i\lambda}$ relative to its real reference configuration. Define N and q by the exact identities

$$D^* a^{e^{i\lambda}} = D^* a - D^* D\lambda + N(a,\lambda),$$
$$T^{-1} i^{-1} \log \mathcal{R}(a; e^{i\lambda}) = Q'\lambda + q(a,\lambda).$$

Here $\mathcal{R}$ is the induced gauge at the retained block tops, and T is the orthogonal transport in its linearization; these are the maps in the proof of Lem. YM3_I:30.4. Their linear scalar system is precisely the compression $\mathbb{K}^{[2]}$ of Section YM3_I:23.2, in the covariant form of Section YM3_I:25:

$$\mathbb{K}^{[2]}(\lambda, w, \sigma) = (\Delta'_a \lambda + w,\ \Delta'_a w - Q'^{*}\sigma,\ Q'\lambda), \qquad \Delta'_a = \Delta_D + Q'^{*}\mathsf{a}'Q'.$$

Here Δ_D is the scalar Dirichlet Laplacian; the boundary identity is $RD^*D = R\Delta_D$. Let $\mathcal{S}(f,b)$ be the first component of the solution with data $(f + Q'^{*}\mathsf{a}'b, 0, b)$. Its first and third rows give $w = f - \Delta_D\lambda$, while its second row says $Rw = 0$. Consequently the two desired gauge conditions are exactly the fixed point $\lambda = \mathcal{S}(D^*a + N(a,\lambda), -q(a,\lambda))$. The mass term $Q'^{*}\mathsf{a}'b$ is necessary when $b \neq 0$. Put $\|\lambda\|_{\mathcal{Z}_h} := |\lambda|_{0,h} + \|(D\lambda, 0)\|_{\mathcal{X}_h}$. For $\lambda, \tilde\lambda$ in one common small ball for $\mathcal{Z}_e$, and every admitted envelope h,

$$\begin{aligned} \|\mathcal{S}(N(a,\lambda) - N(a,\tilde\lambda), -q(a,\lambda) + q(a,\tilde\lambda))\|_{\mathcal{Z}_h} &\le \tfrac{1}{2}\|\lambda - \tilde\lambda\|_{\mathcal{Z}_h}, \\ \|(a^{e^{i\lambda}} - a, 0)\|_{\mathcal{X}_h} &\le C\|\lambda\|_{\mathcal{Z}_h}. \end{aligned} \tag{5.xix}$$

The scalar fixed-point map preserves a sufficiently small common complex ball for $\mathcal{Z}_e$. The estimates also hold for differences about a given solution and for a cutoff repair with its commutator forcing, in the exponentially weighted envelopes. All constants are independent of the volume and number of layers. ■

Proof. The divergence remainder. The divergence remainder has a uniform complex estimate. Let a, λ and $\widetilde{\lambda}$ lie in one sufficiently small ball for the unlocalized norms $\mathcal{X}_e$, $\mathcal{Z}_e$, and let the sum of their norms be at most r. Then

$$|N(a,\lambda) - N(a,\widetilde{\lambda})|_{2,h} \le C(L) r \sup e \, \|\lambda - \widetilde{\lambda}\|_{\mathcal{Z}_h}. \tag{5.xx}$$

Here a localized norm measures the difference; the other factor in each product is bounded on the common unlocalized ball.

For completeness, put $k(l) = (e^{i \operatorname{ad} l} - 1)/(i \operatorname{ad} l)$, with its value at zero defined by the power series. At a site x, use the outgoing and incoming bond logarithms

$$\Psi_{\mathrm{out}}(l;u,d) = i^{-1} \log(e^{il} e^{iu} e^{-i(l+d)}), \qquad \Psi_{\mathrm{in}}(l;u,d) = i^{-1} \log(e^{i(l-d)} e^{iu} e^{-il}).$$

On a fixed complex ball these are holomorphic and have the form

$$\Psi(l;u,d) = \mathrm{Ad}(e^{il})u - k(l)d + \Psi^{(2)}(l;u,d), \qquad |\Psi^{(2)}| \le 14|d| \max\{|u|,|d|\}.$$

The signed bond sum therefore gives the exact identity

$$\begin{aligned} N(a,\lambda)(x) = {} & [\mathrm{Ad}(e^{i\lambda(x)}) - 1] D^* a(x) - [k(\lambda(x)) - 1] D^* D\lambda(x) \\ & + \xi^{-2} \sum_{b \ni x} \epsilon_{x,b} \Psi^{(2)}_{x,b}(\lambda(x); \xi a_{x,b}, \xi D\lambda_{x,b}). \end{aligned}$$

Incoming values are transported to x by the real reference configuration, and $\epsilon_{x,b}$ is the divergence sign. The holomorphic remainder vanishes for $d = 0$ and has no terms of degree less than two in (u,d). On a smaller fixed ball, Cauchy's estimate bounds its differential by products containing one variation and at least one additional small factor, with a constant independent of the lattice. Subtract the formula at λ and $\widetilde{\lambda}$. Every term contains one difference, among $\lambda - \widetilde{\lambda}$, $D(\lambda - \widetilde{\lambda})$ and $D^*D(\lambda - \widetilde{\lambda})$, and a small factor bounded by Cre in its scaled degree. Adjacent scales and envelopes are comparable with constants depending only on L. Dividing by h and multiplying by the degree-two scale gives eq. 5.xx.

The induced top factors. The complex tower estimates of Prop. YM3_I:30.2 give the top remainder a common complex domain. To verify this without iterating estimates for complex group angles, fix a retained top y of level j and work on its tower. Here $\xi_j = \ell_j = \xi L^j$ is the layer scale, $B^j(y)$ is the block beneath y, and $\mathsf{P}^j(y)$ is its tower bond set, as in the cited proposition. Write $\alpha^{(j)}$ for the local bound on the reference plaquette angles at that scale. Put

$$a_y = \xi_j \sup_{\mathsf{P}^j(y)} |a|, \qquad d_y(\lambda) = \max\left\{ \sup_{B^j(y)} |\lambda|, \xi_j \sup_{\mathsf{P}^j(y)} |D\lambda| \right\}.$$

For $a_y + d_y(\lambda)$ below a fixed number of L, the complex bond logarithm for $a^{e^{i\lambda}}$ is defined and obeys

$$\xi_j \sup_{\mathsf{P}^j(y)} |a^{e^{i\lambda}}| \le C\{a_y + d_y(\lambda)\}.$$

Use the straightened averages and their accumulated factors $v_i[a]$, for both a and $a^{e^{i\lambda}}$. Their complex tower estimates give, with constants independent of j,

$$|v_j[a]^{\pm 1} - 1| \le C(L) a_y, \qquad |v_j[a^{e^{i\lambda}}]^{\pm 1} - 1| \le C(L)\{a_y + d_y(\lambda)\}.$$

In particular the expression

$$r_j(a,\lambda)(y) = v_j[a^{e^{i\lambda}}](y)^{-1} e^{i\lambda(y)} v_j[a](y)$$

and its small logarithm are holomorphic on a common complex ball independent of the depth. The factorization identity for the mean scheme identifies r_j with the induced gauge at real arguments. The identity continues to the complex ball. Apply the same bounded-factor formula at every intermediate level $i \le j$. Factorization identifies the straightened averages of $a^{e^{i\lambda}}$ with those of

a transformed by r_i, so every intermediate factor and small logarithm used by the recursion is admitted on this ball.

At $\lambda = 0$ one has $r_j = 1$. The derivative at $(a, \lambda) = (0, 0)$ of $T^{-1} i^{-1} \log r_j$ in λ is $Q' + E'$, where E' is the comparison of the scalar mean and comb averages. Its local bound is

$$|E'\lambda(y)| \le \kappa_F \alpha^{(j)} \sup_{B^j(y)} |\lambda|.$$

The Taylor expansion of $q(a, \lambda) - E'\lambda$ contains only terms involving λ and of total degree at least two in (a, λ). Cauchy's estimate on a smaller common complex ball therefore gives

$$|q(a, \lambda)(y) - q(a, \widetilde{\lambda})(y)| \le C(L)\{\alpha^{(j)} + a_y + d_y(\lambda) + d_y(\widetilde{\lambda})\} d_y(\lambda - \widetilde{\lambda}).$$

For the weighted setting, the layer index is constant on $B^j(y)$ and is j at a retained top. Consequently $e = e(y)$ there and the internal bond scales are ξ_j. Its diameter in the weighted distance is bounded by a number depending only on L. The envelope condition therefore gives $h(x) \le C(L)h(y)$ for $x \in B^j(y)$. The local reference curvature parameter obeys $\alpha^{(j)} \le C(L)e(y)$: use the bound on F_U in the same pillar and the comparison between the small real plaquette angle and its vector part. For a, λ and $\widetilde{\lambda}$ in the same unlocalized radius-r ball as above, division by $h(y)$ now yields

$$|q(a, \lambda) - q(a, \widetilde{\lambda})|_{0,h} \le C(L)(1 + r) \sup e \, \|\lambda - \widetilde{\lambda}\|_{\mathcal{Z}_h}. \tag{5.xxi}$$

At level zero $q = 0$.

The scalar inverse and repaired field. The scalar inverse has the stronger norm required here. Its three equations are

$$\Delta'_a \lambda + w = f + Q'^* \mathsf{a}' b, \qquad \Delta'_a w - Q'^* \sigma = 0, \qquad Q'\lambda = b.$$

The scalar solution seminorms and forcing degrees are those of equations YM3_I:(327)–(328); the regional inverse is Thm. YM3_I:G, with its scalar compression. The solution seminorm retains $|\lambda|_{0;y}$, $|D\lambda|_{-1;y}$ and $|w|_{-2;y}$; its forcing seminorm retains $|f_2|_{-2;y}$, $|f_3|_{-4;y}$ and $|f_4|_{0;y}$. The regional inverse estimate is $|\mathbb{E}^{[2]} g|_y \le B \sup_z e^{-\delta d(y,z)} |g|_z$. Apply it with $g = (f + Q'^* \mathsf{a}' b, 0, b)$. At a bare site of the cell of a level-j top y, $Q'^* \mathsf{a}' b$ is the value $\varkappa_j \xi_j^{-2} b(y)$, transported by the real reference configuration, with $\varkappa_j \le 1$. The input therefore has the claimed degrees, uniformly in the cell size. The envelope inequality eq. 5.viii, with its exponent strictly below δ, bounds the weighted supremum in the regional inverse estimate. It gives

$$|\lambda|_{0,h} + |D\lambda|_{1,h} + |w|_{2,h} \le C_{\mathrm{sc}}(L)(|f|_{2,h} + |b|_{0,h}).$$

The first and third rows therefore imply

$$\Delta_D \lambda = f - w, \qquad |\Delta_D \lambda|_{2,h} \le C(L)(|f|_{2,h} + |b|_{0,h}).$$

At a site meeting the boundary of the first region, $\Delta_D \lambda - D^* D\lambda$ is the missing-neighbour mass $\xi^{-2} n\lambda$. Such sites have layer zero and layer scale ξ, so this term has the same bound.

To add the Hölder terms, put $B = D\lambda$. The exact covariant curl of this gradient is reference curvature acting on a transported value of λ, and its divergence has just been bounded. Apply the local curl–divergence estimate proved in the proof of Lem. 5.19 to B in the same comparison gauges. It gives the degree-one Hölder estimate for $D\lambda$, uniformly in the bare spacing. The pointwise gradient bound also makes λ locally Lipschitz after transport. The reference curvature is Hölder, so the exact formula for $D_1 D\lambda$ gives its degree-two Hölder norm by the product rule. The first two layers require only supremum bounds. No additional regularity of the scalar forcing is used. We have proved

$$\|\mathcal{S}(f, b)\|_{\mathcal{Z}_h} \le K_{\mathrm{sc}}(L)(|f|_{2,h} + |b|_{0,h}). \tag{5.xxii}$$

The regional solution seminorm also retains $|Dw|_{3,h}$ and $|\sigma|_{4,h}$. For general three-row forcing $g = (g_1, g_2, g_3)$, the same inverse and the same local regularity calculation give

$$\|[\mathbb{E}^{[2]} g]_\lambda\|_{\mathcal{Z}_h} + |[\mathbb{E}^{[2]} g]_w|_{2,h} + |D[\mathbb{E}^{[2]} g]_w|_{3,h} + |[\mathbb{E}^{[2]} g]_\sigma|_{4,h} \le K_{\mathrm{sc}}(L)(|g_1|_{2,h} + |g_2|_{4,h} + |g_3|_{0,h}),$$

after enlarging K_{sc}. Indeed its first and third rows bound $\Delta_D\lambda = g_1 - w - Q'^{*}\mathsf{a}'g_3$; the rest of the Hölder argument is unchanged.

The field repaired by this scalar has the full stated norm. The bond logarithm is a holomorphic function of λ, ξa and $\xi D\lambda$, with its linear term $\xi(a - D\lambda)$. Its product and difference estimates on a fixed smaller complex ball give

$$\|a^{e^{i\lambda}} - a\|_{1,1/2,h} \le C(L)\|\lambda\|_{\mathcal{Z}_h}.$$

For its divergence use the exact formula for N above. For its curl use the exact plaquette identity

$$\partial(e^{i\xi a}U)^{e^{i\lambda}}(p) = e^{i\lambda(x_p)}\partial(e^{i\xi a}U)(p)e^{-i\lambda(x_p)},$$

before taking differences. The one-plaquette expansion in the complex potential has the form

$$F_{e^{i\xi a}U}(p) = F_U(p) + M_U(a)(p)D_1a(p) + R_U(a)(p),$$

where $M_U(a)$ and its inverse are bounded on the common ball. More explicitly, with the notation of the one-plaquette calculation, one can take $M_U(a) = c_p\mathsf{f}(-i\xi\,\mathrm{ad}\,a_1)^{-1}$, where $c_p = \frac{1}{2}\operatorname{tr}\partial U(p)$, $\mathsf{f}(z) = (e^z - 1)/z$, $\mathsf{f}(0) = 1$, and a_1 is the potential at the first positively oriented bond of the plaquette based at x_p; the terms left in R_U are linear in a with coefficient $O(\xi F_U)$, or of order at least two in the four potential values. The reference plaquette is small, so c_p stays away from zero.

Write $a' = a^{e^{i\lambda}}$ and $B = a' - a$. Subtracting the last identity and using plaquette conjugation gives

$$\begin{aligned} M_U(a')D_1B = {} & (\mathrm{Ad}(e^{i\lambda(x_p)}) - 1)F_{e^{i\xi a}U} \\ & + (M_U(a) - M_U(a'))D_1a - R_U(a') + R_U(a). \end{aligned}$$

Every term on the right contains λ or B. The quadratic holomorphic remainder has difference bound $C(|a| + |a'|)|B|$, while its curvature term costs $C\xi|F_U||B|$. The same estimates hold in the scaled Hölder norm by the product rule on the fixed comparison cubes. In these gauges a reference bond transport differs from the identity by $O_L(\alpha\xi/\ell)$; its scaled Hölder seminorm is bounded by $C(L)\alpha$, using the supremum bound at distances at least ξ. For the curvature norm on the larger cube, cover it by a fixed number of smaller comparison cubes. On overlaps the two real gauges have relative first difference $O_L(\alpha/\ell)$, because both transformed reference bond fields have the stated small bound. Product norms therefore pass between these gauges with a uniform constant. Pairs of points farther apart than a fixed fraction of ℓ are controlled by the supremum norm. The preceding bound for B and the reference curvature and D_1a bounds therefore give the degree-two supremum and Hölder estimates for D_1B. No second difference of λ is estimated by separately differencing a bond logarithm. Together with the divergence identity this proves

$$\|(a^{e^{i\lambda}} - a, 0)\|_{\mathcal{X}_h} \le C(L)\|\lambda\|_{\mathcal{Z}_h}. \tag{5.xxiii}$$

Subtracting the formulas at two scalars proves the corresponding localized difference bound on the same ball.

Combining eq. 5.xx, eq. 5.xxi and eq. 5.xxii, the nonlinear scalar map has Lipschitz constant at most $C(L)K_{\mathrm{sc}}(1 + r)\sup e$. Reduce the envelope ceiling so that this is at most $1/2$. On the unlocalized ball of radius $2K_{\mathrm{sc}}|D^*a|_{2,e}$ the same map is invariant, since $N(a, 0) = q(a, 0) = 0$; reduce the admitted radius of a so that this ball lies in the common complex ball above. Its fixed point exists and is holomorphic. For localization, retain the auxiliary variables of the scalar system. If a already lies in the slice, put $f = D^*a$. Then $Rf = 0$ and there is a top field σ_0 with $\Delta'_a f = Q'^{*}\sigma_0$. Thus $\Psi_0 = (0, f, \sigma_0)$ satisfies $\mathbb{K}^{[2]}\Psi_0 = (f, 0, 0)$. The retained inverse seminorm gives $|Df|_{3,h} + |\sigma_0|_{4,h} \le C(L)|f|_{2,h}$. Multiply the scalar and top slots by the same slow cutoff, read at their own base points. Subtract the resulting $\chi\Psi_0$ from the scalar system with first datum $D^*(\chi a)$. Its forcing is

$$([D^*, \chi]a, 0, 0) - [\mathbb{K}^{[2]}, \chi]\Psi_0.$$

It is supported in the enlarged collar where the cutoff varies: the Laplacian reads adjacent sites, and every scalar averaging row reads one tower block of bounded weighted diameter. If the cutoff has scaled first and second differences bounded by C/w and C/w^2, its commutator is bounded in degrees $(2,4,0)$ by $C(L)(w^{-1}+w^{-2})$ times the retained solution seminorm on that collar. In particular its second row uses both Df and σ_0. This is the local scalar system used in the cutoff comparison, with coefficients agreeing where the compared backgrounds agree. Use the general three-row inverse estimate above for this forcing, whose second row need not vanish. The localized inverse and the nonlinear estimates therefore localize the scalar correction. Subtracting a given small solution uses the difference estimates on the same complex ball. Together with eq. 5.xxiii, this proves the cutoff and difference assertions. □

Lemma 5.22 (The re-filing of the counterterms)**.** Let $1 \le j \le K$, $m' < m+K-j$, $b_0, b_1, \kappa' > 0$ and let c' be a complex function of the plaquettes. For a site z of the level j let $\square_z$ be the cube of π_j (of side M in level-j units) that holds z, and $X_z = \square_z^{\sim}$ the block of the $3\times3\times3$ cubes of π_j about it, the box of the site z: $X_z \in \mathcal{D}_j$, $z \in X_z$, $d_j(X_z) \le 7$ for the linear size d_j of YM3_I:72 (the definitions of $\mathcal{D}_j$ and d_j at its beginning) (Section YM3_I:72 (the definitions of $\mathcal{D}_j$ and d_j at its beginning)), every plaquette p with $h_z^{(j)}(p) \ne 0$ has its four corners in X_z and its four bonds in $\mathrm{Bd}(X_z)$, and at most M^3 sites z' have $X_{z'} = X_z$, namely those of $\square_z$ (the tent domains and support assertions are those preceding equation YM3_I:(762) and in Lem. YM3_I:49.7).

Here $h_z^{(j)}$ is the tent of the level j at z, the product of the hat functions $\max\{1-|u_\mu|,0\}$ of the coordinates in level-j units, read at the barycentres (equation YM3_I:(760)): $0 \le h_z^{(j)} \le 1$, $h_z^{(j)}(y) \ne 0$ exactly when $|y-z|_\infty < 1$ in level-j units, $\sum_z h_z^{(j)} = 1$, and $\sum_p h_z^{(j)}(p) = 3\eta_j^{-3}$ (Prop. YM3_I:49.6 and the plaquette-count calculation in the proof of Lem. YM3_I:49.7). For a complex weight c and a complex bare configuration $\mathbf{U}$ put $A^\eta(c,\mathbf{U}) = \eta^{-1}\sum_p c(p)a_p(\mathbf{U})$ (equations YM3_I:(6)–(7)), and for a triple $P = (U, A', J)$ of a level q let $\mathbf{U}_P = e^{i\eta_q A'}U$ be its configuration. The tent family at the weight c',

$$A^t[c'](X,P,z) = A^{\eta_j}(c'h_z^{(j)}, \mathbf{U}_P) = \eta_j^{-1}\sum_p c'(p)h_z^{(j)}(p)a_p(\mathbf{U}_P) \ \text{ if } X = X_z, \qquad A^t[c'](X,P,z) = 0 \ \text{ for the other } X \in \mathcal{D}_j,$$

$z \in X$ a site of the level j, is, on the cores $\mathcal{U}_j^\circ(X;b_0,b_1)$, a pointed family of the level j at (b_0,b_1) (Defs. YM3_I:49.4 and YM3_I:49.5), complex linear in c' and 0 at the trivial triple; its size YM3_I:(759), the supremum of $e^{\kappa' d_j(X)}|A^t[c'](X,P,z)|$ over the $X \in \mathcal{D}_j$, the sites $z \in X$ and the triples P of the core, obeys $\|A^t[c']\|_{j,\kappa'} \le \frac{3}{2}|c'|_\infty b_0^2 e^{7\kappa'}$; at $c' = 1$ it is the tent family of equation YM3_I:(762); and $A^{\eta_j}(c',\mathbf{U}) = \sum_z A^{\eta_j}(c'h_z^{(j)},\mathbf{U})$ for every complex $\mathbf{U}$. Consequently, let $1 \le n \le K$ with $m' < m+K-n$, and let $E^{(i)}$, $1 \le i \le n$, be pointed families of the levels i at (α_0,α_1) with complex counterterms e_i, let an be a complex weight with $|\mathrm{an}|_\infty = s$, and $\hat E^{(i)} = E^{(i)} - e_i A_i^t[\mathrm{an}]$, A_i^t the tent family of the level i; let P be a triple of one level $k \ge n$ on the whole torus whose restrictions $\rho_X P = \rho_X^{k\to i}P$ to the domains X of the levels $i \le n$ lie in the cores; since $\eta_i L^{-(k-i)} = \eta_k$, the configuration of $\rho_X P$ is $\mathbf{U}_P$ on X. Then, identically in P,

$$\sum_i\Big[\sum_z\sum_{X\ni z}\big(E^{(i)}(X,\rho_X P,z) - E^{(i)}(X,\mathbf{1},z)\big) - e_i A^{\eta_i}(1+\mathrm{an},\mathbf{U}_P)\Big] = \sum_i\Big[\sum_z\sum_{X\ni z}\big(\hat E^{(i)}(X,\rho_X P,z) - \hat E^{(i)}(X,\mathbf{1},z)\big) - e_i A^{\eta_i}(1,\mathbf{U}_P)\Big],$$

$\hat E^{(i)}(X,\mathbf{1},z) = E^{(i)}(X,\mathbf{1},z)$, and $\|\hat E^{(i)} - E^{(i)}\|_{i,\kappa'} \le \frac{3}{2}s|e_i|\alpha_0^2 e^{7\kappa'}$.

On the label $\mathcal{S}$ of a history $h \in \mathcal{H}_k$ with regular regions Λ_j, assume $m' < m+K-j$ at every level used. The same identity then holds in the following form. Let φ_j be the profile of Λ_j (equation YM3_I:(666) and the definition of φ_j following it): $0 \le \varphi_j \le 1$, $\varphi_j h_z^{(j)} = h_z^{(j)}$ at the

sites z of $\Lambda_j^{\sim -1}$, which is Λ_j less one layer of its cells, and $\varphi_j(x(p)) \neq 0$ only if all the corners of p lie in Λ_j (the tent identity in Section YM3_I:44.4; the corner assertion also follows since a nonzero profile is more than one quarter of a cell width from Z_j, while every plaquette corner is less than one bare spacing from its center). The exponent on the label (equation YM3_I:(1476)) holds, besides its Wilson term and its vacuum constant, the regular sums in equation YM3_I:(1474), over the set $\mathsf{lx}_j^{\mathrm{rs}}(\mathcal{S})$ of the places (z, X) with $X \subset \Lambda_j$ and $z \in \Lambda_j^{\sim -1}$, with the counterterms $-e_j A^{\eta_j}(\varphi_j, \cdot)$, and the boundary sums in equation YM3_I:(1475), which hold the made boundary entries $\mathscr{B}(j, X, \mathrm{mk})$ at the filing places $X \in \mathsf{lx}_j^{\mathrm{mk}}(\mathcal{S})$ and the edge parts of equation YM3_I:(1388) of the families, the members with base points outside $\Lambda_j^{\sim -1}$, at the places $X \in \mathsf{lx}_j^{\mathrm{ed}}(\mathcal{S})$ (the index sets and domains preceding Prop. YM3_I:62.2); each member is filed at exactly one place (by the separate level, domain and origin indexing in Prop. YM3_I:62.2); the entries are boundary terms of their filing places, additive in the exponent, of the sizes $\le \mathsf{B}\, e^{-\kappa d_j(X)}$ of a well filed system (Prop. YM3_I:62.2, on its stated layered output spaces); with the anisotropy the counterterms are $-e_j A^{\eta_j}((1 + \mathrm{an})\varphi_j, \mathbf{U}_P)$.

For a site y of the level j outside $\Lambda_j^{\sim -1}$ put $\epsilon_y = 1$ if $X_y \subset \Lambda_j$ and $\epsilon_y = 0$ otherwise, and

$$r_{(j,y)} = -e_j \eta_j^{-1} \sum_p \mathrm{an}(p)\,[\varphi_j(x(p)) - \epsilon_y]\, h_y^{(j)}(p)\, a_p(\mathbf{U}_P).$$

Then the exponent formed with the families $E^{(j)}$, the entries $\mathscr{B}$ and the counterterms $-e_j A^{\eta_j}((1 + \mathrm{an})\varphi_j, \mathbf{U}_P)$ equals, identically, the exponent formed with the families $\hat{E}^{(j)}$ at the regular places and in the edge parts, the entries $\mathscr{B}(j, X, \mathrm{mk}) + \sum\{r_{(j,y)} : y \notin \Lambda_j^{\sim -1},\ X_y = X\}$ and the counterterms $-e_j A^{\eta_j}(\varphi_j, \mathbf{U}_P)$. Here $r_{(j,y)} = 0$ unless the tent of y meets the set where $\varphi_j \neq \epsilon_y$, a collar of the boundary of Λ_j, and then $X_y \in \mathsf{lx}_j^{\mathrm{mk}}(\mathcal{S})$; $r_{(j,y)}$ is linear in an, reads $\mathbf{U}_P$ in X_y alone and is 0 at the trivial triple; and on the space of the filing place (j, X_y, mk), namely the layered output space at level j specified before Prop. YM3_I:62.2, where the plaquettes concerned have their plaquette variables within $r_0 \eta_j^2$ of $\mathbf{1}$, $r_0 = \mathsf{r}_0((j, \mathrm{out}); j)$ being the prescribed output-layer curvature radius at level j (the same radius used for the edge-tent estimate in the proof of Prop. YM3_I:58.22), $|r_{(j,y)}| \le \frac{3}{2} s |e_j| r_0^2$; in a frame of Thms. YM3_I:AJ and YM3_I:AD the radius of the level is at most $\alpha_0/(1 + \beta_{\mathrm{s}}) < \alpha_0$. The values at the trivial triple are unchanged. ∎

Proof. The room condition gives $2L^{m+K-j-m'} \ge 2L$ cubes of π_j in each coordinate direction. The three consecutive cubes in each projection of X_z therefore determine its middle cube. Thus $X_{z'} = X_z$ exactly for the M^3 level-j sites $z' \in \square_z$.

On the cores $|\partial \mathbf{U}_P(p) - \mathbf{1}| < b_0 \eta_j^2$ at the plaquettes with their bonds in $\mathrm{Bd}(X_z)$, and $a_p = \frac{1}{2} \det(\partial \mathbf{U}_P(p) - \mathbf{1})$ for a matrix of determinant 1, so $|a_p| \le \frac{1}{2} b_0^2 \eta_j^4$; with $h_z^{(j)} \ge 0$ and $\sum_p h_z^{(j)}(p) = 3\eta_j^{-3}$ this gives $|A^t[c'](X_z, P, z)| \le \eta_j^{-1} |c'|_\infty \cdot 3\eta_j^{-3} \cdot \frac{1}{2} b_0^2 \eta_j^4 = \frac{3}{2} |c'|_\infty b_0^2$, and $d_j(X_z) \le 7$ gives the factor $e^{7\kappa'}$. Each member is a polynomial in the entries of the $U(b)^{\pm 1}$ and $e^{\pm i \eta_j A'(b)}$ at the bonds of $\mathrm{Bd}(X_z)$, entire in (A', J), a function of the pair, gauge invariant, and 0 at the trivial triple (Lem. YM3_I:49.7).

The tents form a partition of unity, so $A^{\eta_j}(c', \mathbf{U}) = \sum_z A^{\eta_j}(c' h_z^{(j)}, \mathbf{U})$; at $\mathbf{U} = \mathbf{U}_P$ the counterterm $e_i A^{\eta_i}(\mathrm{an}, \mathbf{U}_P)$ is the sum over the sites z of the members $e_i A_i^t[\mathrm{an}](X_z, \rho_{X_z} P, z)$, which vanish at the trivial triple, and moving each from the counterterm into the family of its box is the display, term by term: the two members differ by $\sum_i e_i \eta_i^{-1} \sum_p \mathrm{an}(p)[a_p(\mathbf{U}_P) - a_p(\mathbf{U}_P)] = 0$, the counterterm and the families being read at the one configuration $\mathbf{U}_P$. On a label, by the partition of unity and $\varphi_j h_z^{(j)} = h_z^{(j)}$ on $\Lambda_j^{\sim -1}$, for every rule $\epsilon_y \in \{0, 1\}$,

$$A^{\eta_j}(\mathrm{an}\,\varphi_j, \mathbf{U}) = \sum_{z \in \Lambda_j^{\sim -1}} A^{\eta_j}(\mathrm{an}\, h_z^{(j)}, \mathbf{U}) + \sum_{y \notin \Lambda_j^{\sim -1}} \epsilon_y A^{\eta_j}(\mathrm{an}\, h_y^{(j)}, \mathbf{U}) + \sum_{y \notin \Lambda_j^{\sim -1}} A^{\eta_j}(\mathrm{an}\,(\varphi_j - \epsilon_y) h_y^{(j)}, \mathbf{U}).$$

For $z \in \Lambda_j^{\sim -1}$ the place (z, X_z) is a regular place, and for $y \notin \Lambda_j^{\sim -1}$ the domain X_y is an edge place exactly when $X_y \subset \Lambda_j$; the exponent with $\hat{E}^{(j)}$ holds $-e_j A_j^t[\mathrm{an}]$ at the regular places and in the edge parts, which are the first sum and the members of the second with $\epsilon_y = 1$: this is why $\epsilon_y = 1$ at the edge places and $\epsilon_y = 0$ elsewhere, and $-e_j$ times the third sum is $\sum_y r_{(j,y)}$, the contribution to the made boundary entries. This identity also holds at $e_j = 0$. If $\mathrm{an}(p)[\varphi_j(x(p)) - \epsilon_y] h_y^{(j)}(p) \neq 0$, the corners of p lie in X_y, and in Λ_j (by the profile corner property just stated if $\varphi_j(x(p)) \neq 0$; and if $\varphi_j(x(p)) = 0$, then $\epsilon_y = 1$ and $X_y \subset \Lambda_j$); as $\Lambda_j \subset \Omega_j$, its bonds lie in the bonds of the label in X_y, and its base point, a corner, is a site of X_y and of Ω_j, so that X_y and Ω_j have a bare cube in common; and as $y \in X_y$ lies outside $\Lambda_j^{\sim -1}$, $X_y \in \mathsf{lx}_j^{\mathrm{mk}}(\mathcal{S})$: this index set consists precisely of domains sharing a bare cube with Ω_j and not contained in $\Lambda_j^{\sim -1}$, as defined before Prop. YM3_I: 62.2.

The size: $|\varphi_j - \epsilon_y| \leq 1$, the mass $3\eta_j^{-3}$ of the tent and $|a_p| \leq \frac{1}{2} r_0^2 \eta_j^4$ on the space of the filing place give $\frac{3}{2} s |e_j| r_0^2$; in a frame of Thms. YM3_I:AJ and YM3_I:AD, $r_0 \leq \alpha_0/(1 + \beta_{\mathrm{s}})$. The tent families and φ_j, ϵ_y read no history beyond the label, and the values at the trivial triple are unchanged, the tent families vanishing there. □

5.1.8. *Proofs of the theorems.*

Proof of Thms. M–P, Q and R, and of the remarks. Step 1: Thm. M(a)–(c). (a) is Lem. 5.7 for the rough entries $F_a = \mathcal{O}(w_a)$, and its second bound is the second decomposition in the sketch of that lemma, applied to the weights $u_a \mathbf{1}\{\lambda(\cdot) = \lambda_a\}$; for a weight that vanishes the cumulant is 0. (b) Let $\ell' = 2\mathsf{l}$, $\mathsf{d} = \min\{\mathsf{d}_0, \frac{\mathsf{l}}{4n}, \frac{1}{2}\}$, and let K_1 be the least $K \geq K_0(r, \mathsf{d})$, $2 \leq r \leq 8n$, with $a_K \leq \frac{\mathsf{l}}{4n}$. For $K \geq K_1$ cut each f_i into at most $(\frac{R}{\rho} + 2)^3$ pieces in balls of radius $\rho = \frac{\mathsf{l}}{8n}$; each choice of pieces lies, for a suitable offset (a pigeonhole on the n centres), in cells of a board of side l at sup distance $\geq \frac{\mathsf{d}}{2}$ from the walls, so that the mirror copies stay $\geq \mathsf{d}$ apart, and Lem. 5.8 with (a) gives

$$\Big\| \prod_{a \leq r} (\mathcal{O}(w_a) - c_a) \Big\|_{\mathrm{ch}} \leq e^{G_g \mathsf{l}^3} \Big[\sum_{\pi} \prod_{I \in \pi} k(\#I, \mathsf{d}) \Big]^{1/8} \prod_a \sup |w_a|,$$

π over the partitions of $8r$ points into blocks of at least two; the moment–cumulant formula, $\langle F_1; \dots; F_n \rangle = \sum_\pi (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} \langle \prod_{a \in B} (F_a - c_a) \rangle$, gives a bound $T_n(\mathsf{d})$. For $K < K_1$ we use $|\mathcal{O}(f)| \leq m_K(R) \sup |f|$, $m_K(R) = 6 a_K^{-1} (\frac{2R}{a_K} + 1)^3$, increasing in K; so $C_n = \max\{(\frac{R}{\rho} + 2)^{3n} T_n(\mathsf{d}), m_{K_1}(R)^n \sum_\pi (\#\pi - 1)!\}$. On T_ℓ the same board is placed with $\ell \geq L\ell'$, and a periodic Gibbs state is dominated through its conditional laws (Lem. 5.8). The bound on $\nu' - \nu$ is the case $n = 1$ with the centring at the mean of the torus. (c) The bounds of (b) give equicontinuity in the sup norm, and a diagonal argument over a countable dense family of tuples gives a convergent subsequence; the lattice symmetries and the reflection positivity of the states pass to the limit, at the three coordinate mirrors in both cases and, for periodic Gibbs states, also at the six diagonal mirrors; the translations by all $v \in \mathbb{R}^3$ follow through $v_j \in a_{K_j} \mathbb{Z}^3$, $v_j \to v$, and the Lipschitz continuity of the limits; the growth in n follows by cutting into $\lceil \frac{2\mathsf{l}}{\mathsf{d}_0} \rceil^3$ cells and counting, with $B = 2e^{3+p} (\frac{8R}{\mathsf{l}} + 2)^3 \mathsf{d}_*^{-p} \max_{r \leq P} [e^{G_g \mathsf{l}^3} \mathrm{Bl}_{8r}^{1/8}]^{1/r}$, $\mathsf{d}_* = \min\{\mathsf{d}_0, \frac{\mathsf{l}}{4}, \frac{1}{2}\}$, $P = \lceil \frac{2\mathsf{l}}{\mathsf{d}_0} \rceil^3$; the sum over the planes is the scalar cumulant by linearity.

Step 2: Thm. M(d). Put $\sigma = \lfloor K/(1260n) \rfloor$. (i) Let $s = \lfloor \sigma/(8n + 3) \rfloor$ be the coarse depth, $T = K - s$, and $\nu = \sigma - s \geq (8n + 2)s$ the number of the levels between the fine depth σ and the coarse one. By forgetting, with the never-large term and the sources switched on at the level $K - \sigma$, each $\mathcal{O}(w_a)$ is exchanged for the weighted energy, plane by plane, of the background over the block field $\mathcal{M}^{K-\sigma} U$: what a source creates at the levels between $K - \sigma$ and T is forgotten by $L^{-\nu/2}$ over these ν levels. For the n sources, at mutual corner distances $\geq \mathsf{d}' = \frac{1}{2}\mathsf{d}$, the cumulants of the $\mathcal{O}(w_a)$

and of the energies differ by at most

$$\mathsf{K}_n\big[2L^{4n}\mathsf{V}_\ell(b)L^{-(b+1)s}+n\,C_{\mathrm{fg}}(\nu+1)^{2\mathfrak{p}_0+2}L^{-\nu/2}+2e^{3-L^s\mathsf{d}'/M}\big]\prod_a y_a,\qquad y_a\le 4\ell^3L^{4s}\sup|w_a|,$$

for every real $b\ge 4n$, with numbers K_n, C_{fg} and $\mathsf{V}_\ell(b)$ that read no K: the sup norms of the sources are paid at the coarse depth, which brings the factor L^{4ns}, and the histories with a first creation of a large field up to the level T bring $\mathsf{V}_\ell(b)L^{-(b+1)s}$; the member $2e^{3-L^s\mathsf{d}'/M}$ is the factor of the distance of the mixed parts at the level T (Lem. 5.6(a), (e)), read under the conditions $L^s\mathsf{d}'\ge 3M$ and $\mathsf{h}_2(s)\le 1$ of Lem. 5.6(e) at $t=T$, which hold for $K\ge K_1$, since s grows with K and $\mathsf{h}_2(s)\to 0$ as $s\to\infty$. At $b=4n+1$, $L^{4ns}L^{-(4n+1)s}=L^{-s}$; and $L^{4ns}L^{-\nu/2}\le L^{-s}$ since $\nu\ge(8n+2)s$; so this error is at most $C\ell^{3n}K^{2\mathfrak{p}_0+2}L^{-s}\prod_a\sup|w_a|$, the factor ℓ^{3n} of $\ell^{3(n+1)}$ in the statement coming from $\prod_a y_a$ (the further ℓ^3 comes from (iii)), and $s\ge\theta_nK-2$. (ii) For one block field at the level t and the cutoffs K, $K+1$ at equal label the plane-wise energies of the two backgrounds, formed with the coefficient function c of the weights, differ per block site by $Ct^{1/2}L^{-t/2}(\sup|\mathsf{c}|+L^{t-K}\mathrm{Lip}_\infty\mathsf{c})$.

For the sum of the three planes this is equation YM3_I:(2156). For one plane, retain the parallel-plaquette support in the proof of Prop. YM3_I:15.1. More precisely, for a set S of coarse plaquettes and a finer real field W with $\varpi=\varpi_{\mathcal{N}(S)}(W)\le\varpi'_L$,

$$\Big(\sum_{p'\in S}a_{p'}(\mathcal{M}(W))\Big)^{1/2}\le L^{1/2}\Big(\sum_{p\in\mathcal{N}^{\mathsf{W}}(S)}a_p(W)\Big)^{1/2}+\sqrt{12}\,C_0^{\mathrm{SO}}\,\varpi\Big(\sum_{p\in\mathcal{N}(S)}a_p(W)\Big)^{1/2},$$

where $\mathcal{N}(S)$ consists of the fine plaquettes in the cones of S, and $\mathcal{N}^{\mathsf{W}}(S)$ consists of those parallel to a corresponding plaquette of S. To verify the estimate, put $f_W(p)=|\partial W(p)-1|$. Lem. YM3_I:14.9 bounds $f_{\mathcal{M}(W)}(p')$ by $(\mathsf{W}f_W)(p')+C_0^{\mathrm{SO}}\varpi_{p'}^2$. The kernel in Lem. YM3_I:14.8 uses only parallel plaquettes, so its squared-norm bound is $L\|f_W\|^2_{\mathcal{N}^{\mathsf{W}}(S)}$. Equation YM3_I:(146) gives

$$\sum_{p'\in S}\varpi_{p'}^4\le 12\varpi^2\|f_W\|^2_{\mathcal{N}(S)}.$$

The triangle inequality and $f_W(p)^2=2a_p(W)$ give the displayed energy bound. Applied to the plaquettes of one plane at which the coefficient exceeds a level λ, and integrated over λ, this bounds the plane-wise energy of the block average by that of the finer background up to the factor $1+C_{\mathrm{en}}\varpi$ and up to the variation of the coefficient over a block; this is the one place where the Lipschitz constants enter, with the factor $L^{-\sigma}$. (iii) The laws of the block fields of depth at most $\lceil K/1260\rceil$ at the two cutoffs at equal label differ, on every bounded Borel gauge invariant function F, by at most $C_{\mathrm{bf}}\ell^3K^{p'}L^{-K/10}\sup|F|$: this is Cor. 2.8(a), (2.xv), with the number C_{bf} of Def. 2.7(5) and the exponent $p'=2\mathfrak{p}_0+\frac12$, the exponent defined before Thm. YM3_I:AR at $n=0$.

With $\sigma=\lfloor K/(1260n)\rfloor$ and $s=\lfloor\sigma/(8n+3)\rfloor$ the three errors are at most $L^{-s}\le L^2L^{-\theta_nK}$, $L^{4n\sigma}L^{-(K-\sigma)/2}\le L^{-K/3}$ and $L^{4n\sigma}L^{-K/10}\le L^{-K/11}$, which gives θ_n and the exponent $\max\{2\mathfrak{p}_0+2,p',1\}$ of K; it equals $2\mathfrak{p}_0+2$, since $p'=2\mathfrak{p}_0+\frac12$ and $\mathfrak{p}_0\ge1$. The comparison of two lattices at equal label, Thm. YM3_I:AR in the setting (E) of §2.3.1, applies: it asks $\mathsf{b}\ge\gamma^{-2}$, which holds because $\mathsf{b}>g^{-2}+\bar B>\gamma^{-2}$, and $MR(\mathsf{b})\le L^{m-1}$, which holds because $L^{m\mathfrak{L}(g)}\ge LM\,R(L(g^{-2}+\bar B))$ by the definition of the sides (equation YM3_I:(1528)), $\mathsf{b}\le L(g^{-2}+\bar B)$ and R increases.

For $\mathsf{b}\ge 2\max\{\gamma^{-2},x_{\mathrm{J}}\}$ and $g=(\mathsf{b}/2)^{-1/2}$ we have $g^{-2}=\frac{\mathsf{b}}{2}\ge x_{\mathrm{J}}$ and $\mathsf{b}\in I(g)$. On a compact interval of labels the numbers are monotone in b; the dilation is the change of units $x\mapsto L^zx$, which carries $\mu_{K,\ell,\beta}$ to $\mu_{K-z,L^z\ell,\beta}$ and multiplies the label by L^z. For continuous ϕ_a with supports at sup distances $\ge 2\mathsf{d}$, uniform approximation by Lipschitz functions with supports at distances $\ge\mathsf{d}$ and the bound of (b), uniform in K, make (G_K) a Cauchy sequence. With one plane field among action densities, (ii) is needed for the plane field alone, the action densities being compared at the two cutoffs by the comparison of two lattices at equal label (Thm. YM3_I:AR).

The plane field is a connected function at both cutoffs: with the tent $\theta_K(y) = \max\{0, 1 - (2/a_{K+1})\mathrm{dist}_\infty(y, \mathrm{Bc}_\lambda)\}$ about the barycentres Bc_λ of the plaquettes normal to λ at the cutoff $K+1$, $k_0 = k\,\theta_K$ is Lipschitz, $\mathrm{supp}\, k_0 \subset \mathrm{supp}\, k$, $\sup|k_0| \le \sup|k|$, and $k_0(x(p)) = k(x(p))\mathbf{1}\{\lambda(p) = \lambda\}$ at the plaquettes of both cutoffs, L being odd; so $\langle \mathcal{O}_\lambda(k); \mathcal{O}(f.)\rangle = S^T_{n+1}(f_1, \dots, f_n, k_0)$ at both cutoffs. The comparison statements on two cutoffs at equal label (Thm. YM3_I:AR with the lemmas on which it rests, Def. 2.7 and Cor. 2.8) hold, with the same proofs, at any two depths $u < s$ in the places of the depths there: their proofs read of the depths only conditions on the cutoff, which hold for $K \ge K_0$ when u and s are fixed fractions of K.

Take $u = \lceil K/(80(3n+8))\rceil$, at whose level $t_1 = K - u$ the sources are carried by the run, and $s = \lceil K/20\rceil$, at whose level $K - s$ the plane field is compared. Three members are left, each with the factor L^{3nu} of Cauchy's estimates in the n sources at the level t_1. What the plane field makes below the level $K - s$ is forgotten over the $s - u$ levels up to t_1, by $L^{-(s-u)}$, against the volume L^{3u} of the level t_1 and the size $L^{s/2}$ of its first variation at the level $K - s$: the exponent $(3n+4)u - \frac{1}{2}s \le -\frac{1}{80}K + 3n + 4$. The forgetting of the run between the two depths, against the size of the first variation at the level t_1, gives the exponent $(3n + \frac{15}{2})u - s \le -\frac{3}{80}K + 3n + 8$.

The classical comparison of the plane-wise energies of the two backgrounds at the level $K - s$, as in (ii), with the factor $K^{1/2}L^{-(K-s)/2}(\sup|k| + L^{-s}\mathrm{Lip}_\infty k)$ against the volume L^{3s}, bounds the difference of the two first variations at the real points of the polydisc of the sources; at its complex points the first variations are bounded crudely, and the estimate with two constants for holomorphic functions, with the exponents $\frac{3}{4}$ and $\frac{1}{4}$, gives the exponent $3nu + \frac{23}{4}s - \frac{3}{8}K \le -\frac{6}{80}K + 3n + 6$. The histories with a first creation of a large field up to the level t_1, and the mixed derivatives of the constants of the run in the sources, which are small by locality, cost less than every power of L^{-K}.

The largest of the three exponents gives the rate $L^{-K/80}$, and the powers of K of Cauchy's estimates in the n sources give $K^{2\mathfrak{p}_0(n+2)+2}$. For the sharpness: L being odd, the barycentres of $a_K\mathbb{Z}^3$ are barycentres of $a_{K+1}\mathbb{Z}^3$ in the same plane, of density L^{-3} among them; so the indicator of the points of a ball that are barycentres at an even cutoff and at no smaller cutoff samples to a weight of density close to 1 in the ball at the even cutoffs and close to L^{-3} at the odd ones, and for suitable smooth spectators the cumulants along the even and along the odd cutoffs have different limits.

Step 3: Thm. M(e)–(g). (e) $\mathcal{O}_\lambda(f) = \mathcal{O}(f\chi_\lambda)$, χ_λ the indicator of the plaquettes normal to λ. Prop. YM3_I:126.4, the two-point law on a torus of small side against the lattice kernel, is stated for every test pair of $\mathfrak{T}_{T_\ell}(\mathsf{d}; \vartheta)$ and every shape parameter $\vartheta \in (0,1)$, with a tolerance $\epsilon_g(s(\mathsf{d}))$ that depends neither on ϑ nor on a_K/d. A pair of non-negative Lipschitz functions, neither identically 0, carried by balls of radius d whose supports are between d and $6\mathsf{d}$ apart, is a test pair for every $\vartheta \in (0,1)$ below four positive numbers formed from their masses, their sup norms and their Lipschitz constants ([7, Lem. I.1]).

For non-negative weights w_1, w_2 carried by balls of radius $\mathsf{d} - a_K$ about x_0, y_0 with $3\mathsf{d} \le |x_0 - y_0| \le 4\mathsf{d}$, the functions $\sum_p w_a(p)\,\mathrm{pk}_p$, pk_p the tent of height 1 and sup radius $\frac{1}{2}a_K$ at $x(p)$, are Lipschitz, take the values w_a at the barycentres, so that $\mathcal{O}(\sum_p w_a(p)\mathrm{pk}_p) = \mathcal{O}(w_a)$, and form such a pair; so Prop. YM3_I:126.4 applies to them with its tolerance; real weights are differences of two non-negative ones (four terms), complex weights have real and imaginary parts (sixteen terms). This applies to the weights $f\chi_\lambda$, $h\chi_{\lambda'}$, which depend on the plane, and the kernel bounds of equations YM3_I:(2245)–(2246), with the bounds that enter the template sums of Prop. YM3_I:121.2 and Lem. 6.33, compare the lattice kernel with $\mathbb{K}_3$ plane by plane at the Lipschitz pair (f, h).

The lower bound on $\mathcal{K}_{\lambda\lambda'}$ follows from the masses and the Lipschitz constants of test pairs. For $\mathcal{O}_1-\mathcal{O}_2$: with $\hat u=u/|u|$ and $\mathbb{K}_3(u)_{\lambda\mu}=(3\hat u_\lambda\hat u_\mu-\delta_{\lambda\mu})/(4\pi|u|^3)$,

$$[\mathbb{K}_3(u)_{11}]^2-2[\mathbb{K}_3(u)_{12}]^2+[\mathbb{K}_3(u)_{22}]^2=\frac{(3\hat u_1^2-1)^2-18\hat u_1^2\hat u_2^2+(3\hat u_2^2-1)^2}{16\pi^2|u|^6}$$
$$=\frac{9(\hat u_1^2-\hat u_2^2)^2-6(\hat u_1^2+\hat u_2^2)+2}{16\pi^2|u|^6},$$

which is at least $(32\pi^2|u|^6)^{-1}$ where $\hat u_1^2+\hat u_2^2\le\frac14$, the bracket being then at least $2-\frac32$; this holds at all $u\in\operatorname{supp}f-\operatorname{supp}h$ for the test pairs of the statement, for instance for those carried by balls of radius d whose centres are $4\mathsf{d}e_3$ apart. With $|u|\le10\mathsf{d}$ there ([7, Lem. I.1](a)) and the masses $\ge\vartheta\mathsf{d}^3\|f\|_\infty$, $\vartheta\mathsf{d}^3\|h\|_\infty$ of the non-negative pair ([7, Lem. I.1]), the main term of the two-point function of $\mathcal{O}_1-\mathcal{O}_2$ is at least $\frac32\mathsf{b}^{-2}(32\pi^2)^{-1}10^{-6}\vartheta^2\|f\|_\infty\|h\|_\infty$, and its error is at most 4 times the error of the display, one for each of the four two-point functions of the planes $1,2$; so it is positive for small d, whereas a plane field that is a multiple of $\mathcal{O}$ would make $\mathcal{O}_1-\mathcal{O}_2$ vanish.

The combination $\mathcal{O}-2\mathcal{O}_m$ has coefficients of modulus 1 in three planes, whence the factor 9 for two entries. Multilinearity gives one further factor b in both the main term and the error for each entry multiplied by b. (f) For free photons, expansion of $\log\det(1-\mathsf{b}^{-1}M_\lambda)^{-3/2}$, where $M_\lambda=\sum_a\lambda_aQ^{\mathrm T}\mathrm{diag}(w_a)Q$, gives $3\mathsf{b}^{-n}\mathcal{K}_K^{(n)}$ exactly. For the interacting weights the lattice comparison is $(\mathrm{Q_n})$. The kernel comparison and bound YM3_I:(2245)–(2246), applied to the factors of each ring and telescoped plane by plane, give the displayed C_n' error; multilinearity gives the factors for plane combinations and rescaled entries. (g)(i) Lem. 5.8 on T_{ℓ_g}, applied to the Wick monomials of the tuple, whose eight mirror copies have as mean a mixture over histories, with jets that are nearly constant at the working level and with large fields that are rare on the small side.

A real multilinear form that is invariant under the lattice translations and bounded by a at every choice of signs of a packing of $N_c\sim\mathsf{d}^{-3(1-\sigma)}$ translates of the tuple in one cube of side $\sim\mathsf{d}^\sigma$ is bounded by a/N_c; this gives the factor $\mathsf{d}^{3(1-\sigma)}$. Each derivative in a source is paid by Cauchy's estimate on the polydisc of the height, which costs $(\frac43\bar{\mathsf{b}}/\mathsf{d})^{[\frac n2-1]_+}$, $\bar{\mathsf{b}}=L(g^{-2}+\bar B)$; the sizes at the working depth give a factor $\Lambda(\mathsf{d})\le C(\log\frac1{\mathsf{d}})^c$; so $\mathsf{e}_n'=2^{n-1}(\frac{42L}{M})^3(\frac43\bar{\mathsf{b}})^{[\frac n2-1]_+}[c_n^\circ Q_0\Lambda(\mathsf{d})]^n\mathsf{d}^{3(1-\sigma)-[\frac n2-1]_+}$ with $c_n^\circ=n2^{3n}(1+n4^{n-1})$ and Q_0 a constant of the stability bounds on T_{ℓ_g}.

Every periodic Gibbs state is a state at g. (ii) For real weights, the mean of the product of the eight mirror copies of the Wick polynomials Ξ_B, centred at the cumulants of $T_{\ell'}$, is exactly the sum over the partitions of the copies all of whose blocks link different copies of the products of the cumulants of the blocks (an identity of formal power series). Hypotheses $(\mathrm{Q_m})$, $2\le m\le8n$, apply to the real plaquette weights without an upper bound on their mutual distances. Together with the following Gaussian ring bound, they show that a linking cumulant is at most $\mathsf{e}_m+\mathsf{v}_m$, where $\mathsf{v}_m=\frac32(m-1)!(81(1+3C_{\mathrm{lat}}))^m(\frac{4\mathsf{d}}{\ell'})^6$ bounds the linking lattice rings, two copies in different cells being $\ge\frac14\ell'$ apart; with $\mathrm{Bl}_N(k)=\sum_\pi\prod_{I\in\pi}k_{\#I}$ over the partitions of N points into blocks of at least two, Lem. 5.8 gives the second display with $\eta^\sharp=2^ne^{G_g|^3}\max_{m\le n}\mathrm{Bl}_{8m}(\mathsf{e}+\mathsf{v})^{1/8}$ and $\mathsf{e}_n^\sharp=\mathsf{e}_n+\eta^\sharp(1+\eta^\sharp)^{n-1}$.

Step 4: Thm. M(h). (i) On the least torus a pair of narrow bumps, of width ρ at the distance d, is treated at the level $k_1=\max\{K-s(\mathsf{d}),2\}$ of the distance d, the working level of Lem. 5.6 at the depth $s(\mathsf{d})$: there the jet of each bump at the background is a constant plus an explicit quadratic form of the curvature, up to errors that the bounds of Thm. YM3_I:D make summable over the levels. The induction over the levels with the two sources runs upward from the level k_1 to the last level K, at which the block field is trivial and the quadratic form vanishes, and carries this to the Gaussian value, the explicit part $\frac32\mathsf{b}^{-2}\sum f_1f_2(C_K^{\mathrm{free}})^2$, with the error $\epsilon_2^{\mathrm{nar}}(g,\mathsf{d})(\rho/\mathsf{d})^6$ in natural sizes, $\epsilon_2^{\mathrm{nar}}\le C(g)(\log\frac1{\mathsf{d}})^{19\mathfrak{p}_0}\mathsf{d}^{1/2}$.

The remainder of the two-point function beyond the Gaussian value is bounded by placing a dummy bump at the distance ρ from the first bump, applying the bound on the remainder of three

sources of (iii) at the lengths (ρ,ρ,d), and setting the parameter of the dummy bump to 0 in the mixture over the histories; it is $\le C\mathsf{b}^2\mathsf{d}^{3/2}(\rho/\mathsf{d})^6$; the chessboard bound, read at two lengths (the fold at two lengths), carries it to every state with a further error $\le C(g)(\rho/\mathsf{d})^6\mathsf{d}^{3/2}$; the comparison of $\sum_{\lambda(p)=\lambda,\lambda(q)=\mu}\varphi_1\varphi_2(C_K^{\mathrm{free}})^2$ with $\mathcal{K}_{\lambda\mu}$ costs $C[\frac{a_K}{\vartheta\rho}+(\frac{\mathsf{d}}{\ell_g})^3](\rho/\mathsf{d})^6$.

The change of units $x\mapsto L^z x$ multiplies the label, the side and the distances by L^z, and for z large it carries a first-passage datum at $g\le\gamma$ to one at a coupling below γ_σ; this gives the narrow form at every $0<g\le\gamma$. (ii) For bumps of radius ρ the bounds of (i) are $A\rho^6\prod\|\phi_a\|_\infty$, uniform in j; a bilinear form with such bounds at all small radii has a Borel kernel with $|k|\le(\frac{3}{\pi})^2A$; the subsequence is taken at all pairs through partitions of unity; every periodic Gibbs state at $\mathfrak{D}(g)$ is invariant under the lattice translations, and the tori too, so the kernels depend on $x-y$; $\sum_\lambda\mathcal{O}_\lambda=\mathcal{O}$ gives $\sum\tau_{\lambda\mu}=\mathsf{s}_2^T$. (iii) With three sources at two scales, an induction over the levels shows: let $k_1=\max\{K-s(\mathsf{d}),2\}$ be the level of the distance d, $\mathsf{b}_{k_1}=A_1(\log x_{k_1})^{\mathfrak{p}_0}$ its threshold and $\delta_{k_1}=\delta(x_{k_1})=A_1x_{k_1}^{-1/2}(\log x_{k_1})^{\mathfrak{p}_0}\le A_1Y_{s(\mathsf{d})}^{\mathfrak{p}_0}(\mathsf{d}/\mathsf{b})^{1/2}$ (equation YM3_I:(634)), and $Z=Z_0(1+\mathsf{b}_{k_1})^2$ with a number Z_0 formed with the constants among the construction results, so that $Z^m\le C\,Y_{s(\mathsf{d})}^{2m\mathfrak{p}_0}$; if at the level k_1 each jet of the sources is a constant plus an explicit quadratic form of the curvature, up to $\epsilon_\circ\mathsf{v}_\beta$ with weights v_β whose products over partitions do not exceed v_β, then at every later level, up to the last level K, it is the Gaussian value, up to $\mathsf{a}_mZ^m\varpi\mathsf{v}_\beta$ with $(\mathsf{a}_m)=(1,14,285)$ and $\varpi\le\epsilon_\circ+C\delta_{k_1}$, provided $(K-k_1)\varpi\le1$, which holds for d below a threshold depending on g.

As $Y_{s(\mathsf{d})}\le2\log\frac{1}{\mathsf{d}}+C(g)$, the factors Z^m and $\mathsf{b}_{k_1}^2\delta_{k_1}$ give the powers of $\log\frac{1}{\mathsf{d}}$ below. The explicit Gaussian part is the ring $3\mathsf{b}^{-3}\mathcal{K}_K^{(3)}$, with the error $C(g)(\log\frac{1}{\mathsf{d}})^{39\mathfrak{p}_0+2}\mathsf{d}^{1/2}$ times the natural size $\rho^9r^{-3}\mathsf{d}^{-6}\mathsf{b}^{-3}\prod\|f_a\|_\infty$; the remainder $S_3^T-\mathsf{J}_3$ is at most $C\mathsf{b}^3\mathsf{d}^{3/2}$ times the natural size, with no power of r; the passage over the least torus costs $(\mathsf{d}/\ell_g)^3$, and the comparison of the lattice and continuum rings at two scales is explicit. As in (i), the fold at two lengths carries the three-point law at narrow bumps from the least torus to every state, with the further error $C_3^{\mathrm{tr3}}(g)\,\mathsf{d}^{3/2}\rho^9r^{-3}\mathsf{d}^{-6}\prod_a\|f_a\|_\infty$, $C_3^{\mathrm{tr3}}(g)$ a number formed with g and the constants among the construction results; E_3 contains this term, the labels at g lying in a compact interval and $\mathsf{d}^{3/2}\le\mathsf{d}^{1/2}$, and the law of (h)(iii) along the states s_j rests on it. (iv) follows from (ii), (iii) and the free identity

$$\big|\tfrac{2\pi}{3}\mathsf{b}r^3\cdot3\mathsf{b}^{-3}k_3(c+\tfrac{r}{2}n,c-\tfrac{r}{2}n,z)-\mathsf{t}^\circ(n,c-z)\big|\le2.66\,(r/\mathsf{d})^2\mathsf{b}^{-2}\mathsf{d}^{-6},$$

with $|t_\lambda(c,z)-\mathsf{t}^\circ(e_\lambda,c-z)|\le6\epsilon_\sigma(g,|c-z|)\mathsf{b}^{-2}|c-z|^{-6}$ almost everywhere by (ii) and the identity for t° of the setting; the null set lies in (c,z) alone because s_3^T is continuous. (v) The limit in K exists at disjoint supports by (d): its clause on one plane field among action densities gives the limit of $\langle\mathcal{O}_\lambda(\phi_1);\mathcal{O}(\phi_2)\rangle$ at the fixed label, and $\langle\mathcal{O}(\phi_1);\mathcal{O}(\phi_2)\rangle$ converges by the convergence of S_2^T at a fixed label. For the identification, the narrow bumps are the $f_\pm^c$ of the statement, the radial tents of radius $\rho=\frac{r}{8}$, of mass 1 and of height $3/(\pi\rho^3)=1536/(\pi r^3)$ about $c\pm\frac{r}{2}e_\lambda$; the threshold on K is $128a_K\le r$, that is $16a_K\le\rho$, and $s(\rho)\le K$, which gives $s(\frac{r}{2})\le K$, the depth function being non-increasing.

The number r_* is taken so that for $0<r\le r_*$ the depth $s=s(\frac{r}{2})$ obeys $s\ge3$ and $L^2C_W(s)Y_s^{3\mathfrak{p}_0}\le L^{s-3}$, the condition under which the family near the pair is formed, and $\frac{r}{2}\le\min\{r_0(\mathsf{b}),r_1(\mathsf{b})\}$, the radii up to which the induction of (iii) and the frame of the pair are formed, which read g and b; all three hold for small r, since $s(\frac{r}{2})\to\infty$ as $r\to0$ and the second holds at all large s; and $r_*\le\frac{1}{6}\mathsf{d}$. Write $S_3^T=\mathsf{J}_3+(S_3^T-\mathsf{J}_3)$, J_3 the explicit Gaussian part, cut ϕ_2 into pieces carried by cubes of side $\frac{1}{4}r$, of the natural size $\mathsf{b}^{-2}\mathsf{d}^{-6}r^3\|\phi_2\|_\infty$, and compare, at each $c\in\operatorname{supp}\phi_1$ and for each piece, the pair of narrow bumps at $c\pm\frac{r}{2}e_\lambda$, weighted by $\frac{2\pi}{3}\mathsf{b}r^3$, with the plane field $\mathcal{O}_\lambda-\frac{1}{3}\mathcal{O}$ at c.

For the remainder: for almost every block field, at every level n with $a_{K-n}\ge r^{1/2}$, the difference jet at the background, the jet of the weighted pair minus that of the plane field, is a constant up to $C[r^{\beta_H/72}+(a_K/r)^{1/18}]\varepsilon_n^2a_{K-n}^{-4}$, ε_n the radius of regularity of the level n and β_H the Hölder exponent; this comes from the Hölder regularity, with the exponent β_H, of the backgrounds at the

scale of the pair, carried over the levels by the induction of (iii). Integrating over the mixture of the histories on the least torus, whose moments of the scale of regularity are bounded uniformly in K, gives for the remainder the difference estimate $C(g,\mathsf{b})\mathsf{d}^{-5/2}[r^{\beta_H/72}+(a_K/r)^{1/18}]$ times the natural size.

The explicit Gaussian parts are compared on the lattice of T_{ℓ_g}, with $C(g,\mathsf{b},\vartheta_0)\mathsf{d}^{-9/2}$ and the same bracket. In the ring of the left member the near leg, between the two tents, whose plaquettes are between $\frac{3}{4}r$ and $\frac{5}{4}r$ apart, is $a_K^3\mathbb{K}_3(re_\lambda)$ up to the cost $(a_K/r)^2+(r/\ell_g)^3$, by the comparison YM3_I:(2245) of the lattice kernel with $\mathbb{K}_3$ in the setting. The two far legs a, a', from the two tents to the support of ϕ_2, enter through $\mathsf{aa}'=\frac{1}{2}(\mathsf{a}^2+\mathsf{a}'^2)-\frac{1}{2}(\mathsf{a}-\mathsf{a}')^2$: the term $(\mathsf{a}-\mathsf{a}')^2$ costs $(r/\mathsf{d})^2$ by the bound on the differences of C_K^{free} along one plane of the setting, its two plaquettes being parallel, equally oriented and at most $\frac{5}{4}r\le\frac{1}{4}(\mathsf{d}-\frac{5}{8}r)$ apart; and in each square the far leg reads one tent alone.

There the lattice sums over the tents are Riemann sums, at the cost $a_K/\rho=8a_K/r$; the entries of $\mathbb{K}_3$ are harmonic off 0, so its mean over the two radial tents, over the one through the sum over its plaquettes and over the other through the integral over c, is its value at re_λ; and the integral over c moves the argument of ϕ_1 by at most r, at the cost $r\,\mathrm{Lip}\,\phi_1\le r\|\phi_1\|_\infty/(\vartheta_0\mathsf{d})$. Finally $\frac{2\pi}{3}r^3\cdot 3\mathbb{K}_3(re_\lambda)=\frac{3}{2}(E_{\lambda\lambda}-\frac{1}{3}\mathbf{1})$, so the weighted pair becomes the plane field $\mathcal{O}_\lambda-\frac{1}{3}\mathcal{O}$ at c in the explicit part of the two-point function of the right member, whose legs are lattice kernels as well.

Each of these costs is a number formed with $g,\mathsf{b},\vartheta_0,\mathsf{d}$ times the bracket, as $a_K/r\le\frac{1}{128}$ and $r\le 1$. Summing over the at most $9^3(\mathsf{d}/r)^3$ pieces and integrating over c, against the mass $\le\frac{4\pi}{3}\mathsf{d}^3\|\phi_1\|_\infty$, gives the display of (h)(v), uniformly in K. Letting $K\to\infty$ at a fixed r, which is allowed since $128a_K\le r$ and $s(\rho)\le K$ from some K on, and then $r\to 0$ gives the identification at the fixed d.

Step 5: Thm. N(a), (b). By Lem. 5.7, $-\beta\sum_p h_{\omega,\chi}=\mathsf{b}\,\mathrm{Pf}(k^h)$, with

$$\|\mathrm{Pf}(k^h)\|\le 156|\omega|\,\mathrm{Lip}(\chi)R_1(\chi)\le 468\ell'|\omega|\,\mathrm{Lip}_\infty(\chi)$$

, and (i) is Lem. 5.7 at $r=n+1$, with $C_n^\flat(\rho)=468\ell' k_n^\flat(\rho)$, $k_n^\flat(\rho)=\max\{k'(n+1,\frac{\rho}{2}),\text{the crude bound below } K_0'\}$. Both clauses follow the one rule of the statement for a rough entry F: its norm $\|F\|$ times a number, and, free of the side and of the state, times the counts N_F and N_i of the cubes of the mesh that meet its carrier and the supports. For (ii) we put the rough entry alone in its cell of a board and the Lipschitz fields in other cells, in any mutual position.

For groups G_j of entries, $R_G=\partial_G[e^{\sum_{e\in G}z_eX_e}/\nu(e^{\sum_{e\in G}z_eX_e})]_{z=0}$ has $\nu(\prod_j R_{G_j})=\sum_\pi\prod_{I\in\pi}\kappa(I)$ over the partitions all of whose blocks meet two groups; and, for integers $n\ge 0$, $p\ge 1$ and numbers $s_1,\dots,s_n>0$, $\rho_*>0$, $Q\ge 1$, $\alpha_1,\dots,\alpha_p\ge 0$: if $A_1,\dots,A_p,B$ are holomorphic on the domain $\mathsf{D}=\{z\in\mathbb{C}^n:\sum_r|z_r|s_r<\rho_*\}$, with $B(0)=1$, $|A_j(z)|\le\alpha_jQ^{\#z}$ and $|B(z)|\le Q^{\#z}$ on D, $\#z$ being the number of the non-zero coordinates of z, then $A_1\cdots A_pB^{-p}$ is holomorphic near 0 and $|\partial_{[n]}(A_1\cdots A_pB^{-p})(0)|\le\binom{n+p-1}{n}n!\prod_j\alpha_j\prod_r(2eQs_r/\rho_*)$.

The chessboard bound of Lem. 5.8 and the bounds of Thms. YM3_I:AJ and YM3_I:AD on tori for the scalar fields, through their cumulant polynomial, with $e^{nc_V\ell'^3}$, conclude. On the larger sides the chessboard seminorm of a centred rough entry is at most $e^{G_g|^3}k_{N_0}(\mathsf{d})\|F\|$, $k_N(\mathsf{d})=[\sum_\pi\prod_{I\in\pi}k'(\#I,\mathsf{d})]^{1/N}$ over the partitions of N points into blocks of at least two, by its N_0 mirror copies on the reference side; a partition of unity by tents of the mesh $h(n,\rho)$ cuts the shell and the fields, which gives the counts N_χ,N_i; the domination of the periodic Gibbs states (Lem. 5.8) gives the states, and the change of units $x\mapsto L^zx$ gives every label and side.

The whole flux insertion is the rough entry $\mathsf{b}\,\mathrm{Pf}(k^h)+\mathcal{O}(v^{\mathfrak{B}})$, of the norm $13\mathsf{b}|k^h|_1+|v^{\mathfrak{B}}|_\infty\le(158\mathsf{b}+3a)|\omega|\,\mathrm{Lip}(\chi)R'(\chi)$, $R'(\chi)=\max\{R_1(\chi),R_1'(\chi)\}$ (Lem. 5.7), and the same rule bounds it. For $\Theta_\beta[\eta]$ we use $\Theta_\beta[\eta]=\mathsf{b}\,\mathrm{Pf}(k^\eta)$; for $\mathfrak{K}^{\mathrm{v}}[\beta\eta]$, Lem. 5.9(a) gives $\mathfrak{K}^{\mathrm{v}}[\beta\eta]=N_\beta[\eta]-\Theta_\beta[\eta]+\mathsf{b}\,\mathcal{O}(\mathrm{div}^s\eta)-\frac{3}{2}a\,\mathcal{O}(e^\eta)$, and the null term drops beside fields that do not depend on the bonds near $\mathrm{supp}\,\eta$. (b) The spin densities written in petals give $T_{lr}(\varphi)=\mathsf{b}(\mathrm{Pf}(k^{lr})+\mathcal{O}(v^{lr}))$; for two entries the two-source mixed

derivatives of Lem. 5.6(a), (e) give the bound first at $0 < g \le \gamma$ with $g^{-2} \ge x_{\mathrm{J}}$, the setting of that lemma, with the factor $(2\ell/\mathsf{d}'')^6 k_2(g, s(\mathsf{d}''))g^4$.

Every $0 < g \le \gamma$ is then reached by the change of units $x \mapsto L^z x$, which carries $\mu_{K,\ell,\beta}$ to $\mu_{K-z,L^z\ell,\beta}$ and multiplies the label by L^z: take z with $L^z\gamma^{-2} \ge 2\max\{\gamma^{-2}, x_{\mathrm{J}}\}$. The labels of $\mathfrak{D}(g)$ lie in the compact interval $[\gamma^{-2}, X_g + \bar{B}]$, whose image under the multiplication by L^z lies in $[2\max\{\gamma^{-2}, x_{\mathrm{J}}\}, \infty)$; every b' there lies inside $I(g')$ at $g' = (\mathsf{b}'/2)^{-1/2} \le \gamma$, where $g'^{-2} = \frac{1}{2}\mathsf{b}' \ge x_{\mathrm{J}}$, so finitely many intervals $I(g'_i)$ cover the image, and at the cutoffs $K > z$ the bound at the couplings g'_i applies; the cutoffs $K \le z$ are bounded crudely, by the suprema of the entries.

The limits are formed as in Step 1. For the symmetry in (l, r) of the limits: for Lipschitz φ, by Lem. 5.9(e), $(T_{lr} - T_{rl})(\varphi) = \beta \sum_b \sum_{p \ni b} (\varphi(x(p)) - \varphi(x_b))a^{-1}[(x_b - x(p))_l \mathfrak{d}_{b,p,r} - (x_b - x(p))_r \mathfrak{d}_{b,p,l}]$, which is b times a rough entry of norm $\le 15a \operatorname{Lip}\varphi$; so, by the two-entry bounds just proved, at every first-passage datum and every $0 < g \le \gamma$, both $\langle (T_{lr} - T_{rl})(\varphi); \mathcal{O}(f)\rangle$ and $\langle (T_{lr} - T_{rl})(\varphi); T_{l'r'}(\varphi')\rangle$ are $O(a)$ at separated supports. The limits are continuous in the sup norms of the entries, by the same bounds, and the Lipschitz functions are dense in the continuous ones in the sup norm; so every limit point is symmetric in (l, r).

Step 6: Thm. N(c), (d). (c) By Lem. 5.9(a) the null term drops, and $a\,\mathcal{O}(e^\eta)$ contributes $O(a)$ by Thm. M(a). By Lem. 5.9(b), $\mathfrak{K}^{\mathrm{v}}[\beta\eta] = -\mathsf{b}\sum_\rho P_\rho(\nabla^+_\rho \eta^\rho) + \beta\mathfrak{K}^\sharp[\eta]$, $P_\rho(\mathsf{k}) = a^{-1}\sum_p \mathsf{k}(x(p))\Psi_\rho(p)$: a bilinear half and a cubic half. In the bilinear half an exact identity in U, direction by direction and with one integration by parts, is used. For a real weight W of the plaquettes and a direction ρ, let (μ, ν, ρ) be the cyclic permutation of $(1, 2, 3)$ that ends with ρ; for a bare site y let $p(y; \alpha, \alpha')$ be the plaquette with the corners y, $y + ae_\alpha$, $y + ae_\alpha + ae_{\alpha'}$, $y + ae_{\alpha'}$, and $w_0(y) = W(p(y; \mu, \nu))$, the weight of the plaquette normal to ρ at y; let $g_\mu(y) = G_\rho(\langle y, y + ae_\mu\rangle)$ and $g_\nu(y) = G_\rho(\langle y, y + ae_\nu\rangle)$, the transported differences of the field strengths of the setting, and $E_\mu(y) = \sum_{p \ni b} \mathbf{w}(W_{b,p})$, the sum over the four plaquettes that hold the bond $b = \langle y, y + ae_\mu\rangle$, the lattice equation of motion at b, and $E_\nu(y)$ in the same way. Then

$$-4\sum_p W(p)\Psi_\rho(p) = \sum_y w_0(y)\big[\langle g_\mu, E_\mu\rangle + \langle g_\nu, E_\nu\rangle\big] + \mathrm{Bil} + \mathrm{Cub} + \mathrm{Q4},$$

where Bil comes from the differences of the weight W, Cub from the defect of the linear Bianchi identity and from commutators (factors $\mathrm{Ad}(W_{b,p}) - \mathbf{1}$ of plaquette holonomies), and Q4 from $a_p^2 + a_{p_+}^2$. For each plaquette $p = p(y; \mu, \alpha')$ through the bond $b = \langle y, y + ae_\mu\rangle$, the one-bond term $\beta\langle \mathbf{w}(W_{b,p}), E_\mu(y)\rangle - 3(1 - a_p)$ has vanishing cumulants with every entry that does not depend on $U(b)$ (an integration by parts in $U(b)$, as in Lem. 5.9(a)), and $g_\mu(y)$ is the sum of the $\mathbf{w}(W_{b,p})$ over the two plaquettes through b in the directions $\pm\rho$; so β times the first sum is, up to a number, $-3a\,\mathcal{O}(v)$ inside cumulants whose other entries avoid the bonds concerned, with the weight $v = w_0(y) + w_0(y + ae_\rho)$ on the plaquettes $p(y; \rho, \mu)$ and $p(y; \rho, \nu)$ and $v = 0$ on the plaquettes normal to ρ; $|v| \le 2\sup|W|$ gives the factor $\frac{3}{2}$ below; β Bil is $a\mathsf{b}\operatorname{Lip}_\infty\mathsf{k}$ times a bounded petal form; Cub and Q4 are block forms all of whose monomials have an order ≥ 3, whose sizes fall like L^{-2j} with the level j and give $L^{-(K-s-1)/4}$ at the working depth s (Lem. 5.6(c), (d)).

So, for $r \ge 1$ spectators F_q of the sizes ω_q (block forms with weights, or $\mathcal{O}(v_q)$) whose carriers are at Euclidean distances $\ge \mathsf{d}$ from one another and from $\{\mathsf{k} \ne 0\}$, at all first-passage data with $36a \le \mathsf{d}$, $K \ge s(\mathsf{d}''_1)$ and $K \ge s(\mathsf{d}'''_1)$, $\mathsf{d}_1 = \frac{3}{4}\mathsf{d}$,

$$|\mathsf{b}\langle P_\rho(\mathsf{k} \circ x); F_1; \dots; F_r\rangle| \le \Big\{a\big[\tfrac{3}{2}\sup|\mathsf{k}| + 190\,\mathsf{b}\operatorname{Lip}_\infty\mathsf{k}\big]e_2 + 50\,\mathsf{b}\,\Gamma L^{-(K-s-1)/4}e_3\sup|\mathsf{k}|\Big\}\prod_q \omega_q,$$

where $e_2 = e_2(r{+}1, \mathsf{d}_1)$ and $e_3 = e_3(r{+}1, \mathsf{d}_1)$, the cumulant having $r{+}1$ entries; for a number $r' \ge 2$ of entries and a distance d_1, $e_2(r', \mathsf{d}_1) = (3\ell'/\mathsf{d}''_1)^{3r'}k^{\mathrm{loc}}_{r'}(g, s(\mathsf{d}''_1); \ell')g^{2r'}$, $\mathsf{d}''_1 = \min\{\frac{4}{5}\mathsf{d}_1, L^{-\bar{s}(r')}\}$, with the depth $\bar{s}(r')$ of the proof of Lem. 5.7, the least integer $s \ge 3$ with the condition $2\mathsf{T}_{r'}(3)\theta(L^{s-2}g^{-2}) \le 1$ of Lem. 5.6(b), read with the series $\mathsf{T}_{r'}$ of the mixed derivatives of the order r' and the size θ stated there, and the number $k^{\mathrm{loc}}_{r'}$ of Lem. 5.6(b); $e_3(r', \mathsf{d}_1) = (3\ell'/\mathsf{d}'''_1)^{3r'}\tilde{k}_{r'}(g, s(\mathsf{d}'''_1); \ell')g^{2r'}$ is the like

number at the second depth $\mathsf{d}_1''' = \min\{\frac{3}{4}\mathsf{d}_1, L^{-\tilde{s}(r')}\}$, $\tilde{s}(r')$ the least integer $s \geq 3$ with the condition of Lem. 5.6(c) at $p = r'$, and $\tilde{k}_{r'}$ the number of the bound of Lem. 5.6(c); $\Gamma = \max\{\Gamma^\circ_{\mathrm{Cub}}, \Gamma^\circ_{\mathrm{Q4}}\}$, the larger of the comparison weights Γ°_c of Lem. 5.6(d) of the two block forms Cub and Q4; and $s = s(\mathsf{d}_1''')$ at $r' = r + 1$.

In (c) this is used with $r = n$ spectators $\mathcal{O}(f_i)$ and $\frac{1}{2}\mathsf{d}$ in the place of d, so that the numbers are formed at $(n+1, \frac{3}{8}\mathsf{d})$. These numbers depend on g, ℓ', r, d and on no K or β. The cubic half is Lem. 5.11, the spread being $\geq \mathsf{d}$; the norm $\|\cdot\|_\sharp$ and the rate $c < \frac{1}{14}$ come from it alone. (d) For the cubic half in sup norms we couple on one probability space the chain of the first-passage datum and the chain along $x_j = L^{K-j}\mathsf{b}$ of Lem. 5.10, which share only the block map, the Haar measure and the bare density, and use the law of total covariance given the block field of the level $t = K - \lceil \phi K \rceil$:

$$\langle X; F\rangle = \int (r_{XF} - r_X r_F)\, \mathsf{dm}_t + \int (r_X - \mathsf{a})(r_F - f_0)\, \mathsf{dm}_t + (\text{large fields}),$$

r_X, r_F, r_{XF} the densities at the never-large history. The first term is a mixed derivative of two separated sources, $\leq C_2 e^{3 - L^u \mathsf{d}'/M} y_X y_F$, $\mathsf{d}' = \frac{1}{2}\mathsf{d}$, $y_X = 4\ell'^3 L^{4u} \cdot 608\, \mathsf{b} L^K \sup|\eta|$, by Lem. 5.6(e) at the level t, whose conditions $L^u \mathsf{d}' \geq 3M$ and $\mathsf{h}_2(u) \leq 1$ hold for large K; the second is the oscillation of Lem. 5.10(b) times $3C_1 y_F$, $y_F = 4\ell'^3 L^{4u} \sup|w|$, and $4\phi < c_0'$; the rest is at most $\mathsf{V}_{\ell'}(\mathsf{b}) L^{-11K}$. In words (Lem. 5.14): a spectator acts on the cubic insertion only through a bounded function of the block field, and the cubic insertion is uncorrelated with every function of the block field up to its oscillation, at the price $L^{4\phi K}$, which a small ϕ pays.

For n spectators the mixed derivatives of every order of the never-large term are bounded by Cauchy's estimates, locality and a count, $e^{-L^u \mathsf{d}/2M}$ paying every power of L^K. In the bilinear half in sup norms, $\mathsf{b}\, a^{-1} \sum_p w(p) \Psi_{\lambda(p)}(p)$, summed over the plaquettes of one normal, is a dominated local source of the weight $76\mathsf{b} \sup|w|$ (Lem. 5.9(b)), and a source cut at a level, with the Hölder bound [7, Thm. BT], gives $L^{-K/200}$. For two cubic insertions the same argument is used twice, and the factor L^{4u} does not enter; two orbital insertions and two stress insertions follow with the bilinear half and the null identity.

The numbers of Lem. 5.10(b) are uniform on $[\gamma^{-2}, X_g + \bar{B}]$, which holds every label of $\mathfrak{D}(g)$. The several-source statements used above hold clause by clause in their own ranges: Lem. 5.6(a), (e), which the clauses on the cubic insertion alone read, at $0 < g \leq \gamma$ with $g^{-2} \geq x_{\mathrm{J}}$; Lem. 5.6(b), (c), which the bilinear half reads, at $0 < g \leq \gamma_W^{\mathrm{ref}}$, and so does Thm. M(a), which bounds the term $a\,\mathcal{O}(e^\eta)$. So at every first-passage datum at g on every side of $\mathfrak{L}(g)$, with numbers that depend on g and on no label, the clauses on the cubic insertion alone hold for $0 < g \leq \gamma$ with $g^{-2} \geq x_{\mathrm{J}}$, and the other clauses for $0 < g \leq \gamma_W^{\mathrm{ref}}$, where $g^{-2} \geq x_{\mathrm{J}}$ holds of itself; and $x \mapsto L^z x$ gives every fixed label and side.

Step 7: Thm. N(e)–(g). In the mean of the mirror copies $\mathrm{St}_K[\eta]$ may be replaced by $-\mathrm{Rm}_K[\eta]$ (Lem. 5.9(a) and the proof of Lem. 5.12). A partition of unity cuts η, η', H' into at most $C(R/\mathsf{d})^3$ pieces of sup diameter $\leq \frac{1}{8}\mathsf{d}$, at the cost $C\mathsf{d}^{-2}$ in the C^2 norms. For two sets of sup diameter $\leq \frac{1}{8}\delta_0$ that are δ_0 apart along one direction there is an offset in $h\mathbb{Z}^3$, $h \leq \frac{1}{64}\delta_0$, of a board of side $\frac{1}{2}\ell_g$ for which they lie in different cells at sup distance $\geq \frac{1}{16}\delta_0$ from the walls; Lems. 5.8 and 5.12 give $|\nu(A^\nu[F_1] A^\nu[F_2])| \leq 2e^{2G_g|^3} \|F_1 - c_1\|_{\mathrm{ch}} \|F_2 - c_2\|_{\mathrm{ch}}$.

Two stress insertions give $L^{-K/28}$ each, hence $L^{-K/14}$; for two conserved tensors $L^{-K/36}$ each gives $L^{-K/18} \leq L^{-K/20}$; against a plain tensor entry, which costs $L^{4u} = L^{K/100}$, the factor $L^{-K/36}$ leaves $L^{-K/60}$; and $\mathsf{b}\,\mathcal{T}_K(H^{\mathrm{cons}}[\eta]) = \mathrm{St}_K[\eta] - \Delta_K[\eta]$. For the mean of the orbital insertion, cut into the pieces of the partition of unity, $\nu(\mathfrak{K}^{\mathrm{v}}[\beta\eta]) = \nu(F_\eta)$ with $F_\eta = \mathfrak{K}^{\mathrm{v}}[\beta\eta] - \frac{1}{4}\mathfrak{N}[\mathsf{k}_\eta]$, $\mathsf{k}_\eta^\rho = \nabla_\rho^+ \eta^\rho$, since $\nu(\mathfrak{N}[\mathsf{k}_\eta]) = 0$ by Lem. 5.9(a); Lem. 5.12 at $m = 1$, $\phi = \frac{1}{400}$ gives $\|F_\eta\|_{\mathrm{ch}} \leq CK^p L^{-K/28} \|\eta\|_{C^2}$, F_η having the mean 0 on $T_{\ell'}$, and Lem. 5.8 at $q = 1$, $c_1 = 0$ gives $|\nu(\mathfrak{K}^{\mathrm{v}}[\beta\eta])| \leq e^{G_g|^3} CK^p L^{-K/28} \|\eta\|_{C^2}$. (f) n sets of sup diameter $\leq \frac{1}{4n}$ lie at sup distance $> \frac{1}{8n}$ from all the walls of some board with offset on a grid of mesh $< \frac{1}{2n}$ (a pigeonhole); several slots may share a cell (Lem. 5.12); Lem. 5.8 holds with any

number of cells; recentring from the torus mean to ν costs 2^n terms; and $4\phi(n-1) < \min\{c_0', \frac{1}{28} - 3\phi\}$ pays the factors L^{4u}.

At $n_1 = 1$, $c = \frac{1}{56}$, a fixed $\mathsf{d} > 0$ satisfies $\mathsf{d}L^u \geq K^2$ for $K \geq K_0(\mathsf{d})$. (g) $\sigma(H^{\mathrm{cons}}[\eta], H_2, \dots, H_n)$ is the limit of the left member of (f) at $n_1 = 1$, which tends to 0 at every fixed $\mathsf{d} > 0$, and disjoint compact supports have a positive sup distance. By reflection positivity of σ at the mirror θ, $\sigma(H^{\mathrm{cons}}[\theta\eta], H^{\mathrm{cons}}[\eta])$ is the square of the norm of the vector of $\mathcal{T}(H^{\mathrm{cons}}[\eta])$ in the reconstruction, which therefore vanishes. The equivalence along the sequence is Lem. 5.9(a) with the bound on Δ_K in (e), the labels lying in $[\gamma^{-2}, X_g + \bar{B}]$.

For $0 < g \leq \gamma_W^{\mathrm{sd}}$ the cited bound on centred products of tensor entries and a diagonal argument give the convergent subsequences; the covariance and the reflection positivity pass to the limits from the periodic Gibbs states, and $\mathcal{T}_K(f\mathbf{1}) = \mathcal{O}(f)$ gives the trace part.

Step 8: Thm. O(a), (b). (a) We write $\mathfrak{U}_{\omega,\chi} = (\mathfrak{U}_{\omega,\chi} - \mathfrak{U}^\sharp[\chi\,\omega y]) + \mathfrak{U}^\sharp[\chi\,\omega y]$, where $\chi\,\omega y$ is a field with $\mathsf{H}(\chi\,\omega y) \leq (2 + 3\ell)|\omega|\|\chi\|_{C^{1,1}}$; since χ covers the f_i, $f_i\chi = f_i$, and the cubic member of the bulk term is $\mathfrak{U}^\sharp[\chi\,\omega y](f)$. The first part has $2(n+1)$ members, a spin member and a bilinear orbital member for each $\varphi \in \{\chi, f_1, \dots, f_n\}$. The spin members are $-\beta\langle \mathfrak{S}_\omega[\chi]; \mathcal{O}(f_\cdot)\rangle$, with $\beta\,\mathfrak{S}_\omega[\chi] = \mathsf{b}\,a^{-1}\mathfrak{S}_\omega[\chi]$, and $\langle a^{-1}\mathfrak{S}_\omega[f_i]; (\mathcal{O}(f_l))_{l\neq i}\rangle$. By Lem. 5.9(e), $\sum_{p\ni b} a^{-1}(x_b - x(p))_\lambda \mathfrak{d}_{b,p,\rho}$ is symmetric in (λ,ρ) and ω is antisymmetric, so that

$$a^{-1}\mathfrak{S}_\omega[\varphi] = \sum_b \sum_{p\ni b} \sum_\rho k^\varphi_{b,p,\rho}\,\mathfrak{d}_{b,p,\rho}, \qquad k^\varphi_{b,p,\rho} = a^{-1}(\varphi(x(p)) - \varphi(x_b)) \sum_\lambda \omega_{\rho\lambda}\, a^{-1}(x_b - x(p))_\lambda :$$

an entry of the order 2 with real weights $|k^\varphi| \leq \frac{1}{4}|\omega|\mathrm{Lip}\,\varphi$, which vanish unless $\varphi(x(p)) \neq \varphi(x_b)$; $\beta\,\mathfrak{S}_\omega[\chi]$ is b times such an entry. By Lem. 5.9(b), and since $\nabla^+_\rho(\varphi\,(\omega y)_\rho) = (\omega y)_\rho \nabla^+_\rho\varphi$, $(\omega y)_\rho$ being free of y_ρ, the bilinear members are $\langle \mathfrak{J}^{\mathrm{bl}}[\mathsf{k}^\chi]; \mathcal{O}(f_\cdot)\rangle$ and $-\mathsf{b}^{-1}\langle \mathfrak{J}^{\mathrm{bl}}[\mathsf{k}^{f_i}]; (\mathcal{O}(f_l))_{l\neq i}\rangle$, where $\mathfrak{J}^{\mathrm{bl}}[\mathsf{k}] = \beta\sum_{p,\rho} \mathsf{k}^\rho(x(p))\Psi_\rho(p)$ and $\mathsf{k}^{\varphi,\rho}(y) = (\omega y)_\rho\nabla^+_\rho\varphi(y)$; these vector fields are Lipschitz, with norm $\leq 5|\omega|(1+\ell)\|\varphi\|_{C^{1,1}}$, which is where the norms $\|f_i\|_{C^{1,1}}$ enter, and supported in the a-neighbourhood of $\mathrm{supp}\,\varphi$.

By Lem. 5.9(c), $\mathfrak{J}^{\mathrm{bl}}[\mathsf{k}] = -\frac{1}{4}\mathfrak{N}[\mathsf{k}] + \sum_{v=1}^5 \mathfrak{J}_v[\mathsf{k}] - \frac{1}{4}c_{\mathsf{k}}$: a null term, entries of the orders $2,2,2,3,4$ with weights of the order of the norm of k and, for the orders $3,4$, of β times it, and a number, which drops from the cumulants. In (i) let $D \geq \mathsf{d}$ be the spread of the supports. Two rules on spreads are used. A family of sets has a spread at least that of each of its subfamilies of two or more sets. And if $2r' < D$ and a set T lies within the sup distance r' of one of the sets and of each set of an index set Q, then T is empty when these are all the sets, and otherwise T and the remaining sets have a spread $\geq D - r'$: points of T and of the remaining sets that were closer would give points of all the sets that are pairwise closer than D.

With χ in the first entry, the carrier of the entry and the supports have a spread $\geq D$. With f_i in the first entry, the carrier lies within $\frac{3}{2}a$ of $\mathrm{supp}\,f_i$ and the cumulant is taken with the $\mathcal{O}(f_l)$, $l \neq i$: the spread is $\geq D - \frac{3}{2}a$. The null terms meet the sources: the covering fixes χ at the barycentres and at the bond midpoints of the plaquettes of the supports, while k^χ reads χ at the points $y + ae_\rho$ as well, so that $\mathrm{supp}\,\mathsf{k}^\chi$ may meet a support; and $\mathrm{supp}\,\mathsf{k}^{f_i}$ may meet $\mathrm{supp}\,f_l$. They are taken apart by the contact formula of Lem. 5.9(f): by (N3) the cumulant of $\mathfrak{N}[\mathsf{k}]$ with the sources of a set A is a sum of cumulants of the entries $\mathfrak{N}[f_{q_1}\cdots f_{q_{j-1}}\mathsf{k}; f_{q_j}]$ with the remaining sources, and by (N4) each such entry is a number, the density a_p and densities of the order 2, with no factor β, carried within the sup distance $2a$ of $\mathrm{supp}\,\mathsf{k}$ and of the supports that it absorbs.

The spread of $\mathrm{supp}\,\mathsf{k}^\chi$ and the supports is $\geq D$, and that of $\mathrm{supp}\,\mathsf{k}^{f_i}$ and the $\mathrm{supp}\,f_l$, $l \neq i$, is $\geq D - a$; once these exceed $4a$, the terms that absorb all the sources vanish, the numbers drop from the others, and by the merging of sets of Lem. 5.9(f) every remaining entry has with the remaining supports a spread $\geq D - 2a$, resp. $\geq D - 3a$. So in every term that is left the spread is $\geq \mathsf{d} - 3a$, which is $\geq \frac{1}{2}\mathsf{d}$ once $6a \leq \mathsf{d}$, and Lem. 5.15 gives $L^{-K+(\frac{1}{2}-c_*)K} \leq L^{-c_*K}$ for the orders 2

and $L^K \cdot L^{-\frac{3}{2}K+(\frac{1}{2}-c_*)K} = L^{-c_*K}$ for the orders $3, 4$. The cubic part is Lem. 5.11, whose spread condition holds because the support of the field $\chi\,\omega y$ is added to the supports, whose spread is D.

In (ii), $n = 1$: the members with f_1 in the first entry are means, of the spin insertion and of the bilinear density $\mathfrak{K}^{\mathrm{v}} - \mathfrak{K}^\sharp$, and vanish by the mirror symmetries of the measure (Lem. 5.9(b)); in the members with χ the weights vanish unless χ varies across the plaquette or the bond, so, χ being 1 on the d-neighbourhood of $\operatorname{supp} f_1$, their carriers are at sup distance $\geq \mathsf{d} - 2a$ from $\operatorname{supp} f_1$, the null term meets no source, and Lem. 5.15 applies to the pair. The cubic member is $\mathfrak{U}^\sharp[\chi\,\omega y](f_1) = -\beta\langle\mathfrak{K}^\sharp[\chi\,\omega y]; \mathcal{O}(f_1)\rangle$, the contact member $a^{-1}\langle\mathfrak{K}^\sharp[f_1\chi\,\omega y]\rangle$ being a mean of $\mathfrak{K}^\sharp$, which vanishes (Lem. 5.9(b)).

Here $\operatorname{supp} f_1$ lies in the set where the field does not vanish, and a periodic field is subtracted: let s be the ℓ-periodic continuous function with $\mathsf{s}(\tau) = \tau$ for $|\tau| \leq \frac{\ell}{2} - \delta_0$ that is affine on $[\frac{\ell}{2} - \delta_0, \frac{\ell}{2} + \delta_0]$, and $\zeta^\rho(y) = \sum_\lambda \omega_{\rho\lambda}\mathsf{s}(y_\lambda)$. Then $\zeta = \omega y$ on the closure of the cube of the margin, ζ^ρ does not depend on y_ρ, $\sup|\zeta^\rho| \leq |\omega|\ell$ and $\operatorname{Lip}_\infty \zeta^\rho \leq |\omega|\ell/\delta_0$. As $\zeta^\rho(y + se_\rho) = \zeta^\rho(y)$ for all y, s and ρ, the clause of Lem. 5.9(b) on such fields gives $\langle\mathfrak{K}^\sharp[\zeta]; \mathcal{O}(f_1)\rangle = 0$: for each direction ρ and each plaquette q, the mirror normal to ρ through the barycentre of q keeps the measure, a_q and ζ^ρ, and changes the sign of $\mathfrak{K}^\sharp[\zeta^\rho e_\rho]$.

The field $\eta' = \chi\,\omega y - \zeta$ vanishes on the d-neighbourhood of $\operatorname{supp} f_1$, where $\chi = 1$ and $\zeta = \omega y$; so $f_1\eta' = 0$, $\mathfrak{U}^\sharp[\chi\,\omega y](f_1) = \mathfrak{U}^\sharp[\eta'](f_1)$, and Lem. 5.11 applies, η' vanishing within d of the support, with $\mathsf{H}(\eta') \leq |\omega|[(\ell+2)\sup|\chi| + \ell\operatorname{Lip}_\infty\chi + \ell + \ell/\delta_0]$: here the margin δ_0 enters C. In (iii), $\chi - \chi'$ vanishes within d of all the supports, so the exact identity gives $\langle\mathfrak{B}_{\omega,\chi} - \mathfrak{B}_{\omega,\chi'}; \mathcal{O}(f_\cdot)\rangle = \beta\langle\mathfrak{S}_\omega[\chi - \chi'] + \mathfrak{K}^{\mathrm{v}}[(\chi - \chi')\omega y]; \mathcal{O}(f_\cdot)\rangle$, an insertion that vanishes near the sources, and no condition on the supports is needed.

The limit of the flux follows from the exact identity, $\nabla^s_\omega f = (\omega y)\cdot\nabla f + D_f$ with $\|D_f\|_\sharp \leq C_0 M|\omega|(1+\ell)a\|f\|_{C^3}$ by Taylor's formula, the bound eq. 5.i on connected functions at separated supports, and the convergence of S^T_n at a fixed label with the rate $K^p L^{-K/10}$ (Cor. YM3_I:117.1). For the quarter turn: $e^{\frac{\pi}{2}\omega_{13}}$ is a signed permutation of the axes, an exact symmetry of the lattice measure on T_ℓ, so $S^T_n(f^{\pi/2}) = S^T_n(f)$; since $\frac{d}{dt}f^t = -(\omega y)\cdot\nabla f^t$, $\int_0^{\pi/2}\sum_i S^T_n(\dots, (\omega y)\cdot\nabla f^t_i, \dots)dt = 0$ at every K; with the exact identity and $\nabla^s_\omega f^t - (\omega y)\cdot\nabla f^t = O(a)$ this is the first display, and in the limit the second. (b) In $\mathfrak{U} - \mathfrak{U}^\sharp$ the entries sit where χ_R varies, in $\{\frac{4}{3}R \leq |y| \leq \frac{5}{3}R\}$ up to $2a$, or near one of the supports, which lie in $\{|y| < \frac{R}{2}\}$ at mutual distances $\geq \mathsf{d}_0$; all the boxes and supports lie in the sup ball of radius $2R$ about 0, and in each cumulant the carrier of the entry is at a sup distance $\geq \min\{\frac{5R}{6\sqrt{3}}, \frac{\mathsf{d}_0}{\sqrt{3}}\} - 2a \geq \mathsf{d}_{\mathrm{b}} := \frac{1}{2}\min\{\frac{5R}{6\sqrt{3}}, \frac{\mathsf{d}_0}{\sqrt{3}}\}$ from every source of that cumulant, for K large, the cumulants with f_i in the first entry being taken with the other sources.

So Lem. 5.15 in its form on all the sides, at the radius $2R$, the distance d_{b} and on the sides $\ell \geq \ell'$ with $\ell' \geq 4R + 2\mathsf{d}_{\mathrm{b}}$, gives the bracket with a number free of the side, the finitely many smaller sides being covered by the form on one torus; the spin part comes with $L^{-(\frac{1}{2}+c_*)K}$. The cubic member $\mathfrak{U}^\sharp[\chi_R\,\omega y](f)$ has its entry on top of the sources, so no wall parts them; it is Lem. 5.16, with Lem. 5.11 for the finitely many smaller sides. Weak limit points of the torus laws inherit the bounds. Along a sequence with converging connected functions, the exact identity gives $\mathfrak{U}^{(j)} + \langle\mathfrak{B}; \mathcal{O}(f_\cdot)\rangle^{(j)} = \sum_q S^T_n(f_1, \dots, \nabla^s_\omega f_q, \dots) \to -\frac{d}{dt}S_n(f(e^{-t\omega}\cdot))|_{t=0}$, and $\mathfrak{U}^{(j)} \to 0$.

For the volume-first limits Thm. YM3_I:AP gives invariance under O(3), so the flux tends to 0. With the limits one after the other: at a label $\mathsf{b}' \in I(g)$, on each torus T_ℓ, $\ell \in \mathfrak{L}(g)$, (a) gives the limit in K of the flux, $\sum_i S^{(\ell,\mathsf{b}')}_n(f_1, \dots, (\omega y)\cdot\nabla f_i, \dots, f_n)$, through every enclosing shell. Along the sides $\ell_k \to \infty$ the convergence $S^{(\ell_k,\mathsf{b}')}_n \to S^\circ_n$ is the hypothesis of the clause; the functions $(\omega y)\cdot\nabla f_i$ are Lipschitz, with the supports of the f_i, so the limit in k is $\sum_i S^\circ_n(f_1, \dots, (\omega y)\cdot\nabla f_i, \dots, f_n)$; and this is $-\frac{d}{dt}S^\circ_n(f(e^{-t\omega}\cdot))|_{t=0}$, as S°_n is multilinear and bounded per ball in the norms $\|\cdot\|_\sharp$, the per-ball bound of Prop. YM3_I:91.13 passing to the limit once $\ell \geq 16n(R+1)$ and the fixed supports lie in

the central cube, and $t \mapsto f_i(e^{-t\omega}\cdot)$ is differentiable in these norms with the derivative $-(\omega y)\cdot\nabla f_i$ at $t = 0$.

The integers $K_k^* \ge k$ are chosen so that for $K \ge K_k^*$ and $n \le k$ the connected functions on T_{ℓ_k} at the label b' lie within $\frac{1}{k}$ of $S_n^{(\ell_k,\mathsf{b}')}$ at all tuples of norms $\|\cdot\|_\sharp \le 1$ with supports in the ball of radius $R_k = \min\{k, \frac{1}{8}\ell_k\}$ about 0 at mutual distances $\ge \frac{1}{k}$, by the rate $K^p L^{-K/10}$ of Cor. YM3_I:117.1; every tuple of the clause lies, up to a factor, in these classes from some k on, so along every sequence $K_k \ge K_k^*$ the connected functions converge to S_n°. For given f_i and χ_k, raising K_k^* makes the bulk term on T_{ℓ_k} at most $\frac{1}{k}$ as well, by (a), and the flux converges along such sequences to the limit of the display.

For slowly growing tori, the threshold in the cutoff comparison must also be uniform in the side. Fix $n \ge 2$ and a positive separation d. Take the label interval

$$J = [\mathsf{b}_-, \mathsf{b}_+], \qquad \mathsf{b}_- = g^{-2} + B_\sharp - L^{-1}\bar{B} \ge g^{-2}, \qquad \mathsf{b}_+ = L(g^{-2} + \bar{B}).$$

It contains all first-passage labels by Cor. YM3_I:74.2. For large sides $\ell = 2L^m$, $MR(\mathsf{b}_+) \le L^m$, so the interval is admitted by Thm. YM3_I:AR. Its cutoff comparison uses

$$\phi_1 = \frac{1}{180(3n+4)}, \qquad u = \lceil \phi_1 K \rceil, \qquad s = \lceil K/9 \rceil.$$

All its smallness conditions are products of powers of ℓ and quantities decaying exponentially or faster in K, except for the volume factor in the discarded mass. The proof of that theorem writes the latter majorant as

$$\bar{\mathsf{R}} = 2M^{-3}\ell^3 L^{3u} \exp\Big[-c' A_0^2 (\log(L^u \mathsf{b}_-))^{2\mathfrak{p}_0} + C_3 \ell^3\Big], \qquad C_3 = \mathsf{C}^\Sigma_{\mathrm{maj}} + 2 + C_{\mathrm{low}}(\mathsf{b}_+).$$

Put

$$c_\ell(n,g) = \left[\frac{c' A_0^2 (\phi_1 \log L)^{2\mathfrak{p}_0}}{2C_3}\right]^{1/3}.$$

If $\ell \le c_\ell(n,g) K^{2\mathfrak{p}_0/3}$, the positive volume exponent uses at most half of the negative exponent, since $\mathsf{b}_- \ge \gamma^{-2} \ge 1$ and $\log(L^u \mathsf{b}_-) \ge \phi_1 K \log L$. As $2\mathfrak{p}_0 > 1$, the remaining bound decays faster than every exponential in K, uniformly over these sides. The other comparison conditions then hold beyond a threshold independent of the side; they continue to hold at every $K' \ge K$ with the side fixed. Thm. YM3_I:AR and Cor. YM3_I:117.1 give

$$\sum_{K' \ge K} |S^T_{n;K'+1,\ell} - S^T_{n;K',\ell}| \le C\ell^{3(n+1)} K^{\mathfrak{p}_n} L^{-K/10} \prod_i \|f_i\|_\sharp, \qquad \mathfrak{p}_n = 2\mathfrak{p}_0(n+1) + \tfrac{1}{2}.$$

Under the stated side bound this is at most $C' K^{2\mathfrak{p}_0(n+1)+\mathfrak{p}_n} L^{-K/10} \prod_i \|f_i\|_\sharp$, which tends to zero. Thus the finite-cutoff connected functions and their spacing limits on those tori have the same limit, and increasing the cutoffs changes the values by at most the same tail.

For completeness, the label bounds follow directly from the canonical flow, as in Prop. YM3_I:74.1 and Cor. YM3_I:74.2. Put $\mathcal{W}_K(g) = \{\beta : (K,\beta) \in \mathfrak{D}(g)\}$ and $X_g = L(g^{-2} + B_\sharp)$. The flow has increments of modulus at most $B_\sharp$ and satisfies

$$|x_j(\beta) - L^{-j}\beta| \le \bar{B}(1 - L^{-j}).$$

At its first passage, $x_K(\beta) \le X_g < x_{K-1}(\beta)$. Thus, for $\mathsf{b} = L^{-K}\beta$,

$$L\mathsf{b} > X_g - \bar{B}, \qquad \mathsf{b} \le X_g + \bar{B} - \bar{B}L^{-K} \le X_g + \bar{B}.$$

These imply the outer inclusion because $L^{-1}(X_g - \bar{B}) = g^{-2} + B_\sharp - L^{-1}\bar{B}$ and $X_g + \bar{B} = L(g^{-2} + \bar{B})$. To prove the inner inclusion, take $\mathsf{b} \in I(g) = (L^{-1}(X_g + \bar{B}), X_g - \bar{B}]$ and put $\beta = L^K \mathsf{b}$. The first passage exists at some $K' \ge 1$, since $\beta > X_g$. Its label $\mathsf{b}' = L^{K-K'}\mathsf{b}$ obeys the preceding two inequalities. If $K' \ge K+1$, then $L\mathsf{b}' \le \mathsf{b} \le X_g - \bar{B}$, a contradiction. If $K' \le K-1$, then

$\mathsf{b}' \geq L\mathsf{b} > X_g + \bar{B}$, also a contradiction. Hence $K' = K$. Existence and uniqueness of the first passage also show that the sets $\mathcal{W}_K(g)$ are disjoint and have union (X_g, ∞).

Step 9: Thm. O(c)–(e). (c) By Lem. 5.9(e), $T_{mm}(\varphi) - \mathsf{b}\check{D}_m(\varphi)$ is β times orbital potentials in the plane, which are $-\frac{1}{4}|G_m(b)|^2$ up to densities of the order 4, plus β times densities of the order 4 with weights $\leq \sup|\varphi|$, plus $\mathsf{b}\,\mathcal{O}(\tilde{\varphi}_m)$ with $|\tilde{\varphi}_m| \leq \frac{1}{2}a \operatorname{Lip}\varphi$. Lem. 5.17(c) bounds the first by $CK^pL^{-c_nK}\sup|\varphi|\prod_i \|f_i\|_\sharp$, Lem. 5.17(a) the second by $CK^pL^{-\lambda_L K/4}$, and Lem. 5.17(b) the third by $C_\epsilon a^{1-\epsilon}\operatorname{Lip}\varphi$. The sum over m gives the trace. By Lem. 5.9(e), $\sum_{p\ni b} a^{-1}(x_b - x(p))_\lambda \mathfrak{d}_{b,p,\rho}$ is symmetric in (λ,ρ), so $(T_{lr} - T_{rl})(\varphi) = \beta\sum_b \sum_{p\ni b}(\varphi(x(p)) - \varphi(x_b))a^{-1}[(x_b - x(p))_l \mathfrak{d}_{b,p,r} - (x_b - x(p))_r \mathfrak{d}_{b,p,l}]$, an insertion of the order 2 with weights $\leq \beta a \operatorname{Lip}\varphi$, and Lem. 5.17(b) at $\sigma = 0$ gives $O(a^{1-\epsilon})\operatorname{Lip}\varphi$; at $\sigma = 1$ it gives $C_\epsilon a^{-\epsilon}\sup|\varphi|$ for T_{lr} itself.

By Lem. 5.9(b), $\mathfrak{K}^{\mathrm{v}}[\beta\eta] - \beta\mathfrak{K}^\sharp[\eta] = -\mathsf{b}\sum_\rho P_\rho(\nabla^+_\rho \eta^\rho)$, whose in-plane potentials are $-\frac{1}{4}|G|^2$ up to the order 4 and whose normal potentials are $\Psi_{\lambda(p)}(p)$: Lem. 5.17(a), (c). The limit at a fixed label follows from the clause of Thm. M(d) on one plane field among action densities, with the convergence of the connected functions of $\mathcal{O}$ at a fixed label, since $\check{D}_m = \mathcal{O} - 2\mathcal{O}_m$. (d) One lattice step along e_r is $U(b) \mapsto W(\pm r, \mu_b)U(b)$ up to a gauge, and for every gauge invariant Φ

$$a\, v_{e_r}\Phi = \tfrac{1}{2}(\mathrm{Sh}_r^+ - \mathrm{Sh}_r^-)\Phi + \mathcal{P}_r\Phi - \mathrm{Sh}_r^-\mathcal{P}_r\Phi + \mathcal{R}_r\Phi, \qquad |\mathcal{P}_r\Phi| \leq C\bar{a}, \quad |\mathcal{R}_r\Phi| \leq C\bar{a}^{3/2},$$

$\mathrm{Sh}_r^\pm$ the shifts by $\pm a e_r$ and $\bar{a}$ the largest plaquette term near the support of Φ. With the null identity of Lem. 5.9(a), its contact form and (c), every member of the Ward identity for the field $\chi\xi$ but $\mathfrak{U}^\sharp[\chi\xi](f)$ is its stated limit up to $O(K^pL^{-c_nK})$, and $\mathfrak{U}^\sharp[\chi\xi](f)$ is bounded by Lem. 5.11, the spread being positive (the field $\chi\xi$ vanishes near the supports of the f_i with $i \notin I$, and near the boundary of the chart for $\xi = \omega y, y$). For $\xi = y$, $\partial_l y^r = \delta_{lr}$, so the contraction with T_{lr} contains the trace, which by (c) is $\mathsf{b}\,\mathcal{O}$; this gives the terms $|I|S_n - \mathsf{b}S_{n+1}(\chi, f)$. For $\xi = e_r$ and χ a function of y_3 alone, $\sum_l T_{lr}(\xi^r \partial_l \chi) = T_{3r}(\chi')$. (e) The equality is the exact identity for the constant field e_ρ (Lem. 5.9(a) with the contact terms; the stress functional of a constant field vanishes). Put $(\tau_\pm f)(y) = f(y \pm a e_\rho)$, $g_i = \nabla^s_\rho f_i$, $f_i^e = \frac{1}{2}(\tau_+ f_i + \tau_- f_i)$ and $h_i = (f_i^e - f_i)/a^2$. The measure is invariant under the lattice translations, so $S_n^T(\tau_+ f) = S_n^T(\tau_- f)$; with $\tau_\pm f_i = f_i^e \pm a g_i$ this gives $0 = 2\sum_{|J|\ \mathrm{odd}} a^{|J|} S_n^T(g_J, f^e_{J^c})$, and with $f^e = f + a^2 h$,

$$\sum_i S_n^T(g_i, (f_l)_{l\neq i}) = -\sum_{|J|\geq 3\ \mathrm{odd}} a^{|J|-1} S_n^T(g_J, f^e_{J^c}) - \sum_i \sum_{J'\neq\emptyset} a^{2|J'|} S_n^T(g_i, h_{J'}, f_{\mathrm{rest}}),$$

at most $2^n + n2^n$ terms, each with a factor a^k, $k \geq 2$. The functions g_i and h_i are averages of translates of $\partial_\rho f_i$ and of $\frac{1}{2}\partial_\rho^2 f_i$, so their norms $\|\cdot\|_\sharp$ are bounded by those of these derivatives, and their supports lie in the a-neighbourhoods of the supp f_i; the bound eq. 5.i on connected functions at separated supports bounds each term. At $n = 2$, $\langle \mathcal{O}(\tau_\pm f_1); \mathcal{O}(f_2)\rangle = \langle \mathcal{O}(f_1); \mathcal{O}(\tau_\mp f_2)\rangle$ by the invariance, and the two members cancel exactly: the statement has content for $n \geq 3$ and expresses the invariance under lattice translations and the failure of the Leibniz rule for ∇^s_ρ. For the second display, the contact form of Lem. 5.9(a) at the field χe_ρ, where $\Theta_{f_i}[\chi e_\rho] = 0$ because $\chi(x_b) = \chi(x(p))$ at every plaquette p at which f_i does not vanish and at its four bonds b, and the vector weight $f_i\chi e_\rho$ is $f_i e_\rho$ for $i \in I$ and 0 otherwise, gives the left member as $-\sum_{i\in I} S_n^T(\ldots, \nabla^s_\rho f_i, \ldots)$; and $\nabla^s_\rho f - \partial_\rho f$ is an average of translates of $\partial^3_\rho f$ with the total weight $\frac{a^2}{6}$.

Step 10: Remark 5.3. For a trilinear form that is invariant under the lattice translations along e_ρ, write the functions through their Fourier coefficients at $k_1 + k_2 + k_3 = 0$ and put $z_i = e^{iak_i}$; then ∇^s_ρ and ∇^+_ρ act by $(z - z^{-1})/2a$ and $(z-1)/a$, and since $z_1z_2z_3 = 1$,

$$\prod_{i=1}^3 (z_i - 1) = \sum_i z_i - \sum_{i<j} z_i z_j = \sum_i (z_i - z_i^{-1}),$$

which is the displayed identity. Prop. 5.18(c) gives $a^3 S_3^T(g_1,g_2,g_3) \to 6\mathsf{b}'^{-3}\int g_1 g_2 g_3$ for Lipschitz functions, without a separation assumption. Thus the two sides can have order a^{-1} when the supports meet. In the rotational identity the flux through a shell separated from the supports stays bounded, by Thm. N(a), whereas the difference between the symmetric and continuous generators contains three derivatives.

We first give the free-photon calculation. Its local part requires a difference estimate for the curvature kernel. For equally oriented parallel plaquettes p, p', put $u = x(p) - x(q)$, using a least lift. If $|x(p) - x(p')| \le |u|/4$ and $|u| \ge 8a$, then

$$|C_K^{\mathrm{free}}(p,q) - C_K^{\mathrm{free}}(p',q)| \le C_\nabla a^3 |x(p) - x(p')|\,|u|^{-4}, \qquad C_\nabla = 256\sqrt{3} C_{\mathrm{h}}(1 + 3C_{\mathrm{lat}}). \tag{5.xxiv}$$

Here C_{h} is an absolute constant in the interior gradient estimate for a scalar discrete harmonic function; Thm. 6.3.8(a) of [5] gives this estimate, with the finitely many small radii absorbed into the constant. Indeed, $dd^* + d^*d$ acts on each plane component as the scalar lattice Laplacian and commutes with the projection onto exact 2-forms. Applied to the column $C_K^{\mathrm{free}}(\cdot, q)$, it gives $dd^*\delta_q$, which vanishes beyond the plaquettes sharing a bond with q. The column is therefore harmonic at distance at least $2a$ from $x(q)$. Join $x(p)$ to $x(p')$ by a lattice path of length at most $\sqrt{3}|x(p) - x(p')|$. This path stays at distance at least $|u|/2$ from $x(q)$. On balls of radius $|u|/4$ about its points, equation YM3_I:(2246) bounds the column by $4^3(1 + 3C_{\mathrm{lat}})a^3|u|^{-3}$. The gradient estimate bounds each lattice step by $4^4 C_{\mathrm{h}}(1 + 3C_{\mathrm{lat}})a^4|u|^{-4}$. Summing along the path proves eq. 5.xxiv.

Let the real functions f_i have compact supports inside the margin and bounded derivatives through order five, and let $\chi = 1$ on a neighbourhood of these supports. Write θ_p for the three-colour Gaussian curvature of covariance $\beta^{-1} C_K^{\mathrm{free}}$. In this model

$$a_p = \tfrac{1}{2}|\theta_p|^2, \qquad \mathbf{w}(W_{b,p}) = \pm\theta_p, \qquad \Psi_\rho(p) = -\tfrac{1}{4}|\theta_{p_+} - \theta_p|^2.$$

The sign records the plaquette orientation. The spin densities and the other insertions are quadratic. The linear Bianchi identity gives $\mathfrak{r}^\sharp = 0$, hence $\mathfrak{U}^\sharp = 0$, and Wick's formula gives the connected functions as rings. The exact Ward identity remains valid after removing the terms $\frac{3}{2} a\,\mathcal{O}(\cdot)$ coming from the curvature of $SU(2)$; against separated sources these terms are $O(a)$. The expression with the continuous generator is therefore, up to $O(a)$,

$$\mathfrak{U}_{\omega,\chi}(f) - \sum_i S_n^T(\ldots, \nabla_\omega^s f_i - (\omega y)\cdot\nabla f_i, \ldots). \tag{5.xxv}$$

The terms with χ in the first slot have their quadratic insertion separated from the supports. The ring formula makes them $O(a)$. The terms with f_i in that slot are local. For the bilinear part their kernel is

$$M_\rho(p,q) = [C_K^{\mathrm{free}}(p_+,q) - C_K^{\mathrm{free}}(p,q)]^2.$$

Equation eq. 5.xxiv bounds it by $Ca^8|u|^{-8}$ for $|u| \ge 8a$; at the remaining distances use the kernel bound and the projection bound on the diagonal. The spin part has one additional factor a and kernels $C_K^{\mathrm{free}}(q,q')C_K^{\mathrm{free}}(q',p)$, where the plaquettes q meet a bond of p.

Expand the test functions about the midpoint of $x(p), x(p_+)$ in the first case and about x_b in the second. In lattice units these kernels have uniformly bounded moments through orders four and two, respectively; the third-order tail in the second case is even and has degree -6. Together with

$$\nabla_\omega^s f - (\omega y)\cdot\nabla f = \frac{a^2}{6}\sum_\rho (\omega y)_\rho \partial_\rho^3 f + O(a^4),$$

the expansion writes eq. 5.xxv as local forms with factors $a^{-3}, a^{-2}, a^{-1}, a^0$, plus an $O(a)$ remainder for $n = 2$ and an $O(a\log(1/a))$ remainder for $n = 3$. The infinite-lattice coefficients are independent of a by scaling. The torus correction changes each term by $O(a)$. At order three, configurations with two nearby entries and a third at distance $|u|$ use the fourth moment of a kernel of degree -7. Their contribution is bounded by a times the sum of $a^3|u|^{-3}$ over $a \le |u| \le 1$, which gives the logarithm.

Five derivatives suffice for these Taylor remainders and for the $O(a^4)$ error in replacing the lattice sum of $(\omega y)\cdot\nabla\prod_i f_i$ by its integral. The constants use the suprema of derivatives through order five.

All four local terms vanish. Inversion through the centre of the chart preserves the Gaussian measure and the expression, and removes the odd orders in this local expansion, namely the coefficients of a^{-2} and a^0. The coefficient of a^{-3} is a multiple of

$$\int (\omega y)\cdot\nabla\prod_i f_i = 0,$$

since $\operatorname{div}(\omega y)=0$. For the coefficient of a^{-1}, use the other expression for the same quantity: the variation of S_n^T under continuous rotations minus the bounded shell flux. The potentially divergent terms are rotation derivatives of cubic-invariant local forms with at most two derivatives. A cubic-invariant symmetric tensor of rank two is a scalar multiple of the identity, so these forms are rotation invariant. At $n=2$ the terms can also be written as combinations of

$$\int (\omega y)_\rho \partial_\rho(\partial_\rho f_1\partial_\rho f_2)=0, \qquad \int\big[(\omega y\cdot\nabla f_1)\Delta f_2+(\omega y\cdot\nabla f_2)\Delta f_1\big]=0.$$

The first identity uses that $(\omega y)_\rho$ is independent of y_ρ. The first cubic-invariant local form that need not be rotation invariant has four derivatives. Thus only the stated remainders remain. Finally, $a^3S_3^T\to 6\mathsf{b}^{-3}\int$ in the Gaussian model and

$$\sum_i\int(\omega y)_\rho\partial_\rho^3 f_i\prod_{l\neq i}f_l = 3\int(\omega y)_\rho\prod_i\partial_\rho f_i.$$

Multiplying the difference of the generators by a gives the asserted limit of $a\mathfrak{U}_{\omega,\chi}$.

At order one the neighbourhood condition has a concrete role. Take

$$S=[-1,1]\times[-2,2]\times[-1,1], \qquad \kappa(y)=3-y_1y_2, \qquad f_1=\kappa\ \operatorname{dist}(\cdot,\partial S)\text{ on }S, \qquad f_1=0\text{ off }S.$$

The faces of S are lattice planes at every cutoff, and f_1 is Lipschitz with a kink at its boundary. Let $\chi=\varsigma(\operatorname{dist}(\cdot,S))$, where $\varsigma\in C^{1,1}$ equals $1-\mu s^2/2$ for $0\le s\le s_0$, with $\mu>0$, and decreases to zero inside the margin. Then $\chi=1$ on S but on no neighbourhood of it. It covers f_1: the barycentre and bond midpoints of every contributing plaquette lie in S. Choose $\omega_{21}=-\omega_{12}=1$, with the other entries zero. The contact terms are means of the spin and vector insertions and vanish by reflection, as in Lem. 5.9(b). Hence

$$\mathfrak{U}_{\omega,\chi}(f_1)=-\beta\langle\mathfrak{S}_\omega[\chi]+\mathfrak{K}^{\mathrm{v}}[\chi\omega y];\mathcal{O}(f_1)\rangle.$$

Its bilinear part is

$$-\tfrac34\mathsf{b}^{-1}\sum_{p,q,\rho}(\omega x(p))_\rho\nabla^+_\rho\chi(x(p))f_1(x(q))M_\rho(p,q), \qquad M_\rho\ge 0.$$

Away from the edges of a face with normal $n_F=\pm e_\nu$, only $\rho=\nu$ contributes. The factor $-(n_F)_\nu\nabla^+_\nu\chi$ is μa times a nonnegative function of the exterior lattice distance; f_1 is κa times the interior lattice distance. Summing the kernel over pairs across the plane and then over the a^{-2} plaquettes per unit area gives

$$\tfrac34\mathsf{b}^{-1}\mu c_{\mathrm{lat}}\int_{\partial S}(\omega y)\cdot n_F\,\kappa\,\mathrm{d}S.$$

The positive constant c_{lat} is the pair sum per unit lattice face. Cubic symmetry makes it the same for every face; its convergence is bounded by $\int_0^\infty\int_0^\infty st(s+t+1)^{-6}\,\mathrm{d}s\,\mathrm{d}t$. Pairs a fixed distance apart contribute $O(a^2)$. The contribution near the edges vanishes with the width of the edge neighbourhood; the spin part is $O(a\log(1/a))$ by its $a^6|u|^{-6}$ kernel, and the torus correction is $O(a)$. The divergence theorem gives

$$\int_{\partial S}(\omega y)\cdot n_F\,\kappa\,\mathrm{d}S=\int_S(\omega y)\cdot\nabla\kappa=\int_S(y_2^2-y_1^2)=16.$$

Consequently $\mathfrak{U}_{\omega,\chi}(f_1) \to 12\mathsf{b}^{-1}\mu c_{\mathrm{lat}} > 0$.

For the $SU(2)$ example, choose an even smooth function $\psi : \mathbb{R} \to [0,1]$, equal to one on $[-1,1]$ and zero off $[-2,2]$, and put $\mathsf{c} = 100$ and

$$\Pi_{(r_1,r_2,r_3)}(y) = \psi((y_1 - \mathsf{c})/r_1)\psi(y_2/r_2)\psi(y_3/r_3).$$

Set $g_3 = \Pi_{(1,1,1)}$, $g_2 = \Pi_{(3,3,3)}$, $g_1 = \Pi_{(7,7,7)}$, and

$$G_i(y) = \int_{-\infty}^{y_2} g_i(y_1, t, y_3)\,\mathrm{d}t, \qquad f_i = G_i(\cdot + 30e_2) + G_i - 2G_i(\cdot - 30e_2) \quad (i \le 3).$$

For $3 < j \le n$ take $f_j = \Pi_{(15,45,15)}$ and take $\chi = \Pi_{(31,91,31)}$, with the same ω as above and $\delta_0 = \mathsf{d} = 1$. These fixed smooth data have compact supports with a common point. The cutoff is one on their closed unit neighbourhood and vanishes outside $\{|y|_\infty \le 182\}$, inside the margin of every admitted side, since its exponent m is at least seven.

Reflection in $y_1 = \mathsf{c}$ removes the rotation about the axis through $\mathsf{c}e_1$; the remaining vector field is $\mathsf{c}e_2$. Translation invariance therefore writes the left side of the exact identity as

$$\frac{\mathsf{c}}{2a} \sum_{|J| \ge 3} (|J| - 2) a^{|J|} S_n^T((\nabla_2^+ f_i)_{i \in J}, (f_j)_{j \notin J}).$$

The bulk term is this expression minus the flux. The latter has a quadratic insertion, weights $O(\beta)$ and distance at least $1/2$ from the supports. Lem. 5.15, at $c_* = 1/4$, bounds it by $CK^{(2n+1)\mathfrak{p}_0} L^{K/4}$. If J contains an index above three, its derivative entry vanishes near the support of f_3, so the same bound applies. For $J = \{1,2,3\}$ the three derivatives are translates of $g_i^K = a^{-1}\int_0^a g_i(\cdot + te_2)\,\mathrm{d}t$ with coefficients $(1,1,-2)$. Replace the spectators by one, discard triples with different translates, and replace g_1^K, g_2^K by one. Each error has an entry separated by at least $1/2$ and is again covered by that lemma. The equal translates sum to $1 + 1 - 8 = -6$. With $I_K = (a^3/3)\sum_p g_3^K(x(p))$, translation invariance gives

$$S_n^T(g_3^K, 1, \dots, 1) = 3a^{-n-3} I_K (-1)^{n-1} E^{(n-1)}(\beta).$$

Thus, up to $CK^{(2n+1)\mathfrak{p}_0} L^{-3K/4}$,

$$a_K \mathfrak{U}_{\omega,\chi}(f) = 900 I_K (-1)^n \mathsf{b}'^{-n} \beta^n E^{(n-1)}(\beta).$$

The inner and outer boxes of g_3^K give $1 \le I_K \le 125$, and its Lipschitz approximation gives $I_K \to \int g_3$. Prop. 5.18(b) gives $(-1)^n \beta^n E^{(n-1)}(\beta) = -(n-1)! + O(K^{2n+3} L^{-K/20})$, uniformly in $\mathsf{b}' \in I(g)$. Finally $900/2 = 450$ and $900 \cdot 125 \cdot 2 \le 2.3 \cdot 10^5$ give the asserted two-sided bounds.

Step 11: Thm. P. Let a board of side $a_i \le \frac{\ell'}{2}$ be placed, $\{x_3 = c\}$ one of its wall planes, θ_c the mirror at it, $\tau_{a_i e_3}$ the translation by $a_i e_3$, and let $\tilde{w}$ be a real weight carried by the cell between $\{x_3 = c\}$ and its translate by a_i that vanishes within $\frac{1}{2}\mathsf{d}$ of the walls. The centred entry $D = \mathcal{O}_K(\tilde{w}) - c_0$, c_0 its mean on $T_{\ell'}$, has the chessboard seminorm $\|D\|_{\mathrm{ch}} \le \mathsf{q}_i(\mathsf{d}) \sup|\tilde{w}|$ for $K \ge K^{\#}(i,\mathsf{d})$, with $\mathsf{q}_i = e^{G_g a_i^3} \mathrm{Bl}_{N_i'}(k(\cdot,\mathsf{d}))^{1/N_i'}$, $N_i' = (\ell' L^i)^3$ and $\mathrm{Bl}_N(k) = \sum_\pi \prod_{I \in \pi} k(\#I, \mathsf{d})$ over the partitions of N points into blocks of at least two: the mean on $T_{\ell'}$ of the product of its N_i' mirror copies is, by the moment–cumulant formula, the sum over these partitions of products of cumulants of copies that are $\ge \mathsf{d}$ apart, each bounded by Lem. 5.7 at the orders $\le N_i'$ (Thm. M(a)), and the passage between the sides costs $e^{G_g a_i^3}$ (Lem. 5.8). A law ν of the lattice that is invariant under its symmetries, reflection positive at the wall planes of the boards, and dominated by the torus measures through its conditional laws obeys the reflection inequality

$$|\nu(\overline{\theta_c F'}\, D\, \tau_{a_i e_3} F)| \le \|D\|_{\mathrm{ch}}\, \|F'\|\, \|F\|, \qquad \|F\| = \nu(\overline{\theta_c F} F)^{1/2},$$

for bounded local functions F, F' of the bonds in the half space $\{x_3 > c\}$: the form $(F,G) \mapsto \nu(\overline{\theta_c F} G)$ is positive, and the Cauchy–Schwarz inequality for it, repeated at the walls of the board on the dominating tori (the chessboard estimate, Prop. YM3_I:86.3, with the Cauchy–Schwarz inequalities of Prop. YM3_I:91.4), bounds the left member by the chessboard seminorm of D read along these

tori. Every periodic Gibbs state has the three properties: it is invariant under the lattice symmetries, it is reflection positive at all the families of mirrors (Prop. YM3_I:103.4), and it is dominated by the torus measures (Prop. YM3_I:91.9). Recentring from c_0 to $\nu(\mathcal{O}_K(\tilde w))$ costs $\mathsf{q}_i(\mathsf{d})\sup|\tilde w|\,\|F'\|\,\|F\|$ more, by Lem. 5.8 at $q=1$ and $|\nu(\overline{\theta_c F'}\,\tau_{a_i e_3}F)|\le\|F'\|\,\|F\|$; whence the factor 2:

$$|\nu(\overline{\theta_c F'}\,[\mathcal{O}_K(\tilde w)-\nu(\cdot)]\,\tau_{a_i e_3}F)|\le 2\mathsf{q}_i(\mathsf{d})\sup|\tilde w|\,\|F'\|\,\|F\|\qquad(K\ge K^{\#}(i,\mathsf{d}));$$

eight shifted boards cover all positions. Along subsequences the limits exist (Thm. M(c)), and in the reconstruction the bound is what Lem. 5.13 asks, and that lemma gives $B^{(m)}$, the density and the L^1 bound. The reconstruction of the limits of Gibbs sequences at $0<g\le\gamma$ has $H\ge 0$ (Thm. YM3_I: AP), the hypothesis of Lem. 5.13 in the direction x_3; and it has $H\ge|P_k|$, $k=1,2$, for the commuting momenta P_k that generate the translations $U(\vec x)$ in the directions orthogonal to e_3, the hypothesis of the last sentence of that lemma in the directions x_1,x_2: every periodic Gibbs state obeys the five-point inequality ([7, eq. 241]) of Lem. 4.46 at the mirror normal to e_3 and the direction e_k, which passes to the limit and puts the spectral measure of every class on $\{\cosh(tE)+\cos(tq_k)\ge 2\}$ for a dense set of $t>0$, hence on $\{E\ge|q_k|\}$, as $\cosh(tE)+\cos(tq_k)=2+\frac12 t^2(E^2-q_k^2)+O(t^4)$.

The same holds for the reconstruction of a tensor limit. So Lem. 5.13 applies in the three directions, which gives the density in all three coordinates and the L^1 bound in them. For the bound on $\|B^{(m)}\|$: for $g\le\gamma_W^{\mathrm{sd}}$ the chessboard seminorm of a centred tensor entry in a cell of side a_i is at most $2\mathsf{C}_T Y_i^{\mathsf{q}_T} a_i^{-11/2}\mathsf{b}^{-1}|H|_\infty$, uniformly in the cutoff, the datum, the board, the side and the state ([7, §2], [7, eq. 4] of [7, Thm. B](a)); with Lem. 5.13 every tensor limit has the insertion bound with every $p>\frac{17}{2}$, and $B^{(m)}$ is the compression to the scalar subspace of the point operator at $e=\mathbf{1}-2E_{mm}$, $|e|=\sqrt3$; that every limit with converging labels arises from a tensor limit along a further subsequence is the extraction of Thm. N(g).

For the behaviour under turns: by [7, §5], every tensor limit at $0<g\le\gamma_W^{\mathrm{sd}}$ obeys $\sigma(RX)=\sigma(X)$, $R\in\mathrm{O}(3)$, at tuples with one non-scalar entry among scalar ones ([7, Thm. U], which gives [7, Thm. R](c) at these tuples); $R(\mathbf{1}-2E_{33})R^{\mathrm{T}}=\mathbf{1}-2nn^{\mathrm{T}}$ with $n=Re_3$, and the sum over $\pm\alpha$ cancels the off-diagonal part of nn^{T}, leaving $2\cos^2\alpha(\mathbf{1}-2E_{33})+2\sin^2\alpha(\mathbf{1}-2E_{ii})$; the entries of nn^{T} are combinations of 1, $\cos2\alpha$ and $\sin2\alpha$, which gives the form $c_0+c_1\cos2\alpha+c_2\sin2\alpha$ and the second derivative.

Step 12: Thm. Q. (a) Put $\lambda=L^{-(k-j)}$, $d=d_j(Z)$ and $S=\sup|E|$; the left member is at most $2S$, and the numbers a_3, C_2^Q, $\bar C_2=\max\{C_2^Q,1\}$, c_3^Q, $c_s=c_s^Q$ and λ_Q are those of the extended list. The trivial cases come first: if $\lambda>\lambda_Q$, then $2S\le 2\lambda_Q^{-4}\lambda^4 S$; if $(d+4)\lambda>c_s$, then $2S\le 2((d+4)\lambda/c_s)^4 S\le 32c_s^{-4}(d+2)^4\lambda^4 S$, as $d+4\le 2(d+2)$. Let now $\lambda\le\lambda_Q$ and $(d+4)\lambda\le c_s$. Then $\lambda\le 1/(4a_3)$, and on a cube that holds the box of Z, whose sites lie within the ℓ^1-distance $l=3(d+4)M$ (level-j units) of its least site, the definition of the cores gives a gauge in which the real part of the potential obeys $|A|\le\lambda$ and $|\nabla A|\le\lambda^2$ in level-j units; with the clause on the complex part this gives a complex potential A^0 of the sizes λ and λ^2, up to the complex gauge transformations $g_t=e^{itm}$, $0\le t\le 1$, with $|m|\le l\lambda a_3\le 3Ma_3c_s\le\frac14$, as $c_s\le 1/(12Ma_3)$, so of the size $e^{1/4}$.

The constant part, of the first order, is removed by g_1: a term is invariant under real gauge transformations and holomorphic, so it is invariant under every complex gauge transformation that is reached by a path inside its domain, and the path $t\mapsto g_t$ stays there, its triples lying in the cores and below the third radius. What is left, $\tilde A$, vanishes at the least site of the cube and has the gradient $|\nabla\tilde A|\le C_2^Q\lambda^2\le\bar C_2\lambda^2$, so that $|\tilde A|\le l\bar C_2\lambda^2=3(d+4)M\bar C_2\lambda^2$ on the cube; the disc $\zeta\mapsto(\mathbf{1},\zeta\tilde A,\zeta\tilde J)$ of the radius $R=c_3^Q/(l\bar C_2\lambda^2)=c_3^Q/(3(d+4)M\bar C_2\lambda^2)$, of the order $((d+2)\lambda^2)^{-1}$, stays in the domain, because the Hölder number depends only on the curl of the potential, up to squares, and scales with the power $\frac52>2$ under $\rho_Z^{k\to j}$; at the trivial field the first derivative of a gauge invariant term vanishes, since the adjoint representation of SU(2) has no invariant vector; and the Schwarz

lemma gives $|E((\mathbf{1},\tilde{A},\tilde{J}))-E(\mathbf{1})|\le 2SR^{-2}=2l^2\bar{C}_2^2(c_3^Q)^{-2}\lambda^4 S\le 72M^2\bar{C}_2^2(c_3^Q)^{-2}(d+2)^2\lambda^4 S$, since $l\le 6(d+2)M$ (if $R\le 1$ the bound $2S\le 2SR^{-2}$ holds already).

This is the clause with $(d+2)^2$, and with the trivial cases it gives

$$C_Q=\max\{2\lambda_Q^{-4},32c_s^{-4},72M^2\bar{C}_2^2(c_3^Q)^{-2}\}$$

and the power $(d+2)^4$. For $\mathrm{U}(1)^3$ the field strength scales like λ^2 and the linear term survives. The power 4 is attained by a term of the second order in the field strength at one plaquette, against a triple with a constant field strength of the largest admitted size. (b) A block of the level k holds $L^{3(k-j)}$ regions of the level j, each constant up to $L^{-4(k-j)}$ by (a); the count of regions among the construction results with the rate $\kappa_{\mathrm{lf}}\ge 4\log 18432$ of (b) gives the bound, uniformly in k. (c) The complex centre of each Gaussian integral is kept as a free argument, one 1-form for every earlier level, in a polydisc.

The first order of a plane weight gives $\sup|w|L^{-5/2}\mathrm{hol}$, hence the third radius, which is small at the true points and stable under the shift system among the construction results; the weight and the trees enter as potentials, so that the action of the step carries no explicit weight; a tree of the level j weighs L^{k-j+1} seen from the level k, there are $L^{3(k-j)}$ of them and each is smaller by $L^{-5(k-j)}$; the non-classical terms stay at their levels and are lifted afresh at every step by (b), at the rate r_1, which (b) admits since $r_1>\mathsf{C}_{\mathrm{cs}}>4\log 18432$. Each step is then the identity of the statement, classical times ideal times the exponential of a tiny term, with $c_k(W)\Xi_0[V_{\mathsf{m}}](Q)=\Phi_{k+1}[\omega](W)$; at $k=0$ the step is written with bond labels.

On the plain cores the Hölder number is at most α_0 times a factor that grows like $L^{n/2}$ with the level, while $\alpha_2/\alpha_0\ge 4.99\cdot 10^6 e^{L/4}$; so the condition of the third radius follows from the plain core at the levels $n\le(\frac{L}{2}+29.39)/\log L$. At $k=0$ the weight enters linearly through the bond labels, and the centre, the solution of the linear system of the step with a source of the size σ, is at most $K_Z\sigma$. The extended list, stated before Thm. Q, places each new constant after the constants on which it depends and before γ; its four new lower bounds are laid on the chosen constants C_{ex}, κ, $\bar{\kappa}$ and M, which are free upward, each by a function of earlier constants, its new conditions on γ are ceilings whose left members tend to 0, and it lays no condition on L itself; so it keeps the admissible shape, its choices exist, and they restrict to choices of the lists of Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD.

At every choice $L>4\cdot 10^{19}$: the retained floor $L\ge L_\Pi=2\mathsf{C}_{\mathrm{ex}}(1+4\beta_{\mathrm{s}})/(1-2\delta_{\mathrm{w}})\ge 2\mathsf{C}_{\mathrm{ex}}$ (equation YM3_I:(904), with $\beta_{\mathrm{s}}>0$ and $0<\delta_{\mathrm{w}}<\frac{1}{2}$), the new lower bound $\mathsf{C}_{\mathrm{ex}}\ge 2\mathsf{C}_{\mathrm{cs}}$, and $\mathsf{C}_{\mathrm{cs}}\ge 24C_{\mathrm{E}}^2e^{24}$ by the choice preceding equation YM3_I:(1332), $C_{\mathrm{E}}=4096$, give $L\ge 4\mathsf{C}_{\mathrm{cs}}\ge 96C_{\mathrm{E}}^2e^{24}>4\cdot 10^{19}$. The further conditions named in the statement are of three kinds. The tower hypotheses of §5.2 and the inequality $\alpha_0+\alpha_1\le a^\sharp$, even $2(\alpha_0+\alpha_1)\le a^\sharp$, are closed inequalities between constants that hold at every choice of the list of Thm. YM3_I:D, and so at every choice of the extended list, which restricts to such a choice.

The standing conditions of the frame of [7, Def. B.14] follow, condition by condition, from members of the lists of Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD, to which the choices of the extended list restrict. The locality for all pairs of regions and the gauge covariance at all decoupling parameters are properties of the Gaussian structure among the construction results as it is formed, at all admitted constants: its structures form a local system, and they are covariant under real gauge transformations at all decoupling parameters s of the closed polydisc (Thm. YM3_I:U(2) and the covariance of the operators in YM3_I:(765): restrictions commute with the unitary gauge actions, as do inversion and the functional calculus used to form the covariance and coupling).

Step 13: Thm. R. (a) Prop. 5.5 proves the uniform critical-background and Hessian assertions. We give the remaining comparisons and check the actual step data.

At $U_0=1$ with all regions the whole torus, the quadratic form for plane weights is

$$\langle B,Z_\epsilon B\rangle=\min_{Qd=B}\{(\partial d,\mathsf{a}\,\partial d)+\|RD^{\mathsf{T}}d\|^2\},\qquad 1-\epsilon\le\mathsf{a}\le 1+\epsilon.$$

Each form before minimization lies between $1-\epsilon$ and $1+\epsilon$ times the unweighted form, including its nonnegative slice term. Taking the infimum gives the claimed comparison of block Hessians. For the single mixed tensor term, $2|\langle f_{\mu\nu}, f_{\mu\lambda}\rangle| \le |f_{\mu\nu}|^2 + |f_{\mu\lambda}|^2$ gives the same form comparison in the stated normalization. Adding the square of the linearized block averages preserves it. Thus the linearization of the constrained system is invertible at zero. On each fixed finite lattice the analytic implicit-function theorem gives the stated family near zero.

The reason a uniform Hölder estimate needs the Hölder norm of the weight can be seen in this same quadratic problem. Fix $B \in Y_0$, $B \neq 0$, and write $h_\epsilon = d_b A \cdot B$ at $b = 0$. Take the torus side $N = 2L^{m+K-k}$ in level units, as in the construction; it has the even number $N/\eta_k = 2L^{m+K}$ of bare sites in each direction. Choose a plane $(\mu\nu)$ in which $F^0 = \partial h_0$ has largest component norm, and put $v_{\mu\nu}(x) = \epsilon(-1)^{x_\mu/\eta_k}$, with all other plane weights zero. Then

$$Qh_\epsilon = B, \quad RD^{\mathsf{T}} h_\epsilon = 0, \quad \partial^{\mathsf{T}} \mathsf{a}\, \partial h_\epsilon = Q^{\mathsf{T}} Z_\epsilon B.$$

The supremum of the source on the right is at most $(1+\epsilon) C_{\mathrm{tr}} c_Z \|B\|$, with $c_Z \ge \|Z_0\|$ and the row-to-supremum norm C_{tr} of Q^{T} depending only on L. Since $\delta = h_\epsilon - h_0$ has $Q\delta = 0$, subtracting the two variational equations and testing by δ gives

$$\|\partial\delta\|^2 + \|RD^{\mathsf{T}}\delta\|^2 = -\epsilon\big((-1)^{x_\mu/\eta_k}(\partial h_\epsilon)_{\mu\nu}, (\partial\delta)_{\mu\nu}\big).$$

It follows that

$$\|(\partial h_\epsilon)_{\mu\nu}\| \ge (1+\epsilon)^{-1}\|F^0_{\mu\nu}\| \ge (1+\epsilon)^{-1}(\gamma^{\mathrm{pc}}/3)^{1/2}\|B\|.$$

On the torus of side N in level units, its supremum f_ϵ is therefore at least $(1+\epsilon)^{-1}(\gamma^{\mathrm{pc}}/(3N^3))^{1/2}\|B\|$. At two neighbouring sites in direction μ, use the bond equation at the site carrying the factor $1+\epsilon$. It expresses $2\epsilon(\partial h_\epsilon)_{\mu\nu}$ as η_k times the source minus two first differences of curvature components, with coefficients at most $1+\epsilon$ and 1. Consequently

$$2\epsilon f_\epsilon \le (2+\epsilon)\,\mathrm{osc}(\partial h_\epsilon) + \eta_k(1+\epsilon) C_{\mathrm{tr}} c_Z \|B\|.$$

Here osc is the largest modulus of the change of one curvature component between neighbouring bare sites. For $\eta_k \le c_\eta \epsilon N^{-3/2}$ this gives

$$[\partial h_\epsilon]_{\beta_H} \ge \frac{1}{6}\Big(\frac{\gamma^{\mathrm{pc}}}{3N^3}\Big)^{1/2} \epsilon \eta_k^{-\beta_H} \|B\|.$$

Here the seminorm includes neighbouring sites. At fixed N this diverges as $k \to \infty$, although the weight has supremum ϵ. Since the background vanishes at zero block datum, the example concerns its derivative there. Cauchy's estimate would bound that derivative if the backgrounds had a uniform Hölder bound on a common block-datum ball.

The constant-weight assertion follows from the exact identity

$$F + T_{(h,v+c)} = (1+c)(F + T_{(h,v)/(1+c)}).$$

Multiplying the critical equation multiplies its multiplier by $1+c$ and leaves its constraint and gauge unchanged. Uniqueness in Prop. 5.5 then gives the asserted identity of critical fields.

For the history application use the collared sequence before the step and its completion by the next large-field region. The background of the run is its real unweighted critical configuration at the prescribed averages. The block field A_k is read about its finer average in the axial gauge, and its row datum is $b = JA_k$, extended by zero to the remaining rows. The regular-background and chart estimates for these actual step fields, Lem. YM3_I:37.5, Prop. YM3_I:53.12, and the fibre-chart bounds in [7, §B.2], give

$$\alpha \le 2C_{\mathfrak{P}}\mathfrak{M}B_3\varepsilon(x_t), \qquad \sup|b| \le 2C_\chi \mathsf{c}_{\mathrm{fl}}\delta(x_k), \quad C_{\mathfrak{P}} = 504L^2B_1^{\mathrm{p}}.$$

For the natural-class assertion, apply Lem. YM3_I:37.5 with the separated witness $m = \mathfrak{M}$, increasing the common block floor if necessary. The layerwise regularity radius of the real background

is at most $2B_3\varepsilon(x_t)$, so that lemma gives the displayed natural-class radius on the completed sequence. The chart converts the fluctuation field to the row datum with the displayed factor $2C_\chi$. Choose

$$a_N \le \min\{a_*, a^{\mathrm{pc}}, \gamma^{\mathrm{pc}}/(4e_0)\}, \quad \rho_N^+ \le \frac{\gamma^{\mathrm{pc}}\rho_*}{4\kappa_0}, \quad z_N^+ \le \frac{\gamma^{\mathrm{pc}} z_*}{4\kappa_0},$$

also keeping $\rho_N^+ < \rho_*$ and $z_N^+ < z_*$. Impose the ceilings, for all $x \ge \gamma^{-2}$,

$$2C_{\mathfrak{P}}\mathfrak{M}B_3\varepsilon(x) \le a_N, \qquad 2C_\chi \mathsf{c}_{\mathrm{fl}}\delta(x) \le \rho_N^+.$$

Their left sides tend to zero with γ. Thus $e_0\alpha \le \gamma^{\mathrm{pc}}/4$ and $\kappa_0 s \le \gamma^{\mathrm{pc}}/4$ at every step $t \ge 2$, which gives the asserted half-floor.

At the first step $k = 0$, the lattice spacing is one, the block averages are the identity and the slice projection vanishes. There is no minimization at this step. At the real regular reference the fluctuation floor differs from the action Hessian by the second derivative of the block constraint paired with the current. Its loss is at most $8c_{\mathrm{pc}}\alpha$, where $c_{\mathrm{pc}} = \sqrt{2}C_K n_{\mathrm{pc}}$ and $n_{\mathrm{pc}} = 36L^3$. For a complex block field $A^{(0)}$ with $r_A = \sup|A^{(0)}|$, differentiating the four link factors in each plaquette and summing their incident bonds gives the following three operator-norm bounds: the change in the plain action Hessian is at most $128 r_A e^{4r_A}$; the tensor term has norm at most $512 e^{8r_A}\|h\|_\infty$; and the plane term has norm at most $32 e^{4r_A}\sup|v|$. Therefore

$$\begin{aligned}\mathrm{Re}(F + T_{\mathbf{h}})''(JA^{(0)})[J\bar{B}, JB] \ge (\gamma^{\mathrm{pc}} - 8c_{\mathrm{pc}}\alpha - 128 r_A e^{4r_A} \\ - 512 e^{8r_A}\|h\|_\infty - 32 e^{4r_A}\sup|v|)\|B\|^2.\end{aligned}$$

This uses the real natural pair on the two-region sequence (Ω_0, Ω_1) and $\mathfrak{M} \ge M^{\mathrm{pc}}$, exactly as in the unweighted positivity argument above. Take

$$a_N^1 \le \min\{a^{\mathrm{pc}}, \varpi_\chi, \gamma^{\mathrm{pc}}/(64c_{\mathrm{pc}})\}, \quad r_N^1 = \min\{1/4, \gamma^{\mathrm{pc}}/(1024e)\}, \quad z_N^1 = \gamma^{\mathrm{pc}}/(2048e^2).$$

The ceilings

$$2C_{\mathfrak{P}}\mathfrak{M}B_3\varepsilon(x) \le a_N^1, \qquad 2C_\chi \mathsf{c}_{\mathrm{fl}}\delta(x) \le \min\{r_N^1, r_\chi\}, \qquad x \ge \gamma^{-2},$$

put the first-step data in these ranges. The losses through α, through r_A, and through the weights are at most $\gamma^{\mathrm{pc}}/8$, $\gamma^{\mathrm{pc}}/8$, and $\gamma^{\mathrm{pc}}/4$ respectively. This leaves $\gamma^{\mathrm{pc}}/2$ for the stated complex data.

(b) Lem. 5.36 identifies the perturbation as $D^{(1)\mathsf{T}} v D^{(1)} + Z_{1,v}$. The complete mean-square argument is Prop. 5.37: its proof first controls the divergence-input system with the actual averaging rows, inserts $vD^{(1)}A$ as a source, and then localizes with a divergence-form cutoff. It proves both inverse existence and the termwise walk bounds without taking a difference of v.

The grouping by visited pieces after Prop. 5.37 gives the connected-set factor in (b). (c) The label supplies its background regular at the radius $C_m\varepsilon_k$ with the envelope $e(x) = C_m\varepsilon_{j(x)}$. Two of the three ceilings of the statement are the label's: $\sup_{x\ge\gamma^{-2}} C_m\varepsilon(x) \le \alpha_\star$ bounds the radius of regularity and the envelope at every level by the number $\alpha_\star$ of L, and $\sup_{x\ge\gamma^{-2}} \max\{w_{\mathrm{G}}, w_{\mathrm{lin}}\}(\log x)^{-r} \le 1$ gives $MR(\gamma^{-2}) \ge \max\{w_{\mathrm{G}}, w_{\mathrm{lin}}\}$, $w_{\mathrm{G}} = 4c_U^{-1}(7\log L + \log 2)$; under these, and with $t(\mathrm{an}) < t_0 \le t_{\mathrm{lin}}(L) \le \frac{1}{4}$, Lem. 5.19 and the locality below are applied; the third, the membership ceiling, is read below.

Use Lems. 5.19 and 5.21. The pair (A, ω) is the fixed point of the critical-point contraction built on eq. 5.x and the local Hölder bounds. To localize it, write $G' = (\Delta_D + Q'^{*}\mathsf{a}'Q')^{-1}$ in the scalar notation and retain $Q'Q'^{*} = 1$. Since $\ker R = \mathrm{ran}(G'Q'^{*})$, the slice condition is equivalently the local equation $(1 - Q'^{*}Q')\Delta_D D^* A = 0$. Cut the far field off with a slow cutoff ch of width $w(L)$ and bounded scaled first and second differences. In the augmented scalar system the cutoff error is supported where ch varies. Its inverse is the actual compression $\mathbb{K}^{[2]}$ described in Lem. 5.21; the two nonlinear remainders are precisely N and q there. Estimate eq. 5.xix therefore makes the scalar correction a contraction in the exponentially weighted norm. It repairs the slice and the induced block gauge and retains exponential decay away from the cutoff. The repaired field nearly solves

the critical-point system about U. Applying its contraction gives the stated locality estimate with c_U and $K_U = 7K_{\mathrm{lin}}^2(1+2L^3)^2 e^{c_U w_*}$, after the permitted enlargement of K_{lin}, where $w_* = w + 21L$.

Continuity: a uniform limit of iterates, with the error $\frac{2}{3}\,4^{-n}$. Uniqueness is in the normalized class: the gauge functions preserve the block averages and induce the identity at the tops. If $v_j[A']$ denotes the accumulated averaging gauge, this means

$$u(y) = v_j[A'^u](y)\,v_j[A'](y)^{-1} \quad \text{at every top } y \text{ of level } j.$$

The scalar transfer carries a sufficiently small critical point into the slice. Its multiplier is absorbed in the fixed-point equation using the same small-radius condition, so the resulting pair lies in the contraction ball and equals the constructed pair. Two normalized gauges with this property agree at every top. Their quotient preserves W, hence is covariantly constant along its bonds. It is therefore the identity along every pillar from a top, and those pillars cover the bare sites. This proves the stated uniqueness. Energy: apply eq. 5.xvi to the fixed point, with its actual row forcing $-C_2(U,A)$. Since $|\xi^2 F_U(p)|^2 \le 2a_p(U)$, the local plaquette product estimate gives $\sum_p |\partial U(p)^{-1}\partial W(p) - 1|^2 \le C(L)|\mathrm{an}|_\infty^2 \sum_p a_p(U)$. Cauchy–Schwarz with the unperturbed plaquette defects gives the stated action comparison. Full covariant gradient: let x_* lie in a layer $n \ge 2$, of the scale l_n, and pass to the axial gauge in the cube of radius $\frac{1}{4}l_n$ about x_*, built from the transports of U along coordinate paths; the transports differ from $\mathbf{1}$ there by $O(e(x_*)l_n^{-1})$ in modulus and in Hölder-$\frac{1}{2}$ quotient.

In this gauge the potential A solves a flat system for its curl and its divergence, $dA = \Phi$, $d^*A = \varpi$, where Φ is the curvature and ϖ the divergence of A, each up to products of A with the transports minus $\mathbf{1}$; by the curvature norm of Lem. 5.19 and the slice estimate for the divergence below, Φ and ϖ are bounded and Hölder-$\frac{1}{2}$ in the scale of the layer, with the sizes $Kt(\mathrm{an})e(x_*)l_n^{-2}$ and, for the Hölder quotients, $Kt(\mathrm{an})e(x_*)l_n^{-5/2}$, K a constant of L alone. The divergence is even Lipschitz: on the slice $RD^*A = 0$, so $D^*A = G'Q'^*m$ for some m, and the range of $G'Q'^*$ has first differences bounded in the scale of the layer, $|\eta_k^3(DD^*A)(b)| \le Kt(\mathrm{an})e(x_b)l(b)^{-3}$.

After a cut-off at the radius $\frac{1}{8}l_n$ whose first and second differences are $O(l_n^{-1})$ and $O(l_n^{-2})$, the identity $\Delta = d^*d + dd^*$ writes each first difference of A as second differences of the lattice Green function of the Laplacian applied to the cut data; an interior estimate of Schauder type for these kernels against Hölder-$\frac{1}{2}$ data, on a cube of radius $\sim l_n$, bounds all nine covariant first differences, $l(x)^2|\eta_k^2(\nabla_\kappa A_\nu)(x)| \le K_5\, t(\mathrm{an})\, e(x)$ with K_5 a function of L alone; in the layers $0, 1$ the bound $|\nabla_\kappa A_\nu(x)| \le |A_\nu(x+e_\kappa)| + |A_\nu(x)|$ suffices. Membership: the plain triple of U lies in the cores at half the radii (Cor. YM3_I:58.4, under its support, radius and smallness hypotheses); the moved triple is the plain triple with the displacement substituted, and in the units of each layer λ the displacement, its first differences, the change of the plaquette variables and the change of the current are at most $K_5'\, t(\mathrm{an})\, C_m\varepsilon_\lambda$, by Lem. 5.19 and the gradient estimate, where $C_m\varepsilon_{j(x)}$ is the envelope with which the label supplies its background, K_5' is a number formed with L through K_{lin} and K_5, and $t(\mathrm{an}) < t_0 \le t_{\mathrm{lin}}(L) \le \frac{1}{4}$, Lem. 5.19 being applied at the anisotropy an.

These are small against the radii, as the substitution asks, under the third ceiling of the statement,

$$\sup_{x \ge \gamma^{-2}} 2K_5' C_m\, \varepsilon(x) \le b_2, \qquad \varepsilon(x) = A_0 x^{-1/2}(\log x)^{\mathfrak{p}_0} \text{ (equation YM3_I:(634))},$$

b_2 being the radius of Prop. 5.27 among the construction results, with which the substitution with three pairs of radii carries the cores at half their radii into the cores at three quarters of them: under this ceiling they are at most $\frac{1}{8}b_2$, and the ceiling is of the admissible shape, its left member tending to 0 as $\gamma \to 0$. Then this substitution puts the moved triple in the cores at three quarters of the radii, all the tower clauses included. Frames retain the following six admissions, supplied by the ordered choice of Prop. YM3_I:69.1 and its source continuation in Section YM3_I:82: (1) the geometric and background hypotheses used in that choice; (2) consistent data and a coupling sequence with flow bound $B_\sharp$; (3) at every level n, $x_n > \bar{B}$, $y_n > 2\mathfrak{p}_0$, $g_n \le \gamma$ and $2\varepsilon_n \le \bar{\varepsilon}$; (4)

$\varepsilon_q \le \varepsilon_\flat$ at every level q; (5) at every level t,

$$C_{\mathrm{reg}}\varepsilon_t \le \varepsilon_{\mathrm{v}}, \qquad L^2 C_{\mathrm{reg}}\varepsilon_t \le c_2^{\mathrm{SO}}, \qquad 2L^2 C_{\mathrm{reg}}\varepsilon_t \le \bar{\varepsilon};$$

and (6) $x_t \ge 2B_\sharp$ at every level. The layer and support conclusions used here are those of Cor. YM3_I:58.4, with these actual admissions; the pivot evaluation additionally retains the ceilings preceding Lem. YM3_I:66.1.

The real and sourced constructions use the same coupling sequence, tests and canonical real domains, by Lems. YM3_I:82.4 and YM3_I:82.5. Condition (1) is part of the geometric choice in Prop. YM3_I:69.1; condition (2) is the admissible-sequence and room hypothesis of Thm. YM3_I:D. The proof of Prop. YM3_I:69.1 imposes $x \ge 2B_\sharp$, $x > \bar{B}$ and $\log x > 2\mathfrak{p}_0$ uniformly for $x \ge \gamma^{-2}$. The additional ceilings in the proof of Lem. YM3_I:82.5 give $2\varepsilon_t \le \bar{\varepsilon}$, $\varepsilon_t \le \varepsilon_\flat$ and $L^2 C_{\mathrm{reg}}\varepsilon_t \le c_2^{\mathrm{SO}}$ at every level, as well as the remaining graded-field conditions there. These are (3)–(6). (d) The Hessian of the Lagrangian at a critical background annihilates, against every field, a gauge direction whose gauge transformation preserves the block averages of every configuration. For a gauge direction whose linearized block average vanishes only at the background, it annihilates the fields in $N(Q)$; against other fields the remaining term lies in the range of Q^{T} and is absorbed by changing the multiplier. Thus Lem. 5.19, applied on the longer sequence in its own gauge, inverts the linearized system in the axial gauge. Its weighted maximum bound is that lemma. Its square-sum bound follows from Lem. 5.20; the multiplier is bounded by testing against a row lift $H\varsigma$ in

$$\langle \omega, \varsigma \rangle = \langle T_\tau A - f - D_1^{\mathsf{T}} Y, H\varsigma \rangle.$$

All constants depend only on L.

With Cauchy's inequalities for the remainder of the system this gives exactly one centre in the polydisc $|\mathsf{n}(b)| < \theta_3 \mathsf{b}_k$ of the statement, $\mathsf{b}_k = A_1(\log x_k)^{\mathfrak{p}_0}$ the threshold of the level k, and it lies in the half polydisc, $|\mathsf{n}(b)| \le \frac{1}{2}\theta_3 \mathsf{b}_k$, once the following hold. The envelope of the step is that of (c) with C'/L in the place of C_m, where $C' = \max\{LC_m, 2LC_\chi(1+2\theta_3)A_1/(\rho_{\mathrm{u}} A_0)\}$ is the number of the statement and $\rho_{\mathrm{u}} = \rho_{\mathrm{u}}(L)$ the radius of the ball in which the centre is unique: the first member keeps the envelope at least that of (c), and the second puts every complex block datum of the box of the step, read in the chart of the fibre, inside that ball in the norm of the envelope; this is the one number that is enlarged, and it is read by the two conditions that follow.

The size of the anisotropies is below the number of the statement, $t_\theta = \min\{t_0, t_{\mathrm{u}}(L), c_{\mathrm{ce}}\theta_3 A_1/A_0\}$, where $t_{\mathrm{u}}(L)$ is the number below which the moved backgrounds with a complex block datum are formed and $c_{\mathrm{ce}} > 0$ is formed with L and C'; so $t_\theta \in (0, t_0]$ is formed with θ_3 and with constants placed before γ. And γ is below ceilings of the shape $\sup_{x \ge \gamma^{-2}} C'\varepsilon(x) \le c(L)$, of the admissible shape, among them the ceiling $C_\chi(1+2\theta_3)\delta(x) < \frac{1}{8} r_c$, $x \ge \gamma^{-2}$, of the statement, with the numbers C_χ, r_c of L: it keeps the box $|\mathrm{Re}\, B_1(b)| + |\mathrm{Im}\, B_1(b)| < (1+2\theta_3)\mathsf{b}_k$ of the complex block data of the step, of the unscaled radius $(1+2\theta_3)\delta(x_k)$, and with it the polydisc $|\mathsf{n}(b)| < \theta_3 \mathsf{b}_k$, of the unscaled radius $\theta_3\delta(x_k)$, inside the ball of the radius r_c, in the unscaled norm, in which the system of the centre has exactly one solution.

The centre is $\mathsf{n} = g_k^{-1} a''$, $g_k = x_k^{-1/2}$, where a' is the finer block average of $W[\mathrm{an}; b_0]$ read about that of $W[0; b_0]$, the two backgrounds having the same row datum, and a'' is this relative field turned into the block axial gauge; n is holomorphic in an, real at real an and real b_0, and 0 if an is one number. By the bound of (c) on the displacement in the units of the level, read with the envelope $(C'/L)\varepsilon_j$, and the passage to the block axial gauge, $|a''(b)| \le K'' t(\mathrm{an})(C'/L)\varepsilon_k$ with a number K'' of L; as $x_k^{1/2}\varepsilon_k = (A_0/A_1)\mathsf{b}_k$, this is $|\mathsf{n}(b)| \le C_\theta t(\mathrm{an})\mathsf{b}_k$ with $C_\theta = K''C'A_0/(LA_1)$, a number of L, θ_3, A_1/A_0 and C_m, and $C_\theta t(\mathrm{an}) \le \frac{1}{2}\theta_3$ at $t(\mathrm{an}) < t_\theta$ once $c_{\mathrm{ce}} \le L/(2K''C')$.

The square sum for $1 \le k < K$, over all the bonds of the step, follows in mean square, in three inequalities. The passage to the axial gauge: $|a''(c)|^2$ is at most a number of L times $|a'(c)|^2$ plus the sum of $|a'(c')|^2$ over the bonds c' inside the two blocks that c meets, a bond c' being counted at most $12L^3$ times. The averages in the mean square: the square sum of the block averages a' over a set of

bonds is at most a number of L times the square sum of the displacement A' over the bonds beneath them, each in the units of its level. The displacement is the difference $A' = A[\mathrm{an}; b_0] - A[0; b_0]$ at the same complex row datum. Its energy is bounded by eq. 5.xvii. At $b_0 = 0$ this is at most $C|\mathrm{an}|_\infty^2 \xi^3 \sum_p |F_U(p)|^2$; in bare units it is the corresponding bound by $C|\mathrm{an}|_\infty^2 \sum_p |\tilde{F}_U(p)|^2$, where $\tilde{F}_U = \xi^2 F_U$ and $|\tilde{F}_U(p)|^2 \le 2a_p(U)$.

At $b_0 = 0$, since $|\mathsf{n}|^2 = x_k |a''|^2$ and, in a frame, $x_k L^k \le \frac{37}{11}\beta_t(y)$ at every point y, this gives $\sum_b |\mathsf{n}(b)|^2 \le C^2 |\mathrm{an}|_\infty^2 A(\beta_t, U)$, C a number of L; at $k = 0$ the same holds at the external bonds of the step. The centres $\mathsf{n}^{(i)}$ of the earlier steps $0 \le i < k$ are formed by the same rule about the moved background of the label, so that $g_i \mathsf{n}^{(i)}$, $g_i = x_i^{-1/2}$, is the block average at the level i of its displacement, turned into the block axial gauge; by the bound of (c) on the displacement in the units of each layer, each obeys $|g_i \mathsf{n}^{(i)}(b)| \le Ct(\mathrm{an})\varepsilon_i$, that is $|\mathsf{n}^{(i)}(b)| \le Ct(\mathrm{an})(A_0/A_1)\mathsf{b}_i$, as $x_i^{1/2}\varepsilon_i = A_0 (\log x_i)^{\mathsf{p}_0} = (A_0/A_1)\mathsf{b}_i$, with one C of L for all the levels; this is at most $\frac{1}{2}\theta_3 \mathsf{b}_i$ at $t(\mathrm{an}) < t_\theta$ once $c_{\mathrm{ce}} \le 1/(2C)$, and $c_{\mathrm{ce}} = \min\{L/(2K''C'), 1/(2C)\}$ is a number formed with L and C'.

The stored bonds, which are the external bonds of these steps, lie for the level i within $2L^i$ (bare units) of the layer i; the square sums of the block averages of all the levels over the bonds near their own layers are bounded together, with one number of L, by the square sum of the displacement weighted by the scales of the layers, because the layers are wide and the bonds near the own layers of different levels overlap boundedly; and the ℓ^2 bound of the linear system, as for the centre of the step, then gives $\sum_{i<k} \sum_b |\mathsf{n}^{(i)}(b)|^2 \le C^2 |\mathrm{an}|_\infty^2 A(\beta_k, U)$, b running over the stored bonds, with one C of L, $A(\beta_k, U)$ being the action of the background with the couplings of the level k. (e) The room hypothesis of Lem. 5.22 holds at every level: $L^m \ge MR(x_K) \ge LM$ gives $m > m'$, hence $m' < m + K - j$ for $j \le K$. The identity is that lemma on a label, for any list of families with counterterms, with the explicit boundary terms of its clause on a label; the trivial values do not change because the tent families vanish at the trivial triple, and, at a fixed triple with the original families and counterterms held fixed, the linear changes in the re-filed local and boundary entries are those of the tent families.

The size bounds require the stated margin: by Lem. 5.22 with $\kappa' = \kappa + 1$ and $|e_j| \le B_\sharp$, $\|\hat{E}^{(j)} - E^{(j)}\|_{j,\kappa+1} \le \frac{3}{2}|\mathrm{an}|_\infty B_\sharp \alpha_0^2 e^{7(\kappa+1)} \le \frac{3}{2}\sigma_3 \|\mathrm{an}\| B_\sharp \alpha_0^2 e^{7(\kappa+1)} \le \theta E_0 \|\mathrm{an}\|$, and the triangle inequality, with the bound $(1-\theta)E_0$ of the families before the re-filing, gives $\|\hat{E}^{(j)}\|_{j,\kappa+1} \le E_0$ and the distance from the family at zero source; the boundary terms are at most $\frac{3}{2}M^3 |\mathrm{an}|_\infty B_\sharp \alpha_0^2 e^{7\kappa} e^{-\kappa d_j(X)} \le \theta M^3 E_0 e^{-7} \|\mathrm{an}\| e^{-\kappa d_j(X)}$ in a frame of Thms. YM3_I:AJ and YM3_I:AD, where the radius of the level is at most α_0. □

5.1.9. *Scope of the conclusions.* Thms. M(d), N(c), (d) and O(a), (c), (d) live on one torus, with constants that depend on its side; so does Thm. O(e), whose numbers are free of the side when the closed a-neighbourhoods of the supports lie in balls of radius R and $\ell \ge 16n(R+1)$. On $\mathbb{R}^3$ the statements concern limit points and tensor limits, at separated or disjoint supports. Thm. O(d) concerns cumulants with one stress entry among action densities on one torus, at a fixed label. The kernels of Thm. M(h) and the operators of Thm. P are those of each limit, taken along the subsequence that gives it. Thms. Q and R concern one step, the never-large term, and the backgrounds, centres and counterterms of the construction.

Along growing tori and in the periodic Gibbs states the section asserts the following. The cumulants of the plane fields have limit points along every sequence of first-passage data at g with $K_j \to \infty$, in periodic Gibbs states and on tori T_{ℓ_j} with $\ell_j \to \infty$, whatever the relation between K_j and ℓ_j, with the bounds, the invariance and the reflection positivity of Thm. M(c). The bulk term of the rotational Ward identity vanishes with a rate and with numbers that read no side, so that a limit along tori is invariant under rotations exactly when the flux through a shell about the supports tends to 0; this flux tends to 0 for the volume-first scalar limits; and, with the

limits taken one after the other, at a label $\mathsf{b} \in I(g)$ the limit in K of the flux on each torus T_ℓ is $\sum_i S_n^{(\ell,\mathsf{b})}(f_1, \dots, (\omega y)\cdot\nabla f_i, \dots, f_n)$ through every enclosing shell, and along sides $\ell_k \to \infty$ in $\mathfrak{L}(g)$ on which the limits $S_n^\circ = \lim_k S_n^{(\ell_k,\mathsf{b})}$ exist, the limit in k of these limits, through any shell that covers the supports, a fixed one or a far one, is $-\frac{d}{dt} S_n^\circ(f(e^{-t\omega}\cdot))|_{t=0}$, and there are integers K_k^* such that S_n° is the limit of the connected functions along $(K_k, \ell_k, L^{K_k}\mathsf{b})$ for every sequence $K_k \geq K_k^*$ (Thm. O(b)).

For sides up to $a^{-\theta}$, $\theta < \frac{1}{30}$, a limit along tori at first-passage labels is the limit taken spacing first, then volume, and raising the cutoffs at fixed sides and labels does not change it (Thm. O(b)). With numbers that read no side, in every periodic Gibbs state the stress insertion collapses onto its classical tensor and the conserved tensor decorrelates from any number of tensor entries at separated supports (Thm. N(e), (f)); every tensor limit is conserved at disjoint supports (Thm. N(g)); and the translation identity of Thm. O(e), with its null term and its contact terms, holds per ball with numbers free of the side.

5.1.10. *The conjugation specific to two by two matrices.* The complex conjugation used for SU(2) preserves the matrix norm. This fact is needed when comparing the domains of a local term and its conjugate; it is not a general property of $X \mapsto (X^*)^{-1}$ in higher matrix dimension.

Lemma 5.23. Put $\epsilon = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ and $\Theta(X) = \epsilon\overline{X}\epsilon^{-1}$. Then Θ is a conjugate-linear multiplicative involution on complex two by two matrices and

$$|\Theta(X)| = |X|, \qquad \Theta(X) = \overline{\operatorname{tr} X}\, I - X^*.$$

For $U \in \mathrm{SL}(2,\mathbb{C})$ and traceless A,

$$\Theta(U) = (U^*)^{-1}, \qquad \Theta(iA) = iA^*.$$

In particular $|(U^*)^{-1} - I| = |U - I|$. ■

Proof. Entrywise conjugation and conjugation by ϵ preserve products. Since ϵ is real and $\epsilon^2 = -I$, their composite squares to the identity. Entrywise conjugation preserves the operator norm, as does conjugation by the unitary matrix ϵ. Direct multiplication gives $\epsilon Y^{\mathsf{T}}\epsilon^{-1} = (\operatorname{tr} Y)I - Y$. Take $Y = X^*$ to obtain the displayed formula for $\Theta(X)$. For determinant-one Y, the same two by two identity gives $Y^{-1} = (\operatorname{tr} Y)I - Y$; apply it to $Y = U^*$. For traceless A, the formula gives $\Theta(iA) = -(iA)^* = iA^*$. Finally apply norm preservation to $U - I$. □

Lemma 5.24. Let j be a level with $m' < m + K - j$, let $D_j = L^{j-K}$, and put $W = MD_j$. Thus the cubes of level j have physical side W and the torus has side $\ell = 2L^m$. Let $\mathrm{pr} : \mathbb{R}^3 \to T_\ell$ be the covering map. A finite localization domain $\widehat{X}$ in the infinite level-j cube lattice is called near if it lies in a cuboid whose three side lengths are strictly less than $\ell/2$. A torus domain is non-wrapping if it is the image of a near set. Then the following hold.

(1) The image of every infinite localization domain belongs to $\mathcal{D}_j$. Every near lift of a non-wrapping domain belongs to the infinite domain family. Its cubes correspond bijectively to those of the torus domain, adjacency is preserved in both directions, and $d_j(\widehat{X}) = d_j(X)$.
(2) All near lifts of one domain are its translates by $\ell\mathbb{Z}^3$. They are pairwise disjoint; specifying one lift of a point of the domain specifies exactly one near lift containing that point.
(3) Every infinite domain $\widehat{X}$ lies in a cuboid with side lengths at most $W(d_j(\widehat{X}) + 2)$. A torus domain with $W(d_j(X) + 2) < \ell/2$ is non-wrapping. Consequently a wrapping domain, or an infinite domain which is not near, has size at least $\ell/(2W) - 2$.

■

Proof. The hypothesis gives $\ell/W = 2L^{m+K-j-m'} \geq 2L$. Projection carries cubes to cubes and shared faces to shared faces. It therefore carries a finite wall-connected union to a wall-connected union. On a near set projection is injective. The coordinate differences of two of its points have

magnitude less than $\ell/2$; a different lift of such a difference differs by a period. Each cube of the image lifts to a whole cube: the difference between the inverse chart on that cube and any one of its cube lifts is continuous and period-valued, hence constant. Thus the inverse chart preserves the faces, edges and corners of its cubes, as well as their adjacency. Lifting a wall chain shows that a near lift is a localization domain. Connected edge sets meeting every cube are carried in both directions with their number of edges unchanged. Taking the minimum of that number proves the equality of the linear sizes.

To compare two near lifts, translate one by a period so that one pair of points over the same torus point agrees. For any other pair over the same point, their displacement is a period. Each coordinate of this displacement is the difference of two displacements from the matched points, each of magnitude less than $\ell/2$. It has magnitude less than ℓ, so is zero. The lifts therefore agree after that translation. Distinct period translates are disjoint, since two points in a near cuboid cannot differ by a nonzero period. Translating a chosen lift to contain the prescribed point proves existence and uniqueness in the second assertion.

For an infinite domain choose a connected edge set with $d = d_j(\widehat{X})$ edges meeting every cube. If $d = 0$ this set is one corner. Otherwise enumerate its edges so that each edge after the first meets an earlier edge. Adding an edge of length W increases each coordinate span by at most W. The edge set is therefore contained in a cuboid with side lengths at most dW. Every cube of the domain meets it and lies within sup distance W of it. Enlarging the cuboid by W on both sides proves the bound $W(d+2)$.

The same construction lifts a small torus domain. Enumerate a minimal connected edge set as above, lift the first edge, and translate each later edge by a period so that it meets the chosen lift of an earlier edge. Their union projects onto the whole edge set and lies in a cuboid of side lengths at most dW. For every cube of the domain choose a lift containing a lifted point where that cube meets the edge set. The union of these cubes projects onto the domain and lies in the W-enlargement of the cuboid. If $W(d+2) < \ell/2$, this is a near lift. The last two lower bounds are the contrapositives of the cuboid bounds. □

The brackets of equation YM3_I:(806) are alternating finite differences. Their useful support property has an elementary proof which we record to make the localization of a shifted local term explicit.

Lemma 5.25. Let P be a finite set of cubes and let $\mathsf{N}_0 \subset \mathsf{P}$ be nonempty and wall-connected. Suppose that, at every vertex $\mathbf{1}_W$, $F(\mathbf{1}_W)$ depends only on the wall-components of W meeting N_0: deleting all other components does not change its value. For $\mathsf{N}_0 \subset W \subset \mathsf{P}$,

$$F(\mathbf{1}_W) - F(0) = \sum_{\mathcal{K} \subset W \setminus \mathsf{N}_0} F^*[\mathcal{K} \mid \mathsf{N}_0]. \tag{5.xxvi}$$

Every term whose union $\mathcal{K} \cup \mathsf{N}_0$ is not wall-connected is zero. ■

Proof. The resummation is equation YM3_I:(809): expand the alternating sums, and use $\sum_{A \subset B} (-1)^{\#A} = 0$ for nonempty B. For the second assertion choose a wall-component c of $\mathcal{K} \cup \mathsf{N}_0$ disjoint from N_0 and a cube $\Delta \in c$. For every $S \subset \mathcal{K} \setminus \{\Delta\}$, all components within c are unrooted, even after Δ is added. Thus $F(\mathbf{1}_{S \cup \mathsf{N}_0}) = F(\mathbf{1}_{S \cup \{\Delta\} \cup \mathsf{N}_0})$. The two terms have opposite signs in the bracket. Pairing them proves its vanishing. The same cancellation applies to $F - F(0)$. □

5.2. **Local analytic cores and substitution.** We use the simple domains and hereditary analytic cores of Def. YM3_I:49.1, with the geometric and regularity thresholds of Prop. YM3_I:51.9. These thresholds, together with the positivity of the local gauge constants, will be called the tower hypotheses below. In particular, the four geometric parameters are powers of L, $L \geq 13$ is odd, $R_1 \geq L$, $M_1 \geq R_1 \mathfrak{M}$, and $M \geq L^4 M_1$; the background and response thresholds are those fixed at the beginning of Section YM3_I:51.5. Write $a^\sharp \leq 1/60$ and $K_{\mathrm{tw}}^\sharp \geq 1$ for its tower radius and bound.

Every use below keeps the domain and subtower restrictions in that proposition. A domain on the infinite lattice is finite; the same statement there is read in a containing lattice chart, with the chart compatibility established in Section YM3_I:52.5, equation YM3_I:(926).

The triple is $P = (U, A', \mathbf{J})$, where U is real and $\mathbf{p}(P) = (e^{i\eta_j A'}U, \mathbf{J})$ is its represented pair. Its independent bond coordinates and gauge action are those preceding Def. YM3_I:49.1. For a real bare configuration V, its pair at level j is $(V, J^{(j)}(V))$ restricted to the incident bonds of the domain. A local term is evaluated at this pair only when that pair belongs to its core. This convention does not assert that an arbitrary real configuration belongs to the core.

For a block field W, the background $U_j(W)$ denotes an element of the unique regular critical orbit over its determining datum, when that orbit exists (Defs. YM3_I:9.3 and YM3_I:9.5). On the full torus the reduced determining set consists exactly of the terminal bonds. Gauge invariant expressions have the same value at every representative. The total of a pointed family is equation YM3_I:(935); it is defined only if the orbit exists and every pair read by the sum belongs to the respective core.

Lemma 5.26 (Open pair domains)**.** Under the tower hypotheses, let X be a localization domain and $b_0, b_1, c_0 > 0$, $b_0 + b_1 \le a^\sharp$. For fixed real U, the set

$$\mathcal{O}_U = \{(A', \mathbf{J}) : (U, A', \mathbf{J}) \in \mathcal{U}_j^\circ(X; b_0, b_1, c_0)\}$$

is open in the independent bond coordinates. The map $\Psi_U(A', \mathbf{J}) = (e^{i\eta_j A'}U, \mathbf{J})$ is biholomorphic from this set onto an open set of pairs. The union of these pair domains over U is open. A function on the core is analytic and depends only on the represented pair if and only if it descends to a holomorphic function on that union. ■

Proof. The conditions on the real factor and its real subtowers are fixed when U is fixed. If any fails, the slice is empty. Otherwise the simple-domain conditions on A' and $\mathbf{J}$ are finitely many strict inequalities between continuous functions. On this open set, Prop. YM3_I:51.9(f) constructs every required complex subtower holomorphically. There are finitely many subtowers, bonds and plaquettes in X. Their strict curvature and current inequalities therefore define an open subset. The holomorphic-continuation clause of the core is preserved on this subset by the same tower construction.

On the slice, $\eta_j|A'(b)| < \eta_j b_1 < 1/2 < \log 2$. Thus the logarithm at I gives the holomorphic inverse

$$(\mathbf{U}, \mathbf{J}) \longmapsto \big((i\eta_j)^{-1}\log(\mathbf{U}U^{-1}), \mathbf{J}\big)$$

of Ψ_U. The exponential and this logarithm identify open neighbourhoods in the bond-coordinate spaces, so the image of the open slice is open. Their union is open as well. Composition proves one direction of the final assertion. Conversely a function of the pair gives a single well-defined function on the union. On each open image it is the analytic slice function composed with the displayed inverse. These local holomorphic functions agree on intersections, and hence define the required holomorphic function. □

The chart at the trivial pair is $P[v] = (I, A, \mathbf{J})$, where $\|v\|_X = \max\{\sup|A|, \sup|\nabla A|, \sup|\mathbf{J}|\}$ and only differences defined on the incident bonds enter the maximum. Equation YM3_I:(933), proved in Section YM3_I:52.6, gives

$$r_{\mathrm{us}} = \min\{b_1, a^\sharp/4, b_0/(4K_{\mathrm{tw}}^\sharp)\}, \qquad P[\{\|v\|_X < r_{\mathrm{us}}\}] \subset \mathcal{U}_j^\circ(X; b_0, b_1).$$

In particular this is a genuine ball in the full core, not just in the simple domain.

Proposition 5.27 (Three pairs of radii)**.** Assume the tower hypotheses and let X be a level-$(j-1)$ domain, with lengths measured in level-j units. Let positive radii satisfy

$$\beta_0 < b_0 \le \gamma_0, \quad \beta_1 < b_1 \le \gamma_1, \quad \gamma_0 + \gamma_1 \le a^\sharp,$$
$$\beta_0 + \tfrac{9}{4}s \le \gamma_0, \quad \beta_1 + \tfrac{11}{10}s \le \gamma_1,$$
$$\beta_0 + \tfrac{9}{4}b_2 \le b_0, \quad \beta_1 + \tfrac{11}{10}b_2 \le b_1,$$
$$2K_{\mathrm{tw}}^\sharp(\gamma_0 + \gamma_1^2)b_2 \le (b_0 - \beta_0)s.$$

If $P = (U, A', \mathbf{J}) \in \mathcal{U}_j^\circ(X; \beta_0, \beta_1, c_0)$, $\mathbf{U} = e^{i\eta_j A'}U$, and $|\mathbf{A}|, |\nabla_{\mathbf{U}}\mathbf{A}| < b_2$, $|\mathbf{J} + \mathbf{K}| < c_0'$, then

$$P \dotplus (\mathbf{A}, \mathbf{K}) = (U, (i\eta_j)^{-1}\log(e^{i\eta_j \mathbf{A}}e^{i\eta_j A'}), \mathbf{J} + \mathbf{K}) \in \mathcal{U}_j^\circ(X; b_0, b_1, c_0').$$

Its pair is $(e^{i\eta_j \mathbf{A}}\mathbf{U}, \mathbf{J} + \mathbf{K})$. ∎

Proof. The logarithm and the simple-domain bounds follow from the pointwise increment calculation in the proof of Prop. YM3_I:51.10: the potential and its first derivative increase by less than $11b_2/10$, and the plaquette bound by less than $9b_2/4$. The real factor does not change. Put $R = s/b_2$. The last inequality and $K_{\mathrm{tw}}^\sharp \ge 1$ imply $R > 2$. For $|z| < R$ the same increment calculation, with $z\mathbf{A}$ in place of $\mathbf{A}$, places the substituted field in the simple domain at outer radii (γ_0, γ_1). Prop. YM3_I:51.9(f) therefore constructs all its subtowers holomorphically on that disc.

Consider a level-p subtower and set $\lambda_p = L^{-(j-p)}$. After its curvature and current are multiplied by L^{2p} and $L^{-2(n-p)}$, respectively, each is bounded on the disc by $\lambda_p K_{\mathrm{tw}}^\sharp(\gamma_0 + \gamma_1^2)$. At $z = 0$ its norm is strictly below $\lambda_p\beta_0$. The Schwarz estimate for the difference from its value at zero, as used in the same proof of Prop. YM3_I:51.10, bounds its increment at $z = 1$ by

$$\frac{2}{R}\lambda_p K_{\mathrm{tw}}^\sharp(\gamma_0 + \gamma_1^2) \le \lambda_p(b_0 - \beta_0).$$

This proves the strict target inequality for each complex subtower. The real subtower inequalities are unchanged and their old radius β_0 is smaller than b_0. All clauses of the target core follow. □

For the passage from half the fixed radii to three quarters, take $\beta_i = \alpha_i/2$, $b_i = 3\alpha_i/4$, $\gamma_i = \alpha_i$, and $s = \min\{2\alpha_0/9, 5\alpha_1/11\}$. A positive b_2 satisfying

$$b_2 \le \min\left\{\alpha_0/9, 5\alpha_1/22, \frac{\alpha_0 s}{8K_{\mathrm{tw}}^\sharp(\alpha_0 + \alpha_1^2)}\right\}$$

satisfies all four displayed conditions. This choice depends only on the fixed radii and the tower constants.

5.2.1. *Normalizers for overlapping entries.* In the setting of [7, Lem. N.1], let $A_g = C_{\mathrm{S}} + C_g + C_{\mathrm{gap}} = G_g$. We use the following normalizer and cumulant constants:

$$\begin{aligned} \mathsf{Q}_1(L) &= \exp\{A_g L^3 + \Phi_1(2, 1; L)\}, \\ Q^\natural(n, R) &= \exp\{A_g h(n, R)^3 + \Phi_1(R, R+1; M)\}, \\ C_n^{\mathrm{per}}(d; R) &= (n/\varrho)^n\big[\log 2\,(2Q^\natural(n, R)^n + 2)^n + \mathsf{U}(R, d)\big]. \end{aligned} \tag{5.xxvii}$$

Here $h(n, R)$ is the least power of L at least $8n(R+1)$; Φ_1 and U are the functions in Section YM3_I:89, with source-energy constants (C_{S}, κ). These are equations YM3_I:(1844) and YM3_I:(1847) with the floor and gap constants of [7, Lem. N.1].

We will use one part of the normalizer proof for tuples whose supports may overlap. Fix a board of side l, and suppose each entry is supported in a ball of radius R whose centre is at distance at least $R + D$ from the walls, where $D > 0$. On the least admissible torus of side at least $2l$, form the

cell normalizers and set $E(z)=\sum_c E_c(z)$. Each E_c reads only source coordinates assigned to cell c. The construction in the proof of Prop. YM3_I:91.12 gives, on the source polydisc of radius ϱ,

$$|e^{-E(z)}|\Pi_\ell(z)\le \exp\{A_g l^3+\Phi_1(R,D;l)\}^{\#B(z)} \qquad (\ell\in\mathfrak{L}(g),\ \ell\ge 2l). \tag{5.xxviii}$$

Indeed, the cell bound gives one factor $\exp\{A_g l^3+\#B_c(z)\Phi_1(R,D;l)\}$ for each cell containing an active coordinate. The number of such cells is at most $\#B(z)$, and $\sum_c \#B_c(z)=\#B(z)$. Multiplying proves the inequality. Neither this count nor the holomorphy of E uses separation of the supports. If at least two cells are occupied, its full mixed derivative vanishes since every summand omits a coordinate. Separation is needed only for the additional bound on that derivative when all entries occupy one cell. The same normalizer works for periodic Gibbs states by Prop. YM3_I:91.11.

5.3. **Background estimates and the normalized step.** The background estimates used here are those for reduced determining sets. The region sequence may have an arbitrary first member. It is separated, its terminal lattice has at least $6R_1\mathfrak{M}$ sites in each direction, and the geometric, regular-reference and datum thresholds are those of Section YM3_I:51.5 and the localization construction in Section YM3_I:53.4. In particular we retain

$$0<\varepsilon_{\mathrm r}\le\bar\varepsilon, \qquad C_{\mathfrak P}\mathfrak{M}B_3\varepsilon_{\mathrm r}\le a_{\mathrm R}, \qquad \varrho=\min\{r_{\mathrm B},\varrho_{\mathrm R}^0\}, \qquad B_3\varepsilon_{\mathrm r}\le L^{-2}a_{\mathsf m}.$$

Here $a_{\mathsf m}$ is the radius of Prop. YM3_I:30.2; the last inequality ensures its curvature hypothesis for the regular backgrounds, whose bare curvature is at most $B_3\varepsilon_{\mathrm r}\eta_n^2$. All depth thresholds chosen from this regularity radius use the decreased value. Thm. YM3_I:E gives a single regular critical orbit and its holomorphic axial representative. Lem. YM3_I:51.6 writes it as $(e^{i\eta_n\mathcal H(B)}U_0)^{u_B}$ on $\sup|B|<\varrho$, with $\mathcal H(0)=0$, $u_0=I$ and the stated Landau condition.

For the response constants in this paper we take $B_5^{\mathrm e}\ge\max\{8,B_5\}$ and $\delta_0^{\mathrm e}=\delta_{\mathrm{lc}}$ from equations YM3_I:(996) and YM3_I:(997). Thus, with $m_y(B)=\sum_{c>0}e^{-\delta_0^{\mathrm e}d(y,y_c)}|B(c)|$,

$$\xi_y|\mathcal H(B)(b)|\le B_5^{\mathrm e}m_y(B), \quad \xi_y^2|\nabla_{U_0}\mathcal H(B)(x)|\le B_5^{\mathrm e}m_y(B), \quad |u_B(y)-I|\le 4\kappa_{\mathrm{av}}B_5^{\mathrm e}m_y(B).$$

The first two bounds also hold for differences at two real data, with $m_y(B-B')$ on the right. The derivative estimate YM3_I:(838) supplies the corresponding linear response and its second covariant difference, where all incident plaquettes are inner. These estimates hold on the specified datum ball; they are not assertions for arbitrary block configurations. Here κ_{av} denotes the joint contour constant called κ_6 in the localization construction. When a bound for the derivative of the gauge at a top is written with a single coefficient k_6, we take $k_6=4\kappa_{\mathrm{av}}$.

The localization estimate itself is Prop. YM3_I:53.5. Its hypotheses include the one- or two-component complement, the four-layer collars and their level assignments, the separation of different components, the joined sequence and its regular datum, and $\varepsilon_{\mathrm r}\le\epsilon_{\mathrm{lc}}$. The definitions and all those conditions are equations YM3_I:(998)–YM3_I:(1000) and the paragraphs preceding the proposition. We keep its wide three-layer collars and its weighted distance to the tower rows.

Lemma 5.28 (Iterated plaquette bound)**.** Let p' be a plaquette at level $q\ge1$. At level $j\le q$, take the union of the four $(q-j)$-blocks below its corners and let ϖ_j be the largest plaquette curvature of $\mathcal M^jU$ within that union, in all three orientations. Set

$$\theta_L=\exp\frac{1}{2(L^2-1)}, \qquad c_2^{\mathrm{SO}}=\min\{\varpi_L'/2,L^2/(4C_0^{\mathrm{SO}})\}.$$

The constants in the latter formula are those of Lem. YM3_I:14.9. If $\beta=L^{2q}\varpi_0\le c_2^{\mathrm{SO}}$, then every average in this cone is defined and

$$\varpi_j\le\theta_L L^{2(j-q)}\beta \quad (0\le j\le q), \qquad 1<\theta_L<1.003.$$

■

Proof. The cone at level $j+1$ is a union of blocks contained in the cone at level j. Consequently the one-step plaquette estimate in Lem. YM3_I:14.9 applies there whenever $\varpi_j \le \varpi'_L$, and gives

$$\varpi_{j+1} \le L^2 \varpi_j (1 + C_0^{\mathrm{SO}} L^{-2} \varpi_j).$$

Put $\theta_0 = 1$ and $\theta_j = \prod_{i<j} (1 + \frac{1}{2} L^{2(i-q)})$. The geometric series and $1 + x \le e^x$ give $\theta_j \le \theta_L < 2$. If $\varpi_j \le \theta_j L^{2(j-q)} \beta$ and $j < q$, then $\varpi_j \le 2 L^{2(j-q)} c_2^{\mathrm{SO}} \le \varpi'_L$, so the small-curvature branch of the real average applies. Moreover $C_0^{\mathrm{SO}} L^{-2} \varpi_j \le \frac{1}{2} L^{2(j-q)}$ by the definition of c_2^{SO}. The recurrence proves the next induction step. The case $j = 0$ is the definition of β. Finally $L \ge 13$ gives $\theta_L \le e^{1/336} \le 336/335 < 1.003$. □

Lemma 5.29 (Averaging along a straight closed line)**.** Let U be a real SU(2)-valued configuration and $q \ge 1$. Suppose every bare plaquette has curvature at most ω_0 and $L^{2q} \omega_0 \le c_2^{\mathrm{SO}}$, with the constant of Lem. 5.28. Let Γ be a straight closed line of N bare bonds through a level-q site, where L^q divides N. Let Γ_q be the same line in the level-q lattice. Then

$$\bigl| \mathcal{M}^q U(\Gamma_q) - U(\Gamma) \bigr| \le 13.3\, N L^q \omega_0.$$

The assertion also holds on a rectangular periodic lattice under these same local hypotheses. ■

Proof. We first keep the curvature parameter in the proof of Lem. YM3_I:4.4. Write $t = L^2 \omega \le 1/200$ at one step. The contour diameter is at most $d = e^{9t/8} - 1$, and the rearrangement bound for the middle path is $e^{25t/8} - 1$ (Lem. YM3_I:4.1). Prop. YM3_I:4.3 writes the contour mean relative to any chosen contour as e^Y, with Y skew-Hermitian and $|Y| \le 2d$. Since $|e^Y - 1| \le |Y|$, the two contour replacements and the middle-path replacement, telescoped between unitary matrices, give

$$|W_x - 1| \le 4(e^{9t/8} - 1) + (e^{25t/8} - 1) < 8t.$$

Indeed $e^u - 1 \le u/(1-u)$ and $t \le 1/200$ give the coefficient $7200/1591 + 200/63 < 8$. Hence $|\log W_x| \le 8t/(1-8t) < 12t$. The average in YM3_I:(37) is the exponential of the mean of these skew-Hermitian logarithms times the straight transport. It differs from that transport by at most $12 L^2 \omega$.

At level j, Lem. 5.28 gives $\omega_j \le \theta_L L^{2j} \omega_0$. The small-curvature admission in that lemma also gives $L^2 \omega_j \le L^2 \varpi'_L < 1/200$ for $j < q$. The line has N/L^{j+1} bonds after the next average. A product of unitary matrices changes by at most the sum of the changes of its factors, so summing the one-step errors gives

$$\sum_{j=0}^{q-1} 12 \frac{N}{L^{j+1}} L^2 \omega_j \le 12 \theta_L N \omega_0 \sum_{j=0}^{q-1} L^{j+1} \le \frac{12 L \theta_L}{L-1} N L^q \omega_0 < 13.3\, N L^q \omega_0.$$

Here $L \ge 13$ and $\theta_L < 1.003$. All paths and exchanges are local to their double blocks. The proof is unchanged when the periods in the three directions differ. □

Lemma 5.30 (Derivatives and remainder of the plain average)**.** Take $a_Q = a_{\mathrm{plain}}$, $c_Q = c_{\mathrm{plain}}$ and C_Q from Prop. YM3_I:51.1, with the smallness choices made in its proof, and put $C'_Q = 4 C_Q / c_Q$. On its tower over a double block, let U be real with $\varpi(U) \le a_Q L^{-2j}$. For a complex bond form a, write

$$Q_j(U,a)(c) = \frac{1}{i} \log\bigl[\mathcal{M}^j(e^{ia} U)(c) \mathcal{M}^j(U)(c)^{-1} \bigr], \qquad \tau_c = L^j \sup_{\mathsf{D}_c^j} |a| < c_Q.$$

The form is defined on the positive bare bonds inside the endpoint double-block tower; negative bonds are extended covariantly. Then

$$|\dot{Q}_j(U) a(c)| \le C_Q \tau_c, \qquad |Q_j(U,a)(c) - \dot{Q}_j(U) a(c)| \le C'_Q \tau_c^2. \tag{5.xxix}$$

At $U = 1$, $\dot{Q}_j(1) = \mathbf{Q}_j^{\mathrm{lin}}$, the actual iterated linear average of Lem. YM3_I:4.6. On a whole real periodic lattice satisfying the same curvature bound, the left-trivialized derivative obeys

$$\|D\mathcal{M}^j(U)w\| \le \sqrt{6}C_Q[w]_j, \qquad [w]_j^2 = \sum_y \left(L^j \max_{b_- \in B^j(y)} |w(b)| \right)^2, \tag{5.xxx}$$

where both sums defining the lattice ℓ^2 norms use positive bonds and counting measure. ■

Proof. Fix the row c. Prop. YM3_I:51.1 gives a holomorphic function $F(a) = Q_j(U, a)(c)$ on the ball $\tau_c < c_Q$, with $F(0) = 0$ and $|F(a)| \le C_Q\tau_c$. On every complex radial line the Cauchy bound gives $|F'(0)a| \le C_Q\tau_c$. If $\tau_c \le c_Q/2$, its Taylor remainder is bounded by

$$\sum_{n\ge 2} C_Q c_Q (\tau_c/c_Q)^n = \frac{C_Q\tau_c^2}{c_Q - \tau_c} \le \frac{2C_Q}{c_Q}\tau_c^2.$$

The coefficients can first be estimated on circles of radius less than c_Q and then passed to their supremum. If $c_Q/2 < \tau_c < c_Q$, the bounds for F and $F'(0)$ instead give $|F(a) - F'(0)a| \le 2C_Q\tau_c \le 4C_Q\tau_c^2/c_Q$. The derivative of the logarithm at 1 is the identity, so the linearization of the iterated average in Lem. YM3_I:4.6 identifies the derivative when $U = 1$.

For the final assertion put $m_y = L^j \max_{b_- \in B^j(y)} |w(b)|$. Every bond in the determining double block of c has its initial site in one of its two endpoint blocks. Differentiating the same row estimate at $a = 0$ therefore gives

$$|D\mathcal{M}^j(U)w(c)|^2 \le C_Q^2 \max\{m_{c_-}, m_{c_+}\}^2 \le C_Q^2(m_{c_-}^2 + m_{c_+}^2).$$

Each site is an endpoint of six positive bonds. Summing proves eq. 5.xxx. For $j = 0$ the derivative is the identity and the same bound follows by the same count. □

5.3.1. *The normalized fluctuation measure.* Consider a real canonical step of height t. Retain its actual array-size bounds, the full-core radii and numerical ceilings of Thm. YM3_I:W, and the whole-chart hypotheses of Prop. YM3_I:52.3. For the tame class $\mathsf{O}_{t+1} = \{\varpi(W) < 2\varepsilon_{t+1}\}$ these hypotheses follow under the coupling ceiling used in the chain; explicitly it is chosen so that

$$2\varepsilon_{t+1} \le \bar{\varepsilon}, \quad 2\varepsilon_{t+1} \le \varepsilon_{\mathcal{W}}\alpha_0, \quad \delta_t \le \varepsilon_{\mathcal{W}}, \quad 4\varepsilon_0 < \varepsilon_{\mathrm{v}}.$$

All geometric room and response requirements of that proposition are retained. At a tame W choose U_+ in its regular orbit so that $\mathcal{M}^t U_+$ is in the axial gauge, and put

$$P_W = (U_+, 0, J^{(t+1)}(U_+)), \qquad \mathsf{Rg}_{x_t} = \{B \text{ real on nonpivot bonds} : |B(b)| < \mathfrak{b}_t\}.$$

Thm. YM3_I:E gives regularity at radius $2B_3\varepsilon_{t+1}$, including the current bound in level-$(t+1)$ units. Prop. YM3_I:52.3 places every restriction of P_W in its full core and supplies the whole chart $V_W(B)$ and its lower backgrounds. The complete Gaussian structure has real positive covariance $C_W = (\mathsf{B}_W + \widehat{\mathsf{G}}_W)^{-1}$, as in Prop. YM3_I:52.2 and Thm. YM3_I:W.

Let S_t be the action of the canonical bare array, $S_0 = 0$, and set

$$F_t(V) = \exp[-x_t A^{\eta_t}(U_t(V)) + S_t(U_t(V))]$$

on its defined set. The chart only reads this set. Define

$$\mathsf{E}_W(B) = \sum_{Y \in \mathcal{D}_t} (\mathbb{V}^{\mathrm{J}} + \mathbb{V}^{\mathrm{G}} + \mathbb{V}^{\mathrm{A}} + \mathbb{V}'[S_t])(Y; P_W|_Y, B|_Y), \tag{5.xxxi}$$

$$\Xi_W = \int_{\mathsf{Rg}_{x_t}} e^{\mathsf{E}_W(B)}\, \mathrm{d}\mu_{C_W}(B), \qquad d\nu_W = \Xi_W^{-1}\mathbf{1}_{\mathsf{Rg}_{x_t}} e^{\mathsf{E}_W}\, \mathrm{d}\mu_{C_W}. \tag{5.xxxii}$$

Proposition 5.31. The integral Ξ_W converges absolutely and is nonzero. Thus ν_W is a complex Borel measure of finite variation and mass one. Its pushforward $\kappa_{t+1}(W,\cdot) = (V_W)_*\nu_W$ is supported on the fibre over W and, for bounded Borel G, satisfies

$$\int G \,\mathrm{d}\kappa_{t+1}(W,\cdot) = \frac{\mathsf{I}_t[GF_t](W)}{\mathsf{I}_t[F_t](W)}. \tag{5.xxxiii}$$

The two potentials $\mathbb{V}^{\mathrm{int}} = \mathbb{V}^{\mathrm{J}} + \mathbb{V}^{\mathrm{G}} + \mathbb{V}^{\mathrm{A}}$ and $\mathbb{V}'[S_t]$, read on real fields, have quadratic-growth norms bounded respectively by

$$\tfrac{7}{2}K_P\delta_t \le \tfrac{1}{2}w_\star, \qquad K_\Lambda s_t^{\mathrm{ab}} N_\star \le \tfrac{1}{4}w_\star, \qquad s_t^{\mathrm{ab}} = (1+\beta_{\mathrm{s}})\delta_t/\alpha_0.$$

They are therefore each at most a constant times δ_t. On the actual all-small-field history measures and with their step factor h_{t+1}, the quotient has the disintegration identity

$$\int H(\mathcal{M}V)G(V)\mathsf{h}_{t+1}(V)\,\mathrm{d}m_t(V) = \int H(W)\left[\int G\,\mathrm{d}\kappa_{t+1}(W,\cdot)\right]\mathrm{d}m_{t+1}(W).$$

All statements use the prescribed tests and gauge factors of the canonical history. ∎

Proof. The actual construction in Thm. YM3_I:W supplies the Gaussian structure and all four local potentials on the full core, with Def. YM3_I:52.1's bounds. Restricting the field polydisc to the real cutoff box gives the quadratic-growth norm. The triangle inequality and the calculation in the proof of Thm. YM3_I:V give the displayed two bounds (its interaction bound is in fact $7w_\star/16$). The fourth norm uses the admitted bare-array bound $N(\mathfrak{f}_t) \le N_\star$. The exact integral identity YM3_I:(902) at $x = \mathsf{t} = x_t$, hence $\tau = 1$, is precisely Ξ_W: the additional quadratic potential is zero there. It proves absolute convergence and nonvanishing. The sum defining E_W is finite and continuous on the box. Division by Ξ_W gives finite variation and mass one.

To identify the quotient, the action and gauge remainders in the chart satisfy

$$e^{-x_t G_W(B)+\mathsf{J}_W(B)}F_t(V_W(B)) = e^{v_W}e^{-(B,C_W^{-1}B)/2}e^{\mathsf{E}_W(B)}, \qquad v_W = -x_t A^{\eta_t}(U_+) + S_t(U_+).$$

This is the sum of the definitions of the Jacobian, gauge and action remainders; their quadratic parts are $\widehat{\mathsf{G}}_W$ and B_W. The physical chart identity of [7, Lem. B.3] therefore gives

$$\mathsf{I}_t[GF_t](W) = k_W \int_{\mathsf{Rg}_{x_t}} G(V_W(B))e^{\mathsf{E}_W(B)}\,\mathrm{d}\mu_{C_W}(B)$$

with $k_W \neq 0$ independent of G. Divide by its value at $G = 1$. The chart is continuous and $\mathcal{M}V_W(B) = W$, proving the Borel and fibre assertions. The averaged axial field and the complete operators are independent of the representative U_+; the action and the array are gauge invariant. Thus the measure is independent of that choice as well.

For disintegration, use the actual density and passing-test identity proved in Prop. 4.42 at zero source. On the passing input set, $dm_t = e^{E_{(t)}}F_t\,\mathrm{d}V$; in the step factor the compatibility tests equal one on the whole small-field chart. The physical chart formula consequently gives

$$\int H(\mathcal{M}V)G(V)\mathsf{h}_{t+1}(V)\,\mathrm{d}m_t(V) = e^{E_{(t)}}\int \chi_{t+1}(W)H(W)\mathsf{I}_t[GF_t](W)\,\mathrm{d}W.$$

At $G = 1$ this is the density formula for m_{t+1}. Insert eq. 5.xxxiii to obtain the assertion. Null sets in the density identities do not affect these integrals. Bounded Borel insertions are allowed by the absolute chart formula; their measurability and integrability follow from that same formula. At $t = 0$ the bare density is $F_0 = \rho_0$ on the passing set, using $4\varepsilon_0 < \varepsilon_{\mathrm{v}}$, so the argument is unchanged. □

5.3.2. *Recovering the datum from the response.* The response estimates give an upper bound for the fine field in terms of its datum. For the comparison of metrics we also need the reverse bound. Only the constant frame is needed: the bare spacing is $\eta = L^{-j}$, the terminal spacing is one, all regions through level j are the whole torus and the next region is empty. Thus every constraint row has scale one and there are no interface rows. All norms in the following statement use counting measure on positive bonds. Retain the geometric and response hypotheses of §5.3.

Lemma 5.32. Let U be a regular real background in this constant frame, with $|\partial U(p) - 1| \le \alpha_0 \eta^2$ and $\alpha_0 \le L^{-2} a_{\mathsf{m}}$, where a_{m} is the radius in Prop. YM3_I:30.2. Let $\mathcal{Q}(A) = a_j(A)$ be its centered straightened row logarithm, and set $K_0 = D\mathcal{Q}(0)$. Then

$$\sup_c \sum_b |K_0(c,b)| \le 2, \qquad \sup_b \sum_c |K_0(c,b)| \le 2L^{-3j}, \qquad \|K_0\|_{2\to 2} \le 2L^{-3j/2}. \tag{5.xxxiv}$$

For the normalized response of equation YM3_I:(837), $\mathcal{Q}(\mathcal{H}_U(\zeta)) = \zeta$ near zero. Consequently, with $\mathsf{Z} = L^{-j} D\mathcal{H}_U(0)$,

$$\|\xi\| \le 2L^{-j/2}\|\mathsf{Z}\xi\| \le C_{\mathcal{Q}} L^{-j/2}\|\mathsf{Z}\xi\|, \qquad C_{\mathcal{Q}} := 34(L^3+2). \tag{5.xxxv}$$

At $j = 0$ the same conclusions hold with $K_0 = \mathsf{Z} = 1$. ■

Proof. The row estimate is the zero-field estimate for Q_j^{m} in Prop. YM3_I:30.2, with $\xi_j = 1$ and $\vartheta_c = 1$. Its level-j column estimate is equation YM3_I:(468), again with $\xi_j = 1$ and $\eta = L^{-j}$. These are bounds on sums of the norms of the actual derivative kernels. For clarity, the counting-measure Schur inequality here is

$$\sum_c \Big| \sum_b K_0(c,b) A(b) \Big|^2 \le 2 \sum_{c,b} |K_0(c,b)|\,|A(b)|^2 \le 4L^{-3j} \sum_b |A(b)|^2.$$

It gives the asserted norm bound.

The straightening recursion gives

$$\mathcal{M}^j(e^{i\eta A} U) = (e^{i\mathcal{Q}(A)} V)^{v_j[A]}, \qquad V = \mathcal{M}^j U.$$

This is the accumulated one-step identity used in Prop. YM3_I:30.2 and in the gauge-removed row formula before Lem. YM3_I:30.6. For the response, equation YM3_I:(837) says that its gauge factor u_ζ satisfies $u_\zeta(y) v_j[\mathcal{H}_U(\zeta)](y) = 1$ at every terminal site. Its actual average is $e^{i\zeta} V$. Gauge covariance and the displayed cancellation therefore give $e^{i\mathcal{Q}(\mathcal{H}_U(\zeta))} V = e^{i\zeta} V$. The two logarithms lie on the branch through zero, proving their equality. Differentiating at zero yields $K_0 D\mathcal{H}_U(0)\xi = \xi$, or $L^j K_0 \mathsf{Z}\xi = \xi$. Equation eq. 5.xxxv follows from eq. 5.xxxiv; the retained constant $34(L^3+2)$ is larger than two. At $j = 0$ no averaging is performed, so both maps are the identity. □

Lemma 5.33 (Continuity on the full core)**.** Let $j \ge 1$ and let X be an admitted finite domain. On the hereditary core of Def. YM3_I:49.1, take $0 < b_1 \le \pi L^j$. Suppose F depends only on the represented pair and is holomorphic on every slice with fixed real factor. Then F is continuous on the full core, with its topology of matrix entries and bond-form coordinates. ■

Proof. Put $\xi = L^{-j}$ and let $P_\nu = (U_\nu, A_\nu, J_\nu) \to P_0$ in the core. Write $\mathbf{U}_\nu = e^{i\xi A_\nu} U_\nu$. We move the real factor to U_0, keeping this represented pair fixed. For each positive bond, $|i\xi A_0(b)| < \pi$. The differential of the exponential at this matrix is invertible: after a left translation it is the entire function $(1 - e^{-\mathrm{ad}Z})/(\mathrm{ad}Z)$, whose only possible zeros have nonzero eigenvalues in $2\pi i\mathbb{Z}$; here $\|\mathrm{ad}Z\| < 2\pi$. A local inverse Λ_b therefore exists at $e^{i\xi A_0(b)}$. For all large ν set

$$A'_\nu(b) = (i\xi)^{-1} \Lambda_b(\mathbf{U}_\nu(b) U_0(b)^{-1}), \qquad \widetilde{P}_\nu = (U_0, A'_\nu, J_\nu).$$

There are finitely many bonds, so these choices hold together, $A'_\nu \to A_0$, and $\widetilde{P}_\nu$ represents the pair $(\mathbf{U}_\nu, J_\nu)$ of P_ν.

We check that $\widetilde{P}_\nu$ remains in the core. Its real conditions hold at U_0, and its pair-only conditions hold because they hold for P_ν. The finitely many strict bounds on A'_ν and its covariant differences follow by continuity at fixed U_0. For each tower, its complex averaged datum is a continuous function of $\mathbf{U}_\nu$ on the admitted analytic branch. Its real datum is now the fixed datum of U_0. Their logarithmic quotient thus converges to the quotient for P_0. At P_0 that quotient is strictly inside the response ball; the same ball contains the quotients for all large ν. Holomorphy of the tower response gives convergence of its configuration, curvature and current. The strict bounds defining that tower consequently persist. There are only finitely many subdomains and tower tests on X, so all these assertions hold simultaneously. This proves $\widetilde{P}_\nu$ lies in the core for all large ν.

Both $\widetilde{P}_\nu$ and P_0 are now on the slice with real factor U_0. Slice holomorphy and pair dependence give

$$F(P_\nu) = F(\widetilde{P}_\nu) \longrightarrow F(P_0).$$

Since the core is a subspace of a finite-dimensional metric space, sequential continuity proves the assertion. □

5.3.3. *The half-Hölder estimate for the ordinary response.* The ordinary response to a block datum has a half-Hölder first covariant difference. This assertion concerns ordinary forcing in the constrained system. It does not give the same-exponent estimate for a variable anisotropy in Lem. 5.19. We retain the natural-pair geometry, actual averaging rows, local operators and decoupling parameters of Thm. YM3_I:M. All constants below depend only on L. The common regularity radius and block-number floor must meet the local inverse conditions used below.

For $0 < \beta < 1$ use the transported Hölder seminorm of Prop. YM3_I:34.1, with exponent β:

$$[f]_{\mathbf{U},\beta;y} = \sup_{i \neq i'} \left(\frac{\xi_y}{|i - i'|_1} \right)^\beta |\operatorname{Ad}(\mathbf{U}(\Gamma_{x,x'}))f(i') - f(i)|.$$

The indices have the same kind, their base points lie in the prescribed enlargement $\Delta^{3/2}(y)$, and the coordinate comb $\Gamma_{x,x'}$ stays in the local domain. At a boundary only admissible indices and combs are used. Write $|f|^\beta_{s;y} = \xi_y^{-s}(\sup|f| + [f]_{\mathbf{U},\beta;y})$ and $|\Psi|^{+\beta}_y = \max\{|\Psi|_y, |\nabla_{\mathbf{U}} A|^\beta_{-2;y}\}$ for the enhanced solution norm. The forcing norms and their divergence injection are those of eq. 5.xliii, with suprema instead of mean squares; only the first-slot divergence tensor receives the additional Hölder norm.

Lemma 5.34. There are a radius $a_H > 0$, a block floor M_H, a decay rate $\delta_H > 0$ and constants $\mathsf{B}_\sharp(0), \mathsf{B}_\sharp(1/2)$ such that the following holds. Take the local response setting above, with the four local-system radii at most a_H and block number $\mathfrak{M} \geq M_H$. Choose the common decay rate $\mathring{\delta}_0 \leq \delta_H$ in the decoupled construction. Then its normalized operators $H_s, K_s, \mathfrak{G}_s$ satisfy

$$\begin{aligned} \mathfrak{n}_{y,\zeta}(H_s X), \quad \mathfrak{n}_{y,\zeta}(K_s X) &\leq 6L^3 \mathsf{B}_\sharp(1/2) e^{8\varkappa} \sum_z e^{-\mu d(y,z)} |X|_z, \\ \mathfrak{n}_{y,\zeta}(\mathfrak{G}_s f) &\leq 6L^3 \mathsf{B}_\sharp(1/2) e^{8\varkappa} \sum_z e^{-\mu d(y,z)} |f|^\sharp_z, \qquad \mu = 3\mathring{\delta}_0/16, \end{aligned} \tag{5.xxxvi}$$

where $|f|^\sharp_z = \xi_z^3 \sup_{\mathsf{y}(b)=z} |f(b)|$ and

$$\mathfrak{n}_{y,\zeta}(A) = \frac{\xi_y^{5/2}[\zeta \nabla_{\mathbf{U}} A]_{1/2;\xi_y}}{\xi_y^{1/2}[\zeta]_{1/2;\xi_y} + |\zeta|}.$$

Here $\zeta \neq 0$ is a smooth scalar cutoff supported in the local comparison cube, whose intersection with the ambient region lies in W. The subscript $1/2;\xi_y$ restricts pairs to physical Manhattan distance at most ξ_y, and uses the same transports and domain restriction. In this display $[\cdot]_{1/2;\xi_y}$ is the physical quotient, with denominator $|x - x'|_1^{1/2}$; thus $\xi_y^{1/2}[\zeta]_{1/2;\xi_y}$ is dimensionless. The constants are

uniform for all admitted local domains and $|s(\Delta)| \le e^{\varkappa}$ and for arbitrary subsums of the localized expansion. ■

Proof. We give the extension of the two local estimates in the proof of Prop. YM3_I:34.1. Only the fixed exponents $1/2$ and $3/4$ are needed.

At the identity, eliminate the scalar and row components as in Section YM3_I:23.4. The potential is the vector Green operator applied to

$$f_1' + \nabla^{\mathsf{T}} X^{(p)} + \nabla^{\mathsf{T}} X_1 + Q^{\mathsf{T}} \omega.$$

The term X_1 is absent for ordinary forcing; $Q^{\mathsf{T}}\omega$ is absent in system [4]. To extend the cutoff proof there from $\beta = 1/2$ to a fixed $0 < \beta < 1$, take $\beta' = (1+\beta)/2$. The scalar elimination bounds $X^{(p)}$ in $C^{\beta'}$ with weight -2, and f_1' and $Q^{\mathsf{T}}\omega$ with weight -3. Apply the general Green-kernel Hölder bounds YM3_I:(225) and YM3_I:(227), with output exponent β and, for $X^{(p)}$, input exponent β'; their gap is $(1-\beta)/2 > 0$. The same cutoff calculation therefore bounds the first difference of the potential in C^{β} for ordinary forcing. We use this extension only at $\beta = 3/4$; the case $\beta = 1/2$ is equation YM3_I:349 itself. For the other estimate needed below, that same equation YM3_I:349 already bounds the $C^{1/2}$ first difference for a $C^{3/4}$ divergence source. Its exponent gap is $1/4$ and is not set to zero.

For completeness, the passage to a small regular field does not require absorption in the enhanced norm. In a local real gauge write $\mathbb{K}_0 - \mathbb{K} = \iota_0 V$. The already proved ordinary inverse controls $|\Psi|_y$, and the local coefficient estimate gives

$$|V\Psi|_y \le C\epsilon \max_{d(y,z)\le\rho} |\Psi|_z.$$

For $\Psi = \mathbb{E}\iota_0 g$ the identity $\Psi = \mathbb{E}_0\iota_0(g + V\Psi)$ puts only ordinary forcing into the flat enhanced estimate. For a divergence datum g^+ one uses $\Psi = \mathbb{E}_0\iota_0^+(g^+ + (V\Psi, 0))$ instead; the ordinary bound for that solution first uses the positive-exponent divergence estimate. The extra term has no divergence component, so no Hölder norm of $V\Psi$ is required. Converting the injection and the transported norms between the flat and small gauges costs a fixed local factor. These are the same coefficient, shift and transport comparisons used to prove equation YM3_I:519. At layer zero, every comparison distance is a bounded number of bare spacings and the massive boundary inverse gives the estimates directly. This proves the three required local estimates, with one radius and floor chosen for the two fixed exponents.

The cutoff commutator has ordinary forcing. Consequently in each walk the enhanced estimate is needed only at its output factor; the remaining factors use the ordinary strong inverse. More explicitly, if T_κ is the local inverse with output cutoff and $R^\sharp_\kappa$ is its commutator forcing, the ordinary walk is

$$T_{\kappa_0} R^\sharp_{\kappa_1} \cdots R^\sharp_{\kappa_m}.$$

The output factor is bounded in $|\cdot|^{+\beta}$, and each of the other factors costs $q = C\mathfrak{M}^{-1}e^{-\delta D_0}$. The enlarged comparison cube and its two difference shifts lie in the common $[5/6]$ region, so the rows, transports and seminorms used by the local and global systems agree. This is the same identification used in Prop. 5.37. A finite number of pieces can follow a given piece. Increasing the common block floor until their number times q is at most $1/2$ gives the summed bound and the termwise walk bound.

For H_s and K_s, apply this estimate to the last-row forcing. For $\mathfrak{G}_s$ use the doubled inverse in the proof of Prop. YM3_I:34.1. Its coupling is $D(D^{\mathsf{T}}A' - w')$, a divergence source plus the explicitly retained layer-zero boundary correction. Apply the ordinary $3/4$ estimate to the second copy. The transported product and shift formulas in that proof bound the coupling tensor in $C^{3/4}$, with weight -2. Apply the divergence estimate with exponents $1/2, 1/4$ to the first copy. This bounds its first difference in $C^{1/2}$. The doubled cutoff commutator in equation YM3_I:526 again has ordinary forcing, so the same walk multiplication and summation apply. The local multiplier insertions see

only the weak norm, already bounded by these output norms; their tree factors and summability are unchanged. This identifies the estimate for the actual projected operator, rather than for a different scalar Green function.

Finally insert the parameters. The carrier count in the decoupled construction gives

$$\#\mathsf{P}(\omega) \le 8 + 68 D_\omega/\mathsf{w}, \qquad |s^\omega| e^{-aD_\omega} \le e^{8\varkappa} e^{-(a-68\varkappa/\mathsf{w})D_\omega}.$$

The standing lower bound on w leaves rate $\mathring{\delta}_0/4$. Fix $\mathsf{B}_\sharp(1/2)$ large enough to include the preceding local constants, the absolute tree masses and the bounded number of output-window cells. The normalized column bounds are therefore

$$[\zeta \nabla H_s e_c]_{1/2;\xi_y}, \quad [\zeta \nabla K_s e_c]_{1/2;\xi_y} \le \mathsf{B}_\sharp(1/2) e^{8\varkappa} (\xi_y^{1/2} [\zeta]_{1/2;\xi_y} + |\zeta|) \xi_y^{-3/2} \xi_z^{-1} e^{-\mathring{\delta}_0 d(y,z)/4} |e|,$$
$$[\zeta \nabla \mathfrak{G}_s f]_{1/2;\xi_y} \le \mathsf{B}_\sharp(1/2) e^{8\varkappa} (\xi_y^{1/2} [\zeta]_{1/2;\xi_y} + |\zeta|) \xi_y^{1/2} e^{-\mathring{\delta}_0 d(y,z)/4} \sup |f|,$$

when c is at z and f is supported over z. The ordinary product inequality $[\zeta v]_\beta \le |\zeta| [v]_\beta + [\zeta]_\beta |v|$ explains the cutoff factor. After multiplying by $\xi_y^{5/2}$, the scale ratios are ξ_y/ξ_z and $(\xi_y/\xi_z)^3$ respectively. The layer comparison absorbs them with a factor L^3 and a loss $\mathring{\delta}_0/16$ in the rate. There are at most six rows at each top. Summing the columns or source cells proves eq. 5.xxxvi. All steps use absolute bounds on individual terms, so arbitrary subsums obey the same estimate. □

Proposition 5.35. Retain the hypotheses of Lem. 5.34 and the datum radius of Thm. YM3_I:M. Choose $\mathsf{B}_\sharp(0)$ as the ordinary kernel constant in that theorem, $\mathsf{B}_\sharp(1/2) \ge \mathsf{B}_\sharp(0)$ as above, and put

$$\mathsf{b}_0 = 6L^3 \mathsf{B}_\sharp(0), \qquad \mathsf{b}_{\mathrm{D}} = \mathsf{b}_0 c_S(\mathring{\delta}_0/16), \qquad \widehat{\mathsf{b}} = \mathsf{b}_{\mathrm{D}} e^{8\varkappa}.$$

For its ordinary response $\mathbf{H}_s$ and any two admissible block data,

$$\sup_y w(y) \mathfrak{n}_{y,\zeta}(\mathbf{H}_s(B_1) - \mathbf{H}_s(B_2)) \le 3\widehat{\mathsf{b}} \frac{\mathsf{B}_\sharp(1/2)}{\mathsf{B}_\sharp(0)} |B_1 - B_2|_w. \tag{5.xxxvii}$$

Here w is any of the admissible exponential weights of Thm. YM3_I:M. In particular, taking $B_2 = 0$ and $w = 1$ gives the same bound for the response itself. ■

Proof. Write $A_i' = K_s B_i + \mathbf{A}_0(B_i)$. The response equations and constraint correction give the exact difference identity

$$\mathbf{H}_s(B_1) - \mathbf{H}_s(B_2) = K_s(B_1 - B_2) - \mathfrak{G}_s(W_s(A_1') - W_s(A_2')) - H_s(D_s(A_1') - D_s(A_2')).$$

The proof of Thm. YM3_I:J, with the radius in Thm. YM3_I:M, gives

$$|W_s(A_1') - W_s(A_2')|_w^\sharp \le |B_1 - B_2|_w, \qquad |D_s(A_1') - D_s(A_2')|_w \le \tfrac{1}{2} |B_1 - B_2|_w.$$

Indeed $\|A_i'\| \le 2\widehat{\mathsf{b}} \sup |B_i|$ and $\|A_1' - A_2'\|_w \le 2\widehat{\mathsf{b}} |B_1 - B_2|_w$; the first bound uses $8C_W \widehat{\mathsf{b}}^4 \sup |B_i| \le 1$, and the second uses $32 C_2 \widehat{\mathsf{e}\mathsf{b}} \varepsilon_3 \le 1$, exactly the radius inequalities in equation YM3_I:494. Multiplication by w in eq. 5.xxxvi leaves decay rate $\mathring{\delta}_0/16$ for the top sum. Each operator is thus bounded by $\widehat{\mathsf{b}} \mathsf{B}_\sharp(1/2)/\mathsf{B}_\sharp(0)$ in the indicated seminorm. The triangle inequality gives the factor $1 + 1 + 1/2 \le 3$. Finally $\mathbf{H}_s(0) = 0$. □

5.3.4. *The local perturbation of the Hessian.* A plaquette weight changes the principal part of the Hessian. The remainder is a local curvature term. This elementary fact identifies the perturbation that a weighted inverse estimate must control.

Lemma 5.36. Let A be a finite ambient region of a three-dimensional lattice of spacing ξ, and let $\mathbf{U}$ be an $\mathrm{SL}(2,\mathbb{C})$ configuration on its inner bonds. Give one-forms the complex bilinear pairing $(A,B)_\xi = \xi^3 \sum_b \frac{1}{2} \operatorname{tr}(A(b)B(b))$, where the sum uses positive inner bonds. For a complex function w on the positive inner plaquettes, define

$$F_w(A) = \xi^{-1} \sum_p w(p) (1 - \tfrac{1}{2} \operatorname{tr} \partial(e^{i\xi A} \mathbf{U})(p)).$$

There is a unique symmetric Hessian Δ_w with $F_w''(0)[E,A] = (E, \Delta_w A)_\xi$. Its entries depend linearly on w and polynomially on the matrices $\mathbf{U}(b)^{\pm 1}$.

Write ∇_μ for the covariant forward difference, set $D^{(1)}A = \nabla_\mu A_\nu - \nabla_\nu A_\mu$ on a $\mu\nu$ plaquette, and use the ξ^3-weighted plaquette pairing to define its transpose. Then

$$\Delta_w = D^{(1)\mathsf{T}} w D^{(1)} + Z_{1,w}. \tag{5.xxxviii}$$

Suppose that $|\partial\mathbf{U}(p) - \mathbf{1}| \le 1$ and $|\operatorname{Ad}(\mathbf{U}(b'))^{\pm 1}| \le 2$ on every plaquette containing a given bond b and on its four bonds. Put

$$w_b = \max_{p \ni b} |w(p)|, \qquad f_b = \xi^{-2} \max_{p \ni b} |\partial\mathbf{U}(p) - \mathbf{1}|, \qquad |A|_{(b)} = \max_{p \ni b} \max_{b' \in p} |A(b')|.$$

With absolute constants, the kernel of Δ_w satisfies $|\Delta_w(b,b')| \le C w_b \xi^{-2}$, and

$$|(Z_{1,w}A)(b)| \le C w_b f_b |A|_{(b)}. \tag{5.xxxix}$$

Both kernels vanish unless b, b' belong to a common plaquette. The same assertions hold after restriction to a subregion, with fields extended by zero and with the ambient plaquettes retained. ■

Proof. We give the weighted version of the plaquette calculation in Lem. YM3_I:25.3. At $p = p_{\mu\nu}(x)$ put $Y_p = \partial\mathbf{U}(p)$, $y_0 = \frac{1}{2}\operatorname{tr} Y_p$, $\mathsf{U}_\mu = \operatorname{Ad}(\mathbf{U}_\mu)$ and

$$\begin{aligned} a_1 &= A_\mu(x), & a_2 &= \mathsf{U}_\mu(x) A_\nu(x + \xi e_\mu), \\ a_3 &= \operatorname{Ad}(Y_p)\mathsf{U}_\nu(x) A_\mu(x + \xi e_\nu), & a_4 &= \operatorname{Ad}(Y_p) A_\nu(x). \end{aligned}$$

For $\epsilon = (1,1,-1,-1)$ and $S_p(A) = \sum_i \epsilon_i a_i$, transporting the four factors to x gives

$$\partial(e^{it\xi A}\mathbf{U})(p) = e^{it\xi a_1} e^{it\xi a_2} e^{-it\xi a_3} e^{-it\xi a_4} Y_p.$$

The quadratic term of an ordered product is $\frac{1}{2}(\sum_i X_i)^2 + \frac{1}{2}\sum_{i<j}[X_i, X_j]$. Since a traceless two-by-two matrix S has $S^2 = \frac{1}{2}\operatorname{tr}(S^2)\mathbf{1}$, and $Y_p = y_0\mathbf{1} + i\xi^2 F_{\mathbf{U}}(p)$, differentiation yields

$$(A, \Delta_w A)_\xi = \xi^3 \sum_p w(p) \left\{ y_0 \langle \xi^{-1} S_p(A), \xi^{-1} S_p(A) \rangle + i \sum_{i<j} \epsilon_i \epsilon_j \langle [a_i, a_j], F_{\mathbf{U}}(p) \rangle \right\}. \tag{5.xl}$$

The pairing is nondegenerate; polarizing this finite quadratic form therefore defines exactly one symmetric operator. The formula also proves its asserted dependence on w and $\mathbf{U}$.

The exact identity

$$\xi^{-1} S_p(A) = D^{(1)}A(p) + \rho_p(A), \qquad \rho_p(A) = \xi^{-1}(I - \operatorname{Ad}(Y_p))\{\mathsf{U}_\nu(x) A_\mu(x + \xi e_\nu) + A_\nu(x)\}$$

shows that the quadratic form of $Z_{1,w}$ is the sum, multiplied by $\xi^3 w(p)$, of

$$\begin{aligned} &(y_0 - 1)\langle D^{(1)}A, D^{(1)}A \rangle + 2y_0 \langle D^{(1)}A, \rho_p(A) \rangle + y_0 \langle \rho_p(A), \rho_p(A) \rangle \\ &\qquad + i \sum_{i<j} \epsilon_i \epsilon_j \langle [a_i, a_j], F_{\mathbf{U}}(p) \rangle. \end{aligned}$$

Here all values in one summand are at p. If $\kappa_p = |Y_p - \mathbf{1}| \le 1$, the two-by-two determinant identity gives

$$|y_0 - 1| \le \kappa_p, \quad \xi^2 |F_{\mathbf{U}}(p)| \le \kappa_p, \quad |I - \operatorname{Ad}(Y_p^{\pm 1})| \le 3\kappa_p.$$

Consequently $|D^{(1)}A(p)| \le C\xi^{-1}|A|_{(p)}$ and $|\rho_p(A)| \le C\xi^{-1}\kappa_p |A|_{(p)}$. Polarize against a one-form supported at b, of norm at most one. Each of the at most four contributing plaquettes then costs $C|w(p)|\xi^{-2}\kappa_p |A|_{(p)}$; the factor ξ^3 cancels against that of the one-bond pairing. The nondegenerate finite dimensional pairing gives eq. 5.xxxix. The same polarization of eq. 5.xl, with the second field supported on one bond, gives the kernel bound. Every summand involves only the four bonds of its plaquette. This proves the locality assertions, including restriction by zero extension. □

In particular the perturbation at $w = 1 + v$ is

$$\Delta_{1+v} - \Delta_1 = D^{(1)\mathsf{T}} v D^{(1)} + Z_{1,v}.$$

The principal insertion contains the divergence source $vD^{(1)}A$. Multiplication by v costs at most $|v|_\infty$ in L^2. This calculation identifies the insertion; it does not assert the uniform inverse or the endpoint Hölder estimate for that insertion.

5.3.5. *Mean-square bounds for the weighted system.* The weight in Lem. 5.36 need not have bounded differences. We keep its principal contribution in divergence form. The relevant inverse estimate is therefore a mean-square estimate with a divergence source. We give the extension of the proof of Thm. YM3_I:G that is needed for this purpose.

We use precisely the separated regions, inner positive bonds, reduced coarse rows and four systems of Sections YM3_I:23 and YM3_I:27. In particular the row space consists of the positive level-n bonds with neither endpoint in Ω_{n+1}. The bilinear pairings have weights ξ^3 on fine sites and bonds, ξ_y^3 on block tops, and ξ_c^3 on a coarse row c. All transposes below use these pairings. The averaging row is the derivative of the physical constraint,

$$QA(c) = \frac{d}{ds}\Big|_{s=0} \xi_c^{-1} \mathcal{Q}_n(\mathbf{U}; s\xi_c A)(c),$$

with the convention of equation YM3_I:370; Q' is the ordered scalar average. At $\mathbf{U} = \mathbf{1}$, these are the straight bond average and the arithmetic scalar average. At nontrivial fields we retain their actual first and second derivatives. In particular,

$$(A, \Delta^{(2)}[\mathsf{j}]B)_\xi = 2(C^{(2)}(A, B), \mathsf{j})_{\mathrm{rows}}.$$

Thus the result below does not replace the physical constraint by its linearization at the identity.

For a fine-indexed field f define the local mean-square seminorm

$$\|f\|_{s;y} = \xi_y^{-s-3/2} \left(\xi^3 \sum_{\mathsf{y}(i)=y} |f(i)|^2 \right)^{1/2}. \tag{5.xli}$$

For a field on tops or rows, use $\|f\|_{s;y} = \xi_y^{-s} \max_{\mathsf{y}(i)=y} |f(i)|$. Values are zero outside the region. The system differences omit translated indices outside their inner-bond or site index set. The ambient Hessian retains boundary plaquettes of the zero-extended field; its additional boundary terms are treated explicitly in the proof below. For $\Psi = (A; v, w, \sigma; p, q, r; \omega)$ put

$$\begin{aligned} \|\Psi\|_y = \max\{&\|A\|_{-1;y}, \|\nabla A\|_{-2;y}, \|v\|_{0;y}, \|Dv\|_{-1;y}, \|w\|_{-2;y}, \|Dw\|_{-3;y}, \|\sigma\|_{-4;y}, \\ &\|p\|_{-2;y}, \|q\|_{0;y}, \|Dq\|_{-1;y}, \|r\|_{-4;y}, \|\omega\|_{-3;y}\}. \end{aligned} \tag{5.xlii}$$

Absent components are omitted for a compressed system. Write $g^+ = (f_1, \dots, f_8; X_5, X_1)$ and set

$$\begin{aligned} \|g^+\|_y = \max\{&\|f_1\|_{-3;y}, \|f_2\|_{-2;y}, \|f_3\|_{-4;y}, \|f_4\|_{0;y}, \|f_5\|_{-4;y}, \|f_6\|_{-2;y}, \|f_7\|_{0;y}, \\ &\|f_8\|_{-1;y}, \|X_5\|_{-3;y}, \|X_1\|_{-2;y}\}. \end{aligned} \tag{5.xliii}$$

Here X_1 has one fine bond index and one difference direction. The source map ι^+ replaces f_1 by $f_1 + \nabla^{\mathsf{T}} X_1$ and f_5 by $f_5 + D^{\mathsf{T}} X_5$. The original map ι is the case $X_1 = 0$.

Proposition 5.37. There are constants $a_2, r_2, B_2, \delta_2 > 0$ and a power M_2 of L, depending only on L, with the following properties. Take the full geometric hypotheses and the natural-pair class of Thm. YM3_I:G, with block number $\mathfrak{M} \geq M_2$ and all four radii $\alpha_0, \alpha_1, \gamma_0, \gamma'$ at most a_2. For any complex plaquette function v with $|v|_\infty < r_2$, replace Δ_Y by Δ_{1+v} in the first slot of each applicable system; retain the other coefficients and rows. The resulting four systems $\mathbb{K}_v^{[i]}$, $i = 1, 2, 4, 5$, are invertible. For $i = 4, 5$,

$$\|(\mathbb{K}_v^{[i]})^{-1} \iota^+ g^+\|_y \leq B_2 \sup_z e^{-\delta_2 d(y,z)} \|g^+\|_z. \tag{5.xliv}$$

The two scalar systems are unchanged. Their bounds are the corresponding compressed bounds. Constants are independent of the volume, the number of layers and the differences of v.

The inverse has a sum over walks of the pieces of Section YM3_I:27.1. If $\varpi = (\kappa_0, \dots, \kappa_m)$, its term $E_{v,\varpi}\iota^+$ vanishes unless consecutive enlarged pieces meet. It is supported in $\kappa_0^{[2/3]}$ at its output and depends only on the coefficients and weights in the union of the local regions of the visited pieces. With $y_0 = y$ it satisfies

$$\|E_{v,\varpi}\iota^+ g^+\|_y \le B_2 q_2^m \sup\left\{e^{-\delta_2 \sum_{a=0}^m d(y_a, y_{a+1})}\|g^+\|_{y_{m+1}}\right\}, \qquad q_2 = \frac{C_2}{\mathfrak{M}} e^{-\delta_2 D_0}. \tag{5.xlv}$$

The intermediate tops belong to $\kappa_{a-1}^{[5/6]} \cap \kappa_a^{[5/6]}$ and the final top to $\kappa_m^{[5/6]}$. The number D_0 is the separation between the input cutoff and the commutator support in the same piece construction. An empty supremum is zero. The rate can be fixed first below the unweighted local rate; increasing M_2 and decreasing a_2, r_2 then preserves that rate for all weights under consideration. ■

Proof. We retain the proof's order: the flat system, a small field, and then localization to regular pieces. This also checks that the mean-square estimate uses the same rows as the nonlinear problem.

The flat inverse. Write

$$G' = (\Delta'_a)^{-1}, \quad C' = (Q'G'^2Q'^{\mathsf{T}})^{-1}, \quad N = G'Q'^{\mathsf{T}}C', \quad P = NQ'G', \quad R = I - P,$$
$$G_0 = (\Delta_Y(\mathbf{1}) + DRD^{\mathsf{T}} + Q^{\mathsf{T}}\mathsf{m}^{-2}Q)^{-1}, \qquad Z = QG_0Q^{\mathsf{T}}.$$

These are the scalar, vector and row operators in Section YM3_I:23.2. In the notation $T : s \Rightarrow s'$, the arrow means an estimate $\|Tf\|_{s';y} \le C \sup_z e^{-\delta d(y,z)}\|f\|_{s;z}$. At a common sufficiently small rate, their mean-square bounds are

$$\begin{gathered} O : s \Rightarrow s+2, \qquad \nabla O, O\nabla^{\mathsf{T}} : s \Rightarrow s+1, \qquad \nabla O\nabla^{\mathsf{T}} : s \Rightarrow s, \\ \nabla\nabla G_0 : s \Rightarrow s, \qquad C' : 0 \Rightarrow -4, \qquad Z^{-1} : s \Rightarrow s-2, \qquad Q, Q^{\mathsf{T}}, Q', Q'^{\mathsf{T}} : s \Rightarrow s. \end{gathered} \tag{5.xlvi}$$

Here $O = G', G_0$, and only weights $-4 \le s \le 2$ are used. For completeness, these bounds follow from the local L^2 estimates in the proof of Prop. YM3_I:21.3, the six bounds of Prop. YM3_I:21.1, and the top and row kernel estimates in Sections YM3_I:21 and YM3_I:22, as follows. If $N_y(f) = (\xi^3 \sum_{\mathsf{y}(i)=y} |f(i)|^2)^{1/2}$, the local bounds have the form

$$N_y(T\mathbf{1}_z f) \le C\xi_y^a e^{-\delta_0 d(y,z)} N_z(f), \qquad a = 2, 1, 0.$$

Normalization introduces $(\xi_z/\xi_y)^{s+3/2}$. The layer-gap estimate of Lem. YM3_I:21.2 bounds this factor by $L^6 e^{\delta_0 d(y,z)/4}$ after increasing the separation threshold. Summation over z spends another quarter of the decay; the remaining half gives the arrow. The same argument applies to the differentiated scalar energy estimate: pairing $\Delta'_a u = \nabla^{\mathsf{T}} X$ with an exponentially weighted u bounds its weighted gradient by $C\|X\|_2$, using the coercive conjugated form in that proof. Thus the mixed scalar arrow costs no difference of X.

The top and row seminorms are suprema, so their arrows are their counting-kernel bounds. For an average, Cauchy–Schwarz gives

$$|Qf(c)|^2 \le \Big(\sum_b |Q(c,b)|\Big) \sum_b |Q(c,b)|\,|f(b)|^2.$$

The straight average has row sum one and $|Q(c,b)| \le (\xi/\xi_c)^3$; Q' is the arithmetic mean. A double block meets only $C(L)$ cells at comparable scales. This proves their mean-square arrows. The weighted transposes have bounded row sums, and top or row input is already measured by its supremum; there are at most nine fine components per site. This proves the two transpose arrows with no mesh-dependent factor. These observations also justify compositions of the arrows: the supremum estimate composes by the triangle inequality for d.

The elimination formulas are unchanged by adding X_1. Set

$$p' = RG' f_5 + P f_6 + N f_7, \qquad p'' = RG' D^{\mathsf T} X_5, \qquad \varphi = P f_2 + RG' f_3 - N f_4,$$
$$f_0 = f_1 + D\varphi + Dp' + Dp'', \qquad f_* = f_0 + \nabla^{\mathsf T} X_1.$$

Direct substitution in the eight equations gives

$$\begin{aligned} p &= p' + p'', & q &= G'(f_6 - p), \\ r &= C'\{Q'G'^2(f_5 + D^{\mathsf T} X_5) - Q'G' f_6 - f_7\}, & & \\ A &= G_0 f_* & &(i=4), \\ \omega &= Z^{-1}(f_8 - QG_0 f_*), & A &= G_0(f_* + Q^{\mathsf T}\omega) \quad (i=5), \\ w &= PD^{\mathsf T} A + \varphi, & v &= G'(f_2 + D^{\mathsf T} A - w), \\ \sigma &= C'\{Q'G'(f_2 + D^{\mathsf T} A) - Q'G'^2 f_3 - f_4\}. & & \end{aligned} \tag{5.xlvii}$$

For example, p', p'', φ have weight -2 and their gradients have weight -3; for Dp'' use $DRG'D^{\mathsf T} = DG'D^{\mathsf T} - (DP)G'D^{\mathsf T}$. Hence f_0 has weight -3. The mixed arrow in eq. 5.xlvi gives A of weight -1 and ∇A of weight -2 even when X_1 is only in mean square. The row expression gives ω of weight -3.

To finish the remaining slots, write $D^{\mathsf T} A = T_1 \nabla A + T_2 A$. The first matrix has at most three entries per row and one per column, each of size one. The second is supported on the layer of bare cells meeting the boundary, where $\xi_y = \xi$, and has at most three entries per row and two per column, each of size ξ^{-1}. Cauchy–Schwarz therefore bounds $D^{\mathsf T} A$ with weight -2. The last line of eq. 5.xlvii now gives exactly the remaining entries of eq. 5.xlii. This proves the flat estimate. When $X_1 = 0$ the same calculation also bounds $\nabla\nabla A$ with weight -3.

Small backgrounds and the actual rows. The defect factorization of Section YM3_I:25.4 is $\mathbb{K}_0 - \mathbb{K} = \iota_0 V$. Its coefficients satisfy the same bound in the mean-square norms,

$$\|V\Psi\|_y \le C\epsilon \max_{d(y,z)\le\rho_0} \|\Psi\|_z, \qquad \rho_0 = C(L). \tag{5.xlviii}$$

Here is the kernel check needed to pass from the displayed pointwise defect to this assertion. The local difference and current terms have uniformly bounded row and column multiplicities, scale factors ξ_y^{-a}, and coefficients $O(\epsilon)$; their second-order part is expressed using one difference of A. Cauchy–Schwarz gives eq. 5.xlviii for these terms. For the averaging terms, the proof of Lem. YM3_I: 25.2 gives

$$\sum_b |\delta Q(c,b)| \le C\epsilon, \qquad |\delta Q(c,b)| \le C\epsilon(\xi/\xi_c)^3,$$

with double-block support; the same estimates hold for Q'. The preceding Cauchy–Schwarz calculation proves their mean-square bounds. Their transposes use the specified physical pairings and the bounded number of incident rows. Products such as $\delta(Q^{\mathsf T}\mathsf{m}^{-2}Q)$ are expanded with one changed factor, so satisfy the required weighted bound.

Finally $\Delta^{(2)}[\mathrm{j}]$ has a symmetric bilinear kernel supported in the same double blocks. The bound for $C^{(2)}$ in that lemma, polarized against a field supported at one bond, bounds its absolute row sum by $C\epsilon\xi_y^{-2}$. In Pauli coordinates the bilinear transpose has norm at most twice the original matrix norm; therefore the column sum obeys the same bound with a fixed extra factor. The weighted Schur inequality proves its mean-square bound. These are all the types of terms in V. In particular this argument uses the Hessian of the actual nonlinear row, not merely a coincident row label.

Choose ϵ so that the norm of $g^+ \mapsto (V\mathbb{E}_0\iota_0^+ g^+, 0)$ is at most $1/4$ at the chosen decay rate. Then

$$\mathbb{E}\iota_0^+ = \mathbb{E}_0\iota_0^+[I - (V\mathbb{E}_0\iota_0^+, 0)]^{-1}$$

converges in these norms. The difference $\iota^+ - \iota_0^+$ is a local source with coefficients $O(\epsilon)$, since it contains only the changed transports in the two divergences. Covariant and ordinary first differences

differ by local zeroth-order terms of the same size. This proves the small-field estimate in the covariant seminorms. Gauge changes by SU(2) matrices preserve all the mean-square norms, so the conclusion also holds for a triple small in a local real gauge.

Inserting the weight. Only the first equation changes. By Lem. 5.36,

$$(\mathbb{K}_v - \mathbb{K})\Psi = e_1\{D^{(1)\mathsf{T}} v D^{(1)} A + Z_{1,v} A\} = \iota^+ V_v \Psi.$$

To read the last expression in the system's source space, put the antisymmetric components of $vD^{(1)}A$ whose indices belong to that space into X_1. Put $Z_{1,v}A$ and the omitted boundary contributions into f_1; the other entries are zero. Indeed, the ambient plaquette form retains plaquettes meeting the boundary, whereas the system's ∇ omits a translated bond outside its inner-bond index set. Their difference is supported on bonds a bounded number of bare steps from the boundary. Every such cell is in the bare layer, where $\xi_y = \xi$. Each difference has bounded row and column multiplicity and size at most $C|v|_\infty \xi^{-2}|A|$ on neighboring bonds. The weighted Schur estimate therefore gives it source weight -3 from the potential weight -1. This is the same boundary treatment used above for $D^{\mathsf{T}}A$. In the interior the identification with the transpose of $D^{(1)}A = \nabla_\mu A_\nu - \nabla_\nu A_\mu$ is exact. The curvature bound of the natural pair and the local plaquette support consequently give

$$\|V_v \Psi\|_y \le C|v|_\infty \max_{d(y,z)\le\rho_0} \|\Psi\|_z. \tag{5.xlix}$$

No derivative of v occurs. For a small-field local inverse, $\|V_v \mathbb{E}\iota^+\| \le 1/4$ follows by decreasing r_2. Consequently

$$\mathbb{E}_v \iota^+ = \mathbb{E}\iota^+ [I + V_v \mathbb{E}\iota^+]^{-1}$$

converges at the same fixed local rate. It is a right inverse on all sources: ι^+ is surjective, since the plain source components alone span the equations. Finite dimensionality therefore gives invertibility. The scalar compressions do not contain the changed first equation. This proves the local weighted conclusion for all four systems.

Localization. Use the pieces, $\widetilde{h}_\kappa$, θ_κ and local sequences of Lem. YM3_I:27.5. Lem. YM3_I:27.6 identifies the global and local physical averaging germs, including $C^{(2)}$, on $\kappa^{[5/6]}$. The pair on a regular piece has a real gauge in which its size is at most $8L^2(\alpha_0 + \alpha_1 + \gamma_0 + \gamma')$. Choose a_2 to place this below the small-field threshold above. A boundary piece has only the bare layer; its mean-square norms are finite-component Euclidean norms, uniformly equivalent to the local supremum norms. The boundary-piece inverse in the proof of Thm. YM3_I:G, followed by the same bounded insertion, gives its local estimate. Thus every local weighted inverse has a uniform mean-square bound at a rate $2\delta_2$.

The source cutoff must respect divergence form. Define $h \star g^+$ by multiplying all components by h and subtracting $[\nabla^{\mathsf{T}}, h]X_1$ from the first component and $[D^{\mathsf{T}}, h]X_5$ from the fifth. Then

$$\iota^+(h \star g^+) = h\, \iota^+ g^+. \tag{5.l}$$

The additional commutators have coefficients bounded by the first scaled difference of h, so are bounded in mean square. For the system commutator write

$$[\mathbb{K}_v, h] = \iota^+ C^+_{v,h}.$$

The unweighted terms have the same three kernel types checked in eq. 5.xlviii, with ϵ replaced by the scaled first or second difference of h. For example, $[Q,h](c,b) = Q(c,b)(h(b) - h(c))$; the quadratic-row term is treated by its symmetric Schur bounds. The weighted principal term is exactly

$$[D^{(1)\mathsf{T}} v D^{(1)}, h]A = D^{(1)\mathsf{T}}\{v[D^{(1)}, h]A\} + [D^{(1)\mathsf{T}}, h] v D^{(1)} A.$$

Keep the admitted indices of its first term as a divergence source. Its omitted boundary indices are placed in the ordinary first slot, by the same bare-layer calculation as for V_v. Commuting their local kernel with h introduces the difference of h across at most a fixed number of bare steps. Thus both the interior and boundary terms cost a first scaled difference of h and $|v|_\infty$; no difference of

v is taken. The local remainder $[Z_{1,v}, h]$ has the same bound by its plaquette support. The piece cutoffs consequently give

$$\|C^{+}_{v,\widetilde{h}_\kappa}\Psi\|_y \le C\mathfrak{M}^{-1}\max_{d(y,z)\le 2\rho_0}\|\Psi\|_z.$$

The commutator is supported where the cutoff is nonconstant, at distance at least D_0 from the input cutoff. All these norms coincide under the local/global identification: the values, differences and one-step shifts that occur here lie in $\kappa^{[5/6]}$. This is the same margin used in Lem. YM3_I:27.6; raising M_2 to ensure $\mathfrak{M} \ge 32L^4$ also includes the two shifts in the differentiated norms.

Define maps on divergence-form sources by

$$T^{+}_{\kappa} = \widetilde{h}_\kappa \mathbb{E}_{v,\kappa}\iota^{+}(\theta_\kappa \star \cdot), \qquad R^{+}_{\kappa} = -C^{+}_{v,\widetilde{h}_\kappa}\,\mathbb{E}_{v,\kappa}\iota^{+}(\theta_\kappa \star \cdot).$$

The identity $\sum_\kappa \theta_\kappa = 1$ gives

$$\mathbb{K}_v \sum_\kappa T^{+}_{\kappa} = \iota^{+}\left(I - \sum_\kappa R^{+}_{\kappa}\right).$$

Use half the local exponential decay for the input-to-commutator distance. The remaining half yields $\|T^{+}_{\kappa}\| \le C$ and $\|R^{+}_{\kappa}\| \le q_2$ with the supports in the statement. The local geometry bounds the number of possible next pieces by a number $N = N(L)$; at a fixed output top at most N first pieces contribute. Increase M_2 until $Nq_2 \le 1/2$. Then

$$\mathbb{E}_v\iota^{+} = \sum_{m\ge 0}\sum_{\kappa_0,\dots,\kappa_m} T^{+}_{\kappa_0}R^{+}_{\kappa_1}\cdots R^{+}_{\kappa_m}.$$

Composition gives eq. 5.xlv; summing the geometric series $\sum_m N^{m+1}q_2^m$ gives eq. 5.xliv. The local construction also proves the stated dependence on the visited regions. This completes the proof. □

A convenient precise meaning of a connected walk term is now available. Make a graph whose vertices are the pieces and whose edges join pieces with meeting [5/6] enlargements. For a finite vertex set S, sum the walks whose set of visited pieces is exactly S. There is no such walk when S is disconnected. If $|S| = s$, every contributing walk has at least $s-1$ transitions. Given $0 < \epsilon_* < 1$, increase M_2 until $Nq_2 \le \min\{1/2, \epsilon_*\}$. At each fixed output top its sum is then bounded, in eq. 5.xliv, by

$$C\sum_{m\ge s-1}(Nq_2)^m \le 2C\epsilon_*^{s-1}.$$

Locality follows from the union of the corresponding local regions. The same grouping applies to a difference of two local inverses: $E_v - E_0 = -E_0(\mathbb{K}_v - \mathbb{K}_0)E_v$ and the displayed local vertex identify its support. This is the connected-set factor used for the weighted fluctuation form; its cubes are these pieces, with the indicated enlargements, rather than an arbitrary partition.

The proposition also permits an envelope h with $h(z)/h(y) \le C_h e^{\delta_2 d(y,z)/2}$. Dividing eq. 5.xliv by $h(y)$ gives the corresponding weighted mean-square estimate, with constant B_2C_h and at least half the decay rate. This quantitative ratio is what is meant here when an envelope varies slowly.

Corollary 5.38. Under the hypotheses of Prop. 5.37, let the only forcing be the first-slot divergence $D^{(1)\mathsf{T}}Y$, with homogeneous row data. The potential component of the solution satisfies

$$\xi^3\sum_b \xi^{-2}_{y(b)}|A(b)|^2 + \xi^3\sum_{b,\mu}|\nabla_\mu A(b)|^2 \le C\xi^3\sum_p |Y(p)|^2. \tag{5.li}$$

Ambient boundary plaquettes are included on the right. The constant depends only on L. ■□

Proof. Decompose the divergence source by its cell positions and apply the local estimate to one cell at a time. The antisymmetric embedding and its bare-layer boundary correction have uniformly bounded multiplicities, as in the proof of eq. 5.xlix. Thus the block norm of the map from Y to $(A, \nabla A)$ is at most $Ce^{-\delta_2 d(y,z)}$, with weights -2 on Y and ∇A and -1 on A. The squares of these

normalized cell norms acquire the common factor ξ_y^{-1} when expressed in the energy norm of eq. 5.li. On the resulting ordinary sequence ℓ^2 spaces the block bound is therefore

$$Ce^{-\delta_2 d(y,z)}(\xi_z/\xi_y)^{1/2} \le C' L^{1/2} e^{-\delta_2 d(y,z)/2},$$

after the same increase of the layer separation used in the flat estimate. The row and column sums of this nonnegative matrix are bounded by the cell-summation estimate of Lem. YM3_I:21.2. Cauchy–Schwarz, first with its row sum and then with its column sum, proves its boundedness on ℓ^2. This is eq. 5.li. □

6. The two-term short-distance law

The leading action-density covariance is the free kernel of degree -6. Its first correction has degree -5. A test pair of scale d and shape $\vartheta \in (0,1)$ consists of nonzero Lipschitz functions $f, h \ge 0$, carried by two closed balls of radius d whose distance lies in $[d, 6d]$, with

$$\int f(x)\,\mathrm{d}x \ge \vartheta |f|_\infty d^3, \qquad \operatorname{Lip} f \le |f|_\infty/(\vartheta d),$$

and the same bounds for h. On a torus require $20d \le \ell$. Write this class as $\mathfrak{T}(d;\vartheta)$ and put

$$\mathcal{K}_q(f,h) = \iint f(x)h(y)|x-y|^{-q}\,\mathrm{d}x\,\mathrm{d}y, \qquad L_b^+ = c_2 b^{-2}(\mathcal{K}_6 + c_1 b^{-1}\mathcal{K}_5),$$

where

$$c_2 = \frac{9}{16\pi^2}, \qquad c_1 = \frac{8}{3\pi}.$$

On the least admissible torus the distance is read in the small Euclidean lift containing the test pair.

Sketch E (The first correction). At sufficiently small reference coupling, fixed label $b \in I(g)$ and sufficiently small d, the connected two-point function satisfies

$$|S_{2;K} - L_b^+|(f,h) \le \left[\bar\varepsilon(d) b^{-3} d + C_{\mathrm{lat}} b^{-2} \frac{a_K}{d}\right] |f|_\infty |h|_\infty, \qquad \bar\varepsilon(d) \longrightarrow 0.$$

The working depth of d must lie below K. After removal of the cutoff, the first error can be bounded by $Cb^{-3}d^{3/2}|f|_\infty|h|_\infty$. The law transfers, in the separately stated ranges, to other admitted tori, periodic states, volume-first limits and their components. ■□

Sketch of the argument. There are two jobs: identify the coefficient and control the interacting remainder. A finite-box perturbative calculation does only the first. To connect it with the full correlation, introduce two source parameters and differentiate the local normalizing energies twice. Sum these mixed derivatives over the renormalisation levels. An auxiliary complex field scale λ, multiplying every fluctuation field, makes this source-energy sum an even holomorphic function. The free contribution is explicit; the coefficient of λ^6 contains the first correction. Parity removes the half-integer orders in the equivalent Laplace expansion.

The coefficient is compared through a sequence of finite objects. First truncate the levels at a working scale, with a bound for the omitted tail. At zero field scale, the fluctuation laws are Gaussian laws restricted to their small-field boxes. Their precision matrices have a uniform positive lower bound. Gaussian tails therefore control the error made by removing the boxes. The coefficient is now a finite Gaussian expression with the coarsest background fixed at the identity. Compare these expressions in different boxes, then exhaust the lattice and compare the resulting lattice kernel with its homogeneous continuum part. Each comparison has its own geometry and error; the auxiliary box and cutoff are chosen so those errors are all small relative to $b^{-3}d$.

The last comparison is where the numerical value enters. The six two-loop graph classes are reduced to a finite residue functional. Their sum gives

$$\kappa^{\mathrm{lat}} = \frac{3}{2\pi^3} = c_2 c_1.$$

The finite-box coefficient is independent of the intermediate gauge parameter. Its infinite-lattice limit is obtained by taking the large-inverse-coupling expansion first at fixed box, then exhausting the lattice. This order does not establish a uniform remainder for growing boxes. The renormalisation comparisons above are needed precisely to return from that coefficient to the interacting measure.

Initially the expansion is proved in a cutoff window $L^K \le d^{-P}$. At fixed label, consecutive-cutoff comparison allows one to stop at a sufficiently fine $K_\sharp(d)$ inside that window and estimate every larger cutoff by it. Thus the window disappears from the fixed-label conclusion. The two errors in the result remain conceptually different: a_K/d is a lattice approximation error, whereas $\bar\varepsilon(d)b^{-3}d$ controls the correction beyond the two retained continuum kernels.

This is a two-term short-distance statement. It neither gives the arbitrary-order free law left open in the preceding sketch nor permits the cutoff and short-distance limits to be exchanged without their displayed scale restrictions.

6.1. **The two-term short-distance law on the torus.** For the action density, the first correction to the free two-point law is of relative order d/b. More precisely, the two-term approximation is

$$c_2 b^{-2}[\mathcal{K}_6 + c_1 b^{-1}\mathcal{K}_5](f,h), \qquad c_2 = \frac{9}{16\pi^2}, \quad c_1 = \frac{8}{3\pi},$$

The error is $[\bar\varepsilon(d)b^{-3}d + C_{\mathrm{lat}}b^{-2}a_K/d]\|f\|_\infty\|h\|_\infty$. Thm. S states the cutoff and label ranges, the improvement to a remainder of order $d^{3/2}$, and the averaged logarithmic correction. Cor. 6.3 passes the law to every periodic state and to the continuum limits and their components.

The proof first expresses the covariance as an average of source-energy jets over translated pairs. A complex field scale separates the free term, of order four, from the correction, of order six. A composed block-field chart identifies the latter with a constrained lattice Laplace coefficient. Comparison of two bare tori and removal of the flat block constraint then connect it to the infinite-lattice coefficient. The lattice calculation is proved in §7.1.

That calculation first takes the weak-coupling limit in a fixed box and then exhausts the lattice. Its box-dependent remainder is not a bound in the simultaneous continuum and large-volume limit. The renormalisation and finite-torus comparisons below provide the separate passage needed here.

6.1.1. *Parameters and test pairs.* We use the Wilson $G = \mathrm{SU}(2)$ model in three dimensions. Its Lie algebra consists of the hermitian traceless 2×2 matrices with $\langle X, Y\rangle = \frac{1}{2}\operatorname{tr} XY$, extended complex bilinearly. The trace is the ordinary matrix trace, and $|X|$ is the operator norm on $\mathbb{C}^2$. Near the identity we use $\log X = -\sum_{n\ge1}(I-X)^n/n$, for $|X-I|<1$. For $U \in \mathrm{SU}(2)\setminus\{-I\}$ the intrinsic branch is $\log U = iY$, where Y is Hermitian and traceless and $|Y|<\pi$; these branches agree on their common domain. Fix a mollifier, a source radius $2\varrho>0$, and one common choice of the constants in Prop. YM3_I:69.1 and Section YM3_I:82. In particular $L \ge 13$ is odd and $a_j = L^{-j}$. We retain all the real-step conditions when imposing the sourced-step conditions: restricting to the real construction leaves every retained constant unchanged. The two radius conditions are

$$2\varrho \le \sigma^{\mathrm{S}}(x-\bar B), \qquad 2\varrho x^{-1/2}(\log x)^{\mathfrak{p}_0} \le \sigma^{\mathrm{S}}_{\mathrm{L}} \qquad (x \ge \gamma^{-2}).$$

They are imposed by Prop. YM3_I:82.2, after the radius-independent choices of Section YM3_I:82. Thus all uses of the real flow, the sourced progression and the sourced chain below refer to the same numerical assignment and the same canonical data.

The running couplings and the data at a reference coupling, the sides, the torus measures and the states, the weighted progression, the sourced chain on the least torus and its histories are therefore formed with this one choice, and every statement of this section is made at it; where Thms. YM3_I:AJ and YM3_I:AD enter, they are read at the radius ϱ and at the choice of the longer list. Among the constants are $B_\sharp > 0$, with $\bar B = B_\sharp L/(L-1)$, the powers M, C_L of L in the geometric setting of Section YM3_I:52.7, $A_0, A_1 > 0$ and $p_0 \in \mathbb{N}$. The real and source hypotheses are those imposed jointly in Section YM3_I:82. A number is *determined by the constants* if it depends on them alone.

For $\ell \in \mathfrak{L} = \{2L^m : m \in \mathbb{N}\}$ the torus $T_\ell = \mathbb{R}^3/\ell\mathbb{Z}^3$ carries the lattice of spacing a_K. The *lattice* (ℓ, K), a side with a cutoff, carries the lattices $\mathbb{T}^{(j)} = a_{K-j}\mathbb{Z}^3/\ell\mathbb{Z}^3$ of the levels $0 \le j \le K$, of spacing $D_j = a_{K-j}$; the level 0 is the bare lattice, which in bare units is $\Lambda_N = \mathbb{Z}^3/N\mathbb{Z}^3$, $N = \ell L^K$. The *height* i is the averaging step from the level i to the level $i+1$. Blocks, the bonds of the coarser lattices and the non-pivot bonds are those of Section YM3_I:4 and the pivot definition preceding Lem. YM3_I:40.1, contours those of Section YM3_I:4 (coordinate-order contours), and $\mathcal{M}$ is the block-averaging map (equations YM3_I:(37)–(39)).

For a plaquette p, $a_p(U) = 1 - \frac{1}{2}\operatorname{tr}\partial U(p)$ and $x(p)$ is its barycentre; $A = \sum_p a_p$, $\mu_{K,\ell,\beta}(dU) = Z^{-1}e^{-\beta A(U)}dU$, $\mathcal{O}(f) = a_K^{-1}\sum_p f(x(p))a_p$, for a complex function w on the bare plaquettes, a *source*, $\mathcal{O}(w) = a_K^{-1}\sum_p w(p)a_p$, so that $\mathcal{O}(f) = \mathcal{O}(f \circ x)$, and $S^T_{2;K,\ell,\beta}(f,h)$ is the covariance of $\mathcal{O}(f)$ and $\mathcal{O}(h)$ under $\mu_{K,\ell,\beta}$. The lattice isometries preserve $\mu_{K,\ell,\beta}$ and carry $\mathcal{O}(f)$ to $\mathcal{O}$ of the carried function ([7, Lem. G.1]: plaquette holonomies are conjugated or inverted, and the positive-bond variables are permuted or inverted under product Haar measure); so $S^T_{2;K,\ell,\beta}(\tau_v f, \tau_v h) = S^T_{2;K,\ell,\beta}(f,h)$ for every $v \in a_K\mathbb{Z}^3$ and all real Lipschitz f, h, whatever their supports, τ_v being the translation by v.

For a reference coupling $0 < g \le \gamma$ (γ the last constant) put $X_g = L(g^{-2} + B_\sharp)$, $b_+ = L(g^{-2} + \bar{B})$ and $I(g) = (g^{-2} + \bar{B}, X_g - \bar{B}]$. The sides at g form $\mathfrak{L}(g) = \{2L^m : m \ge m_{\mathfrak{L}}(g)\}$ (equation YM3_I:(1528)), with least member ℓ_g; so $\ell_g L^s \in \mathfrak{L}(g)$ for every integer $s \ge 0$. A *datum* at g is a first-passage datum (K, β) at g (Cor. YM3_I:74.2): its running couplings $x_0 = \beta > x_1 > \cdots > x_K$ obey $x_{j+1} = L^{-1}x_j - e_{j+1}$ with $|e_{j+1}| \le B_\sharp$, and K is the least index with $x_K \le X_g$; its label is $b = a_K\beta$. The running couplings depend on (K, β) alone.

Every label at g lies in $[g^{-2}, b_+]$ YM3_I:(1531), and for $b \in I(g)$ the pair $(K, L^K b)$ is a datum at g for every $K \in \mathbb{N}$ (Cor. YM3_I:74.2). The *witness depth* is $s(d) = \min\{s \ge 0 : L^s d \ge C_W(s)\}$, $C_W(s) = C_{W,0}A_0^2(\log[L^{s+1}(g^{-2} + \bar{B})])^{2p_0}$, $C_{W,0} = \max\{1, 4L^2M/A_0^2\}$, as in equation YM3_I:(2321) and Thm. YM3_I:AV; it depends on g, d and the constants, and on no cutoff, label or side. As $C_W(s) \ge 4L^2M$, for $d \le L^{-1}$ and $K > s(d)$ one has $L^{s(d)} \ge 4L^3M$, $s(d) \ge 2$, $K \ge 3$ and $a_K \le L^{-1-s(d)} < d/13$; moreover $L^{s(d)} \ge 1/d$, so $C_W(s(d)) \ge (\log\frac{1}{d})^2$ and $d/a_K \ge LC_W(s(d)) \ge L(\log\frac{1}{d})^2$.

Finally $\gamma_W^{\mathrm{sd}} \le \gamma_W^{\mathrm{ref}} \le \gamma$ are the coupling thresholds of the choices in equations YM3_I:(2326)–(2327); $(\gamma_W^{\mathrm{sd}})^{-2} \ge 4\bar{B}$.

For $\vartheta \in (0,1)$, $d > 0$, and $\mathbb{E} = \mathbb{R}^3$ or $\mathbb{E} = T_\ell$ with $20d \le \ell$, $\mathfrak{T}_{\mathbb{E}}(d;\vartheta)$ is the following class of test pairs: Lipschitz functions $f, h \ge 0$ on $\mathbb{E}$, neither 0, with supports in closed balls of radius d at a distance in $[d, 6d]$, $\int f \ge \vartheta\|f\|_\infty d^3$, $\operatorname{Lip} f \le \|f\|_\infty/(\vartheta d)$, and the same for h. Then both supports lie in one ball of radius $5d$, and $d \le |x - y| \le 10d$ for $x \in \operatorname{supp} f$, $y \in \operatorname{supp} h$. The conditions involve distances alone. So the class is invariant under translations; and for $36d < \ell$ the periodisation of a pair of $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$ is a pair of $\mathfrak{T}_{T_\ell}(d;\vartheta)$ with the same sup norms and the same kernels $\mathcal{K}_q$ below, every pair of $\mathfrak{T}_{T_\ell}(d;\vartheta)$ being the periodisation of one such pair, its *lift*, since all the distances that occur are at most $10d < \ell/2$.

For two sides $\ell \le \check{\ell}$, the *pair with the same lift* as a pair (f,h) on T_ℓ is the periodisation on $T_{\check{\ell}}$ of the lift of (f,h).

We put $\eta = \|f\|_\infty\|h\|_\infty$, $\mathcal{K}_q(f,h) = \iint f(x)h(y)|x-y|^{-q}\,dx\,dy$, $\varkappa = c_1c_2 = \frac{3}{2\pi^3}$ and $L_b^+ = c_2b^{-2}[\mathcal{K}_6 + c_1b^{-1}\mathcal{K}_5]$. On $\mathbb{Z}^3$, $\mathsf{B}_1^{\mathrm{lat}}(p,q)$ is the coefficient of β^{-3} in the covariance of a_p and a_q (Def. 7.10), and $\mathsf{P}_1^{\mathrm{lat}}(f,h;a) = a\sum_{p,q} f(ax(p))h(ax(q))\mathsf{B}_1^{\mathrm{lat}}(p,q)$. There are a pure number C_{T} and a constant κ^{lat} with

$$|\mathsf{P}_1^{\mathrm{lat}}(f,h;a) - \kappa^{\mathrm{lat}}\mathcal{K}_5(f,h)| \le C_{\mathrm{T}}\vartheta^{-1}a\lg(d/a)\,\eta \qquad ((f,h) \in \mathfrak{T}_{\mathbb{R}^3}(d;\vartheta),\ 0 < a \le d),$$

$\lg t = 1 + |\log t|$ (Cor. 7.37), and $\kappa^{\mathrm{lat}} = \frac{3}{2\pi^3} = c_1c_2$ (Thms. W, X). For real Lipschitz f, h on T_ℓ, $\ell \in \mathfrak{L}(g)$, with disjoint supports and $b \in \overline{I(g)}$, $S_2^{(\ell,b)}(f,h) = \lim_{K\to\infty} S^T_{2;K,\ell,L^Kb}(f,h)$ is the limit of Thm. YM3_I:A. That theorem gives existence for every $b > 0$, continuity in b on $(0,\infty)$, and

bilinearity on real Lipschitz functions supported in two fixed disjoint closed sets. A function on $\mathbb{R}^3$ with support in a ball of radius $< \ell/4$ is read on T_ℓ by periodisation, and $f_\rho := f(\cdot/\rho)$.

Lemma 6.1 (Kernel bounds for test pairs)**.** For $(f,h) \in \mathfrak{T}_{\mathbb{E}}(d;\vartheta)$ and $q > 0$,

$$10^{-q}\vartheta^2 d^{6-q}\|f\|_\infty\|h\|_\infty \le \mathcal{K}_q(f,h) \le (4\pi/3)^2 d^{6-q}\|f\|_\infty\|h\|_\infty.$$

Moreover $d \le \mathcal{K}_5(f,h)/\mathcal{K}_6(f,h) \le 10d$. ■

Proof. All cross-distances of the two supports lie between d and $10d$. Positivity gives

$$(10d)^{-q}\int f \int h \le \mathcal{K}_q(f,h) \le d^{-q}\int f \int h.$$

The defining mass bounds give the lower estimate, and the volumes of the two radius-d balls give the upper estimate. On a torus the balls have the same volumes because $20d \le \ell$. Finally $|x-y|^{-5} = |x-y|\,|x-y|^{-6}$ on the product of the supports; integration gives the ratio bound. Its denominator is positive by the lower estimate. □

Definition 6.2. Let N_3 be the threefold convolution of $\mathbf{1}_{[-1/2,1/2)}$, $\Phi(x) = \prod_{\mu=1}^3 N_3(x_\mu)$, $\zeta = 1$ on $[0,7]$, $\zeta(t) = 1-3(t-7)^2+2(t-7)^3$ on $[7,8]$, $\zeta = 0$ on $[8,\infty)$, $P(y) = |y|^2 - \frac{3}{4}$ and $\Xi(y) = P(y)\zeta(|y|)$. For odd $q \ge 1$ let $J_q = \{-\frac{q-1}{2},\dots,\frac{q-1}{2}\}$, $c_j^q = \prod_\mu q^{-2}\#\{\iota \in J_q^3 : \iota_1+\iota_2+\iota_3 = j_\mu\}$ for $j \in \mathbb{Z}^3$, and

$$\Upsilon^{(q)}(y) = q^{-1}\sum_{j\in\mathbb{Z}^3} c_j^q\,\Xi((y+j)/q) - \Xi(y).$$

With $e(u) = \int \Phi(x)\Xi(x+u)\,\mathrm{d}x$ let $\mathsf{k}_6 = \int_0^\infty (4\pi - t^{-2}\int_{S^2} e(t\omega)\,\mathrm{d}\omega)t^{-2}\,\mathrm{d}t$. ■

6.1.2. *The scale of the remainder.* The two terms in the covariance have different homogeneities. For a test pair of scale d, the kernel $|x-y|^{-6}$ integrates to a quantity of order one, while $|x-y|^{-5}$ integrates to a quantity of order d. Thus the leading contribution is of size b^{-2} and its first correction is of size $b^{-3}d$. To identify that correction, an error merely tending to zero relative to the leading term is inadequate. The error must be $o(b^{-3}d)$ after the mesh error has been separated.

The nonnegativity and mass bounds in the test-pair definition give

$$10^{-q}\vartheta^2 d^{6-q}|f|_\infty|h|_\infty \le \mathcal{K}_q(f,h) \le (4\pi/3)^2 d^{6-q}|f|_\infty|h|_\infty. \tag{6.i}$$

These floors do more than normalize the functions. They show that the second term is detected by every pair in the class, and they permit division by $\mathcal{K}_5$ when recovering its coefficient. The lattice comparison retains the Lipschitz bound, which measures the error from replacing an integral by samples at plaquette barycentres. These sampling errors and the interacting remainder are kept as separate terms in theorem S.

The coefficient calculation later in the section is finite dimensional. Its remainder depends on the box. The present renormalization argument supplies the estimates needed to use that coefficient in an interacting continuum limit. This distinction determines the order of the comparison: begin with the actual covariance, express it through the sourced progression, and only then compare the coefficient of that progression with a lattice Laplace coefficient.

6.1.3. *The even expansion and its two retained orders.* Let n_j be the mixed derivative in the two sources of the local normalizing energy at level j, summed over its domains. Its sum is J_2. A packing of translated pairs gives

$$\left| S_2^T(f,h) - \frac{1}{\#\mathsf{Pk}}\sum_{v\in\mathsf{Pk}} \mathsf{J}_2(\tau_v f,\tau_v h)\right| \le C(g)\vartheta^{-1}d^2b^{-2}|f|_\infty|h|_\infty.$$

The covariance is unchanged by the lattice translations. The packing therefore allows the contributions left by the conditioning procedure to be averaged while keeping the original quantity. In the units $b^{-3}d|f|_\infty|h|_\infty$, its cost is $C(g)\vartheta^{-1}bd$, which tends to zero at fixed admitted labels.

Multiply every fluctuation field by one complex parameter λ and move the quadratic coefficients of the action by

$$\xi_j(\lambda) = L^{-j}\beta + \lambda^2(x_j - L^{-j}\beta).$$

Write $\mathsf{J}_2(\lambda)$ for the deformed sum, so that $\mathsf{J}_2 = \mathsf{J}_2(1)$, and put $\mathsf{h}_r = [\lambda^r]\mathsf{J}_2(\lambda)$ for $r = 4, 6$. This deformation preserves the identities relating the steps. The simultaneous analyticity domains in proposition 6.5 justify using the same parameter at every height. The jets are even holomorphic functions of λ, and their constant and quadratic coefficients vanish. At the quadratic order the possible term is a covariance of linear colour forms at the identity; invariance under constant gauge rotations makes those linear forms zero. Cauchy's estimate then gives

$$|\mathsf{J}_2 - \mathsf{h}_4 - \mathsf{h}_6| \le Cb^{-4}d^2\bigl(\max\{1, \log(2b/d)\}\bigr)^{8p_0}|f|_\infty|h|_\infty. \tag{6.ii}$$

The orders four and six are the leading covariance and its first correction. The power d^2 in the remaining orders makes eq. 6.ii small on the scale required for identifying the second term.

The order-four tower is compared with the free curvature template. Its covariance is a sum of nested curvature projections, so the telescoping slice identity identifies the free kernel. The supplied estimate retains the error from the truncated Gaussian boxes, the coarsest background and the lattice-to-integral comparison. In particular the order-four error is bounded by $C(g)b^{-3}d^{3/2} + C_{\mathrm{lat}}b^{-2}a_K/d$, multiplied by the two suprema. The slowest part of the eventual continuum error comes from this comparison.

For the sixth order, let $\mathsf{h}_6^{(n)}$ be the sum of the sixth-order coefficients through level n, and stop at a level with spacing $D_n \ge d$. The tail estimate is

$$|\mathsf{h}_6 - \mathsf{h}_6^{(n)}| \le Cb^{-3}d\bigl(\max\{1, \log(2b/d)\}\bigr)^{6p_0}(d/D_n)^{3/2}|f|_\infty|h|_\infty.$$

This is the freedom used to compare finite objects while keeping the discarded coarse levels below the second-term scale.

6.1.4. *The comparison of the sixth-order coefficient.* The composed block-field chart puts all earlier fluctuation fields on the fibre where the coarsest block field is the identity. At real field scale its exact density contains the Wilson action, the gauge-fixing terms and the chart Jacobian. The run equation gives the identity

$$\lambda^{-2}\xi_j(\lambda)L^j + \sum_{i\le j} L^i e_i = \lambda^{-2}\beta. \tag{6.iii}$$

Thus the intermediate counterterms disappear from the total phase. The field changes between matched charts are triangular colour rotations. They preserve Lebesgue measure and the Euclidean balls used for the fluctuation cutoffs. Consequently the sixth-order coefficient agrees exactly with the second Laplace coefficient in that chart, apart from the separately bounded removal of the Gaussian boxes. Coordinate cubes in the colour components would not be preserved by this change; the shape of the boxes enters the argument here.

Set $\mathsf{H}^{(n)} = \beta^3 a_K^2 \mathsf{h}_6^{(n)}$ and let $\mathsf{T}[N, n]$ be the coefficient on a bare torus of N sites per direction, with the n-fold block constraint. The same $\mathsf{T}[N, n]$ can be reached from different running couplings, because eq. 6.iii leaves no dependence on that run. Let D be the original datum on a bare torus of side N_0, and let D' be the comparison datum on an enlarged bare torus of side N'. The truncation depths n_0, n_1 are chosen with the geometric and error inequalities of the detailed comparison below. In the case requiring the enlarged torus, the comparison is:

$$\begin{aligned}\mathsf{H}^{(K)}[D] \longrightarrow \mathsf{H}^{(n_0)}[D] \longrightarrow \mathsf{T}[N_0, n_0] \longrightarrow \mathsf{T}[N', n_0] \longrightarrow \mathsf{H}^{(n_0)}[D'] \\ \longrightarrow \mathsf{H}^{(n_1)}[D'] \longrightarrow \mathsf{T}[N', n_1] \longrightarrow \mathsf{T}^{\mathrm{lat}} \longrightarrow a_K^{-1}\kappa^{\mathrm{lat}}\mathcal{K}_5.\end{aligned} \tag{6.iv}$$

Each kind of comparison addresses a different obstruction. The first and fifth arrows discard coarse tails using the preceding estimate. Passing between H and T removes Gaussian boxes; their precisions have a uniform positive lower bound, so the error is an exponential in minus the squared

cutoff radius, times the stated polynomial in the finite size. The change of bare torus uses locality of the chart and connected Wick pairings. A pairing reaching the remote part must contain a long line, whose exponential bound pays for the polynomial count. The lattice anchor removes the remaining block constraint, and the last arrow replaces the infinite-lattice kernel by its homogeneous part of degree -5. The numerical value of c_1 enters only at this last arrow.

The choices of K', n_0, n_1 in the proof make all these costs small in the common units $r|f|_\infty|h|_\infty$, where $r = d/a_K$. Their dependence is displayed because it is what permits the box-removal exponential to beat its finite-size polynomial. An unqualified passage from finite-box perturbation theory to the continuum covariance would omit this comparison.

6.1.5. *Removing the cutoff window and accumulating scales.* The initial expansion holds in a window $L^K \le d^{-P}$. For a fixed label, proposition 6.18 supplies a cutoff $K_\sharp(d)$ inside that window and a bound for every finer cutoff by the quantity at $K_\sharp(d)$. Its error is $O(d^{178})$ with the stated shape factor. Apply the expansion at that one cutoff, then compare all larger cutoffs with it. The full-sequence limit follows from the already supplied cutoff comparison. This procedure removes the window without exchanging a short-distance limit with an uncontrolled finite-box expansion. The separate same-datum comparisons pass the result to other sides and states, and the conditioned-state limits pass it to the canonical components with their common label.

There is also a reason to retain the improved error, beyond identification of c_1. The spline identity in lemma 6.23 writes the logarithmic observable as a sum of differences at scales $\varrho_k = L^{-k} r_0$. After subtracting the two explicit kernels, the kth term is bounded by a constant times $\varrho_k^{1/2} b^{-3}$. These bounds are summable. They give the uniform limit of Δ_n and its geometric tail. A tolerance only tending to zero, without a summable rate, would not justify that accumulation over all scales.

6.1.6. *The quantitative two-term law.*

Theorem S. There is $\gamma_{\mathrm{e}} \in (0, \frac{1}{2}]$, determined by the constants, with the following property. Let $0 < g \le \gamma_{\mathrm{e}}$, $\vartheta \in (0,1)$, $\ell = \ell_g$, $P \ge P_\sharp := 3601$, and put $C_{\mathrm{lat}} = \frac{3}{2} C_{\mathrm{tm}} \vartheta^{-1}$ and $C_{\mathrm{lat}}^\flat = \frac{9}{2} C_{\mathrm{tm}} \vartheta^{-1} + 1$, where C_{tm}, determined by the constants, is the constant of the template sums of test pairs (Prop. YM3_I:121.2 and Lem. 6.33). There are $d_{\mathrm{e}} > 0$, $c_{1/2} < \infty$, $c_{\mathrm{r}} < \infty$ and a non-decreasing $\bar\varepsilon : (0, d_{\mathrm{e}}] \to [0, \infty)$ with $\bar\varepsilon(d) \to 0$ as $d \to 0$, depending on g, ϑ, P and the constants, and on no cutoff, label, pair or position, such that for $0 < d \le d_{\mathrm{e}}$ and $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$ the following hold.

(a) At every datum (K,β) at g with $s(d) < K$ and $L^K \le d^{-P}$, with $b = a_K \beta$,

$$|S^T_{2;K,\ell,\beta} - L^+_b|(f,h) \le [\bar\varepsilon(d)\, b^{-3} d + C_{\mathrm{lat}}\, b^{-2} a_K/d]\eta. \tag{6.v}$$

(b) For every $b \in I(g)$, eq. 6.v holds at $(K, L^K b)$ for every $K > s(d)$, and $|S_2^{(\ell,b)} - L_b^+|(f,h) \le \min\{\bar\varepsilon(d), c_{1/2} d^{1/2}\}\, b^{-3} d\, \eta$.

(c) If moreover $g \le L^{-1/4} \min\{\gamma_{\mathrm{e}}, \gamma_W^{\mathrm{sd}}\}$, there are $d'_{\mathrm{e}} > 0$ and $\bar\varepsilon'$ with $\bar\varepsilon'(d) \to 0$, depending on g, ϑ, P, the radius ϱ and the choice of the longer list of which the constants are the restriction, and on no cutoff, label, pair or position, such that eq. 6.v, with $\bar\varepsilon'$ in the place of $\bar\varepsilon$, holds for $0 < d \le d'_{\mathrm{e}}$, $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$ and every datum (K,β) at g with $K > s(d)$.

(d) Let $k_0 \in \mathbb{Z}$ with $44 L^{-k_0} < \ell_g$, $r_0 = L^{-k_0}$, and for $n \ge 1$ let $r_1 = L^{-n} r_0$ and

$$\Delta_n(b) = r_1^{-1} S_2^{(\ell,b)}(\Phi_{r_1}, \Upsilon_{r_1}^{(L^n)}) - c_2 b^{-2} \mathsf{k}_6\, (r_1^{-1} - r_0^{-1}) - \frac{6}{\pi^2}\, b^{-3} \log \frac{r_0}{r_1}$$

(the pair has disjoint supports, so that the limit is defined at it). Then Δ_n converges uniformly on $I(g)$ to a bounded continuous function Δ_∞, and there are $C_{\mathrm{co}} < \infty$ and $n_{\mathrm{co}} \in \mathbb{N}$, depending on g, k_0 and the constants, with $\sup_{I(g)} |\Delta_n| \le C_{\mathrm{co}}$ for all $n \ge 1$ and $\sup_{I(g)} |\Delta_n - \Delta_\infty| \le C_{\mathrm{co}} L^{-n/2}$ for $n \ge n_{\mathrm{co}}$.

(e) For every $b \in I(g)$ and every $K > s(d)$,

$$|S^T_{2;K,\ell,L^K b} - L^+_b|(f,h) \le [c_{\mathrm{r}}\, b^{-3} d^{3/2} + C^\flat_{\mathrm{lat}}\, b^{-2} a_K/d]\eta; \tag{6.vi}$$

and eq. 6.vi, with β for $L^K b$, holds at every datum (K,β) at g with $s(d) < K \le K_\sharp(d)$ and $b = a_K\beta$, where $K_\sharp(d)$ is the cutoff of Prop. 6.18 at the side ℓ_g.

(f) Under the assumptions of (c) there are $d''_{\mathrm{e}} > 0$ and $c'_{\mathrm{r}} < \infty$, depending on g, ϑ, the radius ϱ and the choice of the longer list of which the constants are the restriction, and on no P, which enters them as $P_\sharp$, and no cutoff, label, pair or position, such that eq. 6.vi, with c'_{r} for c_{r} and β for $L^K b$, holds for $0 < d \le d''_{\mathrm{e}}$, $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$ and every datum (K,β) at g with $K > s(d)$, $b = a_K\beta$.

■□

As $|x-y| \le 10d$ on the supports, $\mathcal{K}_5(f,h) \ge (10d)^{-5} \int f \int h \ge 10^{-5}\vartheta^2 d\,\eta$ by the nonnegativity and integral floors in [7, Lem. I.1]. So clause (b) determines the number:

$$\Big|\frac{(S_2^{(\ell,b)} - c_2 b^{-2}\mathcal{K}_6)(f,h)}{c_2 b^{-3}\mathcal{K}_5(f,h)} - \frac{8}{3\pi}\Big| \le 10^5 c_2^{-1}\vartheta^{-2}\min\{\bar\varepsilon(d), c_{1/2}d^{1/2}\}.$$

The continuum limit on T_{ℓ_g} at a label of $I(g)$ determines first the label, since

$$S_2^{(\ell,b)}(f,h)/(c_2\mathcal{K}_6(f,h)) \to b^{-2}$$

along pairs of the class with $d \to 0$, and then the number $\frac{8}{3\pi}$. The number enters the proof at one place, the last link of the chain of Step 3 below, through $\kappa^{\mathrm{lat}} = c_1 c_2$; it depends on no L, g, side or label.

Corollary 6.3 (Every state; every side; the limits on $\mathbb{R}^3$)**.** Let g, ϑ and P be as in clause (c) of Thm. S, with $d'_{\mathrm{e}}, \bar\varepsilon'$ of (c) and $d''_{\mathrm{e}}, c'_{\mathrm{r}}$ of (f); let $C_2^{\mathrm{T}}(g)$ and $\mathsf{d}_\star(g,\vartheta)$ be the numbers of Prop. 6.19, and put

$$d_\infty = \min\{d'_{\mathrm{e}}, d''_{\mathrm{e}}, \tfrac{1}{180}, \mathsf{d}_\star(g,\vartheta), L^{-1}\}, \qquad \bar\varepsilon_\infty(d) = \min\{\bar\varepsilon'(d), c'_{\mathrm{r}}d^{1/2}\} + C_2^{\mathrm{T}}(g)\vartheta^{-1}b_+ d.$$

The numbers d_∞ and $\bar\varepsilon_\infty(d)$ depend on g, ϑ, P, the radius ϱ and the choice of the longer list of which the constants are the restriction, and on no cutoff, label, side, state, sequence, component, pair or position. Let $0 < d \le d_\infty$ and $(\tilde f, \tilde h) \in \mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$, with $\eta = \|\tilde f\|_\infty \|\tilde h\|_\infty$.

(i) For every state $\mathfrak{s}$ at g at a datum (K,β) at g with $K > s(d)$ (Prop. 6.19), $b = a_K\beta$:

$$|S_2^T[\mathfrak{s}] - L^+_b|(\tilde f,\tilde h) \le [(\bar\varepsilon'(d) + C_2^{\mathrm{T}}(g)\vartheta^{-1}b_+ d)\, b^{-3} d + C_{\mathrm{lat}} b^{-2} a_K/d]\eta,$$

and the same with $c'_{\mathrm{r}}d^{1/2}$ in the place of $\bar\varepsilon'(d)$ and $C^\flat_{\mathrm{lat}}$ in the place of C_{lat}. In particular this holds for the torus measure of every side of $\mathfrak{L}(g)$ at the periodisations of the pair, and for every periodic Gibbs state at (a_K,β).

(ii) For every side $\ell' \in \mathfrak{L}(g)$, every $b \in I(g)$ and every $(f,h) \in \mathfrak{T}_{T_{\ell'}}(d;\vartheta)$: $|S_2^{(\ell',b)} - L^+_b|(f,h) \le \bar\varepsilon_\infty(d)\, b^{-3} d\,\eta$.

(iii) Let $\mathfrak{s}_j$ be states at g at data (K_j,β_j) at g with $K_j \to \infty$ and labels $b_j = a_{K_j}\beta_j \to b$; the torus measures of sides $\ell_j \in \mathfrak{L}(g)$ that grow with j are among them. Then $b \in [g^{-2}, b_+]$, and every limit point S of the sequence $S_2^T[\mathfrak{s}_j](\tilde f,\tilde h)$ satisfies $|S - L^+_b(\tilde f,\tilde h)| \le \bar\varepsilon_\infty(d)\, b^{-3} d\,\eta$. The labels are determined by the two-point functions: if $\mathfrak{s}_j$ are states at g at data (K_j,β_j) at g with $K_j \to \infty$, and $S_2^T[\mathfrak{s}_j](F)$ converges at every pair F of a set that holds, for arbitrarily small $d \in (0, d_\infty]$, a pair of $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$, then the labels $a_{K_j}\beta_j$ converge, to a number $b \in [g^{-2}, b_+]$.

Consequently the labels of every convergent Gibbs sequence at g (a sequence (K_j,β_j,ν_j) of data at g with $K_j \to \infty$ and periodic Gibbs states ν_j at (a_{K_j},β_j)) converge, to a number b; and the connected two-point function S_2^σ of the action density of its limit σ, and that of every component of a canonical decomposition of σ, satisfy $|S_2^\sigma - L^+_b|(\tilde f,\tilde h) \le \bar\varepsilon_\infty(d)\, b^{-3} d\,\eta$

whenever $\tilde{f}$ and $\tilde{h}$ are of class C^1. Such a pair is a separated tuple, at which these functionals are defined, and $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$ holds pairs of class C^1 for every $d>0$ and $\vartheta\in(0,1)$.

Every volume-first limit at g has one label b, the same along all its defining sequences, and the same inequality holds for it and for its ergodic components ([7, §K.1] and Def. 13.4(2)) at the pairs of $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$ of class C^1. Every $b\in I(g)$ is the label of such limits.

(iv) In the cases (ii) and (iii), with S the two-point function in question at a pair at which (ii) or (iii) states the inequality, $|(S-c_2b^{-2}\mathcal{K}_6)/(c_2b^{-3}\mathcal{K}_5)-\frac{8}{3\pi}|\le 10^5c_2^{-1}\vartheta^{-2}\bar{\varepsilon}_\infty(d)$: one number $\frac{8}{3\pi}$ holds for all the limits on the tori of the sides at g, the limits of states at g and their components, and each of these limits determines its label and the number. In the case (i), at a state at a finite cutoff, the same quotient differs from $\frac{8}{3\pi}$ by at most $10^5c_2^{-1}\vartheta^{-2}[\bar{\varepsilon}'(d)+C_2^{\mathrm{T}}(g)\vartheta^{-1}b_+d+C_{\mathrm{lat}}\,b\,a_K/d^2]$, and by at most the same with $c_{\mathrm{r}}'d^{1/2}$ in the place of $\bar{\varepsilon}'(d)$ and $C_{\mathrm{lat}}^{\flat}$ in the place of C_{lat}.

■□

6.1.7. *Renormalisation and coefficient inputs.* Thm. YM3_I:D stands behind all the statements on the renormalisation group that we use, and Thms. YM3_I:AJ and YM3_I:AD behind the existence of the limit on a torus, the rate of the cutoffs and the agreement of states, into which it enters through the bounds on the partition functions of the tori at the source 0. We use the following construction results: the flow of the coupling and its data (Section YM3_I:74, YM3_I:74, §6.2, Prop. 6.31, equation YM3_I:(1528), YM3_I:74.2, Section YM3_I:82, Prop. YM3_I:74.1, Cor. YM3_I:74.2, the ordered real choice in Prop. YM3_I:69.1 and the common source choice of Section YM3_I:82, the coupling-sequence quantities in equation YM3_I:(634) and the function R of equation YM3_I:(634)); the averaging chart and gauge factor of [7, §B.2], with the measure identity of [7, Lem. B.3] and the cube-form sum in equation YM3_I:(828), and the free response on tori (the construction and periodization in Section YM3_I:52.6, and the curvature bound YM3_I:(2234)); the localisation domains of a torus and of the infinite lattice, their near lifts and their linear sizes (Section YM3_I:72 (the definitions of $\mathcal{D}_j$ and d_j at its beginning), [7, Lem. O.1], Lem. 5.24); the sizes YM3_I:(759) of families (Def. YM3_I:49.5) and the two sizes $|v|_\infty$ and $\mathrm{sw}_k(v)$ of a plaquette weight in §6.2.1; the bracket localisation of a bare array and the octet property (Props. YM3_I:51.14 and YM3_I:51.16, with equation YM3_I:(877)); the one-step theorem (Thm. YM3_I:V); the naturality of one step under the charts between two lattices with one cutoff (Section YM3_I:52.5, in particular equation YM3_I:(928)); the regular progression on all the lattices of a level (Thm. YM3_I:Y); the weighted progression, its steps at a complex field scale and its chain classes ([7, Def. B.7] and equation [7, eq. 200], [7, Def. B.16], the definitions preceding [7, eq. 228], the definitions preceding equation YM3_I:(2276), Section YM3_I:124, eq. 2.iv, 6.25, the bracket-localization constants in (Wp4), [7, eq. 196], [7, eq. 196], [7, eq. 198], [7, eq. 199], [7, eq. 228], equation YM3_I:(2286), Prop. YM3_I:122.1, equation YM3_I:(2258), [7, Lem. B.10], equations YM3_I:(2286)–(2287), YM3_I:124.4, YM3_I:124.3, equation YM3_I:(2291), Props. YM3_I:122.2, YM3_I:123.1, [7, Prop. B.21], YM3_I:124.2, YM3_I:124.5, 2.4, 6.26, [7, Cor. B.9], Lem. YM3_I:126.2), with the restricted Gaussian measures of one step (Lem. 4.53); the level-wise sourced density and packed-source amplitudes (Prop. 4.42 and Lem. 4.44), the first-creation decomposition and its coefficient bounds (Prop. 4.43, Cor. 4.45 and Prop. 4.54), and the least-torus comparison proved in Prop. 6.19; the free template and the template sums (Section YM3_I:120, Section YM3_I:120, YM3_I:(2231), Lem. YM3_I:120.1, the test-pair bounds eq. 8.i, Prop. YM3_I:120.3, YM3_I:120.5 and YM3_I:120.6, YM3_I:121.2 and Lem. 6.33, Prop. YM3_I:120.3); Laplace's method and the lattice coefficient (Defs. 7.7, 7.10, Lems. 7.6, 7.9, 7.11, 7.24, 7.26, 7.27, 7.38, (7.xviii), (7.xix), (7.xxxi), Cor. 7.37, Thms. W, X); the rate of the cutoffs on a torus (the constants and thresholds in Thm. YM3_I:AR, YM3_I:(2206), the two threshold inequalities at the end of the proof of Thm. YM3_I:AR, Cor. YM3_I:117.1); the limit at every label on a torus (Thm. YM3_I:AS, Prop. YM3_I:118.3); the separated tuples and the test pairs of class C^1 ([7, §K.1], the convention in equation YM3_I:(2343),

[7, Lem. I.1] and equation YM3_I:(2247)); the sequences of pairs of laws, their convergence and the connected functions of their limit functionals (Prop. YM3_I:92.1 and its moment–cumulant identity YM3_I:(1856)); the Gibbs sequences, their limits and the components of these (Def. 13.4(2), Thm. YM3_I:AP); and the common construction choice of Section YM3_I:82, with the torus limits of Thm. YM3_I:A and the growing-torus and volume-first limits of Prop. YM3_I:92.1.

6.1.8. *Source-energy jets and the field-scale expansion. Notation.* Throughout, $D = (K, \beta)$ is a datum at g with label b, $\ell = 2L^m \in \mathfrak{L}(g)$, and f, h are real bounded functions on T_ℓ; $w_1 = f \circ x$ and $w_2 = h \circ x$ are the lattice sources on the bare plaquettes. We write $N = \ell L^K$, $\mathsf{N} = N + \beta$, $r = d/a_K$, $l = \max\{1, \log(2b/d)\}$, $\mathfrak{b}(x) = A_1(\log x)^{p_0}$, $\delta(x) = \mathfrak{b}(x)x^{-1/2}$ and $\langle z\rangle = \max\{1, |z|\}$. With $E_\mu^{(j)}$ the local terms of the level j of the weighted progression ([7, eq. 200]) at the datum, with the weight $\mu_1 w_1 + \mu_2 w_2$ of the plaquettes, and $\mathcal{D}_j$ its domains of the level j, the second jet of the level j is

$$\mathsf{n}_j = \sum_{X \in \mathcal{D}_j} \sum_{z \in X \cap \mathbb{T}^{(j)}} \partial_{\mu_1}\partial_{\mu_2} E_\mu^{(j)}(X, \mathbf{1}, z)\big|_{\mu=0}, \qquad \mathsf{J}_2 = \sum_{j=1}^{K} \mathsf{n}_j.$$

The same construction at a complex field scale λ (every fluctuation field multiplied by λ, the coefficient of the action at the height j moved to $\xi_j(\lambda) = L^{-j}x_0 + \lambda^2(x_j - L^{-j}x_0)$) gives the local terms $E_\mu^{(j),\lambda}$ of the level j at the field scale λ, with $E_\mu^{(j),1} = E_\mu^{(j)}$, and their second jets $\mathsf{n}_j^\lambda = \sum_{X \in \mathcal{D}_j} \sum_{z \in X \cap \mathbb{T}^{(j)}} \partial_{\mu_1}\partial_{\mu_2} E_\mu^{(j),\lambda}(X, \mathbf{1}, z)|_{\mu=0} = \sum_{m \geq 0} \mathsf{n}_{j,[2m]} \lambda^{2m}$ (Prop. 6.5); $\mathsf{h}_{2m}^{(n)} = \sum_{j \leq n} \mathsf{n}_{j,[2m]}$, $\mathsf{h}_{2m} = \mathsf{h}_{2m}^{(K)}$, and $\mathsf{H}^{(n)}[D] = \beta^3 a_K^2 \mathsf{h}_6^{(n)}$. As $\beta^{-3}a_K^{-2} = b^{-3}a_K$ and $a_K r = d$, a size $r\eta$ of H is a size $b^{-3}d\eta$ of h_6. Where the side matters we write $\mathsf{J}_2[\ell]$, $\mathsf{h}_6[\ell]$, $\mathsf{H}^{(n)}[D, \ell]$.

The free template (Section YM3_I:120) is the free abelian model of the curvature: cochains, plaquette averages and nested projections. At the height $0 \leq i < K$, $\mathsf{A}_i^{\mathrm{cv}} = \mathsf{B}_i^{\mathrm{cv}} + \hat{\mathsf{G}}$ is its one-step quadratic form, a sum of two real symmetric matrices ≥ 0 on the non-pivot bonds of $\mathbb{T}^{(i)}$, and $\mathsf{A}_i^{\mathrm{cv}} \geq \gamma^{\mathrm{tr}}$ with a lower bound $\gamma^{\mathrm{tr}} > 0$ (Prop. YM3_I:120.3); T_i is its curvature map onto the bare plaquettes (Prop. YM3_I:120.3). Π_0^{tp} is the orthogonal projection onto the exact bare curvatures and Π_n^{tp}, $n \geq 1$, the projection onto those that the plaquette averages of the level n determine, so that $\Pi_0^{\mathrm{tp}} - \Pi_n^{\mathrm{tp}}$ is the orthogonal projection onto the curvatures whose plaquette averages of the level n vanish; $\mathcal{C}_{(j)} = \Pi_{j-1}^{\mathrm{tp}} - \Pi_j^{\mathrm{tp}}$ are the curvature slices, and $C_{K,K-n}^{\mathrm{fl}} = \Pi_0^{\mathrm{tp}} - \Pi_n^{\mathrm{tp}} = \sum_{1 \leq j \leq n} \mathcal{C}_{(j)}$ is the fluctuation kernel, the conditional covariance of the free lattice curvature given its plaquette averages of the level n; on a lattice with t levels we write $\Pi_t^{\mathrm{fl}} = \Pi_0^{\mathrm{tp}} - \Pi_t^{\mathrm{tp}}$.

The numbers $\delta_{\mathrm{tp}} \in (0, \frac{1}{12}]$ and $C_{\mathrm{tp}} \geq 1$ of the decay of the slices (Props. YM3_I:120.5 and YM3_I:120.6), and the number $C_{\mathrm{tp}}^{\mathrm{row}}$ of the row sums of the curvature maps,

$$\sum_b |\mathsf{T}_i(p, b)| \leq C_{\mathrm{tp}}^{\mathrm{row}} L^{-2i} \qquad \text{for every bare plaquette } p,$$

uniformly in the volume, at the height $i = 0$ and at every height at which the side ℓL^{K-i} of the level i, in its units, is at least N_{H}, so at every height if $\ell \geq N_{\mathsf{H}}$ (Prop. YM3_I:120.6), depend on L and on the least admissible side N_{H} fixed with C_{H} and δ_{H} in Section YM3_I:52.6 and the beginning of Section YM3_I:120.3; they, γ^{tr} and N_{H} are determined by the constants.

The Gaussian measures of the heights. Put $\tau_i = L^{-i}x_0/x_i$ and $\mathsf{M}_i = \tau_i \mathsf{B}_i^{\mathrm{cv}} + \hat{\mathsf{G}}$. ν_i° is the centred Gaussian measure with the precision $\mathsf{M}_i \otimes 1$ on the real $\mathfrak{g}$-valued 1-forms B_i on the non-pivot bonds of $\mathbb{T}^{(i)}$, and ν_i^{r} is ν_i° conditioned on the box $\{|B_i(b)| < \mathfrak{b}(x_i)\ \forall b\}$, $|\cdot|$ being the Euclidean norm of $\mathfrak{g} \cong \mathbb{R}^3$: the boxes are products of Euclidean balls. As $\tau_i \in [\frac{3}{4}, \frac{4}{3}]$ (Prop. 6.4) and both matrices are ≥ 0, $\mathsf{M}_i \geq \min\{\tau_i, 1\}\mathsf{A}_i^{\mathrm{cv}} \geq \frac{3}{4}\gamma^{\mathrm{tr}}$, so every variance in ν_i° is at most $\frac{4}{3\gamma^{\mathrm{tr}}}$.

The measures ν_i° are tied to the steps by Lem. 6.12(iii): at $\lambda = 0$, at the trivial configuration, the density of the step of the height i is proportional to $\exp[-\frac{1}{2}(B_i, (\mathsf{M}_i \otimes 1)B_i)]$ on the box of the height, the height 0 included, where $\tau_0 = 1$ and $\mathsf{M}_0 = \mathsf{A}_0^{\mathrm{cv}}$. For polynomials $X_{a,m}$ ($a = 1, 2$,

$m = 0, 1, 2$), W_1, W_2 in these fields and a probability measure ν, with the joint cumulants $\langle \cdot; \dots; \cdot \rangle$ taken in ν,

$$\gamma_2[\nu] = \langle X_{1,2}; X_{2,0} \rangle + \langle X_{1,0}; X_{2,2} \rangle + \langle X_{1,1}; X_{2,1} \rangle - \langle X_{1,1}; X_{2,0}; W_1 \rangle \\ - \langle X_{1,0}; X_{2,1}; W_1 \rangle - \langle X_{1,0}; X_{2,0}; W_2 \rangle + \tfrac{1}{2} \langle X_{1,0}; X_{2,0}; W_1^2 \rangle; \tag{6.vii}$$

$\gamma_2^{\mathrm{r}} = \gamma_2[\bigotimes_{0 \le i < n} \nu_i^{\mathrm{r}}]$ and $\gamma_2^{\circ} = \gamma_2[\bigotimes_{0 \le i < n} \nu_i^{\circ}]$, with the polynomials of Prop. 6.11. If ν is invariant under $B \mapsto -B$, the $X_{a,0}$ are even and W_1 is odd, then $\frac{1}{2}\langle X_{1,0}; X_{2,0}; W_1^2 \rangle = \frac{1}{2}\langle X_{1,0}; X_{2,0}; W_1; W_1 \rangle$: a cumulant with a product entry is $\langle X; Y; W^2 \rangle = \langle X; Y; W; W \rangle + 2\langle X; W \rangle \langle Y; W \rangle + 2\langle W \rangle \langle X; Y; W \rangle$, and the last two terms vanish by parity.

Laplace's method (Lem. 7.11). Let ϕ, ψ, X, Y be real and smooth near $0 \in \mathbb{R}^m$, $\phi(0) = 0$, $\nabla\phi(0) = 0$, with a positive definite Hessian H at 0, and let X, Y vanish at 0 with their gradients. Let $\phi_j, \psi_j, X_j', Y_j'$ be their homogeneous Taylor polynomials of order j at 0, and $W_j = \phi_{j+2} + \psi_j$, $X_{1,j} = X_{j+2}'$, $X_{2,j} = Y_{j+2}'$. Then the covariance of X and Y under the density proportional to $e^{-\beta\phi - \psi}$ on a small window about 0 is

$$\beta^{-2}\gamma_0^{\mathrm{L}} + \beta^{-3}\gamma_2^{\mathrm{L}} + O(\beta^{-4}) \qquad (\beta \to \infty), \tag{6.viii}$$

where $\gamma_0^{\mathrm{L}} = \langle X_{1,0}; X_{2,0} \rangle$ and γ_2^{L} is eq. 6.vii, both in the centred Gaussian measure of covariance H^{-1}. In the variable $\beta^{1/2}x$ the density is the Gaussian one times $e^{-\beta^{-1/2}W_1 - \beta^{-1}W_2 - \cdots}$ and the observables are $\beta^{-1}(X_{a,0} + \beta^{-1/2}X_{a,1} + \beta^{-1}X_{a,2} + \dots)$; the coefficient of β^{-3} of the covariance, expanded in joint cumulants, is eq. 6.vii, written with the parity just noted, and the half-integer orders vanish by parity. We write $\gamma_{m'}^{\mathrm{L}}[\phi, \psi, X, Y]$, $m' = 0, 2$. The numbers $\mathsf{g}_{m'}[N, t](p, q)$ of a flat top are set in Prop. 6.13; for functions w_1, w_2 on the plaquettes of Λ_N, $\mathsf{T}[N, t](w) = \sum_{p,q} w_1(p) w_2(q) \mathsf{g}_2[N, t](p, q)$, and for functions $\tilde{w}_1, \tilde{w}_2$ of finite support on the plaquettes of $\mathbb{Z}^3$, $\mathsf{T}^{\mathrm{lat}}(\tilde{w}) = \sum \tilde{w}_1(\tilde{p}) \tilde{w}_2(\tilde{q}) \mathsf{B}_1^{\mathrm{lat}}(\tilde{p}, \tilde{q})$.

If $\tilde{w}_1(\tilde{p}) = \tilde{f}(a_K x(\tilde{p}))$, $\tilde{w}_2(\tilde{p}) = \tilde{h}(a_K x(\tilde{p}))$, then $a_K \mathsf{T}^{\mathrm{lat}}(\tilde{w}) = \mathsf{P}_1^{\mathrm{lat}}(\tilde{f}, \tilde{h}; a_K)$.

Proposition 6.4 (The setting of the expansion)**.** There is $\gamma_{\mathrm{s}} \in (0, \min\{\gamma_W^{\mathrm{ref}}, \frac{1}{2}\}]$, determined by the constants, such that for $0 < g \le \gamma_{\mathrm{s}}$, every datum (K, β) at g with $K \ge 3$, every $\ell = 2L^m \in \mathfrak{L}(g)$, and real bounded f, h with supports in balls of radius d at a distance $\ge d$, $8a_K \le d$, at every position of (f, h):

(i) $2L^{m+K-j+1} \ge C_L M$ for $1 \le j \le K$, in particular $\ell L \ge C_L M$; and $\ell \ge N_{\mathsf{H}}$;

(ii) $x_0 > x_1 > \cdots > x_K > g^{-2}$, $x_{K-1} > Lg^{-2} \ge 4L$, and for $0 \le j < K$: $\frac{3}{4} b/D_j \le x_j \le 2b/D_j$ and $\frac{3}{4} \le \tau_j \le \frac{4}{3}$;

(iii) at every level $j < K$ the constants, the lattice of the step $j \to j+1$ and x_j are data under (Im) of [7, Def. B.2];

(iv) the data of the weighted progression ([7, Def. B.7]) are given with $k = K - 1$, the sources w_1, w_2 and the running couplings as thresholds; the setting (J) of [7, Def. B.16] holds at $k = K - 1$; $\log x_{K-1} \ge \max\{2p_0, 1 + \log L\}$; and $\delta(x_{K-1})$ lies below the threshold of the chain classes (equations YM3_I:(2285) and eq. 2.vi) and below the smallness that Prop. 6.5 asks of its discs.

No condition is put on the witness depth, on the signs or on the Lipschitz constants of f and h. ■

Proof sketch. (i) For a side at g the lattice of a datum is large at every level, by the flow of the coupling (Section YM3_I:74, YM3_I:74) and Lem. YM3_I:126.2; at $j = K$ this is $\ell L \ge C_L M$. The condition on N_{H} and its two companion numbers is a member of the collective condition, which holds at the fixed constants and under which the geometric hypotheses of Section YM3_I:52.7 hold; it is a condition on these three numbers, and no side occurs in it. The bound $\ell \ge N_{\mathsf{H}}$ is a property of the sides at g: for $\ell = 2L^m \in \mathfrak{L}(g)$ the flow of the coupling gives $LM\, R(x_K) \le L^m$ (Prop. YM3_I: 74.1; R is the function of equation YM3_I:(634)), so $M\, R(x_K) \le L^m$, and with this Lem. YM3_I: 126.2 gives $\ell \ge N_{\mathsf{H}}$, together with the frames of the datum on T_ℓ; the setting (J) at $k = K - 1$ follows in (iv).

(ii) The run equation gives $|x_j - L^{-j}x_0| \le B_\sharp \sum_{i\ge 0} L^{-i} = \bar B$, and $L^{-j}x_0 = b/D_j$. As K is the first passage, $x_j \ge x_{K-1} > X_g \ge Lg^{-2}$ for $j < K$, and $x_K = L^{-1}x_{K-1} - e_K > g^{-2}$; $\gamma_{\rm s}$ is so small that $X_g \ge 12\bar B$. Hence $|\tau_j - 1| \le \bar B/x_j \le \frac{1}{12}$, and $\frac{12}{13} b/D_j \le x_j \le b/D_j + \bar B \le 2b/D_j$. (iii), (iv) $\gamma_{\rm s} \le \gamma_W^{\rm ref}$ is the least of two thresholds determined by the constants. For $g \le \gamma_{\rm s}$, [7, Cor. B.9], [7, Prop. B.21] and Lem. YM3_I:126.2 give the data of the progression and the setting (J) at $k = K-1$; in that setting, at every level, the constants, the lattice of the step and x_j are data under (Im), so that in particular the ratio A_0/A_1 lies above the number that (Im) asks.

As $x_{K-1} > L\gamma_{\rm s}^{-2}$ and δ decreases for large arguments, $\delta(x_{K-1}) \le \delta(L\gamma_{\rm s}^{-2})$, which lies below both thresholds for $\gamma_{\rm s}$ small, and $\log x_{K-1} \ge \log(L\gamma_{\rm s}^{-2}) \ge \max\{2p_0, 1 + \log L\}$. The sources enter through their supports alone. □

Proposition 6.5 (One field scale)**.** Under the assumptions of Prop. 6.4 the progression is formed at one complex field scale for all its steps. There are $R_j > 1$ $(0 \le j < K)$ with $R_j^{-2} \le c_0 (\log x_j)^{2p_0} D_j/b$, and $\rho \in (0, \frac12]$ with $(\rho R_j)^2 \ge 2$, such that, with $c_0, \rho, C_{\mathsf n}, C_{\rm E}, C_{{\rm E},m}, C_{\rm R}, C_{\rm J}$ determined by the constants (they depend on no cutoff, side, run, scale, position or pair), the following hold.

(i) $\lambda \mapsto \mathsf n_{j+1}^\lambda$ is holomorphic and even on $|\lambda| < R_j$, $\mathsf n_{j+1}^1 = \mathsf n_{j+1}$, its Taylor coefficients $\mathsf n_{j+1,[0]}$, $\mathsf n_{j+1,[2]}$ of the orders 0 and 2 vanish, and for $|\lambda| \le \rho R_j$

$$|\mathsf n_{j+1}^\lambda| \le C_{\mathsf n} |\lambda|^2 R_j^{-2} D_j^{-2} \min\{1, (d/D_j)^{5/2}\}\, \Pi_{j+1}\, \eta, \qquad \Pi_{j+1} = \max\{1, (d/D_{j+1})^3\} e^{-[d/(4MD_{j+1})-2]_+}.$$

(ii) $|\mathsf J_2 - \mathsf h_4 - \mathsf h_6| \le C_{\rm E} b^{-4} d^2 l^{8p_0} \eta$ and $|\mathsf h_{2m}| \le C_{{\rm E},m} b^{-m} d^{m-2} l^{2mp_0} \eta$ for $m = 2, 3$.

(iii) For $0 \le n \le K-1$ with $D_n \ge d$ and $m = 2, 3$:

$$\sum_{j=n}^{K-1} |\mathsf n_{j+1,[2m]}| \le C_{\rm R} b^{-m} l^{2mp_0} d^{m-2} (d/D_n)^{\frac52 - (m-2)} \eta.$$

In particular, for $n \le n' \le K$ with $L^n \ge r$: $|\mathsf H^{(n')}[D] - \mathsf H^{(n)}[D]| \le C_{\rm R} l^{6p_0} (r/L^n)^{3/2} r\, \eta$.

(iv) Let $\dot E_a^{(i+1),\lambda} = \partial_{\mu_a} E_\mu^{(i+1),\lambda}|_{\mu=0}$ $(a = 1, 2)$ and $\ddot E^{(i+1),\lambda} = \partial_{\mu_1}\partial_{\mu_2} E_\mu^{(i+1),\lambda}|_{\mu=0}$ be the first and the second jets of the level $i+1$ at the field scale λ, pointed families of the level $i+1$, and let $\psi_i = \min\{1, (d/D_i)^3\}$, $\chi_i = \min\{1, (d/D_i)^{5/2}\}$. At every λ at which they are formed, the member of $\dot E_a^{(i+1),\lambda}$ on a domain X vanishes unless X holds a plaquette of the support of w_a, and the member of $\ddot E^{(i+1),\lambda}$ on X vanishes unless X holds a plaquette of each of the two supports. For $0 \le i \le j < K$ and $|\lambda| \le \rho R_j$, their sizes YM3_I:(759) at the rate $\kappa^+ = (1 + 4\beta_{\rm s})\kappa$, at which the parts of the local terms that depend on the sources decay in the linear size of the domain (Prop. YM3_I:122.1), obey

$$\|\dot E_a^{(i+1),\lambda}\|_{i+1,\kappa^+} \le C_{\rm J} |\lambda|^2 R_i^{-2} D_i^{-1} \psi_i \|w_a\|_\infty, \qquad \|\ddot E^{(i+1),\lambda}\|_{i+1,\kappa^+} \le C_{\rm J} |\lambda|^2 R_i^{-2} D_i^{-2} \chi_i\, \eta.$$

If w_a vanishes identically, all the jets in the direction w_a vanish; so (i)–(iii) hold trivially if w_1 or w_2 vanishes, and so do the bounds of (iv) that involve a vanishing source. ■

Proof sketch. One step at a complex field scale is Prop. 6.26, applied with a parameter set (Def. 6.25) that holds the field scales of the earlier steps; the scale is put on the diagonal afterwards. A lower calibration $\mathsf t_\flat$, fixed with the constants, gives room for the size of the deformed bare array, the bound of the step being uniform in the parameter. The counterterms e_i are kept, and the coefficient of the action is moved along $\xi_j(\lambda)$, which stays in the admitted disc about x_j for $|\lambda| < R_j := \delta(\mathsf t_\flat)/(2\delta(x_j))$, because $|\xi_j(\lambda) - x_j| = |\lambda^2 - 1|\, |x_j - L^{-j}x_0| < 2R_j^2 \bar B$. To make the last bound smaller than $\sigma_* x_j$, decrease $\gamma_{\rm s}$, if necessary, so that

$$\bar B \delta(\mathsf t_\flat)^2 < 2\sigma_* A_1^2 (\log x_j)^{2p_0}$$

for every coupling of the run. This is possible by the lower bound on x_j in Prop. 6.4; substitution of the formula for R_j gives $2R_j^2\bar{B} < \sigma_* x_j$. Thus the simultaneous substitution in the field scale and the quadratic coefficient stays within the product domain of Prop. 6.26.

By Prop. 6.4(ii), $R_j^{-2} = 4\delta(x_j)^2/\delta(\mathsf{t}_\flat)^2 \le \frac{16}{3}A_1^2\delta(\mathsf{t}_\flat)^{-2}(\log x_j)^{2p_0}D_j/b$. Induction over the heights gives the holomorphy, the evenness, $\mathsf{n}^1_{j+1} = \mathsf{n}_{j+1}$ and the locality in the weight of (iv): the member of the level $i+1$ on a domain X is formed from the members of the lower levels on the domains in X and from the weight on the plaquettes of X, at every field scale as at the field scale 1 (Prop. YM3_I: 122.1(c)). The derivatives in the weights obey two linear recursions over the levels, by the chain rule and Cauchy's estimate in the direction of the bare array: one step at the field scale λ multiplies the size of the derivative in the direction of a bare array by at most $c\,|\lambda|^2 R_i^{-2}$ (clauses (cx3), (cx8) of Prop. 6.26, uniformly in the parameter), and the weight member of the level i in the direction w_a has the size at most $36M^3\alpha_0^2 D_i^{-1}\psi_i\|w_a\|_\infty$, its source meeting at most $35(d/a_K)^3$ plaquettes; so the first derivatives follow the profile $\psi_i = \min\{1,(d/D_i)^3\}$, the second the profile $\chi_i = \min\{1,(d/D_i)^{5/2}\}$.

They close on these profiles for $|\lambda| \le \rho R_j$ by the locality in the weight, the bare array of a source with support of diameter $\le 2d$ meeting at most $77\max\{1,(d/D_i)^3\}$ cubes of the level i, the cubes of side MD_i of the partition π_i of the torus whose wall-connected unions are the domains of the level i (the level-i localization partition of Section YM3_I:45); ρ is fixed by the two inequalities that close the induction, and the closed recursions are the two bounds of (iv). A domain that meets both supports holds two points at a distance at least $d/4$, so that its linear size in the units of its level is at least $[d/(4MD_{j+1}) - 2]_+$; with the count of such domains this gives the display of (i).

At the order 2 the earlier members are taken at the field scale 0, where they do not depend on the sources, and the coefficient is a covariance of two linear forms ℓ_Y of the field at the trivial configuration. A constant gauge rotation u fixes the trivial configuration, so $\ell_Y(\mathrm{Ad}(u)B) = \ell_Y(B)$, and $\int_G \mathrm{Ad}(u)Y\,\mathrm{d}u = 0$ for $Y \in \mathfrak{g}$ gives $\ell_Y = 0$.

(ii), (iii): an even holomorphic H with $|H| \le \mathsf{B}$ on $|\lambda| < R$ has $|H_{[2m]}| \le \mathsf{B}R^{-2m}$ and $|H(1) - \sum_{m\le p} H_{[2m]}| \le \mathsf{B}R^{-2p-2}/(1-R^{-2})$. Applied on $|\lambda| < \rho R_j$, where $(\rho R_j)^2 \ge 2$, with the bound of (i), each level gives at most $c\,b^{-m}(\log x_j)^{2mp_0}D_j^{m-2}\chi_j\Pi_{j+1}\eta$ at the order $2m$, and $m = 4$ for the remainder. The levels with $D_j \ge d$ have $\Pi_{j+1} = 1$, $\log x_j \le l$ and $D_j^{m-2}\chi_j = d^{m-2}(d/D_j)^{\frac{5}{2}-(m-2)}$: a geometric series over the levels. The other levels are summed against Π_{j+1}, which is summable against every power of the logarithm. For (iii) at $m = 3$: $\mathsf{h}_6^{(n')} - \mathsf{h}_6^{(n)} = \sum_{n\le j<n'} \mathsf{n}_{j+1,[6]}$, $D_n \ge d$ if and only if $L^n \ge r$, $d/D_n = r/L^n$ and $\beta^3 a_K^2 b^{-3} d = r$. □

Proposition 6.6 (The front on the least torus)**.** There are $\gamma_{\mathrm{C}} > 0$, determined by the constants, and for $0 < g \le \gamma_{\mathrm{C}}$ numbers $d_{\mathrm{A}}(g) > 0$ and $C_{\mathrm{A}}(g) < \infty$, depending on g and the constants and on no K, β, b, ϑ, d, pair or position, such that for every datum (K,β) at g, $\ell = \ell_g$, $0 < d \le d_{\mathrm{A}}(g)$ with $s(d) \le K$, $\vartheta \in (0,1)$ and $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$: $K \ge 3$ and $a_K \le d$; the set

$$\mathsf{Pk} = \{v \in \delta\mathbb{Z}^3 :\ |v_\mu| \le 0.575 - 9d\ (\mu = 1,2,3)\}, \qquad \delta = a_K\lceil 18d/a_K\rceil,$$

which lies in $a_K\mathbb{Z}^3$ and depends on a_K and d alone, consists of vectors of length < 1, has $\#\mathsf{Pk} \ge (30d)^{-3}$, and $(\tau_v f, \tau_v h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$ for $v \in \mathsf{Pk}$; and

$$\Big|S^T_{2;K,\ell,\beta}(f,h) - \frac{1}{\#\mathsf{Pk}}\sum_{v\in\mathsf{Pk}} \mathsf{J}_2(\tau_v f, \tau_v h)\Big| \le C_{\mathrm{A}}(g)\vartheta^{-1}d^2b^{-2}\eta.$$

■

Proof. Decrease $\gamma_{\mathrm{C}} \le \gamma_{\mathrm{s}}$ to satisfy the source and radius conditions of §4.2. These are conditions on the fixed constants and on the lower coupling threshold. Choose $d_{\mathrm{A}}(g)$ small enough for the packing and amplitude bounds below.

If $f \circ x = 0$, every lattice translate of this bare weight is zero. Both the covariance and its source-energy mixed derivative therefore vanish; the assertion follows. The same applies to

$h \circ x$. Henceforth neither is zero. Choose x_1 with $\operatorname{supp} f \subset \mathbb{B}(x_1, d)$. Both supports lie in $\mathbb{B}(x_1, 9d)$. The balls $\mathbb{B}(x_1 + v, 9d)$, $v \in \mathsf{Pk}$, have disjoint interiors, since $\delta \geq 18d$, and lie in $\mathbb{B}(x_1, 1)$. Indeed, $|v| \leq \sqrt{3}(0.575 - 9d)$ and $\sqrt{3}(0.575 - 9d) + 9d < 1$. The condition $s(d) \leq K$ gives $L^K d \geq C_W(s(d)) \geq 4L^2 M$, so $a_K \leq d$; for d small it also gives $K \geq 3$. Now $\delta \leq 18d + a_K \leq 19d$. Thus, for d small,

$$\#\mathsf{Pk} = \left(2\left\lfloor \frac{0.575 - 9d}{\delta} \right\rfloor + 1\right)^3 \geq (30d)^{-3}.$$

The translates preserve the support separation, amplitude and Lipschitz conditions defining the test class.

Use the norm eq. 4.viii and the holomorphic total source energy $\mathcal{E}$ of Prop. 4.42. Lem. 4.44 bounds a disjoint sum $w = (\sum_\iota \zeta_\iota u_\iota) \circ x$ by

$$\|w\|_{\mathrm{lw}} \leq \max_\iota |\zeta_\iota| \mathsf{A}_\iota / \mathsf{A}_\star(d), \qquad \mathsf{A}_\star(d) = \min\left\{ \sigma_{\mathrm{src}} b^\vee, \frac{\mu_\star}{K_{\mathrm{L}}^\vee} \frac{b^\vee}{(b^\wedge)^{1/2}} \left(\frac{\vartheta d}{4\sqrt{3} M} \right)^{1/2} \right\},$$

where $\|u_\iota\|_\infty \leq \mathsf{A}_\iota$, $\operatorname{Lip} u_\iota \leq \mathsf{A}_\iota / (\vartheta d)$, $b^\vee = g^{-2} - \bar{B}$ and $b^\wedge = b_+$. The total-energy identity in Prop. 4.42 shows that J_2 is the mixed derivative of $\mathcal{E}$ at zero. Therefore $S_2^T - \mathsf{J}_2$ is the mixed derivative of $\log(Z[\cdot]/Z[0]) - \mathcal{E}$.

For signs $\epsilon \in \{\pm 1\}^{\mathsf{Pk}}$ put

$$F^\epsilon = \sum_v \epsilon_v \tau_v f, \qquad H^\epsilon = \sum_v \epsilon_v \tau_v h, \qquad w_z = (z_1 F^\epsilon + z_2 H^\epsilon) \circ x, \qquad \Omega_\epsilon(z) = Z[w_z] e^{-\mathcal{E}(w_z)} / Z[0].$$

The nonzero sets of the individual bumps are pairwise disjoint. With $u_1 = \|f\|_\infty / \mathsf{A}_\star(d)$ and $u_2 = \|h\|_\infty / \mathsf{A}_\star(d)$, the amplitude bound gives $\|w_z\|_{\mathrm{lw}} \leq \max_a |z_a| u_a$. Thus Ω_ϵ is holomorphic on $\mathbb{P} = \{|z_1| u_1 < 1,\ |z_2| u_2 < 1\}$ and $\Omega_\epsilon(0) = 1$.

Split the sourced partition by its first large-field creation, using eq. 4.xvi:

$$\Omega_\epsilon = \Omega_\epsilon^\emptyset + \sum_{t=1}^K \Omega_\epsilon^t, \qquad \Omega_\epsilon^\emptyset = e^{-\mathcal{E}(w_z)} m_K^{w_z}(\mathcal{V}_K) / Z[0], \qquad \Omega_\epsilon^t = e^{-\mathcal{E}(w_z)} \mathsf{Cr}_t(w_z) / Z[0].$$

Every source here is supported in the one ball $\mathbb{B}(x_1, 1)$. Prop. 4.43 therefore gives

$$|\Omega_\epsilon^\emptyset| \leq e^{\mathsf{G}_K} \quad \text{on } \mathbb{P}, \qquad |\Omega_\epsilon^t| \leq e^{\|w_z\|_{\mathrm{lw}} (\mathsf{G}_{t-1} + \mathsf{H}_{t-1})} \mathsf{w}_t.$$

Choose $\mathsf{C}_{\mathrm{cnt}} = C_{\mathrm{cnt}}(g) \geq 1$ in eq. 4.xv, and set $\mathsf{Cnt}_t = \mathsf{C}_{\mathrm{cnt}} L^{4(K-t+1)}$. On $\mathbb{P}_t = \{|z_a| u_a \mathsf{Cnt}_t < 1, a = 1, 2\}$ the second bound is at most $e\mathsf{w}_t$. For $\emptyset \neq B \subset \{1, 2\}$ write $[z^B]\Omega = (\prod_{a \in B} \partial_{z_a}) \Omega(0)$ and $u^B = \prod_{a \in B} u_a$. Cauchy's formula on these polydiscs gives

$$|[z^B]\Omega_\epsilon^\emptyset| \leq e^{\mathsf{G}_K} u^B, \qquad |[z^B]\Omega_\epsilon^t| \leq e \mathsf{w}_t \mathsf{Cnt}_t^{|B|} u^B \leq e(\mathsf{C}_{\mathrm{cnt}} L^4)^2 L^{8(K-t)} \mathsf{w}_t u^B.$$

By eq. 4.xviii, the latter coefficients sum to at most $e(\mathsf{C}_{\mathrm{cnt}} L^4)^2 C_8(g) u^B$. Choose $\mathsf{Q}(g) \geq 1$ above this coefficient bound plus $e^{\sup \mathsf{G}_K}$. It is finite and independent of the datum, the positions and the signs. Thus

$$|[z^B]\Omega_\epsilon| \leq \mathsf{Q}(g) u^B, \qquad |\partial_{12} \log \Omega_\epsilon(0)| \leq (\mathsf{Q} + \mathsf{Q}^2) u_1 u_2 \leq 2\mathsf{Q}^2 \mathsf{A}_\star(d)^{-2} \eta.$$

The logarithm is needed only near zero, where $\Omega_\epsilon(0) = 1$ ensures its existence.

By bilinearity the last mixed derivative is

$$\sum_{v, v'} \epsilon_v \epsilon_{v'} (S_2^T - \mathsf{J}_2)(\tau_v f, \tau_{v'} h).$$

Average over the independent signs. The terms $v \neq v'$ vanish and the diagonal terms remain. Translation invariance of S_2^T and the packing count then give

$$\left| S_2^T(f, h) - \frac{1}{\#\mathsf{Pk}} \sum_v \mathsf{J}_2(\tau_v f, \tau_v h) \right| \leq (30d)^3 \, 2\mathsf{Q}^2 \mathsf{A}_\star(d)^{-2} \eta.$$

For $d \le 4\sqrt{3}Mg^{-2}(K_{\mathrm{L}}^{\vee}/K_{\mathrm{Q}}^{\vee})^2$ the second entry in the minimum defining $\mathsf{A}_\star$ is the smaller one. Substitution bounds the last expression by $C(g)\vartheta^{-1}d^2\eta$. Since the label satisfies $b \le b_+$, enlarging this fixed-g constant by b_+^2 gives the asserted $C_{\mathrm{A}}(g)\vartheta^{-1}d^2b^{-2}\eta$. The finitely many ceilings on d determine $d_{\mathrm{A}}(g)$. □

Proposition 6.7 (The order four)**.** Let $0 < g \le \gamma_{\mathrm{s}}$, (K,β) a datum at g, $\ell \in \mathfrak{L}(g)$, $0 < d \le L^{-1}$, $s(d) < K$, $20d \le \ell$, $\vartheta \in (0,1)$ and $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$, at any position. Then $K \ge 3$, $8a_K \le d$, $d/a_K \ge L(\log\frac{1}{d})^2$, and

$$|\mathsf{h}_4 - c_2 b^{-2}\mathcal{K}_6|(f,h) \le [\bar\varepsilon_4(g,d)\, b^{-3}d + C_{\mathrm{lat}}\, b^{-2}a_K/d]\eta,$$

where, with $l' = \max\{1, \log\frac{2b'}{d}\}$,

$$\bar\varepsilon_4(g,d) = \sup_{g^{-2}<b'\le b_+} \frac{b'}{d}\Big\{3C_{\mathrm{tm}}d^3 + C_{\mathrm{Bx}}\big[d^{3/2} + e^{-\mathsf{c}_\chi(\log\frac{3b'}{4d^{1/2}})^2}\big] + C_{\mathrm{Bt}}\, l'^{\,4p_0} e^{-\mathsf{c}_\flat L(\log\frac{1}{d})^2}\Big\},$$

$C_{\mathrm{Bx}}, C_{\mathrm{Bt}}, \mathsf{c}_\chi > 0$ being determined by the constants and $\mathsf{c}_\flat = \min\{\frac{1}{8ML}, \frac{\delta_{\mathrm{tp}}}{2\sqrt{3}}\}$. The function $\bar\varepsilon_4(g,\cdot)$ depends on g and the constants alone and is the same for all sides and data at g; and $\bar\varepsilon_4(g,d) \le C_4(g)d^{1/2}$, with $C_4(g)$ depending on g and the constants. ■

Proof sketch. The first three assertions are those of the setting on the witness depth. Let $2 \le n \le K$ with $D_n \ge d$.

(1)

The tower at a real field scale. Let $0 < \lambda \le 1$, $g_i = x_i^{-1/2}$, and let $c_{\mathfrak{h}}$ be the constant of the shift systems of the steps. The chain of the levels at the field scale λ is formed downwards from the flat top of the level n: $A_n^\lambda := 0$ and, for $i = n-1, \dots, 1$,

$$A_i^\lambda := A^{\mathrm{lw}}(v^{[i+1]}_{A_{i+1}^\lambda}, \lambda g_i B_i), \qquad \tilde U_i^\lambda := e^{iL^{-i}A_i^\lambda},$$

with the lowered 1-form A^{lw} of equation YM3_I:(2286) and the chart point $v_A^{[i+1]}$ of a chain class (equation eq. 2.iv). A *chart point* of a level k is a pair $v = (A, \mathbf{J})$ of 1-forms on the bare bonds; $P[v] = (\mathbf{1}, A, \mathbf{J})$ is its triple in the chart at the trivial pair (Section YM3_I:124), and $P[0]$ is the trivial triple; its norm $\|v\|$ is the largest of the numbers $|A(b)|$, $|\mathbf{J}(b)|$ over the bare bonds b and of the moduli of the difference quotients of A at the spacing L^{-k}; and $\mathsf{Cc}_k(r)$ is the chain class of the level k and the radius r. The A_i^λ are real 1-forms on the bare bonds, and $\tilde U_i^\lambda$, a configuration on the bare bonds, is the *background of the level* i, as in the proof of Prop. 6.11(i): up to a gauge transformation it is a background of the level i in the sense of the actual-background condition in eq. 2.iv; $\mathcal{O}(w_a)$ and the a_p are taken at it.

By Prop. YM3_I:124.2, equation YM3_I:(2288), read at the field λB_i, which lies in the box, as the box is a product of Euclidean balls, and by the bounds of equation YM3_I:(2287) and its domain verification at $\sup|\lambda g_i B_i| \le \lambda\delta(x_i)$, $A_i^\lambda \in \mathsf{Cc}_i(\lambda r_i)$, with $r_n := 0$ and $r_i := L^{-1}r_{i+1} + \frac{11}{10}c_{\mathfrak{h}}\delta(x_i) \le \frac{8}{5}c_{\mathfrak{h}}\delta(x_i)$; the conditions of that Proposition hold at every height, and $r_1 \le \frac{1}{2}r_\alpha$, r_α being the usable radius of the fixed radii (equations YM3_I:(933) and YM3_I:(2285)), as $\delta(x_{K-1})$ lies below the threshold of the chain classes (Prop. 6.4(iv)).

The same Proposition gives $v^{[i]}_{A_i^\lambda} = v^{\mathrm{lw}}(v^{[i+1]}_{A_{i+1}^\lambda}, \lambda g_i B_i)$, the lowered point of equation YM3_I:(2286); the lowered points are holomorphic (the holomorphy following equation YM3_I:(2286)) and obey the bounds of equation YM3_I:(2287) and its domain verification at complex points as well, so the same recursion, with the lowered points, continues the chain to the complex field scales $|\lambda| \le 1$, where the A_i^λ take complex values and the $\tilde U_i^\lambda$ lie in $\mathrm{SL}(2,\mathbb{C})$; one name, $\tilde U_i^\lambda$, serves for both. For $\mu \in \mathbb{C}^2$ near 0 put $w_\mu = \mu_1 w_1 + \mu_2 w_2$ and

$$D_i^{\lambda,\mu} = \mathcal{O}(w_\mu) + \sum_{i'\le i} \mathsf{Sum}\big[E_\mu^{(i'),\lambda} - E_0^{(i'),\lambda}\big],$$

$\mathsf{Sum}[\cdot]$ being the sum of a pointed family at a configuration or at a triple (the definitions preceding equation YM3_I:(2276)). Each step at a height $1 \le i < n$ obeys an identity of its own at the field scale λ, whose case $\lambda = 1$ uses the quotient argument of Prop. 2.4: for $r \le r_{i+1}$, $A \in \mathsf{Cc}_{i+1}(r)$, $v = v_A^{[i+1]}$, $\tilde U = e^{iL^{-i-1}A}$ and $\tilde U^{\mathrm{lw}}(B) = e^{iL^{-i}A^{\mathrm{lw}}(v,\lambda g_i B)}$,

$$\exp D_{i+1}^{\lambda,\mu}(\tilde U) = \int \exp D_i^{\lambda,\mu}(\tilde U^{\mathrm{lw}}(B))\, \mathrm{d}\nu_i^\lambda[v](B).$$

Here $\nu_i^\lambda[v]$ is the complex measure of mass 1 on the box of the height i whose density is the integrand of the integral form of the step at the field scale λ, at the source 0 and at the triple $P[v]$ (clause (cx10) of Prop. 6.26), divided by its integral, which is not 0: the Gaussian density of the covariance $\tau_i(\lambda)^{-1}\mathbf{C}_*(P[v])$, with $\tau_i(\lambda) = \xi_i(\lambda)/x_i$, positive for real $|\lambda| \le 1$, and the whole covariance $\mathbf{C}_*$ of the step, times $\exp[-\frac{1-\tau_i(\lambda)}{2}\sum_\square(B, \hat{\mathsf{G}}_\square(P[v]|_\square)B)]$, with the gauge-fixing operators $\hat{\mathsf{G}}_\square$ of the fine cubes (equation YM3_I:(823)), times the exponential of the potentials of the step at the field scale λ. At complex λ this is the holomorphic continuation supplied by the same one-step theorem, with its nonvanishing normalizer; no positivity is asserted there.

The identity is proved as follows. The integral form of the step at the triple $P[v]$, at the source μ and at the source 0 (Prop. 6.26(cx10), at the data of Prop. 6.5), and the quotient of the two give

$$\exp \mathsf{Sum}[E_\mu^{(i+1),\lambda} - E_0^{(i+1),\lambda}](P[v]) = \int \exp[\mathbb{V}'[\mathfrak{g}]^{\mathrm{tot}}(P[v]; \lambda B)]\, \mathrm{d}\nu_i^\lambda[v](B),$$

because μ stands in the rule of the step alone and the rule is linear in the bare array, $\mathfrak{g}$ being the difference of the bare arrays of the height i at μ and at 0, whose members are the $E_\mu^{(i'),\lambda} - E_0^{(i'),\lambda}$, $i' \le i$, and the weight member of w_μ. Prop. YM3_I:124.2, equation YM3_I:(2289), which holds for every bare array and every field of the box, read at λB, in the plaquette form of equation YM3_I:(2290) and by linearity, gives $\mathbb{V}'[\mathfrak{g}]^{\mathrm{tot}}(P[v]; \lambda B) = D_i^{\lambda,\mu}(\tilde U^{\mathrm{lw}}(B)) - D_i^{\lambda,\mu}(\tilde U)$; and the proof of Prop. YM3_I:124.2 gives $\mathsf{Sum}[\cdot](P[v]) = \mathsf{Sum}[\cdot](\tilde U)$ for the family of the level $i+1$. Multiplying by $\exp D_i^{\lambda,\mu}(\tilde U)$ gives the identity. Iterating it from the height $n-1$, where $A = 0$ and $\tilde U = \mathbf{1}$, down to the height 1 along the chain, and taking the mixed derivative in μ at 0 of the logarithms, gives

$$\sum_{j=2}^n \mathsf{n}_j^\lambda = \langle \mathcal{F}_1^\lambda(\tilde U_1^\lambda); \mathcal{F}_2^\lambda(\tilde U_1^\lambda)\rangle_{\mathbb{Q}_n^\lambda} + \int [\mathsf{F}_1^\lambda(\tilde U_1^\lambda) - \mathsf{F}_1^\lambda(\mathbf{1})]\, d\mathbb{Q}_n^\lambda,$$

where $\mathbb{Q}_n^\lambda(dB) = \prod_{1\le i<n} \nu_i^\lambda[v_{A_{i+1}^\lambda}^{[i+1]}](dB_i)$ is a complex measure of mass 1 on the product of the boxes, with a bounded continuous density, each factor being formed at the chart point that the chain gives above its height; $\mathcal{F}_a^\lambda = \mathcal{O}(w_a) + \mathsf{Sum}[\dot E_a^{(1),\lambda}]$ is $\mathcal{O}(w_a)$ plus the first jet of the level 1 in the direction w_a, and $\mathsf{F}_1^\lambda = \mathsf{Sum}[\ddot E^{(1),\lambda}]$ is the second jet of the level 1, with $\mathsf{F}_1^\lambda(\mathbf{1}) = \mathsf{n}_1^\lambda$ (the jets of Prop. 6.5(iv)).

(2)

The limit $\lambda \to 0$. The lattice being fixed, let $\lambda \downarrow 0$ in the identity of (1) divided by λ^4. Its left side tends to $\sum_{j=2}^n \mathsf{n}_{j,[4]}$, by Prop. 6.5(i).

The measures. The chart points of the chain have $\sup_B \|v_{A_{i+1}^\lambda}^{[i+1]}\| \le \lambda r_{i+1} \to 0$, and $\tau_i(\lambda) \to \tau_i(0) = \tau_i$. Two facts tie the step at the trivial configuration to the free template: the whole covariance of the step at the trivial triple is $(\mathsf{A}_i^{\mathrm{cv}})^{-1} \otimes 1$ (equation YM3_I:(2297); Prop. YM3_I:120.3); and the gauge-fixing operators of the fine cubes at the trivial triple add up to the gauge-fixing operator at the block field $\mathbf{1}$, which is $\hat{\mathsf{G}} \otimes 1$, as the fine cubes partition the sites of the level $i+1$ and the operator of a cube is the gauge-fixing form restricted to the blocks of that cube (equation YM3_I:(823); Prop. YM3_I:120.3).

The covariance and the gauge-fixing operators at the triple $P[v]$ tend to these, entry by entry, as $v \to 0$ (equation YM3_I:(2296); the kernels of the $\hat{\mathsf{G}}_\square$ are holomorphic in the chart point), and the potentials at the field scale λ are at most $c\lambda$ times a function bounded on the box (eq. 6.xiii of

Def. 6.25). So the density of $\mathbb{Q}_n^\lambda$ tends, uniformly on the product of the boxes, to the product over the heights of the densities proportional to

$$e^{-\frac12(B_i,(\tau_i\mathsf{A}_i^{\mathrm{cv}}+(1-\tau_i)\hat{\mathsf{G}})B_i)}=e^{-\frac12(B_i,\mathsf{M}_iB_i)};$$

as the boxes have finite volume and the limits of the normalisations are positive, $\mathbb{Q}_n^\lambda\to\mathbb{Q}_n^0:=\bigotimes_{1\le i<n}\nu_i^{\mathrm{r}}$ in total variation.

The plaquette terms. At every height the linear part of one step of the chain in the field B_i, read in the plaquette angle, is $\mathsf{sg}_i(\mathsf{T}_i\otimes 1)$ with a sign $\mathsf{sg}_i\in\{1,-1\}$ (Lem. YM3_I:124.3, which reads $\ell\ge N_{\mathsf{H}}$ of Prop. 6.4(i); its map $\mathsf{T}_i^{\mathrm{cv}}$ is T_i, the definitions preceding Lem. YM3_I:124.3). With the expansion of the plaquette terms and of the plaquette angles to the second order at one step (Lem. YM3_I:124.4), summed along the chain, this gives, uniformly on the boxes and for every bare plaquette p,

$$|a_p(\tilde U_1^\lambda)-\tfrac12\lambda^2|\Theta_p(B)|^2|\le c_\Theta\lambda^3,\qquad \Theta_p(B):=\sum_{1\le i<n}\mathsf{sg}_i\,x_i^{-1/2}(\mathsf{T}_iB_i)(p),$$

with c_Θ determined by the constants. Hence $\lambda^{-2}\mathcal{O}(w_a)(\tilde U_1^\lambda)\to q_a(B):=\frac12 a_K^{-1}\sum_p w_a(p)|\Theta_p(B)|^2$ uniformly on the boxes.

The level 1. The local terms of the level 1 at $\tilde U_1^\lambda$ take the values of their members at the restrictions of the triple $P[v]$, $v=v^{[1]}_{A_1^\lambda}$, as in the proof of Prop. YM3_I:124.2. For a pointed family f of the level 1, a domain X of the level 1, a site z of X and $\kappa'>0$, $d_1(X)$ being the linear size of X in the units of the level 1,

$$\big|\mathsf{f}(X,P[v]|_X,z)-\mathsf{f}(X,\mathbf{1},z)\big|\le 2\|\mathsf{f}\|_{1,\kappa'}e^{-\kappa' d_1(X)}\lambda^2(r_1/r_\alpha)^2:$$

the function $\zeta\mapsto\mathsf{f}(X,P[\zeta v]|_X,z)$ is holomorphic on $|\zeta|<r_\alpha/\|v\|$, where $r_\alpha/\|v\|\ge r_\alpha/(\lambda r_1)\ge 2$, and bounded there by $\|\mathsf{f}\|_{1,\kappa'}e^{-\kappa' d_1(X)}$; its derivative at 0 vanishes, because a constant gauge rotation u fixes the trivial configuration and acts on the chart point by $\mathrm{Ad}(u)$, the local terms are invariant, and $\int_G\mathrm{Ad}(u)Y\,\mathrm{d}u=0$ for $Y\in\mathfrak{g}$; Cauchy's estimate bounds the rest of the series. At $\mathsf{f}=\dot E_a^{(1),\lambda}$, whose size at the rate κ^+ is at most $C_{\mathrm{J}}\lambda^2R_0^{-2}a_K^{-1}\|w_a\|_\infty$ (Prop. 6.5(iv) at $i=0$, where $\psi_0=1$), this shows that the first jet of the level 1 changes along the chain by $O(\lambda^4)$, uniformly on the boxes; as a covariance does not change when constants are added, $\lambda^{-4}\langle\mathcal{F}_1^\lambda(\tilde U_1^\lambda);\mathcal{F}_2^\lambda(\tilde U_1^\lambda)\rangle_{\mathbb{Q}_n^\lambda}\to\langle q_1;q_2\rangle_{\mathbb{Q}_n^0}$.

At $\mathsf{f}=\ddot E^{(1),\lambda}$, whose size is at most $C_{\mathrm{J}}\lambda^2R_0^{-2}a_K^{-2}\eta$ and whose members vanish on the domains that do not meet both supports (Prop. 6.5(iv) at $i=0$), the count of the domains of the level 1 that meet both supports, weighted with $e^{-\kappa^+d_1(X)}$ ([7, Lem. B.10]), brings the factor Π_1, as in the proof of Prop. 6.5(i), and gives

$$|\mathsf{F}_1^\lambda(\tilde U_1^\lambda)-\mathsf{F}_1^\lambda(\mathbf{1})|\le\lambda^4\,\mathrm{Er}_\flat,\qquad \mathrm{Er}_\flat=c\,R_0^{-2}a_K^{-2}(r_1/r_\alpha)^2\,\Pi_1\,\eta.$$

So the limit $\mathsf{e}_\flat$ of $\lambda^{-4}\int[\mathsf{F}_1^\lambda(\tilde U_1^\lambda)-\mathsf{F}_1^\lambda(\mathbf{1})]\,\mathrm{d}\mathbb{Q}_n^\lambda$ exists, $|\mathsf{e}_\flat|\le\mathrm{Er}_\flat$, the total variation of $\mathbb{Q}_n^\lambda$ tending to 1, and

$$\sum_{j=2}^n\mathsf{n}_{j,[4]}=\langle q_1;q_2\rangle_{\mathbb{Q}_n^0}+\mathsf{e}_\flat.$$

The covariance $\langle q_1;q_2\rangle_{\mathbb{Q}_n^0}$ does not depend on the signs sg_i: every ν_i^{r} is invariant under $B_i\mapsto -B_i$, its box being symmetric, so that the law of $(\mathsf{sg}_iB_i)_i$ under $\mathbb{Q}_n^0$ is $\mathbb{Q}_n^0$. By Prop. 6.5(i) at $j=0$ and Cauchy's estimate on $|\lambda|=\rho R_0$, $|\mathsf{n}_{1,[4]}|\le C_{\mathsf{n}}\rho^{-2}R_0^{-4}a_K^{-2}\Pi_1\eta$. As $x_0=\beta=b/a_K$, $R_0^{-2}\le c_0(\log x_0)^{2p_0}a_K/b$, $r_1\le\frac85c_\flat\delta(x_1)$ and $\delta(x_1)^2\le\frac43A_1^2(\log x_0)^{2p_0}La_K/b$ (Prop. 6.4(ii)), both $|\mathsf{e}_\flat|$ and $|\mathsf{n}_{1,[4]}|$ are at most $c\,b^{-2}(\log\frac{b}{a_K})^{4p_0}\Pi_1\eta$.

With $u_0=d/a_K\ge 8$, $\log\frac{b}{a_K}\le l\,(1+\log u_0)$ and $\Pi_1\le\max\{1,u_0^3\}e^{2-u_0/(4ML)}$; as $(1+\log u)^{4p_0}\max\{1,u^3\}e^{2-u/(8ML)}$ is bounded on $u\ge 8$ and $\mathsf{c}_\flat\le\frac{1}{8ML}$, this is below $C\,b^{-2}l^{4p_0}e^{-\mathsf{c}_\flat d/a_K}\eta$.

(3)

The boxes out at the order four. The cumulants of the orders 2 and 4 of the components $B_i^\alpha(b)$ in $\nu_i^{\mathrm r}$ and in ν_i° differ by at most $C_\chi e^{-\mathsf{c}_\chi(\log x_i)^{2p_0}}$, uniformly in the volume, with C_χ, c_χ determined by the constants (Lem. 4.53, read with the one quadratic system of the height and the threshold $\mathfrak{b}(x_i)$); the odd cumulants vanish in both measures by the symmetry $B \mapsto -B$, and the Gaussian ones are $\delta_{\alpha\beta}\mathsf{M}_i^{-1}(b,b')$ at the order 2 and 0 at the order 4.

(4)

Wick's rule and the slices. The B_i are independent in $\mathbb{Q}_n^0$ and each $\nu_i^{\mathrm r}$ is even. With $F_i^\alpha(p) = \mathsf{sg}_i\, x_i^{-1/2}\sum_b \mathsf{T}_i(p,b)B_i^\alpha(b)$, so that $|\Theta_p|^2 = \sum_\alpha\big(\sum_i F_i^\alpha(p)\big)^2$ in an orthonormal basis (t_α) of $\mathfrak{g}$, $c_i^{\alpha\beta}(p,q) = \langle F_i^\alpha(p)F_i^\beta(q)\rangle_{\nu_i^{\mathrm r}}$ and $\kappa_i^{\alpha\beta}(p,q)$ the joint fourth cumulant of $F_i^\alpha(p), F_i^\alpha(p), F_i^\beta(q), F_i^\beta(q)$,

$$\langle q_1;q_2\rangle_{\mathbb{Q}_n^0} = \tfrac14 a_K^{-2}\sum_{p,q} w_1(p)w_2(q)\sum_{\alpha,\beta}\Big[2\Big(\sum_i c_i^{\alpha\beta}(p,q)\Big)^2 + \sum_i \kappa_i^{\alpha\beta}(p,q)\Big].$$

By the slice identity $\mathsf{T}_i\mathsf{M}_i^{-1}\mathsf{T}_i^{\mathsf T} = L^{-i}\tau_i^{-1}\mathcal{C}_{(i+1)}$ (Prop. YM3_I:120.3, valid at every $\tau_i \neq 0$) and

$$x_i^{-1}L^{-i}\tau_i^{-1} = x_0^{-1},$$

the running couplings drop out: in the Gaussian measures $c_i^{\alpha\beta} = \delta_{\alpha\beta}x_0^{-1}\mathcal{C}_{(i+1)}$. As $a_K x_0 = b$, the Gaussian value of $\langle q_1;q_2\rangle$ is $\frac32 b^{-2}\sum_{p,q} w_1(p)w_2(q)\Sigma(p,q)^2$ with

$$\Sigma = \sum_{1\le i<n}\mathcal{C}_{(i+1)} = C^{\mathrm{fl}}_{K,K-n} - \mathcal{C}_{(1)}.$$

The differences of (3) are carried to the F_i^α through the row sums of T_i, which are at most $C_{\mathrm{tp}}^{\mathrm{row}}L^{-2i}$ at every height, uniformly in the volume, as $\ell \ge N_{\mathsf H}$ (Prop. YM3_I:120.6; Prop. 6.4(i)). By multilinearity, with $\epsilon_i = C_\chi e^{-\mathsf{c}_\chi(\log x_i)^{2p_0}}$, $\mathsf{sg}_i^2 = 1$ and $x_i^{-1}L^{-4i} = \tau_i x_0^{-1}L^{-3i} \le \frac43 x_0^{-1}L^{-3i}$,

$$|c_i^{\alpha\beta}(p,q) - \delta_{\alpha\beta}x_0^{-1}\mathcal{C}_{(i+1)}(p,q)| \le \bar e_i := \tfrac43 (C_{\mathrm{tp}}^{\mathrm{row}})^2 x_0^{-1}L^{-3i}\epsilon_i, \qquad |\kappa_i^{\alpha\beta}(p,q)| \le \tfrac{16}{9}(C_{\mathrm{tp}}^{\mathrm{row}})^4 x_0^{-2}L^{-6i}\epsilon_i.$$

There are at most $35^2(d/a_K)^6$ pairs (p,q) with $w_1(p)w_2(q)\neq 0$, and for them $|x(p)-x(q)|_\infty \ge d/\sqrt3$; with $u_i = d/D_i$ one has $L^{-3i}(d/a_K)^3 = u_i^3$, and Prop. YM3_I:120.6 gives $(d/a_K)^3(|\Sigma|+|\mathcal{C}_{(1)}|)(p,q) \le C_{\mathrm{tp}}S_\Sigma$, $S_\Sigma = \sup_{u>0}\sum_{s\in\mathbb{Z}}(uL^s)^3 e^{-\delta_{\mathrm{tp}}uL^s/\sqrt3}$. Expanding the square, with $\mathsf E_k = \sum_{1\le i<n} u_i^k\epsilon_i$ for $k = 3, 6$,

$$\Big|\langle q_1;q_2\rangle_{\mathbb{Q}_n^0} - \tfrac32 b^{-2}\sum_{p,q} w_1(p)w_2(q)\Sigma(p,q)^2\Big| \le \tfrac{35^2}{4}\, b^{-2}\big[16C_{\mathrm{tp}}S_\Sigma(C_{\mathrm{tp}}^{\mathrm{row}})^2\mathsf E_3 + 32(C_{\mathrm{tp}}^{\mathrm{row}})^4\mathsf E_3^2 + 16(C_{\mathrm{tp}}^{\mathrm{row}})^4\mathsf E_6\big]\eta.$$

The numbers $\mathsf E_k$ are bounded in three ranges of levels. As $\log x_i \ge \log\frac{3b}{4D_i} \ge 0$ (Prop. 6.4(ii), and $\frac34 b \ge 1 \ge D_i$), $\log x_i \ge 1$ and $2p_0 \ge 2$, one has $\epsilon_i \le C_\chi e^{-\mathsf{c}_\chi(\log\frac{3b}{4D_i})^2}$. At the levels with $D_i < d$, $\log\frac{3b}{4D_i} = \Lambda_b + t_i$ with $\Lambda_b = \log\frac{3b}{4d} \ge \log\frac{3b}{4d^{1/2}}$ and $t_i = \log u_i > 0$, the t_i being spaced by $\log L$, and $(\Lambda_b + t)^2 \ge \Lambda_b^2 + t^2$, so that $u_i^k\epsilon_i \le C_\chi e^{-\mathsf{c}_\chi\Lambda_b^2}e^{6t_i - \mathsf{c}_\chi t_i^2}$ for $k \le 6$; so these levels give at most $C_\chi S_\chi e^{-\mathsf{c}_\chi\Lambda_b^2}$, with $S_\chi = \sup_{t_0\ge0}\sum_{s\ge0} e^{6(t_0+s\log L)-\mathsf{c}_\chi(t_0+s\log L)^2}$. At the levels with $d \le D_i \le d^{1/2}$, $\sum u_i^3 \le 2$ and $\epsilon_i \le C_\chi e^{-\mathsf{c}_\chi(\log\frac{3b}{4d^{1/2}})^2}$. At the levels with $D_i > d^{1/2}$, $\sum u_i^3 \le 2d^{3/2}$ and $\epsilon_i \le C_\chi$. So

$$\mathsf E_3,\ \mathsf E_6 \le C_\chi(S_\chi+2)\big[d^{3/2} + e^{-\mathsf{c}_\chi(\log\frac{3b}{4d^{1/2}})^2}\big] \le 2C_\chi(S_\chi+2),$$

and the bracket above gives the member $C_{\mathrm{Bx}}b^{-2}[d^{3/2} + e^{-\mathsf{c}_\chi(\log(3b/(4d^{1/2})))^2}]\eta$, with C_{Bx} formed with C_{tp}, S_Σ, $C_{\mathrm{tp}}^{\mathrm{row}}$, C_χ and S_χ, and so determined by the constants. The missing first slice is small: by Prop. YM3_I:120.6, $(d/a_K)^3|\mathcal{C}_{(1)}(p,q)| \le C_{\mathrm{tp}}u_0^3 e^{-\delta_{\mathrm{tp}}u_0/\sqrt3}$, $u_0 = d/a_K$, at $|x(p)-x(q)| \ge d/\sqrt3$; with the terms of the level 1 of (2) it gives the member $C_{\mathrm{Bt}}l^{4p_0}e^{-\mathsf{c}_\flat d/a_K}b^{-2}\eta$.

(5)

The free value. Lem. 6.33 (the template sums) gives $|\sum_{p,q} w_1 w_2 (C^{\mathrm{fl}}_{K,K-n})^2 - \frac{3}{8\pi^2}\mathcal{K}_6| \le C_{\mathrm{tm}}[\frac{a_K}{\vartheta d} + (\frac{d}{\ell})^3 + (\frac{d}{D_n})^3]\eta$, and $\frac{3}{2}\cdot\frac{3}{8\pi^2} = c_2$. Take $n = K$, $D_K = 1$: $(d/\ell)^3 + d^3 \le 2d^3$, the member $\frac{3}{2}C_{\mathrm{tm}}\vartheta^{-1}b^{-2}a_K/d$ is $C_{\mathrm{lat}}b^{-2}a_K/d$, and $b^{-2} = \frac{b}{d}\,b^{-3}d$; the supremum over the labels at g and $d/a_K \ge L(\log\frac{1}{d})^2$ give $\bar\varepsilon_4$. Finally $\log\frac{3b'}{4d^{1/2}} \ge \frac{1}{2}\log\frac{1}{d}$, as $\frac{3}{4}b' \ge 1$; so $\frac{b'}{d}e^{-\mathsf{c}_\chi(\cdot)^2} \le b_+ e^{\log\frac{1}{d} - \frac{1}{4}\mathsf{c}_\chi(\log\frac{1}{d})^2}$, $\frac{b'}{d}C_{\mathrm{Bx}}d^{3/2} \le C_{\mathrm{Bx}}b_+ d^{1/2}$, and the other members are smaller: $\bar\varepsilon_4(g,d) \le C_4(g)d^{1/2}$. □

Remark 6.8. With Prop. 6.5(ii) at $m = 3$, Prop. 6.7 gives, on every side $\ell \in \mathfrak{L}(g)$ and at every position,

$$|\mathsf{J}_2 - c_2 b^{-2}\mathcal{K}_6|(f,h) \le [(C_{\mathrm{E},3}l^{6p_0} + C_{\mathrm{E}}b^{-1}d\,l^{8p_0} + \bar\varepsilon_4(g,d))\,b^{-3}d + C_{\mathrm{lat}}b^{-2}a_K/d]\eta,$$

and on T_{ℓ_g}, under the assumptions of Prop. 6.6 besides ($g \le \gamma_{\mathrm{C}}$, $d \le d_{\mathrm{A}}(g)$), by that Proposition, the same for $S^T_{2;K,\ell_g,\beta}(f,h)$ with $C_{\mathrm{A}}(g)\vartheta^{-1}bd$ added to the coefficient of $b^{-3}d$. The member $\bar\varepsilon_4(g,d)\,b^{-3}d\,\eta$ holds the two finite-size terms of the free value, $\frac{3}{2}C_{\mathrm{tm}}b^{-2}[(d/\ell)^3 + (d/D_K)^3]\eta$, of the torus and of the constraint at the top level $D_K = 1$, which the proof of Prop. 6.7 bounds by $3C_{\mathrm{tm}}b^{-2}d^3\eta = 3C_{\mathrm{tm}}bd^2\cdot b^{-3}d\,\eta$: they are at most $3C_{\mathrm{tm}}\,b^{-3}d\,\eta$ for $d^2 \le 1/b$, and of the order bd^2 relative to the first correction beyond.

The other members of $\bar\varepsilon_4(g,d)\,b^{-3}d\,\eta$ are at most $C_4(g)\,d^{1/2}\,b^{-3}d\,\eta$ (Prop. 6.7), and $C_{\mathrm{E}}b^{-1}d\,l^{8p_0} \le C_{\mathrm{E}}c_{\mathfrak{p}}\,l^{6p_0}$ with $c_{\mathfrak{p}} = \sup_{0<t\le 1} t\max\{1,\log\frac{2}{t}\}^{2p_0}$. So, beside the lattice term $C_{\mathrm{lat}}b^{-2}a_K/d$ and the two finite-size terms, the deviation from the free law is at most $C\,l^{6p_0}\,b^{-3}d\,\eta$, with a C formed with g and the constants (and with ϑ on T_{ℓ_g}): of the order of the first correction, up to the factor l^{6p_0}; for $d^2 \le 1/b$ the two finite-size terms are of that order too. The coefficient of the first correction is fixed by the order six.

Lemma 6.9 (Gaussian measures conditioned on a convex set)**.** Let ν be a centred Gaussian measure on $\mathbb{R}^I$, I finite, with variances $\le \sigma_i^2$, and $Q \supset \prod_i(-c_i,c_i)$ open, bounded, convex and symmetric; put $t_i = c_i/\sigma_i$ and $\epsilon = \sum_i e^{-t_i^2/2}$. Let $F_1,\dots,F_r$ ($r = 2,3$) be complex polynomials of orders $\le k_j$ with $|F_j| \le S_j$ on Q, and $k = \sum_j k_j$. If $t_i^2 \ge \max\{4,2k\}$ for all i and $2^k\epsilon \le 1$, the joint cumulants of $F_1,\dots,F_r$ in $\nu(\cdot\,|\,Q)$ and in ν differ by at most $q_r(1+2^k)\epsilon\prod_j S_j$, with $q_2 = 5$, $q_3 = 37$. ■

Proof sketch. For $z \notin Q$ let $s = \max_i |z_i|/c_i \ge 1$. The segment from $-z/s$ to z/s lies in $\bar Q$, so $\tau \mapsto F_j(\tau z/s)$ is a polynomial of order $\le k_j$ bounded by S_j on $[-1,1]$, and the extremal property of Chebyshev's polynomials gives $|F_j(z)| \le S_j(2s)^{k_j}$. On Q^{c}, $s^k \le \sum_i(|z_i|/c_i)^k\mathbf{1}\{|z_i| \ge c_i\}$, and for a centred Gaussian variable of variance $\le \sigma_i^2$, $\mathbb{E}[(|z_i|/c_i)^k; |z_i| \ge c_i] \le e^{-t_i^2/2}$ once $t_i^2 \ge \max\{4,2k\}$; so

$$\nu(Q^{\mathrm{c}}) \le \epsilon, \qquad \int_{Q^{\mathrm{c}}} \prod_j |F_j|\,\mathrm{d}\nu \le 2^k\epsilon\prod_j S_j.$$

Let F be a product of some of the F_j, S the product of their S_j, and m, m^{r} its moments in ν and in $\nu(\cdot\,|\,Q)$. Then $m^{\mathrm{r}} - m = m^{\mathrm{r}}\nu(Q^{\mathrm{c}}) - \int_{Q^{\mathrm{c}}} F\,\mathrm{d}\nu$ and $|m^{\mathrm{r}}| \le S$, so $|m^{\mathrm{r}} - m| \le (1+2^k)\epsilon S$; and $|m| \le S + 2^k\epsilon S \le 2S$. A joint cumulant of order r is a sum, over the partitions into p blocks, of $(-1)^{p-1}(p-1)!$ times products of p moments; telescoping each product, with the factors bounded by S and by $2S$, bounds its change by $p\,2^{p-1}(1+2^k)\epsilon\prod_j S_j$. So $q_r = \sum_p \mathrm{S}(r,p)\,p!\,2^{p-1}$, S the Stirling numbers of the second kind: $q_2 = 1+4$, $q_3 = 1+12+24$. □

Lemma 6.10 (The coefficients of a logarithm)**.** Let G be holomorphic on $|\lambda| < 1$ with $G(0) = 0$ and $\mathrm{Re}\,G \le \mathsf{E}$. Then every Taylor coefficient $G_{[m]}$, $m \ge 1$, has $|G_{[m]}| \le 2\mathsf{E}$. Consequently, if F is holomorphic on $|\lambda| < 1$, $F(0) = 1$, $F \ne 0$, $|F| \le e^{\mathsf{E}}$, and $F = 1 + \lambda\mathsf{R}_1 + \lambda^2\mathsf{R}_2 + O(\lambda^3)$, then $W_1 = -\mathsf{R}_1$ and $W_2 = \frac{1}{2}\mathsf{R}_1^2 - \mathsf{R}_2$ have $|W_1|, |W_2| \le 2\mathsf{E}$. The condition $F \ne 0$ cannot be dropped. ■

Proof sketch. For $0 < \rho < 1$ and $m \ge 1$, $G_{[m]}\rho^m = -\pi^{-1}\int_0^{2\pi}(\mathsf{E} - \mathrm{Re}\,G(\rho e^{i\phi}))e^{-im\phi}\,\mathrm{d}\phi$; the factor $\mathsf{E} - \mathrm{Re}\,G$ is ≥ 0 and has the mean E over the circle, as $\mathrm{Re}\,G(0) = 0$; let $\rho \to 1$. For

F, the function $G = \log F$ is defined on the disc, as F has no zero there; $\operatorname{Re} G = \log|F| \le \mathsf{E}$ and $G = -\lambda W_1 - \lambda^2 W_2 + O(\lambda^3)$. For $F = 1 + \mathsf{c}\lambda$ with $\mathsf{c} \ge 3$, which has a zero in the disc, $|\mathsf{R}_1| = \mathsf{c} > 2\log(1+\mathsf{c}) = 2\mathsf{E}$. □

Proposition 6.11 (The order six at one level)**.** Under the assumptions of Prop. 6.4, let $2 \le n \le K$ and $\mathsf{Bx} = \prod_{0\le i<n}\{B_i : |B_i(b)| < \mathfrak{b}(x_i)\ \forall b\}$, a product of Euclidean balls in $\mathfrak{g}$, one for each non-pivot bond of the heights $0 \le i < n$.

(i) There are functions O_1^λ, O_2^λ, D^λ on Bx, continuous and bounded on $\{|\lambda| \le 1\} \times \mathsf{Bx}$ and holomorphic in $|\lambda| < 1$, with $\mathsf{O}_a^\lambda(B) = \mathsf{o}_a(\lambda B)$ for a function o_a holomorphic near 0, $\mathsf{O}_a^{-\lambda}(-B) = \mathsf{O}_a^\lambda(B)$, $\mathsf{D}^{-\lambda}(-B) = \mathsf{D}^\lambda(B)$, $\int_{\mathsf{Bx}} \mathsf{D}^\lambda dB = 1$, $\mathsf{D}^0 dB = \bigotimes_{0\le i<n} \nu_i^{\mathrm{r}}$, and

$$\sum_{j\le n} \mathsf{n}_j^\lambda = \langle \mathsf{O}_1^\lambda; \mathsf{O}_2^\lambda \rangle_{\mathsf{D}^\lambda dB} \qquad (|\lambda| < 1).$$

Here D^λ is the product over the heights $0 \le i < n$ of the normalised densities of the steps along the chain of the backgrounds, which reads (λ, B) through λB; for $0 \le i < n$, the height 0 included, the density of the step of the height i at $\lambda = 0$, at the trivial configuration, is proportional to $\exp[-\frac12(B_i, (\mathsf{M}_i \otimes 1)B_i)]$ on the box of the height, with the real symmetric $\mathsf{M}_i \ge \frac34 \gamma^{\mathrm{tr}}$ of the measures ν_i° (Lem. 6.12(iii)); and for real $0 < \lambda \le 1$, $\log(\mathsf{D}^\lambda/\mathsf{D}^0)$ is the sum of a term that does not depend on B, of terms $\lambda^{-2}F(\lambda, \lambda B)$ and of terms $F(\lambda, \lambda B)$, with functions $F(\lambda, \zeta)$ holomorphic near $(0,0)$, those of the first kind with $F = O(|\zeta|^2)$.

Uniformly on Bx, $\mathsf{O}_a^\lambda = \lambda^2(X_{a,0} + \lambda X_{a,1} + \lambda^2 X_{a,2}) + O(\lambda^5)$ and $\mathsf{D}^\lambda = \mathsf{D}^0 \exp[-\lambda W_1 - \lambda^2 W_2 + O(\lambda^3)]$, where $X_{a,m}$ is homogeneous of order $m+2$, $X_{a,0} = q_a + q_a^{(0)}$ with the q_a of the proof of Prop. 6.7, formed with the signs sg_i of Lem. YM3_I:124.3 there, and a quadratic form $q_a^{(0)}$ that vanishes where $B_0 = 0$, W_1 is odd of order ≤ 3, and W_2 is, up to the constant that the term without B gives, even of order ≤ 4. These are polynomials on the whole space, with Taylor coefficients at the trivial configuration formed with L and the running couplings, so that the cumulants of eq. 6.vii are formed in $\bigotimes_{0\le i<n} \nu_i^\circ$ as well.

Hence $\mathsf{h}_4^{(n)} = \langle X_{1,0}; X_{2,0} \rangle$ and $\mathsf{h}_6^{(n)} = \gamma_2^{\mathrm{r}}$, the cumulants being taken in $\bigotimes_{0\le i<n} \nu_i^{\mathrm{r}}$.

(ii) If $\gamma^{\mathrm{tr}}\mathfrak{b}(x_{n-1})^2 \ge 80$ and $10240 N^3 e^{-\gamma^{\mathrm{tr}}\mathfrak{b}(x_{n-1})^2/8} \le 1$, then

$$\beta^3 a_K^2 |\gamma_2^{\mathrm{r}} - \gamma_2^\circ| \le \mathcal{U}(x_{n-1}, \mathsf{N}) \|w_1\|_\infty \|w_2\|_\infty, \qquad \mathcal{U}(x, \mathsf{N}) = C_{\mathrm{X}} \mathsf{N}^{36} \exp[-\tfrac18 \gamma^{\mathrm{tr}} A_1^2 (\log x)^{2p_0}],$$

with C_{X} determined by the constants.

■

Proof sketch. (i)

The chain of the backgrounds. For $|\lambda| \le 1$ and $B \in \mathsf{Bx}$ put $v_n^\lambda := 0$ and, downwards for $i = n-1, \dots, 1$, let v_i^λ be the lowered point (equation YM3_I:(2286)) that the step $i+1$ forms of the point v_{i+1}^λ of the level $i+1$ and of the fluctuation $\lambda x_i^{-1/2} B_i$. Its first member A_i^λ is a 1-form on the bare bonds, and $\tilde U_i^\lambda := \exp(iL^{-i} A_i^\lambda)$, a configuration on the bare bonds, is the *background of the level i*. The maps of the step are holomorphic on their domain, at real and at complex points (the holomorphy following equation YM3_I:(2286)); by their bounds (equation YM3_I:(2287) and its domain verification), as $\sup|\lambda x_i^{-1/2} B_i| \le |\lambda| \delta(x_i)$ on the box, $\|v_i^\lambda\| \le |\lambda| r_i$ at every height, with the radii $r_n = 0$, $r_i = L^{-1} r_{i+1} + \frac{11}{10} c_{\mathfrak{h}} \delta(x_i) \le \frac85 c_{\mathfrak{h}} \delta(x_i)$ of equation YM3_I:(2291), which keep the chain in the domain of the maps, as $\delta(x_{K-1})$ lies below the threshold of the chain classes (Prop. 6.4(iv)).

So the chain is formed for $|\lambda| \le 1$, it reads (λ, B) through λB, and for real $\lambda \in [0,1]$ it stays in the chain classes (Prop. YM3_I:124.2, equation YM3_I:(2288)). At complex λ the background has values in $\mathrm{SL}(2,\mathbb{C})$, and a_p is read there as $1 - \frac12 \operatorname{tr}$ of the product of the variables around p; on $\mathrm{SU}(2)$ this is a_p. The observable is $\mathsf{O}_a^\lambda(B) := a_K^{-1} \sum_p w_a(p)\, a_p(\tilde U_1^\lambda) + \mathsf{w}_a^\lambda(B_0)$: its first member is $\mathcal{O}(w_a)$ at the background of the level 1, which carries the fields of the heights $1 \le i < n$, and its

second, the source term of the height 0, is the sum over the domains of the level 0 of the action term of the source w_a at the point v_1^λ and the field $\lambda x_0^{-1/2} B_0$; the field of the height 0 enters O_a^λ through this term alone.

The density is $\mathsf{D}^\lambda(B) := \prod_{0\le i<n} \rho_i^\lambda / Z_i^\lambda$, with ρ_i^λ the density of the step of the height i at the field scale λ at the point v_{i+1}^λ, the exponential of the quadratic forms of the step in B_i and of its potentials, and Z_i^λ its integral over the box of the height.

The identity. At real $0 < \lambda \le 1$ each step at a height $1 \le i < n$ obeys the identity of one step at the field scale λ of step (1) of the proof of Prop. 6.7, whose case $\lambda = 1$ uses the quotient argument of Prop. 2.4, and at the height 0 the source enters linearly, through the action term of the sources. Iterating from the height $n-1$ down,

$$\exp D_n^{\lambda,\mu} = \int_{\mathsf{Bx}} \exp[\mu_1 \mathsf{O}_1^\lambda + \mu_2 \mathsf{O}_2^\lambda] \mathsf{D}^\lambda \, \mathrm{d}B,$$

$D_n^{\lambda,\mu}$ the exponent of the progression at the trivial configuration with the sources $\mu_1 w_1 + \mu_2 w_2$. Both sides are holomorphic in μ near 0, and holomorphic in $|\lambda| < 1$, as the maps of the chain are holomorphic at complex points and the normalisations Z_i^λ have no zero (Lem. 6.12(ii)); they agree on $(0,1]$. The mixed derivative in μ at 0 of the logarithms is the covariance.

At $\lambda = 0$. The chain is trivial and the potentials vanish. At the heights $1 \le i < n$ the density of the step is $\exp[-\frac12 (B_i, ((\tau_i \mathsf{A}_i^{\mathrm{cv}} + (1-\tau_i)\hat{\mathsf{G}}) \otimes 1) B_i)] = \exp[-\frac12 (B_i, (\mathsf{M}_i \otimes 1) B_i)]$, and at the height 0, where $\tau_0 = 1$, it is $\exp[-\frac12 (B_0, (\mathsf{A}_0^{\mathrm{cv}} \otimes 1) B_0)]$, $\mathsf{M}_0 = \mathsf{A}_0^{\mathrm{cv}}$; the sources of these two identities are given with Lem. 6.12(iii). So $\mathsf{D}^0 dB = \bigotimes_{0\le i<n} \nu_i^{\mathrm{r}}$.

The orders. The orders 0 and 1 of O_a^λ vanish, the order 1 by the constant gauge rotations, as in Prop. 6.5. As $\mathsf{O}_a^\lambda = \mathsf{o}_a(\lambda B)$ with o_a holomorphic near 0, the coefficient of λ^{m+2} is a homogeneous polynomial of order $m+2$. For real $0 < \lambda \le 1$ the integral identity of the step at a complex field scale (Prop. 6.26), read at the points of the chain, writes each Z_i^λ as $\det^{1/2}$ of the covariance of the step times the exponential of a sum of local terms of the step; in the product of the quotients the families of the progression drop out, by the sum rule of the chain classes (Prop. YM3_I:124.2, equation YM3_I:(2289)), which telescopes over the heights, and $\mathsf{D}^\lambda(B)$ is a number $c_n(\lambda)$, which does not depend on B, times the exponential of a sum of terms $\lambda^{-2} F(\lambda, \lambda B)$ with $F = O(|\zeta|^2)$, namely the quadratic forms of the steps, λ^{-2} times forms in λB_i whose coefficients read λB through the chain, and the potentials of the marks of the gauge fixing and of the action, $x_i \lambda^{-2}$ times functions of $\lambda x_i^{-1/2} B_i$, and of terms $F(\lambda, \lambda B)$, namely the other potentials, the logarithms of the determinants of the covariances along the chain, the localised Gaussian families of the steps and the counterterms at the backgrounds; in all of them $F(\lambda, \zeta)$ is holomorphic near $(0,0)$.

This is the form of $\log(\mathsf{D}^\lambda / \mathsf{D}^0)$ stated in (i), the term without B being $\log(c_n(\lambda)/c_n(0))$. With $F = \sum_{k,\alpha} f_{k\alpha} \lambda^k \zeta^\alpha$ the coefficient of λ^m of $\lambda^{-2} F(\lambda, \lambda B)$ is $\sum_{k+|\alpha|=m+2} f_{k\alpha} B^\alpha$, a polynomial of order $\le m+2$, and that of $F(\lambda, \lambda B)$ one of order $\le m$; this gives the orders of W_1 and W_2, and the identity between holomorphic functions of λ passes from $(0,1]$ to a disc about 0. Odd functions have the mean 0 in $\mathsf{D}^0 dB$, the mass 1 removes the mean of the order 2 of $\mathsf{D}^\lambda / \mathsf{D}^0$, and the coefficient of λ^2 in $\lambda^{-4} \langle \mathsf{O}_1^\lambda; \mathsf{O}_2^\lambda \rangle$ is eq. 6.vii; at λ^0 it is $\langle X_{1,0}; X_{2,0} \rangle$.

By Prop. 6.5(i), $\lambda^{-4} \sum_{j\le n} \mathsf{n}_j^\lambda = \mathsf{h}_4^{(n)} + \lambda^2 \mathsf{h}_6^{(n)} + O(\lambda^4)$.

(ii) By Lem. 6.12, on Bx the $X_{a,m}$ are at most $C_{\mathrm{S}} \mathsf{N}^8 \|w_a\|_\infty$ and W_1, W_2 at most $2 C_{\mathrm{S}} \mathsf{N}^7$; a cumulant of order ≥ 2 does not see the constant in W_2. By (iii) of that Lemma the variances of ν_i° are at most $\frac{4}{3\gamma^{\mathrm{tr}}}$; Bx holds the cube of the half-sides $\mathfrak{b}(x_{n-1})/\sqrt{3}$ in the components, as $x_i \ge x_{n-1}$ for $i < n$; so $t_i^2 \ge \frac14 \gamma^{\mathrm{tr}} \mathfrak{b}(x_{n-1})^2 \ge 20$ and $\epsilon \le 10 N^3 e^{-\gamma^{\mathrm{tr}} \mathfrak{b}(x_{n-1})^2/8}$. Lem. 6.9 at the seven cumulants, whose entries have the total order $k \le 10$, and $\beta^3 a_K^2 \le \mathsf{N}^3$ give (ii). □

Lemma 6.12 (The suprema on the boxes)**.** In the setting of Prop. 6.11(i) there is $C_{\mathrm{S}} \ge 1$, determined by the constants, such that for $|\lambda| < 1$ and $B \in \mathsf{Bx}$:

(i) $|\mathsf{O}_a^\lambda(B)| \le C_{\mathrm{S}}\mathsf{N}^8\|w_a\|_\infty$ for $a = 1, 2$;
(ii) $\mathsf{D}^\lambda(B) \ne 0$ and $|\mathsf{D}^\lambda(B)| \le e^{C_{\mathrm{S}}\mathsf{N}^7}|\mathsf{D}^0(B)|$;
(iii) for $0 \le i < n$, the height 0 included, the density of the step of the height i at $\lambda = 0$, at the trivial configuration, is proportional to $\exp[-\frac{1}{2}(B_i, (\mathsf{M}_i \otimes 1)B_i)]$ on the box of the height: its Gaussian part is ν_i°, the centred Gaussian measure with the real symmetric precision $\mathsf{M}_i \otimes 1 \ge \frac{3}{4}\gamma^{\mathrm{tr}}$, and its normalised restriction to the box is ν_i^{r}, a probability measure; at the height 0, $\tau_0 = 1$ and $\mathsf{M}_0 = \mathsf{A}_0^{\mathrm{cv}}$.

Consequently, on Bx, $|X_{a,m}| \le C_{\mathrm{S}}\mathsf{N}^8\|w_a\|_\infty$ for $m = 0, 1, 2$, and $|W_1|, |W_2| \le 2C_{\mathrm{S}}\mathsf{N}^7$. ■

Proof sketch. (i) The first member of O_a^λ is $a_K^{-1}\sum_p w_a(p)a_p(\tilde{U}_1^\lambda)$, at the background $\tilde{U}_1^\lambda$ of the level 1 of the chain of the proof of Prop. 6.11(i), a configuration on the bare bonds whose values stay near $\mathbf{1}$ in $\mathrm{SL}(2,\mathbb{C})$ by the radii of the chain, with at most $3N^3$ plaquettes and $a_K^{-1} \le N$; the second, the source term of the height 0, is a sum over at most $8N^3$ domains of members of the family of the source, each bounded on the box by the size bounds of the step at a complex field scale (Prop. 6.26) and by $\mathfrak{b}(\beta)^2 \le C\beta$.

(ii) $\mathsf{D}^\lambda = \prod_{0\le i<n}\rho_i^\lambda/Z_i^\lambda$ along the chain at the complex field scale λ, a product over the at most $K \le N$ heights. Each ρ_i^λ is the exponential of a finite sum, so $\rho_i^\lambda \ne 0$; its exponent is a sum over at most $8N^3$ places of quadratic forms in the field, whose coefficients are holomorphic along the chain, and of potentials, and on the box each of them is bounded by the size bounds of the step at a complex field scale (Prop. 6.26) and by $\mathfrak{b}(x_i)^2 \le C\beta$, the radii of the chain keeping the points of all the heights in the discs where these bounds hold.

By the integral identity of the step at a complex field scale, Z_i^λ is $\det^{1/2}$ of the covariance of the step times the exponential of a sum of local terms of the step, each bounded by the size bounds of the step; so $Z_i^\lambda \ne 0$, and $|Z_i^\lambda|$ is bounded below by a determinant bound for its Gaussian part. Hence $\mathsf{D}^\lambda \ne 0$, and, with at most CN^3 terms a height, each at most a number determined by the constants times $\mathfrak{b}(\beta)^2 \le C\beta \le C\mathsf{N}$, this gives the bound with N^7. (iii) For a height $1 \le i < n$, at $\lambda = 0$ the chain is trivial and the potentials vanish, and the density of the step at the trivial triple is $\exp[-\frac{\tau_i}{2}(B, \mathbf{C}_*^{-1}B) - \frac{1-\tau_i}{2}\sum_\square(B, \hat{\mathsf{G}}_\square B)]$.

Here the whole covariance of the step at the trivial triple is $\mathbf{C}_* = (\mathsf{A}_i^{\mathrm{cv}})^{-1} \otimes 1$ (equation YM3_I:(2297), from Prop. YM3_I:120.3 at the place of the height i), and the sum over the cubes of the local gauge-fixing forms is $\hat{\mathsf{G}}\otimes 1$ (equation YM3_I:(828) at the trivial block field, with Prop. YM3_I:120.3); so the density is $\exp[-\frac{1}{2}(B, ((\tau_i\mathsf{A}_i^{\mathrm{cv}} + (1-\tau_i)\hat{\mathsf{G}}) \otimes 1)B)] = \exp[-\frac{1}{2}(B, (\mathsf{M}_i \otimes 1)B)]$, as $\mathsf{A}_i^{\mathrm{cv}} = \mathsf{B}_i^{\mathrm{cv}} + \hat{\mathsf{G}}$. At the height 0, $\tau_0 = 1$, because $\xi_0 \equiv x_0$, and the gauge-fixing term drops out.

The provision of the step 1 is the canonical provision of the step at $(x_0; \mathfrak{a}_0^{\mathrm{abs}})$, the quadruple $(0, x_0, \mathfrak{a}_0^{\mathrm{abs}}, \mathfrak{w})$ being a place by the setting (J) of [7, Def. B.16] at $k = K - 1$ (Prop. 6.4(iv)); by Prop. YM3_I:120.3 and Prop. YM3_I:120.6 at $k = 0$, its whole covariance at the trivial triple is the scalar operator with the real matrix $(\mathsf{A}_0^{\mathrm{cv}})^{-1}$. So the density of the height 0 is $\exp[-\frac{1}{2}(B_0, (\mathsf{A}_0^{\mathrm{cv}} \otimes 1)B_0)]$, and $\mathsf{M}_0 = \mathsf{A}_0^{\mathrm{cv}}$. At every height $\mathsf{B}_i^{\mathrm{cv}}$ and $\hat{\mathsf{G}}$ are real symmetric matrices ≥ 0 (equation YM3_I:(2231)), $\mathsf{A}_i^{\mathrm{cv}} \ge \gamma^{\mathrm{tr}}$ (Prop. YM3_I:120.3) and $\frac{3}{4} \le \tau_i \le \frac{4}{3}$ (Prop. 6.4(ii)); so M_i is real and symmetric, $\mathsf{M}_i \ge \min\{\tau_i, 1\}\mathsf{A}_i^{\mathrm{cv}} \ge \frac{3}{4}\gamma^{\mathrm{tr}}$, the Gaussian part of the density is ν_i°, and ν_i^{r} is a probability measure.

The consequences: Cauchy's estimate on $|\lambda| < 1$ for $\lambda \mapsto \mathsf{o}_a(\lambda B)$, whose coefficient of λ^{m+2} is $X_{a,m}(B)$, and Lem. 6.10 for $\lambda \mapsto \mathsf{D}^\lambda(B)/\mathsf{D}^0(B)$. □

Proposition 6.13 (The composed chart; the lattice number of a flat top)**.** Let $N = \ell L^K$, $\ell = 2L^m$, and $1 \le t \le K$ with $2L^{m+K-t+1} \ge C_L M$; the levels are those of the lattice (ℓ, K), $\mathcal{V}_j$ is the space of the configurations of the level j, and Y the space of the $y = (\mathcal{A}_j)_{j<t}$, $\mathcal{A}_j$ a real $\mathfrak{g}$-valued 1-form on the non-pivot bonds of the level j (Section YM3_I:4 and the pivot definition preceding Lem. YM3_I:40.1).

(i) There are an open neighbourhood O_t of $\mathbf{1}$ in $\mathcal{V}_t$, $\rho_2 \in (0,1]$ and $C_{\mathrm{H}} > 0$ such that, with $\mathsf{Y}_{\rho_2} = \{y \in \mathsf{Y} : |\mathcal{A}_j| < \rho_2,\ j < t\}$: the chain $W_t = W$, $W_j = V_{W_{j+1}}[\mathcal{A}_j]$ of the charts of [7, §B.2] is defined on $\mathsf{O}_t \times \mathsf{Y}_{\rho_2}$, with $\mathcal{M}(W_j) = W_{j+1}$; $\Psi(W, y) = W_0$ is a real analytic diffeomorphism of $\mathsf{O}_t \times \mathsf{Y}_{\rho_2}$ onto an open set $\mathsf{U} \ni \mathbf{1}$; and there is a real continuous θ, real analytic near $(\mathbf{1}, 0)$, with $\int H \,\mathrm{d}U = C_{\mathrm{H}} \int_{\mathsf{O}_t} \mathrm{d}W \int_{\mathsf{Y}_{\rho_2}} \mathrm{d}y\, e^{-\theta} H \circ \Psi$ for every bounded Borel H that vanishes off U.

(ii) For $\kappa \in (0,\infty)^t$ let $F_\kappa = A + \sum_{j<t} \kappa_j G_j(\mathcal{M}^j \cdot)$, G_j the gauge-fixing function YM3_I:(54) at the level j, and $\phi_\kappa = F_\kappa \circ \Psi(\mathbf{1}, \cdot)$. Then $\phi_\kappa \geq 0$, $\phi_\kappa(0) = 0$, $\nabla\phi_\kappa(0) = 0$, and the Hessian of ϕ_κ at 0 is block diagonal for the decomposition of Y by the levels and positive definite.

(iii) The numbers $\mathsf{g}_{m'}[N,t](p,q) = \gamma^{\mathrm{L}}_{m'}[\phi_\kappa, \theta(\mathbf{1},\cdot), a_p \circ \Psi(\mathbf{1},\cdot), a_q \circ \Psi(\mathbf{1},\cdot)]$, $m' = 0, 2$, do not depend on κ, and they are the same in every adapted chart. An *adapted chart* of $(\mathcal{V}_0, dU, \mathrm{Lg}\,\mathcal{M}^t)$, Lg being the Lie algebra logarithm of the intrinsic logarithm on $G \setminus \{-1\}$, is $(\mathsf{E}', \Psi', \theta', C', \varrho')$ with: E' an open neighbourhood of $(0,0)$ in $\mathsf{W}_t \times \mathbb{R}^{\dim \mathsf{Y}}$, W_t the space of the $\mathfrak{g}$-valued 1-forms w on the bonds of the level t; Ψ' a C^∞ diffeomorphism of E' onto an open subset of the domain of $\mathrm{Lg}\,\mathcal{M}^t$, with $\Psi'(0,0) = \mathbf{1}$ and $\mathrm{Lg}\,\mathcal{M}^t\Psi'(w,y) = w$; θ' a real continuous function on E', with $\theta'(0,\cdot)$ of class C^∞; $C' > 0$; $\varrho' > 0$ a continuous function of w alone; and $\int H \,\mathrm{d}U = C' \int_{\mathsf{E}'} \varrho'(w)\, e^{-\theta'(w,y)} H(\Psi'(w,y))\,\mathrm{d}w\,\mathrm{d}y$ for every bounded Borel H that vanishes off $\Psi'(\mathsf{E}')$.

In it the numbers are $\gamma^{\mathrm{L}}_{m'}[F_\kappa \circ \Psi'(0,\cdot), \theta'(0,\cdot), a_p \circ \Psi'(0,\cdot), a_q \circ \Psi'(0,\cdot)]$. They depend on N, t, L, p, q alone.

(iv) One may take $\mathsf{O}_t = \{e^{iw} : |w| < \rho_2\}$ for every $\rho_2 \in (0, \rho_\star]$, with a $\rho_\star \in (0,1]$. Then W_j, $V_j^{\mathrm{bf}}(W_{j+1})$, $G_j(W_j)$, θ and $a_p \circ \Psi$ are real analytic in (w,y) on $\mathsf{O}_t \times \mathsf{Y}_{\rho_2}$, $W_j(\mathbf{1}, 0) = \mathbf{1}$, $0 < e^{-\theta} \leq \prod_{j<t} (27L^9)^{\#\Gamma_j}$, and $C_{\mathrm{H}} = \prod_{j<t} (2\pi^2)^{-\#\mathfrak{X}_j^+} L^{-3\#\Gamma_j}$, where $\mathfrak{X}_j^+$ are the positively oriented non-pivot bonds of the level j and Γ_j the positively oriented bonds of the level $j+1$. One may also take $\mathsf{O}_t = \{W \in \mathcal{V}_t : \max_p |\partial W(p) - \mathbf{1}| < \varepsilon_*\}$, p running over the plaquettes of the level t, and $\rho_2 = \rho_*$, with numbers $\varepsilon_*, \rho_* \in (0,1]$ determined by L and the constants and independent of t, K and m; (i)–(iii) hold for this choice as well.

■

Proof sketch. (i), (iv) One real number $\mathfrak{t}_* > 1$ serves at all levels: with it the constants, the lattice of each step and $\mathfrak{t}_*$ are data under (Im) of [7, Def. B.2]. For such data [7, Lem. B.3] gives, on an open set of block fields W and for $|\mathcal{A}|$ below the bound that belongs to $\mathfrak{t}_*$: the averaged configuration $V_j^{\mathrm{bf}}(W)$ is defined and $\mathcal{M}(V_j^{\mathrm{bf}}(W)) = W$; $\mathcal{M}(V_W[\mathcal{A}]) = W$; and $\mathrm{Lg}(V_W[\mathcal{A}](b) V_j^{\mathrm{bf}}(W)(b)^{-1}) = \mathcal{A}(b)$ on the non-pivot bonds. [7, Lem. B.3] gives the one-step integral formula, with the constant $(2\pi^2)^{-\#\mathfrak{X}_j^+} L^{-3\#\Gamma_j}$, a Jacobian factor in $(0, (27/8)^{\#\Gamma_j}]$ and a normalisation in $(0, (8L^9)^{\#\Gamma_j}]$.

Neighbourhoods of $\mathbf{1}$ chosen downwards, by continuity in $(W, \mathcal{A})$, keep the chain where these statements apply; by induction over the levels the integral formula holds for every Borel subset of the natural domain of the chain, with C_{H} the product of the constants. The map

$$U \mapsto \Big(\mathcal{M}^t U,\ \big(\mathrm{Lg}(\mathcal{M}^j U(b)\, V_j^{\mathrm{bf}}(\mathcal{M}^{j+1}U)(b)^{-1})\big)_{j<t} \Big)$$

is a real analytic left inverse of Ψ with injective differential, and $\dim \mathcal{V}_t + \dim \mathsf{Y} = \dim \mathcal{V}_0$; the inverse function theorem gives (i) after shrinking, and (iv) with $W = e^{iw}$, $dW = \varrho_{\mathrm{H}}(w)\,\mathrm{d}w$, $\varrho_{\mathrm{H}} > 0$ the continuous density of the Haar measure of $\mathcal{V}_t$ in the variable w, on one set that carries (i)–(iii). For the second choice in (iv) one follows, level by level, a regularity number $\mathfrak{r}_j(W_j)$, which measures the plaquette variables of the averaged configuration of the level j and is comparable with $\max_p |\partial W_j(p) - \mathbf{1}|$: one step gives $\mathfrak{r}_j(W_j) \leq L^{-2}\mathfrak{r}_{j+1}(W_{j+1}) + C_{\mathfrak{r}}|\mathcal{A}_j|$, with $C_{\mathfrak{r}}$ determined by L and the constants.

So a threshold $\mathfrak{r}_\flat$ for $\mathfrak{r}$, with ε_* a fixed fraction of it and $\rho_* \leq (1 - L^{-2})\mathfrak{r}_\flat / C_{\mathfrak{r}}$, keeps every level in the sets where the statements of one step apply; nothing in this sees t, K or m. (ii) $a_p \geq 0$ and

$G_j \geq 0$; at $\mathbf{1}$, $a_p(e^{iA}) = \frac{1}{2}|\mathrm{d}A(p)|^2 + O(|A|^3)$, G_j is half the sum of the squares of the linearised contour variables up to $O(|A|^3)$, and a fibre variable whose linearised curvature and contour variables vanish is 0. At the linear order the composed chart is the chart of the free template: the bare curvature of the field of the level j is T_j applied to it, with values in the slice $\mathcal{C}_{(j+1)}$ (Prop. YM3_I: 120.3), the slices are mutually orthogonal, and the linearised contour variables of the level j are those of $\mathcal{A}_j$ alone; this makes the Hessian block diagonal.

(iii) By (i) and (iv), near $(0,0)$ the composed chart $(w,y) \mapsto \Psi(e^{iw},y)$, with C_{H}, ϱ_{H} and $\theta(e^{iw},y)$, is an adapted chart: it is a real analytic diffeomorphism onto an open set, $\Psi(\mathbf{1},0) = \mathbf{1}$ ($W_j(\mathbf{1},0) = \mathbf{1}$), $\operatorname{Lg}\mathcal{M}^t\Psi(e^{iw},y) = w$, and θ is real analytic. Let $(\mathsf{E}',\Psi',\theta',C',\varrho')$ be an adapted chart, and E_1 an open neighbourhood of $(0,0)$ that the composed chart maps into $\Psi'(\mathsf{E}')$. Then $\sigma := \Psi'^{-1} \circ \Psi$ is a C^∞ diffeomorphism of E_1 onto an open set of the form $\sigma(w,y) = (w, s(w,y))$, both charts carrying $\operatorname{Lg}\mathcal{M}^t$ to the first variable, and $\sigma(0,0) = (0,0)$, as $\Psi(\mathbf{1},0) = \mathbf{1} = \Psi'(0,0)$.

The two formulas for dU give

$$C_{\mathrm{H}}\,\varrho_{\mathrm{H}}(w)\,e^{-\theta(e^{iw},y)} = C'\varrho'(w)\,e^{-\theta'(\sigma(w,y))}\,|\det D_y s(w,y)|$$

almost everywhere on E_1, and, both sides being continuous, at every point of E_1; in particular on the fibre $w = 0$, a null set, where the numbers are read. So $y \mapsto s(0,y)$, a diffeomorphism near 0 that fixes 0, carries the phase $F_\kappa \circ \Psi(\mathbf{1},\cdot)$, the observables and the density $e^{-\theta(\mathbf{1},y)}dy$ into $F_\kappa \circ \Psi'(0,\cdot)$, the observables and $e^{-\theta'(0,y')}dy'$, up to the constant factor $C'\varrho'(0)/(C_{\mathrm{H}}\varrho_{\mathrm{H}}(0))$; the Taylor polynomials are taken at the minimum 0 in both charts. The two coefficients of Laplace's method in either chart are those of the expansion of one quantity, the covariance of a_p and a_q under $e^{-\beta F_\kappa}$ on the fibre $\mathcal{M}^tU = \mathbf{1}$ near $\mathbf{1}$, and so they are the same in both.

The gauge transformations equal to $\mathbf{1}$ on the top lattice preserve the fibre, A and the a_p, and the normalised gauge-fixing factor (Def. YM3_I:6.1 and equations YM3_I:(51)–(52)) integrates to one over their orbits at every strength (Lem. YM3_I:5.3 and equation YM3_I:(55)); integrating over the orbits, the finest sites first, the covariance does not depend on κ up to errors $e^{-c\beta}$, and neither do its coefficients. Near $(0,0)$ the composed chart is itself an adapted chart. □

Proposition 6.14 (The identification of the order six)**.** Under the assumptions of Prop. 6.11, $\beta^3 a_K^2 \gamma_2^\circ = \mathsf{T}[N,n](w)$, the plaquettes being read in bare units on Λ_N; $\mathsf{T}[N,n](w)$ depends on N, n, L and the lattice sources alone. The identity has no error term and holds at every size of the boxes. ■

Proof sketch. By Prop. 6.4(i), Prop. 6.13 applies at $t = n$ on the lattice of the datum. For real $0 < \lambda \leq 1$ the exact formula of the steps (Prop. YM3_I:123.1), read at the fields λB, and the sum rules of the chain classes (Prop. YM3_I:124.2, equation YM3_I:(2289)) add the exponents of the steps of Prop. 6.11(i) up to

$$\mathsf{D}^\lambda(B) = c_n(\lambda)\exp\Big[-\lambda^{-2}\Big(x_0 A(W_0) + \sum_{j<n} x_j G_j(W_j)\Big) - \theta\Big],$$

where $c_n(\lambda)$ does not depend on B and (W_j) is the composed chart over the flat top $\mathbf{1}$ of the level n at the fields $\mathcal{A}_i = \lambda x_i^{-1/2}\check{B}_i$, $\check{B}_i = \mathrm{Ad}(u_{i+1})B_i$, with $u_{i+1} \in G$ a function of the fields of the heights $> i$ alone: the chain of Prop. 6.11 runs through configurations in another gauge than the composed chart. The observables become $a_K^{-1}\sum_p w_a(p)a_p(W_0)$ at the same fields. No τ_i and no counterterm e_i is left, because

$$\lambda^{-2}\xi_j(\lambda)L^j + \sum_{i\leq j} L^i e_i = \lambda^{-2}x_0$$

by the run equation; this is what ξ_j is chosen for. So the density is the Wilson weight at $\beta\lambda^{-2}$ with the gauge-fixing strengths $\kappa_j = x_j/\beta$ on the fibre $\mathcal{M}^nU = \mathbf{1}$. The triangular change $B \mapsto \check{B}$ preserves the volume and the boxes, which are Euclidean balls $|B_i(b)| < \mathfrak{b}(x_i)$ and are preserved by

the rotations $\mathrm{Ad}(u)$; its linear part at 0 is a constant rotation of the colours, so it preserves the Gaussian part $\bigotimes_{i<n}\nu_i^\circ$; and the two numbers of Laplace's method are kept by a diffeomorphism that carries phase, observables and density into each other.

So γ_2° is the second coefficient of Laplace's method eq. 6.viii at the parameter $\beta\lambda^{-2}$, and $\beta^3 a_K^2\gamma_2^\circ = \sum_{p,q} w_1(p)w_2(q)\gamma_2^{\mathrm{L}}$ in the chart of Prop. 6.13, which does not depend on κ. The dilation $x \mapsto L^{K-n}x$ carries the lattice (ℓ, K) with its flat top at the level n to the lattice $(\ell L^{K-n}, n)$, with the same bare torus, plaquettes to plaquettes with their barycentres; so the number is $\mathsf{g}_2[N,n](p,q)$. The argument uses that the boxes are Euclidean balls; it would not apply to boxes that are cubes in the components. □

Lemma 6.15 (An implicit function, Stokes's formula in a box, Wick's rule)**.**

(i) Let V be a finite-dimensional complex normed space, $0 < \varsigma \le \frac{1}{120}$, and F holomorphic from $\{\|y\| < \varsigma\} \times \{\lambda \in \mathfrak{g}^{\mathbb{C}} : |\lambda| < \varsigma\}$ to $\mathfrak{g}^{\mathbb{C}}$ with $|F| \le \frac{1}{40}$, $F(0,0) = 0$ and $D_\lambda F(0,0) = 1$. Then for $\|y\| < \varsigma^3$ there is exactly one $h(y)$ with $|h(y)| \le \varsigma^2$ and $F(y, h(y)) = 0$; h is holomorphic, $|h(y)| \le \|y\|/(15\varsigma)$, and h is $\mathfrak{g}$-valued at the real points if F is.

(ii) Let B be the injective image in a bare torus of a box of $\mathbb{Z}^3$. For every closed bare path γ in B there is an integer function k_γ on the plaquettes with all corners in B, $|\mathsf{k}_\gamma| \le |\gamma|$ (the number of bonds of γ), with $a(\gamma) = \sum_p \mathsf{k}_\gamma(p)\,(\mathrm{d}a)(p)$ for every 1-cochain a.

(iii) Let (ζ_ι) be a centred Gaussian family with covariance C, and $P_1, \dots, P_r$ homogeneous polynomials in it of orders $n_1, \dots, n_r$, $n = \sum n_a$, $\|P\|_1$ the sum of the moduli of the coefficients. For monomials the joint cumulant $\langle P_1; \dots; P_r\rangle$ is the sum, over the pairings of the factors that are connected (the graph on $\{1, \dots, r\}$ with an edge for every pair that joins two of the P_a is connected), of the products of the covariances, and by multilinearity this extends to polynomials; the cumulant vanishes if n is odd. With $\mathsf{c} \ge \sup|C|$ on the supports, $|\langle P_1; \dots; P_r\rangle| \le (n-1)!! \prod_a \|P_a\|_1 \mathsf{c}^{n/2}$; if every connected pairing has a pair with $|C| \le \varepsilon$, the bound holds with $\mathsf{c}^{n/2-1}\varepsilon$ for $\mathsf{c}^{n/2}$; and for two covariances C, C' bounded by c the two cumulants differ by at most $(n-1)!!\frac{n}{2}\prod_a \|P_a\|_1 \mathsf{c}^{n/2-1}\sup|C - C'|$.

■

Proof sketch. (i) Put $g(y,\lambda) = \lambda - F(y,\lambda)$, so $|g| \le \frac{1}{40} + \varsigma \le \frac{1}{30}$, $g(0,0) = 0$, $D_\lambda g(0,0) = 0$. Cauchy's estimate and Schwarz's lemma give $|D_\lambda g(y,\lambda)| \le \frac{2}{15}\varsigma^{-2}\max\{\|y\|, |\lambda|\}$ for $\max\{\|y\|, |\lambda|\} \le \frac{1}{2}\varsigma$, and $|g(y,0)| \le \|y\|/(30\varsigma)$. So for $\|y\| < \varsigma^3$ the map $\lambda \mapsto g(y,\lambda)$ sends $\{|\lambda| \le \varsigma^2\}$ into itself and contracts by $\frac{2}{15}$; its fixed point is $h(y)$, the uniform limit of holomorphic iterates, and $|h| \le |g(y,0)| + \frac{2}{15}|h|$. (ii) With the paths P_z from a corner o of the box to z that run first in the direction 1, then 2, then 3, the quantity $a(P_z) + a(e) - a(P_{z+e_\mu})$ of a bond $e = \langle z, z+e_\mu\rangle$ is the sum of $\pm(\mathrm{d}a)$ over a strip of plaquettes of the box; along a closed path the terms $a(P_\cdot)$ telescope.

(iii) By Wick's theorem the moments are sums over all pairings; a pairing determines the partition of the factors into the components of its graph and is connected on each block, which identifies the sums over connected pairings with the joint cumulants. The bounds count the $(n-1)!!$ pairings and use $|\prod_{i\le m} c_i - \prod_{i\le m} c_i'| \le m\,\mathsf{c}^{m-1}\max|c_i - c_i'|$. □

Proposition 6.16 (Two bare tori under one flat top)**.** Let $N = 2L^{\mathsf{m}}$, $N' = L^s N$ $(s \ge 1)$, $1 \le t < \mathsf{m}$, $\pi : \Lambda_{N'} \to \Lambda_N$ the reduction, and

$$\nu = N/L^t \ge \nu_{\mathrm{W}} = \max\Big\{192,\ \frac{48(3\log L + \log 2)}{\delta_{\mathrm{tp}} L(L-1)},\ \frac{C_L M}{L}\Big\}.$$

For plaquettes p', q' of $\Lambda_{N'}$ whose base points lie within $N/8$ of each other (the base point of the plaquette $p_{\mu\nu}(x)$ being x, the distance being the sup distance of the bare torus, in bare units) and $m' = 0, 2$,

$$|\mathsf{g}_{m'}[N,t](\pi p', \pi q') - \mathsf{g}_{m'}[N',t](p',q')| \le C'_{\mathrm{W}} N'^{237} e^{-\delta_{\mathrm{tp}} L\nu/48},$$

with C'_{W} determined by L and C_{tp}. Hence, for real w_1, w_2 on the plaquettes of Λ_N with supports in balls of radius $r \geq 8$, all base points within $N/8$ of each other, and $\breve{w}_a$ equal to $w_a \circ \pi$ on one lift of a ball of radius $N/4$ that holds the supports and 0 elsewhere,

$$|\mathsf{T}[N,t](w) - \mathsf{T}[N',t](\breve{w})| \leq 900 r^6 C'_{\mathrm{W}} N'^{237} e^{-\delta_{\mathrm{tp}} L\nu/48} \|w_1\|_\infty \|w_2\|_\infty.$$

■

Proof sketch. Signs. Let $S \in \{N, N'\}$, and read the objects of the lattice $(S/L^t, t)$ in bare units on Λ_S; distances are those of the bare torus, $|\cdot|_\infty$ its sup distance, and $x(p)$ is the barycentre of a plaquette p, its base point being x if $p = p_{\mu\nu}(x)$. For a bond c of the top lattice $\mathbb{T}^{(t)}$, $B^t(c_\pm)$ are the blocks of order t of its end points, its double block is their union, and its corridor $\mathrm{cor}(c)$ is the set of the bare bonds b with $b_- \in B^t(c_-)$, $b_+ \in B^t(c_+)$; $\mathsf{1}_c$ is the bare 1-cochain that is 1 on $\mathrm{cor}(c)$, -1 on the reversed bonds and 0 elsewhere. As $\nu \geq 8$, every bare bond lies in one top block or in exactly one corridor, and no bond of the corridor of c' has both end points in the double block of c unless $c' \in \{c, \bar{c}\}$; the bonds of the double block of c are those with both end points in it.

For a bare site $x \notin \mathbb{T}^{(t)}$, $j(x) = \max\{j : x \in \mathbb{T}^{(j)}\} < t$, its parent $\hat{x} \in \mathbb{T}^{(j(x)+1)}$ is the centre of the block of order 1 that holds x, and $\mathrm{anc}(z) = \{z, \hat{z}, \hat{\hat{z}}, \dots\} \setminus \mathbb{T}^{(t)}$; the sums over x below run over the bare sites off $\mathbb{T}^{(t)}$. Q_k is the linear part of the k-fold block average, and $\xi_A(x) = (\mathsf{Q}_{j(x)}A)(\hat{x}, x)$ is the linearised contour variable of the site x. A $\mathfrak{g}$-valued 1-form y has the coordinates $y(\iota)$, $\iota = (b, \alpha)$, $b > 0$ a bare bond and α a colour; $\iota + \gamma$ is the translate of ι by γ, and $\mathrm{dist}(\iota, \iota') = |b_- - b'_-|_\infty$.

By Prop. 6.13(iii) the numbers do not depend on κ; we take $\kappa \equiv 1$.

A chart made for locality. Q_k is a mean over bare paths: for every bond c of $\mathbb{T}^{(k)}$ there is a probability measure on the bare paths from c_- to c_+ in the double block of c, with at most $(4L)^k$ bonds, whose mean of $A(\gamma)$ is $(\mathsf{Q}_k A)(c)$; and $\mathsf{Q}_t \mathsf{1}_c$ is 1 at c and 0 at the other top bonds. $\mathcal{M}^t(U)(c)$ depends on U on the double block of the top bond c alone. So, for y in $Y_S := \ker \mathsf{Q}_t$ near 0, the constraint $\mathcal{M}^t(e^{i\hat{y}}) = \mathbf{1}$ for $\hat{y} := y + \sum_{c>0} h_c(y) \mathsf{1}_c$ is solved one top bond at a time, with one element $h_c(y) \in \mathfrak{g}$ (three unknowns) for each: Lem. 6.15(i) at $\varsigma = \varsigma_t := (75L)^{-t}/(60L)$, applied to $(y, \lambda) \mapsto \mathrm{Lg}\, \mathcal{M}^t(e^{i(y+\lambda \mathsf{1}_c)})(c)$, which is holomorphic and bounded by $\frac{1}{40}$ on the polydisc of the Lemma, vanishes at $(0,0)$ and has the derivative 1 in λ there, gives exactly one holomorphic h_c with $|h_c(y)| \leq \|y\|_\infty/(15\varsigma_t)$ for $\|y\|_\infty < \varsigma_t^3$, $\mathfrak{g}$-valued at real y; it reads y on the double block of c alone, $\hat{y} = y$ on the top blocks, and $\hat{y} = y + O(y^2)$ on Y_S.

The map $A \mapsto (\mathrm{Lg}\, \mathcal{M}^t(e^{iA}), A - \sum_{c>0} (\mathsf{Q}_t A)(c) \mathsf{1}_c)$ is a real analytic diffeomorphism near 0, as its differential at 0 has the inverse $(w, y) \mapsto y + \sum_c w(c) \mathsf{1}_c$; its inverse, followed by $A \mapsto e^{iA}$, is an adapted chart Ψ' in the sense of Prop. 6.13(iii), with $\varrho' = 1$, a real analytic θ' and $\Psi'(0, y) = e^{i\hat{y}}$ for $y \in Y_S$; so both numbers may be computed in it. In it the action, the gauge terms, the Haar density (the exponential-coordinate Haar formula following equation YM3_I:(911)) and the Jacobian, the product over the top bonds c of the determinants of the derivatives of $\mathrm{Lg}\, \mathcal{M}^t(e^{i\cdot})(c)$ in the direction $\mathsf{1}_c$, are sums of terms, each holomorphic and bounded by 1 on the polydisc $\|y\|_\infty < \varsigma_t^3$ and depending on y on at most $12L^{3t}$ bonds with end points in a set of diameter $\leq 2L^t$; for a plaquette p these are the top blocks that hold a corner of p, within L^t of its base point.

The reduction π carries the lattices, blocks, contours, corridors and the block average of $\Lambda_{N'}$ to those of Λ_N on every set of diameter $\leq N/2$, so these are the same functions on both tori after the renaming of the variables by π.

The lines. The gauge terms and the Haar terms of the chart read y on the bonds, and not its curvature alone; so the lines of Wick's rule in this chart are the entries of the covariance C_S of the chart variable $y \in Y_S$ under the centred Gaussian measure with the density proportional to $e^{-\frac{1}{2}\mathsf{H}_S(y,y)}$, $\frac{1}{2}\mathsf{H}_S(y,y)$ being the second Taylor polynomial at 0 of the action with the gauge terms,

$$\mathsf{H}_S(A, A') = (\mathrm{d}A, \mathrm{d}A') + \sum_x \langle \xi_A(x), \xi_{A'}(x) \rangle.$$

Let $\mathcal{X}_S$ be the space of the $\mathfrak{g}$-valued functions on the bare sites that vanish on $\mathbb{T}^{(t)}$. Then $\mathrm{d}\mathcal{X}_S \subset Y_S$, as $\mathsf{Q}_t \mathrm{d}\chi = \mathrm{d}(\chi|_{\mathbb{T}^{(t)}})$, and $\chi \mapsto \xi_{\mathrm{d}\chi} = (\chi(x) - \chi(\hat{x}))_x$ is a bijection of $\mathcal{X}_S$ onto the functions on the sites off $\mathbb{T}^{(t)}$, with the inverse $\chi(x) = \sum_{x'' \in \mathrm{anc}(x)} \xi(x'')$. So Y_S is the direct sum of its axial part $Y_S^{\mathrm{ax}} = \{A \in Y_S : \xi_A = 0\}$ and of $\mathrm{d}\mathcal{X}_S$, $y = y^{\mathrm{ax}} + \mathrm{d}\chi_y$, and

$$\mathsf{H}_S(y,y) = \|\mathrm{d}y^{\mathrm{ax}}\|^2 + \sum_x |\xi_y(x)|^2;$$

d maps Y_S^{ax} one to one onto the curvatures whose plaquette averages of the order t vanish, and H_S is positive definite on Y_S. In the Gaussian measure y^{ax} and the $\xi(x)$ are independent: the $\xi(x)$, one for each site off the top lattice, are independent with the variance 1 in each colour, $(\mathrm{d}\chi)(b) = \sum_x \sigma(b,x)\xi(x)$ with $\sigma(b,x) = \mathbf{1}[x \in \mathrm{anc}(b_+)] - \mathbf{1}[x \in \mathrm{anc}(b_-)]$, and $\mathrm{d}y^{\mathrm{ax}}$ has the covariance $\Pi_t^{\mathrm{fl}} = \Pi_0^{\mathrm{tp}} - \Pi_t^{\mathrm{tp}}$ in each colour. On Y_S^{ax} the means of y over two families of bare paths vanish: over the paths from the centre of the top block of a site x to x, inside that block and with at most $(4L)^t$ bonds, whose mean of $y(\gamma)$ is $\sum_{x'' \in \mathrm{anc}(x)} \xi_y(x'') = 0$ (the contour averages of the levels composed from the top down), and over the paths of Q_t along a top bond, as $y \in \ker \mathsf{Q}_t$.

So $y(b)$ is the mean of the values of y on closed bare paths of at most $4(4L)^t$ bonds in the box of b, the top block that holds b, or the double block of c if $b \in \mathrm{cor}(c)$, and by Lem. 6.15(ii), the box being the injective image of a box of $\mathbb{Z}^3$, $y(b) = \sum_p \tau(b,p)\,(\mathrm{d}y)(p)$ with real Stokes coefficients, $|\tau(b,p)| \le 4(4L)^t$, $\tau(b,p) = 0$ unless all the corners of p lie in the box of b. Hence $C_S((b,\alpha),(b',\alpha')) = \delta_{\alpha\alpha'}[C_S^{\mathrm{ax}} + C_S^{\mathrm{g}}](b,b')$,

$$C_S^{\mathrm{ax}}(b,b') = \sum_{p,p'} \tau(b,p)\tau(b',p')\,\Pi_t^{\mathrm{fl}}(p,p'), \qquad C_S^{\mathrm{g}}(b,b') = \sum_x \sigma(b,x)\sigma(b',x),$$

and $C_S^{\mathrm{g}}(b,b') = 0$ unless an end point of b and one of b' lie in one top block. A box holds at most $6L^{3t}$ plaquettes, $|\Pi_t^{\mathrm{fl}}| \le 1$ (the orthogonal projections in YM3_I:(2226)), and $\sigma(b,\cdot) \ne 0$ at no more than $2t$ sites; so $\sup|C_S| \le \mathsf{c}_\tau^2 + 2t$, $\mathsf{c}_\tau = 24L^{3t}(4L)^t$. By Prop. YM3_I:120.6 on the lattice $(S/L^t, t)$ at $s = 0$, uniformly in the volume,

$$|\Pi_t^{\mathrm{fl}}(p,q)| \le 2L^3 C_{\mathrm{tp}} L^{-3t} e^{-\delta_{\mathrm{tp}} L|x(p)-x(q)|_\infty/L^t} \qquad \text{if } \delta_{\mathrm{tp}} L(L-1)|x(p)-x(q)|_\infty \ge (3\log L + \log 2)L^t,$$

the barycentres being taken in bare units. The barycentre of a plaquette in the box of b lies within $2L^t$ of b_-; so if $\mathrm{dist}(\iota,\iota') \ge R \ge 8L^t$ and $\delta_{\mathrm{tp}} L(L-1)(R-4L^t) \ge (3\log L + \log 2)L^t$, then $C_S^{\mathrm{g}}(b,b') = 0$ and

$$|C_S(\iota,\iota')| \le \mathsf{c}_\tau^2\, 2L^{3-3t} C_{\mathrm{tp}} \exp\bigl[-\delta_{\mathrm{tp}} L(R-4L^t)/L^t\bigr].$$

At $R = N/24$ these conditions hold, and the exponent is at least $\delta_{\mathrm{tp}} L\nu/48$: $R = \frac{\nu}{24}L^t \ge 8L^t$ and $R - 4L^t \ge \frac{\nu}{48}L^t$ as $\nu \ge 192$, and $\delta_{\mathrm{tp}} L(L-1)\frac{\nu}{48} \ge 3\log L + \log 2$ by the second member of ν_{W}. Finally the covariance on Λ_N is the periodisation of that on $\Lambda_{N'}$, exactly: with $\Gamma = N\mathbb{Z}^3/N'\mathbb{Z}^3$, $|\Gamma| = (N'/N)^3$,

$$C_N(\pi\iota,\pi\iota') = \sum_{\gamma\in\Gamma} C_{N'}(\iota,\iota'+\gamma).$$

For π^* commutes with d, the Q_k and ξ, and π is $|\Gamma|$ to one on the plaquettes and on the sites, so that $\mathsf{H}_{N'}(\pi^*A,\pi^*A') = |\Gamma|\,\mathsf{H}_N(A,A')$ and $\pi^*Y_N = Y_{N'} \cap \pi^*\mathcal{E}_N$, $\mathcal{E}_N$ the 1-forms on Λ_N; $u = \sum_\gamma C_{N'}(\cdot,\iota'+\gamma)$ is the one solution in $Y_{N'}$ of $\mathsf{H}_{N'}(u,v) = \sum_\gamma v(\iota'+\gamma)$ for all $v \in Y_{N'}$, a problem invariant under Γ, so $u = \pi^*u_0$ with $u_0 \in Y_N$, and the test functions $v = \pi^*v_0$ give $\mathsf{H}_N(u_0,v_0) = v_0(\pi\iota')$ for all $v_0 \in Y_N$, that is $u_0 = C_N(\cdot,\pi\iota')$.

Near and far. By Laplace's method in the chart, each of the seven cumulants of $\mathsf{g}_{m'}$ is, by multilinearity, a sum over at most $64N'^6$ tuples of places of cumulants of homogeneous Taylor polynomials of the terms, of total order ≤ 10, in the Gaussian family with the covariance C_S; Lem. 6.15(iii) writes each as a sum over connected pairings of products of lines. A place is near if the set on which its term reads y lies within $N/4$ of the base point of p', and far otherwise; π maps the near places of $\Lambda_{N'}$ one to one onto those of Λ_N, and p', q' are near, as $L^t \le N/192$. A tuple of

near places gives the same polynomials on both tori, renamed by π, so that its cumulant on Λ_N is that of the polynomials of $\Lambda_{N'}$ in the covariance $C_N(\pi\iota, \pi\iota')$; its indices lie within $N/2$ of each other, so that by the periodisation the two covariances differ by $\sum_{\gamma\neq 0} C_{N'}(\iota, \iota'+\gamma)$, a sum of fewer than $|\Gamma|$ lines of length $\geq N/2$.

If a tuple has a far place, every connected pairing of it contains a line of length $\geq N/24$, on either torus: a path without repetitions in the graph of the pairing leads from the place p' to the far place in at most three edges, the polynomials of one place depend on y on a set of diameter $\leq 2L^t \leq \frac{2}{192}N$, and so the lengths of the at most three lines of the path add up to more than $\frac{N}{4} - 7L^t > \frac{N}{8}$. By the bound of the lines at $R = N/24$, such a line is at most $\varepsilon_C = 2L^{3-3t} C_{\mathrm{tp}} \mathsf{c}_\tau^2 e^{-\delta_{\mathrm{tp}} L\nu/48}$, and $\sup|C_S| \leq 2\mathsf{c}_\tau^2$; the bounds of Lem. 6.15(iii) then bound the far tuples on each torus by a multiple of ε_C and the difference at a near tuple by a multiple of $|\Gamma|\varepsilon_C$, with Cauchy's estimate on the polydisc for the coefficients of the polynomials, at most $(108L^{3t}\varsigma_t^{-3})^k$ in the sum of their moduli at the order k.

The powers of $L^t \leq N'$, the factor $|\Gamma| \leq N'^3$ and the number of the tuples are collected in N'^{237}, with C'_{W} determined by L and C_{tp}. A ball of radius $r \geq 8$ holds at most $30r^3$ barycentres of plaquettes, which gives the second display. □

Proposition 6.17 (The anchor)**.** Let $\ell = 2L^m \geq N_{\mathsf{H}}$ with $\ell L \geq C_L M$, and $K \geq 1$. For all plaquettes p, q of $\Lambda_{\ell L^K}$ and all plaquettes $\tilde{p}, \tilde{q}$ of $\mathbb{Z}^3$ over them whose base points are at sup-distance $\leq \frac{1}{8}\ell L^K$,

$$|\mathsf{g}_2[\ell L^K, K](p,q) - \mathsf{B}_1^{\mathrm{lat}}(\tilde{p},\tilde{q})| \leq C_\star[\ell^{29/2} K L^{-K/2} + \ell^{18} L^{-2K} + \ell^{19} K^2 L^{-5K}] \leq 3C_\star \ell^{19} K L^{-K/2},$$

with $C_\star$ determined by the constants and independent of the side; if $\ell^3 \leq L^K$, the left member is at most $3C_\star \ell^{29/2} K L^{-K/2}$. ■

Proof sketch. Write $\mathsf{N} = \ell L^K$. Distances are those of the bare torus Λ_{N}, in bare units, $|\cdot|$ its Euclidean distance and $|\cdot|_\infty$ its sup distance, $\langle z\rangle = \max\{1, |z|\}$; x_p is the base point of the plaquette p, $\langle b - b'\rangle := \langle b_- - b'_-\rangle$, $\langle x_p - b\rangle := \langle x_p - b_-\rangle$; and o is a fixed site of the top lattice $\mathbb{T}^{(K)}$.

The covariant gauge below the flat top. By Prop. 6.13(iii) and its proof, $\mathsf{g}_2[\mathsf{N}, K](p,q)$ is the coefficient of β^{-3} of one gauge-invariant quantity, the covariance of a_p and a_q under $e^{-\beta A}$ on the fibre $\mathcal{M}^K U = \mathbf{1}$ near $\mathbf{1}$; so every normalised gauge-fixing factor may be used to compute it. We take the covariant one with the gauge parameter 1 of Def. 7.7, with its Faddeev–Popov determinant as in Lem. 7.6, for the whole bare gauge group modulo the constant transformations, and widen the flat top to its orbit under the gauge transformations of the top lattice: the top field runs over the pure gauges $w(\mu')(c) = \mathrm{Lg}(e^{i\mu'(c_-)} e^{-i\mu'(c_+)})$, μ' a $\mathfrak{g}$-valued function on $\mathbb{T}^{(K)}$ with $\mu'(o) = 0$.

The constraint $\mathrm{Lg}\,\mathcal{M}^K(e^{i\mathsf{a}}) = w(\mu')$ is linearised in straight frames: the linear part of $\mathrm{Lg}\,\mathcal{M}^K(e^{i\cdot})$ is $\mathbf{Q}_K^s \mathsf{a} - \mathrm{d}\vartheta_K$, where the straight averages $\mathbf{Q}_K^s$ of the free template (Section YM3_I:120) map the bare 1-forms to the 1-forms of the top lattice, with $\mathrm{d}\mathbf{Q}_K^s = Q_K^{\mathrm{pl}}\mathrm{d}$ (Lem. YM3_I:120.1), Q_K^{pl} the plaquette averages of the level K, and the exact difference $\mathrm{d}\vartheta_K$ on the top lattice goes into μ'. So the linearised constraint is $\mathbf{Q}_K^s \mathsf{a} + \mathrm{d}\mu' = 0$.

Laplace's method under the constraint, with the harmonic right inverse $\omega \mapsto (\mathsf{a}_\omega, \mu'_\omega)$ of the linearised constraint, which parametrises the constraint set up to the order three as a graph over the kernel

$$\hat{E} = \{(\mathsf{a}, \mu') :\ \mathbf{Q}_K^s \mathsf{a} + \mathrm{d}\mu' = 0,\ \mu'(o) = 0\},$$

gives the exact splitting $\mathsf{g}_2[\mathsf{N}, K] = \mathsf{B}_1^{\hat{E}} + \sum_{i\leq 11} \mathsf{t}_i$. Here $\mathsf{B}_1^{\hat{E}}$ collects the terms without the inverse: it is the formula (7.xviii) of $\mathsf{B}_1^{\mathrm{lat}}$ at the gauge parameter 1, with the same vertices and measure terms, its kernels $\mathsf{C}, \mathsf{E}, \Pi$ being those of the centred Gaussian field (a, μ') on $\hat{E} \otimes \mathfrak{g}$ with the form $\frac{1}{2}\|\mathrm{d}\mathsf{a}\|^2 + \frac{1}{2}\|\mathrm{d}^*\mathsf{a}\|^2$, diagonal in the colours, and its kernels g, D being formed with the Green's function $\mathsf{G}^{\mathrm{tor}}$ of $\mathrm{d}^*\mathrm{d}$ on the torus off the constants.

The t_i hold the inverse: $\mathsf{t}_1,\dots,\mathsf{t}_5$ come from the observables, t_6, t_7 from the cubic vertex, t_8 from the quadratic form at the correction, t_9, t_{10} from the traces of the Jacobian of the constraint, and t_{11} from the measure of μ'; each is a cumulant of polynomials in the Gaussian field, in which the quadratic and cubic terms of the block average enter through the response $\omega \mapsto \mathsf{a}_\omega$ of the field to a source ω on the top lattice.

The lines. Let $\mathsf{C}^{\mathrm{tor}}$ be the inverse of $\mathrm{d}^*\mathrm{d}+\mathrm{d}\mathrm{d}^*$ off the constant 1-forms and 0 on them, and $\mathsf{S}_0 := \mathsf{C}^{\mathrm{tor}}\mathrm{d}^*$ on the exact curvatures: $\mathsf{S}_0 F$ is the primitive of F in the range of d^*, and $\mathrm{d}\mathsf{S}_0 = 1$. The a-members of $\hat{E}$ are the $\mathsf{a} = \mathsf{S}_0 F + \mathrm{d}\lambda$ with F a curvature whose plaquette averages of the level K vanish; in the Gaussian measure F and λ are independent, with the covariances $\Pi_0^{\mathrm{tp}} - \Pi_K^{\mathrm{tp}}$ and $(\mathsf{G}^{\mathrm{tor}})^2$. So the covariance of a is $C_{\hat{E}} = \mathsf{C}^{\mathrm{tor}} - \mathsf{S}_0\Pi_K^{\mathrm{tp}}\mathsf{S}_0^{\mathsf{T}}$, with $\mathrm{d}C_{\hat{E}}\,\mathrm{d}^* = \Pi := \Pi_0^{\mathrm{tp}} - \Pi_K^{\mathrm{tp}}$; and the response is $\mathsf{a}_\omega = \mathsf{S}_0\mathsf{H}_K^{\mathrm{cv}}\omega$ plus a constant 1-form, with $\mathrm{d}\mathsf{a}_\omega = \mathsf{H}_K^{\mathrm{cv}}\omega$, the curvature response of the level K, and $\mathrm{d}^*\mathsf{a}_\omega = 0$.

The kernels $\mathsf{C}^{\mathrm{tor}}$, $\mathrm{d}\mathsf{C}^{\mathrm{tor}}$ and $\Pi_0^{\mathrm{tp}} = \mathrm{d}\mathsf{C}^{\mathrm{tor}}\mathrm{d}^*$ are Fourier sums $\mathsf{N}^{-3}\sum_{k\neq 0} e^{ik\cdot z}\sigma(k)$ over the non-zero modes of the torus, z the difference of the base points, with the symbols σ of the lines of $\mathbb{Z}^3$ (Lem. 7.24) at the gauge parameter 1, of the classes $\mathcal{S}^{-2}$, $\mathcal{S}^{-1}$, $\mathcal{S}^0$; by Lem. 7.38(a) they are at most $c_{\mathrm{T}}\langle z\rangle^{-1}$, $c_{\mathrm{T}}\langle z\rangle^{-2}$, $c_{\mathrm{T}}\langle z\rangle^{-3}$, with one number c_{T} for all sides. With $|\Pi_K^{\mathrm{tp}}(p,q)| \le CL^{-3K} = C\ell^3\mathsf{N}^{-3}$ uniformly in the volume (Prop. YM3_I:120.5) and $\sum_z\langle z\rangle^{-2} \le 14\mathsf{N}$ over the torus, the corrections $\mathsf{S}_0\Pi_K^{\mathrm{tp}}\mathsf{S}_0^{\mathsf{T}}$, $\Pi_K^{\mathrm{tp}}\mathsf{S}_0^{\mathsf{T}}$ and Π_K^{tp} are at most $c\,\ell^3\mathsf{N}^{-1}$, $c\,\ell^3\mathsf{N}^{-2}$ and $c\,\ell^3\mathsf{N}^{-3}$, every distance of the torus is below N, and $|\Pi| \le 1$, Π being the kernel of an orthogonal projection (the orthogonal projections in YM3_I:(2226)).

So, with $\mathsf{c}_0 = c\,\ell^3$, the lines obey

$$|C_{\hat{E}}(b,b')| \le \mathsf{c}_0\langle b-b'\rangle^{-1}, \qquad |\mathrm{d}C_{\hat{E}}(p,b)| \le \mathsf{c}_0\langle x_p - b\rangle^{-2}, \qquad |\Pi(p,q)| \le \min\{1, \mathsf{c}_0\langle x_p - x_q\rangle^{-3}\},$$

and the kernels g,D formed with $\mathsf{G}^{\mathrm{tor}}$ obey the majorants of (7.xxxi) in the same way. The response has $|\mathsf{a}_\omega| \le C\ell L^{-K}$ and $|\mathrm{d}\mathsf{a}_\omega| \le CL^{-2K}$ in the sup norm of ω: the latter from the row sums $\le CL^{-2K}$ of the response kernel at every admissible side (equation YM3_I:(2234), whose row sum follows by summing its exponential kernel bound), the former from these and the row sums $\le c\,\mathsf{N}$ of S_0.

The Gaussian member against $\mathbb{Z}^3$. Let $\mathcal{L}_{\mathsf{N}}$ be a Fourier sum over the non-zero modes of the torus of side N with a symbol of the class $\mathcal{S}^s$, $s > -3$, and $\mathcal{L}_\infty$ the kernel of $\mathbb{Z}^3$ with the same symbol. Then $\sum_{v\in\{0,1\}^3}\mathcal{L}_{2\mathsf{N}}(z+\mathsf{N}v) = \mathcal{L}_{\mathsf{N}}(z)$ exactly, as $\sum_v e^{ik\cdot\mathsf{N}v}$ is 8 on the modes of the smaller torus and 0 on the others; for $|z|_\infty \le \frac{3}{4}\mathsf{N}$ the seven images with $v\neq 0$ are at most $C\mathsf{N}^{-3-s}$ each, by Lem. 7.38(a), and telescoping over the sides $2^i\mathsf{N}$, with Lem. 7.38(b), gives $|\mathcal{L}_{\mathsf{N}}(z) - \mathcal{L}_\infty(z)| \le C\mathsf{N}^{-3-s}$.

With the three corrections from Π_K^{tp}, the difference at $|z|_\infty \le \frac{3}{4}\mathsf{N}$ between a line of $\mathsf{B}_1^{\hat{E}}$ and the line of $\mathbb{Z}^3$ at the same base points is at most $C\mathsf{N}^{-1/2}$ times the majorant $\mathsf{c}_0\langle z\rangle^{-n}$ of the line with its exponent n lowered by $\frac{1}{2}$. The two-loop sums of (7.xviii), written out over monomials, connected pairings and places by Lems. 7.9 and 7.26, converge absolutely with these majorants, uniformly in the side, also with one line weakened by $\frac{1}{2}$, as in Lem. 7.27(ii), the sums over a torus being bounded by those over $\mathbb{Z}^3$ at the nearest images of the arguments: the sum is at most $C\mathsf{c}_0^5$.

The terms all of whose places lie within $\frac{1}{4}\mathsf{N}$ of x_p in the sup distance correspond one to one on the torus and on $\mathbb{Z}^3$, with the same monomials, pairings and coefficients, and together they differ by at most $C\mathsf{N}^{-1/2}\mathsf{c}_0^5$. Every other term has a place that is the end of a line whose other end is p or q, of length $\ge \frac{1}{16}\mathsf{N}$, as $|x_p - x_q|_\infty \le \frac{1}{8}\mathsf{N}$; so the sum of the other terms is at most $(16/\mathsf{N})^{1/2}C\mathsf{c}_0^5$ on each side. Hence $|\mathsf{B}_1^{\hat{E}} - \mathsf{B}_1^{\mathrm{lat}}| \le C\ell^{15}(\ell L^K)^{-1/2}$, $\mathsf{B}_1^{\mathrm{lat}}$ being taken at the gauge parameter 1, as in (7.xix).

The other terms. Each t_i is at most $C\ell^{q_i}$ times one of the rates KL^{-K}, K^2L^{-2K}, $L^{-3K/2}$, $KL^{-K/2}$, $L^{-K/2}$, $L^{-5K/2}$, KL^{-3K}, L^{-2K}, K^2L^{-5K}, none worse than $KL^{-K/2}$, and $\sum_i|\mathsf{t}_i| \le C[\ell^{13}KL^{-K/2} + \ell^{18}L^{-2K} + \ell^{19}K^2L^{-5K}]$. The side enters only through c_0, through paths of at most $\frac{3}{2}\ell$ bonds on the top lattice, through one sum of $\langle\cdot\rangle^{-2}$ over the torus, through the number of plaquettes and through the dimension $9\ell^3$ of the top fields; no inverse on the top lattice is bounded

by a norm, and so $C_\star$ does not depend on the side. For $\ell^3 \le L^K$ the last two members are below the first. □

Proposition 6.18 (One lattice is enough)**.** Let $0 < g \le \gamma$ and $\ell \in \mathfrak{L}(g)$. There are integers $K_0, K_\flat$ and numbers $C_\diamond < \infty$, $d_{\mathrm{K}} > 0$, depending on g, ℓ and the constants, such that, with $K_\sharp(d) = \max\{K_0, K_\flat, \lceil 3600 \log_L(\sqrt{3}M/d)\rceil\}$: for $0 < d \le 1$, $20d \le \ell$, $\vartheta \in (0,1)$, $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$, $b \in [\gamma^{-2}, L^2(g^{-2}+\bar B)]$ and $K' \ge K \ge K_\sharp(d)$,

$$|S^T_{2;K',\ell,L^{K'}b} - S^T_{2;K,\ell,L^K b}|(f,h) \le C_\diamond \vartheta^{-2} d^{178}\eta,$$

also with $S_2^{(\ell,b)}$ for the first member; and for $0 < d \le d_{\mathrm{K}}$, $s(d) < K_\sharp(d)$ and $(\sqrt{3}M/d)^{3600} \le L^{K_\sharp(d)} \le d^{-3601}$. ■

Proof sketch. Cor. YM3_I:117.1, equation YM3_I:(2207) gives, for $K' \ge K \ge K_1(\mathsf{d}_\infty)$ and the labels of the interval $[\gamma^{-2}, L^2(g^{-2}+\bar B)]$,

$$|S^T_{2;K'} - S^T_{2;K}|(f,h) \le \mathsf{C}'\ell^9 K^{\mathsf{p}_2} L^{-K/10}\|f\|_\sharp\|h\|_\sharp, \qquad \|f\|_\sharp = \max\{\|f\|_\infty, 40M \operatorname{Lip} f\} \le 40M\|f\|_\infty/(\vartheta d),$$

with C' and $\mathsf{p}_2 = 6p_0 + \frac12$ the constants of that statement; the limit $K' \to \infty$ exists along the full sequence. Here $\mathsf{d}_\infty \ge d/\sqrt{3}$ is the sup distance of the supports, and $K_1(\mathsf{d}_\infty)$ (the constants and thresholds in Thm. YM3_I:AR) is the least integer $\ge K_0$ from which on, with $u = \lceil K/1800\rceil$, the two threshold inequalities at the end of the proof of Thm. YM3_I:AR hold: $\mathsf{d}_\infty L^u \ge 6M$, and $c_9\ell^3 L^{3u} e^{-\mathsf{d}_\infty L^u/M} \le \frac12 \mathsf{C}\ell^9 K^{\mathsf{p}_2} L^{-K/10}$, with the numbers c_9, C of the two threshold inequalities at the end of the proof of Thm. YM3_I:AR at $n = 2$: C is the constant YM3_I:(2206), and $c_9\ell^3 L^{3u}$ is the factor of $e^{-\mathsf{d}_\infty L^u/M}$ in the left member of the second condition.

Let $K \ge K_\sharp(d)$ and $y = K/1800$. The second condition follows, in logarithms, from $\mathsf{d}_\infty L^u/M \ge c_{10} + (3u + \frac{K}{10})\log L$, $c_{10} = \max\{0, \log(2c_9/\mathsf{C})\}$, and this from the two inequalities $L^{y/2} \ge c_{10} + 3\log L + 200y \log L$, which defines $K_\flat = \lceil 1800 y_0\rceil$ through the least y_0 from which on it holds, and $L^{y/2} \ge \sqrt{3}M/d$, which is $K \ge 3600 \log_L(\sqrt{3}M/d)$; the first condition follows from the same product. So $K_1(\mathsf{d}_\infty) \le K_\sharp(d)$, and $K^{\mathsf{p}_2} L^{-K/10} \le cL^{-K/20} \le c(d/\sqrt{3}M)^{180}$, which gives the factor $C_\diamond\vartheta^{-2}d^{178}$. By the minimality of $s(d)$, $L^{s(d)}d < LC_{W,0}A_0^2(\log L^{s(d)} + \log b_+)^{2p_0}$, so that $L^{s(d)} \le d^{-2}$ for small d; and $(\sqrt{3}M/d)^{3600} \le L^{K_\sharp(d)} < L(\sqrt{3}M/d)^{3600} \le d^{-3601}$ for small d. □

Proposition 6.19 (The agreement of states)**.** Let $0 < g \le \gamma_W^{\mathrm{sd}}$, (K,β) a datum at g with $K \ge 3$ and label b, $\vartheta \in (0,1)$, and $0 < d \le \frac{1}{180}$ with $a_K \le 9d$. A *state at g at (K,β)* is the periodically extended torus measure $\tilde\mu_{K,\ell',\beta}$ of a side $\ell' \in \mathfrak{L}(g)$, or a periodic Gibbs state at (a_K,β); $S_2^T[\tilde\mu_{K,\ell',\beta}](\tilde f,\tilde h)$ is $S^T_{2;K,\ell',\beta}$ at the periodisations of $\tilde f, \tilde h$ on $T_{\ell'}$. Put $\mathsf{d}_\star(g,\vartheta) = 4\sqrt{3}M\vartheta^{-1}(K_{\mathrm{L}}^\vee/K_{\mathrm{Q}}^\vee)^2 g^{-2}$, with the pair $K_{\mathrm{L}}^\vee, K_{\mathrm{Q}}^\vee$ of equation YM3_I:(2166). There are $\mathsf{Q}_2^{\mathrm{T}}(g) \ge 1$ and $C_2^{\mathrm{T}}(g) < \infty$, depending on g, on the fixed radius ϱ and the common sourced choice of Section YM3_I:82 specified at the beginning of this section, and on no K, β, side, state, d, ϑ, pair or position, with the following property.

Let $(\tilde f, \tilde h)$ be real Lipschitz functions on $\mathbb{R}^3$, neither 0, each vanishing off a closed ball of radius d, with disjoint sets $\{\tilde f \ne 0\}$, $\{\tilde h \ne 0\}$, both vanishing off one ball of radius $9d$, and with $\operatorname{Lip}\tilde f \le \|\tilde f\|_\infty/(\vartheta d)$, $\operatorname{Lip}\tilde h \le \|\tilde h\|_\infty/(\vartheta d)$; no sign and no lower bound of the integrals is asked. Then for every state $\mathfrak{s}$ at g at (K,β)

$$|S_2^T[\mathfrak{s}] - S_2^T[\tilde\mu_{K,\ell_g,\beta}]|(\tilde f,\tilde h) \le (30d)^3 \cdot 2\,\mathsf{Q}_2^{\mathrm{T}}(g)^2\,\mathsf{A}_\star(d)^{-2}\eta,$$

with a number $\mathsf{A}_\star(d)$ of the order $(b\vartheta d)^{1/2}$ formed with the datum; and for $d \le \mathsf{d}_\star(g,\vartheta)$, in particular for the pairs of $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$,

$$|S_2^T[\mathfrak{s}] - S_2^T[\tilde\mu_{K,\ell_g,\beta}]|(\tilde f,\tilde h) \le C_2^{\mathrm{T}}(g)\vartheta^{-1}d^2 b^{-2}\eta. \tag{6.ix}$$

Any two states at g at (K,β) differ at such pairs by at most twice this. ■

Proof sketch. Let $\ell' = \ell_g = 2L^m$, $\mu' = \tilde{\mu}_{K,\ell',\beta}$ and $\mathsf{l} = \ell'/2 = L^m$. Let x_1 be the centre of the ball of radius $9d$ off which $\tilde{f}$ and $\tilde{h}$ vanish, and let $o \in a_K\mathbb{Z}^3$ be a site with $|o_\mu - (x_1)_\mu| \le \frac{1}{2}a_K$ for $\mu = 1,2,3$. Consider the board of the least torus: the eight cells $\square_{\mathbf{k}} = o + \mathsf{l}\mathbf{k} + [-\frac{\mathsf{l}}{2}, \frac{\mathsf{l}}{2}]^3$, $\mathbf{k} \in \{0,1\}^3$, with the reflections $x_\mu \mapsto 2o_\mu + \mathsf{l} - x_\mu$ in the planes $x_\mu = o_\mu \pm \frac{\mathsf{l}}{2}$, which carry one cell into another. As $o \in a_K\mathbb{Z}^3$ and l is a multiple of a_K, these reflections are maps of the lattice, and they carry plaquettes to plaquettes.

The source families used below are those of Prop. 4.42, at the common sourced choice fixed in this section. Its numerical source conditions are the ones retained in §4.2. The radius condition for Prop. 4.43 holds at the present coupling: $g \le \gamma_W^{\mathrm{sd}} \le \gamma_W^{\mathrm{ref}}$ and equation YM3_I:(2326) give $\varepsilon(x_t) \le c_A\alpha_0/(8MB_3)$ at every level. Also $\mathsf{l} \ge L^2M$ by the least-side definition, so both entries in the minimum defining $d_*(t) + 1$ in eq. 4.xiii are at least 4. The calibrated rarity hypotheses are those of the same real history, as in Prop. 4.54. Thus Cor. 4.45 applies without a further restriction on this datum.

As $1 + \frac{1}{2}a_K < \frac{1}{2}\mathsf{l}$, the ball $\mathbb{B}(x_1, 1)$ lies in the interior of $\square_0$; the signed sums of (3) below vanish off it. Its images under the reflections are eight balls of radius 1, one in each cell, with the centres $o_\mu + \mathsf{l}k_\mu + (-1)^{k_\mu}(x_1 - o)_\mu$ ($\mu = 1,2,3$); any two of these centres differ, in a coordinate μ in which their cells differ, by $\pm(\mathsf{l} - 2(x_1 - o)_\mu)$ modulo $2\mathsf{l}$, so that they lie at a sup distance at least $\mathsf{l} - 2|(x_1 - o)_\mu| \ge \mathsf{l} - a_K$ on $T_{\ell'}$.

(1)

One cell against the least torus. For a bounded complex Borel function F of the inner bonds of one cell, the reflected copies, conjugated on odd reflection parity, satisfy

$$\|F\|_{\mathrm{ch}}^8 = \mu'\Big(\prod_{\mathbf{k}\in\{0,1\}^3} F^{[\mathbf{k}]}\Big) \ge 0.$$

For $\ell \in \mathfrak{L}(g)$, [7, Cor. B.1] gives

$$|\tilde{\mu}_{K,\ell,\beta}(F)| \le \|F\|_{\mathrm{ch},\ell} \le \|F\|_{\mathrm{ch}} \exp[\mathsf{l}^3(p_{\ell'} - p_\ell)].$$

[7, Lem. N.1] bounds $p_{\ell'} - p_\ell$ by $C_{\mathrm{gap}} + C_{\mathrm{S}} + C_g \le 2C_{\mathrm{S}} + C_g$. Thus one may take $\mathsf{Q}_0 = \exp[\mathsf{l}^3(2C_{\mathrm{S}} + C_g)] \ge 1$. For a periodic Gibbs state, Prop. YM3_I:91.9 gives

$$|\mathfrak{s}(F)| \le \liminf_{r\to\infty} \|F\|_{\mathrm{ch},\ell_g L^r} \le \mathsf{Q}_0\|F\|_{\mathrm{ch}}.$$

The constants are those of [7, Lem. N.1], with $C_{\mathrm{S}} \ge C_{\mathrm{gap}}$ chosen there. In particular Q_0 is independent of the cutoff and the larger side.

This is the one place where Thms. YM3_I:AJ and YM3_I:AD enter, and the place through which $\mathsf{Q}_2^{\mathrm{T}}(g)$ and $C_2^{\mathrm{T}}(g)$ depend on the radius and on the choice of the longer list.

(2)

Only linking cumulants are left. With $X_a = \mathcal{O}(F_a)$, $\mathcal{P}_a = X_a - \mu'(X_a)$ and $\mathcal{P}_{12} = \mathcal{P}_1\mathcal{P}_2 - S_2^T[\mu'](F_1, F_2)$ one has $S_2^T[\mathfrak{s}](F_1,F_2) - S_2^T[\mu'](F_1,F_2) = \mathfrak{s}(\mathcal{P}_{12}) - \mathfrak{s}(\mathcal{P}_1)\mathfrak{s}(\mathcal{P}_2)$. Expanding $\mu'(\prod_{\mathbf{k}} \mathcal{P}_{12}^{[\mathbf{k}]})$ in joint cumulants of the sixteen copies $X_{a,\mathbf{k}}$, the blocks that stay in one cell cancel exactly, because $\mathcal{P}_{12}$ is centred by the covariance of μ' itself; what is left is a sum, over the at most Bl_{16} partitions into linking blocks (blocks that meet two cells), of products of linking cumulants, Bl_n being the Bell numbers.

So if every linking cumulant of the copies is at most $C_{\mathrm{LK}}\prod u_a$, then $|S_2^T[\mathfrak{s}] - S_2^T[\mu']|(F_1,F_2) \le [\mathsf{Q}_0\mathsf{Bl}_{16}^{1/8} + \mathsf{Q}_0^2\mathsf{Bl}_8^{1/4}]C_{\mathrm{LK}}u_1u_2 \le 2\mathsf{Q}_1^2u_1u_2$, $\mathsf{Q}_1 = \mathsf{Q}_0\mathsf{Bl}_{16}^{1/8}C_{\mathrm{LK}}$.

(3)

The packing. Apply (2) to the signed sums $F_a^\epsilon = \sum_v \epsilon_v\tau_v f_a$, $f_1 = \tilde{f}$, $f_2 = \tilde{h}$, over a packing of translates $\mathbb{B}(x_1 + v, 9d)$, $v \in a_K\mathbb{Z}^3$, in the ball $\mathbb{B}(x_1, 1)$, with at least $(30d)^{-3}$ members; the mean over the signs and the invariance of every state under the lattice translations give $|S_2^T[\mathfrak{s}] - S_2^T[\mu']|(\tilde{f},\tilde{h}) \le (30d)^3 \cdot 2\mathsf{Q}_1^2u_1u_2$.

(4)

The linking cumulants under μ'. Their generating function is $Z[W_z]/Z$ on $T_{\ell'}$, with the source $W_z = \sum_{\mathbf{k},a} z_{\mathbf{k},a} F^{\epsilon}_{a,\mathbf{k}} \circ x$ read on the torus, $F^{\epsilon}_{a,\mathbf{k}}$ the copy in the cell $\mathbf{k}$. By the property of the level-wise class recalled in the proof of Prop. 6.6, with $b^{\pm}_K = x_K \pm \bar{B}$ in the place of $b^{\vee}$, $b^{\wedge}$, the source lies in the class with norm at most $\max |z_{\mathbf{k},a}| u_a$, $u_a = \|f_a\|_\infty / \mathsf{A}_\star(d)$. Write $\log(Z[W_z]/Z) = \log \Omega_8(z) + [\mathcal{E}(W_z) - \sum_{\mathbf{k}} \mathcal{E}(w^{(\mathbf{k})})] + \sum_{\mathbf{k}} \mathcal{E}(w^{(\mathbf{k})})$, $\Omega_8 = Z[W_z] e^{-\mathcal{E}(W_z)}/Z$, $w^{(\mathbf{k})}$ the part of W_z in the cell $\mathbf{k}$.

A linking derivative kills the last sum. Apply Cor. 4.45 to the eight reflected balls and their sixteen variables. Its hypotheses on the bumps hold because reflections preserve their amplitude, Lipschitz bounds and disjoint nonzero sets. The same amplitude calculation in Lem. 4.44, now with $b^{-}_K = x_K - \bar{B}$ and $b^{+}_K = x_K + \bar{B}$, applies: these bound $a_{K-j} x_j$ below and above. Thus the square-free derivatives of Ω_8 at zero are at most $\mathsf{Q}_{\mathrm{D}} \prod u_a$, with $\mathsf{Q}_{\mathrm{D}} = Q_{8,16}(g)$.

We estimate the bracket in the generating function by using the local energies of Prop. 4.42:

$$e_j(X; w) = \sum_{z \in X \cap \mathbb{T}^{(j)}} \Delta F_j[w](X, \mathbf{1}, z).$$

They depend only on the source plaquettes in X and vanish at zero. The counting calculation in [7, eq. 213], applied to eq. 4.ix, gives

$$|e_j(X; w)| \le 16 M^3 E_{\mathrm{s}} \|w\|_{\mathrm{lw}} e^{-\kappa d_j(X)}.$$

Here E_{s} is the fixed family ceiling of §4.2. If X meets source plaquettes in at most one reflected ball, its contribution to $\mathcal{E}(W_z) - \sum_{\mathbf{k}} \mathcal{E}(w^{(\mathbf{k})})$ is zero: locality identifies the one possible nonzero term, and all other terms vanish. Only domains meeting two balls remain.

The centres of the reflected balls are at sup distance at least $\mathsf{l} - a_K$. Their source barycentres are therefore at distance at least $\mathsf{l} - a_K - 2$, and a corner is within $a_K/2$ of its barycentre. Thus a domain that contributes contains two points at distance at least $\mathsf{l} - 3$, since $K \ge 3$ and $L \ge 13$ imply $2a_K \le 1$. Lem. 5.24 gives

$$d_j(X) \ge \frac{\mathsf{l} - 3}{M D_j} - 2 \ge 12 L^{K-j} - 2, \qquad D_j = L^{j-K}.$$

For a wrapped domain the bound follows instead from $M(d_j(X) + 2) \ge \ell'/(2D_j) = \mathsf{l}/D_j$, which is stronger. In the last inequality we used $\mathsf{l} \ge L^2 M$ and $M \ge 5$. The least-side formula YM3_I:(1528) gives this side bound: its maximum includes $L^{\lceil \log_L C_{\mathrm{s}} \rceil} \ge L$. In particular the displayed lower bound is positive also at $j = K$.

Put $r = \max_{\mathbf{k},a} |z_{\mathbf{k},a}| u_a < 1$. The full source and each cell source have level-wise norm at most r, so the contribution of one domain is at most $9 \cdot 16 M^3 E_{\mathrm{s}} r e^{-\kappa d_j(X)}$. A ball of radius 1 meets source plaquettes whose corners lie in a cube of side $2 + a_K \le 3$. There are at most $(3L^{K-j}/M + 2)^3 \le 27 L^{3(K-j)}$ level-j cubes containing these corners. The eight balls therefore require at most $216 L^{3(K-j)}$ such cubes. Apply [7, Lem. B.10] with $t = j$, $q = 0$ and $D = 12 L^{K-j} - 2$. Its rate condition holds because $\delta_{\mathrm{w}} \kappa > \kappa_{\mathcal{D}} + 2$ and $0 < \delta_{\mathrm{w}} < 1/2$ in equation YM3_I:(877). The extra factor $\#(X \cap \mathbb{T}^{(j)})$ in that lemma is at least one. Summing first in X and then in j gives

$$\left| \mathcal{E}(W_z) - \sum_{\mathbf{k}} \mathcal{E}(w^{(\mathbf{k})}) \right| \le \mathsf{C}_\Delta r, \qquad \mathsf{C}_\Delta = 248832 M^6 E_{\mathrm{s}} C_{\mathrm{E},1} \sum_{n \ge 0} L^{3n} e^{2 - 12 L^n}.$$

The series converges, since the ratio of successive terms is $L^3 e^{-12(L-1)L^n} \to 0$. Cauchy's formula on the source polydisc bounds each nonempty square-free derivative of this bracket by $\mathsf{C}_\Delta \prod u_a$.

So $C_{\mathrm{LK}} = \mathsf{cu}_{16} \mathsf{Q}_{\mathrm{D}}^{16} + \mathsf{C}_\Delta$, with a combinatorial number cu_{16}, and $\mathsf{Q}_2^{\mathrm{T}}(g) := \mathsf{Q}_1$ gives the first display.

(5) For $d \le \mathsf{d}_\star(g, \vartheta)$ the second member of the minimum that defines $\mathsf{A}_\star(d)$ is the smaller one, so that $\mathsf{A}_\star(d)^{-2} = (K_{\mathrm{L}}^{\vee}/\mu_\star)^2 (b^{+}_K/(b^{-}_K)^2)\, 4\sqrt{3} M/(\vartheta d)$, and $b^{+}_K/(b^{-}_K)^2 \le b^{-2} (X_g + \bar{B})^3/(g^{-2} - \bar{B})^2$. With

$30^3 \cdot 2 = 54000$ this is eq. 6.ix with $C_2^{\mathrm{T}}(g) = 54000[\mathsf{Q}_2^{\mathrm{T}}(g)(K_{\mathrm{L}}^\vee/\mu_\star)(X_g+\bar{B})^{3/2}(g^{-2}-\bar{B})^{-1}(4\sqrt{3}M)^{1/2}]^2$. The last assertion is the triangle inequality through μ'. Of this proposition the proof of clause (c) uses eq. 6.ix alone, for one torus measure against the least torus. □

Proposition 6.20 (One datum on two sides)**.** Let $0 < g \le \gamma_{\mathrm{s}}$, (K, β) a datum at g with $K \ge 3$ and label b, $8a_K \le d \le L^{-2}$, $\vartheta \in (0,1)$, $s \ge 0$, $(f,h) \in \mathfrak{T}_{T_{\ell_g}}(d;\vartheta)$, and $(\breve{f},\breve{h})$ the pair on $T_{\ell_g L^s}$ with the same lift; the datum, with its one run, is taken on both tori. Put $\dagger = \log(10LMb/d)$.

(i) For every level $1 \le j \le K$, $\mathsf{n}_j[\ell_g](f,h) - \mathsf{n}_j[\ell_g L^s](\breve{f},\breve{h}) = \mathsf{R}_j[\ell_g] - \mathsf{R}_j[\ell_g L^s]$, where $\mathsf{R}_j[\ell]$ is the part of the sum that defines n_j on T_ℓ over the domains X that meet both supports and are not the image of a domain of the infinite lattice lying in a cuboid of sides $< \frac{1}{2}\ell_g$; each such domain has linear size $\ge \ell_g/(2MD_j) - 2$.

(ii) On every side of $\mathfrak{L}(g)$, at every level j with $D_j > Ld/8$,

$$|\mathsf{n}_j(f,h)| \le C'_{\mathsf{n}}\, \dagger^{3p_0} \Big(\frac{d}{D_j}\Big)^3 [b^{-3/2} D_j^{-1/2} + b^{-1} d^3 D_j^{-4}]\eta.$$

(iii) $|\mathsf{J}_2[\ell_g](f,h) - \mathsf{J}_2[\ell_g L^s](\breve{f},\breve{h})| \le C_{\mathrm{V}}\, \dagger^{3p_0}\, b^{-1} d^3\, \eta$.

C'_{n} and C_{V} are determined by the constants; the bounds depend on no s, K, ϑ or pair beyond η, and no bound on L^K and no witness depth is asked. ■

Proof sketch. If a source vanishes, all the terms vanish. (i) A domain of the level j of the infinite lattice that lies in a cuboid of sides $< \frac{1}{2}\ell_g$ and meets the lifted supports projects onto exactly one domain of each of the two lattices (ℓ_g, K) and $(\ell_g L^s, K)$, with their levels; these do not wrap, and it is a near lift of each. Every domain of either lattice that meets both supports and is not of this kind either wraps the torus or lies in no such cuboid, and in both cases its linear size is $\ge \ell_g/(2MD_j) - 2$. These are the following facts on the localisation domains (Lem. 5.24). The projection of a domain of the infinite lattice is a domain of the lattice (ℓ, K); a domain of the lattice (ℓ, K) does not wrap if and only if it is the projection of a domain of the infinite lattice that lies in a cuboid of lengths less than half the side; such domains are its near lifts, and they are translates of one another by periods and pairwise disjoint; the projection keeps the linear size $d_j(X)$, the number of edges of a least edge tree of X in the units of the level j (Section YM3_I:72 (the definitions of $\mathcal{D}_j$ and d_j at its beginning)); every domain of the infinite lattice lies in a cuboid of lengths $\le MD_j(d_j(X)+2)$; and on T_ℓ a domain with $M(d_j(X)+2) < \ell/(2D_j)$ does not wrap.

A cuboid of lengths $< \frac{1}{2}\ell_g$ has lengths less than half the side of either lattice. On a near lift the two pairs of lattice sources agree: the lifted supports lie in one ball of radius $6d$, and their translates by non-zero periods are more than $\frac{1}{2}\ell_g$ away. On a domain that does not wrap the local terms of the weighted progression are then the same number on the two tori. This is proved by induction over the levels: the regular progression without sources is one family for all the lattices (ℓ, K) of the level (Thm. YM3_I:Y, including its common-companion assertion); one step of the progression is natural under the chart of the domain from one lattice to the other, the two lattices having one K: bare arrays of the two lattices that correspond under the chart (the common-input condition at the beginning of Section YM3_I:52.5) give local terms of the next level that are defined together and are equal (equation YM3_I:(928), including its separately equal three summands); and the bare arrays of the two lattices correspond under this chart, their members of the lower levels by the induction over the levels and their weight members because the sources agree on the plaquettes of the domain.

So the terms of these domains cancel in the difference, which is (i).

(ii) The first jets of the coarse levels gain the factor $(d/D_j)^3$. Here $\dot{E}_a^{(j)}$ and $\ddot{E}^{(j)}$ are the derivatives $\partial_{\mu a} E_\mu^{(j)}$ and $\partial_{\mu_1}\partial_{\mu_2} E_\mu^{(j)}$ at $\mu = 0$ of the local terms of the weighted progression, and $\|\cdot\|$ is the size YM3_I:(759) of a family of the level j at the rate $\kappa^+ = (1+4\beta_{\mathrm{s}})\kappa$, the rate at which the part of the local terms that depends on the sources decays in the linear size of the domain (Prop. YM3_I:122.1);

κ is the decay rate and $\beta_{\mathrm{s}} \in (0, \frac{1}{5}]$ the enlargement fraction among the constants ([7, Def. B.7]), and $\|\cdot\|_{j,\kappa} \le \|\cdot\|$ is the size at the rate κ.

The members of $\dot{E}_a^{(i)}$ vanish on the domains that hold no plaquette of $\operatorname{supp} w_a$, and those of $\ddot{E}^{(i)}$ on the domains that do not hold a plaquette of each support (Prop. YM3_I:122.1(c)). The cubes of the level i are the cubes of side MD_i of the partition π_i of the torus whose wall-connected unions are the domains of the level i (the level-i localization partition of Section YM3_I:45); for a set $\mathcal{S}$ of plaquettes, $\mathsf{N}_i(\mathcal{S})$ is the number of the cubes of the level i in which a corner of a plaquette of $\mathcal{S}$ lies.

The level j_1. Let r be the least sup distance between a corner of a plaquette of $\operatorname{supp} w_1$ and a corner of a plaquette of $\operatorname{supp} w_2$. The barycentres of the plaquettes of $\operatorname{supp} w_1$ and of $\operatorname{supp} w_2$ lie in $\operatorname{supp} f$ and $\operatorname{supp} h$, which lie in balls of radius d with centres at most $8d$ apart and are at a Euclidean distance at least d, and a corner lies at the sup distance $\frac{1}{2}a_K \le \frac{1}{16}d$ from the barycentre; so $\frac{2}{5}d \le \mathsf{r} < 11d$. Let j_1 be the least level $j \ge 1$ with $\mathsf{r} < 3MD_j$. As $3MD_{K-1} = 3M/L > 11d$, it exists and $j_1 \le K-1$; and $\mathsf{r} \ge 3MD_j$ at every level $1 \le j < j_1$.

As a consequence of this definition, D_{j_1} lies in the interval $(\frac{2d}{15M}, \frac{Ld}{8}]$: $D_{j_1} > \mathsf{r}/3M \ge \frac{2d}{15M}$; and $D_{j_1} \le \frac{Ld}{8}$, since $D_{j_1} = LD_{j_1-1} \le L\mathsf{r}/3M < \frac{11Ld}{3M} \le \frac{Ld}{8}$ if $j_1 \ge 2$, and $D_{j_1} = La_K \le \frac{Ld}{8}$ if $j_1 = 1$. The level j_1 is formed on each of the two tori; the bounds below are the same on both. All the corners of the plaquettes of $\operatorname{supp} w_a$ lie in a cube of side $\frac{17}{8}d$, so that $\mathsf{N}_i(\operatorname{supp} w_a) \le (\frac{17d}{8MD_i} + 2)^3$; as $d/MD_{j_1} < \frac{15}{2}$, this is less than 18^3 for $i \ge j_1$, and less than $18^3 L^{3(j_1-i)}$ for $i \le j_1$.

So the bare array of the first derivative in the direction w_a sits at few places at the levels from j_1 on. For such an array the size of the family that one step forms is bounded with the weights $L^{-4(j-i)}$ times the number of the places, where the general size has $L^{3(j-i)}$ places times $L^{-4(j-i)}$: if the counterterms of a bare array are 0, and its members $G^{(i)}$, $i \le j$, and its weight member $H^{(j)}$ vanish on the domains that hold no plaquette of a set $\mathcal{S}$ of plaquettes, then the family that the rule of the step $j+1$ forms of it has a size at most

$$c_\sharp \Big[\sum_{i \le j} L^{-4(j-i)}\, \mathsf{N}_i(\mathcal{S})\, \|G^{(i)}\|_{i,\kappa} + \mathsf{N}_j(\mathcal{S})\, \|H^{(j)}\|_{j,\kappa} \Big],$$

$H^{(j)}$ being the weight member and $c_\sharp$ a number determined by the constants. The proof is that of the bound of the bracket localisation of a bare array (the proof of Prop. YM3_I:51.16, especially the rooted-domain count and equation YM3_I:(881)), with the count of the places changed: as the counterterms vanish, the members enter the rule as they stand; a member of the level i enters the rule of the height j either on a domain whose linear size is at least $\max\{2L^{j-i-1} - 2, 0\}$, which costs $e^{-\max\{2L^{j-i-1}-2,0\}} \le L^{4-4(j-i)}$, or through its octet function, which lies within $\mathsf{C}_{\mathrm{oc}} \|G^{(i)}\|_{i,\kappa} (d_i(X)+2)^2 L^{4-4(j-i)}$ of a constant (Prop. YM3_I:51.14, at fixed radii), where C_{oc} is the larger of 1 and its constant C_{oc}; the weight member enters at the level j alone; and the sum over the domains that hold a plaquette of $\mathcal{S}$ is bounded by [7, Lem. B.10], which gives the factor $\mathsf{N}_i(\mathcal{S})$ in the place of the $L^{3(j-i)}$ places of the general bound.

Two bare arrays are of this kind. The bare array of the first derivative in the direction w_a at the height j has the members $\dot{E}_a^{(i)}$, $i \le j$, the counterterms 0, since the counterterms do not depend on the sources (Prop. YM3_I:122.1(e)), and as weight member that of the source $L^{K-j} w_a$, whose members sit at the domains $X(p)$ of the plaquettes $p \in \operatorname{supp} w_a$, each of which holds its plaquette; it is of this kind with $\mathcal{S} = \operatorname{supp} w_a$. The bare array of the second derivative has the members $\ddot{E}^{(i)}$, $i \le j$, the counterterms 0 and the weight member 0; it is of this kind with $\mathcal{S} = \operatorname{supp} w_1$.

The step depends on its bare array through the size of this family: by the Lipschitz bound and the holomorphy of the step in the family that the rule forms (Thm. YM3_I:V, including equation YM3_I:(899)), the first derivative of the step at a bare array of the progression in the direction of an array is at most θ_j times the size of the family formed of that array, and the mixed second derivative in the directions of two arrays at most $2\theta_j/N_\star^{[j]}$ times the product of the two sizes,

by Cauchy's estimate; here $\theta_j = \theta^{[j]} = K_{\mathbf{M}}(1+\beta_{\mathrm{s}})\delta(x_j)/\alpha_0$ and $N_\star^{[j]}$ are the numbers of [7, eq. 196], $K_{\mathbf{M}} \geq 1$ is the step constant and α_0 the first fixed radius in [7, Def. B.7].

Every running coupling of the datum lies above the threshold x_{J} of the definitions preceding [7, eq. 228] (Lem. YM3_I:126.2, as in the proof of Prop. 6.4), and the conditions [7, eq. 228] that define x_{J} hold at every $x \geq x_{\mathrm{J}}$, as δ decreases there; so $C_0^\flat \mathsf{S}_{\mathrm{A}}\theta_j \leq 1$ and $\frac{4}{3}C_1^\flat L\mathcal{T}(3)\theta_j \leq 1$ at every height $j < K$, with the numbers $C_0^\flat$, $C_1^\flat$ of [7, §B.5], S_{A} of [7, eq. 198] and the function $\mathcal{T}(u) = \sum_{n\geq 1}(n+1)^{2p_0+1}e^{-4\beta_{\mathrm{s}}\kappa[L^{n-1}u-2]_+}$, written T in [7, eq. 199], all determined by the constants.

The first jets. Fix a; let $\dot{\mathsf{B}}_j = \sum_{i\leq j} L^{-4(j-i)}\mathsf{N}_i(\operatorname{supp} w_a)\|\dot{E}_a^{(i)}\|$ be the weighted sum of the sizes of the first jets of the levels up to j, and v_j the size of the weight member of the level j. For $j_1 \leq j \leq K-1$ the bound just stated, with $\mathsf{N}_j(\operatorname{supp} w_a) < 18^3$, gives the family formed of the bare array of the first derivative at the height j a size at most $c_\sharp[\dot{\mathsf{B}}_j + 18^3\mathsf{v}_j]$; the first derivative of the step gives $\|\dot{E}_a^{(j+1)}\|$ at most θ_j times this size; and $\dot{\mathsf{B}}_{j+1} = L^{-4}\dot{\mathsf{B}}_j + \mathsf{N}_{j+1}(\operatorname{supp} w_a)\|\dot{E}_a^{(j+1)}\|$.

The power $d^3D_j^{-4}$ comes from two inputs. The first is the inhomogeneous term: the weight member of the level j is the action term $\eta_j^{-1}\sum_p L^{K-j}w_a(p)\,a_p$, $\eta_j = L^{-j}$ being the bare spacing in the units of the level j; as $\operatorname{supp} w_a$ has at most $3(\frac{17}{8})^3(d/a_K)^3 < 29(d/a_K)^3$ plaquettes, as $|a_p| < \frac{1}{2}\alpha_0^2\eta_j^4$ at the fields of the radii (α_0, α_1), and as the domains $X(p)$ have the linear size 0, its size is $\mathsf{v}_j \leq \frac{29}{2}\alpha_0^2 d^3 D_j^{-4}|w_a|_\infty$. The second is the start: by the general bound of the first jets, $\|\dot{E}_a^{(i)}\| \leq C_0^\flat\mu_{i-1,a}$ with $\mu_{i,a} = 12M^3\alpha_0^2\theta_i|w_a|_\infty/D_i$ (Prop. YM3_I:122.1, equation YM3_I:(2258), [7, eq. 198]), by $\mu_{i-1,a} \leq (\frac{4}{3})^{1/2}L^{n/2}(1+n)^{p_0}\mu_{j_1-1,a}$ for $n = j_1 - i \geq 0$ (the running-coupling comparison following equation YM3_I:(2258)) and by $\mathsf{N}_i(\operatorname{supp} w_a) < 18^3L^{3n}$, the powers of L leave the convergent series $\sum_n(1+n)^{p_0}L^{-n/2}$, and $\dot{\mathsf{B}}_{j_1} \leq C\mu_{j_1-1,a} \leq C'\alpha_0^2|w_a|_\infty/D_{j_1} \leq C'(\frac{L}{8})^3\alpha_0^2|w_a|_\infty d^3D_{j_1}^{-4}$, as $\theta_{j_1-1} \leq (C_0^\flat\mathsf{S}_{\mathrm{A}})^{-1}$ and $D_{j_1} \leq \frac{Ld}{8}$.

With $\hat{\mathsf{v}} = \frac{29}{2}\alpha_0^2 d^3 D_{j_1}^{-4}|w_a|_\infty$, so that $L^{4(j-j_1)}\mathsf{v}_j \leq \hat{\mathsf{v}}$ for $j \geq j_1$, the three relations give

$$L^{4(j+1-j_1)}\dot{\mathsf{B}}_{j+1} + 18^3\hat{\mathsf{v}} \leq (1 + L^4\cdot 18^3 c_\sharp\theta_j)\,(L^{4(j-j_1)}\dot{\mathsf{B}}_j + 18^3\hat{\mathsf{v}}) \qquad (j_1 \leq j \leq K-2).$$

So it is the quantity $L^{4(j-j_1)}\dot{\mathsf{B}}_j + 18^3\hat{\mathsf{v}}$, and not the weighted sum itself, that grows from the level j_1 on by at most the factor $\prod_i(1 + L^4\cdot 18^3 c_\sharp\theta_i) \leq \exp[L^4\cdot 18^3 c_\sharp\sum_i\theta_i]$; and $\sum_{i<K}\theta_i \leq \mathsf{S}_{\mathrm{B}}\theta_{K-1} \leq \mathsf{S}_{\mathrm{B}}/(C_0^\flat\mathsf{S}_{\mathrm{A}})$ (the running-coupling comparison following equation YM3_I:(2258), with the number S_{B} of [7, eq. 198]). Hence $\dot{\mathsf{B}}_j + 18^3\mathsf{v}_j \leq L^{-4(j-j_1)}(L^{4(j-j_1)}\dot{\mathsf{B}}_j + 18^3\hat{\mathsf{v}}) \leq Cd^3D_{j_1}^{-4}L^{-4(j-j_1)}\alpha_0^2|w_a|_\infty = Cd^3D_j^{-4}\alpha_0^2|w_a|_\infty$; the family formed of the bare array of the first derivative at the height j has a size at most $C_{\mathrm{F}}\alpha_0^2|w_a|_\infty d^3D_j^{-4}$, and the recursion of the first jets closes as $\|\dot{E}_a^{(j+1)}\| \leq \theta_j C_{\mathrm{F}}\alpha_0^2|w_a|_\infty d^3D_j^{-4}$ for $j_1 \leq j \leq K-1$, with a number C_{F} determined by the constants.

The second jets. At the levels up to j_1 the general bound holds: $\|\ddot{E}^{(i)}\| \leq C_1^\flat\varpi_{i-1}$ for $1 \leq i \leq j_1$, where $\varpi_i = c_\nu\delta(x_i)^2|w_1|_\infty|w_2|_\infty D_i^{-2}$ is the number written ν_i in [7, eq. 199] and c_ν is determined by the constants. This is the induction of Prop. YM3_I:122.2, carried up to the level j_1: its condition $\frac{4}{3}C_1^\flat L\theta_j\mathcal{T}(\mathsf{r}/MD_j) \leq 1$ holds at the levels $1 \leq j < j_1$, as $\mathsf{r}/MD_j \geq 3$ there and $\mathcal{T}$ decreases (the monotonicity in [7, eq. 199]). Let $\mathsf{B}_j = \sum_{i\leq j} L^{-4(j-i)}\mathsf{N}_i(\operatorname{supp} w_1)\|\ddot{E}^{(i)}\|_{i,\kappa}$ collect the second jets of the levels up to j, at the rate κ.

By the two derivatives of the step, at the arrays of the first and of the second derivative, the second jets obey $\|\ddot{E}^{(j+1)}\| \leq \nu_j^\sharp + \theta_j c_\sharp\mathsf{B}_j$, where $\nu_j^\sharp \leq C\delta(x_j)^2 d^6 D_j^{-8}|w_1|_\infty|w_2|_\infty$ for $j \geq j_1$ comes from the product of the two first jets; and $\mathsf{B}_{j+1} = L^{-4}\mathsf{B}_j + \mathsf{N}_{j+1}(\operatorname{supp} w_1)\|\ddot{E}^{(j+1)}\|_{j+1,\kappa}$. The recursion has the same form as that of the first jets. The start: at a level $i < j_1$ a domain that holds plaquettes of both supports has a linear size at least $\mathsf{r}/MD_i - 2$, so that $\|\ddot{E}^{(i)}\|_{i,\kappa} \leq e^{-4\beta_{\mathrm{s}}\kappa[\mathsf{r}/MD_i-2]_+}\|\ddot{E}^{(i)}\|$ (equation YM3_I:(2259)), with $\mathsf{r}/MD_i \geq 3L^{j_1-1-i}$; with $\varpi_{i-1} \leq \frac{4}{3}L^n(1+n)^{2p_0}\varpi_{j_1-1}$ and $\mathsf{N}_i(\operatorname{supp} w_1) < 18^3L^{3n}$, $n = j_1 - i$, the powers of L cancel and the series $\sum_{n\geq 1}(1+$

$n)^{2p_0} e^{-4\beta_s\kappa[3L^{n-1}-2]_+}$ converges, so that $\mathsf{B}_{j_1} \le C\varpi_{j_1-1} \le C'(\log x_{j_1-1})^{2p_0}|w_1|_\infty|w_2|_\infty/(bD_{j_1})$, by $x_{j_1-1} \ge \frac{3}{4}b/D_{j_1-1}$.

The quantity $L^{4(j-j_1)}\mathsf{B}_j$ grows from one level to the next by at most the factor $1 + L^4 \cdot 18^3 c_\sharp \theta_j$, beside the inhomogeneous terms $L^{4(j+1-j_1)}18^3\nu_j^\sharp$, whose sum over $j \ge j_1$ is at most

$$C(\log x_{j_1-1})^{2p_0} b^{-1}|w_1|_\infty|w_2|_\infty d^6 D_{j_1}^{-7} \le C'(\log x_{j_1-1})^{2p_0}|w_1|_\infty|w_2|_\infty/(bD_{j_1}),$$

as $\delta(x_j)^2 \le \frac{4}{3}A_1^2(\log x_j)^{2p_0}D_j/b$, $\sum_{j\ge j_1} D_j^{-3} \le D_{j_1}^{-3}/(1-L^{-3})$ and $d/D_{j_1} < \frac{15}{2}M$. So, as for the first jets,

$$\mathsf{B}_j \le C(\log x_{j_1-1})^{2p_0} b^{-1}|w_1|_\infty|w_2|_\infty d^3 D_j^{-4}$$

for $j \ge j_1$, by $D_{j_1} \le \frac{Ld}{8}$; and $\|\ddot E^{(j+1)}\| \le \nu_j^\sharp + \theta_j c_\sharp \mathsf{B}_j$, with $\theta_j \le C(\log x_j)^{p_0}(D_j/b)^{1/2}$, gives

$$\|\ddot E^{(j)}\| \le C\dagger^{3p_0}[b^{-3/2}d^3D_j^{-7/2} + b^{-1}d^6D_j^{-7}]|w_1|_\infty|w_2|_\infty$$

for $j > j_1$, as $\log x_j \le \log x_{j_1-1} < \dagger$ for $j \ge j_1 - 1$, by $x_{j_1-1} \le \frac{12}{11}b/D_{j_1-1}$ (proof of Prop. 6.4(ii)) and $D_{j_1-1} = D_{j_1}/L > \frac{2d}{15LM}$.

A level with $D_j > \frac{Ld}{8} \ge D_{j_1}$ lies above j_1, and the count of the domains that meet a support ([7, Lem. B.10], at most 18^3 cubes of the level j holding a corner of a plaquette of $\operatorname{supp} w_1$) gives (ii).

(iii) By (i) the difference is at most $\sum_j(|\mathsf{R}_j[\ell_g]| + |\mathsf{R}_j[\ell_g L^s]|)$. On each torus the local second jets decay like $e^{-\kappa^+ d_j(X)}$ in the linear size $d_j(X)$, with the rate $\kappa^+ = (1+4\beta_s)\kappa$ of (ii) (Prop. YM3_I: 122.1); and the count of the domains of linear size at least R that meet a support, $\sum_X \#(X \cap \mathbb{T}^{(j)})\, e^{-\kappa^+ d_j(X)} \le c\,\nu\, e^{-R}$ over the domains X with $d_j(X) \ge R$ that meet the ν cubes of the level j in which a corner of a plaquette of the support lies ([7, Lem. B.10], as κ^+ exceeds the number $\kappa_{\mathcal D} + 2$ that it asks), taken at $R = A/D_j - 2$, gives $|\mathsf{R}_j| \le C\|\ddot E^{(j)}\| e^{2-A/D_j}$, $A = \ell_g/(2M) \ge L$.

For the levels above j_1, (ii) and $\sum_{n\ge 0} L^{7n}e^{-AL^n} \le \frac{8}{5}e^{-A}$ give $C\dagger^{3p_0}b^{-1}d^3\eta$. For the levels $j \le j_1$ the second jets grow at most like $L^n(1+n)^{2p_0}$, $n = j_1 - j$, by their general bound $C_1^\flat\varpi_{j-1} \le \frac{4}{3}C_1^\flat L^n(1+n)^{2p_0}\varpi_{j_1-1}$ up to the level j_1 in the proof of (ii), against $e^{-A/D_j} \le e^{-8/d}e^{-8L^2(L^n-1)}$ and $e^{-8/d} \le \frac{3}{512}d^4$; this gives again at most $C\dagger^{2p_0}b^{-1}d^3\eta$. Double for the two tori. □

Lemma 6.21 (Two sides at the order six)**.** Let $0 < g \le \gamma_s$, (K,β) a datum at g, $\ell \le \breve\ell$ in $\mathfrak L(g)$, $d \le L^{-1}$, $s(d) < K$, $20d \le \ell$, $(f,h) \in \mathfrak T_{T_\ell}(d;\vartheta)$, and $(\breve f, \breve h)$ the pair on $T_{\breve\ell}$ with the same lift. Then

$$|\mathsf{h}_6[\ell](f,h) - \mathsf{h}_6[\breve\ell](\breve f,\breve h)| \le |\mathsf{J}_2[\ell](f,h) - \mathsf{J}_2[\breve\ell](\breve f,\breve h)| + \Big[(2C_{\mathrm E}b^{-1}d\,l^{8p_0} + 2\bar\varepsilon_4(g,d))b^{-3}d + 3C_{\mathrm{tm}}\vartheta^{-1}b^{-2}\frac{a_K}{d}\Big]\eta.$$

■

Proof sketch. On each side, $\mathsf{h}_6 = \mathsf{J}_2 - \mathsf{h}_4 - [\mathsf{J}_2 - \mathsf{h}_4 - \mathsf{h}_6]$. The bracket is at most $C_{\mathrm E}b^{-1}dl^{8p_0}\cdot b^{-3}d\,\eta$ on each side (Prop. 6.5(ii)), and h_4 is $c_2b^{-2}\mathcal K_6$ up to $[\bar\varepsilon_4 b^{-3}d + C_{\mathrm{lat}}b^{-2}a_K/d]\eta$ on each side (Prop. 6.7, with one $\bar\varepsilon_4$ for all sides and data at g); $\mathcal K_6$ and η are the same for the two pairs. Subtract. □

Proposition 6.22 (Two lattices with one bare torus)**.** Let $0 < g \le \gamma_s$; $D_0 = (K_0,\beta_0)$ a datum at g with label b; $d \le L^{-1}$, $s(d) < K_0$; $(f,h) \in \mathfrak T_{T_{\ell_g}}(d;\vartheta)$, $81d \le \ell_g$, $r = dL^{K_0}$; $b_* \in I(g)$; and integers $K' \ge K_0$ and $2 \le n_0 \le n_1$ with $n_0 \le K_0$, $n_1 \le K'$, $L^{n_0} \ge r$. Put $\breve\ell = \ell_g L^{K'-K_0}$, $N' = \ell_g L^{K'} = \breve\ell L^{K_0}$, $D' = (K', L^{K'}b_*)$, $l' = \max\{1, \log(2b_*L^{K'}/r)\}$, $\mathsf N' = N' + \max\{\beta_0, L^{K'}b_*\}$, and let $(\breve f,\breve h)$ be the pair on $T_{\breve\ell}$ with the same lift. Let the two conditions of Prop. 6.11(ii) hold at the three places $(D_0,\breve\ell;n_0)$, $(D',\ell_g;n_0)$ and $(D',\ell_g;n_1)$, and let $x_\flat$ denote the running coupling x_{n-1} of the place; and let $\sigma \ge 0$ be a number with $|\mathsf{J}_2[\ell_g](f,h) - \mathsf{J}_2[\breve\ell](\breve f,\breve h)| \le \sigma b^{-3}d\eta$ at D_0. Then, with $\tilde w$ the lattice sources of

the lift on $\mathbb{Z}^3$,

$$|\mathsf{H}^{(K_0)}[D_0,\ell_g]-\mathsf{T}^{\mathrm{lat}}(\tilde w)|\le\Big\{[\sigma+2C_{\mathrm E}b^{-1}dl^{8p_0}+2\bar\varepsilon_4(g,d)+C_{\mathrm R}(l^{6p_0}+l'^{6p_0})(r/L^{n_0})^{3/2}]r \\ +\frac{3C_{\mathrm{tm}}b}{\vartheta d}+\sum_{\text{three places}}\mathcal{U}(x_\flat,\mathsf{N}')+2700C_\star(\ell_gL^{K'-n_1})^{19}n_1L^{-n_1/2}r^6\Big\}\eta.$$

■

Proof sketch. If $\tilde w_1$ or $\tilde w_2$ vanishes, both numbers are 0. Consider three places: (1) the datum D_0 on T_{ℓ_g} with (f,h); (2) the datum D_0 on $T_{\breve\ell}$ with $(\breve f,\breve h)$, where $\breve\ell\in\mathfrak{L}(g)$; (3) the datum D' on T_{ℓ_g} with $f'=\breve f\circ\mathsf{D}^{-1}$, $h'=\breve h\circ\mathsf{D}^{-1}$, $\mathsf{D}:T_{\breve\ell}\to T_{\ell_g}$ the dilation $x\mapsto L^{-(K'-K_0)}x$. D' is a datum at g, as $b_*\in I(g)$; f',h' are real and bounded, with the norms of f,h, with supports in balls of radius $d'=rL^{-K'}$ at a distance in $[d',6d']$, and $8a_{K'}\le d'$ as $r\ge8$; so Prop. 6.4 applies at the three places, as it asks no depth.

D carries the bare lattice of $(\breve\ell,K_0)$ onto that of (ℓ_g,K'), plaquettes to plaquettes and barycentres to barycentres: in bare units the sources of (2) and of (3) are one pair of functions on the plaquettes of $\Lambda_{N'}$, the periodisation of $\tilde w$, and r is the same at the three places. The chain has seven links. 1. $\mathsf{H}^{(K_0)}[D_0,\ell_g]\to\mathsf{H}^{(K_0)}[D_0,\breve\ell]$: Lem. 6.21 with σ, multiplied by $\beta_0^3a_{K_0}^2=b^3/a_{K_0}$. 2. $\to\mathsf{H}^{(n_0)}[D_0,\breve\ell]$: Prop. 6.5(iii) at (2). 3. $\to\mathsf{T}[N',n_0]$: Prop. 6.11(i), (ii), with Lem. 6.12, and Prop. 6.14 at (2), as for the link X of Step 3 below. 4.

$\to\mathsf{H}^{(n_0)}[D',\ell_g]$: the same at (3); the number $\mathsf{T}[N',n_0]$ is one, as it depends on N', n_0, L and the lattice sources alone, and the two runs that meet in it leave no trace, by Prop. 6.13(iii). 5. $\to\mathsf{H}^{(n_1)}[D',\ell_g]$: Prop. 6.5(iii) at (3), where l is l' and $d'/D'_{n_0}=r/L^{n_0}$, $D'_{n_0}=a_{K'-n_0}$ being the spacing of the level n_0 of the lattice (ℓ_g,K'). 6. $\to\mathsf{T}[N',n_1]$: Prop. 6.11(i), (ii), with Lem. 6.12, and Prop. 6.14 at (3). 7. $\to\mathsf{T}^{\mathrm{lat}}(\tilde w)$: Prop. 6.17 on the lattice with $\ell'=N'/L^{n_1}=\ell_gL^{K'-n_1}$ top sites and n_1 levels, where $\ell'\ge\ell_g\ge N_{\mathsf H}$, $\ell'L\ge C_LM$, and all base points lie within $10r+1$ of each other, $10r+1\le\frac18N'$ as $81d\le\ell_g$ and $r\ge8$; at most $900r^6$ pairs.

In link 1, $b^3a_{K_0}^{-1}\cdot b^{-3}d=r$ and $b^3a_{K_0}^{-1}\cdot3C_{\mathrm{tm}}\vartheta^{-1}b^{-2}a_{K_0}/d=3C_{\mathrm{tm}}b/(\vartheta d)$. Add the errors of the seven links. □

Lemma 6.23 (The splines of Def. 6.2)**.** Let $q,q'\ge1$ be odd.

(i) $N_3(t)=\frac34-t^2$ for $|t|\le\frac12$, $N_3(t)=\frac12(\frac32-|t|)^2$ for $\frac12\le|t|\le\frac32$, and $N_3(t)=0$ beyond; Φ is of class C^1, $0\le\Phi\le(\frac34)^3$ and $\{\Phi\ne0\}=(-\frac32,\frac32)^3$, so $|x|<2.6$ on $\operatorname{supp}\Phi$; $\int\Phi=1$, $\int\Phi x_\mu=0$ and $\int\Phi|x|^2=\frac34$.

(ii) $\Phi(x/q)=\sum_j c_j^q\Phi(x-j)$, with $c_j^q\ge0$ and $c_j^q=0$ unless $|j_\mu|\le\frac32(q-1)$ for all μ; $\sum_j c_j^q=q^3$, $\sum_j c_j^qj=0$, $\sum_j c_j^q|j|^2=\frac34q^3(q^2-1)$; hence $q^{-1}\sum_j c_j^qP(\frac{y+j}{q})=P(y)$ for all y.

(iii) For $q\ge3$, $\Upsilon^{(q)}$ is of class C^1 with support in $\{7\le|y|\le11q\}$, and the distance of the supports of Φ and $\Upsilon^{(q)}$ exceeds 4.4; $0\le\Upsilon^{(q)}\le64q^2$, and $\Upsilon^{(q)}=63.25$ on $|y|=8$.

(iv) $e(u)=|u|^2$ for $|u|\le4.4$, $e(u)=0$ for $|u|\ge10.6$, $0\le e(u)\le|u|^2$ otherwise, and $\int\Phi(x)\Upsilon^{(q)}(x+u)\,\mathrm dx=q^2e(u/q)-e(u)$.

(v) $\mathsf k_6\in[\frac{4\pi}{10.6},\frac{4\pi}{4.4}]$, and for $q\ge3$, on $\mathbb{R}^3$: $\mathcal{K}_5(\Phi,\Upsilon^{(q)})=4\pi\log q$ and $\mathcal{K}_6(\Phi,\Upsilon^{(q)})=(1-q^{-1})\mathsf k_6$.

(vi) $\Upsilon^{(qq')}(y)=\Upsilon^{(q)}(y)+q^{-1}\sum_j c_j^q\Upsilon^{(q')}(\frac{y+j}{q})$.

■

Proof sketch. (i) N_3 is the density of the sum of three independent variables uniform on $[-\frac12,\frac12)$: integral 1, mean 0, variance $\frac14$. (ii) The intervals $[i-\frac12,i+\frac12)$, $i\in J_q$, tile $[-\frac q2,\frac q2)$, so $\mathbf1_{[-1/2,1/2)}(\cdot/q)=\sum_{i\in J_q}\mathbf1_{[-1/2,1/2)}(\cdot-i)$, and the threefold convolution gives $q^2N_3(x/q)=\sum_{i_1,i_2,i_3\in J_q}N_3(x-i_1-i_2-i_3)$; the moments of the c_j^q are those of the sum of three variables uniform on J_q, and the identity for P

follows by expanding the square. (iii) If $|y| \le 4.4q + 2.6$ and $c_j^q \neq 0$, then $|y + j| \le 7q$, so all the $\Xi(\frac{y+j}{q})$ are $P(\frac{y+j}{q})$, and by (ii) $\Upsilon^{(q)}(y) = P(y)(1 - \zeta(|y|))$, which is 0 for $|y| \le 7$ and ≥ 0 beyond.

If $|y| \ge 4.4q + 2.6$, then $\Xi(y) = 0$ and $|y + j|/q > 1.8$, where $\Xi \ge 0$. If $|y| > 10.6q - 2.6$ all the terms vanish. The upper bound follows from $\Xi \le 64 - \frac{3}{4}$. (iv) For $|u| \le 4.4$, $|x + u| < 7$ on $\operatorname{supp}\Phi$, and (i) gives $e(u) = \frac{3}{4} + |u|^2 - \frac{3}{4}$; the last formula is (ii) after the change of variables $x \mapsto x - j$. (v) With $\varepsilon(t) = t^{-2}\int_{S^2} e(t\omega)\,\mathrm{d}\omega$, which is 4π on $(0, 4.4]$ and 0 on $[10.6, \infty)$, polar coordinates and (iv) give $\mathcal{K}_p(\Phi, \Upsilon^{(q)}) = \int_0^\infty t^{4-p}[\varepsilon(t/q) - \varepsilon(t)]\,\mathrm{d}t$. At $p = 5$ the integral over $t \ge \delta$ is $\int_{\delta/q}^{\delta} \varepsilon(s)\frac{ds}{s} = 4\pi \log q$ for small δ; at $p = 6$ the substitution $t = qs$ gives $(1 - q^{-1})\mathsf{k}_6$.

(vi) $(i, i') \mapsto i + qi'$ is a bijection of $J_q \times J_{q'}$ onto $J_{qq'}$, so $c_J^{qq'} = \sum_{j+qj'=J} c_j^q c_{j'}^{q'}$. □

6.1.9. *Proof of Thm. S. Step 1: the front and the expansion.* We take $\gamma_{\mathrm{e}} = \min\{\gamma_{\mathrm{s}}, \gamma_{\mathrm{C}}\}$. The number d_{e} is the least of L^{-13}, $\frac{1}{200}$, $d_{\mathrm{A}}(g)$, d_{K} of Prop. 6.18 and of the thresholds of Steps 2, 4 and 8: the smallness of d needed for $2 \le n_0 < n_1$ and $12r + 2 \le N_0/8$, for the two conditions of Prop. 6.11(ii) at the places of the chain, and for the three inequalities on d stated at the head of Step 8. Let $D = (K, \beta)$ be a datum at g with $s(d) < K$ and $L^K \le d^{-P}$, and $(f, h) \in \mathfrak{T}_{T_\ell}(d; \vartheta)$; then $K \ge 3$, $8a_K \le d$ and $r \ge L^{s(d)}d \ge C_W(s(d)) \ge (\log \frac{1}{d})^2 \ge 8$. If w_1 or w_2 vanishes, all the jets vanish and only the last link of Step 3 is used. Props. 6.6, 6.5(ii) and 6.7, the last two at every translate $(\tau_v f, \tau_v h)$, $v \in \mathsf{Pk}$, give, as L_b^+ depends on distances alone,

$$|S_2^T - L_b^+|(f, h) \le \Big[\sum_{i=1}^{3} E_i\, b^{-3}d + C_{\mathrm{lat}} b^{-2}\frac{a_K}{d}\Big]\eta + b^{-3}a_K \max_{v\in\mathsf{Pk}} |\mathsf{H}^{(K)}[D] - \varkappa\mathcal{K}_5/a_K|(\tau_v f, \tau_v h), \quad (6.\mathrm{x})$$

with $E_1 = C_{\mathrm{A}}(g)\vartheta^{-1}bd$, $E_2 = C_{\mathrm{E}}b^{-1}dl^{8p_0}$ and $E_3 = \bar{\varepsilon}_4(g, d)$; here $\mathsf{h}_6 = b^{-3}a_K\mathsf{H}^{(K)}[D]$, and $b^{-3}a_K \cdot r\eta = b^{-3}d\eta$. It remains to bound $|\mathsf{H}^{(K)}[D] - \varkappa\mathcal{K}_5/a_K|$ in multiples of $r\eta$ at every position; we write (f, h) for the translate.

Step 2: the levels and a second datum. Put

$$K' = \max\{K, \lceil 24 \log_L r\rceil\}, \quad N_0 = \ell L^K, \quad N' = \ell L^{K'}, \quad n_1 = K' - \lceil K'/78\rceil,$$

so that $r^{24} \le L^{K'} \le \max\{d^{-P}, Lr^{24}\} \le Ld^{-24P}$. If $K' = K$ (Case A) let $D' = D$. If $K' > K$ (Case B), then $r^{24} > L^K = r/d$, that is $r > d^{-1/23}$; fix $b_* \in I(g)$ once, let $D' = (K', L^{K'}b_*)$, a datum at g (Cor. YM3_I:74.2), and let n_0 be the level with $D_{n_0} = a_{K-n_0} \in [d^{1/2}, Ld^{1/2})$. Then $L^{n_0} \in [rd^{-1/2}, Lrd^{-1/2})$; $2 \le n_0 \le K$ for $d \le L^{-2}$, as $L^{n_0} \ge 8L$ and $D_{n_0} < 1$; and $n_0 < n_1$ for small d, as $L^{n_0} < Lrd^{-1/2} < Lr^{12.5}$ and $L^{n_1} \ge L^{77K'/78-1} \ge r^{23.6}/L$. In bare units the base points of the plaquettes of the two supports lie within $12r + 2$ of each other, and $12r + 2 \le N_0/8 = \ell r/(8d)$ for small d.

The lifts $\breve{w}_a$ of Prop. 6.16 on $\Lambda_{N'}$ are the lattice sources, at D', of real bounded functions on T_ℓ with supports in balls of radius $d' = ra_{K'}$ at a distance $\ge d'$, $8a_{K'} \le d'$; Prop. 6.4 applies to them at D', as it puts no condition on the depth, the signs or the Lipschitz constants. The unit $r\eta$ is the same at D and at D', since $d'L^{K'} = r$.

Step 3: the chain. In Case B,

$$\begin{aligned}\mathsf{H}^{(K)}[D] &\xrightarrow{\mathrm{R}} \mathsf{H}^{(n_0)}[D] \xrightarrow{\mathrm{X}} \mathsf{T}[N_0, n_0](w) \xrightarrow{\mathrm{W}} \mathsf{T}[N', n_0](\breve{w}) \xrightarrow{\mathrm{X}} \mathsf{H}^{(n_0)}[D'] \\ &\xrightarrow{\mathrm{R}} \mathsf{H}^{(n_1)}[D'] \xrightarrow{\mathrm{X}} \mathsf{T}[N', n_1](\breve{w}) \xrightarrow{\mathrm{A}} a_K^{-1}\mathsf{P}_1^{\mathrm{lat}}(\tilde{f}, \tilde{h}; a_K) \xrightarrow{\mathrm{T}} \varkappa\mathcal{K}_5(f, h)/a_K. \end{aligned} \quad (6.\mathrm{xi})$$

Here R is Prop. 6.5(iii). X is Prop. 6.11(i), (ii), with Lem. 6.12, and Prop. 6.14: $\mathsf{H}^{(n)}[D] = \beta^3 a_K^2 \mathsf{h}_6^{(n)} = \beta^3 a_K^2 \gamma_2^{\mathrm{r}}$ by Prop. 6.11(i), the cumulants being taken in $\mathsf{D}^0 dB = \bigotimes_{0\le i<n} \nu_i^{\mathrm{r}}$; $\beta^3 a_K^2 \gamma_2^\circ = \mathsf{T}[N, n](w)$ by Prop. 6.14; and $\beta^3 a_K^2 |\gamma_2^{\mathrm{r}} - \gamma_2^\circ| \le \mathcal{U}(x_{n-1}, \mathsf{N})\eta$ by Prop. 6.11(ii) under the two conditions there, its proof reading the bounds of Lem. 6.12 on the boxes and the precisions $\mathsf{M}_i \ge \frac{3}{4}\gamma^{\mathrm{tr}}$ of the Gaussian measures of the heights.

So $|\mathsf{H}^{(n)}[D] - \mathsf{T}[N,n](w)| \le \mathcal{U}(x_{n-1},\mathsf{N})\eta$ under the two conditions; the two middle members $\mathsf{T}[N',n_0](\breve{w})$ are one number, as it depends on no datum. W is Prop. 6.16 at $t=n_0$, $\nu = \ell/D_{n_0} > \ell d^{-1/2}/L \ge \nu_{\mathrm{W}}$, with the base points within $12r+2 \le N_0/8$. A is Prop. 6.17 on the lattice with $\ell' = N'/L^{n_1} = \ell L^{\lceil K'/78\rceil}$ top sites and n_1 levels: $\ell' \ge \ell \ge N_{\mathsf{H}}$ and $\ell' L \ge C_L M$ by Prop. 6.4(i), the base points lie within $12r+2 \le \frac{1}{8}N' - 1$, and the sum runs over at most $900r^6$ pairs; with $\frac{19}{78} + \frac{1}{156} - \frac{1}{2} = -\frac{1}{4}$ and $r^5 \le L^{5K'/24}$ the error is E_{10} below, and $a_K \mathsf{T}^{\mathrm{lat}}(\tilde{w}) = \mathsf{P}_1^{\mathrm{lat}}(\tilde{f},\tilde{h};a_K)$ for the lift $(\tilde{f},\tilde{h}) \in \mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$.

T is Cor. 7.37 for the lift: $|\mathsf{P}_1^{\mathrm{lat}}(\tilde{f},\tilde{h};a) - \kappa^{\mathrm{lat}}\mathcal{K}_5(f,h)| \le C_{\mathrm{T}}\vartheta^{-1} a \lg(d/a)\eta$ for $0 < a \le d$, with $\kappa^{\mathrm{lat}} = \varkappa = c_1c_2$ (Thms. W, X); only here enters $c_1 = \kappa^{\mathrm{lat}}/c_2$, which depends on no L, g or label. The errors of the links, in multiples of $r\eta$, are

$$E_4 = C_{\mathrm{R}} l^{6p_0} d^{3/4}, \qquad E_5 = \mathcal{U}(x_{n_0-1},\mathsf{N}_0)/r, \qquad E_6 = 900 r^5 C'_{\mathrm{W}} N'^{237} e^{-\delta_{\mathrm{tp}}\ell d^{-1/2}/48},$$
$$E_7 = \mathcal{U}(x'_{n_0-1},\mathsf{N}')/r, \qquad E_8 = C_{\mathrm{R}} l'^{6p_0} d^{3/4}, \qquad E_9 = \mathcal{U}(x'_{n_1-1},\mathsf{N}')/r,$$
$$E_{10} = 2700 C_\star \ell^{19} L^{20} K' L^{-K'/24}, \qquad E_{11} = C_{\mathrm{T}}\vartheta^{-1}\lg(r)/r, \tag{6.xii}$$

where x, x' are the running couplings of D, D'; $\mathsf{N}_0 = N_0 + \beta \le (\ell + b_+)d^{-P}$ and $\mathsf{N}' = N' + L^{K'}b_* \le (\ell+b_+)Lr^{24}$; by Prop. 6.4(ii), $x_{n_0-1} \ge \frac{3}{4}bL/D_{n_0} \ge \frac{3}{4}d^{-1/2}$, $x'_{n_0-1} \ge \frac{3}{4}r^{23}d^{1/2} \ge \frac{3}{4}\max\{r^{11.5}, d^{-1/2}\}$ and $x'_{n_1-1} \ge L^{K'/78}$; $l' \le \log(2b_+) + K'\log L$; and $(r/L^{n_0})^{3/2} \le d^{3/4}$. In Case A the chain is $\mathsf{H}^{(K)}[D] \to \mathsf{H}^{(n_1)}[D]$ by R, with the error $C_{\mathrm{R}} l^{6p_0} L^{\frac{3}{2}(1-0.94K)}$, as $r/L^{n_1} \le L^{1-0.94K}$ and $l \le \log(2b_+) + K\log L$, followed by the last three links with $D' = D$.

Step 4: clause (a). At each place of X the two conditions of Prop. 6.11(ii) hold once $\frac{1}{8}\gamma^{\mathrm{tr}}A_1^2(\log x_\flat)^{2p_0} \ge 10 + \log(10240\mathsf{N}^3)$, $x_\flat$ being the lower bound of the running coupling at that place in Step 3; and $\log\mathsf{N} \le C_1 + C_2\log x_\flat$ with $C_1 = \log((\ell+b_+)L) + P$ and $C_2 = 2P$ at the first place ($\log\frac{1}{d} \le 2\log x_\flat + 0.6$), $C_2 = 3$ at the second ($\log x_\flat \ge 11\log r$ for $r \ge 2$), $C_2 = 78$ at the third ($\log x_\flat \ge \frac{K'}{78}\log L$). As $2p_0 \ge 2$,

$$\mathcal{U}(x_\flat,\mathsf{N}) \le C_{\mathrm{X}} \exp\bigl[36C_1 + 36C_2\log x_\flat - \tfrac{1}{8}\gamma^{\mathrm{tr}}A_1^2(\log x_\flat)^{2p_0}\bigr]$$

tends to 0 faster than every power of $x_\flat^{-1}$, and $x_\flat \to \infty$ uniformly as $d \to 0$: $x_\flat \ge \frac{3}{4}d^{-1/2}$ at the first two places and $x_\flat \ge L^{s(d)/78}$ at the third. Further $E_6 \le 900 C'_{\mathrm{W}}(\ell L)^{237} d^{-5P-5688P} e^{-\delta_{\mathrm{tp}}\ell d^{-1/2}/48}$; $E_4, E_8 \le C_{\mathrm{R}}[\log(2b_+L) + 24P\log\frac{1}{d}]^{6p_0} d^{3/4}$; E_{10} and the error of Case A tend to 0 as $K' \ge K > s(d) \to \infty$; and $E_{11} \le C_{\mathrm{T}}\vartheta^{-1}\sup_{\rho \ge (\log 1/d)^2}\lg(\rho)/\rho$. As $b \in [g^{-2}, b_+]$, each E_i is bounded by a function of (g,ϑ,P,d) alone that tends to 0 with d. With $\bar{\varepsilon}_{\mathrm{a}}(d)$ the supremum over $d'' \le d$ of the sum of these bounds, eq. 6.x gives (a).

Step 5: clause (b) and the rate. Let $b \in I(g)$; every $(K, L^K b)$ is a datum at g, and $I(g) \subset [\gamma^{-2}, L^2(g^{-2} + \bar{B})]$. For $s(d) < K \le K_\sharp(d)$, (a) applies, as $L^K \le d^{-3601} \le d^{-P}$ (Prop. 6.18). For $K > K_\sharp = K_\sharp(d)$, and for the limit, Prop. 6.18 and (a) at $K_\sharp$ give, with $a_{K_\sharp} \le (d/\sqrt{3}M)^{3600}$,

$$|S^T_{2;K,\ell,L^K b} - L^+_b|(f,h) \le [\bar{\varepsilon}_{\mathrm{a}}(d) + C_\diamond\vartheta^{-2}b_+^3 d^{177} + C_{\mathrm{lat}}b_+ d^{3598}]b^{-3}d\,\eta,$$

so that $\bar{\varepsilon}(d) = \bar{\varepsilon}_{\mathrm{a}}(d) + \sup_{d''\le d}[C_\diamond\vartheta^{-2}b_+^3 d''^{177} + C_{\mathrm{lat}}b_+ d''^{3598}]$ serves in (a) and (b). For the limit we apply (a) at the single cutoff $K_\sharp$, where $r \ge d^{-3599}$ and Case B holds: E_5, E_6, E_7, E_9 are below every power of d; $E_{10}, E_{11} \le C d^{3598}\log\frac{1}{d}$; $E_1 + E_2 \le Cd(\log\frac{1}{d})^{8p_0}$; $E_4 + E_8 \le Cd^{3/4}(\log\frac{1}{d})^{6p_0}$; and $E_3 \le C_4(g)d^{1/2}$ (Prop. 6.7). So the error of the limit is at most $c_{1/2}d^{1/2}b^{-3}d\,\eta$; its slowest member is that of the order four.

Step 6: clause (c). Let $g_{\mathrm{a}} = L^{1/4}g$ and $g_{\mathrm{b}} = L^{-1/4}g$. For $g'' \in \{g, g_{\mathrm{a}}, g_{\mathrm{b}}\}$, $g'' \le \min\{\gamma_{\mathrm{e}}, \gamma_W^{\mathrm{sd}}\}$, and $\mathsf{x} := g^{-2} \ge L^{1/2}(\gamma_W^{\mathrm{sd}})^{-2} \ge 4L^{1/2}\bar{B} \ge 5\bar{B}$. All these reference couplings use the single assignment fixed at the beginning of the section. The real constants are its unchanged restriction; hence the canonical flow and every sourced construction are evaluated at the same values. This is the

common-choice convention of Section YM3_I:82, with its radius conditions imposed uniformly for $x \geq \gamma^{-2}$.

So there is one flow of the coupling: at every coupling the running couplings of a datum, which depend on (K,β) alone, the data, the histories of the sourced chain and the torus measures are the same objects in the real restriction of the common assignment and in its sourced construction, as fixed in Section YM3_I:82; and $C_{W,0}$ and P are fixed before the couplings. Hence (a) and (b), which hold at this choice, hold at the three couplings g'', and these clauses and Prop. 6.19 speak of the same objects. The labels at g lie in $[\mathsf{x}, L(\mathsf{x}+\bar B)]$, and $I(g)$ leaves out $[\mathsf{x}, \mathsf{x}+\bar B]$ and $(L(\mathsf{x}+B_\sharp)-\bar B, L(\mathsf{x}+\bar B)]$.

As $\mathsf{x} \geq 5\bar B$ and $L \geq 13$,

$$L^{-1/2}\mathsf{x}+\bar B < \mathsf{x}, \quad \mathsf{x}+\bar B \leq L^{1/2}\mathsf{x}+LB_\sharp-\bar B, \quad L^{1/2}\mathsf{x}+\bar B \leq L(\mathsf{x}+B_\sharp)-\bar B, \quad L(\mathsf{x}+\bar B) \leq L(L^{1/2}\mathsf{x}+B_\sharp)-\bar B,$$

which place the first interval in $I(g_{\mathrm{a}}) = (L^{-1/2}\mathsf{x}+\bar B, L^{1/2}\mathsf{x}+LB_\sharp-\bar B]$ and the second in $I(g_{\mathrm{b}}) = (L^{1/2}\mathsf{x}+\bar B, L^{3/2}\mathsf{x}+LB_\sharp-\bar B]$. Let d'_{e} be the least, over the three couplings g'', of $\frac{1}{180}$, L^{-2}, $d_{\mathrm{e}}(g'')$, $\mathsf{d}_\star(g'',\vartheta)$ and $(LC_w(g''))^{-2}$, where $C_w(g'') = \sup_{s\geq 0} C_W(s)L^{-s/2}$ is formed at g''; for $d \leq (LC_w(g''))^{-2}$ the witness depth at g'' has $L^{s(d)} \leq d^{-3}$, since $L^{s-1}d < C_W(s-1) \leq C_w L^{(s-1)/2}$ for $s = s(d) \geq 1$. Let (K,β) be a datum at g with $K > s(d)$. If $L^K \leq d^{-P}$, (a) applies; if $b \in I(g)$, (b) applies.

Otherwise $b \in I(g_\bullet)$ with $g_\bullet \in \{g_{\mathrm{a}}, g_{\mathrm{b}}\}$, and $(K,\beta) = (K, L^K b)$ is a datum at $g_\bullet$ (Cor. YM3_I:74.2), with $K \geq 3$, $a_K \leq d$ and $K > s(d)$ at $g_\bullet$, since $L^K > d^{-P} \geq d^{-3}$. Clause (b) at $g_\bullet$ gives the law on $T_{\ell_{g_\bullet}}$ for the pair with the same lift as (f,h), which has the same η, $\mathcal{K}_5$ and $\mathcal{K}_6$. Of the sides ℓ_g, $\ell_{g_\bullet}$, both of the form $2L^m$, let $g_\circ \in \{g, g_\bullet\}$ be the coupling of the smaller; the larger lies in $\mathfrak{L}(g_\circ)$, so both torus measures are states at $g_\circ$ at the datum (K,β), one of them on the least torus, and eq. 6.ix at $g_\circ$ bounds their difference at the lift by

$$C_2^{\mathrm{T}}(g_\circ)\vartheta^{-1}d^2b^{-2}\eta \leq C_2^{\mathrm{T}}(g_\circ)\vartheta^{-1}b_+d\cdot b^{-3}d\,\eta.$$

So (c) holds with $\bar\varepsilon'(d) = \max_{g''}\bar\varepsilon_{g''}(d) + \max_{g''} C_2^{\mathrm{T}}(g'')\vartheta^{-1}b_+d$, $\bar\varepsilon_{g''}$ being the function of (a) and (b) at g''.

Step 7: clause (d). (1) By Lem. 6.23(ii), $\Phi(x/q) = \sum_j c_j^q \Phi(x-j)$; so every bilinear form S invariant under the translations by $\rho\mathbb{Z}^3$ satisfies

$$(q\rho)^{-1}\mathsf{S}(\Phi_{q\rho}, \Xi_{q\rho}) - \rho^{-1}\mathsf{S}(\Phi_\rho, \Xi_\rho) = \rho^{-1}\mathsf{S}(\Phi_\rho, \Upsilon_\rho^{(q)}).$$

At a cutoff $K \geq k_0 + n$ the covariance $S^T_{2;K,\ell,L^K b}$ is bilinear and invariant under the translations by $a_K\mathbb{Z}^3 \supset \varrho_k\mathbb{Z}^3$, $\varrho_k = L^{-k}r_0$, whatever the supports (Setting and notation). Taken at $\rho = \varrho_k$, $q = L$, summed over $1 \leq k \leq n$, and compared with the identity at $\rho = r_1$, $q = L^n$, it gives

$$r_1^{-1}S(\Phi_{r_1}, \Upsilon_{r_1}^{(L^n)}) = \sum_{k=1}^n \varrho_k^{-1} S(\Phi_{\varrho_k}, \Upsilon_{\varrho_k}^{(L)});$$

these pairs have disjoint supports (Lem. 6.23(iii)), which lie in balls of radius $11r_0 < \ell/4$, and $K \to \infty$ is taken member by member, by the existence and the bilinearity of the limit. (2) By Lem. 6.23(v), $\mathcal{K}_5(\Phi, \Upsilon^{(q)}) = 4\pi \log q$ and $\mathcal{K}_6(\Phi, \Upsilon^{(q)}) = (1-q^{-1})\mathsf{k}_6$; with $\mathcal{K}_q(f_\rho, h_\rho) = \rho^{6-q}\mathcal{K}_q(f,h)$ and $4\pi c_1 c_2 = \frac{6}{\pi^2}$, the subtracted members of Δ_n are $\sum_{k\leq n}\varrho_k^{-1}L_b^+(\Phi_{\varrho_k}, \Upsilon_{\varrho_k}^{(L)})$, so that

$$\Delta_n = \sum_{k=1}^n \delta_k, \qquad \delta_k(b) = \varrho_k^{-1}\big[S_2^{(\ell,b)} - L_b^+\big](\Phi_{\varrho_k}, \Upsilon_{\varrho_k}^{(L)}).$$

(3) The tents $\chi_z(x) = \prod_\mu \max\{0, 1-|2x_\mu - z_\mu|\}$, $z \in \mathbb{Z}^3$, form a partition of unity and split Φ and $\Upsilon^{(L)}$, which is ≥ 0 (Lem. 6.23(iii)), into finitely many nonnegative Lipschitz pieces. Each pair

of pieces $(\chi_z\Phi, \chi_{z'}\Upsilon^{(L)})$, neither 0, lies in $\mathfrak{T}_{\mathbb{R}^3}(d_{zz'};\vartheta_0)$ with $d_{zz'} = \max\{1, \frac{1}{6}\,\mathrm{dist}\} \in [1, 2L]$, the distance of the two supports lying in $(4.4, 11L + 2.6)$, and with $\vartheta_0 \in (0, \frac{1}{2}]$ depending on L alone; scaled by ϱ_k and read on T_ℓ, it lies in $\mathfrak{T}_{T_\ell}(\varrho_k d_{zz'};\vartheta_0)$, since all the points of its two supports lie within $22L\varrho_k \le 22r_0 < \ell/2$ of each other and $20\varrho_k d_{zz'} \le 40r_0 < \ell$. Clause (b) at $(g, \vartheta_0, P_\sharp)$, with its d^0_{e} and $c^0_{1/2}$, gives for $2L\varrho_k \le d^0_{\mathrm{e}}$

$$|\delta_k(b)| \le C_\Psi c^0_{1/2}(2L\varrho_k)^{1/2} b^{-3}, \qquad C_\Psi = \sum_{z,z'} d_{zz'} \|\chi_z\Phi\|_\infty \|\chi_{z'}\Upsilon^{(L)}\|_\infty,$$

C_Ψ depending on L alone. (4) For the finitely many k with $2L\varrho_k > d^0_{\mathrm{e}}$, δ_k is continuous on the compact set $\overline{I(g)} \subset (0,\infty)$, by the continuity of the limit in b, so it is bounded there. (5) As $b^{-3} < g^6$ on $I(g)$, $\sum_k \sup_{I(g)} |\delta_k| < \infty$; so $\Delta_n \to \Delta_\infty = \sum_k \delta_k$ uniformly on $I(g)$, Δ_∞ is bounded and continuous, $\sup_{I(g)} |\Delta_n| \le \sum_k \sup_{I(g)} |\delta_k|$, and for n beyond the coarse scales

$$\sup_{I(g)} |\Delta_n - \Delta_\infty| \le C_\Psi g^6 c^0_{1/2}(2Lr_0)^{1/2} \frac{L^{-(n+1)/2}}{1 - L^{-1/2}}.$$

Of the rate in (b) the argument uses only that the tolerances at the scales $2L\varrho_k$ are summable over k; a tolerance of the order $1/\log(1/d)$ would not give it.

Step 8: clause (e). Here the levels are cut at the scale $d^{1/5}$, and the bare torus is changed through a change of the side at one datum. Let $\Lambda = \log\frac{1}{d}$, $l_+ = \log(2b_+) + \Lambda$, $l'_+ = \log(2b_+L) + 23(P_\sharp - 1)\Lambda$, $A_\ell = 2700 C_\star \ell_g^{19} L^{39/2}$,

$$\mathsf{N}_{\max}(d) = (\ell_g + b_+) L d^{-24(P_\sharp - 1)}, \qquad E_{\mathrm{X}}(d) = C_{\mathrm{X}} \mathsf{N}_{\max}(d)^{36} \exp\big[-\tfrac{1}{8}\gamma^{\mathrm{tr}} A_1^2 (\Lambda/78)^{2p_0}\big],$$

and recall that the number $d_{\mathrm{e}} \le \frac{1}{200}$ of Step 1 is so small that for $0 < d \le d_{\mathrm{e}}$: $\gamma^{\mathrm{tr}} A_1^2 (\Lambda/78)^{2p_0} \ge 80$, $10240\, \mathsf{N}_{\max}(d)^3 \exp[-\frac{1}{8}\gamma^{\mathrm{tr}} A_1^2 (\Lambda/78)^{2p_0}] \le 1$, and

$$\epsilon_{\mathrm{r}}(d) := g^2 d\big[A_\ell (1 + 24(P_\sharp - 1)\Lambda/\log L) + C_{\mathrm{T}} \vartheta^{-1}(1 + (P_\sharp - 1)\Lambda)\big] \le 1.$$

Let $0 < d \le d_{\mathrm{e}}$, $(f, h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$, and let (K, β) be a datum at g with $s(d) < K \le K_\sharp(d)$; for the first assertion, $\beta = L^K b$ with $b \in I(g)$. Write $K_0 = K$, $D_0 = (K_0, \beta)$ and $r = dL^{K_0}$; then $L^{K_0} \le d^{-P_\sharp}$ (Prop. 6.18), $K_0 \ge 3$, $8a_{K_0} \le d$ and $r \ge L\Lambda^2$ (Prop. 6.7). As in Step 1,

$$\Big|S_2^T - c_2 b^{-2}\mathcal{K}_6 - \frac{1}{\#\mathsf{Pk}} \sum_v \mathsf{h}_6(\tau_v f, \tau_v h)\Big| \le \Big[(C_{\mathrm{A}}\vartheta^{-1} b d + C_{\mathrm{E}} b^{-1} d l^{8p_0} + \bar\varepsilon_4)\, b^{-3} d + C_{\mathrm{lat}} b^{-2} \frac{a_{K_0}}{d}\Big]\eta,$$

and, at every position, by the last link T,

$$|\mathsf{h}_6 - \varkappa b^{-3}\mathcal{K}_5| \le b^{-3} a_{K_0}\, |\mathsf{H}^{(K_0)}[D_0, \ell_g] - \mathsf{T}^{\mathrm{lat}}(\tilde w)| + b^{-3} C_{\mathrm{T}} \vartheta^{-1} a_{K_0} \lg(r)\, \eta.$$

Let n_0 be the level with $D_{n_0} \in [d^{1/5}, Ld^{1/5})$; then $4 \le n_0 \le K_0 - 1$ and $r/L^{n_0} \le d^{4/5}$.

Case A, $r^{24} \le L^{K_0}$. With $e = \lceil K_0/78 \rceil$ and $n_1 = K_0 - e$: $2 \le n_1 \le K_0 - 1$, $(r/L^{n_1})^{3/2} \le d^{6/5}$, $L^e \ge d^{-1/78}$ and $r^6 \le L^{K_0/4}$. Props. 6.5(iii) from K_0 to n_1, 6.11(i), (ii) and 6.14 at n_1, and 6.17 on the lattice with $\ell_g L^e$ top sites and n_1 levels give $|\mathsf{H}^{(K_0)} - \mathsf{T}^{\mathrm{lat}}| \le \{C_{\mathrm{R}} l^{6p_0} d^{6/5} r + E_{\mathrm{X}}(d) + A_\ell K_0\}\eta$, since $19e - \frac{1}{2}n_1 < \frac{39}{2} - \frac{K_0}{4}$ gives $(\ell_g L^e)^{19} n_1 L^{-n_1/2} r^6 \le \ell_g^{19} L^{39/2} K_0$.

Case B, $r^{24} > L^{K_0}$. With $K' = \lceil 24 \log_L r \rceil$, $e = \lceil K'/78 \rceil$ and $n_1 = K' - e$: $K' > K_0$, $r^{23} > 1/d$, $r^{24} \le L^{K'} < Lr^{24} \le Ld^{-24(P_\sharp - 1)}$, $n_0 < n_1 \le K' - 1$, $L^e \ge d^{-1/78}$ and $K' \le 1 + 24(P_\sharp - 1)\Lambda/\log L$. Prop. 6.22, with $b_* = b$ if $b \in I(g)$ and a fixed $b_* \in I(g)$ otherwise, and with $\sigma = C_{\mathrm{V}} \dagger^{3p_0} b^2 d^2$ from Prop. 6.20(iii) at D_0 and $s = K' - K_0$, gives

$$|\mathsf{H}^{(K_0)} - \mathsf{T}^{\mathrm{lat}}| \le \Big\{\big[\sigma + 2C_{\mathrm{E}} b^{-1} d l^{8p_0} + 2\bar\varepsilon_4 + C_{\mathrm{R}}(l^{6p_0} + l'^{6p_0}) d^{6/5}\big] r + \frac{3C_{\mathrm{tm}} b}{\vartheta d} + 3E_{\mathrm{X}}(d) + A_\ell K'\Big\}\eta,$$

with $l' \le \log(2b_+ L r^{23}) \le l'_+$.

The places of Prop. 6.11(ii). At every one of them $\mathsf{N} \le \mathsf{N}_{\max}(d)$, and the running coupling $x_\flat = x_{n-1}$ of the place exceeds $d^{-1/78}$: $x_\flat \ge \frac{3}{4} b L^{K_0 - n_0 + 1} > d^{-1/5}$ at the level n_0, on either side, and

$x_\flat \geq \frac{3}{4} b_* L^{e+1} \geq L^e$ at the level n_1, as $\frac{3}{4} b \geq 1$. So the two conditions hold by the choice of d_e, and every $\mathcal{U}$ is at most $E_\mathrm{X}(d)$, which is smaller than every power of d, as $2p_0 \geq 2$.

Collecting. Multiply by $b^{-3} a_{K_0}$: $b^{-3} a_{K_0} r = b^{-3} d$; $b^{-3} a_{K_0} \cdot 3E_\mathrm{X} \leq \frac{3}{8} E_\mathrm{X} b^{-3} d$, as $r \geq 8$; and $b^{-3} a_{K_0} \cdot 3C_\mathrm{tm} b/(\vartheta d) = 3C_\mathrm{tm} \vartheta^{-1} b^{-2} a_{K_0}/d$. The two members that grow with r go into the lattice term: as $\lg r \leq 1 + (P_\sharp - 1)\Lambda$ and $b \geq g^{-2}$,

$$b^{-3} a_{K_0} \bigl[A_\ell K' + C_\mathrm{T} \vartheta^{-1} \lg r \bigr] = \frac{d}{b} \bigl[A_\ell K' + C_\mathrm{T} \vartheta^{-1} \lg r \bigr] \, b^{-2} \frac{a_{K_0}}{d} \leq \epsilon_\mathrm{r}(d) \, b^{-2} \frac{a_{K_0}}{d} \leq b^{-2} \frac{a_{K_0}}{d},$$

with K_0 for K' in Case A. So for $s(d) < K \leq K_\sharp(d)$

$$|S_2^T - L_b^+|(f,h) \leq \Bigl[\bar\varepsilon_\flat(d) \, b^{-3} d + (\tfrac{3}{2} + 3) C_\mathrm{tm} \vartheta^{-1} b^{-2} \frac{a_K}{d} + b^{-2} \frac{a_K}{d} \Bigr] \eta = \Bigl[\bar\varepsilon_\flat(d) \, b^{-3} d + C_\mathrm{lat}^\flat b^{-2} \frac{a_K}{d} \Bigr] \eta,$$

where $C_\mathrm{E} b^{-1} d l^{8p_0}$ and $\bar\varepsilon_4$ enter once through the front and twice through the first link of Prop. 6.22, and

$$\bar\varepsilon_\flat(d) = C_\mathrm{A} \vartheta^{-1} b_+ d + 3 C_\mathrm{E} g^2 d \, l_+^{8p_0} + 3 \bar\varepsilon_4(g,d) + C_\mathrm{V} [\log(10 L M b_+/d)]^{3p_0} b_+^2 d^2 + C_\mathrm{R} \bigl[l_+^{6p_0} + l_+'^{6p_0} \bigr] d^{6/5} + \tfrac{3}{8} E_\mathrm{X}(d).$$

This is the second assertion of (e). For $b \in I(g)$ and $K > K_\sharp(d)$, and for the limit, Prop. 6.18 at $K_\sharp(d)$ adds $\epsilon_\diamond(d) b^{-3} d\,\eta$, $\epsilon_\diamond(d) = C_\diamond \vartheta^{-2} b_+^3 d^{177} + C_\mathrm{lat}^\flat b_+ d^{3598}$, and the lattice term may be dropped, as in Step 5. Finally every member is at most a multiple of $d^{1/2}$: $d \, l_+^{8p_0} \leq C d^{1/2}$; $\bar\varepsilon_4(g,d) \leq C_4(g) d^{1/2}$; the member of the two sides is $O(d^2 \Lambda^{3p_0})$; $d^{6/5} l_+'^{6p_0} \leq C d$; E_X and $\epsilon_\diamond$ are smaller. So $\sup_{d'' \leq d} [\bar\varepsilon_\flat(d'') + \epsilon_\diamond(d'')] \leq c_\mathrm{r} d^{1/2}$, with c_r depending on g, ϑ and the constants; this is (e). The slowest member is again that of the order four.

Step 9: clause (f). Let g, g_a and g_b be as in Step 6, and let d_e'' be the least, over the three couplings g'', of the number $d_\mathrm{e}(g'')$ of Step 1 at $P = P_\sharp$, $\frac{1}{180}$, L^{-2}, $\mathsf{d}_\star(g'', \vartheta)$ and $(L C_w(g''))^{-2}$. Let (K, β) be a datum at g with $K > s(d)$ and label b. If $K \leq K_\sharp(d)$, the second assertion of (e) at g applies. If $K > K_\sharp(d)$ and $b \in I(g)$, the first assertion of (e) at g applies. Otherwise $b \in I(g_\bullet)$ with $g_\bullet \in \{g_\mathrm{a}, g_\mathrm{b}\}$ (Step 6), (K, β) is a datum at $g_\bullet$, and $K > s(d)$ at $g_\bullet$, since $L^K > L^{K_\sharp(d)} \geq (\sqrt{3} M/d)^{3600} > d^{-3}$; the first assertion of (e) at $g_\bullet$ gives the law with the rate on $T_{\ell_{g_\bullet}}$ at the pair with the same lift, and eq. 6.ix, as in Step 6, bounds the difference of the two torus measures at the lift by $C_2^\mathrm{T}(g_\circ) \vartheta^{-1} b_+ d \cdot b^{-3} d\,\eta$.

So (f) holds with $c_\mathrm{r}' = \max_{g''} c_\mathrm{r}(g'') + \max_{g''} C_2^\mathrm{T}(g'') \vartheta^{-1} b_+ (d_\mathrm{e}'')^{1/2}$.

6.1.10. *Proof of Cor. 6.3.* (i) Let $\mathfrak{s}$ be a state at g at a datum (K, β) with $K > s(d)$; as $d \leq L^{-1}$, $K \geq 3$ and $a_K < d/13 < 9d$. By eq. 6.ix,

$$|S_2^T[\mathfrak{s}] - S_2^T[\tilde\mu_{K,\ell_g,\beta}]|(\tilde f, \tilde h) \leq C_2^\mathrm{T}(g) \vartheta^{-1} d^2 b^{-2} \eta \leq C_2^\mathrm{T}(g) \vartheta^{-1} b_+ d \cdot b^{-3} d\,\eta,$$

and $S_2^T[\tilde\mu_{K,\ell_g,\beta}](\tilde f, \tilde h) = S_{2;K,\ell_g,\beta}^T(f,h)$ at the periodisation $(f,h) \in \mathfrak{T}_{T_{\ell_g}}(d;\vartheta)$, to which clauses (c) and (f) of Thm. S apply; $L_b^+(f,h) = L_b^+(\tilde f, \tilde h)$. The torus measure of a side of $\mathfrak{L}(g)$, periodically extended, and every periodic Gibbs state at (a_K, β) are states at g at the datum. (ii) For $b \in I(g)$ every $(K, L^K b)$ is a datum at g. Apply (i) to the torus measure of the side ℓ' at the lift of (f,h), and let $K \to \infty$: the lattice term tends to 0 and $S_{2;K,\ell',L^K b}^T(f,h) \to S_2^{(\ell',b)}(f,h)$. (iii) The labels at g lie in $[g^{-2}, b_+]$, and so does their limit. The witness depth depends on no cutoff, so $K_j > s(d)$ for large j; apply (i) at (K_j, β_j) and let $j \to \infty$: $a_{K_j} \to 0$ and $L_{b_j}^+ \to L_b^+$.

The labels. Let the states $\mathfrak{s}_j$ and the set of pairs be as in the second assertion of (iii), and let b, b' be two limit points of the labels, along two subsequences. Along each of them the first assertion applies, and at a pair $F = (\tilde f, \tilde h)$ of the set that lies in $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$, $d \leq d_\infty$, the limit $S(F)$ of $S_2^T[\mathfrak{s}_j](F)$ is a limit point along both; so $|L_b^+(F) - L_{b'}^+(F)| \leq \bar\varepsilon_\infty(d)(b^{-3} + b'^{-3}) d\,\eta$. On the class, $\mathcal{K}_6(F) \geq (10d)^{-6} \int \tilde f \int \tilde h \geq 10^{-6} \vartheta^2 \eta$, and $\mathcal{K}_5(F) \leq d^{-5} \int \tilde f \int \tilde h \leq (\frac{4\pi}{3})^2 d\,\eta$, as $|x - y| \geq d$ on the

supports and each support lies in a ball of radius d. Hence

$$10^{-6}\vartheta^2 c_2\,|b^{-2}-b'^{-2}| \le \Big[\bar\varepsilon_\infty(d)(b^{-3}+b'^{-3}) + (\tfrac{4\pi}{3})^2 c_1 c_2\,|b^{-3}-b'^{-3}|\Big]d,$$

and $d\to 0$ along the scales of the set gives $b=b'$. The labels lie in the compact interval $[g^{-2},b_+]$ and have one limit point, so they converge.

Limits and components. Let (K_j,β_j,ν_j) be a convergent Gibbs sequence at g and σ its limit (Def. 13.4(2)). Then $S_2^\sigma(X)=\lim_j S_2^T[\nu_j](X)$ at every separated tuple $X\in\mathfrak{S}_2(\mathbb{R}^3)$ (Thm. YM3_I:AP). A pair of $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$ whose entries are of class C^1 is a separated tuple: its entries lie in $C^1_c(\mathbb{R}^3)$ and have disjoint supports ([7, §K.1] and the test-pair convention preceding equation YM3_I:(2343)); and for every $d>0$ and $\vartheta\in(0,1)$ the class $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$ holds a pair of C^∞ functions ([7, Lem. I.1](c)).

So the second assertion of (iii), with the states ν_j and the C^1 pairs in the class, shows that the labels converge, to a number b. For a component σ_ξ of a canonical decomposition of σ, Thms. YM3_I:AL and YM3_I:AP show that σ_ξ is a limit in the frame of the sequence along all j: $\sigma_\xi(X)=\lim_j \rho_j(X_{K_j}^{\nu_j})$ for every $X\in\mathfrak{S}(\mathbb{R}^3)$, $X_{K_j}^{\nu_j}$ being the product of the centred observables $\mathcal{O}(f_q)-\nu_j(\mathcal{O}(f_q))$ of the entries of X, with the approximants $\rho_j(F)=\nu_j(\mathbf{1}_{E_j}F)/\nu_j(E_j)$, where E_j is a Borel set invariant under the translations by $a_{K_j}\mathbb{Z}^3$ with $\nu_j(E_j)>0$; and ρ_j is a periodic Gibbs state at (a_{K_j},β_j) (Thm. YM3_I:AP).

The convergence on every separated tuple also gives convergence of all connected functions. Indeed, put $\sigma_j(X)=\rho_j(X_{K_j}^{\nu_j})$. The inverse moment–cumulant identity in the proof of Prop. YM3_I:92.1 gives, for $n\ge 2$,

$$S_n^T[\rho_j](X)=\sum_{\pi\in\mathrm{Part}(n)}(-1)^{|\pi|-1}(|\pi|-1)!\prod_{B\in\pi}\sigma_j(X_B).$$

Centering changes only the first cumulant. Every subtuple X_B remains separated, so each factor tends to $\sigma_\xi(X_B)$. Passing to the limit in this finite sum proves $S_n^{\sigma_\xi}(X)=\lim_j S_n^T[\rho_j](X)$ for every $n\ge 2$ and $X\in\mathfrak{S}_n(\mathbb{R}^3)$. For σ itself $\rho_j=\nu_j$. In both cases $S_2^{\sigma'}$, $\sigma'\in\{\sigma,\sigma_\xi\}$, is at a C^1 pair in the class the limit of $S_2^T[\rho_j]$ along states at g at the data (K_j,β_j), whose labels converge to b, and the first assertion of (iii) gives the law.

Volume-first limits. A defining sequence of states of a volume-first limit S at g is a Gibbs sequence at g; it converges, and its limit is the centred functional σ_S of S (Prop. YM3_I:92.1 and its moment–cumulant identity); the ergodic components of S are the components of the canonical decompositions of σ_S ([7, §K.1] and Def. 13.4(2)). So the labels along every defining sequence converge, and the law holds for S and its ergodic components at the pairs of class C^1. Two defining sequences give one label: the one function $S_2=S_2^{\sigma_S}$ obeys the law with both limits at the C^1 pairs in the class at every small d, and the inequality displayed above gives their equality.

This is the unique label of Thm. YM3_I:AW. For $b\in I(g)$ the pairs $(K,L^K b)$, $K\in\mathbb{N}$, are data at g (Cor. YM3_I:74.2), and Prop. YM3_I:92.1, applied to torus-limit states at these data, gives a volume-first limit at g with a defining sequence of states (K_j,β_j,ν_j) with $a_{K_j}\beta_j=b$ for every j; its label is b.

(iv) $\mathcal{K}_5\ge 10^{-5}\vartheta^2 d\,\eta$ on the class; divide by $c_2 b^{-3}\mathcal{K}_5$. In the case (i) the lattice term $C_{\mathrm{lat}}b^{-2}(a_K/d)\,\eta$, divided by $c_2b^{-3}\cdot 10^{-5}\vartheta^2 d\,\eta$, gives the member $10^5 c_2^{-1}\vartheta^{-2}C_{\mathrm{lat}}\,b\,a_K/d^2$, which is small only if a_K is small against d^2/b. In the cases (ii) and (iii), $|S-c_2b^{-2}\mathcal{K}_6|\le[(\frac{4\pi}{3})^2c_1c_2+\bar\varepsilon_\infty(d)]\,b^{-3}d\,\eta$ and $\mathcal{K}_6\ge 10^{-6}\vartheta^2\eta$, so that $S/(c_2\mathcal{K}_6)\to b^{-2}$ as $d\to 0$ along the pairs at which the law holds: each of these limits determines its label, and then the quotient determines the number $\frac{8}{3\pi}$. A law of this form with a tolerance that tends to 0 with d thus fixes both its label and its coefficient.

Remark 6.24. At a finite cutoff the tolerance of Thm. S is $\bar\varepsilon(d)$, or $c_{\mathrm{r}}d^{1/2}$, together with the lattice term $C_{\mathrm{lat}}b^{-2}a_K/d$, which is larger than the correction $c_1c_2b^{-3}\mathcal{K}_5$, of the order $b^{-3}d$, unless a_K is small against d^2/b; the number c_1 is read off at such cutoffs or in the continuum limit. Clause (d) is

weaker than clause (b): a mean over scales does not see an oscillation in the scale. The arguments use one finite lattice or one torus at a time, at scales below the least side; the agreement of states uses besides one uniformity in the volume, which comes from reflection positivity and from the bounds of Thms. YM3_I:AJ and YM3_I:AD on the partition functions of the tori at the source 0, and no property of the correlations at large distances.

6.2. **A complex scale for one fluctuation step.** The coefficient arguments in §§6.1 and 8.1 require a disc of fluctuation scales whose radius increases as the running coupling increases. Analyticity near the real step alone does not give this disc. We obtain it by constructing the potentials at a fixed lower real coupling, where the unscaled field has a larger admissible range, and then evaluating the same functions at $\lambda \mathtt{t}^{-1/2} B$.

We use the local input and the localized expansion of Def. YM3_I:52.1 and Thm. YM3_I:V. In particular $k+1$ is a single admitted step, $0 \le k < K$, $M = L^{m'}$ satisfies $m' < m + K - k - 1$, and $P = (U, A', \mathbf{J})$ lies in the output core $\mathcal{U}^\circ(X)$. On a fixed real-U slice its variables $(A', \mathbf{J})$ lie in the open set $\mathcal{O}_U(X)$. The Gaussian structure, its restriction consistency, the gauge-fixing operators and their bounds are those in that definition. We retain the constants and numerical assumptions of Thm. YM3_I:V. Write

$$\delta(t) = A_1 t^{-1/2} (\log t)^{\mathfrak{p}_0}, \qquad \mathfrak{b}(t) = t^{1/2}\delta(t), \qquad s_{\mathfrak{a}}(t) = \delta(t)/a_{0,k+1},$$

and let $\mathbb{D}^{\mathrm{f}}_t(Y)$ be the complex field polydisc $\sup_b |B(b)| < C_{\mathcal{A}} \mathfrak{b}(t)$ on the nonpivot bonds in Y. The potential norm uses the weight

$$Q_Y(B) = 1 + M^{-3} \sum_{b \in \mathfrak{X}^{\circ +}\langle Y \rangle} |B(b)|^2.$$

All parameter sets below are either points or open subsets of a finite-dimensional complex vector space; a point factor is omitted from a product domain.

Definition 6.25 (Scaled input)**.** Fix $\mathtt{t} > 1$ in the admitted range and a radius $R > 0$. Suppose that four potentials $V^{J,\lambda}, V^{G,\lambda}, V^{A,\lambda}, V'^{,\lambda}_\theta$ are given for $|\lambda| < R$, $\theta \in \Theta$, on every local core and field polydisc. They have the pair dependence and gauge covariance of Def. YM3_I:52.1, and are jointly holomorphic on

$$D(0,R) \times \Theta \times \mathcal{O}_U(Y) \times \mathbb{D}^{\mathrm{f}}_{\mathtt{t}}(Y).$$

The first three are independent of θ. The J and fourth potentials vanish at $B = 0$; the G and A potentials vanish through degree two in B. Suppose also that

$$\begin{aligned} |V^{o,\lambda}(Y;P,B)| &\le |\lambda| k_{\mathrm{int}} e^{-\lambda_V d_k(Y)} Q_Y(B) \quad (o = J, G, A), \\ |V'^{,\lambda}_\theta(Y;P,B)| &\le |\lambda| k_{\mathrm{iv}} e^{-\lambda_V d_k(Y)} Q_Y(B), \\ R k_{\mathrm{int}} \le w_*/8, &\qquad R k_{\mathrm{iv}} \le w_*/4, \end{aligned} \tag{6.xiii}$$

and $V^{o,-\lambda}(Y;P,-B) = V^{o,\lambda}(Y;P,B)$ for all four potentials. The Gaussian structure and gauge-fixing operators are fixed as λ, θ vary. The coefficient x is an independent parameter in $D(\mathtt{t}, \sigma_* \mathtt{t})$, and $\tau = x/\mathtt{t}$. ■

Use exactly the localized interpolation and the three bond terms in the proof of Thm. YM3_I:V: $\mathbf{G}_\tau$ is the Gaussian term, E^χ the cutoff term, and $E^{\bullet,\lambda}_\theta$ the interpolated interaction term. For the last term the interpolation is

$$V^{\mathrm{int}}_v(B) = V^{J,\lambda}(vB) + v^{-2}[V^{G,\lambda} + \tau V^{A,\lambda}](vB), \qquad V'_v(B) = V'^{,\lambda}_\theta(vB).$$

The value at $v = 0$ is its removable continuation. Distribute bond terms to sites with the same nonnegative tents h_z:

$$\mathbf{G}_\tau(X,P,z) = \sum_b h_z(b) \mathbf{G}_\tau(X,b;P),$$

$$F_\theta^\lambda(X,P,z)=\sum_b h_z(b)(E^\chi+E_\theta^{\bullet,\lambda})(X,b;P),\qquad F^\chi(X,P,z)=\sum_b h_z(b)E^\chi(X,b;P).$$

Thus $F^\lambda-F^\chi$ is the interaction-dependent part. It is this difference that vanishes at $\lambda=0$.

Proposition 6.26 (The step on the full complex disc)**.** For the data of Def. 6.25, put $M_\bullet = K_\Pi R(32k_{\mathrm{int}}+9k_{\mathrm{iv}}) \le (25/4)K_\Pi w_*$ and $d=d_{k+1}(X)$. The following assertions hold on the full product domain

$$D(\mathsf{t},\sigma_*\mathsf{t})\times D(0,R)\times\Theta\times\mathcal{O}_U(X). \tag{6.xiv}$$

(cx1) All three bond terms are defined and vanish unless $b^\sharp\subset X$. Their tent sums are pointed local families.

(cx2) For fixed x,P, $\mathbf{G}_\tau$ and E^χ are independent of λ,θ and of all four potentials. They have the Gaussian and cutoff bounds of Thm. YM3_I:V.

(cx3) The interaction term obeys

$$|E_\theta^{\bullet,\lambda}(X,b;P)|\le K_\Pi|\lambda|(32k_{\mathrm{int}}+9k_{\mathrm{iv}})e^{-\lambda_{\mathrm{out}}d}.$$

For $F^\lambda-F^\chi$ the same bound holds with the factor $3L^3$. In particular the respective bounds on the disc are $M_\bullet e^{-\lambda_{\mathrm{out}}d}$ and $3L^3M_\bullet e^{-\lambda_{\mathrm{out}}d}$.

(cx4) $E_\theta^{\bullet,0}=0$ and $F_\theta^0=F^\chi$.

(cx5) All terms are jointly holomorphic on eq. 6.xiv; the terms in (cx2) are holomorphic in x and the slice variables alone.

(cx6) Each individual bond term $E_\theta^{\bullet,\lambda}$ and each site term F_θ^λ is even in λ.

(cx7) The terms depend only on the local pair and are gauge invariant, with the same restriction and locality properties as the original step. No sum over domains is needed for these properties.

(cx8) The Taylor series starts at order two:

$$F_\theta^\lambda=F^\chi+\sum_{m\ge1}\lambda^{2m}F_{[2m],\theta},\qquad |F_{[2m],\theta}(X,P,z)|\le 3L^3M_\bullet R^{-2m}e^{-\lambda_{\mathrm{out}}d}.$$

The coefficients are jointly holomorphic in all remaining parameters and inherit locality and gauge invariance. The corresponding bond bounds omit $3L^3$.

(cx9) If $R>1$, then

$$|F^1-F^\chi|\le 3L^3M_\bullet R^{-2}e^{-\lambda_{\mathrm{out}}d},\qquad |F^1-F^\chi-F_{[2]}|\le\frac{3L^3M_\bullet R^{-4}}{1-R^{-2}}e^{-\lambda_{\mathrm{out}}d}.$$

(cx10) The local sum rules and the whole-torus integral identity of Thm. YM3_I:V hold for these potentials. More explicitly, put $V^{\mathrm{int},\lambda}=V^{J,\lambda}+V^{G,\lambda}+\tau V^{A,\lambda}$ and $V^{\mathrm{q}}=-(1-\tau)\sum_\square(B,\widehat{G}_\square B)/2$. Then

$$\begin{aligned}\Xi_\theta^\lambda(x;P_*)&:=\int d\mu_{\tau^{-1}C_*(P_*)}(B)\prod_{b\in\mathfrak{X}^+}\mathbf{1}_{\{|B(b)|<\mathfrak{b}(\mathsf{t})\}}\exp\sum_Y(V^{\mathrm{q}}+V^{\mathrm{int},\lambda}+V_\theta'^{,\lambda})(Y;P_*|_Y,B|_Y)\\&=\exp\sum_{X,z}(\mathbf{G}_\tau+F_\theta^\lambda)(X,P_*|_X,z)\neq0.\end{aligned}$$

The inverse covariance has positive real part, the integral converges absolutely, and its integrand is defined as zero where the cutoff product vanishes. The Gaussian measure and integral are complex when their data are complex.

(cx11) If $R>1$ and the four potentials at $\lambda=1$ are the admitted potentials of the original step, the terms at $\lambda=1$ are exactly its terms, with the same localization.

■

Proof. We check the quantitative bounds in the construction rather than increasing the constant K_P in its global smallness conditions. The Gaussian scaling calculation in Thm. YM3_I:V depends only on x, the fixed structure and the numerical constants. Thus on its stated x disc the scaled structure $(\tau^{-1}C, \tau^{1/2}\Gamma)$ is admitted, the whole covariance has inverse with positive real part, and $\|V^{\mathrm{q}}\| \le w_*/2$.

Put $\nu_{\mathrm{int}} = |\lambda| k_{\mathrm{int}}$ and $\nu_{\mathrm{iv}} = |\lambda| k_{\mathrm{iv}}$. The interpolation estimate, Prop. YM3_I:48.4, uses only the potential norms and their indicated orders of vanishing. It gives

$$\|V_v^{\mathrm{int}}\| \le \tfrac{7}{2}\nu_{\mathrm{int}} \le 7w_*/16, \quad \|V_v'\| \le \nu_{\mathrm{iv}} \le w_*/4, \quad \|\dot V_{v,b}^{\mathrm{int}}\| \le 32\nu_{\mathrm{int}}, \quad \|\dot V_{v,b}'\| \le 9\nu_{\mathrm{iv}}.$$

These are precisely the margins used in the proof of Thm. YM3_I:V. Its insertion expansion therefore defines $E_\theta^{\bullet,\lambda}$ and gives (cx3), with its support. The first two terms use no interaction potentials, proving (cx2). The tent mass at one site is at most $3L^3$, giving the site bounds. At $\lambda = 0$ all four potentials vanish by eq. 6.xiii; hence so do the interpolation and its insertion, proving (cx4).

The same insertion majorant is uniform on the full product domain. Each refined integrand is holomorphic in its parameters on the slice and fixed field polydisc. The common majorant permits parameter integration and locally uniform summation, so the terms are jointly holomorphic. The cutoff terms use their fixed-density formula, whose differentiated masses have the same summable majorant, exactly as in that theorem. This proves (cx5).

For parity apply $B \mapsto -B$ to each refined Gaussian integral. Its covariance, every quadratic gauge term and every cutoff are unchanged. The hypothesis on the four potentials gives $V_v^{o,-\lambda}(-B) = V_v^{o,\lambda}(B)$ for the localized interpolations, including their bond velocities: multiplication by the interpolation field commutes with -1, and v^{-2} is unchanged. In a refined derivative the signs from differentiating the potentials are matched by those from the corresponding Gaussian derivatives; equivalently both are conjugated by the orthogonal map $-I$ on the bond-coordinate space. The insertion expansion is covariant under this map. Thus equality holds for each domain and each marked bond before summation. It proves (cx6). Pair dependence, gauge covariance and restriction commute with the same construction, proving (cx7).

For fixed remaining parameters, $E_\theta^{\bullet,\lambda}$ is even, holomorphic, zero at zero and bounded by $M_\bullet e^{-\lambda_{\mathrm{out}} d}$. Cauchy's formula on $|\lambda| = r < R$ bounds its degree-$2m$ coefficient by $M_\bullet r^{-2m} e^{-\lambda_{\mathrm{out}} d}$; let $r \uparrow R$. The odd coefficients vanish. Coefficient extraction by this contour integral preserves holomorphy in the remaining parameters and all the established invariances. Finite tent summation gives (cx8). Schwarz's lemma applied after the order-two zero gives the first bound in (cx9); summing the coefficient bounds for $m \ge 2$ gives its second bound.

For (cx10), integrate the insertion along v: the sum of its bond velocities is the derivative of the interpolated potential. Thus its sum is the difference of the full cluster sum and the one with only V^{q}. The cutoff term supplies the difference between the finite-threshold and infinite-threshold sums for V^{q}, and the Gaussian term supplies the latter sum. The tents have total mass one when summed over their sites. Restriction consistency identifies the domain terms with those computed on the whole torus. The complete Gaussian expansion therefore gives exactly the displayed exponential identity. Its common majorant proves absolute convergence; its exponential form proves nonvanishing. Finally the formulas defining every term are unchanged at $\lambda = 1$, which proves (cx11). □

We now construct this input on a large disc. The required one-variable estimate is included to distinguish the gain from the mere vanishing of a potential at the origin.

Lemma 6.27 (Rescaling a holomorphic potential)**.** Let $E_r = \{B : \sup|B| < r\}$, $0 < r_\flat \le r$, and $|\mu| \le r_\flat/r$. Suppose f is holomorphic on $E_{r_\flat}$. If $f(0) = 0$ and $|f(B')| \le aQ_Y(B')$, then

$$|f(\mu B)| \le a|\mu|(r/r_\flat)Q_Y(B), \qquad B \in E_r.$$

If the bound is a instead, omit Q_Y. If f vanishes through degree two, then the stronger bound is

$$|f(\mu B)| \le a|\mu|^3(r/r_\flat)(1 + M^3/r_\flat^2)(Q_Y(B) - 1).$$

■

Proof. For the first assertion apply Schwarz's lemma to $z \mapsto f(z(r_\flat/r)B)$ on the unit disc. Its bound is $aQ_Y(B)$, since $Q_Y(cB) = 1 + |c|^2(Q_Y(B) - 1)$. For the third-order assertion let $\beta = \sup|B| > 0$; the case $B = 0$ is immediate. For $|\mu| \le s < r_\flat/\beta$, the function $z \mapsto f(zsB)$ vanishes to order three and is bounded on the unit disc by $a[1 + s^2(Q_Y(B) - 1)]$. Schwarz's lemma of order three gives

$$|f(\mu B)| \le a|\mu|^3[s^{-3} + s^{-1}(Q_Y(B) - 1)].$$

Let $s \uparrow r_\flat/\beta$ and use $\beta^2 \le M^3(Q_Y(B) - 1)$ and $\beta < r$. This proves the displayed estimate. □

Definition 6.28 (A lower calibration)**.** Retain one lattice, one system of radii, its open slices and one Gaussian structure and gauge-fixing family. Let $1 < \mathfrak{t}_\flat \le \mathfrak{t}$ and $\delta(\mathfrak{t}) \le \delta(\mathfrak{t}_\flat)$. All numerical and radius conditions of Thm. YM3_I:V are required at $\mathfrak{t}_\flat$, with $\varepsilon(\mathfrak{t}_\flat), \delta(\mathfrak{t}_\flat) \le 1$, $\mathfrak{b}(\mathfrak{t}_\flat) \ge \mathfrak{b}_*$ and $s_{\mathfrak{a}}(\mathfrak{t}_\flat) \le s^*$. Suppose four functions of the unscaled field $\mathcal{A}$ are defined for $\sup|\mathcal{A}| < C_{\mathcal{A}}\delta(\mathfrak{t}_\flat)$. For $s \in \{\mathfrak{t}_\flat, \mathfrak{t}\}$ set

$$V^J[s](B) = v^J(s^{-1/2}B), \quad V^o[s](B) = -s\,v^o(s^{-1/2}B)\ (o = G, A), \quad V'_\theta[s](B) = v'_\theta(s^{-1/2}B).$$

The dependence on Y, P is suppressed. At $s = \mathfrak{t}_\flat$ require the full local input of Def. YM3_I:52.1, jointly in $\theta \in \Theta$, with fourth-family size at most $n_\flat \le N_*$. No bound at a larger unscaled field is assumed. ■

Proposition 6.29 (The disc supplied by a lower calibration)**.** For Def. 6.28, put

$$R = \frac{\delta(\mathfrak{t}_\flat)}{2\delta(\mathfrak{t})}, \qquad k_{\mathrm{int}} = 2K_P\delta(\mathfrak{t}), \qquad k_{\mathrm{iv}} = K_\Lambda s_{\mathfrak{a}}(\mathfrak{t})n_\flat.$$

On $|\lambda| < 2R$, define

$$\begin{aligned} V^{J,\lambda}(B) &= v^J(\lambda\mathfrak{t}^{-1/2}B), \\ V'^{,\lambda}_\theta(B) &= v'_\theta(\lambda\mathfrak{t}^{-1/2}B), \\ V^{o,\lambda}(B) &= -\mathfrak{t}\lambda^{-2}v^o(\lambda\mathfrak{t}^{-1/2}B) \quad (o = G, A). \end{aligned} \tag{6.xv}$$

with the last two values at zero defined as zero. Restricted to $|\lambda| < R$, these satisfy Def. 6.25. In particular Prop. 6.26 holds on its full product domain, with

$$M_\bullet = K_\Pi[32K_P\delta(\mathfrak{t}_\flat) + \tfrac{9}{2}K_\Lambda s_{\mathfrak{a}}(\mathfrak{t}_\flat)n_\flat] \le \tfrac{41}{8}K_\Pi w_*.$$

If $R > 1$ and the input at $\mathfrak{t}$ is the original admitted input, the value at $\lambda = 1$ is its original step. The order-two gain and fourth-order remainder carry respectively the factors $R^{-2} = 4\delta(\mathfrak{t})^2/\delta(\mathfrak{t}_\flat)^2$ and $R^{-4} = 16\delta(\mathfrak{t})^4/\delta(\mathfrak{t}_\flat)^4$. ■

Proof. Let $r = C_{\mathcal{A}}\mathfrak{b}(\mathfrak{t})$, $r_\flat = C_{\mathcal{A}}\mathfrak{b}(\mathfrak{t}_\flat)$ and $\mu = \lambda(\mathfrak{t}_\flat/\mathfrak{t})^{1/2}$. Then $r_\flat \le r$ and $|\mu|r/r_\flat = |\lambda|/(2R)$. Thus μB lies in the lower field polydisc for $|\lambda| < 2R$, and eq. 6.xv equals

$$V^J[\mathfrak{t}_\flat](\mu B), \qquad V'_\theta[\mathfrak{t}_\flat](\mu B), \qquad \mu^{-2}V^o[\mathfrak{t}_\flat](\mu B).$$

The last functions extend holomorphically through $\lambda = 0$: their numerators vanish through order two, so division by λ^2 has a removable zero, also jointly in the other parameters by Cauchy's coefficient formula.

Apply Lem. 6.27 at the lower place. For J the bound is $|\lambda|K_P\delta(\mathfrak{t})Q_Y(B)e^{-\lambda_V d_k(Y)}$. For G, A, division by μ^2 leaves the bound

$$|\lambda|K_P\delta(\mathfrak{t})(1 + M^3/r_\flat^2)(Q_Y(B) - 1)e^{-\lambda_V d_k(Y)}.$$

The threshold floor gives $\mathfrak{b}(\mathfrak{t}_\flat)^2 > (LM)^3/2$, and $C_{\mathcal{A}} \ge 3$, so $1 + M^3/r_\flat^2 < 11/9 < 2$. The constant-bound version of the lemma gives for the fourth potential $|\lambda|K_\Lambda s_{\mathfrak{a}}(\mathfrak{t})n_\flat e^{-\lambda_V d_k(Y)}$. Moreover

$$Rk_{\mathrm{int}} = K_P\delta(\mathfrak{t}_\flat) \le w_*/8, \qquad Rk_{\mathrm{iv}} = \tfrac{1}{2}K_\Lambda s_{\mathfrak{a}}(\mathfrak{t}_\flat)n_\flat \le w_*/8.$$

These prove all the size requirements. Pair dependence and gauge covariance are preserved by scalar multiplication of the field. The products $(-\lambda)(-B)$ and λB agree, as do $(-\lambda)^{-2}$ and λ^{-2}, proving parity. The orders of vanishing in B are unchanged. At $\lambda = 1$ the defining formulas are precisely those at t. The coefficient and remainder claims now follow from Prop. 6.26. □

Proposition 6.30 (The actual construction at two calibrations)**.** Fix the constants, the tower and shift systems, and one lag-weight case (ω, K) of the actual step Thm. YM3_I:W. Suppose its geometric, core, shift and octet hypotheses hold at a lower real calibration $\mathsf{t}_\flat > 1$, with the same radii $\mathfrak{a}$, and choose positive lag weights satisfying

$$\varrho_j^2 L^{-(k-j)} \le c_\omega \omega_{k+1-j} \quad (1 \le j \le k), \qquad \mathsf{K} = K_\Lambda c_\omega.$$

Retain the rate bounds and all numerical hypotheses of Thm. YM3_I:V at $\mathsf{t}_\flat$, with K_Λ replaced by this K. Choose $\check{s} > 0$ such that

$$\check{s} \le s^*, \qquad 8K_P\check{s} \le w_*, \qquad 4\mathsf{K}N_*\check{s} \le w_*, \qquad s_{\mathfrak{a}}(\mathsf{t}_\flat) \le \check{s}, \quad \mathfrak{b}(\mathsf{t}_\flat) \ge \mathfrak{b}_*.$$

Let $\mathsf{t} \ge \mathsf{t}_\flat$ with $\delta(\mathsf{t}) \le \delta(\mathsf{t}_\flat)$. For a bare array $f_\circ$ on these same cores, assume $N_{\mathfrak{a},\omega}(f_\circ) \le N_*$; at height zero the array is empty. Then the canonical Gaussian structure and the actual four potentials give Def. 6.28, with K_Λ replaced by K and $n_\flat \le N_{\mathfrak{a},\omega}(f_\circ)$. The step is consequently defined on

$$x \in D(\mathsf{t}, \sigma_*\mathsf{t}), \qquad |\lambda| < \delta(\mathsf{t}_\flat)/(2\delta(\mathsf{t})).$$

If the latter radius exceeds one, its terms at one coincide with the actual canonical step terms. This statement concerns $\mathbf{G}_\tau, F^\lambda$; the separate local normalizer is unchanged and is not included in F^λ.

The same conclusion holds jointly for a holomorphic family of arrays with this uniform size bound. If an array has a reserve $N_{\mathfrak{a},\omega}(f_\circ) + 2q \le N_*$ with $q > 0$, its first and mixed second array derivatives, in directions h_1, h_2, satisfy

$$\begin{aligned} |D_f F^\lambda[h_1](X,P,z)| &\le Cq^{-1}\|h_1\| \frac{(|\lambda|/R)^2}{1-(|\lambda|/R)^2} e^{-\lambda_{\mathrm{out}} d}, \\ |D_f^2 F^\lambda[h_1,h_2](X,P,z)| &\le Cq^{-2}\|h_1\|\|h_2\| \frac{(|\lambda|/R)^2}{1-(|\lambda|/R)^2} e^{-\lambda_{\mathrm{out}} d}, \end{aligned} \tag{6.xvi}$$

where $R = \delta(\mathsf{t}_\flat)/(2\delta(\mathsf{t}))$, $\|h\| = N_{\mathfrak{a},\omega}(h)$, and C depends only on the fixed constants and $\check{s}$. The derivatives retain the local dependence on the array and weights. ■

Proof. We verify both the common objects and their domains. The radius and octet conditions of the actual step involve only the chosen radii, geometric constants and shift systems. Their coupling-dependent inequalities involve $\delta(t)$, $s_{\mathfrak{a}}(t)$ and the lower bound on $\mathfrak{b}(t)$. The first two decrease between the two specified calibrations and the third increases. Thus every such hypothesis valid at $\mathsf{t}_\flat$ is valid at t. The lag ratios and their weight inequalities do not involve t.

The actual covariance, its square root, the local backgrounds and the gauge-fixing matrices are built from the pair, the lattice, the fixed shifts and the radii. Changing t with these data fixed changes none of them. In the Jacobian formula the field occurs as $\mathcal{A} = t^{-1/2}B$. In the gauge and action potentials the same field occurs inside their cubic remainders, with the sole additional factor $-t$. In the fourth-potential formula the substitutions, finite vertex differences and octet evaluations are fixed; the field again enters only as $\mathcal{A}$. These facts follow directly from the actual formulas in the proofs of Thm. YM3_I:W and Prop. YM3_I:51.16. They supply the four unscaled functions in Def. 6.28 on $\sup|\mathcal{A}| < C_{\mathcal{A}}\delta(\mathsf{t}_\flat)$: this is exactly the image of the lower field polydisc under $B \mapsto \mathsf{t}_\flat^{-1/2}B$.

At that lower calibration Thm. YM3_I:W gives the three canonical interaction bounds and the Gaussian input. Prop. YM3_I:51.16 bounds the fourth potential by

$$\mathsf{K}s_{\mathfrak{a}}(\mathsf{t}_\flat)N_{\mathfrak{a},\omega}(f_\circ)e^{-\lambda_V d_k(Y)}.$$

It is linear in the array, holomorphic, and zero at zero field. Thus its lower size is at most $N_{\mathfrak{a},\omega}(f_\circ)$. All premises of Def. 6.28 are verified, with K as its fourth-potential constant. Prop. 6.29 proves the asserted domain and the identification at one. It also gives the uniform bound

$$M_\bullet \le K_\Pi(32K_P + \tfrac{9}{2}\mathsf{K}N_*)\check{s}.$$

Uniform holomorphic array parameters are permitted throughout the localized construction, so the parameter statement follows.

For the derivative estimates, vary the array by $z_1h_1+z_2h_2$ on the two discs $|z_i| < q/\|h_i\|$, omitting a zero direction. The reserve keeps their sum within the admitted ball. The term F^χ is independent of the array. For each even coefficient of $F^\lambda - F^\chi$, (cx8) and Cauchy's formula in the z_i give respectively the additional factors $q^{-1}\|h_1\|$ and $q^{-2}\|h_1\|\|h_2\|$. Sum the coefficient bounds for $m \ge 1$ to obtain eq. 6.xvi. The finite local substitutions use only input domains contained in the output domain, and taking these derivatives does not change that dependence. □

When the step is subsequently evaluated at an even holomorphic $x = \xi(\lambda)$, this substitution is legitimate only while $|\xi(\lambda) - \mathsf{t}| < \sigma_*\mathsf{t}$. It preserves holomorphy and evenness. Although the Gaussian and cutoff terms may then vary with λ through ξ, they remain independent of the bare array; the array-derivative argument is unchanged.

6.2.1. *Source families and the local step.* An insertion changes the potentials integrated in a step. It does not change its Gaussian structure. We specify the norm in which this change is small and the parameter dependence needed to differentiate the resulting local terms.

Use one admitted step of Def. YM3_I:52.1 and Thm. YM3_I:V, including their lattice, core, Gaussian and numerical hypotheses. At height k write P for a background triple and let B range over the complex nonpivot-field polydisc $\sup_b |B(b)| < C_{\mathcal{A}}\mathfrak{b}(\mathsf{t})$. For a family V define

$$\mathsf{n}^{\mathrm{f}}(V) = \sup_{X,P} \sup_{Y^\sharp \subset X,\, B} \frac{e^{\lambda_V d_k(Y)}|V(Y;P|_Y,B)|}{1+q_Y(B)}, \qquad q_Y(B) = M^{-3} \sum_{b \in \mathfrak{X}^{\circ+}\langle Y\rangle} |B(b)|^2. \tag{6.xvii}$$

Keeping X, P fixed defines $\mathsf{n}^{\mathrm{f}}_{X,P}$; keeping only X fixed defines $\mathsf{n}^{\mathrm{f}}_X$. A family in the fourth place is holomorphic on the open real-U slices and the field polydiscs, depends on the background pair, is gauge invariant, vanishes at $B = 0$, and has finite norm eq. 6.xvii. Holomorphic families of such inputs are required to be jointly holomorphic in their parameters and these variables, with a uniform norm bound.

The three ordinary interactions $V^{\mathrm{J}}, V^{\mathrm{G}}, V^{\mathrm{A}}$ are also families in the fourth place. Indeed their analyticity, pair dependence and gauge invariance are part of the local input; $V^{\mathrm{J}}(0) = 0$, and $V^{\mathrm{G}}, V^{\mathrm{A}}$ vanish through degree two. Their pointwise bounds give

$$\mathsf{n}^{\mathrm{f}}_X(V^\natural) \le \mathsf{n}^{\mathrm{f}}(V^\natural) \le K_P\delta(\mathsf{t}), \qquad \natural = \mathrm{J}, \mathrm{G}, \mathrm{A}. \tag{6.xviii}$$

The same bound holds for their even and odd parts, since $B \mapsto -B$ preserves the polydisc and q_Y.

For the canonical absolute-radius step, the source potentials are defined on the larger unscaled-field sets

$$\mathfrak{D}^{\mathrm{src}}(Y) = \{(P,\mathcal{A}) : P \in \mathcal{U}^\circ(Y), \quad \sup_b |\mathcal{A}(b)| < \varepsilon_{\mathrm{src}}\}, \qquad \varepsilon_{\mathrm{src}} = \begin{cases} \min\{\varepsilon_{\mathfrak{h}}, c_{\mathrm{os}}\alpha_0/(2c_{\mathfrak{h}})\}, & k \ge 1, \\ r_P^\circ, & k = 0. \end{cases} \tag{6.xix}$$

Retain $C_{\mathcal{A}}\delta(\mathsf{t}) \le \varepsilon_{\mathrm{src}}/2$ and $0 < \lambda_V \le \bar{\kappa} - 2$, as in Lem. YM3_I:114.1. Thus $\mathcal{A} = \mathsf{t}^{-1/2}B$ lies strictly inside this source domain. Its radius is fixed by the cores and does not shrink with t.

Let $\mathbb{W}^{\mathrm{src}}[v]$ be the direct weighted-action family of that lemma, with the reference linear term subtracted bond by bond. Choose $K_{\mathrm{Q}}^{\vee}, K_{\mathrm{L}}^{\vee}$ satisfying equation YM3_I:(2166), and put

$$K^{\mathrm{d}} = \tfrac{351}{2} L^3 K_{\Pi}, \qquad \mu_\star = \min\{w_\star/32, E_0/(16K_{\mathbf{M}}^{\mathrm{f}})\}, \qquad K^{\mathrm{d}} \le K_{\mathbf{M}}^{\mathrm{f}},$$
$$\sigma_{\mathrm{src}} = \mu_\star/K_{\mathrm{Q}}^{\vee}, \qquad o_{\mathrm{src}}(\mathsf{t}) = \mathsf{t}^{-1/2}\mu_\star/K_{\mathrm{L}}^{\vee}, \tag{6.xx}$$
$$\mathsf{Wgt}_k^{\mathrm{src}}(\mathsf{t}) = \{v : |v|_\infty < \sigma_{\mathrm{src}}, \quad \mathrm{sw}_k(v) < o_{\mathrm{src}}(\mathsf{t})\}.$$

Here E_0 denotes the size E_0^{s} in the recursion preceding Prop. YM3_I:114.2, and $K_{\mathbf{M}}^{\mathrm{f}}$ is its constant $3L^3 \max\{K_E e^{4\lambda_1}, 32K_\Pi K_P, 117K_\Pi/2\}$. The oscillation compares plaquettes whose barycentres are at sup distance at most M in level-k units. Write

$$\mathsf{sz}_X(v) = K_{\mathrm{Q}}^{\vee}(1 + 2C_{\mathcal{A}}\delta/\varepsilon_{\mathrm{src}})|v|_X + K_{\mathrm{L}}^{\vee}\mathsf{t}^{1/2}\,\mathrm{sw}_k(v).$$

Proposition 6.31 (Changing the fourth family)**.** Retain the preceding admitted step. At the real field scale $x = \mathsf{t}$, let the three internal families have total norm at most $3w_\star/8$. The pointed integration map $\mathbf{F}$ is holomorphic in a holomorphic fourth-family input on its admitted norm ball and is local in that input. It preserves pair dependence, gauge invariance, conjugation and the lattice covariances of the data. If V_0, V_1 have fourth-family norms at most $w_\star/4$ and $\mathsf{n}^{\mathrm{f}}(V_1 - V_0) \le w_\star/2$, then

$$|\mathbf{F}[V_1] - \mathbf{F}[V_0]|(X, P, z) \le K^{\mathrm{d}}\mathsf{n}_X^{\mathrm{f}}(V_1 - V_0)e^{-\lambda_{\mathrm{out}}d_{k+1}(X)}.$$

In particular let $\mathfrak{f}$ be a canonical old array satisfying $8K_\Lambda s_{\mathfrak{a}} N_{\mathfrak{a}}(\mathfrak{f}) \le w_\star$, with the empty array at $k = 0$. For $v \in \mathsf{Wgt}_k^{\mathrm{src}}$, the fourth family $V'[\mathfrak{f}] - \mathbb{W}^{\mathrm{src}}[v]$ has norm less than $7w_\star/32$. Its output $E_v = \mathbf{N} + \mathbf{F}[V'[\mathfrak{f}] - \mathbb{W}^{\mathrm{src}}[v]]$ is therefore defined on the full output core, jointly holomorphic in the weight and any holomorphic old-array parameters, and

$$|E_v - E_{v=0}|(X, P, z) \le K^{\mathrm{d}}\mathsf{sz}_X(v)e^{-\lambda_{\mathrm{out}}d_{k+1}(X)}.$$

The maps obey the exact local and whole-torus sum rules of Thm. YM3_I:V. On an actual real fibre admitted by [7, Lem. B.3], the source term sums to the weighted-action difference of equation YM3_I:(2168). ■

Proof. At a fixed background the four inputs are families on the field polydiscs. Lem. YM3_I:77.1 constructs their bond insertion terms, and Prop. YM3_I:77.2 compares two fourth families. In its second estimate use $n_{\mathrm{int}} \le 3w_\star/8$, $n_{\mathrm{iv}} \le w_\star/4$ and $n_W \le w_\star/2$. Its coefficient is at most

$$9(3L^3 - 3)K_\Pi\{1 + 8(3/8 + 1/4)\} < 162L^3K_\Pi < K^{\mathrm{d}}.$$

The same proof at fixed X gives the local norm. Holomorphy follows from the parameter-dependent insertion theorem with its common summable majorant. Restriction identifies every Gaussian matrix, potential and insertion on a smaller domain, and hence identifies the localized terms. The covariance assertions follow from the corresponding changes of variables in those same terms.

Prop. YM3_I:51.16 bounds the old fourth family by $K_\Lambda s_{\mathfrak{a}} N_{\mathfrak{a}}(\mathfrak{f}) \le w_\star/8$. Lem. YM3_I:114.1 gives $\mathsf{n}_X^{\mathrm{f}}(\mathbb{W}^{\mathrm{src}}[v]) \le \mathsf{sz}_X(v)$. The half-radius condition and eq. 6.xx imply

$$\mathsf{sz}_{T_\ell}(v) < 2\mu_\star + \mu_\star \le 3w_\star/32.$$

Their sum is strictly below $7w_\star/32 < w_\star/4$. The old-potential map preserves holomorphic parameters and the source family is linear in v, so the preceding parameter result applies. Here $E_{v=0}$ is the output with the zero weight and the same old array. The source-independent normalizer cancels in $E_v - E_{v=0}$. Finally the local sum rule is precisely the pointed insertion identity. The whole-torus identity follows by restriction and summation of the tents. For the source summation the reference linear terms cancel bond by bond by criticality of the real background; the remaining terms are its weighted-action difference, as proved in Lem. YM3_I:114.1. □

Proposition 6.32 (Holomorphic parameters and parity)**.** Use the same local Gaussian and numerical hypotheses, allowing $x \in D(\mathtt{t}, \sigma_\star \mathtt{t})$ and $\tau = x/\mathtt{t}$. Let four families depend jointly holomorphically on $(\mu, \zeta) \in D(0,\rho) \times \mathcal{O}$, where $\mathcal{O}$ is an open finite-dimensional parameter set or a point. The Gaussian structure, τ and the gauge-fixing operators do not depend on μ. Suppose each internal family has norm at most $\bar\nu_{\mathrm{int}} \le w_\star/8$, the fourth has norm at most $\bar\nu' \le w_\star/4$, and all four retain the stated vanishing orders at $B = 0$. Put

$$\bar B = K_\Pi(32\bar\nu_{\mathrm{int}} + 9\bar\nu'), \qquad \Phi(\mu,\zeta) = \sum_b h_z(b) E^\bullet(X, b; \mu, \zeta), \qquad d = d_{k+1}(X).$$

Then the Gaussian and cutoff terms do not depend on μ, Φ is holomorphic, and $|\Phi| \le 3L^3 \bar B e^{-\lambda_{\mathrm{out}} d}$. If each of the four families satisfies $V^\natural_{-\mu,\zeta}(Y; -B) = V^\natural_{\mu,\zeta}(Y; B)$, then $\Phi(-\mu,\zeta) = \Phi(\mu,\zeta)$. Its odd Taylor coefficients vanish. With $\Phi_{(n)} = (n!)^{-1}\partial_\mu^n \Phi(0,\zeta)$ one has

$$|\Phi_{(n)}| \le 3L^3 \bar B \rho^{-n} e^{-\lambda_{\mathrm{out}} d}, \qquad \left| \Phi - \sum_{n \le N} \Phi_{(n)} \mu^n \right| \le 3L^3 \bar B \frac{(|\mu|/\rho)^{N+1}}{1 - |\mu|/\rho} e^{-\lambda_{\mathrm{out}} d}.$$

For $\rho > 1$, parity improves the remainder after degree two at $\mu = 1$ to $3L^3 \bar B \rho^{-4}(1-\rho^{-2})^{-1} e^{-\lambda_{\mathrm{out}} d}$. ■

Proof. Use the interpolation displayed in §6.2. The bond insertions from the three internal families have total norm at most $32\bar\nu_{\mathrm{int}}$, and that from the fourth has norm at most $9\bar\nu'$. The insertion theorem, Thm. YM3_I:S, bounds their interpolated integral by $\bar B e^{-\lambda_{\mathrm{out}} d}$. The nonnegative tents have total at most $3L^3$, giving the asserted site bound. The common majorant also permits integration of the holomorphic parameter dependence. The Gaussian and cutoff terms use none of the four families.

For parity take the orthogonal map $O_b = -I$ at every nonpivot bond. It preserves the field polydiscs and cutoffs. The covariance, coupling and gauge-fixing operators are unchanged, since $(-I)T(-I) = T$ for each of these linear operators. Their quadratic potentials and marked quadratic insertions are unchanged as well. The interpolation commutes with $B \mapsto -B$, by Prop. YM3_I:48.4. Thus the families and bond insertions at μ are carried to those at $-\mu$. The orthogonal covariance of the insertion and cutoff expansions, Thms. YM3_I:S and YM3_I:T, identifies their individual localized terms. Integration of these identical terms and then summation with the fixed tents proves that Φ is even. This argument compares the localized terms, not only their total partition function.

Cauchy's formula on circles of radius increasing to ρ gives the coefficient bounds. Summing their geometric majorant gives the remainder estimate. Evenness removes the odd coefficients; at $\mu = 1$ the remaining tail is bounded by the geometric series $\rho^{-4} + \rho^{-6} + \cdots$. □

Lemma 6.33. The free-kernel comparison of Prop. YM3_I:121.2 also holds for $k = C^{\mathrm{fl}}_{K,s}$ when $a_s \ge d$, with the additional error $e = (d/a_s)^3$. Its constant may be enlarged by a fixed number depending only on the constants of the flat curvature kernels. In particular, its statement for $k = C^{\mathrm{fl}}_{K,K-n}$ has $e = (d/a_{K-n})^3$ whenever $a_{K-n} \ge d$. ■

Proof. The orthogonal decomposition YM3_I:(2227) gives $C^{\mathrm{fl}}_{K,s} = C^{\mathrm{free}}_K - C^{\mathrm{bg}}_{K,s}$. The all-side background estimate of Prop. YM3_I:120.5 is $|C^{\mathrm{bg}}_{K,s}(p,q)| \le C^{\mathrm{bg}}_{\mathrm{tp}}(a_K/a_s)^3$. On the separated supports put $t = d/a_s \le 1$. Thus the two kernels differ by at most $C^{\mathrm{bg}}_{\mathrm{tp}} t^3 a_K^3 d^{-3}$, whereas YM3_I:(2246) bounds the free kernel by $(1 + 3C_{\mathrm{lat}})a_K^3 d^{-3}$. The squared-kernel estimate in the proof of Prop. YM3_I:121.2, obtained by applying $|u^2 - v^2| \le |u - v|(2|v| + |u - v|)$ to its plaquette sum, consequently gives

$$|\mathcal{K}[C^{\mathrm{fl}}_{K,s}] - \mathcal{K}[C^{\mathrm{free}}_K]| \le 324 C^{\mathrm{bg}}_{\mathrm{tp}}(2 + 6C_{\mathrm{lat}} + C^{\mathrm{bg}}_{\mathrm{tp}}) t^3 \|f\|_\infty \|h\|_\infty.$$

Adding its free-kernel conclusion proves the assertion with the stated fixed enlargement, under the same $8a_K \le d$ and $20d \le \ell$ hypotheses. □

When explicit kernel constants are needed, we use

$$C_\star^{\mathrm{fl}} = C_{\mathrm{tp}}^{\mathrm{fl}}(1/\sqrt{3}), \qquad C_\flat^{\mathrm{tm}} = \left(\frac{3\sqrt{3}}{e\delta_{\mathrm{tp}}L}\right)^3. \tag{6.xxi}$$

The first constant is the convergent slice sum of Prop. YM3_I:120.6; the second bounds $\sup_{t\geq 0} t^3 e^{-\delta_{\mathrm{tp}}Lt/\sqrt{3}}$, whose maximum is at $t = 3\sqrt{3}/(\delta_{\mathrm{tp}}L)$. The comparison constant C_{tm} is chosen to cover both Prop. YM3_I:121.2 and Lem. 6.33.

7. The lattice two-loop coefficient

The first correction in the action-density two-point function can be computed in a fixed finite box before studying the interacting continuum remainder. Let $a_p = 1 - \frac{1}{2}\operatorname{tr} U(\partial p)$, and let $\mathsf{B}_0^{[n]}(p,q)$ be the leading Gaussian coefficient in a box Λ_n. The next coefficient is obtained from

$$\mathsf{B}_1^{\mathrm{lat}}(p,q) = \lim_{n\to\infty}\lim_{\beta\to\infty} \beta^3 \left[\langle a_{p+o_n}; a_{q+o_n}\rangle_{\Lambda_n,\beta} - \beta^{-2}\mathsf{B}_0^{[n]}(p+o_n, q+o_n)\right],$$

where the translations o_n put the fixed plaquettes in the interior of the boxes. The semicolon denotes a connected expectation. The order of these two limits is part of the definition.

Sketch F (The coefficient of degree -5). For plaquettes with normals i, j, the large-distance coefficient has the form

$$\mathsf{B}_1^{\mathrm{lat}}(p,q) = \mathsf{U}_{ij}(x(q)-x(p)) + O(r^{-6}\log r), \qquad r = |x(q)-x(p)| \longrightarrow \infty,$$

where U_{ij} is homogeneous of degree -5. Its scalar sum and the relative short-distance coefficient are

$$\sum_{i,j}\mathsf{U}_{ij}(u) = \kappa^{\mathrm{lat}}|u|^{-5}, \qquad \kappa^{\mathrm{lat}} = \frac{3}{2\pi^3}, \qquad c_1 = \frac{\kappa^{\mathrm{lat}}}{c_2} = \frac{8}{3\pi}, \quad c_2 = \frac{9}{16\pi^2}.$$

■□

Sketch of the argument. Expand the Wilson density and the plaquette insertions in a fixed box, retaining all terms that contribute to the coefficient of β^{-3}. A gauge may be used to evaluate the Gaussian integrals; the connected Wilson expectation defining the coefficient is gauge independent. The source organizes the answer into finitely many contraction kernels. The propagators converge as the box grows, so these kernels define $\mathsf{B}_1^{\mathrm{lat}}$.

The important cancellation occurs before estimating their sum. Terms localized at one of the two plaquettes have symbols whose value and first derivatives vanish at zero momentum. Consequently they contribute only at order r^{-6} or smaller. The remaining convolutions have homogeneous terms of degree -5; subtracting those terms leaves the stated logarithmic remainder. Thus the coefficient comes from a finite homogeneous calculation, rather than from a termwise estimate of all lattice sums.

The six scalar homogeneous contributions reduce to a finite residue calculation. Their normalized values sum to $-2/3$, giving

$$\kappa^{\mathrm{lat}} = -\frac{3}{32\pi^3}\,24\left(-\frac{2}{3}\right) = \frac{3}{2\pi^3}.$$

The contraction kernels and their residues are calculated below. The interacting error estimate is developed separately in §6.

For two nonnegative test functions supported in balls of radius d at distance comparable to d, the lattice-to-continuum comparison has relative error $O((a_K/d)(1+|\log(a_K/d)|))$, with the test shapes fixed. This identifies the same c_1 after smearing. It does not interchange the fixed-box weak-coupling expansion with a continuum limit of the interacting measure. That interchange requires the renormalization estimates described in Sketch E.

7.1. **The lattice two-loop coefficient.** This section concerns the truncated two-point function $\langle a_p; a_q\rangle_{n,\beta}$ of the actions of two plaquettes p, q under the Wilson measure of $SU(2)$ lattice gauge theory in a box $\Lambda_n \subset \mathbb{Z}^3$ with Dirichlet boundary data, at large β. We expand it up to order β^{-3}, pass to the infinite lattice in the coefficients of the expansion, determine the behaviour of the coefficient $\mathsf{B}_1^{\mathrm{lat}}(p,q)$ of order β^{-3} at large separation of p and q, and compute the constant κ^{lat} that governs this behaviour. The argument has four stages.

First, in the box Λ_n the change of variables of Faddeev and Popov and Laplace's method give a constant C, depending on n, τ_{gf}, p and q, such that

$$\big|\langle a_p; a_q\rangle_{n,\beta} - \beta^{-2}\,\mathsf{B}_0^{[n]}(p,q) - \beta^{-3}\,\mathsf{B}_1^{[n]}(p,q;\tau_{\mathrm{gf}})\big| \le C\beta^{-4} \qquad (\beta \ge 1),$$

and neither coefficient depends on the gauge parameter $\tau_{\mathrm{gf}} > 0$ (Prop. 7.12).

Second, the coefficients $\mathsf{B}_0^{\mathrm{lat}}$ and $\mathsf{B}_1^{\mathrm{lat}}$ of the lattice $\mathbb{Z}^3$ are defined by the same cumulants of a Gaussian family as those of a box (Def. 7.10). Here $\mathsf{B}_0^{\mathrm{lat}}(p,q) = \frac{\mathsf{n}}{2}\,\Pi^{\mathrm{lat}}(p,q)^2$ by eq. 7.xvii, while $\mathsf{B}_1^{\mathrm{lat}}(p,q)$ is an absolutely convergent sum of lattice graphs with two loops (Lem. 7.27). By the method of images, the coefficients of the plaquettes $p + o_n$, $q + o_n$, translated to the centre o_n of the box, converge to $\mathsf{B}_j^{\mathrm{lat}}(p,q)$ as $n \to \infty$ (Prop. 7.39). Consequently

$$\mathsf{B}_1^{\mathrm{lat}}(p,q) = \lim_{n\to\infty}\ \lim_{\beta\to\infty}\ \beta^3\big[\langle a_{p+o_n}; a_{q+o_n}\rangle_{n,\beta} - \beta^{-2}\,\mathsf{B}_0^{[n]}(p+o_n, q+o_n)\big],$$

the two limits being taken one after the other, first in β and then in n (Thm. X(c)).

Third, a calculus of lattice kernels, combined with identities at zero momentum for the self-energy and for the vertex function, yields functions $\mathsf{U}_{ij} \in C^1(\mathbb{R}^3 \setminus \{0\})$, $i, j \in \{1,2,3\}$, homogeneous of degree -5, and pure constants C, r_0 such that

$$|\mathsf{B}_1^{\mathrm{lat}}(p,q) - \mathsf{U}_{\lambda(p)\lambda(q)}(u)| \le C\,r^{-6}\lg r \quad (r \ge r_0), \qquad \sum_{i,j} \mathsf{U}_{ij}(u) = \kappa^{\mathrm{lat}}|u|^{-5},$$

where $u = x(q) - x(p)$ joins the barycentres of the plaquettes, $r = |u|$, $\lambda(p)$ and $\lambda(q)$ are their dual directions, and the sum is taken over the nine pairs of dual directions (Thm. V). From this we obtain a rate for the convergence of the smeared coefficient $\mathsf{P}_1^{\mathrm{lat}}(f,h;a)$ to $\kappa^{\mathrm{lat}}\mathcal{K}_5(f,h)$ on test pairs (f,h) (Cor. 7.37).

Fourth, we compute κ^{lat}. Six two-loop integrands of a continuum model carry the terms of degree -5 of the lattice classes of graphs; the constant κ^{lat} is tied to their coefficients of $\mathsf{p}^2 \log|\mathsf{p}|$, and these are evaluated by means of finite parts, an expansion by regions and a functional $\mathcal{R}$. The outcome is $\kappa^{\mathrm{lat}} = \frac{3}{2\pi^3}$ (Thm. W), and hence $c_1 := \kappa^{\mathrm{lat}}/\mathsf{c}_2 = \frac{8}{3\pi}$ (Thm. X). The relation of this coefficient calculation to the interacting continuum two-point function is stated separately in §6.1.

The proof below includes the gauge-fixed finite-box expansion, convergence of the lattice graph sums, comparison with the six continuum integrands, and the residue calculation giving the coefficient. All finite-box remainder constants retain their stated volume dependence. The final scope statement records the order of the limits.

7.1.1. *Notation and hypotheses.* We collect the conventions, the standing data and the facts on $SU(2)$ that are used throughout the section, and then summarise its notation.

Definition 7.1 (Standing setting)**.** The following conventions hold throughout this section.

(1) *The group.* We take $G = SU(2)$ and $N = 2$, and we put $\mathsf{n} := N^2 - 1 = 3$. We write tr for the trace of 2×2 matrices, not divided by 2, and for a complex 2×2 matrix W we denote by $|W|$ its operator norm. The group G is a compact subset of $\mathbb{C}^{2\times 2}$ for the distance of this norm. Its normalised Haar measure du is the Borel probability measure on G that is invariant under left translations; it is unique, and it is also invariant under right translations and under $g \mapsto g^{-1}$. We denote by $\mathfrak{g}$ the real vector space of Hermitian traceless 2×2 matrices, with the inner product $\langle X, Y\rangle := \frac{1}{2}\operatorname{tr} XY$ and the norm $|X| := \langle X, X\rangle^{1/2}$, which

on $\mathfrak{g}$ coincides with the operator norm; dY is the Lebesgue measure of the Euclidean space $(\mathfrak{g}, \langle\cdot,\cdot\rangle)$.

The Pauli matrices

$$\mathsf{e}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \mathsf{e}_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \mathsf{e}_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

form an orthonormal basis of $\mathfrak{g}$, and $\mathsf{e}_a\mathsf{e}_b = \delta_{ab}\mathbf{1} + i\sum_c \epsilon_{abc}\mathsf{e}_c$, where ϵ_{abc} is totally antisymmetric in a, b, c with $\epsilon_{123} = 1$: it is the sign of the permutation (a, b, c) of $(1, 2, 3)$, and it vanishes when two of its indices agree. Writing X^a for the coordinates of $X \in \mathfrak{g}$ in this basis, we define the cross product $X \times Y := \sum_{a,b,c} \epsilon_{abc} X^a Y^b \mathsf{e}_c$. For $X, Y \in \mathfrak{g}$ we then have

$$XY = \langle X, Y\rangle\mathbf{1} + i\, X \times Y, \qquad [X, Y] = 2i\, X \times Y, \qquad e^{iX} = \cos|X|\,\mathbf{1} + i\,\frac{\sin|X|}{|X|}\,X,$$
$$\tfrac{1}{2}\operatorname{tr} e^{iX} = \cos|X|;$$

the third identity follows from $X^2 = |X|^2\mathbf{1}$, its right side being $\mathbf{1}$ at $X = 0$. Colour indices $a, a', c, c', \ldots$, as well as the index b in ϵ_{abc}, take their values in $\{1, 2, 3\}$.

(2) *Facts on G; smooth maps.* Put $\sigma_{\mathrm{H}} = \frac{1}{2\pi^2}$; the entire function $\sigma_1(w) = \big(\sum_{n\geq 0}(-w)^n/(2n+1)!\big)^2$, which satisfies $\sigma_1(|Y|^2) = (\sin|Y|/|Y|)^2 \in (0, 1]$ for $|Y| < \pi$; and, for $Y \in \mathfrak{g}$, the linear map $\mathsf{dE}(Y)H = \int_0^1 e^{isY} H e^{-isY}\,\mathrm{d}s$ of $\mathfrak{g}$. For $\varrho > 0$ we write $\mathbb{B}_\varrho = \{Y \in \mathfrak{g} : |Y| < \varrho\}$. The following formulas will be used in the finite-box change of variables. Their proofs follow this definition.

- For $X_1, \ldots, X_n \in i\mathfrak{g}$ one has $|e^{X_1}\cdots e^{X_n} - \mathbf{1}| \leq e^{|X_1|+\cdots+|X_n|} - 1$, all norms being operator norms as in (1). The map $Y \mapsto e^{iY}$ is a homeomorphism of $\mathbb{B}_\pi$ onto $G \setminus \{-\mathbf{1}\}$, whose inverse is denoted by $\frac{1}{i}\log$.
- For $Y, H \in \mathfrak{g}$, $\frac{d}{ds}e^{i(Y+sH)}\big|_{s=0} = i\big(\mathsf{dE}(Y)H\big)e^{iY}$; more generally, $\frac{d}{ds}e^{iZ(s)}\big|_{s=0} = i\big(\mathsf{dE}(Z(0))Z'(0)\big)e^{iZ(0)}$ for every curve $s \mapsto Z(s)$ in $\mathfrak{g}$ that is differentiable at $s = 0$. Moreover $\det \mathsf{dE}(Y) = \sigma_1(|Y|^2)$.
- For every bounded Borel function F on G, $\int_G F\,\mathrm{d}u = \sigma_{\mathrm{H}} \int_{\mathbb{B}_\pi} F(e^{iY})\,\sigma_1(|Y|^2)\,\mathrm{d}Y$. Further, $\int_{\mathfrak{g}} e^{-\frac{1}{2}|Y|^2}\,\mathrm{d}Y = (2\pi)^{3/2}$.

Smooth means C^∞. A map f between subsets of finite-dimensional real vector spaces is smooth if every point of its domain has an open neighbourhood O in the ambient space and a C^∞ map $\tilde{f}$ on O such that $f = \tilde{f}$ on the intersection of O with the domain of f; it is a diffeomorphism if it is bijective and both f and f^{-1} are smooth. Composites of smooth maps are smooth, and for maps between open sets these are the usual notions. The group G and its powers G^m lie in powers of $\mathbb{C}^{2\times 2}$; the product of G and the inversion $g \mapsto g^{-1} = g^*$ are smooth, since they are restrictions of real polynomial maps. Finally, $Y \mapsto e^{iY}$ is a diffeomorphism of $\mathbb{B}_\pi$ onto $G \setminus \{-\mathbf{1}\}$; the proof is given after the present definition.

(3) *Standard facts.* The following standard facts are used without further reference.

- *Measure and integration:* the existence of Haar measure on G; the theorems of Fubini and Tonelli and the monotone and dominated convergence theorems, for integrals and for sums; the Borel measurability of partial integrals; the change of variables formula for a diffeomorphism between open subsets of $\mathbb{R}^m$, in particular for translations, reflections, rotations, dilations and polar coordinates in $\mathbb{R}^3$ and in $\mathbb{R}^6$; the Cauchy–Schwarz inequality; the continuity of translations on $L^2(\mathbb{T})$; Gauss' divergence theorem on a ball and on a cube less a ball. An integer matrix of determinant ± 1 acting on $(\mathbb{R}^3)^{\mathsf{m}}$ induces a measure-preserving bijection of $\mathbb{T}^{\mathsf{m}}$.

  For a function that is continuous on an interval $[a, b]$ or on a compact cube, the Riemann sums over equal subintervals or equal subcubes converge to the integral as the mesh tends to 0. Integration by parts holds in one variable; on $\mathbb{R}^3$, when one of the two

factors has a compact support or lies in the Schwartz space $\mathcal{S}(\mathbb{R}^3)$ of rapidly decreasing smooth functions; and over a period, for smooth periodic functions, where no boundary terms arise.

- *Calculus and topology:* the Leibniz rule, the chain rule, the mean value theorem, the intermediate value theorem and the fundamental theorem of calculus; Taylor's formula with integral remainder, in one and in several variables; the inverse function theorem for C^∞ maps; the Heine–Borel theorem; the density of $\mathbb{Q}^3$ in $\mathbb{R}^3$. A function with continuous partial derivatives is of class C^1. Differentiation under the integral sign with respect to a real or a complex parameter is allowed when the integrand and its derivative have integrable majorants that are locally uniform in the parameter, and a series of C^j functions may be differentiated termwise up to the order j when the suprema of the derivatives of orders $\le j$ are summable. A continuous function on a compact set is bounded and uniformly continuous and attains its minimum, and a finite product of compact metric spaces and a closed subset of a compact metric space are compact.
- *Series:* the power series of exp, cos and sin, the series of $\log(1+w)$ and the binomial series of $(1+w)^{-t}$ for $|w|<1$, and the Cauchy product; the inequalities $\frac{2}{\pi}t \le \sin t \le t$ for $t \in [0, \frac{\pi}{2}]$. A function holomorphic on a disc about 0 is the sum of its Taylor series on that disc, and such a series may be evaluated at a square matrix of norm smaller than its radius of convergence, sums and products of series then giving sums and products of the matrices. For $\|\mathsf{A}\|<1$ we have the Neumann series $(\mathbf{1}+\mathsf{A})^{-1} = \sum_{n\ge 0}(-\mathsf{A})^n$ and the expansion $\log\det(\mathbf{1}+\mathsf{A}) = \sum_{n\ge 1}\frac{(-1)^{n-1}}{n}\operatorname{tr}\mathsf{A}^n$.
  Formal power series in finitely many variables form a ring, in which division by a series with constant term 1 is possible, and exp, applied to series without constant term, and log, applied to series with constant term 1, are mutually inverse. Real analytic functions on open subsets of $\mathbb{R}^m$, with real, complex or matrix values, are given locally by absolutely convergent power series, are smooth, are locally the sums of their homogeneous Taylor polynomials, and are stable under sums, products, composites and quotients with non-vanishing denominator; examples are the polynomials, the restrictions of holomorphic functions to real arguments, log on $(0,\infty)$, and a power series as above evaluated at a matrix with real analytic entries and norm smaller than the radius of convergence.
- *Linear algebra:* trace, determinant, adjoint, orthonormal bases and the polarisation of a quadratic form. A positive definite operator is invertible; the determinant is an affine function of a rank-one summand; a polynomial vanishing on a non-empty open set is zero; and Lagrange's identity $\sum_{\mu<\nu}|v_\mu w_\nu - v_\nu w_\mu|^2 = |v|^2|w|^2 - |v^*w|^2$ holds for $v, w \in \mathbb{C}^3$. The Cayley map $\mathsf{A} \mapsto (\mathbf{1}-\mathsf{A})(\mathbf{1}+\mathsf{A})^{-1}$ is continuous from the real antisymmetric 3×3 matrices onto the rotations without eigenvalue -1, and these rotations are dense in $SO(3)$.
- *Fourier analysis:* Plancherel's theorem on $\mathbb{R}^3$ and for the map $L^2(\mathbb{T}, d^\circ k) \to \ell^2(\mathbb{Z}^3)$; the uniqueness of the Fourier coefficients of an integrable function on a torus; the Poisson summation formula for functions in $C_c^\infty(\mathbb{R}^D)$, the smooth functions on $\mathbb{R}^D$ with a compact support; and the finite inversion formula

$$f(z) = (2n)^{-3} \sum_{k \in \frac{\pi}{n}\mathbb{Z}^3 \cap (-\pi,\pi]^3} e^{ik\cdot z} \sum_{z'} e^{-ik\cdot z'} f(z')$$

  for functions f on $(\mathbb{Z}/2n)^3$, where z' ranges over $(\mathbb{Z}/2n)^3$. The inverse Fourier transform $\mathcal{F}^{-1}$ of (5) maps $\mathcal{S}(\mathbb{R}^3)$ into itself, and $\mathcal{F}^{-1}(x_j\phi) = -i\partial_{p_j}\mathcal{F}^{-1}\phi$ when ϕ is written as a function of x and $\mathcal{F}^{-1}\phi$ as a function of p; further, $\widehat{\partial_j w} = ip_j\hat{w}$ for every continuously

differentiable function w with a compact support, and $\int_{\mathbb{R}^3} e^{-tp^2+ip\cdot x}\,\mathrm{d}p = (\pi/t)^{3/2}e^{-x^2/4t}$ for $t>0$.

- *Probability:* for every positive semidefinite covariance C there is a centred Gaussian family (ϕ_i) with covariance C; its law is determined by C, and $\langle e^{\sum_i s_i\phi_i}\rangle = e^{\frac12\sum_{i,j}s_is_j\mathsf{C}(i,j)}$ for real s_i, finitely many of them non-zero. If H is a real symmetric positive definite $m\times m$ matrix, then under the probability measure on $\mathbb{R}^m$ with a density proportional to $e^{-\frac12(z,Hz)}$ the coordinates of $\mathbb{R}^m$ form a centred Gaussian family with the covariance H^{-1}.
- *Complex analysis:* the identity theorem on connected open sets, and Euler's Gamma function, which we write Γ_{E}: it is given by $\Gamma_{\mathrm{E}}(z)=\int_0^\infty t^{z-1}e^{-t}\,\mathrm{d}t$ and is holomorphic for $\operatorname{Re}z>0$, it satisfies $\Gamma_{\mathrm{E}}(z+1)=z\,\Gamma_{\mathrm{E}}(z)$ and $\Gamma_{\mathrm{E}}(\frac12)=\sqrt{\pi}$, and $1/\Gamma_{\mathrm{E}}$ extends to an entire function.

(4) *Three functions.* For $x\in\mathbb{R}^D$ we put $\langle x\rangle := \max\{1,|x|\}$. For $t>0$ we put $\lg t := 1+|\log t|$; then $\lg t\ge 1$, $\lg(1/t)=\lg t$, and $\lg(st)\le \lg s\,\lg t$ for $s,t>0$. Finally $\log_+ t := \max\{0,\log t\}$, $\log_+\infty:=\infty$, and $t/0:=\infty$ for $t>0$.

(5) *Fourier conventions.* We put $\mathbb{T}:=\mathbb{R}^3/2\pi\mathbb{Z}^3$ and identify functions on $\mathbb{T}$ with the functions on $\mathbb{R}^3$ that are periodic with the periods $2\pi e_\mu$, $\mu=1,2,3$; integrals over $\mathbb{T}$ are integrals over $[-\pi,\pi]^3$. We write $|k|_{\mathbb{T}} := \operatorname{dist}(k,2\pi\mathbb{Z}^3)$, which equals $|k|$ on $[-\pi,\pi]^3$, and $d^\circ k := (2\pi)^{-3}dk$. We put $\mathsf{k}_\mu(k):=e^{ik_\mu}-1$ and $\mathsf{k}(k):=(\mathsf{k}_\mu(k))_\mu\in\mathbb{C}^3$. Then

$$|\mathsf{k}(k)|^2=\sum_\mu(2-2\cos k_\mu),\qquad \frac{4}{\pi^2}|k|^2\le|\mathsf{k}(k)|^2\le|k|^2\quad\text{on }[-\pi,\pi]^3,$$

the inequalities following from $2-2\cos k_\mu=4\sin^2\frac{k_\mu}{2}$ and the bounds on $\sin t$ in (3). The *kernel with the symbol* $\sigma\in L^1(\mathbb{T})$ is the function $K(x):=\int_{\mathbb{T}}e^{ik\cdot x}\sigma(k)\,d^\circ k$ of $x\in\mathbb{Z}^3$. A bound on a symbol or on an integrand that is said to hold "on $[-\pi,\pi]^3$" or "for $|\kappa|\le c_0$" holds almost everywhere if the function is an L^1-class, and at every point $\ne0$ if the function is given by a formula. For a function ϕ on $\mathbb{R}^D$ that lies in the Schwartz space $\mathcal{S}(\mathbb{R}^D)$, or is continuous with a compact support, we put $\hat\phi(\kappa):=\int e^{-i\kappa\cdot\zeta}\phi(\zeta)\,\mathrm{d}\zeta$, and on $\mathbb{R}^3$ we put $(\mathcal{F}^{-1}\phi)(x):=\int e^{ip\cdot x}\phi(p)\,d^\circ p$. On $\mathbb{Z}^D$ we use the differences $(\nabla_j f)(\zeta):=f(\zeta+e_j)-f(\zeta)$ and, for a multi-index γ, $\nabla^\gamma:=\prod_j\nabla_j^{\gamma_j}$.

(6) *Data and constants.* The data of a statement are the real numbers and integers quantified in it, and each of them may take every value in its range. The ranges are $\tau_{\mathrm{gf}}>0$ for the gauge parameter (abbreviated τ where this is said), $\beta>0$, $a>0$, $\mathsf{b}>0$, $\mathsf{d}>0$, $\vartheta\in(0,1)$, $\Lambda>1$, $L>1$ for the real number L of Lem. 7.53, and $n\ge2$ for the side of the box Λ_n. The *tools* are the eight statements marked (K), (S), (P), (L), (D), (E), (V) and (T), namely Lems. 7.17–7.22, Prop. 7.23 and Lem. 7.38; the ranges of their data are those stated in their hypotheses.

A relation between data holds only where a statement imposes it; thus $a\le\mathsf{d}$ is assumed in Cor. 7.37 and in Thm. X(b). For each constant whose existence a statement asserts, such as C, C_γ, c_0, r_0, ϱ_1, ϱ_2, ϱ_3, the statement specifies the quantities on which the constant depends. A *pure constant* is a constant that depends on nothing. In proofs, $C, C', c, \dots$ denote positive constants whose values may change from one occurrence to the next and which depend only on the quantities on which the statement in question allows its constants to depend.

(7) *Length scale.* We work on the lattice $\mathbb{Z}^3$ of spacing 1. For a function f on $\mathbb{R}^3$, $\operatorname{supp}f$ is the closure of $\{f\ne0\}$, $\|f\|_\infty=\sup|f|$, $\operatorname{Lip}f=\sup_{x\ne y}|f(x)-f(y)|/|x-y|$, $\int f$ is the Lebesgue integral, and $\mathbb{B}(x,\rho)=\{y:|y-x|\le\rho\}$.

(8) *Brackets, indices and notations with several meanings.* On complex forms, the bracket $(\cdot,\cdot)$ of Def. 7.2(2) is Hermitian and antilinear in its first argument: it is the sum of the

sesquilinear extension of $\langle\cdot,\cdot\rangle$, evaluated at the values of the two forms. On $\mathfrak{g}\otimes\mathbb{C}$, on the other hand, $\langle X,Y\rangle$ denotes the complex bilinear extension. For two sites x,x', $\langle x,x'\rangle$ is a bond. The brackets $\langle\cdot\rangle_{n,\beta}$, $\langle\cdot\rangle_{\mathcal{W},\beta}$ and $\langle\cdot\rangle_0$ (Def. 7.3, Lem. 7.11 and Def. 7.8) are means; the same brackets with entries separated by semicolons are cumulants; $\langle x\rangle$ is the function of (4).

When a,a',c,c' occur as upper indices of forms or of legs, as indices of e, δ, ϵ or ∂, or as the second entry of a pair (b,a), they are colour indices; b,b' denote bonds. Forms and site functions are $\mathfrak{g}$-valued unless they are said to be $\mathbb{R}$-valued or complex. Finally, D denotes a dimension, equal to 3 or to 6.

Several notations have more than one meaning in this section, and the following rules tell the meanings apart. Without index, Γ ranges over $\{1,\dots,6\}$ and numbers the integrands F_Γ of Def. 7.32; Γ_q is the kernel of Def. 7.13(5), and Γ_{E} is Euler's function of (3). With the index n, Λ_n is the box; in Lem. 7.20 and wherever that lemma is applied, Λ is the function $\Lambda(\kappa)$ defined there, also when its argument is not written; otherwise Λ is a real number >1. The momenta are $\kappa,\kappa_1,\kappa_2,\dots$, the κ_J are the cumulants of Def. 7.8(2), κ^{lat} and $\kappa^{\mathrm{lat}}_\Gamma$ are numbers, and in the statement of Lem. 7.36 κ alone is a constant.

Before Def. 7.16, σ_1 is the entire function of (2); from that Definition on, $\sigma,\sigma_1,\sigma_2,\dots$ denote symbols, and an s or s_i that accompanies a symbol or a line is its degree, whereas $(s_1,s_2,s_3,s_4)=(+,+,-,-)$ are the signs of the four bonds of a plaquette (Lem. 7.5). Further, n is the number 3 of (1), except in Lem. 7.26(a), (b) and in its proof, where it is a matrix $(n_{vv'})$ of integers. As an index, β is a direction or a multi-index; in Prop. 7.46 and in the proof of Prop. 7.50 it is a real coefficient; otherwise it is the coupling $\beta>0$.

As a datum, $\vartheta\in(0,1)$, whereas the argument ϑ of the function j of Lem. 7.14 is a real variable. As a datum, the integer $n\geq 2$ is the side of the box Λ_n; other integers n, such as the degree $n=s+3$ of a line, the index of the class $\mathfrak{K}(n)$, the exponent $n\in\{4,5\}$ of the sums s_n and the index $n\in\mathbb{Z}$ of the spaces $\mathcal{L}_n$, are introduced where they occur. Besides the kernel q of Lem. 7.22 and the kernels derived from it, q is a vector of $\mathbb{R}^3$ where it is the argument of ρ or of a symbol, and $\mathsf{q}(t)$ is the polynomial of Lem. 7.48.

In Def. 7.47(2), Lem. 7.48 and Lem. 7.49, $\mathsf{a},\mathsf{b},\mathsf{c}$ are integers, while the datum $\mathsf{b}>0$ of (6) is a real number; in $\mathsf{d}_{b,a}$ (Lem. 7.28(I)) the d belongs to the name of a polynomial; and τ is a variable of integration in the proofs of Lems. 7.49 and 7.53. With an argument, $\mathcal{S}(\mathbb{R}^D)$ is the Schwartz space; with an upper index, $\mathcal{S}^s$ is a class of symbols (Def. 7.16(2)).

(9) *Further definitions.* The lattice and its forms are defined in Def. 7.2, the finite-box measure in Def. 7.3, and Gaussian cumulants in Def. 7.8. The test pairs and kernels are defined in §6.1; their bounds are Lem. 6.1.

■

Proof of the group formulas in (2). The product bound follows by writing $e^{X_j}=1+Z_j$, using $|Z_j|\leq e^{|X_j|}-1$, and expanding the product: the sum of the norms of its nonconstant terms is at most $\prod_j(1+|Z_j|)-1\leq e^{\sum_j|X_j|}-1$. Every element of $SU(2)$ has the unique form $a_0\mathbf{1}+i\sum_a a_a\mathsf{e}_a$ with real coefficients and $a_0^2+\sum_a a_a^2=1$. The exponential formula in (1) therefore gives a bijection from $\mathbb{B}_\pi$ to $G\setminus\{-\mathbf{1}\}$. Its smooth inverse is verified below.

For matrices A,B, differentiate $s\mapsto e^{s(A+rB)}e^{(1-s)A}$ and integrate to obtain

$$e^{A+rB}-e^A=r\int_0^1 e^{s(A+rB)}Be^{(1-s)A}\,\mathrm{d}s.$$

Dividing by r and passing to zero proves the stated differential formula. If $\vartheta=|Y|>0$, conjugation by e^{isY} fixes the line spanned by Y and acts on its orthogonal plane as a rotation through $2s\vartheta$. Indeed, for $H\perp Y$ the Pauli identities give $HY=-YH$ and $e^{isY}He^{-isY}=\cos(2s\vartheta)H+\sin(2s\vartheta)iYH/\vartheta$.

Consequently the determinant of its average is

$$\det \mathsf{dE}(Y) = \left(\frac{\sin(2\vartheta)}{2\vartheta}\right)^2 + \left(\frac{1-\cos(2\vartheta)}{2\vartheta}\right)^2 = \left(\frac{\sin\vartheta}{\vartheta}\right)^2.$$

The value at zero follows by continuity.

For completeness, this determinant also identifies the Haar density. Let μ be the pushforward of $\sigma_1(|Y|^2)dY$ on $\mathbb{B}_\pi$ under the exponential. Right translation by $u \in G$ induces $\tau_u(Y) = i^{-1}\log(e^{iY}u)$ wherever its argument is not $-\mathbf{1}$. It is a diffeomorphism after removing at most one point from each chart, with inverse $\tau_{u^{-1}}$. Differentiating $e^{i\tau_u(Y)} = e^{iY}u$ gives

$$\mathsf{dE}(\tau_u(Y))D\tau_u(Y) = \mathsf{dE}(Y), \qquad \sigma_1(|\tau_u(Y)|^2)|\det D\tau_u(Y)| = \sigma_1(|Y|^2).$$

The change of variables formula, with those null points omitted, shows that μ is right invariant. Its mass is $4\pi\int_0^\pi \sin^2\vartheta\,\mathrm{d}\vartheta = 2\pi^2$. Uniqueness of normalized Haar measure proves the integration formula. The Gaussian integral is the product of the three one-dimensional Gaussian integrals in the orthonormal Pauli basis. □

Smoothness of the inverse exponential in (2). Let $\mathrm{cs}(w) := \sum_{n\geq 0}(-w)^n/(2n)!$ and $\mathrm{sn}(w) := \sum_{n\geq 0}(-w)^n/(2n+1)!$. These series define entire functions which are real on $\mathbb{R}$, and comparing them term by term with the series of cos and sin, and differentiating term by term, we obtain

$$\mathrm{cs}(\varphi^2) = \cos\varphi, \qquad \mathrm{sn}(\varphi^2) = \frac{\sin\varphi}{\varphi}, \qquad \sigma_1 = \mathrm{sn}^2, \qquad \mathrm{cs}' = -\tfrac{1}{2}\,\mathrm{sn},$$

for real φ, with $\varphi \neq 0$ in the second identity; moreover $\mathrm{sn}(0) = 1$. By (1), $e^{iY} = \mathrm{cs}(|Y|^2)\mathbf{1} + i\,\mathrm{sn}(|Y|^2)\,Y$ for $Y \in \mathfrak{g}$, so the map $Y \mapsto e^{iY}$ is smooth on $\mathfrak{g}$. On the interval $(-\infty, \pi^2)$ the function sn is positive: for $w < 0$ all terms of its series are positive, and for $0 \leq w < \pi^2$ it equals $\sin\sqrt{w}/\sqrt{w} > 0$, with the value 1 at $w = 0$. Hence cs is strictly decreasing on $(-\infty,\pi^2)$ and its derivative does not vanish there. Moreover $\mathrm{cs}(w) \to +\infty$ as $w \to -\infty$, and $\mathrm{cs}(\pi^2) = \cos\pi = -1$. By the intermediate value theorem cs maps $(-\infty,\pi^2)$ onto $(-1,\infty)$, and by the inverse function theorem the inverse map $\mathrm{cs}^{-1} : (-1,\infty) \to (-\infty,\pi^2)$ is smooth.

Now let $W = e^{iY}$ with $Y \in \mathbb{B}_\pi$. Since $\operatorname{tr} Y = 0$, $\operatorname{tr}\mathbf{1} = 2$ and $Y^* = Y$, the formula above gives $\frac{1}{2}\operatorname{tr} W = \mathrm{cs}(|Y|^2)$ and $\frac{1}{2i}(W - W^*) = \mathrm{sn}(|Y|^2)\,Y$. As $|Y|^2 \in [0,\pi^2)$, the first identity yields $|Y|^2 = \mathrm{cs}^{-1}(\frac{1}{2}\operatorname{Re}\operatorname{tr} W)$, and the second then yields

$$\tfrac{1}{i}\log W = \Big[\mathrm{sn}\big(\mathrm{cs}^{-1}(\tfrac{1}{2}\operatorname{Re}\operatorname{tr} W)\big)\Big]^{-1}\,\tfrac{1}{2i}(W - W^*) \qquad (W \in G\setminus\{-\mathbf{1}\}).$$

The right side is defined and smooth on the open set $\{W \in \mathbb{C}^{2\times 2} : \frac{1}{2}\operatorname{Re}\operatorname{tr} W > -1\}$: there cs^{-1} is smooth and takes its values in $(-\infty,\pi^2)$, where sn is positive. This open set contains $G\setminus\{-\mathbf{1}\}$, because $\frac{1}{2}\operatorname{tr} e^{iY} = \cos|Y| > -1$ for $Y \in \mathbb{B}_\pi$. Hence the inverse map $\frac{1}{i}\log$ is smooth, and the bijection $Y \mapsto e^{iY}$ of $\mathbb{B}_\pi$ onto $G\setminus\{-\mathbf{1}\}$ given by (2) is a diffeomorphism. □

For reference we summarise the notation introduced in the rest of the section, together with the places where it is defined.

Lattice and boxes (Def. 7.2). We write $(x,\mu) := \langle x, x+e_\mu\rangle$ for the positively oriented bonds and, for a 1-form A, $A_\mu(x) := A((x,\mu))$, with components $A^a_\mu(x)$. Bonds b and plaquettes p have the base points b_- and x_p; the plaquette $p = p_{\mu\nu}(x)$ has the barycentre $x(p) = x + \frac{1}{2}(e_\mu + e_\nu)$ and the dual direction $\lambda(p) \in \{1,2,3\}\setminus\{\mu,\nu\}$; $p_{[i]}(x)$ is the plaquette with base point x and dual direction i; and $b+y$, $p+y$ are translates. The differences are

$$(\mathrm{d}\lambda)(b) := \lambda(b_+) - \lambda(b_-), \qquad (\mathrm{d}A)(p) := A(b_1) + A(b_2) - A(b_3) - A(b_4), \qquad (\mathrm{d}^*A)(x) := \sum_\mu \big[A_\mu(x-e_\mu) - A_\mu(x)\big],$$

and $(\cdot,\cdot)$, $\|\cdot\|$ are the sums of $\langle\cdot,\cdot\rangle$ over sites, over positively oriented bonds or over plaquettes; further $\bar\lambda(b) := \frac12(\lambda(b_-)+\lambda(b_+))$ and $(A\times\bar\lambda)(b) := A(b)\times\bar\lambda(b)$. The box is $\Lambda_n := \{0,\dots,n\}^3$, with $\Lambda_n^\circ := \{1,\dots,n-1\}^3$ and $\partial\Lambda_n := \Lambda_n\setminus\Lambda_n^\circ$. Attached to it are the set Bd_n of *free bonds*, which are the positively oriented bonds with both end points in Λ_n and at least one of them in the interior Λ_n°, the plaquettes Pl_n, the configurations $\mathcal{U}_n \cong G^{\mathsf{Bd}_n}$, the forms Form_n, the site functions X_n, the group $\mathcal{G}_n := \{u:\Lambda_n\to G,\ u=\mathbf{1} \text{ on } \partial\Lambda_n\}$, and the $\mathbb{R}$-valued analogues $\mathsf{Form}_n^{\mathbb{R}}$, $\mathsf{X}_n^{\mathbb{R}}$ (Def. 7.2(3)).

Finally $|A|_\infty := \max_b|A(b)|$, $|\lambda|_\infty := \max_x|\lambda(x)|$, and the centre of the box is $o_n := (\lfloor n/2\rfloor,\lfloor n/2\rfloor,\lfloor n/2\rfloor)$.

Measure (Def. 7.3). The Wilson measure of the box is

$$d\mu_{n,\beta}(U) := Z_{n,\beta}^{-1}\exp\Big[-\beta\sum_{p\in\mathsf{Pl}_n}a_p(U)\Big]\prod_{b\in\mathsf{Bd}_n}dU(b),$$

with the mean $\langle\cdot\rangle_{n,\beta}$ and the truncated two-point function $\langle X;Y\rangle_{n,\beta} = \langle XY\rangle_{n,\beta} - \langle X\rangle_{n,\beta}\langle Y\rangle_{n,\beta}$.

Chart (Def. 7.4, Lem. 7.5). The coefficients $\mathsf{a}_p^{(m)}(A)$ are given by $a_p(e^{itA}) = \sum_{m\ge0}t^m\,\mathsf{a}_p^{(m)}(A)$. We put $A^\lambda(b) := \frac1i\log(e^{i\lambda(b_-)}e^{iA(b)}e^{-i\lambda(b_+)})$; the gauge function is $A\mapsto \mathrm{d}^*A$, and the Faddeev–Popov operator is $\mathsf{M}(A)\lambda := \frac{d}{ds}\mathrm{d}^*(A^{s\lambda})\big|_{s=0}$, with $\mathsf{M}(tA)=\sum_{j\ge0}t^j\mathsf{M}^{(j)}(A)$. The signs of the four bonds of a plaquette are $(s_1,s_2,s_3,s_4) := (+,+,-,-)$. With $Z := \frac1i\log\partial(e^{itA})(p)$ we put $\mathfrak{z}_j := \frac{1}{(j+1)!}\partial_t^{j+1}Z\big|_{t=0}$, and $\mathfrak{z}_j(p)$ denotes its value at the values of A on the bonds of p. Finally $\mathsf{a}_Y := -2\,Y\times\cdot$, and $\mathsf{r}(Y)$ is the remainder of fourth order in eq. 7.vi.

Slice (Lem. 7.6). We write $S(U) := \sum_{p\in\mathsf{Pl}_n}a_p(U)$, $\mathsf{T}_n := \{\langle x-e_1,x\rangle : x\in\Lambda_n^\circ\}$, $\mathcal{U}_n^{\mathsf{T}} := \{V\in\mathcal{U}_n : V=\mathbf{1} \text{ on } \mathsf{T}_n\}$ and $\mathsf{Form}_n^{\mathsf{T}} := \{A\in\mathsf{Form}_n : A=0 \text{ on } \mathsf{T}_n\}$, and j for the map $(u,V)\mapsto V^u$. The lemma provides the radii $\varrho_1,\varrho_2,\varrho_3$, the set $\mathfrak{D}$, the map $\Xi:(\lambda,A)\mapsto A^\lambda$ and the set $\mathcal{W} := \Xi(\mathfrak{D})$.

Action, propagators and vertices (Def. 7.7). The upper index lat refers to $\mathbb{Z}^3$ and the upper index $[n]$ to Λ_n. We put $\mathsf{S}_2(A) := \frac12\|\mathrm{d}A\|^2 + \frac{1}{2\tau_{\mathrm{gf}}}\|\mathrm{d}^*A\|^2$, $\mathsf{C}_{\tau_{\mathrm{gf}}} = (\mathrm{d}^*\mathrm{d}+\tau_{\mathrm{gf}}^{-1}\mathrm{d}\mathrm{d}^*)^{-1}$ and $\mathsf{G} = (\mathrm{d}^*\mathrm{d})^{-1}$, and

$$\mathsf{E}_p(b) := \sum_i s_i\,\mathsf{C}_{\tau_{\mathrm{gf}}}(b_i(p),b),\qquad \Pi(p,q) := \sum_j s_j\,\mathsf{E}_p(b_j(q)),\qquad \mathsf{g}(b) := \mathsf{G}(b_+,b_+)+\mathsf{G}(b_-,b_-)-2\mathsf{G}(b_+,b_-),$$

$$\mathsf{D}(b;b') := \tfrac12\sum_{x\in\{b'_-,b'_+\}}[\mathsf{G}(x,b_+)-\mathsf{G}(x,b_-)].$$

The vertices are $\mathsf{S}_3 := \sum_p\mathsf{a}_p^{(3)}$, the quartic Wilson vertex $\mathsf{S}_4 := \sum_p\mathsf{a}_p^{(4)}$, and

$$\mathsf{V}_2 := \mathsf{S}_4 + \frac13\sum_b|A(b)|^2 + \frac23\sum_b\mathsf{g}(b)|A(b)|^2 - 4\sum_{b,b'}\mathsf{D}(b;b')\mathsf{D}(b';b)\langle A(b),A(b')\rangle,$$

whose last three terms are the *Haar term*, which is the quadratic term contributed by the Haar density in the exponential chart, the ghost tadpole and the ghost loop.

Gaussian calculus (Def. 7.8, Lem. 7.26). The mean $\langle\cdot\rangle_0$ belongs to the centred Gaussian family with $\langle A^a(b)A^{a'}(b')\rangle_0 = \delta_{aa'}\mathsf{C}^\Omega_{\tau_{\mathrm{gf}}}(b,b')$, and $\langle X_1;\dots;X_r\rangle_0$ is the joint cumulant of polynomials $X_1,\dots,X_r$ in this family. A *leg* is a linear form that occurs as a factor of a monomial of a vertex: a *plain leg* is a variable $A^a(b)$, and a *curl leg* is $(\mathrm{d}A)^a(p) = \sum_i s_iA^a(b_i(p))$. The *standard writing* of $\mathsf{a}_p^{(2)}$, $\mathsf{a}_p^{(3)}$ and $\mathsf{a}_p^{(4)}$ is a fixed way of writing them as sums of monomials in plain and curl legs, chosen so that the counting of powers takes the curl legs into account.

On $\mathbb{Z}^3$, S_3 and V_2 are *formal sums of local terms*: the local terms of S_3 are the $\mathsf{a}_p^{(3)}$, and those of V_2 are the $\mathsf{a}_p^{(4)}$, the terms of the sums over bonds b and the terms of the sum over pairs (b,b'); the cumulants of formal sums are defined term by term under a condition of absolute convergence (Def. 7.8(3)). Further $\mathsf{F}[\![X_1,\dots,X_m]\!] := \partial_{s_1}\cdots\partial_{s_m}\mathsf{F}(\sum_i s_iX_i)$; for $\mathsf{i}=(b,a)$, δ_{i} is the form equal to e_a at b and to 0 at the other positively oriented bonds; $\mathsf{C}(\mathsf{i},\mathsf{i}') := \delta_{aa'}\mathsf{C}(b,b')$ and $\mathsf{E}_p^a := \mathsf{E}_p(\cdot)\mathsf{e}_a$; the

values $\mathsf{val}(\mathsf{n})$ and the *classes* $\mathsf{T}_\mathrm{I}, \mathsf{T}_1, \mathsf{T}_4, \mathsf{T}_{4'}, \mathsf{T}_5, \mathsf{T}_6$, which are the topologies of the connected graphs of B_1, are those of Lem. 7.26; and $\mathsf{t}_P(X,Y,Z) := -\langle \mathrm{d}X(P), \sum_{i<j} s_i s_j\, Y(b_i) \times Z(b_j)\rangle$.

Coefficients (Def. 7.10). We put $\mathsf{B}_0^\Omega(p,q) := \langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(2)}\rangle_0$; $\mathsf{B}_1^\Omega(p,q;\tau_\mathrm{gf})$ is the combination of seven cumulants in eq. 7.xviii; $\mathsf{B}_j^{[n]}$ and $\mathsf{B}_j^\mathrm{lat}$ are these coefficients for $\Omega = \Lambda_n$ and for $\Omega = \mathbb{Z}^3$; and

$$\mathsf{B}_1^\mathrm{lat}(p,q) := \mathsf{B}_1^\mathrm{lat}(p,q;1), \qquad \mathsf{P}_1^\mathrm{lat}(f,h;a) := a \sum_{p,q} f(a\,x(p))\, h(a\,x(q))\, \mathsf{B}_1^\mathrm{lat}(p,q).$$

Laplace's method (Lem. 7.11). Here $\langle\cdot\rangle_{\mathcal{W},\beta}$ is the mean for the density proportional to $e^{-\beta\Phi-\Theta}$ on $\mathcal{W}$; we put $W_j := \Phi_{j+2} + \Theta_j$, $X_j := X'_{j+2}$ and $Y_j := Y'_{j+2}$, and γ_0, γ_2 are the two coefficients defined in the lemma.

Zero momentum (Def. 7.13, Lems. 7.14 and 7.15). We write $\partial_{b,a}$ for the derivative with respect to $A^a(b)$. The matrices $\mathsf{N}(B)$ and $\mathsf{h}_q(B)$ are the Hessians at $A = B$ of $\sum_p a_p(e^{iA}) + \frac{1}{2\tau_\mathrm{gf}} \sum_x |\mathrm{d}^* A(x)|^2$ and of $a_q(e^{iA})$, respectively. The constant abelian field is $Y_\gamma((x,\mu)) := \gamma_\mu \mathsf{e}_3$. We put $\mathsf{e}_\pm := 2^{-1/2}(\mathsf{e}_1 \pm i\mathsf{e}_2)$ and $\mathsf{e}_0 := \mathsf{e}_3$, call the index $e \in \{+1,-1,0\}$ of e_e its *charge*, which is the eigenvalue of $\frac{1}{2}\operatorname{ad}\mathsf{e}_3$ on e_e, and set $\upsilon := 2e\gamma$.

The symbols at Y_γ are $\hat{\mathsf{N}}_e$, $\hat{\mathsf{h}}_{q,e}$ and $\hat{\mathsf{M}}_e$; an additional index j, as in $\hat{\mathsf{N}}_{e,j}$, $\hat{\mathsf{h}}_{q,e,j}$ and $\hat{\mathsf{M}}_{e,j}$, marks the part that is homogeneous of degree j in γ; and $\hat{\mathsf{C}} := \hat{\mathsf{N}}_{e,0}^{-1}$. The integrands $\mathfrak{s}^{(i)}$ and the numbers $c_{\mu\nu}^{(i)}(\tau_\mathrm{gf})$, $i \le 5$, describe the self-energy, and $Z_0 := \int_\mathbb{T} d^\circ k/|\mathsf{k}(k)|^2$; the integrands $\mathfrak{v}_q^{(j)}$ and the numbers $\mathsf{g}_{q;\mu\nu}^{(j)}(\tau_\mathrm{gf})$, $j \le 4$, describe the vertex function; and Σ_tot, Γ_q are the kernels of Def. 7.13(5).

Moreover $j(\vartheta) := (e^{i\vartheta}-1)/i\vartheta$ for real $\vartheta \neq 0$ and $j(0) := 1$, $J := \mathrm{diag}(j(\upsilon_\mu))$, $P(k) := |\mathsf{k}(k)|^2 \mathbf{1} - \mathsf{k}(k)\mathsf{k}(k)^*$, and, for $q = p_{\mu\nu}(x)$, $D_q(k)\alpha = \mathsf{k}_\mu(k)\alpha_\nu - \mathsf{k}_\nu(k)\alpha_\mu$ and $\varpi_q(k) := (|\mathsf{k}_\mu(k)|^2 + |\mathsf{k}_\nu(k)|^2)/|\mathsf{k}(k)|^2$. Finally $R_c : x \mapsto 2c - x$ is the point reflection at c, and $(x,\mu)^R = (2c - x - e_\mu, \mu)$.

Kernel calculus (Def. 7.16 and the tools). The cut-offs are $\theta_0(t) := e^{-1/t}$ for $t > 0$, $\theta_0(t) := 0$ for $t \le 0$,

$$\theta(s) := \frac{\theta_0(1-s)}{\theta_0(1-s) + \theta_0(s - \frac{1}{2})}, \qquad \chi(\kappa) := \theta(|\kappa|/2).$$

The symbol classes are the $\mathcal{S}^s$. A *line symbol* of degree s, an integer ≥ -2, is a symbol in $\mathcal{S}^s$ with a smooth leading part σ^h that is homogeneous of degree s and of parity $(-1)^s$, the remainder $\sigma - \chi\sigma^h$ lying in $\mathcal{S}^{s+1}$; its kernel is an *elementary line* of degree $n = s+3$, which decays like $|x|^{-n}$. The *line class* $\mathfrak{K}(n)$ is defined by (K1)–(K3): its elements are kernels that have a homogeneous leading symbol and a homogeneous leading kernel, up to one order and a logarithm, and an element K has its *leading data* (σ^h, K^h) and its *constants*.

A *slowly varying function* is a lattice function whose differences of orders up to ten each gain a factor R^{-1}, as measured by $\|g\|_{(R)} := \max_{|\gamma|\le 10} R^{|\gamma|} \sup_\zeta |\nabla^\gamma g(\zeta)|$; with it we use $\check{g}(\kappa) := \sum_\zeta g(\zeta) e^{-i\kappa\cdot\zeta}$ and $\|G\|_{C^{10}}$. The *majorants* of degree $s > -D$ are the Borel functions on $\mathbb{R}^D$ with values in $[0,\infty]$ that are homogeneous of degree s and integrable over the ball $\{|\kappa| \le 1\}$; they form the class $\mathcal{H}_s(D)$. The kernel $\mathcal{F}_{\neq 0}^{-1}\sigma^h$ of a homogeneous symbol σ^h is defined in Lem. 7.17(b).

For the one-loop functions of Lem. 7.20, which are built from $\mathsf{m} \in \{2,3\}$ line symbols $\sigma_1, \dots, \sigma_\mathsf{m}$ of the degrees $s_1, \dots, s_\mathsf{m}$, from signs $\epsilon_i \in \{\pm 1\}$ and from linear maps $\lambda_i : \mathbb{R}^D \to \mathbb{R}^3$ with $D = 3(\mathsf{m}-1)$, we use

$$\mathsf{I}(\kappa) := \int_\mathbb{T} \prod_i \sigma_i(\epsilon_i l + \lambda_i(\kappa))\, d^\circ l, \qquad l_i(\kappa) := -\epsilon_i \lambda_i(\kappa), \qquad \omega := 3 + \sum_i s_i, \qquad \Lambda(\kappa) := 1 + \sum_{i<j} \log_+ \frac{|\kappa|}{|l_i(\kappa) - l_j(\kappa)|},$$

together with *the polynomial* Q and *the homogeneous part* I^h of I, which are its Taylor part and its non-analytic homogeneous part at small κ. In Lem. 7.22, a *chain* is a convolution of lines, $(X * Y)(u) :=$

$\sum_z X(z)Y(u-z)$, and a *bubble* is a product $K_\mathrm{a}K_\mathrm{b}$ of two lines, that is, a one-loop two-point graph; with the polynomial Q of the symbol of the bubble we put $\tilde{\mathsf{Q}}(k) := \mathsf{Q}(0) + \sum_j \partial_j \mathsf{Q}(0) \sin k_j$, and q is the kernel with the symbol $\tilde{\mathsf{Q}}$. The *kite* of Prop. 7.23 is the two-loop two-point graph with five elementary lines $K_1, \dots, K_5$ of the degrees $n_1, \dots, n_5$,

$$\mathsf{T}(u) := \sum_{z,w} K_1(z)K_2(w)K_3(w-z)K_4(u-z)K_5(u-w), \qquad \Delta := \sum_i n_i - 6,$$

and with it we use the functions $\hat{\mathsf{T}}_\mathsf{p}, \hat{\mathsf{T}}_\mathsf{q}$, the numbers $\omega_\mathsf{p}, \omega_\mathsf{q}$, the polynomials $\mathsf{Q}_\bullet$ with the corresponding $\tilde{\mathsf{Q}}_\bullet$ and kernels $\mathsf{q}_\bullet$, the function U, and the *short* and the *long lines*, which are the lines of the kite cut off to $|y| \lesssim \varepsilon r$ and to $|y| \gtrsim \varepsilon r$, respectively. On discrete tori (Lem. 7.38) we use $\mathbb{T}_n := \frac{\pi}{n}\mathbb{Z}^3 \cap (-\pi, \pi]^3$, $|z|_n$, $\langle z \rangle_n$ and $K^{(n)}(z) := (2n)^{-3} \sum_{k \in \mathbb{T}_n \setminus \{0\}} e^{ik\cdot z} \sigma(k)$.

Graphs on $\mathbb{Z}^3$. For $p_{[\iota]} = p_{\alpha\beta}$ we put $\mathsf{c}_{\iota,\nu}(k) := \mathsf{k}_\alpha(k)\delta_{\nu\beta} - \mathsf{k}_\beta(k)\delta_{\nu\alpha}$, and $\hat{\mathsf{C}}_{\mu\nu}(k)$ denotes the bracket in the Fourier representation of $\mathsf{C}^{\mathrm{lat}}_{\tau_\mathrm{gf}}$ in Def. 7.7(1) (Lem. 7.24). For $p = p_{\mu\nu}(x)$, $\epsilon_p := -1$ if $(\mu, \nu) = (1, 3)$ and $\epsilon_p := +1$ otherwise, and $p^\flat := (x, \lambda(p))$ (Lem. 7.25). Lem. 7.28 uses $u' := x_q - x_p$, the index sets $\mathfrak{G}_\bullet(\iota, \iota')$, the coefficients c_G of the *elementary terms* G, each of them the contribution of one monomial and one pairing to one of the classes $\mathsf{T}_\bullet$, the lines $K^{\rightarrow}_{\iota\mu}(y) := \mathsf{E}_{p_{[\iota](0)}}((y, \mu))$ and $K^{\leftarrow}_{\nu\iota'}(y) := \mathsf{E}_{p_{[\iota']}(y)}((0, \nu))$, the kernel $\Sigma_{\mu\nu}(y) := \Sigma_{\mathrm{tot}}((x, \mu), (x + y, \nu))$, the signs $\mathsf{s}_P(b)$, the decomposition $\partial_{b,a}\mathsf{S}_3 = \mathsf{d}_{b,a} + \mathsf{u}_{b,a}$, the kernel $\mathsf{q}^0_{\mu\nu}$ of finite support, and the difference operators $\mathsf{f}_G(\nabla)$ of degree d_G.

Lems. 7.29 and 7.35 use $\mathsf{q}_X, \mathsf{q}_Y$, $X' := X - \mathsf{q}_X$, $Y' := Y - \mathsf{q}_Y$ and the functions U_G. Lem. 7.31 uses the kernel q_G of the bubble $K_\mathrm{a}K_\mathrm{b}$ of each term G of $\Sigma_{\mu\nu}$, and it splits $\Sigma_{\mu\nu} = \mathsf{q}^\Sigma_{\mu\nu} + \mathsf{W}_{\mu\nu}$ into a kernel $\mathsf{q}^\Sigma_{\mu\nu}$ of finite support and a remainder $\mathsf{W}_{\mu\nu} \in \mathfrak{K}(4)$,

$$\mathsf{q}^\Sigma_{\mu\nu} := \mathsf{q}^0_{\mu\nu} + \sum_G c_G\, \mathsf{f}_G(\nabla)\mathsf{q}_G, \qquad \mathsf{W}_{\mu\nu} := \sum_G c_G\, \mathsf{f}_G(\nabla)\big[K_\mathrm{a}K_\mathrm{b} - \mathsf{q}_G\big];$$

it also introduces a symmetric kernel γ_q of finite range and the kernels $\gamma_{q;\mu\nu}$ derived from it. The *local parts* are kernels of finite support that carry the Taylor polynomials, of degree at most 1, of one-loop subgraphs, such as q, $\mathsf{q}_\bullet$, $\mathsf{q}^\Sigma_{\mu\nu}$ and γ_q; by Lem. 7.31, the symbols of $\mathsf{q}^\Sigma_{\mu\nu}$ and of $\gamma_{q;\mu\nu}$ vanish to second order at zero momentum. In a second sense of the words, distinct from the local terms of the formal sums S_3 and V_2, the proofs of Lem. 7.35 and Thm. V call local terms those terms of the expansions of the elementary terms that contain one of the kernels q_X, q_Y, q_p, q_q, together with the two halves of $\mathsf{T}_{4'}$; they are grouped into the local terms at q and those at p.

Two integrals over a loop momentum l agree *integrand by integrand* if their integrands coincide before integration, up to substitutions $l \to \pm l + \lambda(\kappa)$ with λ linear.

Model (Def. 7.32, Lems. 7.34, 7.35 and 7.36, Thm. V). The tensors are

$$V_2(k_1, k_2)_{\mu\nu} := -(k_1 \cdot k_2)\delta_{\mu\nu} + k_{1\nu}k_{2\mu}, \qquad V_3(k_1, k_2, k_3)_{\alpha\beta\gamma} := \delta_{\alpha\gamma}(k_1 - k_3)_\beta + \delta_{\alpha\beta}(k_2 - k_1)_\gamma + \delta_{\beta\gamma}(k_3 - k_2)_\alpha,$$

and the variables are $\mathsf{x}_0 := \mathsf{p}^2$, $\mathsf{x}_1 := k^2$, $\mathsf{x}'_1 := (k - \mathsf{p})^2$, $\mathsf{x}_2 := l^2$, $\mathsf{x}'_2 := (l - \mathsf{p})^2$, $\mathsf{x}_3 := (k - l)^2$. The six integrands $F_1, \dots, F_6$ are those of eq. 7.xxxv, each with its *lines*, which are the vectors whose squares occur in its denominator, counted with multiplicity. With $\rho_0 := \frac{\pi}{16}$ and $\rho(\mathsf{q}) := \theta(|\mathsf{q}|/2\rho_0)$, R_Γ is the product of ρ over the lines of Γ, and

$$\Psi_\Gamma(\mathsf{p}) := \iint R_\Gamma F_\Gamma \, d^\circ k \, d^\circ l, \qquad \mathsf{T}^{\mathrm{mod}}_\Gamma(u) := 24 \int_{\mathbb{R}^3} e^{i\mathsf{p}\cdot u} \Psi_\Gamma(\mathsf{p}) \, d^\circ \mathsf{p}.$$

The *model* consists of the continuum two-loop integrals Ψ_Γ, with the smooth ultraviolet cut-off ρ on every line, and of the functions $\mathsf{T}^{\mathrm{mod}}_\Gamma$. Lem. 7.34 introduces the *leading integrand* F_G of an elementary term G, which is the product of the leading parts of the symbols of its lines, the sets $\mathfrak{G}_2, \mathfrak{G}_3$ and the symbols $\sigma_\Pi, \sigma_\rightarrow, \sigma_\leftarrow$; Lem. 7.35 the functions U_Γ, I_Γ and $\Pi^h := \mathcal{F}^{-1}_{\neq 0}\sigma^h_\Pi$; Lem. 7.36

the numbers $c_{4,\Gamma}$ and $\kappa^{\mathrm{lat}}_\Gamma$, given by $\mathsf{I}_\Gamma = c_{4,\Gamma}|u|^{-4}$ and $\mathsf{U}_\Gamma(u) = \kappa^{\mathrm{lat}}_\Gamma|u|^{-5}$; and Thm. V the functions U_{ij} and the constant $\kappa^{\mathrm{lat}} := \sum_{\Gamma=1}^{6} \kappa^{\mathrm{lat}}_\Gamma$.

Value. Lem. 7.41 introduces $c(s) := \dfrac{2^s\,\Gamma_{\mathrm{E}}(\frac{3+s}{2})}{\pi^{3/2}\,\Gamma_{\mathrm{E}}(-\frac{s}{2})}$, and Lem. 7.42 the sums $\mathsf{s}_n(\mathsf{p}) := \sum_{u\in\mathbb{Z}^3\setminus\{0\}} |u|^{-n} e^{-i\mathsf{p}\cdot u}$ and the functions $\mathsf{e}_4, \mathsf{e}_5$. Lem. 7.43 uses the numbers $\mathsf{a}_0, \mathsf{a}_1, \mathsf{a}_2$, the function o, and $R^\Lambda_\Gamma(k,l,\mathsf{p}) := R_\Gamma(k/\Lambda, l/\Lambda, \mathsf{p}/\Lambda)$. Def. 7.44 introduces $\mathbb{D} := \{\mathrm{Re}\,\mathsf{z} > -\frac12\} \setminus \{\frac12, \frac32, \dots\}$, the homogeneous factors of degree d, the class $\mathfrak{P}$, the finite part $\mathrm{f.p.}\int \mathsf{g}\,\mathrm{d}k$, that is, the value at $\mathsf{z}=0$ of the continuation of $\int |k|^{2\mathsf{z}}\mathsf{g}\,\mathrm{d}k$, the cut-off $\eta_c(k) := \theta(4|k|/c)$, the class $\mathfrak{P}_2(c)$ defined by (Φ0)–(Φ3), the finite part $\mathrm{f.p.}\iint$, the class $\mathcal{Y}(c)$ defined by (Y1), (Y2), the scaled cut-offs Υ^Λ, and

$$\overline{G_2}(k,l) := \tfrac16\,\Delta_{\mathsf{p}} G(k,l,\mathsf{p})\big|_{\mathsf{p}=0}, \qquad \Xi_\Gamma[\Upsilon](\mathsf{p}) := \iint \Upsilon F_\Gamma\, d^\circ k\, d^\circ l.$$

The homogeneous factors and the classes $\mathfrak{P}$, $\mathfrak{P}_2(c)$ and $\mathcal{Y}(c)$ describe the domain of the finite part in one and in two variables and the admissible cut-offs, which vanish near the origin. Lem. 7.45(d) treats a *soft line*, that is, a factor $S(k,\mathsf{p})$ of degree -2 that is singular at $k=0$ and at $k=\mathsf{p}$, and it uses $[SJ]_i(k;\mathsf{p}) := \sum_{|\alpha|=i} \frac{\mathsf{p}^\alpha}{\alpha!}\partial^\alpha_{\mathsf{p}}(SJ)(k,0)$ and the function h_S; Prop. 7.46 defines $\mathfrak{f}_\Gamma[\Upsilon] := (2\pi)^{-6}\,\mathrm{f.p.}\iint \overline{(\Upsilon F_\Gamma)_2}\,\mathrm{d}k\,\mathrm{d}l$.

Def. 7.47 introduces the spaces $\mathcal{L}_n$, the coefficient r_{111} of $(\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3)^{-1}$, the space $\mathfrak{F}$ and the functional $\mathcal{R}[F] := r_{111}(\overline{F_2})$, which takes the coefficient of $(\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3)^{-1}$ in the angular mean of the Taylor coefficient of order p^2 of F; Lem. 7.48 uses $\mathsf{q}(t) := \frac13 t(2t-1)$; Prop. 7.50, the *bridge*, expresses $\kappa^{\mathrm{lat}}_\Gamma$ through $\mathcal{R}[F_\Gamma]$ and defines $d_{2,\Gamma} := \mathcal{R}[F_\Gamma]/16\pi^2$; Def. 7.52 defines $c_1 := \kappa^{\mathrm{lat}}/\mathsf{c}_2$; and Lem. 7.53 uses the *shell*, that is, the annulus $\mathsf{Ann}_r := \{u\in\mathbb{R}^3 : r \le |u| < Lr\}$ with a real number $L>1$.

The lattice conventions below specify the oriented cells, gauge transformations and differences. A subscript D on a bond or plaquette set denotes restriction to the domain D.

The joint cumulants are given by the partition formulas in Def. 7.8. The Gibbs kernel and the dilation of the smeared observable are written explicitly where they enter the finite-box argument.

The continuum test pairs $\mathfrak{T}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$ and the kernels $\mathcal{K}_q(f,h)$ are those of §6.1. The leading constant is $\mathsf{c}_2 = 9/(16\pi^2)$. The curl projection and its Fourier kernel are specified below.

7.1.2. *Lattice, boxes, chart.*

Definition 7.2 (Lattice, forms, boxes with Dirichlet data)**.** (1)

The lattice. We work on the lattice $\mathbb{Z}^3$ of spacing 1. A bond $b = \langle b_-, b_+\rangle$ is an ordered pair of neighbouring sites, $\bar b := \langle b_+, b_-\rangle$ is its reverse, and the positively oriented bonds are

$$(x,\mu) := \langle x, x+e_\mu\rangle \qquad (x\in\mathbb{Z}^3,\ \mu\in\{1,2,3\});$$

the base point of (x,μ) is $x = b_-$. For $x\in\mathbb{Z}^3$ and $1\le\mu<\nu\le 3$ the plaquette $p = p_{\mu\nu}(x)$ has the base point $x_p := x$, the four corners x, $x+e_\mu$, $x+e_\mu+e_\nu$, $x+e_\nu$, and the four bonds

$$b_1 = (x,\mu), \qquad b_2 = (x+e_\mu,\nu), \qquad b_3 = (x+e_\nu,\mu), \qquad b_4 = (x,\nu).$$

A plaquette is determined by the triple (x,μ,ν).

Let $\mathcal{B}$ be a set of bonds which contains the reverse of each of its elements. A G-valued configuration on $\mathcal{B}$ is a map U on $\mathcal{B}$ with $U(\bar b) = U(b)^{-1}$; the configuration with the constant value $\mathbf{1}$ is again denoted by $\mathbf{1}$. If the bonds of a plaquette p belong to $\mathcal{B}$, we set

$$\partial U(p) := U(b_1)\,U(b_2)\,U(b_3)^{-1}\,U(b_4)^{-1},$$

and a G-valued function u on a set of sites containing the end points of the bonds of $\mathcal{B}$ acts on U by the gauge transformation $U^u(b) := u(b_-)\,U(b)\,u(b_+)^{-1}$. A $\mathfrak{g}$-valued 1-form is a map A on bonds with $A(\bar b) = -A(b)$, and e^{iA} denotes the configuration $b\mapsto e^{iA(b)}$. The configurations on the set of all bonds form the space $\mathcal{V}(\mathbb{L}_1)$; a configuration on a set B of positively oriented bonds, extended to

the reverses by $U(\bar b) = U(b)^{-1}$, is an element of G^B. We write $A_\mu(x) := A((x,\mu))$, and $A^a_\mu(x)$ for the components of $A_\mu(x)$ in the basis $\mathsf{e}_1, \mathsf{e}_2, \mathsf{e}_3$.

The barycentre of $p = p_{\mu\nu}(x)$ is $x(p) = x + \frac12(e_\mu + e_\nu)$, and its *dual direction* is the index $\lambda(p) \in \{1,2,3\} \setminus \{\mu,\nu\}$. For $x \in \mathbb{Z}^3$ and $i \in \{1,2,3\}$ we denote by $p_{[i]}(x)$ the plaquette with base point x and dual direction i. The plaquettes correspond to the pairs (z,λ), the barycentre of (z,λ) being $z + \frac12(e_1+e_2+e_3-e_\lambda)$: the map $(z,\lambda) \mapsto p_{\mu\nu}(z)$, where $\{\mu,\nu\} = \{1,2,3\} \setminus \{\lambda\}$, is a bijection which preserves base point, barycentre and dual direction, and it sends (x,i) to $p_{[i]}(x)$. For $y \in \mathbb{Z}^3$ the translates $b+y$ and $p+y$ are the images of b and p under the translation τ_y.

For $U \in \mathcal{V}(\mathbb{L}_1)$ the plaquette action is $a_p(U) = 1 - \frac12 \operatorname{tr} \partial U(p)$. It depends on the values of U at the four bonds of p only, and we use the same formula for every configuration U on a set of bonds which is stable under reversal and contains the bonds of p. For every such U, and every G-valued u on a set of sites containing the end points of the bonds on which U is given,

$$a_p(U) \in [0,2], \qquad a_p(U^u) = a_p(U).$$

For the first relation we recall that every $g \in G$ can be written as e^{iY} with $Y \in \mathfrak{g}$, $|Y| \le \pi$: for $g \ne -\mathbf{1}$ this is the exponential chart in Def. 7.1(2), and $-\mathbf{1} = e^{i\pi \mathsf{e}_3}$. Since $\frac12 \operatorname{tr} e^{iY} = \cos|Y|$, the choice $e^{iY} = \partial U(p)$ gives $a_p(U) = 1 - \cos|Y| \in [0,2]$. For the second relation let x be the base point of p; in the product

$$\begin{aligned}\partial U^u(p) = {} & u(x)\, U(b_1)\, u(x+e_\mu)^{-1} \cdot u(x+e_\mu)\, U(b_2)\, u(x+e_\mu+e_\nu)^{-1} \\ & \cdot u(x+e_\mu+e_\nu)\, U(b_3)^{-1}\, u(x+e_\nu)^{-1} \cdot u(x+e_\nu)\, U(b_4)^{-1}\, u(x)^{-1}\end{aligned}$$

the inner factors cancel, so that $\partial U^u(p) = u(x)\, \partial U(p)\, u(x)^{-1}$, and the trace is invariant under conjugation.

(2)

Differences. For a site function λ, $\mathfrak{g}$-valued or complex, and a 1-form A we set

$$(\mathrm{d}\lambda)(b) := \lambda(b_+) - \lambda(b_-), \qquad (\mathrm{d}A)(p) := A(b_1) + A(b_2) - A(b_3) - A(b_4),$$
$$(\mathrm{d}^* A)(x) := \sum_\mu [A_\mu(x-e_\mu) - A_\mu(x)].$$

The relation $\mathrm{dd}\lambda = 0$ holds: at $p = p_{\mu\nu}(x)$,

$$\begin{aligned}(\mathrm{dd}\lambda)(p) = {} & [\lambda(x+e_\mu) - \lambda(x)] + [\lambda(x+e_\mu+e_\nu) - \lambda(x+e_\mu)] \\ & - [\lambda(x+e_\mu+e_\nu) - \lambda(x+e_\nu)] - [\lambda(x+e_\nu) - \lambda(x)] = 0.\end{aligned}$$

On $\mathfrak{g}$-valued λ and A the three operators act componentwise in the basis e_a. By $(\cdot,\cdot)$ and $\|\cdot\|$ we denote the sums of $\langle\cdot,\cdot\rangle$ over the sites, over the positively oriented bonds or over the plaquettes, according to the arguments; on complex forms $(\cdot,\cdot)$ is the Hermitian sum, antilinear in its first argument. If λ and A have finite support, a shift of the summation variable gives

$$\begin{aligned}(\mathrm{d}\lambda, A) &= \sum_{x,\mu} \big\langle \lambda(x+e_\mu) - \lambda(x), A_\mu(x) \big\rangle \\ &= \sum_x \Big\langle \lambda(x), \sum_\mu [A_\mu(x-e_\mu) - A_\mu(x)] \Big\rangle = (\lambda, \mathrm{d}^* A).\end{aligned}$$

On plaquette functions, d^* denotes the adjoint of d on 1-forms, which is again a difference operator. To write it down, we put $F_{\mu\nu}(x) := F(p_{\mu\nu}(x))$ for a plaquette function F and $\mu < \nu$, and $F_{\nu\mu}(x) := -F_{\mu\nu}(x)$, $F_{\mu\mu}(x) := 0$. Since $(\mathrm{d}A)(p_{\mu\nu}(x)) = [A_\nu(x+e_\mu) - A_\nu(x)] - [A_\mu(x+e_\nu) - A_\mu(x)]$, the antisymmetry of $F_{\mu\nu}$ in (μ,ν) and a shift of the summation variable give, for A and F of finite

support,

$$(\mathrm{d}A, F) = \sum_x \sum_{\mu,\nu} \langle A_\nu(x+e_\mu) - A_\nu(x), F_{\mu\nu}(x)\rangle$$
$$= \sum_{x,\rho} \Big\langle A_\rho(x), \sum_\nu [F_{\nu\rho}(x-e_\nu) - F_{\nu\rho}(x)] \Big\rangle,$$

so that

$$(\mathrm{d}^* F)((x,\rho)) = -\sum_\nu [F_{\rho\nu}(x-e_\nu) - F_{\rho\nu}(x)];$$

by this formula we apply d^* to plaquette functions of arbitrary support. Finally we set $\bar\lambda(b) := \frac{1}{2}(\lambda(b_-) + \lambda(b_+))$ and $(A \times \bar\lambda)(b) := A(b) \times \bar\lambda(b)$.

(3)

Boxes. Let $n \geq 2$ be an integer, and let

$$\Lambda_n := \{0, \dots, n\}^3, \qquad \Lambda_n^\circ := \{1, \dots, n-1\}^3, \qquad \partial\Lambda_n := \Lambda_n \setminus \Lambda_n^\circ. \tag{7.i}$$

A positively oriented bond is *free* if both its end points lie in Λ_n and at least one of them lies in Λ_n°; Bd_n denotes the set of free bonds, and Pl_n the set of plaquettes all of whose corners lie in Λ_n. A bond with both end points in Λ_n which is neither free nor the reverse of a free bond has both end points in $\partial\Lambda_n$.

- $\mathcal{U}_n$ is the set of configurations on the bonds with both end points in Λ_n which take the value $\mathbf{1}$ at every bond that is neither free nor the reverse of a free bond; restriction to Bd_n identifies $\mathcal{U}_n$ with G^{Bd_n}.
- Form_n is the space of 1-forms on the same bonds which vanish at every bond that is neither free nor the reverse of a free bond. X_n is the space of $\mathfrak{g}$-valued functions on Λ_n which vanish on $\partial\Lambda_n$, and
$$\mathcal{G}_n := \{u : \Lambda_n \to G : \ u = \mathbf{1} \text{ on } \partial\Lambda_n\}.$$
- For $A \in \mathsf{Form}_n$, $\mathrm{d}A$ is considered on Pl_n, and d^*A on Λ_n°, where it involves only bonds with both end points in Λ_n; we extend d^*A by 0 to $\partial\Lambda_n$, so that $\mathrm{d}^*A \in \mathsf{X}_n$. For $\lambda \in \mathsf{X}_n$ we have $\mathrm{d}\lambda \in \mathsf{Form}_n$, because λ vanishes at both end points of a bond in Λ_n which is not free. The relation $(\mathrm{d}\lambda, A) = (\lambda, \mathrm{d}^*A)$ for $\lambda \in \mathsf{X}_n$, $A \in \mathsf{Form}_n$ follows from (2), applied to the extensions of λ and A by zero. For $u \in \mathcal{G}_n$ and $U \in \mathcal{U}_n$ we have $U^u \in \mathcal{U}_n$, since $U^u(b) = u(b_-)u(b_+)^{-1} = \mathbf{1}$ at a bond in Λ_n which is not free.
- $\mathsf{Form}_n^{\mathbb{R}}$ and $\mathsf{X}_n^{\mathbb{R}}$ denote the real-valued analogues of Form_n and X_n; an element of Form_n, resp. of X_n, has three components in $\mathsf{Form}_n^{\mathbb{R}}$, resp. in $\mathsf{X}_n^{\mathbb{R}}$. We write $|A|_\infty := \max_b |A(b)|$ and $|\lambda|_\infty := \max_x |\lambda(x)|$, and $o_n := (\lfloor n/2 \rfloor, \lfloor n/2 \rfloor, \lfloor n/2 \rfloor)$ is the centre of the box.
- Geometrically, the plaquette $p_{\mu\nu}(x)$ is the square $\{x + se_\mu + te_\nu : \ s,t \in [0,1]\}$ with the four corners listed in (1), and the set of a bond b is $[b_-, b_+] := \{(1-s)b_- + sb_+ : \ s \in [0,1]\}$. Write $\mathsf{Pl}(D)$ and $\mathsf{Bd}(D)$ for the plaquettes and positively oriented bonds contained in a set D. Then $\mathsf{Pl}_n = \mathsf{Pl}([0,n]^3)$, and Bd_n is the set of the bonds of $\mathsf{Bd}([0,n]^3)$ with an end point in Λ_n°. Indeed, the square of a plaquette is the convex hull of its four corners, and $[b_-, b_+]$ is the convex hull of b_- and b_+; since the cube $[0,n]^3$ is convex, it contains the square, resp. the set $[b_-, b_+]$, if and only if it contains these sites, and the sites of $\mathbb{Z}^3$ in $[0,n]^3$ are the points of Λ_n.

■

Definition 7.3 (Wilson measure of a box; truncated two-point function)**.** Let $n \geq 2$ and $\beta > 0$. The *Wilson measure* of the box Λ_n is the probability measure

$$d\mu_{n,\beta}(U) := Z_{n,\beta}^{-1} \exp\Big[-\beta \sum_{p \in \mathsf{Pl}_n} a_p(U)\Big] \prod_{b \in \mathsf{Bd}_n} dU(b) \tag{7.ii}$$

on $\mathcal{U}_n \cong G^{\mathsf{Bd}_n}$, where $dU(b)$ is the normalised Haar measure in the variable $U(b)$ and $Z_{n,\beta} > 0$ is the integral of the exponential with respect to the product measure, so that $\mu_{n,\beta}$ has total mass 1; its mean is denoted by $\langle\cdot\rangle_{n,\beta}$. For bounded Borel functions X, Y on $\mathcal{U}_n$ define their joint cumulant by

$$\langle X;Y\rangle_{n,\beta} = \partial_s\partial_t \log\langle e^{sX+tY}\rangle_{n,\beta}\Big|_{s=t=0}.$$

The exponential mean is holomorphic and equals 1 at the origin, so its logarithm is defined near the origin. Differentiating the logarithm gives

$$\langle X;Y\rangle_{n,\beta} = \langle XY\rangle_{n,\beta} - \langle X\rangle_{n,\beta}\langle Y\rangle_{n,\beta}.$$

The measure $\mu_{n,\beta}$ is the Gibbs kernel of the set of free bonds, evaluated at the configuration $\mathbf{1}$, in the following sense. For a finite set B of positively oriented bonds define its Gibbs kernel by

$$(\gamma_B F)(U) = \frac{\int F(V \vee U_{B^c}) e^{H_B(V\vee U_{B^c})}\,\mathrm{d}V_B}{\int e^{H_B(V\vee U_{B^c})}\,\mathrm{d}V_B}, \qquad H_B = -\beta\sum_{p\in\mathsf{Pl}[B]} a_p,$$

where $V \vee U_{B^c}$ equals V on B and U off B, $\mathsf{Pl}[B]$ consists of the plaquettes containing a bond of B, and dV_B is product Haar measure on G^B. Take $B = \mathsf{Bd}_n$, and for $V \in G^B$ let $V \vee \mathbf{1}_{B^c}$ denote the configuration which equals V on B and $\mathbf{1}$ at all other positively oriented bonds. Then, for every bounded Borel function F on $\mathcal{V}(\mathbb{L}_1)$,

$$(\gamma_B F)(\mathbf{1}) = \int F(V \vee \mathbf{1}_{B^c})\,\mathrm{d}\mu_{n,\beta}(V).$$

Indeed, by definition the left side is the quotient of $\int F e^{H_B}\,\mathrm{d}V_B$ by $\int e^{H_B}\,\mathrm{d}V_B$, both integrands being evaluated at $V \vee \mathbf{1}_{B^c}$. Here dV_B is the product, over the bonds $b \in B$, of the normalised Haar measures in the variables $V(b)$, and $H_B = -\beta\sum_{p\in\mathsf{Pl}[B]} a_p$, the sum running over the set $\mathsf{Pl}[B]$ of the plaquettes which contain a bond of B. Every plaquette of $\mathsf{Pl}[B]$, that is, every plaquette containing a free bond (x,μ), lies in Pl_n: one end point of (x,μ) lies in Λ_n°, so both end points have the coordinates $x_\nu \in \{1,\dots,n-1\}$ for $\nu \neq \mu$, and the corners of such a plaquette are the end points of (x,μ) and their translates by one of the vectors $\pm e_\nu$, $\nu \neq \mu$; all of them lie in Λ_n.

Conversely, the bonds of a plaquette of $\mathsf{Pl}_n \setminus \mathsf{Pl}[B]$, which contains no free bond, have both end points in Λ_n and are not free, so every configuration of $\mathcal{U}_n$ equals $\mathbf{1}$ on them, and $a_p = 0$ on $\mathcal{U}_n$. Hence $H_B(V \vee \mathbf{1}_{B^c}) = -\beta\sum_{p\in\mathsf{Pl}_n} a_p(V)$, and the quotient is the integral of $F(V \vee \mathbf{1}_{B^c})$ with respect to $\mu_{n,\beta}$. ∎

Definition 7.4 (Exponential chart, gauge function, Faddeev–Popov operator)**.** Let Ω be $\mathbb{Z}^3$ or a box Λ_n; in the second case forms are taken in Form_n, site functions in X_n and plaquettes in Pl_n.

(1) For a plaquette p, a 1-form A and $m \geq 0$, the number $\mathsf{a}_p^{(m)}(A)$ is the m-th Taylor coefficient of the plaquette action along the curve $t \mapsto e^{itA}$:

$$a_p(e^{itA}) = \sum_{m\geq 0} t^m\,\mathsf{a}_p^{(m)}(A) \qquad (t \in \mathbb{R}); \tag{7.iii}$$

the series converges for every real t by Lem. 7.5(a).

(2) If a 1-form A and a site function λ satisfy $|A(b)| + |\lambda(b_-)| + |\lambda(b_+)| < 1$ for every bond b, we set

$$A^\lambda(b) := \tfrac{1}{i}\log(e^{i\lambda(b_-)}e^{iA(b)}e^{-i\lambda(b_+)}), \qquad \text{that is,} \qquad e^{iA^\lambda} = (e^{iA})^u, \quad u = e^{i\lambda}.$$

This is well defined: by the product estimate in Def. 7.1(2),

$$|e^{i\lambda(b_-)}e^{iA(b)}e^{-i\lambda(b_+)} - \mathbf{1}| \leq e^{|\lambda(b_-)|+|A(b)|+|\lambda(b_+)|} - 1 < e - 1 < 2 = |-\mathbf{1}-\mathbf{1}|,$$

so the argument of the logarithm lies in $G \setminus \{-\mathbf{1}\}$, where $\frac{1}{i}\log$, the inverse of $Y \mapsto e^{iY}$ on $\mathbb{B}_\pi$, is defined (the exponential chart in Def. 7.1(2)).

(3) The *gauge function* is the map $A \mapsto \mathrm{d}^* A$. For a 1-form A with $|A(b)| < 1$ at every bond b, the Faddeev–Popov operator $\mathsf{M}(A)$ acts on site functions λ by

$$\mathsf{M}(A)\lambda := \frac{d}{ds}\,\mathrm{d}^*(A^{s\lambda})\Big|_{s=0}, \qquad \mathsf{M}(tA) = \sum_{j\ge 0} t^j\,\mathsf{M}^{(j)}(A); \tag{7.iv}$$

the second formula is the expansion at $t = 0$ of the real-analytic function $t \mapsto \mathsf{M}(tA)$ (Lem. 7.5(d)), and it defines the operators $\mathsf{M}^{(j)}(A)$; since $\mathsf{M}(t(cA)) = \mathsf{M}((tc)A)$, the operator $\mathsf{M}^{(j)}(A)$ is homogeneous of degree j in A. The derivative is taken site by site: $\mathrm{d}^*(A^{s\lambda})(x)$ involves the six bonds with end point x, and for each of them $A^{s\lambda}(b)$ is defined and smooth in s for $|s|$ small. In particular the definition applies to every site function λ, bounded or not. ■

Lemma 7.5 (Coefficients of the chart)**.** Let p be a plaquette with bonds $b_1, \dots, b_4$, let A be a 1-form, $A_i := A(b_i)$, and $(s_1, s_2, s_3, s_4) := (+, +, -, -)$, so that $\mathrm{d}A(p) = \sum_i s_i A_i$.

(a) The series eq. 7.iii converges for every $t \in \mathbb{R}$. Each $\mathsf{a}_p^{(m)}$ is a homogeneous polynomial of degree m in the twelve components of $A_1, \dots, A_4$, with real coefficients which do not depend on p. Moreover $\mathsf{a}_p^{(0)} = \mathsf{a}_p^{(1)} = 0$, and $\mathsf{a}_p^{(m)}(gAg^{-1}) = \mathsf{a}_p^{(m)}(A)$ for every $g \in G$, where gAg^{-1} denotes the form $b \mapsto gA(b)g^{-1}$.

(b) On the set $\{(t, A_1, \dots, A_4) \in \mathbb{R} \times \mathfrak{g}^4 :\ |t| \sum_i |A_i| < \log 3\}$ the function $Z := \frac{1}{i} \log \partial(e^{itA})(p) \in \mathbb{B}_\pi$ is defined and smooth. For $j = 0, 1, 2$ the maps $\mathfrak{z}_j := \frac{1}{(j+1)!}\partial_t^{j+1} Z|_{t=0}$ are homogeneous polynomial maps $\mathfrak{g}^4 \to \mathfrak{g}$ of degree $j+1$, which do not depend on p; $\mathfrak{z}_0 = \mathrm{d}A(p)$, and

$$\begin{gathered} \mathsf{a}_p^{(2)} = \tfrac{1}{2}|\mathrm{d}A(p)|^2, \qquad \mathsf{a}_p^{(3)} = \langle \mathrm{d}A(p), \mathfrak{z}_1 \rangle, \\ \mathsf{a}_p^{(4)} = \tfrac{1}{2}|\mathfrak{z}_1|^2 + \langle \mathrm{d}A(p), \mathfrak{z}_2 \rangle - \tfrac{1}{24}|\mathrm{d}A(p)|^4, \qquad \mathfrak{z}_1 = -\sum_{i<j} s_i s_j\, A_i \times A_j. \end{gathered} \tag{7.v}$$

We write $\mathfrak{z}_j(p)$ for the value of $\mathfrak{z}_j$ at $(A(b_1(p)), \dots, A(b_4(p)))$.

(c) The function $Y \mapsto -\log \sigma_1(|Y|^2)$ is real analytic and even on $\mathbb{B}_\pi$, and its Taylor polynomial of degree 3 at 0 is $\frac{1}{3}|Y|^2$.

(d) For $Y \in \mathfrak{g}$ let $\mathsf{a}_Y := -2\,Y \times \cdot$ on $\mathfrak{g}$. Then $e^{isY} X e^{-isY} = e^{s\mathsf{a}_Y} X$ for $s \in \mathbb{R}$ and $X \in \mathfrak{g}$. If $|A(b)| < 1$ and λ is a site function, then

$$\begin{aligned} \frac{d}{ds} A^{s\lambda}(b)\Big|_{s=0} &= \mathsf{dE}(A(b))^{-1}\big(\lambda(b_-) - e^{\mathsf{a}_{A(b)}}\lambda(b_+)\big) \\ &= -\mathrm{d}\lambda(b) + 2A(b) \times \bar\lambda(b) - \tfrac{1}{3}A(b) \times \big(A(b) \times \mathrm{d}\lambda(b)\big) + \mathsf{r}(A(b))\,\mathrm{d}\lambda(b), \end{aligned} \tag{7.vi}$$

where $\mathsf{r} : \mathbb{B}_\pi \to \operatorname{End}\mathfrak{g}$ is real analytic and vanishes at 0 to fourth order. Consequently, if $|A(b)| < 1$ at every bond, $\mathsf{M}(tA)$ is real analytic in t, and

$$\begin{gathered} \mathsf{M}^{(0)} = -\mathrm{d}^*\mathrm{d}, \qquad \mathsf{M}^{(1)}(A)\lambda = 2\,\mathrm{d}^*(A \times \bar\lambda), \\ \mathsf{M}^{(2)}(A)\lambda = -\tfrac{1}{3}\,\mathrm{d}^*(A \times (A \times \mathrm{d}\lambda)), \qquad \mathsf{M}^{(3)}(A) = 0. \end{gathered}$$

■

Proof. Step 1: claim (a). Since $U(b_3)^{-1} = e^{-itA_3}$ and $U(b_4)^{-1} = e^{-itA_4}$ for $U = e^{itA}$, we have $\partial(e^{itA})(p) = \prod_{i=1}^4 e^{its_iA_i}$, a product of four exponential series in t which converge for every t; their Cauchy product converges for every t, and hence so does eq. 7.iii. For $m \ge 1$ the coefficient of t^m is

$$\mathsf{a}_p^{(m)}(A) = -\frac{1}{2} \sum_{m_1+\dots+m_4=m} \operatorname{tr}\Big[\prod_{i=1}^4 \frac{(is_iA_i)^{m_i}}{m_i!}\Big],$$

that is, $-\frac{1}{2}\operatorname{tr}$ of a sum of products of m factors is_iA_i. It is therefore a homogeneous polynomial of degree m in the components of $A_1, \dots, A_4$, given by the same formula for every plaquette. It takes real values on real arguments, since a_p is real; hence, if we write it as $\sum_\alpha c_\alpha y^\alpha$ in the twelve real

components y of $A_1, \dots, A_4$, the polynomial $\sum_\alpha (\operatorname{Im} c_\alpha)\, y^\alpha$ vanishes on $\mathbb{R}^{12}$, so that $\operatorname{Im} c_\alpha = 0$ and the coefficients are real. Replacing every A_i by gA_ig^{-1} conjugates each product of the factors is_iA_i by g, which does not change the trace. The relations $\mathsf{a}_p^{(0)} = \mathsf{a}_p^{(1)} = 0$ follow from (b), see Step 5.

Step 2: existence and smoothness of Z. Let $X_i := its_iA_i$ and $\mathsf{x} := \partial(e^{itA})(p) = e^{X_1}e^{X_2}e^{X_3}e^{X_4}$. Since $|X_i| = |t|\,|A_i|$, the product estimate in Def. 7.1(2) gives on the set of (b)

$$|\mathsf{x} - \mathbf{1}| \le e^{|t|\sum_i |A_i|} - 1 < e^{\log 3} - 1 = 2 = |-\mathbf{1} - \mathbf{1}|;$$

hence $\mathsf{x} \neq -\mathbf{1}$, and $Z = \frac{1}{i}\log \mathsf{x} \in \mathbb{B}_\pi$ is defined. The matrix x depends smoothly on $(t, A_1, \dots, A_4)$, and $W \mapsto \frac{1}{i}\log W$ is smooth on $G \setminus \{-\mathbf{1}\}$, the inverse of the diffeomorphism $Y \mapsto e^{iY}$ of $\mathbb{B}_\pi$ onto $G \setminus \{-\mathbf{1}\}$ (Def. 7.1); thus Z is smooth. At $t = 0$ we have $\mathsf{x} = \mathbf{1}$ and $Z = 0$. The maps $\mathfrak{z}_j$ are t-derivatives of Z at $t = 0$, hence smooth maps $\mathfrak{g}^4 \to \mathfrak{g}$, given by the same formula for all plaquettes.

Step 3: homogeneity. For $c > 0$ the substitution $(t, A) \mapsto (t/c, cA)$ preserves the domain of Z and leaves x unchanged. Hence $Z(t/c, cA) = Z(t, A)$, that is, $Z(s, cA) = Z(cs, A)$, and differentiating $j+1$ times at $s = 0$ we find $\mathfrak{z}_j(cA) = c^{j+1}\mathfrak{z}_j(A)$. It remains to use the following fact:

a smooth function f on a finite-dimensional real vector space with $f(cA) = c^m f(A)$ for all $c > 0$ is a homogeneous polynomial of degree m. To see it, let f_i denote the homogeneous Taylor polynomials of f at 0; Taylor's formula at the points cA gives

$$c^m f(A) = f(cA) = \sum_{i \le m} c^i f_i(A) + O(c^{m+1}) \qquad (c \to 0),$$

and comparing the powers of c as $c \to 0$ we obtain $f_i(A) = 0$ for $i < m$ and $f(A) = f_m(A)$. Applied to the components of $\mathfrak{z}_j$ in a basis of $\mathfrak{g}$, this shows that $\mathfrak{z}_j$ is a homogeneous polynomial map of degree $j+1$.

Step 4: $\mathfrak{z}_0$ and $\mathfrak{z}_1$. At fixed A, Taylor's formula in t gives $Z = t(\mathfrak{z}_0 + t\mathfrak{z}_1 + t^2\mathfrak{z}_2) + O(t^4)$. We compare the two expansions of $e^{iZ} = \mathsf{x}$,

$$e^{iZ} = \mathbf{1} + iZ - \tfrac{1}{2}Z^2 + O(t^3), \qquad \mathsf{x} = \mathbf{1} + \sum_i X_i + \tfrac{1}{2}\sum_i X_i^2 + \sum_{i<j} X_iX_j + O(t^3).$$

The terms of order t give $it\mathfrak{z}_0 = \sum_i X_i$, hence $\mathfrak{z}_0 = \sum_i s_iA_i = \mathrm{d}A(p)$. Since $-\frac{1}{2}Z^2 = \frac{1}{2}(it\mathfrak{z}_0)^2 + O(t^3) = \frac{1}{2}\big(\sum_i X_i\big)^2 + O(t^3)$, the terms of order t^2 give

$$\begin{aligned} it^2\mathfrak{z}_1 &= \tfrac{1}{2}\sum_i X_i^2 + \sum_{i<j} X_iX_j - \tfrac{1}{2}\Big(\sum_i X_i\Big)^2 = \tfrac{1}{2}\sum_{i<j}[X_i, X_j] \\ &= -\tfrac{1}{2}t^2 \sum_{i<j} s_is_j[A_i, A_j] = -it^2 \sum_{i<j} s_is_j\, A_i \times A_j, \end{aligned}$$

where $[X, Y] = 2i\, X \times Y$ was used in the last step. This is the formula for $\mathfrak{z}_1$ in eq. 7.v.

Step 5: the coefficients $\mathsf{a}_p^{(m)}$, $m \le 4$. Since $\frac{1}{2}\operatorname{tr} e^{iZ} = \cos|Z|$, we have $a_p(e^{itA}) = 1 - \cos|Z| = \frac{1}{2}|Z|^2 - \frac{1}{24}|Z|^4 + O(t^6)$. By Step 4,

$$|Z|^2 = t^2\Big[|\mathfrak{z}_0|^2 + 2t\langle\mathfrak{z}_0, \mathfrak{z}_1\rangle + t^2\big(|\mathfrak{z}_1|^2 + 2\langle\mathfrak{z}_0, \mathfrak{z}_2\rangle\big) + O(t^3)\Big],$$

and therefore

$$a_p(e^{itA}) = \tfrac{1}{2}t^2|\mathfrak{z}_0|^2 + t^3\langle\mathfrak{z}_0, \mathfrak{z}_1\rangle + t^4\Big(\tfrac{1}{2}|\mathfrak{z}_1|^2 + \langle\mathfrak{z}_0, \mathfrak{z}_2\rangle - \tfrac{1}{24}|\mathfrak{z}_0|^4\Big) + O(t^5).$$

Comparing with the Taylor series eq. 7.iii we obtain $\mathsf{a}_p^{(0)} = \mathsf{a}_p^{(1)} = 0$ and eq. 7.v.

Step 6: claim (c). From its power series, $\sigma_1(w) = \big(1 - \frac{w}{6} + O(w^2)\big)^2 = 1 - \frac{w}{3} + O(w^2)$, and $\sigma_1(|Y|^2) = (\sin|Y|/|Y|)^2 > 0$ on $\mathbb{B}_\pi$. Thus $Y \mapsto -\log\sigma_1(|Y|^2)$ is the composite of the real analytic maps $Y \mapsto |Y|^2$, σ_1 at real arguments, and log on $(0, \infty)$, hence real analytic. It depends on $|Y|^2$ only and is therefore even, and $-\log(1 - \frac{1}{3}|Y|^2 + O(|Y|^4)) = \frac{1}{3}|Y|^2 + O(|Y|^4)$ shows that its Taylor polynomial of degree 3 at 0 is $\frac{1}{3}|Y|^2$.

Step 7: conjugation. By $[Y,X] = 2iY \times X$ we have $i[Y,X] = -2Y \times X = \mathsf{a}_Y X$. The function $x(s) := e^{isY} X e^{-isY}$ takes values in $\mathfrak{g}$ and satisfies $x'(s) = i[Y, x(s)] = \mathsf{a}_Y x(s)$, so that $\frac{d}{ds}(e^{-s\mathsf{a}_Y} x(s)) = 0$; together with $x(0) = X$ this gives $x(s) = e^{s\mathsf{a}_Y} X$. Consequently

$$\mathsf{dE}(Y) = \int_0^1 e^{s\mathsf{a}_Y}\, \mathrm{d}s = \mathsf{f}(\mathsf{a}_Y), \qquad \mathsf{f}(w) := \frac{e^w - 1}{w} = \sum_{n\geq 0} \frac{w^n}{(n+1)!}.$$

Step 8: first equality in eq. 7.vi. Let $Y := A(b)$ and $\delta := \frac{d}{ds} A^{s\lambda}(b)|_{s=0}$. For $|s|$ small, $A^{s\lambda}(b)$ is defined, depends smoothly on s, equals Y at $s = 0$, and satisfies $e^{iA^{s\lambda}(b)} = e^{is\lambda(b_-)} e^{iY} e^{-is\lambda(b_+)}$. We differentiate this identity at $s = 0$, using the differential formula in Def. 7.1(2) on the left and Step 7 on the right:

$$i\big(\mathsf{dE}(Y)\delta\big) e^{iY} = i\lambda(b_-) e^{iY} - i e^{iY} \lambda(b_+) = i\big(\lambda(b_-) - e^{\mathsf{a}_Y} \lambda(b_+)\big) e^{iY}.$$

By the determinant formula in Def. 7.1(2), $\det \mathsf{dE}(Y) = \sigma_1(|Y|^2) > 0$, so $\mathsf{dE}(Y)$ is invertible, and the first equality follows.

Step 9: expansion. The double cross product formula $X \times (Y \times Z) = \langle X, Z\rangle Y - \langle X, Y\rangle Z$ gives $\mathsf{a}_Y Y = 0$ and $\mathsf{a}_Y^2 = -4|Y|^2$ on the orthogonal complement of Y; so a_Y has the eigenvalues 0 and $\pm 2i|Y|$, and Lagrange's identity $|Y \times X|^2 = |Y|^2|X|^2 - \langle Y, X\rangle^2$ gives $\|\mathsf{a}_Y\| \leq 2|Y| < 2\pi$ on $\mathbb{B}_\pi$. The function $1/\mathsf{f}(w) = w/(e^w - 1)$ is holomorphic for $|w| < 2\pi$, and since $w/(e^w - 1) + w/2$ is even,

$$\frac{1}{\mathsf{f}(w)} = 1 - \frac{w}{2} + \frac{w^2}{12} + w^4 \mathsf{f}_4(w), \qquad \frac{we^w}{e^w - 1} = w + \frac{w}{e^w - 1} \qquad (|w| < 2\pi),$$

with f_4 holomorphic for $|w| < 2\pi$. Evaluating these power series at a_Y, Steps 7 and 8 give

$$\begin{aligned} \delta &= \tfrac{1}{\mathsf{f}}(\mathsf{a}_Y)\lambda(b_-) - \big(\mathsf{a}_Y + \tfrac{1}{\mathsf{f}}(\mathsf{a}_Y)\big)\lambda(b_+) = -\tfrac{1}{\mathsf{f}}(\mathsf{a}_Y)\, \mathrm{d}\lambda(b) - \mathsf{a}_Y \lambda(b_+) \\ &= -\mathrm{d}\lambda(b) - \mathsf{a}_Y\big(\lambda(b_+) - \tfrac{1}{2}\mathrm{d}\lambda(b)\big) - \tfrac{1}{12}\mathsf{a}_Y^2\, \mathrm{d}\lambda(b) - \mathsf{a}_Y^4 \mathsf{f}_4(\mathsf{a}_Y)\, \mathrm{d}\lambda(b). \end{aligned}$$

Now $\lambda(b_+) - \frac{1}{2}\mathrm{d}\lambda(b) = \bar\lambda(b)$, $-\mathsf{a}_Y \bar\lambda(b) = 2Y \times \bar\lambda(b)$ and $\mathsf{a}_Y^2 = 4\, Y \times (Y \times \cdot)$; with $\mathsf{r}(Y) := -\mathsf{a}_Y^4 \mathsf{f}_4(\mathsf{a}_Y)$ this is the second equality in eq. 7.vi. The map r is real analytic on $\mathbb{B}_\pi$, being the sum of a power series at an operator which depends linearly on Y and has norm $< 2\pi$, and it vanishes at 0 to fourth order because of the factor a_Y^4. Since d^* involves at each site finitely many bonds, $\mathsf{M}(A)\lambda = \mathrm{d}^*\delta$, where δ is the form given at every bond by eq. 7.vi. If A is replaced by tA, the first three terms of eq. 7.vi are of degree 0, 1, 2 in t, and the last one is real analytic in t and of order t^4. This proves that $t \mapsto \mathsf{M}(tA)$ is real analytic, and it gives the formulas for $\mathsf{M}^{(0)}, \dots, \mathsf{M}^{(3)}$. □

Lemma 7.6 (Free gauge action, absence of zero modes, a slice, and the Faddeev–Popov change of variables in a box)**.** Let $n \geq 2$. On $\mathcal{U}_n$ let $S(U) := \sum_{p \in \mathsf{Pl}_n} a_p(U)$, and let

$$\mathsf{T}_n := \{\langle x - e_1, x\rangle :\ x \in \Lambda_n^\circ\} \subset \mathsf{Bd}_n,$$

$$\mathcal{U}_n^{\mathsf{T}} := \{V \in \mathcal{U}_n :\ V = \mathbf{1} \text{ on } \mathsf{T}_n\}, \qquad \mathsf{Form}_n^{\mathsf{T}} := \{A \in \mathsf{Form}_n :\ A = 0 \text{ on } \mathsf{T}_n\}.$$

(a) A positively oriented bond (x, μ) with both end points in Λ_n is free if and only if $x_\nu \in \{1, \dots, n-1\}$ for both $\nu \neq \mu$; otherwise it lies in a face $\{x_\nu = 0\}$ or $\{x_\nu = n\}$ of the box, for some $\nu \neq \mu$.

(b) The map $\mathsf{j} : \mathcal{G}_n \times \mathcal{U}_n^{\mathsf{T}} \to \mathcal{U}_n$, $(u, V) \mapsto V^u$, is a diffeomorphism, and for every bounded Borel function F on $\mathcal{U}_n$

$$\int F(U) \prod_{b \in \mathsf{Bd}_n} \mathrm{d}U(b) = \int \prod_{x \in \Lambda_n^\circ} \mathrm{d}u(x) \int \prod_{b \in \mathsf{Bd}_n \setminus \mathsf{T}_n} \mathrm{d}V(b)\ F(V^u). \tag{7.vii}$$

The group $\mathcal{G}_n$ acts freely on $\mathcal{U}_n$. One has $S(U) = 0$ if and only if $U = \mathbf{1}^u$ for some $u \in \mathcal{G}_n$; the only zero of S in $\mathcal{U}_n^{\mathsf{T}}$ is $\mathbf{1}$.

(c) If $A \in \mathsf{Form}_n$ satisfies $\mathrm{d}A = 0$ on Pl_n and $\mathrm{d}^*A = 0$, then $A = 0$. Hence, for every $\tau_{\mathrm{gf}} > 0$, the quadratic form $\frac{1}{2}\|\mathrm{d}A\|^2 + \frac{1}{2\tau_{\mathrm{gf}}}\|\mathrm{d}^*A\|^2$ is positive definite on Form_n, and $\mathrm{d}^*\mathrm{d}$ is positive definite on X_n.

(d) There are numbers $0 < \varrho_2, \varrho_3 \le \varrho_1 \le \frac{1}{20}$, depending on n, with the following properties, where

$$\mathfrak{D} := \{(\lambda, A) \in \mathsf{X}_n \times \mathsf{Form}_n^{\mathsf{T}} :\ |\lambda|_\infty < \varrho_1,\ |A|_\infty < \varrho_3\}.$$

(i) The map $\Xi : (\lambda, A) \mapsto A^\lambda$ extends to a diffeomorphism of a neighbourhood of $\overline{\mathfrak{D}}$ onto an open subset of $\{A' \in \mathsf{Form}_n :\ |A'|_\infty < 1\}$.

(ii) $\det \mathsf{M}(A') \neq 0$ for every $A' \in \overline{\mathcal{W}}$, where $\mathcal{W} := \Xi(\mathfrak{D})$.

(iii) If $A \in \mathsf{Form}_n^{\mathsf{T}}$ and $|A|_\infty \le \varrho_3$, then $\lambda \mapsto \mathrm{d}^*(A^\lambda)$ is injective on $\{\lambda \in \mathsf{X}_n :\ |\lambda|_\infty \le \varrho_1\}$, and the image of $\{\lambda \in \mathsf{X}_n :\ |\lambda|_\infty < \varrho_1\}$ contains $\{c \in \mathsf{X}_n :\ |c|_\infty < \varrho_2\}$.

(iv) For every bounded Borel function Φ on $\mathcal{W}$ and every bounded Borel function $w \ge 0$ on X_n,

$$\begin{aligned}\int_{\mathcal{W}} \Phi(A')\, w(\mathrm{d}^*A')\, |\det \mathsf{M}(A')| \prod_{b \in \mathsf{Bd}_n} \sigma_1(|A'(b)|^2)\, \mathrm{d}A' \\ = \int_{|A|_\infty < \varrho_3} \prod_{b \in \mathsf{Bd}_n \setminus \mathsf{T}_n} \sigma_1(|A(b)|^2)\, \mathrm{d}A \int_{\{\mathrm{d}^*(A^\lambda):\, |\lambda|_\infty < \varrho_1\}} w(c)\, \Phi\big(A^{\lambda(c)}\big)\, \mathrm{d}c,\end{aligned} \tag{7.viii}$$

where $\lambda(c)$ is the solution of $\mathrm{d}^*(A^\lambda) = c$ with $|\lambda|_\infty < \varrho_1$, unique by (iii), and dA', dA, dc are the Lebesgue measures of the Euclidean spaces Form_n, $\mathsf{Form}_n^{\mathsf{T}}$, X_n, with the scalar product $(\cdot,\cdot)$ of Def. 7.2(2). Thus $dA' = \prod_{b \in \mathsf{Bd}_n} dA'(b)$, where $dA'(b)$ is the Lebesgue measure dY of $(\mathfrak{g}, \langle\cdot,\cdot\rangle)$ in the variable $A'(b)$; likewise dA is the product of these measures over the bonds of $\mathsf{Bd}_n \setminus \mathsf{T}_n$, and dc the product over the sites of Λ_n°.

■

Proof. Step 1: claim (a). Let (x, μ) be a positively oriented bond with both end points in Λ_n; then $x_\mu \in \{0, \dots, n-1\}$, and the end points x, $x + e_\mu$ have the same coordinates x_ν, $\nu \neq \mu$. If both these coordinates lie in $\{1, \dots, n-1\}$, then, as $n \ge 2$, one of the numbers x_μ, $x_\mu + 1$ lies in $\{1, \dots, n-1\}$, and the corresponding end point lies in Λ_n°: the bond is free. If $x_\nu \in \{0, n\}$ for some $\nu \neq \mu$, then both end points have this coordinate, neither lies in Λ_n°, and the bond lies in the face $\{x_\nu = 0\}$ or $\{x_\nu = n\}$.

Step 2: j *is bijective and smooth.* Let $U \in \mathcal{U}_n$. For $u' \in \mathcal{G}_n$ the condition $U^{u'} = \mathbf{1}$ on T_n is equivalent to

$$u'(x) = u'(x - e_1)\, U(\langle x - e_1, x\rangle) \qquad (x \in \Lambda_n^\circ).$$

Since $u' = \mathbf{1}$ at the sites with first coordinate 0, which lie in $\partial\Lambda_n$, this recursion in $x_1 = 1, \dots, n-1$ determines u' uniquely: $u'(x)$ is the ordered product of the values of U at the bonds $\langle y - e_1, y\rangle$ on the line from $(0, x_2, x_3)$ to x. Let $V := U^{u'} \in \mathcal{U}_n^{\mathsf{T}}$ and $u := u'^{-1} \in \mathcal{G}_n$; since $(U^v)^w = U^{wv}$, we have $V^u = U$. Conversely, if $U = V^u$ with $(u, V) \in \mathcal{G}_n \times \mathcal{U}_n^{\mathsf{T}}$, then $U^{u^{-1}} = V$ equals $\mathbf{1}$ on T_n, so $u^{-1} = u'$ and $V = U^{u'}$. Hence j is bijective. Both j and j^{-1} are composed of products and inverses of the values of u, V, U, and are therefore smooth (Def. 7.1).

Step 3: the action is free. If $U^u = U$ for some $u \in \mathcal{G}_n$, then $u(b_+) = U(b)^{-1} u(b_-) U(b)$ at every bond $b \in \mathsf{T}_n$, i.e. $u(x) = U(\langle x - e_1, x\rangle)^{-1} u(x - e_1)\, U(\langle x - e_1, x\rangle)$ for $x \in \Lambda_n^\circ$; the recursion of Step 2, started from $u = \mathbf{1}$ at $x_1 = 0$, gives $u = \mathbf{1}$.

Step 4: the identity eq. 7.vii. We fix $u \in \mathcal{G}_n$ on the right side of eq. 7.vii and substitute $U(b) := u(b_-) V(b) u(b_+)^{-1}$ for $b \in \mathsf{Bd}_n \setminus \mathsf{T}_n$. By Fubini's theorem and the invariance of the Haar measure under the maps $g \mapsto hgh'^{-1}$, $h, h' \in G$ (left and right invariance of Haar measure), the inner integral becomes $\int \prod_{b \in \mathsf{Bd}_n \setminus \mathsf{T}_n} \mathrm{d}U(b)\, F(U)$, where U takes the values $U(\langle x - e_1, x\rangle) = u(x - e_1) u(x)^{-1}$ on T_n. In the integral over u we now replace, on each line parallel to e_1 and for $x_1 = n-1, n-2, \dots, 1$ in this order, the variable $u(x)$ by the variable $U(\langle x - e_1, x\rangle) = u(x - e_1) u(x)^{-1}$, at fixed $u(x - e_1)$.

When this is done at x, the variable $u(x)$ enters no other value of U on T_n (the value at $\langle x, x+e_1\rangle$, if $x+e_1 \in \Lambda_n^\circ$, has already been replaced), and the map $u(x) \mapsto u(x-e_1)u(x)^{-1}$, an inversion followed by a left translation, preserves the Haar measure (left and right invariance of Haar measure). After the last substitution we arrive at the left side of eq. 7.vii.

Step 5: zeros of S. If $\partial U(p) = e^{iY}$ with $|Y| \le \pi$, then $a_p(U) = 1 - \cos|Y|$; hence $a_p(U) = 0$ if and only if $\partial U(p) = \mathbf{1}$, and since $a_p \ge 0$, $S(U) = 0$ if and only if $\partial U(p) = \mathbf{1}$ for every $p \in \mathsf{Pl}_n$. By the gauge invariance of a_p (Def. 7.2(1)), $S(\mathbf{1}^u) = S(\mathbf{1}) = 0$ for $u \in \mathcal{G}_n$. Now let $V \in \mathcal{U}_n^{\mathsf{T}}$ with $S(V) = 0$. We show that V equals $\mathbf{1}$ at every positively oriented bond with both end points in Λ_n, direction by direction.

- Bonds $(y,1)$ with $y_1 \le n-2$: if $y + e_1 \in \Lambda_n^\circ$, then $(y,1) \in \mathsf{T}_n$; otherwise one of y_2, y_3 lies in $\{0, n\}$, and $(y,1)$ is not free by (a). In both cases $V((y,1)) = \mathbf{1}$.
- Bonds (y,ν), $\nu \in \{2,3\}$: for $y_1 \le n-2$ the plaquette $p_{1\nu}(y)$ lies in Pl_n, and $\partial V(p_{1\nu}(y)) = \mathbf{1}$ together with $V((y,1)) = V((y+e_\nu,1)) = \mathbf{1}$ gives $V((y+e_1,\nu)) = V((y,\nu))$. The bonds (y,ν) with $y_1 = 0$ are not free, so $V((y,\nu)) = \mathbf{1}$ for $y_1 = 0$, and by induction for $y_1 \le n-1$; the bonds with $y_1 = n$ are not free.
- Bonds $(y,1)$ with $y_1 = n-1$: for $y_2 \le n-1$ the plaquette $p_{12}(y)$ gives $V((y+e_2,1)) = V((y,1))$, because $V = \mathbf{1}$ at the bonds in direction 2. The bonds with $y_2 = 0$ are not free, and induction in y_2 gives $V((y,1)) = \mathbf{1}$ for $y_2 \le n-1$; the bonds with $y_2 = n$ are not free.

Hence $V = \mathbf{1}$. Finally let $U \in \mathcal{U}_n$ with $S(U) = 0$. By (b), $U = V^u$ with $(u, V) \in \mathcal{G}_n \times \mathcal{U}_n^{\mathsf{T}}$, and $S(V) = S(U^{u^{-1}}) = S(U) = 0$; thus $V = \mathbf{1}$ and $U = \mathbf{1}^u$.

Step 6: claim (c). Let $A \in \mathsf{Form}_n$ with $\mathrm{d}A = 0$ on Pl_n and $\mathrm{d}^*A = 0$. We define $\lambda \in \mathsf{X}_n$ by $\lambda = 0$ on $\partial\Lambda_n$ and $\lambda(x) = \lambda(x-e_1) + A(\langle x-e_1, x\rangle)$ for $x \in \Lambda_n^\circ$, recursively in x_1. Then $A' := A - \mathrm{d}\lambda \in \mathsf{Form}_n$ vanishes on T_n, and $\mathrm{d}A' = \mathrm{d}A - \mathrm{d}\mathrm{d}\lambda = 0$ on Pl_n. The walk of Step 5, in additive form, shows $A' = 0$: A' vanishes at the bonds $(y,1)$ with $y_1 \le n-2$, which lie in T_n or are not free; for $\nu \in \{2,3\}$ and $y_1 \le n-2$,

$$0 = \mathrm{d}A'(p_{1\nu}(y)) = A'((y+e_1,\nu)) - A'((y,\nu)),$$

so $A' = 0$ at the bonds in the directions 2 and 3; and for $y_1 = n-1$,

$$0 = \mathrm{d}A'(p_{12}(y)) = A'((y,1)) - A'((y+e_2,1)),$$

so $A' = 0$ at the remaining bonds. Hence $A = \mathrm{d}\lambda$, and $0 = (\lambda, \mathrm{d}^*A) = (\lambda, \mathrm{d}^*\mathrm{d}\lambda) = \|\mathrm{d}\lambda\|^2$, so that $A = \mathrm{d}\lambda = 0$. Consequently, for $\tau_{\mathrm{gf}} > 0$ the quadratic form $\frac{1}{2}\|\mathrm{d}A\|^2 + \frac{1}{2\tau_{\mathrm{gf}}}\|\mathrm{d}^*A\|^2$ is non-negative and vanishes only at $A = 0$; it is therefore positive definite. Similarly, $(\lambda, \mathrm{d}^*\mathrm{d}\lambda) = \|\mathrm{d}\lambda\|^2 \ge 0$ for $\lambda \in \mathsf{X}_n$, and $\mathrm{d}\lambda = 0$ implies $\lambda(x) = \lambda(x-e_1)$ for $x \in \Lambda_n^\circ$, hence $\lambda = 0$, since $\lambda = 0$ at $x_1 = 0$; so $\mathrm{d}^*\mathrm{d}$ is positive definite on X_n.

Step 7: size. If $|\lambda|_\infty < \frac{1}{3}$ and $|A|_\infty < \frac{1}{3}$, then $|A(b)| + |\lambda(b_-)| + |\lambda(b_+)| < 1$ at every bond, and A^λ is defined. If $|\lambda|_\infty \le \frac{1}{10}$ and $|A|_\infty \le \frac{1}{10}$, then

$$|A^\lambda(b)| \le \tfrac{\pi}{2}\,|e^{iA^\lambda(b)} - \mathbf{1}| \le \tfrac{\pi}{2}(e^{3/10} - 1) < 1.$$

The second inequality is the product estimate in Def. 7.1(2). For the first one, let $Y \in \mathbb{B}_\pi$; then $(e^{iY} - \mathbf{1})^*(e^{iY} - \mathbf{1}) = (2 - 2\cos|Y|)\mathbf{1}$, so that $|e^{iY} - \mathbf{1}| = 2\sin\frac{|Y|}{2} \ge \frac{2}{\pi}|Y|$, by the inequality $\sin t \ge \frac{2}{\pi}t$ on $[0, \frac{\pi}{2}]$.

Step 8: Ξ in the exponential charts. The maps $\lambda \mapsto e^{i\lambda}$, $A \mapsto e^{iA}$, $A' \mapsto e^{iA'}$, taken site by site and bond by bond, are diffeomorphisms of the sets $\{|\lambda|_\infty < \pi\} \subset \mathsf{X}_n$, $\{|A|_\infty < \pi\} \subset \mathsf{Form}_n^{\mathsf{T}}$, $\{|A'|_\infty < \pi\} \subset \mathsf{Form}_n$ onto relatively open subsets of $\mathcal{G}_n$, $\mathcal{U}_n^{\mathsf{T}}$, $\mathcal{U}_n$ (Def. 7.1). On $\{|\lambda|_\infty < \frac{1}{3},\ |A|_\infty < \frac{1}{3}\}$ we have $e^{iA^\lambda} = \mathsf{j}(e^{i\lambda}, e^{iA})$ and $|A^\lambda|_\infty < \pi$; thus, in these charts, Ξ is the map j, and (b) shows that Ξ is a diffeomorphism of $\{|\lambda|_\infty < \frac{1}{3},\ |A|_\infty < \frac{1}{3}\}$ onto an open subset of Form_n, with $\Xi(0,0) = 0$. We note that $\dim \mathsf{X}_n + \dim \mathsf{Form}_n^{\mathsf{T}} = \dim \mathsf{Form}_n$, since T_n has as many elements as Λ_n°.

Let $\mathfrak{D}' \subset \{|\lambda|_\infty < \frac{1}{3},\ |A|_\infty < \frac{1}{3}\}$ be open, and let Φ' be a bounded Borel function on $\Xi(\mathfrak{D}')$. We apply eq. 7.vii to the function F on $\mathcal{U}_n$ which equals $\Phi'(A')$ at $U = e^{iA'}$, $A' \in \Xi(\mathfrak{D}')$, and 0 elsewhere, and we express every Haar integral by the Haar formula in Def. 7.1(2) at $u_0 = \mathbf{1}$, bond by bond and site by site. Each of these integrals carries the measure dY of $(\mathfrak{g}, \langle\cdot,\cdot\rangle)$ in the variable of its bond or site; since $(\cdot,\cdot)$ is the sum of $\langle\cdot,\cdot\rangle$ over the bonds, resp. the sites, the products of these measures are the Lebesgue measures dA', dA, $d\lambda$ of the Euclidean spaces Form_n, $\mathsf{Form}_n^{\mathsf{T}}$, X_n, as in (iv).

On the left side this produces the factor $\sigma_{\mathrm{H}}^{\#\mathsf{Bd}_n}$ and an integral over $\Xi(\mathfrak{D}')$. On the right side, the injectivity of j and of the exponential charts shows that, for (λ, A) in the domain of the charts, $(e^{iA})^{e^{i\lambda}} = e^{iA'}$ with $A' \in \Xi(\mathfrak{D}')$ holds if and only if $(\lambda, A) \in \mathfrak{D}'$, and then $A' = A^\lambda$; this produces the factor $\sigma_{\mathrm{H}}^{\#\Lambda_n^\circ + \#(\mathsf{Bd}_n \setminus \mathsf{T}_n)} = \sigma_{\mathrm{H}}^{\#\mathsf{Bd}_n}$ and an integral over $\mathfrak{D}'$. The powers of σ_{H} cancel, and

$$\int_{\Xi(\mathfrak{D}')} \Phi'(A') \prod_{b \in \mathsf{Bd}_n} \sigma_1(|A'(b)|^2)\, \mathrm{d}A' \\ = \int_{\mathfrak{D}'} \Phi'(A^\lambda) \prod_{x \in \Lambda_n^\circ} \sigma_1(|\lambda(x)|^2) \prod_{b \in \mathsf{Bd}_n \setminus \mathsf{T}_n} \sigma_1(|A(b)|^2)\, \mathrm{d}\lambda\, \mathrm{d}A.$$

Step 9: the derivative of the gauge function in λ. Let $\mathsf{c}(\lambda, A) := \mathrm{d}^*(A^\lambda)$ for $\lambda \in \mathsf{X}_n$ and $A \in \mathsf{Form}_n^{\mathsf{T}}$ with $|\lambda|_\infty < \frac{1}{10}$, $|A|_\infty < \frac{1}{10}$. For $H \in \mathsf{X}_n$ and $|s|$ small let $\mu(s) \in \mathsf{X}_n$ be defined, site by site and with small values, by $e^{i\mu(s)} := e^{i(\lambda+sH)}e^{-i\lambda}$. Then $\mu(0) = 0$, and μ is smooth in s. Differentiating $e^{i\mu(s)} = e^{i(\lambda+sH)}e^{-i\lambda}$ at $s = 0$ with the differential formula in Def. 7.1(2) on both sides, and using $\mathsf{dE}(0) = \mathbf{1}$, we find at every site

$$i\mu'(0) = i\big(\mathsf{dE}(\lambda)H\big)e^{i\lambda}e^{-i\lambda}, \qquad \text{that is,} \qquad \mu'(0) = \vartheta_\lambda H, \quad \vartheta_\lambda := \bigoplus_x \mathsf{dE}(\lambda(x)).$$

Since $e^{i(\lambda+sH)} = e^{i\mu(s)}e^{i\lambda}$, we have $(e^{iA})^{e^{i(\lambda+sH)}} = \big((e^{iA})^{e^{i\lambda}}\big)^{e^{i\mu(s)}}$, that is, $A^{\lambda+sH} = (A^\lambda)^{\mu(s)}$ for $|s|$ small, both sides lying in $\mathbb{B}_\pi$ at every bond and having the same exponential. By Step 7, $|A^\lambda|_\infty < 1$, and the definition eq. 7.iv of M with the chain rule gives

$$\partial_\lambda \mathsf{c}(\lambda, A)H = \frac{d}{ds}\, \mathrm{d}^*\big((A^\lambda)^{\mu(s)}\big)\Big|_{s=0} = \mathsf{M}(A^\lambda)\, \vartheta_\lambda H.$$

By the determinant formula in Def. 7.1(2), $\det \vartheta_\lambda = \prod_x \sigma_1(|\lambda(x)|^2) > 0$. At $(\lambda, A) = (0, 0)$ we have $\vartheta_0 = \mathbf{1}$ and $\partial_\lambda \mathsf{c}(0,0) = \mathsf{M}(0) = \mathsf{M}^{(0)} = -\mathrm{d}^*\mathrm{d}$, which is invertible by (c).

Step 10: choice of the radii; claims (i)–(iii). The map $\Theta_* : (\lambda, A) \mapsto (\mathsf{c}(\lambda, A), A)$, defined on the set of the pairs $(\lambda, A) \in \mathsf{X}_n \times \mathsf{Form}_n^{\mathsf{T}}$ with $|\lambda|_\infty < \frac{1}{10}$, $|A|_\infty < \frac{1}{10}$ and taking values in $\mathsf{X}_n \times \mathsf{Form}_n^{\mathsf{T}}$, is smooth, and its differential at $(0, 0)$ is block triangular with the invertible diagonal blocks $-\mathrm{d}^*\mathrm{d}$ and $\mathbf{1}$. By the inverse function theorem there is an open neighbourhood $\mathcal{N}$ of $(0, 0)$ on which Θ_* is a diffeomorphism onto an open set; on $\mathcal{N}$ the differential of Θ_* is invertible, and therefore so is $\partial_\lambda \mathsf{c}$. We first choose $\varrho_1 \le \frac{1}{20}$ with $\{|\lambda|_\infty \le 2\varrho_1,\ |A|_\infty \le 2\varrho_1\} \subset \mathcal{N}$, and then $\varrho_2, \varrho_3 \le \varrho_1$ with

$$\{c \in \mathsf{X}_n :\ |c|_\infty < \varrho_2\} \times \{A \in \mathsf{Form}_n^{\mathsf{T}} :\ |A|_\infty \le \varrho_3\} \subset \Theta_*(\{|\lambda|_\infty < \varrho_1,\ |A|_\infty < 2\varrho_1\});$$

this is possible because the right side is an open set containing $\Theta_*(0, 0) = (0, 0)$. These numbers depend on n only.

For (iii), let $A \in \mathsf{Form}_n^{\mathsf{T}}$ with $|A|_\infty \le \varrho_3$. Since Θ_* is injective on $\mathcal{N}$ and keeps the second component, $\lambda \mapsto \mathsf{c}(\lambda, A)$ is injective on $\{|\lambda|_\infty \le \varrho_1\}$. If $|c|_\infty < \varrho_2$, then $(c, A) = \Theta_*(\lambda, A'')$ with $|\lambda|_\infty < \varrho_1$, and necessarily $A'' = A$; thus $c = \mathrm{d}^*(A^\lambda)$ with $|\lambda|_\infty < \varrho_1$. For (i): since $\varrho_1, \varrho_3 \le \frac{1}{20}$, the closure $\overline{\mathfrak{D}} = \{|\lambda|_\infty \le \varrho_1,\ |A|_\infty \le \varrho_3\}$ lies in the open set $\{|\lambda|_\infty < \frac{1}{10},\ |A|_\infty < \frac{1}{10}\}$, on which Ξ is a diffeomorphism onto an open set (Step 8) with values in $\{|A'|_\infty < 1\}$ (Step 7).

For (ii): $\overline{\mathfrak{D}}$ is compact, so $\Xi(\overline{\mathfrak{D}})$ is compact; it contains $\mathcal{W}$ and is contained in $\overline{\mathcal{W}}$, hence equals $\overline{\mathcal{W}}$. For $(\lambda, A) \in \overline{\mathfrak{D}} \subset \mathcal{N}$ the operator $\partial_\lambda \mathsf{c}(\lambda, A) = \mathsf{M}(A^\lambda)\vartheta_\lambda$ is invertible and $\det \vartheta_\lambda \neq 0$; hence $\det \mathsf{M}(A^\lambda) \neq 0$.

Step 11: claim (iv). By eq. 7.vi, the entries of $\mathsf{M}(A')$ are continuous functions of A' for $|A'|_\infty < 1$; hence $|\det \mathsf{M}|$ is bounded on the compact set $\overline{\mathcal{W}}$. We apply the identity of Step 8 with $\mathfrak{D}' := \mathfrak{D}$ and the bounded Borel function $\Phi' := \Phi \cdot (w \circ \mathrm{d}^*) \cdot |\det \mathsf{M}|$ on $\mathcal{W}$. Its left side is the left side of eq. 7.viii. On its right side we integrate first over λ, at fixed A with $|A|_\infty < \varrho_3$ (Fubini's theorem). By (iii) and Step 9, $\lambda \mapsto c = \mathsf{c}(\lambda, A)$ is an injective smooth map on $\{|\lambda|_\infty < \varrho_1\}$ with the invertible differential $\mathsf{M}(A^\lambda)\vartheta_\lambda$; hence it is a diffeomorphism onto its open image $\{\mathrm{d}^*(A^\lambda) : |\lambda|_\infty < \varrho_1\}$, and

$$dc = |\det \mathsf{M}(A^\lambda)| \prod_{x \in \Lambda_n^\circ} \sigma_1(|\lambda(x)|^2)\, \mathrm{d}\lambda.$$

Since $\Phi'(A^\lambda) = \Phi(A^\lambda)\, w(\mathsf{c}(\lambda, A))\, |\det \mathsf{M}(A^\lambda)|$, the substitution $c = \mathsf{c}(\lambda, A)$ turns the inner integral into $\int w(c)\, \Phi(A^{\lambda(c)})\, \mathrm{d}c$ over this image, and eq. 7.viii follows. □

7.1.3. *A coefficient defined before the continuum limit.* The quantity computed here is a coefficient of the Wilson covariance in a fixed Dirichlet box. Gauge fixing provides coordinates for its Gaussian evaluation. It does not change the observable whose asymptotic coefficient is being defined. For each fixed box, the construction gives

$$\langle a_p; a_q \rangle_{n,\beta} = \beta^{-2} \mathsf{B}_0^{[n]}(p, q) + \beta^{-3} \mathsf{B}_1^{[n]}(p, q) + O_n(\beta^{-4}).$$

The constant in this remainder retains its box dependence. The infinite-lattice coefficient is obtained by exhausting the boxes after extracting the coefficient. The two limits are therefore iterated. The relation of this quantity to an interacting continuum covariance is supplied by the preceding section's renormalization comparisons.

The finite-box calculation starts with a gauge slice on which the quadratic action has no zero modes. The Faddeev–Popov change of variables supplies its determinant, and exponential coordinates supply the Haar density. Both contribute to the coefficient. Write $t = \beta^{-1/2}$. If X_j, Y_j are the homogeneous insertion terms of degree $j + 2$, and W_j is the sum of the phase term of degree $j + 2$ and the density term of degree j, the second connected Laplace coefficient is

$$\begin{aligned} \gamma_2 = {} & \langle X_1; Y_1 \rangle_0 + \langle X_0; Y_2 \rangle_0 + \langle X_2; Y_0 \rangle_0 - \langle X_1; Y_0; W_1 \rangle_0 \\ & - \langle X_0; Y_1; W_1 \rangle_0 - \langle X_0; Y_0; W_2 \rangle_0 + \tfrac{1}{2} \langle X_0; Y_0; W_1; W_1 \rangle_0. \end{aligned} \tag{7.ix}$$

Parity makes the intervening odd coefficients zero. This formula is useful before any graph notation: it states exactly which Taylor terms of the action, density and two insertions must be retained. Omitting the density terms changes the answer.

For Gaussian polynomials, a joint cumulant is the sum of Wick pairings whose graph connects all its polynomial vertices. The partition formula for cumulants removes the disconnected pairings. Applying that rule to eq. 7.ix gives the finite graph classes and their numerical multiplicities. The source retains the counting by incidence matrices, so each factor can be checked without relying on a picture or an implicit convention for equivalent contractions.

7.1.4. *Cancellations before long-distance estimates.* The plain, mixed and curl lines have different decay. In Feynman gauge their leading orders are respectively $|u|^{-1}$, $|u|^{-2}$ and $|u|^{-3}$. Products and convolutions of these lines generate local contributions as well as the desired homogeneous kernel. Estimating every local contribution before adding the relevant classes would retain terms that cancel. The zero-momentum identities are what remove them.

For the self-energy, the five contributions comprise the Wilson vertices, the ghost terms and the Haar term. Evaluating the Hessian at a constant abelian background expresses their sum as a

second derivative of a periodic logarithm. The integral of that derivative vanishes: opposite faces of a period cell cancel, and the flux through a sphere of radius ε is $O(\varepsilon)$. Put

$$Z_0 = \int_{[-\pi,\pi]^3} \left(\sum_{\mu=1}^{3} 4 \sin^2(k_\mu/2) \right)^{-1} \frac{\mathrm{d}k}{(2\pi)^3}.$$

At gauge parameter one, the five diagonal coefficients are

$$-20Z_0 + \tfrac{8}{3}, \quad 16Z_0 - \tfrac{28}{9}, \quad 4Z_0 - \tfrac{2}{3}, \quad \tfrac{4}{9}, \quad \tfrac{2}{3}.$$

Their sum is zero. The plaquette vertex has a corresponding four-term identity. Point reflections supply the required vanishing first derivatives. Together these facts say that the symbols of the combined local insertion terms vanish at zero momentum to the required order.

The kernel lemmas then use these moment cancellations. A local insertion with vanishing low moments gains differences of the external lines instead of contributing its unweighted mass. In the two-point coefficient this places the local pieces at order $|u|^{-6}$ or smaller. The surviving chain, bubble and kite terms have homogeneous pieces of degree -5. Their subtractions and borderline convolution sums give the remainder $O(|u|^{-6}(1 + \log|u|))$. The proof retains the individual kernel estimates because absolute convergence, existence of a homogeneous term and the error after subtraction are distinct claims.

The six continuum integrands reproduce precisely these homogeneous pieces. Their rotationally invariant scalar sum determines κ^{lat}; the individual plane kernels remain the functions U_{ij} of theorem V. The method-of-images estimates separately show that the finite-box coefficients converge to the lattice sums. They complete the definition of the lattice coefficient, rather than supplying a joint bound for growing boxes and couplings.

7.1.5. *Why a residue determines the number.* For a kernel of degree -5 in dimension three, the relevant Fourier term is $\mathsf{p}^2 \log|\mathsf{p}|$. The normalization is fixed by lemma 7.41. With the Fourier convention used there, analytic continuation of the transform of $|p|^s$ gives

$$c(s) = \frac{2^s \Gamma_{\mathrm{E}}((3+s)/2)}{\pi^{3/2} \Gamma_{\mathrm{E}}(-s/2)}, \qquad c(2) = 0, \qquad c'(2) = \frac{3}{2\pi}.$$

Differentiation at $s = 2$ converts $p^2 \log|p|$ into $(3/2\pi)|x|^{-5}$ away from the origin. Stating this constant before the graph evaluation fixes the Fourier normalization and the eventual sign.

For the six integrands $F_\Gamma(k, l, \mathsf{p})$, introduce

$$A = k^2, \quad A' = (k - \mathsf{p})^2, \quad B = l^2, \quad B' = (l - \mathsf{p})^2, \quad C = (k - l)^2, \quad P = \mathsf{p}^2.$$

All contracted numerators become polynomials in these quantities. Take the quadratic term in p, average its angular dependence as in definition 7.47, and express the result as Laurent monomials in A, B, C. The functional $\mathcal{R}$ extracts the coefficient of $(ABC)^{-1}$. Lemma 7.48 provides an explicit rule on each monomial, so the remaining six evaluations are finite algebraic calculations with recorded symmetry factors.

This coefficient is singled out by the finite-part integral under a dilation of the cutoff. For a smooth radial cutoff ϕ equal to one near the origin, the contributing integral is

$$\iint \frac{\phi((k,l)/\Lambda) - \phi(k,l)}{k^2 l^2 (k-l)^2} \, \mathrm{d}k \, \mathrm{d}l = 4\pi^4 \log \Lambda.$$

In six-dimensional polar coordinates, the radial factor is the integral of the cutoff difference against $\mathrm{d}s/s$, hence $\log\Lambda$. The angular integral is $4\pi^4$, computed in lemma 7.49. The other monomials allowed by the graph numerators have zero finite-part dilation difference. After the two Fourier measures this leaves $\mathcal{R}[F_\Gamma]/(16\pi^2)$ as the coefficient of $\log\Lambda$.

The expansion by regions justifies using that finite part: it separates the small-momentum regions, subtracts their specified singular terms and bounds the remainder. Comparing the cutoff-dilation expansion with the Fourier expansion then gives the bridge

$$\kappa_\Gamma^{\mathrm{lat}} = -\frac{3}{32\pi^3}\, 24\, \mathcal{R}[F_\Gamma].$$

The six values are

$$-\frac{4}{3}, \quad -\frac{13}{12}, \quad -\frac{1}{12}, \quad 0, \quad \frac{13}{6}, \quad -\frac{1}{3}.$$

Their sum is $-2/3$, and therefore $\kappa^{\mathrm{lat}} = 3/(2\pi^3)$ and $c_1 = \kappa^{\mathrm{lat}}/c_2 = 8/(3\pi)$. The intermediate cancellations and residue calculation explain the number; the final arithmetic alone does not constitute its derivation.

For separated test pairs, the kernel remainder and the lattice sampling estimate give an error of order $a(1 + |\log(d/a)|)$ in the smeared coefficient. The lower bound for $\mathcal{K}_5$ then gives the relative error in theorem X. This conclusion concerns separated supports. It supplies no coincident-plaquette term and does not extend the coefficient calculation to constant or overlapping sources.

7.1.6. *The coefficient.*

Definition 7.7 (Quadratic form, covariances, vertices)**.** Let Ω be either $\mathbb{Z}^3$, indicated by the upper index lat, or a box Λ_n, $n \geq 2$, indicated by the upper index $[n]$, and let $\tau_{\mathrm{gf}} > 0$; the index is omitted when Ω is fixed.

(1) We set

$$\begin{gathered} \mathsf{S}_2(A) := \tfrac{1}{2}\|\mathrm{d}A\|^2 + \tfrac{1}{2\tau_{\mathrm{gf}}}\|\mathrm{d}^* A\|^2, \\ \mathsf{C}_{\tau_{\mathrm{gf}}} = \left(\mathrm{d}^*\mathrm{d} + \tau_{\mathrm{gf}}^{-1}\mathrm{d}\mathrm{d}^*\right)^{-1}, \qquad \mathsf{G} = (\mathrm{d}^*\mathrm{d})^{-1} \text{ on site functions.} \end{gathered} \tag{7.x}$$

In the box, $\mathrm{d}^*\mathrm{d} + \tau_{\mathrm{gf}}^{-1}\mathrm{d}\mathrm{d}^*$ is positive definite on $\mathsf{Form}_n^{\mathbb{R}}$ and $\mathrm{d}^*\mathrm{d}$ is positive definite on $\mathsf{X}_n^{\mathbb{R}}$, by Lem. 7.6(c) applied to forms and functions with a single non-vanishing component. $\mathsf{C}_{\tau_{\mathrm{gf}}}^{[n]}$ and $\mathsf{G}^{[n]}$ are the inverses of these operators, with the matrices $\mathsf{C}_{\tau_{\mathrm{gf}}}^{[n]}(b, b')$, $b, b' \in \mathsf{Bd}_n$, and $\mathsf{G}^{[n]}(x, y)$, $x, y \in \Lambda_n^\circ$; we set $\mathsf{C}_{\tau_{\mathrm{gf}}}^{[n]}(b, b') := 0$ if b or b' is not a free bond, and $\mathsf{G}^{[n]}(x, y) := 0$ if x or y does not lie in Λ_n°. On $\mathbb{Z}^3$ we define, with $\mathsf{k} = \mathsf{k}(k)$,

$$\begin{aligned} \mathsf{C}_{\tau_{\mathrm{gf}}}^{\mathrm{lat}}((x,\mu),(y,\nu)) &:= \int_{\mathbb{T}} e^{ik\cdot(x-y)} \Big[\frac{\delta_{\mu\nu}}{|\mathsf{k}|^2} + (\tau_{\mathrm{gf}} - 1)\frac{\mathsf{k}_\mu \overline{\mathsf{k}_\nu}}{|\mathsf{k}|^4}\Big] d^\circ k, \\ \mathsf{G}^{\mathrm{lat}}(x,y) &:= \int_{\mathbb{T}} \frac{e^{ik\cdot(x-y)}}{|\mathsf{k}|^2}\, d^\circ k. \end{aligned}$$

The matrix in brackets is Hermitian, with the eigenvalues $|\mathsf{k}|^{-2}$ (twice) and $\tau_{\mathrm{gf}}|\mathsf{k}|^{-2}$; as $|\mathsf{k}|^2 \geq \frac{4}{\pi^2}|k|^2$ on $[-\pi, \pi]^3$, the integrands are bounded by $\frac{\pi^2}{4}\max\{1, \tau_{\mathrm{gf}}\}|k|^{-2}$, which is integrable. The bracket is positive, and it is replaced by its complex conjugate under $k \to -k$, because $\mathsf{k}_\mu(-k) = \overline{\mathsf{k}_\mu(k)}$. Hence the kernels are real and symmetric. They are also positive semidefinite: for finitely supported complex functions f on the positively oriented bonds, resp. on the sites,

$$\begin{aligned} \sum_{b,b'} \overline{f(b)}\, \mathsf{C}_{\tau_{\mathrm{gf}}}^{\mathrm{lat}}(b,b') f(b') &= \int_{\mathbb{T}} v(k)^* [\cdots]\, v(k)\, d^\circ k \geq 0, \\ \sum_{x,y} \overline{f(x)}\, \mathsf{G}^{\mathrm{lat}}(x,y) f(y) &= \int_{\mathbb{T}} \Big|\sum_y e^{-ik\cdot y} f(y)\Big|^2 |\mathsf{k}|^{-2}\, d^\circ k \geq 0, \end{aligned}$$

where $v_\nu(k) := \sum_y e^{-ik\cdot y} f((y,\nu))$ and $[\cdots]$ is the bracket.

The definition on $\mathbb{Z}^3$ is explained by plane waves. Let $A_\mu(x) = e^{ik\cdot x}\alpha_\mu$ with $\alpha \in \mathbb{C}^3$, let $\lambda(x) = e^{ik\cdot x}c$ with $c \in \mathbb{C}$, and let F be the plaquette function with $F_{\rho\nu}(x) = e^{ik\cdot x} f_{\rho\nu}$ for all ρ, ν,

where $f_{\nu\rho} = -f_{\rho\nu}$; we write $\mathsf{k}^*\alpha := \sum_\mu \overline{\mathsf{k}_\mu}\,\alpha_\mu$. The formulas for d and d^* in Def. 7.2(2), together with $e^{ik_\mu} - 1 = \mathsf{k}_\mu$ and $e^{-ik_\mu} - 1 = \overline{\mathsf{k}_\mu}$, give

$$\mathrm{d}A(p_{\mu\nu}(x)) = e^{ik\cdot x}(\mathsf{k}_\mu\alpha_\nu - \mathsf{k}_\nu\alpha_\mu), \qquad \mathrm{d}^*A(x) = e^{ik\cdot x}\,\mathsf{k}^*\alpha,$$
$$(\mathrm{d}\lambda)((x,\rho)) = e^{ik\cdot x}\,\mathsf{k}_\rho\,c, \qquad (\mathrm{d}^*F)((x,\rho)) = -e^{ik\cdot x}\sum_\nu \overline{\mathsf{k}_\nu}\,f_{\rho\nu}.$$

For $F = \mathrm{d}A$ we have $f_{\rho\nu} = \mathsf{k}_\rho\alpha_\nu - \mathsf{k}_\nu\alpha_\rho$, and for $\lambda = \mathrm{d}^*A$ we have $c = \mathsf{k}^*\alpha$; therefore

$$(\mathrm{d}^*\mathrm{d}A)((x,\rho)) = e^{ik\cdot x}(|\mathsf{k}|^2\alpha_\rho - \mathsf{k}_\rho\,\mathsf{k}^*\alpha), \qquad (\mathrm{d}\mathrm{d}^*A)((x,\rho)) = e^{ik\cdot x}\,\mathsf{k}_\rho\,\mathsf{k}^*\alpha.$$

Thus $\mathrm{d}^*\mathrm{d} + \tau_{\mathrm{gf}}^{-1}\mathrm{d}\mathrm{d}^*$ has the symbol $|\mathsf{k}|^2\mathbf{1} - (1-\tau_{\mathrm{gf}}^{-1})\mathsf{k}\mathsf{k}^*$: it maps the plane wave with the amplitude α to the plane wave with the amplitude $(|\mathsf{k}|^2\mathbf{1} - (1-\tau_{\mathrm{gf}}^{-1})\mathsf{k}\mathsf{k}^*)\alpha$. The inverse of this matrix is the bracket: as $\mathsf{k}^*\mathsf{k} = |\mathsf{k}|^2$,

$$(|\mathsf{k}|^2\mathbf{1} - (1-\tau_{\mathrm{gf}}^{-1})\mathsf{k}\mathsf{k}^*)\Big[\frac{\mathbf{1}}{|\mathsf{k}|^2} + (\tau_{\mathrm{gf}}-1)\frac{\mathsf{k}\mathsf{k}^*}{|\mathsf{k}|^4}\Big]$$
$$= \mathbf{1} + \big[(\tau_{\mathrm{gf}}-1) - (1-\tau_{\mathrm{gf}}^{-1}) - (1-\tau_{\mathrm{gf}}^{-1})(\tau_{\mathrm{gf}}-1)\big]\frac{\mathsf{k}\mathsf{k}^*}{|\mathsf{k}|^2} = \mathbf{1},$$

since $(1-\tau_{\mathrm{gf}}^{-1})(1+(\tau_{\mathrm{gf}}-1)) = \tau_{\mathrm{gf}} - 1$.

(2)

Derived kernels. With $b_i(p)$ and s_i as in Lem. 7.5 we set

$$\mathsf{E}_p(b) := \sum_i s_i\,\mathsf{C}_{\tau_{\mathrm{gf}}}(b_i(p), b), \qquad \Pi(p,q) := \sum_j s_j\,\mathsf{E}_p(b_j(q)),$$
$$\mathsf{g}(b) := \mathsf{G}(b_+,b_+) + \mathsf{G}(b_-,b_-) - 2\mathsf{G}(b_+,b_-), \qquad \mathsf{D}(b;b') := \tfrac{1}{2}\sum_{x\in\{b'_-,b'_+\}}[\mathsf{G}(x,b_+) - \mathsf{G}(x,b_-)].$$

The kernels E_p and Π do not depend on τ_{gf}. On $\mathbb{Z}^3$, for $p = p_{\mu\nu}(x)$ and a bond (y,ρ), the four terms of E_p combine to

$$\sum_i s_i\,\mathsf{C}^{\mathrm{lat}}_{\tau_{\mathrm{gf}}}(b_i(p),(y,\rho)) = \int_{\mathbb{T}} e^{ik\cdot(x-y)}(\mathsf{k}_\mu\,[\cdots]_{\nu\rho} - \mathsf{k}_\nu\,[\cdots]_{\mu\rho})\,d^\circ k,$$

where $[\cdots]_{\alpha\rho}$ denotes the entry (α,ρ) of the bracket in (1); the τ_{gf}-dependent term of the bracket contributes $(\tau_{\mathrm{gf}}-1)(\mathsf{k}_\mu\mathsf{k}_\nu - \mathsf{k}_\nu\mathsf{k}_\mu)\overline{\mathsf{k}_\rho}|\mathsf{k}|^{-4} = 0$. In the box,

$$\mathsf{C}_{\tau_{\mathrm{gf}}} = \mathsf{C}_1 + (\tau_{\mathrm{gf}}-1)\,\mathrm{d}\,\mathsf{G}^2\mathrm{d}^*, \qquad \mathrm{d}\mathrm{d} = 0;$$

since $\mathsf{E}_p = \mathsf{C}_{\tau_{\mathrm{gf}}}\mathrm{d}^*\delta_p$ by the symmetry of $\mathsf{C}_{\tau_{\mathrm{gf}}}$, where δ_p is the plaquette function with the value 1 at p and 0 elsewhere, and $\mathrm{d}^*\mathrm{d}^* = (\mathrm{d}\mathrm{d})^* = 0$, the second term does not contribute to E_p or to Π. To prove the identity for $\mathsf{C}_{\tau_{\mathrm{gf}}}$, let $\tau := \tau_{\mathrm{gf}}$ and $\mathsf{L}_\tau := \mathrm{d}^*\mathrm{d} + \tau^{-1}\mathrm{d}\mathrm{d}^*$. From $\mathrm{d}\mathrm{d} = 0$ we get $\mathsf{L}_\tau\mathrm{d} = \tau^{-1}\mathrm{d}(\mathrm{d}^*\mathrm{d})$, hence $\mathsf{L}_\tau\,\mathrm{d}\mathsf{G}^2\mathrm{d}^* = \tau^{-1}\mathrm{d}\mathsf{G}\mathrm{d}^*$.

Moreover $\mathrm{d}^*\mathsf{L}_1 = \mathrm{d}^*\mathrm{d}\,\mathrm{d}^*$ gives $\mathrm{d}^*\mathsf{C}_1 = \mathsf{G}\mathrm{d}^*$, and therefore $\mathsf{L}_\tau\mathsf{C}_1 = \mathbf{1} + (\tau^{-1}-1)\mathrm{d}\mathrm{d}^*\mathsf{C}_1 = \mathbf{1} + (\tau^{-1}-1)\mathrm{d}\mathsf{G}\mathrm{d}^*$. Adding the first identity multiplied by $\tau - 1$ to the second we obtain $\mathsf{L}_\tau[\mathsf{C}_1 + (\tau-1)\mathrm{d}\mathsf{G}^2\mathrm{d}^*] = \mathbf{1}$.

On $\mathbb{Z}^3$, for $b = (x,\mu)$, the reality and symmetry of $\mathsf{G}^{\mathrm{lat}}$ give

$$\mathsf{g}(b) = \int_{\mathbb{T}} \frac{2 - e^{ik_\mu} - e^{-ik_\mu}}{|\mathsf{k}|^2}\,d^\circ k = \int_{\mathbb{T}} \frac{|\mathsf{k}_\mu|^2}{|\mathsf{k}|^2}\,d^\circ k = \frac{1}{3},$$

since $|\mathsf{k}|^2$ is invariant under permutations of the coordinates of k, so that the three integrals $\int |\mathsf{k}_\mu|^2|\mathsf{k}|^{-2}d^\circ k$ are equal, with sum 1.

(3)

Vertices. With sums over the plaquettes and the positively oriented bonds of Ω, that is, over Pl_n and Bd_n in the box, we set

$$\begin{aligned}\mathsf{S}_3 &:= \sum_p \mathsf{a}_p^{(3)},\\ \mathsf{V}_2 &:= \sum_p \mathsf{a}_p^{(4)} + \frac{1}{3}\sum_b |A(b)|^2 + \frac{2}{3}\sum_b \mathsf{g}(b)|A(b)|^2 - 4\sum_{b,b'} \mathsf{D}(b;b')\mathsf{D}(b';b)\langle A(b), A(b')\rangle.\end{aligned} \tag{7.xi}$$

The four terms of V_2 are called the *quartic Wilson vertex* $\mathsf{S}_4 := \sum_p \mathsf{a}_p^{(4)}$, the *Haar term*, the *ghost tadpole* and the *ghost loop*. On $\mathbb{Z}^3$, S_3 and V_2 are formal sums of local terms in the sense of Def. 7.8(3). In the box,

$$\begin{aligned}\tfrac{2}{3}\sum_b \mathsf{g}(b)|A(b)|^2 &= -\operatorname{tr}\big[(\mathsf{M}^{(0)})^{-1}\mathsf{M}^{(2)}(A)\big],\\ -4\sum_{b,b'} \mathsf{D}(b;b')\mathsf{D}(b';b)\langle A(b), A(b')\rangle &= \tfrac{1}{2}\operatorname{tr}\big[\big((\mathsf{M}^{(0)})^{-1}\mathsf{M}^{(1)}(A)\big)^2\big],\end{aligned} \tag{7.xii}$$

the traces being taken on X_n, and $\operatorname{tr}\big[(\mathsf{M}^{(0)})^{-1}\mathsf{M}^{(1)}(A)\big] = 0$.

To prove these identities we use $(\mathsf{M}^{(0)})^{-1} = -\mathsf{G}$ (Lem. 7.5(d)). For a family $T = (T_b)$ of endomorphisms $T_b \in \operatorname{End}\mathfrak{g}$, acting on forms bond by bond, the cyclicity of the trace gives

$$\operatorname{tr}\big[\mathsf{G}\,\mathrm{d}^* T \mathrm{d}\big] = \sum_b \operatorname{tr} T_b \cdot (\mathrm{d}\mathsf{G}\mathrm{d}^*)(b,b) = \sum_b \mathsf{g}(b)\operatorname{tr} T_b,$$

because the diagonal element of the kernel of $\mathrm{d}\mathsf{G}\mathrm{d}^*$ at b is $\mathsf{G}(b_+,b_+) - \mathsf{G}(b_+,b_-) - \mathsf{G}(b_-,b_+) + \mathsf{G}(b_-,b_-) = \mathsf{g}(b)$. The formula $X\times(Y\times Z) = \langle X,Z\rangle Y - \langle X,Y\rangle Z$ gives $\operatorname{tr}[X\times(Y\times\cdot)] = \langle X,Y\rangle - 3\langle X,Y\rangle = -2\langle X,Y\rangle$. By Lem. 7.5(d), $\mathsf{M}^{(2)}(A) = \mathrm{d}^*T\mathrm{d}$ with $T_b = -\frac{1}{3}A(b)\times(A(b)\times\cdot)$, so $\operatorname{tr} T_b = \frac{2}{3}|A(b)|^2$, and the first identity follows. Next let $\mathsf{R} := \mathsf{G}\,\mathrm{d}^*(A\times\dot{-})$, so that $(\mathsf{M}^{(0)})^{-1}\mathsf{M}^{(1)}(A) = -2\mathsf{R}$. Its kernel is

$$\mathsf{R}(x,y) = \sum_b \big[\mathsf{G}(x,b_+) - \mathsf{G}(x,b_-)\big]\,\tfrac{1}{2}\big[\delta_{y,b_-} + \delta_{y,b_+}\big]\,A(b)\times\cdot \;\in \operatorname{End}\mathfrak{g}.$$

Since $\operatorname{tr}[X\times\cdot] = 0$, we get $\operatorname{tr}\mathsf{R} = 0$, which is the last assertion. Summing first over x and y in $\operatorname{tr}\mathsf{R}^2 = \sum_{x,y}\operatorname{tr}[\mathsf{R}(x,y)\mathsf{R}(y,x)]$ we find

$$\operatorname{tr}\mathsf{R}^2 = \sum_{b,b'} \mathsf{D}(b;b')\mathsf{D}(b';b)\operatorname{tr}\big[A(b)\times(A(b')\times\cdot)\big] = -2\sum_{b,b'} \mathsf{D}(b;b')\mathsf{D}(b';b)\langle A(b),A(b')\rangle,$$

and $\frac{1}{2}\operatorname{tr}[(-2\mathsf{R})^2] = 2\operatorname{tr}\mathsf{R}^2$ gives the second identity. ∎

Definition 7.8 (Gaussian family; cumulants of polynomials; formal sums)**.** Let Ω and τ_{gf} be as in Def. 7.7.

(1) The variables $A^a(b)$, where b ranges over the positively oriented bonds of Ω (over the free bonds if $\Omega = \Lambda_n$) and $a \in \{1,2,3\}$, form a centred Gaussian family with

$$\langle A^a(b)A^{a'}(b')\rangle_0 = \delta_{aa'}\mathsf{C}^{\Omega}_{\tau_{\mathrm{gf}}}(b,b');$$

such a family exists because this covariance is positive semidefinite (Def. 7.7(1)), and its mean is denoted by $\langle\cdot\rangle_0$. In the box we put $A^a(b) := 0$ at the bonds which are not free, in accordance with the definition of Form_n. A *leg* is a variable $A^a(b)$ (*plain leg*) or a variable $(\mathrm{d}A)^a(p) = \sum_i s_i A^a(b_i(p))$ (*curl leg*). By the definitions of E_p and Π,

$$\begin{aligned}\langle A^a(b)A^{a'}(b')\rangle_0 &= \delta_{aa'}\mathsf{C}(b,b'), \qquad \langle(\mathrm{d}A)^a(p)A^{a'}(b)\rangle_0 = \delta_{aa'}\mathsf{E}_p(b),\\ \langle(\mathrm{d}A)^a(p)(\mathrm{d}A)^{a'}(q)\rangle_0 &= \delta_{aa'}\Pi(p,q).\end{aligned} \tag{7.xiii}$$

(2) Let $X_1, \dots, X_r$ be polynomials in the variables of the family, and $m_J := \langle \prod_{i\in J} X_i\rangle_0$ for $\emptyset \neq J \subset \{1,\dots,r\}$. Their *joint cumulants* $\langle X_i; i \in J\rangle_0 := \kappa_J$ are defined by the partition formula

$$\kappa_J = \sum_{\pi\in \mathrm{Part}(J)} (-1)^{|\pi|-1}(|\pi|-1)! \prod_{I\in\pi} m_I. \tag{7.xiv}$$

Here $\mathrm{Part}(J)$ denotes the partitions π of J into nonempty blocks I, and $|\pi|$ is the number of blocks. Equivalently, (κ_J) is the unique family satisfying

$$m_J = \sum_{\pi\in \mathrm{Part}(J)} \prod_{I\in\pi} \kappa_I. \tag{7.xv}$$

The equivalence is proved below. The cumulants are symmetric and multilinear in the X_i: each index belongs to exactly one block of each partition, every moment is linear in each factor, and the formula involves J only as a set. Formula eq. 7.xiv requires only the indicated moments; no exponential moment of a polynomial in the Gaussian family is assumed.

When exponential moments exist, this definition agrees with differentiation of their logarithm. For bounded complex variables, the exponential mean is holomorphic near zero and can be differentiated under the integral. For real variables $Y_i = X_i$, $i \in J = \{1,\dots,n\}$, assume instead that $\nu(e^{t\sum_i |Y_i|}) < \infty$ for some $t > 0$, where $\nu(\cdot) = \langle\cdot\rangle_0$. Then $h(z) := \nu(e^{\sum_i z_i Y_i})$ is holomorphic on $\{|\operatorname{Re} z_i| < t\}$ and $h(0) = 1$. Write $\partial_I = \prod_{i\in I} \partial_{z_i}$. For $|\operatorname{Re} z_i| \le t/2$ and $I \subset J$,

$$\prod_{i\in I} |Y_i|\, |e^{\sum_i z_i Y_i}| \le \Big(\frac{2}{t}\Big)^{|I|} e^{t\sum_i |Y_i|},$$

by $s \le e^s$ at $s = t|Y_i|/2$. This integrable majorant gives $\partial_I h(0) = m_I$. The mixed derivative formula for the logarithm, proved below, is

$$\partial_J \log h(0) = \sum_{\pi\in \mathrm{Part}(J)} (-1)^{|\pi|-1}(|\pi|-1)! \prod_{I\in\pi} \partial_I h(0). \tag{7.xvi}$$

It identifies this derivative with κ_J.

(3) A polynomial in the Gaussian family can be written in several ways as a sum of monomials in legs, since a curl leg is a signed sum of four plain legs. The *standard writing* of $\mathsf{a}_p^{(m)}$, $m = 2, 3, 4$, is obtained from eq. 7.v by multiplying out in components, writing each displayed factor $\mathrm{d}A(p)$ in curl legs $(\mathrm{d}A)^a(p)$ and the polynomials $\mathfrak{z}_1$, $\mathfrak{z}_2$ in plain legs $A^a(b_i(p))$. Thus $\mathsf{a}_p^{(2)} = \frac{1}{2}\sum_a \big((\mathrm{d}A)^a(p)\big)^2$; every monomial of $\mathsf{a}_p^{(3)}$ consists of one curl leg and two plain legs; every monomial of $\mathsf{a}_p^{(4)}$ consists of four plain legs (those coming from $\frac{1}{2}|\mathfrak{z}_1|^2$), or of one curl leg and three plain legs, or of four curl legs. The other local terms of V_2 are written in plain legs.

Formal sums. Let $X_i = \sum_{z\in \mathsf{Z}_i} X_{i,z}$, $i = 1,\dots,r$, be formal sums of polynomials $X_{i,z}$ over countable index sets Z_i. We set

$$\langle X_1;\dots;X_r\rangle_0 := \sum_{z=(z_1,\dots,z_r)} \langle X_{1,z_1};\dots;X_{r,z_r}\rangle_0,$$

provided the following absolute convergence holds: when every entry is in its standard writing and each cumulant of monomials is expanded as the sum over pairings of Lem. 7.9, the sum of the moduli of all the resulting terms is finite. Under this proviso $\sum_z |\langle X_{1,z_1};\dots;X_{r,z_r}\rangle_0| < \infty$, because at fixed z the cumulant is a number, the same in every writing, and it is bounded by the sum of the moduli of its terms; hence the value of the sum does not depend on the writing or on the order of summation. The *local terms* are the $\mathsf{a}_p^{(3)}$ for S_3; for V_2 they are the $\mathsf{a}_p^{(4)}$, the terms of the two sums over b, and the terms of the sum over pairs (b, b'). ∎

Proof of the cumulant identities in Def. 7.8. Fix a finite nonempty index set J_0, and write $z^I = \prod_{i\in I} z_i$ for $I \subset J_0$. For a holomorphic function h of finitely many variables and a one-variable holomorphic function ϕ near $h(0)$, repeated differentiation gives

$$\partial_J(\phi\circ h) = \sum_{\pi\in\mathrm{Part}(J)} \phi^{(|\pi|)}(h)\prod_{I\in\pi}\partial_I h.$$

Indeed, when a new variable is differentiated, it either differentiates $\phi^{(|\pi|)}(h)$ and forms a new singleton block, or differentiates one factor $\partial_I h$ and joins that block. Every partition of the enlarged set arises exactly once. This proves the formula by induction from the one-variable chain rule. At $h(0)=1$, the derivatives $(\log)^{(k)}(1) = (-1)^{k-1}(k-1)!$ give eq. 7.xvi. With $\phi=\exp$ and $h(0)=0$, the same formula gives

$$\partial_J e^h(0) = \sum_{\pi\in\mathrm{Part}(J)}\prod_{I\in\pi}\partial_I h(0).$$

For arbitrary moments m_I, put $H(z) = 1+\sum_{\emptyset\neq I\subset J_0} m_I z^I$ and $\varphi = \log H$ near zero. The logarithm formula gives $\partial_I\varphi(0)=\kappa_I$ as defined by eq. 7.xiv. Since $e^\varphi = H$, the exponential formula proves eq. 7.xv. Conversely, suppose eq. 7.xv holds and put $\varphi(z)=\sum_{\emptyset\neq I\subset J_0}\kappa_I z^I$. The exponential formula gives $\partial_I e^\varphi(0) = m_I$, and applying the logarithm formula to e^φ recovers eq. 7.xiv. This also proves uniqueness. □

Lemma 7.9 (Cumulants of monomials: connected pairings)**.** Let $(\phi_i)_{i\in I}$ be a centred Gaussian family with covariance $\mathsf{C}(i,j)$ and mean $\langle\cdot\rangle_0$, the cumulants of polynomials being those of Def. 7.8(2), and let $\mathsf{m}_v = \prod_{e\in E_v}\phi_{i(e)}$, $v=1,\dots,r$, where the sets E_v of legs are finite and pairwise disjoint and $i(e)\in I$. Then

$$\langle \mathsf{m}_1;\dots;\mathsf{m}_r\rangle_0 = \sum_{\mathsf{P}}\prod_{\{e,e'\}\in\mathsf{P}}\mathsf{C}(i(e),i(e')),$$

where the sum ranges over the pairings P of $E_1\cup\cdots\cup E_r$ whose graph on $\{1,\dots,r\}$ is connected; this graph has an edge vv' for each pair $\{e,e'\}\in\mathsf{P}$ with $e\in E_v$, $e'\in E_{v'}$, $v\neq v'$. ■

Proof. Step 1: moments. Let E be a finite set of legs, with $i(e)\in I$ for $e\in E$, and let $s=(s_e)_{e\in E}$ be real variables. The variables $\phi_{i(e)}$, $e\in E$, form a centred Gaussian family with covariance $\mathsf{C}(i(e),i(e'))$, so that

$$\Big\langle \exp\sum_{e\in E} s_e\phi_{i(e)}\Big\rangle_0 = \exp\Big(\tfrac12\sum_{e,e'\in E} s_e s_{e'}\,\mathsf{C}(i(e),i(e'))\Big).$$

Both sides are real analytic in s, and the left side may be differentiated under the mean. We compare the coefficients of the monomial $\prod_{e\in E}s_e$. On the left it is $\langle\prod_{e\in E}\phi_{i(e)}\rangle_0$. On the right, this monomial occurs only in the term $\frac{1}{m!}\big(\frac12\sum_{e,e'}s_e s_{e'}\mathsf{C}(i(e),i(e'))\big)^m$ with $2m=\#E$; when the power is expanded, each pairing of E arises from $m!\,2^m$ choices, namely the order of its m pairs and the order within each pair. Hence

$$\Big\langle\prod_{e\in E}\phi_{i(e)}\Big\rangle_0 = \sum_{\mathsf{P}}\prod_{\{e,e'\}\in\mathsf{P}}\mathsf{C}(i(e),i(e')),$$

the sum ranging over all pairings P of E; it vanishes if $\#E$ is odd.

Step 2: connected pairings. For $\emptyset\neq J\subset\{1,\dots,r\}$ let $\tilde\kappa_J$ be the sum over the pairings of $\bigcup_{v\in J}E_v$ whose graph on J is connected. A pairing of $\bigcup_{v\in J}E_v$ is the same as a partition π of J, namely into the vertex sets of the connected components of its graph, together with a pairing of $\bigcup_{v\in I}E_v$ with connected graph on I for each block $I\in\pi$; the product of the covariances factorises accordingly. Step 1, applied to $E=\bigcup_{v\in J}E_v$, therefore gives

$$m_J := \Big\langle\prod_{v\in J}\mathsf{m}_v\Big\rangle_0 = \sum_{\pi\in\mathrm{Part}(J)}\prod_{I\in\pi}\tilde\kappa_I,$$

which is eq. 7.xv. By Def. 7.8(2), $\tilde\kappa_J$ is the joint cumulant κ_J of the m_v, $v \in J$; for $J = \{1,\dots,r\}$ this is the claim. □

Definition 7.10 (The lattice coefficients B_0, B_1, $\mathsf{P}_1^{\mathrm{lat}}$)**.** Let Ω and τ_{gf} be as in Def. 7.7, let p, q be plaquettes of Ω, and let $\langle\cdot\rangle_0$ refer to the Gaussian family of Ω and τ_{gf} (Def. 7.8). We set

$$\mathsf{B}_0^{\Omega}(p,q) := \langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(2)} \rangle_0 = \tfrac{\mathsf{n}}{2}\, \Pi^{\Omega}(p,q)^2. \tag{7.xvii}$$

The second equality follows from Lem. 7.9 and eq. 7.xiii, applied to the legs, which form a centred Gaussian family as linear combinations of the variables $A^a(b)$. Indeed, $\mathsf{a}_p^{(2)} = \frac12 \sum_a ((\mathrm{d}A)^a(p))^2$ and multilinearity give $\mathsf{B}_0^{\Omega}(p,q) = \frac14 \sum_{a,a'} \langle ((\mathrm{d}A)^a(p))^2; ((\mathrm{d}A)^{a'}(q))^2\rangle_0$; in each of these cumulants exactly two pairings are connected, namely those which pair each leg at p with a leg at q, and each of them contributes $\delta_{aa'}\Pi^{\Omega}(p,q)^2$. Hence $\mathsf{B}_0^{\Omega}(p,q) = \frac14 \cdot 2 \cdot 3\, \Pi^{\Omega}(p,q)^2 = \frac{\mathsf{n}}{2}\Pi^{\Omega}(p,q)^2$. Next we set

$$\begin{aligned} \mathsf{B}_1^{\Omega}(p,q;\tau_{\mathrm{gf}}) := {} & \langle \mathsf{a}_p^{(3)}; \mathsf{a}_q^{(3)} \rangle_0 + \langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(4)} \rangle_0 + \langle \mathsf{a}_p^{(4)}; \mathsf{a}_q^{(2)} \rangle_0 \\ & - \langle \mathsf{a}_p^{(3)}; \mathsf{a}_q^{(2)}; \mathsf{S}_3 \rangle_0 - \langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(3)}; \mathsf{S}_3 \rangle_0 \\ & - \langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(2)}; \mathsf{V}_2 \rangle_0 + \tfrac12 \langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(2)}; \mathsf{S}_3; \mathsf{S}_3 \rangle_0 . \end{aligned} \tag{7.xviii}$$

We write $\mathsf{B}_j^{[n]}$ for $\Omega = \Lambda_n$ and $\mathsf{B}_j^{\mathrm{lat}}$ for $\Omega = \mathbb{Z}^3$, $j = 0, 1$. On $\mathbb{Z}^3$ the four cumulants containing S_3 or V_2 are formal sums in the sense of Def. 7.8(3); the proviso of absolute convergence is satisfied for every $\tau_{\mathrm{gf}} > 0$ by Lem. 7.27. Finally, for bounded Borel functions f, h on $\mathbb{R}^3$ with compact supports and $a > 0$,

$$\mathsf{B}_1^{\mathrm{lat}}(p,q) := \mathsf{B}_1^{\mathrm{lat}}(p,q;1), \qquad \mathsf{P}_1^{\mathrm{lat}}(f,h;a) := a \sum_{p,q} f(a\,x(p))\, h(a\,x(q))\, \mathsf{B}_1^{\mathrm{lat}}(p,q); \tag{7.xix}$$

the sum over pairs of plaquettes of $\mathbb{Z}^3$ is finite, because $f(a\,x(p)) \neq 0$ only for the finitely many plaquettes with $a\,x(p)$ in the compact support of f, and likewise for h.

The normalisation in eq. 7.xix follows from dilation. On the spacing-a lattice define

$$\mathcal{O}_a(f)(V) = a^{-1} \sum_P f(x(P)) a_P(V),$$

with the sum over its plaquettes. The dilation sends a unit bond b to ab and defines $(\mathsf{dil}_a U)(ab) = U(b)$. It therefore preserves the ordered plaquette product and sends its barycentre $x(p)$ to $ax(p)$. Consequently

$$\mathcal{O}_a(f)(\mathsf{dil}_a U) = a^{-1} \sum_p f(a\,x(p))\, a_p(U)$$

for every configuration $U \in \mathcal{V}(\mathbb{L}_1)$, the sum running over the plaquettes of $\mathbb{L}_1 = \mathbb{Z}^3$. If $\beta = \mathsf{b}/a$ with $\mathsf{b} > 0$, then $a^{-2}\beta^{-3} = a\,\mathsf{b}^{-3}$, and eq. 7.xix gives

$$a^{-2} \sum_{p,q} f(a\,x(p))\, h(a\,x(q))\, \beta^{-3}\, \mathsf{B}_1^{\mathrm{lat}}(p,q) = \mathsf{b}^{-3}\, \mathsf{P}_1^{\mathrm{lat}}(f,h;a).$$

■

7.1.7. *The asymptotic expansion in a box.*

Lemma 7.11 (Laplace's method for a truncated two-point function)**.** Let $m \geq 1$ and $\rho > 0$, let $\mathcal{W} \subset \mathbb{R}^m$ be a bounded Borel set containing $\{|y| < \rho\}$, and let Φ, Θ, X, Y be real C^∞ functions on an open set containing $\overline{\mathcal{W}}$ with the following properties: $\Phi(0) = 0$, $\nabla\Phi(0) = 0$, the Hessian H of Φ at 0 is positive definite, and $\inf\{\Phi(y) : \ y \in \mathcal{W},\ |y| \geq \rho'\} > 0$ for every $\rho' \in (0, \rho]$; X and Y vanish at 0 together with their gradients. For $\beta \geq 1$ let $\langle\cdot\rangle_{\mathcal{W},\beta}$ be the mean with respect to the probability density on $\mathcal{W}$, relative to the Lebesgue measure, proportional to $e^{-\beta\Phi-\Theta}$,

and $\langle X;Y\rangle_{\mathcal{W},\beta} := \langle XY\rangle_{\mathcal{W},\beta} - \langle X\rangle_{\mathcal{W},\beta}\langle Y\rangle_{\mathcal{W},\beta}$. Let Φ_j, Θ_j, X'_j, Y'_j be the homogeneous Taylor polynomials of degree j at 0 of Φ, Θ, X, Y, and

$$W_j := \Phi_{j+2} + \Theta_j \quad (j \geq 1), \qquad X_j := X'_{j+2}, \quad Y_j := Y'_{j+2} \quad (j \geq 0).$$

Let $\langle\cdot\rangle_0$ be the centred Gaussian measure on $\mathbb{R}^m$ with covariance H^{-1}, the cumulants of polynomials being taken as in Def. 7.8(2). Then there is $C < \infty$, depending on the data $m, \rho, \mathcal{W}, \Phi, \Theta, X, Y$, such that for all $\beta \geq 1$

$$\left|\langle X;Y\rangle_{\mathcal{W},\beta} - \beta^{-2}\gamma_0 - \beta^{-3}\gamma_2\right| \leq C\beta^{-4},$$

where $\gamma_0 := \langle X_0;Y_0\rangle_0$ and

$$\begin{aligned}\gamma_2 := {} & \langle X_1;Y_1\rangle_0 + \langle X_0;Y_2\rangle_0 + \langle X_2;Y_0\rangle_0 - \langle X_1;Y_0;W_1\rangle_0 - \langle X_0;Y_1;W_1\rangle_0 \\ & - \langle X_0;Y_0;W_2\rangle_0 + \tfrac{1}{2}\langle X_0;Y_0;W_1;W_1\rangle_0.\end{aligned}$$

■

Proof. Step 1: localisation and scaling. By Taylor's formula at 0 and the positivity of H, there is $\rho_0 \in (0,\rho]$ with $\Phi(y) \geq \frac{1}{4}(y,Hy)$ for $|y| \leq \rho_0$. Let $t := \beta^{-1/2} \in (0,1]$, and for bounded Borel functions f on $\mathcal{W}$ let

$$\mathsf{N}(f) := \int_{\mathcal{W}} f\, e^{-\beta\Phi-\Theta}\,\mathrm{d}y.$$

The contribution of $\{y \in \mathcal{W} : |y| \geq \rho_0\}$ to $\mathsf{N}(f)$ is bounded by $C\sup|f|\,e^{-c_*\beta}$, where $c_* := \inf\{\Phi(y) : y \in \mathcal{W},\ |y| \geq \rho_0\} > 0$. On $\{|y| < \rho_0\} \subset \mathcal{W}$ we substitute $y = tz$. Taylor's formula with integral remainder provides smooth functions Φ_α ($|\alpha| = 3$) and X_α, Y_α ($|\alpha| = 2$) on $\{|y| \leq \rho_0\}$ such that

$$\Phi(y) = \tfrac{1}{2}(y,Hy) + \sum_{|\alpha|=3} y^\alpha \Phi_\alpha(y),$$

$$X(y) = \sum_{|\alpha|=2} y^\alpha X_\alpha(y), \qquad Y(y) = \sum_{|\alpha|=2} y^\alpha Y_\alpha(y).$$

We set

$$\varphi(t,z) := t^{-2}\Phi(tz) - \tfrac{1}{2}(z,Hz) + \Theta(tz) - \Theta(0) = t\sum_\alpha z^\alpha \Phi_\alpha(tz) + \Theta(tz) - \Theta(0),$$

$$\mathsf{x}(t,z) := t^{-2}X(tz) = \sum_\alpha z^\alpha X_\alpha(tz), \qquad \mathsf{y}(t,z) := t^{-2}Y(tz) = \sum_\alpha z^\alpha Y_\alpha(tz).$$

These functions are smooth on $\{(t,z) : |tz| \leq \rho_0\}$, the value $t = 0$ included, and they satisfy $|\partial_t^j\varphi|, |\partial_t^j\mathsf{x}|, |\partial_t^j\mathsf{y}| \leq C_j\langle z\rangle^{j+3}$ there. Since the Taylor coefficients at $t = 0$ of $t \mapsto F(tz)$ are the homogeneous Taylor polynomials $F_j(z)$, their formal Taylor series in t are $\sum_{j\geq1} t^j W_j(z)$, $\sum_{j\geq0} t^j X_j(z)$ and $\sum_{j\geq0} t^j Y_j(z)$. Moreover

$$e^{-\frac{1}{2}(z,Hz)-\varphi(t,z)} = e^{-t^{-2}\Phi(tz)-\Theta(tz)+\Theta(0)} \leq Ce^{-\frac{1}{4}(z,Hz)} \qquad (|tz| \leq \rho_0).$$

Step 2: expansion of the integrals. For $\mathsf{f} \in \{1,\mathsf{x},\mathsf{y},\mathsf{xy}\}$ let $\mathsf{F} := \mathsf{f}e^{-\varphi}$. By Step 1, $|\partial_t^j\mathsf{F}| \leq C_j\langle z\rangle^{4j+6}e^{-\varphi}$ on $\{|tz| \leq \rho_0\}$; if $|tz| \leq \rho_0$, the points (s,z) with $0 \leq s \leq t$ belong to this set as well; and the coefficients $\mathsf{F}_j(z) := \frac{1}{j!}\partial_t^j\mathsf{F}(0,z)$ are polynomials in z, since $\varphi(0,z) = 0$ and the t-derivatives of φ, x, y at $t = 0$ are polynomials. Taylor's formula in t with a remainder of order 4, together with the bound of Step 1 at the points (s,z), gives

$$\left|\mathsf{F}(t,z) - \sum_{j<4} t^j\mathsf{F}_j(z)\right| e^{-\frac{1}{2}(z,Hz)} \leq Ct^4\langle z\rangle^{22}e^{-\frac{1}{4}(z,Hz)} \qquad (|tz| \leq \rho_0).$$

Let $\mathsf{Z} := t^m e^{-\Theta(0)}\int e^{-\frac{1}{2}(z,Hz)}\,\mathrm{d}z$, and let $\nu \in \{0,1,1,2\}$ be the number of factors X, Y in $f \in \{1,X,Y,XY\}$, with the corresponding $\mathsf{f} \in \{1,\mathsf{x},\mathsf{y},\mathsf{xy}\}$. We substitute $y = tz$ in $\mathsf{N}(f)$, use $X(tz) = t^2\mathsf{x}(t,z)$, $Y(tz) = t^2\mathsf{y}(t,z)$ and $e^{-\beta\Phi(tz)-\Theta(tz)} = e^{-\Theta(0)}e^{-\frac{1}{2}(z,Hz)-\varphi(t,z)}$, integrate the last bound over

$|z| < \rho_0/t$, and add the Gaussian tails $\int_{|z|\ge\rho_0/t} |\mathsf{F}_j| e^{-\frac12(z,Hz)}\,\mathrm{d}z = O(e^{-c/t^2})$. Since the Gaussian measure with covariance H^{-1} has a density proportional to $e^{-\frac12(z,Hz)}$, we obtain

$$\mathsf{N}(f) = \mathsf{Z}\, t^{2\nu}\Big[\sum_{j<4} t^j \langle \mathsf{F}_j\rangle_0 + O(t^4)\Big] \qquad (f = 1, X, Y, XY),$$

where the contribution of order e^{-c_*/t^2} of $\{|y| \ge \rho_0\}$ is also absorbed in $O(t^4)$, because $\mathsf{Z}\, t^{2\nu+4} \ge c\, t^{m+8}$. For $f = 1$ we have $\mathsf{F}_0 = 1$; hence there is $t_0 \in (0,1]$ with $\mathsf{N}(1) \ge \frac12 \mathsf{Z}$ for $t \le t_0$.

In the range $t_0 < t \le 1$, that is $1 \le \beta < t_0^{-2}$, the claim holds with $C = t_0^{-8}(2\sup|X|\sup|Y| + |\gamma_0| + |\gamma_2|)$: the functions X and Y are continuous, hence bounded on the compact set $\overline{\mathcal{W}}$ (Heine–Borel), so that $|\langle X;Y\rangle_{\mathcal{W},\beta}| \le 2\sup|X|\sup|Y|$, and $\beta^{-4} > t_0^8$.

In the range $t \le t_0$ we have

$$\langle X;Y\rangle_{\mathcal{W},\beta} = \frac{\mathsf{N}(XY)}{\mathsf{N}(1)} - \frac{\mathsf{N}(X)\,\mathsf{N}(Y)}{\mathsf{N}(1)^2} = t^4\Big[\sum_{j<4} t^j\gamma_j + O(t^4)\Big],$$

where $\sum_j t^j\gamma_j$ is the formal power series

$$\mathsf{e} := \frac{\langle \mathsf{x}\mathsf{y} e^{-\varphi}\rangle_0}{\langle e^{-\varphi}\rangle_0} - \frac{\langle \mathsf{x} e^{-\varphi}\rangle_0\,\langle \mathsf{y} e^{-\varphi}\rangle_0}{\langle e^{-\varphi}\rangle_0^2} \in \mathbb{R}[[t]],$$

in which $\mathsf{x} = \sum_j t^j X_j$, $\mathsf{y} = \sum_j t^j Y_j$ and $\varphi = \sum_{j\ge1} t^j W_j$ are formal series and $\langle\cdot\rangle_0$ acts coefficientwise. Indeed, the formal series $\sum_j t^j\langle \mathsf{F}_j\rangle_0$ are the series $\langle \mathsf{f} e^{-\varphi}\rangle_0$; the series $\langle e^{-\varphi}\rangle_0$ has the constant term 1, so that the quotients are defined in $\mathbb{R}[[t]]$; and, the denominators being at least $\frac12$, the quotients of the truncated expansions differ from the truncations of the formal quotients by $O(t^4)$. Since $t^4 = \beta^{-2}$, $t^6 = \beta^{-3}$ and $t^8 = \beta^{-4}$, it remains to identify γ_0 and γ_2 and to show $\gamma_1 = \gamma_3 = 0$.

Step 3: the coefficients. Let α_1, α_2 be formal variables, and let $Z := \alpha_1\mathsf{x} + \alpha_2\mathsf{y} - \varphi$, a formal series in (α_1, α_2, t) without constant term whose coefficients are polynomials in z. By eq. 7.xv and multilinearity,

$$\langle Z^r\rangle_0 = \sum_{\pi\in\mathrm{Part}(\{1,\dots,r\})} \prod_{I\in\pi} \varkappa_{\#I}, \qquad \varkappa_j := \langle Z;\dots;Z\rangle_0 \quad (j \text{ entries}).$$

The number of partitions of $\{1,\dots,r\}$ with m_j blocks of size j ($\sum_j jm_j = r$) is $r!/\prod_j (j!)^{m_j} m_j!$; hence

$$\langle e^Z\rangle_0 = \sum_{r\ge0}\frac{1}{r!}\langle Z^r\rangle_0 = \sum_{(m_j)}\prod_j \frac{1}{m_j!}\Big(\frac{\varkappa_j}{j!}\Big)^{m_j} = \exp\sum_{j\ge1}\frac{1}{j!}\varkappa_j.$$

We compare the coefficients of $\alpha_1\alpha_2$ in $\log\langle e^Z\rangle_0$. Since $\langle e^Z\rangle_0 = \langle e^{-\varphi}\rangle_0 + \alpha_1\langle \mathsf{x} e^{-\varphi}\rangle_0 + \alpha_2\langle \mathsf{y} e^{-\varphi}\rangle_0 + \alpha_1\alpha_2\langle \mathsf{x}\mathsf{y} e^{-\varphi}\rangle_0$ up to terms containing α_1^2 or α_2^2, this coefficient is e. On the other hand, by multilinearity and symmetry of the cumulants, the coefficient of $\alpha_1\alpha_2$ in $\varkappa_j$ is $j(j-1)\langle \mathsf{x};\mathsf{y};-\varphi;\dots;-\varphi\rangle_0$, with $j-2$ entries $-\varphi$. Therefore

$$\mathsf{e} = \sum_{j\ge2}\frac{j(j-1)}{j!}\langle \mathsf{x};\mathsf{y};-\varphi;\dots;-\varphi\rangle_0 = \sum_{r\ge0}\frac{(-1)^r}{r!}\langle \mathsf{x};\mathsf{y};\underbrace{\varphi;\dots;\varphi}_{r}\rangle_0.$$

Inserting the series for x, y, φ, the coefficient of t^0 is $\langle X_0;Y_0\rangle_0 = \gamma_0$. In the coefficient of t^2 the term $r=0$ contributes $\langle X_1;Y_1\rangle_0 + \langle X_0;Y_2\rangle_0 + \langle X_2;Y_0\rangle_0$, the term $r=1$ contributes $-\langle X_1;Y_0;W_1\rangle_0 - \langle X_0;Y_1;W_1\rangle_0 - \langle X_0;Y_0;W_2\rangle_0$, and the term $r=2$ contributes $\frac12\langle X_0;Y_0;W_1;W_1\rangle_0$; the sum is γ_2. The coefficient of an odd power t^k is a sum of cumulants of polynomials X_{j_1}, Y_{j_2}, $W_{i_1},\dots,W_{i_r}$ with $j_1 + j_2 + i_1 + \cdots + i_r = k$. Each of X_j, Y_j, Φ_{j+2}, Θ_j is homogeneous of a degree with the parity of j; hence, in each term of eq. 7.xiv for such a cumulant, at least one moment is the mean of a polynomial which is odd under $z \mapsto -z$, and it vanishes, because the Gaussian measure is invariant under $z \mapsto -z$. Thus $\gamma_1 = \gamma_3 = 0$. □

Proposition 7.12 (The asymptotic expansion in a box)**.** Let $n \geq 2$ be an integer, $\tau_{\mathrm{gf}} > 0$, and $p, q \in \mathsf{Pl}_n$. There is $C < \infty$, depending on n, τ_{gf}, p, q, such that for all $\beta \geq 1$

$$\left|\langle a_p; a_q\rangle_{n,\beta} - \beta^{-2}\,\mathsf{B}_0^{[n]}(p,q) - \beta^{-3}\,\mathsf{B}_1^{[n]}(p,q;\tau_{\mathrm{gf}})\right| \leq C\beta^{-4}. \tag{7.xx}$$

In particular, $\mathsf{B}_1^{[n]}(p,q;\tau_{\mathrm{gf}})$ does not depend on τ_{gf}. ∎

Proof. We write $\tau := \tau_{\mathrm{gf}}$, and we use S, T_n, ϱ_1, ϱ_2, ϱ_3, $\mathfrak{D}$, Ξ, $\mathcal{W}$ of Lem. 7.6. Throughout, F denotes one of the functions 1, a_p, a_q, $a_p a_q$ on $\mathcal{U}_n$; each of them satisfies $|F| \leq 4$ and, like S, is invariant under $\mathcal{G}_n$ (Def. 7.2(1)). Let $d_0 := \dim \mathsf{X}_n$ and $d_1 := \dim \mathsf{Form}_n^{\mathsf{T}}$. The constants $C, c > 0$ below may depend on n, τ, p, q.

Step 1: axial gauge. By eq. 7.vii and the $\mathcal{G}_n$-invariance of $Fe^{-\beta S}$, and then by the Haar formula in Def. 7.1(2) at $u_0 = \mathbf{1}$, bond by bond,

$$\begin{aligned}\int Fe^{-\beta S} \prod_{b\in\mathsf{Bd}_n} \mathrm{d}U(b) &= \int F(V)\, e^{-\beta S(V)} \prod_{b\in\mathsf{Bd}_n\setminus\mathsf{T}_n} \mathrm{d}V(b) \\ &= \sigma_{\mathrm{H}}^{d_1/3} \int_{\{A\in\mathsf{Form}_n^{\mathsf{T}}:\,|A(b)|<\pi\}} (Fe^{-\beta S})(e^{iA}) \prod_{b\in\mathsf{Bd}_n\setminus\mathsf{T}_n} \sigma_1(|A(b)|^2)\,\mathrm{d}A,\end{aligned}$$

as $\mathsf{Bd}_n \setminus \mathsf{T}_n$ has $d_1/3$ elements. Let $\mathsf{I}(F)$ denote the last integral restricted to $|A|_\infty < \varrho_3$. The space $\mathcal{U}_n^{\mathsf{T}}$ is compact, being a finite power of the compact group G, and the image of $\{|A|_\infty < \varrho_3\}$ under $A \mapsto e^{iA}$ is relatively open in it (Def. 7.1); hence its complement K in $\mathcal{U}_n^{\mathsf{T}}$ is compact, and by the injectivity of the exponential on $\mathbb{B}_\pi$, $e^{iA} \in K$ for the remaining A in the integral. The continuous function S has no zero on K, because the only zero of S in $\mathcal{U}_n^{\mathsf{T}}$ is $\mathbf{1} = e^{i0}$ (Lem. 7.6(b)); so its minimum c_* on K is positive, and the integral over the remaining A is bounded by $Ce^{-c_*\beta}$.

By Lem. 7.5(a),(b), $A \mapsto S(e^{iA})$ is smooth and vanishes at 0 together with its gradient; hence $S(e^{iA}) \leq C|A|_\infty^2$ for $|A|_\infty \leq \varrho_3$, and since $\sigma_1(|A(b)|^2) \geq c$ there,

$$\mathsf{I}(1) \geq c \int_{|A|_\infty < \min\{\varrho_3, \beta^{-1/2}\}} e^{-C\beta|A|_\infty^2}\,\mathrm{d}A \geq c\,\beta^{-d_1/2}.$$

As $|\mathsf{I}(F)| \leq 4\,\mathsf{I}(1)$, it follows that $\langle F\rangle_{n,\beta} = \mathsf{I}(F)/\mathsf{I}(1) + O(e^{-c\beta})$.

Step 2: the gauge fixing term. We apply eq. 7.viii with the bounded Borel functions $\Phi(A') := (Fe^{-\beta S})(e^{iA'})$ and $w(c) := e^{-\beta\|c\|^2/2\tau}$. Since $e^{iA^\lambda} = (e^{iA})^{e^{i\lambda}}$ and $Fe^{-\beta S}$ is $\mathcal{G}_n$-invariant, $\Phi(A^\lambda) = \Phi(A)$; thus the inner integral on the right side of eq. 7.viii is $\Phi(A)$ times the integral of w over $\{\mathrm{d}^*(A^\lambda) : |\lambda|_\infty < \varrho_1\}$, a set which contains $\{|c|_\infty < \varrho_2\}$ by Lem. 7.6(d)(iii). Since $\|c\|^2 = \sum_{x\in\Lambda_n^\circ} |c(x)|^2$ and dc is the product of the measures dY at the $d_0/3$ sites of Λ_n° (Lem. 7.6(d)(iv)), the substitution $c \to (\tau/\beta)^{1/2}c$ and the Gaussian integral in Def. 7.1(2), site by site, give

$$\int_{\mathsf{X}_n} w\,\mathrm{d}c = \Big(\frac{\tau}{\beta}\Big)^{d_0/2} \prod_{x\in\Lambda_n^\circ} \int_{\mathfrak{g}} e^{-\frac12|Y|^2}\,\mathrm{d}Y = \Big(\frac{\tau}{\beta}\Big)^{d_0/2} (2\pi)^{d_0/2} = \Big(\frac{2\pi\tau}{\beta}\Big)^{d_0/2},$$

and the integral of w over $\{|c|_\infty \geq \varrho_2\}$ is at most $Ce^{-c\beta}(2\pi\tau/\beta)^{d_0/2}$. Hence the inner integral equals $\Phi(A)(2\pi\tau/\beta)^{d_0/2}(1-\epsilon(A))$ with $0 \leq \epsilon(A) \leq Ce^{-c\beta}$. The left side of eq. 7.viii is

$$\mathsf{J}(F) := \int_{\mathcal{W}} F(e^{iA'})\, e^{-\beta\Phi_\tau(A') - \Theta(A')}\,\mathrm{d}A',$$

$$\Phi_\tau(A') := S(e^{iA'}) + \tfrac{1}{2\tau}\|\mathrm{d}^*A'\|^2, \qquad \Theta(A') := -\sum_{b\in\mathsf{Bd}_n} \log\sigma_1(|A'(b)|^2) - \log|\det \mathsf{M}(A')|;$$

here Θ is defined on $\overline{\mathcal{W}}$, since $\sigma_1 > 0$ on $\mathbb{B}_\pi$ and $\det \mathsf{M} \neq 0$ on $\overline{\mathcal{W}}$ (Lem. 7.6(d)(ii)). Comparing with the definition of $\mathsf{I}(F)$ we obtain

$$\left|\mathsf{J}(F) - (2\pi\tau/\beta)^{d_0/2}\,\mathsf{I}(F)\right| \leq Ce^{-c\beta}(2\pi\tau/\beta)^{d_0/2}\,\mathsf{I}(1).$$

Together with Step 1, and since $\mathsf{J}(F)/\mathsf{J}(1)$ and $\mathsf{I}(F)/\mathsf{I}(1)$ are bounded by 4, this gives $\langle F\rangle_{n,\beta} = \mathsf{J}(F)/\mathsf{J}(1) + O(e^{-c\beta})$. Consequently, up to $O(e^{-c\beta})$, $\langle a_p; a_q\rangle_{n,\beta}$ is the truncated two-point function $\langle X; Y\rangle_{\mathcal{W},\beta}$ of the functions $X(A') := a_p(e^{iA'})$ and $Y(A') := a_q(e^{iA'})$ for the probability density on $\mathcal{W}$ proportional to $e^{-\beta\Phi_\tau - \Theta}$.

Step 3: the hypotheses of Lem. 7.11. We identify $\mathbb{R}^m$ with the Euclidean space Form_n, so that $m = \dim \mathsf{Form}_n$, by means of an orthonormal basis for the scalar product $(\cdot,\cdot)$ of Def. 7.2(2). Then the Lebesgue measure of $\mathbb{R}^m$ becomes the measure dA' of Lem. 7.6(d)(iv), the norm $|y|$ becomes $|A'| := \|A'\|$, and (y, Hy) becomes (A', HA'). We take the set $\mathcal{W}$ and the functions Φ_τ, Θ, X, Y of Step 2. The set $\mathcal{W}$ is open and bounded and contains $0 = \Xi(0,0)$, hence a ball $\{|y| < \rho\}$; its closure is compact (Heine–Borel), contained in $\{|A'|_\infty < 1\}$, and $\det \mathsf{M} \neq 0$ on it (Lem. 7.6(d)).

The four functions are smooth on an open set containing $\overline{\mathcal{W}}$: the maps $A' \mapsto a_P(e^{iA'})$ are smooth for every plaquette P (Def. 7.1); $Y \mapsto \log\sigma_1(|Y|^2)$ is smooth on $\mathbb{B}_\pi$ (Lem. 7.5(c)); and by eq. 7.vi the entries of $\mathsf{M}(A')$ are smooth on $\{|A'|_\infty < 1\}$, being polynomials in the components of the $A'(b)$ plus entries of the $\mathsf{r}(A'(b))$; hence $\det\mathsf{M}$ is continuous, it does not vanish on an open set containing $\overline{\mathcal{W}}$, and $\log|\det\mathsf{M}|$ is smooth there. By Lem. 7.5(a),(b), X, Y and Φ_τ vanish at 0 together with their gradients, and the homogeneous Taylor polynomials of $A' \mapsto a_P(e^{iA'})$ at 0 are the $\mathsf{a}_P^{(j)}(A')$, $j \geq 0$, by eq. 7.iii and the homogeneity of the $\mathsf{a}_P^{(j)}$.

In particular the Taylor polynomial of degree 2 of Φ_τ is S_2 (by eq. 7.v and eq. 7.x), which is positive definite by Lem. 7.6(c). Since $\frac12(A', HA') = \mathsf{S}_2(A') = \frac12(A', (\mathrm{d}^*\mathrm{d} + \tau^{-1}\mathrm{d}\mathrm{d}^*)A')$, the Hessian H acts in each colour as $\mathrm{d}^*\mathrm{d} + \tau^{-1}\mathrm{d}\mathrm{d}^*$, so that $H^{-1} = \mathsf{C}_\tau^{[n]}$ in each colour, and the Gaussian measure of Lem. 7.11 is the Gaussian family of Λ_n and τ (Def. 7.8(1)). Finally let $\rho' \in (0, \rho]$. The function $\Phi_\tau \circ \Xi$ is continuous on the compact set $\overline{\mathfrak{D}}$, and by the gauge invariance of S,

$$\Phi_\tau(\Xi(\lambda, A)) = S(e^{iA}) + \tfrac{1}{2\tau}\|\mathrm{d}^*(A^\lambda)\|^2.$$

If this vanishes, then $S(e^{iA}) = 0$, hence $e^{iA} = \mathbf{1}$ and $A = 0$ (Lem. 7.6(b)), and $\mathrm{d}^*(0^\lambda) = 0 = \mathrm{d}^*(0^0)$, hence $\lambda = 0$ by the injectivity in Lem. 7.6(d)(iii). Therefore $\Phi_\tau \circ \Xi$ has a positive minimum on the compact set $\overline{\mathfrak{D}} \setminus \Xi^{-1}(\{|A'| < \rho'\})$, which does not contain $(0,0)$, and $\inf\{\Phi_\tau(A') : \ A' \in \mathcal{W},\ |A'| \geq \rho'\} > 0$.

Step 4: the Taylor polynomials. By Step 3, $X_j = \mathsf{a}_p^{(j+2)}$, $Y_j = \mathsf{a}_q^{(j+2)}$, $\Phi_{\tau,3} = \mathsf{S}_3$ and $\Phi_{\tau,4} = \sum_P \mathsf{a}_P^{(4)}$, the sums being over Pl_n. By Lem. 7.5(c), as $t \to 0$,

$$-\sum_b \log\sigma_1(t^2|A(b)|^2) = \tfrac13 t^2 \sum_b |A(b)|^2 + O(t^4).$$

By Lem. 7.5(d), $\mathsf{M}(tA) = \mathsf{M}^{(0)} + t\mathsf{M}^{(1)}(A) + t^2\mathsf{M}^{(2)}(A) + O(t^4)$, where $\mathsf{M}^{(0)} = -\mathrm{d}^*\mathrm{d}$ is invertible. With $\mathsf{A} := (\mathsf{M}^{(0)})^{-1}(\mathsf{M}(tA) - \mathsf{M}^{(0)}) = O(t)$ we have $\det(\mathbf{1} + \mathsf{A}) > 0$ for small t, and the series $\log\det(\mathbf{1} + \mathsf{A}) = \sum_{k\geq1} \frac{(-1)^{k-1}}{k} \operatorname{tr}\mathsf{A}^k$ gives

$$\log|\det\mathsf{M}(tA)| - \log|\det\mathsf{M}^{(0)}| = t\operatorname{tr}[(\mathsf{M}^{(0)})^{-1}\mathsf{M}^{(1)}(A)] + t^2\Big\{\operatorname{tr}[(\mathsf{M}^{(0)})^{-1}\mathsf{M}^{(2)}(A)] - \tfrac12\operatorname{tr}\big[\big((\mathsf{M}^{(0)})^{-1}\mathsf{M}^{(1)}(A)\big)^2\big]\Big\} + O(t^3).$$

By eq. 7.xii we conclude that $\Theta_1 = 0$ and

$$\Theta_2 = \tfrac13\sum_b |A(b)|^2 + \tfrac23\sum_b \mathsf{g}(b)|A(b)|^2 - 4\sum_{b,b'} \mathsf{D}(b;b')\mathsf{D}(b';b)\langle A(b), A(b')\rangle,$$

so that $W_1 = \mathsf{S}_3$ and $W_2 = \mathsf{V}_2$, by eq. 7.xi. Lem. 7.11 now gives eq. 7.xx, with $\gamma_0 = \langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(2)}\rangle_0 = \mathsf{B}_0^{[n]}(p,q)$ by eq. 7.xvii and $\gamma_2 = \mathsf{B}_1^{[n]}(p,q;\tau)$ by eq. 7.xviii; the error $O(e^{-c\beta})$ of Step 2 is bounded by $C\beta^{-4}$.

Step 5: independence of the gauge parameter. Neither $\langle a_p; a_q\rangle_{n,\beta}$ nor $\mathsf{B}_0^{[n]}(p,q) = \frac{\mathfrak{n}}{2}\Pi^{[n]}(p,q)^2$ depends on τ (Def. 7.7(2)). Multiplying eq. 7.xx by β^3 we obtain

$$\beta^3\big[\langle a_p; a_q\rangle_{n,\beta} - \beta^{-2}\mathsf{B}_0^{[n]}(p,q)\big] = \mathsf{B}_1^{[n]}(p,q;\tau) + O(\beta^{-1}),$$

and letting $\beta\to\infty$ we see that $\mathsf{B}_1^{[n]}(p,q;\tau)$ does not depend on τ. □

7.1.8. *Zero momentum.* In this subsection we compute the values at zero momentum of the self-energy and of the vertex function of a plaquette action. Both are expressed through Hessians of the action at constant abelian fields, whose symbols can be computed explicitly.

Definition 7.13 (Hessians; self-energy and vertex function; their values at zero momentum)**.** We work on $\mathbb{Z}^3$ and fix a gauge parameter $\tau_{\mathrm{gf}} > 0$. For a positively oriented bond b and a colour index a we write $\partial_{b,a}$ for the derivative with respect to the variable $A^a(b)$. On a formal sum $\mathsf{F} = \sum_z \mathsf{F}_z$ of local terms (Def. 7.8(3)) the derivative acts term by term, and the terms on which it vanishes are omitted. The derivatives $\partial_{b,a}\mathsf{S}_3$, $\partial_{b,a}\partial_{b',a'}\mathsf{S}_4$ and $\partial_{b,a}\partial_{b',a'}\mathsf{V}_2$ which occur below are then finite sums: a bond (x,μ) belongs to the two plaquettes on its two sides in each of the other two coordinate directions, hence to four plaquettes; of a sum over bonds b'' only the term with $b''=b$ survives; and of the sum over pairs of bonds in V_2 only the terms of the pairs (b,b') and (b',b) survive.

(1) Hessians. Let B be a 1-form on $\mathbb{Z}^3$ with $\sup_b|B(b)|<1$, and let q be a plaquette. We define real symmetric matrices $\mathsf{N}(B)$ and $\mathsf{h}_q(B)$, indexed by the pairs (b,a) of a positively oriented bond b and a colour index a, by

$$\mathsf{N}(B)_{(b,a),(b',a')} := \partial_{b,a}\partial_{b',a'}\Big[\sum_p a_p(e^{iA}) + \frac{1}{2\tau_{\mathrm{gf}}}\sum_x |\mathrm{d}^*A(x)|^2\Big]_{A=B}, \qquad \mathsf{h}_q(B)_{(b,a),(b',a')} : = \partial_{b,a}\partial_{b',a'}\, a_q(e^{iA})\big|_{A=B}.$$

The infinite sums over all plaquettes p and over all sites x are understood as follows: under $\partial_{b,a}\partial_{b',a'}$ only the finitely many terms which depend on $A(b)$ contribute, namely those of the plaquettes containing b and those of the two end points of b. Consequently an entry of either matrix vanishes unless b and b' lie in a common plaquette or have a common end point. For fixed (b,a) only finitely many entries $\mathsf{N}(B)_{(b,a),(b',a')}$ and $\mathsf{h}_q(B)_{(b,a),(b',a')}$ are different from 0, and both matrices act on bounded complex 1-forms A by $(\mathsf{N}(B)A)^a(b) := \sum_{(b',a')}\mathsf{N}(B)_{(b,a),(b',a')}A^{a'}(b')$, and likewise for $\mathsf{h}_q(B)$.

(2) Constant abelian fields. For $\gamma\in\mathbb{R}^3$ with $|\gamma|<1$ let Y_γ be the constant 1-form $Y_\gamma((x,\mu)) := \gamma_\mu \mathsf{e}_3$. We put $\mathsf{e}_\pm := 2^{-1/2}(\mathsf{e}_1 \pm i\mathsf{e}_2)$ and $\mathsf{e}_0 := \mathsf{e}_3$. These three vectors form a Hermitian-orthonormal basis of $\mathfrak{g}\otimes\mathbb{C}$, and

$$[\mathsf{e}_3, \mathsf{e}_e] = 2e\,\mathsf{e}_e \qquad (e\in\{+1,-1,0\});$$

we call e the *charge*, and we write $\upsilon := 2e\gamma \in \mathbb{R}^3$. For $k\in\mathbb{R}^3$ we define two 3×3 matrices $\hat{\mathsf{N}}_e(k;\gamma)$, $\hat{\mathsf{h}}_{q,e}(k;\gamma)$ and a number $\hat{\mathsf{M}}_e(k;\gamma)$ by

$$\mathsf{N}(Y_\gamma)(e^{ik\cdot x}\alpha_\mu\mathsf{e}_e) = e^{ik\cdot x}(\hat{\mathsf{N}}_e\alpha)_\mu\,\mathsf{e}_e, \qquad (e^{ik\cdot x}\alpha'\mathsf{e}_e,\, \mathsf{h}_q(Y_\gamma)\,e^{ik\cdot x}\alpha\,\mathsf{e}_e) = \alpha'^*\hat{\mathsf{h}}_{q,e}\alpha,$$
$$\mathsf{M}(Y_\gamma)(e^{ik\cdot x}\mathsf{e}_e) = \hat{\mathsf{M}}_e\,e^{ik\cdot x}\mathsf{e}_e$$

for all $\alpha,\alpha'\in\mathbb{C}^3$. Here $e^{ik\cdot x}\alpha_\mu\mathsf{e}_e$ stands for the complex 1-form $(x,\mu)\mapsto e^{ik\cdot x}\alpha_\mu\mathsf{e}_e$ and $e^{ik\cdot x}\mathsf{e}_e$ for the complex site function $x\mapsto e^{ik\cdot x}\mathsf{e}_e$; the bracket $(\cdot,\cdot)$ is Hermitian, and the sum which defines it is finite, since $\mathsf{h}_q(Y_\gamma)$ has non-zero entries only at bonds of q; the operator $\mathsf{M}(Y_\gamma)$ of Def. 7.4(3) is extended complex linearly. That the left sides have these forms is shown in Lem. 7.14, which also gives explicit formulas; by these formulas, for fixed k, the three objects are real analytic functions of γ in $\{|\gamma|<1\}$.

A second index j denotes the homogeneous polynomial of degree j in γ in the Taylor expansion at $\gamma=0$, so that $\hat{\mathsf{N}}_e = \sum_j \hat{\mathsf{N}}_{e,j}$, $\hat{\mathsf{h}}_{q,e} = \sum_j \hat{\mathsf{h}}_{q,e,j}$ and $\hat{\mathsf{M}}_e = \sum_j \hat{\mathsf{M}}_{e,j}$. For $k\notin 2\pi\mathbb{Z}^3$ we put $\hat{\mathsf{C}} := \hat{\mathsf{N}}_{e,0}^{-1}$.

By Lem. 7.14 at $\gamma = 0$ we have $\hat{\mathsf{N}}_{e,0}(k) = |\mathsf{k}|^2\mathbf{1} - (1-\tau_{\mathrm{gf}}^{-1})\mathsf{k}\mathsf{k}^*$ with $\mathsf{k} = \mathsf{k}(k)$, so that $\hat{\mathsf{C}}$ does not depend on e and is the matrix with the entries

$$\hat{\mathsf{C}}_{\mu\nu}(k) = \frac{\delta_{\mu\nu}}{|\mathsf{k}|^2} + (\tau_{\mathrm{gf}} - 1)\frac{\mathsf{k}_\mu \overline{\mathsf{k}_\nu}}{|\mathsf{k}|^4}$$

which appears in the formula for $\mathsf{C}^{\mathrm{lat}}_{\tau_{\mathrm{gf}}}$ in Def. 7.7(1); moreover $\hat{\mathsf{M}}_{e,0} = -|\mathsf{k}|^2$.

(3) Five numbers of the self-energy. For $k \in [-\pi,\pi]^3 \setminus \{0\}$ and $\gamma \in \mathbb{R}^3$ we put, with all functions taken at $(k;\gamma)$,

$$\begin{aligned}
\mathfrak{s}^{(1)} &:= -\tfrac{1}{4}\sum_e \mathrm{tr}\big[(\hat{\mathsf{C}}\hat{\mathsf{N}}_{e,1})^2\big] && \text{(gluon loop)},\\
\mathfrak{s}^{(2)} &:= \tfrac{1}{2}\sum_e \mathrm{tr}\big[\hat{\mathsf{C}}\hat{\mathsf{N}}_{e,2}\big] && \text{(gluon tadpole)},\\
\mathfrak{s}^{(3)} &:= \tfrac{1}{2}\sum_e \mathrm{Re}\big[(\hat{\mathsf{M}}_{e,1}/\hat{\mathsf{M}}_{e,0})^2\big] && \text{(ghost loop)},\\
\mathfrak{s}^{(4)} &:= -\sum_e \mathrm{Re}\big[\hat{\mathsf{M}}_{e,2}/\hat{\mathsf{M}}_{e,0}\big] && \text{(ghost tadpole)},
\end{aligned}$$

and $\mathfrak{s}^{(5)}(k;\gamma) := \frac{1}{3}|\gamma|^2$ (Haar term). For fixed k each $\mathfrak{s}^{(i)}(k;\cdot)$ is a homogeneous polynomial of degree 2 in γ, bounded by $C|\gamma|^2|k|^{-2}$ (for $i \le 4$ by Thm. T(i), for $i = 5$ because $|k|$ is bounded on the cube); hence it is integrable over $\mathbb{T}$, and its integral is a quadratic form in γ. For $i = 1,\dots,5$ we let $c^{(i)}_{\mu\nu}(\tau_{\mathrm{gf}})$, $1 \le \mu,\nu \le 3$, be the entries of the symmetric matrix with

$$\frac{1}{2}\sum_{\mu\nu} c^{(i)}_{\mu\nu}(\tau_{\mathrm{gf}})\,\gamma_\mu\gamma_\nu = \int_{\mathbb{T}} \mathfrak{s}^{(i)}(k;\gamma)\,d^\circ k \qquad (\gamma \in \mathbb{R}^3).$$

We also need the number

$$Z_0 := \int_{\mathbb{T}} \frac{d^\circ k}{|\mathsf{k}(k)|^2} = \int_{[-\pi,\pi]^3} \frac{dk}{(2\pi)^3}\Big(\sum_\mu 4\sin^2\frac{k_\mu}{2}\Big)^{-1}. \tag{7.xxi}$$

(4) Four numbers of the vertex function of a_q. For a plaquette q, $k \in [-\pi,\pi]^3 \setminus \{0\}$ and $\gamma \in \mathbb{R}^3$ we put, again with all functions taken at $(k;\gamma)$,

$$\begin{aligned}
\mathfrak{v}^{(1)}_q &:= \tfrac{1}{2}\sum_e \mathrm{tr}\big[\hat{\mathsf{h}}_{q,e,2}\hat{\mathsf{C}}\big], & \mathfrak{v}^{(2)}_q &:= -\tfrac{1}{2}\sum_e \mathrm{tr}\big[\hat{\mathsf{h}}_{q,e,1}\hat{\mathsf{C}}\hat{\mathsf{N}}_{e,1}\hat{\mathsf{C}}\big],\\
\mathfrak{v}^{(3)}_q &:= -\tfrac{1}{2}\sum_e \mathrm{tr}\big[\hat{\mathsf{h}}_{q,e,0}\hat{\mathsf{C}}\hat{\mathsf{N}}_{e,2}\hat{\mathsf{C}}\big], & \mathfrak{v}^{(4)}_q &:= \tfrac{1}{2}\sum_e \mathrm{tr}\big[\hat{\mathsf{h}}_{q,e,0}\hat{\mathsf{C}}\hat{\mathsf{N}}_{e,1}\hat{\mathsf{C}}\hat{\mathsf{N}}_{e,1}\hat{\mathsf{C}}\big].
\end{aligned}$$

These are homogeneous polynomials of degree 2 in γ, bounded by $C|\gamma|^2|k|^{-2}$ (Thm. U). For $j = 1,\dots,4$ we let $\mathsf{g}^{(j)}_{q;\mu\nu}(\tau_{\mathrm{gf}})$ be the entries of the symmetric matrix with

$$\frac{1}{2}\sum_{\mu\nu} \mathsf{g}^{(j)}_{q;\mu\nu}(\tau_{\mathrm{gf}})\,\gamma_\mu\gamma_\nu = \int_{\mathbb{T}} \mathfrak{v}^{(j)}_q(k;\gamma)\,d^\circ k \qquad (\gamma \in \mathbb{R}^3).$$

(5) Kernels. Let $\langle\cdot\rangle_0$ refer to the Gaussian family of $\mathbb{Z}^3$ and τ_{gf} (Def. 7.8(1)), and write $\partial = \partial_{b,a}$, $\partial' = \partial_{b',a'}$. We define

$$\begin{aligned}
\Sigma_{\mathrm{tot}}(b,a;b',a') &:= \langle \partial\partial' \mathsf{V}_2\rangle_0 - \langle \partial\mathsf{S}_3; \partial'\mathsf{S}_3\rangle_0,\\
\Gamma_q(b,a;b',a') &:= \langle \partial\partial' \mathsf{a}^{(4)}_q\rangle_0 - \langle \partial\mathsf{a}^{(3)}_q; \partial'\mathsf{S}_3\rangle_0 - \langle \partial'\mathsf{a}^{(3)}_q; \partial\mathsf{S}_3\rangle_0 - \langle \mathsf{a}^{(2)}_q; \partial\partial'\mathsf{S}_4\rangle_0 + \langle \mathsf{a}^{(2)}_q; \partial\mathsf{S}_3; \partial'\mathsf{S}_3\rangle_0.
\end{aligned}$$

These are means and cumulants (Def. 7.8(2)) of polynomials in the Gaussian family, and both kernels are symmetric under the exchange $(b,a) \leftrightarrow (b',a')$.

We determine the colour structure of the two kernels. For $g \in G$ the map $\mathrm{Ad}\,g : X \mapsto gXg^{-1}$ is orthogonal on $(\mathfrak{g}, \langle\cdot,\cdot\rangle)$; let $(\mathrm{Ad}\,g)_{aa'}$ be its matrix in the basis $\mathsf{e}_1, \mathsf{e}_2, \mathsf{e}_3$. The components $(gA(b)g^{-1})^a = \sum_{a'}(\mathrm{Ad}\,g)_{aa'}A^{a'}(b)$ of the $\mathfrak{g}$-valued family $(gA(b)g^{-1})_b$ are real linear combinations

of the variables of the Gaussian family, so that this family is again centred Gaussian. By Def. 7.8(1) and the orthogonality of $\operatorname{Ad} g$, its covariance is

$$\big\langle (gA(b)g^{-1})^a\,(gA(b')g^{-1})^c\big\rangle_0 = \sum_{a'} (\operatorname{Ad} g)_{aa'}(\operatorname{Ad} g)_{ca'}\,\mathsf{C}^{\mathrm{lat}}_{\tau_{\mathrm{gf}}}(b,b') = \delta_{ac}\,\mathsf{C}^{\mathrm{lat}}_{\tau_{\mathrm{gf}}}(b,b'),$$

which is the covariance of $(A(b))_b$; as a centred Gaussian family is determined in law by its covariance, $(gA(b)g^{-1})_b$ has the law of $(A(b))_b$. By Lem. 7.5(a) the polynomials $\mathsf{a}_p^{(m)}$ are invariant under $A \mapsto gAg^{-1}$, and so are S_3, S_4 and, term by term, V_2, whose remaining terms involve A only through $|A(b)|^2$ and $\langle A(b), A(b')\rangle$. By the chain rule, each of the two kernels, regarded at fixed b, b' as a matrix K in (a,a'), satisfies $K = (\operatorname{Ad} g)^{\mathsf{T}} K \operatorname{Ad} g$, that is, it commutes with every $\operatorname{Ad} g$. For $g = e^{isY}$, $Y \in \mathfrak{g}$, we have $\operatorname{Ad} g = e^{s\mathsf{a}_Y}$ (Lem. 7.5(d)), and differentiation at $s = 0$ shows that K commutes with $\mathsf{e}_c \times \cdot$ for $c = 1, 2, 3$.

Such a matrix is a multiple of $\mathbf{1}$: it preserves $\ker(\mathsf{e}_c \times \cdot) = \mathbb{R}\mathsf{e}_c$, so it is diagonal in the basis $\mathsf{e}_1, \mathsf{e}_2, \mathsf{e}_3$, with entries m_1, m_2, m_3 say; applying it to $\mathsf{e}_1 \times \mathsf{e}_2 = \mathsf{e}_3$ and to $\mathsf{e}_3 \times \mathsf{e}_1 = \mathsf{e}_2$ gives $m_3 = m_2$ and $m_2 = m_1$. Hence

$$\Sigma_{\mathrm{tot}}(b,a;b',a') = \delta_{aa'}\,\Sigma_{\mathrm{tot}}(b,b'), \qquad \Gamma_q(b,a;b',a') = \delta_{aa'}\,\Gamma_q(b,b'),$$

and these identities define the kernels $\Sigma_{\mathrm{tot}}(b,b')$ and $\Gamma_q(b,b')$. Finally, the kernels are covariant under the translations of $\mathbb{Z}^3$. Indeed, $\mathsf{C}^{\mathrm{lat}}_{\tau_{\mathrm{gf}}}((x,\mu),(y,\nu))$ and $\mathsf{G}^{\mathrm{lat}}(x,y)$ depend on the sites x, y only through $x-y$ (Def. 7.7(1)), so that the law of the Gaussian family is invariant under translations; the coefficients of the polynomials $\mathsf{a}_p^{(m)}$ do not depend on p (Lem. 7.5(a)); and $\mathsf{g}(b) = \frac{1}{3}$ as well as $\mathsf{D}(b;b')$, which are built from $\mathsf{G}^{\mathrm{lat}}$ (Def. 7.7(2)), do not change when b and b' are translated together. Hence the formal sums S_3, S_4, V_2 are invariant under translations as well, and $\Sigma_{\mathrm{tot}}(b+y,b'+y) = \Sigma_{\mathrm{tot}}(b,b')$ and $\Gamma_{q+y}(b+y,b'+y) = \Gamma_q(b,b')$ for all $y \in \mathbb{Z}^3$. ∎

The symbols of Def. 7.13(2) can be computed explicitly.

Lemma 7.14 (The Hessian at a constant abelian field)**.** Let $\gamma \in \mathbb{R}^3$ with $|\gamma| < 1$, let $e \in \{\pm 1, 0\}$, $\upsilon = 2e\gamma$ and $k \in \mathbb{R}^3$. Let $j(\vartheta) := (e^{i\vartheta} - 1)/i\vartheta$ for real $\vartheta \neq 0$ and $j(0) := 1$, and put

$$J := \operatorname{diag}\big(j(\upsilon_1), j(\upsilon_2), j(\upsilon_3)\big), \qquad P(k) := |\mathsf{k}(k)|^2\mathbf{1} - \mathsf{k}(k)\mathsf{k}(k)^*;$$

the matrix J is invertible, since $|\upsilon_\mu| < 2$ and $j(\vartheta) \neq 0$ for $|\vartheta| < 2\pi$. For a plaquette $q = p_{\mu\nu}(x)$ let $D_q(k)$ be the 1×3 matrix with $D_q(k)\alpha = \mathsf{k}_\mu(k)\alpha_\nu - \mathsf{k}_\nu(k)\alpha_\mu$, $\alpha \in \mathbb{C}^3$. Then the left sides of the three identities in Def. 7.13(2) have the forms stated there, with

$$\hat{\mathsf{N}}_e(k;\gamma) = J^*P(k+\upsilon)J + \tau_{\mathrm{gf}}^{-1}\mathsf{k}(k)\mathsf{k}(k)^*, \qquad \hat{\mathsf{h}}_{q,e}(k;\gamma) = J^*D_q(k+\upsilon)^*D_q(k+\upsilon)J,$$
$$\hat{\mathsf{M}}_e(k;\gamma) = -\mathsf{k}(k)^*J^{-1}\mathsf{k}(k+\upsilon).$$

Moreover the following hold.

(a) We have

$$\det \hat{\mathsf{N}}_e(k;\gamma) = \tau_{\mathrm{gf}}^{-1}\,|\det J|^2\,|\mathsf{k}(k+\upsilon)|^2\,|\hat{\mathsf{M}}_e(k;\gamma)|^2, \qquad \prod_e |\det J| = \prod_\mu \sigma_1(\gamma_\mu^2), \tag{7.xxii}$$

where in the second identity J is taken at $\upsilon = 2e\gamma$ for each $e \in \{+1, -1, 0\}$.

(b) If $\mathsf{k}(k+\upsilon) \neq 0$ and $\hat{\mathsf{M}}_e(k;\gamma) \neq 0$, then $\hat{\mathsf{N}}_e(k;\gamma)$ is invertible, and for $q = p_{\mu\nu}(x)$

$$\operatorname{tr}\big[\hat{\mathsf{h}}_{q,e}(k;\gamma)\hat{\mathsf{N}}_e(k;\gamma)^{-1}\big] = \varpi_q(k+\upsilon), \qquad \varpi_q(k) := \frac{|\mathsf{k}_\mu(k)|^2 + |\mathsf{k}_\nu(k)|^2}{|\mathsf{k}(k)|^2}. \tag{7.xxiii}$$

(c) With $\mathsf{k} = \mathsf{k}(k)$,

$$\hat{\mathsf{M}}_{e,1} = -\sum_\mu \upsilon_\mu \sin k_\mu, \qquad \hat{\mathsf{M}}_{e,2} = \frac{1}{12}\sum_\mu \upsilon_\mu^2|\mathsf{k}_\mu|^2, \qquad \operatorname{tr}\hat{\mathsf{N}}_{e,2} = \sum_\mu \upsilon_\mu^2\Big[2\cos k_\mu - \frac{1}{12}\big(|\mathsf{k}|^2 - |\mathsf{k}_\mu|^2\big)\Big].$$

∎

Proof. Step 1: second variation. Let a be a 1-form of finite support, and let $u_\mu := e^{i\gamma_\mu \mathsf{e}_3}$; these three matrices commute with each other. For a bond b and small real s the product $e^{i(Y_\gamma(b)+sa(b))}e^{-iY_\gamma(b)}$ lies close to $\mathbf{1}$, in particular in $G\setminus\{-\mathbf{1}\}$. By Def. 7.1(2) the map $Y\mapsto e^{iY}$ is a diffeomorphism of $\mathbb{B}_\pi$ onto $G\setminus\{-\mathbf{1}\}$, with the smooth inverse $\frac{1}{i}\log$; we may therefore write the product as $e^{i\eta_s(b)}$, where $\eta_s(b) := \frac{1}{i}\log(e^{i(Y_\gamma(b)+sa(b))}e^{-iY_\gamma(b)}) \in \mathfrak{g}$ is a smooth function of s with $\eta_0(b)=0$. The differential formula in Def. 7.1(2), applied with $Y=Y_\gamma(b)$ and $H=a(b)$, shows that the derivative of the product at $s=0$ equals $i\,\mathsf{dE}(Y_\gamma(b))a(b)$; hence

$$\eta_s(b) = s\,\mathsf{dE}(Y_\gamma(b))\,a(b) + O(s^2).$$

Let $p=p_{\mu\nu}(x)$ have the bonds $b_1,\dots,b_4$, and put $\eta_i := \eta_s(b_i)$. Since $Y_\gamma(b_1)=Y_\gamma(b_3)=\gamma_\mu\mathsf{e}_3$ and $Y_\gamma(b_2)=Y_\gamma(b_4)=\gamma_\nu\mathsf{e}_3$, since u_μ and u_ν commute, and since $ue^{iX}u^{-1}=e^{iuXu^{-1}}$, we find

$$\partial(e^{i(Y_\gamma+sa)})(p) = e^{i\eta_1}u_\mu\,e^{i\eta_2}u_\nu\,u_\mu^{-1}e^{-i\eta_3}\,u_\nu^{-1}e^{-i\eta_4} = e^{i\eta_1}\,e^{iu_\mu\eta_2u_\mu^{-1}}\,e^{-iu_\nu\eta_3u_\nu^{-1}}\,e^{-i\eta_4} = e^{iZ},$$

where $Z\in\mathfrak{g}$ is the image of this product under $\frac{1}{i}\log$; for small s the product is close to $\mathbf{1}$, and by the diffeomorphism property recalled above Z is a smooth function of s, with $Z=0$ at $s=0$. To first order

$$Z = \eta_1 + u_\mu\eta_2u_\mu^{-1} - u_\nu\eta_3u_\nu^{-1} - \eta_4 + O(s^2).$$

Since $a_p(U) = 1-\frac{1}{2}\operatorname{tr}\partial U(p)$ and $\frac{1}{2}\operatorname{tr}e^{iZ}=\cos|Z|$, we have $a_p(e^{i(Y_\gamma+sa)}) = 1-\cos|Z| = \frac{1}{2}|Z|^2 + O(|Z|^4)$. We put $(\mathsf{J}_\gamma a)(b) := \mathsf{dE}(Y_\gamma(b))a(b)$ and, for a 1-form a', $(\mathsf{D}_\gamma a')(p) := a'(b_1) + u_\mu a'(b_2)u_\mu^{-1} - u_\nu a'(b_3)u_\nu^{-1} - a'(b_4)$; by the expansion of $\eta_s(b)$, $Z = s\,(\mathsf{D}_\gamma\mathsf{J}_\gamma a)(p) + O(s^2)$, and therefore

$$a_p(e^{i(Y_\gamma+sa)}) = \tfrac{1}{2}s^2\big|(\mathsf{D}_\gamma\mathsf{J}_\gamma a)(p)\big|^2 + O(s^3).$$

The constant form Y_γ satisfies $\mathrm{d}^*Y_\gamma=0$, so that $\frac{1}{2\tau_{\mathrm{gf}}}\sum_x|\mathrm{d}^*(Y_\gamma+sa)(x)|^2 = \frac{s^2}{2\tau_{\mathrm{gf}}}\|\mathrm{d}^*a\|^2$. Taking second derivatives in s at $s=0$ we obtain the quadratic forms of the two Hessians:

$$(a,\mathsf{h}_p(Y_\gamma)a) = \big|(\mathsf{D}_\gamma\mathsf{J}_\gamma a)(p)\big|^2, \qquad (a,\mathsf{N}(Y_\gamma)a) = \sum_p\big|(\mathsf{D}_\gamma\mathsf{J}_\gamma a)(p)\big|^2 + \tau_{\mathrm{gf}}^{-1}\|\mathrm{d}^*a\|^2,$$

the sum over p being finite. A real symmetric matrix of the kind considered in Def. 7.13(1) is determined by its quadratic form on the forms of finite support; by polarisation, therefore,

$$\mathsf{h}_p(Y_\gamma) = (\delta_p\mathsf{D}_\gamma\mathsf{J}_\gamma)^*(\delta_p\mathsf{D}_\gamma\mathsf{J}_\gamma), \qquad \mathsf{N}(Y_\gamma) = (\mathsf{D}_\gamma\mathsf{J}_\gamma)^*\mathsf{D}_\gamma\mathsf{J}_\gamma + \tau_{\mathrm{gf}}^{-1}\mathrm{dd}^*,$$

where δ_p is the evaluation at p and the adjoints are taken with respect to the brackets $(\cdot,\cdot)$.

Step 2: symbols. We extend all operators complex linearly. From $[\mathsf{e}_3,\mathsf{e}_e] = 2e\,\mathsf{e}_e$ we get $u_\mu\mathsf{e}_eu_\mu^{-1} = e^{2ie\gamma_\mu}\mathsf{e}_e = e^{iv_\mu}\mathsf{e}_e$, and hence

$$\mathsf{dE}(\gamma_\mu\mathsf{e}_3)\,\mathsf{e}_e = \int_0^1 e^{isv_\mu}\,\mathrm{d}s\;\mathsf{e}_e = j(v_\mu)\,\mathsf{e}_e.$$

So J_γ maps the form $e^{ik\cdot x}\alpha\,\mathsf{e}_e$ to $e^{ik\cdot x}J\alpha\,\mathsf{e}_e$, and for $\alpha'\in\mathbb{C}^3$

$$\mathsf{D}_\gamma(e^{ik\cdot x}\alpha'\mathsf{e}_e)(p_{\mu\nu}(x)) = e^{ik\cdot x}\big[\alpha'_\mu + e^{i(v_\mu+k_\mu)}\alpha'_\nu - e^{i(v_\nu+k_\nu)}\alpha'_\mu - \alpha'_\nu\big]\mathsf{e}_e = e^{ik\cdot x}D_{p_{\mu\nu}(x)}(k+v)\,\alpha'\,\mathsf{e}_e.$$

Thus J_γ, D_γ and dd^* commute with the translations of $\mathbb{Z}^3$ and map a plane wave of momentum k and charge e to a plane wave of the same momentum and charge, the coefficient vector being multiplied by J, by the 1×3 matrices $D_{p_{\mu\nu}(x)}(k+v)$ $(\mu<\nu)$, and by $\mathsf{k}(k)\mathsf{k}(k)^*$, respectively; the last symbol follows from the plane-wave formulas of Def. 7.7(1). The adjoint of such an operator acts on plane waves through the adjoint matrix. By Lagrange's identity,

$$\sum_{\mu<\nu}|v_\mu\alpha_\nu - v_\nu\alpha_\mu|^2 = |v|^2|\alpha|^2 - |v^*\alpha|^2 = \alpha^*\big(|v|^2\mathbf{1} - vv^*\big)\alpha \qquad (v,\alpha\in\mathbb{C}^3),$$

so that $\sum_{\mu<\nu}D_{p_{\mu\nu}(x)}(k')^*D_{p_{\mu\nu}(x)}(k') = P(k')$ for $k'\in\mathbb{R}^3$. Together with Step 1 this gives the formula for $\hat{\mathsf{N}}_e$; for $\hat{\mathsf{h}}_{q,e}$ only the plaquette q enters, and the factors $e^{ik\cdot x}$ have modulus 1.

For $\hat{\mathsf{M}}_e$ we use the middle expression in eq. 7.vi, which is real linear in λ and is extended complex linearly, together with $e^{\mathsf{a}_Y}\mathsf{e}_e = u_\mu \mathsf{e}_e u_\mu^{-1}$ for $Y = \gamma_\mu \mathsf{e}_3$ (Lem. 7.5(d)). At $A = Y_\gamma$, $\lambda = e^{ik\cdot x}\mathsf{e}_e$ and the bond (x,μ) it gives

$$\frac{d}{ds}A^{s\lambda}((x,\mu))\Big|_{s=0} = \mathsf{dE}(\gamma_\mu \mathsf{e}_3)^{-1}\big(e^{ik\cdot x}\mathsf{e}_e - e^{iv_\mu}e^{ik\cdot(x+e_\mu)}\mathsf{e}_e\big) = j(v_\mu)^{-1}\big(1-e^{i(v_\mu+k_\mu)}\big)e^{ik\cdot x}\mathsf{e}_e = \\ - j(v_\mu)^{-1}\mathsf{k}_\mu(k+v)\,e^{ik\cdot x}\mathsf{e}_e.$$

Since $\mathsf{M}(A)\lambda$ is d^* of this derivative and d^* multiplies plane waves by $\mathsf{k}(k)^*$ (Def. 7.7(1)), we obtain $\hat{\mathsf{M}}_e = -\mathsf{k}(k)^* J^{-1}\mathsf{k}(k+v)$. Here we have used that $j(\vartheta)\neq 0$ for $|\vartheta|<2\pi$.

Step 3: proof of (a). Let $v := \mathsf{k}(k+v)$ and $\tilde{u} := (J^*)^{-1}\mathsf{k}(k)$. Then

$$\hat{\mathsf{N}}_e = J^*\big(P(k+v) + \tau_{\mathrm{gf}}^{-1}\tilde{u}\tilde{u}^*\big)J, \qquad v^*\tilde{u} = (J^{-1}v)^*\mathsf{k}(k) = -\overline{\hat{\mathsf{M}}_e},$$

and $P(k+v) = |v|^2\mathbf{1} - vv^*$. For $v,w\in\mathbb{C}^3$ we have $\det\big(|v|^2\mathbf{1} - vv^* + ww^*\big) = |v|^2|v^*w|^2$: for $v=0$ both sides vanish; for $v\neq 0$ we use an orthonormal basis whose first vector is $v/|v|$; in it the matrix is $|v|^2\,\mathrm{diag}(0,1,1) + ww^*$, the determinant is an affine function of the rank-one summand, and its value is $0 + |v|^4|w_1|^2$ with $w_1 = v^*w/|v|$. With $w = \tau_{\mathrm{gf}}^{-1/2}\tilde{u}$ this gives $\det\hat{\mathsf{N}}_e = |\det J|^2\,|v|^2\,\tau_{\mathrm{gf}}^{-1}|v^*\tilde{u}|^2$, the first identity in eq. 7.xxii. For the second, $|j(\vartheta)|^2 = (2-2\cos\vartheta)/\vartheta^2 = \big(\sin\frac{\vartheta}{2}/\frac{\vartheta}{2}\big)^2$; since $\det J = \prod_\mu j(2e\gamma_\mu)$, the product over the three charges contributes, for each μ, the factor $|j(2\gamma_\mu)|^2 = (\sin\gamma_\mu/\gamma_\mu)^2 = \sigma_1(\gamma_\mu^2)$.

Step 4: proof of (b). By (a), $\hat{\mathsf{N}}_e$ is invertible. By the cyclicity of the trace,

$$\mathrm{tr}\big[\hat{\mathsf{h}}_{q,e}\hat{\mathsf{N}}_e^{-1}\big] = D_q\big(P + \tau_{\mathrm{gf}}^{-1}\tilde{u}\tilde{u}^*\big)^{-1}D_q^*,$$

with D_q and P taken at $k+v$. Now $D_q v = v_\mu v_\nu - v_\nu v_\mu = 0$, hence $v^*D_q^* = 0$ and $PD_q^* = |v|^2 D_q^*$; moreover $Pv = 0$, and $\tilde{u}^*v \neq 0$ because $v^*\tilde{u} = -\overline{\hat{\mathsf{M}}_e} \neq 0$. The vector

$$\mathsf{X} := |v|^{-2}D_q^* - \frac{\tilde{u}^*D_q^*}{|v|^2\,\tilde{u}^*v}\,v$$

satisfies $P\mathsf{X} = D_q^*$ and $\tilde{u}^*\mathsf{X} = 0$, hence $(P+\tau_{\mathrm{gf}}^{-1}\tilde{u}\tilde{u}^*)\mathsf{X} = D_q^*$. Therefore the trace equals $D_q\mathsf{X} = |v|^{-2}D_qD_q^* = |v|^{-2}(|v_\mu|^2 + |v_\nu|^2) = \varpi_q(k+v)$.

Step 5: proof of (c). We expand in powers of ϑ and of v; the dots stand for terms of degree at least 3:

$$j(\vartheta) = 1 + \frac{i\vartheta}{2} - \frac{\vartheta^2}{6} + \dots, \qquad j(\vartheta)^{-1} = 1 - \frac{i\vartheta}{2} - \frac{\vartheta^2}{12} + \dots, \qquad |j(\vartheta)|^2 = 1 - \frac{\vartheta^2}{12} + \dots,$$
$$\mathsf{k}_\mu(k+v) = \mathsf{k}_\mu + e^{ik_\mu}\big(iv_\mu - \tfrac{1}{2}v_\mu^2 + \dots\big), \qquad |\mathsf{k}_\mu(k+v)|^2 = 2 - 2\cos(k_\mu+v_\mu) = |\mathsf{k}_\mu|^2 \\ + 2v_\mu\sin k_\mu + v_\mu^2\cos k_\mu + \dots.$$

In $\hat{\mathsf{M}}_e = -\sum_\mu \overline{\mathsf{k}_\mu}\,j(v_\mu)^{-1}\mathsf{k}_\mu(k+v)$ the terms of degree 1 are

$$-i\sum_\mu v_\mu\overline{\mathsf{k}_\mu}\big[e^{ik_\mu} - \tfrac{1}{2}\mathsf{k}_\mu\big] = -i\sum_\mu v_\mu\overline{\mathsf{k}_\mu}\,\tfrac{1}{2}(e^{ik_\mu}+1) = -\sum_\mu v_\mu\sin k_\mu,$$

because $\overline{\mathsf{k}_\mu}(e^{ik_\mu}+1) = -2i\sin k_\mu$, and the terms of degree 2 are

$$-\sum_\mu v_\mu^2\,\overline{\mathsf{k}_\mu}\big[-\tfrac{1}{12}\mathsf{k}_\mu + \tfrac{1}{2}e^{ik_\mu} - \tfrac{1}{2}e^{ik_\mu}\big] = \frac{1}{12}\sum_\mu v_\mu^2|\mathsf{k}_\mu|^2.$$

The term $\tau_{\mathrm{gf}}^{-1}\mathsf{k}\mathsf{k}^*$ of $\hat{\mathsf{N}}_e$ does not depend on γ. Hence $\mathrm{tr}\,\hat{\mathsf{N}}_{e,2}$ is the homogeneous polynomial of degree 2 in the expansion of

$$\mathrm{tr}\big[J^*P(k+v)J\big] = \sum_\mu |j(v_\mu)|^2\sum_{\nu\neq\mu}|\mathsf{k}_\nu(k+v)|^2,$$

namely

$$\sum_\mu \Big[-\frac{1}{12} \upsilon_\mu^2 \sum_{\nu\neq\mu} |\mathsf{k}_\nu|^2 + \sum_{\nu\neq\mu} \upsilon_\nu^2 \cos k_\nu \Big] = \sum_\mu \upsilon_\mu^2 \Big[2\cos k_\mu - \frac{1}{12}\big(|\mathsf{k}|^2 - |\mathsf{k}_\mu|^2\big)\Big].$$

This is the asserted formula for $\operatorname{tr} \hat{\mathsf{N}}_{e,2}$, and (c) is proved. □

With these formulas we evaluate the self-energy at zero momentum.

Theorem T (The five contributions to the self-energy at zero momentum)**.** Let $\tau_{\mathrm{gf}} > 0$.

(i) For $i = 1, \dots, 4$, all $k \in [-\pi,\pi]^3 \setminus \{0\}$ and all $\gamma \in \mathbb{R}^3$ we have $|\mathfrak{s}^{(i)}(k;\gamma)| \le C|\gamma|^2 |k|^{-2}$, with $C = C(\tau_{\mathrm{gf}})$.

(ii) $\sum_{i=1}^5 c^{(i)}_{\mu\nu}(\tau_{\mathrm{gf}}) = 0$ for all μ, ν.

(iii) At $\tau_{\mathrm{gf}} = 1$,

$$(c^{(1)}_{\mu\nu}, \dots, c^{(5)}_{\mu\nu})(1) = \delta_{\mu\nu} \cdot \Big(-20Z_0 + \tfrac{8}{3},\ \ 16Z_0 - \tfrac{28}{9},\ \ 4Z_0 - \tfrac{2}{3},\ \ \tfrac{4}{9},\ \ \tfrac{2}{3} \Big). \tag{7.xxiv}$$

■□

Proof. Step 1: bounds. By the formulas of Lem. 7.14, the entries of $J^{\pm 1}$ and of $\mathsf{k}(k+\upsilon)$ are real analytic functions of (k,γ). On $[-\pi,\pi]^3$ we have $P(k) = O(|k|^2)$ and $\partial_k P(k) = O(|k|)$. The terms of degree 1 in γ of $J^* P(k+\upsilon) J$ are terms containing the factor $P(k)$ and the term $\sum_\mu \upsilon_\mu \partial_\mu P(k)$; hence $|\hat{\mathsf{N}}_{e,1}| \le C|\gamma||k|$, while $|\hat{\mathsf{N}}_{e,2}| \le C|\gamma|^2$. Every term of $\hat{\mathsf{M}}_e$ contains the factor $\mathsf{k}(k)^*$, so that $|\hat{\mathsf{M}}_{e,j}| \le C|\gamma|^j |k|$. Finally $|\hat{\mathsf{C}}| \le C|k|^{-2}$ by the formula for $\hat{\mathsf{C}}$ in Def. 7.13(2), and $|\hat{\mathsf{M}}_{e,0}|^{-1} = |\mathsf{k}(k)|^{-2} \le C|k|^{-2}$, as $|\mathsf{k}(k)|^2 \ge \frac{4}{\pi^2}|k|^2$.

Inserting these bounds into the definitions of $\mathfrak{s}^{(1)}, \dots, \mathfrak{s}^{(4)}$ we obtain $|\hat{\mathsf{C}}\hat{\mathsf{N}}_{e,1}|^2 \le C|\gamma|^2|k|^{-2}$, $|\hat{\mathsf{C}}\hat{\mathsf{N}}_{e,2}| \le C|\gamma|^2|k|^{-2}$, $|\hat{\mathsf{M}}_{e,1}/\hat{\mathsf{M}}_{e,0}|^2 \le C|\gamma|^2|k|^{-2}$ and $|\hat{\mathsf{M}}_{e,2}/\hat{\mathsf{M}}_{e,0}| \le C|\gamma|^2|k|^{-1} \le C'|\gamma|^2|k|^{-2}$, which proves (i).

Step 2: identity of the integrands. Fix $k \in [-\pi,\pi]^3 \setminus \{0\}$, and let γ be so close to 0 that $\mathsf{k}(k+\upsilon) \neq 0$ and $\hat{\mathsf{M}}_e(k;\gamma) \neq 0$ for all e. We divide the first identity in eq. 7.xxii by its value at $\gamma = 0$, take one half of its logarithm, sum over e and use the second identity in eq. 7.xxii:

$$\sum_e \Big\{ \tfrac{1}{2} \log\det\big[\hat{\mathsf{C}}\hat{\mathsf{N}}_e(k;\gamma)\big] - \log\big|\hat{\mathsf{M}}_e(k;\gamma)/\hat{\mathsf{M}}_{e,0}\big| \Big\} - \sum_\mu \log\sigma_1(\gamma_\mu^2) = \tfrac{1}{2}\sum_e \log \frac{|\mathsf{k}(k+2e\gamma)|^2}{|\mathsf{k}(k)|^2}.$$

Both sides are real analytic in γ near 0, and we compare their terms of degree 2. On the left, the expansion $\log\det(\mathbf{1}+\mathsf{A}) = \operatorname{tr}\mathsf{A} - \frac{1}{2}\operatorname{tr}\mathsf{A}^2 + \dots$ with $\mathsf{A} = \hat{\mathsf{C}}(\hat{\mathsf{N}}_{e,1} + \hat{\mathsf{N}}_{e,2} + \dots)$ yields $\mathfrak{s}^{(2)} + \mathfrak{s}^{(1)}$; the expansion $\log|1+z| = \operatorname{Re}(z - \frac{1}{2}z^2 + \dots)$ with $z = (\hat{\mathsf{M}}_{e,1} + \hat{\mathsf{M}}_{e,2} + \dots)/\hat{\mathsf{M}}_{e,0}$ yields, with the sign in front, $\mathfrak{s}^{(4)} + \mathfrak{s}^{(3)}$; and by Lem. 7.5(c), applied to $Y = \gamma_\mu \mathsf{e}_3$, the term $-\sum_\mu \log\sigma_1(\gamma_\mu^2)$ yields $\frac{1}{3}|\gamma|^2 = \mathfrak{s}^{(5)}$. On the right, with $\mathsf{l}(k) := \log|\mathsf{k}(k)|^2$, the terms of degree 2 are $\frac{1}{2}\sum_{e=\pm1}\frac{1}{2}\sum_{\mu\nu} 4\gamma_\mu\gamma_\nu\,\partial_\mu\partial_\nu \mathsf{l}(k)$. Hence

$$\sum_{i=1}^5 \mathfrak{s}^{(i)}(k;\gamma) = 2\sum_{\mu\nu} \gamma_\mu\gamma_\nu\,\partial_\mu\partial_\nu \mathsf{l}(k) \qquad (k \neq 0),$$

an identity between homogeneous polynomials of degree 2 in γ, valid for all $\gamma \in \mathbb{R}^3$.

Step 3: flux argument. The function l is smooth on $\mathbb{R}^3 \setminus 2\pi\mathbb{Z}^3$ and periodic, and on the cube $|\partial\mathsf{l}| \le C|k|^{-1}$ and $|\partial\partial\mathsf{l}| \le C|k|^{-2}$. For $0 < \varepsilon < 1$ we apply Gauss' divergence theorem on the cube $[-\pi,\pi]^3$ less the ball $\{|k| \le \varepsilon\}$; the contributions of opposite faces of the cube cancel by periodicity, and

$$\int_{[-\pi,\pi]^3 \cap \{|k|>\varepsilon\}} \partial_\mu\partial_\nu \mathsf{l}\,\mathrm{d}k = -\int_{|k|=\varepsilon} \frac{k_\mu}{\varepsilon}\,\partial_\nu \mathsf{l}\,\mathrm{d}S = O(\varepsilon).$$

Letting $\varepsilon \to 0$ we get $\int_{\mathbb{T}} \partial_\mu\partial_\nu \mathsf{l}\, d^\circ k = 0$. By Step 2 and the definition of the numbers $c^{(i)}_{\mu\nu}$, $\frac{1}{2}\sum_{\mu\nu}(\sum_i c^{(i)}_{\mu\nu})\gamma_\mu\gamma_\nu = 0$ for all γ; as the matrices are symmetric, (ii) follows.

Step 4: elementary integrals. We write $\mathsf{k} = \mathsf{k}(k)$. By the symmetry under permutations of the coordinates, $\int_{\mathbb{T}} |\mathsf{k}_\mu|^2 |\mathsf{k}|^{-2} d^\circ k = \frac{1}{3}$. As $\cos k_\mu = 1 - \frac{1}{2}|\mathsf{k}_\mu|^2$, we get $\int_{\mathbb{T}} \cos k_\mu |\mathsf{k}|^{-2} d^\circ k = Z_0 - \frac{1}{6}$. Since $\partial_\mu |\mathsf{k}|^2 = 2 \sin k_\mu$,

$$\partial_\mu (\sin k_\mu |\mathsf{k}|^{-2}) = \cos k_\mu |\mathsf{k}|^{-2} - 2 \sin^2 k_\mu |\mathsf{k}|^{-4};$$

the function $\sin k_\mu |\mathsf{k}|^{-2}$ is periodic and $O(|k|^{-1})$, so the flux argument of Step 3 shows that the integral of the left side over $\mathbb{T}$ vanishes, and $\int_{\mathbb{T}} \sin^2 k_\mu |\mathsf{k}|^{-4} d^\circ k = \frac{1}{2}(Z_0 - \frac{1}{6})$. Finally $\int_{\mathbb{T}} \sin k_\mu \sin k_\nu |\mathsf{k}|^{-4} d^\circ k = 0$ for $\mu \neq \nu$, the integrand being odd under $k_\mu \to -k_\mu$.

Step 5: values. We use Lem. 7.14(c) with $\upsilon = 2e\gamma$, and $\hat{\mathsf{M}}_{e,0} = -|\mathsf{k}|^2$. For $e = 0$ we have $\upsilon = 0$, so that $\hat{\mathsf{N}}_{0,j} = 0$ and $\hat{\mathsf{M}}_{0,j} = 0$ for $j \geq 1$; the charge 0 contributes nothing, and for $e = \pm 1$ we have $\upsilon_\mu^2 = 4\gamma_\mu^2$. We write δ for the matrix $(\delta_{\mu\nu})$.

- $\mathfrak{s}^{(5)} = \frac{1}{3}|\gamma|^2$, whose integral is $\frac{1}{3}|\gamma|^2$; hence $c^{(5)} = \frac{2}{3}\delta$.
- $\mathfrak{s}^{(4)} = \sum_{e=\pm 1} \frac{4}{12} \sum_\mu \gamma_\mu^2 |\mathsf{k}_\mu|^2 |\mathsf{k}|^{-2}$, whose integral is $\frac{2}{9}|\gamma|^2$ by Step 4; hence $c^{(4)} = \frac{4}{9}\delta$.
- $\mathfrak{s}^{(3)} = 4\big(\sum_\mu \gamma_\mu \sin k_\mu\big)^2 |\mathsf{k}|^{-4}$, whose integral is $4|\gamma|^2 \cdot \frac{1}{2}(Z_0 - \frac{1}{6})$; hence $c^{(3)} = (4Z_0 - \frac{2}{3})\delta$.
- At $\tau_{\mathrm{gf}} = 1$ we have $\hat{\mathsf{C}} = |\mathsf{k}|^{-2}\mathbf{1}$ and

$$\mathfrak{s}^{(2)} = \tfrac{1}{2} \sum_{e=\pm 1} |\mathsf{k}|^{-2} \operatorname{tr} \hat{\mathsf{N}}_{e,2} = 4 \sum_\mu \gamma_\mu^2 \Big[2 \cos k_\mu - \frac{1}{12}\big(|\mathsf{k}|^2 - |\mathsf{k}_\mu|^2\big) \Big] |\mathsf{k}|^{-2},$$

  whose integral is $4|\gamma|^2 [2Z_0 - \frac{1}{3} - \frac{1}{12} \cdot \frac{2}{3}] = |\gamma|^2 (8Z_0 - \frac{14}{9})$; hence $c^{(2)} = (16Z_0 - \frac{28}{9})\delta$.
- By (ii), $c^{(1)} = -\sum_{i \geq 2} c^{(i)} = -(20Z_0 - \frac{28}{9} - \frac{2}{3} + \frac{4}{9} + \frac{2}{3})\delta = (-20Z_0 + \frac{8}{3})\delta$.

This proves (iii). □

The same method applies to the vertex function of a plaquette action.

Theorem U (The vertex function at zero momenta)**.** Let $\tau_{\mathrm{gf}} > 0$ and let q be a plaquette. Then $|\mathfrak{v}_q^{(j)}(k;\gamma)| \leq C|\gamma|^2 |k|^{-2}$ for $j = 1, \dots, 4$, all $k \in [-\pi, \pi]^3 \setminus \{0\}$ and all $\gamma \in \mathbb{R}^3$, where C depends on τ_{gf} only; and

$$\sum_{j=1}^4 \mathsf{g}_{q;\mu\nu}^{(j)}(\tau_{\mathrm{gf}}) = 0 \qquad \text{for all } \mu, \nu.$$

■□

Proof. Step 1: bound. The 1×3 matrix $D_q(k)$ is smooth in k, satisfies $D_q(k) = O(|k|)$, and depends on q only through the two directions of q. In the expansion of $\hat{\mathsf{h}}_{q,e} = J^* D_q(k + \upsilon)^* D_q(k + \upsilon) J$ in powers of γ, each term of degree $j \leq 1$ retains at least $2 - j$ factors $D_q(k)$ or $D_q(k)^*$; hence $|\hat{\mathsf{h}}_{q,e,j}| \leq C|\gamma|^j |k|^{2-j}$ for $j = 0, 1, 2$, with C independent of q. As in Step 1 of the proof of Thm. T, Lem. 7.14 gives $|\hat{\mathsf{N}}_{e,1}| \leq C|\gamma||k|$ and $|\hat{\mathsf{N}}_{e,2}| \leq C|\gamma|^2$, and Def. 7.7(1) gives $|\hat{\mathsf{C}}| \leq C|k|^{-2}$. Inserted into the definitions of $\mathfrak{v}_q^{(1)}, \dots, \mathfrak{v}_q^{(4)}$, these bounds give the asserted bound.

Step 2: identity of the integrands. Fix $k \in [-\pi, \pi]^3 \setminus \{0\}$. For small γ we write $\hat{\mathsf{N}}_e = \hat{\mathsf{N}}_{e,0}(\mathbf{1} + \hat{\mathsf{C}}(\hat{\mathsf{N}}_e - \hat{\mathsf{N}}_{e,0}))$, and the Neumann series gives

$$\hat{\mathsf{N}}_e^{-1} = \hat{\mathsf{C}} - \hat{\mathsf{C}}\hat{\mathsf{N}}_{e,1}\hat{\mathsf{C}} - \hat{\mathsf{C}}\hat{\mathsf{N}}_{e,2}\hat{\mathsf{C}} + \hat{\mathsf{C}}\hat{\mathsf{N}}_{e,1}\hat{\mathsf{C}}\hat{\mathsf{N}}_{e,1}\hat{\mathsf{C}} + O(|\gamma|^3).$$

Hence the terms of degree 2 in γ of $\frac{1}{2}\sum_e \operatorname{tr}[\hat{\mathsf{h}}_{q,e}\hat{\mathsf{N}}_e^{-1}]$ are $\sum_{j=1}^4 \mathfrak{v}_q^{(j)}(k;\gamma)$. On the other hand, for small γ the hypotheses of Lem. 7.14(b) hold, and by eq. 7.xxiii the same expression equals $\frac{1}{2}\sum_e \varpi_q(k + 2e\gamma)$, whose terms of degree 2 are $2\sum_{\mu\nu} \gamma_\mu \gamma_\nu \partial_\mu \partial_\nu \varpi_q(k)$. Therefore

$$\sum_{j=1}^4 \mathfrak{v}_q^{(j)}(k;\gamma) = 2 \sum_{\mu\nu} \gamma_\mu \gamma_\nu \, \partial_\mu \partial_\nu \varpi_q(k) \qquad (k \neq 0, \ \gamma \in \mathbb{R}^3).$$

Step 3: flux argument. The function ϖ_q is smooth on $\mathbb{R}^3 \setminus 2\pi\mathbb{Z}^3$ and periodic, and on the cube $|\partial\varpi_q| \le C|k|^{-1}$ and $|\partial\partial\varpi_q| \le C|k|^{-2}$, since it is a quotient of smooth functions vanishing to second order at 0 whose denominator is at least $\frac{4}{\pi^2}|k|^2$. As in Step 3 of the proof of Thm. T,

$$\int_{[-\pi,\pi]^3\cap\{|k|>\varepsilon\}} \partial_\mu\partial_\nu\varpi_q \,\mathrm{d}k = -\int_{|k|=\varepsilon} \frac{k_\mu}{\varepsilon}\,\partial_\nu\varpi_q \,\mathrm{d}S = O(\varepsilon),$$

hence $\int_{\mathbb{T}} \partial_\mu\partial_\nu\varpi_q \, d^\circ k = 0$. By Step 2, $\frac{1}{2}\sum_{\mu\nu}(\sum_j \mathsf{g}^{(j)}_{q;\mu\nu})\gamma_\mu\gamma_\nu = 0$ for all γ, and the symmetry of the matrices gives the claim. □

We next record the behaviour of these objects under point reflections.

Lemma 7.15 (Point reflections)**.** Let $c \in \frac{1}{2}\mathbb{Z}^3$, and let $R_c : x \mapsto 2c - x$ on $\mathbb{Z}^3$ (note that $2c \in \mathbb{Z}^3$). We let R_c act on bonds by $R_c\langle x, x'\rangle := \langle R_c x, R_c x'\rangle$, on plaquettes through their corners (the corners of $R_c p$ are the images of the corners of p), and on configurations U, 1-forms A and site functions λ by

$$(R_cU)(b) := U(R_cb), \qquad (R_cA)(b) := A(R_cb), \qquad (R_c\lambda)(x) := \lambda(R_cx);$$

thus $(R_cA)_\mu(x) = -A_\mu(2c - x - e_\mu)$. Since $2c \in \mathbb{Z}^3$, R_c preserves the lattice. The following hold.

(a) For all plaquettes p, configurations U, 1-forms A on $\mathbb{Z}^3$ and $m \ge 0$: $a_p(R_cU) = a_{R_cp}(U)$ and $\mathsf{a}^{(m)}_p(R_cA) = \mathsf{a}^{(m)}_{R_cp}(A)$. Moreover $\mathrm{d}^*R_c = R_c\mathrm{d}^*$ on 1-forms, and $\mathrm{d}R_c = R_c\mathrm{d}$ on site functions.

(b) For every $\tau_{\mathrm{gf}} > 0$ the family $\big((R_cA)^a(b)\big)_{b,a}$ has the law of the Gaussian family of $\mathbb{Z}^3$ and τ_{gf}. Moreover $\mathsf{D}(R_cb; R_cb') = \mathsf{D}(b; b')$ for all bonds b, b', where D is taken with oriented arguments, that is, it is given by the formula of Def. 7.7(2) for arbitrary bonds; in particular $\mathsf{D}(\bar{b}; b') = -\mathsf{D}(b; b')$ and $\mathsf{D}(b; \bar{b}') = \mathsf{D}(b; b')$.

(c) For a positively oriented bond b let b^R be the positively oriented one of the bonds R_cb and $\overline{R_cb}$, that is, $(x,\mu)^R = (2c - x - e_\mu, \mu)$. Then for all positively oriented bonds b, b' and all plaquettes q

$$\Sigma_{\mathrm{tot}}(b^R, b'^R) = \Sigma_{\mathrm{tot}}(b, b'), \qquad \Gamma_{R_cq}(b^R, b'^R) = \Gamma_q(b, b').$$

■

Proof. The actions in the statement are defined by the same involution on sites; in particular R_c^2 is the identity on sites, bonds, plaquettes and configurations.

Step 1: proof of (a). Let $p = p_{\mu\nu}(x)$. The loop $\partial p = (b_1, b_2, \bar{b}_3, \bar{b}_4)$ passes through x, $x + e_\mu$, $x + e_\mu + e_\nu$, $x + e_\nu$. The map R_c carries it to the loop through $y + e_\mu + e_\nu$, $y + e_\nu$, y, $y + e_\mu$, where $y := 2c - x - e_\mu - e_\nu$ is the base point of $R_cp = p_{\mu\nu}(y)$; this image loop is $\partial(R_cp)$, traversed in the same sense but started at the opposite corner. Since $\partial(R_cU)(p)$ is the product of the values of U along the image loop, it is conjugate in G to $\partial U(R_cp)$, and the two matrices have the same trace; hence $a_p(R_cU) = a_{R_cp}(U)$. As $R_c(e^{itA}) = e^{itR_cA}$, comparison of the Taylor coefficients in eq. 7.iii gives $\mathsf{a}^{(m)}_p(R_cA) = \mathsf{a}^{(m)}_{R_cp}(A)$.

Next, $(\mathrm{d}^*A)(x) = -\sum_{b:\, b_- = x} A(b)$, the sum over the six bonds starting at x, and R_c maps the bonds starting at x bijectively onto those starting at R_cx; therefore $\mathrm{d}^*R_cA = R_c\mathrm{d}^*A$. Finally $(\mathrm{d}R_c\lambda)(b) = \lambda(R_cb_+) - \lambda(R_cb_-) = (\mathrm{d}\lambda)(R_cb)$.

Step 2: proof of (b). The variables $(R_cA)_\mu(x) = -A_\mu(2c - x - e_\mu)$ form a centred Gaussian family, with the same colour structure as A. By Def. 7.7(1) the covariance of $(R_cA)_\mu(x)$ and $(R_cA)_\nu(y)$, for equal colours, is

$$\mathsf{C}\big((2c - x - e_\mu, \mu), (2c - y - e_\nu, \nu)\big) = \int_{\mathbb{T}} e^{-ik\cdot(x-y)}\, e^{-i(k_\mu - k_\nu)}\, \hat{\mathsf{C}}_{\mu\nu}(k)\, d^\circ k.$$

Since $e^{-ik_\mu}\mathsf{k}_\mu(k) = -\mathsf{k}_\mu(-k)$, and correspondingly $e^{ik_\nu}\overline{\mathsf{k}_\nu(k)} = -\overline{\mathsf{k}_\nu(-k)}$, we have $e^{-i(k_\mu - k_\nu)}\hat{\mathsf{C}}_{\mu\nu}(k) = \hat{\mathsf{C}}_{\mu\nu}(-k)$, also for $\mu = \nu$. The substitution $k \to -k$ shows that the covariance equals $\mathsf{C}((x,\mu),(y,\nu))$,

and a centred Gaussian family is determined in law by its covariance. Moreover $\mathsf{G}(R_c x, R_c y) = \mathsf{G}(x,y)$, again by the substitution $k \to -k$ in the formula for $\mathsf{G}^{\mathrm{lat}}$; with $(R_c b)_\pm = R_c(b_\pm)$ the formula for D in Def. 7.7(2) gives $\mathsf{D}(R_c b; R_c b') = \mathsf{D}(b; b')$. The same formula shows that reversing the first argument changes the sign of D, while reversing the second does not change it.

Step 3: proof of (c). By (a) and (b), S_3 and V_2 are invariant under $A \to R_c A$, term by term: the term of a plaquette p is carried to that of $R_c p$, and the terms of a bond b and of a pair (b, b') in V_2 to those of b^R and of (b^R, b'^R); here $\mathsf{g}(b) = \frac{1}{3}$ is constant on $\mathbb{Z}^3$, $|A(R_c b)| = |A(b^R)|$, and the signs coming from reversed arguments of D cancel in pairs. Likewise $\mathsf{a}_q^{(m)}(A) = \mathsf{a}_{R_c q}^{(m)}(R_c A)$. Since $(R_c A)(b^R) = A(\bar{b}) = -A(b)$, the chain rule gives

$$\partial_{b,a}(\mathsf{F} \circ R_c) = -(\partial_{b^R,a}\mathsf{F}) \circ R_c$$

for every polynomial F. Each term of the kernels in Def. 7.13(5) contains exactly two derivatives, so that the signs cancel, and by (b) the means and cumulants do not change under the substitution $A \to R_c A$. Hence $\Sigma_{\mathrm{tot}}(b, a; b', a') = \Sigma_{\mathrm{tot}}(b^R, a; b'^R, a')$ and $\Gamma_q(b, a; b', a') = \Gamma_{R_c q}(b^R, a; b'^R, a')$, which is (c). □

7.1.9. *Tools on kernels.* We now develop a calculus of kernels on $\mathbb{Z}^3$ given by symbols on $\mathbb{T}$; it serves to determine the behaviour of lattice graphs at large distances. Its basic objects are the lines of Def. 7.16(4): kernels whose symbols near 0 and whose values at large distances are given, up to one order and a logarithm, by homogeneous functions. The lemmas after the definition relate these two homogeneous functions to each other; they compare lattice sums with integrals, and a kernel tested against a slowly varying function with its leading part; they expand one-loop integrals of line symbols at small external momenta; and they show that, under conditions on the degrees, convolutions of lines and, up to local terms, products of two lines are again lines.

Definition 7.16 (Symbols, lines, slowly varying functions, majorants)**.** *(1) Two cut-offs.* Let $\theta_0(t) = e^{-1/t}$ for $t > 0$ and $\theta_0(t) = 0$ for $t \le 0$. This function is smooth. To check the only issue, at zero, induction gives $\theta_0^{(n)}(t) = P_n(1/t)e^{-1/t}$ for $t > 0$, where $P_0 = 1$ and $P_{n+1}(u) = u^2(P_n(u) - P_n'(u))$. Every polynomial times e^{-u} tends to zero as $u \to \infty$, as does $uP_n(u)e^{-u}$. Thus each derivative extends continuously by zero and has derivative zero at the origin. For $s \ge 0$ put

$$\theta(s) := \frac{\theta_0(1-s)}{\theta_0(1-s) + \theta_0(s - \frac{1}{2})};$$

the denominator is positive, since one of the numbers $1 - s$, $s - \frac{1}{2}$ is positive. Hence $\theta \in C^\infty([0,\infty))$, $0 \le \theta \le 1$, $\theta = 1$ on $[0, \frac{1}{2}]$ and $\theta = 0$ on $[1, \infty)$. For $\kappa \in \mathbb{R}^D$, $D \in \{3, 6\}$, let $\chi(\kappa) := \theta(|\kappa|/2)$. Then χ is smooth, $\chi = 1$ for $|\kappa| \le 1$ and $\chi = 0$ for $|\kappa| \ge 2$; since $2 < \pi$, we also regard χ as a function on $\mathbb{T}^{D/3}$. The functions θ and χ are fixed by formulas, so constants which depend on θ and χ alone are pure constants.

(2) Symbol classes. For real s let

$$\mathcal{S}^s := \Big\{\sigma \in C^\infty(\mathbb{T} \setminus \{0\}) : \sup_{k \in [-\pi,\pi]^3 \setminus \{0\}} |k|^{|\alpha| - s} |\partial^\alpha \sigma(k)| < \infty \text{ for every multi-index } \alpha\Big\}.$$

By Leibniz' rule $\mathcal{S}^s \mathcal{S}^{s'} \subset \mathcal{S}^{s+s'}$, and $\mathcal{S}^{s+1} \subset \mathcal{S}^s$ because $|k|$ is bounded on the cube.

(3) Line symbols. Let $s \ge -2$ be an integer. A *line symbol of degree* s is a $\sigma \in \mathcal{S}^s$ for which there is a *leading part* $\sigma^h \in C^\infty(\mathbb{R}^3 \setminus \{0\})$ with

$$\sigma^h(tk) = t^s \sigma^h(k) \quad (t > 0), \qquad \sigma^h(-k) = (-1)^s \sigma^h(k), \qquad \sigma - \chi\sigma^h \in \mathcal{S}^{s+1};$$

here $\chi\sigma^h$ is regarded as a function on $\mathbb{T} \setminus \{0\}$, which is possible since χ vanishes near the boundary of the cube. The kernel K with the symbol σ is then called an *elementary line of degree* $n = s + 3$. Sums of line symbols of the same degree are line symbols, and their leading parts add; the product

of two line symbols whose degrees have a sum ≥ -2 is a line symbol, and the leading parts multiply. For products this follows from

$$\sigma_1\sigma_2 - \chi\sigma_1^h\sigma_2^h = (\sigma_1 - \chi\sigma_1^h)\sigma_2 + \chi\sigma_1^h(\sigma_2 - \chi\sigma_2^h) + (\chi^2 - \chi)\sigma_1^h\sigma_2^h,$$

whose last term vanishes for $|k| \leq 1$ and for $|k| \geq 2$. We list some examples.

- $e^{ik\cdot y}$, $y \in \mathbb{Z}^3$: degree 0, leading part 1.
- k_μ and $\overline{\mathsf{k}_\mu}$: degree 1, leading parts ik_μ and $-ik_\mu$.
- $|\mathsf{k}|^{-2}$: degree -2, leading part $|k|^{-2}$. Indeed, on the cube less 0 we write $|\mathsf{k}|^2 = |k|^2(1+\mathsf{w})$ with $\mathsf{w} := |\mathsf{k}|^2/|k|^2 - 1$; the numerator of w, $|\mathsf{k}|^2 - |k|^2$, vanishes to fourth order at 0, so that w is smooth with the bounds of $\mathcal{S}^2$, and $1 + \mathsf{w} \geq \frac{4}{\pi^2}$; then
$$|\mathsf{k}|^{-2} - \chi|k|^{-2} = \chi|k|^{-2}((1+\mathsf{w})^{-1} - 1) + (1-\chi)|\mathsf{k}|^{-2} \in \mathcal{S}^0.$$
- The periodic continuation of $\rho\sigma^h$, where σ^h is as above and $\rho \in C_c^\infty(\mathbb{R}^3)$ satisfies $\operatorname{supp}\rho \subset \{|k| < \pi\}$ and $\rho = 1$ near 0: its leading part is σ^h.

(4) Lines. For an integer $n \geq 2$ let $\mathfrak{K}(n)$ be the set of functions $K : \mathbb{Z}^3 \to \mathbb{C}$ with the following properties.

(K1) K is the kernel with a symbol $\sigma \in L^1(\mathbb{T})$.

(K2) There are a continuous function $\sigma^h : \mathbb{R}^3 \setminus \{0\} \to \mathbb{C}$, homogeneous of degree $n-3$, and a number c with $|\sigma(k) - \chi(k)\sigma^h(k)| \leq c\,|k|^{n-2}\lg|k|$ on $[-\pi,\pi]^3$.

(K3) There are a function $K^h \in C^\infty(\mathbb{R}^3 \setminus \{0\})$, homogeneous of degree $-n$, and numbers c_γ with $|\nabla^\gamma(K - K^h)(x)| \leq c_\gamma|x|^{-n-1-|\gamma|}\lg|x|$ for all multi-indices γ and all $x \in \mathbb{Z}^3$ with $|x| \geq |\gamma| + 1$.

The function K determines the pair (σ^h, K^h), which we call the *leading data* of K. Indeed, σ is determined almost everywhere by K (uniqueness of Fourier coefficients). If σ^h and σ'^h both satisfy (K2), then $|\sigma^h - \sigma'^h|(k) \leq C|k|^{n-2}\lg|k|$ for $0 < |k| \leq 1$, first almost everywhere and then everywhere by continuity; at the points tk with $t \to 0$ this gives $t^{n-3}|\sigma^h(k) - \sigma'^h(k)| \leq Ct^{n-2}|k|^{n-2}\lg(t|k|)$, whence $\sigma^h = \sigma'^h$. If K^h and K'^h both satisfy (K3), we take $\gamma = 0$ and $x = j\mathsf{m}$ with $\mathsf{m} \in \mathbb{Z}^3 \setminus \{0\}$ and $j \to \infty$: from $j^{-n}|K^h(\mathsf{m}) - K'^h(\mathsf{m})| \leq Cj^{-n-1}\lg(j|\mathsf{m}|)$ we get $K^h = K'^h$ on $\mathbb{Z}^3 \setminus \{0\}$, hence on $\mathbb{Q}^3 \setminus \{0\}$ by homogeneity, and everywhere by density and continuity.

The *constants of* K are c, the numbers c_γ, $\|\sigma\|_{L^1}$, and the suprema of $|\sigma^h|$ and of the derivatives of K^h on S^2.

(5) Slowly varying functions. For $D \in \{3, 6\}$, $R \geq 1$ and $g : \mathbb{Z}^D \to \mathbb{C}$ we put

$$\|g\|_{(R)} := \max_{|\gamma|\leq 10} R^{|\gamma|} \sup_\zeta |\nabla^\gamma g(\zeta)|,$$

and if g has finite support, $\check{g}(\kappa) := \sum_\zeta g(\zeta)e^{-i\kappa\cdot\zeta}$. For $G \in C_c^\infty(\mathbb{R}^D)$ we put $\|G\|_{C^{10}} := \max_{|\gamma|\leq 10}\sup|\partial^\gamma G|$.

(6) Majorants. For real $s > -D$ let $\mathcal{H}_s(D)$ be the set of Borel functions $\mathsf{H} : \mathbb{R}^D \to [0,\infty]$ which are homogeneous of degree s and integrable over $\{|\kappa| \leq 1\}$. If $s < 9 - D$ and $\mathsf{H} \in \mathcal{H}_s(D)$, then $\int_{\mathbb{R}^D} \mathsf{H}(\kappa)\lg|\kappa|\,\langle\kappa\rangle^{-10}d\kappa < \infty$, as one sees in polar coordinates. ■

Lemma 7.17 ((K) From a symbol to its kernel)**.** (a) Let $s > -3$ be real, $\sigma \in \mathcal{S}^s$, and let K be the kernel with the symbol σ. Then for every multi-index γ and all $x \in \mathbb{Z}^3$

$$|\nabla^\gamma K(x)| \leq C_\gamma\langle x\rangle^{-3-s-|\gamma|},$$

where C_γ depends on finitely many of the suprema in Def. 7.16(2).

(b) Let $s > -3$ be real and let $\sigma^h \in C^\infty(\mathbb{R}^3 \setminus \{0\})$ be homogeneous of degree s. There is exactly one continuous function K^h on $\mathbb{R}^3 \setminus \{0\}$, homogeneous of degree $-3-s$, such that for every $\rho \in C_c^\infty(\mathbb{R}^3)$ with $\rho = 1$ near 0 and every N

$$\sup_{|x|\geq 1} |x|^N \Big| \int_{\mathbb{R}^3} e^{ik\cdot x} \rho(k) \sigma^h(k)\, d^\circ k - K^h(x) \Big| < \infty.$$

The function K^h is smooth. If $\sigma^h(-k) = \pm\sigma^h(k)$, then $K^h(-x) = \pm K^h(x)$; replacing σ^h by $(ik)^\gamma \sigma^h$ replaces K^h by $\partial^\gamma K^h$; and replacing σ^h by $\sigma^h \circ O^{-1}$, O a rotation, replaces K^h by $K^h \circ O^{-1}$. We write $K^h =: \mathcal{F}_{\neq 0}^{-1} \sigma^h$.

(c) Let K be an elementary line of degree n with the symbol σ, and $K^h := \mathcal{F}_{\neq 0}^{-1}\sigma^h$. Then

$$|\nabla^\gamma (K - K^h)(x)| \leq C_\gamma |x|^{-n-1-|\gamma|} \qquad (|x| \geq |\gamma| + 1),$$

with $C_\gamma = C_\gamma(\gamma, \sigma)$. In particular, if $n \geq 2$, then $K \in \mathfrak{K}(n)$ with the leading data $(\sigma^h, \mathcal{F}_{\neq 0}^{-1}\sigma^h)$.

■

Proof. Step 1: proof of (a). The kernel $\nabla^\gamma K$ has the symbol $\mathsf{k}^\gamma \sigma := \prod_j \mathsf{k}_j^{\gamma_j} \sigma$, which lies in $\mathcal{S}^{s+|\gamma|}$ since $\mathsf{k}_j \in \mathcal{S}^1$; as $s + |\gamma| > -3$, it suffices to treat $\gamma = 0$. Always $|K(x)| \leq \|\sigma\|_{L^1}$, which settles $x = 0$. Let $|x| \geq 1$, and split $\sigma = \sigma' + \sigma''$ with $\sigma' := \chi(|x|k)\sigma$, where $\chi(|x|k)$ vanishes for $|k| \geq 2/|x|$ and is regarded as a function on $\mathbb{T}$. Then

$$\int_{\mathbb{T}} |\sigma'|\, d^\circ k \leq C \int_{|k|\leq 2/|x|} |k|^s \, \mathrm{d}k \leq C|x|^{-3-s}.$$

The function σ'' is smooth and periodic, vanishes for $|k| \leq 1/|x|$, and satisfies $|\partial^\alpha \sigma''(k)| \leq C|k|^{s-|\alpha|}$: a derivative falling on $\chi(|x|k)$ produces a factor $|x|$, and $|x| \leq 2/|k|$ on the support of the derivatives of $\chi(|x|k)$. We choose j with $|x_j| \geq |x|/\sqrt{3}$ and an integer $N > s+3$. After N integrations by parts in k_j, which produce no boundary terms by periodicity,

$$\Big| \int_{\mathbb{T}} e^{ik\cdot x} \sigma''(k)\, d^\circ k \Big| \leq |x_j|^{-N} \int_{\mathbb{T}} |\partial_j^N \sigma''|\, d^\circ k \leq C|x|^{-N} \int_{|k|\geq 1/|x|} |k|^{s-N}\, \mathrm{d}k \leq C|x|^{-3-s}.$$

These bounds involve only the finitely many suprema in Def. 7.16(2) with $|\alpha| \leq N$.

Step 2: uniqueness in (b). If K^h and K'^h both have the property in (b), then, with one and the same ρ, their difference is homogeneous of degree $-3-s$ and $O(|x|^{-N})$ on $\{|x| \geq 1\}$ for every N; taking $N > 3 + s$ and letting $x = t\omega$, $|\omega| = 1$, $t \to \infty$, we see that it vanishes.

Step 3: existence in (b). Let ρ be as in (b), $x \neq 0$, and let N be an integer with $N > \frac{1}{2}(s+3)$. We put

$$K_\rho^h(x) := \int e^{ik\cdot x} \rho \sigma^h\, d^\circ k + \mathsf{E}_\rho(x), \qquad \mathsf{E}_\rho(x) := |x|^{-2N} \int e^{ik\cdot x} (-\Delta_k)^N \big[(1-\rho)\sigma^h\big]\, d^\circ k,$$

with integrals over $\mathbb{R}^3$. The function $(1-\rho)\sigma^h$ is smooth, and its derivatives of order j are $O(|k|^{s-j})$; as $2N > s+3$, the second integral converges absolutely. Since $|x|^2 e^{ik\cdot x} = -\Delta_k e^{ik\cdot x}$, two integrations by parts over balls (Gauss' theorem) lead from N to $N+1$; the boundary terms on the sphere of radius r are $O((1+|x|)\mathsf{r}^2 \cdot \mathsf{r}^{s-2N})$, which tends to 0 as $\mathsf{r} \to \infty$ because $2N > s+3$. Hence E_ρ does not depend on N, so that $|\mathsf{E}_\rho(x)| \leq C_N |x|^{-2N}$ for every N. For the same reason E_ρ is smooth on $\mathbb{R}^3 \setminus \{0\}$: given an order j, we take N with $2N > s + 3 + j$; since $(-\Delta_k)^N[(1-\rho)\sigma^h] = O(|k|^{s-2N})$ and each derivative of $e^{ik\cdot x}$ in x brings a factor of modulus at most $|k|$, the derivatives of order $\leq j$ in x may be taken under the integral sign, with an integrable majorant independent of x, and $|x|^{-2N}$ is smooth on $\mathbb{R}^3 \setminus \{0\}$.

If ρ' is a second function as in (b), then $(\rho - \rho')\sigma^h \in C_c^\infty(\mathbb{R}^3 \setminus \{0\})$, and integration by parts gives $\int e^{ik\cdot x}(-\Delta_k)^N[(\rho-\rho')\sigma^h]\, d^\circ k = |x|^{2N} \int e^{ik\cdot x}(\rho - \rho')\sigma^h\, d^\circ k$; therefore $K_\rho^h = K_{\rho'}^h =: K^h$. This function is smooth on $\mathbb{R}^3 \setminus \{0\}$, and it has the property in (b), since $\int e^{ik\cdot x} \rho\sigma^h\, d^\circ k - K^h(x) = -\mathsf{E}_\rho(x)$.

The substitution $k \to k/t$, $t > 0$, gives $K^h_\rho(tx) = t^{-3-s} K^h_{\rho(\cdot/t)}(x)$, so that K^h is homogeneous of degree $-3-s$. The parity follows from the substitution $k \to -k$ with an even ρ, the behaviour under rotations from $k \to Ok$ with a radial ρ.

Finally, $\mathsf{E}_\rho(x) = \lim_{t\to\infty} \int e^{ik\cdot x}(1-\rho)\sigma^h \rho(k/t)\, d^\circ k$, together with all derivatives in x and locally uniformly in x, as one sees by integrating by parts as above; so we may differentiate under the integrals, and this gives the rule for $(ik)^\gamma \sigma^h$, a smooth function homogeneous of degree $s + |\gamma| > -3$.

Step 4: proof of (c). Let $s := n-3$. By (b) with $\rho = \chi$ and by (a) applied to $\sigma - \chi\sigma^h \in \mathcal{S}^{s+1}$,

$$K(x) - K^h(x) = \int_{\mathbb{T}} e^{ik\cdot x}(\sigma - \chi\sigma^h)\, d^\circ k + \Big[\int_{\mathbb{R}^3} e^{ik\cdot x}\chi\sigma^h\, d^\circ k - K^h(x)\Big] = O(|x|^{-4-s}) \qquad (x \in \mathbb{Z}^3 \setminus \{0\}).$$

The same argument applies to the line symbol $\mathsf{k}^\gamma\sigma$ of degree $s+|\gamma|$: its kernel is $\nabla^\gamma K$, its leading part is $(ik)^\gamma\sigma^h$, and $\mathcal{F}^{-1}_{\neq 0}[(ik)^\gamma\sigma^h] = \partial^\gamma K^h$ by (b); hence $|\nabla^\gamma K - \partial^\gamma K^h| \le C|x|^{-4-s-|\gamma|}$. By Taylor's formula and the homogeneity of the derivatives of K^h, $|\nabla^\gamma K^h(x) - \partial^\gamma K^h(x)| \le C|x|^{-4-s-|\gamma|}$ for $|x| \ge |\gamma|+1$, because the points at which K^h and its derivatives are evaluated have modulus at least $|x|/(|\gamma|+1)$. As $-4-s = -n-1$, this is the asserted bound. For $n \ge 2$, (K1) holds as $\sigma \in \mathcal{S}^{n-3} \subset L^1(\mathbb{T})$, (K2) holds as $|\sigma - \chi\sigma^h| \le C|k|^{n-2}$ on the cube, and (K3) holds since $\lg \ge 1$. □

Lemma 7.18 ((S) Sums)**.** Let $D \in \{3, 6\}$ and $R \ge 1$.

(a) For real $a > D$, $\sum_{\zeta\in\mathbb{Z}^D}(R+|\zeta|)^{-a} \le C_a R^{D-a}$. If $y, \zeta \in \mathbb{R}^D$ and $|y| \le R$, then $R + |\zeta - y| \ge \frac{1}{2}(R+|\zeta|)$.

(b) If $F \in C^1(\mathbb{R}^D)$ and $\sum_\zeta \sup_{\zeta+[0,1]^D}(|F| + |\nabla F|) < \infty$, then

$$\Big|\sum_{\zeta\in\mathbb{Z}^D} F(\zeta) - \int_{\mathbb{R}^D} F\Big| \le \sqrt{D}\sum_\zeta \sup_{\zeta+[0,1]^D} |\nabla F|.$$

(c) For real $a, a' \ge 0$ with $a + a' > 3$ and for $x \in \mathbb{Z}^3$,

$$\sum_{z\in\mathbb{Z}^3} \langle z\rangle^{-a}\langle x-z\rangle^{-a'} \le C\langle x\rangle^{-\min\{a,a',a+a'-3\}} \lg\langle x\rangle, \qquad C = C(a,a').$$

■

Proof. (a) For $\zeta' \in \zeta + [0,1]^D$ we have $R + |\zeta'| \le R + |\zeta| + \sqrt{D} \le (1+\sqrt{D})(R+|\zeta|)$, as $R \ge 1$; so the factor $R+|\zeta|$ changes at most by the factor $1+\sqrt{D}$ over a cube of side 1, and the sum is at most

$$(1+\sqrt{D})^a \int_{\mathbb{R}^D} (R+|\zeta|)^{-a}\, \mathrm{d}\zeta = (1+\sqrt{D})^a R^{D-a} \int_{\mathbb{R}^D} (1+|\zeta|)^{-a}\, \mathrm{d}\zeta,$$

where the last integral is finite since $a > D$. If $|y| \le R$, then $R + |\zeta - y| \ge R$ and $R + |\zeta - y| \ge R + |\zeta| - |y| \ge |\zeta|$, so that $R + |\zeta - y| \ge \max\{R, |\zeta|\} \ge \frac{1}{2}(R+|\zeta|)$.

(b) The cube $\zeta + [0,1]^D$ has volume 1 and diameter $\sqrt{D}$; by the mean value theorem $|F(\zeta) - \int_{\zeta+[0,1]^D} F| \le \sqrt{D}\sup_{\zeta+[0,1]^D}|\nabla F|$. Summing over ζ, with absolutely convergent series, we obtain the claim.

(c) Let first $|x| < 2$. Then $|z| \le |x-z| + |x| < |x-z| + 2 \le 3\langle x-z\rangle$, so that $\langle x-z\rangle \ge \frac{1}{3}\langle z\rangle$ for all z, and the sum is at most $3^{a'}\sum_z \langle z\rangle^{-a-a'}$, which is finite as $a + a' > 3$. On the other hand, the exponent $m := \min\{a, a', a+a'-3\}$ is not negative, and $1 \le \langle x\rangle < 2$, $\lg\langle x\rangle \ge 1$; so the right side of the claim is at least $2^{-m}C$, and the claim holds for $|x| < 2$ once $C \ge 2^m 3^{a'}\sum_z \langle z\rangle^{-a-a'}$. Let now $|x| \ge 2$. On $\{|z| \le |x|/2\}$ we have $\langle x-z\rangle \ge |x|/2$ and $\sum_{|z|\le|x|/2}\langle z\rangle^{-a} \le C\max\{1, |x|^{3-a}\}\lg|x|$; the sum over this set is therefore at most $C|x|^{-a'}\max\{1,|x|^{3-a}\}\lg|x| \le C|x|^{-\min\{a',a+a'-3\}}\lg|x|$.

The sum over $\{|x-z| \le |x|/2\}$ is treated in the same way, with a and a' exchanged. On the remaining set we have $|z| \ge |x|/2$ and $|x-z| \ge |x|/2$, hence $|z| \le |x| + |x-z| \le 3|x-z|$ and $\langle x-z\rangle \ge \frac{1}{3}\langle z\rangle$; the sum over it is at most $3^{a'}\sum_{|z|\ge|x|/2}\langle z\rangle^{-a-a'} \le C|x|^{3-a-a'}$. □

Lemma 7.19 ((P) A kernel against a slowly varying function)**.** Let $D \in \{3,6\}$, $R \ge 1$, and let $g : \mathbb{Z}^D \to \mathbb{C}$ vanish outside $\{|\zeta| \le R\}$. We write $d^\circ\kappa := (2\pi)^{-D} d\kappa$.

(a) $|\check g(\kappa)| \le C\, R^D \langle R\kappa\rangle^{-10} \|g\|_{(R)}$ for $\kappa \in [-\pi,\pi]^D$, with a pure constant C.

(b) Let $\hat{\mathsf T} \in L^1(\mathbb{T}^{D/3})$, $0 < c_0 \le 1$, $-D < s < 9-D$, $\mathsf H \in \mathcal{H}_s(D)$ and $\iota \in \{0,1\}$, and assume $|\hat{\mathsf T}(\kappa)| \le \mathsf H(\kappa)(\lg|\kappa|)^\iota$ for $|\kappa| \le c_0$. Then

$$\Big|\int_{\mathbb{T}^{D/3}} \hat{\mathsf T}(\kappa)\, \check g(-\kappa)\, d^\circ\kappa\Big| \le C\, R^{-s} (\lg R)^\iota \|g\|_{(R)}, \qquad C = C(\mathsf H, c_0, \|\hat{\mathsf T}\|_{L^1}).$$

(c) Let, besides the hypotheses of (b), $s < 8-D$. Let $\hat{\mathsf T}^h$ be a Borel function on $\mathbb{R}^D$, homogeneous of degree s, with $|\hat{\mathsf T}^h| \le \mathsf H$, and let $|\hat{\mathsf T} - \hat{\mathsf T}^h| \le \mathsf H' \lg|\kappa|$ on $\{|\kappa| \le c_0\}$ for some $\mathsf H' \in \mathcal{H}_{s+1}(D)$. Let $a \in \mathbb{R}$, $A \ge 0$ and $G \in C_c^\infty(\mathbb{R}^D)$ with $\operatorname{supp} G \subset \{|\zeta| \le 1\}$, and assume $\|g - R^{-a} G(\cdot/R)\|_{(R)} \le A R^{-a-1} \lg R$. Then

$$\Big|\int_{\mathbb{T}^{D/3}} \hat{\mathsf T}(\kappa)\, \check g(-\kappa)\, d^\circ\kappa - R^{-a-s} \int_{\mathbb{R}^D} \hat{\mathsf T}^h(\kappa)\, \hat G(-\kappa)\, d^\circ\kappa\Big| \le C(A + \|G\|_{C^{10}}) R^{-a-s-1} \lg R$$

and $\big|\int_{\mathbb{R}^D} \hat{\mathsf T}^h(\kappa) \hat G(-\kappa)\, d^\circ\kappa\big| \le C\|G\|_{C^{10}}$, where $C = C(\mathsf H, \mathsf H', c_0, \|\hat{\mathsf T}\|_{L^1})$ only, independent of g, G, R, a and A.

■

Proof. Step 1: proof of (a). For $j \in \{1,\dots,D\}$ we have $\sum_\zeta g(\zeta + e_j) e^{-i\kappa\cdot\zeta} = e^{i\kappa_j} \check g(\kappa)$; hence the function $\nabla_j^N g$, which again has finite support, satisfies

$$\sum_\zeta (\nabla_j^N g)(\zeta)\, e^{-i\kappa\cdot\zeta} = (e^{i\kappa_j} - 1)^N \check g(\kappa).$$

For $N \le 10$ the function $\nabla_j^N g$ vanishes outside $\{|\zeta| \le 11R\}$, a set of at most CR^D points, and $|\nabla_j^N g| \le R^{-N}\|g\|_{(R)}$. For $N = 0$ this gives $|\check g| \le CR^D \|g\|_{(R)}$. Choosing j with $|\kappa_j| \ge |\kappa|/\sqrt{D}$ and using $|e^{i\kappa_j} - 1| \ge \frac{2}{\pi}|\kappa_j|$ for $|\kappa_j| \le \pi$, we obtain from $N = 10$ that $|\check g(\kappa)| \le C|\kappa|^{-10} R^{D-10} \|g\|_{(R)}$. The two bounds together give (a).

Step 2: proof of (b). We split the integral at $|\kappa| = c_0$. For $|\kappa| \ge c_0$, (a) gives $|\check g(-\kappa)| \le CR^{D-10} c_0^{-10} \|g\|_{(R)}$; since $D - 10 < -s$ and $R \ge 1$, the integral over $\{|\kappa| \ge c_0\}$ is at most $C\|\hat{\mathsf T}\|_{L^1} c_0^{-10} R^{-s} \|g\|_{(R)}$. For $|\kappa| \le c_0$ we use the hypothesis and (a), and substitute $\kappa = \kappa'/R$; as $\mathsf H(\kappa'/R) = R^{-s}\mathsf H(\kappa')$ and $\lg(|\kappa'|/R) \le \lg|\kappa'| \lg R$,

$$\int_{|\kappa|\le c_0} \mathsf H(\kappa)(\lg|\kappa|)^\iota R^D \langle R\kappa\rangle^{-10}\, \mathrm{d}\kappa \le R^{-s}(\lg R)^\iota \int_{\mathbb{R}^D} \mathsf H(\kappa')(\lg|\kappa'|)^\iota \langle\kappa'\rangle^{-10}\, \mathrm{d}\kappa',$$

and the last integral is finite by Def. 7.16(6), because $s < 9 - D$.

Step 3: proof of (c). Let $g^h := R^{-a} G(\cdot/R)$ on $\mathbb{Z}^D$. It vanishes outside $\{|\zeta| \le R\}$, and $\|g^h\|_{(R)} \le R^{-a}\|G\|_{C^{10}}$, since a difference of order γ of $G(\cdot/R)$ is an integral of a derivative of order $|\gamma|$ of $G(\cdot/R)$, which is bounded by $R^{-|\gamma|}\|G\|_{C^{10}}$. On $\{|\kappa| \le c_0\}$ we have $|\hat{\mathsf T}| \le \mathsf H + \mathsf H' \lg|\kappa| \le \mathsf H + \mathsf H'/|\kappa|$, because $t \lg t \le 1$ for $0 < t \le 1$; and $\mathsf H + \mathsf H'/|\kappa| \in \mathcal{H}_s(D)$, as one sees in polar coordinates. The function $h := g - g^h$ vanishes outside $\{|\zeta| \le R\}$, and assertion (b), applied to h in place of g, with this majorant and with $\iota = 0$, gives

$$\Big|\int \hat{\mathsf T}(\kappa)\, \check h(-\kappa)\, d^\circ\kappa\Big| \le CR^{-s}\|h\|_{(R)} \le CA\, R^{-s-a-1} \lg R.$$

Next, the function $\hat{\mathsf T} - \chi(\cdot/c_0)\hat{\mathsf T}^h$ is integrable on the cube and bounded by $\mathsf H' \lg|\kappa|$ on $\{|\kappa| \le c_0\}$, where $\chi(\kappa/c_0) = 1$. Assertion (b), applied with $s+1$ in place of s (admissible, since $s + 1 < 9 - D$), with the majorant $\mathsf H'$, with $\iota = 1$ and with g^h in place of g, gives

$$\Big|\int (\hat{\mathsf T} - \chi(\cdot/c_0)\hat{\mathsf T}^h)(\kappa)\, \check g^h(-\kappa)\, d^\circ\kappa\Big| \le CR^{-s-1} \lg R\, \|g^h\|_{(R)} \le CR^{-s-1-a} \lg R\, \|G\|_{C^{10}}.$$

It remains to treat $\int \chi(\kappa/c_0)\hat{\mathsf{T}}^h(\kappa)\check{g}^h(-\kappa)\,d^\circ\kappa$. By the Poisson summation formula,

$$\check{g}^h(\kappa) = R^{D-a}\sum_{\mathsf{m}\in\mathbb{Z}^D}\hat{G}(R(\kappa+2\pi\mathsf{m})), \qquad \text{and} \qquad |\hat{G}(\xi)| \le C\langle\xi\rangle^{-10}\|G\|_{C^{10}}.$$

For $|\kappa|\le 2c_0\le 2$ and $\mathsf{m}\neq 0$ we have $|\kappa+2\pi\mathsf{m}|\ge\pi|\mathsf{m}|$; hence the terms with $\mathsf{m}\neq 0$ contribute at most $CR^{D-a-10}\|G\|_{C^{10}}\le CR^{-a-s-1}\|G\|_{C^{10}}$, as $D-10<-s-1$. The term with $\mathsf{m}=0$ gives, after the substitution $\kappa=\kappa'/R$,

$$R^{-a-s}\int_{\mathbb{R}^D}\chi(\kappa'/c_0R)\,\hat{\mathsf{T}}^h(\kappa')\,\hat{G}(-\kappa')\,d^\circ\kappa'.$$

Since $\int_{|\kappa'|\ge c_0R}\mathsf{H}(\kappa')\langle\kappa'\rangle^{-10}d\kappa' \le CR^{s+D-10}\le CR^{-1}$, replacing $\chi(\kappa'/c_0R)$ by 1 changes this expression by at most $CR^{-a-s-1}\|G\|_{C^{10}}$. The bound $|\hat{\mathsf{T}}^h(\kappa')\hat{G}(-\kappa')|\le C\mathsf{H}(\kappa')\langle\kappa'\rangle^{-10}\|G\|_{C^{10}}$ also gives the last assertion of (c). □

Lemma 7.20 ((L) One-loop functions)**.** Let $\mathsf{m}\in\{2,3\}$ and $D:=3(\mathsf{m}-1)$. Let $\sigma_1,\dots,\sigma_\mathsf{m}$ be line symbols of degrees $s_1,\dots,s_\mathsf{m}$, let $\epsilon_i\in\{\pm1\}$, let $\lambda_i:\mathbb{R}^D\to\mathbb{R}^3$ be linear, and put $l_i(\kappa):=-\epsilon_i\lambda_i(\kappa)$. We assume:

- there is a number $c_\lambda\ge 1$ with $\max_i|l_i(\kappa)|\le c_\lambda|\kappa|\le c_\lambda^2\max_{i<j}|l_i(\kappa)-l_j(\kappa)|$ for all $\kappa\in\mathbb{R}^D$;
- $l_i\neq l_j$ as maps, for $i\neq j$;
- the linear map $(l,\kappa)\mapsto(\epsilon_i l+\lambda_i(\kappa))_{i=1,\dots,\mathsf{m}}$ of $(\mathbb{R}^3)^\mathsf{m}=\mathbb{R}^3\times\mathbb{R}^D$ has an integer matrix of determinant ±1;
- if $\mathsf{m}=3$, then $s_i+s_j\ge -3$ for $i\neq j$;
- $\omega:=3+\sum_i s_i\ge -1$.

We put

$$\mathsf{I}(\kappa):=\int_{\mathbb{T}}\prod_{i=1}^{\mathsf{m}}\sigma_i(\epsilon_i l+\lambda_i(\kappa))\,d^\circ l, \qquad \Lambda(\kappa):=1+\sum_{i<j}\log_+\frac{|\kappa|}{|l_i(\kappa)-l_j(\kappa)|} \quad (\kappa\neq 0).$$

The function Λ is homogeneous of degree 0 and even, and $\Lambda(\kappa)=\infty$ exactly where two of the points $l_i(\kappa)$ agree.

(0) The integral defining $\mathsf{I}(\kappa)$ converges absolutely for almost every κ, and $\|\mathsf{I}\|_{L^1(\mathbb{T}^{D/3})}\le\prod_i\|\sigma_i\|_{L^1(\mathbb{T})}$. It converges absolutely for every κ with $0<c_\lambda|\kappa|\le\frac12$ at which the points $l_i(\kappa)$ are pairwise different.

Moreover, there are numbers $c_0\in(0,1]$ and C, depending on the σ_i, ϵ_i, λ_i and c_λ, a polynomial Q in κ of degree $\le\omega$, with $\mathsf{Q}=0$ if $\omega=-1$, and a Borel function I^h on $\mathbb{R}^D$, vanishing at $\kappa=0$ and wherever two of the points $l_i(\kappa)$ agree, with the following properties.

(a) I^h is homogeneous of degree ω, $\mathsf{I}^h(-\kappa)=(-1)^{\omega+1}\mathsf{I}^h(\kappa)$, and $|\mathsf{I}^h(\kappa)|\le C|\kappa|^\omega\Lambda(\kappa)$. It depends only on the σ_i^h, ϵ_i and λ_i, and multilinearly on the σ_i^h. If O is a rotation acting on each of the $\mathsf{m}-1$ arguments in $\mathbb{R}^3$ of κ, and the λ_i commute with this action, then replacing every σ_i^h by $\sigma_i^h\circ O^{-1}$ replaces I^h by $\mathsf{I}^h\circ O^{-1}$. If $\mathsf{m}=2$, then I^h is continuous on $\mathbb{R}^D\setminus\{0\}$.

(b) $|\mathsf{I}(\kappa)-\mathsf{Q}(\kappa)-\mathsf{I}^h(\kappa)|\le C|\kappa|^{\omega+1}\lg|\kappa|\,\Lambda(\kappa)$ for $0<|\kappa|\le c_0$ whenever the points $l_i(\kappa)$ are pairwise different.

(c) If $\omega\ge 1$, then $\prod_i\sigma_i(\epsilon_i l)$ is integrable over $\mathbb{T}$, and $\mathsf{Q}(0)=\int_\mathbb{T}\prod_i\sigma_i(\epsilon_i l)\,d^\circ l$.

(d) (Uniqueness.) Let $\mathsf{K}^\sharp\subset\mathbb{R}^D\setminus\{0\}$ be open and non-empty, with $t\mathsf{K}^\sharp=\mathsf{K}^\sharp$ for every real $t\neq 0$, such that the l_i are pairwise different at every point of $\mathsf{K}^\sharp$. Let $\mathsf{Q}^\sharp$ be a polynomial of degree $\le\omega$, with $\mathsf{Q}^\sharp=0$ if $\omega=-1$, and let $\mathsf{I}^\sharp$ be a function on $\mathsf{K}^\sharp$, homogeneous of degree ω, with $\mathsf{I}^\sharp(-\kappa)=(-1)^{\omega+1}\mathsf{I}^\sharp(\kappa)$. If $t^{-\omega}|\mathsf{I}(t\kappa)-\mathsf{Q}^\sharp(t\kappa)-\mathsf{I}^\sharp(t\kappa)|\to 0$ as $t\downarrow 0$ for every $\kappa\in\mathsf{K}^\sharp$, then $\mathsf{Q}^\sharp=\mathsf{Q}$, and $\mathsf{I}^\sharp=\mathsf{I}^h$ on $\mathsf{K}^\sharp$.

Thus Q and I^h are determined by I near 0, and they are independent of c_λ; we call them *the polynomial* and *the homogeneous part* of I. If $\tilde{\mathsf{I}}$ is a function of the same kind, with data

$\tilde\sigma_i$, $\tilde\epsilon_i$, $\tilde\lambda_i$ and the same m and ω, and if $\tilde{\mathsf{I}}(\kappa) = \mathsf{I}(-\kappa)$ for all small $\kappa \neq 0$ at which both families of points, the $l_i(\kappa)$ and the $\tilde l_i(\kappa)$, are pairwise different, then the polynomial of $\tilde{\mathsf{I}}$ is the polynomial $\kappa \mapsto \mathsf{Q}(-\kappa)$, and the homogeneous part of $\tilde{\mathsf{I}}$ equals $\mathsf{I}^h(-\kappa)$ at every $\kappa \neq 0$ at which both families of points are pairwise different.

(e) $|\kappa|^s \Lambda(\kappa) \in \mathcal{H}_s(D)$ for every real $s > -D$.

■

Proof. Step 0: two bounds. Let $t_1, \dots, t_{\mathsf{m}} > -3$ be real, with $t_i + t_j \geq -3$ for $i \neq j$ if $\mathsf{m} = 3$; let $\kappa \neq 0$ and $\varepsilon := c_\lambda|\kappa|$. We first show

$$\int_{|l|\leq 2\varepsilon} \prod_i |l - l_i(\kappa)|^{t_i} \,\mathrm{d}l \leq C\varepsilon^{3+\sum_i t_i} \Lambda(\kappa). \tag{7.xxv}$$

By the scaling $l \to \varepsilon l$, $l_i(\kappa) \to \varepsilon l_i(\kappa)$ it suffices to treat $\varepsilon = 1$; then $|l_i| \leq 1$ and $\max_{i<j} |l_i - l_j| \geq c_\lambda^{-2} =: 2c$, and, the scaled distances being $|l_i - l_j| = |l_i(\kappa) - l_j(\kappa)|/(c_\lambda|\kappa|)$, we have $1 + \log \frac{1}{|l_i - l_j|} \leq (1 + \log c_\lambda)\Lambda(\kappa)$ for $i \neq j$. If $\mathsf{m} = 2$, the two singularities lie at distance at least $2c$ from each other, each of them is integrable, and the integral is at most C. Let $\mathsf{m} = 3$, and number the points so that $d := |l_1 - l_2|$ is the least of the three distances. If $d \geq c/2$, the three singularities are apart from each other, and again the integral is at most C.

If $d < c/2$, then $|l_3 - l_1|, |l_3 - l_2| \geq c$, since one of them is at least $2c$ and they differ at most by d. On $\{|l - l_3| \leq c/2\}$ the integrand is at most $C|l - l_3|^{t_3}$; off this set it is at most $C|l - l_1|^{t_1}|l - l_2|^{t_2}$. The integral of the latter over $\{|l - l_1| \leq 2d\}$ is at most $Cd^{3+t_1+t_2} \leq C$, as $t_1 + t_2 \geq -3$ and $d < 1$; over $\{2d \leq |l - l_1| \leq 3\}$, where $\frac12|l - l_1| \leq |l - l_2| \leq \frac32|l - l_1|$, it is at most $C\int_{2d}^3 \varrho^{2+t_1+t_2}\,\mathrm{d}\varrho \leq C(1 + \log\frac1d) \leq C\Lambda(\kappa)$. This proves eq. 7.xxv.

We also prove (e) here. The function $|\kappa|^s\Lambda(\kappa)$ is non-negative, Borel and homogeneous of degree s, and, as $s > -D$, it is integrable over $\{|\kappa| \leq 1\}$ if Λ is integrable over the sphere $\{|\kappa| = 1\}$. For $i < j$ the linear map $l_i - l_j$ is not 0, so there are a vector $v \in \mathbb{R}^D$ with $|v| = 1$ and a number $c' > 0$ with $|l_i(\kappa) - l_j(\kappa)| \geq c'|v \cdot \kappa|$; and $\log_+(1/(c'|v \cdot \kappa|))$ is integrable over the sphere.

Next let $\tau_i \in \mathcal{S}^{t_i}$, $\mathsf{A}(l, \kappa) := \prod_i \tau_i(\epsilon_i l + \lambda_i(\kappa))$, and $0 < \varepsilon \leq \frac12$. For $l \in [-\pi, \pi]^3$ with $|l| \geq 2\varepsilon$ and for $|\kappa'| \leq |\kappa|$ we have $\epsilon_i l + \lambda_i(\kappa') = \epsilon_i(l - l_i(\kappa'))$ and $|l_i(\kappa')| \leq \varepsilon \leq \frac12|l|$, so that $\frac14|l| \leq |\epsilon_i l + \lambda_i(\kappa')|_{\mathbb{T}} \leq \frac32|l|$. With the chain and Leibniz rules this gives

$$|\partial_\kappa^\alpha \mathsf{A}(l, \kappa')| \leq C_\alpha |l|^{\sum_i t_i - |\alpha|} \qquad (|\kappa'| \leq |\kappa|). \tag{7.xxvi}$$

The same bound holds on $\mathbb{R}^3$, for all l with $|l| \geq 2\varepsilon$, if the τ_i are smooth on $\mathbb{R}^3 \setminus \{0\}$ and homogeneous of degree t_i.

We now prove (0). By Fubini's theorem and the substitution $(l, \kappa) \to (\epsilon_i l + \lambda_i(\kappa))_i$, which induces a measure-preserving bijection of $\mathbb{T}^{\mathsf{m}}$ since its matrix is an integer matrix of determinant ±1, we get $\int_{\mathbb{T}^{D/3}} \int_{\mathbb{T}} \prod_i |\sigma_i(\epsilon_i l + \lambda_i(\kappa))|\, d^\circ l\, d^\circ\kappa = \prod_i \|\sigma_i\|_{L^1(\mathbb{T})}$; this gives the first two assertions. For $0 < c_\lambda|\kappa| \leq \frac12$ with pairwise different $l_i(\kappa)$ we have $\Lambda(\kappa) < \infty$; the integral over $\{|l| \leq 2\varepsilon\}$ converges by eq. 7.xxv with $t_i = s_i$, since $|\sigma_i(\epsilon_i l + \lambda_i(\kappa))| \leq C|l - l_i(\kappa)|^{s_i}$ there, and the integral over $\{|l| \geq 2\varepsilon\}$ by eq. 7.xxvi with $\alpha = 0$.

Step 1: crude expansion. Let $\tau_i \in \mathcal{S}^{t_i}$ with integers $t_i \geq -2$ as in Step 0, $\omega' := 3 + \sum_i t_i \geq 0$, A as in Step 0, and $\mathsf{J}(\kappa) := \int_{\mathbb{T}} \mathsf{A}(l, \kappa)\, d^\circ l$. We claim: there is a polynomial Q' of degree $< \omega'$ with

$$|\mathsf{J}(\kappa) - \mathsf{Q}'(\kappa)| \leq C|\kappa|^{\omega'} \lg|\kappa|\, \Lambda(\kappa) \qquad (0 < c_\lambda|\kappa| \leq \tfrac12),$$

and $\mathsf{Q}'(0) = \int_{\mathbb{T}} \mathsf{A}(l, 0)\, d^\circ l$ if $\omega' \geq 1$. Indeed, the integral over $\{|l| \leq 2\varepsilon\}$ is $O(\varepsilon^{\omega'}\Lambda(\kappa))$ by eq. 7.xxv. For $|l| \geq 2\varepsilon$ we expand by Taylor's formula in κ at 0,

$$\mathsf{A}(l, \kappa) = \sum_{|\alpha| < \omega'} \kappa^\alpha \mathsf{A}_\alpha(l) + \mathsf{A}_{\mathrm{rem}}(l, \kappa), \qquad |\mathsf{A}_{\mathrm{rem}}(l, \kappa)| \leq C|\kappa|^{\omega'}|l|^{-3}$$

by eq. 7.xxvi, so that $\int_{|l|\ge 2\varepsilon} |\mathsf{A}_{\mathrm{rem}}|\, d^\circ l = O(|\kappa|^{\omega'} \lg|\kappa|)$. Moreover $|\mathsf{A}_\alpha(l)| \le C|l|^{\omega'-3-|\alpha|}$, which is integrable over $\mathbb{T}$ for $|\alpha| < \omega'$, and $|\kappa|^{|\alpha|} \int_{|l|\le 2\varepsilon} |\mathsf{A}_\alpha|\, d^\circ l \le C|\kappa|^{\omega'}$. Hence the claim holds with $\mathsf{Q}'(\kappa) := \sum_{|\alpha|<\omega'} \kappa^\alpha \int_{\mathbb{T}} \mathsf{A}_\alpha\, d^\circ l$.

Step 2: leading parts. Let $\mathsf{A}^h(l,\kappa) := \prod_i \sigma_i^h(\epsilon_i l + \lambda_i(\kappa))$ on $\mathbb{R}^3 \times \mathbb{R}^D$. It is homogeneous of degree $\omega - 3$, and $\mathsf{A}^h(-l,-\kappa) = (-1)^{\omega+1}\mathsf{A}^h(l,\kappa)$. For $|l| \ge 2\varepsilon$ we expand by Taylor's formula in κ at 0,

$$\mathsf{A}^h(l,\kappa) = \sum_{j\le\omega} \mathsf{A}^h_j(l;\kappa) + \mathsf{A}^h_{\mathrm{rem}}(l;\kappa);$$

here A^h_j is defined for $l \ne 0$, homogeneous of degree j in κ and of degree $\omega-3-j$ in l, with $\mathsf{A}^h_j(-l;\kappa) = (-1)^{\omega+1+j}\mathsf{A}^h_j(l;\kappa)$, and $|\mathsf{A}^h_{\mathrm{rem}}(l;\kappa)| \le C|\kappa|^{\omega+1}|l|^{-4}$ by the bound eq. 7.xxvi for homogeneous factors. With $\varepsilon = c_\lambda|\kappa|$ we define, where $\kappa \ne 0$ and the points $l_i(\kappa)$ are pairwise different,

$$\mathsf{I}^h(\kappa) := \int_{|l|\le 2\varepsilon} \mathsf{A}^h\, d^\circ l + \int_{|l|>2\varepsilon} \mathsf{A}^h_{\mathrm{rem}}\, d^\circ l - \sum_{j<\omega} \int_{|l|\le 2\varepsilon} \mathsf{A}^h_j\, d^\circ l,$$

and $\mathsf{I}^h := 0$ elsewhere, that is, at 0 and on a finite union of linear subspaces. The substitution $l \to tl$ gives the homogeneity, $l \to -l$ the parity, and $l \to Ol$ the behaviour under rotations. Each of the three terms is $O(|\kappa|^\omega \Lambda(\kappa))$, by eq. 7.xxv and eq. 7.xxvi. The function I^h is Borel, as an absolutely convergent integral of a Borel function of (l,κ). At fixed c_λ it is built from the σ_i^h, ϵ_i and λ_i alone, and it is multilinear in the σ_i^h.

Step 2′: continuity for $\mathsf{m} = 2$. Here $l_1(\kappa) \ne l_2(\kappa)$ for all $\kappa \ne 0$, and by homogeneity it suffices to prove continuity on $\{|\kappa| = 1\}$, where $\varepsilon = c_\lambda$ is fixed. The third term in the definition of I^h is a polynomial in κ: $\mathsf{A}^h_j(l;\kappa) = \sum_{|\alpha|=j} \kappa^\alpha \mathsf{A}^h_\alpha(l)$ with $|\mathsf{A}^h_\alpha(l)| \le C|l|^{\omega-3-j}$, a function integrable near 0 for $j < \omega$, so that the coefficients $\int_{|l|\le 2\varepsilon} \mathsf{A}^h_\alpha\, d^\circ l$ are finite. In the second term the integrand is continuous in κ for every l with $|l| > 2\varepsilon$ and has the integrable majorant $C|l|^{-4}$; dominated convergence applies.

In the first term, $\mathsf{A}^h = \sigma_1^h(\epsilon_1(l - l_1(\kappa)))\, \sigma_2^h(\epsilon_2(l - l_2(\kappa)))$, and by the first hypothesis of the lemma $|l_1(\kappa) - l_2(\kappa)| \ge c_\lambda^{-1} =: 4\delta$. We fix κ_0 with $|\kappa_0| = 1$, put $\psi_1(l) := \theta(|l - l_1(\kappa_0)|/2\delta)$ and $\psi_2 := 1 - \psi_1$, and consider κ with $|\kappa| = 1$ so close to κ_0 that $|l_i(\kappa) - l_i(\kappa_0)| \le \frac{1}{2}\delta$ for $i = 1, 2$. As $\theta = 1$ on $[0, \frac{1}{2}]$ and $\theta = 0$ on $[1, \infty)$, we have $\psi_1 = 1$ where $|l - l_1(\kappa_0)| \le \delta$ and $\psi_1 = 0$ where $|l - l_1(\kappa_0)| \ge 2\delta$. Hence, where $\psi_1 \ne 0$,

$$|l - l_2(\kappa)| \ge |l_1(\kappa_0) - l_2(\kappa_0)| - |l - l_1(\kappa_0)| - |l_2(\kappa_0) - l_2(\kappa)| \ge 4\delta - 2\delta - \tfrac{1}{2}\delta = \tfrac{3}{2}\delta,$$

and where $\psi_2 \ne 0$, that is, where $|l - l_1(\kappa_0)| > \delta$, we have $|l - l_1(\kappa)| \ge \delta - \frac{1}{2}\delta = \frac{1}{2}\delta$. We split the first term by the factors ψ_1 and ψ_2. In the integral with the factor ψ_1 we substitute $l \to l + l_1(\kappa)$; since $|l_1(\kappa)| \le c_\lambda|\kappa| = \varepsilon$, it becomes the integral over $\{|l + l_1(\kappa)| \le 2\varepsilon\} \subset \{|l| \le 3\varepsilon\}$ of

$$\psi_1(l + l_1(\kappa))\, \sigma_1^h(\epsilon_1 l)\, \sigma_2^h(\epsilon_2(l + l_1(\kappa) - l_2(\kappa))).$$

The domain of integration moves with κ; we extend the integrand by 0 off this domain and integrate over $\mathbb{R}^3$. Where $\psi_1(l + l_1(\kappa)) \ne 0$, the last factor is evaluated at points of modulus in $[\frac{3}{2}\delta, 4\varepsilon]$, a compact set not containing 0, on which σ_2^h is bounded and continuous. The singular factor $\sigma_1^h(\epsilon_1 l)$ does not depend on κ, and $|\sigma_1^h(\epsilon_1 l)| \le C|l|^{s_1}$ by homogeneity. Hence the extended integrand has the majorant $C|l|^{s_1}$, which does not depend on κ and is integrable over $\{|l| \le 3\varepsilon\}$ as $s_1 \ge -2$; and as $\kappa \to \kappa_0$ it converges at every $l \ne 0$ with $|l + l_1(\kappa_0)| \ne 2\varepsilon$, that is, outside a null set. So dominated convergence applies. The integral with the factor ψ_2 is treated in the same way, with the substitution $l \to l + l_2(\kappa)$ and the majorant $C|l|^{s_2}$.

Step 2″: comparison. Let $\mathsf{X}(l,\kappa) := \prod_i \chi(\epsilon_i l + \lambda_i(\kappa))$, let $\chi_*(l) := \chi(l)^{\mathsf{m}}$, which is radial, and let $\mathsf{I}^{(0)}(\kappa) := \int_{\mathbb{R}^3} \mathsf{X}\mathsf{A}^h\, d^\circ l$ for $\varepsilon \le \frac{1}{4}$. The function $\mathsf{X} - \chi_*$ is smooth and vanishes for $|l| \le \frac{1}{2}$ and for $|l| \ge 3$; therefore $\int (\mathsf{X} - \chi_*)\mathsf{A}^h\, d^\circ l$ is smooth in κ and vanishes at $\kappa = 0$, so it equals a polynomial of

degree $\le \omega$ without constant term up to $O(|\kappa|^{\omega+1})$. Moreover

$$\int \chi_* \mathsf{A}^h \, d^\circ l = \int_{|l|\le 2\varepsilon} \mathsf{A}^h \, d^\circ l + \int_{|l|>2\varepsilon} \mathsf{A}^h_{\mathrm{rem}} \, d^\circ l - \int (1-\chi_*) \mathsf{A}^h_{\mathrm{rem}} \, d^\circ l + \sum_{j\le\omega} \int_{|l|>2\varepsilon} \chi_* \mathsf{A}^h_j \, d^\circ l.$$

The third term is $O(|\kappa|^{\omega+1})$. For $j<\omega$ we have $\int_{|l|>2\varepsilon} \chi_* \mathsf{A}^h_j = \int \chi_* \mathsf{A}^h_j - \int_{|l|\le 2\varepsilon} \mathsf{A}^h_j$, and $\int \chi_* \mathsf{A}^h_j \, d^\circ l$ is a homogeneous polynomial of degree j in κ. For $j=\omega$ the function $\mathsf{A}^h_\omega(\cdot;\kappa)$ is odd, while χ_* and the domain $\{|l|>2\varepsilon\}$ are even, so that this term vanishes. Consequently

$$\mathsf{I}^{(0)} = \mathsf{Q}^{(0)} + \mathsf{I}^h + O(|\kappa|^{\omega+1}),$$

with a polynomial $\mathsf{Q}^{(0)}$ of degree $\le\omega$; if $\omega \ge 1$, then $\mathsf{Q}^{(0)}(0) = \int \chi_* \mathsf{A}^h_0 \, d^\circ l = \int \mathsf{X}(l,0) \mathsf{A}^h(l,0) \, d^\circ l$.

Step 3: proof of (b) and (c). We write $\sigma_i = \chi\sigma_i^h + \sigma_i^\flat$ with $\sigma_i^\flat \in \mathcal{S}^{s_i+1}$ and multiply out the product in the definition of I. For $\varepsilon \le \frac14$ the term in which all factors are leading is $\mathsf{I}^{(0)}$. Every other term is a function J as in Step 1, with $t_i \in \{s_i, s_i+1\}$ and $\omega' \ge \omega+1 \ge 0$, since $\chi\sigma_i^h \in \mathcal{S}^{s_i}$; the terms of degree $>\omega$ of its polynomial Q' are $O(|\kappa|^{\omega+1})$. This proves (b), with Q the sum of $\mathsf{Q}^{(0)}$ and of the polynomials Q' cut at degree ω, and with $c_0 := \frac{1}{4c_\lambda}$, which lies in $(0,\frac14]$ as $c_\lambda \ge 1$; for $0<|\kappa|\le c_0$ we have $\varepsilon \le \frac14$, as Step 2″ requires, and $c_\lambda|\kappa| \le \frac12$, as Step 1 requires.

If $\omega \ge 1$, then $|\prod_i \sigma_i(\epsilon_i l)| \le C|l|^{\omega-3}$ is integrable, and adding the constant terms we obtain $\mathsf{Q}(0) = \int_{\mathbb{T}} \prod_i (\chi\sigma_i^h + \sigma_i^\flat)(\epsilon_i l) \, d^\circ l = \int_{\mathbb{T}} \prod_i \sigma_i(\epsilon_i l) \, d^\circ l$, which is (c).

Step 4: uniqueness. Let $\kappa \in \mathsf{K}^\sharp$. By (b), and since $\Lambda(t\kappa) = \Lambda(\kappa) < \infty$,

$$t^{-\omega} \big|\mathsf{I}(t\kappa) - \mathsf{Q}(t\kappa) - \mathsf{I}^h(t\kappa)\big| \le Ct|\kappa|^{\omega+1} \lg(t|\kappa|) \, \Lambda(\kappa) \to 0 \qquad (t \downarrow 0).$$

Subtracting the hypothesis of (d) and writing Q_j, $\mathsf{Q}_j^\sharp$ for the homogeneous terms of degree j, we get

$$\sum_{j<\omega} t^{j-\omega} (\mathsf{Q}_j - \mathsf{Q}_j^\sharp)(\kappa) + (\mathsf{Q}_\omega - \mathsf{Q}_\omega^\sharp)(\kappa) + (\mathsf{I}^h - \mathsf{I}^\sharp)(\kappa) \to 0.$$

Beginning with the least j, we conclude $(\mathsf{Q}_j - \mathsf{Q}_j^\sharp)(\kappa) = 0$ for $j<\omega$, and then $(\mathsf{Q}_\omega - \mathsf{Q}_\omega^\sharp)(\kappa) + (\mathsf{I}^h - \mathsf{I}^\sharp)(\kappa) = 0$. The same holds at $-\kappa \in \mathsf{K}^\sharp$, where the two terms acquire the factors $(-1)^\omega$ and $(-1)^{\omega+1}$; hence also $(\mathsf{Q}_\omega - \mathsf{Q}_\omega^\sharp)(\kappa) - (\mathsf{I}^h - \mathsf{I}^\sharp)(\kappa) = 0$, and both terms vanish. (If $\omega = -1$, only the term $\mathsf{I}^h - \mathsf{I}^\sharp$ is present.) Polynomials which vanish on a non-empty open set vanish identically; so $\mathsf{Q}^\sharp = \mathsf{Q}$, and $\mathsf{I}^\sharp = \mathsf{I}^h$ on $\mathsf{K}^\sharp$.

The set of all $\kappa \ne 0$ at which the $l_i(\kappa)$ are pairwise different is such a set $\mathsf{K}^\sharp$, being the complement of finitely many proper linear subspaces, and $\mathsf{I}^h = 0$ outside it; therefore the pair $(\mathsf{Q}, \mathsf{I}^h)$ with (a) and (b) is unique and independent of c_λ. As, by Step 2, at fixed c_λ the function I^h is built from the σ_i^h, ϵ_i and λ_i alone, it depends only on these data.

For the last assertion of (d) let $\tilde{\mathsf{Q}}$ and $\tilde{\mathsf{I}}^h$ be the polynomial and the homogeneous part of $\tilde{\mathsf{I}}$, and let $\mathsf{K}^\sharp$ be the set of all $\kappa \ne 0$ at which the $l_i(\kappa)$ are pairwise different and so are the $\tilde{l}_i(\kappa)$. It is again the complement of finitely many proper linear subspaces, hence open, non-empty and invariant under $\kappa \to t\kappa$ for real $t \ne 0$. We put $\mathsf{Q}^\sharp(\kappa) := \tilde{\mathsf{Q}}(-\kappa)$ and $\mathsf{I}^\sharp(\kappa) := \tilde{\mathsf{I}}^h(-\kappa)$ on $\mathsf{K}^\sharp$; they have the degree, the homogeneity and the parity required in (d). For $\kappa \in \mathsf{K}^\sharp$ and small $t>0$ the hypothesis gives $\mathsf{I}(t\kappa) = \tilde{\mathsf{I}}(-t\kappa)$, and (b) for the data of $\tilde{\mathsf{I}}$, at the point $-t\kappa$, gives

$$t^{-\omega} \big|\mathsf{I}(t\kappa) - \mathsf{Q}^\sharp(t\kappa) - \mathsf{I}^\sharp(t\kappa)\big| = t^{-\omega} \big|\tilde{\mathsf{I}}(-t\kappa) - \tilde{\mathsf{Q}}(-t\kappa) - \tilde{\mathsf{I}}^h(-t\kappa)\big| \to 0 \qquad (t \downarrow 0).$$

By the first part of (d), $\mathsf{Q}^\sharp = \mathsf{Q}$ and $\mathsf{I}^\sharp = \mathsf{I}^h$ on $\mathsf{K}^\sharp$. Thus $\tilde{\mathsf{Q}}(\kappa) = \mathsf{Q}(-\kappa)$ for all $\kappa \in \mathbb{R}^D$, as an identity of polynomials, and, $\mathsf{K}^\sharp$ being symmetric, $\tilde{\mathsf{I}}^h(\kappa) = \mathsf{I}^h(-\kappa)$ for all $\kappa \in \mathsf{K}^\sharp$. □

Lemma 7.21 ((D) Differences of an asymptotic behaviour)**.** Let $n>0$ be real and $f : \mathbb{Z}^3 \to \mathbb{C}$. Assume that for every multi-index γ there are a continuous function H_γ on $\mathbb{R}^3 \setminus \{0\}$, homogeneous of degree $-n-|\gamma|$, and numbers c_γ, r_γ with

$$|\nabla^\gamma f(x) - \mathsf{H}_\gamma(x)| \le c_\gamma |x|^{-n-|\gamma|-1} \lg|x| \qquad (x \in \mathbb{Z}^3 \setminus \{0\},\ |x| \ge r_\gamma).$$

Then $\mathsf{H}_0 \in C^\infty(\mathbb{R}^3 \setminus \{0\})$, $\partial^\gamma \mathsf{H}_0 = \mathsf{H}_\gamma$ for every γ, and

$$|\nabla^\gamma (f - \mathsf{H}_0)(x)| \le c'_\gamma |x|^{-n-|\gamma|-1} \lg|x| \qquad (|x| \ge \max\{r_\gamma, |\gamma|+1\}),$$

with $c'_\gamma = c'_\gamma(n, \gamma, c_\gamma, \mathsf{H}_0)$. ■

Proof. Let $y \in \mathbb{R}^3 \setminus \{0\}$, $\mu \in \{1,2,3\}$, and let $t > 0$ be so small that $y + se_\mu \ne 0$ for $0 \le s \le t$. For $R > 0$ let $x_R := (\lfloor R y_\nu \rfloor)_\nu \in \mathbb{Z}^3$ and $N_R := \lfloor Rt \rfloor$. Then

$$\nabla^\gamma f(x_R + N_R e_\mu) - \nabla^\gamma f(x_R) = \sum_{j < N_R} \nabla^{\gamma + e_\mu} f(x_R + j e_\mu).$$

We multiply by $R^{n+|\gamma|}$ and let $R \to \infty$. By the hypothesis, the homogeneity and the continuity of H_γ, the left side tends to $\mathsf{H}_\gamma(y + te_\mu) - \mathsf{H}_\gamma(y)$. The points $x_R + je_\mu$, $j < N_R$, have modulus at least cR for some $c > 0$; by the hypothesis for $\gamma + e_\mu$ and the homogeneity of $\mathsf{H}_{\gamma+e_\mu}$, the right side equals

$$\frac{1}{R} \sum_{j < N_R} \mathsf{H}_{\gamma+e_\mu}\big((x_R + je_\mu)/R\big) + O(R^{-1} \lg R),$$

and this tends to $\int_0^t \mathsf{H}_{\gamma+e_\mu}(y + se_\mu)\,\mathrm{d}s$: it is a Riemann sum, up to the shift $x_R/R - y = O(R^{-1})$ and the uniform continuity of $\mathsf{H}_{\gamma+e_\mu}$ on a compact neighbourhood of $\{y + se_\mu : 0 \le s \le t\}$. Hence $\mathsf{H}_\gamma(y + te_\mu) - \mathsf{H}_\gamma(y) = \int_0^t \mathsf{H}_{\gamma+e_\mu}(y + se_\mu)\,\mathrm{d}s$ for small $t > 0$, and, applied at the point $y - te_\mu$, the corresponding identity holds for small $t < 0$. Therefore $\partial_\mu \mathsf{H}_\gamma$ exists and equals the continuous function $\mathsf{H}_{\gamma+e_\mu}$. By induction on $|\gamma|$, all partial derivatives of H_0 exist, are continuous, and $\partial^\gamma \mathsf{H}_0 = \mathsf{H}_\gamma$; thus $\mathsf{H}_0 \in C^\infty(\mathbb{R}^3 \setminus \{0\})$.

Finally, Taylor's formula and the homogeneity of the derivatives of order $|\gamma| + 1$ of H_0 give $|\nabla^\gamma \mathsf{H}_0(x) - \partial^\gamma \mathsf{H}_0(x)| \le C|x|^{-n-|\gamma|-1}$ for $|x| \ge |\gamma| + 1$, with C depending on n, γ and H_0; together with the hypothesis this proves the bound for $\nabla^\gamma(f - \mathsf{H}_0)$. □

Lemma 7.22 ((E) Chains (convolutions of lines), bubbles)**.** For $X, Y : \mathbb{Z}^3 \to \mathbb{C}$ we write $(X * Y)(u) := \sum_z X(z) Y(u - z)$, and $r := |u|$.

(a) (Bound.) Let X, Y be kernels with symbols $\hat X, \hat Y \in L^1(\mathbb{T})$; let $a_X, a_Y \in (-3, 3)$ and $n_X, n_Y \ge 0$ with $n_X + n_Y > 3$. Assume $|\hat X(k)| \le c|k|^{a_X} \lg|k|$ on $[-\pi, \pi]^3$ and $|\nabla^\gamma X(x)| \le c\langle x\rangle^{-n_X - |\gamma|}$ for $|\gamma| \le 10$ and all x, and the same for Y with a_Y, n_Y. Then the sum defining $(X * Y)(u)$ converges absolutely, and for $r \ge 1$

$$|(X * Y)(u)| \le C\big(r^{-a_X - n_Y} + r^{-a_Y - n_X} + r^{3 - n_X - n_Y}\big) \lg r, \qquad C = C(c, a_X, a_Y, n_X, n_Y).$$

(b) (Class.) Let $X \in \mathfrak{K}(n_X)$ and $Y \in \mathfrak{K}(n_Y)$ with $n_X, n_Y \le 4$ and $n_X + n_Y \ge 5$; we write $\hat X$ and $\hat Y$ for their symbols, as in (K1), and $(\hat X^h, X^h)$ and $(\hat Y^h, Y^h)$ for their leading data in the sense of Def. 7.16(4). Then $X * Y \in \mathfrak{K}(n_X + n_Y - 3)$, with the symbol $\hat X \hat Y$. Its leading data consist of $\hat X^h \hat Y^h$ and of a function which depends only on the leading data of X and Y, bilinearly. The constants of $X * Y$ depend only on those of X and Y. The leading data are rotation covariant: if the leading data of X and of Y are composed with O^{-1}, O a rotation, then so are those of $X * Y$.

(c) (Bubbles.) Let K_a, K_b be elementary lines of degrees $n_\mathrm{a}, n_\mathrm{b} \in \{1,2,3\}$, with symbols σ_a, σ_b; let $X := K_\mathrm{a} K_\mathrm{b}$ (pointwise product) and $n := n_\mathrm{a} + n_\mathrm{b}$. Then X is the kernel with the symbol

$$\mathsf{I}(k) = \int_{\mathbb{T}} \sigma_\mathrm{a}(k - l)\, \sigma_\mathrm{b}(l)\, d^\circ l,$$

a function as in Lem. 7.20 with $\mathsf{m} = 2$ and $\omega = n - 3$. With its polynomial Q and its homogeneous part I^h we put

$$\tilde{\mathsf{Q}}(k) := \mathsf{Q}(0) + \sum_j \partial_j \mathsf{Q}(0) \sin k_j,$$

and we let q be the kernel with the symbol $\tilde{\mathsf{Q}}$; it vanishes outside $\{0, \pm e_1, \pm e_2, \pm e_3\}$. Then $|\nabla^\gamma X(x)| \le C_\gamma \langle x\rangle^{-n-|\gamma|}$ for all x and γ, and $|\mathsf{I}(k) - \tilde{\mathsf{Q}}(k)| \le C|k|^{\min\{n-3,2\}}$ on $[-\pi,\pi]^3$, where C_γ and C depend on γ, σ_{a}, σ_{b}. If $n \le 4$, then $X - \mathsf{q} \in \mathfrak{K}(n)$, with the leading data $(\mathsf{I}^h, K^h_{\mathrm{a}} K^h_{\mathrm{b}})$, where $K^h_{\mathrm{a}} = \mathcal{F}^{-1}_{\neq 0}\sigma^h_{\mathrm{a}}$ and $K^h_{\mathrm{b}} = \mathcal{F}^{-1}_{\neq 0}\sigma^h_{\mathrm{b}}$.

(d) (Uniqueness.) Let $K \in \mathfrak{K}(n)$ with $n \le 6$ and leading data (σ^h, K^h). If $\sigma^h = 0$, then $K^h = 0$. Hence, within $\mathfrak{K}(n)$, K^h is determined by σ^h, and depends linearly on it.

■

Proof. Step 0: splitting. Let $r \ge 8$; for $1 \le r < 8$ the bound in (a) follows from Lem. 7.18(c). For $z \in \mathbb{R}^3$ we put $\theta_1(z) := \theta(4|z|/r)$ and $\theta_2(z) := \theta(4|u-z|/r)$. These are smooth functions on $\mathbb{R}^3$, as $\theta \in C^\infty([0,\infty))$ is constant on $[0, \frac12]$; they vanish for $|z| \ge r/4$, resp. for $|u-z| \ge r/4$, so that their supports are disjoint, as $|u| = r$. We write $X * Y = \mathsf{T}_1 + \mathsf{T}_2 + \mathsf{T}_0$, where the three sums carry the factors θ_1, θ_2 and $1 - \theta_1 - \theta_2$. The function $g_1 := \theta_1\, Y(u - \cdot)$ on $\mathbb{Z}^3$ vanishes outside $\{|z| \le r/4\}$, a set on which $|u - z| \ge \frac34 r$. Since g_1 has finite support and $X(z) = \int_{\mathbb{T}} e^{ik\cdot z}\hat{X}(k)\, d^\circ k$, we may interchange the sum and the integral:

$$\mathsf{T}_1 = \sum_z X(z) g_1(z) = \int_{\mathbb{T}} \hat{X}(k) \sum_z g_1(z)\, e^{ik\cdot z}\, d^\circ k = \int_{\mathbb{T}} \hat{X}(k)\, \check{g}_1(-k)\, d^\circ k.$$

After the substitution $z \to u - z$ the same argument gives $\mathsf{T}_2 = \int_{\mathbb{T}} \hat{Y}(k)\, \check{g}_2(-k)\, d^\circ k$ with $g_2 := \theta(4|\cdot|/r)\, X(u - \cdot)$.

Step 1: proof of (a). The sum converges absolutely by Lem. 7.18(c) with $a = n_X$, $a' = n_Y$. By the discrete Leibniz rule $\nabla_j(fg) = (\nabla_j f)\, g(\cdot + e_j) + f\, \nabla_j g$, a difference of order ≤ 10 of g_1 is a sum of products of differences of θ_1, bounded by $|\nabla^\beta \theta_1| \le C r^{-|\beta|}$, with differences of Y of order ≤ 10 taken at points $u - z'$ with $|z'| \le r/4 + 10$. At these points $\langle u - z'\rangle \ge r/40$: for $r \le 40$ because $\langle u - z'\rangle \ge 1$, and for $r > 40$ because $|u - z'| \ge \frac34 r - 10 \ge r/40$. With the bounds on the differences of Y this gives $\|g_1\|_{(r)} \le C r^{-n_Y}$; in the same way $\|g_2\|_{(r)} \le C r^{-n_X}$.

Lem. 7.19(b), applied with $D = 3$, $R = r$, $g = g_1$, $\hat{\mathsf{T}} = \hat{X}$, the majorant $\mathsf{H} = c|k|^{a_X} \in \mathcal{H}_{a_X}(3)$ (admissible since $-3 < a_X < 3 < 6$), $\iota = 1$ and $c_0 = 1$, gives $|\mathsf{T}_1| \le C r^{-a_X - n_Y} \lg r$; likewise $|\mathsf{T}_2| \le C r^{-a_Y - n_X} \lg r$. On the support of $1 - \theta_1 - \theta_2$ we have $|z| \ge r/8$ and $|u - z| \ge r/8$, hence $\langle z\rangle \ge \frac19(r + |z|)$ and $\langle u - z\rangle \ge \frac19(r + |u - z|)$; together with $r + |u - z| \ge \frac12(r + |z|)$ (Lem. 7.18(a)) this gives

$$|X(z)Y(u-z)| \le C(r + |z|)^{-n_X}(r + |u - z|)^{-n_Y} \le C(r + |z|)^{-n_X - n_Y},$$

and Lem. 7.18(a) with the exponent $n_X + n_Y > 3$ gives $|\mathsf{T}_0| \le C r^{3 - n_X - n_Y}$.

Step 2: proof of (b), conditions (K1) and (K2). As $n_X, n_Y \ge 2$, conditions (K3) and the homogeneity of X^h, Y^h give $|X(x)| \le C\langle x\rangle^{-n_X}$ and $|Y(x)| \le C\langle x\rangle^{-n_Y}$, so that $X, Y \in \ell^2(\mathbb{Z}^3)$; by the uniqueness of Fourier coefficients $\hat{X}, \hat{Y} \in L^2(\mathbb{T})$, and Plancherel's theorem gives $(X * Y)(u) = \int_{\mathbb{T}} e^{ik\cdot u}\hat{X}\hat{Y}\, d^\circ k$, which is (K1) with the symbol $\hat{X}\hat{Y} \in L^1(\mathbb{T})$. For (K2) let $s_X := n_X - 3$ and $s_Y := n_Y - 3$, both ≥ -1; then

$$\hat{X}\hat{Y} - \chi\hat{X}^h\hat{Y}^h = (\hat{X} - \chi\hat{X}^h)\hat{Y} + \chi\hat{X}^h(\hat{Y} - \chi\hat{Y}^h) + (\chi^2 - \chi)\hat{X}^h\hat{Y}^h = O(|k|^{s_X + s_Y + 1} \lg|k|),$$

and $\hat{X}^h\hat{Y}^h$ is continuous and homogeneous of degree $s_X + s_Y = (n_X + n_Y - 3) - 3$.

Step 3: proof of (b), condition (K3) for $\gamma = 0$. Let $n := n_X + n_Y - 3$, and let γ be a multi-index; in this step $\gamma = 0$, in Step 4 it is arbitrary. We take $r \ge 200 + 8|\gamma|$. In T_1 the line Y enters through g_1, and in T_2 the line X through g_2; in the estimates below these lines are evaluated, together with their differences of the orders $\le |\gamma| + 10$ (the order γ of Step 4 and up to ten further differences from the norm $\|\cdot\|_{(r)}$), at points of modulus at least $r/2 \ge |\gamma| + 11$. In T_0 both lines, with their differences of the orders $\le |\gamma|$, are evaluated at points of modulus at least $r/8 \ge |\gamma| + 1$. Thus all these points lie in the range $|x| \ge |\gamma'| + 1$ in which (K3) for X and Y bounds the differences of

order γ'. Let $\hat u := u/r$ and

$$G_1(\zeta) := \theta(4|\zeta|)\, Y^h(\hat u - \zeta), \qquad G_2(\zeta) := \theta(4|\zeta|)\, X^h(\hat u - \zeta);$$

these functions lie in $C_c^\infty(\mathbb{R}^3)$, with support in $\{|\zeta| \le \frac14\}$, and $r^{-n_Y} G_1(z/r) = \theta_1(z) Y^h(u-z)$ by homogeneity. By (K3) for Y, $\|g_1 - r^{-n_Y} G_1(\cdot/r)\|_{(r)} \le C r^{-n_Y-1} \lg r$. We use Lem. 7.19(c) with $D=3$, $R=r$, $g=g_1$, $\hat{\mathsf T} = \hat X$, $\hat{\mathsf T}^h = \hat X^h$, $s = s_X \le 1$, $\iota = 1$, $c_0 = 1$, $a = n_Y$, $G = G_1$, the constant of the preceding bound as A, and the majorants $\mathsf H = 2c|k|^{s_X}$, $\mathsf H' = c|k|^{s_X+1}$, where c bounds both $|\hat X^h|$ on the unit sphere and the number in (K2) for X. The hypotheses of (b) and (c) of that lemma are satisfied: on $\{|k| \le 1\}$ we have $\chi = 1$, so that (K2) for X gives $|\hat X - \hat X^h| \le \mathsf H' \lg|k|$ there, and hence $|\hat X| \le \mathsf H \lg|k|$, while $|\hat X^h| \le \mathsf H$ everywhere. We obtain

$$\mathsf T_1 = r^{-s_X - n_Y} \int \hat X^h(k)\, \hat G_1(-k)\, d^\circ k + O(r^{-s_X-n_Y-1} \lg r),$$

and in the same way $\mathsf T_2 = r^{-s_Y-n_X} \int \hat Y^h(k)\, \hat G_2(-k)\, d^\circ k + O(r^{-s_Y-n_X-1}\lg r)$; note $s_X + n_Y = s_Y + n_X = n$. In $\mathsf T_0$ we replace X, Y by X^h, Y^h; by (K3) the error is at most

$$C \sum_z (r+|z|)^{-n_X-n_Y-1} \lg(r+|z|) \le C r^{2-n_X-n_Y} \lg r,$$

since $\lg(r+|z|) \le \lg r \lg(1+|z|/r) \le \lg r\, (r+|z|)/r$, and by Lem. 7.18(a) with the exponent $n_X + n_Y > 3$. Next we replace the sum by the integral of $\mathsf F(z) := (1-\theta_1-\theta_2)(z) X^h(z) Y^h(u-z)$, $z \in \mathbb{R}^3$. The factor $1-\theta_1-\theta_2$ vanishes for $|z| \le r/8$ and for $|u-z| \le r/8$, that is, near the points 0 and u at which X^h and $Y^h(u-\cdot)$ are singular; so $\mathsf F$, extended by 0 at these two points, is smooth on $\mathbb{R}^3$, in particular $\mathsf F \in C^1(\mathbb{R}^3)$. Since moreover $|\mathsf F(z)| \le C(r+|z|)^{-n_X-n_Y}$ and $|\nabla \mathsf F(z)| \le C(r+|z|)^{-n_X-n_Y-1}$, Lem. 7.18(b) applies, and together with (a) of that lemma it bounds the error by $C r^{2-n_X-n_Y}$. The substitution $z = r\zeta$ gives

$$\int_{\mathbb{R}^3} \mathsf F = r^{3-n_X-n_Y} \int_{\mathbb{R}^3} (1 - \theta(4|\zeta|) - \theta(4|\hat u - \zeta|)) X^h(\zeta)\, Y^h(\hat u - \zeta)\, \mathrm{d}\zeta.$$

Altogether $(X * Y)(u) = \mathsf H_0(u) + O(r^{-n-1} \lg r)$, where

$$\mathsf H_0(u) := r^{-n} \Big[\int \hat X^h(k)\, \hat G_1(-k)\, d^\circ k + \int \hat Y^h(k)\, \hat G_2(-k)\, d^\circ k + \int (1 - \theta(4|\zeta|) - \theta(4|\hat u - \zeta|)) X^h(\zeta)\, Y^h(\hat u - \zeta)\, \mathrm{d}\zeta \Big].$$

The function $\mathsf H_0$ is homogeneous of degree $-n$, the bracket depending on $\hat u$ only. It is continuous: $\hat u \mapsto G_i$ is continuous with respect to $\|\cdot\|_{C^10}$, so that the last bound of Lem. 7.19(c) applies to the first two integrals; and the third integral depends continuously on $\hat u$ by dominated convergence, since on the support of its integrand $|\zeta| \ge \frac18$ and $|\hat u - \zeta| \ge \frac18$, so that the integrand is bounded by $C\langle\zeta\rangle^{-n_X-n_Y}$ uniformly in $\hat u$, and $n_X + n_Y > 3$. It is bilinear in the leading data of X and Y, and it is rotation covariant, by the substitutions $k \to Ok$, $\zeta \to O\zeta$.

Step 4: proof of (b), condition (K3) for all γ. Let γ be a multi-index and $r \ge 200 + 8|\gamma|$. We have $\nabla^\gamma (X * Y)(u) = \sum_z X(z) (\nabla^\gamma Y)(u-z)$, which we split by the factors θ_1, θ_2 and $1-\theta_1-\theta_2$ as before.

In the first sum the function $g_{1,\gamma} := \theta_1 (\nabla^\gamma Y)(u - \cdot)$ takes the place of g_1; it vanishes outside $\{|z| \le r/4\}$. The function $G_{1,\gamma}(\zeta) := \theta(4|\zeta|)(\partial^\gamma Y^h)(\hat u - \zeta)$ lies in $C_c^\infty(\mathbb{R}^3)$, with support in $\{|\zeta| \le \frac14\}$, and $r^{-n_Y-|\gamma|} G_{1,\gamma}(z/r) = \theta_1(z)(\partial^\gamma Y^h)(u-z)$ by homogeneity. By (K3) for Y, and since $|(\nabla^\gamma - \partial^\gamma) Y^h(x)| \le C|x|^{-n_Y-|\gamma|-1}$ at the points concerned, the function $g_{1,\gamma}$ differs from $r^{-n_Y-|\gamma|} G_{1,\gamma}(\cdot/r)$ by at most $C r^{-n_Y-|\gamma|-1} \lg r$ in $\|\cdot\|_{(r)}$.

As in Step 0 the first sum equals $\int_{\mathbb{T}} \hat X(k)\, \check g_{1,\gamma}(-k)\, d^\circ k$, and Lem. 7.19(c), with the data of Step 3 except that now $g = g_{1,\gamma}$, $a = n_Y + |\gamma|$, $G = G_{1,\gamma}$ and A is the constant of the bound just stated, shows that it equals $r^{-n-|\gamma|} \int \hat X^h(k)\, \hat G_{1,\gamma}(-k)\, d^\circ k + O(r^{-n-|\gamma|-1} \lg r)$; here $s_X + n_Y = n$.

In the second sum we sum by parts $|\gamma|$ times, by the rule $\sum_z f(z)(\nabla_j Y)(u-z) = \sum_z (\nabla_j f)(z) Y(u-z)$ for f of finite support, and obtain $\sum_{z'} Y(z') g_{2,\gamma}(z')$ with $g_{2,\gamma}(z') := \nabla^\gamma(\theta_2 X)(u-z')$. This function vanishes outside $\{|z'| \le r/4 + |\gamma|\}$, a subset of $\{|z'| \le r\}$. The function

$$G_{2,\gamma}(\zeta) := \partial_y^\gamma \big[\theta(4|\hat u - y|) X^h(y)\big]_{y=\hat u-\zeta}$$

lies in $C_c^\infty(\mathbb{R}^3)$, with support in $\{|\zeta| \le \frac14\}$, and by (K3) for X, the discrete Leibniz rule and Taylor's formula, as for the first sum, $g_{2,\gamma}$ differs from $r^{-n_X-|\gamma|} G_{2,\gamma}(\cdot/r)$ by at most $C r^{-n_X-|\gamma|-1} \lg r$ in $\|\cdot\|_{(r)}$. The second sum equals $\int_{\mathbb{T}} \hat Y(k)\, \check g_{2,\gamma}(-k)\, d^\circ k$, and Lem. 7.19(c), now taken with $D=3$, $R=r$, $g = g_{2,\gamma}$, $\hat{\mathsf{T}} = \hat Y$, $\hat{\mathsf{T}}^h = \hat Y^h$, $s = s_Y$, $\iota = 1$, $c_0 = 1$, $\mathsf{H} = 2c|k|^{s_Y}$, $\mathsf{H}' = c|k|^{s_Y+1}$ (with c chosen for Y as it was chosen for X in Step 3), $a = n_X + |\gamma|$ and $G = G_{2,\gamma}$, yields the value $r^{-n-|\gamma|} \int \hat Y^h(k)\, \hat G_{2,\gamma}(-k)\, d^\circ k + O(r^{-n-|\gamma|-1} \lg r)$, as $s_Y + n_X = n$.

In the third sum we replace X and $\nabla^\gamma Y$ by X^h and $\partial^\gamma Y^h$, and then the sum by the integral over $\mathbb{R}^3$, exactly as in Step 3; the errors are $O(r^{-n-|\gamma|-1}\lg r)$, and the substitution $z = r\zeta$ produces the factor $r^{-n-|\gamma|}$. Altogether $\nabla^\gamma(X * Y)(u) = \mathsf{H}_\gamma(u) + O(r^{-n-|\gamma|-1}\lg r)$, where

$$\mathsf{H}_\gamma(u) := r^{-n-|\gamma|}\Big[\int \hat X^h(k)\, \hat G_{1,\gamma}(-k)\, d^\circ k + \int \hat Y^h(k)\, \hat G_{2,\gamma}(-k)\, d^\circ k + \int (1 - \theta(4|\zeta|) - \theta(4|\hat u - \zeta|)) X^h(\zeta)\, (\partial^\gamma Y^h)(\hat u - \zeta)\, \mathrm{d}\zeta\Big].$$

For the reasons given for H_0 in Step 3, the function H_γ is homogeneous of degree $-n-|\gamma|$ and continuous on $\mathbb{R}^3 \setminus \{0\}$: the bracket depends on $\hat u$ only; $\hat u \mapsto G_{1,\gamma}$ and $\hat u \mapsto G_{2,\gamma}$ are continuous with respect to $\|\cdot\|_{C^{10}}$, so that the last bound of Lem. 7.19(c) applies; and the third integrand is bounded by $C\langle\zeta\rangle^{-n_X-n_Y}$ uniformly in $\hat u$, so that dominated convergence applies. Lem. 7.21, applied to $f = X * Y$, with the present $n > 0$, the functions H_γ and $r_\gamma = 200 + 8|\gamma|$, gives (K3) with $K^h = \mathsf{H}_0$ for $|u| \ge 200 + 8|\gamma|$.

For the finitely many other u with $|u| \ge |\gamma| + 1$, the left side of (K3) is a finite number, bounded through the constants of X and Y, as H_0 is evaluated only at points different from 0; so (K3) holds with a larger c_γ, which still depends only on the constants of X and Y. This proves (b).

Step 5: proof of (c). We have $X(x) = \iint e^{i(k'+l)\cdot x} \sigma_{\mathrm{a}}(k') \sigma_{\mathrm{b}}(l)\, d^\circ k'\, d^\circ l$, and the substitution $k = k' + l$ shows that X is the kernel with the symbol I. The data of Lem. 7.20 are $\lambda_1(k) = k$, $\epsilon_1 = -1$, $\lambda_2 = 0$, $\epsilon_2 = 1$, so that $l_1(k) = k$, $l_2 = 0$; the hypotheses hold with $c_\lambda = 1$, the map $(l,k) \mapsto (k-l, l)$ having an integer matrix of determinant -1, and $\omega = 3 + (n_{\mathrm{a}} - 3) + (n_{\mathrm{b}} - 3) = n - 3 \ge -1$; moreover $\Lambda = 1$. The bound on $\nabla^\gamma X$ follows from Lem. 7.17(a) and the discrete Leibniz rule. For $|k| \ge c_0$ the function I is bounded, as an integral with two integrable singularities at distance at least c_0, and so is $\tilde{\mathsf{Q}}$. For $0 < |k| \le c_0$, Lem. 7.20(b) gives

$$\mathsf{I} - \tilde{\mathsf{Q}} = (\mathsf{Q} - \tilde{\mathsf{Q}}) + \mathsf{I}^h + O(|k|^{\omega+1} \lg |k|),$$

where $|\mathsf{I}^h(k)| \le C|k|^\omega$, and $\mathsf{Q} - \tilde{\mathsf{Q}} = O(|k|^2)$, even $O(|k|^3)$ if $\deg \mathsf{Q} \le 1$. This gives the bound on $\mathsf{I} - \tilde{\mathsf{Q}}$, and, for $n \le 4$ (where $\omega \le 1$), conditions (K1) and (K2) for $X - \mathsf{q}$ with $\sigma^h = \mathsf{I}^h$, which is continuous on $\mathbb{R}^3 \setminus \{0\}$ by Lem. 7.20(a). For (K3): at points where no difference involves a point of $\operatorname{supp} \mathsf{q}$,

$$X - K_{\mathrm{a}}^h K_{\mathrm{b}}^h = (K_{\mathrm{a}} - K_{\mathrm{a}}^h) K_{\mathrm{b}} + K_{\mathrm{a}}^h (K_{\mathrm{b}} - K_{\mathrm{b}}^h),$$

and Lem. 7.17(a),(c) gives the required bounds; at the finitely many x with $|x| \ge |\gamma| + 1$ at which a difference of order γ involves a point of $\operatorname{supp} \mathsf{q}$, the left side of (K3) is a finite number, which we absorb into c_γ.

Step 6: proof of (d). Let $G \in C_c^\infty(\mathbb{R}^3)$ with support in $\{\frac12 < |\zeta| < 1\}$, let $R \ge 2$ and $g := G(\cdot/R)$ on $\mathbb{Z}^3$. By (K2) with $\sigma^h = 0$ we have $|\sigma(k)| \le c|k|^{n-2} \lg |k|$ on the cube, and Lem. 7.19(b), with

$D=3$, $s=n-2<6$, $\mathsf{H}=c|k|^{n-2}$, $\iota=1$ and $c_0=1$, gives, since $\|g\|_{(R)}\le\|G\|_{C^{10}}$,

$$\sum_x K(x)g(x)=\int_{\mathbb{T}}\sigma(k)\,\check g(-k)\,d^\circ k=O(R^{2-n}\lg R).$$

On the other hand, g vanishes outside $\{R/2<|x|<R\}$, a set of at most CR^3 points; by (K3) and by Lem. 7.18(b), applied to the function $K^hG(\cdot/R)$, whose gradient is bounded by CR^{-n-1},

$$\sum_x K(x)g(x)=\sum_x K^h(x)g(x)+O(R^{2-n}\lg R)=R^{3-n}\int_{\mathbb{R}^3}K^hG\,\mathrm{d}\zeta+O(R^{2-n}\lg R).$$

Letting $R\to\infty$ we obtain $\int K^hG\,\mathrm{d}\zeta=0$ for all such G; as K^h is continuous and homogeneous, $K^h=0$. □

Proposition 7.23 ((V) The kite)**.** Let $K_1,\dots,K_5$ be elementary lines with symbols $\sigma_1,\dots,\sigma_5$ and degrees $n_1,\dots,n_5$, where

$$n_1,n_2,n_4,n_5\in\{2,3\},\quad n_3\in\{1,2,3\},\quad n_1+n_2+n_3\le7,\quad n_3+n_4+n_5\le7,$$

and put $\Delta:=\sum_{i=1}^5 n_i-6$. For $u\in\mathbb{Z}^3$ let $r:=|u|$ and

$$\mathsf{T}(u):=\sum_{z,w\in\mathbb{Z}^3}K_1(z)\,K_2(w)\,K_3(w-z)\,K_4(u-z)\,K_5(u-w).$$

(i) There is a constant C, depending on $n_1,\dots,n_5$ only, such that for every $u\in\mathbb{Z}^3$

$$\sum_{z,w\in\mathbb{Z}^3}\langle z\rangle^{-n_1}\langle w\rangle^{-n_2}\langle w-z\rangle^{-n_3}\langle u-z\rangle^{-n_4}\langle u-w\rangle^{-n_5}\le C.$$

In particular the series defining $\mathsf{T}(u)$ converges absolutely.

(ii) For $\kappa=(\kappa_1,\kappa_2)\in\mathbb{R}^3\times\mathbb{R}^3$ put

$$\hat{\mathsf{T}}_{\mathsf{p}}(\kappa_1,\kappa_2):=\int_{\mathbb{T}}\sigma_1(\kappa_1+l)\,\sigma_2(\kappa_2-l)\,\sigma_3(l)\,d^\circ l,$$
$$\hat{\mathsf{T}}_{\mathsf{q}}(\kappa_1,\kappa_2):=\int_{\mathbb{T}}\sigma_4(\kappa_1-l)\,\sigma_5(\kappa_2+l)\,\sigma_3(l)\,d^\circ l.$$

Both are functions I of the kind treated in Lem. 7.20, with $\mathsf{m}=3$ and with data satisfying the hypotheses of that lemma; their numbers ω are

$$\omega_{\mathsf{p}}:=n_1+n_2+n_3-6\qquad\text{and}\qquad\omega_{\mathsf{q}}:=n_3+n_4+n_5-6,$$

respectively, and both lie in $\{-1,0,1\}$. For every function g of finite support on $\mathbb{Z}^6$

$$\begin{aligned}\sum_{z,w}K_1(z)\,K_2(w)\,K_3(w-z)\,g(z,w)&=\int_{\mathbb{T}^2}\hat{\mathsf{T}}_{\mathsf{p}}(\kappa)\,\check g(-\kappa)\,d^\circ\kappa,\\ \sum_{z,w}K_4(z)\,K_5(w)\,K_3(z-w)\,g(z,w)&=\int_{\mathbb{T}^2}\hat{\mathsf{T}}_{\mathsf{q}}(\kappa)\,\check g(-\kappa)\,d^\circ\kappa.\end{aligned}\tag{7.xxvii}$$

(iii) Let Q_{p} and Q_{q} be the polynomials of $\hat{\mathsf{T}}_{\mathsf{p}}$ and of $\hat{\mathsf{T}}_{\mathsf{q}}$ in the sense of Lem. 7.20(d), let

$$\tilde{\mathsf{Q}}_\bullet(\kappa):=\mathsf{Q}_\bullet(0)+\sum_{j=1}^6\partial_j\mathsf{Q}_\bullet(0)\,\sin\kappa_j\qquad(\bullet\in\{\mathsf{p},\mathsf{q}\}),$$

and let $\mathsf{q}_{\mathsf{p}},\mathsf{q}_{\mathsf{q}}$ be the kernels on $\mathbb{Z}^6$ with the symbols $\tilde{\mathsf{Q}}_{\mathsf{p}},\tilde{\mathsf{Q}}_{\mathsf{q}}$, that is, $\mathsf{q}_\bullet(\zeta)=\int_{\mathbb{T}^2}e^{i\kappa\cdot\zeta}\,\tilde{\mathsf{Q}}_\bullet(\kappa)\,d^\circ\kappa$; they vanish off $\{0,\pm e_1,\dots,\pm e_6\}$. There are constants r_0 and C, depending on $\sigma_1,\dots,\sigma_5$

only, and a function $\mathsf{U} \in C^1(\mathbb{R}^3 \setminus \{0\})$, homogeneous of degree $-\Delta$, such that for every u with $r \geq r_0$

$$\Big|\mathsf{T}(u) - \sum_{z,w} \mathsf{q}_{\mathsf{p}}(z,w)\, K_4(u-z)\, K_5(u-w) \\ - \sum_{z,w} \mathsf{q}_{\mathsf{q}}(z,w)\, K_1(u-z)\, K_2(u-w) - \mathsf{U}(u)\Big| \leq C\, r^{-\Delta-1} \lg r.$$

(iv) The function U depends only on the leading parts $\sigma_1^h, \dots, \sigma_5^h$, and it is linear in each of them. If every σ_i^h is replaced by $\sigma_i^h \circ O^{-1}$, with one and the same rotation O, then U is replaced by $\mathsf{U} \circ O^{-1}$.

■

Proof. Step 1: convergence. The symbol σ_i of K_i is a line symbol of degree $s_i := n_i - 3 \geq -2 > -3$. Lem. 7.17(a), applied with $s = s_i$ and $\gamma = 0$, therefore gives $|K_i(y)| \leq C\langle y\rangle^{-n_i}$ for all $y \in \mathbb{Z}^3$, and the absolute convergence of the series for $\mathsf{T}(u)$ follows from the bound in (i). To prove that bound we estimate $\langle w - z\rangle^{-n_3} \leq 1$; what remains factorises into

$$\Big(\sum_z \langle z\rangle^{-n_1} \langle u - z\rangle^{-n_4}\Big)\Big(\sum_w \langle w\rangle^{-n_2} \langle u - w\rangle^{-n_5}\Big).$$

Lem. 7.18(c) applies to the first factor with $(a, a', x) = (n_1, n_4, u)$, since $a, a' \geq 2$ and $a + a' \geq 4 > 3$; it bounds that factor by $C\langle u\rangle^{-\min\{n_1, n_4, n_1+n_4-3\}} \lg\langle u\rangle \leq C\langle u\rangle^{-1} \lg\langle u\rangle \leq C$, because $t^{-1} \lg t = t^{-1}(1 + \log t) \leq 1$ for $t \geq 1$. The second factor is bounded in the same way, with (n_2, n_5) in place of (n_1, n_4).

Step 2: the hypotheses of Lem. 7.20. For $\hat{\mathsf{T}}_{\mathsf{p}}$ we take $\mathsf{m} = 3$, $D = 6$, the line symbols $\sigma_1, \sigma_2, \sigma_3$ of degrees s_1, s_2, s_3, the signs $(\epsilon_1, \epsilon_2, \epsilon_3) = (1, -1, 1)$ and the linear maps $\lambda_1(\kappa) = \kappa_1$, $\lambda_2(\kappa) = \kappa_2$, $\lambda_3(\kappa) = 0$. The integrand of $\hat{\mathsf{T}}_{\mathsf{p}}$ is then $\prod_i \sigma_i(\epsilon_i l + \lambda_i(\kappa))$, and $(l_1, l_2, l_3)(\kappa) = (-\kappa_1, \kappa_2, 0)$. Up to signs, the pairwise differences of the $l_i(\kappa)$ are $\kappa_1 + \kappa_2$, κ_1 and κ_2. Hence $\max_i |l_i(\kappa)| \leq |\kappa| \leq 2|\kappa|$ and $|\kappa| \leq \sqrt{2}\max\{|\kappa_1|, |\kappa_2|\} \leq 2\max_{i<j} |l_i(\kappa) - l_j(\kappa)|$, so that the condition on the l_i holds with $c_\lambda = 2$; the three maps l_i are pairwise different.

The map $(l, \kappa_1, \kappa_2) \mapsto (\kappa_1 + l, \kappa_2 - l, l)$ of $(\mathbb{R}^3)^3$ acts on the three components in $\mathbb{R}^3$ by the integer matrix

$$\begin{pmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix},$$

whose determinant is 1. Since $n_1, n_2 \geq 2$ and $n_3 \geq 1$, we have $s_1, s_2 \geq -1$ and $s_3 \geq -2$, hence $s_i + s_j \geq -3$ for $i \neq j$. Finally $\omega = 3 + s_1 + s_2 + s_3 = n_1 + n_2 + n_3 - 6 = \omega_{\mathsf{p}}$, and this number lies between $2 + 2 + 1 - 6 = -1$ and $7 - 6 = 1$. For $\hat{\mathsf{T}}_{\mathsf{q}}$ the line symbols are $\sigma_4, \sigma_5, \sigma_3$, the signs are $(-1, 1, 1)$ and the maps are $\kappa \mapsto \kappa_1$, $\kappa \mapsto \kappa_2$, $\kappa \mapsto 0$, so that $(l_1, l_2, l_3)(\kappa) = (\kappa_1, -\kappa_2, 0)$; the matrix of $(l, \kappa_1, \kappa_2) \mapsto (\kappa_1 - l, \kappa_2 + l, l)$ is $\left(\begin{smallmatrix} -1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{smallmatrix}\right)$, again of determinant 1, and the remaining conditions follow as before, now with $n_3 + n_4 + n_5 \leq 7$.

Step 3: proof of eq. 7.xxvii. We insert $K_1(z) = \int_{\mathbb{T}} e^{ik_1 \cdot z} \sigma_1(k_1)\, d^\circ k_1$, $K_2(w) = \int_{\mathbb{T}} e^{ik_2 \cdot w} \sigma_2(k_2)\, d^\circ k_2$ and $K_3(w - z) = \int_{\mathbb{T}} e^{il \cdot (w-z)} \sigma_3(l)\, d^\circ l$. The sum over (z, w) is finite and the three symbols are integrable, so Fubini's theorem gives

$$\sum_{z,w} K_1(z)\, K_2(w)\, K_3(w - z)\, g(z,w) \\ = \iiint \sigma_1(k_1)\, \sigma_2(k_2)\, \sigma_3(l)\, \check{g}(-(k_1 - l), -(k_2 + l))\, d^\circ k_1\, d^\circ k_2\, d^\circ l,$$

since $\sum_{z,w} g(z,w)\, e^{i(k_1 - l)\cdot z + i(k_2 + l)\cdot w} = \check{g}(-(k_1 - l), -(k_2 + l))$. For fixed l we substitute $k_1 = \kappa_1 + l$ and $k_2 = \kappa_2 - l$, which are translations of $\mathbb{T}$, and we then carry out the integration over l first; this

is the first identity. The function $y \mapsto K_3(-y)$ is the kernel with the symbol $l \mapsto \sigma_3(-l)$, because $K_3(-y) = \int e^{-il\cdot y}\sigma_3(l)\,d^\circ l = \int e^{il\cdot y}\sigma_3(-l)\,d^\circ l$. The derivation of the first identity used only the integrability of the three symbols. Applied to K_4, K_5 and $K_3(-\cdot)$ in place of K_1, K_2 and K_3, it represents $\sum_{z,w} K_4(z)K_5(w)K_3(z-w)g(z,w)$ by the function $\kappa \mapsto \int_{\mathbb{T}} \sigma_4(\kappa_1+l)\,\sigma_5(\kappa_2-l)\,\sigma_3(-l)\,d^\circ l$, which becomes $\hat{\mathsf{T}}_{\mathsf{q}}(\kappa)$ after the substitution $l \to -l$.

Step 4: short and long lines. Let $\varepsilon := \frac{1}{8}$ and, for $r > 0$, $\theta^r(y) := \theta(|y|/\varepsilon r)$ and $\bar\theta^r := 1 - \theta^r$, as functions on $\mathbb{R}^3$ and, by restriction, on $\mathbb{Z}^3$. By Def. 7.16(1), $\theta^r(y) = 1$ for $|y| \le \frac{1}{2}\varepsilon r$ and $\theta^r(y) = 0$ for $|y| \ge \varepsilon r$. For the arguments of the five lines we write

$$(y_1, y_2, y_3, y_4, y_5) := (z,\ w,\ w-z,\ u-z,\ u-w).$$

For $\mathsf{S} \subset \{1,\dots,5\}$ let $\mathsf{T}_{\mathsf{S}}(u)$ be the series defining $\mathsf{T}(u)$, with the additional factor

$$\prod_{i\in\mathsf{S}} \theta^r(y_i) \prod_{i\notin\mathsf{S}} \bar\theta^r(y_i)$$

in each term; the lines with an index in S are called short, the others long. These series converge absolutely by (i), the additional factors taking values in $[0,1]$, and multiplying out $\prod_{i=1}^5 (\theta^r(y_i) + \bar\theta^r(y_i)) = 1$ we obtain $\mathsf{T} = \sum_{\mathsf{S}} \mathsf{T}_{\mathsf{S}}$.

The arguments satisfy

$$u = y_1 + y_4 = y_2 + y_5 = y_1 + y_3 + y_5 = y_2 - y_3 + y_4.$$

If S contains one of the sets $\{1,4\}$, $\{2,5\}$, $\{1,3,5\}$, $\{2,3,4\}$, then a non-vanishing term of $\mathsf{T}_{\mathsf{S}}(u)$ requires $|y_i| < \varepsilon r$ for the indices i of that set, hence $r = |u| < 3\varepsilon r < r$; thus $\mathsf{T}_{\mathsf{S}} = 0$. There are three possibilities for $\mathsf{S} \cap \{1,4\}$, three for $\mathsf{S} \cap \{2,5\}$ and two for $\mathsf{S} \cap \{3\}$; of the resulting $3 \cdot 3 \cdot 2 = 18$ sets, two, namely $\{1,3,5\}$ and $\{2,3,4\}$, contain one of the forbidden triples, and 16 sets remain.

The substitutions $(z,w) \to (w,z)$ and $(z,w) \to (u-z, u-w)$ are bijections of $\mathbb{Z}^6$; they carry $(y_1,\dots,y_5)$ into $(y_2, y_1, -y_3, y_5, y_4)$ and into $(y_4, y_5, -y_3, y_1, y_2)$, respectively. As θ^r is even, they show that T_{S}, computed for the lines $(K_1, K_2, K_3, K_4, K_5)$, equals $\mathsf{T}_{\mathsf{S}'}$ computed for $(K_2, K_1, K_3(-\cdot), K_5, K_4)$, where S' is the image of S under the permutation $(12)(45)$, and that it equals $\mathsf{T}_{\mathsf{S}''}$ computed for $(K_4, K_5, K_3(-\cdot), K_1, K_2)$, where S'' is the image of S under $(14)(25)$. The new families satisfy the hypotheses of the proposition: $K_3(-\cdot)$ is an elementary line of degree n_3, its symbol $\sigma_3(-k)$ being a line symbol with leading part $\sigma_3^h(-k)$ (the cut-off χ is even), and the conditions on the degrees are invariant under both permutations.

Under the group generated by $(12)(45)$ and $(14)(25)$ the sixteen sets fall into seven classes: the set $\emptyset$ alone; the sets $\{1\},\{2\},\{4\},\{5\}$; the set $\{3\}$ alone; the sets $\{1,2\},\{4,5\}$; the sets $\{1,5\},\{2,4\}$; the sets $\{1,3\},\{2,3\},\{3,4\},\{3,5\}$; and the sets $\{1,2,3\},\{3,4,5\}$. We choose the representatives

$$\emptyset, \quad \{1\}, \quad \{3\}, \quad \{1,2\}, \quad \{1,5\}, \quad \{1,3\}, \quad \{1,2,3\}.$$

It therefore suffices to expand T_{S} for these seven sets, for an arbitrary family of lines obeying the hypotheses; for each of the remaining nine sets we use the expansion of its representative, computed for the transformed family.

Step 5: long factors. Let $L_i := \bar\theta^r K_i$ on $\mathbb{Z}^3$, let $K_i^h := \mathcal{F}_{\neq 0}^{-1}\sigma_i^h$, which by Lem. 7.17(b) is smooth on $\mathbb{R}^3 \setminus \{0\}$ and homogeneous of degree $-3 - s_i = -n_i$, and let $L_i^h := \bar\theta^r K_i^h$, extended by 0 at the origin. Since $\bar\theta^r$ vanishes on $\{|y| \le \frac{1}{2}\varepsilon r\}$, the function L_i^h is smooth on $\mathbb{R}^3$, and by homogeneity $L_i^h(y) = r^{-n_i}\,\bar\theta^1(y/r)\,K_i^h(y/r)$. We claim that there are r_0 and C such that for $r \ge r_0$, $i = 1,\dots,5$ and $|\gamma| \le 10$

$$\begin{aligned} |\nabla^\gamma L_i(y)| + |\partial^\gamma L_i^h(y)| &\le C\,(r+|y|)^{-n_i-|\gamma|}, \\ |\nabla^\gamma (L_i - L_i^h)(y)| &\le C\,(r+|y|)^{-n_i-1-|\gamma|}; \end{aligned} \tag{7.xxviii}$$

here $y \in \mathbb{Z}^3$, except in the term $\partial^\gamma L_i^h(y)$, where $y \in \mathbb{R}^3$. Indeed, Lem. 7.17(a) gives $|\nabla^\gamma K_i(x)| \le C\langle x\rangle^{-n_i-|\gamma|}$, and Lem. 7.17(c) gives $|\nabla^\gamma (K_i - K_i^h)(x)| \le C|x|^{-n_i-1-|\gamma|}$ for $|x| \ge |\gamma| + 1$. The

derivatives of order γ of K_i^h are homogeneous of degree $-n_i-|\gamma|$, hence bounded by $C|x|^{-n_i-|\gamma|}$; the derivatives of order $j\ge 1$ of $\bar\theta^r$ are bounded by Cr^{-j} and vanish outside $\{\frac12\varepsilon r\le|y|\le\varepsilon r\}$. The discrete Leibniz rule $\nabla_j(fg)=(\nabla_j f)\,g(\cdot+e_j)+f\,\nabla_j g$ distributes ∇^γ over the two factors, the arguments being shifted by at most $|\gamma|\le 10$.

Only points y' with $|y'|\ge\frac12\varepsilon r$ contribute, and at such points, for $r\ge r_0$ with r_0 large enough, we have $|y'|\ge|\gamma|+1$ and $r+|y|\le C|y'|$ whenever $|y-y'|\le 10$. These facts together give eq. 7.xxviii.

Step 6: the scheme for $\mathsf S\neq\emptyset$. For each of the six non-empty representatives we write

$$\mathsf T_{\mathsf S}(u)=\sum_{\zeta\in\mathbb Z^D}\mathsf T^{\mathsf S}(\zeta)\,g_{\mathsf S}(\zeta),$$

where the kernel $\mathsf T^{\mathsf S}$ on $\mathbb Z^D$ collects the short lines, and we let s be the degree of the leading part $\hat{\mathsf T}^{\mathsf S,h}$ of the symbol of $\mathsf T^{\mathsf S}$. The data are as follows.

- $\mathsf S=\{1\}$, $D=3$, $\zeta=z$: $\mathsf T^{\mathsf S}(z)=K_1(z)$, with symbol $\sigma_1(\kappa)$ and $s=n_1-3$; and $g_{\mathsf S}(z)=\theta^r(z)\,L_4(u-z)\sum_w L_2(w)\,L_3(w-z)\,L_5(u-w)$.
- $\mathsf S=\{3\}$, $D=3$, $\zeta=y$: $\mathsf T^{\mathsf S}(y)=K_3(y)$, with symbol $\sigma_3(\kappa)$ and $s=n_3-3$; and $g_{\mathsf S}(y)=\theta^r(y)\sum_z L_1(z)\,L_2(z+y)\,L_4(u-z)\,L_5(u-z-y)$.
- $\mathsf S=\{1,2\}$, $D=6$, $\zeta=(z,w)$: $\mathsf T^{\mathsf S}(z,w)=K_1(z)K_2(w)$, with symbol $\sigma_1(\kappa_1)\sigma_2(\kappa_2)$ and $s=n_1+n_2-6$; and $g_{\mathsf S}(z,w)=\theta^r(z)\,\theta^r(w)\,L_3(w-z)\,L_4(u-z)\,L_5(u-w)$.
- $\mathsf S=\{1,5\}$, $D=6$, $\zeta=(z,w')$: $\mathsf T^{\mathsf S}(z,w')=K_1(z)K_5(w')$, with symbol $\sigma_1(\kappa_1)\sigma_5(\kappa_2)$ and $s=n_1+n_5-6$; and $g_{\mathsf S}(z,w')=\theta^r(z)\,\theta^r(w')\,L_2(u-w')\,L_3(u-w'-z)\,L_4(u-z)$.
- $\mathsf S=\{1,3\}$, $D=6$, $\zeta=(z,y)$: $\mathsf T^{\mathsf S}(z,y)=K_1(z)K_3(y)$, with symbol $\sigma_1(\kappa_1)\sigma_3(\kappa_2)$ and $s=n_1+n_3-6$; and $g_{\mathsf S}(z,y)=\theta^r(z)\,\theta^r(y)\,L_2(z+y)\,L_4(u-z)\,L_5(u-z-y)$.
- $\mathsf S=\{1,2,3\}$, $D=6$, $\zeta=(z,w)$: $\mathsf T^{\mathsf S}(z,w)=K_1(z)K_2(w)K_3(w-z)$, with symbol $\hat{\mathsf T}_{\mathsf p}(\kappa)$ (by eq. 7.xxvii, applied to $g=\delta_\zeta$) and $s=\omega_{\mathsf p}$; and $g_{\mathsf S}(z,w)=\theta^r(z)\,\theta^r(w)\,\theta^r(w-z)\,L_4(u-z)\,L_5(u-w)$.

In the cases $\{1,5\}$ and $\{1,3\}$ we have substituted $w=u-w'$, resp. $w=z+y$, in the sum over w; in the cases $\{1\}$ and $\{3\}$ the inner sums converge absolutely by Step 1. Each $g_{\mathsf S}$ vanishes off $\{|\zeta|\le r\}$, since the short factors confine every component of ζ to $\{|\cdot|<\varepsilon r\}$ and $\sqrt2\,\varepsilon<1$.

Let $g^h_{\mathsf S}$ be the function on $\mathbb R^D$ obtained from $g_{\mathsf S}$ by replacing every L_i by L_i^h and, in the cases $\{1\}$ and $\{3\}$, the inner sum by the integral over $\mathbb R^3$. Since $\theta^r(y)=\theta^1(y/r)$ and $L_i^h(y)=r^{-n_i}(\bar\theta^1K_i^h)(y/r)$, the substitution of r times the variable in the inner integrals gives

$$g^h_{\mathsf S}(\zeta)=r^{-a}\,G_{\mathsf S}(\zeta/r),\qquad\text{where}\quad a:=\begin{cases}\sum_{i\notin\mathsf S}n_i & \text{if } D=6,\\ \sum_{i\notin\mathsf S}n_i-3 & \text{if } \mathsf S=\{1\},\{3\},\end{cases}$$

so that $a+s=\Delta$ in all six cases. Here $G_{\mathsf S}$ is the function $g^h_{\mathsf S}$ computed with $r=1$ and with $\hat u:=u/r$ in place of u; it belongs to $C^\infty_c(\mathbb R^D)$ and has its support in $\{|\zeta|\le 1\}$. We claim

$$\|g_{\mathsf S}-g^h_{\mathsf S}\|_{(r)}\le C\,r^{-a-1}\qquad(r\ge r_0),\tag{7.xxix}$$

where $g^h_{\mathsf S}$ is restricted to $\mathbb Z^D$ and $\|\cdot\|_{(r)}$ is the norm of Def. 7.16(5).

In the cases $\{1,2\}$, $\{1,5\}$, $\{1,3\}$, $\{1,2,3\}$ the functions $g_{\mathsf S}$ and $g^h_{\mathsf S}$ are products of the same short factors with the long factors L_i, resp. L_i^h, $i\notin\mathsf S$, evaluated at linear functions of ζ. We write the difference as a sum of terms in each of which one factor L_i has been replaced by $L_i-L_i^h$, and we distribute the differences ∇^γ, $|\gamma|\le 10$, by the Leibniz rule. By eq. 7.xxviii a difference falling on a long factor gains a factor $C(r+|y_i|)^{-1}\le Cr^{-1}$, a difference falling on a short factor gains Cr^{-1}, and the factor $L_i-L_i^h$ gains $(r+|y_i|)^{-1}\le r^{-1}$ in comparison with L_i; moreover $\prod_{i\notin\mathsf S}(r+|y_i|)^{-n_i}\le r^{-a}$. This gives $r^{|\gamma|}|\nabla^\gamma(g_{\mathsf S}-g^h_{\mathsf S})|\le Cr^{-a-1}$.

In the case $\mathsf{S}=\{1\}$ let

$$\mathsf{w}(z) := \sum_w L_2(w)\, L_3(w-z)\, L_5(u-w),$$
$$\mathsf{w}^h(z) := \int_{\mathbb{R}^3} L_2^h(w)\, L_3^h(w-z)\, L_5^h(u-w)\,\mathrm{d}w.$$

For $|z|\le r$ the differences in z fall on $L_3(w-z)$ only. By eq. 7.xxviii, by $r+|u-w|\ge\frac12(r+|w|)$ (second assertion of Lem. 7.18(a), as $|u|=r$) and by the first assertion of Lem. 7.18(a) with $D=3$, $R=r$ and exponent $n_2+n_5\ge 4>3$, we obtain

$$|\nabla^\gamma \mathsf{w}(z)| \le Cr^{-n_3-|\gamma|}\sum_w (r+|w|)^{-n_2-n_5} \le Cr^{3-n_2-n_3-n_5-|\gamma|}.$$

Each replacement of an L_i by L_i^h in w costs a factor Cr^{-1}, by the second inequality of eq. 7.xxviii. The replacement of the sum over w by the integral costs a factor Cr^{-1} as well: we apply Lem. 7.18(b), with $D=3$, to $F(w):=L_2^h(w)\,(\nabla_z^\gamma L_3^h)(w-z)\,L_5^h(u-w)$, whose gradient is bounded by $Cr^{-n_3-|\gamma|-1}(r+|w|)^{-n_2-n_5}$ on each cube $w+[0,1]^3$. Since $g_{\{1\}}(z)=\theta^r(z)L_4(u-z)\mathsf{w}(z)$, since $|\nabla^\gamma L_4(u-\cdot)|\le Cr^{-n_4-|\gamma|}$ on $\{|z|\le r\}$ by eq. 7.xxviii, and since the differences of order γ of θ^r are bounded by $Cr^{-|\gamma|}$, the Leibniz rule gives eq. 7.xxix with $a=n_2+n_3+n_4+n_5-3$.

In the case $\mathsf{S}=\{3\}$ let $\mathsf{w}(y):=\sum_z L_1(z)\,L_2(z+y)\,L_4(u-z)\,L_5(u-z-y)$, and let w^h be the integral over $\mathbb{R}^3$ of the corresponding product of the L_i^h. For $|y|\le r$ the differences in y fall on L_2 and on L_5 (and, in $g_{\{3\}}$, on the factor $\theta^r(y)$), each difference gaining a factor Cr^{-1}. By Lem. 7.18(a) we have $r+|z+y|\ge\frac12(r+|z|)$, $r+|u-z|\ge\frac12(r+|z|)$, and, with $R=2r\ge|u-y|$, also $r+|u-y-z|\ge\frac14(r+|z|)$; the first assertion of that lemma, with exponent $n_1+n_2+n_4+n_5\ge 8$, gives $\sum_z(r+|z|)^{-n_1-n_2-n_4-n_5}\le Cr^{3-n_1-n_2-n_4-n_5}$. Hence $|\nabla^\gamma\mathsf{w}(y)|\le Cr^{3-n_1-n_2-n_4-n_5-|\gamma|}$, and, as in the case $\{1\}$, the replacements $L_i\to L_i^h$ and the replacement of the sum by the integral (Lem. 7.18(b)) cost a factor Cr^{-1} each. This proves eq. 7.xxix in all six cases.

Step 7: symbols, and the pairing with slowly varying functions. We verify the hypotheses of Lem. 7.19(c) for $\hat{\mathsf{T}}$, the symbol of T^{S}, and $\hat{\mathsf{T}}^h:=\hat{\mathsf{T}}^{\mathsf{S},h}$. In the cases $\{1\}$ and $\{3\}$ ($D=3$, $i=1$, resp. $i=3$, $s=s_i$) we take $c_0=1$. On $\{|\kappa|\le 1\}$ we have $\chi=1$, so $\sigma_i-\chi\sigma_i^h\in\mathcal{S}^{s+1}$ gives $|\sigma_i-\sigma_i^h|\le C|\kappa|^{s+1}$ there; the majorants are $\mathsf{H}:=C|\kappa|^s\in\mathcal{H}_s(3)$ for σ_i^h and $\mathsf{H}':=C|\kappa|^{s+1}\in\mathcal{H}_{s+1}(3)$, which belong to these classes since $s>-3$; the required bound $|\sigma_i-\sigma_i^h|\le\mathsf{H}'\lg|\kappa|$ holds on $\{|\kappa|\le1\}$, as $\lg|\kappa|\ge1$. In the cases $\{1,2\}$, $\{1,5\}$, $\{1,3\}$ ($D=6$) the symbol is $\sigma_i(\kappa_1)\sigma_j(\kappa_2)$ with $(i,j)=(1,2),(1,5),(1,3)$, its leading part is $\sigma_i^h(\kappa_1)\sigma_j^h(\kappa_2)$, and $s=s_i+s_j$ with $s_i,s_j>-3$. Then $|\sigma_i^h(\kappa_1)\sigma_j^h(\kappa_2)|\le C|\kappa_1|^{s_i}|\kappa_2|^{s_j}$, a function in $\mathcal{H}_{s_i+s_j}(6)$, and on $\{|\kappa|\le1\}$

$$\big|\sigma_i(\kappa_1)\sigma_j(\kappa_2)-\sigma_i^h(\kappa_1)\sigma_j^h(\kappa_2)\big| \le C\big(|\kappa_1|^{s_i+1}|\kappa_2|^{s_j}+|\kappa_1|^{s_i}|\kappa_2|^{s_j+1}\big),$$

a function in $\mathcal{H}_{s_i+s_j+1}(6)$; here we wrote the difference as $(\sigma_i-\sigma_i^h)\sigma_j+\sigma_i^h(\sigma_j-\sigma_j^h)$ and used $|\sigma_j(\kappa_2)|\le C|\kappa_2|^{s_j}$. In the case $\{1,2,3\}$ we use Lem. 7.20, whose hypotheses hold by Step 2, with the number c_0 given there. By part (b) of that lemma, $|\hat{\mathsf{T}}_{\mathsf{p}}-\mathsf{Q}_{\mathsf{p}}-\mathsf{I}_{\mathsf{p}}^h|\le C|\kappa|^{\omega_{\mathsf{p}}+1}\lg|\kappa|\,\Lambda(\kappa)$ for almost every κ with $|\kappa|\le c_0$, where $\mathsf{I}_{\mathsf{p}}^h$ is the homogeneous part of $\hat{\mathsf{T}}_{\mathsf{p}}$. The polynomial Q_{p} has degree at most $\omega_{\mathsf{p}}\le1$, so $\mathsf{Q}_{\mathsf{p}}-\tilde{\mathsf{Q}}_{\mathsf{p}}=\sum_j\partial_j\mathsf{Q}_{\mathsf{p}}(0)(\kappa_j-\sin\kappa_j)=O(|\kappa|^3)$, and $|\kappa|^3\le|\kappa|^{\omega_{\mathsf{p}}+1}\lg|\kappa|\,\Lambda(\kappa)$ for $|\kappa|\le1$.

We therefore take $\hat{\mathsf{T}}:=\hat{\mathsf{T}}_{\mathsf{p}}-\tilde{\mathsf{Q}}_{\mathsf{p}}$, which is integrable on $\mathbb{T}^2$: Lem. 7.20 bounds the L^1 norm of $\hat{\mathsf{T}}_{\mathsf{p}}$ by the product of the L^1 norms of $\sigma_1,\sigma_2,\sigma_3$, and $\tilde{\mathsf{Q}}_{\mathsf{p}}$ is a trigonometric polynomial. We further take $\hat{\mathsf{T}}^h:=\mathsf{I}_{\mathsf{p}}^h$ and the majorants $\mathsf{H}:=C|\kappa|^{\omega_{\mathsf{p}}}\Lambda$ (Lem. 7.20(a)) and $\mathsf{H}':=C|\kappa|^{\omega_{\mathsf{p}}+1}\Lambda$. They belong to $\mathcal{H}_{\omega_{\mathsf{p}}}(6)$ and to $\mathcal{H}_{\omega_{\mathsf{p}}+1}(6)$, respectively, since the same lemma asserts that $|\kappa|^s\Lambda(\kappa)$ belongs to $\mathcal{H}_s(6)$ for every real $s>-6$, and since $\omega_{\mathsf{p}}\ge-1$. In all cases $-D<s<8-D$, as Lem. 7.19(c) requires: for $D=3$ we have $s\in\{-2,-1,0\}\subset(-3,2)$, and for $D=6$ we have $s\in\{-3,\dots,1\}\subset(-6,2)$.

Inserting the Fourier integrals of the short lines, we have $\mathsf{T}_\mathsf{S}(u) = \int_{\mathbb{T}^{D/3}} \hat{\mathsf{T}}^\mathsf{S}(\kappa)\, \check{g}_\mathsf{S}(-\kappa)\, d^\circ\kappa$, where $\hat{\mathsf{T}}^\mathsf{S}$ is the symbol of T^S: for the product kernels this follows from Fubini's theorem as in Step 3, and for $\{1,2,3\}$ it is eq. 7.xxvii. We apply Lem. 7.19(c) with $R = r$, $g = g_\mathsf{S}$, which vanishes off $\{|\zeta| \le r\}$, $G = G_\mathsf{S}$, the number a of Step 6, and $A = C$, which is admissible by eq. 7.xxix because $\lg r \ge 1$. The norm $\|G_\mathsf{S}\|_{C^{10}}$ is bounded uniformly in $\hat{u} \in S^2$, by the smoothness in $(\zeta, \hat{u})$ established in Step 9 and the compactness of S^2. In the first five cases, $\{1\}$, $\{3\}$, $\{1,2\}$, $\{1,5\}$, $\{1,3\}$, we obtain

$$\mathsf{T}_\mathsf{S}(u) = \mathsf{U}_\mathsf{S}(u) + O(r^{-\Delta-1} \lg r), \qquad \mathsf{U}_\mathsf{S}(u) := r^{-\Delta} \int_{\mathbb{R}^D} \hat{\mathsf{T}}^{\mathsf{S},h}(\kappa)\, \hat{G}_\mathsf{S}(-\kappa)\, d^\circ\kappa,$$

because $a + s = \Delta$. In the case $\{1,2,3\}$, where $\hat{\mathsf{T}}^{\mathsf{S},h} := \mathsf{I}^h_\mathsf{p}$, the lemma is applied to $\hat{\mathsf{T}}_\mathsf{p} - \tilde{\mathsf{Q}}_\mathsf{p}$, and the remaining term is $\int_{\mathbb{T}^2} \tilde{\mathsf{Q}}_\mathsf{p}(\kappa) \check{g}_\mathsf{S}(-\kappa)\, d^\circ\kappa = \sum_{z,w} \mathsf{q}_\mathsf{p}(z,w)\, g_\mathsf{S}(z,w)$; thus

$$\mathsf{T}_{\{1,2,3\}}(u) = \sum_{z,w} \mathsf{q}_\mathsf{p}(z,w)\, g_{\{1,2,3\}}(z,w) + \mathsf{U}_{\{1,2,3\}}(u) + O(r^{-\Delta-1} \lg r).$$

On the support of q_p, that is, for $(z,w) \in \{0, \pm e_1, \dots, \pm e_6\}$, we have $|z|, |w|, |w-z| \le 1 \le \frac{1}{2}\varepsilon r$ and $|u-z|, |u-w| \ge r - 1 \ge \varepsilon r$ as soon as $r \ge r_0 \ge 16$; hence $g_{\{1,2,3\}}(z,w) = K_4(u-z)K_5(u-w)$ there, which produces the first local term in (iii). The image of $\{1,2,3\}$ under $(z,w) \to (u-z, u-w)$ is $\{3,4,5\}$, and the transformed family $(K_4, K_5, K_3(-\cdot), K_1, K_2)$ has, in place of $\hat{\mathsf{T}}_\mathsf{p}$, the function $\int_\mathbb{T} \sigma_4(\kappa_1 + l)\sigma_5(\kappa_2 - l)\sigma_3(-l)\, d^\circ l = \hat{\mathsf{T}}_\mathsf{q}(\kappa)$; its local kernel is therefore q_q, and $\mathsf{T}_{\{3,4,5\}}$ produces the second local term in (iii).

Step 8: the case $\mathsf{S} = \emptyset$. Here $\mathsf{T}_\emptyset(u) = \sum_{z,w} \prod_{i=1}^5 L_i(y_i)$. By eq. 7.xxviii with $\gamma = 0$, by $(r + |y_3|)^{-n_3} \le r^{-n_3}$ and by Lem. 7.18(a) (as in Step 6, with $r + |u-z| \ge \frac{1}{2}(r + |z|)$ and $r + |u-w| \ge \frac{1}{2}(r + |w|)$),

$$\sum_{z,w} \prod_{i=1}^5 (r + |y_i|)^{-n_i} \le C\, r^{-n_3}\, r^{3-n_1-n_4}\, r^{3-n_2-n_5} = C\, r^{-\Delta}.$$

Replacing each L_i by L_i^h costs a factor Cr^{-1}, by the second inequality of eq. 7.xxviii; replacing the sum over $(z,w) \in \mathbb{Z}^6$ by the integral over $\mathbb{R}^6$ costs a factor Cr^{-1} by Lem. 7.18(b) with $D = 6$, applied to $F(z,w) := \prod_i L_i^h(y_i)$, whose gradient is bounded by $Cr^{-1} \prod_i (r + |y_i|)^{-n_i}$ on each cube of side 1. Hence $\mathsf{T}_\emptyset(u) = \mathsf{U}_\emptyset(u) + O(r^{-\Delta-1})$ with

$$\mathsf{U}_\emptyset(u) := \iint_{\mathbb{R}^3 \times \mathbb{R}^3} \prod_{i=1}^5 L_i^h(y_i)\, \mathrm{d}z\, \mathrm{d}w,$$

which, after the substitution $(z,w) = r(z',w')$, is seen to be homogeneous of degree $6 - \sum_i n_i = -\Delta$.

Step 9: the function U. Let U be the sum of the sixteen functions U_S, where for a set S which is not a representative, U_S is the function of its representative computed for the transformed family (Step 4). By Steps 4, 7 and 8, the inequality in (iii) holds for $r \ge r_0$. Every U_S is homogeneous of degree $-\Delta$, and so is U. We let a vector $v \in \mathbb{R}^3$ take the place of $\hat{u}$ in the definition of G_S, and we write $G_\mathsf{S}(\zeta; v)$. In the factors $\bar{\theta}^1 K_i^h$ the argument of K_i^h has modulus at least $\frac{1}{2}\varepsilon$ wherever $\bar{\theta}^1$ does not vanish, so these factors are smooth on $\mathbb{R}^3$; the inner integrals in the cases $\{1\}$ and $\{3\}$ may be differentiated under the integral sign, with integrable majorants that are uniform for (ζ, v) in compact sets.

Hence $(\zeta, v) \mapsto G_\mathsf{S}(\zeta; v)$ is smooth, with support in $\{|\zeta| \le 1\}$ for every v, and $v \mapsto G_\mathsf{S}(\cdot\,; v)$ is of class C^1 with respect to the norm $\|\cdot\|_{C^{10}}$. The last bound of Lem. 7.19(c), $|\int \hat{\mathsf{T}}^h \hat{G}(-\kappa)\, d^\circ\kappa| \le C\|G\|_{C^{10}}$, which is linear in G, then shows that $\Phi_\mathsf{S}(v) := \int \hat{\mathsf{T}}^{\mathsf{S},h}(\kappa)\, \hat{G}_\mathsf{S}(-\kappa; v)\, d^\circ\kappa$ is of class C^1 in v. Since $\mathsf{U}_\mathsf{S}(u) = |u|^{-\Delta}\, \Phi_\mathsf{S}(u/|u|)$, the function U_S is of class C^1 on $\mathbb{R}^3 \setminus \{0\}$.

For $\mathsf{U}_\emptyset$ we differentiate under the integral sign.

The functions U_S are built from the leading parts σ_i^h of the short lines, from the functions $K_i^h = \mathcal{F}_{\neq 0}^{-1}\sigma_i^h$ of the long lines, which depend linearly on σ_i^h (Lem. 7.17(b)), from the homogeneous parts I_p^h and I_q^h, which depend only on the leading parts and multilinearly on them (Lem. 7.20(a)), and from radial cut-offs; every line enters each U_S exactly once. Hence U depends only on $\sigma_1^h, \dots, \sigma_5^h$ and is linear in each of them. If every σ_i^h is replaced by $\sigma_i^h \circ O^{-1}$, then K_i^h is replaced by $K_i^h \circ O^{-1}$ (Lem. 7.17(b)), and I_p^h, I_q^h are replaced by their compositions with O^{-1} acting on both arguments, since the linear maps λ_i of Step 2 commute with this action (Lem. 7.20(a)).

The substitutions $\zeta \to O\zeta$ in the integrals defining G_S and $\mathsf{U}_\emptyset$, and $\kappa \to O\kappa$ in the integral defining U_S, together with the invariance of the radial cut-offs, show that U is replaced by $\mathsf{U} \circ O^{-1}$. This proves (iii) and (iv). □

7.1.10. *Classes of pairings; convergence.*

Lemma 7.24 (The lines of the lattice)**.** Let $\tau_{\mathrm{gf}} > 0$; we work on $\mathbb{Z}^3$. For $\iota \in \{1,2,3\}$ write $p_{[\iota]} = p_{\alpha\beta}$ with $\alpha < \beta$, and put

$$\mathsf{c}_{\iota,\nu}(k) := \mathsf{k}_\alpha(k)\,\delta_{\nu\beta} - \mathsf{k}_\beta(k)\,\delta_{\nu\alpha} \qquad (\nu \in \{1,2,3\},\ k \in \mathbb{R}^3),$$

so that for $v \in \mathbb{C}^3$ the form $(x,\nu) \mapsto e^{ik\cdot x}v_\nu$ satisfies $\mathrm{d}(e^{ik\cdot x}v_\nu)(p_{[\iota]}(x)) = e^{ik\cdot x}\sum_\nu \mathsf{c}_{\iota,\nu}(k)\,v_\nu$. Let $\hat{\mathsf{C}}_{\mu\nu}(k)$ be the bracket in the formula for $\mathsf{C}^{\mathrm{lat}}_{\tau_{\mathrm{gf}}}$ in Def. 7.7(1),

$$\hat{\mathsf{C}}_{\mu\nu}(k) = \frac{\delta_{\mu\nu}}{|\mathsf{k}|^2} + (\tau_{\mathrm{gf}} - 1)\,\frac{\mathsf{k}_\mu \overline{\mathsf{k}_\nu}}{|\mathsf{k}|^4}.$$

Then for every $x \in \mathbb{Z}^3$, all directions μ,ν and all ι,ι', with $p_{[\iota']} = p_{\alpha'\beta'}$, $\alpha' < \beta'$, the following functions of $y \in \mathbb{Z}^3$ are elementary lines, with the symbols, the degrees n and the leading parts of the symbols listed:

- $y \mapsto \mathsf{C}\big((x,\mu),(x+y,\nu)\big)$: symbol $\hat{\mathsf{C}}_{\nu\mu}(k)$; $n=1$; leading part $\delta_{\mu\nu}|k|^{-2} + (\tau_{\mathrm{gf}}-1)\,k_\mu k_\nu |k|^{-4}$;
- $y \mapsto \mathsf{E}_{p_{[\iota]}(x)}\big((x+y,\nu)\big)$: symbol $\sum_\mu \overline{\mathsf{c}_{\iota,\mu}(k)}\,\hat{\mathsf{C}}_{\nu\mu}(k) = \overline{\mathsf{c}_{\iota,\nu}(k)}\,|\mathsf{k}|^{-2}$; $n=2$; leading part $-i(k_\alpha\delta_{\nu\beta} - k_\beta\delta_{\nu\alpha})|k|^{-2}$;
- $y \mapsto \Pi\big(p_{[\iota]}(x), p_{[\iota']}(x+y)\big)$: symbol $\sum_\nu \overline{\mathsf{c}_{\iota,\nu}(k)}\,\mathsf{c}_{\iota',\nu}(k)\,|\mathsf{k}|^{-2}$; $n=3$; leading part $\sum_\nu (k_\alpha\delta_{\nu\beta} - k_\beta\delta_{\nu\alpha})(k_{\alpha'}\delta_{\nu\beta'} - k_{\beta'}\delta_{\nu\alpha'})|k|^{-2}$;
- $y \mapsto \mathsf{D}\big((x,\mu);(x+y,\nu)\big)$: symbol $\frac{1}{2}(1+e^{ik_\nu})\,\overline{\mathsf{k}_\mu(k)}\,|\mathsf{k}|^{-2}$; $n=2$; leading part $-ik_\mu|k|^{-2}$.

If $K(y)$ is an elementary line with symbol $\sigma(k)$ and $y_0 \in \mathbb{Z}^3$, then $K(-y)$ and $K(y+y_0)$ are elementary lines of the same degree, with the symbols $\sigma(-k)$ and $e^{ik\cdot y_0}\sigma(k)$. Consequently, the contraction eq. 7.xiii of a leg at a bond or a plaquette with base point $x+\xi$ with a leg at a bond or a plaquette with base point $x+y+\xi'$ equals $\delta_{aa'}$ times an elementary line in y of degree $1 + \#\{\text{curl legs among the two}\}$. ■

Proof. The matrix $\hat{\mathsf{C}}(k)$ is Hermitian, and $\hat{\mathsf{C}}(-k) = \overline{\hat{\mathsf{C}}(k)}$ because $\mathsf{k}_\mu(-k) = \overline{\mathsf{k}_\mu(k)}$. By Def. 7.7(1) and the substitution $k \to -k$,

$$\begin{aligned}\mathsf{C}\big((x,\mu),(x+y,\nu)\big) &= \int_{\mathbb{T}} e^{-ik\cdot y}\,\hat{\mathsf{C}}_{\mu\nu}(k)\,d^\circ k \\ &= \int_{\mathbb{T}} e^{ik\cdot y}\,\overline{\hat{\mathsf{C}}_{\mu\nu}(k)}\,d^\circ k = \int_{\mathbb{T}} e^{ik\cdot y}\,\hat{\mathsf{C}}_{\nu\mu}(k)\,d^\circ k.\end{aligned}$$

The plaquette $p = p_{[\iota]}(x) = p_{\alpha\beta}(x)$ has the bonds $(x,\alpha),(x+e_\alpha,\beta),(x+e_\beta,\alpha),(x,\beta)$, with the signs $+,+,-,-$. By the formula just proved, $\mathsf{C}\big((x+\xi,\mu'),(x+y,\nu)\big) = \int e^{ik\cdot(y-\xi)}\hat{\mathsf{C}}_{\nu\mu'}(k)\,d^\circ k$, and

therefore

$$\begin{aligned}\mathsf{E}_p((x+y,\nu)) &= \sum_i s_i\,\mathsf{C}(b_i(p),(x+y,\nu))\\ &= \int_{\mathbb{T}} e^{ik\cdot y}\Big[\hat{\mathsf{C}}_{\nu\alpha}(k)(1-e^{-ik_\beta})+\hat{\mathsf{C}}_{\nu\beta}(k)(e^{-ik_\alpha}-1)\Big]\,d^\circ k.\end{aligned}$$

As $\overline{\mathsf{k}_\mu(k)} = e^{-ik_\mu}-1$, the bracket equals $\sum_\mu \overline{\mathsf{c}_{\iota,\mu}(k)}\,\hat{\mathsf{C}}_{\nu\mu}(k)$. In this sum the part of $\hat{\mathsf{C}}_{\nu\mu}$ containing $\mathsf{k}_\nu\overline{\mathsf{k}_\mu}$ drops out, since $\sum_\mu \overline{\mathsf{c}_{\iota,\mu}}\,\mathsf{k}_\nu\overline{\mathsf{k}_\mu} = \mathsf{k}_\nu\,\overline{\sum_\mu \mathsf{c}_{\iota,\mu}\mathsf{k}_\mu}$ and $\sum_\mu \mathsf{c}_{\iota,\mu}\mathsf{k}_\mu = \mathsf{k}_\alpha\mathsf{k}_\beta - \mathsf{k}_\beta\mathsf{k}_\alpha = 0$; what remains is $\overline{\mathsf{c}_{\iota,\nu}}\,|\mathsf{k}|^{-2}$. The kernel Π arises from E_p by the operation d in the second argument (Def. 7.7(2)).

With $q = p_{[\iota']}(x+y) = p_{\alpha'\beta'}(x+y)$, the four bonds $(x+y,\alpha')$, $(x+y+e_{\alpha'},\beta')$, $(x+y+e_{\beta'},\alpha')$, $(x+y,\beta')$ of q contribute, with the signs $+,+,-,-$, the symbols $\overline{\mathsf{c}_{\iota,\alpha'}}|\mathsf{k}|^{-2}$, $e^{ik_{\alpha'}}\overline{\mathsf{c}_{\iota,\beta'}}|\mathsf{k}|^{-2}$, $e^{ik_{\beta'}}\overline{\mathsf{c}_{\iota,\alpha'}}|\mathsf{k}|^{-2}$, $\overline{\mathsf{c}_{\iota,\beta'}}|\mathsf{k}|^{-2}$, and their sum is

$$(\mathsf{k}_{\alpha'}\,\overline{\mathsf{c}_{\iota,\beta'}} - \mathsf{k}_{\beta'}\,\overline{\mathsf{c}_{\iota,\alpha'}})|\mathsf{k}|^{-2} = \sum_\nu \overline{\mathsf{c}_{\iota,\nu}}\,\mathsf{c}_{\iota',\nu}\,|\mathsf{k}|^{-2};$$

that is, d in the second argument multiplies the symbol by $\mathsf{c}_{\iota',\nu}(k)$ and sums over ν. Finally, with $\mathsf{G}(x',x'') = \int e^{ik\cdot(x'-x'')}|\mathsf{k}|^{-2}\,d^\circ k$,

$$\begin{aligned}\mathsf{D}((x,\mu);(x+y,\nu)) &= \tfrac12 \sum_{x'\in\{x+y,\,x+y+e_\nu\}} [\mathsf{G}(x',x+e_\mu)-\mathsf{G}(x',x)]\\ &= \int_{\mathbb{T}} e^{ik\cdot y}\,\tfrac12(1+e^{ik_\nu})(e^{-ik_\mu}-1)\,|\mathsf{k}|^{-2}\,d^\circ k.\end{aligned}$$

All four symbols are sums of products of the examples of Def. 7.16(3), namely $e^{ik\cdot y}$ (degree 0, leading part 1), k_μ and $\overline{\mathsf{k}_\mu}$ (degree 1, leading parts ik_μ and $-ik_\mu$) and $|\mathsf{k}|^{-2}$ (degree -2, leading part $|k|^{-2}$), and no product of degree below -2 occurs: the term $\mathsf{k}_\nu\overline{\mathsf{k}_\mu}|\mathsf{k}|^{-4}$ is the product $(\mathsf{k}_\nu|\mathsf{k}|^{-2})(\overline{\mathsf{k}_\mu}|\mathsf{k}|^{-2})$ of two line symbols of degree -1. By the rules for sums and products in Def. 7.16(3), the four symbols are line symbols of the degrees $-2,-1,0,-1$, so the kernels are elementary lines of the degrees $1,2,3,2$; their leading parts arise by replacing k_μ, $\overline{\mathsf{k}_\mu}$, $|\mathsf{k}|^{-2}$, e^{ik_ν} by ik_μ, $-ik_\mu$, $|k|^{-2}$, 1, which gives the leading parts listed.

If K has the symbol σ, then $K(-y) = \int e^{-ik\cdot y}\sigma(k)\,d^\circ k = \int e^{ik\cdot y}\sigma(-k)\,d^\circ k$ and $K(y+y_0) = \int e^{ik\cdot y}e^{ik\cdot y_0}\sigma(k)\,d^\circ k$. The function $\sigma(-k)$ is a line symbol of the same degree, with leading part $\sigma^h(-k)$, as χ is even; and $e^{ik\cdot y_0}\sigma(k)$ is the product of σ with a line symbol of degree 0 and leading part 1. For the consequence we recall from eq. 7.xiii that the contraction of two plain legs is $\delta_{aa'}\mathsf{C}$, that of a curl leg with a plain leg is $\delta_{aa'}\mathsf{E}$, and that of two curl legs is $\delta_{aa'}\Pi$. These kernels are invariant under simultaneous translations of their arguments, by their Fourier representations. Hence, by the list above and by the two rules just proved (which take care of the order of the legs and of the shifts ξ,ξ'), each contraction is an elementary line in y of degree 1, 2, 3, respectively. □

Lemma 7.25 (Π^{lat} is the curvature projection Π_∞ up to signs)**.** For a plaquette $p = p_{\mu\nu}(x)$, $\mu<\nu$, of $\mathbb{Z}^3$, let $\epsilon_p = -1$ for $(\mu,\nu) = (1,3)$ and $\epsilon_p = 1$ otherwise. This converts the positive orientation to the cyclic orientation indexed by the dual direction. Write $p^\flat = (x,\lambda(p))$ in these cyclic coordinates, and let Π_∞ be the orthogonal projection onto the closed range of curl in ℓ^2 of cyclic plaquette cochains. Its Fourier formula is equation YM3_I:(2242). Then for all plaquettes p,q of $\mathbb{Z}^3$

$$\Pi^{\mathrm{lat}}(p,q) = \epsilon_p\,\epsilon_q\,\Pi_\infty(p^\flat,q^\flat), \qquad \text{and hence} \qquad \mathsf{B}_0^{\mathrm{lat}}(p,q) = \tfrac{\mathsf{n}}{2}\,\Pi_\infty(p^\flat,q^\flat)^2.$$

■

Proof. Step 1. Write $p = p_{[\iota]}(x)$ and $q = p_{[\iota']}(x+y)$, and let $\epsilon_\iota := -1$ for $\iota = 2$ and $\epsilon_\iota := 1$ for $\iota\in\{1,3\}$. Since $p_{[2]} = p_{13}$, we have $\epsilon_p = \epsilon_\iota$ and $\epsilon_q = \epsilon_{\iota'}$. By the entry for Π in Lem. 7.24,

$$\Pi^{\mathrm{lat}}(p,q) = \int_{\mathbb{T}} e^{ik\cdot y}\sum_\nu \overline{\mathsf{c}_{\iota,\nu}(k)}\,\mathsf{c}_{\iota',\nu}(k)\,|\mathsf{k}|^{-2}\,d^\circ k.$$

Step 2. We show that

$$\sum_\nu \overline{\mathsf{c}_{\iota,\nu}}\,\mathsf{c}_{\iota',\nu} = \epsilon_\iota\,\epsilon_{\iota'}\big(\delta_{\iota\iota'}\,|\mathsf{k}|^2 - \mathsf{k}_\iota\,\overline{\mathsf{k}_{\iota'}}\big).$$

For $\iota = \iota'$, with $p_{[\iota]} = p_{\alpha\beta}$, the left side is $|\mathsf{k}_\alpha|^2 + |\mathsf{k}_\beta|^2 = |\mathsf{k}|^2 - |\mathsf{k}_\iota|^2$, and $\epsilon_\iota^2 = 1$. Let $\iota \neq \iota'$, and let ι'' be the third direction. As $\mathsf{c}_{\iota,\nu} = 0$ for $\nu = \iota$ and $\mathsf{c}_{\iota',\nu} = 0$ for $\nu = \iota'$, only $\nu = \iota''$ contributes. By definition $\mathsf{c}_{\iota,\iota''} = \pm\mathsf{k}_{\iota'}$, with the sign $+$ if and only if $\iota' < \iota''$, and $\mathsf{c}_{\iota',\iota''} = \pm\mathsf{k}_\iota$, with the sign $+$ if and only if $\iota < \iota''$. So the left side is $\pm\overline{\mathsf{k}_{\iota'}}\,\mathsf{k}_\iota$, with the sign $-$ if and only if ι'' lies between ι and ι', that is, if and only if $\iota'' = 2$. On the right side the factor of $\mathsf{k}_\iota\overline{\mathsf{k}_{\iota'}}$ is $-\epsilon_\iota\epsilon_{\iota'}$, which equals -1 if and only if neither ι nor ι' is 2, that is, if and only if $\iota'' = 2$.

Step 3. By Fubini's theorem, $\int_{\mathbb{T}} e^{ik\cdot y}\,d^\circ k = \prod_\mu \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{ity_\mu}\,\mathrm{d}t = \delta_{y,0}$, because $\int_{-\pi}^{\pi} e^{itm}\,\mathrm{d}t = 0$ for integers $m \neq 0$. The substitution $k \to -k$, together with $\mathsf{k}_\mu(-k) = \overline{\mathsf{k}_\mu(k)}$, gives

$$\int_{\mathbb{T}} e^{ik\cdot y}\,\frac{\mathsf{k}_\iota\,\overline{\mathsf{k}_{\iota'}}}{|\mathsf{k}|^2}\,d^\circ k = \int_{[-\pi,\pi]^3} e^{-ik\cdot y}\,\frac{\overline{\mathsf{k}_\iota(k)}\,\mathsf{k}_{\iota'}(k)}{|\mathsf{k}(k)|^2}\,\frac{dk}{(2\pi)^3} = T_{\iota\iota'}(-y),$$

where $T_{\lambda\lambda'}(y) = \int_{[-\pi,\pi]^3} e^{ik\cdot y}\,\overline{\mathsf{q}_\lambda}\,\mathsf{q}_{\lambda'}\,|\mathsf{q}|^{-2}\,\frac{dk}{(2\pi)^3}$ and $\mathsf{q}_\mu(k) := \mathsf{k}_\mu(k) = e^{ik_\mu} - 1$. Together with Steps 1 and 2 we obtain

$$\Pi^{\mathrm{lat}}(p,q) = \epsilon_\iota\,\epsilon_{\iota'}\,[\delta_{\iota\iota'}\,\delta_{y,0} - T_{\iota\iota'}(-y)].$$

The Fourier formula in equation YM3_I:(2242) states that $\Pi_\infty((z,\lambda),(z',\lambda')) = \delta_{pq} - T_{\lambda\lambda'}(z-z')$ for the plaquettes $p = (z,\lambda)$, $q = (z',\lambda')$ of $\mathbb{Z}^3$. We apply it to $p^\flat = (x,\iota)$ and $q^\flat = (x+y,\iota')$, for which $z - z' = -y$ and $\delta_{p^\flat q^\flat} = \delta_{\iota\iota'}\delta_{y,0}$; this is the first formula. The second follows from eq. 7.xvii, as $\epsilon_p^2 = \epsilon_q^2 = 1$. □

Lemma 7.26 (The list of the pairings)**.** Let Ω and τ_{gf} be as in Def. 7.7, and let $\langle\cdot\rangle_0$ refer to the Gaussian family of Ω and τ_{gf}. For a homogeneous polynomial F of degree m in the variables $A^a(b)$ and for $\mathfrak{g}\otimes\mathbb{C}$-valued forms $X_1,\dots,X_m$ we put

$$\mathsf{F}[\![X_1,\dots,X_m]\!] := \partial_{s_1}\cdots\partial_{s_m}\,\mathsf{F}\Big(\sum_{i=1}^m s_i X_i\Big);$$

this expression is symmetric and multilinear in $X_1,\dots,X_m$, and $\mathsf{F}[\![A,\dots,A]\!] = m!\,\mathsf{F}(A)$. For an index $\mathsf{i} = (b,a)$, where b is a positively oriented bond of Ω (a free bond if $\Omega = \Lambda_n$) and $a \in \{1,2,3\}$, let δ_{i} be the form which equals e_a at b and 0 at all other positively oriented bonds. We put $\mathsf{C}(\mathsf{i},\mathsf{i}') := \delta_{aa'}\,\mathsf{C}(b,b')$ for $\mathsf{i}' = (b',a')$, and $\mathsf{E}^a_p := \mathsf{E}_p(\cdot)\,\mathsf{e}_a$.

(a) (Symmetry factors.) Let $\mathsf{F}_1,\dots,\mathsf{F}_r$ be homogeneous polynomials of degrees $m_1,\dots,m_r$. Then $\langle\mathsf{F}_1;\dots;\mathsf{F}_r\rangle_0 = \sum_{\mathsf{n}}\mathsf{val}(\mathsf{n})$. The sum extends over the symmetric matrices $\mathsf{n} = (n_{vv'})_{v,v'=1}^r$ of non-negative integers such that $2n_{vv} + \sum_{v'\neq v} n_{vv'} = m_v$ for every v and such that the graph on $\{1,\dots,r\}$ with $n_{vv'}$ edges between v and $v' \neq v$ is connected; and

$$\mathsf{val}(\mathsf{n}) = \Big[\prod_{v<v'} n_{vv'}!\prod_v 2^{n_{vv}}\,n_{vv}!\Big]^{-1}\sum\prod_{v=1}^r \mathsf{F}_v[\![\delta_{\mathsf{i}(v,1)},\dots,\delta_{\mathsf{i}(v,m_v)}]\!] \times \prod_{\text{lines }\{e,e'\}}\mathsf{C}(\mathsf{i}(e),\mathsf{i}(e')).$$

Here, for every such n, we fix one pairing of the slots $(v,1),\dots,(v,m_v)$, $v = 1,\dots,r$, with exactly $n_{vv'}$ pairs between the slots of v and those of v', and n_{vv} pairs among the slots of v; its pairs $\{e,e'\}$ are called lines, and the sum extends over all indices $\mathsf{i}(e)$ attached to the slots e. In (a), in (b) and in the proof of the lemma the symbol n stands for such a matrix; in the formula for T_{I} and in eq. 7.xxx below it is the number $\mathsf{n} = 3$ of the setting.

(b) (Colour.) For every plaquette P, every real kernel c on pairs of bonds, symmetric or not, and every form X,

$$\sum_{\mathsf{i},\mathsf{i}'} \mathsf{a}_P^{(3)}[\![\delta_{\mathsf{i}}, \delta_{\mathsf{i}'}, X]\!]\, \mathsf{c}(b,b')\, \delta_{aa'} = 0 \qquad (\mathsf{i} = (b,a),\ \mathsf{i}' = (b',a')).$$

Consequently $\mathsf{val}(\mathsf{n}) = 0$ if $n_{vv} \geq 1$ for some v with $\mathsf{F}_v = \mathsf{a}_P^{(3)}$, and also if $n_{vv'} = 2$ for some v, v' with $\mathsf{F}_v = \mathsf{a}_p^{(2)}$ and $\mathsf{F}_{v'} = \mathsf{a}_P^{(3)}$.

(c) (Classes.) Let Σ_{tot} and Γ_q be defined by the formulas of Def. 7.13(5), with the Gaussian family and the vertices of Ω. Then

$$\mathsf{B}_1^{\Omega}(p,q;\tau_{\mathrm{gf}}) = \mathsf{T}_{\mathrm{I}} + \mathsf{T}_1 + \mathsf{T}_4 + \mathsf{T}_{4'} + \mathsf{T}_5 + \mathsf{T}_6,$$

where, with P, P' denoting plaquettes and $\mathsf{i}, \mathsf{i}', \mathsf{j}, \mathsf{j}', \dots$ indices of Ω,

$$\mathsf{T}_{\mathrm{I}} := -\mathsf{n}\,\Pi(p,q) \sum_{b,b'} \mathsf{E}_p(b)\, \Sigma_{\mathrm{tot}}(b,b')\, \mathsf{E}_q(b'),$$

$$\mathsf{T}_1 := \tfrac{1}{2} \sum_{P,P'} \sum_{a,a'} \sum_{\mathsf{i},\mathsf{j}} \mathsf{a}_P^{(3)}[\![\mathsf{E}_p^a, \mathsf{E}_q^{a'}, \delta_{\mathsf{i}}]\!]\, \mathsf{C}(\mathsf{i},\mathsf{j})\, \mathsf{a}_{P'}^{(3)}[\![\mathsf{E}_p^a, \mathsf{E}_q^{a'}, \delta_{\mathsf{j}}]\!],$$

$$\mathsf{T}_4 := -\tfrac{1}{4} \sum_{P} \sum_{a,a'} \mathsf{a}_P^{(4)}[\![\mathsf{E}_p^a, \mathsf{E}_p^a, \mathsf{E}_q^{a'}, \mathsf{E}_q^{a'}]\!],$$

$$\mathsf{T}_{4'} := \tfrac{1}{4} \sum_{a} \sum_{\mathsf{i},\mathsf{j}} \mathsf{C}(\mathsf{i},\mathsf{j}) \{ \mathsf{a}_q^{(4)}[\![\mathsf{E}_p^a, \mathsf{E}_p^a, \delta_{\mathsf{i}}, \delta_{\mathsf{j}}]\!] + \mathsf{a}_p^{(4)}[\![\mathsf{E}_q^a, \mathsf{E}_q^a, \delta_{\mathsf{i}}, \delta_{\mathsf{j}}]\!] \},$$

$$\begin{aligned} \mathsf{T}_5 := -\tfrac{1}{2} \sum_{P} \sum_{a} \sum_{\mathsf{i},\mathsf{i}',\mathsf{j},\mathsf{j}'} \mathsf{C}(\mathsf{i},\mathsf{j})\, \mathsf{C}(\mathsf{i}',\mathsf{j}') \{ & \mathsf{a}_q^{(3)}[\![\mathsf{E}_p^a, \delta_{\mathsf{i}}, \delta_{\mathsf{i}'}]\!]\, \mathsf{a}_P^{(3)}[\![\mathsf{E}_p^a, \delta_{\mathsf{j}}, \delta_{\mathsf{j}'}]\!] \\ & + \mathsf{a}_p^{(3)}[\![\mathsf{E}_q^a, \delta_{\mathsf{i}}, \delta_{\mathsf{i}'}]\!]\, \mathsf{a}_P^{(3)}[\![\mathsf{E}_q^a, \delta_{\mathsf{j}}, \delta_{\mathsf{j}'}]\!] \}, \end{aligned}$$

$$\mathsf{T}_6 := \tfrac{1}{6} \sum \mathsf{a}_p^{(3)}[\![\delta_{\mathsf{i}_1}, \delta_{\mathsf{i}_2}, \delta_{\mathsf{i}_3}]\!]\, \mathsf{a}_q^{(3)}[\![\delta_{\mathsf{j}_1}, \delta_{\mathsf{j}_2}, \delta_{\mathsf{j}_3}]\!] \prod_{\varsigma=1}^{3} \mathsf{C}(\mathsf{i}_\varsigma, \mathsf{j}_\varsigma),$$

the last sum extending over all indices $\mathsf{i}_1, \mathsf{i}_2, \mathsf{i}_3, \mathsf{j}_1, \mathsf{j}_2, \mathsf{j}_3$. By the first half of $\mathsf{T}_{4'}$, resp. of T_5, we mean the term with the first summand in braces, by the second half the term with the second summand. On $\Omega = \mathbb{Z}^3$ the formulas hold provided the sums converge absolutely, which is shown in Lem. 7.27. Moreover, if the summands of Γ_q are taken in the order of Def. 7.13(5), the second and the third being grouped together, then

$$\begin{aligned} \tfrac{\mathsf{n}}{2} \sum_{b,b'} \Gamma_q(b,b')\, \mathsf{E}_p(b)\, \mathsf{E}_p(b') = {} & [\text{first half of } \mathsf{T}_{4'}] \\ & + [\text{first half of } \mathsf{T}_5] + \mathsf{T}_4 + \mathsf{T}_1 \end{aligned} \tag{7.xxx}$$

term by term, the four groups of summands of Γ_q producing the four terms on the right in this order; the same holds with p and q exchanged and with the second halves of $\mathsf{T}_{4'}$ and T_5.

(d) (Vertices.) For forms X, Y, Z we have $\mathsf{a}_P^{(3)}[\![X,Y,Z]\!] = \sum \mathsf{t}_P(X',Y',Z')$, the sum extending over the six orderings (X',Y',Z') of (X,Y,Z), where

$$\mathsf{t}_P(X,Y,Z) := -\Big\langle \mathrm{d}X(P), \sum_{i<j} s_i s_j\, Y(b_i) \times Z(b_j) \Big\rangle,$$

with $b_i = b_i(P)$ and with the complex bilinear extensions of $\langle \cdot, \cdot \rangle$ and of $\times$. The standard writing of $\mathsf{a}_P^{(4)}$ consists of monomials of three kinds: with four plain legs (these come from $\frac{1}{2}|\mathfrak{z}_1|^2$), with one curl leg and three plain legs, and with four curl legs.

■

Proof. Step 1: proof of (a). Writing $A_{\mathsf{i}} := A^a(b)$ for $\mathsf{i} = (b,a)$, we have $A = \sum_{\mathsf{i}} A_{\mathsf{i}}\,\delta_{\mathsf{i}}$, and multilinearity together with $\mathsf{F}_v(A) = \frac{1}{m_v!}\mathsf{F}_v[\![A,\dots,A]\!]$ gives the finite sum

$$\mathsf{F}_v = \frac{1}{m_v!}\sum_{\mathsf{i}_1,\dots,\mathsf{i}_{m_v}} \mathsf{F}_v[\![\delta_{\mathsf{i}_1},\dots,\delta_{\mathsf{i}_{m_v}}]\!]\,A_{\mathsf{i}_1}\cdots A_{\mathsf{i}_{m_v}}.$$

We use the multilinearity of cumulants and Lem. 7.9, applied to the monomials $A_{\mathsf{i}(v,1)}\cdots A_{\mathsf{i}(v,m_v)}$, with the slots $(v,1),\dots,(v,m_v)$ as legs and with the covariance $\langle A_{\mathsf{i}}A_{\mathsf{i}'}\rangle_0 = \mathsf{C}(\mathsf{i},\mathsf{i}')$ of Def. 7.8(1). We find that $\langle \mathsf{F}_1;\dots;\mathsf{F}_r\rangle_0$ is $\prod_v \frac{1}{m_v!}$ times the sum, over all indices and over all pairings of the slots with connected graph on $\{1,\dots,r\}$, of $\prod_v \mathsf{F}_v[\![\cdots]\!]\prod_{\text{pairs}}\mathsf{C}$. As the forms $\mathsf{F}_v[\![\cdot]\!]$ are symmetric, the contribution of a pairing, summed over the indices, depends only on its matrix $(n_{vv'})$; and the graph of a pairing is connected if and only if that of its matrix is.

It remains to count the pairings with a given matrix n. At each vertex v we decide which slots are joined to which $v' \neq v$ and which are paired among themselves: there are $m_v!/[\prod_{v'\neq v} n_{vv'}!\,(2n_{vv})!]$ possibilities. For each pair $v < v'$ we match the chosen slots of v with those of v': $n_{vv'}!$ possibilities. At each v we pair the $2n_{vv}$ slots reserved for this purpose: $(2n_{vv})!/(2^{n_{vv}}n_{vv}!)$ possibilities. Since $\prod_v\prod_{v'\neq v} n_{vv'}! = \prod_{v<v'}(n_{vv'}!)^2$, the number of pairings is

$$\prod_v m_v!\Big/\Big[\prod_{v<v'} n_{vv'}!\prod_v 2^{n_{vv}}\,n_{vv}!\Big],$$

and multiplication by $\prod_v \frac{1}{m_v!}$ gives the factor in $\mathsf{val}(\mathsf{n})$.

Step 2: proof of (d) and (b). By eq. 7.v, $\mathsf{a}_P^{(3)}(A) = \langle \mathrm{d}A(P), \mathfrak{z}_1(P)\rangle$ with $\mathfrak{z}_1 = -\sum_{i<j} s_i s_j\, A(b_i)\times A(b_j)$; thus $\mathsf{a}_P^{(3)}(A) = \mathsf{t}_P(A,A,A)$ with the trilinear form t_P, and

$$\partial_{s_1}\partial_{s_2}\partial_{s_3}\,\mathsf{t}_P\big(\textstyle\sum s_iX_i, \sum s_iX_i, \sum s_iX_i\big)$$

is the sum of t_P over the six orderings of X_1,X_2,X_3. The three kinds of monomials of $\mathsf{a}_P^{(4)}$ are those of the three terms $\frac12|\mathfrak{z}_1|^2$, $\langle \mathrm{d}A(P),\mathfrak{z}_2\rangle$, $-\frac{1}{24}|\mathrm{d}A(P)|^4$ of eq. 7.v, written as in Def. 7.8(3).

For (b) we insert (d): $\mathsf{a}_P^{(3)}[\![\delta_{(b,a)},\delta_{(b',a)},X]\!]$ is a sum of terms, each the product of a real number which depends on b, b' and on the ordering, but not on a, with one of $\langle \mathsf{e}_a, \mathsf{e}_a\times X'\rangle$, $\langle \mathsf{e}_a, X'\times \mathsf{e}_a\rangle$, $\langle X', \mathsf{e}_a\times \mathsf{e}_a\rangle$, where X' is a value of X or of $\mathrm{d}X$. Summed over a, each of these three expressions vanishes, since $Y\times X'$ is orthogonal to Y and $Y\times Y = 0$; multiplying by $\mathsf{c}(b,b')$ and summing over b,b' we obtain (b). A line between two slots of a vertex $\mathsf{a}_P^{(3)}$ produces exactly the sum in (b) with $\mathsf{c} = \mathsf{C}$.

By the rule for $\mathsf{a}_p^{(2)}$ in Step 3 below, two lines from $\mathsf{a}_p^{(2)} = \frac12\sum_a((\mathrm{d}A)^a(p))^2$ to the same vertex $\mathsf{a}_P^{(3)}$ produce the sum in (b) with $\mathsf{c}(b,b') = \mathsf{E}_p(b)\,\mathsf{E}_p(b')$. In both cases $\mathsf{val}(\mathsf{n}) = 0$.

Step 3: rules for (c). From $\mathsf{a}_p^{(2)} = \frac12\sum_a((\mathrm{d}A)^a(p))^2$ we obtain $\mathsf{a}_p^{(2)}[\![\delta_{\mathsf{i}},\delta_{\mathsf{i}'}]\!] = \delta_{aa'}\,\mathsf{s}_p(b)\,\mathsf{s}_p(b')$, where $\mathsf{s}_p(b_i(p)) = s_i$ and $\mathsf{s}_p(b) = 0$ for the bonds b which do not belong to p. Since $\sum_b \mathsf{s}_p(b)\,\mathsf{C}(b,b'') = \mathsf{E}_p(b'')$, a line from a slot of $\mathsf{a}_p^{(2)}$ to a slot of another vertex inserts E_p^a into that slot, the colour a being summed together with the other slot of $\mathsf{a}_p^{(2)}$. A line from $\mathsf{a}_p^{(2)}$ to $\mathsf{a}_q^{(2)}$ produces $\Pi(p,q)$ and identifies the colours of the two remaining slots. A vertex $\mathsf{a}^{(2)}$ carrying a line between its own two slots is not connected to the other vertices.

Step 4: enumeration for (c). We go through the seven cumulants of eq. 7.xviii and list the matrices n with connected graph whose value does not vanish by (b) or by Step 3; in particular there are no lines between two slots of a vertex $\mathsf{a}^{(2)}$ or $\mathsf{a}^{(3)}$. We index these matrices by names of the vertices: the vertices $\mathsf{a}_p^{(m)}$ and $\mathsf{a}_q^{(m)}$ are called p and q, the vertices $\mathsf{a}_P^{(m)}$ and $\mathsf{a}_{P'}^{(3)}$ coming from S_3 or from V_2 are called P and P', and a quadratic local term F of V_2, taken as a vertex, is called F. Thus n_{pq} is the number of lines between the slots of p and those of q, n_{qq} the number of lines joining two slots of q, and similarly for the other entries.

- $\langle \mathsf{a}_p^{(3)}; \mathsf{a}_q^{(3)} \rangle_0$: here $n_{pq} = 3$, with the factor $\frac{1}{3!} = \frac{1}{6}$; this is T_6.
- $\langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(4)} \rangle_0$: here $n_{pq} = 2$ and $n_{qq} = 1$, with the factor $\frac{1}{2\cdot 2}$; this is the first half of $\mathsf{T}_{4'}$. In the same way $\langle \mathsf{a}_p^{(4)}; \mathsf{a}_q^{(2)} \rangle_0$ gives the second half.
- $\langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(3)}; \mathsf{a}_P^{(3)} \rangle_0$, from the term $-\langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(3)}; \mathsf{S}_3 \rangle_0$: the conditions $n_{pq}+n_{pP} = 2$, $n_{pq}+n_{qP} = 3$, $n_{pP} + n_{qP} = 3$ force $n_{pq} = n_{pP} = 1$ and $n_{qP} = 2$, with the factor $\frac{1}{2}$; with the sign of eq. 7.xviii and the sum over P this is the first half of T_5. In the same way $-\langle \mathsf{a}_p^{(3)}; \mathsf{a}_q^{(2)}; \mathsf{S}_3 \rangle_0$ gives the second half.
- $\langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(2)}; \mathsf{F} \rangle_0$ with a quadratic local term F of V_2: here $n_{pq} = n_{p\mathsf{F}} = n_{q\mathsf{F}} = 1$, with the factor 1, and the value is $\Pi(p,q) \sum_a \mathsf{F}[\![\mathsf{E}_p^a, \mathsf{E}_q^a]\!] = \Pi(p,q) \sum_a \sum_{b,b'} \mathsf{E}_p(b)\, \mathsf{E}_q(b')\, \partial_{b,a}\partial_{b',a}\mathsf{F}$.
- $\langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(2)}; \mathsf{a}_P^{(4)} \rangle_0$: either $n_{pq} = n_{pP} = n_{qP} = n_{PP} = 1$, with the factor $\frac{1}{2}$ and the value

$$\begin{aligned}&\Pi(p,q) \sum_a \tfrac{1}{2} \sum_{\mathsf{i},\mathsf{j}} \mathsf{C}(\mathsf{i},\mathsf{j})\, \mathsf{a}_P^{(4)}[\![\mathsf{E}_p^a, \mathsf{E}_q^a, \delta_\mathsf{i}, \delta_\mathsf{j}]\!] \\ &\quad = \Pi(p,q) \sum_a \sum_{b,b'} \mathsf{E}_p(b)\, \mathsf{E}_q(b')\, \langle \partial_{b,a}\partial_{b',a} \mathsf{a}_P^{(4)} \rangle_0\end{aligned}$$

  (by (a) for the quadratic polynomial $\partial_{b,a}\partial_{b',a}\mathsf{a}_P^{(4)}$, whose polarisation is $\mathsf{a}_P^{(4)}[\![\delta_{(b,a)}, \delta_{(b',a)}, \cdot, \cdot]\!]$); or $n_{pP} = n_{qP} = 2$, with the factor $\frac{1}{4}$, which with the sign of eq. 7.xviii and the sum over P gives T_4. The matrix with $n_{pq} = 2$ has a disconnected graph.
- $\langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(2)}; \mathsf{a}_P^{(3)}; \mathsf{a}_{P'}^{(3)} \rangle_0$: subtracting the conditions at p and q from those at P and P' we get $n_{PP'} = 1 + n_{pq}$. For $n_{pq} = 2$ the graph is disconnected. For $n_{pq} = 1$ we get $n_{PP'} = 2$ and either $n_{pP} = n_{qP'} = 1$ or $n_{pP'} = n_{qP} = 1$, each with the factor $\frac{1}{2}$; summed over P, P' and multiplied by the $\frac{1}{2}$ of eq. 7.xviii, they give

$$\begin{aligned}&\Pi(p,q) \sum_a \tfrac{1}{2} \sum \mathsf{a}_P^{(3)}[\![\mathsf{E}_p^a, \delta_\mathsf{i}, \delta_{\mathsf{i}'}]\!]\, \mathsf{a}_{P'}^{(3)}[\![\mathsf{E}_q^a, \delta_\mathsf{j}, \delta_{\mathsf{j}'}]\!]\, \mathsf{C}(\mathsf{i},\mathsf{j})\, \mathsf{C}(\mathsf{i}',\mathsf{j}') \\ &\quad = \Pi(p,q) \sum_a \sum_{b,b'} \mathsf{E}_p(b)\, \mathsf{E}_q(b')\, \langle \partial_{b,a}\mathsf{S}_3; \partial_{b',a}\mathsf{S}_3 \rangle_0,\end{aligned}$$

  where the sum on the left extends over P, P' and all indices, and the equality is (a) for two quadratic polynomials. For $n_{pq} = 0$ we get $n_{PP'} = 1$ and $n_{pP} + n_{qP} = 2 = n_{pP'} + n_{qP'}$; the possibilities $(n_{pP}, n_{qP}) = (2,0)$ and $(0,2)$ vanish by (b), and there remains $n_{pP} = n_{qP} = n_{pP'} = n_{qP'} = 1$, with the factor 1; with the $\frac{1}{2}$ of eq. 7.xviii this is T_1.
- We collect the terms with $n_{pq} = 1$ found above, with the signs of eq. 7.xviii. By the definition of Σ_{tot} in Def. 7.13(5) they add up to $-\Pi(p,q) \sum_a \sum_{b,b'} \mathsf{E}_p(b)\, \mathsf{E}_q(b')\, \Sigma_{\mathrm{tot}}(b,a;b',a)$. The argument on the colour structure given there applies in the box as well and shows $\Sigma_{\mathrm{tot}}(b,a;b',a) = \Sigma_{\mathrm{tot}}(b,b')$ for every colour a; summing over the three colours we obtain T_I.

Step 5: proof of eq. 7.xxx. Let $\partial = \partial_{b,a}$, $\partial' = \partial_{b',a'}$, $\delta = \delta_{(b,a)}$, $\delta' = \delta_{(b',a')}$. By (a),

$$\begin{aligned}\langle \partial\partial' \mathsf{a}_q^{(4)} \rangle_0 &= \tfrac{1}{2} \sum_{\mathsf{i},\mathsf{j}} \mathsf{C}(\mathsf{i},\mathsf{j})\, \mathsf{a}_q^{(4)}[\![\delta, \delta', \delta_\mathsf{i}, \delta_\mathsf{j}]\!], \\ \langle \partial \mathsf{a}_q^{(3)}; \partial' \mathsf{S}_3 \rangle_0 &= \tfrac{1}{2} \sum_P \sum \mathsf{C}(\mathsf{i},\mathsf{j})\, \mathsf{C}(\mathsf{i}',\mathsf{j}')\, \mathsf{a}_q^{(3)}[\![\delta, \delta_\mathsf{i}, \delta_{\mathsf{i}'}]\!]\, \mathsf{a}_P^{(3)}[\![\delta', \delta_\mathsf{j}, \delta_{\mathsf{j}'}]\!], \\ \langle \mathsf{a}_q^{(2)}; \partial\partial' \mathsf{S}_4 \rangle_0 &= \tfrac{1}{2} \sum_P \sum_{a''} \mathsf{a}_P^{(4)}[\![\delta, \delta', \mathsf{E}_q^{a''}, \mathsf{E}_q^{a''}]\!], \\ \langle \mathsf{a}_q^{(2)}; \partial \mathsf{S}_3; \partial' \mathsf{S}_3 \rangle_0 &= \sum_{P,P'} \sum_{a''} \sum_{\mathsf{i},\mathsf{j}} \mathsf{C}(\mathsf{i},\mathsf{j})\, \mathsf{a}_P^{(3)}[\![\delta, \mathsf{E}_q^{a''}, \delta_\mathsf{i}]\!] \\ &\qquad \times \mathsf{a}_{P'}^{(3)}[\![\delta', \mathsf{E}_q^{a''}, \delta_\mathsf{j}]\!];\end{aligned}$$

in the last line the three polynomials are quadratic, the only connected matrix has one line between any two of them, and its factor is 1. We contract these expressions, and the one obtained from the second by exchanging (b,a) and (b',a'), with $\frac{1}{2}\sum_a \mathsf{E}_p^a \otimes \mathsf{E}_p^a$; that is, we set $a'=a$, multiply by $\frac{1}{2}\,\mathsf{E}_p(b)\,\mathsf{E}_p(b')$ and sum over b,b',a. Since $\Gamma_q(b,a;b',a)=\Gamma_q(b,b')$, this turns Γ_q into the left side of eq. 7.xxx. By multilinearity, and with the signs of Def. 7.13(5), the first summand of Γ_q produces the first half of $\mathsf{T}_{4'}$; the second and the third, whose contributions coincide after the symmetric contraction, together produce the first half of T_5; the fourth produces T_4; and the fifth produces T_1. Exchanging p and q gives the statement for the second halves. □

Lemma 7.27 (Convergence of the graph sums on $\mathbb{Z}^3$)**.** Let $\tau_{\mathrm{gf}}>0$.

(i) On $\mathbb{Z}^3$, for all plaquettes p,q, the sums over the positions of the local terms of S_3 and V_2 in eq. 7.xviii, taken monomial by monomial in the standard writing and pairing by pairing, converge absolutely. Consequently $\mathsf{B}_1^{\mathrm{lat}}(p,q;\tau_{\mathrm{gf}})$ is defined, the formulas of Lem. 7.26(c) hold on $\mathbb{Z}^3$, and $\sup_{p,q}|\mathsf{B}_1^{\mathrm{lat}}(p,q;\tau_{\mathrm{gf}})|<\infty$.

(ii) (Uniform form.) Let $\mathsf{c}_0\geq 1$, let $\Omega=\mathbb{Z}^3$ or $\Omega=\Lambda_n$, and substitute for $\mathsf{C},\mathsf{E},\Pi,\mathsf{D},\mathsf{g}$ arbitrary real kernels which satisfy, with the base points b_- of the bonds and x_p of the plaquettes,

$$|\mathsf{C}(b,b')|\leq\frac{\mathsf{c}_0}{\langle b_- - b'_-\rangle},\qquad |\mathsf{E}_p(b)|\leq\frac{\mathsf{c}_0}{\langle x_p-b_-\rangle^2},$$
$$|\mathsf{D}(b;b')|\leq\frac{\mathsf{c}_0}{\langle b_- - b'_-\rangle^2},\qquad |\Pi(p,q)|\leq\frac{\mathsf{c}_0}{\langle x_p-x_q\rangle^3},\qquad |\mathsf{g}(b)|\leq\mathsf{c}_0. \tag{7.xxxi}$$

Then the sum of the absolute values of all terms of the seven cumulants of eq. 7.xviii, over monomials, pairings and positions, is at most $C\,\mathsf{c}_0^5$, with a pure constant C.

■

Proof. Step 1. By Lem. 7.24, the kernels C, E, Π, D of $\mathbb{Z}^3$ are, as functions of the difference of the base points, elementary lines of the degrees $1,2,3,2$, with finitely many different symbols. Lem. 7.17(a), applied with $s=-2,-1,0,-1$ and $\gamma=0$, bounds them by constants times $\langle\cdot\rangle^{-1}$, $\langle\cdot\rangle^{-2}$, $\langle\cdot\rangle^{-3}$, $\langle\cdot\rangle^{-2}$, and $\mathsf{g}=\frac{1}{3}$ on $\mathbb{Z}^3$ by Def. 7.7(2). Hence these kernels satisfy eq. 7.xxxi with some $\mathsf{c}_0=\mathsf{c}_0(\tau_{\mathrm{gf}})$, and (i) follows from (ii): the bound $C\mathsf{c}_0^5$, which does not depend on p,q, is the absolute convergence required in Def. 7.8(3), so that $\mathsf{B}_1^{\mathrm{lat}}$ is defined and those rearrangements made in the proof of Lem. 7.26(c) are allowed whose terms are bounded by the majorants of Step 2 below, such as regroupings of whole terms of positions and regroupings which split one curl leg or, as in Step 5 of that proof and for $n_{pq}=n_{pP}=n_{qP}=n_{PP}=1$ in its Step 4, curl legs at P or P' joined to p or q, whose lines then have the degree 2 which these majorants have there.

For the kernels of $\mathbb{Z}^3$ we obtain the equality of the display for $n_{pq}=1$ in Step 4 of that proof, whose right side splits the curl legs at P and at P' joined to p and to q, as in $\Pi(p,P)=\sum_b \mathsf{s}_P(b)\,\mathsf{E}_p(b)$, by regrouping one vertex at a time, with u,z,w as in Step 2. At P we write $\mathsf{a}_P^{(3)}[\![\mathsf{E}_p^a,\cdot,\cdot]\!]=\sum_b \mathsf{E}_p(b)\,\mathsf{a}_P^{(3)}[\![\delta_{(b,a)},\cdot,\cdot]\!]$, split the quadratic polynomial $\partial_{b,a}\mathsf{a}_P^{(3)}$, whose polarisation is the last factor, into its two terms as in eq. 7.xxxii, which follows from eq. 7.v by the Leibniz rule, and keep the vertex at P' in the standard writing; the resulting terms, indexed by (P,b,P'), the monomials and the pairings, are bounded by $C\langle z\rangle^{-2}\langle w-z\rangle^{-n_3}\langle u-w\rangle^{-n_5}$, with a constant C which depends only on τ_{gf}, as the kernels of $\mathbb{Z}^3$ satisfy eq. 7.xxxi with $\mathsf{c}_0=\mathsf{c}_0(\tau_{\mathrm{gf}})$, and with the four pairs (n_3,n_5) of Step 2, for which the computation there holds for every $n_1\geq 2$; so they may be grouped by (b,P'), and the sum over $P\ni b$ turns the first entry into $\partial_{b,a}\mathsf{S}_3$.

Next, with z,w the positions of b,P' relative to the base point of p and with $C=C(\tau_{\mathrm{gf}})$, we have $|\langle\partial_{b,a}\mathsf{S}_3;\partial_{b',a}\mathsf{a}_{P'}^{(3)}\rangle_0|\leq C\langle w-z\rangle^{-3}$: by eq. 7.xxxii, a monomial of $\mathsf{u}_{b,a}$ has a curl leg and a plain leg, so that the two lines joining it to a monomial of $\partial_{b',a}\mathsf{a}_{P'}^{(3)}$ have degrees of sum at least 3, and eq. 7.xxxi applies; and $\mathsf{d}_{b,a}$ contributes $\sum_{P\ni b}\mathsf{s}_P(b)\,\mathsf{F}(P)$ with $\mathsf{F}(P):=\langle\langle\mathsf{e}_a,\mathfrak{z}_1(P)\rangle;\partial_{b',a}\mathsf{a}_{P'}^{(3)}\rangle_0$, by

Lem. 7.26(a) and Lem. 7.24 a finite sum of products $K_{\mathrm{a}}K_{\mathrm{b}}$ of elementary lines in the base point of P with $n_{\mathrm{a}}+n_{\mathrm{b}}\geq 2$; since the four plaquettes containing b form two pairs, each of one dual direction, with base points differing by one of the vectors e_1,e_2,e_3 and with opposite signs, this is a sum of two first differences in the base point of P, which Lem. 7.22(c) with $|\gamma|=1$ bounds by $C\langle w-z\rangle^{-3}$.

At P', by Lem. 7.26(a) for two quadratic polynomials, the term of (b,P') so obtained is the sum over b' of $\mathsf{E}_p(b)\,\mathsf{E}_q(b')\,\langle\partial_{b,a}\mathsf{S}_3;\partial_{b',a}\mathsf{a}^{(3)}_{P'}\rangle_0$; these terms, indexed by (b,P',b'), are bounded by $C\langle z\rangle^{-2}\langle w-z\rangle^{-3}\langle u-w\rangle^{-2}$, the pair $(3,2)$ of Step 2; so they may be grouped by (b,b'), and the sum over $P'\ni b'$ gives the right side of that display. Thus this equality holds on $\mathbb{Z}^3$, and the sum over b,b' in the formula for T_{I} converges absolutely: for the summand $-\langle\partial_{b,a}\mathsf{S}_3;\partial_{b',a}\mathsf{S}_3\rangle_0$ of Σ_{tot} by these majorants, and for the summands of $\langle\partial_{b,a}\partial_{b',a}\mathsf{V}_2\rangle_0$, which by eq. 7.xi and eq. 7.xxxi are bounded and of finite range, or bounded by $8\mathsf{c}_0^2\langle b_--b'_-\rangle^{-4}$, by the sums of Step 2.

Step 2. We prove (ii). We write every vertex in the standard writing and count a curl leg as one leg (Lem. 7.26(d)); at every position there are finitely many monomials, with coefficients independent of the position. We follow the enumeration in the proof of Lem. 7.26(c); at fixed positions all sums there are finite. By eq. 7.xxxi a term is bounded, up to a bounded factor, by $\prod_\ell \mathsf{c}_0\langle y_\ell\rangle^{-n_\ell}$, where ℓ ranges over its lines, y_ℓ is the difference of the positions of the two vertices joined by ℓ, and $n_\ell=1+\#(\text{curl legs at }\ell)$; the bounded factor accounts for shifts of length at most 3 between the base points of the legs and the positions of the vertices, since $\langle y\rangle\leq 4\langle y+v\rangle$ for $|v|\leq 3$.

Every term has at most five lines, $\mathsf{c}_0\geq 1$, and the numbers of monomials and pairings are pure constants; it therefore suffices to bound the sums over the positions of the products $\prod_\ell\langle y_\ell\rangle^{-n_\ell}$ by pure constants. Let u be the base point of q minus that of p, and let z,w be the positions of P,P' (or of the bonds of the local terms of V_2) relative to the base point of p.

- T_1: the lines starting at p or ending at q carry a curl leg of p or of q, so their degrees are at least 2, and the line between P and P' has degree at least 1. As $\langle\cdot\rangle^{-n}$ decreases in n, the sum is at most the sum in Prop. 7.23(i) with $(n_1,\dots,n_5)=(2,2,1,2,2)$, which satisfies the hypotheses of that proposition.
- The terms of T_{I} with two cubic vertices, or with the ghost loop, have sums of the form
$$\langle u\rangle^{-3}\sum_{z,w}\langle z\rangle^{-n_1}\langle w-z\rangle^{-n_3}\langle u-w\rangle^{-n_5},$$
$$n_1,n_5\in\{2,3\},\qquad n_3\geq 2,\qquad n_1+n_3+n_5=8;$$
the factor $\langle u\rangle^{-3}$ comes from $\Pi(p,q)$, and $\langle w-z\rangle^{-n_3}$ from the two lines between the cubic vertices, resp. from $\mathsf{D}(b;b')\mathsf{D}(b';b)$. The possible pairs are $(n_3,n_5)=(2,3),(3,2),(3,3),(4,2)$, and for each of them Lem. 7.18(c) gives $\sum_w\langle w-z\rangle^{-n_3}\langle u-w\rangle^{-n_5}\leq C\langle u-z\rangle^{-2}\lg\langle u-z\rangle$. Next, $\sum_z\langle z\rangle^{-2}\langle u-z\rangle^{-2}\lg\langle u-z\rangle\leq C$: by $\lg t\leq 2t^{1/2}$ for $t\geq 1$ and by Lem. 7.18(c) with $(a,a')=(2,\frac{3}{2})$ this sum is at most $C\langle u\rangle^{-1/2}\lg\langle u\rangle\leq C$. Finally $\langle u\rangle^{-3}\leq 1$.
- The other terms of T_{I} (with a line between two slots of the quartic vertex at P, with the Haar term, with the ghost tadpole) have sums bounded by $\sum_z\langle z\rangle^{-2}\langle u-z\rangle^{-2}\leq C$, by Lem. 7.18(c).
- T_4 leads to $\sum_z\langle z\rangle^{-4}\langle u-z\rangle^{-4}$, and T_5 to $\sum_z\langle z\rangle^{-n_X}\langle u-z\rangle^{-n_Y}$ with $n_X,n_Y\geq 2$, the line between p and q being bounded by c_0; both sums are bounded by Lem. 7.18(c). The sums of $\mathsf{T}_{4'}$ and T_6 are finite, as these classes have no vertex besides p and q.
- The graphs which vanish by colour in Lem. 7.26(b) vanish pairing by pairing, so they contribute no terms. Indeed $\langle\mathrm{d}A,A_i\times A_j\rangle=\sum\epsilon_{abc}(\mathrm{d}A)^aA_i^bA_j^c$, so every monomial of $\mathsf{a}^{(3)}_P$ has three legs of three different colours, while $((\mathrm{d}A)^a(p))^2$ has two legs of one colour; since the covariance is diagonal in the colours, a line between two legs of one monomial of $\mathsf{a}^{(3)}_P$, or two lines from $((\mathrm{d}A)^a(p))^2$ to one monomial of $\mathsf{a}^{(3)}_P$, give the factor 0.

In the box the positions range over subsets of $\mathbb{Z}^3$, and the same bounds apply. □

7.1.11. *Elementary terms, their expansions, the local parts.*

Lemma 7.28 (Elementary terms and their degrees)**.** Let $\tau_{\mathrm{gf}} > 0$, let us work on $\mathbb{Z}^3$, let $p = p_{[\iota]}(x_p)$ and $q = p_{[\iota']}(x_q)$ be plaquettes, and put $u' := x_q - x_p$. We write every vertex occurring in the formulas of Lem. 7.26(c) in the standard writing (Def. 7.8(3), Lem. 7.26(d)), we carry out the sums over the colours, and we write every contraction as an elementary line by Lem. 7.24. Then there are finite index sets $\mathfrak{G}_\bullet = \mathfrak{G}_\bullet(\iota, \iota')$, real numbers c_G and elementary lines $K^G_\bullet$, all independent of x_p and x_q, such that the following representations hold; every G is called an elementary term. We omit the index G on the lines, and $*$ denotes the convolution of Lem. 7.22.

(1) $\mathsf{T}_1 = \sum_{G\in\mathfrak{G}_1} c_G\, \mathsf{T}^G(u')$, where every T^G is a kite as in Prop. 7.23, with degrees satisfying the hypotheses of that proposition and with $\Delta = 5$. Here K_1 and K_2 are contractions of $(\mathrm{d}A)(p)$ with legs at P and at P', K_4 and K_5 are contractions of legs at P and at P' with $(\mathrm{d}A)(q)$, and K_3 is the contraction of a leg at P with a leg at P'.

(4) $\mathsf{T}_4 = \sum_{G\in\mathfrak{G}_4} c_G\, (X * Y)(u')$, where $X = K_1K_2$ and $Y = K_3K_4$ with $n_i \in \{2,3\}$, and $(n_X, n_Y) := (n_1+n_2, n_3+n_4) \in \{(4,4),(5,4),(4,5),(6,6)\}$, with $(n_X, n_Y) = (4,4)$ for the monomials of $\frac{1}{2}|\mathfrak{z}1|^2$.

(4′) The first half of $\mathsf{T}_{4'}$ equals $\sum_{G\in\mathfrak{G}_{4'}} c_G\, K_1(u')\, K_2(u')$, where $n_i \in \{2,3\}$ and K_1, K_2 are contractions of $(\mathrm{d}A)(p)$ with legs at q.

(5) The first half of T_5 equals $\sum_{G\in\mathfrak{G}_5} c_G\, K_0(u')\, (X * Y)(u')$ with $Y = K_{\mathrm{a}}K_{\mathrm{b}}$. Here K_0 and X are contractions of $(\mathrm{d}A)(p)$ with a leg at q, resp. at P, and $n_0, n_X \in \{2,3\}$; $K_{\mathrm{a}}, K_{\mathrm{b}}$ are contractions of legs at P with legs at q, and $n_{\mathrm{a}}, n_{\mathrm{b}} \in \{1,2,3\}$; finally $n_0 + n_X + n_{\mathrm{a}} + n_{\mathrm{b}} = 8$, so that $n_Y := n_{\mathrm{a}} + n_{\mathrm{b}} \in \{2,3,4\}$.

(6) $\mathsf{T}_6 = \sum_{G\in\mathfrak{G}_6} c_G\, K_1(u')\, K_2(u')\, K_3(u')$, with $n_1 + n_2 + n_3 = 5$.

(I) $\mathsf{T}_{\mathrm{I}} = -\mathsf{n}\,\Pi(p,q) \sum_{\mu,\nu} (K^{\rightarrow}_{\iota\mu} * \Sigma_{\mu\nu} * K^{\leftarrow}_{\nu\iota'})(u')$, where $K^{\rightarrow}_{\iota\mu}(y) := \mathsf{E}_{p_{[\iota](0)}}((y,\mu))$ and $K^{\leftarrow}_{\nu\iota'}(y) := \mathsf{E}_{p_{[\iota']}(y)}((0,\nu))$ are elementary lines of degree 2, and $\Sigma_{\mu\nu}(y) := \Sigma_{\mathrm{tot}}((x,\mu),(x+y,\nu))$, which does not depend on x. With $\mathsf{s}_P(b) := s_i$ if $b = b_i(P)$ and $\mathsf{s}_P(b) := 0$ otherwise,

$$\partial_{b,a}\mathsf{S}_3 = \mathsf{d}_{b,a} + \mathsf{u}_{b,a},$$
$$\mathsf{d}_{b,a} := \sum_{P\ni b} \mathsf{s}_P(b)\, \langle \mathsf{e}_a, \mathfrak{z}1(P)\rangle, \qquad \mathsf{u}_{b,a} := \sum_{P\ni b} \langle \mathrm{d}A(P), \partial_{b,a}\mathfrak{z}1(P)\rangle; \tag{7.xxxii}$$

the monomials of $\mathsf{d}_{b,a}$ have two plain legs, those of $\mathsf{u}_{b,a}$ a curl leg and a plain leg. Moreover

$$\Sigma_{\mu\nu} = \mathsf{q}^0_{\mu\nu} + \sum_{G\in\mathfrak{G}_{\mathrm{I}}(\mu,\nu)} c_G\, \mathsf{f}_G(\nabla)[K_{\mathrm{a}}K_{\mathrm{b}}], \tag{7.xxxiii}$$

where $\mathsf{q}^0_{\mu\nu}$ has finite support (it comes from the line between two slots of the quartic Wilson vertex, from the Haar term and from the ghost tadpole); $\mathsf{f}_G(k)$ is the product of two trigonometric polynomials, each a line symbol of degree 0 or 1, of total degree $d_G \in \{0,1,2\}$; $\mathsf{f}_G(\nabla)$ is the finite difference operator which multiplies symbols by f_G; $n_{\mathrm{a}}, n_{\mathrm{b}} \in \{1,2,3\}$; and $n_{\mathrm{a}} + n_{\mathrm{b}} + d_G = 4$. The terms G are the ghost loop $-8\,\mathsf{D}(b;b')\,\mathsf{D}(b';b)$, with $d_G = 0$, and the terms of $-\langle\partial\mathsf{S}_3;\partial'\mathsf{S}_3\rangle_0$ multiplied out by eq. 7.xxxii, with d_G the number of entries d.

The second halves of $\mathsf{T}_{4'}$ and T_5 have the same forms, with p and q exchanged and u' replaced by $-u'$. ■

Proof. We use two counting rules: every monomial of $\mathsf{a}^{(3)}$ has exactly one curl leg and $\mathsf{a}^{(2)}$ has two (Def. 7.8(3)); and the degree of a line is $1 + \#$(its curl legs) (Lem. 7.24). By the consequence stated in Lem. 7.24, every contraction is an elementary line in the difference of the base points of the two vertices which it joins, its symbol depending only on the two legs; the shifts between the base points of the legs and those of the vertices are absorbed into the lines.

(1) A kite has five lines. There are four curl legs at p and q and one at each of P and P', hence $\sum_i n_i = 5 + 6 = 11$ and $\Delta = 5$. For $i \in \{1,2,4,5\}$ one end of the line i is a curl leg of p or of q, so

that $n_i \geq 2$, and $n_i \leq 3$. Besides the two curl legs of p, the lines $1,2,3$ carry at most two further curl legs, one of P and one of P'; hence $n_1+n_2+n_3 \leq 2+2+1+2 = 7$, and the same count applies to the lines $3,4,5$. With the base points x_p+z and x_p+w of P and P', the sum over P, P' becomes the sum over z, w and over the dual directions of P, P', and the five lines become functions of z, w, $w-z$, $u'-z$, $u'-w$.

(4) Each of the four lines carries a curl leg at p or at q; it has degree 2, or degree 3 if its leg at P is a curl leg. The three kinds of monomials of $\mathsf{a}_P^{(4)}$ (Lem. 7.26(d)) give $(n_X, n_Y) = (4,4)$; $(5,4)$ or $(4,5)$; and $(6,6)$. With the base point x_p+z of P, the two lines from p make up $X(z)$, the two lines to q make up $Y(u'-z)$, and the sum over z is $(X*Y)(u')$.

(4′) There are two lines from p to q, each with its curl leg at p; the line between two slots of the vertex at q is a number, independent of the positions, and it is absorbed into c_G.

(5) There are four lines, and there are $2+1+1$ curl legs, at p, q and P; hence $\sum n = 4+4 = 8$. With the base point x_p+z of P, the line from p to P is $X(z)$, the two lines from P to q make up $Y(u'-z)$, and the line from p to q is $K_0(u')$.

(6) There are three lines, and there are $1+1$ curl legs, at p and q; hence $\sum n = 3+2 = 5$.

(I) In the formula for T_I we write $b = (x,\mu)$ and $b' = (x',\nu)$ with $x = x_p+z$, $x' = x_p+w$. By translation invariance $\mathsf{E}_p(b) = K^{\rightarrow}_{\iota\mu}(z)$ and $\mathsf{E}_q(b') = K^{\leftarrow}_{\nu\iota'}(u'-w)$, and by the translation covariance in Def. 7.13(5) $\Sigma_\mathrm{tot}(b,b') = \Sigma_{\mu\nu}(w-z)$; this gives the formula for T_I. The functions $K^{\rightarrow}_{\iota\mu}$ and $K^{\leftarrow}_{\nu\iota'}$ are elementary lines of degree 2 by Lem. 7.24, the second because $K^{\leftarrow}_{\nu\iota'}(y) = \mathsf{E}_{p_{[\iota']}(0)}((-y,\nu))$. By Def. 7.13(5) and eq. 7.xi, with $\partial = \partial_{b,a}$ and $\partial' = \partial_{b',a}$,

$$\Sigma_\mathrm{tot}(b,b') = \Big[\langle \partial\partial' \mathsf{S}_4\rangle_0 + (\tfrac{2}{3} + \tfrac{4}{3}\mathsf{g}(b))\delta_{bb'}\Big] \\ - 4\big[\mathsf{D}(b;b')\mathsf{D}(b';b) + \mathsf{D}(b';b)\mathsf{D}(b;b')\big] - \langle \partial\mathsf{S}_3; \partial'\mathsf{S}_3\rangle_0.$$

The bracket has finite range and is translation invariant; it gives $\mathsf{q}^0_{\mu\nu}$. The middle term is the ghost loop $-8\,\mathsf{D}(b;b')\,\mathsf{D}(b';b)$, a product of two elementary lines of degree 2 in y (Lem. 7.24), with $\mathsf{f}_G = 1$. From eq. 7.v and the Leibniz rule we have

$$\partial_{b,a}\langle \mathrm{d}A(P), \mathfrak{z}_1(P)\rangle = \mathsf{s}_P(b)\,\langle \mathsf{e}_a, \mathfrak{z}_1(P)\rangle + \langle \mathrm{d}A(P), \partial_{b,a}\mathfrak{z}_1(P)\rangle,$$

which gives eq. 7.xxxii. For a function F on the plaquettes of one dual direction, the map $b \mapsto \sum_{P\ni b} \mathsf{s}_P(b)\,\mathsf{F}(P)$ is a difference operator whose symbol is $\pm\overline{\mathsf{k}_\alpha}$ up to a factor $e^{ik\cdot y_0}$, a line symbol of degree 1; this operator yields the factor of f_G which belongs to an entry d. An entry u we split into its monomials and treat each of them separately. By the consequence stated in Lem. 7.24, the contraction of a leg with base point $x+\xi$ with a leg with base point $x+y+\xi'$ is an elementary line in y in which the shifts ξ and ξ' are already contained; hence the factor of f_G which belongs to an entry u is 1, a line symbol of degree 0 with leading part 1, just as for the ghost loop.

By Lem. 7.26(a), the cumulant of two quadratic monomials is a sum of two products of two lines. Hence every term of $-\langle \partial\mathsf{S}_3; \partial'\mathsf{S}_3\rangle_0$, multiplied out by eq. 7.xxxii, has the form $c_G\,\mathsf{f}_G(\nabla)[K_\mathrm{a}K_\mathrm{b}]$, with d_G the number of entries d and with $n_\mathrm{a}+n_\mathrm{b} = 2+\#(\text{curl legs on the two lines})$. As d has no curl leg and u has one, $n_\mathrm{a}+n_\mathrm{b} = 2+(2-d_G)$. The second halves of $\mathsf{T}_{4'}$ and T_5 are treated in the same way, the roles of p and q being exchanged. □

Lemma 7.29 (Expansions of the elementary terms, with errors)**.** Let the setting be that of Lem. 7.28, let $r' := |u'|$ be large, and let $K^h := \mathcal{F}^{-1}_{\neq 0}\sigma^h$ for every line K with symbol σ. For the bubbles X and Y let q_X and q_Y be the kernels q of Lem. 7.22(c), and put $X' := X - \mathsf{q}_X$, $Y' := Y - \mathsf{q}_Y$. Then the elementary terms of Lem. 7.28 have the following expansions.

(6) $K_1K_2K_3(u') = K_1^hK_2^hK_3^h(u') + O(r'^{-6})$.

(5) $K_0(u')\,(X{*}Y)(u') = K_0(u')\,(\mathsf{q}_Y{*}X)(u') + K_0^h(u')\,(X{*}Y')^h(u') + O(r'^{-6}\lg r')$, where $(X{*}Y')^h$ is the second component of the leading data of $X*Y' \in \mathfrak{K}(n_X+n_Y-3)$.

(4) $(X*Y)(u') = (\mathsf{q}_X * Y)(u') + (\mathsf{q}_Y * X)(u') + \mathsf{U}_G(u') + O(r'^{-6}\lg r')$, where $\mathsf{U}_G := (X'*Y')^h$ if $(n_X, n_Y) = (4,4)$ and $\mathsf{U}_G := 0$ otherwise.
(1) The expansion of Prop. 7.23(iii) holds, with the error $O(r'^{-6}\lg r')$.
(∗) In all four cases the function of degree -5 in the expansion (that is, $K_1^h K_2^h K_3^h$, $K_0^h (X*Y')^h$, U_G, resp. the function U of Prop. 7.23) is determined by the leading parts σ^h of the lines alone; it is multilinear in them, it belongs to $C^1(\mathbb{R}^3 \setminus \{0\})$, and it is rotation covariant as in Prop. 7.23(iv).

■

Proof. (6) By Lem. 7.17(a),(c), $K_i(u') = O(r'^{-n_i})$, $K_i^h(u') = O(r'^{-n_i})$ and $K_i(u') = K_i^h(u') + O(r'^{-n_i-1})$. As $n_1+n_2+n_3 = 5$, the product $K_1K_2K_3(u')$ differs from $K_1^hK_2^hK_3^h(u')$ by $O(r'^{-6})$, and $K_1^hK_2^hK_3^h$ is homogeneous of degree -5.

(5) The line X is an elementary line of degree $n_X \in \{2,3\}$, so $X \in \mathfrak{K}(n_X)$ by Lem. 7.17(c). The bubble $Y = K_\mathrm{a}K_\mathrm{b}$ has $n_\mathrm{a}, n_\mathrm{b} \in \{1,2,3\}$ and $n_Y \le 4$, so $Y' \in \mathfrak{K}(n_Y)$ by Lem. 7.22(c). We write $X*Y = \mathsf{q}_Y*X + X*Y'$. Lem. 7.22(b) applies to X and Y', as $n_X, n_Y \le 4$ and $n_X+n_Y = 8-n_0 \ge 5$; it gives $X*Y' \in \mathfrak{K}(n_X+n_Y-3)$, and (K3) with $\gamma = 0$ gives $X*Y' = (X*Y')^h + O(r'^{-(n_X+n_Y-3)-1}\lg r')$. Moreover $K_0 = K_0^h + O(r'^{-n_0-1})$ by Lem. 7.17(c), and K_0, K_0^h, $X*Y'$, $(X*Y')^h$ are bounded by the corresponding powers of r'. As $n_0 + (n_X+n_Y-3) = 8-3 = 5$, the claim follows.

(4) Multiplying out $X*Y = (\mathsf{q}_X + X') * (\mathsf{q}_Y + Y')$ we obtain

$$X*Y = \mathsf{q}_X * Y + \mathsf{q}_Y * X - \mathsf{q}_X * \mathsf{q}_Y + X'*Y',$$

and $\mathsf{q}_X * \mathsf{q}_Y$ vanishes off $\{|u'| \le 2\}$, because q_X and q_Y vanish off $\{0, \pm e_j\}$. If $(n_X, n_Y) = (4,4)$, then $X', Y' \in \mathfrak{K}(4)$ by Lem. 7.22(c), and Lem. 7.22(b) gives $X'*Y' \in \mathfrak{K}(5)$ and $X'*Y' = (X'*Y')^h + O(r'^{-6}\lg r')$. Otherwise we apply Lem. 7.22(a) to X' and Y', with $a_X = \min\{n_X - 3, 2\}$ and $a_Y = \min\{n_Y - 3, 2\}$. Its hypotheses hold by Lem. 7.22(c), which bounds the differences of a bubble of degree n by $C_\gamma \langle x\rangle^{-n-|\gamma|}$ (the kernels X', Y' differ from X, Y at finitely many points only) and the symbol of X', resp. Y', by $|\mathsf{I} - \tilde{\mathsf{Q}}| \le C|k|^{\min\{n-3,2\}}$. The three exponents in the bound of Lem. 7.22(a) are $-2-4$, $-1-5$, $3-9$ for $(n_X, n_Y) = (5,4)$, the same by symmetry for $(4,5)$, and at most -8 for $(6,6)$; so $X'*Y' = O(r'^{-6}\lg r')$ in these cases.

(1) This is Prop. 7.23(iii) with $\Delta = 5$ (Lem. 7.28(1)), so that $r^{-\Delta-1}\lg r = r'^{-6}\lg r'$.

(∗) In the case (6) this follows from Lem. 7.17(b). In the cases (5) and (4), Lem. 7.22(b) shows that the second component of the leading data of a convolution depends only on the leading data of the two factors, bilinearly and rotation covariantly. By Lem. 7.22(c) the leading data of Y', and of X' in the case (4), are $(\mathsf{I}^h, K_\mathrm{a}^h K_\mathrm{b}^h)$, where I^h depends only on $\sigma_\mathrm{a}^h, \sigma_\mathrm{b}^h$, bilinearly and rotation covariantly, by Lem. 7.20(a); the leading data of the lines X and K_0 are (σ^h, K^h) with $K^h = \mathcal{F}_{\neq 0}^{-1}\sigma^h$, by Lem. 7.17(b),(c). The functions K^h and the second components of leading data are smooth off the origin. In the case (1) the assertions are Prop. 7.23(iii),(iv). □

Lemma 7.30 (A local insertion with vanishing moments)**.**

(a) Let $\gamma_{\mu\nu}$, $1 \le \mu, \nu \le 3$, be functions of finite support on $\mathbb{Z}^3 \times \mathbb{Z}^3$ whose symbols

$$\hat\gamma_{\mu\nu}(\kappa_1, \kappa_2) := \sum_{y,y'} \gamma_{\mu\nu}(y,y')\, e^{-i(\kappa_1\cdot y + \kappa_2 \cdot y')}$$

vanish at $\kappa = 0$ together with their gradients, and let K_1, K_2, K_3 be elementary lines of degree 2. Then

$$\Big|\sum_{\mu,\nu}\sum_{y,y'} \gamma_{\mu\nu}(y,y')\, K_\mu(u+y)\, K_\nu(u+y')\Big| \le C\,|u|^{-6} \qquad (u \in \mathbb{Z}^3 \setminus \{0\}),$$

where C depends on the $\gamma_{\mu\nu}$ and on the symbols of the K_μ.

(b) Let q be a function of finite support on $\mathbb{Z}^3$ with $\hat{\mathsf{q}}(k) := \sum_y \mathsf{q}(y)\, e^{-ik\cdot y} = O(|k|^2)$ at $k = 0$, and let K_1, K_2 be elementary lines of degree 2. Then $|(K_1 * \mathsf{q} * K_2)(u)| \le C\langle u\rangle^{-3}$ for all $u \in \mathbb{Z}^3$, where C depends on q and on the symbols of K_1, K_2.

■

Proof. (a) As the $\gamma_{\mu\nu}$ have finite support, only points y, y' in a fixed ball occur; let $|u|$ be large. By Lem. 7.17(a) with $s = -1$, $K_\mu(u) = O(|u|^{-2})$, $\nabla K_\mu(u) = O(|u|^{-3})$, and the second differences of K_μ near u are $O(|u|^{-4})$. We write $K_\mu(u+y) - K_\mu(u)$ as a sum of first differences along a nearest-neighbour path from u to $u+y$ and compare each of them with $\nabla_j K_\mu(u)$; this gives

$$K_\mu(u+y) = K_\mu(u) + \sum_j y_j\, \nabla_j K_\mu(u) + O(|u|^{-4}).$$

We insert this expansion for $K_\mu(u+y)$ and for $K_\nu(u+y')$. The terms of the orders $|u|^{-4}$ and $|u|^{-5}$ then carry the coefficients

$$\sum_{y,y'} \gamma_{\mu\nu}(y,y') = \hat\gamma_{\mu\nu}(0), \qquad \sum_{y,y'} \gamma_{\mu\nu}(y,y')\, y_j, \qquad \sum_{y,y'} \gamma_{\mu\nu}(y,y')\, y'_j,$$

and the last two are i times the derivatives of $\hat\gamma_{\mu\nu}$ with respect to $\kappa_{1,j}$, resp. $\kappa_{2,j}$, at 0. All three vanish by hypothesis, and the remaining terms are $O(|u|^{-6})$. For $u \neq 0$ in a bounded set the left side is bounded and $|u|^{-6}$ is bounded below, so the bound holds for all $u \neq 0$ after enlarging C.

(b) The trigonometric polynomial $\hat{\mathsf{q}}$ is smooth and vanishes at 0 together with its gradient, hence $\hat{\mathsf{q}} \in \mathcal{S}^2$. The kernel $\mathsf{q} * K_2$, a finite combination of translates of K_2, has the symbol $\hat{\mathsf{q}}\sigma_2 \in \mathcal{S}^1$, where $\sigma_2 \in \mathcal{S}^{-1}$ is the symbol of K_2; by Lem. 7.17(a) it is $O(\langle x\rangle^{-4})$ and lies in $\ell^2(\mathbb{Z}^3)$, as does $K_1 = O(\langle x\rangle^{-2})$. By Plancherel's theorem $K_1 * (\mathsf{q} * K_2)$ is the kernel with the symbol $\sigma_1\hat{\mathsf{q}}\sigma_2 \in \mathcal{S}^0$, and Lem. 7.17(a) with $s = 0$ gives the bound $C\langle u\rangle^{-3}$. □

Lemma 7.31 (The local parts vanish to the second order)**.** Let $\tau_{\mathrm{gf}} > 0$; we work on $\mathbb{Z}^3$.

(a) (Self-energy.) In eq. 7.xxxiii let q_G be the kernel q of the bubble $K_{\mathrm{a}}K_{\mathrm{b}}$ in the sense of Lem. 7.22(c), and put

$$\mathsf{q}^{\Sigma}_{\mu\nu} := \mathsf{q}^{0}_{\mu\nu} + \sum_G c_G\, \mathsf{f}_G(\nabla)\mathsf{q}_G, \qquad \mathsf{W}_{\mu\nu} := \sum_G c_G\, \mathsf{f}_G(\nabla)[K_{\mathrm{a}}K_{\mathrm{b}} - \mathsf{q}_G],$$

so that $\Sigma_{\mu\nu} = \mathsf{q}^{\Sigma}_{\mu\nu} + \mathsf{W}_{\mu\nu}$. Then $\mathsf{q}^{\Sigma}_{\mu\nu}$ has finite support. Each $\mathsf{f}_G(\nabla)[K_{\mathrm{a}}K_{\mathrm{b}} - \mathsf{q}_G]$ belongs to $\mathfrak{K}(4)$, and the first component of its leading data is $\mathsf{f}^h_G\, \mathsf{l}^h_G$, where f^h_G is the leading part of f_G and l^h_G is the homogeneous part of the symbol l_G of the bubble (Lem. 7.22(c), Lem. 7.20(d)). Hence $\mathsf{W}_{\mu\nu} \in \mathfrak{K}(4)$, and its leading data are functions of the leading parts σ^h and of the f^h_G occurring in eq. 7.xxxiii alone. Moreover the symbol of $\mathsf{q}^{\Sigma}_{\mu\nu}$ satisfies $\hat{\mathsf{q}}^{\Sigma}_{\mu\nu}(k) = O(|k|^2)$ at $k = 0$.

(b) (Vertex.) Let $q = p_{[\iota']}(x_q)$. Let $\gamma_q(b,b')$ be the symmetric kernel of finite range defined below; then, for every plaquette p with $|u'|$ large, $\frac{\mathsf{n}}{2}\sum_{b,b'} \gamma_q(b,b')\, \mathsf{E}_p(b)\, \mathsf{E}_p(b')$ is the sum of the following terms of Lem. 7.29: the first half of $\mathsf{T}_{4'}$; the terms $c_G\, K_0(u')\, (\mathsf{q}_Y * X)(u')$ of the first half of T_5; the terms $c_G\, (\mathsf{q}_Y * X)(u')$ of T_4; and the terms $c_G \sum_{z,w} \mathsf{q}_{\mathsf{q}}(z,w)\, K_1(u'-z)\, K_2(u'-w)$ of T_1. The kernel γ_q is defined as follows. Each of K_0, X, K_1, K_2, at the points where it is evaluated in these terms, is either a value $\mathsf{E}_p(b)$ with b a bond near q, or a value $\Pi(p,P) = \sum_i s_i\, \mathsf{E}_p(b_i(P))$ with P a plaquette near q.

Substituting this term by term in the four sums we obtain a quadratic expression in the values $\mathsf{E}_p(b)$, and $\frac{\mathsf{n}}{2}\gamma_q$ is its coefficient kernel, symmetrised. This kernel does not depend on p. Indeed, by the rule for $\mathsf{a}^{(2)}_p$ in the proof of Lem. 7.26(c), the plaquette p, with its base point and its dual direction, enters the four sums only through the forms E^a_p inserted into the two slots joined to p, that is, only through the values of K_0, X, K_1, K_2 just described. An elementary term G of these sums is given by the monomials at the vertices at q, P, P'

and by the choice of their legs which are contracted with p, with q and with one another, so that the numbers c_G do not involve p; and the kernels q_Y, q_q and the line between two slots of the vertex at q are built from lines among q, P, P' alone.

We define γ_q through this way of writing, since the numbers $\sum_{b,b'} \gamma_q(b,b')\mathsf{E}_p(b)\mathsf{E}_p(b')$ alone would not fix a kernel. Let $\gamma_{q;\mu\nu}(y,y') := \gamma_q((x_q+y,\mu),(x_q+y',\nu))$. Then the symbols $\hat\gamma_{q;\mu\nu}$, defined as in Lem. 7.30(a), vanish at $\kappa=0$ together with their gradients.

■

Proof. Step 1: a trace formula. Let $\mathsf{m}^1,\dots,\mathsf{m}^\mathsf{r}$, $\mathsf{r}\le 3$, be real symmetric matrices indexed by $\mathsf{i}=(x,\mu,a)$, that is, by a bond (x,μ) and a colour a, each with finitely many non-zero entries. For $y\in\mathbb{Z}^3$ let m_y be the translate of m by y, $(\mathsf{m}_y)_{(x,\mu,a),(x',\mu',a')} := \mathsf{m}_{(x-y,\mu,a),(x'-y,\mu',a')}$, and let $\hat{\mathsf{m}}(l',l)$ be the 9×9 matrix, indexed by (μ,a) and (μ',a'), with the entries

$$\sum_{x,x'} e^{-il'\cdot x}\,\mathsf{m}_{(x,\mu,a),(x',\mu',a')}\,e^{il\cdot x'}.$$

Then $\hat{\mathsf{m}}_y(l',l)=e^{-i(l'-l)\cdot y}\,\hat{\mathsf{m}}(l',l)$, and $\hat{\mathsf{m}}(l,l)$ is the symbol of the translation invariant matrix $\sum_y \mathsf{m}_y$, in the sense that $\sum_y \mathsf{m}_y$ maps $e^{il\cdot x}\alpha$ to $e^{il\cdot x}\,\hat{\mathsf{m}}(l,l)\,\alpha$. We insert $\mathsf{C}(\mathsf{i},\mathsf{i}') = \delta_{aa'}\int_{\mathbb{T}} e^{il\cdot(x-x')}\,\hat{\mathsf{C}}_{\mu\mu'}(l)\,d^\circ l$ and let $\hat{\mathsf{C}}(l)$ act on the nine-dimensional space as $\hat{\mathsf{C}}(l)$ in the directions and as the identity in the colours. For $\mathsf{r}=3$ we obtain

$$\begin{aligned}&\operatorname{Tr}\big[\mathsf{m}^1\mathsf{C}\,\mathsf{m}^2_{y_2}\mathsf{C}\,\mathsf{m}^3_{y_3}\mathsf{C}\big]\\ &\quad=\iint e^{i(\kappa_2\cdot y_2+\kappa_3\cdot y_3)}\Big\{\int \operatorname{tr}\big[\hat{\mathsf{m}}^1(l_3,l)\,\hat{\mathsf{C}}(l)\,\hat{\mathsf{m}}^2(l,l_2)\,\hat{\mathsf{C}}(l_2)\\ &\qquad\hat{\mathsf{m}}^3(l_2,l_3)\,\hat{\mathsf{C}}(l_3)\big]\,d^\circ l\Big\}\,d^\circ\kappa_2\,d^\circ\kappa_3,\end{aligned}\tag{7.xxxiv}$$

with $l_2:=l+\kappa_2$, $l_3:=l+\kappa_2+\kappa_3$, all integrals being over $\mathbb{T}$. Indeed, the sums over the indices are finite and the integrals over the variables l,l_2,l_3 of the three covariances converge absolutely; Fubini's theorem allows us to carry out the sums first and then to pass to the variables $\kappa_2=l_2-l$, $\kappa_3=l_3-l_2$. For $\mathsf{r}=2$ one omits m^3, the last factor $\hat{\mathsf{C}}$ and κ_3; for $\mathsf{r}=1$ one has $\operatorname{Tr}[\mathsf{m}^1\mathsf{C}]=\int\operatorname{tr}[\hat{\mathsf{m}}^1(l,l)\,\hat{\mathsf{C}}(l)]\,d^\circ l$. For the quadratic forms $\mathsf{F}_j:=\frac12(A,\mathsf{m}^jA)$, Lem. 7.26(a) gives

$$\langle\mathsf{F}_1\rangle_0=\tfrac12\operatorname{Tr}[\mathsf{m}^1\mathsf{C}],\qquad \langle\mathsf{F}_1;\mathsf{F}_2\rangle_0=\tfrac12\operatorname{Tr}[\mathsf{m}^1\mathsf{C}\,\mathsf{m}^2\mathsf{C}],$$
$$\langle\mathsf{F}_1;\mathsf{F}_2;\mathsf{F}_3\rangle_0=\operatorname{Tr}[\mathsf{m}^1\mathsf{C}\,\mathsf{m}^2\mathsf{C}\,\mathsf{m}^3\mathsf{C}].$$

The formula eq. 7.xxxiv also identifies integrands. For a leg ℓ of colour a we define $\mathsf{v}_\ell(k)\in\mathbb{C}^3$ by $\ell(e^{ik\cdot x}\alpha\,\mathsf{e}_a)=\mathsf{v}_\ell(k)\cdot\alpha$, where $e^{ik\cdot x}\alpha\,\mathsf{e}_a$ denotes the form $(x,\mu)\mapsto e^{ik\cdot x}\alpha_\mu\mathsf{e}_a$; thus $\mathsf{v}_\ell(k)=e^{ik\cdot\xi}e_\nu$ for the plain leg at the bond (ξ,ν), and $\mathsf{v}_\ell(k)=e^{ik\cdot\xi}\big(\mathsf{c}_{\iota,\nu}(k)\big)_{\nu=1,2,3}$ for the curl leg at $p_{[\iota]}(\xi)$. The contraction of ℓ with the translate by y of a leg ℓ' of the same colour is $\int e^{-ik\cdot y}\,\mathsf{v}_\ell(k)^\mathsf{T}\,\hat{\mathsf{C}}(k)\,\mathsf{v}_{\ell'}(-k)\,d^\circ k$, and the Hessian m of the product $\ell_1\ell_2$ of two legs has

$$\hat{\mathsf{m}}(l',l)=\mathsf{v}_{\ell_1}(-l')\,\mathsf{v}_{\ell_2}(l)^\mathsf{T}+\mathsf{v}_{\ell_2}(-l')\,\mathsf{v}_{\ell_1}(l)^\mathsf{T},$$

the colour indices being those of the legs. The matrices m^j used below are sums of Hessians of monomials in legs. Multiplying out, we find that for all (l,κ) the integrand in braces in eq. 7.xxxiv is a sum of products of the symbols of r lines evaluated at $\pm l,\pm l_2,\pm l_3$, and that these products are, term by term, the integrands of the one-loop functions (Lem. 7.20, Lem. 7.22(c), Prop. 7.23) of the elementary terms of Lem. 7.28, up to substitutions $l\to\pm l+\lambda(\kappa)$ with λ linear. At $\kappa=0$ the integrand in braces is the sum of the integrands of these one-loop functions at $\kappa=0$, up to the substitution $l\to -l$ in single terms. We express this by saying that the two sides agree integrand by integrand.

Step 2: the Hessians at a constant abelian field. By eq. 7.iii with $t=1$, $a_p(e^{iA})=\sum_m \mathsf{a}_p^{(m)}(A)$. Let $\mathsf{N}_j(B)$ and $\mathsf{h}_q^{(j)}(B)$ be the parts of degree j in B of the Hessians $\mathsf{N}(B)$ and $\mathsf{h}_q(B)$ of Def. 7.13(1).

Since S_3 is cubic and S_4 quartic, $\mathsf{N}_1(B)$ is the Hessian of the quadratic polynomial $\sum_{b,a} B^a(b)\,\partial_{b,a}\mathsf{S}_3$, and $\mathsf{N}_2(B)$ is that of $\frac12\sum B^a(b)\,B^{a'}(b')\,\partial_{b,a}\partial_{b',a'}\mathsf{S}_4$; the matrix $\mathsf{h}_q^{(j)}(B)$ is obtained in the same way from $\mathsf{a}_q^{(2+j)}$.

Let H_b be the Hessian of $\partial_{b,3}\mathsf{S}_3$, and let $\mathsf{H}_b^{\mathsf{u}}$ and $\mathsf{H}_b^{\mathsf{d}}$ be those of $\mathsf{u}_{b,3}$ and $\mathsf{d}_{b,3}$ in eq. 7.xxxii; these matrices are translation covariant, $\mathsf{H}_{b+y}=(\mathsf{H}_b)_y$. For the constant field Y_γ of Def. 7.13(2), with $\mu(b)$ the direction of b, we have

$$\sum_b \gamma_{\mu(b)}\,\mathsf{d}_{b,3}=\sum_P\langle \mathrm{d}Y_\gamma(P),\mathfrak{z}_1(P)\rangle=0$$

(as a quadratic form, every entry of its Hessian being a finite sum), because $\sum_b\gamma_{\mu(b)}\,\mathsf{s}_P(b)\,\mathsf{e}_3=\mathrm{d}Y_\gamma(P)$ and $\mathrm{d}Y_\gamma=0$. Hence $\mathsf{N}_1(Y_\gamma)=\sum_b\gamma_{\mu(b)}\,\mathsf{H}_b^{\mathsf{u}}=\sum_\mu\gamma_\mu\sum_y(\mathsf{H}^{\mathsf{u}}_{(0,\mu)})_y$, and by Step 1 and Def. 7.13(2) its symbol is

$$\sum_\mu\gamma_\mu\,\hat{\mathsf{H}}^{\mathsf{u}}_{(0,\mu)}(l,l)=\bigoplus_e\hat{\mathsf{N}}_{e,1}(l;\gamma)$$

in the basis e_e of the colours.

Step 3: the structure in (a). Let $K\in\mathfrak{K}(n)$ have the symbol σ and the leading data (σ^h,K^h), and let $y_0\in\mathbb{Z}^3$. The kernels with the symbols $e^{ik\cdot y_0}\sigma$, $\mathsf{k}_\mu\sigma$ and $\overline{\mathsf{k}_\mu}\sigma$ lie in $\mathfrak{K}(n)$, $\mathfrak{K}(n+1)$ and $\mathfrak{K}(n+1)$, respectively, with the leading data (σ^h,K^h), $(ik_\mu\sigma^h,\partial_\mu K^h)$ and $(-ik_\mu\sigma^h,-\partial_\mu K^h)$. Indeed, the symbol $\mathsf{k}_\mu\sigma$ belongs to the kernel $\nabla_\mu K$. For (K2) we write

$$\mathsf{k}_\mu\sigma-\chi\,ik_\mu\sigma^h=\mathsf{k}_\mu(\sigma-\chi\sigma^h)+(\mathsf{k}_\mu-ik_\mu)\,\chi\sigma^h=O(|k|^{n-1}\lg|k|),$$

and for (K3), by Taylor's formula and the homogeneity of the derivatives of K^h,

$$\nabla^\gamma(\nabla_\mu K-\partial_\mu K^h)=\nabla^{\gamma+e_\mu}(K-K^h)+\nabla^\gamma(\nabla_\mu-\partial_\mu)K^h=O(|x|^{-n-2-|\gamma|}\lg|x|);$$

at the finitely many points with $|\gamma|+1\le|x|<|\gamma|+2$ the left side is a finite number, which is absorbed into the constant. Next, $\overline{\mathsf{k}_\mu}=-e^{-ik_\mu}\mathsf{k}_\mu$. For the shift, $(e^{ik\cdot y_0}-1)\,\chi\sigma^h=O(|k|^{n-2})$ and $\nabla^\gamma[K^h(\cdot+y_0)-K^h]=O(|x|^{-n-1-|\gamma|})$. By Lem. 7.22(c), $K_{\mathrm{a}}K_{\mathrm{b}}-\mathsf{q}_G\in\mathfrak{K}(n_{\mathrm{a}}+n_{\mathrm{b}})$, with the leading data $(\mathsf{l}_G^h,K_{\mathrm{a}}^hK_{\mathrm{b}}^h)$; that lemma applies, since $n_{\mathrm{a}},n_{\mathrm{b}}\in\{1,2,3\}$ and $n_{\mathrm{a}}+n_{\mathrm{b}}=4-d_G\le4$. A trigonometric polynomial f is a finite sum of terms $c\,e^{ik\cdot y_0}$.

As a line symbol of degree 0 it has the leading part $\mathsf{f}(0)$. As a line symbol of degree 1 it has $\mathsf{f}(0)=0$, its leading part is its linear part at 0, and f is a finite sum of terms $c\,e^{ik\cdot y_0}\mathsf{k}_\mu$: indeed f is a linear combination of the functions $e^{ik\cdot y}-1$, and each of these is a telescoping sum, along a nearest-neighbour path from 0 to y, of terms $\pm e^{ik\cdot y_0}\mathsf{k}_\mu$ (with $\overline{\mathsf{k}_\mu}=-e^{-ik_\mu}\mathsf{k}_\mu$ for the steps in the negative direction). Since $\mathfrak{K}(n)$ is a linear space and the leading data depend linearly on the kernel, multiplication of the symbol by f maps $\mathfrak{K}(n)$ into $\mathfrak{K}(n)$, resp. $\mathfrak{K}(n+1)$, and multiplies the first component of the leading data by f^h.

Applying this once for each of the two factors of f_G, we obtain $\mathsf{f}_G(\nabla)[K_{\mathrm{a}}K_{\mathrm{b}}-\mathsf{q}_G]\in\mathfrak{K}(n_{\mathrm{a}}+n_{\mathrm{b}}+d_G)=\mathfrak{K}(4)$, with $\mathsf{f}_G^h\,\mathsf{l}_G^h$ as the first component of its leading data. Hence $\mathsf{W}_{\mu\nu}\in\mathfrak{K}(4)$. The first component of its leading data, $\sum_G c_G\,\mathsf{f}_G^h\,\mathsf{l}_G^h$, depends only on the leading parts σ^h (Lem. 7.20(a)) and on the f_G^h; by Lem. 7.22(d) the second component is determined by the first, linearly. Finally $\mathsf{q}_{\mu\nu}^\Sigma$ has finite support, since $\mathsf{q}_{\mu\nu}^0$ and the q_G have and $\mathsf{f}_G(\nabla)$ is a finite difference operator.

Step 4: the value of $\hat{\mathsf{q}}_{\mu\nu}^\Sigma$ at 0. The symbol of q_G is $\tilde{\mathsf{Q}}_G$, whose value at 0 is $\mathsf{Q}_G(0)$, and $\mathsf{f}_G(0)=0$ if $d_G\ge1$. Hence

$$\hat{\mathsf{q}}_{\mu\nu}^\Sigma(0)=\hat{\mathsf{q}}_{\mu\nu}^0(0)+\sum_{G:\,d_G=0}c_G\,\mathsf{f}_G(0)\,\mathsf{Q}_G(0).$$

For $d_G=0$ we have $n_{\mathrm{a}}+n_{\mathrm{b}}=4$, so that the bubble symbol has $\omega=1$, and Lem. 7.20(c), with the data of the bubble given in the proof of Lem. 7.22(c), yields $\mathsf{Q}_G(0)=\int_{\mathbb{T}}\sigma_{\mathrm{a}}(-l)\,\sigma_{\mathrm{b}}(l)\,d^\circ l$. The terms with $d_G=0$ are the ghost loop and the terms of $-\langle\mathsf{u}_{b,3};\mathsf{u}_{b',3}\rangle_0=-\frac12\operatorname{Tr}[\mathsf{H}_b^{\mathsf{u}}\mathsf{C}\,\mathsf{H}_{b'}^{\mathsf{u}}\mathsf{C}]$ (Step 1). We compute $\frac12\sum_{\mu,\nu}\gamma_\mu\gamma_\nu$ times each contribution to $\hat{\mathsf{q}}_{\mu\nu}^\Sigma(0)$.

- Gluon loop. By eq. 7.xxxiv, $-\frac12 \operatorname{Tr}[\mathsf{H}^{\mathsf{u}}_{(0,\mu)} \mathsf{C}\, \mathsf{H}^{\mathsf{u}}_{(y,\nu)} \mathsf{C}]$ is the kernel in $y = b'_- - b_-$ with the symbol $\kappa_2 \mapsto -\frac12 \int \operatorname{tr}[\hat{\mathsf{H}}^{\mathsf{u}}_{(0,\mu)}(l_2,l)\, \hat{\mathsf{C}}(l)\, \hat{\mathsf{H}}^{\mathsf{u}}_{(0,\nu)}(l,l_2)\, \hat{\mathsf{C}}(l_2)]\, d^\circ l$, and this symbol is $\sum c_G\, \mathsf{f}_G\, \mathsf{I}_G$, summed over the corresponding G, integrand by integrand. At zero momentum the integrand is $-\frac12 \operatorname{tr}[\hat{\mathsf{H}}^{\mathsf{u}}_{(0,\mu)}(l,l)\, \hat{\mathsf{C}}(l)\, \hat{\mathsf{H}}^{\mathsf{u}}_{(0,\nu)}(l,l)\, \hat{\mathsf{C}}(l)]$, and by Step 2
$$\tfrac12 \sum_{\mu,\nu} \gamma_\mu\gamma_\nu \Big(-\tfrac12 \int \operatorname{tr}[\hat{\mathsf{H}}^{\mathsf{u}}_{(0,\mu)}(l,l)\, \hat{\mathsf{C}}\, \hat{\mathsf{H}}^{\mathsf{u}}_{(0,\nu)}(l,l)\, \hat{\mathsf{C}}]\, d^\circ l\Big)$$
$$= -\tfrac14 \sum_e \int \operatorname{tr}[(\hat{\mathsf{C}}\hat{\mathsf{N}}_{e,1})^2]\, d^\circ l = \int_{\mathbb{T}} \mathfrak{s}^{(1)}(l;\gamma)\, d^\circ l.$$
- Gluon tadpole. Let $\mathsf{H}_{bb'}$ be the Hessian of $\partial_{b,3}\partial_{b',3}\mathsf{S}_4$, and let $\mathsf{m} := \frac12 \sum_{\mu,\nu} \sum_y \gamma_\mu\gamma_\nu\, \mathsf{H}_{(0,\mu),(y,\nu)}$, a finite sum. By Step 2 and translation covariance $\mathsf{N}_2(Y_\gamma) = \sum_x \mathsf{m}_x$, hence $\hat{\mathsf{m}}(l,l) = \bigoplus_e \hat{\mathsf{N}}_{e,2}(l;\gamma)$, and Step 1 with $\mathsf{r}=1$ gives
$$\tfrac12 \sum_{\mu,\nu} \gamma_\mu\gamma_\nu \sum_y \langle \partial_{(0,\mu),3}\partial_{(y,\nu),3}\mathsf{S}_4\rangle_0 = \tfrac12 \operatorname{Tr}[\mathsf{m}\,\mathsf{C}]$$
$$= \tfrac12 \int \operatorname{tr}[\hat{\mathsf{m}}(l,l)\, \hat{\mathsf{C}}(l)]\, d^\circ l = \int_{\mathbb{T}} \mathfrak{s}^{(2)}(l;\gamma)\, d^\circ l.$$
- Haar term: $\frac12 \cdot \frac23 |\gamma|^2 = \frac13 |\gamma|^2 = \int \mathfrak{s}^{(5)}$. Ghost tadpole: $\frac12 \cdot \frac43 \cdot \frac13 |\gamma|^2 = \frac29 |\gamma|^2 = \int \mathfrak{s}^{(4)}$, since by Lem. 7.14(c) $\int \mathfrak{s}^{(4)} = \sum_{e=\pm1} \frac{4}{12} \sum_\mu \gamma_\mu^2 \int |\mathsf{k}_\mu|^2 |\mathsf{k}|^{-2}\, d^\circ k$, and $\int |\mathsf{k}_\mu|^2 |\mathsf{k}|^{-2}\, d^\circ k = \frac13$.
- Ghost loop. Here $\mathfrak{s}^{(3)} = 4\big(\sum_\mu \gamma_\mu \sin k_\mu\big)^2 |\mathsf{k}|^{-4}$. By Lem. 7.24 the symbols of $y \mapsto \mathsf{D}((x,\mu);(x+y,\nu))$ and of $y \mapsto \mathsf{D}((x+y,\nu);(x,\mu))$ are $\frac12 (1 + e^{ik_\nu})\, \overline{\mathsf{k}_\mu}\, |\mathsf{k}|^{-2}$ and its image under $\mu \leftrightarrow \nu$, $k \to -k$. Since $(1+e^{ik_\nu})\, \overline{\mathsf{k}_\nu} = -2i \sin k_\nu$, the ghost loop G, for which $c_G = -8$ and $\mathsf{f}_G = 1$, has $\mathsf{Q}_G(0) = \int \frac14 (-2i \sin l_\mu)(-2i \sin l_\nu)\, |\mathsf{k}(l)|^{-4}\, d^\circ l$, and $\frac12 \sum_{\mu,\nu} \gamma_\mu\gamma_\nu\, (-8)\, \mathsf{Q}_G(0) = \int \mathfrak{s}^{(3)}$.

Summing up, we find $\frac12 \sum_{\mu,\nu} \gamma_\mu\gamma_\nu\, \hat{\mathsf{q}}^\Sigma_{\mu\nu}(0) = \sum_{i=1}^5 \int \mathfrak{s}^{(i)} = \frac12 \sum_{\mu,\nu} (\sum_i c^{(i)}_{\mu\nu}) \gamma_\mu\gamma_\nu$ for small γ (Def. 7.13(3)); hence $\hat{\mathsf{q}}^\Sigma_{\mu\nu}(0) + \hat{\mathsf{q}}^\Sigma_{\nu\mu}(0) = 2\sum_i c^{(i)}_{\mu\nu}$, and this number vanishes, because by Thm. T the five matrices $c^{(i)}$ add up to zero, $\sum_{i=1}^5 c^{(i)}_{\mu\nu} = 0$ for all μ,ν. It remains to show that $\hat{\mathsf{q}}^\Sigma_{\mu\nu}(0) = \hat{\mathsf{q}}^\Sigma_{\nu\mu}(0)$. Let $\hat\Sigma_{\mu\nu} := \hat{\mathsf{q}}^\Sigma_{\mu\nu} + \hat{\mathsf{W}}_{\mu\nu}$ be the symbol of $\Sigma_{\mu\nu}$.

Since Σ_{tot} is symmetric and translation covariant, $\Sigma_{\mu\nu}(y) = \Sigma_{\nu\mu}(-y)$, and the uniqueness of Fourier coefficients gives $\hat\Sigma_{\mu\nu}(k) = \hat\Sigma_{\nu\mu}(-k)$ for almost every k. As $\hat{\mathsf{q}}^\Sigma$ is continuous and $\hat{\mathsf{W}}(k) \to 0$ for $k \to 0$, by (K2) for $\mathsf{W} \in \mathfrak{K}(4)$, letting $k \to 0$ gives the symmetry, and therefore $\hat{\mathsf{q}}^\Sigma_{\mu\nu}(0) = 0$.

Step 5: the gradient of $\hat{\mathsf{q}}^\Sigma_{\mu\nu}$ at 0. The function $\hat\Sigma_{\mu\nu}$ is integrable on $\mathbb{T}$, with the Fourier coefficients $\Sigma_{\mu\nu}(y)$. Lem. 7.15(c) with $c=0$, for which $(x,\mu)^R = (-x - e_\mu, \mu)$, gives
$$\Sigma_{\mu\nu}(y) = \Sigma_{\mathrm{tot}}((-e_\mu,\mu),(-y-e_\nu,\nu)) = \Sigma_{\mu\nu}(-y + e_\mu - e_\nu),$$
hence $\hat\Sigma_{\mu\nu}(k) = e^{-ik\cdot(e_\mu - e_\nu)}\, \hat\Sigma_{\mu\nu}(-k)$ for almost every k, by the uniqueness of Fourier coefficients. With $\varphi(k) := \frac12 (k_\mu - k_\nu)$ the function $\tilde\Sigma(k) := e^{i\varphi(k)} \hat\Sigma_{\mu\nu}(k)$ is therefore even. By (K2) for $\mathsf{W}_{\mu\nu} \in \mathfrak{K}(4)$, $\hat{\mathsf{W}}_{\mu\nu} = \hat{\mathsf{W}}^h + O(|k|^2 \lg|k|)$ near 0, where $\hat{\mathsf{W}}^h = \sum_G c_G\, \mathsf{f}^h_G\, \mathsf{I}^h_G$ is homogeneous of degree 1 and even: by Lem. 7.20(a) the parity of $\mathsf{f}^h_G \mathsf{I}^h_G$ is $(-1)^{d_G}(-1)^{\omega_G+1}$, where $\omega_G = n_{\mathrm{a}} + n_{\mathrm{b}} - 3$, and $\omega_G + d_G = 1$.

Consequently the odd part of the smooth function $e^{i\varphi} \hat{\mathsf{q}}^\Sigma_{\mu\nu} = \tilde\Sigma - e^{i\varphi} \hat{\mathsf{W}}_{\mu\nu}$ equals the odd part of $-e^{i\varphi} \hat{\mathsf{W}}^h + O(|k|^2 \lg|k|)$, which is $O(|k|^2 \lg|k|)$, because $\hat{\mathsf{W}}^h$ is even and $(e^{i\varphi} - 1)\hat{\mathsf{W}}^h = O(|k|^2)$. So the gradient of $e^{i\varphi} \hat{\mathsf{q}}^\Sigma_{\mu\nu}$ at 0 vanishes; as $\hat{\mathsf{q}}^\Sigma_{\mu\nu}(0) = 0$ by Step 4, the gradient of $\hat{\mathsf{q}}^\Sigma_{\mu\nu}$ at 0 vanishes as well. Since $\hat{\mathsf{q}}^\Sigma_{\mu\nu}$ is a trigonometric polynomial, this proves $\hat{\mathsf{q}}^\Sigma_{\mu\nu}(k) = O(|k|^2)$ and completes the proof of (a).

Step 6: set-up for (b). Let $\Gamma_{q;\mu\nu}(y,y') := \Gamma_q((x_q+y,\mu),(x_q+y',\nu))$. In the rest of the proof c_G denotes $\frac{2}{\mathsf{n}}$ times the number c_G of Lem. 7.28; the reason is that the sums with the numbers c_G of that lemma equal $\frac{\mathsf{n}}{2}\Gamma_q$, by eq. 7.xxx, resp. $\frac{\mathsf{n}}{2}\gamma_q$, by the definition of γ_q in (b). By the first, second, third and fourth summand we mean the four groups of summands of Γ_q as in eq. 7.xxx. In each summand the derivatives $\partial_{b,a}$ and $\partial_{b',a'}$ of Def. 7.13(5) act on two legs of the vertices at q, P, P'; we call these two legs the outer legs.

In the terms on the right of eq. 7.xxx the outer legs are the legs which are contracted with the two curl legs $(\mathrm{d}A)(p)$, namely the legs at which the lines K_1, K_2 of Lem. 7.28(4′), (4) and (1), resp. the lines K_0, X of Lem. 7.28(5), end. An outer leg is either plain or a curl leg; in the second case we speak of an outer curl leg. The symbol of a kernel on $\mathbb{Z}^6$, such as $\Gamma_{q;\mu\nu}$, is defined as in Lem. 7.30(a), so that κ_1 is the variable dual to y and κ_2 the variable dual to y'; accordingly we say that the outer leg on which $\partial_{b,a}$ acts, $b=(x_q+y,\mu)$, carries the momentum κ_1, and that the outer leg on which $\partial_{b',a'}$ acts, $b'=(x_q+y',\nu)$, carries the momentum κ_2.

By eq. 7.xxx and Lem. 7.28, $\Gamma_{q;\mu\nu}$ is a finite sum of kernels on $\mathbb{Z}^6$ with the symbols

$$c_G\,\mathsf{f}_{G,1}(\kappa_1)\,\mathsf{f}_{G,2}(\kappa_2)\,\mathsf{J}_G(\kappa),$$

symmetrised under $(\mu,\kappa_1)\leftrightarrow(\nu,\kappa_2)$. Here the $\mathsf{f}_{G,i}$ are trigonometric polynomials, line symbols of degree 0 if the corresponding outer leg is plain and of degree 1 if it is a curl leg. Further, $\mathsf{J}_G=1$ for the first summand; J_G is the bubble symbol $\mathsf{I}_G(\pm\kappa_2)$ of Y for the second summand, the outer leg at q carrying the momentum κ_1; $\mathsf{J}_G=\mathsf{I}_G(\pm(\kappa_1+\kappa_2))$ for the third summand, both outer legs sitting at P; and $\mathsf{J}_G=\hat{\mathsf{T}}_{\mathsf{q}}(\pm\kappa)$ for the fourth summand, the Fourier representation being eq. 7.xxvii.

By the definition of γ_q in (b), the symbol $\hat\gamma_{q;\mu\nu}$ is the same sum with J_G replaced by $\tilde{\mathsf{Q}}_G$, where $\tilde{\mathsf{Q}}_G:=1$ for the first summand, which is entirely local (the line between two slots of the vertex at q is a number, a factor of c_G), and otherwise $\tilde{\mathsf{Q}}_G$ is the function $\tilde{\mathsf{Q}}$ of Lem. 7.22(c), resp. $\tilde{\mathsf{Q}}_{\mathsf{q}}$ of Prop. 7.23, taken at the argument of J_G. Indeed, each of the local terms listed in (b) arises from the corresponding full term of Lem. 7.28 by replacing a single kernel by its local kernel, while the lines which end at the outer legs are left as they are.

In the first half of T_5 the full term is $K_0(u')\sum_z Y(z)\,X(u'-z)$, and the local term has q_Y in place of Y, in accordance with $X*Y=\mathsf{q}_Y*X+X*Y'$; in T_4 the full term is $\sum_z Y(z)\,(K_1K_2)(u'-z)$, and again Y is replaced by q_Y. In T_1 the substitution $(z,w)\to(u'-z,u'-w)$ turns the full term into

$$\sum_{z,w}\big[K_4(z)\,K_5(w)\,K_3(z-w)\big]\,K_1(u'-z)\,K_2(u'-w),$$

where the bracket, a kernel in (z,w), has the symbol $\hat{\mathsf{T}}_{\mathsf{q}}$ by the second identity of eq. 7.xxvii; the local term has $\mathsf{q}_{\mathsf{q}}(z,w)$, with the symbol $\tilde{\mathsf{Q}}_{\mathsf{q}}$, in place of the bracket. The first half of $\mathsf{T}_{4'}$ is entirely local and is its own local term. In each case the symbol J_G of the replaced kernel turns into $\tilde{\mathsf{Q}}_G$, while the number c_G, the factors $\mathsf{f}_{G,1}$, $\mathsf{f}_{G,2}$ and the argument of J_G remain the same. Let $\hat\Gamma:=\hat\Gamma_{q;\mu\nu}$ be the sum of the symbols of $\Gamma_{q;\mu\nu}$; in $\hat\Gamma-\hat\gamma_{q;\mu\nu}$ the first summand cancels.

Step 7: counting degrees. Let $d_{G,i}\in\{0,1\}$ be the degree of $\mathsf{f}_{G,i}$ and $d_G:=d_{G,1}+d_{G,2}$, the number of outer curl legs; for G in the second, third and fourth summand let ω_G be the number ω of J_G. We use again that the degree of a line is $1+\#$(its curl legs) (Lem. 7.24), and that $\mathsf{a}^{(2)}$ has two curl legs and every monomial of $\mathsf{a}^{(3)}$ one (Def. 7.8(3)). In the second summand the vertices at q and P have two curl legs, of which $2-d_G$ lie on the two bubble lines, so that $\omega_G=2+(2-d_G)-3=1-d_G$. In the fourth summand the vertices at q, P, P' have four curl legs, of which $4-d_G$ lie on the three lines of $\hat{\mathsf{T}}_{\mathsf{q}}$, so that $\omega_G=3+(4-d_G)-6=1-d_G$.

In the third summand the bubble lines carry the two curl legs at q: for the monomials of $\frac12|\mathfrak{z}_1|^2$ we get $(d_G,\omega_G)=(0,1)$; for the monomials of $\mathsf{a}_P^{(4)}$ with one curl leg we get $(1,1)$ or $(0,2)$, according as this leg is outer or not; for those with four curl legs we get $(2,3)$. Thus $d_G+\omega_G=1$, except for these last three kinds, for which $d_G+\min\{\omega_G,2\}\geq 2$ and $\omega_G\geq 1$.

Step 8: expansion of $\hat\Gamma - \hat\gamma_{q;\mu\nu}$. We show that for almost every κ with $|\kappa| \le c$, c small,

$$\hat\Gamma - \hat\gamma_{q;\mu\nu} = \hat\Gamma^h + O(|\kappa|^2 \lg|\kappa|\, \Lambda(\kappa)),$$
$$\hat\Gamma^h := \sum_{G:\, d_G+\omega_G=1} c_G\, \mathsf{f}^h_{G,1}\, \mathsf{f}^h_{G,2}\, \mathsf{J}^h_G, \qquad |\hat\Gamma^h| \le C|\kappa|\, \Lambda(\kappa).$$

Here J^h_G is the homogeneous part of J_G (Lem. 7.20(d)), taken at the argument of J_G, and $\Lambda \ge 1$ is the function Λ of Lem. 7.20 for $\hat{\mathsf{T}}_{\mathsf{q}}$. The function $\hat\Gamma^h$ is homogeneous of degree 1 and even, each of its terms having the parity $(-1)^{d_G}(-1)^{\omega_G+1} = 1$. Let κ' denote the argument of J_G; then $|\kappa'| \le \sqrt{2}|\kappa|$. For $d_G + \omega_G \ge 2$, Lem. 7.22(c) gives

$$|\mathsf{f}_{G,1}\mathsf{f}_{G,2}\,(\mathsf{J}_G - \tilde{\mathsf{Q}}_G)| \le C|\kappa|^{d_G}\, |\kappa'|^{\min\{\omega_G,2\}} \le C|\kappa|^2.$$

For the other G we have $\omega_G \le 1$, hence $\deg \mathsf{Q}_G \le 1$ and $\mathsf{Q}_G - \tilde{\mathsf{Q}}_G = O(|\kappa'|^3)$, and Lem. 7.20(b) gives

$$\mathsf{J}_G - \tilde{\mathsf{Q}}_G = \mathsf{J}^h_G + O(|\kappa'|^{\omega_G+1} \lg|\kappa'|\, \Lambda_G(\kappa')) + O(|\kappa'|^3),$$

where Λ_G is bounded for the bubbles and equals Λ for $\hat{\mathsf{T}}_{\mathsf{q}}$. Moreover $\mathsf{f}_{G,i}(\kappa_i) = \mathsf{f}^h_{G,i}(\kappa_i) + O(|\kappa_i|^{d_{G,i}+1})$, and the functions $t\lg t$ and $t^2\lg t$ are increasing on $(0,1]$, their derivatives being $-\log t$ and $t(1-2\log t)$. We go through the summands.

- Third summand, $\omega_G = 1$, $d_G = 0$: $|\mathsf{J}^h_G| \le C|\kappa'| \le C|\kappa|$, and the errors are $O(|\kappa|^2\lg|\kappa|)$.
- Fourth summand: $\kappa' = \pm\kappa$, $|\mathsf{J}^h_G| \le C|\kappa|^{\omega_G}\Lambda$ and $|\mathsf{f}_{G,1}\mathsf{f}_{G,2}| \le C|\kappa|^{d_G}$; the leading term is $O(|\kappa|\Lambda)$, and the errors are $O(|\kappa|^2\lg|\kappa|\,\Lambda)$.
- Second summand: $\kappa' = \pm\kappa_2$, and $\omega_G = -1$ occurs. We take the factor of the outer leg at P, which carries the momentum κ_2, together with J_G. As $d_{G,2} + \omega_G = 1 - d_{G,1} \ge 0$, we have $|\mathsf{f}^h_{G,2}(\kappa_2)\, \mathsf{J}^h_G| \le C|\kappa_2|^{1-d_{G,1}}$ and $|\mathsf{f}^h_{G,1}\mathsf{f}^h_{G,2}\, \mathsf{J}^h_G| \le C|\kappa_1|^{d_{G,1}}|\kappa_2|^{1-d_{G,1}} \le C|\kappa|$. Replacing $\mathsf{f}_{G,i}$ by its leading part costs a factor $|\kappa_i|$, and

$$|\mathsf{f}_{G,1}\mathsf{f}_{G,2}|\, |\kappa_2|^{\omega_G+1} \lg|\kappa_2| \le C|\kappa_1|^{d_{G,1}}\, |\kappa_2|^{2-d_{G,1}} \lg|\kappa_2| \le C|\kappa|^2\lg|\kappa|.$$

The symmetrised terms are treated in the same way, with the indices 1 and 2 exchanged.

Step 9: the value of $\hat\gamma_{q;\mu\nu}$ *at* 0. We have $\hat\gamma_{q;\mu\nu}(0) = \sum c_G\, \mathsf{f}_{G,1}(0)\, \mathsf{f}_{G,2}(0)\, \tilde{\mathsf{Q}}_G(0)$, the sum extending over the G with two plain outer legs, as an outer curl leg gives $\mathsf{f}_{G,i}(0) = 0$. For such G in the first summand, $\tilde{\mathsf{Q}}_G = 1$, and the line between two slots of the vertex at q, contained in c_G, is the integral of its symbol (the case $\mathsf{r} = 1$ of Step 1). For such G in the other summands, $d_G = 0$, hence $\omega_G \ge 1$ by Step 7, and $\tilde{\mathsf{Q}}_G(0) = \mathsf{Q}_G(0)$ is the integral of the integrand at $\kappa = 0$ (Lem. 7.20(c)). By Step 1 the four summands of $\Gamma_q(b,3;b',3)$ have the trace forms

$$\tfrac12 \operatorname{Tr}[\mathsf{m}_{bb'}\mathsf{C}], \qquad -\tfrac12 \operatorname{Tr}[\mathsf{m}_b \mathsf{C}\, \mathsf{H}_{b'}\mathsf{C}] - (b \leftrightarrow b'),$$
$$-\tfrac12 \operatorname{Tr}[\mathsf{h}^{(0)}_q \mathsf{C}\, \mathsf{H}_{bb'}\mathsf{C}], \qquad \operatorname{Tr}[\mathsf{h}^{(0)}_q \mathsf{C}\, \mathsf{H}_b \mathsf{C}\, \mathsf{H}_{b'}\mathsf{C}],$$

where $\mathsf{m}_{bb'}$, m_b and $\mathsf{H}_{bb'}$ are the Hessians of $\partial\partial'\mathsf{a}^{(4)}_q$, $\partial\mathsf{a}^{(3)}_q$ and $\partial\partial'\mathsf{S}_4$ ($\partial = \partial_{b,3}$, $\partial' = \partial_{b',3}$), H_b is the Hessian of $\partial_{b,3}\mathsf{S}_3$ (Step 2), and $\mathsf{h}^{(0)}_q$ is the Hessian of $\mathsf{a}^{(2)}_q$. By Step 2,

$$\sum_{b,b'} \tfrac12 \gamma_{\mu(b)}\gamma_{\mu(b')}\, \mathsf{m}_{bb'} = \mathsf{h}^{(2)}_q(Y_\gamma), \qquad \sum_b \gamma_{\mu(b)}\, \mathsf{m}_b = \mathsf{h}^{(1)}_q(Y_\gamma),$$
$$\sum_\mu \gamma_\mu\, \hat{\mathsf{H}}_{(0,\mu)}(l,l) = \bigoplus_e \hat{\mathsf{N}}_{e,1}(l;\gamma),$$
$$\sum_{\mu,\nu}\sum_y \tfrac12 \gamma_\mu\gamma_\nu\, \hat{\mathsf{H}}_{(0,\mu),(y,\nu)}(l,l) = \bigoplus_e \hat{\mathsf{N}}_{e,2}(l;\gamma),$$

the sums in the first line being finite; and $\hat{\mathsf{h}}(l,l)$ is the matrix of the form $(e^{il\cdot x}\alpha'\mathsf{e}_e, \mathsf{h}\, e^{il\cdot x}\alpha\, \mathsf{e}_e)$, which for $\mathsf{h} = \mathsf{h}^{(j)}_q(Y_\gamma)$ is $\bigoplus_e \hat{\mathsf{h}}_{q,e,j}(l;\gamma)$ by Def. 7.13(2). The parts of the Hessians which come from an outer curl leg drop out here, since $\mathrm{d}Y_\gamma = 0$; they belong to the G with $\mathsf{f}_{G,1}(0)\, \mathsf{f}_{G,2}(0) = 0$. We

use eq. 7.xxxiv at $\kappa = 0$, integrand by integrand, and write $\hat{\mathsf{h}}_j := \bigoplus_e \hat{\mathsf{h}}_{q,e,j}$ and $\hat{\mathsf{N}}_j := \bigoplus_e \hat{\mathsf{N}}_{e,j}$. The four summands contribute to $\frac{1}{2}\sum_{\mu,\nu}\gamma_\mu\gamma_\nu\,\hat\gamma_{q;\mu\nu}(0)$ the amounts

$$\tfrac{1}{2}\int \mathrm{tr}[\hat{\mathsf{h}}_2\hat{\mathsf{C}}], \qquad \tfrac{1}{2}\cdot 2\cdot(-\tfrac{1}{2})\int \mathrm{tr}[\hat{\mathsf{h}}_1\hat{\mathsf{C}}\hat{\mathsf{N}}_1\hat{\mathsf{C}}],$$
$$-\tfrac{1}{2}\int \mathrm{tr}[\hat{\mathsf{h}}_0\hat{\mathsf{C}}\hat{\mathsf{N}}_2\hat{\mathsf{C}}], \qquad \tfrac{1}{2}\int \mathrm{tr}[\hat{\mathsf{h}}_0\hat{\mathsf{C}}\hat{\mathsf{N}}_1\hat{\mathsf{C}}\hat{\mathsf{N}}_1\hat{\mathsf{C}}]$$

(integrals over $\mathbb{T}$ with respect to $d^\circ l$), and these are $\int \mathfrak{v}_q^{(j)}(l;\gamma)\,d^\circ l$, $j = 1,\dots,4$, by Def. 7.13(4). Hence, by Def. 7.13(4) and Thm. U,

$$\tfrac{1}{2}\sum_{\mu,\nu}\gamma_\mu\gamma_\nu\,\hat\gamma_{q;\mu\nu}(0) = \tfrac{1}{2}\sum_{\mu,\nu}\gamma_\mu\gamma_\nu\sum_{j=1}^{4}\mathsf{g}^{(j)}_{q;\mu\nu} = 0$$

for all small γ. Since γ_q is a symmetric kernel, $\hat\gamma_{q;\mu\nu}(0)$ is symmetric in μ,ν, and therefore $\hat\gamma_{q;\mu\nu}(0) = 0$.

Step 10: the gradient of $\hat\gamma_{q;\mu\nu}$ at 0. Lem. 7.15(c) with $c = x(q)$, for which $R_c q = q$, gives

$$\Gamma_{q;\mu\nu}(y,y') = \Gamma_{q;\mu\nu}(\mathsf{o} - e_\mu - y,\ \mathsf{o} - e_\nu - y'), \qquad \mathsf{o} := 2(x(q) - x_q),$$

and the uniqueness of Fourier coefficients gives $\hat\Gamma(\kappa) = e^{-2i\varphi(\kappa)}\,\hat\Gamma(-\kappa)$ for almost every κ, with the real linear form $\varphi(\kappa) := \frac{1}{2}[\kappa_1\cdot(\mathsf{o}-e_\mu) + \kappa_2\cdot(\mathsf{o}-e_\nu)]$; hence $e^{i\varphi}\hat\Gamma$ is even almost everywhere. Since $\hat\Gamma^h$ is even, the odd part of $e^{i\varphi}\hat\Gamma^h$ is $O(|\kappa|^2\Lambda)$. By Step 8 the odd part of the smooth function $e^{i\varphi}\hat\gamma_{q;\mu\nu}$ is therefore $O(|\kappa|^2\lg|\kappa|\,\Lambda)$ almost everywhere near 0. The function Λ is bounded on an open cone. Indeed, for $\hat{\mathsf{T}}_{\mathsf{q}}$ we have $(l_1,l_2,l_3)(\kappa) = (\kappa_1,-\kappa_2,0)$ (Step 2 of the proof of Prop. 7.23), so that the differences $l_i(\kappa) - l_j(\kappa)$, $i<j$, are $\kappa_1+\kappa_2$, κ_1 and $-\kappa_2$. Let

$$\mathsf{K} := \{\kappa\in\mathbb{R}^6 :\ \min\{|\kappa_1|,|\kappa_2|,|\kappa_1+\kappa_2|\} > \tfrac{1}{2}|\kappa|\}.$$

This set is open, it is invariant under $\kappa\to t\kappa$ for real $t\neq 0$, and it is not empty, since it contains (v,v) for every $v\in\mathbb{R}^3\setminus\{0\}$: there $|\kappa| = \sqrt{2}\,|v|$, while $|\kappa_1| = |\kappa_2| = |v|$ and $|\kappa_1+\kappa_2| = 2|v|$. On K each of the three terms $\log_+(|\kappa|/|l_i(\kappa)-l_j(\kappa)|)$ in the definition of Λ is at most $\log 2$, so that $\Lambda\le 1+3\log 2$ there. On K the odd part is therefore $O(|\kappa|^2\lg|\kappa|)$ almost everywhere near 0. The odd part of the smooth function $e^{i\varphi}\hat\gamma_{q;\mu\nu}$ differs from its linear part $\kappa\mapsto\kappa\cdot\nabla(e^{i\varphi}\hat\gamma_{q;\mu\nu})(0)$ by $O(|\kappa|^3)$.

Hence $|\kappa\cdot\nabla(e^{i\varphi}\hat\gamma_{q;\mu\nu})(0)|\le C|\kappa|^2\lg|\kappa|$ for almost every, and by continuity for every, $\kappa\in\mathsf{K}$ with $|\kappa|\le c$. Replacing κ by $t\kappa$ and letting $t\downarrow 0$, we see that this linear function vanishes on K, a non-empty open set, and hence everywhere: the gradient of $e^{i\varphi}\hat\gamma_{q;\mu\nu}$ at 0 vanishes. As $\hat\gamma_{q;\mu\nu}(0) = 0$ by Step 9, the gradient of $\hat\gamma_{q;\mu\nu}$ at 0 vanishes as well. This proves (b). □

7.1.12. *The six terms, the continuum model, the kernel of degree -5.* In this subsection we introduce six continuum two-loop integrands and a model built from them, with a smooth ultraviolet cut-off on every line. We show that the model has the same parts of degree -5 as the lattice classes of Lem. 7.28, that these parts are multiples of $|u|^{-5}$, and we deduce the large-distance asymptotics of $\mathsf{B}_1^{\mathrm{lat}}$ and its consequence for smeared plaquette observables.

Definition 7.32 (The six two-loop integrands; the model)**.** *(1) Tensors and variables.* For $k_1,k_2,k_3\in\mathbb{R}^3$ we put

$$V_2(k_1,k_2)_{\mu\nu} := -(k_1\cdot k_2)\,\delta_{\mu\nu} + k_{1\nu}k_{2\mu},$$
$$V_3(k_1,k_2,k_3)_{\alpha\beta\gamma} := \delta_{\alpha\gamma}(k_1-k_3)_\beta + \delta_{\alpha\beta}(k_2-k_1)_\gamma + \delta_{\beta\gamma}(k_3-k_2)_\alpha.$$

The tensor V_2 is even and the tensor V_3 is odd under the simultaneous change of sign $k_i\to -k_i$ of all arguments, and $|V_2(k_1,k_2)_{\mu\nu}|\le 2|k_1||k_2|$ for all μ,ν. For $k,l,\mathsf{p}\in\mathbb{R}^3$ we write

$$\mathsf{x}_0 := \mathsf{p}^2, \quad \mathsf{x}_1 := k^2, \quad \mathsf{x}_1' := (k-\mathsf{p})^2, \quad \mathsf{x}_2 := l^2, \quad \mathsf{x}_2' := (l-\mathsf{p})^2, \quad \mathsf{x}_3 := (k-l)^2.$$

(2) The six integrands. With summation over repeated tensor indices we define

$$\begin{aligned}
F_1 &:= -\tfrac{1}{2}\, V_2(k,\mathsf{p}-k)_{\mu\nu}\, V_2(-l,l-\mathsf{p})_{\rho\sigma} \\
&\qquad \times V_3(-k,l,k-l)_{\mu\rho\alpha}\, V_3(k-\mathsf{p},\mathsf{p}-l,l-k)_{\nu\sigma\alpha}\,/\,\mathsf{x}_1\mathsf{x}_1'\mathsf{x}_2\mathsf{x}_2'\mathsf{x}_3, \\
F_2 &:= -\tfrac{1}{2}\, V_2(k,\mathsf{p}-k)_{\mu\nu}\, V_2(-k,k-\mathsf{p})_{\rho\nu} \\
&\qquad \times V_3(-k,l,k-l)_{\mu\beta\gamma}\, V_3(k,-l,l-k)_{\rho\beta\gamma}\,/\,\mathsf{x}_1^2\mathsf{x}_1'\mathsf{x}_2\mathsf{x}_3, \\
F_3 &:= -\, V_2(k,\mathsf{p}-k)_{\mu\nu}\, V_2(-k,k-\mathsf{p})_{\rho\nu}\, l_\rho(l-k)_\mu\,/\,\mathsf{x}_1^2\mathsf{x}_1'\mathsf{x}_2\mathsf{x}_3, \\
F_4 &:= -\tfrac{1}{2}\big[V_2(k,\mathsf{p}-k)_{\mu\mu} V_2(l,\mathsf{p}-l)_{\nu\nu} \\
&\qquad -\tfrac{1}{2} V_2(k,\mathsf{p}-k)_{\mu\nu}\big(V_2(l,\mathsf{p}-l)_{\mu\nu}+V_2(l,\mathsf{p}-l)_{\nu\mu}\big)\big]\,/\,\mathsf{x}_1\mathsf{x}_1'\mathsf{x}_2\mathsf{x}_2', \\
F_5 &:= V_3(\mathsf{p}-k,l,k-l)_{\alpha\beta\gamma}\, V_3(k,-l,l-k)_{\sigma\beta\gamma}\, V_2(k-\mathsf{p},-k)_{\alpha\sigma}\,/\,\mathsf{x}_1'\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3, \\
F_6 &:= -\tfrac{1}{6}\, V_3(l,k-l,\mathsf{p}-k)_{\alpha\beta\gamma}\, V_3(-l,l-k,k-\mathsf{p})_{\alpha\beta\gamma}\,/\,\mathsf{x}_2\mathsf{x}_3\mathsf{x}_1'.
\end{aligned} \tag{7.xxxv}$$

We call F_1 the kite, F_2 the gluon loop on a line, F_3 the ghost loop on a line, F_4 the quartic vertex, F_5 the term of $\mathsf{a}^{(3)}$ with S_3, and F_6 the term of $\mathsf{a}^{(3)}$ with $\mathsf{a}^{(3)}$. Each F_Γ, $\Gamma\in\{1,\dots,6\}$, is a rational function of (k,l,p), homogeneous of degree -4, even under $(k,l,\mathsf{p})\to-(k,l,\mathsf{p})$, and invariant under simultaneous rotations of k, l and p. The *lines of* Γ are the vectors whose squares occur in the denominator of F_Γ, counted with multiplicity: for $\Gamma=1$ the vectors $k,\mathsf{p}-k,l,\mathsf{p}-l,k-l$; for $\Gamma=2,3$ the vector k twice and the vectors $\mathsf{p}-k,l,k-l$; for $\Gamma=4$ the vectors $k,\mathsf{p}-k,l,\mathsf{p}-l$; for $\Gamma=5$ the vectors $k,\mathsf{p}-k,l,k-l$; for $\Gamma=6$ the vectors $\mathsf{p}-k,l,k-l$.

(3) The model. Let $\rho_0:=\frac{\pi}{16}$, and let $\rho(\mathsf{q}):=\theta(|\mathsf{q}|/2\rho_0)$ for $\mathsf{q}\in\mathbb{R}^3$, with the function θ of Def. 7.16(1); thus $\rho(\mathsf{q})=1$ for $|\mathsf{q}|\le\rho_0$ and $\rho(\mathsf{q})=0$ for $|\mathsf{q}|\ge2\rho_0$. Let $R_\Gamma(k,l,\mathsf{p})$ be the product of the values of ρ at the lines of Γ. We define

$$\Psi_\Gamma(\mathsf{p}):=\iint R_\Gamma F_\Gamma\, d^\circ k\, d^\circ l \quad (\mathsf{p}\in\mathbb{R}^3\setminus\{0\}), \qquad \mathsf{T}_\Gamma^{\mathrm{mod}}(u):=24\int_{\mathbb{R}^3} e^{i\mathsf{p}\cdot u}\,\Psi_\Gamma(\mathsf{p})\, d^\circ\mathsf{p} \quad (u\in\mathbb{Z}^3),$$

the first integral extending over $\mathbb{R}^3\times\mathbb{R}^3$.

These integrals converge absolutely. Indeed, every entry of $V_3(k_1,k_2,k_3)$ is bounded by $C\max_i|k_i|$; together with the bound on the entries of V_2 and with the denominators this gives, on every bounded set of points (k,l,p) and with a constant C depending on that set,

$$\begin{gathered}
|F_1|\le\frac{C}{|k|\,|\mathsf{p}-k|\,|l|\,|\mathsf{p}-l|\,|k-l|^2}, \qquad |F_2|,\ |F_3|\le\frac{C(|k|+|l|)}{|k|^2|l|^2|k-l|^2}, \\
|F_4|\le\frac{C}{|k|\,|\mathsf{p}-k|\,|l|\,|\mathsf{p}-l|}, \qquad |F_5|\le\frac{C}{|k|\,|\mathsf{p}-k|\,|l|^2|k-l|^2}, \qquad |F_6|\le\frac{C}{|\mathsf{p}-k|^2|l|^2|k-l|^2}.
\end{gathered} \tag{7.xxxvi}$$

We integrate these majorants first over l and then over k. For v,w in a bounded subset of $\mathbb{R}^3$, and with integrals over a bounded set of l, we have the elementary bounds

$$\int|l-v|^{-2}|l-w|^{-2}\,\mathrm{d}l\le C|v-w|^{-1}, \qquad \int|l-v|^{-1}|l-w|^{-2}\,\mathrm{d}l\le C\lg|v-w|,$$

with constants depending on these sets; moreover $|l|^{-1}|\mathsf{p}-l|^{-1}\le|\mathsf{p}|^{-1}\big(|l|^{-1}+|\mathsf{p}-l|^{-1}\big)$, since $|\mathsf{p}|\le|l|+|\mathsf{p}-l|$, and $|k|+|l|\le2|k|+|k-l|$. For $\Gamma=1$ the majorant is at most $C|\mathsf{p}|^{-2}\big(|k|^{-1}+|\mathsf{p}-k|^{-1}\big)\big(|l|^{-1}+|\mathsf{p}-l|^{-1}\big)|k-l|^{-2}$; its integral over l is at most $C|\mathsf{p}|^{-2}\big(|k|^{-1}+|\mathsf{p}-k|^{-1}\big)\big(\lg|k|+\lg|k-\mathsf{p}|\big)$, which is integrable over bounded sets of k, with a bound uniform for p in a bounded set after the factor $|\mathsf{p}|^{-2}$ is removed. For $\Gamma=4$ the same argument applies without the factor $|k-l|^{-2}$.

For $\Gamma=2,3$ the majorant is at most $2C|k|^{-1}|l|^{-2}|k-l|^{-2}+C|k|^{-2}|l|^{-2}|k-l|^{-1}$, whose integral over l is at most $C|k|^{-2}\lg|k|$, a function integrable over bounded sets of k. For $\Gamma=5$ the integral over l is at most $C|k|^{-2}|\mathsf{p}-k|^{-1}$, and the subsequent integral over k is at most $C\lg|\mathsf{p}|$. For $\Gamma=6$ the integral over l is at most $C|k|^{-1}|\mathsf{p}-k|^{-2}$, and again the integral over k is at most $C\lg|\mathsf{p}|$. Hence,

for p in a bounded set, the integral of each majorant over a bounded set of (k,l) is at most $C|\mathsf{p}|^{-2}$, since 1 and $\lg|\mathsf{p}|$ are bounded by $C|\mathsf{p}|^{-2}$ there.

Where $R_\Gamma \neq 0$, every line of Γ has modulus less than $2\rho_0$. Hence $|l| < 2\rho_0$ and $|k| < 4\rho_0$ there (by the lines l and k, and for $\Gamma = 6$ by the lines l and $k-l$), and $|\mathsf{p}| \le |\mathsf{p}-k| + |k| < 6\rho_0$, by the line $\mathsf{p}-k$, which occurs for every Γ. Consequently the integral defining $\Psi_\Gamma(\mathsf{p})$ converges absolutely, $|\Psi_\Gamma(\mathsf{p})| \le C|\mathsf{p}|^{-2}$, and Ψ_Γ is integrable, radial (by the rotation invariance of F_Γ and of R_Γ, and the substitution $(k,l) \to (Ok, Ol)$), and equal to 0 for $|\mathsf{p}| \ge 6\rho_0 = \frac{3\pi}{8}$; in particular $\mathsf{T}_\Gamma^{\mathrm{mod}}(u)$ is well defined. ∎

In order to compare the model with the elementary terms of Lem. 7.28, we write these terms, too, as integrals over momenta.

Lemma 7.33 (Momentum-space forms)**.** (a) Let $K_1, \dots, K_5$ be elementary lines as in Prop. 7.23, with symbols $\sigma_1, \dots, \sigma_5$, and let T be their kite. Then, for every $u \in \mathbb{Z}^3$,

$$\mathsf{T}(u) = \int_{\mathbb{T}} e^{i\mathsf{p}\cdot u} \Big[\iint_{\mathbb{T}^2} \sigma_1(k)\, \sigma_2(\mathsf{p}-k)\, \sigma_3(k-l)\, \sigma_4(l)\, \sigma_5(\mathsf{p}-l)\, d^\circ k\, d^\circ l \Big] d^\circ \mathsf{p},$$

and the triple integral converges absolutely.

(b) If two kernels have symbols $\sigma_\mathrm{a}, \sigma_\mathrm{b} \in L^1(\mathbb{T})$, then their product is the kernel whose symbol is the convolution $k \mapsto \int_{\mathbb{T}} \sigma_\mathrm{a}(k-l)\, \sigma_\mathrm{b}(l)\, d^\circ l$ of the two symbols. If $X, Y \in \ell^2(\mathbb{Z}^3)$ are kernels with symbols $\hat{X}, \hat{Y} \in L^1(\mathbb{T})$, then $X * Y$ (Lem. 7.22) is the kernel with the symbol $\hat{X}\hat{Y}$. ∎

Proof. We begin with (b). Let $K_\mathrm{a}, K_\mathrm{b}$ be the two kernels. The function $(k,l) \mapsto \sigma_\mathrm{a}(k)\sigma_\mathrm{b}(l)$ is integrable over $\mathbb{T}^2$; by Fubini's theorem and the substitution $k \to k-l$ on $\mathbb{T}$,

$$K_\mathrm{a}(x)\, K_\mathrm{b}(x) = \iint e^{i(k+l)\cdot x}\, \sigma_\mathrm{a}(k)\, \sigma_\mathrm{b}(l)\, d^\circ k\, d^\circ l = \int_{\mathbb{T}} e^{ik\cdot x} \Big[\int_{\mathbb{T}} \sigma_\mathrm{a}(k-l)\, \sigma_\mathrm{b}(l)\, d^\circ l \Big] d^\circ k,$$

and the bracket is an integrable function of k. For the second assertion we note that $X \in \ell^2(\mathbb{Z}^3)$ is, by Plancherel's theorem, the sequence of Fourier coefficients of a function in $L^2(\mathbb{T})$, which coincides with $\hat{X}$ by the uniqueness of Fourier coefficients; thus $\hat{X} \in L^2(\mathbb{T})$, and likewise $\hat{Y} \in L^2(\mathbb{T})$. The kernel $z \mapsto Y(u-z)$ has the symbol $k \mapsto e^{-ik\cdot u}\hat{Y}(-k)$, and Plancherel's theorem in the form $\sum_z a(z)b(z) = \int_{\mathbb{T}} \hat{a}(-k)\, \hat{b}(k)\, d^\circ k$, for square summable kernels a, b with symbols $\hat{a}, \hat{b}$, gives $(X*Y)(u) = \int_{\mathbb{T}} e^{ik\cdot u} \hat{X}(k)\hat{Y}(k)\, d^\circ k$.

We turn to (a).

Step 1. For $i \neq 3$ the degree n_i is at least 2, so that $|\sigma_i(\mathsf{q})| \le C|\mathsf{q}|^{-1}$ on $[-\pi,\pi]^3$ (Def. 7.16(2),(3)); hence $\sigma_i \in L^2(\mathbb{T})$ for $i \neq 3$, while $\sigma_3 \in L^1(\mathbb{T})$. The functions

$$\mathsf{A}(v) := \int_{\mathbb{T}} \sigma_1(v+\mathsf{q})\, \sigma_4(\mathsf{q})\, e^{iu\cdot\mathsf{q}}\, d^\circ \mathsf{q}, \qquad \mathsf{B}(v) := \int_{\mathbb{T}} \sigma_2(v+\mathsf{q})\, \sigma_5(\mathsf{q})\, e^{iu\cdot\mathsf{q}}\, d^\circ \mathsf{q}$$

are bounded, by the Cauchy–Schwarz inequality, and continuous, since translations act continuously on $L^2(\mathbb{T})$. Consequently

$$\Phi(\eta_1, \eta_2) := \int_{\mathbb{T}} \sigma_3(\mathsf{q})\, \mathsf{A}(\eta_1+\mathsf{q})\, \mathsf{B}(\eta_2-\mathsf{q})\, d^\circ \mathsf{q}$$

is bounded and continuous on $\mathbb{T}^2$, by dominated convergence, and the integral over $(\mathsf{q}_3, \mathsf{q}_4, \mathsf{q}_5) \in \mathbb{T}^3$ which results from inserting the definitions of A and B (with q_3 in place of q and $\mathsf{q}_4, \mathsf{q}_5$ the variables of A, B) converges absolutely.

Step 2. Let $\phi \in C_c^\infty(\mathbb{R}^3)$ with support in $\{|x| \le 1\}$ and $\phi(0) = 1$, and let $\phi_R := \phi(\cdot/R)$. By Prop. 7.23, the double sum defining $\mathsf{T}(u)$ converges absolutely; by dominated convergence, $\mathsf{T}(u) = \lim_{R\to\infty} \mathsf{T}_R(u)$, where $\mathsf{T}_R(u)$ denotes the same sum with the additional factor $\phi_R(z)\phi_R(w)$. For fixed R we insert $K_i(y) = \int_{\mathbb{T}} e^{i\mathsf{q}_i\cdot y} \sigma_i(\mathsf{q}_i)\, d^\circ \mathsf{q}_i$. The sums over z and w are then finite, and by Fubini's theorem they produce the factor

$$\check{\phi}_R(-(\mathsf{q}_1 - \mathsf{q}_3 - \mathsf{q}_4))\, \check{\phi}_R(-(\mathsf{q}_2 + \mathsf{q}_3 - \mathsf{q}_5))\, e^{iu\cdot(\mathsf{q}_4+\mathsf{q}_5)}.$$

We pass to the variables $\eta_1 := \mathsf{q}_1 - \mathsf{q}_3 - \mathsf{q}_4$, $\eta_2 := \mathsf{q}_2 + \mathsf{q}_3 - \mathsf{q}_5$, q_3, q_4, q_5; this is a linear change of variables with integer matrix of determinant ± 1, hence a measure preserving bijection of $\mathbb{T}^5$, and it gives

$$\mathsf{T}_R(u) = \iint_{\mathbb{T}^2} \check\phi_R(-\eta_1)\, \check\phi_R(-\eta_2)\, \Phi(\eta_1,\eta_2)\, d^\circ\eta_1\, d^\circ\eta_2.$$

Here $\int_{\mathbb{T}} \check\phi_R(-\eta)\, d^\circ\eta = \phi_R(0) = 1$, and $|\check\phi_R(\eta)| \le CR^3\langle R\eta\rangle^{-10}$ by Lem. 7.19(a), applied with $D = 3$ to the restriction of ϕ_R to $\mathbb{Z}^3$; this function vanishes off $\{|\zeta| \le R\}$, and its norm $\|\phi_R\|_{(R)}$ is at most $\|\phi\|_{C^{10}}$, a difference being an integral of a derivative. Thus the functions $\eta \mapsto \check\phi_R(-\eta)$ have integral 1, bounded L^1 norms, and integrals over $\{|\eta| \ge \delta\}$ tending to 0 as $R \to \infty$, for every $\delta > 0$. As Φ is bounded and continuous at $(0,0)$, $\mathsf{T}_R(u) \to \Phi(0,0)$.

Step 3. In the triple integral $\Phi(0,0)$ we change variables to $l = \mathsf{q}_4$, $k = \mathsf{q}_3 + \mathsf{q}_4$, $\mathsf{p} = \mathsf{q}_4 + \mathsf{q}_5$, again a measure preserving bijection of $\mathbb{T}^3$. Then $\mathsf{q}_3 = k - l$, $\mathsf{q}_5 = \mathsf{p} - l$ and $\mathsf{q}_5 - \mathsf{q}_3 = \mathsf{p} - k$, so that the integrand becomes $\sigma_1(k)\,\sigma_2(\mathsf{p}-k)\,\sigma_3(k-l)\,\sigma_4(l)\,\sigma_5(\mathsf{p}-l)\,e^{i\mathsf{p}\cdot u}$. The triple integral converges absolutely by Step 1, and Fubini's theorem yields the formula of (a). □

The next lemma computes the weights with which the leading parts of the elementary terms add up to the six integrands of Def. 7.32.

Lemma 7.34 (The weights of the six terms)**.** Let $\tau_{\mathrm{gf}} = 1$, and let $p = p_{[\iota]}(x_p)$, $q = p_{[\iota']}(x_q)$. To an elementary term G of Lem. 7.28, with its lines named as there, we attach its *leading integrand* $\mathsf{F}_G(k,l,\mathsf{p})$, where σ^h denotes the leading part of the symbol of the line in question:

$$\begin{aligned}
&G \in \mathfrak{G}_1 : && \sigma_1^h(k)\,\sigma_2^h(\mathsf{p}-k)\,\sigma_3^h(k-l)\,\sigma_4^h(l)\,\sigma_5^h(\mathsf{p}-l);\\
&G \in \mathfrak{G}_4,\ (n_X,n_Y) = (4,4) : && \sigma_1^h(k)\,\sigma_2^h(\mathsf{p}-k)\,\sigma_3^h(l)\,\sigma_4^h(\mathsf{p}-l);\\
&G \in \mathfrak{G}_5 : && \sigma_0^h(\mathsf{p}-k)\,\sigma_X^h(k)\,\sigma_{\mathrm{a}}^h(k-l)\,\sigma_{\mathrm{b}}^h(l);\\
&G \in \mathfrak{G}_6 : && \sigma_1^h(\mathsf{p}-k)\,\sigma_2^h(k-l)\,\sigma_3^h(l);\\
&G \in \mathfrak{G}_{\mathrm{I}}(\mu,\nu) : && \sigma_\Pi^h(\mathsf{p}-k)\,\sigma^h_{\rightarrow,\iota\mu}(k)\,\mathsf{f}_G^h(k)\,\sigma_{\mathrm{a}}^h(k-l)\,\sigma_{\mathrm{b}}^h(l)\,\sigma^h_{\leftarrow,\nu\iota'}(k);
\end{aligned}$$

here σ_Π, $\sigma_\rightarrow$ and $\sigma_\leftarrow$ are the symbols of $y \mapsto \Pi(p_{[\iota]}(0), p_{[\iota']}(y))$, of $K^{\rightarrow}_{\iota\mu}$ and of $K^{\leftarrow}_{\nu\iota'}$. Let $\mathfrak{G}_2 = \mathfrak{G}_2(\iota,\iota')$, resp. $\mathfrak{G}_3 = \mathfrak{G}_3(\iota,\iota')$, be the set of the triples (μ,ν,G) such that $G \in \mathfrak{G}_{\mathrm{I}}(\mu,\nu)$ is a term of $-\langle \partial\mathsf{S}_3; \partial'\mathsf{S}_3\rangle_0$, resp. of the ghost loop, in eq. 7.xxxiii; such a triple carries the coefficient $-\mathsf{n}\,c_G$, which takes the place of c_G in the sums below. Then, with sums over $\iota,\iota' \in \{1,2,3\}$ and over $G \in \mathfrak{G}_\Gamma(\iota,\iota')$, where for $\Gamma = 4$ only the terms with $(n_X,n_Y) = (4,4)$ are taken,

$$\sum_{\iota,\iota'}\sum_G c_G\,\mathsf{F}_G = 24\,F_\Gamma \quad (\Gamma = 1,2,3,4,6), \qquad \sum_{\iota,\iota'}\sum_{G\in\mathfrak{G}_5} c_G\,\mathsf{F}_G = 12\,F_5.$$

■

Proof. Step 1 (Feynman rules). Let $K(y) = \int e^{i\mathsf{q}\cdot y}\sigma(\mathsf{q})\, d^\circ\mathsf{q}$ be a line, where y is the base point of its second leg minus that of its first leg; we say that the line carries the momentum q from the first vertex to the second. The momentum entering a vertex through this line is then q at the second vertex and $-\mathsf{q}$ at the first. Leading parts of line symbols are additive and multiplicative (Def. 7.16(3)); hence the sum over the monomials of a vertex may be carried out on the leading parts. By Lem. 7.24 at $\tau_{\mathrm{gf}} = 1$, every line contributes a factor $\delta_{\mu\nu}|\mathsf{q}|^{-2}$, where at a plain leg the index of the line is the direction of the bond of that leg; a curl leg at a plaquette $p_{\alpha\beta}$ contributes the factor $i(\mathsf{q}'_\alpha\delta_{\nu\beta} - \mathsf{q}'_\beta\delta_{\nu\alpha})$, where q' is the momentum entering the vertex; and shifts of base points contribute the factor 1, the leading part of $e^{ik\cdot y_0}$.

Step 2 (the vertex $\mathsf{a}^{(2)}$). Two curl legs at $p_{[\iota]}$, with entering momenta k_1, k_2 and line indices μ, ν, give after summation over ι, that is, over $\alpha < \beta$,

$$i^2 \sum_{\alpha<\beta} (k_{1\alpha}\delta_{\mu\beta} - k_{1\beta}\delta_{\mu\alpha})(k_{2\alpha}\delta_{\nu\beta} - k_{2\beta}\delta_{\nu\alpha})$$
$$= -\big[(k_1 \cdot k_2)\,\delta_{\mu\nu} - k_{1\nu}k_{2\mu}\big] = V_2(k_1, k_2)_{\mu\nu}.$$

Step 3 (the vertex $\mathsf{a}^{(3)}$). By Lem. 7.26(d), $\mathsf{a}^{(3)}_P[\![X, Y, Z]\!]$ is the sum of t_P over the six orders of X, Y, Z. Let $P = p_{\mu\nu}$. To leading order, $(Y(b_1), \dots, Y(b_4))$ is replaced by $(Y_\mu, Y_\nu, Y_\mu, Y_\nu)$; hence

$$\sum_{i<j} s_i s_j\, Y(b_i) \times Z(b_j) \to Y_\mu \times Z_\nu - Y_\nu \times Z_\mu - Y_\mu \times Z_\mu - Y_\nu \times Z_\nu,$$

and, by the antisymmetry of the cross product,

$$\mathsf{t}_P(X, Y, Z) + \mathsf{t}_P(X, Z, Y) \to -2\langle (\mathrm{d}X)_{\mu\nu},\, Y_\mu \times Z_\nu - Y_\nu \times Z_\mu\rangle.$$

Both factors are antisymmetric in $\mu\nu$. Summing over $\mu < \nu$ and inserting $(\mathrm{d}X)_{\mu\nu} \to i(k_{1\mu}X_\nu - k_{1\nu}X_\mu)$, we obtain

$$-2i \sum_{\mu,\nu} k_{1\mu} \langle X_\nu,\, Y_\mu \times Z_\nu - Y_\nu \times Z_\mu\rangle.$$

For legs of colours a, b, c, with indices α, β, γ and entering momenta k_1, k_2, k_3, this equals $-2i\,\epsilon_{abc}[\delta_{\alpha\gamma}k_{1\beta} - \delta_{\alpha\beta}k_{1\gamma}]$. We add the terms in which the second, resp. the third, leg is the curl leg; they arise by cyclic permutation, under which ϵ_{abc} is invariant. The leading part of $\sum_\iota \mathsf{a}^{(3)}_{p_{[\iota]}}[\![\cdot,\cdot,\cdot]\!]$ is therefore

$$-2i\,\epsilon_{abc}\, V_3(k_1, k_2, k_3)_{\alpha\beta\gamma}.$$

Step 4 ($\Gamma = 1$). By Lem. 7.33(a) the five lines $1, \dots, 5$ of the kite carry k (from p to P), $\mathsf{p} - k$ (from p to P'), $k - l$ (from P to P'), l (from P to q) and $\mathsf{p} - l$ (from P' to q). The entering momenta are $(-k, k - \mathsf{p})$ at p; $(k, -l, l - k)$ at P, for the legs towards p, q, P'; $(\mathsf{p} - k, l - \mathsf{p}, k - l)$ at P'; and $(l, \mathsf{p} - l)$ at q. At both cubic vertices the colours are (a, a', c), and $\sum_{a,a',c} \epsilon^2_{aa'c} = 6$. With the factor $\frac{1}{2}$ of T_1 in Lem. 7.26(c) we obtain

$$\tfrac{1}{2}(-2i)^2 \cdot 6 \cdot V_2(-k, k - \mathsf{p})_{\mu\nu}\, V_2(l, \mathsf{p} - l)_{\rho\sigma}$$
$$\times\, V_3(k, -l, l - k)_{\mu\rho\alpha}\, V_3(\mathsf{p} - k, l - \mathsf{p}, k - l)_{\nu\sigma\alpha} \,/\, \mathsf{x}_1\mathsf{x}'_1\mathsf{x}_2\mathsf{x}'_2\mathsf{x}_3,$$

and $\frac{1}{2}(-4) \cdot 6 = 24 \cdot (-\frac{1}{2})$. Since V_2 is even, and the product of the two factors V_3 is even under the change of sign of all their arguments, this is $24F_1$.

Step 5 ($\Gamma = 6$). The factor is $\frac{1}{6}(-2i)^2 \cdot 6 = 24 \cdot (-\frac{1}{6})$. The entering momenta are $-(\mathsf{p} - k)$, $-(k - l)$, $-l$ at p and the opposite ones at q. The tensor V_3 is invariant under the cyclic permutations of its arguments together with its indices, and it changes sign under a transposition, like ϵ_{abc}; this gives $24F_6$.

Step 6 ($\Gamma = 5$). The line K_0 carries $\mathsf{p} - k$ from p to q, the line X carries k from p to P, and K_a, K_b carry $k - l$ and l from P to q. The factor is $-\frac{1}{2}(-2i)^2 \cdot 6 = 12$, with the factor $-\frac{1}{2}$ of T_5, and the tensors are

$$V_2(k - \mathsf{p}, -k)_{\alpha\sigma}\, V_3(\mathsf{p} - k, l, k - l)_{\alpha\beta\gamma}\, V_3(k, -l, l - k)_{\sigma\beta\gamma};$$

with the factors $|\mathsf{q}|^{-2}$ of the four lines this is $12F_5$.

Step 7 ($\Gamma = 2$). By Lem. 7.26(a), applied to two quadratic polynomials,

$$-\langle \partial_{b,3}\mathsf{S}_3; \partial_{b',3}\mathsf{S}_3\rangle_0 = -\tfrac{1}{2} \sum \mathsf{a}^{(3)}_P[\![\delta_{b,3}, \delta_\mathsf{i}, \delta_{\mathsf{i}'}]\!]\, \mathsf{a}^{(3)}_{P'}[\![\delta_{b',3}, \delta_\mathsf{j}, \delta_{\mathsf{j}'}]\!]\, \mathsf{C}(\mathsf{i}, \mathsf{j})\, \mathsf{C}(\mathsf{i}', \mathsf{j}'),$$

the sum extending over P, P' and over the indices $\mathsf{i}, \mathsf{i}', \mathsf{j}, \mathsf{j}'$. Here $\sum_{c,c'} \epsilon^2_{3cc'} = 2$, and together with the factor $-\mathsf{n}$ of T_I the total factor is $-\mathsf{n} \cdot (-\frac{1}{2})(-2i)^2 \cdot 2 = -12 = 24 \cdot (-\frac{1}{2})$. The line Π carries $\mathsf{p} - k$ from p to q, and the chain carries k. The curl legs give $V_2(-k, k - \mathsf{p})_{\mu\nu}$ at p and $V_2(k, \mathsf{p} - k)_{\rho\nu}$ at q, where ν is the index of the line Π, and the cubic vertices give $V_3(k, -l, l - k)_{\mu\beta\gamma}\, V_3(-k, l, k - l)_{\rho\beta\gamma}$.

With the factors of the five lines, whose squares give the denominator $\mathsf{x}_1^2\mathsf{x}_1'\mathsf{x}_2\mathsf{x}_3$, and with the parities of V_2 and V_3, this is $24F_2$.

Step 8 ($\Gamma=3$). The ghost loop enters Σ_{tot} through the kernel $-8\,\mathsf{D}(b;b')\,\mathsf{D}(b';b)$. By Lem. 7.24, the leading parts of the symbols of $y\mapsto \mathsf{D}((x,\mu);(x+y,\rho))$ and of $y\mapsto \mathsf{D}((x+y,\rho);(x,\mu))$ are $-i\mathsf{q}_\mu|\mathsf{q}|^{-2}$ and $+i\mathsf{q}_\rho|\mathsf{q}|^{-2}$. With $\mathsf{q}=k-l$, resp. $\mathsf{q}=l$, and with the factor $-\mathsf{n}$ of T_{I} we obtain

$$-\mathsf{n}\,(-8)\,(k-l)_\mu\,l_\rho\,/\,\mathsf{x}_3\mathsf{x}_2 = 24\cdot(-1)\,l_\rho(l-k)_\mu\,/\,\mathsf{x}_2\mathsf{x}_3;$$

together with the factors at p and q and on the lines Π, $K^{\rightarrow}$, $K^{\leftarrow}$, as in Step 7, this is $24F_3$.

Step 9 ($\Gamma=4$). By Lem. 7.28(4), the terms $G\in\mathfrak{G}_4$ with $(n_X,n_Y)=(4,4)$ are exactly those which arise from the monomials of $\frac12|\mathfrak{z}_1|^2$ in the standard writing of $\mathsf{a}_P^{(4)}$ (Lem. 7.26(d)); here, by eq. 7.v in Lem. 7.5(b), with $A_i:=A(b_i(P))$,

$$\mathsf{a}_P^{(4)}=\tfrac12|\mathfrak{z}_1|^2+\langle \mathrm{d}A(P),\mathfrak{z}_2\rangle-\tfrac{1}{24}|\mathrm{d}A(P)|^4,\qquad \mathfrak{z}_1=-\sum_{i<j}s_is_j\,A_i\times A_j.$$

Thus the vertex of the class $\Gamma=4$ is $\frac12|\mathfrak{z}_1|^2$. At $P=p_{\alpha\beta}$ the replacement of Step 3, $(A_1,\dots,A_4)\to(A_\alpha,A_\beta,A_\alpha,A_\beta)$, gives to leading order

$$\mathfrak{z}_1\to-(A_\alpha\times A_\beta-A_\beta\times A_\alpha-A_\alpha\times A_\alpha-A_\beta\times A_\beta)=-2A_\alpha\times A_\beta;$$

since $\frac12|2A_\alpha\times A_\beta|^2=2|A_\alpha\times A_\beta|^2$ and the terms with $\alpha=\beta$ vanish, the sum over ι, that is, over $\alpha<\beta$, gives $\sum_\iota\frac12|\mathfrak{z}_1|^2\to\mathsf{Q}(A):=\sum_{\alpha,\beta}|A_\alpha\times A_\beta|^2$, where $|Z|^2:=\langle Z,Z\rangle$ with the complex bilinear extension of $\langle\cdot,\cdot\rangle$. For $X=\xi\mathsf{e}_a$ and $Y=\eta\mathsf{e}_{a'}$ with $\xi,\eta\in\mathbb{C}^3$, that is, $X_\alpha=\xi_\alpha\mathsf{e}_a$ and $Y_\alpha=\eta_\alpha\mathsf{e}_{a'}$, we have, with the bilinear dot product on $\mathbb{C}^3$,

$$\mathsf{Q}(sX+tY)=s^2t^2(1-\delta_{aa'})\sum_{\alpha,\beta}(\xi_\alpha\eta_\beta-\xi_\beta\eta_\alpha)^2,$$

$$\sum_{\alpha,\beta}(\xi_\alpha\eta_\beta-\xi_\beta\eta_\alpha)^2=2\big[(\xi\cdot\xi)(\eta\cdot\eta)-(\xi\cdot\eta)^2\big].$$

Polarising, with $X_i=\xi_i\mathsf{e}_a$ and $Y_j=\eta_j\mathsf{e}_{a'}$ $(i,j\in\{1,2\})$, we find

$$\mathsf{Q}[\![X_1,X_2,Y_1,Y_2]\!]=4(1-\delta_{aa'})\cdot2\big[(\xi_1\cdot\xi_2)(\eta_1\cdot\eta_2)-\tfrac12(\xi_1\cdot\eta_1)(\xi_2\cdot\eta_2)-\tfrac12(\xi_1\cdot\eta_2)(\xi_2\cdot\eta_1)\big].$$

Since $\sum_{a,a'}(1-\delta_{aa'})=6$, the factor is $-\frac14\cdot4\cdot2\cdot6=-12=24\cdot(-\frac12)$, with the factor $-\frac14$ of T_4. The pairs of legs contracted with p and with q give $\xi_{1\mu}\xi_{2\nu}\to V_2(k,\mathsf{p}-k)_{\mu\nu}/\mathsf{x}_1\mathsf{x}_1'$ and $\eta_{1\mu}\eta_{2\nu}\to V_2(l,\mathsf{p}-l)_{\mu\nu}/\mathsf{x}_2\mathsf{x}_2'$, and the result is $24F_4$. □

We now show that the model $\mathsf{T}_\Gamma^{\mathrm{mod}}$ of Def. 7.32 has, up to a part of degree -4, the same part of degree -5 as the corresponding lattice terms. For a kernel K in a class $\mathfrak{K}(n)$ we write K^h, or $(K)^h$ when K is given by a longer expression, for the second component of its leading data (Def. 7.16(4)), that is, for the function K^h of condition (K3), and we call it the *leading kernel* of K; it is smooth off 0 and homogeneous of degree $-n$.

Lemma 7.35 (The model has the degree -5 parts of the lattice classes)**.** Let $\tau_{\mathrm{gf}}=1$. For $G\in\mathfrak{G}_1\cup\mathfrak{G}_4\cup\mathfrak{G}_5\cup\mathfrak{G}_6$ let U_G be the function of degree -5 attached to G in Lem. 7.29. For $(\mu,\nu,G)\in\mathfrak{G}_2\cup\mathfrak{G}_3$ let

$$\mathsf{U}_G:=\Pi^h\cdot\big(K^{\rightarrow}_{\iota\mu}*\mathsf{f}_G(\nabla)[K_{\mathrm{a}}K_{\mathrm{b}}-\mathsf{q}_G]*K^{\leftarrow}_{\nu\iota'}\big)^h,\qquad \Pi^h:=\mathcal{F}^{-1}_{\neq0}\sigma^h_\Pi,$$

where the chain lies in $\mathfrak{K}(2)$ by Lem. 7.31(a) and two applications of Lem. 7.22(b). Let

$$\mathsf{U}_\Gamma:=\sum_{\iota,\iota'}\sum_{G\in\mathfrak{G}_\Gamma}c_G\,\mathsf{U}_G\quad(\Gamma\neq5),$$

$$\mathsf{U}_5(u):=\sum_{\iota,\iota'}\sum_{G\in\mathfrak{G}_5}c_G\big[\mathsf{U}_G(u)+\mathsf{U}_G(-u)\big],$$

with the coefficients $-\mathsf{n}\,c_G$ of Lem. 7.34 for $\Gamma = 2, 3$. Then for each $\Gamma = 1, \dots, 6$ there are a continuous function l_Γ on $\mathbb{R}^3 \setminus \{0\}$, homogeneous of degree -4, and a pure constant C such that

$$|\mathsf{T}_\Gamma^{\mathrm{mod}}(u) - \mathsf{l}_\Gamma(u) - \mathsf{U}_\Gamma(u)| \le C|u|^{-6} \lg|u| \qquad (u \in \mathbb{Z}^3 \setminus \{0\}).$$

■

Proof. Step 0 (model lines). For a line whose symbol has the leading part σ^h, of degree s, let σ^{mod} be the periodic continuation of $\rho\sigma^h$, with ρ of Def. 7.32(3). As $\rho \in C_c^\infty(\mathbb{R}^3)$, $\operatorname{supp}\rho \subset \{|k| \le 2\rho_0\} \subset \{|k| < \pi\}$ and $\rho = 1$ near 0, the function σ^{mod} is a line symbol with the same leading part (Def. 7.16(3)), and its parity is exact: $\sigma^{\mathrm{mod}}(-k) = (-1)^s\sigma^{\mathrm{mod}}(k)$. Its kernel $K^{\mathrm{mod}}(x) = \int_{\mathbb{R}^3} e^{ik\cdot x}\rho\,\sigma^h\, d^\circ k$ satisfies $K^{\mathrm{mod}} = K^h + O(|x|^{-N})$ for every N, with $K^h = \mathcal{F}_{\neq 0}^{-1}\sigma^h$, by Lem. 7.17(b). For $(\mu, \nu, G) \in \mathfrak{G}_2 \cup \mathfrak{G}_3$ the elementary term is, by Lem. 7.28(I), the value at $u = u'$ of the function

$$u \mapsto K_\Pi(u)\,(K_{\iota\mu}^{\rightarrow} * \mathsf{f}_G(\nabla)[K_{\mathrm{a}}K_{\mathrm{b}}] * K_{\nu\iota'}^{\leftarrow})(u), \qquad K_\Pi(y) := \Pi(p_{[\iota]}(0), p_{[\iota']}(y)) \quad (y \in \mathbb{Z}^3);$$

here K_Π is the elementary line of degree 3 of Lem. 7.24, with the symbol σ_Π of Lem. 7.34, and $\Pi(p, q) = K_\Pi(u')$ by translation invariance. For an elementary term G let $G^{\mathrm{mod}}(u)$ be the same expression with every line K replaced by K^{mod}, in particular K_Π by K_Π^{mod}; for G of type (I) we moreover drop the operator $\mathsf{f}_G(\nabla)$ and replace $K^{\rightarrow}$ by the elementary line K_1' whose symbol is the periodic continuation of $\rho\,\sigma_{\rightarrow}^h\mathsf{f}_G^h$; it has degree $2 + d_G$.

Step 1 (the model in momentum space). We show that $\mathsf{T}_\Gamma^{\mathrm{mod}} = \sum c_G\, G^{\mathrm{mod}}$, the sum extending over ι, ι' and G as in Lem. 7.34, where for $\Gamma = 5$ the term $G^{\mathrm{mod}}(u)$ is replaced by $G^{\mathrm{mod}}(u) + G^{\mathrm{mod}}(-u)$. By Lem. 7.33,

$$G^{\mathrm{mod}}(u) = \int_{\mathbb{T}} e^{i\mathsf{p}\cdot u}\Big[\iint_{\mathbb{T}^2} \prod \sigma^{\mathrm{mod}}\, d^\circ k\, d^\circ l\Big] d^\circ \mathsf{p},$$

where the product extends over the lines of G and the arguments of the symbols are those in F_G. For the kites this is assertion (a) of that lemma, the model lines having the degrees of the lattice lines. For the other types we apply assertion (b) to the products and convolutions of which G^{mod} is built, one at a time, and we verify its hypotheses. A model line of degree n is bounded by $C\langle x\rangle^{-n}$ (Lem. 7.17(a)), and its symbol is integrable. A kernel in $\ell^2(\mathbb{Z}^3)$ with an integrable symbol has its symbol in $L^2(\mathbb{T})$, as in the proof of Lem. 7.33(b); hence the symbol of the convolution of two such kernels, a product of two functions in $L^2(\mathbb{T})$, is integrable, and so is the symbol of the product of two kernels with integrable symbols, a convolution of two integrable functions.

- For $G \in \mathfrak{G}_6$, G^{mod} is a product of three model lines, and the rule for products is used twice.
- For $G \in \mathfrak{G}_4$ with $(n_X, n_Y) = (4, 4)$, $G^{\mathrm{mod}} = X * Y$, where X and Y are products of two model lines of degree 2. Their symbols are given by the rule for products, and $|X(x)|, |Y(x)| \le C\langle x\rangle^{-4}$, so that $X, Y \in \ell^2(\mathbb{Z}^3)$ and the rule for $X * Y$ applies.
- For $G \in \mathfrak{G}_5$, $G^{\mathrm{mod}}(u) = K_0(u)\,(X * Y)(u)$ with model lines K_0, X and a product Y of two model lines, where $n_X \in \{2, 3\}$ and $n_Y \in \{2, 3, 4\}$. Thus $|X(x)| \le C\langle x\rangle^{-n_X}$ and $|Y(x)| \le C\langle x\rangle^{-n_Y}$ with exponents at least 2, so that $X, Y \in \ell^2(\mathbb{Z}^3)$; the kernel $X * Y$ has the symbol $\hat{X}\hat{Y}$, and the product with K_0 is treated by the rule for products.
- For G of type (I), $G^{\mathrm{mod}} = K_\Pi^{\mathrm{mod}} \cdot (K_1' * X * K^{\leftarrow,\mathrm{mod}})$ with $X := K_{\mathrm{a}}^{\mathrm{mod}}K_{\mathrm{b}}^{\mathrm{mod}}$; the degrees of K_1', X and $K^{\leftarrow,\mathrm{mod}}$ are $2 + d_G$, $4 - d_G$ and 2, all at least 2. Hence $K_1' * X$ has the symbol $\hat{K}_1'\hat{X}$, and Lem. 7.18(c), with $(a, a') = (2 + d_G, 4 - d_G)$ and $a + a' - 3 = 3$, gives

  $$|(K_1' * X)(x)| \le C\langle x\rangle^{-\min\{2+d_G,\, 4-d_G,\, 3\}} \lg\langle x\rangle;$$

  the exponent is at least 2, so that $K_1' * X \in \ell^2(\mathbb{Z}^3)$, the logarithm included. The rule for convolutions applies once more to $(K_1' * X) * K^{\leftarrow,\mathrm{mod}}$, and the product with K_Π^{mod} is treated by the rule for products.

For $k, l, \mathsf{p} \in [-\pi, \pi]^3$ the periodic continuations may be replaced by $\rho\sigma^h$. We use the following observation: if $v \in [-\pi, \pi]^3$, $|w| < 4\rho_0$ and $|v - w|_{\mathbb{T}} < 2\rho_0$, then $|v - w| < 2\rho_0$, because the distance

of $v-w$ from any point $2\pi\mathsf{m}$ with $\mathsf{m}\in\mathbb{Z}^3\setminus\{0\}$ is at least $\pi-4\rho_0>2\rho_0$. Let $k,l,\mathsf{p}\in[-\pi,\pi]^3$, and assume first that every line q of Γ satisfies $|\mathsf{q}|_{\mathbb{T}}<2\rho_0$; this is the case wherever all the factors σ^{mod} are non-zero. Then $|l|=|l|_{\mathbb{T}}<2\rho_0$, since $l\in[-\pi,\pi]^3$. For $\Gamma\neq 6$ we have likewise $|k|<2\rho_0$; then $|k-l|<4\rho_0<\pi$, so that $|k-l|=|k-l|_{\mathbb{T}}<2\rho_0$ wherever $k-l$ is a line ($\Gamma=1,2,3,5$), and the observation with $v=\mathsf{p}$ and $w=k$, resp. $w=l$, gives $|\mathsf{p}-k|<2\rho_0$, resp. $|\mathsf{p}-l|<2\rho_0$, wherever these vectors are lines.

For $\Gamma=6$ the observation with $(v,w)=(k,l)$ gives $|k-l|<2\rho_0$, hence $|k|<4\rho_0$, and with $(v,w)=(\mathsf{p},k)$ it gives $|\mathsf{p}-k|<2\rho_0$. Thus every line satisfies $|\mathsf{q}|<2\rho_0$, and $\sigma^{\mathrm{mod}}(\mathsf{q})=\rho(\mathsf{q})\sigma^h(\mathsf{q})$. If, on the other hand, some line satisfies $|\mathsf{q}|_{\mathbb{T}}\geq 2\rho_0$, then $\sigma^{\mathrm{mod}}(\mathsf{q})=0$ and also $\rho(\mathsf{q})=0$, since $|\mathsf{q}|\geq|\mathsf{q}|_{\mathbb{T}}$. In either case the product of the factors σ^{mod} equals $R_\Gamma\mathsf{F}_G$ at (k,l,p), the lines of G being the lines of Γ (for type (I) the factor $\rho\,\sigma^h_{\rightarrow}\mathsf{f}^h_G$ of K'_1 provides the factors $\sigma^h_{\rightarrow}(k)\,\mathsf{f}^h_G(k)$ of F_G).

Since R_Γ vanishes unless $k,l,\mathsf{p}\in[-\pi,\pi]^3$ (Def. 7.32(3)), we obtain

$$G^{\mathrm{mod}}(u)=\int_{\mathbb{R}^3}e^{i\mathsf{p}\cdot u}\Big[\iint R_\Gamma\,\mathsf{F}_G\,d^\circ k\,d^\circ l\Big]d^\circ\mathsf{p}.$$

By Lem. 7.34, $\sum c_G\iint R_\Gamma\mathsf{F}_G\,d^\circ k\,d^\circ l=24\Psi_\Gamma$ for $\Gamma\neq 5$, whence $\sum c_G G^{\mathrm{mod}}=\mathsf{T}^{\mathrm{mod}}_\Gamma$. For $\Gamma=5$ the term $G^{\mathrm{mod}}(-u)$ carries the factor $e^{-i\mathsf{p}\cdot u}$, and $\sum c_G\iint R_5\mathsf{F}_G\,d^\circ k\,d^\circ l=12\Psi_5$ is even; hence $\sum c_G[G^{\mathrm{mod}}(u)+G^{\mathrm{mod}}(-u)]=12\int(e^{i\mathsf{p}\cdot u}+e^{-i\mathsf{p}\cdot u})\Psi_5\,d^\circ\mathsf{p}=\mathsf{T}^{\mathrm{mod}}_5(u)$.

Step 2 (expansions). Lem. 7.29 applies to G^{mod}, the model lines being elementary lines of the same degrees as the lattice lines; the functions of degree -5 which it provides are determined by the leading parts of the symbols of the lines alone (Lem. 7.29), and therefore coincide with the U_G. For G of type (I) let $X:=K^{\mathrm{mod}}_{\mathrm{a}}K^{\mathrm{mod}}_{\mathrm{b}}$, of degree $n_X=4-d_G$, and write $X=\mathsf{q}_X+X'$ with $X'\in\mathfrak{K}(n_X)$ (Lem. 7.22(c)). Since K'_1 is a line of degree $2+d_G\leq 4$, two applications of Lem. 7.22(b) give $K'_1*X'*K^{\leftarrow,\mathrm{mod}}\in\mathfrak{K}(2)$, and the first component of its leading data is $\sigma^h_{\rightarrow}\mathsf{f}^h_G\mathsf{I}^h\sigma^h_{\leftarrow}$, where the homogeneous part I^h of the bubble depends only on σ^h_{a}, σ^h_{b} (Lem. 7.20(a)).

The lattice chain $K^{\rightarrow}_{\iota\mu}*\mathsf{f}_G(\nabla)[K_{\mathrm{a}}K_{\mathrm{b}}-\mathsf{q}_G]*K^{\leftarrow}_{\nu\iota'}$ has the same first component of its leading data, by Lem. 7.31(a) and by Lem. 7.22(b), used for each of its two convolutions. Within $\mathfrak{K}(2)$ the leading kernel is determined by the first component of the leading data (Lem. 7.22(d)); hence the two chains have the same leading kernel. Now $G^{\mathrm{mod}}=K^{\mathrm{mod}}_\Pi\cdot(K'_1*\mathsf{q}_X*K^{\leftarrow,\mathrm{mod}})+K^{\mathrm{mod}}_\Pi\cdot(K'_1*X'*K^{\leftarrow,\mathrm{mod}})$, and the first term is local. In the second term, $K^{\mathrm{mod}}_\Pi=\Pi^h+O(|u|^{-N})$ with $\Pi^h=O(|u|^{-3})$, and $K'_1*X'*K^{\leftarrow,\mathrm{mod}}$ differs from its leading kernel, which is $O(|u|^{-2})$, by $O(|u|^{-3}\lg|u|)$, by condition (K3).

We obtain in all cases

$$G^{\mathrm{mod}}=[\text{local terms}]+\mathsf{U}_G+O(|u|^{-6}\lg|u|).$$

Step 3 (local terms). In this step all lines are model lines, and we omit the superscript mod. Because of the exact parity of the model symbols, every one-loop function I built from model lines satisfies $\mathsf{I}(-\kappa)=(-1)^{\omega+1}\mathsf{I}(\kappa)$, as one sees by the substitution $l\to-l$; its homogeneous part I^h has the same parity (Lem. 7.20(a)). By Lem. 7.20(b) the polynomial $\mathsf{Q}(\kappa)-(-1)^{\omega+1}\mathsf{Q}(-\kappa)$, of degree at most ω, is therefore $O(|\kappa|^{\omega+1}\lg|\kappa|\,\Lambda(\kappa))$, hence equal to 0. Consequently $\mathsf{Q}=0$ if $\omega\leq 0$, and $\mathsf{Q}=\mathsf{Q}(0)$ if $\omega=1$; in both cases $\mathsf{q}=\mathsf{Q}(0)\,\delta_0$.

The local terms are thus: $\mathsf{Q}_{\mathsf{p}}(0)K_4K_5(u)+\mathsf{Q}_{\mathsf{q}}(0)K_1K_2(u)$ for the kites, where $n_4+n_5=4$ if $\omega_{\mathsf{p}}=1$ and $n_1+n_2=4$ if $\omega_{\mathsf{q}}=1$; $\mathsf{Q}_Y(0)K_0X(u)$, with $n_0+n_X=4$, for $\mathfrak{G}_5$; $\mathsf{Q}_X(0)Y(u)+\mathsf{Q}_Y(0)X(u)$, with $n_X=n_Y=4$, for $\mathfrak{G}_4$; and, for type (I) with $d_G=0$, $\mathsf{Q}_X(0)K_\Pi(u)\,(K'_1*K^{\leftarrow})(u)$, where $K'_1*K^{\leftarrow}$ is, by Lem. 7.33(b), the elementary line of degree 1 whose symbol is the periodic continuation of $\rho^2\sigma^h_{\rightarrow}\mathsf{f}^h_G\sigma^h_{\leftarrow}$ (a line symbol by Def. 7.16(3), with ρ^2 in place of ρ), $\mathsf{f}^h_G=\mathsf{f}_G(0)$ being a number.

By Step 0, and by Lem. 7.17(b) with ρ^2 in place of ρ for the line $K'_1*K^{\leftarrow}$, each of these local terms is, up to $O(|u|^{-N})$, its factor $\mathsf{Q}_\bullet(0)$ times the product of the functions $K^h=\mathcal{F}^{-1}_{\neq 0}\sigma^h$ of its lines, a function homogeneous of degree -4 and smooth off 0. We let I_Γ be the sum of these functions, each taken with the coefficient with which its term G^{mod} enters $\mathsf{T}^{\mathrm{mod}}_\Gamma$ in Step 1. For $\Gamma=5$, where

$\mathsf{T}_5^{\mathrm{mod}}(u) = \sum c_G\big[G^{\mathrm{mod}}(u) + G^{\mathrm{mod}}(-u)\big]$, the expansion of Step 2 is used at u and at $-u$; accordingly, every such function h of a term $G \in \mathfrak{G}_5$ enters I_5 together with its reflection $u \mapsto \mathsf{h}(-u)$, just as U_5 contains $\mathsf{U}_G(u)$ and $\mathsf{U}_G(-u)$.

Adding the expansions of Step 2 with these coefficients, we obtain the claimed bound for large $|u|$; for small $|u|$ all terms are bounded. All data being fixed, the constant C is pure. □

The functions I_Γ and U_Γ are in fact multiples of $|u|^{-4}$ and $|u|^{-5}$, since $\mathsf{T}_\Gamma^{\mathrm{mod}}$ is radial.

Lemma 7.36 (Rotation invariance of the homogeneous parts; definition of $\kappa_\Gamma^{\mathrm{lat}}$)**.** Let $\mathsf{t}\colon [0,\infty) \to \mathbb{C}$, let I, U be continuous functions on $\mathbb{R}^3 \setminus \{0\}$, homogeneous of degrees -4 and -5 respectively, and assume that for some C

$$|\mathsf{t}(|u|) - \mathsf{I}(u) - \mathsf{U}(u)| \le C|u|^{-6} \lg|u| \qquad (u \in \mathbb{Z}^3 \setminus \{0\}).$$

Then $\mathsf{I}(u) = c_4|u|^{-4}$ and $\mathsf{U}(u) = \kappa|u|^{-5}$ with constants c_4, κ. In particular the functions of Lem. 7.35 satisfy $\mathsf{I}_\Gamma(u) = c_{4,\Gamma}|u|^{-4}$ and

$$\mathsf{U}_\Gamma(u) = \kappa_\Gamma^{\mathrm{lat}}\,|u|^{-5} \qquad (\Gamma = 1,\dots,6);$$

this defines the numbers $c_{4,\Gamma}$ and $\kappa_\Gamma^{\mathrm{lat}}$. ■

Proof. Let O be a rotation with rational entries, N a common denominator of these entries, and $\mathsf{m} \in \mathbb{Z}^3 \setminus \{0\}$. For integers $j \ge 1$ the points $j\mathsf{N}\mathsf{m}$ and $j\mathsf{N}O\mathsf{m}$ lie in $\mathbb{Z}^3$ and have the same norm jr_1, $r_1 := \mathsf{N}|\mathsf{m}|$. Let $\hat{\mathsf{m}} := \mathsf{m}/|\mathsf{m}|$. Subtracting the hypothesis at these two points, at which t takes the same value, we obtain by homogeneity

$$\Big|(jr_1)^{-4}\big[\mathsf{I}(\hat{\mathsf{m}}) - \mathsf{I}(O\hat{\mathsf{m}})\big] + (jr_1)^{-5}\big[\mathsf{U}(\hat{\mathsf{m}}) - \mathsf{U}(O\hat{\mathsf{m}})\big]\Big| \le 2C(jr_1)^{-6}\lg(jr_1).$$

Multiplying by $(jr_1)^4$ and letting $j \to \infty$ we obtain $\mathsf{I}(\hat{\mathsf{m}}) = \mathsf{I}(O\hat{\mathsf{m}})$; then, multiplying the remaining difference by $(jr_1)^5$, we obtain $\mathsf{U}(\hat{\mathsf{m}}) = \mathsf{U}(O\hat{\mathsf{m}})$. The directions $\mathsf{m}/|\mathsf{m}|$ are dense in S^2, since $\mathbb{Q}^3$ is dense in $\mathbb{R}^3$. The rational rotations are dense in $SO(3)$: the Cayley map $\mathsf{A} \mapsto (\mathbf{1}-\mathsf{A})(\mathbf{1}+\mathsf{A})^{-1}$ is continuous from the real antisymmetric 3×3 matrices onto the rotations without eigenvalue -1, which are dense in $SO(3)$; the antisymmetric matrices with rational entries are dense among the real ones, and by the formula they are mapped to rotations with rational entries, since the inverse of an invertible matrix with rational entries has rational entries.

By continuity, I and U are invariant under all rotations; as $SO(3)$ acts transitively on S^2, they are constant on S^2, and homogeneity gives the claim. For the last assertion we note that $\mathsf{T}_\Gamma^{\mathrm{mod}}$ is the restriction to $\mathbb{Z}^3$ of a radial function, since Ψ_Γ is radial (Def. 7.32(3)); thus $\mathsf{T}_\Gamma^{\mathrm{mod}}(u) = \mathsf{t}(|u|)$ for a function t, and Lem. 7.35 provides the hypothesis. □

We can now describe $\mathsf{B}_1^{\mathrm{lat}}(p,q)$ at large distances of the two plaquettes. The local parts of the elementary terms drop out by the zero-momentum identities of Lem. 7.31, and the parts of degree -5 add up, by Lems. 7.35 and 7.36, to a multiple of $|u|^{-5}$.

Theorem V (The kernel of degree -5: large-distance asymptotics of $\mathsf{B}_1^{\mathrm{lat}}$)**.** There are functions $\mathsf{U}_{ij} \in C^1(\mathbb{R}^3 \setminus \{0\})$, $i,j \in \{1,2,3\}$, homogeneous of degree -5, and pure constants C, r_0 such that for all plaquettes p, q of $\mathbb{Z}^3$, with $u := x(q) - x(p)$ and $r := |u| \ge r_0$,

$$\begin{aligned} |\mathsf{B}_1^{\mathrm{lat}}(p,q) - \mathsf{U}_{\lambda(p)\lambda(q)}(u)| &\le C\,r^{-6}\lg r, \\ \sum_{i,j} \mathsf{U}_{ij}(u) = \kappa^{\mathrm{lat}}|u|^{-5}, &\qquad \kappa^{\mathrm{lat}} := \sum_{\Gamma=1}^{6} \kappa_\Gamma^{\mathrm{lat}}, \end{aligned} \tag{7.xxxvii}$$

with the numbers $\kappa_\Gamma^{\mathrm{lat}}$ of Lem. 7.36. Moreover $\mathsf{B}_1^{\mathrm{lat}}$ is bounded. ■□

Proof. We work at $\tau_{\mathrm{gf}} = 1$; recall that $\mathsf{B}_1^{\mathrm{lat}}(p,q) = \mathsf{B}_1^{\mathrm{lat}}(p,q;1)$ by eq. 7.xix. Boundedness is contained in Lem. 7.27, which gives $\sup_{p,q} |\mathsf{B}_1^{\mathrm{lat}}(p,q;\tau_{\mathrm{gf}})| < \infty$ for every $\tau_{\mathrm{gf}} > 0$. Let $p = p_{[\iota]}(x_p)$, $q = p_{[\iota']}(x_q)$ and $u' := x_q - x_p$; then $|u - u'| \le \sqrt{2}$, and we take r large. By Lem. 7.26(c) and Lem. 7.28, $\mathsf{B}_1^{\mathrm{lat}}(p,q)$ is the sum of T_{I} and of finitely many elementary terms, to which Lem. 7.29 applies.

Step 1 (local terms at q). These are the first half of $\mathsf{T}_{4'}$ and the terms with q_Y in the first half of T_5, with q_Y in T_4 and with q_{q} in T_1, in the expansions of Lem. 7.29. By Lem. 7.31(b), and since $\mathsf{E}_p((x_q + y, \mu)) = K^{\rightarrow}_{\iota\mu}(u' + y)$ by translation invariance, their sum equals

$$\tfrac{\mathsf{n}}{2} \sum_{b,b'} \gamma_q(b,b')\, \mathsf{E}_p(b)\, \mathsf{E}_p(b') = \tfrac{\mathsf{n}}{2} \sum_{\mu,\nu} \sum_{y,y'} \gamma_{q;\mu\nu}(y,y')\, K^{\rightarrow}_{\iota\mu}(u' + y)\, K^{\rightarrow}_{\iota\nu}(u' + y').$$

The kernels $\gamma_{q;\mu\nu}$ have finite support, and their symbols vanish at $\kappa = 0$ together with their gradients (Lem. 7.31(b)); the $K^{\rightarrow}_{\iota\mu}$ are elementary lines of degree 2. Lem. 7.30(a), applied with $\gamma_{\mu\nu} := \gamma_{q;\mu\nu}$ and $K_\mu := K^{\rightarrow}_{\iota\mu}$, shows that the sum is $O(|u'|^{-6}) = O(r^{-6})$. By the translation invariance in Lem. 7.28, the kernels $\gamma_{q;\mu\nu}$ depend on q only through ι', so that only finitely many constants occur.

Step 2 (local terms at p). These are the second half of $\mathsf{T}_{4'}$ and the terms with q_Y in the second half of T_5, with q_X in T_4 and with q_{p} in T_1. We show that they are the local terms at the second plaquette of the ordered pair (q,p); by Step 1, applied to this pair, their sum is then $\frac{\mathsf{n}}{2} \sum_{b,b'} \gamma_p(b,b')\, \mathsf{E}_q(b)\, \mathsf{E}_q(b') = O(r^{-6})$. For the second halves of $\mathsf{T}_{4'}$ and T_5 this holds because they are the first halves for (q,p) (Lem. 7.28). The formulas for T_1 and T_4 in Lem. 7.26(c) are symmetric under $p \leftrightarrow q$.

An elementary term G of T_1 or T_4 is given by the monomials at P, P' and by the choice of the legs contracted with p, with q, and with one another; taken for the ordered pair (q,p), it becomes a term $\tilde{G}$ with $c_{\tilde{G}} = c_G$, and $G \mapsto \tilde{G}$ is a bijection $\mathfrak{G}_\bullet(\iota,\iota') \to \mathfrak{G}_\bullet(\iota',\iota)$. In $\tilde{G}$ the base points are counted from x_q, u' is replaced by $-u'$, and the contractions, which are symmetric in their two legs, are taken from the other end: for the kite

$$\tilde{K}_1 = K_4(-\cdot), \quad \tilde{K}_2 = K_5(-\cdot), \quad \tilde{K}_3 = K_3, \quad \tilde{K}_4 = K_1(-\cdot), \quad \tilde{K}_5 = K_2(-\cdot),$$

and for T_4 we have $\tilde{X} = Y(-\cdot)$, $\tilde{Y} = X(-\cdot)$. Since $K(-\cdot)$ has the symbol $\sigma(-k)$, the function $\hat{\mathsf{T}}_{\mathsf{q}}$ of $\tilde{G}$ is

$$\int_{\mathbb{T}} \sigma_1(-\kappa_1 + l)\, \sigma_2(-\kappa_2 - l)\, \sigma_3(l)\, d^\circ l = \hat{\mathsf{T}}_{\mathsf{p}}(-\kappa),$$

and the symbol of $\tilde{Y}$ is $\mathsf{I}_X(-k)$, after the substitution $l \to -l$. By the last assertion of Lem. 7.20(d), whose hypothesis holds here for all κ, the corresponding polynomials are $\mathsf{Q}_{\mathsf{p}}(-\kappa)$ and $\mathsf{Q}_X(-k)$. The passage from a polynomial Q to $\tilde{\mathsf{Q}}(\kappa) = \mathsf{Q}(0) + \sum_j \partial_j \mathsf{Q}(0) \sin \kappa_j$ in Prop. 7.23 and Lem. 7.22(c) commutes with the change of sign of the argument, the sine being odd. Hence the kernel q_{q} of $\tilde{G}$ is $(z,w) \mapsto \mathsf{q}_{\mathsf{p}}(-z,-w)$, and $\mathsf{q}_{\tilde{Y}} = \mathsf{q}_X(-\cdot)$, so that

$$\sum_{z,w} \mathsf{q}_{\mathsf{p}}(-z,-w)\, \tilde{K}_1(-u' - z)\, \tilde{K}_2(-u' - w) = \sum_{z,w} \mathsf{q}_{\mathsf{p}}(z,w)\, K_4(u' - z)\, K_5(u' - w),$$

$$(\mathsf{q}_{\tilde{Y}} * \tilde{X})(-u') = (\mathsf{q}_X * Y)(u').$$

These are the local terms at p in T_1 and in T_4.

Step 3 (the class T_{I}). By Lem. 7.31(a),

$$K^{\rightarrow} * \Sigma_{\mu\nu} * K^{\leftarrow} = K^{\rightarrow} * \mathsf{q}^{\Sigma}_{\mu\nu} * K^{\leftarrow} + K^{\rightarrow} * \mathsf{W}_{\mu\nu} * K^{\leftarrow}.$$

The first term is $O(r^{-3})$ by Lem. 7.30(b), since $\hat{\mathsf{q}}^{\Sigma}_{\mu\nu}(k) = O(|k|^2)$. The second lies in $\mathfrak{K}(2)$, by Lem. 7.22(b) applied to each of its two convolutions, and it equals its leading kernel, of degree -2, up to $O(r^{-3} \lg r)$; by the linearity of the leading data, this leading kernel is the sum over G, with the coefficients c_G, of the leading kernels of the chains in the definition of U_G in Lem. 7.35. Moreover

$\Pi(p,q) = \Pi^h(u') + O(r^{-4})$, with Π^h homogeneous of degree -3 (Lem. 7.24, Lem. 7.17(c)). Therefore

$$\mathsf{T}_{\mathrm{I}} = \sum_{(\mu,\nu,G)\in\mathfrak{G}_2\cup\mathfrak{G}_3} (-\mathsf{n}\, c_G)\, \mathsf{U}_G(u') + O(r^{-6}\lg r).$$

Step 4 (assembly). Let

$$\mathsf{U}_{\iota\iota'}(u) := \sum_{\Gamma\neq 5} \sum_{G\in\mathfrak{G}_\Gamma(\iota,\iota')} c_G\, \mathsf{U}_G(u) + \sum_{G\in\mathfrak{G}_5(\iota,\iota')} c_G\, \mathsf{U}_G(u) + \sum_{G\in\mathfrak{G}_5(\iota',\iota)} c_G\, \mathsf{U}_G(-u),$$

with the coefficients $-\mathsf{n}\, c_G$ for $\Gamma = 2,3$; the last sum comes from the second half of T_5, which is the first half for (q,p) with u' replaced by $-u'$. The function $\mathsf{U}_{\iota\iota'}$ is homogeneous of degree -5 and of class C^1 off 0, by Lem. 7.29 and since the leading kernels of the classes $\mathfrak{K}(n)$ are smooth off 0. By Steps 1–3 and Lem. 7.29, $\mathsf{B}_1^{\mathrm{lat}}(p,q) = \mathsf{U}_{\iota\iota'}(u') + O(r^{-6}\lg r)$. The gradient of $\mathsf{U}_{\iota\iota'}$ is homogeneous of degree -6, and all points $u' + t(u-u')$, $0 \le t \le 1$, have modulus at least $r/2$; hence $|\mathsf{U}_{\iota\iota'}(u') - \mathsf{U}_{\iota\iota'}(u)| \le C r^{-6}$.

With $\mathsf{U}_{ij} := \mathsf{U}_{\iota\iota'}$ for $(i,j) = (\iota,\iota') = (\lambda(p),\lambda(q))$ we obtain the bound in eq. 7.xxxvii; as $\tau_{\mathrm{gf}} = 1$, the finite sets $\mathfrak{G}_\bullet(\iota,\iota')$, the coefficients c_G and the lines are fixed once and for all, the constants C, r_0 are pure. Finally, exchanging ι and ι' in the last sum and using Lems. 7.35 and 7.36,

$$\sum_{\iota,\iota'} \mathsf{U}_{\iota\iota'} = \sum_{\Gamma=1}^{6} \mathsf{U}_\Gamma = \sum_{\Gamma=1}^{6} \kappa_\Gamma^{\mathrm{lat}}\, |u|^{-5}.$$

□

For the smeared observables of eq. 7.xix the theorem gives the following rate of convergence on test pairs. Here $\mathfrak{T}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$ is the set of test pairs at the scale d with the shape parameter ϑ defined in §6.1, with $\mathbb{E} = \mathbb{R}^3$.

Corollary 7.37 (Rate of convergence of $\mathsf{P}_1^{\mathrm{lat}}$ to $\kappa^{\mathrm{lat}}\mathcal{K}_5$)**.** There is a pure constant C such that the following holds for all $\mathsf{d} > 0$, $\vartheta \in (0,1)$, $(f,h) \in \mathfrak{T}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$ and $0 < a \le \mathsf{d}$. For the kernel of §6.1, at $q = 5$,

$$\mathcal{K}_5(f,h) = \iint f(x)\, h(y)\, |x-y|^{-5}\, \mathrm{d}x\, \mathrm{d}y,$$

the integrand being 0 where $f(x)h(y) = 0$. Then

$$|\mathsf{P}_1^{\mathrm{lat}}(f,h;a) - \kappa^{\mathrm{lat}}\, \mathcal{K}_5(f,h)| \le C\, \vartheta^{-1}\, a\, \lg(\mathsf{d}/a)\, \|f\|_\infty \|h\|_\infty. \tag{7.xxxviii}$$

■□

We recall three properties of a test pair $(f,h) \in \mathfrak{T}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$: the supports of f and h lie in closed balls $\mathbb{B}(x_f,\mathsf{d})$ and $\mathbb{B}(x_h,\mathsf{d})$ of radius d; $\mathsf{d} \le |v-w| \le 10\,\mathsf{d}$ for $v \in \operatorname{supp} f$ and $w \in \operatorname{supp} h$ (the support separation in §6.1); and $\operatorname{Lip} f \le \|f\|_\infty/\vartheta\mathsf{d}$, $\operatorname{Lip} h \le \|h\|_\infty/\vartheta\mathsf{d}$ (by the definition of that test class).

Proof. Let $\varrho := \mathsf{d}/a \ge 1$, $\mathsf{A} := \|f\|_\infty\|h\|_\infty$ and $\varrho_0 := \max\{r_0, 8\}$, with r_0 of Thm. V. A ball of radius $\varrho + 2$ contains at most $C\varrho^3$ points of $\frac{1}{2}\mathbb{Z}^3$. The barycentres $x(p)$ lie in $\frac{1}{2}\mathbb{Z}^3$, distinct plaquettes have distinct barycentres, and $f(ax(p)) \neq 0$ implies $x(p) \in \mathbb{B}(x_f/a, \varrho)$; likewise for h.

Step 1 ($\varrho \le \varrho_0$). Then the sum defining $\mathsf{P}_1^{\mathrm{lat}}(f,h;a)$ has at most C non-zero terms, and since $\mathsf{B}_1^{\mathrm{lat}}$ is bounded (Thm. V), $|\mathsf{P}_1^{\mathrm{lat}}(f,h;a)| \le Ca\mathsf{A}$. Moreover $|x-y| \ge \mathsf{d}$ where $f(x)h(y) \neq 0$, whence $|\mathcal{K}_5(f,h)| \le (\frac{4\pi}{3}\mathsf{d}^3)^2\mathsf{d}^{-5}\mathsf{A} \le C\varrho_0 a\mathsf{A}$. As $\vartheta^{-1} > 1$ and $\lg(\mathsf{d}/a) \ge 1$, eq. 7.xxxviii holds in this case.

Step 2 ($\varrho > \varrho_0$, first error). If $f(ax(p))h(ax(q)) \neq 0$, then $r = |x(q) - x(p)| \in [\varrho, 10\varrho]$ by the separation of the supports; in particular $r \ge r_0$. There are at most $C\varrho^6$ such pairs, and $r^{-6}\lg r \le$

$\varrho^{-6}\lg(10\varrho)$ for these r. By eq. 7.xxxvii, the replacement of $\mathsf{B}_1^{\mathrm{lat}}(p,q)$ by $\mathsf{U}_{\lambda(p)\lambda(q)}(x(q)-x(p))$ in $\mathsf{P}_1^{\mathrm{lat}}(f,h;a)$ costs at most

$$a\cdot C\varrho^6\cdot C\varrho^{-6}\lg(10\varrho)\,\mathsf{A}\le Ca\lg\varrho\,\mathsf{A}.$$

Step 3 (second error). We write $p=p_{[i]}(x)$, $q=p_{[j]}(y)$, so that $x(p)=x+\mathsf{o}_i$ with $\mathsf{o}_i:=x(p_{[i]}(0))$, $|\mathsf{o}_i|=2^{-1/2}$; all terms below are set equal to 0 where the product of the values of f and h vanishes. After Step 2 we have to compare

$$\mathfrak{R}_1:=a\sum_{i,j}\sum_{x,y\in\mathbb{Z}^3}f(a(x+\mathsf{o}_i))\,h(a(y+\mathsf{o}_j))\,\mathsf{U}_{ij}(y-x+\mathsf{o}_j-\mathsf{o}_i)$$

with

$$\mathfrak{R}_2:=a\sum_{x,y}f(ax)\,h(ay)\sum_{i,j}\mathsf{U}_{ij}(y-x)=a\,\kappa^{\mathrm{lat}}\sum_{x,y}\frac{f(ax)\,h(ay)}{|y-x|^5},$$

the last equality by eq. 7.xxxvii. We replace f, h and U_{ij} one after the other. In every non-zero term, the arguments of U_{ij} which occur, and the points between them, have modulus at least $\varrho-2\ge\varrho/2$, and x,y lie in balls of radius $\varrho+1$. By homogeneity, $|\mathsf{U}_{ij}|\le C\varrho^{-5}$ and $|\nabla\mathsf{U}_{ij}|\le C\varrho^{-6}$ at these points, and the Lipschitz bound gives $|f(a(x+\mathsf{o}_i))-f(ax)|\le a\|f\|_\infty/\vartheta\mathsf{d}$, and likewise for h. Therefore

$$|\mathfrak{R}_1-\mathfrak{R}_2|\le a\,C\varrho^6\Big[\frac{a}{\vartheta\mathsf{d}}\,\varrho^{-5}+\varrho^{-6}\Big]\mathsf{A}\le Ca\vartheta^{-1}\mathsf{A}.$$

Step 4 (third error). Let $\mathsf{F}(v,w):=f(v)h(w)|v-w|^{-5}$, equal to 0 where $f(v)h(w)=0$. Then $\mathfrak{R}_2=\kappa^{\mathrm{lat}}\sum_{x,y}a^6\mathsf{F}(ax,ay)$ and $\mathcal{K}_5(f,h)=\iint\mathsf{F}$. Let $(v,w)\in(ax,ay)+[0,a)^6$. Unless F vanishes at both points (v,w) and (ax,ay), we have $|v-w|\ge\mathsf{d}-2\sqrt{3}a\ge\mathsf{d}/2$ at both points and at all points $(1-t)(ax,ay)+t(v,w)$, $0\le t\le1$ (recall $\varrho>8$), and the Lipschitz bounds for f,h together with the bound $5|z|^{-6}$ for the gradient of $|z|^{-5}$ give

$$|\mathsf{F}(v,w)-\mathsf{F}(ax,ay)|\le\sqrt{3}a\Big[\frac{2}{\vartheta\mathsf{d}}\Big(\frac{\mathsf{d}}{2}\Big)^{-5}+10\Big(\frac{\mathsf{d}}{2}\Big)^{-6}\Big]\mathsf{A}.$$

At most $C\varrho^6$ of the cubes $(ax,ay)+[0,a)^6$, of volume a^6, contain points where $\mathsf{F}\ne0$. Hence

$$|\mathfrak{R}_2-\kappa^{\mathrm{lat}}\mathcal{K}_5(f,h)|\le C\varrho^6a^6\cdot a\vartheta^{-1}\mathsf{d}^{-6}\mathsf{A}=Ca\vartheta^{-1}\mathsf{A}.$$

Adding the bounds of Steps 2–4 and using $\vartheta^{-1}>1$, $\lg\varrho\ge1$, we obtain eq. 7.xxxviii. □

7.1.13. *The boxes exhaust the lattice.* We now show that the coefficients $\mathsf{B}_j^{[n]}$ of the boxes converge to the coefficients $\mathsf{B}_j^{\mathrm{lat}}$ of the lattice $\mathbb{Z}^3$; together with Prop. 7.12, this shows that $\mathsf{B}_1^{\mathrm{lat}}$ is independent of the gauge parameter. The kernels of a box are expressed by the method of images through kernels on a discrete torus, which we first bound uniformly in the size of the torus.

Lemma 7.38 (Kernels on discrete tori, uniformly)**.** For an integer $n\ge2$ let $\mathbb{T}_n:=\frac{\pi}{n}\mathbb{Z}^3\cap(-\pi,\pi]^3$; for $z\in\mathbb{Z}^3$ let $|z|_n$ be the least norm of the points of $z+2n\mathbb{Z}^3$, and $\langle z\rangle_n:=\max\{1,|z|_n\}$. For real $s>-3$ and $\sigma\in\mathcal{S}^s$ let

$$K^{(n)}(z):=(2n)^{-3}\sum_{k\in\mathbb{T}_n\setminus\{0\}}e^{ik\cdot z}\,\sigma(k)\qquad(z\in\mathbb{Z}^3).$$

(a) $|K^{(n)}(z)|\le C\langle z\rangle_n^{-3-s}$ for all $n\ge2$ and $z\in\mathbb{Z}^3$, where C depends on s and on finitely many of the suprema in Def. 7.16(2), but not on n and z.

(b) For every $z\in\mathbb{Z}^3$, $K^{(n)}(z)\to K(z)=\int_{\mathbb{T}}e^{ik\cdot z}\sigma(k)\,d^\circ k$ as $n\to\infty$. ■

Proof. Let $\mathsf{h}:=\pi/n$, so that $(2n)^{-3}=(2\pi)^{-3}\mathsf{h}^3$.

Step 0. If $0 \neq k \in \mathsf{h}\mathbb{Z}^3$ and k' lies in the cube of side h centred at k, then $(1-\frac{\sqrt{3}}{2})|k| \le |k'| \le (1+\frac{\sqrt{3}}{2})|k|$, as $|k'-k| \le \frac{\sqrt{3}}{2}\mathsf{h} \le \frac{\sqrt{3}}{2}|k|$. Comparing every term with the integral over its cube, we obtain

$$(2n)^{-3} \sum_{k\in\mathsf{h}\mathbb{Z}^3,\ 0<|k|\le\varrho} |k|^t \le C\varrho^{3+t} \quad (t>-3),$$
$$(2n)^{-3} \sum_{k\in\mathsf{h}\mathbb{Z}^3,\ |k|\ge\varrho} |k|^t \le C\varrho^{3+t} \quad (t<-3),$$

with C depending on t only.

Step 1 (proof of (b)). Let $0<\varepsilon<1$, and split $\sigma = \chi(k/\varepsilon)\sigma + (1-\chi(k/\varepsilon))\sigma$ on $[-\pi,\pi]^3$. The first summand is bounded by $C|k|^s$ and vanishes for $|k| \ge 2\varepsilon$; by Step 0 its contributions to $K^{(n)}(z)$ and to $K(z)$ are $O(\varepsilon^{3+s})$, uniformly in n. The second summand, set equal to 0 at $k=0$, is continuous on $[-\pi,\pi]^3$; its contribution to $K^{(n)}(z)$ is a Riemann sum over the $(2n)^3$ subcubes of side h, which converges to its contribution to $K(z)$. Letting first $n\to\infty$ and then $\varepsilon\to 0$ we obtain (b).

Step 2 (proof of (a)). Since $K^{(n)}$ is $2n$-periodic in each coordinate, we may take the representative of z with $|z| = |z|_n$; then $|z_j| \le n$ for all j. For $z=0$ the bound follows from Step 0. Let $|z| \ge 1$, and write $\sigma = \sigma' + \sigma''$ with $\sigma' := \chi(|z|k)\sigma$ on $[-\pi,\pi]^3$. As σ' is bounded by $C|k|^s$ and vanishes for $|k| \ge 2/|z|$, Step 0 shows that the contribution of σ' is at most $C|z|^{-3-s}$. The function σ'' is smooth and periodic, it vanishes where $|k|_{\mathbb{T}} \le 1/|z|$, and $|\partial^\alpha \sigma''(k)| \le C|k|_{\mathbb{T}}^{s-|\alpha|}$, because every derivative falling on $\chi(|z|k)$ produces a factor $|z| \le 2/|k|$ on its support. Since $\mathbb{T}_n$ represents $\mathsf{h}\mathbb{Z}^3/2\pi\mathbb{Z}^3$, we have $\sum_k e^{ik\cdot z}\sigma''(k-\mathsf{h}e_j) = e^{i\mathsf{h}z_j}\sum_k e^{ik\cdot z}\sigma''(k)$; with $(\delta_j \mathsf{F})(k) := \mathsf{F}(k) - \mathsf{F}(k-\mathsf{h}e_j)$ it follows that

$$(1-e^{i\mathsf{h}z_j})^N \sum_{k\in\mathbb{T}_n} e^{ik\cdot z}\,\sigma''(k) = \sum_{k\in\mathbb{T}_n} e^{ik\cdot z}\,\delta_j^N\sigma''(k)$$

for every integer $N \ge 1$; the term $k=0$ vanishes, as $\sigma''(0)=0$. Here $|1-e^{i\mathsf{h}z_j}| = 2|\sin\frac{\pi z_j}{2n}| \ge 2|z_j|/n$, since $\sin t \ge \frac{2}{\pi}t$ on $[0,\frac{\pi}{2}]$ and $|z_j| \le n$; moreover $|\delta_j^N\sigma''(k)| \le \mathsf{h}^N \sup_{0\le t\le N\mathsf{h}} |\partial_j^N \sigma''(k-te_j)|$. We fix an integer $N > s+3$; as $|z| \le \sqrt{3}n$, we have $N\mathsf{h} \le C/|z|$. For $|k| \ge 2N\mathsf{h}$ the points $k-te_j$, $0\le t\le N\mathsf{h}$, satisfy $|k-te_j|_{\mathbb{T}} \in [\frac{1}{2}|k|, \frac{3}{2}|k|]$; they may leave $[-\pi,\pi]^3$, but their distance from $2\pi\mathbb{Z}^3$ equals $|k|$ at $t=0$ and changes by at most $t \le \frac{1}{2}|k|$. Hence for these k the supremum is at most $C|k|^{s-N}$, and it vanishes unless $|k| \ge \frac{2}{3|z|}$. For $|k| < 2N\mathsf{h}$ the supremum is at most $C|z|^{N-s}$, as σ'' vanishes where $|k|_{\mathbb{T}} \le 1/|z|$ and $s-N<0$; there are at most C such points k, and $(2n)^{-3} \le C|z|^{-3}$. Together with Step 0 we find

$$(2n)^{-3} \sum_{k\in\mathbb{T}_n} |\delta_j^N\sigma''(k)| \le C\mathsf{h}^N|z|^{N-s-3}.$$

Choosing j with $|z_j| \ge |z|/\sqrt{3}$, we conclude that the contribution of σ'' is at most

$$(n/2|z_j|)^N(\pi/n)^N\, C|z|^{N-s-3} \le C|z|^{-3-s}.$$

This proves (a). □

Proposition 7.39 (The boxes exhaust the lattice)**.** Let $\tau_{\mathrm{gf}} > 0$, let p, q be plaquettes of $\mathbb{Z}^3$ and $j \in \{0,1\}$, and write $\mathsf{B}_0^\Omega(p,q;\tau_{\mathrm{gf}}) := \mathsf{B}_0^\Omega(p,q)$. The numbers $\mathsf{B}_j^{[n]}(p+o_n, q+o_n;\tau_{\mathrm{gf}})$ are defined for all $n \ge 2$ with $p+o_n, q+o_n \in \mathsf{Pl}_n$, that is, for all but finitely many n, and

$$\lim_{n\to\infty} \mathsf{B}_j^{[n]}(p+o_n, q+o_n;\tau_{\mathrm{gf}}) = \mathsf{B}_j^{\mathrm{lat}}(p,q;\tau_{\mathrm{gf}}). \tag{7.xxxix}$$

■

Proof. We use the method of images. Fix n and write $\tau := \tau_{\mathrm{gf}}$.

Step 1 (reflections). We regard forms and functions on the torus $(\mathbb{Z}/2n)^3$ as $2ne_\nu$-periodic forms and functions on $\mathbb{Z}^3$, and we write $(\cdot,\cdot)_{\mathsf{t}}$ for the sums over one period. Let ϱ_ν be the reflection

$z_\nu \to -z_\nu$, acting on bonds by $\varrho_\nu\langle z,z'\rangle = \langle \varrho_\nu z, \varrho_\nu z'\rangle$, and put $(\varrho_\nu^* \tilde A)(b) := \tilde A(\varrho_\nu b)$ and $\varrho_\nu^*\lambda := \lambda\circ\varrho_\nu$; these maps commute with d and d^*. Let F^{as} and X^{as} be the spaces of real periodic forms $\tilde A$ and functions λ with $\varrho_\nu^*\tilde A = -\tilde A$ and $\varrho_\nu^*\lambda = -\lambda$ for all ν; in components, $\tilde A_\mu(\varrho_\nu z) = -\tilde A_\mu(z)$ for $\nu\neq\mu$, and $\tilde A_\mu(\varrho_\mu z - e_\mu) = \tilde A_\mu(z)$.

We claim that restriction to the box is a bijection of F^{as} onto $\mathsf{Form}_n^{\mathbb{R}}$ and of X^{as} onto $\mathsf{X}_n^{\mathbb{R}}$. If $z_\nu\in\{0,n\}$, then $\varrho_\nu z \equiv z$ modulo $2n$, hence $\tilde A_\mu(z)=0$ for $\nu\neq\mu$, and $\lambda(z)=0$; by Lem. 7.6(a), $\tilde A$ vanishes on the non-free bonds of the box, and λ vanishes on $\partial\Lambda_n$. For $\varepsilon\in\{0,1\}^3$ let $R_\varepsilon := \prod_\nu \varrho_\nu^{\varepsilon_\nu}$, $|\varepsilon| := \sum_\nu \varepsilon_\nu$, and $R_\varepsilon(z,\mu) := (R_\varepsilon z - \varepsilon_\mu e_\mu, \mu)$, the positively oriented one among the image bond and its reverse.

Every positively oriented bond of the torus has the form $R_\varepsilon(x,\mu)$ with $x_\mu\in\{0,\dots,n-1\}$ and $x_\nu\in\{0,\dots,n\}$ for $\nu\neq\mu$; here (x,μ) is unique, and ε is unique if (x,μ) is free. Hence every $A\in\mathsf{Form}_n^{\mathbb{R}}$ has the unique continuation $\tilde A(R_\varepsilon b) = (-1)^{|\varepsilon|+\varepsilon_\mu}A(b)$, $b=(x,\mu)$, and every $\lambda\in\mathsf{X}_n^{\mathbb{R}}$ the unique continuation $\tilde\lambda(R_\varepsilon x) = (-1)^{|\varepsilon|}\lambda(x)$.

Step 2 (operators). The eight boxes $R_\varepsilon[0,n]^3$ cover the torus, and a plaquette lying in two of them lies in a hyperplane $\{z_\nu=0\}$ or $\{z_\nu=n\}$, on whose bonds $\tilde A=0$. The functions $|\mathrm{d}\tilde A|$ and $|\mathrm{d}^*\tilde A|$ are invariant under the ϱ_ν, and $\mathrm{d}^*\tilde A\in\mathsf{X}^{\mathrm{as}}$ vanishes on these hyperplanes. Hence

$$(\tilde A,\tilde A')_{\mathsf{t}} = 8(A,A'),\qquad \|\mathrm{d}\tilde A\|_{\mathsf{t}}^2 = 8\|\mathrm{d}A\|^2,\qquad \|\mathrm{d}^*\tilde A\|_{\mathsf{t}}^2 = 8\|\mathrm{d}^*A\|^2.$$

The operator $\mathsf{L}_\tau := \mathrm{d}^*\mathrm{d} + \tau^{-1}\mathrm{d}\mathrm{d}^*$ on the torus commutes with the ϱ_ν^* and preserves F^{as}. By polarisation, $(\tilde A',\mathsf{L}_\tau\tilde A)_{\mathsf{t}} = 8(A',\mathsf{L}_\tau^{[n]}A)$ for all $A,A'\in\mathsf{Form}_n^{\mathbb{R}}$, where $\mathsf{L}_\tau^{[n]}$ is the operator on $\mathsf{Form}_n^{\mathbb{R}}$ of the quadratic form in Def. 7.7(1); hence $\mathsf{L}_\tau\tilde A$ is the continuation of $\mathsf{L}_\tau^{[n]}A$. The forms in F^{as} have mean 0, and on periodic forms of mean 0 the inverse of L_τ is, by the finite Fourier inversion formula, the convolution with

$$\mathsf{C}_{\mu\nu}^{(n)}(z) := (2n)^{-3}\sum_{k\in\mathbb{T}_n\setminus\{0\}} e^{ik\cdot z}\,\hat{\mathsf{C}}_{\mu\nu}(k).$$

The continuation of $\delta_{b'}$, for a free bond $b'=(x',\mu')$, is $\sum_\varepsilon(-1)^{|\varepsilon|+\varepsilon_{\mu'}}\delta_{R_\varepsilon b'}$. For functions we have in the same way $(\tilde\lambda,\tilde\lambda')_{\mathsf{t}} = 8(\lambda,\lambda')$ and $\|\mathrm{d}\tilde\lambda\|_{\mathsf{t}}^2 = 8\|\mathrm{d}\lambda\|^2$, since $\mathrm{d}\tilde\lambda\in\mathsf{F}^{\mathrm{as}}$ is the continuation of $\mathrm{d}\lambda$; the elements of X^{as} have mean 0, and the continuation of $\delta_{x'}$, $x'\in\Lambda_n^\circ$, is $\sum_\varepsilon(-1)^{|\varepsilon|}\delta_{R_\varepsilon x'}$. By polarisation, as for forms, $\mathrm{d}^*\mathrm{d}\tilde\lambda$ is the continuation of $\mathrm{d}^*\mathrm{d}\lambda$, and on periodic functions of mean 0 the inverse of $\mathrm{d}^*\mathrm{d}$ is the convolution with $\mathsf{G}^{(n)}(z) := (2n)^{-3}\sum_{k\in\mathbb{T}_n\setminus\{0\}} e^{ik\cdot z}|\mathsf{k}(k)|^{-2}$.

We apply this to $B := (\mathsf{L}_\tau^{[n]})^{-1}\delta_{b'}$, whose value at a free bond b is $\mathsf{C}_\tau^{[n]}(b,b')$; its continuation $\tilde B$ has mean 0 and satisfies $\mathsf{L}_\tau\tilde B = \tilde\delta_{b'}$, the continuation of $\delta_{b'}$, so that $\tilde B$ is the convolution of $\mathsf{C}^{(n)}$ with $\tilde\delta_{b'}$. In the same way we treat $(\mathrm{d}^*\mathrm{d})^{-1}\delta_{x'}$ on $\mathsf{X}_n^{\mathbb{R}}$, whose value at $x\in\Lambda_n^\circ$ is $\mathsf{G}^{[n]}(x,x')$. Consequently, for free bonds $(x,\mu),(x',\mu')$ and for $x,x'\in\Lambda_n^\circ$,

$$\begin{aligned}\mathsf{C}_\tau^{[n]}((x,\mu),(x',\mu')) &= \sum_\varepsilon(-1)^{|\varepsilon|+\varepsilon_{\mu'}}\,\mathsf{C}_{\mu\mu'}^{(n)}(x-R_\varepsilon x'+\varepsilon_{\mu'}e_{\mu'}),\\ \mathsf{G}^{[n]}(x,x') &= \sum_\varepsilon(-1)^{|\varepsilon|}\,\mathsf{G}^{(n)}(x-R_\varepsilon x').\end{aligned}\tag{7.xl}$$

In the first argument the right sides are values of elements of F^{as}, resp. X^{as}, and therefore vanish at non-free bonds, resp. on $\partial\Lambda_n$; in the second argument the same follows by symmetry. Since the box kernels vanish there by definition, eq. 7.xl holds for all bonds and sites of Λ_n.

Step 3 (uniform bounds). The kernels $\mathsf{E}^{[n]},\Pi^{[n]},\mathsf{D}^{[n]},\mathsf{g}^{[n]}$ arise from eq. 7.xl by the differences and means of Def. 7.7(2). A difference in the first argument acts on $\mathsf{C}^{(n)}$ and $\mathsf{G}^{(n)}$. In the second argument, R_ε maps the four bonds of q onto those of $R_\varepsilon q$, and $\sum_j s_j(-1)^{\varepsilon_{\mu_j}}\delta_{R_\varepsilon b_j(q)} = \pm\sum_j s_j\delta_{b_j(R_\varepsilon q)}$, with μ_j the direction of $b_j(q)$, since d commutes with the ϱ_ν^*; the same holds for the end points of a bond. Hence each of these kernels is a sum of eight terms $\pm K^{(n)}(x-R_\varepsilon x'+y_0)$ with $|y_0|\le 3$, where x and x' are the base points of its two arguments, as in eq. 7.xxxi (the base point b_- of a bond b, the base

point x_p of a plaquette p; for $\mathsf{g}^{[n]}(b)$ both are b_-), and where $K^{(n)}$ is built as in Lem. 7.38 with the symbol of the corresponding line of Lem. 7.24, an element of $\mathcal{S}^{n_\ell-3}$, n_ℓ being the degree of the line.

For $x, x' \in \Lambda_n$ we have $|x - R_\varepsilon x'|_n \geq |x - x'|$: if $\varepsilon_\nu = 0$, then $|x_\nu - x'_\nu| \leq n$, and if $\varepsilon_\nu = 1$, then $\min\{x_\nu + x'_\nu, 2n - x_\nu - x'_\nu\} \geq |x_\nu - x'_\nu|$. By Lem. 7.38(a), the kernels of all boxes therefore satisfy the bounds eq. 7.xxxi with one constant $\mathsf{c}_0 = \mathsf{c}_0(\tau_{\mathrm{gf}})$.

Step 4 (pointwise limits). The arguments are translates by o_n of fixed bonds, sites and plaquettes. For $\varepsilon = 0$ the term $K^{(n)}(x - x' + y_0)$ has a fixed argument and converges to the corresponding kernel of $\mathbb{Z}^3$, by Lem. 7.38(b) and Lem. 7.24. If $\varepsilon_\nu = 1$, the ν-th coordinate of the argument is $\tilde{x}_\nu + \tilde{x}'_\nu + 2\lfloor n/2 \rfloor + O(1)$, where $\tilde{x} = x - o_n$ and $\tilde{x}' = x' - o_n$ are fixed; its distance from $2n\mathbb{Z}$ is therefore at least $n - C$, and the term tends to 0 by Lem. 7.38(a).

Step 5 (conclusion). For $j = 0$, $\mathsf{B}_0^{[n]} = \frac{\mathsf{n}}{2}(\Pi^{[n]})^2$ by eq. 7.xvii, which converges by Step 4 to $\frac{\mathsf{n}}{2}(\Pi^{\mathrm{lat}})^2 = \mathsf{B}_0^{\mathrm{lat}}$. For $j = 1$, the formulas of Lem. 7.26(c) in the box, which are finite sums, express $\mathsf{B}_1^{[n]}(p + o_n, q + o_n; \tau_{\mathrm{gf}})$ as a sum, over monomials, pairings and positions, of products of box kernels. We write the positions in the box as translates $P + o_n$, $P' + o_n$, $b + o_n, \dots$ of plaquettes P, P' and bonds $b, \dots$ of $\mathbb{Z}^3$, and we index the terms by the monomial, the pairing and the positions $P, P', b, \dots$ in $\mathbb{Z}^3$; this set of indices does not depend on n, and a term whose positions do not all lie in the box is set equal to 0.

For fixed $P, P', b, \dots$ all positions lie in the box for every large n, and by Step 4 the term converges, as $n \to \infty$, to the term of the formulas of Lem. 7.26(c) on $\mathbb{Z}^3$ with the same monomial, the same pairing and the positions $P, P', b, \dots$. By Step 3, the box kernels satisfy eq. 7.xxxi with one constant c_0; these bounds depend only on differences of base points, which the translation by o_n does not change, and the uniform form of Lem. 7.27, valid for any kernels with these bounds, therefore dominates the moduli of the terms, uniformly in n, by a family of majorants over the same set of indices whose sum is at most $C\mathsf{c}_0^5$.

By dominated convergence, $\mathsf{B}_1^{[n]}(p + o_n, q + o_n; \tau_{\mathrm{gf}})$ tends to the sum of the terms on $\mathbb{Z}^3$, which converges absolutely and equals $\mathsf{B}_1^{\mathrm{lat}}(p, q; \tau_{\mathrm{gf}})$ by Lem. 7.27. This is eq. 7.xxxix. □

Corollary 7.40 (Independence of the gauge parameter)**.** For every $\tau_{\mathrm{gf}} > 0$ and all plaquettes p, q of $\mathbb{Z}^3$,

$$\mathsf{B}_1^{\mathrm{lat}}(p, q; \tau_{\mathrm{gf}}) = \mathsf{B}_1^{\mathrm{lat}}(p, q).$$

■□

Proof. By Prop. 7.12, $\mathsf{B}_1^{[n]}(p+o_n, q+o_n; \tau_{\mathrm{gf}}) = \mathsf{B}_1^{[n]}(p+o_n, q+o_n; 1)$ for every n with $p+o_n, q+o_n \in \mathsf{Pl}_n$. Letting $n \to \infty$, Prop. 7.39 gives $\mathsf{B}_1^{\mathrm{lat}}(p, q; \tau_{\mathrm{gf}}) = \mathsf{B}_1^{\mathrm{lat}}(p, q; 1)$, and the right side is $\mathsf{B}_1^{\mathrm{lat}}(p, q)$ by eq. 7.xix. □

7.1.14. *The value: Fourier constants, coefficient of* $\mathsf{p}^2 \log|\mathsf{p}|$. To compute κ^{lat} we relate each $\kappa_\Gamma^{\mathrm{lat}}$ to the coefficient of $\mathsf{p}^2 \log|\mathsf{p}|$ in the expansion of Ψ_Γ at small p. This requires two Fourier constants and the corresponding expansions of two lattice sums. In this subsection $(\mathcal{F}^{-1}\phi)(p) = \int e^{ip\cdot x}\phi(x)\, d^\circ x$ for a function ϕ of $x \in \mathbb{R}^3$.

Lemma 7.41 (Two Fourier constants)**.** Let $\phi \in \mathcal{S}(\mathbb{R}^3)$ vanish near 0.

(a) For $s \in \mathbb{C}$ with $\operatorname{Re} s > -3$,

$$\int_{\mathbb{R}^3} |p|^s\, (\mathcal{F}^{-1}\phi)(p)\, \mathrm{d}p = c(s) \int_{\mathbb{R}^3} |x|^{-3-s}\, \phi(x)\, \mathrm{d}x,$$
$$c(s) := \frac{2^s\, \Gamma_{\mathrm{E}}(\frac{3+s}{2})}{\pi^{3/2}\, \Gamma_{\mathrm{E}}(-\frac{s}{2})}, \tag{7.xli}$$

where $1/\Gamma_{\mathrm{E}}$ is entire. Moreover $c(s) = 0$ for $s = 0, 2, 4, \dots$, and $c(1) = -\pi^{-2}$.

(b) We have

$$\int_{\mathbb{R}^3} p^2 \log|p|\,(\mathcal{F}^{-1}\phi)(p)\,\mathrm{d}p = c'(2)\int_{\mathbb{R}^3}|x|^{-5}\,\phi(x)\,\mathrm{d}x, \qquad c'(2)=\frac{3}{2\pi}. \tag{7.xlii}$$

■

Proof. Step 1. Let $\psi := \mathcal{F}^{-1}\phi \in \mathcal{S}(\mathbb{R}^3)$, and assume first $-3 < \operatorname{Re} s < 0$. The substitution $t \to t/p^2$ in Euler's integral gives $\int_0^\infty t^{-s/2-1}e^{-tp^2}\,\mathrm{d}t = \Gamma_{\mathrm{E}}(-\frac{s}{2})\,|p|^s$ for $p \neq 0$. Since

$$\iint t^{-\operatorname{Re} s/2-1}e^{-tp^2}\,|\psi(p)|\,\mathrm{d}t\,\mathrm{d}p = \Gamma_{\mathrm{E}}(-\tfrac{1}{2}\operatorname{Re} s)\int |p|^{\operatorname{Re} s}|\psi(p)|\,\mathrm{d}p < \infty,$$

Fubini's theorem yields

$$\int |p|^s\psi(p)\,\mathrm{d}p = \Gamma_{\mathrm{E}}(-\tfrac{s}{2})^{-1}\int_0^\infty t^{-s/2-1}\Big[\int e^{-tp^2}\psi(p)\,\mathrm{d}p\Big]\,\mathrm{d}t.$$

By Fubini's theorem and the Gaussian integral $\int_{\mathbb{R}^3} e^{-tp^2+ip\cdot x}\,\mathrm{d}p = (\pi/t)^{3/2}e^{-x^2/4t}$,

$$\int e^{-tp^2}\psi(p)\,\mathrm{d}p = \int \phi(x)\Big[\int e^{-tp^2+ip\cdot x}\,d^\circ p\Big]\,\mathrm{d}x = \int (4\pi t)^{-3/2}e^{-x^2/4t}\,\phi(x)\,\mathrm{d}x.$$

For $x \neq 0$ the substitution $t \to x^2/4t$ gives

$$\int_0^\infty t^{-(s+3)/2-1}e^{-x^2/4t}\,\mathrm{d}t = \Gamma_{\mathrm{E}}\big(\tfrac{s+3}{2}\big)\Big(\frac{x^2}{4}\Big)^{-(s+3)/2},$$

as $\operatorname{Re} s + 3 > 0$. The same identity with $\operatorname{Re} s$ in place of s,

$$\int_0^\infty t^{-(\operatorname{Re} s+3)/2-1}e^{-x^2/4t}\,\mathrm{d}t = \Gamma_{\mathrm{E}}\big(\tfrac{\operatorname{Re} s+3}{2}\big)\Big(\frac{x^2}{4}\Big)^{-(\operatorname{Re} s+3)/2},$$

together with $\int |x|^{-\operatorname{Re} s-3}|\phi(x)|\,\mathrm{d}x < \infty$, which holds as ϕ vanishes near 0 and lies in $\mathcal{S}(\mathbb{R}^3)$, shows that the modulus $(4\pi)^{-3/2}t^{-(\operatorname{Re} s+3)/2-1}e^{-x^2/4t}|\phi(x)|$ of the integrand $t^{-s/2-1}(4\pi t)^{-3/2}e^{-x^2/4t}\phi(x)$ is integrable over $(0,\infty)\times\mathbb{R}^3$. Hence Fubini's theorem applies once more, and

$$\int |p|^s\psi(p)\,\mathrm{d}p = \Gamma_{\mathrm{E}}(-\tfrac{s}{2})^{-1}(4\pi)^{-3/2}\,2^{s+3}\,\Gamma_{\mathrm{E}}\big(\tfrac{s+3}{2}\big)\int |x|^{-s-3}\phi(x)\,\mathrm{d}x,$$

with $(4\pi)^{-3/2}2^{s+3} = 2^s\pi^{-3/2}$. This is eq. 7.xli for $-3 < \operatorname{Re} s < 0$.

Step 2. The left side of eq. 7.xli is holomorphic in s on $\{\operatorname{Re} s > -3\}$, since $|p|^{\operatorname{Re} s}|\psi|$ provides integrable majorants locally uniformly in s; the function $s \mapsto \int |x|^{-3-s}\phi(x)\,\mathrm{d}x$ is entire, since ϕ vanishes near 0 and lies in $\mathcal{S}(\mathbb{R}^3)$; and c is holomorphic on $\{\operatorname{Re} s > -3\}$, since Γ_{E} is holomorphic on the right half plane and $1/\Gamma_{\mathrm{E}}$ is entire. Both sides agree for $-3 < \operatorname{Re} s < 0$, hence on the connected set $\{\operatorname{Re} s > -3\}$ by the identity theorem. The entire functions $1/\Gamma_{\mathrm{E}}(z)$ and $z/\Gamma_{\mathrm{E}}(z+1)$ agree for $\operatorname{Re} z > 0$, hence for all z; therefore $1/\Gamma_{\mathrm{E}}$ vanishes at 0, and then successively at $-1, -2, \ldots$, which gives $c(s) = 0$ for $s = 0, 2, 4, \ldots$.

Moreover $1/\Gamma_{\mathrm{E}}(-\frac{1}{2}) = -\frac{1}{2}/\Gamma_{\mathrm{E}}(\frac{1}{2}) = -\frac{1}{2\sqrt{\pi}}$, whence $c(1) = \frac{2\,\Gamma_{\mathrm{E}}(2)}{\pi^{3/2}}\cdot(-\frac{1}{2\sqrt{\pi}}) = -\pi^{-2}$.

Step 3. For (b) we differentiate eq. 7.xli with respect to s at $s = 2$. On the left we obtain $\int p^2\log|p|\,\psi(p)\,\mathrm{d}p$, by differentiation under the integral; on the right $c'(2)\int|x|^{-5}\phi - c(2)\int|x|^{-5}\log|x|\,\phi$, where $c(2) = 0$. From $1/\Gamma_{\mathrm{E}}(z) = z(z+1)/\Gamma_{\mathrm{E}}(z+2)$ the derivative of $1/\Gamma_{\mathrm{E}}$ at $z = -1$ is $(2z+1)/\Gamma_{\mathrm{E}}(z+2)|_{z=-1} = -1$; with $z = -\frac{s}{2}$ this gives $\partial_s\big[1/\Gamma_{\mathrm{E}}(-\frac{s}{2})\big]_{s=2} = \frac{1}{2}$. As $1/\Gamma_{\mathrm{E}}(-1) = 0$, only this derivative contributes, and $c'(2) = 2^2\,\Gamma_{\mathrm{E}}(\frac{5}{2})\,\pi^{-3/2}\cdot\frac{1}{2}$ with $\Gamma_{\mathrm{E}}(\frac{5}{2}) = \frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi} = \frac{3}{4}\sqrt{\pi}$; thus $c'(2) = \frac{3}{2\pi}$. □

Lemma 7.42 (Two Fourier series: Epstein-type sums near $\mathsf{p} = 0$)**.** For $n \in \{4, 5\}$ and $\mathsf{p} \in \mathbb{R}^3$ let $\mathsf{s}_n(\mathsf{p}) := \sum_{u\in\mathbb{Z}^3\setminus\{0\}}|u|^{-n}e^{-i\mathsf{p}\cdot u}$. There are functions $\mathsf{e}_4, \mathsf{e}_5 \in C^\infty(\{|\mathsf{p}| < 1\})$, invariant under the 48 signed permutations of the coordinates, such that

$$\mathsf{s}_4(\mathsf{p}) = -\pi^2|\mathsf{p}| + \mathsf{e}_4(\mathsf{p}), \qquad \mathsf{s}_5(\mathsf{p}) = \tfrac{2\pi}{3}\,\mathsf{p}^2\log|\mathsf{p}| + \mathsf{e}_5(\mathsf{p}) \qquad (0 < |\mathsf{p}| < 1).$$

■

Proof. We treat the two cases $(\varpi, n, \mathsf{c}) = (|p|, 4, -\pi^{-2})$ and $(\varpi, n, \mathsf{c}) = (p^2 \log|p|, 5, \frac{3}{2\pi})$ together. The function ϖ is continuous, radial and smooth off 0, and its derivatives of order j are $O(|p|^{2-j} \lg|p|)$ at infinity. For an integer $\mathsf{N} \geq 3$ let

$$\mathsf{K}(x) := \int e^{ip\cdot x} \chi(p)\, \varpi(p)\, d^\circ p, \qquad \mathsf{E}(x) := |x|^{-2\mathsf{N}} \int e^{ip\cdot x} (-\Delta_p)^{\mathsf{N}} [(1-\chi)\varpi](p)\, d^\circ p \quad (x \neq 0).$$

Step 1. The integrand of E vanishes for $|p| \leq 1$ and is $O(|p|^{2-2\mathsf{N}} \lg|p|)$, with $2\mathsf{N} - 2 > 3$; hence the integral converges absolutely. The value of $\mathsf{E}(x)$ does not depend on N: we write $|x|^2 e^{ip\cdot x} = -\Delta_p e^{ip\cdot x}$ and integrate by parts twice over balls (Gauss' theorem); the boundary terms on the sphere of radius r are $O((1+|x|)\mathsf{r}^2 \cdot \mathsf{r}^{2-2\mathsf{N}} \lg \mathsf{r})$, which tends to 0 as $\mathsf{r} \to \infty$ because $\mathsf{N} \geq 3$; hence N and $\mathsf{N}+1$ give the same value. Consequently E is continuous off 0 and radial, and $|\mathsf{E}(x)| \leq C_{\mathsf{N}} |x|^{-2\mathsf{N}}$ for every $\mathsf{N} \geq 3$.

Step 2. We show $\mathsf{K} + \mathsf{E} = \mathsf{c}\,|x|^{-n}$ on $\mathbb{R}^3 \backslash \{0\}$. Let $\phi \in C_c^\infty(\mathbb{R}^3 \backslash \{0\})$. By Fubini's theorem, $\int \mathsf{K}\phi\, dx = \int \chi\varpi\, \mathcal{F}^{-1}\phi\, dp$. Let $\phi_{\mathsf{N}} := |x|^{-2\mathsf{N}}\phi \in C_c^\infty(\mathbb{R}^3 \setminus \{0\})$. By Fubini's theorem and $2\mathsf{N}$ integrations by parts, which are permitted as $\mathcal{F}^{-1}\phi_{\mathsf{N}} \in \mathcal{S}(\mathbb{R}^3)$, and by $(-\Delta_p)^{\mathsf{N}} \mathcal{F}^{-1}\phi_{\mathsf{N}} = \mathcal{F}^{-1}(|x|^{2\mathsf{N}}\phi_{\mathsf{N}}) = \mathcal{F}^{-1}\phi$,

$$\int \mathsf{E}\phi\, dx = \int (-\Delta_p)^{\mathsf{N}} [(1-\chi)\varpi]\, \mathcal{F}^{-1}\phi_{\mathsf{N}}\, dp = \int (1-\chi)\varpi\, \mathcal{F}^{-1}\phi\, dp.$$

The sum is $\int \varpi\, \mathcal{F}^{-1}\phi\, dp = \mathsf{c} \int |x|^{-n}\phi\, dx$, by Lem. 7.41(a) at $s = 1$ in the first case and by Lem. 7.41(b) in the second. As $\mathsf{K} + \mathsf{E} - \mathsf{c}|x|^{-n}$ is continuous on $\mathbb{R}^3 \setminus \{0\}$, it vanishes there.

Step 3. The periodic continuation of $\chi\varpi$, whose support lies in $\{|p| \leq 2\}$ and thus in the interior of $[-\pi, \pi]^3$, is continuous, and its Fourier coefficients are the numbers $\mathsf{K}(u)$, $u \in \mathbb{Z}^3$. By Steps 1 and 2, $\sum_u |\mathsf{K}(u)| < \infty$; hence the continuous function $\sum_u \mathsf{K}(u) e^{-i\mathsf{p}\cdot u}$ has the same Fourier coefficients and, both functions being continuous, coincides with the periodic continuation of $\chi\varpi$. For $|\mathsf{p}| < 1$ we have $\chi(\mathsf{p}) = 1$, and therefore

$$\varpi(\mathsf{p}) = \mathsf{K}(0) + \mathsf{c}\,\mathsf{s}_n(\mathsf{p}) - \sum_{u \neq 0} \mathsf{E}(u)\, e^{-i\mathsf{p}\cdot u}.$$

We put $\mathsf{e}_n(\mathsf{p}) := \mathsf{c}^{-1}\big[\sum_{u\neq 0} \mathsf{E}(u) e^{-i\mathsf{p}\cdot u} - \mathsf{K}(0)\big]$. This function is smooth, by the rapid decay of E, and invariant under the signed permutations of the coordinates, since E is radial. Finally $\mathsf{c}^{-1} = -\pi^2$, resp. $\mathsf{c}^{-1} = \frac{2\pi}{3}$, which gives the claim. □

Lemma 7.43 (Coefficient of $\mathsf{p}^2 \log|\mathsf{p}|$ in Ψ_Γ; scaling of the cut-off)**.** Let $\Gamma \in \{1, \dots, 6\}$.

(a) There are numbers $\mathsf{a}_0, \mathsf{a}_1, \mathsf{a}_2$ and a function o on $\{|\mathsf{p}| < 1\}$ with $\mathsf{o}(\mathsf{p})/\mathsf{p}^2 \to 0$ as $\mathsf{p} \to 0$, depending on Γ only, such that for almost every p with $|\mathsf{p}| < 1$

$$\Psi_\Gamma(\mathsf{p}) = \mathsf{a}_0 + \mathsf{a}_1|\mathsf{p}| + \mathsf{a}_2\mathsf{p}^2 + \tfrac{\pi}{36}\,\kappa_\Gamma^{\mathrm{lat}}\, \mathsf{p}^2 \log|\mathsf{p}| + \mathsf{o}(\mathsf{p}).$$

(b) For real $\Lambda > 1$ let $R_\Gamma^\Lambda(k, l, \mathsf{p}) := R_\Gamma(k/\Lambda, l/\Lambda, \mathsf{p}/\Lambda)$. Then $\iint R_\Gamma^\Lambda F_\Gamma\, d^\circ k\, d^\circ l = \Lambda^2 \Psi_\Gamma(\mathsf{p}/\Lambda)$ for every $\mathsf{p} \neq 0$, and for almost every p with $|\mathsf{p}| < 1$

$$\iint (R_\Gamma^\Lambda - R_\Gamma) F_\Gamma\, d^\circ k\, d^\circ l = (\Lambda^2 - 1)\mathsf{a}_0 + (\Lambda - 1)\mathsf{a}_1|\mathsf{p}| - \tfrac{\pi}{36}\,\kappa_\Gamma^{\mathrm{lat}} \log\Lambda\, \mathsf{p}^2 + \Lambda^2 \mathsf{o}(\mathsf{p}/\Lambda) - \mathsf{o}(\mathsf{p}).$$

■

Proof. Step 1. By Lems. 7.35 and 7.36,

$$\mathsf{T}_\Gamma^{\mathrm{mod}}(u) = c_{4,\Gamma}|u|^{-4} + \kappa_\Gamma^{\mathrm{lat}}|u|^{-5} + \mathsf{r}(u),$$
$$|\mathsf{r}(u)| \leq C|u|^{-6} \lg|u| \qquad (u \in \mathbb{Z}^3 \setminus \{0\});$$

we put $\mathsf{r}(0) := \mathsf{T}_\Gamma^{\mathrm{mod}}(0)$. Hence $\sum_u |\mathsf{T}_\Gamma^{\mathrm{mod}}(u)| < \infty$. The function Ψ_Γ is integrable and vanishes for $|\mathsf{p}| \geq \frac{3\pi}{8}$ (Def. 7.32(3)), that is, outside a ball contained in $[-\pi, \pi]^3$. Therefore the periodic function which equals $24\Psi_\Gamma$ on $[-\pi, \pi]^3$ has the Fourier coefficients $24 \int_{[-\pi,\pi]^3} e^{i\mathsf{p}\cdot u} \Psi_\Gamma(\mathsf{p})\, d^\circ \mathsf{p} = \mathsf{T}_\Gamma^{\mathrm{mod}}(u)$, which

are also the Fourier coefficients of the continuous function $\sum_u \mathsf{T}_\Gamma^{\mathrm{mod}}(u)e^{-i\mathsf{p}\cdot u}$. By the uniqueness of Fourier coefficients the two periodic functions agree almost everywhere; as the first one coincides with $24\Psi_\Gamma$ on $[-\pi,\pi]^3$, we obtain, for almost every $\mathsf{p}\in[-\pi,\pi]^3$,

$$24\,\Psi_\Gamma(\mathsf{p}) = c_{4,\Gamma}\,\mathsf{s}_4(\mathsf{p}) + \kappa_\Gamma^{\mathrm{lat}}\,\mathsf{s}_5(\mathsf{p}) + \hat{\mathsf{r}}(\mathsf{p}),$$
$$\hat{\mathsf{r}}(\mathsf{p}) := \sum_u \mathsf{r}(u)\,e^{-i\mathsf{p}\cdot u}.$$

Step 2. Since $\sum_u |u|^2|\mathsf{r}(u)| < \infty$, the function $\hat{\mathsf{r}}$ is of class C^2. As $\mathsf{T}_\Gamma^{\mathrm{mod}}$ is the restriction to $\mathbb{Z}^3$ of a radial function, r and $\hat{\mathsf{r}}$ are invariant under the signed permutations of the coordinates. A function f of class C^2 near 0 with this invariance has vanishing gradient at 0 (invariance under $-\mathbf{1}$), a Hessian at 0 with vanishing off-diagonal entries (change of sign of one coordinate) and equal diagonal entries (permutations); hence $\mathsf{f}(\mathsf{p}) = \mathsf{f}(0) + \mathsf{a}\,\mathsf{p}^2 + o(\mathsf{p}^2)$. We apply this to $\hat{\mathsf{r}}$, e_4 and e_5 and insert Lem. 7.42; this gives (a), the coefficient of $\mathsf{p}^2\log|\mathsf{p}|$ being $\frac{1}{24}\cdot\frac{2\pi}{3}\,\kappa_\Gamma^{\mathrm{lat}} = \frac{\pi}{36}\,\kappa_\Gamma^{\mathrm{lat}}$, and o being $\frac{1}{24}$ times the sum of the remainders $o(\mathsf{p}^2)$ of $c_{4,\Gamma}\mathsf{e}_4$, $\kappa_\Gamma^{\mathrm{lat}}\mathsf{e}_5$ and $\hat{\mathsf{r}}$.

Step 3. For (b) we substitute $k\to\Lambda k$, $l\to\Lambda l$ and use $F_\Gamma(\Lambda k,\Lambda l,\Lambda\mathsf{q}) = \Lambda^{-4}F_\Gamma(k,l,\mathsf{q})$ with $\mathsf{q} = \mathsf{p}/\Lambda$; this produces the factor $\Lambda^{6-4} = \Lambda^2$ and gives the first assertion. For the second we insert (a) at p/Λ and at p, both of modulus less than 1. The terms with a_2 cancel, the terms with a_0, a_1 give $(\Lambda^2-1)\mathsf{a}_0 + (\Lambda-1)\mathsf{a}_1|\mathsf{p}|$, and

$$\Lambda^2\,\tfrac{\pi}{36}\,\kappa_\Gamma^{\mathrm{lat}}\Lambda^{-2}\mathsf{p}^2(\log|\mathsf{p}| - \log\Lambda) - \tfrac{\pi}{36}\,\kappa_\Gamma^{\mathrm{lat}}\,\mathsf{p}^2\log|\mathsf{p}| = -\tfrac{\pi}{36}\,\kappa_\Gamma^{\mathrm{lat}}\log\Lambda\;\mathsf{p}^2.$$

□

7.1.15. *The value: finite parts and regions.* The coefficient $\frac{\pi}{36}\kappa_\Gamma^{\mathrm{lat}}$ of Lem. 7.43 will be computed by an expansion by regions of two-loop integrals with cut-offs vanishing near the origin. We first define the finite parts which occur in this expansion, by analytic regularisation, and the classes of cut-offs.

Definition 7.44 (Finite parts (analytic regularisation); cut-offs vanishing near the origin)**.** Let $\mathbb{D} := \{\mathsf{z}\in\mathbb{C} : \mathrm{Re}\,\mathsf{z} > -\frac{1}{2}\}\setminus\{\frac{1}{2},\frac{3}{2},\frac{5}{2},\dots\}$; this set is open and connected, and it contains 0.

(1) One variable. A *homogeneous factor of degree* $d\in\mathbb{Z}$ is a function $\mathsf{h}\in C^\infty(\mathbb{R}^3\setminus\{0\})$ with $\mathsf{h}(tk) = t^d\mathsf{h}(k)$ for $t>0$ and $\mathsf{h}(-k) = (-1)^d\mathsf{h}(k)$. Let $\mathfrak{P}$ be the set of functions on $\mathbb{R}^3\setminus\{0\}$ of the form $\mathsf{g} = \sum_\iota \mathsf{h}_\iota\mathsf{g}_\iota$, a finite sum, with homogeneous factors h_ι of degrees d_ι and with $\mathsf{g}_\iota\in C_c^\infty(\mathbb{R}^3)$. For such g the integral $\int|k|^{2\mathsf{z}}\mathsf{g}(k)\,\mathrm{d}k$ converges absolutely and is holomorphic in z for $2\,\mathrm{Re}\,\mathsf{z} + \min_\iota d_\iota > -3$; it has a holomorphic continuation to $\mathbb{D}$ (Lem. 7.45(a)), which is unique. The *finite part* $\mathrm{f.p.}\int\mathsf{g}\,\mathrm{d}k$ is the value of this continuation at $\mathsf{z}=0$; it depends on g only, and it is linear in g.

(2) Two variables. Let $c>0$ and $\eta_c(k) := \theta(4|k|/c)$, so that $\eta_c(k) = 1$ for $|k|\le\frac{c}{8}$ and $\eta_c(k) = 0$ for $|k|\ge\frac{c}{4}$. Let $\mathfrak{P}_2(c)$ be the set of Borel functions Φ on $\{(k,l) : k\neq 0,\ l\neq 0,\ k\neq l\}$ for which there is a constant C_Φ with the following properties.

- (Φ0) $\Phi(k,l) = 0$ if $|k|+|l| < c$ or $|k|+|l| > C_\Phi$.
- (Φ1) For $0<|k|<\frac{c}{3}$, $\Phi(k,l) = \sum_\iota \mathsf{h}_\iota(k)\Phi_\iota(k,l)$, a finite sum with homogeneous factors h_ι and with functions $\Phi_\iota\in C^\infty(\{|k|<\frac{c}{3}\}\times\mathbb{R}^3)$ which vanish where $|k|+|l|<c$ or $|k|+|l|>C_\Phi$.
- (Φ2) For $0<|l|<\frac{c}{3}$, $\Phi(k,l) = \sum_\iota \mathsf{h}'_\iota(l)\Phi'_\iota(k,l)$, with the same properties after the exchange of k and l.
- (Φ3) $|\Phi(k,l)|\le C_\Phi|k-l|^{-2}$ for $|k|,|l|\ge\frac{c}{8}$.

For $\Phi\in\mathfrak{P}_2(c)$ the function

$$k\mapsto\eta_c(k)\int\Phi(k,l)\,\mathrm{d}l = \sum_\iota \mathsf{h}_\iota(k)\,\eta_c(k)\int\Phi_\iota(k,l)\,\mathrm{d}l$$

lies in $\mathfrak{P}$, since η_c vanishes for $|k| \geq \frac{c}{4}$ and the Φ_ι are smooth and vanish for $|l| > C_\Phi$; likewise $l \mapsto \eta_c(l) \int \Phi(k,l)\,\mathrm{d}k$ lies in $\mathfrak{P}$. We define

$$\begin{aligned}\mathrm{f.p.}\iint \Phi\,\mathrm{d}k\,\mathrm{d}l := {} & \iint (1-\eta_c(k)-\eta_c(l))\Phi\,\mathrm{d}k\,\mathrm{d}l \\ & + \mathrm{f.p.}\int \Big[\eta_c(k)\int \Phi(k,l)\,\mathrm{d}l\Big]\,\mathrm{d}k + \mathrm{f.p.}\int \Big[\eta_c(l)\int \Phi(k,l)\,\mathrm{d}k\Big]\,\mathrm{d}l.\end{aligned}$$

On the support of Φ, $|k| < \frac{c}{8}$ implies $|l| > \frac{7}{8}c$, hence $\eta_c(k) = 1$ and $\eta_c(l) = 0$; the same holds with k and l exchanged. Therefore the first integrand vanishes unless $|k|, |l| \geq \frac{c}{8}$, and the first integral converges absolutely, by (Φ0) and (Φ3): its integrand is bounded by $C_\Phi |k-l|^{-2}$, which is integrable over bounded sets of (k,l), and it vanishes for $|k|+|l| > C_\Phi$. This finite part is linear on $\mathfrak{P}_2(c)$ and symmetric under the exchange of k and l.

(3) Cut-offs. Let $\mathcal{Y}(c)$ be the set of functions $\Upsilon \in C^\infty(\mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}^3)$ for which there is a constant C_Υ with the following properties.

(Y1) $\Upsilon(k,l,\mathsf{p}) = 0$ if $|\mathsf{p}| < c$ and ($|k|+|l| < c$ or $|k|+|l| > C_\Upsilon$).

(Y2) $\Upsilon(-k,-l,-\mathsf{p}) = \Upsilon(k,l,\mathsf{p}) = \Upsilon(Ok, Ol, O\mathsf{p})$ for every rotation O.

Then $\mathcal{Y}(c') \subset \mathcal{Y}(c)$ for $c < c'$. For $\Lambda > 1$ and a function Υ of (k,l,p) or of (k,l), let Υ^Λ denote Υ evaluated at the arguments divided by Λ; if $\Upsilon \in \mathcal{Y}(c)$, then $\Upsilon^\Lambda \in \mathcal{Y}(\Lambda c) \subset \mathcal{Y}(c)$. For a function $G(k,l,\mathsf{p})$ which is of class C^2 in p near 0 we put

$$\overline{G_2}(k,l) := \tfrac{1}{6}\,\Delta_{\mathsf{p}} G(k,l,\mathsf{p})\big|_{\mathsf{p}=0};$$

this is the mean over the directions of p of the term of second order in the Taylor expansion of G in p, divided by p^2, since the mean of $\mathsf{p}_i\mathsf{p}_j$ over $\{|\mathsf{p}| = t\}$ is $\frac{1}{3}t^2\delta_{ij}$. Finally, for $\Upsilon \in \mathcal{Y}(c)$ and $0 < |\mathsf{p}| < c$ we put

$$\Xi_\Gamma[\Upsilon](\mathsf{p}) := \iint \Upsilon F_\Gamma\, d^\circ k\, d^\circ l;$$

this integral converges absolutely by eq. 7.xxxvi and the estimates following it in Def. 7.32(3), since by (Y1) the integrand vanishes for $|k|+|l| > C_\Upsilon$. ∎

Lemma 7.45 (Properties of finite parts)**.**

(a) (Continuation with a parameter.) Let h be a homogeneous factor of degree d, and let $\mathsf{g}\colon \mathbb{R}^3 \times \mathbb{C} \to \mathbb{C}$ have the following properties: for every $\mathsf{z} \in \mathbb{C}$ the function $\mathsf{g}(\cdot;\mathsf{z})$ is smooth, and its support lies in a compact set which does not depend on z; for every multi-index α the derivative $\partial_k^\alpha \mathsf{g}$ is continuous on $\mathbb{R}^3 \times \mathbb{C}$ and, for each fixed k, an entire function of z. Then the function

$$\mathsf{z} \longmapsto \int |k|^{2\mathsf{z}}\, \mathsf{h}(k)\, \mathsf{g}(k;\mathsf{z})\,\mathrm{d}k,$$

which is defined for $2\,\mathrm{Re}\,\mathsf{z} + d > -3$, has a holomorphic continuation to $\mathbb{D}$, and the value of this continuation at $\mathsf{z} = 0$ is $\mathrm{f.p.}\int \mathsf{h}\,\mathsf{g}(\cdot;0)\,\mathrm{d}k$.

(b) If $\mathsf{g} \in \mathfrak{P}$ is integrable, then $\mathrm{f.p.}\int \mathsf{g}\,\mathrm{d}k = \int \mathsf{g}\,\mathrm{d}k$.

(c) Let h be a homogeneous factor of degree $d \leq -4$, and let $\eta(k) = \theta_1(|k|)$, where $\theta_1 \in C^\infty([0,\infty))$ has compact support and is equal to 1 near 0. Then

$$\mathrm{f.p.}\int \eta\,\mathsf{h}\,\mathrm{d}k = -\int (1-\eta)\,\mathsf{h}\,\mathrm{d}k.$$

(d) (A soft line.) Let $\mu_0 > 0$ and $\eta(k) := \theta(|k|/2\mu_0)$. Let

$$S(k,\mathsf{p}) = \mathsf{P}(k,\mathsf{p})\,|k|^{-2\mathsf{a}}\,|k-\mathsf{p}|^{-2\mathsf{a}'},$$

where $\mathsf{a}, \mathsf{a}' \geq 0$ are integers and P is a polynomial, homogeneous of degree $2\mathsf{a}+2\mathsf{a}'-2$, and assume that $|S(k,\mathsf{p})| \leq C_S\big(|k|^{-2} + |k-\mathsf{p}|^{-2}\big)$. Let $J \in C^\infty\big(\{|k| < 3\mu_0\} \times \{|\mathsf{p}| < \mu_0\}\big)$ satisfy

$J(-k,-\mathsf{p}) = J(k,\mathsf{p})$, and for $k \neq 0$ and $i \in \{0,1,2\}$ put

$$[SJ]_i(k;\mathsf{p}) := \sum_{|\alpha|=i} \frac{\mathsf{p}^\alpha}{\alpha!}\, \partial_{\mathsf{p}}^\alpha (SJ)(k,0).$$

Then $\eta\,[SJ]_i(\cdot;\mathsf{p}) \in \mathfrak{P}$. Moreover, there are a Borel function h_S on $\mathbb{R}^3$, which depends on S only, is bounded on the unit sphere and satisfies $\mathsf{h}_S(t\mathsf{p}) = t\,\mathsf{h}_S(\mathsf{p})$ for $t>0$, and a constant $C = C(\mu_0, S, J)$, such that for $0 < |\mathsf{p}| \le \frac{1}{2}\mu_0$

$$\Big| \int \eta SJ\,\mathrm{d}k - \mathrm{f.p.} \int \eta\,[SJ]_0\,\mathrm{d}k - \mathrm{f.p.} \int \eta\,[SJ]_2(k;\mathsf{p})\,\mathrm{d}k - J(0,0)\,\mathsf{h}_S(\mathsf{p}) \Big| \le C\,|\mathsf{p}|^3 \lg|\mathsf{p}|.$$

(e) (Two variables.) Let $\Phi \in \mathfrak{P}_2(c)$ and $(\epsilon,\epsilon') \in \{(1,1),(1,0),(0,1)\}$. If $\epsilon = 0$, assume that all degrees of the homogeneous factors in (Φ1) are ≥ -2; if $\epsilon' = 0$, assume that all degrees of the homogeneous factors in (Φ2) are ≥ -2. Then the integral

$$\iint |k|^{2\epsilon\mathsf{z}}\,|l|^{2\epsilon'\mathsf{z}}\,\Phi\,\mathrm{d}k\,\mathrm{d}l$$

converges absolutely for large $\mathrm{Re}\,\mathsf{z}$, and it has a holomorphic continuation to $\mathbb{D}$ whose value at $\mathsf{z}=0$ is $\mathrm{f.p.}\iint \Phi\,\mathrm{d}k\,\mathrm{d}l$. If $0 < c' < c$, then $\mathfrak{P}_2(c) \subset \mathfrak{P}_2(c')$, and every $\Phi \in \mathfrak{P}_2(c)$ has the same finite part whether it is regarded as an element of $\mathfrak{P}_2(c)$ or of $\mathfrak{P}_2(c')$. If $\Phi \in \mathfrak{P}_2(c)$ is integrable, then $\mathrm{f.p.}\iint \Phi\,\mathrm{d}k\,\mathrm{d}l = \iint \Phi\,\mathrm{d}k\,\mathrm{d}l$.

■

Proof. Proof of (a). Put $m := \max\{0, -d-3\}$; then $d+m+1 \ge -2$. For the proof of (a) let $\eta(k) := \theta(|k|)$, with the function θ of Def. 7.16(1), so that $\eta = 1$ for $|k| \le \frac{1}{2}$ and $\eta = 0$ for $|k| \ge 1$, and let

$$T\mathsf{g}(k;\mathsf{z}) := \sum_{|\alpha|\le m} \frac{k^\alpha}{\alpha!}\, \partial_k^\alpha \mathsf{g}(0;\mathsf{z})$$

be the Taylor polynomial of degree m of $\mathsf{g}(\cdot;\mathsf{z})$ at $k=0$. For $2\,\mathrm{Re}\,\mathsf{z} + d > -3$ we decompose

$$\int |k|^{2\mathsf{z}} \mathsf{h}\,\mathsf{g}\,\mathrm{d}k = \int |k|^{2\mathsf{z}} \mathsf{h}\,[\mathsf{g} - \eta\, T\mathsf{g}]\,\mathrm{d}k + \sum_{|\alpha|\le m} \frac{\partial_k^\alpha \mathsf{g}(0;\mathsf{z})}{\alpha!} \int |k|^{2\mathsf{z}}\,\eta(k)\,\mathsf{h}(k)\,k^\alpha\,\mathrm{d}k.$$

The integrand of the first integral on the right vanishes outside a compact set which does not depend on z. For $|k| \le \frac{1}{2}$ we have $\eta = 1$, and Taylor's formula, together with the continuity of the derivatives of order $m+1$ of g on $\mathbb{R}^3 \times \mathbb{C}$, bounds the modulus of the integrand by $C|k|^{2\,\mathrm{Re}\,\mathsf{z}+d+m+1}$, with a constant C that stays bounded as long as z ranges over a compact set. Since $d+m+1 \ge -2$, the exponent is larger than -3 as soon as $\mathrm{Re}\,\mathsf{z} > -\frac{1}{2}$. For each $k \neq 0$ the integrand is an entire function of z; differentiating under the integral sign, we see that the first integral is holomorphic on $\{\mathrm{Re}\,\mathsf{z} > -\frac{1}{2}\}$.

In the integrals of the sum we pass to polar coordinates and integrate by parts once. With

$$\mathsf{m}_\alpha := \int_{S^2} \mathsf{h}(\omega)\,\omega^\alpha\,\mathrm{d}\omega$$

we obtain, for $2\,\mathrm{Re}\,\mathsf{z} + d + |\alpha| > -3$,

$$\int |k|^{2\mathsf{z}}\,\eta\,\mathsf{h}\,k^\alpha\,\mathrm{d}k = \mathsf{m}_\alpha \int_0^\infty \theta(s)\,s^{2\mathsf{z}+d+|\alpha|+2}\,\mathrm{d}s = -\frac{\mathsf{m}_\alpha}{2\mathsf{z}+d+|\alpha|+3} \int_0^\infty \theta'(s)\,s^{2\mathsf{z}+d+|\alpha|+3}\,\mathrm{d}s.$$

As θ' is supported in $[\frac{1}{2},1]$, the last integral is an entire function of z. If $d+|\alpha|$ is odd, then $\omega \mapsto \mathsf{h}(\omega)\omega^\alpha$ is odd and $\mathsf{m}_\alpha = 0$. If $d+|\alpha|$ is even, the denominator vanishes only at $\mathsf{z} = -\frac{1}{2}(d+|\alpha|+3)$, which is half an odd integer and hence not a point of $\mathbb{D}$. Consequently the right side of the decomposition is holomorphic on $\mathbb{D}$; it is the asserted continuation, which is unique because $\mathbb{D}$ is connected.

The value of this continuation at $\mathsf{z}=0$ involves g only through $\mathsf{g}(\cdot;0)$: it contains the function $\mathsf{g}(\cdot;0)-\eta\,T\mathsf{g}(\cdot;0)$ and the numbers $\partial_k^\alpha \mathsf{g}(0;0)$. It is therefore equal to the value at 0 of the same expression built with the function $\mathsf{g}(\cdot;0)$, which does not depend on z. By the argument just given, applied to this function, that expression continues $\mathsf{z}\mapsto \int |k|^{2\mathsf{z}}\mathsf{h}\,\mathsf{g}(\cdot;0)\,\mathrm{d}k$ to $\mathbb{D}$, and by Def. 7.44(1) its value at 0 is $\mathrm{f.p.}\int \mathsf{h}\,\mathsf{g}(\cdot;0)\,\mathrm{d}k$.

Proof of (b). Let $\mathsf{g}\in\mathfrak{P}$ be integrable. Like every element of $\mathfrak{P}$, the function g vanishes outside a bounded set; hence for every $\mathsf{N}>0$ there is C_N with $|k|^{2\,\mathrm{Re}\,\mathsf{z}}|\mathsf{g}(k)|\le C_\mathsf{N}|\mathsf{g}(k)|$ whenever $0\le \mathrm{Re}\,\mathsf{z}\le \mathsf{N}$. By differentiation under the integral sign and by dominated convergence, $\int |k|^{2\mathsf{z}}\mathsf{g}\,\mathrm{d}k$ is holomorphic on $\{\mathrm{Re}\,\mathsf{z}>0\}$ and continuous on $\{\mathrm{Re}\,\mathsf{z}\ge 0\}$. The open set $\mathbb{D}\cap\{\mathrm{Re}\,\mathsf{z}>0\}$ is connected, and on it this function coincides with the continuation of Def. 7.44(1), because the two functions agree for large $\mathrm{Re}\,\mathsf{z}$ (identity theorem). Letting $\mathsf{z}\downarrow 0$ along the real axis, we obtain $\mathrm{f.p.}\int \mathsf{g}\,\mathrm{d}k=\int \mathsf{g}\,\mathrm{d}k$.

Proof of (c). Let $\mathsf{m}_0:=\int_{S^2}\mathsf{h}\,\mathrm{d}\omega$. For $2\,\mathrm{Re}\,\mathsf{z}+d>-3$, polar coordinates and one integration by parts give

$$\int |k|^{2\mathsf{z}}\,\eta\,\mathsf{h}\,\mathrm{d}k=\mathsf{m}_0\int_0^\infty \theta_1(s)\,s^{2\mathsf{z}+d+2}\,\mathrm{d}s=-\frac{\mathsf{m}_0}{2\mathsf{z}+d+3}\int_0^\infty \theta_1'(s)\,s^{2\mathsf{z}+d+3}\,\mathrm{d}s.$$

The support of θ_1' is a compact subset of $(0,\infty)$, so the last integral is an entire function of z. As in the proof of (a), either d is odd and $\mathsf{m}_0=0$, or the zero of the denominator is half an odd integer and lies outside $\mathbb{D}$. Hence the right side is the continuation of the left side to $\mathbb{D}$, and its value at $\mathsf{z}=0$ is

$$\mathrm{f.p.}\int \eta\,\mathsf{h}\,\mathrm{d}k=-\frac{\mathsf{m}_0}{d+3}\int_0^\infty \theta_1'(s)\,s^{d+3}\,\mathrm{d}s.$$

On the other hand, $1-\theta_1$ vanishes near 0 and $d+3<0$; an integration by parts without boundary terms therefore yields

$$\int (1-\eta)\,\mathsf{h}\,\mathrm{d}k=\mathsf{m}_0\int_0^\infty (1-\theta_1(s))\,s^{d+2}\,\mathrm{d}s=\frac{\mathsf{m}_0}{d+3}\int_0^\infty \theta_1'(s)\,s^{d+3}\,\mathrm{d}s.$$

Comparing the last two displays, we obtain (c).

Proof of (d)*: preparation.* The function S is smooth on $\{k\ne 0,\ k\ne \mathsf{p}\}$, homogeneous of degree -2 in (k,p) jointly, and even, since the polynomial P is homogeneous of even degree. The set $\{(k,\mathsf{q}):|k|=1,\ |\mathsf{q}|\le\tfrac12\}$ is compact, and S is smooth in a neighbourhood of it; together with homogeneity this gives, for every multi-index α,

$$|\partial_\mathsf{p}^\alpha S(k,\mathsf{q})|\le C_\alpha\,|k|^{-2-|\alpha|}\qquad (|\mathsf{q}|\le\tfrac12|k|).$$

For every multi-index β the function $\partial_\mathsf{p}^\beta S(\cdot,0)$ is a homogeneous factor of degree $-2-|\beta|$: it is smooth off 0, homogeneous of this degree, and of parity $(-1)^{|\beta|}$ because S is even. By the Leibniz rule, $\eta\,[SJ]_i(\cdot;\mathsf{p})$ is a finite sum of products of such factors with the functions $\eta\,\partial_\mathsf{p}^\gamma J(\cdot,0)$; these belong to $C_c^\infty(\mathbb{R}^3)$, since η vanishes for $|k|\ge 2\mu_0$. Hence $\eta\,[SJ]_i(\cdot;\mathsf{p})\in\mathfrak{P}$.

Since J is even, the terms of odd degree in its Taylor expansion at $(0,0)$ vanish, and its Taylor polynomial of degree 3 is $J(0,0)+J_2$ with a quadratic form J_2 in (k,p). Thus $J=J(0,0)+J_2+J_4$, and Taylor's formula with integral remainder, applied to J and to its derivatives, gives

$$|\partial_\mathsf{p}^\alpha J_4(k,\mathsf{p})|\le C\,(|k|+|\mathsf{p}|)^{4-|\alpha|}\qquad (|\alpha|\le 3)\quad\text{on }\{|k|\le 2\mu_0,\ |\mathsf{p}|\le\tfrac12\mu_0\}.$$

Each of the four expressions inside the modulus in the asserted estimate, namely $\int \eta SJ\,\mathrm{d}k$, $\mathrm{f.p.}\int \eta\,[SJ]_0\,\mathrm{d}k$, $\mathrm{f.p.}\int \eta\,[SJ]_2(k;\mathsf{p})\,\mathrm{d}k$ and $J(0,0)\,\mathsf{h}_S(\mathsf{p})$, depends linearly on J, because $J\mapsto[SJ]_i$ and the finite part are linear. We insert $J=J(0,0)+J_2+J_4$. The quadratic form J_2 and the function J_4 are even and vanish at $(0,0)$, so that for them the last of the four expressions is 0. It therefore suffices to consider, in place of SJ, the three functions

$$N:=S,\qquad N:=SJ_2,\qquad N:=SJ_4,$$

and to show that the difference $\int \eta N\,\mathrm{d}k-\mathrm{f.p.}\int \eta\,[N]_0\,\mathrm{d}k-\mathrm{f.p.}\int \eta\,[N]_2\,\mathrm{d}k$ is equal to $\mathsf{h}_S(\mathsf{p})+O(|\mathsf{p}|^3)$ for $N=S$, the case which enters with the factor $J(0,0)$, and that it is $O(|\mathsf{p}|^3\lg|\mathsf{p}|)$ for $N=SJ_2$

and for $N = SJ_4$. For each of the three functions we write $N_i := [N]_i$ $(i = 0, 1, 2)$, and for $|k| \geq 2|\mathsf{p}|$ we put $N_{\mathrm{r}} := N - N_0 - N_1 - N_2$; moreover, $\mathsf{B} := \{|k| \leq 2|\mathsf{p}|\}$. Since $|\mathsf{p}| \leq \frac{1}{2}\mu_0$, we have $\eta = 1$ on B. In all three cases $N(-k, -\mathsf{p}) = N(k, \mathsf{p})$, as S, J_2 and J_4 are even; differentiating with respect to p at $\mathsf{p} = 0$, we see that N_1 is odd in k while N_0 and N_2 are even.

The case $N = S$. By the bound on the derivatives of S, $|N_i| \leq C|\mathsf{p}|^i|k|^{-2-i}$. On the complement B^c of B, Taylor's formula in p, applied at the points $t\mathsf{p}$ with $0 \leq t \leq 1$, all of which satisfy $|t\mathsf{p}| \leq \frac{1}{2}|k|$, gives $|N_{\mathrm{r}}| \leq C|\mathsf{p}|^3|k|^{-5}$. We write

$$\int \eta N \,\mathrm{d}k = \int_{\mathsf{B}} N \,\mathrm{d}k + \int_{\mathsf{B}^c} \eta\,(N_0 + N_1 + N_2 + N_{\mathrm{r}})\,\mathrm{d}k$$

and consider the four terms of the second integral one after the other. First, $\int_{\mathsf{B}^c} \eta N_0 = \int \eta N_0 - \int_{\mathsf{B}} N_0$. Second, $\int_{\mathsf{B}^c} \eta N_1 = 0$, because N_1 is odd and integrable over $\mathsf{B}^c \cap \operatorname{supp}\eta$, while η and B^c are invariant under $k \to -k$. Third, $N_2(\cdot\,; \mathsf{p})$ is a homogeneous factor of degree -4; assertion (c), applied with $\theta_1(s) := \theta(s/2\mu_0)$, and the vanishing of $1 - \eta$ on B give

$$\int_{\mathsf{B}^c} \eta N_2 = \int_{\mathsf{B}^c} N_2 - \int (1 - \eta) N_2 = \int_{\mathsf{B}^c} N_2 + \mathrm{f.\,p.} \int \eta N_2.$$

Fourth, $\int_{\mathsf{B}^c} \eta N_{\mathrm{r}} = \int_{\mathsf{B}^c} N_{\mathrm{r}} - \int (1 - \eta) N_{\mathrm{r}}$, and

$$\Big| \int (1 - \eta) N_{\mathrm{r}} \Big| \leq C|\mathsf{p}|^3 \int_{|k| \geq \mu_0} |k|^{-5} \,\mathrm{d}k \leq C|\mathsf{p}|^3.$$

Collecting the terms, we arrive at

$$\int \eta S \,\mathrm{d}k = \int \eta N_0 \,\mathrm{d}k + \mathrm{f.\,p.} \int \eta N_2 \,\mathrm{d}k + \mathsf{h}_S(\mathsf{p}) + O(|\mathsf{p}|^3), \qquad \mathsf{h}_S(\mathsf{p}) := \int_{|k| \leq 2|\mathsf{p}|} (S - N_0)\,\mathrm{d}k + \int_{|k| > 2|\mathsf{p}|} (N_2 + N_{\mathrm{r}})\,\mathrm{d}k.$$

Both integrals in the definition of h_S converge absolutely, because $|S| + |N_0| \leq C(|k|^{-2} + |k - \mathsf{p}|^{-2})$, and $|N_2 + N_{\mathrm{r}}| \leq C|\mathsf{p}|^2|k|^{-4}$ on B^c. The substitution $k \to tk$, together with $N_i(tk; t\mathsf{p}) = t^{-2} N_i(k; \mathsf{p})$, yields $\mathsf{h}_S(t\mathsf{p}) = t\,\mathsf{h}_S(\mathsf{p})$ for $t > 0$; the same bounds, taken at $|\mathsf{p}| = 1$, show that h_S is bounded on the unit sphere $\{|\mathsf{p}| = 1\}$, so that $|\mathsf{h}_S(\mathsf{p})| \leq C|\mathsf{p}|$; and h_S is a Borel function by Fubini's theorem. By its definition h_S depends on S only. Finally, $\eta N_0 \in \mathfrak{P}$ is integrable, so that $\int \eta N_0 \,\mathrm{d}k$ is its finite part by (b).

The case $N = SJ_2$. Now N is homogeneous of degree 0 in (k, p), and as above $|\partial_{\mathsf{p}}^{\alpha} N(k, \mathsf{q})| \leq C|k|^{-|\alpha|}$ for $|\mathsf{q}| \leq \frac{1}{2}|k|$. Hence $|N_i| \leq C|\mathsf{p}|^i|k|^{-i}$, which is integrable near $k = 0$ for $i \leq 2$; further $|N_{\mathrm{r}}| \leq C|\mathsf{p}|^3|k|^{-3}$ on B^c, and $|N| \leq C|\mathsf{p}|^2(|k|^{-2} + |k - \mathsf{p}|^{-2})$ on B. Therefore

$$\int \eta N \,\mathrm{d}k = \sum_{i \leq 2} \int \eta N_i \,\mathrm{d}k + \Big[\int_{\mathsf{B}} (N - N_0 - N_1 - N_2)\,\mathrm{d}k + \int_{\mathsf{B}^c} \eta N_{\mathrm{r}} \,\mathrm{d}k \Big],$$

and the bracket is bounded in modulus by

$$C|\mathsf{p}|^3 + C|\mathsf{p}|^3 \int_{2|\mathsf{p}| \leq |k| \leq 2\mu_0} |k|^{-3} \,\mathrm{d}k \leq C|\mathsf{p}|^3 \lg|\mathsf{p}|.$$

The term with $i = 1$ vanishes because N_1 is odd, and the terms with $i = 0$ and $i = 2$ are finite parts by (b).

The case $N = SJ_4$. By the Leibniz rule and the bounds on the derivatives of S and of J_4,

$$|\partial_{\mathsf{p}}^{\alpha} N(k, \mathsf{q})| \leq C|k|^{2-|\alpha|} \qquad (|\alpha| \leq 3,\ |\mathsf{q}| \leq \tfrac{1}{2}|k|,\ |\mathsf{q}| \leq \tfrac{1}{2}\mu_0,\ |k| \leq 2\mu_0).$$

Hence $|N_i| \leq C|\mathsf{p}|^i|k|^{2-i}$, $|N_{\mathrm{r}}| \leq C|\mathsf{p}|^3|k|^{-1}$ on $\mathsf{B}^c \cap \operatorname{supp}\eta$, and $|N| \leq C|\mathsf{p}|^4(|k|^{-2} + |k - \mathsf{p}|^{-2})$ on B. Arguing as in the previous case, we find $\int \eta N \,\mathrm{d}k = \int \eta N_0 \,\mathrm{d}k + \int \eta N_2 \,\mathrm{d}k + O(|\mathsf{p}|^3)$, and the two integrals on the right are finite parts by (b).

Adding the three cases, the first one multiplied by $J(0,0)$, and using $|\mathsf{p}|^3 \le |\mathsf{p}|^3 \lg|\mathsf{p}|$, we obtain the estimate of (d), with a constant depending on μ_0, S and J.

Proof of (e). We split $1 = \big(1-\eta_c(k)-\eta_c(l)\big)+\eta_c(k)+\eta_c(l)$ and treat the three resulting integrals separately; each of them converges absolutely for large $\operatorname{Re}\mathsf{z}$, as the following discussion shows.

On the support of Φ, $|k| < \frac{c}{8}$ implies $|l| > \frac{7}{8}c$ by (Φ0), hence $\eta_c(k) = 1$ and $\eta_c(l) = 0$; the same holds with k and l exchanged. Therefore the first integrand, which contains the factor $1-\eta_c(k)-\eta_c(l)$, vanishes unless $|k|, |l| \ge \frac{c}{8}$, and by (Φ0) it vanishes unless $|k|+|l| \le C_\Phi$. On this bounded set the factor $|k|^{2\epsilon\mathsf{z}}|l|^{2\epsilon'\mathsf{z}}$ is an entire function of z, bounded for z in compact sets, while $|\Phi| \le C_\Phi |k-l|^{-2}$ by (Φ3). Hence the first integral is an entire function of z, whose value at $\mathsf{z}=0$ is $\iint \big(1-\eta_c(k)-\eta_c(l)\big)\Phi \,\mathrm{d}k\,\mathrm{d}l$.

In the second integral, $\eta_c(k) \ne 0$ forces $|k| \le \frac{c}{4} < \frac{c}{3}$, so that ($\Phi$1) applies, and Fubini's theorem turns this integral into

$$\sum_\iota \int |k|^{2\epsilon\mathsf{z}}\, \mathsf{h}_\iota(k)\, \mathsf{g}_\iota(k;\mathsf{z})\,\mathrm{d}k, \qquad \mathsf{g}_\iota(k;\mathsf{z}) := \eta_c(k) \int |l|^{2\epsilon'\mathsf{z}}\, \Phi_\iota(k,l)\,\mathrm{d}l.$$

Since Φ_ι vanishes where $|k|+|l| < c$ or $|k|+|l| > C_\Phi$, the integrand defining g_ι vanishes unless $\frac{3}{4}c \le |l| \le C_\Phi$; consequently g_ι has the properties required in (a). If $\epsilon = 1$, assertion (a) provides a holomorphic continuation to $\mathbb{D}$ whose value at $\mathsf{z}=0$ is $\mathrm{f.p.}\int \mathsf{h}_\iota\, \mathsf{g}_\iota(\cdot;0)\,\mathrm{d}k$. If $\epsilon = 0$, all degrees of the h_ι are ≥ -2; the k-integral then converges absolutely for every z and defines an entire function, whose value at 0 is again the finite part, by (b). Summing over ι, we obtain at $\mathsf{z}=0$ the number $\mathrm{f.p.}\int \big[\eta_c(k)\int \Phi(k,l)\,\mathrm{d}l\big]\,\mathrm{d}k$.

The hypotheses of (e) are symmetric under the exchange of k with l, of ϵ with ϵ' and of (Φ1) with (Φ2): the set $\{(1,1),(1,0),(0,1)\}$ is invariant under $(\epsilon,\epsilon') \mapsto (\epsilon',\epsilon)$, conditions ($\Phi$0) and ($\Phi$3) do not change when k and l are interchanged, and the assumptions on the degrees in (Φ1) and (Φ2) are interchanged. Under this exchange the third integral, which contains the factor $\eta_c(l)$, becomes the second one, and the argument just given applies to it. In detail: $\eta_c(l) \ne 0$ forces $|l| \le \frac{c}{4} < \frac{c}{3}$, so that ($\Phi$2) applies, and Fubini's theorem turns the third integral into

$$\sum_\iota \int |l|^{2\epsilon'\mathsf{z}}\, \mathsf{h}'_\iota(l)\, \mathsf{g}'_\iota(l;\mathsf{z})\,\mathrm{d}l, \qquad \mathsf{g}'_\iota(l;\mathsf{z}) := \eta_c(l) \int |k|^{2\epsilon\mathsf{z}}\, \Phi'_\iota(k,l)\,\mathrm{d}k.$$

Since Φ'_ι vanishes where $|k|+|l| < c$ or $|k|+|l| > C_\Phi$, the integrand defining g'_ι vanishes unless $\frac{3}{4}c \le |k| \le C_\Phi$, and g'_ι has the properties required in (a). If $\epsilon' = 1$, assertion (a) provides a holomorphic continuation to $\mathbb{D}$ whose value at $\mathsf{z}=0$ is $\mathrm{f.p.}\int \mathsf{h}'_\iota\, \mathsf{g}'_\iota(\cdot;0)\,\mathrm{d}l$. If $\epsilon' = 0$, all degrees of the h'_ι are ≥ -2; the l-integral then converges absolutely for every z and defines an entire function, whose value at 0 is the finite part, by (b). Summing over ι, we obtain at $\mathsf{z}=0$ the number $\mathrm{f.p.}\int \big[\eta_c(l)\int \Phi(k,l)\,\mathrm{d}k\big]\,\mathrm{d}l$. The sum of the three values at $\mathsf{z}=0$ is the right side of the definition of $\mathrm{f.p.}\iint \Phi\,\mathrm{d}k\,\mathrm{d}l$ in Def. 7.44(2).

Change of the parameter c. Let $0 < c' < c$ and $\Phi \in \mathfrak{P}_2(c)$. Conditions ($\Phi$0), ($\Phi$1) and ($\Phi$2) at c' follow from the same conditions at c by restriction. For (Φ3) at c', let $|k|, |l| \ge \frac{c'}{8}$ and $\Phi(k,l) \ne 0$. If $|k|, |l| \ge \frac{c}{8}$, (Φ3) at c applies. Both $|k|$ and $|l|$ smaller than $\frac{c}{8}$ is impossible by (Φ0). If $\frac{c'}{8} \le |k| < \frac{c}{8}$, we use $\Phi = \sum_\iota \mathsf{h}_\iota \Phi_\iota$ from (Φ1): the h_ι are bounded on $\{\frac{c'}{8} \le |k| \le \frac{c}{8}\}$, the Φ_ι are bounded there as well (they are continuous on $\{|k| \le \frac{c}{8}\} \times \mathbb{R}^3$ and vanish for $|l| > C_\Phi$), and $|k-l| \le |k|+|l| \le C_\Phi$; hence $|\Phi| \le C \le C\, C_\Phi^2 |k-l|^{-2}$.

The case $\frac{c'}{8} \le |l| < \frac{c}{8}$ is symmetric, with (Φ2). Thus $\mathfrak{P}_2(c) \subset \mathfrak{P}_2(c')$. The integral with $(\epsilon,\epsilon') = (1,1)$ does not involve c, so the value at $\mathsf{z}=0$ of its continuation, which is the finite part with respect to c and also the finite part with respect to c', is the same.

Finally, let Φ be integrable. By Fubini's theorem the functions $k \mapsto \eta_c(k)\int \Phi(k,l)\,\mathrm{d}l$ and $l \mapsto \eta_c(l)\int \Phi(k,l)\,\mathrm{d}k$ are integrable elements of $\mathfrak{P}$, so by (b) their finite parts are their integrals, and the definition of $\mathrm{f.p.}\iint \Phi$ reduces to $\iint \Phi\,\mathrm{d}k\,\mathrm{d}l$. □

Proposition 7.46 (Expansion by regions of $\Xi_\Gamma[\Upsilon]$ at small p)**.** Let $\Gamma \in \{1,\dots,6\}$, $c>0$ and $\Upsilon \in \mathcal{Y}(c)$. Then $\overline{(\Upsilon F_\Gamma)_2} \in \mathfrak{P}_2(c)$, and there are numbers α, β and C, depending only on Γ, c and Υ, such that

$$|\Xi_\Gamma[\Upsilon](\mathsf{p}) - \alpha - \beta\,|\mathsf{p}| - \mathfrak{f}_\Gamma[\Upsilon]\,\mathsf{p}^2| \le C\,|\mathsf{p}|^3 \lg|\mathsf{p}| \qquad (0<|\mathsf{p}|\le \tfrac{c}{16}),$$

where

$$\mathfrak{f}_\Gamma[\Upsilon] := (2\pi)^{-6}\,\mathrm{f.\,p.}\iint \overline{(\Upsilon F_\Gamma)_2}\,\mathrm{d}k\,\mathrm{d}l.$$

The map $\Upsilon \mapsto \mathfrak{f}_\Gamma[\Upsilon]$ is linear on $\mathcal{Y}(c)$, and $\mathfrak{f}_\Gamma[\Upsilon^\Lambda] = \mathfrak{f}_\Gamma[\Upsilon]$ for every $\Lambda>1$. ■

Proof. Step 1: two factorisations of F_Γ. We write the integrand F_Γ of eq. 7.xxxv in two ways,

$$F_\Gamma = \sum S\cdot H = \sum S'\cdot H',$$

where the sums extend over the tensor indices carried by S, resp. by S', the factor S depends on (k,p) and S' on (l,p) as in the following table, and H, resp. H', denotes the product of all the other factors of F_Γ:

Γ	$S(k,\mathsf{p})$	$S'(l,\mathsf{p})$
1	$V_2(k,\mathsf{p}-k)_{\mu\nu}/\mathsf{x}_1\mathsf{x}'_1$	$V_2(-l,l-\mathsf{p})_{\rho\sigma}/\mathsf{x}_2\mathsf{x}'_2$
2, 3	$V_2(k,\mathsf{p}-k)_{\mu\nu}V_2(-k,k-\mathsf{p})_{\rho\nu}/\mathsf{x}_1^2\mathsf{x}'_1$	$1/\mathsf{x}_2$
4	$V_2(k,\mathsf{p}-k)_{\mu\nu}/\mathsf{x}_1\mathsf{x}'_1$	$V_2(l,\mathsf{p}-l)_{\mu\nu}/\mathsf{x}_2\mathsf{x}'_2$
5	$V_2(k-\mathsf{p},-k)_{\alpha\sigma}/\mathsf{x}_1\mathsf{x}'_1$	$1/\mathsf{x}_2$
6	$1/\mathsf{x}'_1$	$1/\mathsf{x}_2$

For $\Gamma=4$ this means

$$H_{\mu\nu} = -\tfrac12\big[\delta_{\mu\nu}V_2(l,\mathsf{p}-l)_{\lambda\lambda} - \tfrac12 V_2(l,\mathsf{p}-l)_{\mu\nu} - \tfrac12 V_2(l,\mathsf{p}-l)_{\nu\mu}\big]/\mathsf{x}_2\mathsf{x}'_2,$$

and H' has the same form, with $V_2(k,\mathsf{p}-k)$ and $\mathsf{x}_1\mathsf{x}'_1$ in place of $V_2(l,\mathsf{p}-l)$ and $\mathsf{x}_2\mathsf{x}'_2$. Every component of S has the form required in Lem. 7.45(d): it is a homogeneous polynomial divided by $|k|^{2\mathsf{a}}|k-\mathsf{p}|^{2\mathsf{a}'}$, with $(\mathsf{a},\mathsf{a}')=(1,1)$ for $\Gamma\in\{1,4,5\}$, $(2,1)$ for $\Gamma\in\{2,3\}$ and $(0,1)$ for $\Gamma=6$, and since $|V_2(k_1,k_2)|\le 2|k_1||k_2|$ its modulus is at most $2/|k||k-\mathsf{p}| \le |k|^{-2}+|k-\mathsf{p}|^{-2}$ for $\Gamma\in\{1,4,5\}$, at most $4|k|^{-2}$ for $\Gamma\in\{2,3\}$, and equal to $|k-\mathsf{p}|^{-2}$ for $\Gamma=6$.

The same holds for the components of S', as functions of l. The factor S contains all factors $\mathsf{x}_1,\mathsf{x}'_1$ of the denominator of F_Γ, and S' all factors $\mathsf{x}_2,\mathsf{x}'_2$. Hence H is a polynomial divided by powers of $\mathsf{x}_2,\mathsf{x}'_2,\mathsf{x}_3$, and H' a polynomial divided by powers of $\mathsf{x}_1,\mathsf{x}'_1,\mathsf{x}_3$; in both, x_3 occurs at most to the first power. Moreover, H and H' are even under the reflection of all their arguments: they are quotients of homogeneous polynomials of even degree, because besides factors V_2 each of them contains two factors V_3, or the factor $l_\rho(l-k)_\mu$, or neither.

Step 2: where the lines are long. Put $\mu_0 := \frac{c}{8}$ and $\eta := \eta_c = \theta(|\cdot|/2\mu_0)$. Let $|\mathsf{p}|<\mu_0$. By (Y1), $\Upsilon(k,l,\mathsf{p})\neq 0$ implies $|k|+|l|\ge 8\mu_0$. If, in addition, $|k|<3\mu_0$, then

$$|l|>5\mu_0, \qquad |l-\mathsf{p}|>4\mu_0, \qquad |k-l|>2\mu_0;$$

these are the lengths whose squares $\mathsf{x}_2,\mathsf{x}'_2,\mathsf{x}_3$ occur in the denominator of H. Consequently the product ΥH, extended by 0 where $\Upsilon=0$, is smooth on $\{|k|<3\mu_0\}\times\mathbb{R}^3\times\{|\mathsf{p}|<\mu_0\}$: this set is covered by the open set $\{|k|+|l|<8\mu_0,\ |\mathsf{p}|<c\}$, on which Υ vanishes identically, and by the open set $\{|k|<3\mu_0,\ |l|>5\mu_0,\ |\mathsf{p}|<\mu_0\}$, on which H is smooth. Symmetrically, if $\Upsilon(k,l,\mathsf{p})\neq 0$ and $|l|<3\mu_0$, then $|k|\ge 8\mu_0-|l|$, and

$$|k|>5\mu_0, \qquad |k-\mathsf{p}|>4\mu_0, \qquad |k-l|>2\mu_0;$$

these are the lengths whose squares $\mathsf{x}_1,\mathsf{x}'_1,\mathsf{x}_3$ occur in the denominator of H'. The set $\mathbb{R}^3\times\{|l|<3\mu_0\}\times\{|\mathsf{p}|<\mu_0\}$ is covered by the open set $\{|k|+|l|<8\mu_0,\ |\mathsf{p}|<c\}$, on which Υ vanishes identically, and by the open set $\{|k|>5\mu_0,\ |l|<3\mu_0,\ |\mathsf{p}|<\mu_0\}$, on which H' is smooth; hence $\Upsilon H'$, extended by 0 where $\Upsilon=0$, is smooth on it. Finally, if $|k|,|l|\ge\mu_0$ and $|\mathsf{p}|\le\frac12\mu_0$, then $\mathsf{x}_1,\mathsf{x}'_1,\mathsf{x}_2,\mathsf{x}'_2\ge\frac14\mu_0^2$, and therefore $F_\Gamma = F^\sharp/\mathsf{x}_3$ with a function $F^\sharp$ which is smooth on this set.

Step 3: $\Phi := \overline{(\Upsilon F_\Gamma)_2}$ *belongs to* $\mathfrak{P}_2(c)$. Condition (Φ0) follows from (Y1): where $|k|+|l|<c$ or $|k|+|l|>C_\Upsilon$, the function ΥF_Γ vanishes for all $|\mathsf{p}|<c$, and so does Φ. For (Φ1) let $0<|k|<\frac{c}{3}$; then $|k|<3\mu_0$, and

$$\Phi = \frac{1}{6}\sum_j \sum \Big[\partial_{\mathsf{p}_j}^2 S\cdot \Upsilon H + 2\,\partial_{\mathsf{p}_j} S\cdot \partial_{\mathsf{p}_j}(\Upsilon H) + S\cdot \partial_{\mathsf{p}_j}^2(\Upsilon H)\Big]_{\mathsf{p}=0}$$

by the Leibniz rule. Here $\partial_{\mathsf{p}}^\beta S(\cdot,0)$ is a homogeneous factor of degree $-2-|\beta|$, and $\partial_{\mathsf{p}}^\beta(\Upsilon H)(k,l,0)$ is smooth by Step 2. The latter function vanishes wherever $\Upsilon(\cdot,\cdot,\mathsf{p})=0$ for all small p, that is, by (Y1), where $|k|+|l|<c$ or $|k|+|l|>C_\Upsilon$; this is the support condition of (Φ1), with any $C_\Phi \ge C_\Upsilon$. Condition (Φ2) is obtained from the factorisation $F_\Gamma = \sum S'\cdot H'$ by the same argument with the roles of k and l exchanged: for $0<|l|<\frac{c}{3}$ the Leibniz rule writes Φ as the corresponding sum of products of the functions $\partial_{\mathsf{p}}^\beta S'(\cdot,0)$, which are homogeneous factors of degree $-2-|\beta|$ in l, with the functions $\partial_{\mathsf{p}}^\beta(\Upsilon H')(k,l,0)$, which are smooth on $\mathbb{R}^3\times\{|l|<3\mu_0\}$ by Step 2 and vanish where $|k|+|l|<c$ or $|k|+|l|>C_\Upsilon$.

For (Φ3) let $|k|,|l|\ge\frac{c}{8}=\mu_0$; by Step 2,

$$\Phi = \tfrac{1}{6}\,\Delta_{\mathsf{p}}(\Upsilon F^\sharp)\big|_{\mathsf{p}=0}/\mathsf{x}_3,$$

and the numerator is bounded, being smooth with compact support in (k,l).

Step 4: three regions. For $0<|\mathsf{p}|\le\frac12\mu_0$ we split $\Xi_\Gamma[\Upsilon]=\Xi_{\mathrm{H}}+\Xi_{\mathrm{A}}+\Xi_{\mathrm{B}}$, inserting into the integral the factors $1-\eta(k)-\eta(l)$, $\eta(k)$ and $\eta(l)$, respectively. If $\Upsilon\ne0$ and $|k|<\mu_0$, then $|l|>7\mu_0$, hence $\eta(k)=1$ and $\eta(l)=0$; thus on the support of Υ the first factor vanishes unless $|k|,|l|\ge\mu_0$, and the same holds with k and l exchanged. Each of the three terms is invariant under rotations of p, by (Y2), by the rotation invariance of F_Γ, and because η is radial.

Step 5: the hard region. The integrand of Ξ_{H} is $(1-\eta(k)-\eta(l))\,\Upsilon F^\sharp/\mathsf{x}_3$; it is supported in $\{\mu_0\le|k|,|l|\le C_\Upsilon\}$, and all its derivatives with respect to p are bounded by $C|k-l|^{-2}$. Hence Ξ_{H} is smooth on $\{|\mathsf{p}|<\frac12\mu_0\}$; it is even, by (Y2) and the evenness of F_Γ, and rotation invariant. At $\mathsf{p}=0$ its gradient and its third derivatives therefore vanish, and its Hessian is a multiple of the identity, so that

$$\Xi_{\mathrm{H}}(\mathsf{p}) = \Xi_{\mathrm{H}}(0) + \tfrac16\,\Delta\Xi_{\mathrm{H}}(0)\,\mathsf{p}^2 + O(|\mathsf{p}|^4), \qquad \tfrac16\,\Delta\Xi_{\mathrm{H}}(0) = (2\pi)^{-6}\iint (1-\eta(k)-\eta(l))\,\Phi\,\mathrm{d}k\,\mathrm{d}l.$$

Step 6: k soft. Let $J(k,\mathsf{p}) := \int \Upsilon H\, d^\circ l$, one function for each value of the tensor indices of S. By Step 2, and since ΥH vanishes for $|l|>C_\Upsilon$, the function J is smooth on $\{|k|<3\mu_0\}\times\{|\mathsf{p}|<\mu_0\}$, and it is even, as the substitution $l\to -l$ shows. By Fubini's theorem $\Xi_{\mathrm{A}} = \sum\int \eta S J\, d^\circ k$. We apply Lem. 7.45(d), with the present μ_0, to each component of S and the corresponding component of J; for $0<|\mathsf{p}|\le\frac12\mu_0$ this gives

$$(2\pi)^3\,\Xi_{\mathrm{A}}(\mathsf{p}) = \sum\Big\{\text{f. p.}\int \eta\,[SJ]_0\,\mathrm{d}k + \sum_{i,j}\mathsf{p}_i\mathsf{p}_j\,\tfrac12\,\text{f. p.}\int \eta\,\partial_{\mathsf{p}_i}\partial_{\mathsf{p}_j}(SJ)(k,0)\,\mathrm{d}k + J(0,0)\,\mathsf{h}_S(\mathsf{p})\Big\} + O(|\mathsf{p}|^3\lg|\mathsf{p}|).$$

We take the mean of both sides over the sphere $|\mathsf{p}|=t$. The left side is constant there, by Step 4. The mean of h_S is t times its mean over the unit sphere, by homogeneity, and the mean of $\mathsf{p}_i\mathsf{p}_j$ is $\frac13 t^2\delta_{ij}$. For $k\ne0$, differentiation under the integral sign gives $\sum\frac16\Delta_{\mathsf{p}}(SJ)(k,0) = \int\Phi(k,l)\,d^\circ l$. Since the finite part is linear, we conclude that

$$\Xi_{\mathrm{A}}(\mathsf{p}) = \alpha_{\mathrm{A}} + \beta_{\mathrm{A}}|\mathsf{p}| + (2\pi)^{-6}\,\mathsf{p}^2\,\text{f. p.}\int\Big[\eta(k)\int\Phi(k,l)\,\mathrm{d}l\Big]\,\mathrm{d}k + O(|\mathsf{p}|^3\lg|\mathsf{p}|)$$

with numbers α_{A}, β_{A}.

Step 7: l soft, and the sum. For Ξ_{B} we exchange the roles of k and l in Step 6. We use the factorisation $F_\Gamma=\sum S'\cdot H'$ of Step 1 and put $J'(l,\mathsf{p}) := \int\Upsilon H'\,d^\circ k$, one function for each value of

the tensor indices of S'. By Step 2, and since $\Upsilon H'$ vanishes for $|k| > C_\Upsilon$, the function J' is smooth on $\{|l| < 3\mu_0\} \times \{|\mathsf{p}| < \mu_0\}$, and it is even, as the substitution $k \to -k$ shows; by Fubini's theorem $\Xi_{\mathrm{B}} = \sum \int \eta S' J' \, d^\circ l$. We apply Lem. 7.45(d), with l in place of k, to each component of S' and the corresponding component of J', and we take the mean over the sphere $|\mathsf{p}| = t$ as in Step 6, now with $\sum \frac{1}{6}\Delta_{\mathsf{p}}(S'J')(l,0) = \int \Phi(k,l)\, d^\circ k$ for $l \neq 0$. This gives

$$\Xi_{\mathrm{B}}(\mathsf{p}) = \alpha_{\mathrm{B}} + \beta_{\mathrm{B}}|\mathsf{p}| + (2\pi)^{-6}\,\mathsf{p}^2\,\mathrm{f.p.}\int \Big[\eta(l)\int \Phi(k,l)\,\mathrm{d}k\Big]\,\mathrm{d}l + O(|\mathsf{p}|^3 \lg|\mathsf{p}|) \qquad (0 < |\mathsf{p}| \le \tfrac{1}{2}\mu_0)$$

with numbers α_{B}, β_{B}. Since $\eta = \eta_c$ and $\Phi \in \mathfrak{P}_2(c)$ by Step 3, Def. 7.44(2) shows that the sum of the three coefficients of p^2 obtained in Steps 5–7 is $\mathfrak{f}_\Gamma[\Upsilon]$. Adding the three expansions, and noting that $\frac{1}{2}\mu_0 = \frac{c}{16}$, we obtain the asserted estimate, with $\alpha := \Xi_{\mathrm{H}}(0) + \alpha_{\mathrm{A}} + \alpha_{\mathrm{B}}$ and $\beta := \beta_{\mathrm{A}} + \beta_{\mathrm{B}}$.

Step 8: linearity and scaling. The map $\Upsilon \mapsto \overline{(\Upsilon F_\Gamma)_2}$ and the finite part are linear; hence so is $\mathfrak{f}_\Gamma$. Let $\Lambda > 1$. The substitution $k \to \Lambda k$, $l \to \Lambda l$, together with the homogeneity of degree -4 of F_Γ, gives $\Xi_\Gamma[\Upsilon^\Lambda](\mathsf{p}) = \Lambda^2\,\Xi_\Gamma[\Upsilon](\mathsf{p}/\Lambda)$, so that by the estimate already proved

$$\Xi_\Gamma[\Upsilon^\Lambda](\mathsf{p}) = \Lambda^2\alpha + \Lambda\beta\,|\mathsf{p}| + \mathfrak{f}_\Gamma[\Upsilon]\,\mathsf{p}^2 + O(|\mathsf{p}|^3 \lg|\mathsf{p}|).$$

On the other hand $\Upsilon^\Lambda \in \mathcal{Y}(c)$, and the estimate applied to Υ^Λ gives $\Xi_\Gamma[\Upsilon^\Lambda](\mathsf{p}) = \alpha' + \beta'|\mathsf{p}| + \mathfrak{f}_\Gamma[\Upsilon^\Lambda]\,\mathsf{p}^2 + O(|\mathsf{p}|^3\lg|\mathsf{p}|)$ with numbers α', β'. If $\mathsf{a} + \mathsf{b}t + \mathsf{c}t^2 = o(t^2)$ as $t \downarrow 0$, then $\mathsf{a} = \mathsf{b} = \mathsf{c} = 0$; comparing the two expansions, we obtain in particular $\mathfrak{f}_\Gamma[\Upsilon^\Lambda] = \mathfrak{f}_\Gamma[\Upsilon]$. □

7.1.16. *The value: the functional $\mathcal{R}$, the residue, the bridge.* By Prop. 7.46, the coefficient of p^2 in the expansion of $\Xi_\Gamma[\Upsilon]$ at small p is the number $\mathfrak{f}_\Gamma[\Upsilon]$, which is given by a finite part. We now compute it for the differences $R_\Gamma^\Lambda - R_\Gamma$ of the cut-offs of Lem. 7.43(b). The result is $d_{2,\Gamma}\log\Lambda$ with $d_{2,\Gamma} = \mathcal{R}[F_\Gamma]/16\pi^2$, where $\mathcal{R}$ is a linear functional defined below; the comparison with Lem. 7.43 then expresses $\kappa_\Gamma^{\mathrm{lat}}$ through $\mathcal{R}[F_\Gamma]$.

Definition 7.47 (The functional $\mathcal{R}$)**.** We use the variables $\mathsf{x}_0, \mathsf{x}_1, \mathsf{x}_1', \mathsf{x}_2, \mathsf{x}_2', \mathsf{x}_3$ of Def. 7.32(1).

(1) The scalar products of k, l, p are linear in these variables:

$$k\cdot l = \tfrac{1}{2}(\mathsf{x}_1 + \mathsf{x}_2 - \mathsf{x}_3), \qquad k\cdot\mathsf{p} = \tfrac{1}{2}(\mathsf{x}_1 - \mathsf{x}_1' + \mathsf{x}_0), \qquad l\cdot\mathsf{p} = \tfrac{1}{2}(\mathsf{x}_2 - \mathsf{x}_2' + \mathsf{x}_0).$$

(2) For $n \in \mathbb{Z}$ let $\mathcal{L}_n$ be the real linear span of the functions $\mathsf{x}_1^{-\mathsf{a}}\mathsf{x}_2^{-\mathsf{b}}\mathsf{x}_3^{-\mathsf{c}}$, where $\mathsf{a},\mathsf{b},\mathsf{c} \in \mathbb{Z}$ and $\mathsf{a}+\mathsf{b}+\mathsf{c} = -n$, on the set $\{(k,l) : k \neq 0,\ l \neq 0,\ k \neq l\}$. These functions are linearly independent. Indeed, the image of the map $(k,l) \mapsto (\mathsf{x}_1,\mathsf{x}_2,\mathsf{x}_3)$ contains a non-empty open set: for $k = (t_1,t_2,0)$ and $l = (0,t_3,0)$ the map becomes

$$(t_1,t_2,t_3) \longmapsto (t_1^2 + t_2^2,\ t_3^2,\ t_1^2 + (t_2 - t_3)^2),$$

whose differential at $(1,0,1)$ has determinant 8, so that the inverse function theorem applies. A vanishing finite linear combination of the functions, multiplied by a sufficiently high power of $\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3$, becomes a polynomial in $(\mathsf{x}_1,\mathsf{x}_2,\mathsf{x}_3)$ which vanishes on a non-empty open set, and such a polynomial is zero. We denote by $r_{111}\colon \mathcal{L}_{-3} \to \mathbb{R}$ the linear map which assigns to an element of $\mathcal{L}_{-3}$ its coefficient of $(\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3)^{-1}$.

(3) Let $\mathfrak{F}$ be the real linear span of the monomials

$$\mathsf{x}_0^n\,\mathsf{x}_1^{-a_1}\mathsf{x}_1'^{-a_2}\mathsf{x}_2^{-a_3}\mathsf{x}_2'^{-a_4}\mathsf{x}_3^{-a_5}, \qquad \text{integers } n \ge 0,\ a_1,\dots,a_5, \quad \textstyle\sum_i a_i = 2 + n;$$

these monomials are homogeneous of degree -4. The numerators of the integrands in eq. 7.xxxv are complete contractions of tensors built from k, l, p and Kronecker symbols; hence they are polynomials in the scalar products of k, l, p, and by (1) polynomials in the six variables. Since moreover x_3 occurs in the denominators of eq. 7.xxxv at most to the first power, and not at all in F_4, each F_Γ belongs to $\mathfrak{F}$: it is a linear combination of monomials with $a_5 \le 1$, and F_4 is a linear combination of monomials with $a_5 \le 0$.

(4) For $F \in \mathfrak{F}$ we put

$$\mathcal{R}[F] := r_{111}(\overline{F_2}), \tag{7.xliii}$$

with $\overline{F_2}$ as in Def. 7.44(3); by Lem. 7.48 below, $\overline{F_2} \in \mathcal{L}_{-3}$.

■

Lemma 7.48 (Values of $\mathcal{R}$ on monomials)**.** For every $F \in \mathfrak{F}$ the function $\overline{F_2}$ belongs to $\mathcal{L}_{-3}$; hence $\mathcal{R}$ is well defined on $\mathfrak{F}$, and it is linear. If F is a linear combination of monomials with $a_5 \le 1$, then $\overline{F_2}$ is a linear combination of functions $\mathsf{x}_1^{-\mathsf{a}}\mathsf{x}_2^{-\mathsf{b}}\mathsf{x}_3^{-\mathsf{c}}$ with $\mathsf{c} \le 1$. For integers $n \ge 0$ and $a_1, \dots, a_5$ with $\sum_i a_i = 2 + n$, put

$$(\alpha_1, \alpha_2, \alpha_3) := (a_1 + a_2,\ a_3 + a_4,\ a_5), \qquad \mathsf{q}(t) := \tfrac{1}{3}t(2t-1).$$

Then

$$\mathcal{R}\big[\mathsf{x}_0^n\, \mathsf{x}_1^{-a_1}\mathsf{x}_1'^{-a_2}\mathsf{x}_2^{-a_3}\mathsf{x}_2'^{-a_4}\mathsf{x}_3^{-a_5}\big] = \begin{cases} 1 & n = 1,\ (\alpha_1,\alpha_2,\alpha_3) = (1,1,1), \\ \mathsf{q}(a_2) + \frac{2}{3}a_2a_4 & n = 0,\ (\alpha_1,\alpha_2,\alpha_3) = (0,1,1), \\ \mathsf{q}(a_4) + \frac{2}{3}a_2a_4 & n = 0,\ (\alpha_1,\alpha_2,\alpha_3) = (1,0,1), \\ -\frac{2}{3}a_2a_4 & n = 0,\ (\alpha_1,\alpha_2,\alpha_3) = (0,0,2), \\ 0 & \text{otherwise.} \end{cases} \tag{7.xliv}$$

■

Proof. The map $F \mapsto \overline{F_2}$ is linear; it therefore suffices to consider a monomial F as on the left of eq. 7.xliv. We write $x := k \cdot \mathsf{p}$ and $y := l \cdot \mathsf{p}$; then $\mathsf{x}_1' = \mathsf{x}_1 - 2x + \mathsf{x}_0$ and $\mathsf{x}_2' = \mathsf{x}_2 - 2y + \mathsf{x}_0$. For an integer t, the binomial series $(1+w)^{-t} = 1 - tw + \frac{1}{2}t(t+1)w^2 + O(w^3)$, taken at $w = (\mathsf{x}_0 - 2x)/\mathsf{x}_1$, gives for fixed (k,l) and small p

$$\mathsf{x}_1'^{-t} = \mathsf{x}_1^{-t}\Big[1 + 2t\,\frac{x}{\mathsf{x}_1} - t\,\frac{\mathsf{x}_0}{\mathsf{x}_1} + 2t(t+1)\,\frac{x^2}{\mathsf{x}_1^2} + O(|\mathsf{p}|^3)\Big],$$

and $\mathsf{x}_2'^{-t}$ has the same expansion, with y and x_2 in place of x and x_1. We use these expansions at $t = a_2$ and at $t = a_4$. On the terms of second order the operator $\frac{1}{6}\Delta_{\mathsf{p}}$ acts by

$$\tfrac{1}{6}\Delta_{\mathsf{p}}\mathsf{x}_0 = 1, \qquad \tfrac{1}{6}\Delta_{\mathsf{p}}x^2 = \tfrac{1}{3}\mathsf{x}_1, \qquad \tfrac{1}{6}\Delta_{\mathsf{p}}y^2 = \tfrac{1}{3}\mathsf{x}_2, \qquad \tfrac{1}{6}\Delta_{\mathsf{p}}(xy) = \tfrac{1}{3}\,k\cdot l = \tfrac{1}{6}(\mathsf{x}_1 + \mathsf{x}_2 - \mathsf{x}_3),$$

the last equality by Def. 7.47(1).

If $n \ge 2$, then $F = O(|\mathsf{p}|^4)$ and $\overline{F_2} = 0$. If $n = 1$, the term of second order in p of F is $\mathsf{x}_0\,\mathsf{x}_1^{-\alpha_1}\mathsf{x}_2^{-\alpha_2}\mathsf{x}_3^{-\alpha_3}$, and $\overline{F_2} = \mathsf{x}_1^{-\alpha_1}\mathsf{x}_2^{-\alpha_2}\mathsf{x}_3^{-\alpha_3}$. If $n = 0$, the terms of second order are

$$\mathsf{x}_1^{-\alpha_1}\mathsf{x}_2^{-\alpha_2}\mathsf{x}_3^{-\alpha_3}\Big\{\Big[-a_2\frac{\mathsf{x}_0}{\mathsf{x}_1} + 2a_2(a_2+1)\frac{x^2}{\mathsf{x}_1^2}\Big] + \Big[-a_4\frac{\mathsf{x}_0}{\mathsf{x}_2} + 2a_4(a_4+1)\frac{y^2}{\mathsf{x}_2^2}\Big] + 4a_2a_4\,\frac{xy}{\mathsf{x}_1\mathsf{x}_2}\Big\},$$

and since $-t + \frac{2}{3}t(t+1) = \mathsf{q}(t)$, the rules above give

$$\overline{F_2} = \mathsf{x}_1^{-\alpha_1}\mathsf{x}_2^{-\alpha_2}\mathsf{x}_3^{-\alpha_3}\Big\{\mathsf{q}(a_2)\,\mathsf{x}_1^{-1} + \mathsf{q}(a_4)\,\mathsf{x}_2^{-1} + \tfrac{2}{3}a_2a_4\big[\mathsf{x}_2^{-1} + \mathsf{x}_1^{-1} - \mathsf{x}_3\mathsf{x}_1^{-1}\mathsf{x}_2^{-1}\big]\Big\}.$$

In each case $\overline{F_2} \in \mathcal{L}_{-3}$, because $\alpha_1 + \alpha_2 + \alpha_3 = 2 + n$; and the exponent c of x_3 in the terms obtained is a_5 or $a_5 - 1$, which proves the assertion on $\mathsf{c} \le 1$. Finally, the monomial $(\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3)^{-1}$ occurs for $n = 1$ exactly when $(\alpha_1,\alpha_2,\alpha_3) = (1,1,1)$, with coefficient 1. For $n = 0$ it arises from the terms containing x_1^{-1} exactly when $(\alpha_1,\alpha_2,\alpha_3) = (0,1,1)$, from the terms containing x_2^{-1} exactly when $(\alpha_1,\alpha_2,\alpha_3) = (1,0,1)$, and from the last term exactly when $(\alpha_1,\alpha_2,\alpha_3) = (0,0,2)$. Collecting the coefficients, we obtain eq. 7.xliv. □

Lemma 7.49 (The residue)**.** Let $\mathsf{M} = \mathsf{x}_1^{-\mathsf{a}}\mathsf{x}_2^{-\mathsf{b}}\mathsf{x}_3^{-\mathsf{c}}$ with integers $\mathsf{a},\mathsf{b},\mathsf{c}$ such that $\mathsf{a}+\mathsf{b}+\mathsf{c}=3$ and $\mathsf{c}\le 1$. Let $\phi \in C_c^\infty(\mathbb{R}^3\times\mathbb{R}^3)$ be equal to 1 near 0, let $\Lambda>1$, and let ϕ^Λ be as in Def. 7.44(3). Then $(\phi^\Lambda-\phi)\mathsf{M} \in \mathfrak{P}_2(c)$ for all sufficiently small $c>0$, and

$$\text{f. p.}\iint (\phi^\Lambda-\phi)\,\mathsf{M}\,\mathrm{d}k\,\mathrm{d}l = \begin{cases} 4\pi^4\log\Lambda & (\mathsf{a},\mathsf{b},\mathsf{c})=(1,1,1),\\ 0 & \text{otherwise;}\end{cases}$$

$$\oint_{S^5}\frac{\mathrm{d}\omega}{\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3} = 4\pi^4, \tag{7.xlv}$$

$$\frac{4\pi^4}{(2\pi)^6} = \frac{1}{16\pi^2},$$

where $d\omega$ denotes the surface measure of the sphere $S^5\subset\mathbb{R}^3\times\mathbb{R}^3$. ■

Proof. Step 0: the class. Let $\Upsilon_0\in C_c^\infty(\mathbb{R}^6)$ vanish for $|k|+|l|<c$. We show that $\Upsilon_0\mathsf{M}\in\mathfrak{P}_2(c)$. Condition ($\Phi$0) holds, since Υ_0 vanishes for $|k|+|l|<c$ and has compact support. If $|k|<\frac{c}{3}$ and $\Upsilon_0(k,l)\neq 0$, then $|l|\ge c-|k|>\frac{2}{3}c$ and $|k-l|\ge|l|-|k|>\frac{c}{3}$. Hence $\Upsilon_0\,\mathsf{x}_2^{-\mathsf{b}}\mathsf{x}_3^{-\mathsf{c}}$, extended by 0 where Υ_0 vanishes, is smooth on $\{|k|<\frac{c}{3}\}\times\mathbb{R}^3$ (it vanishes on the open set $\{|k|+|l|<c\}$, and it is smooth on the open set $\{|k|<\frac{c}{3},\ |l|>\frac{2}{3}c\}$), and it vanishes where $|k|+|l|<c$ or where (k,l) lies outside the support of Υ_0.

Thus (Φ1) holds with the single homogeneous factor $|k|^{-2\mathsf{a}}$, of degree $-2\mathsf{a}$. Symmetrically, if $|l|<\frac{c}{3}$ and $\Upsilon_0(k,l)\neq 0$, then $|k|\ge c-|l|>\frac{2}{3}c$ and $|k-l|\ge|k|-|l|>\frac{c}{3}$. Hence $\Upsilon_0\,\mathsf{x}_1^{-\mathsf{a}}\mathsf{x}_3^{-\mathsf{c}}$, extended by 0 where Υ_0 vanishes, is smooth on $\mathbb{R}^3\times\{|l|<\frac{c}{3}\}$ (it vanishes on the open set $\{|k|+|l|<c\}$, and it is smooth on the open set $\{|k|>\frac{2}{3}c,\ |l|<\frac{c}{3}\}$), and it vanishes where $|k|+|l|<c$ or where (k,l) lies outside the support of Υ_0. Thus (Φ2) holds with the single homogeneous factor $|l|^{-2\mathsf{b}}$, of degree $-2\mathsf{b}$.

Condition (Φ3) holds because $\mathsf{c}\le 1$ and Υ_0 has compact support. We apply this to $\Upsilon_0=\phi^\Lambda-\phi$, which vanishes near 0, since ϕ and ϕ^Λ are both equal to 1 there.

Step 1: independence of the cut-off. Let Υ_0 be as in Step 0. For large $\mathrm{Re}\,\mathsf{z}$ the substitution $(k,l)\to(\Lambda k,\Lambda l)$ and the homogeneity of degree -6 of M give

$$\iint(|k||l|)^{2\mathsf{z}}\,\Upsilon_0^\Lambda\,\mathsf{M}\,\mathrm{d}k\,\mathrm{d}l = \Lambda^{4\mathsf{z}+6-6}\iint(|k||l|)^{2\mathsf{z}}\,\Upsilon_0\,\mathsf{M}\,\mathrm{d}k\,\mathrm{d}l.$$

Since Υ_0^Λ also vanishes for $|k|+|l|<c$, Lem. 7.45(e), with $(\epsilon,\epsilon')=(1,1)$, continues both sides holomorphically to $\mathbb{D}$, and at $\mathsf{z}=0$ we obtain $\text{f. p.}\iint\Upsilon_0^\Lambda\mathsf{M} = \text{f. p.}\iint\Upsilon_0\mathsf{M}$. If $\tilde\phi$ is a second function with the properties of ϕ, we take $\Upsilon_0:=\phi-\tilde\phi$; then

$$(\phi^\Lambda-\phi)-(\tilde\phi^\Lambda-\tilde\phi)=\Upsilon_0^\Lambda-\Upsilon_0,$$

and the finite part of $(\phi^\Lambda-\phi)\mathsf{M}$ does not depend on the choice of ϕ. We may therefore choose ϕ conveniently in each case. In the following steps we use the function $\varphi(s):=\theta(s)$ of one variable $s\ge 0$; it is equal to 1 on $[0,\frac{1}{2}]$ and to 0 on $[1,\infty)$.

Step 2: the case $\mathsf{c}\le 0$. We choose $\phi(k,l):=\varphi(|k|)\varphi(|l|)$. Since $\mathsf{x}_3^{-\mathsf{c}}=(k^2-2k\cdot l+l^2)^{|\mathsf{c}|}$ is a polynomial, M is a finite sum

$$\mathsf{M}=\sum c_{\alpha\beta}\,|k|^{-2\mathsf{a}_\alpha}k^\alpha\cdot|l|^{-2\mathsf{b}_\beta}l^\beta$$

with numbers $c_{\alpha\beta}$, multi-indices α,β and integers $\mathsf{a}_\alpha,\mathsf{b}_\beta$. For large $\mathrm{Re}\,\mathsf{z}$ put

$$\mathsf{Z}(\mathsf{z}):=\iint(|k||l|)^{2\mathsf{z}}\,\phi\,\mathsf{M}\,\mathrm{d}k\,\mathrm{d}l=\sum c_{\alpha\beta}\int|k|^{2\mathsf{z}-2\mathsf{a}_\alpha}k^\alpha\varphi(|k|)\,\mathrm{d}k\int|l|^{2\mathsf{z}-2\mathsf{b}_\beta}l^\beta\varphi(|l|)\,\mathrm{d}l.$$

As M is homogeneous of degree -6, we have $\iint(|k||l|)^{2\mathsf{z}}\phi^\Lambda\mathsf{M}\,\mathrm{d}k\,\mathrm{d}l=\Lambda^{4\mathsf{z}}\mathsf{Z}(\mathsf{z})$. The functions $|k|^{-2\mathsf{a}_\alpha}k^\alpha$ and $|l|^{-2\mathsf{b}_\beta}l^\beta$ are homogeneous factors, so by Lem. 7.45(a) each factor in the expression

for Z continues holomorphically to $\mathbb{D}$. Hence the continuation of $\iint(|k||l|)^{2\mathsf{z}}(\phi^\Lambda-\phi)\mathsf{M}\,\mathrm{d}k\,\mathrm{d}l$ is $(\Lambda^{4\mathsf{z}}-1)\mathsf{Z}(\mathsf{z})$, which vanishes at $\mathsf{z}=0$; by Lem. 7.45(e) this value is the finite part.

Step 3: the case $\mathsf{c}=1$, $(\mathsf{a},\mathsf{b})\neq(1,1)$. Now $\mathsf{a}+\mathsf{b}=2$. The finite part is symmetric under $k\leftrightarrow l$ (Def. 7.44(2)), and ϕ may be chosen freely; we may therefore assume $\mathsf{b}\le 0$ and $\mathsf{a}\ge 2$. The degree of the homogeneous factor in (Φ2) is then $-2\mathsf{b}\ge 0$, so that Lem. 7.45(e) applies with $(\epsilon,\epsilon')=(1,0)$. We choose $\phi(k,l):=\varphi(|k|)\varphi(|l-k|)$ and write $\mathsf{m}:=l-k$. Then

$$\mathsf{M}=|k|^{-2\mathsf{a}}\,|k+\mathsf{m}|^{2|\mathsf{b}|}\,|\mathsf{m}|^{-2}=\sum c_{\alpha\beta}\,|k|^{-2\mathsf{a}}k^\alpha\cdot\mathsf{m}^\beta|\mathsf{m}|^{-2},$$

and for large $\operatorname{Re}\mathsf{z}$

$$\mathsf{Z}(\mathsf{z}):=\iint|k|^{2\mathsf{z}}\,\phi\,\mathsf{M}\,\mathrm{d}k\,\mathrm{d}l=\sum c_{\alpha\beta}\int|k|^{2\mathsf{z}-2\mathsf{a}}k^\alpha\varphi(|k|)\,\mathrm{d}k\int\mathsf{m}^\beta|\mathsf{m}|^{-2}\varphi(|\mathsf{m}|)\,\mathrm{d}\mathsf{m},$$

while $\iint|k|^{2\mathsf{z}}\phi^\Lambda\mathsf{M}\,\mathrm{d}k\,\mathrm{d}l=\Lambda^{2\mathsf{z}}\mathsf{Z}(\mathsf{z})$. The integrals over m converge absolutely, and the integrals over k continue to $\mathbb{D}$ by Lem. 7.45(a). The continuation $(\Lambda^{2\mathsf{z}}-1)\mathsf{Z}(\mathsf{z})$ vanishes at $\mathsf{z}=0$.

Step 4: the case $(\mathsf{a},\mathsf{b},\mathsf{c})=(1,1,1)$. We choose $\phi(k,l):=\varphi((k^2+l^2)^{1/2})$. All degrees of the homogeneous factors in (Φ1) and (Φ2) are now -2; hence $(\phi^\Lambda-\phi)\mathsf{M}$ is integrable, and by Lem. 7.45(e) its finite part is its integral. In polar coordinates of $\mathbb{R}^6$,

$$\iint(\phi^\Lambda-\phi)\,\mathsf{M}\,\mathrm{d}k\,\mathrm{d}l=\oint_{S^5}\mathsf{M}\,\mathrm{d}\omega\cdot\int_0^\infty\bigl[\varphi(s/\Lambda)-\varphi(s)\bigr]\frac{\mathrm{d}s}{s}.$$

The integrand on the right vanishes for $s\le\frac12$, and for $0<\varepsilon\le\frac12$

$$\int_\varepsilon^\infty\bigl[\varphi(s/\Lambda)-\varphi(s)\bigr]\frac{ds}{s}=\int_{\varepsilon/\Lambda}^{\varepsilon}\varphi(s)\,\frac{ds}{s}=\log\Lambda.$$

Together with the value of the integral over S^5, computed in Step 5, this gives $4\pi^4\log\Lambda$.

Step 5: the integral over the sphere. We parametrise S^5 by $k=\cos\psi\,\omega_1$, $l=\sin\psi\,\omega_2$ with $\psi\in(0,\frac{\pi}{2})$ and $\omega_1,\omega_2\in S^2$. In $\mathbb{R}^6$ we have $dk\,\mathrm{d}l=\mathsf{r}_1^2d\mathsf{r}_1\,\mathrm{d}\omega_1\,\mathsf{r}_2^2d\mathsf{r}_2\,\mathrm{d}\omega_2$, with $\mathsf{r}_1=s\cos\psi$, $\mathsf{r}_2=s\sin\psi$ and $d\mathsf{r}_1\,\mathrm{d}\mathsf{r}_2=s\,\mathrm{d}s\,\mathrm{d}\psi$; hence

$$d\omega=\cos^2\psi\,\sin^2\psi\,\mathrm{d}\psi\,\mathrm{d}\omega_1\,\mathrm{d}\omega_2.$$

On S^5 we have $\mathsf{x}_1\mathsf{x}_2=\cos^2\psi\sin^2\psi$ and $\mathsf{x}_3=1-\sin2\psi\;\omega_1\cdot\omega_2$. For Borel functions $g\ge0$ on $[-1,1]$ we have $\iint_{S^2\times S^2}g(\omega_1\cdot\omega_2)\,\mathrm{d}\omega_1\,\mathrm{d}\omega_2=4\pi\cdot2\pi\int_{-1}^1g(\tau)\,\mathrm{d}\tau$, and for $0\le z<1$

$$\int_{-1}^1\frac{d\tau}{1-z\tau}=\sum_{j\ge0}z^j\int_{-1}^1\tau^j\,\mathrm{d}\tau=2\sum_{n\ge0}\frac{z^{2n}}{2n+1}.$$

Applying this with $z=\sin2\psi$ for $\psi\neq\frac{\pi}{4}$, and interchanging sum and integral, which is allowed since all terms are non-negative, we find

$$\oint_{S^5}\frac{d\omega}{\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3}=8\pi^2\int_0^{\pi/2}2\sum_{n\ge0}\frac{\sin^{2n}2\psi}{2n+1}\,\mathrm{d}\psi=16\pi^2\sum_{n\ge0}\frac{\mathsf{w}_n}{2n+1},\qquad\mathsf{w}_n:=\int_0^{\pi/2}\sin^{2n}2\psi\,\mathrm{d}\psi$$
$$=\int_0^{\pi/2}\sin^{2n}\psi\,\mathrm{d}\psi.$$

An integration by parts gives $2n\,\mathsf{w}_n=(2n-1)\,\mathsf{w}_{n-1}$, and $\mathsf{w}_0=\frac{\pi}{2}$. Hence $\mathsf{w}_n=\frac{\pi}{2}\,\mathsf{b}_n$ with

$$\mathsf{b}_n:=\prod_{j=1}^n\frac{2j-1}{2j},$$

the coefficients of the binomial series $(1-t^2)^{-1/2}=\sum_n\mathsf{b}_nt^{2n}$, $|t|<1$. Integrating this series over $[0,x]$ and letting $x\uparrow1$ (monotone convergence), we obtain

$$\sum_n\frac{\mathsf{b}_n}{2n+1}=\int_0^1(1-t^2)^{-1/2}\,\mathrm{d}t=\frac{\pi}{2},$$

where we have substituted $t = \sin\psi$, $0 \le \psi < \frac{\pi}{2}$, so that $(1-t^2)^{-1/2}dt = d\psi$. Therefore the integral over S^5 equals $16\pi^2 \cdot \frac{\pi}{2} \cdot \frac{\pi}{2} = 4\pi^4$, and $4\pi^4/(2\pi)^6 = 1/16\pi^2$. □

Proposition 7.50 (The bridge: $\kappa_\Gamma^{\mathrm{lat}}$ from $\mathcal{R}[F_\Gamma]$)**.** Let $\Gamma \in \{1,\dots,6\}$ and $\Lambda > 1$; let $\mathfrak{f}_\Gamma$ be as in Prop. 7.46, R_Γ^Λ as in Lem. 7.43(b), and $\kappa_\Gamma^{\mathrm{lat}}$ as in Lem. 7.36. Then

$$\mathfrak{f}_\Gamma[R_\Gamma^\Lambda - R_\Gamma] = d_{2,\Gamma}\log\Lambda, \quad d_{2,\Gamma} := \frac{\mathcal{R}[F_\Gamma]}{16\pi^2}; \qquad \kappa_\Gamma^{\mathrm{lat}} = -\frac{3}{2\pi}\cdot 24\cdot d_{2,\Gamma} = -\frac{3}{32\pi^3}\cdot 24\cdot\mathcal{R}[F_\Gamma]. \tag{7.xlvi}$$

■

Proof. Step 1: cut-offs. The function R_Γ is smooth, even and rotation invariant. If $|k|+|l| < \frac12\rho_0$ and $|\mathsf{p}| < \frac12\rho_0$, then each of the vectors k, l, $k-l$, $\mathsf{p}-k$, $\mathsf{p}-l$, among which are the lines of Γ, has length less than ρ_0, and therefore $R_\Gamma = 1 = R_\Gamma^\Lambda$ there. If $R_\Gamma \neq 0$, then $|l| < 2\rho_0$ and $|k| < 4\rho_0$: for $\Gamma \neq 6$ because l and k are lines of Γ, and for $\Gamma = 6$ because l and $k-l$ are. Put $\phi(k,l) := \theta((k^2+l^2)^{1/2})$; then $\phi = 1$ for $|k|+|l| \le \frac12$. Consequently the functions

$$\Theta := R_\Gamma^\Lambda - R_\Gamma, \qquad \Upsilon_0 := R_\Gamma - \phi, \qquad \phi^\Lambda - \phi,$$

regarded as functions of (k,l,p), belong to $\mathcal{Y}(c)$ with $c := \frac12\rho_0$. Indeed, they are smooth and satisfy (Y2); for $|\mathsf{p}| < c$ and $|k|+|l| < c$ they vanish, because R_Γ, R_Γ^Λ, ϕ and ϕ^Λ are all equal to 1 there ($c < \frac12$); and they vanish when $|k|+|l|$ is large, because R_Γ, R_Γ^Λ, ϕ and ϕ^Λ do. We note that, by Lem. 7.45(e), the finite part, and with it $\mathfrak{f}_\Gamma$, does not depend on the parameter c with respect to which an element of $\mathcal{Y}(c)$ is considered.

Step 2: $\mathfrak{f}_\Gamma[\Theta] = -\frac{\pi}{36}\kappa_\Gamma^{\mathrm{lat}}\log\Lambda$. The function $\Xi_\Gamma[\Theta](\mathsf{p})$ is the left side of the identity in Lem. 7.43(b). We thus have two expansions of it: that of Lem. 7.43(b), valid for almost every p with $|\mathsf{p}| < 1$, and that of Prop. 7.46, valid for every p with $0 < |\mathsf{p}| \le \frac{c}{16}$. Since the remainder $\Lambda^2\mathsf{o}(\mathsf{p}/\Lambda) - \mathsf{o}(\mathsf{p})$ of the former is $o(\mathsf{p}^2)$, they give

$$(\Lambda^2-1)\,\mathsf{a}_0 + (\Lambda-1)\,\mathsf{a}_1|\mathsf{p}| - \tfrac{\pi}{36}\,\kappa_\Gamma^{\mathrm{lat}}\log\Lambda\;\mathsf{p}^2 + o(\mathsf{p}^2) = \alpha + \beta\,|\mathsf{p}| + \mathfrak{f}_\Gamma[\Theta]\,\mathsf{p}^2 + o(\mathsf{p}^2).$$

Along a sequence $\mathsf{p} \to 0$ of points at which both expansions hold, we compare first the constant terms, then the coefficients of $|\mathsf{p}|$, and then those of p^2.

Step 3: $\mathfrak{f}_\Gamma[\Theta] = \mathfrak{f}_\Gamma[\phi^\Lambda - \phi]$. We have $\Theta = (\phi^\Lambda - \phi) + (\Upsilon_0^\Lambda - \Upsilon_0)$. By the linearity of $\mathfrak{f}_\Gamma$ and the identity $\mathfrak{f}_\Gamma[\Upsilon_0^\Lambda] = \mathfrak{f}_\Gamma[\Upsilon_0]$ of Prop. 7.46, the second difference contributes nothing.

Step 4: the value. Since ϕ does not depend on p, we have $\overline{((\phi^\Lambda-\phi)F_\Gamma)_2} = (\phi^\Lambda-\phi)\,\overline{(F_\Gamma)_2}$. By Def. 7.47(3) and Lem. 7.48, $\overline{(F_\Gamma)_2} = \sum_{\mathsf{M}} r_{\mathsf{M}}\mathsf{M}$ is a finite linear combination of monomials M as in Lem. 7.49, and the coefficient of $\mathsf{M} = (\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3)^{-1}$ is $r_{\mathsf{M}} = \mathcal{R}[F_\Gamma]$. The finite part is linear and independent of the parameter c (Lem. 7.45(e)); hence eq. 7.xlv yields

$$\mathfrak{f}_\Gamma[\phi^\Lambda - \phi] = (2\pi)^{-6}\cdot 4\pi^4\log\Lambda\cdot\mathcal{R}[F_\Gamma] = d_{2,\Gamma}\log\Lambda.$$

Step 5: conclusion. By Steps 2–4, $\mathfrak{f}_\Gamma[R_\Gamma^\Lambda - R_\Gamma] = \mathfrak{f}_\Gamma[\Theta] = d_{2,\Gamma}\log\Lambda$ and $-\frac{\pi}{36}\kappa_\Gamma^{\mathrm{lat}} = d_{2,\Gamma}$. Hence

$$\kappa_\Gamma^{\mathrm{lat}} = -\frac{36}{\pi}\,d_{2,\Gamma} = -\frac{3}{2\pi}\cdot 24\cdot d_{2,\Gamma},$$

and $\frac{3}{2\pi}\cdot\frac{1}{16\pi^2} = \frac{3}{32\pi^3}$ gives the last expression in eq. 7.xlvi. □

Proposition 7.51 (The six numbers $\mathcal{R}[F_\Gamma]$)**.** For the integrands $F_1,\dots,F_6$ of eq. 7.xxxv,

$$\mathcal{R}[F_1] = -\tfrac43, \quad \mathcal{R}[F_2] = -\tfrac{13}{12}, \quad \mathcal{R}[F_3] = -\tfrac{1}{12}, \quad \mathcal{R}[F_4] = 0, \quad \mathcal{R}[F_5] = +\tfrac{13}{6}, \quad \mathcal{R}[F_6] = -\tfrac13; \qquad \sum_{\Gamma=1}^{6}\mathcal{R}[F_\Gamma] = -\tfrac23. \tag{7.xlvii}$$

■

Proof. Step 0: notation. Throughout this proof we use the abbreviations

$$A,\ A',\ B,\ B',\ C,\ P := \mathsf{x}_1,\ \mathsf{x}_1',\ \mathsf{x}_2,\ \mathsf{x}_2',\ \mathsf{x}_3,\ \mathsf{x}_0,$$

and we put $k' := k - \mathsf{p}$, $l' := l - \mathsf{p}$ and $q := k - l = k' - l'$, so that $A' = k'^2$, $B' = l'^2$ and $C = q^2$. The relations $P = (k-k')^2 = (l-l')^2$, $C = (k-l)^2 = (k'-l')^2$, $k\cdot\mathsf{p} = A - k\cdot k'$ and $l\cdot\mathsf{p} = B - l\cdot l'$ express the remaining scalar products through these variables:

$$\begin{aligned} a &:= k\cdot k' = \tfrac12(A + A' - P),\\ b &:= l\cdot l' = \tfrac12(B + B' - P),\\ e &:= k\cdot l = \tfrac12(A + B - C),\\ e' &:= k'\cdot l' = \tfrac12(A' + B' - C),\\ f &:= k\cdot l' = e - k\cdot\mathsf{p} = \tfrac12(A' + B - C - P),\\ f' &:= k'\cdot l = e - l\cdot\mathsf{p} = \tfrac12(A + B' - C - P). \end{aligned}$$

We shall use two symmetries. The exchange of primes, $(k,l) \leftrightarrow (k',l')$, interchanges $A \leftrightarrow A'$, $B \leftrightarrow B'$, $e \leftrightarrow e'$, $f \leftrightarrow f'$ and fixes a and b. The exchange $(k,k') \leftrightarrow (l,l')$ interchanges $(A,A') \leftrightarrow (B,B')$, $a \leftrightarrow b$, $f \leftrightarrow f'$ and fixes e and e'. Finally, $\mathsf{q}(1) = \frac13$, $\mathsf{q}(0) = 0$ and $\mathsf{q}(-1) = 1$.

Step 1: two identities for V_3. Let

$$V := V_3(k_1,k_2,k_3)_{\alpha\beta\gamma} = \delta_{\alpha\gamma}\mathsf{a}_\beta + \delta_{\alpha\beta}\mathsf{b}_\gamma + \delta_{\beta\gamma}\mathsf{c}_\alpha, \qquad \mathsf{a} := k_1 - k_3, \quad \mathsf{b} := k_2 - k_1, \quad \mathsf{c} := k_3 - k_2.$$

(i) Since $\mathsf{a} + \mathsf{b} + \mathsf{c} = 0$,

$$V_{\alpha\beta\gamma}V_{\alpha\beta\gamma} = 3(\mathsf{a}^2+\mathsf{b}^2+\mathsf{c}^2)+2(\mathsf{a}\cdot\mathsf{b}+\mathsf{a}\cdot\mathsf{c}+\mathsf{b}\cdot\mathsf{c}) = 2(\mathsf{a}^2+\mathsf{b}^2+\mathsf{c}^2), \qquad \mathsf{a}^2+\mathsf{b}^2+\mathsf{c}^2 = 3\sum_i k_i^2 - \Big(\sum_i k_i\Big)^2.$$

(ii) If $k_1 + k_2 + k_3 = 0$, then

$$k_{1\alpha}V_{\alpha\beta\gamma} = (\mathsf{P}_2 - \mathsf{P}_3)_{\beta\gamma}, \qquad \mathsf{P}_i := k_i^2\,\delta_{\beta\gamma} - k_{i\beta}k_{i\gamma}.$$

Indeed, $k_{1\alpha}V_{\alpha\beta\gamma} = k_{1\gamma}(k_1-k_3)_\beta + k_{1\beta}(k_2-k_1)_\gamma + \delta_{\beta\gamma}\,k_1\cdot(k_3-k_2)$; here $k_{1\gamma}(k_1-k_3)_\beta + k_{1\beta}(k_2-k_1)_\gamma = k_{1\beta}k_{2\gamma} - k_{1\gamma}k_{3\beta}$, which equals $k_{3\beta}k_{3\gamma} - k_{2\beta}k_{2\gamma}$ because $k_1 = -k_2 - k_3$, and for the same reason $k_1\cdot(k_3 - k_2) = k_2^2 - k_3^2$.

Step 2: $\Gamma = 4$. By Def. 7.47(3), F_4 is a linear combination of monomials with $a_5 \le 0$, whereas every non-zero case of eq. 7.xliv has $\alpha_3 = a_5 \ge 1$. Hence $\mathcal{R}[F_4] = 0$.

Step 3: $\Gamma = 6$. Since V_3 is odd, $V_3(-l, l-k, k-\mathsf{p}) = -V_3(l, k-l, \mathsf{p}-k)$, and therefore

$$F_6 = +\tfrac16\, V_{\alpha\beta\gamma}V_{\alpha\beta\gamma}/A'BC, \qquad V := V_3(l, q, -k'),$$

where the three arguments of V_3 add up to $l+q-k' = \mathsf{p}$. By Step 1(i), $V_{\alpha\beta\gamma}V_{\alpha\beta\gamma} = 2\big[3(B+C+A')-P\big]$, hence

$$F_6 = \frac{1}{A'C} + \frac{1}{A'B} + \frac{1}{BC} - \frac13\,\frac{P}{A'BC}.$$

We evaluate $\mathcal{R}$ by eq. 7.xliv. The monomial $1/A'C$ has $n = 0$, $(\alpha_1,\alpha_2,\alpha_3) = (1,0,1)$ and $a_2 = 1$, $a_4 = 0$, so its value is $\mathsf{q}(0) = 0$; the monomial $1/A'B$ has $(\alpha_1,\alpha_2,\alpha_3) = (1,1,0)$ and value 0; the monomial $1/BC$ has $n = 0$, $(\alpha_1,\alpha_2,\alpha_3) = (0,1,1)$ and $a_2 = 0$, so its value is $\mathsf{q}(0) = 0$; and $P/A'BC$ has $n = 1$ and $(\alpha_1,\alpha_2,\alpha_3) = (1,1,1)$, so its value is 1. Therefore $\mathcal{R}[F_6] = -\frac13$.

Step 4: $\Gamma = 2, 3$, *a common factor.* Put

$$T_{\mu\nu} := V_2(k, \mathsf{p}-k)_{\mu\nu} = V_2(k,-k')_{\mu\nu} = a\,\delta_{\mu\nu} - k_\nu k'_\mu;$$

as V_2 is even, also $V_2(-k, k-\mathsf{p})_{\rho\nu} = T_{\rho\nu}$. Let

$$H_{\mu\rho} := T_{\mu\nu}T_{\rho\nu}/A' = \big[a^2\delta_{\mu\rho} - a\,(k_\rho k'_\mu + k_\mu k'_\rho) + A\,k'_\mu k'_\rho\big]/A',$$

and $V := V_3(-k, l, q)_{\mu\beta\gamma}$, so that $V_3(k,-l,-q) = -V$. In this notation

$$F_2 = \tfrac12\, H_{\mu\rho}V_{\mu\beta\gamma}V_{\rho\beta\gamma}/A^2BC, \qquad F_3 = -H_{\mu\rho}\,l_\rho(l-k)_\mu/A^2BC,$$

and p enters these expressions through H only. We compute $\overline{(H_{\mu\rho})_2}$. We insert $k' = k - \mathsf{p}$, $a = A - x$ and $A' = A - 2x + P$, with $x := k \cdot \mathsf{p}$, into H and expand: the term of second order in p is a sum of products of $\mathsf{p}_i \mathsf{p}_j$ with tensors built from δ and k, the coefficients being powers of A. Applying $\frac{1}{6}\Delta_{\mathsf{p}}$, we obtain

$$\overline{(H_{\mu\rho})_2} = \mathsf{a}_1\, \delta_{\mu\rho} + \mathsf{a}_2\, k_\mu k_\rho / A$$

with numbers $\mathsf{a}_1, \mathsf{a}_2$, since H is homogeneous of degree 2 in (k, p). Next, $k_\mu T_{\mu\nu} = a\, k_\nu - k_\nu (k \cdot k') = 0$ for all p; hence $k_\mu \overline{(H_{\mu\rho})_2} = 0$, which gives $\mathsf{a}_2 = -\mathsf{a}_1$. To determine a_1 we take the trace. From $T_{\mu\nu} T_{\mu\nu} = 3a^2 - 2a^2 + AA'$ we get $H_{\mu\mu} = a^2/A' + A$, and with $a = A - x$, $A' = A - 2x + P$,

$$\frac{a^2}{A'} = \Big(A - 2x + \frac{x^2}{A}\Big)\Big(1 + \frac{2x}{A} - \frac{P}{A} + \frac{4x^2}{A^2} + O(|\mathsf{p}|^3)\Big),$$

whose term of second order is $-P + 4x^2/A - 4x^2/A + x^2/A$; its image under $\frac{1}{6}\Delta_{\mathsf{p}}$ is $-1 + \frac{1}{3}$. Since the trace of $\overline{(H_{\mu\rho})_2}$ is $3\mathsf{a}_1 + \mathsf{a}_2 = 2\mathsf{a}_1$, we obtain $2\mathsf{a}_1 = -\frac{2}{3}$, that is,

$$\overline{(H_{\mu\rho})_2} = -\tfrac{1}{3}\,(\delta_{\mu\rho} - k_\mu k_\rho / A).$$

Step 5: $\Gamma = 2$. By Step 4,

$$\overline{(F_2)_2} = -\tfrac{1}{6}\,\mathsf{W}/A^2 BC, \qquad \mathsf{W} := V_{\mu\beta\gamma} V_{\mu\beta\gamma} - |k_\mu V_{\mu\beta\gamma}|^2 / A.$$

The function $A\mathsf{W}$ is a polynomial in A, B, C, homogeneous of degree 2, and $\overline{(F_2)_2} = -\frac{1}{6} A\mathsf{W}/A^3 BC$. A monomial $A^i B^j C^m$ of $A\mathsf{W}$ thus contributes a multiple of $A^{i-3} B^{j-1} C^{m-1}$, and only A^2 yields $(ABC)^{-1}$; we may therefore set $B = C = 0$. By Step 1(i), applied to the arguments $-k, l, q$, whose sum is 0, $V_{\mu\beta\gamma} V_{\mu\beta\gamma} = 6(A + B + C)$, which becomes $6A$. By Step 1(ii), applied with $(k_1, k_2, k_3) = (-k, l, q)$, we have $|k_\mu V_{\mu\beta\gamma}|^2 = \operatorname{tr}(\mathsf{P}_l - \mathsf{P}_q)^2$ with $\mathsf{P}_l = B\delta - ll^{\mathsf{T}}$ and $\mathsf{P}_q = C\delta - qq^{\mathsf{T}}$. Here $\operatorname{tr} \mathsf{P}_l^2 = 2B^2$, $\operatorname{tr} \mathsf{P}_q^2 = 2C^2$, $\operatorname{tr} \mathsf{P}_l \mathsf{P}_q = BC + (l \cdot q)^2$ and $l \cdot q = e - B = \frac{1}{2}(A - B - C)$; thus $|k_\mu V_{\mu\beta\gamma}|^2 = 2B^2 + 2C^2 - 2BC - 2(l \cdot q)^2$ is a polynomial in A, B, C, which becomes $-2(\frac{1}{2}A)^2 = -\frac{1}{2}A^2$ at $B = C = 0$.

Hence $A\mathsf{W}$ becomes $6A^2 + \frac{1}{2}A^2$, and

$$\mathcal{R}[F_2] = -\tfrac{1}{6} \cdot \tfrac{13}{2} = -\tfrac{13}{12}.$$

Step 6: $\Gamma = 3$. By Step 4,

$$\overline{(F_3)_2} = +\tfrac{1}{3}\,\mathsf{W}'/A^2 BC, \qquad \mathsf{W}' := (\delta_{\mu\rho} - k_\mu k_\rho / A)\, l_\rho (l - k)_\mu = l \cdot (l - k) - \frac{(k \cdot l)\,(k \cdot (l - k))}{A}$$
$$= B - e - \frac{e(e - A)}{A}.$$

Again $A\mathsf{W}'$ is a polynomial in A, B, C, homogeneous of degree 2, and as in Step 5 only its coefficient of A^2 matters. At $B = C = 0$ we have $e = \frac{1}{2}A$, and $A\mathsf{W}'$ becomes $-\frac{1}{2}A^2 + \frac{1}{4}A^2$. Hence

$$\mathcal{R}[F_3] = \tfrac{1}{3} \cdot (-\tfrac{1}{4}) = -\tfrac{1}{12}.$$

Step 7: values for the denominator $AA'BC$. For a monomial $\mathsf{m} = P^n A^i A'^{i'} B^j B'^{j'} C^m$ of degree 2 we put $\mathsf{L}_5(\mathsf{m}) := \mathcal{R}[\mathsf{m}/AA'BC]$. In the notation of Lem. 7.48, the monomial $\mathsf{m}/AA'BC$ has the exponents $(a_1, \dots, a_5) = (1-i, 1-i', 1-j, -j', 1-m)$, and $(\alpha_1, \alpha_2, \alpha_3) = (2-i-i',\ 1-j-j',\ 1-m)$. By eq. 7.xliv, $\mathsf{L}_5(\mathsf{m}) \neq 0$ requires $m = 0$, and the values are as follows.

- $n = 1$ (then $i + i' = 1$, $j = j' = 0$): $\mathsf{L}_5(PA) = \mathsf{L}_5(PA') = 1$.
- $n = 0$, $(\alpha_1, \alpha_2, \alpha_3) = (0, 1, 1)$ (then $i + i' = 2$, $a_4 = 0$, value $\mathsf{q}(1 - i')$): $\mathsf{L}_5(A^2) = \frac{1}{3}$, $\mathsf{L}_5(AA') = 0$, $\mathsf{L}_5(A'^2) = 1$.
- $n = 0$, $(\alpha_1, \alpha_2, \alpha_3) = (1, 0, 1)$ (then $i + i' = 1$, $j + j' = 1$, value $\mathsf{q}(-j') - \frac{2}{3}(1 - i')j'$): $\mathsf{L}_5(AB) = \mathsf{L}_5(A'B) = 0$, $\mathsf{L}_5(AB') = 1 - \frac{2}{3} = \frac{1}{3}$, $\mathsf{L}_5(A'B') = 1$.

All other monomials of degree 2 have the value 0; in particular, every monomial containing B or C has the value 0.

Step 8: $\Gamma = 5$. Put

$$\tilde{T}_{\alpha\sigma} := V_2(k-\mathsf{p},-k)_{\alpha\sigma} = V_2(k',-k)_{\alpha\sigma} = a\,\delta_{\alpha\sigma} - k_\alpha k'_\sigma.$$

Then $k'_\alpha \tilde{T}_{\alpha\sigma} = 0 = \tilde{T}_{\alpha\sigma} k_\sigma$; we write $\tilde{T}(X,Y) := X_\alpha \tilde{T}_{\alpha\sigma} Y_\sigma = a\,X\cdot Y - (k\cdot X)(k'\cdot Y)$, and we note $\tilde{T}_{\alpha\alpha} = 2a$. Further,

$$\begin{aligned} V_3(\mathsf{p}-k,l,k-l) &= V_3(-k',l,q) \\ &= \delta_{\alpha\gamma}u_{1\beta} + \delta_{\alpha\beta}v_{1\gamma} + \delta_{\beta\gamma}w_{1\alpha}, \\ u_1 = -k'-q, \qquad v_1 &= l+k', \qquad w_1 = q-l, \\ V_3(k,-l,l-k) &= \delta_{\sigma\gamma}u_{2\beta} + \delta_{\sigma\beta}v_{2\gamma} + \delta_{\beta\gamma}w_{2\sigma}, \\ u_2 = k+q, \qquad v_2 &= -(k+l), \qquad w_2 = l-q. \end{aligned}$$

Hence $F_5 = N_5/AA'BC$, where, after multiplying out the nine products and summing over β,γ,

$$\begin{aligned} N_5 = 2a\,(u_1\cdot u_2 + v_1\cdot v_2) + \tilde{T}(v_2,u_1) + \tilde{T}(u_1,w_2) + \tilde{T}(u_2,v_1) + \tilde{T}(v_1,w_2) + \tilde{T}(w_1,u_2) \\ + \tilde{T}(w_1,v_2) + 3\,\tilde{T}(w_1,w_2). \end{aligned}$$

The first argument of $\tilde{T}$ matters only modulo multiples of k', the second only modulo multiples of k. Using $q = k-l$ we find, in the first argument,

$$u_1 \equiv l-k, \qquad v_1 \equiv l, \qquad w_1 = k-2l, \qquad u_2 = 2k-l, \qquad v_2 = -(k+l),$$

and in the second argument

$$u_1 \equiv l-k', \qquad v_1 = l+k', \qquad u_2 \equiv -l, \qquad v_2 \equiv -l, \qquad w_2 \equiv 2l.$$

Therefore

$$\begin{gathered} \tilde{T}(u_1,w_2) = -2\tilde{T}(k,l) + 2\tilde{T}(l,l), \\ \tilde{T}(v_1,w_2) = 2\tilde{T}(l,l), \\ \tilde{T}(w_1,u_2) = \tilde{T}(w_1,v_2) = -\tilde{T}(k,l) + 2\tilde{T}(l,l), \\ 3\,\tilde{T}(w_1,w_2) = 6\tilde{T}(k,l) - 12\tilde{T}(l,l), \\ \tilde{T}(v_2,u_1) + \tilde{T}(u_2,v_1) = -\tilde{T}(k+l,l-k') + \tilde{T}(2k-l,l+k') = \tilde{T}(k,l) - 2\tilde{T}(l,l) + 3\tilde{T}(k,k'), \end{gathered}$$

and the sum of these seven terms is $3\tilde{T}(k,l) - 6\tilde{T}(l,l) + 3\tilde{T}(k,k')$. We regard N_5 as a polynomial in A, A', B, B', C, P; by the last assertion of Step 7 we may set $B = C = 0$ in it. Then $e = \frac{1}{2}A$,

$f' = \frac12(A + B' - P)$, $k\cdot q = A - e = \frac12 A$ and $k'\cdot q = a - f'$, and with the values of Step 7 we find:

$$\begin{aligned}
\tilde T(k,l) = ae - Af' = \tfrac14 A\,(-A + A' - 2B' + P) :\\
\mathsf{L}_5 = \tfrac14(-\tfrac13 + 0 - \tfrac23 + 1) = 0;\\
\tilde T(l,l) = aB - ef' = -\tfrac14 A\,(A + B' - P) :\\
\mathsf{L}_5 = -\tfrac14(\tfrac13 + \tfrac13 - 1) = \tfrac{1}{12};\\
\tilde T(k,k') = a^2 - AA' = \tfrac14(A + A' - P)^2 - AA' :\\
\mathsf{L}_5 = \tfrac14(\tfrac13 + 1 - 2 - 2) = -\tfrac23;\\
u_1\cdot u_2 + v_1\cdot v_2\\
= -(a + k'\cdot q + k\cdot q + C) - (e + B + a + f')\\
= -(3a + A);\\
a\,(3a + A) = \tfrac34(A + A' - P)^2 + \tfrac12 A\,(A + A' - P) :\\
\mathsf{L}_5 = \tfrac34(-\tfrac83) + \tfrac12(\tfrac13 - 1) = -\tfrac73.
\end{aligned}$$

In the third line the four numbers in the bracket come from the monomials A^2, A'^2, $-2PA$, $-2PA'$ of $(A + A' - P)^2$, the others having the value 0; thus $\mathsf{L}_5((A + A' - P)^2) = \frac13 + 1 - 2 - 2 = -\frac83$, the value used in the last line. Consequently

$$\mathcal{R}[F_5] = -2(-\tfrac73) + 3\cdot 0 - 6\cdot\tfrac{1}{12} + 3(-\tfrac23) = \tfrac{14}{3} - \tfrac52 = \tfrac{13}{6}.$$

Step 9: values for the denominator $AA'BB'C$. For a monomial $\mathsf{m} = P^n A^i A'^{i'} B^j B'^{j'} C^m$ of degree 3 we put $\mathsf{L}(\mathsf{m}) := \mathcal{R}[\mathsf{m}/AA'BB'C]$. Now $(a_1,\dots,a_5) = (1-i, 1-i', 1-j, 1-j', 1-m)$ and $(\alpha_1,\alpha_2,\alpha_3) = (2-i-i',\ 2-j-j',\ 1-m)$, and again $\mathsf{L}(\mathsf{m}) \neq 0$ requires $m = 0$. Let $X \in \{A, A'\}$ and $Y \in \{B, B'\}$. The values are as follows.

- $n = 1$: $\mathsf{L}(PXY) = 1$.
- $n = 0$, $(\alpha_1,\alpha_2,\alpha_3) = (0,1,1)$ (then $i + i' = 2$, $j + j' = 1$, value $\mathsf{q}(1-i') + \frac23(1-i')(1-j')$): $\mathsf{L}(A^2B) = \frac13 + \frac23 = 1$, $\mathsf{L}(A^2B') = \frac13$, $\mathsf{L}(A'^2B) = 1 - \frac23 = \frac13$, $\mathsf{L}(A'^2B') = 1$, $\mathsf{L}(AA'Y) = 0$.
- $n = 0$, $(\alpha_1,\alpha_2,\alpha_3) = (1,0,1)$ (then $i + i' = 1$, $j + j' = 2$, value $\mathsf{q}(1-j') + \frac23(1-i')(1-j')$): $\mathsf{L}(AB^2) = 1$, $\mathsf{L}(A'B^2) = \frac13$, $\mathsf{L}(AB'^2) = \frac13$, $\mathsf{L}(A'B'^2) = 1$, $\mathsf{L}(XBB') = 0$.

All other monomials of degree 3 have the value 0: those containing C or P^2, and those of the forms PX_1X_2, PY_1Y_2, $X_1X_2X_3$, $Y_1Y_2Y_3$ with $X_i \in \{A, A'\}$, $Y_i \in \{B, B'\}$. These values are invariant under the exchange of primes and under $(A, A') \leftrightarrow (B, B')$.

Step 10: $\Gamma = 1$, reduction. As in Step 4, $T_{\mu\nu} = V_2(k, \mathsf{p} - k)_{\mu\nu} = a\,\delta_{\mu\nu} - k_\nu k'_\mu$, and $k_\mu T_{\mu\nu} = 0 = T_{\mu\nu}k'_\nu$. Similarly,

$$S_{\rho\sigma} := V_2(-l, l - \mathsf{p})_{\rho\sigma} = V_2(-l, l')_{\rho\sigma} = b\,\delta_{\rho\sigma} - l_\sigma l'_\rho, \qquad l_\rho S_{\rho\sigma} = 0 = S_{\rho\sigma} l'_\sigma.$$

The tensor $V_3(-k, l, q)_{\mu\rho\alpha} = \delta_{\mu\alpha}(l - 2k)_\rho + \delta_{\mu\rho}(k + l)_\alpha + \delta_{\rho\alpha}(k - 2l)_\mu$ is contracted with $T_{\mu\nu}S_{\rho\sigma}$, which annihilates the terms with l_ρ and with k_μ; in this contraction it may therefore be replaced by

$$\mathsf{W}_{\mu\rho\alpha} := -2k_\rho\,\delta_{\mu\alpha} + (k + l)_\alpha\,\delta_{\mu\rho} - 2l_\mu\,\delta_{\rho\alpha}.$$

Likewise, $V_3(k - \mathsf{p}, \mathsf{p} - l, l - k)_{\nu\sigma\alpha} = V_3(k', -l', l' - k')_{\nu\sigma\alpha} = \delta_{\nu\alpha}(2k' - l')_\sigma - \delta_{\nu\sigma}(k' + l')_\alpha + \delta_{\sigma\alpha}(2l' - k')_\nu$, contracted with $T_{\mu\nu}S_{\rho\sigma}$, may be replaced by $-\mathsf{W}'_{\nu\sigma\alpha}$, where W' denotes W with (k', l') in place of (k, l). Hence

$$F_1 = +\tfrac12\, N'/AA'BB'C, \qquad N' := T_{\mu\nu}S_{\rho\sigma}\,\mathsf{W}_{\mu\rho\alpha}\mathsf{W}'_{\nu\sigma\alpha}.$$

We use the bilinear forms

$$T(X,Y) := X_\mu T_{\mu\nu} Y_\nu = a\,X\cdot Y - (k'\cdot X)(k\cdot Y), \qquad S(X,Y) := X_\rho S_{\rho\sigma} Y_\sigma = b\,X\cdot Y - (l'\cdot X)(l\cdot Y),$$

the vectors $\mathsf{s}_\sigma := k_\rho S_{\rho\sigma} = (bk - fl)_\sigma$ and $\mathsf{t}_\rho := S_{\rho\sigma} k'_\sigma = (bk' - f'l')_\rho$, and the identities $T_{\mu\mu} = 2a$, $S_{\rho\rho} = 2b$, $T_{\mu\nu}S_{\mu\nu} = 3ab - ab - ab + ee' = ab + ee'$. Let N'_{ij} be the contribution to N' of the i-th term of W and the j-th term of W'. Then

$$\begin{aligned} N'_{11} &= 4T_{\mu\mu}S(k,k') = 8a(ab - ff'),\\ N'_{33} &= 4T(l,l')S_{\rho\rho} = 8b(ab - ff'),\\ N'_{22} &= (k+l)\cdot(k'+l')\,(ab + ee'),\\ N'_{13} &= 4T(\mathsf{s},l'),\\ N'_{31} &= 4T(l,\mathsf{t}),\\ N'_{12} &= -2T(k'+l',\mathsf{s}),\\ N'_{21} &= -2T(\mathsf{t},k+l),\\ N'_{23} &= -2\,T_{\mu\nu}l'_\nu\,S_{\mu\sigma}(k+l)_\sigma,\\ N'_{32} &= -2\,l_\mu T_{\mu\nu}\,(k'+l')_\rho S_{\rho\nu}. \end{aligned}$$

Under the exchange of primes, $T_{\mu\nu}$ goes over into $T_{\nu\mu}$, so that $T(X,Y)$ becomes the value of T at the images of Y and X, in this order, and s goes over into t; hence N'_{13} goes over into N'_{31}, and N'_{12} into N'_{21}. Under the exchange $(k,k') \leftrightarrow (l,l')$, the tensors $T_{\mu\nu}$ and $S_{\mu\nu}$ are interchanged; hence $N'_{12} = -2(k'+l')_\mu T_{\mu\nu}k_\rho S_{\rho\nu}$ goes over into N'_{32}, $N'_{21} = -2S_{\mu\sigma}k'_\sigma T_{\mu\nu}(k+l)_\nu$ into N'_{23}, and N'_{11} into N'_{33}. By the invariance of the values in Step 9,

$$\mathsf{L}(N'_{33}) = \mathsf{L}(N'_{11}), \qquad \mathsf{L}(N'_{31}) = \mathsf{L}(N'_{13}), \qquad \mathsf{L}(N'_{21}) = \mathsf{L}(N'_{32}) = \mathsf{L}(N'_{23}) = \mathsf{L}(N'_{12}).$$

Step 11: $\Gamma = 1$*, values.* By Step 9 we may set $C = 0$. Then

$$4(ab - ff') = (A + A' - P)(B + B' - P) - (A' + B - P)(A + B' - P) = AB' + A'B - AA' - BB' =: \mathsf{D}.$$

- $N'_{11} = (A + A' - P)\,\mathsf{D}$. Of its monomials only A^2B', A'^2B, $-PAB'$ and $-PA'B$ have non-zero values, and $\mathsf{L}(N'_{11}) = \frac{1}{3} + \frac{1}{3} - 1 - 1 = -\frac{4}{3}$.
- N'_{13}: since $\mathsf{s}\cdot l' = bf - fb = 0$ and $k'\cdot\mathsf{s} = ab - ff'$, we have $N'_{13} = -4f(ab - ff') = -\frac{1}{2}(A' + B - P)\,\mathsf{D}$. In $(A' + B - P)\,\mathsf{D}$ the monomials with non-zero values are A'^2B, $A'B^2$, $-PAB'$ and $-PA'B$; they give $\frac{1}{3} + \frac{1}{3} - 2 = -\frac{4}{3}$, and therefore $\mathsf{L}(N'_{13}) = \frac{2}{3}$.
- N'_{22}: here $(k+l)\cdot(k'+l') = a + f + f' + b = A + A' + B + B' - 2P$, and

$$\begin{aligned} 4(ab + ee') = (A + A' - P)(B + B' - P) + (A + B)(A' + B') = \mathsf{E} + AA' + BB' - P(A\\ + A' + B + B') + P^2, \qquad \mathsf{E} := AB + 2AB' + 2A'B + A'B'. \end{aligned}$$

  We have $\mathsf{L}\big[(A+A'+B+B')\,\mathsf{E}\big] = (1+\frac{2}{3})+(\frac{2}{3}+1)+(1+\frac{2}{3})+(\frac{2}{3}+1) = \frac{20}{3}$, the four contributions coming from $A(AB+2AB')$, $A'(2A'B+A'B')$, $B(AB+2A'B)$ and $B'(2AB'+A'B')$. Further, $\mathsf{L}\big[(A + A' + B + B')(AA' + BB')\big] = 0$, $\mathsf{L}\big[-P(A + A' + B + B')^2\big] = -2\cdot 4 = -8$ and $\mathsf{L}[-2P\,\mathsf{E}] = -2\cdot 6 = -12$, while the remaining monomials, which contain P^2 or have one of the forms PX_1X_2, PY_1Y_2, have the value 0. Hence $\mathsf{L}(N'_{22}) = \frac{1}{4}\big(\frac{20}{3} - 20\big) = -\frac{10}{3}$.
- $N'_{12} = -2\big[a\,(k'+l')\cdot\mathsf{s} - \big(k'\cdot(k'+l')\big)(k\cdot\mathsf{s})\big]$. Since $(k'+l')\cdot\mathsf{s} = ab - ff'$, the first term is $a(ab - ff') = \frac{1}{8}(A + A' - P)\,\mathsf{D}$, with the value $\frac{1}{8}\cdot(-\frac{4}{3}) = -\frac{1}{6}$. Next, $k'\cdot(k'+l') = A' + e' = \frac{1}{2}(3A' + B')$ and

$$k\cdot\mathsf{s} = bA - fe = \tfrac{1}{4}\big[2A(B+B'-P) - (A'+B-P)(A+B)\big] = \tfrac{1}{4}\big[AB+2AB'-AA'-A'B-B^2-PA+PB\big].$$

  In $(3A'+B')\big[AB+2AB'-AA'-A'B-B^2-PA+PB\big]$ the monomials with non-zero values are $-3A'^2B$, $-3A'B^2$, $+3PA'B$, $+2AB'^2$ and $-PAB'$; they give $-1-1+3+\frac{2}{3}-1 = \frac{2}{3}$. The second term, $\frac{1}{8}(3A' + B')\big[\cdots\big]$, therefore has the value $\frac{1}{8}\cdot\frac{2}{3} = \frac{1}{12}$, and $\mathsf{L}(N'_{12}) = -2\big(-\frac{1}{6} - \frac{1}{12}\big) = \frac{1}{2}$.
- Summing, with the identities at the end of Step 10,

$$\mathsf{L}(N') = 2\big(-\tfrac{4}{3}\big) + 2\cdot\tfrac{2}{3} - \tfrac{10}{3} + 4\cdot\tfrac{1}{2} = -\tfrac{8}{3}, \qquad \text{hence} \qquad \mathcal{R}[F_1] = \tfrac{1}{2}\big(-\tfrac{8}{3}\big) = -\tfrac{4}{3}.$$

Step 12: the sum. Adding the six values obtained in Steps 2, 3, 5, 6, 8 and 11, we find

$$\sum_{\Gamma=1}^{6}\mathcal{R}[F_\Gamma]=-\tfrac{4}{3}-\tfrac{13}{12}-\tfrac{1}{12}+0+\tfrac{13}{6}-\tfrac{1}{3}=\frac{-16-13-1+26-4}{12}=-\tfrac{2}{3},$$

which completes the proof of eq. 7.xlvii. □

7.1.17. *The value: κ^{lat} and c_1.* Adding the six contributions of Prop. 7.50 and Prop. 7.51, we now obtain the value of the constant κ^{lat} of Thm. V, and with it the number c_1.

Theorem W (The value of κ^{lat})**.** The number κ^{lat} of Thm. V is

$$\kappa^{\mathrm{lat}}=\frac{3}{2\pi^3}.$$

■□

Proof. In eq. 7.xxxvii, κ^{lat} is the sum of the numbers $\kappa_\Gamma^{\mathrm{lat}}$ of Lem. 7.36. Expressing each of them by eq. 7.xlvi and inserting the sum from eq. 7.xlvii, we obtain

$$\kappa^{\mathrm{lat}}=\sum_{\Gamma=1}^{6}\kappa_\Gamma^{\mathrm{lat}}=-\frac{3}{32\pi^3}\cdot 24\cdot\sum_{\Gamma=1}^{6}\mathcal{R}[F_\Gamma]=-\frac{3}{32\pi^3}\cdot 24\cdot\Big(-\frac{2}{3}\Big)=\frac{48}{32\pi^3}=\frac{3}{2\pi^3},$$

as asserted. □

Definition 7.52 (The number c_1)**.** We put

$$c_1:=\frac{\kappa^{\mathrm{lat}}}{\mathsf{c}_2},$$

where κ^{lat} is the number of Thm. V and $\mathsf{c}_2=\frac{3\dim\mathfrak{g}}{16\pi^2}=\frac{9}{16\pi^2}$ is the leading constant fixed in §6.1; here $\dim\mathfrak{g}=3=\mathsf{n}$.

This fixes the coefficient used in §6.1; it does not by itself give a cutoff-uniform remainder for the interacting measure. ■

Theorem X (The number c_1)**.**

(a) $c_1=\dfrac{8}{3\pi}$, and $\mathsf{c}_2c_1=\dfrac{3}{2\pi^3}$.

(b) There is a pure constant C with the following property. Let $\mathsf{d}>0$, $\vartheta\in(0,1)$, $(f,h)\in\mathfrak{T}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$ and $0<a\le\mathsf{d}$, the class of test pairs and the functional $\mathcal{K}_5$ being those of Cor. 7.37. Then

$$\Big|\frac{\mathsf{P}_1^{\mathrm{lat}}(f,h;a)}{\mathsf{c}_2\,\mathcal{K}_5(f,h)}-c_1\Big|\le C\,\vartheta^{-3}\,\frac{a}{\mathsf{d}}\,\lg\frac{\mathsf{d}}{a}.\tag{7.xlviii}$$

Consequently, $\lim_{a\to0}\mathsf{P}_1^{\mathrm{lat}}(f,h;a)/\mathsf{c}_2\mathcal{K}_5(f,h)=c_1$ for every such pair.

(c) For all plaquettes p,q of $\mathbb{Z}^3$, with n ranging over the integers for which $p+o_n,q+o_n\in\mathsf{Pl}_n$,

$$\mathsf{B}_1^{\mathrm{lat}}(p,q)=\lim_{n\to\infty}\lim_{\beta\to\infty}\ \beta^3\Big[\langle a_{p+o_n};a_{q+o_n}\rangle_{n,\beta}-\beta^{-2}\,\mathsf{B}_0^{[n]}(p+o_n,q+o_n)\Big].$$

The right side is gauge independent: no gauge fixing enters it.

■□

Proof. (a) By Thm. W,

$$c_1=\frac{3}{2\pi^3}\cdot\frac{16\pi^2}{9}=\frac{48}{18\pi}=\frac{8}{3\pi},$$

and $\mathsf{c}_2c_1=\kappa^{\mathrm{lat}}=\frac{3}{2\pi^3}$.

(b) Lem. 6.1 on $\mathbb{E}=\mathbb{R}^3$, with $q=5$, gives

$$\mathcal{K}_5(f,h)\ge 10^{-5}\,\vartheta^2\,\mathsf{d}\,\|f\|_\infty\|h\|_\infty,$$

and the right side is positive, since neither f nor h vanishes identically. We divide eq. 7.xxxviii by $\mathsf{c}_2\mathcal{K}_5(f,h)$ and use $\kappa^{\mathrm{lat}} = \mathsf{c}_2 c_1$. The left side of eq. 7.xlviii is then at most

$$C\,\vartheta^{-1}\,a\,\lg(\mathsf{d}/a)\cdot 10^5\,\mathsf{c}_2^{-1}\,\vartheta^{-2}\,\mathsf{d}^{-1},$$

which is the asserted bound, with a pure constant. Since $\frac{a}{\mathsf{d}}\lg\frac{\mathsf{d}}{a}\to 0$ as $a\to 0$, the limit follows.

(c) We apply eq. 7.xx at $\tau_{\mathrm{gf}}=1$ to the plaquettes $p+o_n, q+o_n\in\mathsf{Pl}_n$: for $\beta\geq 1$ the expression $\beta^3[\cdots]$ differs from $\mathsf{B}_1^{[n]}(p+o_n,q+o_n;1)$ by at most $C\beta^{-1}$, with a constant C depending on n, p and q. This gives the inner limit. By eq. 7.xxxix with $j=1$ and $\tau_{\mathrm{gf}}=1$, the outer limit equals $\mathsf{B}_1^{\mathrm{lat}}(p,q;1)$, which is $\mathsf{B}_1^{\mathrm{lat}}(p,q)$ by eq. 7.xix. The mean $\langle\cdot\rangle_{n,\beta}$ is taken in the Wilson measure of the box, and $\mathsf{B}_0^{[n]}=\frac{\mathsf{n}}{2}(\Pi^{\Lambda_n})^2$ by eq. 7.xvii, where Π^{Λ_n} does not depend on τ_{gf} (Def. 7.7(2)); thus no gauge fixing enters the right side. □

7.1.18. *What one shell adds; scope.* We close the section with two items. The first, Lem. 7.53, concerns a kernel $\mathsf{U}(u)=\mathsf{c}\,|u|^{-5}\varphi(u)$ with $\varphi(Lu)=\varphi(u)$ for a real number $L>1$, for instance $\kappa^{\mathrm{lat}}|u|^{-5}$, the case $\mathsf{c}=\kappa^{\mathrm{lat}}$, $\varphi\equiv 1$; it computes the part of the quadratic form $\frac12\iint w(x)\,w(y)\,\mathsf{U}(x-y)\,\mathrm{d}x\,\mathrm{d}y$ that comes from the shell $r\leq|x-y|<Lr$, which covers the scales from r to Lr. The second, Remark 7.54, delimits the results of the section.

Lemma 7.53 (What one shell adds)**.** Let $L>1$ and $\mathsf{c}\geq 0$ be real numbers, and let $\varphi\colon\mathbb{R}^3\setminus\{0\}\to\mathbb{R}$ be bounded, Borel and even, with $\varphi(Lu)=\varphi(u)$. Put $\mathsf{U}(u):=\mathsf{c}\,|u|^{-5}\varphi(u)$ and, for $r>0$,

$$\mathsf{Ann}_r:=\{u\in\mathbb{R}^3: r\leq|u|<Lr\}.$$

(a) For every $r>0$,

$$\int_{\mathsf{Ann}_r}\mathsf{U}(u)\,|u|^2\,\mathrm{d}u=\mathsf{c}\int_{S^2}\int_1^L\varphi(t\omega)\,\frac{dt}{t}\,\mathrm{d}\omega; \tag{7.xlix}$$

for $\varphi\equiv 1$ this equals $4\pi\,\mathsf{c}\log L$.

(b) Assume moreover that $\int_{\mathsf{Ann}_r}\mathsf{U}(u)\,u_iu_j\,\mathrm{d}u$ is a multiple of δ_{ij}; this is the case if φ is invariant under the signed permutations of the coordinates. Then for every real $w\in C_c^2(\mathbb{R}^3)$

$$\begin{aligned}\tfrac12\iint_{x-y\in\mathsf{Ann}_r} w(x)\,w(y)\,\mathsf{U}(x-y)\,\mathrm{d}x\,\mathrm{d}y=\\ \tfrac12\Big(\int_{\mathsf{Ann}_r}\mathsf{U}\Big)\int w^2\\ -\tfrac{1}{12}\Big(\int_{\mathsf{Ann}_r}\mathsf{U}(u)\,|u|^2\,\mathrm{d}u\Big)\int|\nabla w|^2+\mathrm{Rem},\\ |\mathrm{Rem}|\leq\tfrac{\pi}{24}\,\mathsf{c}\,\|\varphi\|_\infty\,(Lr)^2\,\|\Delta w\|_2^2.\end{aligned} \tag{7.l}$$

(c) If $\mathsf{c}=\mathsf{c}_2c_1\mathsf{b}^{-3}$ with $\mathsf{b}>0$, and $\varphi\equiv 1$, then the middle term on the right of eq. 7.l equals

$$-\frac{1}{2\pi^2}\,\mathsf{b}^{-1}\log L\int|\nabla(w/\mathsf{b})|^2.$$

■

Proof. (a) In polar coordinates the left side of eq. 7.xlix equals

$$\mathsf{c}\int_{S^2}\int_r^{Lr}t^{-5}\varphi(t\omega)\,t^2\,t^2\,\mathrm{d}t\,\mathrm{d}\omega=\mathsf{c}\int_{S^2}\int_r^{Lr}\varphi(t\omega)\,\frac{dt}{t}\,\mathrm{d}\omega.$$

For fixed ω the function $\tau\mapsto\varphi(e^\tau\omega)$ is bounded, Borel and periodic with period $\log L$; its integral over an interval of length $\log L$ is therefore the same for all such intervals, and

$$\int_r^{Lr}\varphi(t\omega)\,\frac{dt}{t}=\int_{\log r}^{\log r+\log L}\varphi(e^\tau\omega)\,\mathrm{d}\tau=\int_1^L\varphi(t\omega)\,\frac{dt}{t}.$$

For $\varphi\equiv 1$ the right side of eq. 7.xlix is $\mathsf{c}\cdot 4\pi\cdot\log L$.

(b) If φ is invariant under the signed permutations of the coordinates, then so are Ann_r and U. Changing the sign of u_i shows that the off-diagonal entries of the matrix $\big(\int_{\mathsf{Ann}_r} \mathsf{U}\, u_i u_j\big)_{ij}$ vanish, and permuting the coordinates shows that its diagonal entries are equal.

We turn to the expansion. The substitution $x = y+u$ (Fubini's theorem and translation invariance) turns the left side of eq. 7.l into

$$\tfrac{1}{2}\int_{\mathsf{Ann}_r} \mathsf{U}(u)\,\mathsf{a}(u)\,\mathrm{d}u, \qquad \mathsf{a}(u) := \int w(y+u)\,w(y)\,\mathrm{d}y.$$

Since $w(\cdot+u)$ has the transform $e^{ip\cdot u}\hat{w}(p)$, Plancherel's theorem gives

$$\mathsf{a}(u) = \int |\hat{w}(p)|^2\, e^{ip\cdot u}\, d^\circ p = \int |\hat{w}(p)|^2 \cos(p\cdot u)\, d^\circ p,$$

the second equality because a is real. For real ϑ', Taylor's formula with integral remainder and $|\cos^{(4)}| \le 1$ give $|\cos\vartheta' - 1 + \tfrac{1}{2}\vartheta'^2| \le \tfrac{1}{24}\vartheta'^4$. The functions w, $\partial_i w$ and Δw belong to L^2, with the transforms $\hat{w}$, $ip_i\hat{w}$ and $-|p|^2\hat{w}$, by the rule for the transform of a derivative, applied to w and to $\partial_i w \in C^1_c(\mathbb{R}^3)$. Plancherel's theorem therefore yields

$$\int |\hat{w}|^2\, d^\circ p = \int w^2, \qquad \int |\hat{w}|^2\, p_i p_j\, d^\circ p = \int \partial_i w\, \partial_j w, \qquad \int |\hat{w}|^2\, |p|^4\, d^\circ p = \|\Delta w\|_2^2,$$

and, with $\vartheta' = p\cdot u$ and $|p\cdot u| \le |p||u|$,

$$\Big|\mathsf{a}(u) - \int w^2 + \tfrac{1}{2}\sum_{i,j} u_i u_j \int \partial_i w\, \partial_j w\Big| \le \tfrac{1}{24}\,|u|^4\,\|\Delta w\|_2^2.$$

By the hypothesis of (b), taking the trace, $\int_{\mathsf{Ann}_r} \mathsf{U}\, u_i u_j = \tfrac{1}{3}\delta_{ij}\int_{\mathsf{Ann}_r} \mathsf{U}\,|u|^2$. Inserting the expansion of a, we obtain the right side of eq. 7.l: its first term is $\tfrac{1}{2}(\int_{\mathsf{Ann}_r}\mathsf{U})\int w^2$, its second term has the coefficient $\tfrac{1}{2}\cdot\tfrac{1}{2}\cdot\tfrac{1}{3} = \tfrac{1}{12}$, and the remainder satisfies

$$|\mathrm{Rem}| \le \tfrac{1}{2}\cdot\tfrac{1}{24}\,\|\Delta w\|_2^2 \int_{\mathsf{Ann}_r} |\mathsf{U}|\,|u|^4\,\mathrm{d}u \le \tfrac{1}{48}\,\|\Delta w\|_2^2\,\mathsf{c}\,\|\varphi\|_\infty\, 4\pi \int_r^{Lr} t\,\mathrm{d}t \le \tfrac{1}{48}\,\|\Delta w\|_2^2\,\mathsf{c}\,\|\varphi\|_\infty\, 4\pi\,\frac{(Lr)^2}{2},$$

which is the bound in eq. 7.l.

(c) By (a) and Thm. X(a), the middle term equals

$$-\tfrac{1}{12}\cdot 4\pi\cdot\frac{3}{2\pi^3}\,\mathsf{b}^{-3}\log L\int|\nabla w|^2 = -\frac{1}{2\pi^2}\,\mathsf{b}^{-3}\log L\int|\nabla w|^2,$$

and $\mathsf{b}^{-3}\int|\nabla w|^2 = \mathsf{b}^{-1}\int|\nabla(w/\mathsf{b})|^2$. □

Remark 7.54 (Scope of the results). The results of this section are delimited as follows.

(i) We make no statement about the cumulants of the smeared plaquette observables $\mathcal{O}_a(f)$ in Wilson measures on tori. In particular, it is not asserted that $\mathsf{B}_1^{\mathrm{lat}}$ is an expansion coefficient of these cumulants when $\beta = \mathsf{b}/a$ and the side of the torus, measured in lattice spacings, grow jointly as $a \to 0$.

(ii) The constant C of eq. 7.xx is not controlled as a function of n. Accordingly, the limits in Thm. X(c) are iterated limits, to be taken in the order written there: first $\beta\to\infty$, then $n\to\infty$.

(iii) Cor. 7.37 concerns test pairs, whose supports are at distance at least d from each other; it gives no information on the terms with $p = q$ of the double sum eq. 7.xix, on constant functions f, h, or on functions f, h with intersecting supports.

(iv) The group is $SU(2)$ throughout; nothing is asserted for $N > 2$.

(v) Of the term of degree -6 in the large-distance behaviour of $\mathsf{B}_1^{\mathrm{lat}}$, we prove only the bound eq. 7.xxxvii.

8. Natural scaling and failure of dilation covariance

The physical label b has the dimension of a length. The natural small parameter in the preceding law is consequently d/b. Writing the error in this variable makes the range of the law independent of the reference coupling used to describe a fixed physical sequence.

For a test pair $\Psi = (f, h)$ put $|\Psi| = |f|_\infty |h|_\infty$ and write $\Psi_r = (f(\cdot/r), h(\cdot/r))$. The normalization of these weights is unchanged: they are not multiplied by r^{-3}. Hence

$$\mathcal{K}_q(\Psi_r) = r^{6-q}\mathcal{K}_q(\Psi).$$

In particular the leading $\mathcal{K}_6$ term is constant in r, while the next term is linear.

Sketch G (The natural-scale law). For every volume-first limit in the stated short-distance coupling range and every canonical component, the two-point function S satisfies

$$|\mathsf{S}(\Psi) - L_b^+(\Psi)| \le \varepsilon^{\mathrm{nat}}(d/b) b^{-3} d|\Psi|, \qquad \varepsilon^{\mathrm{nat}}(\delta) \le C\delta^{1/2},$$

for the fixed-shape pairs at $0 < d \le \delta_c b$. The constants depend on the admitted construction and the shape, not on the component. On a fixed torus there is the corresponding law at every positive label, in its stated range. None of these two-point functions is dilation covariant. ■□

Sketch of the argument. The transfer to natural scales uses the exact dilation of the lattices and Wilson measures. Dilating a sequence moves its label and its observation scale by the same factor. One may therefore move to an admitted reference window where the short-distance estimate is available, apply it there, and carry the result back. The ratio d/b is unchanged. The comparison with the least admissible torus then transfers the law to the periodic-state limit and its components. The error constants must be uniform in the component for this last assertion.

For a fixed nonzero positive test pair at scale d_1, put

$$A = c_2 b^{-2}\mathcal{K}_6(\Psi), \qquad B = \frac{3}{2\pi^3} b^{-3}\mathcal{K}_5(\Psi) > 0.$$

The law gives

$$\mathsf{S}(\Psi_r) = A + Br + O(r^{3/2}).$$

This exhibits two distinct homogeneities with nonzero coefficients. Dilation covariance, in the sense $\mathsf{S}(\Phi_r) = r^{\varpi}\mathsf{S}(\Phi)$ for all separated pairs and all $r > 0$, is incompatible with them. The nonzero constant limit fixes the possible exponent, and the nonzero linear term excludes that exponent.

A three-scale combination isolates the correction without subtracting A separately. For $t > 1$, define

$$\mathcal{N}_t(\Psi; r) = \mathsf{S}(\Psi_{t^2 r}) - (1 + t^6)\mathsf{S}(\Psi_{tr}) + t^6\mathsf{S}(\Psi_r).$$

Its weights annul constants and terms proportional to r^6. For the linear correction they give $-t(t-1)(t^5-1)Br$. At $t = 2$ this reads

$$\mathsf{S}(\Psi_{4r}) - 65\mathsf{S}(\Psi_{2r}) + 64\mathsf{S}(\Psi_r) = -62Br + O(r^{3/2}).$$

It is negative at all sufficiently small r. The estimate retains a quantitative tolerance, so this is a conclusion about the correlation function, not merely a formal comparison of degrees. The same test works in every canonical component because the coefficient and the tolerance survive the component transfer.

Failure of dilation covariance does not by itself establish interaction. A quadratic observable of a Gaussian field can have the same two-term law; the construction gives such an example. Its three-point behaviour distinguishes it from the torus limits under discussion. A claim of non-Gaussianity must therefore use that additional information, rather than the second two-point coefficient alone.

8.1. **Natural scaling and failure of dilation covariance.** We now express the two-term law in the natural parameter d/b. The resulting tolerance is uniform over the admitted reference couplings. The law holds at every positive label on a fixed torus and, on $\mathbb{R}^3$, in every volume-first limit at the stated coupling ceiling and in each of its canonical components. The nonzero second term excludes dilation covariance of the two-point function.

We give two passages to this conclusion: a direct transfer of the law between labels and an alternative using blocking factors with incommensurable logarithms. We also retain the leading law for signed test functions, with its sharp linear rate. These conclusions do not by themselves establish interaction. Prop. 8.18 constructs a quadratic Gaussian model with the same two-term law; its three-point behavior distinguishes it from the torus limits considered here.

8.1.1. *Parameters and test pairs.* We retain the construction setting of §6.1, and write b and d for the label b and scale d. A *choice* $\mathfrak{c}$ consists of a mollifier, a radius $\varrho > 0$, the common real and sourced numerical assignment of Prop. YM3_I:69.1 and Section YM3_I:82 at source radius 2ϱ, and the canonical auxiliary data for these constructions. Every constant in the real restriction keeps its value. The choice fixes an odd $L \geq 13$, $B_\sharp \geq 0$, $\bar{B} = B_\sharp L/(L-1)$, M, and $0 < \gamma \leq 1/2$. The short-distance ceilings γ_W of Thm. YM3_I:AV and γ_W^{sd} of the choices in equations YM3_I:(2326)–(2327) are both at most γ. The ceiling $\gamma_{\mathrm{e}} \in (0, \gamma]$, depending only on the constants, is fixed at the beginning of the subsection on the least torus, and $\Gamma := \min\{\gamma_{\mathrm{e}}, \gamma_W^{\mathrm{sd}}\}$.

We write $a_K = L^{-K}$, $X_g = L(g^{-2} + B_\sharp)$, $I(g) = (g^{-2} + \bar{B}, X_g - \bar{B}]$ for the interval of labels at g (Section YM3_I:74) and $I^\circ(g)$ for its interior, $\mathfrak{L}(g) = \{2L^m : m \geq m_{\mathfrak{L}}(g)\}$ for the set of the sides at g (equation YM3_I:(1528)) and ℓ_g for its least member, and $T_\ell = \mathbb{R}^3/\ell\mathbb{Z}^3$. A triple (g, b, ℓ) is *admissible* if $0 < g \leq \gamma_W^{\mathrm{sd}}$, $b \in I^\circ(g)$ and $\ell \in \mathfrak{L}(g)$. Since $LB_\sharp = (L-1)\bar{B}$, we have $I^\circ(g) = (g^{-2} + \bar{B},\, Lg^{-2} + (L-2)\bar{B})$, an interval of length $(L-1)g^{-2} + (L-3)\bar{B} > 0$.

We use the lattices, first-passage data, labels and torus cumulants of §6.1, with the same fixed choice of construction data. The continuum torus cumulants $S_n^{(\ell,b)}$ exist for every $b > 0$ on Lipschitz tuples with no common point, by Thm. YM3_I:A. A volume-first limit S at g is defined by a sequence (K_j, β_j, ν_j) as in [7, §K.1]; for $g \leq \gamma_W$ it has one label $\mathsf{b}(S) = \lim_j a_{K_j}\beta_j$. $(\mathsf{X}, \lambda, (\sigma_\xi))$ denotes a canonical decomposition of its functional σ_S, with the probability measure λ on X; S_2 and S_2^ξ are the two-point functions of σ_S and σ_ξ, and $\mathsf{N} \subset \mathsf{X}$ is the set of the ξ for which S_2^ξ is not dilation covariant.

The class $\mathfrak{T}_{\mathbb{E}}(d;\vartheta)$ is the test-pair class of §6.1. $\mathfrak{T}^1(d;\vartheta)$ is the set of its pairs on $\mathbb{R}^3$ of class C^1. Over $\mathbb{R}^3$ the pairs of real C^1 functions with compact disjoint supports are called *separated*; the pairs of $\mathfrak{T}^1(d;\vartheta)$ are separated. On a torus we admit real Lipschitz entries with disjoint supports. Every $\mathfrak{T}^1(d;\vartheta)$, $d > 0$, $\vartheta \in (0,1)$, contains a pair ([7, Lem. I.1](c)); for $\vartheta \leq 3\sqrt{3}/8$ one may take $f(u) = (1 - |u-x|^2/d^2)_+^2$, $h(u) = (1 - |u-y|^2/d^2)_+^2$ with $3d \leq |x-y| \leq 8d$. For a pair we write $|\Psi| = \|f\|_\infty \|h\|_\infty$, $\Psi_r = (f(\cdot/r), h(\cdot/r))$ for $r > 0$, $\Psi^{\langle\ell\rangle}$ for the periodisation, and

$$\mathcal{K}_q(\Psi) = \iint f(x)h(y)\,|x-y|^{-q}\,\mathrm{d}x\,\mathrm{d}y,$$

on T_ℓ with the Euclidean distance of T_ℓ; further $T^{(\ell,b)}(\Psi) = S_2^{(\ell,b)}(\Psi^{\langle\ell\rangle})$, $c_2 = \frac{9}{16\pi^2}$, $c_1 = \frac{8}{3\pi}$, $\varkappa = c_1c_2 = \frac{3}{2\pi^3}$ and $L_b^+ = c_2 b^{-2}[\mathcal{K}_6 + c_1 b^{-1}\mathcal{K}_5]$. By [7, Lem. I.1] and equation YM3_I:(2247) and [7, Lem. I.1](b): $\Psi_r \in \mathfrak{T}^1(rd;\vartheta)$ for $\Psi \in \mathfrak{T}^1(d;\vartheta)$, with the same ϑ, $|\Psi_r| = |\Psi|$ and $\mathcal{K}_q(\Psi_r) = r^{6-q}\mathcal{K}_q(\Psi)$; translations keep the classes, the suprema and the $\mathcal{K}_q$; for $d < \ell/20$ the periodisation maps $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$ into $\mathfrak{T}_{T_\ell}(d;\vartheta)$, keeps the suprema, and $\mathcal{K}_q(\Psi^{\langle\ell\rangle}) = \mathcal{K}_q(\Psi)$ for every real $q > 0$. As $d \leq |x-y| \leq 10d$ on the supports of a pair of the class, $(10d)^{-q}\int f \int h \leq \mathcal{K}_q(\Psi) \leq d^{-q}|\Psi|(\frac{4\pi}{3}d^3)^2$, that is,

$$10^{-5}\vartheta^2 d\,|\Psi| \leq \mathcal{K}_5(\Psi) \leq \left(\tfrac{4\pi}{3}\right)^2 d\,|\Psi|, \qquad 10^{-6}\vartheta^2|\Psi| \leq \mathcal{K}_6(\Psi) \leq \left(\tfrac{4\pi}{3}\right)^2|\Psi|, \qquad \varkappa\left(\tfrac{4\pi}{3}\right)^2 = \tfrac{8}{3\pi}; \quad (8.\mathrm{i})$$

note $10^{-5}\varkappa = 4.837\ldots \cdot 10^{-7}$.

A functional σ on separated tuples over $\mathbb{R}^3$ *obeys the law* (L) *with* $(b, c, \varepsilon, d_7, \vartheta)$, where ε is non-decreasing on $(0, d_7]$ and $\varepsilon(d) \to 0$ as $d \to 0$, if

$$|S_2^\sigma(\Psi) - c_2 b^{-2}[\mathcal{K}_6 + cb^{-1}\mathcal{K}_5](\Psi)| \le \varepsilon(d)\, b^{-3} d\, |\Psi| \qquad (0 < d \le d_7,\ \Psi \in \mathfrak{T}^1(d;\vartheta)).$$

S_2^σ is *dilation covariant* if there is a real ϖ with $S_2^\sigma(\Phi_r) = r^\varpi S_2^\sigma(\Phi)$ for all $r > 0$ and all separated pairs Φ. For a function S on pairs, $t > 1$ and $r > 0$ we call

$$\mathcal{N}_t^{\mathsf{S}}(\Psi; r) = \mathsf{S}(\Psi_{t^2 r}) - (1 + t^6)\, \mathsf{S}(\Psi_{tr}) + t^6\, \mathsf{S}(\Psi_r)$$

the t-scale number; at $t = 2$ it is the *three-scale number* $\mathsf{S}(\Psi_{4r}) - 65\, \mathsf{S}(\Psi_{2r}) + 64\, \mathsf{S}(\Psi_r)$. The weights annul constants and multiples of r^6, and give the function $r \mapsto r$ the factor $t^2 - (1+t^6)t + t^6 = -t(t-1)(t^5 - 1)$, which is -62 at $t = 2$. For a triple $\tau = (g, b, \ell)$ and a pair Ψ we put $E_\tau = 2C_2^{\mathrm{T}}(g)\vartheta^{-1}b^2$, with the number C_2^{T} of Prop. 8.3, and $Y_\tau(\delta) = b^2\, T^{(\ell,b)}(\Psi_{\delta b})$.

8.1.2. *Changing units and keeping the label.* The natural short-distance variable is d/b. The label b has the dimension of length, and under a lattice dilation the mesh, side and label change together. At powers of the blocking factor, the exact identity is

$$T^{(\ell,b)}(\Psi) = L^{2s} T^{(L^s\ell, L^s b)}(\Psi_{L^s}), \qquad \Psi_r = (f(\cdot/r), h(\cdot/r)). \tag{8.ii}$$

It follows at finite cutoff by transporting the same Wilson measure to the dilated lattice, keeping the bare inverse coupling fixed, and using the factor L^s in each smeared observable. The full-sequence cutoff limits then give eq. 8.ii. This identity compares theories at different labels. Dilation covariance of one two-point function would instead keep its label fixed and require one power of r for every separated pair.

The two explicit kernels transform compatibly with eq. 8.ii: $\mathcal{K}_6(\Psi_r) = \mathcal{K}_6(\Psi)$ and $\mathcal{K}_5(\Psi_r) = r\mathcal{K}_5(\Psi)$. Their common natural parameter is consequently d/b. The purpose of theorem Y is to make the tolerance uniform in that parameter across the admitted labels and canonical components. A bound $C(g)d^{1/2}$ on the least torus does not yet have this uniformity.

The supplied proof obtains it by putting labels into one multiplicative period. Choose Λ once from the fixed construction constants. Every $b > 0$ has a unique dilate $b' = L^z b$ in $[\Lambda, L\Lambda)$. Two fixed reference couplings have admissible label intervals covering that period. The least-torus law is therefore needed at only these two couplings. The state comparison of lemma 8.11 bounds the difference between a continuum component and the corresponding torus function by a constant times $(d/b)^2$ after multiplication by b^2. At the dilated label this comparison constant is bounded throughout the fixed period.

The torus tolerance contributes a constant times $(d/b)^{1/2}$ in the normalized remainder, and the state comparison contributes a constant times d/b. Taking their nondecreasing majorant gives $\varepsilon^{\mathrm{nat}}(d/b) \le C(d/b)^{1/2}$. For components, the comparison is justified by their conditioned periodic-Gibbs approximants. Centring affects only the first cumulant; the covariance is the limit of those approximants. Thus the same transfer applies to each component with the same recovered label. No averaging over components is used to obtain a pointwise assertion about an individual component.

8.1.3. *The nonzero linear term.* Fix a test pair Ψ of scale d_1 in the positive class. The natural-scale law gives

$$\mathsf{S}(\Psi_r) = A + Br + O(r^{3/2}), \qquad A = c_2 b^{-2}\mathcal{K}_6(\Psi) > 0, \qquad B = \kappa^{\mathrm{lat}} b^{-3}\mathcal{K}_5(\Psi) > 0, \tag{8.iii}$$

for the volume-first two-point function and for each canonical component. Positivity of A and B follows from the mass floors of the test class, not merely from the signs of their displayed numerical coefficients. The error is uniform in the small-radius range specified by the natural variable.

If S were dilation covariant with exponent ϖ, its value at Ψ_r would be $r^\varpi \mathsf{S}(\Psi)$. The positive finite limit A as $r \to 0$ forces $\varpi = 0$. Covariance would then make the value constant in r, whereas dividing eq. 8.iii minus A by r gives the positive limit B. This is the direct contradiction in lemma 8.12.

The scale difference used in the source gives a quantitative version. For $t > 1$, the combination

$$\mathcal{N}_t^{\mathsf{S}}(\Psi; r) = \mathsf{S}(\Psi_{t^2 r}) - (1 + t^6)\mathsf{S}(\Psi_{tr}) + t^6 \mathsf{S}(\Psi_r)$$

annuls constants and terms proportional to r^6. Its value on the linear term is $-t(t-1)(t^5-1)Br$. At $t = 2$ this is $-62Br$, and the sum of the three remainder bounds has coefficient 198. Hence the three-scale number is negative for sufficiently small r, with an explicit threshold given by the normalized tolerance. The test records a property of the two-point function at fixed label, rather than the exact change-of-units identity between labels.

The alternative proof in lemma 8.15 and proposition 8.17 transfers this obstruction from a single admitted triple. Choose a second blocking factor whose logarithm is incommensurable with the first. Sequences of powers of the two factors approximate any positive ratio of scales. On each pair of corresponding lattices, transport the same bare periodic Gibbs state. Its two rescaled covariances agree exactly after multiplication by the squares of their labels. Comparing each with its torus limit leaves an error $O((d/b)^2)$. Thus a hypothetical quadratic-order agreement with the free law in one limit would transfer to the triple where the nonzero linear correction is already known. It cannot absorb that correction. This second route retains the original comparison estimates and continuity in the varying label and test pair; density of scale ratios alone would not justify passage to the limit.

8.1.4. *Signed pairs and what the two-term law detects.* The positivity conditions on the test class are used to identify the coefficient and its sign. A leading short-distance estimate for arbitrary signed separated Lipschitz functions follows by a finite decomposition into signed nonnegative pieces. A common test shape serves the finitely many pairs of pieces. Apply the two-term estimate to each pair and use bilinearity. This gives the $O(r)$ error in corollary 8.1 uniformly in position and on compact label intervals. A nonzero positive reflected pair has $\mathcal{K}_5 > 0$, so its linear correction shows that this rate is attained. The label derivative identity is retained separately: inserting the constant function differentiates the coupling, and integration over a compact interval of labels gives the stated bound involving $S_3(1, f_r, h_r)$.

The failure of dilation covariance is not by itself a test of interaction. Proposition 8.18 supplies a concrete comparison model. On T_ℓ, take three independent Gaussian fields with covariance symbol

$$\widehat{G}_b(k) = (b|k|^2 + \alpha|k| + \upsilon)^{-1}, \qquad \upsilon = \upsilon_0/\ell > 0.$$

Its positive spectral representation is

$$(bu^2 + \alpha u + \upsilon)^{-1} = \int_0^\infty \frac{\rho_{\mathrm{sp}}(m)}{u^2 + m^2}\,\mathrm{d}m, \qquad \rho_{\mathrm{sp}}(m) = \frac{2\alpha}{\pi}\frac{m^2}{(\upsilon - bm^2)^2 + \alpha^2 m^2}.$$

Reflection positivity follows by integrating the reflection-positive massive covariances against this positive density. The model's observable is a quadratic functional of mollified fields, combining $-\frac{1}{2}{:}|\nabla\varphi|^2{:}$ with $-{:}\varphi\Delta\varphi{:}$ as specified there. Its separated cumulants have limits independent of the mollifier, although its variance at a nonzero single test function diverges. Consequently it is important to retain the separated-cumulant formulation and not claim an L^2 limiting observable.

Let $\mathcal{C}$ be its covariance kernel and $\mathcal{H} = \Delta\mathcal{C}$ off the origin. Wick's rule gives the two-point kernel

$$\frac{3}{2}\left[\mathrm{tr}(\nabla\nabla^{\mathsf{T}}\mathcal{C})^2 + 4\nabla\mathcal{C}\cdot\nabla\mathcal{H} + 2\mathcal{H}^2 + 2\mathcal{C}\Delta\mathcal{H}\right].$$

The supplied expansion of $\mathcal{C}$ makes its first two singular terms

$$c_2 b^{-2}\left[|x|^{-6} + \frac{4\alpha}{3\pi}b^{-1}|x|^{-5}\right].$$

At $\alpha = 2$ the relative coefficient is exactly $8/(3\pi)$. After smearing, the remaining error is $O(d^2(1 + \log(\ell/d)))$, which is smaller than the linear correction. This model therefore reproduces the same two-term law.

The source distinguishes it using a three-point property of the torus limits. Their cutoff estimates imply convergence to zero for tuples with uniformly bounded supremum norms and Lipschitz constants, fixed positive spread, and a product of supremum norms tending to zero. Here the spread of supports $S_1, \dots, S_n$ is $\inf_{x_a \in S_a} \max_{a,a'} |x_a - x_{a'}|_\infty$. For $f_{\mathbf{q}} = |\mathbf{q}|^{-1} \psi \cos(\mathbf{q} \cdot x)$ with ψ separated from a nonzero $h \geq 0$, these conditions hold. The first two supports may coincide; the third stays separated from them. The quadratic Gaussian model instead has the limit

$$\mathcal{S}_3(f_{\mathbf{q}}, f_{\mathbf{q}}, h) \longrightarrow -\frac{3}{2b} \iint \psi(x)^2 h(y) \big[|\nabla \mathcal{C}|^2 + 2\mathcal{C} \Delta \mathcal{C} \big](x-y) \, \mathrm{d}x \, \mathrm{d}y < 0.$$

The spectral representation makes the bracket positive off the origin. This is the supplied test that excludes that comparison model. It also explains why the two-point coefficient, despite detecting a scale, cannot alone identify the full field theory.

8.1.5. *The law at natural scales.*

Theorem Y (The components)**.** Let $\mathfrak{c}$ be a choice and $\vartheta \in (0,1)$. There are $\delta_{\mathrm{c}} > 0$ and a non-decreasing function $\varepsilon^{\mathrm{nat}}$ on $(0, \delta_{\mathrm{c}}]$ with $\varepsilon^{\mathrm{nat}}(\delta) \leq C\delta^{1/2}$, where δ_{c}, $\varepsilon^{\mathrm{nat}}$ and C depend only on $\mathfrak{c}$ and ϑ, such that for every $0 < g \leq \gamma_W$, every volume-first limit S at g with $b = \mathsf{b}(S)$, every canonical decomposition $(\mathsf{X}, \lambda, (\sigma_\xi))$ of σ_S, and for $\mathsf{S} = S_2$ and $\mathsf{S} = S_2^\xi$, $\xi \in \mathsf{X}$, the following holds.

(1) $|\mathsf{S}(\Psi) - L_b^+(\Psi)| \leq \varepsilon^{\mathrm{nat}}(d/b)\, b^{-3} d\, |\Psi|$ for $0 < d \leq \delta_{\mathrm{c}} b$ and $\Psi \in \mathfrak{T}^1(d; \vartheta)$. So σ_S and every σ_ξ obey (L) with $(b, \frac{8}{3\pi}, \varepsilon^S, d^S, \vartheta)$, where $\varepsilon^S(d) = \varepsilon^{\mathrm{nat}}(g^2 d)$ and $d^S = \delta_{\mathrm{c}} g^{-2}$ depend only on $\mathfrak{c}$, g and ϑ. They obey (L) also with a tolerance formed at any dilated label: if z is an integer with $L^z b \geq 2\Gamma^{-2}$ and $g' = (2/(L^z b))^{1/2}$, then (L) holds with $(b, \frac{8}{3\pi}, \varepsilon', d', \vartheta)$, where $\varepsilon'(d) = \varepsilon^L_{g',\vartheta}(L^z d) + 2C_2^{\mathrm{T}}(g')\vartheta^{-1} L^{2z} b\, d$ and $d' = L^{-z} \min\{d^L, d_\flat\}(g', \vartheta)$, with the objects of Thm. Z (a) and Prop. 8.3.

(2) Let $d_1 > 0$, $\Psi \in \mathfrak{T}^1(d_1; \vartheta)$ and $B_S = \varkappa b^{-3} \mathcal{K}_5(\Psi)$, so that $B_S \geq 10^{-5} \varkappa \vartheta^2 b^{-3} d_1 |\Psi| > 0$; let $t > 1$. Then

$$|\mathsf{S}(\Psi_r) - c_2 b^{-2} \mathcal{K}_6(\Psi) - B_S r| \leq \varepsilon^{\mathrm{nat}}(r d_1/b)\, b^{-3} d_1 |\Psi|\, r \qquad (r d_1 \leq \delta_{\mathrm{c}} b),$$

$$|\mathcal{N}_t^{\mathsf{S}}(\Psi; r) + t(t-1)(t^5-1) B_S r| \leq (t^2 + t + t^6 + t^7)\, \varepsilon^{\mathrm{nat}}(t^2 r d_1/b)\, b^{-3} d_1 |\Psi|\, r \qquad (t^2 r d_1 \leq \delta_{\mathrm{c}} b).$$

So $\mathsf{S}(\Psi_r) = c_2 b^{-2} \mathcal{K}_6(\Psi) + B_S r + O(r^{3/2})$; $|r^{-1}[\mathsf{S}(\Psi_{tr}) - \mathsf{S}(\Psi_r)] - (t-1) B_S| \leq (t+1)\, \varepsilon^{\mathrm{nat}}(t r d_1/b)\, b^{-3} d_1 |\Psi|$, so that $r^{-1}[\mathsf{S}(\Psi_{tr}) - \mathsf{S}(\Psi_r)] \to (t-1) B_S$; $r^{-1} \mathcal{N}_t^{\mathsf{S}}(\Psi; r) \to -t(t-1)(t^5-1) B_S$, and so do the means $n^{-1} \sum_{j<n} (t^{-j} r_0)^{-1} \mathcal{N}_t^{\mathsf{S}}(\Psi; t^{-j} r_0)$ as $n \to \infty$, for every r_0 with $t^2 r_0 d_1 \leq \delta_{\mathrm{c}} b$; and $\mathcal{N}_t^{\mathsf{S}}(\Psi; r) < 0$ as soon as $(t^2 + t + t^6 + t^7)\, \varepsilon^{\mathrm{nat}}(t^2 r d_1/b) < t(t-1)(t^5-1) \cdot 10^{-5} \varkappa \vartheta^2$. At $t = 2$: the three-scale number is $-62 B_S r + O(r^{3/2})$, within $198\, \varepsilon^{\mathrm{nat}}(4 r d_1/b)\, b^{-3} d_1 |\Psi|\, r$ of $-62 B_S r$, and negative as soon as $198\, \varepsilon^{\mathrm{nat}}(4 r d_1/b) < 62 \cdot 10^{-5} \varkappa \vartheta^2$, that is, for all small r.

(3) Neither S_2 nor any S_2^ξ is dilation covariant: $\mathsf{N} = \mathsf{X}$ and $\lambda(\mathsf{N}) = 1$.

■□

Theorem Z (The tori)**.** Let $\mathfrak{c}$ be a choice and $\vartheta \in (0,1)$; let d_{e}, $\bar{\varepsilon} = \bar{\varepsilon}_{g,\vartheta}$, $\bar{\varepsilon}_\infty = \bar{\varepsilon}_{\infty;g,\vartheta}$ and C_{cut} be the objects of Prop. 8.10 and the depth $s(d)$ that of the subsection on the least torus, and $d_\flat$, C_2^{T} those of Prop. 8.3.

(a) *The least torus.* For $0 < g \leq \gamma_{\mathrm{e}}$ there are $d^L(g, \vartheta) > 0$ and a non-decreasing function $\varepsilon^L_{g,\vartheta}$ on $(0, d^L(g,\vartheta)]$ with $\varepsilon^L_{g,\vartheta}(d) \leq C(g,\vartheta) d^{1/2}$, all depending only on $\mathfrak{c}$, g and ϑ, such that

$$|T^{(\ell_g, b)}(\Psi) - L_b^+(\Psi)| \leq \varepsilon^L_{g,\vartheta}(d)\, b^{-3} d\, |\Psi| \qquad (b \in I(g),\ 0 < d \leq d^L(g,\vartheta),\ \Psi \in \mathfrak{T}_{\mathbb{R}^3}(d; \vartheta)).$$

(b) *Every torus of the family; every state at a finite cutoff.* Let $0 < g \leq \Gamma$, $b \in I(g)$, $\ell \in \mathfrak{L}(g)$, $0 < d \leq \min\{d_{\mathrm{e}}, \ell_g/20, d_\flat\}(g, \vartheta)$ and $\Psi \in \mathfrak{T}_{\mathbb{R}^3}(d; \vartheta)$. Then

$$|T^{(\ell,b)}(\Psi) - L_b^+(\Psi)| \leq \big[\bar{\varepsilon}_\infty(d) + 2C_2^{\mathrm{T}}(g) \vartheta^{-1} b d \big]\, b^{-3} d\, |\Psi|,$$

and the bracket is at most $C(g,\vartheta)(1+b)d^{1/2}$. Moreover, for every cutoff $K > s(d)$ and every state $\mathfrak{s}$ at the datum $(K, L^K b)$ in the sense of Prop. 8.3,

$$|S_2^T[\mathfrak{s}](\Psi) - L_b^+(\Psi)| \le \Big[(\bar\varepsilon(d) + C_2^{\mathrm{T}}(g)\vartheta^{-1}bd)\, b^{-3}d + C_{\mathrm{cut}}\, b^{-2}\frac{a_K}{d}\Big]|\Psi|,$$

and the same holds with $C_{\mathrm{r}}d^{1/2}$ in the place of $\bar\varepsilon(d)$ and C'_{cut} in the place of C_{cut}, with the numbers of Prop. 8.10 (c).

(c) *Every side, every label.* Let $\ell = 2L^m$, $m \ge 1$, $b > 0$, $s \ge 0$ an integer and $0 < g'' \le \Gamma$ a coupling for which $(g'', L^s b, L^s \ell)$ is admissible; this holds for all large s with $g'' = (2/(L^s b))^{1/2}$. Then for $0 < d \le L^{-s}\min\{d_{\mathrm{e}}, \ell_{g''}/20, d_{\mathrm{b}}\}(g'', \vartheta)$ and $\Psi \in \mathfrak{T}^1(d;\vartheta)$

$$|T^{(\ell,b)}(\Psi) - L_b^+(\Psi)| \le \big[\bar\varepsilon_\infty(L^s d) + 2C_2^{\mathrm{T}}(g'')\vartheta^{-1}L^{2s}bd\big]\, b^{-3}d\,|\Psi|,$$

with $\bar\varepsilon_\infty$ formed at (g'', ϑ); the bracket is at most $Cd^{1/2}$, with C depending on $\mathfrak{c}$, g'', s, b and ϑ.

(d) *Exact dilation by powers of L.* $T^{(\ell,b)}(\Psi) = L^{2s}\, T^{(L^s\ell, L^s b)}(\Psi_{L^s})$ for $\ell = 2L^m$, $b > 0$, integers $s \ge 0$ and pairs Ψ of real Lipschitz functions with compact supports whose periodisations to T_ℓ have disjoint supports. In particular, for $0 < g \le \gamma_{\mathrm{e}}$, $b \in I(g)$ and every integer $z \ge 0$, the law of (a) holds on the side $L^z\ell_g$ at the label $L^z b$ with no comparison of states:

$$|T^{(L^z\ell_g, L^z b)}(\Psi) - L^+_{L^z b}(\Psi)| \le \varepsilon^L_{g,\vartheta}(L^{-z}d)\,(L^z b)^{-3}d\,|\Psi| \qquad (0 < d \le L^z d^L(g,\vartheta),\ \Psi \in \mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)).$$

(e) *One coefficient.* If at one side $2L^m$ and one label the inequality of (c) holds with a number c and with a number c' in the place of $\frac{8}{3\pi}$, each with a tolerance that tends to 0, then $c = c'$.

(f) *Compact sets of labels.* At a fixed side, one tolerance, one range of d and one constant C serve in (c) for all the labels of a compact subset of $(0,\infty)$.

Let now $0 < g \le \gamma_{\mathrm{e}}$, $b \in I(g)$, $T = T^{(\ell_g, b)}$, $\varepsilon^L = \varepsilon^L_{g,\vartheta}$, $d^L = d^L(g,\vartheta)$, $d_1 > 0$, $\Psi \in \mathfrak{T}^1(d_1;\vartheta)$, $\mathsf{A} = c_2 b^{-2}\mathcal{K}_6(\Psi)$, $\mathsf{B} = \varkappa b^{-3}\mathcal{K}_5(\Psi)$ and $\mathsf{u} = b^{-3}d_1|\Psi|$. Then $10^{-5}\varkappa\vartheta^2\mathsf{u} \le \mathsf{B} \le \frac{8}{3\pi}\mathsf{u}$, and for $0 < rd_1 \le d^L$

$$|T(\Psi_r) - \mathsf{A} - \mathsf{B}r| \le \varepsilon^L(rd_1)\,\mathsf{u}\,r \le C(g,\vartheta)\, d_1^{1/2}\,\mathsf{u}\,r^{3/2}. \tag{8.iv}$$

(i) For $t > 1$ and $0 < trd_1 \le d^L$: $|r^{-1}[T(\Psi_{tr}) - T(\Psi_r)] - (t-1)\mathsf{B}| \le (t+1)\,\varepsilon^L(trd_1)\,\mathsf{u} \to 0$; for $0 < t^2 rd_1 \le d^L$: $|r^{-1}\mathcal{N}_t^T(\Psi;r) + t(t-1)(t^5-1)\mathsf{B}| \le (t^2+t+t^6+t^7)\,\varepsilon^L(t^2rd_1)\,\mathsf{u} \to 0$. At $t = 2$: $|r^{-1}[T(\Psi_{2r}) - T(\Psi_r)] - \mathsf{B}| \le 3\varepsilon^L(2rd_1)\mathsf{u}$.

(ii) $r^{-2}|T(\Psi_r) - \mathsf{A}| \ge r^{-1}\mathsf{u}\,[10^{-5}\varkappa\vartheta^2 - \varepsilon^L(rd_1)] \to \infty$ as $r \to 0$.

(iii) For all $d_n \to 0$ and all $\Psi_n \in \mathfrak{T}^1(d_n;\vartheta)$: $|T(\Psi_n) - c_2 b^{-2}\mathcal{K}_6(\Psi_n)|/(d_n^2|\Psi_n|) \ge b^{-3}[10^{-5}\varkappa\vartheta^2 - \varepsilon^L(d_n)]/d_n \to \infty$.

(iv) Let $g \le \Gamma$ and $b \in I^\circ(g)$. For every $0 < C < \infty$, every $d \le d^L$ with $\varepsilon^L(d) \le \frac12\cdot 10^{-5}\varkappa\vartheta^2$ and $d/b < \frac12\cdot 10^{-5}\varkappa\vartheta^3/C$, and every $\Psi \in \mathfrak{T}^1(d;\vartheta)$: $b^2|T(\Psi) - c_2 b^{-2}\mathcal{K}_6(\Psi)| > C\vartheta^{-1}(d/b)^2|\Psi|$.

■□

Corollary 8.1 (The leading term with the rate r, for signed functions)**.** Let $\mathfrak{c}$ be a choice, $\ell = 2L^m$ with $m \ge 1$, $b > 0$, $p \in T_\ell$ with a lift $\tilde p$, let f_0, h_0 be real Lipschitz functions on $\mathbb{R}^3$ with compact disjoint supports, and put $f_r = (f_0((\cdot - \tilde p)/r))^{\langle\ell\rangle}$ and $h_r = (h_0((\cdot - \tilde p)/r))^{\langle\ell\rangle}$ for $r > 0$.

(a) There are $C, r_* > 0$, independent of p and the same for all the labels of a compact subset of $(0,\infty)$, with

$$|S_2^{(\ell,b)}(f_r, h_r) - c_2 b^{-2}\mathcal{K}_6(f_0, h_0)| \le Cr \qquad (0 < r \le r_*).$$

In particular, for $\theta_\mu x = x - 2x_\mu e_\mu$ and every real Lipschitz function ψ_0 with compact support in $\{x_\mu > 0\}$, the pair $(f_0, h_0) = (\psi_0\circ\theta_\mu, \psi_0)$ gives $S_2^{(\ell,b)}(f_r, h_r) = c_2 b^{-2}\mathcal{K}_6(\psi_0\circ\theta_\mu, \psi_0) + O(r)$.

(b) The rate r is attained: if $\psi_0 \geq 0$ is of class C^1, not 0, with support in the ball $\mathbb{B}(e_\mu, \frac{1}{2})$, then

$$S_2^{(\ell,b)}(f_r, h_r) - c_2 b^{-2} \mathcal{K}_6(\psi_0 \circ \theta_\mu, \psi_0) = \varkappa\, b^{-3} \mathcal{K}_5(\psi_0 \circ \theta_\mu, \psi_0)\, r + O(r^{3/2}), \qquad \mathcal{K}_5(\psi_0 \circ \theta_\mu, \psi_0) > 0;$$

so no bound $o(r)$ holds in (a).

(c) For $0 < b_1 < b_2$, with the constant C of (a) for the labels of $[b_1, b_2]$ and $0 < r \leq r_*$,

$$\Big| \int_{b_1}^{b_2} [b^2 S_3^{(\ell,b)}(1, f_r, h_r) - 2b\, S_2^{(\ell,b)}(f_r, h_r)]\, \mathrm{d}b \Big| \leq (b_1^2 + b_2^2)\, C\, r,$$

where 1 is the constant function on T_ℓ.

■□

8.1.6. *Construction inputs.* The proof rests on Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD among the construction results, with its conditions on the constants, through the statements cited below. Among them are Section YM3_I:74, [7, result G.2], YM3_I:(1528) and YM3_I:82, the real-sequence conditions and smallness floor YM3_I:(972) preceding Thm. YM3_I:Y, the bounds YM3_I:(1531) on labels, Cor. YM3_I:74.2 and [7, Lem. G.4] on couplings and labels, which we use in the form of Lem. 8.2 below, the exact dilation by powers of L YM3_I:(2215), Thm. YM3_I:AS on the limits at a fixed label (their existence, the rate of removal of the cutoff, the bounds YM3_I:(2214) uniform in the cutoff and in the labels of a compact set, the continuity in the label and the convergence along converging labels), Prop. YM3_I:118.3 on the derivatives in the label, the uniform cutoff estimate of Thm. YM3_I:AR on the removal of the cutoff, the weighted progression and its source energies ([7, Def. B.7] and [7, Cor. B.9]), its setting (J) for the canonical objects ([7, Def. B.16] and equation [7, eq. 228] and [7, Prop. B.21]), the places with a lower place of a step at a complex field scale (Def. 6.28 and Prop. 6.30), the sourced renormalisation-group chain and its source energies ([7, Prop. B.12] and equations [7, eq. 212]–[7, eq. 214]), the histories, the sum over the level of the first creation of a large-field region and the rarity of large fields (Section YM3_I:123, equation eq. 4.xxxi and Prop. 4.54), the reflection positivity of the limits on the tori (proved in Prop. 8.18(c) from Thm. YM3_I:A and finite-cutoff positivity, for every side $2L^m$, every $b > 0$ and C^1 entries at every coordinate mirror) and that of the Wilson measures on the tori at the site and bond mirrors (Prop. YM3_I:91.4), which, with the existence of the limits along the full sequence of the cutoffs, gives it for real Lipschitz entries at the mirrors $\{x_\mu = c\}$, $c \in \bigcup_{K \in \mathbb{N}} \frac{1}{2} L^{-K} \mathbb{Z}$, of the lattices of the cutoffs, the block averages ([7, Lem. B.3]), the template sum of the free lattice curvature (Prop. YM3_I:121.2 and Lem. 6.33), Laplace's method (Lem. 7.11) and the coefficients of lattice perturbation theory (Def. 7.10, Cor. 7.37, Thms. W and X), the chessboard estimate ([7, Cor. B.1]), the representation of the components by conditioned periodic Gibbs states (Thm. YM3_I:AP), and, for $g \leq \gamma_W$, the leading law of Thm. YM3_I:AW for σ_S and, using the conditioned approximants in Thm. YM3_I:AP, for its components: for $0 < d \leq \mathsf{d}_0(g, \vartheta)$ and the C^1 pairs of $\mathfrak{T}_{\mathbb{R}^3}(d; \vartheta)$, $\mathsf{S}(\Psi) = c_2 b^{-2} \mathcal{K}_6(\Psi)(1 + \eta)$ with $|\eta| \leq \eta_{g,\vartheta}(d) \leq \frac{1}{3}$ and $\eta_{g,\vartheta}(d) \to 0$ as $d \to 0$.

Two further statements are proved below from these: the agreement of all states at one datum at two points, which uses no decay of correlations (Prop. 8.3), and the next-order law on the least torus (Prop. 8.10).

Lemma 8.2 (Couplings and labels)**.**

(a) $\gamma^{-2} > \bar{B}$. For every real $b' > 2\bar{B}$, in particular for every $b' \geq 2\gamma^{-2}$, the coupling $g' = (2/b')^{1/2}$ satisfies $b' \in I^\circ(g')$; if $b' \geq 2\gamma^{-2}$, then $g' \leq \gamma$.

(b) Let $0 < g' \leq \gamma$, and let (K_j, β_j) be pairs of integers $K_j \geq 1$ and reals $\beta_j > 0$ with $a_{K_j} \beta_j \to b' \in I^\circ(g')$. Then (K_j, β_j) is a first-passage datum at g' for all large j.

■

Proof sketch. (a) Cor. YM3_I:74.2 and [7, Lem. G.4] at the label $2\gamma^{-2}$, where the coupling is γ, puts $2\gamma^{-2}$ in $I(\gamma) = (\gamma^{-2} + \bar{B}, X_\gamma - \bar{B}]$; so $\gamma^{-2} > \bar{B}$. With $g'^{-2} = b'/2$ the formula for I° in the

setting gives $I^\circ(g') = (\frac12 b' + \bar B,\ \frac12 Lb' + (L-2)\bar B)$. Its upper end exceeds b', as $L \ge 3$, and its lower end is less than b' if and only if $b' > 2\bar B$, which holds for $b' \ge 2\gamma^{-2}$; and then $g' \le (2/(2\gamma^{-2}))^{1/2} = \gamma$. (b) As $I^\circ(g')$ is open, $a_{K_j}\beta_j \in I(g')$ for all large j, and Cor. YM3_I:74.2 at the label $a_{K_j}\beta_j$ says that $(K_j, L^{K_j}a_{K_j}\beta_j) = (K_j, \beta_j)$ is a first-passage datum at g'. □

8.1.7. *Agreement of states at two points.*

Proposition 8.3 (Agreement of two states at two points)**.** Let $\mathfrak c$ be a choice, $0 < g' \le \gamma_W^{\rm sd}$, (K,β) a first-passage datum at g' with $K \ge 3$ and label $b = a_K\beta$, $\vartheta \in (0,1)$, $0 < d \le d_\flat(g',\vartheta)$ and $a_K \le 9d$. A *periodic Gibbs state* of the Wilson action at β on $a_K\mathbb Z^3$ is a Gibbs state at (a_K,β) that is invariant under the translations of a subgroup of finite index of $a_K\mathbb Z^3$ (Section YM3_I:91); these states form the set $\mathcal G^{\rm per}_{K,\beta}$. For every K and every $\beta > 0$ this set is the set $\mathcal G^{\rm inv}_{K,\beta}$ of the periodic Gibbs states that are invariant under all the translations of $a_K\mathbb Z^3$ (Prop. YM3_I:103.4; [7, §K.1]), and it contains the states of the tori, the weak limit points of the periodic extensions of the Wilson measures on the tori T_{2L^m}, $m \to \infty$ (Prop. YM3_I:91.2; [7, §K.1]).

A *state at the datum* is the periodic extension to $a_K\mathbb Z^3$ of the Wilson measure on T_ℓ at (K,β), for some $\ell \in \mathfrak L(g')$, or a periodic Gibbs state of the Wilson action at β on $a_K\mathbb Z^3$: a state at g' at (K,β) in the sense of [7, §K.1]. Every state at the datum is invariant under the translations of $a_K\mathbb Z^3$, and so are its cumulants: $S_n^T[\mathfrak s](\tau_v f_1,\dots,\tau_v f_n) = S_n^T[\mathfrak s](f_1,\dots,f_n)$ for all $v \in a_K\mathbb Z^3$ ([7, §K.1]). Let μ' be the periodic extension of the Wilson measure on the least torus $T_{\ell_{g'}}$ at (K,β).

(a) Every state $\mathfrak s$ at the datum satisfies $|S_2^T[\mathfrak s](\Psi) - S_2^T[\mu'](\Psi)| \le C_2^{\rm T}(g')\vartheta^{-1}d^2b^{-2}|\Psi|$ for all $\Psi \in \mathfrak T_{\mathbb R^3}(d;\vartheta)$; hence any two states $\mathfrak s, \mathfrak s'$ at the datum have

$$|S_2^T[\mathfrak s](\Psi) - S_2^T[\mathfrak s'](\Psi)| \le 2C_2^{\rm T}(g')\,\vartheta^{-1}d^2(a_K\beta)^{-2}|\Psi|.$$

(b) Here $d_\flat(g',\vartheta) = \min\{\frac{1}{180}, d_\star(g',\vartheta)\}$ with $d_\star(g',\vartheta) = c_{\rm a}\vartheta^{-1}g'^{-2}$, and

$$C_2^{\rm T}(g') = C_{\rm a}\big[Q_0(g')\,Q_1(g')\big]^2\,\frac{(X_{g'}+\bar B)^3}{(g'^{-2}-\bar B)^2}, \qquad Q_0(g') = \exp\big[\tfrac18\ell_{g'}^3(2C_{\rm S}+C_{g'})\big],$$

where $C_{\rm S}$ and $C_{g'}$ are the numbers of the bounds of Thms. YM3_I:AJ and YM3_I:AD on the partition functions of the tori, $Q_1(g') \ge 1$ is the bound on the linking cumulants of step (4) of the proof, and $c_{\rm a}, C_{\rm a} > 0$ depend only on the constants. So $d_\star > 0$ and $C_2^{\rm T}(g') < \infty$ depend only on the radius, the constants and g' (and on ϑ, for $d_\star$), and on no K, β, side, state, pair or position.

■

Proof sketch. Let $\ell' = \ell_{g'}$, let x_1 be a centre of a ball of radius d that holds $\operatorname{supp} f$, and consider on $a_K\mathbb Z^3$ the board of cubic cells of side $\ell'/2$ one of which, $\square_0$, is centred within $\frac12 a_K$ of x_1; on $T_{\ell'}$ there are eight cells, the mirror images of $\square_0$ under lattice reflections.

(1)

One cell against the least torus. For a bounded function F of the configuration that depends only on the bonds in the interior of $\square_0$ put $\|F\|_{\rm ch}^8 = \mu'(\prod_{\mathbf k\in\{0,1\}^3} F^{[\mathbf k]}) \ge 0$, the $F^{[\mathbf k]}$ being the mirror copies of F in the eight cells, conjugated when the number of reflections is odd. This is the chessboard seminorm $\|F\|_{{\rm ch},\ell'}$ of the side ℓ'. Then $|\mathfrak s(F)| \le Q_0\|F\|_{\rm ch}$ for every state at the datum. For the torus of a side $\ell \ge \ell'$ of $\mathfrak L(g')$ the chessboard estimate at the mirrors of the cells gives $|\mathfrak s(F)| \le \|F\|_{{\rm ch},\ell}$ (clause (c) of [7, Cor. B.1]), and the comparison of two sides gives $\|F\|_{{\rm ch},\ell} \le \|F\|_{{\rm ch},\ell'}\,e^{(\ell'/2)^3(p_{\ell'}-p_\ell)}$ with the pressures $p_\ell = \ell^{-3}\log Z_{K,\ell,\beta}$ (clause (e) there); Thms. YM3_I:AJ and YM3_I:AD at the source 0 bound $p_{\ell'} - p_\ell$ by $2C_{\rm S} + C_{g'}$, so that $e^{(\ell'/2)^3(p_{\ell'}-p_\ell)} \le Q_0$.

A periodic Gibbs state $\mathfrak s$ is dominated through tori of growing sides: by the domination of the means of a periodic Gibbs state by the chessboard seminorms of the tori (Prop. YM3_I:

91.9, for the board above, the one cell $\square_0$ and a sequence of sides $\ell_k \in \mathfrak{L}(g')$ with $\ell_k \to \infty$), $|\mathfrak{s}(F)| \le \liminf_k \|F\|_{\mathrm{ch},\ell_k}$, and every $\|F\|_{\mathrm{ch},\ell_k}$ is at most $Q_0\|F\|_{\mathrm{ch}}$ by the comparison of two sides.

(2)

Normalised moments; only linking cumulants remain. For bounded real F_1, F_2 localised in $\square_0$ let $X_a = \mathcal{O}_K(F_a)$, $M'(z) = \mu'(e^{z_1X_1+z_2X_2})$, $M_{\mathfrak{s}}(z) = \mathfrak{s}(e^{z_1X_1+z_2X_2})$, and $c_I = \partial_I(1/M')(0)$, $I \subset \{1,2\}$, which are real. For $\emptyset \ne B \subset \{1,2\}$ the function $P_B = \sum_{I\subset B} c_{B\setminus I} \prod_{a\in I} X_a$ has $\mathfrak{s}(P_B) = \partial_B(M_{\mathfrak{s}}/M')(0)$. Its mirror copies are the same polynomials in the mirror images $X_{a,\mathbf{k}} = \mathcal{O}_K(F_a \circ \hat{\mathsf{r}}_{\mathbf{k}}^{-1})$, and as μ' is invariant under the lattice isometries,

$$\|P_B\|_{\mathrm{ch}}^8 = \mu'\Big(\prod_{\mathbf{k}} P_B^{[\mathbf{k}]}\Big) = \sum_{\pi} \prod_{\gamma\in\pi} \langle X_{a,\mathbf{k}};\, (\mathbf{k},a)\in\gamma\rangle_{\mu'},$$

the sum running over the partitions π of $\{0,1\}^3 \times B$ into *linking* blocks, blocks that hold images in two different cells: all cumulants within one cell cancel. So if every linking cumulant is at most $Q_1 \prod_{(\mathbf{k},a)\in\gamma} u_a$, then $|\partial_B(M_{\mathfrak{s}}/M')(0)| \le Q_0 \mathsf{Bl}_{16}^{1/8} Q_1^{|B|/2} \prod_{a\in B} u_a$, Bl_N the number of partitions of a set of N elements, and the two cumulants $\partial_1\partial_2 \log M_{\mathfrak{s}}(0)$, $\partial_1\partial_2 \log M'(0)$ differ by $|\partial_1\partial_2 \log(M_{\mathfrak{s}}/M')(0)| \le 2(Q_0\mathsf{Bl}_{16}^{1/8}Q_1)^2 u_1u_2$.

(3)

Packing and signs. Let $\iota = a_K\lceil 18d/a_K\rceil$ and $\mathsf{Pk} = \{v \in \iota\mathbb{Z}^3 : |v_\mu| \le 0.575 - 9d\}$. The balls $\mathbb{B}(x_1+v, 9d)$, $v \in \mathsf{Pk}$, are disjoint and lie in $\mathbb{B}(x_1,1) \subset \square_0$, both supports of Ψ lie in $\mathbb{B}(x_1,9d)$, and $\#\mathsf{Pk} \ge (0.3/9d)^3 = (30d)^{-3}$. For signs $\epsilon = (\epsilon_v)$ apply (2) to $F_1 = \sum_v \epsilon_v\tau_v f$, $F_2 = \sum_v \epsilon_v\tau_v h$. The difference $\mathsf{D} = S_2^T[\mathfrak{s}] - S_2^T[\mu']$ is bilinear, its mean over the signs at (F_1,F_2) is $\sum_v \mathsf{D}(\tau_v f, \tau_v h)$, and both cumulants are invariant under the translations of $a_K\mathbb{Z}^3$ ([7, §K.1]); so $|\mathsf{D}(f,h)| \le (30d)^3 \sup_\epsilon |\mathsf{D}(F_1,F_2)|$.

(4)

The linking cumulants on the least torus. For a complex source w on the plaquettes of $T_{\ell'}$ let $Z[w] = \int \rho_0\, e^{\mathcal{O}(w)}\, \mathrm{d}U$ be the partition function of $T_{\ell'}$ with the source w, ρ_0 the Wilson density at (K,β) and $\mathcal{O}(w)(U) = L^K \sum_p w(p)\, a_p(U)$, and $Z = Z[0]$. The generating function of the sixteen mirrored, packed and signed observables is $Z[W_z]/Z$, with the source $W_z = \sum_{\mathbf{k},a} z_{\mathbf{k},a}(F_a \circ \hat{\mathsf{r}}_{\mathbf{k}}^{-1})^{\langle\ell'\rangle} \circ x$; it vanishes off the plaquettes whose barycentres lie in the eight mirror images of $\mathbb{B}(x_1,1)$, closed balls of radius 1 whose centres are at pairwise sup distances $\ge \frac12\ell' - a_K$. Write

$$\log\frac{Z[W]}{Z} = \log\frac{Z[W]e^{-\mathcal{E}(W)}}{Z} + \Big[\mathcal{E}(W) - \sum_{\mathbf{k}} \mathcal{E}(w^{(\mathbf{k})})\Big] + \sum_{\mathbf{k}} \mathcal{E}(w^{(\mathbf{k})}),$$

$\mathcal{E}$ the source energy of $T_{\ell'}$, the sum over the levels and localisation domains of the source energies [7, eq. 212] of the sourced renormalisation-group chain [7, eq. 205], and $w^{(\mathbf{k})}$ the part of W in the cell $\mathbf{k}$.

(4a) *The setting on $T_{\ell'}$.* Let $\ell' = 2L^m$, $m = m_{\mathfrak{L}}(g')$. By YM3_I:(1528), $L^m = LM\max\{R(L(g'^{-2}+\bar B)), L^{\lceil \log_L C_{\mathfrak{s}}\rceil}\}$, and the values of the cube-size function R are powers L^{u} with $\mathsf{u} \ge 1$; so the cells have the side $\mathsf{l} := \frac12\ell' \ge L^2M$. As $g' \le \gamma_W^{\mathrm{sd}}$, the least side is small over the run: $MR(x_K) \le L^m$ (Lem. YM3_I:126.3). As $K \ge 3$, Lem. YM3_I:126.2, whose head asks $K \ge 3$, gives $x_j \ge x_{\mathrm{J}}$ for $0 \le j \le K$ and makes the cuts of the run, on the lattice (ℓ', K), frames of the construction; so the setting (J) holds with the last height $K-1$ ([7, Prop. B.21]), and the canonical sourced chain of $T_{\ell'}$ is formed, its conditions holding by [7, Cor. B.13], at the sources of a class of small sources, an open neighbourhood of 0.

By the running estimate YM3_I:(1530) at the index K and the endpoint bounds YM3_I:(1529), $b_- \le a_{K-j}x_j \le b_+$ for $0 \le j \le K$, with $b_\pm = x_K \pm \bar B$; and $g'^{-2} - \bar B < b_-$, $b_+ \le X_{g'} + \bar B$. The places, the classes of the sum rule, the image sets and the chart integrals of the steps, the backgrounds and the minimal orbits that the setting (J) names ([7, Lem. B.17] and [7, Prop. B.18]) are the objects that the sourced chain names ([7, Prop. B.15] and its one-step proof): both use the same definitions

(among them [7, Def. B.2] and equation [7, eq. 191] of the chart integral and [7, Def. B.2] of the image set), at the same lattice (ℓ', K), the same height j and the same coupling $\mathsf{t} = x_j$.

(4b)

The level-wise norm; the families and the energies. For a source w let $v_j^w = -L^{K-j} w/x_j$ be its weight at the level j. The *level-wise norm* $\|w\|_{\mathrm{lw}}$ is the largest, over $0 \le j < K$, of $|v_j^w|_\infty/\sigma_{\mathrm{a}}$ and of $\mathrm{sw}_j(v_j^w)/(\mu_{\mathrm{a}} x_j^{-1/2})$, where $\mathrm{sw}_j(v)$ is the largest difference of the values of v at two plaquettes whose barycentres are at a sup distance $\le M a_{K-j}$, and $\sigma_{\mathrm{a}} = \sigma_{\mathrm{src}}$, $\mu_{\mathrm{a}} x_j^{-1/2} = o_{\mathrm{src}}(x_j)$ are the radii of the class of the weights of eq. 6.xx, σ_{a} and μ_{a} depending only on the constants.

It is a norm; its open unit ball $\mathsf{Src}^{\mathrm{lw}}$ is convex, consists of the sources whose weights lie at every level in the class of the weights at which Prop. 6.31 is stated, and contains the class of small sources of (4a); and $|w|_\infty \le \sigma_{\mathrm{a}} b_+ \|w\|_{\mathrm{lw}}$, as $|v_j^w|_\infty = |w|_\infty/(a_{K-j} x_j)$. [7, Prop. B.12] runs the induction on $\mathsf{Src}^{\mathrm{lw}}$; consequently the sourced families of all levels are formed there, holomorphic in the source, bounded by E_0, a constant of the canonical list, in the norm of the localised terms at the rate $\kappa + 1$, and the family on a domain reads the source on the plaquettes of that domain alone.

So the source energies $e_j(X; w)$ of the domains X of the level j are holomorphic on $\mathsf{Src}^{\mathrm{lw}}$, vanish at $w = 0$ and read w on the plaquettes of X alone (equation [7, eq. 212] and its proof), and, by [7, eq. 213] at the bound $2E_0$ of the differences of the families and by Schwarz's lemma for $\zeta \mapsto e_j(X; \zeta w/\|w\|_{\mathrm{lw}})$ on the unit disc,

$$|e_j(X; w)| \le 16 M^3 E_0 \, \|w\|_{\mathrm{lw}} \, e^{-\kappa d_j(X)},$$

$d_j(X)$ the linear size of X in the units of the level j. Put $\mathcal{E}_t(w) = \sum_{j \le t} \sum_{X \in \mathcal{D}_j} e_j(X; w)$, $\mathcal{D}_j$ the localisation domains of the level j; so $\mathcal{E} = \mathcal{E}_K$, and these functions are holomorphic on $\mathsf{Src}^{\mathrm{lw}}$ and vanish at 0.

(4c)

The term without large fields, with a source. For $0 \le t \le K$ let h_t be the history without large-field regions up to the level t and $\tau_{h_t}[\cdot]$ its term (Section YM3_I:123), a complex Borel measure on the configurations $\mathcal{V}_t$ of the level t, and $\mathsf{m}_t = \tau_{h_t}[\rho_0]$: a positive measure of mass at most Z (equation eq. 4.xxxi), equal to $\rho_0 \, \mathrm{d}U$ at $t = 0$ and carried, for $t \ge 1$, by the open set $\mathcal{V}_t^\chi$ of the configurations that pass the density tests of the level (Prop. YM3_I:123.1). For $w \in \mathsf{Src}^{\mathrm{lw}}$ and $0 \le t \le K$,

$$\tau_{h_t}\big[\rho_0 e^{\mathcal{O}(w)}\big] = e^{\mathcal{E}_t(w)} \, e^{\mathsf{R}_t(w,\cdot)} \, \mathsf{m}_t,$$

an identity of complex Borel measures on $\mathcal{V}_t$, with $\mathsf{R}_0(w, \cdot) = \mathcal{O}(w)$ and, for $t \ge 1$ and $W \in \mathcal{V}_t^\chi$,

$$\mathsf{R}_t(w, W) = \mathcal{O}(w)(U) + S[\mathfrak{f}_t[w]](U) - S[\mathfrak{f}_t[0]](U),$$

U an element of the minimal orbit over W and $S[\mathfrak{f}_t[w]]$ the action of the sourced bare array of the level t; this value does not depend on the element of the orbit, the domain of definition of the action being a condition on the configuration alone, and $\mathsf{R}_t(w, \cdot)$ is a bounded Borel function on $\mathcal{V}_t^\chi$. For the small sources of (4a) and $1 \le t \le K$, the density of $\tau_{h_t}[\rho_0 e^{\mathcal{O}(w)}]$ is

$$\exp\Big[\mathsf{C}_t(w) - A^{\eta_t}(x_t^w, U_t(W)) + S[\mathfrak{f}_t[w]](U_t(W))\Big] \quad \text{on } \mathcal{V}_t^\chi, \text{ and 0 off it,} \qquad \mathsf{C}_t(w) = \sum_{1 \le j \le t} \mathsf{c}_j + \mathcal{E}_t(w),$$

with numbers c_j formed with the source 0 alone, the weight $x_t^w = x_t - L^{K-t} w$ of the weighted action A^{η_t}, $\eta_t = L^{-t}$, and the background $U_t(W)$ over W. This follows by induction over t. At $t = 0$ the term is $\rho_0 e^{\mathcal{O}(w)} dU$, and $\rho_0 e^{\mathcal{O}(w)} = e^{-A^{\eta_0}(x_0^w, \cdot)}$ (equation [7, eq. 216]). From t to $t+1$: the term of the history is carried by one step, $\tau_{h_{t+1}}[\cdot] = \mathbf{T}(\mathsf{g}_{t+1} \tau_{h_t}[\cdot])$ with the factor $\mathsf{g}_{t+1} \ge 0$ of the step, and a measure whose density on $\mathcal{V}_t^\chi$ is e^{e} times a bounded Borel function F is carried to the density $e^{\mathsf{e}} \, \mathsf{I}[F]$ on $\mathcal{V}_{t+1}^\chi$, and 0 off it, $\mathsf{I}[F]$ being the chart integral of the step ([7, Prop. B.18]); here, for $t \ge 1$, $\mathsf{e} = \mathsf{C}_t(w)$ and F is the function F_t^w in the proof of [7, Prop. B.15] at $k = t$, which on $\mathcal{V}_t^\chi$ is $\exp[-A^{\eta_t}(x_t^w, U_t(V)) + S[\mathfrak{f}_t[w]](U_t(V))]$ (the definition preceding [7, eq. 218]), and for $t = 0$, $\mathsf{e} = 0$

and $F = \rho_0 e^{\mathcal{O}(w)}$, whose chart integral is that of F_0^w on the class of the sum rule (the height-zero case in that proof); $\mathcal{V}_{t+1}^{\chi}$ lies in the class of the sum rule of the step and in the set on which the chart integral is formed ([7, Lem. B.17]); and there the one-step identity of the sourced chain [7, eq. 218] gives

$$\mathsf{I}[F_t^w](W) = \exp\Big[\mathsf{c}_{t+1} + \sum_{X\in\mathcal{D}_{t+1}} e_{t+1}(X;w) - A^{\eta_{t+1}}(x_{t+1}^w, U_{t+1}(W)) + S[\mathsf{f}_{t+1}[w]](U_{t+1}(W))\Big].$$

The weighted action is linear in the weight, $A^{\eta_t}(x_t^w, U) = A^{\eta_t}(x_t, U) - \mathcal{O}(w)(U)$; so the quotient of the density by the same density at $w = 0$, which is the density of m_t, is $e^{\mathcal{E}_t(w)+\mathsf{R}_t(w,\cdot)}$ on $\mathcal{V}_t^{\chi}$: the identity holds for the small sources. For $w \in \mathsf{Src}^{\mathrm{lw}}$, $w \neq 0$, and a bounded Borel function ψ on $\mathcal{V}_t$, the function $\zeta \mapsto \int \psi \,\mathrm{d}\tau_{h_t}[\rho_0 e^{\zeta\mathcal{O}(w)}]$ is entire, and $\zeta \mapsto e^{\mathcal{E}_t(\zeta w)} \int \psi\, e^{\mathsf{R}_t(\zeta w,\cdot)}\,\mathrm{d}\mathsf{m}_t$ is holomorphic on $\{|\zeta|\,\|w\|_{\mathrm{lw}} < 1\}$, by (4b); they agree for small ζ, hence at $\zeta = 1$.

(4d)

The remainders. Let $w \in \mathsf{Src}^{\mathrm{lw}}$ vanish off the plaquettes whose barycentres lie in N_{b} closed balls of radius 1. Then

$$|\mathsf{R}_K(w,\cdot)| \le \|w\|_{\mathrm{lw}} N_{\mathrm{b}}\, \mathsf{g}_\emptyset \ \text{ on } \mathcal{V}_K^{\chi}, \qquad |\mathsf{R}_{n-1}(w,\cdot)| + |\mathcal{E}(w) - \mathcal{E}_{n-1}(w)| \le \|w\|_{\mathrm{lw}} N_{\mathrm{b}}\, C_{\mathrm{cnt}}\, L^{4(K-n+1)} \ \text{ on } \mathcal{V}_{n-1}^{\chi} \quad (1 \le n \le K),$$

on all of $\mathcal{V}_0$ for $n = 1$, with numbers $\mathsf{g}_\emptyset$ and $C_{\mathrm{cnt}} \ge 1$ that depend only on g' and the constants. Indeed, let $W \in \mathcal{V}_s^{\chi}$. The background over W has $0 \le a_p(U_s(W)) < 2B_3^2\varepsilon_s^2 L^{-4s}$ at every bare plaquette [7, eq. 221], $\varepsilon_s = \varepsilon(x_s) \le \varepsilon(g'^{-2})$ being the threshold of the small fields at the coupling x_s and B_3 a constant of the canonical list; at most $81N_{\mathrm{b}}L^{3K}$ plaquettes have their barycentres in the balls; and $|w|_\infty \le \sigma_{\mathrm{a}} b_+ \|w\|_{\mathrm{lw}}$. So $|\mathcal{O}(w)(U_s(W))| \le 162 N_{\mathrm{b}} B_3^2 \varepsilon_s^2 \sigma_{\mathrm{a}} b_+ L^{4(K-s)} \|w\|_{\mathrm{lw}}$, and at $s = 0$, where $0 \le a_p \le 2$, $|\mathcal{O}(w)| \le 162 N_{\mathrm{b}} \sigma_{\mathrm{a}} b_+ L^{4K} \|w\|_{\mathrm{lw}}$.

The difference of the two actions is the sum, over the levels $j \le s$, the domains X of the level j that meet the support of w and their points z, of the differences between the values at U and at $\mathbf{1}$ of the local terms $(E^{(j)}[w] - E^{(j)}[0])(X,\cdot,z)$, each of a norm at most $2E_0\|w\|_{\mathrm{lw}} e^{-(\kappa+1)d_j(X)}$ by (4b). Every element of the minimal orbit over W is regular of the level s at the radius $2B_3\varepsilon_s$ ([7, Lem. B.17]); at such a configuration a local term of the level j moves from its value at $\mathbf{1}$ by at most its norm times $C(\varepsilon_s/\alpha_0)^2 L^{-4(s-j)}(1 + d_j(X))^2$ on the domains that are small against the radius, and by at most twice its norm on the other ones, whose linear sizes are large (equation YM3_I: (2263) and its proof), α_0 being the fixed radius of the localised terms; the bound in [7, Lem. B.10], with at most $27N_{\mathrm{b}}L^{3(K-s)}$ cubes of the level s holding a corner of a plaquette of the support, sums these differences to at most $CN_{\mathrm{b}}L^{3(K-s)}\|w\|_{\mathrm{lw}}$.

The energies of the levels $> s$ are bounded in the same way, by (4b) and (q3), by $CN_{\mathrm{b}}L^{3(K-s)}\|w\|_{\mathrm{lw}}$. With $b_+ \le X_{g'} + \bar{B}$ this gives the two bounds.

(4e) *The coefficients of $Z[w]e^{-\mathcal{E}(w)}/Z$.* By eq. 4.xxxi at level K, applied to the bounded gauge invariant function $e^{\mathcal{O}(w)}$, with the canonical history measures of Section YM3_I:123,

$$Z[w] = Z_\emptyset[w] + \sum_{n=1}^{K} Z_n[w], \qquad Z_\emptyset[w] = \tau_{h_K}[\rho_0 e^{\mathcal{O}(w)}](\mathcal{V}_K), \qquad Z_n[w] = \int \mathsf{g}_n^{\mathrm{w}}\,\mathrm{d}\tau_{h_{n-1}}[\rho_0 e^{\mathcal{O}(w)}],$$

where $\mathsf{g}_n^{\mathrm{w}} \ge 0$ is the factor of the step n at which a large-field region is first created, and $\mathsf{w}_n = Z^{-1}\int \mathsf{g}_n^{\mathrm{w}}\,\mathrm{d}\mathsf{m}_{n-1}$ is the weight of that event; each term is an entire function of w. By (4c),

$$Z_\emptyset[w]e^{-\mathcal{E}(w)} = \int e^{\mathsf{R}_K(w,\cdot)}\,\mathrm{d}\mathsf{m}_K, \qquad Z_n[w]e^{-\mathcal{E}(w)} = e^{-(\mathcal{E}-\mathcal{E}_{n-1})(w)} \int \mathsf{g}_n^{\mathrm{w}}\, e^{\mathsf{R}_{n-1}(w,\cdot)}\,\mathrm{d}\mathsf{m}_{n-1},$$

so that, by (4d) and as m_K has a mass at most Z,

$$\big|Z_\emptyset[w]e^{-\mathcal{E}(w)}\big| \le e^{\|w\|_{\mathrm{lw}} N_{\mathrm{b}} \mathsf{g}_\emptyset}\, Z, \qquad \big|Z_n[w]e^{-\mathcal{E}(w)}\big| \le \exp\big[\|w\|_{\mathrm{lw}} N_{\mathrm{b}} C_{\mathrm{cnt}} L^{4(K-n+1)}\big]\, \mathsf{w}_n\, Z.$$

At $w = W_z$ take $N_{\mathrm{b}} = 8$ and the polydisc $\mathbb{P} = \{|z_{\mathbf{k},a}| < r_a\}$, $r_a = A_\star(d)/\|f_a\|_\infty$, on which $\|W_z\|_{\mathrm{lw}} \le \max_{\mathbf{k},a} |z_{\mathbf{k},a}|/r_a < 1$ by (4g). The function $\Phi_{\mathrm{D}}(z) = Z[W_z]e^{-\mathcal{E}(W_z)}/Z$ is holomorphic on $\mathbb{P}$, $Z[w]$ being entire, and $\Phi_{\mathrm{D}}(0) = 1$; its term without large fields is at most $e^{8\mathsf{g}_\emptyset}$ on the whole of $\mathbb{P}$, and its term of the level n at most $e\,\mathsf{w}_n$ on the polydisc smaller by the factor $8C_{\mathrm{cnt}}L^{4(K-n+1)}$. Cauchy's estimate on each polydisc gives, for every set J of at most sixteen of the variables,

$$|\partial_J \Phi_{\mathrm{D}}(0)| \le \Big[e^{8\mathsf{g}_\emptyset} + e\,(8C_{\mathrm{cnt}}L^4)^{16} \sum_{n=1}^{K} L^{64(K-n)}\mathsf{w}_n\Big] \prod_{i\in J} r_i^{-1}.$$

By eq. 4.xxxi, $\mathsf{w}_n \le \tilde{\nu}_K\{Z_n \ne \emptyset\}$. Equation eq. 4.xxxiv, at $q = 64$ and $t = K$, therefore bounds $\sum_n L^{64(K-n)}\mathsf{w}_n$ by

$$4M^{-3}(\ell')^3 e^{\mathsf{B}(X_{g'})(\ell')^3}\mathsf{C}_{\mathrm{ag}}(65)e^{-c'p_0(g_K)^2/2}.$$

The assumptions hold by the canonical choice of the fixed history thresholds in Prop. 4.54; the ceiling $g' \le \gamma_W^{\mathrm{sd}}$ preserves them. Since ℓ' depends only on g' and the constants, the square bracket is bounded by a number $Q_{\mathrm{D}}(g') \ge 1$ with the same dependence.

(4f)

The linking cumulants. The last sum of the decomposition is a sum of functions each of which reads the variables of one cell; it is annulled by every linking derivative. In the bracket only the domains X that meet the supports of two parts $w^{(\mathbf{k})}$ contribute, the source energies being local and vanishing at 0. Such a domain of the level j holds corners of plaquettes whose barycentres lie in two different balls, and two such corners are at a sup distance $\ge \mathsf{l}-3$, while two points of a domain of the level j are at a sup distance at most $M(d_j(X)+2)a_{K-j}$; so $d_j(X) \ge (\mathsf{l}-3)L^{K-j}/M - 2 \ge 166L^{K-j}$, as $\mathsf{l} \ge L^2M$ and $L \ge 13$. The bound of (4b) at the nine sources W, $w^{(\mathbf{k})}$ and the bound in [7, Lem. B.10] give

$$\Big|\mathcal{E}(W) - \sum_{\mathbf{k}} \mathcal{E}(w^{(\mathbf{k})})\Big| \le C_\Delta \max_{\mathbf{k}}\{\|W\|_{\mathrm{lw}}, \|w^{(\mathbf{k})}\|_{\mathrm{lw}}\},$$

this exponential decay of the source energies in the size of their domains summing over the levels to a number C_Δ depending only on the constants. On $\mathbb{P}$ the bracket is holomorphic and at most C_Δ, so that Cauchy's estimate bounds its squarefree coefficients by $C_\Delta \prod_i r_i^{-1}$. A derivative ∂_J at 0 of the logarithm of a function equal to 1 at 0 is the sum, over the partitions π of J, of $(-1)^{|\pi|-1}(|\pi|-1)!$ times the product over the blocks of its derivatives; so the squarefree coefficients of $\log \Phi_{\mathrm{D}}$ are at most $\mathsf{cu}_{|J|}Q_{\mathrm{D}}^{|J|}\prod_{i\in J} r_i^{-1}$, with $\mathsf{cu}_N = \sum_\pi (|\pi|-1)!$ over the partitions of N elements.

(4g)

The packed bumps. For a source made of bumps with pairwise disjoint sets of non-vanishing, of height at most A and Lipschitz constant at most $A/(\vartheta d)$, the supremum of the weight of the level j is at most A/b_-, and its oscillation at most $(A/b_-)\min\{2, 2\sqrt{3}Ma_{K-j}/(\vartheta d)\}$; as $a^{-1/2}\min\{2, ca\} \le (2c)^{1/2}$ for all $a > 0$, its level-wise norm is at most $A/A_\star(d)$ at every level, with

$$A_\star(d) = \min\Big\{\sigma_{\mathrm{a}} b_-,\ \mu_{\mathrm{a}}\frac{b_-}{b_+^{1/2}}\Big(\frac{\vartheta d}{4\sqrt{3}M}\Big)^{1/2}\Big\} \ge c\,(g'^{-2} - \bar{B})(X_{g'} + \bar{B})^{-3/2}(\vartheta d)^{1/2}\,b \qquad (d \le d_\star(g', \vartheta)),$$

$d_\star$ being the scale below which the second member of the minimum is the smaller one. The bumps of W_z are the mirror images of the $z_{\mathbf{k},a}\epsilon_v\tau_v f_a$; their sets of non-vanishing are pairwise disjoint, their heights are at most $A = \max_{\mathbf{k},a}|z_{\mathbf{k},a}|\,\|f_a\|_\infty$ and their Lipschitz constants at most $A/(\vartheta d)$, as $\mathrm{Lip} f_a \le \|f_a\|_\infty/(\vartheta d)$; so $\|W_z\|_{\mathrm{lw}} \le \max_{\mathbf{k},a}|z_{\mathbf{k},a}|/r_a$. So, by (4f), every linking cumulant is at most $(\mathsf{cu}_{16}Q_{\mathrm{D}}^{16} + C_\Delta)\prod_{(\mathbf{k},a)\in\gamma} r_a^{-1}$: the linking cumulants obey the bound of (2) with $u_a = \|f_a\|_\infty/A_\star(d)$ and $Q_1 = \mathsf{cu}_{16}Q_{\mathrm{D}}^{16} + C_\Delta$.

(5)

Conclusion. By (1)–(4),

$$|S_2^T[\mathfrak{s}](\Psi) - S_2^T[\mu'](\Psi)| \le (30d)^3\,2(Q_0\mathsf{BI}_{16}^{1/8}Q_1)^2 A_\star(d)^{-2}|\Psi| \le C_2^{\mathrm{T}}(g')\vartheta^{-1}d^2b^{-2}|\Psi|$$

for $d \le d_\flat$, which is (a) with the constants of (b); two states differ by at most twice this. □

Remark 8.4. The proof uses only the following properties of the pair: real Lipschitz functions, neither 0, each vanishing off a ball of radius d, whose sets of non-vanishing are disjoint and lie in one ball of radius $9d$, with $\mathrm{Lip} f_a \le \|f_a\|_\infty/(\vartheta d)$. So Prop. 8.3 holds also for such pairs of either sign. With n entries in the place of two, every pair of which lies in $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$, the same argument gives a difference of at most $2C_n^{\mathrm{T}}(g')\vartheta^{-n/2}d^{3-n/2}b^{-n}\prod_a\|f_a\|_\infty$, which is small against the natural size $b^{-n}\prod_a\|f_a\|_\infty$ for $n \le 5$; we use only $n = 2$.

8.1.8. *The coefficient on the least torus.* Here the constants are an admissible assignment of the canonical list. The statements used in Lems. 8.5–8.8 are made for the data (Wp1)–(Wp6) of a weighted progression ([7, Def. B.7]), in its setting (J) ([7, Def. B.16]) and, where a step is continued to a complex field scale, at a place with a lower place (Def. 6.28); we call this the setting of the weighted progression (the statements used in Lem. 8.9, the constants and thresholds in Thm. YM3_I:AR, Thm. YM3_I:AR, equation YM3_I:(2203), Cor. YM3_I:117.1, Lem. 2.5 and Cor. YM3_I:118.4, are made in the setting (Th1), (Th2), whose datum is the choice of the constants).

For the canonical objects at every admissible assignment of the canonical list it holds: the data are given by [7, Cor. B.9]; the setting (J) holds by [7, Prop. B.21] as soon as the couplings are at least the threshold x_{J} of the definitions preceding [7, eq. 228], a number depending only on the constants, and at a first-passage datum at g this is the case whenever $g^{-2} \ge x_{\mathrm{J}}$, all the couplings of the run exceeding g^{-2} (Lem. 8.5); and the places with a lower place are those of Prop. 6.30. Besides the constants the setting involves two auxiliary real numbers: the lower place $\mathsf{t}_\flat \ge \gamma^{-2}$, and a room $\mathsf{q} > 0$, such that a bound on the sizes of the bare arrays formed with the constants alone, increased by $2\mathsf{q}$, does not exceed the size ceiling at the lower place.

For the canonical objects $\mathsf{q} = 1$, and $\mathsf{t}_\flat$ is a number depending only on the constants. No further condition on the constants enters. $\gamma_{\mathrm{e}} \in (0,\gamma]$ is the largest number under which the setting holds and all the conditions on g of these statements hold (each has the form $g \le \gamma'$ or $\log g^{-2} \ge Y$ with numbers γ', Y depending only on the constants), together with $g^{-2} \ge 4\bar{B}$; for $0 < g \le \gamma_{\mathrm{e}}$, $d_{\mathrm{e}}(g,\vartheta) \in (0, L^{-13}]$ is the largest number below which their conditions on d and those of the proof of Prop. 8.10 hold (the arithmetic of the levels there uses $d \le L^{-13}$, and the removal of the cutoff there uses $d \le d_{\mathrm{K}}$, the number of Lem. 8.9 at $\ell = \ell_g$).

The depth of the scale d at g is the witness depth $s(d) = \min\{s \ge 0 : L^s d \ge C_W(s)\}$, $C_W(s) = \mathsf{c}_W(\log[L^{s+1}(g^{-2}+\bar{B})])^{2p_0}$, with a number $\mathsf{c}_W \ge 1$ depending only on the constants (equation YM3_I:(2321)); so $C_W(s(d)) \le L^{s(d)}d < LC_W(s(d))$, $C_W(s(d)) \ge (\log\frac{1}{d})^2$, $s(d) \to \infty$ as $d \to 0$, and $K > s(d)$ entails $d/a_K \ge L(\log\frac{1}{d})^2$, hence $K \ge 3$ and $8a_K \le d$. We write $D_j = a_{K-j}$ (the spacing of the level j), $\mathsf{l} = \max\{1, \log(2b/d)\}$, $\mathfrak{b}(x) = A_1(\log x)^{p_0}$ for the field cut-off, A_1 and the integer $p_0 \ge 1$ being constants of the canonical list, $\|u\|_\sharp = \max\{\|u\|_\infty, 40M\,\mathrm{Lip}\,u\}$ and $\mathbb{Z}^3_N = \mathbb{Z}^3/N\mathbb{Z}^3$; C denotes numbers that depend only on the constants, and on g and ϑ where this is shown.

Lemma 8.5 (The run)**.** Let $0 < g \le \gamma_{\mathrm{e}}$ and (K,β) a first-passage datum at g with $K \ge 3$ and label b. Then $|x_j - L^{-(j-i)}x_i| \le \bar{B}$ for $0 \le i \le j \le K$, $g^{-2} < x_K \le X_g$, and $\frac{3}{4}b/D_j \le x_j \le 2b/D_j$ for $0 \le j \le K$. On every $\ell \in \mathfrak{L}(g)$ the setting of the weighted progression holds with the sources $f\circ x$, $h\circ x$ for all real bounded f, h on T_ℓ supported in closed balls of radius d at a distance $\ge d$, with $8a_K \le d$, neither $f\circ x$ nor $h\circ x$ vanishing identically; no other condition on d enters. If $f\circ x$ or $h\circ x$ vanishes identically, then so does $(\tau_v f)\circ x$, respectively $(\tau_v h)\circ x$, for every $v \in a_K\mathbb{Z}^3$, and $S^T_{2;K,\ell,\beta}(f,h)$, the jets $\mathsf{n}_j(f,h)$ and $\mathsf{n}^{\mathsf{z}}_j(f,h)$ of Lem. 8.7 and every sum $\sum_{p,q}(f\circ x)(p)\,(h\circ x)(q)\,\mathsf{k}(p,q)$ against a kernel k on the plaquettes vanish.

So, for $b_* \in I(g)$, $K' \ge 3$ and real functions w_1, w_2 on the plaquettes of $\mathbb{Z}^3_{\ell L^{K'}}$, neither identically 0, supported in balls of radius $r \ge 8$ at a distance $\ge r$, $w_a = f'_a \circ x$ for such a pair (f'_1, f'_2) at the scale $ra_{K'}$ and the first-passage datum $(K', L^{K'}b_*)$. ■

Proof sketch. The first two assertions are the running and endpoint estimates YM3_I:(1529)–(1530). As $L^{-j}x_0 = b/D_j$ and $x_j \geq x_K > g^{-2} \geq 4\bar{B}$: $b/D_j \leq x_j + \bar{B} \leq \frac{4}{3}x_j$, and $x_j \leq b/D_j + \bar{B} \leq 2b/D_j$ because $b/D_j \geq b > g^{-2} \geq \bar{B}$. The data of the weighted progression are those of the canonical objects ([7, Cor. B.9]) at the calibrations $x_0, \dots, x_{K-1}$ of the run, with the lower place $\mathfrak{t}_\flat$ and the room $\mathfrak{q} = 1$; the lattice (ℓ, K), with the run, is admissible over every cut of the run, for every $\ell \in \mathfrak{L}(g)$ (Prop. YM3_I:74.1); and the setting (J) holds, as $x_j \geq x_K > g^{-2} \geq x_{\mathrm{J}}$ ([7, Prop. B.21]).

The other conditions of the setting are lower bounds on x_{K-1} and on $\log x_K$ by numbers formed with the constants and with $\mathfrak{t}_\flat$; they hold as $x_{K-1} > X_g \geq Lg^{-2}$ and $x_K > g^{-2}$ once g is small, and they enter γ_{e}. Of the sources the setting reads only that they are real and bounded, supported in balls of radius d at a distance $\geq d$, with $8a_K \leq d$, and not identically 0; a translation by $v \in a_K\mathbb{Z}^3$ permutes the barycentres of the plaquettes. If $f \circ x \equiv 0$, then $\mathcal{O}_K(f) = 0$, the source $\mu_1 f \circ x + \mu_2 h \circ x$ does not read μ_1, so that the source energies of the weighted progression do not read it at any field scale, and a sum of the values $(f \circ x)(p)$ against anything vanishes.

For the last sentence let f'_a take the values of w_a at the barycentres of the plaquettes of spacing $a_{K'}$ and vanish elsewhere: they are real and bounded, supported in balls of radius $ra_{K'}$ at a distance $\geq ra_{K'}$, with $8a_{K'} \leq ra_{K'}$, and no condition on the depth of the scale $ra_{K'}$ enters; $(K', L^{K'}b_*)$ is a first-passage datum since $b_* \in I(g)$. □

Lemma 8.6 (The chart of the iterated block averages)**.** Let $\ell = 2L^m$, $1 \leq n \leq K$ and $2L^{m+K-n+1} \geq C_L M$, C_L a constant of the canonical list. For a block field W of the level n and fluctuation fields $y = (\mathcal{A}_j)_{j<n}$, $\mathcal{A}_j$ an $\mathfrak{su}(2)$-valued 1-form on the non-pivot bonds of the level j, let $W_n = W$, $W_j = V_{W_{j+1}}[\mathcal{A}_j]$ ([7, §B.2]), $\Upsilon(W, y) = W_0$, and $\theta_{\mathrm{ch}} = -\sum_{j<n}[\mathsf{J}_{V_j}[\mathcal{A}_j] + \log z^{(j)}_{V_j}]$, with the Jacobian term J and the normalisation $z^{(j)}$ of [7, Lem. B.3], $V_j = V_j^{\mathrm{bf}}(W_{j+1})$ being the averaged background over the block field W_{j+1} ([7, §B.2]).

There are an open neighbourhood O_n of $\mathbf{1}$ among the block fields of the level n and a number $\rho_2 \in (0, 1]$, both defined by bounds that depend only on the constants and not on n, K or m (they are written out in the proof), such that, with $\Omega = \{|\mathcal{A}_j| < \rho_2,\ j < n\}$ and

$$C_{\mathrm{H}} = \prod_{j<n} (2\pi^2)^{-\#\mathfrak{X}_j^+} L^{-3\#\mathfrak{B}_{j+1}^+},$$

$\mathfrak{X}_j^+$ the positively oriented non-pivot bonds of the level j and $\mathfrak{B}_{j+1}^+$ the positively oriented bonds of the level $j+1$:

(i) on $\mathsf{O}_n \times \Omega$ the chain and θ_{ch} are formed, W_{j+1} is the block average of W_j, and θ_{ch} is real analytic;
(ii) Υ is a real analytic diffeomorphism of $\mathsf{O}_n \times \Omega$ onto an open set $\mathcal{U} \ni \mathbf{1}$;
(iii) $\int H \,\mathrm{d}U = C_{\mathrm{H}} \int_{\mathsf{O}_n} \mathrm{d}W \int_\Omega \mathrm{d}y\, e^{-\theta_{\mathrm{ch}}} H(\Upsilon(W, y))$ for every bounded Borel function H that vanishes off $\mathcal{U}$, and the same holds with every open subset of $\mathsf{O}_n \times \Omega$ and its image.

■

Proof sketch. The numbers and the good sets. For a configuration V of a level let $\varpi(V) = \max_p |\partial V(p) - \mathbf{1}|$, the maximum over the plaquettes of that level, and $\varpi_c(V)$ the same maximum over the plaquettes of the double block of a bond c of the next level; $\mathcal{M}$ is the block average and Lg the $\mathfrak{su}(2)$-valued logarithm. With the numbers ϖ_χ, $\varpi_1 = 2\varpi_\chi$, r_χ, C_χ of the chart of the fibre ([7, §B.2]), which depend only on L, the small-curvature radius c_2^{SO} of Prop. YM3_I:14.1 and the response constant K_{B} of [7, Lem. P.5], and the constants B_3, $\bar{\varepsilon}$, ε_{v}, r_{B} of the canonical list, put

$$\mu_\flat = \min\Big\{c_2^{\mathrm{SO}},\ \tfrac{1}{2}\bar{\varepsilon},\ \tfrac{1}{4}L^2\varpi_\chi,\ \tfrac{1}{8}\varepsilon_{\mathrm{v}}\Big\}, \qquad \rho_2 = \min\Big\{\frac{r_\chi}{C_\chi},\ \frac{r_{\mathrm{B}}}{4C_\chi},\ \frac{(1-L^{-2})\,\mu_\flat}{4K_{\mathrm{B}}C_\chi},\ \frac{\varepsilon_{\mathrm{v}}}{64C_\chi}\Big\}, \qquad \mathsf{O}_n = \Big\{W :\ \varpi(W) < \frac{\mu_\flat}{B_3}\Big\};$$

they read neither n nor K nor m, and $\rho_2 \le r_\chi/C_\chi \le 1$. For $1 \le i \le n$ let Gd_i be the set of the block fields W of the level i over which the background exists and has a regularity number $\mu_i(W) < \mu_\flat$ ([7, Def. B.2]; the number is the same at all the elements of the minimal orbit, by unitary gauge invariance), and $\mathsf{Gd}_0 = \{V : \varpi(V) < \frac{1}{4}\varepsilon_{\mathrm{v}}\}$.

(1)

The averaged background over a good block field. Gd_i is open and $\{\varpi < \mu_\flat/B_3\} \subset \mathsf{Gd}_i \subset \{\varpi < 2\mu_\flat\}$ for $1 \le i \le n$: over a block field with $\varpi < \mu_\flat/B_3$ the background exists and has a regularity number $< \mu_\flat$ (Thm. YM3_I:E in the setting of Section YM3_I:52.7), a regularity number $< \mu_\flat$ gives $\varpi < 2\mu_\flat \le \bar\varepsilon$ (Lem. 5.28), and the representative of the background is a real analytic function of the block field (Thm. YM3_I:E and [7, Lem. P.5]); so $\mathbf{1} \in \mathsf{O}_n \subset \mathsf{Gd}_n$. Let $0 \le j < n$ and $W \in \mathsf{Gd}_{j+1}$. Then $V_j(W) = V_j^{\mathrm{bf}}(W)$ is defined and $\mathcal{M}V_j(W) = W$: it is the average $\mathcal{M}^j U$ of an element U of the minimal orbit over W, carried to the axial gauge by a gauge transformation (the definition of $p(W)$ in [7, Def. B.2]), which conjugates the plaquette variables; so Lem. 5.28 gives $\varpi(V_j(W)) \le 2L^{-2}\mu_{j+1}(W) < \frac{1}{2}\varpi_\chi$.

The map $W \mapsto V_j(W)$ is real analytic on Gd_{j+1}: $\mathcal{M}^j U^\circ(W)$, with the representative $U^\circ(W)$ of the background, is real analytic by Thm. YM3_I:E and Lem. 5.28, on the good-set and actual-average domains just established, and the gauge transformation to the axial gauge is $v(x) = (\mathcal{M}^j U^\circ(W))(y,x)$, x in the block of y, an averaged contour variable, a real analytic function of the configuration where it is formed. And $V_j(\mathbf{1}) = \mathbf{1}$: $\mathbf{1}$ is flat, lies in the minimal orbit over $\mathbf{1}$, and every path product and its average are $\mathbf{1}$; uniqueness of the minimal orbit in Thm. YM3_I:E identifies its axial representative with $\mathbf{1}$.

(2)

One step. Let O be an open subset of Gd_{j+1}, $0 < \rho \le r_\chi/C_\chi$, $G(\rho) = \{u \in SU(2) : |\operatorname{Lg} u| < \rho\}$, and

$$\mathcal{N} = \{V : \ \mathcal{M}V \in \mathsf{O}, \ \ V(\mathsf{e})\, V_j(\mathcal{M}V)(\mathsf{e})^{-1} \in G(\rho) \text{ at every non-pivot bond } \mathsf{e}\}.$$

At $W' \in \mathsf{O}$ the configuration $V_j(W')$, with $\varpi_c(V_j(W')) \le \varpi_\chi$, is a reference configuration of the chart of the fibre, and Prop. YM3_I:53.2, with empty exterior, gives for $|\mathcal{A}| < \rho$: $V_j(W')$ is real and admissible, with invertible pivot coefficients; the chart is formed at $\mathcal{A}$, $V_{W'}[\mathcal{A}] = e^{iB'} V_j(W')$ with a real B', $\sup|B'| \le 2C_\chi|\mathcal{A}|$; $\mathcal{M}V_{W'}[\mathcal{A}] = W'$; $\operatorname{Lg}(V_{W'}[\mathcal{A}](\mathsf{e})V_j(W')(\mathsf{e})^{-1}) = \mathcal{A}(\mathsf{e})$ at the non-pivot bonds; $\varpi_c(V_{W'}[\mathcal{A}]) < \varpi_1$; and $\mathsf{J}_{V_j}[\mathcal{A}]$, $z_{V_j}^{(j)}$ are formed, real and continuous (the continuity statement and the local bounds in its proof).

By the converse in that proposition, $\Phi_j(W', \mathcal{A}) = V_{W'}[\mathcal{A}]$ is a bijection of $\mathsf{O} \times \{|\mathcal{A}| < \rho\}$ onto the open set $\mathcal{N}$, with the inverse $\Xi_j(V) = (\mathcal{M}V, (\operatorname{Lg}(V(\mathsf{e})V_j(\mathcal{M}V)(\mathsf{e})^{-1}))_{\mathsf{e}})$. The map Ξ_j is real analytic on $\mathcal{N}$: $\mathcal{M}$ is real analytic at configurations whose plaquettes in the double blocks are small, V_j is by (1), and Lg is the logarithmic series where $|u - \mathbf{1}| < 1$. Its differential is injective: if the derivatives of $\mathcal{M}V$ and of the second member of $\Xi_j(V)$ vanish along a tangent vector $\dot a$, then $\dot a = 0$ at the non-pivot bonds, and then the derivative of $(\mathcal{M}V)(c)$ at a bond c of the level $j+1$ is $\mathsf{n}_c\, \dot a(b_0(c))$, with the invertible kernel n_c at the pivot bond $b_0(c)$ (the pivot differential preceding equation YM3_I:(911)), so that $\dot a = 0$.

The dimensions agree, the positively oriented bonds of the level j being the non-pivot ones and one pivot bond for each positively oriented bond of the level $j+1$. By the inverse function theorem Ξ_j is a local real analytic diffeomorphism; being injective, it has the real analytic inverse Φ_j, a real analytic diffeomorphism of $\mathsf{O} \times \{|\mathcal{A}| < \rho\}$ onto $\mathcal{N}$. The integral identity of the chart of the fibre (equation YM3_I:(986), with $e^{\mathsf{z}_c(V)} = L^{-3} z_V^{(j)}(c)$ because $\mathsf{n}_c = L\mathsf{m}_c$ acts on a three-dimensional pivot space) gives, for every Borel function $H \ge 0$ on the configurations of the level j,

$$\int_{\mathcal{N}} H \,\mathrm{d}V = c_j \int_{\mathsf{O}} dW'\, z_{V_j}^{(j)} \int_{|\mathcal{A}|<\rho} e^{\mathsf{J}_{V_j}[\mathcal{A}]}\, H(V_{W'}[\mathcal{A}])\,\mathrm{d}\mathcal{A}, \qquad V_j = V_j(W'), \quad c_j = (2\pi^2)^{-\#\mathfrak{X}_j^+} L^{-3\#\mathfrak{B}_{j+1}^+},$$

and the exponent has the closed form

$$\mathsf{J}_{V_j}[\mathcal{A}] + \log z_{V_j}^{(j)} = \sum_{\mathsf{e}\in\mathfrak{X}_j^+} \log\sigma_1(|\mathcal{A}(\mathsf{e})|^2) + \sum_c \mathsf{z}_c(V_{W'}[\mathcal{A}]) + 3\,\#\mathfrak{B}_{j+1}^+ \log L, \qquad \sigma_1(s^2) = \Big(\frac{\sin s}{s}\Big)^2,$$

where $\mathsf{z}_c = -\log\det \mathsf{n}_c$ is a real analytic function of the configuration on $\{\varpi_c < \varpi_1\}$.

(3)

The step keeps the good sets, with constants alone. Let $W' \in \mathsf{Gd}_{j+1}$, $|\mathcal{A}| < \rho_2$ and $W'' = V_{W'}[\mathcal{A}]$. For $j \geq 1$: $V_j(W') = \mathcal{M}^j U_+$ with an element U_+ of the minimal orbit over W', which lies in the minimal orbit over $V_j(W')$ ([7, Lem. B.4](2)), so that $\mu_j(V_j(W')) = L^{-2}\mu_{j+1}(W')$ (the level scaling in [7, Def. B.2]); and $W'' = e^{iB'}V_j(W')$ with $\sup|B'| \leq 2C_\chi|\mathcal{A}| < \frac{1}{2}r_{\mathrm{B}}$ moves the representative of the background so that its regularity number moves by at most $2K_{\mathrm{B}}\sup|B'|$ ([7, Lem. P.5]): the background over W'' exists and

$$\mu_j(W'') \leq L^{-2}\mu_{j+1}(W') + 4K_{\mathrm{B}}C_\chi|\mathcal{A}| < L^{-2}\mu_\flat + (1-L^{-2})\mu_\flat = \mu_\flat.$$

For $j = 0$: every bond moves by $|e^{iB'(\mathsf{e})} - \mathbf{1}| \leq |B'(\mathsf{e})|$, and a plaquette variable by at most the sum over its four bonds, so that $\varpi(W'') \leq \varpi(V_0(W')) + 8C_\chi|\mathcal{A}| \leq 2L^{-2}\mu_1(W') + 8C_\chi|\mathcal{A}| < \frac{1}{4}\varepsilon_{\mathrm{v}}$. So the step maps $\mathsf{Gd}_{j+1} \times \{|\mathcal{A}| < \rho_2\}$ into Gd_j.

(4)

Composition. Put $\mathcal{U}_n = \mathsf{O}_n$ and, downwards, $\mathcal{U}_j = \Phi_j(\mathcal{U}_{j+1} \times \{|\mathcal{A}_j| < \rho_2\})$. By (2) at $\mathsf{O} = \mathcal{U}_{j+1}$, which lies in Gd_{j+1} by (3), every $\mathcal{U}_j$ is open and lies in Gd_j, and $\Phi_j \times \mathrm{id}$ is a real analytic diffeomorphism. Υ is their composite: a real analytic diffeomorphism of $\mathsf{O}_n \times \Omega$ onto the open set $\mathcal{U} = \mathcal{U}_0$, injective on all of $\mathsf{O}_n \times \Omega$, with the inverse $U \mapsto (\mathcal{M}^n U, (\mathrm{Lg}(\mathcal{M}^j U(\mathsf{e})\, V_j(\mathcal{M}^{j+1}U)(\mathsf{e})^{-1}))_{\mathsf{e},j})$, and $\Upsilon(\mathbf{1}, 0) = \mathbf{1}$ by (1). Along the chain the backgrounds exist (at level 0 the background is $U_0(W_0) = W_0$, and $|J(W_0)| \leq 4\varpi(W_0) < \varepsilon_{\mathrm{v}}$, as in the height-zero part of the proof of Prop. YM3_I: 52.3), $\mathcal{M}W_j = W_{j+1}$ and θ_{ch} is formed; this is (i) and (ii), and by the closed form of (2)

$$\theta_{\mathrm{ch}} = -\sum_{j<n}\Big[\sum_{\mathsf{e}\in\mathfrak{X}_j^+} \log\sigma_1(|\mathcal{A}_j(\mathsf{e})|^2) + \sum_c \mathsf{z}_c(W_j)\Big] - 3\log L \sum_{j<n} \#\mathfrak{B}_{j+1}^+,$$

a real analytic function on $\mathsf{O}_n \times \Omega$. The identities of the steps, applied downwards with Tonelli's theorem, give (iii) for Borel $H \geq 0$ that vanish off $\mathcal{U}$, with $C_{\mathrm{H}} = \prod_j c_j$, and for bounded complex Borel H by linearity. As Υ is injective on all of $\mathsf{O}_n \times \Omega$, a function that vanishes off the image of an open subset E gives an integrand that vanishes off E: this is (iii) for E. The density is fixed by (iii): if a continuous θ' satisfied (iii) as well, the function H equal to the sign of $e^{-\theta'} - e^{-\theta_{\mathrm{ch}}}$, carried to $\mathcal{U}$ by Υ, would give $\int |e^{-\theta'} - e^{-\theta_{\mathrm{ch}}}|\,\mathrm{d}W\,\mathrm{d}y = 0$, and a continuous function with this property vanishes. □

Lemma 8.7 (The jets of the weighted progression)**.** Let $0 < g \leq \gamma_{\mathrm{e}}$, (K, β) and b as in Lem. 8.5, $\ell \in \mathfrak{L}(g)$, and f, h as there, with $d < 1$; $\Psi = (f, h)$. $\mathsf{n}_j(f, h)$ is the mixed second derivative at 0 in (μ_1, μ_2) of the source energy of the level j of the weighted progression with the source $\mu_1 f \circ x + \mu_2 h \circ x$, summed over the localisation domains and points of the level j; $\mathsf{J}_2 = \sum_{j\leq K} \mathsf{n}_j$. When all fluctuation fields are multiplied by a complex field scale z and each x_i is replaced by $L^{-i}\beta + \mathsf{z}^2(x_i - L^{-i}\beta)$, n_{j+1} becomes an even function $\mathsf{n}_{j+1}^{\mathsf{z}} = \sum_m \mathsf{n}_{j+1,[2m]}\mathsf{z}^{2m}$, holomorphic for $|\mathsf{z}| < R_j$ with $R_j^{-2} \leq C(\log x_j)^{2p_0} D_j/b$, and $\mathsf{n}_{j+1}^1 = \mathsf{n}_{j+1}$; put $\mathsf{h}_{2m}^{(n)} = \sum_{j<n} \mathsf{n}_{j+1,[2m]}$ and $\mathsf{h}_{2m} = \mathsf{h}_{2m}^{(K)}$. Then:

(i) $\mathsf{n}_{j,[0]} = \mathsf{n}_{j,[2]} = 0$, and $|\mathsf{J}_2 - \mathsf{h}_4 - \mathsf{h}_6| \leq Cb^{-4}d^2|^{8p_0}|\Psi|$;

(ii) for $0 \leq n \leq K-1$ with $D_n \geq d$: $|\mathsf{h}_6 - \mathsf{h}_6^{(n)}| \leq Cb^{-3}|^{6p_0} d\,(d/D_n)^{3/2}|\Psi|$;

(iii) for $(f, h) \in \mathfrak{T}_{T_\ell}(d; \vartheta)$, $20d \leq \ell$, $d \leq L^{-1}$ and $K > s(d)$: $|\mathsf{h}_4 - c_2 b^{-2}\mathcal{K}_6(f, h)| \leq [\bar{\varepsilon}_4(g, d)\, b^{-3}d + \frac{3}{2}C_{\mathrm{tm}}\vartheta^{-1}b^{-2}a_K/d]|\Psi|$, with $\bar{\varepsilon}_4(g, d) \leq C(g)d^{1/2}$ and the constant C_{tm} of Prop. YM3_I:121.2 and Lem. 6.33;

(iv) for $\ell = \ell_g$, $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$, $d \le d_{\mathrm{e}}(g,\vartheta)$ and $s(d) \le K$: there is a set $\mathsf{Pk} = \mathsf{Pk}(a_K, d) \subset a_K\mathbb{Z}^3$, depending on a_K and d alone, with $|v| < 1$ on it and $\#\mathsf{Pk} \ge (30d)^{-3}$, such that

$$\Big| S^T_{2;K,\ell,\beta}(f,h) - \frac{1}{\#\mathsf{Pk}} \sum_{v\in\mathsf{Pk}} \mathsf{J}_2(\tau_v f, \tau_v h) \Big| \le C_{\mathrm{A}}(g)\, \vartheta^{-1} d^2 b^{-2} |\Psi|,$$

τ_v the translation by v and $C_{\mathrm{A}}(g)$ a number depending only on g and the constants.

■

Proof sketch. (i), (ii). By Lem. 8.5, if $f \circ x$ or $h \circ x$ vanishes identically, all the jets vanish; let neither vanish.

The steps at one field scale. When the fluctuation fields of all the steps are multiplied by one complex field scale z and the coefficients of the action are moved as in the statement, the step of the level $j+1$ is a step at the place x_j with the lower place $\mathsf{t}_\flat$ (Def. 6.28; Prop. 6.30 for the canonical objects), the deformed activities of all the earlier levels standing in its bare array: the room q bounds their size uniformly in z, so that one induction over the levels forms all the steps at once. So the activities of the level $j+1$ are pointed families, holomorphic and even in z on $|\mathsf{z}| < R_j$, $R_j = \delta(\mathsf{t}_\flat)/(2\delta(x_j))$ with $\delta(\mathsf{t}) = \mathsf{t}^{-1/2}\mathfrak{b}(\mathsf{t})$, equal at $\mathsf{z} = 1$ to the activities of the weighted progression, and local in the weights; and $R_j^{-2} = 4\delta(x_j)^2/\delta(\mathsf{t}_\flat)^2 \le C(\log x_j)^{2p_0} D_j/b$, as $x_j \ge \frac{3}{4}b/D_j$ (Lem. 8.5).

One step at the field scale z, as a map of its bare array (the activities of the earlier levels and the weight of the level), has first and second derivatives at most $C\,q(|\mathsf{z}|/R_j)$ times the sizes of the increments, $q(t) = t^2/(1-t^2)$, by Cauchy's estimate, its order 0 in z reading no bare array (Prop. 6.26 with Prop. 6.29); the chain rule gives the derivatives in the two parameters μ_1, μ_2 of the activities of the level $j+1$ from those of the earlier levels and from the weight.

The derivatives in the sources. The weight of the source a at the level i is $L^{K-i}w_a$, $w_1 = f \circ x$, $w_2 = h \circ x$, of size $\|f_a\|_\infty/D_i$ at a plaquette, $f_1 = f$, $f_2 = h$. The weight member of the bare array of the level i is L^i times its sum against the a_p over the plaquettes of a domain, a_p being at most of the order L^{-4i} on the analyticity core; a domain of one cube holds at most of the order L^{3i} plaquettes, and at most $35(d/a_K)^3$ plaquettes carry w_a; as $L^{-3i}(d/a_K)^3 = (d/D_i)^3$, the size of this member is at most $C(\|f_a\|_\infty/D_i)\psi_i$, $\psi_i = \min\{1, (d/D_i)^3\}$. An activity of an earlier level $i-n$ that vanishes off the domains meeting a set of sites enters the bare array of the level i with a size at most $C[\varpi_n + (n+2)L^{-4n}\min\{L^{3n}, N_{\mathrm{cb}}\}]$ times its norm, $\varpi_n = L^{3n}e^{-\max\{2L^{n-1}-2,0\}}$ and N_{cb} the number of the cubes of the level $i-n$ that hold such a site; for the supports of the sources $N_{\mathrm{cb}} \le 77\max\{1, (d/D_{i-n})^3\}$.

Along the run $\delta(x_{i-n}) \le C(1+n)^{p_0}L^{-n/2}\delta(x_i)$, so $R_{i-n}^{-2} \le C(1+n)^{2p_0}L^{-n}R_i^{-2}$. By induction over $i \le j$ one obtains, for $|\mathsf{z}| \le \eta_0 R_j$, $\eta_0 \in (0, \frac{1}{2}]$ a number depending only on the constants: the first derivative in μ_a of the activity of the level $i+1$ has a norm at most $C|\mathsf{z}|^2R_i^{-2}(\|f_a\|_\infty/D_i)\psi_i$, and the mixed second derivative a norm at most $C|\mathsf{z}|^2R_i^{-2}(\|f\|_\infty/D_i)(\|h\|_\infty/D_i)\chi_i$, $\chi_i = \min\{1, (d/D_i)^{5/2}\}$; here $\psi_i^2 \le \chi_i$ bounds the product of two first derivatives, and the second derivatives of the earlier levels, carried up with the sizes above, sum to at most $C\chi_i$, the levels below the scale d through the decay of the activities across the distance of the two supports, those above it through the factor $L^{-n/2}$ that is left when the factor L^{-4n} of the size is set against the growth L^n of the radii and weights and $L^{5n/2}$ of the profile.

The jet. $\mathsf{n}^{\mathsf{z}}_{j+1}$ is the sum, over the domains of the level $j+1$ and their points, of the mixed second derivative at the trivial configuration. Only the domains that meet both supports contribute, the activities being local in the weights; the cubes of the level $j+1$ that hold a corner of a plaquette of a support number at most $77\max\{1, (d/D_{j+1})^3\}$; two corners of plaquettes of the two supports are at a sup distance $\ge d/4$, so that such a domain has a linear size $\ge [d/(4MD_{j+1}) - 2]_+$ in the units of the level $j+1$, and the decay of the localised terms in the size of their domains, summed by the

bound in [7, Lem. B.10], gives the factor $e^{-[d/(4MD_{j+1})-2]_+}$. Hence, on $|\mathsf{z}| \le \eta_0 R_j$,

$$|\mathsf{n}^{\mathsf{z}}_{j+1}| \le C|\mathsf{z}|^2 R_j^{-2} D_j^{-2} \chi_j \, \Xi_{j+1} |\Psi|, \qquad \chi_j = \min\{1, (d/D_j)^{5/2}\}, \qquad \Xi_{j+1} = \max\{1, (d/D_{j+1})^3\} \, e^{-[d/(4MD_{j+1})-2]_+};$$

χ_j records that a source of width d is seen at the spacing D_j through its mean, and Ξ_{j+1} counts the cubes of the level $j+1$ that meet the supports and the decay of the activities across the distance of the two supports; the bound is local, and it reads no K, side, position or pair.

The orders. At $\mathsf{z} = 0$ the steps add nothing that reads the sources, so the order 0 in z vanishes. The order z^2 comes from the step of the level $j+1$ alone, the earlier levels being taken at the field scale 0, where they read no source; it is the covariance, in the measure of the step, of the parts linear in the fluctuation field of the two source potentials at the trivial configuration. These linear forms are invariant under the adjoint action of the constant gauge transformations, which leave the trivial configuration and the gauge invariant terms unchanged, and the mean of $\mathrm{Ad}(u)$ over $u \in SU(2)$ is 0 on $\mathfrak{su}(2)$; so they vanish, and the order 2 vanishes. Cauchy's estimate on $|\mathsf{z}| = \eta_0 R_j$ gives

$$|\mathsf{n}_{j+1,[2m]}| \le C\eta_0^{-2m} R_j^{-2m} \, D_j^{-2} \chi_j \, \Xi_{j+1} |\Psi| \le C b^{-m} (\log x_j)^{2mp_0} D_j^{m-2} \chi_j \, \Xi_{j+1} |\Psi|,$$

and the same with $m = 4$ for the sum of the orders ≥ 8 at $\mathsf{z} = 1$, as $\eta_0^2 R_j^2 \ge 2$ for $g \le \gamma_{\mathrm{e}}$ (geometric summation of the even coefficient bounds in Prop. 6.26(cx8)).

The sums over the levels. For $D_j \ge d$: $\Xi_{j+1} = 1$, $D_j^{m-2}\chi_j = d^{m-2}(d/D_j)^{9/2-m}$ is geometric in j with the ratio $L^{-(9/2-m)} \le L^{-1/2}$ for $m \le 4$, and $\log x_j \le \log(2b/D_j) \le \mathsf{l}$. For $D_j < d$: $D_j^{m-2}\chi_j \le d^{m-2}$, $\log x_j \le \mathsf{l}\,(1 + (r+1)\log L)$ at the r-th such level counted from above, and $\Xi_{j+1} \le \sup_{u \in [L^{r-1}, L^r]} \max\{1, u^3\} e^{-[u/4M-2]_+}$, which is summable against every power of r. Summing over all the levels gives (i), with $|\mathsf{h}_{2m}| \le C b^{-m} d^{m-2} \mathsf{l}^{2mp_0} |\Psi|$ for $m = 2, 3$; summing over the levels $j \ge n$ alone, where $D_j \ge D_n \ge d$, gives $\sum_{j \ge n} |\mathsf{n}_{j+1,[6]}| \le C b^{-3} \mathsf{l}^{6p_0} d \, (d/D_n)^{3/2} |\Psi|$, which is (ii).

(iii). If $f \circ x$ or $h \circ x$ vanishes identically, $\mathsf{h}_4 = 0$ and the template sum below vanishes, so that its bound alone gives the assertion; let neither vanish.

At a real field scale. The coefficients $\xi_i(\mathsf{z}) = L^{-i}\beta + \mathsf{z}^2(x_i - L^{-i}\beta)$ of the action are chosen so that, the counterterms $e_{i'}$ of the progression being kept, $\mathsf{z}^{-2}\xi_i(\mathsf{z}) L^i + \sum_{1 \le i' \le i} L^{i'} e_{i'} = \mathsf{z}^{-2}\beta$, by the run equation $L^i x_i + \sum_{1 \le i' \le i} L^{i'} e_{i'} = \beta$; and the sum rules of the steps are stated for every field of the box. So at a real field scale $0 < \mathsf{z} \le 1$ the chain of the weighted progression is the chain of the chart integrals of the Wilson weight at β/z^2, read at the fields $\mathsf{z}\zeta$, $\zeta = (\zeta_i)_i$ the fluctuation fields of the levels in their boxes $|\zeta_i| < \mathfrak{b}(x_i)$, and $\sum_{2 \le j \le K} \mathsf{n}^{\mathsf{z}}_j$ is, up to a term of the first level, the covariance of the first source derivatives of the exponent of the level 1, taken at the background that the chain builds at the level 1 from these fields, in a complex measure of mass 1 on the product of the boxes, the product over the levels $1 \le i < K$ of normalised measures on the box of the level i, each of which depends on the fields of the higher levels.

As $\mathsf{z} \to 0$ at a fixed lattice, the backgrounds tend to the trivial configuration uniformly on the boxes, all the potentials vanish, and the coefficient of the action of the level i tends to $L^{-i}x_0$, so that this measure tends in total variation to the product $\mathbb{Q}$ of the measures ν_i^{r} below; dividing by z^4,

$$\sum_{2 \le j \le K} \mathsf{n}_{j,[4]} = \langle q_1; q_2 \rangle_{\mathbb{Q}} + \mathsf{e}_1, \qquad q_a = \tfrac{1}{2} a_K^{-1} \sum_p w_a(p) \, |\Theta_p|^2,$$

Θ_p the linear curvature of the plaquette p in the fields ζ_i, $\mathbb{Q} = \bigotimes_{1 \le i < K} \nu_i^{\mathrm{r}}$, ν_i^{r} the centred Gaussian measure of the quadratic part of the step of the level i at the coefficient $L^{-i}x_0$ of the action, restricted to the box of the level i and normalised, and e_1 a term of the first level. With $\mathsf{n}_{1,[4]}$ and the slice of the level 1 below, e_1 costs at most $C\, b^{-2} \mathsf{l}^{4p_0} e^{-c_\flat d/a_K} |\Psi|$, $c_\flat = \min\{1/(8ML), \delta_{\mathrm{tp}}/(2\sqrt{3})\}$, δ_{tp} the rate of Props. YM3_I:120.5 and YM3_I:120.6, as the supports are d/a_K bare spacings apart.

The Gaussian values; the running couplings drop out. Under the Gaussian measure of the level i at the coefficient $L^{-i}x_0$ the curvature of the fluctuation field of the level i has the covariance

$x_0^{-1}\mathcal{C}_{(i+1)}$, $\mathcal{C}_{(i+1)}$ the slice of the level $i+1$ of the free lattice curvature (Prop. YM3_I:120.3): the coupling x_i enters the scaled field and the coefficient of the action in inverse proportions and drops out. The slices $\mathcal{C}_{(i+1)}$, $1 \le i < K$, add up to $C^{\mathrm{fl}}_{K,0} - \mathcal{C}_{(1)}$, where $C^{\mathrm{fl}}_{K,0} = \sum_{1\le j\le K} \mathcal{C}_{(j)}$ is the fluctuation kernel of depth 0 of Section YM3_I:120 (Props. YM3_I:120.5 and YM3_I:120.6), the kernel of the orthogonal projection of the plaquette 2-cochains, in each colour, onto the curvatures of the fields whose K-fold linear block average vanishes, that is, onto $\mathsf{Fl}_K = \mathrm{d}(\ker \mathbf{Q}^s_K)$, $\mathbf{Q}^s_K$ the straight average of Section YM3_I:120; it has no dimension and $|C^{\mathrm{fl}}_{K,0}| \le 1$; the slice $\mathcal{C}_{(1)}$ of the level 1 goes into e_1.

The variables of different levels are independent and every ν^{r}_i is even; by Wick's rule, with three colours, the Gaussian value of $\langle q_1; q_2\rangle$ is, up to that slice,

$$\tfrac{1}{4}a_K^{-2}\cdot 2\cdot 3\, x_0^{-2}\sum_{p,q} w_1(p)\, w_2(q)\, C^{\mathrm{fl}}_{K,0}(p,q)^2 = \tfrac{3}{2}\, b^{-2}\, \mathcal{K}[C^{\mathrm{fl}}_{K,0}](f,h), \qquad b = a_K x_0,$$

with the template sum $\mathcal{K}[\mathsf{k}](f,h) = \sum_{p,q} f(x(p))\, h(x(q))\, \mathsf{k}(p,q)^2$; no further power of a_K enters.

The boxes. Every joint cumulant of order 2 or 4 of the components of the field of the level i under ν^{r}_i differs from the Gaussian one by at most $C_{\mathrm{bx}} e^{-\frac{1}{2}\gamma_2 \mathfrak{b}(x_i)^2}$, uniformly in the volume and in the cutoff (Lem. 4.53, the components entering as insertions), with numbers C_{bx}, γ_2 depending only on the constants; the odd cumulants vanish. The map from the fields of the level i to the curvature has row sums at most CL^{-2i}, and at most $35^2(d/a_K)^6$ pairs of plaquettes have $w_1(p)w_2(q) \ne 0$; with $L^{-3i}(d/a_K)^3 = (d/D_i)^3$ the boxes change $\langle q_1; q_2\rangle$ by at most $Cb^{-2}[\mathsf{E}^{(3)} + (\mathsf{E}^{(3)})^2 + \mathsf{E}^{(6)}]|\Psi|$, $\mathsf{E}^{(q)} = \sum_{1\le i<K} u_i^q \epsilon_i$, $u_i = d/D_i$, $\epsilon_i = e^{-\frac{1}{2}\gamma_2 \mathfrak{b}(x_i)^2} \le \phi_{\mathrm{bx}}(\frac{3}{4}b/D_i)$, where $\phi_{\mathrm{bx}}(s) = e^{-c_{\mathrm{bx}}(\log s)^2}$, $c_{\mathrm{bx}} = \frac{1}{2}\gamma_2 A_1^2$, since $\mathfrak{b}(x_i)^2 \ge A_1^2(\log x_i)^2$ and $x_i \ge \frac{3}{4}b/D_i$ (Lem. 8.5).

For $s, s' \ge 1$, ϕ_{bx} decreases, $\phi_{\mathrm{bx}}(ss') \le \phi_{\mathrm{bx}}(s)\phi_{\mathrm{bx}}(s')$ and $s^q\phi_{\mathrm{bx}}(s) \le e^{q^2/(4c_{\mathrm{bx}})}$. Cut the levels at $D_i \approx d^{1/2}$: the levels with $D_i < d$ give at most $C\phi_{\mathrm{bx}}(\frac{3}{4}b)\phi_{\mathrm{bx}}(d^{-1/2})$, their u_i being spaced by the factor L; those with $d \le D_i \le d^{1/2}$ have $u_i \le 1$, $\sum u_i^3 \le 2$ and $\epsilon_i \le \phi_{\mathrm{bx}}(\frac{3}{4}b)\phi_{\mathrm{bx}}(d^{-1/2})$; those with $D_i > d^{1/2}$ have $\sum u_i^3 \le 2d^{3/2}$ and $\epsilon_i \le \phi_{\mathrm{bx}}(\frac{3}{4}b)$. As $\phi_{\mathrm{bx}}(d^{-1/2}) \le Cd^{3/2}$ and $\phi_{\mathrm{bx}}(\frac{3}{4}b) \le C/b$, $\mathsf{E}^{(3)}, \mathsf{E}^{(6)} \le Cb^{-1}d^{3/2}$, and also $\le C$; so the boxes cost at most $Cb^{-3}d^{3/2}|\Psi|$, with C depending only on the constants.

The bound is uniform in K and in the side through the locality of the cumulant bounds; no count of the restrictions enters.

The template sum. Lem. 6.33 at the depth 0, where $a_0 = 1 \ge d$:

$$\Big|\mathcal{K}[C^{\mathrm{fl}}_{K,0}](f,h) - \frac{3}{8\pi^2}\mathcal{K}_6(f,h)\Big| \le C_{\mathrm{tm}}\Big[\frac{a_K}{\vartheta d} + \Big(\frac{d}{\ell}\Big)^3 + d^3\Big]|\Psi|,$$

and $\frac{3}{2}\cdot\frac{3}{8\pi^2} = c_2$. Collecting,

$$|\mathsf{h}_4 - c_2 b^{-2}\mathcal{K}_6(f,h)| \le b^{-2}\Big\{\tfrac{3}{2}C_{\mathrm{tm}}\Big[\frac{a_K}{\vartheta d} + \Big(\frac{d}{\ell}\Big)^3 + d^3\Big] + C\,\frac{d}{b}\,d^{1/2} + C\,\mathsf{I}^{4p_0}e^{-c_\flat d/a_K}\Big\}|\Psi|.$$

As $\ell \ge 1$, $d/a_K \ge L(\log\frac{1}{d})^2$ for $K > s(d)$, and $g^{-2} < b \le L(g^{-2} + \bar{B})$, the members other than $\frac{3}{2}C_{\mathrm{tm}}b^{-2}a_K/(\vartheta d)$ are at most $\bar{\varepsilon}_4(g,d)\, b^{-3}d\,|\Psi|$ with

$$\bar{\varepsilon}_4(g,d) = \sup_{g^{-2}<b\le L(g^{-2}+\bar{B})}\Big\{3C_{\mathrm{tm}}\, b\, d^2 + C\, d^{1/2} + C\,\frac{b}{d}\,\mathsf{I}^{4p_0}e^{-c_\flat L(\log\frac{1}{d})^2}\Big\} \le C(g)\, d^{1/2},$$

$e^{-c(\log\frac{1}{d})^2}$ being below every power of d.

(iv). Let $Z[w]$ be the generating function of T_ℓ, the partition function with the source w, and $\mathcal{E}(w)$ its source energy, the sum over the levels and localisation domains of the source energies [7, eq. 212] of the sourced chain on T_ℓ [7, eq. 205]. On the least torus the setting and the identity of (4a)–(4c) in the proof of Prop. 8.3 hold, the conditions on g of the statements read there entering γ_{e}: the term without large-field regions in the sum over the level of the first creation of a large-field region (equation eq. 4.xxxi, with the histories of Section YM3_I:123) is $e^{\mathcal{E}(w)}\int e^{\mathsf{R}_K(w,\cdot)}\,\mathsf{dm}_K$ for the

sources of level-wise norm < 1, with $|\mathsf{R}_K(w,\cdot)| \le \|w\|_{\mathrm{lw}}\mathsf{g}_\emptyset$ for a source carried by one ball of radius 1, and with a measure m_K of mass at most $Z[0]$ (steps (4c), (4d) there).

The weighted progression ([7, Def. B.7]) and the sourced chain are made level by level by the same renormalisation step, with the same couplings, the same auxiliary data and the same lattice, and the source enters both only through the weights $x_j - L^{K-j}w$ of the plaquettes. At the trivial configuration every localised term is taken at the trivial pair, and $a_p(\mathbf{1}) = 0$, so that there the relative exponent of the weighted progression after the level K is the sum over the levels and localisation domains of its source energies; following the two constructions level by level, this sum is $\mathcal{E}(w)$ for all sources near 0, with no condition on the depth.

So the mixed second derivative of $\mathcal{E}$ at $(\tau_v f, \tau_v h)$ is $\mathsf{J}_2(\tau_v f, \tau_v h)$, and $\log(Z[w]/Z[0]) - \mathcal{E}(w)$ is holomorphic near 0 with the mixed second derivative $\mathsf{R} = S_2^T - \mathsf{J}_2$. Take $\iota = a_K\lceil 18d/a_K\rceil$ and $\mathsf{Pk} = \{v \in \iota\mathbb{Z}^3 : |v_\mu| \le 0.575{-}9d\}$; the balls of radius $9d$ about the $x_1{+}v$, x_1 a centre of a ball of radius d that holds $\operatorname{supp} f$, are disjoint and lie in $\mathbb{B}(x_1, 1)$. For signs ϵ_v the source $z_1\sum_v \epsilon_v\tau_v f + z_2\sum_v \epsilon_v\tau_v h$ has level-wise norm < 1 for $|z_1|\|f\|_\infty, |z_2|\|h\|_\infty < A_\star$, with $A_\star \ge c(g)(\vartheta d)^{1/2}b$, and on this polydisc the squarefree coefficients of $Z[w]e^{-\mathcal{E}(w)}/Z[0]$ in (z_1, z_2) are at most $Q_{\mathrm{D}}(g)\prod_a \|f_a\|_\infty/A_\star$, with the number $Q_{\mathrm{D}}(g)$ of (4e) in the proof of Prop. 8.3 formed with one ball and two variables: split at the level of the first creation of a large field, bound the part in which no field is large, and apply Cauchy's estimate on a polydisc for each level, as in (4e) there.

So $|\mathsf{R}(\sum_v \epsilon_v\tau_v f, \sum_v \epsilon_v\tau_v h)| \le 2Q_{\mathrm{D}}^2 A_\star^{-2}|\Psi|$; its mean over the signs is $\sum_v \mathsf{R}(\tau_v f, \tau_v h)$; S_2^T is invariant under the translations of $a_K\mathbb{Z}^3$; divide by $\#\mathsf{Pk} \ge (30d)^{-3}$. □

For $N = 2L^{\mathsf{M}}$, $1 \le n < \mathsf{M}$ with $2L^{\mathsf{M}-n+1} \ge C_L M$, and plaquettes p, q of $\mathbb{Z}^3_N$, let $\mathsf{g}_2[N,n](p,q)$ be the coefficient of β^{-3} in the expansion by Laplace's method (Lem. 7.11), as $\beta \to \infty$, of the covariance of a_p and a_q on the fibre $\{\mathcal{M}^n U = \mathbf{1}\}$ of $\mathbb{Z}^3_N$ under $\exp[-\beta(A + \sum_{j<n}\mathsf{k}_j G_j \circ \mathcal{M}^j)]$, written in the chart $\Upsilon(\mathbf{1},\cdot)$ of Lem. 8.6; here $A = \sum_p a_p$ is the Wilson action, G_j the gauge-fixing function YM3_I: (54) of the level j, and $\mathsf{k} \in (0,\infty)^n$ are strengths. This is the number $\mathsf{g}_2[N,n]$ of Prop. 6.13(iii): both definitions take the coefficient of the covariance on the same fibre, and that proposition proves independence from the adapted chart and from the gauge-fixing strengths. In particular, it depends only on the bare torus, the number of blocking levels, L and the plaquettes, with no running coupling or cutoff remaining. $\mathsf{B}_1^{\mathrm{lat}}$ and $\mathsf{P}_1^{\mathrm{lat}}(f,h;a) = a\sum_{\tilde p,\tilde q} f(ax(\tilde p))h(ax(\tilde q))\mathsf{B}_1^{\mathrm{lat}}(\tilde p,\tilde q)$, sums over the plaquettes of $\mathbb{Z}^3$, are the coefficients of Def. 7.10; Cor. 7.37 and Thms. W, X (a) give, for $(f,h) \in \mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$, $0 < a \le d$ and $\lg(\cdot) = 1 + |\log(\cdot)|$,

$$|\mathsf{P}_1^{\mathrm{lat}}(f,h;a) - \varkappa\mathcal{K}_5(f,h)| \le C_\varkappa\vartheta^{-1}a\lg(d/a)\,\|f\|_\infty\|h\|_\infty, \qquad \varkappa = \tfrac{3}{2\pi^3} = c_1c_2, \tag{8.v}$$

with an absolute number $C_\varkappa$.

Lemma 8.8 (Lattice perturbation theory)**.**

(i) In the setting of Lem. 8.7 let $N = \ell L^K$, $w_1 = f\circ x$ and $w_2 = h\circ x$ on $\mathbb{Z}^3_N$, $2 \le n \le K$,

$$\mathsf{H}^{(n)} = \beta^3 a_K^2 \mathsf{h}_6^{(n)}$$

and

$$\mathsf{T}[N,n](w) = \sum_{p,q} w_1(p)w_2(q)\mathsf{g}_2[N,n](p,q).$$

If $\gamma^{\mathrm{tr}}\mathfrak{b}(x_{n-1})^2 \ge 80$ and

$$10240\,N^3 e^{-\gamma^{\mathrm{tr}}\mathfrak{b}(x_{n-1})^2/8} \le 1,$$

$\gamma^{\mathrm{tr}} > 0$ a number depending only on the constants, then

$$|\mathsf{H}^{(n)} - \mathsf{T}[N,n](w)| \le C(N+\beta)^{36}e^{-\gamma^{\mathrm{tr}}\mathfrak{b}(x_{n-1})^2/8}\,\|w_1\|_\infty\|w_2\|_\infty.$$

(ii) Let $N = 2L^{\mathsf{M}}$, $N' = L^s N$ with $s \geq 1$, $1 \leq n < \mathsf{M}$ with $N/L^n \geq \max\{192, 48(3\log L + \log 2)/(\delta_{\mathrm{tp}} L(L-1)), C_L M/L\}$, $\delta_{\mathrm{tp}} > 0$ the rate of Props. YM3_I:120.5 and YM3_I:120.6, and π the reduction $\mathbb{Z}^3_{N'} \to \mathbb{Z}^3_N$. For plaquettes p', q' of $\mathbb{Z}^3_{N'}$ whose base points have a distance $\leq N/8$: $|\mathsf{g}_2[N,n](\pi p', \pi q') - \mathsf{g}_2[N',n](p',q')| \leq C N'^{237} e^{-\delta_{\mathrm{tp}} N/(48L^{n-1})}$.

(iii) Let $\ell' = 2L^{m'} \geq N_{\mathsf{H}}$, the side threshold fixed in Section YM3_I:52.6 and retained for the kernel bounds in Section YM3_I:120.3, $\ell' L \geq C_L M$ and $K \geq 1$. For plaquettes p, q of $\mathbb{Z}^3_{\ell' L^K}$ and plaquettes $\tilde p, \tilde q$ of $\mathbb{Z}^3$ over them with $|x_{\tilde p} - x_{\tilde q}|_\infty \leq \ell' L^K/8$: $|\mathsf{g}_2[\ell' L^K, K](p,q) - \mathsf{B}_1^{\mathrm{lat}}(\tilde p, \tilde q)| \leq 3 C_\star \ell'^{19} K L^{-K/2}$, with a number $C_\star$ independent of the side.

■

Proof sketch. (i) In the chart of Lem. 8.6 at the trivial block field of the level n, an exact identity, made one level at a time along matched chains of fields, shows that at a real field scale z the sum of the jets of the levels $\leq n$ is the covariance of two observables in the probability measure $\propto \exp[-\mathsf{z}^{-2}\mathcal{Q}(\mathsf{z} g\zeta) - \mathcal{Q}'(\mathsf{z} g\zeta)] d\zeta$ on the product of the boxes, $g\zeta = (x_i^{-1/2}\zeta_i)_i$. Here $\mathcal{Q} \geq 0$ has a positive definite Hessian, block diagonal over the levels; $\mathcal{Q}$, $\mathcal{Q}'$ and the observables are real analytic near 0 and contain no z, no counterterm and no box; and, up to colour rotations at each level, which keep the boxes and the Lebesgue measure, they are β times the action with the gauge fixing of the strengths $\mathsf{k}_j = x_j/\beta$, the function θ_{ch}, and $\mathcal{O}(w_a) \circ \Upsilon$.

The chart is read there on the product of the boxes $|\mathcal{A}_i| < x_i^{-1/2}\mathfrak{b}(x_i)$ of the fields $\mathcal{A}_i = x_i^{-1/2}\zeta_i$, and these boxes lie in Ω once $\delta(x_i) = x_i^{-1/2}\mathfrak{b}(x_i) < \rho_2$; as ρ_2 depends only on the constants (it is written out in the proof of Lem. 8.6), and as δ decreases on $[e^{2p_0}, \infty)$ while all the couplings of the run exceed the inverse square of the coupling g of the datum (Lem. 8.5), this is a condition on the coupling g alone, and it enters γ_{e}. So $\mathsf{h}_6^{(n)}$ is the second Laplace coefficient computed with the Gaussian measures restricted to the boxes.

Without the boxes, $\beta^3 a_K^2$ times this coefficient is $\mathsf{T}[N,n](w)$, by the intrinsic fibre-coefficient identity of Prop. 6.13(iii) and because the second Laplace coefficient of $c\,\mathcal{Q}$ scales like c^{-3}. For these matched chains, Prop. 6.11(ii) bounds the change on restricting the Gaussian cumulants to the boxes: the total polynomial order is at most 10, there are at most N^3 places, and its Gaussian precision and normalisation bounds give the stated $(N+\beta)^{36}$ times exponential error. (ii) Apply Prop. 6.16 with $t = n$ and $m' = 2$. Its lower bound on N/L^n is the one in this statement, and $\delta_{\mathrm{tp}} L(N/L^n)/48 = \delta_{\mathrm{tp}} N/(48L^{n-1})$. The coefficients are the same intrinsic fibre coefficients by Prop. 6.13(iii), so its estimate gives the assertion.

(iii) This is the coarser bound already stated in Prop. 6.17, at the side ℓ'. The side threshold, room condition and separation of the lifted plaquettes are exactly those of that proposition. Its detailed proof keeps both the torus-to-lattice kernel comparison and the correction from the flat top. □

Lemma 8.9 (Removal of the cutoff at a fixed label)**.** Let $0 < g \leq \gamma_{\mathrm{e}}$, $\ell \in \mathfrak{L}(g)$ and $\vartheta \in (0,1)$. Let $\mathsf{J}_g = [\gamma^{-2}, L(X_g + \bar B)]$, let $\mathsf{C}' = \mathsf{C}'_2(\mathsf{J}_g)$ be the number of the definition in Cor. YM3_I:117.1 at $n = 2$ on the side ℓ, $\mathsf{p}_2 = 6p_0 + \frac12$ the exponent the exponent defined before Thm. YM3_I:AR at $n = 2$, $c_\diamond = \sup_{k \geq 1} k^{\mathsf{p}_2} L^{-k/20} < \infty$, and

$$\epsilon_\diamond(d) = (X_g + \bar B)^3\, \mathsf{C}'\, \ell^9\, c_\diamond \Big(\frac{40M}{\vartheta}\Big)^2 d^{87}.$$

There are $d_{\mathrm{K}} \in (0, \frac{1}{200}]$ and integers $K_\sharp(d) > s(d)$ with $d^{-1800} < L^{K_\sharp(d)} \leq d^{-1801}$ for $0 < d \leq d_{\mathrm{K}}$, such that the following holds at every label $b \in I(g)$, at which every pair $(K, L^K b)$, $K \in \mathbb{N}$, is a first-passage datum at g, for $0 < d \leq d_{\mathrm{K}}$ and $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$.

(a) For every $K \geq K_\sharp(d)$

$$|S^T_{2;K,\ell,L^K b}(f,h) - S^T_{2;K_\sharp(d),\ell,L^{K_\sharp(d)} b}(f,h)| \leq \epsilon_\diamond(d)\, b^{-3} d\, \|f\|_\infty \|h\|_\infty,$$

and the same with $S_2^{(\ell,b)}(f,h)$ in the place of the first term.

(b) If, for numbers $\epsilon_{\mathrm{h}}, C_{\mathrm{h}} \geq 0$, the cutoff $K_\sharp(d)$ has $|S^T_{2;K_\sharp(d),\ell,L^{K_\sharp(d)}b} - L_b^+|(f,h) \leq [\epsilon_{\mathrm{h}}\, b^{-3}d + C_{\mathrm{h}}\, b^{-2} a_{K_\sharp(d)}/d]\, \|f\|_\infty \|h\|_\infty$, then for every $K \geq K_\sharp(d)$

$$|S^T_{2;K,\ell,L^K b} - L_b^+|(f,h) \leq \big[\epsilon_{\mathrm{h}} + \epsilon_\diamond(d) + C_{\mathrm{h}}(X_g + \bar{B})\, d^{1798}\big]\, b^{-3} d\, \|f\|_\infty \|h\|_\infty,$$

with no term in a_K, and the same with $S_2^{(\ell,b)}$ in the place of $S^T_{2;K,\ell,L^K b}$.

■

Proof sketch. The rate. the constants and thresholds in Thm. YM3_I:AR and Thm. YM3_I:AR, equation YM3_I:(2203) are made in the setting (Th1), (Th2), whose datum is the choice of the constants and whose numbers are formed from it. The interval J_g contains $I(g)$, and it is an interval of labels under (Tor) on the side ℓ (Lem. 2.5). Let $\mathsf{C} = \mathsf{C}_2(\mathsf{J}_g)$, $K_0 = K_0(\ell, 2, \mathsf{J}_g)$ and $K_1(\cdot) = K_1(\ell, 2, \mathsf{J}_g, \cdot)$ be the other numbers of the constants and thresholds in Thm. YM3_I:AR at $n = 2$; so $\mathsf{C}' = \mathsf{C} \sum_{j\geq 0} (1+j)^{\mathsf{p}_2} L^{-j/10}$. For real Lipschitz functions f, h on T_ℓ, neither of them 0, whose supports have the spread $\mathsf{d}_\infty(f,h) \geq d' > 0$ in the sup distance, the uniform cutoff estimate of Thm. YM3_I:AR gives, at every label $b \in \mathsf{J}_g$ and every integer $K \geq K_1(d')$, the inequality YM3_I:(2203) at $n = 2$,

$$|S^T_{2;K+1,\ell,L^{K+1}b}(f,h) - S^T_{2;K,\ell,L^K b}(f,h)| \leq \mathsf{C}\, \ell^9 K^{\mathsf{p}_2} L^{-K/10}\, \|f\|_\sharp \|h\|_\sharp,$$

and its sum over the cutoffs, with $(K+j)^{\mathsf{p}_2} \leq K^{\mathsf{p}_2}(1+j)^{\mathsf{p}_2}$, is the inequality YM3_I:(2207) of Cor. YM3_I:117.1:

$$|S^T_{2;K',\ell,L^{K'}b}(f,h) - S^T_{2;K,\ell,L^K b}(f,h)| \leq \mathsf{C}'\, \ell^9 K^{\mathsf{p}_2} L^{-K/10}\, \|f\|_\sharp \|h\|_\sharp \qquad (K' \geq K \geq K_1(d')).$$

The supports of a pair $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$ are at the Euclidean distance $\geq d$, so at the sup distance $\geq d/\sqrt{3}$, both distances of T_ℓ being minima over the lifts and $|v|_\infty \geq |v|/\sqrt{3}$ on $\mathbb{R}^3$; so $\mathsf{d}_\infty(f,h) \geq d/\sqrt{3}$ (the definition preceding Lem. YM3_I:114.6), and $d' = d/\sqrt{3}$ serves; put $K_\diamond(d) = K_1(d/\sqrt{3})$. Moreover $\|f\|_\sharp \leq 40M\|f\|_\infty/(\vartheta d)$, as $\mathrm{Lip}\, f \leq \|f\|_\infty/(\vartheta d)$ and $40M \geq \vartheta d$, and the same for h.

The threshold. By the two threshold inequalities at the end of the proof of Thm. YM3_I:AR at $n = 2$, $K_1(d')$ is the least integer $k \geq K_0$ such that every integer $K \geq k$ has

$$d' L^u \geq 6M \qquad \text{and} \qquad A_\diamond\, N_{[u]}\, e^{-d' L^u/M} \leq \tfrac{1}{2}\mathsf{C}\, \ell^9 K^{\mathsf{p}_2} L^{-K/10},$$

with the depth $u = \lceil K/1800 \rceil$ of the depth choice preceding Thm. YM3_I:AR at $n = 2$, $N_{[u]} = \ell^3 L^{3u}$, and $A_\diamond = 8e^3 E_0 C_{\mathrm{E},0} \varrho^{-2}$ the factor in front of $N_{[u]}$ in the two threshold inequalities at the end of the proof of Thm. YM3_I:AR, ϱ being the radius of the class of sources in §2.3.1 that stands in the formulas of the constants and thresholds in Thm. YM3_I:AR, not the radius of the choice. The first condition at $K = K_\diamond(d)$, with $u < K/1800 + 1$, gives $L^{K_\diamond(d)} > (6\sqrt{3}M/(Ld))^{1800}$; as $6\sqrt{3}M > L$ (the cube size is $M = L^{m'}$ with $m' \geq 5$), $a_{K_\diamond(d)} < d^{1800}$. Conversely let $\Lambda_\diamond(d) = 4000 + 366 \log(\sqrt{3}M/d)$, and let K be an integer whose number $v = L^u$ has $v \geq (\sqrt{3}M/d)\Lambda_\diamond(d)$.

The first condition holds at $d' = d/\sqrt{3}$, as $\Lambda_\diamond(d) \geq 6$. As $K \leq 1800u$, $L^{-K/10} \geq v^{-180}$; with $K^{\mathsf{p}_2} \geq 1$ and $\ell \geq 1$ the second condition follows from $A_\diamond v^{183} e^{-d'v/M} \leq \frac{1}{2}\mathsf{C}$, that is, from $\varphi_\diamond(v) = d'v/M - 183 \log v \geq \log(2A_\diamond/\mathsf{C})$, and $2A_\diamond < \mathsf{C}$: at $n = 2$, as $\log(L \cdot L(X_g + \bar{B})) \geq 1$ and $\bar{\mathsf{C}}_{\mathsf{M}} \geq \varrho^{-1}$ in the majorant definitions in the proof of Thm. YM3_I:AR, the number C of YM3_I:(2206) is at least $4\,(16\varrho^{-1})^2\, 2^3 (640/\pi)^2 (1 + 8E_0 C_{\mathrm{E},0}) > 2A_\diamond$. So $\varphi_\diamond(v) \geq 0$ suffices. The function $\varphi_\diamond$ increases for $v \geq 183M/d'$, and at $v = (M/d')\Lambda_\diamond(d) = (\sqrt{3}M/d)\Lambda_\diamond(d)$, since $183 \log\lambda < \frac{1}{2}\lambda$ for $\lambda \geq 4000$, $\varphi_\diamond > \frac{1}{2}\Lambda_\diamond(d) - 183 \log(\sqrt{3}M/d) = 2000$; so $\varphi_\diamond(v) > 0$ for all these v.

Every $K \geq 1800 \log_L(\sqrt{3}M\Lambda_\diamond(d)/d)$ has such a v, and $L^{\lceil x \rceil} < L \cdot L^x$. So

$$\Big(\frac{6\sqrt{3}M}{Ld}\Big)^{1800} < L^{K_\diamond(d)} \leq \max\Big\{L^{K_0},\ L\Big(\frac{\sqrt{3}M\,\Lambda_\diamond(d)}{d}\Big)^{1800}\Big\}.$$

(a). For $K \ge K_\diamond(d)$: $K^{\mathsf{p}_2} L^{-K/10} \le c_\diamond L^{-K/20} \le c_\diamond\, a_{K_\diamond(d)}^{1/20} < c_\diamond d^{90}$. So for $b \in I(g) \subset \mathsf{J}_g$, $K' \ge K \ge K_\diamond(d)$ and $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$

$$|S^T_{2;K',\ell,L^{K'}b}(f,h) - S^T_{2;K,\ell,L^K b}(f,h)| \le \mathsf{C}'\ell^9 c_\diamond \Big(\frac{40M}{\vartheta}\Big)^2 d^{88}\,\|f\|_\infty\|h\|_\infty \le \epsilon_\diamond(d)\, b^{-3} d\,\|f\|_\infty\|h\|_\infty,$$

as $d^{88} = b^{-3}d \cdot b^3 d^{87}$ and $b \le X_g + \bar B$ on $I(g)$. Put $K_\sharp(d) = \max\{K_\diamond(d), s(d)+1\}$; then $L^{K_\sharp(d)} \ge L^{K_\diamond(d)} > d^{-1800}$. From $L^{s(d)} d < L C_W(s(d)) = L\mathsf{c}_W(\log L^{s(d)} + \log(X_g + \bar B))^{2p_0}$ one gets $L^{s(d)} \le d^{-2}$, so $L^{s(d)+1} \le d^{-3}$, for small d; and, K_0 being fixed with g and ℓ, $L^{K_0} \le L(\sqrt{3}M\Lambda_\diamond(d)/d)^{1800} \le d^{-1801}$ for small d. Let $d_{\mathrm{K}} \in (0, \frac{1}{200}]$ be a number below which these hold; then $L^{K_\sharp(d)} \le d^{-1801}$ for $0 < d \le d_{\mathrm{K}}$. The last display, with $K \ge K_\sharp(d)$ in the place of K' and $K_\sharp(d)$ in the place of K, is (a) at the finite cutoffs. At the labels of $I(g)$ the limit $S_2^{(\ell,b)}(f,h)$ exists along the full sequence of the cutoffs (Cor. YM3_I:118.4), and a bound that holds at all large K with one right member holds for the limit.

(b). For $b \in I(g)$ the pairs $(K, L^K b)$, $K \in \mathbb{N}$, are the first-passage data at g with the label b (Cor. YM3_I:118.4). The main term $L_b^+(f,h)$ reads the label and the pair, not the cutoff. By the triangle inequality, (a) and the inequality assumed at $K_\sharp(d)$, the left member at $K \ge K_\sharp(d)$, and at the limit, is at most $[\epsilon_{\mathrm{h}} b^{-3}d + \epsilon_\diamond(d) b^{-3}d + C_{\mathrm{h}} b^{-2} a_{K_\sharp(d)}/d]\|f\|_\infty\|h\|_\infty$, and $b^{-2}a_{K_\sharp(d)}/d = b^{-3}d \cdot b\, a_{K_\sharp(d)} d^{-2} < b^{-3}d \cdot (X_g + \bar B)\, d^{1798}$, as $a_{K_\sharp(d)} \le a_{K_\diamond(d)} < d^{1800}$. □

Proposition 8.10 (The law on the least torus)**.** Let $0 < g \le \gamma_{\mathrm{e}}$, $\vartheta \in (0,1)$, $\ell = \ell_g$ and $C_{\mathrm{cut}} = \frac{3}{2} C_{\mathrm{tm}} \vartheta^{-1}$. There is a non-decreasing function $\bar\varepsilon = \bar\varepsilon_{g,\vartheta}$ on $(0, d_{\mathrm{e}}(g,\vartheta)]$, $\bar\varepsilon(d) \to 0$, depending on g, ϑ and the constants and on no K, β, label, pair or position, such that for $0 < d \le d_{\mathrm{e}}(g,\vartheta)$ and $\Psi = (f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$:

(a) at every first-passage datum (K,β) at g with $s(d) < K$ and $L^K \le d^{-1801}$, whatever its label $b = a_K\beta$,
$$|S^T_{2;K,\ell,\beta} - L_b^+|(f,h) \le [\bar\varepsilon(d)\, b^{-3}d + C_{\mathrm{cut}}\, b^{-2} a_K/d]|\Psi|;$$

(b) for $b \in I(g)$ the same holds at $(K, L^K b)$ for every $K > s(d)$, and $|S_2^{(\ell,b)} - L_b^+|(f,h) \le \bar\varepsilon_\infty(d)\, b^{-3} d\,|\Psi|$ with a non-decreasing $\bar\varepsilon_\infty = \bar\varepsilon_{\infty;g,\vartheta} \le \bar\varepsilon$ and $\bar\varepsilon_\infty(d) \le C(g,\vartheta) d^{1/2}$;

(c) with $C'_{\mathrm{cut}} = \frac{9}{2} C_{\mathrm{tm}} \vartheta^{-1} + 1$ in the place of C_{cut}, the tolerance has the rate $d^{1/2}$ at every cutoff: there is $C_{\mathrm{r}}(g,\vartheta) < \infty$ with $|S^T_{2;K,\ell,L^K b} - L_b^+|(f,h) \le [C_{\mathrm{r}}\, b^{-3} d^{3/2} + C'_{\mathrm{cut}}\, b^{-2} a_K/d]|\Psi|$ for all $b \in I(g)$ and $K > s(d)$, and $|S_2^{(\ell,b)} - L_b^+|(f,h) \le C_{\mathrm{r}}\, b^{-3} d^{3/2} |\Psi|$.

■

Proof sketch. Before the chain. If $f \circ x$ or $h \circ x$ vanishes identically, then $S^T_{2;K,\ell,\beta}(f,h)$, all the jets and every lattice sum of the sources vanish (Lem. 8.5), and Lem. 8.7 (iii) holds by the template sum alone; in the chain below every arrow but the last then reads $0 = 0$, and the last is eq. 8.v as it stands, with $\mathsf{P}_1^{\mathrm{lat}} = 0$. Let $\mathsf{r} = d/a_K$, so that $\mathsf{r} \ge L(\log \frac{1}{d})^2$. Translations keep the class and the $\mathcal{K}_q$, so Lem. 8.7 (iv), (i), (iii) at the pairs $(\tau_v f, \tau_v h)$ give

$$|S^T_{2;K,\ell,\beta} - L_b^+|(f,h) \le \Big[C(g,\vartheta)(d^{1/2} + d(\log\tfrac{1}{d})^{8p_0}) b^{-3} d + C_{\mathrm{cut}}\, b^{-2} \frac{a_K}{d}\Big]|\Psi| + \max_{v \in \mathsf{Pk}} |\mathsf{h}_6 - \varkappa b^{-3} \mathcal{K}_5|(\tau_v f, \tau_v h),$$

the packing mean costing $C_{\mathrm{A}}(g)\vartheta^{-1} b d$ and the orders ≥ 8 costing $C b^{-1} d\, \mathsf{I}^{8p_0}$ in the unit $b^{-3}d|\Psi|$. As $\mathsf{h}_6 = b^{-3} a_K \mathsf{H}^{(K)}$ and $b^{-3} a_K \mathsf{r} = b^{-3} d$, it remains to show $|\mathsf{H}^{(K)} - \varkappa \mathcal{K}_5/a_K| \le \eta(d)\,\mathsf{r}\,|\Psi|$ with $\eta(d) \to 0$ uniformly; all the costs below are in the unit $\mathsf{r}|\Psi|$.

The levels. Let $K' = \max\{K, \lceil 24 \log_L \mathsf{r} \rceil\}$, so that $L^{K'} \ge \mathsf{r}^{24}$ and $K' \le C \log \frac{1}{d}$; let $n_1 = K' - \lceil K'/78 \rceil$, $N_0 = \ell L^K$ and $N' = \ell L^{K'}$. If $K' > K$, let n_0 be the level with $D_{n_0} \in [d^{1/2}, L d^{1/2})$, $b_* \in I(g)$ fixed, $D' = (K', L^{K'} b_*)$, a first-passage datum at g, and $\breve w_a$ the sources lifted to one copy

in $\mathbb{Z}^3_{N'}$ of a ball of radius $N_0/4$ that holds the supports; by Lem. 8.5 the lifted sources are a pair at the datum D', with no condition on its depth. The chain is

$$\mathsf{H}^{(K)} \to \mathsf{H}^{(n_0)} \to \mathsf{T}[N_0, n_0] \to \mathsf{T}[N', n_0] \to \mathsf{H}^{(n_0)}[D'] \to \mathsf{H}^{(n_1)}[D'] \to \mathsf{T}[N', n_1] \to \mathsf{T}^{\mathrm{lat}} \to \varkappa \mathcal{K}_5/a_K,$$

with $\mathsf{T}^{\mathrm{lat}} = \sum \tilde{w}_1(\tilde{p})\tilde{w}_2(\tilde{q})\mathsf{B}_1^{\mathrm{lat}}(\tilde{p},\tilde{q})$ for the sources of the lifts $\tilde{f}, \tilde{h}$ to $\mathbb{R}^3$. By Lem. 8.7 (ii) at $n = n_0$, the first arrow costs at most $C\mathsf{I}^{6p_0}(d/D_{n_0})^{3/2} \le C\mathsf{I}^{6p_0}d^{3/4}$. By Lem. 8.8 (i), with $x_{n_0-1} \ge \frac{3}{4}b/D_{n_0-1} \ge \frac{3}{4}d^{-1/2}$ (Lem. 8.5) and $N_0 + \beta \le (\ell + X_g + \bar{B})d^{-1801}$, as $b \le L(g^{-2} + \bar{B}) = X_g + \bar{B}$, the second costs at most $e^{-c(\log \frac{1}{d})^{2p_0}}$ times a power of $1/d$. By Lem. 8.8 (ii), applied pair by pair to the at most $900\mathsf{r}^6$ pairs of plaquettes of the supports, with $N_0/L^{n_0} = \ell/D_{n_0} \ge \ell d^{-1/2}/L$, the third costs at most $900\mathsf{r}^5 C N'^{237} e^{-\delta_{\mathrm{tp}}\ell d^{-1/2}/48}$.

By Lem. 8.8 (i) at (D', n_0), the number $\mathsf{T}[N', n_0]$ being free of β, and with $x'_{n_0-1} \ge \frac{3}{4}\mathsf{r}^{11}$, the fourth costs less than every power of d. By Lem. 8.7 (ii) at D' the fifth costs at most $C\mathsf{I}'^{6p_0}d^{3/4}$, $\mathsf{I}' \le \log(2X_g + 2\bar{B}) + K' \log L$. By Lem. 8.8 (i) at (D', n_1), with $x'_{n_1-1} \ge L^{K'/78}$, the sixth costs less than every power of d. By Lem. 8.8 (iii) on the torus of the side $N'/L^{n_1} = \ell L^{\lceil K'/78 \rceil}$ with n_1 levels, over at most $900\mathsf{r}^6$ pairs, the seventh costs at most $C\ell^{19}K'\mathsf{r}^5 L^{-K'/4} \le CK'/\mathsf{r}$, since $\mathsf{r}^{24} \le L^{K'}$.

By eq. 8.v at $a = a_K$, as $a_K\mathsf{T}^{\mathrm{lat}} = \mathsf{P}_1^{\mathrm{lat}}(\tilde{f}, \tilde{h}; a_K)$, the eighth costs at most $C_\varkappa \vartheta^{-1} \lg(\mathsf{r})/\mathsf{r}$. If $K' = K$, that is $\mathsf{r}^{24} \le L^K$, there is no second datum and no second torus: the chain is $\mathsf{H}^{(K)} \to \mathsf{H}^{(n_1)} \to \mathsf{T}[N_0, n_1] \to \mathsf{T}^{\mathrm{lat}} \to \varkappa\mathcal{K}_5/a_K$, its first arrow by Lem. 8.7 (ii) at $n = n_1$, with the cost $C\mathsf{I}^{6p_0}(\mathsf{r}/L^{n_1})^{3/2}$, $\mathsf{r}/L^{n_1} \le L^{1-0.94K}$ and $\mathsf{I} \le \log(2X_g + 2\bar{B}) + K \log L$, and the other three as the last three above with $D' = D$ and $N' = N_0$.

The conditions and the limits. At the three places where Lem. 8.8 (i) is used its two conditions hold for $d \le d_{\mathrm{e}}$, because $\frac{1}{8}\gamma^{\mathrm{tr}}\mathfrak{b}(X)^2 = \frac{1}{8}\gamma^{\mathrm{tr}}A_1^2(\log X)^{2p_0}$ with $2p_0 \ge 2$ outgrows $10 + \log(10240N^3)$, the logarithm of N being at most a constant plus a fixed multiple of $\log X$ at the lower bounds X of the three places. Every cost tends to 0 with d, uniformly ($\mathsf{r} \to \infty$, $K' \le C \log \frac{1}{d}$), and depends only on g, ϑ and the constants; $\bar{\varepsilon}(d)$ is the supremum over $d'' \le d$ of their sum. This is (a).

Removal of the cutoff. For $b \in I(g)$ the pairs $(K, L^K b)$ are first-passage data at g, and for $s(d) < K \le K_\sharp(d)$, (a) applies, as $L^K \le d^{-1801}$. At $K_\sharp(d)$ the inequality of (a) is the one that Lem. 8.9 (b) asks there, with $\epsilon_{\mathrm{h}} = \bar{\varepsilon}(d)$ and $C_{\mathrm{h}} = C_{\mathrm{cut}}$; so for $K \ge K_\sharp(d)$ and for the limit the inequality holds with the tolerance $\bar{\varepsilon}(d) + \epsilon_\diamond(d) + C_{\mathrm{cut}}(X_g + \bar{B})d^{1798}$ and without the second member, and $\bar{\varepsilon} + \sup_{d'' \le d}[\epsilon_\diamond(d'') + C_{\mathrm{cut}}(X_g + \bar{B})d''^{1798}]$ serves in (a) and (b). At $K_\sharp$ one has $\mathsf{r} \ge d^{-3}$, so the second, third, fourth and sixth costs are below every power of d, K'/r and $\lg(\mathsf{r})/\mathsf{r}$ are at most Cd^2, the packing mean and the orders ≥ 8 give $Cd(\log \frac{1}{d})^{8p_0}$, the order six $C\mathsf{I}^{6p_0}d^{3/4}$, and the slowest member is the order four, $\bar{\varepsilon}_4(g, d) \le C(g)d^{1/2}$: this is the rate of $\bar{\varepsilon}_\infty$.

The rate at every cutoff. Run the same chain with the cut at $D_{n_0} \in [d^{1/5}, Ld^{1/5})$; the arithmetic of the levels goes through as before for $d \le L^{-13}$. The first and fifth arrows now cost at most $C\mathsf{I}^{6p_0}d^{6/5}$ and $C\mathsf{I}'^{6p_0}d^{6/5}$, the third $e^{-c\ell d^{-1/5}}$ times a power of $1/d$, and the second, fourth and sixth stay below every power of d. The seventh and eighth members, which near the smallest r decay only like a power of $\log \frac{1}{d}$, contribute at most $C(K' + \lg \mathsf{r})\mathsf{r}^{-1}\, b^{-3}d|\Psi| = C(K' + \lg \mathsf{r})(d/b)\, b^{-2}(a_K/d)|\Psi| \le \eta(d)\, b^{-2}(a_K/d)|\Psi|$, with $\eta(d) = Cg^2 d \log \frac{1}{d} \le 1$ for $d \le d_{\mathrm{e}}$; they go into the second member, whose constant becomes at most $C_{\mathrm{cut}} + 1 \le C'_{\mathrm{cut}}$.

All the remaining members are $O(d\,(\log \frac{1}{d})^{8p_0})$ but the order four, which is at most $C(g)d^{1/2}$. Beyond $K_\sharp(d)$, Lem. 8.9 (b), with $C_{\mathrm{h}} = C'_{\mathrm{cut}}$, removes the second member at the price $\epsilon_\diamond(d) + C'_{\mathrm{cut}}(X_g + \bar{B})d^{1798}$, at most a constant times $d^{1/2}$. This is (c). □

The coefficient $\frac{8}{3\pi}$ comes from $\varkappa = c_1 c_2$ in eq. 8.v alone; in particular it does not depend on L, g or the label.

8.1.9. *Transfer to continuum components.*

Lemma 8.11 (From a limit at a fixed label to the components)**.** Let $\mathfrak{c}$ be a choice, $0 < g \le \gamma_W$, S a volume-first limit at g, $b = \mathsf{b}(S)$, S the function S_2 or a function S_2^ξ of a canonical decomposition of σ_S, $\vartheta \in (0,1)$, $z \in \mathbb{Z}$, and $\tau' = (g', b', \ell')$ an admissible triple with $b' = L^z b$. For $L^z d \le d_\flat(g', \vartheta)$ and $\Psi \in \mathfrak{T}^1(d;\vartheta)$,

$$|b^2 \mathsf{S}(\Psi) - b'^2 T^{(\ell',b')}(\Psi_{L^z})| \le E_{\tau'}\,(d/b)^2 |\Psi|.$$

■

Proof sketch. Let (K_j, β_j, ν_j) define S; then $a_{K_j}\beta_j \to b$. S is the limit of $S_2^T[\rho_j]$ for periodic Gibbs states ρ_j at (K_j, β_j), the class of Prop. 8.3, which is $\mathcal{G}^{\mathrm{per}}_{K_j,\beta_j} = \mathcal{G}^{\mathrm{inv}}_{K_j,\beta_j}$ and holds the states of the tori: $\rho_j = \nu_j$ for S_2 ([7, §K.1]), and for S_2^ξ the conditioned states of Thms. YM3_I:AL and YM3_I:AP. By Lem. 8.13 (a) with $s = L^z$ the dilates $\rho_j^{(z)}$ are such states at $(K_j - z, \beta_j)$, and $S_2^T[\rho_j](\Phi) = L^{2z} S_2^T[\rho_j^{(z)}](\Phi_{L^z})$. As $a_{K_j - z}\beta_j = L^z a_{K_j}\beta_j \to b' \in I^\circ(g')$, the pairs $(K_j - z, \beta_j)$ are first-passage data at g' for large j (Lem. 8.2 (b)). Prop. 8.3 there, the $\rho_j^{(z)}$ being states at the datum, at the scale $L^z d \le \frac{1}{180} < \ell'/20$ and with $a_{K_j - z} \le 9 L^z d$ for large j, gives

$$|S_2^T[\rho_j^{(z)}](\Psi_{L^z}) - S^T_{2;K_j - z, \ell', \beta_j}((\Psi_{L^z})^{\langle \ell' \rangle})| \le 2 C_2^{\mathrm{T}}(g') \vartheta^{-1} (L^z d)^2 (a_{K_j - z}\beta_j)^{-2} |\Psi|.$$

Let $j \to \infty$, using the convergence along converging labels (Thm. YM3_I:AS), and multiply by $b'^2 = L^{2z} b^2$; then $2 C_2^{\mathrm{T}}(g') \vartheta^{-1} L^{2z} d^2 = E_{\tau'}(d/b)^2$. □

Lemma 8.12 (A second term excludes covariance)**.** Let a functional σ on separated tuples over $\mathbb{R}^3$ obey (L) with $(b, c, \varepsilon, d_7, \vartheta)$, $c \ne 0$; let $d_1 > 0$, $\Psi \in \mathfrak{T}^1(d_1;\vartheta)$, $\mathsf{B} = c\, c_2 b^{-3} \mathcal{K}_5(\Psi)$, $\mathsf{u} = b^{-3} d_1 |\Psi|$ and $t > 1$. Then

$$|S_2^\sigma(\Psi_r) - c_2 b^{-2} \mathcal{K}_6(\Psi) - \mathsf{B} r| \le \varepsilon(r d_1)\, \mathsf{u}\, r \quad (r d_1 \le d_7), \qquad |\mathcal{N}_t^{S_2^\sigma}(\Psi; r) + t(t-1)(t^5 - 1)\mathsf{B} r| \le (t^2 + t + t^6 + t^7)\, \varepsilon(t^2 r d_1)\, \mathsf{u}\, r \quad (t^2 r d_1 \le d_7);$$

$\mathcal{N}_t^{S_2^\sigma}(\Psi; r) \ne 0$, with the sign of $-c$, as soon as $(t^2 + t + t^6 + t^7)\varepsilon(t^2 r d_1) < t(t-1)(t^5 - 1) \cdot 10^{-5} |c| c_2 \vartheta^2$; and S_2^σ is not dilation covariant. ■

Proof sketch. (L) at $\Psi_r \in \mathfrak{T}^1(r d_1; \vartheta)$, with $|\Psi_r| = |\Psi|$, $\mathcal{K}_6(\Psi_r) = \mathcal{K}_6(\Psi)$ and $\mathcal{K}_5(\Psi_r) = r \mathcal{K}_5(\Psi)$, gives the first bound. In $\mathcal{N}_t$ the weights $1, -(1 + t^6), t^6$ at $t^2 r, t r, r$ annul the constant $c_2 b^{-2} \mathcal{K}_6(\Psi)$, give $\mathsf{B} r$ the factor $t^2 - (1 + t^6) t + t^6 = -t(t-1)(t^5 - 1)$, and give the remainders a sum at most $(t^2 + t + t^6 + t^7)\varepsilon(t^2 r d_1)\mathsf{u} r$, ε being non-decreasing; at $t = 2$ these are -62 and 198. By eq. 8.i, $|\mathsf{B}| \ge 10^{-5} |c| c_2 \vartheta^2 \mathsf{u}$, which gives the sign. If $S_2^\sigma(\Phi_r) = r^\varpi S_2^\sigma(\Phi)$ for all $r > 0$ and all separated Φ, take $\Phi = \Psi$: by the first bound the left side tends to $c_2 b^{-2} \mathcal{K}_6(\Psi) \in (0, \infty)$ as $r \to 0$, so $S_2^\sigma(\Psi) \ne 0$ and r^ϖ has a finite non-zero limit, whence $\varpi = 0$; then $S_2^\sigma(\Psi_r) = S_2^\sigma(\Psi) = c_2 b^{-2} \mathcal{K}_6(\Psi)$ for all r, and the first bound gives $|\mathsf{B}| \le \varepsilon(r d_1)\mathsf{u} \to 0$, against $|\mathsf{B}| \ge 10^{-5} |c| c_2 \vartheta^2 \mathsf{u} > 0$. □

8.1.10. *An alternative transfer using two blocking factors.*

Lemma 8.13 (Three facts on the lattices)**.**

(a) For $a, s, \beta > 0$ the map $x \mapsto s x$ carries the Gibbs states of the Wilson action at β on $a\mathbb{Z}^3$, and the periodic ones, onto those on $sa\mathbb{Z}^3$. For the image $D_s\nu$ of ν, $S_n^T[\nu](f_1, \dots, f_n) = s^n S_n^T[D_s\nu](f_1(\cdot/s), \dots, f_n(\cdot/s))$; so $(sa\beta)^2 S_2^T[D_s\nu](\Psi_{sr}) = (a\beta)^2 S_2^T[\nu](\Psi_r)$.

(b) For $\ell = 2L^m$, a compact set $J' \subset (0, \infty)$ and $d_* > 0$ there is C with $|S^T_{2;K,\ell,L^K b}(u, w)| \le C \|u\|_\sharp \|w\|_\sharp$ for all K, all $b \in J'$ and all real Lipschitz u, w on T_ℓ at a sup-distance $\ge d_*$. Hence, if $K_j \to \infty$, $a_{K_j}\beta_j \to b > 0$ and $r_j \to r > 0$, then $S^T_{2;K_j,\ell,\beta_j}((\Psi_{r_j})^{\langle \ell \rangle}) \to T^{(\ell,b)}(\Psi_r)$ for every separated pair Ψ whose dilate Ψ_r has both supports in one ball of radius $< \ell/16$; for $\Psi \in \mathfrak{T}^1(d_1;\vartheta)$ it suffices that $80 r d_1 < \ell$.

(c) If $\log L_1 / \log L_2 \notin \mathbb{Q}$ and $\sigma_0 > 0$, there are integers $K_j, K_j' \to \infty$ with $L_1^{K_j} L_2^{-K_j'} \to \sigma_0$.

■

Proof sketch. (a) The action and the Haar measure depend on the lattice as a graph, and $a^{-1}\sum_p f(x(p))a_p = s\,(sa)^{-1}\sum_p f(sx(p)/s)\,a_p$, the plaquette p of $a\mathbb{Z}^3$ going to the plaquette sp of $sa\mathbb{Z}^3$; cumulants are multilinear. (b) The bound is Thm. YM3_I:AS and its uniform bound YM3_I:(2214). For C^1 entries with compact supports $\|f(\cdot/r_j) - f(\cdot/r)\|_\sharp \to 0$, and the dilated supports stay at a distance bounded below; so, by bilinearity, the pair may be fixed at Ψ_r, and the convergence along converging labels, Thm. YM3_I:AS, applies. A pair of $\mathfrak{T}^1(d_1;\vartheta)$ dilated by r lies in two balls of radius rd_1 whose centres are at most $8rd_1$ apart, so in one ball of radius $5rd_1$. (c) The multiples $K\log L_1$ are dense modulo $\log L_2$ (Kronecker's theorem). □

Lemma 8.14 (A second choice)**.** Let $\varrho > 0$, values of the constants of $\mathcal{T}^{\mathrm{S}}[2\varrho]$ with the blocking factor L, and an odd integer $Q \geq 1$ be given. There is j_0 such that for every $j \geq j_0$ there is an admissible assignment of $\mathcal{T}^{\mathrm{S}}[2\varrho]$ in which the constants before L keep their values and the blocking factor is $\tilde{L} = (LQ)^j + 2$. $\tilde{L}$ is odd and prime to L and to Q; $\log L/\log\tilde{L} \notin \mathbb{Q}$, and $\log Q/\log\tilde{L} \notin \mathbb{Q}$ if $Q > 1$. With the radius and the mollifier of a choice $\mathfrak{c}$ it is a *second choice* $\tilde{\mathfrak{c}}$ of $\mathfrak{c}$; at $Q = L$, $\tilde{L} = L^{2j} + 2$. ■

Proof sketch. By the ordered choices in the proofs of Prop. YM3_I:69.1 and Lem. YM3_I:82.1, for an admissible assignment σ_0 of the constants before L there is an odd integer $L_\uparrow(\sigma_0)$ such that (σ_0, L') is admissible for every odd $L' \geq L_\uparrow(\sigma_0)$: the conditions attached to the blocking factor are oddness and floors formed with the earlier constants; and every admissible assignment up to a place of the order of the constants is the truncation of a full admissible assignment. For large j, $\tilde{L} \geq L_\uparrow$. $\tilde{L}$ is odd; a prime that divides $\tilde{L}$ and L (or Q) divides 2, which is impossible. If $\log L/\log\tilde{L} = p/q$ with integers $p, q \geq 1$, then $L^q = \tilde{L}^p$ with coprime $L, \tilde{L} > 1$, which is impossible; the same for $Q > 1$. □

Lemma 8.15 (Two blocking factors)**.** Let $\mathfrak{c}_1, \mathfrak{c}_2$ be choices with $\log L_1/\log L_2 \notin \mathbb{Q}$, $\tau_i = (g_i, b_i, \ell_i)$ an admissible triple of $\mathfrak{c}_i$ (for $i = 1$, $b_1 \in I(g_1)$ suffices), $\vartheta \in (0,1)$, $d_1 > 0$, $\Psi \in \mathfrak{T}^1(d_1;\vartheta)$, and $\delta > 0$ with $\delta b_1 d_1 \leq d_\flat^{(1)}(g_1,\vartheta)$ and $\delta b_2 d_1 < d_\flat^{(2)}(g_2,\vartheta)$, the thresholds of Prop. 8.3 at $\mathfrak{c}_1$ and $\mathfrak{c}_2$. With T, E and Y formed at the respective choice,

$$|Y_{\tau_1}(\delta) - Y_{\tau_2}(\delta)| \leq (E_{\tau_1} + E_{\tau_2})\, d_1^2 |\Psi|\, \delta^2.$$

■

Proof sketch. Lem. 8.13 (c) gives $K_j, K_j' \to \infty$ with $s_j = L_2^{K_j'} L_1^{-K_j} \to b_1/b_2$. Put $\beta_j = L_1^{K_j} b_1$: (K_j, β_j) is a first-passage datum of $\mathfrak{c}_1$ at g_1 with the label b_1, and $b_{2,j} = L_2^{-K_j'}\beta_j = b_1/s_j \to b_2 \in I^\circ(g_2)$, so that (K_j', β_j) is a first-passage datum of $\mathfrak{c}_2$ at g_2 for large j (Lem. 8.2 (b) at $\mathfrak{c}_2$). Both data have the bare coupling β_j on the lattice $\mathbb{Z}^3$ in lattice units. Let ρ_j be a periodic Gibbs state at β_j there, and $\rho_j^{(1)}$, $\rho_j^{(2)}$ its images on $L_1^{-K_j}\mathbb{Z}^3$ and $L_2^{-K_j'}\mathbb{Z}^3$; then $\rho_j^{(1)} = D_{s_j}\rho_j^{(2)}$, and by Lem. 8.13 (a)

$$b_1^2 S_2^T[\rho_j^{(1)}](\Psi_{\delta b_1}) = b_{2,j}^2 S_2^T[\rho_j^{(2)}](\Psi_{\delta b_{2,j}}).$$

Prop. 8.3 at $\mathfrak{c}_1$ and at $\mathfrak{c}_2$ compares $\rho_j^{(1)}$ and $\rho_j^{(2)}$ with the Wilson measures on T_{ℓ_1} and T_{ℓ_2}; after multiplication by b_1^2 and $b_{2,j}^2$ the errors are $E_{\tau_1} d_1^2|\Psi|\delta^2$ and, in the limit, $E_{\tau_2} d_1^2|\Psi|\delta^2$. As $j \to \infty$ the torus terms tend to $Y_{\tau_1}(\delta)$, by the limit at a fixed label along the full sequence, and to $Y_{\tau_2}(\delta)$, by Lem. 8.13 (b), since $80\,\delta b_2 d_1 < 1 < \ell_2$. □

Corollary 8.16 (Two labels; all real dilations)**.** Let $\mathfrak{c}$ be a choice, $\tilde{\mathfrak{c}}$ a second choice of $\mathfrak{c}$ (Lem. 8.14 with $Q = L$) with an admissible triple $\tau_2 = (g_2, b_2, \ell_2)$ and the threshold $\tilde{d}_\flat$, and $\vartheta \in (0,1)$.

(i) For admissible triples $\tau = (g, b, \ell)$ and $\tau' = (g', b', \ell')$ of $\mathfrak{c}$, $d_1 > 0$, $\Psi \in \mathfrak{T}^1(d_1; \vartheta)$ and $\delta > 0$ with $\delta b d_1 \le d_\flat(g, \vartheta)$, $\delta b' d_1 \le d_\flat(g', \vartheta)$ and $\delta b_2 d_1 < \tilde{d}_\flat(g_2, \vartheta)$,

$$|Y_\tau(\delta) - Y_{\tau'}(\delta)| \le (E_\tau + E_{\tau'} + 2E_{\tau_2})\, d_1^2 |\Psi|\, \delta^2.$$

Equivalently, with $s = b'/b$, for $\Phi \in \mathfrak{T}^1(d; \vartheta)$ with $d \le d_\flat(g, \vartheta)$, $sd \le d_\flat(g', \vartheta)$ and $(b_2/b)d < \tilde{d}_\flat(g_2, \vartheta)$: $|b'^2 T^{(\ell', b')}(\Phi_s) - b^2 T^{(\ell, b)}(\Phi)| \le (E_\tau + E_{\tau'} + 2E_{\tau_2})(d/b)^2 |\Phi|$. So, up to errors $O((d/b)^2)$, the limits on the tori at the labels b and b' are carried into each other by the real dilation by b'/b; for b'/b a power of L and $\ell'/\ell = b'/b$ this holds exactly, by Thm. Z (d).

(ii) If at one admissible triple $\tau_* = (g_*, b_*, \ell_*)$ of $\mathfrak{c}$ a function $T_* = T^{(\ell_*, b_*)}$ satisfies $|T_*(\Phi) - c_2 b_*^{-2}[\mathcal{K}_6 + c b_*^{-1} \mathcal{K}_5](\Phi)| \le \varepsilon(d) b_*^{-3} d |\Phi|$ for $0 < d \le d_7$ and $\Phi \in \mathfrak{T}^1(d; \vartheta)$, with a real c and a non-decreasing $\varepsilon \to 0$, then at every admissible triple $\tau = (g, b, \ell)$ of $\mathfrak{c}$ the same inequality holds with the same c, with b in the place of b_* and with the tolerance $\varepsilon(b_* d/b) + (E_\tau + E_{\tau_*} + 2E_{\tau_2}) b^{-1} d$, for d below the least of $(b/b_*)d_7$, $(b/b_*) d_\flat(g_*, \vartheta)$, $d_\flat(g, \vartheta)$ and $\frac{1}{2}(b/b_2)\tilde{d}_\flat(g_2, \vartheta)$.

In particular the law of Thm. Z (a) at one admissible triple $(g^\circ, b^\circ, \ell_{g^\circ})$ with $g^\circ \le \Gamma$ gives the law with the same $\frac{8}{3\pi}$ at every admissible triple of $\mathfrak{c}$, on every side of $\mathfrak{L}(g)$, with the tolerance $\varepsilon^L_{g^\circ, \vartheta}(b^\circ d/b) + C b^{-1} d$.

(iii) Let S and S' be volume-first limits at couplings g and g' with $0 < g, g' \le \gamma_W$, with the labels $b = \mathsf{b}(S)$ and $b' = \mathsf{b}(S')$; let S be S_2 or a function S_2^ξ of a canonical decomposition of σ_S, and S' be S_2' or a function $S_2'^\eta$ of a canonical decomposition of $\sigma_{S'}$; let $z, z' \in \mathbb{Z}$, and let $\tau = (g_\tau, L^z b, \ell_\tau)$ and $\tau' = (g_{\tau'}, L^{z'} b', \ell_{\tau'})$ be admissible triples of $\mathfrak{c}$. For $d_1 > 0$, $\Psi \in \mathfrak{T}^1(d_1; \vartheta)$ and $\delta > 0$ with $\delta L^z b d_1 \le d_\flat(g_\tau, \vartheta)$, $\delta L^{z'} b' d_1 \le d_\flat(g_{\tau'}, \vartheta)$ and $\delta b_2 d_1 < \tilde{d}_\flat(g_2, \vartheta)$,

$$|b^2 \mathsf{S}(\Psi_{\delta b}) - b'^2 \mathsf{S}'(\Psi_{\delta b'})| \le (2E_\tau + 2E_{\tau'} + 2E_{\tau_2})\, d_1^2 |\Psi|\, \delta^2.$$

Equivalently, with $s = b'/b$, for $\Phi \in \mathfrak{T}^1(d; \vartheta)$ with $L^z d \le d_\flat(g_\tau, \vartheta)$, $L^{z'} sd \le d_\flat(g_{\tau'}, \vartheta)$ and $(b_2/b)d < \tilde{d}_\flat(g_2, \vartheta)$: $|b'^2 \mathsf{S}'(\Phi_s) - b^2 \mathsf{S}(\Phi)| \le (2E_\tau + 2E_{\tau'} + 2E_{\tau_2})(d/b)^2 |\Phi|$. So, up to errors $O((d/b)^2)$, the two-point functions of the volume-first limits and of all their components at the couplings $g \le \gamma_W$ are carried into each other by the real dilation by the quotient of their labels. The law of Thm. Y (1) at the two labels gives the same comparison with the error $2\varepsilon^{\mathrm{nat}}(d/b)(d/b)|\Phi|$, which is only $O((d/b)^{3/2})|\Phi|$.

■□

Proof sketch. (i) Lem. 8.15 for the pair $(\mathfrak{c}, \tilde{\mathfrak{c}})$, whose blocking factors have incommensurable logarithms, at (τ, τ_2) and at (τ', τ_2), and the triangle inequality. For the second form put $\Phi = \Psi_{\delta b}$ and $d = \delta b d_1$; then $\Psi_{\delta b'} = \Phi_{b'/b}$ and $d_1^2 \delta^2 = (d/b)^2$. (ii) Let $\Phi \in \mathfrak{T}^1(d; \vartheta)$ and $\Phi_* = \Phi_{b_* d/b} \in \mathfrak{T}^1(b_* d/b; \vartheta)$. By (i), $b^2 T^{(\ell, b)}(\Phi)$ is within $(E_\tau + E_{\tau_*} + 2E_{\tau_2})(d/b)^2 |\Phi|$ of $b_*^2 T_*(\Phi_*)$, and by the assumption of (ii) $b_*^2 T_*(\Phi_*)$ is within $\varepsilon(b_* d/b)(d/b)|\Phi|$ of $c_2[\mathcal{K}_6(\Phi_*) + c b_*^{-1} \mathcal{K}_5(\Phi_*)] = c_2[\mathcal{K}_6(\Phi) + c b^{-1} \mathcal{K}_5(\Phi)]$; divide by b^2.

(iii) Lem. 8.11, for S with z and τ at the pair $\Psi_{\delta b} \in \mathfrak{T}^1(\delta b d_1; \vartheta)$, and for S' with z' and τ' at the pair $\Psi_{\delta b'}$, gives $|b^2 \mathsf{S}(\Psi_{\delta b}) - Y_\tau(\delta)| \le E_\tau d_1^2 |\Psi| \delta^2$ and $|b'^2 \mathsf{S}'(\Psi_{\delta b'}) - Y_{\tau'}(\delta)| \le E_{\tau'} d_1^2 |\Psi| \delta^2$, as $(\Psi_{\delta b})_{L^z} = \Psi_{\delta L^z b}$ and $(d/b)^2 = d_1^2 \delta^2$ at $d = \delta b d_1$; (i) at (τ, τ') and the triangle inequality give the bound, and the second form follows as in (i). By Thm. Y (1) at Φ and at $\Phi_s \in \mathfrak{T}^1(sd; \vartheta)$, $b^2 \mathsf{S}(\Phi)$ and $b'^2 \mathsf{S}'(\Phi_s)$ lie within $\varepsilon^{\mathrm{nat}}(d/b)(d/b)|\Phi|$ of the one number $c_2[\mathcal{K}_6(\Phi) + \frac{8}{3\pi} b^{-1} \mathcal{K}_5(\Phi)]$, as $sd/b' = d/b$ and $b'^{-1} \mathcal{K}_5(\Phi_s) = b^{-1} \mathcal{K}_5(\Phi)$. □

Proposition 8.17 (Non-covariance from the law at one triple)**.** Let $\mathfrak{c}_*$ and $\mathfrak{c}^\circ$ be choices, equal or not, $\vartheta \in (0, 1)$, and $\tau^\circ = (g^\circ, b^\circ, \ell^\circ)$ an admissible triple of $\mathfrak{c}^\circ$ such that, with $T^\circ = T^{(\ell^\circ, b^\circ)}$ formed at $\mathfrak{c}^\circ$, there are $d_n \to 0$ and $\Psi_n \in \mathfrak{T}^1(d_n; \vartheta)$ with $|T^\circ(\Psi_n) - c_2 b^{\circ -2} \mathcal{K}_6(\Psi_n)| / (d_n^2 |\Psi_n|) \to \infty$. Then for every $0 < g \le \gamma_W$ of $\mathfrak{c}_*$, every volume-first limit S at g and every canonical decomposition $(\mathsf{X}, \lambda, (\sigma_\xi))$ of σ_S:

(i) neither S_2 nor any S_2^ξ is dilation covariant: $\mathsf{N} = \mathsf{X}$ and $\lambda(\mathsf{N}) = 1$;

(ii) there is no $d_* > 0$ such that the three-scale number $S_2(\Psi_{4r}) - 65S_2(\Psi_{2r}) + 64S_2(\Psi_r)$ vanishes for all d, r with $4rd \le d_*$ and all $\Psi \in \mathfrak{T}^1(d;\vartheta)$.

■

Proof sketch. Let $b = \mathsf{b}(S)$ and $\mathsf{S} = S_2$ or S_2^ξ. Both assertions follow once a bound

$$|\mathsf{S}(\Phi) - c_2 b^{-2}\mathcal{K}_6(\Phi)| \le C\, d^2 |\Phi| \qquad (0 < d \le d_0,\ \Phi \in \mathfrak{T}^1(d;\vartheta)) \tag{8.vi}$$

is excluded for every C and $d_0 > 0$. If S is dilation covariant with the exponent ϖ, the leading laws YM3_I:(2353) and YM3_I:(2356) gives $\mathsf{S}(\Phi_r) \to c_2 b^{-2}\mathcal{K}_6(\Phi) \in (0,\infty)$ as $r \to 0$, so $\varpi = 0$ and $\mathsf{S} = c_2 b^{-2}\mathcal{K}_6$ on the pairs of all the classes $\mathfrak{T}^1(d;\vartheta)$: eq. 8.vi with $C = 0$. If the three-scale number vanishes as in (ii), we may take $d_* \le 4\mathsf{d}_0(g,\vartheta)$; for $\Phi \in \mathfrak{T}^1(d';\vartheta)$ with $d_*/8 < d' \le d_*/4$ the numbers $u_k = S_2(\Phi_{2^{-k}})$ satisfy $u_k - 65u_{k+1} + 64u_{k+2} = 0$, so $u_k = \alpha + \alpha' 64^{-k}$; by the leading law $\alpha = \lim_k u_k = c_2 b^{-2}\mathcal{K}_6(\Phi)$ and $|\alpha'| = |u_0 - \alpha| \le \frac{1}{3} c_2 b^{-2}\mathcal{K}_6(\Phi) \le C|\Phi|$.

Every pair of $\mathfrak{T}^1(d;\vartheta)$ with $d \le d_*/4$ is $\Phi_{2^{-k}}$ for such a Φ and some $k \ge 0$, so $|S_2(\Phi_{2^{-k}}) - c_2 b^{-2}\mathcal{K}_6(\Phi_{2^{-k}})| \le C|\Phi|\, 2^{-6k} \le C(8/d_*)^6 d^6 |\Phi|$: eq. 8.vi with d^6 in the place of d^2.

Suppose now eq. 8.vi. Take z with $b' = L^z b \ge 2(\gamma_W^{\mathrm{sd}})^{-2}$ and $g' = (2/b')^{1/2}$; then $b' \in I^\circ(g')$ by Lem. 8.2 (a), as $b' \ge 2(\gamma_W^{\mathrm{sd}})^{-2} \ge 2\gamma^{-2}$, and $\tau' = (g', b', \ell_{g'})$ is an admissible triple of $\mathfrak{c}_*$. For $\Psi \in \mathfrak{T}^1(d_1;\vartheta)$, eq. 8.vi at $\Phi = \Psi_{\delta b}$ and Lem. 8.11 give

$$|Y_{\tau'}(\delta) - c_2\mathcal{K}_6(\Psi)| \le (Cb^4 + E_{\tau'})\, d_1^2 |\Psi|\, \delta^2$$

for small δ. Let $\tilde{\mathfrak{c}}$ be a second choice of $\mathfrak{c}_*$ with Q the blocking factor of $\mathfrak{c}^\circ$ (Lem. 8.14), and τ_2 an admissible triple of $\tilde{\mathfrak{c}}$. Lem. 8.15, for the pair $(\mathfrak{c}_*, \tilde{\mathfrak{c}})$ at (τ', τ_2) and for the pair $(\mathfrak{c}^\circ, \tilde{\mathfrak{c}})$ at (τ°, τ_2), gives for small δ

$$|Y_{\tau^\circ}(\delta) - c_2\mathcal{K}_6(\Psi)| \le C' d_1^2 |\Psi|\, \delta^2, \qquad C' = Cb^4 + 2E_{\tau'} + 2E_{\tau_2} + E_{\tau^\circ}.$$

At $d_1 = 1$, $\delta = d_n/b^\circ$ and $\Psi = (\Psi_n)_{1/d_n}$ this reads $|T^\circ(\Psi_n) - c_2 b^{\circ -2}\mathcal{K}_6(\Psi_n)| \le C' b^{\circ -4} d_n^2 |\Psi_n|$, against the assumption on T°. As λ is a probability measure, $\lambda(\mathsf{N}) = \lambda(\mathsf{X}) = 1$. □

8.1.11. *A Gaussian model satisfying the same two-term law.*

Proposition 8.18 (A quadratic functional of a Gaussian field)**.** Let $\ell > 0$, $\alpha > 0$, $\upsilon_0 > 0$, $\upsilon = \upsilon_0/\ell$ and $b > 0$. Let $\varphi_1, \varphi_2, \varphi_3$ be independent centred Gaussian fields on T_ℓ with the covariance operator $\hat{G}_b(D)$, $\hat{G}_b(k) = (b|k|^2 + \alpha|k| + \upsilon)^{-1}$ for $k \in \frac{2\pi}{\ell}\mathbb{Z}^3$, and let $\mathcal{C}$ be the kernel of $\hat{G}_b(D)$. A *mollifier* is a real radial function $\jmath \in C_c^\infty(\mathbb{B}(0,1))$ with $\int \jmath = 1$. For a mollifier $\jmath$ and $0 < \varepsilon_{\mathrm{m}} < \frac{\ell}{2}$ put $\jmath_{\varepsilon_{\mathrm{m}}} = \varepsilon_{\mathrm{m}}^{-3}\jmath(\cdot/\varepsilon_{\mathrm{m}})$ and $\varphi_{a,\varepsilon_{\mathrm{m}}} = \varphi_a * \jmath_{\varepsilon_{\mathrm{m}}}$, and for a real Lipschitz function f on T_ℓ let

$$\mathcal{O}_{\varepsilon_{\mathrm{m}}}(f) = \sum_a \int f\, [-\tfrac{1}{2}{:}|\nabla\varphi_{a,\varepsilon_{\mathrm{m}}}|^2{:} - {:}\varphi_{a,\varepsilon_{\mathrm{m}}}\Delta\varphi_{a,\varepsilon_{\mathrm{m}}}{:}]$$

be the observable of the mollified field. For $f \ne 0$ the variance of $\mathcal{O}_{\varepsilon_{\mathrm{m}}}(f)$ tends to ∞ as $\varepsilon_{\mathrm{m}} \to 0$, so that there is no L^2 limit observable. For $n \ge 2$ and real Lipschitz functions $f_1, \dots, f_n$ on T_ℓ whose supports have no common point we put

$$\mathcal{S}_n^{(\ell,b)}(f_1, \dots, f_n) := \lim_{\varepsilon_{\mathrm{m}} \to 0} \langle \mathcal{O}_{\varepsilon_{\mathrm{m}}}(f_1); \dots; \mathcal{O}_{\varepsilon_{\mathrm{m}}}(f_n) \rangle,$$

the limit of the joint cumulants of the observables of the mollified field; d_∞ is the spread of such a tuple, as in Thm. YM3_I:AS.

(a) $(bu^2 + \alpha u + \upsilon)^{-1} = \int_0^\infty \rho_{\mathrm{sp}}(m)\, (u^2 + m^2)^{-1} dm$ for $u \ge 0$, with $\rho_{\mathrm{sp}}(m) = \frac{2\alpha}{\pi}\, m^2 \big((\upsilon - bm^2)^2 + \alpha^2 m^2\big)^{-1} > 0$ and $\int_0^\infty \rho_{\mathrm{sp}} = \frac{1}{b}$.

(b) The limit exists, depends on no mollifier, and is symmetric and multilinear, as long as the tuples keep supports without a common point; at tuples with pairwise disjoint supports it is the Wick integral of the kernels of the vertices and of $\mathcal{C}$. For every $n \ge 2$, every compact subset of $(0,\infty)$ and every $d_0 > 0$ there is c with $|\mathcal{S}_n^{(\ell,b)}(f_1, \dots, f_n)| \le c \prod_i \|f_i\|_\sharp$ for all labels

b of that subset and all such tuples with $\mathsf{d}_\infty \ge d_0$. At pairs (f,h) of real Lipschitz functions on T_ℓ with disjoint supports, $\mathcal{S}_2^{(\ell,b)}(f,h) = \iint f(x)h(y)\,s_2(x-y)\,\mathrm{d}x\,\mathrm{d}y$ with a kernel s_2 smooth off 0 and

$$s_2(x) = c_2 b^{-2}\Big[|x|^{-6} + \frac{4\alpha}{3\pi}\,b^{-1}|x|^{-5}\Big] + O\big(|x|^{-4}(1+\log^+(\ell/|x|))\big) \qquad (x\to 0).$$

So at every label b, for $(f,h)\in\mathfrak{T}_{T_\ell}(d;\vartheta)$ and small d,

$$|\mathcal{S}_2^{(\ell,b)}(f,h) - c_2 b^{-2}[\mathcal{K}_6 + \frac{4\alpha}{3\pi}b^{-1}\mathcal{K}_5](f,h)| \le C(\ell,\alpha,v_0,b)\,d^2(1+\log(\ell/d))\,\|f\|_\infty\|h\|_\infty$$

: the two-term law with the coefficient $\frac{4\alpha}{3\pi}$, which is $\frac{8}{3\pi}$ at $\alpha=2$, and with a remainder $O(d^2\log\frac{1}{d})$.

(c) Let $\{x_\mu = s\}$ be a coordinate mirror of T_ℓ, s any real number, θ the reflection in it and $\Sigma_+ = \{s < x_\mu < s+\frac{\ell}{2}\}$ one of its open sides. Then $(w\circ\theta, \mathcal{C}w)\ge 0$ for every real function w carried by Σ_+; hence the Gaussian measure is reflection positive. For a tuple $\mathcal{F} = (f_i)_{i\in I}$, I a finite set, let $\mathfrak{m}(\mathcal{F}) = \sum_\pi \prod_{J\in\pi}\mathcal{S}_{|J|}^{(\ell,b)}((f_i)_{i\in J})$, the sum running over the partitions π of I into blocks of at least two elements, and $\mathfrak{m}(\varnothing)=1$; let $\theta\mathcal{F} = (f_i\circ\theta)_{i\in I}$, and let $\theta\mathcal{F}\cdot\mathcal{F}'$ be the tuple of the entries of $\theta\mathcal{F}$ and of $\mathcal{F}'$. Then

$$\sum_{i,i'} \bar z_i z_{i'}\,\mathfrak{m}(\theta\mathcal{F}_i\cdot\mathcal{F}_{i'}) \ge 0$$

for every finite family of complex numbers z_i and of tuples $\mathcal{F}_i$, the empty tuple among them, of real Lipschitz functions with pairwise disjoint supports in compact subsets of Σ_+; the same holds for entries that overlap on Σ_+, as long as every cumulant that occurs stands at a tuple whose supports have no common point. The limits on the tori have the same property at the mirrors of the lattices of the cutoffs: for $\ell = 2L^m$, every $b>0$, $\mu\in\{1,2,3\}$ and every $c\in\bigcup_{K\in\mathbb{N}}\frac12 L^{-K}\mathbb{Z}$, with θ the reflection in $\{x_\mu = c\}$ and $\Sigma_+ = \{c<x_\mu<c+\frac{\ell}{2}\}$, the inequality above holds with the $S_n^{(\ell,b)}$ in the place of the $\mathcal{S}_n^{(\ell,b)}$, for every finite family of complex numbers z_i and of tuples $\mathcal{F}_i$, the empty tuple among them, of real Lipschitz functions with pairwise disjoint supports in compact subsets of Σ_+.

For entries of class C^1 the same inequality holds at every mirror $\{x_\mu = c\}$, c any real number (the moving-mirror limit argument in the proof below, using Thm. YM3_I:A). As the model is reflection positive at every coordinate mirror, it is so at all the mirrors at which the limits are. On the tuples whose supports have no common point, the family $(\mathcal{S}_n^{(\ell,b)})_n$ is invariant under the isometries of T_ℓ, and

$$\mathcal{S}_n^{(r\ell,rb)}(f_1(\cdot/r),\dots,f_n(\cdot/r)) = r^{-n}\mathcal{S}_n^{(\ell,b)}(f_1,\dots,f_n) \qquad (r>0),$$

the number v_0 being kept.

(d) For every such tuple $\underline{f} = (f_1,\dots,f_n)$ the map $b\mapsto\mathcal{S}_n^{(\ell,b)}(\underline{f})$ extends holomorphically to the half plane $\operatorname{Re} b>0$, and

$$\partial_b\mathcal{S}_n^{(\ell,b)}(\underline{f}) = -\mathcal{S}_{n+1}^{(\ell,b)}(1,\underline{f}), \qquad (-\partial_b)^{\mathsf{j}}\mathcal{S}_n^{(\ell,b)}(\underline{f}) = \mathcal{S}_{n+\mathsf{j}}^{(\ell,b)}(1,\dots,1,\underline{f}) \quad (\mathsf{j}\ge 0),$$

with j entries equal to the constant function 1 on T_ℓ: the rule that Prop. YM3_I:118.3, equation YM3_I:(2217) gives for the limits on the tori. Consequently, for $0<b_1<b_2$ and every pair (f,h) of real Lipschitz functions on T_ℓ with disjoint supports,

$$\int_{b_1}^{b_2}[b^2\mathcal{S}_3^{(\ell,b)}(1,f,h) - 2b\,\mathcal{S}_2^{(\ell,b)}(f,h)]\,\mathrm{d}b = b_1^2\mathcal{S}_2^{(\ell,b_1)}(f,h) - b_2^2\mathcal{S}_2^{(\ell,b_2)}(f,h),$$

the identity on which Cor. 8.1 (c) rests.

(e) Let $\ell \in \{2L, 2L^2, 2L^3, \dots\}$, $b > 0$ and $n \geq 2$. The limits on the tori have the following property: if $\underline{f}^{(j)}$, $j \in \mathbb{N}$, are tuples of n real Lipschitz functions on T_ℓ, none of them 0, whose supports have no common point, with spreads bounded below by a positive number, norms $\|f_i^{(j)}\|_\sharp$ bounded in j, and $\prod_i \|f_i^{(j)}\|_\infty \to 0$, then $S_n^{(\ell,b)}(\underline{f}^{(j)}) \to 0$. The model fails this at $n = 3$ and at every label: if $\psi \in C^\infty(T_\ell)$ is real and $h \geq 0$ is Lipschitz, neither of them 0, with $\operatorname{supp}\psi \cap \operatorname{supp} h = \varnothing$, and $f_{\mathbf{q}} = |\mathbf{q}|^{-1}\psi\cos(\mathbf{q}\cdot x)$ for $\mathbf{q} \in \frac{2\pi}{\ell}\mathbb{Z}^3 \setminus \{0\}$, then the triples $(f_{\mathbf{q}}, f_{\mathbf{q}}, h)$ satisfy these conditions as $|\mathbf{q}| \to \infty$, and

$$\mathcal{S}_3^{(\ell,b)}(f_{\mathbf{q}}, f_{\mathbf{q}}, h) \to -\frac{3}{2b}\iint \psi(x)^2 h(y)\, \mathsf{p}(x-y)\,\mathrm{d}x\,\mathrm{d}y < 0 \qquad (|\mathbf{q}| \to \infty),$$

with $\mathsf{p} = |\nabla\mathcal{C}|^2 + 2\,\mathcal{C}\Delta\mathcal{C}$, which is positive off 0.

■

Proof sketch. (a) With $P(\zeta) = b\zeta^2 + \alpha\zeta + \upsilon$, whose roots have negative real parts, $\rho_{\mathrm{sp}}(m) = \frac{1}{\pi i}\, m\,[P(-im)^{-1} - P(im)^{-1}]$, and closing the contour of $\int_{\mathbb{R}} m\,\mathrm{d}m/((m^2+u^2)P(im))$ in the lower half plane, where the only pole is $m = -iu$, gives $1/P(u)$; letting $u \to \infty$ in $u^2/P(u)$ gives the mass $\frac{1}{b}$. (b)

The limit and the bound. For smooth u, v the vertex $\mathsf{V}_f(u,v) = \int f\,\nabla u\cdot\nabla v + \int \nabla f\cdot\nabla(uv)$ equals $\int f[-\nabla u\cdot\nabla v - u\Delta v - v\Delta u]$, so the observable of the mollified field is, in each colour, the Wick ordered half of V_f at that field. The operator V_f with $(u, V_f v) = \mathsf{V}_f(u,v)$ maps H^1 to H^{-1} with a norm at most $\|f\|_\infty + \operatorname{Lip} f \leq c\|f\|_\sharp$, and $\hat{G}_b(D)$ maps H^{-1} to H^1, uniformly for b in a compact subset of $\operatorname{Re} b > 0$. The joint cumulant of half squares of a Gaussian vector is half the sum, over the cyclic orders of the entries, of the traces of the products of the forms against the covariance; so, the three colours giving the factor 3, the joint cumulant of the $\mathcal{O}_{\varepsilon_{\mathrm{m}}}(f_i)$ is $\frac{3}{2}$ times the sum, over the $(n-1)!$ cyclic orders of $f_1, \dots, f_n$, of the traces of the products of the operators $V_{f_i}|\hat{\jmath}(\varepsilon_{\mathrm{m}}D)|^2\hat{G}_b(D)$ in these orders, $\hat{\jmath}$ the Fourier transform of $\jmath$.

If the supports have no common point, split every propagator into a part of range less than d_∞/n and a smooth remainder. The product of the short parts vanishes: the vertices are local, and a closed chain of n short propagators through the n supports would give points of all the supports at mutual distances less than the spread. Every other product contains a smooth propagator, a trace class operator, among bounded ones. So the products are trace class, with trace norms at most $c\prod_i \|f_i\|_\sharp$ for b in a compact subset of $\operatorname{Re} b > 0$ and $\mathsf{d}_\infty \geq d_0$, and their traces are continuous under the convergence of the propagators as $\varepsilon_{\mathrm{m}} \to 0$.

This gives the limit, its independence of the mollifier, its symmetry and multilinearity, and the bound; at pairwise disjoint supports the traces are the integrals of $\prod_i f_i(x_i)$ against the products of the kernels off the diagonal, the Wick integrals.

The variance. At $n = 2$ and $f_1 = f_2 = f$ the same trace formula is the variance of $\mathcal{O}_{\varepsilon_{\mathrm{m}}}(f)$, and it is a sum of terms ≥ 0. In the orthonormal basis $e_k = \ell^{-3/2}e^{ik\cdot x}$, $k \in \frac{2\pi}{\ell}\mathbb{Z}^3$, the matrix of V_f has the entries $\int \bar{e}_p V_f e_q = \ell^{-3}v(p,q)\hat{f}(p-q)$, with $v(p,q) = |p|^2 + |q|^2 - p\cdot q = p\cdot q + |p-q|^2$ and $\hat{f}(k) = \int f(x)e^{-ik\cdot x}\,\mathrm{d}x$: the first member of V_f gives $p\cdot q$, and the second, after one integration by parts, $|p-q|^2$. As f is real, the entries at (p,q) and at (q,p) are conjugate; as $\hat{\jmath}$ is real, the mollified field has the covariance $\hat{\jmath}(\varepsilon_{\mathrm{m}}D)^2\hat{G}_b(D)$, and

$$\operatorname{Var}\mathcal{O}_{\varepsilon_{\mathrm{m}}}(f) = \frac{3}{2}\,\ell^{-6}\sum_{p,q\in\frac{2\pi}{\ell}\mathbb{Z}^3} \hat{\jmath}(\varepsilon_{\mathrm{m}}p)^2\,\hat{\jmath}(\varepsilon_{\mathrm{m}}q)^2\,\hat{G}_b(p)\,\hat{G}_b(q)\,v(p,q)^2\,|\hat{f}(p-q)|^2,$$

every term of which is ≥ 0. Let $f \neq 0$, and let k_0 have $\hat{f}(k_0) \neq 0$. Without the mollifier, the terms with $p = q + k_0$ tend to $b^{-2}|\hat{f}(k_0)|^2 > 0$ as $|q| \to \infty$, since $v(q+k_0, q) = |q|^2 + q\cdot k_0 + |k_0|^2$ and $|q|^2\hat{G}_b(q) \to b^{-1}$; so their series diverges. As $\hat{\jmath}(\varepsilon_{\mathrm{m}}k) \to \hat{\jmath}(0) = \int \jmath = 1$ at every k when $\varepsilon_{\mathrm{m}} \to 0$, Fatou's lemma for sums of non-negative terms gives $\operatorname{Var}\mathcal{O}_{\varepsilon_{\mathrm{m}}}(f) \to \infty$.

The kernel. By (a) the covariance kernel is $\mathcal{C} = \int_0^\infty \rho_{\mathrm{sp}}(m)\mathcal{C}_m \,\mathrm{d}m$,

$$\mathcal{C}_m(x) = \sum_{j\in\mathbb{Z}^3} e^{-m|x+\ell j|}/(4\pi|x+\ell j|) > 0,$$

and $\mathcal{H} = \Delta\mathcal{C} = \int \rho_{\mathrm{sp}}(m)m^2\mathcal{C}_m \,\mathrm{d}m$ off 0. Since $m^2\rho_{\mathrm{sp}}(m) = \frac{2\alpha}{\pi b^2} + O((1+m^2)^{-1})$, the Laplace transform $\hat\rho_{\mathrm{sp}}$ of ρ_{sp} has $\hat\rho_{\mathrm{sp}}(0) = \frac{1}{b}$ and $\hat\rho''_{\mathrm{sp}}(|x|) = \frac{2\alpha}{\pi b^2}|x|^{-1} + O(1)$, so that, with $\omega_0 = \frac{1}{4\pi b}$ and $\omega_1 = \frac{\alpha}{2\pi^2 b^2}$,

$$\begin{aligned}
\mathcal{C} &= \frac{\omega_0}{|x|} + O(\lg(\ell/|x|)),\\
\nabla\mathcal{C} &= -\omega_0\frac{\hat x}{|x|^2} + O\Big(\frac{\lg(\ell/|x|)}{|x|}\Big),\\
\nabla\nabla^{\mathsf T}\mathcal{C} &= \omega_0\frac{3\hat x\hat x^{\mathsf T}-1}{|x|^3} + \omega_1\frac{1-2\hat x\hat x^{\mathsf T}}{|x|^2} + O(|x|^{-1}),\\
\mathcal{H} &= \frac{\omega_1}{|x|^2} + O(|x|^{-1}),\\
\nabla\mathcal{H} &= -\frac{2\omega_1\hat x}{|x|^3} + O(|x|^{-2}),\\
\Delta\mathcal{H} &= \frac{2\omega_1}{|x|^4} + O(|x|^{-2}),
\end{aligned}$$

with $\hat x = x/|x|$ and $\lg(\ell/|x|) = 1 + \log(\ell/|x|)$, as $|x| < \ell$ on T_ℓ, the images under $\ell\mathbb{Z}^3$ being smooth near 0. The observable is half the quadratic form of the vertex in each colour, so Wick's rule, with three colours, gives

$$s_2 = \frac{3}{2}[\mathrm{tr}(\nabla\nabla^{\mathsf T}\mathcal{C})^2 + 4\nabla\mathcal{C}\cdot\nabla\mathcal{H} + 2\mathcal{H}^2 + 2\mathcal{C}\Delta\mathcal{H}]$$

off 0. The leading terms give $\frac{3}{2}\cdot 6\,\omega_0^2|x|^{-6} = c_2 b^{-2}|x|^{-6}$; the terms of the order $|x|^{-5}$ are $-8\omega_0\omega_1$, $+8\omega_0\omega_1$ and $+4\omega_0\omega_1$, in all

$$\frac{3}{2}\cdot 4\,\omega_0\omega_1|x|^{-5} = \frac{3\alpha}{4\pi^3}b^{-3}|x|^{-5} = c_2\frac{4\alpha}{3\pi}b^{-3}|x|^{-5};$$

the rest is $O(\lg(\ell/|x|)\,|x|^{-4})$. Integrating the remainder against a pair of the class, on whose supports $d \le |x-y| \le 10d$, gives the bound.

(c) Write $x = (x_\mu, x_\perp)$, and let w be carried by Σ_+. A Fourier series in $x_\perp$, with the frequency $k_\perp$ and $m_\perp = (|k_\perp|^2 + m^2)^{1/2}$, writes $(w\circ\theta, \mathcal{C}_m w)$ as a sum over $k_\perp$ of double integrals, in the distances $\mathsf{v}, \mathsf{v}' \in (0, \frac{\ell}{2})$ of two points of Σ_+ to the mirror, against the kernel $\cosh m_\perp(\frac{\ell}{2} - \mathsf{v} - \mathsf{v}')/(2m_\perp \sinh\frac{m_\perp\ell}{2})$, and $\cosh m_\perp(\frac{\ell}{2} - \mathsf{v} - \mathsf{v}') = \cosh m_\perp(\frac{\ell}{4} - \mathsf{v})\cosh m_\perp(\frac{\ell}{4} - \mathsf{v}') + \sinh m_\perp(\frac{\ell}{4} - \mathsf{v})\sinh m_\perp(\frac{\ell}{4} - \mathsf{v}')$ makes each of them a sum of two squares; integrating against $\rho_{\mathrm{sp}} \ge 0$ gives $(w\circ\theta, \mathcal{C}w) \ge 0$, and with it the reflection positivity of the Gaussian measure, as for every centred Gaussian measure with a reflection positive covariance.

For ε_{m} less than the distance of the supports from the mirror, the observables of the mollified field of functions carried by compact subsets of Σ_+ are functions of the field on Σ_+, and, the mollifier being radial, the reflection carries $\mathcal{O}_{\varepsilon_{\mathrm{m}}}(f)$ to $\mathcal{O}_{\varepsilon_{\mathrm{m}}}(f\circ\theta)$. The moments of centred variables are the sums, over the partitions without blocks of one element, of the products of their cumulants; so $\sum \bar z_i z_{i'}\mathfrak{m}_{\varepsilon_{\mathrm{m}}}(\theta\mathcal{F}_i\cdot\mathcal{F}_{i'})$, with $\mathfrak{m}_{\varepsilon_{\mathrm{m}}}$ formed with the cumulants of the mollified field, is the expectation of $\overline{\Theta F}\,F$ for $F = \sum_{i'} z_{i'}\prod_{f\in\mathcal{F}_{i'}}\mathcal{O}_{\varepsilon_{\mathrm{m}}}(f)$, Θ the reflection of the field, and it is ≥ 0.

As $\varepsilon_{\mathrm{m}} \to 0$, each cumulant that occurs stands at a tuple whose supports have no common point and converges by (b). For the limits on the tori, let $2c \in a_{K_0}\mathbb{Z}$. For every $K \ge K_0$ the plane $\{x_\mu = c\}$ is a site or a bond mirror of the lattice of the spacing a_K, at which the Wilson measure on T_ℓ at $\beta = L^K b$ is reflection positive (Prop. YM3_I:91.4). For large K the function

$F_K = \sum_{i'} z_{i'} \prod_{f\in\mathcal{F}_{i'}} (\mathcal{O}_K(f) - \langle \mathcal{O}_K(f)\rangle)$ reads only bonds in Σ_+, the measure is invariant under the reflection, and $\langle \overline{\Theta F_K}\, F_K\rangle \geq 0$ is the inequality with the cumulants $S^T_{n;K,\ell,L^K b}$ in the place of the $\mathcal{S}^{(\ell,b)}_n$.

At the tuples that occur, whose supports are pairwise disjoint, the cumulants converge along the full sequence of cutoffs by Thm. YM3_I:A; hence the finite positive matrix converges to the asserted matrix. For a real mirror position c, choose $c_j \in \bigcup_K \frac{1}{2}L^{-K}\mathbb{Z}$ with $c_j \to c$. The fixed compact positive-side supports remain in the positive half-torus for all large j, and their reflected supports remain in disjoint compact neighbourhoods. For C^1 entries the translated reflections converge in the sharp norm. The multilinear bound in Thm. YM3_I:AS, with the all-side dilation of Cor. YM3_I:119.2, therefore passes each cumulant, and hence each centred moment, to the limit. Positivity is closed under this finite matrix limit. This is the moving-mirror argument of Prop. YM3_I:92.4, here on the two half-tori. It proves the assertion at every real c, for every $b > 0$ and every side $2L^m$ allowed by Thm. YM3_I:A.

The invariance is that of $\hat{G}_b$ and of the mollifiers, which are radial; the dilation identity follows from $\hat{G}_{rb}(k/r)$ on $T_{r\ell}$ being $r\hat{G}_b(k)$ on T_ℓ, with $\upsilon = \upsilon_0/\ell$: the field on $T_{r\ell}$ has the law of $x \mapsto r^{-1}\varphi(x/r)$, the mollifier at the scale ε_{m} goes to that at the scale $r\varepsilon_{\mathrm{m}}$, and each observable, with its two derivatives, scales by r^{-1}; the identity passes to the limits.

(d) The trace formula of (b) defines $\mathcal{S}^{(\ell,b)}_n$ for $\operatorname{Re} b > 0$, and its terms are holomorphic in b, locally uniformly, as $\hat{G}_b$ is; so $\mathcal{S}^{(\ell,b)}_n(\underline{f})$ is holomorphic there. The vertex of the constant function 1 is $V_1 = -\Delta$, and $\partial_b \hat{G}_b = -|k|^2\hat{G}_b^2$, that is, $\partial_b \hat{G}_b(D) = -\hat{G}_b(D)V_1\hat{G}_b(D)$: the derivative of a propagator inserts the vertex of 1 between its ends. Summing over the n propagators of each product and over the cyclic orders of $f_1,\dots,f_n$ gives the cyclic orders of $(1,f_1,\dots,f_n)$; hence $\partial_b\mathcal{S}^{(\ell,b)}_n(\underline{f}) = -\mathcal{S}^{(\ell,b)}_{n+1}(1,\underline{f})$, and the higher derivatives follow by induction, the tuples with entries 1 having supports without a common point.

At the entry 1 the observable of the mollified field is $\frac{1}{2}\sum_a \int :|\nabla\varphi_{a,\varepsilon_{\mathrm{m}}}|^2:$, since $\int \varphi\Delta\varphi = -\int|\nabla\varphi|^2$ on the torus. The integral identity is the integral over $[b_1,b_2]$ of $\partial_b[-b^2\mathcal{S}^{(\ell,b)}_2(f,h)] = b^2\mathcal{S}^{(\ell,b)}_3(1,f,h) - 2b\,\mathcal{S}^{(\ell,b)}_2(f,h)$.

(e) For the limits, let $d_0 > 0$ bound the spreads from below. By Thm. YM3_I:AS, at the interval $\{b\}$ and with d_0 in the place of d, for all large K the term of index K of the sequence of the cutoffs differs from $S^{(\ell,b)}_n(\underline{f}^{(j)})$ by at most $\prod_i \|f_i^{(j)}\|_\sharp$ times a number that tends to 0 as $K\to\infty$ and does not depend on j; and, as $|\mathcal{O}_K(f)| \leq 6\ell^3L^{4K}\|f\|_\infty$, the term of index K is at most a number depending only on n times $\prod_i(6\ell^3L^{4K}\|f_i^{(j)}\|_\infty)$, as in the first member of YM3_I:(2214). Let $j\to\infty$ and then $K\to\infty$.

For the model, $\|f_{\mathbf{q}}\|_\infty \leq |\mathbf{q}|^{-1}\|\psi\|_\infty$, $\operatorname{Lip} f_{\mathbf{q}}$ is bounded in $\mathbf{q}$, and the spread of $(f_{\mathbf{q}}, f_{\mathbf{q}}, h)$ is at least the distance of $\operatorname{supp}\psi$ and $\operatorname{supp} h$ in the norm $|\cdot|_\infty$; so the conditions hold. By (b), $\mathcal{S}^{(\ell,b)}_3(f_{\mathbf{q}}, f_{\mathbf{q}}, h)$ is three times the trace of $V_{f_{\mathbf{q}}}\hat{G}_b(D)V_{f_{\mathbf{q}}}\hat{G}_b(D)V_h\hat{G}_b(D)$, the two cyclic orders giving equal traces. Now $V_{f_{\mathbf{q}}}u = -f_{\mathbf{q}}\Delta u - \nabla\cdot(u\nabla f_{\mathbf{q}})$; the first member tends to 0, and the second, of the order 1 in H^{-1}, carries the frequencies $\pm\mathbf{q} + O(1)$, at which $\hat{G}_b(k)kk^{\mathsf{T}} \to b^{-1}\hat{\mathbf{q}}\hat{\mathbf{q}}^{\mathsf{T}}$, $\hat{\mathbf{q}} = \mathbf{q}/|\mathbf{q}|$.

So $V_{f_{\mathbf{q}}}\hat{G}_b(D)V_{f_{\mathbf{q}}}$ tends to the multiplication by $(2b)^{-1}\psi^2$, the mean of $b^{-1}\psi^2\sin^2(\mathbf{q}\cdot x)$; as the supports of ψ and h are disjoint, $\hat{G}_b(D)V_h\hat{G}_b(D)$ has a smooth kernel near $\operatorname{supp}\psi$, and the traces tend to $\frac{3}{2b}\operatorname{Tr}[\psi^2\hat{G}_b(D)V_h\hat{G}_b(D)] = -\frac{3}{2b}\iint \psi(x)^2h(y)\,\mathsf{p}(x-y)\,dx\,dy$, since $\mathsf{V}_h(u,u) = -\int h(|\nabla u|^2 + 2u\Delta u)$ at $u = \mathcal{C}(x-\cdot)$. Finally $\mathsf{p} = |\nabla\mathcal{C}|^2 + 2\mathcal{C}\mathcal{H} > 0$ off 0, as $\mathcal{C} > 0$ and $\mathcal{H} > 0$ there, by the representation of $\mathcal{C}$ above. □

8.1.12. *Proof of Thm. Z. Step 1: the choice.* At $\pi^+(\varsigma^{\mathrm{S}})$ every constant keeps its value (the unchanged restriction of Section YM3_I:82); so L, $I(g)$, $\mathfrak{L}(g)$, ℓ_g, the first-passage data and the functions $S^T_{2;K,\ell,\beta}$ formed there are those of $\mathfrak{c}$, and Prop. 8.10 speaks of $\mathfrak{c}$.

Step 2: (a). Put $d^L = \min\{d_{\mathrm{e}}, \ell_g/21\}$. For $d \le d^L$ and $\Psi \in \mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$ the periodisation $\Psi^{\langle \ell_g \rangle}$ lies in $\mathfrak{T}_{T_{\ell_g}}(d;\vartheta)$ and has the suprema and the kernels $\mathcal{K}_q$ of Ψ, and Prop. 8.10 (b) gives

$$|T^{(\ell_g,b)}(\Psi) - L_b^+(\Psi)| \le \bar\varepsilon_\infty(d)\, b^{-3} d\, |\Psi| \qquad (b \in I(g)).$$

Let $\varepsilon^L_{g,\vartheta}(d)$ be the supremum, over $0 < d'' \le d$, $b \in I(g)$ and $\Psi \in \mathfrak{T}_{\mathbb{R}^3}(d'';\vartheta)$, of $|T^{(\ell_g,b)}(\Psi) - L_b^+(\Psi)|/(b^{-3}d''|\Psi|)$: the least non-decreasing tolerance; it is at most $\bar\varepsilon_\infty \le C(g,\vartheta)d^{1/2}$.

Step 3: the consequences (i)–(iv). (a) at $\Psi_r \in \mathfrak{T}^1(rd_1;\vartheta)$, with $\mathcal{K}_6(\Psi_r) = \mathcal{K}_6(\Psi)$, $\mathcal{K}_5(\Psi_r) = r\mathcal{K}_5(\Psi)$ and $|\Psi_r| = |\Psi|$, gives eq. 8.iv: a term of the order r against an error $O(r^{3/2})$ and the threshold r^2 of a dilation covariant function; the bounds on B are eq. 8.i. Write $T(\Psi_r) = \mathsf{A} + \mathsf{B}r + \rho(r)$ with $|\rho(r)| \le \varepsilon^L(rd_1)\mathsf{u}r$; then $T(\Psi_{tr}) - T(\Psi_r) = (t-1)\mathsf{B}r + \rho(tr) - \rho(r)$ with $|\rho(tr) - \rho(r)| \le (t+1)\varepsilon^L(trd_1)\mathsf{u}r$, and $\mathcal{N}_t^T$ is treated as in Lem. 8.12: (i). For $\Phi \in \mathfrak{T}^1(d;\vartheta)$ and $d \le d^L$, by eq. 8.i,

$$|T(\Phi) - c_2 b^{-2}\mathcal{K}_6(\Phi)| \ge \varkappa b^{-3}\mathcal{K}_5(\Phi) - \varepsilon^L(d)\, b^{-3} d|\Phi| \ge [10^{-5}\varkappa\vartheta^2 - \varepsilon^L(d)]\, b^{-3} d\, |\Phi|;$$

at $\Phi = \Psi_r$, $d = rd_1$, divided by r^2, this is (ii), and at $\Phi = \Psi_n$, divided by $d_n^2|\Psi_n|$, it is (iii). Under $\varepsilon^L(d) \le \frac12 \cdot 10^{-5}\varkappa\vartheta^2$ the same line gives $b^2|T(\Psi) - c_2 b^{-2}\mathcal{K}_6(\Psi)| \ge \frac12 \cdot 10^{-5}\varkappa\vartheta^2 b^{-1} d|\Psi|$, which exceeds $C\vartheta^{-1}(d/b)^2|\Psi|$ if and only if $d/b < \frac12 \cdot 10^{-5}\varkappa\vartheta^3/C$; such d exist, as $\varepsilon^L \to 0$: (iv).

Step 4: (b). As $d_\flat \le \frac{1}{180} < \ell_g/21$, Step 2 applies. At the data $(K, L^K b)$ with $a_K \le 9d$ the periodic extensions of the Wilson measures on T_ℓ and T_{ℓ_g} are two states at the datum; Prop. 8.3 and $K \to \infty$, along the full sequence, give

$$|T^{(\ell,b)}(\Psi) - T^{(\ell_g,b)}(\Psi)| \le 2C_2^{\mathrm{T}}(g)\vartheta^{-1}d^2b^{-2}|\Psi| = 2C_2^{\mathrm{T}}(g)\vartheta^{-1}bd \cdot b^{-3}d\,|\Psi|,$$

and Prop. 8.10 (b) on T_{ℓ_g} gives the first assertion; $\bar\varepsilon_\infty(d) \le C(g,\vartheta)d^{1/2}$ and $d < 1$ bound the bracket. At a cutoff $K > s(d)$ one has $K \ge 3$ and $8a_K \le d$, so $(K, L^K b)$ is a first-passage datum at g at which Prop. 8.3 (a) compares every state with the periodic extension μ' of the Wilson measure on T_{ℓ_g}; and $S_2^T[\mu'](\Psi) = S^T_{2;K,\ell_g,L^K b}(\Psi^{\langle \ell_g\rangle})$, to which Prop. 8.10 (b), respectively (c), applies.

Step 5: (d). This is YM3_I:(2215) at $n = 2$: the dilation by L^s carries T_ℓ with the spacing a_K onto $T_{L^s\ell}$ with the spacing a_{K-s}, keeps $\beta = L^K b = L^{K-s}(L^s b)$, and $\mathcal{O}_K(f) = L^s\mathcal{O}_{K-s}(f(\cdot/L^s))$; so $S^T_{2;K,\ell,L^K b}(\Psi^{\langle\ell\rangle}) = L^{2s}S^T_{2;K-s,L^s\ell,L^{K-s}L^s b}((\Psi_{L^s})^{\langle L^s\ell\rangle})$, and we let $K \to \infty$. For the second sentence let $\Psi \in \mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$ with $d \le L^z d^L$; then $\Psi_{L^{-z}} \in \mathfrak{T}_{\mathbb{R}^3}(L^{-z}d;\vartheta)$, $T^{(L^z\ell_g, L^z b)}(\Psi) = L^{-2z}T^{(\ell_g,b)}(\Psi_{L^{-z}})$, $L^{-2z}L_b^+(\Psi_{L^{-z}}) = L^+_{L^z b}(\Psi)$, and $L^{-2z}b^{-3}L^{-z}d = (L^z b)^{-3}d$; apply (a) at $\Psi_{L^{-z}}$.

Step 6: (c). For large s the coupling $g'' = (2/(L^s b))^{1/2}$ is $\le \Gamma$, $L^s b \in I^\circ(g'')$ by Lem. 8.2 (a), as $L^s b \ge 2\Gamma^{-2} \ge 2\gamma^{-2}$ for large s, and $L^s\ell \in \mathfrak{L}(g'')$ (equation YM3_I:(1528)), as $m_{\mathfrak{L}}(g'')$ grows only like a multiple of $\log\log g''^{-2}$ while $m + s$ grows like s. Apply (b) at $(g'', L^s b, L^s\ell)$ to $\Psi_{L^s} \in \mathfrak{T}^1(L^s d;\vartheta)$, and (d), with

$$\mathcal{K}_6(\Psi_{L^s}) = \mathcal{K}_6(\Psi), \qquad (L^s b)^{-1}\mathcal{K}_5(\Psi_{L^s}) = b^{-1}\mathcal{K}_5(\Psi), \qquad L^{2s}(L^s b)^{-3}L^s d = b^{-3}d;$$

the bracket of (b) at $(g'', L^s b, L^s d)$ is $\bar\varepsilon_\infty(L^s d) + 2C_2^{\mathrm{T}}(g'')\vartheta^{-1}L^{2s}bd \le C(g'',\vartheta)(1 + L^s b)(L^s d)^{1/2}$.

Step 7: (e), (f). Two inequalities as in (e), at a pair $\Psi \in \mathfrak{T}^1(d;\vartheta)$, give $|c - c'|\, c_2 b^{-3}\mathcal{K}_5(\Psi) \le (\varepsilon + \varepsilon')(d)\, b^{-3} d|\Psi|$, and $\mathcal{K}_5(\Psi) \ge 10^{-5}\vartheta^2 d|\Psi|$; so $|c - c'|\, c_2\, 10^{-5}\vartheta^2 \le (\varepsilon + \varepsilon')(d) \to 0$. For a compact set J of labels one s serves for all $b \in J$, and the compact set $L^s J$ is covered by finitely many open intervals $I^\circ(g_i)$, $g_i \le \Gamma$, $L^s\ell \in \mathfrak{L}(g_i)$, since each point b'' of $L^s J$ lies in $I^\circ((2/b'')^{1/2})$ (Lem. 8.2 (a)); the least of the ranges and the largest of the tolerances and constants of the g_i serve for all $b \in J$. □

8.1.13. *Proof of Thm. Y. Step 1: a period of labels.* Let $\gamma_o = \min\{\gamma_{\mathrm{e}}, \gamma_W, \gamma_W^{\mathrm{sd}}\}$, $y = \gamma_o^{-2}$, $\Lambda = 2y$, $\hat g_1 = \gamma_o$ and $\hat g_2 = \gamma_o/\sqrt{2}$. As $y \ge \gamma_{\mathrm{e}}^{-2} \ge 4\bar B$ and $y \ge 4$, we have $y > \bar B$, so that

$$I^\circ(\hat g_1) = (y + \bar B,\, Ly + (L-2)\bar B) \supset [2y, Ly), \qquad I^\circ(\hat g_2) = (2y + \bar B,\, 2Ly + (L-2)\bar B) \supset [3y, 2Ly),$$

and, as $L \geq 3$, $[\Lambda, L\Lambda) \subset I^\circ(\hat{g}_1) \cup I^\circ(\hat{g}_2)$. Put $\delta_{\mathrm{c}} = (L\Lambda)^{-1} \min_i \min\{d^L, d_\flat\}(\hat{g}_i, \vartheta)$ and, for $0 < \delta \leq \delta_{\mathrm{c}}$,

$$\varepsilon^{\mathrm{nat}}(\delta) = \max_{i=1,2} \varepsilon^L_{\hat{g}_i,\vartheta}(L\Lambda\delta) + 2 \max_{i=1,2} C_2^{\mathrm{T}}(\hat{g}_i)\, \vartheta^{-1} L^2\Lambda^2\delta :$$

non-decreasing, depending only on $\mathfrak{c}$ and ϑ, and $\varepsilon^{\mathrm{nat}}(\delta) \leq C\delta^{1/2}$ by Thm. Z (a).

Step 2: a dilated label. For S with the label b there is exactly one $z \in \mathbb{Z}$ with $b' = L^z b \in [\Lambda, L\Lambda)$; take i with $b' \in I^\circ(\hat{g}_i)$. As $\hat{g}_i \leq \gamma_W^{\mathrm{sd}}$, the triple $\tau' = (\hat{g}_i, b', \ell_{\hat{g}_i})$ is admissible, and $E_{\tau'} = 2C_2^{\mathrm{T}}(\hat{g}_i)\vartheta^{-1}b'^2 \leq 2C_2^{\mathrm{T}}(\hat{g}_i)\vartheta^{-1}L^2\Lambda^2$.

Step 3: (1). Let $d \leq \delta_{\mathrm{c}} b$, $\Psi \in \mathfrak{T}^1(d;\vartheta)$ and $d' = L^z d = b'd/b \leq L\Lambda\delta_{\mathrm{c}}$. Thm. Z (a) at $(\hat{g}_i, b')$, where $\hat{g}_i \leq \gamma_{\mathrm{e}}$ and $b' \in I(\hat{g}_i)$, for the pair $\Psi_{L^z} \in \mathfrak{T}^1(d';\vartheta)$, with $\mathcal{K}_6(\Psi_{L^z}) = \mathcal{K}_6(\Psi)$, $b'^{-1}\mathcal{K}_5(\Psi_{L^z}) = b^{-1}\mathcal{K}_5(\Psi)$ and $b'^{-1}d' = b^{-1}d$, together with Lem. 8.11, gives

$$\left|b^2\mathsf{S}(\Psi) - c_2[\mathcal{K}_6 + \tfrac{8}{3\pi}b^{-1}\mathcal{K}_5](\Psi)\right| \leq \Big[\varepsilon^L_{\hat{g}_i,\vartheta}\Big(L\Lambda\frac{d}{b}\Big) + 2C_2^{\mathrm{T}}(\hat{g}_i)\vartheta^{-1}L^2\Lambda^2\frac{d}{b}\Big]\frac{d}{b}\,|\Psi| \leq \varepsilon^{\mathrm{nat}}(d/b)\, b^{-1}d\,|\Psi|;$$

divide by b^2. The labels of first-passage data at g exceed g^{-2} by YM3_I:(1531), so $b \geq g^{-2}$, $d/b \leq g^2 d$, and $d \leq d^S$ implies $d \leq \delta_{\mathrm{c}} b$: the law with ε^S and d^S. For the form at a dilated label, let $L^z b \geq 2\Gamma^{-2}$ and $g' = (2/(L^z b))^{1/2}$; then $g' \leq \Gamma$ and $L^z b \in I^\circ(g')$ by Lem. 8.2 (a), as $2\Gamma^{-2} \geq 2\gamma^{-2}$, so $\tau' = (g', L^z b, \ell_{g'})$ is admissible; the same computation, with Thm. Z (a) at $(g', L^z b)$ and Lem. 8.11, gives the tolerance $\varepsilon^L_{g',\vartheta}(L^z d) + E_{\tau'}b^{-1}d$ for $L^z d \leq \min\{d^L, d_\flat\}(g', \vartheta)$, and $E_{\tau'}b^{-1} = 2C_2^{\mathrm{T}}(g')\vartheta^{-1}L^{2z}b$.

Step 4: (2). Lem. 8.12 with $c = \frac{8}{3\pi}$, $\varepsilon(d) = \varepsilon^{\mathrm{nat}}(d/b)$ and $d_7 = \delta_{\mathrm{c}} b$; there $\mathsf{B} = cc_2 b^{-3}\mathcal{K}_5(\Psi) = B_S$, and the lower bound on B_S is eq. 8.i. The first bound at tr and at r gives the bound on $r^{-1}[\mathsf{S}(\Psi_{tr}) - \mathsf{S}(\Psi_r)]$; $\varepsilon^{\mathrm{nat}}(\delta) \leq C\delta^{1/2}$ turns the remainders $o(r)$ into $O(r^{3/2})$; the means converge because their terms do; and at $t = 2$ the numbers $t(t-1)(t^5-1)$ and $t^2 + t + t^6 + t^7$ are 62 and 198.

Step 5: (3). By Lem. 8.12, S_2 and every S_2^ξ fail to be dilation covariant; so $\mathsf{N} = \mathsf{X}$, and $\lambda(\mathsf{N}) = 1$ as λ is a probability measure.

Step 6: (3) from the law at a single triple of any choice. Let $b^\circ \in I^\circ(\Gamma)$. The triple $(\Gamma, b^\circ, \ell_\Gamma)$ is admissible, and clause (iii) of Thm. Z holds there, as $\Gamma \leq \gamma_{\mathrm{e}}$. Prop. 8.17 with $\mathfrak{c}_* = \mathfrak{c}^\circ = \mathfrak{c}$ gives (3) again, and moreover the failure of the three-scale number of S_2 to vanish at pairs and r as small as is wished. This second proof uses the law at one triple and one ϑ only, and no law at the labels of the limits; the triple may belong to any choice. □

Remark 8.19 (The constants in the second proof of Thm. Y (3)). The departure of clause (iv) of Thm. Z dominates explicitly the errors of the comparisons by which the proof of Prop. 8.17 passes from a limit to the least torus, when the label of the limit is first dilated into one period $[\Lambda_W, L\Lambda_W)$. Let $g \leq \Gamma$ and $b \in I^\circ(g)$; let $\tilde{\mathfrak{c}}$ be a second choice of $\mathfrak{c}$ (Lem. 8.14 with $Q = L$), with its number $\tilde{C}_2^{\mathrm{T}}$ and an admissible triple $\tau_2 = (g_2, b_2, \ell_2)$; put $\Lambda_W = 2(\gamma_W^{\mathrm{sd}})^{-2}$, $g_1' = \gamma_W^{\mathrm{sd}}$ and $g_2' = \gamma_W^{\mathrm{sd}}/\sqrt{2}$. Since $(\gamma_W^{\mathrm{sd}})^{-2} \geq \gamma^{-2} > \bar{B}$ (Lem. 8.2 (a)), the computation of Step 1 above gives $[\Lambda_W, L\Lambda_W) \subset I^\circ(g_1') \cup I^\circ(g_2')$; every $b' > 0$ has exactly one dilate $L^z b'$ in $[\Lambda_W, L\Lambda_W)$. The number

$$c_{\max} = \max_{i=1,2} \sup\{2C_2^{\mathrm{T}}(g_i')\, b'^2 :\ b' \in [\Lambda_W, L\Lambda_W) \cap I^\circ(g_i')\} \leq 2L^2\Lambda_W^2 \max_i C_2^{\mathrm{T}}(g_i')$$

is finite, and so is $\mathsf{C}_{\mathrm{pair}} = 2C_2^{\mathrm{T}}(g)b^2 + 4\tilde{C}_2^{\mathrm{T}}(g_2)b_2^2 + 2c_{\max}$. For every admissible triple $\tau' = (g_i', b', \ell')$ with $b' \in [\Lambda_W, L\Lambda_W) \cap I^\circ(g_i')$,

$$\vartheta^{-1}\mathsf{C}_{\mathrm{pair}} \geq 2E_{\tau'} + 2E_{\tau_2} + E_{(g,b,\ell_g)},$$

the sum of the constants of the three comparisons by which the proof of Prop. 8.17, with τ' of this kind and $\tau^\circ = (g, b, \ell_g)$, passes from a limit at the label b', through the second choice, to $T^{(\ell_g, b)}$. Clause (iv) of Thm. Z, at $C = \mathsf{C}_{\mathrm{pair}}$, says that for small d/b the departure $b^2|T^{(\ell_g,b)}(\Psi) - c_2 b^{-2}\mathcal{K}_6(\Psi)|$ exceeds $(2E_{\tau'} + 2E_{\tau_2} + E_{(g,b,\ell_g)})(d/b)^2|\Psi|$ at every pair $\Psi \in \mathfrak{T}^1(d;\vartheta)$.

8.1.14. *Proof of Cor. 8.1. Step 1: pieces.* Let $\delta_0 = \mathrm{dist}(\mathrm{supp} f_0, \mathrm{supp} h_0) > 0$ and $\rho_0 = \delta_0/4$, and let $\phi_i, \chi_k \geq 0$, finitely many, be Lipschitz functions supported in closed balls of radius ρ_0 that lie within $\delta_0/4$ of $\mathrm{supp} f_0$, respectively of $\mathrm{supp} h_0$, with $\sum_i \phi_i = 1$ on $\mathrm{supp} f_0$ and $\sum_k \chi_k = 1$ on $\mathrm{supp} h_0$. With $M_0 = \|f_0\|_\infty + 1$, $f_0 = \sum_i [(f_0 + M_0)\phi_i - M_0 \phi_i]$ is a signed sum of non-negative Lipschitz pieces, and likewise h_0; we drop the pieces that vanish. A piece F of f_0 and a piece H of h_0 lie in balls of radius ρ_0 at the distance $\delta' = \mathrm{dist}(\mathrm{supp} F, \mathrm{supp} H) \geq \delta_0/2 = 2\rho_0$; so $(F, H) \in \mathfrak{T}_{\mathbb{R}^3}(\delta'/2; \vartheta_0)$ for one $\vartheta_0 \in (0,1)$ that serves for all the finitely many pairs of pieces.

Step 2: one pair of pieces. Let s and g'' be as in Step 6 of the proof of Thm. Z. Clauses (b) at $(g'', L^s b, L^s \ell)$ and (d), which hold on all of $\mathfrak{T}_{\mathbb{R}^3}$, applied to the translate by $\tilde{p}$ of $(F, H)_r \in \mathfrak{T}_{\mathbb{R}^3}(r\delta'/2; \vartheta_0)$, with $\mathcal{K}_6((F,H)_r) = \mathcal{K}_6(F,H)$, $\mathcal{K}_5((F,H)_r) = r\,\mathcal{K}_5(F,H)$ and eq. 8.i, give for small r, with a tolerance ε independent of the position,

$$\big|T^{(\ell,b)}((F,H)_r(\cdot - \tilde{p})) - c_2 b^{-2} \mathcal{K}_6(F,H)\big| \leq \Big[\frac{8}{3\pi} + \varepsilon(r\delta'/2)\Big]\, b^{-3} r\, \frac{\delta'}{2}\, \|F\|_\infty \|H\|_\infty .$$

Step 3: (a). $S_2^{(\ell,b)}$ is bilinear on the functions supported in two disjoint closed sets, and so is $\mathcal{K}_6$; summing over the pairs of pieces, with their signs, gives $|S_2^{(\ell,b)}(f_r, h_r) - c_2 b^{-2} \mathcal{K}_6(f_0, h_0)| \leq Cr$, with C independent of p and, by the covering argument of Step 7 of the proof of Thm. Z, the same for the labels of a compact set. For $(\psi_0 \circ \theta_\mu, \psi_0)$ the supports are compact subsets of $\{x_\mu < 0\}$ and $\{x_\mu > 0\}$, so at a positive distance.

Step 4: (b). The supports of $\psi_0 \circ \theta_\mu$ and ψ_0 lie in $\mathbb{B}(-e_\mu, \frac{1}{2})$ and $\mathbb{B}(e_\mu, \frac{1}{2})$, at a distance in $[1,3]$; so $(\psi_0 \circ \theta_\mu, \psi_0) \in \mathfrak{T}^1(\frac{1}{2}; \vartheta_1)$ for some $\vartheta_1 \in (0,1)$, and its translate by $\tilde{p}$ dilated by r lies in $\mathfrak{T}^1(\frac{r}{2}; \vartheta_1)$. Clause (c) of Thm. Z, whose bracket is at most $C(r/2)^{1/2}$, gives the expansion, and $\mathcal{K}_5(\psi_0 \circ \theta_\mu, \psi_0) > 0$ by eq. 8.i.

Step 5: (c). At a finite cutoff $\beta = L^K b$, so that $\partial_b = L^K \partial_\beta$, and ∂_β of a cumulant of observables under $e^{-\beta A}$ is minus the cumulant with the action $A = \sum_p a_p = a_K \mathcal{O}_K(1)$ added; hence $\partial_b S^T_{2;K,\ell,L^K b}(f,h) = -S^T_{3;K,\ell,L^K b}(1,f,h)$, which is the finite-cutoff differentiation identity in the proof of Prop. YM3_I:118.3 for the first derivative. By Prop. YM3_I:118.3, equation YM3_I:(2217), applied to the pair (f_r, h_r), whose supports are disjoint, $b \mapsto S_2^{(\ell,b)}(f_r, h_r)$ is infinitely differentiable on $(0,\infty)$ and $\partial_b S_2^{(\ell,b)}(f_r, h_r) = -S_3^{(\ell,b)}(1, f_r, h_r)$, the derivatives at the finite cutoffs converging to it uniformly on $[b_1, b_2]$; so the integrand is $-\partial_b[b^2 S_2^{(\ell,b)}(f_r, h_r)]$, and the integral equals

$$b_1^2[S_2^{(\ell,b_1)}(f_r,h_r) - c_2 b_1^{-2}\mathcal{K}_6(f_0,h_0)] - b_2^2[S_2^{(\ell,b_2)}(f_r,h_r) - c_2 b_2^{-2}\mathcal{K}_6(f_0,h_0)],$$

each bracket being at most Cr by (a). □

8.1.15. *Scope of the conclusions.* The theorems concern short distances at a fixed coupling; on $\mathbb{R}^3$ they hold for $0 < g \leq \gamma_W$. Not being dilation covariant is not being interacting: three free photon fields together with an independent generalised free field of two-point kernel $\varkappa b^{-3} |x-y|^{-5}$, whose spectral density $\varkappa b^{-3} \frac{2\pi}{3} m^3$ is positive, obey (L) with $c = \frac{8}{3\pi}$ and remainder 0 and have $\mathsf{N} = \mathsf{X}$, and the Gaussian model of Prop. 8.18 has the same two-term law on a torus. The numbers γ_{e}, d_{e}, d^L, δ_{c} and the tolerances are given by existence.

9. Compact gauge groups

Let $G \subset U(N)$ be compact, connected and simply connected, with simple Lie algebra $\mathfrak{g}$. Use the representation to normalize $\langle X, Y\rangle = N^{-1}\,\mathrm{tr}(XY)$ on the Hermitian form of $\mathfrak{g}$, and choose an orthonormal basis $(t_a)_{a=1}^n$, where $n = \dim \mathfrak{g}$. Put $\Omega = \sum_a t_a^2$. The two Casimir numbers in the calculation are

$$C_A = \mathrm{tr}_{\mathfrak{g}^{\mathbb{C}}}(\mathrm{ad}\, H)^2, \qquad C_\rho = N^{-1}\,\mathrm{tr}(\Omega H^2), \qquad \langle H, H\rangle = 1.$$

For a simple algebra these numbers do not depend on H.

Sketch H (Extension to a simple compact group). The proposed extension of the $SU(2)$ construction has the same lattice geometry, with constants depending on G and its representation. Its leading action-density two-point coefficient and relative next perturbative coefficient are

$$c_2(G) = \frac{3n}{16\pi^2}, \qquad c_1(G) = \frac{\varphi_G}{3\pi n}, \qquad \varphi_G = nC_A.$$

The coefficient calculation is local. The full extension still requires the uniform construction at nontrivial regular backgrounds described below. ■□

Sketch of the argument. At the identity configuration, a linear operator of the construction has the form $K \otimes 1_{\mathfrak{g}^{\mathbb{C}}}$, where K depends only on the lattice. The quadratic Gaussian theory therefore consists of n identical colour components. This gives the factor n in $c_2(G)$. At one loop, summing the colour indices reduces the commutator terms to C_A and the representation terms to C_ρ. Let Ψ_k and Ψ_k^G denote the one-loop functionals for $SU(2)$ and G, and write $b[\Psi]$ for the quadratic coefficient. Colour contraction gives

$$b[\Psi_k^G] = \frac{C_A}{8} b[\Psi_k] + \left(\frac{C_\rho}{3} - \frac{C_A}{8}\right)(L^{-k} - L^{-4k}).$$

This also identifies the increments of the Gaussian normalizer. For $SU(3)$ in its defining representation, $(n, C_A, C_\rho) = (8, 18, 8)$, and hence $c_2 = 3/(2\pi^2)$ and $c_1 = 6/\pi$.

The nonlinear construction needs more than this colour count. The logarithm must lie in $\mathfrak{g}^{\mathbb{C}}$, the Haar density must be retained in the local chart, and projection and transpose bounds must be read in their correct norms. Complex domains must control both $|X - 1|$ and $|X^{-1} - 1|$. These changes give local formulas; they do not establish uniform inverses at every regular background.

The nonlinear step is to construct these inverses, critical orbits, localized responses and Gaussian families with bounds uniform in depth and volume, with radii satisfying all the estimates. These uniform constructions remain to be supplied. Once they are available, the multiscale source stability and subsequential limits follow by the same induction. The finite-box perturbative calculation remains an iterated limit: expand at fixed box before increasing its size. For separately acting products of $SU(2)$ factors, the Wilson measures factor exactly, giving a special case without this general-group transfer problem.

9.1. **Compact gauge groups.** For a general compact gauge group, the lattice geometry of the construction is unchanged, but the colour algebra and the matrix chart are not. At the trivial field every linear operator is a scalar lattice operator tensored with the colour space. This observation determines the leading two-point coefficient. At one loop the dependence on the representation is carried by two Casimir numbers. Away from the trivial field, the construction also requires uniform estimates for propagators, critical configurations and localised Gaussian integrals.

We first state the local calculations and a conditional extension of the multiscale construction. The hypotheses for that extension are given explicitly below. They separate the local group calculations from the estimates at general regular backgrounds. The finite-box calculation of the next perturbative coefficient is independent of this extension. For products acting on separate blocks, the measures factor, and the corresponding conclusions follow directly from the $SU(2)$ construction. For quotients, the preliminary-gauge and quantitative local assertions of Prop. 9.33(iii),(iv) retain their separately stated analytic hypotheses.

9.1.1. *Notation and hypotheses.*

Definition 9.1 (The pair)**.** Let $N \geq 2$ and let $G \subset U(N)$ be nontrivial, compact and connected; the *pair* is G with its representation on $\mathbb{C}^N$. Let $\mathfrak{g} = \{Y = Y^* : e^{itY} \in G \text{ for all } t \in \mathbb{R}\}$, $n = \dim \mathfrak{g}$, $\mathfrak{g}^c = \mathfrak{g} + i\mathfrak{g}$, and G^c the group generated by $\exp \mathfrak{g}^c$; $\langle X, Y\rangle = \frac{1}{N}\operatorname{tr} XY$ with the norm $\|X\| = (\frac{1}{N}\operatorname{tr} X^*X)^{1/2}$, and $|\cdot|$ the operator norm; P the projection of $\mathbb{C}^{N\times N}$ onto $\mathfrak{g}^c$

along $\{M : \operatorname{tr} MZ = 0 \text{ for all } Z \in \mathfrak{g}^c\}$; $(t_a)_{a\le n}$ an orthonormal basis of $\mathfrak{g}$ and $\Omega = \sum_a t_a^2$; $\kappa_G = \max_{0\neq Y\in\mathfrak{g}} |Y|/\|Y\|$ and $\tau_G = \sup_{\mathbf{1}\neq V\in G} |V-\mathbf{1}|/\|V-\mathbf{1}\|$;

$G^\pi = \exp(i\{Y \in \mathfrak{g} : |Y| < \pi\})$; r_G the largest r with $\{u \in G : |u-\mathbf{1}| < r\} \subset G^\pi$, and $\rho_G = 2\arcsin\frac{r_G}{2} \in (0,\pi]$, the largest number such that every $u \in G$ whose eigen-angles are below ρ_G in modulus lies in G^π; $z_G = \min\{1, \operatorname{dist}(\mathbf{1}, Z(G)\setminus\{\mathbf{1}\})\}$, and $z_G = 1$ if $Z(G) = \{\mathbf{1}\}$.

With log the power series on $|X-\mathbf{1}| < 1$, let $D^c = \{X \in G^c : |X-\mathbf{1}| < 1,\ \log X \in \mathfrak{g}^c\}$ and $D = D^c \cap G$; further $X^c = (X^*)^{-1}$, $m(X) = \max\{|X-\mathbf{1}|, |X^{-1}-\mathbf{1}|\}$, $c(X) = \frac12(X+X^{-1})$ and $H_X = \frac{1}{2i}(X-X^{-1})$; $|P|$ is the norm of P for $|\cdot|$, and $\varsigma_G = \sup\{|PH_V|/|V-\mathbf{1}| : \mathbf{1} \neq V \in G\}$, $\varsigma_G^D = \sup\{|PH_V|/|V-\mathbf{1}| : \mathbf{1} \neq V \in D\}$, $\kappa' = \kappa'(G) = 4\sup\{|(1-P)H_V|/|V-\mathbf{1}|^3 : \mathbf{1} \neq V \in D\}$, which Lem. 9.12(ii) bounds, and $j_G = \max\{1+\kappa' z_G^2, 3\varsigma_G^D\}$ are numbers of the pair, $(1-P)H_V$ being of third order in $V-\mathbf{1}$.

The action density is $a_p = a(\partial U(p))$ with $a(X) = 1 - \frac{1}{2N}\operatorname{tr}(X+X^{-1})$, which is $1 - \frac1N \operatorname{Re}\operatorname{tr} X$ on G; $F(p) = \xi^{-2} P H_{\partial U(p)}$ and $J = D^{\mathrm T}F$, with ∂U, D and ξ of Def. YM3_I:9.1; the *Casimir density* is $k_p = n - \frac1N \operatorname{Re}\operatorname{tr}(\Omega\,\partial U(p))$.

For simple $\mathfrak{g}$: $C_A = \operatorname{tr}(\operatorname{ad} H)^2$, the trace taken on $\mathfrak{g}^c$, and $C_\rho = \frac1N \operatorname{tr}(\Omega H^2)$ for $H \in \mathfrak{g}$ with $\|H\| = 1$, numbers that do not depend on H (Lem. 9.15); $\varphi_G = \sum_a \operatorname{tr}(\operatorname{ad} t_a)^2 = nC_A$; $\gamma_G = \frac12 C_\rho - \frac{3}{16}C_A$; $c_2(G) = \frac{3n}{16\pi^2}$ and $c_1(G) = \frac{\varphi_G}{3\pi n}$.

For semisimple $\mathfrak{g} = \bigoplus_{i\le r}\mathfrak{g}_i$ with simple ideals $\mathfrak{g}_i$: $n_i = \dim\mathfrak{g}_i$, P_i the orthogonal projection onto $\mathfrak{g}_i$, $\kappa_i = \max_{0\neq Y\in\mathfrak{g}_i}|Y|/\|Y\|$, and $C_A^{(i)}$, θ_i are formed as C_A, C_ρ with $H \in \mathfrak{g}_i$. ■

If $\mathfrak{g}$ is semisimple, then $G \subset SU(N)$: det restricted to G is a character, and its differential tr vanishes on $[\mathfrak{g},\mathfrak{g}] = \mathfrak{g}$. For every G, $|P| \le N^{1/2}$: $\mathfrak{g}^c$ being closed under adjoints, P is the orthogonal projection for $\operatorname{tr} X^*Y$, so that $|PZ| \le (\operatorname{tr} Z^*Z)^{1/2} \le N^{1/2}|Z|$. For $SU(N)$: $n = N^2-1$, $\kappa_G^2 = \tau_G^2 = N-1$, $r_G = 2\sin\frac{\pi}{N}$, $\rho_G = \frac{2\pi}{N}$, $z_G = \min\{1, r_G\}$, which is 1 exactly for $N \le 6$, $\Omega = n\mathbf{1}$, $|P| \le 2$, $\varsigma_G \le 2 - \frac{2}{N}$ (Remark 9.6), $\kappa' \le 2 - \pi/\sqrt3$ (Lem. 9.12(ii)), $\varsigma_G^D = 1$ and $j_G = 3$ (on D, where $|V-\mathbf{1}| < 1$, $|PH_V| \le |H_V| + |(1-P)H_V| \le |V-\mathbf{1}| - \frac18|V-\mathbf{1}|^3 + \frac14\kappa'|V-\mathbf{1}|^3 < |V-\mathbf{1}|$, and the quotient tends to 1 at $\mathbf{1}$), $(C_A, C_\rho) = (2N^2, N^2-1)$, $\gamma_G = \frac{N^2-4}{8}$ and $c_1(G) = \frac{2N^2}{3\pi}$; at $SU(2)$ this is $(C_A, C_\rho) = (8,3)$, $\gamma_G = 0$, $c_1 = \frac{8}{3\pi}$, and at $SU(3)$ it is $(n, C_A, C_\rho, \gamma_G) = (8, 18, 8, \frac58)$, $r_G = \sqrt3$, $z_G = 1$, $\kappa_G^2 = 2$. The blocking factor L, the cutoff K, the levels, $\eta = L^{-k}$ and $a_K = L^{-K}$ are those of Section YM3_I:3; $B_\sharp$ bounds the increments $L^{-1}x_{k-1} - x_k$ of the run, $\bar B = B_\sharp\frac{L}{L-1}$, $\mathsf{G}_\sharp$ bounds the coefficient, and $x_k \ge \gamma^{-2}$, as in Thm. YM3_I:D; the reference coupling and first-passage notation are those of Prop. YM3_I:74.1: $0 < g \le \gamma$, $X_g = L(g^{-2} + B_\sharp)$ and $I(g) = (g^{-2} + \bar B, X_g - \bar B]$, with the admitted sides determined by the geometric room condition there. The short-distance ceiling $\gamma_W \le \gamma$ and the scale and cutoff restrictions are those of Thm. YM3_I:AV, transferred under Def. 9.5 and the construction hypotheses below. Set $\mathcal{K}_q(f,h) = \iint f(x)h(y)|x-y|^{-q}\,\mathrm dx\,\mathrm dy$.

The averaging radius of the $SU(2)$ construction is $\omega_M = \frac{1}{200L^2}$. A *number of L and the pair* is a number that depends on L and on G with its representation, and on nothing else.

Definition 9.2 (Regular classes at the local scale)**.** For a separated sequence $\Omega_0 \supset \cdots \supset \Omega_k$, put $\eta = L^{-k}$. The local index of a bare bond or plaquette $\mathfrak{e}$ meeting Ω_0 is the largest $j \le k$ for which one of its corners lies in Ω_j, and $\mathsf{L}(\mathfrak{e}) = L^{j-k}$. A plaquette is inner if all its corners lie in Ω_0; a bond is full if its four incident plaquettes are inner. These are the scales of Def. YM3_I:9.3.

Both $\mathcal{U}'(r) = \mathfrak{U}'_k(\mathbf{B}, r)$ and $\mathcal{U}(r) = \mathfrak{U}_k(\mathbf{B}, r)$ consist of real G-valued configurations on the inner bare bonds with

$$|\partial U(p) - \mathbf{1}| < r\eta^2 \mathsf{L}(p)^{-2} \qquad (p \text{ inner}).$$

The additional condition, at every full bond b, is respectively

$$|D_U^{\mathrm T}\partial U(b)| < r\eta^2\mathsf{L}(b)^{-3}, \qquad |J(U)(b)| < r\mathsf{L}(b)^{-3}.$$

Here the divergence and current use bare spacing η, and J is the projected current of Def. 9.1. The definitions themselves impose no inclusion between the two classes. ■

Lemma 9.3 (Transpose bound and depth zero)**.** For a complex linear map T of $\mathfrak{g}^c$, let T^{T} be its transpose for $\langle X, Y\rangle = N^{-1}\mathrm{tr}(XY)$. With operator norms induced by $|\cdot|$,

$$|T^{\mathrm{T}}| \le N|T|.$$

At depth zero the determining set consists of all inner bare bonds, and its constraint fibre over a real datum V is $\{V\}$. If each inner plaquette of V has curvature less than ε_1, then

$$|J(V)(b)| < 4\varsigma_G\varepsilon_1 \quad (b \text{ full}), \qquad V \in \mathcal{U}(4\varsigma_G\varepsilon_1).$$

In particular, if $B_3 \ge 4\varsigma_G$ and $\varepsilon_0 \ge B_3\varepsilon_1$, this singleton is the set of critical configurations in the regularity class of radius ε_0. ■

Proof. In the real orthonormal basis (t_a), the trace pairing is the ordinary bilinear coordinate pairing and $\|X\|^2 = \sum_a |x_a|^2$. Singular values give $\|X\| \le |X| \le \sqrt{N}\|X\|$. The Hilbert operator norm is unchanged by transposition, since the transpose is the complex conjugate of the adjoint. Applying the two norm comparisons, first to the output and then to the input, gives

$$|T^{\mathrm{T}}X| \le \sqrt{N}\|T^{\mathrm{T}}\|_{\mathrm{op}}\|X\| = \sqrt{N}\|T\|_{\mathrm{op}}\|X\| \le N|T|\,|X|.$$

At depth zero there are no inward bonds and the averaging map is the identity. Thus the fibre is a singleton and every curve in it is constant. In bare units the current at a full bond is a sum of four transported terms $\pm PH_{\partial V(p)}$. Each has norm less than $\varsigma_G\varepsilon_1$; unitary transport preserves this norm. This proves the current bound. Since G has positive dimension, the ratio defining ς_G tends to one along e^{itY} for any nonzero $Y \in \mathfrak{g}$; hence $\varsigma_G \ge 1$, and the curvature bound also has the claimed radius. The final assertion follows by increasing that radius to ε_0. □

Lemma 9.4 (The averaging row constant)**.** Suppose that the linearized average at a positive-level row c has the counting kernel

$$(\hat{Q}A)(c) = \sum_{b \in \mathsf{P}_c} k(c,b)\xi_c A(b), \qquad |k(c,b)| \le C_{\mathrm{av}}(\xi/\xi_c)^3,$$

where ξ is the bare spacing and $C_{\mathrm{av}} \ge 1$. Use the actual double blocks and neighbourhoods of the averaging map: each P_c has at most $6(\xi_c/\xi)^3$ positive bonds; each bare bond lies in at most eighteen such blocks; and the scale ratios used in either row or column sum are at most L. Then the two linear averaging bounds in equation YM3_I:(1045) hold with

$$C_Q = 18NL^3C_{\mathrm{av}}.$$

At level zero the average is the identity. The choice $C_{\mathrm{av}} = 3$ therefore gives $C_Q = 54NL^3$, including $108L^3$ at $SU(2)$. For a general pair this numerical choice requires the displayed kernel estimate with $C_{\mathrm{av}} = 3$ on the admitted averaging domain; it is a quantitative averaging input in the transferred construction. ■

Proof. For a row over y, the row sum is bounded by $6LC_{\mathrm{av}} \max_{N_L(y)} |A|$, with the dimensionless field norms of equation YM3_I:(1045). The transpose uses the bare weight ξ^3 and the row weight ξ_c^{-1}. The latter cancels the factor ξ_c in the counting kernel, giving

$$(\hat{Q}^{\mathrm{T}}\varphi)(b) = \xi^{-3} \sum_{c : b \in \mathsf{P}_c} k(c,b)^{\mathrm{T}}\varphi(c).$$

If b is over y, Lem. 9.3 and the eighteen-block count give

$$\xi_y^3|\hat{Q}^{\mathrm{T}}\varphi(b)| \le \sum_{c : b \in \mathsf{P}_c} NC_{\mathrm{av}}(\xi_y/\xi_c)^3|\varphi(c)| \le 18NL^3C_{\mathrm{av}} \max_{N_L(y)} |\varphi|.$$

The same constant dominates the row bound and the identity rows. This is the row/column count following YM3_I:(1045), with the transpose factor N retained. □

Definition 9.5 (Transfer to G)**.** The transfer to G of a definition or a statement of the $SU(2)$ construction is made by kind.

(1) What stands for one object is replaced alike at all places: $N \times N$ matrices, $\frac{1}{N}\mathrm{tr}$, G, G^c, $\mathfrak{g}$, $\mathfrak{g}^c$ for 2×2 matrices, $\frac{1}{2}\mathrm{tr}$, $SU(2)$, $SL(2,\mathbb{C})$ and their algebras; colour indices run to n; a_p, F and J are those of Def. 9.1, and $a(X)$ stands for every trace of a group-valued argument at complex fields, the gauge-fixing term included; $P[\frac{1}{2}\{c(X),\theta\}]$ stands where $SU(2)$ has $\frac{1}{2}(\mathrm{tr}\, X)\theta$; $|\cdot|$ stands in suprema, radii, series, domains and tests, $\|\cdot\|$ and $\langle\cdot,\cdot\rangle$ in Hilbert sums, pairings and quadratic forms; the cut-off of the bond shift (the cutoff preceding Lem. YM3_I:53.3), $\vartheta_{\mathrm{co}}(|Y|/\delta_k)$, is $\vartheta_{\mathrm{co}}(\frac{4}{3} + \lambda(\nu(Y)/\delta_k - \frac{4}{3}))$, with $\nu(Y) = (\frac{1}{N}\mathrm{tr}\, Y^{2p})^{1/2p}$, p an integer with $w = N^{1/2p} < \frac{5}{4}$, and $\lambda = \frac{w}{5-4w}$, and $\lambda\|\vartheta'_{\mathrm{co}}\|_\infty$ stands for $\|\vartheta'_{\mathrm{co}}\|_\infty$ in every condition and bound; as $\nu \le |\cdot| \le w\nu$, it is smooth and invariant, 0 on $|Y| \le \frac{4}{3}\delta_k$ and 1 on $|Y| \ge \frac{5}{3}\delta_k$ (Lem. 9.14(ii)); and "a constant depending only on L and the pair" stands for "a number of L".

(2) A numeral that counts colours, compares $|\cdot|$ with $\|\cdot\|$, comes from a transpose or comes from the projection P takes its value for G: n, $\frac{3n}{2}$, $3n$ for 3, $\frac{9}{2}$, 9; the mean over n colours replaces the mean over three colours in the quadratic coefficient of a local term; N replaces the 2 of $|T^{\mathrm{T}}| \le 2|T|$ (Lem. 9.3); $\sqrt{N}$ for $\sqrt{2}$; a factor $N^{1/2}$ where an inverse bounded in ℓ^2 is read in $\sup|\cdot|$; $4\varsigma_G$ for the 4 at depth 0 (Lem. 9.3), which is at most $8(1 - \frac{1}{N})$ for $SU(N)$; $3|P|$ for the 3 of the bound $3\alpha_0$ of the current at an inner bond that is not full (Prop. YM3_I:30.1), which is at most 6 for $SU(N)$, and, under $\alpha_0 \le z_G$, j_G for the 3 of that lemma, which is 3 for $SU(N)$ (Def. 9.1, Remark 9.6, Lem. 9.12(ii)); $6\hat{\varsigma}_G$ for the 12 of the bound $\xi^3|J_W(U)(b)| \le 12a$ of the $SU(2)$ construction on the current J_W of a region W at a bond b, at G^c-valued fields whose plaquette variables at b lie within a of $\mathbf{1}$ and whose bond variables there have condition number $|u||u^{-1}| \le 2$, $\hat{\varsigma}_G = \sup|PH_g|/|g - \mathbf{1}|$ over the plaquette variables $g \ne \mathbf{1}$ of such fields, which is at most 2 at $SU(2)$ and at most $\frac{17}{2}|P|$ for semisimple $\mathfrak{g}$ (Remark 9.6); the factor $|P|$ in the kernel and Weitzenböck bounds of the Hessian in $|\cdot|$ (Lem. 9.19); for the averaging row constant of Lem. 9.4, $C_Q = 18NL^3C_{\mathrm{av}}$, which is $54NL^3$ when the normalized kernel bound $C_{\mathrm{av}} = 3$ is supplied, and $108L^3$ at $SU(2)$; the constants formed from $6LC$ and $12LC$ move with this value; $z < (\frac{5}{4})^n L^{3n}$ for the bound $z < 2L^9$ of Lem. YM3_I:65.3; and an inequality that absorbs a colour factor, such as $\frac{3}{4} \le 1$ in the determinant counting argument of Lem. 9.25, must hold anew with the new numbers.

(3) The following statements of the $SU(2)$ construction are false for G as worded, and their transfers are the forms named, each with the statement of this section that bears it. The inclusion $\mathcal{U}'(r) \subset \mathcal{U}(r)$ of the two regular classes $\mathcal{U}'(r) = \mathfrak{U}'_k(\mathbf{B}, r)$ and $\mathcal{U}(r) = \mathfrak{U}_k(\mathbf{B}, r)$ of Def. 9.2 (the $SU(2)$ inclusions are Lem. YM3_I:9.4 and equation YM3_I:(465)), the first bounding $D_U^{\mathrm{T}}\partial U$ and the second the current, becomes $\mathcal{U}'(r) \subset \mathcal{U}(\kappa_G r)$, and for $r \le z_G$ the sharper $\mathcal{U}'(r) \subset \mathcal{U}((1 + \kappa' r^2)r)$, the form that Lem. 9.12(ii) states for $SU(N)$ (Remark 9.6). The second inclusion of the two classes in the $SU(2)$ construction, $\mathcal{U}(r) \subset \mathcal{U}'(r + r^2)$, becomes $\mathcal{U}(r) \subset \mathcal{U}'(r + r^2 + \kappa' r^3)$ for every G and $r \le z_G$, which at $SU(2)$, where $\kappa' = 0$ and $z_G = 1$, is the inclusion of the $SU(2)$ construction for $r \le 1$: the plaquette variables of a field of $\mathcal{U}(r)$ or of $\mathcal{U}'(r)$ lie within z_G of $\mathbf{1}$, so in D (Lem. 9.9(iii)), where $|(1-P)H_V| \le \frac{1}{4}\kappa'|V - \mathbf{1}|^3$ (Def. 9.1), and the two bounds $\eta|(D^{\mathrm{T}}\partial U)(b)| \le \rho^2 + \eta^3|J(b)| + \kappa'\rho^3$ and $\eta^3|J(b)| \le \eta|(D^{\mathrm{T}}\partial U)(b)| + \kappa'\rho^3$ at a bond b whose plaquette variables lie in D within $\rho \le z_G$ of $\mathbf{1}$, which Lem. 9.12(ii) states for every G, give the two inclusions at each full bond, as in the proof of that lemma for $SU(N)$.

The $SU(2)$ null-set exception $G \setminus G^\pi = \{-\mathbf{1}\}$ becomes the indicator of G^π in the integrand (Lem. 9.14(i)); and where the $SU(2)$ construction uses $|\log j| < 1$ on the ball of radius $\frac{3}{8}$, the bound $|\log j_G(Y)| \le 0.173\, n|Y|^2$ stands, its constant entering a maximum (Lem. 9.14(i)).

"$|X-\mathbf{1}|<1$" becomes "$X\in D^c$" wherever a series logarithm is to lie in $\mathfrak{g}^c$ (Lem. YM3_I:5.1; Lem. 9.9(iii)). The mean of a family of diameter below $\frac{1}{40}$ (Prop. YM3_I:4.3) takes the proviso that the quotients lie in D^c, which holds if the diameter is below z_G (Lem. 9.9(iv)). The second assertion of Lem. YM3_I:14.2 takes the proviso that every eigen-angle of each W_x is below ρ_G in modulus, or that W_x is a product of exponentials $e^{i\gamma_j}$, $\gamma_j\in\mathfrak{g}^c$, $\sum_j|\gamma_j|<\ln 2$ (Lem. 9.11(ii)). "$u\neq-\mathbf{1}$" becomes "$u\in G^\pi$" (Lems. 9.9(ii), 9.11(ii)). $|X-\mathbf{1}|$ becomes $m(X)$ in conditions on complex fields that are to be kept by $X\mapsto X^c$ and wherever $|X^{-1}-\mathbf{1}|=|X-\mathbf{1}|=|X^*-\mathbf{1}|$ was used; "the conjugate point lies in the domain if and only if the point does" (Lem. 5.23) and "the function z_c of the elimination factor is formed at $(U^*)^{-1}$ if and only if at U" take the two-sided guard $m(X)$; and the symmetric part Q^s of a complex domain is replaced by its real-connected part Q^r, or moved into one, the conditions of Lem. 9.21(iii) on the radii standing then among the conditions on the constants of §9.1.1(K) (Lems. 9.20(ii), 9.21(iii), 9.14(iv)).

The averaging radius ω_M (equation YM3_I:(30)) becomes $\omega^G=\min\{\omega_M,z_G\}$, or the floor $200L^2z_G\geq 1$ is laid once (Lem. 9.11(i)).

The radius $\frac{1}{2}$ that is to keep $\det T$ off $(-\infty,0]$ (the three-dimensional determinant argument in the proof of Prop. YM3_I:50.2) becomes $r_n=\sin\frac{\pi}{2n}$, with the numbers of Lem. 9.14(iv) in the chains in the proof of Prop. YM3_I:50.2 and in the elimination-factor estimates of Lem. 9.14(iii), (iv). The full constant C_{mob} in the local determinant bound takes the factor $\max\{1,n2^{-11}\}$, as shown in Lem. 9.25; its trace-dependent part is evaluated with n colours.

The comb gauge (Lem. YM3_I:14.4) takes the guard $\varpi<z_G$ (Lem. 9.12(iii)), and Lem. 9.10 the guard $\varpi_y(W)<z_G$, under which its contour quotients lie in D^c and their means in G (Lem. 9.9(iii), (iv)). The quantitative cube-gauge input (Lem. 9.12(iii)) takes constants c, B of the pair (Lem. 9.12(iii)). The orthogonal field maps in the covariance argument following Prop. YM3_I:47.4 are required here to keep $|\cdot|$, as $\mathrm{Ad}(G)$ does (Prop. 9.24). The membership in $\mathcal{U}'(2s)$, $s=2\alpha_0+3\alpha_2$, of the curvature/current conversion in Lem. 9.12(ii), with both of its quantitative hypotheses, takes the guard $\alpha_0+3\alpha_2<z_G$, under which the plaquette variables there, within $\alpha_0+3\alpha_2$ of $\mathbf{1}$, lie in D (Lem. 9.9(iii)); the bound of Lem. 9.12(ii) at a full bond b, with $\rho=(\alpha_0+3\alpha_2)\eta^2\mathsf{L}(b)^{-2}$ and the bound $s\mathsf{L}(b)^{-3}$ of the current there, then gives for every G the membership in $\mathcal{U}'(s+s^2+\kappa's^3)$, which lies in $\mathcal{U}'(2s)$ as soon as $s(1+\kappa's)\leq 1$, so at $SU(N)$ for $s\leq\frac{1}{3}$, κ' being at most $2-\pi/\sqrt{3}$ there.

The bound $|\mathrm{Bl}(\mathsf{g},\mathsf{f})|\leq\|\mathsf{g}\|_2\|\mathsf{f}\|_2$ that the $SU(2)$ construction draws by polarisation from $|\mathrm{Qf}(\mathsf{a})|\leq\|\mathsf{a}\|_2^2$, Qf a quadratic form on $\mathfrak{g}^c$-valued functions a on bonds, Bl its polar form, $\|\mathsf{a}\|_2^2=\sum_b|\mathsf{a}(b)|^2$, at real g, f, takes the factor κ_G^2, and keeps the factor 1 where $|X+iY|^2\leq|X|^2+|Y|^2$ for all $X,Y\in\mathfrak{g}$ (Remark 9.6, Lem. 9.9(i)).

For $G\subset SU(N)$ the energy of block averages takes $\frac{1}{4\tau_G^2}$ and the threshold $e_*=e_L/(2\tau_G^2)$, or $N-1$ in the place of τ_G^2, and the lower bounds of the action on large-field regions take the factor τ_G^{-2} (Lem. 9.13).

The assembly of the bond maps (Lem. YM3_I:53.3) takes r_G for 2, with the condition $\mathsf{c}_L c_4\leq 2Lr_G$ (Prop. 9.22(iv)).

The regularity radius ε_{v} of the critical configurations (Def. YM3_I:9.5) takes the ceilings $\varepsilon_{\mathrm{v}}\leq a_G$, with a fixed admitted positive orbit radius a_G from §9.1.6, $\varepsilon_{\mathrm{v}}\leq\varepsilon_G$, ε_G the number of the uniqueness clause of §9.1.1(V), and $\varepsilon_{\mathrm{v}}\leq L^2r_G$ (Prop. 9.26(f)).

The π of the ceiling $\alpha_1<\pi L^j$, $1\leq j\leq k$, under which the local terms are continuous on the analyticity cores (Lem. 5.33), which a proof of the chain from level k to level $k+1$ quotes, becomes ρ_G, the bound on the eigen-angles under which an element of G lies in G^π (Def. 9.1); the ceiling reads $\alpha_1<\rho_GL^j$ and is met by the row $\alpha_1<\rho_GL$ of the table (Prop. 9.24).

These rules define the transferred statement. They do not, by themselves, prove it. When no correction in (3) applies, the convention uses (1) and (2) alone. ■

Remark 9.6 (Where the literal transfer fails). A rule that replaced the signs alike and kept every numeral would not transfer the proofs, as the following instances show. Each names the statement of the $SU(2)$ construction that it concerns, the least N at which the literal transfer fails for $SU(N)$, or the least n, and the statement of this section that proves the form that holds.

Numerals. The 2 of $|T^{\mathrm{T}}| \le 2|T|$ (Lem. 9.3) is N, sharply at $SU(N)$, $\kappa^2 = N-1$ being attained by general kernels (Prop. 9.26(g)). The 4 of Lem. 9.3 at $SU(2)$, without its general-group factor ς_G, fails from $N = 6$: let Ω_0 hold a bare bond b with its four plaquettes, and let the datum be $X_N = \operatorname{diag}(e^{ia}, e^{-ia}, \dots, e^{-ia}) \in SU(N)$, $a = \frac{2\pi}{N-2}$, at b and $\mathbf{1}$ at the other bonds; with $\varpi = |X_N - \mathbf{1}| = 2\sin\frac{\pi}{N-2}$ its plaquette variables are within ϖ of $\mathbf{1}$, while its current at b is $4PH_{X_N}$ up to sign, of norm $4r_N\varpi$, $r_N = (2-\frac{2}{N})\cos\frac{\pi}{N-2} \ge 1.178$ for $N \ge 6$; so for $\varepsilon_1 \in (\varpi, r_N\varpi]$ the datum is ε_1-regular and does not lie in $\mathfrak{U}(\mathbf{B}; 4\varepsilon_1)$. The current at b is a sum of four transported terms $\pm PH_{\partial V(p)}$, so the lemma holds for every G with $4\varsigma_G$ for 4. For $SU(N)$ this is at most $8(1-\frac{1}{N})$: for $V \in SU(N)$ with eigen-angles θ_j the eigenvalues of PH_V are $\sin\theta_j - \frac{1}{N}\sum_l \sin\theta_l$, so that $|PH_V| \le (2-\frac{2}{N})\max_j|\sin\theta_j| \le (2-\frac{2}{N})|V-\mathbf{1}|$ (Prop. 9.26(g)). The value $8(1-\frac{1}{N})$ is that of $SU(N)$. For the spin representation of $Spin(2m+1)$ on $\mathbb{C}^{2^m}$, simple, simply connected and with $z_G = 1$, let $(h_a)_{a\le m}$ be the orthonormal basis of a Cartan subalgebra that acts on a basis of weight vectors by the diagonal matrices of signs; P commutes with the adjoint action of the maximal torus, so it maps the diagonal matrices to the Cartan subalgebra, and at $V = \exp(i\sum_a x_a h_a)$

$$PH_V = \sum_a \sin x_a \prod_{b\ne a} \cos x_b \; h_a, \qquad |PH_V| = \sum_a |\sin x_a| \prod_{b\ne a} |\cos x_b|,$$

every sign vector being a weight. At $x_a = x$, $\tan^2 x = \frac{1}{m-1}$, this is $\sqrt{m}(1-\frac{1}{m})^{(m-1)/2}$, which exceeds 4 from $m = 43$; as $|V-\mathbf{1}| \le 2$, $\varsigma_G > 2$ there. The inequality $\frac{3}{4} \le 1$ by which the proof of Lem. 9.25 absorbs a colour factor reads $\frac{n}{4} \le 1$, false from $n = 5$, and $(\frac{5}{4})^3 < 2$ behind the bound $2L^9$ of Lem. YM3_I:65.3 fails from $n = 4$ (Prop. 9.26(d)). The 12 of the bound $\xi^3|J_W(U)(b)| \le 12a$ on the current of a region at G^c-valued fields whose bond variables have condition number at most 2 (Def. 9.5(2)) fails for every $N \ge 5$: on the thirteen bonds of the four plaquettes at b a field with values in $SL(N,\mathbb{R})$ of condition number at most 2, products of plane rotations and of diagonal matrices $\operatorname{diag}(d_1, d_2, d_3, d_3, d_3, 1, \dots, 1)$ with rational entries, and $\mathbf{1}$ at the other bonds, has its four plaquette variables at b within $a = \frac{103}{100}$ of $\mathbf{1}$ and $\xi^3|J_W(U)(b)| > 12.98 > 12a$. The current at b is ξ^{-3} times a sum over the two directions ν of the plaquettes at b, each term being $\pm PH_g$ at the plaquette variable g of one plaquette of direction ν and $\mp u^{-1}PH_{g'}u$ at that of the other, u a bond variable, with $|u^{-1}Xu| \le |u||u^{-1}||X| \le 2|X|$; so the bound holds for every semisimple $\mathfrak{g}$ with $2(1+2)\hat{\varsigma}_G = 6\hat{\varsigma}_G$ for 12. At $SU(2)$, $g - g^{-1} = 2(g-\mathbf{1}) - \operatorname{tr}(g-\mathbf{1})\mathbf{1}$ gives $|H_g| \le 2|g-\mathbf{1}|$, so that $\hat{\varsigma}_G \le 2$ and $6\hat{\varsigma}_G \le 12$; for semisimple $\mathfrak{g}$, $G^c \subset SL(N,\mathbb{C})$ (Lem. 9.9(iii)), so a bond variable u of condition number at most 2 has $|u| \le 2$ and $|u^{-1}| \le 2$, a plaquette variable g has $|g^{-1}| \le 16$, and $|PH_g| \le \frac{1}{2}|P|(|g-\mathbf{1}| + |g^{-1}||g-\mathbf{1}|) \le \frac{17}{2}|P||g-\mathbf{1}|$.

One matrix and the logarithm. The identity $X + X^{-1} = (\operatorname{tr} X)\mathbf{1}$ of YM3_I:(70), which (1) transfers to $X + X^{-1} = \frac{2}{N}(\operatorname{tr} X)\mathbf{1}$, holds for every $X \in G$ if and only if $X + X^{-1}$ is scalar for every $X \in G$, that is, if and only if G is conjugate to $\{u \otimes \mathbf{1}_k : u \in SU(2)\}$ (for $n \ge 2$; Lem. 9.9(i)); with it fail at $SU(3)$ the identity $a_p = \frac{1}{2}|\partial U(p) - \mathbf{1}|^2$, the membership $H_X \in \mathfrak{g}^c$, $|Y| = \|Y\|$ on $\mathfrak{g}$ and $|X^{-1} - \mathbf{1}| = |X - \mathbf{1}|$ on G^c (at $X = \operatorname{diag}(\sqrt{2}, \sqrt{2}, \frac{1}{2})$); P, the two norms, $m(X)$ and G^π take their places (Lems. 9.9(i), 9.20(ii)). "$|X-\mathbf{1}| < 1$ gives $\log X \in \mathfrak{g}^c$" (Lem. YM3_I:5.1) fails for $N \ge 7$, at $X = e^{2\pi i/N}\mathbf{1}$; "$X \in D^c$" takes its place (Lem. 9.9(iii)). "A family in G of diameter less than $\frac{1}{40}$ has its mean in G" (Prop. YM3_I:4.3) fails for $N \ge 252$: the mean of $\mathbf{1}$ and $e^{2\pi i/N}\mathbf{1}$ is $e^{\pi i/N}\mathbf{1}$, of determinant -1; the quotients in D^c take its place (Lem. 9.9(iv)).

The averaging radius ω_M of equation YM3_I:(30) fails for $N \geq 212372$ at $L = 13$; ω^G, or the floor $200L^2 z_G \geq 1$, takes its place (Lem. 9.11(i)). "$|\mathbf{1} - T| \leq \frac{1}{2}$ keeps $\det T$ off $(-\infty, 0]$" (the three-dimensional determinant argument in the proof of Prop. YM3_I:50.2) fails for $n \geq 6$; the radius r_n takes its place (Lem. 9.14(iv)).

Fields. The inclusion $\mathcal{U}'(r) \subset \mathcal{U}(r)$ of the two regular classes (Lem. YM3_I:9.4) fails for $N \geq 3$ at every small r (Lem. 9.12(ii)). For every G it holds at the radius $\kappa_G r$: the current is $J(U) = \eta^{-2} PT$ with $T = \frac{1}{2i}(S - S^*)$ and $S = D_U^{\mathrm{T}} \partial U$, the transports commuting with the adjoint, and P is the orthogonal projection onto $\mathfrak{g}$ on hermitian matrices, so that $|PT| \leq \kappa_G \|PT\| \leq \kappa_G |T| \leq \kappa_G |S|$; and for $r \leq z_G$ it holds at the radius $(1 + \kappa' r^2) r$, by the second bound at a bond of Lem. 9.12(ii), the plaquette variables lying in D (Def. 9.5(3)). The bound $3\alpha_0$ of the current at an inner bond that is not full (Prop. YM3_I:30.1) fails for $N \geq 8$ (Lem. 9.12(ii)). It holds for every G with $3|P|\alpha_0$, as $|PH_V| \leq |P||H_V| \leq |P||V - \mathbf{1}|$ at unitary V, which is at most $6\alpha_0$ at $SU(N)$; and under $\alpha_0 \leq z_G$, where the plaquette variables lie in D, the bound of that lemma holds with $j_G \alpha_0$: at a full bond by the inclusion at the radius $(1 + \kappa' \alpha_0^2)\alpha_0$, and at an inner bond b that is not full, where $\mathsf{L}(b)^3$ times the current is a sum of at most three transported terms $\pm P H_{\partial U(p)}$, with $3\varsigma_G^D \alpha_0$. The 3 fails under $\alpha_0 \leq z_G$ at the pair of $SU(2)$ on $\mathbb{C}^4 \oplus (\mathbb{C}^2)^{100}$, spin $\frac{3}{2}$ and a hundred copies of spin $\frac{1}{2}$, which is simply connected, with simple Lie algebra and $z_G = 1$: with J_3 the image of $\frac{1}{2}\sigma_3$, $\operatorname{tr} J_3^2 = 55$, and $w = e^{itJ_3}$ one has $PH_w = c(t) J_3$, $c(t) = \operatorname{tr}(J_3 \sin t J_3)/\operatorname{tr} J_3^2 = \frac{1}{55}(3 \sin \frac{3t}{2} + 101 \sin \frac{t}{2})$, so that $|PH_w| = \frac{3}{2} c(t) = \frac{3}{2} t - 0.1034\, t^3 + O(t^5)$ exceeds $|w - \mathbf{1}| = 2 \sin \frac{3t}{4} = \frac{3}{2} t - 0.1406\, t^3 + O(t^5)$ for small $t > 0$; with Ω_0 not the whole torus, the field U_0 equal to w at an inner bond b in the outermost layer of Ω_0 along a face, three of whose four plaquettes are inner, and to $\mathbf{1}$ at the other bonds lies in $\mathfrak{U}'_k(\mathbf{B}, \alpha_0)$ for $\alpha_0 > |w - \mathbf{1}|$, and $\mathsf{L}(b)^3 |J_{\Omega_0}(U_0)(b)| = 3|PH_w|$, above $3\alpha_0$ for $\alpha_0 < |PH_w|$.

The comb gauge of a box of Lem. YM3_I:14.4 needs $\varpi < z_G$ for $N \geq 7$, and the form of that clause with some gauge of the box in the place of the comb, that $2s\varpi < 2$ alone gives a gauge in which $W^h = e^{ia^0}$ with a^0 in $\mathfrak{g}$ and $\sup |a^0| \leq \pi s \varpi$, fails for $N \geq 14$; the gauge on a cube of Thm. YM3_I:F has the quantitative general-group counterpart assumed in Lem. 9.12(iii). That $G \setminus G^\pi$ is a null set, as it is for $SU(2)$, fails for $N \geq 3$; the indicator of G^π takes its place (Lem. 9.14(i)). The cut-off $\vartheta_{\mathrm{co}}(|Y|/\delta_k)$ of the cutoff preceding Lem. YM3_I:53.3 is not differentiable for rank at least 2; the cut-off $\vartheta_{\mathrm{co}}(\frac{4}{3} + \lambda(\nu(Y)/\delta_k - \frac{4}{3}))$ of Def. 9.5(1) takes its place, with $\lambda \|\vartheta'_{\mathrm{co}}\|_\infty$ for $\|\vartheta'_{\mathrm{co}}\|_\infty$; at $SU(3)$, $p = 3$ and $\lambda = 6.1194$ (Lem. 9.14(ii)). The orthogonal colour maps in the covariance argument following Prop. YM3_I:47.4 keep the ball of $|\cdot|$ for $SU(2)$, and general rotations of $\mathfrak{g}$ do not keep it for rank at least 2; the rotations are to keep $|\cdot|$, as $\mathrm{Ad}(G)$ does, and covariance under $\mathrm{Ad}(G)$ is what the proofs of the induction use (Prop. 9.24).

The bound $|\mathrm{Bl}(\mathsf{g}, \mathsf{f})| \leq \|\mathsf{g}\|_2 \|\mathsf{f}\|_2$ that the $SU(2)$ construction draws by polarisation from $|\mathrm{Qf}(\mathsf{a})| \leq \|\mathsf{a}\|_2^2$ (Def. 9.5(3)) fails at $SU(3)$, of rank 2: on two bonds b_1, b_2 let x, y, z be the diagonal entries of $\mathsf{a}(b_1)$, $\zeta_\pm = \frac{1}{2}(\mathsf{a}(b_2)_{11} - \mathsf{a}(b_2)_{22} \pm i\mathsf{a}(b_2)_{12} \pm i\mathsf{a}(b_2)_{21})$, $\psi_1 = \frac{1}{2}(e^{-i\pi/4}\zeta_+ + e^{i\pi/4}\zeta_-)$, $\psi_2 = \frac{1}{2i}(e^{i\pi/4}\zeta_- - e^{-i\pi/4}\zeta_+)$ and $\mathrm{Qf}(\mathsf{a}) = \frac{1}{2}(x - z)\psi_1 + \frac{3}{4}\psi_1^2 + \frac{3\sqrt{6}}{4} y \psi_2 - \frac{1}{8}\psi_2^2$; then $|\mathrm{Qf}(\mathsf{a})| \leq |\mathsf{a}(b_1)|^2 + |\mathsf{a}(b_2)|^2$ for all $\mathfrak{sl}(3, \mathbb{C})$-valued a, while the real $\mathsf{g} = (\frac{3}{5}\mathrm{diag}(1, -1, 0), \frac{4}{5}\sigma_3 \oplus 0)$ and $\mathsf{f} = (\frac{3}{5}\mathrm{diag}(0, 1, -1), \frac{4}{5}\sigma_1 \oplus 0)$ at (b_1, b_2), σ_1 and σ_3 the Pauli matrices, have $\|\mathsf{g}\|_2 = \|\mathsf{f}\|_2 = 1$ and $\mathrm{Bl}(\mathsf{g}, \mathsf{f}) = \frac{1}{25}(7 + 3\sqrt{2} + 9\sqrt{3}) > 1.07$. For every G the bound holds with κ_G^2: as $\|\cdot\| \leq |\cdot| \leq \kappa_G \|\cdot\|$ on $\mathfrak{g}$ (Lem. 9.9(i)) and $\|\cdot\|$ obeys the parallelogram law, $4|\mathrm{Bl}(\mathsf{g}, \mathsf{f})| \leq \|\mathsf{g} + \mathsf{f}\|_2^2 + \|\mathsf{g} - \mathsf{f}\|_2^2 \leq 2\kappa_G^2(\|\mathsf{g}\|_2^2 + \|\mathsf{f}\|_2^2)$, and $\mathsf{g} \mapsto t\mathsf{g}$, $\mathsf{f} \mapsto t^{-1}\mathsf{f}$ gives $|\mathrm{Bl}(\mathsf{g}, \mathsf{f})| \leq \kappa_G^2 \|\mathsf{g}\|_2 \|\mathsf{f}\|_2$, which is the bound of the $SU(2)$ construction at $SU(2)$; and where $|X + iY|^2 \leq |X|^2 + |Y|^2$ for all $X, Y \in \mathfrak{g}$ it holds with 1, by $4i\,\mathrm{Bl}(\mathsf{g}, \mathsf{f}) = \mathrm{Qf}(\mathsf{g} + i\mathsf{f}) - \mathrm{Qf}(\mathsf{g} - i\mathsf{f})$ and $|X - iY| = |X + iY|$ for hermitian X, Y.

Complex fields. The bare trace of the gauge-fixing term (the definition of Ψ^{G} in Section YM3_I:51, p. 393) is not real at real fields for $N \geq 3$; $a(X)$ takes its place (Prop. 9.23). "The conjugate point lies in the domain if and only if the point does", which for $SU(2)$ rests on the isometry Θ of Lem. 5.23, fails for $N \geq 3$; the two-sided guard $m(X)$ takes its place (Lem. 9.20(ii), (iv)).

For $SU(N)$ with $N \le 6$ the forms with D^c coincide with the statements of the $SU(2)$ construction, and the proviso on means of Lem. 9.9(iv) is void for $N \le 251$.

What remains in the general-group construction. Let G be simply connected with $\mathfrak{g}$ simple, $L \ge 13$ odd and $200L^2 z_G \ge 1$; constants are constants depending only on L and the pair, uniform in the volume. The following discussion identifies the analytic work still needed to extend the complete multiscale construction. The local matrix identities do not establish these estimates.

Propagators and covariances (P).. One must establish the uniform linear estimates at regular backgrounds required below. Their $SU(2)$ forms are Thms. YM3_I:G, YM3_I:I and YM3_I:L, and Prop. YM3_I:31.1: the constrained inverse and its local expansion, the real regional inverse, the fluctuation covariance, and the current multiplier. For G these are assumptions on the natural field classes and on every regional geometry used by the construction; the cited reduced-row theorems alone do not assert their extension to other determining sets. The quantitative local gauges in Lem. 9.12(iii), and the local current, difference, commutator, action-remainder and layer estimates used to construct these inverses, are also required wherever used. The required inverse bounds include: for a separated sequence of regions with M above a floor and fields of the natural class with small parameters, the operator K of the random-walk expansion is invertible, $\mathsf{K}^{-1} = \sum_w E_w$ over finite sequences w of pieces, local and holomorphic, with

$$|E_w \iota g|_y \le 8 B_4 \, q_M^{|w|} \, \sup e^{-\delta_2 \sum_i d(y_i, y_{i+1})} |g|, \qquad q_M = C_q M^{-1} e^{-\delta_2 M/(8L)},$$

the rate δ_2 being that of the $SU(2)$ construction and the floor on M a constant depending only on L and the pair; and for G-valued U in the class $\mathfrak{U}'_k(\mathbf{B}, \alpha)$ of Def. 9.2, α small, $\Delta_U + Q'^{*} \mathsf{a} Q'$ is positive definite, with an inverse G' that obeys $|G' f|_y \le B_{\mathrm{ps}} \xi_y^2 e^{-\delta_{\mathrm{ps}} d(y,y')} \sup |f|$, and the like for its covariant derivative, for $(Q' G'^2 Q'^{*})^{-1}$ and for the gauge-fixed operator.

Critical configurations off the trivial field (V).. One must construct a regular critical orbit over every admitted regular datum, prove uniqueness of that orbit at one fixed positive regularity radius and improvement to a constant times the datum radius, and establish holomorphic response with exponential decay uniformly in depth and volume. The $SU(2)$ reduced-data models are Thm. YM3_I:E, Prop. YM3_I:37.1, Thms. YM3_I:K and YM3_I:M, and Lem. YM3_I:51.6. The precise existence requirement is stated in §9.1.8. Restrictions, joined data and localised data must remain in the admitted classes; their critical backgrounds must satisfy the comparison in Prop. 9.22(iii). These estimates remain to be established for G, including the uniform near-trivial background, tower and analyticity-core bounds of Prop. 9.22(ii), with the response constant $C_\partial = 12\kappa_6 + 4L$, on the objects of a step at the fields over a regular base (the global and the localised background of the step and the minimal orbits of its three data), and on the towers off the trivial field. Among them, the uniqueness of the regular critical orbit near the trivial field and its normalized representatives as stated in Prop. 9.17, for the full regional geometry of §9.1.6 and every reversal-invariant set $\mathfrak{F}$ of removed inward bonds. The radii and bounds there are uniform in depth, sub-domain, cutoff and volume.

Conjugation and domain preservation (C).. With $a(X)$ for every trace of a group-valued argument at complex fields, the gauge-fixing term included, and with the domains written with $m(X)$ and with D^c at the logarithms of the averages, every map between sets of complex fields that the induction uses commutes with $p \mapsto p^c$ on the real-connected parts of its domains (Lem. 9.21) and maps them into the real-connected parts of its targets. The covariance assertion follows from Prop. 9.23(i) for these maps wherever they are formed; the source and target domain memberships required to apply it remain among (V), (II), and (K).

The nonlinear step (II).. One must verify that the transfers to G of the statements of the second part of the $SU(2)$ construction that the induction cites hold, in the forms of Prop. 9.23(ii), every proposition that the proof of Thm. YM3_I:D quotes among them, and the objects of the definitions that these statements read, formed for G, have the properties that the $SU(2)$ construction proves

for them; so do the statements of the second part that use the averaging map off the small fields, the pivot bond or the measure of $G \setminus G^{\pi}$, the uniform box-gauge bound of Lem. 9.12(iv), and the large-field chain of Lem. 9.13(ii); and at every citation in the proofs named in Prop. 9.24 the assumptions of the cited statement, in its transferred form, hold at the data of the citation. Among them: the pivot bond (Prop. YM3_I:14.11) holds for G in all its clauses with the constants of equation YM3_I:(128), no numeral changed; and the localised Gaussian family (Prop. YM3_I:50.2) is a pointed family of its level on the analyticity cores of G, holomorphic on balls of one radius, with the contraction bound of its expansion.

Compatible constants (K).. One must show that every condition on the constants read for G, the places where z_G enters included, passes its test in Prop. 9.26(b): it holds on a large set, and at each two-sided constant the set of the earlier choices at which its window meets all its ends together is large. The conditions read for G are every condition of the $SU(2)$ construction transferred to G, every condition of Prop. 9.26(c)–(g), the conditions $\varepsilon_{\mathrm{v}} \le \varepsilon_G$ and $c_{11} \le r_G/(2C_{\mathrm{pg}})$ of the uniqueness clause of (V), c_{11} the radius of §9.1.6 and C_{pg} the admitted preliminary-gauge constant of §9.1.6, the ceiling $\mathsf{c}_L c_4 \le 2Lr_G$ of Prop. 9.22(iv), the row $\alpha_1 < \rho_G L$ of Prop. 9.24, and, at the memberships reached from memberships of restrictions had apart, the conditions of Lem. 9.21(iii): the moved first radius $b_0' = b_0 + 2e^{4\xi b_1} b_1^2$, for the regular pairs $\alpha_0 + 2\vartheta^2 e^{4\vartheta\alpha_1}\alpha_1^2$ with $\vartheta \ge 1$ the largest ratio of the spacings around a plaquette, at most the first radius B_0 at which the statement read there is had, and for the analyticity cores besides $b_1 \le B_1$, $K(b_0' + b_1^2) \le B_0$ and $b_0' + b_1 \le a^\sharp$, with $K \ge 16L^2$ and $a^\sharp$ the constant and the radius of that lemma, which are positive ceilings on the second radius where $b_0 < B_0$, $Kb_0 < B_0$ and $b_0 < a^\sharp$. Each number of G that stands in such a condition has a place before γ in the order of the table and is a function of the constants before that place and of the pair, which reads no radius.

The near-trivial uniqueness clause of (V) is reduced in Prop. 9.17 to the transferred uniform slice and contraction estimates; their uniform nonlinear bounds remain to be established there. The orbit and energy hypothesis of §9.1.6, with a fixed admitted positive radius a_G, is also required wherever its ceiling is used in (K). For quotients, the sector statements are those of Prop. 9.33.

9.1.2. *What changes with the gauge group.* The representation of the group fixes the normalization of the action. For $G \subset U(N)$ we use Hermitian Lie-algebra elements with scalar product $\langle X, Y\rangle = N^{-1}\operatorname{tr}(XY)$ and action

$$a(U) = 1 - \frac{1}{2N}\operatorname{tr}(U + U^{-1}).$$

On G this is the real Wilson action. Its displayed form also specifies its holomorphic continuation. The pair consisting of G and its representation is therefore part of the data. A numerical Casimir computed with another scalar product cannot be inserted without changing the coupling normalization.

Let $n = \dim \mathfrak{g}$ and choose an orthonormal basis (t_a). At the trivial background the differential operators act independently on the n colour coordinates: a scalar lattice operator K becomes $K \otimes 1_{\mathfrak{g}}$. This explains the factor n in the leading action-density coefficient. The lattice projection, the Gaussian covariance and their decay estimates at this reference are the same spatial operators as before. The two-point Gaussian contraction therefore gives

$$c_2(G) = \frac{3n}{16\pi^2}.$$

This calculation identifies a local coefficient; it does not construct the interacting measure at a nontrivial background.

For simple $\mathfrak{g}$, invariance reduces the next colour contractions to two scalars. Put $\Omega = \sum_a t_a^2$ and, for $\|H\| = 1$, define

$$C_A = \operatorname{tr}_{\mathfrak{g}^c}(\operatorname{ad} H)^2, \qquad C_\rho = N^{-1}\operatorname{tr}(\Omega H^2).$$

These numbers are independent of H in the simple case. The source's one-loop coefficient identity is

$$b[\Psi_k^G] = \frac{C_A}{8} b[\Psi_k] + \left(\frac{C_\rho}{3} - \frac{C_A}{8} \right) (L^{-k} - L^{-4k}).$$

Here $b[\Psi_k]$ is the quadratic colour coefficient in the SU(2) calculation, and $b[\Psi_k^G]$ is its counterpart for the pair. The second term is explicit in k. Thus the convergence and bounds of the original coefficient sequence transfer through an algebraic identity, rather than through a new interacting construction. The detailed contraction below also gives

$$\varphi_G = \sum_a \operatorname{tr}(\operatorname{ad} t_a)^2 = nC_A, \qquad c_1(G) = \frac{\varphi_G}{3\pi n}.$$

At complex fields the matrix changes are consequential. For SU(2), special conjugation identities simplify the trace and the domains of logarithms. They fail in higher matrix dimension. The holomorphic action above replaces a bare trace where the latter ceases to be real on real fields, and the control of

$$m(X) = \max\{|X - 1|, |X^{-1} - 1|\}$$

keeps the domain compatible with $X \mapsto (X^*)^{-1}$. The projection onto $\mathfrak{g}^c$, the chart radius, and the distance to nontrivial central elements also enter the estimates. These changes are part of defining the maps on the correct domains, before one can attempt the induction.

The general-group sketch must then address regular backgrounds. The first task is to invert the covariant operators with uniform local decay. The second is to construct a critical orbit over each admitted datum, control its response, and compare restrictions and localized backgrounds. The nonlinear fluctuation integral must stay in those domains at every step, and its Gaussian family must satisfy the required contraction bounds. Finally, the constants must meet their upper and lower constraints simultaneously. The colour calculation at the identity gives the local coefficients. The detailed reductions below specify the separate slice, contraction and orbit estimates needed for the nonlinear construction.

There is an exact special case that avoids this transfer problem. For separately acting SU(2) factors with their stated positive weights, the action splits and the Wilson measure factors. A cumulant of the sum of the factor observables is the sum of their cumulants with the corresponding weights. This is a product construction using already supplied factors. A quotient changes the global topology and sectors, while a representation tying several factors together changes the analytic operators. Those cases need the separate local and global analysis retained below; they do not follow merely by adding colour dimensions.

9.1.3. *Construction and local coefficients.*

Detailed sketch AA (Construction programme and local coefficients). Let $G \subset U(N)$ be compact and connected with $\mathfrak{g}$ simple, and let $L \geq 13$ be odd with $200L^2 z_G \geq 1$ (for $SU(N)$ at $L = 13$ this is $N \leq 212371$). Every constant below is a positive number of L and the pair, uniform in the volume. The construction clauses in (a), the second half of (b), the existence assertion of (d), and the second half of (e) describe the proposed extension. Their regular-background and nonlinear multiscale estimates remain to be established as described in §9.1.1. The fixed-lattice identities, one-loop coefficients, finite-box expansion and assertions about an already convergent sequence have separate supplied arguments.

(a) For simply connected G, the construction aims to establish that Thm. YM3_I:D holds for G at every choice of the constants of the $SU(2)$ construction that meets the conditions on the constants read for G (§9.1.1(K)): its conditions transferred to G and the conditions of Prop. 9.26, among them $\varepsilon_{\mathrm{v}} \leq a_G$, $\varepsilon_{\mathrm{v}} \leq \varepsilon_G$ and $\varepsilon_{\mathrm{v}} \leq L^2 r_G$ on the regularity radius ε_{v} of the critical configurations (Def. YM3_I:9.5), a_G a fixed positive radius admitted by the hypothesis of §9.1.6 and ε_G the number of the uniqueness clause of §9.1.1(V), and

the ceiling $\mathsf{c}_L c_4 \le 2Lr_G$ of Prop. 9.22(iv); such choices exist, in the order of the $SU(2)$ construction with γ last (§9.1.1(K) and Prop. 9.26(b)). The run is the one sequence $x_0 = \beta$, $x_k = L^{-1}x_{k-1} - b_k$, b_k the mean of the n quadratic colour coefficients of the local terms of second order of step k: it is real, does not depend on the torus, and $|b_k| \le \mathsf{G}_\sharp \le B_\sharp$.

(b) Let $A(v,V) = \sum_p v(p)a_p(V)$ for a complex weight v of the plaquettes, and $\mathcal{O}(w) = a_K^{-1}A(w,\cdot)$ for a source w. For every compact $G \ne \{\mathbf{1}\}$ in $U(N)$: the chessboard norm of a source is $\|e^{\mathcal{O}(w)}\|_{\mathrm{ch}}^{\mathsf{N}} = Z_\beta[w^{\mathrm{ch}}]/Z_\beta \ge 0$ (equation YM3_I:(1759), transferred to G); a source w with $\operatorname{Re} w \le (1-c_{\mathrm{pr}})a_K\beta$, $c_{\mathrm{pr}} \in (0,1]$, leaves every plaquette at least the fraction c_{pr} of its price, $|e^{-A(\beta - a_K^{-1}w, V)}| \le e^{-c_{\mathrm{pr}}\beta\sum_p a_p(V)}$, as $a_p \ge 0$ on G; and on a torus with $|\Lambda|$ sites the Wilson measure with one coupling $s \ge 1$ at all plaquettes, which at $s = c_{\mathrm{pr}}\beta$ dominates the weight with the source by the price, has

$$\langle a_p\rangle_s \le \frac{C}{s} + \frac{n\log s}{|\Lambda|\, s}$$

at every plaquette, C a number of the pair, while for plaquette couplings $v(p)$ with $s \le v(p) \le C's$, $C' \ge 1$, the mean $\frac{1}{3|\Lambda|}\sum_p \langle a_p\rangle_v$ over the plaquettes of the torus is at most $(C + n\log C')/s + n\log s/(|\Lambda|s)$, and at couplings invariant under the lattice symmetries every plaquette has this mean; for $c_{\mathrm{pr}}\beta \ge 1$ the couplings $\beta - a_K^{-1}\operatorname{Re} w(p)$ of a source with $0 \le \operatorname{Re} w \le (1-c_{\mathrm{pr}})a_K\beta$ are of this kind, with $s = c_{\mathrm{pr}}\beta$ and $C' = 1/c_{\mathrm{pr}}$. On the tori of the continuum theorem at its first-passage data, which have at least $2L^K$ sites a side and couplings at most $L^{K+1}(g^{-2} + \bar{B})$, $n|\Lambda|^{-1}\log s$ is below a number of g, the pair and the constants. For simply connected G, the intended sourced-density and stability conclusions are those of Thms. YM3_I:AJ and YM3_I:AD with the run of (a): the effective densities with a source have at every level the structure that the $SU(2)$ construction proves, and the stability holds through the whole induction with complex plaquette weights, uniformly in the volume; writing $\widetilde{Z}(w) = e^{-E_{(K)} - \mathcal{E}_K(w)}Z_{K,\ell,\beta}[w]$, with $\mathcal{E}_K$ defined by [7, eq. 212], gives

$$\int e^{-A(\beta - a_K^{-1}w, V)}\,\mathrm{d}V = \exp\Big(E_{(K)} + \sum_{j=1}^{K}\sum_{X\in\mathcal{D}_j} e_j(X;w)\Big)\,\widetilde{Z}(w),$$

$E_{(K)}$ the vacuum constant at the source 0, $e_j(X;w)$ the source energy of level j on the domain X of level j [7, eq. 212] and $\widetilde{Z}(w)$ the normalised partition function with the source, with the bounds of Thm. YM3_I:AD on them. With it the connected functions of the action density obey the bound per ball of Prop. YM3_I:89.4: for every reference coupling $0 < g \le \gamma$, every order $\mathsf{m} \ge 2$, every $R > 0$ and every $\delta > 0$, at every first-passage datum (K,β) at g, on every side ℓ admitted at g with $\ell \ge 16\mathsf{m}(R+1)$ and for real Lipschitz functions $f_1, \dots, f_{\mathsf{m}}$ on the torus, each supported in a ball of radius R, the supports pairwise at a distance at least δ, $|\langle \mathcal{O}(f_1); \dots; \mathcal{O}(f_{\mathsf{m}})\rangle|$ is at most the constant $C_{\mathsf{m}}^{\infty}(\delta;R)$ of that proposition times the product of the norms of the f_i in that proposition, $C_{\mathsf{m}}^{\infty}(\delta;R)$ being formed from g, m, δ, R and the constants of Thm. YM3_I:AD for G alone, free of K, β and ℓ: the connected functions are derivatives at 0 of the logarithm of the normalised partition function with sources, and Cauchy's inequality on the polydiscs of sources on which the stability holds gives the bound, as in Prop. YM3_I:89.4.

(c) For $k \ge 1$ and on every torus, the one-loop functional Ψ_k^G of Thm. YM3_I:O for G is tempered (Section YM3_I:43.1), with the coefficient

$$b[\Psi_k^G] = \frac{C_A}{8}\,b[\Psi_k] + \Big(\frac{C_\rho}{3} - \frac{C_A}{8}\Big)(L^{-k} - L^{-4k}),$$

Ψ_k that of $SU(2)$; Thms. YM3_I:O and AC hold for G with constants of L, C_A and C_ρ; and near the trivial block field the quadratic coefficient of the logarithm of the Gaussian

normaliser of the step to level l is the one-loop increment (Thm. YM3_I:O)

$$b_l^0(G) = \frac{C_A}{8}\, b_l^0 + \Big(\frac{C_\rho}{3} - \frac{C_A}{8}\Big)(L^3-1)L^{-4l}.$$

(d) Let the run have real increments that depend on no torus and are bounded by $B_\sharp$, as in (a). Let $0 < g \le \gamma$, $X_g = L(g^{-2} + B_\sharp)$, $K(\beta, g)$ the least k with $x_k \le X_g$, and let $K_j \to \infty$, let $l_j \to \infty$ be sides admitted at g, and $K(\beta_j, g) = K_j$. For $\mathsf{m} \ge 2$ and real C^1 functions f_i on $\mathbb{R}^3$ of compact pairwise disjoint supports let $S^{(j)}_{\mathsf{m}}(f_1, \dots, f_{\mathsf{m}}) = \langle \mathcal{O}(f_1); \dots; \mathcal{O}(f_{\mathsf{m}})\rangle$ on the torus of side l_j at (K_j, β_j), $\mathcal{O}(f) = a_{K_j}^{-1} \sum_p f(x(p)) a_p$, f periodised. Then along a subsequence $a_{K_j}\beta_j$ converges in $[g^{-2} + B_\sharp - \bar{B}/L,\ L(g^{-2} + \bar{B})]$. For simply connected G, after completion of the construction tasks, with the run of (a), the $S^{(j)}_{\mathsf{m}}$ obey the bound per ball of (b) uniformly in j, which holds at every first-passage datum and on every side admitted at g and at least $16\mathsf{m}(R+1)$ with one constant for given m, R and δ, each of the compact pairwise disjoint supports of the f_i lying in a ball of some radius R and the supports lying at mutual distances at least some $\delta > 0$; so along a further subsequence every $S^{(j)}_{\mathsf{m}}$ converges. For simply connected G, the following are the transferred conclusions sought from the uniform construction in §9.1.1, with the same parameter restrictions. On a fixed torus the fixed-label and moving-label cutoff limits have the properties of Thm. YM3_I:A. Growing-torus and periodic-Gibbs sequences have the subsequential compactness, multilinear bounds and inherited coordinate reflection positivity of Props. YM3_I:92.1, YM3_I:92.3 and YM3_I:92.4. For periodic-Gibbs limits the canonical component and reconstruction conclusions are those of Thms. YM3_I:AL and YM3_I:AP: each component is a conditioned periodic-Gibbs limit, is Euclidean invariant and reflection positive, clusters, and has a unique vacuum; the undecomposed limit has the same reconstruction, with zero-energy space $L^2(\mathsf{X}, \lambda)$. For a general growing-torus limit, full rotation invariance retains the Ward criterion in (g). The short-distance conclusions retain the restrictions in (e) and use $c_2(G)$ in place of c_2. For every compact connected G and along every sequence of tori on which all $S^{(j)}_{\mathsf{m}}$ converge, each limit S_{m} is m-linear, real and symmetric, obeys $|S_{\mathsf{m}}(f_1, \dots, f_{\mathsf{m}})| \le C(A_1, \dots, A_{\mathsf{m}}) \prod_i \|f_i\|_{C^1}$ for supports in fixed disjoint compact sets A_i, with C depending on the sequence, is invariant under the translations of $\mathbb{R}^3$ and the 48 signed permutations of the axes, and is reflection positive at every coordinate mirror. On one torus of side $l = 2L^m$ with $K_j \to \infty$ the same holds with the translations of the torus, the signed permutations and the pairs of mirrors $\{y_\mu = s\} \cup \{y_\mu = s + \frac{l}{2}\}$.

(e) The Gaussian measure defined by the quadratic action at $\mathbf{1}$ is the product of n identical one-colour measures. Its leading two-point coefficient is

$$c_2(G) = \frac{3n}{16\pi^2}.$$

If G is simply connected and satisfies all the construction hypotheses, the finite-cutoff estimate of Thm. YM3_I:AV and its continuum version, Thm. YM3_I:AW, transfer to its tori and their limits:

$$\langle \mathcal{O}(f); \mathcal{O}(h)\rangle = c_2(G)\mathsf{b}^{-2}\mathcal{K}_6(f,h)(1+\varepsilon'), \qquad |\varepsilon'| \le \bar{\varepsilon}_g(\mathsf{d}),$$

At finite cutoff this means $0 < g \le \gamma_W$, $(K, \beta) \in \mathfrak{d}(g)$, $0 < \vartheta < 1$, $0 < \mathsf{d} \le d_0(g, \vartheta)$, $K > s(\mathsf{d})$, $\ell \in \mathfrak{L}(g)$, and a test pair $(f, h) \in \mathfrak{T}_{T_\ell}(\mathsf{d}; \vartheta)$; then $\mathsf{b} = a_K\beta$ and the tolerance is the transferred $\bar{\varepsilon}_{g,\vartheta}(\mathsf{d}) \le 1/3$ of Thm. YM3_I:AV. In a continuum limit the pairs are C^1, the cutoffs tend to infinity, and b is the limiting label of Thm. YM3_I:AW. For block sums of $SU(2)$, the corresponding statement is Thm. AB(iii).

(f) For every compact connected G, the fixed-box expansion is that of Lem. 9.28. Its coefficient of β^{-3} has an infinite-box limit; after summing over orientations, its leading part at separation

r is $\varphi_G \mathsf{r}^{-5}/(16\pi^3)$. Relative to the coefficient in (e), its ratio is

$$c_1(G) = \frac{\varphi_G}{3\pi n}.$$

The expansion and the limit are iterated, with the box fixed during the expansion in β^{-1}.

(g) At every cutoff, on every torus and for every $\beta > 0$, the finite-cutoff rotational Ward identity used in §5.1 holds for G with $\frac{1}{2}k_p$ in the place of $\frac{3}{2}a_p$; and $k_p = na_p$ if and only if $\Omega = n\mathbf{1}$, as for an irreducible representation or a sum of copies of one. Let (K_j, l_j, β_j) be a sequence of tori with $l_j \to \infty$ along which all connected functions converge, S_{m} their limits as in (d), and let $(\omega, \mathsf{m}, \mathcal{R}, f)$ be an admitted quadruple (Section YM3_I:112, before equation YM3_I:(2115)): $\omega \in \mathfrak{so}(3)$, $\mathsf{m} \geq 2$, $\mathcal{R} > 0$, and $f = (f_1, \dots, f_{\mathsf{m}})$ smooth functions with compact pairwise disjoint supports in $\{|y| < \frac{1}{2}\mathcal{R}\}$. For the j with $2\mathcal{R} + 2a_{K_j} < \frac{1}{2}l_j$ the bulk term and the flux term of the identity at the cut-off radius $\mathcal{R}$ are defined, and their sum has a limit, which equals

$$\sum_i S_{\mathsf{m}}(f_1, \dots, L_\omega f_i, \dots, f_{\mathsf{m}}) = -\frac{d}{dt} S_{\mathsf{m}}(e^{t\omega} f)\Big|_{t=0},$$

L_ω the rotation field of ω, $t \mapsto S_{\mathsf{m}}(e^{t\omega} f)$ being differentiable, and which does not depend on $\mathcal{R}$. The limit of the connected functions is invariant under $O(3)$ at all C^1 tuples of (d) if and only if this limit of the sum vanishes for one $\omega_0 \neq 0$ and every admitted quadruple $(\omega_0, \mathsf{m}, \mathcal{R}, f)$, and if and only if it vanishes at every admitted quadruple: the conjugates of ω_0 under the signed permutations span $\mathfrak{so}(3)$, and the invariance passes from smooth tuples to C^1 tuples by mollifying, with the bound of (d). It is the limit of the flux term alone wherever the bulk term tends to 0. If the limit is invariant, the rotations and reflections about the third axis act on the Hilbert space of the reflection positive form of the limit by a strongly continuous unitary representation U of $O(2)$ that fixes the vacuum vector, and $\mathsf{U}(R)[\tau_{te_3} F] = [\tau_{te_3}((R \oplus 1)F)]$ for $t \geq 0$ on the classes of the functions F supported in $\{y_3 > 0\}$, τ_{te_3} being the translation by te_3.

(h) Let instead $\mathfrak{g} = \bigoplus_{i \leq r} \mathfrak{g}_i$ be semisimple, $r \geq 2$.

(h1) The mean of the n quadratic colour coefficients of a local term of second order is $\mathsf{cf} = \sum_i \frac{n_i}{n} \mathsf{cf}_i$, where cf_i is its coefficient on the i-th ideal; moreover, $|\mathsf{cf}_i| \leq \kappa_i^2 C_{\mathrm{am}} r_0^{-2} \|E\| = (\kappa_i/\kappa_G)^2 \mathsf{G}_\sharp \|E\|/E_0$ for the terms E of Thm. YM3_I:D, C_{am} the constant of the coefficient bound YM3_I:(940), $r_0 = a^{\mathrm{us}}(\alpha_0, \alpha_1)$ the radius of the $SU(2)$ construction and $\mathsf{G}_\sharp = \kappa_G^2 C_{\mathrm{am}} r_0^{-2} E_0$ (Prop. 9.24), C_{am} and r_0 reading constants of the $SU(2)$ construction that come before them in its order. If $|e_k^{(i)}| \leq \Gamma$, the sequences $x_0^{(i)} = \beta$, $x_k^{(i)} = L^{-1} x_{k-1}^{(i)} - e_k^{(i)}$ and their mean $x_k = \sum_i \frac{n_i}{n} x_k^{(i)}$ satisfy $x_k = L^{-1} x_{k-1} - \sum_i \frac{n_i}{n} e_k^{(i)}$ and $|x_k^{(i)} - x_k| \leq 2(1 - \frac{n_i}{n})\Gamma \frac{L}{L-1}(1 - L^{-k})$.

(h2) The one-loop theorem holds in block form at every semisimple pair, the ideals being tied by the representation or not: on $\mathfrak{g}_i$, $b^{(i)}[\Psi_k^G] = \frac{C_A^{(i)}}{8} b[\Psi_k] + (\theta_i - \frac{3}{8} C_A^{(i)}) t_k$, $t_k = \frac{1}{3}(L^{-k} - L^{-4k})$, $b^{(i)}[\Psi_k^G]$ being the coefficient on $\mathfrak{g}_i$ of the one-loop functional Ψ_k^G of the pair and Ψ_k that of $SU(2)$, with one set of sequences for all ideals; here $\theta_i = \frac{1}{N}\mathrm{tr}(\Omega H^2)$ for a unit $H \in \mathfrak{g}_i$, Ω being summed over a basis of all of $\mathfrak{g}$, so that $\theta_i = \sum_j n_j \tau_{ij}$ with numbers τ_{ij} of the pair (Prop. 9.18(iv)). Equivalently, $b^{(i)}[\Psi_k^G]$ is the coefficient of the simple pair of $\mathfrak{g}_i$ plus $(\sum_{j \neq i} n_j \tau_{ij})$ times the coefficient t_k of one scalar form Υ_k of the lattice, which is tempered. For a unit $H \in \mathfrak{g}_i$ the numbers $\mathrm{tr}(\mathrm{ad}\, H)^2$ and $\frac{1}{N}\mathrm{tr}(\Omega H^2)$ that Def. 9.1 forms for simple $\mathfrak{g}$ are $C_A^{(i)}$ and θ_i; they depend on the ideal wherever these pairs are not all equal, as at $SU(2) \times SU(3)$ on $\mathbb{C}^2 \oplus \mathbb{C}^3$, where $C_A^{(i)}$ is 20 and 30 (Lem. 9.15(i)), and the coefficient is then given ideal by ideal, as above. The r

coefficients are bounded ideal by ideal, $|b^{(i)}[\Psi_k^G]| \le \frac{C_A^{(i)}}{8}\sup_k|b[\Psi_k]| + |\frac{\theta_i}{3} - \frac{C_A^{(i)}}{8}|L^{-1}$, and so by the largest of these r bounds (Prop. 9.18(iv)).

(h3) For the tied action, the missing transfer task is to establish open slices for the backgrounds in the setting preceding Def. YM3_I:52.1, the Gaussian structure of Thm. YM3_I:U and normaliser of Prop. YM3_I:50.2 with their stated properties, localisation as in Prop. YM3_I:51.16, and one step as in Thm. YM3_I:V, with the counterterm numbers free. Thm. YM3_I:X must also be verified at the simple pairs (G_i, Λ) on $\mathbb{C}^N$, with the fixed weight $\Lambda = \frac{1}{N}\mathbf{1}$ of the action of G, which lies in the commutant (Prop. 9.24). This is the transfer task (T). Once this transfer is established, the intended one-run conclusion uses the mean $x_k = L^{-1}x_{k-1} - \sum_i \frac{n_i}{n}e_k^{(i)}$, $e_k^{(i)}$ the coefficient on $\mathfrak{g}_i$ of the local terms of second order of step k: the sequences $x_k^{(i)}$ of (h1) stay within $2(1-\frac{n_i}{n})\mathsf{G}_\sharp\frac{L}{L-1}(1-L^{-k})$ of x_k, by (h1) with $\Gamma = \mathsf{G}_\sharp$, as $|e_k^{(i)}| \le (\kappa_i/\kappa_G)^2\mathsf{G}_\sharp \le \mathsf{G}_\sharp$, and all clauses of the $SU(2)$ construction on the increments hold, the increment theorem, the regular progression, the sequence and the bounds on polydiscs among them. At the least tied pair, the additional propagator assertions require the regular-domain and relative-smallness hypotheses stated in Prop. 9.30(v); under them the kinematic and interaction parts of the first step, the Hessian lemma and the propagators at every level hold (Prop. 9.30(v)).

(h4) If $G = G_1 \times \cdots \times G_r$ acts on $\mathbb{C}^{N_1} \oplus \cdots \oplus \mathbb{C}^{N_r}$ by blocks, the Wilson measure at β is the product of those of the G_i at $\beta N_i/N$, cumulants add with the weights $(N_i/N)^{\mathsf{m}}$, and for simply connected G_i with simple Lie algebras and $200L^2 z_{G_i} \ge 1$, each satisfying the construction hypotheses, clauses (a)–(g) hold factor by factor at these couplings, with precisely the qualifications stated there.

■□

Remark 9.7 (Leading law and perturbative coefficients). *The constants.* The Hessian of a at $\mathbf{1}$ is $\langle\cdot,\cdot\rangle$, so to leading order the field consists of n independent copies of the free field with this form, colour by colour, each contributing $\frac{3}{16\pi^2}$ as at $SU(2)$, where $3\cdot\frac{3}{16\pi^2} = \frac{9}{16\pi^2}$; this is $c_2(G) = \frac{3n}{16\pi^2}$, the constant of the law of leading order. For $SU(2)^r$ the product $c_2(G)\,\mathsf{b}^{-2}$ is the sum $\sum_i \lambda_i^2\frac{9}{16\pi^2}(\lambda_i\mathsf{b})^{-2} = r\frac{9}{16\pi^2}\mathsf{b}^{-2}$ that Step 3 of the proof of Thm. AB forms from the laws of the factors. The next term is governed by the coefficient of β^{-3} of $\langle a_p; a_q\rangle$; by Lem. 9.28 its leading part, summed over orientations, is $\frac{\varphi_G}{16\pi^3}\mathsf{r}^{-5}$ at distance r, and $c_1(G) = \frac{\varphi_G}{16\pi^3}/c_2(G) = \frac{\varphi_G}{3\pi n}$ is the ratio of the next perturbative coefficient to the leading coefficient. At $SU(2)$, $\varphi = 24$, and this is $\frac{3}{2\pi^3}/\frac{9}{16\pi^2} = \frac{8}{3\pi}$, the coefficient c_1 of Thm. X.

The law of leading order. Let G be simply connected with $\mathfrak{g}$ simple, $L \ge 13$ odd and $200L^2 z_G \ge 1$, and assume the construction hypotheses. On the tori of the continuum theorem and in their limits along tori, under the finite-cutoff restrictions of Thm. YM3_I:AV and the continuum restrictions of Thm. YM3_I:AW, as stated in Thm. AA(e),

$$\langle\mathcal{O}(f);\mathcal{O}(h)\rangle = c_2(G)\,\mathsf{b}^{-2}\mathcal{K}_6(f,h)\,(1+\varepsilon'), \qquad |\varepsilon'| \le \bar{\varepsilon}_g(\mathsf{d}),$$

$\mathsf{b} = a_K\beta$ the label and $\bar{\varepsilon}_g$ the number of that law, which tends to 0 with d; this is the law of leading order of Thm. AA(e). For $SU(2)^r$ the law holds with one label (Thm. AB(iii)). For simply connected G with $\mathfrak{g}$ simple its proof is the proof of the law of Thms. YM3_I:AV and YM3_I:AW, with the colour sum running to n. That proof rests on Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD and on the count of colours: Thm. AA(a) gives Thm. YM3_I:D for G along the one run; Thms. YM3_I:AJ and YM3_I:AD for G with that run (Thm. AA(b)) gives the stability with sources; the Gaussian measure that the quadratic part of the action at the trivial field defines is the product of n copies of that of one colour of $SU(2)$, with the same normalisation (Thm. AA(e)); and the remainders are bounded by the same terms of the effective action as for $SU(2)$. The passage to limits along tori is that of Thm. AA(d). The law concerns the tori and their limits along tori.

The fixed-box expansion in Lem. 9.28 has an error constant depending on the box. Its large-box limit is taken after the expansion at fixed box. Thus the calculation of $c_1(G)$ in this lemma is an iterated perturbative limit.

Corollary 9.8 ($SU(3)$)**.** Let $G = SU(3)$ and $L \geq 13$ odd. Then $z_G = 1$, so that no floor on L is needed and the forms with D^c are the statements of the $SU(2)$ construction; no number of the averaging layer depends on the group; $r_G = \sqrt{3}$, $\kappa_G^2 = \tau_G^2 = 2$, $n = 8$; the least admissible p of the cut-off is 3, with $\lambda = 6.1194$; and the extra conditions of Lem. 9.11(v) and Prop. 9.22(iv) are void. The local coefficients in Thm. AA have the following values. The construction conclusions require the separate tasks of §9.1.1:

$$b[\Psi_k^G] = \frac{9}{4}\, b[\Psi_k] + \frac{5}{12}(L^{-k} - L^{-4k}), \qquad c_2(G) = \frac{3}{2\pi^2}, \qquad c_1(G) = \frac{6}{\pi}.$$

Besides, $\mathfrak{su}(3)$ has a basis, orthonormal for $\frac{1}{2}\mathrm{tr}\, XY$, whose eight members have the eigenvalues $1, -1, 0$, as the Pauli matrices, with the eigenvalues $1, -1$, are such a basis of $\mathfrak{su}(2)$. ■□

Sketch of proof. The numbers are those of $SU(N)$ at $N = 3$ (Def. 9.1); $\frac{C_A}{8} = \frac{9}{4}$, $\frac{C_\rho}{3} - \frac{C_A}{8} = \frac{5}{12}$, $\frac{3\cdot 8}{16\pi^2} = \frac{3}{2\pi^2}$, $\frac{C_A}{3\pi} = \frac{6}{\pi}$. The cut-off asks $3^{1/2p} < \frac{5}{4}$, first met at $p = 3$, and $\lambda = w/(5 - 4w)$ at $w = 3^{1/6}$. With $r_G = \sqrt{3}$ the condition $\varpi_L < r_G$ and the condition $\mathsf{c}_L c_4 \leq 2Lr_G$ are met by the ceilings of the $SU(2)$ construction. For the basis: the rotation R by $\frac{2\pi}{5}$ about e_3 acts on the real symmetric traceless matrices, an orthogonal sum $V_0 \oplus V_1 \oplus V_2$ of the multiples of $\mathrm{diag}(-1, -1, 2)$, of the span of $e_1 e_3^{\mathrm{T}} + e_3 e_1^{\mathrm{T}}$ and $e_2 e_3^{\mathrm{T}} + e_3 e_2^{\mathrm{T}}$, and of the span of $\mathrm{diag}(1, -1, 0)$ and $e_1 e_2^{\mathrm{T}} + e_2 e_1^{\mathrm{T}}$, as the identity, the rotation by $\frac{2\pi}{5}$ and the rotation by $\frac{4\pi}{5}$. So for X of unit norm with squared components (c_0, c_1, c_2), $\frac{1}{2}\mathrm{tr}(R^j X R^{-j} R^k X R^{-k}) = c_0 + c_1 \cos\frac{2\pi(j-k)}{5} + c_2 \cos\frac{4\pi(j-k)}{5}$, which vanishes for $j \not\equiv k$ exactly when $(c_0, c_1, c_2) = (\frac{1}{5}, \frac{2}{5}, \frac{2}{5})$. For $X = ab^{\mathrm{T}} + ba^{\mathrm{T}}$ with orthonormal $a, b \in \mathbb{R}^3$, which has the eigenvalues $1, -1, 0$, one finds $c_0 = 3a_3^2 b_3^2$ and $c_2 = (1 - a_3^2)(1 - b_3^2)$; the two conditions say that a_3^2, b_3^2 are the roots $\frac{5 \pm \sqrt{10}}{15}$ of $15x^2 - 10x + 1 = 0$, whose sum $\frac{2}{3}$ leaves $\frac{1}{3}$ for the square of the third component of $a \times b$, so such a, b exist. The five $R^k X R^{-k}$ and the three imaginary antisymmetric Gell-Mann matrices, which are orthogonal to all real symmetric matrices, have the eigenvalues $1, -1, 0$ and are pairwise orthogonal. □

Theorem AB (Sums of copies of $SU(2)$)**.** Let $G = SU(2)^r$ act on $\mathbb{C}^2 \oplus \cdots \oplus \mathbb{C}^2$ by blocks, $a_p = \sum_i \lambda_i (1 - \frac{1}{2}\mathrm{tr}\, \partial U^{(i)}(p))$ with $\lambda_i > 0$, $\sum_i \lambda_i = 1$ (a block repeated m_i times in $\mathbb{C}^N$ gives $\lambda_i = 2m_i/N$), one coupling β, $n = 3r$, $\lambda_\pm$ the largest and the least λ_i, and let the constants of the $SU(2)$ construction, $\mathsf{G}_\sharp$ among them with its value for $SU(2)$ (Prop. 9.24), obey besides $\lambda_- B_\sharp \geq \lambda_+ \mathsf{G}_\sharp$.

(i) On every finite lattice and for $\beta > 0$ the Wilson measure is the product of those of $SU(2)$ at $\lambda_i \beta$, and $\langle \mathcal{O}(f_1); \dots; \mathcal{O}(f_{\mathsf{m}})\rangle_\beta = \sum_i \lambda_i^{\mathsf{m}} \langle \mathcal{O}(f_1); \dots; \mathcal{O}(f_{\mathsf{m}})\rangle_{\lambda_i \beta}^{SU(2)}$.

(ii) For $\lambda_- \beta \geq \gamma^{-2}$ the mean run $x_0 = \beta$, $x_k = L^{-1} x_{k-1} - \frac{1}{r}\sum_i \lambda_i^{-1} c_k^{(i)}$, with $c_k^{(i)}$ the coefficient of step k of $SU(2)$ along the couplings $\lambda_i x_0, \dots, \lambda_i x_{k-1}$, is defined and real while $\lambda_- x_{k-1} \geq \gamma^{-2}$. Along it Thm. YM3_I:D holds in every factor, the step of the factor i at level j being the step of $SU(2)$ at $\lambda_i x_j$ with its own thresholds and histories; and $x_0^{(i)} = \beta$, $x_k^{(i)} = L^{-1} x_{k-1}^{(i)} - \lambda_i^{-1} c_k^{(i)}$ have the mean x_k and $|x_k^{(i)} - x_k| \leq 2\lambda_-^{-1} \mathsf{G}_\sharp \frac{L}{L-1}$.

(iii) Let $\lambda_- g^{-2} \geq \gamma_W^{-2} + \bar{B}$, K the least k with $x_k \leq L(g^{-2} + \frac{\mathsf{G}_\sharp}{r}\sum_i \lambda_i^{-1})$, $\mathsf{b} = a_K \beta$, and let the hypotheses of the law of two points of Thm. YM3_I:AV on the scale d, the cutoff and the test pair (f, h) hold separately at the $2r$ reference couplings $g_{i,1}$, $g_{i,2}$ with $g_{i,1}^{-2} = \lambda_i g^{-2} - \bar{B}$, $g_{i,2}^{-2} = L\lambda_i g^{-2} - 3\bar{B}$, and let the tori have sides admitted at each of the $2r$ couplings $g_{i,s}$, that is, sides $2L^m$ with m at least the largest of the $2r$ least exponents admitted at them,

an unbounded set. Then

$$\langle \mathcal{O}(f); \mathcal{O}(h) \rangle = \frac{3n}{16\pi^2}\, \mathsf{b}^{-2} \mathcal{K}_6(f,h)\,(1+\varepsilon'), \qquad |\varepsilon'| \le \max_{i,s} \bar{\varepsilon}_{g_{i,s}}(\mathsf{d}) + \tfrac{r-1}{12}\varepsilon^{\times}(\mathsf{d}),$$

uniformly in K and in β in its window, on these tori, in their limits along such tori, and in every Gibbs state of $e^{-\beta \sum_p a_p}$ whose marginals on the factors are periodic; $\bar{\varepsilon}_g$ is the number of that law, and $\varepsilon^{\times}(\mathsf{d})$, a number of g, the λ_i, the parameter of the test pairs and the constants, is at most a constant times d and vanishes on the tori and in product states. ■□

9.1.4. *Inputs from the $SU(2)$ construction.* We start from Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD for $SU(2)$ and from their proofs: the inductive step, Prop. YM3_I:69.3 and Thm. YM3_I:AC; the flow bound YM3_I:(1468) and the intrinsic coupling construction in Section YM3_I:52.10; the large-field chain, Prop. YM3_I:44.4, Prop. YM3_I:44.5 and Lem. YM3_I:65.2 and the local-deposit argument in the proof of Prop. YM3_I:65.4; and the verification of the conditions on the constants, the ordered choice and its numerical verification in Prop. YM3_I:69.1. From the first part we use the mean and the block average (equation YM3_I:(25) and YM3_I:(37)–(39), Prop. YM3_I:4.3), the propagators in a background (the precise input (P) of §9.1.1), the critical configurations and their localisation (input (V) there and Prop. 9.22(iii)), the bond map (Lem. YM3_I:53.3), the elimination factor (Lem. 9.14(iii)), the one-loop Thm. YM3_I:O, with the determinant representation and telescoping argument in Section YM3_I:43.3 and Lem. 9.40, and its limit, Thm. AC; and the continuum theorems with their proofs. Of G we use the closed subgroup theorem, the spectral theorem, Schur's lemma, the Haar measure, the Peter–Weyl theorem, and for groups other than $SU(N)$ maximal tori and roots.

9.1.5. *Matrix charts, averages and gauges.*

Lemma 9.9 (Norms, the domain D^c and means)**.**

(i) $\|Y\| \le |Y| \le \kappa_G \|Y\|$ on $\mathfrak{g}$, with $\kappa_G^2 \le N-1$ for semisimple $\mathfrak{g}$; $a(V) = \frac{1}{2}\|V-\mathbf{1}\|^2$ on $U(N)$; on $SU(N)$, $\frac{1}{2(N-1)}|V-\mathbf{1}|^2 \le a(V) \le \frac{1}{2}|V-\mathbf{1}|^2$ and $|V-\mathbf{1}| \le (N-1)^{1/2}\|V-\mathbf{1}\|$, all three sharp. So $a(V) \ge \frac{1}{2}\tau_G^{-2}|V-\mathbf{1}|^2$ on G, with $\tau_G^2 \le N-1$ for $G \subset SU(N)$. For $n \ge 2$, $X + X^{-1}$ is scalar for every $X \in G$ if and only if G is conjugate to $\{u \otimes \mathbf{1}_k : u \in SU(2)\}$.

(ii) $Y \mapsto e^{iY}$ maps $\{Y \in \mathfrak{g} : |Y| < \pi\}$ bijectively onto G^{π}, which is the component of $\mathbf{1}$ in $\{u \in G : -1 \notin \operatorname{spec} u\}$. If $A_j \in \mathfrak{g}$ and $\sum_j |A_j| < \pi$, then $e^{iA_1}\cdots e^{iA_m} = e^{iA}$ with $A \in \mathfrak{g}$, $|A| \le \sum_j |A_j|$. For hermitian Y, Y' with $|Y|, |Y'| \le t < \pi$: $|Y - Y'| \le \frac{t}{\sin t}|e^{iY} - e^{iY'}|$. For a matrix X with $s = |X-\mathbf{1}| < 1$: $|X^{-1}-\mathbf{1}| \le \frac{s}{1-s}$; and $|X^{-1}| \le |\operatorname{Ad} X|$ for $X \in SL(N,\mathbb{C})$, $\operatorname{Ad} X$ being read on all $N \times N$ matrices, where $|\operatorname{Ad} X| = |X|\,|X^{-1}|$; with the norm of $\operatorname{Ad} X$ on $\mathfrak{g}^c$ alone in its place the inequality fails for some pairs, as the proof of Lem. 9.19 shows.

(iii) For compact connected $G \subset U(N)$ the statements $\mathfrak{g}^c = \{Z : \operatorname{tr} Z = 0\}$, $G^c = \{X : \det X = 1\}$ and $G = SU(N)$ are equivalent; for semisimple $\mathfrak{g}$, $\operatorname{tr} Z = 0$ on $\mathfrak{g}^c$ and $\det X = 1$ on G^c, and either inclusion the other way holds only at $SU(N)$. For every compact connected G: if $Z_1, \dots, Z_q \in \mathfrak{g}^c$ and $\sum_j |Z_j| < \ln 2$, then $|e^{Z_1}\cdots e^{Z_q} - \mathbf{1}| < 1$ and $\log(e^{Z_1}\cdots e^{Z_q}) \in \mathfrak{g}^c$, and it lies in $i\mathfrak{g}$ if the Z_j do. Let $\mathfrak{g}$ be semisimple. Then $Z(G^c) = Z(G)$ is a finite set of unitary matrices; for $X \in G^c$ with $|X-\mathbf{1}| < 1$, $z(X) = \exp((1-P)\log X)$ lies in $Z(G)$, $|z(X)-\mathbf{1}| \le |X-\mathbf{1}|$, $\log X = \log z(X) + P \log X$, and z is locally constant. So $D^c = \{z(X) = \mathbf{1}\}$ is the connected component of $\mathbf{1}$ in $\{X \in G^c : |X-\mathbf{1}| < 1\}$ and the trace on G^c of an open set of matrices; it contains $\{X \in G^c : |X-\mathbf{1}| < z_G\}$, and no larger ball if $z_G < 1$; it is kept by the powers $X^t = e^{t \log X}$, $0 \le t \le 1$, by continuous paths in the unit ball that begin in it, by products of elements X_j of D^c with $\prod_j (1 + |X_j - \mathbf{1}|) < 2$, by the products $e^{Z_1}\cdots e^{Z_q}$ with $Z_j \in \mathfrak{g}^c$, $\sum_j |Z_j| < \ln 2$, by conjugation with G, by inverses in the unit ball and by adjoints; $\{X \in G^c : |\log X| < 2 \arcsin \frac{z_G}{2}\} \subset D^c$; and $D = \exp(i\{Y \in \mathfrak{g} : |Y| < \frac{\pi}{3}\}) = G^{\pi} \cap \{|X-\mathbf{1}| < 1\}$. Let $0 < c_j \le 1$. Within the domain cut

out in stages by $|X_j - \mathbf{1}| < c_j$ for continuous G^c-valued maps, retain these radius conditions and add $X_j \in D^c$ at each stage. The resulting domain is open and closed in the original domain, and equals it if $z_G = 1$.

(iv) Let $V_1, \dots, V_q$ be invertible, $d = \max_{i,j} |V_i V_j^{-1} - \mathbf{1}| \le \frac{1}{40}$. Then

$$\Phi(Y) = Y + \frac{1}{q} \sum_i \log(V_i V_1^{-1} e^{-Y})$$

is a contraction with factor $\frac{2}{5}$ of $\{|Y| \le \frac{1}{8}\}$; its fixed point has $|Y_*| \le 2d$, and $\mathrm{av} = e^{Y_*} V_1$ satisfies $\sum_i \log(V_i \mathrm{av}^{-1}) = 0$ and $|V_i \mathrm{av}^{-1} - \mathbf{1}| \le 4d$, is holomorphic in (V_i), and is the only unitary mean of a unitary family. Let $\mathfrak{g}$ be semisimple, let the V_i lie in G^c, and let the quotients $V_i V_1^{-1}$ lie in D^c, which holds if $d < z_G$. Then $Y_* \in \mathfrak{g}^c$, $\mathrm{av} \in G^c$ and $V_i \mathrm{av}^{-1} \in D^c$ for all i; and $\mathrm{av} \in G$ for a family in G. Further,

$$\mathrm{av}(W \mathbf{V} W') = W \,\mathrm{av}(\mathbf{V})\, W'$$

for $W, W' \in G$. For semisimple $\mathfrak{g}$, if $z_G \le \frac{1}{40}$ and $z \in Z(G)$ has $|z - \mathbf{1}| = z_G$, the family $(\mathbf{1}, z)$ has no mean in G.

■

Sketch of proof. (i) For V with the eigen-angles θ_j, $|V - \mathbf{1}| = \max_j 2|\sin\frac{\theta_j}{2}|$ and $a(V) = \frac{2}{N}\sum_j \sin^2\frac{\theta_j}{2} = \frac{1}{2}\|V - \mathbf{1}\|^2$. On $SU(N)$, $\sum_j \theta_j \in 2\pi\mathbb{Z}$, so $|\sin\frac{\theta_1}{2}| \le \sum_{j\ge 2} |\sin\frac{\theta_j}{2}|$, and Cauchy's inequality gives $\sin^2\frac{\theta_1}{2} \le (N-1)\sum_{j\ge 2}\sin^2\frac{\theta_j}{2}$, which is $|V - \mathbf{1}|^2 \le (N-1)\|V - \mathbf{1}\|^2$; the same argument with the eigenvalues of a traceless hermitian Y gives $\kappa_G^2 \le N - 1$. The constants are approached along $\exp(it\,\mathrm{diag}(N-1, -1, \dots, -1))$, $t \to 0$, and $\frac{1}{2}$ is attained at $e^{2\pi i/N}\mathbf{1}$. If $X + X^{-1}$ is scalar on G, then $2\cos(tY)$ is scalar for $Y \in \mathfrak{g}$ and all t, so $Y^2 = \|Y\|^2\mathbf{1}$ and orthogonal elements of $\mathfrak{g}$ anticommute; for orthonormal $X, Y \in \mathfrak{g}$ the element $Z = iXY = \frac{i}{2}[X, Y]$ lies in $\mathfrak{g}$, and an element W orthogonal to X, Y, Z would anticommute with X and Y, hence commute with Z, and anticommute with Z, which is absurd as $Z^2 = W^2 = \mathbf{1}$; so $\mathfrak{g}$ is spanned by X, Y, Z, these generate a quaternion algebra, and $\mathbb{C}^N$ is a multiple of its two-dimensional representation.

(ii) The path $s \mapsto e^{isA_1} \cdots e^{isA_m}$ has speed at most $\sum_j |A_j|$, so its eigen-angles stay below π, and its logarithm, continuous along it and in $\mathfrak{g}$ near $\mathbf{1}$ by the closed subgroup theorem, stays in $\mathfrak{g}$; the Lipschitz bound is the spectral calculus of $Y \mapsto e^{iY}$; the bound on X^{-1} is the series $\sum_j (\mathbf{1} - X)^j$; and for $X \in SL(N, \mathbb{C})$ with singular values $\sigma_1 \ge \cdots \ge \sigma_N$, $\prod \sigma_j = 1$, one has $|X^{-1}| = \sigma_N^{-1} \le \sigma_1 \sigma_N^{-1} = |X|\,|X^{-1}| = |\mathrm{Ad}\, X|$, the norm of $\mathrm{Ad}\, X$ on all $N \times N$ matrices being attained at vw^* for unit vectors v, w with $|Xv| = |X|$ and $|(X^{-1})^* w| = |X^{-1}|$.

(iii) $G^c = G\exp(\mathfrak{g}) = \{ue^H : u \in G,\ H \in \mathfrak{g}\}$, e^H being positive (a polar decomposition got by integrating e^{tZ} inside the set $\{ue^H\}$), and $\det e^Z = e^{\mathrm{tr}\, Z}$; for semisimple $\mathfrak{g}$ every element of $\mathfrak{g}^c$ is a sum of brackets, so its trace on every invariant subspace vanishes. The rule for products holds for every compact connected G: along $X_t = e^{tZ_1} \cdots e^{tZ_q}$, $0 \le t \le 1$, $|X_t - \mathbf{1}| \le e^{t\sum_j |Z_j|} - 1 < 1$, and $X_t' X_t^{-1} = \sum_j \mathrm{Ad}(e^{tZ_1} \cdots e^{tZ_{j-1}}) Z_j$ lies in $\mathfrak{g}^c$, as $\mathrm{Ad}(G^c)$ keeps $\mathfrak{g}^c$; $W_t = \log X_t$ has $X_t' X_t^{-1} = \Xi(\mathrm{ad}\, W_t) W_t'$ with $\Xi(w) = (e^w - 1)/w$, and $\Xi(\mathrm{ad}\, W_t)$ is invertible, the eigenvalues of $\mathrm{ad}\, W_t$ being differences of logarithms of eigenvalues of X_t, with imaginary parts in $(-\pi, \pi)$; so W_t solves $W' = \Xi(\mathrm{ad}\, W)^{-1}(X' X^{-1})$ from $W_0 = 0$. For a real Lie algebra $\mathfrak{h}$ of matrices that holds W, $\Xi(\mathrm{ad}\, W)$, a power series in $\mathrm{ad}\, W$ with real coefficients, maps $\mathfrak{h}$ into itself and, being invertible, onto itself; so the equation keeps $\mathfrak{h} = \mathfrak{g}^c$, and $\mathfrak{h} = i\mathfrak{g}$ if the Z_j lie there, and W_t stays in $\mathfrak{h}$. With $W = \log X$ the eigenvalues of $\mathrm{ad}\, W$ have imaginary parts in $(-\pi, \pi)$, so $\mathrm{ad}\, W$ is a polynomial in $e^{\mathrm{ad}\, W} = \mathrm{Ad}(X)$ and keeps $\mathfrak{g}^c$; hence $(1 - P)W$ commutes with $\mathfrak{g}^c$, it is a scalar on each isotypic component, the mean there of the eigenvalues of W, and $z(X) = Xe^{-PW} \in G^c$ is central. A mean c of numbers $\log\lambda_j$ with $|\lambda_j - 1| < r \le 1$ has $|e^c - 1| < r$, the set $\{|e^{x+iy} - 1| < r,\ |y| < \frac{\pi}{2}\}$ being convex; this gives $|z(X) - \mathbf{1}| \le |X - \mathbf{1}|$, and z, continuous with values in the finite set $Z(G)$, is locally constant.

The products are joined to $\mathbf{1}$ inside the unit ball by $t \mapsto e^{tZ_1}\cdots e^{tZ_q}$ and $t \mapsto X_1^t \cdots X_q^t$; the inner ball comes from $|\log X| \geq \theta(z(X))$, $\theta(V)$ the largest modulus of the argument of an eigenvalue of V.

(iv) $|\log(\mathbf{1}+E)| \leq -\log(1-|E|)$ and the bound on the derivative of log for $|E| \leq \frac{1}{10}$ give the contraction; if the quotients lie in D^c, so does $V_i V_1^{-1} e^{-Y}$ for $Y \in \mathfrak{g}^c$, $|Y| \leq \frac{1}{8}$, by the rule for products of (iii), so Φ keeps $\mathfrak{g}^c$, $Y_* \in \mathfrak{g}^c$ and $V_i \mathrm{av}^{-1} = V_i V_1^{-1} e^{-Y_*} \in D^c$; and $\mathrm{av} = e^{Y_*} V_1 \in G^c$. For unitary V_i, V' with $\sum_i \log(V_i V'^{-1}) = 0$ the spectral theorem gives that every angle of $V_j V'^{-1}$ is at most $\frac{2\pi}{3\sqrt{3}} d$, whence uniqueness. At $(\mathbf{1}, z)$, $\mathrm{av} = e^{\frac{1}{2}\log z}$, which is not in G^c (for $SU(N)$ its determinant is -1). □

Lemma 9.10 (A bond in the contour gauge)**.** Let W be a real G-valued field on the bonds of a block $B(y)$, and let $\varpi_y(W)$ be the largest plaquette distance from $\mathbf{1}$ in that block. Suppose

$$\tfrac{3}{4}(L-1)^2 \varpi_y(W) < \omega_{\mathrm{c}}, \qquad \varpi_y(W) < z_G,$$

where $\omega_{\mathrm{c}} := 1/40$, the contour-mean radius admitted by Prop. YM3_I:4.3. Then the contour means $W(y,x)$ exist in G, and every bond b inside the block satisfies

$$|W(y,b_-)W(b)W(y,b_+)^{-1} - \mathbf{1}| \leq 2\kappa_1^{\mathrm{SO}} \varpi_y(W).$$

Here $\kappa_1^{\mathrm{SO}} = \pi(L-1)^2/(2\sqrt{3})$. ■

Proof. Two coordinate-order paths from y to x differ by at most $\frac{3}{4}(L-1)^2$ plaquettes. Telescoping their transports gives the displayed diameter bound for the contour family. Its quotients are products of conjugated plaquette variables. The guard $\varpi_y(W) < z_G$ and Lem. 9.9(iii) put their logarithms in $\mathfrak{g}$; Lem. 9.9(iv) therefore gives the mean in G. Apply the inverse-exponential inequality of Lem. 9.9(ii) to the intrinsic logarithms of these quotients, then the zero-sum-logarithm argument of Lem. YM3_I:14.5. This gives

$$|W(\Gamma)W(y,x)^{-1} - \mathbf{1}| \leq \kappa_1^{\mathrm{SO}} \varpi_y(W)$$

for every coordinate-order path Γ from y to x. Reverse b if necessary so that its step moves away from y in its coordinate direction. Choose that direction last in the coordinate order. The resulting path Γ' from y to b_+ ends with b, and deleting this last bond gives a coordinate-order path Γ to b_-. Thus

$$W(y,b_-)W(b)W(y,b_+)^{-1} = [W(\Gamma)W(y,b_-)^{-1}]^{-1}[W(\Gamma')W(y,b_+)^{-1}].$$

The two factors lie in G and have distance at most $\kappa_1^{\mathrm{SO}}\varpi_y(W)$ from $\mathbf{1}$. The triangle inequality and unitary invariance prove the claim. □

Lemma 9.11 (Block averages)**.** Let $L \geq 13$ be odd and let c be a coarse bond.

(i) Let U be G-valued with $|\partial U(p) - \mathbf{1}| < \min\{\omega_M, r_G\}$ at the plaquettes of the double block of c. Then the contour quotients lie in D within $\frac{1}{160}$ of $\mathbf{1}$, each contour family (Section YM3_I:4 (coordinate-order contours)) has exactly one mean in G, within $\frac{1}{40}$, the W_x of equations YM3_I:(37)–(39) lie in D within $\frac{1}{14}$, and $\mathcal{M}(U)(c) = \exp(L^{-3}\sum_x \log W_x)\, U[c] \in G$ is real-analytic in U. This holds for every compact connected G; with ω_M in the place of $\min\{\omega_M, r_G\}$ it fails for every G with $r_G < \omega_M$. Let now the centre of $\mathfrak{g}$ be 0. Then $Z(G)$ is finite and $\min\{1, r_G\} = z_G$, so that the statement holds with ω_M in the place of $\min\{\omega_M, r_G\}$ if and only if $\omega_M \leq z_G$. At G^c-valued fields the same formula is holomorphic in the matrix entries where the contour quotients and the W_x lie in D^c; without this condition it leaves G^c for every such G with $z_G < 1$.

(ii) The averaging map of equations YM3_I:(37)–(39) transferred to G, its three cases read with "$u \in G^\pi$" and $\mathrm{Log}\, e^{iY} = iY$ for $Y \in \mathfrak{g}$, $|Y| < \pi$, is G-valued and Borel on all fields, depends on U on the double block of c alone, is compatible with the reversal of bonds, and $\mathcal{M}(U^h)(c) = h(c_-)\mathcal{M}(U)(c)h(c_+)^{-1}$ with the same case; so $\mathcal{M}^q(U^h) = (\mathcal{M}^q U)^h$. As a map it needs no condition on z_G. For $SU(N)$ with $\omega^G = \min\{\omega_M, 2\sin\frac{\pi}{N}\}$ the averaging

map and the lemmas of the $SU(2)$ construction on it, Lem. YM3_I:14.2 among them, hold with their constants but for one clause: the second assertion of Lem. YM3_I:14.2, that $L^{-3}\sum_x \log W_x \in i\mathfrak{g}$ and $\mathcal{M}(W)(c) \in G$ for every G-valued W of its domain, holds for every G with $\rho_G \geq \frac{\pi}{3}$, that is $r_G \geq 1$, and fails where $\rho_G < \frac{\pi}{3}$, that is $r_G < 1$, so for $SU(N)$ exactly when $N \geq 7$. It holds for every G with the proviso that every eigen-angle of each W_x is below ρ_G in modulus, or that W_x is a product of exponentials $e^{i\gamma_j}$, $\gamma_j \in \mathfrak{g}^c$, with $\sum_j |\gamma_j| < \ln 2$; the proviso is met under the averaging radius.

(iii) If $|\partial U(p) - \mathbf{1}| \leq \alpha_0 L^{-2k}$ at all bare plaquettes and $\alpha_0 \leq c(G) = \min\{3.9\cdot 10^{-9} L^{-2}, \frac{1}{4} r_G\}$, then for $0 \leq q \leq k$ the plaquette variables of $\mathcal{M}^q U$ are within $2\alpha_0 L^{2(q-k)}$ of $\mathbf{1}$, $\mathcal{M}^k U(c) U(\Gamma_c)^{-1}$ is a product of exponentials e^{iA_j}, $A_j \in \mathfrak{g}$, of total size at most $12\alpha_0$, so that it lies in G^π, and $|\mathcal{M}^k U(c) - U(\Gamma_c)| \leq 12\alpha_0$, Γ_c the straight bare path from c_- to c_+. The same holds for every G-valued averaging map that is the formula of (i) on small fields, and in a local form, for fields given beneath the double block of c alone.

(iv) Let $G = \tilde{G}/\mathsf{Z}$ with $\tilde{G}$ simply connected. On a torus a field as in (iii) has a sector $\zeta(U) \in \mathsf{Z}^3$, the class of the Z-valued cocycle of the central parts of the plaquette variables of a lift of U to $\tilde{G}$, and $\mathcal{M}^k U$ lies in the sector of U. More generally, a coarse field V with $|V(c) - U(\Gamma_c)| \leq \epsilon$ over a fine field with plaquettes within κ of $\mathbf{1}$ lies in the sector of U as soon as $\mathcal{L}^2\kappa + 4\epsilon < r_G$, $\mathcal{L}$ the ratio of the spacings; in angles the threshold is π, and it is sharp for $SO(3)$. Fields whose block averages are all $\mathbf{1}$ lie in the trivial sector, and the block averages alone fix the sector.

(v) Let $1 \leq j \leq K$, let Y be the two end points of a bond of level j and $\mathsf{Y}_i = B^{j-i}(\mathsf{Y})$, $0 \leq i \leq j$, the sites of level i in their blocks of order $j - i$; let U_0 be G-valued on the bare bonds with both end points in Y_0, with $|\partial U_0(p) - \mathbf{1}| \leq \alpha L^{-2j}$ at the bare plaquettes with all corners in Y_0, where $0 < \alpha \leq c_2^{\mathrm{g}}$, and let A_0 be $\mathfrak{g}$-valued on these bonds with $L^j \sup|A_0| < c_4$; here $c_2^{\mathrm{g}} \leq \frac{1}{4}\min\{\omega_M, r_G\}$ and the radius $c_4 \leq \frac{1}{4} r_L$ of the averages over a box, r_L the radius of the one step below, are numbers of L and r_G, so small that $\sum_l C_{\mathrm{P}} \varpi_l \leq \ln 2$ and $\sum_l C\lambda_l \leq \ln 2$ in the proof. Then, uniformly in the number j of steps, for $0 \leq i \leq j$ the averages $\mathcal{M}^i U_0$ and $\mathcal{M}^i(e^{iA_0} U_0)$ are formed on the bonds with both end points in Y_i, by the formula of (i) at every step, the plaquette variables of $\mathcal{M}^i U_0$ lie within $2\alpha L^{-2(j-i)}$ of $\mathbf{1}$, and there are $\mathfrak{g}$-valued A_i and $f_i : \mathsf{Y}_i \to G$, $f_0 = \mathbf{1}$, with

$$\mathcal{M}^i(e^{iA_0} U_0)(c) = f_i(c_-)\, e^{iA_i(c)}\, (\mathcal{M}^i U_0)(c)\, f_i(c_+)^{-1}, \qquad |A_i| \leq 2L^i \sup|A_0|,$$

$f_{i+1}(y) = f_i(y) e^{i\sigma_i(y)}$ with $\sigma_i(y) \in \mathfrak{g}$, $|\sigma_i(y)| \leq \kappa_6 \sup|A_i|$, and $f_i \in G^\pi$ with $|\frac{1}{i}\log f_i| \leq 6L^i \sup|A_0|$, so that the bound $|f_i - \mathbf{1}| \leq C_v L^i \sup|A_0|$ of the averages over a box holds with $C_v = 6$. One step is governed by the numbers $r_L = \frac{1}{1600 L^2}$, $\varpi_L = \frac{1}{2}\min\{\omega_M, r_G\}$ and $\kappa_6 = 2L$: if W_0 is G-valued on the double block of a bond c with plaquette variables within ϖ_L of $\mathbf{1}$, and a is $\mathfrak{g}$-valued there with $\sup|a| < r_L$, then the contour variables of $W = e^{ia} W_0$ and of W_0 are formed, $|W(y,x) W_0(y,x)^{-1} - \mathbf{1}| < 1$, and its logarithm lies in $i\mathfrak{g}$ with $|\frac{1}{i}\log(W(y,x) W_0(y,x)^{-1})| \leq \kappa_6 \sup|a|$. The condition $\varpi_L < r_G$ cannot be spared. At complex arguments there is a number $a_{\mathrm{c}} > 0$ of L and the pair such that, at every real background U_0 beneath a bond c of level q whose bare plaquette variables lie within $a_{\mathrm{c}} L^{-2q}$ of $\mathbf{1}$, and for $\mathfrak{g}^c$-valued a with $L^q \sup|a| < a_{\mathrm{c}}$, $\mathcal{M}^q(e^{ia} U_0)(c) = e^{iQ}\, \mathcal{M}^q U_0(c)$ with $Q \in \mathfrak{g}^c$ holomorphic in a, in $\mathfrak{g}$ for real a, and $|Q| \leq 167\, L^q \sup|a|$, uniformly in q; and $|\frac{1}{i}\log \mathcal{M}^q(e^{ia})| \leq 9.2\, L^q \sup|a|$, uniformly in q and N. At the trivial background, for $\xi = L^{-k}$ and a $\mathfrak{g}^c$-valued form f beneath the double block of c with $\sup|f| < 4\cdot 10^{-7}$, the averages $\mathcal{M}^j(e^{i\xi f})$, $0 \leq j \leq k$, are e^{ia_j} with $\mathfrak{g}^c$-valued a_j, holomorphic in f, real for real f, and $\sup|a_j| \leq 8L^{j-k} \sup|f|$. For $SU(3)$ no number changes.

■

Sketch of proof. (i) A contour quotient is a product of at most $\frac{9}{8}L^2$ conjugated plaquette variables, those of the area between two shortest paths in a block, so it lies within $(1+\omega_M)^{9L^2/8}-1 \le e^{9/1600}-1 \le \frac{1}{160}$ of $\mathbf{1}$. A plaquette variable within r_G of $\mathbf{1}$ is e^{iF} with $F\in\mathfrak{g}$, $|F|\le\frac{\pi}{2}|\partial U(p)-\mathbf{1}|$, and so is each of its conjugates by G; so the quotient is e^{iA} with $A\in\mathfrak{g}$, $|A|\le\frac{9\pi}{16}L^2\omega_M<\frac{1}{100}$, and lies in D (Lem. 9.9(ii)). Lem. 9.9(iv) gives exactly one unitary mean, within $4d\le\frac{1}{40}$, and its map Φ keeps $i\mathfrak{g}$, each $V_iV_1^{-1}e^{-Y}$ with $Y\in i\mathfrak{g}$, $|Y|\le\frac{1}{8}$, being a product of two exponentials of elements of $i\mathfrak{g}$ of total size below $\ln 2$ (Lem. 9.9(iii)); so the fixed point lies in $i\mathfrak{g}$ and the mean in G. $P_x=U(\Gamma)U([x,x'])U(\Gamma')^{-1}U[c]^{-1}$ takes $\frac{25}{8}L^2$ plaquettes and lies within $e^{1/64}-1\le\frac{1}{60}$; then $|W_x-\mathbf{1}|\le\frac{6541}{96000}<\frac{1}{14}$, and W_x, the product of P_x with conjugates of the quotients of two contour variables by their means, is again e^{iB} with $B\in\mathfrak{g}$ small, so it lies in D. Holomorphy holds on the open sets of matrices cut out by the conditions with D^c, on which av and log are holomorphic. If $r_G<\omega_M$, one bond set to an element $u_*\in G\setminus G^\pi$ with $|u_*-\mathbf{1}|=r_G$ gives a contour family without a mean in G, as in (v). Let the centre of $\mathfrak{g}$ be 0. A central $z\ne\mathbf{1}$ lies outside G^π, since $z=e^{iY}$ with $Y\in\mathfrak{g}$, $|Y|<\pi$, would give $e^{i\operatorname{ad}Y}=1$ with the eigenvalues of $\operatorname{ad}Y$ below 2π in modulus, so $\operatorname{ad}Y=0$ and $Y=0$; and an element $u\in G\setminus G^\pi$ with $|u-\mathbf{1}|<1$ has the central factor $z(u)\ne\mathbf{1}$, with $|z(u)-\mathbf{1}|\le|u-\mathbf{1}|$ (Lem. 9.9(iii)); so $\min\{1,r_G\}=z_G$. If $z_G<\omega_M$, one bond set to a central z with $|z-\mathbf{1}|=z_G$ gives the contour family $(z,\mathbf{1})$, which has no mean in G (Lem. 9.9(iv)).

(ii) The three cases are read with G^π, on which Log inverts the chart of Lem. 9.9(ii); each is covariant, as $\operatorname{av}(h[V]h')=h\operatorname{av}([V])h'$ and $\operatorname{Log}(huh^{-1})=h(\operatorname{Log}u)h^{-1}$; the sets on which the cases apply are open, closed and Borel, and the map is continuous on each. The domain of Lem. YM3_I: 14.2 asks $|W_x-\mathbf{1}|<1$, so the eigen-angles of W_x are below $\frac{\pi}{3}$ in modulus; if $\rho_G\ge\frac{\pi}{3}$, then $\log W_x\in i\mathfrak{g}$ by the definition of ρ_G and Lem. 9.9(ii), and under the proviso the same holds by Lem. 9.9(ii), (iii). If $\rho_G<\frac{\pi}{3}$, there is $V\in G$ whose eigen-angles are at most ρ_G in modulus and whose hermitian logarithm Y does not lie in $\mathfrak{g}$, for $SU(N)$ the element $e^{2\pi i/N}\mathbf{1}$; with V on one bond of the corridor of c other than the pivot bond and $\mathbf{1}$ at the others, $L^{-3}\sum_x\log W_x=iL^{-2}Y\notin i\mathfrak{g}$.

(iii) We never take the logarithm of an element that is merely near $\mathbf{1}$: every object of a step is written as a product of exponentials e^{iA_j}, $A_j\in\mathfrak{g}$ small, and $\sum_j|A_j|$ is added up (Lem. 9.9(ii)). One step gives $\mathcal{M}(W)(c)=e^{iY}W[c]$ with $Y\in\mathfrak{g}$, $|Y|\le\kappa_3\max_p|\partial W(p)-\mathbf{1}|$, $\kappa_3<6L^2$; the plaquette variables of an average are within $L^2\varpi+C_0\varpi^2$ of $\mathbf{1}$ if those beneath are within ϖ, C_0 a number of L; and the corrections sum over the levels as geometric series, with the factor $\frac{2\kappa_3}{L(L-1)}<11.99$; the number $c(G)$ reads of the pair only r_G.

(iv) If $\sum_j|X_j|<\pi$ in the norm of $\mathbb{C}^N$, then $\prod_j e^{i\tilde{X}_j}$ is the canonical lift of $\prod_j e^{iX_j}$; by (iii) the plaquette variables of every average are such products of small exponentials of those beneath, so the central parts of the plaquette variables of the lift multiply under blocking. A field with trivial block averages is a gauge transform of e^{ia}, $4\sup|a|<\pi$ (Prop. 9.33), whose lift $e^{i\tilde{a}}$ has central parts $\mathbf{1}$. Four bonds turned by $\frac{\pi}{4}+\delta$ about one axis around one plaquette of $SO(3)$ give a field near the trivial one in a twisted sector.

(v) One step, relative to the background. On the double block of c a tree gauge, the comb of Lem. YM3_I:14.4, writes $W_0=e^{iA_0}$ and $W=e^{ia}W_0=e^{iA}$ with $\mathfrak{g}$-valued A_0, A of size at most $2\pi L\varpi+\sup|a|$ and $\sup|A-A_0|\le1.01\sup|a|$ (Lem. 9.9(ii)). In this gauge every contour variable, every W_x and the average are exponentials e^{iX} of elements $X\in\mathfrak{g}$ formed from A by means and by logarithms of products of small exponentials, and the Lipschitz bound of the remainder of Campbell and Hausdorff, $|F(X)-F(X')|\le4S\sum_j|X_j-X_j'|$ for $F(X)=\log(e^{X_1}\cdots e^{X_r})-\sum_jX_j$ and $S=\sum_j\max\{|X_j|,|X_j'|\}\le\frac{1}{4}$, compares each of them with its linear part, a straight average with real weights. Let $\xi(y,x)=\frac{1}{i}\log(W(y,x)W_0(y,x)^{-1})$, so that $|\xi|\le\kappa_6\sup|a|$, and let $\sigma(y)=L^{-3}\sum_{x\in B(y)}\xi(y,x)$. Then $\mathcal{M}(W)(c)=e^{i\sigma(c_-)}e^{ia'(c)}\mathcal{M}(W_0)(c)e^{-i\sigma(c_+)}$ with $a'(c)\in\mathfrak{g}$: in the linear part the means of $A-A_0$ over the contours from $c_\pm$ cancel against $\sigma(c_\pm)$, and what is left is $L^{-3}\sum_{x\in B(c_-)}(A-A_0)([x,x'])$, a sum over the L bonds of each $[x,x']$, whence the exact factor L;

the rest is bounded by the Lipschitz bound, so that $|a'(c)| \le L \sup|a|\,(1+C\lambda)$ with λ a number of L times $\varpi + \sup|a|$. In the same way the plaquette variables of the average lie within $L^2\varpi(1+C_{\mathrm{P}}\varpi)$ of $\mathbf{1}$ if those beneath lie within ϖ, C_{P} a number of L. Over the levels the plaquette bounds $\varpi_i = 2\alpha L^{-2(j-i)}$ and the numbers λ_i, a number of L times $\varpi_i + 2c_4 L^{i-j}$, decrease geometrically in $L^{-(j-i)}$, with $\sum_{l<j}\varpi_l \le \frac{2c_2^{\mathrm{g}}}{L^2-1}$ and $\sum_{l<j} 2c_4 L^{l-j} \le \frac{2c_4}{L-1}$. At level i the one step applies, as $\varpi_i \le 2c_2^{\mathrm{g}} \le \varpi_L$ and $\sup|A_i| \le 2L^i \sup|A_0| < 2c_4 L^{i-j} \le \frac12 r_L$; and the two conditions on c_2^{g} and c_4, $\sum_l C_{\mathrm{P}}\varpi_l \le \ln 2$ and $\sum_l C\lambda_l \le \ln 2$, keep the products $\prod_{l<i}(1+C_{\mathrm{P}}\varpi_l)$ and $\prod_{l<i}(1+C\lambda_l)$ at most 2, as $1+x \le e^x$; this closes the induction on i and gives $|A_i| \le 2L^i \sup|A_0|$ and the plaquettes within $2\alpha L^{-2(j-i)}$. Further $f_{i+1} = f_i e^{i\sigma_i}$ with $\sum_{l<i}|\sigma_l| \le \kappa_6 \sum_{l<i} 2L^l \sup|A_0| \le 6L^i \sup|A_0| < \pi$, so $f_i \in G^\pi$ (Lem. 9.9(ii)), and $|f_i - \mathbf{1}| \le |\frac{1}{i}\log f_i|$, as $|e^{i\theta}-1| \le |\theta|$ for real θ. At complex arguments the same recursion runs with $\mathfrak{g}^c$-valued exponents: the bound of Campbell and Hausdorff holds for complex matrices, the products $e^{Z_1}\cdots e^{Z_q}$ with $Z_k \in \mathfrak{g}^c$, $\sum_k |Z_k| < \ln 2$, have their logarithms in $\mathfrak{g}^c$ (Lem. 9.9(iii)), and the averages, composites of exponentials, products, means and logarithms, are holomorphic in a; this gives $Q \in \mathfrak{g}^c$ and the bounds with 167, 9.2 and 8, uniformly in q. If $\varpi_L \ge r_G$, one bond set to $u_* \in G \setminus G^\pi$ with $|u_* - \mathbf{1}| = r_G$ gives two contours without a mean in G. □

Lemma 9.12 (Regular fields and gauges)**.**

(i) For every G the gradient of $\xi^{-1}\sum_p a_p$ for the pairing of Def. YM3_I:9.1 is $J = D^{\mathrm{T}}F$, at G- and at G^c-valued fields.

(ii) Let κ' be the number of the pair of Def. 9.1. Then $4|(1-P)H_V| \le \kappa'|V-\mathbf{1}|^3$ for $V \in D$ with $|V-\mathbf{1}| \le z_G$; $|(1-P)H_V| \le (\frac{\pi}{3} - \frac{\sqrt3}{2})|1-P|\,|V-\mathbf{1}|^3$ for every $V \in D$, $|1-P|$ the norm of $1-P$ for $|\cdot|$, so that $\kappa' \le 4(\frac{\pi}{3} - \frac{\sqrt3}{2})|1-P|$; at $SU(2)$, $\kappa' = 0$; and at $SU(N)$, for every N, $4|(1-P)H_V| \le (2-\pi/\sqrt3)|V-\mathbf{1}|^3$ for every $V \in D$, so that $\kappa' \le 2-\pi/\sqrt3$, $|(1-P)H_V| \le \frac15|V-\mathbf{1}|^3$ and $|PH_V| \le (1+\frac15|V-\mathbf{1}|^2)|V-\mathbf{1}|$ on D. For every G, if the plaquette variables at the four plaquettes of a bond b lie in D within $\rho \le z_G$ of $\mathbf{1}$, then

$$\eta|(D^{\mathrm{T}}\partial U)(b)| \le \rho^2 + \eta^3|J(b)| + \kappa'\rho^3, \qquad \eta^3|J(b)| \le \eta|(D^{\mathrm{T}}\partial U)(b)| + \kappa'\rho^3.$$

For every G and every r, $\mathcal{U}'(r) \subset \mathcal{U}(\kappa_G r)$, the classes being those of Def. 9.2; for every G and $r \le z_G$, $\mathcal{U}(r) \subset \mathcal{U}'(r + r^2 + \kappa' r^3)$; Moreover, let $\alpha_0, \alpha_2 > 0$ and let E be a real field whose plaquette variables obey $|\partial E(p) - \mathbf{1}| \le (\alpha_0 + 3\alpha_2)\eta^2 \mathsf{L}(p)^{-2}$ and whose current at every full bond obeys $|J(E)(b)| \le s\mathsf{L}(b)^{-3}$, where $s = 2\alpha_0 + 3\alpha_2$ and $\alpha_0 + 3\alpha_2 < z_G$. Then $E \in \mathcal{U}'(s + s^2 + \kappa' s^3)$, and hence $E \in \mathcal{U}'(2s)$ if $s(1+\kappa' s) \le 1$. For $SU(2)$ the two hypotheses are supplied by the action expansion under equations YM3_I:(466)–YM3_I:(467). Their general-group counterparts are quantitative inputs in (P), wherever this conversion is applied to a solved field. The bound $\mathsf{L}(b)^3|J_{\Omega_0}(U_0)(b)| \le 3\alpha_0$ of Prop. YM3_I:30.1 at an inner bond that is not full holds for every G with $3|P|\alpha_0$ in the place of $3\alpha_0$. Let now $G = SU(N)$ and $r \le z_G$. Then $\mathcal{U}'(r) \subset \mathcal{U}((1+\kappa' r^2)r)$; at $SU(2)$, where $\kappa_G = 1$, $\kappa' = 0$ and $z_G = 1$, the inclusions are the two inclusions of the $SU(2)$ construction. Under the same curvature and current hypotheses, the membership in $\mathcal{U}'(2s)$ holds for $s \le \frac13$; and the bound at an inner bond that is not full holds with $6\alpha_0$, with $3\alpha_0$ exactly when $N \le 7$, and for every N with $3\alpha_0$ if $\alpha_0 \le 2\sin\frac{\pi}{N}$, so if $\alpha_0 \le z_G$. If $|\partial U(p) - \mathbf{1}| \le \rho \le 1$, $\rho < z_G$, at the four plaquettes of a bond b, then $J(U)(b) \in \mathfrak{g}$ and

$$|(D^{\mathrm{T}}\partial U)(b) - i\eta^2 J(U)(b)| \le \eta^{-1}\rho^2(1+\tfrac35\rho^2)^{1/2} \le \tfrac{13}{10}\eta^{-1}\rho^2.$$

The inclusion $\mathcal{U}'(r) \subset \mathcal{U}(r)$ is false for $N \ge 3$ at every small r, and the guard $\rho < z_G$ cannot be dropped.

(iii) The following quantitative cube-gauge assertion is an additional local analytic input when it is used for a general closed $G \subset U(N)$ of positive dimension. It requires constants $c, B > 0$ depending only on the pair: if Q is a cube of l^3 blocks at level k, the concentric cube

$\tilde{Q}$ of side $l+4$ is shorter than half the torus, $l\alpha \le c$, and $|\partial U(p)-\mathbf{1}|$ and the covariant divergence of ∂U are below $\alpha\eta^2$ on $\tilde{Q}$, then on $\tilde{Q}$ the field is gauge equivalent to $e^{i\eta A}$ with $|A|, |\nabla A|, |d^*dA|, |\Delta A| < Bl\alpha$ on Q. For $SU(2)$ all four bounds hold. The first three are Thm. YM3_I:F; the Laplacian bound follows from the same construction, as proved below, after enlarging B.

The comb gauge of a box of side s (Lem. YM3_I:14.4) gives $|W^h(b)-\mathbf{1}| \le \min\{2s, |x-y|_1\}\varpi$, and $W^h = e^{ia^0}$ with $\sup|a^0| \le \pi s\varpi$ if besides $2s\varpi < 2$ also $\varpi < z_G$; at $SU(N)$ this guard bites from $N=7$ on. The form of that clause with some gauge of the box in the place of the comb, that $2s\varpi < 2$ alone gives a gauge in which $W^h = e^{ia^0}$ with a^0 in $\mathfrak{su}(N)$ and $\sup|a^0| \le \pi s\varpi$, holds for $SU(N)$ exactly when $N \le 13$. The quantitative regular-cube assertion for $SU(N)$ uses the input just stated; the comb estimate alone does not supply its derivative bounds.

(iv) For $SU(2)$ the following box-gauge estimate is Prop. 15.8. For other G with centre-free Lie algebra it is an additional analytic input, required in (II): there is C such that for every s and $0 < \kappa \le C^{-2}$ every G-valued field on $\{0,\dots,s\}^3$ with $|\partial U(p)-\mathbf{1}| \le \kappa$ has a gauge transform within $C\sqrt{\kappa}$ of $\mathbf{1}$ at all bonds. The general-group assertion requires a uniform quantitative filling estimate. For a closed G that is not connected this holds if and only if it holds for the identity component, with $C(G) = \max\{C(G_0), (2/r_G)^{1/2}\}$. It fails for every G whose Lie algebra has a centre, and on periodic lattices for $SO(3)$.

■

Sketch of proof. (i) For $\theta \in \mathfrak{g}^c$ and every $X \in G^c$, near $\mathbf{1}$ or not, $\frac{d}{dt}a(e^{it\theta}X)|_{t=0} = \frac{1}{N}\mathrm{tr}(\theta H_X) = \frac{1}{N}\mathrm{tr}(\theta P H_X) = \langle\theta, PH_X\rangle$, since $(1-P)M$ is trace-orthogonal to $\mathfrak{g}^c$ by the definition of P (Def. 9.1). A change $e^{it\theta}$ of one bond changes the plaquette variables of its four plaquettes by conjugates of $e^{\pm it\theta}$ under transports, and $P\,\mathrm{Ad}(g) = \mathrm{Ad}(g)P$ for $g \in G^c$, as $\mathrm{Ad}(g)$ keeps $\mathfrak{g}^c$ and the trace form; summing over the four plaquettes, with the factors ξ of Def. YM3_I:9.1, gives $J = D^{\mathrm{T}}F$ with $F = \xi^{-2}PH_{\partial U}$.

(ii) For $V = e^{iY} \in D$, $Y \in \mathfrak{g}$, $t = |Y| < \frac{\pi}{3}$: $H_V = \sin Y$ and $(1-P)Y = 0$, so $(1-P)H_V = (1-P)(\sin Y - Y)$, of norm at most $|1-P|(t-\sin t) \le (\frac{\pi}{3} - \frac{\sqrt{3}}{2})|1-P|\,|V-\mathbf{1}|^3$, as $|V-\mathbf{1}| = 2\sin\frac{t}{2}$ and $(t-\sin t)/(2\sin\frac{t}{2})^3$ increases on $(0, \frac{\pi}{3}]$. For $V \in G$ with $0 < |V-\mathbf{1}| < z_G$, $V \in D$ (Lem. 9.9(iii)), so that κ' is at most four times the supremum of $|(1-P)H_V|/|V-\mathbf{1}|^3$ over $D \setminus \{\mathbf{1}\}$; and at $V = e^{iY} \in D$ with $|V-\mathbf{1}| = z_G$ the bound $4|(1-P)H_V| \le \kappa'|V-\mathbf{1}|^3$ is the limit of those at e^{isY}, $s \uparrow 1$, which lie in G within z_G of $\mathbf{1}$. At $SU(2)$, H_V lies in $\mathfrak{g}^c$. At $SU(N)$, $|1-P| = 1$ and $(1-P)H_V = \frac{1}{N}\mathrm{Im}\,\mathrm{tr}\,V \cdot \mathbf{1} = \frac{1}{N}\sum_j(\sin\theta_j - \theta_j)\mathbf{1}$, the eigen-angles θ_j of V being at most $t < \frac{\pi}{3}$ in modulus and summing to 0. With $f(x) = x - \sin x$, which is odd, convex on $[0,t]$ and concave on $[-t,0]$, the sum is extreme with a share $p \le \frac{1}{2}$ of the angles at t and the rest equal, at $-\frac{pt}{1-p}$: $\frac{1}{N}|\sum_j f(\theta_j)| \le \max_{0\le p\le 1/2}\psi(p)$, $\psi(p) = pf(t) - (1-p)f(\frac{pt}{1-p})$. The share $p = \frac{1}{3}$, which is extreme to third order in t, gives $\psi(\frac{1}{3}) = \frac{2}{3}\sin\frac{t}{2}\,(1-\cos\frac{t}{2})$, that is $4\psi(\frac{1}{3}) = \frac{|V-\mathbf{1}|^3}{3(1+\cos(t/2))}$, at most $0.1787\,|V-\mathbf{1}|^3$. The supremum of $4\psi(p)/|V-\mathbf{1}|^3$ over p and $t \le \frac{\pi}{3}$ is below 0.179, so that $4|(1-P)H_V| \le 0.179\,|V-\mathbf{1}|^3$, below $(2-\pi/\sqrt{3})|V-\mathbf{1}|^3$. The bound with $2-\pi/\sqrt{3}$ comes from the tangent of ψ at $p = \frac{1}{3}$: ψ is concave, $\psi''(p) = -t^2(1-p)^{-3}\sin\frac{pt}{1-p}$; with $s = \frac{t}{2}$, $\psi'(\frac{1}{3}) = 3s\cos s - \sin s - \sin 2s \le 0$, as $2\sin s + \tan s \ge 3s$; so $\psi \le \psi(\frac{1}{3}) - \frac{1}{3}\psi'(\frac{1}{3}) = \sin s - s\cos s$ on $[0, \frac{1}{2}]$, and $4(\sin s - s\cos s)/(2\sin s)^3$ increases in s, its derivative having the sign of $x(2+\cos x) - 3\sin x \ge 0$ at $x = 2s$, to $2-\pi/\sqrt{3}$ at $s = \frac{\pi}{6}$. At a bond b, $S = \eta(D^{\mathrm{T}}\partial U)(b)$ is the sum of the four plaquette variables transported to b, two with the sign $+$ and two with the sign $-$, and $S = R + iT$ with R, T the like signed sums of the transported $c(\partial U(p))$ and $H_{\partial U(p)}$, both hermitian at real fields, and $PT = \eta^3 J(U)(b)$, P commuting with the transports. The four constants $\mathbf{1}$ cancel and each $c(\partial U(p)) - \mathbf{1}$ lies between $-\frac{1}{2}\rho^2$ and 0, so $|R| \le \rho^2$; and $|(1-P)T| \le \kappa'\rho^3$, the four plaquette variables lying in D within $\rho \le z_G$ of $\mathbf{1}$. Hence

$|S| \le |R|+|PT|+|(1-P)T| \le \rho^2+\eta^3|J(b)|+\kappa'\rho^3$, and $\eta^3|J(b)| = |PT| \le |T|+|(1-P)T| \le |S|+\kappa'\rho^3$. At G-valued fields, for every G, T is hermitian and P commutes with taking adjoints, $\mathfrak{g}^c$ being closed under them, so $PT \in \mathfrak{g}$ and $|PT| \le \kappa_G\|PT\| \le \kappa_G\|T\| \le \kappa_G|T| \le \kappa_G|S|$, P being orthogonal for $\langle\cdot,\cdot\rangle$ (Def. 9.1, Lem. 9.9(i)); at a full bond of a field of $\mathcal{U}'(r)$ this gives $|J(U)(b)| < \kappa_G r\,\mathsf{L}(b)^{-3}$, and as $\kappa_G \ge 1$, $\mathcal{U}'(r) \subset \mathcal{U}(\kappa_G r)$. For every G and $r \le z_G$ the plaquette variables of the two classes lie within z_G of $\mathbf{1}$, so in D (Lem. 9.9(iii)); at a full bond b they are within $\rho = r\eta^2\mathsf{L}(b)^{-2}$ of $\mathbf{1}$ at the four plaquettes p of b, as $\mathsf{L}(p) \ge \mathsf{L}(b)$, each p containing the end points of b (Def. 9.2); with $\eta \le \mathsf{L}(b)$ the first bound gives $|(D^{\mathrm{T}}\partial U)(b)| < (r + r^2\frac{\eta}{\mathsf{L}(b)} + \kappa' r^3\frac{\eta^3}{\mathsf{L}(b)^3})\eta^2\mathsf{L}(b)^{-3} \le (r + r^2 + \kappa' r^3)\eta^2\mathsf{L}(b)^{-3}$ on $\mathcal{U}(r)$, which is the second inclusion, and the second bound gives $|J(U)(b)| < (1+\kappa' r^2)r\,\mathsf{L}(b)^{-3}$ on $\mathcal{U}'(r)$, which is the inclusion $\mathcal{U}'(r) \subset \mathcal{U}((1+\kappa' r^2)r)$. Under the stated curvature and current hypotheses the plaquette variables at a full bond b lie within $\rho = (\alpha_0 + 3\alpha_2)\eta^2\mathsf{L}(b)^{-2}$ of $\mathbf{1}$, so in D, and the current there is at most $s\,\mathsf{L}(b)^{-3}$; the first bound gives $|(D^{\mathrm{T}}\partial U)(b)| \le (s+s^2+\kappa' s^3)\eta^2\mathsf{L}(b)^{-3}$, as $\alpha_0+3\alpha_2 \le s$ and $\eta \le \mathsf{L}(b)$. As $\eta^3 J = PT$ and $|PT| \le |P|\,|T|$, the bound $3\alpha_0$ at an inner bond that is not full holds for every G with $3|P|\alpha_0$, its proof bounding T as at $SU(2)$, where $PT = T$; at $SU(N)$, $|P| \le 2$. At $SU(N)$, $(1-P)T$ is a real multiple of $\mathbf{1}$, so $|R+i(1-P)T|^2 \le |R|^2 + |(1-P)T|^2 \le \rho^4(1+\kappa'^2\rho^2)$, and $(D^{\mathrm{T}}\partial U)(b) - i\eta^2 J(U)(b) = \eta^{-1}(R + i(1-P)T)$ gives the display, as $\kappa'^2 \le \frac{3}{5}$. One excited bond with all other variables $\mathbf{1}$ refutes $\mathcal{U}'(r) \subset \mathcal{U}(r)$; the central plaquette variables $z\mathbf{1}, \mathbf{1}, \mathbf{1}, z\mathbf{1}$, $z = e^{2\pi i/N}$, have $J = 0$ and the left member $2\eta^{-1}z_G$. So each later proof that uses the bound of the regular class goes through with $\kappa_G\alpha_0$, and at $SU(N)$ under $\alpha_0 \le z_G$ with $(1+\kappa')\alpha_0$, each threshold on α_0 read as a ceiling being divided by κ_G, or by $1+\kappa'$.

(iii) For $SU(2)$, use the gauge constructed in the proof of Thm. YM3_I:F. We record the ordinary Laplacian bound as well. In its box notation, put $\epsilon = s\alpha$, $s = l + 4 \le 5l$, and let $d(x)$ be the depth of a site in the box. The smoothed section defines a quaternion-valued one-form ω and a smooth map Θ with

$$A = \eta^{-1}\Theta(\mathbf{1} + \eta\omega), \qquad |\omega| \le 20\epsilon, \qquad |\nabla^\eta\omega| \le C_M\epsilon \quad (d \ge 1).$$

Its equation and the interior estimate in that proof give

$$|\Delta^\eta\omega_\mu(x)| \le 55\epsilon + 120\epsilon \max_{x'\sim x} \eta^{-1}|\omega_\mu(x') - \omega_\mu(x)| \le (55 + C_M)\epsilon \quad (d(x) \ge 1 + \eta),$$

provided $120\epsilon \le 1$. Choose c_Θ to bound the first two derivatives of Θ on $|q - \mathbf{1}| \le 1/2$. Taylor's formula at each of the six neighbours, with the constant terms cancelling and the linear terms combining into the Laplacian of ω, yields

$$|\Delta^\eta A_\mu(x)| \le c_\Theta|\Delta^\eta\omega_\mu(x)| + 6c_\Theta\eta \max_{x'\sim x}(\eta^{-1}|\omega_\mu(x') - \omega_\mu(x)|)^2 \le c_\Theta(55 + C_M)\epsilon + 6c_\Theta\eta C_M^2\epsilon^2.$$

All arguments of Θ lie in this ball by $20\eta\epsilon \le 1/2$. Since $\eta \le 1/13$ and ϵ has a fixed upper bound, the last expression is at most $C\epsilon$. Bonds with an endpoint in Q have base depth at least $2 - \eta \ge 3/2 > 1 + \eta$; hence the bound is at most $5Cl\alpha$ there. Enlarging B proves the fourth estimate without a further hypothesis for $SU(2)$.

For general G, the quantitative assertion also requires the corresponding derivative bounds for the retracted smoothed section; it remains the local input stated in (iii). For the elementary comb estimate: under $\varpi < z_G$ the bond variables of the comb are products of at most $2s$ conjugated plaquette variables, with logarithms in $\mathfrak{g}$ of size at most $\pi s\varpi$ (Lem. 9.9(ii)); at $SU(N)$, $N \le 13$, a box with $z_G \le \varpi$ and $2s\varpi < 2$ has $s = 1$, $N \ge 7$, or $s = 2$, $N = 13$, since $2\sin\frac{\pi}{13} < \frac{1}{2} < 2\sin\frac{\pi}{12}$ and $\frac{1}{3} < 2\sin\frac{\pi}{18}$. For $s = 1$, $7 \le N \le 13$ and $\varpi < 1$, plaquette variables within ϖ of $\mathbf{1}$ at the six faces of $\{0,1\}^3$ give $U^v = e^{iA}$ with $|A| \le 6\arcsin\frac{\varpi}{2} \le \pi\varpi$, even near a central element; at $N = 13$ and $\varpi = z_G$ this is $\frac{6\pi}{13} = \frac{\pi}{2}\cdot\frac{12}{13}$, the least that a gauge can give. For $s = 2$, $N = 13$ and $\varpi \in [2\sin\frac{\pi}{13}, \frac{1}{2})$, a gauge of $\{0,1,2\}^3$ has $|A(b)| \le 6\theta \le 2\pi\varpi$ at its 54 bonds, $\theta = 2\arcsin\frac{\varpi}{2}$ being the largest eigen-angle of a plaquette variable. For $N \ge 14$ one plaquette variable $e^{2\pi i/N}\mathbf{1}$, at the

distance $2\sin\frac{\pi}{N}<1$, refutes the clause at $s=1$: $e^{iA_1}\cdots e^{iA_4}=e^{\pm 2\pi i/N}\mathbf{1}$ with $A_j\in\mathfrak{su}(N)$ forces $\sum_j|A_j|\geq 2\pi\frac{N-1}{N}$, so $\max_j|A_j|\geq\frac{\pi}{2}\frac{N-1}{N}>2\pi\sin\frac{\pi}{N}$.

(iv) The $SU(2)$ proof uses a quantitative filling of boundary maps in the quaternionic sphere. For general simply connected G, avoidance of the singular set $\{u:\dim\ker(\operatorname{Ad}u-1)\geq\operatorname{rk}G+2\}$ alone does not give a logarithm or extension with a uniform Lipschitz bound. Such a quantitative filling bound, together with its lifting to quotients, is needed to deduce the asserted estimate; it is retained as part of the analytic input in (II). For $G=U(1)$ the box estimate fails at constant field strength: let U have the plaquette angle κ at every plaquette of one plane direction and 0 at the others, as $U(\langle x,x+e_2\rangle)=e^{i\kappa x_1}$ and $\mathbf{1}$ at the other bonds does. Its plaquette variables are within κ of $\mathbf{1}$. In a gauge with all bonds within $C\sqrt{\kappa}$ of $\mathbf{1}$, $\kappa\leq\frac{1}{4}C^{-2}$, the angles $a(b)$ of the bonds obey $|a(b)|\leq\frac{\pi}{2}C\sqrt{\kappa}$, so $da=\kappa$ exactly at those plaquettes, and summing over a square of side l in that plane gives $\kappa l^2\leq 4l\cdot\frac{\pi}{2}C\sqrt{\kappa}$, which is false for $l>2\pi C\kappa^{-1/2}$; so no C serves for all s, and the same field strength in a central direction gives the failure whenever $\mathfrak{g}$ has a centre. With the lemma fail the vanishing of invariant linear forms (the constant-gauge argument following equation YM3_I:(943)) and further statements, among them a step of the proof of Prop. YM3_I:42.1, so the theorems ask $\mathfrak{g}$ semisimple. A flat $SO(3)$-field in a twisted sector of a torus is no gauge transform of $\mathbf{1}$. □

Lemma 9.13 (Energy of block averages and large fields)**.**

(i) Let $G\subset SU(N)$ be compact, $L\geq 13$ odd, $\mathcal{M}$ a G-valued averaging map that equals the formula of Lem. 9.11(i) where $|\partial U(p)-\mathbf{1}|\leq\varpi'(L,r_G)\leq\frac{1}{2}r_G$, and $e_*=e_L/(2\tau_G^2)$, where $e_L>0$ is a constant depending only on L and the pair with $(2e_L/L)^{1/2}\leq\varpi'(L,r_G)$ and $(N-1)^{1/2}C_{\mathrm{en}}(L)(2e_L)^{1/2}\leq\frac{1}{2}(L^{1/2}-1)\ln 2$, $C_{\mathrm{en}}(L)$ being the number of L in the cost of one step of the proof. For all K, tori, $1\leq t\leq K$, sets S of plaquettes of level t and fields U, with $\mathcal{N}(S)$ the bare plaquettes beneath S, of all three pairs of directions (the cones defined at the beginning of Section YM3_I:15),

$$L^t\sum_{\mathcal{N}(S)}a_p(U)\geq\min\Big\{\frac{1}{4\tau_G^2}\sum_{p'\in S}|\partial\mathcal{M}^tU(p')-\mathbf{1}|^2,\ e_*\Big\};$$

as $\tau_G^2\leq N-1$, the same holds with $N-1$ in the place of τ_G^2. No number above $\frac{1}{2\kappa_G^2}$ serves for $\frac{1}{4\tau_G^2}$, and $\frac{1}{4}$ fails where $\kappa_G^2>2$, so at $SU(N)$ for $N\geq 4$. At $SU(3)$, $\frac{1}{4}$ serves for every set S in which no site is a corner of two plaquettes of S of different pairs of directions, with a threshold $e(L)>0$ in the place of e_*; and $\frac{1}{4(1+\delta)}$ serves for every S and every $\delta>0$, with a threshold of L and δ.

(ii) Let $\mathfrak{g}$ be semisimple, so that $G\subset SU(N)$ and (i) applies, $L\geq 13$ odd, and let the averaging radius $\min\{\omega_M,r_G\}$ stand for ω_M. Assume the transferred witness and local-term estimates in (II). Then the large-field chain holds for G with the following prices. Its first link is Prop. YM3_I:44.4 and Prop. YM3_I:44.5: a failed plaquette test or a failed approximate fluctuation test of a test cube (equations YM3_I:(636)–(637); conditions in $|\cdot|$ on plaquette variables and on averaged contour variables) forces a plaquette variable or an averaged contour variable farther than the numbers of the $SU(2)$ construction from $\mathbf{1}$, or an averaged contour variable that is not formed. The fluctuation-test implication is used with $V_t=\mathcal{M}(V_{t-1})$ on the inner level-t bonds of the tested region, as required by Prop. YM3_I:44.5; the successive block fields satisfy this local averaging identity. Its second link is Lem. YM3_I:65.2, on the witnesses of the cells created at one step, with "$\notin G^\pi$" for "$=-\mathbf{1}$"; The local-deposit argument in the proof of Prop. YM3_I:65.4 uses its alternatives for a failed plaquette or fluctuation test, a failed density test, and a large or undefined stored bond. Its third link is the local-deposit argument in the proof of Prop. YM3_I:65.4, the lower bounds of the amount of action near a created cell, which hold with the expressions of the $SU(2)$ construction, taken in constants of the pair, times τ_G^{-2}. Here an energy pair (c,e) means

that the action beneath one level-t plaquette p' obeys

$$L^t \sum_{p\in\mathcal{N}(p')} a_p(U) \geq \min\{c|\partial\mathcal{M}^t U(p') - \mathbf{1}|^2, e\}.$$

Taking $S = \{p'\}$ in (i) gives the pair $(\frac{1}{4\tau_G^2}, e_*)$; and the deposit constant

$$c_\star = 5^{-3}\min\{\kappa_{\mathrm{W}}, \kappa_{\mathrm{P}}, \kappa_{\mathrm{Q}}, \kappa_{\mathrm{F}}\}$$

of equation YM3_I:(1429), where

$$\kappa_{\mathrm{W}} = \tfrac{1}{2}, \qquad \kappa_{\mathrm{P}} = \tfrac{1}{2}c_{\mathrm{P}}, \qquad \kappa_{\mathrm{Q}} = \tfrac{1}{2}c_{\mathrm{Q}}(A_1/A_0)^2, \qquad \kappa_{\mathrm{F}} = \tfrac{1}{4}\gamma_{\mathrm{kp}}\varkappa_{\mathrm{w}}^2(A_1/A_0)^2,$$

formed from the four prices per created cell, so that each created cell of level t gives the small factor $e^{-c_\star p_0(g_t)^2}$, is at least τ_G^{-2} times its expression in the $SU(2)$ construction, taken in constants of the pair.

■

Sketch of proof. (i) Sizes are followed in $|\cdot|$ and energies in $\|\cdot\|$, where $a = \frac{1}{2}\|V-\mathbf{1}\|^2$ exactly, and at one plaquette $|V-\mathbf{1}|^2 \leq 2\tau_G^2 a(V)$ (Def. 9.1, Lem. 9.9(i)). Let E_j be the energy $\sum a_p(\mathcal{M}^j U)$ over the plaquettes of level j beneath S, of all three pairs of directions, and ϖ_j the largest $|\partial\mathcal{M}^j U(p)-\mathbf{1}|$ among them. Where $\varpi_j \leq \varpi'(L, r_G)$ the averages are given by the formula of Lem. 9.11(i), and one step costs, between square roots of energies, the factor $L^{1/2}(1+(N-1)^{1/2}C_{\mathrm{en}}(L)\varpi_j)$, the exact $L^{1/2}$ carrying the comparison of two cutoffs. Let $L^t E_0 < e_*$, the other case being plain. By induction on j, $E_j \leq 2L^j E_0$: if this holds up to $j<t$, then $\varpi_i^2 \leq 2\tau_G^2 E_i \leq 4\tau_G^2 L^i E_0 < 2e_L L^{-(t-i)}$ for $i \leq j$, so that $\varpi_i \leq (2e_L/L)^{1/2} \leq \varpi'(L, r_G)$, and the product of the factors up to j is at most $\exp((N-1)^{1/2}C_{\mathrm{en}}(L)(2e_L)^{1/2}\sum_{m\geq 1} L^{-m/2}) \leq \sqrt{2}$, which gives $E_{j+1} \leq 2L^{j+1}E_0$. At $j=t$, $\sum_{p'\in S}|\partial\mathcal{M}^t U(p')-\mathbf{1}|^2 \leq 2\tau_G^2 E_t \leq 4\tau_G^2 L^t E_0$, which is the inequality. The constant $\frac{1}{2\kappa_G^2}$ is approached by abelian fields $e^{i\theta Y}$ of small constant field strength in one plane along a direction $Y \in \mathfrak{g}$ with $|Y| = \kappa_G\|Y\|$: the averages of such a field are exact, L^t times the bare energy beneath a coarse plaquette, carried by L^{3t} parallel bare plaquettes of angle φ, is $\frac{1}{2}L^{4t}\varphi^2\|Y\|^2$ to leading order, and the coarse plaquette variable is within $L^{2t}\varphi|Y|$ of $\mathbf{1}$. At $SU(3)$ let S have one pair of directions, and let $E_\parallel$ and $E_\perp$ be the sums of $a_p(U)$ over the bare plaquettes beneath S parallel and transverse to that pair. Below a threshold of L, $\sum_{p'\in S} a_{p'}(\mathcal{M}^t U) \leq L^t(E_\parallel + w_\perp E_\perp)$ with a weight $w_\perp \leq C(L)\big(L^t\sum_{\mathcal{N}(S)} a_p(U)\big)^{1/2} \leq \frac{1}{2}$. For $W = e^{i\epsilon A}$ with constant field strengths $F_{\mu\nu} = dA$ in $\mathfrak{g}$, the defect of one step at one coarse plaquette of the pair 12 contains the cubic term $-\epsilon^3\frac{L^2-1}{12}L^4\langle F_{12}, i[F_{13}, F_{23}]\rangle$, which has either sign; it stands at every coarse plaquette, coming from the variance $\frac{L^2-1}{12}$ of the transverse coordinate of the bare plaquettes beneath, and it is paid by a small part of the energy of the transverse plaquettes beneath, whose weights stay below 1 over the t steps, the errors summing as a geometric series in $L^{-1/2}$. If no site is a corner of two plaquettes of S of different pairs of directions, the cones of members of S of different pairs are disjoint, so the transverse energy spent for one pair is claimed by no plaquette of S, and $\sum_{p'\in S} a_{p'}(\mathcal{M}^t U) \leq L^t\sum_{\mathcal{N}(S)} a_p(U)$. For general S the plaquettes beneath members of two pairs add at most $2C(L)\big(L^t\sum_{\mathcal{N}(S)} a_p(U)\big)^{1/2}$ times $L^t\sum_{\mathcal{N}(S)} a_p(U)$, which below a threshold of L and δ is at most δ times it. With $a(V) \geq \frac{1}{4}|V-\mathbf{1}|^2$ on $SU(3)$ this gives the two forms.

(ii) A test is a condition in $|\cdot|$ on plaquette variables and on averaged contour variables. A test passed at all plaquettes of a double block puts the field under the averaging radius $\min\{\omega_M, r_G\}$, where every contour family has exactly one mean, within $4d$ (Lems. 9.9(iv), 9.11(i)); by contraposition a failed test forces a plaquette variable or an averaged contour variable farther than the numbers of the $SU(2)$ construction from $\mathbf{1}$, or an averaged contour variable that is not formed: this is the first link, with the numerals of the $SU(2)$ construction, the straightening of Lem. 9.11(iii) holding for

every closed G. The geometric and running-coupling statements used in the local-deposit argument of Prop. YM3_I:65.4, besides Lem. YM3_I:65.2 involve cubes, regions and runs of couplings alone, read no group and hold for G as written. The witness and local-deposit estimates of Lem. YM3_I:65.2 and the local-deposit argument in the proof of Prop. YM3_I:65.4 are the second-part inputs assumed in (II); their inputs ask that the terms of each level be bounded in norm by E_0 and the increments of the run by $B_\sharp$, separately, and the bounds of a step add these sizes; and a family of local terms splits off from the effective action of a large-field region, with a contribution bounded by a small factor times its norm times a factor that decays exponentially in the size of the region.

The group enters the prices alone, where the local-deposit argument in the proof of Prop. YM3_I:65.4 turns the witnesses of Lem. YM3_I:65.2 into lower bounds of the action: at one plaquette through $a(V) \geq \frac{1}{2}\tau_G^{-2}|V-\mathbf{1}|^2$ (Lem. 9.9(i)), and at averaged variables through (i), whose pair $(\frac{1}{4\tau_G^2}, \frac{e_L}{2\tau_G^2})$ takes the place of the pair $(\frac{1}{4}, \frac{e_L}{2})$ of the $SU(2)$ construction. So every price of the local-deposit argument in the proof of Prop. YM3_I:65.4 is the expression of the $SU(2)$ construction, in constants of the pair, times τ_G^{-2}, each of the four prices per created cell is at least τ_G^{-2} times its expression, and so is $c_\star$. For simply connected G with $\mathfrak{g}$ simple this chain is among the statements of §9.1.1(II). □

Lemma 9.14 (The Haar measure in the chart, the cut-off and determinants)**.**

(i) For $u_0 \in G$ and Borel $F \geq 0$, $\int_{G^\pi u_0} F \,\mathrm{d}u = \mathrm{vol}(G)^{-1} \int_{|Y|<\pi} F(e^{iY}u_0)\, j_G(Y)\,\mathrm{d}Y$, $j_G = \prod_\alpha (\frac{\sin(\alpha/2)}{\alpha/2})^2$ over the positive eigenvalues α of $\mathrm{ad}\, Y$; $|\log j_G(Y)| \leq (n - \mathrm{rk}\, G)(-\log \frac{\sin|Y|}{|Y|})$ on $|Y| < \pi$, and $|\log j_G(Y)| \leq 0.173\, n|Y|^2$ on $|Y| \leq 1$; the quadratic part of $\log j_G$ is $-\frac{1}{24}\sum_i C_A^{(i)} \|P_i Y\|^2$. The identities of the $SU(2)$ construction that integrate in the chart hold for G with the indicator of $e^{iA}e^{iH} \in G^\pi$ in the integrand, $A, H \in \mathfrak{g}$ being the two exponents of the product that such an identity integrates in the chart. $G \setminus G^\pi$ is a null set if and only if $r_G = 2$; otherwise it has inner points.

(ii) Let ϑ_{co} be the profile of the cut-off of the bond shift (the cutoff preceding Lem. YM3_I:53.3), smooth, 0 on $(-\infty, \frac{4}{3}]$ and 1 on $[\frac{5}{3}, \infty)$, with the absolute constant $\|\vartheta'_{\mathrm{co}}\|_\infty \geq 3$, and $\delta \in (0,1]$ its scale. If $w = N^{1/2p} < \frac{5}{4}$ and $\lambda = \frac{w}{5-4w}$, then, with $\vartheta_G(s) := \vartheta_{\mathrm{co}}(\frac{4}{3} + \lambda(s - \frac{4}{3}))$, the function $\vartheta_G(\nu(Y)/\delta)$ is smooth and invariant, 0 on $|Y| \leq \frac{4}{3}\delta$ and 1 on $|Y| \geq \frac{5}{3}\delta$, and $\|\vartheta'_G\|_\infty = \lambda\|\vartheta'_{\mathrm{co}}\|_\infty \geq 3$; the cut-off $\vartheta_{\mathrm{co}}(|Y|/\delta)$ itself is not differentiable for rank at least 2.

(iii) If $T_j = L^3 \mathrm{Ad}(g_j)\, \mathsf{m}\, \mathrm{Ad}(g'_j)^{-1}$, $g_j, g'_j \in G$, $|T_j - 1| \leq r < 1$ $(j = 1, 2)$, then m is invertible, $|\mathrm{tr}\log T_j| \leq -n\log(1-r)$ and $\mathrm{tr}\log T_1 = \mathrm{tr}\log T_2$. For every compact connected G the elimination factor $\mathsf{m}_c(V)$ of Lem. 9.14(iii) is formed, as a complex linear map of $\mathfrak{g}^c$ into itself, at every V of the domain of the complex average at c: the G^c-valued V on P_c whose contour families have diameter below $\frac{1}{40}$ and whose contour quotients and variables W_x lie in D^c; at G-valued V it maps $\mathfrak{g}$ into $\mathfrak{g}$. For $\mathfrak{g}^c$-valued a on P_c with $\sup|a| < \frac{3}{4}r_*$,

$$|\mathsf{m}_c(e^{ia}) - L^{-3}| \leq L^{-1} C_S \sup|a|,$$

where $0 < r_* \leq \frac{1}{600L}$, $r_1 = \frac{r_*}{40}$, $m_1 = \max\{\frac{1}{r_1}, \frac{4}{r_1^2}\}$ and $C_S \geq \frac{8m_1}{3r_*}$: numbers of L, free of k, N and G.

(iv) Let $n \geq 2$. If T acts on an n-dimensional space and $|1 - T| \leq r_n := \sin\frac{\pi}{2n}$, then $\det T \notin (-\infty, 0]$, $\mathrm{Log}\det T = \mathrm{tr}\log T$ and $|\mathrm{Log}\det T| \leq -n\log(1-r_n) =: \rho_n$, with $\frac{\pi}{2} < \rho_n < \frac{5}{2}$; and $|\mathrm{Log}\det(1+E)| \leq c_n|E| \leq \rho_n$ for $|E| \leq r_n$, where $c_n := \rho_n/r_n = -\frac{n}{r_n}\log(1-r_n)$, which is below 5 for $n \leq 3$ and above 5 for $n \geq 4$.

The radius $\frac{1}{2}$ keeps $\det T$ off $(-\infty, 0]$ if and only if $n \leq 5$. The following two forms of the elimination estimate use this radius; their three-dimensional counterparts are the proof of Prop. YM3_I:50.2 and the argument preceding equation YM3_I:(1393). Let

$K := LC_S C''_{\mathrm{ia}}$, with C_S, r_* as in (iii) and $0 < c_\sharp \le 4\cdot 10^{-7}$ and $C''_{\mathrm{ia}} \ge 8$, using the last, trivial-background estimate of Lem. 9.11(v), read at the trivial background. In the chain of the localised Gaussian family, whose radius $a_{\mathsf N}$ is taken with $0 < a_{\mathsf N} \le \min\{\frac12, \frac{Lc_\sharp}{8}, \frac{Lr_*}{8C''_{\mathrm{ia}}}, \frac{r_n}{4K}\}$, the normalised factor T stays within r_n of 1 and the function z_c of the elimination factor obeys $|z_c - 2n\log L| \le \rho_n$, for every G. The corresponding linear estimate is $|z_c - 2n\log L| \le C_z a$ on fields of size a for every $C_z \ge \max\{1, 4c_n K\}$ and every a with

$$0 < a \le \min\Big\{\frac12,\ \frac{Lc_\sharp}{8},\ \frac{3Lr_*}{32C''_{\mathrm{ia}}},\ \frac{r_n}{4K},\ \frac{L\varrho_G^{\log}}{4}\Big\},$$

$\varrho_G^{\log} \in (0, \log 2]$ being the radius of the logarithm on G^c, the largest $\varrho \le \log 2$ such that $X \in G^c$, $|X - \mathbf 1| < 1$ and $|\log X| < \varrho$ give $\log X \in \mathfrak g^c$; $\varrho_G^{\log}$ is positive, $G^c = G\exp(\mathfrak g)$ being a closed subgroup of $GL(N,\mathbb C)$ with the Lie algebra $\mathfrak g^c$, it is at least $\min\{\log 2, 2\arcsin\frac{z_G}{2}\}$ for semisimple $\mathfrak g$ (Lem. 9.9(iii)), and at $SU(N)$ and at $U(N)$ the last member of the minimum is implied by the others. So all these constants are explicit in L and n at $SU(N)$ and $U(N)$, and for the other G the radius of the second chain reads the pair through $\varrho_G^{\log}$.

■

Sketch of proof. (i) For $SU(2)$, write $U = \cos r\,\mathbf 1 + i\sin r\,\omega$, where $0 \le r \le \pi$ and $\omega \in \mathfrak{su}(2)$ has norm one. This identifies $SU(2)$ with the unit sphere in $\mathbb R\mathbf 1 + i\mathfrak{su}(2)$. Left and right multiplication preserve its Euclidean norm, so its normalized surface measure is Haar measure. In these coordinates it is $(2\pi^2)^{-1}\sin^2 r\,\mathrm dr\,\mathrm d\omega$, whereas $dY = r^2 dr\,\mathrm d\omega$ for $Y = r\omega$. Therefore

$$\int_{SU(2)} F(U)\,\mathrm dU = \frac{1}{2\pi^2}\int_{|Y|<\pi} F(e^{iY})\left(\frac{\sin|Y|}{|Y|}\right)^2 dY.$$

The omitted point $-\mathbf 1$ has surface measure zero. For general G, the differential of $\exp(iY)$, after translation back to the identity, is $i\int_0^1 e^{-it\,\mathrm{ad}Y}\,\mathrm dt$. The paired eigenvalues $\alpha, -\alpha$ of $\mathrm{ad}Y$ contribute $(\sin(\alpha/2)/(\alpha/2))^2$ to its real Jacobian; a zero eigenvalue contributes one. The principal logarithm is unique on $|Y| < \pi$, so change of variables on $G^\pi u_0$ gives the formula in (i), with the normalization $\mathrm{vol}(G)^{-1}$. The range of the chart is the one in Lem. 9.9(ii); the eigenvalues of $\mathrm{ad}\,Y$ are at most $2|Y|$ in modulus and at least $\mathrm{rk}\,G$ of them vanish, and on $|Y| \le 1$ the bound $0.173\,n|Y|^2$ follows because $x^{-2}(-\log\frac{\sin x}{x})$ increases and is 0.1726 at $x = 1$. For the nonnegative function F in (i), the same formula with $\int_G$ on the left holds whenever F vanishes Haar-almost everywhere off $G^\pi u_0$. Conversely, if the integral over $G^\pi u_0$ is finite, equality requires this vanishing. The $SU(2)$ construction bounds the one-bond functions ls_b defined before equation YM3_I:(1381), the logarithms $-\log\sigma_1$ of the density at the arguments $g_k\mathsf a(b)$, by 1 on their domain $\{g_k|\mathsf a(b)| < 1\}$ (the Haar-term calculation in the proof of Prop. YM3_I:62.1), and evaluates them with room at arguments of modulus below $\frac38$ (the same calculation: $g_k|\mathsf a + \mathsf e| < 1/8 + 2\rho_{\mathrm{bd}} \le 3/8$). For G the logarithm of j_G is formed and holomorphic on $\{Y \in \mathfrak g^c : |Y| < \pi\}$, so the domain stays, and with $\lambda(x) = -\log\frac{\sin x}{x}$ its modulus is at most $(n - \mathrm{rk}\,G)\lambda(1)$, a number of the pair, on $\{Y \in \mathfrak g^c : |Y| < 1\}$. On $\mathfrak g$ at $SU(N)$, λ being convex and increasing, the supremum of $|\log j_G|$ on a ball $|Y| \le r$ is taken with the eigenvalues of Y at $\pm r$ (and one at 0 for odd N): it is at most $\frac{N^2}{2}\lambda(r)$, with equality for even N, so about $0.0863N^2$ at $r = 1$ and $0.0118N^2$ at $r = \frac38$, and at $r = \frac38$ it is below 1 exactly when $N \le 9$. The bound enters Prop. YM3_I:53.15, the transfer of these one-bond functions, linearly, as the size of the function transferred in equation YM3_I:(1039); so a constant of the pair takes the place of 1 there, the constants that use this transfer bound absorb it in a maximum, and neither the domain nor the radius $\frac38$ moves.

(ii) ν is smooth off 0, invariant, and $\nu \le |\cdot| \le w\nu$; so $\nu(Y)/\delta \le \frac43$ on $|Y| \le \frac43\delta$, and on $|Y| \ge \frac53\delta$ the argument of ϑ_{co} is at least $\frac43 + \lambda(\frac{5}{3w} - \frac43) = \frac53$.

(iii) T_2 is conjugate to $T_1 W^{-1}$ with $W = \mathrm{Ad}(h)$; $\operatorname{tr}\log X - \operatorname{tr}\log(XW^{-1})$ lies in $2\pi i\mathbb{Z}$, is constant on the convex set $\{|X-1| \le r, |X-W| \le r\}$, and vanishes at $\frac{1}{2}(1+W)$, the eigen-angles of $\mathrm{Ad}(h)$ being symmetric. The elimination factor is read off the relative average $B \mapsto \frac{1}{i}\log[\mathcal{M}(e^{iB}V)(c)\mathcal{M}(V)(c)^{-1}]$: $L\,\mathsf{m}_c(V)X$ is the derivative at $s=0$ of its value at the field equal to sX on the pivot bond and to 0 on the other bonds of P_c, so that $L\,\mathsf{m}_c(V)$ is the coefficient of the pivot bond in its derivative at $B=0$. For V in the domain and B small, $e^{iB}V$ stays in it, its quotients and its W_x staying in D^c by the rule for products of Lem. 9.9(iii), so the relative average is formed by the series, holomorphic and $\mathfrak{g}^c$-valued, $\mathfrak{g}$-valued at G-valued V and real B, and $\mathsf{m}_c(V)$ maps $\mathfrak{g}^c$ into itself. At $V = e^{ia}$ with $\sup|a| < \frac{3}{4}r_*$ and $\sup|B| < r_1$ the relative average is at most 0.55 and holomorphic in (a,B); so, by Cauchy's estimate in B, $|L\,\mathsf{m}_c(e^{ia})| \le \frac{1}{r_1}$, and $L\,\mathsf{m}_c(e^{ia})$ is holomorphic in a. At $V=\mathbf{1}$ the relative average is $\frac{1}{i}\log\mathcal{M}(e^{iB})(c)$, whose derivative at $B=0$ is the linear average with real weights, which gives the pivot bond the weight L^{-2}; so $L\,\mathsf{m}_c(\mathbf{1}) = L^{-2}$, that is $\mathsf{m}_c(\mathbf{1}) = L^{-3}$. Finally $h(a) = L\,\mathsf{m}_c(e^{ia}) - L^{-2}$ is holomorphic on $\sup|a| < \frac{3}{4}r_*$, vanishes at 0 and has $|h| \le \frac{1}{r_1} + L^{-2} \le 2m_1$, so Schwarz's lemma gives $|h(a)| \le \frac{8m_1}{3r_*}\sup|a|$; as $\mathsf{m}_c(e^{ia}) - L^{-3} = L^{-1}h(a)$, this is the bound for every $C_S \ge \frac{8m_1}{3r_*}$.

(iv) Each eigenvalue of T lies within r_n of 1 and so has an argument of modulus at most $\arcsin r_n = \frac{\pi}{2n}$; the arguments add up to at most $\frac{\pi}{2}$; at $\cos\frac{\pi}{n}\,e^{i\pi/n}1$ the radius $\frac{1}{2}$ fails for $n \ge 6$. As $-\log(1-x)$ is convex on $[0,r_n]$, $|\operatorname{Log}\det(1+E)| \le -n\log(1-|E|) \le \frac{|E|}{r_n}\rho_n = c_n|E|$; and $c_2 < 3.48$, $c_3 = 6\log 2 < 4.16$, $c_n > 5$ for $n \ge 4$. In the chains the radii are bounded from above only and the bounds from below only. What the chains need at G is $4Ka \le r_n$, which keeps T within r_n of 1 through $|1 - L^3\mathsf{m}_c| \le L^2 C_S C''_{\mathrm{ia}}\sup|\varphi|$ at potentials with $\sup|\varphi| < \frac{4a}{L}$, and $C_z \ge 4c_nK$; for $n \ge 4$ the radii of the $SU(2)$ construction, which give $4Ka \le \frac{1}{2}$, fall by the factor $2r_n$, and its $C_z \ge 20K$ rises by the factor $\frac{c_n}{5}$, which gives these two inequalities, the two of the $SU(2)$ construction holding besides as $r_n < \frac{1}{2}$ and $c_n > 5$; for $n = 2, 3$ the constants of the $SU(2)$ construction serve, as $r_n \ge \frac{1}{2}$ and $c_n < 5$. The member $\frac{L\varrho_G^{\log}}{4}$ of the second radius keeps $\frac{4a}{L} \le \varrho_G^{\log}$, so that the logarithms of that chain lie in $\mathfrak{g}^c$. The one input "z_c is formed at $(U^*)^{-1}$ if and only if at U" fails from $SU(3)$ on (at $\operatorname{diag}(1.9, 1.9, \frac{1}{3.61})$); the chain needs only conjugate values where both are formed (Lem. 9.20), and with the two-sided guard $m(X)$ the clause holds for every G. □

9.1.6. *Scalar kernels and the one-loop calculation.*

Lemma 9.15 (Colour at the trivial field)**.**

(i) Let $\mathfrak{g}$ be semisimple. Every $\mathrm{Ad}(G)$-invariant complex bilinear form on $\mathfrak{g}^c$ is $q = \sum_i q_i\langle P_i\cdot, P_i\cdot\rangle$, $q_i = \frac{1}{n_i}\sum_a q(t_a^{(i)}, t_a^{(i)})$ over an orthonormal basis of $\mathfrak{g}_i$; every invariant form is a multiple of $\langle\cdot,\cdot\rangle$ if and only if $\mathfrak{g}$ is simple, and then C_A and C_ρ do not depend on H; invariant linear forms vanish, and a vector fixed by $\mathrm{Ad}(G)$ is 0. For semisimple $\mathfrak{g}$ and a unit H, $C_A(H) = \operatorname{tr}(\operatorname{ad}H)^2 = \sum_i C_A^{(i)}\|P_iH\|^2$, which does not depend on H if and only if all $C_A^{(i)}$ agree: for $SU(2)\times SU(3)$ on $\mathbb{C}^2\oplus\mathbb{C}^3$ the $C_A^{(i)}$ are 20 and 30, for $SU(2)\times SU(2)$ on $\mathbb{C}^2\oplus\mathbb{C}^2$ both are 16. For simple $\mathfrak{g}$ and $z > 0$ the Hessian at $U=\mathbf{1}$ of $\ln\det(D^{\mathrm{T}}D + z) - \frac{1}{2}\ln\det(A'' + DD^{\mathrm{T}} + z)$, A'' the Hessian of the action, depends on the pair only through C_A and C_ρ, exactly, on every finite lattice.

(ii) At $U = \mathbf{1}$ every linear operator of a step (averaging operators, Laplacians, gauge-fixing projections, propagators, the fluctuation operator) is $K\otimes 1_{\mathfrak{g}^c}$, K a real matrix formed from the lattice alone; so the linear average is $Q_k\otimes 1$, the Hessian is $d^*d\otimes 1$, the projection R is $R_{\mathrm{real}}\otimes 1$ and the propagator is $\mathsf{M}^{-1}\otimes 1$, $\mathsf{M} = d^*d + dR_{\mathrm{real}}d^* + Q_k^*Q_k$. For such operators: $K \mapsto K\otimes 1_V$ respects sums, products, inverses and adjoints; for pairwise disjoint sets P_j that cover the support of $K(x',\cdot)$ and numbers $w_j \ge 0$, $|(K_Vf)(x')| \le \sum_j w_j\sup_{P_j}|f|$ holds for one normed space V if and only if $\sum_{x\in P_j}|K(x',x)| \le w_j$, so for one V if and only if for all;

the best constants of ℓ^2 bounds and of lower bounds for inner product norms do not depend on V; and $|l_0(f)| \le \Phi(|l_1(f)|, \dots, |l_m(f)|)$, with real linear forms l_j and non-decreasing Φ, holds for functions with values in one normed space if and only if it holds for real functions. So the bounds of the $SU(2)$ construction at the trivial field hold for G with their constants, each in the norms to which its kind passes: the bounds in suprema, by row sums and by non-decreasing functions of real linear forms in $|\cdot|$ and in $\|\cdot\|$ alike; the ℓ^2 bounds, the quadratic forms and the lower bounds, the lower bound of the fluctuation operator among them, with its number of L alone, in the inner-product norm $\|\cdot\|$, the sign $|X|^2$ of the $SU(2)$ construction for $X \in \mathfrak{g}$ being read as $\langle X, X\rangle$. A bound of a sum of squares of $|\cdot|$ retains its constant where it follows from Schur's test on $|K|$ or from convexity. Otherwise the comparison $\|X\| \le |X| \le N^{1/2}\|X\|$ costs at most $N^{1/2}$ in an unsquared ℓ^2 norm estimate and at most N in its squared form.

■

Sketch of proof. (i) is Schur's lemma for $\mathrm{Ad}(G)$ on each simple ideal; the ideals are pairwise orthogonal for every invariant form, and invariant linear forms vanish because $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$. For $\mathfrak{su}(m)$, $\mathrm{tr}(\mathrm{ad}\, H)^2 = 2m\,\mathrm{tr}\, H^2$, and $\|H\| = 1$ in $\frac{1}{5}\mathrm{tr}$ means $\mathrm{tr}\, H^2 = 5$; this gives 20 and 30. The statement on the Hessian is the computation of Prop. 9.18, made on one finite lattice.

(ii) The averages of $\mathbf{1}$ are $\mathbf{1}$; by the chain rule the first-order part of the average at $\mathbf{1}$ is the linear average, the straight average plus the gradient of the block mean of the contour sums, and that of the straightened averages is the straight average, the contour terms cancelling; all have real weights formed from the lattice alone; the second variation of the action at $\mathbf{1}$ is $\sum_p \|dA(p)\|^2$, colour by colour; and a holomorphic map that is $\mathfrak{g}$-valued at real points is $\mathfrak{g}^c$-valued. The transfer of bounds is linear algebra, and a norming functional reduces a normed space to $\mathbb{R}$; for the row sums, f equal to the sign of $K(x', \cdot)$ times a unit vector on one P_j gives the converse. In an orthonormal basis (t_a) of $\mathfrak{g}$, $(F, (X \otimes 1)F) = \sum_a \langle F^a, XF^a\rangle$ for real F, so quadratic forms and their lower bounds pass between inner-product norms; $|\cdot|$ on $\mathfrak{g}$ is no inner-product norm for rank at least 2, and read with the operator norm a clause of this kind can fail: at $SU(3)$ the floor of the $SU(2)$ construction for a datum $X\delta_b$, $X = \mathrm{diag}(2, -1, -1)$, gives $4\|X\|^2 = 8$, against $4|X|^2 = 16$. Where a sharp linear bound for G is wanted, it is read off a nonlinear bound of the $SU(2)$ construction along $f = tg\sigma_3$, $t \to 0$, with Cauchy's inequality for the rest. □

The uniqueness argument needs a uniform gauge slice and a contraction estimate in that slice. We state the required input explicitly, including the extra rows at the interfaces. The result remains conditional for a general group. The reduced $SU(2)$ problem is treated in Props. YM3_I:30.9 and YM3_I:37.3; those statements do not assert the additional interface-row estimates below.
The slice and nonlinear estimates still required. We specify the regional geometry and row maps for the near-trivial uniqueness reduction. The uniform slice and nonlinear contraction estimates described here remain to be proved for the general pair. Fix the pair, an odd $L \ge 13$ with $200L^2 z_G \ge 1$, and the separated regional geometry, with

$$\mathfrak{M} \ge M_{\mathrm{ps}}, \qquad R_1 \ge L, \qquad R_1\mathfrak{M} \ge w_{\mathrm{S}}, \qquad \#\mathbb{T}^{(k)} \text{ in each direction} \ge 6LM_{\mathrm{ps}}.$$

The separation condition is that of $\mathfrak{r}_1$ for the determining set $\mathbf{B}$. Permit every reversal-invariant set $\mathfrak{F}$ of removed inward bonds. Lengths are level-k lengths: $\eta = L^{-k}$ and $\xi_j = L^j\eta$. All constants sought in these estimates must depend only on L and the pair, uniformly in this geometry, $\mathfrak{F}$, depth, cutoff and volume.

The reference is $U_0 = \mathbf{1}$. Let $a_j[A]$ and $v_j[A]$ be the centered averaging logarithms and the accumulated straightening factors, so that, on a bond b of level j,

$$\mathcal{M}^j(e^{i\eta A})(b) = v_j[A](b_-)\,\mathsf{x}_A(b)\,v_j[A](b_+)^{-1}, \qquad \mathsf{x}_A(b) = e^{ia_j[A](b)}. \tag{9.i}$$

The value at a reversed bond is its inverse. At the outer endpoint z of an outward bond the factors satisfy $v_j[A](z) = v_{j-1}[A](z)\,\mathsf{v}(a_{j-1}[A])(z)$. The last factor depends only on interior bonds in the preceding block; write $\kappa_6 \geq 1$ for its uniform small-field bound.

There is one row for each positively oriented interior or outward bond. At a seam node x, order the retained inward bonds ending there as $b_1(x), \dots, b_m(x)$ and add the rows $(b_i(x), b_1(x))$, $2 \leq i \leq m$. Their scale is ξ_j if the bonds have level j. Let $\mathfrak{W}$ be the space of $\mathfrak{g}$-valued row functions, with norm $\sup|\cdot|$ and inner product

$$(w, w')_1 = \sum_{\iota} \xi_\iota^{-1} \langle w(\iota), w'(\iota)\rangle.$$

Here $\langle\cdot,\cdot\rangle$ is the inner product of the pair; it is not replaced by the operator norm. Fine-field adjoints use the η^3-weighted inner product. The nonlinear row map is

$$\begin{aligned} \mathcal{Q}^\natural(A)(c) &= a_j[A](c) && (c \text{ interior or outward}),\\ \mathcal{Q}^\natural(A)(b_i, b_1) &= \frac{1}{i}\log\{\mathsf{x}_A(b_i)\mathsf{x}_A(b_1)^{-1}\} && (\text{seam rows}). \end{aligned} \tag{9.ii}$$

For $\sup|B| \leq c_{11}/2$, the products defining the datum in eq. 9.viii lie on the intrinsic branch and obey eq. 9.ix. All logarithms below are on the intrinsic branch through $\mathbf{1}$. The construction also needs these logarithms and maps to exist and be analytic on neighbourhoods of the closed balls used below.

The uniform slice. The required estimate would provide $c_{11} > 0$, $\mathsf{B}_1 \geq 1$ and $C_\sharp \geq 1 + \kappa_6$. A regular real configuration U with regularity parameter α_0 and datum e^{iB}, where $\alpha_1 = \sup|e^{iB} - \mathbf{1}|$ and $\alpha_0 + \alpha_1 \leq c_{11}$, has a chart and gauge satisfying

$$U^g = e^{i\eta A}, \qquad g = v_j[A] \text{ on } \Lambda_j, \qquad \mathsf{R}D^*A = 0, \qquad \|A\|_{X,\mathbf{B}} \leq \tfrac{1}{4}\mathsf{B}_1(\alpha_0 + C_\sharp\alpha_1). \tag{9.iii}$$

At every retained inward bond b ending at a seam node x,

$$\mathsf{x}_A(b)\, v_j[A](x)^{-1} g(x) = e^{iB_j(b)}. \tag{9.iv}$$

The preliminary gauge used to obtain this chart bounds the discrepancy from $\mathbf{1}$ at every level by $C_{\mathrm{pg}}(\alpha_0 + \alpha_1)$; its radius is included in c_{11}, and $c_{11} \leq r_G/(2C_{\mathrm{pg}})$. The regular class here bounds curvature by $\alpha_0\eta^2\mathsf{L}(p)^{-2}$ and the current on full bonds by $\alpha_0\mathsf{L}(b)^{-3}$.

The nonlinear contraction. The required estimates would provide positive regularity ceilings c_8, c_9, a chart radius $r_\circ$ and constants $K_1 \geq 4$, $K_7 \geq 1$. Set $K_5 = 2(K_7 + K_1)$ and $K_8 = K_1(16 + 9K_1^3)$. For regularity parameters at most $\min\{c_8, c_9\}$, the following operators and estimates hold on $\|A\|_{X,\mathbf{B}} \leq r_\circ$. Let $F(A)$ be the real action in the chart, $F'(0) = 0$, and write

$$\Delta_{\mathsf{R}} = F''(0) + D\mathsf{R}D^*, \qquad F'(A) + D\mathsf{R}D^*A = \Delta_{\mathsf{R}}A + \mathcal{V}(A).$$

There are real linear maps $Q, H, \widetilde{G}$ and an isometric row map T preserving both the row supremum norm and its inner product, with

$$QH = 1, \quad Q\widetilde{G} = 0, \quad \widetilde{G}Q^* = 0, \quad \widetilde{G}\Delta_{\mathsf{R}} = 1 - HQ, \quad \ker\widetilde{G} = \operatorname{ran} Q^*. \tag{9.v}$$

Define

$$\mathcal{C}(A) = T^{-1}\mathcal{Q}^\natural(A) - QA, \qquad K_A = D\mathcal{Q}^\natural(A), \qquad S_A = 1 + H^*\mathcal{C}'(A)^*.$$

The map S_A is invertible, with $\|S_A^{-1}\|_{\infty\to\infty} \leq 8/7$. The differential of the actual constraint, after removing the gauge factors, is $K_A^\flat = \Xi'_A K_A$, with Ξ'_A invertible. In particular a configuration $e^{i\eta A}$ is critical for its actual averaging constraint precisely when $F'(A)$ annihilates $\ker K_A$. For $B^\natural \in \mathfrak{W}$, put

$$\begin{aligned} w(A) &= S_A^{-1}H^*(F'(A) + D\mathsf{R}D^*A),\\ N(A) &= -\mathcal{V}(A) + \mathcal{C}'(A)^*w(A),\\ \Psi_{B^\natural}(A) &= \widetilde{G}N(A) + H\{T^{-1}B^\natural - \mathcal{C}(A)\}. \end{aligned} \tag{9.vi}$$

Whenever $\beta = \sup|B^\natural|$, $K_5\beta \le r \le r_\circ$ and $\|A\|, \|A_1\|, \|A_2\| \le r$, the uniform bounds are

$$\|\Psi_{B^\natural}(A)\| \le \tfrac{1}{2}K_5\beta + \tfrac{1}{2}r, \qquad \|\Psi_{B^\natural}(A_1) - \Psi_{B^\natural}(A_2)\| \le \tfrac{1}{2}\|A_1 - A_2\|. \tag{9.vii}$$

The norms in this display are $\|\cdot\|_{X,\mathbf{B}}$. The radii are also chosen so that the chart and row transformations just described remain on their intrinsic logarithm branches.

The uniform slice and nonlinear estimates remain to be verified. Colour-scalar linear operators at the trivial reference do not by themselves establish either task.

We describe explicitly which row datum corresponds to e^{iB}. At an interior bond put $B^\natural(c) = B_j(c)$. For an outward bond with outer endpoint z, put $v_z[B] = \mathsf{v}(B_{j-1})(z)$ and

$$\begin{aligned} B^\natural(c) &= \frac{1}{i}\log(e^{iB_j(c)}v_z[B]) && (z = c_+),\\ B^\natural(c) &= \frac{1}{i}\log(v_z[B]^{-1}e^{iB_j(c)}) && (z = c_-),\\ B^\natural(b_i, b_1) &= \frac{1}{i}\log(e^{iB_j(b_i)}e^{-iB_j(b_1)}) && (\text{seam rows}). \end{aligned} \tag{9.viii}$$

The uniform small-field bounds included in the hypotheses give

$$\sup|B^\natural| \le (1+\kappa_6)\sup|B| \le C_\sharp \sup|B|. \tag{9.ix}$$

Indeed the outward products have logarithm norm at most $|B_j(c)| + \kappa_6 \sup|B|$; the seam products have norm at most $2\sup|B|$, and $\kappa_6 \ge 1$. These products are taken within the specified branch, as are the one-step factors defining $v_z[B]$.

Lemma 9.16. The following reduction describes the consequence of establishing the slice estimates in §9.1.6: the row derivative K_A is onto for $\|A\|_{X,\mathbf{B}} \le r_\circ$. For $K_5\sup|B^\natural| \le r_\circ$ the system

$$F'(A) + D\mathsf{R}D^*A = K_A^*\omega, \qquad \mathcal{Q}^\natural(A) = B^\natural \tag{9.x}$$

has exactly one solution with $\|A\|_{X,\mathbf{B}} \le r_\circ$. The multiplier is unique without any further bound on its size. ∎

Proof. Write $\widehat{K}_A = T^{-1}K_A = Q + \mathcal{C}'(A)$ and $\widehat{\omega} = T^{-1}\omega$. Since $H^*Q^* = 1$, $H^*\widehat{K}_A^* = S_A$. The invertibility of S_A implies the injectivity of K_A^*, and hence the surjectivity of K_A in these finite-dimensional real inner-product spaces.

Set $\mathcal{G}(A) = F'(A) + D\mathsf{R}D^*A$. The first equation in eq. 9.x implies $S_A\widehat{\omega} = H^*\mathcal{G}(A)$, so $\widehat{\omega} = w(A)$. With this value of the multiplier the two equations become

$$\Delta_{\mathsf{R}}A - N(A) = Q^*w(A), \qquad QA = T^{-1}B^\natural - \mathcal{C}(A).$$

Apply $\widetilde{G}$ and use eq. 9.v: these equations imply $A = \Psi_{B^\natural}(A)$. Conversely, a fixed point has the second equation after applying Q. Applying $\widetilde{G}$ shows that $\Delta_{\mathsf{R}}A - N(A)$ belongs to $\operatorname{ran}Q^*$. Its coefficient is $w(A)$, because applying H^* gives $H^*\mathcal{G}(A) - H^*\mathcal{C}'(A)^*w(A) = w(A)$. Thus the fixed-point equation is equivalent to the system.

By eq. 9.vii, $\Psi_{B^\natural}$ maps the closed radius-$r_\circ$ ball into itself and contracts distances by $1/2$. This ball is complete, so it has one fixed point. The formula $\omega = Tw(A)$ gives the only multiplier. The same argument on the radius-$K_5\beta$ ball, or directly the fixed-point bound when $\beta = 0$, gives $\|A\|_{X,\mathbf{B}} \le K_5\beta$. □

Proposition 9.17 (Uniqueness from uniform slice estimates)**.** Let G be compact and connected and let $L \ge 13$ be odd with $200L^2z_G \ge 1$. The supplied reduction shows what would follow once the uniform slice and nonlinear estimates in §9.1.6 are established: there would be $\varepsilon_G, \varrho_G > 0$, depending only on L and the pair, with the following property in every geometry described there. For any $\mathfrak{g}$-valued datum B on the positively oriented bonds of $\mathbf{B}_\mathfrak{F}$ with $\sup|B| < \varrho_G$, any two real critical configurations in $C(\mathbf{B}_\mathfrak{F}, e^{iB})$ satisfying

$$|\partial U(p) - \mathbf{1}| < \varepsilon_G\eta^2\mathsf{L}(p)^{-2}, \qquad |J(U)(b)| < \varepsilon_G\mathsf{L}(b)^{-3}$$

at inner plaquettes and full bonds, respectively, belong to the same residual gauge orbit. Thus $U' = U^u$, where $u = \mathbf{1}$ at every node of $\mathbf{B}_{\mathfrak{F}}$. Each configuration has a representative $U^g = e^{i\eta A}$ with $g = v_j[A]$ on the tops, $\mathsf{R}D^*A = 0$ and

$$\|A\|_{X,\mathbf{B}} \le \tfrac{1}{4}\mathsf{B}_1(\varepsilon_G + C_\sharp \sup|B|).$$

The two radii are uniform in depth, sub-domain, cutoff and volume. ■

Proof. Take the constants of §9.1.6 and put

$$\varepsilon_G = \min\{c_{11}/2, c_8, c_9, r_\circ/(K_5 L\mathsf{B}_1)\}, \qquad \varrho_G = \min\{c_{11}/2, r_\circ/(K_5 L\mathsf{B}_1 C_\sharp)\}.$$

For two configurations U_a, U_b as in the statement set $\alpha_0 = \varepsilon_G$ and $\alpha_1 = \sup|e^{iB} - \mathbf{1}| \le \sup|B| < \varrho_G$. Hypothesis (S) gives A_*, g_*, $* = a, b$, with

$$U_*^{g_*} = e^{i\eta A_*}, \qquad \mathsf{R}D^*A_* = 0, \qquad \|A_*\|_{X,\mathbf{B}} \le \tfrac{1}{4}\mathsf{B}_1(\alpha_0 + C_\sharp\alpha_1) \le \frac{r_\circ}{2K_5 L}.$$

This gives the asserted representatives.

Both have the row datum eq. 9.viii. To check this, at an interior bond the two endpoint gauges in eq. 9.i cancel, giving $e^{ia_j[A_*](c)} = e^{iB_j(c)}$ and hence $a_j[A_*](c) = B_j(c)$. At an outward bond the uncancelled factor is $\mathsf{v}(a_{j-1}[A_*])(z)$; it equals $v_z[B]$, since it reads only the interior bonds of the preceding block. Moving this factor to the other side gives the first two formulas in eq. 9.viii. Finally, eq. 9.iv at b_i and b_1 gives

$$\mathsf{x}_{A_*}(b_i)\mathsf{x}_{A_*}(b_1)^{-1} = e^{iB_j(b_i)}e^{-iB_j(b_1)}.$$

Taking the stipulated intrinsic logarithms proves $\mathcal{Q}^\natural(A_a) = B^\natural = \mathcal{Q}^\natural(A_b)$. By eq. 9.ix, $K_5 \sup|B^\natural| \le K_5 C_\sharp \sup|B| < r_\circ/(L\mathsf{B}_1) \le r_\circ$.

Criticality is preserved by gauge transformations. The criticality clause of (N), together with the surjectivity in Lem. 9.16, gives a multiplier ω_* with $F'(A_*) = K_{A_*}^*\omega_*$. Since $\mathsf{R}D^*A_* = 0$, both pairs solve eq. 9.x for the same datum. Lem. 9.16 therefore gives $A_a = A_b$. Consequently $U_b = U_a^{g_b^{-1}g_a}$. The two gauges agree at every top by eq. 9.iii. At a seam node, one retained inward bond in eq. 9.iv determines $g_*(x)$ from A_* and B, so they agree there as well. Every node of $\mathbf{B}_{\mathfrak{F}}$ is a top or such a seam node. Thus $g_b^{-1}g_a$ is a residual gauge transformation. □

The orbit and energy estimates still required. Let $G \subset U(N)$ be compact and connected and $L \ge 13$ odd. Let $\mathcal{M}$ be a G-valued averaging map that, at every coarse bond c, is the formula of Lem. 9.11(i) at every field whose plaquettes on the double block of c are within $\omega_G = \min\{\omega_M, r_G\}$ of $\mathbf{1}$, the radius below which Lem. 9.11(i) forms that formula in G for every compact connected G, also where $\mathfrak{g}$ has a centre and $z_G = 0$; where the centre of $\mathfrak{g}$ is 0, ω_G is the radius $\omega^G = \min\{\omega_M, z_G\}$ of Def. 9.5(3). The radial reduction below reads $\mathcal{M}$ at such fields alone. Let $\Lambda_0, \dots, \Lambda_t$ be the lattices of the tower, Λ_0 the bare one and Λ_t the block lattice. In this definition and its reductions, in exception to Def. 9.5(1), for functions f on the bonds or on the plaquettes of one lattice $\|f\|^2 = \sum|f(\cdot)|^2$, $\|f\|_1 = \sum|f(\cdot)|$ and $|f|_\infty = \max|f(\cdot)|$, the values being measured in the operator norm; the energy norm of a $\mathfrak{g}$-valued 1-form δ on Λ_0 is $\|\delta\|_E^2 = \|d\delta\|^2 + L^{-2t}\|\delta\|^2$, in bare units; and the two matrix norms are compared only through $\|X\| \le |X| \le N^{1/2}\|X\|$. The orbit-and-energy task is to obtain $a_G, \varepsilon_* \in (0,1]$, constants depending only on L and the pair that read ω_G, such that for every $t \ge 1$, on every torus whose lattice Λ_t has at least 12 sites a side, all regions being the whole torus:
Flatness of a small critical field. The target is that a configuration U with $\mathcal{M}^tU = \mathbf{1}$ that is critical for the action on $\{\mathcal{M}^tU = \mathbf{1}\}$ and has $|\partial U(p) - \mathbf{1}| < \epsilon L^{-2t}$ at all bare plaquettes, $\epsilon \le a_G$, is flat;
Flat fields over the trivial datum. Independently of that critical-orbit estimate, flat fields U with $\mathcal{M}^tU = \mathbf{1}$ are the $\mathbf{1}^u$ with $u = \mathbf{1}$ on Λ_t; once the preceding flatness target is established, therefore, the small critical configurations over $\mathbf{1}$ with these plaquettes are exactly this orbit;

Nearby critical orbits. One must also show that for each t, each torus and each $\epsilon' \le a_G$ there is a neighbourhood of $\mathbf{1}$, depending on them, at each point V of which two critical configurations of $\{\mathcal{M}^t U = V\}$ with $|\partial U(p) - \mathbf{1}| < \epsilon' L^{-2t}$ lie in one orbit of the gauge functions equal to $\mathbf{1}$ on Λ_t, which depends holomorphically on V;
The energy comparison. The required estimate is that if U is critical for the action on $\{\mathcal{M}^t U' = V\}$, $V = \mathcal{M}^t U$, and $|\partial U(p) - \mathbf{1}| \le \varepsilon_* L^{-2t}$, then

$$\sum_p |\partial U(p) - \mathbf{1}|^2 \le 4N\varepsilon_* L^{-t} \sum_{p'} \Big| \frac{1}{i} \log \partial V(p') \Big|,$$

the sums running over the plaquettes of Λ_0 and of Λ_t; in particular over a flat block field the critical configurations with these plaquettes are flat, and over a block field near $\mathbf{1}$ the regular critical configurations form one gauge orbit.

The required estimates must make a_G and ε_* to be independent of t and the torus. If all the uniform analytic estimates in the reduction below hold, its radial argument permits $a_G = \min\{\varepsilon_5/2, [4N(68C_sC_\alpha + 132C_R)]^{-1}\}$ with the numbers ε_5, C_s, C_α, C_R of those estimates. Write $a_*(G)$ for the supremum of the admitted orbit radii, taking it to be 0 if no positive radius is admitted. The construction uses a fixed admitted positive a_G, not an assumed maximiser. Positivity of this radius for general G, and a lower bound uniform in N for $Na_*(SU(N))$, are not established by the matrix estimates alone.

Flat fields and conditional reductions. The flatness reduction requires a uniform gauge g, equal to $\mathbf{1}$ on Λ_t, in which the potentials of U and of all its averages are small, $(\mathcal{M}^i U)^g = e^{i\alpha_i}$ with $|\alpha_i|_\infty \le C_s \epsilon L^{-(t-i)}$ and $\|\alpha_i\| \le C_\alpha L^{t-i/2} \mathcal{E}^{1/2}$, $\mathcal{E} = \sum_p |\partial U(p) - \mathbf{1}|^2$. Let $F(a')$ be the action at $e^{ia'}$, $a = \alpha_0$, and T the derivative at a of the t-fold average read in the chart. The required first-variation estimate is $dF(a) \ge (\frac{5}{6N} - 68C_sC_\alpha\epsilon)\mathcal{E}$, from $\langle \theta, H_{e^{i\theta}} \rangle \ge \frac{5}{6}\|e^{i\theta} - \mathbf{1}\|^2$ for $|\theta| \le 1$. Along the ray $s \mapsto e^{isa}$ an abelian field would stay in the constraint set; for G it leaves it at second order, and we correct it. It also requires the derivative to have the form $T\delta = d\lambda_\delta + K\delta$ with $K = (Q' + E_{t-1}) \cdots (Q' + E_0)$, $\|Q'\rho\| \le L^{-1/2}\|\rho\|$, $\|E_i\rho\| \le C(|\alpha_i|_\infty + |\alpha_{i+1}|_\infty)\|\rho\|$; the interpolation H_0 of the straight average, composed with the inverse of its average, has $Q'^t H_0 = 1$ and $\|H_0\psi\|_E \le 33L^{-t/2}\|\psi\|$. The number ε_5 is the last of a falling chain $\varepsilon_1 \ge \varepsilon_2 \ge \cdots \ge \varepsilon_5$ in $(0, 1]$; existence of these positive thresholds uniformly in depth and torus is part of the analytic input: below ε_1 all the averages $\mathcal{M}^i U$ are formed by the formula, their plaquettes staying within $2\epsilon L^{-2(t-i)}$ of $\mathbf{1}$, below ω_G, and $\|\partial \mathcal{M}^i U - \mathbf{1}\| \le 2L^{i/2}\mathcal{E}^{1/2}$; below ε_2 the gauge g exists with the two bounds above; below ε_3 the first variation obeys the bound above; below ε_4 the derivative T has the structure above, and the radial direction the form below; and below ε_5, besides, $\|(K - Q'^t)H_0\| \le \frac{1}{2}$. For $\epsilon \le \varepsilon_5$, $KH_0 = 1 + \Delta$ with $\|\Delta\| \le \frac{1}{2}$, so K has the right inverse $H = H_0(1 + \Delta)^{-1}$ with $\|H\psi\|_E \le 66L^{-t/2}\|\psi\|$, and T is onto; and $Ta = d\lambda_* + R$ with $\|R\| \le C_R \epsilon L^{t/2}\mathcal{E}^{1/2}$. The direction $a - \delta_{\lambda_*} a - HR + \delta_{\lambda'} a$, δ_λ the infinitesimal gauge transformations and $\lambda' = \lambda_{HR}$, lies in $\ker T$, and dF vanishes on gauge directions; so the critical equation gives

$$\Big(\frac{5}{6N} - 68C_sC_\alpha\epsilon\Big)\mathcal{E} \le dF(a) = dF(HR) \le 2\mathcal{E}^{1/2}\|HR\|_E \le 132\,\mathcal{E}^{1/2}L^{-t/2}\|R\| \le 132\,C_R\epsilon\,\mathcal{E},$$

and $\mathcal{E} = 0$ for $\epsilon \le a_G$. The same argument proves flatness with $2a_G$ in place of a_G. Indeed, $2a_G \le \varepsilon_5$ and

$$(68C_sC_\alpha + 132C_R)2a_G \le \frac{1}{2N} < \frac{5}{6N}.$$

This proves the radial reduction under the displayed uniform estimates. For the independent flat-field assertion, a flat field has trivial transports around closed paths in boxes, and its averages are the transports along straight paths. This gives the flat-field assertion without the uniform analytic input: lifting to $\mathbb{Z}^3$ and using the comb gauge on growing cubes gives $U = \mathbf{1}^u$, and the averages $\mathbf{1}$ force $u = \mathbf{1}$ on Λ_t. For nearby critical orbits, this margin also excludes other critical branches near

the trivial datum, once the gauge-fixed constrained Hessian at the flat orbit is invertible. Fix the torus, t and $0 < \epsilon' \le a_G$. The implicit function theorem gives a critical orbit near the flat orbit for all data V near $\mathbf{1}$, with uniqueness in a fixed neighbourhood of that orbit. Suppose that critical configurations U_m over $V_m \to \mathbf{1}$, with $|\partial U_m - \mathbf{1}| < \epsilon' L^{-2t}$, escape this neighbourhood. Compactness of the finite product of copies of G gives a subsequence $U_m \to U_\infty$. The averaging map is smooth near U_∞: the bounds through ε_5 keep every average in its small-field formula. At U_∞ the differential is onto, by the right inverse in the radial reduction. It remains onto on a neighbourhood of U_∞. The tangent kernels therefore vary continuously along the subsequence, and constrained criticality passes to U_∞. This limit lies over $\mathbf{1}$ and has curvature at most $a_G L^{-2t} < 2a_G L^{-2t}$, so it is flat. By the flat-field assertion, a residual gauge transformation carries it to $\mathbf{1}$. Applying this fixed transformation to U_m places the subsequence in the neighbourhood where the implicit function theorem gives uniqueness, a contradiction. Thus the nearby-orbit conclusion follows from the radial estimates and this local invertibility; it requires no separate exclusion of remote branches. The neighbourhood obtained in this way may depend on the torus, t and ϵ'. The terminal-field energy bound is an additional analytic input. Its descending construction would require the following estimates. Let $V_j = \mathcal{M}^j U$, $\phi_V(p) = |\partial V(p) - \mathbf{1}|$, $\Phi_V(p) = \frac{1}{i}\log \partial V(p)$, $D_j = \max_p \phi_{V_j}(p)$, and D_V the derivative of the plaquette variables along a tangent vector in the left chart. The required hierarchical axial-gauge descent solves, one step down, $\mathcal{M}'(V_j)X_j = X_{j+1}$ with $\|X_j\| \le L(1+CD_j)\|X_{j+1}\| + C\|\phi_{V_j}\|$, starting from $X_t = 0$, and the defect $E_j = D_{V_j}X_j - \Phi_{V_j}$ obeys $\|E_j\|_1 \le L\|E_{j+1}\|_1 + C\|\phi_{V_j}\|(\|\phi_{V_j}\| + \|X_{j+1}\|)$; X_0 is tangent to the constraint set, so $0 = \sum_p \langle \Phi_U(p) + E_0(p), \sin \Phi_U(p)\rangle$, and $\langle \Phi, \sin\Phi\rangle \ge \frac{1}{2N}|\Phi|^2$ does not by itself provide the uniform aggregate error bound needed for (iv); that bound is part of the terminal-field input. At $SU(N)$ a central element $e^{2\pi i/N}\mathbf{1}$ on one bond off the block axes gives a nonflat configuration with zero current and plaquettes at distance $2\sin(\pi/N)$. If the chosen extension of $\mathcal{M}$ averages this field to $\mathbf{1}$, it obstructs an orbit radius above $2L^2\sin(\pi/N)$ at the first level. Agreement with the small-field averaging formula alone does not impose this behaviour on the extension. □

Proposition 9.18 (One loop)**.** Let $L \ge 13$ be odd and $k \ge 1$.

(i) Let $\mathfrak{g}$ be simple. On every torus Ψ_k^G is tempered: real-analytic near $\mathbf{1}$, zero at $\mathbf{1}$, invariant under gauge transformations and lattice isometries, with the symbol $\delta_{aa'}\hat{\pi}^G_{\mu\nu}$, $\hat{\pi}^G = \frac{C_A}{8}\hat{\pi}^{\Psi_k} + \gamma_G \hat{\pi}^{\mathfrak{B}_k}$, holomorphic and bounded on a strip; here $\mathfrak{B}_k = (\mathfrak{B}_{k,s})_s$ is the scalar functional of the tadpole, formed for $SU(2)$ on the tower of height k and side s (Section YM3_I:42.1),

$$\mathfrak{B}_{k,s}(W) = \sum_p w_{k,s}(p)\, a_p(c_k(W)) \qquad (W \text{ near } \mathbf{1} \text{ on } \Lambda_k),$$

c_k the critical branch (Thm. YM3_I:N) and $w_{k,s}(p) \in [0,1]$ the diagonal of the orthogonal projection of the plaquette functions of Λ_0 onto $d(\ker \mathsf{Q}^{[k]})$, $\mathsf{Q}^{[k]}$ the k-fold linear average, which reads L, k, s and no group; $\mathfrak{B}_0 = 0$, and with the weights 1 in the place of $w_{k,s}$ it would be $\mathcal{S}_k = A \circ c_k$. The functional $\mathfrak{B}_k$ is tempered (Section YM3_I:43.1), with $b[\mathfrak{B}_k] = j_k := \frac{2}{3}(L^{-k} - L^{-4k})$; the coefficient of Ψ_k^G is that of Thm. AA(c), and the increments $b[\Psi_l^G] - L^{-1}b[\Psi_{l-1}^G]$ are the coefficients of tempered functionals and telescope as in Section YM3_I:43.3. Thm. YM3_I:O holds for G with the remainder constant $\frac{C_A}{8}C_{\mathrm{rem}}(L) + \frac{2}{3}L^{-4}|\gamma_G|\delta_0^2$, $C_{\mathrm{rem}}(L)$ and δ_0 those of $SU(2)$, the least that the argument gives, and

$$\sup_k |b[\Psi_k^G]| \le \frac{C_A}{8}\sup_k |b[\Psi_k]| + \Big|\frac{C_\rho}{3} - \frac{C_A}{8}\Big| L^{-1};$$

so the condition $B_\sharp \ge (1 + L^{-1})c_2 + 1$ of the $SU(2)$ construction, c_2 that of Thm. YM3_I:O, and Lem. 9.40 stand with this number, which depends on L, C_A, C_ρ alone. For $SU(N)$ the two factors are $\frac{N^2}{4}$ and $\frac{N^2-4}{12}$.

(ii) Let $\mathfrak{g}$ be simple. Thm. AC holds for G in every clause: the numbers $b_l^0(G)$ converge geometrically,

$$\Big|b_l^0(G) - \frac{C_A}{8} b_\infty^0\Big| \le \frac{C_A}{8} C_{\mathrm{geo}} L^{-l/6} + \frac{2}{3}|\gamma_G| L^{3-4l},$$

the functionals of that theorem for G are $\frac{C_A}{8}$ times those of $SU(2)$ plus γ_G times one scalar functional bounded by $5C_h^2 L^{-k}$, $C_{\mathrm{re}}(G) \le \frac{C_A}{8} C_{\mathrm{re}} + 10|\gamma_G| C_h^2$, and $C_{\mathrm{inc}}(G) \le 17(2C_{\mathrm{rem}}(G) C_{\mathrm{re}}(G))^{1/2} \delta_0^{-2}$ with $C_{\mathrm{rem}}(G) = \frac{C_A}{8} C_{\mathrm{rem}} + 5|\gamma_G| C_h^2/L$ in this clause, a larger number than the remainder constant of (i).

(iii) Let G be compact and connected, $l \ge 1$ and, in this clause, $k = l - 1 \ge 0$. On the tower of height l and side s, formed for G, let $p_j(W)$ be the critical point near $\mathbf{1}$ of $\mathcal{S}_{j-1} + \mathrm{GF}_j$ on the fibre $\{\mathcal{M}(V) = W\}$, $c_l = p_1 \circ \cdots \circ p_l$ the critical branch, with $\mathcal{M}^l c_l(W) = W$ and $c_l(\mathbf{1}) = \mathbf{1}$, and $\mathcal{S}_l = A \circ c_l$ (Thm. YM3_I:N); let $\mathsf{Z}_l(W)$ be the normaliser of the first part formed for G, the Gaussian integral of the Hessian of $\mathcal{S}_{l-1} + \mathrm{GF}_l$ on the fibre at $p_l(W)$ (Thm. YM3_I:N), whose logarithm has the quadratic coefficient $b_l^0(G)$ for simple $\mathfrak{g}$ (Thm. YM3_I:O, Thm. AA(c)); and let $\mathsf{Z}^{(k)}[\mathbf{P}] = \prod_c |\det \mathsf{m}_c(V)|^{-1} \int \mathrm{d}B\, e^{-\frac{1}{2}(B, \mathsf{A}^\sharp(\mathbf{P})B)_1}$, $V = \mathcal{M}^k U$, be the Gaussian integral of the step of the second part at a triple $\mathbf{P} = (U, 0, J')$ (Prop. YM3_I:50.2, with $\dim \mathfrak{g}$ real coordinates per positive nonpivot bond), with the fluctuation operator $\mathsf{A}^\sharp$ of the step, whose value at the plain triples over W is the Gaussian normaliser $Z^{(k)}(W) = Z_k[P_W]$ preceding equation YM3_I:(957). Near $W = \mathbf{1}$ it is formed at the plain triple $\mathbf{P}_W = (c_l(W), 0, L^3 J(c_l(W)))$ of the branch, is positive, is the same along the gauge orbit of this specified branch, and identification with every regular critical configuration over W requires the orbit hypothesis of §9.1.6, or (V) in its regular domain. At the specified branch one has

$$\mathsf{Z}^{(k)}[\mathbf{P}_W] = L^{nn_l} L^{-kd_l/2} \mathsf{Z}_l(W), \qquad n_l = 3s^3, \quad d_l = 2n(L^3 - 1)s^3;$$

so the quadratic coefficient of $\log Z^{(l-1)}$ is $b_l^0(G)$ for simple $\mathfrak{g}$, and for semisimple $\mathfrak{g}$ it is, on each ideal $\mathfrak{g}_i$, $b_l^{0,(i)} = \frac{C_A^{(i)}}{8} b_l^0 + (\theta_i - \frac{3}{8} C_A^{(i)}) \frac{1}{3}(L^3 - 1) L^{-4l}$. The localised Gaussian family (Prop. YM3_I:50.2), formed for G, has its members formed and holomorphic near the trivial triple, and there the same Hessian as $\log Z^{(k)}$; its coefficient there is one number for all data: for simple $\mathfrak{g}$

$$\frac{C_A}{8} b_{k+1}^0 + \gamma_G (j_{k+1} - L^{-1} j_k) = b_{k+1}^0(G),$$

with $j_k = \frac{2}{3}(L^{-k} - L^{-4k})$ and b_{k+1}^0 the increment of $SU(2)$, and for semisimple $\mathfrak{g}$, on $\mathfrak{g}_i$, $\frac{C_A^{(i)}}{8} b_{k+1}^0 + \gamma^{(i)}(j_{k+1} - L^{-1} j_k) = b_{k+1}^{0,(i)}$, $\gamma^{(i)} = \frac{1}{2}\theta_i - \frac{3}{16} C_A^{(i)}$. That this family is a pointed family of its level on the analyticity cores of G, holomorphic on balls of one radius, with the contraction bound of its expansion, is part of (II) of §9.1.1 (Prop. 9.23(iii)), and therefore is asserted here under hypothesis (II) for simply connected G with simple $\mathfrak{g}$ and $200 L^2 z_G \ge 1$.

(iv) Let $\mathfrak{g} = \bigoplus_i \mathfrak{g}_i$ be semisimple, Ω_i the sum of t_a^2 over an orthonormal basis of $\mathfrak{g}_i$, $\tau_{ij} = \frac{1}{N}\mathrm{tr}(t^2 s^2)$ for unit $t \in \mathfrak{g}_i$, $s \in \mathfrak{g}_j$, $j \ne i$, and $\tau_{ii} = \frac{1}{n_i N}\mathrm{tr}(\Omega_i t^2)$ for unit $t \in \mathfrak{g}_i$, numbers of the pair; so $\theta_i = \frac{1}{N}\mathrm{tr}(\Omega H^2) = \sum_j n_j \tau_{ij}$ for a unit $H \in \mathfrak{g}_i$, and $n_i \tau_{ii}$ is the number C_ρ of the simple pair of $\mathfrak{g}_i$, the subgroup with Lie algebra $\mathfrak{g}_i$ on $\mathbb{C}^N$. With one set of sequences and functions for all ideals and all groups, the one-loop coefficients on $\mathfrak{g}_i$ are $\frac{C_A^{(i)}}{8}$ times those of $SU(2)$ plus $(\theta_i - \frac{3}{8} C_A^{(i)})$ times one scalar sequence: at every semisimple pair, the ideals being tied by the representation or not,

$$b^{(i)}[\Psi_k^G] = \frac{C_A^{(i)}}{8} b[\Psi_k] + \Big(\theta_i - \frac{3}{8} C_A^{(i)}\Big) t_k, \qquad t_k = \tfrac{1}{3}(L^{-k} - L^{-4k}),$$

$b^{(i)}[\Psi_k^G]$ being the coefficient on $\mathfrak{g}_i$ of the one-loop functional Ψ_k^G of the pair and Ψ_k that of $SU(2)$. The same holds with weights, for an action $\sum_m w_m$ times the Wilson actions in representations ρ_m, $w_m > 0$, with $C_A^{(i)}(w)$, $\theta_i(w)$ formed with the weighted form $\langle X, Y\rangle_w = \sum_m w_m N_m^{-1} \mathrm{tr}\, \rho_m(X)\rho_m(Y)$ in the place of $\langle \cdot, \cdot\rangle$, a rational weight being a multiplicity; the limits of the coefficients, $\frac{C_A^{(i)}(w)}{8} b_\infty^0$, read the weights through this normalisation alone, and the limit as a form on $\mathfrak{g}$, $\frac{b_\infty^0}{8}\mathrm{tr}(\mathrm{ad}\, X)^2$, does not read them. Equivalently, for a unit vector t of $\mathfrak{g}_i$, with $\Pi^F = F''(\mathbf{1})$ the polarisation of a functional F (Section YM3_I:43.1),

$$\Pi^{\Psi_k^G}[tf, tf'] = \Pi^{\Psi_k^{(i)}}[tf, tf'] + \Big(\sum_{j\neq i} n_j \tau_{ij}\Big)\Upsilon_k[f, f'], \qquad 0 \leq \Upsilon_k[f, f] \leq \tfrac{1}{2}\mathsf{h}_k(f, f),$$

$\Psi_k^{(i)}$ that of the simple pair of $\mathfrak{g}_i$, $\mathsf{h}_k = \mathcal{S}_k''(\mathbf{1})$ the Hessian at $\mathbf{1}$, for one colour, of $\mathcal{S}_k = A \circ \mathsf{c}_k$ (Thm. YM3_I:N), and $\Upsilon_k[f, f'] = \frac{1}{2}\sum_p \mathsf{v}_k(p)(d\mathsf{c}_k f)(p)(d\mathsf{c}_k f')(p)$ one scalar form of the lattice that reads no group, $\mathsf{v}_k(p) \in [0, 1]$ the diagonal of the orthogonal projection onto $d(\ker \mathsf{Q}^{[k]})$, a variance of the scalar Gaussian theory on $\ker \mathsf{Q}^{[k]}$, and c_k the linear part of the chain at $\mathbf{1}$. The form $\Upsilon_k = \frac{1}{2}\Pi^{\mathfrak{B}_k}$ is the polarisation of the tempered functional $\frac{1}{2}\mathfrak{B}_k$ of (i), with $b[\Upsilon_k] = t_k \in [0, \frac{1}{3}L^{-k}]$; so $b^{(i)}[\Psi_k^G] = b[\Psi_k^{(i)}] + (\sum_{j\neq i} n_j\tau_{ij})\, t_k$. For block sums $\tau_{ij} = 0$ for $j \neq i$. The form of the coefficients on $\mathfrak{g}$ is $\sum_i b^{(i)}[\Psi_k^G]\langle P_i \cdot, P_i \cdot\rangle$, with r numbers, which are bounded ideal by ideal,

$$|b^{(i)}[\Psi_k^G]| \leq \frac{C_A^{(i)}}{8} \sup_k |b[\Psi_k]| + \Big|\frac{\theta_i}{3} - \frac{C_A^{(i)}}{8}\Big| L^{-1},$$

and so by the largest of these r bounds. For a unit $H \in \mathfrak{g}_i$ the numbers $\mathrm{tr}(\mathrm{ad}\, H)^2$ and $\frac{1}{N}\mathrm{tr}(\Omega H^2)$ that Def. 9.1 forms for simple $\mathfrak{g}$ are $C_A^{(i)}$ and θ_i. The pairs $(C_A^{(i)}, \theta_i)$ may differ between the ideals, as at $SU(2) \times SU(3)$ on $\mathbb{C}^2 \oplus \mathbb{C}^3$, where $C_A^{(i)}$ is 20 and 30 (Lem. 9.15(i)), the coefficient being then given ideal by ideal, as above, or agree, as $(16, 6)$ on both ideals of $SU(2) \times SU(2)$ on $\mathbb{C}^2 \oplus \mathbb{C}^2$ and $(8, 6)$ on both on $\mathbb{C}^2 \otimes \mathbb{C}^2$; and $b^{(1)}[\Psi_k^G] - b^{(2)}[\Psi_k^G] = \frac{1}{8}(C_A^{(1)} - C_A^{(2)})(b[\Psi_k] - 3t_k) + (\theta_1 - \theta_2)t_k$.

■

Sketch of proof. (i) By invariance the background may be taken, to second order, along one unit $H \in \mathfrak{g}$, and the fluctuations split into the eigenspaces of $\mathrm{ad}\, H$. Every object of a step but the action is built from products, logarithms and means of group elements, which commute with homomorphisms of Lie algebras; with a parameter whose cube vanishes this shows that the sector of eigenvalue c of $\mathrm{ad}\, H$ in the field θH is the charged sector of $SU(2)$ in the field $\frac{c}{2}\theta t_3$, $\mathrm{ad}\, t_3$ having the eigenvalues $0, \pm 2$. Summing over the sectors, the group enters through $\sum c^2 = C_A$. The action adds one term: $\cos(fH) = \mathbf{1} - \frac{1}{2}f^2H^2 + O(f^4)$ gives the tadpole $-\frac{1}{2}d^{\mathrm{T}}(d\theta)^2 d \otimes M_H$, $M_H Y = \frac{1}{2}P\{H^2, Y\}$, $\mathrm{tr}\, M_H = C_\rho$, whose propagator is the orthogonal projection onto $d(\ker Q^{[k]})$, $Q^{[k]}$ the k-fold linear average; its trace is $2s^3(\eta^{-3} - 1)$, two thirds of the plaquettes less the block lattice. Eliminating the group-free second differential,

$$\Xi^{G\prime\prime}(\mathbf{1})(\theta H, \theta H) = \frac{C_A}{8}\,\Xi''(\mathbf{1})(\theta t_3, \theta t_3) + \frac{1}{2}\Big(C_\rho - \frac{3}{8}C_A\Big)\sum_p w_k(p)(d\theta)(p)^2,$$

Ξ the logarithm of the normaliser and w_k the diagonal of the projection; the count of the trace gives the formula for $b[\Psi_k^G]$. The functional $\mathfrak{B}_k$, formed here, is tempered by a step of its own. It is real-analytic near $\mathbf{1}$ and 0 at $\mathbf{1}$; it is gauge invariant, as $c_k(W^v) = c_k(W)^{\tilde{v}}$ for v near $\mathbf{1}$ or constant and a_p is a class function; and it is invariant under the isometries of the torus that map Λ_k onto itself, which keep $d(\ker \mathsf{Q}^{[k]})$ and so $w_{k,s}$. On every torus the orthogonal

projection onto $d(\ker \mathsf{Q}^{[k]})$ is the matrix of the periodic operator $\mathcal{P} = d(\mathsf{X}^{-1} - \mathsf{X}^{-1}Q^{\mathrm{T}}C^{-1}Q\mathsf{X}^{-1})d^{\mathrm{T}}$, $\mathsf{X} = d^{\mathrm{T}}d + d\varrho d^{\mathrm{T}} + Q^{\mathrm{T}}Q$, $\varrho = 1 - R^{\mathrm{T}}R$, $C = Q\mathsf{X}^{-1}Q^{\mathrm{T}}$, formed with the operators Q, R of the response of the $SU(2)$ construction (Section YM3_I:43.2), which is real, acts trivially on the colour and has an exponentially decaying kernel: its quadratic form at a plaquette function g is the maximum of $2(dB, g) - |dB|^2 - |\varrho d^{\mathrm{T}}B|^2$ over $B \in \ker Q$, and a gauge direction $d\lambda$, $\lambda \in \ker R$, removes the last term, which leaves $|\mathsf{P}_{\mathcal{C}}g|^2$, $\mathsf{P}_{\mathcal{C}}$ the projection onto $\mathcal{C} = d(\ker Q) = d(\ker \mathsf{Q}^{[k]})$. So $w_{k,s}(p) - w_k(p) = \sum_{m\neq 0} \mathcal{P}(p, p + sm)$ tends to 0 exponentially in s, w_k the diagonal at the infinite side, and the symbol $\hat{\pi}^{\mathfrak{B}_k}_{\mu\nu}(q) = \eta^4 \sum_{p \bmod \mathbb{Z}^3} w_k(p)\, (\hat{f}(q)e_\mu)(p)\, (\hat{f}(-q)e_\nu)(p)$, $\hat{f} = \hat{d}\hat{h}$ with the symbol $\hat{h}$ of the response (Section YM3_I:43.2), is holomorphic and bounded on the strip of width $2\delta_0$. The trace $\sum_p w_{k,s}(p) = \dim d(\ker \mathsf{Q}^{[k]}) = 2s^3(\eta^{-3} - 1)$, a third on each orientation, gives $b[\mathfrak{B}_k] = \eta^4 \cdot \frac{2}{3}(\eta^{-3} - 1) = j_k$. With this the theorems of the $SU(2)$ construction on tempered functionals apply to $\mathfrak{B}_k$: the coefficient (Section YM3_I:43.1); the composition with one step of the branch (Section YM3_I:43.3), $b[\mathfrak{B}_{l-1} \circ p_l] = L^{-1}b[\mathfrak{B}_{l-1}]$, hence the increments and their telescoping sum (Section YM3_I:43.3); and the remainder, the term of $\mathfrak{B}_k$ adding at most $\frac{2}{3}|\gamma_G|L^{-4}\delta_0^2$ to the constant. Temperedness, the telescoping sum, the remainder and the uniform bound of Ψ^G_k follow from the theorems for $SU(2)$ applied to the two terms, with $0 \leq L^{-k} - L^{-4k} \leq L^{-1}$. The gauge-fixing term is a trace too; it enters at its zero, the branch being in the axial gauge, with the Hessian $|\mathsf{L}_U B|^2$, L_U being built from products, logarithms and means, so it brings no second tadpole.

A second proof walks the chain of the $SU(2)$ construction for G: at the trivial configuration all operators are scalar times $1_{\mathfrak{g}^c}$ (Lem. 9.15(ii)), so the numbers δ_0, C_0, C_h pass from three colours to n.

(ii) Along one colour direction every block of the limit theorem is a scalar kernel tensor a polynomial in $\operatorname{ad} H$, but for the tadpole, and a $\log\det$ sees $\operatorname{tr}(\operatorname{ad} H)^2$ and $\operatorname{tr} M_H$ alone; the two scalar contributions in the proof of Thm. AC have the same lattice kernels. Their estimates add, with factors $C_A/8$ and $|\gamma_G|$, giving the bounds in (ii).

(iii) For the Gaussian identity, work at the specified local critical branch. Identification of all regular critical configurations with this branch requires §9.1.6, or (V) on its regular domain. The difference $\Theta = A^\eta - L^k\mathcal{S}_k \circ \mathcal{M}^k$ of the action $A^\eta = \eta^{-1}A = L^kA$, written in the units of level k, and the effective action of the averaged field is gauge invariant, vanishes on the branch and is stationary there. Its second-order expansion identifies the quadratic form at this branch with the Hessian of $L^k\mathcal{S}_k$ plus the gauge-fixing term on the fibre $\{\mathcal{M} = W\}$, a Schur complement. At $W = \mathbf{1}$ the pivot coefficients are $L^{-3}\,\mathrm{id}$ and the fluctuation operator plus $Q^{\mathrm{T}}Q$ is positive definite, so everything is formed near $\mathbf{1}$; the Gaussian integral at the point of the branch gives the powers of L, with nn_l for $3n_l$ and $d_l = 2n(L^3 - 1)s^3$; $\operatorname{Log}\det$ is taken at the radius r_n (Lem. 9.14(iv)); and the value is the same on the orbit, the mean being covariant under all of $G \times G$ and the pivot determinants being those of their restrictions to $\mathfrak{g}$. The step of the $SU(2)$ construction that $|1 - T| \leq \frac{1}{2}$ keeps $\det T$ off the negative axis (the three-dimensional determinant argument in the proof of Prop. YM3_I:50.2) fails from $n = 6$ and is replaced by the radius r_n.

The localised Gaussian family is formed for G near the trivial triple from the operators $\mathsf{A}^\sharp_Y$ of the regions Y, of whose $\operatorname{tr}\log$ its members are the differences over regions; at complex fields near $\mathbf{1}$ these operators are holomorphic germs, built from means, averages, logarithms, the action and its derivatives, and so are the members. The coefficient of the family reads of it the terms of second order at the trivial triple alone, in the directions $f\,t_a$, f scalar; these read the lattice, L and the pair and none of the data, so the coefficient is one number for all data. Along an abelian direction θH every operator of which $\mathsf{A}^\sharp_Y$ is made is $o_0 \otimes 1 + o_1 \otimes \operatorname{ad}(\theta H) + o_2 \otimes (\operatorname{ad}(\theta H))^2 + o_\tau \otimes M_{\theta H} + O(\theta^3)$ with matrices o that read no group, as in the proof of (i); so the term of second order for G is $\frac{C_A}{8}$ times that of $SU(2)$ plus γ_G times that of one gauge invariant family of $SU(2)$, formed with the Wilson action with the numerical weights of the tadpole on the plaquettes, to which

the statements of the second part at $SU(2)$ apply as they stand. The same splitting holds for $\log Z^{(k)}$ by (i) and the identity above; and at $SU(2)$ the coefficient of the localised family is that of $\log Z^{(k)}$ (equation YM3_I:(790) identifies their totals up to a constant, and Thm. YM3_I:X identifies the coefficient of such totals), for both families. This gives the Hessian of the family at the trivial triple and its coefficient, $\frac{C_A}{8}b^0_{k+1}+\gamma_G(j_{k+1}-L^{-1}j_k)$, the scalar family having the coefficient $j_{k+1}-L^{-1}j_k=b[\mathfrak{B}_{k+1}]-L^{-1}b[\mathfrak{B}_k]$ by (i); this is the germ of second order of the family at the trivial triple, which the proof of Prop. 9.23(iii) computes in the same way, by the sectors of (i); on an ideal $C_A^{(i)}$, $\gamma^{(i)}$ take the places of C_A, γ_G, by the splitting of (iv).

(iv) A basis split along the ideals. For weights the form $\langle\cdot,\cdot\rangle_w$, the Hessian of the weighted action at $\mathbf{1}$, takes the place of $\langle\cdot,\cdot\rangle$, and $C_A^{(i)}$, θ_i, $\gamma^{(i)}=\frac12\theta_i-\frac{3}{16}C_A^{(i)}$ become $C_A^{(i)}(w)$, $\theta_i(w)$, $\gamma^{(i)}(w)$; a rational weight is a multiplicity of a representation, which brings the weighted action to an unweighted one in a larger representation, at a coupling multiplied by a number, and both members of the display are analytic in the weights, the rationals being dense: so it holds at all positive weights. The term with $\gamma^{(i)}(w)$ drops in the limit, $t_k\to0$, and $C_A^{(i)}(w)\langle P_iX,P_iX\rangle_w=\operatorname{tr}(\operatorname{ad}P_iX)^2$, whose sum over i is $\operatorname{tr}(\operatorname{ad}X)^2$. For G_i-valued W near $\mathbf{1}$ the chain of the pair is that of the simple pair of $\mathfrak{g}_i$; the linear averages and the second variation split over the orthogonal sum $\bigoplus_l\mathfrak{g}_l\otimes\mathbb{R}^{\mathsf{b}_0}$, $\mathbb{R}^{\mathsf{b}_0}$ the real functions on the bonds of Λ_0, the ideals $j\neq i$ seeing the plaquettes of U only through scalar weights; so the normaliser factorises into that of the simple pair times n_j-th powers of scalar Gaussian integrals, whose second derivative is the display. These scalar integrals are those of the tadpole of (i), whose propagator is the projection onto $d(\ker\mathsf{Q}^{[k]})$ with the diagonal $\mathsf{v}_k=w_k$; so $\Upsilon_k=\frac12\Pi^{\mathfrak{B}_k}$, which is real, symmetric, vanishes on closed forms, is invariant under the isometries and is tempered by (i), with the coefficient $\frac12 j_k=t_k$, $j_k=\frac23(L^{-k}-L^{-4k})$ being that of $\mathfrak{B}_k$ by (i), (iii) and Thm. AA(c); $0\le\mathsf{v}_k\le1$ gives $0\le\Upsilon_k\le\frac12\mathsf{h}_k$, with $b[\mathsf{h}_k]=L^{-k}$, the composite $A\circ p_1\circ\cdots\circ p_k$ having the coefficient L^{-k}. Adding the coefficient $\frac{C_A^{(i)}}{8}b[\Psi_k]+(n_i\tau_{ii}-\frac38C_A^{(i)})t_k$ of the simple pair and $(\sum_{j\neq i}n_j\tau_{ij})t_k$ gives the formula with θ_i. The bound ideal by ideal is that of (i) with $C_A^{(i)}$, θ_i for C_A, C_ρ, as $0\le t_k\le\frac13L^{-1}$. At $SU(2)\times SU(2)$ a unit H of an ideal has a factor $h\in\mathfrak{su}(2)$ and $C_A^{(i)}=\operatorname{tr}(\operatorname{ad}h)^2=4\operatorname{tr}h^2$, which is 16 on $\mathbb{C}^2\oplus\mathbb{C}^2$, where $\operatorname{tr}h^2=4$, and 8 on $\mathbb{C}^2\otimes\mathbb{C}^2$, where $\operatorname{tr}h^2=2$; in both $\Omega=6\cdot\mathbf{1}$, so $\theta_i=6$ on both ideals. □

9.1.7. *Hessians and conjugation at complex fields.*

Lemma 9.19 (The Hessian at complex fields)**.** Let U be $SL(N,\mathbb{C})$-valued, p a plaquette with its bonds $b_1,\dots,b_4$ numbered as in the $SU(2)$ construction, $U_i=U(b_i)$ and $Y_p=\partial U(p)=U_1U_2U_3^{-1}U_4^{-1}$; for an $\mathfrak{sl}(N,\mathbb{C})$-valued form A let $a_1,\dots,a_4$ be its values on the bonds of p transported to the base point along the plaquette in its order,

$$a_1=A(b_1),\quad a_2=\operatorname{Ad}(U_1)A(b_2),\quad a_3=\operatorname{Ad}(U_1U_2U_3^{-1})A(b_3),\quad a_4=\operatorname{Ad}(Y_p)A(b_4),$$

so that $\partial(e^{it\xi A}U)(p)=e^{it\xi a_1}e^{it\xi a_2}e^{-it\xi a_3}e^{-it\xi a_4}Y_p$; with the signs $\epsilon=(1,1,-1,-1)$ let $S_p=\sum_i\epsilon_ia_i$. Here and below $|\operatorname{Ad}g|=|g|\,|g^{-1}|$ is the norm of $X\mapsto gXg^{-1}$ on all $N\times N$ matrices. With no condition on U, the Hessian of the action of $SU(N)$ is

$$(A,\Delta(U)A)=\xi^3\sum_p\Big\{\tfrac1N\operatorname{tr}\big(\xi^{-1}S_p\,c(Y_p)\,\xi^{-1}S_p\big)+i\sum_{i<j}\epsilon_i\epsilon_j\langle[a_i,a_j],F(p)\rangle\Big\}.$$

Let $f^\circ(b)=\xi^{-2}\max_{p\ni b}|Y_p-\mathbf{1}|$, and $f(b)$ the same maximum over the plaquettes with a corner within sup distance ξ of b_-. If $|\operatorname{Ad}(U(b))^{\pm1}|\le2$ at all bonds and $|Y_p-\mathbf{1}|\le\frac12$, the kernel $\Delta(U)(b,b')$ vanishes unless b and b' lie in a common plaquette, $\xi^2|\Delta(U)(b,b')|\le259$, and on forms on a region

R interior to the ambient one

$$\Delta_{\mathsf{R}}(U) + DD^{\mathrm{T}} = \Delta^\circ + Z(U), \qquad |(Z(U)A)(b)| \le 1346 f^\circ(b) \max|A| \le 3456 f(b) \max|A|, \quad Z(\mathbf{1}) = 0,$$

Δ_{R} the compression of $\Delta(U)$ to the forms on R and Δ° the covariant Dirichlet Laplacian, $(\Delta^\circ\phi)(x) = \xi^{-2}[6\phi(x) - \sum_\nu (\mathrm{Ad}_\nu(x)\phi(x+\xi e_\nu) + \mathrm{Ad}_\nu(x-\xi e_\nu)^{-1}\phi(x-\xi e_\nu))]$, $\mathrm{Ad}_\nu(x) = \mathrm{Ad}(U(x, x+\xi e_\nu))$. Under the guard $|Y_p - \mathbf{1}| \le 1$ of the $SU(2)$ construction the kernel is at most $180\xi^{-2}$ and $|(Z(U)A)(b)| \le 868 f(b) \max|A|$; so the numerals 1536 and 3456 of the $SU(2)$ construction serve in all its clauses. If U is $SU(N)$-valued with $|Y_p - \mathbf{1}| \le \sqrt{2}$, then $-12 \sup f^\circ (A, A) \le (A, \Delta(U)A) \le (16\xi^{-2} + 12 \sup f^\circ)(A, A)$ for $\mathfrak{su}(N)$-valued A. The constants are absolute, the same for every $N \ge 2$. Let $\mathfrak{g}$ be semisimple, so that $G \subset SU(N)$ and $G^c \subset SL(N, \mathbb{C})$, and let U be G^c-valued. Then the Hessian of the action of G is this form restricted to $\mathfrak{g}^c$-valued forms, and its operator and its Weitzenböck term are those of $SU(N)$ followed by P, the orthogonal projection onto $\mathfrak{g}^c$ for $\operatorname{tr} X^*Y$: the bounds on the quadratic form and the bounds in the normalised trace norm stay, and the kernel and Weitzenböck bounds in $|\cdot|$ hold with the factor $|P| \le N^{1/2}$, the norm of P for $|\cdot|$, the condition $|\mathrm{Ad}(U(b))^{\pm 1}| \le 2$ being read on all $N \times N$ matrices. Read with Ad on $\mathfrak{g}^c$ alone, the clauses under $|Y_p - \mathbf{1}| \le \frac{1}{2}$ stand, with $|Y_p^{-1}| \le 2$ by the Neumann series, and under the guard $|Y_p - \mathbf{1}| \le 1$ the inverse obeys $|Y_p^{-1}| \le |Y_p|^{N-1} \le 2^{N-1}$, a number of the pair, in the place of 16. ■

Sketch of proof. The identity is the second derivative of $a(e^{it\xi a_1} \cdots)$ written with $c(Y_p)$ and H_{Y_p}; in operator form the bilinear maps N_{ij}, such as $X \mapsto \frac{1}{2}(XY_p + Y_p^{-1}X)$, have norm at most $\frac{1}{2}(|Y_p| + |Y_p^{-1}|)$ and differ from the identity by at most $\frac{1}{2}(|Y_p - \mathbf{1}| + |Y_p^{-1} - \mathbf{1}|)$.

The inverse of a plaquette variable is the one new quantity, and $|X^{-1}| \le |\mathrm{Ad}\, X|$ on $SL(N, \mathbb{C})$, Ad acting on all $N \times N$ matrices (Lem. 9.9(ii)), bounds it with no N: $|\mathrm{Ad}(U(b))^{\pm 1}| \le 2$ at four bonds gives $|Y_p^{-1}| \le |\mathrm{Ad}\, Y_p| \le 16$; on $\mathfrak{g}^c$ alone the inequality $|X^{-1}| \le |\mathrm{Ad}\, X|$ fails, as at the representation of spin 2 of $SU(2)$ at $X = e^{(\log 2)J_3}$, where $|X^{-1}| = 4$ and the norm of $\mathrm{Ad}\, X$ on $\mathfrak{g}^c$ is 2.

Before the projection P the count of the kernel is the same in $|\cdot|$ and in $\|\cdot\|$: $|N_{ij}| \le \frac{1}{2}(\frac{3}{2} + 2) = \frac{7}{4}$, the transports give $|R_i^{-1}||R_i| = 1$ at the first position of a bond in a plaquette and at most $(3 \cdot 2)^2 = 36$ at the third, $R_3 = \mathrm{Ad}(Y_p U_4)$, and the four plaquettes of a bond hold it at most at the positions $1, 3, 1, 3$, as for a bond of the first direction: so $\xi^2 |\Delta(U)(b, b')| \le \frac{7}{4} \cdot 74 = 129.5$ before P; P has the norm 1 for $\|\cdot\|$ and $2 - \frac{2}{N} < 2$ for $|\cdot|$, whence 259 in $|\cdot|$, with no passage between the two norms; the formula of Weitzenböck, with the covariant Laplacian, which is the operator of U itself and not the free one, gives 1346, and $f^\circ \le f$.

Under the guard $|Y_p - \mathbf{1}| \le 1$ the finer count, in which $\mathrm{Ad}(Y_p)$ drops out of every block, gives 180 and 868.

With the eigen-angles of a unitary Y_p the bounds on the form follow as in the $SU(2)$ construction. For semisimple $\mathfrak{g}$ the action of G is the restriction of that of $SU(N)$ to G^c-valued fields, $\mathrm{Ad}(U(b))$, DD^{T} and the covariant Laplacian keep $\mathfrak{g}^c$-valued forms, and $\langle [a_i, a_j], F(p)\rangle = \langle [a_i, a_j], PF(p)\rangle$, $\langle A, PB\rangle = \langle A, B\rangle$ for $A \in \mathfrak{g}^c$; so the form is the restriction, and the operator and Z are P times those of $SU(N)$. As P is orthogonal for $\operatorname{tr} X^*Y$, the form bounds and the bounds in the trace norm stay; in $|\cdot|$, $|PX| \le (\operatorname{tr}(PX)^*PX)^{1/2} \le N^{1/2}|X|$. □

Lemma 9.20 (Conjugation)**.**

(i) $X \mapsto X^c = (X^*)^{-1}$ is an antiholomorphic involutive automorphism of $GL(N, \mathbb{C})$ whose fixed points are the unitary matrices; it keeps G^c; $X^c - \mathbf{1} = (X^{-1} - \mathbf{1})^*$; $\partial(U^c)(p) = \partial U(p)^c$, $H_{X^c} = H_X^*$, and F, J and covariant differences are covariant likewise; $a(X^c) = \overline{a(X)}$.

(ii) For a set Q of points, $Q^s = Q \cap Q^c$ is the largest subset kept by the conjugation; it holds the real points, commutes with intersections and is open with Q; if $Q = \{|X_\alpha(p) - \mathbf{1}| < r_\alpha\}$ with covariant X_α of group type, then $Q^s = \{m(X_\alpha) < r_\alpha\}$. $m(X^{-1}) = m(X^*) = m(X^c) = m(gXg^{-1}) = m(X)$ for unitary g, and $x \le m(X) \le \frac{x}{1-x}$ for $x = |X - \mathbf{1}| < 1$. If $|Y - \mathbf{1}| \le \varphi$

on a set kept by the conjugation, Y covariant and φ symmetric, then $m(Y) \le \varphi$. If $m(X) < 1$, then $\log X^{-1} = -\log X$ and $\frac{1}{i}\log X^c = (\frac{1}{i}\log X)^*$; and $\mathrm{av}(V^c) = \mathrm{av}(V)^c$.

(iii) The chain of objects of a step from the fields to the fluctuation operator commutes with the conjugation wherever both sides are formed, $A^\sharp(p^c) = \theta A^\sharp(p)\theta^{-1}$ with $\theta Z = Z^*$, on the domain of the $SU(2)$ construction written with $m(X) < 1$ and with $X \in D^c$ at the logarithms of the averages, the latter condition being void for $SU(N)$ with $N \le 6$. On this domain the symmetric action $a(X)$ suffices while the proviso on means of Lem. 9.9(iv), the contour quotients in D^c, is void, that is for $z_G > \frac{1}{40}$, at $SU(N)$ exactly for $N \le 251$; from $N = 252$ on the quotients in D^c are added to the domain. This applies in particular to the intrinsic cube response of Prop. YM3_I:54.2, with the uniqueness ball and radii of Prop. YM3_I:54.1; to the unparameterized response obtained there by setting $s = \mathbf{1}$ and $\varkappa = 0$; and to the potential and axial gauge (A, h) of Thm. YM3_I:E, constructed in Section YM3_I:38. Each is taken on its stated domain. For another group the same covariance assertion applies wherever the corresponding construction and uniqueness bounds are supplied under (V); it does not supply those bounds.

(iv) For $SU(N)$, $N \ge 3$, the moduli are not kept: with $H = \mathrm{diag}(1, \dots, 1, -(N-1))$ and $s > 0$, e^{-sH} is the conjugate of e^{sH} and $|e^{sH} - \mathbf{1}| = \max\{e^s - 1, 1 - e^{-(N-1)s}\} < e^{(N-1)s} - 1 = |e^{-sH} - \mathbf{1}|$, at $SU(3)$ with $H = \mathrm{diag}(1, 1, -2)$; so neither the analyticity cores of the $SU(2)$ construction nor the moduli of the quantities of its towers are kept by $X \mapsto X^c$ unless the conditions are written with $m(X)$.

■

Sketch of proof. (i) is direct, G^c being closed under adjoints and inverses; all of it says that $X \mapsto (X, X^{-1})$ carries the conjugation to the involution $\Theta(E, F) = (F^*, E^*)$ of the ring $M_N \times M_N^{\mathrm{op}}$, of which m, a and H_X are a norm, a trace and the odd part. (ii) The adjoint keeps the operator norm, so $m(X^c) = \max\{|X^{-1} - \mathbf{1}|, |X - \mathbf{1}|\}$; the bound by $\frac{x}{1-x}$ is Lem. 9.9(ii); for a covariant Y the bound at the point and at the conjugate point gives both members of m; the logarithms follow from the series, and the mean from its uniqueness (Lem. 9.9(iv)). (iii) Products, inverses, means, logarithms, the action a and its derivatives commute with the stated conjugation by (i), (ii). For the regional response retain the two equations YM3_I:(1051), with the regional substitutions YM3_I:(1052). In particular the identity in $S - I$ remains untruncated. The current is extended by zero, and the nonlinear maps and every surviving walk term use only the pair in the region. The proof of Prop. YM3_I:54.2 identifies the regional first-difference norm on the response carrier with that for any continuation. Thus conjugation carries both equations and their uniqueness ball to the equations and ball at the conjugate data:

$$\max\{\|T\|, |\lambda - H^{\natural}(s)^{\mathrm{T}}\mathbf{J}^0|_\infty\} \le C_{\mathrm{D,s}}\varepsilon_{\mathrm{D,s}}e^{-8\varkappa}.$$

The parameter and datum restrictions remain $|s| \le e^{\varkappa}$ and $|B'|_\infty, \gamma_0 \le \varepsilon_{\mathrm{D,s}}e^{-16\varkappa}$. Here the field and multiplier are conjugated by $Z \mapsto Z^*$, which preserves $|Z|$ and intertwines the covariant differences; the group-valued domain restrictions use m. Uniqueness gives the claimed response identity. The same argument at $s = \mathbf{1}$ uses the unparameterized bounds, with $\varkappa = 0$.

For the canonical pair, the inverse datum twist YM3_I:(580) commutes with conjugation. Its constrained response is the one of Thm. YM3_I:K, and the inverse formula for its multiplier in Section YM3_I:38 consists of covariant operations. Finally the pillar recursion YM3_I:(584) constructs h from the response by products, inverses and means. These operations give $A(B^*) = A(B)^*$ and $h(B^*) = h(B)^c$ at the real reference of that construction. Every radius used here is the radius of the cited construction, restricted further by this lemma's two-sided guards. For G, the same calculation applies to the supplied counterparts under (V). (iv) is a computation with the eigenvalues. □

Lemma 9.21 (Reflection on the real-connected part)**.** Let Q be a set of points $p = (\varrho, v)$, ϱ real parameters and v complex fields and numbers; let $p \mapsto p^c$ act by $X \mapsto X^c$ on group-valued members, by $Z \mapsto Z^*$ on $\mathfrak{g}^c$-valued ones and by complex conjugation on numbers; p is real if $p^c = p$. Let $Q^s = \{p \in Q : p^c \in Q\}$, and let Q^r, the *real-connected part*, be the set of $p \in Q^s$ whose connected component in the slice $(Q^s)_\varrho$ holds a real point.

(i) Q^r is kept by the conjugation, holds the real points of Q, is monotone in Q, has open slices if Q has, and is the largest subset of Q^s every slice component of which holds a real point; polydiscs and products pass; $Q^r = Q^s$ if the slices of Q^s are star-shaped about real points, or if the slices of Q are convex and have no group-valued member, the conjugation being then an antilinear involution; the convex clause asks for this last condition, as at $G = U(2)$, where the segment from $\mathrm{diag}(2, 3)$ to $\mathrm{diag}(\frac{1}{2}, \frac{1}{3})$ has conjugate ends and keeps the distance 0.08 from $U(2)$, its open 0.05-neighbourhood Q in $|\cdot|$, which is convex, lies in $GL(2, \mathbb{C})$ and holds no unitary matrix, has $Q^s \neq \emptyset$ and $Q^r = \emptyset$; and $(\bigcap_i Q_i)^r \subset \bigcap_i Q_i^r$, possibly strictly.

(ii) Let the slices of Q_1^r be open and let Φ be holomorphic in v there, with real members constant on slice components. Then $\Phi(p^c) = \Phi(p)^c$ on Q_1^r if and only if $\Phi(p)$ is real for every real $p \in Q_1$; and then $\Phi(Q_1^r) \subset Q_2$ implies $\Phi(Q_1^r) \subset Q_2^r$. Composites of such maps and of continuous covariant maps are covariant; a composite of maps built from products, inverses, means and logarithms and of continuations on convex subsets of a vector space kept by the conjugation (slices without group-valued member) is covariant on all of Q^s, and so are the towers wherever they exist at both points. If the slices of Q are open, the function equal to i on $Q^s \setminus Q^r$ and to 0 on Q^r is holomorphic, real at real points, and not covariant at any point of $Q^s \setminus Q^r$.

(iii) Let $\mathcal{Q}(b_0, b_1, c_0)$ be the set of the $SU(2)$ construction of the triples $(U, A', \mathbf{J})$, U real with $|\partial U(p) - \mathbf{1}| < b_0\xi^2$, A' with $|A'|$ and its covariant differences below b_1, $|\partial(e^{i\xi A'}U)(p) - \mathbf{1}| < b_0\xi^2$ at the plaquettes and $|\mathbf{J}| < c_0$; its symmetric part has $m(\partial(e^{i\xi A'}U)(p))$ in the place of the one-sided distance. Then

$$\mathcal{Q}(b_0, b_1, c_0)^s \subset \mathcal{Q}\big(b_0 + 2e^{4\xi b_1} b_1^2, b_1, c_0\big)^r,$$

the ray $(U, tA', t\mathbf{J})$, $0 \leq t \leq 1$, joining each point to the real point $(U, 0, 0)$ inside the larger set. For the classes of regular pairs the same holds with $\alpha_0 + 2\vartheta^2 e^{4\vartheta\alpha_1}\alpha_1^2$ in the place of α_0, $\vartheta \geq 1$ the largest ratio of the spacings around a plaquette, and the like holds for the natural classes. For the analyticity cores, $\mathcal{U}^\circ(b_0, b_1, c_0)^s \subset \mathcal{U}^\circ(B_0, B_1, c_0)^r$ whenever $b_0' = b_0 + 2e^{4\xi b_1} b_1^2 \leq B_0$, $b_1 \leq B_1$, $K(b_0' + b_1^2) \leq B_0$ and $b_0' + b_1 \leq a^\sharp$, where $K \geq 16L^2$ and $a^\sharp$ are the constant and the radius of the inclusion of the simple sets in the cores, in its symmetric form, assumed for G through Prop. 9.22(ii) and the uniqueness clause of §9.1.1(V). With Q^r in the place of Q^s no number moves but at a membership reached from memberships of restrictions had apart: such a membership is reached in Q^s, and the inclusions above carry it into a real-connected part, at the radii named there. The symmetric parts of the domains of the $SU(2)$ construction are cut out by conditions $m(X) < r$ and commute with intersections (Lem. 9.20(ii)); the real-connected parts pass to intersections as the inclusion $(\bigcap_i Q_i)^r \subset \bigcap_i Q_i^r$ of (i).

■

Sketch of proof. (i) is set topology. In the instance at $U(2)$ the distance in $|\cdot|$ of a positive diagonal matrix from $U(2)$ is the largest distance of its entries from 1, and from the singular matrices its least entry, singular values moving by at most the norm of a difference; on the segment, $\mathrm{diag}(2 - \frac{3}{2}t, 3 - \frac{8}{3}t)$ with $0 \leq t \leq 1$, the first is least at $t = \frac{18}{25}$, where it is 0.08, and the second is at least $\frac{1}{3}$. (ii) $p \mapsto \Phi(p^c)^c$ is holomorphic in v, as a composite of two antiholomorphic maps, and agrees with Φ at the real points, which form in each slice a totally real submanifold of maximal dimension; by the identity theorem the two agree on every component that holds a real point. No formula of Φ is

used, so sums of group elements, complex constants and maps defined as the one solution in a ball are covered. Conversely covariance at a real point says that its image is real. (iii) Along the ray tA' a product of four exponentials differs from its chord by

$$|X_t - (1-t)X_0 - tX_1| \le 8\,t(1-t)\,\xi^2 d^2 e^{4\xi b_1},$$

where d is the maximum operator norm of the four transported parameter increments A' in this plaquette product, so $d < b_1$; the same estimate holds for X_t^{-1}. Thus $m(X_t) \le (1-t)m(X_0) + tm(X_1) + 8t(1-t)\xi^2 d^2 e^{4\xi b_1}$, and with $X_0 = \partial U(p)$, $m(X_0) = |\partial U(p) - \mathbf{1}| < b_0\xi^2$, $m(X_1) < b_0\xi^2$ at a point of the symmetric part and $d < b_1$ this gives $m(X_t) < b_0\xi^2 + 2e^{4\xi b_1} b_1^2 \xi^2$, as $8t(1-t) \le 2$, the bounds on tA' and $t\mathbf{J}$ being smaller than those on A', $\mathbf{J}$; so the ray from $(U, A', \mathbf{J})$ to the real point $(U, 0, 0)$ stays in the larger set. For the regular pairs the ray is the same, the factor ϑ entering through the spacings of the plaquettes. For the cores, $\mathcal{U}^\circ(b_0, b_1, c_0)^s \subset \mathcal{Q}(b_0, b_1, c_0)^s$, and the ray lies in $\mathcal{Q}(b_0', b_1, c_0)^s$, which lies in $\mathcal{U}^\circ(B_0, B_1, c_0)^s$ by the inclusion of the simple sets under the conditions named, and it ends at a real point. □

Proposition 9.22 (Fields off the trivial field)**.** Let $\mathfrak{g}$ be simple, $L \ge 13$ odd, $200L^2 z_G \ge 1$; constants are constants depending only on L and the pair, uniform in the volume. In (i), the completed propagator construction assumes (P). The uniform critical-background, tower and core assertions in (ii), as well as (iii)–(v), assume (V). The explicit Hessian identities and bounds, the averaging formulas on their guarded domains, and holomorphic germs at each fixed finite tower are local statements. A fixed finite-dimensional germ does not assert a radius or a nonlinear majorant uniform in depth and volume.

(i) Let $G = SU(N)$, $N \ge 2$ arbitrary. The Hessian is given by Lem. 9.19, the current by Lem. 9.12(i), and the trivial-field operators by Lem. 9.15. Their gauge covariance follows from their transport formulas. The remaining quantitative inputs—the local action remainders, difference and commutator estimates in the natural norms, their variable-layer forms, the local gauges, and the order-zero inverse at rate $\frac{1}{4}\log(37/36)$—are included in (P). For $SU(2)$ the local nonlinear estimates are supplied by Props. YM3_I:30.1 and YM3_I:33.1; the completed inverse is Thm. YM3_I:G. The following Wilson-Hessian bounds are proved directly below. In the Wilson part $\Delta_U^W = D^{\mathrm{T}}D + \Delta'(U)$ the operator $\Delta'(U)$ is local, symmetric and real on real forms, with row sums at most $18\alpha \mathsf{L}_{\mathbf{B}}(b)^{-2}$ in $|\cdot|$ and $20\alpha \mathsf{L}_{\mathbf{B}}(b)^{-2}$ in $\|\cdot\|$ for U in the class $\mathfrak{U}_k(\mathbf{B}, \alpha)$ of Def. 9.2 with $\alpha \le \frac{3}{8}$, and, for U in that class at every α, $(A, \Delta_U^W A) \ge \eta^3 \sum \|DA\|^2 - 32\alpha\eta^3 \sum_b \mathsf{L}_{\mathbf{B}}(b)^{-2} \|A(b)\|^2$, with no N, $\mathsf{L}_{\mathbf{B}}$ being the local block side of the determining set $\mathbf{B}$ (Def. YM3_I:9.3): $\mathsf{L}_{\mathbf{B}}(\mathfrak{e}) = L^{j-k}$, j the largest index for which the bond or plaquette $\mathfrak{e}$ has an end point or a corner in Ω_j, so that $\eta^2 \mathsf{L}_{\mathbf{B}}(\mathfrak{e})^{-2} = L^{-2j}$; and $|\tau^* - U(\Gamma)| \le \kappa_7 \alpha_0$ for the transports of the chains, with the transport-comparison constant $\kappa_7(L)$ in the proof of Lem. YM3_I:30.3. The same path and mean comparisons apply on the guarded domains of Lem. 9.11. The completed propagator and covariance estimates are hypothesis (P) of §9.1.1; bounds of them in $|\cdot|$ give the same bounds in the normalised trace norm, with the factors $(N-1)^{1/2}$ and $N^{1/2}$ between the two (Lem. 9.9(i)). For the other simple G of the proposition Lem. 9.19 gives the Hessian among these inputs, with the factor $|P|$ in its kernel and Weitzenböck bounds.

(ii) For every G of the proposition, averages exist, with logarithms in $\mathfrak{g}^c$, at every regular real background. Over every real e^{iB} with $\sup|B|$ small there is a regular critical configuration with a holomorphic continuation $W(B)$ on a polydisc whose radius depends on no scale, and $|\partial W - \mathbf{1}| \le C \sup|B|\, \eta^2 \mathsf{L}_{\mathbf{B}}^{-2}$, $|J(W)| \le C \sup|B|\, \mathsf{L}_{\mathbf{B}}^{-3}$ at the plaquettes and the bonds, $\mathbf{B}$ the determining set and $\mathsf{L}_{\mathbf{B}}$ its local block side, as in (i); the flatness L^{-2q} of towers of depth q, with plaquettes within cL^{-2q} of $\mathbf{1}$ at every truncated depth, is included in this hypothesis. The linearised tower kernel is independently the same real scalar kernel for all G; control of the nonlinear remainder at this scale is required in (V). Under the uniform

estimates of Prop. 9.17, small fields lie in the analyticity cores of G uniformly in the scale, the sub-domain, the cutoff and the volume.

(iii) The localisation input in (V) is the following counterpart of Prop. YM3_I:53.5. Use exactly the separated joined sequence, the one or two disjoint wide collars, the regular original and joined data, and the torus-size hypothesis of Section YM3_I:53.4. For arbitrary elements U, U' of the two minimal orbits, require gauges w, w' and a real odd field a such that $(U|_\Omega)^w = e^{ia}(U')^{w'}$, and, outside the wide collars,

$$|a(b)| \le \mathsf{C}_{\rm lc}\varepsilon_{\rm r} L^{-j(b)} e^{-\delta_{\rm lc} D(b)/2}, \qquad w'(y) = 1, \quad |w(y) - 1| \le \mathsf{C}_{\rm lc}\varepsilon_{\rm r} e^{-\delta_{\rm lc} D(y)/2}$$

at the nodes y. Here $j(b)$ is the largest level containing an endpoint, and D is the weighted distance to the endpoints of the joined tower bonds, with $D = +\infty$ when there are none. The constants are independent of depth and volume. Retain all five entries of the smallness ceiling in equation YM3_I:(1000), evaluated at the admitted constants of the pair, as well as its geometric floors and response hypotheses. This is the comparison used in (iv); no localisation estimate for a different joining geometry is inferred.

(iv) Let the localisation of (iii) hold for G, and let the objects of a step be formed for G at the fields over a regular base, which is part of (V): the global and the localised background of the step are defined there; the minimal orbits of the three data of the step exist and have elements whose k-fold averages are in the axial gauge; their elements U' have $|\partial U'(p) - \mathbf{1}| < 2B_3\varepsilon_t L^{-2j}$ at the inner bare plaquettes p, j the largest index for which p has a corner in Ω'_j; and the data lie in the regular sets of radius $2\varepsilon_t$. For simply connected G they follow under (V), by (iii). The zone-bond conditions use the following explicit ceilings. The background constants are those admitted in (V), and $c_{\rm Q}$, $\mathsf{c}_{\rm s}$ and $\mathsf{c}_{\rm fl}$ have the formulas in equation YM3_I:(1026), evaluated at the constants of the pair. Put

$$\begin{aligned}
\epsilon_{\rm Q} &= \min\{\varepsilon_{\flat}, \bar\varepsilon/2, c_2^{\rm SO}/(2B_3), \omega_{\rm p}/(2 + 4B_3 L^{-2}), 1/520\},\\
\epsilon_{\rm fl} &= \min\{\epsilon_{\rm Q}, (1 + \mathsf{c}_{\rm s})^{-1}\}, \qquad \mathsf{c}_{\rm H} = \pi\kappa_{\mathsf{c}}\sqrt{2L}\max\{2\bar C_{\rm reg}, 4B_3\},\\
\epsilon_{\rm H} &= \min\{\epsilon_{\rm fl}, c_2^{\rm SO}/(2B_3), \pi/[\pi(1 + \mathsf{c}_{\rm s}) + \mathsf{c}_{\rm H}]\},\\
\epsilon_{\rm cu} &= \min\{\epsilon_{\rm lc}/2, c_2^{\rm g}/(2B_3), c_4/(4\mathsf{C}_{\rm lc})\}, \qquad \mathsf{c}_L = 4 + 2C_v + 4\kappa_6,\\
\epsilon_{\rm xp} &= \min\{\epsilon_{\rm H}, \epsilon_{\rm cu}, \epsilon_{\rm lc}/2, c_2^{\rm g}/(2L^2B_3), c_4/(4L\mathsf{C}_{\rm lc}), (2\mathsf{c}_L\mathsf{C}_{\rm lc})^{-1}, (2 + \mathsf{c}_{\rm s})^{-1}\}.
\end{aligned}$$

Here $\bar C_{\rm reg}$ is the coupling-independent regularity bound, $\varepsilon_{\flat}$ is the background radius, κ_{c} is the contour-comparison constant, and $c_2^{\rm g}, c_4, C_v$ are the curvature, perturbation and gauge-factor constants of the centered averaging estimate in Lem. 9.11(v). Assume $\varepsilon_t \le \epsilon_{\rm xp}$; $A_1/A_0 \le c_{\rm Q}$; $\gamma \le \gamma^G_{\rm xp}$, where $\gamma^G_{\rm xp} = e^{-y^G_{\rm xp}/2}$ and $y^G_{\rm xp}$ is formed by equation YM3_I:(1016) with $\|\vartheta'_G\|_\infty = \lambda\|\vartheta'_{\rm co}\|_\infty$ of Lem. 9.14(ii) in the place of $\|\vartheta'_{\rm co}\|_\infty$ in Υ_1, the one formula of the chain that changes; and $C_\chi \mathsf{c}_{\rm fl}\delta_k \le r_\chi \le 1 \le C_\chi$, with the chart constants r_χ, C_χ of [7, §B.2], whose transferred bounds are among the construction hypotheses. Here ε_t and δ_k are the thresholds of the levels t and $k = t - 1$ of the step, g_k the running coupling and $\check R_t$ the cube size (equation YM3_I:(634), YM3_I:(634)), $\mathfrak{b}_k = \delta_k/g_k$ is the radius of the scaled fluctuation field (equation YM3_I:(634)), $\mathsf{c}_{\rm fl}$ is the constant of the fluctuation range (equation YM3_I:(1026)), $\mathsf{C}_{\rm lc}$, $\delta_{\rm lc}$ are those of the localisation, and $\mathsf{c}_L = 4 + 2C_v + 4\kappa_6$ as above, with the radius c_4 and the constants C_v, κ_6 of the averages over a box and of the contour variables (Lem. 9.11(v), with the one-step contour estimate of Lem. YM3_I:14.7), which for G are constants depending only on L and the pair with $|f_i - \mathbf{1}| \le C_v L^i \sup|A_0|$ whenever $L^j \sup|A_0| < c_4$, in the notation of Lem. 9.11(v). Put $\varsigma = 2\mathsf{C}_{\rm lc}\varepsilon_t \exp(-\frac14\delta_{\rm lc} M\check R_t)$; the fifth member of $\epsilon_{\rm xp}$, $4L\mathsf{C}_{\rm lc}\varepsilon_t \le c_4$, gives $\varsigma < \frac{c_4}{2L}$. Let finally $\mathsf{c}_L c_4 \le 2Lr_G$, which is void at $SU(3)$. Then at each bond b of the cut-off zone the quotient E_b of the global and the localised background has $|E_b - \mathbf{1}| \le \mathsf{c}_L\varsigma < r_G$; $H_b = \frac1i \log E_b \in \mathfrak{g}$, the shift of YM3_I:(989), has

$|H_b| \le \frac{\pi}{2}\mathsf{c}_{L\varsigma}$ and $8\|\vartheta'_G\|_\infty|H_b| = 8\lambda\|\vartheta'_{\mathrm{co}}\|_\infty|H_b| \le \delta_k$; with $\theta(W) = \vartheta_G(\nu(Y(W))/\delta_k)$ where -1 is no eigenvalue of W, $Y(W)$ the principal hermitian logarithm, and $\theta(W) = 1$ elsewhere, the bond map $W \mapsto We^{-i\theta(W)H_b}e^{iH_b}$ of Lem. YM3_I:53.3 is a diffeomorphism of G onto itself, its inverse at W being the fixed point of the contraction $W' \mapsto We^{-iH_b}e^{i\theta(W')H_b}$ of factor at most $\frac{\pi}{16}$, with Jacobian in $[\frac{7}{8}, \frac{9}{8}]$; and as $g_k\mathsf{c}_{\mathfrak{fl}}\mathfrak{b}_k + |H_b| = \mathsf{c}_{\mathfrak{fl}}\delta_k + |H_b| \le 1 + \frac{1}{24} < \pi$, the substitution $W \mapsto We^{-i\theta(W)H_b}$, $W = e^{ig_kY(b)}$, is a homeomorphism between open sets each holding $\{g_k|Y(b)| + |H_b| < \pi\}$, outside of which the integrands vanish.

(v) The domains of analyticity are defined for G, are gauge stable, and are the same on all large tori and on $\mathbb{R}^3$; the totals do not depend on the chosen background; and at the trivial real factor the continuations and the bounds of the towers are those of (ii). The continuations and the towers off the trivial field are those of (V), which are assumed in (V) for the simply connected construction.

■

Sketch of proof. (i) The local identities follow from the current and Hessian formulas. The quantitative inverse and local-remainder bounds specified in (P) are assumptions here. The following calculation proves the asserted Wilson-Hessian bounds without using those assumptions.

The group enters them through the Hessian; through the commutators, $|[X,Y]| \le 2|X||Y|$; through sums, products and Neumann series of decaying kernels in $|\cdot|$, which are sums over the lattice and keep the constants of the $SU(2)$ construction for every N; through the transpose of a kernel, which takes N for the 2 of $|T^{\mathrm{T}}| \le 2|T|$, and the bounds of Schur's kind, whose numeral 3 becomes n (Def. 9.5(2), Prop. 9.26(g)); through the table at the trivial field (Lem. 9.15); and through the small-field theorem in the gauge of the class (Lem. 9.12(iii)).

The Wilson part $\Delta'(U) = \Delta(U) - D^{\mathrm{T}}D$ is read off the identity of Lem. 9.19, with $\xi = \eta$: at a unitary Y_p its terms are $\xi^{-2}\frac{1}{N}\mathrm{tr}(S_p(c(Y_p) - \mathbf{1})S_p)$ and $i\sum_{i<j}\epsilon_i\epsilon_j\langle[a_i,a_j], F(p)\rangle$, where $|c(Y_p) - \mathbf{1}| = \frac{1}{2}x_p^2$ with $x_p = |Y_p - \mathbf{1}|$, and $\langle[a_i,a_j], F(p)\rangle = \xi^{-2}\langle[a_i,a_j], H_{Y_p}\rangle$ with $|H_{Y_p}| \le x_p$, as $\langle A, PB\rangle = \langle A, B\rangle$ for $A \in \mathfrak{g}^c$. A bond lies in four plaquettes and a plaquette has four bonds. So, with x the largest x_p at the plaquettes of b, which is $\xi^2 f^\circ(b)$, the row sums of $\Delta'(U)$ at b are at most $(12+8x)f^\circ(b)$ in $\|\cdot\|$ and $(12+16x)f^\circ(b)$ in $|\cdot|$, where $|P| \le 2$ at $SU(N)$; in the class $\mathfrak{U}_k(\mathbf{B}, \alpha)$ one has $f^\circ(b) \le \alpha\mathsf{L}_{\mathbf{B}}(b)^{-2}$ and $x \le \alpha$, and for $\alpha \le \frac{3}{8}$ this gives $18\alpha\mathsf{L}_{\mathbf{B}}(b)^{-2}$ in $|\cdot|$ and $15\alpha\mathsf{L}_{\mathbf{B}}(b)^{-2} \le 20\alpha\mathsf{L}_{\mathbf{B}}(b)^{-2}$ in $\|\cdot\|$. As $\Delta(U)$ and $D^{\mathrm{T}}D$ are Hessians of real functions at G-valued U, $\Delta'(U)$ is real on real forms. With $\|S_p\|^2 \le 4\sum_i\|a_i\|^2$ the same count gives, bond by bond, $(A, \Delta'(U)A) \ge -\eta^3\sum_b(12+8x)f^\circ(b)\|A(b)\|^2$, x taken at b, and $12 + 8x \le 28 \le 32$ at every unitary Y_p, where $x \le 2$: this is the lower bound. The transport-comparison constant $\kappa_7(L)$ in the proof of Lem. YM3_I:30.3 reads no N (Prop. 9.26(g)).

(ii) The averages: Lems. 9.9(iii) and 9.11 with the rule D^c.

At a fixed finite tower the implicit function theorem gives a local critical germ over e^{iB}. The uniform radius and nonlinear majorants asserted in (ii) are part of (V), not consequences of this finite-dimensional theorem. The linearisation is formed from the operators of Lem. 9.15(ii), with Lem. 9.19; a holomorphic map that is $\mathfrak{g}$-valued at real points is $\mathfrak{g}^c$-valued. On a tower of depth q, truncated at a level $q' \le q$, the towers ask at this configuration the flatness L^{-2q}, which a bound of first order in $\sup|B|$ gives only up to the factor $L^{q-q'}$. The linear term has a kernel $\hat{D}$ that is real, scalar and the same for all G: for a real scalar 1-form ψ, $|\hat{D}[\psi]| \le K_{\mathrm{lin}}\sup_p \mathsf{L}_-(p)^2|d\psi(p)|$, where $\mathsf{L}_-(p)$ is the least of the local scales $L^{\lambda(x)-k}$ at the corners x of p, $\lambda(x)$ being the level of the sequence of regions at x and $\eta = L^{-k}$, and the constant K_{lin} of the $SU(2)$ construction is read off its nonlinear bound of the towers along ψ times a fixed unit of $\mathfrak{su}(2)$, as the factor tends to 0 (Lem. 9.15(ii)); Cauchy's estimate controls higher orders only on an admitted polydisc with a uniform majorant. The uniform nonlinear tower-flatness estimate itself remains part of (V). To lie in the analyticity core a small field asks, on every sub-domain, for both tower rules and at every depth, that its averages exist with logarithms in $\mathfrak{g}^c$ and that over each real datum near $\mathbf{1}$ the regular

critical configurations form one gauge orbit whose representative continues holomorphically with the tower bounds. These uniform assertions are the near-trivial input in (V); the slice and contraction reduction of Prop. 9.17 gives the one-orbit conclusion when its uniform hypotheses hold. The local matrix and colour-scalar linear estimates do not establish those hypotheses by themselves.

(iii) The $SU(2)$ statement, including the choice of one common triple for both the bond and node estimates, is proved in Section YM3_I:53.4. Its general-group counterpart with the same domains is the explicit localisation input of (V). The logarithms used to align the collar data are formed from products of small exponentials under Lem. 9.9(ii).

(iv) At a bond b of the cut-off zone E_b is a product of seven factors, and their chords add: the localisation of (iii) bounds the relative field of the two backgrounds beneath b by $L^{-k}\varsigma$ and gives two factors at the ends within ς of $\mathbf{1}$; Lem. 9.11(v), as L^k times that field is below $\varsigma < c_4$, gives the averaged exponent, at most 2ς, the two factors f_k at the ends, within $C_v\varsigma$ of $\mathbf{1}$, and the two exponents of the contour variables at the ends, at most $2\kappa_6\varsigma$ each; so $|E_b - \mathbf{1}| \le (4+2C_v+4\kappa_6)\varsigma = \mathsf{c}_L\varsigma$. The fifth member of ϵ_{xp} gives $\varsigma < \frac{c_4}{2L}$, so $\mathsf{c}_L c_4 \le 2Lr_G$ gives $|E_b - \mathbf{1}| < r_G$, $E_b \in G^\pi$ and $|H_b| \le \frac{\pi}{2}|E_b - \mathbf{1}|$; the bound $8\|\vartheta'_G\|_\infty |H_b| \le \delta_k$ comes from $A_1/A_0 \le c_{\mathrm{Q}}$ and $\gamma \le \gamma^G_{\mathrm{xp}}$, and $\mathsf{c}_{\mathrm{fl}}\delta_k \le 1$ from $C_\chi \mathsf{c}_{\mathrm{fl}}\delta_k \le r_\chi \le 1$; the assembly is that of the $SU(2)$ construction with r_G for 2.

A further argument gives $E_b \in G^\pi$ from the block constraints. On the double block of the coarse bond c of b let V and V' be the k-fold averages of the global and of the localised background. They have the same contour variables $X_\pm$ at the two end blocks, both being in the axial gauge at the sites of the region of level $k+1$ and both being given by the datum at the others, as an element of a minimal orbit satisfies its block constraints and the joined datum is the datum at the common bonds; and they have the same average X_0 at c. So $V(b) = X_-^{-1}e^{i\beta}X_0X_+$ and $V'(b) = X_-^{-1}e^{i\beta'}X_0X_+$ with $\beta, \beta' \in \mathfrak{g}$, $|\beta| + |\beta'| < \pi$, and $E_b = V(b)V'(b)^{-1} = X_-^{-1}e^{i\beta}e^{-i\beta'}X_-$, which lies in G^π by Lem. 9.9(ii), G^π being kept by conjugation with G. The statement that $G \setminus G^\pi$ is a null set (the $SU(2)$ case of Lem. 9.14(i) and Prop. YM3_I:53.4) is false for $SU(N)$, $N \ge 3$, at every $H_b \ne 0$, and it is not needed: the substitution enters only integrands that vanish outside the chart (the chart-supported substitution in the proof of Prop. YM3_I:53.4).

(v) The proofs of the $SU(2)$ construction stand; "trace 0" becomes $\mathrm{Ad}(g)\mathfrak{g}^c = \mathfrak{g}^c$, and $F = \xi^{-2}PH$ with $P\,\mathrm{Ad}(g) = \mathrm{Ad}(g)P$; a holomorphic map that is $\mathfrak{g}$-valued at real points is $\mathfrak{g}^c$-valued; and $G^c = G\exp(\mathfrak{g})$ is closed. □

9.1.8. *Dependence on regular backgrounds.* The construction hypotheses have different roles. The propagator estimates control inverses at general regular fields. The variational estimates then construct and localise the constrained critical orbit. The second-part hypotheses provide the holomorphic Gaussian families and the one-step estimates on their full domains. Finally, the constant conditions must be compatible in the prescribed order. The following reductions identify the group dependence in these requirements.

Propagators. Prop. 9.22(i) proves the local Hessian and current identities and separates them from the quantitative inputs in (P). For other simple G the Hessian bounds include the factor $|P|$. Changing the colour norm, taking a transpose, and summing components may introduce the factors κ_G, $N^{1/2}$ and N in Def. 9.5(2). The constants of an entire inverse or tree expansion must be chosen from its admitted quantitative bounds; these norm comparisons alone do not certify a particular numerical choice. Hölder and ℓ^2 seminorms of $\mathfrak{g}^c$-valued functions compare with those of real functions by the factors $N^{1/2}$ and $(2n)^{1/2}$, since $\|\cdot\| \le |\cdot| \le N^{1/2}\|\cdot\|$ and such a function has $2n$ real components; the estimate required in (P) keeps the rate δ_2, and the floor on M moves with N through the natural bound B_{nat}, of the order $N^{1/2}$. The covariance numbers, thresholds, rates and cell sizes are formed from those of the expansion.

Variational estimates. The variational estimates in (V) require the background estimates of (P) with the changes of Def. 9.5, among which these concern the variational part: a proof that uses the current bound of the regular class runs with the class $\mathcal{U}(r)$, into which $\mathcal{U}'(r)$ falls at the radius

$\kappa_G r$, and for $r \le z_G$ at $(1+\kappa' r^2)r$, thresholds on α_0 read as ceilings being divided by κ_G, or by $1+\kappa'$ (Remark 9.6, Lem. 9.12(ii)); the bound $3\alpha_0$ of Prop. YM3_I:30.1 becomes $3|P|\alpha_0$, at most $6\alpha_0$ at $SU(N)$, and $j_G\alpha_0$ under $\alpha_0 \le z_G$, which is $3\alpha_0$ at $SU(N)$; the second inclusion becomes $\mathcal{U}(r) \subset \mathcal{U}'(r + r^2 + \kappa' r^3)$ for every G and $r \le z_G$, and the membership in $\mathcal{U}'(2s)$, $s = 2\alpha_0 + 3\alpha_2$, of the curvature/current conversion in Lem. 9.12(ii), with both of its quantitative hypotheses, holds for every G under $\alpha_0 + 3\alpha_2 < z_G$ and $s(1+\kappa' s) \le 1$, so for $SU(N)$ under $\alpha_0 + 3\alpha_2 < z_G$ and $s \le \frac{1}{3}$, both by the bound at a bond of Lem. 9.12(ii), for the membership at each full bond b with $\rho = (\alpha_0 + 3\alpha_2)\eta^2 \mathsf{L}(b)^{-2}$ (Def. 9.5(3)); Lem. 9.10, whose averaged contour variables are means of contour families (Prop. YM3_I:4.3), takes the guard $\varpi_y(W) < z_G$, under which the contour quotients, products of at most $\frac{9}{8}L^2$ conjugated plaquette variables within $\varpi_y(W)$ of $\mathbf{1}$, lie in D^c by the rule for products, so that the means lie in G (Lem. 9.9(iii), (iv)), and the comb gauge of Lem. YM3_I:14.4 takes the guard $\varpi < z_G$ (Lem. 9.12(iii)); and wherever the towers use $|X^c - \mathbf{1}| = |X - \mathbf{1}|$, an equality of moduli at the plaquettes alone, they take the form with $m(X)$, with $|\mathrm{Ad}(g) - 1| \le m(g)(2 + m(g))$, and a logarithm is guarded by $X \in D^c$.

The existence assertion required in (V) has the following uniform content: there must be a power $\mathfrak{r}(G)$ of L, a number $B_3(G) \ge 1$ and, for every power $\rho \ge \mathfrak{r}(G)$ of L, a number $a_1(\rho, G) > 0$, numbers of L, ρ and the pair with $\kappa_G B_3(G) a_1(\rho, G) \le a_0(G)$, $a_0(G) > 0$ the fixed uniqueness radius required in (V), such that in every setting of the $SU(2)$ construction at the separation ρ, for every reversal-invariant set $\mathfrak{F}$ of inward bonds and over every real datum V on $\mathbf{B}_{\mathfrak{F}}$ whose plaquette variables, and on a sequence of regions whose holonomies at the tested plaquettes, lie within $\varepsilon_1 \le a_1(\rho, G)$ of $\mathbf{1}$, the set $C(\mathbf{B}_{\mathfrak{F}}, V)$ holds a critical configuration in $\mathcal{U}'(B_3(G)\varepsilon_1)$. Together with uniqueness of the regular critical orbit this assumption gives one nonempty regular critical orbit, contained in $\mathcal{U}'(B_3(G)\varepsilon_1)$; near the trivial datum it is the conditional assertion of Prop. 9.22(ii). Its route is the corresponding $SU(2)$ proof, the existence of a critical configuration in the class and the improvement of the regularity radius (the $SU(2)$ model is Prop. YM3_I:37.1) being run for G with the bounds that the variational part takes from the earlier sections at a regular reference configuration $U_0 \ne \mathbf{1}$: those of the linear operators at U_0, which rest on the propagators of (P) at regular backgrounds, and those of the straightened averages, the action and the gauge transformations in the chart, the rows and the gauge functional at U_0.

Conjugation. Prop. 9.23(i) gives (C) for the maps listed there and for their composites, finite sums and continuations on convex subsets of a vector space kept by the conjugation (slices without group-valued member, Lem. 9.21(i), (ii)). The maps of the induction are of these kinds: each is a chain of factors, each factor being built from products, inverses, means, logarithms in D^c, the action a and its derivatives, which commute with the conjugation by Lem. 9.20(i), (ii), or one of the three implicit systems of Lem. 9.20(iii), or a tower, or a continuation on a convex subset of a vector space kept by the conjugation, or a map holomorphic on the open slices of the real-connected part of its domain that takes real fields to real values, every trace of a group-valued argument at complex fields being $a(X)$ (Def. 9.5(1)); and the domains of the $SU(2)$ construction pass into real-connected parts by Lem. 9.21(iii). A map that is holomorphic on the open slices of the real-connected part of its domain, its real members being constant on slice components, is of them as soon as it takes real fields to real values, and it then maps the real-connected part of its domain into that of its target as soon as it maps it into the target (Lem. 9.21(ii)).

The pivot bond and Gaussian coefficient. For the pivot bond, the unknown element $x \in G$ at the pivot bond enters the quotient $R = W[c]G_1[c]^{-1}$ through a homeomorphism of G onto itself; in each of the three cases of the averaging map a solution has the eigen-angles of R below $\frac{1}{272L^2}$ in modulus, so that it lies among the fields with plaquette variables within ω_M of $\mathbf{1}$ on the double block, where the average is the formula of Lem. 9.11(i) and the other two cases have no solution; every element that occurs is a product of conjugated plaquette variables within $\omega_M \le z_G \le r_G$ of $\mathbf{1}$, so that its logarithm lies in $i\mathfrak{g}$ (Lem. 9.9(ii), (iii)); and in the chart the equation reads $r = L^2(t - \bar{s} - E(r))$ for $r = \log R$, with t and $\bar{s}$ averages over the block of logarithms that do not read x and with E

holomorphic, the right member being a contraction of $\{r \in i\mathfrak{g} : |r| \le \frac{1}{50}\}$ into itself with constant below $\frac{1}{121L^2}$; its fixed point gives the one solution and the bound at the pivot bond, with the numerals of the $SU(2)$ construction. The large-field chain is Lem. 9.13(ii). The localised Gaussian family is formed from the fluctuation operators of (P) by the formulas of the $SU(2)$ construction, and at the trivial triple its Hessian is that of the normaliser (Prop. 9.18(iii)).

Choice of constants. Prop. 9.26 gives the shape of the table and the criterion: one test for each condition and one joint test for each two-sided constant. A floor on a constant free upward and a positive ceiling on a constant free downward are admissible. A condition whose last constant is determined must be checked from its formula. The conclusion that all the conditions pass these tests is hypothesis (K).

The conditions of Lem. 9.21(iii) at the memberships reached from memberships of restrictions had apart are positive ceilings on the second radius where the first radius lies below B_0, and for the cores $Kb_0 < B_0$ and $b_0 < a^\sharp$; the ceiling $\mathsf{c}_L c_4 \le 2Lr_G$ is a positive ceiling on c_4, and the row $\alpha_1 < \rho_G L$ one on α_1 (Props. 9.22(iv), 9.24).

The comparison of the two norms, the regularity window and the central-element guards are specified in Prop. 9.26(c), (e), (f). The numbers of the pair must occur before γ in the order of choice; they may depend only on the constants preceding them and on the pair.

9.1.9. *Multiscale induction and the choice of constants.*

Proposition 9.23 (Conjugation and the inputs of the induction)**.** The conjugation assertions apply to the maps wherever they are formed. The global second-part assertions and the pointed-family assertion in (iii) assume (II) of §9.1.1. Let $\mathfrak{g}$ be simple, $L \ge 13$ odd, $200L^2 z_G \ge 1$; constants are constants depending only on L and the pair, uniform in the volume.

(i) Let $a(X)$ stand for every trace of a group-valued argument at complex fields, the gauge-fixing term included, and let the domains be written with $m(X)$ and with D^c at the logarithms of the averages. Then plaquettes, path products, gauge transforms, covariant differences, F, J, means, averages and their iterates, the chain to the fluctuation operator and the three implicit systems of Lem. 9.20(iii), the towers, and the composites and finite sums of these maps and their continuations on convex subsets of a vector space kept by the conjugation (slices without group-valued member) commute with $p \mapsto p^c$ on the real-connected parts of their domains and map them into the real-connected parts of their targets. A map holomorphic on the open slices of Q_1^r, with real members constant on slice components, commutes with $p \mapsto p^c$ there if and only if it takes the real points of Q_1 to real points, and then it maps Q_1^r into Q_2^r as soon as it maps Q_1^r into Q_2 (Lem. 9.21(ii)).

(ii) In the second-part statements assumed in (II), the clauses that are false for G as worded are required in the following corrected forms, among them these: $m(X)$ where $|X^{-1} - \mathbf{1}| = |X - \mathbf{1}| = |X^* - \mathbf{1}|$ was used, which is false at $SU(3)$; the radius r_n where $|1 - T| \le \frac{1}{2}$ was to keep $\det T$ off $(-\infty, 0]$ (the three-dimensional determinant argument in the proof of Prop. YM3_I:50.2), which is false from $n = 6$; the factor $\max\{1, n2^{-11}\}$ in the bound eq. 9.xii (Prop. 9.26(d)); the bound $z < (\frac{5}{4})^n L^{3n}$ in the place of $z < 2L^9$ in Lem. YM3_I:65.3; orthogonal colour maps that keep $|\cdot|$ in the covariance argument following Prop. YM3_I:47.4; the indicator of G^π where $G \setminus G^\pi$ is used as a null set, and integrands that vanish outside the chart in the substitution of Prop. YM3_I:53.4; $a(X)$ for the bare trace of the gauge-fixing term of the definition of Ψ^{G} in Section YM3_I:51, p. 393; $E^{\mathrm{fam}} \ge e_n$ in the bound for the local Gaussian constants, as derived in Prop. 9.26(d); and the two-sided guard $m(X)$ in the elimination-factor estimate of Lem. 9.14(iv). Of the clauses in which a proof of the second part works with one colour, those at the trivial field and in the averaging layer hold for G with the constants of the $SU(2)$ construction; those that concern the background near trivial block data, and the radius $r_0 = a^{\mathrm{us}}(\alpha_0, \alpha_1)$ in $\mathsf{G}_\sharp$ (Prop. 9.24), hold as soon as on every truncated tower the background exists, is one orbit, is holomorphic on a polydisc

of uniform radius and has there the bounds of Prop. 9.22(ii), whose uniform conclusions here are assumed in (V), including uniqueness of the critical orbit; and one more ceiling, $\varepsilon_{\mathrm{v}} \le a_G$, enters on the regularity radius ε_{v} of the critical configurations (Def. YM3_I:9.5), with a fixed admitted positive radius a_G from §9.1.6 (Prop. 9.26(f)).

(iii) Near $W = \mathbf{1}$ the localised Gaussian family of the second part (Prop. YM3_I:50.2) has at the trivial triple the Hessian of the Gaussian normaliser of Prop. 9.18(iii); its quadratic coefficient there is one number for all data, $b_l^0(G)$. Under (II), for simply connected G the family is a pointed family of its level on the analyticity cores of G, holomorphic on balls of one radius, with the contraction bound of its expansion (§9.1.1(II)).

■

Sketch of proof. (i) A map is covariant on the real-connected part if it is a chain of factors each of which is either built from products, inverses, means, logarithms in D^c, the action and its derivatives, and so covariant by Lem. 9.20, or holomorphic on open slices and real at real fields, and so covariant by Lem. 9.21(ii), or a continuation on a convex subset of a vector space kept by the conjugation (a slice without group-valued member); a second way is to check that each of its formulas is real and holds at the doubled pair $X \mapsto \mathrm{diag}(X, (X^{\mathrm{T}})^{-1})$, for which $|\mathrm{diag}(X, (X^{\mathrm{T}})^{-1}) - \mathbf{1}| = m(X)$ and the action, the norm and the form restrict to those of G. The maps listed in (i) are of these kinds: plaquettes, path products, gauge transforms, covariant differences, F, J, means, averages and their iterates and the chain to the fluctuation operator are built from products, inverses, means, logarithms in D^c, the action and its derivatives (Lem. 9.20(i)–(iii)); the three implicit systems are covariant by Lem. 9.20(iii); the towers and the continuations on convex subsets of a vector space kept by the conjugation by Lem. 9.21(ii); and composites and finite sums of covariant maps are covariant. Lem. 9.21(iii) moves the domains of the $SU(2)$ construction into real-connected parts at the cost named there. Two statements of the $SU(2)$ construction take another form for G: the gauge-fixing term of the definition of Ψ^{G} in Section YM3_I:51, p. 393, set with a bare trace at complex fields, whose imaginary part $\frac{t}{N}\mathrm{tr}\sin(tH)$ at $SU(3)$ does not vanish at a real field, takes $a(X)$, which keeps its quadratic part and its bounds; and the statement that the conjugate point lies in the domain if and only if the point does, which for $SU(2)$ rests on the isometry Θ of Lem. 5.23 and is false beyond $SU(2)$, takes the two-sided guard $m(X)$.

(ii) The forms that hold are those of Lems. 9.20(ii), 9.14(iv) and Prop. 9.26(d) for the first three, and those of Def. 9.5(2), (3), Remark 9.6, (i) above, Props. 9.22(iv), 9.26(d) and Lem. 9.14(iv) for the others. At the trivial pair every operator is scalar times $1_{\mathfrak{g}^c}$ (Lem. 9.15(ii)), with the lower bound of the fluctuation operator and only 3 replaced by n, and the averaging layer is Lem. 9.11; near trivial block data the clauses need the existence, one orbit, holomorphy on a polydisc of uniform radius and the bounds of the background on every truncated tower, of which Prop. 9.22(ii) gives these conclusions under (V). The ceiling uses a fixed admitted orbit radius from §9.1.6; the central-field obstruction described there applies only when the chosen averaging extension sends that field to the trivial block datum. It does not establish existence or a sharp value of a positive orbit radius for every averaging extension.

(iii) At the trivial triple the germ of second order of the localised family is computed as that of the normaliser in Prop. 9.18(i): by invariance the background is taken, to second order, along one unit $H \in \mathfrak{g}$; the fluctuations split into the eigenspaces of $\mathrm{ad}\,H$, each of which is a charged sector of $SU(2)$; and the action adds the one tadpole, with $\mathrm{tr}\,M_H = C_\rho$. So the coefficient of the family there is $\frac{C_A}{8}$ times that of the localised family of $SU(2)$, which is one number for all data and equals the coefficient of $\log Z^{(k)}$ (equation YM3_I:(790) identifies their totals up to a constant, and Thm. YM3_I:X identifies the coefficient of such totals), the increment b_{k+1}^0, plus $\gamma_G(j_{k+1} - L^{-1}j_k) = (\frac{C_\rho}{3} - \frac{C_A}{8})(L^3 - 1)L^{-4(k+1)}$, $j_k = \frac{2}{3}(L^{-k} - L^{-4k})$; this is $b_{k+1}^0(G)$, the coefficient of the normaliser (Thm. AA(c), Prop. 9.18(iii)), and it reads no datum. □

Proposition 9.24 (Induction for the proposed construction)**.** For a simple Lie algebra, the construction remains to be completed through the tasks (P)–(K) of §9.1.1. For a semisimple Lie algebra, the transferred one-step statements must likewise be established and their applicability checked at the data of each citation below. For tied ideals, this also requires property (T) of Thm. AA(h3). The proposition records the induction that would then give the construction conclusions. The coefficient estimates below apply whenever the indicated terms and their analyticity bounds are given. Let $\mathfrak{g}$ be semisimple. Each of the following proofs of the $SU(2)$ construction derives its transferred statement from the transferred statements that it cites, the hypotheses of each citation, those that the forms (3) of Def. 9.5 add to a cited statement among them, being met at its data, as in §9.1.1(II): the regular progression in Thm. YM3_I:Y; the actual one-step estimates of Thm. YM3_I:W, their quadratic coefficient in Thm. YM3_I:X, and the canonical coupling construction in Section YM3_I:52.10, with $\mathsf{G}_\sharp$ as its coefficient bound; the historical step in Prop. YM3_I:69.3, its iteration in Thm. YM3_I:AC and Cor. YM3_I:69.4; and the history bounds of Section YM3_I:70, which give Thm. YM3_I:D. In the historical step the proof explicitly checks the coordinate changes and tests, the separate on-shell identities for the old exponent and retained factors, the conditional Gaussian and cluster identity, the normalizer, the kept and rejected forms, the strip sum, and the canonical identification of the new exponents. The positive change-of-variables theorem used there is Thm. YM3_I:AB; the boundary and retained-factor bounds are those of Section YM3_I:62. Their transferred hypotheses, including their domains and measurability assertions, remain required here. Of the group these proofs use only that $\mathrm{Ad}(G)$ keeps $\langle\cdot,\cdot\rangle$ and $|\cdot|$, that invariant linear forms vanish, that $|t_a| \le \kappa_G$, and that $a(wuw^{-1}) = a(u)$. The part of second order of the effective terms enters them as one real number, the mean coefficient cf, the mean of its n quadratic colour coefficients: it is defined, bounded, additive, real and holomorphic in the parameters, never a multiple of the Hessian of the action, with $|\mathsf{cf}| \le \mathsf{G}_\sharp = \kappa_G^2 C_{\mathrm{am}} r_0^{-2} E_0 \le B_\sharp$, where C_{am} is the constant of the coefficient bound YM3_I:(940) and $r_0 = a^{\mathrm{us}}(\alpha_0, \alpha_1)$ the radius of the $SU(2)$ construction; C_{am} and r_0 read constants of the $SU(2)$ construction that come before them in its order, $\mathsf{G}_\sharp$ reads besides the pair, through κ_G, and E_0, and none of them reads γ, $B_\sharp$, a level, the cutoff, β or a field. On an ideal $\mathfrak{g}_i$, $|\mathsf{cf}_i| \le \kappa_i^2 C_{\mathrm{am}} r_0^{-2} E_0 = (\kappa_i/\kappa_G)^2 \mathsf{G}_\sharp$. The numbers $\frac{9}{2}$ and 3 of Thm. YM3_I:D become $\frac{3n}{2}$ and n; the π of the ceiling $\alpha_1 < \pi L^j$, $1 \le j \le k$, under which the local terms are continuous on the analyticity cores, which a proof of the chain from level k to level $k+1$ quotes, becomes ρ_G of Def. 9.1, the bound on the eigen-angles under which an element of G lies in G^π, the domain of the chart, so that the ceiling is $\alpha_1 < \rho_G L$, one more row of the table of Prop. 9.26(a), a positive ceiling on α_1, which holds of itself for $\rho_G > \frac{1}{13}$, as $\alpha_1 \le 1$ and $L \ge 13$; as $\rho_G \le \pi$, this row gives $\alpha_1 < \pi L^j$, under which the exponential of $\mathfrak{g}^c$ is a local diffeomorphism at every Z_0 with $|Z_0| < \pi$, the eigenvalues of $\mathrm{ad}\, Z_0$ being at most $2|Z_0| < 2\pi$ in modulus for matrices of every size; the $\frac{3}{2}$ in the size of an array stays if the analyticity condition is read with $m(\partial U)$, since $|a(u)| \le \frac{1}{2}|u - \mathbf{1}||u^{-1} - \mathbf{1}|$; and the transferred statements of these proofs differ from those of the $SU(2)$ construction only in these numbers and in "of L and the pair" for "of L", the sign $\mathsf{G}_\sharp$ keeping its places with its value for G. The large-field chain of the $SU(2)$ construction, Prop. YM3_I:44.4, Prop. YM3_I:44.5 and Lem. YM3_I:65.2 and the local-deposit argument in the proof of Prop. YM3_I:65.4, is transferred by Lem. 9.13(ii), its statements taking the prices of the $SU(2)$ construction times τ_G^{-2} and the averaging radius $\min\{\omega_M, r_G\}$. The same holds for an action $a(u) = 1 - \frac{1}{2}\mathrm{tr}\,\Lambda(u + u^{-1})$ with a weight $\Lambda > 0$ in the commutant of G, $\mathrm{tr}\,\Lambda = 1$, the form $\mathrm{tr}(\Lambda XY)$ and $\kappa_\Lambda = \max|Y|/\mathrm{tr}(\Lambda Y^2)^{1/2}$ in the places of $\langle\cdot,\cdot\rangle$ and κ_G. ∎

Sketch of proof. The construction hypotheses provide the cited one-step estimates and their applicability on the transferred domains. There are five additional guards to retain. Logarithms of group variables are taken in D^c; a logarithm in the real group is taken in G^π; complex domains use $m(X)$; determinant logarithms use the radius r_n; and local terms on the analyticity cores satisfy

$\alpha_1 < \rho_G L^j$. Lems. 9.9, 9.14, 9.20 and Prop. 9.22(iv) supply the first four at their stated data. The row $\alpha_1 < \rho_G L$ supplies the last because $L^j \geq L$. Compatibility of these rows is part of (K).

The colour algebra in the induction consists of invariance of the trace form and of the operator norm, vanishing of invariant linear forms, $|t_a| \leq \kappa_G$, and conjugation invariance of a. The comparison of the two volume exponents in Thm. YM3_I:D becomes $n \leq 3n/2$. With the one-step bounds and domain conditions assumed, the induction proceeds on the same histories.

Terms of first order in the fluctuation field are bounded by Schwarz's lemma: if h is holomorphic on $|\lambda| < \rho$ with $\rho > 1$, $h(0) = 0$ and $|h| \leq c$, then $|h(1)| \leq c/\rho$, nothing being asked of $h'(0)$. The damping $L^{-(k-j)}$ of old levels at $U = \mathbf{1}$ comes from the vanishing of invariant linear forms. In the nine proofs of the chain from level k to level $k+1$ the inputs ask $\|E^{(j)}\| \leq E_0$ and $|b_j| \leq B_\sharp$ separately, and the bounds of a step add these sizes. The increments enter only as the real numbers $e_j = b_j$ with $|b_j| \leq B_\sharp$, no relation between them and the entries of a term or their derivatives is used, and the one stationarity used is that of the one action. The bounds of the coefficient: by Cauchy's estimate on the ball of radius r_0 the part q of second order of a term E obeys $|q(X,X)| \leq C_{\mathrm{am}} r_0^{-2} \|E\| |X|^2$, as in the $SU(2)$ construction, where the orthonormal basis has $|t_a| = 1$; with $|t_a| \leq \kappa_G$, and $|t_a| \leq \kappa_i$ on $\mathfrak{g}_i$, this gives $|\mathsf{cf}_i| \leq \kappa_i^2 C_{\mathrm{am}} r_0^{-2} E_0$ and $|\mathsf{cf}| \leq \sum_i \frac{n_i}{n} \kappa_i^2 C_{\mathrm{am}} r_0^{-2} E_0 \leq \mathsf{G}_\sharp$. The factor κ_i^2 against C_{am} cannot be dropped: for $SU(2)^r$ on blocks, $r \geq 2$, one has $\kappa_i^2 = r$, and a term that reads one factor has on its ideal r times its coefficient for $SU(2)$. □

The colour factor in the determinant estimate is most easily checked before it is absorbed into the constant. The following calculation records the precise contraction and closed-chain bounds it requires. Their availability for the transferred operators remains part of the construction hypotheses.

Lemma 9.25 (The colour factor in a local determinant bound)**.** Let $n \geq 1$, $c > 0$, $0 < q < 1$, $Q > 1$ and $\delta_1 > 0$. Put

$$\delta = \frac{\delta_1}{128} \frac{\log(1/q)}{\log Q + \log(1/q)},$$
$$C_{\mathrm{a}} = n \left(|\log c| + \log \frac{1}{1-q} \right),$$
$$C_{\mathrm{d}} = \frac{Q}{Q-1} + \frac{1}{1-q}, \qquad C_{\mathrm{mob}} = 2^7 C_{\mathrm{a}} + C_{\mathrm{d}}.$$

Assume $\delta M \geq 9 \log 2$. Let X be a finite set of coarse cubes, $d = d_t(X) \geq 0$, and let $\emptyset \neq S \subset X$ satisfy $d_t(S) = 0$ and $\#X \leq 8(d+1)$. For each $S \subset Y \subset X$ let T_Y be an operator on bond fields with n colour coordinates, with $\|T_Y\| \leq q$, and with a common bond b in its domain. Define

$$f_b(Y) = n \log c - \sum_{\nu \geq 1} \frac{1}{\nu} \operatorname{tr} T_Y^\nu(b,b), \qquad E(X,b) = \sum_{S \subset Y \subset X} (-1)^{\#X - \#Y} f_b(Y).$$

The trace is over the colour coordinates. When $d > 0$, suppose the closed-chain sums

$$\Sigma_\nu = \sum_{S \subset Y \subset X} (-1)^{\#X - \#Y} \operatorname{tr} T_Y^\nu(b,b)$$

satisfy, with $\mathcal{N} = 2^{\#X - \#S}$,

$$|\Sigma_\nu| \leq n \mathcal{N} q^\nu, \qquad |\Sigma_\nu| \leq n \mathcal{N} Q^\nu e^{-\delta_1 M (d+1)/64}. \tag{9.xi}$$

Then

$$|E(X,b)| \leq 2^{-d} \max\{1, n 2^{-11}\} C_{\mathrm{mob}} e^{-\delta M d}. \tag{9.xii}$$

■

Proof. The contraction bound makes each series absolutely convergent and gives $|f_b(Y)| \leq C_{\mathrm{a}}$. There are at most $2^{8(d+1)-1}$ sets Y. If $d = 0$, their sum is bounded by $2^7 C_{\mathrm{a}} \leq C_{\mathrm{mob}}$. For $d > 0$, $X \neq S$, so the alternating sum of the constant $n \log c$ vanishes. Thus $E(X,b) = -\sum_{\nu \geq 1} \Sigma_\nu / \nu$.

Put

$$\theta = \frac{\delta_1}{64(\log Q + \log(1/q))}, \qquad \nu_* = \lfloor \theta M(d+1) \rfloor .$$

Use the second majorant up to ν_* and the first afterwards. Since $\theta \log(1/q) = 2\delta$ and $\theta \log Q - \delta_1/64 = -2\delta$, the two geometric sums give

$$|E(X,b)| \le n\mathcal{N} \left(\frac{Q}{Q-1} + \frac{1}{1-q} \right) e^{-2\delta M(d+1)}.$$

One exponential absorbs the region count:

$$|E(X,b)| \le \frac{n}{4}\, 2^{-d} C_{\mathrm{d}} e^{-\delta M(d+1)} \le n 2^{-11}\, 2^{-d} C_{\mathrm{d}} e^{-\delta M d}.$$

Here the second inequality retains the further factor $e^{-\delta M} \le 2^{-9}$. Together with the case $d = 0$ and $C_{\mathrm{d}} \le C_{\mathrm{mob}}$, this proves the claim. □

Proposition 9.26 (The constants)**.** Let $L \ge 13$ be odd and $200 L^2 z_G \ge 1$. The constants are chosen in the order of the $SU(2)$ construction, γ last, each chosen constant ranging, given the earlier ones, over the range that the $SU(2)$ construction declares for it; each is free upward, free downward, two-sided (its admissible values, given the earlier ones, form a window) or determined by a formula from earlier ones. The table of the constants of the $SU(2)$ construction carries besides a radius $\rho > 0$, which is not chosen and stands in the rows on γ alone. A set of choices of all the constants is *large* if it is large at the empty choice, where, from the last constant down to the first, a set is large at a choice of the constants before the i-th one if the values of the i-th constant at which it is large at the longer choice hold, for some t and some $c > 0$, the part $[t, \infty)$ of its range when this constant is free upward, the interval $(0, c]$ when it is free downward, all its admissible values when it is two-sided, and its value when it is determined; at a choice of all the constants a set is large if it holds that choice. A set that reads γ or ρ is large if so is the set of the choices of the constants before γ at which, for every $\rho > 0$, some interval $(0, c]$ consists of admissible values of γ at which it holds.

(a) For $SU(N)$, every $N \ge 2$, the table of the constants of the $SU(2)$ construction, formed with the numbers of G as data (any positive values of these numbers in the ranges that the $SU(2)$ construction states for them serve for its shape), with the changes of Def. 9.5 and with those conditions of (c)–(g) that are a floor on a constant free upward or a positive ceiling on a constant free downward laid as rows, has admissible shape, the shape under which, at every admissible choice of the earlier constants, the admissible values of a constant free upward hold a half-line of its range, those of a constant free downward an interval $(0, c]$, and the window of a two-sided constant is not empty; γ is last, and choices exist at every value of the radius ρ, the admissible choices of the constants before γ being the same for all ρ; a further condition read for G is laid on it as a row when it is large. For a semisimple pair with a weight in the commutant, the table of the second part of the $SU(2)$ construction, its chosen constants, their order, kinds and ranges kept, is, as the table of the $SU(2)$ construction is, of admissible shape and without cap, with γ last, and choices exist, given that the numbers of the first part of the pair exist with the ranges that the $SU(2)$ construction states for them. In both, a floor on a constant free upward and a positive ceiling on a constant free downward never obstruct; L is free upward, with floors alone, and a floor on L may be laid once at the root, every other constant keeping the same admissible set. This is the shape of the table; that every condition read for G is large, so that all of them are rows of a table of admissible shape and hold at each of its choices, is (K).

(b) Large sets form a filter: the intersection of two sets of choices is large if and only if both are. Finitely many conditions are each large if and only if there is a table obtained from that of (a) by floors on free upward constants, positive ceilings on free downward constants, and further constants determined by formulas from earlier ones, at every choice of which all the

conditions hold. These additions retain every original two-sided window whole; one floor or ceiling on each free constant suffices. Separately, further ends on a two-sided constant are jointly feasible at an earlier choice exactly when its original window meets all those ends together. Requiring a large set of such earlier choices is a compatibility test for the restricted windows, not largeness of the ends as subsets of the original full table. Hypothesis (K) requires both the largeness conditions and the joint-window compatibility tests used by the construction.

(c) For $SU(N)$ the two-sided conditions that compare $|\cdot|$ with $\|\cdot\|$ hold together for every $N \geq 2$ at the ratio $\frac{1}{N-1} = \kappa_G^{-2}$ of the squares of the two norms, with the kernel bounds of (P) read in the normalised trace norm, which the bounds in $|\cdot|$ give with the factors $(N-1)^{1/2}$ and $N^{1/2}$ between the two norms (Prop. 9.22(i)). The two-sided condition on the regularity radius ε_{v}, which comes from the inclusions $\mathcal{U}'(r) \subset \mathcal{U}((1+\kappa' r^2)r)$ and $\mathcal{U}(r) \subset \mathcal{U}'(r + r^2 + \kappa' r^3)$, $r \leq z_G$, of Lem. 9.12(ii), is not empty while $\frac{1}{400L^2} < 2\sin\frac{\pi}{N}$, a constant that precedes ε_{v} in the order being multiplied by $1+\kappa'$, with the numbers of the critical configuration of (V), and one floor on the separation constant R_1 makes its ends compatible; for the two-sided conditions that a central element near $\mathbf{1}$ enters when $z_G < 1$ see (e).

(d) The chains that use $\operatorname{Log}\det$ (the proof of Prop. YM3_I:50.2 and the elimination-factor estimates of Lem. 9.14(iii), (iv)) hold at the radius r_n, with $4Ka \leq r_n$ and $C_z \geq 4c_n K$, $c_n = \rho_n / r_n$, the pair $4Ka \leq \frac{1}{2}$, $C_z \geq 20K$ of the $SU(2)$ construction holding besides, and with the further member of the radius of the second chain that Lem. 9.14(iv) names, which keeps its logarithms in $\mathfrak{g}^c$; their constants are constants depending only on L and the pair, explicit in L and n at $SU(N)$, that obey the conditions in which they enter: there the radii are bounded from above only and the bounds from below only, so the ordered choice closes again provided every subsequent use of the normalizer and elimination constants requires only these upper bounds on radii and lower bounds on estimate constants. This compatibility with the later inequalities remains part of (K). Write E^{fam} for the local Gaussian bound constant, the counterpart of E_b in Prop. YM3_I:50.2; here E_b already denotes the quotient of Prop. 9.22(iv). Require

$$E^{\mathrm{fam}} \geq e_n := \frac{2(2n\log L + \rho_n)}{L^3 + 1}.$$

Indeed, Lem. 9.14(iv) then gives

$$|z_c| \leq 2n\log L + \rho_n \leq \tfrac{1}{2}(L^3+1)E^{\mathrm{fam}},$$

which is the bound needed for the local Gaussian constant at an elimination term. The chosen constants must satisfy this guard; it is not inferred merely from a dimension-independent lower bound on E^{fam}. The inequality $\frac{3}{4} \leq 1$ by which the proof of Lem. 9.25 absorbs a colour factor reads $\frac{n}{4} \leq 1$, which fails from $n = 5$; a factor 2^{-9} of the same estimate, which that proof does not use, restores it for $n \leq 2048$, and for every n the precise bound is eq. 9.xii. Here C_{mob} is the full constant, including the zero-size term, with its trace contribution evaluated in dimension n. $(\frac{5}{4})^3 < 2$ becomes $(\frac{5}{4})^n$, which fails from $n = 4$, and the bound $z < 2L^9$ of Lem. YM3_I:65.3 becomes $z < (\frac{5}{4})^n L^{3n}$. The devices by which the $SU(2)$ construction fixes the branch of a logarithm or of a root or power of a determinant ask something of the operator and not of the value of its determinant, and serve in every dimension, but for those that read the value: the principal logarithm of the elimination factor (Lem. 9.14(iii)); the power series of the logarithm about 1 in the Jacobian of the exponential map (Lem. 9.14(i)); the power series of the logarithm about 1 at the Jacobian $1 - \mathsf{jc}$ of the bond shift with its cut-off (the cutoff preceding Lem. YM3_I:53.3), jc being the derivative of the cut-off in the direction of the shift, so that $1 - \mathsf{jc}$ is the determinant of a perturbation of rank one of a map of determinant 1, whose value is computed outright and in which no count of the dimension

enters; and the power series of the logarithm about 1 at a quotient $\det(\mathsf{T}(0)^{-1}\mathsf{T}(\zeta))$ of the determinants of a holomorphic family $\mathsf{T}(\zeta)$ of operators on $\mathfrak{g}^c$-valued fields, formed near $\zeta = 0$, where the quotient is near 1 by continuity. Among the devices that ask something of the operator is the number $2n \log L - \operatorname{tr} \log T$ of the elimination, $T = L^3 \operatorname{Ad}(g)\, \mathsf{m}\, \operatorname{Ad}(g')^{-1}$ with $g, g' \in G$ as in Lem. 9.14(iii): it asks $|T - 1| < 1$ at some g, g', and it does not depend on their choice among those with $|T - 1| < 1$, for every n (Lem. 9.14(iii)). Where a count of the dimension, $n = 3$ against π or 2π or $n - \operatorname{rk} G = 2$ against 1, fixes a branch, as in the determinant argument in Prop. YM3_I:50.2, the elimination estimates of Lem. 9.14(iii), (iv), the auxiliary-frame independence proved in Lem. 9.14(iii) and the bound for the Gaussian constants just proved, the radius r_n and the numbers of Lem. 9.14(iv) take its place, or, where it compares $\operatorname{tr} \log T$ at two choices of g, g', the equality of Lem. 9.14(iii); and where the sign $(-1)^3 = (-1)^{\dim \mathfrak{g}}$ stands twice in one identity, in the adjoint determinant calculation, it cancels for every n. That the branch is fixed in one of these ways at every place where the $SU(2)$ construction forms a logarithm or a root or power of a determinant over $\mathfrak{g}^c$ is part of (II) and (K).

(e) If $z_G < 1$, the conditions that fall to z_G include a ceiling on γ, a floor on L and the radii that the $SU(2)$ construction defines as minima with the averaging radius. For the rows that fall to z_G the floor on L is $L^2 \geq \frac{1}{200 z_G}$ together with the row $\alpha_0 \leq z_G^2/(4C^2)$, C of Lem. 9.12(iv), or $L > \frac{1}{z_G}$; the floor $L > 2 z_G^{-2}$, which implies $L > \frac{1}{z_G}$ and $200 L^2 z_G \geq 1$, carries with a ceiling on γ all the relations in N of the table. The numbers $\frac{1}{200}$, $\frac{1}{2}$, $\frac{1}{3}$, the radius 1 and the like do not fall, for there "$X \in D^c$" is supplied by a product or a path; the mean and the block average keep their numerals under $\omega_M \leq z_G$, which the floor $200 L^2 z_G \geq 1$ gives; so do the charts, the gluing of gauges and the gauges at fields whose plaquette variables lie in D under the averaging radius, while the gauges that Def. 9.5(3) names take there the forms named: the comb gauge of Lem. YM3_I:14.4 takes the guard $\varpi < z_G$ and Lem. 9.10 the guard $\varpi_y(W) < z_G$, which bite at $SU(N)$ from $N = 7$, and the quantitative cube-gauge input of Lem. 9.12(iii) takes constants c, B of the pair; the box clause of Lem. YM3_I:14.4 read with some gauge of the box in the place of the comb holds at $SU(N)$ exactly for $N \leq 13$ (Lem. 9.12(iii)); an element known by a chord $|V - V'|$ alone takes z_G in its radius; and the statements that place a logarithm in $\mathfrak{g}^c$ by "$|X - \mathbf{1}| < 1$" are false for $z_G < 1$ as worded and hold with "$X \in D^c$". Among the conditions that the centre enters, a condition can bound N only through an inequality with no free constant, $b < 2 \sin \frac{\pi}{N}$, where a proof concludes from "X is within b of $\mathbf{1}$" that $\log X$ is traceless; read with D^c, such a conclusion asks instead for a product or a path, while a ball of radius z_G in the place of D^c would let the number $\frac{1}{14}$ of Lem. 9.11(i) cap N at 87. The other inequalities with no free constant that bound n or N take the forms named at their places: $\frac{n}{4} \leq 1$, $(\frac{5}{4})^n < 2$ and $|\mathbf{1} - T| \leq \frac{1}{2}$ in (d), $|\log j| < 1$ on the ball of radius $\frac{3}{8}$ in Lem. 9.14(i), and the ratio of the two norms in (c). A field whose bond variables lie within r of $\mathbf{1}$, $(1 + r)^4 < 2$, is ζe^{iA} with $\zeta(b) = z(U(b)) \in Z(G)$ and $\mathfrak{g}$-valued A, and $\partial\zeta(p) = z(\partial U(p))$; so ζ is flat exactly when every plaquette variable lies in D^c, and a flat ζ is a central gauge transform of $\mathbf{1}$ on a box and is fixed by its three holonomies up to such a transform on a torus. A gauge function h on a connected set with $|h(x) - \mathbf{1}| < 1$ and $h(x)h(y)^{-1} \in D^c$ at neighbouring sites x, y, as when h carries a field with bond variables in D to another such field, is a central constant times $e^{i\psi}$, ψ being $\mathfrak{g}$-valued. The sets of fields cut out by plaquette tests on a block, a double block or a tower are connected to the field $\mathbf{1}$ inside themselves, and a field that passes the plaquette test t on a box is joined to $\mathbf{1}$ inside the set cut out by the test $t + 4.49t^2$; and at these places no two-sided condition becomes empty. That "$X \in D^c$" holds at every place where a logarithm is taken, the places being closed under the statements that inherit a radius, is part of (K).

(f) For every G, also at $z_G = 1$, the conditions include $\varepsilon_{\mathrm{v}} \le L^2 r_G$ and $\varepsilon_{\mathrm{v}} \le a_G$, where a positive orbit radius a_G admitted by §9.1.6 is fixed as part of the analytic input. No unconditional positivity or two-sided $SU(N)$ estimate for the supremum $a_*(G)$ is used. At $SU(N)$, $N \ge 3$, the central-field obstruction at $\varepsilon_{\mathrm{v}} > 2L^2 \sin(\pi/N)$ applies if the chosen averaging extension sends the displayed central field to $\mathbf{1}$: the nonflat zero-current field of §9.1.6 is then admitted at the first level. The conditions also include $\varepsilon_{\mathrm{v}} \le \varepsilon_G$ and $c_{11} \le r_G/(2C_{\mathrm{pg}})$ of the uniqueness clause of §9.1.1(V), c_{11} the radius of §9.1.6 and C_{pg} the admitted preliminary-gauge constant of §9.1.6; the ceiling $\mathsf{c}_L c_4 \le 2Lr_G$ of Prop. 9.22(iv), a positive ceiling on c_4, void at $SU(3)$; the row $\alpha_1 < \rho_G L$ of Prop. 9.24, a positive ceiling on α_1; and, at the memberships reached from memberships of restrictions had apart, the conditions of Lem. 9.21(iii): the moved first radius $b_0' = b_0 + 2e^{4\xi b_1} b_1^2$, for regular pairs $\alpha_0 + 2\vartheta^2 e^{4\vartheta\alpha_1}\alpha_1^2$, at most the first radius B_0 at which the statement read there is had, and for the analyticity cores besides $b_1 \le B_1$, $K(b_0' + b_1^2) \le B_0$ and $b_0' + b_1 \le a^\sharp$, with $K \ge 16L^2$ and $a^\sharp$ of that lemma; where $b_0 < B_0$, and for the cores $Kb_0 < B_0$ and $b_0 < a^\sharp$, these are positive ceilings on the second radius.

(g) The constants in the matrix product, logarithm and contour comparisons of Lems. 9.9(ii), 9.11(iii) and 9.19 have the stated dependence on the pair. The $SU(2)$ contour coefficients are those of Lems. YM3_I:14.5 and YM3_I:14.6; the accumulated transport coefficient is constructed in the proof of Lem. YM3_I:30.3. Exact numerical choices for a general-group Hessian or nonlinear-remainder bound must be supplied as part of (P). In particular, this paragraph does not deduce the values $12, 1536, 3456$, the action radius and bound $1/128, 808$, or the gauge radius and bound $1/64, 93$ from matrix norm comparison alone. Of the constants of the first part of the $SU(2)$ construction these move with N: the averaging radius; the 2 of $|T^{\mathrm{T}}| \le 2|T|$ (Lem. 9.3), which becomes N, sharply; a numeral 3 that counts colours in a bound of Schur's kind of the opening lemmas of the propagator construction (Prop. 9.22(i)), which becomes n (Def. 9.5(2)); the unmodified $SU(2)$ coefficient 4 in Lem. 9.3 at depth 0, which becomes $4\varsigma_G$, at most $8(1 - \frac{1}{N})$ for $SU(N)$; the averaging constant $C_Q = 18NL^3 C_{\mathrm{av}}$ of Lem. 9.4, with its admitted kernel bound. When $C_{\mathrm{av}} = 3$ is supplied, its value is $54NL^3$, compared with $108L^3$ at $SU(2)$; the constants formed from $6LC$ and $12LC$ then move by $N/2$ and their reciprocal thresholds by $2/N$; and one constant, which takes the factor κ_G, for $SU(N)$ and radii $r \le z_G$ the factor $1 + \kappa'$, the thresholds on α_0 being divided by it. Two bounds of the $SU(2)$ construction take new numbers besides (Def. 9.5(2), (3), Remark 9.6): the 12 of the bound $\xi^3 |J_W(U)(b)| \le 12a$ on the current of a region at complex fields whose bond variables have condition number at most 2 becomes $6\hat{\varsigma}_G$, the 12 failing at $SU(N)$ from $N = 5$; and the factor 1 of the bound $|\mathrm{Bl}(\mathsf{g}, \mathsf{f})| \le \|\mathsf{g}\|_2 \|\mathsf{f}\|_2$ drawn by polarisation becomes κ_G^2, and stays 1 where $|X + iY|^2 \le |X|^2 + |Y|^2$ on $\mathfrak{g}$. Further numbers move in Def. 9.5(2) and Lems. 9.12(ii), (iii), 9.13(i) and 9.14(i), (ii). The constants of the propagators and the covariances, those of the critical configurations and those of the sections, which rest on both, are set by the statements of Prop. 9.22 and of (P), (V); and all relations among the constants follow from the form of the displays. A condition of the table need not stay true when a threshold that stands in a conclusion is lowered: the floor $\delta_{2,\mathrm{b}} R_1 \mathfrak{M} \ge 64(3 \log L + \log 2)$ in the required separation conditions of (K) fails when the rate $\delta_{2,\mathrm{b}}$ is lowered with R_1 and $\mathfrak{M}$ fixed, already at $SU(2)$; the ordered choice of (a) uses no such monotonicity.

■

Sketch of proof. (a), (b) The table of (a) follows the ordered choice of Prop. YM3_I:69.1, with the numbers of G as data and the compatibility required in §9.1.1(K): the constants of $SU(2)$ are kept, the value for G stands behind each function symbol, and new letters enter only where a formula needs them; a floor on a constant free upward or a positive ceiling on one free downward keeps the shape admissible and leaves the admissible sets of the other constants as they are, and a condition whose last constant is determined is read off its formula. The recursion that defines large sets

commutes with intersections, so the intersection of two sets is large exactly when both are; a large set holds, at each choice of the earlier constants at which it is large, a half-line, an interval $(0, c]$ or the whole window of the next constant, so one floor or ceiling on each free constant, laid in the order of the choice, makes every admissible choice lie in it, and a set in which every choice of such a table lies is large, provided the original two-sided windows have been retained whole. Further ends on a two-sided constant instead restrict its window. Their joint feasibility at an earlier choice means exactly that the original window meets their intersection; this does not make the ends large in the original table. The radius stands in rows on γ alone, so the tables at two radii differ in rows on γ alone and have the same admissible choices before γ; a set that reads γ or ρ and is large is met by laying its set of earlier choices as rows before γ, which read no ρ, and then at each ρ a ceiling on γ. (c) On $\mathfrak{su}(N)$, $|\cdot| \le (N-1)^{1/2}\|\cdot\|$ (Lem. 9.9(i)), that is $\|\cdot\|^2 \ge \frac{1}{N-1}|\cdot|^2$, and the conditions that compare the norms hold together at this ratio of the squares. The two-sided condition on ε_{v} is that of the inclusions of Lem. 9.12(ii), valid for $r \le z_G$: its window, which uses the two inclusions, is not empty with the numbers of the critical configuration of (V), and its ends are made compatible by one floor on R_1. (d) Lem. 9.14(iv): the chains need $4Ka \le r_n$, which keeps the normalised factor within r_n of 1, and $C_z \ge 4c_nK$; for $n \ge 4$ the radii of the $SU(2)$ construction fall by the factor $2r_n$ and its constant C_z rises by the factor $\frac{c_n}{5}$, which gives both, the pair $4Ka \le \frac12$, $C_z \ge 20K$ of the $SU(2)$ construction holding besides as $r_n < \frac12$ and $c_n > 5$; for $n = 2, 3$ the constants of the $SU(2)$ construction serve, as $r_n \ge \frac12$ and $c_n < 5$; and the further member of the radius of the second chain keeps the logarithms of that chain in $\mathfrak{g}^c$. The other devices, a trace series, an integral along a segment, a Gaussian integral, positivity along a path, uniqueness by continuity and the number $2n \log L - \operatorname{tr} \log T$ of the elimination, ask their conditions of the operator or of each of its eigenvalues by itself, and so read no dimension. The last asks $|T - 1| < 1$ at one choice of g, g'; at two choices T_1, T_2 with $|T_j - 1| \le r < 1$, T_2 is conjugate to $T_1 W^{-1}$, $W = \mathrm{Ad}(h)$ with $h \in G$, and $\operatorname{tr} \log X - \operatorname{tr} \log(XW^{-1})$ lies in $2\pi i\mathbb{Z}$, is constant on the convex set $\{|X - 1| \le r,\ |X - W| \le r\}$ and vanishes at $\frac12(1 + W)$, the eigen-angles of W being symmetric, so that the number is the same at both, for every n (Lem. 9.14(iii)). Of the devices that read the value, the series at the Jacobian of the bond shift and the series at the quotient of determinants read it without a count: $\det(A + u \otimes v) = \det A\,(1 + v(A^{-1}u))$ makes the Jacobian of the bond shift, a map of determinant 1 plus the rank-one term that the cut-off brings, the scalar $1 - \mathsf{jc}$, and the bound $8\|\vartheta'_G\|_\infty |H_b| \le \delta_k$ on the shift keeps $|\mathsf{jc}|$ below 1, the Jacobian lying in $[\frac78, \frac98]$ (Prop. 9.22(iv)); and $\det(\mathsf{T}(0)^{-1}\mathsf{T}(\zeta))$ is 1 at $\zeta = 0$, so within 1 of 1 near $\zeta = 0$. (e) The central factor $z(X)$ is a local homomorphism, invariant under conjugation in G^c, and $|\log X| \ge \theta(z(X))$; so the inequality by which a product is shown to lie near $\mathbf{1}$ shows its membership in D^c, and $\{|\log X| < 2 \arcsin \frac{z_G}{2}\} \subset D^c$ exactly (Lem. 9.9(iii)); a radius known only through a chord meets the central elements at distance z_G. Around a plaquette whose bond variables lie within r of $\mathbf{1}$, $(1 + r)^4 < 2$, the partial products stay in the unit ball, so $z(\partial U(p)) = \partial\zeta(p)$; and a gauge function with $h(x)h(y)^{-1} \in D^c$ at neighbours has $z(h(x)) = z(h(y))$ there, so $z \circ h$ is constant on a connected set. (f) Use the admitted orbit radius of §9.1.6; for an averaging extension that sends its central field to $\mathbf{1}$: at $SU(N)$ the field with $e^{2\pi i/N}\mathbf{1}$ on one bond off the block axes is critical, without current, averages to $\mathbf{1}$ and is no pure gauge, and its plaquettes are at distance $2 \sin \frac{\pi}{N}$ from $\mathbf{1}$, so it is admitted at the first level as soon as $2 \sin \frac{\pi}{N} < \varepsilon_{\mathrm{v}} L^{-2}$; the further conditions of (f) are those that §9.1.1(V), Props. 9.22(iv), 9.24 and Lem. 9.21(iii) set. (g) Each constant is given by the statement that sets it, in which N enters only as named; $|T^{\mathrm{T}}| \le N|T|$ is sharp, $\kappa^2 = N - 1$ being attained by general kernels; $4\varsigma_G$, $6\hat{\varsigma}_G$ and κ_G^2 are those of Remark 9.6, where $8(1 - \frac1N)$ is the bound at $SU(N)$ and $4\varsigma_G$ exceeds 8 at the spin representation of $Spin(2m+1)$ from $m = 43$. The displayed separation floor bounds from below a product in which $\delta_{2,\mathrm{b}}$ is a factor, so it is lost when $\delta_{2,\mathrm{b}}$ alone is lowered; in the ordered choice each constant is chosen after those it depends on. □

Proposition 9.27 (Reflection positivity and Gibbs states)**.** Let $G \neq \{\mathbf{1}\}$ be a closed subgroup of $U(N)$ and $\beta \geq 0$, and consider the Wilson measure of a periodic lattice.

(i) With $\hat{g} = \mathrm{diag}(g, \bar{g})$,

$$e^{-\beta a(gg'^{-1})} = e^{-\beta} \sum_{\mathsf{n} \geq 0} \frac{(\beta/2N)^{\mathsf{n}}}{\mathsf{n}!} \sum_{I,J} \hat{g}_{IJ}\, \overline{\hat{g}'_{IJ}},$$

absolutely and uniformly convergent, with positive coefficients, where $I = (i_1, \dots, i_{\mathsf{n}})$ and $J = (j_1, \dots, j_{\mathsf{n}})$ run over $\{1, \dots, 2N\}^{\mathsf{n}}$ and $\hat{g}_{IJ} = \prod_k \hat{g}_{i_k j_k}$, the empty product being 1; so the measure is reflection positive at every mirror $v \cdot x = c$, $v = e_\mu$ with $2c \in \mathbb{Z}$ or $v = e_\mu \pm e_\nu$ with $c \in \mathbb{Z}$, whose period lattice, with periods at least 3, splits along v, for every bounded Borel function of the bonds on one side, gauge invariant or not; and for $\beta > 0$ every $\lambda_\pi(\beta) = d_\pi^{-1} \int e^{-\beta a} \chi_\pi$ obeys $\lambda_\pi(\beta) \geq e^{-\beta} \frac{(\beta/2N)^{\mathsf{n}(\pi)}}{\mathsf{n}(\pi)!\, d_\pi} > 0$, $\mathsf{n}(\pi)$ being an n for which π occurs in the n-th tensor power of $\mathbb{C}^N \oplus \overline{\mathbb{C}^N}$. At a mirror through the midpoints of bonds reflection positivity holds if and only if the crossing operator K is positive; K is the average over gauge transformations times the convolution with the weight on every bond, with the eigenvalues $\prod_b c_{\pi_b}$, $c_\pi = d_\pi^{-1} \int k\, \chi_\pi$ for a real even class weight k in the place of $e^{-\beta a}$, so that $c_\pi = \lambda_\pi(\beta)$ at $k = e^{-\beta a}$; two site mirrors ask nothing of the weight, a bond mirror asks these products to be non-negative, and a diagonal pair asks every $c_\pi \geq 0$.

(ii) The two chessboard estimates of the $SU(2)$ construction hold, with mirror copies conjugated at odd reflection parity in the slab products and in Fold:

$$|\mu(\prod_{j \leq 2M} f_j^{[j]})| \leq \prod_j \mu(\prod_i f_j^{[i]})^{1/2M}$$

for slabs and

$$|\langle \prod_\kappa F_\kappa \rangle_\beta| \leq \prod_\kappa \langle \mathrm{Fold}\, F_\kappa \rangle_\beta^{1/\mathsf{N}}$$

for cubic cells, and so do the statements of the $SU(2)$ construction that follow them: for sources $\|e^{\mathcal{O}(w)}\|_{\mathrm{ch}}^{\mathsf{N}} = Z_\beta[w^{\mathrm{ch}}]/Z_\beta \geq 0$ (equation YM3_I:(1759)); subadditivity,

$$Z_{\underline{\mathsf{n}}}[\mathrm{Fold}\, F] \leq Z_{\underline{\mathsf{n}}'}[\mathrm{Fold}\, F]^{\prod_\mu \mathsf{n}_\mu / \mathsf{n}'_\mu}$$

for boards with even numbers $\mathsf{n}'_\mu \leq \mathsf{n}_\mu$ of cells along the axes and one function F of a cell on both (Prop. YM3_I:86.5); and the volume (Cor. YM3_I:86.6): on tori of sides $\ell \geq \ell' \geq 2\ell_{\mathrm{c}}$, ℓ_{c} the side of the cells, with $p_\ell = \ell^{-3} \log Z_\beta$ on the torus of side ℓ, $p_\ell \leq p_{\ell'} \leq 0$ and $\|F\|_{\mathrm{ch},\ell} \leq \|F\|_{\mathrm{ch},\ell'} e^{\ell_{\mathrm{c}}^3 (p_{\ell'} - p_\ell)}$. The comparison of the Gibbs kernel of a box with the torus measure (Prop. YM3_I:91.8) holds for every such G, as it reads of the group only $0 \leq a_p \leq 2$. The Wilson measure on a torus and on the periodic configurations of the infinite lattice are one measure. Given $\mathsf{K} \geq 0$ in the three directions, reflection positivity and the two chessboard estimates hold for every real bounded even class function in the place of βa, and so do the statements that follow them, with $p_\ell \leq p_{\ell'}$, and $p_{\ell'} \leq 0$ for a non-negative function.

(iii) For representations ρ_i whose sum is the inclusion and all $\beta_i > 0$, every periodic Gibbs state of $\prod_p \exp[-\sum_i \beta_i (1 - \frac{1}{N_i} \mathrm{Re}\, \mathrm{tr}\, \rho_i(\partial U(p)))]$ is invariant under the reflections in bond mirrors and under the lattice isometries, and is reflection positive at the nine families of mirrors; for one block this holds also at $\beta = 0$; and $\nu(\mathrm{Im}\, \chi_\pi(\partial U(p))) = 0$ at every plaquette. The evenness $k(g^{-1}) = k(g)$ of the weight cannot be dropped.

(iv) Let G be compact, connected and semisimple, $w(V) = P[\frac{1}{2i}(V - V^*)]$, and W the holonomy of a plaquette $p \ni b$ based at b. Then $\sum_a \frac{d}{ds}\big|_{s=0} \langle t_a, w(e^{ist_a} W) \rangle = n - k_p$. With the isotypic decomposition $\mathbb{C}^N = \bigoplus_s W_s \otimes \mathbb{C}^{k_s}$, $d_s = \dim W_s$, Ω acting on W_s by ω_s and $w_s = 1 - \mathrm{Re}\, \chi_s / d_s$:

$a_p = \sum_s \frac{d_s k_s}{N} w_s(\partial U(p))$ and $k_p = \sum_s \omega_s \frac{d_s k_s}{N} w_s(\partial U(p))$; so k_p is a multiple of a_p if and only if the ω_s of the non-trivial constituents agree, and then $k_p = \frac{nN}{N-k_0} a_p$, k_0 the multiplicity of the trivial constituent, on which $w_s = 0$; under this condition, $k_p = n a_p$ if no constituent is trivial, as for an irreducible representation and for sums of copies of one, and $k_p = \frac{9}{2} a_p$ at $\{\mathrm{diag}(u,1) : u \in SU(2)\} \subset U(3)$. For $SU(2)$ on $\mathbb{C}^2 \oplus \mathbb{C}^3$, with $a_p = 1 - \frac{1}{5}\mathrm{Re}\,\mathrm{tr}\,\partial U(p)$, no multiple of a_p serves for k_p. With the weight $\Lambda = \mathrm{diag}(0.26, 0.26, 0.16, 0.16, 0.16)$ in the commutant, at first order of strong coupling, the integrated identity holds with the Casimir density, both sides being 0.206, and fails with $\frac{n}{2} a_p$ in the place of $\frac{1}{2} k_p$, the side that carries it becoming 0.2796 against 0.206. The same holds for an action with a weight Λ in the commutant: the bonds move by the vector part w, the action is differentiated by the gradient part, which is $P[N\Lambda(V - V^*)/2i]$ for the pairing $\langle \cdot, \cdot \rangle$, w itself at $\Lambda = \frac{1}{N}\mathbf{1}$, and $P[\Lambda(V - V^*)/2i]$ for the pairing $\mathrm{tr}\, XY$, and the flux carries the same Casimir density.

■

Sketch of proof. (i) We expand $e^{\frac{\beta}{2N}\mathrm{tr}(\hat{g}\hat{g}'^*)}$; every term is $f(g)\overline{f(g')}$ with a positive coefficient, which gives reflection positivity, and one term bounds λ_π; the square root of the weight for the convolution, $k(gh^{-1}) = \int_G q(gx^{-1}) q(xh^{-1})\, \mathrm{d}x$ with q a real continuous even class function whose coefficients are the $\lambda_\pi^{1/2}$, makes each plaquette cut by a mirror a square, $\partial U(p) = g_p(U) g_p(\theta U)^{-1}$; at a bond mirror the partition function is a trace of powers of $\mathsf{K}^{1/2}\mathsf{S}\mathsf{K}^{1/2}$, K holding the crossing plaquettes alone and S, the slab operator, the plaquettes inside a layer.

(ii) The partition function with insertions is the trace of a product in which the crossing operator alternates with the slab operators, which carry the plaquettes inside a layer and the insertions; the eigenvalues of the crossing operator are products of the $\lambda_\pi \geq 0$, so $\mathsf{K} \geq 0$, and Hölder's inequality for traces, $|\mathrm{Tr}(A_{2M} \cdots A_1)| \leq \prod_j \|A_j\|_{2M}$, gives the estimates; where the proofs of the chessboard estimates of the $SU(2)$ construction (Prop. YM3_I:86.3) use $\mathrm{tr}\, X^{-1} = \mathrm{tr}\, X$, false at $i \in U(1)$, they need only $a(X^{-1}) = a(X) = a(hXh^{-1})$, that is, that flipping a plaquette keeps its action. Reflections and translations keep action, observable and measure; folded integrals on two tori are traces of powers of one positive operator, so that $\mathrm{Tr}\, B^{\mathsf{p}} \leq (\mathrm{Tr}\, B^{\mathsf{p}'})^{\mathsf{p}/\mathsf{p}'}$ for $B \geq 0$ and integers $\mathsf{p} \geq \mathsf{p}' \geq 1$ gives subadditivity and the comparison of the volumes for every weight with $\mathsf{K} \geq 0$, and $Z_\beta \leq 1$, that is $p_{\ell'} \leq 0$, where the weight is at most 1; the proof of Prop. YM3_I:91.8 reads of the group only $0 \leq a_p \leq 2$, which holds on every $G \subset U(N)$; and the cell estimate also follows from the slab estimate in three directions, with no trace formula.

(iii) $\log(Z_\Gamma[t\mathbb{A}]/Z_\Gamma[0]) \geq t|\mathbb{A}|\nu(A)$ minus a surface term; the trace formula with insertions has modulus at most $\prod \|X\|$; at $X = e^{\pm\varepsilon \mathsf{D}}$, D real antisymmetric, $\nu(A_{\mathsf{D}}) = 0$, so neighbouring layers are exchangeable, and two layers fix the chain of layers because the kernel is injective on measures, all λ_π being positive. For the nine families, reflection positivity on adapted tori, Schwarz's inequality with a longest run and a separation argument identify every periodic Gibbs state with a reflection positive one.

(iv) The derivative of $\langle t_a, w(e^{ist_a} W) \rangle$ at 0 is $\frac{1}{N}\mathrm{Re}\,\mathrm{tr}(t_a^2 W)$; summing gives $\frac{1}{N}\mathrm{Re}\,\mathrm{tr}(\Omega W) = n - k_p$. The decomposition follows from Schur's lemma. At the trivial constituent $w_s = 0$; a constituent s and its conjugate s^* have $w_s = w_{s^*}$ and $\omega_s = \omega_{s^*}$; and the w_s of the non-trivial constituents, one for each pair $\{s, s^*\}$, are linearly independent functions on G, as 1 and the real parts of their characters are, by the orthogonality of characters. So $k_p = c a_p$ if and only if $\omega_s = c$ at every non-trivial constituent, and $\frac{1}{N}\mathrm{tr}\,\Omega = \sum_s \omega_s \frac{d_s k_s}{N} = n$ gives $c = \frac{nN}{N-k_0}$. For the action $1 - \frac{1}{2}\mathrm{tr}\,\Lambda(u + u^{-1})$ with Λ in the commutant, $\frac{d}{ds}\big|_{s=0}(1 - \frac{1}{2}\mathrm{tr}\,\Lambda(e^{isX} W + W^{-1} e^{-isX})) = \mathrm{tr}(X\Lambda H_W) = \langle X, P[N\Lambda H_W] \rangle$ for $X \in \mathfrak{g}$, as Λ commutes with W. □

Lemma 9.28 (The coefficient of β^{-3} in a box)**.** Let $G \subset U(N)$ be compact and connected, $n \geq 1$, semisimple or not, and consider the box $\{0, \ldots, s\}^3$ with trivial Dirichlet data, $\beta \geq 1$. Then $|\langle a_p; a_q \rangle - \beta^{-2} B_0^{[s]}(p,q) - \beta^{-3} B_1^{[s]}(p,q)| \leq C\beta^{-4}$, with C depending on the box, as in Prop. 7.12 for

$SU(2)$, and

$$B_1^{[s]}(G) = \frac{\varphi_G}{24}\, B_1^{[s]}(SU(2)) + c_G\, Y^{[s]}, \qquad c_G = \frac{1}{N}\operatorname{tr}\Omega^2 - \frac{3}{8}\varphi_G,$$

$\varphi_G = \sum_a \operatorname{tr}(\operatorname{ad} t_a)^2$ and $Y^{[s]}$ the coefficient for $U(1)$, $0 \le Y^{[s]}(p,q) \le C\langle x_q - x_p\rangle^{-6}$. As $s \to \infty$ the coefficient converges, the points being recentred at the centre of the box, and summed over orientations its part at distance r is $\frac{\varphi_G}{16\pi^3}\mathsf{r}^{-5}$ up to $C_G\mathsf{r}^{-6}(1 + |\log \mathsf{r}|)$. ■

Sketch of proof. Laplace's method about the gauge orbit of $\mathbf{1}$ in the box gives the expansion. The Wick pairings that decay like the fifth inverse power fall into six classes, and in each the structure constants enter through $\sum_{abc} f_{abc}^2 = \varphi_G$: the numeral 24 of $SU(2)$ becomes φ_G in five classes and 12 becomes $\frac{1}{2}\varphi_G$ in one. The representation enters through Ω only in terms that decay like the sixth power. The limit and the leading part are those of $SU(2)$ (Thm. V), whose constant $\frac{3}{2\pi^3}$ is multiplied by $\frac{\varphi_G}{24}$. □

Lemma 9.29 (Coefficients on the simple ideals)**.** Let $\mathfrak{g} = \bigoplus_{i\le r}\mathfrak{g}_i$ be semisimple. The mean of the n quadratic colour coefficients of a local term of second order is $\sum_i \frac{n_i}{n}\mathsf{cf}_i$, cf_i its coefficient on $\mathfrak{g}_i$, and $|\mathsf{cf}_i| \le \kappa_i^2 C_{\mathrm{am}} r_0^{-2} E_0 = (\kappa_i/\kappa_G)^2 \mathsf{G}_\sharp$ for the terms of Thm. YM3_I:D, with C_{am}, r_0 and $\mathsf{G}_\sharp$ of Prop. 9.24. If $|e_k^{(i)}| \le \Gamma$, the sequences $x_0^{(i)} = \beta$, $x_k^{(i)} = L^{-1}x_{k-1}^{(i)} - e_k^{(i)}$ and $x_k = \sum_i \frac{n_i}{n} x_k^{(i)}$ satisfy $x_k = L^{-1}x_{k-1} - \sum_i \frac{n_i}{n} e_k^{(i)}$ and

$$|x_k^{(i)} - x_k| \le 2\Big(1 - \frac{n_i}{n}\Big)\Gamma\frac{L}{L-1}(1 - L^{-k}).$$

■

Sketch of proof. The first claim is the splitting of an orthonormal basis along the ideals; the bound is that of the $SU(2)$ construction with $|t_a| \le \kappa_i$ on $\mathfrak{g}_i$ (Prop. 9.24). Further $x_k^{(i)} - L^{-k}\beta = -\sum_{s\le k} L^{-(k-s)} e_s^{(i)}$, so $|x_k^{(i)} - x_k^{(m)}| \le 2\Gamma\frac{L}{L-1}(1 - L^{-k})$, and $x_k^{(i)} - x_k = \sum_{m\ne i}\frac{n_m}{n}(x_k^{(i)} - x_k^{(m)})$. □

Proposition 9.30 (Semisimple Lie algebras and tied ideals)**.** Let $\mathfrak{g} = \bigoplus_{i\le r}\mathfrak{g}_i$ be semisimple, $r \ge 2$, and G_i the subgroup with Lie algebra $\mathfrak{g}_i$.

(i) The averages $E_j(M) = \int_{G_j} hMh^{-1}\,\mathrm{d}h$ and their products over $j \notin I$ are conditional expectations; so the thresholds of the $SU(2)$ construction pass down to the ideals with the constant 1, and back up with the factor r.

(ii) For $u_i = e^{iZ_i} \in G_i$, $a(u_1\cdots u_r) = \sum_i a(u_i) - t(u)$ with the tie $t = \frac{1}{4}\sum_{i<j}\tau_{ij}\langle Z_i, Z_i\rangle\langle Z_j, Z_j\rangle + O(Z^6)$, τ_{ij} of Prop. 9.18(iv); the tie vanishes exactly for block sums. In the Hessian of the action at $u = u_i v$, $v = \prod_{j\ne i} u_j$, the block of the ideal $\mathfrak{g}_i$ is $\frac{1}{N}\operatorname{Re}\operatorname{tr}(Sc(u)T)$, $S, T \in \mathfrak{g}_i$, with $c(u) = c(u_i)c(v) - H_{u_i}H_v$: it is the Hessian of the simple pair of $\mathfrak{g}_i$ with the weight $c(v)/N$, which is hermitian, lies in the commutant of G_i, is positive near $\mathbf{1}$ and reads the field v of the other factors, less the term $\frac{1}{N}\operatorname{Re}\operatorname{tr}(H_v S H_{u_i} T)$, at most $|S||T||u_i - \mathbf{1}||v - \mathbf{1}|$. This term vanishes when on each constituent $W_s \otimes M_s$ of $\mathbb{C}^N$, W_s irreducible under G_i and M_s its space of multiplicity under the other factors, one of W_s and M_s is self-conjugate, so at every ideal $SU(2)$ and at $SU(2)\times SU(3)$ on $\mathbb{C}^2\otimes\mathbb{C}^3$; at $SU(3)\times SU(3)$ on $\mathbb{C}^3\otimes\mathbb{C}^3$ it is not zero in general. The gradient on $\mathfrak{g}_i$ carries the like term $P_i[c(u_i)H_v]$ beside $P_i[c(v)H_{u_i}]$. Between two ideals stands one mixed block $\frac{1}{N}\operatorname{Re}\operatorname{tr}(SS'(c(u) - \mathbf{1}))$, $S \in \mathfrak{g}_i$, $S' \in \mathfrak{g}_j$, $i \ne j$, at most $|S||S'||u_i - \mathbf{1}||u_j - \mathbf{1}|$, which vanishes at $u = \mathbf{1}$ and need not vanish at a background (at $SU(2)\times SU(2)$ on $\mathbb{C}^2\otimes\mathbb{C}^2$, with $S = \sigma_3\otimes\mathbf{1}$, $S' = \mathbf{1}\otimes\sigma_3$ and $u = e^{it\sigma_3}\otimes e^{it\sigma_3}$, it is $-\sin^2 t$). The block averages on small fields and the exponential map are products over the factors, the density $j_G = \prod_i j_{G_i}\circ P_i$ is a product, and the quadratic part of the gauge-fixing term is a sum over the ideals; the gauge-fixing term $a(X)$ itself is $\sum_i a(X_i) - t(X)$, with the tie of fourth order; and the domain $\{|Y| < \pi\}$ of the chart, G^π with its indicator and the cut-off ν

are compared with those of the factors through (i), with the constant 1 down to the ideals and the factor r back up.

(iii) Let $F_i = P_i F$ be the part of the field strength on $\mathfrak{g}_i$, and $Q_i = \sum_p \frac{1}{2}\langle P_i H_{\partial U(p)}, P_i H_{\partial U(p)}\rangle$, the sum running over the bare plaquettes. By the transferred statements of the step on its lists, its potentials and their edge parts (Prop. 9.24), the leftover $\sum_i(\mathsf{cf}_i - \mathsf{cf})\frac{1}{2}\langle F_i, F_i\rangle$ of a level splits off at the level of the potentials, with size at most $\frac{7}{4}\alpha_0^2 \max_i |\mathsf{cf}_i - \mathsf{cf}| \le \frac{7}{2}(1 - \min_i \frac{n_i}{n})\alpha_0^2 \max_i |\mathsf{cf}_i|$, its edge parts $8M^3$ times this; its terms of first order are of third order in the field strength and are bounded by Cauchy's estimate whether or not they vanish; in dimension three the part of second order of an earlier level is bounded and is not taken out; the leftover of level j is $L^j \sum_i (e_j^{(i)} - b_j) Q_i$, $e_j^{(i)}$ the coefficient on $\mathfrak{g}_i$ and b_j their mean; and the leftovers of all levels add up to $-L^k \sum_i (x_k^{(i)} - x_k) Q_i$, which is bounded uniformly in k, as $L^k Q_i$ is of the order α_0^2 on a block of level k and $|x_k^{(i)} - x_k|$ is bounded by Lem. 9.29.

(iv) If the action of G has property (T) of Thm. AA(h3) at a choice of the constants, then at that choice one run, the mean, suffices: the r couplings $x_k^{(i)}$ stay within $2(1 - \frac{n_i}{n})\mathsf{G}_\sharp \frac{L}{L-1}$ of it (Lem. 9.29 with $\Gamma = \mathsf{G}_\sharp$, as $|e_k^{(i)}| \le (\kappa_i/\kappa_G)^2 \mathsf{G}_\sharp \le \mathsf{G}_\sharp$, $\kappa_i \le \kappa_G$ since $\mathfrak{g}_i \subset \mathfrak{g}$), and all clauses on the increments hold, the increment theorem, the regular progression, the sequence and the bounds on polydiscs among them; a format with one exponent, one integrand, one total and a carried term is the format of one run.

(v) At the least tied pair, $SU(2) \times SU(2)$ acting on $\mathbb{C}^4 \oplus \mathbb{C}^2$ by $(u_1, u_2) \mapsto (u_1 \otimes u_2) \oplus u_2$ with every weight, that is, with the action $\lambda_1(1 - \frac{1}{4}\mathrm{Re}\,\mathrm{tr}(u_1 \otimes u_2)) + (1 - \lambda_1)(1 - \frac{1}{2}\mathrm{Re}\,\mathrm{tr}\,u_2)$ for every $\lambda_1 \in (0, 1)$, one has $1 - a = (1 - a^{(2)})(1 - \lambda_1 a^{(1)})$, $a^{(i)} = 1 - \frac{1}{2}\mathrm{tr}\,u_i$; so the gradient and the Hessian of the action are the weighted ones of the $SU(2)$ construction, with the plaquette weights $\lambda_1(1 - a_p^{(2)})$ on the first factor and $1 - \lambda_1 a_p^{(1)}$ on the second, plus the block $-\lambda_1(\nabla a^{(1)} \otimes \nabla a^{(2)} + \nabla a^{(2)} \otimes \nabla a^{(1)})$, which annihilates infinitesimal gauge directions. The fluctuation operator of the first step is $A^\sharp(p_1) \oplus A^\sharp(p_2) + C^{\mathrm{T}} T C$, where p_1, p_2 are the points of the two factors, each with its plaquette weights, $A^\sharp$ is the fluctuation operator of the $SU(2)$ construction, C is the linear map from the fluctuation fields to the forms on the bonds through which the $SU(2)$ construction forms $A^\sharp$ from the Hessian of the action, and T, the Hessian of the action less the direct sum of the Hessians of the two factors, is symmetric, of the range of one plaquette, vanishes at $\mathbf{1}$ and at fields with values in one factor, and obeys $|(TA)(b)| \le 36300 \max_{p \ni b} |\partial U(p) - \mathbf{1}|^2 \, |A|_{(b)}$, $|A|_{(b)}$ the largest $|A|$ on the bonds of the plaquettes that hold b. To obtain the propagator conclusion one still has to establish at each level a regular domain on which the diagonal weighted operator $A_0 = A^\sharp(p_1) \oplus A^\sharp(p_2)$ has the required inverse and decay bounds, and $\|A_0^{-1} C^{\mathrm{T}} T C\| < 1$ in the corresponding forcing and decay norm, uniformly in the level. That relative bound would justify the Neumann inverse. The kinematic, interaction and Hessian conclusions require the corresponding regular-domain estimates; the algebraic identities alone do not establish them or property (T).

■

Sketch of proof. (i) E_j is the average of the conjugation over a compact group, so it is a positive unital projection onto the commutant of G_j, a conditional expectation, and does not increase $|\cdot|$, also at complex arguments; the E_j commute, and their product over $j \notin I$ is the average over the product of these G_j, a conditional expectation onto its commutant. On $\mathfrak{g}^c$, E_j is 0 on $\mathfrak{g}_j^c$, a vector of the simple ideal $\mathfrak{g}_j$ fixed by $\mathrm{Ad}(G_j)$ being 0 (Lem. 9.15(i)), and the identity on the other ideals, which commute with G_j; so $\prod_{j \ne i} E_j = P_i$ there, whence $|P_i X| \le |X|$ and $|X| \le \sum_i |P_i X| \le r \max_i |P_i X|$.

(ii) For the commuting Z_i, $1 - \frac{1}{N}\mathrm{Re}\,\mathrm{tr}(u_1 \cdots u_r) = 1 - \frac{1}{N}\mathrm{tr}\cos(\sum_i Z_i)$. Its terms of second order are those of the $a(u_i)$, as $\mathrm{tr}\, Z_i Z_j = 0$ for $i \ne j$; of its terms of fourth order those in which some Z_j stands

to the first power vanish, the trace of an element of a simple ideal being 0 on every invariant subspace, and the cross terms that remain are $-\frac{1}{4N}\sum_{i<j}\operatorname{tr}(Z_i^2Z_j^2)$, where $\frac{1}{N}\operatorname{tr}(Z_i^2Z_j^2)=\tau_{ij}\langle Z_i,Z_i\rangle\langle Z_j,Z_j\rangle$, an invariant quadratic form on a simple ideal being a multiple of $\langle\cdot,\cdot\rangle$. They vanish exactly when $Z_i^2Z_j^2=0$ for all $i\neq j$, a product of two commuting positive matrices with trace 0 being 0, that is, when the G_i act on separate blocks. At $u=u_iv$, $v=\prod_{j\neq i}u_j$, the elements u_i and v commute, so that $c(u_iv)=c(u_i)c(v)-H_{u_i}H_v$ and $H_{u_iv}=H_{u_i}c(v)+c(u_i)H_v$, and $c(v)$ and H_v commute with G_i and with $\mathfrak{g}_i$. So the block $\frac{1}{N}\operatorname{Re}\operatorname{tr}(Sc(u)T)$ of $\mathfrak{g}_i$, the second derivative along $S,T\in\mathfrak{g}_i$, is $\frac{1}{N}\operatorname{Re}\operatorname{tr}(c(v)\,Sc(u_i)T)$, the Hessian of the simple pair of $\mathfrak{g}_i$ with the weight $c(v)/N$, less $\frac{1}{N}\operatorname{Re}\operatorname{tr}(H_vSH_{u_i}T)$, whose modulus is at most $|S||T||H_{u_i}||H_v|\le|S||T||u_i-\mathbf{1}||v-\mathbf{1}|$, as $|H_w|\le|w-\mathbf{1}|$ for unitary w; and the gradient on $\mathfrak{g}_i$ is $P_iH_u=P_i[c(v)H_{u_i}]+P_i[c(u_i)H_v]$. On a constituent $W_s\otimes M_s$ the elements S, T and H_{u_i} act on W_s and H_v on M_s, with $\operatorname{tr}_{M_s}H_v=\operatorname{Im}\chi_{M_s}(v)$, and $\operatorname{Re}\operatorname{tr}(SHT)=\operatorname{tr}(\frac{1}{2}\{S,T\}H)$ for hermitian S, H, T; so the odd term is $\frac{1}{N}\sum_s\operatorname{tr}_{W_s}(\frac{1}{2}\{S,T\}H_{u_i})\operatorname{Im}\chi_{M_s}(v)$. An antiunitary J that commutes with the group on a self-conjugate side has $JSJ^{-1}=-S$ for the elements S of its Lie algebra and $JH_wJ^{-1}=-H_w$ for its elements w, and $\operatorname{tr}(JXJ^{-1})=\overline{\operatorname{tr}X}$; so the factor of that side, a real number, equals its negative and vanishes. Every representation of $SU(2)$ is self-conjugate, and at $SU(2)\times SU(3)$ on $\mathbb{C}^2\otimes\mathbb{C}^3$ the side of the other factor of the ideal of $SU(3)$ is the $\mathbb{C}^2$ of $SU(2)$; at $SU(3)\times SU(3)$ on $\mathbb{C}^3\otimes\mathbb{C}^3$, with $S=T=D$, $Z_i=\epsilon D$ and $v=e^{i\epsilon'D}$, $D=\operatorname{diag}(1,1,-2)$, the two factors are $\epsilon\operatorname{tr}D^3+O(\epsilon^3)$ and $-\frac{\epsilon'^3}{6}\operatorname{tr}D^3+O(\epsilon'^5)$, with $\operatorname{tr}D^3=-6$. Between two ideals the second derivative along $S\in\mathfrak{g}_i$, $S'\in\mathfrak{g}_j$ is $\frac{1}{N}\operatorname{Re}\operatorname{tr}(SS'u)=\frac{1}{N}\operatorname{Re}\operatorname{tr}(SS'(c(u)-\mathbf{1}))$, as $\operatorname{tr}SS'=0$; on a constituent of the factor i, the factor j and the rest, $\operatorname{tr}(SS'u)$ is $\operatorname{tr}(S(u_i-\mathbf{1}))\operatorname{tr}(S'(u_j-\mathbf{1}))$ times the trace of the rest, S and S' having trace 0 there, which gives the bound; and the example is a computation. The block averages on small fields and the exponential map are products over the factors, the mean of a family of products of commuting factors being the product of their means and the logarithm of a product of small commuting factors the sum of their logarithms; $j_G(Y)=\prod_i j_{G_i}(P_iY)$, the eigenvalues of $\operatorname{ad}Y$ being those of the $\operatorname{ad}P_iY$ on the $\mathfrak{g}_i$; the gauge-fixing term is a trace $a(X)$, to which the expansion above applies at complex fields, so that its quadratic part is a sum over the ideals; and the chart, G^π with its indicator and ν are compared with those of the factors by the bounds of (i).

(iii) The leftover is a sum over the ideals of the difference of their coefficient and the mean times their part $\frac{1}{2}\langle F_i,F_i\rangle$ of the Yang–Mills term. By the transferred statements of the step on its lists, its potentials and their edge parts (Prop. 9.24) it splits off as a named term of the potentials of each level, as the $SU(2)$ construction splits off a term of the potentials, with the size $\frac{7}{4}\alpha_0^2\max_i|\mathsf{cf}_i-\mathsf{cf}|$ and its edge parts $8M^3$ times this; and $\mathsf{cf}_i-\mathsf{cf}=\sum_m\frac{n_m}{n}(\mathsf{cf}_i-\mathsf{cf}_m)$ gives $|\mathsf{cf}_i-\mathsf{cf}|\le2(1-\frac{n_i}{n})\max_m|\mathsf{cf}_m|$, whence the second bound. Its terms of first order in the fluctuation field are of third order in the field strength, and Cauchy's estimate bounds them, as Schwarz's lemma bounds such terms in the proof of Prop. 9.24, whether or not they vanish. In dimension three the part of second order of an earlier level is bounded, as in the $SU(2)$ construction, and stays in the potentials. At level j the leftover is $L^j\sum_i(e_j^{(i)}-b_j)Q_i$; as $x_k^{(i)}-x_k=-\sum_{j\le k}L^{-(k-j)}(e_j^{(i)}-b_j)$, the leftovers of the levels up to k add up to $-L^k\sum_i(x_k^{(i)}-x_k)Q_i$, which is bounded uniformly in k, L^kQ_i being of the order α_0^2 on a block of level k and $|x_k^{(i)}-x_k|$ bounded by Lem. 9.29.

(iv) The proof of the $SU(2)$ construction meets the colour through one theorem on a whole class of families of terms, Thm. YM3_I:X, so a term of the tied action may be restricted to an ideal, and the leftover of (iii) is carried as a term; with property (T) the remaining steps are those of the $SU(2)$ construction, and Lem. 9.29 with $\Gamma=\mathsf{G}_\sharp$ gives the distance of the couplings, as $|e_k^{(i)}|=|\mathsf{cf}_i|\le(\kappa_i/\kappa_G)^2\mathsf{G}_\sharp\le\mathsf{G}_\sharp$ (Prop. 9.24), $\kappa_i\le\kappa_G$ since $\mathfrak{g}_i\subset\mathfrak{g}$. A format with one exponent, one integrand and one total, the leftover of (iii) being the carried term, is the format of the $SU(2)$ construction for one run.

(v) At this pair $1-a=(1-a^{(2)})(1-\lambda_1 a^{(1)})$, $a^{(1)}$ and $a^{(2)}$ being the plaquette functions of the two factors, so that $1-a$ is a product of two factors each of which reads one group; hence the gradient and the Hessian of the action are the weighted ones of the $SU(2)$ construction, one weight on each ideal, $\lambda_1(1-a_p^{(2)})$ and $1-\lambda_1 a_p^{(1)}$, plus the block $-\lambda_1(\nabla a^{(1)}\otimes\nabla a^{(2)}+\nabla a^{(2)}\otimes\nabla a^{(1)})$, which annihilates infinitesimal gauge directions, the $a_p^{(i)}$ being gauge invariant. The fluctuation operator of the first step is the direct sum $A^\sharp(p_1)\oplus A^\sharp(p_2)$ plus $C^{\mathrm T}TC$, T being this block: it is symmetric, of the range of one plaquette and of second order in the plaquettes, each gradient being of first order in its plaquette variable, and it vanishes at $\mathbf{1}$ and at fields with values in one factor, where the gradient of the factor whose field is $\mathbf{1}$ vanishes. If the regular-domain inverse hypotheses of (v) hold, $(A_0+C^{\mathrm T}TC)^{-1}=\sum_{j\ge0}(-A_0^{-1}C^{\mathrm T}TC)^jA_0^{-1}$ in the stated norm. A pointwise second-order bound on T does not establish this relative-smallness condition after the maps C, scale changes and gauge fixing, nor does it supply the all-level regular-domain estimates. These remain analytic inputs for the additional assertions of (v). □

Lemma 9.31 (Block sums)**.** Let $G=G_1\times\cdots\times G_r$ act on $\mathbb{C}^{N_1}\oplus\cdots\oplus\mathbb{C}^{N_r}$ by blocks, $G_i\neq\{\mathbf{1}\}$ closed.

(i) $a(u)=\sum_i\frac{N_i}{N}a_i(u_i)$ and the Haar measure is a product; so on every finite lattice the Wilson measure at β is the product of those of the G_i at $\beta N_i/N$, the fields of the factors are independent, and $\langle\mathcal{O}(f_1);\dots;\mathcal{O}(f_{\mathsf m})\rangle_\beta=\sum_i(\frac{N_i}{N})^{\mathsf m}\langle\mathcal{O}^{(i)}(f_1);\dots;\mathcal{O}^{(i)}(f_{\mathsf m})\rangle_{\beta N_i/N}$; with a weight $\Lambda=\bigoplus\Lambda_i$, $\frac{N_i}{N}$ is replaced by $\operatorname{tr}\Lambda_i$.

(ii) Let ν be a Gibbs state of the product weight. Then $\nu(\prod_i F_i(U^{(i)}))=\nu(\prod_i\bar F_i(U^{(i)}))$ for bounded Borel functions F_i, $\bar F_i$ any representative of the conditional expectation of F_i given the tail field $\mathcal{T}$ in the marginal ν_i: the factors are independent given the tails. For a Gibbs state ν_1 of one factor and $\mathsf m\ge1$ there is exactly one probability on $\mathsf m$ copies whose value at $F_1\otimes\cdots\otimes F_{\mathsf m}$ is $\nu_1(\prod_a\mathbb{E}_{\nu_1}[F_a|\mathcal{T}])$; it is a Gibbs state of the product weight of $\mathsf m$ copies at equal couplings, all its marginals are ν_1, and it is periodic if ν_1 is. Let the marginals ν_i of ν be periodic for the factors i of a set I', and for $i\in I'$ let $F_{\mathbf k}^{(i)}\in\mathcal{A}(\square_{\mathbf k}^{(i)})$, $\mathbf k$ in a finite set $\mathcal{K}_0^{(i)}$, be functions of the cells of a board of $\mathbb{L}_K$, each factor with its own board and its own sequence of sides $\ell_j^{(i)}\to\infty$. Then the chessboard domination of the $SU(2)$ construction (Prop. YM3_I:91.9) holds for their product:

$$\Big|\nu\Big(\prod_{i\in I'}\prod_{\mathbf k}F_{\mathbf k}^{(i)}(U^{(i)})\Big)\Big|\le\prod_{i\in I'}\liminf_{j\to\infty}\prod_{\mathbf k}\|F_{\mathbf k}^{(i)}\|^{(i)}_{\mathrm{ch},\ell_j^{(i)}},$$

$\|\cdot\|^{(i)}_{\mathrm{ch},\ell}$ being the chessboard seminorm of the Wilson measure of G_i at its coupling.

(iii) Let $G=SU(2)^r$ in $\mathbb{C}^2\oplus\cdots\oplus\mathbb{C}^2$, $0<g_i\le\gamma$, $(K;(\beta_i))$ a first-passage datum at (g_i) with the labels $\mathsf b_i=a_K\beta_i$, ν a Gibbs state whose marginals ν_i, ν_j are periodic, $i\neq j$, $\vartheta\in(0,1)$, $a_K\le\mathsf d\le\frac{1}{20}$, and (f,h) a test pair of scale $\mathsf d$ and parameter ϑ as defined in §6.1. Then

$$|\langle\mathcal{O}^{(i)}(f);\mathcal{O}^{(j)}(h)\rangle_\nu|\le\frac14\varepsilon_{ij}^{\times}(\mathsf d)\,\frac{3}{16\pi^2}(\mathsf b_i\mathsf b_j)^{-1}\mathcal{K}_6(f,h),$$

with $\varepsilon_{ij}^{\times}(\mathsf d)$ at most a number of g_i, g_j, ϑ and the constants times $\mathsf d$, reading none of K, β_i, ν, f, h. In every Gibbs state of the product weight whose marginals are periodic in all the factors $i_1,\dots,i_{\mathsf m}$ of a mixed cumulant $\langle\mathcal{O}^{(i_1)}(f_1);\dots;\mathcal{O}^{(i_{\mathsf m})}(f_{\mathsf m})\rangle$, $\mathsf m\ge2$, the i_q not all equal, this cumulant obeys the per-ball and factorial growth estimates obtained by the normalizer argument in the proof of Prop. YM3_I:91.13, with constants depending on the factor ceilings and the radii of the balls containing the supports. No separation of the supports is required: the mixed derivative of every single-factor normalizing exponent is zero, as checked below. ■

Sketch of proof. (i) is the product structure; cumulants of sums of independent families add, and $\mathcal{O} = \sum_i \frac{N_i}{N} \mathcal{O}^{(i)}$ gives the powers.

(ii) The Gibbs kernels of the product weight are products of those of the factors, and the marginal ν_i is a Gibbs state of the factor i. For finite sets of bonds $B_1 \subset B_2 \subset \cdots$ that exhaust the lattice, $\nu(\prod_i F_i) = \nu(\prod_i \gamma^{(i)}_{B_l} F_i)$, $\gamma^{(i)}_{B_l}$ the Gibbs kernel of the factor i; $\gamma^{(i)}_{B_l} F_i$ is the conditional expectation of F_i in ν_i given the bonds off B_l, these σ-fields decrease to $\mathcal{T}$, and the downward martingale theorem gives the convergence to $\bar{F}_i$ off a ν_i-null set; dominated convergence gives the first statement. The space of configurations being compact and metrisable, ν_1 has a regular conditional probability π given $\mathcal{T}$, and the m-copy state is $\int \nu_1(\,\mathrm{d}\omega)\, \pi(\omega,\cdot)^{\otimes \mathsf{m}}$: its values at products fix it; it obeys the Gibbs equations because $\mathcal{T}$ lies in the σ-field of the bonds off every finite set; and a translation that keeps ν_1 keeps $\mathcal{T}$ and so keeps it. The domination is the proof of Prop. YM3_I:91.9 run with the factors, first for a periodic Gibbs state of the product weight and functions that are products over the factors: the comparison of the Gibbs kernel of a box with the torus measure (Prop. YM3_I:91.8) holds factor by factor, by Fubini, the exponents adding, and reads of a group only $0 \le a_p \le 2$; the master inequality (the proof of Prop. YM3_I:91.9) holds with the diagonal translations, the geometry of boards (the proof of Prop. YM3_I:91.9) reading no group; and after the expansion of the product of the translates each term is a product over the factors of functions of one factor alone, whose mean under the product of the torus measures is the product of the means, each bounded by the chessboard estimate of its factor (Prop. 9.27(ii)). Periodic marginals suffice: with $\bar{F}^{(i)}$ the conditional expectation given $\mathcal{T}$ of the product $F^{(i)}$ of the functions of the factor i, the first statement and Hölder's inequality give $|\nu(\prod_i F^{(i)})| \le \prod_{i \in I'} \nu_i(|\bar{F}^{(i)}|^{\mathsf{m}})^{1/\mathsf{m}}$, m the least even integer at least $\#I'$; and $\nu_i(|\bar{F}^{(i)}|^{\mathsf{m}})$ is the value of the m-copy state of ν_i, a periodic Gibbs state of the product weight of m copies, at the product of $\frac{\mathsf{m}}{2}$ copies of $F^{(i)}$ and $\frac{\mathsf{m}}{2}$ copies of its conjugate, which the domination for the product weight bounds, the conjugate having the seminorm of the function.

(iii) By (ii) the mixed covariance is the covariance of the conditional expectations given the tails, so that $|\langle \mathcal{O}^{(i)}(f); \mathcal{O}^{(j)}(h)\rangle_\nu|^2 \le \mathsf{V}_{\nu_i}(f)\, \mathsf{V}_{\nu_j}(h)$, $\mathsf{V}_{\nu_i}(f)$ the variance of $\mathbb{E}_{\nu_i}[\mathcal{O}(f)|\mathcal{T}]$ in ν_i. Let $\mathsf{D}(f_1, f_2)$ be the covariance of $\mathbb{E}_{\nu_i}[\mathcal{O}(f_1)|\mathcal{T}]$ and $\mathbb{E}_{\nu_i}[\mathcal{O}(f_2)|\mathcal{T}]$ in ν_i, a real bilinear form; it is the mixed covariance of the two copies in the two-copy state of ν_i. *The unit scale.* Let F_1, F_2 vanish off $\mathbb{B}(0,1)$. The systems of the $SU(2)$ construction give at the first-passage datum (K, β_i) the normalisers (Nr1) of the one-entry tuples on the board of side L (equation YM3_I:(1844), Prop. YM3_I:91.12 (1)); they pass to the two-copy state, on the product of the polydiscs of the two copies, by the domination of (ii), as in the transfer principle (Prop. YM3_I:91.11); at this mixed word the derivative of the sum of the two normalising exponents vanishes, and Cauchy's estimate gives $|\mathsf{D}(F_1, F_2)| \le C'_V(g_i) \|F_1\|_\sharp \|F_2\|_\sharp$, with $C'_V(g)$ a number of g and the constants and $\|F\|_\sharp = \max\{\sup|F|, 40M \operatorname{Lip} F\}$. *The factor of the radius.* ν_i is invariant under all translations of $a_K\mathbb{Z}^3$ (Prop. 9.27(iii)), so $\mathsf{D}(\tau_v f_1, \tau_v f_2) = \mathsf{D}(f_1, f_2)$ for $v \in a_K\mathbb{Z}^3$. Let f_1, f_2 vanish off a ball $\mathbb{B}(x_0, \mathsf{r})$, $a_K \le \mathsf{r} \le \frac{1}{20}$. There is a packing of at least $(0.3/\mathsf{r})^3$ translates of this ball in $\mathbb{B}(0,1)$ by vectors of $a_K\mathbb{Z}^3$ (the packing argument in the proof of Prop. YM3_I:127.1); the signed sums $\sum_v \epsilon_v \tau_v f_l$ over the packing, $\epsilon_v = \pm 1$, have the norms $\|f_l\|_\sharp$, and the mean over the signs of D at these sums is the sum over the packing of the $\mathsf{D}(\tau_v f_1, \tau_v f_2)$, the cross terms cancelling (the packing argument in the proof of Prop. YM3_I:127.1). So $|\mathsf{D}(f_1, f_2)| \le C'_V(g_i)(\mathsf{r}/0.3)^3 \|f_1\|_\sharp \|f_2\|_\sharp$, and $\mathsf{V}_{\nu_i}(f) \le C'_V(g_i)(\mathsf{r}/0.3)^3 \|f\|_\sharp^2$. *The test pairs.* The two functions of a test pair of scale d and parameter ϑ vanish off balls of radius d, so $\mathsf{r} = \mathsf{d}$; as $\vartheta < 1$, $\mathsf{d} \le \frac{1}{20}$ and $M \ge 1$, $\|f\|_\sharp \le \frac{40M}{\vartheta \mathsf{d}} \|f\|_\infty$, and the same for h; $\|f\|_\infty \|h\|_\infty \le 10^6 \vartheta^{-2} \mathcal{K}_6(f,h)$ and $\mathsf{b}_i \le L(g_i^{-2} + \bar{B})$; this gives the bound, with $\varepsilon^{\times}_{ij}(\mathsf{d})$ a number of g_i, g_j, ϑ and the constants times d. *The higher cumulants.* For each factor of the word, the graded normaliser that the systems give to the sub-tuple of its entries (equation eq. 5.xxviii), with no separation of the supports, passes to the state by the domination of (ii), on the product of the polydiscs of the factors; at a mixed word the derivative of the sum of the exponents vanishes, each exponent reading the entries of its own factor alone, so that Cauchy's estimate bounds the cumulant itself, and the per-ball bound and the growth

bound follow from the placement and tent-decomposition calculations in the proof of Prop. YM3_I: 91.13. The separation-dependent contribution there is the mixed derivative of the normalizing exponent; here that contribution vanishes because every exponent depends on the variables of only one factor. The remaining Cauchy and counting estimates require no separation. □

Lemma 9.32 (Change of units and one-step normalization)**.**

(i) Remove the last level of a regional sequence, and denote its length convention and determining set by $\eta^- = L\eta$ and $\mathbf{B}^-$. For a real form A^- put $A_0 = LA^-$. Then

$$\eta A_0 = \eta^- A^-, \qquad \|A^-\|_{X,\mathbf{B}^-} \le \|A_0\|_{X,\mathbf{B}} \le L^2 \|A^-\|_{X,\mathbf{B}^-}.$$

This holds for every closed matrix group and every background.

(ii) Let $k \ge 1$ and suppose all regions through level k are the whole torus. Let U_0, U' be regular backgrounds with parameter α_0, and let (A_0, g_0) be a real chart pair, $U'^{g_0} = e^{i\eta A_0} U_0$, normalized on the preceding top: $g_0 = v_{k-1}[A_0]$ on the level-k sites. Assume the gauge-correction estimate of Lem. YM3_I:30.4 and the slice estimate of Lem. YM3_I:30.10 hold at these backgrounds, with their admitted radii and constants, and assume the one-step factor bound

$$\sup_{\Lambda_k} |\mathsf{v}_{k-1}[A_0] - \mathbf{1}| \le 2\kappa_6 L^{-1} \|A_0\|_{X,\mathbf{B}}.$$

Here the bounds may be the transferred bounds when a transfer is explicitly supplied. In particular this statement does not assert an unconditional transfer to an arbitrary group. If α_0 and $\|A_0\|_{X,\mathbf{B}}$ are below the thresholds specified in the proof, and

$$\sup_c |\mathcal{M}^k U'(c) \mathcal{M}^k U_0(c)^{-1} - \mathbf{1}| \le \gamma,$$

there is a new chart pair (A, g) with

$$\mathsf{R} D^* A = 0, \qquad g = v_k[A] \text{ on } \Lambda_k, \qquad \sup_c |a_k[A](c)| \le \tfrac{\pi}{2}\gamma,$$

and

$$|A|_{(-1)} \le k_1 \|A_0\|_{X,\mathbf{B}}, \qquad \|A\|_{X,\mathbf{B}} + |D^* D A|_{(-3)} \le C_{\mathrm{sl}}(\alpha_0 + \gamma).$$

The constants and the thresholds depend only on those of the stated inputs and on L.

■

Proof. (i) If $j(b)$ is the level of a bare bond, its weighted field values in the two conventions are

$$L^{j(b)-k} L |A^-(b)|, \qquad L^{\min\{j(b),k-1\}-(k-1)} |A^-(b)|.$$

They agree unless $j(b) = k$, when their ratio is L. A covariant difference gains two factors of L, one from the form and one from the inverse bare spacing. Its weighted values therefore agree below level k and have ratio L^2 at level k. The background transports are identical. Taking the maxima gives both inequalities. In particular, on the constant sequence this computation is independent of the sector.

(ii) Write $\alpha_3 = \|A_0\|_{X,\mathbf{B}}$ and prescribe $r = \mathsf{v}_{k-1}[A_0]$ on the top sites. Let a_{gc} be the radius admitted by gauge correction and a_{sl} the field radius admitted by the slice estimate, decreased to stay in the averaging chart. Set

$$k_1 = 1 + 2C_{\mathrm{gc}} K_3 (3 + 2\kappa_6 L^{-1}).$$

Require α_0 below both input regularity ceilings and require

$$\alpha_3 \le a_{\mathrm{gc}}, \qquad 2\kappa_6 L^{-1} \alpha_3 \le a_{\mathrm{gc}}, \qquad k_1 \alpha_3 \le a_{\mathrm{sl}}.$$

Gauge correction gives $w = e^{i\lambda}$ with $\mathsf{R} D^* A_0^w = 0$ and induced top factor r. Since $|D^* A_0|_{(-2)} \le 3\alpha_3$, its bounds give

$$\|\lambda\|_Y \le K_3(3 + 2\kappa_6 L^{-1})\alpha_3, \qquad |A_0^w|_{(-1)} \le k_1 \alpha_3.$$

Put $A = A_0^w$ and $g = wg_0$. The initial residual factor at a level-k site is $v_k[A_0]^{-1}g_0 = \mathsf{v}_{k-1}[A_0]^{-1}$. Under the correction it is multiplied on the left by the prescribed r, so it becomes $\mathbf{1}$. Thus $g = v_k[A]$ on the top.

Every retained row on the constant sequence is an interior row at level k. The normalized factorization of the actual average gives

$$e^{ia_k[A](c)}\mathcal{M}^k U_0(c) = \mathcal{M}^k U'(c).$$

For a unitary matrix on the intrinsic branch, $|\log V| \le (\pi/2)|V - \mathbf{1}|$, by the scalar spectral formula; hence $\sup|a_k[A]| \le \pi\gamma/2$. Gauge invariance preserves regularity of $e^{i\eta A}U_0 = U'^g$. The slice estimate now applies with row bound $\pi\gamma/2$, proving the two final bounds after increasing C_{sl}. For the $SU(2)$ constants of Lem. YM3_I:30.10, one may take $C_{\mathrm{sl}} = 20\pi K_1$. This proof uses only the stated local bounds and group identities, so the same calculation applies whenever those bounds have been supplied in another sector. □

Proposition 9.33 (Quotients)**.** Let $G = \tilde{G}/\mathsf{Z}$ with $\tilde{G}$ simply connected and Z finite and central, and $L \ge 13$ odd. For $g = e^{iY} \in G^\pi$, $Y \in \mathfrak{g}$, $|Y| < \pi$, the *canonical lift* $s(g) \in \tilde{G}$ is $e^{i\tilde{Y}}$, $\tilde{Y}$ the element of the Lie algebra of $\tilde{G}$ that the covering carries to Y; for a lift $\tilde{U}$ to $\tilde{G}$ of a field U whose plaquette variables lie in G^π, the *central part* of $\tilde{U}$ at a plaquette p is $z_{\tilde{U}}(p) = \partial\tilde{U}(p)\, s(\partial U(p))^{-1} \in \mathsf{Z}$.

(i) The sector of Lem. 9.11(iv) is defined, gauge invariant and locally constant on the fields with small plaquettes, and kept by blocking. Let a separated sequence of regions have the whole torus at the bottom. Write the finite sequence as $\Omega_0, \dots, \Omega_k$, with $\Omega_{-1} = \Omega_0$ and $\Omega_{k+1} = \varnothing$. In bare units its members Ω_q are unions of cubes of side $\mathfrak{M}L^q$, with $\mathfrak{M}$ and R_1 powers of L, $R_1 \ge L$, $R_1\mathfrak{M} \ge L^2$, and $\mathrm{dist}_\infty(\Omega_q^c, \Omega_{q+1}) > R_1\mathfrak{M}L^q$. Let U be a field with $|\partial U(p) - \mathbf{1}| < \varepsilon L^{-2\mathrm{j}(p)}$ at every bare plaquette, $\mathrm{j}(p)$ the level of the sequence at p, where $L^2\varepsilon \le c(G)$ of Lem. 9.11(iii), and let V be its block datum. For a coordinate plane P through a site of the top lattice, the *plane number* of U at P is the product of the central parts $z_{\tilde{U}}(p)$ over the bare plaquettes p of P, for a lift $\tilde{U}$ of U; the *cells* S of P are defined as follows. Write $Q_r(p)$ for the square of side L^r in P containing the bare plaquette p, with corners on the level-r lattice. Its cell is $Q_q(p)$ for the largest $q \le k$ at which a corner lies in Ω_q. These cells tile P. A side of a level-q cell having an end in Ω_q is one determining bond of level q; a side with both ends outside Ω_q consists of L determining bonds of level $q-1$. Call these bonds the elements of the cell. Thus each cell has at most $4L$ elements, and $\partial_S V$ is their ordered product, with the signs of a walk around the boundary of S. Then every $\partial_S V$ lies in G^π; for any lifts $\hat{V}$ of the values of V, $z_{\hat{V}}(S) := \partial_S\hat{V}\, s(\partial_S V)^{-1}$ lies in Z; and the plane number of U at P is $\prod_S z_{\hat{V}}(S)$ over the cells S of P. So the plane numbers depend on V alone, and two such fields over one datum lie in one sector; over $V = e^{iB}$ with $4L\sup|B| < \pi$ the sector is trivial.

(ii) Let $G \subset U(N)$ be closed and let the averaging map agree with the formula of Lem. 9.11(i) on small fields. Let the sequence have the whole torus at the bottom and satisfy the following stronger separation conditions in bare units: Ω_q is a union of cubes of side L^{q+1}, and $\mathrm{dist}_\infty(\Omega_q^c, \Omega_{q+1}) > L^{q+2}$. Empty and full members are allowed. Then a field U with $|\partial U(p) - \mathbf{1}| < \varepsilon L^{-2\mathrm{j}(p)}$ at every bare plaquette, $\mathrm{j}(p)$ the level of the sequence at p, and block datum $\mathbf{1}$, $\varepsilon \le \varepsilon_2 = \min\{\varepsilon_2(L), \frac{1}{106}r_G\}$, has a gauge transform, by a gauge function equal to $\mathbf{1}$ at the tops and within $C\varepsilon$ of $\mathbf{1}$ at the nodes, of the form e^{ia} with $|a(b)| \le C\varepsilon L^{-\mathrm{j}(b)}$ and $\sum_b L^{-2\mathrm{j}(b)}|a(b)|^2 \le C\sum_p |\partial U(p) - \mathbf{1}|^2$, C a number of L; and a flat field with block datum $\mathbf{1}$ is a pure gauge. So these statements on the whole torus hold as worded; their hypothesis, block datum $\mathbf{1}$, selects the trivial sector.

(iii) Suppose that all regions are the whole torus. For every closed $G \subset U(N)$, the preliminary gauge, the change of level (Lem. 9.32(i)) and the section of a datum (Lem. YM3_I:36.3) hold in every sector. More precisely, two configurations in $\mathcal{U}'(\alpha_0)$ whose terminal data differ

by at most α_1 admit a gauge equal to $\mathbf{1}$ at the top sites that identifies their averaged contour variables and obeys

$$|\mathcal{M}^n(U^{\hat{g}})(b)\mathcal{M}^n U_0(b)^{-1} - \mathbf{1}| \le C_{\mathrm{pg}}(\alpha_0 + \alpha_1)$$

at every level. Here C_{pg} depends only on L, and the ceiling on α_0 is $c(G)$ of Lem. 9.11(iii).

(iv) Let $G = SO(3) \subset U(3)$, all regions being the whole torus, and let U_0 be a background of the regular class $\mathfrak{U}'_k(\mathbf{B}, \alpha_0)$ of the $SU(2)$ construction, in any sector. There are two numbers of L, a ceiling on α_0 and a floor $6LM$ on the number of sites of the top lattice along each axis, $M \ge M_{\mathrm{ps}}$ a power of L, in the place of the floor $6LM_{\mathrm{ps}}$ of the $SU(2)$ construction (Thm. YM3_I:I), such that under them the operator $\Delta^{\mathrm{W}}_{U_0} + D\mathsf{R}D^* + \mathsf{Q}^*\mathsf{Q}$ of the $SU(2)$ construction is formed on the whole torus and is positive definite, and its inverse G_0, the operator $\mathsf{H}_0 = \mathsf{G}_0\mathsf{Q}^*(\mathsf{Q}\mathsf{G}_0\mathsf{Q}^*)^{-1}$ and $\mathsf{G}_0 - \mathsf{H}_0\mathsf{Q}\mathsf{G}_0$ obey the bounds of the $SU(2)$ construction with numbers of L in the places of its constants.

The clauses of the $SU(2)$ construction that read the background on pieces of bounded size hold for $SO(3)$ in every sector. A bound of such a clause that is of degree d in sizes of forms and of degree e in chords of values in the group, chords at most α, $\alpha \in (0, 1]$ the parameter of the regular class $\mathfrak{U}'_k(\mathbf{B}, \alpha)$ of the background, keeps its number C of the $SU(2)$ construction where $d + e \ge 2$, where $d = 1$ and $e = 0$ at real forms, and, for a linear map whose kernel is real, in bounds on the norms of its kernel and on their sums along rows, and takes $(1 + 0.04\alpha)C$ where $d = 0$ and $e = 1$, as the constant of the gauge on a cube in the improvement of the regularity radius does, at most $\sqrt{2}C$ in the other bounds with $d = 1$ and $e = 0$ at complex forms, and $2C$ where $d = e = 0$, while a bound on the norm from suprema to suprema of an operator on complex forms takes a factor at most $\sqrt{2}$.

The constant $C_{4,\mathrm{ec}} = 808$ of the $SU(2)$ normalization in the bounds of the action of Prop. YM3_I:30.1 gives way to an absolute number, which serves at real and at complex forms and at every scale of that clause, a scale being a number $\xi \ge \eta$ at which the plaquette variables at the bond meet the condition of the clause. The comparison of the two regular classes at complex fields holds with $r + 5r^2$ in the place of $r + 4r^2$ at the fields $e^{i\eta A}U_0$ with $\eta \sup |A| \le \frac{1}{8}$.

For an averaging map that is covariant at every field and the formula of Lem. 9.11(i) only below a smaller radius, the same holds with three ceilings, each the least of the number of the $SU(2)$ construction and a fixed multiple of that radius ϖ: on α_0, the ceiling $L^{-2}c_2^{\mathrm{g}}$ of the global chart with $\frac{1}{4}L^{-2}\varpi$; on the size $|A|_{(-1)}$ of the potential A of a chart pair (A, g) of the global chart, the ceiling $L^{-1}c_4$, c_4 the number of the local chart estimates, with $\frac{1}{32}L^{-1}\varpi$; and on the radius of the one-step chart maps, the maps $a \mapsto \xi_\star(W_0, a)$, $\frac{1}{i}\log \mathsf{v}_\star(W_0, a)$ and $\mathcal{Q}^1_\star(W_0, a)$ of the one-step chart, the chart radius r_L with $\frac{1}{8}\varpi$. The backgrounds of these maps then lie within $\frac{1}{2}\varpi$ of $\mathbf{1}$, and the fields of a step within ϖ, in the way in which Lem. 9.11(v) keeps them within $\min\{\omega_M, r_G\}$.

With these ceilings, the gauge correction (Lem. YM3_I:30.4) holds in every sector with constants K_3 and C_{gc} that are numbers of L; and one step (Lem. 9.32(ii)) and one step of the ascent (Prop. YM3_I:37.2) hold in every sector, with constants of L in the places of the numerals of the $SU(2)$ construction, the improvement of the regularity radius (Prop. YM3_I:37.1) holding with free constants. In particular, over every datum whose plaquette variables lie within ε' of $\mathbf{1}$ and which lies within ε' of the averages of a configuration of the regular class one level down, ε' below a number of L, there is a critical configuration of the regular class one level up. The sector does enter objects on the whole torus: at a flat $SO(3)$-field the covariant difference on $\mathfrak{g}$-valued site functions has a non-zero kernel in the trivial sector and none in a twisted one.

■

Sketch of proof. (i) The squares $Q_r(p)$ are nested, so the largest-level rule assigns the same cell to every bare plaquette of that cell. A level-q cell has a corner in Ω_q and no corner in Ω_{q+1}. The block structure then puts no level-q site of the cell in Ω_{q+1}. Separation places the cell and the double blocks of its boundary elements in Ω_{q-1}: their distance from that corner is at most $\frac{3}{2}L^q < L^{q+1}$. On a side with no end in Ω_q, the level-$(q-1)$ sites are outside Ω_q, so the L subdivisions are indeed determining bonds.

An element of level r has an end in Ω_r and neither end in Ω_{r+1}. The cell across it has level at least r and at most $r+1$: a cell of level $q' \geq r+2$ would put an endpoint within $L^{q'}$ of $\Omega_{q'}$, hence in $\Omega_{q'-1} \subset \Omega_{r+1}$, a contradiction. At level r the element is a whole side; at level $r+1$ it is one of the L subdivisions of a side. Consequently each element belongs to exactly two cell boundaries, with opposite orientations.

If $\sum_j |X_j| < \pi$ in the norm of $\mathbb{C}^N$, $X_j \in \mathfrak{g}$, then $\prod_j e^{i\tilde{X}_j}$ is the canonical lift of $\prod_j e^{iX_j}$, and a product of small exponentials that lies in Z is $\mathbf{1}$. By Lem. 9.11(iii) in its local form, under $L^2\varepsilon \leq c(G)$, the value of the datum at a bond is the transport of U along the straight bare path of the bond times a product of exponentials of elements of $\mathfrak{g}$ of small total size; the paths of the sides of a cell S compose its boundary, so that $\partial_S V$ is the transport of U around S, a product of conjugated plaquette variables of U in S, times such a product, of total size below π. So $\partial_S V \in G^\pi$, and, the values of V being lifted along the lifted paths, $\partial_S \hat{V} = \prod_{p \subset S} z_{\tilde{U}}(p) \cdot s(\partial_S V)$. Another choice of the lifts of the values of V changes the $z_{\hat{V}}(S)$ by central factors that cancel in the product over the cells of P, each bond of the datum in P bounding two cells with opposite orientations; and the product over the cells of the central parts of their plaquettes is the plane number. The plane numbers, with the cube numbers, which are $\mathbf{1}$ for small plaquettes, fix the sector. Over $V = e^{iB}$ the product around a cell has at most $4L$ factors $e^{\pm iB(e)}$, and their lifts $e^{\pm i\tilde{B}(e)}$ give $z_{\hat{V}}(S) = \mathbf{1}$.

(ii) We construct the gauge by descending through the layers. The interpolation used in the descent is needed both for the pointwise bound and for the energy estimate; setting every contour variable equal to the identity would not give the latter estimate.

Write n for the last level, put $\Omega_{-1} = T_\ell$ and $\Omega_{n+1} = \emptyset$, and set

$$j(x) = \max\{q : x \in \Omega_q\}, \qquad j(e) = \max_{x \text{ a corner of } e} j(x).$$

Here e may be a bond or a plaquette of any level. Let $\Lambda_q = \mathbb{T}^{(q)} \cap (\Omega_q \setminus \Omega_{q+1})$; these are the tops. The datum prescribes the averages on the level-q bonds with an endpoint in Λ_q, with any of the inward bonds omitted as permitted by the given bond set. In particular it prescribes all such bonds with no endpoint in Ω_{q+1}. A node is an endpoint of a retained bond. All bond and plaquette sums below use positive orientations.

We first record the geometric consequences of the stated separation. The blocks $B^q(y)$, $y \in \Lambda_q$, partition the bare sites, and $j = q$ throughout each such block. Indeed, membership in Ω_q, a union of cubes of side L^{q+1}, is constant on each level-q block; assign a site to the block of the largest level whose region contains it. Furthermore,

$$|x - x'|_\infty \leq L^{j(x)+1} \quad \Longrightarrow \quad j(x') \geq j(x) - 1. \tag{9.xiii}$$

For $j(x) \geq 1$ this follows from the distance between $\Omega_{j(x)}$ and $\Omega^c_{j(x)-1}$; for $j(x) = 0$ it is automatic. A node which is not a top is the inner endpoint of a retained inward bond whose other endpoint is a top.

For level i, let $\mathcal{B}_i$ and $\mathcal{P}_i$ be the bonds and plaquettes with all corners in Ω_{i-1}, and put

$$V_i = \mathcal{M}^i U|_{\mathcal{B}_i}, \qquad u_i(p) = |\partial V_i(p) - \mathbf{1}|.$$

The bare cone $Y_i(p)$ of a plaquette is the union of the level-i blocks of its four corners. The double block beneath a bond of $\mathcal{B}_{i+1}$ lies in Ω_i, so each averaging step below is local to the indicated sets.

The two estimates for an averaging step. We use constants depending only on L. If v is the largest plaquette distance on the four blocks of a coarse plaquette P, then, for sufficiently small v,

$$u_{i+1}(P) \le (\mathsf{W}_i u_i)(P) + C_0 v^2, \qquad (\mathsf{W}_i f)(P) = L^{-3} \sum_{x \in B(P_-)} \sum_{p \in S_x(P)} f(p). \tag{9.xiv}$$

Here P_- is the base point of P, and $S_x(P)$ is the $L \times L$ square of fine plaquettes parallel to P with base point x. The kernel is nonnegative and has row sum L^2 and column sum L^{-1}; consequently $\|\mathsf{W}_i\|_{2\to 2} \le L^{1/2}$.

For completeness, the matrix calculation behind this estimate is the one-step calculation of Lem. YM3_I:14.9. Put the field in the comb gauge on the four blocks. Each resulting bond is a product of at most C_L conjugated plaquette variables, so its logarithm has size at most $C_L v$. Differentiating the mean formula at the identity gives

$$(Qa)(c) = L^{-3} \sum_{x \in B(c_-)} \big[\bar a(c_-, x) + a([x, x']) - \bar a(c_+, x')\big], \qquad x' = x + c_+ - c_-,$$

where $\bar a(y, x)$ is the arithmetic mean of the line sums along the coordinate-order contours. On going around P, the contour terms at its four corners cancel. The remaining sum is $L^{-3} \sum_x \sum_{p \in S_x(P)} da(p)$. The product-exponential estimate and the quadratic Taylor remainder of the mean give $C_L v^2$ for the discarded terms. This proves eq. 9.xiv. All estimates are matrix norm estimates. Each logarithm is intrinsic to G: the comb products and the contour quotients are products of small exponentials in $i\mathfrak{g}$, and Lems. 9.9(ii), (iv) and 9.11(i) apply. Thus the argument uses no $SU(2)$ identity. Counting L^3 translated squares, each with L^2 plaquettes, gives the row sum; a fixed fine plaquette has total multiplicity L^2 in these squares as the coarse plaquette varies, giving the column sum L^{-1}.

These facts imply the two multiscale bounds

$$u_i(p) \le 2\varepsilon L^{2(i-k)} \qquad \text{if } Y_i(p) \subset \Omega_k, \quad k \ge \max\{i-1, 0\}, \tag{9.xv}$$

$$\sum_{p \in \mathcal{P}} u_i(p)^2 \le 2L^i \sum_{p_0 \subset \bigcup_{p \in \mathcal{P}} Y_i(p)} u_0(p_0)^2 \qquad (\mathcal{P} \subset \mathcal{P}_i). \tag{9.xvi}$$

The bare plaquettes on the right are counted once. To see the first bound, take maxima P_r on the nested cones beneath p. Equation eq. 9.xiv gives $P_{r+1} \le L^2 P_r + C_0 P_r^2$. Write $P_r = \varepsilon L^{2(r-k)} p_r$. Starting with $p_0 \le 1$, the induction $p_r \le 2$ closes because

$$p_r \le \exp\left(2C_0 \varepsilon L^{-2} \sum_{t<r} L^{2(t-k)}\right) < 2$$

when ε is below a number of L. For the second bound, restrict at each level to the cones of $\mathcal{P}$. A fine plaquette belongs to the four-block set of at most twelve coarse plaquettes. Hence the square root of the plaquette energy increases at one step by at most

$$L^{1/2}(1 + 4C_0 L^{-1/2} v_r), \qquad v_r \le 2\varepsilon L^{2(r-i+1)} \quad (r < i).$$

Since $\sum_{r<i} v_r < 4\varepsilon$, the product of the parentheses is at most $\sqrt{2}$, after decreasing the same ceiling on ε. This proves eq. 9.xvi.

Interpolation and its error. Measure a fine spacing as one, so the coarse lattice is $L\mathbb{Z}^3$. For $b = \langle x, x + e_\mu\rangle$, write $x = LY + r$, $0 \le r_\nu < L$, and put

$$(I\alpha)(b) = L^{-1} \sum_{s_\mu = 0,\ s_\nu \in \{0,1\}} \prod_{\nu \ne \mu} (1 - r_\nu/L)^{1-s_\nu} (r_\nu/L)^{s_\nu} \alpha\big(\langle L(Y+s), L(Y+s) + Le_\mu\rangle\big).$$

This is periodic and independent of the chosen coarse origin. Its weights are nonnegative and sum to one. In particular

$$|I\alpha| \le I|\alpha| \le L^{-1} \sup|\alpha|, \qquad \|I\alpha\|_2 \le L^{1/2} \|\alpha\|_2.$$

For the second inequality apply convexity of the square: the total weight of the fine bonds reading any fixed coarse bond is L^3, which cancels two powers of the prefactor L^{-1}. The line sum of $I\alpha$ along a straight coarse bond is $\alpha(c)$. Directly taking transverse differences also gives $dI\alpha = L^{-2}$ times a convex combination of the two adjacent parallel values of $d\alpha$.

Suppose $s = \sup|\alpha| \le r_*/2$ on the coarse bonds within one coarse spacing of a double block, and v bounds the plaquette distances of $e^{i\alpha}$ there. Choose $r_* = r_*(L) > 0$ small enough that the logarithm of the mean at the identity is holomorphic and bounded by one on the complex ball of radius r_*, and $r_* \le 1/(200L)$. Such a choice follows from the last, trivial-background estimate of Lem. 9.11(v). For $\beta = I\alpha$, the preceding identities and the elementary product estimates give

$$\begin{aligned} \sup|\beta| &\le L^{-1}s, \\ \varpi_c(e^{i\beta}) &\le L^{-2}(v + 51s^2), \\ |\mathcal{M}(e^{i\beta})(c)e^{-i\alpha(c)} - \mathbf{1}| &\le v + (35 + 2r_*^{-2})s^2. \end{aligned} \tag{9.xvii}$$

Here is the error calculation. On a coarse plaquette, $|d\alpha| \le v + 35s^2$ by expanding its four exponentials. The curl identity for I and the same expansion on a fine plaquette add at most $16L^{-2}s^2$, giving the second line. The logarithm of the mean differs from $Q\beta$ by at most $2r_*^{-2}s^2$, by Cauchy's estimate. Moving each of the L straight steps past a coordinate-order contour uses at most $L(L-1)$ plaquettes. Thus $|Q\beta(c) - \alpha(c)| \le v + 35s^2$, proving the last line.

We also use the following rigidity estimate on a double block. If W, W_0 have identical averaged contour variables on its two blocks, then

$$|W(b)W_0(b)^{-1} - \mathbf{1}| \le |\mathcal{M}(W)(c)\mathcal{M}(W_0)(c)^{-1} - \mathbf{1}| + C_L \max\{\varpi_c(W), \varpi_c(W_0)\}.$$

Inside either block, choose a coordinate-order path whose last step is the bond. The two quotients of its transports by the endpoint contour means have size at most $C_L\varpi$, by the path comparison in Lem. 9.11(i); applying this to both fields gives the bound. For a bond between the blocks, insert the two endpoint contour means; move its straight crossing to the central coarse path and then replace that path by its mean. Each replacement costs C_L times the largest plaquette distance, and the endpoint means of the two fields cancel. This is precisely the double-block reconstruction calculation of Lem. YM3_I:44.3. It proves the displayed estimate with a constant of L for the present G as well. In addition, the quotients in this calculation are products of conjugated small plaquette exponentials and the mean corrections. Their intrinsic logarithms therefore lie in $\mathfrak{g}$ when the total exponent size is less than π, by Lem. 9.9(ii). For a closed group which is not connected these products lie in its identity component; the unrestricted transports themselves need not do so.

The descending gauge. Set $g = \mathbf{1}$ at the tops. The other bare sites form the disjoint sets

$$N_i = (\mathbb{T}^{(i)} \setminus \mathbb{T}^{(i+1)}) \cap \Omega_{i+1}, \qquad i = n-1, \dots, 0.$$

When g is known on $\mathbb{T}^{(i+1)}$, set

$$W_{i+1} = V_{i+1}^{\cdot,g}, \quad a_{i+1} = \frac{1}{\mathrm{i}} \log W_{i+1}, \quad \beta_i = I a_{i+1}, \quad W_{0,i} = e^{i\beta_i},$$

and, for $x \in B(y) \setminus \{y\}$, $y \in \mathbb{T}^{(i+1)} \cap \Omega_{i+1}$, define

$$g(x) = W_{0,i}(y,x)^{-1} g(y) V_i(y,x).$$

The expressions $W(y,x)$ are averaged contour variables. This choice makes $W_i = V_i^g$ and $W_{0,i}$ have the same contour variables on each such block; gauge covariance gives $\mathcal{M}(W_i) = W_{i+1}$.

A bond of $\mathcal{B}_i$ with no endpoint in Ω_{i+1} has both endpoints at tops, and $W_i = \mathbf{1}$ there. If one endpoint lies in Ω_i, this follows from the trivial datum. Otherwise its double block consists of bonds of the level-$i-1$ datum, all equal to $\mathbf{1}$, so its average is also $\mathbf{1}$. In particular $W_n = \mathbf{1}$ and $a_n = 0$. This observation also treats the exterior block of a double block crossing the boundary of Ω_{i+1}: both its contour variables and the interpolated reference are trivial.

Here are the estimates that justify every step of the construction. For a bond b of level i with an endpoint in Ω_{i+1}, choose a coarse double block containing it, with one centre $y \in \Omega_{i+1}$, and put $k = j(y) - 1 \geq i$. Enlarge the bare double block by $2L^{i+1}$. Equation eq. 9.xiii shows that $k \leq j \leq k+2$ throughout this enlargement. Let $v_i(b)$ be the maximum of u_i on the double block and of u_{i+1} within one coarse spacing of it, and let $s_{i+1}(b)$ be the maximum of $|a_{i+1}|$ on the coarse bonds in that neighbourhood. Inductively assume

$$|a_q(c)| \leq C_A \varepsilon L^{q-j(c)} \quad (q > i).$$

Equation eq. 9.xv gives

$$v_i(b) \leq 2\varepsilon L^{2(i+1-k)}, \qquad s_{i+1}(b) \leq C_A \varepsilon L^{i+1-k}, \qquad s_{i+1}(b)^2 \leq \varepsilon L^{2(i+1-k)}$$

if $C_A^2 \varepsilon \leq 1$. Hence the reference plaquettes are at distance at most 53ε from $\mathbf{1}$, by eq. 9.xvii. Rigidity and that equation give

$$W_i(b) = e^{iz_i(b)} e^{i\beta_i(b)}, \qquad |z_i(b)| \leq C_z(v_i(b) + s_{i+1}(b)^2). \tag{9.xviii}$$

The constant $C_z = C_z(L) \geq 1$ may be enlarged to control the total size of the small exponentials just described. Thus the logarithm defining z_i is in $\mathfrak{g}$, independently of the size of r_G. On bonds with no endpoint in Ω_{i+1} put $z_i = v_i = s_{i+1} = 0$.

Choose

$$N_2 = 3(8L)^3, \qquad C_A \geq \max\{4C_z L^8, 375 C_z^2 N_2 L\},$$

and choose $\varepsilon_2(L) > 0$ below the ceilings used in eq. 9.xv–eq. 9.xvi and such that

$$C_A^2 \varepsilon_2(L) \leq 1, \qquad C_A L \varepsilon_2(L) \leq r_*/2, \qquad 53\varepsilon_2(L) \leq \omega_M/2, \qquad 2C_z C_A N_2^{1/2} L^{3/2} \varepsilon_2(L) \leq \log 2.$$

We may decrease it further by constants depending only on L to keep the contour means in their stated domains. The additional ceiling $\varepsilon \leq r_G/106$ gives $53\varepsilon \leq r_G/2$. In particular every interpolated reference has the small plaquettes required by Lem. 9.11(i). The actual intermediate fields do as well: $C_A \geq 4L^8$ implies $\varepsilon_2(L) \leq L^{-16}/16 < 3.9 \cdot 10^{-9} L^{-2}$, so $\varepsilon \leq c(G)$. The local version of Lem. 9.11(iii) on each required double block gives intermediate plaquette distances at most 2ε. Thus each chosen averaging map agrees with the small-field formula throughout this construction.

The small exponential product in eq. 9.xviii and Lem. 9.9(ii) give $|a_i| \leq I|a_{i+1}| + |z_i|$. Iterating, using positivity of I and $a_n = 0$, gives

$$|a_i| \leq \sum_{r \geq i} I^{r-i} |z_r|.$$

For a fixed bond b of level i, put $J = j(b)$. A nonzero term at level r can influence b only if $r \leq J$; its endpoints lie within $2L^r$ of those of b, and its layer index is at least $J - 1$, by eq. 9.xiii. The local bound above implies $|z_r(b')| \leq 3C_z L^6 \varepsilon L^{2r-2j(b')}$. Since each interpolation costs L^{-1} in supremum norm,

$$|a_i(b)| \leq 3C_z L^8 \varepsilon L^{i-2J} \sum_{r=i}^{J} L^r \leq 4C_z L^8 \varepsilon L^{i-J} \leq C_A \varepsilon L^{i-J}.$$

This closes the induction and justifies all logarithms in it. At $i = 0$ it proves the claimed pointwise bound. At a node which is not a top, the retained inward bond of level q joining it to a top has $V_q = \mathbf{1}$. Its transformed value is $g(z)$ or $g(z)^{-1}$, and the same bound gives $|g(z) - \mathbf{1}| \leq C_A \varepsilon$.

The energy estimate. Keep the term quadratic in the coarse potential in eq. 9.xviii. With all functions extended by zero off their indicated bond sets, it gives

$$|a_r| \leq T_r |a_{r+1}| + \zeta_r, \qquad \zeta_r = C_z v_r, \qquad T_r = I + B_r,$$

where, on an active bond b,

$$(B_r f)(b) = C_z C_A \varepsilon L^{r+3-j(b)} \max_{\mathcal{N}_b} f.$$

The set $\mathcal{N}_b$ is the coarse neighbourhood used in the definition of $s_{r+1}(b)$; on an inactive bond $B_r f = 0$. The maps T_r are monotone and subadditive on nonnegative functions. Split ζ_r according to its layer: $\zeta_r^J = \mathbf{1}_{\{j=J\}}\zeta_r$, which is zero unless $r < J$, and put

$$f^J = \sum_{r<J} T_0 \cdots T_{r-1}\zeta_r^J. \qquad \text{Then} \qquad |a| \le \sum_J f^J.$$

Each step spreads support by at most $3L^{r+1}$. Therefore all the descendants of ζ_r^J are within $4L^{J-1}$ of a site of layer J, and their layer indices lie in $[J-1, J+1]$, by eq. 9.xiii.

A coarse bond belongs to $\mathcal{N}_b$ for at most N_2 fine bonds: their base points lie in a cube of side $6L^{r+1}$, with three bond directions at each site. Thus, on such a layer-supported descendant,

$$\|T_t f\|_2 \le (L^{1/2} + C_z C_A N_2^{1/2} \varepsilon L^{t+4-J})\|f\|_2.$$

Since $t < r < J$, the sum of $L^{t+7/2-J}$ over $t < r$ is less than $2L^{3/2}$. The chosen smallness ceiling therefore gives

$$\|T_0 \cdots T_{r-1}\zeta_r^J\|_2 \le 2L^{r/2}\|\zeta_r^J\|_2.$$

Let

$$E^J = \sum_{p:\,|j(p)-J|\le 2} u_0(p)^2.$$

Every plaquette entering $v_r(b)$ for $j(b) = J$ has its bare cone in these layers. The same multiplicity bound N_2, now for plaquettes, and eq. 9.xvi give

$$\|\zeta_r^J\|_2^2 \le 4C_z^2 N_2 L^{r+1} E^J, \qquad \sum_J E^J \le 5\sum_p u_0(p)^2.$$

Consequently

$$\|f^J\|_2 \le 4C_z N_2^{1/2} L^{1/2} (E^J)^{1/2} \sum_{r<J} L^r \le 5C_z N_2^{1/2} L^{J-1/2} (E^J)^{1/2}.$$

At a bare bond at most three f^J are nonzero, those with $|J - j(b)| \le 1$. Cauchy–Schwarz now yields

$$\begin{aligned}\sum_b L^{-2j(b)}|a(b)|^2 &\le 3\sum_J L^{-2(J-1)}\|f^J\|_2^2 \\ &\le 75C_z^2 N_2 L \sum_J E^J \le 375 C_z^2 N_2 L \sum_p |\partial U(p) - \mathbf{1}|^2.\end{aligned}$$

This is the asserted energy bound, with $C = C_A$.

Finally suppose the field is flat. In each pillar let $t(x)$ be transport from its top to x. Flatness makes this independent of the path inside the pillar. On a determining bond the straight transport equals its block average, hence is $\mathbf{1}$. For a bare bond within one pillar, its transport and the two pillar paths form a contractible loop. For a bond joining pillars of equal level, close the loop by their determining bond. For adjacent pillars of levels q and $q+1$, first pass through the level-$(q+1)$ double block, using its outer determining bond and the level-q determining bonds in the outer block. All closing transports are $\mathbf{1}$; the separation and block nesting place the loop inside that double block. These three cases give $t(b_-)U(b)t(b_+)^{-1} = \mathbf{1}$ on every bare bond. The same closing path at a retained inward bond gives $t = \mathbf{1}$ at its inner node, as at every top. Thus $U = \mathbf{1}^{t^{-1}}$, with the residual gauge equal to $\mathbf{1}$ at all nodes. In the quotient case the constructed potential also satisfies $4\sup|a| \le 4C_A\varepsilon \le 4/C_A < \pi$; lifting its small exponentials therefore has trivial central plaquette parts. No sector condition was assumed in the construction; the trivial datum enforces it.

(iii) We give the preliminary gauge argument, including its estimate. Write $W_n = \mathcal{M}^n U$ and $W_{0,n} = \mathcal{M}^n U_0$. The local straightening estimate of Lem. 9.11(iii) keeps every mean used below in its

small-field domain. Its contour means lie in G and are gauge covariant, by Lems. 9.9(iv) and 9.11(i). Set $g_k = \mathbf{1}$ on the top lattice and descend through the blocks. If $x \in B(y)$ at level n, set

$$g_n(x) = W_{0,n}(y,x)^{-1} g_{n+1}(y) W_n(y,x),$$

where $W_n(y,x)$ is the averaged contour variable from y to x. It equals $\mathbf{1}$ at $x = y$, so g_n extends g_{n+1}. Thus $\hat{g} = g_0$ is a gauge, equal to $\mathbf{1}$ at the top sites. Gauge covariance gives

$$(\mathcal{M}^n U^{\hat{g}})(y,x) = W_{0,n}(y,x).$$

These identities also determine $\hat{g}$ uniquely, by downward induction.

For the quantitative assertion, put

$$\delta_n(b) = |\mathcal{M}^n(U^{\hat{g}})(b) W_{0,n}(b)^{-1} - \mathbf{1}|.$$

At the terminal level $\delta_k \le \alpha_1$. Within a single block, the equality of the contour variables and the contour straightening bound give $\delta_n \le C_L \alpha_0$. Across two adjacent blocks, with coarse bond c, the same comparison gives

$$\delta_n(b) \le \delta_{n+1}(c) + C_L \alpha_0 L^{2(n-k)}.$$

Indeed, a path product changes by at most the sum of the changes of its factors, since the other factors are unitary. Replacing the paths by their means costs at most a constant of L times the largest plaquette discrepancy in the two blocks; the two end-point contour discrepancies vanish by the identities above. The straightening estimate bounds this plaquette discrepancy by $2\alpha_0 L^{2(n-k)}$. Iterating until the two end points enter one block or reach the top and summing $\sum_{j \ge 1} L^{-2j} = (L^2 - 1)^{-1}$ proves the asserted bound. This argument uses only multiplication, inversion, unitarity and the mean estimates, so applies to every closed G. Change of level is a rescaling; the section of a datum is constructed from its group values by the same contour identities. Neither uses a lift or a choice of sector.

(iv) *What the operators read.* The operators of the $SU(2)$ construction read the background through $\mathrm{Ad}(U_0(b))$ at the bonds and through the plaquette functions, on pieces of bounded size. Unrolled to the covering lattice the field is periodic; on a box, which is simply connected, every sector is trivial, and the field has a lift $\tilde{U}$ to $SU(2)$ with $\partial\tilde{U}(p) = s(\partial U_0(p))$, which lies in the regular class of $SU(2)$ at the parameter $\frac{\alpha}{2}(1 + 0.04\alpha)$, $\alpha \le 1$ the parameter of U_0: at the plaquettes by the ratio $\frac{1}{2\cos(\theta/4)}$ of the chords of $s(g)$ and of g, θ the angle of g, which is at most $\frac{1}{2}(1 + 0.04\alpha)$ for chords at most α; and at the bonds, as the scalar part of the divergence of $\partial\tilde{U}$ at a bond b is at most $0.3\alpha^2\eta^3$ and the norm of its part $i\eta^2 J(\tilde{U})(b)$ at most half that of the divergence of ∂U_0 at b plus $0.3\alpha^3\eta^5$, which keeps the divergence of $\partial\tilde{U}$ below $\frac{\alpha}{2}(1 + 0.002\alpha^2)\eta^2$ at $\eta \le \frac{1}{13}$. At the lift the operators of $SO(3)$ are those of $SU(2)$ conjugated by the differential of the covering, which doubles the operator norm on $\mathfrak{su}(2)$, the pairings changing by the one factor $\frac{8}{3}$; and the action of the pair is $a_3 = a^\flat(1 - \frac{3}{16}a^\flat)$, with $a^\flat = \frac{8}{3}\, a_2 \circ s$ and a_2 the action of $SU(2)$.

The operator on the whole torus. The boxes are taken as the first members of separated sequences of larger tori, at growing radii; there the propagator statements of the $SU(2)$ construction for $SU(2)$ (Thm. YM3_I:I and its local inverse construction), read at the lift, give the operators with the Hessian $\Delta^{\mathrm{W},\flat}$ of $\sum_p a^\flat$, positive and with inverses that decay exponentially, uniformly in the box.

Their limits are kernels on the covering lattice, invariant under the periods of the torus, and their periodisations are, on the whole torus and in every sector, the operators with $\Delta^{\mathrm{W},\flat}_{U_0}$: positive definite, with inverses that decay exponentially. Since the distance d_T on the torus is the least distance d of two lifts, the sum over the periods keeps with its constant a bound on the sums along rows weighted by $e^{\delta d}$, and it carries a pointwise bound $Be^{-\delta d}$ of a kernel, d in the spacing of the top lattice, to $9Be^{-\delta d_T}$ once $LM\delta \ge 2$, which the floor $6LM$ gives for M large. This periodisation factor must be at least 8 on a torus of even side, where the point antipodal to a site has eight images at one distance from that site.

The Hessian of a_3 differs from $\Delta_{U_0}^{\mathrm{W},\flat}$ by a local symmetric operator E, $|(\mathsf{E}A)(b)| \le 96\,\alpha_0^2\eta^2 \max|A|$, the maximum over the bonds of the four plaquettes at b, of the relative order $|\partial U_0(p) - \mathbf{1}|^2$; $\mathsf{G}_0 = (1 - \mathsf{G}_0^\flat\mathsf{E})^{-1}\mathsf{G}_0^\flat$, $\mathsf{G}_0^\flat$ the inverse with $\Delta_{U_0}^{\mathrm{W},\flat}$, by a Neumann series in norms weighted by the decay, for α_0 below the ceiling. Positivity holds along $\Delta_{U_0}^{\mathrm{W},\flat} - t\mathsf{E}$, $0 \le t \le 1$, on which the operator stays invertible.

The bounds take numbers of L in the places of the constants of the $SU(2)$ construction. With $q < 1$ the norm of $\mathsf{G}_0^\flat\mathsf{E}$ for the sums along rows weighted by the decay, kept below 1 by the ceiling on α_0, a bound B of these sums for $\mathsf{G}_0^\flat$ becomes $B/(1-q)$ for G_0, and its pointwise bound $9Be^{-\delta d_T}$, with the factor 9 of the sum over the periods, becomes $9B(1-q)^{-1}e^{-\delta d_T}$ at the same rate, and the bounds of H_0 and of $\mathsf{G}_0 - \mathsf{H}_0\mathsf{Q}\mathsf{G}_0$, formed from those of G_0, take numbers of L in the same way.

The local clauses. The same lifts, on boxes shorter than half the torus: the clause of the $SU(2)$ construction for $SU(2)$ is read at the lift and carried back by the covering. Its differential doubles $|\cdot|$ on $\mathfrak{su}(2)$ and multiplies it on $\mathfrak{su}(2)^c = \mathfrak{sl}(2,\mathbb{C})$ by a factor between $\sqrt{2}$ and 2, as $|z\cdot\sigma|^2 = |z|^2 + 2|x\times y|$ and $|2z\cdot T| = 2|z|$ for $z = x + iy \in \mathbb{C}^3$, σ the Pauli matrices and T the generators of $\mathfrak{so}(3)$ to which it takes $\frac{1}{2}\sigma$. The chords at most α of values in the group are halved at the lift up to the factor $1 + 0.04\alpha$, as above. The way back multiplies the size of a form by at most 2 and a chord by $2\cos(\theta/4) \le 2$, keeps the norms of operators on real forms, and keeps on complex forms the norms of a real kernel and their sums along rows, a real 3×3 matrix having one norm on $\mathbb{R}^3$ and on $\mathbb{C}^3$.

Thus a bound $C\prod_i \sigma_i^{d_i}$ of a form or a chord, of degree d in sizes of forms and e in such chords, comes back with $2^{1-d-e}(1+0.04\alpha)^e C$ at real forms and with $2^{1-d/2-e}(1+0.04\alpha)^e C$ at complex ones: with its constant C at $d + e \ge 2$, and at $d = 1$, $e = 0$ for real forms; with $(1+0.04\alpha)C$ at $d = 0$, $e = 1$, the gauge on a cube among these, of degree one in α and none in forms, whose bound $Bl\alpha$ comes back as $2Bl\cdot\frac{\alpha}{2}(1+0.04\alpha)$; with $\sqrt{2}C$ at $d = 1$, $e = 0$ for complex forms, and with C there for a linear map whose kernel is real, in bounds on the norms of its kernel or on their sums along rows; and with $2C$ at $d = e = 0$. A bound on the norm from suprema to suprema of an operator on complex forms comes back with a factor at most $\sqrt{2}$, and a clause of $SU(2)$ on the forms below r serves for the complex forms below $\sqrt{2}r$ and the real ones below $2r$. The bounds of the action are taken anew for a_3, whence the absolute number in the place of $C_{4,\mathrm{ec}}$ and the form with $r + 5r^2$. The three ceilings keep the fields of a step below the radius under which the averaging map is the formula.

The three lemmas. The gauge correction, one step and the ascent are the proofs of the $SU(2)$ construction read at such backgrounds with the operator and these clauses, and so are the statements that these proofs cite and those that these cite in turn. Among them the estimate in the slice (Lem. YM3_I:30.10) and the solution of the system (Prop. YM3_I:30.9) are statements on the whole torus, whose proofs read the background only through the estimates on G_0 and H_0 and through clauses on pieces of bounded size, so that they too hold in every sector. The improvement of the regularity radius is the proof of Prop. YM3_I:37.1 read so with all its numerals free.

The constants are fixed in this order: K_3, C_{gc} and the constant of the gauge on a cube, at most 1.04 times its value in the $SU(2)$ construction, as numbers of L; then the numbers that the $SU(2)$ construction forms from them by its formulas, those of Prop. YM3_I:37.1 among them; and the ceilings on α_0 and on ε' last, so that the inequalities by which the improvement and the ascent bring the radius back into the regular class, hold with the constants as moved, each being a ceiling on a constant chosen after the numbers that it reads; the rate δ of the decay stays that of the $SU(2)$ construction, and the floor $6LM$ is laid after it.

Flat fields. At a flat field the equation $D_U\lambda = 0$ says $\lambda(b_-) = \mathrm{Ad}(U(b))\lambda(b_+)$ at all bonds, and a twist leaves no invariant vector. □

9.1.10. *Proof of Thm. AA.* For the construction clauses, the argument reduces to the tasks (P)–(K). The following steps describe the construction once those estimates are established. The local calculations have the separate scope specified in the statement.

Step 1: the objects of a step. The following definitions of the first part of the $SU(2)$ construction, transferred to G by Def. 9.5, define objects with the properties that the $SU(2)$ construction proves: the logarithm on D^c, the mean, the block average on small fields and on all fields and its iterates (Lems. 9.9, 9.11); the gradient (Lem. 9.12(i)); the regular classes, with the inclusions $\mathfrak{U}'_k(\mathbf{B},r)\subset\mathfrak{U}_k(\mathbf{B},\kappa_G r)$ for every G (Remark 9.6) and $\mathfrak{U}_k(\mathbf{B},r)\subset\mathfrak{U}'_k(\mathbf{B},r+r^2+\kappa' r^3)$ for every G and $r\le z_G$ (Def. 9.5(3), by the bound at a bond of Lem. 9.12(ii)), and for $SU(N)$ and $r\le z_G$ the sharper $\mathfrak{U}'_k(\mathbf{B},r)\subset\mathfrak{U}_k(\mathbf{B},(1+\kappa' r^2)r)$ (Lem. 9.12(ii)); the gauges (Lem. 9.12(iii), (iv)); the chart with the indicator of G^π, the Haar density and the smooth cut-off (Lem. 9.14(i), (ii)); the elimination factor of Lem. 9.14(iii), a complex linear map of $\mathfrak{g}^c$ into itself at every V of the domain of the complex average at c, the G^c-valued V on P_c whose contour families have diameter below $\frac{1}{40}$ and whose contour quotients and variables W_x lie in D^c, with its branch of $\operatorname{tr}\log$ and $|\mathsf{m}_c(e^{ia})-L^{-3}|\le L^{-1}C_S\sup|a|$ for $\mathfrak{g}^c$-valued a with $\sup|a|<\frac{3}{4}r_*$ (Lem. 9.14(iii)); Log det at the radius r_n (Lem. 9.14(iv)); and the domains of analyticity with their structural facts (Prop. 9.22(v)). The propagators, the covariances, the critical configurations and the towers off the trivial field are the objects of (P) and (V) of §9.1.1. Every logarithm is taken of an element of D^c, the averages being those of Lem. 9.11(i) with $W_x\in D$; integrals over $G\setminus G^\pi$, which has inner points wherever $r_G<2$, so for $SU(N)$ with $N\ge 3$ (Lem. 9.14(i)), carry the indicator of G^π; and the uses of the averaging map off the small fields, of the pivot bond and of the measure of $G\setminus G^\pi$ in the second part are those of (II) in §9.1.1. Among them the pivot bond (Prop. YM3_I:14.11) holds for G in all its clauses with all its constants: the quotients that it reads are products of conjugated plaquette variables within $\omega_M\le z_G\le r_G$ of $\mathbf{1}$, so in G^π, the averaged contour variables have one mean each (Lem. 9.9(iv)), and the equation for the variable at the pivot bond is solved in the chart of G^π by a contraction, the nonlinear part being bounded by Cauchy's estimate.

Step 2: the trivial field; clause (c). By Lem. 9.15 every linear operator of a step at $\mathbf{1}$ is $K\otimes 1_{\mathfrak{g}^c}$, K a real matrix formed from the lattice alone, and every bound of the $SU(2)$ construction there holds for G with its constant, each in the norms to which its kind passes (Lem. 9.15(ii)). The bounds in suprema, by row sums and by non-decreasing functions of real linear forms hold in $|\cdot|$ and in $\|\cdot\|$ alike:

$$|l_0(f)|\le\Phi(|l_1(f)|,\dots,|l_m(f)|)\text{ for all }f:\Lambda\to\mathfrak{g}^c\iff\text{the same for all }f:\Lambda\to\mathbb{R}.$$

The ℓ^2 bounds, the quadratic forms and the lower bounds, the lower bound of the fluctuation operator among them, hold in $\|\cdot\|$, the sign $|X|^2$ of the $SU(2)$ construction being read as $\langle X,X\rangle$ for $X\in\mathfrak{g}$; sums of squares in $|\cdot|$ retain their constants under Schur's test on $|K|$ or convexity. Otherwise the comparison $\|X\|\le|X|\le N^{1/2}\|X\|$ costs $N^{1/2}$ for an unsquared ℓ^2 estimate and N for its squared form. Identification of all small critical configurations with the near-trivial branch requires the orbit input of §9.1.6; the local coefficient calculation in Prop. 9.18(i)–(iii) gives (c).

Step 3: fields off the trivial field. Lem. 9.19 gives the Hessian at $SL(N,\mathbb{C})$-valued fields with the absolute numerals 12, 1536, 3456 of the $SU(2)$ construction, the inverse of a plaquette variable being bounded by $|X^{-1}|\le|\operatorname{Ad}X|$, Ad acting on all $N\times N$ matrices (Lem. 9.9(ii)). For another G, that is, for semisimple $\mathfrak{g}$, so that $G\subset SU(N)$, and G^c-valued U, the Hessian of the action of G is the restriction of this form to $\mathfrak{g}^c$-valued forms; its two-sided bound and its bounds in the normalised trace norm stay, and the kernel and Weitzenböck bounds in $|\cdot|$ carry the norm $|P|\le N^{1/2}$ of the projection of the pair, the condition $|\operatorname{Ad}(U(b))^{\pm1}|\le 2$ being read on all $N\times N$ matrices. Prop. 9.22(i) gives the inputs of the propagator construction for $SU(N)$, and, for the other simple G, the Hessian among them with the factor $|P|$; (ii) the averages and the critical configuration near the trivial field; (iii) the localisation, assuming (V); and (iv) the bond maps, with $|E_b-\mathbf{1}|<r_G$ at every bond of the cut-off zone under the ceiling $\mathsf{c}_L c_4\le 2Lr_G$ of Prop. 9.22(iv). This is a positive

ceiling on the constant c_4, which is free downward, so that it never obstructs the ordered choice (Prop. 9.26(a)); it stands among the conditions on the constants read for G under which clause (a) holds. With (P) and (V) of §9.1.1 these are the propagators, the covariances, the minimisers over several scales with their towers, the localisation and the bond maps of the $SU(2)$ construction, transferred to G.

Step 4: conjugation. Each symmetric domain Q^s of the $SU(2)$ construction is read as its real-connected part Q^r (Lem. 9.21): no number moves, but at the memberships reached from memberships of restrictions had apart. The analyticity of the maps is had on the open domains cut out by conditions $m(X) < r$ and by $X \in D^c$ at the logarithms of the averages, and the clauses on the conjugation are stated on the real-connected parts; Q^r is the part of Q^s whose slice components hold real points, and it passes to intersections as the inclusion $(\bigcap_i Q_i)^r \subset \bigcap_i Q_i^r$. A map of the induction is a composite of maps built from products, inverses, means, logarithms in D^c, the action and its derivatives, which commute with $p \mapsto p^c$ by Lem. 9.20, of maps holomorphic on open slices and real at real points, and of continuations on convex subsets of a vector space kept by the conjugation (slices without group-valued member); by Lem. 9.21(ii) and Prop. 9.23(i) such a map commutes with $p \mapsto p^c$ on Q_1^r and maps Q_1^r into Q_2^r as soon as it maps Q_1^r into Q_2; that every map of the induction is of this kind is (C) of §9.1.1. A membership that is reached from memberships of restrictions had apart is reached in Q^s, and there Lem. 9.21(iii) carries it into a real-connected part: $\mathcal{Q}(b_0, b_1, c_0)^s \subset \mathcal{Q}(b_0 + 2e^{4\xi b_1} b_1^2, b_1, c_0)^r$, the like for regular pairs and the natural classes, and for the analyticity cores $\mathcal{U}^\circ(b_0, b_1, c_0)^s \subset \mathcal{U}^\circ(B_0, B_1, c_0)^r$ under the four conditions of Lem. 9.21(iii), $b_0' = b_0 + 2e^{4\xi b_1} b_1^2 \le B_0$, $b_1 \le B_1$, $K(b_0' + b_1^2) \le B_0$ and $b_0' + b_1 \le a^\sharp$, where $K \ge 16L^2$ and $a^\sharp$ are the constant and the radius of the inclusion of the simple sets in the cores. The moved first radius, which is to be at most the first radius B_0 at which the statement read at such a membership is had, and these conditions stand among the conditions on the constants read for G (Prop. 9.26): where the first radius of the membership lies below B_0, and for the cores $Kb_0 < B_0$ and $b_0 < a^\sharp$, they are positive ceilings on the second radius, which is free downward.

Step 5: large fields; sectors. Lem. 9.13 gives the energy of block averages and the large-field chain, the prices being those of the $SU(2)$ construction times τ_G^{-2}, by $a(V) = \frac{1}{2}\|V - \mathbf{1}\|^2 \ge \frac{1}{2}\tau_G^{-2}|V - \mathbf{1}|^2$. Lem. 9.13(i) holds with the pair $(\frac{1}{4\tau_G^2}, e_*)$, $e_* = e_L/(2\tau_G^2)$: at one plaquette its proof reads the pair through $|V - \mathbf{1}|^2 \le 2\tau_G^2 a(V)$ alone; and $\tau_G^2 \le N - 1$. The chain has three links, those of Lem. 9.13(ii). The first link is Prop. YM3_I:44.4 and Prop. YM3_I:44.5, on the failed tests. A failed small-field test forces a plaquette variable or an averaged contour variable farther than the constants of the $SU(2)$ construction from $\mathbf{1}$: by contraposition, a test passed at all plaquettes of a double block puts the field under the averaging radius $\min\{\omega_M, r_G\}$, where the contour families have one mean within $4d$ (Lems. 9.9(iv), 9.11(i)), the straightening of Lem. 9.11(iii) holding for every closed G. The second link is Lem. YM3_I:65.2, on the witnesses of the cells created at one step, with "$\notin G^\pi$" for "$= -\mathbf{1}$". The third link is the local-deposit argument in the proof of Prop. YM3_I:65.4, the lower bounds of the amount of action near a created cell, which cites clauses of the second. Beside these, the proof of the third link cites statements on cubes, regions and runs of couplings alone, which read no group and hold for G as written. The second and the third link hold for G by Lem. 9.13(ii), by which Prop. 9.24 transfers the chain and which (II) of §9.1.1 includes: their proofs derive them, by the device of the proof of Prop. 9.24, from the transferred statements that they cite, their inputs asking $\|E^{(j)}\| \le E_0$ and $|b_j| \le B_\sharp$ separately; and the prices follow from these three links and Lem. 9.13(i): the group enters them where the third link turns the witnesses of the second into lower bounds of the action, at one plaquette through $a(V) \ge \frac{1}{2}\tau_G^{-2}|V - \mathbf{1}|^2$ and at averaged variables through the pair $(\frac{1}{4\tau_G^2}, e_*)$ of Lem. 9.13(i), so that the clauses of the third link hold with the prices of the $SU(2)$ construction divided by τ_G^2. As G is simply connected, the fields on a torus with small plaquettes form one sector, and the statements on the whole torus that the induction uses hold as for $SU(2)$; $SU(N)$ is simply connected, and for quotients Prop. 9.33 gives what holds in each sector.

Step 6: clause (a). Let G be simply connected. Assume the five construction hypotheses of §9.1.1. By Steps 1–5, Prop. 9.23 and (II), the statements cited in the proofs of the induction hold for G, with $m(X)$ where $|X^{-1}-\mathbf{1}| = |X-\mathbf{1}| = |X^*-\mathbf{1}|$ was used, with the radius r_n where $|1-T| \le \frac{1}{2}$ was to keep $\det T$ off $(-\infty, 0]$, with $\max\{1, n2^{-11}\}$ in the bound eq. 9.xii, with the ceilings $\varepsilon_{\mathrm{v}} \le a_G$, $\varepsilon_{\mathrm{v}} \le \varepsilon_G$ and $\varepsilon_{\mathrm{v}} \le L^2 r_G$ on the regularity radius (Prop. 9.23(ii), Prop. 9.26(f), §9.1.1(V)), the member $r_G/(2C_{\mathrm{pg}})$ of the radius c_{11} of §9.1.6, the ceiling $\mathsf{c}_L c_4 \le 2Lr_G$ of Step 3 and the conditions of Step 4 on the radii at the memberships reached from memberships of restrictions had apart among the conditions on the constants (§9.1.1(K)), and with the colour clauses near trivial block data from Prop. 9.22(ii) and (V). The ceiling $b_1 \le \pi L^j$ under which the local terms are continuous on the analyticity cores (Lem. 5.33), which a proof of the chain from level k to level $k+1$ quotes at the fixed radii as $\alpha_1 < \pi L^j$, $1 \le j \le k$, takes ρ_G of Def. 9.1 in the place of π (Prop. 9.24): $\alpha_1 < \rho_G L^j$, which the row $\alpha_1 < \rho_G L$ of the table gives, as $L^j \ge L$. This row is a positive ceiling on α_1 and holds of itself for $\rho_G > \frac{1}{13}$, as $\alpha_1 \le 1$ and $L \ge 13$, so for $SU(N)$ with $N \le 81$, where $\rho_G = \frac{2\pi}{N}$. As $\rho_G \le \pi$, it gives $\alpha_1 < \pi L^j$, under which the exponential of $\mathfrak{g}^c$ is a local diffeomorphism at the Z_0 with $|Z_0| < \pi$ that the continuity on the cores reads, for matrices of every size, the eigenvalues of $\operatorname{ad} Z_0$ being at most $2|Z_0| < 2\pi$ in modulus. The maps of the induction commute with the conjugation by (C). By Prop. 9.24, the hypotheses of each citation, those that the forms (3) of Def. 9.5 add among them, being met at its data by (II), the proofs of the induction then give Thm. YM3_I:D for G, the mean coefficient being the one number of second order that they read, so that, with b_k the mean of the n quadratic colour coefficients of step k, the run is

$$x_k = L^{-1}x_{k-1} - b_k, \qquad |b_k| \le \mathsf{G}_\sharp \le B_\sharp,$$

real and free of the torus as for $SU(2)$. By (K) and Prop. 9.26(b) the conditions on the constants read for G, those of Prop. 9.26(c)–(g) included, hold together on a large set, so choices exist, in the order of the $SU(2)$ construction with γ last.

Step 7: clause (b). The chessboard norm of a source is Prop. 9.27(ii), for every compact $G \ne \{\mathbf{1}\}$. With $A(v,V) = \sum_p v(p)a_p(V)$ for a complex weight v of the plaquettes and $\mathcal{O}(w) = a_K^{-1}A(w,\cdot)$ for a source w, the weight with the source is $e^{-A(\beta - a_K^{-1}w, V)}$, of modulus $e^{-\sum_p (\beta - a_K^{-1}\operatorname{Re} w(p))a_p(V)}$, with $a_p \ge 0$ on G; so $\operatorname{Re} w \le (1-c_{\mathrm{pr}})a_K\beta$, $c_{\mathrm{pr}} \in (0,1]$, gives $|e^{-A(\beta - a_K^{-1}w,V)}| \le e^{-c_{\mathrm{pr}}\beta\sum_p a_p(V)}$: every plaquette keeps at least the fraction c_{pr} of its price, and the Wilson measure at the one coupling $s = c_{\mathrm{pr}}\beta$ dominates the weight with the source.

The bound on the mean plaquette reads of the group its dimension n and that a vanishes on G at $\mathbf{1}$ alone, through $\frac{1}{2}\tau_G^{-2}|V-\mathbf{1}|^2 \le a(V) \le \frac{1}{2}|V-\mathbf{1}|^2$ (Lem. 9.9(i)) and through the Haar measure of balls, $cr^n \le \mathrm{Haar}\{|V-\mathbf{1}| < r\} \le C''r^n$, the first for $r \le 1$, c and C'' numbers of the pair, which the chart of Lem. 9.14(i) gives, its density being at most 1 and bounded below on $|Y| \le \frac{\pi}{3}$. With the first of the two bounds on a they give $\int_G e^{-\frac{s}{2}a(W)}\,\mathrm{d}W \le C_1 s^{-n/2}$ for $s > 0$, C_1 a number of the pair; in the chart, with $a(e^{iY}) \ge \frac{2}{\pi^2}\|Y\|^2$ for $|Y| < \pi$ and $a \ge \frac{1}{2}\tau_G^{-2}r_G^2$ off G^π, $\int_G e^{-ya(u)}\,\mathrm{d}u$ lies between two numbers of the pair times $y^{-n/2}$, the lower bound for $y \ge 1$ and the upper one for $y > 0$. Let the couplings of the plaquettes of a torus with $|\Lambda|$ sites satisfy $s \le v(p) \le C's$, $s \ge 1$, $C' \ge 1$, let $Z(v) = \int e^{-A(v,V)}\,\mathrm{d}V$, and let $\langle\cdot\rangle_v$ be the Wilson measure at the couplings v. As $\log Z(tv)$ is convex in t with the derivative $-\sum_p v(p)\langle a_p\rangle_{tv}$, this weighted sum does not increase with t, its derivative being minus a variance, and $\frac{s}{2}\sum_p\langle a_p\rangle_v \le \frac{1}{2}\sum_p v(p)\langle a_p\rangle_v \le \log Z(\frac{v}{2}) - \log Z(v)$.

From below: in the gauge in which a maximal tree carries $\mathbf{1}$ the $2|\Lambda|+1$ bonds off the tree are independent Haar variables, and restricting them to within $(C's)^{-1/2}$ of $\mathbf{1}$ gives $a_p \le \frac{1}{2}(4(C's)^{-1/2})^2$ by the second bound on a, so $v(p)a_p \le 8$ at each of the $3|\Lambda|$ plaquettes, and $Z(v) \ge (c(C's)^{-n/2})^{2|\Lambda|+1}e^{-24|\Lambda|}$.

From above: of all the bonds, $2|\Lambda|-2$ can be ordered and matched to distinct plaquettes so that each lies in its own plaquette and in none of the plaquettes of the earlier ones; dropping

the other plaquettes, as $a_p \geq 0$, and integrating these bonds one by one, the last first, by the invariance of the Haar measure, gives $Z(\frac{v}{2}) \leq (C_1 s^{-n/2})^{2|\Lambda|-2}$. So the mean $\frac{1}{3|\Lambda|}\sum_p \langle a_p\rangle_v$ is at most $(C + n\log C')/s + n\log s/(|\Lambda|s)$, C a number of the pair. At couplings invariant under the lattice symmetries every plaquette has this mean, so at the one coupling s, where $C' = 1$, $\langle a_p\rangle_s \leq C/s + n\log s/(|\Lambda|s)$ at every plaquette. The term $n\log s/(|\Lambda|s)$ comes from the three holonomies of the torus, $2|\Lambda|+1$ bonds off the tree against $2|\Lambda|-2$ matched bonds; on the tori of the continuum theorem, with at least $2L^{K+1}$ sites a side, at couplings at most $L^{K+1}(g^{-2}+\bar{B})$, that of its first-passage data, $n|\Lambda|^{-1}\log s$ is below a number of g and the constants. At equal couplings equipartition gives the mean plaquette $\frac{n}{3s}(1+o(1))$ on large tori as $s\to\infty$; so the number C in the bound reads the pair, and it is at least $\frac{n}{3}$.

Let G be simply connected and assume the construction hypotheses. The sourced-density and stability assertions of Thms. YM3_I:AJ and YM3_I:AD hold for G with the run of (a), by the corresponding $SU(2)$ proof run for G: a source enters as a weight of the plaquette function a_p, which for simple $\mathfrak{g}$ is the same class function as in the action, so the steps of the induction of (a) run for the weighted action, whose gradient and Hessian are the weighted ones, through complex plaquette weights near β and uniformly in the volume, the price of a plaquette keeping the prices of the large fields; the effective densities with a source have at every level the structure that the $SU(2)$ construction proves, and writing $\widetilde{Z}(w) = e^{-E_{(K)}-\mathcal{E}_K(w)}Z_{K,\ell,\beta}[w]$, with $\mathcal{E}_K$ defined by [7, eq. 212], gives the factorisation of the partition function with a source that (b) displays, with the bounds of Thm. YM3_I:AD on its factors. With it the bound per ball of (b) is the passage of the $SU(2)$ construction from the stability to Prop. YM3_I:89.4, run for G, at every reference coupling $0 < g \leq \gamma$, at every first-passage datum (K,β) at g and on every side ℓ admitted at g with $\ell \geq 16\mathsf{m}(R+1)$, for supports in balls of radius R pairwise at a distance at least δ: the tuple of test functions is placed on a board by a lattice translation (Lem. YM3_I:89.3); the stability gives on each cell a normaliser without zeros, with its bound per cell (Prop. YM3_I:89.1, Prop. YM3_I:89.1, Prop. YM3_I:89.2); the expectation with sources is dominated by the chessboard norms of the occupied cells, through subadditivity and the comparison of two volumes (equation YM3_I:(1759), Prop. YM3_I:86.5, Cor. YM3_I:86.6, which hold for G by Prop. 9.27(ii)), the price of a plaquette and the bound on the mean plaquette entering as in the $SU(2)$ construction; and the graded Cauchy inequality on the polydisc of sources (Lems. YM3_I:87.1 and YM3_I:87.2) gives the bound per ball, uniformly in the torus, with the constant $C^{\infty}_{\mathsf{m}}(\delta;R)$ of Prop. YM3_I:89.4, which these sections form from g, m, δ, R and the constants of Thm. YM3_I:AD for G alone, free of K, β and ℓ. These sections of the $SU(2)$ construction read the group through the statements named alone.

Step 8: clause (d). The increments of the run are real, depend on no torus and are bounded by $B_\sharp$, by (a); so $|x_j - L^{-(j-i)}x_i| \leq \bar{B}(1-L^{-(j-i)})$, for $\beta > X_g$ there is exactly one first passage K, and

$$g^{-2} < x_K \leq X_g, \qquad \frac{\beta}{L(g^{-2}+\bar{B})} \leq L^K < \beta g^2, \qquad g^{-2} + B_\sharp - \frac{\bar{B}}{L} < a_K\beta \leq L(g^{-2}+\bar{B});$$

the window $I(g)$ has the length $(L-1)g^{-2} + B_\sharp\frac{L(L-3)}{L-1} > 0$, and $L^K I(g)$ consists of first-passage couplings; so $a_{K_j}\beta_j$ has a convergent subsequence in the closed interval. For simply connected G satisfying the construction hypotheses, by (b) the bound per ball holds uniformly in j, at the first-passage data (K_j,β_j) and on the sides l_j admitted at g, from the j with $l_j \geq 16\mathsf{m}(R+1)$ on: $|S^{(j)}_{\mathsf{m}}(f)| \leq C^{\infty}_{\mathsf{m}}(\delta;R)\prod_i \|f_i\|$, in the norms of Lipschitz functions of Prop. YM3_I:89.4, each at most a constant times $\|f_i\|_{C^1}$, for supports in balls of radius R pairwise at a distance at least δ, $C^{\infty}_{\mathsf{m}}(\delta;R)$ being formed from g, m, δ, R and the constants of Thm. YM3_I:AD for G alone; on one torus of side l the bound $C_{\mathsf{m}}(\delta)e^{\mathsf{m}c_V l^3}\prod_i \|f_i\|_{C^1}$ holds (Prop. YM3_I:88.3); with it, a diagonal subsequence over a countable dense set of tuples and m-linearity give convergence. The properties of the limits need convergence alone and hold for every compact connected G: the bound by uniform

boundedness for m-linear forms on Banach spaces, which is why C depends on the sequence; the translations and the signed permutations from the lattice symmetries, which keep the measure, the action and the observables; reflection positivity from Prop. 9.27(i); and on one torus its translations and the pairs of parallel mirrors at the distance $\frac{l}{2}$. For the additional conclusions assume throughout that G is simply connected and that (P), (V), (C), (II) and (K) hold. The transferred construction in (a), with the sourced estimates in (b), supplies the inputs of the fixed-torus comparison in Thm. YM3_I:A; its proof uses these estimates at the unchanged label and admitted sides. The bounds in (b) give the compactness and distributional limits of Props. YM3_I:92.1–YM3_I:92.4 by the same diagonal extraction and passage of finite positive matrices used above. For Gibbs limits, Prop. 9.27(iii) supplies periodic Gibbs states, their lattice symmetries and all the required reflection families. Conditioning on a translation-invariant event preserves the Gibbs specification by the conditional-expectation argument of Prop. YM3_I:106.1, which uses no special identity of the group. Together with the transferred cumulant and one-point bounds, these are exactly the hypotheses used for the component construction in Thm. YM3_I:AL and its application in Thm. YM3_I:AP. The reflection-Hilbert-space and reconstruction arguments concern the resulting scalar functionals and use no further identity of $SU(2)$. Thus they give the Gibbs conclusions stated in (d). For growing-torus limits this argument supplies coordinate reflection positivity; their full rotational invariance is the criterion proved in Step 10. The colour coefficient in the short-distance law is computed in Step 9.

Step 9: the leading law and the perturbative ratio. The quadratic part of the action at $\mathbf{1}$ is $\frac{1}{2}\sum_p \|dA(p)\|^2$, colour by colour (Lem. 9.15(ii)). In an orthonormal basis of $\mathfrak{g}$, its Gaussian measure is therefore the product of n copies of the one-colour Gaussian measure for $SU(2)$, with the same normalisation. Each colour contributes $3/(16\pi^2)$, and hence

$$c_2(G) = \frac{3n}{16\pi^2}.$$

Under the construction hypotheses, the estimates in (a) and (b) give the effective-action remainders used in the finite-cutoff leading law of Thm. YM3_I:AV. Its colour sum now runs to n, and the label and continuum passage use Thm. YM3_I:AW. The passage to limits is the one in (d). For $SU(2)^r$ the product argument is given in Thm. AB(iii).

For (f), Lem. 9.28 gives the coefficient of β^{-3} first at fixed box and then at infinite volume. Its leading part at large separation is $\varphi_G \mathsf{r}^{-5}/(16\pi^3)$, after summing over orientations. Dividing this coefficient by the leading one gives

$$c_1(G) = \frac{\varphi_G}{16\pi^3} \Big/ \frac{3n}{16\pi^2} = \frac{\varphi_G}{3\pi n}.$$

For $SU(2)$, $\varphi_G = 24$, so that $c_1 = 8/(3\pi)$.

Step 10: clause (g). Prop. 9.27(iv) replaces the formula of the $SU(2)$ construction for each generator. The rest of its proof of the Ward identity at finite cutoff (Prop. YM3_I:112.1) is an integration by parts on G at every cutoff, $\int v_{\mathcal{X}}\Phi \,\mathrm{d}V = -\int \Phi \,\mathrm{div}\, v_{\mathcal{X}} \,\mathrm{d}V$ for a field of directions $\mathcal{X}$, and it is unchanged but for the divergence of the rotation field, which carries $\frac{1}{2}k_p$ where the $SU(2)$ construction has $\frac{3}{2}a_p$. For $\omega \in \mathfrak{so}(3)$ let $(L_\omega f)(y) = \sum_\rho (\omega y)_\rho \,\partial_\rho f(y)$, let $e^{t\omega}f$ be the tuple of the functions $f_i(e^{-t\omega}\,\cdot\,)$, and let $\nabla^{\mathrm{s}}_\omega$ be the symmetric lattice difference along ωy, $(\nabla^{\mathrm{s}}_\omega \phi)(y) = \sum_\rho (\omega y)_\rho (2a)^{-1}[\phi(y+ae_\rho) - \phi(y-ae_\rho)]$. A quadruple $(\omega, \mathsf{m}, \mathcal{R}, f)$ is admitted (Section YM3_I:112, before equation YM3_I:(2115)) if $\omega \in \mathfrak{so}(3)$, $\mathsf{m} \geq 2$, $\mathcal{R} > 0$ and the f_i are smooth, with compact pairwise disjoint supports in $\{|y| < \frac{1}{2}\mathcal{R}\}$. Let (K_j, l_j, β_j) be a sequence of tori as in (d), $K_j \to \infty$, with $l_j \to \infty$. For the j with $2\mathcal{R} + 2a_{K_j} < \frac{1}{2}l_j$ the ball $\{|y| < \frac{5}{3}\mathcal{R} + a_{K_j}\}$, which carries the cut-off of radius $\mathcal{R}$ and its shell, has a diameter below l_j and lies in the torus without wrapping, so the bulk term and the flux term of the identity at the cut-off radius $\mathcal{R}$ are defined; $j_1(\mathcal{R}, f)$ is the least index from which on $2\mathcal{R} + 2a_{K_j} < \frac{1}{2}l_j$, $2\sqrt{3}\,a_{K_j} \leq \frac{1}{2}\mathcal{R}$ and $a_{K_j} \leq \frac{1}{4}\min\{1, \mathrm{sep}(f)\}$, $\mathrm{sep}(f)$ the least distance of two of the supports. For $j \geq j_1(\mathcal{R}, f)$ the identity at the cutoff K_j, with the

radial cut-off $\chi_{\mathcal{R}}(y) = \chi_0(|y|/\mathcal{R})$ (choose χ_0 smooth, equal to 1 on $[0, 4/3]$ and zero on $[5/3, \infty)$, as permitted in Section YM3_I:112), which equals 1 on $\{|y| \le \frac{4}{3}\mathcal{R}\}$ and so covers the f_i, reads

$$\sum_i S^{(j)}_{\mathsf{m}}(f_1, \dots, \nabla^{\mathrm{s}}_\omega f_i, \dots, f_{\mathsf{m}}) = \mathfrak{U}^{(j)}_{\omega,\mathcal{R}}(f) + \mathrm{Fl}^{(j)}_{\omega,\mathcal{R}}(f),$$

the bulk term $\mathfrak{U}^{(j)}_{\omega,\mathcal{R}}(f)$ collecting the spin and orbital insertions, and the flux $\mathrm{Fl}^{(j)}_{\omega,\mathcal{R}}(f)$ being the connected function of the $\mathcal{O}(f_i)$ with the shell flux, a sum over the plaquettes p with $\frac{4}{3}\mathcal{R} - a < |x(p)| < \frac{5}{3}\mathcal{R} + a$ that carries $\frac{1}{2}k_p$ (Prop. YM3_I:112.1, under YM3_I:(2116)). In the limit, along such a sequence on which all connected functions converge,

$$\lim_j [\mathfrak{U}^{(j)}_{\omega,\mathcal{R}}(f) + \mathrm{Fl}^{(j)}_{\omega,\mathcal{R}}(f)] = \sum_i S_{\mathsf{m}}(f_1, \dots, L_\omega f_i, \dots, f_{\mathsf{m}}) = -\frac{d}{dt} S_{\mathsf{m}}(e^{t\omega} f)\Big|_{t=0},$$

independently of $\mathcal{R}$ among the radii that admit f: the bound of (d) (Step 8), on the spaces of C^1 functions with supports in fixed disjoint compact neighbourhoods of the supports, with $\|\nabla^{\mathrm{s}}_\omega f_i - L_\omega f_i\|_\sharp \le c\, a^2_{K_j}$ and $\|t^{-1}(e^{t\omega} f_i - f_i) + L_\omega f_i\|_\sharp \le c|t|$, c a number of ω and f_i, gives the limit and the differentiability of $t \mapsto S_{\mathsf{m}}(e^{t\omega} f)$ (Prop. YM3_I:112.3). The conjugates of one $\omega_0 \neq 0$ under the signed permutations span $\mathfrak{so}(3)$, and the one-parameter group of ω_0 with the cubic group generates $O(3)$ (Lem. YM3_I:112.4 and the proof of Thm. YM3_I:AQ), which gives the equivalence at the admitted quadruples (Thm. YM3_I:AQ); the invariance under $O(3)$ passes to the C^1 tuples of (d) by mollifying, with the bound of (d): the mollified entries and their gradients converge uniformly, with supports in fixed disjoint compact sets, on whose C^1 functions that bound is continuous. The limit of $\mathrm{Fl}^{(j)}_{\omega,\mathcal{R}}(f)$ alone is this limit exactly when $\mathfrak{U}^{(j)}_{\omega,\mathcal{R}}(f) \to 0$. When the limit is invariant, the rotations about the third axis and the reflections in planes through it keep the half-space $\{y_3 > 0\}$, commute with the reflection in $\{y_3 = 0\}$ and keep the reflection positive form, so $\mathsf{U}(R)[F] = [(R \oplus 1)F]$, on the classes of the functions F supported in $\{y_3 > 0\}$, is well defined modulo the null space, unitary and fixes the vacuum vector; as $R \oplus 1$ commutes with the translation by te_3, $\mathsf{U}(R)[\tau_{te_3} F] = [\tau_{te_3}((R \oplus 1)F)]$ for $t \ge 0$. Strong continuity comes from the continuity of the S_{m}.

Step 11: clause (h). (h1) is Lem. 9.29 with the bound of Prop. 9.24; (h2) is Prop. 9.18(iv); (h3) is Prop. 9.30(iv), with Lem. 9.29 at $\Gamma = \mathsf{G}_\sharp$, as $|e^{(i)}_k| \le (\kappa_i/\kappa_G)^2 \mathsf{G}_\sharp \le \mathsf{G}_\sharp$ (Prop. 9.24), which gives

$$|x^{(i)}_k - x_k| \le 2\Big(1 - \frac{n_i}{n}\Big)\mathsf{G}_\sharp \frac{L}{L-1}(1 - L^{-k}),$$

and so the same with $\kappa^2_G \mathsf{G}_\sharp$ in the place of $\mathsf{G}_\sharp$, as $\kappa_G \ge 1$; and (h4) is Lem. 9.31(i) with clauses (a)–(g) for each factor, in the forms stated there. □

9.1.11. *Proof of Thm. AB. Step 1: clause (i).* $a(u) = \sum_i \lambda_i a_i(u_i)$ and the Haar measure is a product, so the fields of the factors are independent (Lem. 9.31(i)); cumulants of sums of independent families add, and $\mathcal{O} = \sum_i \lambda_i \mathcal{O}^{(i)}$ gives the factor λ^{m}_i.

Step 2: clause (ii). Prop. YM3_I:52.4 proves covariance of the pointed totals for the actual arrays admitted by Thm. YM3_I:W. Its proof uses the earlier array through the sum in equation YM3_I:(963): earlier pointed totals, constants, and multiples of the plaquette action. The constants and the action are invariant, independently of the counterterm coefficients. Thus the covariance induction applies to these arrays whenever the earlier pointed totals are covariant. So symmetry passes from level to level along every good sequence, $|L^{-1}x_{j-1} - x_j| \le B_\sharp$ and $x_j \ge \gamma^{-2}$, and along it the coefficient of $SU(2)$ is defined, real and at most $\mathsf{G}_\sharp$, and Thm. YM3_I:D holds whatever its quadratic leftovers. The sequences $\lambda_i x_j$ are good: their increments are $\lambda_i \frac{1}{r} \sum_m \lambda^{-1}_m c^{(m)}_j$, at most

$\lambda_+\lambda_-^{-1}\mathsf{G}_\sharp \le B_\sharp$, and $\lambda_i x_j \ge \lambda_- x_j \ge \gamma^{-2}$. Then

$$|x_k^{(i)} - x_k| \le \mathsf{G}_\sharp\Big(\lambda_i^{-1} + \frac{1}{r}\sum_m \lambda_m^{-1}\Big)\frac{L}{L-1}(1-L^{-k}) \le 2\lambda_-^{-1}\mathsf{G}_\sharp\frac{L}{L-1}.$$

Step 3: clause (iii). The first-passage arithmetic of Prop. YM3_I:74.1 and Cor. YM3_I:74.2 applies to the mean run, whose increments are real and bounded by $B_\sharp$. The windows $I(g_{i,1}) = (\lambda_i g^{-2}, L\lambda_i g^{-2} - 2\bar{B}]$ and $I(g_{i,2}) = (L\lambda_i g^{-2} - 2\bar{B}, L^2\lambda_i g^{-2} - 2(L+1)\bar{B}]$ are adjacent, and $\lambda_i a_K \beta$ lies in one of them, $I(g_{i,s(i)})$, so $(K, \lambda_i\beta)$ is a first-passage datum of the factor i at the reference coupling $g_{i,s(i)}$, $s(i)$ being read off $(K, \lambda_i\beta)$. The law of the continuum theorem for the factor i is cited at $g_{i,s(i)}$, with the hypotheses on the scale, the cutoff and the test pair that (iii) lays at the $2r$ couplings, and on tori whose side is admitted at $g_{i,s(i)}$, which the hypothesis that the sides are admitted at all $2r$ couplings $g_{i,s}$ provides; these laws and (i) give

$$\sum_i \lambda_i^2 \frac{9}{16\pi^2}(\lambda_i \mathsf{b})^{-2}\mathcal{K}_6 = \frac{3n}{16\pi^2}\,\mathsf{b}^{-2}\mathcal{K}_6.$$

In a Gibbs state with periodic marginals the factors are independent given the tails, the diagonal entries are the laws of the factors in their periodic marginals, and the mixed entries are bounded by Lem. 9.31(iii), whose range $a_K \le \mathsf{d} \le \frac{1}{20}$ holds, as the hypotheses of the law on the scale and the cutoff give $a_K < \mathsf{d} \le \frac{1}{200}$; as $\sum_{i\ne j}\lambda_i\lambda_j\frac{1}{4}\varepsilon^\times\frac{3}{16\pi^2}(\lambda_i\lambda_j\mathsf{b}^2)^{-1} = \frac{r-1}{12}\varepsilon^\times\cdot\frac{9r}{16\pi^2}\mathsf{b}^{-2}$, this gives the error term $\frac{r-1}{12}\varepsilon^\times(\mathsf{d})$, the divergent parts never mixing the factors. □

9.1.12. *Scope of the conclusions.* The full construction for a simply connected simple group is conditional on §9.1.1. Its two principal outputs are the source stability in Thm. AA(b) and the resulting bound per ball. The latter yields subsequential limits. In contrast, the algebraic coefficients and the finite-box perturbative expansion require no global construction hypothesis. The error constant in that expansion depends on the box, so its stated limit is iterated.

For any compact connected group, convergence of the connected functions along a sequence of tori gives the translation symmetry, cubic symmetry and reflection positivity stated in Thm. AA(d). Its rotational Ward identity gives the criterion in (g): full $O(3)$ invariance is equivalent to the vanishing of the indicated bulk-plus-flux limit. No vanishing assertion is included in that criterion.

Products acting on separate blocks are governed by exact factorisation. For tied simple ideals, the one-loop calculation is block diagonal at the trivial field, while the nonlinear action couples the ideals. The assertion that the mean run suffices has property (T) as its explicit hypothesis. Quotients require the sector distinctions in Prop. 9.33. The regular-field construction here assumes a semisimple Lie algebra: a nonzero centre obstructs the box gauge in Lem. 9.12(iv), and invariant linear forms then need not vanish.

9.2. **Geometric convergence of the Gaussian one-loop coefficient.** The bounded Gaussian coefficient has a limit as the number of averaging steps increases. We prove this by comparing two successive meshes in units in which the terminal lattice has spacing one. Throughout this section the group is SU(2), $L \ge 13$ is odd, and the averages, Gaussian normalisers and coefficient functional are those of Sections YM3_I:42–YM3_I:43. In particular, Thm. YM3_I:O gives

$$c_k = \mathsf{b}[\Psi_k] = \sum_{l=1}^{k} L^{-(k-l)} b_l^0, \qquad c_0 = 0, \qquad c_2^{\mathrm{loop}} := \sup_{k\ge 0}|c_k| < \infty.$$

We write c_k here for the numbers denoted by c_k elsewhere.

Theorem AC. There are real numbers c_∞ and $b_\infty^0 = (1-L^{-1})c_\infty$, depending only on L, and a finite constant $C_{\mathrm{geo}} = C_{\mathrm{geo}}(L)$ such that

$$|c_k - c_\infty| \le \tfrac{1}{2}C_{\mathrm{geo}}L^{-k/6} \quad (k \ge 0), \qquad |b_l^0 - b_\infty^0| \le C_{\mathrm{geo}}L^{-l/6} \quad (l \ge 1).$$

Moreover

$$|b^0_{l+1} - b^0_l| \le (1 - L^{-1/6}) C_{\mathrm{geo}} L^{-l/6} \quad (l \ge 1).$$

■□

The local determinant analysis already gives a uniformly holomorphic remainder. The additional issue is convergence of that remainder on changing Fourier spaces. We first remove the explicit massive term, then compare the remaining finite expression. The comparison will be made at real momentum; a separate finite-difference argument will recover the derivative defining the coefficient.

9.2.1. *The explicit massive term.* Put $\eta = L^{-k}$ and $\psi(q) = \sum_\mu (2 - 2\cos q_\mu)$, with normalised integration over $[-\pi, \pi]^3$. The massive determinant calculation in Prop. YM3_I:43.3 gives fixed real trigonometric polynomials g_1, g_2 such that

$$\widetilde{\beta}(z) = I_2[g_2](z) + I_1[g_1](z), \qquad I_n[g](z) = \int (\psi(q) + z)^{-n} g(q)\, \mathrm{d}q.$$

More explicitly, $g_2 = 4 \operatorname{tr} \sigma_1^2 - \psi_{11}\psi_{22}/3$ and $g_1 = -8 \operatorname{tr} \sigma_2 + (\psi^{(1)} + \psi^{(2)})/2$ in that calculation. Define

$$E(\eta) = \eta[\widetilde{\beta}(\eta^2) - \eta^2 \widetilde{\beta}'(\eta^2) + \tfrac{1}{2}\eta^4 \widetilde{\beta}''(\eta^2)].$$

With $\delta = \delta_{\mathrm{loop}} \le 1/64$, let

$$F_k(p) = \widehat{\pi}^0_{12}(p) - \mathcal{L}^\sharp_k(p), \qquad r_k = c_k - E(L^{-k}).$$

The determinant representation in Section YM3_I:43.3, Prop. YM3_I:43.2 and the massive calculation give

$$r_k = -\partial_1 \partial_2 F_k(0), \qquad |F_k(p)| \le C_{\mathrm{rem}} \quad (|p_\mu| < \delta), \tag{9.xix}$$

with C_{rem} independent of k.

Lemma 9.34. There is an absolute real number E_∞ such that $C_E := \sup_{0<\eta\le 1} \eta^{-1} |E(\eta) - E_\infty| < \infty$. ■

Proof. Differentiation under the integral gives $I_n' = -nI_{n+1}$. Writing $z = \eta^2$, we obtain

$$E(\eta) = z^{1/2}(I_2 + 2zI_3 + 3z^2 I_4)[g_2](z) + z^{1/2}(I_1 + zI_2 + z^2 I_3)[g_1](z).$$

For a fixed trigonometric polynomial g, put $i_n = (2\pi)^{-3} \int_{\mathbb{R}^3} (1 + |y|^2)^{-n}\, \mathrm{d}y$. We claim

$$|I_1[g](z)| \le C_g, \qquad |z^{n-3/2} I_n[g](z) - g(0) i_n| \le C_g \sqrt{z} \quad (n = 2, 3, 4; 0 < z \le 1). \tag{9.xx}$$

Indeed, on the integration cube,

$$a|q|^2 \le \psi(q) \le |q|^2, \qquad 0 \le |q|^2 - \psi(q) \le |q|^4/12, \qquad a = 4/\pi^2.$$

The first bound follows by integrating $C|q|^{-2}$ over a fixed ball. For the others replace g by its even part, whose difference from $g(0)$ is $O(|q|^2)$. The error is bounded by $C \int_{|q| \le \sqrt{3}\pi} |q|^2 (a|q|^2 + z)^{-n}\, \mathrm{d}q$. Replacing ψ by $|q|^2$ costs at most $C \int_{|q| \le \sqrt{3}\pi} |q|^4 (a|q|^2 + z)^{-n-1}\, \mathrm{d}q$. For either integral, multiplication by $z^{n-3/2}$ gives $O(\sqrt{z})$ when $n = 2$, by the integrable radial majorant $C|q|^{-2}$, and $O(z)$ when $n = 3, 4$, by $q = \sqrt{z}\, y$. Extending the cube to $\mathbb{R}^3$ costs $O(z^{n-3/2})$, since its complement has $|q| \ge \pi$. The same substitution then gives i_n exactly. This proves eq. 9.xx.

It follows that $E_\infty = (i_2 + 2i_3 + 3i_4) g_2(0)$ and $E(\eta) - E_\infty = O(\eta)$. The contribution of g_1 is $O(\sqrt{z})$ by the same estimates. Reality follows from that of the determinant coefficient. Since E is defined by $\widetilde{\beta}$, its limit and C_E do not depend on a choice of the polynomial representation. □

9.2.2. *A finite expression for the remainder.* We use the physical pairings of weight ℓ_j^3 on $\ell_j\mathbb{Z}^3$, $\ell_j = L^{j-k}$, and the scaled exterior derivative. The operators $D_U, Q_U, R_U, K_U, \Delta_U, \Xi_U, Y_U$ are those defined before Section YM3_I:43.2. Thus $Q_U = \eta\mathring{\mathcal{Q}}_U^{[k]}$ and $R_U = \mathfrak{a}_U^{[k]}$. Unsubscripted d, K, Δ, Q, R denote their free values, and $G = (K+1)^{-1}$, $g = (\Delta+1)^{-1}$. The response is $h = \Xi^{-1}Q^{\mathsf{T}}(Q\Xi^{-1}Q^{\mathsf{T}})^{-1}$. Along

$$\zeta_\pm = \eta e^{\pm ip\cdot x}\alpha_\pm, \qquad \alpha_\pm = \widehat{h}(\pm p)e_{\mu_\pm}t_3,$$

bold letters denote the jets with components $00, 10, 01, 11$. The two nonzero field directions and their curls are bounded uniformly in k on the fixed strip, by Section YM3_I:43.2. In this subsection either pair of directions is allowed; we later take $(\mu_+, \mu_-) = (1, 2)$.

For completeness, the real domains for these jets are $K^{00} = K = [-\pi-\delta, \pi+\delta]^3$, $K^{11} = [-\pi, \pi]^3$ and $K^{10} = K \cap (K-p)$, $K^{01} = K \cap (K+p)$. Their norm is the sum of the four supremum operator norms. Products are the jet products: if $a, b \in \{0, 1\}$, then

$$(\mathbf{T}^1 \cdots \mathbf{T}^s)_{ab}(\varkappa) = \sum_{\sum a_i = a,\, \sum b_i = b} \prod_{i=1}^{s} T^i_{a_i b_i}\left(\varkappa + \sum_{j>i}(a_j - b_j)p\right). \tag{9.xxi}$$

Every argument lies in its stated domain. Consequently the norm is submultiplicative, and a product containing three factors with zero 00 component vanishes. Transposes use the bilinear pairing; their arguments are $-\varkappa$, $-\varkappa - p$, $-\varkappa + p$, $-\varkappa$, respectively, and preserve the norm. Put

$$\mathsf{t}(\mathbf{T}) = \int \operatorname{tr} T_{11}(\varkappa)\,\mathrm{d}\varkappa,$$

and, when C_{00} is invertible,

$$\ell(\mathbf{C}) = \int \operatorname{tr}[C_{00}^{-1}C_{11} - C_{00}(\varkappa)^{-1}C_{10}(\varkappa - p)C_{00}(\varkappa - p)^{-1}C_{01}(\varkappa)]\,\mathrm{d}\varkappa.$$

The integrals are normalised over $[-\pi, \pi]^3$.

The local Hessian formula of Section YM3_I:43.4 reads $H_U = (d + E_U)^{\mathsf{T}} z_U (d + E_U) + f_U$. Write $\mathbf{e} = \mathbf{D} - d$, and define

$$\kappa = G^{1/2}(\mathbf{K} - K)G^{1/2}, \quad x = \mathbf{Q}G^{1/2}, \quad y = G^{1/2}\mathbf{D}\mathbf{R}^{\mathsf{T}}, \qquad \kappa_g = g^{1/2}(\mathbf{\Delta} - \Delta)g^{1/2}, \quad x_g = \mathbf{R}g^{1/2}.$$

All of these are jets. For either datum $(\mathsf{G}, \kappa, x, y) = (G, \kappa, x, y)$ or $(g, \kappa_g, x_g, 0)$, put

$$\omega = x^{\mathsf{T}}x - yy^{\mathsf{T}}, \quad B_t = (1 + \kappa + t\omega)^{-1}, \quad B_\tau^\flat = (1 + \kappa + \omega - (1-\tau)\mathsf{G})^{-1}.$$

The strip estimate, Prop. YM3_I:43.1, and the bounded jets in Section YM3_I:43.4 bound all these jets and inverses by one constant $A = A(L)$. Indeed the four free 00 inverses are precisely the sandwiched inverses in that proposition; the inverse jet follows by multiplying out its $10, 01, 11$ components, each involving at most three such inverses. The 00 inverse of $xB_0^\flat x^{\mathsf{T}}$ is bounded by a constant ι, also uniformly in k. These statements hold for $0 \le t, \tau \le 1$, continuously in those parameters.

For these data define the following explicit expression:

$$\begin{aligned}
\Phi[\mathsf{G}, \kappa, x, y] = {} & \int_0^1 [\mathsf{t}(xB_t x^{\mathsf{T}}) - \mathsf{t}(y^{\mathsf{T}}B_t y)]\,\mathrm{d}t + \ell(xB_0^\flat x^{\mathsf{T}}) \\
& - \int_0^1 (1-\tau)^2 \mathsf{t}(\mathsf{G}B_0\mathsf{G}B_0\mathsf{G}B_\tau^\flat)\,\mathrm{d}\tau \\
& + \int_0^1 [\mathsf{t}(xB_\tau^\flat\mathsf{G}B_0 x^{\mathsf{T}}) - \mathsf{t}(y^{\mathsf{T}}B_\tau^\flat\mathsf{G}B_0 y)]\,\mathrm{d}\tau \\
& + \int_0^1 (1-\tau)[\mathsf{t}(xB_\tau^\flat\mathsf{G}B_0\mathsf{G}B_0 x^{\mathsf{T}}) - \mathsf{t}(y^{\mathsf{T}}B_\tau^\flat\mathsf{G}B_0\mathsf{G}B_0 y)]\,\mathrm{d}\tau.
\end{aligned} \tag{9.xxii}$$

Lemma 9.35. For real $|p_\mu| \le \delta$,

$$\widehat{\pi}^0_{\mu+\mu-}(p) - \mathcal{L}^\sharp_k(p) = -\tfrac{1}{2}\Phi[G,\kappa,x,y] + \Phi[g,\kappa_g,x_g,0].$$

■

Proof. Set $V = \mathbf{Q}^{\mathsf{T}}\mathbf{Q} - \mathbf{D}\mathbf{R}^{\mathsf{T}}\mathbf{R}\mathbf{D}^{\mathsf{T}}$. The Schur factorisation of the vector saddle matrix gives $\ell(\mathbf{\Xi}) + \ell(\mathbf{Q}\mathbf{\Xi}^{-1}\mathbf{Q}^{\mathsf{T}})$; the scalar version gives $\ell(\mathbf{Y}) + \ell(\mathbf{R}\mathbf{Y}^{-1}\mathbf{R}^{\mathsf{T}})$. The triangular factors have determinant one and hence zero polarisation. The coarse factors are invertible by the strip estimate, so these identities take place inside the same jet algebra.

For either pair $(K_\circ, V_\circ) = (\mathbf{K}, V)$ or $(\mathbf{\Delta}, \mathbf{R}^{\mathsf{T}}\mathbf{R})$, let $G_U = (K_\circ + 1)^{-1}$ and $P_\tau = (K_\circ + V_\circ + \tau)^{-1}$. Integration of the logarithmic derivative along $tV_\circ$ and then along τ gives

$$\ell(K_\circ + V_\circ) = \ell(K_\circ + 1) + \int_0^1 \mathsf{t}((K_\circ + 1 + tV_\circ)^{-1}V_\circ)\,\mathrm{d}t - \int_0^1 \mathsf{t}(P_\tau)\,\mathrm{d}\tau.$$

Twice using the resolvent identity yields exactly

$$P_\tau = G_U + (1-\tau)G_U^2 + (1-\tau)^2 G_U^2 P_\tau - G_U V_\circ P_\tau - (1-\tau)G_U^2 V_\circ P_\tau.$$

The first two terms give $[1 - \partial_z + \frac{1}{2}\partial_z^2]\ell(K_\circ + z)|_{z=1}$. Substitute $G_U = \mathsf{G}^{1/2}B_0\mathsf{G}^{1/2}$, $P_\tau = \mathsf{G}^{1/2}B_\tau^\flat\mathsf{G}^{1/2}$ and $\mathsf{G}^{1/2}V_\circ\mathsf{G}^{1/2} = \omega$ in the remaining terms. Cyclicity of t gives the four integrals in eq. 9.xxii, and the coarse Schur factor gives its last logarithmic term. Finally take minus one half of the vector identity plus the scalar identity. This also explains why the only fine-space trace left has three propagator factors. □

9.2.3. *Comparing the two Fourier spaces.* We compare height k with height $k+1$; primes indicate the latter. Their bare spacings are η and $\eta' = \eta/L$, and all later lattices agree: $\Lambda'_{j+1} = \Lambda_j$. Let $N = (\eta^{-1} - 1)/2$ and $\mathfrak{m} = \{m \in \mathbb{Z}^3 : |m|_\infty \le N\}$, with $\mathfrak{m}'$ defined at η'. Fourier coefficients identify each fine site, bond or plaquette fibre isometrically with $\ell^2(\mathfrak{m};\mathsf{V})$, where V is its colour and component space. The bilinear pairing pairs m with $-m$. Let J extend coefficients by zero from $\mathfrak{m}$ to $\mathfrak{m}'$, and let $J = 1$ on the common coarse fibres. Then $J^{\mathsf{T}} = J^*$ and $J^*J = 1$. For $1 \le \Lambda \le N$ let P_Λ, P'_Λ retain the modes $|m|_\infty \le \Lambda$. These projections commute with J in the corresponding sense and are their own transposes.

For two jets between corresponding fibres define $\mathrm{df}(T', T) = \|T'J - JT\|$ in the real jet norm. The product rule eq. 9.xxi gives

$$\begin{aligned}
\mathrm{df}(T'S', TS) &\le \|T'\|\,\mathrm{df}(S', S) + \mathrm{df}(T', T)\|S\|,\\
\mathrm{df}(T'^{\mathsf{T}}, T^{\mathsf{T}}) &\le \mathrm{df}(T', T) + \|T'(1 - JJ^*)\|,\\
\mathrm{df}((1+S')^{-1}, (1+S)^{-1}) &\le \|(1+S')^{-1}\|\,\mathrm{df}(S', S)\|(1+S)^{-1}\|.
\end{aligned} \tag{9.xxiii}$$

The first and third follow by inserting J between the factors. For the second, insert $JJ^* + (1 - JJ^*)$ before transposing. This extra tail will be retained throughout the comparison.

Here are the elementary multiplier estimates we need. For $h = \eta, \eta'$ and $q = \varkappa + 2\pi m$, write

$$d^h_\mu(q) = h^{-1}(e^{ihq_\mu} - 1), \quad D_h(q) = \sum_\mu |d^h_\mu(q)|^2, \quad s_h(t) = h \sum_{|j| \le (h^{-1}-1)/2} e^{ihjt},$$

$$v_h(q) = \prod_\nu s_h(q_\nu), \qquad u_{h,\mu}(q) = e^{i(1-h)q_\mu/2}s_h(q_\mu)v_h(q), \qquad g_h(q) = (1 + D_h(q))^{-1/2}.$$

Define $s_1(t) = L^{-1}\sum_{|j| \le (L-1)/2} e^{i\eta' jt}$, $v_1 = \prod s_1$ and $u_{1,\mu} = e^{i(\eta - \eta')q_\mu/2}s_1(q_\mu)v_1$. Division with centred remainder gives

$$s_{\eta'} = s_\eta s_1, \qquad v_{\eta'} = v_\eta v_1, \qquad u_{\eta',\mu} = u_{\eta,\mu}u_{1,\mu}. \tag{9.xxiv}$$

The free finest-step averages therefore act by

$$(Q^{\mathrm{f}}\varphi')_{\mu,m}=\sum_{m'\equiv m}u_{1,\mu}(q_{m'})\varphi'_{\mu,m'},\qquad (R^{\mathrm{f}}\varphi')_{m}=\sum_{m'\equiv m}v_{1}(q_{m'})\varphi'_{m'},\tag{9.xxv}$$

where congruence is modulo $\eta^{-1}\mathbb{Z}^3$. They have norm at most one. The transpose of the second map sends φ_m to $v_1(q_{m'})\varphi_m$ at every such alias. These formulas follow by averaging each plane wave over the centred block and, for Q^{f}, over its straight path.

Put $w(n)=\max(1,2|n|-1)^{-1}$, $w^{(3)}(m)=\prod_\nu w(m_\nu)$ and $w_3(m)=w^{(3)}(m)/\max(1,|m|_\infty)$. For $\varkappa\in K$ and the respective mode sets,

$$\begin{gathered}D_h(q)\geq\tfrac{9}{25}|q|^2,\quad |u_{h,\mu}(q)|,|v_h(q)|\leq 2w^{(3)}(m),\quad |q|\leq 11(1+|m|_\infty),\\ |g_\eta-g_{\eta'}|\leq 6\eta,\quad |g_\eta^2-g_{\eta'}^2|\leq 3\eta^2,\quad |d_\mu^{\eta'}g_{\eta'}-d_\mu^{\eta}g_{\eta}|\leq 8\eta|q|,\\ \sum_{|m|_\infty\geq\Lambda}w_3(m)\leq 252\Lambda^{-1/2},\quad \sum_m w_3(m)\leq 253,\quad \eta(3+\log\eta^{-1})^3\leq 49\eta^{1/2}.\end{gathered}\tag{9.xxvi}$$

For the middle line the modes are common modes. To verify the bounds, use $h^{-1}|\sin(ht/2)|\geq 3|t|/10$ for $|t|\leq\pi/h+1$, and $h^{-1}\sin(ht/2)s_h(t)=\sin(t/2)$. Taylor's formula gives $|d^\eta-d^{\eta'}|\leq\eta|q|^2$ and $|D_\eta-D_{\eta'}|\leq\eta^2|q|^4/12$; the root estimates follow by subtracting the two positive square roots. Also

$$|1-v_1|\leq\eta^2|q|^2/8,\quad |1-u_{1,\mu}|\leq\eta|q_\mu|/2+\eta^2|q|^2/4,$$

and $1-|u_{1,\mu}|^2,1-v_1^2\leq\eta^2|q|^2/2$. For the sums, $\sum_n w(n)^2\leq 4$ and its tail from n_0 is at most $4/n_0$. On the shell $|m|_\infty=n$,

$$\sum w_3(m)\leq 6n^{-2}(3+\log 2n)^2\leq 84n^{-3/2}.$$

Summing the shells proves the displayed tail. The final bound follows by maximising $e^{-u/2}(3+u)^3$ at $u=3$.

9.2.4. *The backgrounds at the two heights.* The comparison must include the response fields themselves. Write their components as $\alpha_{\pm,\mu}=a_\mu^{\pm}t_3$ and $f_{\pm,\mu\nu}=f_{\mu\nu}^{\pm}t_3$, where $d(e^{\pm ipx}\alpha_\pm)=e^{\pm ipx}f_\pm$. For any one of these scalar functions write $\widehat{a}_m$ for its Fourier coefficient. Constants below depend only on L.

Lemma 9.36. For common modes,

$$|\widehat{a}_m|,|\widehat{a}'_m|\leq Cw_3(m),\qquad |\widehat{a}'_m-\widehat{a}_m|\leq C\eta w^{(3)}(m).$$

The first bound also holds on all primed modes. Consequently

$$\sum|\widehat{a}_m|\leq C,\quad \sum_{|m|_\infty\geq\Lambda}|\widehat{a}_m|\leq C\Lambda^{-1/2},\quad \sum_{m\in\mathfrak{m}}|\widehat{a}'_m-\widehat{a}_m|+\sum_{m\in\mathfrak{m}'\setminus\mathfrak{m}}|\widehat{a}'_m|\leq C\eta^{1/2}.\tag{9.xxvii}$$

Moreover $Q^{\mathrm{f}}(e^{\pm ipx}\alpha'_\pm)$ differs from $e^{\pm ipx}\alpha_\pm$ in supremum norm by $C\eta^{1/2}$, and $e^{\pm ipx}a'^{\pm}_\mu(x)$ has oscillation at most $C\eta^{1/2}$ between sites at distance at most η. ∎

Proof. It suffices to use the plus sign. For $b=\widehat{h}(p)\varepsilon$, $n=\widehat{\nu}(p)\varepsilon$ and $\gamma=\widehat{R}(p)\widehat{d^{\mathsf{T}}}(p)b$, Section YM3_I: 43.2 gives

$$D_\eta(q_m)b_{\mu,m}=u_{\eta,\mu}(-q_m)n_\mu+d_\mu^\eta(q_m)v_\eta(q_m)\gamma,\qquad \|b\|,|n|,|\gamma|\leq C_h|\varepsilon|.\tag{9.xxviii}$$

For $m\neq 0$, $|q_m|\geq\pi|m|_\infty$. The multiplier bounds give $|b_m|\leq Cw_3(m)$. Taking a curl cancels the two γ terms and gives

$$f_{\mu\nu,m}=\frac{d_\mu(q_m)u_\nu(-q_m)n_\nu-d_\nu(q_m)u_\mu(-q_m)n_\mu}{D_\eta(q_m)},$$

which obeys the same bound. The zero modes are bounded by the response norm.

To compare zero modes without dividing by $|p|$, use $Qh = 1$, $d^{\mathsf{T}}\nu = 0$ and $\sum_m v(q_m)^2 = 1$. They imply

$$\gamma = v(p)^{-1}\sum_\nu d_\nu(-p)b_{\nu,0}, \qquad n_\mu = u_\mu(-p)^{-1}[D_\eta(p)b_{\mu,0} - d_\mu(p)\sum_\nu d_\nu(-p)b_{\nu,0}].$$

Here $v(p) \geq (2/\pi)^3$ and $|u_\mu(p)| \geq (2/\pi)^4$. Set $A_\mu = \sum_{m\neq 0}|u_\mu(q_m)|^2/D_\eta(q_m)$ and $A_0 = \sum_{m\neq 0} v(q_m)^2/D_\eta(q_m)$. Write $d_\mu = d^\eta_\mu$ and $d^{\mathrm{c}}_\mu(p) = e^{ip_\mu} - 1$. The constraint becomes $Hb_0 = \varepsilon$, where

$$H_{\mu\nu} = \delta_{\mu\nu}[u_\mu(p) + A_\mu D_\eta(p)/u_\mu(-p)] + [d^{\mathrm{c}}_\mu(p)A_0/v(p) - A_\mu d_\mu(p)/u_\mu(-p)]d_\nu(-p).$$

This holds for every ε, so H is invertible and $\|H^{-1}\| \leq C_h$ by the response bound. Factorisation eq. 9.xxiv gives $|A'_\mu - A_\mu| + |A'_0 - A_0| \leq C\eta^2$: on common nonzero modes the reciprocal-denominator and averaging defects are $O(\eta^2)$ times $|u|^2$ or v^2, whose sums are at most one; on added modes $|q_m|^{-2} \leq C\eta^2$. All other entries in H change by $O(\eta)$, with denominators bounded away from zero. Thus $H' - H = O(\eta)$ and $b'_0 - b_0, n' - n, \gamma' - \gamma = O(\eta)$.

Subtract eq. 9.xxviii and the curl formula at a common nonzero mode, one factor at a time. The defects of u, v, d, D^{-1} are bounded respectively by $C\eta|q|w^{(3)}$, $C\eta^2|q|^2w^{(3)}$, $\eta|q|^2$ and $C\eta^2$. Since $|q| \geq 1$ and $\eta|q| \leq 6$, each resulting term is $O(\eta w^{(3)}(m))$. This proves the coefficient comparison. Summation uses eq. 9.xxvi; in particular $\eta\sum_{|m|_\infty\leq N} w^{(3)}(m) \leq \eta(3 + \log\eta^{-1})^3$. The added-mode tail starts at $N + 1 \geq (2\eta)^{-1}$.

Finally use the alias formula eq. 9.xxv. The principal alias contributes the coefficient difference and $(u_1 - 1)\widehat{a}'_m$, bounded by $C\eta(1 + |m|_\infty)w_3(m)$; other aliases are controlled by the added-mode tail. For the oscillation, multiply the coefficients by $\min(2, 19\eta(1 + |m|_\infty))$ and split at N. The same two sums give $C\eta^{1/2}$. □

9.2.5. *The actual averaging maps.* We now compare the field-dependent averages. This requires their actual local definitions: equality of the free multipliers alone would not identify their derivatives. Let ρ_*, C_* be the uniform field radius and bounds from Section YM3_I:43.4, and C_h the response bound from Section YM3_I:43.2. Enlarge C_* to be at least 16, so it also bounds $\ell_j^{-1}|\mathcal{M}^j(e^{i\eta a}) - 1|_\infty/|a|_\infty$. Decrease ρ_* if necessary so that $\rho_* \leq 1$, and put

$$r_1 = \frac{\rho_*}{5C_*}, \qquad r_2 = \frac{\rho_*}{2C_*L}, \qquad r_0 = \frac{r_1}{2C_h}.$$

For $|s_\pm| < r_0$ the fields $a(s) = s_+e^{ipx}\alpha_+ + s_-e^{-ipx}\alpha_-$ and $a'(s)$ have supremum norm less than r_1.

A dilation taking one averaging step to another takes centred blocks, coordinate contours and their orders to the corresponding blocks and contours. It therefore preserves their products and the equations defining the local mean. The holomorphic mean is the branch through the identity, so uniqueness on its defining polydisc preserves this branch as well. Differentiating these identities preserves the corrected differential average and the scalar transport average. In the operator norms the factor λ^3 in the physical pairing occurs on both sides and cancels. Thus the estimates for a step with terminal spacing one apply to every step. Splitting the product after the first step is consequently the exact identity

$$\mathbf{Q}' = \mathbf{Q}^\sharp\mathbf{Q}^{\mathrm{f}}, \qquad \mathbf{R}' = \mathbf{R}^\sharp\mathbf{R}^{\mathrm{f}}. \tag{9.xxix}$$

Here the sharp maps are the remaining height-k averages evaluated at $\mathcal{M}(e^{i\eta' a'(s)})$.

We record explicitly how field bounds give bounds on these jets. If a local operator function $T(s)$ is holomorphic on $|s_\pm| < r_0$ and $\|T(s) - T(0)\| \leq M$, pairing with two finitely supported unit vectors and applying Cauchy's formula bounds its $10, 01, 11$ derivatives by $M/r_0, M/r_0, M/r_0^2$. For real p, the first derivative kernels become periodic after multiplication at the output by $e^{\mp ipx}$, an isometry; the mixed derivative kernel is already periodic. The Fourier symbol norm is bounded by the corresponding operator norm. This proves the same bounds in the real jet norm. Applying this

argument to $T(s) - \widetilde{T}(s)$ also proves the difference version, including the 00 component when it is nonzero.

For a single step and an absorbed field A, the uniform bound and Schwarz's lemma on $|A| < \rho_*/L$ give

$$\|L^{-1}(\mathring{\mathcal{Q}}_{e^{iA}} - \mathring{\mathsf{Q}})\| + \|\mathfrak{a}_{e^{iA}} - \mathfrak{a}\| \le 4C_* L\rho_*^{-1}|A|_\infty.$$

Indeed each difference vanishes at zero and is bounded by $2C_*$ on that ball; apply the scalar Schwarz lemma after pairing. For the intermediate logarithm $A_{j-1} = i^{-1}\log \mathcal{M}^{j-1}(e^{i\eta a})$, the uniform mean bound gives $|A_{j-1}|_\infty \le 2C_*\ell_{j-1}|a|_\infty$. The preceding estimate and the jet Cauchy estimate therefore give

$$\mathbf{A}_j = \mathfrak{a} + \mathbf{a}_j, \quad (\mathbf{a}_j)_{00} = 0, \quad \|\mathbf{a}_j\| \le c_a\ell_{j-1}, \qquad \|\mathbf{Q}^{\mathrm{f}} - Q^{\mathrm{f}}\| + \|\mathbf{R}^{\mathrm{f}} - R^{\mathrm{f}}\| \le C\eta, \tag{9.xxx}$$

where $\mathbf{A}_j$ is the jet of the jth scalar step and $c_a = (8C_*^2 L/\rho_*)r_1(2r_0^{-1} + r_0^{-2})$. All these constants depend only on L.

The scalar average also suppresses high Fourier modes:

$$\|\mathbf{R}(1 - P_\Lambda)\|, \|(1 - P_\Lambda)\mathbf{R}^{\mathsf{T}}\| \le 28(1 + c_a)^2\Lambda^{-1/2}. \tag{9.xxxi}$$

To prove this, the product $a^{[n]}$ of the first n free steps has multiplier

$$v^{(n)}(q) = \prod_\nu L^{-n} \sum_{|j|\le (L^n-1)/2} e^{i\eta j q_\nu}$$

. Its output modes have aliases $m = \bar{m} + \ell_n^{-1}e$. The sine identity for this finite sum gives $|v^{(n)}(q_m)|^2 \le 2w^{(3)}(e)^2$. If $|m|_\infty > \Lambda$ and $\ell_n\Lambda \ge 1$, some $|e_\nu| > \ell_n\Lambda - 1/2$; thus the sum of the squares in each alias class is at most $768/(\ell_n\Lambda)$. Cauchy–Schwarz gives $\|a^{[n]}(1 - P_\Lambda)\| \le 28(\ell_n\Lambda)^{-1/2}$. When $\ell_n\Lambda < 1$ this follows instead from the contraction bound. Expand $\mathbf{R} = \prod_{j=k}^1(\mathfrak{a} + \mathbf{a}_j)$. There are only the free term, terms with one insertion, and terms with two insertions: three zero-00 jets have zero product. Applying the free tail bound to the rightmost free segment gives

$$\|\mathbf{R}(1 - P_\Lambda)\| \le 28\Lambda^{-1/2}\left(1 + c_a\sum_j \ell_{j-1}^{1/2} + c_a^2\sum_{i<j}\ell_{j-1}\ell_{i-1}^{1/2}\right).$$

Since $\sum_j \ell_{j-1}^{1/2} \le (\sqrt{L} - 1)^{-1} \le 1$, this proves the first bound; transposition proves the second.

It remains to compare the sharp maps with the unprimed maps. For the finest step define

$$\mathcal{N}(A) = i^{-1}\log \mathcal{M}(e^{iA}) - \mathsf{Q}A.$$

On $|A|_\infty \le r_2$, the mean differs from the identity by at most $\rho_*/2$, and hence the logarithm is defined. The logarithm and the linear average bound $\mathcal{N}$ by 3. Its value and first derivative vanish at zero because $\mathcal{M}'(1) = \mathsf{Q}$. Schwarz's lemma for a double zero gives $|\mathcal{N}(A)|_\infty \le 3r_2^{-2}|A|_\infty^2$. The relation between the ordinary and corrected linear averages is $\mathsf{Q} = \mathring{\mathsf{Q}} - d_0\mathsf{n}$, with d_0 the unscaled coarse difference. Therefore, for $A = \eta' a'(s)$,

$$a^\sharp = Q^{\mathrm{f}}a'(s) - \eta^{-1}d_0\mathsf{n}(\eta' a'(s)) + \eta^{-1}\mathcal{N}(\eta' a'(s)). \tag{9.xxxii}$$

The operator n annihilates constant forms; its paths have at most $3L/2$ bonds within one block. Subtract a constant value on each such block and use the oscillation bound in Lem. 9.36. The middle term in eq. 9.xxxii is bounded by $C\eta^{1/2}$; the last is bounded by $C\eta$. The first differs from $a(s)$ by $C\eta^{1/2}$. Thus $|a^\sharp - a(s)|_\infty \le C\eta^{1/2}$ on the fixed bidisc. Both fields remain within ρ_*, and $|a(s)| < \rho_*/2$.

The uniformly bounded full averages are Lipschitz between such fields, with constant $4C_*/\rho_*$. To see this, restrict to the complex line from the first field to the second. If their distance is at most

$\rho_*/2$, Schwarz's lemma on the disc of radius $\rho_*/(2|a_1-a_0|)$ gives the bound; for a larger distance the bound $2C_*$ suffices. Applying the difference Cauchy estimate on the bidisc now proves

$$\|\mathbf{Q}^\sharp-\mathbf{Q}\|+\|\mathbf{R}^\sharp-\mathbf{R}\|\le C\eta^{1/2}. \tag{9.xxxiii}$$

9.2.6. *The local multiplication terms.* We next compare the curvature and covariant-derivative jets. The comparison uses their actual first derivatives, including the lattice shifts. For a periodic scalar function a, let

$$M_h[a;\varpi;\chi_1,\chi_2]_{m,\widetilde{m}}=\widehat{a}_{m-\widetilde{m}}e^{ih(\varpi+2\pi(m-\widetilde{m}))\cdot\chi_1}e^{ihq_{\widetilde{m}}\cdot\chi_2},$$

where the coefficient index is read modulo $h^{-1}\mathbb{Z}^3$, $\varpi=\pm p$ and $\chi_1,\chi_2\in\{-1,0,1\}^3$. In position space this is multiplication by $e^{i\varpi\cdot(x+h\chi_1)}a(x+h\chi_1)$ followed by the shift $x\mapsto x+h\chi_2$. Schur's inequality gives $\|M_h\|\le\sum_m|\widehat{a}_m|$. If $3\Lambda\le N$, it also gives

$$\begin{aligned}\|(1-P_{2\Lambda})M_\eta P_\Lambda\|&\le\sum_{|m|_\infty\ge\Lambda}|\widehat{a}_m|,\\ \|P'_{2\Lambda}(M_{\eta'}J-JM_\eta)P_\Lambda\|&\le\sum_{|m|_\infty\le3\Lambda}|\widehat{a}'_m-\widehat{a}_m|+90\eta\Lambda\sum_m|\widehat{a}_m|.\end{aligned} \tag{9.xxxiv}$$

For the first inequality an output above 2Λ and an input below Λ have coefficient index at least Λ, even after periodic reduction, since $N+1-\Lambda\ge\Lambda$. For the second the coefficient indices lie below 3Λ, so no reduction is needed. Subtract the two phase factors and use $|e^{iu}-e^{iv}|\le|u-v|$; their difference is at most $90\eta\Lambda$. The row and column sums prove the claim.

Here are the particular multiplication terms in the Hessian. Write $\theta=\operatorname{ad}t_3$, so $\|\theta\|=2$ and $\theta^{\mathsf{T}}=-\theta$. On the commuting background $U=\exp(ihat_3)$,

$$(e_U\lambda)(b)=h^{-1}(e^{iha(b)\theta}-1)\lambda(b_+).$$

Thus its first derivative is $ia_\mu\theta$ followed by the shift he_μ. Its transpose has shifts $(-e_\mu,-e_\mu)$ and coefficient $-i\theta$. The mixed derivative is $-ha^+_\mu a^-_\mu\theta^2$ followed by that shift, hence $O(h)$.

On a positively oriented $\mu\nu$ plaquette enumerate the bonds as bottom, right, top, left, with signs $(1,1,-1,-1)$ and origins $o_1=o_4=0$, $o_2=e_\mu$, $o_3=e_\nu$. For their field values a_i, put $\vartheta_1=0$, $\vartheta_2=a_1$, $\vartheta_3=a_1+a_2-a_3$, $\vartheta_4=h\,\mathrm{d}a$. Then

$$E_Ub=h^{-1}\sum_i\varsigma_i(e^{ih\vartheta_i\theta}-1)b_i,\qquad z_U=\cos(h^2da).$$

The first derivative of E in this plane is

$$i\theta\{a_\mu(x)b_\nu(x+he_\mu)-[a_\mu(x)+a_\nu(x+he_\mu)-a_\mu(x+he_\nu)]b_\mu(x+he_\nu)-h\,\mathrm{d}a(p)b_\nu(x)\}.$$

This is a list of five multiplication terms per plane. Its mixed derivative is $-h\sum_i\varsigma_i\vartheta_i^+\vartheta_i^-\theta^2b_i$, which is $O(h)$. The first derivatives of z vanish and $z_{11}=-h^4f_+f_-$.

Finally the remaining curvature term has quadratic form

$$(b,f_Ub)=-ih^3\sum_p h^{-2}\sin(h^2da(p))\sum_{i<j}\varsigma_i\varsigma_j\langle b_i,\theta e^{ih(\vartheta_j-\vartheta_i)\theta}b_j\rangle.$$

Its first derivative is the symmetrisation of the matrix with entries $-i\varsigma_i\varsigma_j da\,\theta$ from b_j to b_i. Its multiplication terms have offsets $(-o_i,o_j-o_i)$, together with the transposed offsets $(-o_j,o_i-o_j)$, and coefficients of norm at most one. There are 36 terms over the three planes. The mixed derivative comes from differentiating the exponential once and da once; it is a sum of $h(da)(\vartheta_j-\vartheta_i)\theta^2$ and its polarisation, so is $O(h)$. The sine factor has no quadratic term.

Consequently the first jets of e,e^{T},E,f are finite matched lists of the operators in eq. 9.xxxiv, with coefficients $a^\pm_\mu$, $f^\pm_{\mu\nu}$ or $hf^\pm_{\mu\nu}$. The lists and shifts are identical at both heights. Lem. 9.36 controls their coefficient sums and differences. Their mixed jets are $O(\eta)$ and $O(\eta')$; this last observation is what permits the comparison without introducing additional second-order background coefficients.

9.2.7. *Defects of the determinant data.* Let $1 \le \Lambda$ and $3\Lambda \le N$, and put $\epsilon_1 = \Lambda^{-1/2} + \eta\Lambda + \eta^{1/2}$. All constants in the following comparison are independent of k, p, Λ in their stated ranges.

Lemma 9.37. For each of the vector and scalar data in eq. 9.xxii,

$$\mathrm{df}(\kappa', \kappa) + \mathrm{df}(x', x) + \|x'(1 - JJ^*)\| + \mathrm{df}(y', y) \le C\epsilon_1.$$

■

Proof. The free roots $G^{1/2}, g^{1/2}$ have diagonal entries $g_\eta(q_m)$. Denote this diagonal matrix by g_*, and the matrix on the same modes with entries $g_{\eta'}(q_m)$ by $g_\flat$. The elementary bounds imply

$$\|(1 - P_\Lambda)g_*\|, \|(1 - P_\Lambda)g_\flat\| \le 2/\Lambda, \quad \|G'^{1/2}(1 - JJ^*)\| \le 4\eta, \quad \|g_\flat - g_*\| \le 6\eta; \tag{9.xxxv}$$

the corresponding scalar statements hold as well. Indeed a nonzero mode has $|q_m| \ge \pi|m|_\infty/2$, and an added mode has $|m|_\infty \ge N + 1 \ge (2\eta)^{-1}$.

Consider the free blocks $D_1 = dG^{1/2}$, $D_0 = d^\mathsf{T}G^{1/2}$, $D_g = dg^{1/2}$ and their transposes. They are diagonal in the Fourier modes, with matrices of at most 3×3 entries 0 or $\pm d^\eta_\mu(\pm q)g_\eta(q)$ and the same positions and signs at both heights. Consequently

$$\|D\|, \|D'\| \le 3, \quad D'J - JD = J\delta_D, \quad \|\delta_D\| \le 6, \quad \|\delta_D P_\Lambda\| \le 528\eta\Lambda. \tag{9.xxxvi}$$

The last estimate follows from $|d^{\eta'}_\mu g_{\eta'} - d^\eta_\mu g_\eta| \le 8\eta|q|$ and $|q| \le 22\Lambda$ on these modes.

For a multiplication jet $M = E, e^\mathsf{T}, f$ or e, let $X = MG^{1/2}$ or $Mg^{1/2}$ with the appropriate spaces. We claim

$$\|X'(1 - JJ^*)\| \le C\eta, \quad \|(1 - P_{2\Lambda})X\| \le C\epsilon_1, \quad \mathrm{df}(X', X) \le C\epsilon_1. \tag{9.xxxvii}$$

The first bound is the added-mode root bound times the bounded multiplication jet. For the second split the input at P_Λ: the high input costs C/Λ, the low input followed by the high output costs $C\Lambda^{-1/2}$ by eq. 9.xxxiv and the coefficient tail, and the mixed jet costs $C\eta$. For the third use

$$X'J - JX = (M'J - JM)g_\flat + JM(g_\flat - g_*).$$

The second term is $O(\eta)$. Split the first at P_Λ on the input and $P'_{2\Lambda}$ on the output. The high input costs C/Λ; the low input and high output are bounded by the preceding tail argument at the two heights; on the remaining modes eq. 9.xxxiv costs $C(\eta^{1/2} + \eta\Lambda)$. This proves eq. 9.xxxvii.

Now put $X_1 = EG^{1/2}$, $X_0 = e^\mathsf{T}G^{1/2}$, $X_f = fG^{1/2}$ and $X_g = eg^{1/2}$. The Hessian formula gives

$$\begin{aligned}\kappa &= D_1^\mathsf{T}(z-1)D_1 + D_1^\mathsf{T}zX_1 + X_1^\mathsf{T}zD_1 + X_1^\mathsf{T}zX_1 + G^{1/2}X_f \\ &\quad + D_0^\mathsf{T}X_0 + X_0^\mathsf{T}D_0 + X_0^\mathsf{T}X_0, \\ \kappa_g &= D_g^\mathsf{T}X_g + X_g^\mathsf{T}D_g + X_g^\mathsf{T}X_g.\end{aligned}$$

The terms containing $z - 1$ are $O(\eta^4)$ at either height, since its only nonzero component is 11. For $D^\mathsf{T}X$, split X at $P_{2\Lambda}$ and use eq. 9.xxxvi on the low modes and the output tail in eq. 9.xxxvii on the other modes. This gives defect $C\epsilon_1$. For $X^\mathsf{T}D$ apply the transpose rule to $D^\mathsf{T}X$, retaining its added-mode tail, bounded by $3\|X'(1 - JJ^*)\|$. Products $X^\mathsf{T}X$ and $G^{1/2}X_f$ follow from the product and transpose rules. These observations prove the claims for κ and κ_g.

For x the exact split eq. 9.xxix gives

$$\begin{aligned}x'J - x &= (\mathbf{Q}^\sharp - \mathbf{Q})\mathbf{Q}^\mathrm{f}G'^{1/2}J \\ &\quad + \mathbf{Q}(\mathbf{Q}^\mathrm{f} - Q^\mathrm{f})G'^{1/2}J + \mathbf{Q}(Q^\mathrm{f}G'^{1/2}J - G^{1/2}).\end{aligned}$$

The first two terms cost $C\eta^{1/2}$ and $C\eta$. The last difference is diagonal, with entries $u_{1,\mu}g_{\eta'} - g_\eta$, of modulus at most 11η: use the bounds for $u_1 - 1$, $g_{\eta'}$ and $g_{\eta'} - g_\eta$. The scalar version uses $v_1 g_{\eta'} - g_\eta$, bounded by 8η. The added-mode tails of x' and x'_g are bounded by 4η times the uniform averaging bound. This proves the assertions concerning x.

For y, set $X' = G'^{1/2}\mathbf{D}'\mathbf{R}^{\mathrm{f}\mathsf{T}}$ and $X = G^{1/2}\mathbf{D}$. Then

$$\begin{aligned} y' - Jy &= X'(\mathbf{R}^{\sharp} - \mathbf{R})^{\mathsf{T}} + (X' - JX)P_{\Lambda}\mathbf{R}^{\mathsf{T}} \\ &\quad + (X' - JX)(1 - P_{\Lambda})\mathbf{R}^{\mathsf{T}}. \end{aligned}$$

The first term costs $C\eta^{1/2}$; the last costs $C\Lambda^{-1/2}$ by eq. 9.xxxi. To estimate the middle term, expand $X' - JX$ as

$$\begin{aligned} &[G'^{1/2}d'R^{\mathrm{f}\mathsf{T}} - JG^{1/2}d] \\ &\quad + G'^{1/2}\mathbf{D}'(\mathbf{R}^{\mathrm{f}} - R^{\mathrm{f}})^{\mathsf{T}} \\ &\quad + G'^{1/2}e'(R^{\mathrm{f}\mathsf{T}} - J) + [G'^{1/2}e'J - JG^{1/2}e]. \end{aligned}$$

The second line is $O(\eta)$ by eq. 9.xxx. For the free aliases at $|m|_{\infty} \le \Lambda$, the principal coefficient satisfies $|v_1(q_m) - 1| \le 11\eta\Lambda$. For any other alias, some component of $q_{m'}$ has modulus at least $\pi\eta^{-1}$; the sine quotient defining s_1 then gives $|v_1(q_{m'})| \le 6\eta\Lambda$. There are at most L^3 aliases. Hence

$$\|(R^{\mathrm{f}\mathsf{T}} - J)P_{\Lambda}\| \le 11L^{3/2}\eta\Lambda.$$

The same calculation, with the normalized derivative, bounds the first line by $187\sqrt{3}L^{3/2}\eta\Lambda$; on the principal alias use $|d^{\eta'}g_{\eta'} - d^{\eta}g_{\eta}| \le 8\eta|q|$, and on the others use $|d^{\eta'}g_{\eta'}| \le 1$. The third line's first term is thus $O(\eta\Lambda)$. For its last term, split the output at $P'_{2\Lambda}$. On the low output, the multiplication comparison and the root difference give $C(\eta^{1/2} + \eta\Lambda + \eta)$; on the high output the left root gives C/Λ. This proves $\mathrm{df}(y', y) \le C\epsilon_1$ and completes the proof. □

9.2.8. *Stability of the finite expression.* The coarse traces in eq. 9.xxii have dimension at most nine. Its one remaining fine trace is controlled instead by the three propagators in it. Write $\|\cdot\|_3$ for the Schatten norm of exponent three on a Fourier fibre.

Lemma 9.38. Suppose two data for eq. 9.xxii satisfy the following bounds, with $A, C_3, \iota \ge 1$ and $\epsilon > 0$. At every real $\varkappa \in K$ the diagonal propagators satisfy

$$\|\mathsf{G}\|_3, \|\mathsf{G}'\|_3 \le C_3, \qquad \|\mathsf{G}'(1 - JJ^*)\|_3, \|(\mathsf{G}'J - J\mathsf{G})J^*\|_3 \le \epsilon.$$

The jets $\kappa, x, y, B_t, B^{\flat}_{\tau}$ have norms at most A at both heights, for $t, \tau \in [0,1]$, and the 00 inverses of $xB^{\flat}_0x^{\mathsf{T}}$ have norms at most ι. Finally

$$\mathrm{df}(\kappa', \kappa), \quad \mathrm{df}(x', x), \quad \|x'(1 - JJ^*)\|, \quad \mathrm{df}(y', y) \le \epsilon.$$

Then, with $a = \max(A, C_3)$,

$$|\Phi[\mathsf{G}', \kappa', x', y'] - \Phi[\mathsf{G}, \kappa, x, y]| \le 2600\iota^3a^{11}\epsilon.$$

■

Proof. The propagator operator norms are at most C_3, and their defect is at most ϵ. The transpose defects of x, y are at most $2\epsilon, \epsilon$, respectively; for y its input is coarse and there is no added input space. The product and inverse rules therefore give

$$\mathrm{df}(\omega', \omega) \le 5A\epsilon, \quad \mathrm{df}(B'_t, B_t) \le 6A^3\epsilon, \quad \mathrm{df}(B^{\flat\prime}_{\tau}, B^{\flat}_{\tau}) \le 7A^3\epsilon.$$

A product of n displayed factors has defect at most $7na^{n+2}\epsilon$.

For an endomorphism of a coarse fibre, the trace of its 11 component is bounded by nine times its jet norm. The six coarse-trace products in eq. 9.xxii have respectively $3, 3, 5, 5, 7, 7$ factors. Remembering that $\int_0^1 (1 - \tau)\,\mathrm{d}\tau = 1/2$, their total difference is at most

$$9(42a^5 + 70a^7 + 49a^9)\epsilon \le 1449a^9\epsilon.$$

For the logarithmic term put $C = xB^{\flat}_0x^{\mathsf{T}}$. Then $\|C\|, \|C'\| \le a^3$ and $e := \|C' - C\| \le 21a^5\epsilon$. The inverse identity bounds the difference of C_{00}^{-1} by $\iota^2 e$. Subtracting the two products in the definition of ℓ one factor at a time gives at most $6\iota^3a^6e$ before the trace, hence $1134\iota^3a^{11}\epsilon$ after integration.

It remains to compare $W = \mathsf{G}B_0\mathsf{G}B_0\mathsf{G}B_\tau^\flat$. Insert the projection onto the common modes in the trace:

$$\operatorname{tr} W'_{11} - \operatorname{tr} W_{11} = \operatorname{tr}[W'_{11}(1 - JJ^*)] + \operatorname{tr}[(W'J - JW)_{11}J^*].$$

In the first term cycle the projection next to a propagator. Hölder's inequality for the three Schatten-three factors bounds it by $\epsilon C_3^2 A^3 \le a^5\epsilon$. In the second term telescope the six factors. Insert $J^*J = 1$ between common-space factors so that each term is a product on the primed space. The three propagator-difference terms are bounded by $3a^5\epsilon$ in total. The three inverse-jet differences, bounded by $6A^3\epsilon, 6A^3\epsilon, 7A^3\epsilon$, contribute at most $19a^8\epsilon$. In each case the other three propagator factors, including a difference when present, are estimated in Schatten norm; the jet product norm sums the finitely many 11 terms. The remaining integral has weight $(1-\tau)^2$, whose integral is $1/3$. Its contribution is therefore at most $8a^8\epsilon$. Adding the three bounds and using $a, \iota \ge 1$ proves the lemma. □

Proposition 9.39. There is a finite $C_{\mathrm{re}} = C_{\mathrm{re}}(L)$ such that

$$|F_{k+1}(p) - F_k(p)| \le C_{\mathrm{re}} L^{-k/3} \quad (k \ge 1,\ p \in \mathbb{R}^3,\ |p_\mu| \le \delta).$$

■

Proof. Choose $\Lambda = \lfloor \eta^{-2/3}/4 \rfloor$. For $\eta^{-1} \ge 13$ this is at least one, is at least $\eta^{-2/3}/8$, and satisfies $3\Lambda \le N$. For the last assertion the function $(y-1)/2 - 3y^{2/3}/4$ is positive at $y = 13$ and increasing thereafter. Thus

$$\epsilon_1 \le (\sqrt{8} + 1/4 + 1)\eta^{1/3} \le 5\eta^{1/3}.$$

Lem. 9.37 supplies the operator defects in Lem. 9.38 with $\epsilon = C\eta^{1/3}$. The uniform inverse and jet bounds were established before eq. 9.xxii.

We verify the required Schatten bounds explicitly. Both propagators have eigenvalues $g_h(q_m)^2$, each repeated at most nine times. Since $g_h \le \min(1, 2/|m|_\infty)$ away from zero and the shell $|m|_\infty = n$ has $24n^2 + 2 \le 26n^2$ points,

$$9\sum_m g_h(q_m)^6 \le 9\left(1 + 64 \cdot 26 \sum_{n\ge 1} n^{-4}\right).$$

This gives a uniform C_3. The sum over added modes starts beyond $N \ge 6\eta^{-1}/13$, and its tail is $O(\eta^3)$. On common modes eq. 9.xxvi gives $|g_{\eta'}^2 - g_\eta^2| \le 3\eta^2$; there are η^{-3} such modes, so their cubed Schatten difference is at most $9\eta^{-3}(3\eta^2)^3 = 243\eta^3$. Both relevant Schatten norms are therefore $O(\eta)$, and hence at most $C\eta^{1/3}$.

Apply Lem. 9.38 to the vector and scalar data and use Lem. 9.35. This proves the stated bound. In particular the choice

$$C_{\mathrm{re}} := \sup_{k\ge 1,\ |p_\mu|\le\delta,\ p\in\mathbb{R}^3} L^{k/3}|F_{k+1}(p) - F_k(p)|$$

is finite and depends only on L. □

9.2.9. *Recovering the coefficient and summing the increments.* The real-momentum bound does not by itself bound derivatives. We use the uniform complex bound from eq. 9.xix as follows. If f is holomorphic on $|p_\mu| < \delta$, bounded by M there and by ϵ on the real cube $|p_\mu| \le \delta/4$, then

$$|\partial_1\partial_2 f(0)| \le 17\sqrt{\epsilon M}\,\delta^{-2}. \tag{9.xxxviii}$$

Indeed expand $f(s,t,0) = \sum c_{ij}s^it^j$, with $|c_{ij}| \le M\delta^{-i-j}$. The centred mixed difference at the four points $(\pm h, \pm h, 0)$ divided by $4h^2$ retains exactly the odd powers of s, t. For $h \le \delta/4$ the terms other than c_{11} total at most $3Mh^2\delta^{-4}$. Therefore

$$|c_{11}| \le \epsilon h^{-2} + 3Mh^2\delta^{-4}.$$

When $0 < \epsilon < M$, choose $h^2 = \delta^2\sqrt{\epsilon/M}/16$. When $\epsilon \geq M$, the ordinary Cauchy bound suffices; when $\epsilon = 0$, let h tend to zero. These cases prove eq. 9.xxxviii.

Proof of Thm. AC. Apply eq. 9.xxxviii to $F_{k+1} - F_k$, with $M = 2C_{\mathrm{rem}}$ and $\epsilon = C_{\mathrm{re}}L^{-k/3}$. By eq. 9.xix,

$$C_{\mathrm{inc}} := \sup_{k\geq 1} L^{k/6}|r_{k+1} - r_k| \leq 17(2C_{\mathrm{rem}}C_{\mathrm{re}})^{1/2}\delta^{-2} < \infty.$$

Define

$$C_\circ = \max\{C_{\mathrm{inc}} + 2C_E, c_2^{\mathrm{loop}}\}, \qquad C_{\mathrm{geo}} = \frac{2C_\circ}{1 - L^{-1/6}}. \tag{9.xxxix}$$

These are functions of L alone: each is defined from the Gaussian coefficients, their fixed massive term and their fixed determinant remainder. Lem. 9.34 gives

$$|c_{k+1} - c_k| \leq C_\circ L^{-k/6} \qquad (k \geq 1).$$

The same bound holds at $k = 0$ because $c_0 = 0$ and $|c_1| \leq c_2^{\mathrm{loop}}$. For $n > k \geq 0$ it follows that

$$|c_n - c_k| \leq \sum_{j\geq k} C_\circ L^{-j/6} = \tfrac{1}{2}C_{\mathrm{geo}}L^{-k/6}.$$

Thus c_k converges to a real c_∞, with the asserted error, and $|c_\infty| \leq c_2^{\mathrm{loop}}$. Also $r_k \to c_\infty - E_\infty$.

The defining identity for c_k gives $b_l^0 = c_l - L^{-1}c_{l-1}$ for $l \geq 1$. Therefore

$$|b_{l+1}^0 - b_l^0| \leq C_\circ(1 + L^{-5/6})L^{-l/6} \leq (1 - L^{-1/6})C_{\mathrm{geo}}L^{-l/6}.$$

This includes $l = 1$ by the already verified $k = 0$ bound. The limit is $(1 - L^{-1})c_\infty$; summing this last increment estimate from l to infinity gives $|b_l^0 - b_\infty^0| \leq C_{\mathrm{geo}}L^{-l/6}$, as claimed. □

9.2.10. *The bound used in the coupling flow.* The coupling recursion needs a bound on the increments, not only on their accumulated coefficients. The following elementary form also applies to the coefficients of a general group once their stated bound has been established.

Lemma 9.40. Let $L \geq 13$, let $c_0 = 0$ and $|c_k| \leq c_2^{\mathrm{loop}}$ for all $k \geq 0$, and put $b_l^0 = c_l - L^{-1}c_{l-1}$. If $B_\sharp \geq (1 + L^{-1})c_2^{\mathrm{loop}} + 1$ and $\bar{B} = LB_\sharp/(L - 1)$, then

$$B_\sharp \geq 1, \qquad |b_l^0| \leq B_\sharp - 1 \quad (l \geq 1), \qquad B_\sharp < \bar{B} \leq \tfrac{13}{12}B_\sharp.$$

■

Proof. The triangle inequality gives $|b_l^0| \leq (1 + L^{-1})c_2^{\mathrm{loop}} \leq B_\sharp - 1$. Since $c_2^{\mathrm{loop}} \geq 0$, we have $B_\sharp \geq 1$. Finally $1 < L/(L - 1) \leq 13/12$, which proves the bounds on $\bar{B}$. □

10. Renormalized Wilson loops

Fix a physical torus T_ℓ and a label $b > 0$, with $a_K = L^{-K}$ and $\beta = L^K b$. Let C be a self-avoiding closed polygon with finitely many axis-parallel pieces, corners in a fixed coarse lattice $a_{i_0}\mathbb{Z}^3/\ell\mathbb{Z}^3$, and contained in a ball of radius less than $\ell/4$. Write $|C|$ for its physical perimeter and $W_C = \frac{1}{2}\operatorname{tr}U(C)$. The same notation applies to a finite family of polygons at positive mutual distances, with the product of their traces and the sum of their perimeters.

Sketch I (A perimeter renormalization). If $\kappa = 3|C|/(4\pi b) < 1/2$, then

$$a_K^{-\kappa}\langle W_C\rangle_K \longrightarrow \mathcal{W}(C)$$

along the full cutoff sequence on the fixed torus. For coordinate rectangles the limit is strictly positive. This is a statement about expectations; it does not assert convergence of the renormalized sharp loop as a random variable. ■□

Sketch of the argument. Keep the line insertion when integrating one fluctuation scale. The leading factor is the expectation of the corresponding Gaussian line, and can be written

$$\lambda_n(C) = \exp\left[-\frac{3}{2b} a_K(X_C, \mathcal{C}_{(n)} X_C)\right].$$

Here $d^{\mathrm{T}} X_C$ is the loop current and $\mathcal{C}_{(n)}$ is the curvature covariance removed at the nth block step. The orthogonal covariance decomposition makes these factors multiply. Their logarithmic self-energy divergence has coefficient $\kappa \log(1/a_K)$; the remaining Coulomb energy has a finite part. Thus one power $a_K^{-\kappa}$ removes the divergence.

The remaining insertion is compared with the holonomy of the critical background. An ordered-product estimate controls the change of a noncommuting line when its individual bonds change. The fluctuation step gives an error with a summable power a_d^{ϑ} for every $\vartheta < 1/2$. After renormalization its sum has exponent $\vartheta - \kappa$, explaining the perimeter restriction.

To take the cutoff limit of the background loop, first fix a block depth d. The compatible block laws converge in total variation. A thread is a family of coarse fields compatible with the block averaging maps. The large-field tails are summable over d, so almost every thread is eventually in the regular domain. Over a regular block field at this fixed depth, the critical backgrounds on successively finer cutoff lattices give a smoothed loop in the limit $K \to \infty$. The line-integration estimate then compares the sharp loop expectation with a fixed-depth smoothed expectation. Take $K \to \infty$ first and $d \to \infty$ afterwards. Reflection positivity and bounds for small rectangles give the positive lower bound for rectangles.

Smoothed loops at different depths use different critical refinements of the thread, and are different random functions. The argument gives limits of means, not an almost sure limit as the depth increases. Also, the unrenormalized sharp trace has a semicircle limit law, whereas its mean tends to zero. Dividing by that mean does not produce a tight family. Mixed functions with local fields can be treated by keeping the field sources in the line integration, but require the stronger perimeter and support-separation conditions of that argument.

10.1. **Wilson loops on a fixed torus.** We consider three limits of lattice Wilson loops. Background-smoothed loops converge along almost every block-field thread. Sharp polygonal loops have renormalised expectation limits in the stated perimeter range. The unrenormalised trace has a semicircle limit, and division by its vanishing mean produces divergent variance.

These are different limiting statements. The sharp-loop theorem concerns expectations on a fixed torus; its stronger mixed-function conclusions have their own perimeter and separation restrictions. The proof retains the ordered-product estimate, the inserted-line renormalisation step and the summability restriction on the normalised errors.

10.1.1. *Notation and hypotheses. Constants.* Use a common real numerical choice as in Prop. YM3_I:69.1. We use the odd blocking factor $L \geq 13$; $M = L^{m'}$ with $m' \geq 5$; $r \in \mathbb{N}$; $\gamma \in (0, \frac{1}{2}]$ with $\log \gamma^{-2} \geq \log 4 > 1$; $A_0 > 0$, $\mathfrak{p}_0 \in \mathbb{N}$ and the threshold $\varepsilon(t) = A_0 t^{-1/2} (\log t)^{\mathfrak{p}_0}$, $t > 1$, of YM3_I:(634), which decreases for $\log t > 2\mathfrak{p}_0$; $B_3 \geq 1$; $c_\rho = \max\{2B_3, C_H A_1/((1 - L^{-3/2})A_0)\}$ with the constants C_H, A_1 of this formula (§2.3.1), so that $2 \leq 2B_3 \leq c_\rho < \infty$; $c' \in (0, \frac{1}{250})$, $\mathsf{C}^{\Sigma}_{\mathrm{maj}} \geq 0$ and $c_{\mathrm{low}} \geq 24$.

The number $t_\natural \geq \gamma^{-2}$ is the least real number from which on the three smallness conditions (cut1)–(cut3) of §2.3.1 on $\varepsilon(t)$ hold; they hold at every $t \geq t_\natural$, and there

$$2c_\rho \varepsilon(t) \leq \varepsilon_{\mathcal{W}} \alpha_0 \leq \bar{\varepsilon} \leq 1$$

, $\bar{\varepsilon}$ the radius of Thm. YM3_I:E in the setting of Section YM3_I:52.7 and $\varepsilon_{\mathcal{W}}$, α_0 the numbers of (cut1), formed from the constants alone (equation YM3_I:(2134)). $\Theta_\star \geq t_\natural$ is the least real number from which on the five conditions of the comparison of two consecutive cutoffs (Prop. YM3_I:113.9)

hold; both exist (the threshold choices in Sections YM3_I:113 and YM3_I:115, with [7, Cor. B.13]), and $2c_\rho\varepsilon(t) < 1/200L^2$ for $t \geq \Theta_\star$.

The lowered choice of the constants (Def. 4.47: A_1 multiplied by $\lambda'_\natural$, $B_\sharp$ not smaller, A_0 and the constants fixed before A_1 kept) gives the number $c'_\star$, which is $c_\star$ of eq. 4.xx at that choice, and the numbers $\mathsf{Y}_1^\natural, \dots, \mathsf{Y}_4^\natural$ of eq. 4.xxi formed from the two choices; we put $c^\natural = c'_\star/(4 \cdot 49^3) > 0$ and $Y^\natural = \max_{i \leq 4} \mathsf{Y}_i^\natural$. All these numbers depend on the constants alone.

The one-torus setting: a label $\mathsf{b} \geq \gamma^{-2}$ and a side $\ell = 2L^m$, $m \in \mathbb{N}$, with $L^m \geq M(\log \mathsf{b})^r$, equivalently $M\,Q(\mathsf{b}) \leq L^m$ (Lem. 10.1).

Fields and laws. $T_\ell = (\mathbb{R}/\ell\mathbb{Z})^3$, $a_d = L^{-d}$ and $\Lambda_d = a_d\mathbb{Z}^3/\ell\mathbb{Z}^3$ (Section YM3_I:3). $\mathcal{V}_d$ is the set of the fields on the bonds of Λ_d with values in $G = \mathrm{SU}(2)$, $V(\bar e) = V(e)^{-1}$, with the gauge transformations $V^v(e) = v(e_-)V(e)v(e_+)^{-1}$ (Section YM3_I:3); $V(\Gamma)$ is the ordered product along a path Γ, $\partial V(p)$ the ordered product along the boundary of a plaquette p, and $\varpi(V) = \max_p |\partial V(p) - \mathbf{1}|$, in the operator norm . $\mathcal{M} : \mathcal{V}_{d+1} \to \mathcal{V}_d$ is the block averaging map (equations YM3_I:(37)–(39)); it is Borel and gauge covariant, and one map serves all cutoffs.

The Wilson measure $\langle\cdot\rangle_K$ on $\mathcal{V}_K$ has the density $Z^{-1}e^{-\beta\sum_p a_p}$ with respect to the Haar measure dU, where $a_p = 1 - \frac12 \operatorname{tr} \partial U(p)$ and $\beta = L^K\mathsf{b}$. μ_d^K is the law of $\mathcal{M}^{K-d}U$ under $\langle\cdot\rangle_K$, and $\mu_d = \lim_K \mu_d^K$ in total variation along the full sequence of cutoffs (Cor. 2.8). $\mathcal{X} = \prod_{d\geq 0}\mathcal{V}_d$ carries the product topology and the coordinates π_d; $\mathsf{Thr} = \{V_\bullet : V_d = \mathcal{M}(V_{d+1})\ \forall d\}$ is the set of the threads; $\mathcal{F}_d = \sigma(\pi_0, \dots, \pi_d)$; $(\theta_v V_\bullet)_d = V_d^{v|\Lambda_d}$ for $v : \bigcup_d \Lambda_d \to G$.

Finally μ denotes the Borel probability measure on $\mathcal{X}$ whose image under $V_\bullet \mapsto (V_0, \dots, V_D)$ is, for every D, the image of μ_D under $V \mapsto (\mathcal{M}^D V, \dots, \mathcal{M}V, V)$; it exists and is unique (Prop. 10.2(a)).

Depths. We write $x_{[d]} = L^d\mathsf{b}$ and $\varepsilon_{[d]} = \varepsilon(x_{[d]})$, with brackets, so as to tell these depth quantities from the run quantities x_t, ε_t of the frame below. In $\mathcal{V}_d$ let $I_d = \{\varpi < 2\varepsilon_{[d]}\}$ and $J_d = \{\varpi < c_\rho\varepsilon_{[d]}/B_3\}$; $I_d \subset J_d$ since $c_\rho \geq 2B_3$. Put

$$R_d = \frac{2\ell^3 L^{3d}}{M^3}\, e^{-c' A_0^2 (\log x_{[d]})^{2\mathfrak{p}_0} + (\mathsf{C}^\Sigma_{\mathrm{maj}} + 2 + c_{\mathrm{low}} + \frac92 \log \mathsf{b})\ell^3}, \qquad R_d^\natural = \ell^3 L^{3d} e^{-c^\natural A_0^2 (\log x_{[d]})^{2\mathfrak{p}_0}}.$$

A depth d is admitted if $x_{[d]} \geq \Theta_\star$. $d_\natural$ and d_Θ are the least integers $d \geq 1$ with $x_{[d]} \geq t_\natural$, resp. $x_{[d]} \geq \Theta_\star$, and $s_\natural$ is the least integer $s \geq 0$ with $\log x_{[s]} \geq Y^\natural$; they depend on b and the constants, not on ℓ, and $d_\Theta \geq d_\natural$. $\tau_0(V_\bullet)$ is the least $D \geq d_\natural$ with $V_d \in I_d$ for all $d \geq D$ (∞ if there is none); the starting depth is $\tau = \max\{\tau_0, d_\Theta\}$, and $\Omega = \mathsf{Thr} \cap \{\tau < \infty\}$.

Loops and their smoothing. A lattice loop C of $\Lambda_{d'}$ with N bonds has the length $|C| = Na_{d'}$; for $j \geq d'$, $C^{\langle j\rangle}$ arises by replacing $j - d'$ times every bond by the straight path of L bonds of the next lattice, and $W_C(V) = \frac12 \operatorname{tr} V(C^{\langle j\rangle})$ for $V \in \mathcal{V}_j$. For $W \in \mathcal{V}_d$ and $K \geq d$ the backgrounds over W form the set $\mathcal{C}_K(W)$ of the $U \in \mathcal{V}_K$ with $\mathcal{M}^{K-d}U = W$ that are critical for $\sum_p a_p$ on this constraint set and regular at the radius ε_{v} (Defs. YM3_I:9.3 and YM3_I:9.5); $\mathcal{G}_K^d$ is the group of the gauge transformations of Λ_K that equal $\mathbf{1}$ on Λ_d, and $w_C^K(W) = W_C(U)$ for $U \in \mathcal{C}_K(W)$.

For $X \in \mathcal{V}_s$ let $H_s^K(X) = \mathcal{M}^{K-s-1}U_+ \in \mathcal{V}_{s+1}$, U_+ an element of $\mathcal{C}_K(X)$ for which the average $\mathcal{M}^{K-s-1}U_+$ is in the axial gauge on Λ_s ([7, §B.2]); $H_s^\infty = \lim_K H_s^K$ and $H_{d,n}^\bullet = H_{d+n-1}^\bullet \circ \cdots \circ H_d^\bullet$. The smoothed loop is $W_C^{[d]}(W) = \lim_n W_C(H_{d,n}^\infty(W))$; $F_C^{[d]}$ is $W_C^{[d]}$ on J_d and 0 off J_d; $Y_{d,C} = F_C^{[d]} \circ \pi_d$ and $Y_{d,C}^K = (\mathbf{1}_{J_d} w_C^K) \circ \pi_d$. A curve is a continuous $\hat\gamma : [0,1] \to \mathbb{R}^3$ of finite length $|\gamma|$ with $\hat\gamma(1) - \hat\gamma(0) \in \ell\mathbb{Z}^3$; a shadow $\Gamma^{(j)}$ of it on Λ_j cuts it into at most $|\gamma|L^j + 2$ pieces of sup-diameter $\leq a_j$, takes sites of Λ_j within $\frac12 a_j$ of the cut points and joins them by shortest paths of Λ_j; $W_\gamma^{[d]}(W) = \lim_n W_{\Gamma^{(d+n)}}(H_{d,n}^\infty(W))$, $w_\gamma^K(W) = W_{\Gamma^{(K)}}(U)$ for $U \in \mathcal{C}_K(W)$, and F_γ is $W_\gamma^{[d]}$ on J_d and 0 off it.

Polygons. A polygon is a closed self-avoiding lattice path with finitely many axis-parallel pieces (maximal straight parts) and corners in some Λ_{i_0}, lying in a ball of radius $< \ell/4$. A family $\mathsf{C} = (C_1, \dots, C_{\#\mathsf{C}})$ consists of polygons at pairwise positive distances; $|\mathsf{C}| = \sum_i |C_i|$ and

$W_{\mathsf C}=\prod_i W_{C_i}$. For a polygon or a family, $n_{\mathsf C}$ is the number of pieces, $s_{\mathsf C}$ the least length of a piece, $\delta_{\mathsf C}$ the least distance of two different parallel pieces of one polygon, $\mathfrak s_{\mathsf C}=n_{\mathsf C}^2+|{\mathsf C}|^2/\delta_{\mathsf C}^2$, $\mathfrak d_{\mathsf C}=\min\{s_{\mathsf C},\delta_{\mathsf C}\}$, $c_{\mathsf C}\in(0,1]$ a chord-arc constant ($|x-y|\ge c_{\mathsf C}\operatorname{arc}(x,y)$ for x,y on one C_i, arc the shorter arc), $\delta'_{\mathsf C}$ the least sup-distance of two different polygons (∞ for one polygon), and $\rho_{\mathsf C}>0$ the least $|x-y|_\infty$ over the pairs of points that lie on two different polygons, or on one polygon with both arcs between them of length $\ge\frac12 s_{\mathsf C}$.

We put $k=\frac{3}{4\pi}$, $\kappa=k|{\mathsf C}|/\mathsf b$ and $\mathcal W_K({\mathsf C})=a_K^{-\kappa}\langle W_{\mathsf C}\rangle_K$; for a square $n_{\mathsf C}=4$ and one may take $c_{\mathsf C}=\frac12$, numbers that dilations do not move.

Numbers of the free field. By this we mean numbers formed from the Gaussian field alone, in which no measure of the interacting theory enters. For loops C, D with corners in Λ_K, $\mathcal I_a(C,D)=a\sum_\mu\sum_{y,y'}j^C_\mu(y)\,g_N(\frac{y-y'}{a})\,j^D_\mu(y')$ is the lattice Coulomb energy at the spacing $a=a_K$, $N=\ell/a$, j^C the current of C and g_N the Green function of the lattice Laplacian on $\mathbb Z^3/N\mathbb Z^3$ without its zero mode. G_ℓ is the Green function of $-\Delta$ on T_ℓ without its zero mode, $\mathcal I_\ell(C,D)=\oint_C\oint_D G_\ell(x-y)\,\mathrm dx\cdot dy$ for loops at positive distance, $\mathcal J_\ell(C)$ the limit as $u\to0$ of the part $|x-y|>u$ of $\oint_C\oint_C G_\ell(x-y)\,\mathrm dx\cdot dy$ less $\frac{|C|}{2\pi}\log\frac1u$, and $\mathcal J(C)$ the same with $\frac{1}{4\pi|x|}$ in the place of G_ℓ; $\mathrm{Lg}(s)=(1+\log(e+s))^{3/2}$. With Euler's constant γ_E, $S(z)=\frac{2\sin(z/2)}{z}$, $p=q+2\pi n$, $\mathsf N=\sum_{n\in\mathbb Z^2}\frac{S(p_2)S(p_3)}{p_2^2+p_3^2}$ and $\mathsf D$ the same with S^2 in the place of S:

$$c_{\mathrm{lat}}=\frac{5\log2+2\gamma_E}{4\pi}=0.36766\ldots,\quad c_r=\frac{1}{4\pi^2}\int_{[-\pi,\pi]^2}\Big[\frac{\mathsf N^2}{\mathsf D}-\frac{1}{4\sin^2\frac{q_2}{2}+4\sin^2\frac{q_3}{2}}\Big]dq_2\,\mathrm dq_3=$$
$$-0.03149\ldots,\quad c_Q=c_{\mathrm{lat}}+c_r=0.33617\ldots$$

Action density; mirrors. $\mathcal O(f)=a_K^{-1}\sum_p f(x(p))a_p$ for real Lipschitz f, $x(p)$ the centre of p, and $\mathcal W_K({\mathsf C};f_1,\ldots,f_n)=a_K^{-\kappa}\langle W_{\mathsf C}\prod_q(\mathcal O(f_q)-\langle\mathcal O(f_q)\rangle_K)\rangle_K$. $S^{\mathsf T}_{m,K}(f_1,\ldots,f_m)$ is the joint cumulant of $\mathcal O(f_1),\ldots,\mathcal O(f_m)$ under $\langle\cdot\rangle_K$, the truncated m-point function of the action density; for pairwise disjoint supports its limit $S_m(f_1,\ldots,f_m)$ exists along the full sequence YM3_I:(2212). A coordinate mirror is $\theta x=x-2(x_\mu-c)e_\mu$ with $c\in\frac12 a_{i_0}\mathbb Z$; it fixes the planes $x_\mu=c$ and $x_\mu=c+\frac{\ell}{2}$, it is a site mirror at the cutoff K if $c\in a_K\mathbb Z$ and a bond mirror otherwise, and $S^+_\theta=\{c\le x_\mu\le c+\frac{\ell}{2}\}$. It acts on functions of the field by $(\theta F)(U)=F(\theta U)$ and on test functions by $(\theta f)(x)=f(\theta x)$, so that $\theta W_C=W_{\theta C}$ and $\theta\mathcal O(f)=\mathcal O(\theta f)$.

Three free photon fields. X_C is the exact bare 2-cochain with $d^{\mathsf T}X_C=j_C$; in lattice units $a_K(X_C,X_D)=\mathcal I_{a_K}(C,D)$. The three free photon fields at the label $\mathsf b$ are the curvatures $F^a=\beta^{-1/2}\Pi\xi^a$, $a=1,2,3$, ξ^a independent standard Gaussian variables on the bare plaquettes and Π the orthogonal projection onto the exact 2-cochains; $\phi^a_C=(F^a,X_C)$ is the flux of F^a through C, so that $\mathbb E\,\phi^a_C\phi^{a'}_D=\delta_{aa'}\mathcal I_{a_K}(C,D)/\mathsf b$; their loop is $W^{\mathrm{fr}}_C=\prod_{a=1}^3\cos\phi^a_C$, their action density $\mathcal O^{\mathrm{fr}}(f)=a_K^{-1}\sum_p f(x(p))\,\frac12\sum_a(F^a(p))^2$, and their block curvatures $F^{(d)}$ are the curvatures of the block averages at the depth d, linear images of F.

The frame of the renormalisation group (Thm. AE and Thm. AD(5)). A reference coupling $g\le\gamma$ with $g^{-2}\ge4\bar B$, $\log g^{-2}\ge\max\{6\mathfrak p_0,A_0^{-1/\mathfrak p_0},(1+\log LM)^{1/2}\}$ and $g(\log g^{-2})^{\mathfrak p_0}\le c_{\mathrm{th}}$, $c_{\mathrm{th}}>0$ formed from the constants (true for small g); the flow bound $B_\sharp$ of the run and $\bar B=\frac{L}{L-1}B_\sharp$; $\mathsf I(g)=(g^{-2}+\bar B,L(g^{-2}+B_\sharp)-\bar B]$ (Section YM3_I:74); a side $\ell\in\mathfrak L(g)$, the set of the sides at g (equation YM3_I:(1528)); a compact $\mathsf J\subset\mathsf I(g)$, $\mathsf b\in\mathsf J$, $\mathsf b_-=\min\mathsf J$, $\mathsf b_+=\max\mathsf J$.

The run is $x_0=\beta$, $x_t=x_{t-1}/L-u_t$ with $|u_t|\le B_\sharp$, so that $|x_t-L^{K-t}\mathsf b|\le\bar B$ and $\frac23 x_{[d]}\le x_{K-d}\le x_{[d]}+\bar B$; $g_t=x_t^{-1/2}$, $\varepsilon_t=\varepsilon(x_t)$, $\mathfrak b_t=A_1(\log x_t)^{\mathfrak p_0}$ and $\delta_t=g_t\mathfrak b_t$. The level t is the lattice Λ_{K-t}, with the tame set $\mathsf O_t=\{\varpi<2\varepsilon_t\}$. Thm. YM3_I:D, the inductive description of the effective densities, with the actual-history conclusion of Cor. YM3_I:69.4, provides the following objects. Finite positive Borel measures m_t on $\mathcal V_{K-t}$, carried by $\mathsf O_t$ for $t\ge1$, with $m_0=e^{-\beta\sum_p a_p}dU$; bounded Borel functions $\mathsf h_t,\mathsf h_t^{\mathrm w}\ge0$ on $\mathcal V_{K-t+1}$, the weights of the step t without and with a new large

field, with $\int G\,\mathrm{d}m_t = \int (G\circ\mathcal{M})\,\mathsf{h}_t\,\mathrm{d}m_{t-1}$; for every bounded Borel F on $\mathcal{V}_K$ quotients $\Phi^{(t)}[F]$ with $\Phi^{(0)}[F]=F$, $|\Phi^{(t)}[F]|\le \sup|F|$ and $\int G\,\Phi^{(t)}[F]\,\mathrm{d}m_t = \int (G\circ\mathcal{M})\,\Phi^{(t-1)}[F]\,\mathsf{h}_t\,\mathrm{d}m_{t-1}$ for bounded Borel G, out of which functions of coarser block fields pull, $\Phi^{(t)}[F\cdot(G\circ\mathcal{M}^{t'})] = \Phi^{(t)}[F]\cdot(G\circ\mathcal{M}^{t'-t})$ for $t\le t'$; and, for gauge invariant F and every last level $0\le \bar t\le K$,

$$Z\langle F\rangle_K = \int \Phi^{(\bar t)}[F]\,\mathrm{d}m_{\bar t} + \sum_{t\le \bar t}\int \Phi^{(t-1)}[F]\,\mathsf{h}_t^{\mathrm{w}}\,\mathrm{d}m_{t-1}, \qquad \mathsf{w}_t := \frac{1}{Z}\int \mathsf{h}_t^{\mathrm{w}}\,\mathrm{d}m_{t-1} \le \mathsf{V}_\ell C_{\mathrm{r}}(p) L^{-p(K-t)} \quad (p\ge 0), \tag{10.i}$$

the last inequality being the *rarity bound*; it gives $\sum_{t'\le t} L^{p(K-t')}\mathsf{w}_{t'} \le 2\mathsf{V}_\ell C_{\mathrm{r}}(p+1)L^{-(K-t)}$. Here $C_{\mathrm{r}}(p)$ is formed from the constants and p, and V_ℓ, free of K, is at most $2M^{-3}\ell^3 e^{B\ell^3}$ on every side of $\mathfrak{L}(g)$, with B formed from the constants and g, and equals 1 on the least sides of $\mathfrak{L}(g)$ when g is small. We take for $\mathfrak{su}(2)$ the real space of the Hermitian traceless 2×2 matrices with the scalar product $\langle X,Y\rangle = \frac12 \operatorname{tr} XY$, whose norm $|X|$ is the operator norm on it; t_1,t_2,t_3 is an orthonormal basis, with $t_a t_b + t_b t_a = 2\delta_{ab}\mathbf{1}$ and $\sum_a t_a^2 = 3\cdot\mathbf{1}$, the components of X are $X^a = \langle t_a, X\rangle$, a complex 2×2 matrix is written $x = x^0\mathbf{1} + \sum_a x^a\, i t_a$, and $\mathrm{R}(h)\in SO(3)$, $\mathrm{R}(h)_{ab} = \langle t_a, h t_b h^{-1}\rangle$, is the rotation of $h\in \mathrm{SU}(2)$.

For $W\in \mathsf{O}_{t+1}$ there are a chart $B\mapsto V_W(B)$, from the real $\mathfrak{su}(2)$-valued 1-forms B on a set of bonds of Λ_{K-t} (the fluctuation field of the step) with $|B|<\mathfrak{b}_t$ at every bond into $\mathcal{V}_{K-t}$, with $\mathcal{M}V_W(B) = W$ and $V_W(B)(b) = e^{ig_t B(b)}V_W(0)(b)$ at the fluctuation bonds, and the kernel ν_W of one step, a complex measure of mass 1 on $\{|B|<\mathfrak{b}_t\}$, with the *disintegration*

$$\int H(\mathcal{M}V)\,G(V)\,\mathsf{h}_{t+1}(V)\,m_t(\mathrm{d}V) = \int H(W)\Big[\int G(V_W(B))\,\nu_W(\mathrm{d}B)\Big] m_{t+1}(\mathrm{d}W)$$

for bounded Borel G, H; so $\Phi^{(t+1)}[F](W) = \int \Phi^{(t)}[F](V_W(B))\,\nu_W(\mathrm{d}B)$ for m_{t+1}-almost every W. On $\{|B|<\mathfrak{b}_t\}$ the kernel is $\nu_W = \Xi_W^{-1} e^{\mathsf{E}_W}\mu_{C_W}$: the Gaussian measure μ_{C_W} of the covariance C_W of the step at the background triple of W, which is formed from a background U over W with $\mathcal{M}^t U$ in the axial gauge and from the current of U (the triple defined before equation YM3_I:(827) and §5.3.1), times e^{E_W}, E_W the sum of two local potentials, the interaction and the part of the bare array of couplings, each with a norm with quadratic growth at most $c\,\delta_t$, c formed from the constants, and Ξ_W the number that makes the mass 1 (eq. 5.xxxii and Prop. 5.31). $W_{\mathsf{C}}^{(t)}(V) = W_{\mathsf{C}}(U)$ for a background U over $V\in\mathcal{V}_{K-t}$, and U_+ denotes a background over $W\in\mathsf{O}_{t+1}$. Π_n is the orthogonal projection of the bare 2-cochains onto the linearised curvatures of the backgrounds of n block steps (Π_0: onto the exact ones), $\mathcal{C}_{(n)} = \Pi_{n-1} - \Pi_n$, and

$$\lambda_n = \prod_i \exp\Big[-\frac{3}{2\mathfrak{b}}\,a_K(X_{C_i}, \mathcal{C}_{(n)}X_{C_i})\Big] \in (0,1], \qquad \Xi_t = \prod_{s\le t}\lambda_s, \qquad v_{K,d}(C) = a_K(X_C, \Pi_{K-d}X_C).$$

10.1.2. *The perimeter factor and the summable error.* Fix the physical torus and a permitted polygon C. At each fluctuation step the loop insertion has a leading Gaussian factor. If X_C is the exact plaquette cochain whose boundary is the loop current and $\mathcal{C}_{(n)}$ is the covariance of that step, this factor is

$$\lambda_n(C) = \exp\left[-\frac{3}{2b}a_K(X_C, \mathcal{C}_{(n)}X_C)\right].$$

The covariance pairing is the line's self-energy at that scale. Its singular part accumulates along the straight segments. Multiplying the one-step factors gives the perimeter power a_K^{κ}, with $\kappa = 3|C|/(4\pi b)$. Corners and interactions between separated segments enter the finite comparison terms and the errors, rather than changing this leading power.

The existence argument must control the remainder after dividing by that vanishing factor. Choose $\vartheta\in(\kappa,1/2)$. The inserted-line estimate gives a one-step error bounded by a constant times

$L^{-\vartheta j}$ at depth j. The accumulated normalization can cost $L^{\kappa j}$. Thus the error in the normalized expectation is controlled by

$$C \sum_{j \geq d} L^{-(\vartheta - \kappa) j},$$

where d is the retained coarse depth. This sum tends to zero as d increases. The strict restriction $\kappa < 1/2$ supplies room to choose ϑ; it is used at the step that sums the normalized errors. The source then compares successive cutoffs with these summable errors. A pointwise Gaussian approximation without this normalized summability would not establish the claimed renormalized limit.

The background loop provides the coarse comparison. The laws of the block fields are compatible under averaging, so they define a law on sequences of compatible fields. The large-field probability at depth d is bounded by R_d, with $\sum_d R_d < \infty$. The event that a sequence is regular at every depth above D has complement of probability at most $\sum_{d \geq D} R_d$. Almost every sequence is therefore eventually regular. One may then choose the minimal background over its field at depth D and study its loop as the finer cutoff is removed. The convergence at this fixed depth is uniform on the admitted regular set.

Two different limits occur here. At fixed D, the background loop becomes a bounded function of the block field V_D. Different choices of D give different functions of the sequence. The full limiting thread law permits their expectations and joint expectations to be compared. The estimates for these means do not state that the functions themselves converge almost surely as D increases. The sharp-loop calculation, meanwhile, removes its perimeter factor and produces a limit of expectations. It does not provide a random holonomy obtained by dividing a sharp trace by its vanishing mean.

Localization of the background uses gauge invariance in a concrete way. If the loop lies a positive distance inside a region, compare its background with the one constructed using only data in that region. In suitable gauges, each bond discrepancy is exponentially small in the distance to the boundary measured in coarse lattice units. For unitary matrices,

$$\left| \operatorname{tr} \prod_i A_i - \operatorname{tr} \prod_i B_i \right| \leq 2 \sum_i |A_i - B_i|.$$

Summing over the $|C|/a_K$ fine bonds cancels the fine spacing in the background estimate. The remaining error depends on the coarse depth and decays exponentially with the boundary distance. This is why the comparison survives removal of the fine cutoff while retaining a fixed loop and region.

The free-field loop, the background loop, and the sharp Wilson trace consequently play different roles in the calculation. The free field computes the self-energy; the background carries the coarse dependence; the inserted-line estimate controls their comparison with the interacting expectation. The detailed formulas below retain these roles and the stronger restrictions needed when local fields are inserted as well.

10.1.3. *Main results.*

Theorem AD (Smoothed loops along almost every thread)**.** Assume the one-torus setting.

(1) *The starting depth.* τ is a Borel function on $\mathcal{X}$ with values in $\{d_\Theta, d_\Theta + 1, \dots\} \cup \{\infty\}$; for every integer $D \geq d_\Theta$, $\{\tau \leq D\} = \bigcap_{d \geq D} \pi_d^{-1}(I_d)$, a countable intersection of open sets; Ω is Borel and $\mu(\Omega) = 1$. For $D \geq d_\Theta$: $\mu(\tau > D) \leq \sum_{d \geq D} R_d$, which is at most $\frac{L}{L-1} R_D$ if $c' A_0^2 \log x_{[D]} \geq 2$; both bounds exceed 1 as long as $c' A_0^2 (\log x_{[D]})^{2\mathfrak{p}_0} \leq 26 \ell^3$. For $D \geq \max\{d_\Theta, s_\natural\}$: $\mu(\tau > D) \leq \sum_{d \geq D} R_d^\natural$, which is at most $\frac{L}{L-1} R_D^\natural$ if $c^\natural A_0^2 \log x_{[D]} \geq 2$; and $R_D^\natural \leq \eta \in (0, 1]$ as soon as

$$D \log L \geq \max\Big\{ \frac{6}{c^\natural A_0^2}, \Big[\frac{2(3 \log \ell + \log \frac{1}{\eta})}{c^\natural A_0^2} \Big]^{1/2} \Big\}.$$

So τ is at most of the order $(\log \ell)^{1/2}$ on events of probability at least $1 - \frac{L}{L-1}\eta$.

(2) *Loops from the starting depth on.* If $K > d$ and $\varpi(W) < \bar\varepsilon$, $\bar\varepsilon$ the radius of Thm. YM3_I:E in the setting of Section YM3_I:52.7, then $\mathcal{C}_K(W)$ is one non-empty orbit of $\mathcal{G}_K^d$; and $\{\varpi < 2c_\rho \varepsilon_{[d]}\} \subset \{\varpi < \bar\varepsilon\}$ if $x_{[d]} \geq t_\natural$. Hence for $d \geq \tau_0(V_\bullet)$ and $K \geq d+1$ the backgrounds over V_d form one non-empty orbit, and $w_C^K(V_d)$ is one number for every closed path C of $\Lambda_{d'}$, $d' \leq K$. If $d \geq \tau(V_\bullet)$, then d is admitted, $V_d \in I_d \subset J_d$, and for every lattice loop C of $\Lambda_{d'}$, $d' \geq d$,

$$\lim_{K\to\infty} w_C^K(V_d) = W_C^{[d]}(V_d) = \lim_{n\to\infty} W_C(H_{d,n}^\infty(V_d)), \qquad |w_C^K(V_d) - W_C^{[d]}(V_d)| \leq |C| L^d \mathsf{c}_{\mathrm{f}} L^{-(K-d)/3} \quad (K \geq K_2),$$

with the numbers c_{f}, K_2 of Prop. 10.4. For a curve the same holds with w_γ^K, $W_\gamma^{[d]}$ and the bound $(|\gamma| L^d + 2)\mathsf{c}_{\mathrm{g}} L^{-(K-d)/3}$, and $W_\gamma^{[d]}$ does not depend on the shadows. At these V_d the smoothed loop, as a function on J_d, is continuous and gauge invariant with values in $[-1,1]$, and $0 \leq 1 - W_C^{[d]}(V_d) \leq \frac{1}{2}(2B_3 \varpi(V_d) L^{2d} s_1 s_2)^2$ if C bounds a coordinate rectangle with sides s_1, s_2. Ω depends on none of d, C, γ, K. The statements at finite cutoffs hold at every depth $d \geq \tau_0$, the limits as $K \to \infty$ at every depth $d \geq \tau$. No step of (2) and (3) uses the thread property $V_d = \mathcal{M}(V_{d+1})$: they hold at every point of $\{\tau < \infty\} \subset \mathcal{X}$.

(3) *Random variables.* Let d be admitted and C a lattice loop of $\Lambda_{d'}$, $d' \geq d$. Then $Y_{d,C}$ is Borel on $\mathcal{X}$, $[-1,1]$-valued, $\sigma(\pi_d)$-measurable, θ_v-invariant, and continuous on the open set $\pi_d^{-1}(J_d)$, where it is $W_C^{[d]}(V_d)$; for $K \geq \max\{2, d+1, d'\}$, $Y_{d,C}^K$ is Borel and $\sup_{\mathcal{X}} |Y_{d,C}^K - Y_{d,C}| \to 0$; $\{\tau \leq d\}$ is Borel, θ_v-invariant and contained in $\pi_d^{-1}(I_d)$, and $Y_{d,C} = \lim_K w_C^K(V_d)$ on it. With $Y_{d,C} := Y_{d,C^{\langle d\rangle}}$ for $d > d'$, $(Y_{d,C})_{d \geq d_\Theta}$ is an $(\mathcal{F}_d)$-adapted sequence with values in $[-1,1]$ on $(\mathcal{X}, \mu)$ whose members of index $\geq \tau$ are, on Ω, the smoothed loops along C. The same holds for rectifiable closed curves, with one shadow fixed on each Λ_j, with $F_\gamma \circ \pi_d$ and w_γ^K, $K \geq \max\{2, d+1\}$.

(4) *Means.* For admitted d,

$$\int Y_{d,C}\, \mathrm{d}\mu = \int F_C^{[d]}\, \mathrm{d}\mu_d = \lim_K \langle F_C^{[d]}(\mathcal{M}^{K-d}U)\rangle_K = \lim_K \langle (\mathbf{1}_{J_d} w_C^K)(\mathcal{M}^{K-d}U)\rangle_K,$$

with the rates of Prop. 10.4(d); the same holds for curves. For $d \geq d_\Theta$,

$$\Big| \int Y_{d,C}\, \mathrm{d}\mu - \int \mathbf{1}\{\tau \leq d\} Y_{d,C}\, \mathrm{d}\mu \Big| \leq \sum_{d' \geq d} R_{d'},$$

which is at most $\frac{L}{L-1} R_d$ if $c' A_0^2 \log x_{[d]} \geq 2$; if G_d is Borel on $\mathcal{V}_d$ with $|G_d| \leq 1$ and $G_d = W_C^{[d]}$ on J_d, then $G_d(V_d) = Y_{d,C}$ for all $d \geq \tau$ and $\int |G_d(V_d) - Y_{d,C}|\, \mathrm{d}\mu \leq R_d$. For admitted depths $d_1, \dots, d_n$ and loops C_i of depths $\geq d_i$, $\langle \prod_i F_{C_i}^{[d_i]}(\mathcal{M}^{K-d_i}U)\rangle_K \to \int \prod_i Y_{d_i,C_i}\, \mathrm{d}\mu$.

(5) *Locality.* In the frame, let d be admitted and at least the depth of the corners of C, $M_1 a_d \leq \frac{\delta_1}{8}$, $K > d$ and $2\varepsilon_{K-d} \leq \epsilon_{\mathrm{lc}}$; let Ω' be a slab of the depth d with faces in $M_1 a_d(\mathbb{Z} + \frac{1}{2})$, or the intersection of two such slabs, with $\mathrm{dist}(C, T_\ell \setminus \Omega') \geq \frac{\delta_1}{2}$, and let $Y_{K,d}(V) = W_C(U')$, U' the localised background formed from the bonds of V in Ω' (M_1, δ_{lc}, C_{lc}, ϵ_{lc} are the constants of the localisation of backgrounds, Prop. YM3_I:53.5 and Section YM3_I:53.4). Then for every $V \in \mathsf{O}_{K-d}$

$$|w_C^K(V) - Y_{K,d}(V)| \leq 2\mathsf{C}_{\mathrm{lc}} \hat\varepsilon_d |C| L^d e^{-\delta_{\mathrm{lc}}\delta_1 L^d/8}, \qquad \hat\varepsilon_d = A_0 (\tfrac{2}{3} x_{[d]})^{-1/2} \log^{\mathfrak{p}_0}(x_{[d]} + \bar B),$$

a bound free of K that falls faster than every power of a_d; so two fields of J_d that lie in O_{K-d} for all large K and agree on the bonds of Ω' have smoothed loops within twice this bound.

(6) *Every label.* Let $\mathsf{b} > 0$, $\ell = 2L^m$ with $m \in \mathbb{N}$, and let $s \geq 0$ be an integer with $L^s\mathsf{b} \geq \gamma^{-2}$ and $M\,Q(L^s\mathsf{b}) \leq L^{m+s}$ (Lem. 10.1). Then (1)–(4) hold at (ℓ, b) for the depths $d \geq s$, the objects of the depth d being those of $(L^s\ell, L^s\mathsf{b})$ at the depth $d-s$, carried back by $x \mapsto L^{-s}x$, with $R_{d-s}[L^s\ell, L^s\mathsf{b}]$ in the place of R_d and with the depths $d_\natural$, d_Θ, $s_\natural$ of $(L^s\ell, L^s\mathsf{b})$ increased by s; the backgrounds so obtained are backgrounds of the lattices of side ℓ. Indeed, $x \mapsto L^{-s}x$ maps Λ_{d-s} of side $L^s\ell$ onto Λ_d of side ℓ, one graph with $2L^{m+d}$ sites in each direction, and the cutoff $K-s$ onto the cutoff K, with the same number $K-d$ of block steps above the depth d; it carries the Wilson measure at $(K-s, L^s\ell, L^s\mathsf{b})$ onto the one at (K, ℓ, b), as $\beta = L^{K-s}(L^s\mathsf{b})$.

The block averaging map with its averaging radius $1/200L^2$ (equations YM3_I:(37)–(39)); the constraint sets, the action $\sum_p a_p$, criticality and the regularity at the radius ε_{v}, measured in the units of the level, with the bare spacing $L^{-(K-d)}$ (Defs. YM3_I:9.3 and YM3_I:9.5); the residual gauge groups and the minimal orbit (§5.2); the axial gauge of the averaged backgrounds ([7, §B.2]); the radius $\bar{\varepsilon}$ and the lattices of Thm. YM3_I:E in the setting of Section YM3_I:52.7, whose lattice hypotheses are imposed through the number of the sites in each direction at the levels used; and the plaquette size ϖ: all are formed from these graphs, from $K-d$ and from the constants, and hold neither the side, nor the label, nor the spacing.

And $x_{[d]} = L^d\mathsf{b} = L^{d-s}(L^s\mathsf{b})$, so that $\varepsilon_{[d]}$, I_d and J_d at (ℓ, b) are the sets of $(L^s\ell, L^s\mathsf{b})$ at the depth $d-s$. So $\mathcal{C}_K(W)$, H_d^K, w_C^K and $W_C^{[d]}$ at (ℓ, b) are the carried objects.

■□

Theorem AE (The renormalised sharp loop)**.** Work in the frame; C is a family of polygons and $\kappa = k|\mathsf{C}|/\mathsf{b}$.

(1) *Ceiling.* If $\kappa < \frac{1}{2}$, that is $|\mathsf{C}| < \frac{2\pi}{3}\mathsf{b} = 2.094\ldots\mathsf{b}$, then $\sup_{K>s_0} |\mathcal{W}_K(\mathsf{C})| < \infty$; precisely, for $\vartheta \in (\kappa, \frac{1}{2})$ and $K > d \geq s_0$,

$$\Big|\mathcal{W}_K(\mathsf{C}) - a_K^{-\kappa}\Xi_{K-d}\frac{1}{Z}\int W_{\mathsf{C}}^{(K-d)}\,\mathrm{d}m_{K-d}\Big| \leq \mathsf{E}'_d := \epsilon_1 L^{-(\vartheta-\kappa)(d+1)} + (c_\Xi L^\kappa + \epsilon_1)\,2\mathsf{V}_\ell C_{\mathrm{r}}(\kappa+1)L^{-d}, \tag{10.ii}$$

where $\epsilon_1 = c_\Xi C_\eta/(1-L^{\kappa-\vartheta})$, C_η and s_0 are the numbers of Prop. 10.6(a) at the exponent ϑ, and c_Ξ is the number of Prop. 10.7(c). So $\sup_{K>s_0}|\mathcal{W}_K(\mathsf{C})| \leq c_\Xi L^{\kappa s_0} + \mathsf{E}'_{s_0}$, and $|\Phi^{(t)}[W_{\mathsf{C}}]| \leq (c_\Xi + \epsilon_1)L^{-\kappa t}$ m_t-almost everywhere for $t \leq K - s_0$. On tori with $\ell \leq 1.48\,\mathsf{b}$ every coordinate rectangle has $\kappa < \frac{1}{2}$. If $\kappa \geq \frac{1}{2}$, then $|\langle W_{\mathsf{C}}\rangle_K| \leq c_\vartheta a_K^\vartheta$ for every $\vartheta < \frac{1}{2}$.

(2) *The limit.* If $\kappa < \frac{1}{2}$, then $\mathcal{W}(\mathsf{C}) = \lim_K \mathcal{W}_K(\mathsf{C})$ exists along the full sequence of cutoffs. Let $F_{\mathsf{C}}^{[d]} = \prod_i W_{C_i}^{[d]}$ on J_d and 0 off J_d, and

$$\mathsf{I}_d = \lim_K a_K^{-\kappa}\Xi_{K-d} = a_d^{-\kappa}\exp\Big[-\frac{3}{2\mathsf{b}}\big((c_{\mathrm{lat}} - c_Q)|\mathsf{C}| + O_{\mathsf{C},\ell}(a_d(1+\log^+(1/a_d)))\big)\Big].$$

Then for all $d \geq \max\{s_0, d_0\}$, d_0 the depth of Prop. 10.4(e),

$$\Big|\mathcal{W}(\mathsf{C}) - \mathsf{I}_d\int F_{\mathsf{C}}^{[d]}\,\mathrm{d}\mu_d\Big| \leq \mathsf{E}_d := \mathsf{E}'_d + 2c_\Xi\mathsf{V}_\ell C_{\mathrm{r}}(\kappa+1)L^{-d} + 2c_\Xi L^{\kappa d}R_d \to 0,$$

and so $|a_d^{-\kappa}\int F_{\mathsf{C}}^{[d]}\,\mathrm{d}\mu_d - \mathcal{W}(\mathsf{C})\,e^{\frac{3}{2\mathsf{b}}(c_{\mathrm{lat}}-c_Q)|\mathsf{C}|}| \leq c\,(\mathsf{E}_d + a_d(1+\log^+(1/a_d))/\mathsf{b})$, whether or not $\mathcal{W}(\mathsf{C})$ vanishes; here $c_{\mathrm{lat}} - c_Q = 0.03149\ldots$ and $\frac{3}{2}(c_{\mathrm{lat}} - c_Q) = 0.047235\ldots$. One power of the spacing renormalises; no further divergent counterterm is needed. The ball on which $F_{\mathsf{C}}^{[d]}$ is formed may be taken at the label b or at the label $\mathsf{b} - \bar{B}$: the two means differ by at most R_d. If $k|\mathsf{C}|/\mathsf{b}_- < \frac{1}{2}$, then $\mathcal{W}_K \to \mathcal{W}$ uniformly on J, $\mathcal{W}$ is continuous on J, and $\mathcal{W}_K$ formed at labels $\mathsf{b}_K \in \mathsf{J}$ with $\mathsf{b}_K \to \mathsf{b}$, b_K in the exponent too, tends to $\mathcal{W}$ at b.

(3) *Every torus, every label.* For every integer m, $\ell = 2L^m$, every $\mathsf{b} > 0$ and every family with $\kappa < \frac{1}{2}$ the limit $\mathcal{W}^{(\ell,\mathsf{b})}(\mathsf{C})$ exists along the full sequence, and $\mathcal{W}^{(L^z\ell,L^z\mathsf{b})}(L^z\mathsf{C}) = L^{-z\kappa}\mathcal{W}^{(\ell,\mathsf{b})}(\mathsf{C})$ for every integer z.

(4) *Positivity; perimeter law.* Wherever they exist, the limits have the properties of limits listed in Prop. 10.9. For every coordinate rectangle C with $\kappa < \frac{1}{2}$ there are $0 < c \le c_2$ with $c \le \mathcal{W}_K(C) \le c_2$ for all large K, uniformly in $\mathsf{b} \in \mathsf{J}$ if $k|C|/\mathsf{b}_- < \frac{1}{2}$; so $\mathcal{W}(C) > 0$. For rectangles $C_i = n_1 a_i \times n_2 a_i$ with corners in Λ_i, n_1, n_2 fixed, $d_i = \lceil (\frac{4}{3} - \frac{u}{2})i \rceil$ with $u \in (0, \frac{1}{3})$, and a fixed $\vartheta \in (0, \frac{1}{2})$, let $s_0(i) = \max\{i, s_0'\}$ and $C_\eta(i) = C(\mathsf{J})L^{\vartheta i}$ be the numbers of the one-step bound linear in $|C_i|/\mathsf{b}$ (Prop. 10.6(b)) for C_i, $C(\mathsf{J})$ depending on the constants, ϑ, J, n_1 and n_2, and let i be so large that $(n_1^2 + n_2^2)a_i^2 < \ell^2/4$ (so that C_i is a polygon), $\kappa(C_i) \le k|C_i|/\mathsf{b}_- < \vartheta$, $s_0(i) \le d_i$ and $\theta_{d_i} \le 1$, with the number θ_{d_i} below, which tends to 0 as $i \to \infty$ (so that the factor $1 - \frac{1}{2}\theta_{d_i}^2$ is at least $\frac{1}{2}$). Then

$$e^{-\frac{3}{2\mathsf{b}_-}\eta_i}(1 - \tfrac{1}{2}\theta_{d_i}^2)(1 - 2\mathsf{V}_\ell C_{\mathrm{r}}(1)L^{-d_i}) - \mathsf{E}'_{d_i} \le \mathcal{W}_K(C_i) \le c_\Xi L^{\kappa d_i} + \mathsf{E}'_{d_i} \qquad (K > d_i,\ \mathsf{b} \in \mathsf{J}),$$

where E'_{d_i} is the number of eq. 10.ii formed with $C_\eta(i)$ and $s_0(i)$ in the place of the numbers of Prop. 10.6(a), $\eta_i = \sup_{K \ge i} |c_{\mathrm{lat}}|C_i| + \mathcal{J}_\ell(C_i) + E_{a_K}(C_i)|$ with E_a the remainder of the lattice Coulomb energy (Prop. 10.7(a)), and $\theta_{d_i} = 2B_3 \sup_{K > d_i, \mathsf{b} \in \mathsf{J}} \varepsilon_{K-d_i} L^{2d_i} n_1 n_2 a_i^2$ (Prop. 10.8(d)); both ends tend to 1: $\sup\{|\mathcal{W}_K(C_i) - 1| : \mathsf{b} \in \mathsf{J},\ K > d_i\} \to 0$ as $i \to \infty$. For the squares Q_i of side a_i with corners in Λ_i there are i_2 and n_Q with $\mathcal{W}_K(Q_i) \ge \frac{1}{8}$ for all $\mathsf{b} \in \mathsf{J}$, $i \ge i_2$ and $K > i + n_Q$. On sides with $\mathsf{V}_\ell > 1$ these small rectangles must be so small that $\mathsf{V}_\ell L^{-i}$ is small. If $s \times T$ and $s \times 2T$ are rectangles with $\kappa < \frac{1}{2}$, then $\liminf_K \mathcal{W}_K(s \times nT) \ge \mathcal{W}(s \times 2T)[\mathcal{W}(s \times 2T)/\mathcal{W}(s \times T)]^{n-2}$ for every $n \ge 2$ with $nT < \ell/2$.

(5) *Block fields; a static source.* Let $\kappa < \frac{1}{2}$ and H be a bounded Borel gauge invariant function on $\mathcal{V}_{s_1}$. For $K > d' \ge \max\{s_0, s_1\}$,

$$\Big| a_K^{-\kappa} \langle W_{\mathsf{C}} H(\mathcal{M}^{K-s_1} U) \rangle_K - a_K^{-\kappa} \Xi_{K-d'} \frac{1}{Z} \int W_{\mathsf{C}}^{(K-d')} \, (H \circ \mathcal{M}^{d'-s_1}) \, \mathrm{d}m_{K-d'} \Big| \le \sup |H| \, \mathsf{E}'_{d'};$$

so $\lim_K a_K^{-\kappa} \langle W_{\mathsf{C}} H(\mathcal{M}^{K-s_1} U) \rangle_K$ exists along the full sequence and equals $\lim_d \mathsf{l}_d \int F_{\mathsf{C}}^{[d]} \, (H \circ \mathcal{M}^{d-s_1}) \, \mathrm{d}\mu_d$. For a rectangle C the quotients $\langle W_C H(\mathcal{M}^{K-s_1} U) \rangle_K / \langle W_C \rangle_K$ converge; their limit $\omega(H)$ is a linear functional on the bounded gauge invariant Borel functions of all block fields, real on the real ones, normalised, $\omega(1) = 1$, consistent between the depths, $\omega(H \circ \mathcal{M}) = \omega(H)$, and bounded at each depth: $|\omega(H)| \le \sup |H| \, (c_\Xi L^{\kappa d'} + \mathsf{E}'_{d'})/\mathcal{W}(C)$ with $d' = \max\{s_0, s_1\}$, $\mathcal{W}(C) > 0$ by (4); $c^{-1} a_d^\kappa \le \int Y_{d,C} \, \mathrm{d}\mu \le c \, a_d^\kappa$ for large d, and $\int H(V_{s_1}) Y_{d,C} \, \mathrm{d}\mu / \int Y_{d,C} \, \mathrm{d}\mu$ tends to $\omega(H)$ as $d \to \infty$. For H bounded Borel but not gauge invariant all of this holds with the gauge average $\bar{H}(V) = \int H(V^v) \, \mathrm{d}v$ (Haar measure on the gauge transformations of Λ_{s_1}) in the place of H, the Wilson measure, μ and $Y_{d,C}$ being gauge invariant.

(6) *Mixed functions.* Let a coordinate mirror θ part C, at distance $\ge \delta_1 > 0$ from its planes, from the supports of real Lipschitz $f_1, \dots, f_n$, at distance $\ge \delta > 0$ from the planes, and let $2\kappa < \frac{1}{2}$. Then $Q_K(z) = a_K^{-\kappa} \langle W_{\mathsf{C}} e^{\mathcal{O}(w_z)} \rangle_K / \langle e^{\mathcal{O}(w_z)} \rangle_K$, $w_z = \sum_q z_q f_q$, obeys $|Q_K(z)| \le \sqrt{6} \, e^{\Delta(\delta)/2} \mathcal{W}_K(\mathsf{C}, \theta\mathsf{C})^{1/2}$ for $K > s_0$ on a polydisc free of K, and converges locally uniformly there to a holomorphic Q; the supports need not be disjoint. If they are pairwise disjoint, then $\lim_K \mathcal{W}_K(\mathsf{C}; f_1, \dots, f_n)$ exists along the full sequence and equals $\sum_{A \subset \{1,\dots,n\}} \partial_A Q(0) \sum_\pi \prod_{\beta \in \pi} S_{|\beta|}(f_\beta)$, π running over the partitions of the complement of A into blocks of at least two elements, $f_\beta = (f_q)_{q \in \beta}$.

The same holds for two orthogonal coordinate mirrors θ_1, θ_2 if $4\kappa < \frac{1}{2}$, C lies on one side of both at distance $\ge \delta_1$ from their planes, and the f_q fall into two groups, the first with supports on the other side of θ_1, the second with supports on the side of C of θ_1 and on

the other side of θ_2, the two groups at sup-distance $\ge 2\delta$ and all supports at distance $\ge \delta$ from the planes. For $n = 1$ it holds, if $2\kappa < \frac12$ and C has extent $< \frac{\ell}{2}$ in each coordinate direction, for every real Lipschitz f whose support misses the least closed coordinate box that holds C. Where no such mirror parts them, but N equal coordinate boxes between equally spaced walls in $\frac12 a_i\mathbb{Z}$, for an index i, hold each polygon in one box, the number of the boxes being even in every direction that the walls cut, so that both fixed planes c and $c + \frac{\ell}{2}$ of every wall mirror are walls (on $\ell = 2L^m$, $2L^j$ or $4L^j$ slabs in each cut direction), each f_q is supported in one box, the supports lie off the walls and no box meets both a polygon and a support, one has $\langle W_{\mathsf{C}^{\mathrm{f}}_\iota}\rangle_K \ge 0$ and $|Q_K(z)| \le M_S \prod_\iota \mathcal{W}_K(\mathsf{C}^{\mathrm{f}}_\iota)^{1/N}$ for $\sum_q |z_q| \|f_q\|_\sharp < \rho_D/N$, a polydisc free of K, with $\|f\|_\sharp$, ρ_D and M_S of Prop. 10.10, $\mathsf{C}^{\mathrm{f}}_\iota$ being the family of the images under the wall reflections of the polygons C_ι of the box ι, which has exactly one image in every box, with $\kappa(\mathsf{C}^{\mathrm{f}}_\iota) = N\kappa(\mathsf{C}_\iota)$; so, if $N\kappa(\mathsf{C}_\iota) < \frac12$ for every box ι, $\sup_K |\partial_A Q_K(0)| < \infty$ for every $A \subset \{1,\dots,n\}$, and this holds whatever the supports off the walls, a function whose support misses the walls being the sum of pieces each supported in one box; and $\sup_K |\mathcal{W}_K(\mathsf{C}; f_1,\dots,f_n)| < \infty$ when the supports are pairwise disjoint, the truncated functions of the action density being bounded in K there YM3_I:(2212); for $n = 1$, $\mathcal{W}_K(\mathsf{C}; f_1) = \partial_1 Q_K(0)$, with no further condition.

The limits obey the law of dilations $\mathcal{W}^{(L^z\ell, L^z\mathsf{b})}(L^z\mathsf{C}; f^{(z)}) = L^{-z\kappa - zn}\mathcal{W}^{(\ell,\mathsf{b})}(\mathsf{C}; f)$, $f_q^{(z)} = f_q(L^{-z}\cdot)$, and the positive form of Prop. 10.9(b) with mixed entries: let θ' be the coordinate mirror $\theta' x = x - 2(x_\mu - \hat c)e_\mu$ with a place $\hat c \in \frac12 a_{i_0}\mathbb{Z}$, let families $G_1,\dots,G_s$ lie in $\{\hat c < x_\mu < \hat c + \frac{\ell}{4}\}$ and tuples $f_1,\dots,f_s$ of real Lipschitz functions, each tuple with pairwise disjoint supports, have their supports in $\{\hat c + \frac{\ell}{4} < x_\mu < \hat c + \frac{\ell}{2}\}$, all at positive distance from the planes $x_\mu \in \hat c + \frac{\ell}{4}\mathbb{Z}$, and let $2[\kappa(G_i) + \kappa(G_j)] < \frac12$ for all i, j; then each entry $\mathcal{W}(\theta' G_i, G_j; \theta' f_i, f_j)$ is a limit of the kind above, the coordinate mirror with the place $\hat c + \frac{\ell}{4}$, which lies in $\frac12 a_{i_0}\mathbb{Z}$ as $\frac{\ell}{4} = \frac12 L^m$, parting the family $(\theta' G_i, G_j)$ from the supports of $\theta' f_i$ and f_j, and $\sum_{i,j} \bar c_i c_j \mathcal{W}(\theta' G_i, G_j; \theta' f_i, f_j) \ge 0$ for all complex $c_1,\dots,c_s$.

(7) *The action density beside a loop.* Let C be a polygon with $\psi = \langle W_C\rangle_K > 0$ (every coordinate rectangle has it), f bounded and real, and θ a site mirror with C in one open half torus and every plaquette p with $f(x(p)) \ne 0$ with its bonds in the other closed half S. Write $\langle X; Y\rangle_K = \langle XY\rangle_K - \langle X\rangle_K\langle Y\rangle_K$. At every cutoff, for every bounded real Borel function H_0 of the bonds of S and $H = \mathcal{O}(f) - H_0$: $\langle W_C; W_{\theta C}\rangle_K \ge 0$, $\langle \theta H; H\rangle_K \ge 0$, and

$$\Big|\frac{\langle W_C; \mathcal{O}(f)\rangle_K}{\psi} - \frac{\langle W_C; H_0\rangle_K}{\psi}\Big| \le \Big[\frac{\langle W_C W_{\theta C}\rangle_K}{\psi^2} - 1\Big]^{1/2} \langle \theta H; H\rangle_K^{1/2};$$

at $H_0 = 0$ the left side is $|\langle W_C \mathcal{O}(f)\rangle_K/\psi - \langle \mathcal{O}(f)\rangle_K|$. Let moreover, in the frame, $g \le \gamma_W$, the ceiling of equation YM3_I:(2346); let $\vartheta_{\mathrm{sh}} \in (0,1)$ be a shape parameter, $\mathsf{d} \in (0, \mathsf{d}_0]$ with the threshold $\mathsf{d}_0 = \mathsf{d}_0(g, \vartheta_{\mathrm{sh}}) \in (0, \frac{1}{200}]$ of equation YM3_I:(2350), and $K - s(\mathsf{d}) \ge 1$, $s(\mathsf{d})$ the witness depth at g of equation YM3_I:(2321); and let $(\theta f, f)$ be a test pair of the class $\mathfrak{T}_{T_\ell}(\mathsf{d}; \vartheta_{\mathrm{sh}})$ of [7, Lem. I.1], on a torus with $20\mathsf{d} \le \ell$: f is non-negative, Lipschitz and not $\equiv 0$, the supports of θf and f lie in closed balls of radius d and are at a distance in $[\mathsf{d}, 6\mathsf{d}]$, $\int f \ge \vartheta_{\mathrm{sh}} \|f\|_\infty \mathsf{d}^3$ and $\mathrm{Lip}\, f \le \|f\|_\infty/\vartheta_{\mathrm{sh}}\mathsf{d}$.

As $(K, L^K\mathsf{b})$ is a first-passage datum at g with the label $\mathsf{b} \in \mathsf{l}(g)$ (Cor. YM3_I:74.2), on the side $\ell \in \mathfrak{L}(g)$ of the frame, the law of the truncated two-point function of the action density, Thm. YM3_I:AV, gives $\langle \theta\mathcal{O}(f); \mathcal{O}(f)\rangle_K = \mathsf{c}_2 \mathsf{b}^{-2} \iint f(\theta x) f(y) |x-y|^{-6}\, \mathrm{d}x\, \mathrm{d}y\, (1+\epsilon')$, with $\mathsf{c}_2 = \frac{9}{16\pi^2}$ the constant of YM3_I:C, the double integral the free kernel YM3_I:(2240) of the test pair, and $|\epsilon'| \le \epsilon_{\mathrm{tp}} := \bar\varepsilon_{g,\vartheta_{\mathrm{sh}}}(\mathsf{d})$, the tolerance of equation YM3_I:(2349), which is

at most $\frac{1}{3}$ in that theorem; with the bound above and $\mathsf{c}_2^{1/2} = \frac{3}{4\pi}$ this gives

$$\Big|\frac{\langle W_C\mathcal{O}(f)\rangle_K}{\psi} - \langle\mathcal{O}(f)\rangle_K\Big| \le \frac{3}{4\pi\mathsf{b}}\Big[(1+\epsilon_{\mathrm{tp}})\iint\frac{|f(\theta x)f(y)|}{|x-y|^6}\,\mathrm{d}x\,\mathrm{d}y\Big]^{1/2}\Big[\frac{\langle W_CW_{\theta C}\rangle_K}{\psi^2}-1\Big]^{1/2};$$

and, as the free kernel of a test pair is at most $(\frac{4\pi}{3})^2\|f\|_\infty^2$ (equation YM3_I:(2247); two balls of radius d at a distance $\ge\mathsf{d}$), the same two-point estimate gives on this whole class $\langle\theta\mathcal{O}(f);\mathcal{O}(f)\rangle_K \le \frac{4}{3}(\frac{4\pi}{3})^2\mathsf{c}_2\mathsf{b}^{-2}\|f\|_\infty^2 = \frac{4}{3}\mathsf{b}^{-2}\|f\|_\infty^2$, so that, with no condition beyond this class,

$$\Big|\frac{\langle W_C\mathcal{O}(f)\rangle_K}{\psi} - \langle\mathcal{O}(f)\rangle_K\Big| \le \frac{2}{\sqrt{3}}\,\mathsf{b}^{-1}\|f\|_\infty\Big[\frac{\langle W_CW_{\theta C}\rangle_K}{\psi^2}-1\Big]^{1/2}.$$

Besides, at every cutoff,

$$\frac{\langle W_C\mathcal{O}(f)\rangle_K}{\psi} - \langle\mathcal{O}(f)\rangle_K = -\frac{1}{a_K}\sum_p f(x(p))\langle W_{\partial p}\rangle_K\Big[\frac{\langle W_CW_{\partial p}\rangle_K}{\psi\langle W_{\partial p}\rangle_K}-1\Big],$$

$$\frac{d}{d\mathsf{b}}\log\langle W_C\rangle_K = -\Big[\frac{\langle W_C\mathcal{O}(1)\rangle_K}{\psi} - \langle\mathcal{O}(1)\rangle_K\Big].$$

For a rectangle with $2\kappa < \frac{1}{2}$ at positive distance from the planes of θ and Lipschitz f with support at positive distance from them, $\langle W_C;\mathcal{O}(f)\rangle_K/\psi = \mathcal{W}_K(C;f)/\mathcal{W}_K(C)$, $\langle W_CW_{\theta C}\rangle_K/\psi^2 = \mathcal{W}_K(C,\theta C)/\mathcal{W}_K(C)^2$ and $\langle\theta\mathcal{O}(f);\mathcal{O}(f)\rangle_K = S^{\mathsf{T}}_{2,K}(\theta f,f)$ converge as $K\to\infty$.

(8) *Short distances.* Let $0<\vartheta<\frac{1}{2}$ and $p\ge1$. (a) For rectangles $C_i = n_1a_i\times n_2a_i$ with corners in Λ_i, n_1, n_2 fixed, $\varrho_i = \max\{n_1,n_2\}a_i$, $\mathsf{n} = \max\{L,n_1,n_2\}$, all large i and all $K>i$ with $\Delta_i\le\frac{1}{2}$,

$$\Big|\log\mathcal{W}_K(C_i) + \frac{3}{2\mathsf{b}}\big[c_{\mathrm{lat}}|C_i| + \mathcal{J}_\ell(C_i) + E_{a_K}(C_i)\big]\Big| \le \kappa\log L + c\frac{|C_i|}{\mathsf{b}} + 2\Delta_i,$$

$$\Delta_i = 4B_3^2A_0^2\mathsf{n}^3\frac{\varrho_i}{\mathsf{b}}\Big(1+\log\frac{\mathsf{n}\mathsf{b}}{\varrho_i}\Big)^{2\mathsf{p}_0} + C_1\frac{|C_i|}{\mathsf{b}}\Big(\frac{\varrho_i}{\mathsf{b}}\Big)^{\vartheta/2} + C_3\mathsf{V}_\ell C_{\mathrm{r}}(p)\varrho_i^{p-1/2},$$

E_a the remainder of Prop. 10.7(a) and c, C_1, C_3 depending on the constants, ϑ, n_1, n_2 and J: an error of the order of the main term, which gives the two-sided bound of (4) at these rectangles. (b) For rectangles C with sides in $[c_0\varrho,10\varrho]$ and corners in Λ_i, i the integer with $a_i\le\frac{1}{2}c_0\varrho<La_i$, put $a_*=\varrho(\mathsf{b}/\varrho)^{\vartheta/(3+\vartheta)}$, a length coarser than the loop. At every level t with $a_{K-t}\le a_*$ the mean over the block field of the defect of one step, $\bar{\mathsf{D}}_{t+1}$ of Prop. 10.8(b) for C, has modulus at most $C_*\frac{\varrho}{\mathsf{b}}(\frac{a_{K-t}}{\mathsf{b}})^\vartheta$, C_* depending on the constants, c_0, ϑ and J alone: at the levels finer than the corners, $K-t>s_0=\max\{i,s_0'\}$, by Prop. 10.6(b), and at the levels with $a_i\le a_{K-t}\le a_*$, about $\frac{\vartheta}{3+\vartheta}\log_L\frac{\mathsf{b}}{\varrho}$ of them, where the loop passes a number of the cubes of the level that is bounded in terms of L and c_0, by the argument of one step with this bounded number of segments. Hence, as $\varrho\to0$, with a constant depending on the constants, c_0, ϑ, J and ℓ,

$$\log\mathcal{W}(C) = -\frac{3}{2\mathsf{b}}\big[\mathcal{J}(C)+c_{\mathrm{lat}}|C|\big] + O\Big(\Big(\frac{\varrho}{\mathsf{b}}\Big)^{1+\frac{3\vartheta}{3+\vartheta}}\Big(1+\log\frac{\mathsf{b}}{\varrho}\Big)^{2\mathsf{p}_0} + \frac{\varrho^4}{\mathsf{b}\,\ell^3}\Big);$$

the exponent $1+\frac{3\vartheta}{3+\vartheta}$ reaches $\frac{10}{7}$ as $\vartheta\to\frac{1}{2}$, the length a_* balancing the step errors, $\frac{\varrho}{\mathsf{b}}(\frac{a_*}{\mathsf{b}})^\vartheta$, against the last level, $\frac{\varrho^4}{\mathsf{b}a_*^3}$, and the remainder, which holds $\log(\mathsf{b}/\varrho)$ and not $\log\varrho$, is invariant under dilations.

■□

Theorem AF (Without the renormalisation group)**.** Let C be a coordinate rectangle on a torus of side ℓ, $\mathsf{b} > 0$, a the spacing, $\beta = \mathsf{b}/a \geq e^3$, $\Lambda = \max\{1, |C|/\mathsf{b}\}$, $\rho = \lfloor (\frac{\beta}{\Lambda \log^3 \beta})^{1/4} \rfloor \geq 2$, the sides of C at least $(4\rho+1)a$, and the points within $(\rho+1)a$ of C in a cube of side $\leq \frac{\ell}{2}$. Then

$$|\langle W_C \rangle| \leq 6 e^{3.1\mathsf{b}/\ell}\, 1.58^{|C|/\mathsf{b}} \Big[\Lambda \frac{a}{\mathsf{b}} \log^3 \frac{\mathsf{b}}{a}\Big]^{k|C|/4\mathsf{b}}.$$

With a dyadic $\rho = 2^S$ instead, sides at least $(4\rho+1)a$ and $k|C|/\mathsf{b} < 3$, the ceiling of the tube with the profiles of the scales $2, 4, \dots, 2^S$ reads

$$|\langle W_C \rangle| \leq e^{\delta_0 + 2.2144 k|C|/\mathsf{b}} \rho^{-k|C|/\mathsf{b}} \Big[1 + \frac{72\, \bar{a}\, \rho^3 \log^2 \rho}{2^{3-k|C|/\mathsf{b}} - 1}\Big] + 35\, \bar{a}, \qquad \delta_0 = \frac{16 k \rho \log \rho + k^2 (|C|/\mathsf{b}) \log^2 \rho}{\beta},$$

with the plaquette mean $\bar{a} = \langle a_p \rangle \leq \beta^{-1} \log \mathfrak{M}_\nu$ (Lem. 10.12(f)), the condition $k|C|/\mathsf{b} < 3$ keeping the denominator positive; at $\rho \in (\frac{1}{2}\beta^{1/3}, \beta^{1/3}]$, where $\bar{a}\rho^3 \log^2 \rho$ is at most a multiple of $\log^3 \beta$, and $\beta \geq \beta_2$, a threshold depending on $k|C|/\mathsf{b}$ alone (the term $192\frac{\mathsf{b}}{\ell}$ of $\log \mathfrak{M}_\nu$ is at most $384\mathsf{b}/|C|$, as $|C| < 2\ell$): $|\langle W_C \rangle| \leq C(\frac{k|C|}{\mathsf{b}})(\frac{a}{\mathsf{b}})^{|C|/4\pi\mathsf{b}} \log^3 \frac{\mathsf{b}}{a}$, as $\rho^{-k|C|/\mathsf{b}} \leq 2^{k|C|/\mathsf{b}} (\frac{a}{\mathsf{b}})^{k|C|/3\mathsf{b}}$ and $\frac{k}{3} = \frac{1}{4\pi}$. The first bound holds in the same form, with a constant depending also on the number of pieces, for polygons whose pieces have at least $4\rho+1$ steps and whose pieces without a common corner are at sup-distance at least $(2\rho+1)a$, for families of such polygons in which also any two pieces of different polygons are at sup-distance at least $(2\rho+1)a$, also with a further factor G, $|G| \leq 1$, a Borel function of the bonds farther than ρa from the polygons; and for the characters χ_j of spin j, with $\kappa_j = \frac{4}{3} j(j+1) \leq 20 \log \beta$ and $A = 6e^{3.1\mathsf{b}/\ell}$:

$$|\langle \chi_j(U(C)) \rangle| \leq (2j+1)(A\, 1.58^{|C|/\mathsf{b}})^{\kappa_j} e^{\kappa_j^2/100} \Big[\Lambda \frac{a}{\mathsf{b}} \log^3 \frac{\mathsf{b}}{a}\Big]^{\kappa_j k|C|/4\mathsf{b}}.$$

These bounds hold with 3 in the place of $6e^{3.1\mathsf{b}/\ell}$ for every periodic Gibbs state of the Wilson action at β on $\mathbb{Z}^3$ that is invariant under the isometries of $\mathbb{Z}^3$ and reflection positive in the lattice planes. So, as $a \to 0$ with b in a compact set, uniformly in $\ell \geq \ell_0$, for polygons: $\langle h(W_C) \rangle \to \sigma(h) = \frac{2}{\pi} \int_{-1}^{1} h(w) \sqrt{1-w^2}\, \mathrm{d}w$ for continuous h, $\langle W_C^2 \rangle \to \frac{1}{4}$, and for polygons C_i at pairwise positive distances, continuous h_i and Borel G, $|G| \leq 1$, depending only on the bonds at distance $\geq \delta > 0$ from them, $\langle \prod_i h_i(W_{C_i})\, G \rangle - \prod_i \sigma(h_i) \langle G \rangle \to 0$.

For a coordinate rectangle, whose mean $\langle W_C \rangle$ is positive (Prop. 10.9(c)), the normalised sharp loop has the variance $\mathrm{Var}(W_C/\langle W_C \rangle) = (1 + \langle \chi_1(U(C)) \rangle)/(4\langle W_C \rangle^2) - 1$, which tends to infinity as $a \to 0$, at least like a positive constant times $[\Lambda \frac{a}{\mathsf{b}} \log^3 \frac{\mathsf{b}}{a}]^{-k|C|/2\mathsf{b}}$; and no multiple $c_a W_C$ of the sharp loop has both a limit law and a mean $c_a \langle W_C \rangle$ that stays away from 0. In the other direction, $\langle W_C \rangle \geq \langle W_{\partial p} \rangle^{A_C/a^2} \geq \exp[-\frac{3}{2} \frac{A_C}{a\mathsf{b}} (192\frac{\mathsf{b}}{\ell} + 56.6 + 12 \log \beta)]$ for a rectangle of area A_C and sides $< \ell/2$, the second inequality when $192\frac{\mathsf{b}}{\ell} + 56.6 + 12 \log \beta \leq \frac{\beta}{2}$; for the Gibbs states above and rectangles with even numbers of steps, $\langle W_C \rangle \geq \exp[-\frac{A_C}{a\mathsf{b}} (43 + 9 \log \frac{\mathsf{b}}{a})]$ when $28.3 + 6 \log \beta \leq \frac{\beta}{2}$.

Without the renormalisation group the floor falls like the exponential of the area in units of $a\mathsf{b}$. ■□

10.1.4. *Construction estimates.* Thm. YM3_I:D with Cor. YM3_I:69.4 on the actual admissible histories enters through the objects of the frame and through several of the following statements, cited by their keys, which we use: the limit laws of the block fields, with one block map for all cutoffs, with their rate and with the two depths of their proof (Cor. 2.8, (2.xv), §2.3.1), the numbers C_{bf}, K_{bf} of the rate being formed at an interval of labels under the condition (Tor) (Defs. 2.3.1 and 2.7(5)); the bound on the history without large-field regions (Prop. YM3_I:115.2, YM3_I:(2187)); the rarity of large-field regions (Prop. 4.54); the bound at one place at the fixed and at the lowered constants (Lem. 4.51, Def. 4.47); the three smallness conditions that define $t_\natural$ and the facts on them (§2.3.1, §2.3.1, §2.3.1); the comparison of two consecutive cutoffs (Prop. YM3_I:113.9, YM3_I:(2154)) with the competitors, the lower bound on their excess, the chart of the fibre at the

pivot bonds and the numbers of the comparison (Lem. YM3_I:113.8 and Prop. YM3_I:113.5, the chart bounds in equation YM3_I:(2138), the global hypotheses preceding equation YM3_I:(2132) and the constants in and before equation YM3_I:(2133)), the averaged backgrounds (Prop. YM3_I:52.3, the axial-normalization argument in the proof of Prop. YM3_I:52.3), and the response of the background to a change of its datum with the averages of a moved field (Thm. YM3_I:E and Lem. YM3_I:51.6, with the thresholds of §5.3, Lem. 9.11(v)), which give the comparison of the averaged backgrounds of two consecutive cutoffs over one block field and along the tower, and the Lipschitz bound of the tower in the block field (Lem. 10.3); the existence of the background over a small field as one orbit (Thm. YM3_I:E in the setting of Section YM3_I:52.7, §5.2); the localisation of backgrounds (Prop. YM3_I:53.5, Section YM3_I:53.4, including equations YM3_I:(998)–YM3_I:(1000)), with the localised background of a region (equation YM3_I:(77)) and the arguments that give its hypotheses for a slab of M_1-cubes (the geometric verification in Step 5 of the proof of Lem. 10.12, using Prop. YM3_I:44.2); the covariance of the block averages and of the backgrounds under the lattice isometries ([7, §G.1], [7, Lem. G.1]); for one step, the triples, the analyticity cores, the chart at the trivial pair and the ball about the trivial triple in the core (§5.2, Def. YM3_I:49.1, and equation YM3_I:(933)), the background triple and its regularity (the triple defined before equation YM3_I:(827) and §5.3.1), the kernel of one step (eq. 5.xxxii and Prop. 5.31), the provision of the step on the whole core (Def. YM3_I:52.1, Thm. YM3_I:W), the shift of Prop. YM3_I:51.5 with their brackets (equations YM3_I:(806)–YM3_I:(809) and Lem. 5.25), the chart and the pivot values at $t = 0$ (equation YM3_I:(804) and the bounds and covariance proved with it), the potential on a cube of any side (Thm. YM3_I:F), the cluster expansion of the step with up to four potential families (Thm. YM3_I:R, Cor. YM3_I:47.3 and Prop. YM3_I:47.4, with the following covariance argument), applied to the actual Gaussian structure and the two potentials of Prop. 5.31, the constrained variational equation for critical curves at the trivial field (the proof of Lem. YM3_I:124.1, through its gauge normalization), and its curvature identification (equation YM3_I:(2230)), the curvature response (the curvature-response construction preceding Lem. YM3_I:120.2, the curvature-response calculation preceding Lem. YM3_I:120.2), the slice of the free Coulomb energy at the trivial triple (Prop. YM3_I:120.3, Prop. YM3_I:120.3) and the linearised shift system (Lem. YM3_I:124.3, whose proof, with the two changes named in the proof of Prop. 10.6(a), runs for the objects of the frame on every side of $\mathfrak{L}(g)$); and the limits of the truncated functions of the action density YM3_I:(2212).

Thms. YM3_I:AJ and YM3_I:AD, the bound for a history with a source on a polydisc uniformly in K, is used in this section only in Prop. 10.10, hence in Thm. AE(6) and in the convergence of the three quantities of Thm. AE(7); the explicit ceiling of Thm. AE(7) uses besides the law of the truncated two-point function of the action density, Thm. YM3_I:AV, with its test pairs ([7, Lem. I.1]), the free kernel YM3_I:(2240) and its bound on a test pair (equation YM3_I:(2247)), the constant c_2 YM3_I:C, its ceiling γ_W equation YM3_I:(2346), its threshold d_0 equation YM3_I:(2350), its witness depth $s(\mathsf{d})$ in equation YM3_I:(2321) and its tolerance equation YM3_I:(2349), and the first-passage data (Cor. YM3_I:74.2).

The facts on the constants (Prop. YM3_I:69.1, the calibration in §2.3.1, and [7, Cor. B.13]) make fixed constants satisfy the conditions of the setting. Besides these we use standard facts: reflection positivity of the Wilson measure on the torus at site and bond mirrors, the chessboard estimate, Kolmogorov's extension theorem, Vitali's theorem, Schwarz's lemma and Cauchy's estimates. In Thm. AD(1)–(5) and in Props. 10.2 and 10.4(a)–(d) one torus and one label are fixed, and the limit is $K \to \infty$, with the limit $n \to \infty$ of the tower inside it; in the frame the numbers of the rate of Prop. 10.4(d) serve all the labels of J.

Thm. AE takes besides the limit $d \to \infty$ of means after $K \to \infty$ ((2), (5)), lets the label move in a compact set J and along labels $\mathsf{b}_K \to \mathsf{b}$ ((2)), lets rectangles shrink, $i \to \infty$ and $\varrho \to 0$ ((4), (8)), and moves the torus and the label together by dilations ((3), as Thm. AD(6) does). Thm. AF takes

$a \to 0$ uniformly in $\ell \geq \ell_0$ with b in a compact set, and holds also for periodic Gibbs states on $\mathbb{Z}^3$. In Thms. AD and AE the side ℓ of the torus is fixed, or moves together with the label, by dilations.

10.1.5. *Auxiliary estimates.*

Lemma 10.1. Let $L \geq 2$, $m' \in \mathbb{Z}_{\geq 0}$, $r, m \in \mathbb{N}$, $M = L^{m'}$, b real with $\log \mathsf{b} > 1$, and $Q(\mathsf{b}) = \min\{L^j : j \in \mathbb{Z}_{\geq 0},\ L^j \geq (\log \mathsf{b})^r\}$. Then $M\,Q(\mathsf{b}) \leq L^m$ if and only if $L^m \geq M(\log \mathsf{b})^r$. The hypothesis $\log \mathsf{b} > 1$ is needed: at $L = 3$, $m' = 2$, $m = 1$, $r = 2$, $\mathsf{b} = 1.2$ the second condition holds and the first fails. In the setting it holds since $\gamma \leq \frac{1}{2}$. ■

Sketch of proof. $Q \geq (\log \mathsf{b})^r$ gives $M(\log \mathsf{b})^r \leq MQ \leq L^m$. Conversely $L^{m-m'} \geq (\log \mathsf{b})^r > 1$, so $j = m - m'$ is a positive integer, L^j lies in the set whose least element is Q, and $Q \leq L^{m-m'}$. In the example $(\log 1.2)^2 = 0.033\ldots$, so $Q = 1$ and $MQ = 9 > 3 = L^m \geq 9 \cdot 0.033$. □

Proposition 10.2 (Laws on the threads; small plaquettes)**.** In the one-torus setting: (a) there is exactly one Borel probability measure μ on $\mathcal{X}$ whose image under $V_\bullet \mapsto (V_0, \ldots, V_D)$ is, for every D, the image of μ_D under $V \mapsto (\mathcal{M}^D V, \ldots, \mathcal{M}V, V)$; $\mu(\mathsf{Thr}) = 1$, the image of μ under π_d is μ_d, μ is θ_v-invariant, and the law of $(\mathcal{M}^K U, \ldots, \mathcal{M}^{K-D} U)$ under $\langle\cdot\rangle_K$ tends in total variation to the image of μ under $(\pi_0, \ldots, \pi_D)$; (b) each I_d is open and gauge invariant; for $d \geq d_\natural$ and $K > d$, $\mu_d^K(\mathcal{V}_d \setminus I_d) \leq R_d$ and $\mu_d(\mathcal{V}_d \setminus I_d) \leq R_d$; for $D \geq d_\natural$, $\mu(\tau_0 > D) \leq \sum_{d \geq D} R_d$, which is at most $\frac{L}{L-1} R_D$ if $c' A_0^2 \log x_{[D]} \geq 2$; and R_d tends to 0 faster than every L^{-pd}; (c) for $d \geq s_\natural$, $K > d$ and every set S of sites of Λ_d, the μ_d^K- and the μ_d-probability that some plaquette with a corner in S has $|\partial V(p) - \mathbf{1}| \geq 2\varepsilon_{[d]}$ are at most $\#S\, e^{-c^\natural A_0^2 (\log x_{[d]})^{2\mathfrak{p}_0}}$; (d) with the constants $C_5, \mathsf{C}_{\mathrm{maj}} \geq 0$ of Thm. YM3_I:D, $s_d = \frac{1}{4}\min\{\bar{\varepsilon}, B_3^{-1}\varepsilon_{[d]}\}$,

$$\mathsf{p}^{[d]} = \exp\bigl[-(c_{\mathrm{low}} + \tfrac{9}{2}\log x_{[d]} + \mathsf{C}_{\mathrm{maj}} + 2)(\ell L^d)^3\bigr], \qquad \mathsf{q}^{[d]} = \exp\Bigl[-\Bigl(C_5 + \mathsf{C}_{\mathrm{maj}} + 2 + 3\log\frac{7}{s_d^3} + 24 B_3^2 s_d^2 x_{[d]}\Bigr)(\ell L^d)^3\Bigr],$$

one has at every admitted depth d and for $K \geq \max\{2, d+1\}$: $\mu_d^K(J_d) \geq \mu_d^K(I_d) \geq \mathsf{q}^{[d]} > 0$, and $\mu_d^K(I_d) \geq \max\{1 - R_d, \mathsf{p}^{[d]}, \mathsf{q}^{[d]}\}$ if $d \geq 1$; the same holds for μ_d. The floors hold no K and fall like $e^{-c(\ell L^d)^3}$. ■

Sketch of proof. (a) By one block map, $\mu_{d+1} \circ \mathcal{M}^{-1} = \mu_d$ (Cor. 2.8(c)), so the images of the μ_D form a projective family of Borel probabilities on the finite products of the compact metric spaces $\mathcal{V}_d$, which has exactly one extension to $\mathcal{X}$ (Kolmogorov's theorem). Each marginal is carried by the Borel set $\{V_d = \mathcal{M}V_{d+1},\ d < D\}$, whence $\mu(\mathsf{Thr}) = 1$; gauge invariance of the μ_d and covariance of $\mathcal{M}$ give the invariance; the joint law at the cutoff K is the image of the single law μ_D^K under $V \mapsto (\mathcal{M}^D V, \ldots, V)$, and $\mu_D^K \to \mu_D$ in total variation. (b) ϖ is continuous and $\varpi(V^v) = \varpi(V)$.

Here Thm. YM3_I:D is applied along the line of couplings $x_{[d']} = L^{d'}\mathsf{b}$, $0 \leq d' \leq K$ (the run with all $u_t = 0$, at the label b), with the inner ball $I_{d'} = \{\varpi < 2\varepsilon_{[d']}\}$ at the depth d': the partition function is a sum over the histories of large-field regions, and the small-field chain of this line (Prop. YM3_I:115.2), which at the depths $d' > d$ cuts the fields at the radii $c_\rho \varepsilon_{[d']}$, measured on their backgrounds (the definitions preceding Lem. YM3_I:113.10), is a bounded Borel weight $\mathfrak{w} \geq 0$ on the bare fields with three properties: the image of $\mathfrak{w}\, e^{-\beta\sum_p a_p} dU$ under $\mathcal{M}^{K-d}$ is carried by I_d (clause (a) there); $\int F(1-\mathfrak{w}) e^{-\beta\sum_p a_p}\, \mathrm{d}U \geq 0$ for every gauge invariant bounded Borel $F \geq 0$ (clause (b), which places $\mathfrak{w}$ between the history without large-field regions and the full density); and $\int (1-\mathfrak{w}) e^{-\beta\sum_p a_p}\, \mathrm{d}U \leq R_d Z$, R_d being YM3_I:(2187) at the depth d and the label b (clause (c)).

As I_d is gauge invariant, $\mu_d^K(\mathcal{V}_d \setminus I_d) = Z^{-1} \int \mathbf{1}_{\mathcal{V}_d \setminus I_d}(\mathcal{M}^{K-d}U)(1-\mathfrak{w}) e^{-\beta\sum_p a_p}\, \mathrm{d}U \leq R_d$, uniformly in K. This chain of the line is another object than the measures m_t of the frame, which follow the

run, with $|x_{K-d} - x_{[d]}| \le \bar{B}$ and the thresholds $2\varepsilon(x_t)$: the tame set O_{K-d} that carries m_{K-d} is placed by Prop. 10.4(e) in the larger ball J'_d. The bound for μ_d^K passes to μ_d in total variation. For $D \ge d_\natural$, $\{\tau_0 > D\} = \bigcup_{d\ge D} \pi_d^{-1}(\mathcal{V}_d \setminus I_d)$, and the image of μ under π_d is μ_d by (a); countable subadditivity gives $\mu(\tau_0 > D) \le \sum_{d\ge D} R_d$.

As $(y+\log L)^{2\mathfrak{p}_0} - y^{2\mathfrak{p}_0} \ge 2y \log L$ for $y \ge 1$, and $\log x_{[d]} \ge \log \mathsf{b} > 1$, $R_{d+1}/R_d \le L^{3-2c'A_0^2 \log x_{[d]}}$, which is at most L^{-1} for all $d \ge D$ if $c'A_0^2 \log x_{[D]} \ge 2$; then $\sum_{d\ge D} R_d \le \frac{L}{L-1} R_D$. R_d falls faster than every L^{-pd} because $(\log x_{[d]})^{2\mathfrak{p}_0} \ge (d \log L)^2$. (c) One witness function of a large plaquette at one place has a mean at most $e^{-c^\natural A_0^2 (\log x_{[d]})^{2\mathfrak{p}_0}}$, free of the volume: Lem. 4.51, which applies Thm. YM3_I:D at the fixed constants and at the lowered choice and frees the bound of the volume by the chessboard estimate for the reflection positive Wilson measure.

Sum over the corners in S; the bound passes to μ_d in total variation. (d) For $\mathsf{p}^{[d]}$: Thm. YM3_I:D, along the line of couplings of (b), bounds, at the level of Λ_d, the mass of the history without large-field regions below by $e^E e^{-(c_{\mathrm{low}} + \frac{9}{2}\log x_{[d]})(\ell L^d)^3}$ and the partition function above by $e^E e^{(\mathsf{C}_{\mathrm{maj}}+2)(\ell L^d)^3}$, with one number E, and that history is carried by fields whose block field of the depth d lies in the inner ball I_d of the line. For $\mathsf{q}^{[d]}$: Thm. YM3_I:D along the line bounds the density of the block field of the depth d below, as a measure, by $e^{E - C_5(\ell L^d)^3}$ times the small-field indicator times the Boltzmann factor of the background over V; on the ball of the fields with $|V(b) - \mathbf{1}| < s_d$ at every bond, whose plaquettes are $< 4s_d \le \varepsilon_{[d]}$ so that it lies in I_d, the integral of this density is at least $\exp[-(3\log \frac{7}{s_d^3} + 24 B_3^2 s_d^2 x_{[d]})(\ell L^d)^3]$; with the upper bound on the partition function this gives $\mathsf{q}^{[d]}$.

The floors hold no K; let $K \to \infty$. □

Lemma 10.3 (Two consecutive cutoffs over one block field; the Lipschitz bound of the tower)**.** In the one-torus setting put $\varepsilon_{\mathrm{r}}(s) = 2c_\rho \varepsilon_{[s]}$; for two fields V, V' on one lattice put $\mathrm{dist}(V,V') = \max_b \theta(V(b)V'(b)^{-1})$, $\theta(X) \in [0,\pi]$ the angle of $X \in G$, $\cos\theta(X) = \frac{1}{2}\operatorname{tr} X$; for a real 1-form B on the bonds of a lattice (its values $B(b)$ hermitian and traceless, $B(\bar{b}) = -B(b)$) put $\sup|B| = \max_b |B(b)|$ and $(e^{iB}X')(b) = e^{iB(b)}X'(b)$ at the positively oriented bonds b. With the chart constant $C_{\mathrm{db}} = 8L^3$ and the constant $\gamma_{\mathrm{r}} \in (0,1]$ of the lower bound on the excess of a competitor (the global hypotheses preceding equation YM3_I:(2132), Prop. YM3_I:113.5), the constant K_u of the comparison of the background actions at two consecutive cutoffs (equation YM3_I:(2155), YM3_I:(2154)), the numbers $\varepsilon_{\mathrm{cp}}$, $\varrho_{\mathrm{no}} = \min\{r_{\mathrm{B}}, \varrho_{\mathrm{R}}^0\}$ and S_{e} of the definitions preceding and in equation YM3_I:(2133), the radius $c_4 \in (0,1]$ and the positive curvature ceiling $c_2' = c_2^{\mathrm{g}}$ of Lem. 9.11(v), decreasing c_4 to 1 if necessary, and the constant $B_5^{\mathrm{e}} \ge 8$ of the response of the background to a change of its datum (§5.3), we put

$$\mathsf{c}_{\mathrm{A}} = 16\, C_{\mathrm{db}}\, c_\rho\, (\mathsf{K}_u/\gamma_{\mathrm{r}})^{1/2}, \qquad a_s(K) = \mathsf{c}_{\mathrm{A}}\, \varepsilon_{[s]}\, (\ell L^s)^{3/2} L^{-(K-s+1)/2}, \qquad r_s(K) = \sum_{K'\ge K} a_s(K') = \frac{a_s(K)}{1-L^{-1/2}} = r_s(s) L^{-(K-s)/2},$$

$$\varepsilon_{\mathrm{L}} = \min\{\varepsilon_{\mathrm{cp}}, c_2'/B_3\}, \qquad \beta_{\mathrm{L}} = \min\Big\{1,\ \tfrac{1}{2}\varrho_{\mathrm{no}},\ \frac{c_4}{2B_5^{\mathrm{e}}\mathsf{S}_{\mathrm{e}}}\Big\}, \qquad \mathsf{c}_9 = \frac{2L}{L^{1/2}-1}\, B_5^{\mathrm{e}}\mathsf{S}_{\mathrm{e}};$$

c_{A}, ε_{L}, β_{L} and c_9 depend on the constants alone, $a_s(\cdot)$ decreases, and $\varepsilon_{\mathrm{r}}(s) \le \varepsilon_{\mathrm{L}}$ at every admitted depth s, by the conditions of the comparison of two consecutive cutoffs that define $\Theta_\star$. (a) *One depth.* Let s be admitted, $K \ge \max\{2, s+1\}$ and $X \in \mathcal{V}_s$ with $\varpi(X) < \varepsilon_{\mathrm{r}}(s)$. Then $H_s^K(X)$ and $H_s^{K+1}(X)$ are defined; each is in the axial gauge on Λ_s, has the average X and has plaquettes at most $2B_3 L^{-2}\varpi(X)$; and

$$\mathrm{dist}\big(H_s^{K+1}(X), H_s^K(X)\big) \le a_s(K).$$

Hence $H_s^\infty(X) = \lim_K H_s^K(X)$ exists, $\mathrm{dist}(H_s^\infty(X), H_s^K(X)) \le r_s(K)$ uniformly on $\{\varpi < \varepsilon_{\mathrm{r}}(s)\}$, H_s^∞ is continuous there, $\mathcal{M}H_s^\infty(X) = X$, $\varpi(H_s^\infty(X)) \le 2B_3 L^{-2}\varpi(X)$, and $H_s^\infty(X^u) = H_s^\infty(X)^{u\circ\pi}$

for every gauge transformation u of Λ_s, π mapping each site of Λ_{s+1} to the site of Λ_s whose block holds it. (b) *Along the tower.* Let d be admitted, $K \geq \max\{2, d+1\}$, $W \in J_d$, and $Y_j = H_{d,j}^{K+1}(W)$, $0 \leq j \leq K+1-d$, the tower over W at the cutoff $K+1$. Then for $0 \leq j < K-d$

$$\operatorname{dist}\bigl(Y_{j+1}, H_{d+j}^{K}(Y_j)\bigr) \leq L^{-j/2} a_d(K),$$

whereas (a) at the depth $d+j$ gives the number $a_{d+j}(K) = (\varepsilon_{[d+j]}/\varepsilon_{[d]}) L^{2j} a_d(K)$, up to $L^{2j} a_d(K)$. (c) *The Lipschitz bound in the block field.* Let $K \geq 2$, $d' < K$, $X' \in \mathcal{V}_{d'}$ with $\varpi(X') < \varepsilon_{\mathrm{L}}$, B a real 1-form on the bonds of $\Lambda_{d'}$ with $\sup|B| \leq \beta_{\mathrm{L}}$, and $X = e^{iB} X'$. Then $\mathcal{C}_K(X')$ and $\mathcal{C}_K(X)$ are non-empty, each one orbit of $\mathcal{G}_K^{d'}$, and for $U' \in \mathcal{C}_K(X')$, $U \in \mathcal{C}_K(X)$, every depth d'' with $d' \leq d'' \leq K$ and every closed path Γ of $\Lambda_{d''}$ with N bonds

$$\Bigl| \tfrac{1}{2} \operatorname{tr}(\mathcal{M}^{K-d''} U)(\Gamma) - \tfrac{1}{2} \operatorname{tr}(\mathcal{M}^{K-d''} U')(\Gamma) \Bigr| \leq 2 B_5^{\mathrm{e}} \mathsf{S}_{\mathrm{e}}\, N\, L^{-(d''-d')} \sup|B|;$$

the constant holds neither K, nor d', nor the number $d''-d'$ of the depths between. (d) *Two consecutive cutoffs: the loops at the depth $d+n$.* Let d be admitted, $n \geq 1$, $K \geq \max\{2, d+n+1\}$ with $a_d(K) \leq \beta_{\mathrm{L}}$, $W \in J_d$, Γ a closed path of Λ_{d+n} with N bonds and $W_\Gamma(V) = \frac{1}{2} \operatorname{tr} V(\Gamma)$. Then

$$|W_\Gamma(H_{d,n}^{K+1} W) - W_\Gamma(H_{d,n}^{K} W)| \leq \mathsf{c}_9\, N\, L^{-n/2} a_d(K), \qquad |W_\Gamma(H_{d,n}^{\infty} W) - W_\Gamma(H_{d,n}^{K} W)| \leq \mathsf{c}_9\, N\, L^{-n/2} r_d(K),$$

$H_{d,n}^{\infty} W$ being the limit of the towers $H_{d,n}^{K} W$ (Prop. 10.4(a)). ■

Sketch of proof. (a) Let $t = K - s$ and $\tilde{U}$ a background over X at the cutoff $K+1$ (it exists by Thm. YM3_I:E in the setting of Section YM3_I:52.7, as $\varepsilon_{\mathrm{r}}(s) \leq \bar{\varepsilon}$). Its average $U' = \mathcal{M}\tilde{U}$, a field of the bare lattice of the cutoff K with $\mathcal{M}^t U' = X$, is a competitor there (Lem. YM3_I:113.8): its action, in the units of the level t, exceeds the action of the backgrounds over X at the cutoff K by at most $2\mathsf{K}_u \varepsilon_{\mathrm{r}}(s)^2 L^{-t} (\ell L^s)^3$, the comparison of the background actions at the cutoffs K and $K+1$ (equation YM3_I:(2155)) and the energy of the background (equation YM3_I:(2149)) bounding the two parts of this excess.

The excess is at least $\frac{1}{8}\gamma_{\mathrm{r}} \sum_{k<t} L^{t-k} \|\mathcal{A}_k\|^2$ (Prop. YM3_I:113.5), where $\|\cdot\|$ is the ℓ^2 norm over the bonds and $\mathcal{A}_k$ the real 1-form by which the average $\mathcal{M}^k U'$, in the gauge of its averaged contour variables, differs from the averaged background over $\mathcal{M}^{k+1} U'$; the term $k = t-1$ alone gives $\|\mathcal{A}_{t-1}\| \leq 4(\mathsf{K}_u/\gamma_{\mathrm{r}})^{1/2} \varepsilon_{\mathrm{r}}(s) (\ell L^s)^{3/2} L^{-(t+1)/2}$. In that gauge $\mathcal{M}^{t-1} U'$ is the averaged background over X at the cutoff $K+1$, that is $H_s^{K+1}(X)$ (the axial-normalization argument in the proof of Prop. YM3_I:52.3), and the chart of the fibre over X writes it as $e^{iB'} H_s^K(X)$ with $\sup|B'| \leq 2C_{\mathrm{db}} \sup|\mathcal{A}_{t-1}|$ (equation YM3_I:(2138)); with $\varepsilon_{\mathrm{r}}(s) = 2c_\rho \varepsilon_{[s]}$ this is $a_s(K)$.

The axial gauge, the average and the plaquettes are equation YM3_I:(953). Summed over the cutoffs, a geometric series of ratio $L^{-1/2}$, the bound makes $(H_s^K(X))_K$ a Cauchy sequence in a compact space, uniformly in X; each H_s^K is continuous (Prop. YM3_I:52.3) and covariant, a bare gauge transformation equal to $u \circ \pi$ on Λ_{s+1} carrying the elements of the minimal orbit over X whose averages are in the axial gauge to those over X^u; $\mathcal{M}$ is continuous on the fields with plaquettes $< 1/200L^2$, which holds the fields $H_s^K(X)$; and the limit inherits all this. (b) The same argument with $\tilde{U} = Y_{t+1}$, $t = K-d$, the bare field at the end of the tower of the cutoff $K+1$: it is a background over W and over every Y_j, with $\mathcal{M}^{t+1-j} \tilde{U} = Y_j$ (the minimal orbit over $H_{d+j}^{K+1}(Y_j)$ lies in the one over Y_j, [7, Lem. B.4](2)).

The competitor $U' = \mathcal{M}\tilde{U} = Y_t$ has the averages $\mathcal{M}^k U' = Y_{t-k}$, and its excess is bounded once, at the depth d, by $2\mathsf{K}_u \varepsilon_{\mathrm{r}}(d)^2 L^{-t} (\ell L^d)^3$. At $k = t-1-j$ the term of the lower bound has the weight $L^{t-k} = L^{j+1}$, so $\|\mathcal{A}_k\| \leq 4(\mathsf{K}_u/\gamma_{\mathrm{r}})^{1/2} \varepsilon_{\mathrm{r}}(d) (\ell L^d)^{3/2} L^{-(t+1)/2} L^{-j/2}$; at the point Y_j the chart of the fibre and the axial-normalization argument in the proof of Prop. YM3_I:52.3 give $Y_{j+1} = e^{iB'} H_{d+j}^K(Y_j)$ with $\sup|B'| \leq 2C_{\mathrm{db}} \sup|\mathcal{A}_k| \leq L^{-j/2} a_d(K)$. The single bound on the excess at the depth d pays for all the depths of the tower together. (c) The response of the background

to a change of its datum (Thm. YM3_I:E and Lem. YM3_I:51.6, with the thresholds of §5.3, for the $n = K - d'$ averaging steps on the whole torus, the datum X' and the radius ε_{L}): $\mathcal{C}_K(X')$ and $\mathcal{C}_K(e^{iB}X')$ are single orbits of $\mathcal{G}_K^{d'}$, and a background over X is, up to a bare gauge transformation, $e^{iL^{-n}\mathcal{H}(B)}U^\circ$ with $U^\circ \in \mathcal{C}_K(X')$ and $|\mathcal{H}(B)(b)| \le \sigma := B_5^{\mathrm{e}}\mathsf{S}_{\mathrm{e}} \sup|B|$ at every bare bond b, the exponentially decaying weights of the response summing to at most S_{e}.

The averages of every order $i \le n$ of a field so moved are, up to gauge, $e^{ia_i}\mathcal{M}^i U^\circ$ with $|a_i| \le 2L^{-(n-i)}\sigma$ (Lem. 9.11(v)). Along a closed path a gauge transformation conjugates the holonomy; the product bound $|\prod_i A_i - \prod_i A_i'| \le \sum_i |A_i A_i'^{-1} - \mathbf{1}|$ on G, $|e^{iY} - \mathbf{1}| \le |Y|$ and $|\operatorname{tr} Y| \le 2|Y|$ give the bound at $i = K - d''$. (d) A telescope over the depths $d + j$, $0 \le j < n$, along the tower $Y_j = H_{d,j}^{K+1}(W)$ of the finer cutoff. For a field Y of a depth between d and $d+n$ and $K' \in \{K, K+1\}$ let $\mathsf{u}^{K'}(Y)$ be the loop along Γ of the average down to Λ_{d+n} of a background over Y at the cutoff K': one number on the orbit, with $\mathsf{u}^{K'}(Y) = \mathsf{u}^{K'}(H_s^{K'}(Y))$ at the depth s of Y ([7, Lem. B.4](2)) and $\mathsf{u}^{K'}(Y) = W_\Gamma(Y)$ at the depth $d + n$.

With $\Delta_j = \mathsf{u}^{K+1}(Y_j) - \mathsf{u}^K(Y_j)$ one has $\Delta_n = 0$, Δ_0 the left side of the first bound, and $\Delta_j - \Delta_{j+1} = \mathsf{u}^K(Y_{j+1}) - \mathsf{u}^K(H_{d+j}^K(Y_j))$. By (b), $H_{d+j}^K(Y_j) = e^{iB}Y_{j+1}$ with $\sup|B| \le L^{-j/2}a_d(K) \le \beta_{\mathrm{L}} < \pi$, and $\varpi(Y_{j+1}) \le 2B_3\varpi(W)L^{-2(j+1)} < \varepsilon_{\mathrm{r}}(d+j+1) \le \varepsilon_{\mathrm{L}}$; so (c) at $d' = d+j+1$ and $d'' = d+n$, $n-j-1$ depths apart, gives $|\Delta_j - \Delta_{j+1}| \le 2B_5^{\mathrm{e}}\mathsf{S}_{\mathrm{e}}NL^{-(n-j-1)}L^{-j/2}a_d(K)$. The roles of the two fields are so chosen because (c) asks a bound on the plaquettes of X' alone, which the tower of the finer cutoff has. The sum over j is a geometric series of the ratio $L^{1/2}$, $\sum_{j<n} L^{j/2} \le L^{n/2}/(L^{1/2} - 1)$, which gives $\mathsf{c}_9 N L^{-n/2}a_d(K)$.

The conditions persist for $K' \ge K$, as a_d decreases, and the sum over $K' \ge K$ with $\sum_{K' \ge K} a_d(K') = r_d(K)$ gives the second bound. □

Proposition 10.4 (Smoothed loops over a fixed field)**.** In the one-torus setting let d be admitted, $W \in J_d$ and C a lattice loop of $\Lambda_{d'}$, $d' \ge d$. Put $\varepsilon_{\mathrm{r}} = 2c_\rho\varepsilon_{[d]}$; let $\mathsf{c}_{\mathrm{A}} > 0$ be the constant with which the comparison of two consecutive cutoffs (Prop. YM3_I:113.9) bounds the distance of the averaged backgrounds of two consecutive cutoffs over one block field (Lem. 10.3(a)), and $\mathsf{c}_9, \beta_{\mathrm{L}} > 0$ the constant and the radius of the Lipschitz bound of the tower in the block field (Lem. 10.3(c), (d)), the numbers formed in Lem. 10.3; all three depend on the constants alone. Put $a_s(K) = \mathsf{c}_{\mathrm{A}}\varepsilon_{[s]}(\ell L^s)^{3/2}L^{-(K-s+1)/2}$ and $r_d(K) = \sum_{K' \ge K} a_d(K') = r_d(d)L^{-(K-d)/2}$; let K_2 be the least $K \ge 2$ with $\lfloor \frac{K-d}{3} \rfloor \ge \max\{1, d' - d\}$ and $a_d(K) \le \beta_{\mathrm{L}}$; and put $\mathsf{c}_{\mathrm{f}} = \frac{28L\varepsilon_{\mathrm{r}}}{1-L^{-1}} + \mathsf{c}_9 r_d(d)$ and $\mathsf{c}_{\mathrm{g}} = \frac{1544L^2\varepsilon_{\mathrm{r}}}{1-L^{-1}} + 6\mathsf{c}_9 r_d(d)$, numbers that depend on d, ℓ, b and the constants (K_2 also on $d' - d$). (a) $V_n = H_{d,n}^\infty(W)$ satisfies $\mathcal{M}V_{n+1} = V_n$, $\varpi(V_n) \le 2B_3\varpi(W)L^{-2n}$ and, for $n \ge d' - d$, $|W_C(V_n) - W_C(V_{n+1})| \le 28B_3|C|L^d\varpi(W)L^{-n}$. So $W_C^{[d]}$ exists uniformly on J_d, with $|W_C^{[d]} - W_C(V_n)| \le \frac{14}{1-L^{-1}}|C|L^d\varepsilon_{\mathrm{r}}L^{-n}$ there; it is continuous and gauge invariant on J_d with values in $[-1, 1]$; $F_C^{[d]}$ is Borel; and $0 \le 1 - W_C^{[d]}(W) \le \frac{1}{2}(2B_3\varpi(W)L^{2d}s_1s_2)^2$ for a coordinate rectangle with sides s_1, s_2. (b) For $K \ge \max\{2, d+1, d'\}$, $\mathcal{C}_K(W)$ is one non-empty orbit of $\mathcal{G}_K^d$; w_C^K is continuous and gauge invariant on J_d; $\sup_{J_d}|w_C^K - W_C^{[d]}| \to 0$, and this supremum is at most $|C|L^d\mathsf{c}_{\mathrm{f}}L^{-(K-d)/3}$ for $K \ge K_2$. (c) For curves and any shadows, $|W_{\Gamma^{(d+n)}}(V_n) - W_{\Gamma^{(d+n+1)}}(V_{n+1})| \le 772(|\gamma|L^{d+n+1} + 2)L^{-2n}\varepsilon_{\mathrm{r}}$; so $W_\gamma^{[d]}$ exists uniformly on J_d, does not depend on the shadows, equals $W_C^{[d]}$ along a lattice loop C, and is continuous and gauge invariant, and F_γ is Borel. $w_\gamma^K(W)$ is one number on $\mathcal{C}_K(W)$; two shadows on the bare lattice change it by at most $1496(|\gamma|L^K + 2)L^{-2(K-d)}\varepsilon_{\mathrm{r}}$; and $\sup_{J_d}|w_\gamma^K - W_\gamma^{[d]}| \le (|\gamma|L^d + 2)\mathsf{c}_{\mathrm{g}}L^{-(K-d)/3}$ for $K \ge K_2$, taken at $d' = d$. (d) For $K \ge K_{\mathrm{bf}}$ with

$\lceil K/1260\rceil \ge d$,

$$\Big|\langle F_C^{[d]}(\mathcal{M}^{K-d}U)\rangle_K - \int F_C^{[d]}\,\mathrm{d}\mu_d\Big| \le \mathfrak{r}_K := \mathsf{C}_{\mathrm{bf}} S_L \ell^3 K^{2\mathfrak{p}_0+1/2} L^{-K/10}, \qquad S_L = \sum_{j\ge 0}(1+j)^{2\mathfrak{p}_0+1/2} L^{-j/10},$$

a bound free of C and d, where C_{bf} and K_{bf} are the numbers of Def. 2.7(5), formed at an interval of labels under the condition (Tor) on the side ℓ (§2.3.1) that holds b, and depend on the label only through that interval (Cor. 2.8(a)); in the frame $[\mathsf{b}_-, \mathsf{b}_+]$ is such an interval, and it gives one pair of numbers for all $\mathsf{b} \in \mathsf{J}$, with the range $K \ge K_{\mathrm{bf}}$, $\lceil K/1260\rceil \ge d$; for $K \ge K_2$ besides, $|\langle(\mathbf{1}_{J_d} w_C^K)(\mathcal{M}^{K-d}U)\rangle_K - \int F_C^{[d]}\,\mathrm{d}\mu_d| \le |C| L^d \mathsf{c}_{\mathrm{f}} L^{-(K-d)/3} + \mathfrak{r}_K$. The same holds for curves, with F_γ, w_γ^K and $(|\gamma|L^d + 2)\mathsf{c}_{\mathrm{g}}$. (e) In the frame put $\mathsf{b}' = \mathsf{b} - \bar{B}$ and $J'_d = \{\varpi < c_\rho \varepsilon(L^d \mathsf{b}')/B_3\}$, and let $d \ge d_0$, the least integer $d_0 \ge 1$ with $L^{d_0}\gamma^{-2} \ge \Theta_\star$ and $\log(L^{d_0}\gamma^{-2}) > 2\mathfrak{p}_0$. Then the one-torus setting holds at the labels b and b' and d is admitted at both; $I_d \subset J_d \subset J'_d$ and $\mathsf{O}_{K-d} \subset J'_d$ for every $K > d$, so that m_{K-d} gives no mass to $\mathcal{V}_d \setminus J'_d$, and $\mu_d^K(\mathcal{V}_d \setminus J'_d) \le R_d$; over every $V \in J'_d$ and every $K > d$ the backgrounds form one non-empty orbit, w_C^K is Borel, and (a), (b) hold on J'_d, also for families: $\sup_{J'_d} |w_{\mathsf{C}}^K - \prod_i W_{C_i}^{[d]}| \to 0$. ■

Sketch of proof. (a) Over W the backgrounds of the finer cutoffs are built one block step at a time, $H_{d,n}^K(W)$, and all of them are, up to gauge, block averages $\mathcal{M}^{K-d-n}U$ of one bare background U, whose plaquettes are $< B_3\varpi(W)(a_K/a_d)^2$ (Thm. YM3_I:E in the setting of Section YM3_I:52.7); so the factor $2B_3$ is paid once, not at every step: $\varpi(\mathcal{M}^{K-d-n}U) \le 2B_3\varpi(W)L^{-2n} < \varepsilon_{\mathrm{r}}L^{-2n} \le L^{-3n/2}\,2c_\rho\varepsilon_{[d+n]}$. Induction on n, uniform continuity on the compact set $\{\varpi \le L^{-3n/2}2c_\rho\varepsilon_{[d+n]}\}$ and Lem. 10.3(a) give $H_{d,n}^K \to H_{d,n}^\infty$ uniformly on J_d; the bound passes to V_n, which lies in $\{\varpi < 2c_\rho\varepsilon_{[d+n]}\}$, so that $H_{d+n}^\infty(V_n)$ is defined.

The ball $\{\varpi < 2c_\rho\varepsilon_{[d]}\}$, larger than J_d by the factor $2B_3 \ge 2$, is the domain of the single step H_d^∞; the whole tower needs room for the factor $2B_3$ that the backgrounds pay, which is why the proposition asks $W \in J_d$, and why the smaller ball $I_d \subset J_d$, and not the larger one, enters the starting depth. One refinement costs $|W_\Gamma(\mathcal{M}V) - W_\Gamma(V)| \le 14L^2N\varpi(V)$ for a loop of N bonds if $\varpi(V) < 1/200L^2$ (in a gauge fixed along the blocks the average of a nearly flat field moves each bond by $O(L^2\varpi(V))$), which holds as $2c_\rho\varepsilon_{[d]} < 1/200L^2$; with $N = |C|L^{d+n}$ the number of bonds grows like L^n while the plaquettes fall like L^{-2n}: a geometric series, whose tail with $\varpi(W) < \varepsilon_{\mathrm{r}}/2B_3$ is the stated bound.

For a rectangle, $a_p = \frac{1}{2}|\partial U(p) - \mathbf{1}|^2$ and the holonomy of ∂R is an ordered product of conjugated plaquette variables, so $1 - W_{\partial R} \le (\sum_{p\subset R}\sqrt{a_p})^2$ (Prop. 10.9(c)), and the bare plaquettes of the background are $< 2B_3\varpi(W)(a_K/a_d)^2$. (b) The tower ends in the non-empty set $\mathcal{C}_K(W)$, one orbit by Thm. YM3_I:E in the setting of Section YM3_I:52.7 and §5.2. The comparison of two consecutive cutoffs (Prop. YM3_I:113.9) gives $\mathrm{dist}(H_s^{K+1}(X), H_s^K(X)) \le a_s(K)$ for $\varpi(X) < 2c_\rho\varepsilon_{[s]}$ (Lem. 10.3(a)). At a depth $d+n$ we split into three terms. $|w_C^K(W) - W_C(\mathcal{M}^{K-d-n}U)|$ and $|W_C^{[d]}(W) - W_C(V_n)|$ are at most $\frac{14}{1-L^{-1}}|C|L^d\varepsilon_{\mathrm{r}}L^{-n}$, uniformly in K, by the series of (a); and $|W_C(H_{d,n}^K W) - W_C(V_n)| \le |C|L^{d+n}\sup\mathrm{dist}(H_{d,n}^K, H_{d,n}^\infty) \to 0$ at fixed n. Choose n, then K. For the rate we use the Lipschitz bound of the tower in the block field (Lem. 10.3(d)): for $n \ge 1$, $K \ge \max\{2, d+n+1\}$ with $a_d(K) \le \beta_{\mathrm{L}}$, $W \in J_d$ and every closed path Γ of Λ_{d+n} with N bonds, $|W_\Gamma(H_{d,n}^{K+1}W) - W_\Gamma(H_{d,n}^K W)| \le \mathsf{c}_9 N L^{-n/2} a_d(K)$.

It is a telescope over the depths $d+j$, $0 \le j < n$, along the tower $Y_j = H_{d,j}^{K+1}(W)$ of the finer cutoff: at the depth $d+j$ the one-depth smoothings H_{d+j}^{K+1} and H_{d+j}^K of the two cutoffs, applied to Y_j, give fields of the depth $d+j+1$ at distance at most $L^{-j/2}a_d(K)$ (Lem. 10.3(b): the excess of the action of the averaged background of the finer cutoff is bounded once, at the depth d, and

the term of the depth $d+j$ in the lower bound on the excess carries the weight L^{j+1}; the bound $a_{d+j}(K)$ of one depth, applied at Y_j, is up to L^{2j} times larger), and the loop along Γ of the average down to Λ_{d+n} of the background over such a field, $n-j-1$ depths below it, moves by at most $2B_5^{\mathrm{e}}\mathsf{S}_{\mathrm{e}}NL^{-(n-j-1)}$ times that distance, a constant free of the cutoff and of the number of the depths between (Lem. 10.3(c)); the sum over j is a geometric series of the ratio $L^{1/2}$, which gives c_9.

Summed over $K'\ge K$, with $\sum_{K'\ge K}a_d(K')=r_d(K)$, at $\Gamma=C^{\langle d+n\rangle}$ it bounds the middle term by $\mathsf{c}_9|C|L^dL^{n/2}r_d(K)$; at $n=\lfloor\frac{K-d}{3}\rfloor$ both L^{-n} and $L^{n/2}L^{-(K-d)/2}$ are at most $L\cdot L^{-(K-d)/3}$, which gives c_{f}. Continuity and gauge invariance of w_C^K on J_d come with the tower. (c) Two shadows of consecutive lattices, evaluated at V_{n+1}, differ by a band of $O(|\gamma|L^{d+n+1}+2)$ plaquettes of size $O(L^{-2n}\varepsilon_{\mathrm{r}})$; two shadows on the bare lattice, evaluated at a background over W, differ by a band of $O(|\gamma|L^K+2)$ bare plaquettes of size $O(L^{-2(K-d)}\varepsilon_{\mathrm{r}})$; then argue as in (a), (b). (d) By Cor. 2.8 and (2.xv), two consecutive cutoffs differ on a gauge invariant bounded Borel function F of the block field of a depth at most $\lceil K/1260\rceil$ ($1260=180(3N+4)$ at $N=1$ in §2.3.1) by at most $\mathsf{C}_{\mathrm{bf}}\ell^3K^{2\mathfrak{p}_0+1/2}L^{-K/10}\sup|F|$; the comparison there is made through the small-field chains of the two cutoffs at that depth, and its numbers, formed at an interval of labels under (Tor), serve every label of the interval (Cor. 2.8(a)). In the frame $[\mathsf{b}_-,\mathsf{b}_+]$ is such an interval on the side ℓ: $\mathsf{b}_->g^{-2}\ge\gamma^{-2}$, and $MQ(\mathsf{b}_+)\le L^m$, with Q of Lem. 10.1, holds on the sides of $\mathfrak{L}(g)$ at every label of $\mathsf{I}(g)$, as in (e). Sum over $K'\ge K$ with $F=F_C^{[d]}$; the second mean adds the rate of (b). (e) $\gamma^{-2}\le g^{-2}<\mathsf{b}'<\mathsf{b}<L(g^{-2}+\bar B)$, so the sides at g satisfy the condition on the side at both labels, and $L^d\mathsf{b}'\ge L^d\gamma^{-2}\ge\Theta_\star$; the constraint sets, the action and the regularity radius hold no label, so (a), (b) apply at the label b' on its ball J'_d. $x_{K-d}\ge L^d\mathsf{b}-\bar B\ge L^d\mathsf{b}'$, ε decreases for $\log t>2\mathfrak{p}_0$ and $c_\rho\ge 2B_3$; so $2\varepsilon(x_{K-d})\le 2\varepsilon(L^d\mathsf{b}')\le c_\rho\varepsilon(L^d\mathsf{b}')/B_3$, that is $\mathsf{O}_{K-d}\subset J'_d$, and $J_d\subset J'_d$ likewise. $\mu_d^K(\mathcal{V}_d\setminus J'_d)\le\mu_d^K(\mathcal{V}_d\setminus I_d)\le R_d$ by Prop. 10.2(b). For families $|\prod_i x_i-\prod_i y_i|\le\sum_i|x_i-y_i|$ on $[-1,1]$. □

The new estimate is one renormalisation step with lines inserted. Its soft part rests on three facts about ordered products of matrices under a measure.

Lemma 10.5 (Ordered products under a measure)**.** (a) Let $\mathfrak{A}$ be a unital normed algebra, I a finite totally ordered set, and for $j\in I$ let At_j be a finite set of atoms α, each with an element $\tau_\alpha\in\mathfrak{A}$, $|\tau_\alpha|\le 1$. A block is a non-empty set of atoms of pairwise different indices, a packing π a set of blocks with pairwise disjoint sets of indices, and $\tau^{\rightarrow}(\pi)$ the product of the τ_α, α in the blocks of π, in the order of I. For complex weights w on the blocks put $Z[w]=\sum_\pi\prod_{\beta\in\pi}w(\beta)\,\tau^{\rightarrow}(\pi)$ and $|w|_1=\sum_\beta|w(\beta)|$. Then $|Z[w]|\le e^{|w|_1}$ and $|Z[w_1+w_2]-Z[w_1]|\le|w_2|_1e^{|w_1|_1+|w_2|_1}$. If bounded functions f_α on a space with a complex measure ν of mass 1 and finite total variation have $\int e^{\sum_\alpha z_\alpha f_\alpha}\,\mathrm{d}\nu=e^{F(z)}$ with F holomorphic near 0 and $F(0)=0$, then

$$\int\prod_{j\in I}^{\rightarrow}\Big(\mathbf{1}+\sum_{\alpha\in\mathsf{At}_j}f_\alpha\tau_\alpha\Big)\,\mathrm{d}\nu=Z[\kappa],\qquad\kappa(\beta)=\Big(\prod_{\alpha\in\beta}\partial_{z_\alpha}\Big)F(0),$$

the joint cumulants of the f_α, $\alpha\in\beta$. *Generators.* Here and below $\mathfrak{su}(2)$ is the real space of the Hermitian traceless 2×2 matrices, as the chart $e^{ig_tB(b)}V_W(0)(b)$ of the frame asks, with the scalar product $\langle X,Y\rangle=\frac12\operatorname{tr}XY$, and t_1,t_2,t_3 is an orthonormal basis of it: $t_at_b+t_bt_a=2\delta_{ab}\mathbf{1}$, $t_a^2=\mathbf{1}$, $|t_a|=1$ in the operator norm and $\sum_a t_a^2=3\cdot\mathbf{1}$ (the Pauli matrices up to a rotation). The components of $X\in\mathfrak{su}(2)$ are $X^a=\langle t_a,X\rangle$, so that $X=\sum_aX^at_a$ and $|X|=(\sum_a(X^a)^2)^{1/2}$; this is used for the variables y_j, the fluctuation field B, the test forms φ and the rotations Υ below, and $(\varphi,B)=\sum_b\sum_a\varphi^a(b)B^a(b)$. A complex 2×2 matrix is written $x=x^0\mathbf{1}+\sum_a x^a\,it_a$, with $x^0=\frac12\operatorname{tr}x$, $x^a=-\frac{i}{2}\operatorname{tr}(xt_a)$ and $|x^0|,|x^a|\le|x|$; as $(\sum_a x^at_a)^2=\sum_a(x^a)^2\mathbf{1}$, $\det x=(x^0)^2+\sum_a(x^a)^2$. For $h\in\mathrm{SU}(2)$, $\mathrm{R}(h)\in SO(3)$ is the rotation with $ht_bh^{-1}=\sum_a\mathrm{R}(h)_{ab}t_a$, $\mathrm{R}(h)_{ab}=\langle t_a,ht_bh^{-1}\rangle$. (b) Let $\mathfrak{A}=M_2(\mathbb{C})^{\otimes r}$ with the operator norm and $I=I_1<\dots<I_r$ a union of intervals; let c_j ($j\in I$) and

p_e ($e = \{j < j'\}$ in a set of pairs, each inside one I_i) be complex numbers, and w_{g} the weight that is c_j at one block $\{\alpha^0_j\}$ with $\tau = \mathbf{1}$, p_e at the three blocks $\{\alpha^a_j, \alpha^a_{j'}\}$, $a = 1,2,3$, with $\tau = it_a$ in the i-th factor, and 0 elsewhere. Two pairs cross if their end points interlace. With $\mathfrak{T} = \sum_j |c_j| + 3\sum_e |p_e|$, $\mathfrak{X} = \sum 9|p_e||p_{e'}|$ over the crossing pairs of pairs, and $\mathfrak{O} = \sum |\hat w(\beta)||\hat w(\beta')|$ over the ordered pairs (equal ones included) of overlapping sets β, β' among the $\{j\}$ and the e, $\hat w = c_j$ resp. $-3p_e$:

$$\Big|Z[w_{\mathrm{g}}] - \exp\Big(\sum_j c_j - 3\sum_e p_e\Big)\mathbf{1}\Big| \le (2\mathfrak{X} + \tfrac12\mathfrak{O})\, e^{\mathfrak{T}}.$$

(c) Let $J = J_1 \sqcup \cdots \sqcup J_r$ be finite cyclically ordered sets (the lines), $g_j \in \mathrm{SU}(2)$, and $w(y) = \prod_i \frac12 \operatorname{tr} \prod_{j\in J_i}(e^{i\sum_a y^a_j t_a} g_j)$ for $y = (y_j) \in (\mathbb{R}^3)^J$; every derivative of w of every order has modulus at most 1. Let ϱ be a complex Borel measure of mass 1 on $\{|y_j| \le \bar y\ \forall j\}$ whose logarithmic Laplace transform $\ell(z) = \log \int e^{(z,y)} d\varrho$ is holomorphic on a polydisc of radius $r_1 > 1$, with the cumulants $\kappa_\gamma = \partial^\gamma \ell(0)$, and let $s_{jj'} = s_{j'j}$ be real numbers, for the pairs $\{j, j'\}$ of one line and for $j = j'$, with $S = \sum_j s_{jj} + 2\sum_{\{j,j'\}} s_{jj'} \ge 0$. For a pair of one line let $j \to j'$ be the arc from j forward to j', $G_{j\to j'} = g_j g_{j+1} \cdots g_{j'-1}$, $\mathrm{R}(h) \in SO(3)$ the rotation with $h t_b h^{-1} = \sum_a \mathrm{R}(h)_{ab} t_a$, $\kappa^{(2)}_{jj'} = (\kappa_{e_{ja}+e_{j'b}})_{a,b}$, $d_{jj'} = \kappa^{(2)}_{jj'} - s_{jj'}\mathrm{R}(G_{j\to j'})$, $d_{jj} = \kappa^{(2)}_{jj} - s_{jj}\mathbf{1}$, and $|m|_1 = \sum_{a,b}|m^{ab}|$. Then

$$\Big|\int w\,\mathrm{d}\varrho - e^{-\frac32 S} w(0)\Big| \le |\varrho|\,(E_1 + E_2 + E_3), \qquad E_1 = \sum_\alpha |\kappa_{e_\alpha}|, \quad E_2 = \sup_{|y_j|\le\bar y} \mathsf{e}(y), \quad E_3 = \sum_{|\gamma|\ge 3} \frac{|\gamma|}{\gamma!}|\kappa_\gamma|,$$

$|\varrho|$ the total variation, with

$$\mathsf{e}(y) = \sum_{\{j,j'\}} \Big[|d_{jj'}|_1 + 6|s_{jj'}| \sum_{i\in(j\to j')} |y_i| + |\kappa^{(2)}_{jj'}|_1(|y_j| + |y_{j'}| + |y_j||y_{j'}|)\Big] + \sum_j \big[\tfrac12|d_{jj}|_1 + |\kappa^{(2)}_{jj}|_1|y_j|\big] + \sum_{\text{two lines}} |\kappa^{(2)}_{jj'}|_1,$$

the first sum over the pairs of one line, i over the indices strictly between j and j' on the arc. If $\ell = \sum_X \ell_X$, $|\ell_X| \le A_X$ on the polydisc, ℓ_X depending on at most N_X of the variables, and $r_1 \ge 2$, then $E_3 \le 3\sum_X A_X (N_X/r_1)^3 e^{2(N_X+3)/r_1}$. ∎

Sketch of proof. (a) $|\tau^{\to}(\pi)| \le 1$, and a packing with n blocks arises from $n!$ of the n-tuples of blocks, so $\sum_\pi \prod |w(\beta)| \le e^{|w|_1}$; expanding $\prod(w_1 + w_2)(\beta)$ and isolating one block of w_2 gives the second bound. For the identity, multiply out: the left side is $\sum_{S\subset I}\sum_{(\alpha_j)_{j\in S}} \int \prod_{j\in S} f_{\alpha_j}\,\mathrm{d}\nu \prod^{\to} \tau_{\alpha_j}$; the moment is $\prod \partial_{z_{\alpha_j}} e^F$ at 0, and by induction from $\partial e^F = (\partial F)e^F$ the derivatives of e^F in pairwise different variables are, at 0, sums over the partitions of the variables of products of derivatives of F; a choice of S, of the atoms and of a partition is a packing. (b) A packing in the support of w_{g} is a set of singletons, a set m of disjoint pairs and a colouring of m.

If no two pairs of m cross, the sum over the colours is $(-3)^{\#\mathsf{m}}\mathbf{1}$: take a shortest pair; between its two factors τ_a stand only factors $\mathbf{1}$, and $\sum_a \tau_a\tau_a = -\sum_a t_a^2 = -3$; induct. The packings with a crossing contribute at most $2\mathfrak{X}e^{\mathfrak{T}}$; and $\exp\sum \hat w - Z_{\mathrm{nc}}$, Z_{nc} the sum over the packings with the values $(-3)^{\#\mathsf{m}}$, is a sum over tuples of sets with an overlap, at most $\frac12\mathfrak{O}e^{\mathfrak{T}}$. (c) Let $f(\tau) = \int w(\tau y)\,\mathrm{d}\varrho$. The cumulant formula $\int y_\alpha G\,\mathrm{d}\varrho = \sum_\beta \frac{1}{\beta!}\kappa_{\beta+e_\alpha}\int \partial^\beta G\,\mathrm{d}\varrho$, for entire G with bounded derivatives at 0 (from $\partial_\alpha Z = Z\,\partial_\alpha \ell$ for $Z = e^\ell$, coefficient by coefficient, then by Taylor polynomials and dominated convergence), gives $f'(\tau) = \sum_{\alpha,\beta} \frac{\tau^{|\beta|}}{\beta!}\kappa_{\beta+e_\alpha}\int(\partial^\beta\partial_\alpha w)(\tau y)\,\mathrm{d}\varrho$.

The terms $|\beta| = 0$ are at most $|\varrho|E_1$, those with $|\beta| \ge 2$ at most $|\varrho|E_3$. For $|\beta| = 1$ the matrix identity $\sum c^{ab}_{jj'}\partial_{y^a_j}\partial_{y^b_{j'}} w = -3Sw + \mathcal{R}$, $|\mathcal{R}| \le 2\mathsf{e}(y)$, at $c = \kappa^{(2)}$ holds: by cyclicity the

trace of the line of a pair is $\frac12\operatorname{tr}(u_jHu_{j'}H')$ with $u_j=e^{i\sum_a y_j^a t_a}$; moving $e^{iy\cdot t}$ across t_a costs $|y|$; $\sum_{a,b}\mathrm{R}(G)_{ab}t_aHt_b=\sum_b(Gt_bG^{-1})Ht_b=3H+\rho$ with $|\rho|\le 6|H-G|\le 6\sum_{i\in(j\to j')}|y_i|$ (a telescoping product of unitaries; $\sum_b t_b^2=3$); on the diagonal $\partial_a\partial_b u=-\delta_{ab}u$ up to $2|y|$. So $f'=-3\tau Sf+\mathcal{E}$ with $|\mathcal{E}|\le|\varrho|(E_1+2\tau E_2+E_3)$, and $f(1)=e^{-\frac32 S}f(0)+\int_0^1 e^{-\frac32 S(1-\tau^2)}\mathcal{E}\,\mathrm{d}\tau$, $S\ge0$.

The bound on E_3: Cauchy's estimates for each ℓ_X and $\sum_m m\binom{m+N-1}{m}x^m=Nx(1-x)^{-N-1}$, $x=1/r_1\le\frac12$. □

Proposition 10.6 (One step with lines inserted)**.** In the frame let C be a family with corners in Λ_{i_0}. (a) There are $C_\eta<\infty$ and an integer $s_0\ge i_0$, depending on the constants, g, J and on C only through $\#\mathsf{C}$, $|\mathsf{C}|$, s_C, ρ_C and i_0, not on K, $\mathsf{b}\in\mathsf{J}$, ℓ, t, W, such that for $\mathsf{b}\in\mathsf{J}$, $K>s_0$, $0\le t<K-s_0$ and every $W\in\mathsf{O}_{t+1}$

$$\Big|\int W_\mathsf{C}^{(t)}(V_W(B))\,\nu_W(\,\mathrm{d}B)-\lambda_{t+1}W_\mathsf{C}^{(t+1)}(W)\Big|\le C_\eta L^{-(K-t)/4};$$

more precisely, with $s=K-t$ (in (a) and its proof), $\Lambda_\mathsf{C}=\frac{3|\mathsf{C}|}{2M\mathsf{b}_-}$, $\mathsf{g}_s=(\frac{3}{2\mathsf{b}_-})^{1/2}L^{-s/2}$ and $\mathsf{y}_s=\log(2\mathsf{b}_+)+s\log L$, the left side is at most

$$C'(1+\#\mathsf{C}+\Lambda_\mathsf{C})e^{C''\Lambda_\mathsf{C}}\big[\mathsf{y}_s^{2\mathfrak{p}_0}(\mathsf{g}_s+L^{-s/2})+e^{-\rho_\mathsf{C}L^s/LM}\big]+\tfrac32\,\mathsf{c}\,C_X\mathsf{b}_-^{-2}|\mathsf{C}|L^{-s},\qquad \mathsf{c}=\tfrac54\bar B,$$

with C', C'', C_X depending on the constants alone; so for every $\vartheta<\frac12$ the same holds with $L^{-\vartheta(K-t)}$ and another C_η. For one polygon the same bound holds for the holonomy matrix, $|\int\mathrm{Hol}_C(e^{i\eta_t\mathbf{A}(B)}U_+)\,\nu_W(\,\mathrm{d}B)-\lambda_{t+1}\mathrm{Hol}_C(U_+)|\le C_\eta L^{-(K-t)/4}$ in the operator norm, from a fixed base point, $e^{i\eta_t\mathbf{A}(B)}U_+$ being the configuration of the shift formula of step 1 of the proof (at $t=0$, where $\eta_0=1$, the chart $e^{iB'}U_+$ itself). No positivity of ν_W is used. (b) If $\nu_W\ge0$, which is the case for m_{t+1}-almost every W, and $K-t>s_0$, the left side is at most

$$C_\sharp\frac{|\mathsf{C}|}{\mathsf{b}}\Big[(1+\mathfrak{b}_t)^2(\varepsilon_{t+1}+\delta_t+L^{-(K-t)})+e^{-\mathsf{D}_t}+\frac{\bar B}{\mathsf{b}}L^{-(K-t)}\Big],\qquad \mathsf{D}_t=\frac{\delta'_\mathsf{C}L^{K-t}}{LM}-2,$$

where $C_\sharp$ depends only on the constants, n_C and c_C, and s_0 is the least integer $\ge i_0$ such that every $x\ge\frac23\mathsf{b}_-L^{s_0-1}$ satisfies explicit smallness conditions formed from the constants and n_C, so that $s_0=\max\{i_0,s_0'\}$ with s_0' formed from the constants, b_- and n_C alone; with the Euclidean distance in the place of the sup-distance in δ'_C, D_t would carry a further factor $1/\sqrt3$. At every $W\in\mathsf{O}_{t+1}$ a bound of the same form holds with the total variation of ν_W as a further factor. The right side is at most $C_\eta L^{-\vartheta(K-t)}$ for every $\vartheta<\frac12$, with C_η depending on the constants, ϑ, J, $|\mathsf{C}|$, n_C, c_C, δ'_C, and on g through J alone. For the squares Q_i of side a_i with corners in Λ_i one has $s_0=\max\{i,s_0'\}=i+O(1)$, so $s_0=i$ for $i\ge s_0'$, and the bound $C(\mathsf{J})\frac{a_i}{\mathsf{b}}L^{-\vartheta(K-t)}\le C(\mathsf{J})L^{-\vartheta(K-t-i)}$, which depends on the scale of the step only through a_{K-t}/a_i. (c) (The drift along a small square.) Let C be a square of side a_s in a coordinate plane with corners in Λ_s, let $\sigma_1,\dots,\sigma_N$ be its $N=4L^{K-t-s}$ bonds of Λ_{K-t}, and let $\Upsilon_i(B)=\frac1i\log[\mathrm{hol}_{\sigma_i}(e^{i\eta_t\mathbf{A}(B)}U_+)\,\mathrm{hol}_{\sigma_i}(U_+)^{-1}]\in\mathfrak{su}(2)$ be the rotation of σ_i in the shift formula of step 1 of the proof of (a) (at $t=0$ the σ_i are bare bonds, $\Upsilon_i=B'(\sigma_i)$ at a positively oriented σ_i and $\Upsilon_i=-\operatorname{Ad}(U_+(\sigma_i))B'(\bar\sigma_i)$ at a reversed one). For every $\alpha<\frac12$ there are $c_1(\alpha)$ and s_A, depending on the constants, J, α and a lower bound of ℓ, not on g, K, t, s, b, W, such that for $s\ge s_A$, $K>s$, $0\le t<K-s$ and every $W\in\mathsf{O}_{t+1}$

$$\sum_{i,a}\Big|\int\Upsilon_i^a\,\mathrm{d}\nu_W\Big|\le c_1(\alpha)L^{-s}L^{-\alpha(K-t)}.$$

These are the rotations of the shift formula; the loop, being a trace, does not see the difference to the rotations of the background over $V_W(B)$ in any other gauge. (d) (The mean of the fluctuation field.) For every $W\in\mathsf{O}_{t+1}$ and every real 1-form φ on the fluctuation bonds, $|\int(\varphi,B)\,\nu_W(\,\mathrm{d}B)|\le C_4\delta_t(\varepsilon_{t+1}+a_{K-t}/\ell)\sum_b\max_a|\varphi^a(b)|$, C_4 depending on the constants: the mean of the fluctuation field is of the order of the coupling with its logarithm, δ_t, times the bound ε_{t+1} of the curvature of

a tame W, the term a_{K-t}/ℓ coming from the domains that do not fit into the torus; on tame W it is so of the order of the coupling squared, up to logarithms. The order g_t^2 times the curvature, up to a logarithm, is that of the mean of one rotation Υ_i of (c), which holds the further factor g_t of $g_t B$. (e) (Levels at least as coarse as the corners.) Let $0 < \vartheta < \frac{1}{2}$, $c_0 > 0$ and $\varrho > 0$, let C be a rectangle with sides in $[c_0\varrho, 10\varrho]$ and corners in Λ_i, i the integer with $a_i \le \frac{1}{2}c_0\varrho < La_i$, and put $a_* = \varrho(\mathsf{b}/\varrho)^{\vartheta/(3+\vartheta)}$, a length coarser than the loop for small ϱ/b. There is C_*, depending on the constants, c_0, ϑ and J alone, such that at every cutoff $K \ge i$, at every level $0 \le t < K$ with $a_i \le a_{K-t} \le a_*$ and at every $W \in \mathsf{O}_{t+1}$ with $\nu_W \ge 0$, hence for m_{t+1}-almost every W,

$$\Big| \int W_C^{(t)}(V_W(B))\, \nu_W(\,\mathrm{d}B) - \lambda_{t+1} W_C^{(t+1)}(W) \Big| \le C_* \frac{\varrho}{\mathsf{b}} \Big(\frac{a_{K-t}}{\mathsf{b}} \Big)^{\vartheta};$$

so the mean of the defect over the block field, $\bar{\mathsf{D}}_{t+1}$ of Prop. 10.8(b) for C, has modulus at most $C_* \frac{\varrho}{\mathsf{b}} (\frac{a_{K-t}}{\mathsf{b}})^{\vartheta}$, as $Z^{-1} m_{t+1}$ has mass at most 1 (Prop. 10.8(c)). At the levels finer than the corners, $K - t > s_0 = \max\{i, s_0'\}$, (b) gives the same bound with another C_* of the same kind: there $|C| \le 40\varrho$, $n_C = 4$, the chord-arc constant of C is bounded below in terms of c_0, and the bracket of (b), whose distance term vanishes for one polygon ($\delta_C' = \infty$), is at most a multiple of $(a_{K-t}/\mathsf{b})^{\vartheta}$ formed from the constants, ϑ and J. The levels $i < K - t \le s_0'$ occur only when $i < s_0'$, and then $\varrho > 2a_{s_0'}/c_0$ and $a_{K-t} \ge a_{s_0'}$, so that there the bound 2 of the defect at $\nu_W \ge 0$ is at most $C_* \frac{\varrho}{\mathsf{b}} (\frac{a_{K-t}}{\mathsf{b}})^{\vartheta}$ for C_* of the same kind. So the bound of one step holds at every level with $a_{K-t} \le a_*$; this is the bound of one step in Thm. AE(8)(b). ∎

Sketch of proof. (a) *0. The objects of the step.* A *triple* on a domain X is $P = (U, A', \mathbf{J})$, U an SU(2)-valued configuration and A', $\mathbf{J}$ complex 1-forms on the bare bonds of X; it enters through its pair $\mathbf{p}(P) = (e^{i\eta_{t+1}A'}U, \mathbf{J})$, and a gauge transformation u acts by $P^u = (U^u, \mathrm{Ad}(u)A', \mathrm{Ad}(u)\mathbf{J})$ (§5.2). The *background triple* of $W \in \mathsf{O}_{t+1}$ is $P_W = (U_+, 0, J^{(t+1)}(U_+))$, U_+ a background over W whose average $\mathcal{M}^t U_+$ is in the axial gauge and $J^{(t+1)}(U_+)$ its current (the background triple defined before equation YM3_I:(827), §5.3.1); by Thm. YM3_I:E in the setting of Section YM3_I:52.7 at the radius $2\varepsilon_{t+1}$, $|\partial U_+(p) - \mathbf{1}|$ and $|(D^{\mathsf{T}}_{U_+}\partial U_+)(b)|$ are below $\alpha\eta_{t+1}^2$ at all bare plaquettes and bonds, $\alpha := 2B_3\varepsilon_{t+1}$, and $|J^{(t+1)}(U_+)| < \alpha$ (§5.3.1 at the level $t+1$).

The *trivial triple* is $(\mathbf{1}, 0, 0)$; the constant gauge transformations fix it. For a pair $v = (A, \mathbf{J})$ of 1-forms on X, $P[v] = (\mathbf{1}, A, \mathbf{J})$, and $\|v\|_X$ is the largest of $|A|$, $|\nabla A|$, $|\mathbf{J}|$ on the bonds of X (§5.2). The *analyticity core* $\mathcal{U}^\circ(X)$ is a set of complex triples on X given by bounds whose radii are formed from the constants alone (Def. YM3_I:49.1); it is carried onto itself by $P \mapsto P^u$, kept by restriction to smaller domains, holds $P_W|_X$ for tame W (Prop. 5.31), and holds the ball of the $P[v]$, $\|v\|_X < r_\alpha$, about the trivial triple, r_α formed from the constants (equation YM3_I:(933)).

Three cited facts enter. (i) *The kernel.* On $\{|B| < \mathfrak{b}_t\}$, $\nu_W = \Xi_W^{-1} e^{\mathsf{E}_W} \mu_{C_W}$: the Gaussian measure μ_{C_W} of the covariance C_W of the step at P_W, times the exponential of two local potentials, the interaction and the part of the bare array of couplings, each with a norm with quadratic growth at most $c\,\delta_t$ and below the admitted norm $w_\star$ of the cluster expansion, normalised to the mass 1 (eq. 5.xxxii and Prop. 5.31). (ii) *The provision on the core.* At every triple P of $\mathcal{U}^\circ(X)$, complex ones included, the Gaussian structure and both potentials of the step are given, holomorphic in $(A', \mathbf{J})$ at a fixed U, functions of the pair, gauge covariant, local (their restrictions to a smaller domain are the objects formed there) and with the same bounds as at P_W (Def. YM3_I:52.1 and Thm. YM3_I:W); so the cluster expansion for data with up to four potentials (Thm. YM3_I:R for the logarithm and bound; Cor. YM3_I:47.3 for an added potential and the comparison with the Gaussian part; Prop. YM3_I:47.4 for holomorphy and locality, and the covariance argument following it) bounds every cluster term on the whole core, and not only at P_W. (iii) *A potential on a cube.* If $|\partial U - \mathbf{1}|$ and $|D^{\mathsf{T}}_U \partial U|$ are below $\alpha\eta^2$ near a cube Q of integer side l in level units, with $\mathsf{l}\alpha \le c_*$ and $\mathsf{l} + 4$ less than half the side of the torus, then $U^{u^{-1}} = e^{i\eta A}$ on Q for a gauge transformation u and a real 1-form A with $|A|, |\nabla A| < B_*\mathsf{l}\alpha$, B_* and c_* formed from the constants and free of l

(Thm. YM3_I:F). *1. The exact formula.* For $t \geq 1$ the shift of Prop. YM3_I:51.5, taken at the background triple and the 1-form $g_t B$, gives a 1-form $\mathbf{A}(B)$ on the bare bonds such that $e^{i\eta_t \mathbf{A}(B)} U_+$ is a gauge transform of a background over $V_W(B)$ (equation YM3_I:(827)); the loop being gauge invariant, $W_{\mathsf{C}}^{(t)}(V_W(B)) = W_{\mathsf{C}}(e^{i\eta_t \mathbf{A}(B)} U_+)$ exactly, $\eta_t = a_K / a_{K-t}$. $\mathbf{A}$ is holomorphic in B and in the triple, at most $\mathsf{c}_{\mathfrak{h}} \sup |g_t B|$, local through the decoupling parameters, gauge covariant and real at real data (the shift properties proved in Prop. YM3_I:51.5). At $t = 0$ there is no shift system and no decoupling parameter: $\eta_0 = 1$, $W_{\mathsf{C}}^{(0)} = W_{\mathsf{C}}$, and the chart itself gives $V_W(B) = e^{iB'} U_+$, $U_+ = V_W(0)$, with $B' = g_0 B$ on the fluctuation bonds and, at the pivot bond $b_0(c)$ of each block bond c, $B'(b_0(c)) = \beta_c(P_W, g_0 B)$, a pivot value that reads the double block of c alone, vanishes at $B = 0$, is at most $2 C_P \sup |g_0 B|$, is holomorphic in the triple and in B, covariant at its first end point and $\mathfrak{su}(2)$-valued at real data (equation YM3_I:(804) and the bounds and covariance proved with it); we put $\mathbf{A}(B) := B'$.

In both cases $U' := e^{i\eta_t \mathbf{A}(B)} U_+$ is the configuration of the shift formula. *2. Segments.* The trace does not see where a closed path begins: each C_i is cut at one site, its base point, and run from there. For $s \geq i_0 + m' + 1$, which s_0 ensures, the corners are centres of cubes of side M of the level t, and the cubes that C passes are all different; we cut C into $N = |\mathsf{C}|/(M a_{K-t})$ segments, one for each such cube, the segment j being the steps of the path that start in the j-th cube. The segments of C_i form an interval I_i of indices, in the order of the path from the base point, and $I = I_1 < \cdots < I_r$. In the frames R_x carried along the loop by U_+, R_x the product of the $U_+(b')$ over the steps of C_i from the base point to x, the holonomy of U' from the base point is $\prod_{j \in I_i}^{\rightarrow} (\mathbf{1} + k_j) \cdot \mathrm{Hol}_{C_i}(U_+)$, where $\mathbf{1} + k_j$ is the ordered product over the steps b of the segment j of $R_b e^{i\eta_t \sigma_b \mathbf{A}(b^+)} R_b^{-1}$, b^+ the positively oriented bond of the step b, $\sigma_b = \pm 1$ its orientation and R_b the frame at the first end point of b^+ (a step against the orientation gives $U'(b) = U_+(b) e^{-i\eta_t \mathbf{A}(b^+)}$); $\det(\mathbf{1} + k_j) = 1$.

For $t \geq 1$ the brackets in the decoupling parameters of the shift system (equations YM3_I:(806)–YM3_I:(809) and Lem. 5.25) make each k_j a sum of local terms over the domains of the level t that hold the cube of j, each with a factor $e^{-c\,d}$ in its size d, $c > 1$ formed from the constants; at $t = 0$, k_j is one local term, on the cube of the segment with its six neighbours. The linear part of k_j in B, which has no component along $\mathbf{1}$ (its exponents have trace 0), and its remainder have, in the norm with quadratic growth of the cluster expansion, the sizes $\mathsf{n}_\ell \leq c\, g_t$ and $\mathsf{n}_{\mathrm{r}} \leq c\, g_t^2$, c formed from the constants, and not $c\,\delta_t$, which is what keeps every sum below of the order $|\mathsf{C}|/\mathsf{b}$; a sum of them over any set of segments has the same norms up to the factors 3 and 4.

In components, $k_j = \sum_a \ell_j^a\, it_a + \sum_{\mu=0}^3 r_j^\mu \tau_\mu$, $\tau_0 = \mathbf{1}$, $\tau_a = it_a$, with the linear parts ℓ_j^a and the remainders r_j^μ; so the atoms of j are three linear ones, with $\tau = it_a$, and four remainder ones, with $\tau = \mathbf{1}$ and $\tau = it_a$, and $k_j^0 = r_j^0$, $k_j^a = \ell_j^a + r_j^a$. *3. Cumulants.* The linear parts and the remainders are entered as two further potentials into the cluster expansion of the step (Thm. YM3_I:R, with four potential families, and Prop. YM3_I:47.4 for holomorphy and locality; the two families already in ν_W are those of Prop. 5.31), the linear parts as potentials given at every real B, the remainders as potentials given below the threshold $\mathfrak{b}_t$: $\log \int \exp(\sum_\alpha z_\alpha f_\alpha)\, \mathrm{d}\nu_W = \sum_X F_X(z)$, the sum over the domains X of the level $t+1$, with F_X holomorphic, $F_X(0) = 0$, $|F_X| \leq c\, e^{-\lambda d(X)}$, c and $\lambda > 2$ constants of the cluster expansion ($\lambda - 1$ exceeding the growth constant of the number of domains) and $d(X)$ the linear size of X, and F_X depending on the variables of the segments in X alone.

The polydisc is the one on which the inserted potentials stay below the admitted norm: its radii are $\rho_\ell = w_\star / (3\mathsf{n}_\ell)$, of the order $1/g_t$, in the variables of the linear atoms and $\rho_{\mathrm{r}} = w_\star / (4\mathsf{n}_{\mathrm{r}})$, of the order $1/g_t^2$, in those of the remainder atoms, the inverses of the sizes of step 2; they do not depend on N. Cauchy's estimates give all joint cumulants $\kappa(\beta) = \partial_\beta \sum_X F_X(0)$, with tree decay: a block β with n_ℓ linear and n_{r} remainder atoms has $|\kappa(\beta)| \leq \sum_X c\, \rho_\ell^{-n_\ell} \rho_{\mathrm{r}}^{-n_{\mathrm{r}}} e^{-\lambda d(X)}$, of the order $g_t^{n_\ell + 2n_{\mathrm{r}}}$, the sum over the X that hold the segments of β.

We write $E(X)$ for the X-term of this expansion with one or two insertions, that is a first or a second derivative of F_X at $z=0$, regarded as a function of the triple at which the step is formed. *4. The ordered product.* By Lem. 10.5(a), with $\mathfrak{A}=M_2(\mathbb{C})^{\otimes r}$, one tensor factor for each polygon, the integral of the ordered product $\bigotimes_i \prod^{\rightarrow}_{j\in I_i}(\mathbf{1}+k_j)$ is $Z[\kappa]$; the functional $T\mapsto 2^{-r}\operatorname{tr}(T\cdot\bigotimes_i \operatorname{Hol}_{C_i}(U_+))$ has norm at most 1, sends $\mathbf{1}$ to $W_{\mathsf{C}}^{(t+1)}(W)$ and the integral of the ordered product to $\int W_{\mathsf{C}}^{(t)}(V_W(B))\,\nu_W(\mathrm{d}B)$, so that the left side of (a) is at most $|Z[\kappa]-\lambda_{t+1}\mathbf{1}|$.

For all $j\le j'$ in I let $\delta_{cc'}p_{jj'}$ be the second cumulant of the linear parts ℓ_j^c, $\ell_{j'}^{c'}$ formed at the trivial triple with the frames $\mathbf{1}$, under the measure of the step there; it is a multiple of $\delta_{cc'}$ by step 5, it depends neither on W nor on the colour, and $p_{j'j}:=p_{jj'}$. Let $c_j:=\kappa(\{(j;\mathrm{r},0)\})=\int k_j^0\,\mathrm{d}\nu_W$, the first cumulant of the central remainder atom. A pair $j<j'$ of one polygon is *near* if $j'-j\le N_\sharp:=\lfloor s_{\mathsf{C}}/(2Ma_{K-t})\rfloor$; the arc from the segment j to the end of the segment j' then has length at most $\frac12 s_{\mathsf{C}}+Ma_{K-t}\le s_{\mathsf{C}}$ (s_0 ensures $N_\sharp\ge 1$), so that it holds at most one corner, and it does not pass the base point.

It is *straddling* if it is not near and $N_i-(j'-j)\le N_\sharp$, $N_i=\#I_i$ (near across the base point), and *far* otherwise; pairs on two polygons are far. A far pair is at sup-distance at least $\rho_{\mathsf{C}}/a_{K-t}$ in level units, so that the domains that hold it have $d(X)\ge\rho_{\mathsf{C}}/(LMa_{K-t})-2$. We write $\kappa=w_{\mathrm{g}}+w_{\mathrm{b}}$, w_{g} the weight of Lem. 10.5(b) with these c_j and with $p_e=p_{jj'}$ at the near pairs $e=\{j<j'\}$, and $w_{\mathrm{b}}=\kappa-w_{\mathrm{g}}$; w_{b} holds the coloured singletons, the colour defects $\kappa(\{(j;\ell,a),(j';\ell,a')\})-\delta_{aa'}p_{jj'}$ of the near pairs, the other blocks with a remainder atom (all but the central singletons), the blocks of three or more atoms, and the far and the straddling pairs.

With $\mathsf{g}_s\ge g_t$, $N\mathsf{g}_s^2=\Lambda_{\mathsf{C}}$, and the bounds of steps 3 and 5, $|w_{\mathrm{b}}|_1\le c(1+\Lambda_{\mathsf{C}})\mathsf{y}_s^{2\mathfrak{p}_0}(\mathsf{g}_s+L^{-s/2})+c\Lambda_{\mathsf{C}}e^{2-\rho_{\mathsf{C}}L^s/LM}+c\,\#\mathsf{C}\,\mathsf{g}_s^2$, the far pairs giving the term with ρ_{C} and the straddling ones, of the order g_t^2 for each polygon, the term with $\#\mathsf{C}$. Further $\mathfrak{T}\le c\Lambda_{\mathsf{C}}$, and $\mathfrak{X},\mathfrak{O}\le c\Lambda_{\mathsf{C}}\mathsf{g}_s^2$: a near pair that crosses a near pair $\{j_1<j_2\}$ has an end point among the j_2-j_1 indices between them, and $\sum_{j'}|p_{jj'}|$ and $\sum_{j'}|j'-j||p_{jj'}|$, the latter over the near pairs, are at most $c\,g_t^2$.

Lem. 10.5(a), (b) give $Z[\kappa]=\exp(\sum_j c_j-3\sum_e p_e)\mathbf{1}$ up to these errors. *5. Symmetry.* What is not small by size, the first cumulants of the colour components and the colour defects of the near pairs and of the second cumulants $\langle\ell_j^a;\ell_j^{a'}\rangle$ of one segment, is small by symmetry. By fact (ii) each cluster term $E(X)$ is defined at every triple of $\mathcal{U}^\circ(X)$, holomorphic there in the sense of the slices, and bounded there as at P_W. Let X be small, $d(X)\le\frac12 s\log L$, and let Q_X be the cube of side $\mathsf{l}_X=2M(d(X)+2)+1$ (level-$(t+1)$ units) about a cube of X; it holds X and the arc of every near pair whose segments lie in X.

For $s\ge s_0$ the conditions of fact (iii) hold on Q_X for every small X ($\mathsf{l}_X\alpha\le cMs\log L\,\varepsilon_{t+1}$ is small and l_X+4 is far below the side of the torus), and fact (iii) at the α of step 0 gives u and a real A with $U_+^u=e^{i\eta_{t+1}A}$ on Q_X and $|A|,|\nabla A|<B_*\mathsf{l}_X\alpha$; with $v_X:=(A,\operatorname{Ad}(u)J^{(t+1)}(U_+))$ on the bonds of X, $\|v_X\|_X<B'_*\mathsf{l}_X\alpha$, $B'_*=\max\{B_*,1\}$, and $(P_W|_X)^u$ and $P[v_X]$ have one pair. So the triple lies, up to the gauge transformation u, on the disc $\zeta\mapsto P[\zeta v_X]$, $|\zeta|<\rho_X$, about the trivial triple, inside the core by equation YM3_I:(933); its radius $\rho_X=\min\{r_\flat,1/(3\mathsf{l}_X)\}/(B'_*\mathsf{l}_X\alpha)$, $r_\flat$ formed from the constants, has $1/\rho_X\le cM^2\varepsilon_{t+1}(d(X)+2)^2$, c formed from the constants, keeps the frames along the arcs in Q_X within a factor e of $\mathbf{1}$, and is at least 2 for large s.

The locality, the covariance and the dependence on the pair alone of the cluster terms (Prop. YM3_I:47.4 and the covariance argument following it, applied with fact (ii)) turn $E(X)$ at P_W, formed on the torus, into $E(X)$ at $P[v_X]$, formed on X, the frames being changed by one constant rotation, so that the colours are turned by one $O_X\in SO(3)$. At $\zeta=0$ the constant gauge rotations fix the trivial triple: a colour vector that they turn, the first derivative of a term in one coloured atom, vanishes there, and a bilinear form in the colours that commutes with all of them is a multiple of δ_{ab}, $\mathbb{C}^3$ being irreducible under $SO(3)$. Schwarz' lemma for $\zeta\mapsto E(X)[P[\zeta v_X]]-E(X)[P[0]]$ on the disc gives the factor $1/\rho_X\le cM^2\varepsilon_{t+1}(d(X)+2)^2$, paid by the decay $e^{-\lambda d(X)}$; the domains with $d(X)>\frac12 s\log L$ cost $e^{-\frac12 s\log L}=L^{-s/2}$.

The size that this factor multiplies is the size of step 3, except for the first cumulant of a linear part, whose size is smaller: the Gaussian part of a cluster term, the same term formed with the linear insertions alone under the Gaussian measure of the step without threshold, is even under $B \mapsto -B$ (apply the covariance argument following Prop. YM3_I:47.4 to the reflection $B \mapsto -B$), so that its derivative in one linear atom vanishes at $z = 0$; the first cumulant of a linear part is therefore the derivative of the difference of the term and its Gaussian part, which is at most $c\,\delta_t\, e^{-\lambda d(X)}$ (Cor. YM3_I:47.3, equation YM3_I:(724)): it has the amplitude $c\,\delta_t \mathsf{n}_\ell$, that of the non-Gaussian potentials.

With $\alpha \le c\,\mathsf{g}_s \mathsf{y}_s^{\mathfrak{p}_0}$, $\delta_t \le A_1 \mathsf{g}_s \mathsf{y}_s^{\mathfrak{p}_0}$ and the sums over the N segments, these are the bounds of step 4.

6. Singletons. $\det(\mathbf{1} + k_j) = 1$ and $\det x = (x^0)^2 + \sum_a (x^a)^2$ give $(1 + k_j^0)^2 + \sum_a (k_j^a)^2 = 1$, that is $k_j^0 = -\frac{1}{2}\sum_a (k_j^a)^2 - \frac{1}{2}(k_j^0)^2$; integrated with $\int f^2 \,\mathrm{d}\nu_W = \langle f; f\rangle + (\int f \,\mathrm{d}\nu_W)^2$, $\langle f; f\rangle$ the second cumulant, and with $\langle \ell_j^a; \ell_j^a\rangle = p_{jj}$ plus a diagonal defect, this gives $c_j = -\frac{3}{2} p_{jj}$ up to errors of the size of step 4 (the diagonal defects, the mixed second cumulants with a remainder, the squares of first cumulants, c_j^2).

As $\sum_{j,j' \in I_i} p_{jj'} = \sum_{j \in I_i} p_{jj} + 2\sum_{j<j'} p_{jj'}$,

$$\sum_j c_j - 3\sum_e p_e = -\frac{3}{2}\sum_i \sum_{j,j' \in I_i} p_{jj'} + \sum_j (c_j + \tfrac{3}{2} p_{jj}) + 3 \sum_{j<j' \text{ in one } I_i, \text{ not near}} p_{jj'},$$

and the last two sums are of the size of the errors of step 4.

7. The flat value. At the trivial triple with the frames $\mathbf{1}$ the linear part of a segment is linear in B, and the sum over the segments of C_i of its colour component a is $g_t(j_{C_i}, \eta_t(\dot{\mathbf{A}}B)^a) = g_t(X_{C_i}, \eta_t\,\mathrm{d}(\dot{\mathbf{A}}B)^a)$, $\dot{\mathbf{A}}$ the derivative at 0 of $\mathcal{A} \mapsto \mathbf{A}$ formed at the trivial triple (at $t = 0$ the derivative $\mathsf{C}_{\mathbf{1}} = \mathsf{C}_{\mathbf{1},\mathbb{R}} \otimes 1$ of the chart), j_C the current of C and $\mathrm{d}^{\mathsf{T}} X_C = j_C$. The linear part of the shift system at the trivial triple produces, up to one sign, the curvature response of the step: on every side of $\mathfrak{L}(g)$ and for $1 \le t < K$ there is $\mathsf{sg}_t \in \{1, -1\}$ with $\eta_t\,\mathrm{d}(\dot{\mathbf{A}}\mathcal{A}) = \mathsf{sg}_t(\mathsf{T}_t^{\mathrm{cv}} \otimes 1)\mathcal{A}$ for every 1-form $\mathcal{A}$ on the fluctuation bonds, $\mathsf{T}_t^{\mathrm{cv}} = \mathsf{H}_t^{\mathrm{cv}} \mathsf{C}_{\mathbf{1},\mathbb{R}}$, $\mathsf{H}_t^{\mathrm{cv}}$ the curvature response of the level t (the curvature-response construction preceding Lem. YM3_I:120.2).

The proof follows that of Lem. YM3_I:124.3, which is stated for data with two real sources and on large sides only; two changes make it run for the objects of the frame on every side of $\mathfrak{L}(g)$. First, the block field $\mathbf{1}$ lies in O_{t+1}, so that it has the properties that the kernel lemma of fact (i) (Prop. 5.31) gives to every tame block field and on which the hypotheses of (F7) rest; the configuration $\mathbf{1}$ is a background over it, its average $\mathcal{M}^t\mathbf{1} = \mathbf{1}$ is in the axial gauge and its current vanishes, so that its background triple is the trivial triple; and the value of the shift on the curve below is taken from (F7) at the block field $\mathbf{1}$.

Second, we use only the constrained variational equation in the proof of Lem. YM3_I:124.1 and the curvature identity YM3_I:(2230). These calculations hold on every side; the large-side identification with the full Landau inverse is not needed. Constant gauge rotations fix the trivial triple, so covariance makes the shift derivative commute with every $\mathrm{Ad}(g_0)$, $g_0 \in \mathrm{SU}(2)$. It is therefore one real kernel times the identity on colours. For real $\mathcal{A}$, (F7) makes $U_\tau = e^{i\eta_t \mathbf{A}(\tau\mathcal{A})}$ a curve of configurations critical over their own averages. Write $\dot{A} = \dot{\mathbf{A}}\mathcal{A}$ and $\mathcal{M}^t U_\tau = \mathbf{1} + i\tau\dot{V} + o(\tau)$. The cited variational calculation gives a real fine-site function θ and a multiplier ω such that

$$h = \dot{A} + \nabla\theta, \qquad D_{\mathbf{1}}^{\mathsf{T}} D_{\mathbf{1}} h = Q^{\mathsf{T}}\omega, \qquad Qh = \dot{V}.$$

Indeed, differentiation of the action on the moving constraint kernel makes its flat Hessian pairing with $\dot{A}$ vanish on the kernel of the linearized average. Factoring through that average gives the multiplier; its value on gradients is zero, and the gauge normalization restores the straight average $Qh = \dot{V}$. Equation YM3_I:(2230) now gives

$$\eta_t\,\mathrm{d}\dot{A} = \eta_t^2 D_{\mathbf{1}} h = (\mathsf{H}_t^{\mathrm{cv}} \otimes 1)\dot{V}.$$

The chart over $\mathbf{1}$ identifies $\dot V$, up to a constant rotation $\mathrm{Ad}(g_*)$, with $\mathsf{C}_{\mathbf{1}}\mathcal{A}$ minus a gradient. The curvature response annihilates this gradient. On a single colour the remaining rotation gives a sign; linearity in $\mathcal{A}$ makes it the same sign for all $\mathcal{A}$, as in the last paragraph of the proof of Lem. YM3_I: 124.3.

At $t=0$, $\mathsf{H}_0^{\mathrm{cv}}=\mathrm{d}$ (the curvature-response calculation preceding Lem. YM3_I:120.2) and the sign is $+$. At the trivial triple the covariance of the step is $\mathbf{C}_*(P_{\mathbf{1}})=(\mathsf{A}_t^{\mathrm{cv}})^{-1}\otimes 1$, and $\mathsf{T}_t^{\mathrm{cv}}(\mathsf{A}_t^{\mathrm{cv}})^{-1}(\mathsf{T}_t^{\mathrm{cv}})^{\mathsf{T}}=\eta_t\,\mathcal{C}_{(t+1)}$ (Prop. YM3_I:120.3, Prop. YM3_I:120.3); so the Gaussian variance of the sum of the linear parts along C_i is $g_t^2\eta_t(X_{C_i},\mathcal{C}_{(t+1)}X_{C_i})$ in each colour, the slice $(X_{C_i},\mathcal{C}_{(t+1)}X_{C_i})$ of the free Coulomb energy, and the sign drops out. The true pair sum differs from the Gaussian one, as in step 5, by the difference of the terms and their Gaussian parts: $\sum_{j,j'\in I_i}p_{jj'}=g_t^2\eta_t(X_{C_i},\mathcal{C}_{(t+1)}X_{C_i})$ up to $cN_i\delta_t g_t^2$, N_i the number of segments of C_i.

With step 6, $\sum_j c_j-3\sum_e p_e=-\frac{3}{2}S$ up to the errors of step 4, where

$$S:=g_t^2\eta_t\sum_i(X_{C_i},\mathcal{C}_{(t+1)}X_{C_i})\ge 0,$$

$\mathcal{C}_{(t+1)}$ being an orthogonal projection: three colours and $\frac{1}{2}\,\mathrm{tr}$ give $\frac{3}{2}$. Between different polygons all pairs are far, and

$$|a_K(X_{C_i},\mathcal{C}_{(t+1)}X_{C_{i'}})|\le C_X|\mathsf{C}|e^{2-\rho_{\mathsf{C}}/LMa_{K-t}}$$

for $i\ne i'$; and $0\le a_K(X_C,\mathcal{C}_{(t+1)}X_C)\le C_X|C|$, C_X formed from the constants. *8. The label.* $g_t^2\eta_t=a_K/(a_{K-t}x_t)$ and $|1/(a_{K-t}x_t)-1/\mathsf{b}|\le \mathsf{c}\,\mathsf{b}^{-2}L^{-(K-t)}$, $\mathsf{c}=\frac{5}{4}\bar B$, turn $e^{-\frac{3}{2}S}$ into λ_{t+1} up to $\frac{3}{2}\mathsf{c}C_X\mathsf{b}_-^{-2}|\mathsf{C}|L^{-s}$, both exponents being ≤ 0.

Steps 4 to 8 together give the bound, with C', C'' formed from the constants; $\sup_s \mathsf{y}_s^{2\mathfrak{p}_0}L^{-s/4}<\infty$ and $\sup_s L^{s/4}e^{-\rho_{\mathsf{C}}L^s/LM}<\infty$ give C_η. The matrix form is step 4 without the trace, for one polygon from its base point.

(b) The variables are now the rotations $\Upsilon_\sigma(B)=\frac{1}{i}\log[(e^{i\eta_t\mathbf{A}(B)}U_+)(\sigma)U_+(\sigma)^{-1}]$ of the $|\mathsf{C}|/a_{K-t}$ bonds σ of Λ_{K-t} along C, oriented as C runs (at $t=0$ the bare bonds of C, with $\Upsilon_\sigma=B'(\sigma)$ at a positively oriented σ and $-\mathrm{Ad}(U_+(\sigma))B'(\bar\sigma)$ at a reversed one), $|\Upsilon_\sigma|\le c\delta_t$, so that $W_{\mathsf{C}}^{(t)}(V_W(B))=w(\Upsilon(B))$, $W_{\mathsf{C}}^{(t+1)}(W)=w(0)$, with w as in Lem. 10.5(c), $g_\sigma=U_+(\sigma)$. By the brackets each Υ_σ is a sum of local terms, gauge covariant at σ_- ($\Upsilon_\sigma\mapsto\mathrm{Ad}(u(\sigma_-))\Upsilon_\sigma$); their linear parts and remainders enter the cluster expansion as in (a), with $\log\nu_W(e^{(z,\Upsilon)})=\sum_X\ell_X(z)$ on a polydisc of radius $r_1\ge c/(n_{\mathsf{C}}g_t)$, one variable $z_\sigma\in\mathbb{C}^3$ for the whole of Υ_σ.

Lem. 10.5(c), with ϱ the image of $\nu_W\ge 0$ under $B\mapsto\Upsilon(B)$, and with $s_{\sigma\sigma'}\delta_{aa'}$ the covariance of the linear parts of Υ_σ^a, $\Upsilon_{\sigma'}^{a'}$ formed at the trivial triple under the Gaussian measure of the step there (real, symmetric, free of W; a multiple of $\delta_{aa'}$ by the constant rotations), gives the bound with $E_1\le C_1\mathsf{l}g_t^2(\mathfrak{b}_t+\mathfrak{b}_t^2)\epsilon_t^\circ$, $E_2\le C_2\mathsf{l}g_t^2[\epsilon_t^\circ+(1+\mathfrak{b}_t)^2\delta_t+e^{-\mathsf{D}_t}]$ and $E_3\le C_3\mathsf{l}g_t^3$, $\mathsf{l}=|\mathsf{C}|/a_{K-t}$, where $\epsilon_t^\circ\le c(\varepsilon_{t+1}+L^{-(K-t)})$ is the price of the symmetry argument: the first cumulants and the defects of the second cumulants from $s_{\sigma\sigma'}\mathrm{R}(U_+(\sigma_-\to\sigma'_-))$ are small by the symmetry of (a), step 5, the transport running along the shorter arc, whose length the chord-arc constant bounds; the pairs on two polygons have $d(X)\ge\mathsf{D}_t$. $\sum_{\sigma,\sigma'\in C_i}s_{\sigma\sigma'}=g_t^2\eta_t(X_{C_i},\mathcal{C}_{(t+1)}X_{C_i})\ge 0$ as in (a), step 7, and $\mathsf{l}g_t^2\le\frac{3}{2}|\mathsf{C}|/\mathsf{b}$; the label costs $\frac{9}{4}C_2\bar B|\mathsf{C}|\mathsf{b}^{-2}L^{-(K-t)}$.

For a complex ν_W of mass 1 Lem. 10.5(c) holds with the total variation as a factor. Positivity: by the disintegration with m_t, m_{t+1}, $\mathsf{h}_{t+1}\ge 0$, $\int H(W)[\int G(V_W(B))\nu_W(\,\mathrm{d}B)]m_{t+1}(\,\mathrm{d}W)\ge 0$ for all bounded Borel $H,G\ge 0$; testing a countable set of continuous $G\ge 0$ dense among them, the image of ν_W under $B\mapsto V_W(B)$ is a positive measure of mass 1 for m_{t+1}-almost every W; as $V_W(B)(b)=e^{ig_tB(b)}V_W(0)(b)$ at the fluctuation bonds and $|g_tB(b)|<\delta_t<\log 2$, the chart is injective with a Borel inverse on $\{|B|<\mathfrak{b}_t\}$, so ν_W itself is ≥ 0 for almost every W. The exponent $\frac{1}{2}$ is the order of the error: five of its members are of the order $\frac{|\mathsf{C}|}{\mathsf{b}}x_t^{-1/2}$ up to logarithms, two of them of first order in the curvature of W.

For Q_i: $n_{Q_i} = 4$ and $c_{Q_i} = \frac{1}{2}$ are free of i, there is no distance term, and $|Q_i|/\mathsf{b} = 4a_i/\mathsf{b}$.

(c), (d) A covariant insertion based at a site x_0 is a family of local terms of the step that turns under gauge transformations as a vector at x_0. For such an insertion each term $D(X;P)$ of the first cumulant, formed at the triple P on a domain X (the derivative at 0 of the cluster term of X with the insertion multiplied by z), is holomorphic in P on the analyticity core and bounded there by $\mathsf{A}(\mathbb{W})e^{-\lambda d(X)}$ (fact (ii) in step 0 of the proof of (a)), is a function of the pair, turns as a vector under constant rotations, and so vanishes at the trivial triple; a tame background on X is gauge equivalent to a potential of norm $\le c\,M(d(X)+2)\varepsilon_{t+1}$, c formed from the constants (fact (iii) on the cube of side $2M(d(X)+2)+1$ about X), as long as this cube fits into the torus and $c\,M(d(X)+2)\varepsilon_{t+1}$ stays below a number formed from the constants, the other domains costing a_{K-t}/ℓ, resp. ε_{t+1}, with the factor $d(X)+2$; Schwarz' lemma gives $\sum_a |\int \mathbb{W}_a^{\mathrm{tot}}\,\mathrm{d}\nu_W| \le c\,\mathsf{A}(\mathbb{W})\varsigma_t$, $\varsigma_t = c\varepsilon_{t+1} + 6LMa_{K-t}/\ell$, where $\mathbb{W}_a^{\mathrm{tot}}$ is the sum of the local terms of the insertion and $\mathsf{A}(\mathbb{W})$ its amplitude in the cluster expansion.

For the fluctuation field, an odd insertion linear in B, the amplitude is that of the non-Gaussian potentials, $\lesssim \delta_t$, because the Gaussian part is even: this is (d). For the rotations Υ_i the linear part is an odd insertion of norm $\le cg_t$ and the remainder an insertion of norm $\le cg_t^2$, so each bond costs $cg_t^2(\log x_t)^{\mathfrak{p}_0}\varsigma_t$; and $Ng_t^2 \le c\,a_s/\mathsf{b}$ for the $N = 4L^{K-t-s}$ bonds, which gives (c). At $t=0$ the rotations are those of the chart, step 1 of the proof of (a), and the same argument runs.

(e) As $a_{K-t} \ge a_i > c_0\varrho/2L$ and $|C| \le 40\varrho$, $|C|/a_{K-t} < 80L/c_0$, and C lies in a cube of side less than $20L/c_0$ in the units of the level. If ϱ/b is at least a number $\eta_0 > 0$ formed from the constants, c_0, ϑ and J, the bound 2 of the defect at $\nu_W \ge 0$ is at most $C_*\frac{\varrho}{\mathsf{b}}(\frac{c_0\varrho}{2L\mathsf{b}})^{\vartheta} \le C_*\frac{\varrho}{\mathsf{b}}(\frac{a_{K-t}}{\mathsf{b}})^{\vartheta}$ for $C_* \ge 2(2L/c_0)^{\vartheta}\eta_0^{-1-\vartheta}$; so let $\varrho/\mathsf{b} < \eta_0$, and with it $a_{K-t}/\mathsf{b} \le a_*/\mathsf{b} = (\varrho/\mathsf{b})^{3/(3+\vartheta)}$, g_t, δ_t and ε_{t+1} be small, and $K-t$ large.

The proof of (b) runs with the pieces of C in the cubes of the level in the place of the bonds of the level. *The exact formula.* Step 1 of the proof of (a) holds for every bare loop, the corners entering nowhere: $W_C^{(t)}(V_W(B)) = W_C(U')$, $U' = e^{i\eta_t\mathbf{A}(B)}U_+$, at every level. *Pieces.* From a base point, C is cut into the maximal runs $C_1,\dots,C_n$ of consecutive steps that start in one of the half-open cubes of side a_{K-t} about the sites of Λ_{K-t}, the cubes of the level that C passes; each of the four sides of C gives at most its length over a_{K-t} plus 2 of them, so that $n < 80L/c_0 + 8$, a number formed from L and c_0. With $g_j = U_+(C_j)$, the holonomy of U_+ along the piece, and $\Upsilon_j(B) = \frac{1}{i}\log[U'(C_j)\,U_+(C_j)^{-1}] \in \mathfrak{su}(2)$, one has $|\Upsilon_j| \le c\,\delta_t|C_j|/a_{K-t}$, c formed from the constants (the shift is at most $\mathsf{c}_{\mathfrak{h}}\sup|g_tB|$, at $t=0$ at most $2C_P\sup|g_0B|$, at each of the $|C_j|/a_K$ bare steps, and $\eta_t|C_j|/a_K = |C_j|/a_{K-t}$), and $W_C^{(t)}(V_W(B)) = w(\Upsilon(B))$, $W_C^{(t+1)}(W) = w(0)$, with w as in Lem. 10.5(c) for one line.

Insertions. By the brackets each Υ_j is a sum of local terms of the level, gauge covariant at the first end point of C_j, with a linear part in B of norm at most $c\,g_t|C_j|/a_{K-t}$ and a remainder of norm at most $c\,g_t^2\frac{|C_j|}{a_{K-t}}(1+\frac{|C_j|}{a_{K-t}})$ in the norm of the cluster expansion; each piece is one insertion, of norm of the order g_t times its length over a_{K-t}. Entered into the cluster expansion of the step as in step 3 of the proof of (a), they give $\log\nu_W(e^{(z,\Upsilon)}) = \sum_X \ell_X(z)$, one variable $z_j \in \mathbb{C}^3$ for the whole of Υ_j, on a polydisc of radius $r_1 \ge c\,a_{K-t}/(g_t|C|) \ge c\,c_0/(Lg_t) \ge 2$, and Cauchy's estimates give a cumulant of order m the factor $(c\,g_t|C|/a_{K-t})^m$ and the decay of its domains.

Symmetry. Let $s_{jj'}\delta_{aa'}$ be the covariance of the linear parts of Υ_j^a, $\Upsilon_{j'}^{a'}$ formed at the trivial triple under the Gaussian measure of the step there (real, symmetric, free of W; a multiple of $\delta_{aa'}$ by the constant rotations). At the trivial triple the first cumulants, colour vectors, vanish, and the pair cumulants are multiples of $\delta_{aa'}$, which differ from $s_{jj'}\delta_{aa'}$ by parts of the relative size $c\,\delta_t$, from the non-Gaussian potentials, and $c\,g_t$, from the remainders (step 5 of the proof of (a)). For a domain X that holds pieces of C, with $d(X) \le \frac{1}{2}(K-t)\log L$, fact (iii) of step 0 of the proof of (a), on a cube of side at most $2M(d(X)+2)+1+40L/c_0$ that holds X and C, where its conditions hold for $\varrho/\mathsf{b} < \eta_0$, puts the background triple, up to a gauge transformation, on a disc about the trivial triple

whose radius ρ_X has $1/\rho_X \le c\,\varepsilon_{t+1}(d(X)+2)^2$, c formed from the constants, L and c_0, the frames along C staying within a factor e of $\mathbf{1}$; Schwarz' lemma on this disc bounds the terms of X of the first cumulants, and of the pair cumulants less their values at the trivial triple turned by the frames, by $1/\rho_X$ times their size, paid by the decay $e^{-\lambda d(X)}$, and the domains with $d(X) > \frac12(K-t)\log L$ cost $L^{-(K-t)/2}$.

So the first cumulants and the defects $d_{jj'}$ of the pair cumulants from $s_{jj'}\mathrm{R}(G_{j\to j'})$ carry, relative to their size, the factor $c\,(\varepsilon_{t+1} + L^{-(K-t)/2} + \delta_t + g_t)$, and the first cumulant of a linear part carries besides the amplitude $c\,\delta_t$ of the non-Gaussian potentials, its Gaussian part being even. *The pair term.* The argument of step 7 of the proof of (a) holds for every bare loop: at the trivial triple with the frames $\mathbf{1}$ the sum of the linear parts of the Υ_j is $g_t\eta_t(j_C, \dot{\mathbf{A}}B)$, so that $S := \sum_{j,j'} s_{jj'} = g_t^2\eta_t(X_C, \mathcal{C}_{(t+1)}X_C) \ge 0$, the slice of the free Coulomb energy at the step, at the depths coarser than the corners as at the finer ones.

By Prop. 10.7(c), (d), which hold at every depth, $0 \le a_K(X_C, \mathcal{C}_{(t+1)}X_C) = v_{K,K-t}(C) - v_{K,K-t-1}(C) \le C^{\mathrm{bg}}A_C^2/a_{K-t}^3 \le c\,|C|$, with the area $A_C \le 100\varrho^2$ of C, $|C| \ge 4c_0\varrho$ and c formed from L and c_0; so step 8 of the proof of (a) turns $e^{-\frac32 S}$ into λ_{t+1} up to $c\,|C|\mathsf{b}^{-2}a_{K-t}$. *The bound.* Lem. 10.5(c), its measure being the image of $\nu_W \ge 0$ under $B \mapsto \Upsilon(B)$, a probability measure on $\{|y_j| \le c\,\delta_t|C|/a_{K-t}\}$, bounds the defect by $E_1 + E_2 + E_3$ and this term of the label, the sum over two lines being empty. With $g_t^2 \le \frac32 a_{K-t}/\mathsf{b}$, $\delta_t = g_t\mathfrak{b}_t$, ε_{t+1} and $L^{-(K-t)/2}$ of the order $(a_{K-t}/\mathsf{b})^{1/2}$ up to logarithms, and the sizes above, each of E_1, E_2, E_3 is at most $\frac{|C|}{\mathsf{b}}(\frac{a_{K-t}}{\mathsf{b}})^{1/2}y_t^{\mathfrak{p}}$, $y_t = \log x_t$, $\mathfrak{p}$ formed from $\mathfrak{p}_0$, times a polynomial in $|C|/a_{K-t}$ with coefficients formed from the constants and J, and the term of the label is at most $c\frac{|C|}{\mathsf{b}}\frac{a_{K-t}}{\mathsf{b}}$.

As $|C|/a_{K-t} < 80L/c_0$, $|C| \le 40\varrho$ and $(\frac{a_{K-t}}{\mathsf{b}})^{1/2}y_t^{\mathfrak{p}} \le c\,(\frac{a_{K-t}}{\mathsf{b}})^{\vartheta}$ for $\vartheta < \frac12$, c formed from the constants, ϑ and J, the defect is at most $C_*\frac{\varrho}{\mathsf{b}}(\frac{a_{K-t}}{\mathsf{b}})^{\vartheta}$. The mean over the block field follows, $|\bar{\mathsf{D}}_{t+1}|$ being at most the bound of the defect at m_{t+1}-almost every W times $Z^{-1}m_{t+1}(\mathcal{V}_{K-t-1}) \le 1$. The levels finer than the corners are treated in the statement of (e) itself. □

Proposition 10.7 (Numbers of the free field)**.** Let $L \ge 3$ be odd, $\ell = 2L^m$, and C a polygon with corners in Λ_i; no measure and no SU(2) enter. (a) For $K \ge i$ and $a = a_K$: $\mathcal{I}_a(C,C) = \frac{|C|}{2\pi}\log\frac1a + c_{\mathrm{lat}}|C| + \mathcal{J}_\ell(C) + E_a(C)$ with $|E_a(C)| \le c[n_C a(1+\log\frac{|C|}{a}) + a^2|C|^2\mathrm{Lg}(\ell/a)(\delta_C^{-3} + \ell^{-3})]$, c absolute; pieces in perpendicular directions do not interact in the sum; $|\mathcal{J}_\ell(C) - \mathcal{J}(C)| \le c|C|^2\mathrm{diam}(C)^2\ell^{-3}$; for a rectangle $s \times t$,

$$\mathcal{J}(C) = \frac1\pi\Big[s\log s + t\log t - s\,\mathrm{asinh}\,\frac{s}{t} - t\,\mathrm{asinh}\,\frac{t}{s} + 2\sqrt{s^2+t^2} - 2(s+t)\Big].$$

For polygons C, D at distance $\delta > 0$: $|\mathcal{I}_a(C,D) - \mathcal{I}_\ell(C,D)| \le ca^2|C||D|\,\mathrm{Lg}(\ell/a)(\delta^{-3} + \ell^{-3})$. (b) For $0 \le d \le K$ and every closed path C' of Λ_K, $v_{K,d}(C')$ is the largest value of $2\mathcal{I}_{a_K}(j_{C'}, Q^{\mathsf{T}}\iota) - \mathcal{I}_{a_K}(Q^{\mathsf{T}}\iota, Q^{\mathsf{T}}\iota)$ over the closed currents ι of Λ_d without zero mode, Q the linearisation of $\mathcal{M}^{K-d}$ at the trivial field (the mean over a block of the straight bare paths): the energy of the best smeared current. For the polygon C and $i \le d \le K$,

$$v_{K,d}(C) = \frac{|C|}{2\pi}\log\frac{1}{a_d} + (c_{\mathrm{lat}} + c_{r,K-d})|C| + \mathcal{J}_\ell(C) + \mathrm{Rem}_{K,d}(C),$$

$|\mathrm{Rem}_{K,d}(C)| \le |E_{a_d}(C)| + c[n_C^2 a_d + n_C a_d(1+\log\frac{|C|}{a_d}) + a_d|C|^2/\delta_C^2]$, a bound free of K. Here the numbers $c_{r,j}$, $j \ge 0$ an integer, are formed as c_r in the setting, with $S_{(j)}(z) = \frac{\sin(z/2)}{L^j\sin(z/2L^j)}$ in the place of $S(z)$, $E_{(j)}(p_\nu) = 4L^{2j}\sin^2\frac{p_\nu}{2L^j}$ in the place of p_ν^2 in the denominators, and the sums over the $n \in \mathbb{Z}^2$ with $|n_2|, |n_3| \le \frac{L^j-1}{2}$ in the places of the sums over $\mathbb{Z}^2$, in N and in D; at $j = K-d$, $L^j = a_d/a_K$ is the ratio of the two spacings. The number $c_{r,j}$ depends on L^j alone, $c_{r,0} = 0$ (then $S_{(0)} = 1$ and only $n = 0$ enters), $|c_{r,j}| \le c_3$, and $c_{r,j} \to c_r$ as $j \to \infty$. The limit $v_d = \lim_K v_{K,d}$

exists and obeys the same remainder bound:

$$\Big|v_d - \frac{|C|}{2\pi}\log\frac{1}{a_d} - c_Q|C| - \mathcal{J}_\ell(C)\Big| \le |E_{a_d}(C)| + c\Big[n_C^2 a_d + n_C a_d\Big(1+\log\frac{|C|}{a_d}\Big) + \frac{a_d|C|^2}{\delta_C^2}\Big].$$

For a fixed polygon the right side is $O_{C,\ell}(a_d(1+\log^+(1/a_d)))$ as $d\to\infty$. The constants c and c_3 are free of the side; c_Q is a number of the plane, free of L, ℓ and C. (c) $\prod_{n\le K-d}\lambda_n = \exp[-\frac{3}{2\mathsf{b}}(\mathcal{I}_{a_K}(C,C) - v_{K,d}(C))]$ for all $0\le d\le K$. Hence for $i\le d\le K$

$$a_K^{-\kappa}\prod_{n\le K-d}\lambda_n = L^{\kappa d}\exp\Big[\frac{3}{2\mathsf{b}}\big(c_{r,K-d}|C| + \mathrm{Rem}_{K,d}(C) - E_{a_K}(C)\big)\Big],$$

and one step, $n=K-d$ with $d\ge i$, gives $\lambda_n = L^{-\kappa}\exp[-\frac{3}{2\mathsf{b}}((c_{r,n-1}-c_{r,n})|C| + \mathrm{Rem}_{K,d+1} - \mathrm{Rem}_{K,d})]$: each step takes away $\kappa\log L$ and a multiple of the length; the steps sum to $\sum_{n\ge1}(c_{r,n-1}-c_{r,n}) = -c_r = c_{\mathrm{lat}} - c_Q$, every term depending on L and the sum not. For families, $c_\Xi^{-1}L^{\kappa d} \le a_K^{-\kappa}\Xi_{K-d} \le c_\Xi L^{\kappa d}$ for $i_0\le d\le K$, with c_Ξ free of d and K and uniform for b in a compact set, and $\mathsf{I}_d = \lim_K a_K^{-\kappa}\Xi_{K-d} = a_d^{-\kappa}\exp[-\frac{3}{2\mathsf{b}}((c_{\mathrm{lat}}-c_Q)|\mathsf{C}| + O_{\mathsf{C},\ell}(a_d(1+\log^+(1/a_d))))]$, so $a_d^\kappa \mathsf{I}_d \to e^{-0.047235\ldots|\mathsf{C}|/\mathsf{b}}$. (d) For a rectangle C of area A with corners in Λ_K, in any position, and $0\le d\le K$: $0\le v_{K,d}(C)\le C^{\mathrm{bg}}A^2/a_d^3$, C^{bg} absolute. ■

Sketch of proof. (a) Fourier analysis of g_N: a component of the current couples only to the same component, so perpendicular pieces do not meet; the sum of g_N along an axis gives the logarithm and the constant c_{lat}, a computation with the lattice Green function on a line; the torus gives $\mathcal{J}_\ell$; distant pieces are compared with the continuum with an error $O(a^2)$, the corners with an error $O(a)$ each. The formula for the rectangle is the double integral of $\frac{1}{4\pi|x-y|}$ over its four sides, the parallel pieces interacting with the sign -1 and the perpendicular ones not at all. (b) The range of Π_{K-d} is spanned by the curvatures of the bare fields $\Delta^{-1}Q^{\mathsf{T}}\iota$, ι a closed current of Λ_d (the Lagrange multiplier of the linearised background problem); as $(X,\Pi X)$ is the largest value of $2(X,Y)-(Y,Y)$ over the range and $(X_{C'},\mathrm{d}A) = (j_{C'},A)$, $v_{K,d}$ is the stated maximum, at every depth and for every closed bare path.

Since the corners of the polygon lie in Λ_d for $d\ge i$, the Fourier transform of the sharp loop at the momenta $(q+2\pi n)/a_d$ is the transform $\hat\jmath(q)$ of its chain on Λ_d times the form factor of one bond; the quadratic form is diagonal in q and in the direction of the current, and the maximum is solved mode by mode. Without the constraint that ι be closed one gets $a_d N^{-3}\sum_{q\ne0}\sum_\mu |\hat\jmath_\mu(q)|^2 \mathsf{m}_\mu^{(K-d)}(q)$, $N=\ell/a_d$, where $\mathsf{m}_\mu^{(j)} = (\mathsf{N}_\mu^{(j)})^2/\mathsf{D}_\mu^{(j)}$ is the propagator that the block averages see: with $p=q+2\pi n$, $s_{(j)}(p) = \prod_\nu S_{(j)}(p_\nu)$ and $\Delta_{(j)}(p) = \sum_\nu E_{(j)}(p_\nu)$, $\mathsf{N}_\mu^{(j)}$ sums, over the aliases (n in the L^{3j} classes $|n_\nu|\le\frac{L^j-1}{2}$ at a finite cutoff, $j=K-d$, and in $\mathbb{Z}^3$ at $K=\infty$, where $S_{(\infty)}=S$ and $E_{(\infty)}(z)=z^2$), the bare propagator $1/\Delta_{(j)}(p)$ times the form factors $S_{(j)}(p_\mu)$ of the sharp bond and $S_{(j)}(p_\mu)s_{(j)}(p)$ of the smeared bond, and $\mathsf{D}_\mu^{(j)}$ the same with the smeared bond twice; at $q_\mu=0$ only $n_\mu=0$ survives, and they reduce to the sums N, D of $c_{r,j}$ (of c_r at $K=\infty$).

The constraint costs a non-negative term at most $cn_C^2a_d$: with $\partial_\mu(q) = e^{iq_\mu}-1$ and $\langle z,z'\rangle = \sum_\mu \bar z_\mu z'_\mu$, $\langle\partial(q),\hat\jmath(q)\rangle = 0$ for a closed C (the sum over the steps telescopes), $|\bar\partial_\mu\hat\jmath_\mu|$ is at most twice the number of pieces in the direction μ, $\mathsf{N}_\mu^{(j)}/\mathsf{D}_\mu^{(j)}$ differs from $1/s_{(j)}(q)$, the inverse form factor of the block, which does not depend on μ, by $O(|q|^2)$, and $\sum_\mu|\partial_\mu|^2/\mathsf{D}_\mu^{(j)} \ge c|q|^4$. With $\Delta_1(q) = \sum_\nu 4\sin^2\frac{q_\nu}{2}$, the symbol $\mathsf{m}_\mu^{(j)} - 1/\Delta_1$ is bounded with derivatives at most $c|q|^{-|\alpha|}$, c free of j; its kernel falls like $|x|^{-2}$; along one straight piece of n bonds its sum gives $c_{r,j}$ per unit length (the sum along an axis is the integral over the plane $q_\mu=0$, whence the two-dimensional formula) up to $O(a_d(1+\log n))$, and parallel pieces at distance $\ge\delta_C$ interact through it by $O(a_d|C|^2/\delta_C^2)$.

With (a) at the spacing a_d this gives the law at every finite cutoff, with a remainder free of K; $c_{r,j}\to c_r$ by dominated convergence, and $|c_{r,j}|\le c_3$ by the bound on the symbol. (c)

$\sum_{n\le K-d}\mathcal{C}_{(n)} = \Pi_0 - \Pi_{K-d}$ at every d, and $a_K(X_C,\Pi_0X_C) = \mathcal{I}_{a_K}(C,C)$; insert (a) and (b): $\frac{3}{2\mathsf{b}}\frac{|\bar C|}{2\pi}\log\frac{a_d}{a_K} = \kappa\log\frac{a_d}{a_K}$; one step is $a_K(X_C,\mathcal{C}_{(K-d)}X_C) = v_{K,d+1} - v_{K,d}$.

For families the factors multiply, and one may take $c_\Xi = \prod_i \exp[\frac{3}{2\mathsf{b}}(c_3|C_i| + 2\mathsf{r}_i)]$, r_i the supremum over $d \ge i_0$ of the bound on $|\mathrm{Rem}_{K,d}(C_i)|$ in (b), which also bounds $|E_{a_K}(C_i)|$ for $K \ge d$. (d) $v_{K,d} \ge 0$, Π_{K-d} being an orthogonal projection. It vanishes on the 2-cochains Y with $\mathrm{d}^\mathsf{T}Y = 0$, as its range consists of exact cochains; so $v_{K,d}(C) = a_K(X,\Pi_{K-d}X)$ for every 2-cochain X with $\mathrm{d}^\mathsf{T}X = j_C$, in particular for X the indicator of the bare plaquettes of the rectangle, with $\|X\|_1 = \sum_p |X(p)| = A/a_K^2$. The maximum of (b) holds at every depth $0 \le d \le K$ for this bare current, wherever the corners lie (corners in Λ_d are needed only to factor $\hat\jmath$ in the proof of (b)).

Drop the constraint that ι be closed: the maximum can only grow. With $\mathfrak{q} = a_d/a_K = L^{K-d}$, the bare momenta $\varkappa = (q + 2\pi n)/\mathfrak{q}$, $q \in (2\pi/N)\mathbb{Z}^3$, $|n_\nu| \le \frac{\mathfrak{q}-1}{2}$, and $p = \mathfrak{q}\varkappa$, the transform of $Q^\mathsf{T}\iota$ at $\varkappa$ is $\hat\iota_\mu(q)\varphi_\mu(\varkappa)$, $|\varphi_\mu(\varkappa)| = \mathfrak{q}|S_{(K-d)}(p_\mu)s_{(K-d)}(p)|$ the form factor of the smeared bond, and $\Delta_1(\varkappa) = \mathfrak{q}^{-2}\Delta_{(K-d)}(p)$; the maximum over all ι, taken mode by mode, gives

$$v_{K,d}(C) \le a_K(N\mathfrak{q})^{-3}\sum_{q\ne0}\sum_\mu \frac{|\mathcal{B}_\mu(q)|^2}{\mathfrak{q}^4\mathsf{D}_\mu^{(K-d)}(q)}, \qquad \mathcal{B}_\mu(q) = \sum_n \frac{\hat\jmath_\mu(\varkappa)\,\overline{\varphi_\mu(\varkappa)}}{\Delta_1(\varkappa)},$$

$\hat\jmath$ the transform of the bare current j_C. Three bounds hold uniformly in $\mathfrak{q}$: $|\hat\jmath_\mu(\varkappa)| \le \Delta_1(\varkappa)^{1/2}\|X\|_1$, as $j_C = \mathrm{d}^\mathsf{T}X$ and $\sum_\nu |e^{i\varkappa_\nu} - 1|^2 = \Delta_1(\varkappa)$; $\sum_n |S_{(K-d)}(p_\mu)s_{(K-d)}(p)|\Delta_{(K-d)}(p)^{-1/2} \le c/|q|$, the term $n = 0$ being at most $(0.3|q|)^{-1}$ and, for $n \ne 0$, each one-dimensional factor at most $51/\max\{1,|n_\nu|\}$ and $\Delta_{(K-d)}(p) \ge 0.2|n|^2$, so that the terms $n \ne 0$ add up to at most $c\sum_n\prod_\nu \max\{1,|n_\nu|\}^{-4/3}$; and $\mathsf{D}_\mu^{(K-d)}(q) \ge 0.3^8|q|^{-2}$ ($S_{(j)} \ge 0.3$ on $[-\frac32\pi,\frac32\pi]$ and $E_{(j)}(z) \le z^2$). So $|\mathcal{B}_\mu(q)| \le c\,\mathfrak{q}^2\|X\|_1/|q|$, each of the N^3 modes contributes at most $c\|X\|_1^2$, and $v_{K,d}(C) \le c\,a_K\mathfrak{q}^{-3}\|X\|_1^2 = c\,A^2/a_d^3$, as $a_K\mathfrak{q} = a_d$: C^{bg} is absolute, for every odd $L \ge 3$, every $0 \le d \le K$ and every position. Equivalently, the kernel of Π_{K-d} on the bare plaquettes is at most $c\,(a_K/a_d)^3$ in modulus: at one plaquette this is the bound on the diagonal, and Schwarz' inequality gives the rest. □

Proposition 10.8 (The tower)**.** In the frame let $F = W_\mathsf{C}$ and let $0 \le \bar t \le K$ be the last level of the tower. (a) Suppose $|\int W_\mathsf{C}^{(t)}(V_W(B))\,\nu_W(\,\mathrm{d}B) - \lambda_{t+1}W_\mathsf{C}^{(t+1)}(W)| \le \eta_{t+1}$ for m_{t+1}-almost every W and $0 \le t < \bar t$, and put $\mathsf{e}_t = \sum_{s\le t}\Xi_{s-1}\eta_s$. Then $|\Phi^{(t)}[F] - \Xi_tW_\mathsf{C}^{(t)}| \le \mathsf{e}_t$ m_t-almost everywhere for $t \le \bar t$, and

$$\Big|\langle F\rangle_K - \Xi_{\bar t}\frac{1}{Z}\int W_\mathsf{C}^{(\bar t)}\,\mathrm{d}m_{\bar t}\Big| \le \mathsf{e}_{\bar t} + \sum_{t\le\bar t}(\Xi_{t-1} + \mathsf{e}_{t-1})\mathsf{w}_t;$$

the same holds with a further factor $H(\mathcal{M}^{K-s_1}U)$ in $\langle F\rangle_K$ and $H\circ\mathcal{M}^{K-\bar t-s_1}$ in the integral, for H bounded Borel gauge invariant on $\mathcal{V}_{s_1}$ and $K - \bar t \ge s_1$, with $\sup|H|$ multiplying the right side. No positivity of ν_W is used. (b) Without any hypothesis, let $\mathsf{A}_t = Z^{-1}\int W_\mathsf{C}^{(t)}\,\mathrm{d}m_t$ and $\bar{\mathsf{D}}_t = Z^{-1}\int[\int W_\mathsf{C}^{(t-1)}(V_W(B))\,\nu_W(\,\mathrm{d}B) - \lambda_tW_\mathsf{C}^{(t)}(W)]\,m_t(\,\mathrm{d}W)$, the mean of the defect of the step over the block field. Then $\mathsf{A}_{t-1} = \lambda_t\mathsf{A}_t + \bar{\mathsf{D}}_t + \omega_t$ with $|\omega_t| \le \mathsf{w}_t$, and so

$$\langle W_\mathsf{C}\rangle_K = \Xi_{\bar t}\mathsf{A}_{\bar t} + \sum_{t\le\bar t}\Xi_{t-1}(\bar{\mathsf{D}}_t + \omega_t)\,:$$

as an identity between numbers, only the means of the defects over the block field enter. (c) On gauge invariant Borel sets $\mu_{K-\bar t}^K - Z^{-1}m_{\bar t}$ is a positive measure of mass $\sum_{t\le\bar t}\mathsf{w}_t$, and $1 - 2\mathsf{V}_\ell C_\mathrm{r}(1)L^{-(K-\bar t)} \le Z^{-1}m_{\bar t}(\mathcal{V}_{K-\bar t}) \le 1$. (d) For a coordinate rectangle C with sides s_1, s_2: $1 - \frac12\theta_d^2 \le W_C^{(K-d)} \le 1$ on O_{K-d}, with $\theta_d = 2B_3\varepsilon_{K-d}L^{2d}s_1s_2$. ■

Sketch of proof. (a) For m_{t+1}-almost every W,

$$\Phi^{(t+1)}[F](W) - \Xi_{t+1} W_{\mathsf{C}}^{(t+1)}(W) = \int [\Phi^{(t)}[F] - \Xi_t W_{\mathsf{C}}^{(t)}](V_W(B))\, \nu_W(\,\mathrm{d}B) + \Xi_t \Big[\int W_{\mathsf{C}}^{(t)}(V_W(B))\, \nu_W(\,\mathrm{d}B) - \lambda_{t+1} W_{\mathsf{C}}^{(t+1)}(W) \Big].$$

The second member is at most $\Xi_t \eta_{t+1}$ almost everywhere by hypothesis. The first, $\tilde\Psi(W) := \int \Psi_t(V_W(B))\, \nu_W(\,\mathrm{d}B)$ with $\Psi_t := \Phi^{(t)}[F] - \Xi_t W_{\mathsf{C}}^{(t)}$, is at most e_t in modulus for m_{t+1}-almost every W whenever $|\Psi_t| \le \mathsf{e}_t$ m_t-almost everywhere; this duality step reads only $m_t, m_{t+1}, \mathsf{h}_{t+1} \ge 0$. The disintegration at $G = \Psi_t$, a bounded Borel function, and the transport identity $\int G'\,\mathrm{d}m_{t+1} = \int (G' \circ \mathcal{M})\, \mathsf{h}_{t+1}\,\mathrm{d}m_t$ at $G' = |H|$ give, for every bounded Borel H,

$$\Big| \int H \tilde\Psi\,\mathrm{d}m_{t+1} \Big| = \Big| \int (H \circ \mathcal{M})\, \Psi_t\, \mathsf{h}_{t+1}\,\mathrm{d}m_t \Big| \le \mathsf{e}_t \int |H \circ \mathcal{M}|\, \mathsf{h}_{t+1}\,\mathrm{d}m_t = \mathsf{e}_t \int |H|\,\mathrm{d}m_{t+1};$$

at $H = \overline{\operatorname{sgn}\tilde\Psi}\, \mathbf{1}\{|\tilde\Psi| > \mathsf{e}_t + \epsilon\}$ this says $(\mathsf{e}_t + \epsilon)\, m_{t+1}(|\tilde\Psi| > \mathsf{e}_t + \epsilon) \le \mathsf{e}_t\, m_{t+1}(|\tilde\Psi| > \mathsf{e}_t + \epsilon)$, so the set has measure 0, for every $\epsilon > 0$. No law of $V_W(B)$, no null set of m_t under ν_W and no positivity of ν_W enter. Induction on t, with $\mathsf{e}_{t+1} = \mathsf{e}_t + \Xi_t \eta_{t+1}$; then eq. 10.i with $|W_{\mathsf{C}}^{(t)}| \le 1$ and $|\Phi^{(t-1)}[F]| \le \Xi_{t-1} + \mathsf{e}_{t-1}$. With H: $\Phi^{(t)}[W_{\mathsf{C}}\, H \circ \mathcal{M}^{K-s_1}] = \Phi^{(t)}[W_{\mathsf{C}}]\, H \circ \mathcal{M}^{K-t-s_1}$ for $K - t \ge s_1$. (b) For gauge invariant bounded Borel G on $\mathcal{V}_{K-t+1}$, eq. 10.i at $F = G \circ \mathcal{M}^{t-1}$ for $\bar t = t-1$ and $\bar t = t$, with $\Phi^{(t-1)}[G \circ \mathcal{M}^{t-1}] = G$ (functions pull out and $\Phi^{(t-1)}[1] = 1$), gives $\int G\,\mathrm{d}m_{t-1} = \int G\,(\mathsf{h}_t + \mathsf{h}_t^{\mathrm{w}})\,\mathrm{d}m_{t-1}$.

At $G = W_{\mathsf{C}}^{(t-1)}$ the disintegration with $H = 1$ turns $Z^{-1} \int W_{\mathsf{C}}^{(t-1)} \mathsf{h}_t\,\mathrm{d}m_{t-1}$ into $\lambda_t \mathsf{A}_t + \bar{\mathsf{D}}_t$, and $\omega_t = Z^{-1} \int W_{\mathsf{C}}^{(t-1)} \mathsf{h}_t^{\mathrm{w}}\,\mathrm{d}m_{t-1}$; $\mathsf{A}_0 = \langle W_{\mathsf{C}} \rangle_K$. (c) eq. 10.i at $F = \mathbf{1}_E \circ \mathcal{M}^{\bar t}$, E gauge invariant: $Z \mu_{K-\bar t}^K(E) - m_{\bar t}(E) = \sum_{t \le \bar t} \int (\mathbf{1}_E \circ \mathcal{M}^{\bar t - t+1}) \mathsf{h}_t^{\mathrm{w}}\,\mathrm{d}m_{t-1} \ge 0$, of total mass $Z \sum_{t \le \bar t} \mathsf{w}_t$; the rarity bound at $p = 0$ in its summed form. (d) Over a tame field the background has bare plaquettes $< 2B_3 \varepsilon_{K-d} (a_K/a_d)^2$ (Thm. YM3_I:E in the setting of Section YM3_I:52.7), and $1 - W_{\partial R} \le (\sum_{p \subset R} \sqrt{a_p})^2$ over the $s_1 s_2 / a_K^2$ plaquettes of the rectangle (Prop. 10.9(c)). □

Proposition 10.9 (Identities and inequalities at every cutoff)**.** For every $\beta > 0$, every torus and every cutoff: (a) $\mathcal{W}_{K-z}^{(L^z \ell, L^z \mathsf{b})}(L^z \mathsf{C}; f^{(z)}) = L^{-z\kappa - zn} \mathcal{W}_K^{(\ell, \mathsf{b})}(\mathsf{C}; f_1, \dots, f_n)$, $f_q^{(z)} = f_q(L^{-z}\cdot)$; (b) if families $G_1, \dots, G_s$ lie in the interior of S_θ^+ for a coordinate mirror θ and the supports of the tuples f_i lie in S_θ^+ (and do not meet the fixed planes if θ is a bond mirror), then $\sum_{i,j} \bar c_i c_j \mathcal{W}_K(\theta G_i, G_j; \theta f_i, f_j) \ge 0$ for all complex c_i; so $\mathcal{W}_K(G_1, \theta G_2)^2 \le \mathcal{W}_K(G_1, \theta G_1)\, \mathcal{W}_K(G_2, \theta G_2)$ for real entries; if θ cuts a loop C_{12} into two halves and C_{11}, C_{22} are the loops made of each half and its mirror image, then $\mathcal{W}_K(C_{12})^2 \le \mathcal{W}_K(C_{11})\, \mathcal{W}_K(C_{22})$; (c) the plaquette mean $\langle W_{\partial p} \rangle_K$ is > 0; for bare rectangles $p \times q$ with $p, q < \ell/2a_K$, $w(p,q) = \langle W_{p \times q} \rangle_K \ge w(1,1)^{pq} > 0$, and $\log w$ is convex in p and in q; and $0 \le 1 - W_{\partial R} \le (\sum_{p \subset R} \sqrt{a_p})^2$ for every field and every rectangle R. For the rectangles $s \times T$ of one coordinate plane, s and T lengths in $\bigcup_i a_i \mathbb{Z}$, put $\mathcal{W}(s \times T) := \lim_K \mathcal{W}_K(s \times T) = \lim_K a_K^{-\kappa} \langle W_{s \times T} \rangle_K$, $\kappa = \kappa(s \times T)$, wherever the limit exists. A *range* of the limits is a set of such rectangles on which they exist and are finite, closed downwards (with $s \times T$ it holds every $s' \times T'$ with $s' \le s$, $T' \le T$) and open upwards (with $s \times T$ it holds a rectangle with both sides larger); in the frame, and by dilation on every torus and at every label, the polygons $s \times T$ with $\kappa < \frac12$ form one (Thm. AE(1)–(3)). Every limit inherits (a), (b), $\mathcal{W} \ge 0$ and the midpoint inequality $\mathcal{W}(s \times \frac{T'+T''}{2})^2 \le \mathcal{W}(s \times T')\, \mathcal{W}(s \times T'')$, and the same in s, whenever the three rectangles lie in a range, the powers of a_K matching because $2|s \times \frac{T'+T''}{2}| = |s \times T'| + |s \times T''|$; a limit that is > 0 at one rectangle of a range is > 0 at all rectangles of that range. Strict positivity of a limit is not inherited; it comes from the floor of Thm. AE(4). ■

Sketch of proof. (a) The Wilson measure at (K, ℓ, b) is the one at $(K - z, L^z \ell, L^z \mathsf{b})$ relabelled, as $\beta = L^K \mathsf{b} = L^{K-z}(L^z \mathsf{b})$: lengths are multiplied by L^z, κ is unchanged, $a_K^{-\kappa}$ gains $L^{-z\kappa}$ and each $\mathcal{O}(f)$ the factor L^{-z}. (b) Reflection positivity of the Wilson measure at site and bond mirrors,

for functions of the bonds of one closed (resp. open) half torus, and Schwarz' inequality: each $a_K^{-\kappa(G_i)} W_{G_i} \prod_q (\mathcal{O}(f_{iq}) - \langle \mathcal{O}(f_{iq})\rangle_K)$ lives on one side of θ, and the powers of a_K split because lengths add. For the cut loop, $W_{C_{12}} = \frac{1}{2}\operatorname{tr}(U(\gamma_1)U(\gamma_2))$ with the two halves γ_1, γ_2, and reflection positivity for matrix-valued functions with Schwarz' inequality gives the bound; $|C_{11}| + |C_{22}| = 2|C_{12}|$. (c) $s \mapsto \log \int e^{s \sum_p \frac{1}{2}\operatorname{tr} \partial U(p)}\, \mathrm{d}U$ has the derivative 0 at $s = 0$ (the Haar mean of a non-trivial character vanishes) and is strictly convex, so its derivative at $s = \beta$, the number of plaquettes times the plaquette mean, is > 0.

Reflection positivity in the two directions of the plane of the rectangle gives $w(p,q)^2 \le w(p,q-1)\, w(p,q+1)$ and the same in p; with $w(p,0) = 1$ the ratios $w(p,q)/w(p,q-1)$ increase, so $w(p,q) \ge w(p,1)^q \ge w(1,1)^{pq}$. For $g \in \mathrm{SU}(2)$, $1 - \frac{1}{2}\operatorname{tr} g = \frac{1}{2}|g - \mathbf{1}|^2$; the holonomy of ∂R is an ordered product of conjugated plaquette variables, and $|\prod_i g_i - \mathbf{1}| \le \sum_i |g_i - \mathbf{1}|$. Limits keep ≥ 0, the positive forms and the inequalities; the midpoint inequality of $\mathcal{W}_K$ is the convexity of $\log w$ in each side, multiplied by $a_K^{-2\kappa(s\times\frac{T'+T''}{2})} = a_K^{-\kappa(s\times T')-\kappa(s\times T'')}$, at every cutoff at which the three rectangles have their corners in Λ_K.

Spreading in a range R, the limits being finite there (Thm. AE(1)): if $\mathcal{W}(s \times T_0) > 0$ and $s \times (T_0 \pm \epsilon) \in \mathsf{R}$, the product $\mathcal{W}(s \times (T_0 - \epsilon))\, \mathcal{W}(s \times (T_0 + \epsilon)) \ge \mathcal{W}(s \times T_0)^2 > 0$ of two finite numbers ≥ 0 makes both positive; going down from the lowest positive point with an upper point of R fixed, the distance to it doubles at each step, so that every T below T_0 is reached, and upwards in the same way with a lower point; then the same in s; and two rectangles of R are joined through the one with the smaller sides, which lies in R. The values of ϵ are taken in $\bigcup_i a_i\mathbb{Z}$. □

Proposition 10.10 (A source off the loop)**.** In the frame let $f_1, \dots, f_n$ be real Lipschitz, $\|f\|_\sharp = \max\{\sup|f|, 40M \operatorname{Lip} f\}$, and z in the polydisc $\sum_q |z_q| \|f_q\|_\sharp < \rho_D$, where $\rho_D = 2\varrho_S/(2e^{(C_S+C_g)\ell^3}+2)$ is formed from the radius ϱ_S and the constants C_S, C_g of Thms. YM3_I:AJ and YM3_I:AD; ρ_D depends on neither K nor $\mathsf{b} \in \mathsf{I}(g)$. Write $w_z = \sum_q z_q f_q$. The radius estimates below are stated for a family with $\max_q \|f_q\|_\sharp > 0$. If every f_q vanishes, the source is zero, its derivatives of positive order vanish, and the assertions reduce directly to the zero-source statements. (a) $\langle e^{\mathcal{O}(w_z)}\rangle_K = e^{\mathcal{E}(w_z)} h_K(z)$ with $\frac{1}{2} \le |h_K(z)| \le \frac{3}{2}$, whatever the supports, $\mathcal{E}(w)$ a sum of local energies of the source; the energies of sources $v_i = \sum_q z_q^{(i)} f_q^{(i)}$, $i = 1, \dots, s$, $f_q^{(i)}$ real Lipschitz, with $\sum_i \sum_q |z_q^{(i)}| \|f_q^{(i)}\|_\sharp < \rho_D$ and supports at pairwise sup-distance $\ge 2\delta$, add up to an error: $|\mathcal{E}(\sum_i v_i) - \sum_i \mathcal{E}(v_i)| \le \Delta_s(\delta) = c(s+1)(\ell/2\delta)^3$, c formed from the constants of Thms. YM3_I:AJ and YM3_I:AD, free of K; so, with $\mathsf{e}(v) = |\langle e^{\mathcal{O}(v)}\rangle_K|$, $\frac{1}{2}(\frac{2}{3})^s e^{-\Delta_s(\delta)} \prod_i \mathsf{e}(v_i) \le \mathsf{e}(\sum_i v_i) \le \frac{3}{2}\, 2^s e^{\Delta_s(\delta)} \prod_i \mathsf{e}(v_i)$; we write $\Delta(\delta) = \Delta_2(\delta)$. In particular $\langle e^{\mathcal{O}(w_z)}\rangle_K \ne 0$ on the polydisc, for all K. (b) The objects of the frame exist with the weight $e^{\mathcal{O}(w)}$. For complex bounded w and bounded Borel F on $\mathcal{V}_K$ let $m_t^w[F] = \Phi^{(t)}[F e^{\mathcal{O}(w)}]\, m_t$, a complex measure on $\mathcal{V}_{K-t}$, and $m_t^w = m_t^w[1]$. Then $\int G\, \mathrm{d}m_t^w[F] = \int (G \circ \mathcal{M})\, \mathsf{h}_t\, \mathrm{d}m_{t-1}^w[F]$ for bounded Borel G, and, for gauge invariant F and G and $0 \le \bar{t} \le K$, the sum over the level of the first creation of a large field

$$Z\big\langle F\, G(\mathcal{M}^{\bar{t}} U)\, e^{\mathcal{O}(w)}\big\rangle_K = \int G\, \mathrm{d}m_{\bar{t}}^w[F] + \sum_{t \le \bar{t}} \int (G \circ \mathcal{M}^{\bar{t}-t+1})\, \mathsf{h}_t^{\mathrm{w}}\, \mathrm{d}m_{t-1}^w[F].$$

At real w only, $\Phi^{(t)}[e^{\mathcal{O}(w)}] > 0$ m_t-almost everywhere, and the quotients

$$\Phi^{(t),w}[F] = \Phi^{(t)}[F e^{\mathcal{O}(w)}]/\Phi^{(t)}[e^{\mathcal{O}(w)}],$$

with $|\Phi^{(t),w}[F]| \le \sup|F|$, and the kernel

$$\nu_W^w(dB) = \Phi^{(t)}[e^{\mathcal{O}(w)}](V_W(B))\, \nu_W(dB)/\Phi^{(t+1)}[e^{\mathcal{O}(w)}](W),$$

a probability measure for m_{t+1}-almost every W, obey eq. 10.i and the disintegration for (m^w, ν^w), with $Z\langle e^{\mathcal{O}(w)}\rangle_K$ in the place of Z; m_t^w is then a positive measure with the null sets of m_t, and on gauge invariant sets $\mu_{K-\bar t}^{K,w} - m_{\bar t}^w/(Z\langle e^{\mathcal{O}(w)}\rangle_K)$ is a positive measure of mass $\sum_{t\le\bar t}\mathsf{w}_t^w$, $\mathsf{w}_t^w = (Z\langle e^{\mathcal{O}(w)}\rangle_K)^{-1}\int \mathsf{h}_t^{\mathrm{w}}\,\mathrm{d}m_{t-1}^w$, $\mu_d^{K,w}$ being the law of $\mathcal{M}^{K-d}U$ under $e^{\mathcal{O}(w)}\langle\cdot\rangle_K/\langle e^{\mathcal{O}(w)}\rangle_K$.

The rarity bound holds with a source, at complex z for the integral of $\mathsf{h}_t^{\mathrm{w}}$:

$$|Z\langle e^{\mathcal{O}(w)}\rangle_K|^{-1}\Big|\int \mathsf{h}_t^{\mathrm{w}}\,\mathrm{d}m_{t-1}^w\Big| \le \mathsf{V}_S C_{\mathrm{r}}^S(p) L^{-p(K-t)},$$

with V_S, $C_{\mathrm{r}}^S(p)$ formed from the constants of Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD, p and ℓ; at real z this bounds w_t^w, and $\sum_{t\le\bar t} L^{\kappa(K-t)}\mathsf{w}_t^w \le 2\mathsf{V}_S C_{\mathrm{r}}^S(\kappa+1)L^{-(K-\bar t)}$. (c) For bounded Borel gauge invariant F on $\mathcal{V}_d$,

$$\mu_d^{K,w_z}(F) = \langle F(\mathcal{M}^{K-d}U)e^{\mathcal{O}(w_z)}\rangle_K/\langle e^{\mathcal{O}(w_z)}\rangle_K$$

obeys $|\mu_d^{K,w_z}(F)| \le B_d \sup|F|$, with B_d formed from the constants of Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD, d, b and ℓ, and free of K. The limit $\mu_d^{w_z}(F) = \lim_K \mu_d^{K,w_z}(F)$ exists for z in the polydisc $|z_q| < \rho_* := \rho_D/(n\max_q \|f_q\|_\sharp)$, which lies in the polydisc of the head and is free of K, uniformly on its half $|z_q| \le \frac12\rho_*$ and in $\sup|F| \le 1$; it is holomorphic in z, a probability measure at real z, and $\mu_{d+1}^w(\mathcal{M}^{-1}E) = \mu_d^w(E)$. No condition on the supports is asked; so the derivatives at $z=0$, the mixed functions of $F(\mathcal{M}^{K-d}U)$ and of the $\mathcal{O}(f_q)$, converge too. (d) If a coordinate mirror θ parts C, at distance $\ge\delta_1$ from its planes, from the supports, at distance $\ge\delta$, and z lies in half the polydisc, then for every bounded function X of the bonds on the side of C

$$|\langle X e^{\mathcal{O}(w_z)}\rangle_K| \le M_0\|X\|_\theta\,|\langle e^{\mathcal{O}(w_z)}\rangle_K|, \qquad \|X\|_\theta^2 = \langle X\,\overline{\theta X}\rangle_K \ge 0, \qquad M_0 = \sqrt{6}\,e^{\Delta(\delta)/2};$$

so $|Q_K(z)| \le M_0\,\mathcal{W}_K(\mathsf{C},\theta\mathsf{C})^{1/2}$. For two orthogonal mirrors, the sources in two groups as in Thm. AE(6), and z in a quarter of the polydisc, $\sum_q |z_q|\|f_q\|_\sharp < \rho_D/4$ (for one mirror, half the polydisc is $\sum_q |z_q|\|f_q\|_\sharp < \rho_D/2$),

$$|\langle X e^{\mathcal{O}(w_z)}\rangle_K|^4 \le M_0'^4\langle X\,\overline{\theta_1 X}\,\overline{\theta_2 X}\,\theta_1\theta_2 X\rangle_K\,|\langle e^{\mathcal{O}(w_z)}\rangle_K|^4$$

for X on the side of C of both mirrors, $M_0'^4 = 24\cdot 36\cdot(\frac92)^4 e^{\Delta_4(\delta)+6\Delta_2(\delta)}$, the mean on the right holding no source and being ≥ 0. Where equally spaced coordinate walls part the polygons from the supports as in Thm. AE(6), with an even number of boxes in each direction that the walls cut, so that both fixed planes c and $c+\frac{\ell}{2}$ of every wall mirror are walls (N boxes in all; on $\ell = 2L^m$, with walls in $\frac12 a_i\mathbb{Z}$, the number of slabs in a cut direction divides $4L^{m+i}$, and it is taken even, $2L^j$ or $4L^j$), and with each f_q supported in one box: $\langle W_{\mathsf{C}_\iota^{\mathrm{f}}}\rangle_K \ge 0$, $\kappa(\mathsf{C}_\iota^{\mathrm{f}}) = N\kappa(\mathsf{C}_\iota)$, and

$$|Q_K(z)| \le M_S\prod_\iota \mathcal{W}_K(\mathsf{C}_\iota^{\mathrm{f}})^{1/N} \quad \text{for } \sum_q |z_q|\|f_q\|_\sharp < \rho_D/N, \qquad M_S = 2(\tfrac32)^{n_{\mathrm{b}}}e^{\Delta_{n_{\mathrm{b}}}(\delta)}\big[\tfrac32\,2^N e^{\Delta_N(\delta)}\big]^{n_{\mathrm{b}}/N},$$

n_{b} the number of boxes that meet the supports and δ the distance of the supports to the walls; a function whose support misses the walls is the sum of such pieces, one for each box. (e) $\mathcal{W}_K(\mathsf{C};f_1,\dots,f_n) = \sum_A \partial_A Q_K(0)\,\partial_{A^c}E_K(0)$, $E_K(z) = \langle\exp\sum_q z_q(\mathcal{O}(f_q) - \langle\mathcal{O}(f_q)\rangle_K)\rangle_K$, and $\partial_{A^c}E_K(0) = \sum_\pi\prod_{\beta\in\pi} S^{\mathsf{T}}_{|\beta|,K}(f_\beta)$, π over the partitions of A^c into blocks of at least two elements: only the derivatives of Q_K in pairwise different variables enter. (f) Let $0<\vartheta'\le 1$, $\bar t(K) = K - \lceil\vartheta' K\rceil$, $\eta = (24\ell^3\sum_q \sup|f_q|)^{-1}$, and let $q_{K,\bar t}(z) = a_K^{-\kappa}\int\Phi^{(\bar t)}[W_{\mathsf{C}}e^{\mathcal{O}(w_z)}]\,\mathrm{d}m_{\bar t}/\int\Phi^{(\bar t)}[e^{\mathcal{O}(w_z)}]\,\mathrm{d}m_{\bar t}$ be the same quotient formed with the measure $m_{\bar t}$ without large fields. For every multi-index α, with $p = (4|\alpha|+\kappa+1)/\vartheta'$, and all K with $2\mathsf{V}_\ell C_{\mathrm{r}}(p)L^{-p\vartheta' K} \le 0.16$: $|\partial^\alpha Q_K(0) - \partial^\alpha q_{K,\bar t(K)}(0)| \le 14\,\alpha!\,\eta^{-|\alpha|}\mathsf{V}_\ell C_{\mathrm{r}}(p)L^{-K}$: the Taylor coefficients of Q_K at 0 are those of the chain without large fields. ∎

Sketch of proof. (a) Thms. YM3_I:AJ and YM3_I:AD for the Wilson measure with the source w_z writes $\langle e^{\mathcal{O}(w_z)}\rangle_K$ as $e^{\mathcal{E}(w_z)}h$, $\mathcal{E}(w)=\sum_{j\le K}\sum_X e_j(X;w)$ a sum of local terms $e_j(X;w)$ over the levels j and the domains X of the level j, which read w near X alone and vanish at $w=0$, with $|e_j(X;w)|\le C_S e^{-\kappa_S d_j(X)}$, κ_S the decay number of Thms. YM3_I:AJ and YM3_I:AD and $d_j(X)$ the size of X in cubes of side M of the level j; and h holomorphic with $h(0)=1$ and $|h|\le e^{(C_S+C_g)\ell^3}$. With $\|z\|=\sum_q|z_q|\|f_q\|_\sharp$, Schwarz' lemma for $\zeta\mapsto h(w_{\zeta z})-1$ on $|\zeta|<2\varrho_S/\|z\|$ gives $|h-1|\le\frac12$ on the polydisc.

In $\mathcal{E}(\sum_i v_i)-\sum_i\mathcal{E}(v_i)$ a local term that sees one support alone cancels; one that sees two supports at sup-distance $\ge 2\delta$ belongs to a domain X with $d_j(X)\ge 2\delta L^{K-j}/M-4$ and is at most $(s+1)C_S e^{-\kappa_S d_j(X)}$; the sum over such domains and levels is at most $c(s+1)(\ell/2\delta)^3$. The bounds on $\mathsf{e}(\sum_i v_i)$ follow, as $\frac12 e^{\operatorname{Re}\mathcal{E}(v)}\le\mathsf{e}(v)\le\frac32 e^{\operatorname{Re}\mathcal{E}(v)}$ for every source v of the polydisc. (b) A history is linear in the bare density, so $\Phi^{(t)}[Fe^{\mathcal{O}(w)}]$ is the density of the history without large fields with the source, at complex w as well. The transport identity is that of the frame at $Fe^{\mathcal{O}(w)}$, and the sum over the first creation is eq. 10.i at the gauge invariant function $F\,(G\circ\mathcal{M}^{\bar t})\,e^{\mathcal{O}(w)}$, the factor $G\circ\mathcal{M}^{\bar t}$ pulling out of $\Phi^{(\bar t)}$ and of $\Phi^{(t-1)}$ as G and as $G\circ\mathcal{M}^{\bar t-t+1}$.

The quotients and the kernel are formed at real w: there $e^{\mathcal{O}(w)}\ge e^{-\sup|\mathcal{O}(w)|}>0$ gives, at every level t, $\Phi^{(t)}[e^{\mathcal{O}(w)}]\ge e^{-\sup|\mathcal{O}(w)|}$ off one m_t-null set $\mathcal{N}_t$, so m_t^w has the null sets of m_t. The kernel ν_W^w is ν_W times the density $B\mapsto\Phi^{(t)}[e^{\mathcal{O}(w)}](V_W(B))/\Phi^{(t+1)}[e^{\mathcal{O}(w)}](W)$, and it is a probability measure for m_{t+1}-almost every W: $\nu_W\ge0$ for m_{t+1}-almost every W (Prop. 10.6(b)); the denominator is $\ge e^{-\sup|\mathcal{O}(w)|}$ for $W\notin\mathcal{N}_{t+1}$; the numerator is $\ge e^{-\sup|\mathcal{O}(w)|}$ off $V_W^{-1}\mathcal{N}_t$, and $\nu_W(V_W^{-1}\mathcal{N}_t)=0$ for m_{t+1}-almost every W, the one set $\mathcal{N}_t$ being fixed first and the exceptional set of W depending on it: the disintegration at $G=\mathbf{1}_{\mathcal{N}_t}$ gives $\int H(W)\,\nu_W(V_W^{-1}\mathcal{N}_t)\,m_{t+1}(\,\mathrm{d}W)=\int(H\circ\mathcal{M})\,\mathbf{1}_{\mathcal{N}_t}\,\mathsf{h}_{t+1}\,\mathrm{d}m_t=0$ for every bounded Borel H; and the mass is 1 by the disintegration at $F=e^{\mathcal{O}(w)}$, $\Phi^{(t+1)}[e^{\mathcal{O}(w)}](W)=\int\Phi^{(t)}[e^{\mathcal{O}(w)}](V_W(B))\,\nu_W(\,\mathrm{d}B)$ for m_{t+1}-almost every W.

The disintegration for (m^w,ν^w) is that of the frame at $G\,\Phi^{(t)}[e^{\mathcal{O}(w)}]$. The wild mass is a sum over the histories of the level K in which a large field is created at the step t; Thms. YM3_I:AJ and YM3_I:AD bounds every history with a source as Thm. YM3_I:D bounds it without one, with the constants of Thms. YM3_I:AJ and YM3_I:AD, and (a) bounds the denominator below; the rarity bound then follows as in the square-completion estimate eq. 4.xxxv and the geometric-series calculation in the proof of Prop. 4.54, with the constants of Thms. YM3_I:AJ and YM3_I:AD. (c) Thms. YM3_I:AJ and YM3_I:AD at one depth bounds $|\langle F(\mathcal{M}^{K-d}U)e^{\mathcal{O}(w_z)}\rangle_K|$ by $\sup|F|$ times a number free of K times $|\langle e^{\mathcal{O}(w_z)}\rangle_K|$: this is B_d.

For the limit let $\sup|F|\le1$ and $G_K(z)=\mu_d^{K,w_z}(F)$, holomorphic and bounded by B_d on the polydisc $|z_q|<\rho_*$. We use an interpolation inequality: if G is holomorphic on the polydisc of radius ρ in n variables, $|G|\le B$ there and $|G|\le\sigma$ on the real cube $[-r,r]^n$, $0<r\le\rho/2$, then its Taylor coefficients at 0 obey $|c_\alpha(G)|\le C_{n,j}r^{-|\alpha|}(\sigma+2B(r/\rho)^{j+1})$ for $|\alpha|\le j$ (the Taylor polynomial of order j is within $2B(r/\rho)^{j+1}$ of G on the cube, by Cauchy's estimates on the complex lines through 0, and on the polynomials of order at most j in n variables the largest coefficient and the supremum on a cube are two norms).

It suffices that every coefficient $c_\alpha(G_K)$ converges, uniformly in F: Cauchy's estimate $|c_\alpha(G_K)|\le B_d\rho_*^{-|\alpha|}$ makes the tails of the Taylor series small on $|z_q|\le\frac12\rho_*$, uniformly in K and F. For $\alpha=0$ this is Cor. 2.8. Fix $j=|\alpha|\ge1$. The proof of the limit laws of the block fields compares two consecutive cutoffs through the small-field chains at the depth $u=\lceil\phi_1K\rceil$, $\phi_1=1/(180(3N'+4))$, N' the number of the test functions (§2.3.1); we prolong the tuple f by copies of f_1 to $N'=\max\{n,j^2\}$ functions, the new variables being 0, so that $u\ge d$ for large K. Let G_K^{sf} be the quotient of F formed with the weight of the history without large-field regions down to the depth u, and $D_K=G_K-G_K^{\mathrm{sf}}$; then $c_\alpha(G_{K+1})-c_\alpha(G_K)=c_\alpha(D_{K+1})-c_\alpha(D_K)+c_\alpha(G_{K+1}^{\mathrm{sf}}-G_K^{\mathrm{sf}})$, and the interpolation inequality is used twice, on two polydiscs. (i) On the polydisc of radius $\rho'=(24n\ell^3L^{4(K+1)}\max_q\sup|f_q|)^{-1}$,

on which $|\mathcal{O}(w_z)| \le \frac14$ at the cutoffs K and $K+1$ (there are $3(\ell/a_K)^3$ plaquettes and $0 \le a_p \le 2$), $|G_K|, |G_K^{\mathrm{sf}}| < 1.8$ and $|D_K| \le 3.6$; at its real points the measure without large-field regions is below the full one with a mass defect of the order of the rarity at the depth u, so that $|D_K| \le C\exp[3K\log L - cK^2]$, below every power of L^K; with $r = L^{-(4j+1)K}\rho'$ the coefficients of D_K are at most $C(L^{(4j^2+5j+3)K}e^{-cK^2} + L^{-K})$, summable in K. (ii) On the polydisc on which the proof of the limit laws bounds the small-field chains, of radii ρ'' with $1/\rho'' \le cK^{2\mathfrak{p}_0}L^{3u}$, so that $\rho''^{-j} \le (cK^{2\mathfrak{p}_0})^j L^{3j+K/(180j)}$ by $3\phi_1 j \le 1/(180j)$, which is where $N' \ge j^2$ enters, $|G_K^{\mathrm{sf}}| \le 1.8$; at its real points that proof goes through with the source, the contact terms of the source standing in a factor that drops out of the quotient, and gives $|G_{K+1}^{\mathrm{sf}} - G_K^{\mathrm{sf}}| \le cK^{2\mathfrak{p}_0+1/2}L^{-K/10}$; with $r = L^{-K/(90j)}\rho''$ the coefficients are at most a power of K times $L^{K(-\frac{1}{10}+\frac{1}{90}+\frac{1}{180j})} + L^{K(-\frac{1}{90j}+\frac{1}{180j})}$, summable in K, both exponents being negative.

So every $c_\alpha(G_K)$ converges with a summable rate, uniformly in $\sup|F| \le 1$, and G_K converges uniformly on $|z_q| \le \frac12\rho_*$; the limit is holomorphic, a complex measure of mass 1 in F (a limit, uniform in $\sup|F| \le 1$, of such measures), a probability measure at real z, where the $\mu_d^{K,w}$ are, and $\mathcal{M}$-compatible, as $\mu_{d+1}^{K,w}(\mathcal{M}^{-1}E) = \mu_d^{K,w}(E)$ at every K. (d) Reflection positivity at θ and Schwarz' inequality give $|\langle XH\rangle_K|^2 \le \|X\|_\theta^2\langle\overline{\theta H}H\rangle_K$ for H on the other side; at $H = e^{\mathcal{O}(w)}$, $\overline{\theta H}H = e^{\mathcal{O}(w+w^\theta)}$ with $w^\theta = \overline{\theta w} = \sum \bar z_q \theta f_q$.

The mean, not the energy, is invariant: $\langle e^{\mathcal{O}(w^\theta)}\rangle_K = \overline{\langle e^{\mathcal{O}(w)}\rangle_K}$, as θ is a symmetry of the Wilson measure and the a_p are real. By (a) for w, w^θ and $w+w^\theta$, which lie in the polydisc of the head as z lies in its half and whose supports lie 2δ apart, $\langle e^{\mathcal{O}(w+w^\theta)}\rangle_K \le \frac32 e^{\Delta}e^{\mathrm{Re}\,\mathcal{E}(w)+\mathrm{Re}\,\mathcal{E}(w^\theta)}$, and $e^{\mathrm{Re}\,\mathcal{E}(v)} \le 2|\langle e^{\mathcal{O}(v)}\rangle_K|$ at $v = w$ and $v = w^\theta$; so $\langle e^{\mathcal{O}(w+w^\theta)}\rangle_K \le 6e^{\Delta}|\langle e^{\mathcal{O}(w)}\rangle_K|^2$. At $X = W_{\mathsf{C}}$, $\|W_{\mathsf{C}}\|_\theta^2 = \langle W_{\mathsf{C}}W_{\theta\mathsf{C}}\rangle_K = a_K^{2\kappa}\mathcal{W}_K(\mathsf{C},\theta\mathsf{C})$.

Two mirrors: Schwarz at θ_1 with $Xe^{\mathcal{O}(w_2)}$ on the side of C, then at θ_2; the sources that arise hold four images of the second group and two of the first, pairwise 2δ apart, and together they stay in the polydisc of (a), $\sum_q |z_q|\|f_q\|_\sharp < \rho_D$ with each z_q counted as often as its images, as soon as z lies in a quarter of it; then (a) bounds the source factors. Walls: in a cut direction with an even number n of slabs the reflections in the walls map walls to walls, both fixed planes of each being walls, and generate the $n/2$ translations by even multiples of the width and $n/2$ reflections, a group of n elements that acts simply transitively on the slabs; so each box has exactly one image in every box, an image has a parity (the number of reflections modulo 2), and the folds below are defined. (With an odd number of slabs the translation by one width lies in the group, and so does the reflection of a box in its own mid-plane, which is no wall; the evenness excludes this.) The chessboard estimate $|\langle\prod_\iota F_\iota\rangle_K| \le \prod_\iota \langle \mathrm{Fold}\,F_\iota\rangle_K^{1/N}$ for functions F_ι of the bonds of the boxes, $\mathrm{Fold}\,F_\iota$ the product of the N images of F_ι under the group (conjugated at the odd ones), each $\langle\mathrm{Fold}\,F_\iota\rangle_K$ being ≥ 0, is applied with $F_\iota = W_{\mathsf{C}_\iota}$ for a box with polygons, $F_\iota = e^{\mathcal{O}(w_\iota)}$ for a box with a support, w_ι the part of w in it, and $F_\iota = 1$ otherwise. $\mathrm{Fold}\,W_{\mathsf{C}_\iota} = W_{\mathsf{C}_\iota^{\mathrm{f}}}$, with $\kappa(\mathsf{C}_\iota^{\mathrm{f}}) = N\kappa(\mathsf{C}_\iota)$, so that the powers of a_K split. $\mathrm{Fold}\,e^{\mathcal{O}(w_\iota)} = e^{\mathcal{O}(\sum_g w_\iota^g)}$, the N images w_ι^g lying in different boxes, pairwise 2δ apart, each with $\mathsf{e}(w_\iota^g) = \mathsf{e}(w_\iota)$ and together in the polydisc of (a) for $\sum_q |z_q|\|f_q\|_\sharp < \rho_D/N$; by (a) at $s = N$, $\langle\mathrm{Fold}\,e^{\mathcal{O}(w_\iota)}\rangle_K^{1/N} \le [\frac32\,2^N e^{\Delta_N(\delta)}]^{1/N}\mathsf{e}(w_\iota)$.

In the denominator, (a) at $s = n_{\mathrm{b}}$ gives $|\langle e^{\mathcal{O}(w)}\rangle_K| \ge \frac12(\frac23)^{n_{\mathrm{b}}}e^{-\Delta_{n_{\mathrm{b}}}(\delta)}\prod_\iota \mathsf{e}(w_\iota)$; the factors $\mathsf{e}(w_\iota)$ cancel and M_S remains. (e) Leibniz' rule for $\partial_{\{1,\dots,n\}}[Q_K(z)E_K(z)]$ at 0; the derivatives of E_K at 0 in pairwise different variables are sums over the partitions of products of joint cumulants, the blocks of one element vanishing because the $\mathcal{O}(f_q)$ are centred. (f) eq. 10.i at $F = W_{\mathsf{C}}e^{\mathcal{O}(w_z)}$ and at $F = e^{\mathcal{O}(w_z)}$, on the polydisc $|z_q| \le \eta a_K^4$, where $|\mathcal{O}(w_z)| \le \frac14$ (there are $3(\ell/a_K)^3$ plaquettes and $0 \le a_p \le 2$), together with the rarity bound at p, bounds the difference of the two quotients

by $c\mathsf{V}_\ell C_{\mathrm{r}}(p) a_K^{-\kappa} L^{-p(K-\bar t)}$; Cauchy's estimate on this polydisc costs $\alpha!(\eta a_K^4)^{-|\alpha|}$, and $p(K-\bar t) \geq (4|\alpha|+\kappa+1)K$. □

Proposition 10.11 (Three free photon fields)**.** (a) Uniformly in K, $\mathbb{P}(\max_p |F^{(d)}(p)| \geq 2\varepsilon_{[d]}) \leq 18\ell^3 L^{3d} e^{-\frac{2}{3}A_0^2(\log x_{[d]})^{2\mathfrak{p}_0}}$, and the probability that some plaquette with a corner in a set S of sites of Λ_d has $|F^{(d)}(p)| \geq 2\varepsilon_{[d]}$ is at most $72\,\#S\, e^{-\frac{2}{3}A_0^2(\log x_{[d]})^{2\mathfrak{p}_0}}$. These are bounds at one depth, uniform in K: the first has the form of $R_d^\natural$, with 18 and $\frac{2}{3}$ in the place of 1 and $c^\natural$, the second the local form of Prop. 10.2(c), with 72 and $\frac{2}{3}$ in the place of 1 and $c^\natural$; summed over the depths $d \geq D$ (a union bound) they bound the probability that some depth $d \geq D$ has a plaquette with $|F^{(d)}(p)| \geq 2\varepsilon_{[d]}$. There the smoothing is linear and needs no ball. (b) For a family $\mathsf{C} = (C_1, \dots, C_r)$, at every length,

$$a_K^{-\kappa}\, \mathbb{E} \prod_i W_{C_i}^{\mathrm{fr}} \to \prod_i e^{-\frac{3}{2\mathsf{b}}[c_{\mathrm{lat}}|C_i| + \mathcal{J}_\ell(C_i)]} \Big[2^{-r} \sum_{\epsilon \in \{\pm 1\}^r} e^{-\frac{1}{\mathsf{b}} \sum_{i<j} \epsilon_i \epsilon_j \mathcal{I}_\ell(C_i, C_j)} \Big]^3 > 0;$$

at $r = 2$ the bracket is $\cosh(\mathcal{I}_\ell(C_1, C_2)/\mathsf{b})$. (c) For one loop the step is exact: the conditional mean of $W_C^{\mathrm{fr},(t)}$, the loop of the conditional mean of the field given the block field of the level t, given the block field of the level $t+1$ is $\lambda_{t+1} W_C^{\mathrm{fr},(t+1)}$; for a family the defect is at most $(e^{3x}-1)\lambda_{t+1}$, $x = \frac{1}{\mathsf{b}} \sum_{i<j} |a_K(X_{C_i}, \mathcal{C}_{(t+1)} X_{C_j})|$, which for polygons at fixed distances falls faster than every power of a_{K-t}; so for families the factor without cross terms is the right one. The exponent κ, the perimeter constant $c_{\mathrm{lat}}|C| + \mathcal{J}_\ell(C)$ and the factor λ_{t+1} of one step are those of three free photon fields. (d) At every cutoff K, for every polygon C with corners in Λ_K and every bounded real f,

$$\frac{\mathbb{E}[W_C^{\mathrm{fr}} \mathcal{O}^{\mathrm{fr}}(f)]}{\mathbb{E}\, W_C^{\mathrm{fr}}} - \mathbb{E}\, \mathcal{O}^{\mathrm{fr}}(f) = -\frac{3}{2\mathsf{b}^2}\, a_K^{-1} \sum_p f(x(p))\, \mathcal{I}_{a_K}(C, \partial p)^2.$$

If C has its corners in a fixed Λ_i and f is Lipschitz with support at positive distance from C, then as $K \to \infty$

$$-\frac{3}{2\mathsf{b}^2}\, a_K^{-1} \sum_p f(x(p))\, \mathcal{I}_{a_K}(C, \partial p)^2 \;\longrightarrow\; -\frac{3}{2\mathsf{b}^2} \int_{T_\ell} f(x)\, |B_{C,\ell}(x)|^2 \, \mathrm{d}x, \qquad B_{C,\ell}(x) = \mathrm{curl}_x \oint_C G_\ell(x-y)\, \mathrm{d}y,$$

the field of C on the torus; in the ball of radius $< \ell/4$ that holds the polygon, $|B_{C,\ell}(x) - B_C(x)| \leq c\,|C|\,\mathrm{diam}(C)\,\ell^{-3}$, c absolute, with $B_C(x) = \oint_C \frac{dy \times (x-y)}{4\pi|x-y|^3}$ the Biot–Savart field of C in $\mathbb{R}^3$. For a bounded f that is not Lipschitz the sums over the plaquette centres need not tend to the integral: an f that vanishes at the plaquette centres of all cutoffs, a countable set, gives 0 at every cutoff. ■

Sketch of proof. (a) $F^{(d)}$ is a linear image of F; each component at a plaquette of Λ_d is a centred Gaussian variable of variance at most $1/x_{[d]}$ (the smeared self-energy of the boundary of one plaquette of Λ_d, divided by β). $|F^{(d)}(p)| \geq 2\varepsilon$ forces a component $\geq 2\varepsilon/\sqrt{3}$, of probability at most $2e^{-\frac{2}{3}\varepsilon^2 x_{[d]}}$, and $\frac{2}{3}\varepsilon_{[d]}^2 x_{[d]} = \frac{2}{3}A_0^2(\log x_{[d]})^{2\mathfrak{p}_0}$; a union bound over the $3(\ell L^d)^3$ plaquettes and the three components gives 18, over the 12 plaquettes at a site 72. (b) $\mathbb{E}\prod_i \cos\phi_{C_i}^a = 2^{-r}\sum_\epsilon \mathbb{E}\, e^{i\sum_i \epsilon_i \phi_{C_i}^a} = 2^{-r}\sum_\epsilon \exp[-\frac{1}{2\mathsf{b}}\sum_{i,j}\epsilon_i\epsilon_j \mathcal{I}_{a_K}(C_i, C_j)]$; three independent colours; Prop. 10.7(a) for the diagonal terms and for the distant pairs. (c) Given the block field of the level $t+1$, the curvature is its conditional mean, the curvature $\Pi_{t+1}F$ of the linear background, plus an independent Gaussian fluctuation; between the levels t and $t+1$ the fluctuation of the flux through C has the variance $\frac{a_K}{\mathsf{b}}(X_C, \mathcal{C}_{(t+1)} X_C)$ per colour, and $\mathbb{E}[\cos(\phi+\xi) \mid \phi] = e^{-\frac{1}{2}\mathrm{Var}\,\xi}\cos\phi$; three colours give λ_{t+1}.

For a family the cross covariances $\frac{a_K}{\mathsf{b}}(X_{C_i}, \mathcal{C}_{(t+1)} X_{C_j})$ enter as factors $e^{\mp \frac{a_K}{\mathsf{b}}(X_{C_i}, \mathcal{C}_{(t+1)} X_{C_j})}$, at most e^x for each colour. (d) For jointly Gaussian ϕ, F: $\mathbb{E}[\cos\phi\, F^2]/\mathbb{E}\cos\phi - \mathbb{E}F^2 = -\mathrm{Cov}(\phi, F)^2$; $\mathrm{Cov}(\phi_C^a, F^a(p)) = \beta^{-1}(X_C, X_{\partial p}) = \mathcal{I}_{a_K}(C, \partial p)/\mathsf{b}$; $\frac{1}{2}$ and three colours give $\frac{3}{2}$. The identity holds

for every bounded f, as nothing in it is a limit. For the limit let the corners of C lie in a fixed Λ_i: away from C, uniformly on the support of f, $\mathcal{I}_{a_K}(C,\partial p) = a_K^2(B_{C,\ell}(x(p))\cdot n_p + O(a_K \mathrm{Lg}(\ell/a_K)))$, the mutual energy of C with a small loop being the flux through it of the field of C formed with the Green function of the torus: Prop. 10.7(a) for the two polygons C and ∂p at positive distance compares the lattice energy with $\mathcal{I}_\ell(C,\partial p) = \oint_{\partial p}\oint_C G_\ell(x-y)\,\mathrm{d}y\cdot dx$ up to $O(a_K^3 \mathrm{Lg}(\ell/a_K))$, and Stokes' formula makes $\mathcal{I}_\ell(C,\partial p)$ the flux of $B_{C,\ell}$ through p; summing $a_K^3 f(x(p))(B_{C,\ell}(x(p))\cdot n_p)^2$ over the three orientations of the plaquettes gives a Riemann sum of $f|B_{C,\ell}|^2$, which converges for Lipschitz f.

Finally $G_\ell(x) = \frac{1}{4\pi|x|} + H_\ell(x)$ with $H_\ell(x) = \ell^{-1}H_1(x/\ell)$ smooth on $|x|<\ell$; as $\oint_C dy = 0$,

$$\operatorname{curl}\oint_C H_\ell(x-y)\,\mathrm{d}y = \oint_C [\nabla H_\ell(x-y) - \nabla H_\ell(x-y_0)]\times dy,$$

of modulus at most

$$|C|\operatorname{diam}(C)\sup_{|u|\le \ell/2}|\nabla^2 H_\ell(u)| = |C|\operatorname{diam}(C)\,\ell^{-3}\sup_{|u|\le 1/2}|\nabla^2 H_1(u)|.$$

□

The proof of Thm. AF uses the following tools; we state them in lattice units, with spacing 1 and $\beta = \mathsf{b}/a$.

Lemma 10.12 (Tools of the tube)**.** Let ν be the Wilson measure at β on a torus $(\mathbb{Z}/N\mathbb{Z})^3$, N even, or a periodic Gibbs state as in Thm. AF. For a step $c = \langle x_-, x_+\rangle$ of a closed path let $c(y)$, $y\in\mathbb{Z}^2$, be its translates in the lattice plane through x_- orthogonal to c; for an integer $\rho\ge 2$ let $T_\rho(c) = \{c(y) : |y|<\rho\}$ be the slab and $K_\rho(c) = \{x_- + y + se : |y|_\infty \le \rho,\ 0\le s\le 1\}$ the box of c, e its direction; let $\phi_\rho(y) = \min\{1, \log^+(\rho/|y|)/\log\rho\}$, $\phi_\rho(0)=1$, so that $\phi_\rho = 0$ at $|y|\ge\rho$, $E_\rho = (2\pi\log\rho + 7.63)/\log^2\rho$, and $c_j = j(j+1)$. (a) (Admissible steps.) A set I of steps of closed paths $C_1,\dots,C_r$ is admissible at ρ if no other step of the family runs through a bond of $T_\rho(c_i)$, $i\in I$, and no bond of $T_\rho(c_{i'})$, $i'\ne i$, lies in $K_\rho(c_i)$. For closed paths without self-intersection with s straight pieces in all, each of at least $4\rho+1$ steps, whose pieces without a common corner are at sup-distance at least $2\rho+1$, the steps $2\rho\le j\le l'-1-2\rho$ of each piece of l' steps form an admissible set of $n-4\rho s$ steps, n the number of steps; for a rectangle, $n-16\rho$. (b) (The profile.) $\sum_{\langle x,y\rangle}(\phi_\rho(x)-\phi_\rho(y))^2 \le E_\rho$, the sum over the bonds of $\mathbb{Z}^2$, and $(\log\rho - 1.2144)/2\pi \le 1/E_\rho \le \log\rho/2\pi$. (c) (An information inequality, three colours at once.) Let $\varrho>0$ be a C^2 density on $\mathrm{SU}(2)^Y$ with $0\in Y$, $\phi: Y\to[0,\infty)$ with $\phi_0=1$, frames $P_y\in\mathrm{SU}(2)$ with $P_0=\mathbf{1}$, and for a unit vector $\mathbf{n}=(\mathbf{n}_1,\mathbf{n}_2,\mathbf{n}_3)\in\mathbb{R}^3$ the moves $v_y\mapsto \exp[is\phi_y P_y^{-1}(\frac12\sum_a \mathbf{n}_a\sigma_a)P_y]v_y$, $\sigma_1,\sigma_2,\sigma_3$ the Pauli matrices, with the derivative $D_{\mathbf{n}}$ at $s=0$; let $\bar I = \sup_{|\mathbf{n}|=1}\mathbb{E}[-D_{\mathbf{n}}^2\log\varrho]$. Then $\|\mathbb{E}\,D^j(v_0)\|^2 \le \bar I/(\bar I + c_j)$ in the operator norm for every spin $j>0$, where D^j is the representation of spin j. The spin-zero representation has norm 1. (d) (The second variation.) For these moves on the bonds $c(y)$ of a slab, with any frames, a bond $e'=\langle x,y\rangle$ of the plane, $p(e')$ the plaquette spanned by e' and c, and $H_{e'} = P_y U(e')^{-1}P_x^{-1}$: $D_{\mathbf{n}}^2 a_{p(e')} \le \frac14(\phi_x-\phi_y)^2 + \phi_x\phi_y(3a(H_{e'}) + a_{p(e')})$, $a(g) = 1-\frac12\operatorname{tr} g$. (e) (Ceilings multiply over cells.) Let $B = \bigsqcup_l B_l$ be a set of bonds made of cells B_l, no plaquette having bonds in two cells, and $C=\gamma_1\delta_1\cdots\gamma_s\delta_s$ a closed path with γ_l through no bond of $B\setminus B_l$ and δ_l through no bond of B. If Q_l is the conditional mean under ν of $D^j(U(\gamma_l))$ given the bonds off B_l, then $\nu(\chi_j(U(C))\,G) = \nu(\operatorname{tr}[\prod_l Q_l D^j(U(\delta_l))]\,G)$ for every bounded Borel G of the bonds off B, and $|\nu(\chi_j(U(C))\,G)| \le (2j+1)\sup|G|\,\nu(\prod_l\|Q_l\|)$. (f) (One plaquette.) With $\mathfrak{M}_\nu = \sup_p \nu(e^{\beta a_p})$: $\nu(a_p)\le \beta^{-1}\log\mathfrak{M}_\nu$ and $\nu(a_p\ge s)\le \mathfrak{M}_\nu e^{-\beta s}$; for $\beta\ge1$, $\log\mathfrak{M}_\nu \le 192\frac{\mathsf{b}}{\ell} + 4(14.15 + 3\log\beta)$ on the torus of side ℓ, and $\log\mathfrak{M}_\nu \le 28.3 + 6\log\beta$ for the Gibbs states. (g) (The ceiling of a tube.) Let $\beta>0$, $\rho\ge2$, C_l closed paths with spins j_l, I admissible at ρ, $c_i = c_{j_l}$ for a step of C_l, G a bounded Borel function of the bonds off $B=\bigcup_{i\in I}T_\rho(c_i)$, $\mathcal{G} = \sum_{i\in I}\frac{2c_i}{\beta E_\rho}(1-\frac{2c_i}{\beta E_\rho})$ and $\omega_\rho = 40\rho^4\sum_{i\in I}c_i/(\beta E_\rho)^2$. If $\omega_\rho\le1$, then $|\nu(G\prod_l\chi_{j_l}(U(C_l)))| \le \sup|G|\prod_l(2j_l+1)\,e^{-\mathcal{G}}\mathfrak{M}_\nu^{\omega_\rho}$; and

$2c_j/E_\rho \ge \frac{4}{3}c_j k(\log\rho - 1.2144)$: the number per scale is k. (h) (The ceiling with dyadic profiles.) Let $\rho = 2^S \ge 2$ and C a coordinate rectangle of n steps with sides of at least $4\rho+1$ steps and, on the torus, with the points within $\rho+1$ of C in a cube of side at most $\frac{N}{2}$; let $W_C = \frac{1}{2}\operatorname{tr} U(C)$, $p = kn/\beta < 3$, $\delta_0 = (16k\rho\log\rho + pk\log^2\rho)/\beta$ and $\bar a = \nu(a_p)$, the plaquette mean. Then, with the profiles ϕ_r of the scales $r = 2, 4, \dots, 2^S$,

$$|\nu(W_C)| \le e^{\delta_0 + 2.2144p}\rho^{-p}\Big[1 + \frac{72\,\bar a\rho^3\log^2\rho}{2^{3-p}-1}\Big] + 35\,\bar a;$$

the plaquette mean enters linearly, and $p < 3$ keeps the denominator positive. ■

Sketch of proof. (a) A step at distance $\ge 2\rho$ from the ends of its piece has a slab that meets no other step, and the slabs of two steps of pieces without a common corner at sup-distance $\ge 2\rho+1$ do not reach each other's boxes. (b) The sum is a lattice form of $\int|\nabla\phi_\rho|^2 = 2\pi\int_1^\rho (r\log\rho)^{-2} r\,\mathrm{d}r = 2\pi/\log\rho$, with explicit corrections at the centre and at the rim. (c) The moves are one-parameter groups of maps of $\mathrm{SU}(2)^Y$ that keep the Haar measure; under the move of v_0, $D^j(v_0)$ is multiplied by $e^{isJ_{\mathbf n}}$, $J_{\mathbf n}$ the generator of spin j, the image of $\frac{1}{2}\sum_a \mathbf{n}_a\sigma_a$, and $\sum_{\mathbf n} J_{\mathbf n}^2 = c_j$ over an orthonormal basis of directions.

With $M = \mathbb{E}\,D^j(v_0)$ and a unit vector u, integration by parts gives $iJ_{\mathbf n}Mu = -\mathbb{E}[(D^j(v_0) - M)u\,D_{\mathbf n}\log\varrho]$. Put $s_a = D_{e_a}\log\varrho$ and $s = (s_1, s_2, s_3)$. Since the supremum defining $\bar I$ runs over every unit direction, integration by parts gives $\mathbb{E}(ss^{\mathsf T}) \le \bar I\mathbf{1}$. The vector form of Schwarz's inequality therefore gives $\sum_a |\mathbb{E}[Xs_a]|^2 \le \bar I\mathbb{E}|X|^2$ for $X = (D^j(v_0) - M)u$. Using $\sum_a |J_{e_a}Mu|^2 = c_j|Mu|^2$ and $\mathbb{E}|X|^2 = 1 - |Mu|^2$ yields $c_j|Mu|^2 \le \bar I(1 - |Mu|^2)$. (d) Expansion of $a_{p(e')}$ in s: the abelian part gives $\frac{1}{4}(\phi_x - \phi_y)^2$, and the commutator terms are bounded by the defect $a(H_{e'})$ of the frames and by $a_{p(e')}$. (e) The cells interact with the rest only through plaquettes with bonds in one cell and off B, so conditioning cell by cell replaces $D^j(U(\gamma_l))$ by Q_l, the δ_l and G being functions of the bonds off B; $|\operatorname{tr}\prod A_i| \le (2j+1)\prod\|A_i\|$. (f) Jensen's and Chebyshev's inequalities; the chessboard estimate bounds $\nu(e^{\beta a_p})$ by a power of a ratio of partition functions, and $\int\exp[-\beta\sum_{p\in Q} a_p]\,\mathrm{d}U \ge (\mathsf{K}\beta^3)^{-|Q|}$, $\mathsf{K} = 1.39\cdot 10^6$, for a box Q of sites ($\log\mathsf{K} < 14.15$); the torus adds $192\mathsf{b}/\ell$. (g) In each slab of I we use comb frames: in the plane through x_- orthogonal to c_i a coordinate axis through x_- is the base, the lines of the plane orthogonal to it are the teeth, and P_y is the transport from x_- along the base and then along the tooth to $x_- + y$, a product over bonds of the plane, which lie off B.

Then $H_{e'} = \mathbf{1}$ on the base and on the teeth, and at a rung e', a bond parallel to the base at the height $t_{e'} \ne 0$, $H_{e'}$ is conjugate to the holonomy of the boundary of the strip $S_{e'}$ of the $|t_{e'}|$ plaquettes between the rung and the base, so that $a(H_{e'}) \le |t_{e'}|\sum_{q\in S_{e'}} a_q$ (Prop. 10.9(c) and Schwarz' inequality). Write $\nu_B(F)$ for the conditional mean of F under ν given the bonds off B. As no plaquette has bonds in two cells, given the bonds off B the cells are independent, and the conditional law of the bonds of the slab of c_i given the bonds off it is their conditional law given the bonds off B. The bonds of the slab enter the action only through the plaquettes $p(e')$, e' a bond of the plane, so that $-D_{\mathbf n}^2$ of the logarithm of this conditional density is $\beta\sum_{e'} D_{\mathbf n}^2 a_{p(e')}$, with ϕ taken as 0 off the slab, whose bonds alone move; by (b), (d) its conditional mean is $\bar I \le \frac{\beta}{4}(E_\rho + X_i)$, with

$$X_i = 4\sum_{e'}\phi_x\phi_y\big[3a(H_{e'}) + \nu_B(a_{p(e')})\big],$$

a function of the bonds off B. It is at most $\sum_q w_q^{(i)}\,\nu_B(a_q)$, a sum over plaquettes with weights $w_q^{(i)} \ge 0$: $4\phi_x\phi_y$ at the plaquette $p(e')$ of every bond e' of the disc, and $12|t_{e'}|\phi_x\phi_y$ at each plaquette of the strip of every rung e', where $\nu_B(a_q) = a_q$, the bonds of the strip lying in the box $K_\rho(c_i)$ and so off B. The total weight $\sum_q w_q^{(i)}$ is at most $20\rho^4$: the strips give about $12\int_{|y|<\rho} y_2^2\,\mathrm{d}y = 3\pi\rho^4$, the plaquettes $p(e')$ a multiple of ρ^2. (c) gives $\|Q_i\|^2 \le (1 + 4c_i/(\beta(E_\rho + X_i)))^{-1}$, so, by $-\frac{1}{2}\log(1+z) \le$

$-\frac{z}{2}+\frac{z^2}{4}$ and $\frac{1}{E+X}\ge\frac{1}{E}-\frac{X}{E^2}$, $\log\|Q_i\|\le-\frac{2c_i}{\beta E_\rho}(1-\frac{2c_i}{\beta E_\rho})+\frac{2c_iX_i}{\beta E_\rho^2}$; (e) multiplies these over the disjoint cells.

Put $w_q=\sum_i 2c_i w_q^{(i)}/(\beta E_\rho^2)$. Jensen's inequality for the conditional mean ν_B gives $\exp\sum_i\frac{2c_iX_i}{\beta E_\rho^2}\le\nu_B(\prod_q e^{w_qa_q})$; as $\sum_q w_q/\beta\le 40\rho^4\sum_i c_i/(\beta E_\rho)^2=\omega_\rho\le 1$, Hölder's inequality with the exponents β/w_q and (f), with $\mathfrak{M}_\nu\ge 1$, give $\nu(\prod_q e^{w_qa_q})\le\prod_q\nu(e^{\beta a_q})^{w_q/\beta}\le\mathfrak{M}_\nu^{\omega_\rho}$. The last inequality is (b) with $\frac{4}{3}k=\frac{1}{\pi}$. (h) Keep the cells $T_\rho(c_i)$ of the $n-16\rho$ steps of the admissible set of (a) (spin $\frac12$, $c_{1/2}=\frac34$). On each cell and for every dyadic $2\le r\le\rho$ the moves with the profile ϕ_r, which vanishes at $|y|\ge r$, rotate bonds of the cell alone, and (b)–(e), as in the proof of (g), give at every field $\prod_i\|Q_i\|\le\min_r e^{-\mathcal{G}_r+Y_r}$, with $\mathcal{G}_r$ the number $\mathcal{G}$ of (g) formed with E_r, $Y_r=\sum_i 2c_{1/2}X_i^{(r)}/(\beta E_r^2)$ and $X_i^{(r)}$ the X_i of the proof of (g) formed with ϕ_r; by (b), $e^{-\mathcal{G}_r}\le e^{\delta_0+1.2144p}r^{-p}$, δ_0 paying for the 16ρ omitted steps and for the second-order term of $\mathcal{G}_r$, so that $e^{-\mathcal{G}_r+Y_r}\le e^{\delta_0+2.2144p}r^{-p}$ where $Y_r\le p$.

The sharper estimate of the non-abelian member of the second variation bounds the mean of Y_r in the place of an exponential moment: in comb frames $a(H_{e'})$ at a rung e' of height $t_{e'}$ is $1-W$ of the boundary of the strip $S_{e'}$ of $1\times|t_{e'}|$ plaquettes, whose mean is $1-w(1,|t_{e'}|)\le 1-w(1,1)^{|t_{e'}|}\le|t_{e'}|\,\bar a$ by Prop. 10.9(c) (for the Gibbs states the strips of even height by the reflections in the planes of sites, those of odd height by $\sqrt{a(gh)}\le\sqrt{a(g)}+\sqrt{a(h)}$), so that $\nu(X_i^{(r)})\le c\,r^3\bar a$ and $\nu(Y_r)\le c\,p\,r^3\log^2 r\,\bar a$: the plaquette mean enters linearly, and this bound of a mean takes the place of the condition $\omega_\rho\le 1$ and of the weight $40\rho^4$ of (g).

Let r_* be the largest dyadic $r\le\rho$ with $Y_r\le p$; for $r<\rho$ the event $\{r_*=r\}$ lies in $\{Y_{2r}>p\}$, of probability at most $\nu(Y_{2r})/p$ by Markov's inequality, and on the event that $Y_r>p$ at every r the bound 1 serves. The sum over the dyadic $r<\rho$ of $r^{-p}(2r)^3\log^2(2r)$ is geometric and led by its last term exactly when $p<3$, which gives the display. □

10.1.6. *Proof of Thm. AD. Step 1: (1).* For $D\ge d_\Theta\ge d_\natural$ (as $\Theta_\star\ge t_\natural$), $\{\tau\le D\}=\{\tau_0\le D\}=\bigcap_{d\ge D}\pi_d^{-1}(I_d)$: if $\tau_0\le D$, then $V_d\in I_d$ for all $d\ge\tau_0$, so for all $d\ge D$; if $V_d\in I_d$ for all $d\ge D$, then D is one of the integers of which τ_0 is the least. Each member is open (I_d is open, π_d continuous), $\{\tau\le D\}=\emptyset$ for $D<d_\Theta$, and $\{\tau=\infty\}=\{\tau_0=\infty\}$; a function into a countable set whose sublevel sets are Borel is Borel. As $\pi_d(\mu)=\mu_d$, Prop. 10.2(b) and countable subadditivity give the tail by the sum, and the same clause of Prop. 10.2(b) the bound $\frac{L}{L-1}R_D$ when $c'A_0^2\log x_{[D]}\ge 2$, as then $R_{d+1}\le L^{-1}R_d$ for all $d\ge D$; $\sum_d R_d<\infty$ and $\mu(\mathsf{Thr})=1$ give $\mu(\Omega)=1$, and for this the bound $\mu_d(\mathcal{V}_d\setminus I_d)\le R_d$ with $R_d\to 0$ faster than every power of L^{-d} suffices.

If $c'A_0^2(\log x_{[D]})^{2\mathfrak{p}_0}\le 26\ell^3$, the exponent in R_D is at least $(\mathsf{C}_{\mathrm{maj}}^{\Sigma}+2+c_{\mathrm{low}}-26)\ell^3\ge 0$ by $c_{\mathrm{low}}\ge 24$, and $2\ell^3L^{3D}M^{-3}\ge 16$ as $\ell\ge 2M$; so $R_D>1$, and the two bounds, $\sum_{d\ge D}R_d$ and $\frac{L}{L-1}R_D$, both at least R_D, exceed 1. Next, $\{\tau>D\}=\bigcup_{d\ge D}\pi_d^{-1}(\mathcal{V}_d\setminus I_d)$, and $\mathcal{V}_d\setminus I_d$ is the event of Prop. 10.2(c) for the set of all ℓ^3L^{3d} sites, which gives the bound by $\sum_{d\ge D}R_d^\natural$. As $(y+\log L)^{2\mathfrak{p}_0}-y^{2\mathfrak{p}_0}\ge 2y\log L$ for $y\ge 1$, $R_{d+1}^\natural/R_d^\natural\le L^{3-2c^\natural A_0^2\log x_{[d]}}$, which is at most L^{-1} for $d\ge D$ if $c^\natural A_0^2\log x_{[D]}\ge 2$.

With $y=D\log L\le\log x_{[D]}$, $\log R_D^\natural\le 3\log\ell+3y-c^\natural A_0^2y^2$; the first member of the condition gives $3y\le\frac12 c^\natural A_0^2y^2$ and $c^\natural A_0^2\log x_{[D]}\ge 6$, the second $\frac12 c^\natural A_0^2y^2\ge 3\log\ell+\log\frac{1}{\eta}$; so $R_D^\natural\le\eta$.

Step 2: (2). Thm. YM3_I:E in the setting of Section YM3_I:52.7 and §5.2 give the orbit for $\varpi(W)<\bar\varepsilon$; the inclusion $\{\varpi<2c_\rho\varepsilon_{[d]}\}\subset\{\varpi<\bar\varepsilon\}$ for $x_{[d]}\ge t_\natural$ is the condition (cut1) at $t=x_{[d]}$, which gives $2c_\rho\varepsilon(t)\le\varepsilon_{\mathcal{W}}\alpha_0$, with $\varepsilon_{\mathcal{W}}\alpha_0\le\bar\varepsilon$ (equation YM3_I:(2134)). For $d\ge\tau_0\ge d_\natural$, $x_{[d]}\ge t_\natural$ and $V_d\in I_d\subset\{\varpi<2c_\rho\varepsilon_{[d]}\}$. If $d\ge\tau(V_\bullet)$, then $d\ge d_\Theta$ is admitted and $d\ge\tau_0$, so $V_d\in I_d\subset J_d$. Each clause of Prop. 10.4 asks of W only that it lie in J_d at an admitted depth, not that it be a block average or a member of a thread; we substitute $W:=V_d$, and Prop. 10.4(a)–(c) give the limits, the rate, the curves, the continuity, the gauge invariance and the bound for rectangles. The thread property enters no step, and Ω is defined without d, C, γ, K.

Step 3: (3). J_d is open and $W_C^{[d]}$ is continuous and gauge invariant on it; the inverse image of a Borel set under $F_C^{[d]}$ is a Borel subset of J_d, joined with the closed set $\mathcal{V}_d \setminus J_d$ if it contains 0; π_d is continuous; so $Y_{d,C}$ is Borel and $\sigma(\pi_d)$-measurable, $\sigma(\pi_d) \subset \mathcal{F}_d$, and its restriction to the Borel set Thr is a Borel function on the threads. $\sup_{\mathcal{X}} |Y_{d,C}^K - Y_{d,C}| \le \sup_{J_d} |w_C^K - W_C^{[d]}| \to 0$ by Prop. 10.4(b). θ_v is a bijection, with inverse $\theta_{v^{-1}}$, that maps every $\pi_j^{-1}(I_j)$ onto itself, and $Y_{d,C}(\theta_v V_\bullet) = F_C^{[d]}(V_d^{v|\Lambda_d}) = Y_{d,C}(V_\bullet)$. $(C^{\langle d\rangle})^{\langle j\rangle} = C^{\langle j\rangle}$, so $W_{C^{\langle d\rangle}} = W_C$ on $\mathcal{V}_j$, and every $d \ge d_\Theta$ is admitted. For curves Prop. 10.4(c) replaces (a), (b); the shadows are fixed lattice by lattice because w_γ^K depends on the shadow on the bare lattice, and a curve has no refinement.

Step 4: (4). $\pi_d(\mu) = \mu_d$ gives the first equality, Prop. 10.4(b), (d) the others and the rates. $|Y_{d,C}| \le 1$ and Step 1 give $|\int Y_{d,C}\,\mathrm{d}\mu - \int \mathbf{1}\{\tau \le d\} Y_{d,C}\,\mathrm{d}\mu| \le \mu(\tau > d) \le \sum_{d' \ge d} R_{d'}$, at most $\frac{L}{L-1} R_d$ if $c' A_0^2 \log x_{[d]} \ge 2$ (Step 1). $G_d - F_C^{[d]}$ vanishes on J_d and has modulus at most 1 off it, and $\mu_d(\mathcal{V}_d \setminus J_d) \le \mu_d(\mathcal{V}_d \setminus I_d) \le R_d$; for $d \ge \tau$, $V_d \in J_d$ and so $G_d(V_d) = Y_{d,C}$. For the joint means, with $D = \max_i d_i$: the law of $(V_0, \dots, V_D)$ is the image of μ_D under $V \mapsto (\mathcal{M}^D V, \dots, V)$, $\prod_i F_{C_i}^{[d_i]} \circ \mathcal{M}^{D-d_i}$ is one bounded Borel function on $\mathcal{V}_D$, $\mathcal{M}^{K-d_i} = \mathcal{M}^{D-d_i}\mathcal{M}^{K-D}$, and $\mu_D^K \to \mu_D$ in total variation; no measure on an infinite product is needed here. The means depend on the single laws μ_d; what μ adds to them are the events that involve infinitely many depths, $\{\tau \le D\}$ and Ω.

Step 5: (5). We use the localisation of backgrounds (Prop. YM3_I:53.5) at $n = K - d$, for the constant sequence $\Omega_0 = \cdots = \Omega_{K-d} = T_\ell$, whose datum is V on the bonds of Λ_d and whose minimal orbit is the set of the backgrounds over V; for the region Ω', a region for the partition into the M_1-cubes of the depth d, with the one part $\mathsf{Y}_1 = T_\ell \setminus \Omega'$ of its complement and the index $l_1 = K - d$; at $\varepsilon_{\mathrm{r}} = 2\varepsilon_{K-d} \le \epsilon_{\mathrm{lc}}$; and at $V \in \mathsf{O}_{K-d}$. We verify its hypotheses for this region of M_1-cubes, using the joined sequence defined in equation YM3_I:(999): the joined sequence is the $\mathfrak{r}_2$-tower of Ω', whose minimal orbit over the datum induced by V defines the localised background U' (equation YM3_I:(77)); it is separated; its tower bonds are the bonds of its members of index $< K - d$, with both end points in the collar $\mathsf{Y}_1^{\sim 2} \setminus \mathsf{Y}_1$, which lies in the wide collar $\mathsf{Y}_1^{\sim 3} \setminus \mathsf{Y}_1$ (Section YM3_I:53.4), the points of Ω' within three layers of M_1-cubes of the depth d of $T_\ell \setminus \Omega'$; the core $\Omega'^{\sim -2}$ lies in its last member, so that $j(b) = K - d$ at the bonds of the core; and both data are regular at the radius $2\varepsilon_{K-d}$, as $V \in \mathsf{O}_{K-d}$ (Prop. YM3_I:44.2).

So, after gauge transformations, the background over V and U' differ by factors $e^{i\mathsf{a}(b)}$ at the bare bonds b with no end point in the wide collar, with $|\mathsf{a}(b)| \le \mathsf{C}_{\mathrm{lc}}\, 2\varepsilon_{K-d} L^{d-K} e^{-\delta_{\mathrm{lc}} D(b)/2}$, $D(b)$ the distance of b from the tower bonds in the weighted distance of the tower (Section YM3_I:53.4). The bare bonds of C are such bonds of the core, as $\mathrm{dist}(C, T_\ell \setminus \Omega') \ge \frac{\delta_1}{2} \ge 4M_1 a_d$. The tower-bond endpoints lie within the two-layer collar: a row below the last level has both endpoints outside the next tower member, which contains the core $\Omega'^{\sim -2}$. Moreover every local spacing in the joined sequence is at most a_d. For any connected path Γ joining two points, the layer pieces cover Γ, so its weighted length is at least $a_d^{-1}\,\mathrm{length}(\Gamma)$, and hence at least a_d^{-1} times the periodic sup-distance between those points. Taking the infimum over paths proves the same bound for weighted distance; the sup-distance to the complement of a slab, or of the intersection of two slabs, is the distance; so on C, $D(b) \ge \mathrm{dist}(C, T_\ell \setminus \Omega')/a_d - 2M_1 \ge \frac{\delta_1}{2} L^d - 2M_1 \ge \frac{\delta_1}{4} L^d$, as $M_1 a_d \le \frac{\delta_1}{8}$.

The loop is gauge invariant, $W_C = \frac{1}{2}\,\mathrm{tr}$, and $|\,\mathrm{tr}\prod_i A_i - \mathrm{tr}\prod_i B_i| \le 2\sum_i |A_i - B_i|$ for unitary matrices; summing $|e^{i\mathsf{a}(b)} - \mathbf{1}| \le |\mathsf{a}(b)|$ over the $|C|/a_K$ bare bonds of C gives $|w_C^K(V) - Y_{K,d}(V)| \le \frac{|C|}{a_K}\, 2\mathsf{C}_{\mathrm{lc}}\varepsilon_{K-d} L^{d-K} e^{-\delta_{\mathrm{lc}}\delta_1 L^d/8} = 2\mathsf{C}_{\mathrm{lc}}\varepsilon_{K-d}|C| L^d e^{-\delta_{\mathrm{lc}}\delta_1 L^d/8}$, and $\varepsilon_{K-d} \le \hat{\varepsilon}_d$ because $\frac{2}{3} x_{[d]} \le x_{K-d} \le x_{[d]} + \bar{B}$. Two fields that agree on the bonds of Ω' have the same $Y_{K,d}$; let $K \to \infty$ with Prop. 10.4(b), (e).

Step 6: (6). The Wilson measure at (K, ℓ, b) is the one at $(K - s, L^s\ell, L^s\mathsf{b})$ carried by $x \mapsto L^s x$, since $\beta = L^K \mathsf{b} = L^{K-s}(L^s\mathsf{b})$, and Λ_d goes to Λ_{d-s}; so the limit laws, the measure on the threads from the depth s on and Prop. 10.2 hold at every $\mathsf{b} > 0$ with $R_{d-s}[L^s\ell, L^s\mathsf{b}]$. At $(L^s\ell, L^s\mathsf{b})$ the

one-torus setting holds (Lem. 10.1), and (1)–(4) hold there at the depth $d-s$, with the depths $d_\natural$, d_Θ, $s_\natural$ of $(L^s\ell, L^s\mathsf{b})$; increased by s, these depths take the places of $d_\natural$, d_Θ, $s_\natural$ in (1)–(4) at (ℓ,b), the depth $d-s$ of $(L^s\ell, L^s\mathsf{b})$ being the depth d of (ℓ,b).

The dilation maps Λ_j of T_ℓ onto Λ_{j-s} of $T_{L^s\ell}$, one graph with the same $\ell L^j = 2L^{m+j}$ sites in each direction, bond onto bond, block onto block and plaquette onto plaquette, and the cutoff K onto the cutoff $K-s$, with the same number $K-d$ of block steps above the depth d; it leaves β, $x_{[d]} = L^d\mathsf{b} = L^{d-s}(L^s\mathsf{b})$, $\varepsilon_{[d]}$, the balls I_d, J_d and the length in units of the lattice, $|C|L^d$, unchanged. Every object in the definition of the smoothed loop is formed from these graphs, from $K-d$ and from the constants, in the units of the lattices, and holds neither the side, nor the label, nor the spacing. Item by item: the block averaging map with its averaging radius $1/200L^2$ (equations YM3_I:(37)–(39)); the constraint sets $\{\mathcal{M}^{K-d}U = W\}$, the action $\sum_p a_p$, criticality for it on them and the regularity at the radius ε_{v}, a constant, all measured in the units of the level, with the bare spacing $L^{-(K-d)}$ (Defs. YM3_I:9.3 and YM3_I:9.5); the residual gauge groups $\mathcal{G}_K^d$ and the minimal orbit (§5.2); the axial gauge of the averaged backgrounds in the definition of H_s^K ([7, §B.2]); the radius $\bar\varepsilon$ and the lattices of Thm. YM3_I:E in the setting of Section YM3_I:52.7, whose lattice hypotheses are imposed through the number of the sites in each direction at the levels used, so that the backgrounds over a field with $\varpi < \bar\varepsilon$ form one non-empty orbit at (ℓ,b) and at $(L^s\ell, L^s\mathsf{b})$ alike; and the plaquette size ϖ, a maximum over the plaquettes of one graph .

So the transported backgrounds over V_d are the backgrounds over V_d on the lattices of side ℓ, the transported $\mathcal{C}_K(W)$, H_d^K, $H_{d,n}^\infty$, w_C^K, smoothed loops, balls and variables are those of (ℓ,b) at the depth d, and (1)–(4) at $(L^s\ell, L^s\mathsf{b})$ and the depth $d-s$ are (1)–(4) at (ℓ,b) and the depth d.

10.1.7. *Proof of Thm. AE. Step 1: the ceiling.* Fix $\vartheta \in (\kappa, \frac12)$. Prop. 10.6(a) gives the hypothesis of Prop. 10.8(a) with $\eta_s = C_\eta L^{-\vartheta(K-s+1)}$ up to the last level $\bar t = K-d$, $d \ge s_0$. By Prop. 10.7(c), $a_K^{-\kappa}\Xi_{s-1} \le c_\Xi L^{\kappa(K-s+1)}$, the depth $K-s+1 \ge d$ being at least i_0; so

$$a_K^{-\kappa}\mathsf{e}_t \le c_\Xi C_\eta \sum_{j\ge K-t+1} L^{-(\vartheta-\kappa)j} = \epsilon_1 L^{-(\vartheta-\kappa)(K-t+1)}\,:$$

the additive errors of the steps, weighed by $\Xi_t \approx L^{-\kappa t}$, sum after division by a_K^κ exactly when $\vartheta > \kappa$; what ends at $\kappa = \vartheta$ is the additive error, not the tower. The rarity bound at $p = \kappa+1$ gives $\sum_{t\le\bar t} L^{\kappa(K-t)}\mathsf{w}_t \le 2\mathsf{V}_\ell C_{\mathrm{r}}(\kappa+1)L^{-d}$, and $a_K^{-\kappa}(\Xi_{t-1}+\mathsf{e}_{t-1}) \le (c_\Xi L^\kappa + \epsilon_1)L^{\kappa(K-t)}$; Prop. 10.8(a) gives eq. 10.ii. At $d = s_0$, $|W_{\mathsf{C}}^{(\bar t)}| \le 1$ and $Z^{-1}m_{\bar t}(\mathcal{V}_{K-\bar t}) \le 1$ give the supremum, and $|\Phi^{(t)}[W_{\mathsf{C}}]| \le \Xi_t + \mathsf{e}_t \le (c_\Xi+\epsilon_1)L^{-\kappa t}$. A rectangle in a ball of radius $< \ell/4$ has $|C| < \sqrt{2}\,\ell$, which is $< \frac{2\pi}{3}\mathsf{b}$ if $\ell \le 1.48\,\mathsf{b}$. If $\kappa \ge \frac12 > \vartheta$, the same tower with $\Xi_{s-1} \le c\,L^{-\kappa(s-1)}$ gives $\mathsf{e}_{\bar t} \le c\,a_K^\vartheta$ and $\Xi_{\bar t} \le c\,a_K^\kappa$, and the rarity terms are smaller.

Step 2: the limit. Let $d \ge \max\{s_0, d_0\}$. By Prop. 10.4(e), m_{K-d} is carried by $\mathsf{O}_{K-d} \subset J'_d$, where $W_{\mathsf{C}}^{(K-d)} = w_{\mathsf{C}}^K \to F' := \prod_i W_{C_i}^{[d]}$ uniformly. By Prop. 10.8(c), $Z^{-1}m_{K-d}$ and μ_d^K differ on gauge invariant sets by a positive measure of mass $\sum_{t\le K-d}\mathsf{w}_t \le 2\mathsf{V}_\ell C_{\mathrm{r}}(\kappa+1)L^{-(\kappa+1)d}$; $\mu_d^K \to \mu_d$ in total variation; $a_K^{-\kappa}\Xi_{K-d} \to \mathsf{I}_d \le c_\Xi L^{\kappa d}$; and $\mathbf{1}_{J'_d}F'$ differs from $F_{\mathsf{C}}^{[d]}$ only off J_d, a set of μ_d-measure at most R_d, which is also the difference of the two means. With eq. 10.ii, $\limsup_K \mathcal{W}_K - \liminf_K \mathcal{W}_K \le 2\mathsf{E}_d \to 0$, so the limit exists, and $|\mathcal{W}(\mathsf{C}) - \mathsf{I}_d \int F_{\mathsf{C}}^{[d]}\,\mathrm{d}\mu_d| \le \mathsf{E}_d$.

The value of I_d (Prop. 10.7(c)) gives the asymptotics of $a_d^{-\kappa}\int F_{\mathsf{C}}^{[d]}\,\mathrm{d}\mu_d$, its remainder giving $c\,a_d(1+\log^+(1/a_d))/\mathsf{b}$ for a fixed polygon, which tends to zero. Uniformity on J: the numbers of Prop. 10.6 depend on J alone; c_Ξ is uniform on J; $a_K^{-\kappa}\prod_n \lambda_n \to \mathsf{I}_d$ locally uniformly in b; the interval $[\mathsf{b}_-, \mathsf{b}_+]$ is an interval of labels under (Tor) on the side ℓ (§2.3.1: $\mathsf{b}_- \ge \gamma^{-2}$, and $MQ(\mathsf{b}_+) \le L^m$, with Q of Lem. 10.1, which is the function R of that definition, holds on the sides of $\mathfrak{L}(g)$ at every label of $\mathsf{l}(g)$, as in the proof of Prop. 10.4(e)), so the numbers C_{bf}, K_{bf} of Def. 2.7(5) formed at this interval depend on the label only through it, and Cor. 2.8(a) bounds

the difference of the laws of the block field of the depth d at two consecutive cutoffs K', $K'+1$, on the gauge invariant sets, by $\mathsf{C}_{\rm bf}\ell^3K'^{2\mathfrak{p}_0+1/2}L^{-K'/10}$ at every $\mathsf{b}\in\mathsf{J}$ and every $K'\ge K_{\rm bf}$ with $\lceil K'/1260\rceil\ge d$; summed over $K'\ge K$, with Cor. 2.8(b), (c) for the limit and for all Borel sets, $\sup_E|\mu_d^K(E)-\mu_d(E)|\le\sum_{K'\ge K}\mathsf{C}_{\rm bf}\ell^3K'^{2\mathfrak{p}_0+1/2}L^{-K'/10}$ for all $\mathsf{b}\in\mathsf{J}$ and all $K\ge K_{\rm bf}$ with $\lceil K/1260\rceil\ge d$; R_d and the rate of Prop. 10.4(b) are explicit and continuous in b; the balls J'_d may move with b.

So Steps 1 and 2 hold uniformly on J, $\mathcal{W}_K\to\mathcal{W}$ uniformly, each $\mathcal{W}_K$ is continuous in b, and the same bounds hold at labels $\mathsf{b}_K\in\mathsf{J}$.

Step 3: every torus and label. Given $m\in\mathbb{Z}$ and $\mathsf{b}>0$, for large s the number $g_s=(L^s\mathsf{b}-\bar B-1)^{-1/2}$ is a reference coupling below the thresholds of the frame, $L^s\mathsf{b}\in\mathsf{I}(g_s)$, and $2L^{m+s}\in\mathfrak{L}(g_s)$; the exponent of Prop. 10.6 is the same for all g, κ is invariant under dilations, and Prop. 10.9(a) gives $\mathcal{W}_K^{(\ell,\mathsf{b})}(\mathsf{C})=L^{s\kappa}\mathcal{W}_{K-s}^{(L^s\ell,L^s\mathsf{b})}(L^s\mathsf{C})$; so the limit exists and obeys the law of dilations.

Step 4: the floor. For $C_i=n_1a_i\times n_2a_i$, Prop. 10.6(b) applies with $n_{C_i}=4$ and c_{C_i} fixed, both invariant under dilations, with $s_0(i)=\max\{i,s'_0\}=i+O(1)$ and no distance term; its bound is at most $C(\mathsf{J})\frac{a_i}{\mathsf{b}}L^{-\vartheta(K-t)}\le C(\mathsf{J})L^{-\vartheta(K-t-i)}$, an error that depends on the scale of the step only through a_{K-t}/a_i; it holds for m_{t+1}-almost every W, which is all that Prop. 10.8(a) asks, so Step 1 runs with $C_\eta(i)=C(\mathsf{J})L^{\vartheta i}$ and $s_0(i)$ in the place of the numbers of Prop. 10.6(a). Let i be so large that $(n_1^2+n_2^2)a_i^2<\ell^2/4$, so that C_i lies in a ball of radius $<\ell/4$ and is a polygon, that $\kappa(C_i)\le k|C_i|/\mathsf{b}_-<\vartheta$ for all $\mathsf{b}\in\mathsf{J}$, that $s_0(i)\le d_i$, which holds for large i as $d_i\ge\frac76 i$, and that $\theta_{d_i}\le1$, which holds for large i as $\theta_{d_i}\to0$ (below).

We run Prop. 10.8 down to the depth $d_i=\lceil(\frac43-\frac u2)i\rceil$. There

$$\theta_{d_i}=2B_3\sup_{K>d_i,\mathsf{J}}\varepsilon_{K-d_i}L^{2d_i}n_1n_2a_i^2\le c\,L^{\frac32d_i-2i}(1+d_i)^{\mathfrak{p}_0}\to0,$$

as $\frac32d_i\le(2-\frac34u)i+\frac32$ and $\varepsilon_{K-d}\le A_0(2a_d/\mathsf{b}_-)^{1/2}(\log(\mathsf{b}_-/2a_d))^{\mathfrak{p}_0}$ for large d. The numbers of the free field of C_i tend to 0: $\eta_i\to0$ by Prop. 10.7(a), so $e^{-\frac{3}{2\mathsf{b}_-}\eta_i}\le a_K^{-\kappa}\Xi_{K-d_i}$ (as $v_{K,d}\ge0$), $c_\Xi\to1$ and $L^{\kappa(d_i+1)}\to1$ ($\kappa d_i\le c\,a_ii$); and $\mathsf{E}'_{d_i}\to0$, as $\epsilon_1\le2C(\mathsf{J})L^{\vartheta i}/(1-L^{-\vartheta/2})$ and $d_i\ge(\frac43-\frac u2)i$.

Prop. 10.8(a), (c), (d) give the display of (4): by (d) and $\theta_{d_i}\le1$, $W_{C_i}^{(K-d_i)}\ge1-\frac12\theta_{d_i}^2\ge\frac12$ on O_{K-d_i}, which carries m_{K-d_i}, so

$$Z^{-1}\int W_{C_i}^{(K-d_i)}\,\mathrm{d}m_{K-d_i}\ge(1-\frac12\theta_{d_i}^2)Z^{-1}m_{K-d_i}(\mathcal{V}_{d_i})\ge(1-\frac12\theta_{d_i}^2)(1-2\mathsf{V}_\ell C_{\rm r}(1)L^{-d_i})$$

by (c), whatever the sign of the last factor, and this integral, being ≥0, is multiplied by $a_K^{-\kappa}\Xi_{K-d_i}\ge e^{-\frac{3}{2\mathsf{b}_-}\eta_i}$; both ends are $1+o(1)$ uniformly in $K>d_i$ and $\mathsf{b}\in\mathsf{J}$. For the squares Q_i we use the mean form, Prop. 10.8(b), at one i: by Prop. 10.6(b) the mean defects are at most $C(\mathsf{J})L^{-\vartheta(K-t-i)}$.

With $d=i+n_Q$, choose n_Q and then i so large that $\kappa(Q_i)\le\frac\vartheta2$, the free factor $a_K^{-\kappa}\Xi_{K-d}$ lies within a factor $\sqrt2$ of $L^{\kappa d}$, $L^{\kappa(d+1)}\le\sqrt2$, the summed step errors $2C(\mathsf{J})L^{-\vartheta(n_Q+1)}/(1-L^{-\vartheta/2})$ are at most $\frac{1}{32}$, $2B_3L^{2n_Q}\sup_{K>d}\varepsilon_{K-d}\le1$ (so that the loop at the last level is at least $\frac12$ on tame fields), the mass of the measure without large fields is at least $\frac78$, and the rarity terms are at most $\frac{1}{32}$; then $\mathcal{W}_K(Q_i)\ge\frac{1}{\sqrt2}\cdot\frac12\cdot\frac78-\frac{1}{16}>\frac18$. For other rectangles we spread with Prop. 10.9(b): $\rho_K(s,T)=\mathcal{W}_K(s\times T)$ has $\rho_K(s,\frac{T'+T''}{2})^2\le\rho_K(s,T')\rho_K(s,T'')$, the powers of a_K matching because lengths add; so $T\mapsto\log\rho_K(s,T)$ is convex, $\rho_K(s,nT)\ge\rho_K(s,2T)[\rho_K(s,2T)/\rho_K(s,T)]^{n-2}$, and a finite lim sup (Step 1) on a set of rectangles closed downwards together with a positive lim inf at one of them gives a positive lim inf at all rectangles of the range, going down by $\log\rho(T)\ge2\log\rho(T')-\log\rho(2T'-T)$ and up by convexity, in each side.

Step 5: block fields; the static source. $H\circ\mathcal{M}^{K-s_1}$ pulls out of the quotients, $\Phi^{(t)}[W_\mathsf{C}\,H\circ\mathcal{M}^{K-s_1}]=\Phi^{(t)}[W_\mathsf{C}]\,H\circ\mathcal{M}^{K-t-s_1}$ for $K-t\ge s_1$; so Steps 1 and 2 run with the factor H, $\sup|H|$ multiplying the errors (Prop. 10.8(a)), and $Y_{d,C}$-means are those of $F_C^{[d]}$. For a rectangle we divide by $\mathcal{W}(C)>0$

(Step 4): the limit quotients $\omega(H) = \lim_K \langle W_C H(\mathcal{M}^{K-s_1}U)\rangle_K / \langle W_C\rangle_K$ form a real linear functional with $\omega(1) = 1$ on the bounded gauge invariant Borel functions of all block fields, consistent between depths, $\omega(H\circ\mathcal{M}) = \omega(H)$, because $H\circ\mathcal{M}^{d'-s_1}$ is the same function seen at a finer depth; at the depth s_1 it is bounded, $|\omega(H)| \le \sup|H|\,(c_\Xi L^{\kappa d'} + \mathsf{E}'_{d'})/\mathcal{W}(C)$ with $d' = \max\{s_0, s_1\}$, by the display of (5) with $|W_C^{(K-d')}| \le 1$, $Z^{-1} m_{K-d'}(\mathcal{V}_{d'}) \le 1$ and $a_K^{-\kappa}\Xi_{K-d'} \le c_\Xi L^{\kappa d'}$.

On the threads $H(V_{s_1}) = H(\mathcal{M}^{d-s_1}V_d)$ μ-almost surely, so $\int H(V_{s_1})Y_{d,C}\,\mathrm{d}\mu = \int F_C^{[d]}\,(H\circ\mathcal{M}^{d-s_1})\,\mathrm{d}\mu_d$, and Step 2, with $\mathsf{l}_d \asymp a_d^{-\kappa}$, gives the limits in d and the two-sided bound for $\int Y_{d,C}\,\mathrm{d}\mu$. For H not gauge invariant: for a gauge transformation v of Λ_K, $\mathcal{M}^{K-s_1}(U^v) = (\mathcal{M}^{K-s_1}U)^{v|\Lambda_{s_1}}$; averaging v over the Haar measure and using the invariance of the Wilson measure, of W_C, of μ and of $Y_{d,C}$ replaces H by $\bar H$.

Step 6: mixed functions. By Prop. 10.10(d) at $X = W_\mathsf{C}$, $|Q_K(z)| \le M_0\mathcal{W}_K(\mathsf{C},\theta\mathsf{C})^{1/2}$ on half the polydisc; $(\mathsf{C},\theta\mathsf{C})$ is a family (its polygons lie $2\delta_1$ apart) with $\kappa(\mathsf{C},\theta\mathsf{C}) = 2\kappa < \frac12$, so Step 1 bounds the right side uniformly in $K > s_0$. For the limit at real z we take the source off the loop by Prop. 10.10(d) applied to the difference of the loop and a function of the block field on the side of C; no step with a source is used. Let $\bar t = K - d$. As $c \in \frac12 a_{i_0}\mathbb{Z}$, θ is a lattice isometry of Iso_{i_0} ([7, §G.1]); let $d_1 := i_0 + \log_L M_1$, so that from the depth d_1 on θ maps Λ_d and the partition into the M_1-cubes of the depth d, cubes of side $M_1 a_d$, onto themselves ([7, Lem. G.1]).

Let d be at least $\max\{s_0(\mathsf{C}), s_0(\mathsf{C},\theta\mathsf{C}), d_0, d_1\}$, $s_0(\cdot)$ the number of Prop. 10.6(a) at the family named, and so large that $M_1 a_d \le \frac{\delta_1}{8}$ and $2\hat\varepsilon_d \le \epsilon_{\mathrm{lc}}$, so that $2\varepsilon_{K-d} \le \epsilon_{\mathrm{lc}}$ for every $K > d$ and Thm. AD(5) applies at d; let $K > d$. Then θ commutes with the block averages, $\mathcal{M}^j(\theta U) = \theta\,\mathcal{M}^j U$ for $j \le \bar t$, and carries the constraint sets, criticality, regularity and the backgrounds over V to those over θV ([7, Lem. G.1]); ϖ is θ-invariant, so $\mathsf{O}_{\bar t}$ is; and the lattice Green function is θ-invariant and the linearised block averages are θ-covariant ([7, Lem. G.1]), so the projections Π_n commute with θ and the lattice Coulomb energies are θ-invariant.

Hence $W_\mathsf{C}^{(\bar t)}(\theta^{-1}V) = W_{\theta\mathsf{C}}^{(\bar t)}(V)$ on $\mathsf{O}_{\bar t}$ and $\Xi_{\bar t}(\theta\mathsf{C}) = \Xi_{\bar t}(\mathsf{C})$. Let $\mathcal{Y}_{K,d}(V)$ be the loop of C at the background localised to a slab Ω' on the side of C as in Thm. AD(5); then $Y = \Xi_{\bar t}(\mathsf{C})\mathcal{Y}_{K,d}(\mathcal{M}^{\bar t}U)$ is a function of the bonds on the side of C, and $\sup_{\mathsf{O}_{\bar t}}|W_\mathsf{C}^{(\bar t)} - \mathcal{Y}_{K,d}| \le \epsilon_d^{\mathrm{loc}}$, the bound of Thm. AD(5). Now $\|W_\mathsf{C} - Y\|_\theta^2 = \langle W_\mathsf{C} W_{\theta\mathsf{C}}\rangle_K - 2\langle W_\mathsf{C}\,\theta Y\rangle_K + \langle Y\,\theta Y\rangle_K$, and three towers without source evaluate the three means: Prop. 10.8(a) for the family $(\mathsf{C},\theta\mathsf{C})$, whose loop at the background is $W_\mathsf{C}^{(\bar t)}W_{\theta\mathsf{C}}^{(\bar t)}$ and whose factor is $\Xi_{\bar t}^2$; Prop. 10.8(a) for C with the gauge invariant factor $\theta\mathcal{Y}_{K,d}$ of the block field; and Prop. 10.8(c) for the function $\mathcal{Y}_{K,d}\,\theta\mathcal{Y}_{K,d}$ of the block field.

The main terms add up to $\Xi_{\bar t}^2 Z^{-1}\int[W_\mathsf{C}^{(\bar t)}(W_{\theta\mathsf{C}}^{(\bar t)} - \theta\mathcal{Y}_{K,d}) - (W_\mathsf{C}^{(\bar t)} - \mathcal{Y}_{K,d})\,\theta\mathcal{Y}_{K,d}]\,\mathrm{d}m_{\bar t}$, of modulus at most $2\Xi_{\bar t}^2\epsilon_d^{\mathrm{loc}}$. With $a_K^{-\kappa}\Xi_{\bar t} \le c_\Xi L^{\kappa d}$ and the rarity bound at $p = 2\kappa+1$,

$$a_K^{-2\kappa}\|W_\mathsf{C} - Y\|_\theta^2 \le \sigma_d := c_\Xi^2 L^{2\kappa d}\big[2\epsilon_d^{\mathrm{loc}} + 2\mathsf{V}_\ell C_{\mathrm{r}}(2\kappa+1)L^{-(2\kappa+1)d}\big] + \mathsf{E}'_d(\mathsf{C},\theta\mathsf{C}) + 2c_\Xi L^{\kappa d}\mathsf{E}'_d(\mathsf{C}),$$

free of K, and $\sigma_d \to 0$ because $2\kappa < \vartheta < \frac12$ and $\epsilon_d^{\mathrm{loc}}$ falls faster than every power of a_d. At real z, Prop. 10.10(d) at $X = W_\mathsf{C} - Y$ gives $|Q_K(z) - a_K^{-\kappa}\Xi_{\bar t}\,\mu_d^{K,w_z}(\mathcal{Y}_{K,d})| \le M_0\sigma_d^{1/2}$. By Prop. 10.10(b) the probability $\mu_d^{K,w}$ gives at most $\sum_{t\le\bar t}\mathsf{w}_t^w \le 2\mathsf{V}_S C_{\mathrm{r}}^S(\kappa+1)L^{-(\kappa+1)d}$ to the complement of $\mathsf{O}_{\bar t}\cap J'_d$, since the sourced measure without large fields has the null sets of $m_{\bar t}$ and is carried by $\mathsf{O}_{\bar t}\subset J'_d$; on $\mathsf{O}_{\bar t}\cap J'_d$, $|\mathcal{Y}_{K,d} - F'_d| \le \epsilon_d^{\mathrm{loc}} + \sup_{J'_d}|w_\mathsf{C}^K - F'_d|$ with $F'_d = \mathbf{1}_{J'_d}\prod_i W_{C_i}^{[d]}$ (Prop. 10.4(e)). With $a_K^{-\kappa}\Xi_{\bar t} \to \mathsf{l}_d$ and $\mu_d^{K,w} \to \mu_d^w$ (Prop. 10.10(c)),

$$\limsup_K |Q_K(z) - \mathsf{l}_d\,\mu_d^{w_z}(F'_d)| \le M_0\sigma_d^{1/2} + c_\Xi L^{\kappa d}\epsilon_d^{\mathrm{loc}} + 4c_\Xi\mathsf{V}_S C_{\mathrm{r}}^S(\kappa+1)L^{-d} \to 0,$$

so $\lim_K Q_K(z) = \lim_d \mathsf{l}_d\,\mu_d^{w_z}(F'_d)$ exists at every real z of half the polydisc. A bounded family of holomorphic functions that converges on a real cube converges locally uniformly to a holomorphic limit (Vitali's theorem); so $Q_K \to Q$, the derivatives converge, and Prop. 10.10(e) with the limits

S_m of YM3_I:(2212), for pairwise disjoint supports, gives the limit of $\mathcal{W}_K(\mathsf{C};f)$. For two orthogonal mirrors the same argument runs with the four images of C and Prop. 10.10(d) twice, on the quarter $\sum_q |z_q|\|f_q\|_\sharp < \rho_D/4$ of the polydisc, the towers at the subfamilies of $(\mathsf{C},\theta_1\mathsf{C},\theta_2\mathsf{C},\theta_1\theta_2\mathsf{C})$, of κ-value up to $4\kappa<\frac12$, and a region Ω' that is the intersection of two slabs; the limit then exists at the real z with $|z_q|<\rho_D/(4n\max_q\|f_q\|_\sharp)$, a real cube in that quarter, and Vitali's theorem applies there.

For $n=1$, $\mathcal{W}_K(\mathsf{C};f)$ is linear in f; as each coordinate extent of C is $<\frac{\ell}{2}$, every point outside the least closed coordinate box that holds C lies, with a neighbourhood, beyond a coordinate mirror that misses the box, with $c\in\frac12 a_i\mathbb{Z}$ for some i; a Lipschitz partition of unity subordinate to finitely many such mirrors cuts f into pieces, each of which the case of one mirror covers. With walls, the number of boxes is even in every direction that the walls cut and the walls lie in $\frac12 a_i\mathbb{Z}$, so that both fixed planes c and $c+\frac{\ell}{2}$ of every wall mirror are walls and each box has exactly one image in every box; and each f_q is supported in one box, a function whose support misses the walls being the sum of its pieces in the boxes, with the derivatives in its variable the sums of those in the variables of its pieces.

Then Prop. 10.10(d) on the polydisc $\sum_q|z_q|\|f_q\|_\sharp<\rho_D/N$, Step 1 for each family $\mathsf{C}^{\mathrm{f}}_\iota$, of κ-value $N\kappa(\mathsf{C}_\iota)<\frac12$, and Cauchy's estimates bound the derivatives $\partial_A Q_K(0)$ uniformly in K, for all supports off the walls; by Prop. 10.10(e) this bounds $\mathcal{W}_K(\mathsf{C};f_1,\dots,f_n)$ uniformly in K for $n=1$, where $\mathcal{W}_K(\mathsf{C};f_1)=\partial_1 Q_K(0)$, and for pairwise disjoint supports, where the cumulants $S^{\mathsf{T}}_{|\beta|,K}(f_\beta)$ of the blocks of at least two elements converge YM3_I:(2212) and so are bounded in K. The law of dilations and the positive form with mixed entries hold at every cutoff (Prop. 10.9(a), (b)) and pass to the limits; for the latter the plane $x_\mu=\hat c+\frac{\ell}{4}$, $\hat c\in\frac12 a_{i_0}\mathbb{Z}$ the place of the mirror θ' of (6), a fixed plane of the coordinate mirror with the place $\hat c+\frac{\ell}{4}$, which lies in $\frac12 a_{i_0}\mathbb{Z}$ as $\frac{\ell}{4}=\frac12 L^m$, parts polygons and supports, so that every entry converges.

Step 7: beside the loop. Let $X=W_C-\psi$, a function of the bonds on the side of C, and $Y=H-\langle H\rangle_K$, a real function of the bonds of S. Reflection positivity at the site mirror and Schwarz' inequality (as in Prop. 10.9(b)) give $|\langle XY\rangle_K|^2\le\langle X\,\theta X\rangle_K\langle\theta Y\,Y\rangle_K$, with $\langle X\,\theta X\rangle_K=\langle W_C W_{\theta C}\rangle_K-\psi^2\ge0$ by the θ-invariance of the measure, $\langle\theta Y\,Y\rangle_K=\langle\theta H;H\rangle_K\ge0$, and $\langle XY\rangle_K=\langle W_C;\mathcal{O}(f)\rangle_K-\langle W_C;H_0\rangle_K$; divide by ψ. The explicit ceiling uses the law of the truncated two-point function of the action density, Thm. YM3_I:AV.

Let, as in (7), $g\le\gamma_W$ in the frame, γ_W the ceiling of equation YM3_I:(2346); let $\vartheta_{\mathrm{sh}}\in(0,1)$ be a shape parameter, and $\mathsf{d}\in(0,\mathsf{d}_0(g,\vartheta_{\mathrm{sh}})]$ with $K-s(\mathsf{d})\ge1$, $\mathsf{d}_0(g,\vartheta_{\mathrm{sh}})$ the threshold of equation YM3_I:(2350) and $s(\mathsf{d})$ the witness depth of equation YM3_I:(2321) at g. As $\mathsf{b}\in\mathsf{I}(g)$, in the frame $(K,L^K\mathsf{b})$ is a first-passage datum at g with the label b (Cor. YM3_I:74.2), and the side ℓ of the frame lies in $\mathfrak{L}(g)$. Then for every test pair $(h_1,h_2)\in\mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta_{\mathrm{sh}})$ on a torus T_ℓ with $20\mathsf{d}\le\ell$ ([7, Lem. I.1]: non-negative Lipschitz functions, neither 0, with supports in closed balls of radius d at a distance between d and $6\mathsf{d}$, $\int h_j\ge\vartheta_{\mathrm{sh}}\|h_j\|_\infty\mathsf{d}^3$ and $\operatorname{Lip}h_j\le\|h_j\|_\infty/(\vartheta_{\mathrm{sh}}\mathsf{d})$)

$$S^{\mathsf{T}}_{2,K}(h_1,h_2)=a_K^{-2}\sum_{p,q}h_1(x(p))h_2(x(q))\langle W_{\partial p};W_{\partial q}\rangle_K=\frac{9}{16\pi^2}\mathsf{b}^{-2}\iint h_1(x)h_2(y)|x-y|^{-6}\,\mathrm{d}x\,\mathrm{d}y\,(1+\epsilon'),\qquad|\epsilon'|\le\epsilon_{\mathrm{tp}},$$

with $\frac{9}{16\pi^2}$ the constant c_2 of YM3_I:C, the double integral the free kernel YM3_I:(2240), and $\epsilon_{\mathrm{tp}}:=\bar\varepsilon_{g,\vartheta_{\mathrm{sh}}}(\mathsf{d})$, the tolerance equation YM3_I:(2349), one function for all these data, non-decreasing, tending to 0 with d and at most $\frac13$ on $(0,\mathsf{d}_0(g,\vartheta_{\mathrm{sh}})]$ (Thm. YM3_I:AV); so $|\epsilon'|\le\frac13$ and the bound $(\frac{4\pi}{3})^2\|h_1\|_\infty\|h_2\|_\infty$ of the free kernel of a test pair, two balls of radius d at a distance $\ge\mathsf{d}$ (equation YM3_I:(2247)), give $S^{\mathsf{T}}_{2,K}(h_1,h_2)\le\frac43(\frac{4\pi}{3})^2\frac{9}{16\pi^2}\mathsf{b}^{-2}\|h_1\|_\infty\|h_2\|_\infty$, the bound of Thm. YM3_I:AV.

As $\mathcal{O}(f)=a_K^{-1}\sum_p f(x(p))(1-W_{\partial p})$, $\langle\theta\mathcal{O}(f);\mathcal{O}(f)\rangle_K=a_K^{-2}\sum_{p,q}f(\theta x(p))f(x(q))\langle W_{\partial p};W_{\partial q}\rangle_K=S^{\mathsf{T}}_{2,K}(\theta f,f)$, and $\sqrt{9/16\pi^2}=\frac{3}{4\pi}$: for a test pair $(\theta f,f)$ the inequality at $H_0=0$ gives the bound of

(7) with the factor $1+\epsilon_{\mathrm{tp}}$. On the whole class of the law, $\frac{4}{3}(\frac{4\pi}{3})^2\frac{9}{16\pi^2}=\frac{4}{3}$ gives $\langle\theta\mathcal{O}(f);\mathcal{O}(f)\rangle_K\le\frac{4}{3}\mathsf{b}^{-2}\|f\|_\infty^2$, and the inequality at $H_0=0$ gives the ceiling $\frac{2}{\sqrt{3}}\mathsf{b}^{-1}\|f\|_\infty[\langle W_C W_{\theta C}\rangle_K/\psi^2-1]^{1/2}$ with no further condition.

The identity follows from the same expression of $\mathcal{O}(f)$:

$$\langle W_C\mathcal{O}(f)\rangle_K/\psi-\langle\mathcal{O}(f)\rangle_K=a_K^{-1}\sum_p f(x(p))[\langle W_{\partial p}\rangle_K-\langle W_C W_{\partial p}\rangle_K/\psi],$$

and $\langle W_{\partial p}\rangle_K>0$ (Prop. 10.9(c)). The derivative is $\frac{d}{d\mathsf{b}}=L^K\frac{d}{d\beta}$ applied to $\log\langle W_C\rangle_K$. For a rectangle with $2\kappa<\frac{1}{2}$ the three quantities converge by (6) at $n=1$ with (4), by (2) at the family $(C,\theta C)$, and by YM3_I:(2212), the supports of θf and f being disjoint.

Step 8: short distances. (a) We run the tower in its mean form, Prop. 10.8(b), down to the depth i of the corners, that is to the level $t_i=K-i$; when $\max\{n_1,n_2\}<L$ this is the level t_ϱ with $a_{K-t_\varrho}\le\varrho_i<La_{K-t_\varrho}$. The number s_0 of Prop. 10.6(b) for C_i is $\max\{i,s_0'\}$, s_0' formed from the constants, b_- and $n_{C_i}=4$; so $s_0=i$ for all large i, and every step that the tower uses, at a level $t<t_i$, has $K-t>i=s_0$. There Prop. 10.6(b) bounds the steps with a constant free of the size of the loop; weighed by the free factors, whose ratios between the levels are bounded by Prop. 10.7(c), they contribute $C_1\frac{|C_i|}{\mathsf{b}}(\frac{\varrho_i}{\mathsf{b}})^{\vartheta/2}$, and the rarity terms contribute $C_3\mathsf{V}_\ell C_{\mathrm{r}}(p)\varrho_i^{p-1/2}$.

At the depth i a tame background has $1-W_{C_i}^{(t_i)}\le\frac{1}{2}\theta^2=2B_3^2n_1^2n_2^2\varepsilon_{K-i}^2\le4B_3^2A_0^2\mathsf{n}^3\frac{\varrho_i}{\mathsf{b}}(1+\log\frac{\mathsf{nb}}{\varrho_i})^{2\mathfrak{p}_0}$ (Prop. 10.8(d), with $x_{K-i}\ge\frac{2}{3}\mathsf{b}/a_i$, $n_1^2n_2^2a_i\le\mathsf{n}^3\varrho_i$ and $\mathsf{b}/a_i\le\mathsf{nb}/\varrho_i$). The free factor down to that level is $a_K^{-\kappa}\Xi_{t_i}=\exp[-\frac{3}{2\mathsf{b}}(c_{\mathrm{lat}}|C_i|+\mathcal{J}_\ell(C_i)+E_{a_K}(C_i)-v_{K,i}(C_i))]$ (Prop. 10.7(a), (c)), and $0\le\frac{3}{2\mathsf{b}}v_{K,i}(C_i)\le\kappa\log^+\frac{\varrho_i}{a_i}+c\frac{|C_i|}{\mathsf{b}}\le\kappa\log L+c\frac{|C_i|}{\mathsf{b}}$, c depending on n_1, n_2: at the depth of the corners the logarithm of v (Prop. 10.7(b)) and that of $\mathcal{J}_\ell$ cancel up to $\frac{|C_i|}{2\pi}\log\max\{n_1,n_2\}$ and a multiple of $|C_i|$, and $\kappa\log\max\{n_1,n_2\}\le\kappa\log L+k\frac{|C_i|}{\mathsf{b}}\log\frac{\mathsf{n}}{L}$; $|\log(1-z)|\le2|z|$ for $|z|\le\frac{1}{2}$. (b) Let $a_*:=\varrho(\mathsf{b}/\varrho)^{\vartheta/(3+\vartheta)}$ and let d_* be the least depth with $a_{d_*}\le a_*$, so that $a_*/L<a_{d_*}\le a_*$; for small ϱ, $a_{d_*}>a_i$, and the depth d_* is coarser than the corners.

At all the levels with $a_{K-t}\le a_*$ the defect of one step is at most $C_*\frac{\varrho}{\mathsf{b}}(\frac{a_{K-t}}{\mathsf{b}})^\vartheta$ at m_{t+1}-almost every W, hence in the mean, $|\bar{\mathsf{D}}_{t+1}|\le C_*\frac{\varrho}{\mathsf{b}}(\frac{a_{K-t}}{\mathsf{b}})^\vartheta$ with $\bar{\mathsf{D}}_{t+1}$ of Prop. 10.8(b), C_* depending on the constants, c_0, ϑ and J alone: this is the bound of one step of (8)(b), shown at the end of this step. We stop the tower, in its mean form, Prop. 10.8(b), at the depth d_*. Prop. 10.7(c), valid at every depth, and (d) give the factor $e^{-\frac{3}{2\mathsf{b}}(\mathcal{I}_{a_K}(C,C)-v_{K,d_*})}$ with $0\le v_{K,d_*}(C)\le C^{\mathrm{bg}}A_C^2/a_{d_*}^3$, A_C the area of C; on tame fields $0\le1-W_C^{(K-d_*)}\le4B_3^2A_0^2\frac{A_C^2}{\mathsf{b}a_{d_*}^3}(1+\log\frac{\mathsf{b}}{a_{d_*}})^{2\mathfrak{p}_0}$ (Prop. 10.8(d)).

As $A_C\le100\varrho^2$ and $a_*/L<a_{d_*}\le a_*$, these are $O(\varrho^4/a_*^3)$ and $O(\frac{\varrho^4}{\mathsf{b}a_*^3}(1+\log\frac{\mathsf{b}}{a_*})^{2\mathfrak{p}_0})$, with constants that hold 10^4L^3 (and $\log L$ in the logarithm); the step errors add up to $O(\frac{\varrho}{\mathsf{b}}(\frac{a_*}{\mathsf{b}})^\vartheta)$; the terms of the large fields, $\sum_{t\le K-d_*}\Xi_{t-1}\omega_t$ with $|\omega_t|\le\mathsf{w}_t$, are by the rarity bound at $p=2$ at most $c\,\mathsf{V}_\ell C_{\mathrm{r}}(2)\,a_{d_*}^2\le c\,\mathsf{V}_\ell C_{\mathrm{r}}(2)\,a_*^2$ times the free factor, and $a_*^2=\mathsf{b}^2(\frac{\varrho}{\mathsf{b}})^{2-2\alpha}$ with $\alpha=\frac{\vartheta}{3+\vartheta}<\frac{1}{7}$, below the remainder as $2-2\alpha>1+3\alpha$; and $\frac{\varrho}{\mathsf{b}}(\frac{a_*}{\mathsf{b}})^\vartheta=\frac{\varrho^4}{a_*^3\mathsf{b}}=(\frac{\varrho}{\mathsf{b}})^{1+\frac{3\vartheta}{3+\vartheta}}$: the choice of a_* balances the two.

Prop. 10.7(a) and $|\mathcal{J}_\ell-\mathcal{J}|\le c\varrho^4\ell^{-3}$ finish; $\log(\mathsf{b}/\varrho)$, and not $\log\varrho$, is the logarithm that dilations leave invariant. The bound of one step at the levels with $a_{K-t}\le a_*$. At the levels finer than the lattice of the corners, $K-t>s_0=\max\{i,s_0'\}$, which is i for small ϱ, it is Prop. 10.6(b): its constant depends on the constants, $n_C=4$ and the chord-arc constant of C, which is bounded below in terms of c_0, its factor $|C|/\mathsf{b}$ is at most $40\varrho/\mathsf{b}$, and its bracket, without a distance term for one polygon, is at most $c\,(a_{K-t}/\mathsf{b})^\vartheta$ with c depending on the constants, ϑ and J.

At the remaining levels, $a_i\le a_{K-t}\le a_*$, at least as coarse as the lattice of the corners, $a_{K-t}\ge a_i>c_0\varrho/2L$ and $|C|\le40\varrho$, so that the loop passes a number of cubes of the level that is bounded in terms of L and c_0, and $|C|/a_{K-t}<80L/c_0$. There the proof of Prop. 10.6(b) runs

with the segments C_j of C, one for each cube of the level that the loop passes, in the place of the bonds of the level, down to one segment where the cells are larger than the loop. The shift of Prop. YM3_I:51.5 gives $W_C^{(t)}(V_W(B)) = W_C(e^{i\eta_t \mathbf{A}(B)} U_+)$ exactly for every bare loop (step 1 of the proof of Prop. 10.6(a)); so $W_C^{(t)}(V_W(B)) = w(\Upsilon(B))$ and $W_C^{(t+1)}(W) = w(0)$, with w as in Lem. 10.5(c), $g_j = U_+(C_j)$ the holonomy of U_+ along the segment, and the rotations $\Upsilon_j(B) = \frac{1}{i}\log[(e^{i\eta_t \mathbf{A}(B)} U_+)(C_j)\, U_+(C_j)^{-1}]$ of the segments.

Each segment is one insertion into the four-family Gaussian cluster expansion of Thm. YM3_I:R, applied to the two potentials in ν_W as in Step 3 of the proof of Prop. 10.6(a), with a linear part of norm of the order $g_t|C_j|/a_{K-t}$ and a remainder of the order $(g_t|C_j|/a_{K-t})^2$, so that $\log \nu_W(e^{(z,\Upsilon)})$ is a sum of local terms on a polydisc of radius of the order $a_{K-t}/(g_t|C|)$. The first cumulants, and the defects of the pair cumulants from their values $s_{jj'}\mathrm{R}(G_{j\to j'})$ at the trivial triple, are colour vectors and colour defects that vanish at the flat background, and Schwarz' lemma in the background gives them the factor ε_{t+1}, as in step 5 of the proof of Prop. 10.6(a); the pair values at the trivial triple sum to the slice $g_t^2 \eta_t(X_C, \mathcal{C}_{(t+1)} X_C)$ of the free Coulomb energy, as in its step 7, at depths coarser than the corners as well (Prop. 10.7(c) holds at every depth); and $\nu_W \geq 0$ for m_{t+1}-almost every W.

Lem. 10.5(c) then bounds the defect of the step by $E_1 + E_2 + E_3$ and the term of the label, at most $c\,|C|\mathsf{b}^{-2} a_{K-t}$; each of E_1, E_2, E_3 is at most $\frac{|C|}{\mathsf{b}}(\frac{a_{K-t}}{\mathsf{b}})^{1/2} y_t^{\mathfrak{p}}$, $y_t = \log x_t$, $\mathfrak{p}$ formed from $\mathfrak{p}_0$, times a polynomial in $|C|/a_{K-t}$ with coefficients formed from the constants, since a cumulant of order m carries the factor $(g_t|C|/a_{K-t})^m$, the first cumulants besides the amplitude $\delta_t = g_t \mathfrak{b}_t$ of the non-Gaussian potentials, the Gaussian part being even, the variables are at most $c\,\delta_t|C|/a_{K-t}$, g_t^2 is of the order a_{K-t}/b, and the first cumulants and the defects of the pair cumulants carry the factor ε_{t+1}, of the order $(a_{K-t}/\mathsf{b})^{1/2} y_t^{\mathfrak{p}_0}$.

As $|C|/a_{K-t} < 80L/c_0$, $|C| \leq 40\varrho$ and $(\frac{a}{\mathsf{b}})^{1/2} y^{\mathfrak{p}} \leq c\,(\frac{a}{\mathsf{b}})^{\vartheta}$ for $\vartheta < \frac{1}{2}$, the bound is at most $C_* \frac{\varrho}{\mathsf{b}}(\frac{a_{K-t}}{\mathsf{b}})^{\vartheta}$. No level coarser than a_* is used; at the levels with $a_{K-t}^3 \geq \varrho^2 \mathsf{b}$, coarser than a_*, the step bound holds by itself, since there $1 - \lambda_{t+1}$ and $1 - W_C^{(t)}$ on tame fields are at most of the order $\varrho^4/(\mathsf{b} a_{K-t}^3)$ up to logarithms, below $C_* \frac{\varrho}{\mathsf{b}}(\frac{a_{K-t}}{\mathsf{b}})^{\vartheta}$.

10.1.8. *Proof of Thm. AF.* We work in lattice units, $\beta = \mathsf{b}/a$, $n = |C|/a$ steps. By Lem. 10.12(a) the steps of C at distance $\geq 2\rho$ from the corners form an admissible set I of $n - 16\rho$ steps (of $n - 4\rho s$ steps for polygons with s pieces), whose slabs lie in disjoint cells. Lem. 10.12(g) at spin $\frac{1}{2}$, $c_i = \frac{3}{4}$, gives $|\langle W_C \rangle| \leq e^{-\mathcal{G}} \mathfrak{M}_\nu^{\omega_\rho}$: across each admissible step the bonds of the slab that cross the disc of radius ρ about it are rotated together with the profile ϕ_ρ, whose Dirichlet energy is about $2\pi/\log\rho$, at an action cost $\approx 2\pi\beta/\log\rho$; the information inequality in three colours gives the factor $e^{-k\log\rho/\beta}$ per slab, up to the second-order term of $\mathcal{G}$ and the constant 1.2144, and the bounds multiply exactly over the disjoint cells; with n slabs, $e^{-kn\log\rho/\beta} = \rho^{-k|C|/\mathsf{b}}$.

The second variation (Lem. 10.12(d)) limits ρ: its second member, summed over the disc, asks $\omega_\rho = 40\rho^4 \sum_i c_i/(\beta E_\rho)^2 \leq 1$, $E_\rho \sim 2\pi/\log\rho$; with $\rho = \lfloor (\frac{\beta}{\Lambda \log^3 \beta})^{1/4} \rfloor$ one has $\omega_\rho \leq c/\log\beta$, so that $\mathfrak{M}_\nu^{\omega_\rho}$, with Lem. 10.12(f), the corrections of $\mathcal{G}$ and the 16ρ omitted steps give the prefactor $6e^{3.1\mathsf{b}/\ell} 1.58^{|C|/\mathsf{b}}$, and $\rho^{-1} \leq 2(\Lambda \log^3\beta/\beta)^{1/4}$ gives the power $[\Lambda \frac{a}{\mathsf{b}} \log^3 \frac{\mathsf{b}}{a}]^{k|C|/4\mathsf{b}}$, the factor $2^{k|C|/\mathsf{b}}$ going into the prefactor. For the second bound let $\rho = 2^S \in (\frac{1}{2}\beta^{1/3}, \beta^{1/3}]$ and $p = k|C|/\mathsf{b} = kn/\beta < 3$, and keep the cells $T_\rho(c_i)$ of the $n - 16\rho$ admissible steps.

On each cell, for every dyadic $2 \leq r \leq \rho$, the moves with the profile ϕ_r, taken as 0 at $|y| \geq r$, rotate bonds of the cell alone, and Lem. 10.12(b)–(e), as in the proof of (g), give at every field $\prod_i \|Q_i\| \leq \min_r e^{-\mathcal{G}_r + Y_r}$, with $\mathcal{G}_r = \sum_i \frac{2c_i}{\beta E_r}(1 - \frac{2c_i}{\beta E_r})$, $Y_r = \sum_i 2c_i X_i^{(r)}/(\beta E_r^2)$ and $X_i^{(r)}$ the X_i of that proof formed with ϕ_r; by (b), $e^{-\mathcal{G}_r} \leq e^{\delta_0 + 1.2144 p} r^{-p}$, where $\delta_0 = (16k\rho\log\rho + pk\log^2\rho)/\beta$ pays for the 16ρ omitted steps and the second-order term of $\mathcal{G}_r$. The sharper estimate bounds the mean of Y_r instead of an exponential moment: in comb frames $a(H_{e'})$ at a rung e' of height $t_{e'}$ is $1 - W$ of the boundary of the strip $S_{e'}$ of $1 \times |t_{e'}|$ plaquettes, whose mean is $1 - w(1, |t_{e'}|) \leq$

$1-w(1,1)^{|t_{e'}|}\le|t_{e'}|\,\nu(a_p)$ by Prop. 10.9(c), in the place of the bound $|t_{e'}|\sum_{q\in S_{e'}}a_q$ at every field; so $\nu(X_i^{(r)})\le c\,r^3\nu(a_p)$, r^3 in the place of r^4, and $\nu(Y_r)\le c\,p\,r^3\log^2 r\,\bar a$ with $\bar a=\nu(a_p)\le\beta^{-1}\log\mathfrak{M}_\nu$ (Lem. 10.12(f)): the plaquette mean enters linearly, and this bound of a mean takes the place of the condition $\omega_\rho\le1$ and of the weight $40\rho^4$.

Let r_* be the largest dyadic $r\le\rho$ with $Y_r\le p$; on $\{r_*=r\}$, $r<\rho$, $Y_{2r}>p$, an event of probability at most $\nu(Y_{2r})/p$ by Markov's inequality, and on the event that $Y_r>p$ at every r the bound 1 serves. So, as Lem. 10.12(h) states,

$$|\langle W_C\rangle|\le e^{\delta_0+2.2144p}\rho^{-p}\Big[1+\frac{72\,\bar a\rho^3\log^2\rho}{2^{3-p}-1}\Big]+35\,\bar a,$$

the sum over the dyadic $r<\rho$ of $r^{-p}(2r)^3\log^2(2r)$ being geometric and led by its last term exactly when $p<3$. With $\rho^3\le\beta$, so that $\bar a\rho^3\le\log\mathfrak{M}_\nu$, with $\rho^{-p}\le2^p\beta^{-p/3}$, $\bar a\le\beta^{-p/3}\log\mathfrak{M}_\nu$ and $\log\mathfrak{M}_\nu\le192\frac{\mathsf{b}}{\ell}+4(14.15+3\log\beta)$, where $192\frac{\mathsf{b}}{\ell}\le\frac{384k}{p}$ as $|C|<2\ell$, this is at most $C(\frac{k|C|}{\mathsf{b}})(\frac{a}{\mathsf{b}})^{|C|/4\pi\mathsf{b}}\log^3\frac{\mathsf{b}}{a}$ for $\beta\ge\beta_2$, the threshold, depending on p alone, from which on $n\ge16\rho+4$ and $\delta_0\le1$: $\frac{p}{3}=\frac{|C|}{4\pi\mathsf{b}}$, and the third logarithm is $\log^2\rho$ times $\log\mathfrak{M}_\nu$.

For the Gibbs states the strips of even height are treated by the reflections in the planes of sites, those of odd height by $\sqrt{a(gh)}\le\sqrt{a(g)}+\sqrt{a(h)}$. For the spin j, $c_j=j(j+1)$ replaces $\frac34$, that is $k\mapsto\frac43c_jk=\frac{j(j+1)}{\pi}$, and $\kappa_j\le20\log\beta$ keeps ω_ρ admissible. For a family of such polygons whose pieces, those of different polygons included, are at sup-distance at least $(2\rho+1)a$ when they have no common corner, Lem. 10.12(a) applies to the whole family and gives one admissible set of $n-4\rho s$ steps, n and s the numbers of all the steps and of all the pieces, and Lem. 10.12(e), (g) bound $\langle\prod_i W_{C_i}G\rangle$ in the same way, a factor G of the bonds farther than ρa from the polygons being a function of the bonds off B; a fixed family at positive distances has this property as soon as $(2\rho+1)a$ is below these distances, that is for all small a, which is what the limits as $a\to0$ use.

The proof uses of ν only the conditional laws of finite sets of bonds, reflection positivity and invariance (through Lem. 10.12(f), and for the second bound through Prop. 10.9(c) for the strips); for the Gibbs states the bound of (f) gives 3 in the place of $6e^{3.1\mathsf{b}/\ell}$. So all characters of the holonomy of non-zero spin have means tending to 0: its class tends to the Haar law of SU(2), whose image under $\frac12$ tr has the density $\frac{2}{\pi}\sqrt{1-w^2}$ (the χ_j are the Chebyshev polynomials of the second kind in W_C, orthogonal for σ, and polynomials are dense among the continuous h); and $\langle W_C^2\rangle=\frac14(1+\langle\chi_1(U(C))\rangle)\to\frac14$, from $W_C^2=\frac14\chi_{1/2}(U(C))^2=\frac14(1+\chi_1(U(C)))$.

For a rectangle $\langle W_C\rangle>0$ (Prop. 10.9(c)), so $\mathrm{Var}(W_C/\langle W_C\rangle)=\langle W_C^2\rangle/\langle W_C\rangle^2-1=(1+\langle\chi_1(U(C))\rangle)/(4\langle W_C\rangle^2)-1$; as $\langle\chi_1(U(C))\rangle\to0$ and the first bound gives

$$\langle W_C\rangle^{-2}\ge(6e^{3.1\mathsf{b}/\ell}1.58^{|C|/\mathsf{b}})^{-2}[\Lambda\frac{a}{\mathsf{b}}\log^3\frac{\mathsf{b}}{a}]^{-k|C|/2\mathsf{b}}$$

, the variance tends to infinity at least at this rate. If c_aW_C had a limit law while $|c_a\langle W_C\rangle|\ge c>0$, then $|c_a|\ge c/|\langle W_C\rangle|\to\infty$; as the law of W_C tends to the semicircle law, which gives no mass to 0, $\langle\mathbf{1}\{|c_aW_C|>R\}\rangle\to1$ for every R, and the laws of c_aW_C are not tight: a contradiction.

The rotations touch only bonds within $\rho a\to0$ of the polygons, act on several polygons at once and leave G unchanged, so $\langle\prod_i\chi_{j_i}(U(C_i))\,G\rangle\to0$ unless all $j_i=0$, which is the asymptotic independence. The floor: $\langle W_C\rangle\ge\langle W_{\partial p}\rangle^{A_C/a^2}$ by Prop. 10.9(c), and $\langle W_{\partial p}\rangle=1-\langle a_p\rangle\ge1-\beta^{-1}\log\mathfrak{M}_\nu\ge e^{-\frac32\beta^{-1}\log\mathfrak{M}_\nu}$ when $\beta^{-1}\log\mathfrak{M}_\nu\le\frac12$ (Lem. 10.12(f)), with $\frac{A_C}{a^2}\beta^{-1}=\frac{A_C}{a\mathsf{b}}$; for the Gibbs states the same argument, with reflections in planes of sites and even numbers of steps, and $\frac32(28.3+6\log\beta)\le43+9\log\beta$.

10.1.9. *Scope of the conclusions.* At a fixed depth the variable $Y_{d,C}$ of Thm. AD is the smoothed loop on $\{V_d\in J_d\}$ and 0 off it, and Prop. 10.2(d) bounds the probability of $\{V_d\in J_d\}$ below; the starting depth is random and by its form no stopping time, since $\{\tau\le D\}$ involves all the

coordinates of index $\geq D$; the loop of the depth d uses $H^{\infty}_{d,n}(V_d)$ and not the finer members of the thread, so that $W_C^{[d]}(V_d)$ and $W_C^{[d+1]}(V_{d+1})$ are two different functions on the threads, and the limit in the depth is a limit of means and of quotients of means (Thm. AE(2), (5)); for the sharp loop itself no multiple has both a limit law and a mean that stays away from 0, by the non-tightness consequence of the semicircle law (Thm. AF).

The bounds of the convergence in n and in K hold the factor $L^d \varepsilon_{[d]} = A_0 \mathsf{b}^{-1/2} L^{d/2} (\log x_{[d]})^{\mathfrak{p}_0}$, which grows with d. On the side ℓ depend the tails of τ, which hold ℓ^3 or $e^{C\ell^3}$; the floors $\mathsf{p}^{[d]}$, $\mathsf{q}^{[d]}$; the speed of $K \to \infty$, as $a_s(K)$ holds $(\ell L^s)^{3/2}$, $\mathfrak{r}_K$ holds ℓ^3 and K_{bf} depends on ℓ; and V_ℓ; the event $\{V_d \in J_d\}$ asks all the $3(\ell L^d)^3$ plaquettes. Free of ℓ are the bounds in n on the tower, on curves and on small rectangles (Prop. 10.4(a), the first bound of (c)), the locality bound of Thm. AD(5), the numbers c_{lat}, c_Q, c and c_3 (Prop. 10.7), and, for $\ell \geq \ell_0$, all the ceilings of Thm. AF.

Thm. AE concerns $|\mathsf{C}| < \frac{2\pi}{3}\mathsf{b}$ and is a perimeter law, two-sided for rectangles. Three free photon fields at the label b have the same features: their block curvatures obey at each depth, uniformly in K, bounds of the same form as those of Prop. 10.2(c) and of $R_d^{\natural}$ (Prop. 10.11(a)), and their sharp loops have the same exponent κ, the same perimeter constant $c_{\mathrm{lat}}|C| + \mathcal{J}_\ell(C)$ and the same factor λ_{t+1} of one step as the loops of SU(2) in Thm. AE and Prop. 10.6 (Prop. 10.11(b), (c)). Thm. AE(7) is an upper bound: for three free photon fields the response of the action density is $-\frac{3}{2\mathsf{b}^2} \int f |B_{C,\ell}|^2 \, \mathrm{d}x$, with the field $B_{C,\ell}$ of C on the torus, within $c|C| \operatorname{diam}(C) \ell^{-3}$ of its Biot–Savart field B_C in the ball that holds the polygon (Prop. 10.11(d)).

The quotients of Thm. AE(5) formed for three free photon fields, $\lim_K \mathbb{E}[W_C^{\mathrm{fr}} H] / \mathbb{E}\, W_C^{\mathrm{fr}}$ for H a bounded function of a block field, take negative values at functions $H \geq 0$: by Prop. 10.11(c) the conditional mean of W_C^{fr} given the block field of the level t is $\Xi_t W_C^{\mathrm{fr},(t)}$, and $W_C^{\mathrm{fr},(t)}$ is a product of three cosines of independent centred Gaussian variables of the variance $v_{K,K-t}(C)/\mathsf{b}$, which grows like $\frac{|C|}{2\pi\mathsf{b}} \log \frac{1}{a_{K-t}}$ (Prop. 10.7(b)); so $W_C^{\mathrm{fr},(t)}$ is negative with positive probability, and the quotient at the indicator of $\{W_C^{\mathrm{fr},(t)} < 0\}$ is negative.

Alternatively, the levels $\varrho < a_{K-t} \leq a_*$ of the proof of Thm. AE(8)(b), whose cells are larger than the loop, can be traded for a law of one point at the depth of the size of the loop: $1 - W_C^{(t)}$ is, on tame fields, a quadratic form in the curvature, and the law follows from a bound on $|Z^{-1} \int (1 - W_C^{(t)}) \, \mathrm{d}m_t - \frac{3}{2\mathsf{b}} v_{K,K-t}(C)|$ at that level, the mean of one bounded function of the block field against its free value.

11. Centre flux

For $SU(2)$, reverse the plaquette trace on a column winding around a periodic box. A twist class has three signs, one for each coordinate two-torus. On choosing a time direction, write the class as (k, m), where $k \in \mathbb{Z}_2^2$ is electric and $m \in \mathbb{Z}_2$ is magnetic. Let T_m be the transfer operator on a spatial section with magnetic twist m, and let C^k be the product of its two centre-flip operators.

Sketch J (Positive electric-flux sectors). The twisted partition functions have the decomposition

$$Z[(k,m)] = \operatorname{Tr}(T_m^N C^k), \qquad Z_{e,m} = \frac{1}{4} \sum_k (-1)^{e \cdot k} Z[(k,m)] = \operatorname{Tr}(P_e T_m^N) \geq 0,$$

where N is the number of time layers and $P_e = \frac{1}{4} \sum_k (-1)^{e \cdot k} C^k$. Changing the time direction relates these sectors by a finite Fourier transform. Reflection-positive mixtures are obtained when their electric Fourier weights are nonnegative in each time direction. ■□

Sketch of the argument. Split the plaquette action into its spatial and crossing parts. The crossing kernel is positive, and multiplication by the square root of the spatial Boltzmann weight gives $T_m \geq 0$. The centre flips are commuting unitary involutions preserving this operator. Their four

character projections are the P_e. Closing the time direction with the chosen electric twist inserts C^k in the trace, and Fourier inversion gives the displayed formulas.

The same box can be traced in a second coordinate direction. Comparing the two formulas exchanges one electric index with the magnetic index while retaining the common-plane flux. This crossing relation supplies inequalities between the sector traces. The strictly positive transfer kernel has a simple largest eigenvalue in the zero-electric-flux sector. Ratios of eigenvalues then give the electric and magnetic free energies over the chosen spatial section.

For a mixture with weights $w(k,m)$, define its electric-sector coefficients by

$$\widehat{w}(e,m) = \sum_k (-1)^{e\cdot k} w(k,m).$$

Nonnegativity of these coefficients makes the reflected trace positive in that time direction. Requiring it for each direction gives the stated class of mixtures. A single twisted class does not automatically satisfy all these positivity conditions.

The continuum sector ratios require a further comparison of the twisted and untwisted small-field integrations. The source leaves full-Hessian control on doubled domains and a quantitative twisted-window estimate as additional work for its strongest uniform asymptotics. These are steps to establish, not consequences of the scalar Gaussian quadratic estimate. A gap found over one fixed section also does not establish the infinite-volume mass gap.

Finally, the direction of the confinement asymptotic matters. In the finite-size area-law regime the twisted quotient Z^-/Z approaches one. Its defect $1 - Z^-/Z$, or the associated odd-flux weight, is the small quantity. The trace decomposition makes this distinction explicit before any limiting argument is attempted.

11.1. **Centre flux and twisted partition functions.** Centre twists admit a transfer-operator description in terms of electric-flux sectors. We establish the crossing and Fourier relations, the positivity properties of admissible mixtures and the continuum limits that follow from the construction estimates.

A single twisted class need not be reflection positive in every direction. The flux asymptotics remain sketches at the Hessian, window and transport steps specified below. We also distinguish a gap over a fixed small section from a gap of the infinite-volume continuum theory. In the conditional area-law regime the quotient Z^-/Z tends to 1; the small quantity is its defect.

The continuum centre-flux argument requires two estimates: **(Hess)** for the full fluctuation Hamiltonian on a doubled domain, and **(Flux-window)** for the probability of a twisted window. They are described below but are not consequences of the Gaussian quadratic estimate alone.

11.1.1. *Notation and hypotheses. The lattice.* Throughout, $G = \mathrm{SU}(2)$, the directions are $\mu \in \{1,2,3\}$, and tr is the trace of 2×2 matrices, not halved. A box is a triple $N = (N_1, N_2, N_3)$ of integers ≥ 3 with its periodic lattice; $|N| = N_1N_2N_3$, a cube has equal sides, and $N^{\times 3} = (N,N,N)$. $\partial U(p)$ is the plaquette variable of a plaquette p, $t_p = \operatorname{tr}\partial U(p)$, and dU is the product of the Haar probability measures on the bonds. A sign field γ on the bonds acts by $(\gamma U)(b) = \gamma(b)U(b)$, and $d\gamma(p)$ is the product of γ over the four bonds of p. For a sign field η on the plaquettes, $\beta \geq 0$ and a bounded Borel function F of the field,

$$A_\eta(U) = \sum_p (1 - \tfrac{1}{2}\eta(p)\operatorname{tr}\partial U(p)), \qquad Z_\beta[\eta;F] = \int F\, e^{-\beta A_\eta(U)}\,\mathrm{d}U, \tag{11.i}$$

and $A = A_\eta$ at $\eta \equiv 1$.

Classes. A class is $n = (n_{12}, n_{13}, n_{23}) \in \mathbb{Z}_2^3$; a plane $\mu\nu$ with $n_{\mu\nu} = 1$ is a twisted plane of n, and $n_{\mu\nu}$ also names the class whose one twisted plane is $\mu\nu$. A column of the plane $\mu\nu$ is the set of the plaquettes of that plane over one point of it; η^n is -1 exactly on one column of each twisted plane; $Z[n;F] = Z_\beta[\eta^n;F]$, $Z[n] = Z[n;1]$, $r[n] = Z[n]/Z[0]$, $F^n = -\log r[n]$, $\mu_n = Z[n;\cdot\,]/Z[n]$ and $\langle\cdot\rangle = \mu_0$. On a cube, r_i is $r[n]$ for a class with i twisted planes; it depends on i alone. A function of

the field is centre-blind if it is invariant under $U \mapsto \gamma U$ for every sign field γ on the bonds. A sign field s on the plaquettes is closed if its product over the faces of each unit cube is 1; its product over a lattice plane parallel to $\mu\nu$ is then $(-1)^{n_{\mu\nu}}$ for one class $n(s)$, and $[n] \ni \eta^n$ is the set of the closed s with $n(s) = n$. A bounded function $G(U, s)$ of a field and a closed sign field, Borel in U, is covariant if $G(\gamma U, s\,\mathrm{d}\gamma) = G(U, s)$ for all sign fields γ on the bonds; for it $Z_\beta[s; G(\cdot, s)]$ is the same number $\mathsf{M}_n[G]$ for all $s \in [n]$, and $\mu_n(G) = \mathsf{M}_n[G]/Z[n]$.

Transfer operators. With μ as the time and $\nu < \rho$ the other two directions we write $n = (k, m)_\mu$, with the electric part $k = (n_{\mu\nu}, n_{\mu\rho})$ and the magnetic part $m = n_{\nu\rho}$. The section is the periodic lattice $N_\nu \times N_\rho$, and a layer is the field on one section. On L^2 of the fields of the section, $T_m = D_m \mathsf{K} D_m$, where K is the crossing operator defined in Section YM3_I:85 by integrating the time-like bonds between two layers and $D_m^2 = e^{-\beta A_\Sigma^m}$ is the Boltzmann weight of the section Σ, with the plaquette whose corner is a fixed site ξ_* of the section reversed if $m = 1$; a section with $m = 1$ is called twisted. For a site ξ of the section and $j \in \{\nu, \rho\}$, (ξ, j) is the bond from ξ to $\xi + e_j$, and the plaquette with the corner ξ has the bonds (ξ, ν), $(\xi + e_\nu, \rho)$, $(\xi + e_\rho, \nu)$, (ξ, ρ). The flip C_j reverses the bonds (ξ, j) with $\xi_j = (\xi_*)_j$, the bonds of direction j that leave the line of the section through ξ_* normal to j; $C^k = C_\nu^{k_1} C_\rho^{k_2}$ is composition with a map c^k of the fields; $P_e = \frac{1}{4}\sum_k (-1)^{e\cdot k} C^k$ for $e \in \mathbb{Z}_2^2$; and $Z_{e,m} = \mathrm{Tr}(P_e T_m^{N_\mu})$, written $Z_{e,m}^{[\mu]}$ where the time is to be named.

Numbers of a box. $F_0 = \log Z[0]$, $P = P(\beta) = \inf F_0/|N|$ over all boxes, the excess $\delta = F_0 - |N|P$, and $\delta^n = \log Z[n] - |N|P = \log r[n] + \delta$. Over a section and with the height M, $\mathsf{G}(M) = \log(\mathrm{Tr}\, T_0^M / \|T_0\|^M)$ and $\mathsf{d}(M) = 2F_0(M) - F_0(2M)$, the section being fixed. For integers $l \geq 3$, $D_q(l) = q^3 F_0(l^{\times 3}) - F_0((ql)^{\times 3})$, $D_2(l) = 8F_0(l, l, l) - F_0(2l, 2l, 2l)$ and $\mathsf{e}(l) = 2F_0(l, 2l, 2l) - F_0(2l, 2l, 2l)$; over a section (l_1, l_2), $\varepsilon_{\mathrm{v}}(l_1, l_2) = \log(\|T_0\|/\|T_1\|)$, and $\varepsilon_{\mathrm{v}}(l) = \varepsilon_{\mathrm{v}}(l, l)$. For a loop C, $W_C = \frac{1}{2}\mathrm{tr}\, U(C)$; $W_{R\times T}$ is W_C for the boundary of a rectangle $R \times T$ of a coordinate plane.

Gibbs states. For a box N, μ_N is the measure $\langle\cdot\rangle = \mu_0$ of the periodic box N (the index of μ_N is a box, that of μ_n a class); where it stands beside a measure on the fields of the infinite lattice, it is extended periodically. A Gibbs state at β on the infinite lattice $a\mathbb{Z}^3$, $a > 0$, is, as in Section YM3_I:91, a Borel probability measure ν on its fields with $\nu(\gamma_B F) = \nu(F)$ for every finite set B of bonds and every continuous function F of finitely many bonds, γ_B the Gibbs kernel of B with the weight $\exp[-\beta\sum_p (1 - \frac{1}{2}\mathrm{tr}\,\partial U(p))]$, the sum over the plaquettes p that hold a bond of B (equation YM3_I:(1818)); it is periodic if it is invariant under the translations of a subgroup of finite index of $a\mathbb{Z}^3$ (Section YM3_I:91), and every periodic Gibbs state is invariant under all isometries of the lattice (Prop. YM3_I:103.4). In Thm. AG(10), (11) the lattice is $\mathbb{Z}^3$.

Mixtures and boards. A weight $w\colon \mathbb{Z}_2^3 \to [0, \infty)$, $w \neq 0$, is admissible if

$$\hat{w}(e, m) = \sum_k (-1)^{e\cdot k} w((k, m)_\mu) \geq 0$$

for all μ, e, m; then $Z_w = \sum_n w(n) Z[n; \cdot]$, $|Z_w| = Z_w[1]$ and $\mu_w = Z_w/|Z_w|$. A board is a box with $N_\mu = \mathsf{c}\,\mathsf{n}_\mu$, $\mathsf{c} \geq 3$ odd and $\mathsf{n}_\mu \geq 2$ even, cut into $\mathsf{N} = \prod_\mu \mathsf{n}_\mu$ cells by mirrors through bonds, the columns placed where mirrors cross. For a bounded complex Borel function F of one cell, $\mathrm{Fold}\, F$ is the product over the cells of the mirror copies of F, the copy in a cell being complex conjugated if the cell is odd relative to the given one, that is, reached from it by an odd number of reflections in the mirrors (equation YM3_I:(1749), YM3_I:(1749), YM3_I:(1749)); for real F the copies are the mirror images, and for complex F the conjugation makes $Z[0; \mathrm{Fold}\, F] \geq 0$ (with the plain mirror images the constant $e^{i\pi/8}$ on a board of 8 cells would have the fold -1).

$$\|F\|_{\mathrm{ch}}^{\mathsf{N}} = Z[0; \mathrm{Fold}\, F]/Z[0]$$

is the periodic chessboard seminorm (Prop. YM3_I:86.3), and $\|F\|_w^{\mathsf{N}} = Z_w[\mathrm{Fold}\, F]/|Z_w|$.

Cutoffs. L, an odd integer with $L \geq 13$, and $a_K = L^{-K}$ are the blocking factor and the spacings; the label is $b > 0$ and $\beta = L^K b$. A box s of the cutoff K has sides $s_\mu \in a_K\mathbb{N}$, $s_\mu \geq 3a_K$; an index K on a number above denotes that number at $N_\mu = s_\mu/a_K$ and $\beta = L^K b$, as in $Z_K^z(s; b) = Z_K[z](s; b)$,

$r_K[z](s;b)$ and δ_K^0; with $p_\infty(K,b) = a_K^{-3}P(L^K b)$ we put $\delta_K^n(s;b) = \log Z_K[n](s;b) - |s|\,p_\infty(K,b)$. A length is L-adic if it lies in $\mathbb{D} = \bigcup_K a_K\mathbb{N}$. For $K, m \in \mathbb{N}$ the cube $\ell = 2L^m$ is the lattice $\Lambda_{K,\ell}$ of the torus T_ℓ (Section YM3_I:3); for other m it is reached by the dilation $Z_K[n](s;b) = Z_{K-j}[n](L^j s; L^j b)$. $Z_{K,\ell}$ is its periodic partition function, $\Delta_K[\ell,b] = \log Z_{K+1,\ell} - \log Z_{K,\ell}$, and on cubes $t = \ell/b$.

A box s is mixed for the class z if all $s_\mu \in L^{\mathbb{Z}} \cup 2L^{\mathbb{Z}}$ and each twisted plane of z has a side in $L^{\mathbb{Z}}$. We write $\rho^z(N,\beta)$ for $r[z]$ on the box N at β, and, for a fixed class z, $\phi_j(t) = \rho^z((2L^j)^{\times 3}, 2L^j/t)$. The constants γ, $M = L^{m'}$, $r_\star$ and $\mathfrak{p}_0$ are those of Prop. YM3_I:69.1, with $r_\star = r$ and $2\mathfrak{p}_0 > 3r_\star + 2$. Put $R(b) = \min\{L^a : a \in \mathbb{N}_0,\ L^a \geq (\log b)^{r_\star}\}$, $\ell_\flat(b) = 2MR(b)$ and $\ell_\sharp(b) = L\ell_\flat(b)$, and set

$$m_0(b) = 1 + m' + \max\{\log_L R(b + \bar{B}), \lceil \log_L C_{\mathrm{s}} \rceil\}, \qquad b \geq \gamma^{-2}.$$

Here C_{s} is the torus-floor constant, chosen at least C_L as in the paragraph following Prop. YM3_I: 74.5. With σ and $\mathfrak{b}$ of Thm. AG(3) at $\beta = L^K b$, $\sigma_K(l) = a_K^{-2}\sigma(l/a_K)$ and $\mathfrak{b}_K(t,\lambda;l)$ is $\mathfrak{b}$ on the box $(t,\lambda,l)/a_K$.

Levels and block fields. Over a section $l = (l_1, l_2)$ and with a time μ, the levels are $E_i^{K,m,e} = a_K^{-1}\log(\|T_0\|/\lambda_i)$, $\lambda_1 \geq \lambda_2 \geq \cdots > 0$ the eigenvalues of $P_e T_m$, each repeated according to its multiplicity, and $E_i^{K,m,e} = \infty$ beyond them. $\mathcal{M}$ is the averaging map; on the cube ℓ at the cutoff K the block field of depth $d \leq K$ is $\mathcal{M}^{K-d}U$, a field on the block lattice of spacing a_d, and μ_d^K is its law under the periodic measure. A block line is a line of a block lattice.

11.1.2. *Positivity belongs to the electric sectors.* Fix a time direction. A magnetic twist m determines a transfer space and a positive transfer operator T_m. The two independent centre flips commute with T_m and generate a representation $k \mapsto C^k$ of $\mathbb{Z}_2^2$. A temporal twist inserts C^k in the transfer trace. Thus the twisted partition function has the exact finite-cutoff expression

$$Z[(k,m)] = \mathrm{Tr}(T_m^N C^k).$$

The integer N is the number of time steps. Although T_m is positive, the insertion C^k may have both signs. This formula does not imply that a fixed twisted class has all the reflection positivity properties of the untwisted measure.

The character projections isolate the positive quantities. For $e \in \mathbb{Z}_2^2$, put

$$P_e = \frac{1}{4}\sum_{k\in\mathbb{Z}_2^2} (-1)^{e\cdot k} C^k.$$

Character orthogonality makes the P_e orthogonal projections summing to the identity. They commute with T_m, so

$$Z_{e,m} = \mathrm{Tr}(P_e T_m^N) = \frac{1}{4}\sum_k (-1)^{e\cdot k} Z[(k,m)] \geq 0.$$

The source's electric-flux decomposition is therefore a decomposition of one positive transfer operator into invariant subspaces. The positivity is visible before any limit is taken. Changing which coordinate is time gives a second transfer-trace description and the crossing identities between the finite Fourier transforms. Those identities are retained below with their finite-box parity and covering conventions.

The same distinction governs mixtures of classes. In a chosen time direction, nonnegative electric Fourier weights ensure reflection positivity. To obtain reflection positivity in every coordinate direction by this argument, impose the same condition in each description. Positivity in one description alone does not establish the other conditions.

The continuum flux sketch asks for more than these algebraic identities. Its window comparison concerns the actual interacting integral in boxes with different twists. The Hamiltonian differentiated there includes the interaction, the chart Jacobian and the boundary regularization. A lower bound for the Gaussian quadratic form at its reference field does not control that full Hessian. Likewise, a

change of sign in two plane means specifies a candidate window but does not estimate its probability or its transfer between cutoffs. The sketch below specifies these probability and transfer estimates. The finite-cutoff Fourier and transfer arguments do not depend on their completion.

There are also two different large-distance spectral questions. At a fixed small transverse section, the largest transfer eigenvalue and the remaining thermal weights define sector free energies and a transfer gap. An infinite-volume mass gap requires control as the transverse section grows. The finite-size confinement criterion in §17 supplies a separate route to that uniform control; the existence of one fixed-section gap does not supply it.

Finally, the twist ratio in an area-law regime approaches one. The decaying quantity is its defect. Keeping $1 - Z^-/Z$ visible in the finite Fourier formulas prevents the electric-flux suppression from being misread as decay of Z^-/Z itself. The quantitative tube estimates later in the confinement argument concern this small flux weight.

11.1.3. *Principal results.* The untwisted partition function supplies the reference for the centre-flux ratios. We record the stability estimate at a fixed physical label, including the room needed to compare two tori.

Lemma 11.1 (Stability at a fixed label)**.** Use the common construction constants, with $C_{\mathrm{s}} \ge C_L$ as in the torus-floor choice following Prop. YM3_I:74.5, and let $b \ge \gamma^{-2}$, $K \ge 1$ and $m_i \ge m_0(b)$ for $i = 1, 2$. Put $\ell_i = 2L^{m_i}$ and $\theta_j = L^{K-j}b$. The vacuum constants $E_i = E_{(K)}[((\ell_i, K), \theta_\bullet)]$ of the real progression are real, and

$$\begin{aligned} e^{-[c_{\mathrm{low}} + \frac{9}{2}\log b]\ell_i^3} \le e^{-E_i} Z_{K,\ell_i,L^K b} \le e^{(C_{\mathrm{maj}}+2)\ell_i^3}, \\ \left| E_1 \ell_1^{-3} - E_2 \ell_2^{-3} \right| \le 2C_{\mathrm{tt}} C'_{\mathrm{tt}} E_0 e^{-\kappa \min\{\ell_1, \ell_2\}/(5M)}. \end{aligned} \tag{11.ii}$$

Here $C_{\mathrm{tt}} = 4096 \sum_{q \ge 0} e^{-q}$ and $C'_{\mathrm{tt}} = 2$; the rate and size constants are those of Prop. YM3_I: 74.5. In particular, the first logarithmic error is bounded by $(C_{\mathrm{lab}} + \frac{9}{2}\log b)\ell_i^3$, where $C_{\mathrm{lab}} = \max\{c_{\mathrm{low}}, C_{\mathrm{maj}} + 2\}$. ■

Proof. The increments of the sequence are $L^{-1}\theta_{j-1} - \theta_j = 0$, and $\theta_j \ge b \ge \gamma^{-2}$. Thus it is an admissible prescribed sequence in equation YM3_I:(1524), with last radius $R(b)$. The definition of m_0 gives

$$L^{m_i} \ge LMR(b + \bar{B}) \ge LMR(b), \qquad \ell_i = 2L^{m_i} \ge 2LMC_{\mathrm{s}} \ge C_L M.$$

Both tori are therefore admitted, including the stronger room required for their comparison. Apply Prop. YM3_I:74.4 at $k = K$, with $u_0 = L^K b$ and $u_K = b$, and Prop. YM3_I:74.5 to this same sequence on the two tori. These give the two bounds and the reality of the vacuum constants. The vacuum terms are those of the canonical progression of the prescribed sequence; its bare counterterms are its zero increments. Finally $b \ge \gamma^{-2} \ge 4$ implies $\log b > 0$, so the definition of C_{lab} gives the symmetric logarithmic bound. □

Theorem AG (One lattice)**.** Let N be a box, $\beta \ge 0$, and μ a direction, taken as the time, $\nu < \rho$ the other two.

(1) *(Transfer operator; order; momentum; crossing.)* $T_m \ge 0$ is Hilbert–Schmidt with a continuous symmetric kernel > 0; C_ν, C_ρ are commuting unitary involutions that commute with T_m. For every class $Z[(k,m)_\mu] = \mathrm{Tr}(T_m^{N_\mu} C^k)$, and $Z_{e,m} = \frac{1}{4}\sum_k (-1)^{e\cdot k} Z[(k,m)_\mu] = \sum_i \lambda_i(P_e T_m)^{N_\mu} \ge 0$ with eigenvalues $\lambda_i(P_e T_m)$ that do not depend on N_μ; $Z_{e,m} < Z_{00,m}$ for $e \ne 0$, and $Z_{e,m} > 0$ if $\beta > 0$. $\|T_m\|$ is a simple eigenvalue with an eigenfunction > 0 in the range of P_{00}, and $\|T_1\| \le \|T_0\|$. Moreover

$$e^{-2\beta(n_{23}N_1 + n_{13}N_2 + n_{12}N_3)} Z[0] \le Z[n] \le Z[(0,m)_\mu] \le Z[0],$$

the last two inequalities strict if $\beta > 0$ and the classes differ; so $r[n] \le r[n']$ for every one-plane class n' whose plane is twisted in n. Over a twisted section, with the reversed

plaquette at the corner ξ_* and the lines of C_ν, C_ρ through ξ_*, as in the setting, let $\mathcal{T}_\nu$ be the composition with the map $V \mapsto \mathsf{S}_\nu(\gamma_\nu V)$ of the fields, where $(\mathsf{S}_\nu V)_{(\xi,j)} = V_{(\xi+e_\nu,j)}$ is the step by one site in the direction ν and γ_ν reverses the one bond $(\xi_* + e_\nu, \rho)$, which the reversed plaquette shares with its neighbour in the direction ν; and let $\mathcal{T}_\rho$ be formed likewise, with ν and ρ exchanged and the bond $(\xi_* + e_\rho, \nu)$.

On all of L^2, $\mathcal{T}_\nu$ and $\mathcal{T}_\rho$ are unitaries that commute with T_1 and map the gauge invariant functions onto themselves, and $\mathcal{T}_\nu^{N_\nu} = C_\rho$, $\mathcal{T}_\rho^{N_\rho} = C_\nu$; so $\mathcal{T}_\nu$ commutes with C_ρ, and $\mathcal{T}_\rho$ with C_ν. The products $\mathcal{T}_\nu\mathcal{T}_\rho$ and $\mathcal{T}_\rho\mathcal{T}_\nu$ differ by the reversal of the four bonds at the site $\xi_* + e_\nu + e_\rho$, and $\mathcal{T}_\nu C_\nu$ and $C_\nu\mathcal{T}_\nu$, as $\mathcal{T}_\rho C_\rho$ and $C_\rho\mathcal{T}_\rho$, by the gauge transformation that is -1 at the sites of one line of the section and 1 elsewhere: gauge transformations with the values ±1, which act as the identity on the gauge invariant functions.

On the gauge invariant functions, off which T_1 vanishes ($T_1 = \Pi T_1 \Pi$, Π the gauge average) and which for $\beta > 0$ form the closure of the range of T_1, the five operators T_1, $\mathcal{T}_\nu$, $\mathcal{T}_\rho$, C_ν, C_ρ commute pairwise. So the range of P_e, $e = (e_\nu, e_\rho)$, in the gauge invariant functions, on which $C_\nu = (-1)^{e_\nu}$ and $C_\rho = (-1)^{e_\rho}$, has an orthonormal basis of joint eigenvectors of T_1, $\mathcal{T}_\nu$ and $\mathcal{T}_\rho$, and the joint eigenvalues $(e^{i\theta_\nu}, e^{i\theta_\rho})$ of $(\mathcal{T}_\nu, \mathcal{T}_\rho)$ there have $\theta_\nu \in \frac{2\pi}{N_\nu}(\mathbb{Z} + \frac{e_\rho}{2})$ and $\theta_\rho \in \frac{2\pi}{N_\rho}(\mathbb{Z} + \frac{e_\nu}{2})$; and $Z[(k,1)_\mu] = \mathrm{Tr}(T_1^{N_\mu}\mathcal{T}_\nu^{k_2 N_\nu}\mathcal{T}_\rho^{k_1 N_\rho})$.

For the times 1 and 2, with $e = (e_1, e_2)$ dual to (n_{12}, n_{13}) and $m = n_{23}$ for the time 1, and (e_1, e_2') dual to (n_{12}, n_{23}) and $m' = n_{13}$ for the time 2, the crossing relation

$$Z^{[1]}_{e_1 e_2, m} = \tfrac{1}{2} \sum_{m', e_2' \in \{0,1\}} (-1)^{e_2 m' + e_2' m}\, Z^{[2]}_{e_1 e_2', m'}$$

holds, and likewise for every pair of times: the flux of the common plane is kept, and the pairs (e_2, m) and (m', e_2') are exchanged by the Fourier transform of $\mathbb{Z}_2^2$. Hence, for every time, $Z_{e_1 e_2, m} \le \min\{Z_{e_1 0,0}, Z_{0 e_2, 0}\}$, $\|T_m P_{e_1 e_2}\| \le \min\{\|T_0 P_{e_1 0}\|, \|T_0 P_{0 e_2}\|\}$ and $\|\frac{1}{2}(1 - C_\nu)T_0\| = \|T_0 P_{10}\|$: the least level of odd electric flux along ν has no flux along ρ. For $\beta > 0$ and a fixed section, $\varepsilon_{\mathrm{mag}} = \log(\|T_0\|/\|T_1\|) = -\lim_{N_\mu \to \infty} N_\mu^{-1} \log r[n_{\nu\rho}] \in [0, \infty)$ and $\varepsilon_{\mathrm{el}} = \log(\|T_0\|/\|T_0 P_{10}\|) = -\lim_{N_\mu \to \infty} N_\mu^{-1} \log \frac{Z[0] - Z[n_{\mu\nu}]}{Z[0] + Z[n_{\mu\nu}]} \in (0, \infty)$.

(2) *(Two lengths; the excess.)* Each of $\frac{1}{2}(Z[0] \pm Z[n_{12}])$ is a sum of N_1-th powers of numbers ≥ 0 that do not depend on N_1, and a sum of N_2-th powers of numbers ≥ 0 that do not depend on N_2. $F_0/|N|$ does not increase in any side and tends to $P \ge -6\beta$ as all sides tend to infinity. Over every section and for every time, $0 \le \mathsf{G} \le \delta$, G does not increase in the height, and $\mathsf{G}(M) \le \mathsf{d}(M) = 2\delta(M) - \delta(2M) = 2\mathsf{G}(M) - \mathsf{G}(2M) \le 2\mathsf{G}(M)$. For every integer $q \ge 2$, $qF_0(N) - F_0(N^{[\mu]}) = q\delta(N) - \delta(N^{[\mu]}) \ge 0$, $N^{[\mu]}$ the box with N_μ multiplied by q. And $\delta(N')/|N'| \le \min\{1, \sum_\mu N_\mu/N_\mu'\}\, \delta(N)/|N|$ if $N_\mu' \ge N_\mu$ for all μ.

(3) *(Electric flux tube.)* Let $\beta > 0$ and $l \ge 3$. Over the sections of sides λ and l let $E_0 = -\log\|T_0\|$ and $E_{\mathrm{e}} = -\log\|\frac{1}{2}(1 - C')T_0\| = -\log\|T_0 P_{10}\|$, C' the flip of the bonds parallel to the side λ, and $\varepsilon = E_{\mathrm{e}} - E_0$; then $0 \le E_0 \le E_{\mathrm{e}} < \infty$. For $f \in \{E_0, E_{\mathrm{e}}\}$ and $\lambda, \lambda' \ge 3$: $2f(\lambda) \ge f(\lambda - j) + f(\lambda + j)$ for $\lambda - j \ge 3$, $f(\lambda + \lambda') \ge f(\lambda) + f(\lambda')$, $f(\lambda)/\lambda$ does not decrease and tends to a real number s_f, $s_f \le [f(2\lambda) - f(\lambda)]/\lambda$, and $s_f\lambda - f(\lambda) \ge 0$ is convex and does not increase. With $s_0 = s_{E_0}$ and $R(\lambda) = s_0\lambda - E_0(\lambda)$, the tension $\sigma(l) = s_{E_{\mathrm{e}}} - s_0 = \lim \varepsilon(\lambda)/\lambda \in [0, \infty)$ obeys

$$\frac{\varepsilon(\lambda) - R(\lambda)}{\lambda} \le \sigma(l) \le \frac{\varepsilon(2\lambda) - \varepsilon(\lambda) + E_0(2\lambda) - 2E_0(\lambda)}{\lambda} \qquad (\lambda \ge 3).$$

On the boxes (t, λ, l) let $r = r[n_{12}]$, $q = \frac{1}{2}(Z[0] - Z[n_{12}])$, $p = \frac{1}{2}(Z[0] + Z[n_{12}])$, $\Phi = F_0 + t\lambda s_0$ and $\mathfrak{b} = \frac{1}{2}(1 - r)e^{\Phi} = q\, e^{t\lambda s_0}$. Then $-s_0 = \inf_{t', \lambda' \ge 3} F_0(t', \lambda', l)/t'\lambda' \ge lP$, $0 \le \Phi \le \delta$, $\Phi/t\lambda$

does not increase in t nor in λ, $p \ge e^{-t\lambda s_0}$ and so $\mathfrak{b} \le e^{\Phi} - 1$; and for $t \ge t_0 \ge 3$, $\lambda \ge \lambda_0 \ge 3$

$$\frac{1-r}{1+r} = \frac{q}{p} \le \mathfrak{b}(t,\lambda) \le \mathfrak{b}(t_0,\lambda_0)^{t\lambda/t_0\lambda_0} \le \bigl(e^{\Phi(t_0,\lambda_0)} - 1\bigr)^{t\lambda/t_0\lambda_0},$$

so that $\Phi(t_0,\lambda_0) < \log 2$, a number of periodic boxes, is enough for an area law, and $r(t,\lambda,l) \ge (1-x)/(1+x)$ with $x = \mathfrak{b}(t_0,\lambda_0)^{t\lambda/t_0\lambda_0}$. Further $-(t\lambda)^{-1}\log\mathfrak{b}(t,\lambda)$ does not decrease in t nor in λ, and

$$\sigma(l) = \sup_{t,\lambda} \frac{\log(1/\mathfrak{b}(t,\lambda))}{t\lambda} = \lim_{t,\lambda\to\infty} \frac{\log(1/\mathfrak{b}(t,\lambda))}{t\lambda};$$

so $\sigma(l) > 0$ if and only if $\mathfrak{b} < 1$ on one box. For $t > t_0$, $\lambda > \lambda_0$: $-\log\mathfrak{b}(t,\lambda) \ge (t-t_0)[(\lambda - \lambda_0)\sigma(l) - R(\lambda_0)] - (\lambda/\lambda_0)\Phi(t_0,\lambda_0)$. Finally $\varrho = q/Z[0]$ has $\varrho(2t) \le \varrho(t)^2 e^{2F_0(t)-F_0(2t)}$, and the same in λ.

(4) *(Ceilings of the tension; Casimir terms; slabs.)* Let $\beta > 0$ and $l \ge 3$, with the signs of (3). (a) For integer heights $3 \le h_0 < h_1$ and every length $\lambda \ge 3$,

$$(h_1 - h_0)\,\varepsilon(\lambda) \le \log(e^{\mathsf{G}(h_0;\lambda,l)} - 1) + \log\frac{2}{1 - r(h_1,\lambda,l)},$$

$\mathsf{G}(h_0;\lambda,l)$ being G at the height h_0 over the section (λ,l); and for $3 \le \lambda_0 < \lambda_1$, with $\kappa' = \log(2/(1-r(h_1,\lambda_1,l)))$,

$$\sigma(l) \le \frac{1}{h_1 - h_0}\Bigl\{\frac{\Phi(h_0,\lambda_0)}{\lambda_0} + \frac{\kappa' + \Phi(h_1,\lambda_0)}{\lambda_1 - \lambda_0}\Bigr\} + \frac{\Phi(h_1,\lambda_0)}{\lambda_0 h_1},$$

and the same with $\delta(\cdot,\cdot,l)$ in the place of Φ. (b) Over a transverse side $b' \ge 3$, for all integers $a \ge 3$, $k' \ge 2$: $R(a) \le \mathsf{G}(a;b',k'a)/((k'-1)a)$, R formed over the sections (a,b'); and $\delta(M,M,M) \le \mathsf{G}(M;M,M) + \mathsf{G}(M;M,2M) + \frac{1}{2}\mathsf{G}(M;2M,2M) + \frac{1}{4}\mathsf{G}(2M;2M,4M)$ for every $M \ge 3$. (c) For all integers $t,\lambda_1,l \ge 3$,

$$\sigma(l) \ge \frac{1}{\lambda_1}\Bigl\{\frac{1}{t}\Bigl[\log\frac{2}{1-r(t,\lambda_1,l)} - \log\frac{Z[0](t,\lambda_1,l)^2}{Z[0](2t,\lambda_1,l)}\Bigr] - \frac{1}{\lambda_1}\log\frac{Z[0](\lambda_1,2\lambda_1,l)^2}{Z[0](2\lambda_1,2\lambda_1,l)}\Bigr\}:$$

a positive tension is one inequality between five partition functions. (d) On the slab $\{0,\dots,A'\}\times\{0,\dots,B'\}\times\mathbb{Z}_{N'}$, free in the first two directions and periodic in the third, $N' \ge 3$, with the Wilson action on all its plaquettes, $|\langle \operatorname{tr} U(C)\rangle| \le 2(\tanh\beta N')^{RT}$ for the boundary C of every rectangle of $R\times T$ plaquettes in a plane normal to the third direction, at every $\beta \ge 0$. (e) On a box (t,λ,l) let the $t\lambda$ plaquettes of the periodic $t\times\lambda$ lattice of the twisted plane be parted into sets D, each connected through common edges; call a bond inner to D if it is parallel to the twisted plane and lies over an edge both of whose plaquettes lie in D, or normal to it and lies over a site all four of whose plaquettes lie in D. Then $q/p = \int\prod_D \rho_D\,\mathrm{d}\mu$, where μ is a probability measure on the configurations of the bonds that are inner to no D, and $\rho_D = (Z_D^+ - Z_D^-)/(Z_D^+ + Z_D^-) \in (-1,1)$, $Z_D^\pm$ the integrals over the bonds inner to D of the weight of the plaquettes parallel to the twisted plane over D and of the plaquettes normal to it over the inner edges of D, without and with one reversed column over D; Z_D^- does not depend on the place of that column.

(5) *(How floors and ceilings travel.)* The time μ and the section are fixed, the boxes differ in N_μ alone, and all numbers of layers are ≥ 3. (a) For $k \ne 0$ and $m \in \{0,1\}$,

$$Z[(k,m)_\mu](N_\mu)^2 \ge 2Z[(0,m)_\mu](2N_\mu) - Z[(0,m)_\mu](N_\mu)^2;$$

so with $\mathcal{R} = Z[0](N_\mu)^2/Z[0](2N_\mu)$, $r[(k,0)_\mu]^2 \ge 2/\mathcal{R} - 1$, and $1/\mathcal{R} \le \frac{1}{4}\sum_k r[(k,0)_\mu]^2$. With $Z' = Z[(0,m)_\mu]$ and $q_0 = Z_{00,m}/Z' \ge \|T_m\|^{N_\mu}/\operatorname{Tr} T_m^{N_\mu}$: $Z[(k,m)_\mu]/Z' \ge 2q_0 - 1$ for every k; $\sum_{k\ne0} Z[(k,m)_\mu]/Z' = 4q_0 - 1$, so that $\sum_{k\ne0} r[(k,0)_\mu] \ge 4e^{-\mathsf{G}(N_\mu)} - 1 \ge 4e^{-\delta} - 1$; and

$$Z[((1,1),m)_\mu] \ge Z[((1,0),m)_\mu] + Z[((0,1),m)_\mu] - Z'.$$

(b) *A ceiling moves inside a twisted plane.* Let $n = (k,0)_\mu$, $k \neq 0$, and $r(N_\mu) = r[n]$. For $N_\mu - j \geq 3$,

$$(1 - r(N_\mu - j))(1 - r(N_\mu + j)) \geq (1 - r(N_\mu))^2 e^{2\delta(N_\mu) - \delta(N_\mu - j) - \delta(N_\mu + j)}.$$

(c) *A floor moves along its own time.* For $\emptyset \neq S \subset \mathbb{Z}^2_2$, $P_S = \sum_{e \in S} P_e$, $f(N') = \mathrm{Tr}(P_S T_m^{N'})/\mathrm{Tr}\, T_0^{N'}$, $\lambda = \|P_S T_m\|$ and $\Theta = e^{\mathsf{G}}$:

$$(\lambda/\|T_0\|)^{N'}/\Theta(N') \leq f(N') \leq (\lambda/\|T_0\|)^{N' - N''}\Theta(N'')$$

for $3 \leq N'' < N'$; so for $3 \leq M_0 < M_1$ and $M_2 \geq 3$,

$$f(M_2) \geq \Theta(M_2)^{-1}[f(M_1)/\Theta(M_0)]^{M_2/(M_1 - M_0)}.$$

In particular, if $r[n_{\nu\rho}](M_1) \geq \varrho > 0$, then $\varepsilon_{\mathrm{v}} \leq [\mathsf{G}(M_0) + \log(1/\varrho)]/(M_1 - M_0)$ and $r[n_{\nu\rho}](M) \geq e^{-\mathsf{G}(M)}(\varrho e^{-\mathsf{G}(M_0)})^{M/(M_1 - M_0)}$ for every $M \geq 3$. (d) *Floors move inside a twisted plane.* With n and r as in (b): for $3 \leq N_1 < N_\mu < N_2$ and $\vartheta = (N_2 - N_\mu)/(N_2 - N_1)$,

$$1 - r(N_\mu) \leq (1 - r(N_1))^{\vartheta}(1 - r(N_2))^{1-\vartheta} e^{\vartheta\delta(N_1) + (1-\vartheta)\delta(N_2) - \delta(N_\mu)};$$

for $N_\mu \geq N_0 \geq 3$,

$$\frac{1}{2}(1 - r(N_\mu)) \leq [\frac{1}{2}(1 - r(N_0))e^{\mathsf{G}(N_0)}]^{N_\mu/N_0};$$

and for every class z one of whose twisted planes holds μ, all integers $n' \geq 1$, $m_1, m_2 \geq 2$, $m = m_1 + m_2$, and every odd $k' \geq 1$,

$$F^z(k'n' + m) \leq k' F^z(n' + m) + \tfrac{k'-1}{2}[\delta(2m_1) + \delta(2m_2)] + \delta(k'n' + m) - k'\delta(n' + m),$$

the argument being the number of layers in the time μ. (e) *Cubes.* For odd $L' \geq 3$, $N' = 2L'^j$ with $j \geq 2$, every class $z \neq 0$ and every β, on the cubes,

$$F^z(L'N') \leq (2L' + 1)^3 F^z(N') + C_{L'}\,\delta(N'/L'), \qquad C_{L'} = 34L'^5 + 53L'^4 + 27L'^3 + 4L'^2,$$

$\delta(N'/L')$ the excess of the cube of side N'/L'.

(6) *(The overlap; odd functions.)* Let $N_\mu = 2\tau$, $k \neq 0$, Ψ the kernel of T_m^τ, $\mu_2 = \Psi^2 dV\, \mathrm{d}W/Z[(0,m)_\mu]$ the law of two layers half a period apart in the class $(0,m)_\mu$, $\langle\cdot\rangle_2$ the mean under μ_2, $g(V,W) = (V, c^k W)$, $r' = Z[(k,m)_\mu]/Z[(0,m)_\mu]$, $\mathsf{m} = (\Psi^2 - \Psi^2 \circ g)/(\Psi^2 + \Psi^2 \circ g)$, p_1 the law of one layer and $E_c = \{\Psi \circ g \geq c\Psi\}$. Then $r' = \langle \Psi \circ g/\Psi\rangle_2 = \int \sqrt{d\mu_2\, d(g_*\mu_2)} = \langle(1 - \mathsf{m}^2)^{1/2}\rangle_2$; $1 - \mathrm{TV} \leq r' \leq (1 - \mathrm{TV}^2)^{1/2}$, TV the total variation distance of μ_2 and $g_*\mu_2$; $c\,\mu_2(E_c) \leq r' \leq c + \mu_2(E_c)^{1/2}$ for every $c > 0$; $r' \leq 2\mu_2(E)^{1/2}$ for every Borel E with $gE = E^c$; $1 - r' \leq \sup\langle\Phi\rangle_2^2/\langle\Phi^2\rangle_2 = \langle\mathsf{m}^2\rangle_2 \leq 1 - r'^2$, the supremum over the bounded Borel Φ with $\Phi \circ g = -\Phi$ and $\langle\Phi^2\rangle_2 > 0$; $r' \geq 1 - 2\,\mathrm{TV}(\mu_2, p_1 \otimes p_1)$; if $\mu_2 \geq c_0\, p_1 \otimes p_1$ on a Borel set $\mathcal{G}$, then $r' \geq c_0\,(p_1 \otimes p_1)(\mathcal{G} \cap g\mathcal{G})$; $-\log r' \leq \frac{1}{2}\mathrm{KL}(\mu_2 | g_*\mu_2) = \langle \log(\Psi/\Psi \circ g)\rangle_2 < \infty$; and for every bimeasurable bijection Φ of the fields that keeps dU and the layer 0 and maps the layer τ by c^k, with μ' the law of the class $(0,m)_\mu$ and $A' = A_{\eta^{(0,m)_\mu}}$,

$$r' \geq \int \sqrt{\mathrm{d}\mu'\, d(\Phi_*\mu')} \geq \exp[-\tfrac{\beta}{2}\mu'(A' \circ \Phi^{-1} - A')], \qquad \tfrac{1}{2}\mathrm{KL}(\mu_2 | g_*\mu_2) \leq \tfrac{\beta}{2}\mu'(A' \circ \Phi^{-1} - A').$$

Odd functions: let $n = (k,0)_\mu$ have one twisted plane, $N_\mu = 2k'\vartheta$ with integers $\vartheta \geq 2$, $k' \geq 2$, let D, D' be bounded complex Borel functions of the bonds inside one slab of ϑ layers, D' odd under c^k applied to all layers of the slab, and $\mathsf{t}_j D'$ the translate of D' by $j\vartheta$ layers, $1 \leq j \leq 2k' - 1$. Then

$$|\langle D\, \mathsf{t}_j D'\rangle| \leq \sup|D|\, \sup|D'| \sum_{\varsigma = \pm 1} \Big(\frac{1 + \varsigma r[n]}{2}\Big)^{\frac{2k' - j - 1}{2k'}} \Big(\frac{1 - \varsigma r[n]}{2}\Big)^{\frac{j-1}{2k'}},$$

and at $j = k'$, $|\langle D\,\mathrm{t}_{k'} D'\rangle| \le 2 \sup|D| \sup|D'| \,[\frac{1}{4}(1 - r[n]^2)]^{\frac{1}{2} - \frac{1}{2k'}}$.

(7) *(Reflection positivity.)* Let $N_\mu \ge 4$. On centre-blind functions $Z[n;\cdot\,]$ does not depend on the places of the columns, and for $\beta > 0$ it is reflection positive at the mirrors normal to μ if and only if μ lies in no twisted plane of n; when μ lies in a twisted plane, the form is negative at a centre-blind gauge invariant function of one slab, a product of even class functions of the plaquettes through μ in one crossing. For centre-blind F of a slab and each (e,m), $\sum_k (-1)^{e\cdot k} Z[(k,m)_\mu; \overline{\theta F} F] \ge 0$. For each of the eight sets $\mathfrak{s}$ of coordinate planes, $\sum\{Z[n;\cdot\,] : n_{\mu\nu} = 0 \text{ for } \mu\nu \notin \mathfrak{s}\}$ is reflection positive on centre-blind functions at the mirrors normal to every direction λ with $N_\lambda \ge 4$; of the single classes, $n = 0$ alone is so in all three directions; and $Z[0] \le \sum_n Z[n] \le 8 Z[0]$.

If N_μ is even, $2 \le \vartheta \le N_\mu/2$, F is a bounded Borel function of a slab of ϑ layers, and the columns of the planes through μ lie in a crossing outside the slab, then $|Z[(k,m)_\mu; F]|^2 \le Z[(0,m)_\mu]\, Z[(0,m)_\mu; \overline{\theta F} F]$, θ the reflection that leaves $\frac{1}{2}N_\mu - \vartheta + 1$ crossings on each side, and $Z[(k,m)_\mu; \mathbf{1}_E] \le Z[(0,m)_\mu]\, \mu_{(0,m)_\mu}(E)^{1/2}$ for every event E of the slab: what is rare without electric flux is rare with it.

(8) *(Chessboard bounds.)* On a board, for bounded complex Borel functions F_κ of the cells and every class, $|Z[n; \prod_\kappa F_\kappa]| \le Z[0] \prod_\kappa \|F_\kappa\|_{\mathrm{ch}}$, and the same with η^n replaced by any sign field carried by the plaquettes in the mirrors. For admissible w: Z_w is reflection positive at the mirrors of the three directions; $w(0) Z[0] \le |Z_w| = Z[0] \sum_n w(n) r[n] \le 8 w(0) Z[0]$; $|Z_w[\prod_\kappa F_\kappa]| \le |Z_w| \prod_\kappa \|F_\kappa\|_w$; $\|F\|_w^{\mathsf{N}} \le \frac{\sum_n w(n)}{\sum_n w(n) r[n]} \|F\|_{\mathrm{ch}}^{\mathsf{N}} \le 8 \|F\|_{\mathrm{ch}}^{\mathsf{N}}$, and for $F \ge 0$, $\|F\|_w^{\mathsf{N}} \ge \frac{w(0)}{\sum_n w(n) r[n]} \|F\|_{\mathrm{ch}}^{\mathsf{N}} \ge \frac{1}{8} \|F\|_{\mathrm{ch}}^{\mathsf{N}}$; so $|\mu_w(\prod_\kappa F_\kappa)| \le 8 \prod_\kappa \|F_\kappa\|_{\mathrm{ch}}$.

All this holds for covariant functions $G_\kappa(U,s)$ of the cells, with $F_\kappa = G_\kappa(\cdot, 1)$, every $s \in [n]$ and the columns anywhere. The constants are attained: the functions of the cells that cancel the weight of rods of plaquettes through the centres of the cells give equality in the first bound and in the bound of $\|\cdot\|_w$ at $w \equiv 1$; and with the fold taken in a class $n \ne 0$ itself the estimate fails for $\beta > 0$: there are positive functions of the cells with $Z[n; \prod_\kappa F_\kappa] = Z[0]$ and $\prod_\kappa Z[n; \mathrm{Fold}\, F_\kappa]^{1/\mathsf{N}} = Z[n] < Z[0]$. In a class: for real $q \ge 1$, $\mu_n(\prod_\kappa |G_\kappa|) \le r[n]^{-1/q} \prod_\kappa \||F_\kappa|^q\|_{\mathrm{ch}}^{1/q}$; for a covariant event E of one cell, $\mu_n(E) \le r[n]^{-1} \|\mathbf{1}_E\|_{\mathrm{ch}}$; and the law of the inner bonds of one cell under μ_n has a density h_n against their law under $\langle\cdot\rangle$ with $\int h_n \log h_n \le \mathsf{N} \log(1/r[n])$, so that the two laws differ in total variation by at most $(\frac{\mathsf{N}}{2} \log \frac{1}{r[n]})^{1/2}$.

Without a board: let $N_\mu = 2h$, $h \ge 2$, and let G be covariant and depend only on the bonds and signs of the h layers on one side of a mirror normal to μ, $F = G(\cdot, 1)$. If μ lies in every twisted plane of n, then $r[n]|\mu_n(G)| \le \langle \overline{\theta F} F\rangle^{1/2} \le \langle |F|^2\rangle^{1/2}$, the first inequality attained, and

$$|\mu_n(G) - \langle F\rangle| \le (r[n]^{-2} - 1)^{1/2} (\langle |F|^2\rangle - |\langle F\rangle|^2)^{1/2},$$

which is at most $\frac{1}{2}(r[n]^{-2} - 1)^{1/2}$ for events; the condition on μ cannot be dropped, since an event of a half torus may be determined by the flux through a plane inside it. If besides $N_\nu = 2h'$, $h' \ge 2$, and G depends only on a quarter of the torus cut by mirrors normal to μ and to ν, then $r[n]|\mu_n(G)| \le \langle |F|^4\rangle^{1/4}$ for every n, and for every $w \ge 0$, $w \ne 0$, and every covariant event E of the quarter, $\mathsf{M}_w[E]/|Z_w| \le \frac{\sum_n w(n)}{\sum_n w(n) r[n]} \langle \mathbf{1}_E\rangle^{1/4}$, where $\mathsf{M}_w = \sum_n w(n) \mathsf{M}_n$. Pairs: if $N_\mu = 2h$, $h \ge 2$, $w \ge 0$ has $\hat{w} \ge 0$ at the time μ, and $\mathbb{M}_w[G] = \sum_s w(n(s)) Z_\beta[s; G(\cdot, s)]$ over the closed s, then $\mathbb{M}_w[G\, \overline{\theta G}] \ge 0$ for every bounded $G(U,s)$, Borel in U and not supposed covariant, that depends only on the bonds and plaquette signs of the h layers on one side of a plane midway between two layers normal to μ, θ the reflection in it.

The infinite lattice: let $\beta > 0$, $K, m \in \mathbb{N}$, $\ell = 2L^m$, let pr be the projection of $a_K\mathbb{Z}^3$ onto $\Lambda_{K,\ell}$, on sites, bonds and plaquettes, and $\mathrm{per}_\ell U = U \circ \mathrm{pr}$, on the bonds, the periodic extension. For a class n and $s \in [n]$ the sign field $s \circ \mathrm{pr}$ on the plaquettes of $a_K\mathbb{Z}^3$ is closed; closed sign fields on the plaquettes of $a_K\mathbb{Z}^3$ being exact, $s \circ \mathrm{pr} = d\chi$ for a sign field χ on the bonds of $a_K\mathbb{Z}^3$, unique up to a factor $d\epsilon$, ϵ a sign field on the sites and $d\epsilon$ at a bond the product of ϵ at its two ends; and the plaquette variables of the twisted extension $\chi\,\mathrm{per}_\ell U$ are $\partial(\chi\,\mathrm{per}_\ell U)(p) = s_{\mathrm{pr}\,p}\,\partial U(\mathrm{pr}\,p)$.

The law $\tilde\mu_n$ of the class n on $a_K\mathbb{Z}^3$ is the image of $e^{-\beta A_s} dU/Z[n]$ under $U \mapsto \chi\,\mathrm{per}_\ell U$; on the functions invariant under $U \mapsto (d\epsilon)U$ for all ϵ it depends neither on $s \in [n]$ nor on χ (a change $s \mapsto s\,d\gamma$ is undone by $U \mapsto \gamma U$, with $\chi(\gamma \circ \mathrm{pr})$ for χ, and χ is fixed up to $d\epsilon$), and $\tilde\mu_0$ is the periodic measure extended periodically. For admissible w let $\tilde\mu_w = \sum_n w(n) r[n]\,\tilde\mu_n / \sum_n w(n) r[n]$; for a board of $a_K\mathbb{Z}^3$ with cells of side $\mathsf{l} \le \ell/2$ and a function F of a cell let $\|F\|_{w,\ell} = \|F \circ \mathrm{per}_\ell\|_w$, on the board of $\Lambda_{K,\ell}$ under it, and let $\|F\|_{\mathrm{ch},\ell}$, the seminorm of [7, Cor. B.1], be the same number at $w = \delta_0$.

Then on the local functions invariant under all $U \mapsto (d\epsilon)U$ the law $\tilde\mu_w$ satisfies [7, Cor. B.1](a)–(d) with $\|\cdot\|_{w,\ell}$ for $\|\cdot\|_{\mathrm{ch},\ell}$ and Z_w for the periodic partition function, and clause (e) of it with the pressure $p_\ell^w = \ell^{-3}\log(|Z_w|/\underline{w})$ for p_ℓ, where $|Z_w|$ is formed on $\Lambda_{K,\ell}$ and $\underline{w}$ is the least positive value of $\hat w(e,m)$ over all μ, e, m ($\underline{w} = 4$ at $w \equiv 1$, $\underline{w} = 1$ at $w = \delta_0$): for $\ell \ge \ell' \ge 2\mathsf{l}$, $p_\ell^w \le p_{\ell'}^w$ and $\|F\|_{w,\ell} \le \|F\|_{w,\ell'}\, e^{\mathsf{l}^3(p_{\ell'}^w - p_\ell^w)}$; at $w \equiv 1$ the same holds with $\ell^{-3}\log|Z_w|$ for p_ℓ^w. (The division by $\underline{w}$ makes $e^{\mathsf{l}^3 p_\ell^w}$, the root of order $(\ell/\mathsf{l})^3$ of $|Z_w|/\underline{w}$, non-increasing in ℓ: with the time μ, a fold integral of Z_w over n_μ cells in the direction μ is $\sum_{e,m} \hat w(e,m)\,\mathrm{Tr}\,B_{e,m}^{\mathsf{n}_\mu/2}$ with positive operators $B_{e,m}$ that do not depend on n_μ; for $q = \mathsf{n}_\mu/\mathsf{n}'_\mu \ge 1$, $\mathrm{Tr}\,B^{q\mathsf{n}'_\mu/2} \le (\mathrm{Tr}\,B^{\mathsf{n}'_\mu/2})^q$, $\omega \le \omega^q$ for $\omega = \hat w(e,m)/\underline{w} \in \{0\} \cup [1,\infty)$, and $\sum_i x_i^q \le (\sum_i x_i)^q$ for $x_i \ge 0$; the three directions are taken in turn.

At $w = t\delta_0$ the undivided pressure is $p_\ell + \ell^{-3}\log t$, and for small t it is larger at ℓ than at $\ell' < \ell$.) Further $r[n]\,|\tilde\mu_n(\prod_\kappa F_\kappa)| \le \prod_\kappa \|F_\kappa\|_{\mathrm{ch},\ell}$ for such functions F_κ of the cells of one period box. For all local functions F_κ of the cells, invariant or not, with the plain extension and the columns of every class on the edges of the board of $\Lambda_{K,\ell}$, so that the functional μ_w names the board, $|\mu_w(\prod_\kappa F_\kappa \circ \mathrm{per}_\ell)| \le \prod_\kappa \|F_\kappa\|_{w,\ell}$.

(9) *(The density of the mixture.)* Let $\beta > 0$, $\mathsf{w}_{\mathrm{mix}} = \sum_s e^{-\beta A_s}$ over the closed sign fields s, and $\hat{\mathsf{w}}_{\mathrm{mix}} = 2^{2-2|N|}\mathsf{w}_{\mathrm{mix}}$, $|N|$ the number of sites. For every class and every bounded centre-blind Borel F, $\int F \sum_{s\in[n]} e^{-\beta A_s}\,\mathrm{d}U = 2^{2|N|-2} Z[n;F]$, whatever the places of the columns; so $\hat{\mathsf{w}}_{\mathrm{mix}}$ is the density of $\sum_n Z[n;\cdot\,]$ on the centre-blind functions. It is the restriction of an entire function of the matrix entries, centre-blind, gauge invariant and invariant under the isometries of the lattice, and $\hat{\mathsf{w}}_{\mathrm{mix}} \le 8$. With $\bar A = \sum_p (1 - \frac12|t_p|)$, $z_p = e^{-\beta|t_p|}$ and $\mathsf{M}(U)$ the set of the unit cubes on whose faces the signs of the t_p have the product -1, $\mathsf{w}_{\mathrm{mix}} = e^{-\beta\bar A}\sum_\Gamma \prod_{p\in\Gamma} z_p$, the sum over the sets Γ of plaquettes whose boundary, the set of the cubes with an odd number of faces in Γ, is $\mathsf{M}(U)$. $\log \mathsf{w}_{\mathrm{mix}}$ is no sum of functions of the four bonds of single plaquettes.

If the angles $\theta_p = \arccos\frac12|t_p|$ of the faces of a unit cube add up to less than π, no t_p there vanishes and the product of their signs is 1, and π cannot be replaced by a larger number; so on $\mathfrak{S} = \{\theta_p < \frac{\pi}{6} \text{ for all } p\}$ the signs of the t_p form a closed sign field $s(U)$, and $\mathfrak{S}$ is the disjoint union of the eight open, non-empty, centre-blind, gauge invariant sets $\mathfrak{S}_n = \{U \in \mathfrak{S} : n(s(U)) = n\}$. Let now all sides be ≥ 4. On $\mathfrak{S}$, for $\beta \ge 4$,

$$\log\bigl(e^{\beta A_{s(U)}}\mathsf{w}_{\mathrm{mix}}(U)\bigr) = \sum_X E(X), \qquad |E(X)| \le \frac{|X|}{198}\prod_{p\in X}(400 z_p),$$

the sum over the connected sets X of plaquettes that are unions of vortices, where in this item two plaquettes are neighbours if they are faces of one unit cube, that is, dual bonds with a common dual site, and a set is connected if it is connected for this relation (so the consecutive plaquettes of a straight column, opposite faces of one cube, are neighbours; the sets D of (4)(e) are connected through common edges instead), a vortex being a non-empty connected set of plaquettes that every unit cube meets in an even number of faces (it has at least 4 plaquettes, and at least $\min_\mu N_\mu$ if the sign field that is -1 on it is not exact), with $E(X) = \log(1 + \prod_{p\in X} z_p)$ for a vortex X that holds no smaller one: thin vortices are local terms. For $\vartheta \le \frac{\pi}{6}$ with $\zeta = 100e^{-2\beta\cos\vartheta} \le \frac12$ and every centre-blind $F \ge 0$ vanishing off $\mathfrak{S}_n \cap \{\theta_p < \vartheta\ \forall p\}$, $(1-\epsilon)\sum_{n'} Z[n';F] \le Z[n;F] \le \sum_{n'} Z[n';F]$ with $\epsilon = \frac{3|N|}{50}\zeta^{\min_\mu N_\mu}$.

(10) *(Area laws from one inequality.)* Let $\beta > 0$ and $D_2(l_0) \le \frac{1}{20}$ at an integer $l_0 \ge 3$. Then $D_2(l_0) > 0$; put $m_* = l_0^{-1}\log\frac{1}{18D_2(l_0)}$, let l_1 be the least $2^i l_0$ with $m_* l_1 \ge 3$, and $\sigma_* = m_*/l_1$. Then $\sigma_* = m_*/l_0$ if $m_* l_0 \ge 3$ and $\sigma_* > \frac16 m_*^2$ otherwise; $D_2(l) \le \frac{1}{18}e^{-m_* l}$ for $l \in \{2^i l_0\}$; $l\,\varepsilon_{\mathrm{v}}(l) \le e^{-\sigma_* l^2}$ for $l \in \{2^j l_1\}$; for every integer $h \ge 4l_1$ and all $t, \lambda \ge (l_1 h)^{1/2}$, on the box (t,λ,h) whose first two sides span the twisted plane,

$$-\log[\tfrac12(1-r)] \ge \sigma_* t\lambda - \frac{4t\lambda}{l_1 h}\Big(\frac12\log\frac{4h}{l_1} - 0.66\Big),$$

in particular, on every cube of side $N \ge 4l_1$, $-\log[\frac12(1-r_1)] \ge \sigma_* N^2 - \frac{4N}{l_1}(\frac12\log\frac{4N}{l_1} - 0.66)$; over the section (N,N), $N \ge 4l_1$, the norm $\Lambda_-(N)$ of T_0 on the functions odd under one of the two flips has $\log(\|T_0\|/\Lambda_-(N)) \ge \sigma_* N - \frac{2}{l_1}\log\frac{4N}{l_1}$; and on the even cubes of side $N \ge \max\{2R, T+1, 4l_1\}$, $|\mu_{N^{\times 3}}(W_{R\times T})| \le \exp[-\sigma_* RT + \frac{T}{N}\log 8 + \frac{2RT}{l_1 N}\log\frac{4N}{l_1}]$. There is exactly one periodic Gibbs state ν, the measures of the periodic cubes converge to it, and $|\nu(W_C)| \le e^{-\sigma_* |A(C)|}$ for every simple closed path C of a coordinate plane that each line parallel to one fixed axis of the plane, through midpoints of bonds, meets at most twice, $|A(C)|$ the number of plaquettes enclosed; such a C is made of T rows of plaquettes along the axis, and if they lie in a segment of length w' along it, $|\mu_{N^{\times 3}}(W_C)| \le 8^{T/N} e^{1.13 D_2(N/2)}[\frac12(1 - r_1(N))]^{|A(C)|/N^2}$ on the cubes $N \in \{2^j l_0\}$ with $N \ge \max\{4l_1, 2l_0, 2T, 2w'\}$. If $\mathsf{e}(l) \le 0.069$ at an integer $l \ge 3$, then $D_2(8l) \le 0.04736$, so that all of this holds at $l_0 = 8l$; and on every cube of side $N \ge 2l$, $\delta(N^{\times 3}) \le 0.13736\,\theta^{N/2l}$ and $1 - r[n] \le 0.4121\,\theta^{N/2l}$ for every $n \ne 0$, $\theta = 12.9\,\mathsf{e}(l) \le 0.8901$.

(11) *(Rectangular loops.)* Let $\beta > 0$, ν a periodic Gibbs state, $\mathsf{m}(R,T) = \nu(W_{R\times T})$ and $\mathsf{w} = \mathsf{m}(1,1)$. Then m depends on R,T alone, $\mathsf{m}(R,T) = \mathsf{m}(T,R)$, $\mathsf{m}(R,T)^2 \le \mathsf{m}(R,T-1)\,\mathsf{m}(R,T+1)$ with $\mathsf{m}(R,0) = 1$, $\mathsf{w} \ge \frac{\beta}{4(\beta+1)}$, $\mathsf{m}(R,T) \ge \mathsf{w}^{RT} > 0$, and $\mathsf{m}(R,T) \le (8\beta/(8\beta+3))^{2(R+T)-4}$ for $R,T \ge 2$. Hence $V(R) = \lim_T -T^{-1}\log\mathsf{m}(R,T) = \inf_T(-T^{-1}\log\mathsf{m}(R,T))$ exists, is concave and non-decreasing on $\{0,1,2,\dots\}$ with $V(0) = 0$; $\sigma_{\mathrm{W}} = \lim V(R)/R = \inf_R V(R)/R$; $\mathsf{m}(R,T) \le e^{-TV(R)} \le e^{-\sigma_{\mathrm{W}} RT}$; $V(1) \ge \log(1 + \frac{3}{8\beta})$ and $V(R) \ge 2\log(1+\frac{3}{8\beta})$ for $R \ge 2$, while $V(1) < 2\log(1+\frac{3}{8\beta})$ for $\beta < \frac{1}{24}$; and $0 \le \sigma_{\mathrm{W}} \le V(1) \le -\log\mathsf{w} \le \log\frac{4(\beta+1)}{\beta}$.

(12) *(Purity floor.)* Let $N_3 = 2h$, $n = (k,0)_3$ with $k \ne 0$, and $F \in [0,1]$ a Borel function of the fields on the closed half slab of the layers $0,\dots,h$ that is invariant under c^k applied to all its layers; F may be such a function of the block fields of the half slab. On the box $(N_1, N_2, 2ih)$ lay F into each of the $2i$ half slabs, mirrored at the odd places; let $\mathcal{Z}_i$ be the periodic integral of the product, $w_F = \mathcal{Z}_1/Z[0]$. If $\mathcal{Z}_1 > 0$, put $\mathfrak{q}_i = \mathcal{Z}_i/\mathcal{Z}_1^i$. Then, for $i \ge 2$,

$$r[n] \ge w_F\,\frac{\mathfrak{q}_i - 2^{1-i}}{1-2^{1-i}}, \qquad r[n] \ge w_F\big((2\mathfrak{q}_2 - 1)_+\big)^{1/2}, \qquad r[n] \ge w_F\Big(\frac{2\|\mathsf{P}\|}{\operatorname{Tr}\mathsf{P}} - 1\Big),$$

where $\mathsf{P} = \mathsf{Y}^*\mathsf{Y}$, Y the operator of F on the half slab, so that $\mathcal{Z}_i = \operatorname{Tr}\mathsf{P}^i$. If $\mathcal{Z}_1 = 0$, positivity gives $\mathsf{P} = 0$ and $\mathcal{Z}_i = 0$ for every i; no purity quotient is asserted in this case.

(13) *(Small cubes, exactly.)* On every cube $N^{\times 3}$, $N \geq 3$, with $r_1 \geq r_2, r_3$ its flux quotients and every time: $|Z_{e,0}/Z[0] - \frac{1}{4}| \leq \frac{1}{4}(2r_1 + r_2) \leq \frac{3}{4}r_1$ for all e; $\delta(N^{\times 3}) \geq \mathsf{G}(N) \geq \log \frac{4}{1+2r_1+r_2} \geq \log \frac{4}{1+3r_1}$; $D_q(N) \geq q^2(q-1)\,\mathsf{G}(N)$ for $q \geq 2$; $Z[0](N^{\times 3})^2/Z[0](2N, N, N) \geq 4/(1+2r_1^2+r_2^2) \geq 4/(1+3r_1^2)$; if N is even, $N \geq 6$ and $r_1 \leq \frac{1}{3}$, then $\mathsf{e}(N/2) \geq \log 2$; and for $\beta > 0$ the bracket of (3) on the cube has $\mathfrak{b}(N, N) \geq \frac{1}{2}(1 - r_1)e^{\mathsf{G}(N)} \geq 2(1 - r_1)/(1 + 3r_1)$. So on a cube with $r_1 \leq \frac{1}{5}$: $\delta \geq \log \frac{5}{2}$, $\mathfrak{b} \geq 1$, and, when N is even with $N \geq 6$, $\mathsf{e}(N/2) \leq 0.069$ fails.

(14) *(A floor and the bulk by construction.)* Let $\lambda(\beta) = \frac{9}{2}\log\beta + 31.4$. (a) Let $\mu\nu$ be any coordinate plane and ρ the third direction. If $\beta \geq 1$ and $h' \geq 1$ is an integer with $2h' \leq \min\{N_\mu, N_\nu\}$, then

$$\log r[n_{\mu\nu}] \geq -N_\rho[32\beta/(h' - \tfrac{1}{2})^2 + 2432\, h'\lambda(\beta)];$$

if $\min\{N_\mu, N_\nu\} \geq 4 + 2(\beta/\lambda(\beta))^{1/3}$, then $\log r[n_{\mu\nu}] \geq -N_\rho[2464\,\beta^{1/3}\lambda(\beta)^{2/3} + 4864\,\lambda(\beta)]$; and if $\min_\lambda N_\lambda \geq 4h' + 2$, then for every $n \neq 0$, $\log r[n] \geq -[32\beta/(h' - \frac{1}{2})^2 + 2432\, h'\lambda(\beta)] \sum N_\rho$, the sum over the twisted planes of n, ρ normal to the plane. (b) With c_∞ of Prop. 11.14 and $P^{\mathrm{G}}(\beta) = \log \frac{2}{\pi\beta^3} - \frac{3}{2}c_\infty$: $P(\beta) - P^{\mathrm{G}}(\beta) \leq 2.2 \cdot 10^5 (\log\beta)\,\beta^{-1/6}$ for $\beta \geq e^{148}$.

■□

The mixed-box comparison still to be proved. The fixed-cutoff transfer identities do not complete the continuum centre-flux argument. One still needs the following estimates for the actual interacting integrals.

The full interacting Hessian (Hess). For every mixed box, every embedding in a periodic covering torus used in Lem. 11.9, every level of its small-field chain, and every admitted real background, remove the factors independent of the fluctuation field. The remaining real fluctuation integral has the form $\int_{D_r} e^{-H(B)}\,\mathrm{d}B$, where $D_r = \{B : |B_y| < r \text{ for every fluctuation site } y\}$. Here H includes the interaction, Jacobian, and all other field-dependent factors of this integral, and is invariant under the covering group. Its restriction to the fixed fields, with the action divided by the order of the group and with Lebesgue measure at orbit representatives, is the fluctuation integral of the mixed box. There are no additional boundary factors or tests in either integral; alternatively, any such factors have a common invariant convex regularization for which the following bounds hold uniformly and which converges to these integrals.

Let $c_- > 0$, $c_+ > 0$, $\mu > 0$ and $g_0 \geq 0$. The function H extends to a real smooth function on D_{2r}, and, on that whole set,

$$\nabla^2 H \geq c_-, \qquad \sup_y \sum_{y'} \sup_{D_{2r}} |\nabla_y \nabla_{y'} H| e^{\mu d(y,y')} \leq c_+, \qquad \sup_y |\nabla_y H(0)| \leq g_0.$$

The constants can be chosen uniformly over the cutoffs, backgrounds, embeddings and labels in a compact subset of $(0, \infty)$, with the dependence on the level, radius and box controlled so that the right side of Prop. 11.10 is at most

$$C|s|L^{3(K-i)} \exp[-cs_{\min}L^{K-i}]$$

at level i. The same realization and bounds hold for the intermediate small-field integrals and boundary regularizations used in the comparison of two consecutive cutoffs.

The twisted window (Flux-window). For the flip-invariant window F defined in Step 11 of the proof of Thm. AH, one must establish, for a fixed sufficiently large m_0, absolute constants $C_2 > 0$, $\epsilon_F \geq 0$, and $t_0 > 0$ such that, on every cube $\ell = 2L^m$, at every cutoff, for $t = \ell/b \leq t_0$ and the one-plane class $n = n_{13}$,

$$Z_K[0; F] \leq C_2 t Z_K[0], \qquad Z_K[n; 1 - F] \leq \epsilon_F t Z_K[0].$$

The window is read at the block layer next to a mirror at the cutoffs $K \geq K_\sharp$, and at the bare layer specified in that step below $K_\sharp$; the flux columns are placed away from these layers. The same

estimates hold for its reflected window. All constants are independent of K and of the representative obtained by the dilations used in that step.

The remaining analytic task is to control the full interacting Hessian and establish the window probabilities. The Gaussian lower bound (Prop. YM3_I:55.2, equation YM3_I:(1099)) and the flip of the two plane means do not supply these estimates. They enter the mixed-box and spectral-limit argument below; the fixed-cutoff transfer identities and confinement implication from the stated box inequality are independent of them.

Detailed sketch AH (Uniformity in the cutoff and the proposed limits). Let the constants be fixed as in Prop. YM3_I:69.1, and $b > 0$. The mixed-box floor in (2)(a) and two-cutoff comparison in (4) remain targets until the full interacting Hessian estimate described above is proved. The same Hessian estimate is required for the proposed consequences in (2)(b,c), the continuum and full-sequence spectral assertions of (3), (5)(b,c), the continuum-function and zero-free conclusions of (6), the full-sequence limits in (7), the uniform-cutoff and limiting assertions of (8), and (9). In (10), the large-N conclusion of (a) and the normalization by positive limiting class masses in (d) still require that step unless the stated positivity has been established separately.

The ceiling in (5)(a) and its proposed consequence (5)(d) still require the twisted-window estimate described above; the same applies to the strict sector-survival and $v_z < 1$ conclusions of (3) and their uses. For the quantitative windows and ultraviolet transport in (10), one must establish the estimates specified there. Fixed-cutoff identities, reflection positivity and estimates whose hypotheses do not invoke these comparisons are unaffected.

(1) *(Window.)* For every box s of the cutoff K and $n \neq 0$, $e^{-2bL^{2K}(n_{23}s_1+n_{13}s_2+n_{12}s_3)} \leq r_K[n](s;b) < 1$, and $r_K[n] \leq r_K[n']$ for every one-plane class n' contained in n; $\delta_K^0 \geq 0$, $\delta_K^n = \log r_K[n] + \delta_K^0 \leq \delta_K^0$, $P(L^K b) \geq -6L^K b$, and the identities and inequalities of Thm. AG(2) hold for δ_K^0 in physical units. For every compact set $\mathsf{J} \subset (0,\infty)$ of labels and every $s_{\min} > 0$ there are j and C, depending on J, $s_{\min}$ and the constants, with $\delta_K^0(s;b) \leq 2C|s|$ for all $b \in \mathsf{J}$ and $K > j$ on every box whose sides are $\geq s_{\min}$, and $\operatorname{Tr} T_m^{t/a_K}/\|T_0\|^{t/a_K} \leq e^{2Ctl_1l_2}$ over the sections (l_1,l_2) for $t, l_1, l_2 \geq s_{\min}$, $m = 0, 1$, $b \in \mathsf{J}$ and $K > j$: with $\mathsf{J} \subset [b_-, b_+]$ one may take j least with $L^j b_- \geq \gamma^{-2}$ and $L^j s_{\min} \geq 2L^{m_0(L^j b_+)}$, and $C = L^{3j} C_*(L^j b_+)$, where $C_*(b)$ is the constant of the comparison $|\log Z_K[0](s;b) - \varepsilon_K|s|| \leq C_*(b)|s|$, with numbers $\varepsilon_K(b)$, on all boxes with sides $\geq 2L^{m_0(b)}$ at $b \geq \gamma^{-2}$, which Lem. 11.1 and the monotone density of F_0 give; $C_*(b)$ and $m_0(b)$ do not decrease in b.

On the cubes, $0 \leq \delta_K^0((2L^m)^{\times 3}; b) \leq (2L^m)^3[c_R + \frac{9}{2}\log b]$ for $b \geq \gamma^{-2}$, $m \geq m_0(b)$ and $K \geq 1$, with c_R formed from the constants of Lem. 11.1, and $\delta_K^0(\ell'^{\times 3}; b) \leq C_f \ell'^3 \log b$ at $\ell' \in \{\ell_\flat(b)/L, \ell_\flat(b)\}$ for $b \geq b_\sharp$ of (4) and $K \geq 4$, C_f formed from the constants; and the supremum of δ^0 over all lattices and couplings with N_μ/β in a compact subset of $(0,\infty)$ is finite. On every box with L-adic sides $\sup_K \delta_K^n < \infty$, and $\inf_K r_K[n] > 0$ if and only if $\inf_K \delta_K^n > -\infty$.

(2) *(Floor.)* (a) If s is mixed for z, there is $C(s,b) < \infty$, depending on s, b and the constants and bounded for b in compact sets, with $\log Z_K^z(s;b) - |s|\, p_\infty(K,b) \geq -C(s,b)$ at every K at which s is a box of the cutoff K; hence $\inf_K r_K[z](s;b) > 0$, uniformly for b in compact sets. (For b in a compact set J, (4) for z and for the class 0, for which s is mixed too, gives $|\log r_{K+1}[z] - \log r_K[z]| \leq 2CL^{-K/10}$ for $K \geq K_0$, with the C, K_0 of (4) at J; the first sentence of (1) bounds $r_K[z]$ from below, uniformly on J, at the finitely many $K < K_0$; and $\delta_K^0 \geq 0$.) (b) For a class with one twisted plane the same holds on every box with L-adic sides, uniformly for b in compact sets, and $\inf\{\rho^z(N,\beta) : N_\mu \geq 3,\ N_\mu/\beta \in I\ \forall \mu\} > 0$ for every compact interval $I \subset (0,\infty)$. (c) For every class the same holds on every box with sides in $2L^{\mathbb{Z}}$, the cubes $\ell = 2L^m$ among them, uniformly for b in compact sets; $c(I) = \inf\{\rho^z(N^{\times 3},\beta) : N \geq 3,\ N/\beta \in I\} > 0$ for every compact interval $I \subset (0,\infty)$, so

that $\sup_{j\ge1}\sup_{t\in I}(-\log\phi_j(t))\le\log(1/c(I))$; and the infimum of ρ^z over the lattices with N/β in a compact subset of $\{v_z>0\}$ of (3) is positive.

On these boxes the weights $r_K[n]/\sum_{n'}r_K[n']$ of the mixture converge, to numbers in $[v_n/8,\frac12]$ for $n\neq0$.

(3) *(Limit.)* For every class there is a continuous $v_z\colon(0,\infty)^3\to[0,1]$ with $\rho^z(N_j,\beta_j)\to v_z(\tau)$ along every sequence with $\beta_j\to\infty$ and $N_j/\beta_j\to\tau$; in particular $r_K[z](s;b)\to V_z(s;b)=v_z(s/b)$ along the full sequence of the cutoffs on every box with L-adic sides, uniformly for b in compact sets. In each side, at all real values of the two others, v_z is real-analytic on $(0,\infty)$, and $|V_z(s;b)-V_z(s';b)|\le\frac{4}{s_-}e^{\Gamma}|s-s'|_1$ for $s,s'\in[s_-,s_+]^3$, where Γ is the supremum of δ^0 over the lattices with $N_\mu/\beta\in[\tau_-/2,2\tau_+]$ for all μ, $[\tau_-,\tau_+]$ an interval that holds $[s_-/b,s_+/b]$ in its interior; Γ is finite by (1). $v_z>0$ on the boxes of (2), $\{v_z>0\}$ is open, $v_z>0$ everywhere for one twisted plane, and $v_z<1$ on every box with L-adic sides.

For each time and all e, m, the sector weight $\frac14\sum_k(-1)^{e\cdot k}v_{(k,m)_\mu}\ge0$ is >0 on all of $(0,\infty)^3$ or $=0$ on all of it, and so is v_z for a class with one twisted plane; so a positive lower limit in the cutoff on one box gives one on all boxes, uniform for the sides divided by the label in a compact set; the limits obey the order of Thm. AG(1), $\frac14\sum_k(-1)^{e\cdot k}v_{(k,m)_\mu}\ge0$, and $v_{((1,1),m)_\mu}\ge v_{((1,0),m)_\mu}+v_{((0,1),m)_\mu}-v_{(0,m)_\mu}$. For each time, each section $l\in\mathbb{D}^2$ and all m,e,i, $E_i^{K,m,e}\to E_i^{m,e}(l;b)\in[0,\infty]$ along the full sequence; the limits extend to real sections, are continuous in b, and $E_i^{m,e}(\lambda' l;\lambda' b)=E_i^{m,e}(l;b)/\lambda'$ for real $\lambda'>0$.

Let $H_{l,m}$ be the diagonal operator whose eigenvalues are the finite limits, the eigenvectors labelled by (e,i), and $C^k=(-1)^{e\cdot k}$ on the label e. Then $H_{l,m}\ge0$ has finite multiplicities and no finite point of accumulation; $\operatorname{Tr}e^{-tH_{l,1}}\le\operatorname{Tr}e^{-tH_{l,0}}<\infty$ for $t>0$, and for every L-adic height h_0 the number of levels $\le\Lambda$ is at most $\exp[h_0\Lambda+2Ch_0l_1l_2]$, C that of (1) at $\mathsf{J}=\{b\}$ and $s_{\min}=\min\{h_0,l_1,l_2\}$; 0 is an eigenvalue of $H_{l,0}$ with label $e=00$; the space of $H_{l,1}$ and each sector e of $H_{l,0}$ are not $\{0\}$; with $H_{l,m}$ at the label b,

$$v_{(k,m)_\mu}((t,l)/b)=\operatorname{Tr}(e^{-tH_{l,m}}C^k)/\operatorname{Tr}e^{-tH_{l,0}},$$

and, for every time μ, the sector weights $w_{e,m}^{[\mu]}=\frac14\sum_k(-1)^{e\cdot k}v_{(k,m)_\mu}$, functions on $(0,\infty)^3$, are by this formula $w_{e,m}^{[\mu]}(s/b)=\operatorname{Tr}(P_ee^{-s_\mu H_{l,m}})/\operatorname{Tr}e^{-s_\mu H_{l,0}}$ on the boxes s with L-adic sides, l the section of s normal to μ and P_e the projection on the label e, each trace divided by the periodic trace of its own time and section; at every point of $(0,\infty)^3$ they obey the crossing relation of Thm. AG(1) with $w_{e,m}^{[\mu]}$ in the place of $Z_{e,m}^{[\mu]}$, $w_{e_1e_2,m}^{[1]}=\frac12\sum_{m',e_2'}(-1)^{e_2m'+e_2'm}w_{e_1e_2',m'}^{[2]}$, and likewise for every pair of times, since on each lattice the relation of Thm. AG(1) divided by $Z[0]$ is this relation between the numbers $\frac14\sum_k(-1)^{e\cdot k}r[(k,m)_\mu]$; $E_{\mathrm{mag}}(l)=\inf\operatorname{spec}H_{l,1}=\lim_K a_K^{-1}\log(\|T_0\|/\|T_1\|)$ is finite, and $E_{\mathrm{el}}(l)=\min\{E_1^{0,e}:e_1=1\}=-\lim_{t\to\infty}t^{-1}\log\frac{1-v}{1+v}$, $v=v_{((1,0),0)_\mu}((t,l)/b)$, is finite.

For each time, L-adic section l, direction ν of it and integer $q\ge2$, $a_K^{-1}[q\log\|T_0(l)\|-\log\|T_0(l^{[\nu]})\|]$ converges to a finite number ≥0, $l^{[\nu]}$ the section with l_ν multiplied by q. δ_K^0 converges on every box with L-adic sides. On the cubes $\ell=2L^m$, $D(t)=\lim_K[L^3\log Z_{K,\ell}-\log Z_{K,L\ell}]$ exists, depends on $t=\ell/b$ alone, is continuous, and $D(t)-(L^3-1)\log(1/t)$ has a finite limit as $t\to0$; $\delta(t)=\lim_K\delta_K^0(\ell^{\times3};b)=\sum_{r\ge0}L^{-3(r+1)}D(L^rt)$, locally uniformly, and $\lim_K\delta_K^n=\log v_n(t,t,t)+\delta(t)$ is finite for every n. The limits do not depend on the choice of the constants: they are the same for the blocking factor of every other admissible choice.

By (6) and Thm. YM3_I:AD, $t\mapsto v_z(t,t,t)$ and D are real-analytic on $(0,\infty)$.

(4) *(Two consecutive cutoffs.)* If s is mixed for z, $\ell \in 2L^{\mathbb{Z}}$ and $\mathsf{J} \subset (0,\infty)$ is compact, there are C, K_0 with

$$\Big|\log Z^z_{K+1}(s;b) - \log Z^z_K(s;b) - \frac{|s|}{\ell^3}\Delta_K[\ell,b]\Big| \le CL^{-K/10} \qquad (b \in \mathsf{J},\ K \ge K_0);$$

so for two classes z, z' for which s is mixed, $|\log(Z^z_K/Z^{z'}_K)(s;b) - \log(V_z/V_{z'})(s;b)| \le C'L^{-K/10}$, the limits lying in $(0,\infty)$. On boxes with sides in $2L^{\mathbb{Z}}$, in every class, the left member tends to 0. Uniformly in the label: there are numbers $b_\sharp, c_1, C_V, C_G, j_0$, formed from the constants, such that with $\varpi = 3 + 2\mathfrak{p}_0/r_\star$ and $c_2 = 20\varpi$, for $b \ge b_\sharp$, every s mixed for z with sides in $L^{\mathbb{N}} \cup 2L^{\mathbb{N}}$, all $\ge \ell_\flat(b)$, $S = \max_\mu s_\mu$, and every $K \ge c_2 \log_L S$ with $((\lceil K/1260\rceil - j_0)\log L + \log b)^{2\mathfrak{p}_0} \ge C_V|s|\log b$, the left member at $\ell = \ell_\flat(b)$ is at most $c_1 S^\varpi L^{-K/10}$; the last condition is void where $|s| \le (\log b)^{2\mathfrak{p}_0 - 1}/C_V$, so that for every $0 \le q' < 2\mathfrak{p}_0 - 3r_\star - 1$, all large b and $s_\mu = \frac12 \ell_\sharp(b) L^{i_\mu}$ with $L^{i_1+i_2+i_3} \le (\log b)^{q'}$ only $K \ge c_2 \log_L S$ is asked. And $0 \le \frac{|s|}{\ell_\flat^3}\log Z_{K,\ell_\flat} - \log Z^z_K(s;b) \le C_G|s|\log b$ for $b \ge b_\sharp$, these boxes and $K \ge j_0 + 2$.

(5) *(Small cubes.)* (a) There are an absolute constant $C_\square$ and $t_0 > 0$ depending on the constants with $r_K[n] \le C_\square t$ on every cube $\ell = 2L^m$ with $t = \ell/b \le t_0$, at every K, for every $n \ne 0$; so $-\log r_K[n] \ge \log(1/t) - \log C_\square$, and no floor is uniform as $t \to 0$. (b) As $t \to 0$, $v_z(t,t,t)/t \to \mathscr{G}_z$, the number eq. 11.iii, whose numerical evaluation gives 0.10868, 0.08251, 0.07574 for one, two, three twisted planes; so $C^{-1}t \le v_z(t,t,t) \le Ct$ for $t \le t_0$; and for $z, z' \ne 0$, with ϑ and θ^a of Prop. 11.14,

$$\lim_{t\to 0}\frac{v_z(t,t,t)}{v_{z'}(t,t,t)} = \exp\Big\{\frac12\int_0^\infty \sum_a \Big[\prod_\mu \vartheta(u,\theta^a_\mu(z)) - \prod_\mu \vartheta(u,\theta^a_\mu(z'))\Big]\frac{du}{u}\Big\},$$

which for two and three twisted planes against one is 0.75926 and 0.69695. (c) On the odd cubes $s = \frac12\ell_\sharp(b)$, $b \ge b_\sharp$, at every cutoff, $|\log r_K[n](s^{\times 3};b) - \log(t\,C^n(N)/N)| \le \omega_1(t)$, $N = s/a_K$, $t = s/b$, $C^n(N) = \lim_{\beta\to\infty}\beta\rho^n(N^{\times 3},\beta)$, with $\omega_1(t) \to 0$ as $t \to 0$. (d) There is $t_1 > 0$, formed from the constants, such that on every cube $2L^m$ with $t \le t_1$, at every cutoff, $r_1 \le \frac15$; there, by Thm. AG(13), $\delta^0_K \ge \log\frac52$, $\mathfrak{b}_K \ge 1$ on the cube and $\mathsf{e} \le 0.069$ fails at the half side; and $\liminf_{t\to 0}\inf_K \delta^0_K \ge \log 4$, the four electric sectors being equally filled in the limit.

(6) *(The label.)* Let a reference coupling $g \le \gamma$, a side $\ell \in \mathfrak{L}(g)$ (equation YM3_I:(1528)) and a label $b_0 \in I(g)$ (Cor. YM3_I:74.2) be given, with the radius and the constants of Thm. YM3_I:AD. There are $R_0, R_0' > 0$ and $\mathsf{M}_\mathrm{S} < \infty$, formed from these data and the constants and not depending on K, such that for every K the periodic partition function of the cube ℓ has no zero at the complex labels $b_0 - c$, $|c| < R_0$, and $c \mapsto r_K[n](\ell; b_0 - c)$ is holomorphic in $|c| < R_0$ with $|r_K[n]| \le 27\mathsf{M}_\mathrm{S}$ on $|c| \le R_0/8$ for one or two twisted planes, and holomorphic in $|c| < R_0'$ with $|r_K[n]| \le 12600\mathsf{M}_\mathrm{S}$ on $|c| \le R_0'/15$ for three; so $\sup_K |\partial_b^\nu r_K[n](\ell;b_0)| \le 27\mathsf{M}_\mathrm{S}\,\nu!\,(8/R_0)^\nu$ for one or two planes, $\le 12600\mathsf{M}_\mathrm{S}\,\nu!\,(15/R_0')^\nu$ for three.

For every $t' > 0$ there are j', $\varrho > 0$ and M such that for $j \ge j'$ the function ϕ_j extends holomorphically to $\{|t - t'| < \varrho\}$ with $|\phi_j| \le \mathsf{M}$ there; hence $\phi_j \to v_n(t,t,t)$ with all derivatives, locally uniformly on a complex neighbourhood of $(0,\infty)$. There is an open $\Omega \subset \mathbb{C}$ containing $(0,\infty)$ such that for every compact $Q \subset \Omega$ there is j_Q with $Z[n] \ne 0$ on the cube of $2L^j$ sites at $\beta = 2L^j/t$ for $t \in Q$, $j \ge j_Q$. With $m_0 = \inf_K r_K[n](\ell;b_0) > 0$, $\int \exp[-L^K(b_0 - c)A_{\eta^n}]\,\mathrm{d}U \ne 0$ for all complex $|c| < R_1$ and all K, where $R_1 = \min\{R_0/16, m_0R_0/864\mathsf{M}_\mathrm{S}\}$ for one or two twisted planes and $R_1 = \min\{R_0'/30, m_0R_0'/756000\mathsf{M}_\mathrm{S}\}$ for three.

For labels in $[b_1,b_2] \subset I(g)$, with $R_* = R_0(b_1)$, $R_*' = R_0'(b_1)$, m the infimum of $r_K[n](\ell;b)$ over K and $b \in [b_1,b_2]$, $\mathcal{O} = a_K^{-1}A$ under μ_0 and $\mathcal{O}^n = a_K^{-1}A_{\eta^n}$ under μ_n: at every K, for one

or two twisted planes, the means differ by at most $V_1 = 216\mathsf{M}_\mathrm{S}/(R_*\mathsf{m})$ and the variances by at most $3456\mathsf{M}_\mathrm{S}/(R_*^2\mathsf{m})+V_1^2$; for three, by $V_1 = 189000\mathsf{M}_\mathrm{S}/(R'_*\mathsf{m})$ and $5670000\mathsf{M}_\mathrm{S}/(R'^2_*\mathsf{m})+V_1^2$.

(7) *(Block pairs.)* Extend the averaging map $\mathcal{M}$ to pairs (U,s), s closed, by $\mathcal{M}_s(U)(c) = \gamma[c]\,\mathcal{M}(\gamma U)(c)$, where γ is a sign field on the bonds with $d\gamma = s$ on the two blocks at the ends of the block bond c and $\gamma[c]$ is its product over the L bonds of the straight path of c; the block sign field is the product of s over the L^2 plaquettes of a block plaquette, and $s^{(d)}$ is its $(K-d)$-th iterate. Let $w \ge 0$, $w \not\equiv 0$, $\ell = 2L^m$ with $m \ge 1$, $K \ge 2$ and $0 \le d \le K$. Under $\sum_n w(n) \sum_{s\in[n]} e^{-\beta A_s} dU$ the law of the block pairs $(\mathcal{M}_s^{K-d}U, s^{(d)})$ of depth d is reflection positive at the mirrors midway between neighbouring layers of the block lattice of depth d normal to each direction in which $\hat w \ge 0$; for admissible w in all three directions; and so is every weak limit point as $K \to \infty$ of the laws of the towers $((\mathcal{M}_s^{K-d}U, s^{(d)}))_{1\le d\le K}$. At every cutoff the block sign field $s^{(d)}$ of a closed s is closed and $n(s^{(d)}) = n(s)$, its product over a plane of the block lattice being the product of s over the plane of the lattice that holds it: the class is a function of the block pairs. Let $b \ge \gamma^{-2}$ and $MR(b) \le L^m$, let $d_\natural$ be the least $d \in \mathbb{N}$ with $L^d b \ge \mathsf{t}_\natural$, and for $d \ge d_\natural$ let

$$\mathsf{R}(d) = 2M^{-3}(\ell L^d)^3 \exp\bigl[-c' A_0^2 (\log L^d b)^{2\mathfrak{p}_0}\bigr] \exp\bigl[(C + \tfrac{9}{2}\log b)\,\ell^3\bigr] \le C(\ell,b) L^{3d} e^{-cd^2},$$

where $\mathsf{t}_\natural$, c', A_0, C, c are formed from the constants and $C(\ell,b)$ from ℓ, b and the constants; $\mathsf{R}(d)$ does not depend on K. For integers $d \ge d_\natural$ and $K \ge d+1$ let $\hat E_{K,d}$ be the event of the pairs (U,s) that for some depth $d' \in [d,K]$ and some block plaquette P of depth d' the matrix $s_P^{(d')}\,\partial(\mathcal{M}_s^{K-d'}U)(P)$ is farther than $1/400L^2$ from $\mathbf{1}$: a union of covariant local events, one for each block plaquette, which at $s=1$ is the event that some plaquette of the block field $\mathcal{M}^{K-d'}U$ of some depth $d' \in [d,K]$ is farther than $1/400L^2$ from $\mathbf{1}$. Its periodic probability is at most $\mathsf{R}(d)$; off it, at each depth $d' \in [d,K]$, the signs of the traces of the block plaquettes of the block field $\mathcal{M}_s^{K-d'}U$ form the block sign field $s^{(d')}$, whose class is $n(s)$, so that off $\hat E_{K,d}$ the class is a function of each of these block fields alone; and $r_K[n]\,\mu_n(\hat E_{K,d}) \le 16\mathsf{R}(d)^{1/4}$, while under the normalised mixture of weight w with $w(0) > 0$ the probability of $\hat E_{K,d}$ is at most $\frac{\sum_n w(n)}{w(0)} 16\mathsf{R}(d)^{1/4}$.

Other pairs (ℓ,b) are brought into this range by the dilation by a power L^j with $L^j b \ge \gamma^{-2}$ and $MR(L^j b) \le L^{m+j}$; then the same holds at the depths $d \ge j + d_\natural(L^j b)$, a threshold that depends on ℓ and b. Under these conditions, in every weak limit point, with $r_n = \lim_K r_K[n]$ and $\mathbb{P}^n$ the limit law of the towers in the class n: the class is, up to null sets, a function of the tower of block fields, and the classes with $r_n > 0$ are carried by disjoint sets of towers; each class is locally dominated by the periodic limit law, that is, for $d \ge 1$ and every covariant Borel event B of the block pairs of depth d that depends only on a quarter of the torus cut by block mirrors normal to two directions, $r_n \mathbb{P}^n(B) \le \mu_d(B_1)^{1/4}$, where $B_1 = \{V : (V,1) \in B\}$ and μ_d is the periodic limit law of the block field of depth d, and $r_n\mathbb{P}^n(B) \le \mu_d(B_1)^{1/2}$ if B depends only on a half of the torus whose normal lies in every twisted plane of n; and in every class with $r_n > 0$ the towers are consistent under $\mathcal{M}_s$: with (V_d, s_d) the pair of depth d of a tower, almost surely $(V_d, s_d) = (\mathcal{M}_{s_{d+1}} V_{d+1},$ the block sign field of $s_{d+1})$ for all $d \ge 1$.

(8) *(Tension of a slab.)* For $\ell = 2L^j$ and $K+j \ge 2$, with r_1 on the cube ℓ and $\kappa_K = \log(2/(1-r_1))$,

$$\Bigl(1-\frac{1}{L}\Bigr)\ell\, E_{\mathrm{el},K}(\ell,\ell) \le L^2 \delta_K^0((\ell/L)^{\times 3}; b) + \kappa_K, \qquad \sigma_K(\ell)\,\ell^2 \le \Bigl(\frac{L}{L-1}\Bigr)^2 \Bigl[L^3\,\delta_K^0((\ell/L)^{\times 3}; b) + \kappa_K\Bigr],$$

$E_{\mathrm{el},K} = a_K^{-1}\varepsilon_{\mathrm{el}}$ over the section (ℓ,ℓ), and $(L/(L-1))^2 \le 1.1737$ for $L \ge 13$. Both right members are bounded in K; so $\sup_K \sigma_K(\ell) < \infty$; by Thm. AG(13) the lower limits in K of $\delta_K^0(\ell^{\times 3}; b)$, of G over the section of the cube and of every $D_q(\ell/a_K)$ are positive; and in every limit the least level of the sector $e = 10$ of $H_{(\ell,\ell),0}$ is at most $\frac{L}{(L-1)\ell}[L^2 \sup_K \delta_K^0((\ell/L)^{\times 3}; b) +$

$\log\frac{2}{1-v_1(t,t,t)}]$, $t=\ell/b$. For $l\in\mathbb{D}$, $\sigma_K(l)\ge -(t\lambda)^{-1}\log\mathfrak{b}_K(t,\lambda;l)$ at every L-adic (t,λ); for $t>t_0$, $\lambda>\lambda_0$ in $\mathbb{D}$ and $s'=2L^i\le\min\{t_0,\lambda_0,l\}$,

$$-\log\mathfrak{b}_K(t,\lambda;l)\ge(t-t_0)\Big[(\lambda-\lambda_0)\sigma_K(l)-\frac{2\lambda_0 l}{s'^3}\bar\delta(s')\Big]-\frac{t_0\lambda l}{s'^3}\bar\delta(s'),\qquad \bar\delta(s')=\sup_K\delta^0_K(s'^{\times 3};b),$$

at all K at which the boxes and the cube are formed; so $\liminf_K\sigma_K(l)>0$ if and only if $\mathfrak{b}_K(t,\lambda;l)\le c<1$ at one L-adic (t,λ) for all large K.

(9) *(Small twisted section.)* There is $c_0>0$, depending on the constants, such that for $l\in L^{\mathbb{Z}}$ with $t=l/b\le c_0$, over the section (l,l) with the time 1 at every cutoff with $l/a_K\in L^{\mathbb{N}}$, the following hold. In this item $\lambda_0\ge\lambda_1\ge\lambda_2\ge\dots$ are all the eigenvalues of T_1, the four sectors together, each repeated according to its multiplicity, and the levels of T_1 are the numbers $a_K^{-1}\log(\lambda_0/\lambda_i)$, $i\ge0$; as $\lambda_0=\|T_1\|$, they are the levels $E_j^{K,1,e}$ of the setting, the four sectors together, each diminished by $a_K^{-1}\log(\|T_0\|/\|T_1\|)$, and by (3) they converge along the full sequence to the eigenvalues of $H_{(l,l),1}$, each diminished by $E_{\mathrm{mag}}(l,l)=\inf\operatorname{spec}H_{(l,l),1}$, which is the least eigenvalue of the sector 00. (a) The top eigenvalue λ_0 of T_1 is simple, and the gap $a_K^{-1}\log(\lambda_0/\lambda_1)$, λ_1 the next eigenvalue, lies between $1.2142/l$ and $\pi L/((L-1)l)\le 3.4034/l$; for $t\le c_0'$, $c_0'\le c_0$ formed from the constants, it lies between $3.0117/l$ and $3.19/l$ at every such cutoff, and at all large cutoffs every level of T_1 but the least, in all four sectors, is at least $1.2519/l$, with $\sum e^{-lE}\le e^{0.2515}-1<0.2860$ over these levels E. (b) With $E(e)$ the least eigenvalue of $H_{(l,l),1}$ in the sector e minus its least eigenvalue $E_{\mathrm{mag}}(l,l)$, $lE(e)\to\pi|e|$ as $t\to0$: $\pi,\pi,\sqrt{2}\pi$ for $e=10,01,11$; and l times the second eigenvalue of $H_{(l,l),1}$ in the sector 00, counted with multiplicity, minus $E_{\mathrm{mag}}(l,l)$ tends to 2π as $t\to0$: in the normalisation of this item, l times the limit in the cutoff of the least level of T_1 in the sector 00 above the vacuum tends to 2π. (c) Let $t\le c_0'$, and let the cutoffs be those with $l/a_K=L^J$, $J\ge2$.

A plaquette function is a bounded Borel function of the numbers $1-\frac12\eta(p)\operatorname{tr}\partial U(p)$ of finitely many plaquettes p, η a closed sign field of the class; it is a covariant function. Let $\omega_K(G)=\lim_{h\to\infty}\mu_{n_{23}}(G)$ on the boxes $(h,l/a_K,l/a_K)$, for plaquette functions G of finitely many layers: the state of the twisted cylinder $C_l=\mathbb{R}\times(\mathbb{R}/l\mathbb{Z})^2$ at the cutoff K, invariant under the translations of its lattice and under the reflection θ_K in the plane at the time $a_K/2$, midway between two layers. For $f\in\operatorname{Lip}_c(C_l)$ let $\mathcal{O}_K(f)=a_K^{-1}\sum_p f(x(p))(1-\frac12\eta(p)\operatorname{tr}\partial U(p))$, $x(p)$ the barycentre of p, $e_K(f)=a_K^{-1}\sum_p f(x(p))\bar a_{K,p}$, $\bar a_{K,p}$ the mean of $1-\frac12\operatorname{tr}\partial U(p')$ in the periodic measure of the cube $2l/L$ at the cutoff K for a plaquette p' of the orientation of p, $\mathcal{O}_K^-(f)=\mathcal{O}_K(f)-e_K(f)$, and $\|f\|_\sharp=\max\{\sup|f|,40M\operatorname{Lip}f\}$.

For $n'\ge1$ and $d,R>0$ there are $C_{n'}(d,R)$ and $K_{n'}$, formed from n', d, R, l, t and the constants, with $|\omega_K(\prod_{q\le n'}\mathcal{O}_K^-(f_q))|\le C_{n'}(d,R)\prod_q\|f_q\|_\sharp$ for $K\ge K_{n'}$ whenever the supports of $f_1,\dots,f_{n'}$ have mutual distances $\ge d$ and each lies in a ball of radius R. Every sequence of cutoffs tending to infinity has a subsequence along which the levels of T_1 converge, to $E_0\le E_1\le\dots$, the eigenvalues of $H_{(l,l),1}$ diminished by $E_{\mathrm{mag}}(l,l)$, and $S_{n'}(f_1,\dots,f_{n'})=\lim\omega_K(\prod_q\mathcal{O}_K^-(f_q))$ exists for all n' and all tuples of functions of class C^1 with compact and disjoint supports; each $S_{n'}$ is multilinear, symmetric, invariant under the translations of C_l and the reflection of the time, bounded as above, and reflection positive.

For every such limit, the Hilbert space $\mathcal{H}$ formed from the tuples supported at positive times, with the vacuum Ω, carries a contraction semigroup e^{-sH} with $H\ge0$ of pure point spectrum contained in $\{E_i\}$; $E_0=0$ and $3.0117/l\le E_1\le 3.19/l$; $\ker H=\mathbb{C}\Omega$ and $\operatorname{spec}H\subset\{0\}\cup[3.0/l,\infty)$; for tuples F,G supported at positive times and $s\ge0$,

$$|S(\theta F,\tau_s G)-S(F)S(G)|\le e^{-3.0\,s/l}\left(S(\theta F,F)-S(F)^2\right)^{1/2}\left(S(\theta G,G)-S(G)^2\right)^{1/2},$$

θ the reflection of the time and τ_s the translation by s; with the law of (4) used also on the boxes (L^3l,l,l), 3.0 may be replaced by 6.1 in the last two assertions, the action density

being blind to the electric sectors other than 00. Finally $\mathcal{H} \neq \mathbb{C}\Omega$: with $l = L^m$, let $j(t)$ be the least integer for which the reference coupling $g = (L^{j(t)}/2t)^{-1/2}$ is at most the ceiling γ_W of the reference couplings of Thm. YM3_I:AV and $2L^{j(t)-1}$ is a side at g (equation YM3_I:(1528)), an integer formed from t and the constants; there is a number $d_* > 0$, formed from t and the constants, such that the bump $f^\circ(y) = (1 - |y - (2d^\circ, \xi)|^2/d^{\circ 2})_+^2$ of the radius $d^\circ = L^{m-j(t)}d_*$, ξ a point of the section, has $S_2(\theta f^\circ, f^\circ) - S_1(f^\circ)^2 \geq \frac{1}{8}\,10^{-6}c_{\mathrm{sd}}b^{-2} > 0$, b the label and $c_{\mathrm{sd}} = 9/16\pi^2$; so d° is l times a number formed from t and the constants.

(10) *(Even cubes.)* Let $n \neq 0$, $q_K(b) = r_K[n](\ell_\sharp(b)^{\times 3}; b)$, $K_1(b) = \lfloor \frac{1}{68}\log_L b \rfloor$ and $C^n(N) = \lim_{\beta\to\infty} \beta\rho^n(N^{\times 3}, \beta)$. There is $t_{\mathrm{w}} > 0$ such that for $t \leq t_{\mathrm{w}}$ there is a least integer $\mu(t)$ with $b'(t) = 2L^{\mu(t)}/t \geq b_\sharp$ and $\ell_\sharp(b'(t)) = 2L^{\mu(t)}$; then $\log\frac{1}{t} \leq \log b'(t) \leq 2\log\frac{1}{t}$ and $\phi_j(t) = q_{j-\mu(t)}(b'(t))$. (a) $|\log\rho^n(N^{\times 3}, N/t) - \log(tC^n(N)/N)| \leq \omega(t) \to 0$ $(t \to 0)$ for $N \in 2L^{\mathbb{N}}$ with $N \geq N(t)$, and for $N \in 2L^{\mathbb{N}}$ with $N \leq \ell_\sharp(b'(t))\, b'(t)^{1/68}$. (b) To complete the ultraviolet comparison one needs the estimate denoted **(R)**: for some $\kappa > 0$, $|\log q_{K+1} - \log q_K| \leq c_1\ell_\sharp^{\varpi}L^{-\kappa K}$ for $b \geq b_\sharp$, $K \geq c_2\log_L \ell_\sharp$, as (4) gives on mixed boxes.

That estimate would imply $\sup_{b'\geq b}\sum_{K\geq K_1(b')}|\log q_{K+1} - \log q_K|(b') \to 0$ as $b \to \infty$; the bound of (a) holds for all $N \in 2L^{\mathbb{N}}$ at $t \leq t_0$, so $ct \leq r_K[n] \leq Ct$ on the small cubes $2L^m$ at every cutoff; and $\sum_j |\phi_{j+1} - \phi_j| < \infty$ on $(0, t_{\mathrm{w}}]$. For the lower bound $ct \leq r_K[n]$ and for the bounded variation the one-sided form $\log q_{K+1} \geq \log q_K - c_1\ell_\sharp^{\varpi}L^{-\kappa K}$ of (R) is enough. (c) The window estimates in §11.1.5 and their transport between cutoffs remain to be established. The transport estimate sought, denoted **(W)**, is: for one $\Lambda < \lambda_\flat$ and some $c, \eta, t_* > 0$, $\varsigma_K \geq e^{-\Lambda}\varsigma_{K_\sharp} - e^{-c(\log b)^{1+\eta}}$ for $K \geq K_\sharp$, $t \leq t_*$ (Lem. 11.11 and §11.1.5, with F_t or F_t').

Then $r_1(N, N/t) \geq c't$ for small t and all $N \in 2L^{\mathbb{N}}$. $\Lambda \leq 0.55L - 1.1$ is admissible for F_t, and $\Lambda = \frac{L}{2}$, a loss of $\frac{1}{2}$ for each cube of the box $(\ell, \ell, L\ell)$, for F_t', as $L \geq 7$. (W) may be replaced by the same inequality for F_t' on the box $(\ell, \ell, i\ell)$ at one $2 \leq i \leq L$, with L replaced by i in the definition of ς_K and $\Lambda < \log(0.42 \cdot 1.8712^i)$; or by **(W′)**: for some $\gamma' > 0$, $\|\mathsf{P}_K\| \geq \frac{1}{2}e^{\gamma'}\operatorname{Tr}\mathsf{P}_K$ for $K \geq K_\sharp$ and $t \leq t_*$, P_K the operator P of Thm. AG(12) for F_t on the cube ℓ at the cutoff K; under (W′), $r_1(N, N/t) \geq \frac{1}{2}\gamma' c_\flat t$ for small t and all $N \in 2L^{\mathbb{N}}$.

With the window F_t'' of §11.1.5 each of these gives the same for r_2. (d) On a cube $2L^m$ with $b \geq \gamma^{-2}$ and $MR(b) \leq L^m$ the weak limit points of the laws of (7) are, in every class n with $\lim_K r_K[n] > 0$, carried by towers consistent under $\mathcal{M}_s$, by (7). Full convergence of the block-pair laws still requires the summable total-variation estimate denoted **(M)**: $|\mu_{\mathrm{mix}}^{K+1}(H \circ \mathcal{M}_s^{K+1-d}) - \mu_{\mathrm{mix}}^{K}(H \circ \mathcal{M}_s^{K-d})| \leq o_K \sup|H|$, $\sum_K o_K < \infty$, for $\mu_{\mathrm{mix}} = \mathbb{M}_w/\mathbb{M}_w[1]$ at $w \equiv 1$ and all bounded Borel H of the block pairs of depth d with $H(\gamma V, s\,\mathrm{d}\gamma) = H(V, s)$. This estimate would give convergence in total variation of the block-pair laws under μ_n in every class.

■□

Theorem AI (Area laws uniformly in the cutoff, under one inequality)**.** The final small-side exclusion requires **(Hess)** and **(Flux-window)**; the fixed-cutoff confinement implications use only their stated box hypothesis. Let the constants be fixed as in Prop. YM3_I:69.1 and $b > 0$, and assume the hypothesis

(H) there are $s_0 > 0$ and K_0 such that for every $K \geq K_0$ the number $l_K = s_0/2a_K$ is an integer ≥ 3 and $\mathsf{e}(l_K) \leq 0.069$ at $\beta = L^K b$.

At each cutoff $K \geq K_0$, (H) is one inequality between the partition functions of two periodic boxes, $\mathsf{e}(l_K) = 2F_0(l_K, 2l_K, 2l_K) - F_0(2l_K, 2l_K, 2l_K) \leq 0.069$; the theorem is conditional on this hypothesis. Then for every $K \geq K_0$, at $\beta = L^K b$: Thm. AG(10) applies at $l_0 = 8l_K$, with $l_0 a_K = 4s_0$, $D_2(l_0) \leq 0.047355$, $m_* l_0 \geq 0.1597$, $4s_0 \leq l_1 a_K \leq 150.4\, s_0$ and $\sigma_* a_K^{-2} \geq 2.65\cdot 10^{-4} s_0^{-2}$; on every cube of physical

side $\ell \geq 602\, s_0$, $-\log[\frac{1}{2}(1-r_1)] \geq 2.65 \cdot 10^{-4} \ell^2/s_0^2 - (\ell/2s_0)\log(\ell/s_0)$; the periodic Gibbs state ν_K of the lattice $a_K \mathbb{Z}^3$ (Section YM3_I:91) is unique and $|\nu_K(W_{R\times T})| \leq \exp[-2.65 \cdot 10^{-4}(Ra_K)(Ta_K)/s_0^2]$ for every rectangle; on every cube of physical side $\ell \geq s_0$, $1 - r_K[n] \leq 0.4121 \cdot 0.8901^{\ell/s_0}$ for every $n \neq 0$; and $\sigma_K(\ell) \geq 1.9\, \ell^{-2}$ for every $\ell \in \mathbb{D}$ with $\ell \geq s_0$ and $\ell/a_K \in \mathbb{N}$. All these bounds are uniform in $K \geq K_0$. Under **(Hess)** and **(Flux-window)**, with t_1 of Thm. AH(5), (H) fails if $s_0 \leq t_1 b$: it can hold only at physical sides of the order of the label or more. ■□

11.1.4. *Construction estimates.* We use Thm. YM3_I:D, the inductive description of the effective densities, with its constants and their conditions (Prop. YM3_I:69.1 and Cor. YM3_I:69.4), and the lattices and tori T_ℓ (Section YM3_I:3); through Thm. YM3_I:D we use the small-field chain on T_ℓ with its small-field tests and its control of large fields (Prop. 4.54), the comparison of two consecutive cutoffs on T_ℓ (Section YM3_I:116), reflection positivity on tori, the statements on blocks and on the averaging map $\mathcal{M}$ (its covariance under lattice isometries and sign changes), and the convergence in total variation of the laws of the block fields on T_ℓ (Cor. 2.8).

We use the stability bound on periodic cubes (Lem. 11.1): for $b \geq \gamma^{-2}$ and $m_\iota \geq m_0(b)$, $\log Z_{K,\ell_\iota}$ at $\ell_\iota = 2L^{m_\iota}$ lies within a constant times ℓ_ι^3 of a vacuum energy E_ι, and $|E_1/\ell_1^3 - E_2/\ell_2^3|$ is bounded, no constant depending on K or m_ι. We use Thm. YM3_I:AD, the local source-energy and stability theorem, with the windows of labels (Cor. YM3_I:74.2), the sides (equation YM3_I:(1528)), the extents ([7, eq. 321]) and the bound on the number of localization domains ([7, Lem. O.1]); and, of the constants, that the blocking factor L is an odd integer with $L \geq 13$, and that the ordered choice in the proof of Prop. YM3_I:69.1, with the further floors of Section YM3_I:82, permits every sufficiently large odd L with the preceding constants fixed, so that both L and $L+2$ serve.

For Thm. AH(9)(c) we use besides, with the radius and the constants of Thms. YM3_I:AJ and YM3_I:AD, the bounds on the joint cumulants of the action density on periodic tori, uniform in the cutoff ([7, Lem. N.2]), and the short-distance law of its two-point function (Thm. YM3_I:AV), whose leading term is $c_{\mathrm{sd}} b^{-2}$ times the free kernel, $c_{\mathrm{sd}} = 9/16\pi^2$ (Thm. YM3_I:C and equation YM3_I:(2240)), with the witness depth (equation YM3_I:(2321)); and, for its last assertion, the normalisers at the first-passage data (Prop. YM3_I:91.12 (1)), the transfer principle for one law and for two laws under one normaliser (Prop. YM3_I:91.11), the first facts on states ([7, §K.1]), the device of sign averaging over a packing with its count (the packing and sign-averaging argument in the proof of Prop. YM3_I:127.1), and the first-passage data at the labels of the windows (Cor. YM3_I:74.2 and [7, Lem. G.4]).

For Prop. 11.15 we use besides the lemma on rearranged paths (Lem. YM3_I:4.1). On one lattice we use the exact statements among the construction results: the periodic lattice and its means, the crossing operator, slabs and slab operators, boards and the periodic chessboard estimate (Prop. YM3_I:86.3), the chessboard theorem on the infinite lattice ([7, Cor. B.1]), and the statements on periodic Gibbs states (Prop. YM3_I:91.2 and Prop. YM3_I:91.9). A Gibbs state at β on the infinite lattice $a\mathbb{Z}^3$ is, as in Section YM3_I:91, a Borel probability measure ν on its fields with $\nu(\gamma_B F) = \nu(F)$ for every finite set B of bonds and every continuous function F of finitely many bonds, γ_B the conditional kernel of B with the weight $\exp[-\beta \sum_p (1 - \frac{1}{2}\mathrm{tr}\, \partial U(p))]$, the sum over the plaquettes p that hold a bond of B (equation YM3_I:(1818)); it is periodic if it is invariant under the translations of a subgroup of finite index of $a\mathbb{Z}^3$ (Section YM3_I:91), and every periodic Gibbs state is invariant under all isometries of the lattice (Prop. YM3_I:103.4).

For a box N, μ_N is the measure $\langle\cdot\rangle = \mu_0$ of the periodic box N; where it stands beside a measure on the fields of the infinite lattice, it is extended periodically. Thm. AI rests, besides, on its hypothesis (H): at each cutoff $K \geq K_0$, the one inequality $\mathsf{e}(l_K) \leq 0.069$ between the partition functions of the two periodic boxes $(l_K, 2l_K, 2l_K)$ and $(2l_K, 2l_K, 2l_K)$.

11.1.5. *Estimates and proofs.*

Lemma 11.2 (Power sums; lists of levels)**.** Let $T \ge 0$ be compact, Q a projection commuting with T, $z(N) = \operatorname{Tr}(QT^N) < \infty$ for integers $N \ge N_*$, and $\lambda = \|QT\|$. Then $z(N)^2 \le z(N-j)z(N+j)$ for $N - j \ge N_*$; $z(N)^{1/N}$ does not increase and tends to λ; $z(N+N') \le \lambda^{N'} z(N) \le z(N)z(N')$; and, if $\lambda > 0$, $z(N)/\lambda^N \ge 1$ does not increase and is at most $(z(N)^q/z(qN))^{1/(q-1)}$ for every integer $q \ge 2$. (a) If (E_i), (E_i') are non-decreasing sequences in $[0,\infty]$ with $f(t) = \sum_i e^{-tE_i}$ and $f'(t) = \sum_i e^{-tE_i'}$ finite for all $t > 0$, and $f(t_r) = f'(t_r)$ along one sequence $t_r \to \infty$, then $E_i = E_i'$ for all i. (b) If non-decreasing lists $(E_i^K)_i$ in $[0,\infty]$ have $\sum_i e^{-t_0 E_i^K} \le B(t_0)$ for all large K and every t_0 of a set with infimum 0, then $E_i^K \ge t_0^{-1}\log(i/B(t_0))$; every infinite set of indices K holds an infinite subset along which all E_i^K converge in $[0,\infty]$; and along it $\sum_i e^{-tE_i^K} \to \sum_i e^{-tE_i} < \infty$ for every $t > 0$, locally uniformly on $(0,\infty)$, the limit list having finitely many members in every bounded set. ■

Proof sketch. If $QT = 0$, all the unnormalised power sums vanish. Otherwise $\lambda > 0$. With $z(N) = \sum_i \lambda_i^N$, λ_i the eigenvalues of QT, the first assertions are the Cauchy–Schwarz inequality, the monotonicity of ℓ^p norms in p, and $\sum_i \lambda_i^{N+N'} \le \lambda^{N'} \sum_i \lambda_i^N$, $\lambda^{N'} \le z(N')$; with $\mu_i = \lambda_i/\lambda$, $x(N) = \sum_i \mu_i^N \ge 1$ does not increase and $x(qN) \le x(N)$, so $x(N)^q/x(qN) \ge x(N)^{q-1}$, the powers of λ cancelling in $z(N)^q/z(qN)$. For (a), with $f_i(s) = \sum_{p \ge i} e^{-sE_p}$, $e^{-sE_i} \le f_i(s) \le e^{-(s-s_1)E_i} f_i(s_1)$ for $s \ge s_1$, so $E_i = -\lim_r t_r^{-1} \log f_i(t_r)$; by induction f_i and f_i' agree along (t_r), and so do E_i and E_i'. Without the finiteness of f at all $t > 0$ the conclusion fails: $E_i = 1$ and $E_i' = 2$ for all i. For (b), $i\, e^{-t_0 E_i^K} \le \sum_{p \le i} e^{-t_0 E_p^K} \le B(t_0)$; $[0,\infty]^{\mathbb{N}}$ is compact and metrisable; and for $t \ge t_1 > t_0$ the terms are bounded by $\min\{1, (B(t_0)/i)^{t_1/t_0}\}$, summable in i, so dominated convergence applies. □

Lemma 11.3 (The path inequality)**.** Let X be a continuous, strictly positive, symmetric kernel on a compact metric space with a Borel probability measure, u, v continuous and strictly positive, and $a \ge 1$ odd. Then $\langle u, X^a v\rangle \ge \langle u, Xv\rangle^a (\|u\|_2 \|v\|_2)^{1-a}$. In particular, if $t^{(h)}$ is the kernel of T_m^h, g a flip c^k of the fields of a section, $\Delta(h) = \int t^{(h)}(V, gV)\,\mathrm{d}V$ and $\mathsf{Z}(h) = \int t^{(h)}(V,V)\,\mathrm{d}V$, then for integers $n' \ge 1$, $m_1, m_2 \ge 1$, $m = m_1 + m_2$, and odd k',

$$t^{(k'n'+m)}(V, gW) \ge \frac{t^{(n'+m)}(V, gW)^{k'}}{[t^{(2m_1)}(V,V)\, t^{(2m_2)}(W,W)]^{(k'-1)/2}}, \qquad \Delta(k'n'+m) \ge \frac{\Delta(n'+m)^{k'}}{[\mathsf{Z}(2m_1)\mathsf{Z}(2m_2)]^{(k'-1)/2}}.$$

■

Proof sketch. Let $\|u\|_2 = \|v\|_2 = 1$, $c = \langle u, Xv\rangle$, and $\nu(x,y) = u(x)X(x,y)v(y)/c$, a probability density with the marginals $\nu_1 = u\,Xv/c$ and $\nu_2 = v\,Xu/c$ (symmetry). On $(v_0, \dots, v_a)$ the chain $\nu(v_0, v_1)\prod_{i=1}^{a-1} q_i$, with $q_i = \nu(v_{i+1}, v_i)/\nu_2(v_i)$ for odd i and $\nu(v_i, v_{i+1})/\nu_1(v_i)$ for even i, has the i-th marginal ν_1 for even i and ν_2 for odd i; as a is odd, v_a carries an odd index. Writing $\langle u, X^a v\rangle$ as the mean of a quotient under this chain, Jensen's inequality and Gibbs' inequality give $\log\langle u, X^a v\rangle \ge a \log c + \frac{a-1}{2}[\mathrm{KL}(\nu_1|u^2) + \mathrm{KL}(\nu_2|v^2)] \ge a \log c$. For the second part, $X(V,W) = t^{(n')}(V, gW)$ is symmetric, since g commutes with T_m and $g^2 = 1$, and $X^a(V,W) = t^{(an')}(V, g^a W)$; with $u = t^{(m_1)}(\cdot, V)$, $v = t^{(m_2)}(\cdot, W)$ and odd a, $\langle u, X^a v\rangle = t^{(an'+m)}(V, gW)$ and $\|u\|_2^2 = t^{(2m_1)}(V,V)$; apply the first part at $a = k'$ and $a = 1$, put $W = V$, and integrate with Hölder's inequality (exponents k', $k'/(k'-1)$) and the Cauchy–Schwarz inequality. □

Lemma 11.4 (Doubling)**.** Let $\beta \ge 0$, and for a box N and a direction μ let $\mathsf{d}^{(\mu)}(N) = 2F_0(N) - F_0(N^{(\mu)})$, $N^{(\mu)}$ the box with N_μ doubled. Then $\mathsf{d}^{(\mu)} \ge 0$, and $\mathsf{d}^{(\mu)}(N^{(\mu)}) \le \psi(\mathsf{d}^{(\mu)}(N))$ whenever $\mathsf{d}^{(\mu)}(N) < \log 2$, with $\psi(x) = 2\log(1 + \varrho_x^2)$, $\varrho_x = (e^x - 1)/(1 + \sqrt{1 - (e^x - 1)^2})$. Further $D_2(N) \ge 4\mathsf{G}(N) > 0$ for $\beta > 0$, G over the section (N, N) at the height N, and $\mathsf{e}(l) \le D_2(l)$; if $\mathsf{e}(l) \le \frac{1}{20}$, then $D_2(2l) \le 17.72\,\mathsf{e}(l)^2$; if $\mathsf{e}(l) \le 0.069$, then $\mathsf{e}(2l) \le 12.9\,\mathsf{e}(l)^2$, $D_2(2l) \le 20\,\mathsf{e}(l)^2$, and $\delta(M,M,M) \le 0.01717\,\theta^{M/l}$ for $M = 2^j l$, $j \ge 1$, $\theta = 12.9\,\mathsf{e}(l)$. On every cube of side N and for every $n \ne 0$, $1 - r[n] \le 3(1 - e^{-\mathsf{G}(N)})$, G over the section of the cube. ■

Proof sketch. With the time μ, $Z[0] = \|T_0\|^{N_\mu}(1+\mathsf{r})$, $\mathsf{r} = \sum_{i\ge1}(\lambda_i/\lambda_0)^{N_\mu}$, and the doubled side has $\mathsf{r}' \le \mathsf{r}^2$; so $e^{\mathsf{d}^{(\mu)}} = (1+\mathsf{r})^2/(1+\mathsf{r}') \ge 1 + 2\mathsf{r}/(1+\mathsf{r}^2)$, which gives $\mathsf{r} \le \varrho_x$, and the defect at the doubled side is at most $2\log(1+\mathsf{r}^2)$. By Thm. AG(2), $D_2(N)$ is a sum of three defects of doubling, each at least $\mathsf{d} \ge \mathsf{G}$ of its section, the first of them $4\mathsf{d}^{(1)}(N,N,N)$. By the symmetry of the axes $\mathsf{e}(l) = \mathsf{d}^{(1)}(l,2l,2l) = \mathsf{d}^{(2)}(2l,l,2l) = \mathsf{d}^{(3)}(2l,2l,l)$; doubling once in each direction, the bulk cancelling in sums of defects, gives $\mathsf{a} = \mathsf{d}^{(2)}(2l,2l,2l) \le \psi(\mathsf{e})$, $\mathsf{b} = \mathsf{d}^{(3)}(2l,4l,2l) \le \psi(2\mathsf{e}+\mathsf{a})$, $\mathsf{d}^{(1)}(l,4l,4l) \le 4\mathsf{e}+2\mathsf{a}+\mathsf{b}$, $\mathsf{e}(2l) \le \psi(4\mathsf{e}+2\mathsf{a}+\mathsf{b})$, and

$$D_2(2l) = 4\mathsf{a}+2\mathsf{b}+\mathsf{e}(2l), \qquad \delta(M,M,M) = \sum_{s\ge0} 8^{-s-1} D_2(2^s M);$$

with $\kappa(y_0) = \varrho_{y_0}/y_0$, as ϱ_y/y increases and $\log(1+u) \le u$, $\psi(y) \le 2\kappa(y_0)^2 y^2$ on $[0,y_0]$, $y_0 < \log 2$, which gives the constants of the lemma. On a cube, $\sum_k r[(k,0)_\mu] = 4Z_{00,0}/Z[0] \ge 4e^{-\mathsf{G}}$; with $x_i = 1-r_i$ this is $2x_1 + x_2 \le 4(1-e^{-\mathsf{G}})$, and the order of Thm. AG(1) with $Z_{11,0}, Z_{11,1} \ge 0$ gives $x_1 \le x_2 \le 2x_1$ and $x_3 \le 2x_2 - x_1$. □

Lemma 11.5 (The fourth power; loops)**.** Let $\beta > 0$. On a box (t,t',h) whose first two sides span the plane of the class n put $\zeta = \frac12(1-r[n])$, and over the section (t,t'), with the third direction as the time, $\mathsf{r}_h = \operatorname{Tr} T_0^h/\|T_0\|^h - 1$ and $\mathsf{d}(h;t,t') = 2F_0(t,t',h) - F_0(t,t',2h)$. For integers $t,t',h \ge 3$ and $u \ge t$, $u' \ge t'$:

$$\zeta(u,u';h) \le \big[\zeta(t,t';h)\, e^{\delta(t,t',h)}\big]^{uu'/tt'} e^{-\delta(u,u',h)}, \qquad 2\zeta \le h\varepsilon_{\mathrm{v}}(t,t') + \mathsf{r}_h, \qquad h\varepsilon_{\mathrm{v}}(t,t') \le \mathsf{d}(h;t,t') - \log(1-2\zeta(t,t';2h)),$$

and $\zeta(t,t';h) \le 1 - e^{-\mathsf{G}(t;t',h)} \le \delta(t,t',h)$. If $\bar z = [(h\varepsilon_{\mathrm{v}}(t,t') + \frac12 \mathsf{r}_{2h})e^{\delta(t,t',2h)}]^4 < \frac12$, then

$$h\,\varepsilon_{\mathrm{v}}(2t,2t') \le \mathsf{d}(h;2t,2t') - \log(1-2\bar z).$$

If t' is even and C is a rectangle parallel to that plane with sides $R \le t-1$ along t and $T \le t'-1$ along t', then $|\langle W_C\rangle| \le (4\langle \mathsf{PP}'\rangle)^{T/t'}$ and $0 \le \langle \mathsf{PP}'\rangle \le (1-\zeta)^{1-\varrho}\zeta^{\varrho} + \zeta^{1-\varrho}(1-\zeta)^{\varrho} \le 2\zeta^{\min\{\varrho,1-\varrho\}}$, $\varrho = R/t$, P,P' the half traces of the two straight closed lines through the sides of length T; the second chain holds for all $t' \ge 3$. ■

Proof sketch. Along the third side $Z[0] = \|T_0\|^h(1+\mathsf{r}_h)$ and $Z[n] = \operatorname{Tr} T_1^h \ge \|T_1\|^h$, so $2\zeta \le 1 - e^{-h\varepsilon_{\mathrm{v}}}/(1+\mathsf{r}_h) \le h\varepsilon_{\mathrm{v}} + \mathsf{r}_h$. By Thm. AG(2), $q = \frac12(Z[0]-Z[n])$ is a power sum in each length of the plane, so $q(u,u') \le q(t,t')^{uu'/tt'}$ with the volumes taken out by δ: this is the first display, and doubling both short sides raises ζ to the fourth power. Back along the third side, $\|T_1\|^h \ge \operatorname{Tr} T_1^{2h}/\operatorname{Tr} T_1^h \ge (1-2\zeta(2h))Z[0](2h)/Z[0](h)$ and $\|T_0\|^h \le Z[0](h)$ give the third inequality; with the time along t, $1-\zeta \ge Z_{00,0}/Z[0] \ge e^{-\mathsf{G}}$. The fourth power follows by chaining these. For the loop, $Z\langle W_C\rangle = \operatorname{Tr}(T_0^{t'-T}\Gamma^* T_{\mathrm{or}}^T \Gamma)$ with the transfer operator T_{or} with two static charges, acting on functions with values in 2×2 matrices; Hölder's inequality for the double series over the two eigenbases gives the first bound, and $\langle \mathsf{PP}'\rangle$, with the time along t, is a sum of the weights $1-\zeta$ and ζ of the two flux sectors at the distances R and $t-R$. □

Lemma 11.6 (Lipschitz bounds in the sides)**.** Let $\beta > 0$, N a box, λ a direction with $N_\lambda \ge 5$, z a class, $N^\flat$ the box N with $\lceil N_\lambda/2\rceil$ in place of N_λ and $N^\natural$ the box N with $N_\lambda + 1 + \lfloor N_\lambda/2\rfloor$ in place of N_λ. Then

$$|\rho^z(N+e_\lambda,\beta) - \rho^z(N,\beta)| \le \frac{4}{N_\lambda} e^{\delta(N^\flat)}, \qquad |\delta(N+e_\lambda) - \delta(N)| \le \frac{3}{N_\lambda}\max\{\delta(N^\flat),\delta(N^\natural)\}.$$

■

Proof sketch. With λ as the time, $\rho^z = \sum_e (-1)^{e\cdot k}\mathcal{W}(e;m)/\sum_e \mathcal{W}(e;0)$, $\mathcal{W}(e;m) = \sum_i \vartheta_i^{N_\lambda}$ with the eigenvalues $\vartheta_i \in [0,1]$ of $P_e T_m/\|T_0\|$, and the denominator is ≥ 1. One more layer changes each

term by $\vartheta^N(1-\vartheta) \le \vartheta^{\lceil N/2\rceil}/(\lfloor N/2\rfloor+1)$, and $\sum \vartheta_i^{\lceil N/2\rceil}$ over all sectors is at most $e^{\mathsf{G}} \le e^{\delta(N^\flat)}$ for each m, by $\operatorname{Tr} T_1^M \le \operatorname{Tr} T_0^M$. For δ, $n \mapsto \log\sum_i a_i^n$ is convex and $\delta \ge 0$: the increment at N_λ lies between the mean increments over the heights $\lceil N_\lambda/2\rceil \dots N_\lambda$ and $N_\lambda+1 \dots N_\lambda+1+\lfloor N_\lambda/2\rfloor$, and $\lfloor N_\lambda/2\rfloor \ge N_\lambda/3$. □

Lemma 11.7 (Periodic Gibbs states)**.** Let $\beta > 0$. (a) For every periodic Gibbs state ν, every bounded Borel function F of the bonds in a cube of side q and every cube of side $l \in 2q\mathbb{N}$, $|\nu(F) - \mu_{l^{\times 3}}(F)| \le \|F\|_\infty (2\delta(l^{\times 3}))^{1/2}$. (b) Every weak limit point of the measures of the periodic cubes, extended periodically, is a periodic Gibbs state. (c) So if $\delta(l^{\times 3}) \to 0$ as $l \to \infty$, there is exactly one periodic Gibbs state and the measures of the periodic cubes converge to it; this is so if $D_2(l_0) \le \frac{1}{20}$ at one $l_0 \ge 3$ or $\mathsf{e}(l) \le 0.069$ at one $l \ge 3$, and then $|\nu(F) - \mu_{l^{\times 3}}(F)| \le 0.126\|F\|_\infty e^{-m_* l/2}$ for $l \in \{2^i l_0\} \cap 2q\mathbb{N}$ in the first case. ■

Proof sketch. (a) Let A be a real continuous function of the bonds in the cube of side q. For $n \in 2q\mathbb{N}$ let $\mathbb{A}_n$ be the sum of the mirror images of A over the $(n/q)^3$ cells of side q of the cube n, an array periodic under $2q$, and $F_t(n) = \log Z[0; e^{t\mathbb{A}_n}](n^{\times 3})$. By reflection positivity at the mirrors between the cells, F_t is a sum of powers in each side in $2q\mathbb{N}$, so $F_t(n)/n^3$ does not increase (Lem. 11.2) and tends to its infimum $\pi(t)$; $\pi(0) = P$. The convex form of the domination of periodic Gibbs states by the chessboard seminorms (Prop. YM3_I:91.9; its proof, the Gibbs equations of ν on a large box and the chessboard estimate on the periodic cube, holds for cells of every side) gives the tangent property $\pi(t) - \pi(0) \ge t\nu(A)/q^3$.

On the cube l, with $N_c = (l/q)^3$ cells, the N_c mirror images of A have one mean $\mu_{l^{\times 3}}(A)$ under the measure $\mu_{l^{\times 3}}$ of the periodic cube, and $|\mathbb{A}_l| \le N_c\|A\|_\infty$; Hoeffding's lemma gives $F_t(l) \le F_0(l) + tN_c\mu_{l^{\times 3}}(A) + \frac{1}{2}t^2N_c^2\|A\|_\infty^2$. As $l^3\pi(t) \le F_t(l)$ and $F_0(l) = l^3P + \delta(l^{\times 3})$, with $s = tN_c > 0$ this is $\nu(A) - \mu_{l^{\times 3}}(A) \le \delta/s + s\|A\|_\infty^2/2$, whose infimum over s is $\|A\|_\infty(2\delta)^{1/2}$; the same holds for $-A$, and bounded Borel functions follow by the regularity of the two measures. (b) is Prop. YM3_I:91.2. (c) Two periodic Gibbs states agree on F up to $2\|F\|_\infty(2\delta(l^{\times 3}))^{1/2}$ for all large l; with (b) the periodic cubes converge.

Under $D_2(l_0) \le \frac{1}{20}$, Lem. 11.4 and Thm. AG(2) give $\delta(N^{\times 3}) \le 0.14108\,(N/l)^3 D_2(l) \le 8 \cdot 0.14108 \cdot \frac{1}{18}e^{-m_* N/2}$ with $l \le N < 2l$ in $\{2^i l_0\}$; under $\mathsf{e}(l) \le 0.069$, $\delta(N^{\times 3}) \le 0.13736\,\theta^{N/2l}$. At $l \in \{2^i l_0\}$, $(2 \cdot 0.14108 \cdot \frac{1}{18}e^{-m_* l})^{1/2} \le 0.126\,e^{-m_* l/2}$. □

Lemma 11.8 (Chessboard bounds for every number of cells; the cylinder over a twisted section)**.** Let $\beta \ge 0$, $\lambda' \ge 3$ odd and $\mathsf{q} \ge 2$ even, and let Q be the periodic cube of side $\mathsf{q}\lambda'$, a board with q^3 cells of side λ', with the fold and the chessboard seminorm (equation YM3_I:(1749), Prop. YM3_I:86.3): for a bounded complex Borel function ϕ of one cell, $\operatorname{Fold}\phi$ is the product over the cells of Q of the mirror copies of ϕ, the copy in a cell being complex conjugated if the cell is odd relative to the given one, $Z_Q[0; \operatorname{Fold}\phi] \ge 0$, and $\|\phi\|_{\mathrm{ch};Q} = (Z_Q[0; \operatorname{Fold}\phi]/Z_Q[0])^{1/\mathsf{q}^3}$; for real ϕ these are the fold and the seminorm of the setting. A plaquette function of a cell is a bounded complex Borel function ϕ of the numbers $1 - \frac{1}{2}\operatorname{tr}\partial U(p)$ of the plaquettes p of the cell; in a class it is taken with $1 - \frac{1}{2}\eta(p)\operatorname{tr}\partial U(p)$ in the place of these numbers, and $\phi^{(u)}$ is its copy in the cell u.

(a) On every box with $k_\mu\lambda'$ sites in the direction μ, $k_\mu \ge \mathsf{q}$ integers, odd ones admitted, cut into cells of side λ', and in every class n,

$$\Big|Z\Big[n; \prod_u \phi_u^{(u)}\Big]\Big| \le \prod_u Z_Q[0; \operatorname{Fold}\phi_u]^{1/\mathsf{q}^3}; \qquad \text{in particular} \quad Z[n] \le Z_Q[0]^{|N|/|Q|}.$$

(b) Let $\beta > 0$ and $N = k\lambda'$ with an integer $k \ge \mathsf{q}$; let $\omega = \lim_{h\to\infty}\mu_{n_{23}}$ on the boxes (h, N, N), the state of the twisted cylinder $\mathbb{Z} \times (\mathbb{Z}/N)^2$ on the plaquette functions of finitely many layers, $\lambda_0 > \lambda_1 \ge \lambda_2 \ge \dots$ the eigenvalues of T_1 over the twisted section (N, N) with the time 1, and $\Xi = Z_Q[0]^{k^2/\mathsf{q}^3}\lambda_0^{-\lambda'}$. For plaquette functions ϕ_u of the cells $u \in \mathbb{Z} \times (\mathbb{Z}/k)^2$, all

but finitely many equal to 1,

$$\Big|\omega\Big(\prod_u \phi_u^{(u)}\Big)\Big| \le \Xi^{J'} \prod_u \|\phi_u\|_{\mathrm{ch};Q},$$

J' the number of the slabs of cells in time that carry a function other than 1. Moreover $1 \le \Xi \le \exp[\Phi'/(k_0 - \mathsf{q})]$ for every integer $k_0 > \mathsf{q}$, where $\Phi' = \frac{k_0 k^2}{\mathsf{q}^3} \log Z_Q[0] - \log Z_{(k_0\lambda',N,N)}[n_{23}]$, and $\sum_i (\lambda_i/\lambda_0)^{j\lambda'} \le \Xi^j$ for every integer $j \ge \mathsf{q}$. With $N' = \mathsf{q}\lambda'$ the side of Q, $k^2/\mathsf{q}^3 = N^2\lambda'/N'^3$, so that $\log \Xi = \lambda'[(N^2/N'^3) \log Z_Q[0] - \log \lambda_0]$: at a fixed partner Q and a fixed section (N, N), $\log \Xi$ is linear in the number λ' of the sites of a side of a cell.

■

Proof sketch. (a) Without a twist, cut the box in a direction μ into its $k_\mu \ge \mathsf{q}$ slabs of cells. By the trace formula the integral is the trace of the product of the slab operators; Hölder's inequality for traces with q factors, the operator norm being at most the norm of order q, bounds its modulus by the product of the norms of order q of the slab operators. The adjoint of a slab operator Y is the operator of the mirror copy of the slab carrying the complex conjugate function; so the q-th power of the norm of order q, $\mathrm{Tr}\,(Y^*Y)^{\mathsf{q}/2}$, is the integral, over the box with q slabs in the direction μ, of the mirror copies of the function of the slab, complex conjugated at every other place.

The three directions in turn give the bound, the copies composing to the folds on Q, each of which is thus an integral ≥ 0. In a class n place the columns inside cells; on a reversed plaquette $e^{-\beta(1+\frac{1}{2}\mathrm{tr}\,u)} = e^{-\beta(1-\frac{1}{2}\mathrm{tr}\,u)} e^{-\beta\,\mathrm{tr}\,u}$, so the twist is a family of functions of cells and the periodic bound applies. After the fold the reversed plaquettes of the q^3 copies form, q being even, the boundary $d\gamma$ of a sign field γ on the bonds of Q, and $U \mapsto \gamma U$ removes them: the folds are those of the periodic class. (b) By the trace formula with the time 1, $\omega(\prod_u \phi_u^{(u)})$ is λ_0^{-w} times the mean, in the top eigenvector of T_1, of the product of the operators of the slabs of cells, w the number of their layers; a slab with the function 1 has the operator $T_1^{\lambda'}$, of norm $\lambda_0^{\lambda'}$.

The norm of the operator of an occupied slab is at most its norm of order q, whose q-th power is, as in (a), a twisted integral over the box $(\mathsf{q}\lambda', N, N)$ with the mirror copies of the functions of the slab, complex conjugated at every other place in time, bounded by (a), the fold of a mirror copy being the fold of the function itself (Prop. YM3_I:86.4). At $\phi \equiv 1$ this is $\lambda_0^{\lambda'} \le Z_Q[0]^{k^2/\mathsf{q}^3}$, that is $\Xi \ge 1$. By (a), $\mathrm{Tr}\,T_1^{j\lambda'} = Z_{(j\lambda',N,N)}[n_{23}] \le Z_Q[0]^{jk^2/\mathsf{q}^3}$ for $j \ge \mathsf{q}$, which is the last assertion, and with $\mathrm{Tr}\,T_1^{k_0\lambda'} \le \lambda_0^{(k_0-\mathsf{q})\lambda'} \mathrm{Tr}\,T_1^{\mathsf{q}\lambda'}$ it gives $\Xi^{k_0-\mathsf{q}} \le e^{\Phi'}$.

The linearity of $\log \Xi$ in λ' is the identity $k^2/\mathsf{q}^3 = N^2\lambda'/N'^3$, λ_0 and $Z_Q[0]$ reading N, N' and β alone. □

Lemma 11.9 (A box with flux as a sector of a periodic torus)**.** Let $\ell = 2L^m$ with $m \ge 1$, $K \ge 2$, $s_\mu \in \{L^j, 2L^j : 0 \le j \le m\}$, $n_\mu = \ell/s_\mu$, and $w_{\mu\nu} \in \{0, 1\}$ for $\mu \ne \nu$ with $w_{\mu\nu} = 1$ only if n_μ is even; let $z_{\mu\nu} = w_{\mu\nu} + w_{\nu\mu} \bmod 2$, $\mathfrak{T} = \{\sum_\mu k_\mu s_\mu e_\mu\} \subset \mathbb{Z}^3/\ell\mathbb{Z}^3$ and $n = |\mathfrak{T}| = \ell^3/|s|$. On the lattice $L^{-j}\mathbb{Z}^3$, $j \ge 0$, which covers the lattice of T_ℓ of spacing L^{-j}, let $n_\nu(x) = \lfloor [x]_\nu/s_\nu \rfloor$, $[x]$ the nearest point of $\mathbb{Z}^3$, $\varepsilon_\nu(x) = (-1)^{n_\nu(x)}$ and $\varsigma_\nu(\langle x, x'\rangle) = \varepsilon_\nu(x)\varepsilon_\nu(x')$; ς_ν is invariant under the shifts by $s_\mu e_\mu$, $\mu = 1, 2, 3$, and so it is a sign field on the bonds of the lattice of T_ℓ; and for $v = \sum_\mu k_\mu s_\mu e_\mu$

$$(\gamma_v V)(b) = \chi_v(b)\, V(b - v), \qquad \chi_v = \prod_\mu \prod_{\nu \ne \mu} \varsigma_\nu^{w_{\mu\nu} k_\mu}.$$

A twist triple of the class z is $\Omega \in Q_8^3$, $Q_8 = \{\pm\mathbf{1}, \pm i\sigma_1, \pm i\sigma_2, \pm i\sigma_3\}$, with $\Omega_\mu\Omega_\nu = (-1)^{z_{\mu\nu}}\Omega_\nu\Omega_\mu$ for $\mu \ne \nu$ (for the class whose one twisted plane is 12, $\Omega = (i\sigma_1, i\sigma_2, \mathbf{1})$); as every element of Q_8 is its own inverse up to the sign, the conjugations by $\Omega_1, \Omega_2, \Omega_3$ commute. The twisted fields of a box N of the class z are the fields $\tilde{U}$ on the bonds of $\mathbb{Z}^3$ with $\tilde{U}(b + N_\mu e_\mu) = \Omega_\mu \tilde{U}(b) \Omega_\mu^{-1}$, $\mu = 1, 2, 3$, with the Haar measure $d\tilde{U}$ at the bonds that begin in one period box and the action $A'(\tilde{U})$, the sum of $1 - \frac{1}{2}\mathrm{tr}\,\partial\tilde{U}(p)$ over the $3|N|$ plaquettes whose least corner lies in that box; neither depends

on the period box. Then $\int e^{-\beta A'(\tilde U)} d\tilde U = Z[z]$ on the box N for every $\beta \geq 0$; $v \mapsto \gamma_v$ is an action of $\mathfrak{T}$ on configurations that keeps the Haar measure, and the underlying translations act freely on sites, bonds and plaquettes; every operation of the small-field chain on T_ℓ (averaging, gauge fixing, the minimisers about which it expands and the backgrounds, the charts and their Jacobians, the small-field tests and the cuts) commutes with each γ_v; and, with $dU^{\mathfrak{T}}$ the Haar measure at representatives of the bonds modulo $\mathfrak{T}$,

$$\int_{\{U\colon\ \gamma_v U = U\ \forall v\}} e^{-\beta A(U)/n}\, \mathrm{d}U^{\mathfrak{T}} = Z^z_K(s;b) \qquad \text{on } \Lambda_{K,\ell}.$$

Every box that is mixed for z is of this form, for some m and w, after a dilation by a power of L. The construction holds on every periodic torus with multiplicities n_μ: for a box s of the lattice $\mathbb{Z}^3$, integers $n_\mu \geq 1$ and $w_{\mu\nu} \in \{0,1\}$ with $w_{\mu\nu} = 1$ only where n_μ is even, and z formed from w as above, the same ς_ν, χ_v and γ_v, formed on $\mathbb{Z}^3$ with $n_\nu(x) = \lfloor x_\nu / s_\nu \rfloor$, give on the periodic torus of the sides $n_\mu s_\mu$ an action on configurations of $\mathfrak{T} = \{\sum_\mu k_\mu s_\mu e_\mu\}$, of order $n = \prod_\mu n_\mu$, that keeps the Haar measure, with free underlying translations on sites, bonds and plaquettes, and $\int_{\{U\colon\ \gamma_v U = U\ \forall v\}} e^{-\beta A(U)/n}\, \mathrm{d}U^{\mathfrak{T}} = Z[z]$ on the box s for every $\beta \geq 0$. ■

Proof sketch. $\varsigma_\nu = -1$ exactly on the bonds that cross the walls $x_\nu \in s_\nu \mathbb{Z} - \frac{1}{2}$; as L is odd, ε_ν is constant on the blocks of every order and the walls lie between blocks; $n_\nu(x + s_\mu e_\mu) = n_\nu(x) + \delta_{\mu\nu}$, so the two ends of a bond change sign together under these shifts. k_μ is given modulo n_μ, and χ_v reads it modulo 2 only where $w_{\mu\nu} = 1$, that is, where n_μ is even; so χ_v is well defined, $\chi_{v+v'}(b) = \chi_v(b)\chi_{v'}(b-v)$ makes $v \mapsto \gamma_v$ an action, and $\mathfrak{T}$ acts freely on the sites, the bonds and the plaquettes, its least displacement being the least side $s_\mu < \ell$. So γ_v is the translation by v followed by the multiplication by χ_v, a closed central sign field which on the lifts is $\varepsilon_v(x)\varepsilon_v(x')$, $\varepsilon_v = \prod_\mu \prod_{\nu \neq \mu} \varepsilon_\nu^{w_{\mu\nu} k_\mu} = \pm 1$ constant on the blocks of every order: near every block it is a gauge transformation by $\pm\mathbf{1}$.

Every operation of the chain commutes with the translations (its covariance under the lattice isometries), and with the multiplication by $\chi = \chi_v$ for the following reasons. (1) The contours that define $V(y,x)$, x in the block of y, lie inside that block, where $\chi = 1$: $V(y,x)$ is kept, and with it the gauge factors and the axial gauge, which read the family $(V(y,x))$ alone; the straight path $[x,x']$ over a block bond c takes the sign $\chi(c)$, which stands twice in each W_x and each $Y_{x,\Gamma,\Gamma'}$ of the definition of $\mathcal{M}$; so the same case of that definition applies at c, and $\mathcal{M}(\chi V) = \chi\, \mathcal{M}(V)$. (2) χ keeps every plaquette variable and, near every bond, is a gauge transformation by $\pm\mathbf{1}$, which keeps the current; with (1) iterated, the critical regular configurations over V go onto those over χV: the minimisers are covariant, and the cuts and the inner balls, which read the plaquette variables of V and of its minimiser, are invariant. (3) The backgrounds are covariant, $V^{\mathrm{bf}}(\chi W) = \chi V^{\mathrm{bf}}(W)$, defined and stably defined together. (4) In the argument $V(b)V^{\mathrm{bf}}(\mathcal{M}V)(b)^{-1}$ of a small-field test the sign $\chi(b)^2 = 1$ cancels: the tests are invariant; and the charts are covariant, $V_{\gamma_v W}(\gamma_v B) = \gamma_v V_W(B)$, γ_v acting on the fluctuation fields B by the translation alone. (5) The Jacobians of the charts are invariant, $\varrho_{c+v}(\gamma_v V) = \varrho_c(V)$.

Hence every density of the chain is invariant under $\mathfrak{T}$, and the chain of T_ℓ restricted to the fixed fields is a chain for them.

For the class, put $U(b) = \tilde U(b)\Omega_\mu$ at the bonds b of the direction μ that begin in the period box $\prod_\mu [0, N_\mu)$ and end outside it, and $U(b) = \tilde U(b)$ at the other bonds that begin in it: a bijection of the twisted fields onto the periodic fields of the box N that keeps the Haar measure. In a plaquette the factors $\Omega_\mu^{\pm 1}$ cancel, except in the plaquettes of a plane $\mu\nu$ whose least corner has $x_\mu = N_\mu - 1$ and $x_\nu = N_\nu - 1$, where they leave the commutator $\Omega_\nu^{-1}\Omega_\mu^{-1}\Omega_\nu\Omega_\mu = (-1)^{z_{\mu\nu}}$; so $\operatorname{tr} \partial\tilde U(p) = \eta^z(p) \operatorname{tr} \partial U(p)$, η^z being -1 exactly on one column of each twisted plane, and $\int e^{-\beta A'} d\tilde U = Z[z]$. With a twist

triple Ω of z and $g(x) = (-1)^{\sum_{\nu<\mu} w_{\mu\nu} n_\mu(x) n_\nu(x)} \Omega_1^{n_1(x)} \Omega_2^{n_2(x)} \Omega_3^{n_3(x)}$ on the lifts,

$$g(x + s_\mu e_\mu) = \textstyle\prod_{\nu\neq\mu} \varepsilon_\nu(x)^{w_{\mu\nu}}\, g(x)\, \Omega_\mu;$$

so the gauge transformation by g^{-1}, constant on all blocks, maps the fixed fields bijectively onto the fields with $\tilde V(b + s_\mu e_\mu) = \Omega_\mu \tilde V(b) \Omega_\mu^{-1}$, the twisted fields of the class z of the box s measured in the spacing of the lattice, and $dU^{\mathfrak T}$ onto $d\tilde U$, the action of a fixed field being n times that of its image. On $\Lambda_{K,\ell}$, where that box is s/a_K and $\beta = L^K b$, the identity $\int e^{-\beta A'} d\tilde U = Z[z]$ gives the display; and the flat field $\tilde V \equiv \mathbf 1$ gives a flat fixed field. n_μ is even exactly when $s_\mu \in L^{\mathbb Z}$, and $z_{\mu\nu} = 1$ needs $w_{\mu\nu} = 1$ or $w_{\nu\mu} = 1$, that is, a side of the plane in $L^{\mathbb Z}$.

On a periodic torus with the sides $n_\mu s_\mu$ the same computations, on the lattice $\mathbb Z^3$ with $n_\nu(x) = \lfloor x_\nu / s_\nu \rfloor$, read the box s and the multiplicities alone: $n_\nu(x + s_\mu e_\mu) = n_\nu(x) + \delta_{\mu\nu}$, so that ς_ν is a sign field on that torus; χ_v is well defined, as $w_{\mu\nu} = 1$ only where n_μ is even, and $\mathfrak T$ acts freely on sites, bonds and plaquettes; the identity for g gives the bijection of the fixed fields onto the twisted fields of the class z of the box s, the action of a fixed field being n times that of its image; and $\int e^{-\beta A'} d\tilde U = Z[z]$ gives the identity of the integrals. The statements on the blocks and on the chain are those of T_ℓ. □

Proposition 11.10 (The root of an invariant integral)**.** Let $\mathfrak Y$ be a finite set with a pseudo-distance d, $\mathcal A = (\mathbb R^3)^{\mathfrak Y}$, and $\mathfrak T$ a group of order n acting freely on $\mathfrak Y$, keeping d, and on $\mathcal A$ by $(\gamma x)_{\gamma y} = \varrho(\gamma) x_y$, $\varrho(\gamma) \in \mathrm O(3)$; $\sigma = \min\{\mathrm d(y, \gamma y) : \gamma \neq 1\}$ when $n > 1$. For $n = 1$ the two integrals below agree exactly; assume $n > 1$ for the estimate. Let $c_- > 0$, $c_+ > 0$, $\mu > 0$, $g_0 \geq 0$, $r > 0$, $\rho = \frac 3 2 r$, $D_a = \{|x_y| < a \ \forall y\}$, and $H \in C^\infty(D_{2r})$ real and $\mathfrak T$-invariant with $\nabla^2 H \geq c_-$ on D_{2r}, $\sup_y \sum_{y'} \sup_{D_{2r}} |\nabla_y \nabla_{y'} H|\, e^{\mu \mathrm d(y,y')} \leq c_+$, and $|\nabla_y H(0)| \leq g_0$. With $\alpha = \mu c_- / 2c_+$ and $S_\nu = \sup_y \sum_{y'} e^{-\nu \mathrm d(y,y')}$,

$$\Big| \log \int_{D_r \cap \mathcal A^{\mathfrak T}} e^{-H/n} d^{\mathfrak T} x - \frac 1 n \log \int_{D_r} e^{-H} dx \Big| \leq \frac{\#\mathfrak Y}{n} \Big[\frac{4\rho c_+}{c_-} (g_0 + \tfrac 5 2 \rho c_+) S_{\alpha/2} e^{-\alpha\sigma/4} + \rho^2 c_+ e^{-\mu\sigma/2} \Big],$$

$\mathcal A^{\mathfrak T}$ the fixed points and $d^{\mathfrak T} x$ Lebesgue measure at representatives of the orbits. ■

Proof sketch. Interpolate $H_\tau = \frac{c_-}{2}|x|^2 + \tau(H - \frac{c_-}{2}|x|^2)$; at $\tau = 0$ both integrals factor over the sites and agree. The τ-derivatives of the two free energies are sums of means of the same local pieces $\hat u_y(x) = x_y \cdot \int_0^1 \nabla_y H(sx)\, \mathrm d s - \frac{c_-}{2}|x_y|^2$, under the full measure and under the product of n copies of the quotient measure. Fix y_0 and representatives containing the ball of radius $\sigma/2$ about y_0; the gradients of the two Hamiltonians differ only through interactions that leave this ball, and in the norm weighted by $\mathsf W = \mathrm{diag}(e^{-\alpha \mathrm d(y_0, y)})$, in which the Hessian stays $\geq \frac 1 2 c_-$ by Schur's test,

$$\sup |\mathsf W \nabla (h_1 - h_2)| \leq 2\rho c_+ S_{\alpha/2} e^{-\alpha\sigma/4}.$$

Two Langevin dynamics for h_1 and h_2 driven by one Brownian path stay within $2/c_-$ times this in that norm, which bounds the difference of the means of $\hat u_{y_0}$; the walls of D_r are softened by a convex penalty and removed at the end, and the choice of representatives costs the second term. Integrate over τ and sum over the $\#\mathfrak Y / n$ representatives. □

Lemma 11.11 (Block functions at two cutoffs)**.** (a) For $b \geq \gamma^{-2}$, a cube $\ell = 2L^m$ with $MR(b) \leq L^m$, $K \geq 2$ at which the conditions (k0)–(k4) and (k6) of Cor. 2.8(a) hold (§2.3.1 at $n = 1$, whose two depths are $u = \lceil K/1260 \rceil$ and $u' = \lceil K/9 \rceil$, the latter written s there), $0 \leq d \leq u$ and every gauge invariant Borel function F with values in $[0,1]$ of the block field of depth d: $\mu_d^{K+1}(F) \leq e^{m_K} \mu_d^K(F) + \epsilon_K$ and $\mu_d^K(F) \leq e^{m_K} \mu_d^{K+1}(F) + \epsilon_K$, where, with the numbers ϵ_{fg}, ϵ_{cl}, ϵ_{sh} of YM3_I: (2192)–(2194) and $\mathsf R_{t_1}$ of §2.3.1,

$$m_K = 4\epsilon_{\mathrm{fg}} + 2\epsilon_{\mathrm{cl}} + 2\epsilon_{\mathrm{sh}} + 2\mathsf R_{t_1}, \qquad \epsilon_K = \tfrac 4 3 \epsilon_{\mathrm{sh}} + \mathsf R_{t_1}.$$

These are numbers of K, ℓ, b and the constants, free of F and d: ϵ_{fg} is the error of forgetting the start of the small-field flow, ϵ_{cl} that of the classical start, ϵ_{sh} the relative mass where the small-field tests of the two cutoffs differ, and R_{t_1} the relative mass of the large fields of the periodic cube at the depth u, of the form of $\mathsf{R}(u)$ of Thm. AH(7). Up to factors formed from the constants, ϵ_{fg} is $\ell^3 L^{4u-u'}$, ϵ_{cl} is $(\log L^K b)^{2\mathfrak{p}_0}\ell^3 K^{1/2}L^{3u'-(K-u')/2}$, ϵ_{sh} is $(u'-u)\ell^3 L^{3u'}e^{-c_{\mathrm{sh}}(\log L^u b)^{2\mathfrak{p}_0}}$ and R_{t_1} is $\ell^3 L^{3u}e^{-c'A_0^2(\log L^u b)^{2\mathfrak{p}_0}}e^{(C+\frac{9}{2}\log b)\ell^3}$, with numbers $c_{\mathrm{sh}}, c', A_0, C > 0$ formed from the constants. (b) There are $A, c > 0$ and $b_0 \geq \gamma^{-2}$, formed from the constants, such that for $b \geq b_0$, $\ell = 2L^m$ with $MR(b) \leq L^m$ and $\ell \leq 2ML^2(\log b)^{r_\star}$, $K_\sharp = \lceil A \log_L \log b \rceil$, every $K \geq K_\sharp$ and every such F of the block field of depth one,

$$e^{-E}\mu_1^{K_\sharp}(F) - \epsilon \leq \mu_1^K(F) \leq e^E(\mu_1^{K_\sharp}(F) + \epsilon), \qquad E \leq 1/\log b, \quad \epsilon \leq e^{-c(\log b)^{2\mathfrak{p}_0}}.$$

As $2\mathfrak{p}_0 > 3r_\star + 2 > 1$ and $t = \ell/b \geq 2MR(b)/b \geq 2M(\log b)^{r_\star}/b$, for every $k > 0$ there is b_k, formed from k and the constants, with $\epsilon \leq t^k$ for $b \geq b_k$: on these cubes $\epsilon = o(t^k)$ as $b \to \infty$, for every k. ■

Proof sketch. (a) The comparison of two consecutive cutoffs (Section YM3_I:116) restarts the small-field flows of the two lattices from the classical action at the depth u' and compares them at the depth u. For F as in (a), $\mu_d^K(F)$ is the periodic mean of $F(\mathcal{M}^{K-d}U)$, and $F \circ \mathcal{M}^{u-d}$ is a gauge invariant function of the block field of depth u. Split the periodic measure into the small-field measure of the chain at the depth u, carried by the inner ball, and the rest, whose relative mass is at most R_{t_1} by the control of large fields (Prop. 4.54); the gauge factors integrate to at most 1 against gauge invariant functions. The small-field masses at the cutoffs K and $K+1$ are compared pointwise at the fields of depth u': each density differs from that of its classical run by a factor $e^{\pm\epsilon_{\mathrm{fg}}}$, the flow forgetting its start; the two classical starts differ by a factor $e^{\pm\epsilon_{\mathrm{cl}}}$; and the two systems of small-field tests differ on a set whose relative mass is at most ϵ_{sh}. Putting the factors and the masses together for F and for $F = 1$, which compares the two normalisations, gives (a) with the stated m_K and ϵ_K.

(b) Let $\ell \leq 2ML^2(\log b)^{r_\star}$, so that $\ell^3 \leq C(\log b)^{3r_\star}$. As $4u - u' \leq -\frac{K}{10} + 4$ and $3u' - \frac{1}{2}(K - u') \leq -\frac{K}{9} + 4$, $\sum_{K \geq K_\sharp}(\epsilon_{\mathrm{fg}} + \epsilon_{\mathrm{cl}}) \leq C(\log b)^{2\mathfrak{p}_0+3r_\star}K_\sharp^{2\mathfrak{p}_0+1}L^{-K_\sharp/10}$, which is at most $1/(8\log b)$ for $K_\sharp = \lceil A\log_L\log b\rceil$ with A large and b large. In ϵ_{sh} and R_{t_1}, $(u\log L + \log b)^{2\mathfrak{p}_0} \geq (\log b)^{2\mathfrak{p}_0} + 2\mathfrak{p}_0(\log b)^{2\mathfrak{p}_0-1}u\log L$, as $2\mathfrak{p}_0 \geq 1$; with $u \geq K/1260$ the second term gives a factor L^{-K} for large b, which beats $L^{3u'} \leq L^{K/3+3}$ and makes the sum over K converge, and the first leaves $e^{-c(\log b)^{2\mathfrak{p}_0}}$.

The volume factor of R_{t_1} is $e^{(C+\frac{9}{2}\log b)\ell^3} \leq e^{C(\log b)^{3r_\star+1}}$, beaten by $e^{-\frac{1}{2}c'A_0^2(\log b)^{2\mathfrak{p}_0}}$ because $2\mathfrak{p}_0 > 3r_\star + 1$: this is why ℓ is bounded above. So $\sum_{K\geq K_\sharp}(\epsilon_{\mathrm{sh}} + \mathsf{R}_{t_1}) \leq e^{-2c(\log b)^{2\mathfrak{p}_0}}$ for large b, $E := \sum_{K\geq K_\sharp} m_K \leq 1/\log b$ and $\epsilon := \sum_{K\geq K_\sharp}\epsilon_K \leq e^{-c(\log b)^{2\mathfrak{p}_0}}$. The same bounds give the conditions (k0)–(k4) and (k6) at every $K \geq K_\sharp$ for large b ((k0) asks $K \geq 10$, (k1) $b \geq \Theta_\star$, and (k4) a power $\ell^3 L^{-5K/9+4}$ that is small). Now iterate (a) at $d = 1$: $\mu_1^K \leq e^{\sum_j m_j}\mu_1^{K_\sharp} + \sum_j \epsilon_j e^{\sum_{i>j} m_i} \leq e^E(\mu_1^{K_\sharp} + \epsilon)$, and $\mu_1^{j+1} \geq e^{-m_j}(\mu_1^j - \epsilon_j)$ gives $\mu_1^K \geq e^{-E}\mu_1^{K_\sharp} - \epsilon$, the sums running over $K_\sharp \leq j < K$.

Finally $t \geq 2M(\log b)^{r_\star}/b \geq 1/b$ for large b, and $\epsilon \leq b^{-k}$ once $c(\log b)^{2\mathfrak{p}_0-1} \geq k$, which holds for large b as $2\mathfrak{p}_0 > 1$. □

Proposition 11.12 (Rigidity)**.** Let $q \geq 2$ be an integer, $\tau \in \mathbb{D}^3$, $\mathfrak{S} = \prod_\mu \tau_\mu q^{\mathbb{N}_0}$, and $s^{[\mu]}$ the box s with s_μ multiplied by q. For $b \in (0,\infty)$ let $\mathscr{S}_b$ be the class of the sequences $(K_j, b_j)_{j\geq 1}$ of cutoffs and labels with $K_j \to \infty$ and $b_j \to b$; it is closed under subsequences, and the full sequence of the cutoffs at the fixed label b, $K_j = j$ and $b_j = b$, belongs to it. Along a sequence of $\mathscr{S}_b$ a number written with the index K at the label b, as $r_K[n](s;b)$ or $E_i^{K,m,e}$, is taken at the cutoff K_j and the label b_j, that is at $\beta = L^{K_j}b_j$, and it converges if it converges as $j \to \infty$. Along every sequence of $\mathscr{S}_b$ the following are equivalent: **(R1)** for every $s \in \mathfrak{S}$, $r_K[n](s;b)$ converges for $n \neq 0$, and $q\log Z_K[0](s;b) - \log Z_K[0](s^{[\mu]};b)$ converges for each μ; **(R2)** for every $s \in \mathfrak{S}$ and every n, $\delta_K^n(s;b)$

converges in $[-\infty,\infty)$; **(R3)** for each time, every L-adic section and all m,e,i, $E_i^{K,m,e}$ converges in $[0,\infty]$; **(R4)** (R1) with every integer $q'\ge 2$ in the place of q, and (R2), hold on every box with L-adic sides.

If they hold along a sequence of $\mathscr{S}_b$, operators $H_{l,m}$ are formed from the limits along it as in Thm. AH(3), and for each time, L-adic section l, direction ν of it and integer $q'\ge 2$, the difference of the norms $a_K^{-1}[q'\log\|T_0(l)\|-\log\|T_0(l^{[\nu]})\|]$ converges along it to a finite number ≥ 0, $l^{[\nu]}$ the section with l_ν multiplied by q'. If they hold along every sequence of $\mathscr{S}_b$, then all levels, $r_K[z]$ on every L-adic box, δ_K^0 and the differences of the norms converge along each of these sequences, to limits that are the same for all of them and so depend on b alone; and if this is so for every b of an open set $I\subset(0,\infty)$, these limits are continuous in $b\in I$ and are reached uniformly for the labels in every compact subset of I, the levels in a metric of $[0,\infty]$. ■

Proof sketch. Fix a sequence of $\mathscr{S}_b$. Its labels and b form a compact set $\mathsf{J}\subset(0,\infty)$, at which the window of Thm. AH(1) holds with one constant C at all large cutoffs, uniformly along the sequence; the window is the one input of the argument that reads the label. (R2) gives (R1): δ_K^0 is bounded (Thm. AH(1)), $r_K[n]=e^{\delta_K^n-\delta_K^0}$, and $q\log Z_K[0](s)-\log Z_K[0](s^{[\mu]})=q\delta_K^0(s)-\delta_K^0(s^{[\mu]})$. Assume (R1). Over a fixed section and with $\Theta_K(t)=\operatorname{Tr}T_0^{t/a_K}/\|T_0\|^{t/a_K}\in[1,B]$ (Thm. AH(1)), the powers of $\|T_0\|$ cancel in $x_p^K=q\log Z_K[0](q^pt_0,l)-\log Z_K[0](q^{p+1}t_0,l)=q\log\Theta_K(q^pt_0)-\log\Theta_K(q^{p+1}t_0)\in[-\log B,q\log B]$, and

$$\log\Theta_K(t_0)=\sum_{p\le J}q^{-p-1}x_p^K+O(q^{-J})$$

uniformly in K; so Θ_K converges at the heights t_0q^p, and with it the eight sector traces $\Theta_K\cdot\frac14\sum_k(-1)^{e\cdot k}r_K[(k,m)]$. Two limits of the levels along subsequences (Lem. 11.2(b)) have Laplace transforms that agree along $t_0q^p\to\infty$, hence are equal (Lem. 11.2(a), applied to the sector traces, whose coefficients are positive): all levels over the section converge. To turn the box, $q\log Z_K[0](s_0,l)-\log Z_K[0](s_0,l')=s_0\kappa_K+q\log\Theta_K(s_0;l)-\log\Theta_K(s_0;l')$ with $\kappa_K=a_K^{-1}[q\log\|T_0(l)\|-\log\|T_0(l')\|]$, which does not depend on s_0: its convergence at one height gives it at all; three turns of the box reach every L-adic box, which gives (R3).

From (R3) the quotients and the differences of (R1) converge at every L-adic box and every q', and $\delta_K^0(s)/|s|$, telescoped over the boxes q^ps with a tail $\le 6Cq^{-J}$ by the last inequality of Thm. AG(2) and the window, converges: (R4). $\kappa_K\ge 0$ because $\log\|T_0(\Sigma)\|/|\Sigma|=\lim_t F_0(t,\Sigma)/(t|\Sigma|)$ does not increase in a side of Σ. Let now (R1)–(R4) hold along every sequence of $\mathscr{S}_b$. Two of these sequences interlace into a third one, which belongs to $\mathscr{S}_b$; along it the limits exist, so they are the same along the two. Let this be so for every $b\in I$, and $b_i\to b$ in I: choose $K_i\to\infty$ with the values at (K_i,b_i) within $1/i$ of the limits at the label b_i; the sequence (K_i,b_i) belongs to $\mathscr{S}_b$, so the values along it, and with them the limits at the labels b_i, tend to the limits at b, and the limits are continuous on I.

If the convergence were not uniform on a compact set $\mathsf{J}'\subset I$, there would be $K_j\to\infty$ and $b_j\in\mathsf{J}'$, $b_j\to b\in\mathsf{J}'$, with the values at (K_j,b_j) at a distance at least $\epsilon>0$ from the limits at the labels b_j; these limits tend to those at b, by the continuity, and so do the values along (K_j,b_j), a sequence of $\mathscr{S}_b$, so that the distance tends to 0. □

Lemma 11.13 (One-sided recursions)**.** Let $0\le x_K\le 1$ with $x_{K+1}\ge e^{-\varepsilon_K}x_K-x'_K$, $\varepsilon_K,x'_K\ge 0$, $\sum_K(\varepsilon_K+x'_K)=D'<\infty$. Then $\sum_K(x_K-x_{K+1})_+\le D'$, $\sum_K|x_{K+1}-x_K|\le 1+2D'$, and x_K converges. By Thm. AH(6), if $\lim_j\phi_j(t)$ exists for every t of a set with a limit point in $(0,\infty)$, it exists at every $t>0$, locally uniformly with all derivatives. ■

Proof sketch. $x_K-x_{K+1}\le(1-e^{-\varepsilon_K})x_K+x'_K\le\varepsilon_K+x'_K$; the positive and the negative parts of the increments differ by at most 1 in their sums. The second assertion is Vitali's theorem for the normal family of Thm. AH(6). □

Proposition 11.14 (Laplace asymptotics on one lattice)**.** For a box N and $\varphi\in[-\pi,\pi]^3$ let $S_\varphi(N)=\sum\log\hat k^2$ over the momenta $k_\mu\in(2\pi\mathbb{Z}+\varphi_\mu)/N_\mu$ mod 2π with $\hat k^2=\sum_\mu(2-2\cos k_\mu)\ne 0$,

$c_\infty = (2\pi)^{-3}\int_{[-\pi,\pi]^3}\log\hat k^2\,\mathrm{d}k$, and $\vartheta(u,\varphi)=\sum_{n'\in\mathbb{Z}}e^{-u(2\pi n'+\varphi)^2}$. $Z^z_\beta(N)$ is $Z[z]$ on the box N at β, so that $\rho^z(N,\beta)=Z^z_\beta(N)/Z^0_\beta(N)$. For $z\neq 0$ let $\theta^a(z)\in\{0,\pi\}^3$, $a=1,2,3$, be the phases of the three colours under a twist triple Ω of the class z (Lem. 11.9) ($\theta^a_\mu=\pi$ where the a-th diagonal entry of the adjoint action of Ω_μ is -1; for one twisted plane 12 they are $(0,\pi,0)$, $(\pi,0,0)$, $(\pi,\pi,0)$, and none is 0), and $\mathscr{D}^z(N)=\exp[-\frac12\sum_a S_{\theta^a(z)}(N)]$. There are absolute constants C_T, C_P, C_S with the following properties.

(a) For every box, every class $z\neq 0$ and $\beta\geq e$ with $\beta/(\log\beta)^3\geq C_T\bar N^{36}$, $\bar N=\max_\mu N_\mu$:

$$\bigl|\log Z^z_\beta(N)-|N|\log\frac{2}{\pi\beta^3}-\log\mathscr{D}^z(N)\bigr|\leq C_T\bar N^{18}(\log\beta)^{3/2}\beta^{-1/2}.$$

(b) For every $N\geq 3$ and $\beta\geq e$ with $\beta^{1/4}\geq C_P(N\log\beta)^{C_P}$:

$$C_P^{-1}\beta/N\leq Z^0_\beta(N^{\times 3})(\pi\beta^3/2)^{N^3}e^{\frac32 N^3 c_\infty}\leq C_P\beta/N.$$

So for every $c\in(0,1]$ there are β_c and C_c with $C_c^{-1}N/\beta\leq Z^z_\beta(N^{\times 3})/Z^0_\beta(N^{\times 3})\leq C_cN/\beta$ for every $N\geq 3$, $z\neq 0$ and $\beta\geq\max\{\beta_c,\exp(N^c)\}$.

(c) For every cube $N\geq 2$ and $\varphi\in[-\pi,\pi]^3\setminus\{0\}$, $|S_\varphi(N^{\times 3})-N^3c_\infty-\log|\varphi|^2|\leq C_S$ and $|S_0(N^{\times 3})-N^3c_\infty-2\log N|\leq C_S$. The limits $f(\varphi)=\lim_N[S_\varphi(N^{\times 3})-N^3c_\infty]$, $\varphi\neq 0$, and $f_0=\lim_N[S_0(N^{\times 3})-N^3c_\infty-2\log N]$ exist; $f(\varphi)=\log|\varphi|^2+h(\varphi)$ and $f_0=h(0)$ with the continuous bounded function

$$h(\varphi)=\int_0^\infty\Bigl[(4\pi u)^{-3/2}-\prod_\mu\vartheta(u,\varphi_\mu)+e^{-u|\varphi|^2}-e^{-u}\Bigr]\frac{du}{u},$$

even in each φ_μ; and $\lim_N[S_\theta(N^{\times 3})-S_{\theta'}(N^{\times 3})]=\int_0^\infty[\prod_\mu\vartheta(u,\theta'_\mu)-\prod_\mu\vartheta(u,\theta_\mu)]\frac{du}{u}$ for $\theta,\theta'\in\{0,\pi\}^3\setminus\{0\}$.

(d) For $\lambda,R>0$, $\int_{|c_\mu|\leq R}e^{-\lambda\sum_{\mu<\nu}|c_\mu\times c_\nu|^2}d^9c\leq 48\pi^3R\lambda^{-2}$, and the integral is $\geq\pi^3e^{-5}R\lambda^{-2}$ if $\lambda R^4\geq 4$.

(e) With

$$\mathscr{G}_z=\frac{8\pi^2\exp[\frac12 f_0-\frac12\sum_a f(\theta^a(z))]}{\int_{[-\pi,\pi]^3}e^{-f(\varphi)}d\varphi},\tag{11.iii}$$

$\frac{\beta}{N}Z^z_\beta(N^{\times 3})/Z^0_\beta(N^{\times 3})\to\mathscr{G}_z$ as $N\to\infty$ with $\beta\geq\exp(N^c)$, for every $c\in(0,1]$; so $C^z(N)/N\to\mathscr{G}_z$, $C^z(N)=\lim_{\beta\to\infty}\beta\rho^z(N^{\times 3},\beta)$ the number of (g).

(f) (Free values.) For $n\neq 0$ put $d_n(N)=-\frac12\sum_a[S_{\theta^a(n)}(N)-|N|c_\infty]$, so that $\log\mathscr{D}^n(N)=d_n(N)-\frac32|N|c_\infty$. On the boxes $N=N's$ of a fixed integer shape s, $d_n(N's)=\log D_n(s)+d_n^{(1)}(s)N'^{-2}+O(N'^{-4}(\log N')^{7/2})$ with coefficients in closed form; on cubes $\log D_n=-\frac12\sum_a f(\theta^a(n))$ is -0.127453, -0.402863, -0.488493 and $d_n^{(1)}$ is 0.476777, -0.755058, -0.834736 for one, two, three twisted planes; and $\sum_{n\neq 0}\log D_n(s)=-3\log 2$ on every shape. With Thm. AG(14)(b), in the range of (a) with $\beta\geq e^{148}$,

$$\delta^n(N)\geq d_n(N)-C_T\bar N^{18}(\log\beta)^{3/2}\beta^{-1/2}-2.2\cdot 10^5\,|N|(\log\beta)\beta^{-1/6}.$$

(g) (A fixed cube.) On every cube $N\geq 3$, as $\beta\to\infty$, $\beta^{3N^3}Z^z_\beta(N^{\times 3})$ tends to $\mathscr{Z}^z(N)=(\frac{2}{\pi})^{N^3}\mathscr{D}^z(N^{\times 3})$ for $z\neq 0$, and $\beta^{3N^3-1}Z^0_\beta(N^{\times 3})$ tends to

$$\mathscr{Z}(N)=\Bigl(\frac{2}{\pi}\Bigr)^{N^3}\frac{e^{-\frac12 S_0(N^{\times 3})}}{8\pi^2}\int_{[-\pi,\pi]^3}e^{-S_\varphi(N^{\times 3})}\,\mathrm{d}\varphi\in(0,\infty);$$

so for every $z \neq 0$, $\beta\rho^z(N^{\times 3},\beta)$ tends to $C^z(N) = \mathcal{Z}^z(N)/\mathcal{Z}(N) \in (0,\infty)$. There are absolute constants c_3, $x_0 > 0$ such that for $\beta \geq 3$

$$\Big|\frac{\beta\rho^z(N^{\times 3},\beta)}{C^z(N)} - 1\Big| \leq 3c_3\Big(\frac{N^{66}\log\beta}{\beta}\Big)^{1/8} \qquad \text{whenever} \quad c_3\Big(\frac{N^{66}\log\beta}{\beta}\Big)^{1/8} \leq x_0.$$

■

Proof sketch. (a) In a class $z \neq 0$ the minima of the twisted action form exactly two orbits under the gauge group, with stabiliser ± 1 and without zero modes. The Gaussian integral about them is the determinant of the lattice Laplacian with momenta shifted by the colour phases, which is $\mathscr{D}^z$; the cubic and quartic remainders, cut at fields of size $(\log\beta/\beta)^{1/2}$, give the error, and $(2\pi)^3/(2\pi^2)^2 = 2/\pi$ is the Haar normalisation. (b) In the periodic class the nine constant modes $c \in (\mathbb{R}^3)^3$ are kept as coordinates; by

$$1 - \tfrac{1}{2}\mathrm{tr}(ABA^{-1}B^{-1}) = 2\sin^2\alpha\sin^2\alpha'|\mathsf{a}\times\mathsf{a}'|^2 \qquad (A = e^{i\alpha\mathsf{a}\cdot\sigma},\ B = e^{i\alpha'\mathsf{a}'\cdot\sigma},\ |\mathsf{a}| = |\mathsf{a}'| = 1)$$

their action is quartic, between $\frac{32}{\pi^4 N}\sum_{\mu<\nu}|c_\mu \times c_\nu|^2$ and $\frac{2}{N}\sum_{\mu<\nu}|c_\mu\times c_\nu|^2$ for $|c_\mu| \leq \frac{\pi}{2}N$, and (d) gives one power of β/N more than in the twisted classes; the linear term of the fluctuation vanishes, and the determinant of the remaining modes is bounded by (c). The two-sided bound of the quotient follows from (a) and (b) with $\log\beta \geq N^c$. (c) $S_\varphi - N^3 c_\infty$ is a Riemann sum minus its integral; the theta-function representation $\log x = \int_0^\infty (e^{-u} - e^{-ux})\frac{du}{u}$, applied to every momentum, with Poisson summation in each direction, gives the bounds, the limits and the formula for h. (e) Over the moduli of the flat periodic fields, the orbit volume, the three directions $N^{3/2}d^3\theta$ and the determinants $e^{-\frac{1}{2}S_0 - S_\varphi}$ give

$$Z^0_\beta(N^{\times 3}) \sim \Big(\frac{2}{\pi\beta^3}\Big)^{N^3} e^{-\frac{3}{2}N^3 c_\infty}\,\frac{\beta}{N}\,\frac{e^{-f_0/2}}{8\pi^2}\int_{[-\pi,\pi]^3} e^{-f}\,\mathrm{d}\varphi$$

in the range of the limit, and with (a) the quotient tends to $\mathscr{G}_z$; the numerical evaluation of eq. 11.iii uses the integral representation of f and f_0. (f) The expansion is Poisson summation with the lattice factor $\hat{k}^2$ expanded to the fourth order. The eight phase sets $\{0,\pi\}^3$ of the box N together give the momenta of the box $2N$: $\sum_{\varphi\in\{0,\pi\}^3} S_\varphi(N) = S_0(2N)$; so $\sum_{\varphi\neq 0}[S_\varphi(N) - |N|c_\infty] = [S_0(2N) - 8|N|c_\infty - 2\log(2N)] - [S_0(N) - |N|c_\infty - 2\log N] + 2\log 2$ tends to $2\log 2$, and each non-zero φ occurs three times among the colour phases of the seven classes, whence $\sum_{n\neq 0}\log D_n = -\frac{3}{2}\cdot 2\log 2$.

The last inequality is (a) with $P \leq P^{\mathrm{G}} + 2.2\cdot 10^5(\log\beta)\beta^{-1/6}$. (g) Put $y = N^{66}\beta^{-1}\log\beta$. The twisted half is (a) at the fixed cube: for $y \leq 1$ and $\beta \geq 3$, $N^{36}(\log\beta)^3/\beta = y^{6/11}(\log\beta)^{27/11}\beta^{-5/11} \leq 5.4\,y^{6/11}$, the maximum of $(\log\beta)^{27/11}\beta^{-5/11}$ being $(5.4/e)^{27/11} < 5.4$, at $\log\beta = 5.4$; so for $y \leq y_0$, $y_0 \in (0,1]$ absolute, the range of (a) holds and $|\log(\beta^{3N^3}Z^z_\beta(N^{\times 3})/\mathcal{Z}^z(N))| \leq C_T(5.4\,y^{6/11})^{1/2} \leq C_T\sqrt{5.4}\,y^{1/8} \leq 1$, which gives the limit $\mathcal{Z}^z(N)$ and a relative error at most $2C_T\sqrt{5.4}\,y^{1/8}$. In the periodic class the minima of the action are the flat fields; at the flat field with the commuting holonomies $e^{i\phi_\mu \mathsf{a}\cdot\sigma}$ the Gaussian integral over the modes transverse to the flat fields carries the factor $e^{-\frac{1}{2}S_0 - S_\theta}$, $\theta = 2\phi$, from one neutral colour and two charged ones of the phases $\pm\theta$, and the integral over the moduli of the flat fields, in the variable $\theta \in [-\pi,\pi]^3$, with the orbit volume and the quartic action of the constant modes of (b), gives the limit $\mathcal{Z}(N)$.

Its rate comes in four steps; A is the action, $|M|^2 = \frac{1}{2}\mathrm{tr}\,M^*M$ for 2×2 matrices, and $a_\beta = 4N^3\beta^{-1}\log\beta$. *The split.* A gauge transformation sets to $\mathbf{1}$ the bonds of the spanning tree formed by the bonds $\langle x, x+e_3\rangle$ with $x_3 \leq N-2$, $\langle x, x+e_2\rangle$ with $x_3 = 0$, $x_2 \leq N-2$, and $\langle x, x+e_1\rangle$ with $x_2 = x_3 = 0$, $x_1 \leq N-2$, the sites being $x \in \{0,\dots,N-1\}^3$; in integrals of gauge invariant functions this removes the gauge group exactly, without a Jacobian. Then the variable h_μ of the bond $\langle (N-1)e_\mu, Ne_\mu\rangle$ is the holonomy along the axis μ, and the field is bounded by its curvature, through chains of plaquettes along the tree: the variable of every bond of the direction μ outside

the tree lies within $2NA^{1/2}$ of h_μ if the bond starts at $x_\mu = N-1$, and of $\mathbf{1}$ otherwise, and $|h_\mu h_\nu h_\mu^{-1} h_\nu^{-1} - \mathbf{1}| \le N(2A)^{1/2}$.

Let $\mathsf{s}_\mu = 1 - (\frac{1}{2}\mathrm{tr}\, h_\mu)^2$, which is $\sin^2 \frac{\theta_\mu}{2}$ on the flat field of the moduli θ; for $0 < r \le \frac{1}{4}$ let $\psi_\mu = \min\{1, \max\{0, \frac{1}{3}(\mathsf{s}_\mu/r^2 - 1)\}\}$, $\chi_1 = \psi_1$, $\chi_2 = (1-\psi_1)\psi_2$, $\chi_3 = (1-\psi_1)(1-\psi_2)\psi_3$ and $\chi_\mathrm{s} = \prod_\mu (1-\psi_\mu)$, so that $\chi_1 + \chi_2 + \chi_3 + \chi_\mathrm{s} = 1$. χ_i vanishes unless $\mathsf{s}_i \ge r^2$: this is the regular part, at a distance at least $2r$ from $\theta = 0$; χ_s vanishes unless $\mathsf{s}_\mu < 4r^2$ for every μ: this is the part within a distance of the order r of $\theta = 0$, the image of the eight central flat fields $h_\mu = \pm\mathbf{1}$.

The fields with $A \ge a_\beta$ carry at most $e^{-\beta a_\beta} = \beta^{-4N^3}$, which is at most $(C/\beta)^{N^3} \beta^{1-3N^3} \mathcal{Z}(N)$, as $\mathcal{Z}(N) \ge c^{N^3}$ with an absolute $c > 0$. *The regular part.* On the support of χ_1 write $h_1 = e^{i\phi_1 \mathsf{a}\cdot\sigma}$, $|\mathsf{a}| = 1$, $h_\nu = e^{i\alpha_\nu \mathsf{a}\cdot\sigma} e^{iy_\nu\cdot\sigma}$ with $y_\nu \perp \mathsf{a}$, $|y_\nu| < \frac{\pi}{2}$, for $\nu = 2, 3$, and $e^{iX(b)\cdot\sigma}$, $|X(b)| < \pi$, for the variable of every other bond b outside the tree divided by h_μ or by $\mathbf{1}$ as above; the flat fields are $y_2 = y_3 = 0$, $X = 0$, with the coordinates $\phi_1\mathsf{a}$, α_2, α_3 along them, and $\nu = (y_2, y_3, X)$ are the $6N^3 - 2$ coordinates transverse to them.

As $|h_1 h_\nu h_1^{-1} h_\nu^{-1} - \mathbf{1}| = 2|\sin\phi_1| \sin|y_\nu|$ and $|\sin\phi_1| = \mathsf{s}_1^{1/2} \ge r$, the bounds by the curvature give $\|\nu\|^2 \le \Lambda_r A$ with $\Lambda_r = 2\pi^2 N^5 + \frac{\pi^2}{4} N^2 r^{-2}$; the third derivatives of the plaquette terms along the rays through the flat field give $|A - Q| \le 43N^2\|\nu\|^3$, Q the quadratic form of the transverse modes at the flat field, and so $Q \ge \|\nu\|^2/\Lambda_r$. On $\{A < a_\beta\}$ then $\|\nu\| \le \varrho_r = (\Lambda_r a_\beta)^{1/2}$ and $|A - Q| \le \epsilon_r Q$, $\epsilon_r = 43N^2 \Lambda_r \varrho_r$: there the integral of $\chi_1 e^{-\beta A}$ lies between the Gaussian integrals of $(1+\epsilon_r)Q$ and of $(1-\epsilon_r)Q$, which are those of Q times $(1 \pm \epsilon_r)^{1-3N^3}$, while over $\|\nu\| \le \varrho_r$ the Haar density changes by a factor within $e^{\pm 1.2\varrho_r}$ and χ_1 by at most $r^{-2}\varrho_r^2$, and the Gaussian measures give the outside of $\{A < a_\beta\}$ a relative mass at most β^{-N^3} for $\log\beta \ge 3$.

Let $\mathcal{I}_1$ be the Gaussian integral of Q against χ_1 over the flat fields, which is the limit of $\beta^{3N^3-1} \int \chi_1 e^{-\beta A}\, \mathrm{d}U$ as $\beta \to \infty$ at fixed r, and $\mathcal{I}_2$, $\mathcal{I}_3$ the same for χ_2, χ_3; the same Gaussian integral without χ_i over $\{\mathsf{s}_i \ge r^2\}$ is at most $\mathcal{Z}(N)$ by the limit, and so

$$\Big|\beta^{3N^3-1} \int_{\{A<a_\beta\}} \chi_i e^{-\beta A} dU - \mathcal{I}_i\Big| \le \big[e^{6N^3\epsilon_r + 1.2\varrho_r} - 1 + \beta^{-N^3} + 2r^{-2}\varrho_r^2\big] \mathcal{Z}(N),$$

where, if $\Lambda_r \le 3N^2 r^{-2}$, $6N^3\epsilon_r \le 2682\, N^{19/2} r^{-3} (\beta^{-1}\log\beta)^{1/2}$ and $r^{-2}\varrho_r^2 \le 12N^5 r^{-4}\beta^{-1}\log\beta$: the relative error of Laplace's method on the regular part is $N^{19/2} r^{-3} (\beta^{-1}\log\beta)^{1/2}$ up to an absolute factor. *The part about $\theta = 0$.* Where $\chi_\mathrm{s} \ne 0$, one of the eight changes of sign of the bonds of some directions μ that start at $x_\mu = N-1$, which keep the tree, the action and χ_s and change the signs of these h_μ, makes $\mathrm{tr}\, h_\mu > 0$ for every μ, and then $|h_\mu - \mathbf{1}| \le \arcsin 2r \le \pi r$; if besides $A < a_\beta$, the bounds by the curvature give $\sum_b |U(b) - \mathbf{1}|^2 \le \varrho_\mathrm{s}^2 = 6\pi^2 N^2 r^2 + 24N^5 a_\beta$; let $\varrho_\mathrm{s} \le \frac{1}{2}$.

A gauge transformation that maximises $\sum_b \frac{1}{2}\mathrm{tr}\, U(b)$ (Landau's gauge) keeps this bound and writes the field as the transform, by a gauge transformation u with $u(0) = \mathbf{1}$, of the field $(1 - |Y(b)|^2)^{1/2}\mathbf{1} + iY(b)\cdot\sigma$ with $\sum_\mu [Y(\langle x, x+e_\mu\rangle) - Y(\langle x - e_\mu, x\rangle)] = 0$ at every site; $Y = c + w$, $c = (c_\mu)$ the nine constant modes, with $|c_\mu| \le \varrho_\mathrm{s} N^{-3/2}$, and w the $6N^3 - 6$ others. By Kirchhoff's matrix-tree theorem the Jacobian of $(u, Y) \mapsto U$ is at most $(e^{S_0(N^{\times 3})}/N^3)^{3/2}(1+\epsilon_J)^{3N^3}$, $\epsilon_J = 7N^2\varrho_\mathrm{s}$. The action of a plaquette is at least half the square of the vector part of the difference of the products along its two halves, which at constant fields is exactly $-2c_\mu \times c_\nu$; so $A \ge \frac{1}{2}(1-\epsilon_A)\|d_1 w\|^2 + N^3 \sum_{\mu<\nu} |c_\mu \times c_\nu|^2$, $\|d_1 w\|^2$ the sum over the plaquettes of the squares of the circulations of w, $\epsilon_A = 4\mathsf{k}N\varrho_\mathrm{s} + 3\mathsf{k}^2\varrho_\mathrm{s}^2$ with an absolute k.

The area formula, the Gaussian integral over w, whose determinant is $e^{6S_0(N^{\times 3})}$, and (d) at $\lambda = \beta N^3$ and $R = \varrho_\mathrm{s} N^{-3/2}$, which gives the factor $48\pi^3 R\lambda^{-2}$, together give, the powers of $2\pi^2$, of 2π and of $e^{S_0(N^{\times 3})}$ cancelling against those of $\mathcal{Z}(N)$,

$$\beta^{3N^3-1} \int_{\{A<a_\beta\}} \chi_\mathrm{s} e^{-\beta A} dU \le \mathsf{E}_\mathrm{s}\, \mathcal{Z}(N), \qquad \mathsf{E}_\mathrm{s} = \frac{192}{\mathcal{J}_N} N^{-1/2} \varrho_\mathrm{s} \exp\big[21N^5\varrho_\mathrm{s} + \varrho_\mathrm{s}^2 + 6N^3\epsilon_A\big],$$

$\mathcal{J}_N = N^{-2}\int_{[-\pi,\pi]^3} e^{S_0(N^{\times 3})-S_\varphi(N^{\times 3})}d\varphi \geq e^{-2C_S}\int_{[-\pi,\pi]^3}|\varphi|^{-2}d\varphi$ by (c). So, where $N^5\varrho_{\mathrm{s}}$ is bounded and $r^2 \geq N^3 a_\beta$, the mass about $\theta = 0$ is of the order $N^{1/2}r$ against $\beta^{1-3N^3}\mathcal{Z}(N)$, the one power of r being the one power of R in (d). *The choice of* r. As r decreases, the error $N^{19/2}r^{-3}(\beta^{-1}\log\beta)^{1/2}$ of the regular part grows and the mass $N^{1/2}r$ about $\theta = 0$ shrinks; the two are equal at $r^4 = N^9(\beta^{-1}\log\beta)^{1/2}$, that is at $r = (N^{18}\beta^{-1}\log\beta)^{1/8} = N^{-6}y^{1/8}$. The root of order 8 is this balance between the third power of $1/r$ beside $(\beta^{-1}\log\beta)^{1/2}$ and the one power of r from (d); and at this r, $N^5\varrho_{\mathrm{s}}$, of the order $N^6 r = y^{1/8}$, stays bounded, which is why the power in y is $66 = 8\cdot 6 + 18$.

Indeed, with $r = N^{-6}y^{1/8}$, $\beta^{-1}\log\beta = N^{-66}y$ and $y \leq 1$: $\Lambda_r \leq 3N^2r^{-2} = 3N^{14}y^{-1/4}$, $\varrho_r^2 \leq 12N^{-49}y^{3/4}$, $\epsilon_r \leq 447N^{-17/2}y^{1/8}$, $6N^3\epsilon_r \leq 2682N^{-11/2}y^{1/8} \leq 7y^{1/8}$, $r^{-2}\varrho_r^2 \leq 12N^{-37}y^{1/2}$, $\varrho_{\mathrm{s}}^2 \leq 60N^{-10}y^{1/4}$ and $N^5\varrho_{\mathrm{s}} \leq 8y^{1/8}$; so the exponent in E_{s} is at most an absolute multiple of $y^{1/8}$, and $\mathsf{E}_{\mathrm{s}} \leq CN^{-11/2}y^{1/8}$. As $\beta\to\infty$ at fixed r, $\beta^{3N^3-1}\int\chi_{\mathrm{s}}e^{-\beta A}\,\mathrm{d}U$ tends to $\mathcal{I}_{\mathrm{s}} = \mathcal{Z}(N) - \mathcal{I}_1 - \mathcal{I}_2 - \mathcal{I}_3$, by the limit $\mathcal{Z}(N)$, and $\mathcal{I}_{\mathrm{s}} \leq \mathsf{E}_{\mathrm{s}}\mathcal{Z}(N)$, E_{s} being increasing in a_β.

Summing over the four parts and the fields with $A \geq a_\beta$,

$$\Big|\frac{\beta^{3N^3-1}Z^0_\beta(N^{\times 3})}{\mathcal{Z}(N)} - 1\Big| \leq 3\big[e^{6N^3\epsilon_r+1.2\varrho_r} - 1 + \beta^{-N^3} + 2r^{-2}\varrho_r^2\big] + 2\mathsf{E}_{\mathrm{s}} + \Big(\frac{C}{\beta}\Big)^{N^3} \leq c_3 y^{1/8}$$

for $y \leq y_0$, with an absolute c_3, any larger number serving as well, and y_0 taken smaller if need be, so that $\epsilon_r, \epsilon_A \leq \frac12$, $\varrho_r < 1$, $\varrho_{\mathrm{s}} \leq \frac12$, $r \leq \frac14$ and $\beta^{-N^3} + (C/\beta)^{N^3} \leq y^{1/8}$. So $C^z(N) = \mathcal{Z}^z(N)/\mathcal{Z}(N) = 8\pi^2 e^{\frac12 S_0(N^{\times 3})}\mathscr{Z}^z(N^{\times 3})/\int_{[-\pi,\pi]^3} e^{-S_\varphi(N^{\times 3})}d\varphi$, and with (c) and dominated convergence $C^z(N)/N \to \mathscr{G}_z$. Take $c_3 \geq 2C_T\sqrt{5.4}$ and $x_0 \leq \min\{\frac13, c_3 y_0^{1/8}\}$: where the second right member $x = c_3 y^{1/8}$ is at most x_0, $y \leq y_0$, so that the twisted relative error is at most x as well; and a quotient of two numbers within $x \leq \frac13$ of 1 lies within $3x$ of 1. □

Quantitative estimates for the explicit windows. The windows are constructed below. Their moving-window Laplace bounds and the following numerical estimates remain to be proved. The required estimates, referred to as **(Window estimates)**, would provide absolute constants $0 < c_\flat \leq C_\flat$, a number $x_\flat(L) > 0$, $t_* > 0$, and for $0 < t \leq t_*$ Borel functions F_t, F_t' with values in $[0,1]$ of the block field of depth one on a half slab of normal 3, invariant under the flip c^k of the class $n_{13} = ((1,0),0)_3$, and F_t'' invariant under the flip of the class $((1,1),0)_3$, such that on every cube of $N = \mathsf{n}L^j$ sites, $\mathsf{n} \in 2L^{\mathbb{N}}$, $j \geq 1$, at $\beta = N/t \geq 3$, where they are read on the block field $\mathcal{M}^j U$ of n sites a side (at the cutoff K on the cube ℓ, $\mathsf{n} = L\ell$ and $j = K-1$, and $\mathcal{M}^{K-1}U$ is the block field of depth one), the numbers of Thm. AG(12) obey: for F_t, if $N^{66}\beta^{-1}\log\beta \leq x_\flat(L)$, $c_\flat t \leq w_F \leq C_\flat t$ and $\log(2^{L-1}\mathfrak{q}_L) \geq \lambda_\flat = 0.5516L - 0.7162$; for F_t', if $N^{68}\beta^{-1}\log\beta \leq x_\flat(L)$, $w_F \geq c_\flat t$, $\mathfrak{q}_i \geq 0.84\cdot 0.9356^i$ for $i \leq L$, and $\log(2^{L-1}\mathfrak{q}_L) \geq \lambda_\flat = 0.6266L - 0.873$; and for F_t'' in the same range, $w_F \geq c_\flat t$ and $\mathfrak{q}_i \geq 0.84\cdot 0.944^i$ for $i \leq L$.

On a cube $\ell = 2L^m$ with $MR(b) \leq L^m \leq ML^2(\log b)^{r_\star}$, w_K and $\mathfrak{q}_L(K)$ are these numbers at the cutoff K, and $\varsigma_K = w_K^L\mathfrak{q}_L(K)$ is an integral over the box $(\ell,\ell,L\ell)$ divided by $Z^L_{K,\ell}$.

Construction and elementary properties of the windows. The windows. On the cube of $N = \mathsf{n}L^j$ sites with the time 3 let $h = N/2$; the half slab holds the layers $0 \leq x_3 \leq h$, and the block layers inside it are those of the block lattice of $\mathcal{M}^j U$ at the heights $c = L^j y_3$, $1 \leq y_3 \leq \frac{\mathsf{n}}{2} - 1$. In the block layer at the height c, for a block site y of it let $\bar h_1(y)$, $\bar h_2(y)$ be the products of the block field along the straight closed block lines from y in the directions 1 and 2, and let $\bar s_\pm(c)$, $\bar s_1(c)$, $\bar s_2(c)$ be the means over the n^2 sites y of the layer of $\frac12\mathrm{tr}[\bar h_1(y)\bar h_2(y)^{\pm 1}]$, $\frac12\mathrm{tr}\,\bar h_1(y)$ and $\frac12\mathrm{tr}\,\bar h_2(y)$: $\bar s_\pm$ are the means of the half traces of the closed block lines that wind once in the direction 1 and once, forwards or backwards, in the direction 2. With $c_1 = L^j$, the block layer next to the lower end of the half slab, and $c_2 = h - L^j$, the one next to its upper end, $F_t = \Phi_t \circ \mathcal{M}^j$, $F_t' = \Phi_t' \circ \mathcal{M}^j$,

$F_t'' = \Phi_t'' \circ \mathcal{M}^j$ with

$$\Phi_t = \mathbf{1}\{\bar s_2(c_1) > \tfrac12\}\, e^{-\frac4t(\bar s_+(c_1)^2+\bar s_-(c_1)^2)}, \qquad \Phi_t' = \prod_{c\in\{c_1,c_2\}} \mathbf{1}\{\bar s_2(c) > \tfrac12\}\, e^{-\frac2t(\bar s_+(c)^2+\bar s_-(c)^2)},$$

$$\Phi_t'' = \prod_{c\in\{c_1,c_2\}} \mathbf{1}\{\bar s_+(c) < -\tfrac12\}\, e^{-\frac4t(\bar s_1(c)^2+\bar s_2(c)^2)} :$$

one pin for F_t, two pins for F_t' and F_t''. They are Borel, with values in $[0,1]$, and gauge invariant. The flip of the class n_{13}, taken across a line in a wall of the blocks of every order (L is odd), passes the averaging map and becomes the change of sign of the block bonds of the direction 1 that cross that line: $\bar h_1 \mapsto -\bar h_1$ and $\bar h_2$ is kept, so $\bar s_\pm$ change sign, $\bar s_2$ is kept, and F_t, F_t' are invariant; the flip of $((1,1),0)_3$ changes the signs of $\bar h_1$ and of $\bar h_2$, so $\bar s_1$, $\bar s_2$ change sign, $\bar s_+$ is kept, and F_t'' is invariant. On a flat field with the holonomies $e^{i\varphi_\mu \mathsf{a}\cdot\sigma}$, $|\mathsf{a}| = 1$, the means are $\bar s_\pm = \cos(\varphi_1 \pm \varphi_2)$ and $\bar s_\mu = \cos\varphi_\mu$.

The flip of n_{13} acts by $\varphi_1 \mapsto \varphi_1 + \pi$; up to $\varphi \mapsto -\varphi$ its fixed points are $\varphi_1 = \frac\pi2$, $\varphi_2 \in \{0,\pi\}$, and inside $\{\bar s_2 > \frac12\}$ the common zeros of $\bar s_+$ and $\bar s_-$ are the one fixed point $(\frac\pi2, 0)$, at which they vanish transversally and independently: at $(\frac\pi2 + a, b)$, $\bar s_+^2 + \bar s_-^2 = 2(a^2+b^2) + O(|(a,b)|^4)$. Thus both transverse angle variables have scale $t^{1/2}$ in this flat parametrisation. Likewise the flip of $((1,1),0)_3$ acts by $\varphi_\mu \mapsto \varphi_\mu + \pi$, $\mu = 1,2$; the indicator $\{\bar s_+ < -\frac12\}$ singles out its fixed point $(\frac\pi2, \frac\pi2)$, about which $\bar s_1^2 + \bar s_2^2$ cuts both angles at the width $t^{1/2}$: at $(\frac\pi2 + a, \frac\pi2 + b)$, $\bar s_1^2 + \bar s_2^2 = \sin^2 a + \sin^2 b = a^2 + b^2 + O(|(a,b)|^4)$.

The numbers of Thm. AG(12) are $w_F = \mathcal{Z}_1/Z[0]$ and $\mathfrak{q}_i = \mathcal{Z}_i/\mathcal{Z}_1^i$, where $\mathcal{Z}_i$ is the periodic integral over the box (N, N, iN) of the product of the $2i$ mirrored copies of the window. The integral identity in the definition follows from the slab trace formula. The bounds called **(Window estimates)** are the uniform estimates on these inserted integrals that remain to be established.

The uninserted periodic Laplace estimate of Prop. 11.14 does not establish these bounds for the moving pins. In particular, **(Window estimates)** includes the stated weight bounds, numerical flat ratios and growing-lattice ranges. □

Proposition 11.15 (Winding block loops on small tori)**.** Let $\ell = 2L^m$, $b \ge \gamma^{-2}$ with $MR(b) \le L^m$, and $d \ge 1$ an integer with $L^d b \ge \mathsf{t}_\natural$, $\delta_d = \exp[7(\ell L^d)^2 \varepsilon(L^d b)] - 1 \le \frac{1}{20}$ and $\mathsf{R}(d) \le \frac{1}{40}$, where $\mathsf{t}_\natural$, $\varepsilon(x) = A_0 x^{-1/2}(\log x)^{\mathfrak{p}_0}$ and $\mathsf{R}(d)$ (of Thm. AH(7)) are formed from the constants. For a site y of the block lattice of depth d let $\gamma_1, \gamma_2, \gamma_{12}$ be the closed block lines through y that wind once in the direction 1, once in the direction 2, and once in 1 and then once in 2, W_γ the half trace of the block field along γ, t the translation by $\frac\ell2 e_3$, and $\mathsf{m}_\gamma = \langle W_\gamma \cdot \mathsf{t} W_\gamma\rangle$ at the cutoff $K > d$. Then $\mathsf{m}_{\gamma_2} = \mathsf{m}_{\gamma_1}$ and

$$f^K = 4\mathsf{m}_{\gamma_1} + \mathsf{m}_{\gamma_{12}} \ge (1 - \tfrac14\delta_d^2)(1 - \mathsf{R}(d)) - 5(\delta_d + 2\mathsf{R}(d)) \ge 0.4743,$$

while $|f^K| \le 10[\frac14(1 - r_1^2)]^{\frac12 - \frac{1}{\ell L^d}}$ with r_1 on the cube ℓ; so $r_1 \le 1 - 2\,(0.045)^{2/(1-2L^{-d}/\ell)} \le 0.99596$ at every cutoff $K > d$, and there $\delta_K^0 \ge \mathsf{G} \ge 0.003034$ by Thm. AG(13). These conditions hold at $\ell = 2L^m$, $b_m = \exp[L^{(m-m')/r_\star}]$, $d = 1$ for all large m, $M = L^{m'}$, where $\ell/b_m \to 0$. For every $b > 0$, $f^K(b)$ converges as $K \to \infty$ to a number $f(b)$, and, by Thms. YM3_I:AJ and YM3_I:AD, f is real-analytic on $(0,\infty)$. ■

Proof sketch. W_{γ_1} and $W_{\gamma_{12}}$ are real, of modulus ≤ 1, depend only on the bonds of the slab of L^{K-d} layers that holds the block layer of y, and are odd under the flip c^k of the class n_{13} applied to all layers of that slab; Thm. AG(6), with $2k' = \ell L^d$ slabs of that thickness and $j = k'$, gives the upper bound. By the small-field chain and the rarity of its large-field regions on the periodic torus (Prop. 4.54), off an event of periodic probability at most $\mathsf{R}(d)$ every plaquette of the block field V of depth d is within 2ε of $\mathbf{1}$, $\varepsilon = \varepsilon(L^d b)$, in the operator norm $|\cdot|$; let B be the event that this holds, and $\mathsf{n} = \ell L^d$ the number of block bonds of a straight closed block line.

On B the lemma on rearranged paths (Lem. YM3_I:4.1), read on the block lattice of depth d, where a step sequence may wind about the torus, applies to every pair of paths Γ, Γ' from one site whose step sequences, of a length q, are rearrangements of one another: their transports obey $|V(\Gamma')V(\Gamma)^{-1} - \mathbf{1}| \le (1+2\varepsilon)^{q(q-1)/2} - 1 \le e^{\varepsilon q^2} - 1$. (i) Let $\mathsf{H}_i = V(\gamma_i)$, the transport along γ_i from y, so that $W_{\gamma_1} = \frac{1}{2}\mathrm{tr}\,\mathsf{H}_1$, $W_{\gamma_2} = \frac{1}{2}\mathrm{tr}\,\mathsf{H}_2$ and $W_{\gamma_{12}} = \frac{1}{2}\mathrm{tr}\,\mathsf{H}_1\mathsf{H}_2$. The sequences $(e_1^{\mathsf{n}}, e_2^{\mathsf{n}})$ and $(e_2^{\mathsf{n}}, e_1^{\mathsf{n}})$, of the length $q = 2\mathsf{n}$, have the transports $\mathsf{H}_1\mathsf{H}_2$ and $\mathsf{H}_2\mathsf{H}_1$, so that $|\mathsf{H}_2\mathsf{H}_1\mathsf{H}_2^{-1}\mathsf{H}_1^{-1} - \mathbf{1}| \le e^{4\mathsf{n}^2\varepsilon} - 1 \le \delta_d$.

On SU(2), $|g-\mathbf{1}|^2 = 2 - \mathrm{tr}\,g$, and $\mathrm{tr}(ABA^{-1}B^{-1}) = (\mathrm{tr}\,A)^2 + (\mathrm{tr}\,B)^2 + (\mathrm{tr}\,AB)^2 - \mathrm{tr}\,A\,\mathrm{tr}\,B\,\mathrm{tr}\,AB - 2$ (by $X + X^{-1} = (\mathrm{tr}\,X)\mathbf{1}$); so the half traces $x_1 = W_{\gamma_1}$, $x_2 = W_{\gamma_2}$, $x_{12} = W_{\gamma_{12}}$ obey $x_1^2 + x_2^2 + x_{12}^2 - 2x_1x_2x_{12} \ge 1 - \frac{1}{4}\delta_d^2$, and, as $|x_{12}| \le 1$ and $2|x_1x_2| \le x_1^2 + x_2^2$,

$$2W_{\gamma_1}^2 + 2W_{\gamma_2}^2 + W_{\gamma_{12}}^2 \ge 1 - \tfrac{1}{4}\delta_d^2 \qquad \text{on } \mathsf{B}.$$

In the case $\delta_d = 0$, that of two commuting transports $\mathsf{H}_\mu = e^{i\theta_\mu \mathsf{a}\cdot\sigma}$ with one axis a, this inequality is the identity

$$2\cos^2\theta_1 + 2\cos^2\theta_2 + \cos^2(\theta_1+\theta_2) = 1 + (\cos\theta_1 + \cos\theta_2\cos(\theta_1+\theta_2))^2 + \cos^2\theta_2\sin^2(\theta_1+\theta_2) \ge 1.$$

(ii) Let γ be one of the three lines, $\mathfrak{s}$ its step sequence, of the length n or $2\mathsf{n}$, Γ_3 the straight block path of $\frac{\mathsf{n}}{2}$ steps in the direction 3 from y to $\mathsf{t}y$, and v the transport along it. The sequences $(\mathfrak{s}, e_3^{\mathsf{n}/2})$ and $(e_3^{\mathsf{n}/2}, \mathfrak{s})$, of a length $q \le \frac{5}{2}\mathsf{n}$, have the paths γ followed by Γ_3, and Γ_3 followed by $\mathsf{t}\gamma$; so $|\mathsf{v}\,V(\mathsf{t}\gamma)\,\mathsf{v}^{-1}V(\gamma)^{-1} - \mathbf{1}| \le e^{6.25\,\mathsf{n}^2\varepsilon} - 1 \le \delta_d$, which is the place of the numeral 7 in δ_d, and, with $|\frac{1}{2}\mathrm{tr}\,X| \le |X|$, $|VV'^{-1} - \mathbf{1}| = |V - V'|$ for unitary V, V' and the invariance of the trace under conjugation, $|W_{\mathsf{t}\gamma} - W_\gamma| \le \delta_d$ on B. (iii) As $\mathsf{t}W_\gamma = W_{\mathsf{t}\gamma}$ and $|W_\gamma| \le 1$, $W_\gamma\,\mathsf{t}W_\gamma \ge W_\gamma^2 - |W_\gamma - W_{\mathsf{t}\gamma}|$, and $|W_\gamma - W_{\mathsf{t}\gamma}|$ is at most δ_d on B and at most 2 off it; so $\mathsf{m}_\gamma \ge \langle W_\gamma^2\rangle - \delta_d - 2\mathsf{R}(d)$. With the weights $2, 2, 1$, the inequality of (i) on B, $W_\gamma^2 \ge 0$ off B and the periodic probability of B at least $1 - \mathsf{R}(d)$,

$$f^K = 2\mathsf{m}_{\gamma_1} + 2\mathsf{m}_{\gamma_2} + \mathsf{m}_{\gamma_{12}} \ge (1 - \tfrac{1}{4}\delta_d^2)(1 - \mathsf{R}(d)) - 5(\delta_d + 2\mathsf{R}(d)),$$

which decreases in δ_d and in $\mathsf{R}(d)$ and equals $\frac{1599}{1600}\cdot\frac{39}{40} - \frac{1}{2} = 0.47439\ldots$ at $\delta_d = \frac{1}{20}$, $\mathsf{R}(d) = \frac{1}{40}$; $\mathsf{m}_{\gamma_2} = \mathsf{m}_{\gamma_1}$ is the symmetry of the axes, the exchange of the directions 1 and 2 about y keeping the periodic law of the block field and commuting with t. The limit in K is the convergence in total variation of the laws of the block fields on T_ℓ (Thm. YM3_I:D); its analyticity is Thms. YM3_I:AJ and YM3_I:AD, f^K being at every label the mean of one bounded Borel function of the block field. □

Proof of Thms. AG, AH and AI. In this proof A(k) and B(k) denote part (k) of Thms. AG and AH. Thm. AG is proved on one lattice. In Thms. AH and AI, the first sentence of B(1), the inequalities of B(8), the reflection positivity of B(7) at a fixed cutoff and Thm. AI, under its hypothesis (H), at each cutoff follow from Thm. AG and the properties of the averaging map; the rest of B(1) uses Lem. 11.1; everything else uses Thm. YM3_I:D, and B(6), B(9)(c) and the statements on analyticity use Thms. YM3_I:AJ and YM3_I:AD as well, B(9)(c) with the radius and the constants of Thms. YM3_I:AJ and YM3_I:AD, through [7, Lem. N.2] and Thm. YM3_I:AV and, for its last assertion, through the normalisers at the first-passage data (Prop. YM3_I:91.12 (1)), the transfer principle for one law and for two laws under one normaliser (Prop. YM3_I:91.11), the first facts on states ([7, §K.1]), the device of sign averaging over a packing with its count (the packing and sign-averaging argument in the proof of Prop. YM3_I:127.1) and the windows of first passage (Cor. YM3_I:74.2).

Step 1: the trace and the order (A(1)). For a sign field γ on the bonds, -1 being central and Haar measure invariant, $Z_\beta[\eta\,\mathrm{d}\gamma; F] = Z_\beta[\eta; F\circ\gamma]$: columns move, two columns over one plane cancel, and on centre-blind F the places of the columns do not matter. Cut the box along μ into

layers V^τ and crossings S^τ, with the columns through μ in the crossing τ_*:

$$A_{\eta^n}(U) = \sum_\tau A^m_\Sigma(V^\tau) + \sum_{\tau\neq\tau_*} A_\times(V^\tau, S^\tau, V^{\tau+1}) + A_\times(V^{\tau_*}, S^{\tau_*}, c^k V^{\tau_*+1}),$$

$A_\times$ the action of the plaquettes of one crossing; integrating the S^τ gives the trace formula. As $\operatorname{tr}(vw^{-1}) = \sum_{ij} v_{ij}\bar w_{ij}$, $\prod_b e^{-\beta(1-\frac12 \operatorname{tr} V_b W_b^{-1})}$ is a series of squares of monomials with coefficients ≥ 0, so $\mathsf K = \Pi \mathsf K_0 \Pi \ge 0$, Π the gauge average, strictly on gauge invariant functions if $\beta > 0$ (Stone–Weierstrass). A continuous kernel > 0 gives a simple top with a positive eigenfunction, which C^k fixes. $Z[(0,m)_\mu] - Z[(k,m)_\mu] = 2\sum_{e\cdot k=1} Z_{e,m} \ge 0$ removes the planes through μ, and the same with the time in the remaining twisted plane removes it; dually $Z_{00,m} - Z_{e,m} = \frac12 \sum_{e\cdot k=1} Z[(k,m)_\mu] > 0$ for $e \neq 0$, every $Z[n]$ being positive; $A_{\eta^n} \le A + 2\#\{\eta^n = -1\}$ gives the lower bound. $Z_{e,m} > 0$: the product of the traces of straight closed lines of the directions ν, ρ with the exponents e lies in the range of P_e, and Jensen's inequality for its spectral measure bounds $\operatorname{Tr}(P_e T_m^N)$ below. $\|T_1\| \le \|T_0\|$ follows from $\operatorname{Tr} T_1^N \le \operatorname{Tr} T_0^N$ by Lem. 11.2.

A step in the direction ν moves the reversed plaquette of the section by one site, and reversing the bond that it shares with its image moves it back: $\mathcal T_\nu$ keeps dV and A^1_Σ and commutes with $\mathsf K$, so with T_1, and it maps the gauge invariant functions onto themselves, the step and the sign changes carrying gauge transformations to gauge transformations. For a sign field γ on the bonds, $\mathsf S_\nu(\gamma V) = \gamma' \mathsf S_\nu V$ with $\gamma'(\xi, j) = \gamma(\xi + e_\nu, j)$; so the map of $\mathcal T_\nu$ is the step followed by the reversal of the bond (ξ_*, ρ), and by induction $\mathcal T_\nu^q$ is the composition with the step by q sites followed by the reversal of the q bonds $(\xi_* - ie_\nu, \rho)$, $0 \le i < q$, of the direction ρ on the line of the section through ξ_* normal to ρ.

At $q = N_\nu$ the step is the identity and these are the bonds that C_ρ reverses: $\mathcal T_\nu^{N_\nu} = C_\rho$, and likewise $\mathcal T_\rho^{N_\rho} = C_\nu$. In the same way $\mathcal T_\nu \mathcal T_\rho$ and $\mathcal T_\rho \mathcal T_\nu$ differ by the reversal of the four bonds at the site $\xi_* + e_\nu + e_\rho$, the gauge transformation that is -1 at that site. Let $c_\nu^{(s)}$ reverse the bonds (ξ, ν) with $\xi_\nu = s$, so that C_ν is the composition with $c_\nu^{(s)}$ at $s = (\xi_*)_\nu$; then $\mathsf S_\nu c_\nu^{(s)} = c_\nu^{(s-1)} \mathsf S_\nu$, and $c_\nu^{(s-1)} c_\nu^{(s)}$ is the gauge transformation that is -1 at the sites of the line $\xi_\nu = s$ and 1 elsewhere: $\mathcal T_\nu C_\nu$ and $C_\nu \mathcal T_\nu$ differ by it, and $\mathcal T_\rho C_\rho$ and $C_\rho \mathcal T_\rho$ likewise.

Gauge transformations act as the identity on the gauge invariant functions, off which $T_1 = D_1 \mathsf K D_1$ vanishes; for $\beta > 0$ these are the closure of the range of T_1, as $\mathsf K$ is strictly positive on them and D_1 is invertible and gauge invariant. On them T_1, $\mathcal T_\nu$, $\mathcal T_\rho$, C_ν, C_ρ commute pairwise, and on the range of P_e there, where $C_\nu = (-1)^{e_\nu}$ and $C_\rho = (-1)^{e_\rho}$, $\mathcal T_\nu^{N_\nu} = (-1)^{e_\rho}$ and $\mathcal T_\rho^{N_\rho} = (-1)^{e_\nu}$: the commuting unitaries $\mathcal T_\nu$, $\mathcal T_\rho$ have finitely many eigenvalues there, N_ν-th roots of $(-1)^{e_\rho}$ and N_ρ-th roots of $(-1)^{e_\nu}$, which gives the momenta of A(1); their joint eigenspaces are invariant under the compact self-adjoint T_1, and an orthonormal eigenbasis of T_1 in each of them is a joint eigenbasis.

Finally $C^k = C_\nu^{k_1} C_\rho^{k_2} = \mathcal T_\nu^{k_2 N_\nu} \mathcal T_\rho^{k_1 N_\rho}$ gives the trace formula with the momenta. For the crossing relation, expand $Z^{[1]}_{e,m} = \frac14 \sum_k (-1)^{e\cdot k} Z[(k,m)_1]$ and $Z[n] = \sum_{e'} (-1)^{e'\cdot k'} Z^{[2]}_{e',m'}$, $k' = (n_{12}, n_{23})$, $m' = n_{13}$: the sum over n_{12} forces the first entries to agree, and what is left is the display. All $Z^{[2]} \ge 0$, so its modulus is at most its value at $e_2 = 0$, $m = 0$, which is $Z^{[1]}_{e_1 0,0}$; the time 3 in the place of 2 gives $Z_{0e_2,0}$; N_μ-th roots give the norms. $\varepsilon_{\mathrm{mag}}$ and $\varepsilon_{\mathrm{el}}$ are Lem. 11.2 for the sector traces, the top of T_0 being simple with its eigenfunction in the range of P_{00}.

Step 2: lengths (A(2)–(4)). The class n_{12} is electric for the times 1 and 2, so $\frac12(Z[0] \pm Z[n_{12}]) = \operatorname{Tr}(\frac12(1 \pm C) T_0^{N_\mu})$ with either direction of the plane as the time, and Lem. 11.2 applies in each length. $N_\mu^{-1} F_0$ decreases to $\log \|T_0\| \ge |\Sigma| P$, $P \ge -6\beta$ because $A \le 6|N|$, and $\delta(t, \Sigma) = t[\log \|T_0\| - |\Sigma| P] + \mathsf G(t)$ gives $\mathsf G \le \delta$, the identities for $\mathsf d$ and, the bulk cancelling, $qF_0(N) - F_0(N^{[\mu]}) = q\delta(N) - \delta(N^{[\mu]})$; $F_0/|N|$ minus its limit in one side is $\mathsf G/|N|$, and the last inequality of A(2) follows side after side, $\mathsf G$ not increasing in its height.

For A(3), $q(t,\lambda)$ is a power sum in t and in λ, and $p \ge \|T_0\|^t \ge e^{-t\lambda s_0}$, the top eigenfunction lying in the range of $\frac{1}{2}(1+C)$ and $E_0(\lambda) \le s_0\lambda$; so $\mathfrak{b} = q\,e^{t\lambda s_0} \le \mathfrak{b}(t_0,\lambda_0)^{t\lambda/t_0\lambda_0}$, $\mathfrak{b} \le Z[0]e^{t\lambda s_0} - 1 = e^{\Phi} - 1$ and $q/p \le \mathfrak{b}$; $-t^{-1}\log$ and $t \to \infty$ carry the log-convexity and the submultiplicativity in λ to E_0 and E_{e}, and the monotone increments give s_f and the bounds on s_f and on σ. The roots of the power sums in both lengths give the monotonicity of $-(t\lambda)^{-1}\log\mathfrak{b}$ and its limit $s_{E_{\mathrm{e}}} - s_0 = \sigma$; the lower bound on $-\log\mathfrak{b}(t,\lambda)$ follows from the convexity in t continued from t_0, with the Casimir term $R(\lambda_0) \ge R(\lambda)$; and $q(2t) \le q(t)^2$ gives the bound on ϱ.

For A(4)(a), $g(h) = \log(q(h)/\|T_0\|^h)$ is convex in the height h and $g(h)/h \to -\varepsilon(\lambda)$; its chord from h_0 to h_1, continued beyond h_1, lies below it, so $(h_1-h_0)\varepsilon \le g(h_0) - g(h_1)$, with $g(h_0) \le \log(e^{\mathsf{G}(h_0)}-1)$, the top eigenvalue lying in the even part, and $-g(h_1) \le \log(2/(1-r(h_1)))$; (a) in the two lengths, with A(3), gives the bound on σ. For (b), $\Lambda(a,b') = \log\|T_0\|$ over the section (a,b') is symmetric in its two arguments, $F_0(a,b',c) = c\Lambda(a,b') + \mathsf{G}(c;a,b')$ with any side as the time, $\Lambda_1(b') = \lim_a \Lambda(a,b')/a$, and $\varphi(a) = \Lambda(a,b') - a\Lambda_1(b') = R(a) \ge 0$ is convex and does not increase; writing $F_0(a,b',k'a)$ with the time along a gives $\varphi(a) \le \varphi(a)/k' + \mathsf{G}(a;b',k'a)/(k'a)$, which is the bound; the same loop in the second transverse side and $\delta(M^{\times 3}) = \mathsf{G}(M;M,M) + M\varphi(M;M) + M^2[\Lambda_1(M) - MP]$ give the bound on $\delta(M,M,M)$. (c) is the left bound on σ in A(3) with $q(t) \ge e^{-tE_{\mathrm{e}}}$, $Z[0](2t) \le \|T_0\|^t Z[0](t)$ and (b) at $k' = 2$.

For (d), let p run over the plaquettes of the planar rectangle and let $\gamma_{\mathsf{p}} = -1$ exactly on the planar edges crossed by the straight dual path from p to the free boundary, lifted to the bonds of the directions $1,2$ at all heights: $d\gamma_{\mathsf{p}} = -1$ exactly on the stack of the N' plaquettes over p, the plaquettes through the third direction keep their weight, and $\operatorname{tr} U(C)$ takes the product of the signs of the enclosed p. Averaging over all products of these changes, with $h_{\mathsf{p}} = \frac{\beta}{2}\sum_{\text{stack}} \operatorname{tr}\partial U$, $|h_{\mathsf{p}}| \le \beta N'$, gives $\langle \operatorname{tr} U(C)\rangle = \int \operatorname{tr} U(C) \prod_{\text{inside}} \tanh h_{\mathsf{p}}\, d\mu'$ with a probability measure μ'. (e) is the same averaging over the reversal of one column inside each container, at fixed walls.

Step 3: how bounds travel; the overlap (A(5), A(6)). A(5)(a): with $B_\pm = \frac{1}{2}(1 \pm C^k)T_m^N \ge 0$, $B_+B_- = 0$, $\operatorname{Tr} T_m^{2N} = \sum_\pm \operatorname{Tr} B_\pm^2 \le \sum_\pm (\operatorname{Tr} B_\pm)^2 = \frac{1}{2}(x^2+y^2)$, $x = \operatorname{Tr} T_m^N$, $y = \operatorname{Tr}(T_m^N C^k)$. With $z_e = Z_{e,m} \ge 0$ and $z_{00} \ge \|T_m\|^N$: $Z[(k,m)] = \sum_e (-1)^{e\cdot k} z_e \ge Z' - 2(Z' - z_{00})$, $\sum_k Z[(k,m)] = 4z_{00}$, $\operatorname{Tr} T_m^{2N} \le \sum_e z_e^2 = \frac{1}{4}\sum_k Z[(k,m)]^2$, and $Z[((1,0),m)] + Z[((0,1),m)] - Z' = Z[((1,1),m)] - 4z_{11}$. (b): $q(N) = \frac{1}{2}(Z[0] - Z[n])(N) = \operatorname{Tr}(\frac{1}{2}(1-C^k)T_0^N)$ is log-convex in N (Lem. 11.2), $q = \frac{1}{2}Z[0](1-r)$ with $0 \le 1 - r \le 1$, and the volumes cancel in $Z[0](N)^2/Z[0](N-j)Z[0](N+j) = e^{2\delta(N) - \delta(N-j) - \delta(N+j)}$. (c): $\operatorname{Tr}(P_S T_m^{N'}) \ge \lambda^{N'}$, and $\operatorname{Tr}(P_S T_m^{N'}) \le \lambda^{N'-N''}\operatorname{Tr}(P_S T_m^{N''}) \le \lambda^{N'-N''}\operatorname{Tr} T_0^{N''}$ by Lem. 11.2 and A(1), with $\operatorname{Tr} T_0^{N'} = \|T_0\|^{N'}\Theta(N')$. (d): the first inequality is the log-convexity of q between two heights, the second the monotonicity of $q(N)^{1/N}$ (Lem. 11.2) with $Z[0](N) \ge \|T_0\|^N$, and the third is Lem. 11.3 for the class z with the time μ, divided by the periodic numbers, the volumes cancelling: $k'(n'+m) - (k'-1)m - (k'n'+m) = 0$; the denominator $\mathsf{Z} = Z[(0,m')] \le Z[0]$ stands the right way. (e): put $k' = 2L'+1$, $n' = L'^{j-1}(L'-1)$, $m = L'^{j-1}(L'+1)$ and $m_1 = m_2 = m/2$, so that $n' + m = N'$ and $k'n' + m = L'N'$, and enlarge the three sides of the cube N' one after the other.

In a direction that lies in a twisted plane of z the third inequality of (d) applies; for a class with one twisted plane, in the normal of that plane (c) applies at $M_1 = N'$, $M_0 = N'/L'$, with the exponent $L'^2/(L'-1) \le 2L'+1$. A(1) passes between the three expressions, and all excesses of the auxiliary boxes are bounded against the cube N'/L' by the last inequality of A(2). A(6): $T_m^\tau C^k$ has the kernel $\Psi \circ g$, so $Z[(k,m)_\mu] = \iint \Psi\,\Psi \circ g$ by A(1), and μ_2 is the joint law of the layers 0 and τ; $w = 2\Psi\,\Psi\circ g/(\Psi^2 + \Psi^2\circ g)$ is g-invariant, which gives the three expressions of r'. $\int\sqrt{f_1 f_2} \ge \int \min(f_1,f_2) = 1 - \mathrm{TV}$ and the Cauchy–Schwarz inequality give the bracket; on E_c the integrand $\Psi\circ g/\Psi$ is $\ge c$, off it $< c$, and $\langle(\Psi\circ g/\Psi)^2\rangle_2 = 1$; if $gE = E^c$, $r' = 2\int_E \Psi\,\Psi\circ g/Z'$; for odd Φ, $\langle\Phi\rangle_2 = E_{\bar\nu}[\Phi\mathsf{m}]$, $\bar\nu = \frac{1}{2}(\mu_2 + g_*\mu_2)$, and the Cauchy–Schwarz inequality with $1 - \sqrt{1-x} \le x$ gives the next bracket; $g_*(p_1\otimes p_1) = p_1\otimes p_1$ gives the bound by $\mathrm{TV}(\mu_2, p_1\otimes p_1)$; the bound on $\mathcal{G}\cap g\mathcal{G}$ is pointwise; Jensen's inequality for $-\log$ gives the entropy bound; and an affinity does not

decrease under a map, whereas relative entropy does not increase, while Φ intertwines the projection on the two layers with g.

For the odd functions, cut the period into $2k'$ slabs of ϑ layers: $Z[0] = \operatorname{Tr} Y^{2k'}$ and $Z[n] = \operatorname{Tr}(\mathsf{Z} Y^{2k'})$ with the transfer operator Y over ϑ layers and the flip Z in one layer, $\mathsf{Z}Y = Y\mathsf{Z}$; the projections $\frac{1}{2}(1 \pm \mathsf{Z})$ carry the weights $\frac{1}{2}(1 \pm r[n])$, the operator of D' maps each of their ranges into the other, and Hölder's inequality for traces over the $j-1$ and the $2k'-j-1$ slabs between the two functions gives the bound.

Step 4: reflection positivity, chessboard bounds, the mixture (A(7)–(9)). For a slab with α, α' crossings on its two sides and the columns through μ on the side of α', the trace formula gives

$$Z[(k,m)_\mu; \overline{\theta F} F] = \sum_e (-1)^{e\cdot k} \| T_m^{\alpha/2} \hat{F}\, T_m^{\alpha'/2} P_e \|_2^2,$$

$\hat{F}$ the slab operator; this is ≥ 0 at $k = 0$, $\sum_k (-1)^{e\cdot k}$ of it is $4\| T_m^{\alpha/2} \hat{F} T_m^{\alpha'/2} P_e\|_2^2 \geq 0$, and summed over the classes allowed by $\mathfrak{s}$ the signs add up to $\prod_j (1 + (-1)^{e_j}) \geq 0$. For $k \neq 0$, $\beta > 0$, write $e^{\frac{\beta}{2}\operatorname{tr} u} = \cosh \cdot (1 + \tanh)$ at $\frac{\beta}{2} \operatorname{tr} u$, and let one plaquette of the flipped row of one crossing carry $\chi_J / \cosh$, χ_J the character of spin J, and the others a factor concentrated at $\operatorname{tr} u = 0$, through which integer spins alone pass; the part even under the flip falls like $c^J/(2J)!$, while the odd part is a Fourier coefficient of a function with the pole of $\tanh(\beta \cos\theta)$ and is $\geq e^{-cJ}$ for infinitely many J: the form is negative. $\sum_n Z[n] \leq 8 Z[0]$ is A(1).

For the last assertion of A(7), $Z[(k,m)_\mu; F] = \operatorname{Tr}(\hat{F} T_m^{N_\mu - \vartheta + 1} C^k)$ with one slab, and the Cauchy–Schwarz inequality for traces. On a board a reversed plaquette in a mirror is a unitary flip commuting with $\mathsf{K}^{1/2}$, so $Z_\beta[\eta; \prod F_\kappa] = \operatorname{Tr}(C_r \mathfrak{Y}_r \cdots C_1 \mathfrak{Y}_1)$, and Hölder's inequality for traces does not see unitaries; the adjoint of a slab operator is the operator of the mirror copy of the slab that carries the complex conjugate function, so the norms of the slab operators are fold integrals with the copies complex conjugated in the cells odd relative to the given one (equation YM3_I:(1749)), which are ≥ 0.

Put each transverse electric-flip cut on a wall of the board. A cell function reads only bonds with both endpoints in that cell; the flip changes only bonds crossing the wall, so it fixes every inserted cell function, including a complex one. Applying the flip on all layers of a slab preserves its action and Haar measure, hence its inserted slab operator commutes with the flip. The crossing operator and its square root commute as well. Thus each $\mathfrak{Y}_i$, and each $\mathfrak{Y}_i^* \mathfrak{Y}_i$, commutes with the electric-sector projections. In Z_w the flips add up to $\mathsf{D}_m^w = \sum_e \hat{w}(e,m) P_e \geq 0$, which commutes with all $\mathfrak{Y}_i$: $Z_w[\prod F_\kappa] = \sum_m \operatorname{Tr}(\mathsf{D}_m^w \mathfrak{Y}_r \cdots \mathfrak{Y}_1)$, and $|Z_w| = Z[0] \sum w(n) r[n]$ lies between $w(0) Z[0]$ and $8 w(0) Z[0]$ by A(1). Comparing $Z_w[\operatorname{Fold} F]$ with $\sum_n w(n) Z[n; \operatorname{Fold} F]$ and with its term $n = 0$ gives the two bounds of $\| \cdot \|_w$. A covariant function of a cell is $G(U,s) = F(\gamma U)$ with $d\gamma = s$ on the cell and F invariant under the gauge transformations with values ± 1, and $\mathsf{M}_n[\prod G_\kappa] = Z[n; \prod F_\kappa]$ with the columns in the mirrors. The functions of the cells that cancel the weight of the rods of plaquettes through the centres of the cells make $Z[n; \operatorname{Fold} F] = Z[0; \operatorname{Fold} F]$ for every n, which gives equality; the functions that remove the reversed columns from the weight give the failure of the fold in a class, since $Z[n] < Z[0]$.

In a class: Hölder's inequality with the first bound for $|G|^q$; the density bound by the first bound for $h_n^{\mathsf{N}/(\mathsf{N}-1)}$ and the entropy inequality, and Pinsker's inequality. Without a board, one mirror and the Cauchy–Schwarz inequality for $\mathsf{M}_{(k,m)}[\overline{\theta G} G'] = \operatorname{Tr}(Y_m[G]^* C^k Y_m[G'])$ give $r[n] |\mu_n(G)| \leq \langle \overline{\theta F} F \rangle^{1/2}$ when $n = (k,0)_\mu$, that is, when μ lies in every twisted plane of n; the same for $G - \langle F \rangle$ gives the two-sided bound; a second mirror gives the quarter; and the event $\{\prod_{\text{plane}} s = -1\}$ of a plane normal to μ inside the half has periodic probability 0 and probability 1 in a class twisted in that plane.

For pairs, summing over the signs inside each half gives $\mathbb{M}_{\delta_{(k,m)_\mu}}[\overline{\theta G} G'] = \operatorname{Tr}(Y_m[G]^* C^k Y_m[G'])$, and $\sum_k w((k,m)_\mu) C^k = \sum_e \hat{w}(e,m) P_e \geq 0$. On $a_K \mathbb{Z}^3$ a closed sign field is $d\chi$, χ on the bonds, closed sign fields being exact on every box of sites: a class is the image of its measure under

$U \mapsto \chi \operatorname{per}_\ell U$, the plain Wilson measure of a field periodic up to gauge transformations with values ± 1, and its law on the invariant local functions depends neither on s nor on χ; the proof of [7, Cor. B.1] carries over, and for other functions the columns are put on the edges of the board.

Its clause (e) holds with p^w_ℓ: with the time μ and n_μ cells in the direction μ, $Z_w[\operatorname{Fold} F] = \sum_{e,m} \hat w(e,m) \operatorname{Tr} (P_e B_m P_e)^{\mathsf{n}_\mu/2}$, where $B_m = \mathfrak{Y}^*_m \mathfrak{Y}_m \ge 0$ is the operator of a slab of cells and its mirror copy and commutes with P_e, the same operators for every number of cells in the direction μ. For $\mathsf{n}_\mu = q\mathsf{n}'_\mu$, $\operatorname{Tr} B^{q\mathsf{n}'_\mu/2} \le (\operatorname{Tr} B^{\mathsf{n}'_\mu/2})^q$ for $B \ge 0$, $\omega \le \omega^q$ for $\omega = \hat w(e,m)/\underline w \in \{0\} \cup [1,\infty)$, and $\sum_i x_i^q \le (\sum_i x_i)^q$ for $x_i \ge 0$; so $Z_w[\operatorname{Fold} F]/\underline w$ over n_μ cells is at most the q-th power of the same number over n'_μ cells, and its root of the order of the number of cells does not increase when the board grows in one direction.

The three directions in turn carry this from the board of $\Lambda_{K,\ell'}$ to that of $\Lambda_{K,\ell}$; at $F = 1$ it is $p^w_\ell \le p^w_{\ell'}$, and the inequality at F and at $F = 1$ gives $\|F\|_{w,\ell} \le \|F\|_{w,\ell'} e^{|3|(p^w_{\ell'} - p^w_\ell)}$. At $w \equiv 1$, $\hat w(e,m) \in \{0,4\}$, and the same holds without the division by $\underline w$. A(9): for a closed $s = \eta^n d\gamma$, γ is fixed up to the closed sign fields on the bonds and $e^{-\beta A_s}(U) = e^{-\beta A_{\eta^n}}(\gamma U)$; integrating against a centre-blind F gives the first identity, $2^{2|N|-2}$ being the number of the closed sign fields of one class. The mean of $\sum_n e^{-\beta A_{\eta^n}}$ over all sign changes γ is $\hat{\mathsf{w}}_{\mathrm{mix}}$, which is therefore ≤ 8, and centre-blind, gauge invariant and invariant under the isometries.

Writing $e^{-\beta A_s} = e^{-\beta \bar A} \prod_{p:\, s_p \ne \operatorname{sign} t_p} z_p$, the sum over the closed s becomes the sum over the sets $\Gamma = \{p : s_p \ne \operatorname{sign} t_p\}$ with boundary $\mathsf{M}(U)$. At the constant field $U^0(x,\mu) = \cos\alpha + \sin\alpha\, i\sigma_\mu$, $\sin^2\alpha = 2^{-1/2}$, all t_p vanish, and a sixth mixed derivative of $\log \mathsf{w}_{\mathrm{mix}}$ along a hexagon of edges of one cube, two of which lie in no common plaquette, has the modulus $\beta^6(2\sqrt2 - 2)^{1/2} = 0.91018\,\beta^6$, while a sum of functions of single plaquettes has none. On one cube the plaquette variables of the faces, carried to a common corner, multiply to $\mathbf{1}$; if the angles add up to less than π, the product of the signs of the traces is 1, and a cube with six angles $\frac\pi6$ and the product -1 shows that π is sharp.

On $\mathfrak{S}$ the sign field $s(U)$ is locally constant and closed, $\mathsf{w}_{\mathrm{mix}} = e^{-\beta A_{s(U)}} \Xi(z)$ with $\Xi(z) = \sum_\Gamma \prod_{p\in\Gamma} z_p$ over the closed sets Γ, and the expansion of $\log \Xi$ in connected sets at $|z_p| \le \frac{1}{200}$ gives the $E(X)$, which vanish unless X is connected and a union of vortices, connected for the relation of A(9): two plaquettes are neighbours if they are faces of one unit cube, that is, dual bonds with a common dual site. For this relation the faces of one cube are pairwise neighbours, so each cube meets one connected component of a closed set only, the components of a closed set are closed, the closed sets are the unions of vortices pairwise apart, and Ξ is a gas of vortices; for a vortex X that holds no smaller one the only clusters with the support X are the repetitions of X, whose terms add up to $\log(1 + \prod_{p\in X} z_p)$.

The last bound is the smallness, on $\mathfrak{S}_n$, of the weights $\sum_{s\in[n']} e^{-\beta A_s}/\mathsf{w}_{\mathrm{mix}}$ of the other classes, whose sets Γ wind about the torus and so have at least $\min_\mu N_\mu$ plaquettes.

Step 5: one inequality; loops (A(10), A(11)). Under the hypothesis $D_2(l_0) \le \frac{1}{20}$ of A(10), one inequality between the partition functions of the periodic cubes l_0 and $2l_0$, Lem. 11.4 gives $18D_2(2^i l_0) \le (18D_2(l_0))^{2^i}$, hence $\delta(s,u,w) \le 0.14108\,(suw/l^3)D_2(l)$ for $s,u,w \ge l \in \{2^i l_0\}$ and $\mathsf{r}_h(l,l) \le \frac{1}{18} e^{-m_* h}$. Lem. 11.5 then gives, for $\bar\varepsilon \ge \varepsilon_{\mathrm{v}}(l)$ and an integer $h \ge 2l$ with $e^{-m_* h} \le (h\bar\varepsilon)^4$, $h\bar\varepsilon \le \frac{1}{10}$ and $2(h/l)D_2(l) \le \frac14$,

$$2l\,\varepsilon_{\mathrm{v}}(2l) \le 4.84\,(h/l)^3 (l\bar\varepsilon)^4,$$

whose iteration from l_1 gives $l\varepsilon_{\mathrm{v}}(l) \le e^{-\sigma_* l^2}$; the bound $\sigma_* \ge \min\{\frac16 m_*^2, m_*/l_0\}$ is the definition of l_1. With $2\zeta \le h\varepsilon_{\mathrm{v}} + \mathsf{r}_h$ and the doubling of the short sides (Lem. 11.5) this is the bound on the boxes (t,λ,h) and on the cubes; the bound on $\Lambda_-(N)$ is the same chain for the norm of T_0 on the odd functions, which is $\|T_0 P_{10}\|$ by A(1). The loop bounds are Lem. 11.5 on the even cubes N, the factor $8^{T/N}$ tending to 1 as $N \to \infty$; loops made of rows go the same way, row by row, with the norms of the transfer operators with static charges, the excess entering through $e^{1.13 D_2(N/2)}$.

Uniqueness and convergence are Lem. 11.7. Under the hypothesis $\mathsf{e}(l) \le 0.069$, Lem. 11.4 gives $\theta \le 0.8901$, $\mathsf{e}(4l) \le \theta^4/12.9$ and $D_2(8l) \le 20\,\mathsf{e}(4l)^2 \le 20\theta^8/166.41 \le 0.047355$; for a cube $N \ge 2l$ and the largest $M \in \{2^j l\}$, $j \ge 1$, with $M \le N$, $\delta(N^{\times 3}) \le (N/M)^3 \delta(M^{\times 3}) \le 8 \cdot 0.01717\,\theta^{M/l} \le 0.13736\,\theta^{N/2l}$; and $1 - r[n] \le 3(1 - e^{-\mathsf{G}}) \le 3\delta$ (Lem. 11.4), with $3 \cdot 0.13736 \le 0.4121$. A(11): ν is reflection positive at site mirrors and at bond mirrors, the exact statements on periodic Gibbs states and translation invariance comparing it with the periodic tori, where A(7) holds at $k = 0$.

For a rectangle of height T, the paths from a point of the mirror that run k bonds up, R across and k down, with $k = h + 1, h$ at a site mirror for $T = 2h + 1$ and $k = h, h - 1$ at a bond mirror for $T = 2h$, form half loops whose Gram matrix has the entries $2\mathsf{m}(R, T + 1)$, $2\mathsf{m}(R, T)$, $2\mathsf{m}(R, T - 1)$; its positivity is the log-convexity. So $\mathsf{m}(R, T)/\mathsf{m}(R, T - 1)$ does not decrease in T, $\mathsf{m}(R, T) \ge \mathsf{m}(R, 1)^T \ge \mathsf{w}^{RT}$ by the symmetry, V exists, is concave in R by the log-convexity in R, and $V(R)/R$ does not increase; the bounds on w, on m for $R, T \ge 2$ and on $V(1)$, $V(R)$ come from integrating the bonds of the loop in their plaquettes.

Step 6: purity; small cubes; constructions (A(12)–(14)). A(12): $\mathsf{P} = \mathsf{Y}^*\mathsf{Y} \ge 0$ commutes with C^k; with $B_\pm = \frac{1}{2}(1 \pm C^k)\mathsf{P} \ge 0$ and $d = \operatorname{Tr}(C^k\mathsf{P})/\operatorname{Tr}\mathsf{P} \in [0, 1]$,

$$\operatorname{Tr}\mathsf{P}^i = \sum_{\pm} \operatorname{Tr} B_\pm^i \le \sum_{\pm} (\operatorname{Tr} B_\pm)^i = (\operatorname{Tr}\mathsf{P})^i \sum_{\pm} (\tfrac{1 \pm d}{2})^i;$$

convexity in d gives $d \ge (\mathfrak{q}_i - 2^{1-i})/(1 - 2^{1-i})$, at $i = 2$ the display gives $d^2 \ge 2\mathfrak{q}_2 - 1$, and as $i \to \infty$ it gives $\|\mathsf{P}\| \le \frac{1}{2}(1 + d) \operatorname{Tr}\mathsf{P}$; and $r[n] \ge Z[n; F \otimes \theta F]/Z[0] = d\,w_F$. For block fields, the averaging map commutes with the flip in a wall of the blocks of all orders, L being odd. A(13): with the time μ, $Z_{e,0}/Z[0] = \frac{1}{4}\sum_k (-1)^{e \cdot k} r[(k, 0)_\mu]$, and the three classes with $k \ne 0$ have the quotients r_1, r_1, r_2: the first bound; $e^{-\mathsf{G}} = \|T_0\|^N/Z[0] \le Z_{00,0}/Z[0]$; $D_q(N)$ is a sum of defects of q-fold enlargement, the first $q^2[qF_0(N^{\times 3}) - F_0(qN, N, N)] \ge q^2(q - 1)\mathsf{G}(N)$; the bound on $Z[0](N^{\times 3})^2/Z[0](2N, N, N)$ is A(5)(a); if $r_1 \le \frac{1}{3}$, each sector trace at the height N is at most $\frac{1}{2}Z[0]$, and with $\operatorname{Tr}(P_e T_0^N) \le (\operatorname{Tr} P_e T_0^{N/2})^2$ this gives $(\operatorname{Tr} T_0^{N/2})^2 \ge 2 \operatorname{Tr} T_0^N$, that is $\mathsf{e}(N/2) \ge \log 2$; and $\Phi \ge F_0 - N \log \|T_0\| = \mathsf{G}(N)$ on the cube gives the bound on $\mathfrak{b}$.

A(14)(a): there are two Haar-preserving, gauge covariant bijections $\Phi_\pm$ of the fields, moving only the bonds of the directions μ, ν inside a tube of radius h' about the reversed column, which spread the flux of the column over the cross-section of the tube; pointwise, $\frac{1}{2}[A_\eta(\Phi_+ U) + A_\eta(\Phi_- U)] - A(U)$ is at most a sum over the layers of the squares of the spread curvature, of order $(h' - \frac{1}{2})^{-2}$ in total, and of the actions of ladders of plaquettes of length of order h' weighted by the angles of rotation. As $\Phi_\pm$ keep dU, $Z[\eta] = \int e^{-\beta A_\eta(\Phi_\pm U)}\,\mathrm{d}U$ for both signs; the arithmetic mean of the two integrands is at least their geometric mean, and Jensen's inequality in the periodic measure gives $\log r \ge -\beta\langle \frac{1}{2}[A_\eta(\Phi_+ U) + A_\eta(\Phi_- U)] - A(U)\rangle$.

For $\beta \ge 1$ the mean actions $\bar a_{12}, \bar a_{13}, \bar a_{23}$ of the plaquettes of the three orientations in the periodic measure have $\bar a_{12} + \bar a_{13} + \bar a_{23} \le \lambda(\beta)/\beta$, by $\log Z[0] \ge -|N|[\frac{9}{2}\log\beta + 30]$ and the convexity of $\log Z[0]$ in β; and the mean action of a ladder of $1 \times k$ plaquettes of one orientation is k times the mean of that orientation. The second form is the choice of h' of the order of $(\beta/\lambda(\beta))^{1/3}$, and for several twisted planes the tubes are disjoint. (b): $\log Z[0] \ge -|N|[\frac{9}{2}\log\beta + 30]$ by an elementary estimate; the chessboard estimate for $e^{\frac{\beta}{2}(1 - \frac{1}{2}t_p)}$ in one cell bounds its periodic mean by $\beta^{121.5} e^{810}$, so up to a factor 2 in the partition function of the cube all plaquettes lie within $(130\pi^2 \log\beta/\beta)^{1/2}$ of $\mathbf{1}$ once $\log\beta \ge 147.6$; cut the cube into eight free blocks, fix the gauge on a comb in each, and bound the action from below by a Gaussian whose determinant is the product over the inner Neumann momenta, the blocks having a side of the order $\beta^{1/6}$.

Step 7: the window (B(1)). The first sentence of B(1) is A(1) and A(2) at $N_\mu = s_\mu/a_K$, $\beta = L^K b$, with $|s|p_\infty = |N|P$; $\delta_K^0 \ge 0$ is the definition of P. On the cubes $2L^m$, $m \ge m_0(b)$, $b \ge \gamma^{-2}$, Lem. 11.1 places $\log Z_K[0]$ within $C_*(b)\ell^3$ of a vacuum energy whose density varies boundedly from cube to cube, no constant depending on K or m; the monotone density squeezes a box with sides

$\geq 2L^{m_0(b)}$ between two cubes, so $|\log Z_K[0](s) - \epsilon_K|s|| \leq C_*(b)|s|$ and $p_\infty \geq \epsilon_K - C_*(b)$, whence $\delta^0_K \leq 2C_*(b)|s|$ and, on the cubes, the explicit bound. For a compact set $\mathsf{J} \subset [b_-, b_+]$ of labels and $s_{\min} > 0$, all labels of J and all sides $\geq s_{\min}$ are brought into this range at once by

$$Z_K[n](s;b) = Z_{K-j}[n](L^j s; L^j b),$$

with the least j for which

$$L^j b_- \geq \gamma^{-2}$$

and

$$L^j s_{\min} \geq 2L^{m_0(L^j b_+)},$$

which exists because

$$L^{-j} 2L^{m_0(L^j b_+)} \to 0$$

: for $b \in \mathsf{J}$, $L^j b \geq \gamma^{-2}$ and, m_0 not decreasing, $L^j s_{\min} \geq 2L^{m_0(L^j b)}$; and, C_* not decreasing, $C_*(L^j b) \leq C_*(L^j b_+)$, so that $C = L^{3j} C_*(L^j b_+)$ serves for all $b \in \mathsf{J}$ and $K > j$. The bound on

$$\operatorname{Tr} T_m^{t/a_K} / \|T_0\|^{t/a_K}$$

follows from $\operatorname{Tr} T_1^M \leq \operatorname{Tr} T_0^M$ and $\mathsf{G} \leq \delta$. A lattice with N_μ/β in a compact set and β large is a box of a cutoff K at a label in $[1, L)$ with sides in a compact set, where these bounds hold uniformly, by the above at $\mathsf{J} = [1, L]$; for bounded β, $\delta^0 \leq 6\beta|N|$. On a box with L-adic sides the remaining cutoffs are finitely many, each with a finite δ^0_K and $r_K[n] > 0$, and $\delta^n_K - \log r_K[n] = \delta^0_K$ is bounded.

Step 8: floor and two cutoffs on mixed boxes (B(2)(a), B(4)). After a dilation, s is a centre sector of some T_ℓ (Lem. 11.9), and the chain of T_ℓ restricted to the fixed fields is a chain for $Z^z_K(s;b)$. At the level i the fluctuation integral over the fixed fields is, by Prop. 11.10 in the coordinates (requiring **(Hess)** for the full fluctuation Hamiltonian, and with least displacement $s_{\min} L^{K-i}$ in the units of that level), the n-th root of that of the cover up to $\epsilon_i \leq C|s|L^{3(K-i)} e^{-c s_{\min} L^{K-i}}$. With $E(d) = \sum_{i \leq K-d} \epsilon_i$, bounded independently of K, Σ the small-field mass of the cover at the depth d and C_0 formed from the constants,

$$\log Z^z_K(s;b) \geq \frac{\log \Sigma}{n} - E(d) - |s| L^{3d} [C_0 + \tfrac{9}{2} \log(L^d b)], \qquad \Sigma \geq \tfrac{1}{2} Z_{K,\ell}, \qquad \frac{|s|}{\ell^3} \log Z_{K,\ell} \geq |s|\, p_\infty(K, b),$$

the second when the large-field part of the cover is at most $\frac{1}{2}$, the third by A(2); finitely many K remain, at which the integrand is positive. Against the small periodic cube $\ell_\flat(b)$, box and cube being two sectors of one torus, the same chain gives the bound with C_G: the cover drops out. For two cutoffs, the small-field masses at K and $K+1$ are compared pointwise at fixed fields as on T_ℓ, the sector taking $|s|/\ell^3$ times the error of the cover; large fields are rare in the box, since an event of a slab passes by the Cauchy–Schwarz inequality at mirrors, with the power $\frac{1}{8}$, to the small periodic cube $\ell_\flat(b)$, where Prop. 4.54 controls it; box and cube are two sectors of one torus, whose partition function cancels.

The rate $L^{-K/10}$ and the uniformity in the label are those of the cover, with the arithmetic of the constants, which makes the condition on K void for $|s| \leq (\log b)^{2\mathfrak{p}_0 - 1}/C_V$. For two classes the terms $\frac{|s|}{\ell^3}\Delta_K$ cancel in the difference of the two laws, and the increments are summable. So, for a compact set J of labels, B(4) at J for z and for the class 0, for which s is mixed too, gives $|\log r_{K+1}[z] - \log r_K[z]| \leq 2CL^{-K/10}$ for $K \geq K_0(\mathsf{J})$; the first sentence of B(1) bounds $r_K[z]$ from below, uniformly on J, at the finitely many $K < K_0$ at which s is a box of the cutoff K; and as $\delta^0_K \geq 0$, $\log Z^z_K(s;b) - |s|\, p_\infty(K,b) = \log r_K[z] + \delta^0_K \geq \log r_K[z]$: the floor of B(2)(a) holds with $C(s,b)$ bounded on J, and $\inf_K r_K[z] > 0$ uniformly on J.

On the boxes with sides in $2L^{\mathbb{Z}}$ the left member is the increment of $\log r_K[z]$, which tends to 0 by B(2)(c) and B(3), plus the left member of the periodic class, for which every such box is mixed.

Step 9: compactness, no zero, the full sequence (B(2)(b), B(3)). By B(1) at a compact set J of labels that holds the b_j, $i\, e^{-t_0 E_i^K} \leq e^{2Ct_0 l_1 l_2}$: Lem. 11.2(b) applies, so every sequence (K_j, b_j), $b_j \to b$, has a subsequence along which all levels converge, the sector traces divided by $\|T_0\|^{t/a_K}$ tend

to $\mathcal{W}(e;m)(t;l)=\sum_i e^{-tE_i^{m,e}}$, holomorphic in $\operatorname{Re} t>0$, and $v_{(k,m)_\mu}=\sum_e(-1)^{e\cdot k}\mathcal{W}(e;m)/\mathcal{D}$, $\mathcal{D}=\sum_e\mathcal{W}(e;0)\geq 1$, in each of the three expressions. The order of A(1) and A(5)(a) pass to the limit, and so does the crossing relation, for the sector weights $w^{[\mu]}_{e,m}=\frac14\sum_k(-1)^{e\cdot k}v_{(k,m)_\mu}=\mathcal{W}(e;m)/\mathcal{D}$, each the sector trace divided by the periodic trace of its own time and section: on each lattice the relation of A(1) divided by $Z[0]$ is the same relation between the numbers $Z^{[\mu]}_{e,m}/Z[0]=\frac14\sum_k(-1)^{e\cdot k}r[(k,m)_\mu]$, which converge to the $w^{[\mu]}_{e,m}$. A weight $w=\mathcal{W}(e;m)/\mathcal{D}$ vanishes at one height if and only if at all; it is real-analytic in a side x of the section, and if it vanishes at x_0 then, for $q=L^p$ and $\mathsf{c}_q\geq\mathcal{D}(t/q;l_x)$ from B(1),

$$w_t(x)\leq\mathcal{W}(t;l_x)\leq\mathcal{W}(t/q;l_x)^q\leq\left[\mathsf{c}_q\,w_{t/q}(x)\right]^q=O(|x-x_0|^q),$$

so $w\equiv 0$ in that side, and, side after side, on all boxes: this is the alternative of B(3), which passes to the limits of Step 10, and for a class with one twisted plane, whose quotient is the sum of its four sector weights with the time normal to its plane, it holds for v_z. With the floor on the box $(1,1,1)$, which is mixed for every class, this shows that the space of $H_{l,1}$ is not $\{0\}$ and, with the floor along the normal of A(5)(c) and compactness, B(2)(b); with B(5), which makes each electric weight on a small cube at least $\frac14(1-2C_\square t)$, each electric sector is not $\{0\}$, so $v_z<1$. On boxes with sides in $L^{\mathbb{Z}}$, B(4) makes the increments of $\log r_K[z]$ and of $L\log Z_K[0](s)-\log Z_K[0](s^{\mu+})$, $s^{\mu+}$ the box with s_μ multiplied by L, summable uniformly on compacts of labels, the terms Δ_K cancelling in the second: this is (R1) of Prop. 11.12 on the ray of the boxes with sides in $L^{\mathbb{Z}}$, and Prop. 11.12 gives the convergence of all levels along the full sequence, of $r_K[z]$ on every L-adic box, of δ^0_K and of the differences of the norms.

The same holds along every sequence (K_j,b_j) of cutoffs and labels with $K_j\to\infty$ and $b_j\to b\in(0,\infty)$, in the place of the full sequence at the fixed label b; these sequences form a class that is closed under subsequences. (R1) holds along each of them: each $r_K[z](s;\cdot)$ and each difference $L\log Z_K[0](s)-\log Z_K[0](s^{\mu+})$ is continuous in the label at a fixed K, and their increments in K are summable uniformly for the labels in a compact set, so that they converge uniformly on compacts of labels to limits that are continuous in the label, and so converge along (K_j,b_j). The proof of Prop. 11.12 then runs along such a sequence as it stands: its only input, the window, is uniform for the labels in a compact set (B(1): the supremum of δ^0 over the lattices with N_μ/β in a compact set is finite), and Lem. 11.2(a), (b) and the turning of the box do not read the label.

So all levels, $r_K[z]$ on every L-adic box, δ^0_K and the differences of the norms converge along every such sequence. Two such sequences with one limit label b interlace into a third one, along which the limits exist: the limits depend on b alone. They are continuous in b: for $b_i\to b$ a choice of $K_i\to\infty$ with the values at (K_i,b_i) within $1/i$ of the limits at b_i gives a sequence of the class. And the convergence is uniform for the labels in a compact set, since otherwise a compact set of labels would hold a sequence (K_j,b_j) with $b_j\to b$ whose values stay at a fixed distance from the limits at the labels b_j, which tend to those at b, against the convergence along it. E_{mag} and E_{el} are Lem. 11.2 for the limit lists, the space of $H_{l,1}$ not being $\{0\}$.

On the cubes, $L^3\log Z_{K,\ell}-\log Z_{K,L\ell}$ is a difference of periodic numbers of two cubes in which the terms Δ_K of B(4) cancel, so $D(t)$ exists; it depends on t alone by the argument of Step 10; $\delta^0_K(\ell^{\times 3})$, telescoped over the cubes $L^r\ell$, gives $\delta(t)=\sum_r L^{-3(r+1)}D(L^rt)$; and the behaviour of D as $t\to 0$ is the asymptotics of Prop. 11.14(b), (e) for the periodic cubes, the bulk cancelling, carried to every cutoff by B(4) as in Step 11. By Thms. YM3_I:AJ and YM3_I:AD, through Thm. YM3_I:AD, the periodic quotients are normal families in the label, and B(6) gives the analyticity.

Step 10: every lattice; two blocking factors (B(2)(c), B(3)). By Lem. 11.6 and B(1) the quotients are equicontinuous in the sides for N_μ/β in a compact set; with Step 9, in its form for the sequences (K_j,b_j) with a moving label $b_j=L^{-K_j}\beta_j\to b$, this gives a continuous $V^L_z(s;b)$, the limit along every sequence with $L^{-K_j}N_j\to s$, $L^{-K_j}\beta_j\to b$, with the Lipschitz bound of B(3); in each side, at real values of the others, it is $\sum_e(-1)^{e\cdot k}\mathcal{W}(e;m)/\mathcal{D}$, a quotient of Laplace transforms, and

real-analytic. The constants may be chosen with the blocking factors $L_1 = L$ and $L_2 = L + 2$, and $\log L_1/\log L_2 \notin \mathbb{Q}$; so for $b_1, b_2 > 0$ there are $K_j, M_j \to \infty$ with $K_j \log L_1 - M_j \log L_2 \to \log(b_2/b_1)$, and $\beta_j = L_1^{K_j} b_1$, $N_j \approx \tau\beta_j$ is a sequence of both scales, of the label b_1 at the scale L_1 and of the moving label $L_2^{-M_j}\beta_j \to b_2$ at the scale L_2, to which Step 9 applies in its form for sequences:

$$V_z^{L_1}(\tau b_1; b_1) = V_z^{L_2}(\tau b_2; b_2),$$

so $V_z(s; b) = v_z(s/b)$, every sequence with $\beta_j \to \infty$, $N_j/\beta_j \to \tau$ converges ($\beta_j = L^{K_j} b_j$, $b_j \in [1, L)$), and the limits do not depend on the choice of the constants. For $s \in (2L^{\mathbb{Z}})^3$, $V_z(s; b) = V_z(\frac{1}{2}s; \frac{1}{2}b) > 0$, the box $\frac{1}{2}s$ being mixed for every class; positivity of the limit and of each r_K is the floor, uniform for the labels in a compact set, since the convergence is uniform there (Step 9), the limits are continuous and positive, and the first sentence of B(1) bounds the finitely many remaining cutoffs; and an infimum 0 over the cubes with $N/\beta \in I$, or over the lattices with N/β in a compact subset of $\{v_z > 0\}$, would give, along $\beta_j \to \infty$, a point of that set with $v_z = 0$. The same for the levels gives their extension to real sections and their homogeneity. The weights of the mixture converge with the $r_K[n]$, and $1 \le \sum_n v_n \le 8$ with $v_0 = 1 \ge v_n$, so that their limits lie in $[v_n/8, \frac{1}{2}]$ for $n \ne 0$.

Step 11: small cubes (B(5)). (a), the ceiling. By A(1), $r_2, r_3 \le r_1$, and by the symmetry of the axes it is enough to bound $r[n]$ for $n = n_{13} = ((1, 0), 0)_3$, whose twisted plane holds the time 3, on the cubes $\ell = 2L^m$. A dilation by a power of L keeps t and the lattices and brings (ℓ, b) into the range $b \ge b_0$, $MR(b) \le L^m$, $\ell \le 2ML^2(\log b)^{r_\star}$ of Lem. 11.11(b), where small t is large b. In a block layer normal to 3 of the block field of depth one, with $\bar{h}_1(y)$, $\bar{h}_2(y)$ the products of the block field along the straight closed block lines from y in the directions 1 and 2 (as in §11.1.5), let $\bar{s}_1$ and $\bar{s}_+$ be the means over the sites y of the layer of $\frac{1}{2}\mathrm{tr}\,\bar{h}_1(y)$ and of $\frac{1}{2}\mathrm{tr}[\bar{h}_1(y)\bar{h}_2(y)]$. The flip of n changes the signs of both; on a flat field with the holonomies $e^{i\varphi_\mu \mathsf{a}\cdot\sigma}$ they are $\cos\varphi_1$ and $\cos(\varphi_1 + \varphi_2)$, which vanish together exactly at the fixed points $\varphi_1 = \pm\frac{\pi}{2}$, $\varphi_2 \in \{0, \pi\}$ of the flip $\varphi_1 \mapsto \varphi_1 + \pi$, taken up to $\varphi \mapsto -\varphi$. Let

$$F = \mathbf{1}\{\max(|\bar{s}_1|, |\bar{s}_+|) < m_0 t^{1/2}\},$$

with an absolute m_0, be read at the block layer next to a block mirror normal to 3 at the cutoffs $K \ge K_\sharp$ of Lem. 11.11(b), and at the bare layer next to a mirror between two layers below $K_\sharp$; F is gauge invariant, Borel with values in $[0, 1]$ and flip-blind, and its mirror image θF is F read at the layer on the other side of the mirror. Establishing **(Flux-window)** would give, with absolute C_2 and ϵ_F, at every cutoff and with the columns of n away from the layers of F and θF: **(F1)** $\mu(F) \le C_2 t$, and **(F2)** $Z[n; 1 - F] \le \epsilon_F\, t\, Z[0]$.

Both estimates remain to be established in **(Flux-window)**. In particular, (F1) is a bound for a shrinking event of the block means, and does not follow from the uninserted periodic Laplace estimate of Prop. 11.14. The sign reversal of the plane means under the twist likewise does not by itself establish (F2).

Now

$$r[n] = \frac{Z[n; \theta F\, F]}{Z[0]} + \frac{Z[n; 1 - \theta F\, F]}{Z[0]}.$$

The first term is bounded through the overlap representation of the class with flux, in the form that A(6) takes for a function of a slab and its mirror image, the trace formula of Step 4 with α, α' as there: $Z[n; \theta F\, F] = \sum_e (-1)^{e\cdot k} \|T_0^{\alpha/2} \hat{F}\, T_0^{\alpha'/2} P_e\|_2^2 \le Z[0; \theta F\, F] \le Z[0; F] = \mu(F)\, Z[0]$. In the second, $1 - \theta F\, F \le (1 - F) + (1 - \theta F)$, and $Z[n; 1 - \theta F] = Z[n; 1 - F]$ by the reflection, the columns being moved away from the layers of F (Step 1). So $r_1 \le (C_2 + 2\epsilon_F)\, t$ at every cutoff, with the absolute $C_\square = C_2 + 2\epsilon_F$; t_0 and both bounds are as specified in (Flux-window). (b) and (c): on the odd cube $s = \frac{1}{2}\ell_\sharp(b) = LMR(b)$, $b \ge b_\sharp$, $S = s$, B(4) holds uniformly in the label from $K_0 = \lceil c_2 \log_L S \rceil$ on, $c_2 = 20\varpi$: its second condition is void on this cube, since $|s| \le L^6 M^3 (\log b)^{3r_\star} \le (\log b)^{2\mathfrak{p}_0 - 1}/C_V$ for large b, as $2\mathfrak{p}_0 > 3r_\star + 1$, so that the low cutoff of B(4) is K_0.

At K_0 the lattice has $N_0 = SL^{K_0} \in [S^{1+c_2}, LS^{1+c_2}]$ sites a side, so $N_0 \le C(\log b)^{r_\star(1+c_2)}$, at $\beta_0 = bL^{K_0} \ge b$; this lattice and those of the cutoffs $K < K_0$, which are smaller at $\beta \ge b$, lie in the range of Prop. 11.14 for $c < 1/(r_\star(1+c_2))$ and large b. From K_0 on, B(4), uniformly in the label, carries the lattice value to every cutoff with a relative error $o(1)$ as $b \to \infty$, the sum of $c_1 S^{\varpi} L^{-K/10}$ over $K \ge K_0$ being at most a multiple of $S^{\varpi - c_2/10} = S^{-\varpi}$; this is (c), the values $t = LMR(b)/b$ filling $(0, t_0]$; v_z depends on t alone (B(3)), which gives $C^{-1}t \le v_z \le Ct$ and, with Prop. 11.14(e), $v_z(t,t,t)/t \to \mathscr{G}_z$.

The ratios are Prop. 11.14(c), (e) for two classes, the periodic denominators cancelling. (d): with the ceiling, $r_1 \le \frac{1}{5}$ on the cubes $2L^m$ with $t \le t_1 = \min\{t_0, 1/5C_\square\}$ at every cutoff, and A(13) applies; $\delta_K^0 \ge \mathsf{G} \ge \log(4/(1+3r_1)) \ge \log(4/(1+3C_\square t))$.

Step 12: the label (B(6)). A source is a complex function h on T_ℓ in the ball of the norm of Thm. YM3_I:AD (the source w there); for a class n, with $x(p)$ the barycentre of the plaquette p, put

$$Z_K[n;\mathsf{h}] = \int \exp\Big[-\sum_p L^K\big(b_0 - \mathsf{h}(x(p))\big)\big(1 - \tfrac{1}{2}\eta^n(p)\,\mathrm{tr}\,\partial U(p)\big)\Big]\,\mathrm{d}U;$$

at $n = 0$ this is the left member of the identity of Thm. YM3_I:AD at $\beta = L^K b_0$, and at a constant source $\mathsf{h} \equiv c$ it is $Z_K[n]$ at the label $b_0 - c$. Let n have one or two twisted planes, μ a direction that lies in each of them, and θ the reflection $x_\mu \mapsto -x_\mu$, with the mirror planes $x_\mu \in \{0, \frac{\ell}{2}\}$, or $x_\mu \mapsto \frac{\ell}{2} - x_\mu$, with the mirror planes $x_\mu \in \{\frac{\ell}{4}, \frac{3\ell}{4}\}$; the two mirror planes cut T_ℓ into the closed halves H_- and $H_+ = \theta H_-$. Let h be symmetric under θ and real within $\frac{\ell}{16}$ of the two mirror planes, and let its conjugated fold $\mathsf{h}^\flat$ be h on H_- and $\bar{\mathsf{h}}$ on H_+.

With μ as the time and the mirror layers $x_\mu \in \{0, \frac{\ell}{2}\}$, $Z_K[n;\mathsf{h}] = \iint \Psi_-(V,V')\,\Psi_+(V, c^k V')\,\mathrm{d}V\,\mathrm{d}V'$, where V, V' are the fields of the two mirror layers, on whose plaquettes the weights $L^K(b_0 - \mathsf{h}(x(p)))$ are real and positive and give the square roots of their factors to each half, $\Psi_\mp$ are the integrals over the bonds inside $H_\mp$, and c^k is the flip of the electric part of n, whose magnetic part vanishes because μ lies in every twisted plane. The reflection θ keeps the Haar measure and the traces of the plaquettes, so the symmetry of h gives $\Psi_+ = \Psi_-$, while $\overline{\Psi_-}$ is the integral over H_- at the source $\bar{\mathsf{h}}$, that is, by θ, over H_+ at $\bar{\mathsf{h}}$; the Cauchy–Schwarz inequality in the fields of the mirror layers, c^k keeping their Haar measure, gives

$$|Z_K[n;\mathsf{h}]| \le \|\Psi_-\|_2^2 = Z_K[0;\mathsf{h}^\flat].$$

At mirrors midway between two layers the same holds, the crossing kernel at the mirror being positive semidefinite at real positive weights (Step 1). For three twisted planes the same with the time 3, which lies in the planes 13 and 23 and whose section is twisted, gives $|Z_K[n;\mathsf{h}]| \le Z_K[n_{12};\mathsf{h}^\flat]$, the right member being ≥ 0, and a second fold with the time 1, which lies in the plane 12, at mirrors normal to the direction 1 at which the source is symmetric and real as well, gives $Z_K[n_{12};\mathsf{h}^\flat] \le Z_K[0;\mathsf{h}^{\flat\flat}]$, $\mathsf{h}^{\flat\flat}$ being h or $\bar{\mathsf{h}}$ according to the product of the signs of the two halves. The identity of Thm. YM3_I:AD writes $Z_K[0;\mathsf{h}^\flat]/|Z_K[0;\mathsf{h}]|$ as the exponential of the sum of the differences of the real parts of the local energies $e_j(X;\cdot)$ at $\mathsf{h}^\flat$ and at h, times the quotient of the moduli of the two remainders $\widetilde{Z}$.

An energy at $\bar{\mathsf{h}}$ is the conjugate of the energy at h, and it reads the source near its domain; so the real parts agree on every localization domain that does not meet both $H_- \cap \{\mathrm{Im}\,\mathsf{h} \ne 0\}$ and $H_+ \cap \{\mathrm{Im}\,\mathsf{h} \ne 0\}$, two sets at a distance $\ge \frac{\ell}{8}$. A domain of the level j that meets both has an extent of at least $\ell L^{K-j}/8M - 4$ ([7, eq. 321]); the number of the localization domains ([7, Lem. O.1]) and the decay of the energies in the extent bound their sum by a number that does not depend on K, and with the bounds on $\widetilde{Z}$, which also give $Z_K[0;\mathsf{h}] \ne 0$ for small sources,

$$|Z_K[n;\mathsf{h}]| \le Z_K[0;\mathsf{h}^\flat] \le \mathsf{M}_\mathrm{S}|Z_K[0;\mathsf{h}]|, \qquad Z_K[0;\mathsf{h}] \ne 0,$$

and the same with $\mathsf{h}^{\flat\flat}$ for three twisted planes. For the constant source $b_0 \mapsto b_0 - c$, written as $c\chi_1 + c\chi_2$ with a partition of unity $\chi_1 + \chi_2 = 1$ in x_μ, χ_1 vanishing within $\frac{\ell}{16}$ of the planes $x_\mu \in \{0, \frac{\ell}{2}\}$ and χ_2 within $\frac{\ell}{16}$ of the planes $x_\mu \in \{\frac{\ell}{4}, \frac{3\ell}{4}\}$, the quotient $r_K = Z_K[n;\cdot]/Z_K[0;\cdot]$ at the source $\zeta_1\chi_1 + \zeta_2\chi_2$, in which one of the two parts may be complex while the other is real, is bounded by M_S on the two crosses of a bidisc in (ζ_1, ζ_2), and the following lemma bounds it on a smaller bidisc. If F is holomorphic near $\bar D_R^2$ and $|F| \le \mathsf{M}$ wherever one variable lies in $\bar D_R$ and the other in $[-R, R]$, then $|F| \le 27\,\mathsf{M}$ on $\bar D_{R/8}^2$; if F is holomorphic near $\bar D_R^4$ and $|F| \le \mathsf{M}$ wherever one variable lies in $\bar D_R$ and the three others in $[-R, R]$, then $|F| \le 12600\,\mathsf{M}$ on $\bar D_{R/15}^4$ (this form serves for three twisted planes).

For the proof put $\rho_\circ = 2 + \sqrt{3}$ and $\mathsf{j}(u) = \frac{R}{4}(u + u^{-1})$, which maps the annulus $\rho_\circ^{-1} \le |u| \le \rho_\circ$ into $\bar D_R$ and the unit circle onto $[-\frac{R}{2}, \frac{R}{2}]$. Moving the circle of one variable of the Cauchy integral to $|u| = \rho_\circ^{\pm 1}$, the other variables staying on the unit circle, where j is real, bounds the Laurent coefficients of $F \circ \mathsf{j}$ by $\mathsf{M}\rho_\circ^{-\max_i |k_i|}$; summing them on the annuli $1.3^{-1} \le |u_i| \le 1.3$, whose images hold $\bar D_{R/8}$, gives at most $26.2\,\mathsf{M}$, and on $1.15^{-1} \le |u_i| \le 1.15$, whose images hold $\bar D_{R/15}$, at most $12563\,\mathsf{M}$ in four variables. This bounds r_K on a disc whose radius does not depend on K, and Cauchy's estimate gives the derivatives.

In t this is the normal family of the ϕ_j, and Vitali's theorem gives the analyticity of the limit. With $m_0 = \inf_K r_K(b_0) > 0$, Cauchy's estimate keeps r_K away from 0 on a smaller disc, which, $Z[0]$ having no zero there, gives the zero-free discs and Ω; the first two derivatives of $\log r_K$ in the label are differences of means and of variances of the total action, bounded by Cauchy's estimate.

Step 13: block pairs (B(7)). A block mirror lies midway between two layers, L being odd; for G bounded Borel in the block pairs on one side, $(U, s) \mapsto G(\mathcal{M}_s^{K-d} U, s^{(d)})$ depends on the bonds and signs on that side, and the reflection commutes with $\mathcal{M}_s$: the clause on pairs in A(8) gives the positivity, which weak limits keep; at each cutoff $\mathcal{M}_s^{K-d} = \mathcal{M}_s \circ \mathcal{M}_s^{K-d-1}$ makes the tower consistent. Under the conditions of B(7), the small-field chain among the construction results on the periodic torus, with the rarity of its large-field regions (Prop. 4.54), bounds the periodic probability of a rough block plaquette of some depth $d' \in [d, K]$ by $\mathsf{R}(d)$, uniformly in K; the dilation carries this to the other pairs (ℓ, b), the depth d becoming $d - j$.

With γ a chart of s near a block plaquette P, the sign of P in the block sign field times $\partial(\mathcal{M}_s^k U)(P)$ is $\partial(\mathcal{M}^k(\gamma U))(P)$: the event of a rough plaquette is covariant and local, and off it the signs of the block traces are the block sign field. In $\mathbb{Z}/2\lambda$, λ odd, every arc of at most $\frac{1}{2}(\lambda + 1)$ points lies in one of four arcs of λ points, so each local event lies in one of sixteen quarters of the torus cut by mirrors in two directions; the quarter bound of A(8) carries the union of the events of one quarter into the class with the root $\frac{1}{4}$, which gives $16\mathsf{R}(d)^{1/4}$, and the bound for the mixtures.

In a weak limit the class is then, up to null sets, the class of the block sign fields of the tower. For the other statements on limit points pass to a subsequence along which the laws of the towers converge in every class. The quarter bound of A(8) at each cutoff, $r_K[n]\mu_n(B) \le \mu_d^K(B_1)^{1/4}$, and the convergence in total variation of μ_d^K to μ_d (through Thm. YM3_I:D) give $r_n \mathbb{P}^n(B) \le \mu_d(B_1)^{1/4}$ for open B by the portmanteau theorem, and for Borel B by outer regularity, the group of the sign changes being finite; the half bound of A(8) gives the exponent $\frac{1}{2}$. For the consistency of the limit towers the identity at each cutoff is joined to Lusin's theorem, $\mathcal{M}_s$ being a Borel map and not continuous.

The relation between the block sign fields is a closed condition and passes to the limit. For the fields fix a class with $r_n > 0$, a depth $d \ge 1$, a block bond c of depth d and $\epsilon > 0$; $\mathcal{M}(\cdot)(c)$ is a Borel function of the block field of depth $d+1$ near c, and by Lusin's theorem it is continuous on a compact set $\mathcal{K}$ of these local fields of μ_{d+1}-measure at least $1 - \epsilon$. The towers with $V_d(c) = (\mathcal{M}_{s_{d+1}} V_{d+1})(c)$ whose local field near c, read in a chart of s_{d+1}, lies in $\mathcal{K}$ form a closed set; at each cutoff it holds

the tower off a covariant local event whose probability under μ_n is at most $r_K[n]^{-1}\mu_{d+1}^K(\mathcal{K}^c)^{1/4}$ by the quarter bound.

By the portmanteau theorem and the convergence of μ_{d+1}^K in total variation, $\mathbb{P}^n$ gives this closed set a probability at least $1-\epsilon^{1/4}/r_n$, and $\epsilon\to 0$ gives the consistency at c, almost surely; there are finitely many c and countably many d.

Step 14: tension of a slab (B(8)). On one lattice, A(4)(a) at the heights and lengths N/L and N, $N=\ell/a_K$, with $\mathsf{G},\Phi\le\delta$ and the last inequality of A(2), $\delta(N/L,N,N)\le L^2\delta((N/L)^{\times 3})$ and $\delta(N/L,N/L,N)\le L\delta((N/L)^{\times 3})$, gives the two inequalities in lattice units, the three terms of the bound on σ adding up to $(L/(L-1))^2L^3\delta((N/L)^{\times 3})$ exactly; in physical units a product of two lattice lengths carries L^{2K}. The right members are bounded in K: δ_K^0 by B(1), and $\kappa_K\to\log(2/(1-v_1(t,t,t)))<\infty$ by B(3), since $v_1<1$ on the cubes with L-adic sides. In the limit the least level of the sector $e=10$ is $\varepsilon_{\mathrm{el}}$ (A(1)) in physical units. The lower bound on σ_K is A(3) in physical units, and the bound on $-\log\mathfrak{b}_K$ is the last bound of A(3) with $R_K(\lambda_0)\le\mathsf{G}_K(\lambda_0;l,2\lambda_0)/\lambda_0\le\delta_K^0(\lambda_0,l,2\lambda_0)/\lambda_0\le(2\lambda_0 l/s'^3)\bar\delta(s')$ by A(4)(b) and A(2), and $\Phi_K(t_0,\lambda_0)\le(t_0\lambda_0 l/s'^3)\bar\delta(s')$; the equivalence follows by choosing first λ, then t, large.

Step 15: the small twisted section (B(9)). The levels are read in the normalisation of B(9): they are the numbers $a_K^{-1}\log(\lambda_0/\lambda_i)$, $\lambda_0=\|T_1\|$, the four sectors together, so that the least of them, the vacuum, is $E_0=0$; they are the levels of the setting diminished by $a_K^{-1}\log(\|T_0\|/\|T_1\|)$, and their limits are the eigenvalues of $H_{(l,l),1}$ diminished by $E_{\mathrm{mag}}(l,l)$. Over the twisted section (N,N), $N=l/a_K$, at the time 1, $\operatorname{Tr}T_1^h=Z[n_{23}](h,N,N)$; with $x(h)=\sum_{i\ge1}(\lambda_i/\lambda_0)^h$, in

$$L\log\operatorname{Tr}T_1^h-\log\operatorname{Tr}T_1^{Lh}=L\log(1+x(h))-\log(1+x(Lh))$$

the bulk and all that is proportional to the time cancel. The chain that bounds it has these links. (i) A dilation by a power of L keeps t and the lattices and takes (l,b) to (l',b') with $l'=\frac12\ell_\sharp(b')=LMR(b')\ge\ell_\flat(b')$, the values $LMR(b')/b'$ filling $(0,t_0]$ (Step 11); small t is large b'. (ii) The odd boxes (l',l',l'), (Ll',l',l'), (L^2l',l',l') and, for each fixed $j\ge1$, (L^jl',l',l') have their sides in $L^{\mathbb{N}}$ and are mixed for n_{23}; on them B(4) holds uniformly in the label from its low cutoff $\lceil c_2\log_L S\rceil$ on, its second condition being void for large b', since $|s|\le L^{j+6}M^3(\log b')^{3r_\star}\le(\log b')^{2\mathfrak{p}_0-1}/C_V$ as $2\mathfrak{p}_0>3r_\star+1$.

In the left member of the display at $h=N$, which in the dilated units is $L\log Z[n_{23}](l'^{\times 3})-\log Z[n_{23}](Ll',l',l')$, and at $h=LN$, the terms $\frac{|s|}{\ell^3}\Delta_K$ of B(4) cancel, so that B(4) carries its value at the low cutoff to every cutoff with an error summable in K. (iii) At and below the low cutoff the lattices lie in the range of the twisted clause of Prop. 11.14(a), whose volume terms cancel in the display and whose Gaussian term gives the free value: with θ^a the phases of the three colours a of n_{23} and the momenta $p\in((2\pi\mathbb{Z}+\theta_2^a)/N)\times((2\pi\mathbb{Z}+\theta_3^a)/N)$ mod 2π, none of them 0, $\prod_{j<h}(2\cosh\omega-2\cos\frac{2\pi j}{h})=2\cosh h\omega-2$ gives $\log\mathscr{D}^{n_{23}}(h,N,N)=-hE_{\mathrm{free}}+\log(1+x_{\mathrm{free}}(h))$, $E_{\mathrm{free}}=\frac12\sum_{a,p}\omega(p)$, with

$$1+x_{\mathrm{free}}(h)=\prod_{a,p}(1-e^{-h\omega(p)})^{-1},\qquad 2\cosh\omega(p)-2=\hat p^2,$$

and the free value of the display is that of x_{free}. (iv) With $L\ge13$ of the setting, the free value at $h=N$ is at most $0.26(L-1)$: at $N=L=13$ it is 3.0917 against $0.26(L-1)=3.12$, the margin 0.028; at $h=LN$ it is about $4Le^{-\pi L}$. (v) c_0, formed from the constants, is chosen so that for $t\le c_0$ the errors of Laplace's method and of B(4) stay below that margin at $h=N$, and are small enough against the free value at $h=LN$ that $x(LN)\ge0.29693\,e^{-\pi L}$ follows. Then at every cutoff the display at $h=N$ is at most $0.26(L-1)$; as x does not increase in h, $(L-1)\log(1+x(N))\le0.26(L-1)$, so $x(N)\le e^{0.26}-1=0.29693$ and $N\log(\lambda_0/\lambda_1)\ge\log(1/0.29693)\ge1.2142$; at $h=LN$, with this bound on $x(N)$ and $x(LN)\le(\lambda_1/\lambda_0)^{(L-1)N}x(N)$, $N\log(\lambda_0/\lambda_1)\le\pi L/(L-1)$.

The sharper bounds use the same comparison at smaller t with the free values over the twisted section, whose least excitation is ω_N/N with $3.1264 \le \omega_N < \pi$ for $N \ge 13$. Likewise B(4) on the odd boxes $(L^j l', l', l')$, $j \ge 1$, carries the quotients $Z[(k,1)_1]/Z[(0,1)_1]$ to their free values; the sector traces $\frac{1}{4}\sum_k (-1)^{e\cdot k} Z[(k,1)_1]$ at the heights $L^j l$, which are $L^j l'$ in the dilated units, fix the levels (Lem. 11.2); in the normalisation of B(9) the least free level of the sector $e \neq 00$ is $\pi|e|/l$, its momenta lying in $\frac{2\pi}{l}(\mathbb{Z} + \frac{e_\rho}{2}) \times \frac{2\pi}{l}(\mathbb{Z} + \frac{e_\nu}{2})$ by A(1), while in the sector 00, whose least level is the vacuum $E_0 = 0$, the least positive free level, the second level of 00 minus the vacuum, is $2\pi/l$.

For (c), the state of the cylinder has $\omega_K(\overline{\theta_K F}\, \tau_{n'} G) = \langle \psi(G), (T_1/\lambda_0)^{n'} \psi(F) \rangle$ for plaquette functions of finitely many layers, with vectors $\psi(F)$ formed from the slab operators and the top eigenvector of T_1. Lem. 11.8(b) with cells of side l/L, that is $\lambda' = L^{J-1}$ sites and $k = L$ cells a side of the section, and with $\mathsf{q} = 2$, compares the cylinder with the periodic cube Q of side $2l/L$, at the cost of one factor Ξ for each slab of cells in time that carries a function; at $k_0 = L$, $\Xi \le \exp[\Phi'/(L-2)]$, and $\Phi' = \frac{L^3}{8}\delta^0_K(Q) - \delta^{n_{23}}_K(l^{\times 3})$, the bulk cancelling, is bounded uniformly in K by the window of B(1) on the cube $2l/L$ and the floor of B(2) on the odd cube l.

A partition of unity of each f_q into pieces supported inside cells of translates of the board cuts a tuple into finitely many tuples placed in cells, at a cost formed from n', R and l; for a placed tuple the product of the centred observables of one cell is a polynomial in the numbers $1 - \frac{1}{2}\mathrm{tr}\, \partial U(p)$ of its plaquettes, and its seminorm on Q is a centred moment of the observables of the eight mirror copies in the periodic measure of Q, the means e_K being those of Q; the bounds on the joint cumulants of the action density on that periodic cube ([7, Lem. N.2]) then give $C_{n'}(d, R)$. Compactness (Lem. 11.2(b) for the levels, a diagonal choice for the $S_{n'}$) gives the limit points; reflection positivity is A(7) at the mirrors normal to the time, which lies in no twisted plane of n_{23}.

In every limit, $S(\theta F, \tau_s G) = \sum_i e^{-sE_i} c_i(F, G)$ with a positive semidefinite matrix of coefficients, which defines an isometry of $\mathcal{H}$ into the ℓ^2 space of the limit levels carrying e^{-sH} to multiplication by e^{-sE_i}; $E_0 = 0$ is simple and belongs to Ω, and the gap gives the spectral bound and the clustering. The action density is invariant under the flips C^k, so its vectors lie in the sector 00, whose least positive level exceeds $6.1/l$ at small t when the law of (4) is used on the boxes $(L^3 l, l, l)$. Finally the lower bound on $S_2(\theta f^\circ, f^\circ) - S_1(f^\circ)^2$. Let $j(t)$ be the least integer for which $g = (L^{j(t)}/2t)^{-1/2}$ is at most the ceiling γ_W of the reference couplings of Thm. YM3_I:AV and $2L^{j(t)-1}$ is a side at g (equation YM3_I:(1528)); it is formed from t and the constants.

Put $\hat{L} = L^{j(t)-m}$ and read everything in the units dilated by $\hat{L}$, which name the same lattice and the same β: the section has the side $\hat{l} = \hat{L} l = L^{j(t)}$, the label is $\hat{b} = \hat{L} b = L^{j(t)}/t = 2g^{-2}$, the cutoff is $\hat{K} = K - j(t) + m$, the bump has the radius $d_* = \hat{L} d^\circ$, and the partner of the cylinder is the periodic cube of side $2\hat{l}/L$, which is the cube Q of side $2l/L$ above; by Cor. YM3_I:74.2 and [7, Lem. G.4] every $(\hat{K}, L^{\hat{K}}\hat{b})$, $\hat{K} \ge 1$, is a first-passage datum at g with the label $\hat{b}$. (1) The state of the cylinder defines a law $\tilde{\omega}$ on the configurations of $a_{\hat{K}}\mathbb{Z}^3$, carried by the fields that are periodic up to the twist in the two directions of the section, with $\tilde{\omega}(G) = \omega_K(G^{\mathrm{per}})$ for the continuous plaquette functions G of finitely many plaquettes, G^{per} reading each plaquette at its class modulo the section; on these functions $\tilde{\omega}$ is invariant under the translations of $a_{\hat{K}}\mathbb{Z}^3$. (2) Let F_1, F_2 be Lipschitz functions supported in one ball of radius 2 whose centre lies at the distance ≥ 3 from the walls of a board of side L, and $\mathsf{u} = \zeta_1 F_1 + \zeta_2 F_2$, $\zeta \in \mathbb{C}^2$.

Lem. 11.8(b), with cells of side L, that is of $L^{\hat{K}+1}$ sites a side, one slab of cells in time that carries a function, and the same partner Q, gives $|\tilde{\omega}(e^{\mathcal{O}_{\hat{K}}(\mathsf{u})})| \le \hat{\Xi}\, \|e^{\mathcal{O}_{\hat{K}}(\mathsf{u})}\|_{\mathrm{ch};Q}$, where $\hat{\Xi} = \Xi^{L^{2-j(t)}} \le \Xi$, because at a fixed partner $\log \Xi$ is linear in the number of sites of a side of a cell; so $\hat{\Xi}$ is bounded uniformly in K by a number Ξ_* formed from t and the constants. (3) By the normalisers among the construction results at the first-passage data (Prop. YM3_I:91.12 (1)) there is a function $\mathcal{E}$, holomorphic in ζ on the domain $\mathbb{D}_\varrho(F_1, F_2)$ among the construction results and with $\mathcal{E}(0) = 0$, with $\|e^{\mathcal{O}_{\hat{K}}(\mathsf{u})}\|_{\mathrm{ch};\ell''} \le |e^{\mathcal{E}(\zeta)}|\, \mathsf{Q}_1(L)$ for all sides ℓ'' at g, the side $2\hat{l}/L$ among them; so $|e^{-\mathcal{E}(\zeta)}\mathfrak{s}(e^{\mathcal{O}_{\hat{K}}(\mathsf{u})})| \le$

$Q' = \Xi_* \mathsf{Q}_1(L)$ for $\mathfrak{s} = \tilde{\omega}$ and, by the transfer principle (Prop. YM3_I:91.11), for the periodic state $\tilde{\mu}$ of Q, read as a law on $a_{\hat{K}}\mathbb{Z}^3$. (4) The transfer between two laws under one normaliser (Prop. YM3_I:91.11) gives $|S_2^T[\tilde{\omega}](F_1,F_2) - S_2^T[\tilde{\mu}](F_1,F_2)| \le C_D \|F_1\|_\sharp \|F_2\|_\sharp$, $C_D = 2\log 2\,(2Q'+2)^2(2/\varrho)^2$, ϱ the radius of that domain, with no separation of the supports. (5) Put

$$D(f,h) = S_2^T[\tilde{\omega}](f,h) - S_2^T[\tilde{\mu}](f,h).$$

This is a bilinear form invariant under lattice translations, by (1) and the translation invariance of the periodic law. We apply the packing argument in the proof of Prop. YM3_I:127.1; only these two properties and the unit-ball bound in (4) enter that argument. In particular, it does not require the cylinder law to be a periodic Gibbs state.

Let $r = 5\mathsf{d}$, with $a_{\hat{K}} \le r \le 1/20$, and suppose that f_1, f_2 vanish outside one ball of radius r. Set $\delta = a_{\hat{K}}\lceil 2r/a_{\hat{K}}\rceil$, so that $2r \le \delta < 3r$. Translates of this ball whose centres have all coordinates of modulus at most $0.575 - r$ lie in the unit ball and have disjoint interiors when their translation vectors lie in $\delta\mathbb{Z}^3$. Each coordinate interval contains at least $0.3/r$ points of the required translate of $\delta\mathbb{Z}$. We thus obtain a set P of lattice translations with $N = \#P \ge (0.3/r)^3$. For independent signs $\epsilon_v = \pm 1$ put

$$F_{i,\epsilon} = \sum_{v\in P} \epsilon_v \tau_v f_i, \qquad i = 1,2.$$

Their supremum and Lipschitz norms do not exceed those of f_i. For the Lipschitz assertion, each function vanishes on the boundary of its supporting ball; on a segment joining different balls, bound the endpoint values by the distances to the respective exit and entry points. The sum of these distances is no greater than the length of the segment. Consequently $\|F_{i,\epsilon}\|_\sharp \le \|f_i\|_\sharp$. Bilinearity and sign averaging give

$$\mathbb{E} D(F_{1,\epsilon}, F_{2,\epsilon}) = \sum_{v\in P} D(\tau_v f_1, \tau_v f_2) = N D(f_1,f_2).$$

Applying (4) and dividing by N yields $|D(f_1,f_2)| \le C_D (r/0.3)^3 \|f_1\|_\sharp \|f_2\|_\sharp$. If $\operatorname{Lip} f_i \le 2\|f_i\|_\infty/\mathsf{d}$, then $\|f_i\|_\sharp \le 80M\|f_i\|_\infty/\mathsf{d}$, and hence

$$|S_2^T[\tilde{\omega}](f_1,f_2) - S_2^T[\tilde{\mu}](f_1,f_2)| \le C_2' \mathsf{d}\, \|f_1\|_\infty \|f_2\|_\infty, \qquad C_2' = C_D (50/3)^3 (80M)^2.$$

(6) Let f_+ be the bump of radius d_* about the point $x_+ = (2d_*, \hat{L}\xi)$, the dilate of f°, and f_- its mirror image in the plane at the time $a_{\hat{K}}/2$, the bump about the mirror point x_- of x_+.

When $a_{\hat{K}} \le d_*/4$, $3d_* \le |x_+ - x_-| \le 8d_*$, both bumps vanish off one ball of radius $5d_*$ and have Lipschitz constants $\le 2/d_*$, and (f_-, f_+) is a pair of the law of two points at $\vartheta = \frac12$. For the state $\tilde{\mu}$ of the torus of side $2\hat{l}/L$, a state at g, Thm. YM3_I:AV gives, for $d_* \le \mathsf{d}_0(g, \frac12)$ and $s(d_*) < \hat{K}$, $s(\cdot)$ the witness depth (equation YM3_I:(2321)), $S_2^T[\tilde{\mu}](f_-,f_+) = c_{\mathrm{sd}} \hat{b}^{-2} \mathcal{K}_{\mathrm{free}}(1+\varepsilon')$ with $|\varepsilon'| \le \frac13$, where the free kernel of the pair has $\mathcal{K}_{\mathrm{free}} = \mathcal{K}_{\mathrm{free}}(f_-,f_+) \ge \frac14\, 10^{-6}$. (7) With $d_* = \min\{\mathsf{d}_0(g,\frac12),\, c_{\mathrm{sd}}/(24\cdot 10^6 C_2' \hat{b}^2)\}$, a number formed from t and the constants, $C_2' d_* \le \frac16 c_{\mathrm{sd}} \hat{b}^{-2} \cdot \frac14\, 10^{-6} \le \frac16 c_{\mathrm{sd}} \hat{b}^{-2} \mathcal{K}_{\mathrm{free}}$, and so, by (5) at $\mathsf{d} = d_*$,

$$S_2^T[\tilde{\omega}](f_-,f_+) \ge (1 - \tfrac13 - \tfrac16) c_{\mathrm{sd}} \hat{b}^{-2} \mathcal{K}_{\mathrm{free}} \ge \tfrac18\, 10^{-6} c_{\mathrm{sd}} \hat{b}^{-2}$$

at every cutoff with $\hat{K} > s(d_*)$ and $a_{\hat{K}} \le d_*/4$. Back in the original units, $\mathcal{O}_{\hat{K}}(f^{[\hat{L}]}) = \hat{L}^{-1} \mathcal{O}_K(f)$ for $f^{[\hat{L}]}(x) = f(x/\hat{L})$, the barycentres being multiplied by $\hat{L}$ and the factor a^{-1} divided by it; so the covariance of $\mathcal{O}_K(\theta_K f^\circ)$ and $\mathcal{O}_K(f^\circ)$ under ω_K is $\hat{L}^2$ times this one, and $\hat{L}^2 \hat{b}^{-2} = b^{-2}$: the bound $\frac18\, 10^{-6} c_{\mathrm{sd}} b^{-2}$ holds at all large cutoffs. A covariance does not see the centring by e_K, and $\theta_K f^\circ$ is the translate of θf° by a_K in the time; so the bound holds for $S_2(\theta f^\circ, f^\circ) - S_1(f^\circ)^2$ in every limit.

Step 16: even cubes (B(10)). The reduction is arithmetic in $\ell_\sharp$ and R: for $t \le t_{\mathrm{w}}$ the label $b'(t)$ exists, and the cube $2L^j$ at $\beta = 2L^j/t$ is the cube $\ell_\sharp(b') L^{j-\mu(t)}$ at $\beta = L^{j-\mu(t)} b'$. (a) follows for $N \ge N(t)$ from B(3), B(5) and $C^n(N)/N \to \mathscr{G}_n$ (Prop. 11.14(e)); and up to $K_1(b')$ the cube

$\ell_\sharp(b')$ has $N \le \ell_\sharp b'^{1/68}$ sites a side at $\beta = L^K b'$, so that $N^{66}\log\beta/\beta \le 2\ell_\sharp^{66} b'^{-3/68}\log b'$, which tends to 0 as $b' \to \infty$, and Prop. 11.14(g) gives the lattice value. Under (R), $\sum_{K\ge K_1(b)} c_1 \ell_\sharp^{\varpi} L^{-\kappa K} \le C(\log b)^{r_\star \varpi} b^{-\kappa/68} \to 0$, also under $\sup_{b'\ge b}$; summing from $K_1(b')$ gives the bound of (a) at every $N \in 2L^{\mathbb{N}}$, with its lower bound on $C^n(N)/N$ the floor ct, and the bounded variation, $\phi_{j+1} - \phi_j$ being a difference of two consecutive cutoffs on the cubes $\ell_\sharp(b'(t))L^{j-\mu(t)}$; with the one-sided form, $\log q_K \ge \log q_{K_1(b')} - o(1)$ for $K \ge K_1(b')$ gives $ct \le r_K$, and Lem. 11.13 gives the bounded variation.

The proposed transport argument still requires (W) and (Window estimates). At $K_\sharp$ the lattice lies in the range of §11.1.5: $w_{K_\sharp} \ge c_\flat t$ and $\varsigma_{K_\sharp} \ge 2^{1-L} e^{\lambda_\flat}(c_\flat t)^L$, beside which $e^{-c(\log b)^{1+\eta}}$ is $o(1)$; Lem. 11.11(b) puts w_K within $e^{\pm E}$ of $w_{K_\sharp}$ up to ϵ, which is $o(t)$, indeed $o(t^k)$ for every k, as Lem. 11.11(b) states (m_K and ϵ_K being formed from the numbers YM3_I:(2192)–(2194) and R_{t_1} of §2.3.1), and so negligible beside $w_{K_\sharp} \ge c_\flat t$; so $\log(2^{L-1}\mathfrak{q}_L(K)) \ge \lambda_\flat - \Lambda - o(1) > 0$, and A(12) gives

$$r_K[n_{13}] \ge w_K \,\frac{\mathfrak{q}_L(K) - 2^{1-L}}{1 - 2^{1-L}} \ge c' t;$$

smaller lattices lie in the window. On the box $(\ell, \ell, i\ell)$ the same holds with $2^{i-1}\mathfrak{q}_i \ge 0.42 \cdot 1.8712^i$ for F_t'. Under (W′), A(12) gives $r_K[n_{13}] \ge w_K(2\|\mathsf{P}_K\|/\mathrm{Tr}\,\mathsf{P}_K - 1) \ge w_K(e^{\gamma'} - 1)$, with $w_K \ge \frac{1}{2} c_\flat t$ for small t by Lem. 11.11(b), its ϵ being $o(t)$. With F_t'' the same gives the quotient of the class $((1,1),0)_3$, which on the cube is r_2. $0.55L - 1.1 < 0.5516L - 0.7162$, and $\frac{L}{2} < 0.6266L - 0.873$, as $L \ge 13$. In (d), the consistency is that of B(7); as $\{n(s) = n\}$ is invariant under $s \mapsto s\,\mathrm{d}\gamma$, (M) makes the unnormalised laws of each class converge, and B(2)(c) normalises them.

Step 17: Thm. AI. Under the hypothesis (H), at each $K \ge K_0$, Lem. 11.4 at $l = l_K$ gives $\theta \le 0.8901$, $\mathsf{e}(4l_K) \le \theta^4/12.9$ and $D_2(8l_K) \le 20\,\mathsf{e}(4l_K)^2 \le 0.047355$, so A(10) applies at $l_0 = 8l_K$; $m_* l_0 \ge \log \frac{1}{18\cdot 0.047355} \ge 0.1597$, and if $m_* l_0 < 3$ then $l_1 < 6/m_* \le 37.6\, l_0$ and $\sigma_* > m_*^2/6 \ge 0.00425\, l_0^{-2}$, while if $m_* l_0 \ge 3$ then $\sigma_* = m_*/l_0 \ge 3 l_0^{-2}$; with $l_0 a_K = 4s_0$ this is the stated bound in physical length. The cube and loop bounds are A(10) in physical length, with $4N/l_1 \le \ell/s_0$ and $N \ge 4l_1$ for $\ell \ge 602\, s_0$; the uniqueness of ν_K is Lem. 11.7. On a cube of side $N \ge 2l_K$, A(10) gives $\Phi \le \delta \le 0.13736\,\theta^{N/2l_K}$ and $1 - r[n] \le 0.4121\,\theta^{N/2l_K}$; so $\mathfrak{b} \le e^{0.13736} - 1 < 0.1473$ on that cube, and A(3) gives

$$\sigma(N) \ge N^{-2}\log(1/0.1473) \ge 1.9\, N^{-2}, \qquad \text{that is} \qquad \sigma_K(\ell) \ge 1.9\,\ell^{-2}.$$

For the proposed final exclusion, the following reduction requires the separate estimates (Hess) and (Flux-window). Let $s_0 \le t_1 b$. On the cube of physical side $s_0 = 2l_K a_K$, $r_1 \to v_1(s_0/b, s_0/b, s_0/b) \le C_\square t_1 \le \frac{1}{5}$ by B(3) and B(5), so $r_1 \le \frac{1}{3}$ for all large K, and A(13) gives $\mathsf{e}(l_K) \ge \log 2 > 0.069$ there, against the hypothesis (H). □

11.1.6. *Scope.* Thm. AG is a statement on one lattice, for every box and every $\beta \ge 0$, in lattice units; besides the periodic boxes it treats the infinite lattice at one β, in the last assertion of (8), in (10), under its hypothesis, one inequality between the partition functions of two periodic cubes, with its periodic Gibbs state and its loops, and in (11), for every periodic Gibbs state, and, in (4)(d), a slab with free boundaries in two directions. Thm. AH concerns the periodic boxes of a fixed physical size, at a fixed label or at labels in a compact set, at every cutoff and as $K \to \infty$, the Hamiltonians $H_{l,m}$ over fixed sections, and, in (9)(c), the cylinder over one small twisted section.

Thm. AI concerns the infinite lattice $a_K\mathbb{Z}^3$ at each cutoff $K \ge K_0$: under the hypothesis (H), the inequality $\mathsf{e}(l_K) \le 0.069$ between the partition functions of the two periodic boxes $(l_K, 2l_K, 2l_K)$ and $(2l_K)^{\times 3}$, it gives the uniqueness of the periodic Gibbs state ν_K, an area law for the loops of every rectangle with a physical tension at least $2.65 \cdot 10^{-4} s_0^{-2}$, and bounds on the cubes of every physical side $\ge s_0$, resp. $\ge 602\, s_0$, all uniform in K; it is a statement of the kind of confinement at each cutoff, conditional on the hypothesis (H). Completing (Hess) and (Flux-window) would make the final exclusion show that (H) can hold only at physical sides s_0 of the order of the label or more.

The vacuum, the gap and the clustering of Thm. AH(9) belong to one small twisted section, the uniqueness of ν_K is that of a Gibbs state of the lattice of one cutoff, and the tension $\sigma_K(l)$ of Thms. AH(8) and AI is that of a slab of fixed width l. Thm. AH(10)(b)–(d) and Thm. AI remain sketches where they use (R), (W), (W′) or (M), with the genuine box-inequality premise (H) retained for the area-law criterion. The window estimates specified in (10)(c), and (Hess) and (Flux-window) in the final small-side exclusion, are the remaining tasks; and Thm. AG(7), (8) give reflection positivity for the eight sums over sets of coordinate planes and for the admissible mixtures. A box of the class z is a centre sector of a periodic torus, that is, its fields of the class z are the fields of the torus fixed by a group generated by the shifts by the sides of the box, each followed by the multiplication by a sign field χ on the bonds with $d\chi \equiv 1$, if and only if $n_\mu n_\nu$ is even in every twisted plane $\mu\nu$ of z, n_μ being the side of the torus divided by that of the box.

Indeed, if it is such a sector, a fixed field is determined by its values at representatives of the bonds, which are free, its plaquette variables are periodic under the shifts, and on one copy of the box they are those, $\eta^z \partial U'$, of a field U' of the box; for values in the diagonal subgroup of SU(2), which holds $-\mathbf{1}$, the product of the plaquette variables over a plane $\mu\nu$ of the torus is $\mathbf{1}$, each bond entering it twice in opposite senses, and it is $(-1)^{z_{\mu\nu} n_\mu n_\nu}$, the product over the $n_\mu n_\nu$ copies of a plane of the box; the converse is the construction of Lem. 11.9, read on that torus, which asks $w_{\mu\nu} = 1$ only where n_μ is even.

On the tori $T_{\ell'}$, $\ell' = 2L^{m+i}$ with $i \geq 0$, the multiplicities of the cube $2L^m$ are L^i, odd, and a dilation by a power of L does not change them: the cube $2L^m$ with flux is a centre sector of no torus $T_{\ell'}$, and no box of a class with a twisted plane whose two sides lie in $2L^{\mathbb{Z}}$ is a centre sector of a torus $T_{\ell'}$. The cube $2L^m$ of the class z is a centre sector of the periodic tori with one doubled side in each twisted plane of z: of $(4L^m, 2L^m, 2L^m)$, the doubled side lying in the twisted plane, for one twisted plane, and in both twisted planes for two; of $(4L^m, 4L^m, 2L^m)$ or of the cube $4L^m$ for three. The chain and the two-cutoff comparison are stated on the tori T_ℓ, and Lem. 11.11 on periodic cubes.

For the pairs (U, s) of the mixture μ_{mix} of (M), let a chain be run with the averaging map $\mathcal{M}_s$ of Thm. AH(7) and with the small-field tests applied to the signed plaquette variables, $s(p)\, \partial U(p)$ on the lattice and $s^{(d)}(P)\, \partial(\mathcal{M}_s^{K-d} U)(P)$ on the block lattice of depth d. These variables, and so the small-field and the large-field regions of this chain, do not change under $(U, s) \mapsto (\gamma U, s\, \mathrm{d}\gamma)$, γ a sign field on the bonds: it multiplies $\partial U(p)$ and $s(p)$ by $d\gamma(p)$, and, at each depth, $\mathcal{M}_s^{K-d} U$ by the sign field $\bar{\gamma}$ on the block bonds, $\bar{\gamma}(c)$ the product of γ over the straight path of the block bond c, and $s^{(d)}$ by $d\bar{\gamma}$.

When the small-field region of this chain does not wind about the torus, the closed sign field s is $d\gamma$ on it for a sign field γ on the bonds, and the change of variables $(U, s) \mapsto (\gamma U, s\, \mathrm{d}\gamma)$, under which $A_{s\, \mathrm{d}\gamma}(\gamma U) = A_s(U)$ and which the functions H of (M) do not see, makes the integrand on that region the periodic one; $s\, \mathrm{d}\gamma$ is then a closed sign field of the class $n(s)$ that is 1 on that region, and so, for $n(s) \neq 0, -1$ on some plaquette of the large-field region of this chain (the probability estimate for such large-field events is Prop. 4.54).

12. Vanishing latent heat

Let $P(\beta)$ be the pressure per lattice site and let $J(\beta) = P'(\beta+) - P'(\beta-)$ be its derivative jump. Convexity identifies J with the possible variation of the mean action among periodic Gibbs states. The normalization of the physical action density multiplies this variation by a_K^{-4}.

Sketch K (Decay of the pressure jump). For every $c < 1/180$,

$$J(\beta) = O(\beta^{-4-c}) \qquad (\beta \to \infty).$$

Consequently, along the continuum cutoffs, periodic Gibbs states have asymptotically equal one-point action densities. In particular,

$$3a_K^{-4}|\rho(a_p) - \nu(a_p)| \le a_K^{-4} J(\beta) \longrightarrow 0.$$

This does not assert that J vanishes at a finite coupling. ■□

Sketch of the argument. Compare the pressure of the infinite lattice with that on a fixed physical torus. Reflection positivity bounds their difference, and local source estimates control its variations in the physical label. The part of the second derivative concentrated at one label is precisely the rescaled latent heat. It is therefore enough to prevent the limiting pressure response from accumulating an atom.

Repeat the construction with two blocking factors whose logarithms are incommensurable. They act as two translations of the logarithmic coupling variable. The two resulting descriptions concern the same lattice pressure. The two-period argument shows that the limiting response cannot retain a nonconstant periodic contribution common to both descriptions. In its quantitative form, it bounds the mass of the response on a short logarithmic interval by its length and an error inherited from the cutoff comparison.

The error must be estimated with its actual dependence on the torus. For an n-point function the comparison has the form

$$|S_{K'} - S_K| \le C\ell^{3(n+1)} K^{p_n} L^{-K/10}, \qquad K' \ge K \ge k + 54m, \quad \ell = 2L^m.$$

An improved bound for the discarded large-field mass supplies this rate. Choosing the relation between m and K makes the box error decrease while retaining a sufficiently fine cutoff. For every slope $A > 90$, the two-period estimate gives $\beta^4 J(\beta) \le C_A \beta^{-1/(2A)}$. Taking A sufficiently close to 90 yields every exponent $c < 1/180$.

The derivative-jump interpretation then gives the one-point conclusion directly. For a bounded compactly supported test, the difference of its means is bounded by the displayed plaquette estimate times the volume of its supporting cube and its supremum norm. Thus conditioning on translation-invariant events cannot leave different continuum one-point constants.

The threshold $k + 54m$ grows with the box side and must remain in the argument. Replacing it by a cutoff independent of the side would be a stronger estimate, with a stronger conclusion about J. The present reasoning controls the jump asymptotically; it does not prove uniqueness of all finite-coupling Gibbs states.

12.1. **Latent heat and one-point constants.** Let $P = \lim_{N\to\infty} N^{-3} \log Z_N$ be the lattice pressure and $J(\beta) = P'(\beta+) - P'(\beta-)$ its derivative jump. We prove $J(\beta) = O(\beta^{-4-c})$ for every $c < 1/180$. The proof combines a reflection-positive lower bound, comparison with finite tori and an oscillation estimate for two incommensurable blocking factors.

The cutoff threshold depends on the size of the box. We retain this dependence: the result is a decay estimate for the jump and its continuum one-point consequences. The two-period lemma and the improved discarded-mass estimate are the main quantitative steps.

12.1.1. *Notation and hypotheses. One lattice.* For a plaquette p of $\mathbb{Z}^3$ with barycentre $x(p)$ and holonomy $\partial U(p)$ we put $a_p = 1 - \frac{1}{2}\operatorname{tr} \partial U(p) \in [0, 2]$, and $\mathcal{O}_1(f) = \sum_p f(x(p))a_p$ for bounded f of compact support. For an integer $N \ge 3$, S_N is the sum of the a_p over the $3N^3$ plaquettes of the periodic cube with N^3 sites, $Z_N(\beta) = \int e^{-\beta S_N}\, \mathrm{d}U$ with the Haar probability measure of SU(2) on every bond, $\mu_{N,\beta}$ is the Wilson measure of the cube, read as a law on the configurations of $\mathbb{Z}^3$ that is carried by the $N\mathbb{Z}^3$-periodic configurations (the periodic Wilson law in equation YM3_I:(1817), with $\Gamma = N\mathbb{Z}^3$), and $P_N = N^{-3} \log Z_N$, so $P = \lim_N P_N$.

For $x \in \mathbb{Z}^3$ let $c_x = \frac{1}{2}\sum a_p$, the sum over the six faces of the unit cube $x + [0, 1]^3$. A Gibbs state of Wilson's action at the spacing $a > 0$ and the coupling β is a Borel probability measure ν on the configurations of $a\mathbb{Z}^3$ such that $\nu(\gamma_B F) = \nu(F)$ for every finite set B of bonds and every continuous

function F of finitely many bonds; here the Gibbs kernel γ_B is the normalised average over the configurations on B, with the Haar measure of SU(2) on every bond of B and the weight $e^{-\beta\sum_p a_p}$, the sum over the plaquettes p that contain a bond of B, the configuration off B being kept fixed (Section YM3_I:91).

A Gibbs state is periodic if it is invariant under the translations by a subgroup of finite index of $a\mathbb{Z}^3$ (Section YM3_I:91); by Prop. YM3_I:103.4 every periodic Gibbs state is invariant under all isometries of $a\mathbb{Z}^3$, so the periodic Gibbs states are exactly the Gibbs states that are invariant under $a\mathbb{Z}^3$. $\mathcal{G}(\beta)$ is the set of all periodic Gibbs states on $\mathbb{Z}^3$ at β, that is, at the spacing 1 (the set $\mathcal{G}^{\mathrm{per}}_{1,\beta}$ of Section YM3_I:91), and $e_\nu = \nu(a_p)$ for $\nu \in \mathcal{G}(\beta)$; this is one number for all p (Lem. 12.1). For a law $\mathfrak{s}$ that is invariant under $\mathbb{Z}^3$ we write $\langle f; h\rangle_{\mathfrak{s}} = \mathrm{Cov}_{\mathfrak{s}}(\mathcal{O}_1(f), \mathcal{O}_1(h))$ and $\Gamma_{\mathfrak{s}}(z) = \mathrm{Cov}_{\mathfrak{s}}(c_0, c_z)$.

Tents and the reference. $\Lambda_\eta(u) = (1 - |u|/\eta)_+$ for real u and $\eta > 0$. For an integer $\eta \geq 1$ and $X \in \mathbb{Z}^3$ the tent is $\varphi^\eta_X(z) = \prod_{i\leq 3} \Lambda_\eta(z_i - \frac{1}{2} - \eta X_i)$, the hull is $h^\eta = \sum_{|X|_\infty \leq 2} \varphi^\eta_X$, the sum of the 5^3 tents nearest to φ^η_0, and $\varphi^\eta_c(x) = \prod_i \Lambda_\eta(x_i)$ for $x \in \mathbb{Z}^3$. For a law $\mathfrak{s}$ invariant under $\mathbb{Z}^3$ we put $s^\eta_{\mathfrak{s}}(X) = \langle \varphi^\eta_0; \varphi^\eta_X\rangle_{\mathfrak{s}}$, and the near part of the specific heat is $n_\eta(\mathfrak{s}) = \eta^{-3}\langle \varphi^\eta_0; h^\eta\rangle_{\mathfrak{s}}$. For $r_1 > 0$ the reference is

$$\hat{n}(\beta) = \liminf_{N\to\infty} n_{\lceil r_1\beta\rceil}(\mu_{N,\beta}),$$

the lower limit along all integers $N \geq 3$; it is formed from Wilson's action and r_1 alone. $P''(d\beta)$ is the second derivative of the convex function P, a positive measure, and $\mathbb{M}$ is the image under $s = \log\beta$ of the signed measure $\beta^4[P''(d\beta) - \hat{n}(\beta)d\beta]$. Thus $\mathbb{M}(\{s\}) = e^{4s}J(e^s)$, and

$$|\mathbb{M}|([s, s+\delta]) = \int_{[\beta, e^\delta\beta]} x^4\, |P''(dx) - \hat{n}(x)\,\mathrm{d}x| \qquad (\beta = e^s).$$

Settings. A setting consists of a radius $\varrho > 0$, a common choice ς^{S} of the real and source construction constants and auxiliary data in Section YM3_I:82 at the radius required by Thms. YM3_I:AJ and YM3_I:AD, with eq. 2.ii, and the associated Wilson measures and source normalization. Write $\mathcal{T}^{\mathrm{S}}[2\varrho]$ for this ordered choice and $\varsigma = \pi^+(\varsigma^{\mathrm{S}})$ for its restriction to the real construction, denoted by $\mathcal{T}^{\mathrm{F}+}$. The numerical constants retain their values: the real hypotheses are those of Prop. YM3_I:69.1, Prop. YM3_I:69.3 and Cor. YM3_I:69.4; the additional source floors are imposed in their stated order. The coupling ceiling $\gamma \in (0, \frac{1}{2}]$ is chosen last.

Among the constants chosen in Prop. YM3_I:69.1 are the odd block size $L \geq 13$, the cell size M, a power of L, the flow bound $B_\sharp$ with $\bar{B} = B_\sharp L/(L-1)$, the exponent $\mathfrak{p}_0 \in \mathbb{N}$ of the large-field threshold, the exponent r of the cube sizes, with which the cube size at a coupling $x > 1$ is $\mathcal{R}(x) = \min\{L^c : c \in \mathbb{Z}_{\geq 0},\ L^c \geq (\log x)^r\}$ (equation YM3_I:(634); below it is read at couplings $x \geq \gamma^{-2}$ alone), and the amplitudes A_1 and A_0. A lowered choice ς' of a choice ς of $\mathcal{T}^{\mathrm{F}+}$ (Def. 4.47) keeps the values of the constants before A_1 and of A_0, has $B'_\sharp \geq B_\sharp$, and has $A'_1 = \lambda'_\natural A_1$ with $\lambda'_\natural = \lambda_\natural/(2 + 30L)$, where $\lambda_\natural \in (0, \frac{1}{8\pi}]$ is a number of ς; so the small-field thresholds δ_j of the two choices satisfy $\delta'_j = \lambda'_\natural\delta_j$.

For the ordered choices constructed in Lem. 4.48, lowered choices exist: decreasing A_1 preserves the eight inequalities at A_0, and the later constants are reselected in the order proved there. Def. 4.47 attaches to the pair (ς, ς') a floor $\mathsf{Y}^\natural$ for the logarithm of the last coupling and the rate $c_2^\natural = \frac{1}{2}\min\{\frac{1}{8}c_1^\natural, \lambda_\natural^2 A_1^2/2A_0^2\}$, with a number $c_1^\natural > 0$ of the pair. We put $a_K = L^{-K}$, $\mathbb{L}_K = a_K\mathbb{Z}^3$, $\lambda = \log L$, $\ell = 2L^m$ and $T_\ell = \mathbb{R}^3/\ell\mathbb{Z}^3$; the label of a pair (K, β) is $b = a_K\beta$, and

$$a_K^{-4}J(\beta) = b^{-4}\beta^4 J(\beta).$$

The physical scale. For the plaquettes p of $\mathbb{L}_K$ we put $\mathcal{O}_K(f) = a_K^{-1}\sum_p f(x(p))a_p$ and $r_K(f) = \frac{1}{3}a_K^3\sum_p f(x(p))$, a Riemann sum of $\int f$; and $\|f\|_\sharp = \max\{\|f\|_\infty, 40M\,\mathrm{Lip}\, f\}$. $\mathcal{G}_{K,\beta}$ is the set of all periodic Gibbs states on $\mathbb{L}_K$ at β, that is, at the spacing a_K (the set $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ of Section YM3_I:91), a convex set (Props. YM3_I:91.1 and YM3_I:103.4), and its extreme points are the pure phases

(Prop. YM3_I:106.1). $S_n^T[\nu](f_1,\dots,f_n)$ is the joint cumulant of $\mathcal{O}_K(f_1),\dots,\mathcal{O}_K(f_n)$ in a law ν on $\mathbb{L}_K$; for real Lipschitz f_i on T_ℓ, $S^T_{n;K,\ell,\beta}(f_1,\dots,f_n)$ is the joint cumulant under the Wilson measure of T_ℓ at the spacing a_K, that is, of the periodic cube with ℓL^K sites a side. $f^{\langle\ell\rangle}=\sum_{v\in\ell\mathbb{Z}^3} f(\cdot-v)$ is the periodisation. The spread of $(f_1,\dots,f_n)$ is the least sup-diameter of a set of points taken one from each support; for $n=2$ it is the sup-distance of the two supports. On T_ℓ one has $|\mathcal{O}_K(f)|\le 6\ell^3L^{4K}\|f\|_\infty$, so that

$$|S^T_{n;K,\ell,\beta}(f_1,\dots,f_n)|\le \mathsf{op}_n\prod_{i\le n}6\ell^3L^{4K}\|f_i\|_\infty,$$

with a combinatorial number op_n of n; we call this the crude bound.

Labels and windows. For $0<g\le\gamma$ let $X_g=L(g^{-2}+B_\sharp)$ and $I(g)=(g^{-2}+\bar B, X_g-\bar B]$ (Section YM3_I:74), and let $\mathfrak{d}(g)$ be the canonical family at g ([7, §K.1]), the set of the first-passage data at g (Cor. YM3_I:74.2): the pairs (K,β), $K\in\mathbb{N}$, for which K is the least index k at which the running coupling $x_k(\beta)$, the member of index k of the sequence of the increments from $x_0(\beta)=\beta$, is defined, real and at most X_g; in words, the running coupling arrives at the scale of g at the last step. We use two facts on $\mathfrak{d}(g)$. By [7, §K.1], the labels $a_K\beta$ of the pairs of $\mathfrak{d}(g)$ lie in the compact interval $I^{\mathrm{lab}}(g)=[g^{-2}+B_\sharp\frac{L-2}{L-1},\,L(g^{-2}+\bar B)]$ of [7, §K.1], whose lower end is at least g^{-2}; so $a_K\beta\ge g^{-2}$ on $\mathfrak{d}(g)$. By [7, §K.1], $(K,L^Kb)\in\mathfrak{d}(g)$, with the label b, for every $b\in I(g)$ and $K\in\mathbb{N}$. A window is an interval of labels; in a setting we use the window

$$\mathsf{I}=[b_-,b_+],\qquad b_-=\gamma^{-2}+\bar B+\tfrac12,\qquad b_+=L(\gamma^{-2}+3\bar B+1)-2\bar B,$$

and $g_2=(\gamma^{-2}+2\bar B+1)^{-1/2}$. $m_0\ge1$ is the least integer such that the sides $2L^m$, $m\ge m_0$, are admitted at γ and at g_2 (the sides that the section of Section YM3_I:74 attaches to a reference coupling), and $\ell_0=2L^{m_0}$. With $\varepsilon=L^{-1}$, the tent $\mathsf{f}(y)=\prod_i\Lambda_\varepsilon(y_i-\frac12)$ and its hull $\mathsf{h}=\sum_{|X|_\infty\le2}\mathsf{f}(\cdot-\varepsilon X)$, the far part is

$$y_{m,K}(\upsilon)=b^5\varepsilon^{-3}S^T_{2;K,\ell,L^Kb}(\mathsf{f}^{\langle\ell\rangle},1-\mathsf{h}^{\langle\ell\rangle})\big|_{b=e^\upsilon},\qquad \ell=2L^m.$$

When two settings (S_1), (S_2) are used, an index i marks the objects of (S_i): L_i, $\lambda_i=\log L_i$, M_i, $\mathsf{I}_i=[b_{-,i},b_{+,i}]$, $m_{0,i}$, $\ell_{0,i}$, ε_i, $y^{(i)}_{m,K}$, and so on.

Tuples and functionals. A separated tuple is $X=(f_1,\dots,f_n)$ with real $f_i\in C^1_c(\mathbb{R}^3)$ and pairwise disjoint supports; the empty tuple is one of them, (f) is the tuple with the one entry f, $|X|=n$ is the number of entries, $X_A=(f_i)_{i\in A}$ for $A\subset\{1,\dots,n\}$, $(\tau_vf)(x)=f(x-v)$, $\tau_vX=(\tau_vf_i)_i$, and $X\cdot Y$ is the joined tuple, formed when its entries have pairwise disjoint supports. A symmetric functional is a map σ on the separated tuples that is multilinear and symmetric in the entries, with σ of the empty tuple equal to 1. For a law ν on $\mathbb{L}_K$ we put

$$X^\nu_K=\prod_{i\le n}[\mathcal{O}_K(f_i)-\nu(\mathcal{O}_K(f_i))],\qquad \sigma_\nu(X)=\nu(X^\nu_K).$$

A Gibbs sequence at g is $(K_j,\beta_j,\nu_j)_j$ with $(K_j,\beta_j)\in\mathfrak{d}(g)$, $K_j\to\infty$ and $\nu_j\in\mathcal{G}_j=\mathcal{G}_{K_j,\beta_j}$. A limit in its frame is $\sigma'(X)=\lim_j\rho_j(X^{\nu_j}_{K_j})$, for all separated X, with approximants $\rho_j\in\mathcal{G}_j$. The sequence converges if $\sigma(X)=\lim_j\sigma_{\nu_j}(X)$ exists for every separated X; then σ is its limit, and $S^\sigma_2(f,h)=\sigma((f,h))-\sigma((f))\sigma((h))$. For separated $X^1,\dots,X^q$ the iterated plateau of a symmetric functional σ is the iterated limit

$$\lim_{|v_q|\to\infty}\cdots\lim_{|v_2|\to\infty}\sigma\big(X^1\cdot\tau_{v_2}X^2\cdots\tau_{v_q}X^q\big),$$

the translations receding one after the other, v_2 first; at $q=1$ it is $\sigma(X^1)$, and at $q=2$ it is $\lim_{|v|\to\infty}\sigma(X^1\cdot\tau_vX^2)$. A canonical decomposition $(\mathsf{X},\varpi,(\sigma_\xi)_{\xi\in\mathsf{X}})$ of σ consists of (Can1) a compact metrisable space X, (Can2) a Borel probability measure ϖ of full support on it, (Can3) a map $\xi\mapsto\sigma_\xi$ into the symmetric functionals that is continuous for the pointwise convergence and injective, such

that (Can4) σ has iterated plateaux and $\int \prod_{i\le q} \sigma_\xi(X^i)\,\mathrm{d}\varpi$ is the iterated plateau of $(X^1,\dots,X^q)$, for all $q\ge 1$ and all separated $X^1,\dots,X^q$; at $q=1$, $\sigma=\int \sigma_\xi\,\mathrm{d}\varpi$.

The σ_ξ are the components. A symmetric functional has one vacuum if it has a canonical decomposition whose space is a point. By Thm. YM3_I:AP, the limit σ of a convergent Gibbs sequence at g has a canonical decomposition, unique up to isomorphism; every component σ_ξ is a limit in the frame along the full sequence, with the approximants $\nu_j(\cdot\,|B_j)$, B_j invariant with $\nu_j(B_j)>0$; $\sigma_\xi((f))=\mathfrak{c}_\xi\int f$ with a number $\mathfrak{c}_\xi$; $|\sigma_\xi(X\cdot\tau_v Y)-\sigma_\xi(X)\sigma_\xi(Y)|\le C^g(X,Y)/|v|$ for $|v|\ge r(X,Y)$, with the numbers C^g, r of the paragraph on one cutoff below; and $\#\mathsf{X}=\dim\ker H_\sigma$, both finite and equal or both infinite, where $H_\sigma\ge 0$ is the Hamiltonian of the reconstruction of σ in any of the six coordinate directions and $H_\sigma\Omega_\sigma=0$ for its vacuum vector Ω_σ.

So such a limit has one vacuum if and only if $\ker H_\sigma=\mathbb{C}\Omega_\sigma$. We put $\kappa=\int\mathfrak{c}_\xi^2\,\mathrm{d}\varpi$.

Limits along sequences of pairs. A sequence of pairs is $\mathfrak{p}=(K_j,\beta_j)_j$ with $K_j\in\mathbb{N}$, $\beta_j>0$ and $K_j\to\infty$; it is at the label $b>0$ if $a_{K_j}\beta_j\to b$. $\mathsf{K}_\mathfrak{p}$ is the set of the functionals σ for which there are $\nu_j\in\mathcal{G}_{K_j,\beta_j}$ with $\sigma(X)=\lim_j\sigma_{\nu_j}(X)$ for every separated X, along the full sequence; it carries the topology of pointwise convergence. $\mathsf{K}^\bullet_\mathfrak{p}$ is the set of its members with one vacuum. $\mathrm{Lim}[b]$ is the union of the sets $\mathsf{K}_\mathfrak{p}$ over the sequences of pairs at the label b, and $\mathrm{Lim}^\bullet[b]$ the set of its members with one vacuum; for a set $\mathfrak{q}$ of pairs, $\mathrm{Lim}_\mathfrak{q}[b]$ and $\mathrm{Lim}^\bullet_\mathfrak{q}[b]$ are the same unions over the sequences all of whose pairs lie in $\mathfrak{q}$. A sequence of pairs $\mathfrak{p}$ is saturated if $\mathsf{K}_\mathfrak{p}\ne\emptyset$ and $\mathsf{K}_{\mathfrak{p}'}=\mathsf{K}_\mathfrak{p}$ for every subsequence $\mathfrak{p}'$. For a Borel probability measure μ on a compact metrisable set of symmetric functionals, $\mathrm{bar}(\mu)=\int\psi\,\mu(\,\mathrm{d}\psi)$ is its barycentre.

Dilations; the reconstruction at every label. For $z\in\mathbb{Z}$ and a separated tuple $X=(f_1,\dots,f_n)$ put $X^{[z]}=(f_i(L^z\cdot))_i$, again a separated tuple, and for a symmetric functional σ put $\sigma^{[z]}(X)=L^{-z|X|}\sigma(X^{[z]})$. The dilation $x\mapsto L^zx$ carries a law ν on $\mathbb{L}_K$ to a law $\nu^{(z)}$ on $\mathbb{L}_{K-z}$, maps $\mathcal{G}_{K,\beta}$ onto $\mathcal{G}_{K-z,\beta}$ ([7, Lem. K.1]) and turns $\mathcal{O}_{K-z}(f)$ into $L^{-z}\mathcal{O}_K(f(L^z\cdot))$ ([7, Lem. K.1], applied to the function $f(L^z\cdot)$: the dilated function $f^{[z]}$ of that Proposition is $f(L^{-z}\cdot)$, whereas the tuple $X^{[z]}$ above has the entries $f_i(L^z\cdot)$); so $\sigma_{\nu^{(z)}}=(\sigma_\nu)^{[z]}$, and the pair $(K-z,\beta)$ has the label $L^za_K\beta$.

Hence, if $\mathfrak{p}=(K_j,\beta_j)_j$ is a sequence of pairs at the label b and $\sigma\in\mathsf{K}_\mathfrak{p}$, then $\sigma^{[z]}\in\mathsf{K}_{\mathfrak{p}'}$ for the sequence of pairs $\mathfrak{p}'=(K_j-z,\beta_j)_{j\ge j_0}$ at the label L^zb, where $K_j>z$ for $j\ge j_0$. As $\sigma^{[z]}(X\cdot\tau_vY)=L^{-z(|X|+|Y|)}\sigma(X^{[z]}\cdot\tau_{L^{-z}v}Y^{[z]})$, the conditions (Can1)–(Can4) pass through the dilation, the receding translations by v becoming translations by $L^{-z}v$: $(\mathsf{X},\varpi,(\sigma_\xi)_{\xi\in\mathsf{X}})$ is a canonical decomposition of σ if and only if $(\mathsf{X},\varpi,(\sigma_\xi^{[z]})_{\xi\in\mathsf{X}})$ is one of $\sigma^{[z]}$, with the same space X.

For every $b>0$ there are an integer $z\ge 0$ and $0<g\le\gamma$ with $g^{-2}+\bar B<L^zb<X_g-\bar B$, because $L^zb>\gamma^{-2}+\bar B$ for large z and these open intervals overlap one another as g decreases and cover $(\gamma^{-2}+\bar B,\infty)$; we fix such a pair (z,g) for b, with $z=0$ if b itself lies in one of these intervals. Then $L^za_{K_j}\beta_j\in I(g)$ for large j, so $(K_j-z,\beta_j)\in\mathfrak{d}(g)$ by [7, §K.1], and $\sigma^{[z]}$ is the limit of a convergent Gibbs sequence at g over these pairs, to which Thm. YM3_I:AP applies. For $\sigma\in\mathsf{K}_\mathfrak{p}$ we put $H_\sigma=L^zH_{\sigma^{[z]}}$ and $\Omega_\sigma=\Omega_{\sigma^{[z]}}$, on the space of the reconstruction of $\sigma^{[z]}$, in each of the six coordinate directions, which the dilation keeps, as it commutes with the coordinate mirrors: the time t of $\sigma^{[z]}$ is the time $L^{-z}t$ of σ, and the kernel is that of $H_{\sigma^{[z]}}$.

As the canonical decompositions of σ and of $\sigma^{[z]}$ correspond to one another with the same space X, which is unique up to homeomorphism (Thm. YM3_I:AP at $\sigma^{[z]}$), $\dim\ker H_\sigma=\#\mathsf{X}$, both finite and equal or both infinite, and $\ker H_\sigma=\mathbb{C}\Omega_\sigma$ if and only if σ has one vacuum; these statements read X alone, so they do not depend on the choice of (z,g), and at $z=0$ they are those of Thm. YM3_I:AP for σ itself.

One cutoff. $\mathcal{I}$ is the σ-field of the Borel sets invariant under the translations by $a_K\mathbb{Z}^3$. By Prop. YM3_I:106.3, for $(K,\beta)\in\mathfrak{d}(g)$ the set $\mathcal{G}_{K,\beta}$ is a class at the cutoff K that obeys the bound system at g: it is not empty; its members are invariant under the isometries of $\mathbb{L}_K$ and reflection positive at every lattice mirror; with ν it contains $h\nu/\nu(h)$ for every bounded $\mathcal{I}$-measurable $h\ge 0$

with $\nu(h) > 0$, so in particular the conditioned states $\nu(\cdot\,|B)$, $B \in \mathcal{I}$, $\nu(B) > 0$; and it obeys the bound system $\mathsf{C}^g = (\hat{C}_1^g, (\hat{C}_n^g)_{n\geq 2})$ at g (equation YM3_I:(2030); Section YM3_I:100), which has two members: for $n \geq 2$ the cumulants of separated tuples of type $(\mathsf{d}; R)$, that is, with supports at mutual distances $\geq \mathsf{d}$ and each in a ball of radius R, satisfy $|S_n^T[\rho](f_1, \dots, f_n)| \leq \hat{C}_n^g(\mathsf{d}; R) \prod_i \|f_i\|_\sharp$ for $\rho \in \mathcal{G}_{K,\beta}$; and the one-point functions of two members of the class satisfy $|\rho(\mathcal{O}_K(f)) - \nu(\mathcal{O}_K(f))| \leq \hat{C}_1^g(R)\|f\|_\sharp$ for $\rho, \nu \in \mathcal{G}_{K,\beta}$ and every $f \in C_c^1(\mathbb{R}^3)$ with support in a closed ball of radius R.

These numbers are free of K, β and the states. For such a class the plateau theorem, Prop. YM3_I:100.1, provides for separated X and $\bar{R} \in a_K\mathbb{N}$ numbers $\delta(X; \bar{R}) > 0$, formed from X and $\bar{R}$, and $\mathsf{N}(X; \bar{R})$, formed from X, $\bar{R}$ and both members of the bound system at g, and a constant $\mathfrak{k}_0$ (equations YM3_I:(1976)–(1980)); none of them depends on the cutoff, the coupling or the state. Its clustering inequality says: if ν belongs to the class, X, Y are separated with supports in $(-\bar{R}, \bar{R})^3$, $a_K \leq 2\min\{\delta(X; \bar{R}), \delta(Y; \bar{R})\}$, $v \in a_K\mathbb{Z}^3$ and $|v| \geq 4\sqrt{3}\bar{R}$, then $X \cdot \tau_v Y$ is formed and

$$\big|\sigma_\nu(X \cdot \tau_v Y) - \nu\big(E_\nu[X_K^\nu|\mathcal{I}]\, E_\nu[Y_K^\nu|\mathcal{I}]\big)\big| \leq \mathfrak{k}_0\, [\mathsf{N}(X; \bar{R})\mathsf{N}(Y; \bar{R})]^{1/2}\, \bar{R}/|v|.$$

With it comes the statement on iterated plateaux (Prop. YM3_I:100.2, with the conditional means defined immediately before it): the limit σ of a convergent Gibbs sequence at g has iterated plateaux, and the plateau of $(X^1, \dots, X^q)$ is $\lim_j \nu_j\big(\prod_{i\leq q} E_{\nu_j}[(X^i)_{K_j}^{\nu_j}|\mathcal{I}]\big)$. We put $r(X, Y) = 4\sqrt{3}\bar{R}$ and $C^g(X, Y) = \mathfrak{k}_0[\mathsf{N}(X; \bar{R})\mathsf{N}(Y; \bar{R})]^{1/2}\bar{R}$ where $\bar{R}(X, Y)$ is the least L^k, $k \in \mathbb{Z}$, for which $\operatorname{supp} X \cup \operatorname{supp} Y \subset (-L^k/2, L^k/2)^3$, and is 1 when both supports are empty; these are the numbers of the clustering of components above, and they depend on X, Y, the bounds at g, L and M alone.

12.1.2. *Pressure curvature and the two blocking factors.* For the unit lattice, write $P_N(\beta) = N^{-3}\log Z_N(\beta)$ and $P = \lim_N P_N$. The finite-volume identity $P_N'' = N^{-3}\operatorname{Var}(\sum_p a_p) \geq 0$ gives convexity. The one-sided derivatives of P describe the extremal mean action densities of periodic Gibbs states. Its derivative jump

$$J(\beta) = P'(\beta+) - P'(\beta-)$$

is therefore the quantity that bounds differences between those one-point expectations. The problem is to control this possible atomic part of the pressure curvature at large β.

Comparison with finite tori separates the pressure response that is already represented by local covariances from a possible residual response. The proof treats the second derivative of a convex function as a measure; differentiability is not assumed at the point where a jump is to be estimated. After multiplying by β^4 and changing variable to $s = \log\beta$, a jump of size $J(\beta)$ is an atom of size $\beta^4 J(\beta)$ in the corresponding response measure. This normalization matches the physical one-point scale.

One blocking factor compares the response at values of s separated by $\log L$. Such a comparison still permits an oscillation tied to that one logarithmic period. The second admitted construction uses a factor with an incommensurable logarithm. Both describe the same Wilson pressure, so their comparisons constrain the same residual response. The two-period estimate controls the possible concentration that a single period leaves open. In particular, the argument does not infer continuity merely from a bounded second derivative at finite volume; it estimates how a concentrating part could survive both comparisons.

The comparison error has to be made uniform on the boxes used in this argument. The discarded large-field mass is improved before the box side is allowed to grow. A typical cutoff comparison retained in the detailed proof is

$$|S_{K'} - S_K| \lesssim \ell^{3(n+1)} K^{p_n} L^{-K/10}, \qquad \ell = 2L^m, \qquad K \geq k + 54m.$$

Here S_K is the n-point source cumulant at the cutoff under comparison, and p_n depends on n. The power of ℓ competes with the decay in K. The restriction on K is consequently part of using the

estimate, rather than a threshold that may be fixed independently of the box. The two-period argument and the scale balance give

$$J(\beta) = O(\beta^{-4-c}), \qquad c < 1/180.$$

To read this in physical units, set $\beta_K = L^K b$ and $a_K = L^{-K}$. Differences of periodic-state plaquette means are bounded by the pressure jump. Hence

$$3a_K^{-4}|\rho(a_p) - \nu(a_p)| \le a_K^{-4} J(\beta_K) \longrightarrow 0.$$

The strict improvement over β^{-4} is what makes the last limit vanish. It gives agreement of the physical one-point constants in this comparison. It does not assert that J vanishes identically beyond a finite coupling or that all Gibbs states at that coupling coincide. The state-space and subsequence statements in the detailed argument retain these distinctions.

12.1.3. *Principal results.*

Theorem AJ.

(A) *The limit.*

(1) $\beta^4 J(\beta) \to 0$ as $\beta \to \infty$ through the reals; equivalently, $\beta^4 \sup\{|\nu(a_p) - \nu'(a_p)| : \nu, \nu' \in \mathcal{G}(\beta)\} \to 0$.

(2) Let (S_1), (S_2) be settings at one radius with $L_2 = L_1 + 2$ (such pairs exist at every radius), and let $r_1 > 0$ with $13 r_1 b_{+,i} \le 1$, $i = 1, 2$. There is $C < \infty$, formed from the two settings and r_1, such that $\limsup_{s\to\infty} |\mathbb{M}|([s, s+\delta]) \le C\delta$ for every $\delta > 0$.

(3) Let the two settings of (2) have one common value of $\mathfrak{p}_0$; for every radius and every integer $a \ge 1$ there are such pairs with $\mathfrak{p}_0 \ge a$. There are $C_\mathrm{A}, s_\mathrm{A} < \infty$, formed from the two settings and r_1, such that $\beta^4 J(\beta) \le C_\mathrm{A} (\log\beta)^{-\mathfrak{p}_0/3}$ for $\beta \ge 3$, and $|\mathbb{M}|([s, s+\delta]) \le C_\mathrm{A}\delta$ for $s \ge s_\mathrm{A}$ and $C_\mathrm{A} s^{-\mathfrak{p}_0/3} \le \delta \le 1$. Hence for every real q_0 a pair of settings with $\mathfrak{p}_0 \ge 3q_0$ gives $\sup_{\beta\ge 3} (\log\beta)^{q_0} \beta^4 J(\beta) < \infty$, and then, in every setting and for every $0 < g \le \gamma$, $a_K^{-4} J(\beta) \le C' K^{-q_0}$ for all $(K, \beta) \in \mathfrak{d}(g)$, with a number C' free of K and β.

(4) In either of the two settings of (2) put, for $K \in \mathbb{N}$ and $b \in \mathsf{I}$,

$$u_K(b) = a_K^{-3}[P(L^K b) - P_{N_0}(L^K b)], \qquad N_0 = \ell_0 L^K,$$

the free energy per physical volume of the infinite lattice minus that of the torus T_{ℓ_0}. Then $-3C_\mathrm{v} \le u_K \le 0$ on I, with the number C_v of Prop. 12.6, and there are $C_\mathrm{sc}, C_u < \infty$, formed from the two settings and r_1, such that $u_K + \frac{1}{2} C_\mathrm{sc} b^2$ is convex on (b_-, b_+) for every K. Every sequence of cutoffs has a subsequence along which u_K converges uniformly on the compact subsets of (b_-, b_+); every such limit u is continuously differentiable, $|u'(b) - u'(b')| \le C_u |b - b'|$, $-3C_\mathrm{v} \le u \le 0$, $u + \frac{1}{2} C_\mathrm{sc} b^2$ is convex; and along the subsequence the one-sided derivatives $u_K'(b\pm)$ converge to $u'(b)$ uniformly on the compact subsets. For $\nu \in \mathcal{G}_{K, L^K b}$ and bounded f of compact support,

$$\nu(\mathcal{O}_K(f)) - \tilde{\mu}_{K,\ell_0,L^K b}(\mathcal{O}_K(f)) = -\mathsf{u}\, r_K(f) \qquad \text{with a number } \mathsf{u} \in [u_K'(b-), u_K'(b+)],$$

where $\tilde{\mu}_{K,\ell_0,\beta}$ is the Wilson measure of T_{ℓ_0} at the spacing a_K: along the subsequence the energy density of the periodic Gibbs states, counted from that of the torus T_{ℓ_0}, converges to $-u'(b)$, uniformly in the state and in the label on compact sets.

(B) *The power.* Let (S_1), (S_2) and r_1 be as in (A)(2), and let ς_i' be a lowered choice of ς_i, $i = 1, 2$.

(1) For every real $A > 90$ there are $C_\mathrm{h}, C_\mathrm{r}, \beta_1 < \infty$, given by eq. 12.xi, which depend only on A, r_1, the two settings and the pairs $(\varsigma_i, \varsigma_i')$, such that for $\beta \ge \beta_1$, $s = \log\beta$ and $0 \le \delta \le 1$

$$|\mathbb{M}|([s, s+\delta]) \le C_\mathrm{h}\delta + C_\mathrm{r}\beta^{-1/2A}, \qquad \beta^4 J(\beta) \le C_\mathrm{r}\beta^{-1/2A}.$$

At $A = 91$ the power is $\beta^{-1/182}$.

(2) Consequently $c_*(c) = \sup_{\beta\ge 1} \beta^{4+c} J(\beta)$ is finite for every real $c < \frac{1}{180}$; it is a number of Wilson's action alone.
(3) Consequently $\sup_{\beta\ge 3} (\log\beta)^q \beta^4 J(\beta) < \infty$ for every real q, at this one pair of settings.
(4) The two bounds of (1) hold, with other constants formed from the same objects, also when only ς_1' is given; then Thm. YM3_I:D is used at the three choices ς_1, ς_1', ς_2.
(5) Let $A > 90$ and $C_{\rm h}$, $C_{\rm r}$, β_1 be as in (1). For $i = 1, 2$, all $m \ge m_{0,i}$, all $K \in \mathbb{N}$ with $L_i^K b_{-,i} \ge \beta_1$ and all $G \in C^2(\mathbb{R})$ with support in $\log \mathsf{I}_i$,

$$\Big| \int G(v)\, y^{(i)}_{m,K}(v)\, \mathrm{d}v \Big| \le (C_{\rm h} + C_{\rm r} + C_N)\|G\|_1 + (C_{\rm h} + C_{\rm r})(L_i^K b_{-,i})^{-1/2A} \|G'\|_1 + c_{{\rm P},i} L_i^{-m} \mathsf{Q}(G),$$

with $\mathsf{Q}(G) = \|G''\|_1 + 7\|G'\|_1 + 12\|G\|_1$, $c_{{\rm P},i}$ of Prop. 12.6(c) and $C_N = \max_i C_i^{\rm ref}$ of Prop. 12.6(d): in the mean over the labels the far part is bounded on all sides, beyond a cutoff that does not depend on the side.

(C) *The physical scale.* In any setting let $0 < g \le \gamma$, $(K, \beta) \in \mathfrak{d}(g)$, $\rho, \nu \in \mathcal{G}_{K,\beta}$ and $0 \le c < \frac{1}{180}$; $e_\rho = \rho(a_p)$ is one number for all plaquettes p of $\mathbb{L}_K$. For every plaquette p of $\mathbb{L}_K$ and every bounded f with support in $[-R, R]^3$,

$$3a_K^{-4} |\rho(a_p) - \nu(a_p)| \le a_K^{-4} J(\beta) \le g^{8+2c} c_*(c) L^{-cK},$$
$$|\rho(\mathcal{O}_K(f)) - \nu(\mathcal{O}_K(f))| = 3a_K^{-4} |e_\rho - e_\nu|\, |r_K(f)| \le a_K^{-4} J(\beta) (2R + a_K)^3 \|f\|_\infty.$$

(D) *One-point functions.* In any setting let $0 < g \le \gamma$ and let $(K_j, \beta_j, \nu_j)_j$ be a Gibbs sequence at g.
(1) For $(K, \beta) \in \mathfrak{d}(g)$, $a_K^{-4} J(\beta) \le g^8 \sup_{\beta' \ge L^K g^{-2}} \beta'^4 J(\beta')$, which tends to 0 as $K \to \infty$. So $\rho_j(\mathcal{O}_{K_j}(f)) - \nu_j(\mathcal{O}_{K_j}(f)) \to 0$ for all $\rho_j \in \mathcal{G}_j$ and all bounded f of compact support.
(2) Every limit σ' in the frame of the sequence has $\sigma'((f)) = 0$ for all $f \in C_c^1(\mathbb{R}^3)$.
(3) If the sequence converges, with the limit σ, then $\mathfrak{c}_\xi = 0$ for every canonical decomposition of σ and every ξ, so $\kappa = 0$; every component σ_ξ is the limit of the Gibbs sequence $(K_j, \beta_j, \nu_j(\cdot\,|B_j))_j$, along the full sequence, with $B_j \in \mathcal{I}$, $\nu_j(B_j) > 0$, so that Thm. YM3_I:AP applies to it, with its reconstruction, and it has one vacuum. For a component of such a limit the three properties $\mathfrak{c}_\xi = 0$, "σ_ξ is the limit of a convergent Gibbs sequence at g with the pairs (K_j, β_j)" and "σ_ξ is such a limit and has one vacuum" are equivalent, by an argument that does not use (A).
(4) In (3), $|S_2^\sigma(f, \tau_v h)| \le C^g((f), (h))/|v|$ for $f, h \in C_c^1(\mathbb{R}^3)$ and $|v| \ge r((f), (h))$.

(E) *The rate on a torus.* Let ς be a choice of $\mathcal{T}^{\rm F+}$, ς' a lowered choice of it, $n \ge 2$ an integer, $0 < b' \le b''$ and $\mathsf{d} > 0$. There are $k^{\rm rt} \ge 0$ and $C^{\rm rt}, C^{\rm pol} < \infty$, depending only on the values of the constants at ς, on $c_2^\natural$, $\Upsilon^\natural$, n, b', b'' and d, with the following property. Let $m \in \mathbb{N}$, $\ell = 2L^m$, $b \in [b', b'']$, and let $f_1, \dots, f_n$ be real Lipschitz functions on T_ℓ, none of them 0, of spread at least d; put $S_K = S^T_{n;K,\ell,L^K b}(f_1, \dots, f_n)$, $\Pi = \prod_i \|f_i\|_\sharp$ and $\mathsf{p}_n = 2\mathsf{p}_0(n+1) + \frac{1}{2}$. Then:
(a) $|S_{K'} - S_K| \le C^{\rm rt} \ell^{3(n+1)} K^{\mathsf{p}_n} L^{-K/10} \Pi$ for all integers $K' \ge K \ge k^{\rm rt} + 54m$; so $S_\infty = \lim_K S_K$ exists, and $|S_\infty - S_K|$ obeys the same bound;
(b) $|S_K| \le C^{\rm pol} \ell^{219n} \Pi$ for all $K \in \mathbb{N}$, and $|S_\infty| \le C^{\rm pol} \ell^{219n} \Pi$;
(c) for every real $A > 30(n+1)$ and all integers $K' \ge K \ge k^{\rm rt} + Am$,

$$|S_{K'} - S_K| \le 2^{3(n+1)} C^{\rm rt} K^{\mathsf{p}_n} L^{-(\frac{1}{10} - \frac{3(n+1)}{A})K} \Pi,$$

a bound that does not depend on the side and tends to 0;
(d) at $n = 2$, $[b', b''] = [2\gamma^{-2}, L^2\gamma^{-2}]$ and $\mathsf{d} = \frac{1}{4}$, with $k^{\rm rt}$, $C^{\rm rt}$, $C^{\rm pol}$ of these data and

$$C = \max\{11,\ 54/\lambda + k^{\rm rt}/\log 2,\ C^{\rm rt}(110\mathsf{p}_2/e\lambda)^{\mathsf{p}_2},\ C^{\rm pol}\} :$$

for all $m \in \mathbb{N}$, $b \in [2\gamma^{-2}, L^2\gamma^{-2}]$, real Lipschitz f_1, f_2 on T_ℓ whose supports are at sup-distance at least $\frac{1}{4}$, and all integers $K \geq C \log \ell$, $|S_K - S_\infty| \leq C\ell^{438} L^{-K/C} \|f_1\|_\sharp \|f_2\|_\sharp$ and $|S_\infty| \leq C\ell^{438} \|f_1\|_\sharp \|f_2\|_\sharp$.

(F) *The plateau at one cutoff.* In a setting let $0 < g \leq \gamma$, $(K, \beta) \in \mathfrak{d}(g)$ and $\nu \in \mathcal{G}_{K,\beta}$.

(1) There is one bounded $\mathcal{I}$-measurable function e with $\mathsf{e} = E_\nu[a_p|\mathcal{I}]$ ν-almost surely for every plaquette p of $\mathbb{L}_K$; and $-3\mathsf{e} \in [P'(\beta-), P'(\beta+)]$ ν-almost surely.

(2) Let $f, h \in C^1_c(\mathbb{R}^3)$, $\bar{R} \in a_K\mathbb{N}$ with

$$\operatorname{supp} f \cup \operatorname{supp} h \subset (-\bar{R}, \bar{R})^3$$

and

$$a_K \leq 2 \min\{\delta((f); \bar{R}), \delta((h); \bar{R})\},$$

and $v \in a_K\mathbb{Z}^3$ with $|v| \geq 4\sqrt{3}\bar{R}$. Then

$$|S_2^T[\nu](f, \tau_v h) - 9a_K^{-8} r_K(f) r_K(h) \operatorname{Var}_\nu \mathsf{e}| \leq \mathfrak{k}_0[\mathsf{N}((f); \bar{R})\mathsf{N}((h); \bar{R})]^{1/2} \bar{R}/|v|,$$

$$0 \leq 9a_K^{-8} \operatorname{Var}_\nu \mathsf{e} \leq \tfrac{1}{4}(a_K^{-4} J(\beta))^2 \leq \tfrac{1}{4}(g^{8+2c} c_*(c) L^{-cK})^2 \qquad (0 \leq c < \tfrac{1}{180}).$$

(3) There are $\nu_\pm \in \mathcal{G}_{K,\beta}$ with $-3e_{\nu_\pm} = P'(\beta\pm)$. For any two such states and $\nu = \frac{1}{2}(\nu_+ + \nu_-)$, $9a_K^{-8} \operatorname{Var}_\nu \mathsf{e} = \frac{1}{4}(a_K^{-4} J(\beta))^2$. If ν is a pure phase, or $-3e_\nu \in \{P'(\beta-), P'(\beta+)\}$, then $\operatorname{Var}_\nu \mathsf{e} = 0$.

(4) Let X, Y be separated tuples, and $\bar{R}$, v as in (2) with $\delta(X; \bar{R})$, $\delta(Y; \bar{R})$ in the place of $\delta((f); \bar{R})$, $\delta((h); \bar{R})$. Then

$$|\sigma_\nu(X \cdot \tau_v Y) - \sigma_\nu(X)\sigma_\nu(Y) - \operatorname{Cov}_\nu(E_\nu[X_K^\nu|\mathcal{I}], E_\nu[Y_K^\nu|\mathcal{I}])| \leq \mathfrak{k}_0[\mathsf{N}(X; \bar{R})\mathsf{N}(Y; \bar{R})]^{1/2} \bar{R}/|v|,$$

and the modulus of the covariance is at most $\frac{1}{4} \operatorname{osc}(X_K^\nu) \operatorname{osc}(Y_K^\nu)$, where $\operatorname{osc}(F) = \sup_\rho \rho(F) - \inf_\rho \rho(F)$, ρ running through $\mathcal{G}_{K,\beta}$. For every real continuous local function F of the configuration, $\operatorname{osc}(F) \leq D^+\bar{p}^F(0) - D^-\bar{p}^F(0)$, the kink at $t = 0$ of the pressure $\bar{p}^F(t)$ perturbed by tF (Lem. 12.1(b)); at $X = (f)$, $\operatorname{osc}((f)_K^\nu) = a_K^{-4} J(\beta)|r_K(f)|$.

(G) *The limits at a label.* In a setting let $b > 0$, let $\mathfrak{p} = (K_j, \beta_j)_j$ be a sequence of pairs at the label b, $\mathsf{K} = \mathsf{K}_\mathfrak{p}$ and $\mathsf{K}^\bullet = \mathsf{K}^\bullet_\mathfrak{p}$; for $\sigma \in \mathsf{K}$, H_σ and Ω_σ are those of the paragraph on dilations.

(1) K is convex, compact and metrisable; the empty set is not excluded.

(2) Let $\sigma = \lim_j \sigma_{\nu_j} \in \mathsf{K}$. For every canonical decomposition of σ and every ξ there are $B_j \in \mathcal{I}$ with $\nu_j(B_j) > 0$ such that $\sigma_\xi = \lim_j \sigma_{\nu_j(\cdot|B_j)}$ along the full sequence, and σ_ξ has one vacuum; so $\sigma_\xi \in \mathsf{K}^\bullet$. The set $\mathsf{K}^\bullet$ is compact.

(3) $\operatorname{bar} : \operatorname{Prob}(\mathsf{K}^\bullet) \to \mathsf{K}$ is an affine homeomorphism onto K. For $\sigma \in \mathsf{K}$ and every canonical decomposition of σ, the map $\xi \mapsto \sigma_\xi$ is a homeomorphism of X onto the support of the one measure μ_σ with $\operatorname{bar} \mu_\sigma = \sigma$, and it carries ϖ to μ_σ; and $\dim \ker H_\sigma = \#\operatorname{supp} \mu_\sigma$, both finite and equal or both infinite.

(4) For $\sigma \in \mathsf{K}$ these say the same: σ has one vacuum; $\ker H_\sigma = \mathbb{C}\Omega_\sigma$ in one, and then in each, of the six directions; σ is an extreme point of K; $\sigma(X \cdot \tau_v X) \to \sigma(X)^2$ as $|v| \to \infty$, for every separated X; $|\sigma(X \cdot \tau_v Y) - \sigma(X)\sigma(Y)| \leq c_{X,Y}/|v|$ for $|v| \geq r_{X,Y}$, with numbers $c_{X,Y}$, $r_{X,Y}$ free of σ (in a window they are $C^g(X, Y)$, $r(X, Y)$); σ is the limit, along the full sequence, of σ_{ω_j} with pure phases $\omega_j \in \mathcal{G}_{K_j,\beta_j}$; $\operatorname{Var}_{\nu_j}(E_{\nu_j}[X_{K_j}^{\nu_j}|\mathcal{I}]) \to 0$ for every separated X, for one, and then for every, choice of states $\nu_j \in \mathcal{G}_{K_j,\beta_j}$ with $\sigma_{\nu_j} \to \sigma$.
For a subsequence $\mathfrak{p}'$ of $\mathfrak{p}$, K is a closed face of $\mathsf{K}_{\mathfrak{p}'}$; if $\tilde{\mathfrak{p}}$ is a second sequence of pairs at the label b and $\mathfrak{p}''$ interlaces $\mathfrak{p}$ and $\tilde{\mathfrak{p}}$, then $\mathsf{K}_{\mathfrak{p}''} = \mathsf{K} \cap \mathsf{K}_{\tilde{\mathfrak{p}}}$. If $\mathsf{K} \neq \emptyset$, these say the same: K is a point; $\mathsf{K}^\bullet$ is a point; $\mathsf{K} = \mathsf{K}^\bullet$; $\operatorname{Cov}_{\nu_j}(E_{\nu_j}[X_{K_j}^{\nu_j}|\mathcal{I}], E_{\nu_j}[Y_{K_j}^{\nu_j}|\mathcal{I}]) \to 0$ for all separated X, Y and all choices of $\nu_j \in \mathcal{G}_{K_j,\beta_j}$ for which σ_{ν_j} converges. At single fields this covariance tends to 0 in every case:

$$|\operatorname{Cov}_{\nu_j}(E_{\nu_j}[(f)_{K_j}^{\nu_j}|\mathcal{I}], E_{\nu_j}[(h)_{K_j}^{\nu_j}|\mathcal{I}])| \leq \tfrac{1}{4}(a_{K_j}^{-4} J(\beta_j))^2 |r_{K_j}(f) r_{K_j}(h)| \to 0.$$

(5) $\mathrm{Lim}[b]$ has exactly one element if and only if $\mathrm{Lim}^\bullet[b]$ has exactly one element. The same holds for $\mathrm{Lim}_{\mathfrak{q}}[b]$ and $\mathrm{Lim}^\bullet_{\mathfrak{q}}[b]$ for every set $\mathfrak{q}$ of pairs that contains the pairs of a sequence of pairs at the label b; in particular for $\mathfrak{q} = \{(K, L^K b) : K \in \mathbb{N}\}$. If $g^{-2} + \bar{B} < b < X_g - \bar{B}$ for some $0 < g \le \gamma$, a third equivalent statement is that every Gibbs sequence at g with $a_{K_j}\beta_j \to b$ converges; and then the one limit has one vacuum.

(6) $\mathrm{Lim}[b]$ and $\mathrm{Lim}^\bullet[b]$ are non-empty, compact and metrisable. Every sequence of pairs at the label b has a saturated subsequence, and every subsequence of a saturated sequence is saturated, with the same set of limits; $\mathrm{Lim}[b]$ is the union of the sets $\mathsf{K}_{\mathfrak{p}}$, and $\mathrm{Lim}^\bullet[b]$ the union of their sets of extreme points, over the saturated sequences $\mathfrak{p}$ at the label b. Every member of $\mathrm{Lim}[b]$ is the barycentre of exactly one Borel probability measure on $\mathrm{Lim}^\bullet[b]$. The map bar is an affine homeomorphism of $\mathrm{Prob}(\mathrm{Lim}^\bullet[b])$ onto the closed convex hull $\overline{\mathrm{co}}\,\mathrm{Lim}[b]$ of $\mathrm{Lim}[b]$ for the pointwise convergence, and for $\sigma \in \mathrm{Lim}[b]$ these say the same: σ has one vacuum; σ is an extreme point of $\overline{\mathrm{co}}\,\mathrm{Lim}[b]$; σ is an extreme point of $\mathrm{Lim}[b]$.
The following four statements are equivalent, and they hold if $\mathrm{Lim}[b]$ has exactly one element: $\mathrm{Lim}[b]$ is convex; $\mathrm{Lim}[b] = \mathsf{K}_{\mathfrak{q}}$ for one sequence of pairs $\mathfrak{q}$ at the label b; $\mathrm{Lim}[b] = \mathrm{bar}(\mathrm{Prob}(\mathrm{Lim}^\bullet[b]))$; every finite subset of $\mathrm{Lim}^\bullet[b]$ lies in $\mathsf{K}^\bullet_{\mathfrak{p}}$ for one sequence of pairs $\mathfrak{p}$ at the label b.

(7) Let $0 < g \le \gamma_W$, where γ_W is the ceiling of the law of two points, the short-distance law for the two-point function (equation YM3_I:(2346)); $\gamma_W < \gamma$ and $\gamma_W^{-2} \ge L^2\gamma^{-4}$. Let $\mathfrak{p}$ be a sequence of pairs in $\mathfrak{d}(g)$; no label is supposed. If $\mathsf{K}_{\mathfrak{p}} \ne \emptyset$, then $a_{K_j}\beta_j$ converges, to a label b in the interval $I^{\mathrm{lab}}(g)$ of [7, §K.1], and for every $\sigma \in \mathsf{K}_{\mathfrak{p}}$, every $\vartheta \in (0,1)$ and every pair (f,h) of the set $\mathfrak{T}^1_{\mathbb{R}^3}(\mathsf{d}_0(g,\vartheta);\vartheta)$, which is not empty,

$$b^{-2} = \lim_{t\to 0} \frac{S_2^\sigma(f_t, h_t)}{\mathsf{c}_2\, \mathcal{K}_{\mathrm{free}}(f,h)}, \qquad f_t(x) = f(x/t), \qquad h_t(x) = h(x/t).$$

Here $\mathfrak{T}^1_{\mathbb{R}^3}(\mathsf{d};\vartheta)$ is the set of the test pairs of class C^1 at the scale d with the shape parameter ϑ ([7, Lem. I.1] and the test-pair convention preceding equation YM3_I:(2343)): pairs of non-negative C^1 functions f, h, neither of them 0, with supports in balls of radius d at a distance between d and $6\mathsf{d}$, $\int f \ge \vartheta \|f\|_\infty \mathsf{d}^3$ and $\mathrm{Lip}\, f \le \|f\|_\infty/\vartheta\mathsf{d}$, and the same for h; $\mathsf{d}_0(g,\vartheta) > 0$ is the threshold of equations YM3_I:(2349)–(2350); and $\mathsf{c}_2 = \frac{9}{16\pi^2}$ and $\mathcal{K}_{\mathrm{free}}(f,h) = \iint f(x)h(y)|x-y|^{-6}\,\mathrm{d}x\,\mathrm{d}y > 0$ are the constant and the free kernel of Thm. YM3_I:C and equation YM3_I:(2240). So the sets $\mathrm{Lim}_{\mathfrak{d}(g)}[b]$, $b > 0$, are pairwise disjoint and their union is the set of the limits of all convergent Gibbs sequences at g; a sequence of pairs in $\mathfrak{d}(g)$ that interlaces two sequences with different labels has $\mathsf{K}_{\mathfrak{p}} = \emptyset$; and a convex combination $t\sigma + (1-t)\sigma'$, $0 < t < 1$, of limits at two different labels is the limit of no Gibbs sequence at g.

■□

The second theorem has a hypothesis on two periodic boxes. Let $Z(n_1, n_2, n_3)$ be the partition function of Wilson's action at the coupling β on the periodic lattice with n_1, n_2, n_3 sites in the three directions; it does not depend on the spacing. In a setting let $b > 0$, $\ell_* = 2L^{m_*}$ with $m_* \in \mathbb{Z}$, K_* an integer with $K_* + m_* \ge 1$, and $0 < x_0 \le 0.069$. The box inequality $\mathrm{BI}[b, \ell_*, K_*; x_0]$ says: for every integer $K \ge K_*$, with $l_K = \ell_*/2a_K = L^{K+m_*}$,

$$\mathcal{E}(l_K) = 2\log Z(l_K, 2l_K, 2l_K) - \log Z(2l_K, 2l_K, 2l_K) \le x_0 \qquad \text{at } \beta = L^K b.$$

Theorem AK. In a setting suppose $\mathrm{BI}[b, \ell_*, K_*; x_0]$, and let $\underline{K} = \max\{K_*, 1\}$. Then:

(1) for every $K \ge \underline{K}$ the set $\mathcal{G}_{K, L^K b}$ has exactly one element ν_K, which is a pure phase, and $J(L^K b) = 0$;

(2) for every $n \geq 2$ and every tuple X of n real Lipschitz functions on $\mathbb{R}^3$ with pairwise disjoint compact supports, $\lim_K S_n^T[\nu_K](X)$ exists along all $K \geq \underline{K}$ and equals $\lim_{m'\to\infty} \lim_{K\to\infty} S^T_{n;K,\ell',L^K b}(X^{\langle \ell' \rangle})$, $\ell' = 2L^{m'}$;

(3) for $\mathfrak{p} = (K, L^K b)_{K\geq \underline{K}}$ the set $\mathsf{K}_\mathfrak{p}$ is one point $\sigma = \lim_K \sigma_{\nu_K}$, and σ has one vacuum.

For every $z \in \mathbb{Z}$, BI$[L^z b, L^z \ell_*, K_* - z; x_0]$ is the same statement as BI$[b, \ell_*, K_*; x_0]$, so (1)–(3) hold at the label $L^z b$ as well. ■□

12.1.4. *Construction estimates.* The proof starts from Thm. YM3_I:D, with the common numerical choice of Prop. YM3_I:69.1 and its actual-history application in Cor. YM3_I:69.4, and from what is derived from it on the tori: the discarding of the histories with a large-field region (Prop. YM3_I:115.2), with its weight (the definitions preceding Prop. YM3_I:115.2) and with the statement on two nearby positive functionals and their cumulants ([7, Lem. K.2]), by which the discarding holds at every bound on the lost mass that is at most $\frac{1}{2}$; the comparison of two consecutive cutoffs (Thm. YM3_I:AR); and the convergence on each torus (Thm. YM3_I:AS). The last says, at every choice: let $\ell = 2L^m$, $n \geq 2$, let $f_1, \dots, f_n$ be real Lipschitz functions on T_ℓ, none of them 0, whose supports have no common point, let $0 < b' \leq b''$, let $\mathsf{d} > 0$ be at most the spread of $(f_1, \dots, f_n)$, and let $s \geq 0$ be an integer with $L^s b' \geq \gamma^{-2}$ and $M\mathcal{R}(L^s b'') \leq L^{m+s}$; such integers exist (Lem. YM3_I:118.2, equation YM3_I:(2210)), and $s = 0$ is one of them when $b' \geq \gamma^{-2}$ and $M\mathcal{R}(b'') \leq L^m$. Then for all $b \in [b', b'']$ and all integers $K' \geq K \geq K_{1,s}(\mathsf{d}) + s$

$$|S^T_{n;K',\ell,L^{K'}b}(f_1, \dots, f_n) - S^T_{n;K,\ell,L^K b}(f_1, \dots, f_n)| \leq L^{sn} C'_s \, (L^s \ell)^{3(n+1)} (K - s)^{\mathsf{p}_n} L^{-(K-s)/10} \prod_i \|f_i\|_\sharp,$$

so that $S^T_{n;K,\ell,L^K b}(f_1, \dots, f_n)$ converges as $K \to \infty$, uniformly in $b \in [b', b'']$. Here $C'_s = \mathsf{C}'_n([L^s b', L^s b''])$ and $K_{1,s}(\mathsf{d}) = K_1(L^s \ell, n, [L^s b', L^s b''], L^s \mathsf{d})$ are numbers of the constants and thresholds in Thm. YM3_I:AR, which attaches to a side ℓ, to n, to an interval J of labels with $\min \mathsf{J} \geq \gamma^{-2}$ and $M\mathcal{R}(\max \mathsf{J}) \leq L^m$, and to $\mathsf{d} > 0$ the thresholds $K_0(\ell, n, \mathsf{J}) \leq K_1(\ell, n, \mathsf{J}, \mathsf{d})$ and the rate constant $\mathsf{C}'_n(\mathsf{J})$. At $s = 0$ the bound reads $\mathsf{C}'_n([b', b'']) \, \ell^{3(n+1)} K^{\mathsf{p}_n} L^{-K/10} \prod_i \|f_i\|_\sharp$ for $K' \geq K \geq K_1(\ell, n, [b', b''], \mathsf{d})$; for other sides and windows it carries the dilation exponent s.

For a choice and a lowered choice, the setting (Th) gives an admissible lattice by Lem. YM3_I:118.1. §4.3 supplies the common lattice and background definitions, the explicit threshold, the witnesses for each creation, the gauge integration, and the chessboard estimate. Its Prop. 4.52 gives the volume-uniform probability bound. The history weights and masses are those of Thm. YM3_I:D.

From Thms. YM3_I:AJ and YM3_I:AD, it uses the zero-source stability bounds (ii), (iii) of Thm. YM3_I:AD; the comparison of two laws in one ball of radius 1 at a first-passage datum, the common-normalizer comparison in the proof of Prop. YM3_I:91.13, whose hypothesis (Nr1) Prop. YM3_I:91.12 supplies at every pair of $\mathfrak{d}(g)$; the bound at separated supports in every periodic Gibbs state, Prop. YM3_I:91.13; and the family of bounds at g (Prop. YM3_I:106.3, with equation YM3_I:(2030)); for the far part on all sides also the bound for connected functions on the tori at first-passage data, Prop. YM3_I:89.5.

For the limits it uses Thm. YM3_I:AP, Prop. YM3_I:100.1 with equations YM3_I:(1976)–(1980), the statement on iterated plateaux (Prop. YM3_I:100.2), and Prop. YM3_I:106.1 (pure phases drawn from a regular conditional law given $\mathcal{I}$); for (G)(7) the law of two points: its test pairs, its constant and its free kernel (Thm. YM3_I:C and equation YM3_I:(2240), [7, Lem. I.1] and the convention in equation YM3_I:(2343)), with non-empty sets of test pairs ([7, Lem. I.1](c)), its ceiling and its threshold (equations YM3_I:(2346) and YM3_I:(2349)–(2350)), the law in the limits through periodic Gibbs states, with the convergence of the labels (Prop. YM3_I:127.8), and the law in every component (Cor. YM3_I:127.11); for the box inequality the domination theorem (Prop. YM3_I:91.9), the seminorm as a mean over a period box (equation [7, eq. 189]), the chessboard estimate on the tori of $\mathbb{L}_K$ and the comparison of two sides ([7, Cor. B.1](b), (c), (e)), a coarse and a fine

board (the proof of Lem. YM3_I:104.1), the functions of a cell (the plaquette-support argument in [7, §B.1]), the chessboard estimate (Prop. YM3_I:86.3), and the standing choice of an odd blocking factor $L \geq 13$.

The choices used here are supplied by the ordered proof of Prop. YM3_I:69.1 and its sourced extension in Section YM3_I:82. The exponents are fixed before the block size; every sufficiently large odd block size is then admitted, and the remaining constants are chosen in the stated order. Besides, we use the first-passage data at g and the canonical family $\mathfrak{d}(g)$ (Cor. YM3_I:74.2 and [7, result K.1]) with the two facts on its labels ([7, §K.1]), the cube size $\mathcal{R}$ (equation YM3_I:(634)), the dilation of the tori, of their partition functions and connected functions and of the labels (Lem. YM3_I:118.2), the dilation of the lattices $\mathbb{L}_K$, of the laws on them, of their observables and of their periodic Gibbs states ([7, Lem. K.1]), the regions of a history, with $Z_t = \Lambda_t^c$ defined by equation YM3_I:(644), and statements on one lattice that use neither Thm. YM3_I:D nor Thms. YM3_I:AJ and YM3_I:AD: the Gibbs states, the periodic Gibbs states and the pure phases (Section YM3_I:91 and Prop. YM3_I:106.1); periodic laws and their bond representatives (equation YM3_I:(1817) and the construction immediately preceding it); the lower tangent inequality for periodic Gibbs states (Lem. YM3_I:91.6); weak limit points of torus measures are periodic Gibbs states (Prop. YM3_I:91.2), and weak limits of Gibbs states at couplings $\beta_k \to \beta$ are Gibbs states at β (Prop. YM3_I:91.1, with the family of the functions $-a_p$); every periodic Gibbs state is invariant under the isometries of the lattice and reflection positive at every lattice mirror (Prop. YM3_I:103.4), and the periodic Gibbs states at one coupling form a convex set (Props. YM3_I:91.1 and YM3_I:103.4) whose members are integrals of its ergodic members (Prop. YM3_I:106.2); reflection positivity of the torus measures at the coordinate mirrors (Prop. YM3_I:91.4, with $\Gamma = a(N_1\mathbb{Z} \times N_2\mathbb{Z} \times N_3\mathbb{Z})$ and height period $N_\mu a$ in direction e_μ); the reconstruction of the transfer operator with its spectral measure (Lem. YM3_I:99.1); the trace formula for boxes of slabs (Prop. YM3_I:85.2, with the spectral argument in the proof of Prop. YM3_I:86.5 and the slab operators of equation YM3_I:(1744)); and the crossing operator on the Hilbert space of one layer (Section YM3_I:85), which is positive and of trace class (Prop. YM3_I:85.1, with the rank-one expansion in its proof).

Lems. 12.1, 12.2, 12.10, Props. 12.3, 12.5, 12.11, Lem. 12.12(a), (c) and Prop. 12.16(a)–(e) use neither Thm. YM3_I:D nor Thms. YM3_I:AJ and YM3_I:AD, and Prop. 12.16(a)–(e) does not use the supply of choices either. Clause (E) uses Thm. YM3_I:D alone, at a choice and at a lowered choice of it. Clause (A) uses Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD at the choices of its two settings, Thm. YM3_I:D through the convergence on each torus; it uses no lowered choice and no statement on the rarity of a large field. Clause (B), in (1)–(3) and (5) and with the constants eq. 12.xi, uses Thms. YM3_I:AJ and YM3_I:AD at the two choices ς_1^{S}, ς_2^{S} of its settings and Thm. YM3_I:D at the four choices ς_1, ς_2, ς_1', ς_2', at each pair $(\varsigma_i, \varsigma_i')$ through (E).

With other constants, (B)(4) uses Thms. YM3_I:AJ and YM3_I:AD at the same two choices and Thm. YM3_I:D at the three choices ς_1, ς_1', ς_2 alone: at $(\varsigma_1, \varsigma_1')$ through (E), and at ς_2 through the convergence on each torus, by Prop. 12.5. In (A) and (B) Thms. YM3_I:AJ and YM3_I:AD enter only through the stability bounds at zero source (Thm. YM3_I:AD), the comparison of two laws in one unit ball (the common-normalizer comparison in the proof of Prop. YM3_I:91.13, with Prop. YM3_I:91.12) and the bound at separated supports in every periodic Gibbs state (Prop. YM3_I:91.13).

Clause (C) uses, besides (B)(2), only that a setting forms $\mathfrak{d}(g)$; its setting need not be one of the settings of (B). Clauses (D), (F) and (G) use Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD at the choice of the setting, together with (A) where the decay of $a_K^{-4} J(\beta)$ to 0 enters and with (B) where its rate enters; the bound by $\frac{1}{4}(a_K^{-4} J(\beta))^2$ in (F) uses the choice of the setting alone. In Thm. AK, clause (1) uses neither Thm. YM3_I:D nor Thms. YM3_I:AJ and YM3_I:AD nor the supply of choices; clause (2) uses besides Thm. YM3_I:D at the choice of the setting, through the convergence on each torus in Prop. 12.16(f); and clause (3), one vacuum, uses besides Thm. YM3_I:

AP and the family of bounds at g (Prop. YM3_I:106.3) through (G)(4), that is, Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD at the choice of the setting.

It does not use (A): by (1), $J(L^K b) = 0$, so that $a_K^{-4} J(L^K b) = 0$ along the sequence.

12.1.5. *Auxiliary results.*

Lemma 12.1 (One lattice)**.**

(a) For $\beta \geq 0$ the limit $P(\beta) = \lim_N P_N(\beta)$ exists along all integers $N \geq 3$. Each P_N is smooth and convex, $-6\beta \leq P_N \leq 0$, $P_N' = -N^{-3}\mu_{N,\beta}(S_N) \in [-6, 0]$ and $P_N'' = N^{-3}\operatorname{Var}_{\mu_{N,\beta}}(S_N)$. P is finite and convex, its difference quotients lie in $[-6, 0]$, and $0 \leq J \leq 6$.

(b) Let $\beta > 0$ and $\nu \in \mathcal{G}(\beta)$. Then $\nu(a_p)$ is one number e_ν for all p, and $P(\beta+t) - P(\beta) \geq -3e_\nu t$ for $t > -\beta$; so $-3e_\nu \in [P'(\beta-), P'(\beta+)]$, and $3|e_\rho - e_\nu| \leq J(\beta)$ for $\rho, \nu \in \mathcal{G}(\beta)$. There are $\nu_\pm \in \mathcal{G}(\beta)$ with $-3e_{\nu_\pm} = P'(\beta\pm)$; hence $J(\beta) = 3\max\{|e_\rho - e_\nu| : \rho, \nu \in \mathcal{G}(\beta)\}$. More generally, let F be a real continuous function of the bonds in a cube of side $k \in \mathbb{N}$, let F_N be the sum of its translates over a period cell of the cube with N^3 sites, and $p_N^F(t) = N^{-3}\log \mu_{N,\beta}(e^{tF_N})$, $\bar{p}^F = \limsup_N p_N^F$. These functions are convex, $|p_N^F(t)| \leq |t|\|F\|_\infty$, and for every $\nu \in \mathcal{G}(\beta)$, $N \geq 3$ and real t

$$p_N^F(t) \geq t\,\nu(F) - \big[144\beta + 12(2k-1)^3|t|\,\|F\|_\infty\big]/N;$$

so $D^-\bar{p}^F(0) \leq \nu(F) \leq D^+\bar{p}^F(0)$. The same holds on $\mathbb{L}_K$ for $\mathcal{G}_{K,\beta}$.

(c) For laws $\mathfrak{s}$ invariant under $\mathbb{Z}^3$ and integers $\eta \geq 1$: $0 \leq c_x \leq 6$, $|\Gamma_\mathfrak{s}| \leq 9$, $|s_\mathfrak{s}^\eta| \leq 9\eta^6$ and $|n_\eta(\mathfrak{s})| \leq 1125\eta^3$; on the periodic configurations $S_N = \sum_x c_x$ over a period cell, and $P_N'' = \sum_z \Gamma_{\mu_{N,\beta}}(z)$ over a period cell. If η divides N and $N \geq 7\eta$, then, with $\mathcal{D} = \{X \in \mathbb{Z}^3 : -\frac{N}{2\eta} \leq X_i < \frac{N}{2\eta}\}$,

$$P_N''(\beta) = n_\eta(\mu_{N,\beta}) + \eta^{-3} \sum_{X \in \mathcal{D},\ |X|_\infty \geq 3} s^\eta_{\mu_{N,\beta}}(X). \tag{12.i}$$

■

Proof sketch. (a) Removing n plaquette terms changes $\log Z$ by at most $2\beta n$, and a bond that no remaining term reads integrates to 1. A torus and the free box of the same side differ by at most $6N^2$ plaquettes, so their logarithms differ by at most $12\beta N^2$. A free box of side $N' = qN + r'$, $0 \leq r' < N$, holds q^3 disjoint free boxes of side N and at most $9N'^2N + 6N'^3/N$ further plaquettes; so the free energies per site p^{f}_N of free boxes satisfy $|p^{\mathrm{f}}_{N'} - (qN/N')^3 p^{\mathrm{f}}_N| \leq 2\beta(9N/N' + 6/N)$, hence $\limsup_{N'} |p^{\mathrm{f}}_{N'} - p^{\mathrm{f}}_N| \leq 12\beta/N$, and (P_N) is a Cauchy sequence. As $0 \leq a_p \leq 2$, $e^{-6\beta N^3} \leq Z_N \leq 1$; the derivatives are those of the logarithm of a Laplace transform on a compact space; convexity and the bounds on the difference quotients pass to the limit. (b) Periodic Gibbs states are invariant under the lattice isometries, which act transitively on the plaquettes. The lower tangent inequality (Lem. YM3_I:91.6), applied on the cube with N^3 sites, whose boundary layer holds at most $6N^2$ sites, gives $\log \mu_{N,\beta}(e^{tF_N}) \geq tN^3\nu(F) - 144\beta N^2 - 12(2k-1)^3|t|\|F\|_\infty N^2$; divide by N^3. Convexity follows from Hölder's inequality, and the bound by $|t|\|F\|_\infty$ from $|F_N| \leq N^3\|F\|_\infty$; letting $N \to \infty$ gives $\bar{p}^F(t) \geq t\nu(F)$, which is the statement on the one-sided derivatives. At $F = -(a_{p_{12}(0)} + a_{p_{13}(0)} + a_{p_{23}(0)})$, the three plaquettes at the site 0, we have $k = 1$, $\|F\|_\infty \leq 6$, $F_N = -S_N$ and $\mu_{N,\beta}(e^{tF_N}) = Z_N(\beta+t)/Z_N(\beta)$, so $p_N^F(t) = P_N(\beta+t) - P_N(\beta)$ and $\nu(F) = -3e_\nu$:

$$\log Z_N(\beta+t) - \log Z_N(\beta) \geq -3e_\nu t N^3 - 144\beta N^2 - 72|t|N^2,$$

and $-3e_\nu$ is a subgradient of P at β. For the ends take points $\beta_k \downarrow \beta$ (or $\beta_k \uparrow \beta$) at which P is differentiable and $\nu_k \in \mathcal{G}(\beta_k)$ invariant under $\mathbb{Z}^3$, weak limit points of torus measures (Prop. YM3_I:91.2): $-3e_{\nu_k} = P'(\beta_k) \to P'(\beta\pm)$, and weak limit points of (ν_k) are Gibbs states at β (Prop. YM3_I:91.1) invariant under $\mathbb{Z}^3$, so periodic Gibbs states at β. On $\mathbb{L}_K$ use the dilation by L^K, which maps $\mathcal{G}_{K,\beta}$ onto $\mathcal{G}(\beta)$ ([7, Lem. K.1] at $z = K$). (c) $0 \leq c_x \leq 6$ since $0 \leq a_p \leq 2$, and a variable

with values in $[0,6]$ has variance at most 9. Next, $\sum_j \Lambda_\eta(u-\eta j)=1$, $\sum_{x\in\mathbb{Z}}\Lambda_\eta(x)=\eta$, and $\sum_{x\in\mathbb{Z}}\Lambda_\eta(x)\Lambda_1(u-x)=\Lambda_\eta(u)$, the left member being the piecewise affine interpolation of Λ_η at the integers; so $\mathcal{O}_1(\varphi^\eta_X)=\sum_x \varphi^\eta_{\mathrm{c}}(x-\eta X)c_x$, which gives $|s^\eta_{\mathfrak{s}}|\le 9\eta^6$ and, with the 125 points $|X|_\infty\le 2$, $|n_\eta|\le 1125\eta^3$. Every plaquette is a face of exactly two unit cubes, so $S_N=\sum_x c_x$ and $P''_N=N^{-3}\operatorname{Var}(S_N)=\sum_z\Gamma(z)$. On the periodic configurations $\sum_{X\in\mathcal{D}}\mathcal{O}_1(\varphi^\eta_X)=S_N$, $\mathcal{D}$ being a set of representatives of $\mathbb{Z}^3$ modulo N/η; by invariance $\operatorname{Cov}(\mathcal{O}_1(\varphi^\eta_0),S_N)=\eta^3\operatorname{Cov}(c_0,S_N)=\eta^3P''_N$. Split $\mathcal{D}$ into $|X|_\infty\le 2$, which gives the hull (all these points lie in $\mathcal{D}$ as $N/2\eta\ge 3.5$), and the rest. □

Lemma 12.2 (Reflection positivity: the specific heat is at least its near part)**.**

(a) Let $\beta\ge 0$, $\nu\in\mathcal{G}(\beta)$ and $\eta\ge 1$ an integer. There is a finite positive Borel measure m on $[0,1]\times S^1\times S^1$ with

$$s^\eta_\nu(n+2,w)=\int t^{\eta n}\zeta_2^{\eta w_2}\zeta_3^{\eta w_3}\,\mathrm{d}\mathsf{m}(t,\zeta_2,\zeta_3)\qquad(n\ge 0,\ w\in\mathbb{Z}^2);$$

so $s^\eta_\nu(3e_1)\ge 0$, and $\sum_{|X|_\infty\ge 3}e^{-\epsilon|X|_1}s^\eta_\nu(X)\ge -870\,s^\eta_\nu(3e_1)$ for every $\epsilon>0$.

(b) Let $I\subset(0,\infty)$ be an open interval, $\eta\ge 1$ an integer and q a Borel function, bounded on the compact subsets of I, with $q(\beta)\le n_\eta(\nu)-870\eta^{-3}s^\eta_\nu(3e_1)$ for all $\beta\in I$ and $\nu\in\mathcal{G}(\beta)$. Then $P''(d\beta)\ge q(\beta)\,\mathrm{d}\beta$ on I.

(c) Let $\mathfrak{s}$ be invariant under $\mathbb{Z}^3$, $\eta,\eta'\ge 1$ integers, $\bar\eta=\max\{\eta,\eta'\}$, $w=\varphi^\eta_{\mathrm{c}}/\eta^3$, $w'=\varphi^{\eta'}_{\mathrm{c}}/\eta'^3$ (non-negative, each of sum 1), and $\Theta(y)=\prod_i\min\{1,(5-|y_i-\frac12|/\bar\eta)_+\}$. Then

$$n_\eta-n_{\eta'}=\sum_v\Big[w'(v)\frac{\langle\varphi^\eta_0;\Theta-\Theta(\cdot+v)\rangle}{\eta^3}-w(v)\frac{\langle\varphi^{\eta'}_0;\Theta-\Theta(\cdot+v)\rangle}{\eta'^3}\Big]-\frac{\langle\varphi^\eta_0;\Theta-h^\eta\rangle}{\eta^3}+\frac{\langle\varphi^{\eta'}_0;\Theta-h^{\eta'}\rangle}{\eta'^3},\tag{12.ii}$$

all brackets in $\mathfrak{s}$; in each bracket the second entry vanishes within the sup-distance $\min\{\eta,\eta'\}$ of the support of the first, and off $\{|y-(\frac12,\frac12,\frac12)|_\infty\le 6\bar\eta\}$.

(d) Let $N\ge 3$, $\beta\ge 0$, $\delta\ge 0$, and let $\omega_1,\omega_2,\omega_3$ be real, even, summable functions on $\mathbb{Z}$ with non-negative Fourier transforms and $\sum_j\omega_i(jN)\le 1+\delta$. Then $P''_N(\beta)\ge(1+\delta)^{-3}\sum_{z\in\mathbb{Z}^3}\omega_1(z_1)\omega_2(z_2)\omega_3(z_3)\Gamma_{\mu_{N,\beta}}(z)$.

■

Proof sketch. (d) Put $\omega'_i(t)=(1+\delta)^{-1}\sum_j\omega_i(t+jN)$ on $\mathbb{Z}/N$; its Fourier coefficients are non-negative, so $|\omega'_i|\le\omega'_i(0)\le 1$. Reflection positivity of the torus measure at the coordinate mirrors makes every slice sum $t\mapsto\sum_u e^{2\pi i(k_1u_1+k_2u_2)/N}\Gamma(t,u)$, the sum over the two other coordinates, non-negative: for $1\le t\le N-1$ the mirror $\{x_3=\frac{t+1}{2}\}$ carries the cubes of the layer t to those of the layer 0, and at $t=0$ the sum is a square. Writing $1-\omega'_1\omega'_2\omega'_3=(1-\omega'_3)+\omega'_3(1-\omega'_2)+\omega'_3\omega'_2(1-\omega'_1)$ and expanding the factors ω' in characters with non-negative coefficients, each term gives a non-negative sum against Γ; so $\sum_z\Gamma(z)\ge\sum_z\omega'_1\omega'_2\omega'_3\Gamma(z)$, and the left member is P''_N by Lem. 12.1(c). (a) ν is invariant and reflection positive at the mirrors $\{x_1=\frac12\}$ and $\{x_1=1\}$. The reconstruction of the transfer operator T, $0\le T\le 1$, with the step e_1, and of the tangential translations, with their joint spectral measure, gives $\mathsf{m}=\langle[F],\mathfrak{E}(\cdot)[F]\rangle$ for $F=\mathcal{O}_1(\varphi^\eta_{e_1})-\nu(\mathcal{O}_1(\varphi^\eta_{e_1}))$, which reads only bonds in $\{x_1\ge 1\}$ and whose mirror image is the centred $\mathcal{O}_1(\varphi^\eta_{-e_1})$; this is the representation, and $s^\eta_\nu(3e_1)=\int t^\eta d\mathsf{m}\ge 0$. Distribute the X with $|X|_\infty\ge 3$ over the six cones about the coordinate half-axes, with the weights $\kappa(k,w)=1$ inside a cone, $\frac12$ on a face and $\frac13$ on an edge, which sum to one.

By the invariance under the permutations and reflections of the axes the left member becomes $6e^{-2\epsilon}\int\Sigma\,\mathrm{d}\mathsf{m}$, where, with $x=e^{-\epsilon}t^\eta$, $e^{iq_i}=\zeta_i^\eta$ and $\Psi_x(w)=\sum_{k\ge 3}x^{k-2}\kappa(k,w)$, $\Sigma=\sum_w e^{-\epsilon(|w_2|+|w_3|)}e^{iq\cdot w}\Psi_x(w)$. For $x\le\frac12$, $|\Sigma|\le\sum_w\Psi_x(w)=x\sum_{i\ge 0}x^i(4(i+3)^2+\frac13)\le 145x$. For $x\ge\frac12$, $x^2\Psi_x=\Phi_x-\mathsf{R}_x$ with $\Phi_x(w)=x^{|w|_\infty}\big(\frac{1+x}{2(1-x)}-\frac16\mathbf{1}_{\{|w_2|=|w_3|\}}\big)$ and a remainder $\mathsf{R}_x\ge 0$

supported in $|w|_\infty \le 2$ of total mass $\frac{1}{3}+\frac{13}{3}x+\frac{49}{3}x^2$. Two elementary Fourier inequalities for the transforms $\hat E_x$ of $x^{|w|_\infty}$ on $\mathbb{Z}^2$ and $\hat D_x$ of its restriction to the two diagonals, namely $\hat D_x \ge -1$ and $\hat E_x \ge \frac{1-x}{2(1+x)}(\hat D_x+1)$, give $\hat\Phi_x = \frac{1+x}{2(1-x)}\hat E_x - \frac16 \hat D_x \ge \frac{1}{12}\hat D_x + \frac14 \ge \frac16$; the transform of $e^{-\epsilon(|w_2|+|w_3|)}\Phi_x$ is then a convolution of non-negative functions, so $\Sigma \ge -x^{-2}\sum_w \mathsf{R}_x(w) \ge -53x$.

In both cases $\Sigma \ge -145x$, so the left member is at least $-870e^{-3\epsilon}\int t^\eta d\mathsf{m} \ge -870\, s^\eta_\nu(3e_1)$; and $6\cdot 145 = 870$. (b) The window $\tilde w_1(z) = \eta^{-1}\sum_X e^{-\epsilon|X|}\sum_x \Lambda_\eta(x)\Lambda_\eta(x+z-\eta X)$ is even, has values in $[0,1]$ and the transform $\eta^{-1}\sum_u e^{-\epsilon|u|}e^{iu\eta\theta}\,|\sum_x \Lambda_\eta(x)e^{-ix\theta}|^2 \ge 0$, and $\delta_N = \sum_{j\ne 0}\tilde w_1(jN) \to 0$. By (d) with $\omega_i = \tilde w_1$, $P_N''(\beta) \ge (1+\delta_N)^{-3}\eta^{-3}\sum_X e^{-\epsilon|X|_1} s^\eta_{\mu_{N,\beta}}(X)$. Integrate against $0 \le G \in C_c^\infty(I)$, let $N\to\infty$ (dominated convergence for $\int G''P_N$, Fatou's lemma for the right member, and weak limit points of $\mu_{N,\beta}$ lie in $\mathcal{G}(\beta)$), use (a) to bound the terms with $|X|_\infty \ge 3$, and let $\epsilon\to 0$. (c) $\sum_v \varphi_c^{\eta'}(v)\varphi_0^\eta(\cdot - v) = \sum_{v,x}\varphi_c^{\eta'}(v)\varphi_c^\eta(x)\varphi_0^1(\cdot - v - x)$ is symmetric in (η,η'); expand $\eta'^3\langle\varphi_0^\eta;\Theta\rangle$ with it, exchange η and η', divide by $\eta^3\eta'^3$, and write $h = \Theta - (\Theta - h)$. The vectors v with a weight $\ne 0$ have $|v|_\infty < \bar\eta$, so $\Theta - \Theta(\cdot + v)$ vanishes where $|y - (\frac12,\frac12,\frac12)|_\infty \le 3\bar\eta$, and $\Theta - h^\eta$ vanishes where this distance is at most 2η. □

Proposition 12.3 (For data: two incommensurable periods)**.** Let $\beta_0 > 0$, let P be convex on (β_0,∞), $\hat n$ Borel and bounded on the compact subsets of (β_0,∞), and $C_{\text{low}} \ge 0$ with

$$P''(d\beta) \ge \big[\hat n(\beta) - C_{\text{low}}\beta^{-5}\big]d\beta \qquad \text{on } (\beta_0,\infty); \tag{12.iii}$$

let $\mathbb{M}$ be the image under $s = \log\beta$ of $\beta^4[P''(d\beta) - \hat n\,\mathrm{d}\beta]$, and $J_P(\beta) = P'(\beta+) - P'(\beta-)$. For $i = 1,2$ let $\lambda_i > 0$ with $\lambda_1/\lambda_2 \notin \mathbb{Q}$, $\alpha_i \in \mathbb{R}$, $\varkappa_i > 0$, $\mathsf{J}_i = [\alpha_i, \alpha_i + \lambda_i + 2\varkappa_i]$, integers k_i with $\alpha_i + k_i\lambda_i > \log\beta_0$, and integers $m_i^0 \ge 0$; for integers $m \ge m_i^0$ and $K \ge k_i$ let $P^{(i)}_{m,K}$ be a C^2 function and $n^{(i)}_{m,K}$ a bounded Borel function on $W_K^{(i)} = \exp(\mathsf{J}_i + K\lambda_i)$, with numbers $\epsilon_m^{(i)}, C_N \ge 0$ such that on $W_K^{(i)}$

$$\beta^3|P - P^{(i)}_{m,K}| \le \epsilon_m^{(i)}, \qquad \beta^5|n^{(i)}_{m,K} - \hat n| \le C_N; \tag{12.iv}$$

put $y^{(i)}_{m,K}(\upsilon) = \beta^5[(P^{(i)}_{m,K})'' - n^{(i)}_{m,K}](\beta)$ at $\beta = e^{\upsilon + K\lambda_i}$, $\upsilon \in \mathsf{J}_i$, and $\varkappa = \min\{\varkappa_1,\varkappa_2,1\}$.

(a) Let $\sup_{K\ge k^0}\sup_{\mathsf{J}_1}|y^{(1)}_{m^0,K}| \le Y^0$ for some $m^0 \ge m_1^0$ and some integer $k^0 \ge k_1$; let $m_i \ge m_i^0$, integers $\bar K_i \ge k_i$ with $\bar K_1 \ge k^0$, and $\omega_i \ge 0$ satisfy $\sup_{\mathsf{J}_i}|y^{(i)}_{m_i,K} - y^{(i)}_{m_i,K'}| \le \omega_i$ for $K,K' \ge \bar K_i$. Let $\epsilon = \max_i \epsilon^{(i)}_{m_i} \le \varkappa^2/4$,

$$\mathsf{M}_1 = (Y^0 + C_N + C_{\text{low}})\varkappa_1 + \epsilon^{(1)}_{m^0}(15/\varkappa_1 + 14 + 12\varkappa_1), \qquad \mathsf{A}_0 = 7\mathsf{M}_1/\varkappa_1 + C_{\text{low}} + 4C_N + \omega_1 + \omega_2.$$

Then, with $\mathbb{M}^+ = \mathbb{M} + C_{\text{low}}ds$, for $s \ge s_* = \max_i(\alpha_i + 2\varkappa_i + \bar K_i\lambda_i)$: $\mathbb{M}^+([s,s+a]) \le \mathsf{A}_0 a + (\mathsf{A}_0 + 134)\epsilon^{1/2}$ for $0 \le a \le \varkappa/2$; and for $0 \le \delta \le 1$

$$e^{4s}J_P(e^s) \le (\mathsf{A}_0 + 134)\epsilon^{1/2}, \qquad |\mathbb{M}|([s,s+\delta]) \le (2\mathsf{A}_0 + C_{\text{low}} + 134)\delta + (\mathsf{A}_0 + 134)\epsilon^{1/2}.$$

Hence, if such thresholds $\bar K_i$, of whatever size, exist for all $m_i \ge m_i^0$ with one pair ω_1,ω_2, and $\inf_m \epsilon^{(i)}_m = 0$, then $e^{4s}J_P(e^s) \to 0$ and $\limsup_{s\to\infty}|\mathbb{M}|([s,s+\delta]) \le (2\mathsf{A}_0 + C_{\text{low}} + 134)\delta$.

(b) Let $\epsilon^{(i)}_m = c_{\mathrm{P},i}e^{-\lambda_i m}$ with $c_{\mathrm{P},i} > 0$, and let there be $\mathsf{a}_i \ge 0$, $A_i > 0$, $\omega_i \ge 0$ with $\sup_{\mathsf{J}_i}|y^{(i)}_{m,K} - y^{(i)}_{m,K'}| \le \omega_i$ for all $m \ge m_i^0$ and all integers $K,K' \ge \bar K_i(m) = \max\{k_i, \lceil \mathsf{a}_i + A_i m\rceil\}$. Then the hypothesis of (a) on Y^0 holds at $m^0 = m_1^0$, $k^0 = \bar K_1(m_1^0)$, with $Y^0 = \sup_{\mathsf{J}_1}|y^{(1)}_{m_1^0,k^0}| + \omega_1$. Let A_0 be that of (a) at these numbers, $A = \max_i A_i$,

$$c_{3,i} = |\alpha_i| + 2\varkappa_i + \lambda_i(\max\{k_i,0\} + \mathsf{a}_i + 1), \quad c_6 = \max_i c_{\mathrm{P},i}e^{\lambda_i + c_{3,i}/A_i}, \quad T_i(m) = \alpha_i + 2\varkappa_i + \lambda_i\bar K_i(m),$$

$s_\sharp = \max\{0, T_1(m_1^0), T_2(m_2^0), A\log^+(4c_6/\varkappa^2)\}$, $C_{\mathrm{h}} = 2\mathsf{A}_0 + C_{\mathrm{low}} + 134$ and $C_{\mathrm{r}} = (\mathsf{A}_0+134)c_6^{1/2}$. Then for $s \ge s_\sharp$, $\beta = e^s$ and $0 \le \delta \le 1$: $\beta^4 J_P(\beta) \le C_{\mathrm{r}}\beta^{-1/2A}$ and $|\mathbb{M}|([s, s+\delta]) \le C_{\mathrm{h}}\delta + C_{\mathrm{r}}\beta^{-1/2A}$.

(c) (The cut.) Let the hypothesis of (a) on Y^0 hold, and let there be one integer $K_* \ge \max\{k_1, k_2, k^0\}$ and numbers ω_i such that $\sup_{\mathsf{J}_i} |y^{(i)}_{m,K} - y^{(i)}_{m,K'}| \le \omega_i$ for $K, K' \ge K_*$ and every $m \ge m_i^0$, and $\inf_m \epsilon_m^{(i)} = 0$, $i = 1, 2$. Then $J_P(e^s) = 0$ for $s \ge s_* = \max_i(\alpha_i + 2\varkappa_i + K_*\lambda_i)$, and on $[s_*, \infty)$ the measure $\mathbb{M}$ has a density with values in $[-C_{\mathrm{low}}, \mathsf{A}_0 - C_{\mathrm{low}}]$, A_0 being that of (a).

(d) (The converse.) Let $\mathbb{M}$ be any signed Radon measure on (s_0, ∞), $s_2 > s_0$, $\epsilon_1 : [s_2, \infty) \to (0, 1]$ non-increasing, and $C_{\mathrm{h}}, C_{\mathrm{r}} \ge 0$ with $|\mathbb{M}|([s, s+\delta]) \le C_{\mathrm{h}}\delta + C_{\mathrm{r}}\epsilon_1(s)$ for $s \ge s_2$ and $0 \le \delta \le 1$. Then for $s_3 \ge s_2$ and $G \in C^1(\mathbb{R})$ with compact support in $[s_3, \infty)$, $|\langle \mathbb{M}, G\rangle| \le (C_{\mathrm{h}} + C_{\mathrm{r}})(\|G\|_1 + \epsilon_1(s_3)\|G'\|_1)$; so $|\langle \mathbb{M}, G(\cdot - t)\rangle - \langle \mathbb{M}, G\rangle| \le 2(C_{\mathrm{h}} + C_{\mathrm{r}})(\|G\|_1 + \epsilon_1(s_3)\|G'\|_1)$ for every $t \ge 0$.

(e) (The exponent is attained.) For all $\lambda_1, \lambda_2 > 0$ with $\lambda_1/\lambda_2 \notin \mathbb{Q}$, all $\alpha_i \in \mathbb{R}$, $\varkappa_i > 0$, $\mathsf{a}_i \ge 0$ and every $A > 0$ there are data as in (b), with $\beta_0 = 1$, $m_i^0 = 0$, $A_1 = A_2 = A$, $C_N = 0$, $C_{\mathrm{low}} = 7$, $\omega_1 = \omega_2 = 0$ and $c_{\mathrm{P},i} = \frac14 e^{(1+|\alpha_i|)/A}$, such that $\beta_k^4 J_P(\beta_k) = \beta_k^{-1/2A}$ at $\beta_k = e^{2k}$, $k \in \mathbb{N}$. ∎

Proof sketch. $\mathbb{M}^+$ is a positive measure by eq. 12.iii; it has the same atoms as $\mathbb{M}$, $\mathbb{M}(\{s\}) = e^{4s}J_P(e^s)$, and $\mathbb{M} \le \mathbb{M}^+$, $|\mathbb{M}| \le \mathbb{M}^+ + C_{\mathrm{low}}ds$. For $G \in C^2$ supported in $\mathsf{J}_i + K\lambda_i$ put $g(\beta) = \beta^4 G(\log\beta)$, so that $g'' = \beta^2(G'' + 7G' + 12G)(\log\beta)$. Then $\langle \mathbb{M}, G\rangle = \int g''(P - P^{(i)}_{m,K})\,\mathrm{d}\beta + \int g\,(n^{(i)}_{m,K} - \hat n)\,\mathrm{d}\beta + \int g\beta^{-5} y^{(i)}_{m,K}(\log\beta - K\lambda_i)\,\mathrm{d}\beta$, by two integrations by parts against $P^{(i)}_{m,K}$; with eq. 12.iv this is the identity

$$\langle \mathbb{M}, G\rangle = \int G(x)\, y^{(i)}_{m,K}(x - K\lambda_i)\,\mathrm{d}x + r, \qquad |r| \le \epsilon_m^{(i)}\mathsf{Q}(G) + C_N\|G\|_1, \quad \mathsf{Q}(G) = \|G''\|_1 + 7\|G'\|_1 + 12\|G\|_1. \tag{12.v}$$

Let G be supported in $[-\varkappa, \varkappa]$ and $A_G(s) = \langle \mathbb{M}, G(\cdot - s)\rangle$. For $s \ge \alpha_i + \varkappa_i + \bar K_i\lambda_i$ write $s = s_0 + K\lambda_i$ with $s_0 \in [\alpha_i + \varkappa_i, \alpha_i + \varkappa_i + \lambda_i)$, so $K \ge \bar K_i$ and $G(\cdot - s)$ is supported in $\mathsf{J}_i + K\lambda_i$; a shift of s by a multiple of λ_i changes K alone, and eq. 12.v at the two cutoffs, with the oscillation bound, gives $|A_G(s) - A_G(s')| \le 2\epsilon^{(i)}_{m_i}\mathsf{Q}(G) + (2C_N + \omega_i)\|G\|_1$ whenever $s' - s \in \lambda_i\mathbb{Z}$, whatever the number of periods. As $\lambda_1/\lambda_2 \notin \mathbb{Q}$, the set $\{k\lambda_1 - k'\lambda_2 : k, k' \ge 0\} \cap [0, \infty)$ is dense in $[0, \infty)$ (by the pigeon-hole principle), and A_G is continuous; moving first forward by $k\lambda_1$ and then back by $k'\lambda_2$ we get, for $s \ge s_*$ and every $t \ge 0$,

$$|A_G(s+t) - A_G(s)| \le 2(\epsilon^{(1)}_{m_1} + \epsilon^{(2)}_{m_2})\mathsf{Q}(G) + (4C_N + \omega_1 + \omega_2)\|G\|_1;$$

only the irrationality of λ_1/λ_2 is used. At the exponent m^0, eq. 12.v with Y^0 gives $\mathbb{M}^+([s, s+\frac12\varkappa_1]) \le \mathsf{M}_1$ for $s \ge \alpha_1 + \varkappa_1 + k^0\lambda_1$. Now let $G = \Phi_{a,\flat}$ be a plateau function: the indicator of $[-\frac\flat2, a + \frac\flat2]$ smoothed with the kernel $\frac{15}{16}(1-y^2)_+^2$ scaled to the width $\frac\flat2$, which is C^2, has values in $[0, 1]$, equals 1 on $[0, a]$, is supported in $[-\flat, a + \flat]$ and has $\mathsf{Q}(G) \le \frac{15}{2\flat} + 14 + 12(a + \flat)$. Averaging the last display over $t \in [0, \varkappa_1]$ and using $\mathbb{M} \le \mathbb{M}^+$ bounds $A_G(s)$ by $7\mathsf{M}_1\|G\|_1/\varkappa_1$ plus the error; hence, for $0 \le a \le \varkappa/2$ and $\flat = \epsilon^{1/2}$,

$$\mathbb{M}^+([s, s+a]) \le \mathsf{A}_0(a + \flat) + 4\epsilon\left(\tfrac{15}{2\flat} + 26\right) \le \mathsf{A}_0 a + (\mathsf{A}_0 + 134)\epsilon^{1/2}.$$

At $a = 0$ this bounds the atom; covering $[s, s+\delta]$ by at most $2\delta/\varkappa + 1$ intervals of length $\le \varkappa/2$ gives the bound for $|\mathbb{M}|$. (b) $y^{(1)}_{m_1^0,k^0}$ is bounded on J_1, and the oscillation bound at m_1^0 bounds $y^{(1)}_{m_1^0,K}$, $K \ge k^0$, by it plus ω_1. At a given $s \ge s_\sharp$ let m_i be the largest $m \ge m_i^0$ with

$T_i(m) \le s$; as $\bar K_i(m) \le \max\{k_i,0\} + \mathsf{a}_i + A_i m + 1$, $T_i(m_i+1) > s$ gives $\lambda_i m_i > (s - c_{3,i})/A_i - \lambda_i$, so $\epsilon \le c_6 e^{-s/A} \le \varkappa^2/4$, and (a) applies with $\bar K_i = \bar K_i(m_i)$ and $s_* \le s$; A_0 does not depend on m_1, m_2. (c) A_0 and s_* do not depend on m_1, m_2, and the lower bound of ϵ over the admitted pairs is 0; so $\mathbb{M}^+([s, s+a]) \le \mathsf{A}_0 a$ for $s \ge s_*$: there are no atoms, and by outer regularity $\mathbb{M}^+(B) \le \mathsf{A}_0 |B|$ for Borel sets $B \subset [s_*, \infty)$. (d) Cut $[s_3,\infty)$ into intervals I_j of length $\delta = \epsilon_1(s_3)$; then $|\mathbb{M}|(I_j) \le (C_{\mathrm{h}} + C_{\mathrm{r}})\delta$, and $\delta \sup_{I_j} |G| \le \int_{I_j} |G| + \delta \int_{I_j} |G'|$; sum over j. The translate $G(\cdot - t)$ has the same norms and its support lies in $[s_3,\infty)$. (e) Let $s_k = 2k$, $\theta_k = e^{-s_k/2A}$, $\Phi(x) = -\frac12 (1-|x|)_+^2$, so that $\Phi'' = 2\delta_0 - \mathbf{1}_{(-1,1)}$, and $\varphi = \sum_k \frac12 \theta_k^2 \Phi((\cdot - s_k)/\theta_k)$. Put $P(\beta) = \beta^{-3}[\varphi(\log\beta) + \frac{7}{12}]$ and $\hat n(\beta) = 7\beta^{-5}$. For continuous ϕ the image of $\beta^4\, d^2(\beta^{-3}\phi(\log\beta))$ is the distribution $\phi'' - 7\phi' + 12\phi$; so the image of $\beta^4 P''(d\beta)$ is $\sum_k \theta_k \delta_{s_k} + (\rho + 7)dx$ with a density $\rho \ge -\frac12 - \frac72 - 3 = -7$: P is convex, $\mathbb{M} = \sum_k \theta_k \delta_{s_k} + \rho\, \mathrm{d}x$, eq. 12.iii holds with $C_{\mathrm{low}} = 7$, and $\beta_k^4 J_P(\beta_k) = \theta_k = \beta_k^{-1/2A}$. As approximants take $n^{(i)}_{m,K} = \hat n$ and $P^{(i)}_{m,K} = \beta^{-3}[\varphi^{[i,m]}(\log\beta) + \frac{7}{12}]$, where $\varphi^{[i,m]}$ keeps, mollified to within $\epsilon^{(i)}_m$, only the bumps with $s_k + 1 \le \alpha_i + \lambda_i \bar K_i(m)$: the bumps left out have $|\varphi_k| \le \frac14 \theta_k^2 \le \epsilon^{(i)}_m$, and $y^{(i)}_{m,K} = 0$ on J_i for $K \ge \bar K_i(m)$. □

Remark 12.4 (One period does not suffice, and the lower bound cannot be dropped). (1) Let $\lambda > 0$, $0 < \theta < \min\{1, \lambda/2\}$, Φ as in the proof of (e), $\varphi(x) = \sum_{k\in\mathbb{Z}} \frac12 \theta^2 \Phi((x - k\lambda)/\theta)$, $P(\beta) = \beta^{-3}[\varphi(\log\beta) + \frac{7}{12}]$ and $\hat n = 7\beta^{-5}$. With λ-periodic smooth $\varphi^{[m]}$, $\|\varphi - \varphi^{[m]}\|_\infty \le e^{-\lambda m}$, the approximants $\beta^{-3}[\varphi^{[m]}(\log\beta) + \frac{7}{12}]$ and $n_{m,K} = \hat n$ satisfy eq. 12.iii with $C_{\mathrm{low}} = 7$, eq. 12.iv with $C_N = 0$ and $\epsilon_m = e^{-\lambda m}$, the bound on Y^0 at every m^0, and the oscillation bound with $\omega = 0$ at every cutoff, a threshold free of m, since $y_{m,K}$ does not depend on K; yet $\beta^4 J_P(\beta) = \theta$ at every $\beta = e^{k\lambda}$. So the hypotheses at one period, or at two commensurable periods $p\lambda$, $q\lambda$ (with $\varphi^{[pm]}$, $\varphi^{[qm]}$ in the place of $\varphi^{[m]}$, so that $\epsilon^{(i)}_m = e^{-p\lambda m}$, $e^{-q\lambda m}$), give neither (a) nor (b) nor (c). (2) The lower bound eq. 12.iii cannot be dropped from (a), nor from (b), at any two periods. Let $\beta_j = 4^j$ for the integers $j \ge j_0 \ge 1$, numbers w_j with $2\beta_j^{-2} \le w_j \le \frac14 \beta_j$ and $w_j/\beta_j \to 0$, $c_j = \frac16 w_j \beta_j^{-4}$, $\tau(x) = -(1-|x|)_+^3$, $\beta_0 = 1$,

$$P(\beta) = -\log\beta + \sum_{j \ge j_0} c_j\, \tau\Big(\frac{\beta - \beta_j}{w_j}\Big) \qquad \text{and} \qquad \hat n(\beta) = \beta^{-2}.$$

The j-th term vanishes off $[\beta_j - w_j, \beta_j + w_j] \subset [\frac34\beta_j, \frac54\beta_j]$, and these intervals are disjoint. The function τ is C^2 off 0, $\tau'(0\pm) = \pm 3$, and $\tau''(x) = -6(1-|x|)$ for $0 < |x| < 1$; so on the j-th interval, off β_j, $P'' \ge \beta^{-2} - 6c_j w_j^{-2} = \beta^{-2} - \beta_j^{-4} w_j^{-1} \ge (\frac{16}{25} - \frac12)\beta_j^{-2} > 0$, while at β_j the slope of P jumps up by $6c_j/w_j = \beta_j^{-4}$. So P is convex on (β_0, ∞), and $\beta_j^4 J_P(\beta_j) = 1$ for every j. Moreover $\beta^3 c_j |\tau((\beta - \beta_j)/w_j)| \le (\frac54\beta_j)^3 c_j < \frac12 w_j/\beta_j$. For given λ_i, α_i, $\varkappa_i$, k_i as in Prop. 12.3, put $m_i^0 = 0$ and $n^{(i)}_{m,K} = \hat n$, so that $C_N = 0$, and, for all K,

$$P^{(i)}_{m,K}(\beta) = -\log\beta + \sum_{j_0 \le j < j(m)} c_j\, \tau_m\Big(\frac{\beta - \beta_j}{w_j}\Big),$$

with integers $j(m) \to \infty$ and C^2 functions τ_m with support in $[-1,1]$ and $\sup|\tau_m - \tau| \le \epsilon_m := \sup_{j \ge j(m)} w_j/\beta_j$. Then eq. 12.iv holds with $\epsilon^{(i)}_m = \epsilon_m$, and $\epsilon_m \to 0$. On $W^{(i)}_K$ the function $y^{(i)}_{m,K}$ vanishes as soon as $e^{\alpha_i + K\lambda_i} > \frac54 \beta_{j(m)-1}$, because there $(P^{(i)}_{m,K})'' = \beta^{-2} = \hat n$; so the oscillation bound holds with $\omega_i = 0$ beyond a finite threshold at each m, and the bound on Y^0 holds at every m^0, as the cutoffs below the threshold are finitely many. Thus every condition of (a) and of its last sentence holds except eq. 12.iii: near $\log\beta_j$ the measure $\mathbb{M}$ has the density $\beta^5 c_j w_j^{-2} \tau''((\beta - \beta_j)/w_j)$, which tends to $-\beta_j/w_j$ at $\log\beta_j$, so eq. 12.iii fails for every C_{low}; and the conclusion of (a) fails, as $e^{4s} J_P(e^s) = 1$ at every $s = \log\beta_j$.

Let now $w_j = \min\{\frac14\beta_j, \beta_j^{1-1/A}\}$ with $A > \frac13$ and j_0 so large that $w_j \ge 2\beta_j^{-2}$ for $j \ge j_0$, let $\mathsf{a}_i \ge 0$ and $\bar K_i(m) = \max\{k_i, \lceil \mathsf{a}_i + Am\rceil\}$, and at the period λ_i keep in $P^{(i)}_{m,K}$ the terms with $2\beta_j \le e^{\alpha_i+\lambda_i\bar K_i(m)}$, with τ_m within $c_{\mathrm{P},i}e^{-\lambda_i m}$ of τ, where $c_{\mathrm{P},i} = 2^{1/A}e^{|\alpha_i|/A}$. A term left out has $\beta_j > \frac12 e^{\alpha_i+\lambda_i\bar K_i(m)} \ge \frac12 e^{-|\alpha_i|}e^{\lambda_i Am}$, so that $w_j/\beta_j \le \beta_j^{-1/A} < c_{\mathrm{P},i}e^{-\lambda_i m}$; and $y^{(i)}_{m,K} = 0$ on J_i for $K \ge \bar K_i(m)$. So every condition of (b) holds, with $A_1 = A_2 = A$, $\omega_1 = \omega_2 = 0$, $C_N = 0$ and $\epsilon^{(i)}_m = c_{\mathrm{P},i}e^{-\lambda_i m}$, except eq. 12.iii, and the conclusion of (b) fails.

In the form in which the proofs of (a) and (b) use the data, namely a measure with bounded variation on unit intervals whose smoothed translates are nearly invariant, a lower bound is needed as well: $\mathbb{M} = \sum_j(\delta_{s_j} - \delta_{s_j+\eta_j})$, with $s_{j+1} - s_j \ge 1$, $0 < \eta_j < 1$ and $\eta_j \to 0$, has $|\mathbb{M}|(I) \le 4$ on every interval I of length 1 and $\langle \mathbb{M}, G(\cdot - s)\rangle \to 0$ as $s \to \infty$ for every $G \in C^1_c(\mathbb{R})$, but atoms 1 at the points s_j. Its atoms -1 at the points $s_j + \eta_j$ set it apart from the measures $\mathbb{M}$ of data, whose atoms $e^{4s}J_P(e^s)$ are non-negative; in the instance above the density $-\beta_j/w_j$ takes their place. (3) By (d), a bound $|\mathbb{M}|([s, s+\delta]) \le C\delta + Ce^{-cs}$ gives the uniform translation bound of Prop. 12.3(d), with its persistent L^1 term and an exponentially small derivative correction; the proofs of (a) and (b) turn near invariance at such a rate under two incommensurable periods, with eq. 12.iii, into the same bound at the rate $e^{-cs/2}$; and by (e) this loss of one half is real for data.

Proposition 12.5 (For data: only one slope in the exponent)**.** Let the data of Prop. 12.3 be given, with $\epsilon^{(i)}_m = c_{\mathrm{P},i}e^{-\lambda_i m}$, $c_{\mathrm{P},i} > 0$. Suppose:

(i) there are $\mathsf{a}_1 \ge 0$, $A_1 > 0$, $\omega_1 \ge 0$ with $\sup_{\mathsf{J}_1} |y^{(1)}_{m,K} - y^{(1)}_{m,K'}| \le \omega_1$ for all $m \ge m_1^0$ and all integers $K, K' \ge \max\{k_1, \mathsf{a}_1 + A_1 m\}$, and $\sup_{K\ge k_1}\sup_{\mathsf{J}_1}|y^{(1)}_{m_1^0,K}| \le Y^0$;

(ii) there is $\omega_2 \ge 0$ such that for every m in an unbounded set of integers $\ge m_2^0$ there is an integer $\bar K_2(m) \ge k_2$ with $\sup_{\mathsf{J}_2}|y^{(2)}_{m,K} - y^{(2)}_{m,K'}| \le \omega_2$ for $K, K' \ge \bar K_2(m)$.

Put $\mathsf{M}_0 = \varkappa_1(Y^0 + C_N + C_{\mathrm{low}}) + c_{\mathrm{P},1}(4/\varkappa_1 + 14 + 12\varkappa_1)$, $\mathsf{H}_* = \omega_1 + \omega_2 + 4C_N + 120 + 16\mathsf{M}_0/\varkappa_1 + 12\mathsf{M}_0/\lambda_2$ and $C_\vartheta = \max\{1, c_{\mathrm{P},1}^{1/2}\}\exp[(\alpha_1 + \varkappa_1 + \lambda_1(\mathsf{a}_1+1))/2A_1 + \lambda_1/2]$. There is $s_\sharp$ such that for $s \ge s_\sharp$, $\beta = e^s$ and $\delta \ge 0$

$$|\mathbb{M}|([s, s+\delta]) \le (2\mathsf{H}_* + C_{\mathrm{low}})\delta + 2\mathsf{H}_* C_\vartheta \beta^{-1/2A_1}, \qquad \beta^4 J_P(\beta) \le \mathsf{H}_* C_\vartheta \beta^{-1/2A_1}.$$

Only the slope A_1 of the first period enters the exponent; of the second period only one bound ω_2, some finite threshold at each m, and $\epsilon^{(2)}_m \to 0$ are used. ∎

Proof sketch. The identity for tents. For $\varphi_{c,\vartheta}(x) = (1 - |x-c|/\vartheta)_+$ with $[c-\vartheta, c+\vartheta] \subset \mathsf{J}_i + K\lambda_i$ the identity eq. 12.v holds with $\mathsf{Q}(G)$ replaced by $4/\vartheta + 14 + 12\vartheta$: with $Q(s) = e^{3s}P(e^s)$ and $Q_{m,K}$ formed in the same way from $P^{(i)}_{m,K}$, the image of $\beta^4 P''(d\beta)$ is the distribution $Q'' - 7Q' + 12Q$, $D = Q - Q_{m,K}$ has $|D| \le \epsilon^{(i)}_m$, and $\int \varphi_{c,\vartheta}\, d(D'' - 7D' + 12D) = [D(c-\vartheta) - 2D(c) + D(c+\vartheta)]/\vartheta + 7\int \varphi'_{c,\vartheta}D + 12\int \varphi_{c,\vartheta}D$. *Step 1 (mass).* With $c_1 = \alpha_1 + k_1\lambda_1 + \varkappa_1$, every $c \ge c_1$ is $\upsilon_0 + K\lambda_1$ with $K \ge k_1$ and $\upsilon_0 \in [\alpha_1 + \varkappa_1, \alpha_1 + \varkappa_1 + \lambda_1]$; the identity at m_1^0, $\vartheta = \varkappa_1$, with (i), gives $\int \varphi_{c,\varkappa_1} d\mathbb{M}^+ \le \mathsf{M}_0$, so $\mathbb{M}^+([c - \frac{\varkappa_1}{2}, c + \frac{\varkappa_1}{2}]) \le 2\mathsf{M}_0$, and an interval of length l in $[c_1, \infty)$ has $\mathbb{M}^+$-mass at most $2\mathsf{M}_0(l/\varkappa_1 + 1)$. *Step 2 (one period).* For an admitted m, $0 < \vartheta \le \min\{\varkappa_i, 1\}$ with $\epsilon^{(i)}_m \le \vartheta^2$, $\upsilon_0 \in [\alpha_i + \varkappa_i, \alpha_i + \varkappa_i + \lambda_i]$ and cutoffs K, K' beyond the threshold of m, the identity at the two cutoffs gives $|\int \varphi_{\upsilon_0 + K\lambda_i,\vartheta} d\mathbb{M}^+ - \int \varphi_{\upsilon_0 + K'\lambda_i,\vartheta} d\mathbb{M}^+| \le t_i\vartheta$ with $t_i = \omega_i + 2C_N + 60$. *Step 3 (two periods).* Let $m \ge m_1^0$, $\vartheta \le \min\{\varkappa_1, \varkappa_2, 1, \lambda_2/6\}$ with $\vartheta^2 \ge c_{\mathrm{P},1}e^{-\lambda_1 m}$, $c = \upsilon_0 + K\lambda_1$ with $K \ge \max\{k_1, \mathsf{a}_1 + A_1 m\}$, and $\mathsf{H} = \vartheta^{-1}\int \varphi_{c,\vartheta} d\mathbb{M}^+$. Take m_2 in the set of (ii) with $\epsilon^{(2)}_{m_2} \le \vartheta^2$, an integer $K_2 \ge \bar K_2(m_2)$ with $S = \alpha_2 + \varkappa_2 + K_2\lambda_2 \ge c_1 + 1$, $n = \lfloor \lambda_2/3\vartheta \rfloor$, and the intervals $I_r = (S + 3\vartheta(r-1), S + 3\vartheta(r-1) + \vartheta) \subset [S, S + \lambda_2)$, $r \le n$. As $\lambda_1/\lambda_2 \notin \mathbb{Q}$, the points $c + k\lambda_1$, $k \ge k_\flat$, are dense modulo λ_2 for every $k_\flat$; so there are integers $k_r \ge 0$ and $j_r \ge 1$ with $p_r = c + k_r\lambda_1 - j_r\lambda_2 \in I_r$.

Step 2 at $i=1$, with the cutoffs K and $K+k_r$, both beyond the threshold however large k_r is, and then at $i=2$, with the cutoffs K_2+j_r and K_2, gives $\int \varphi_{p_r,\vartheta}\,d\mathbb{M}^+ \geq \vartheta(\mathsf{H}-t_1-t_2)$. The p_r are at least 2ϑ apart, so $\sum_r \varphi_{p_r,\vartheta}\leq 1$, and by Step 1, $n\vartheta(\mathsf{H}-t_1-t_2)\leq \mathbb{M}^+([S-\vartheta,S+\lambda_2+\vartheta])\leq 2\mathsf{M}_0(\frac{4\lambda_2}{3\varkappa_1}+1)$; with $n\vartheta\geq\lambda_2/6$ this gives $\mathsf{H}\leq\mathsf{H}_*$. *Step 4.* For large s let $m(s)=\lfloor((s-\alpha_1-\varkappa_1)/\lambda_1-\mathsf{a}_1-1)/A_1\rfloor$ and $\vartheta(s)=\max\{1,c_{\mathrm{P},1}^{1/2}\}e^{-\lambda_1 m(s)/2}\leq C_\vartheta e^{-s/2A_1}$; choose $s_\sharp$ such that for $s\geq s_\sharp$, $m(s)\geq m_1^0$, $(s-\alpha_1-\varkappa_1)/\lambda_1\geq k_1+1$ and $\vartheta(s)\leq\min\{\varkappa_1,\varkappa_2,1,\lambda_2/6\}$. Then every $c\geq s$ is admitted in Step 3 with $m(s)$ and $\vartheta(s)$. So the atom at s is at most $\int\varphi_{s,\vartheta}\,d\mathbb{M}^+\leq\mathsf{H}_*\vartheta$; $[s,s+\delta]$ is covered by at most $\delta/\vartheta+1$ intervals $[c-\frac{\vartheta}{2},c+\frac{\vartheta}{2}]$, $c\geq s$, each of $\mathbb{M}^+$-mass at most $2\mathsf{H}_*\vartheta$; and $|\mathbb{M}|\leq\mathbb{M}^+ + C_{\mathrm{low}}ds$. □

Proposition 12.6 (One block size)**.** In a setting:

(a) $(K,L^Kb)\in\mathfrak{d}(\gamma)\cup\mathfrak{d}(g_2)$ for $b\in\mathsf{I}$ and $K\in\mathbb{N}$; $b_+-Lb_-=(2L-2)\bar B+\frac{L}{2}>0$, so $\bigcup_K L^K\mathsf{I}=[Lb_-,\infty)$.

(b) Let $K,m\in\mathbb{N}$, $\ell=2L^m$, $N=2L^{m+K}$, $\beta>0$ and $b=a_K\beta$. The partition function of T_ℓ at the spacing a_K is $Z_N(\beta)$, so its free energy per physical volume is $a_K^{-3}P_N(\beta)$; the dilation by L^K maps $\mathcal{G}_{K,\beta}$ onto $\mathcal{G}(\beta)$ and the Wilson measure of T_ℓ to $\mu_{N,\beta}$; for a law ν on $\mathbb{L}_K$ invariant under $a_K\mathbb{Z}^3$, with image $\nu^{(K)}$, and an integer $\eta\geq1$, $\beta^5 n_\eta(\nu^{(K)})=b^5(a_K\eta)^{-3}S_2^T[\nu](\mathsf{f}_\eta,\mathsf{h}_\eta)$, where f_η is the tent of half width $a_K\eta$ about a point of $a_K(\mathbb{Z}+\frac12)^3$ and h_η its hull; and, with $\eta_K=L^{K-1}$,
$$\beta^5P_N''(\beta)=\beta^5 n_{\eta_K}(\mu_{N,\beta})+y_{m,K}(\log b). \tag{12.vi}$$

(c) For $b\in\mathsf{I}$, $K\in\mathbb{N}$, $\beta=L^Kb$, $m\geq m_0$ and $N=2L^{m+K}$,
$$0\leq a_K^{-3}[P_N(\beta)-P(\beta)]\leq 3\ell_0C_{\mathrm{v}}/\ell,\qquad\text{so}\qquad 0\leq\beta^3[P_N(\beta)-P(\beta)]\leq c_{\mathrm{P}}L^{-m},$$
with $c_{\mathrm{P}}=\frac32 b_+^3\ell_0C_{\mathrm{v}}$ and $C_{\mathrm{v}}=2C_{\mathrm{S}}+\max\{C_\gamma,C_{g_2}\}$, where C_{S} is the constant in Thm. YM3_I: AD(ii), (iii) and $C_g=C_{\mathrm{S}}(1+\log g^{-2})$ is its lower-bound constant at fixed g; C_{S} does not depend on ϱ, γ, g, K, β, ℓ, and C_g not on K, β, ℓ.

(d) Let $r_1>0$ with $13r_1b_+\leq1$. Then $\hat n$ is Borel, $|\hat n(\beta)|\leq1125\lceil r_1\beta\rceil^3$, and $\hat n(\beta)=n_{\lceil r_1\beta\rceil}(\nu_\beta)$ for some $\nu_\beta\in\mathcal{G}(\beta)$. With the constant C_{nr} of the comparison of two laws in one ball of radius 1 (the common-normalizer comparison in the proof of Prop. YM3_I:91.13) and the constants $C_{\mathrm{sp}}(\mathsf{d})$ of the bound at separated supports in every periodic Gibbs state (Prop. YM3_I: 91.13), both taken at $g\in\{\gamma,g_2\}$ as in the proof, $\mathsf{d}_1=\min\{r_1b_-,\varepsilon\}$ and
$$C^{\mathrm{ref}}=b_+^5\mathsf{d}_1^{-3}(40\sqrt3M/\mathsf{d}_1)^2[C_{\mathrm{nr}}+8C_{\mathrm{sp}}(\mathsf{d}_1)+C_{\mathrm{sp}}(r_1b_-)],$$
for $b\in\mathsf{I}$, $K\in\mathbb{N}$, $\beta=L^Kb$, $m\geq m_0$ and $N=2L^{m+K}$:
$$|\beta^5P_N''(\beta)-y_{m,K}(\log b)-\beta^5\hat n(\beta)|\leq C^{\mathrm{ref}};$$
$\beta^5|n_{\lceil r_1\beta\rceil}(\nu)-\hat n(\beta)|\leq C^{\mathrm{ref}}$ and $0\leq\beta^5\lceil r_1\beta\rceil^{-3}s_\nu^{\lceil r_1\beta\rceil}(3e_1)\leq C^{\mathrm{ref}}$ for every $\nu\in\mathcal{G}(\beta)$; and $P''(d\beta)\geq[\hat n(\beta)-871C^{\mathrm{ref}}\beta^{-5}]d\beta$ on (Lb_-,∞).

(e) For all $K\in\mathbb{N}$, $m\geq m_0$ and $\upsilon\in\log\mathsf{I}$, $|y_{m,K}(\upsilon)|\leq b_+^5L^3C_3(\varepsilon)(40\sqrt3ML)^2\ell^3$, where $C_3(\mathsf{d})=(40\sqrt3M+1)C_{\mathrm{tt}}(\mathsf{d})$ and $C_{\mathrm{tt}}(\mathsf{d})$ is the larger, over $g\in\{\gamma,g_2\}$, of the numbers $A(\mathsf{d})B(1)^2\,2^{\varkappa_\star}$ of Prop. YM3_I:89.5 at g: the constant of the bound for connected two-point functions on the tori at first-passage data, for two functions each supported in a closed sup-ball of radius 1, at distance at least d; it does not depend on K, β or the side. This is better than the bound by $C^{\mathrm{pol}}\ell^{438}$ that (E)(b) gives.

■

Proof sketch. (a) With $y_1=\gamma^{-2}+\bar B\geq4+\bar B$ and $L\geq13$, $(L-1)y_1\geq4\bar B+1$, which says $y_2=g_2^{-2}+\bar B=y_1+2\bar B+1\leq Ly_1-2\bar B$; so $I(\gamma)\cup I(g_2)=(y_1,Ly_2-2\bar B]\supset\mathsf{I}$, as $b_+=Ly_2-2\bar B$. And $b_+-Lb_-=L(y_2-y_1)-2\bar B-\frac{L}{2}=(2L-2)\bar B+\frac{L}{2}$, so consecutive intervals $L^K\mathsf{I}$ overlap. (b) The dilation by L^K carries bonds, plaquettes and the Haar measures of $\mathbb{L}_K$ to those of $\mathbb{Z}^3$ and $\ell\mathbb{Z}^3$

to $N\mathbb{Z}^3$, and $\mathcal{O}_K(f)$ becomes $L^K\mathcal{O}_1(f(L^{-K}\cdot))$; so $S_2^T[\nu](f,h) = L^{2K}\langle f(L^{-K}\cdot); h(L^{-K}\cdot)\rangle_{\nu^{(K)}}$, and $\beta^5\eta^{-3}L^{-2K} = b^5(a_K\eta)^{-3}$. As L^K is odd, $\frac{1}{2} \in a_K(\mathbb{Z}+\frac{1}{2})$, and $\varepsilon = a_K\eta_K$. At $\eta = \eta_K$, which divides N with $N/\eta_K = 2L^{m+1} \geq 7$, eq. 12.i becomes eq. 12.vi: the rescaled tents are the translates $\mathsf{f}(\cdot - \varepsilon X)$, their sum over the period cell is 1 on T_ℓ, so the far sum is $1 - \mathsf{h}^{\langle\ell\rangle}$. (c) By (a), $(K,\beta) \in \mathfrak{d}(g)$ for some $g \in \{\gamma, g_2\}$. Let $p(j)$, $j \in \mathbb{N}^3$, be the free energy per physical volume of the periodic box with $j_\mu\ell_0/a_K$ sites in the direction μ, at the spacing a_K and the coupling β. With $Z_{K,\ell,\beta} = Z_{\ell L^K}(\beta)$, the partition function of T_ℓ at the spacing a_K, (b) gives $p(k,k,k) = (k\ell_0)^{-3}\log Z_{K,k\ell_0,\beta} = a_K^{-3}P_{k\ell_0 L^K}(\beta)$ for every power k of L. In the first direction such a box consists of $2j_1$ slabs, and the trace formula (Prop. YM3_I:85.2) writes its partition function as $\operatorname{Tr} B^{j_1}$ with one positive trace-class operator B for all j_1, formed at j_2, j_3.

With the top eigenvalue $\|B\|$, $u_k = \frac{1}{k}\log\operatorname{Tr} B^k$ satisfies $0 \leq k(u_k - \log\|B\|) \leq u_1 - \log\|B\|$ and does not increase in k; hence p does not increase in j_1 and $0 \leq p(k',j_2,j_3) - p(k,j_2,j_3) \leq \frac{1}{k'}[p(1,j_2,j_3) - \lim_n p(n,j_2,j_3)]$ for $k' \leq k$. The same argument in the second and in the third direction shows that p does not increase in any of its three arguments, with the same bound in each direction. The bracket is at most C_{v}, for all j_2, j_3. By the monotonicity in each argument, $p(1,j_2,j_3) \leq p(1,1,1)$, and $p(n,j_2,j_3) \geq p(L^i,L^i,L^i)$ whenever $L^i \geq \max\{n,j_2,j_3\}$; so the bracket is at most $\sup_{i\geq 0}[p(1,1,1) - p(L^i,L^i,L^i)]$, where $p(1,1,1)$ and $p(L^i,L^i,L^i)$ belong to the tori with the sides $\ell_0 = 2L^{m_0}$ and $L^i\ell_0 = 2L^{m_0+i}$, both admitted at γ and at g_2.

To (K,β) and to each side ℓ admitted at g, Thm. YM3_I:AD attaches a real number $E_{K,\ell,\beta}$, and its stability bounds say: by the zero-source factorisation and (ii), $-C_g \leq \ell^{-3}\log Z_{K,\ell,\beta} - \ell^{-3}E_{K,\ell,\beta} \leq C_{\mathrm{S}}$, that is, $\ell^{-3}\log Z_{K,\ell,\beta}$ lies within C_{S} above and C_g below the energy per volume $\ell^{-3}E_{K,\ell,\beta}$; by (iii) the energies per volume at two such sides differ by at most C_{S}; and by the sentence following (iii), C_{S} is free of $\varrho, \gamma, g, K, \beta, \ell$. At the two tori this gives $p(1,1,1) - p(L^i,L^i,L^i) \leq C_{\mathrm{S}} + C_{\mathrm{S}} + C_g \leq C_{\mathrm{v}}$ for every i; and the brackets of the directions 2 and 3 obey the same bound.

Changing one direction at a time, from (k',k',k') through (k,k',k') and (k,k,k') to (k,k,k), with $k' = \ell/\ell_0 = L^{m-m_0}$ and a power $k \geq k'$ of L, each step changes p by at most C_{v}/k', so $0 \leq p(k',k',k') - p(k,k,k) \leq 3C_{\mathrm{v}}/k'$, that is, $0 \leq a_K^{-3}[P_N - P_{k\ell_0 L^K}] \leq 3\ell_0 C_{\mathrm{v}}/\ell$, as $N = \ell L^K$; let $k \to \infty$ through the powers of L (Lem. 12.1(a)), and multiply by $b^3 = \beta^3 a_K^3$. (d) Thms. YM3_I:AJ and YM3_I:AD enter through two bounds at first-passage data, with constants free of K, β, m and the state; by (a) the pair (K,β) lies in $\mathfrak{d}(g)$ for some $g \in \{\gamma, g_2\}$. By the common-normalizer comparison in the proof of Prop. YM3_I:91.13 at $n = 2$, whose hypothesis (Nr1) Prop. YM3_I:91.12 supplies at every pair of $\mathfrak{d}(g)$, two laws, each a torus measure on an admitted side $2L^m$, $m \geq m_0$, or a periodic Gibbs state at (K,β), satisfy $|S_2^T[\mathfrak{s}](F_1,F_2) - S_2^T[\mathfrak{s}'](F_1,F_2)| \leq C_{\mathrm{nr}}\|F_1\|_\sharp\|F_2\|_\sharp$ for F_1, F_2 supported in one closed ball of radius 1 about a lattice point; here C_{nr} is the larger, over $g \in \{\gamma, g_2\}$, of the numbers $8\log 2\,(2\mathsf{Q}_1(L)+2)^2\varrho^{-2}$, with $\mathsf{Q}_1(L)$ of equation eq. 5.xxvii at g.

By Prop. YM3_I:91.13 at $n = 2$ and $R = 2$, in every periodic Gibbs state $|S_2^T[\nu](F_1,F_2)| \leq C_{\mathrm{sp}}(\mathsf{d})\|F_1\|_\sharp\|F_2\|_\sharp$ for supports in balls of radius 2 at distance at least d; here $C_{\mathrm{sp}}(\mathsf{d})$ is the larger, over $g \in \{\gamma, g_2\}$, of the numbers $C_2^{\mathrm{per}}(\mathsf{d};2)$ of equation eq. 5.xxvii at g. The physical half widths $a_K\lceil r_1\beta\rceil \in [r_1 b, r_1 b + a_K]$ and $a_K\eta_K = \varepsilon$ lie in $[\mathsf{d}_1, \frac{2}{13}]$, so all tents and hulls involved, and the function Θ of eq. 12.ii rescaled, have $\|\cdot\|_\sharp \leq 40\sqrt{3}M/\mathsf{d}_1$ and live in a ball of radius 1 or 2 about one lattice point.

On each interval on which $\lceil r_1\beta\rceil$ is constant, $\beta \mapsto n_{\lceil r_1\beta\rceil}(\mu_{N,\beta})$ is continuous; so $\hat{n}$ is Borel, and it is attained along a subsequence $\mu_{N_k,\beta} \to \nu_\beta \in \mathcal{G}(\beta)$, the near part being weakly continuous. Write

$$n_{\eta_K}(\mu_{N,\beta}) - \hat{n}(\beta) = [n_{\eta_K}(\mu_{N,\beta}) - n_{\eta_K}(\nu_\beta)] + [n_{\eta_K}(\nu_\beta) - n_{\lceil r_1\beta\rceil}(\nu_\beta)]\,:$$

the first bracket compares two states in one unit ball, and contributes at most $b_+^5\mathsf{d}_1^{-3}(40\sqrt{3}M/\mathsf{d}_1)^2 C_{\mathrm{nr}}$ after multiplication by β^5; this is the one place where m enters, and the bound does not see it. By eq. 12.ii the second bracket is a sum of four means, with weights summing to 1, of covariances in the one state ν_β between a tent and a function H with $\|H\|_\sharp \leq 2\cdot 40\sqrt{3}M/\mathsf{d}_1$ which vanishes

within the distance d_1 of the support of the tent; after multiplication by β^5 each mean is at most $2b_+^5\mathsf{d}_1^{-3}(40\sqrt{3}M/\mathsf{d}_1)^2C_{\mathrm{sp}}(\mathsf{d}_1)$. With eq. 12.vi this gives the first inequality. In the same way $\beta^5|n_{\lceil r_1\beta\rceil}(\nu)-\hat n|$ compares two states in a unit ball, and $\beta^5\lceil r_1\beta\rceil^{-3}s_\nu(3e_1)$ is a covariance at the distance $a_K\lceil r_1\beta\rceil\ge r_1b_-$, non-negative by Lem. 12.2(a). Finally, every $\beta>Lb_-$ is L^Kb with $b\in\mathsf{I}$ by (a); on each open interval of (Lb_-,∞) on which $\lceil r_1\beta\rceil$ is constant, Lem. 12.2(b) applies with $q=\hat n-(1+870)C^{\mathrm{ref}}\beta^{-5}$; the end points of these intervals are countably many, and there P'' has atoms ≥ 0 while $q\,\mathrm{d}\beta$ has none. (e) Let χ_k, $k\in(\mathbb{Z}/\ell\mathbb{Z})^3$, be the periodised products $\prod_i(1-|y_i-k_i|)_+$: they sum to 1 on T_ℓ, each is supported in a closed sup-ball of radius 1, $\operatorname{Lip}\chi_k\le\sqrt{3}$, and so $\|\chi_kw\|_\sharp\le(40\sqrt{3}M+1)\|w\|_\sharp$ and $\operatorname{supp}(\chi_kw)\subset\operatorname{supp}w$. By bilinearity $S_2^T(u,w)=\sum_kS_2^T(u,\chi_kw)$, a sum of ℓ^3 terms, each at most $C_{\mathrm{tt}}(\mathsf{d})(40\sqrt{3}M+1)\|u\|_\sharp\|w\|_\sharp$ when u is supported in a sup-ball of radius 1 and the supports of u and w are at distance at least d. Apply this to $u=\mathsf{f}^{\langle\ell\rangle}$, $w=1-\mathsf{h}^{\langle\ell\rangle}$, $\mathsf{d}=\varepsilon$, with $\|u\|_\sharp,\|w\|_\sharp\le40\sqrt{3}ML$ and $b^5\varepsilon^{-3}\le b_+^5L^3$. □

Lemma 12.7 (Two block sizes)**.** For every $\varrho>0$ and every integer $a\ge1$ there are two settings at the radius ϱ with the block sizes L_1 and $L_2=L_1+2$ and one common value $\mathfrak{p}_0\ge a$. For every odd $L_1\ge3$, $\log L_1/\log(L_1+2)\notin\mathbb{Q}$. ■

Proof sketch. Use the ordered construction in the proof of Prop. YM3_I:69.1: choose $r\ge2$, then an integer $\mathfrak{p}_0\ge\max\{a,6r+1\}$ and $q\ge41\mathfrak{p}_0+1$, followed by the reserve fractions and the other constants preceding L. The proof admits every sufficiently large odd L with these values fixed. The additional floors in Section YM3_I:82 have the same property. Choose an odd L_1 above their common threshold and complete the prescribed construction separately at L_1 and L_1+2; with ϱ and the associated Wilson measures and source normalization, each is a setting. If $\log L_1/\log L_2=u/v$ with positive integers, then $L_1^v=L_2^u$, and a prime factor of L_1 divides $L_2-L_1=2$, which is impossible for odd L_1. Every statement that we use is made for every radius and every choice, and no object of one choice is compared with an object of the other; the two settings meet in P and $\hat n$ alone. □

Lemma 12.8 (The lost mass on tori, without an exponential of the volume)**.** Let ς' be a lowered choice of a choice ς of $\mathcal{T}^{\mathrm{F}+}$, and let the setting (Th) of §2.3.1 hold at ς and at a label $b\ge\gamma^{-2}$ with $\log b\ge\mathsf{Y}^\natural$: $K\ge2$, $\ell=2L^m$ with $M\mathcal{R}(b)\le L^m$, where $\mathcal{R}(x)=\min\{L^c:c\in\mathbb{Z}_{\ge0},\ L^c\ge(\log x)^r\}$ (equation YM3_I:(634)), the line of couplings $x_j=L^{K-j}b$, and a top level. Let $Z=\sum_h\mathsf{t}(h)$ be the sum of the masses $\mathsf{t}(h)$ of the histories (Thm. YM3_I:D), $Z_t(h)=\Lambda_t(h)^c$ the large-field region defined by equation YM3_I:(644), and $N_t=\ell^3L^{3(K-t)}$. Then for $1\le t\le K$

$$\sum_{h:\,Z_t(h)\neq\emptyset}\mathsf{t}(h)\le\mathsf{R}_tZ,\qquad \mathsf{R}_t=\tfrac{L}{L-1}M^{-3}N_t\,e^{-c_2^\natural A_0^2(\log x_t)^{2\mathfrak{p}_0}}.$$

The side enters R_t only through the number N_t of the sites of the level t; and R_t is smaller by the factor 25000 than the number $2\cdot24^3M^{-3}N_te^{-c_2^\natural A_0^2(\log x_t)^{2\mathfrak{p}_0}}$ that the rarity of a large field at a point, Prop. 4.52, gives. ■

Proof. The line $x_j=L^{K-j}b$ is admissible under the stated side and coupling conditions, by Lem. YM3_I:118.1. Def. 4.47 and Lem. 4.48 put it in the common setting of Prop. 4.52. That proposition bounds the probability of a prescribed creation at level t' by $e^{-c_2^\natural p_0(g_{t'})^2}$.

If $Z_t\neq\emptyset$, Prop. YM3_I:67.2 gives a creation at some level $t'\le t$. There are at most $M^{-3}N_{t'}=M^{-3}N_tL^{3(t-t')}$ cells of that level, since $\check R_{t'}\ge1$. Write $a=t-t'$. On this line, $y_{t'}=y_t+a\log L$, and convexity gives

$$y_{t'}^{2\mathfrak{p}_0}-y_t^{2\mathfrak{p}_0}\ge2\mathfrak{p}_0y_t^{2\mathfrak{p}_0-1}a\log L.$$

Condition (Y.7) of eq. 4.xxi therefore bounds the contribution of level $t-a$ by L^{-a} times the contribution of level t. The union bound and the geometric series yield

$$\sum_{h:Z_t(h)\neq\emptyset} \mathsf{t}(h) \le M^{-3} N_t e^{-c_2^\natural p_0(g_t)^2} Z \sum_{a=0}^{t-1} L^{-a} \le \frac{L}{L-1} M^{-3} N_t e^{-c_2^\natural p_0(g_t)^2} Z.$$

Finally $L/(L-1) \le 13/12 < 2\cdot 24^3/25000$. □

Lemma 12.9 (The threshold on a torus is of exact order $\ell^{3/2\mathfrak{p}_0}$)**.** In a setting let $K_0(\ell) = K_0(\ell, 2, \mathsf{I})$ and $K_1(\ell) = K_1(\ell, 2, \mathsf{I}, \varepsilon)$ be the thresholds of the convergence on the torus T_ℓ (Thm. YM3_I:AS at $s=0$, with the constants and thresholds in Thm. YM3_I:AR) at $n=2$, for the labels of I and the spread ε, as they are set there. There are $c_1, c_2 < \infty$ such that for all $m \ge m_0$ with $M\mathcal{R}(b_+) \le L^m$, $\ell = 2L^m$,

$$\frac{1800}{\lambda}\Big[\Big(\frac{\Xi\ell^3}{c' A_0^2}\Big)^{1/2\mathfrak{p}_0} - \log(Lb_-)\Big] \le K_0(\ell) \le K_1(\ell) \le c_1 + c_2\ell^{3/2\mathfrak{p}_0},$$

where $\Xi > 0$ is the number in the factor $e^{\Xi\ell^3}$ that the bound on the lost mass, with which these thresholds are formed, carries, and $c' > 0$ the number in its large-field weight $e^{-c' A_0^2 (\log x)^{2\mathfrak{p}_0}}$. ■

Proof sketch. The conditions that define the thresholds read the side only through $N_{[d]} = \ell^3 L^{3d}$ at depths $d \le K/9 + 1$ and through the factor $e^{\Xi\ell^3}$. At $n=2$ the depth u of the comparison of the cutoffs satisfies $K/1800 \le u \le K/1800 + 1$. One condition asks that the bound on the lost mass, $2M^{-3}\ell^3 L^{3u} \exp[\Xi\ell^3 - c' A_0^2 (\log(L^u b_-))^{2\mathfrak{p}_0}]$, be at most $\frac12$; as $2M^{-3}\ell^3 \ge 16$ and $\log(L^u b_-) \le K\lambda/1800 + \log(Lb_-)$, it fails unless $(K\lambda/1800 + \log(Lb_-))^{2\mathfrak{p}_0} \ge \Xi\ell^3/c' A_0^2$, which is the lower bound. For the upper bound, each condition says that a quantity $c\,\ell^3 L^{-cK}$, or $c\exp[c\log K + c\log\ell + cK + \Xi\ell^3 - c'' K^{2\mathfrak{p}_0}]$ with $c'' > 0$ formed from A_0, A_1 and the constants that enter the definition of the thresholds, is at most a fixed number. For $K \ge c_2\ell^{3/2\mathfrak{p}_0}$ we have $\Xi\ell^3 \le \frac12 c'' K^{2\mathfrak{p}_0}$ and $\log\ell \le \frac{2\mathfrak{p}_0}{3c_2} K$, and all conditions hold once also $K \ge c_1$, because $2\mathfrak{p}_0 \ge 2$. □

Lemma 12.10 (One cutoff: conditional expectations)**.** Let $K \in \mathbb{N}$, and let $\mathcal{G}$ be a set of laws on $\mathbb{L}_K$, each invariant under the isometries of $\mathbb{L}_K$, such that $h\nu/\nu(h) \in \mathcal{G}$ for $\nu \in \mathcal{G}$ and every bounded $\mathcal{I}$-measurable $h \ge 0$ with $\nu(h) > 0$. For $\rho \in \mathcal{G}$, $e_\rho = \rho(a_p)$ is one number for all plaquettes p; put $e_+ = \sup_{\rho\in\mathcal{G}} e_\rho$, $e_- = \inf_{\rho\in\mathcal{G}} e_\rho$, and let $\nu \in \mathcal{G}$.

(a) There is one bounded $\mathcal{I}$-measurable e with $\mathsf{e} = E_\nu[a_p|\mathcal{I}]$ ν-almost surely for every plaquette p; $e_- \le \mathsf{e} \le e_+$ ν-almost surely, and $\mathrm{Var}_\nu \mathsf{e} \le \frac14 (e_+ - e_-)^2$.

(b) For bounded f of compact support, $E_\nu[\mathcal{O}_K(f) - \nu(\mathcal{O}_K(f))\,|\,\mathcal{I}] = 3a_K^{-4} r_K(f)(\mathsf{e} - e_\nu)$ ν-almost surely.

(c) For every bounded real Borel function F, $E_\nu[F|\mathcal{I}]$ takes ν-almost surely values in $[\inf_{\rho\in\mathcal{G}} \rho(F), \sup_{\rho\in\mathcal{G}} \rho(F)]$.

(d) If ν is ergodic, or $e_\nu \in \{e_-, e_+\}$, then e is ν-almost surely constant.

■

Proof sketch. Let k be bounded and $\mathcal{I}$-measurable, $c = 1 + \sup|k|$ and $h = k + c \ge 1$. The laws $h\nu/\nu(h)$ and ν lie in $\mathcal{G}$, so they are invariant, and the isometries act transitively on the plaquettes; so $\nu(ha_p) = \nu(ha_{p'})$ and $\nu(ca_p) = \nu(ca_{p'})$ for all p, p', and subtracting, $\nu(ka_p) = \nu(ka_{p'})$: all a_p have one conditional expectation given $\mathcal{I}$. (c): if $\{E_\nu[F|\mathcal{I}] > \sup_\rho \rho(F) + \epsilon\}$ had positive mass, conditioning on it would give a member of $\mathcal{G}$ in which the mean of F exceeds $\sup_\rho \rho(F)$; likewise below. At $F = a_p$ this gives the range of e, and a variable with values in an interval of length $e_+ - e_-$ has variance at most $\frac14 (e_+ - e_-)^2$. (b): linearity, and $a_K^{-1}\sum_p f(x(p)) = 3a_K^{-4} r_K(f)$. (d): if ν is ergodic, $\mathcal{I}$ is trivial modulo ν; if $\nu(\mathsf{e}) = e_\nu = e_+$, the function $\mathsf{e} \le e_+$ has the mean e_+, so it is constant; likewise at e_-. □

Proposition 12.11 (One cutoff: the plateau for several fields)**.** Let $\mathcal{G}$ be a class at the cutoff K (Section YM3_I:100) that obeys the bound system $\mathsf{C}^g = (\hat{C}_1^g, (\hat{C}_n^g)_{n\ge 2})$ at g, with the properties listed in the paragraph on one cutoff: for $n \ge 2$ and every tuple $X = (f_1, \dots, f_n)$ of type $(\mathsf{d}; R)$, $|S_n^T[\rho](X)| \le \hat{C}_n^g(\mathsf{d}; R) \prod_i \|f_i\|_\sharp$ for $\rho \in \mathcal{G}$, and $|\rho(\mathcal{O}_K(f)) - \rho'(\mathcal{O}_K(f))| \le \hat{C}_1^g(R)\|f\|_\sharp$ for $\rho, \rho' \in \mathcal{G}$ and $f \in C_c^1(\mathbb{R}^3)$ with support in a closed ball of radius R; the numbers $\delta(X; \bar{R})$ and $\mathsf{N}(X; \bar{R})$ are those of the plateau theorem, the second formed with this bound system. Let $\nu \in \mathcal{G}$. Let X, Y be separated, $\bar{R} \in a_K\mathbb{N}$ with all supports in $(-\bar{R}, \bar{R})^3$, $a_K \le 2\min\{\delta(X; \bar{R}), \delta(Y; \bar{R})\}$, $v \in a_K\mathbb{Z}^3$ and $|v| \ge 4\sqrt{3}\bar{R}$. Then

$$\big|\sigma_\nu(X \cdot \tau_v Y) - \sigma_\nu(X)\sigma_\nu(Y) - \mathrm{Cov}_\nu(E_\nu[X_K^\nu|\mathcal{I}], E_\nu[Y_K^\nu|\mathcal{I}])\big| \le \mathfrak{k}_0[\mathsf{N}(X; \bar{R})\mathsf{N}(Y; \bar{R})]^{1/2}\bar{R}/|v|,$$
$$\big|\mathrm{Cov}_\nu(E_\nu[X_K^\nu|\mathcal{I}], E_\nu[Y_K^\nu|\mathcal{I}])\big| \le \tfrac{1}{4}\,\mathrm{osc}_{\mathcal{G}}(X_K^\nu)\,\mathrm{osc}_{\mathcal{G}}(Y_K^\nu), \qquad \mathrm{osc}_{\mathcal{G}}(F) = \sup_{\rho\in\mathcal{G}} \rho(F) - \inf_{\rho\in\mathcal{G}} \rho(F);$$

and $\mathrm{osc}_{\mathcal{G}}((f)_K^\nu) = 3a_K^{-4}|r_K(f)|(e_+ - e_-)$. ∎

Proof sketch. Clause (b) of the plateau theorem at ν compares $\sigma_\nu(X \cdot \tau_v Y)$ with

$$\nu(E_\nu[X_K^\nu|\mathcal{I}]E_\nu[Y_K^\nu|\mathcal{I}]),$$

which is $\sigma_\nu(X)\sigma_\nu(Y)$ plus the covariance, as $\nu(E_\nu[A|\mathcal{I}]) = \nu(A)$. By Lem. 12.10(c) each conditional expectation lies almost surely in an interval of length $\mathrm{osc}_{\mathcal{G}}$, so its variance is at most $\frac{1}{4}\mathrm{osc}_{\mathcal{G}}^2$; then the Cauchy–Schwarz inequality. At $X = (f)$, $\rho((f)_K^\nu) = 3a_K^{-4}r_K(f)(e_\rho - e_\nu)$. □

Lemma 12.12 (Recentring and clustering)**.**

(a) For laws ρ, ν, ν' on $\mathbb{L}_K$ and $X = (f_1, \dots, f_n)$,

$$\rho(X_K^{\nu'}) = \sum_{A\subset\{1,\dots,n\}} \prod_{i\in A} [\nu(\mathcal{O}_K(f_i)) - \nu'(\mathcal{O}_K(f_i))] \\ \rho((X_{A^c})_K^\nu). \tag{12.vii}$$

(b) Let $0 < g \le \gamma$ and $B_n^g(\mathsf{d}; R) = \sum_\pi \prod_{A\in\pi} \hat{C}_{|A|}^g(\mathsf{d}; R)$, the sum over the partitions of $\{1, \dots, n\}$ into blocks of at least two elements ($B_0^g = 1$, $B_1^g = 0$). The set $\mathcal{E}^g$ of the symmetric functionals Φ with $|\Phi(X)| \le B_n^g(\mathsf{d}; R)\prod_i \|f_i\|_\sharp$ for every X of type $(\mathsf{d}; R)$ is convex, and compact and metrisable for pointwise convergence; it contains σ_ν for $(K, \beta) \in \mathfrak{d}(g)$ and $\nu \in \mathcal{G}_{K,\beta}$.

(c) Let T be a compact metrisable set of symmetric functionals that are real-valued on real tuples, with pointwise convergence, such that for all separated X, Y there are $r_{X,Y}, c_{X,Y} < \infty$ with $\sup_{\psi\in\mathsf{T}} |\psi(X \cdot \tau_v Y) - \psi(X)\psi(Y)| \le c_{X,Y}/|v|$ for $|v| \ge r_{X,Y}$. Then for every Borel probability measure μ on T, $(\mathrm{supp}\,\mu, \mu, (\psi)_{\psi\in\mathrm{supp}\,\mu})$ is a canonical decomposition of $\mathrm{bar}(\mu)$, and $\mu \mapsto \mathrm{bar}(\mu)$ is affine, continuous and injective.

(d) Let $0 < g \le \gamma$, $(K, \beta) \in \mathfrak{d}(g)$, $\nu^1, \nu^2 \in \mathcal{G}_{K,\beta}$, $0 \le t \le 1$, $\nu = t\nu^1 + (1-t)\nu^2$, X of n entries and of type $(\mathsf{d}; R)$, and $\eta = a_K^{-4}J(\beta)(2R+1)^3$. Then

$$|\sigma_\nu(X) - t\sigma_{\nu^1}(X) - (1-t)\sigma_{\nu^2}(X)| \le \sum_{k=1}^n \binom{n}{k} \eta^k B_{n-k}^g(\mathsf{d}; R) \prod_i \|f_i\|_\sharp,$$

and at $n = 2$ exactly

$$\sigma_\nu((f,h)) = t\sigma_{\nu^1}((f,h)) + (1-t)\sigma_{\nu^2}((f,h)) + 9t(1-t)a_K^{-8}r_K(f)r_K(h)(e_{\nu^1} - e_{\nu^2})^2.$$

With (C), $\eta \le (2R+1)^3 g^{8+2c} c_*(c) L^{-cK}$ for $0 \le c < \frac{1}{180}$: the defect of convexity of $\nu \mapsto \sigma_\nu$ at one cutoff falls like a power of the spacing.

∎

Proof sketch. (a) Write $\mathcal{O}_K(f_i)-\nu'(\mathcal{O}_K(f_i)) = [\mathcal{O}_K(f_i)-\nu(\mathcal{O}_K(f_i))]+[\nu-\nu'](\mathcal{O}_K(f_i))$ and multiply out. (b) A centred moment $\sigma_\nu(X)$ is the sum, over the partitions without singletons, of the products of the cumulants of the blocks, which the family of bounds at g bounds; the defining conditions are inequalities and equalities at finitely many tuples, so they pass to convex combinations and to pointwise limits, and Tychonoff's theorem gives compactness. Approximating each entry of a tuple in $\|\cdot\|_\sharp$ by functions from a countable family supported in finite unions of rational cubes, and using multilinearity with the bound, one finds a countable set of tuples on which two members of $\mathcal{E}^g$ agree only if they are equal, and on which pointwise convergence implies convergence at every tuple; so $\mathcal{E}^g$ is metrisable. (c) By induction on q, the uniform clustering shows that $\int\prod_{i\le q}\psi(X^i)\,\mathrm{d}\mu$ is the iterated plateau of $\mathrm{bar}(\mu)$ at $(X^1,\dots,X^q)$: this is (Can4); the other conditions are plain.

So $\mathrm{bar}(\mu)$ determines the integrals of the products of the functions $\psi\mapsto\psi(X)$, which are real-valued and separate the points of T; their products and constants form a real unital algebra, so by the Stone–Weierstrass theorem it determines μ. The barycentre is affine in μ, and continuous since the integrands are continuous. (d) $\sigma_\nu(X) = t\nu^1(X_K^\nu)+(1-t)\nu^2(X_K^\nu)$, and by eq. 12.vii with $\rho=\nu^i$, $\nu'=\nu$ and the centring ν^i, $\nu^i(X_K^\nu)-\sigma_{\nu^i}(X)$ is a sum over $A\neq\emptyset$ of products of $|A|$ one-point differences, each at most $a_K^{-4}J(\beta)(2R+a_K)^3\|f\|_\infty\le\eta\|f\|_\sharp$ by (C), times $\sigma_{\nu^i}(X_{A^c})$, which is at most $B^g_{n-|A|}(\mathsf{d};R)\prod_{A^c}\|f_i\|_\sharp$ by (b).

At $n=2$, $\mathrm{Cov}_\nu(F,H) = t\,\mathrm{Cov}_{\nu^1}(F,H)+(1-t)\,\mathrm{Cov}_{\nu^2}(F,H)+t(1-t)[\nu^1-\nu^2](F)\,[\nu^1-\nu^2](H)$ for $F=\mathcal{O}_K(f)$, $H=\mathcal{O}_K(h)$, and $[\nu^1-\nu^2](\mathcal{O}_K(f)) = 3a_K^{-4}r_K(f)(e_{\nu^1}-e_{\nu^2})$. □

Remark 12.13 (The one-point differences and the convexity of the limits). In the proof of (G)(1) the vanishing of the one-point differences, (D)(1), makes the limits of the mixtures the mixtures of the limits; the identity at $n=2$ in (d) shows, for data, what can happen without it. Let, along a sequence of pairs, the class at the j-th pair be the segment $\{s\nu_j^+ + (1-s)\nu_j^- : 0\le s\le 1\}$ between two ergodic laws $\nu_j^\pm$ whose one-point functions differ by an amount that tends to $c\int f$ with $c\neq 0$, and let $\sigma_{\nu_j^\pm}\to\sigma^\pm$. Then the mixtures $\nu_j^s = s\nu_j^+ + (1-s)\nu_j^-$ have

$$\sigma_{\nu_j^s}((f,h)) \to \sigma_s((f,h)) = s\sigma^+((f,h)) + (1-s)\sigma^-((f,h)) + s(1-s)c^2\int f\int h.$$

If moreover $(f,h)\mapsto\sigma^+((f,h))-\sigma^-((f,h))$ is not a multiple of $(f,h)\mapsto\int f\int h$, the limits σ_s, $0\le s\le 1$, form an arc and not a segment, and the set of the limits along the sequence, which is $\{\sigma_s : 0\le s\le 1\}$, does not contain the midpoint of σ^+ and σ^-: it is not convex. For the model, (D)(1) excludes such sequences among the periodic Gibbs states at first-passage data.

Proposition 12.14 (What a rate on the tori gives)**.** Let (S$_1$), (S$_2$) and r_1 be as in (A)(2); no lowered choice is given. Suppose that for $i=1,2$ there are numbers $C_i\ge 1$, $c_{1,i}\ge 0$, integers $m_{1,i}\ge m_{0,i}$ and, for $m\ge m_{1,i}$, $\ell = 2L_i^m$ and $b\in\mathsf{I}_i$, real numbers $S^{(\ell,b)}$ such that for all integers $K\ge C_i\log\ell$

$$|S^T_{2;K,\ell,L_i^K b}(f_1,f_2) - S^{(\ell,b)}| \le C_i\ell^{c_{1,i}}L_i^{-K/C_i}\|f_1\|_\sharp\|f_2\|_\sharp \qquad \text{at the one pair } f_1=\mathsf{f}^{\langle\ell\rangle},\ f_2=1-\mathsf{h}^{\langle\ell\rangle}.$$

Then the oscillation hypothesis of Prop. 12.3(b) holds for $m\ge m_{1,i}$, with $\mathsf{a}_i = C_i\log 2$, $A_i = C_i\max\{\lambda_i,c_{1,i}\}$ and $\omega_i = 2^{1+c_{1,i}}\cdot 4800 b_{+,i}^5 M_i^2 L_i^5 C_i$, and the two bounds of (B)(1) hold with the power $\beta^{-1/2A}$, $A=\max_i A_i$, beyond some β_1. More generally, any oscillation bound for the far parts with thresholds $\mathsf{a}_i + A_i m$ on all sides $m\ge m_{1,i}$ gives $\beta^4 J(\beta)\le C\beta^{-1/2\max_i A_i}$ for large β. ■

Proof sketch. Index i dropped. For $m\ge m_1$ and integers $K,K'\ge \mathsf{a}+Am$ we have $K\ge C\log 2 + C\lambda m = C\log\ell$, the same for K', and $\ell^{c_1}L^{-K/C} = 2^{c_1}L^{c_1 m - K/C}\le 2^{c_1}$, since $K/C\ge c_1 m$. So the two torus functions differ by at most $2C2^{c_1}\|f_1\|_\sharp\|f_2\|_\sharp$; with $b^5\varepsilon^{-3}\le b_+^5 L^3$ and $\|f_1\|_\sharp,\|f_2\|_\sharp\le 40\sqrt{3}ML$, $\sup_{\log\mathsf{I}}|y_{m,K}-y_{m,K'}|\le\omega$. Enter in Prop. 12.3(b) as in Step 1 of the proof of Thm. AJ below, with $m_i^0 = m_{1,i}$; the second sentence is the same argument with the given thresholds. □

Proposition 12.15 (A threshold that does not grow with the side gives $J = 0$ on a half line)**.** Let (S_1), (S_2) and r_1 be as in (A)(2). Suppose that there are an integer K_* and numbers ω_i such that $\sup_{\log \mathsf{I}_i} |y^{(i)}_{m,K} - y^{(i)}_{m,K'}| \le \omega_i$ for all integers $K, K' \ge K_*$ and all $m \ge m_{0,i}$, $i = 1, 2$. Then there is β_* with $J(\beta) = 0$ for $\beta \ge \beta_*$, and on (β_*, ∞) the measure $\beta^4[P''(d\beta) - \hat{n}\,\mathrm{d}\beta]$ has a bounded density. In terms of one lattice: with $Y_N(\beta) = \beta^5[P''_N(\beta) - \hat{n}(\beta)]$, Prop. 12.6(d) puts $Y_N(\beta)$ within C^{ref} of $y_{m,K}(\log b)$ for $N = 2L^{m+K}$, $\beta = L^K b$, $b \in \mathsf{I}$, $m \ge m_0$. Given lowered choices, the oscillation bound of Step 5 of the proof of Thm. AJ gives $|Y_{L^k N}(L^k\beta) - Y_N(\beta)| \le \omega_A + 2C^{\mathrm{ref}}$ for all $k \ge 0$ at these N, β with $N/\beta \le 2b_+^{-1-1/A} L^{-k^{\mathrm{rt}}/A} \beta^{1/A}$; the hypothesis here asks such a bound at all of them. ■

Proof sketch. Enter in Prop. 12.3(c) with the data of Step 1 of the proof of Thm. AJ and with $\max\{K_*, k_1, k_2\}$ in the place of K_*: $\epsilon^{(i)}_m = c_{\mathrm{P},i} L_i^{-m}$ has the lower bound 0, and the hypothesis on Y^0 holds at $m^0 = m_{0,1}$ with $k^0 = \max\{K_*, k_1\}$, since $|y^{(1)}_{m_{0,1},K}| \le |y^{(1)}_{m_{0,1},k^0}| + \omega_1$ for $K \ge k^0$. Since P is the pressure, $J_P = J$. For the last sentence: $L^k N = 2L^{m+K+k}$ and $L^k\beta = L^{K+k}b$, and $K \ge k^{\mathrm{rt}} + Am$ reads $N/\beta \le 2b^{-1-1/A} L^{-k^{\mathrm{rt}}/A} \beta^{1/A}$. □

Proposition 12.16 (The inequality of two boxes at one pair, and along a label)**.** Let $a > 0$ and $\beta > 0$. For the periodic boxes $\underline{n} = (n_1, n_2, n_3)$, $n_\mu \ge 3$, of the lattice $a\mathbb{Z}^3$ at the coupling β let $F(\underline{n}) = \log Z(\underline{n})$, $|\underline{n}| = n_1 n_2 n_3$, $\Delta(\underline{n}) = F(\underline{n}) - |\underline{n}|P(\beta)$, write $F(N^3)$ and $\Delta(N^3)$ for the cube of side N, and $\mathcal{E}(l) = 2F(l, 2l, 2l) - F(2l, 2l, 2l)$.

(a) $\Delta(\underline{n}) \ge 0$; $\Delta(\underline{n}')/|\underline{n}'| \le \Delta(\underline{n})/|\underline{n}|$ for $\underline{n}' \ge \underline{n}$ in each component; and $\mathcal{E}(N) \le 8\Delta(N^3) - \Delta((2N)^3) \le 8\Delta(N^3)$ for $N \ge 3$.

(b) If $\mathcal{E}(l) \le x_0$ for one integer $l \ge 3$ and one number $0 < x_0 \le 0.069$, then, with $\theta = 12.9x_0 \le 0.8901$: $\mathcal{E}(2l) \le 12.9\,\mathcal{E}(l)^2$; $\Delta(N'^3) < 0.13736\,\theta^{N'/2l}$ for every $N' \ge 2l$; and $\mathcal{E}(l') \le 0.743\,\theta^{l'/2l}$ for every $l' \ge l$.

(c) If $\mathcal{E}(l) \le 0.069$ for one integer $l \ge 3$, the lattice $a\mathbb{Z}^3$ has exactly one periodic Gibbs state at β; it is a pure phase, and $J(\beta) = 0$.

(d) If the lattice has two periodic Gibbs states at β, then $\mathcal{E}(l) > 0.069$ for every $l \ge 3$, $\Delta(N^3) > 0.0086$ for every $N \ge 3$, and $\liminf_{j\to\infty} \Delta((2L^j)^3) \ge \log 2$ for every odd integer $L \ge 13$.

(e) At every $\beta > 0$, either $\mathcal{E}(l) > 0.069$ for every $l \ge 3$, or $\mathcal{E}(l') \to 0$ exponentially in l'.

(f) In a setting suppose $\mathrm{BI}[b, \ell_*, K_*; x_0]$, let $\underline{K} = \max\{K_*, 1\}$, $\theta = 12.9x_0$, and let ν_K be the one periodic Gibbs state of (c) at $a = a_K$, $\beta = L^K b$. Let $n \ge 2$, and let $X = (f_1, \dots, f_n)$ be real Lipschitz functions, none of them 0, with pairwise disjoint supports in $(-\frac{\ell''}{4}, \frac{\ell''}{4})^3$ for a side $\ell'' = 2L^{m''} \ge \ell_*$, $m'' \in \mathbb{N}$. There is $C(X) < \infty$, formed from X, ℓ'' and b and from no cutoff and no further side, such that for all $K \ge \underline{K}$ and all sides $\ell' = 2L^{m'} \ge \ell''$

$$|S^T_n[\nu_K](X) - S^T_{n;K,\ell',L^K b}(X^{\langle \ell' \rangle})| \le C(X)\,\theta^{\ell'/4\ell_*}.$$

■

Proof sketch. (a) Fix a direction of a box as the time and the two other sides, $n_\nu \ge 3$, the section Σ, and write $F(t, \Sigma)$ and $Z(t, \Sigma)$ for the box with $t \ge 3$ layers. On the Hilbert space of the configurations of the bonds of one layer (Section YM3_I:85) let C_Σ be the crossing operator of Section YM3_I:85, written K there, and D_Σ the multiplication by $e^{-\beta A_\Sigma}$, the weight of the plaquettes of one layer. C_Σ is positive (Prop. YM3_I:85.1), and it is of trace class: the series the rank-one expansion in the proof of Prop. YM3_I:85.1 in the proof of that clause writes it as $\sum_\alpha c_\alpha |\chi_\alpha\rangle\langle\chi_\alpha|$ with $\|\chi_\alpha\| \le 1$ and $c_\alpha \ge 0$, $\sum_\alpha c_\alpha < \infty$ the rank-one expansion in the proof of Prop. YM3_I:85.1.

So the transfer operator of one layer,

$$T_\Sigma := \mathsf{C}_\Sigma^{1/2} \mathsf{D}_\Sigma \mathsf{C}_\Sigma^{1/2},$$

is positive, as

$$\langle\psi, T_\Sigma\psi\rangle = \int e^{-\beta A_\Sigma} |\mathsf{C}_\Sigma^{1/2}\psi|^2 \,\mathrm{d}V,$$

and of trace class; like C_Σ and D_Σ it does not depend on the number of layers (Prop. YM3_I:85.2). Integrating the variables of the bonds between two layers turns the weight of a slab into a chain of kernels: the slab operator of the function 1 on $\vartheta \geq 2$ layers is $\mathsf{D}_\Sigma(\mathsf{C}_\Sigma\mathsf{D}_\Sigma)^{\vartheta-1}$, so that

$$\mathfrak{Y}[1] = \mathsf{C}_\Sigma^{1/2}\mathsf{D}_\Sigma(\mathsf{C}_\Sigma\mathsf{D}_\Sigma)^{\vartheta-1}\mathsf{C}_\Sigma^{1/2} = T_\Sigma^\vartheta$$

(equation YM3_I:(1744)).

The trace formula, Prop. YM3_I:85.2 with the function 1 on two slabs of 2 and $t-2$ layers, gives $Z(t,\Sigma) = \operatorname{Tr} T_\Sigma^t$ for $t \geq 4$; at $t = 3$ the integral over the three layers is $\operatorname{Tr}(\mathsf{C}_\Sigma\mathsf{D}_\Sigma)^3$, by the Hilbert–Schmidt kernel identity $\operatorname{Tr}(AB) = \iint A(y,x)B(x,y)\,\mathrm{d}x\,\mathrm{d}y$ used in the proof of Prop. YM3_I:85.2, applied to the operators $(\mathsf{C}_\Sigma\mathsf{D}_\Sigma)^2$ and $\mathsf{C}_\Sigma\mathsf{D}_\Sigma$, which have bounded kernels, and this is $\operatorname{Tr} T_\Sigma^3$ by the cyclicity of the trace. With the eigenvalues $\mu_i \geq 0$ of the positive compact operator T_Σ, counted with their multiplicities, the trace of a power is the sum of the powers of the eigenvalues, by computing the trace in an orthonormal eigenbasis, and

$$e^{F(t,\Sigma)} = Z(t,\Sigma) = \operatorname{Tr} T_\Sigma^t = \sum_i \mu_i^t \qquad \text{for every integer } t \geq 3,$$

with numbers $\mu_i \geq 0$ that do not depend on t, odd t and $t = 3$ included. By the spectral argument in the proof of Prop. YM3_I:86.5, $(\operatorname{Tr} T_\Sigma^t)^{1/t}$ does not increase in t; so $F/|\underline{n}|$ does not increase when a side grows, and as $F(N^3)/N^3 \to P$ (Lem. 12.1(a)), $P = \inf F/|\underline{n}|$, and $\Delta \geq 0$ with the monotonicity per site. In particular $F(N,2N,2N) \leq 4F(N^3)$, so $\mathcal{E}(N) \leq 8F(N^3) - F((2N)^3) = 8\Delta(N^3) - \Delta((2N)^3)$. (b) *One direction.* Fix a direction of a box as the time and the two other sides, the section Σ, and write $F(t,\Sigma)$ for the box with $t \geq 3$ layers. By (a), with $\lambda_0 > 0$ the largest of the numbers μ_i and $\mathsf{y}_j \in [0,1]$ the quotients μ_i/λ_0 of the others, $F(t,\Sigma) = t\log\lambda_0 + \log(1+\mathsf{r}_t)$ with $\mathsf{r}_t = \sum_j \mathsf{y}_j^t$. So r_t does not increase in t, $\mathsf{r}_{2t} \leq \mathsf{r}_t^2$, and

$$\mathsf{d}(t) := 2F(t,\Sigma) - F(2t,\Sigma) = 2\log(1+\mathsf{r}_t) - \log(1+\mathsf{r}_{2t}) \;\geq\; \log(1+\mathsf{r}_t) \;\geq 0.$$

Let $\mathsf{d}(t) \leq x < \log 2$, $\eta = e^x - 1 < 1$ and $\rho(\eta) = \eta/(1+\sqrt{1-\eta^2})$. Then $\mathsf{r}_t \leq \eta$, and $1 + 2\mathsf{r}_t/(1+\mathsf{r}_t^2) \leq (1+\mathsf{r}_t)^2/(1+\mathsf{r}_{2t}) = e^{\mathsf{d}(t)} \leq 1+\eta$; as $r \mapsto 2r/(1+r^2)$ increases on $[0,1]$ and takes the value η at $r = \rho(\eta)$, we get $\mathsf{r}_t \leq \rho(\eta)$, and so

$$\mathsf{d}(2t) \leq 2\log(1+\mathsf{r}_{2t}) \leq 2\log(1+\mathsf{r}_t^2) \leq \psi(x) := 2\log(1+\rho(e^x-1)^2).$$

The function ψ increases on $[0,\log 2)$, and, as $(e^y-1)/y$ increases, $\psi(y) \leq 2\kappa(y_0)^2 y^2$ for $0 \leq y \leq y_0 < \log 2$, with $\kappa(y_0) = \frac{e^{y_0}-1}{y_0}\big(1+\sqrt{1-(e^{y_0}-1)^2}\big)^{-1}$; and $\kappa(0.069) \leq 0.51832$, $\kappa(0.14056) \leq 0.53995$, $\kappa(0.29264) \leq 0.59869$.

Doubling. For a box $\underline{n}$ and a direction μ let $\mathsf{d}^{(\mu)}(\underline{n})$ be $2F(\underline{n})$ minus F of the box with n_μ doubled. By the above, read in the direction μ, $\mathsf{d}^{(\mu)} \geq 0$, and $\mathsf{d}^{(\mu)}$ of the box with n_μ doubled is at most $\psi(\mathsf{d}^{(\mu)}(\underline{n}))$ when $\mathsf{d}^{(\mu)}(\underline{n}) < \log 2$. Let $\epsilon = \mathcal{E}(l) \leq 0.069$. As F does not depend on the order of the three sides, $\epsilon = \mathsf{d}^{(1)}(l,2l,2l) = \mathsf{d}^{(2)}(2l,l,2l) = \mathsf{d}^{(3)}(2l,2l,l)$. Then:

(i) $\mathsf{a} := \mathsf{d}^{(2)}(2l,2l,2l) \leq \psi(\epsilon) \leq 2\kappa(0.069)^2\epsilon^2 \leq 0.53732\,\epsilon^2 \leq 0.03708\,\epsilon$;

(ii) since $F(2l,4l,l) \leq 2F(2l,2l,l)$ and $F(2l,4l,2l) = 2F(2l,2l,2l) - \mathsf{a}$, we have $\mathsf{d}^{(3)}(2l,4l,l) \leq 2\,\mathsf{d}^{(3)}(2l,2l,l) + \mathsf{a} = 2\epsilon + \mathsf{a} \leq 2.03708\,\epsilon \leq 0.14056$, so $\mathsf{b} := \mathsf{d}^{(3)}(2l,4l,2l) \leq \psi(2\epsilon+\mathsf{a}) \leq 2.41970\,\epsilon^2 \leq 0.16696\,\epsilon$;

(iii) since $F(l,4l,4l) \leq 4F(l,2l,2l)$ and $F(2l,4l,4l) = 2F(2l,4l,2l) - \mathsf{b} = 4F(2l,2l,2l) - 2\mathsf{a} - \mathsf{b}$, we have $\mathsf{c} := \mathsf{d}^{(1)}(l,4l,4l) \leq 4\epsilon + 2\mathsf{a} + \mathsf{b} \leq 4.24112\,\epsilon \leq 0.29264 < \log 2$;

(iv) $\mathcal{E}(2l) = \mathsf{d}^{(1)}(2l,4l,4l) \leq \psi(\mathsf{c}) \leq 2\cdot 0.59869^2 \cdot 4.24112^2\,\epsilon^2 \leq 12.8943\,\epsilon^2$.

So $\mathcal{E}(2l) \le 12.9\,\mathcal{E}(l)^2$ when $\mathcal{E}(l) \le 0.069$. As $12.9 \cdot 0.069 = 0.8901 < 1$, induction shows that $\epsilon_k = \mathcal{E}(2^k l)$ does not increase in k and that $12.9\,\epsilon_{k+1} \le (12.9\,\epsilon_k)^2$; so $12.9\,\epsilon_k \le (12.9\,\epsilon_0)^{2^k} \le \theta^{2^k}$, that is, $\mathcal{E}(2^k l) \le x_k := \theta^{2^k}/12.9 \le x_0 \le 0.069$.

Cubes. Put $D(N) = 8F(N^3) - F((2N)^3) = 8\Delta(N^3) - \Delta((2N)^3)$. At $n = 2^k l$, with a and b formed at n in the place of l, the three brackets of

$$D(2n) = \big[8F((2n)^3) - 4F(2n,4n,2n)\big] + \big[4F(2n,4n,2n) - 2F(2n,4n,4n)\big] + \big[2F(2n,4n,4n) - F((4n)^3)\big]$$

are $4\mathsf{a}$, $2\mathsf{b}$ and $\mathcal{E}(2n)$; so $D(2n) \le (2.14928 + 4.83940 + 12.8943)\,\epsilon_k^2 \le 20\,\epsilon_k^2$. By (a) the density $F/|\underline{n}|$ decreases to P along the cubes of the sides $2^s M$, $s \to \infty$, so that $\Delta(M^3) = \sum_{s\ge 0} 8^{-(s+1)} D(2^s M)$. At $M = 2^j l$, $j \ge 1$, every $D(2^s M)$ is at most $20\,\epsilon_{s+j-1}^2 \le 20\,\epsilon_{j-1}^2$, and

$$\Delta(M^3) \le \tfrac{20}{7}\,\epsilon_{j-1}^2 \le \tfrac{20}{7\cdot 12.9^2}\,\theta^{2^j} \le 0.017170\,\theta^{M/l}.$$

In particular $\Delta((2n)^3) \le \frac{20}{7}\,\mathcal{E}(n)^2 \le \frac{20}{7}x_k^2$ at $n = 2^k l$. With $\mathcal{E}(N) \le 8\Delta(N^3)$ from (a) and $\frac{20}{7}\cdot 64 < 182.9$ this gives $\Delta((2N)^3) \le 183\,\Delta(N^3)^2$ when $\Delta(N^3) \le 0.008625$, for then $\mathcal{E}(N) \le 0.069$; if moreover $\Delta(N^3) > 0$, the hypothesis of (b) holds at $l = N$ with $x_0 = 8\Delta(N^3)$, so with $\theta = 103.2\,\Delta(N^3)$, as $12.9 \cdot 8 = 103.2$. For $N' \ge 2l$ let $j \ge 1$ with $M = 2^j l \le N' < 2M$; by (a), $\Delta(N'^3) \le (N'/M)^3 \Delta(M^3) < 8 \cdot 0.017170\,\theta^{M/l} \le 0.13736\,\theta^{N'/2l}$, since $M/l > N'/2l$ and $0 < \theta < 1$.

All sides $l' \ge l$. At $n = 2^k l$, the identity $\mathcal{E}(n) = 2\Delta(n,2n,2n) - \Delta((2n)^3)$ and the two bounds $\mathcal{E}(n) \le x_k$, $\Delta((2n)^3) \le \frac{20}{7}x_k^2$ give $\Delta(n,2n,2n) \le \frac{x_k}{2} + \frac{10}{7}x_k^2$. Let $2^k l \le l' < 2^{k+1} l$. As $\Delta \ge 0$, and by (a) applied to the boxes $(2^k l, 2^{k+1} l, 2^{k+1} l) \le (l', 2l', 2l')$, whose volumes have the ratio $(l'/2^k l)^3 < 8$,

$$\mathcal{E}(l') \le 2\Delta(l',2l',2l') \le 16\,\Delta(2^k l, 2^{k+1} l, 2^{k+1} l) \le 8x_k + \tfrac{160}{7}x_k^2 \le \Big(\frac{8}{12.9} + \frac{160\cdot 0.8901}{7\cdot 12.9^2}\Big)\theta^{2^k} \le 0.743\,\theta^{2^k},$$

using $x_k^2 = \theta^{2^{k+1}}/12.9^2 \le 0.8901\,\theta^{2^k}/12.9^2$; and $\theta^{2^k} \le \theta^{l'/2l}$, since $2^k > l'/2l$. (c) Let L be an odd integer ≥ 13, an allowed blocking factor, and $K \in \mathbb{N}$. Wilson's action reads the lattice only as a graph, so the dilation $x \mapsto (a_K/a)x$ carries the periodic boxes and the periodic Gibbs states of $a\mathbb{Z}^3$ at β onto those of $\mathbb{L}_K$, with the same partition functions, and we take $a = a_K$. The cube with $(2L^j)^3$ sites, $j > K$, is then the torus T_ℓ, $\ell = 2L^{j-K}$, at the spacing a_K; we put $p_\ell = \ell^{-3}\log Z_{K,\ell,\beta}$. The domination theorem (Prop. YM3_I:91.9) bounds the number n of pairwise mutually singular periodic Gibbs states by $\liminf_j e^{\Delta((2L^j)^3)}$. Indeed, let Ψ be a function of the cell $[-\frac{\ell}{4}, \frac{\ell}{4}]^3$ of the board of $\mathbb{L}_K$ with the side $\frac{\ell}{2}$, half the side of T_ℓ, and the offset 0, and let ν be a periodic Gibbs state. Prop. YM3_I:91.9 gives $|\nu(\Psi)| \le \liminf_{\ell'\to\infty} \|\Psi\|_{\mathrm{ch},\ell'}$, and the comparison of two sides, [7, Cor. B.1](e), gives $\|\Psi\|_{\mathrm{ch},\ell'} \le \|\Psi\|_{\mathrm{ch},\ell}\, e^{(\ell/2)^3(p_\ell - p_{\ell'})}$ for $\ell' \ge \ell$, where $(\ell/2)^3(p_\ell - p_{\ell'}) \to \frac{1}{8}\Delta$, Δ the excess of the cube with ℓ/a_K sites a side, as $p_{\ell'} \to a_K^{-3}P$; so

$$|\nu(\Psi)| \le e^{\Delta/8}\,\|\Psi\|_{\mathrm{ch},\ell},$$

and $\|\Psi\|_{\mathrm{ch},\ell}^8$ is the mean, in the torus measure $\tilde{\mu}$ of T_ℓ, of the product of the eight mirror copies of Ψ (equation [7, eq. 189]). Mutually singular laws $\nu_1, \dots, \nu_n$ have pairwise disjoint carriers, which local events approximate: for $\epsilon > 0$ there are pairwise disjoint local events A_i with $\nu_i(A_i) \ge 1 - n\epsilon$, and for large ℓ they are events of the bonds of the cell. At their indicators the last bound reads $\nu_i(A_i) \le e^{\Delta/8}\tilde{\mu}(E_i)^{1/8}$, where E_i is the intersection of the eight mirror images of A_i; the E_i are pairwise disjoint, so $n(1-n\epsilon)^8 \le e^{\Delta}$. By (b) the right member tends to 1 as $\ell \to \infty$; as ϵ is arbitrary, $n \le 1$. Every periodic Gibbs state is an integral of ergodic periodic Gibbs states, and two different ergodic ones are mutually singular; so there is one, it is extreme, and $J(\beta) = 0$ by Lem. 12.1(b). (d) The first statement is (c); the second follows from (a), as $8 \cdot 0.008625 = 0.069$; the third from the bound of (c) with $n = 2$, at every odd $L \ge 13$, two different ergodic periodic Gibbs states being

mutually singular. (e) is (b) and (c). (f) Let G be a product of centred $\mathcal{O}_K(f_i)$, $i \in I \subset \{1,\dots,n\}$, centred at the means on $T_{\ell''}$. Its chessboard seminorm $\|G\|_{\mathrm{ch},\ell''}$, on the board of $T_{\ell''}$ whose cell holds the supports, has as its eighth power a centred moment, on the fixed torus $T_{\ell''}$, of $8|I|$ observables with pairwise disjoint supports, the mirror copies of the $\mathcal{O}_K(f_i)$; it is a polynomial in cumulants of orders ≥ 2, each of which converges as $K \to \infty$ by Thm. YM3_I:AS; so it is bounded in K. Let μ' be the measure of $T_{\ell'}$ at the spacing a_K, read on $\mathbb{L}_K$, $Q' = e^{\Delta'/8}$ with Δ' the excess of the cube with ℓ'/a_K sites a side, $Q'' = e^{\Delta''/8}$ likewise at ℓ'', $p_\ell = \ell^{-3} \log Z_{K,\ell,L^K b}$ as in the proof of (c), and $\mathsf{g} = Q'' \|G\|_{\mathrm{ch},\ell''}$. Then

$$|\nu_K(G) - \mu'(G)| \leq 10.7\,[Q'(Q'-1)]^{1/2}\,\mathsf{g} \qquad \text{if } \Delta' \leq 1.81. \tag{12.viii}$$

Four facts enter. (i) For a real function Ψ of the cell $[-\frac{\ell'}{4}, \frac{\ell'}{4}]^3$ of the board of $\mathbb{L}_K$ with the side $\frac{\ell'}{2}$ and the offset 0, the coarse board, $|\nu_K(\Psi)| \leq Q' \|\Psi\|_{\mathrm{ch},\ell'}$, by Prop. YM3_I:91.9 and [7, Cor. B.1](e), as in the proof of (c); G is such a function (the plaquette-support argument in [7, §B.1]). (ii) The board with the side $\frac{\ell''}{2}$ and the offset 0, the fine board, has the cell $[-\frac{\ell''}{4}, \frac{\ell''}{4}]^3$, which holds the supports. As $\ell'/\ell'' = L^{m'-m''}$ is odd, the coarse board is coarser than the fine one, and each of its mirror maps carries the cells of the fine board onto cells of the fine board, with the same parity (the proof of Lem. YM3_I:104.1).

So the mirror copies $G^{[k]}$ of G in the eight cells k of the coarse board of $T_{\ell'}$ are mirror copies of G in eight different cells of a period box of the fine board, and the chessboard estimate on the fine board of $T_{\ell'}$ ([7, Cor. B.1](b), (c), which rests on Prop. YM3_I:86.3) gives $|\mu'(\prod_{k\in S} G^{[k]})| \leq \|G\|_{\mathrm{f}}^{|S|}$ for every set S of these cells, $\|G\|_{\mathrm{f}}$ being the seminorm of G at the side ℓ' on the fine board. (iii) By [7, Cor. B.1](e) on the fine board, at the sides $\ell' \geq \ell''$, and as (a) gives $\Delta \geq 0$ at the cube with ℓ'/a_K sites a side, that is, $p_{\ell'} \geq a_K^{-3} P$: $\|G\|_{\mathrm{f}} \leq \|G\|_{\mathrm{ch},\ell''}\, e^{(\ell''/2)^3 (p_{\ell''} - p_{\ell'})} \leq \mathsf{g}$. (iv) μ' is invariant under the isometries of $T_{\ell'}$, so $\mu'(G^{[k]}) = \mu'(G)$, and $|\mu'(G)| \leq \|G\|_{\mathrm{f}} \leq \mathsf{g}$.

Now let t be real, $\Psi = 1 + tG$, $x = |t|\mathsf{g}$ and $R(x) = (1+x)^8 - 1 - 8x$. By equation [7, eq. 189] on the coarse board, (ii) and (iv),

$$\|\Psi\|_{\mathrm{ch},\ell'}^8 = \mu'\Big(\prod_k (1 + tG^{[k]})\Big) = 1 + 8t\mu'(G) + \sum_{|S|\geq 2} t^{|S|} \mu'\Big(\prod_{k\in S} G^{[k]}\Big) \leq 1 + 8t\mu'(G) + R(x),$$

so $\|\Psi\|_{\mathrm{ch},\ell'} \leq 1 + t\mu'(G) + \frac{1}{8}R(x)$, as $(1+y)^{1/8} \leq 1 + \frac{y}{8}$ for $y \geq -1$; and $1 + t\nu_K(G) = \nu_K(\Psi) \leq Q' \|\Psi\|_{\mathrm{ch},\ell'}$ by (i). Hence $t(\nu_K(G) - \mu'(G)) \leq (Q'-1)(1+x) + \frac{1}{8}Q'R(x)$ for both signs of t, with $Q' \geq 1$ by (a). At $\mathsf{g} = 0$ this gives $\nu_K(G) = \mu'(G)$; at $\mathsf{g} > 0$

$$|\nu_K(G) - \mu'(G)| \leq \mathsf{g}\Big[\frac{(Q'-1)(1+x)}{x} + \frac{Q'R(x)}{8x}\Big] \qquad \text{for every } x > 0.$$

For $Q' > 1$ and $x^2 = (Q'-1)/Q'$ the bracket is $[Q'(Q'-1)]^{1/2}(1 + x + R(x)/8x^2)$, and $1 + x + R(x)/8x^2 \leq 10.7$ for $x \leq 0.45$, that is, for $\Delta' \leq 1.81$; for $Q' = 1$ let $x \to 0$. This is eq. 12.viii. By (b) at $l = l_K = \ell_*/2a_K$, $\Delta' < 0.13736\,\theta^{\ell'/\ell_*}$ and $\Delta'' < 0.13736$; so $Q'' < 1.02$, $[Q'(Q'-1)]^{1/2} \leq 0.14\,\theta^{\ell'/2\ell_*}$, and eq. 12.viii gives $|\nu_K(G) - \mu'(G)| \leq 2\,\theta^{\ell'/2\ell_*} \|G\|_{\mathrm{ch},\ell''}$. A cumulant of order $n \geq 2$ is one polynomial in the means $\mathfrak{s}(G_I)$, $I \subset \{1,\dots,n\}$, for both laws $\mathfrak{s} \in \{\nu_K, \mu'\}$; by (iv) and the last bound these means are bounded in K and ℓ', and as $\theta^{\ell'/2\ell_*} \leq \theta^{\ell'/4\ell_*}$ this gives the display. □

12.1.6. *Proof of Thm. AJ. Step 1: (A)(1) and (A)(2).* Take two settings at one radius with $L_2 = L_1 + 2$ (Lem. 12.7) and $r_1 > 0$ with $13 r_1 b_{+,i} \leq 1$. We enter in Prop. 12.3: $\beta_0 = \max_i L_i b_{-,i}$; $C_{\mathrm{low}} = 871 C_1^{\mathrm{ref}}$; $\lambda_i = \log L_i$, so $\lambda_1/\lambda_2 \notin \mathbb{Q}$ by Lem. 12.7; $\alpha_i = \log b_{-,i}$ and $\varkappa_i = \frac{1}{2}[\log(b_{+,i}/b_{-,i}) - \lambda_i] > 0$, so that $\mathsf{J}_i = \log \mathsf{I}_i$; k_i the least integer ≥ 1 with $L_i^{k_i} b_{-,i} > \beta_0$; $m_i^0 = m_{0,i}$; $P_{m,K}^{(i)} = P_N$ and $n_{m,K}^{(i)} = n_{\eta_K}(\mu_{N,\beta})$ with $N = 2L_i^{m+K}$ and $\eta_K = L_i^{K-1}$. By Prop. 12.6(b)–(d), eq. 12.iv holds with

$\epsilon_m^{(i)} = c_{\mathrm{P},i} L_i^{-m}$ and $C_N = \max_i C_i^{\mathrm{ref}}$, eq. 12.iii holds on (β_0, ∞), and by eq. 12.vi the function $y_{m,K}^{(i)}$ of Prop. 12.3 is the far part of the setting (S_i). By the convergence on each torus (Thm. YM3_I:AS at ς_i; the supports of $\mathsf{f}^{\langle\ell\rangle}$ and $1-\mathsf{h}^{\langle\ell\rangle}$ have no common point), for each m the far part converges as $K \to \infty$, uniformly on J_i; so there are non-decreasing integers $K^{(i)}(m) \ge k_i$, the maxima over $m_{0,i} \le m' \le m$ of the thresholds at the single sides, with

$$\sup_{\mathsf{J}_i} |y_{m,K}^{(i)} - y_{m,K'}^{(i)}| \le 1 \qquad (K, K' \ge K^{(i)}(m)). \tag{12.ix}$$

At $m^0 = m_{0,1}$ and $k^0 = K^{(1)}(m^0)$ this gives $Y^0 = \sup_{\mathsf{J}_1} |y_{m^0,k^0}^{(1)}| + 1 < \infty$, the far part at one cutoff being bounded on J_1. Prop. 12.3(a) with $\omega_i = 1$, large m_i and $\bar K_i = K^{(i)}(m_i)$ gives, with A_0 independent of m_1, m_2 and for $0 \le \delta \le 1$,

$$\limsup_{s\to\infty} e^{4s} J(e^s) \le (\mathsf{A}_0 + 134) \max_i (c_{\mathrm{P},i} L_i^{-m_i})^{1/2}, \quad \limsup_{s\to\infty} |\mathbb{M}|([s, s+\delta]) \le (2\mathsf{A}_0 + C_{\mathrm{low}} + 134)\delta + (\mathsf{A}_0 + 134) \max_i (c_{\mathrm{P},i} L_i^{-m_i})^{1/2}.$$

Letting $m_1, m_2 \to \infty$ gives (A)(1), and (A)(2) with $C = 2\mathsf{A}_0 + C_{\mathrm{low}} + 134$, first for $\delta \le 1$ and then for all $\delta > 0$ by cutting $[s, s+\delta]$ into intervals of length at most 1. The equivalent form of (A)(1) is Lem. 12.1(b).

Step 2: (A)(3). Let the two settings have one value of $\mathfrak{p}_0$; such pairs exist with $\mathfrak{p}_0 \ge a$ by Lem. 12.7. The pair $(\mathsf{f}^{\langle\ell\rangle}, 1-\mathsf{h}^{\langle\ell\rangle})$ has the spread ε and norms $\|\cdot\|_\sharp \le 40\sqrt{3}ML$ (values in $[0,1]$, Lipschitz constants at most $\sqrt{3}L$), and $b^5\varepsilon^{-3} \le b_+^5 L^3$. By the rate of the convergence on each torus and Lem. 12.9, in each setting and for the sides with $M\mathcal{R}(b_+) \le L^m$,

$$\sup_{\log \mathsf{I}} |y_{m,K} - y_{m,K'}| \le 2b_+^5 L^3 (40\sqrt{3}ML)^2 C' \ell^9 K^{\mathfrak{p}_2} L^{-K/10} \le 1 \qquad \text{for } K, K' \ge c(1+\ell^{3/2\mathfrak{p}_0}),$$

with the rate constant $C' = \mathsf{C}_2'(\mathsf{I})$ of Thm. YM3_I:AS at $n = 2$ and $s = 0$, which is admitted as $b_- \ge \gamma^{-2}$ and $M\mathcal{R}(b_+) \le L^m$, and a number c independent of m, because $\log \ell \le \frac{2\mathfrak{p}_0}{3}\ell^{3/2\mathfrak{p}_0}$. We may therefore take $K^{(i)}(m) \le c(1 + (2L_i^m)^{3/2\mathfrak{p}_0})$ in Step 1. At a large s let m_i be the largest integer with $\alpha_i + 2\varkappa_i + \lambda_i K^{(i)}(m_i) \le s$; then $s < c'(1 + L_i^{3(m_i+1)/2\mathfrak{p}_0})$, so $L_i^{-m_i} \le c'' s^{-2\mathfrak{p}_0/3}$, and Prop. 12.3(a) gives $e^{4s}J(e^s) \le Cs^{-\mathfrak{p}_0/3}$ and $|\mathbb{M}|([s,s+\delta]) \le (2\mathsf{A}_0 + C_{\mathrm{low}} + 134)\delta + Cs^{-\mathfrak{p}_0/3}$ for large s; the second is at most $C_{\mathrm{A}}\delta$ when $C_{\mathrm{A}} \ge \max\{2\mathsf{A}_0 + C_{\mathrm{low}} + 135, C\}$ and $C_{\mathrm{A}} s^{-\mathfrak{p}_0/3} \le \delta$. On bounded sets of $\beta \ge 3$ use $J \le 6$. For a real q_0 choose $\mathfrak{p}_0 \ge 3q_0$; as $\log\beta \ge 1$ for $\beta \ge 3$, $(\log\beta)^{-\mathfrak{p}_0/3} \le (\log\beta)^{-q_0}$. In any setting, for $(K,\beta) \in \mathfrak{d}(g)$ we have $b = a_K\beta \ge g^{-2} \ge 4$, so $\log\beta \ge K\lambda$ and $a_K^{-4} J(\beta) = b^{-4}\beta^4 J(\beta) \le g^8 C (K\lambda)^{-q_0}$.

Step 3: (A)(4). In one of the two settings let $b \in \mathsf{I}$, $a = a_K$, $\beta = L^K b$. By Prop. 12.6(c) at $m = m_0$, $0 \le a^{-3}[P_{N_0}(\beta) - P(\beta)] \le 3C_{\mathrm{v}}$, which is $-3C_{\mathrm{v}} \le u_K \le 0$. u_K is the difference of a convex and a smooth function of b; $u_K'(b\pm) = a^{-4}[P'(\beta\pm) - P'_{N_0}(\beta)]$, and as measures on (b_-, b_+), by eq. 12.vi at $m = m_0$,

$$u_K''(db) = b^{-4}\mathbb{M}_K(db) + \vartheta_K(b)\,\mathrm{d}b, \qquad \vartheta_K = -b^{-5}[\beta^5(n_{\eta_K}(\mu_{N_0,\beta}) - \hat n(\beta)) + y_{m_0,K}(\log b)],$$

where $\mathbb{M}_K$ is the image of $\mathbb{M}$ under $s \mapsto e^{s-K\lambda}$. The far parts at m_0 are bounded at each cutoff and converge by eq. 12.ix, so $Y_0^* = \sup_{K\ge1}\sup_{\log\mathsf{I}} |y_{m_0,K}| < \infty$, and by Prop. 12.6(d), $|\vartheta_K| \le b_-^{-5}(C^{\mathrm{ref}} + Y_0^*)$. By the last assertion of Prop. 12.6(d), in the setting at hand, $P''(d\beta) \ge [\hat n(\beta) - 871 C^{\mathrm{ref}}\beta^{-5}]d\beta$ on (Lb_-, ∞), which contains $L^K(b_-, b_+)$ for every $K \ge 1$; so $\mathbb{M} \ge -871 C^{\mathrm{ref}} ds$ on $(\log(Lb_-), \infty)$, and, as $ds = db/b$, $b^{-4}\mathbb{M}_K(db) \ge -871 C^{\mathrm{ref}} b_-^{-5} db$ on (b_-, b_+). So $u_K'' \ge -C_{\mathrm{sc}}\,\mathrm{d}b$ with $C_{\mathrm{sc}} = b_-^{-5}(871C^{\mathrm{ref}} + C^{\mathrm{ref}} + Y_0^*)$, for every $K \ge 1$, the numbers C^{ref}, b_- and Y_0^* being those of the setting at hand.

As $\log(b+h) - \log b \le h/b_-$, (A)(2) gives $\limsup_K \sup\{|u_K''|([b, b+h]) : b_- < b,\ b+h < b_+\} \le C_u h$ for every $h > 0$, with $C_u = b_-^{-5}(C + C^{\mathrm{ref}} + Y_0^*)$, the upper limit of (A)(2) being over the real

$s \to \infty$ and so uniform in b. Let $I' = [\alpha, \alpha'] \subset (b_-, b_+)$. By the last bound and a covering, $\sup_{K \geq K_1} |u_K''|(I') < \infty$ for some K_1; the mean of u_K' over I' is $(u_K(\alpha') - u_K(\alpha))/|I'|$, of modulus at most $3C_{\mathrm{v}}/|I'|$, and $u_K'(\cdot\pm)$ differs from it by at most $|u_K''|(I')$; so the u_K, $K \geq K_1$, are bounded and have one Lipschitz constant on I'. The theorem of Arzelà and Ascoli and a diagonal sequence give the subsequence.

For $b, b+h \in I'$, $|u_K'(b\pm) - h^{-1}(u_K(b+h) - u_K(b))| \leq |u_K''|([b, b+h])$; so along the subsequence $\limsup_{K,K'} \sup_{I'} |u_K'(\cdot\pm) - u_{K'}'(\cdot\pm)| \leq 2C_u h$ for every $h > 0$: the one-sided derivatives converge uniformly on I' to a function w, $u(b') - u(b) = \int_b^{b'} w$, and $|w(b') - w(b)| \leq \limsup_K |u_K''|([b, b']) \leq C_u|b' - b|$; so $u' = w$. The other properties pass to uniform limits. Last, by Lem. 12.1(b) on $\mathbb{L}_K$, $\nu(\mathcal{O}_K(f)) = 3a^{-4}e_\nu r_K(f)$ with $-3e_\nu \in [P'(\beta-), P'(\beta+)]$; the torus measure $\tilde{\mu} = \tilde{\mu}_{K,\ell_0,\beta}$ is invariant under the isometries of $\mathbb{L}_K$, so $\tilde{\mu}(\mathcal{O}_K(f)) = 3a^{-4}e_* r_K(f)$ with $-3e_* = P_{N_0}'(\beta)$; subtract.

Step 4: (E). (i) Let first $b' \geq \max\{\gamma^{-2}, e^{\mathsf{Y}\natural}\}$ and $M\mathcal{R}(b'') \leq L^m$. Discarding (Prop. YM3_I: 115.2, its lost-mass and cumulant bounds) holds with any number ϵ_{R} that bounds the lost mass, $\sum_{h: Z_t(h) \neq \emptyset} \mathsf{t}(h) \leq \epsilon_{\mathrm{R}} Z$, in the place of the bound on the lost mass that its statement takes from the definitions preceding Prop. YM3_I:115.2: if $\epsilon_{\mathrm{R}} \leq \frac{1}{2}$, then the generating function T of the kept histories is entire, $T(0) \geq \frac{1}{2}Z$, and

$$|S^T_{n;K,\ell,\beta}(\underline{f}) - \partial_z^n \log T(0)| \leq 4\, \mathsf{op}_n \epsilon_{\mathrm{R}} \prod_i 6\ell^3 L^{4K} \|f_i\|_\infty .$$

Indeed, with the bare density ρ_0 and the weight $\mathfrak{w}_t^{\mathrm{sf}} \geq 0$ of the definitions preceding Prop. YM3_I: 115.2, $\Lambda(F) = \int F \rho_0 \, \mathrm{d}U$ and $\Lambda'(F) = \int F\, \mathfrak{w}_t^{\mathrm{sf}} \rho_0 \, \mathrm{d}U$ are positive functionals on the gauge-invariant bounded Borel functions, $\Lambda - \Lambda'$ is positive by Prop. YM3_I:115.2, its positive-domination inequality, $\Lambda(1) = Z$, and $T(z) = \Lambda'(e^{\sum_i z_i \mathcal{O}_K(f_i)})$. The proof of equation YM3_I:(2190) first bounds the lost mass by the sum of the histories with $Z_t(h) \neq \emptyset$, before estimating that sum. It therefore gives $\Lambda(1) - \Lambda'(1) \leq \epsilon_{\mathrm{R}} \Lambda(1)$; and the statement on two nearby positive functionals and their cumulants ([7, Lem. K.2], with $|\mathcal{O}_K(f_i)| \leq 6\ell^3 L^{4K} \|f_i\|_\infty$) gives $T(0) = \Lambda'(1) \geq \frac{1}{2}Z$ and the display, the cumulants of Λ and of Λ' being $S^T_{n;K,\ell,\beta}(\underline{f})$ and $\partial_z^n \log T(0)$.

These are the last two steps of the proof of Prop. YM3_I:115.2, made with ϵ_{R} in the place of that bound. The proof of Thm. YM3_I:AR (the comparison of the cutoffs K and $K+1$, with its three lemmas) uses of the bound ϵ_{R} only these two facts, that it bounds the lost mass and that it is at most $\frac{1}{2}$. Its error terms are of three kinds: terms exponentially small in K times $N_{[d]} = \ell^3 L^{3d}$ at depths $d \leq K/9 + 1$; the term $8\, \mathsf{op}_n \epsilon_{\mathrm{R}} \prod_i 6\ell^3 L^{4(K+1)} \|f_i\|_\infty$; and, for $n \geq 2$ and spread d, a contact term at most $c\, \ell^3 L^{3u} e^{-\mathsf{d} L^u / M}$, with the depth $u = \lceil K/180(3n+4) \rceil$.

The bound on the lost mass with which the thresholds of Thm. YM3_I:AS are formed carries a factor $e^{\mathrm{const}\, \ell^3}$. We take instead $\epsilon_{\mathrm{R}} = \mathsf{R}_{t_1}$ of Lem. 12.8 at the level $t_1 = K - u$: since $\log(L^u b) \geq u\lambda \geq K\lambda/180(3n+4)$ and $2\mathfrak{p}_0 \geq 2$, $\mathsf{R}_{t_1} \leq c\, \ell^3 L^{3u} e^{-c'' K^{2\mathfrak{p}_0}}$, which beats the factor ℓ^3 and the products of the crude bound. The most demanding of the remaining conditions is $\ell^3 L^{-K/18} \leq c$. Hence there is an integer $k_\flat$, free of m, ℓ and K, such that for $K \geq k_\flat + 54m$ all conditions hold and

$$|S_{K+1} - S_K| \leq C_n \ell^{3(n+1)} K^{\mathsf{p}_n} L^{-K/10} \Pi .$$

Summing over the cutoffs, with $C_n \sum_{k \geq K} k^{\mathsf{p}_n} L^{-k/10} \leq C_n' K^{\mathsf{p}_n} L^{-K/10}$, which defines C_n', gives (a) with $k_\flat$ and C_n' in this range. For $K \leq k_\flat + 54m$ the crude bound and $L^{4n(k_\flat + 54m)} \leq L^{4nk_\flat} \ell^{216n}$ give $|S_K| \leq \mathsf{op}_n 6^n L^{4nk_\flat} \ell^{219n} \Pi$, and for larger K (a) compares S_K with $S_{k_\flat + 54m}$; this gives (b) with $C_n^{\mathrm{pol}} = \mathsf{op}_n 6^n L^{4nk_\flat} + C_n' \sup_k k^{\mathsf{p}_n} L^{-k/10}$, as $3(n+1) \leq 219n$. (ii) Let $\jmath \geq 0$ be the least integer with $L^\jmath \geq \max\{\gamma^{-2}/b', e^{\mathsf{Y}\natural}/b', b''\}$ and $L^{\jmath/2} \geq M(4r/e)^r$. For $K > \jmath$, Lem. YM3_I:118.2 gives, with $f^{(\jmath)} = f(L^{-\jmath}\cdot)$,

$$S^T_{n;K,\ell,L^K b}(\underline{f}) = L^{\jmath n} S^T_{n;K-\jmath, L^\jmath \ell, L^{K-\jmath}(L^\jmath b)}(\underline{f}^{(\jmath)}) :$$

both members are cumulants of the same variables on the same periodic cube with ℓL^K sites a side, since $\mathcal{O}_K(f) = L^\jmath \mathcal{O}_{K-\jmath}(f^{(\jmath)})$ under the dilation by $L^\jmath$. The dilation keeps the supremum, divides the Lipschitz constant by $L^\jmath$ and multiplies the spread by $L^\jmath$, so $\|f^{(\jmath)}\|_\sharp \le \|f\|_\sharp$. The window $L^\jmath[b', b'']$ lies above both floors; and as $\mathcal{R}(x) < L(\log x)^r$, $\log(L^\jmath b'') \le 2\jmath\lambda$ and $(2\jmath\lambda)^r \le (4r/e)^r L^{\jmath/2}$, the sides $2L^{m+\jmath}$, $m \ge 1$, satisfy $M\mathcal{R}(L^\jmath b'') \le L^{m+\jmath}$. So (i) applies on the dilated data, with numbers $k_{\flat,\jmath}$, $C'_{n,\jmath}$, $C^{\mathrm{pol}}_{n,\jmath}$; as $n + 3(n+1) + \frac{1}{10} \le 4n+4$ and $K - \jmath \ge k_{\flat,\jmath} + 54(m+\jmath)$ for $K \ge k_{\flat,\jmath} + 55\jmath + 54m$, this gives (a) with $k^{\mathrm{rt}} = k_{\flat,\jmath} + 55\jmath$ and $C^{\mathrm{rt}} = L^{\jmath(4n+4)} C'_{n,\jmath}$, and $|S_K| \le L^{220n\jmath} C^{\mathrm{pol}}_{n,\jmath} \ell^{219n} \Pi$ for $K > \jmath$.

For $1 \le K \le \jmath$ the crude bound gives $|S_K| \le \mathsf{op}_n 6^n L^{4n\jmath} \ell^{3n} \Pi$; so (b) holds with $C^{\mathrm{pol}} = \max\{L^{220n\jmath} C^{\mathrm{pol}}_{n,\jmath}, \mathsf{op}_n 6^n L^{4n\jmath}\}$. (c): $A > 30(n+1) \ge 90 > 54$, so (a) applies for $K \ge k^{\mathrm{rt}} + Am$, and $m \le K/A$ gives $\ell^{3(n+1)} \le 2^{3(n+1)} L^{3(n+1)K/A}$. (d): $K \ge C \log \ell$ with $C \ge 54/\lambda + k^{\mathrm{rt}}/\log 2$ gives $K \ge 54m + k^{\mathrm{rt}}$, as $\log \ell \ge \log 2$ and $\log \ell = \log 2 + m\lambda$; with $C \ge 11$, $\frac{1}{10} - \frac{1}{C} \ge \frac{1}{110}$ and $K^{\mathsf{p}_2} L^{-K/110} \le (110\mathsf{p}_2/e\lambda)^{\mathsf{p}_2}$, so (a) gives the first bound, as $\ell^9 \le \ell^{438}$; (b) gives the second.

Step 5: the far part beyond $k^{\mathrm{rt}} + Am$. In a setting apply (E)(a) at $n = 2$ to ς and a lowered choice ς', with $[b', b''] = \mathsf{I}$, $\mathsf{d} = \varepsilon$ and the pair of Step 2. With $(40\sqrt{3}ML)^2 = 4800M^2L^2$, for all $m \in \mathbb{N}$, $\upsilon \in \log \mathsf{I}$ and $K' \ge K \ge k^{\mathrm{rt}} + 54m$,

$$|y_{m,K'}(\upsilon) - y_{m,K}(\upsilon)| \le c_{\mathrm{y}} \ell^9 K^{\mathsf{p}_2} L^{-K/10}, \qquad c_{\mathrm{y}} = 4800 b_+^5 M^2 L^5 C^{\mathrm{rt}}. \tag{12.x}$$

For real $A > 90$ and $K, K' \ge k^{\mathrm{rt}} + Am$ we have $\ell^9 = 2^9 L^{9m} \le 2^9 L^{9K/A}$ and $\frac{1}{10} > \frac{9}{A}$; so

$$\sup_{\log \mathsf{I}} |y_{m,K} - y_{m,K'}| \le \omega_A = 2^9 c_{\mathrm{y}} \sup_{k \ge 1} k^{\mathsf{p}_2} L^{-(\frac{1}{10} - \frac{9}{A})k} < \infty.$$

This is the oscillation hypothesis of Prop. 12.3(b) with the slope A; and $\frac{1}{10} > \frac{9}{A}$ holds exactly for $A > 90$. With it, $\sup_K \sup_{\log \mathsf{I}} |y_{m,K}| < \infty$ at each m, the finitely many cutoffs below the threshold being bounded one by one. (B) uses (E) here and nowhere else.

Step 6: (B). Enter in Prop. 12.3(b) with the data of Step 1, $\mathsf{a}_i = k_i^{\mathrm{rt}}$, $A_i = A$ and $\omega_i = \omega_{A,i}$ of Step 5 at $(\varsigma_i, \varsigma_i')$. At each single (m, K), by eq. 12.vi and Lem. 12.1(c), $|y_{m,K}| \le \beta^5 (P''_N + |n_{\eta_K}(\mu_{N,\beta})|) \le (L^K b_+)^5 (9N^3 + 1125\eta_K^3)$. This gives (B)(1) with the constants

$$\begin{aligned}
k^0 &= \max\{k_1, \lceil k_1^{\mathrm{rt}} + Am_{0,1} \rceil\},\\
Y^0 &\le (L_1^{k^0} b_{+,1})^5 (72 L_1^{3(m_{0,1}+k^0)} + 1125 L_1^{3(k^0-1)}) + \omega_{A,1},\\
\mathsf{M}_1 &= (Y^0 + C_N + C_{\mathrm{low}}) \varkappa_1 + c_{\mathrm{P},1} L_1^{-m_{0,1}} (15/\varkappa_1 + 14 + 12\varkappa_1),\\
\mathsf{A}_0 &= 7\mathsf{M}_1/\varkappa_1 + C_{\mathrm{low}} + 4C_N + \omega_{A,1} + \omega_{A,2},\\
c_{3,i} &= |\log b_{-,i}| + 2\varkappa_i + \lambda_i (k_i + k_i^{\mathrm{rt}} + 1),\\
c_6 &= \max_i c_{\mathrm{P},i} e^{\lambda_i + c_{3,i}/A},\\
T_i(m) &= \log b_{-,i} + 2\varkappa_i + \lambda_i \max\{k_i, \lceil k_i^{\mathrm{rt}} + Am \rceil\},\\
C_{\mathrm{h}} &= 2\mathsf{A}_0 + C_{\mathrm{low}} + 134,\\
C_{\mathrm{r}} &= (\mathsf{A}_0 + 134) c_6^{1/2},\\
\log \beta_1 &= \max\{0, T_1(m_{0,1}), T_2(m_{0,2}), A \log^+(4c_6/\varkappa^2)\},
\end{aligned} \tag{12.xi}$$

since P is the pressure, $J_P = J$, and $|\mathbb{M}|([s, s+\delta])$ is the integral of (B)(1). (2): for $0 < c < \frac{1}{180}$ take $A = \frac{1}{2c} > 90$; then $\beta^{4+c} J(\beta) \le C_{\mathrm{r}}$ for $\beta \ge \beta_1$, and $\beta^{4+c} J \le 6 \max\{1, \beta_1\}^{4+c}$ on $[1, \beta_1]$; for $c \le 0$, $\beta^{4+c} J \le \beta^{4+1/182} J$ on $[1, \infty)$. $c_*(c)$ is defined by P alone. (3): $(\log \beta)^q \beta^{-1/182}$ is bounded on $[3, \infty)$. (4): enter in Prop. 12.5 with the data of Step 1. At $i = 1$, Step 5 at $(\varsigma_1, \varsigma_1')$ gives (i) with $\mathsf{a}_1 = k_1^{\mathrm{rt}}$, $A_1 = A$ and $\omega_1 = \omega_{A,1}$, and $\sup_{K \ge k_1} \sup |y^{(1)}_{m_{0,1},K}|$ is finite, the cutoffs below the

threshold being finitely many; at $i=2$, eq. 12.ix at ς_2 gives (ii) with $\omega_2=1$ for every $m\ge m_{0,2}$, and $\epsilon^{(2)}_m = c_{\mathrm{P},2}L_2^{-m}\to 0$.

Thm. YM3_I:D is used at ς_1 and ς_1' through (E), and at ς_2 through the convergence on each torus. (5): by eq. 12.v with the data of Step 1 and $G_K = G(\cdot - K\lambda_i)$, which is supported in $\mathsf{J}_i + K\lambda_i \subset [s_3,\infty)$, $s_3 = \log(L_i^K b_{-,i}) \ge \log\beta_1$ (and $K\ge k_i$, as $\beta_1>\beta_0$), $\int G\, y^{(i)}_{m,K} = \langle \mathbb{M}, G_K\rangle - r$ with $|r| \le c_{\mathrm{P},i}L_i^{-m}\mathsf{Q}(G) + C_N\|G\|_1$; by (1), the hypothesis of Prop. 12.3(d) holds with $s_2=\log\beta_1$ and $\epsilon_1(s) = e^{-s/2A}$, and it bounds $|\langle \mathbb{M}, G_K\rangle|$ by $(C_{\mathrm{h}}+C_{\mathrm{r}})(\|G\|_1 + (L_i^K b_{-,i})^{-1/2A}\|G'\|_1)$.

Step 7: (C). By the dilation, $\rho^{(K)},\nu^{(K)}\in\mathcal{G}(\beta)$, and $\rho(a_p) = e_{\rho^{(K)}}$; Lem. 12.1(b) gives $3|e_\rho - e_\nu| \le J(\beta)$, and invariance gives $\nu(\mathcal{O}_K(f)) = 3a_K^{-4}e_\nu r_K(f)$. For each of the three orientations the barycentres form a translate of $a_K\mathbb{Z}^3$, of which at most $(2R/a_K+1)^3$ points lie in $[-R,R]^3$; so $|r_K(f)| \le (2R+a_K)^3\|f\|_\infty$. On $\mathfrak{d}(g)$, $b = a_K\beta \ge g^{-2}\ge 4$, so $\beta\ge 1$, and, as $4+c>0$,

$$a_K^{-4}J(\beta) = b^{-4}\beta^4 J(\beta) \le c_*(c)\, b^{-4}\beta^{-c} = c_*(c)\, b^{-4-c}L^{-cK} \le g^{8+2c}c_*(c)L^{-cK}.$$

The setting is used only to form $\mathfrak{d}(g)$; $c_*(c)$ is a number of Wilson's action.

Step 8: (D). (1) On $\mathfrak{d}(g)$, $b\ge g^{-2}$ and $\beta = L^K b \ge L^K g^{-2}$, so $a_K^{-4}J(\beta) = b^{-4}\beta^4 J(\beta) \le g^8 \sup_{\beta'\ge L^K g^{-2}} \beta'^4 J(\beta')$, which tends to 0 by (A)(1); and by (C) with $a_K\le 1$ the difference of the one-point functions is at most $a_K^{-4}J(\beta)(2R+1)^3\|f\|_\infty$. (2) At $X=(f)$ a limit in the frame is $\lim_j[\rho_j(\mathcal{O}_{K_j}(f)) - \nu_j(\mathcal{O}_{K_j}(f))] = 0$. (3) Each component is a limit in the frame along the full sequence, with the approximants $\rho_j = \nu_j(\cdot|B_j)$ (Thm. YM3_I:AP); by (2), $0 = \sigma_\xi((f)) = \mathfrak{c}_\xi\int f$; take $\int f = 1$. As the one-point values of the component vanish, eq. 12.vii with $\rho=\nu'=\rho_j$ and $\nu=\nu_j$ gives $\rho_j(X^{\rho_j}_{K_j}) - \rho_j(X^{\nu_j}_{K_j})\to 0$, every term with $A\ne\emptyset$ having a factor that tends to 0 and a factor that converges; so the Gibbs sequence (K_j,β_j,ρ_j) converges, with the limit σ_ξ.

The component clusters with the numbers C^g, r (Thm. YM3_I:AP); so for its own canonical decomposition $(\mathsf{X}',\varpi',(\sigma'_\zeta))$, (Can4) at $q=2$ gives $\int\sigma'_\zeta(X)^2 d\varpi' = \lim_{|v|\to\infty}\sigma_\xi(X\cdot\tau_v X) = \sigma_\xi(X)^2 = (\int\sigma'_\zeta(X)d\varpi')^2$; so $\zeta\mapsto\sigma'_\zeta(X)$ is constant on the support of ϖ', which is X', and the injectivity in (Can3) makes X' a point. This argument used of (A) only that $\mathfrak{c}_\xi = 0$; conversely, a limit of a convergent Gibbs sequence (K_j,β_j,ρ'_j) has the one-point values $\lim_j \rho'_j((f)^{\rho'_j}_{K_j}) = 0$, so $\mathfrak{c}_\xi = 0$; this is the equivalence. (4) The one-point values vanish; with (Can4) at $q=1$ and the clustering of the components,

$$S_2^\sigma(f,\tau_v h) = \sigma((f)\cdot\tau_v(h)) = \int \sigma_\xi((f)\cdot\tau_v(h))\,\mathrm{d}\varpi, \qquad |\sigma_\xi((f)\cdot\tau_v(h))| \le C^g((f),(h))/|v|.$$

Step 9: (F). (1) By Prop. YM3_I:106.3, $\mathcal{G}_{K,\beta}$ satisfies the hypotheses of Lem. 12.10, and by Lem. 12.1(b) on $\mathbb{L}_K$, $-3e_+ = P'(\beta-)$, $-3e_- = P'(\beta+)$ and $e_+ - e_- = \frac{1}{3}J(\beta)$. (2) The clustering inequality of the plateau theorem at $X=(f)$, $Y=(h)$ compares $\sigma_\nu((f)\cdot\tau_v(h)) = S_2^T[\nu](f,\tau_v h)$ with $\nu(E_\nu[(f)^\nu_K|\mathcal{I}]\,E_\nu[(h)^\nu_K|\mathcal{I}])$, which is $9a_K^{-8}r_K(f)r_K(h)\,\mathrm{Var}_\nu\,\mathsf{e}$ by Lem. 12.10(b), since $\nu(\mathsf{e}) = e_\nu$; and $\mathrm{Var}_\nu\,\mathsf{e}\le\frac{1}{4}(J/3)^2$; the last bound is (C). (3) $\nu_\pm$ exist by Lem. 12.1(b) on $\mathbb{L}_K$. By Lem. 12.10(d), e is almost surely constant in $\nu_\pm$, in every ν with $-3e_\nu$ at an end of the interval, and in a pure phase, which is ergodic. For $\nu = \frac{1}{2}(\nu_+ + \nu_-)$, which lies in the convex set $\mathcal{G}_{K,\beta}$, and bounded F,F',

$$\mathrm{Cov}_\nu(F,F') = \tfrac{1}{2}\mathrm{Cov}_{\nu_+}(F,F') + \tfrac{1}{2}\mathrm{Cov}_{\nu_-}(F,F') + \tfrac{1}{4}(\nu_+(F)-\nu_-(F))(\nu_+(F')-\nu_-(F')).$$

Take $F=\mathcal{O}_K(f)$ and $F'=\mathcal{O}_K(\tau_v f)$, $v\in a_K\mathbb{Z}^3$, with $0\le f\in C^1_c((-\frac{1}{2},\frac{1}{2})^3)$ and $r_K(f)>0$, for which (2) applies at $\bar R = 1$. Then $\nu_+(F)-\nu_-(F) = 3a_K^{-4}r_K(f)(e_{\nu_+}-e_{\nu_-}) = -a_K^{-4}J(\beta)r_K(f)$, the same for F', as $r_K(\tau_v f) = r_K(f)$; (2) in the three states, with $|v|\to\infty$, gives $9a_K^{-8}r_K(f)^2\,\mathrm{Var}_\nu\,\mathsf{e} = \frac{1}{4}(a_K^{-4}J(\beta)r_K(f))^2$. (4) Prop. 12.11 with $\mathcal{G}=\mathcal{G}_{K,\beta}$; X^ν_K is a real continuous local function, a polynomial in finitely many a_p, and by Lem. 12.1(b) every $\rho\in\mathcal{G}_{K,\beta}$ has $D^-\bar p^F(0)\le\rho(F)\le D^+\bar p^F(0)$ for such F; at $X=(f)$, $3a_K^{-4}|r_K(f)|(e_+-e_-) = a_K^{-4}J(\beta)|r_K(f)|$.

Step 10: (G) in a window. Let $g^{-2}+\bar B<b<X_g-\bar B$ for some $0<g\le\gamma$. From some index on $a_{K_j}\beta_j\in I(g)$, so $(K_j,\beta_j)\in\mathfrak d(g)$; we drop the first members, which changes no set of limits. Let $\mathsf S_j=\{\sigma_\nu:\nu\in\mathcal G_{K_j,\beta_j}\}\subset\mathcal E^g$ and fix a metric of $\mathcal E^g$.

(1) $\mathsf K=\{\sigma\in\mathcal E^g:\operatorname{dist}(\sigma,\mathsf S_j)\to0\}$, a closed subset of the compact metrisable space $\mathcal E^g$ (Lem. 12.12(b)). Convexity: if $\sigma^i=\lim_j\sigma_{\nu_j^i}$ and $\nu_j=t\nu_j^1+(1-t)\nu_j^2\in\mathcal G_{K_j,\beta_j}$, then

$$\sigma_{\nu_j}(X)=t\,\nu_j^1(X_{K_j}^{\nu_j})+(1-t)\,\nu_j^2(X_{K_j}^{\nu_j}),$$

and by eq. 12.vii with the centrings ν_j^i and ν_j each bracket is at most $a_{K_j}^{-4}J(\beta_j)(2R+1)^3\|f\|_\infty\to0$ by (D)(1), while the moments $\nu_j^i((X_A)_{K_j}^{\nu_j^i})$ converge; so $\sigma_{\nu_j}\to t\sigma^1+(1-t)\sigma^2$.

(2) $(K_j,\beta_j,\nu_j)_j$ is a convergent Gibbs sequence at g; by (D)(3) each component is $\lim_j\sigma_{\nu_j(\cdot|B_j)}$ and has one vacuum. Components cluster at the rate $C^g(X,Y)/|v|$. A member of $\mathsf K$ that clusters, in the sense $\sigma(X\cdot\tau_vX)\to\sigma(X)^2$ for all X, has one vacuum, by (Can4) at $q=2$ as in Step 8; conversely, a member with one vacuum is its own component and clusters at the rate $C^g/|v|$. As C^g and r do not depend on the member, pointwise limits of members of $\mathsf K^\bullet$ cluster: $\mathsf K^\bullet$ is closed in $\mathsf K$, so compact.

(3) Barycentres of measures on $\mathsf K^\bullet$ lie in $\mathsf K$: for finite supports by (1), and in general by Riemann sums, the uniform continuity of $\psi\mapsto\psi(X)$ on the compact $\mathsf K^\bullet$, and the closedness of $\mathsf K$. Onto: for $\sigma\in\mathsf K$ the image of ϖ under $\xi\mapsto\sigma_\xi$ is a measure on $\mathsf K^\bullet$ by (2), with the barycentre σ by (Can4) at $q=1$. Injective: Lem. 12.12(c) with $\mathsf T=\mathsf K^\bullet$. A continuous bijection of the compact $\operatorname{Prob}(\mathsf K^\bullet)$ onto the Hausdorff space $\mathsf K$ is a homeomorphism; by injectivity the image of ϖ is μ_σ; $\xi\mapsto\sigma_\xi$ is continuous and injective on the compact $\mathsf X$, and its image is the support of μ_σ, ϖ having full support. $\dim\ker H_\sigma=\#\mathsf X=\#\operatorname{supp}\mu_\sigma$ by Thm. YM3_I:AP.

(4) One vacuum is $\ker H_\sigma=\mathbb C\Omega_\sigma$ by $\#\mathsf X=\dim\ker H_\sigma$, in each of the six directions. By (3), the extreme points of $\mathsf K$ are the barycentres of the extreme points of $\operatorname{Prob}(\mathsf K^\bullet)$, the point masses: extreme points are the members with one vacuum. The two forms of clustering are (2). *Pure phases.* If $\sigma=\lim_j\sigma_{\omega_j}$ with pure phases ω_j, each ω_j is ergodic, so $E_{\omega_j}[\cdot\,|\mathcal I]$ is almost surely constant, and by the statement on iterated plateaux $\int\sigma_\xi(X)^2d\varpi-\sigma(X)^2=\lim_j\operatorname{Var}_{\omega_j}(E_{\omega_j}[X_{K_j}^{\omega_j}|\mathcal I])=0$ for every X; so $\xi\mapsto\sigma_\xi(X)$ is constant on $\mathsf X$, and $\mathsf X$ is a point.

Conversely, let $\sigma\in\mathsf K^\bullet$ be the limit of σ_{ν_j}. By the variance statement below, $E_{\nu_j}[X_{K_j}^{\nu_j}|\mathcal I]\to\sigma(X)$ in $L^2(\nu_j)$ for every X. Let $(\nu_j^U)_U$ be a regular conditional law of ν_j given $\mathcal I$; by Prop. YM3_I: 106.1 its members are, off a ν_j-null set, pure phases in $\mathcal G_{K_j,\beta_j}$, and $\nu_j^U(X_{K_j}^{\nu_j})=E_{\nu_j}[X_{K_j}^{\nu_j}|\mathcal I](U)$ for ν_j-almost every U. Take a countable family $X_1,X_2,\dots$ of tuples on which pointwise convergence in $\mathcal E^g$ implies convergence at every tuple (Lem. 12.12(b)). By Chebyshev's inequality there are, for each j, a point U_j outside the null sets and numbers $\epsilon_j\to0$, $k_j\to\infty$ with $|\nu_j^{U_j}((X_k)_{K_j}^{\nu_j})-\sigma(X_k)|\le\epsilon_j$ for $k\le k_j$.

The pure phases $\omega_j=\nu_j^{U_j}$ then have $\omega_j((X_k)_{K_j}^{\nu_j})\to\sigma(X_k)$ for every k; by eq. 12.vii and (D)(1), $\sigma_{\omega_j}(X_k)\to\sigma(X_k)$; and as $\sigma_{\omega_j}\in\mathcal E^g$, $\sigma_{\omega_j}\to\sigma$. *Variances.* By the statement on iterated plateaux, $\operatorname{Var}_{\nu_j}(E_{\nu_j}[X_{K_j}^{\nu_j}|\mathcal I])\to\int\sigma_\xi(X)^2d\varpi-\sigma(X)^2$, which does not depend on the choice of the ν_j, and which vanishes for every X if and only if $\mathsf X$ is a point. *Faces.* For a subsequence $\mathfrak p'$, $\mathsf K\subset\mathsf K_{\mathfrak p'}$. If $\sigma\in\mathsf K$ is $t\sigma_1+(1-t)\sigma_2$ with $\sigma_i\in\mathsf K_{\mathfrak p'}$, $0<t<1$, the uniqueness of the representing measure on $\operatorname{Lim}^\bullet[b]$ (Lem. 12.12(c), with the compact set $\operatorname{Lim}^\bullet[b]$ of (6)) gives $\mu_\sigma=t\mu_{\sigma_1}+(1-t)\mu_{\sigma_2}$; so the μ_{σ_i} live on $\operatorname{supp}\mu_\sigma\subset\mathsf K^\bullet$, and $\sigma_i\in\mathsf K$ by (3).

For the interlacing $\mathfrak p''$, $\mathsf K_{\mathfrak p''}\subset\mathsf K\cap\mathsf K_{\tilde{\mathfrak p}}$, as $\mathfrak p$ and $\tilde{\mathfrak p}$ are subsequences of $\mathfrak p''$; conversely, two Gibbs sequences that converge to σ along $\mathfrak p$ and along $\tilde{\mathfrak p}$ interlace to one that converges to σ along $\mathfrak p''$. *One point.* Let $\mathsf K\ne\emptyset$. If $\mathsf K$ is a point, $\mathsf K^\bullet$ is not empty, as it holds the components of that point, and lies in $\mathsf K$; if $\mathsf K^\bullet$ is a point, so is $\operatorname{Prob}(\mathsf K^\bullet)$ and hence $\mathsf K$; if $\mathsf K$ is a point, $\mathsf K=\mathsf K^\bullet$; if $\mathsf K=\mathsf K^\bullet$ had two members $\sigma_1\ne\sigma_2$, then $\frac12(\sigma_1+\sigma_2)\in\mathsf K=\mathsf K^\bullet$ would be an extreme point.

By the statement on iterated plateaux, $\mathrm{Cov}_{\nu_j}(E_{\nu_j}[X^{\nu_j}_{K_j}|\mathcal{I}], E_{\nu_j}[Y^{\nu_j}_{K_j}|\mathcal{I}]) \to \int \sigma_\xi(X)\sigma_\xi(Y)d\varpi - \sigma(X)\sigma(Y)$, which vanishes for all X, Y if and only if X is a point; for all convergent choices this says $\mathsf{K} = \mathsf{K}^\bullet$. At single fields, Lem. 12.10(b) and (F)(2) give the displayed bound, with $|r_K(f)| \le (2R+1)^3\|f\|_\infty$, and it tends to 0 by (D)(1).

(6) The pairs $(K, L^K b)$, $K \in \mathbb{N}$, lie in $\mathfrak{d}(g)$; as $\mathcal{E}^g$ is compact, a subsequence of (σ_{ν_K}) converges, so $\mathrm{Lim}[b] \neq \emptyset$. $\mathrm{Lim}[b] = \bigcap_k \overline{\bigcup\{\mathsf{S}_{K,\beta} : K \ge k,\ |a_K\beta - b| \le \frac{1}{k}\}}$, with $\mathsf{S}_{K,\beta} = \{\sigma_\nu : \nu \in \mathcal{G}_{K,\beta}\}$, is compact, and $\mathrm{Lim}^\bullet[b]$ is closed in it by the uniform clustering. A saturated subsequence: with a countable base (U_r) of $\mathcal{E}^g$, pass to nested subsequences along which, for each r, $\mathsf{S}_j \cap U_r$ is either non-empty for all members or empty for all members, and take the diagonal; for it and for each of its subsequences a functional is a limit if and only if each U_r that holds it meets S_j for all large j, so all these sets of limits agree, and they are not empty by compactness. Every $\sigma \in \mathsf{K}_\mathfrak{p}$ lies in $\mathsf{K}_{\mathfrak{p}'}$ for a saturated subsequence $\mathfrak{p}'$, and the extreme points are the members with one vacuum by (4); this gives the two unions. Uniqueness of the representing measure on $\mathrm{Lim}^\bullet[b]$: Lem. 12.12(c) with $\mathsf{T} = \mathrm{Lim}^\bullet[b]$; existence: (3).

The hull. $\mathrm{Lim}^\bullet[b]$ is compact and its members cluster uniformly, with the numbers C^g, r; so by Lem. 12.12(c) the map bar is affine, continuous and injective on the compact convex set $\mathrm{Prob}(\mathrm{Lim}^\bullet[b])$, hence an affine homeomorphism onto its image, a compact convex set. The image contains $\mathrm{Lim}[b]$, by the existence just shown, and it is the closure of the image of the finitely supported measures, which is the convex hull of $\mathrm{Lim}^\bullet[b] \subset \mathrm{Lim}[b]$; so it is $\overline{\mathrm{co}}\,\mathrm{Lim}[b]$. Its extreme points are the images of the point masses, that is, the members of $\mathrm{Lim}^\bullet[b]$. Hence a member of $\mathrm{Lim}[b]$ with one vacuum is an extreme point of $\overline{\mathrm{co}}\,\mathrm{Lim}[b]$, and so of $\mathrm{Lim}[b]$; and a member σ that is an extreme point of $\mathrm{Lim}[b]$ is an extreme point of every $\mathsf{K}_\mathfrak{p}$ that contains it, so it has one vacuum by (4).

Convexity. If $\mathrm{Lim}[b] = \mathsf{K}_\mathfrak{q}$, it is convex by (1). If $\mathrm{Lim}[b]$ is convex, it is compact and convex, so it equals $\overline{\mathrm{co}}\,\mathrm{Lim}[b] = \mathrm{bar}(\mathrm{Prob}(\mathrm{Lim}^\bullet[b]))$. If this equality holds and $\psi_1, \dots, \psi_n \in \mathrm{Lim}^\bullet[b]$, then $\frac{1}{n}\sum_i \psi_i$ lies in some $\mathsf{K}_\mathfrak{p}$; by (3) it is the barycentre of a measure on $\mathsf{K}^\bullet_\mathfrak{p}$, which by the uniqueness on $\mathrm{Lim}^\bullet[b]$ is $\frac{1}{n}\sum_i \delta_{\psi_i}$, so that $\psi_1, \dots, \psi_n \in \mathsf{K}^\bullet_\mathfrak{p}$. If every finite subset of $\mathrm{Lim}^\bullet[b]$ lies in one $\mathsf{K}^\bullet_\mathfrak{p}$, let $\psi_1, \psi_2, \dots$ be dense in $\mathrm{Lim}^\bullet[b]$ and $\psi_1, \dots, \psi_n \in \mathsf{K}^\bullet_{\mathfrak{p}_n}$; for each n there is a pair (K'_n, β'_n) of $\mathfrak{p}_n$ with $K'_n \ge n$ and $|a_{K'_n}\beta'_n - b| \le \frac{1}{n}$ which carries, for each $i \le n$, a state ν with $\mathrm{dist}(\sigma_\nu, \psi_i) \le \frac{1}{n}$.

Along $\mathfrak{q} = (K'_n, \beta'_n)_n$, a sequence of pairs at the label b, every ψ_i is a limit; so the closed set $\mathsf{K}_\mathfrak{q}$ contains $\mathrm{Lim}^\bullet[b]$, and by (3) for $\mathfrak{q}$ it contains $\mathrm{bar}(\mathrm{Prob}(\mathrm{Lim}^\bullet[b])) \supset \mathrm{Lim}[b] \supset \mathsf{K}_\mathfrak{q}$. Finally, a set with one element is convex.

(5) Every $\sigma \in \mathrm{Lim}[b]$ lies in some $\mathsf{K}_\mathfrak{q}$; its components lie in $\mathsf{K}^\bullet_\mathfrak{q} \subset \mathrm{Lim}^\bullet[b]$, and $\sigma = \int \sigma_\xi \,\mathrm{d}\varpi$. If $\mathrm{Lim}^\bullet[b] = \{\tau\}$, all components are τ and $\sigma = \tau$; if $\mathrm{Lim}[b] = \{\sigma\}$, then $\mathrm{Lim}^\bullet[b]$ is not empty by (6) and lies in $\{\sigma\}$. The same lines serve for $\mathrm{Lim}_\mathfrak{q}[b]$, the components of a limit along pairs of $\mathfrak{q}$ being limits along the same pairs, and $\mathrm{Lim}_\mathfrak{q}[b]$ being non-empty by (6) applied to a sequence with pairs in $\mathfrak{q}$. For the third statement: a Gibbs sequence at g with the label b has convergent subsequences (Thm. YM3_I:AP), whose limits lie in $\mathrm{Lim}[b]$; if this is one point, every subsequence has a subsequence that converges to it, so the sequence converges; conversely, interlacing two convergent Gibbs sequences with the label b gives one, which converges, so their limits agree. The one limit then has one vacuum, as $\mathrm{Lim}^\bullet[b] \neq \emptyset$.

The clauses (1)–(4) use of (A) only that $a_{K_j}^{-4} J(\beta_j) \to 0$ along $\mathfrak{p}$. Along $\mathfrak{p} = (K, L^K b)_K$ this holds without (A) for every b outside the countable set $\bigcup_K L^{-K}\{\beta : J(\beta) > 0\}$, as the convex function P is differentiable off a countable set; at such labels these clauses use Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD at the choice of the setting alone.

Step 11: (G) at every label. For $z \in \mathbb{Z}$ put, as in the paragraph on dilations, $X^{[z]} = (f_i(L^z\cdot))_i$ and $\sigma^{[z]}(X) = L^{-z|X|}\sigma(X^{[z]})$. The dilation $x \mapsto L^z x$ carries a law ν on $\mathbb{L}_K$ to a law $\nu^{(z)}$ on $\mathbb{L}_{K-z}$, maps $\mathcal{G}_{K,\beta}$ onto $\mathcal{G}_{K-z,\beta}$ and conditioned states to conditioned states, as it carries the invariant events to the invariant events ([7, Lem. K.1]), and turns $\mathcal{O}_{K-z}(f)$ into $L^{-z}\mathcal{O}_K(f(L^z\cdot))$ ([7, Lem. K.1], applied

to the function $f(L^z\cdot)$); so

$$\sigma_{\nu^{(z)}} = (\sigma_\nu)^{[z]}, \qquad a_{K-z}^{-4} J(\beta) = L^{-4z} a_K^{-4} J(\beta).$$

So $\sigma \mapsto \sigma^{[z]}$ is a linear homeomorphism that carries $\mathsf{K}_\mathfrak{p}$, $\mathrm{Lim}[b]$, $\mathrm{Lim}^\bullet[b]$ onto the same sets at the label $L^z b$ for the pairs $(K_j - z, \beta_j)$, and canonical decompositions to canonical decompositions, component by component, with the same space X. It carries the reconstruction as well. Let (z, g) be a pair for b, that is, $z \in \mathbb{Z}$ and $0 < g \le \gamma$ with $g^{-2} + \bar{B} < L^z b < X_g - \bar{B}$; the pair fixed for b in the paragraph on dilations is one, with $z \ge 0$. For $\sigma \in \mathsf{K}_\mathfrak{p}$, $\sigma^{[z]}$ is then, as in that paragraph, the limit of a convergent Gibbs sequence at g over the pairs $(K_j - z, \beta_j)$ of large index, and Thm. YM3_I: AP gives its reconstruction, with the Hamiltonian $H_{\sigma^{[z]}}$ and the vacuum vector $\Omega_{\sigma^{[z]}}$, in each of the six coordinate directions.

For a coordinate direction u, the reflection θ in the plane $\{x \cdot u = 0\}$ commutes with the dilation, so that $\sigma^{[z]}(\theta X \cdot Y) = L^{-z(|X|+|Y|)}\sigma(\theta X^{[z]} \cdot Y^{[z]})$ for the tuples X, Y in the half space $\{x \cdot u > 0\}$; hence the form $\langle [X], [Y] \rangle_\sigma = \sigma(\theta X \cdot Y)$ is positive together with that of $\sigma^{[z]}$, and $W : [X] \mapsto L^{-z|X|}[X^{[z]}]$ is a unitary map of the space of the reconstruction of $\sigma^{[z]}$ onto the space that this form builds from σ, with $W[\emptyset] = [\emptyset]$. As $(\tau_v X)^{[z]} = \tau_{L^{-z}v} X^{[z]}$, W carries the translations by tu to the translations by $L^{-z}tu$; so, with $\hat{H}_\sigma$ the generator of the semigroup $e^{-t\hat{H}_\sigma}[X] = [\tau_{tu} X]$, $t \ge 0$, in the space that the form builds from σ, $\hat{\Omega}_\sigma = [\emptyset]$ the class of the empty tuple there, and $e^{-tH_{\sigma^{[z]}}}[X] = [\tau_{tu} X]$, $\Omega_{\sigma^{[z]}} = [\emptyset]$ in the reconstruction of $\sigma^{[z]}$,

$$H_{\sigma^{[z]}} = L^{-z}\, W^* \hat{H}_\sigma W, \qquad W\Omega_{\sigma^{[z]}} = \hat{\Omega}_\sigma, \qquad \ker \hat{H}_\sigma = W \ker H_{\sigma^{[z]}},$$

in each of the six directions: the kernels and the vacua correspond. A linear homeomorphism carries convex sets, closed sets, extreme points, faces and one-point sets to the same, and the dilation carries the clustering numbers to dilated ones. For every $b > 0$ there are pairs for b, already with $z \ge 0$, because $L^z b > \gamma^{-2} + \bar{B}$ for large z and the open intervals $(g^{-2} + \bar{B}, X_g - \bar{B})$ overlap one another as g decreases and cover $(\gamma^{-2} + \bar{B}, \infty)$. So W carries the operator $L^z H_{\sigma^{[z]}}$ and the vector $\Omega_{\sigma^{[z]}}$, which are H_σ and Ω_σ of the paragraph on dilations at the pair (z, g) fixed for b, to $\hat{H}_\sigma$ and $\hat{\Omega}_\sigma$, the generator of the translations and the class of the empty tuple in the space that the form builds from σ.

These read no window, so another pair (z, g) gives a unitarily equivalent operator with the corresponding vector, and $\dim\ker H_\sigma = \dim\ker H_{\sigma^{[z]}} = \#\mathsf{X}$, both finite and equal or both infinite. Step 10 at $(L^z b, (K_j - z, \beta_j))$ gives (1)–(6) at $(b, \mathfrak{p})$. In particular the kernel clauses of (3) and (4) pass to $(b, \mathfrak{p})$, with $H_\sigma = L^z H_{\sigma^{[z]}}$ and $\Omega_\sigma = \Omega_{\sigma^{[z]}}$ of the paragraph on dilations at the pair (z, g) fixed for b: the dilation carries μ_σ to $\mu_{\sigma^{[z]}}$, so that $\dim\ker H_\sigma = \dim\ker H_{\sigma^{[z]}} = \#\mathsf{X} = \#\operatorname{supp}\mu_\sigma$, and $\ker H_\sigma = \mathbb{C}\Omega_\sigma$ in one, and then in each, of the six directions if and only if σ has one vacuum.

Step 12: (G)(7). Let $0 < g \le \gamma_W$ and let $\mathfrak{p}$ be a sequence of pairs in $\mathfrak{d}(g)$ with $\mathsf{K}_\mathfrak{p} \ne \emptyset$; $\sigma \in \mathsf{K}_\mathfrak{p}$ is the limit of a convergent Gibbs sequence over $\mathfrak{p}$. By Prop. YM3_I:127.8, applied with $\rho_j = \nu_j$ (see also Cor. YM3_I:127.11), along a convergent Gibbs sequence at g the labels $a_{K_j}\beta_j$ converge, to a number $b \in I^{\mathrm{lab}}(g)$, and the two-point function of the limit, $S_2^\sigma = \lim_j S_2^T[\nu_j]$, has the displayed limit with this b, for every $\vartheta \in (0, 1)$ and every pair of $\mathfrak{T}^1_{\mathbb{R}^3}(\mathsf{d}_0(g, \vartheta); \vartheta)$; this set is not empty by [7, Lem. I.1](c).

The right member of the formula reads σ alone; so the sets $\mathrm{Lim}_{\mathfrak{d}(g)}[b]$ are pairwise disjoint, and every convergent Gibbs sequence at g is over a sequence of pairs with a non-empty set of limits, which has a label. An interlacing of two sequences with different labels has no label, so its set of limits is empty. Suppose that $\sigma'' = t\sigma + (1-t)\sigma'$, $0 < t < 1$, with $\sigma \in \mathrm{Lim}_{\mathfrak{d}(g)}[b]$, $\sigma' \in \mathrm{Lim}_{\mathfrak{d}(g)}[b']$, were the limit of a Gibbs sequence over $\mathfrak{p}''$; then $\mathfrak{p}''$ has a label b''. The sets of the limits with one vacuum at the labels b and b', along sequences in $\mathfrak{d}(g)$, are compact, by the diagonal argument of Step 10 (6), and their members cluster uniformly, with the numbers C^g, r; with the representing measures

μ_σ, $\mu_{\sigma'}$ of (G)(6), carried by the components of σ and of σ', Lem. 12.12(c) makes $(\operatorname{supp}\mu'', \mu'', (\psi))$, $\mu'' = t\mu_\sigma + (1-t)\mu_{\sigma'}$, a canonical decomposition of σ''.

By Cor. YM3_I:127.11, the formula in every component of every canonical decomposition, each component of σ'' has the coefficient b''^{-2}, whereas the members of $\operatorname{supp}\mu_\sigma$, which are the components of σ by (G)(3), have b^{-2}, and those of $\operatorname{supp}\mu_{\sigma'}$ have b'^{-2}; so $b = b'' = b'$. □

Proof of Thm. AK. (1) is Prop. 12.16(c) at $a = a_K$, $\beta = L^K b$ and $l = l_K = L^{K+m_*} \geq L$, as $K + m_* \geq K_* + m_* \geq 1$; that P is differentiable at β is $J(\beta) = 0$. (2) Let first the entries be none 0 and their supports lie in $(-\frac{\ell''}{4}, \frac{\ell''}{4})^3$ for a side $\ell'' = 2L^{m''} \geq \ell_*$, $m'' \in \mathbb{N}$; an entry 0 makes every cumulant 0, and every compact set lies in such a cube. The torus cumulants converge as $K \to \infty$ (Thm. YM3_I:AS), so Prop. 12.16(f) gives $\limsup_K S_n^T[\nu_K](X) - \liminf_K S_n^T[\nu_K](X) \leq 2C(X)\theta^{\ell'/4\ell_*}$ for every side $\ell' \geq \ell''$; let $\ell' \to \infty$. The same bound shows that the limit is within $C(X)\theta^{\ell'/4\ell_*}$ of $\lim_K S^T_{n;K,\ell',L^K b}(X^{\langle \ell' \rangle})$. (3) Since $\mathcal{G}_{K,L^K b} = \{\nu_K\}$, the set $\mathsf{K}_\mathfrak{p}$ has at most one point, the limit of σ_{ν_K} if it exists. For a separated X, $\sigma_{\nu_K}(X)$ is a centred moment, so

$$\sigma_{\nu_K}(X) = \sum_\pi \prod_{A \in \pi} S^T_{|A|}[\nu_K](X_A),$$

the sum over the partitions of $\{1, \dots, |X|\}$ into blocks of at least two elements; each factor converges by (2). So $\mathsf{K}_\mathfrak{p} = \{\sigma\}$; the point of a one-point convex set is extreme, so σ has one vacuum by Thm. AJ(G)(4). The last sentence: $l_K = \ell_*/2a_K$ and $\beta = L^K b$ do not change when (b, ℓ_*, K) is replaced by $(L^z b, L^z \ell_*, K - z)$, and $K \geq K_*$ says $K - z \geq K_* - z$. □

12.1.7. *Scope of the conclusions.* The bounds (A)–(C) are upper bounds; they do not show that J vanishes at any finite β. The numbers 54, 90 and $\frac{1}{180}$ are those of sufficient conditions in the estimates: $54 = 3 \cdot 18$ comes from the condition $\ell^3 L^{-K/18} \leq c$ of Step 4 and stands in the threshold $k^{\mathrm{rt}} + 54m$ of (E), which the slope $A > 90$ of Step 5 covers in (B); $90 = 9 \cdot 10$ comes from the comparison of ℓ^9 with $L^{-K/10}$ in Step 5; and $\frac{1}{180} = \sup_{A>90} \frac{1}{2A}$ comes from Prop. 12.3(b) in Step 6. Prop. 12.3(e) is a statement on data and on the map from the slope to the exponent: for every slope $A > 0$ there are data as in (b), with $A_1 = A_2 = A$, whose atoms are $\beta_k^{-1/2A}$, so that the loss of one half is real for data (Remark 12.4(3)); and the data of (e) at $A = 90$ satisfy the hypothesis of (b) at every slope $A' \geq 90$, a larger threshold being a weaker hypothesis, and have the atoms $\beta_k^{-1/180}$.

By Prop. 12.14 the exponent follows the slope that a rate on the tori gives, and by Prop. 12.15 a threshold that does not grow with the side gives $J = 0$ on a half line. Thm. AK assumes the box inequality and speaks of the couplings $L^K b$ of one label. Clause (G)(5) is an equivalence and asserts neither of its two sides; clause (G)(6) asserts none of its four equivalent statements on the convexity of $\mathrm{Lim}[b]$; and (G)(1)–(4) allow $\mathsf{K}_\mathfrak{p} = \emptyset$ for the given sequence, whereas (G)(6) gives non-empty sets along saturated subsequences. Clause (E) concerns $n \geq 2$ and tuples of positive spread, with a threshold $k^{\mathrm{rt}} + 54m$ that grows with the side, and (A)(4) concerns limit points along subsequences of the cutoffs.

13. Infrared bounds

Let S_2 be the two-point function of the centred action density in a continuum limit through periodic Gibbs states, or in one of its components. Euclidean invariance writes its kernel off the diagonal as $s(|x - y|)$. Large cube averages test the small spectral masses of this field without requiring pointwise control of the underlying connection.

Sketch L (Polynomial infrared decay). The radial kernel has a positive representation

$$s(r) = \frac{1}{4\pi r} \int_0^\infty e^{-mr} \, \mathrm{d}\rho(m), \qquad r > 0,$$

and, at large r and small m, respectively,

$$s(r) = O(r^{-2}), \qquad \rho([0,m]) = O(m), \qquad \rho(\{0\}) = 0.$$

These conclusions exclude a zero-mass atom, while allowing spectral mass arbitrarily close to zero. ■□

Sketch of the argument. Average the lattice action density over a cube of side q, normalizing by its volume. Keep a collar of width T at its boundary, so that sources in distinct cells can be reflected and folded without meeting the walls. Source-pressure bounds and the chessboard estimate give, for the resulting cube means,

$$|\operatorname{Cov}(x_c, x_{c'})| \le C(T/q)^2 \qquad (c \ne c').$$

The constants are uniform in the fine cutoff once its threshold for the selected cube size has been reached.

The source-pressure argument first gives a semiconvex response measure. A concentration of this measure would obstruct the uniform cube estimate. Compare the pressure using a second blocking factor with an incommensurable logarithm, as in the latent-heat argument. This comparison excludes concentration and permits the bound to pass to continuum limits. The vanishing one-point differences remove the plateau that could otherwise come from varying component means.

Reflection positivity gives a positive spectral representation of the radial two-point function. Test it with two reflected large-cube profiles. Their positive kernels compare the cube estimate with the value of s at their separation, yielding $s(r) \le C/r^2$. In the positive integral representation, the part with $0 \le m \le 1/r$ is bounded below by $e^{-1}\rho([0,1/r])/(4\pi r)$. Consequently $\rho([0,m]) \le Cm$ near zero. A zero-mass atom would instead contribute a nonzero multiple of $1/r$, so it is absent.

For separated compact tests this yields a two-point bound of order the inverse square of their separation. Reflection positivity and the corresponding reconstructed vectors also bound a connected function with one insertion moved far away by the inverse distance. These are infrared estimates for the Euclidean-invariant Gibbs limits. General growing-torus limits retain weaker direction-dependent conclusions until their rotation problem is settled.

Nothing in this argument supplies a positive mass threshold below the support of ρ. The stronger exponential decay and spectral gap follow from the finite-size criterion in the last sketch. They should not be inferred from the absence of the atom proved here.

13.1. **Cube means and infrared bounds.** We average the action density over cubes with a boundary collar. Chessboard estimates turn control of the source pressure into bounds on correlations of distinct cube means. Comparing two blocking factors with incommensurable logarithms excludes concentration of the pressure response.

The resulting continuum bounds give an $O(r^{-2})$ radial two-point kernel and at most linear spectral mass near zero. They exclude an atom at zero without giving a positive gap. The growing-torus conclusions use only the positivity and symmetry established for those limits. The stronger exponential conclusions explicitly assume the box criterion.

13.1.1. *Notation and hypotheses.* Lengths are measured on the physical scale, on which the bare lattice spacing is $a_K = L^{-K}$; thus ℓ, M, I, T, q and r are lengths on this scale. A floor is a lower threshold, a window a compact set of labels; a quantity is free of another if it does not depend on it. $|\cdot|$ is the Euclidean norm, $|\cdot|_\infty$ the sup-norm and $\operatorname{dist}_\infty$ its distance (on a torus, the distance of the torus); $[t]_+ = \max\{t, 0\}$; $\operatorname{int} I$ and $|I|$ are the interior and the length of an interval I; Lip is the Lipschitz constant for the Euclidean distance.

Definition 13.1 (Frames; the numbers at a reference coupling)**.**

(1) A *frame* consists of a radius $\varrho > 0$, a common real and sourced choice of constants obtained by the ordered construction in the proof of Prop. YM3_I:69.1 and Section YM3_I:82, and the corresponding Wilson measures and source normalization of Thms. YM3_I:AJ

and YM3_I:AD. The choice retains the source-room condition eq. 2.ii and the array radius [7, eq. 194]. A frame determines an odd integer $L \geq 13$; $M = L^{m'}$ with an integer m'; numbers $B_\sharp \geq 0$ and $\bar{B} = B_\sharp L/(L-1)$; $\gamma \in (0, \frac{1}{2}]$; and the upper stability constant C_S and decay rate $\kappa > \log 9216 + 2$ of Thm. YM3_I:AD. Choose C_{st} and $C_S \geq C_{\mathrm{fp}}^{\mathrm{st}}$ as in [7, eq. 288]. [7, Lem. G.2] gives the lower stability bound with $C_g = C_{\mathrm{st}} + \frac{9}{2}\log X_g$ and the estimate $C_g \leq C_S(1 + \log g^{-2})$. We put $S(\kappa) = 2^{12}(1 - 9216\, e^{-\kappa})^{-1}$, the number of the count of domains.

(2) At a reference coupling $0 < g \leq \gamma$ we put

$$X_g = L(g^{-2} + B_\sharp), \quad I(g) = (g^{-2} + \bar{B},\, X_g - \bar{B}], \quad C_g = C_{\mathrm{st}} + \tfrac{9}{2}\log X_g, \quad B' = 2C_S + C_g,$$

and $I^{\mathrm{lab}}(g) = [g^{-2} + B_\sharp \frac{L-2}{L-1},\, L(g^{-2} + \bar{B})]$. A first-passage datum at g is a pair (K, β) of a cutoff and a bare coupling such that K is the first index at which the running inverse coupling started at β is $\leq X_g$, as in Prop. YM3_I:74.1; $\mathfrak{d}(g)$ is the set of these pairs. $\mathfrak{L}(g) = \{2L^m : m \geq m_{\mathfrak{L}}(g)\}$ is the set of the sides at g of equation YM3_I:(1528), $m_{\mathfrak{L}}(g)$ being the integer YM3_I:(1528) there, which the frame and g fix. The set $\mathfrak{L}(g)$ is closed upward among the sides $2L^m$, and by [7, Lem. G.4] and equation YM3_I:(1533), for $0 < g' \leq g \leq \gamma$, $m_{\mathfrak{L}}(g) \leq m_{\mathfrak{L}}(g')$, so that $\mathfrak{L}(g') \subset \mathfrak{L}(g)$. These are the sides of which Thm. YM3_I:AD and the transfer (M7) speak.

(3) A second frame, whose objects wear a hat, is a *partner* of the first if $\log L/\log \hat{L} \notin \mathbb{Q}$.

■

Definition 13.2 (The lattice, the pressures, the sources)**.**

(1) For an integer $K \geq 0$, $a_K = L^{-K}$ and $\mathbb{L}_K = a_K\mathbb{Z}^3$; for $\ell = 2L^m$, $T_\ell = (\mathbb{R}/\ell\mathbb{Z})^3$ carries the lattice of spacing a_K, with $\mathsf{N} = \ell/a_K$ sites a side. For a plaquette p, $a_p = 1 - \frac{1}{2}\operatorname{tr}\partial U(p) \in [0, 2]$, and $x(p)$ is its barycentre. For a function f we put $\mathcal{O}(f) = \mathcal{O}_K(f) = a_K^{-1}\sum_p f(x(p))a_p$ and $\mathrm{Rie}_K(f) = \frac{1}{3}a_K^3\sum_p f(x(p))$, the sums over the plaquettes of $\mathbb{L}_K$ or of the torus.

(2) $Z_{K,\ell,\beta}[w] = \int e^{-\beta\sum_p a_p + \mathcal{O}(w)}\,\mathrm{d}U$, with the Haar measure of mass 1; $Z_{K,\ell,\beta} = Z_{K,\ell,\beta}[0]$ defines the law $\langle\cdot\rangle_{K,\ell,\beta}$. $S^T_{n;K,\ell,\beta}(f_1, \dots, f_n)$ and $S^T_n[\nu](f_1, \dots, f_n)$ are the joint cumulants of $\mathcal{O}(f_1), \dots, \mathcal{O}(f_n)$ under the law of the torus and under a law ν; Cov and Var are covariance and variance. $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ is the set of the periodic Gibbs states on $\mathbb{L}_K$ at β, in the sense of the Definitions of kernels and Gibbs states (Section YM3_I:91).

(3) The label of (K, β) is $b = a_K\beta$. We put $p_{K,\ell}(b) = \ell^{-3}\log Z_{K,\ell,L^K b}$ and $p_{K,\infty}(b) = \inf_\ell p_{K,\ell}(b)$ over the sides $\ell = 2L^m$. For a reference side ℓ': $D_{K,\ell} = p_{K,\ell} - p_{K,\ell'}$, $D_K = p_{K,\infty} - p_{K,\ell'}$, and $\bar{\eta}(\ell) = 3\ell' B'/\ell$. P is the pressure of the Wilson model on $\mathbb{Z}^3$ (Lem. 13.11) and $J(\beta) = P'(\beta+) - P'(\beta-)$. For a function D on an interval such that $D + \frac{1}{2}Cy^2$ is convex, D'' is the signed measure $(D + \frac{1}{2}Cy^2)'' - C\,\mathrm{d}y$; $\Delta_s^2 D(b) = D(b+s) + D(b-s) - 2D(b)$.

(4) A *source* is a complex Lipschitz function w on T_ℓ with $\|w\|_\sharp = \max\{\sup|w|, 40M\,\mathrm{Lip}\,w\} < 2\varrho$; this class lies in the class of the sources of Thm. YM3_I:AD. A real number x in the place of a source is the constant function $x \cdot 1$. Since $\mathcal{O}(x \cdot 1) = L^K x\sum_p a_p$, we have the *shift of the label*

$$Z_{K,\ell,L^K b}[w + x] = Z_{K,\ell,L^K(b-x)}[w].$$

(5) For $(K, \beta) \in \mathfrak{d}(g)$ and $\ell \in \mathfrak{L}(g)$, Thm. YM3_I:AD provides the source energies: a number E_ℓ and local terms $e_j(X; w)$, $1 \leq j \leq K$, X a localisation domain of level j with linear size $d_j(X)$. We put $\mathcal{E}_\ell = \sum_{j,X} e_j(X;\cdot)$ and $\widetilde{Z}_\ell(w) = e^{-E_\ell - \mathcal{E}_\ell(w)}Z_{K,\ell,\beta}[w]$; the index ℓ is dropped where one torus is fixed. G is the finite group generated by the translations of T_ℓ by $a_K\mathbb{Z}^3$ and the reflections $x_\mu \mapsto -x_\mu$, acting on sources by $(\mathsf{r}w)(y) = w(\mathsf{r}^{-1}y)$; it contains the reflections at all mirrors of all boards. We put $\mathcal{E}^{\mathrm{av}}(w) = |\mathsf{G}|^{-1}\sum_{\mathsf{r}\in\mathsf{G}}\mathcal{E}(\mathsf{r}w)$ and $\widetilde{Z}^{\mathrm{av}} = e^{-E-\mathcal{E}^{\mathrm{av}}}Z$.

■

Definition 13.3 (Boards, profiles, cube means)**.**

(1) l and l_1 are the least powers of L with $\mathsf{l} \geq 32M$ and $\mathsf{l}_1 \geq 8M + 16$; $T_0 = 640\sqrt{3}M$. The width of the collar is a real number $T \geq \max\{T_0, \mathsf{l}\}$; the side of a board that carries profiles and cube means is a power $q = L^i \geq 2T$; we write $q_i = L^i$.

(2) For a power $\mathsf{j} = L^i$ with $\mathsf{j} \geq La_K$, a *board* of side j with offset $o \in a_K\mathbb{Z}^3$ is the partition of $\mathbb{L}_K$, or of T_ℓ when $\ell \geq 2\mathsf{j}$, into the cubes $\square_c$ of side j centred at the points $o + \mathsf{j}c$, $c \in \mathbb{Z}^3$ (on T_ℓ, c modulo ℓ/j), the *cells*; c is the *index* of the cell. Since j/a_K is odd, the centres are sites and the *mirrors*, the planes half way between neighbouring centres, pass through midpoints of bonds. On T_ℓ there are $N = (\ell/\mathsf{j})^3$ cells, an even number in each direction, and every cell is even or odd, by the parity of $c_1 + c_2 + c_3$. The mirror copy $F^{[c']}$ of a function F of the cell c in the cell c' is its image under the reflections at the mirrors that carry c to c', complex conjugated if c and c' have different parities; $\|F\|_{\mathrm{ch}} = \langle \prod_{c'} F^{[c']} \rangle^{1/N}$ is the chessboard seminorm (equation YM3_I:(1756)). Profiles and cube means are formed on boards of the sides q of (1); boards of the sides l_1 and l enter the proofs as well.

(3) For a cell $\square_c$ of a board of side q, $\mathfrak{P}_T(\square_c)$ is the set of the real C^1 functions f with $0 \leq f \leq 1$, $f = 1$ at sup-distance $\geq T/2$ from $\partial\square_c$, $f = 0$ off $\square_c$ and at sup-distance $\leq T/8$ from $\partial\square_c$, and $\operatorname{Lip} f \leq 16\sqrt{3}/T$; then $\|f\|_\sharp = 1$. Its members are the *profiles* of the cell.

(4) The *cube mean* of the cell c with the profile f is $x_c = q^{-3}\mathcal{O}(f)$.

(5) $\mathsf{c} = \max\{1, 40\sqrt{3}M(16/T + 4/\mathsf{l})\} \in [1, 1 + 5\sqrt{3}]$; $C_w = 96\mathsf{c}(C_S + C_g)/\varrho$, and C_w^-, C_w^+ are the same number with 1, respectively $1 + 5\sqrt{3}$, in the place of c; $C_2 = 72e^4C_w^2$; $C_I' = 16(C_S + C_g)/(9\varrho^2)$; $x_q = 1/(3C_wTq^2)$ and $I_q(b) = [b - x_q, b + x_q]$.

(6) On T_ℓ, $\ell \geq 2q$, for a profile f of a cell c: F_e and F_o are the sums of the mirror copies of f over the cells of the parity of c and over the others, $F = F_e + F_o$, and $1 - F$ are the *walls*. Copies in different cells have supports at sup-distance $\geq T/4$, and $\|F\|_\sharp = \|F_e - F_o\|_\sharp = 1$. For complex z, $w_z = zF_e + \bar{z}F_o$ is the source of the source formula for the source zf of the cell, and $\Lambda(z) = \log \|e^{z\mathcal{O}(f)}\|_{\mathrm{ch}} = N^{-1} \log(Z[w_z]/Z)$.

(7) The model profile is $f_0 = \prod_\mu \varphi_0(x_\mu)$, with φ_0 even, C^1, $0 \leq \varphi_0 \leq 1$, non-increasing in $|s|$, $\varphi_0 = 1$ for $|s| \leq q/2 - 3T/8$, $\varphi_0 = 0$ for $|s| \geq q/2 - T/4$, $|\varphi_0'| \leq 16/T$; then $f_0(\cdot - y) \in \mathfrak{P}_T(\square_c)$ whenever $|y - y_c|_\infty \leq T/8$, y_c the centre of $\square_c$. With f_0 formed at the side q_i: $w_i = q_i^{-3} f_0(\cdot - \frac{1}{2}q_i e_3)$ and $\theta : x_3 \mapsto -x_3$; for $\mu = 1, 2, 3$, $w_i^{(\mu)} = q_i^{-3} f_0(\cdot - \frac{1}{2}q_i e_\mu)$ and $\theta_\mu : x_\mu \mapsto -x_\mu$.

■

Definition 13.4 (Limits)**.**

(1) $\mathfrak{S}_n(O)$ is the set of the n-tuples of real functions of $C_c^1(O)$ with pairwise disjoint supports, and $\mathfrak{S}(O) = \bigcup_n \mathfrak{S}_n(O)$, the separated tuples of [7, §K.1]. $X \cdot Y$ is the concatenation of tuples, τ_a the translation by a, (h) the tuple with the one entry h; translations and reflections act entry by entry. All assertions below on limits concern such tuples.

(2) A *Gibbs sequence* at g is a sequence (K_j, β_j, ν_j) with $(K_j, \beta_j) \in \mathfrak{d}(g)$, $K_j \to \infty$ and $\nu_j \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$; its labels are $b_j = a_{K_j}\beta_j$. It *converges* if $\sigma(X) = \lim_j \nu_j(\prod_i(\mathcal{O}(f_i) - \nu_j(\mathcal{O}(f_i))))$ exists for every $X = (f_1, \dots, f_n) \in \mathfrak{S}(\mathbb{R}^3)$. By the theorem on their limits (Thm. YM3_I:AP), σ is invariant under $\mathbb{R}^3 \rtimes \mathrm{O}(3)$ and reflection positive at every mirror; its connected functions S_n^σ are the limits of the cumulants $S_n^T[\nu_j]$; $\sigma((f)) = 0$; and in a canonical decomposition $(\mathsf{X}, \lambda, (\sigma_\xi))$ of σ, in the sense of Thm. YM3_I:AL, applied to the classes of periodic Gibbs states in Thm. YM3_I:AP, every component is a limit along the full sequence, $\sigma_\xi(X) = \lim_j \rho_j(\prod_i(\mathcal{O}(f_i) - \nu_j(\mathcal{O}(f_i))))$ with states $\rho_j \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$, is invariant and reflection positive likewise, and has $\sigma_\xi((f)) = \mathfrak{c}_\xi \int f$ with a number $\mathfrak{c}_\xi$, its one-point constant. Both obey the bound tables (Prop. YM3_I:106.3) of Thm. YM3_I:AP.

(3) For $\sigma' \in \{\sigma, \sigma_\xi\}$ there is a real-analytic function $\mathbf{s} \geq 0$ on $(0,\infty)$ such that
$$\sigma'(f,h) = \iint f(x)h(y)\mathbf{s}(|x-y|)\,\mathrm{d}x\,\mathrm{d}y, \qquad (f,h) \in \mathfrak{S}_2(\mathbb{R}^3).$$
Indeed, Thm. YM3_I:AP supplies the reconstruction data, and Prop. YM3_I:108.3 gives their real-analytic two-point kernel off the diagonal. Translation and rotation invariance identify this kernel with $\mathbf{s}(|x-y|)$. At separation re_3, its point-operator formula is
$$\mathbf{s}(r) = \langle \Omega, \mathsf{B}(e_3, \tfrac{r}{2}e_3)\mathsf{B}(\tfrac{r}{2}e_3, e_3)\Omega\rangle = \|\mathsf{B}(\tfrac{r}{2}e_3, e_3)\Omega\|^2 \geq 0.$$
Here the first equality follows by joining the two inner semigroup intervals and using that the vacuum is fixed by translations; the second is the adjoint identity for the point operators. Holomorphy of these operators for $\operatorname{Re} r > 0$ proves the stated real analyticity.

(4) The reflection Hilbert space and translation construction in Prop. YM3_I:94.1 gives, for such a σ' and the direction e_3, a Hilbert space $\mathcal{H}(\sigma', e_3)$ with vectors $[X]$, $X \in \mathfrak{S}(\mathbb{E}_+)$, $\mathbb{E}_+ = \{x_3 > 0\}$, $\langle [X], [Y]\rangle = \sigma'(\theta X \cdot Y)$; a self-adjoint $H \geq 0$ and unitary operators $\mathcal{U}(v)$, $v \perp e_3$, commuting and strongly continuous, with $[\tau_{te_3+v}X] = e^{-tH}\mathcal{U}(v)[X]$; and the joint spectral measure $\mathcal{P}$ of (H, P_1, P_2), carried by $[0,\infty)\times\mathbb{R}^2$, with $e^{-tH}\mathcal{U}(v) = \int e^{-tE+iv\cdot\mathfrak{q}}\,\mathrm{d}\mathcal{P}(E,\mathfrak{q})$. For real $h \in C_c^1(\mathbb{E}_+)$ we put $\psi_h = [(h)]$ and $\mu_h = \langle \psi_h, \mathcal{P}(\cdot)\psi_h\rangle$; $Q_r = [0,r]\times[-r,r]^2$.

(5) A continuum limit S at g *along growing tori* has a defining sequence (K_j, ℓ_j, β_j) with $(K_j, \beta_j) \in \mathfrak{d}(g)$, $\ell_j \in \mathfrak{L}(g)$, $K_j \to \infty$ and $\ell_j \to \infty$, no relation between K_j and ℓ_j being asked, and $S_n(X) = \lim_j S^T_{n;K_j,\ell_j,\beta_j}(X^{\langle \ell_j\rangle})$ for $n \geq 2$, $X^{\langle \ell\rangle}$ the periodisation; σ_S is its centred functional, formed from the S_n with $S_1 = 0$. By Props. YM3_I:92.1, YM3_I:92.3 and YM3_I:92.4, σ_S obeys a bound table with one-point constant 0, is invariant under the translations of $\mathbb{R}^3$ and the cubic group, and is reflection positive at every coordinate mirror; the reconstruction applies to it in each of the directions e_1, e_2, e_3.

■

Lemma 13.5 (Every frame has a partner)**.** For every frame there is a partner frame with the same radius and the same values of the constants that precede L in the list $\mathcal{T}^{\mathrm{S}}[2\varrho]$: one may take $\hat{L} = L^k + 2$ for every large integer k. For every radius there are also partner frames whose block sizes are L_1 and $L_1 + 2$. The cited construction theorems apply to each of these common choices, with its own subsequent constants and coupling ceiling. ■

Sketch of proof. Keep the exponents, reserve fractions and other constants preceding L in the ordered construction that defines the frame. The proof of Prop. YM3_I:69.1 admits every sufficiently large odd value of the new block size with these preceding values fixed. The further floors of Section YM3_I:82 preserve this property. Thus $\hat{L} = L^k + 2$ is admitted for every sufficiently large k. Complete the same construction at $\hat{L}$, choosing all subsequent constants anew and retaining the source-room and array-radius conditions at their respective choices; choose the coupling ceiling last. This gives a second frame with the same radius and the same pre-L data. Since L is odd, $L^k + 2$ is odd and $\gcd(L^k + 2, L) = \gcd(2, L) = 1$; if $L^p = \hat{L}^{p'}$ with positive integers p, p', a prime factor of L would divide $\hat{L}$, so
$$\log L / \log \hat{L} \notin \mathbb{Q}.$$
For an odd L_1 above the threshold of admission, L_1 and $L_1 + 2$ are both admitted, and a common prime factor would divide 2. Thm. YM3_I:AJ and the source-energy bounds of Thm. YM3_I:AD apply separately at both frames. The argument does not require equality of constants chosen after L. □

Definition 13.6 (The Setting; the constants; the floors)**.**

(1) *Frames.* Two frames, the second a partner of the first (Lem. 13.5); a hat marks the objects of the second.

(2) *Windows and reference sides.* In the first frame: $0 < g \le \gamma$; a nonempty compact set $W \subset I(g)$; a number $g_1 \in (0, g]$ with $W \subset \operatorname{int} I(g_1)$, which exists by Lem. 13.24; a number $\varepsilon > 0$ with $W_1 = [\min W - 3\varepsilon, \max W + 3\varepsilon] \subset I(g_1)$; the interval $W' = [\min W - 2\varepsilon, \max W + 2\varepsilon]$; and a reference side $\ell' \in \mathfrak{L}(g_1)$ with $\ell' \ge 2\mathsf{l}_1$. In the second frame: $0 < \hat g \le \hat\gamma$; a compact interval $\hat W = [\hat w_-, \hat w_+] \subset \hat I(\hat g)$ of positive length; and a reference side $\hat\ell' \in \hat{\mathfrak{L}}(\hat g)$ with $\hat\ell' \ge 2\hat{\mathsf{l}}_1$. All numbers of the first frame are formed at g_1 (so C_g stands for C_{g_1}) with the reference side ℓ', and those of the second at $\hat g$ with $\hat\ell'$.

(3) *The constants of the argument.*

- $C_\times(y) = 2592\,[65y^3 + 2]\,C_S S(\kappa)$ for $y > 0$, and $M_\times = 1348488\,C_S S(\kappa)$.
- Q_1 is the constant of the transfer (Subsection 13.1.4, item (M7)); it depends on L and on g_1 and on none of K, β, ℓ.
- $\mathsf{c}_\chi = 40\sqrt{3}ML$; $\mathsf{Q}_2 = \exp[(C_S + C_g)\mathsf{l}_1^3 + C_\times(1/8)]$; $\mathsf{Q}_* = 2\max\{\mathsf{Q}_1, \mathsf{Q}_2\} + 2$; $C_{\mathrm{lg}} = 54L^3 C_S S(\kappa)$; $C'_{\mathrm{lg}} = 27(10m' + 75)C_S S(\kappa)$;

$$h_1 = \frac{\varrho}{250e\,\mathsf{c}_\chi \mathsf{Q}_*}, \qquad \mathsf{A}_1 = 2C_{\mathrm{lg}} + \frac{L^3}{125}(4\log 2 + 2C'_{\mathrm{lg}}).$$

- $r_2 = \varrho/(e^{(C_S+C_g)\ell'^3}+1)$; $C_\omega = 2\log 2/(\ell'^3 r_2^2)$; $C'_2 = \mathsf{c}^2\varrho^{-2}[1152\,C_\times(7/2)\,\mathsf{l}^{-3} + 65536\,(C_S + C_g)]$; $r_1 = \min\{\varrho/(256\mathsf{c}),\, r_2/2,\, h_1/2\}$;

$$C_I = C'_I + C'_2 + C_\omega + 2\mathsf{A}_1/h_1^2 + M_\times T^{-3}\varrho^{-2}, \qquad C_0 = 2C'_I + C_\omega + 2\mathsf{A}_1/h_1^2;$$

  r_1^- and C_I^+ are r_1 and C_I with $1 + 5\sqrt{3}$ in the place of c and T_0 in the place of T.
- With $w_\pm$ the ends of W_1: $\lambda_- = \hat w_-/w_+$, $\lambda_+ = \hat w_+/w_-$, and

$$\delta_{\max} = \min\{\varepsilon/2,\, |\hat W|/(48\lambda_+)\}.$$

- k_7 is the least integer ≥ 1 with $\varsigma_7 = L^{-k_7} \le 2/(3\sqrt{3}\max\{1, \lambda_+\})$; the *tent* is $\chi(y) = \prod_\mu (1 - |y_\mu|/\varsigma_7)_+$ and the *plateau* $\eta(y) = \prod_\mu \min\{1, (3 - |y_\mu|/\varsigma_7)_+\}$; $\mathsf{c}_7 = 40\sqrt{3}M/\varsigma_7$; $c_T = 2\log 2\,(2\mathsf{Q}_1 + 2)^2(2/\varrho)^2$; $c_N = \varsigma_7^{-3} c_T \mathsf{c}_7^2$; $\hat c_N = \hat L^3 \hat c_T (40\sqrt{3}\hat M\hat L)^2$; $\hat{\mathsf{c}}_8 = \max\{1, 40\sqrt{3}\hat M/(\lambda_-\varsigma_7)\}$;

$$c_F = \sup_{K \ge k_7} \sup_{b \in W_1} |S^T_{2;K,\ell',L^K b}(\chi, 1 - \eta)|;$$

  $\hat c_F$ is the bound that the convergence on fixed tori (item (M8)) gives in the second frame at the side $\hat\ell'$ for $n = 2$, the interval $\hat W$, the spread $\lambda_-\varsigma_7$ and norms $\le \hat{\mathsf{c}}_8$; both are finite by (M8);

$$c_2 = \varsigma_7^{-3}[c_T\mathsf{c}_7^2 + \lambda_+^2 \hat c_T \hat{\mathsf{c}}_8^2 + c_F + \lambda_+^2 \hat c_F], \qquad \hat{\mathfrak{m}} = 16\hat B'/|\hat W| + \hat C_0|\hat W|,$$

$$C_\flat = \lambda_+^4[24\lambda_+\hat{\mathfrak{m}}/|\hat W| + 10\lambda_+\hat c_N + 1] + 6c_N + 3(C_0 + c_2) + 1.$$

- $\mu_K = (D_K + \frac{1}{2}C_0 y^2)''$ on $\operatorname{int} I(g_1)$ (a positive measure by Prop. 13.16). For $0 < \delta \le \delta_{\max}$, $K_1(\delta)$ is the least integer $K_1 \ge 1$ such that $\mu_K(I) \le C_\flat(|I| + \delta)$ for all $K \ge K_1$ and all closed intervals or points $I \subset W'$; such integers exist by Prop. 13.23, and $K_1(\delta)$ does not increase in δ.

(4) *The floors.*

$$q_0 = \max\Big\{\frac{4C_\flat + C_I^+}{9(C_w^-)^2 T_0^2},\ (3C_w^- T_0 \min\{r_1^-, \varepsilon/2\})^{-1/2}\Big\}, \qquad \delta^\circ(q) = \min\Big\{\delta_{\max},\ \frac{3C_w^- T_0}{2C_\flat q}\Big\},$$

$$K_0(q) = K_1(\delta^\circ(q)), \qquad \ell_0(q) = \max\Big\{2q,\ \ell',\ \frac{12\ell' B'}{C_\flat \varepsilon^2},\ \frac{16\,C_\flat \ell' B' q^2}{3(C_w T)^2}\Big\}.$$

(5) *Dependence.* All these numbers are formed from the two frames, g_1, $\hat{g}$, W, ε, $\hat{W}$, ℓ', $\hat{\ell}'$, and those that contain c or T (namely c, C_w, C_2, C_2', C_I, r_1, x_q and $\ell_0(\cdot)$) also from T; none depends on K, ℓ, b, a profile, an offset or a state. q_0, C_0, $C_\flat$, $\delta_{\max}$, $K_1(\cdot)$ and $K_0(\cdot)$ do not depend on T, and $K_0(q)$ and $\ell_0(q)$ do not depend on each other. $K_0(q)$ does not decrease in q.

■

13.1.2. *From cube means to low spectral mass.* The infrared argument measures an observable on a large cube while leaving a collar of fixed physical width. Let q be the cube side and T that width. Profiles in different cubes have separated supports, so their source insertions can be compared by chessboard estimates. The normalization by the cube volume turns the pressure estimates into a bound of order $(T/q)^2$ for the covariance of distinct cube means. The collar is retained throughout this comparison. Removing it prematurely would place insertions against the reflection boundaries used to prove the estimate.

There are two uses of positivity. On the lattice, reflection positivity controls the repeated source insertion by one periodic board. In the Euclidean invariant continuum limit, it supplies a positive representation of the radial two-point kernel. The two blocking factors enter between these steps: their pressure comparison prevents a concentrating response from defeating the cube-mean estimate. The pressure mechanism is shared with the preceding section, but its use here is a spatial covariance bound rather than a bound on a one-point jump.

Write $\mathbf{s}(r)$ for the radial two-point kernel after the constant part appropriate to the component has been removed. The representation proved in the detailed argument is

$$\mathbf{s}(r) = \frac{1}{4\pi r} \int_{[0,\infty)} e^{-mr} \, \mathrm{d}\rho(m).$$

Before the cube bound is imposed, the source also keeps a possible nonnegative constant κ' in the kernel. The cube-mean estimate excludes this constant and yields $\mathbf{s}(r) = O(r^{-2})$. To see the role of the cube geometry, take the reflected profiles in adjacent cells. Their separation is bounded above by a constant times q, their normalized integrals stay bounded below, and $\mathbf{s}$ is nonincreasing. Their covariance therefore bounds $\mathbf{s}(Cq)$ from above, up to those integral factors. The $(T/q)^2$ estimate passes to the claimed radial decay. The detailed calculation below keeps the collar correction and interpolates between successive admitted values of q.

The estimate on ρ then follows directly from its positivity. At $r = 1/m$, the integrand is at least e^{-1} on $[0, m]$, so the source obtains

$$\rho([0, m]) \le 4\pi e \, m^{-1} \mathbf{s}(1/m) = O(m).$$

In particular $\rho(\{0\}) = 0$. This is a bound on how much spectral mass can lie near zero. It permits nonzero mass in every interval $(0, m)$.

The source provides an example that attains both orders: $\rho = c \, \mathrm{d}m$ gives $\mathbf{s}(r) = c/(4\pi r^2)$. Thus neither the power bound nor the absence of an atom implies a positive mass gap. The conditional gap estimate belongs to the finite-size criterion later in the manuscript. In addition, the radial representation uses the Euclidean invariant Gibbs limits and their components. The weaker statements for arbitrary growing-torus limits are retained with their actual directional positivity; they are not silently identified with the radial case.

13.1.3. *Principal results.*

Theorem AL (Cube means in every periodic Gibbs state, on large tori and in every continuum limit)**.** Let Thm. YM3_I:D, Thms. YM3_I:AJ and YM3_I:AD and the statements (M1)–(M12) among the construction results listed in Subsection 13.1.4 be granted at each of the two frames of a Setting (Def. 13.6); by Lem. 13.5 every frame is the first frame of a Setting. Let $T \ge \max\{T_0, \mathsf{l}\}$. Then for every $q = L^i \ge \max\{2T, q_0\}$ the following hold.

(a) *Periodic Gibbs states.* Let $K \ge K_0(q)$, $b \in W$ and $\nu, \nu' \in \mathcal{G}^{\mathrm{per}}_{K,L^K b}$. For every board of $\mathbb{L}_K$ of side q, with any offset, and profiles of $\mathfrak{P}_T$ in distinct cells $c, c', c_1, \dots, c_n$ of it:

$$|\operatorname{Cov}_\nu(x_c, x_{c'})| \le C_2 (T/q)^2, \qquad |\nu(x_c) - \nu'(x_c)| \le 12 e^2 C_w T/q, \qquad \Big|\nu\Big(\prod_{i=1}^n (x_{c_i} - \nu(x_{c_i}))\Big)\Big| \le (12 e^2 C_w T/q)^n .$$

(b) *Tori.* Let $K \ge K_0(q)$, $b \in W$ and $\ell \in \mathfrak{L}(g)$ with $\ell \ge \ell_0(q)$. For the law $\langle\cdot\rangle_{K,\ell,L^K b}$ and every board of T_ℓ of side q the first and the third bound of (a) hold.

(c) *Limits of Gibbs sequences.* Let (K_j, β_j, ν_j) be a convergent Gibbs sequence at g whose labels lie in W from some index on (they need not converge), σ its limit, $(\mathsf{X}, \lambda, (\sigma_\xi))$ a canonical decomposition of σ, $\sigma' = \sigma$ or $\sigma' = \sigma_\xi$ for any ξ, $\mathbf{s}$ the radial kernel of σ' and $\mathcal{H}(\sigma', e_3)$ its reconstruction. Let j_0 be an index with $L^{j_0} \ge \max\{2T, q_0\}$, $r_0 = \sqrt{6} L^{j_0}$ and $C_{\mathbf{s}} = 6L^2 (1 - 3T/(4L^{j_0}))^{-6} C_2$. Then:

1. $\mathfrak{c}_\xi = 0$ for every ξ.
2. $S_2^{\sigma'}(\theta w_i, w_i) \le C_2 (T/q_i)^2$ for all $i \ge j_0$.
3. There is exactly one positive Borel measure ρ on $[0, \infty)$ with $\mathbf{s}(r) = \int e^{-mr} (4\pi r)^{-1} \,\mathrm{d}\rho(m)$ for all $r > 0$.
4. $\mathbf{s}(r) \le C_{\mathbf{s}} T^2 / r^2$ for $r \ge r_0$; in particular $r\mathbf{s}(r) \to 0$ as $r \to \infty$.
5. $|S_2^{\sigma'}(f, \tau_a h)| \le C_{\mathbf{s}} T^2 \|f\|_1 \|h\|_1 (|a| - R_f - R_h)^{-2}$ for all $f, h \in C^1_c(\mathbb{R}^3)$ with supports in the balls of radii R_f, R_h about 0 and all a with $|a| \ge R_f + R_h + r_0$.
6. $\rho([0, m]) \le 4\pi e C_{\mathbf{s}} T^2 m$ for $0 < m \le 1/r_0$.
7. $\rho(\{0\}) = 0$.
8. For real $h \in C^1_c(\mathbb{E}_+)$: $\|e^{-tH} \psi_h\|^2 \le C_{\mathbf{s}} T^2 \|h\|_1^2 / (4t^2)$ for $t \ge r_0/2$; ψ_h is orthogonal to $\ker H$; $\int_0^\infty \sigma'(\theta h, \tau_{te_3} h) \,\mathrm{d}t = \|H^{-1/2} \psi_h\|^2 < \infty$; and $\langle \psi_h, 1_{[0,\epsilon]}(H) \psi_h \rangle \le \frac{e^2}{4} C_{\mathbf{s}} T^2 \|h\|_1^2 \epsilon^2$ for $0 < \epsilon \le 2/r_0$.
9. Let $X = (f_1, \dots, f_n) \in \mathfrak{S}_n(\mathbb{R}^3)$ have its supports in the ball $B(0, R_X)$ and $h \in C^1_c(\mathbb{R}^3)$ its support in $B(0, R_h)$; let $\mathsf{N}(X)^2 = \sup_u \sigma'(X \cdot \theta_u X)$, the supremum over the unit vectors u, θ_u the reflection in the plane $\{u \cdot x = R_X + 1\}$; it is finite. Then for $|a| \ge R_X + R_h + 2 + r_0$

$$|\sigma'(X \cdot \tau_a h)| \le \frac{\mathsf{N}(X) \, \|h\|_1 \, C_{\mathbf{s}}^{1/2} T}{2(|a| - R_X - R_h - 2)}, \qquad |S^{\sigma'}_{n+1}(f_1, \dots, f_n, \tau_a h)| \le \frac{C(X) \, \|h\|_1 \, T}{|a| - R_X - R_h - 2},$$

with a number $C(X)$ formed from $C_{\mathbf{s}}$ and from the values of σ' at the subtuples of X and at their concatenations with their reflections: a connected function with one argument moved away tends to 0 at the rate $|a|^{-1}$, in every direction.

10. The converse and the measure that attains the bounds (Prop. 13.25 (d), (e)): for the limit of every convergent Gibbs sequence and every component of it, with the constant κ' and the measure ρ of Prop. 13.25, a bound $\sigma'(\theta w_i, w_i) \le C(T/q_i)^2$ for all large i, a bound $\mathbf{s}(r) = O(r^{-2})$ as $r \to \infty$, the pair $\kappa' = 0$, $\rho([0, m]) = O(m)$ as $m \to 0$, and a bound $\mu_h(Q_r) \le A_\mu \|h\|_1^2 r^2$ for small r are equivalent; the measure $\rho = c_\rho \,\mathrm{d}m$, whose kernel is $\mathbf{s}(r) = c_\rho/(4\pi r^2)$, satisfies them and attains 4 and 6 in order, so that these conclusions do not exclude it; and with no hypothesis on the labels the cube means of σ' obey the bound of the order $1/q$ of Prop. 13.25 (d), and q times the quantity that it bounds tends to 0 if and only if $\rho(\{0\}) = 0$.

(d) *Growing tori.* Let S be a continuum limit at g along growing tori, with no relation between K_j and ℓ_j, whose defining sequence has its labels in W from some index on, and let j_0 be as in (c). Then:

1. $\sigma_S(\theta_\mu w_i^{(\mu)}, w_i^{(\mu)}) \le C_2 (T/q_i)^2$ for all $i \ge j_0$ and $\mu = 1, 2, 3$.

2. For real $h \in C^1_c(\mathbb{E}_+)$, in $\mathcal{H}(\sigma_S, e_3)$:
$$\mu_h(Q_r) \le 12.46\,(1 - 3T/(4q_{j_0}))^{-6} L^2 C_2 \|h\|_1^2 T^2 r^2$$
for $0 < r \le 1/q_{j_0}$, and $\mathcal{P}(\{0\})\psi_h = 0$.
3. For
$$X = (f_1, \dots, f_n) \in \mathfrak{S}_n(\mathbb{R}^3)$$
and $h \in C^1_c(\mathbb{R}^3)$: $\sigma_S(X \cdot \tau_a h) \to 0$ and $S_{n+1}(f_1, \dots, f_n, \tau_a h) \to 0$ as $\min_\mu |a_\mu| \to \infty$.

(e) *Every label and every reference coupling.*

1. For every $b_\infty > 0$ there are an integer $s \ge 0$ and $g_2 \in (0, \gamma]$ with $L^s b_\infty \in \operatorname{int} I(g_2)$.
2. Let $s \ge 0$ be an integer, $0 < g_2 \le \gamma$, $W^{(2)} \subset I(g_2)$ compact, and form the numbers of Def. 13.6 for a Setting built on $(g_2, W^{(2)})$ in the place of (g, W), with the same second frame, and with the width $L^s T$. If $L^s b \in W^{(2)}$, $L^s T \ge \max\{T_0, \mathsf{l}\}$, $L^s q \ge \max\{2L^s T, q_0\}$ and $K - s \ge K_0(L^s q)$, then at $(K, L^K b)$ the three bounds of (a) hold multiplied by L^{8s}, L^{4s} and L^{4ns} respectively, T/q standing as it is; if besides $L^s \ell \in \mathfrak{L}(g_2)$ and $L^s \ell \ge \ell_0(L^s q)$, the bounds of (b) hold on T_ℓ multiplied likewise.
3. For every compact set $\mathsf{B} \subset (0, \infty)$ there are a constant C_B and floors q_B, $K_\mathsf{B}(\cdot)$, $\ell_\mathsf{B}(\cdot)$, none of them depending on T, such that
$$|\operatorname{Cov}_\nu(x_c, x_{c'})| \le C_\mathsf{B}(T/q)^2, \qquad |\nu(x_c) - \nu'(x_c)| \le (2C_\mathsf{B})^{1/2} T/q$$
for all $b \in \mathsf{B}$, $T \ge \max\{T_0, \mathsf{l}\}$, $q = L^i \ge \max\{2T, q_\mathsf{B}\}$, $K \ge K_\mathsf{B}(q)$, $\nu, \nu' \in \mathcal{G}^{\mathrm{per}}_{K, L^K b}$ and cells and profiles as in (a); and the covariance bound alone holds for the law $\langle\cdot\rangle_{K,\ell,L^K b}$ on boards of T_ℓ for all sides $\ell = 2L^m \ge \ell_\mathsf{B}(q)$.
4. At every $g' \in (0, \gamma]$, clauses (c) and (d) hold for every convergent Gibbs sequence at g' and for every continuum limit at g' along growing tori, with no condition on the labels, with C_B for $\mathsf{B} = I^{\mathrm{lab}}(g')$ in the place of C_2 and with q_B in the place of q_0.

■□

Theorem AM (The curvature of the pressure at large cutoffs; mean actions; the second bound of differences of means)**.** Assume the construction inputs listed in Thm. AL, and let a Setting be given.

(a) *No pile of specific heat.* For $0 < \delta \le \delta_{\max}$ and $K \ge K_1(\delta)$: $\mu_K(I) \le C_\flat(|I| + \delta)$ for every closed interval or point $I \subset W'$; $|D''_K|(I) \le (C_\flat + C_0)(|I| + \delta)$; and
$$-C_0 s^2 \le \Delta_s^2 D_K(b) \le C_\flat s^2 + C_\flat \delta s \qquad \text{whenever } [b - s, b + s] \subset W'.$$
With $\delta_0(K) = \inf\{\delta \in (0, \delta_{\max}] : K_1(\delta) \le K\}$, which tends to 0 as $K \to \infty$, these bounds hold with $\delta_0(K)$ in the place of δ. For $\delta' > 0$ let $\mathsf{p} = D_K * \phi_{\delta'}$, $\phi_{\delta'}(t) = \delta'^{-2}(\delta' - |t|)_+$; at every b with $[b - \delta', b + \delta'] \subset W'$: $-C_0 \le \mathsf{p}''(b) \le C_\flat(1 + \delta_0(K)/\delta')$ and $-C_0\delta'^2/12 \le \mathsf{p}(b) - D_K(b) \le C_\flat(\delta'^2/12 + \delta_0(K)\delta'/6)$; at $\delta' = \delta_0(K) > 0$: $-C_0 \le \mathsf{p}'' \le 2C_\flat$ and $|\mathsf{p} - D_K| \le \frac{1}{4}\max\{C_\flat, C_0/3\}\delta_0(K)^2$.

(b) *The atom at one label.* For $0 < \delta \le \delta_{\max}$, $b \in W'$ and $K \ge K_1(\delta)$:
$$\mu_K(\{b\}) = p'_{K,\infty}(b+) - p'_{K,\infty}(b-) = L^{4K} J(L^K b) \le C_\flat \delta.$$
And $\beta^4 J(\beta) \to 0$ as $\beta \to \infty$ through the reals.

(c) *Mean actions.* For every integer $K \ge 0$, every $b > 0$ and $\nu \in \mathcal{G}^{\mathrm{per}}_{K, L^K b}$, the number $\nu(a_p) = \mathsf{a}(\nu)$ is the same for all plaquettes p; with $e(\nu) = 3a_K^{-4}\mathsf{a}(\nu)$ we have $\nu(\mathcal{O}(f)) = \mathrm{Rie}_K(f)\, e(\nu)$ for every bounded f with compact support, and $p_{K,\infty}(b') \ge p_{K,\infty}(b) - (b' - b)e(\nu)$ for all $b' \ge 0$. There are $\nu_\pm \in \mathcal{G}^{\mathrm{per}}_{K, L^K b}$ with $-e(\nu_\pm) = p'_{K,\infty}(b\pm)$. Hence
$$\sup\{|e(\rho) - e(\nu)| : \rho, \nu \in \mathcal{G}^{\mathrm{per}}_{K, L^K b}\} = J_K(b) := p'_{K,\infty}(b+) - p'_{K,\infty}(b-) = L^{4K} J(L^K b),$$
and $J_K(b) = 0$ if and only if all periodic Gibbs states at $(K, L^K b)$ have the same mean action.

(d) *The second bound of differences of means.* For every integer $K \geq 0$, $b > 0$, $\nu, \nu' \in \mathcal{G}^{\mathrm{per}}_{K, L^K b}$, every board of $\mathbb{L}_K$ of side q and every f with $0 \leq f \leq 1$ that vanishes off a cube of side $\leq q - a_K$ (in particular every profile, every q and every cell): $|\nu(x_c) - \nu'(x_c)| \leq J_K(b)$. In the Setting, $J_K(b) = \mu_K(\{b\}) \leq C_\flat \delta$ for $b \in W'$ and $K \geq K_1(\delta)$. So in Thm. AL (a) the second bound may be read as $\min\{12e^2 C_w T/q,\ \mu_K(\{b\})\}$, and $\sup_{b \in W} |\nu(x_c) - \nu'(x_c)|$ tends to 0 as $K \to \infty$ uniformly in q, in the cells and in the profiles.

(e) *The plateau.* In any frame let $0 < g \leq \gamma$, $(K, \beta) \in \mathfrak{d}(g)$, $b = a_K \beta$, $\nu \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$, and let $\mathcal{I}$ be the σ-field of the Borel sets of configurations fixed by all translations by $a_K \mathbb{Z}^3$. There is one bounded real $\mathcal{I}$-measurable function e with $\nu(k a_p) = \nu(k\mathsf{e})$ for every plaquette p and every bounded $\mathcal{I}$-measurable k. For $f, h \in C^1_c(\mathbb{R}^3)$, $\bar{R} \in a_K \mathbb{N}$ with $\operatorname{supp} f \cup \operatorname{supp} h \subset (-\bar{R}, \bar{R})^3$ and $a_K \leq 2 \min\{\delta((f); \bar{R}), \delta((h); \bar{R})\}$, and $v \in a_K \mathbb{Z}^3$ with $|v| \geq \mathfrak{k}_1 \bar{R}$:

$$\left| S_2^T[\nu](f, \tau_v h) - 9 a_K^{-8} \operatorname{Rie}_K(f) \operatorname{Rie}_K(h) \operatorname{Var}_\nu(\mathsf{e}) \right| \leq \mathfrak{k}_0\, \mathsf{N}((f); \bar{R})^{1/2}\, \mathsf{N}((h); \bar{R})^{1/2}\, \frac{\bar{R}}{|v|},$$

where $\mathfrak{k}_0$, $\mathfrak{k}_1 = 4\sqrt{3}$, $\delta(\cdot; \bar{R})$ and $\mathsf{N}(\cdot; \bar{R})$ are the numbers of the plateau statement (Prop. YM3_I:100.1 and equations YM3_I:(1976)–(1980)); and $0 \leq 9 a_K^{-8} \operatorname{Var}_\nu(\mathsf{e}) \leq \frac{1}{4} J_K(b)^2$, with equality at $\nu = \frac{1}{2}(\nu_+ + \nu_-)$, while $\operatorname{Var}_\nu(\mathsf{e}) = 0$ if ν is ergodic or $\nu = \nu_\pm$.

(f) *A power of the spacing, under a rate in the cutoff.* Let two frames at one radius be given whose block sizes are L_1 and $L_2 = L_1 + 2$, which exist by Lem. 13.5; an index $i \in \{1, 2\}$ marks the objects of the frame i. Let Thms. YM3_I:AJ and YM3_I:AD be granted at both frames, and Thm. YM3_I:D at the choices of its constants that the two frames determine and at a lowered choice of each (Def. 4.47). In the frame i let $\mathcal{W}_i = [\gamma_i^{-2} + \bar{B}_i + \frac{1}{2},\ L_i(\gamma_i^{-2} + 3\bar{B}_i + 1) - 2\bar{B}_i]$, let $\chi^{(i)}(y) = \prod_\mu (1 - L_i |y_\mu - \frac{1}{2}|)_+$ and $\eta^{(i)}(y) = \prod_\mu \min\{1, (3 - L_i |y_\mu - \frac{1}{2}|)_+\}$ be the tent and the plateau of half-width $1/L_i$ about the point $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, and for integers $m, K \geq 1$, $\ell = 2L_i^m$ and $b > 0$ let

$$\mathsf{y}^{(i)}_{m,K}(b) = b^5 L_i^3\, S^T_{2; K, \ell, L_i^K b}((\chi^{(i)})^{\langle \ell \rangle}, 1 - (\eta^{(i)})^{\langle \ell \rangle})$$

be the far part of the specific heat of T_ℓ. Let the far part settle in the cutoff at a rate on every torus: there are numbers $\mathsf{k}_i \geq 0$ and $\mathsf{c}_{\mathrm{y},i} < \infty$, free of m, K and b, such that for all integers $m \geq 1$, $\ell = 2L_i^m$, all $b \in \mathcal{W}_i$ and all integers $K' \geq K \geq \mathsf{k}_i + 54m$

$$|\mathsf{y}^{(i)}_{m,K'}(b) - \mathsf{y}^{(i)}_{m,K}(b)| \leq \mathsf{c}_{\mathrm{y},i}\, \ell^9 K^{\mathsf{p}_{2,i}} L_i^{-K/10}, \qquad \mathsf{p}_{2,i} = 6\mathfrak{p}_{0,i} + \tfrac{1}{2},$$

$\mathfrak{p}_{0,i}$ the exponent of the theorem on two consecutive cutoffs (Thm. YM3_I:AR) in the frame i. Then for every real $A > 90$ there are numbers C_{r} and β_1 with $\beta^4 J(\beta) \leq \mathsf{C}_{\mathrm{r}} \beta^{-1/(2A)}$ for all $\beta \geq \beta_1$. Hence for every $0 \leq \alpha < 1/180$ the number $\mathsf{c}_*(\alpha) = \sup_{\beta \geq 1} \beta^{4+\alpha} J(\beta)$, a number of the Wilson model on $\mathbb{Z}^3$, is finite; and in every frame, for $0 < g \leq \gamma$, $(K, \beta) \in \mathfrak{d}(g)$ and $\rho, \nu \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$:

$$3 a_K^{-4} |\rho(a_p) - \nu(a_p)| \leq a_K^{-4} J(\beta) \leq g^{8+2\alpha}\, \mathsf{c}_*(\alpha)\, L^{-\alpha K};$$

$|\rho(\mathcal{O}_K(f)) - \nu(\mathcal{O}_K(f))| \leq a_K^{-4} J(\beta)(2R + a_K)^3 \|f\|_\infty$ for every bounded f with support in $[-R, R]^3$; and $|\rho(x_c) - \nu(x_c)| \leq g^{8+2\alpha} \mathsf{c}_*(\alpha) L^{-\alpha K}$ for every cube mean, free of q and of T, with no floor in the cutoff.

■□

The hypothesis K1 is the following inequality between the free energies of two periodic boxes. Its data are a label $\mathsf{b} > 0$, a side $\ell_\circ = 2L^{m_\circ}$ with $m_\circ \in \mathbb{Z}$, an integer $K_\circ$ with $K_\circ + m_\circ \geq 1$ and a number $0 < x_\circ \leq 0.069$; with $Z(n_1, n_2, n_3; \beta)$ the partition function of Wilson's action on the periodic lattice with n_1, n_2, n_3 sites a side, $l_K = \ell_\circ/(2a_K) = L^{K + m_\circ}$ and $\beta_K = L^K \mathsf{b}$, the hypothesis K1$[\mathsf{b}, \ell_\circ, K_\circ; x_\circ]$ requires

$$2 \log Z(l_K, 2l_K, 2l_K; \beta_K) - \log Z(2l_K, 2l_K, 2l_K; \beta_K) \leq x_\circ \qquad \text{for every integer } K \geq K_\circ.$$

We put $\theta_\circ = 12.9\, x_\circ \le 0.8901$ and $m_* = 2\log(1/\theta_\circ)/\ell_\circ \ge 0.2328/\ell_\circ$.

Theorem AN (The two-point function of the action density under the hypothesis K1 at one label)**.** Assume the construction inputs listed in Thm. AL; let $0 < g \le \gamma$, $\mathsf{b} \in I(g)$, and assume K1$[\mathsf{b}, \ell_\circ, K_\circ; x_\circ]$.

(a) *The gap* (from (M13) of Subsection 13.1.4). For each $K \ge \max\{K_\circ, 1\}$ there is exactly one periodic Gibbs state ν_K at $(K, L^K\mathsf{b})$; the Gibbs sequence $(K, L^K\mathsf{b}, \nu_K)_{K\ge\max\{K_\circ,1\}}$ converges; its limit σ is its own single component, has a unique vacuum Ω and is invariant under $\mathbb{R}^3 \rtimes \mathrm{O}(3)$; the limits at this label in every order of cutoff and volume, that is, the limits of $S^T_{n;K,\ell,L^K\mathsf{b}}(X^{\langle\ell\rangle})$ for $n \ge 2$ and $X \in \mathfrak{S}_n(\mathbb{R}^3)$, at the couplings $L^K\mathsf{b}$ themselves and with ℓ running through the sides $2L^m$, as K and ℓ tend to infinity jointly in any relation, or ℓ first and then K, or K first and then ℓ, are the connected functions $S^\sigma_n(X)$; in particular every continuum limit along growing tori at this label, that is, with a defining sequence in which $\beta_j = L^{K_j}\mathsf{b}$ from an index on, has the centred functional σ; the spectrum of H in $\mathcal{H}(\sigma, e_3)$ lies in $\{0\} \cup [m_*, \infty)$, and for $X, Y \in \mathfrak{S}(\mathbb{E}_+)$ and $t \ge 0$

$$\big|\sigma(\theta X \cdot \tau_{te_3} Y) - \sigma(X)\sigma(Y)\big| \le e^{-m_* t}\, \|(1 - P_\Omega)[X]\|\, \|(1 - P_\Omega)[Y]\|,$$

P_Ω the projection on $\mathbb{C}\Omega$, and likewise in every direction.

(b) *The kernel and the spectral measure.* Let $\mathbf{s}$ and ρ be the radial kernel of σ and its measure of Thm. AL (c) 3, which applies to σ by Thm. AL (e) 4. Then $\mathbf{s}(r) \le e^{-m_*(r-r')}\mathbf{s}(r')$ for $0 < r' \le r$; $\rho([0, m_*)) = 0$; and $\mathbf{s}(r) \le (r'/r)\, e^{-m_*(r-r')}\, \mathbf{s}(r')$ for $0 < r' \le r$. So $\mathbf{s}(r) = O(r^{-3})$ as $r \to \infty$ and $\rho([0, m]) = O(m^2)$ as $m \to 0$.

(c) *Cube means in the limit.* For $T \ge \max\{T_0, 1\}$, $q = L^i \ge 2T$, a board of $\mathbb{R}^3$ of side q (a partition of $\mathbb{R}^3$ into the cubes $\square_c$ of side q centred at the points $o + qc$, $c \in \mathbb{Z}^3$, with an offset $o \in \mathbb{R}^3$) and profiles $f_c \in \mathfrak{P}_T(\square_c)$, $f_{c'} \in \mathfrak{P}_T(\square_{c'})$ of distinct cells:

$$\big|S^\sigma_2(q^{-3} f_c, q^{-3} f_{c'})\big| \le \chi_\circ\, q^{-3}, \qquad \chi_\circ = 4\pi \int_{T/4}^\infty r^2 \mathbf{s}(r)\, \mathrm{d}r < \infty;$$

if the closures of the two cells have no common point, that is $|c - c'|_\infty \ge 2$, the supports of the two profiles are at distance $\ge q + T/4$, and the left member is at most $\mathbf{s}(q)$, which is exponentially small in q by (b).

■□

Proposition 13.7 (Two fixed tori: means, and the convergence of the difference of the pressures)**.** Assume the construction inputs listed in Thm. AL; let one frame and $0 < g \le \gamma$ be given.

(a) *An exact identity.* For $K \ge 1$, a side $\lambda = 2L^m$, $b \ge 0$ and a bounded f with compact support on $\mathbb{R}^3$: $\langle \mathcal{O}(f^{\langle\lambda\rangle})\rangle_{K,\lambda,L^K b} = -\operatorname{Rie}_K(f)\, p'_{K,\lambda}(b)$. For two sides $\ell > \ell'$,

$$A_K(f; b) := \langle \mathcal{O}(f^{\langle\ell\rangle})\rangle_{K,\ell,L^K b} - \langle \mathcal{O}(f^{\langle\ell'\rangle})\rangle_{K,\ell',L^K b} = -\operatorname{Rie}_K(f)\, D'_{K,\ell}(b),$$

where $D'_{K,\ell}(b) = 3a_K^{-4}\big[\langle a_p\rangle_{K,\ell',L^K b} - \langle a_p\rangle_{K,\ell,L^K b}\big] = -L^3 A_K(\chi_0; b)$, $\chi_0(y) = \prod_\mu (1 - L|y_\mu|)_+$.

(b) *Bounded, uniformly in the cutoff and in both sides.* For $b \in I(g)$, $K \ge 1$, $\ell, \ell' \in \mathfrak{L}(g)$ and f with support in a ball of radius ≤ 2: $|A_K(f; b)| \le 2\log 2\, (2\mathsf{Q}_1 + 2)\varrho^{-1}\|f\|_\sharp$; so $|D'_{K,\ell}(b)| \le 2\log 2\, (2\mathsf{Q}_1 + 2)\varrho^{-1} L^3 \mathsf{c}_\chi$, free of K, b, ℓ, ℓ'. For a cube mean whose cell lies in a ball of radius $\le \ell'/8$, its expectations on the two tori differ by $-q^{-3}\operatorname{Rie}_K(f) D'_{K,\ell}(b)$, and $0 \le q^{-3}\operatorname{Rie}_K(f) \le 1$.

(c) *Convergence of the difference, with a rate.* Let $\ell' < \ell$ be two sides of $\mathfrak{L}(g)$, $\mathsf{J} \subset I(g)$ a compact interval and $U_K = p_{K,\ell} - p_{K,\ell'}$. For $b \in \mathsf{J}$ and $K \geq \max\{K_{\mathrm{bf}}(\ell, \mathsf{J}), K_{\mathrm{bf}}(\ell', \mathsf{J})\}$:

$$|U_{K+1}(b) - U_K(b)| \leq 2\,\mathsf{C}_{\mathrm{bf}}(\mathsf{J})\,K^{\mathsf{p}_0} L^{-K/10} + 2\mathsf{v}_K, \qquad \mathsf{v}_K = 2C_{\mathrm{tt}} E_0 \sum_{s \geq u} L^{3s} \exp\Big[- \frac{\kappa \ell' L^s}{5M}\Big], \quad u = \lceil K/1260 \rceil,$$

with $\mathsf{p}_0 = 2\mathfrak{p}_0 + \frac{1}{2}$; here the number $\mathsf{C}_{\mathrm{bf}}(\mathsf{J})$, formed from the constants of the frame and the ends of J and free of the sides, the exponent p_0 and the thresholds $K_{\mathrm{bf}}(\cdot, \mathsf{J})$ are those of the theorem on two consecutive cutoffs (Thm. YM3_I:AR, Def. 2.7 (5), Cor. 2.8 (a)), and C_{tt}, E_0 those of the comparison of the vacuum energies of two tori (Lem. 13.31). So U_K converges as $K \to \infty$, uniformly on J.

(d) *Convergence of the derivative.* Let moreover $\ell' \geq 2\mathsf{l}_1$, and put $r_3 = \min\{h_1, \varrho/(e^{(C_S + C_g)\ell^3} + 1)\}$ and $M_3 = \mathsf{A}_1 + (\ell^{-3} + \ell'^{-3}) \log 2$, both free of K and of b. Then $|\partial_b^n U_K(b)| \leq n!\, M_3 r_3^{-n}$ for all $b \in I(g)$, $K \geq 1$ and $n \geq 1$. Let $\mathsf{W} = [b_-, b_+]$ with $\mathsf{J} = [b_- - r_3, b_+ + r_3] \subset I(g)$, and $\mathsf{o}_K = 4\mathsf{C}_{\mathrm{bf}}(\mathsf{J}) K^{\mathsf{p}_0} L^{-K/10} + 4\mathsf{v}_K$. For $K \geq \max\{K_{\mathrm{bf}}(\ell, \mathsf{J}), K_{\mathrm{bf}}(\ell', \mathsf{J})\}$

$$\sup_{\mathsf{W}} |U'_{K+1} - U'_K| \leq \frac{4\mathsf{o}_K}{r_3}\Big[1 + \log_2^+ \frac{2M_3}{\mathsf{o}_K}\Big].$$

Hence $\sum_K K \sup_{\mathsf{W}} |U'_{K+1} - U'_K| < \infty$, U'_K converges uniformly on W, and $\sup_K \sup_{\mathsf{W}} |U'_K| < \infty$; and for every real Lipschitz $f \neq 0$ with support in a ball of radius $\leq \ell'/8$, the difference between the two tori of the derivatives at zero source of the contact terms that the theorem on two consecutive cutoffs splits off at the cutoffs K and $K+1$ is summable in K, uniformly on W.

■

Proposition 13.8 (Variances, the cost of a source on the walls, and covariances)**.** Assume the construction inputs listed in Thm. AL; let one frame, $0 < g \leq \gamma$, $T \geq \max\{T_0, \mathsf{l}\}$ and $q = L^i \geq 2T$ be given.

(a) *Variances from below.* Let $b \in I(g)$, $K \geq 1$, a board of $\mathbb{L}_K$ of side q with any offset, a profile f of one of its cells c, and $X_\kappa = \mathcal{O}(f^{[\kappa]})$ for the cells κ. For $\ell \in \mathfrak{L}(g)$, $\ell \geq 2q$, with the board of T_ℓ of the same offset, let $\mathsf{e}_\ell = N^{-1} \partial_s^2 \mathcal{E}_\ell^{\mathrm{av}}(s(F_e - F_o))|_{s=0}$ and $\bar{\mathsf{e}} = \limsup_{\ell \to \infty} \mathsf{e}_\ell$, which is finite. Then for every $\nu \in \mathcal{G}^{\mathrm{per}}_{K, L^K b}$ and real numbers y_κ, finitely many of them non-zero,

$$\mathrm{Var}_\nu\Big(\sum_\kappa y_\kappa X_\kappa\Big) \geq (\bar{\mathsf{e}} - 2C'_I q^3) \sum_\kappa y_\kappa^2.$$

Consequently $\mathsf{V}_\nu(\kappa) := \mathrm{Var}_\nu(X_\kappa) - \bar{\mathsf{e}} \geq -2C'_I q^3$, and $|\operatorname{Cov}_\nu(X_\kappa, X_{\kappa'})| \leq \frac{1}{2}[\mathsf{V}_\nu(\kappa) + \mathsf{V}_\nu(\kappa')] + 2C'_I q^3$ for $\kappa \neq \kappa'$.

(b) *The cost of a source on the walls controls the cube means.* Let $b \in I(g)$, $K \geq 1$, $\ell \in \mathfrak{L}(g)$ with $\ell \geq 2q$, a board of T_ℓ of side q, a profile f of a cell c, and for $0 < r \leq \varrho/2$

$$\mathsf{S}_f(r) = N^{-1} \log \frac{\widetilde{Z}^{\mathrm{av}}(rF)\, \widetilde{Z}^{\mathrm{av}}(-rF)}{\widetilde{Z}^{\mathrm{av}}(0)^2}.$$

Then $\mathsf{S}_f(r) \geq -2C'_I q^3 r^2$, and there is a real number m with

$$\|x_c - m\|_{\mathrm{ch}} \leq (rq^3)^{-1} \exp\{\tfrac{1}{2}\mathsf{S}_f(r) + (2C'_I + M_\times T^{-3} \varrho^{-2}) q^3 r^2\}.$$

So for profiles f, f' of distinct cells c, c': $|\operatorname{Cov}_{K,\ell,L^K b}(x_c, x_{c'})| \leq 2(rq^3)^{-2} \exp\{\frac{1}{2}\mathsf{S}_f(r) + \frac{1}{2}\mathsf{S}_{f'}(r) + 2(2C'_I + M_\times T^{-3}\varrho^{-2}) q^3 r^2\}$; in particular, if $\mathsf{S}_f(r), \mathsf{S}_{f'}(r) \leq \mathsf{C}$ at $r = \hat{r} q^{-3+p/2} \leq \varrho/2$ with $p \in [2, 3]$ and $0 < \hat{r} \leq 1$, then $|\operatorname{Cov}_{K,\ell,L^K b}(x_c, x_{c'})| \leq 2\hat{r}^{-2} e^{\mathsf{C} + 2(2C'_I + M_\times T^{-3}\varrho^{-2})\hat{r}^2} q^{-p}$.

(c) *The cost from above.* In a Setting, with the numbers at g_1: for $b \in I(g_1)$, $K \geq 1$, $\ell \in \mathfrak{L}(g_1)$ with $\ell \geq \max\{2q, \ell'\}$, and $0 < r \leq \min\{r_1, b\}$,

$$\mathsf{S}_f(r) \leq q^3 \Delta_r^2 D_{K,\ell}(b) + 6C_w T q^2 r + 2C_I q^3 r^2.$$

In particular, for $b \in W$, $\ell \geq \max\{2q, \ell', 6\ell' B' q^3\}$, $0 < \hat{r} \leq 1$ with $\hat{r} q^{-3/2} \leq \min\{r_1, \varepsilon/2\}$, and $K \geq K_1(\min\{\delta_{\max}, q^{-3/2}\})$:

$$-2C_I' \hat{r}^2 \leq \mathsf{S}_f(\hat{r} q^{-3/2}) \leq 6C_w T \hat{r}\, q^{1/2} + 2C_I \hat{r}^2 + 2C_\flat + 1.$$

(d) *A power of q passes to the kernel.* Let $0 < g' \leq \gamma$, $p \in [2,3]$, and let a convergent Gibbs sequence at g' satisfy the hypothesis of Lem. 13.27 with numbers C_{LH}, c_{LH}, q_{LH} and, besides, for every $q = L^i \geq \max\{2T, q_{\mathrm{LH}}\}$ and $j \geq j(q)$, $|\operatorname{Cov}_\nu(x_c, x_{c'})| \leq C_p q^{-p}$ for all $\nu \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$ and profiles in two distinct cells of one board of side q. Then for σ and every σ_ξ, with $L^{j_0} \geq \max\{2T, q_{\mathrm{LH}}\}$ and $C_p' = (\sqrt{6}L)^p (1 - 3T/(4L^{j_0}))^{-6} C_p$: $\mathfrak{s}(r) \leq C_p' r^{-p}$ for $r \geq \sqrt{6} L^{j_0}$, and $\rho([0,m]) \leq 4\pi e C_p' m^{p-1}$ for $0 < m \leq 1/(\sqrt{6}L^{j_0})$.

(e) *One frame and a growth bound of the specific heat.* Let $W \subset \operatorname{int} I(g)$ be a compact interval, $\ell' \in \mathfrak{L}(g)$ with $\ell' \geq 2\mathsf{l}_1$, $C_{\mathrm{FS}} = 3\ell' B'$, and suppose that there are Y_0 and integers $K_Y(\ell)$ with $D''_{K,\ell} \leq Y_0 \ell^{1/3}$ on W for all $\ell \in \mathfrak{L}(g)$ with $\ell \geq \ell'$ and all $K \geq K_Y(\ell)$. Let $q \geq q_1 = \max\{2T, C_I/(9C_w^2 T^2), (3C_w T r_1)^{-1/2}\}$ with $6C_{\mathrm{FS}} q^3 \geq \max\{2q, \ell', 2L^{m_{\mathfrak{L}}(g)}\}$, let ℓ_q be the least side of $\mathfrak{L}(g)$ with $\ell_q \geq 6C_{\mathrm{FS}} q^3$, $K \geq K_Y(\ell_q)$ and $[b - x_q, b + x_q] \subset W$. Then for all $\nu, \nu' \in \mathcal{G}^{\mathrm{per}}_{K,L^K b}$ and profiles of $\mathfrak{P}_T$ in distinct cells $c, c', c_1, \dots, c_n$ of one board of side q: $|\operatorname{Cov}_\nu(x_c, x_{c'})| \leq 2e^4 \delta^2$, $|\nu(x_c) - \nu'(x_c)| \leq 2e^2 \delta$ and $|\nu(\prod_i (x_{c_i} - \nu(x_{c_i})))| \leq (2e^2\delta)^n$, with

$$\delta \leq \frac{T}{q}\Big[4C_w + \frac{(6LC_{\mathrm{FS}})^{1/3} Y_0}{3C_w T^2}\Big] + C_0 x_q.$$

(f) *The cost bounds the variances in every periodic Gibbs state.* Let $b \in I(g)$, $K \geq 1$, a board of $\mathbb{L}_K$ of side q, a profile f of one of its cells c, $X = \mathcal{O}(f)$, and sides $\ell_k \in \mathfrak{L}(g)$ with $\ell_k \geq 2q$ and $\ell_k \to \infty$ be given, the boards of T_{ℓ_k} having the same offset; let $\bar{\mathsf{e}}$ be the number of (a), $\mathsf{S}_{f,\ell_k}(r)$ the cost of (b) on T_{ℓ_k}, $c_X = M_\times q^3 T^{-3}$ the constant of the cross terms between the cells, and $c_3 = 3C_I' q^3 + 4c_X/(9\varrho^2)$. Then for every $\nu \in \mathcal{G}^{\mathrm{per}}_{K,L^K b}$ and $0 < r \leq \varrho/2$

$$\operatorname{Var}_\nu(X) \leq \bar{\mathsf{e}} + 2c_X \varrho^{-2} + 2r^{-2} \exp\Big[\limsup_{k\to\infty} \mathsf{S}_{f,\ell_k}(r) + c_3 r^2\Big].$$

If, with V_ν as in (a), $\mathsf{V}_\nu(\kappa) \leq C_{\mathrm{var}} q^{6-p}$ and $\mathsf{V}_\nu(\kappa') \leq C_{\mathrm{var}} q^{6-p}$ at two distinct cells κ, κ', with $p \in [2,3]$, then $|\operatorname{Cov}_\nu(q^{-3} X_\kappa, q^{-3} X_{\kappa'})| \leq (C_{\mathrm{var}} + 2C_I') q^{-p}$; at $p = 3$ this is the bound $(C_{\mathrm{var}} + 2C_I') q^{-3}$.

(g) *A one-sided bound on the walls bounds the cost.* In the situation of (b) let

$$\Upsilon_f(x) = N^{-1}\big[\log \widetilde{Z}^{\mathrm{av}}(xF) - \log \widetilde{Z}^{\mathrm{av}}(x)\big], \qquad \upsilon_f(x) = N^{-1} \partial_t \log \widetilde{Z}^{\mathrm{av}}(xF + t(1-F))\big|_{t=0},$$

the second being the derivative at the source xF in the direction of the walls. Then $\Upsilon_f(x) \leq -x\, \upsilon_f(x) + C_I' q^3 x^2$ for real $|x| \leq \varrho/2$. In a Setting, with the numbers at g_1, let $b \in W$, $\ell \in \mathfrak{L}(g_1)$ with $\ell \geq \max\{2q, \ell'\}$, $0 < \delta \leq \delta_{\max}$ and $K \geq K_1(\delta)$, and let $\Upsilon_f(x) \leq \mathsf{s}_1 x + C_\Upsilon$ for all real $|x| \leq \bar{x}$, with real numbers s_1, C_Υ and $\bar{x} > 0$. Then for $0 < r \leq \min\{\bar{x}, r_1, \varepsilon/2\}$

$$\mathsf{S}_f(r) \leq 2C_\Upsilon + q^3\big[C_\flat r^2 + C_\flat \delta r + 2\bar{\eta}(\ell)\big] + 2(C_\omega + 2\mathsf{A}_1/h_1^2) q^3 r^2.$$

■

13.1.4. *Construction estimates.* The proofs use Thm. YM3_I:D and Thms. YM3_I:AJ and YM3_I:AD among the construction results and the following statements (M1)–(M12) among the construction results, each taken at both frames of the Setting. Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD are granted together with these statements, several of which rest on them: among them, the convergence on fixed tori (M8) and the theorem on two consecutive cutoffs in (M12) rest on Thm. YM3_I:D, and the theorems on limits (M10) and Prop. YM3_I:106.3 in (M9) on Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD.

Statements among the construction results that are not cited by number are cited by the names they carry there. Thm. AN uses besides (M13), and clause (f) of Thm. AM uses (M14). (M13) is proved in §17.2 on the hypothesis K1, under that hypothesis. (M14) is proved in §12.1 on the latent heat of the Wilson model: its power is derived there from Thms. YM3_I:AJ and YM3_I:AD at the two frames of (M14), Thm. YM3_I:D at their choices and at a lowered choice of each, and the rate at which the far part of the specific heat settles in the cutoff on every torus, the rate that clause (f) of Thm. AM assumes and that (M14) carries among its hypotheses: for $i = 1, 2$, numbers $\mathsf{k}_i \geq 0$ and $\mathsf{c}_{\mathrm{y},i} < \infty$, free of m, K and b, with $|\mathsf{y}^{(i)}_{m,K'}(b) - \mathsf{y}^{(i)}_{m,K}(b)| \leq \mathsf{c}_{\mathrm{y},i}\, \ell^9 K^{\mathsf{p}_{2,i}} L_i^{-K/10}$ for all integers $m \geq 1$, $\ell = 2L_i^m$, all $b \in \mathcal{W}_i$ and all integers $K' \geq K \geq \mathsf{k}_i + 54m$.

The list is followed by the four statements (i)–(iv) that the proof of Prop. 13.7 (c) derives from (M12).

(M1) *Source energies.* Thm. YM3_I:AD, Prop. YM3_I:83.2, and Prop. YM3_I:74.4 give the following at $(K, \beta) \in \mathfrak{d}(g)$ and $\ell \in \mathfrak{L}(g)$. Each $e_j(X; \cdot)$ is holomorphic along finite families $z \mapsto w_0 + \sum_q z_q f_q$, vanishes at zero, depends on the source only on X with one layer of its level cubes, obeys conjugation reality, and has modulus at most $C_S e^{-\kappa d_j(X)}$. Moreover,

$$|\widetilde{Z}_\ell(w)| \leq e^{C_S \ell^3}, \qquad \widetilde{Z}_\ell(0) \geq e^{-C_g \ell^3}, \qquad |E_\ell \ell^{-3} - E_{\ell'} \ell'^{-3}| \leq C_S.$$

The vacuum splitting is equation YM3_I:(1737). The translation law in Thm. YM3_I:AJ, summed in equation YM3_I:(1736), makes $\mathcal{E}_\ell$ invariant under translations by $M\mathbb{Z}^3$.

(M2) *The stability functional.* Thm. YM3_I:AJ gives the finite history decomposition, its source-independent reference measures and evaluation domains, and one null set outside which both exponents are defined for all sources. Its equation YM3_I:(1728) is the density $\mathsf{Stay}_h^{\mathrm{T}} \chi_K(\Omega_K) e^{\mathbf{\Phi}_h^w + A^w_{K,h}}$. The local, boundary and retained terms are holomorphic in the source on these domains and obey conjugation reality. The real construction supplies the nonnegative Borel test factors and the reference measure; at a real source, the resulting history measure is positive because it is the pushforward of the positive bare density. The bare action is linear in its weight:

$$\rho_0^w(U) = e^{-A(\beta - L^K w, U)} = e^{-\beta \sum_p a_p + \mathcal{O}(w)}, \qquad A(c, U) = \sum_p c(x(p)) a_p(U), \qquad Z[w] = \int \rho_0^w \, \mathrm{d}U.$$

The source-independent coordinate maps and the measure identity are those of Thm. YM3_I:AI and Lem. YM3_I:82.5. Prop. YM3_I:83.2 identifies $E_\ell = E_{(K)}$, the source energies in equation YM3_I:(1736), and $\widetilde{Z}_\ell = \widetilde{Z}_K$ in equation YM3_I:(1737). Its equation YM3_I:(1738) bounds the sum of the normalized total variations of the history measures. This stronger estimate, not merely a bound on $|\widetilde{Z}|$, is used below.

(M3) *The count of domains.* For a domain X of level j and depth $s = K - j$, and sets Y, Y': (a) if X with one layer of cubes meets Y and Y', then $d_j(X) \geq L^s \operatorname{dist}_\infty(Y, Y')/M - 4$; (b) for $\kappa' > \log 9216$ and real D', the sum of $e^{-\kappa' d_j(X)}$ over the X that with one layer meet Y and have $d_j(X) \geq D'$ is $\leq 27 N_s(Y) S(\kappa') e^{-(\kappa' - \log 9216)[D']_+}$; (c) $N_s(Y) \leq (\ell L^s/M)^3$, and $N_s(Y) \leq (2RL^s/M + 2)^3$ if Y lies in a sup-ball of radius R, $N_s(Y)$ being the number of the cubes of side ML^{-s} of the standard partition that meet Y.

(M4) *The isometries.* The translations by $a_K\mathbb{Z}^3$ and the reflections carry the Wilson weight and the Haar measure into themselves ($\operatorname{tr} g^{-1} = \operatorname{tr} g$ on SU(2)), and $\mathcal{O}(\mathsf{r}w) = \mathcal{O}(w) \circ \mathsf{r}^{-1}$; so $Z[\mathsf{r}w] = Z[w]$ and $\|\mathsf{r}w\|_\sharp = \|w\|_\sharp$.

(M5) *The passage and labels.* For $b \in I(g)$ and $K \geq 1$, the pair $(K, L^K b)$ belongs to $\mathfrak{d}(g)$, by Cor. YM3_I:74.2. Prop. YM3_I:74.1 gives the label bounds, in particular $a_K\beta \geq g^{-2}$; their fixed compact interval is $I^{\mathrm{lab}}(g)$, equation YM3_I:1854. *The sides.* The family $\mathfrak{L}(g)$ is closed upward, and $\mathfrak{L}(g') \subset \mathfrak{L}(g)$ for $0 < g' \leq g$, as follows from its lower bound on the side.

(M6) *The chessboard estimate*, at every $\beta \geq 0$, on boards of every side $\mathsf{j} = L^i \leq \ell/2$ (Def. 13.3 (2)): $\|cF\|_{\mathrm{ch}} = |c|\,\|F\|_{\mathrm{ch}}$, $\|1\|_{\mathrm{ch}} = 1$, $\|F\|_{\mathrm{ch}} \leq \sup|F|$, the triangle inequality, and $|\langle \prod_s F_s\rangle| \leq \prod_s \|F_s\|_{\mathrm{ch}}$ for functions F_s of different cells. *The source formula*: for a complex source w that vanishes outside the interior of a cell, $\|e^{\mathcal{O}(w)}\|_{\mathrm{ch}}^N = Z_\beta[w^{\mathrm{ch}}]/Z_\beta \geq 0$, wch the sum of the mirror copies of w (conjugated in the cells of the other parity). *The domination*: for $K \in \mathbb{N}$, $\beta > 0$, $\nu \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$, a board of $\mathbb{L}_K$ of any side $\geq La_K$ and any offset, functions $F_{\mathbf{k}}$ of finitely many different cells and every sequence of sides $\ell_k = 2L^{m_k} \to \infty$: $|\nu(\prod F_{\mathbf{k}})| \leq \liminf_k \prod \|F_{\mathbf{k}}\|_{\mathrm{ch},\ell_k}$.

(M7) *The transfer.* There is Q_1, depending on L and on g and on none of K, β, ℓ, such that for $(K,\beta) \in \mathfrak{d}(g)$, all $n \geq 1$, all $f_1, \dots, f_n \in \mathrm{Lip}_c(\mathbb{R}^3)$ with supports in one ball of radius 2, in any mutual position, and any two of the laws $\nu \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$ and $\langle\cdot\rangle_{K,\ell,\beta}$, $\ell \in \mathfrak{L}(g)$ (the latter carried to $\mathbb{L}_K$, so that its cumulants at f are $S^T_{n;K,\ell,\beta}(f^{\langle\ell\rangle})$), the cumulants of order n differ by at most $2\log 2\,(2\mathsf{Q}_1 + 2)^n (n/\varrho)^n \prod_q \|f_q\|_\sharp$.

(M8) *The convergence on fixed tori.* For every side $\ell = 2L^m$, $n \geq 2$, real Lipschitz $f_1, \dots, f_n$ on T_ℓ, none of them 0, whose supports have no common point and the spread $\geq \mathsf{d} > 0$ (for $n = 2$: the sup-distance of the supports), and every compact interval $\mathsf{J} \subset (0,\infty)$: $S^T_{n;K,\ell,L^K b}(f_1, \dots, f_n)$ converges as $K \to \infty$ uniformly in $b \in \mathsf{J}$, and $\sup_K \sup_{b\in\mathsf{J}} |S^T_{n;K,\ell,L^K b}(f)| \leq \mathsf{c}'(\ell, n, \mathsf{J}, \mathsf{d}) \prod_i \|f_i\|_\sharp$, the right member reading of the functions the spread and the norms alone.

(M9) *Periodic Gibbs states.* The Definitions of kernels and Gibbs states: the Gibbs kernel γ_B of a finite set B of bonds is formed with the exponent $H_B = -\beta \sum_{p\in \mathrm{Pl}[B]} a_p$, $\mathrm{Pl}[B]$ the plaquettes containing a bond of B; ν is a Gibbs state if $\nu(\gamma_B F) = \nu(F)$ for all B and bounded Borel F, and periodic if it is invariant under a subgroup of finite index of the lattice translations. There is at least one periodic Gibbs state at every (a,β), and weak limits of the laws of the tori are such states; every periodic Gibbs state is invariant under the translations and the cubic group of its lattice; weak limits of periodic Gibbs states at couplings $\beta_k \to \beta$ that are invariant under the lattice translations are periodic Gibbs states at β; the tangent inequality for periodic Gibbs states; the exact dilation facts: for real $s > 0$ the dilation $\mathrm{dil}_s : x \mapsto sx$ carries the bonds, the plaquettes, the functions a_p, the product Haar measures and the glued configurations of $a\mathbb{Z}^3$ to those of $sa\mathbb{Z}^3$, with $\mathcal{O}_a(f)(U) = s\,\mathcal{O}_{sa}(f(\cdot/s))(\mathrm{dil}_s U)$, and for s a power of L it maps the periodic Gibbs states onto the periodic Gibbs states; the classes at a cutoff and the bound system at g (equation YM3_I:(2030); Section YM3_I:100), whose two members bound the cumulants of separated tuples in the members of a class and the differences of the one-point functions of two members, free of the cutoff, the coupling and the state; Prop. YM3_I:106.3: for $(K,\beta) \in \mathfrak{d}(g)$ the set $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ is a class at the cutoff K that obeys the bound system at g, and with ν it contains $k\nu/\nu(k)$ for every bounded $\mathcal{I}$-measurable $k \geq 0$ with $\nu(k) > 0$; and the plateau statement for such a class (Prop. YM3_I:100.1, with the numbers of equations YM3_I:(1976)–(1980)), with its statements on conditional expectations given the invariant σ-field $\mathcal{I}$.

(M10) *Limits.* We use the Gibbs limits and their component laws in Thms. YM3_I:AP and YM3_I:AL, and the compactness, translation and cubic invariance, and coordinate reflection positivity of growing-torus limits in Props. YM3_I:92.1, YM3_I:92.3 and YM3_I:92.4. The

radial kernel and the reconstruction, including their bounds and continuity, are those stated in Def. 13.4.

(M11) *The ordered choice of constants.* Use the order in the proof of Prop. YM3_I:69.1, with the source floors of Section YM3_I:82 inserted at their stated places. Every initial segment selected by this procedure extends to a complete choice: the next free constant has either a finite lower bound, a positive upper bound or the nonempty interval specified at that step, and each determined constant is then fixed by its formula. Continue through the finite list and choose γ last. With the constants before L fixed, all sufficiently large odd L are allowed; the remaining constants are subsequently chosen in that same order.

(M12) For Prop. 13.7 alone, the following, on one torus T_λ, $\lambda = 2L^m$, at one label (a hat marks the objects of the classical starts). The theorem among the construction results on two consecutive cutoffs (Thm. YM3_I:AR), with its read-out of the one-point functions, for derivatives in the source and read here at the order 1, up to the derivative at the source 0 of the difference of its contact terms (Section YM3_I:116), and with the bound of its error, summable in K (its clause (3)). The objects of its comparison (Section YM3_I:116) for the lattices of the cutoffs K and $K+1$, marked A and B, with the small numbers ϵ_{fg}, ϵ_{cl}, ϵ_{sh}, $\epsilon_3 = 4\epsilon_{\mathrm{cl}} + 6\epsilon_{\mathrm{sh}}$ and the depth $u = \lceil K/1260 \rceil$ of §2.3.1 at $n = 1$, where $\epsilon_{\mathrm{cl}}, \epsilon_{\mathrm{sh}} \le \frac{1}{20}$ under the conditions of the theorem: on the inner ball In, a set of block fields of the depth u that contains the trivial block field $\mathbf{1}$, the measures of the chains and of the classical starts are a constant times the exponential of an exponent times the Haar measure, and their masses, integrals and constants satisfy $T^\bullet = e^{c^\bullet} I^\bullet$, $\hat{T}^\bullet = e^{\hat{c}^\bullet} \hat{I}^\bullet$ and are holomorphic in the source (Prop. YM3_I:114.2 and the proof of Prop. YM3_I:116.3); the measures of the classical starts are integrals over the block fields of a larger depth, against the weight of the run (§2.3.2), and at real sources the functions so integrated and the measures are non-negative (the proof of Prop. YM3_I:116.1).

The data of its read-out (Section YM3_I:116): $|I^\bullet/\hat{I}^\bullet - 1| \le 2\epsilon_{\mathrm{fg}} \le \frac{1}{4}$ and a bound of the constants of the classical starts (the proof of Prop. YM3_I:116.3), and at real sources $\hat{T}^{\mathrm{A}}, \hat{T}^{\mathrm{B}} > 0$ with $|\hat{T}^{\mathrm{B}}/\hat{T}^{\mathrm{A}} - 1| \le \epsilon_3 \le \frac{1}{2}$ (Prop. YM3_I:116.1). The discarding bound $0 \le Z - T^\bullet(0) \le \mathsf{R}_{t_1} Z$, Z the partition function of the torus at the cutoff of $\bullet$ and R_{t_1} the discarding number, one number for both cutoffs (Prop. YM3_I:115.2, the proof of Thm. YM3_I:AR), with $\mathsf{R}_{t_1} \le \frac{1}{2}$ under the conditions of the theorem.

The number $\mathfrak{o}_K = 8\mathsf{R}_{t_1} + 12\epsilon_{\mathrm{fg}} + 8\epsilon_{\mathrm{cl}} + 12\epsilon_{\mathrm{sh}}$ and, for an interval J of labels of the theorem on the side λ (§2.3.1; every compact interval $\mathsf{J} \subset I(g)$ is one on every side of $\mathfrak{L}(g)$, as $\min \mathsf{J} > g^{-2} + \bar{B} \ge \gamma^{-2}$, $\max \mathsf{J} \le X_g - \bar{B} \le L(g^{-2} + \bar{B})$ and $M\,R(\max \mathsf{J}) \le M\,R(L(g^{-2} + \bar{B})) < L^{m_{\mathfrak{L}}(g)}$ by equation YM3_I:(1528), the cube-size function R being non-decreasing), its number $\mathsf{C}_{\mathrm{bf}}(\mathsf{J})$, formed from the constants and the ends of J, its exponent $\mathfrak{p}_0$ and its thresholds $K_{\mathrm{bf}}(\lambda, \mathsf{J})$: for every label of J and every $K \ge K_{\mathrm{bf}}(\lambda, \mathsf{J})$ the conditions of the theorem hold and $\mathfrak{o}_K \le \mathsf{C}_{\mathrm{bf}}(\mathsf{J}) \lambda^3 K^{\mathfrak{p}_0} L^{-K/10}$, $\mathfrak{p}_0 = 2\mathfrak{p}_0 + \frac{1}{2}$ (Def. 2.7 (4), (5), Cor. 2.8 (a)); with these conditions the setting (Th) among the construction results holds for the run of each of the two cutoffs on the torus (§2.3.1).

Its comparison of the two classical runs: their masses and their densities in L^1 (Prop. YM3_I:116.1 and its proof); the pointwise quotient of the two classical starts at a real source, a factor in $[e^{-\epsilon_{\mathrm{cl}}}, e^{\epsilon_{\mathrm{cl}}}]$ where both weights are positive, the two weights being equal there (Prop. YM3_I:116.1); and the bound, by ϵ_{sh} times its mass, of the part of one classical run where the weight of the other vanishes (the positive-kernel argument in the proof of Prop. YM3_I:116.1), which bounds the tails of a classical run by multiples of its measures as an order of measures, against every bounded Borel function of the block field. The exponent and the constant of a step of a run (Prop. YM3_I:114.2), and the action of a bare array, in which every member enters less its value at the trivial pair (equation YM3_I:(874)); the

one value and the continuity of the weighted action of the background (the gauge-invariance argument preceding equation YM3_I:(2133) and the local analytic-background argument in the proof of Prop. YM3_I:52.3), the continuous choice of the background near a regular block field (Thm. YM3_I:E and its real-analytic continuation), and the one value of the action of the array on the minimal orbit (Props. YM3_I:114.2 and YM3_I:113.13); a local term is a holomorphic, so continuous, function of the pair that it reads, on an open set of pairs (Lem. 5.26), and the pair of a configuration on a domain is a polynomial map of its matrix entries (§5.2, YM3_I:(70), YM3_I:(69)).

The line $x_j = L^{K-j}b$, $0 \le j \le K$, of a label, whose increments vanish, is an admissible finite sequence with last index K, which with the integer m of a side $2L^m$ forms admissible data, and its cut $(x_0, \dots, x_{K-1})$ with the lattice (λ, K) and these couplings forms a frame of level $K-1$ among the construction results (the proof of Lem. YM3_I:118.1). The family of the level j of the chain at the source 0 is the family of the level j of the canonical progression of the cut $(x_0, \dots, x_{K-1})$ of the line $x_j = L^{K-j}b$ of the label (the proof of Lem. YM3_I:118.1); at that frame the members of the levels $j \le K$ of this progression are those of the progression of the line, and they are regular families on every lattice of their level, with their companions (Thm. YM3_I:Y).

The exact step identity eq. 13.ii identifies the logarithm of the Gaussian normalizing factor with the extensive term and the sum of its local Gaussian constants. Adding the values $E^{(j+1)}(X, \mathbf{1}, z)$ of the regular family gives the vacuum number $\mathsf{E}^{(j)}(T_\ell, T_\ell)$ of that step. Their sum through step n is $E_{(n)}$, as in eq. 13.i; these definitions agree with equations YM3_I:(1406)–(1407). Lem. 13.31 proves the exact identity and the comparison needed below.

For the comparison on two tori we use the same admissible sequence and its regular families, with their common companions, on both sides. The real vacuum families and the rate conditions are those stated in Lem. 13.31. The pair of sides is admitted once $LMR(x_K) \le L^{m_i}$ for $i = 1, 2$: the retained room condition [7, eq. 215] gives $C_L \le 2LR(\gamma^{-2}) \le 2LR(x_K)$, so $\ell_i = 2L^{m_i} \ge 2LMR(x_K) \ge C_L M$. For the line $x_j = L^{K-j}b$ of a label $b \in \mathsf{J}$, the last radius is $R(b)$ (equation YM3_I:(634)), and on the sides of $\mathfrak{L}(g)$ the required room is $LMR(b) \le LMR(L(g^{-2} + \bar{B})) \le L^{m_{\mathfrak{L}(g)}} \le L^{m_i}$, by equation YM3_I:(1528).

(M13) For Thm. AN alone: the consequences of the hypothesis K1 proved in §17.2 on one inequality between the partition functions of two periodic boxes. Let $0 < g \le \gamma$, $\mathsf{b} \in I(g)$, let K1$[\mathsf{b}, \ell_\circ, K_\circ; x_\circ]$ hold, and let $K \ge \max\{K_\circ, 1\}$ and $\beta = L^K \mathsf{b}$. (i) For every side $\ell = 2L^m \ge \ell_\circ$: $0 \le \log Z_{K,\ell,\beta} - \ell^3 p_{K,\infty}(\mathsf{b}) < 0.13736\, \theta_\circ^{\ell/\ell_\circ}$, the bound being the same at every such cutoff. (ii) There is exactly one periodic Gibbs state ν_K at (K, β). (iii) Over every cross-section of physical sides $\ge \ell_\circ$ the largest eigenvalue λ_0 of the transfer operator defined in §17.2 is simple, and the next one, λ_1, has $-a_K^{-1} \log(\lambda_1/\lambda_0) > m_*$. (iv) For $n \ge 2$, a side $\ell_1 = 2L^{m_1} \ge \ell_\circ$ and $X = (f_1, \dots, f_n)$ of real Lipschitz functions, none of them 0, with pairwise disjoint compact supports in $(-\ell_1/4, \ell_1/4)^3$, there is a number $C(X)$ with $|S_n^T[\nu_K](X) - S^T_{n;K,\ell_2,\beta}(X^{\langle \ell_2 \rangle})| \le C(X)\, \theta_\circ^{\ell_2/(2\ell_\circ)}$ for all $K \ge \max\{K_\circ, 1\}$ and all sides $\ell_2 = 2L^{m_2} \ge \ell_1$. (v) The Gibbs sequence $(K, L^K \mathsf{b}, \nu_K)_{K \ge \max\{K_\circ, 1\}}$ converges; its limit σ is its own single component; $\ker H = \mathbb{C}\Omega$ in $\mathcal{H}(\sigma, e_3)$; and the spectrum of H lies in $\{0\} \cup [m_*, \infty)$.

(M14) For clause (f) of Thm. AM alone: the power of the jump of the slope of the pressure that §12.1 on the latent heat of the SU(2) Wilson model on $\mathbb{Z}^3$ proves for two frames at one radius with the block sizes L_1 and $L_2 = L_1 + 2$, under the following hypotheses. Thms. YM3_I:AJ and YM3_I:AD are granted at both frames, and Thm. YM3_I:D at their choices and at a lowered choice of each. In each frame $i \in \{1, 2\}$ the far part of the specific heat settles in the cutoff at a rate on every torus: with the window $\mathcal{W}_i$, the tent $\chi^{(i)}$ and the plateau $\eta^{(i)}$ of half-width $1/L_i$ about $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ of clause (f), and the far part

$\mathsf{y}^{(i)}_{m,K}(b) = b^5 L_i^3\, S^T_{2;K,\ell,L_i^K b}((\chi^{(i)})^{\langle\ell\rangle}, 1-(\eta^{(i)})^{\langle\ell\rangle})$ of the specific heat of T_ℓ, $\ell = 2L_i^m$, there are numbers $\mathsf{k}_i \geq 0$ and $\mathsf{c}_{\mathrm{y},i} < \infty$, free of m, K and b, such that for all integers $m \geq 1$, all $b \in \mathcal{W}_i$ and all integers $K' \geq K \geq \mathsf{k}_i + 54m$

$$|\mathsf{y}^{(i)}_{m,K'}(b) - \mathsf{y}^{(i)}_{m,K}(b)| \leq \mathsf{c}_{\mathrm{y},i}\, \ell^9 K^{\mathsf{p}_{2,i}} L_i^{-K/10}, \qquad \mathsf{p}_{2,i} = 6\mathfrak{p}_{0,i} + \tfrac{1}{2},$$

$\mathfrak{p}_{0,i}$ the exponent of the theorem on two consecutive cutoffs (Thm. YM3_I:AR) in the frame i. Under these hypotheses, for every real $A > 90$ there are numbers C_{r} and β_1 with $\beta^4 J(\beta) \leq \mathsf{C}_{\mathrm{r}} \beta^{-1/(2A)}$ for all $\beta \geq \beta_1$.

The proof of Prop. 13.7 (c) derives from (M12), in its Step 3 and on one torus T_λ, four statements: (i) the read-out at the order 0; (ii) the comparison of the two classical starts as measures on the inner ball; (iii) the cancelling of the constants of the two classical starts, from the value 0 and the continuity of their exponents at the trivial configuration; and (iv) the identity of the sum of the constants of the chain at the source 0 with a vacuum energy of the comparison of two tori, a real number. We use the following standard results: Cauchy's formula and estimates, Schwarz's lemma, the maximum principle, Poisson's formula, the inequalities of Jensen, Hölder and Cauchy–Schwarz, the theorems of Tonelli and Fubini, Morera's theorem, Kronecker's theorem (the multiples of an irrational number are dense modulo 1), Helly's selection theorem, Bernstein's theorem on completely monotone functions and his inequality for the derivatives of polynomials, the spectral theorem, Stolz's theorem, and Hölder's inequality in the Schatten classes, which in the one case used is proved in Prop. 13.15.

13.1.5. *Estimates and proofs.* Up to the window inequality the argument runs in one frame. The energies averaged over the isometries and their cross terms (Lem. 13.9) give the energies of a constant source on two sides on a complex disc (Lem. 13.10); finite size (Lem. 13.11) is exact; the stability functional (Lem. 13.12) gives the imaginary direction (Prop. 13.13), which with the cross terms and the chessboard estimate gives the walls (Prop. 13.14); these four give the main inequality and semi-convexity (Prop. 13.16); with Cauchy's estimate in the chessboard seminorm (Lem. 13.17) the main inequality gives the reduction on tori (Prop. 13.18), and with finite size and the domination the window inequality in every periodic Gibbs state (Prop. 13.19). The bound against piles of specific heat (Prop. 13.23) uses semi-convexity and finite size in both frames, the transfer, the convergence on fixed tori and dilations by real factors (Lems. 13.20–13.22). Clauses (a), (b) of Thm. AL follow from the reduction and this bound; clauses (c), (d) from (a), (b) and the theorems on limits through Prop. 13.25 and Lems. 13.26–13.28; clause (e) through Lem. 13.24.

Lemma 13.9 (The energies averaged over the isometries; cross terms)**.** Let $(K, \beta) \in \mathfrak{d}(g)$ and $\ell \in \mathfrak{L}(g)$.

(A1) $\mathcal{E}^{\mathrm{av}}$ is holomorphic along finite families, $\mathcal{E}^{\mathrm{av}}(0) = 0$, $\mathcal{E}^{\mathrm{av}}(\bar{w}) = \overline{\mathcal{E}^{\mathrm{av}}(w)}$, $\mathcal{E}^{\mathrm{av}}(\mathsf{r}w) = \mathcal{E}^{\mathrm{av}}(w)$ for $\mathsf{r} \in \mathsf{G}$, $\mathcal{E}^{\mathrm{av}}(z \cdot 1) = \mathcal{E}(z \cdot 1)$ for complex $|z| < 2\varrho$, and $|\widetilde{Z}^{\mathrm{av}}(w)| = \prod_{\mathsf{r}\in\mathsf{G}} |\widetilde{Z}(\mathsf{r}w)|^{1/|\mathsf{G}|} \leq e^{C_S \ell^3}$.

(A2) Let $\mathcal{F} \in \{\mathcal{E}, \mathcal{E}^{\mathrm{av}}\}$, and let $u, w_1, \dots, w_n$ be complex Lipschitz functions on T_ℓ such that the support of w_r is covered by n_r closed sup-balls of radius R_0, the supports of different w_r are at sup-distance $\geq \Delta \geq 4M$, and $\|u\|_\sharp$, $\|u + w_r\|_\sharp$, $\|u + \sum_r w_r\|_\sharp < 2\varrho$. With $\mathcal{F}'(w) = \mathcal{F}(u+w) - \mathcal{F}(u)$, the cross term $\mathrm{cr} = \mathcal{F}'(\sum_r w_r) - \sum_r \mathcal{F}'(w_r)$ satisfies $|\mathrm{cr}| \leq C_\times (R_0/\Delta) \sum_r n_r$.

(A3) If the w_r depend linearly on $\zeta \in \mathbb{C}^k$ and (A2) applies for all ζ with $\max_i |\zeta_i| < \rho_3$, then $|\mathrm{cr}(\zeta)| \leq C_\times (R_0/\Delta)(\sum_r n_r)(\max_i |\zeta_i|/\rho_3)^2$.

These bounds are free of K; what grows with K is only the bound of $\mathcal{E}$ itself. ■

Sketch of proof. (A1) By (M4), $Z[w] = Z[\mathsf{r}w] = e^{E + \mathcal{E}(\mathsf{r}w)} \widetilde{Z}(\mathsf{r}w)$ for every $\mathsf{r} \in \mathsf{G}$; the geometric mean of the moduli over G gives $|Z[w]| = e^{E + \mathrm{Re}\, \mathcal{E}^{\mathrm{av}}(w)} \prod_{\mathsf{r}} |\widetilde{Z}(\mathsf{r}w)|^{1/|\mathsf{G}|}$, which is the formula for $|\widetilde{Z}^{\mathrm{av}}(w)|$, bounded by (M1). Holomorphy, the value at 0 and the conjugation pass from $\mathcal{E}$ to its mean; the

invariance holds because G is a group; constants are fixed by every r. (A2) It suffices to take $\mathcal{F}=\mathcal{E}$, since the isometries preserve the hypotheses; then $\mathrm{cr}=\sum_{j,X}\mathsf{b}(X)$ with

$$\mathsf{b}(X)=e_j(X;u+\textstyle\sum_r w_r)-e_j(X;u)-\sum_r\big[e_j(X;u+w_r)-e_j(X;u)\big].$$

By locality $\mathsf{b}(X)=0$ unless X with one layer of cubes meets at least two of the supports; then $\mathsf{b}(X)$ has at most $3\#R(X)$ terms that do not cancel, $R(X)$ the set of the supports met, each $\le C_S e^{-\kappa d_j(X)}$, and $d_j(X)\ge y_s-4$ with $y_s=L^s\Delta/M\ge 4L^s$ by (M3)(a). The count of domains at Y one of the n_r balls, with $\kappa-\log 9216>2$, bounds the sum over such X by $n_r\cdot 27S(\kappa)(2R_0L^s/M+2)^3e^{-2(y_s-4)}$; with $(2R_0L^s/M+2)^3\le 32[(R_0/\Delta)^3y_s^3+1]$, $\sum_{s\ge0}y_s^3e^{-2(y_s-4)}\le 65$ and $\sum_s e^{-2(y_s-4)}\le 2$ one obtains $|\mathrm{cr}|\le 3\cdot 27\cdot 32\,[65(R_0/\Delta)^3+2]C_SS(\kappa)\sum_r n_r$, and $3\cdot27\cdot32=2592$. (A3) $\mathrm{cr}(\zeta)$ is holomorphic on the polydisc, vanishes at 0 with its first derivatives, since $D\mathcal{F}'(0)$ is linear; the maximum principle for $t\mapsto \mathrm{cr}(t\zeta^0)/t^2$ gives the claim. □

Lemma 13.10 (Two sides at a constant source, on a complex disc)**.** Let $(K,\beta)\in\mathfrak{d}(g)$, $K\ge1$, and let $\ell_1,\ell_2\in\mathfrak{L}(g)$ with $\ell_1,\ell_2\ge 2\mathsf{l}_1$; let h_1, A_1 be the numbers of Def. 13.6 formed at g. The function

$$d(z)=\ell_1^{-3}\mathcal{E}_{\ell_1}(z\cdot1)-\ell_2^{-3}\mathcal{E}_{\ell_2}(z\cdot1)$$

is holomorphic on $|z|<2\varrho$, real on the real axis, $d(0)=0$, and $|d(z)|\le\mathsf{A}_1$ for all complex $|z|\le h_1$. Hence, with $\mu'=d'(0)\in\mathbb{R}$, $|d(x)-\mu' x|\le 2\mathsf{A}_1x^2/h_1^2$ for real $|x|\le h_1/2$ and $|d''(0)|\le 2\mathsf{A}_1/h_1^2$. The numbers h_1, A_1 are free of K, β, ℓ_1 and ℓ_2. ■

Sketch of proof. Write $z\cdot1=\sum_u z\chi_u$ with the tents χ_u of half-width $\varsigma=1/L$ about the points $u\in\varsigma\mathbb{Z}^3$; $\|\chi_u\|_\sharp\le\mathsf{c}_\chi$. Call a domain of depth s large if $d_j(X)\ge A_s-4$, $A_s=L^{s-1-m'}$, and small otherwise; by (M3)(a) a small domain cannot meet two sets at sup-distance $\ge\varsigma$. The large part has a density bounded on the whole disc, torus by torus: $\ell^{-3}|\mathcal{E}_\ell^{\mathrm{lg}}(z\cdot1)|\le 27C_SS(\kappa)L^3\sum_s A_s^3e^{-2[A_s-4]_+}\le C_{\mathrm{lg}}$. For the small part, the invariance of $\mathcal{E}^{\mathrm{av}}$ under the translations by $\varsigma\mathbb{Z}^3$ and the chain rule give

$$\ell^{-3}\partial_z^n\mathcal{E}_\ell^{\mathrm{sm}}(z\cdot1)\big|_0=L^3\sum_{\mathrm{near}}D^n\mathcal{E}_\ell^{\mathrm{sm,av}}[\chi_0,\chi_{v_2},\dots,\chi_{v_n}],$$

the sum over the at most 125^{n-1} near tuples, $|v_q-v_{q'}|_\infty\le 2\varsigma$, the same index set on both tori (here $\ell_1,\ell_2\ge2\mathsf{l}_1$ is used: the 125 points are distinct points of each torus). At a near tuple $D^n\mathcal{E}^{\mathrm{av}}$ is a joint cumulant minus a derivative of $\log G_\ell$, $G_\ell(\zeta)=\widetilde{Z}_\ell^{\mathrm{av}}(w)/\widetilde{Z}_\ell(0)$, $w=\sum\zeta_q\chi_{v_q}$. The chessboard estimate on the board of side l_1 (an even number ≥2 of cells in each direction), with the source formula and the cross terms of Lem. 13.9 at $R_0/\Delta\le1/8$, spends the extensive bounds of (M1) on one cell and gives $|G_\ell|\le\mathsf{Q}_2$; Schwarz's lemma and Cauchy's estimate give $|\partial^n\log G_\ell(0)|\le\log2\,(n\mathsf{c}_\chi(2\mathsf{Q}_2+2)/\varrho)^n$; the large part at a near tuple is $\le C'_{\mathrm{lg}}(n\mathsf{c}_\chi/\varrho)^n$ by Cauchy; and the cumulants on the two tori differ by the transfer (M7) by at most $2\log2\,(2\mathsf{Q}_1+2)^n(n\mathsf{c}_\chi/\varrho)^n$. Altogether, with $n^n\le e^nn!$,

$$\frac{|\partial_z^nd^{\mathrm{sm}}(0)|}{n!}\le\frac{L^3}{125}(4\log2+2C'_{\mathrm{lg}})\Big(\frac{125e\,\mathsf{c}_\chi\mathsf{Q}_*}{\varrho}\Big)^n=\frac{L^3}{125}(4\log2+2C'_{\mathrm{lg}})(2h_1)^{-n},$$

which sums on $|z|\le h_1$; with the large part, $|d|\le\mathsf{A}_1$. The forms at real x follow from Cauchy's estimate on $|z|=h_1$; d is real on the real axis by the conjugation in (M1). □

Lemma 13.11 (Finite size, exactly)**.**

(a) For even integers $n_1,n_2,n_3\ge2$ and $\beta\ge0$ let $Z(n_1,n_2,n_3)$ be the partition function of Wilson's action on the periodic box $\mathbb{Z}^3/(n_1\mathbb{Z}\times n_2\mathbb{Z}\times n_3\mathbb{Z})$ and $\mathsf{g}=\log Z/(n_1n_2n_3)$. Then g does not increase in each argument; $P(\beta)=\inf\mathsf{g}=\lim_n\mathsf{g}(n,n,n)$, the limit along the even n, is convex in β and $P\ge-6\beta$; and for even $\mathsf{N}\ge e_0\ge2$

$$0\le\mathsf{g}(\mathsf{N},\mathsf{N},\mathsf{N})-P\le3(e_0/\mathsf{N})\big[\mathsf{g}(e_0,e_0,e_0)-P\big].$$

(b) For every block size L, every $K \geq 0$, $b \geq 0$ and sides $\ell \geq \ell'$ of the form $2L^m$: $p_{K,\infty}(b) = \lim_\ell p_{K,\ell}(b) = L^{3K} P(L^K b)$ with the one function P of (a), $D_K \leq D_{K,\ell} \leq 0$, and $0 \leq p_{K,\ell} - p_{K,\infty} \leq 3(\ell'/\ell)(p_{K,\ell'} - p_{K,\infty})$.

(c) In a frame, for $b \in I(g)$, $K \geq 1$ and $\ell \geq \ell'$ in $\mathfrak{L}(g)$: $-B' \leq D_K(b) \leq D_{K,\ell}(b) \leq 0$ and $0 \leq D_{K,\ell}(b) - D_K(b) \leq \bar\eta(\ell)$.

Parts (a) and (b) are exact statements on one lattice, at every $\beta \geq 0$. ■

Sketch of proof. Fix even n_2, n_3 and let $K_n(\phi, \phi')$ be the slab kernel: the integral, over the bonds of the slab $\{0 \leq x_1 \leq n\}$ that lie in neither boundary plane, of $\exp[-\beta(\sum' a_p + \frac12 \sum_{x_1=0} a_p + \frac12 \sum_{x_1=n} a_p)]$, the bonds of the two planes being fixed to ϕ, ϕ'. It is continuous with values in $(0,1]$, symmetric (the reflection $\vartheta : x_1 \mapsto n - x_1$ maps the slab onto itself, preserves the Haar measure and carries a_p to $a_{\vartheta p}$, because $\operatorname{tr} g^{-1} = \operatorname{tr} g$ on SU(2)) and multiplicative, $K_{n+m}(\phi, \phi'') = \int K_n(\phi, \phi') K_m(\phi', \phi'') \, \mathrm{d}\phi'$ (Fubini; the plaquettes of the middle plane receive $\frac12 + \frac12$). With $z(n) = Z(n, n_2, n_3) = \int K_n(\phi, \phi) \, \mathrm{d}\phi$ one has $z(n+m) = \langle K_n, K_m \rangle_{L^2}$ and $\|K_n\|_2^2 = z(2n)$, so the Cauchy–Schwarz inequality gives

$$z(n+m)^2 \leq z(2n) z(2m), \qquad z(4k) \leq z(2k)^2.$$

The first, at $n = e/2 - 1$, $m = e/2 + 1$, makes $\varphi(e) = \log z(e) \leq 0$ convex on the even e; its slopes increase to a finite $\Lambda_\infty \leq 0$, and $\varphi(e)/e \to \Lambda_\infty$ by Stolz. The excess $\operatorname{exc}(e) = \varphi(e) - e\Lambda_\infty$ does not increase, and it is ≥ 0 because $\varphi(2^k e)/2^k e$ does not increase by the second inequality; so $\varphi(e)/e = \Lambda_\infty + \operatorname{exc}(e)/e$ does not increase. This in each direction gives the monotonicity and $P = \inf \mathsf{g} = \lim \mathsf{g}(n,n,n)$, convex as a limit of convex functions and $\geq -6\beta$ since $a_p \leq 2$ and there are three plaquettes to a site. For the rate, $\operatorname{exc}(\mathsf{N}) \leq \operatorname{exc}(e_0) \leq e_0 n_2 n_3 [\mathsf{g}(e_0, e_0, e_0) - P]$ gives $\mathsf{g}(\mathsf{N}, n_2, n_3) \leq \mathsf{g}(n, n_2, n_3) + e_0[\mathsf{g}(e_0, e_0, e_0) - P]/\mathsf{N}$ for all even n; applied in the three directions in turn, and letting $n \to \infty$, this is (a).

For (b), T_ℓ at the cutoff K is the box with $\mathsf{N} = \ell L^K$ sites a side, $p_{K,\ell}(b) = L^{3K} \mathsf{g}(\mathsf{N}, \mathsf{N}, \mathsf{N})$ at $\beta = L^K b$, and $e_0/\mathsf{N} = \ell'/\ell$ for $e_0 = \ell' L^K$; the limit P names no block size. For (c), by the passage and (M1), $p_{K,\ell} = \ell^{-3}[E_\ell + \log \widetilde{Z}_\ell(0)] \in \ell^{-3} E_\ell + [-C_g, C_S]$, and $|\ell^{-3} E_\ell - \ell'^{-3} E_{\ell'}| \leq C_S$ gives $D_{K,\ell} \geq -B'$ for all $\ell \in \mathfrak{L}(g)$, hence $D_K \geq -B'$; enter $-D_K \leq B'$ in (b). □

Lemma 13.12 (The third member of the stability chain as a functional of the source)**.** Let $(K, \beta) \in \mathfrak{d}(g)$ and $\ell \in \mathfrak{L}(g)$. There are a positive σ-finite measure space $(\Xi, \mathfrak{M})$ and functions Ψ_ϖ, $\varpi \in \Xi$, on the class of the sources, neither Ξ nor $\mathfrak{M}$ depending on the source, such that with $h_\varpi = \operatorname{Re} \Psi_\varpi$:

(T1) for every source w, $\varpi \mapsto e^{\Psi_\varpi(w)}$ is measurable for the completion of $\mathfrak{M}$, $\widetilde{Z}(w) = \int e^{\Psi_\varpi(w)} \, \mathrm{d}\mathfrak{M}(\varpi)$, and

$$|\widetilde{Z}(w)| \leq \mathsf{T}(w) := \int e^{h_\varpi(w)} \, \mathrm{d}\mathfrak{M}(\varpi) \leq e^{C_S \ell^3};$$

(T2) $\mathsf{T}(u) = \widetilde{Z}(u) > 0$ at every real source u;

(T3) for every ϖ and every finite family $f_1, \dots, f_n$, the map $z \mapsto e^{\Psi_\varpi(\sum_q z_q f_q)}$ is holomorphic and without zeros on the convex open set $\{z : \|\sum_q z_q f_q\|_\sharp < 2\varrho\}$; so $z \mapsto h_\varpi(\sum_q z_q f_q)$ is the real part of a holomorphic function there;

(T4) for real u, v there is an $\mathfrak{M}$-null set, depending on the pair, off which $h_\varpi(u + \bar\zeta v) = h_\varpi(u + \zeta v)$ for all ζ with $\|u + \zeta v\|_\sharp < 2\varrho$; and $(\varpi, z) \mapsto h_\varpi(w(z))$ is jointly measurable.

The functional T is the normalized total-variation sum in equation YM3_I:(1738), at $k = K$. ■

Sketch of proof. Take $k = K$. By Thm. YM3_I:AJ, there are finitely many histories and a common source-independent null set N_h for each history. Outside N_h, let D_w be its density: it is zero where $\mathsf{Stay}_h^{\mathrm{T}} \chi_K(\Omega_K) = 0$ and otherwise equals $\mathsf{Stay}_h \exp(\mathbf{\Phi}_h^w + A_{K,h}^w)$. Put $\Xi_h = \{\mathsf{D}_0 \neq 0\}$. This set is the same for all sources, since the tests are unchanged and an exponential has no zero. On it put $d\mathfrak{M}_h = \mathsf{Stay}_h \, \mathrm{d}\mathcal{M}_h$. The reference measure is a finite product of Haar and Lebesgue measures, hence

is σ-finite; multiplication by the nonnegative finite Borel test factor preserves this property. Neither this measure nor Ξ_h depends on the source.

Histories with $\mathfrak{M}_h(\Xi_h) = 0$ contribute zero at every source and may be omitted. Take the disjoint union of the remaining Ξ_h, with measure $\mathfrak{M}_h$ on each piece, and set

$$\Psi_{(h,\phi)}(w) = \mathbf{\Phi}^w_h(\phi) + A^w_{K,h}(\phi) - E^w_{(K)}.$$

The total variation of $\mathsf{D}_w\mathcal{M}_h$ is $|\mathsf{D}_w|\mathcal{M}_h$. Thus the integral of $e^{\operatorname{Re}\Psi}$ is exactly the normalized total-variation sum in equation YM3_I:(1738), which proves the two inequalities in (T1). Without moduli, the history decomposition in Thm. YM3_I:AJ and the vacuum splitting in equation YM3_I:(1737) give $\int e^{\Psi(w)} d\mathfrak{M} = e^{-E^w_{(K)}} Z[w] = \widetilde{Z}(w)$.

For a real source, the bare density is positive, so each history measure is positive; the vacuum constant is real by conjugation. This proves (T2). To prove (T3), fix a history point and a source-independent evaluation representative, as supplied by Thm. YM3_I:AJ. Every argument in its exponent is fixed as the source varies. The active action is linear in w, and the regular, boundary and retained terms are finite sums of the holomorphic families in that theorem. The whole-torus vacuum constant has the holomorphic finite-sum splitting in Prop. YM3_I:83.2.

The one member that no clause covers, the constant $c_h(w) = \exp(-\mathcal{E}^w_{K,\mathcal{S}(h)})$ of the label, which does not depend on ϕ, is holomorphic as a quotient: on measurable sets $B_n \uparrow \Xi_h$ of finite measure on which the rest $e^{\Psi^\circ_\varpi}$ of the exponential is bounded near z_0, the map $z \mapsto \int F_0 1_{B_n}\, \mathrm{d}\hat{\tau}^{\mathrm{co}\prime}_h[\rho_0^{w(z)}]$ is holomorphic by the bare-integral formula and holomorphy assertion of Prop. YM3_I:73.5, and it equals $e^{E^{w(z)}_{(K)}}\, c_h(w(z)) I_n(z)$ with $I_n(z) = \int_{B_n} F_0 e^{\Psi^\circ_\varpi(w(z))} d\mathfrak{M}$ holomorphic by Morera and Fubini, and $I_n(z_0) = \int_{B_n} |e^{\Psi^\circ_\varpi(w(z_0))}|\, \mathrm{d}\mathfrak{M} > 0$ for all large n when F_0 is the conjugate phase of the integrand at z_0, because the integrand has no zero, $B_n \uparrow \Xi_h$ and $\mathfrak{M}(\Xi_h) > 0$.

So $e^{\Psi_\varpi(w(z))}$ is holomorphic without zeros on a convex set and has a holomorphic logarithm, whose real part is $h_\varpi(w(z))$. (T4) follows from conjugation of the history measures in Thm. YM3_I:AJ and of the vacuum splitting in Prop. YM3_I:83.2, first at the points ζ with rational coordinates and then by continuity; joint measurability from measurability in ϖ and continuity in z. □

Proposition 13.13 (The imaginary direction; a floor; Harnack; Hölder over several directions)**.** Let $(K,\beta) \in \mathfrak{d}(g)$ and $\ell \in \mathfrak{L}(g)$, and let $u, v, v_1, \dots, v_n$ be real Lipschitz functions on T_ℓ; $\|\cdot\| = \|\cdot\|_\sharp$.

(P1) If $\|u\| + \|v\| < 2\varrho$: $|\widetilde{Z}(u+iv)| \le \mathsf{T}(u+iv) \le \widetilde{Z}(u)^{1-\vartheta} e^{\vartheta C_S \ell^3}$ with $\vartheta = 2\|v\|^2/((2\varrho - \|u\|)^2 + \|v\|^2)$.

(P2) If $\|u\| + \|v\| < 2\varrho$: $\widetilde{Z}(u+v) \le \widetilde{Z}(u)^{1-\vartheta_1} e^{\vartheta_1 C_S \ell^3}$ with $\vartheta_1 = 2\|v\|/(2\varrho - \|u\| + \|v\|)$.

(P3) If $\rho > 0$, $\|u\| + \rho \sum_i \|v_i\| < 2\varrho$ and $0 \le nt \le \rho/4$: $\widetilde{Z}(u + t\sum_i v_i) \le \prod_i \widetilde{Z}(u + ntv_i)^{(1-\vartheta'')/n} e^{\vartheta'' C_S \ell^3}$ with $\vartheta'' = 8n^2t^2/\rho^2$.

For $\|u\| \le \varrho/2$: $\log \widetilde{Z}(u) \ge -(2C_g + C_S)\ell^3$ (the floor); if moreover $\|u\| + \|v\| < 2\varrho$, then

$$\log|\widetilde{Z}(u+iv)| \le \log \widetilde{Z}(u) + C_I' \|v\|^2 \ell^3 \quad \text{(IM)}, \qquad \log \widetilde{Z}(u+v) - \log \widetilde{Z}(u) \le \tfrac{8}{3}(C_S + C_g)\varrho^{-1}\|v\|\ell^3 \quad \text{(the Harnack bound)},$$

and $\partial_t^2 \log \widetilde{Z}(u+tv)|_{t=0} \ge -2C_I'\|v\|^2\ell^3$ (semi-convexity). All of this holds for $\widetilde{Z}^{\mathrm{av}}$ in the place of $\widetilde{Z}$. C_I' is free of K, β, ℓ and of the sources. ■

Sketch of proof. For the Poisson kernel $\mathsf{P}_\rho(\zeta_0;\phi) = (\rho^2 - |\zeta_0|^2)/|\rho e^{i\phi} - \zeta_0|^2$ one has (i) $\frac{1}{2}[\mathsf{P}_\rho(iy;\phi) + \mathsf{P}_\rho(iy;-\phi)] \ge 1 - 2y^2/(\rho^2+y^2)$ for $0 \le y < \rho$; (ii) $\mathsf{P}_\rho(s;\phi) \ge 1 - 2s/(\rho+s)$ for $0 \le s < \rho$; (iii) $\prod_{i\le n} \mathsf{P}_\rho(t;\phi_i) \ge (1 - 8n^2t^2/\rho^2)\, n^{-1} \sum_i \mathsf{P}_\rho(nt;\phi_i)$ for $0 \le nt \le \rho/4$; (i), (ii) are elementary, and (iii) follows from $\log \mathsf{P}_\rho(t;\phi) \ge \log(1-\tau^2) + 2\tau\cos\phi - \tau^2/(1-\tau)$ and $n^{-1}\sum_i \mathsf{P}_\rho(nt;\phi_i) \le (1+\mathsf{y})[1 + \frac{16}{3}n^2\tau^2]$, $\tau = t/\rho$, $\mathsf{y} = 2\tau\sum_i \cos\phi_i$. Thus each kernel is $(1-\vartheta)$ times the reference kernel plus ϑR with $R \ge 0$ of mean 1. For (P1) take $1 < \rho < (2\varrho - \|u\|)/\|v\|$; by (T3), (T4) of Lem. 13.12, off a null

set $G(\zeta) = h_{\varpi}(u + \zeta v)$ is harmonic near $|\zeta| \le \rho$ and $G(\bar{\zeta}) = G(\zeta)$. Poisson's formula at $\zeta = i$, made symmetric, gives $G(i) = (1-\vartheta)G(0) + \vartheta \int R G(\rho e^{i\phi})\,\mathrm{d}m$; Jensen's inequality, integration in ϖ, Hölder's inequality with the exponents $1/(1-\vartheta)$, $1/\vartheta$ and Tonelli give

$$\mathsf{T}(u+iv) \le \mathsf{T}(u)^{1-\vartheta}\Big(\int R\,\mathsf{T}(u+\rho e^{i\phi}v)\,\mathrm{d}m\Big)^{\vartheta} \le \widetilde{Z}(u)^{1-\vartheta} e^{\vartheta C_S \ell^3}$$

by (T1), (T2); let ρ increase to its bound. (P2) is the same at $\zeta = 1$ with (ii), and (P3) the same in n variables with (iii), Hölder's inequality now with n exponents $n/(1-\vartheta'')$ and one $1/\vartheta''$. The floor is (P2) at the base u and the direction $-u$, where $\vartheta_1 = \|u\|/\varrho \le \frac{1}{2}$, with $\widetilde{Z}(0) \ge e^{-C_g\ell^3}$ from (M1). (IM): $\vartheta \le 8\|v\|^2/(9\varrho^2)$ and $C_S\ell^3 - \log\widetilde{Z}(u) \le 2(C_S + C_g)\ell^3$ by the floor; the Harnack bound likewise with $\vartheta_1 \le 4\|v\|/(3\varrho)$. Semi-convexity: $\mathsf{h}(\zeta) = \log\widetilde{Z}(u+\zeta v)$ is holomorphic near 0 and real on the real axis, and $\operatorname{Re}\mathsf{h}(iy) - \mathsf{h}(0) = -\frac{1}{2}\mathsf{h}''(0)y^2 + O(y^4) \le C_I'\|v\|^2 y^2 \ell^3$ by (IM). For $\widetilde{Z}^{\mathrm{av}}$ each inequality is one between products of powers of values of T whose exponents read norms alone; write it at the images under $\mathsf{r} \in \mathsf{G}$ and take the geometric mean, using (A1) of Lem. 13.9 and equality at real sources. □

Proposition 13.14 (The walls)**.** Let $b \in I(g)$, $K \ge 1$, $T \ge \max\{T_0, \mathsf{l}\}$, $q = L^i \ge 2T$, $\ell \in \mathfrak{L}(g)$ with $\ell \ge 2q$, a board of T_ℓ of side q with any offset, and $f \in \mathfrak{P}_T$ a profile of one of its cells, with F as in Def. 13.3. For real x with $|x| \le \varrho/(256\mathsf{c})$ and $x \le b$:

$$\log\widetilde{Z}^{\mathrm{av}}(xF) - \log\widetilde{Z}^{\mathrm{av}}(x) \le N\big[3C_w T q^2|x| + C_2' q^3 x^2\big].$$

■

Sketch of proof. By the passage, $(K, L^K b) \in \mathfrak{d}(g)$. Fix $o_0 \in a_K\mathbb{Z}$ and take eight boards of side l: the board $\iota \in \{0,1\}^3$ has the offset o_ι with $(o_\iota)_\mu = o_0 + \iota_\mu(\mathsf{l} + a_K)/2$, a point of $a_K\mathbb{Z}^3$ as l/a_K is odd. Take the partition of unity $\Theta_\iota = \prod_\mu \mathsf{t}_{\iota_\mu}(y_\mu)$, $\mathsf{t}_1 = 1 - \mathsf{t}_0$, t_0 the l-periodic trapezoid equal to 1 within $\mathsf{l}/8$ of $o_0 + \mathsf{l}\mathbb{Z}$ and to 0 beyond $3\mathsf{l}/8$, with $\operatorname{Lip}\Theta_\iota \le 4\sqrt{3}/\mathsf{l}$; in a cell of the board ι the support of Θ_ι lies in a sup-ball of radius $R_0 \le 7\mathsf{l}/16$ at sup-distance $\ge \mathsf{l}/16$ from the walls of that board.

The walls $\mathsf{g} = 1 - F$ split as $\mathsf{g}\Theta_\iota = \sum_r h_r$ into pieces of the cells of the board ι, with $\|h_r\|_\sharp \le \mathsf{c}$ and mutual sup-distances $\ge \Delta = \mathsf{l}/8 \ge 4M$; since $\mathsf{g} \ne 0$ only within $T/2$ of a mirror of the board of side q, at most $n_\iota \le 9NTq^2/\mathsf{l}^3$ pieces are non-zero. Hölder's inequality (P3) at $u = x$, $n = 8$, $v_\iota = -\operatorname{sgn}(x)\mathsf{g}\Theta_\iota$, $t = |x|$, $\rho = \varrho/(8\mathsf{c})$, with the floor, costs $\vartheta''(2C_S + 2C_g)\ell^3$, $\vartheta'' = 32768\mathsf{c}^2x^2/\varrho^2$ (this is the term in x^2, paid on the whole torus), and leaves the eight numbers $\mathsf{B}_\iota = \log[\widetilde{Z}^{\mathrm{av}}(x + z\mathsf{g}\Theta_\iota)/\widetilde{Z}^{\mathrm{av}}(x)]$, $z = -8x$.

Write $Z_b[w] = Z_{K,\ell,L^K b}[w]$. On the board ι, the shift of the label, $Z_b[x+w]/Z_b[x] = Z_{K,\ell,\beta'}[w]/Z_{K,\ell,\beta'}$ with $\beta' = L^K(b-x) \ge 0$, and the chessboard estimate at β' give $Z_b[x + z\mathsf{g}\Theta_\iota]/Z_b[x] \le \prod_r \|e^{z\mathcal{O}(h_r)}\|_{\mathrm{ch}}$, and by the source formula $\|e^{z\mathcal{O}(h_r)}\|_{\mathrm{ch}}^{N_f} = Z_b[x + zh_r^{\mathrm{per}}]/Z_b[x]$, where h_r^{per}, the sum of the $N_f = (\ell/\mathsf{l})^3$ mirror copies of h_r, is a dense array. The energies, whose bounds grow with K, cancel exactly up to cross terms. Put $\mathcal{E}'(w) = \mathcal{E}^{\mathrm{av}}(x+w) - \mathcal{E}^{\mathrm{av}}(x)$, the function $\mathcal{F}'$ of Lem. 13.9 (A2) for $\mathcal{F} = \mathcal{E}^{\mathrm{av}}$ at the base source $u = x$; then $Z_b[x+w]/Z_b[x] = e^{\mathcal{E}'(w)}\widetilde{Z}^{\mathrm{av}}(x+w)/\widetilde{Z}^{\mathrm{av}}(x)$, and so, the sums being over the r with $h_r \ne 0$,

$$\mathsf{B}_\iota \le -\mathcal{E}'(z\mathsf{g}\Theta_\iota) + \sum_r N_f^{-1}\Big[\mathcal{E}'(zh_r^{\mathrm{per}}) + \log\frac{\widetilde{Z}^{\mathrm{av}}(x + zh_r^{\mathrm{per}})}{\widetilde{Z}^{\mathrm{av}}(x)}\Big].$$

The mirror copies of h_r are its images $\mathsf{r}h_r$ under reflections $\mathsf{r} \in \mathsf{G}$ at mirrors of the board ι, which fix the constant x, so that $\mathcal{E}^{\mathrm{av}}(x + z\mathsf{r}h_r) = \mathcal{E}^{\mathrm{av}}(\mathsf{r}(x + zh_r)) = \mathcal{E}^{\mathrm{av}}(x + zh_r)$ by (A1). By Lem. 13.9 (A3), applied twice with the base source $u = x$, $\mathcal{E}'(z\mathsf{g}\Theta_\iota) = \sum_r \mathcal{E}'(zh_r) + \mathrm{cr}_\iota$ and $\mathcal{E}'(zh_r^{\mathrm{per}}) = N_f\mathcal{E}'(zh_r) + \mathrm{cr}_r$; so the terms in $\mathcal{E}'$ of the bound of B_ι are $-\mathrm{cr}_\iota + \sum_r N_f^{-1}\mathrm{cr}_r$, at most $2C_\times(7/2)n_\iota|z|^2\mathsf{c}^2/\varrho^2$. The dense arrays are bounded by the Harnack bound, $N_f^{-1}\log[\widetilde{Z}^{\mathrm{av}}(x + zh_r^{\mathrm{per}})/\widetilde{Z}^{\mathrm{av}}(x)] \le 4\mathsf{c}(C_S + C_g)\varrho^{-1}\mathsf{l}^3|z|$.

Summing,

$$\mathsf{B}_\iota \le n_\iota\Big[\frac{32\mathsf{c}(C_S+C_g)}{\varrho}\mathsf{I}^3|x| + \frac{128\,C_\times(7/2)\mathsf{c}^2x^2}{\varrho^2}\Big] \le N\Big[3C_wTq^2|x| + \frac{1152\,C_\times(7/2)\mathsf{c}^2}{\varrho^2\mathsf{I}^3}Tq^2x^2\Big],$$

since $72\cdot 4\mathsf{c}(C_S+C_g)/\varrho = 3C_w$; with $Tq^2 \le q^3$ and the cost of Hölder's inequality this is the claim. □

Proposition 13.15 (Slabs; the walls without a term in x^2 for products)**.**

(a) *The slab inequality (exact).* On a periodic box $(\mathbb{Z}/\mathsf{N}\mathbb{Z})^3$ let $c(p)$ be any real weights on the plaquettes and $Z(c) = \int e^{-\sum_p c(p)a_p}\,\mathrm{d}U$. Let $\mathsf{N} = 2k\,t$ with integers $k, t \ge 1$, cut the box in one direction by the $2k$ planes of sites $x_\mu = n_0 + jt$ into the closed slabs S_j, and let c_j^{arr} be the weights that agree with c on S_j and are symmetric under the reflections at all $2k$ planes. Then $Z(c) \le \prod_j Z(c_j^{\mathrm{arr}})^{1/(2k)}$; no sign of c is supposed. For $c(p) = \beta - a^{-1}w(x(p))$ with w real this reads $Z[w] \le \prod_j Z[w_j^{\mathrm{arr}}]^{1/(2k)}$, w_j^{arr} the function that agrees with w on S_j and is even at all the planes.

(b) *The walls for products.* Let $(K,\beta) \in \mathfrak{d}(g)$, $\ell \in \mathfrak{L}(g)$, $T \ge T_0$, $q = L^i \ge 2T$, $\ell \ge 2q$, a board of T_ℓ of side q with mirrors $y_\mu \in s_\mu + q\mathbb{Z}$, and T' the least power of L that is $\ge 2T$. Let $\varphi_\mu : \mathbb{R}/\ell\mathbb{Z} \to [0,1]$ have $\operatorname{Lip}\varphi_\mu \le 16/T$ and $\varphi_\mu = 1$ at distance $\ge T/2$ from $s_\mu + q\mathbb{Z}$, and $F(y) = \prod_\mu \varphi_\mu(y_\mu)$; let x_0, x be real with $|x_0| + |x| \le \varrho/2$, and $C_V = 16L(C_S+C_g)/\varrho + 3C_\times(1)/(4\varrho T_0^3)$. Then

$$\log\widetilde{Z}^{\mathrm{av}}(x_0 + xF) - \log\widetilde{Z}^{\mathrm{av}}(x_0 + x) \le C_V|x|\,Tq^2N;$$

if every φ_μ has the period q and is even at s_μ (as for the mirror copies of the centred f_0), the same holds for $\widetilde{Z}$ itself. No condition $x \le b$ is asked.

■

Sketch of proof. (a) Cut in the direction $\mu = 1$. The kernel $K_j(\phi,\phi')$ of the slab S_j, with half weights on its two faces and the bonds of the faces fixed to ϕ, ϕ', is real, continuous and positive on a compact space, so a Hilbert–Schmidt operator, and Fubini gives $Z(c) = \operatorname{Tr}(K_0K_1\cdots K_{2k-1})$. The reflection at the mid-plane of a slab shows that K_j^T is the kernel of the slab with mirrored weights, so $\operatorname{Tr}((K_jK_j^T)^k) = Z(c_j^{\mathrm{arr}})$; Hölder's inequality in the Schatten classes, $|\operatorname{Tr}(A_1\cdots A_{2k})| \le \prod_i \|A_i\|_{2k}$, is the claim. In the case used it is the classical chessboard argument: $\mathsf{F}(C_1,\dots,C_{2k}) = \operatorname{Tr}(C_1C_2^TC_3C_4^T\cdots C_{2k-1}C_{2k}^T)$ is multilinear, invariant under a cyclic shift by one place with all letters transposed, and satisfies $|\mathsf{F}(C)|^2 \le \mathsf{F}(C_1,\dots,C_k,C_k,\dots,C_1)\,\mathsf{F}(C_{2k},\dots,C_{k+1},C_{k+1},\dots,C_{2k})$ by Cauchy–Schwarz; after the normalisation $\mathsf{F}(C_i,\dots,C_i) = 1$, among the words that maximise $|\mathsf{F}|$ one with the longest run of equal letters must have a run of length $\ge k$, whence the maximum is ≤ 1. (b) Three steps, one direction at a time: $u_0 = x_0 + x$ and $u_\mu = x_0 + x\prod_{\nu\le\mu}\varphi_\nu(y_\nu)$, so that $u_3 = x_0 + xF$.

In the direction μ the planes $y_\mu \in s_\mu + T'/2 + T'\mathbb{Z}$ are planes of sites, because q/a_K and T'/a_K are odd, and they cut T_ℓ into the ℓ/T' closed slabs $\mathsf{S}_j = \{|y_\mu - s_\mu - jT'| \le T'/2\}$, $j \in \mathbb{Z}/(\ell/T')\mathbb{Z}$, an even number of them; as q/T' is an integer, the mirrors are the mid-planes of the ℓ/q *wall slabs*, those with $jT' \in q\mathbb{Z}$. The function $u_{\mu-1}$ does not read y_μ, and $u_\mu - u_{\mu-1} = -x\prod_{\nu<\mu}\varphi_\nu(y_\nu)\,(1-\varphi_\mu(y_\mu))$ vanishes at distance $\ge T/2$ from the mirrors, hence off the wall slabs and near their faces, as $T'/2 \ge T$. For a wall slab S_j put

$$w_j = (u_\mu - u_{\mu-1})\mathbf{1}_{\mathsf{S}_j} = -x\prod_{\nu<\mu}\varphi_\nu(y_\nu)\,(1-\varphi_\mu(y_\mu))\mathbf{1}_{\mathsf{S}_j},$$

so that $u_\mu = u_{\mu-1} + \sum_{\text{wall slabs}} w_j$, and let w_j^{arr}, the sum of the ℓ/T' mirror images of w_j at the planes, one in each slab, be the even continuation of w_j from S_j. The even continuation of u_μ from

a slab is $u_{\mu-1}$ for a slab that is not a wall slab and $u_{\mu-1}+w_j^{\mathrm{arr}}$ for a wall slab S_j; so (a) at the real source u_μ gives

$$\log\frac{Z[u_\mu]}{Z[u_{\mu-1}]}\le\frac{T'}{\ell}\sum_{\text{wall slabs}}\log\frac{Z[u_{\mu-1}+w_j^{\mathrm{arr}}]}{Z[u_{\mu-1}]}.$$

The normalisers cancel exactly. Let $\mathcal{F}=\mathcal{E}^{\mathrm{av}}$ for the first assertion and $\mathcal{F}=\mathcal{E}$ in the periodic even case, $\widetilde{Z}^{\mathcal{F}}=e^{-E-\mathcal{F}}Z$ (that is, $\widetilde{Z}^{\mathrm{av}}$, respectively $\widetilde{Z}$), and $\mathcal{F}'(w)=\mathcal{F}(u_{\mu-1}+w)-\mathcal{F}(u_{\mu-1})$ as in Lem. 13.9 (A2) at the base $u_{\mu-1}$, so that $Z[u_{\mu-1}+w]/Z[u_{\mu-1}]=e^{\mathcal{F}'(w)}\widetilde{Z}^{\mathcal{F}}(u_{\mu-1}+w)/\widetilde{Z}^{\mathcal{F}}(u_{\mu-1})$. The pieces of w_j^{arr} are images $\mathsf{r}w_j$ under reflections $\mathsf{r}\in\mathsf{G}$ at the planes, which fix $u_{\mu-1}$, so $\mathcal{F}'(\mathsf{r}w_j)=\mathcal{F}'(w_j)$ for $\mathcal{F}=\mathcal{E}^{\mathrm{av}}$ by (A1); in the periodic even case $1-\varphi_\mu$ is even at the mid-plane of S_j, the pieces are the translates of w_j by $T'\mathbb{Z}e_\mu$, which lies in $M\mathbb{Z}^3$ as T' and M are powers of L with $T'>M$, and the same holds for $\mathcal{E}$ itself by (M1).

Hence $\mathcal{F}'(w_j^{\mathrm{arr}})=(\ell/T')\mathcal{F}'(w_j)+\mathrm{cr}_j$ and $\mathcal{F}'(\sum_j w_j)=\sum_j\mathcal{F}'(w_j)+\mathrm{cr}$, the sum over the wall slabs, and in $\log[\widetilde{Z}^{\mathcal{F}}(u_\mu)/\widetilde{Z}^{\mathcal{F}}(u_{\mu-1})]=\log(Z[u_\mu]/Z[u_{\mu-1}])-\mathcal{F}'(\sum_j w_j)$ the terms $\mathcal{F}'(w_j)$ cancel, leaving the cross terms. These are bounded by Lem. 13.9 (A3), with x replaced by ζ, $|\zeta|<\varrho$, in the w_j alone: each support lies within $T/2$ of the mid-plane of its slab and is covered by $(\ell/T')^2$ closed sup-balls of radius $R_0=T'/2$, and different supports are at sup-distance $\ge\Delta=T'-T\ge T'/2\ge 4M$, so that $R_0/\Delta\le 1$.

The Harnack bound at the base $u_{\mu-1}$, with $\|u_{\mu-1}\|_\sharp\le|x_0|+|x|\le\varrho/2$, and in the direction w_j^{arr}, with $\|w_j^{\mathrm{arr}}\|_\sharp\le|x|$ as $40M\cdot16\sqrt{3}/T\le1$, costs $\frac{8}{3}(C_S+C_g)\varrho^{-1}|x|\ell^3$ for each wall array, and $\#\{\text{wall slabs}\}\,T'/\ell=T'/q$; a slab needs no partition of unity across it, so (P3) and its term in x^2 do not enter. Summing the three steps with $2T\le T'<2LT$ and $|x|\le\varrho/2$ gives the constant C_V. □

Proposition 13.16 (The main inequality; semi-convexity)**.** Let $\ell'\in\mathfrak{L}(g)$ with $\ell'\ge2\mathsf{l}_1$ be a reference side, $T\ge\max\{T_0,\mathsf{l}\}$, $q=L^i\ge2T$, $\ell\in\mathfrak{L}(g)$ with $\ell\ge\max\{2q,\ell'\}$, $K\ge1$ and $b\in I(g)$, all numbers of Def. 13.6 being formed at g with ℓ'. Put $\omega(z)=\ell'^{-3}\log[\widetilde{Z}_{\ell'}(z)/\widetilde{Z}_{\ell'}(0)]$, let $\mu'=d'(0)$ for the d of Lem. 13.10 at $(\ell_1,\ell_2)=(\ell,\ell')$, and for a profile f let $\epsilon(z)=\mathcal{E}^{\mathrm{av}}_\ell(zf)+q^3[\omega'(0)-\mu']z$, a function holomorphic on $|z|<2\varrho$ with $\epsilon(0)=0$ and $\epsilon'(0)$ real.

(a) For every board of T_ℓ of side q, every offset and every profile $f\in\mathfrak{P}_T$, for $|z|\le r_1$ with $\operatorname{Re}z\le b$:

$$\Lambda(z)-\operatorname{Re}\epsilon(z)\le q^3\big[D_{K,\ell}(b-\operatorname{Re}z)-D_{K,\ell}(b)\big]+3C_wTq^2|\operatorname{Re}z|+C_Iq^3|z|^2.$$

(b) $D''_{K,\ell}(b)\ge-C_0$ at every $b\in I(g)$. So $D_{K,\ell}+\frac12C_0y^2$ and $D_K+\frac12C_0y^2$ are convex on $I(g)$; $\mu_K=(D_K+\frac12C_0y^2)''$ and $\mu_{K,\ell}=(D_{K,\ell}+\frac12C_0y^2)''$ are positive Radon measures on $\operatorname{int}I(g)$; and for both, $\mu([y_1,y_2])\le2B'/t+C_0(y_2-y_1+t)$ whenever $[y_1-t,y_2+t]\subset I(g)$.

■

Sketch of proof. By the passage $(K,L^Kb)\in\mathfrak{d}(g)$, so Lems. 13.9, 13.10, 13.12 and Props. 13.13, 13.14 hold on both tori. The shift of the label gives on each torus, for real $|x|<2\varrho$, $\log\widetilde{Z}(x)-\log\widetilde{Z}(0)=\ell^3[p_{K,\ell}(b-x)-p_{K,\ell}(b)]-\mathcal{E}(x\cdot1)$; dividing by the volumes and subtracting,

$$\ell^{-3}\big[\log\widetilde{Z}_\ell(x)-\log\widetilde{Z}_\ell(0)\big]=\big[D_{K,\ell}(b-x)-D_{K,\ell}(b)\big]+\omega(x)-d(x).$$

$H(z)=\widetilde{Z}_{\ell'}(z)/\widetilde{Z}_{\ell'}(0)$ is holomorphic on $|z|<2\varrho$, $H(0)=1$ and $|H|\le e^{(C_S+C_g)\ell'^3}$ by (M1); Schwarz's lemma gives $|H-1|\le\frac12$ on $|z|\le r_2$, so $|\omega(x)-\omega'(0)x|\le C_\omega x^2$ for $|x|\le r_2/2$ and $|\omega''(0)|\le C_\omega$. For (a), with $z=x+i\tau$, the source formula gives $N\Lambda(z)=\operatorname{Re}\mathcal{E}^{\mathrm{av}}(w_z)+\log|\widetilde{Z}^{\mathrm{av}}(w_z)|-\log\widetilde{Z}(0)$. Then (i) (IM) for $\widetilde{Z}^{\mathrm{av}}$ at $u=xF$, $v=\tau(F_e-F_o)$ removes τ at the cost $C'_I\tau^2\ell^3$; (ii) the walls (Prop. 13.14) replace xF by the constant x at the cost $N[3C_wTq^2|x|+C'_2q^3x^2]$; (iii) the identity, the bound of ω and Lem. 13.10 turn $\log\widetilde{Z}(x)-\log\widetilde{Z}(0)$ into at most $Nq^3\{[D_{K,\ell}(b-x)-D_{K,\ell}(b)]+[\omega'(0)-\mu']x+(C_\omega+2\mathsf{A}_1/h_1^2)x^2\}$; (iv) the cross terms between the cells, Lem. 13.9 (A3) with $R_0=q/2$, $\Delta=T/4$, $\rho_3=\varrho$ and $C_\times(2q/T)\le M_\times(q/T)^3$, give $\operatorname{Re}\mathcal{E}^{\mathrm{av}}(w_z)\le N\operatorname{Re}\mathcal{E}^{\mathrm{av}}(zf)+NM_\times T^{-3}\varrho^{-2}q^3|z|^2$.

Adding and dividing by N gives (a). For (b), semi-convexity in Prop. 13.13 at $u=0$, $v=1$ gives $\mathsf{h}''(0) \geq -2C_I'\ell^3$ for $\mathsf{h}(x) = \log \widetilde{Z}_\ell(x)$; differentiating the identity twice at $x=0$,

$$D''_{K,\ell}(b) = \ell^{-3}\mathsf{h}''(0) - \omega''(0) + d''(0) \geq -2C_I' - C_\omega - 2\mathsf{A}_1/h_1^2 = -C_0.$$

$D_{K,\ell}$ is real-analytic in the label, and D_K is its pointwise limit by Lem. 13.11. For $\Phi = D + \frac{1}{2}C_0 y^2$ with $D \in \{D_K, D_{K,\ell}\}$, $\mu([y_1,y_2]) = \Phi'(y_2+) - \Phi'(y_1-)$ is bounded by difference quotients over the length t, and $D \in [-B', 0]$ by Lem. 13.11 (c). □

Lemma 13.17 (Cauchy's estimate in the chessboard seminorm; moments from the seminorm)**.** On a board of T_ℓ, at any $\beta \geq 0$:

(a) If $X = \mathcal{O}(f)$ is a real function of one cell and $\tilde{\epsilon}$ is holomorphic near the disc $|z| \leq r$ with $\tilde{\epsilon}(0) = 0$, then $\|X - \tilde{\epsilon}'(0)\|_{\mathrm{ch}} \leq r^{-1} \max_{|z|=r} \exp[\Lambda(z) - \operatorname{Re}\tilde{\epsilon}(z)]$.

(b) For functions $X_1, \dots, X_n$ of n different cells and complex numbers m_i: $|\langle \prod_i (X_i - m_i)\rangle| \leq \prod_i \|X_i - m_i\|_{\mathrm{ch}}$; $|\langle X_i\rangle - m_i| \leq \|X_i - m_i\|_{\mathrm{ch}}$; $|\operatorname{Cov}(X_1, X_2)| \leq 2\|X_1 - m_1\|_{\mathrm{ch}}\|X_2 - m_2\|_{\mathrm{ch}}$; and the centred moments satisfy $|\langle \prod_i (X_i - \langle X_i\rangle)\rangle| \leq \prod_i 2\|X_i - m_i\|_{\mathrm{ch}}$.

■

Sketch of proof. (a) Pointwise on the configurations, $X - \tilde{\epsilon}'(0) = (2\pi i)^{-1} \oint_{|z|=r} e^{zX - \tilde{\epsilon}(z)} z^{-2}\,\mathrm{d}z$; the Riemann sums converge uniformly since X is bounded; the triangle inequality and the homogeneity of the seminorm, with $\|F\|_{\mathrm{ch}} \leq \sup|F|$ for the remainder, give $\|X - \tilde{\epsilon}'(0)\|_{\mathrm{ch}} \leq r^{-1}\max_{|z|=r}\|e^{zX-\tilde{\epsilon}(z)}\|_{\mathrm{ch}}$, and $\|e^{zX-\tilde{\epsilon}(z)}\|_{\mathrm{ch}} = e^{-\operatorname{Re}\tilde{\epsilon}(z)}(Z[w_z]/Z)^{1/N}$ by the source formula. (b) The first inequality is the chessboard estimate; the second is the first with one factor and $\|1\|_{\mathrm{ch}} = 1$; the third follows from $\operatorname{Cov}(X_1,X_2) = \langle (X_1 - m_1)(X_2 - m_2)\rangle - (\langle X_1\rangle - m_1)(\langle X_2\rangle - m_2)$; and $\|X_i - \langle X_i\rangle\|_{\mathrm{ch}} \leq \|X_i - m_i\|_{\mathrm{ch}} + |\langle X_i\rangle - m_i| \leq 2\|X_i - m_i\|_{\mathrm{ch}}$. □

Proposition 13.18 (From a bound on increments of the pressure to the cube means, on a torus)**.** In the setting of Prop. 13.16, let s, δ_2, $\tilde{C}$, η_+, x_0 be real numbers with $\delta_2 \geq 0$, $\tilde{C} \geq 0$, $\eta_+ \geq 0$ and $0 < x_0 \leq b$, such that

$$D_{K,\ell}(b-x) - D_{K,\ell}(b) \leq \mathsf{s}x + \delta_2|x| + \tilde{C}x^2 + \eta_+ \qquad \text{for all real } |x| \leq x_0.$$

Put $\delta = 3C_wT/q + \delta_2$, and let $q \geq \max\{(\tilde{C} + C_I)/(9C_w^2T^2), (3C_wT\min\{r_1, x_0\})^{-1/2}\}$. Then on every board of T_ℓ of side q, $\|x_c - m'\|_{\mathrm{ch}} \leq e^{2+q^3\eta_+}\delta$ with the real number $m' = q^{-3}\epsilon'(0) + \mathsf{s}$, which depends on ℓ and on the profile. Hence, for profiles of $\mathfrak{P}_T$ in distinct cells,

$$|\operatorname{Cov}_{K,\ell,L^Kb}(x_c, x_{c'})| \leq 2e^{4+2q^3\eta_+}\delta^2, \qquad \left|\left\langle \prod_{i=1}^n (x_{c_i} - \langle x_{c_i}\rangle)\right\rangle_{K,\ell,L^Kb}\right| \leq \left(2e^{2+q^3\eta_+}\delta\right)^n.$$

■

Sketch of proof. Put $\tilde{\epsilon}(z) = \epsilon(z) + q^3\mathsf{s}z$. For $|z| = r \leq \min\{r_1, x_0\}$, so that $\operatorname{Re} z \leq x_0 \leq b$, the main inequality and the hypothesis at $x = \operatorname{Re} z$ give, since $\delta_2 \geq 0$ and $\tilde{C} \geq 0$ allow $\delta_2|x| \leq \delta_2 r$ and $\tilde{C}x^2 \leq \tilde{C}r^2$,

$$\Lambda(z) - \operatorname{Re}\tilde{\epsilon}(z) \leq q^3\delta r + q^3(\tilde{C} + C_I)r^2 + q^3\eta_+.$$

Take $r = 1/(q^3\delta)$; since $\delta \geq 3C_wT/q$, $r \leq 1/(3C_wTq^2) \leq \min\{r_1, x_0\}$ by the second floor; $q^3\delta r = 1$; and $q^3(\tilde{C} + C_I)r^2 = (\tilde{C} + C_I)/(q^3\delta^2) \leq (\tilde{C} + C_I)/(9C_w^2T^2q) \leq 1$ by the first. Lem. 13.17 (a) gives $\|\mathcal{O}(f) - \tilde{\epsilon}'(0)\|_{\mathrm{ch}} \leq q^3\delta e^{2+q^3\eta_+}$; divide by q^3; for a profile of another cell the same holds with its own ϵ and m'; Lem. 13.17 (b) gives the rest. □

Proposition 13.19 (The window inequality, in one frame, at every cutoff)**.** In the setting of Prop. 13.16, let b be a label with $I_q(b) \subset \operatorname{int} I(g)$, $q \geq q_1 = \max\{2T, C_I/(9C_w^2T^2), (3C_wTr_1)^{-1/2}\}$, $K \geq 1$, and

$$\delta_K(q,b) = 3C_wT/q + \mu_K(I_q(b)).$$

(a) For all $\nu,\nu' \in \mathcal{G}^{\mathrm{per}}_{K,L^K b}$, every board of $\mathbb{L}_K$ of side q with any offset and profiles of $\mathfrak{P}_T$ in distinct cells $c, c', c_1, \dots, c_n$:

$$|\operatorname{Cov}_\nu(x_c, x_{c'})| \le 2e^4 \delta_K(q,b)^2, \qquad |\nu(x_c) - \nu'(x_c)| \le 2e^2 \delta_K(q,b), \qquad \Big|\nu\Big(\prod_i (x_{c_i} - \nu(x_{c_i}))\Big)\Big| \le (2e^2\delta_K(q,b))^n.$$

(b) The same holds with $\delta = 3C_w T/q + C_0 x_q + x_q^{-1}\Delta^2_{x_q} D_K(b)$, which is $\le \delta_K(q,b)$; and for every $\ell \in \mathfrak{L}(g)$ with $\ell \ge \ell'$,

$$x_q^{-1}\Delta^2_{x_q} D_K(b) \le x_q^{-1}\Delta^2_{x_q} D_{K,\ell}(b) + 6C_w C_{\mathrm{FS}} T q^2/\ell \le x_q \sup_{|y-b|\le x_q} D''_{K,\ell}(y) + 6C_w C_{\mathrm{FS}} T q^2/\ell, \qquad C_{\mathrm{FS}} = 3\ell' B' :$$

the bounds in every periodic Gibbs state are governed by the pressures of the two tori T_ℓ, $T_{\ell'}$ at a side ℓ of the order q^3.

■

Sketch of proof. (a) *A window gives the hypothesis of Prop. 13.18.* $\Phi = D_K + \frac12 C_0 y^2$ is convex on $\operatorname{int} I(g) \supset I_q(b)$; with $\sigma_0 = \Phi'(b-)$, for $0 < x \le x_q$, $\Phi(b+x) - \Phi(b) \le \sigma_0 x + x\mu_K([b, b+x))$ and $\Phi(b-x) - \Phi(b) \le -\sigma_0 x + x\mu_K((b-x, b))$; since $D_K(b-x) - D_K(b) = \Phi(b-x) - \Phi(b) + C_0 bx - \frac12 C_0 x^2$, the function D_K has the hypothesis with $\mathsf{s} = C_0 b - \sigma_0$, $\delta_2 = \mu_K(I_q(b)) \ge 0$, $\tilde C = 0$, $\eta_+ = 0$, $x_0 = x_q$. *From the tori to the states.* By Lem. 13.11, $D_{K,\ell} = D_K + \mathsf{w}_\ell$ with $0 \le \mathsf{w}_\ell \le \bar\eta(\ell)$, so $D_{K,\ell}$ has it with the tolerance $\eta_+ = \bar\eta(\ell)$, and Prop. 13.18 gives $\|x_c - m'_\ell\|_{\mathrm{ch},\ell} \le e^{2+q^3\bar\eta(\ell)}\delta_K(q,b)$ for all $\ell \in \mathfrak{L}(g)$, $\ell \ge \max\{2q, \ell'\}$. The real numbers m'_ℓ are bounded at fixed K, since $|m'_\ell - \langle x_c\rangle_\ell| \le e^3\delta_K$ for large ℓ and $|x_c|$ is bounded. For finitely many profiles choose $\ell_k \to \infty$ with $m'_{\ell_k,i} \to m_{*,i}$; then $\limsup_k \|x_{c_i} - m_{*,i}\|_{\mathrm{ch},\ell_k} \le e^2\delta_K$, and the domination gives

$$\Big|\nu\Big(\prod_i (x_{c_i} - m_{*,i})\Big)\Big| \le (e^2\delta_K)^n \qquad \text{for every } \nu \in \mathcal{G}^{\mathrm{per}}_{K,L^K b};$$

at $n = 1$, $|\nu(x_c) - m_*| \le e^2\delta_K$ for ν and for ν', one common centre serving both; expanding as in Lem. 13.17 (b) gives the three bounds. (b) In Prop. 13.18 the hypothesis enters only at $|x| \le r = 1/(q^3\delta) \le x_q$. On $[b-r, b+r]$ the convex Φ lies below its chord, so $\Phi(b-x) - \Phi(b) \le -\mathsf{s}_r x + \frac12\Delta^2_r\Phi(b)$ with the slope s_r of the chord; and $\Delta^2_r\Phi(b)/r = r^{-1}\int_0^r \mu_K((b-u, b+u))\,du$ does not decrease in r, so $\frac12\Delta^2_r\Phi(b) \le r\, x_q^{-1}\Delta^2_{x_q}\Phi(b)$, which is $r(C_0 x_q + x_q^{-1}\Delta^2_{x_q} D_K(b))$ and at most $r\mu_K(I_q(b))$. The comparison with the torus follows from $D_K \le D_{K,\ell} \le D_K + \bar\eta(\ell)$, $2\bar\eta(\ell)/x_q = 6C_w C_{\mathrm{FS}} T q^2/\ell$, and $\Delta^2_s D \le s^2 \sup D''$ for the real-analytic $D_{K,\ell}$. □

Lemma 13.20 (A sum rule; near and far parts; a pile recurs at all large cutoffs)**.** Let one frame, $0 < g \le \gamma$, a compact interval $W \subset I(g)$ of positive length and $\ell' \in \mathfrak{L}(g)$ with $\ell' \ge 2\mathsf{l}_1$ be given, let $k_7 \ge 1$ be an integer with $\varsigma_7 = L^{-k_7} \le 2/(3\sqrt3)$, and let χ, η, c_7, c_T, c_N be formed with this ς_7 as in Def. 13.6.

(i) *Exact.* For $K \ge k_7$, every side $\ell > 6\varsigma_7$ and $b \ge 0$:

$$p''_{K,\ell}(b) = \varsigma_7^{-3} S^T_{2;K,\ell,L^K b}(\chi, 1) = V^{\mathrm{near}}_{K,\ell}(b) + V^{\mathrm{far}}_{K,\ell}(b), \qquad V^{\mathrm{near}} = \varsigma_7^{-3} S^T_2(\chi,\eta), \quad V^{\mathrm{far}} = \varsigma_7^{-3} S^T_2(\chi, 1-\eta).$$

(ii) $|V^{\mathrm{near}}_{K,\ell} - V^{\mathrm{near}}_{K,\ell'}| \le c_N$ on $I(g)$ for all $K \ge k_7$ and $\ell \in \mathfrak{L}(g)$; and for each side ℓ, $\omega_\ell(K, K'') = \sup_W |V^{\mathrm{far}}_{K,\ell} - V^{\mathrm{far}}_{K'',\ell}| \to 0$ as $K, K'' \to \infty$.

(iii) For every closed interval I and $\delta > 0$ with $I^{+\delta} = [\min I - \delta, \max I + \delta] \subset \operatorname{int} W$, all $K, K'' \geq k_7$ and all $\ell \in \mathfrak{L}(g)$ with $\ell \geq \ell'$:

$$\mu_K(I) \leq \mu_{K''}(I^{+\delta}) + (|I| + 2\delta)\big[2c_N + \omega_\ell(K, K'') + \omega_{\ell'}(K, K'')\big] + \frac{24\ell' B'}{\delta \ell}.$$

■

Sketch of proof. (i)

$$p''_{K,\ell} = \ell^{-3} L^{2K} \operatorname{Var}(\sum_p a_p) = a_K^{-5} \mathsf{N}^{-3} \operatorname{Var}(\sum_p a_p),$$

while $S_2^T(\chi, 1) = a_K^{-2} \sum_p \chi(x(p)) \operatorname{Cov}(a_p, \sum_{p'} a_{p'})$. By translation invariance the covariance depends on the orientation of p alone, the three orientations summing to $\mathsf{N}^{-3} \operatorname{Var}(\sum_p a_p)$, and for each orientation $\sum_p \chi(x(p)) = (\varsigma_7/a_K)^3$, the sums of tents of the integer half-width ς_7/a_K; the splitting is $1 = \eta + (1-\eta)$. (ii) The supports of χ and η lie in a ball of radius $3\sqrt{3}\varsigma_7 \leq 2$, and $\|\chi\|_\sharp, \|\eta\|_\sharp, \|1-\eta\|_\sharp \leq \mathsf{c}_7$, so the transfer (M7) at $n = 2$ between the two tori, at the first-passage data given by the passage, bounds the near parts; the supports of χ and $1 - \eta$ are at sup-distance ς_7, so the convergence on fixed tori (M8) gives the far parts. (iii) Take a C^2 function $1_I \leq \eta_I \leq 1_{I^{+\delta}}$ with $\|\eta_I''\|_1 \leq 8/\delta$, the indicator of $[\min I - \delta/2, \max I + \delta/2]$ smoothed by the kernel $\frac{15}{16}(1-t^2)_+^2$ at the scale $\delta/2$. Since $\mu_K \geq 0$, $\mu_K(I) \leq \int \eta_I \, \mathrm{d}\mu_K$ and $\int \eta_I \, \mathrm{d}\mu_{K''} \leq \mu_{K''}(I^{+\delta})$, and

$$\int \eta_I \, d(\mu_K - \mu_{K''}) = \int \eta_I'' \big[(D_K - D_{K,\ell}) - (D_{K''} - D_{K'',\ell})\big] + \int \eta_I \, (D''_{K,\ell} - D''_{K'',\ell}),$$

where the bracket lies in $[-\bar\eta(\ell), \bar\eta(\ell)]$ by Lem. 13.11, and by (i) $D''_{K,\ell} = (V^{\mathrm{near}}_{K,\ell} - V^{\mathrm{near}}_{K,\ell'}) + V^{\mathrm{far}}_{K,\ell} - V^{\mathrm{far}}_{K,\ell'}$, so that $|D''_{K,\ell} - D''_{K'',\ell}| \leq 2c_N + \omega_\ell + \omega_{\ell'}$ on W. In words: a concentration of μ_K at one large cutoff recurs at all large cutoffs. □

Lemma 13.21 (Dilations by real factors)**.** For real $s > 0$, $a > 0$ and $\beta > 0$ the image under dil_s maps the periodic Gibbs states at (a, β) onto the periodic Gibbs states at (sa, β), and $S_n^T[\nu](f_1, \dots, f_n) = s^n \, S_n^T[(\mathrm{dil}_s)_* \nu](f_1(\cdot/s), \dots, f_n(\cdot/s))$. ■

Sketch of proof. The exponent $H_B = -\beta \sum_{p \in \mathrm{Pl}[B]} a_p$ of a Gibbs kernel contains no spacing. By the exact dilation facts (M9), dil_s carries $\mathrm{Pl}[B]$ onto $\mathrm{Pl}[\mathrm{dil}_s B]$, $a_{\mathrm{dil}_s p}(\mathrm{dil}_s U) = a_p(U)$, the product Haar measure on the bonds of B to that on the bonds of $\mathrm{dil}_s B$, and the configuration that is V on B and U off B to the one that is $\mathrm{dil}_s V$ on $\mathrm{dil}_s B$ and $\mathrm{dil}_s U$ off it; hence

$$\mathrm{dil}_s(\gamma_B F) = \gamma_{\mathrm{dil}_s B}(\mathrm{dil}_s F),$$

and $((\mathrm{dil}_s)_* \nu)(\gamma_{\mathrm{dil}_s B} \mathrm{dil}_s F) = \nu(\gamma_B F) = \nu(F)$; as $B \mapsto \mathrm{dil}_s B$ and $F \mapsto \mathrm{dil}_s F$ are onto, the image is a Gibbs state. Invariance under the translations by a subgroup Γ of finite index passes to $s\Gamma$, of the same index; $\mathrm{dil}_{1/s}$ is the inverse. The formula for the cumulants follows from $\mathcal{O}_a(f)(U) = s \, \mathcal{O}_{sa}(f(\cdot/s))(\mathrm{dil}_s U)$. □

Lemma 13.22 (One pressure for both frames; the reference tori compared)**.** In a Setting, let $K \geq k_7$, $\hat K \geq 1$, $\lambda = L^K / \hat L^{\hat K}$ and J a closed interval with $J \subset W_1$ and $\lambda J \subset \hat W$, so that $\lambda \in [\lambda_-, \lambda_+]$. Then $p''_{K,\infty}(J) = \lambda^4 \hat p''_{\hat K, \infty}(\lambda J)$ and

$$D''_K(J) \leq \lambda^4 \hat D''_{\hat K}(\lambda J) + c_2 |J|.$$

■

Sketch of proof. By Lem. 13.11 (b), $p_{K,\infty}(b) = L^{3K} P(L^K b)$ and $\hat p_{\hat K, \infty}(\hat b) = \hat L^{3\hat K} P(\hat L^{\hat K} \hat b)$ with one convex P; for $J = [y_1, y_2]$, $p''_{K,\infty}(J) = L^{4K}[P'(L^K y_2 +) - P'(L^K y_1 -)] = \lambda^4 \hat p''_{\hat K, \infty}(\lambda J)$, since $\hat L^{\hat K} \lambda = L^K$. With $\mathsf{r}_1 = p''_{K,\ell'}$, $\mathsf{r}_2 = \hat p''_{\hat K, \hat \ell'}$ this gives $D''_K(J) - \lambda^4 \hat D''_{\hat K}(\lambda J) = \int_J [\lambda^5 \mathsf{r}_2(\lambda b) - \mathsf{r}_1(b)] \, \mathrm{d}b$, and both $\mathsf{r}_1(b)$ and $\lambda^5 \mathsf{r}_2(\lambda b)$ are $a_K^{-5}(\mathrm{box})^{-3}$ times a variance of the action at the one coupling

$\beta = L^K b = \hat{L}^{\hat{K}} \lambda b$. In the units of the second frame the tent becomes $\chi(\cdot/\lambda)$, $\hat{\mathcal{O}} = \lambda^{-1}\mathcal{O}$ and $S_2 = \lambda^2 \hat{S}_2$; the sum rule of Lem. 13.20 holds on both reference boxes, as $6\lambda\varsigma_7 < \hat{\ell}'$.

The far parts are bounded by c_F and $\hat{c}_F$, the bound of (M8) reading of the functions only the spread $\lambda\varsigma_7 \geq \lambda_-\varsigma_7$ and the norms $\leq \hat{\mathsf{c}}_8$. For the near parts take $\nu \in \mathcal{G}^{\mathrm{per}}$ at (a_K, β), which exists by (M9), and its image $\hat{\nu}$ under the dilation by the real factor λ, a periodic Gibbs state at $(\hat{a}_{\hat{K}}, \beta)$ by Lem. 13.21, with $S_2[\nu](\chi, \eta) = \lambda^2 \hat{S}_2[\hat{\nu}](\chi(\cdot/\lambda), \eta(\cdot/\lambda))$; by the passage in each frame both pairs are first-passage data, and the transfer in each frame between the reference torus and the state $(3\sqrt{3}\lambda\varsigma_7 \leq 2)$ gives

$$|\lambda^5 \mathsf{r}_2(\lambda b) - \mathsf{r}_1(b)| \leq \varsigma_7^{-3}\big[c_T \mathsf{c}_7^2 + \lambda_+^2 \hat{c}_T \hat{\mathsf{c}}_8^2 + c_F + \lambda_+^2 \hat{c}_F\big] = c_2. \qquad \square$$

Proposition 13.23 (No pile of specific heat)**.** In a Setting, with (g_1, W_1, W') and $\operatorname{dist}(W', \partial W_1) = \varepsilon$: for every $0 < \delta \leq \delta_{\max}$ there is an integer K° with

$$\mu_K(I) \leq C_\flat(|I| + \delta) \qquad \text{for all } K \geq K^\circ \text{ and all closed intervals or points } I \subset W'.$$

So $K_1(\delta)$ of Def. 13.6 exists, and it does not increase in δ. Of the convergence on fixed tori (item (M8)) the proof reads two clauses, both at $n = 2$, for pairs with disjoint supports and without a rate: (1) the convergence in the cutoff, uniformly in the label, of the cumulant S_2^T at the pair $(\chi, 1-\eta)$ on the tori T_ℓ, $\ell \in \mathfrak{L}(g_1)$, $\ell \geq \ell'$, over W_1, and at the pair of the tent and the plateau of half-width $1/\hat{L}$ on the tori $T_{\hat{\ell}}$, $\hat{\ell} \in \hat{\mathfrak{L}}(\hat{g})$, $\hat{\ell} \geq \hat{\ell}'$, over $\hat{W}$; (2) the bound clause, which gives c_F, for the one pair $(\chi, 1-\eta)$ on $T_{\ell'}$ over W_1, and $\hat{c}_F$ on $T_{\hat{\ell}'}$, uniformly in $\hat{K} \geq 1$, in $\hat{b} \in \hat{W}$ and in the pairs $(\chi(\cdot/\lambda), 1-\eta(\cdot/\lambda))$, $\lambda \in [\lambda_-, \lambda_+]$, whose spread $\lambda\varsigma_7 \geq \lambda_-\varsigma_7$ and whose norms $\leq \hat{\mathsf{c}}_8$ are all that the clause reads of them. Thus on $T_{\hat{\ell}'}$ two kinds of pairs are read: the pair of half-width $1/\hat{L}$ through the convergence, and the family of the dilated pairs through the bound. ■

Sketch of proof. Thresholds. In the first frame Lem. 13.20 is used with g_1, W_1, ℓ' and k_7 of Def. 13.6. Take $\ell(\delta) \in \mathfrak{L}(g_1)$, $\ell(\delta) \geq \ell'$, with $24\ell' B'/(\delta\ell(\delta)) \leq \delta/2$, and an integer $K^\circ \geq k_7$ with $\omega_{\ell(\delta)}(K, K'') + \omega_{\ell'}(K, K'') \leq 1/6$ for all $K, K'' \geq K^\circ$ (Lem. 13.20 (ii)). In the second frame Lem. 13.20 is used with $\hat{g}$, $\hat{W}$, $\hat{\ell}'$ and $\hat{k}_7 = 1$, so that the tent and the plateau have the half-width $1/\hat{L} \leq 2/(3\sqrt{3})$ there and the constant of the near parts is $\hat{c}_N$. Put $\hat{\delta} = \lambda_+\delta$; take $\hat{\ell} \in \hat{\mathfrak{L}}(\hat{g})$, $\hat{\ell} \geq \hat{\ell}'$, with

$$\frac{24\hat{\ell}'\hat{B}'}{\hat{\delta}\hat{\ell}} \leq \frac{\delta}{2},$$

and an integer $\hat{K}_* \geq 1$ with $\hat{\omega}_{\hat{\ell}}(\hat{K}, \hat{K}'') + \hat{\omega}_{\hat{\ell}'}(\hat{K}, \hat{K}'') \leq 1/(10\lambda_+)$ for all $\hat{K}, \hat{K}'' \geq \hat{K}_*$. *One interval.* Let $I = [y - \delta/2, y + \delta/2]$, $y \in W'$, $J = I^{+\delta} \subset \operatorname{int} W_1$, $K \geq K^\circ$. Let $R = \lfloor |\hat{W}|/(12\lambda_+\delta) \rfloor \geq |\hat{W}|/(24\lambda_+\delta)$ and $t_r = \hat{w}_- + |\hat{W}|/4 + (r - \frac{1}{2})6\lambda_+\delta$, $r = 1, \dots, R$. As $\log L / \log \hat{L}$ is irrational, Kronecker's theorem gives integers $K_r'' \geq K^\circ$ and $\hat{K}_r \geq \hat{K}_*$ with

$$|\lambda_r y - t_r| < \lambda_+\delta/2, \qquad \lambda_r = L^{K_r''}/\hat{L}^{\hat{K}_r};$$

then $\lambda_r \in [\lambda_-, \lambda_+]$, and the intervals $\mathsf{I}_r = (\lambda_r J)^{+\hat{\delta}}$ are pairwise disjoint and lie in the middle half of $\hat{W}$. *The chain.* Lem. 13.20 (iii) in the first frame, for I with the margin δ and the side $\ell(\delta)$, from K to K_r'': $\mu_K(I) \leq \mu_{K_r''}(J) + 3\delta(2c_N + \frac{1}{6}) + \frac{\delta}{2} = \mu_{K_r''}(J) + (6c_N + 1)\delta$. Lem. 13.22 between the frames, with $\hat{D}'' \leq \hat{\mu}$ (as $\hat{C}_0 \geq 0$), $\hat{\mu} \geq 0$ and $\lambda_r \leq \lambda_+$: $\mu_{K_r''}(J) = D''_{K_r''}(J) + 3C_0\delta \leq \lambda_+^4 \hat{\mu}_{\hat{K}_r}(\lambda_r J) + 3(c_2 + C_0)\delta$. Lem. 13.20 (iii) in the second frame, for the interval $\lambda_r J$, of length $3\lambda_r\delta \leq 3\lambda_+\delta$, with the margin $\hat{\delta}$ and the side $\hat{\ell}$, from $\hat{K}_r$ to the one cutoff $\hat{K}_*$, whose added terms are at most $5\lambda_+\delta\,(2\hat{c}_N + \frac{1}{10\lambda_+}) + \frac{\delta}{2} = 10\lambda_+\hat{c}_N\delta + \delta$: $\hat{\mu}_{\hat{K}_r}(\lambda_r J) \leq \hat{\mu}_{\hat{K}_*}(\mathsf{I}_r) + 10\lambda_+\hat{c}_N\delta + \delta$. Together:

$$\mu_K(I) \leq \lambda_+^4\big[\hat{\mu}_{\hat{K}_*}(\mathsf{I}_r) + 10\lambda_+\hat{c}_N\delta + \delta\big] + \big[6c_N + 3(c_2 + C_0) + 1\big]\delta \qquad \text{for every } r.$$

Take the mean over r: the l_r are pairwise disjoint and lie in the middle half of $\hat W$, whose mass for $\hat\mu_{\hat K_*}$ is at most $8\hat B'/|\hat W| + \frac34 \hat C_0|\hat W| \le \hat{\mathfrak m}$ by the mass bound of Prop. 13.16 (b) in the second frame at $t = |\hat W|/4$; so $R^{-1}\sum_r \hat\mu_{\hat K_*}(\mathsf{l}_r) \le \hat{\mathfrak m}/R \le 24\lambda_+\hat{\mathfrak m}\delta/|\hat W|$. This is $\mu_K(I) \le C_\flat\delta$. *Any closed* $I \subset W'$ is covered by at most $|I|/\delta + 1$ such intervals. The property at δ implies it at every $\delta' > \delta$, so K_1 does not increase. □

Lemma 13.24 (The labels; the exact dilation)**.**

(i) If $W \subset I(g)$ is nonempty and compact, then $W \subset \operatorname{int} I(g_1)$ for every $g_1 < g$ with $g_1^{-2} + \bar B < \min W$, and such g_1 exist.

(ii) Every $b > \gamma^{-2} + \bar B$ lies in $\operatorname{int} I(g_2)$ for some $g_2 \le \gamma$; hence for every $b_\infty > 0$ there are an integer $s \ge 0$ and $g_2 \le \gamma$ with $L^s b_\infty \in \operatorname{int} I(g_2)$.

(iii) For integers $0 \le s \le K$ the dilation $x \mapsto L^s x$ carries $\mathbb{L}_K$ onto $\mathbb{L}_{K-s}$, $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ onto $\mathcal{G}^{\mathrm{per}}_{K-s,\beta}$, T_ℓ at the cutoff K with its law onto $T_{L^s\ell}$ at the cutoff $K-s$, boards of side q onto boards of side $L^s q$, $\mathfrak{P}_T(\square_c)$ onto $\mathfrak{P}_{L^sT}(L^s\square_c)$ by $\tilde f = f(\cdot/L^s)$, and the label b into $L^s b$; and $x_c = L^{4s}\tilde x_c$. ■

Sketch of proof. (i) $I(g)$ is open on the left and W is compact, so $\min W > g^{-2} + \bar B$; for $g_1^{-2} \in (g^{-2}, \min W - \bar B)$ one has $g_1 < g$, $g_1^{-2} + \bar B < \min W$ and $X_{g_1} - \bar B > X_g - \bar B \ge \max W$. (ii) Put $g_2^{-2} = b - \bar B - \epsilon$ with $\epsilon = \min\{1, b - \bar B - \gamma^{-2}\}$; then $g_2 \le \gamma$ and $g_2^{-2} + \bar B < b$, and since $L(\bar B - B_\sharp) = \bar B$,

$$X_{g_2} - \bar B > b \iff (L-1)b > 2\bar B + L\epsilon;$$

as $b > 4 + \bar B$ (because $\gamma \le \frac12$), $(L-1)b > 4(L-1) + (L-1)\bar B \ge L + 2\bar B \ge L\epsilon + 2\bar B$. For $b_\infty > 0$ take s with $L^s b_\infty > \gamma^{-2} + \bar B$. (iii) The dilation is a graph isomorphism of the lattices, the Wilson weight reads the graph and β alone, so the laws of the tori correspond, and the periodic Gibbs states correspond by (M9); $\beta = L^K b = L^{K-s}(L^s b)$; $\tilde{\mathcal{O}}_{K-s}(\tilde f) = a_{K-s}^{-1}\sum_p f(x(p))a_p = L^{-s}\mathcal{O}_K(f)$, hence $\tilde x_c = (L^s q)^{-3}\tilde{\mathcal{O}}(\tilde f) = L^{-4s}x_c$; the collars and the Lipschitz constant scale with L^s, and offsets in $a_K\mathbb{Z}^3$ go to offsets in $a_{K-s}\mathbb{Z}^3$. □

Proposition 13.25 (The Källén–Lehmann form)**.** Let σ' be the limit σ of a convergent Gibbs sequence or a component σ_ξ of it, and $\mathbf{s}$ its radial kernel.

(a) $\mathbf{s}$ is completely monotone on $(0,\infty)$; so $\mathbf{s} \ge 0$ does not increase, and $\kappa' = \lim_{r\to\infty}\mathbf{s}(r) \in [0,\infty)$ exists.

(b) $r \mapsto r(\mathbf{s}(r) - \kappa')$ is completely monotone; there is exactly one positive Borel measure ρ on $[0,\infty)$ with $\mathbf{s}(r) = \kappa' + \int e^{-mr}(4\pi r)^{-1}\,\mathrm{d}\rho(m)$.

(c) $\rho(\{0\}) = \lim_{r\to\infty} 4\pi r(\mathbf{s}(r) - \kappa')$, and $\rho([0,m]) \le 4\pi e\, m^{-1}(\mathbf{s}(1/m) - \kappa')$ for $m > 0$.

(d) *Cube means, with no further hypothesis.* Let $T \ge \max\{T_0, \mathsf{l}\}$, $q = L^i \ge 2T$, a board of $\mathbb{R}^3$ of side q (a partition of $\mathbb{R}^3$ into the cubes $\square_c$ of side q centred at the points $o + qc$, $c \in \mathbb{Z}^3$, with an offset $o \in \mathbb{R}^3$; c is the index of the cell) and profiles $f_c \in \mathfrak{P}_T(\square_c)$, $f_{c'} \in \mathfrak{P}_T(\square_{c'})$ of two distinct cells be given. For every $0 < r_0 \le T/4$

$$0 \le q^{-6}\Big[\sigma'(f_c, f_{c'}) - \kappa' \int f_c \int f_{c'}\Big] \le 24\pi\, r_0(\mathbf{s}(r_0) - \kappa')/q,$$

and if the two cells share a face, the middle member is $\ge \rho(\{0\})(1 - T/q)^6/(4\sqrt{6}\,\pi q)$. So q times the middle member tends to 0 as $q \to \infty$, uniformly in the pairs of distinct cells and in the profiles, if and only if $\rho(\{0\}) = 0$.

(e) *The converse; the measure that attains the bounds.* The following are equivalent: (i) there are $T \ge \max\{T_0, \mathsf{l}\}$, C and j_0 with $L^{j_0} \ge 2T$ and $\sigma'(\theta w_i, w_i) \le C(T/q_i)^2$ for all $i \ge j_0$, the w_i of Def. 13.3 (7) being formed with this T; (ii) $\mathbf{s}(r) = O(r^{-2})$ as $r \to \infty$; (iii) $\kappa' = 0$ and $\rho([0,m]) = O(m)$ as $m \to 0$; (iv) there are A_μ and $r_* > 0$ with $\mu_h(Q_r) \le A_\mu \|h\|_1^2 r^2$ for every real $h \in C^1_c(\mathbb{E}_+)$ and $0 < r \le r_*$, μ_h formed in $\mathcal{H}(\sigma', e_3)$. Under them $\rho(\{0\}) = 0$, so that $\rho \ge \mathsf{Z}\delta_0$ with $\mathsf{Z} > 0$ is excluded, and so is $\rho([0,m]) \ge c_\rho m^\vartheta$ for all small m with $c_\rho > 0$ and

$\vartheta < 1$; the measure $\rho = c_\rho \, \mathrm{d}m$ on $[0,\infty)$, $c_\rho > 0$, whose kernel is $\mathbf{s}(r) = c_\rho/(4\pi r^2)$, satisfies (ii) and (iii) and attains both bounds in order.

■

Sketch of proof. We use Bernstein's theorem (a function on $(0,\infty)$ is completely monotone if and only if it is $\int e^{-tE}\,\mathrm{d}\mu(E)$ with a unique positive μ), that $-c'$ is completely monotone with c, and that a pointwise finite limit of completely monotone functions is completely monotone (Helly's theorem applied to the measures $e^{-t_0 E} d\mu_n$). (a) By (M10), σ' is invariant under $\mathbb{R}^3 \rtimes \mathrm{O}(3)$ and reflection positive, so the reconstruction applies: for $f_\epsilon(x) = \eta_\epsilon(x_3)\mathsf{g}(x_1,x_2)$ with $0 \le \eta_\epsilon \in C^1_c((s-\epsilon, s+\epsilon))$ of mass 1 and real $\mathsf{g} \in C^1_c(\mathbb{R}^2)$, $c_\epsilon(t) = \sigma'(\theta f_\epsilon, \tau_{te_3} f_\epsilon) = \int e^{-tE} d\mu_{f_\epsilon}$ is completely monotone, and by the kernel it tends to $\mathsf{P}_{\mathsf{g}}(2s+t)$, $\mathsf{P}_{\mathsf{g}}(u) = \iint \mathsf{g}(y)\mathsf{g}(y')\mathbf{s}((u^2+|y-y'|^2)^{1/2})\,\mathrm{d}y\,\mathrm{d}y'$; letting g tend to a point mass gives (a). (b) For a radial $\mathsf{g} \ne 0$ put $G = \mathsf{g} * \mathsf{g}$, with radial profile $\hat G$, and $\mathsf{g}_R = R^{-1}\mathsf{g}(\cdot/R)$, so that $\mathsf{P}_R(u) = 2\pi \int_0^\infty \hat G(\varrho'/R)\mathbf{s}((u^2+\varrho'^2)^{1/2})\varrho' d\varrho'$; with $\partial_u \mathbf{s}((u^2+\varrho'^2)^{1/2}) = (u/\varrho')\partial_{\varrho'}\mathbf{s}((u^2+\varrho'^2)^{1/2})$ one integration by parts gives

$$-\mathsf{P}'_R(u) = 2\pi u\Big[\hat G(0)\mathbf{s}(u) + \int_0^\infty \hat G'(v)\,\mathbf{s}((u^2+R^2v^2)^{1/2})\,\mathrm{d}v\Big] \;\longrightarrow\; 2\pi\hat G(0)\,u\,(\mathbf{s}(u)-\kappa') \qquad (R\to\infty),$$

and $-\mathsf{P}'_R$ is completely monotone; Bernstein's theorem gives ρ, normalised so that $4\pi r(\mathbf{s}(r)-\kappa') = \int e^{-mr}\,\mathrm{d}\rho(m)$. (c) Monotone convergence, and $\int e^{-mr}\,\mathrm{d}\rho(m) \ge e^{-1}\rho([0,1/r])$. (d) As $\mathbf{s} \ge \kappa'$ and $f_c, f_{c'} \ge 0$, the middle member is $q^{-6}\iint f_c(x) f_{c'}(y)(\mathbf{s}(|x-y|)-\kappa')\,\mathrm{d}x\,\mathrm{d}y \ge 0$. The supports are at distance $\ge T/4 \ge r_0$, where $\mathbf{s}(|x-y|)-\kappa' \le r_0(\mathbf{s}(r_0)-\kappa')/|x-y|$, since $r(\mathbf{s}(r)-\kappa')$ does not increase by (b); and with $0 \le f_c, f_{c'} \le 1$ on the cells, $u = (y-x)/q$ and the tent $\chi_{\mathbf{d}}(u) = \prod_\mu (1-|u_\mu - d_\mu|)_+$ about the difference $\mathbf{d} \ne 0$ of the indices of the two cells, $q^{-6}\iint_{\square_c\times\square_{c'}} |x-y|^{-1}\,\mathrm{d}x\,\mathrm{d}y = q^{-1}\int \chi_{\mathbf{d}}(u)|u|^{-1}\,\mathrm{d}u$, which is $\le q^{-1}$ if $|\mathbf{d}|_\infty \ge 2$ and $\le q^{-1}\int_{|u|\le 2\sqrt3} |u|^{-1} du = 24\pi/q$ otherwise.

For cells that share a face, $4\pi r(\mathbf{s}(r)-\kappa') = \int e^{-mr}\,\mathrm{d}\rho \ge \rho(\{0\})$, the profiles are 1 on the concentric cubes of side $q-T$, and two points of these cubes are at distance $\le \sqrt6 q$. If $\rho(\{0\}) = 0$, then $r(\mathbf{s}(r)-\kappa') \le \epsilon_1$ for $r \ge r_{\epsilon_1}$, and the pairs of points of $\square_c \times \square_{c'}$ at distance $< r_{\epsilon_1}$ have measure $\le 8\pi q^2 r_{\epsilon_1}^4$; so q times the middle member is $\le 24\pi\epsilon_1 + 8\pi(\mathbf{s}(T/4)-\kappa')r_{\epsilon_1}^4 q^{-3}$, for every $\epsilon_1 > 0$. (e) (i) gives (ii): $\theta w_i, w_i \ge 0$ have disjoint supports in a box of diameter $\sqrt6 q_i$, $\int w_i \ge (1-3T/(4q_i))^3$, and $\mathbf{s}$ does not increase, so $C(T/q_i)^2 \ge \sigma'(\theta w_i, w_i) \ge \mathbf{s}(\sqrt6 q_i)(1-3T/(4q_i))^6$; for $\sqrt6 q_i \le r < \sqrt6 q_{i+1}$, $\mathbf{s}(r) \le \mathbf{s}(\sqrt6 q_i)$ and $q_i^{-2} < 6L^2/r^2$. (ii) gives (i): if $\mathbf{s}(r) \le A_{\mathbf{s}} r^{-2}$ for $r \ge r_0$, take $T \ge \max\{T_0, \mathsf{l}, 4r_0\}$; the supports of θw_i and w_i are at distance $\ge T/4 \ge r_0$, so $\sigma'(\theta w_i, w_i) \le A_{\mathbf{s}} q_i^{-2}\int \chi_{e_3}(u)|u|^{-2}\,\mathrm{d}u \le 3\pi A_{\mathbf{s}} q_i^{-2}$, as $\chi_{e_3}(u) \le u_3$ on its support, which lies in $\{|u| \le \sqrt6,\ u_3 > 0\}$. (ii) gives (iii) by (c). (iii) gives (ii): if $\kappa' = 0$ and $\rho([0,m]) \le C_\rho m$ for $0 < m \le m_0$, then $e^{-mr} = e^{-m_0 r} + r\int_m^{m_0} e^{-ur} du$ and Fubini give $\int_{[0,m_0]} e^{-mr} d\rho(m) \le C_\rho m_0 e^{-m_0 r} + C_\rho/r$, and for $r \ge 1$ the rest of $4\pi r\mathbf{s}(r)$ is $\le e^{-m_0(r-1)} 4\pi \mathbf{s}(1)$. (iii) and (iv): for real $h \in C^1_c(\mathbb{E}_+)$ the kernel, the form (b) and the formula

$$\frac{e^{-m\sqrt{\tau^2+|w|^2}}}{4\pi\sqrt{\tau^2+|w|^2}} = \int_{\mathbb{R}^2} e^{i\mathsf{q}\cdot w}\,\frac{e^{-\tau\omega_m(\mathsf{q})}}{2\omega_m(\mathsf{q})}\,\frac{d^2\mathsf{q}}{(2\pi)^2}, \qquad \omega_m(\mathsf{q}) = (m^2+|\mathsf{q}|^2)^{1/2},\ \tau > 0,\ w \in \mathbb{R}^2,$$

give for $t \ge 0$ and $v \perp e_3$

$$\langle \psi_h, e^{-tH}\mathcal{U}(v)\psi_h\rangle = \sigma'(\theta h, \tau_{te_3+v}h) = \kappa'\Big(\int h\Big)^2 + \int \mathrm{d}\rho(m)\int e^{-t\omega_m(\mathsf{q})-iv\cdot\mathsf{q}}\,\frac{|\hat h_m(\mathsf{q})|^2}{2\omega_m(\mathsf{q})}\,\frac{d^2\mathsf{q}}{(2\pi)^2},$$

with $\hat h_m(\mathsf{q}) = \int h(y) e^{-\omega_m(\mathsf{q})y_3 + i\mathsf{q}\cdot y_\perp}\,\mathrm{d}y$; as a finite measure on $[0,\infty)\times\mathbb{R}^2$ is determined by its transform, μ_h is $\kappa'(\int h)^2\delta_{(0,0)}$ plus the images under $\mathsf{q} \mapsto (\omega_m(\mathsf{q}), -\mathsf{q})$ of the measures $|\hat h_m|^2(2\omega_m)^{-1}d^2\mathsf{q}/(2\pi)^2$, integrated in $d\rho(m)$. As $|\hat h_m| \le \|h\|_1$ and $\int_{\omega_m(\mathsf{q})\le r}(2\omega_m(\mathsf{q}))^{-1}d^2\mathsf{q}/(2\pi)^2 = (r-m)_+/(4\pi)$, this gives $\mu_h(Q_r) \le \kappa'\|h\|_1^2 + (4\pi)^{-1} r\,\rho([0,r])\|h\|_1^2$, so (iii) gives (iv). Conversely, take $0 \le h \ne 0$ with support in $\{y_3 \le R_h,\ |y_\perp| \le R_h\}$ and $r \le \min\{r_*, 1/(2R_h)\}$: the points $(\omega_m(\mathsf{q}), -\mathsf{q})$ with $m \le r/2$ and $|\mathsf{q}| \le r/2$ lie in Q_r, and there $|\hat h_m(\mathsf{q})| \ge e^{-1}\cos(\frac12)\|h\|_1$; so

$A_\mu\|h\|_1^2 r^2 \ge \mu_h(Q_r) \ge \kappa'\|h\|_1^2 + e^{-2}\cos^2(\frac12)(32\pi)^{-1} r\,\rho([0,r/2])\|h\|_1^2$, which gives (iii). Under (iii), $\rho(\{0\}) \le \rho([0,m]) \to 0$ as $m \to 0$; and $\int_0^\infty e^{-mr} c_\rho\,\mathrm{d}m/(4\pi r) = c_\rho/(4\pi r^2)$. □

Lemma 13.26 (Three mirrors; the spectral form)**.**

(a) Let σ' be a real, symmetric, normalized functional on $\mathfrak{S}(\mathbb{R}^3)$, multilinear on products of test spaces with disjoint closures, whose one-point function is $\sigma'(f) = c\int f$ and whose connected functions obey a bound table as in Section YM3_I:90. Assume that σ' is invariant under the translations of $\mathbb{R}^3$ and reflection positive at the three coordinate mirrors through 0; let $X, Y \in \mathfrak{S}(\mathbb{R}^3)$ have their supports in $\{|x|_\infty < R\}$ and $G(a) = \sigma'(X \cdot \tau_a Y)$. Then $G(a)$ converges as $\min_\mu a_\mu \to \infty$, jointly in the three coordinates, to $\langle[\theta X_0], \mathcal{P}(\{0\})[Y_0]\rangle$ in $\mathcal{H}(\sigma', e_3)$, with $X_0 = \tau_{-(R+\frac12)e_3} X$ and $Y_0 = \tau_{(R+\frac12)e_3} Y$; likewise in each of the eight orthants, X and Y being exchanged in the directions in which $a_\mu \to -\infty$.

(b) Let σ' obey a bound table with one-point constant 0, be invariant under the translations of $\mathbb{R}^3$ and reflection positive at $(e_3, 0)$, and let $\sigma'(\theta w_i, w_i) \le C_{\mathrm{LH}}(T/q_i)^2$ for $i \ge j_0$, with $q_{j_0} \ge 2T$. Then for real $h \in C_c^1(\mathbb{E}_+)$

$$\mu_h(Q_r) \le 12.46\,(1 - 3T/(4q_{j_0}))^{-6} L^2 C_{\mathrm{LH}} \|h\|_1^2 T^2 r^2 \quad (0 < r \le 1/q_{j_0}), \qquad \mathcal{P}(\{0\})\psi_h = 0.$$

Neither the cubic group nor O(3) nor a kernel is used. ■

Sketch of proof. (a) In the picture e_3, for $t = a_3 - 2R - 1 \ge 0$, translation invariance and the reconstruction give $G(a) = \langle[\theta X_0], e^{-tH}\mathcal{U}(a_\perp)[Y_0]\rangle = \int e^{-tE + i a_\perp\cdot \mathsf{q}}\,\mathrm{d}m_3$, $m_3 = \langle[\theta X_0], \mathcal{P}(\cdot)[Y_0]\rangle$, so $\sup_{a_\perp} |G(a) - G_3(a_\perp)| \le \varepsilon_3(t) = \int_{E>0} e^{-tE} d|m_3| \downarrow 0$, with $G_3(a_\perp)$ the part at $E = 0$; the same in the pictures e_1, e_2. Changing one coordinate at a time between two points with all coordinates $\ge A$,

$$|G(a) - G(a')| \le 2\sum_\mu \varepsilon_\mu(A - 2R - 1) \to 0,$$

so G converges; its limit is the limit of the Cesàro means $A^{-3}\int_{[A,2A]^3} G$, whose three brackets in the picture e_3 tend to $1_{E=0}$, $1_{\mathsf{q}_1=0}$, $1_{\mathsf{q}_2=0}$, giving $m_3(\{(0,0,0)\})$. In the other orthants write $G(a) = \sigma'(\tau_{-a}X \cdot Y)$. (b) Let $w = w_i$, $q = q_i$, $i \ge j_0$; $w \ge 0$ has its support in $\{T/4 \le x_3 \le q - T/4\}$. The convolution $w * h = \int w(y)\tau_y h\,\mathrm{d}y = \int h(y)\tau_y w\,\mathrm{d}y$ is read twice, the Riemann sums converging in $\|\cdot\|_\sharp$ and $h \mapsto \psi_h$ being linear and bounded by the bound table:

$$\psi_{w*h} = \mathsf{G}_w(H, P)\psi_h, \quad \mathsf{G}_w(E, \mathsf{q}) = \int w(y) e^{-y_3 E + i y_\perp\cdot\mathsf{q}}\,\mathrm{d}y, \qquad \psi_{w*h} = \int h(y) e^{-y_3 H}\mathcal{U}(y_\perp)\psi_w\,\mathrm{d}y.$$

The operators $e^{-tH}\mathcal{U}(v)$ are contractions, so

$$\int |\mathsf{G}_w|^2 d\mu_h = \|\psi_{w*h}\|^2 \le \|h\|_1^2 \sigma'(\theta w, w) \le \|h\|_1^2 C_{\mathrm{LH}}(T/q)^2.$$

On $Q_{1/q}$, as φ_0 is even,

$$\mathsf{G}_w = q^{-3}\int \varphi_0(y_3 - \frac{q}{2}) e^{-y_3 E}\,\mathrm{d}y_3 \prod_{k=1,2}\int \varphi_0(y_k)\cos(y_k \mathsf{q}_k)\,\mathrm{d}y_k \ge e^{-1}\cos^2(\tfrac12)(1 - 3T/(4q))^3;$$

and $e^2/\cos^4(\frac12) \approx 12.4577$. For $1/(Lq_i) < r \le 1/q_i$, $\mu_h(Q_r) \le \mu_h(Q_{1/q_i})$ and $q_i^{-2} < L^2 r^2$. Finally $\|\mathcal{P}(\{0\})\psi_h\|^2 = \mu_h(\{0\}) \le \mu_h(Q_r) \to 0$. □

Lemma 13.27 (Limits of Gibbs sequences, with free constants)**.** Let a frame, $0 < g' \le \gamma$, $T \ge \max\{T_0, \mathrm{l}\}$ and numbers $C_{\mathrm{LH}}, c_{\mathrm{LH}}, q_{\mathrm{LH}} > 0$ be given, and let (K_j, β_j, ν_j) be a convergent Gibbs sequence at g' with the following property: for every $q = L^i \ge \max\{2T, q_{\mathrm{LH}}\}$ there is $j(q)$ such that for all $j \ge j(q)$, all $\nu, \nu' \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$, every board of $\mathbb{L}_{K_j}$ of side q with any offset and profiles of $\mathfrak{P}_T$ in two distinct cells c, c': $|\operatorname{Cov}_\nu(x_c, x_{c'})| \le C_{\mathrm{LH}}(T/q)^2$ and $|\nu(x_c) - \nu'(x_c)| \le c_{\mathrm{LH}} T/q$.

Then the conclusions 1–9 of Thm. AL (c) hold for σ and every σ_ξ, with C_{LH} in the place of C_2, $L^{j_0} \ge \max\{2T, q_{\mathrm{LH}}\}$, $r_0 = \sqrt{6}L^{j_0}$ and $C_{\mathbf{s}} = 6L^2(1-3T/(4L^{j_0}))^{-6}C_{\mathrm{LH}}$. ■

Sketch of proof. The pair of cells. Fix $q = q_i \ge \max\{2T, q_{\mathrm{LH}}\}$. On the board of $\mathbb{L}_{K_j}$ of side q with the offset $(0,0,\frac12 q + \frac12 a_{K_j})$, which lies in $a_{K_j}\mathbb{Z}^3$ because q/a_{K_j} is odd, the two cells with the centres $(0,0,\pm\frac12 q + \frac12 a_{K_j})$ have a common face, and $f_0(\cdot \mp \frac12 q e_3)$ is moved from them by $\frac12 a_{K_j} \le T/8$ for large j; so, φ_0 being even, $\mathcal{O}(w_i)$ and $\mathcal{O}(\theta w_i)$ are cube means x_c, $x_{c'}$ of two distinct cells. *The lattice bounds pass to σ and to every σ_ξ*, because they name the pair (K_j, β_j) alone and so hold for the ν_j and for the approximants ρ_j of σ_ξ:

$$\mathfrak{c}_\xi \int w_i = \sigma_\xi((w_i)) = \lim_j (\rho_j - \nu_j)(x_c), \qquad |(\rho_j - \nu_j)(x_c)| \le c_{\mathrm{LH}} T/q, \qquad \int w_i \ge (1 - \tfrac{3T}{4q})^3 \ge (\tfrac58)^3,$$

for every q above the floor, so $\mathfrak{c}_\xi = 0$; this is shown before the kernel is used, so that $S_2^{\sigma'}(f,h) = \sigma'(f,h)$. Further $S_2^{\sigma}(\theta w_i, w_i) = \lim_j \mathrm{Cov}_{\nu_j}(x_{c'}, x_c)$ and $\sigma_\xi(\theta w_i, w_i) = \lim_j [\mathrm{Cov}_{\rho_j}(x_{c'}, x_c) + (\rho_j - \nu_j)(x_{c'})(\rho_j - \nu_j)(x_c)]$, whose second term tends to 0; this is conclusion 2. *From two cells with a common face to the kernel.* By Prop. 13.25, $\mathbf{s} \ge 0$ does not increase; θw_i, $w_i \ge 0$ have disjoint supports in a box of diameter $\sqrt{6}q$, so $C_{\mathrm{LH}}(T/q_i)^2 \ge \sigma'(\theta w_i, w_i) \ge \mathbf{s}(\sqrt{6}q_i)(1 - 3T/(4q_i))^6$; for $r \ge r_0$ take $i \ge j_0$ with $\sqrt{6}q_i \le r < \sqrt{6}Lq_i$, so $q_i^{-2} < 6L^2/r^2$ and $\mathbf{s}(r) \le C_{\mathbf{s}} T^2/r^2$.

Hence $\kappa' = 0$, conclusion 3 is Prop. 13.25 (b), conclusion 5 follows from $|x-y| \ge |a| - R_f - R_h \ge r_0$ on the supports, and 6, 7 from Prop. 13.25 (c). *One field.* For real $h \in C_c^1(\mathbb{E}_+)$, $\|e^{-tH}\psi_h\|^2 = \sigma'(\theta h, \tau_{2te_3} h) \le \|h\|_1^2 \mathbf{s}(2t)$, the supports being at distance $\ge 2t$; so $e^{-tH}\psi_h \to 0$, which kills the projection on $\ker H$; $\int_0^\infty \langle \psi_h, e^{-tH}\psi_h\rangle \, \mathrm{d}t = \|H^{-1/2}\psi_h\|^2$ is finite by the bound at large t; and $e^{-2}\langle \psi_h, 1_{[0,\epsilon]}(H)\psi_h\rangle \le \|e^{-H/\epsilon}\psi_h\|^2$. For conclusion 9 rotate a to the e_3 axis by O(3) and put $X_0 = \tau_{-(R_X+1)e_3} X$, $h_0 = \tau_{(R_h+1)e_3} h$, $t = |a| - R_X - R_h - 2$:

$$\sigma'(X \cdot \tau_a h) = \langle [\theta X_0], e^{-tH}\psi_{h_0}\rangle, \qquad |\sigma'(X \cdot \tau_a h)| \le \|[\theta X_0]\| \, \|e^{-tH}\psi_{h_0}\| \le \mathsf{N}(X)\|h\|_1 C_{\mathbf{s}}^{1/2} T/(2t),$$

where $\|[\theta X_0]\|^2 = \sigma'(X \cdot \theta_{e_3} X) \le \mathsf{N}(X)^2$, finite by the bound tables since the supports of $X \cdot \theta_u X$ lie in one ball and those of X and $\theta_u X$ are at distance ≥ 2; in the moment–cumulant formula for $S_{n+1}^{\sigma'}(f_1, \dots, f_n, \tau_a h)$ exactly one factor holds $\tau_a h$, and it is $\sigma'(X_J \cdot \tau_a h)$ for a subtuple X_J or $\sigma'((\tau_a h)) = 0$. □

Lemma 13.28 (Growing tori, with free constants)**.** Let a frame, $0 < g' \le \gamma$, $T \ge \max\{T_0, \mathsf{l}\}$, numbers $C_{\mathrm{LH}}, q_{\mathrm{LH}} > 0$ and j_0 with $L^{j_0} \ge \max\{2T, q_{\mathrm{LH}}\}$ be given, and let S be a continuum limit at g' along growing tori with a defining sequence (K_j, ℓ_j, β_j) that has the following property: for every $q = L^i \ge \max\{2T, q_{\mathrm{LH}}\}$ there is $j(q)$ such that for all $j \ge j(q)$, $\ell_j \ge 2q$ and $|\mathrm{Cov}_{K_j,\ell_j,\beta_j}(x_c, x_{c'})| \le C_{\mathrm{LH}}(T/q)^2$ for profiles of $\mathfrak{P}_T$ in two distinct cells of every board of T_{ℓ_j} of side q. Then the conclusions 1–3 of Thm. AL (d) hold with C_{LH} in the place of C_2. ■

Sketch of proof. By (M10), σ_S obeys a bound table with one-point constant 0, is invariant under the translations and the cubic group, and is reflection positive at the coordinate mirrors, so Lem. 13.26 applies at $u = e_1, e_2, e_3$. For $q = q_i$ and large j, as in the proof of Lem. 13.27 on the board of T_{ℓ_j} of side q with the offset $(0,0,\frac12 q + \frac12 a_{K_j})$ ($2q$ divides ℓ_j, and the supports lie in $(-\ell_j/4, \ell_j/4)^3$), θw_i and w_i are q^{-3} times profiles of two distinct cells, so

$$\sigma_S(\theta w_i, w_i) = S_2(\theta w_i, w_i) = \lim_j S_{2;K_j,\ell_j,\beta_j}^T(\theta w_i, w_i) \le C_{\mathrm{LH}}(T/q_i)^2;$$

the cubic group carries this to the mirrors $(e_1, 0)$, $(e_2, 0)$. The spectral clauses are Lem. 13.26 (b). For conclusion 3, Lem. 13.26 (a) at $Y = (h)$ gives the limit $\langle [\theta X_0], \mathcal{P}(\{0\})\psi_{h_0}\rangle = 0$, $h_0 = \tau_{(R+\frac12)e_3} h \in C_c^1(\mathbb{E}_+)$, and 0 likewise in the other orthants; in the moment–cumulant formula exactly one factor holds $\tau_a h$, and it is $\sigma_S(X_J \cdot \tau_a h) \to 0$ or $\sigma_S((\tau_a h)) = 0$. □

Proof of Thm. AL. Step 1: the Setting is consistent. g_1 exists by Lem. 13.24 (i), hence ε exists; Prop. 13.23 is applied to (g_1, W_1, W'), with $\operatorname{dist}(W', \partial W_1) = \varepsilon$. Since $C_w^- \le C_w$, $r_1^- \le r_1$, $C_I \le C_I^+$ and $T \ge T_0$, the floor $q \ge q_0$ gives

$$\text{(f1)} \quad q \ge \frac{4C_\flat + C_I}{9C_w^2 T^2}, \qquad \text{(f2)} \quad x_q = \frac{1}{3C_w T q^2} \le \min\{r_1, \varepsilon/2\},$$

so $I_q(b) \subset W' \subset \operatorname{int} I(g_1)$ for $b \in W$, and $\delta = \delta^\circ(q) \le \delta_{\max}$ has $C_\flat \delta \le \frac{3}{2} C_w^- T_0/q \le \frac{3}{2} C_w T/q$. The number $C_\flat$ is formed from λ_+, $\hat{\mathfrak{m}}$, $\hat{c}_N$, c_N, c_2 and C_0, none of which depends on T.

Step 2: clause (a). Let $b \in W$ and $K \ge K_0(q) = K_1(\delta)$. Prop. 13.23 gives $\mu_K(I_q(b)) \le C_\flat(2x_q + \delta)$, and $2C_\flat x_q \le \frac{3}{2} C_w T/q$ is (f1); so

$$\delta_K(q, b) = 3C_w T/q + \mu_K(I_q(b)) \le 6C_w T/q.$$

Prop. 13.19 (a) at g_1 applies, as $q \ge q_1$ by $q \ge 2T$, (f1) and (f2): $2e^4 \cdot 36 = 72e^4$ and $2e^2 \cdot 6 = 12e^2$. The clause names the states at $(K, L^K b)$, which do not depend on the reference coupling.

Step 3: clause (b). Let $\ell \in \mathfrak{L}(g)$, $\ell \ge \ell_0(q) \ge \ell'$; as $\mathfrak{L}(g_1)$ is closed upward, $\ell \in \mathfrak{L}(g_1)$. $\Psi = D_K + \frac{1}{2} C_0 y^2$ and $\Phi = D_{K,\ell} + \frac{1}{2} C_0 y^2$ are convex on W_1 and $0 \le \Phi - \Psi \le \bar\eta = \bar\eta(\ell)$. With $\sigma_0 = \Psi'(b-)$, $t = (\bar\eta/C_\flat)^{1/2} \le \varepsilon/2$ (this is $\ell \ge 12\ell' B'/(C_\flat \varepsilon^2)$) and $0 < x \le \varepsilon/2$, the slopes of a convex function do not decrease:

$$\frac{\Phi(b+x) - \Phi(b)}{x} \le \frac{\Phi(b+x+t) - \Phi(b+x)}{t} \le \frac{\Psi(b+x+t) - \Psi(b+x) + \bar\eta}{t} \le \sigma_0 + C_\flat(x + t + \delta) + \frac{\bar\eta}{t},$$

by Prop. 13.23 on $[b, b+x+t] \subset W'$, and the mirror bound holds on the left; $C_\flat t + \bar\eta/t = 2(C_\flat \bar\eta)^{1/2}$. So $D_{K,\ell}$ has the hypothesis of Prop. 13.18 with $\mathsf{s} = C_0 b - \sigma_0$, $\delta_2 = 2(C_\flat \bar\eta)^{1/2} + C_\flat \delta \ge 0$, $\tilde C = C_\flat \ge 0$, $\eta_+ = 0$, $x_0 = \varepsilon/2 < b$. Its floors are (f1), (f2); $2(3C_\flat \ell' B'/\ell)^{1/2} \le \frac{3}{2} C_w T/q$ is $\ell \ge \frac{16}{3} C_\flat \ell' B' q^2/(C_w T)^2$; so $\delta_2 \le 3C_w T/q$, $\delta \le 6C_w T/q$, and the bounds follow as in Step 2.

Step 4: clause (c). Take $g' = g$ in Lem. 13.27. For $q = L^i \ge \max\{2T, q_0\}$ and large j we have $K_j \ge K_0(q)$ and $b_j \in W$; clause (a), which names the pair (K_j, β_j) alone, gives the hypothesis of Lem. 13.27 with

$$(C_{\mathrm{LH}}, c_{\mathrm{LH}}, q_{\mathrm{LH}}) = (C_2,\ 12e^2 C_w,\ q_0),$$

and its conclusions are clause (c).

Step 5: clause (d). Take $g' = g$ in Lem. 13.28. For $q = L^i \ge \max\{2T, q_0\}$ and large j: $K_j \ge K_0(q)$, $b_j \in W$, and $\ell_j \in \mathfrak{L}(g)$, $\ell_j \ge \ell_0(q) \ge 2q$. The two floors do not depend on each other, so no relation between K_j and ℓ_j is used, and clause (b) gives the hypothesis of Lem. 13.28 with $(C_{\mathrm{LH}}, q_{\mathrm{LH}}) = (C_2, q_0)$; its conclusions are clause (d).

Step 6: clauses (e) 1 and (e) 2. (e) 1 is Lem. 13.24 (ii). For (e) 2, Lem. 13.24 (iii) carries the states at $(K, L^K b)$ and their boards of side q to states at $(K - s, L^{K-s} \cdot L^s b)$ and boards of side $L^s q$, with

$$\operatorname{Cov}_\nu(x_c, x_{c'}) = L^{8s} \operatorname{Cov}_{\tilde\nu}(\tilde x_c, \tilde x_{c'}), \qquad \nu(x_c) - \nu'(x_c) = L^{4s}(\tilde\nu - \tilde\nu')(\tilde x_c),$$

and n-cell moments carrying L^{4ns}; $(L^s T)/(L^s q) = T/q$; apply (a), (b) in the Setting built on $(g_2, W^{(2)})$ at $(L^s T, L^s q, K - s, L^s \ell)$.

Step 7: clause (e) 3. Each $b_\infty \in \mathsf{B}$ has, by (e) 1, a pair (s, g_2) and a compact interval $W^{(2)} \subset \operatorname{int} I(g_2)$ holding $L^s b_\infty$ in its interior; finitely many of the open sets $L^{-s} \operatorname{int} W^{(2)}$ cover B, say $(s_i, g^{(i)}, W^{(i)})$, $i \le n_{\mathsf{B}}$, each with its Setting in the frame L and one second frame for all. Put

$$C_{\mathsf{B}} = \max_i L^{8s_i} 72e^4 (C_w^{+,(i)})^2, \quad q_{\mathsf{B}} = \max_i L^{-s_i} q_0^{(i)}, \quad K_{\mathsf{B}}(q) = \max_i [s_i + K_0^{(i)}(L^{s_i} q)],$$

and let $\ell_{\mathsf{B}}(q)$ be the maximum over i of L^{-s_i} times $\ell_0^{(i)}(L^{s_i} q)$ formed with $C_w^{-,(i)} T_0$ in the place of $C_w T$, which bounds $\ell_0^{(i)}(L^{s_i} q)$ at the width $L^{s_i} T$ because $(L^{s_i} q)^2/(C_w L^{s_i} T)^2 = q^2/(C_w T)^2$. None of these depends on T. A side $L^{s_i} \ell = 2L^{m+s_i} \ge \ell'^{(i)}$ lies in $\mathfrak{L}(g_1^{(i)}) \subset \mathfrak{L}(g^{(i)})$ by (M5), so (b) at $g^{(i)}$

speaks of it; $T \geq \max\{T_0, \mathsf{l}\}$ gives $L^{s_i}T \geq \max\{T_0, \mathsf{l}\}$; and $(2C_{\mathsf{B}})^{1/2} \geq L^{4s_i} 12e^2 C_w^{(i)}$. Step 6 gives (e) 3.

Step 8: clause (e) 4. At $g' \leq \gamma$ the labels of every Gibbs sequence, and of every defining sequence of a limit along growing tori, lie in the compact set $\mathsf{B} = I^{\mathrm{lab}}(g')$ by (M5). For $q = L^i \geq \max\{2T, q_{\mathsf{B}}\}$ and large j, $K_j \geq K_{\mathsf{B}}(q)$ and $\ell_j \geq \ell_{\mathsf{B}}(q) \geq 2q$, so (e) 3 gives the hypotheses of Lems. 13.27 and 13.28 with

$$(C_{\mathrm{LH}}, c_{\mathrm{LH}}, q_{\mathrm{LH}}) = (C_{\mathsf{B}}, (2C_{\mathsf{B}})^{1/2}, q_{\mathsf{B}}),$$

and these lemmas, in which the theorems on limits are taken at g' itself, give (c) and (d) at g'. The lemmas with free constants are needed here: no single window, dilated or not, holds $I^{\mathrm{lab}}(g')$, whose ends have a quotient $\geq L$, while the quotient of the ends of $I(g)$ is at most L with the lower end open; so (e) 4 does not follow from (c), (d) applied to the sequence taken whole, and the only lattice input of (c), (d) is a bound expressed through (K_j, β_j) and ℓ_j. □

Lemma 13.29 (Mean actions are slopes of the pressure; both ends are taken; the plateau)**.** Let $a > 0$ and $\beta > 0$.

(i) For $\nu \in \mathcal{G}^{\mathrm{per}}_{a,\beta}$ the number $\mathsf{a}(\nu) = \nu(a_p)$ is the same for all plaquettes p, and $P(\beta+t) - P(\beta) \geq -3\mathsf{a}(\nu)t$ for all $t > -\beta$; so $-3\mathsf{a}(\nu) \in [P'(\beta-), P'(\beta+)]$.

(ii) There are $\nu_\pm \in \mathcal{G}^{\mathrm{per}}_{a,\beta}$ with $-3\mathsf{a}(\nu_\pm) = P'(\beta\pm)$; so $3\sup\{|\mathsf{a}(\rho) - \mathsf{a}(\nu)| : \rho, \nu \in \mathcal{G}^{\mathrm{per}}_{a,\beta}\} = J(\beta)$.

(iii) At $a = a_K$, for $(K, \beta) \in \mathfrak{d}(g)$ and $\nu \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$, the function e of Thm. AM (e) exists, takes its values ν-almost surely in $[\inf_\rho \mathsf{a}(\rho), \sup_\rho \mathsf{a}(\rho)]$, an interval of length $J(\beta)/3$, and $3a_K^{-4}\,\mathrm{Rie}_K(f)(\mathsf{e} - \mathsf{a}(\nu))$ is a conditional expectation of $\mathcal{O}_K(f) - \nu(\mathcal{O}_K(f))$ given $\mathcal{I}$, for every bounded f with compact support.

■

Sketch of proof. (i) The lattice isometries act transitively on the plaquettes, carry a_p to a_{rp}, and leave every periodic Gibbs state invariant (M9); so $\nu(a_p)$ is one number. The tangent inequality (M9) for the function $A = -(a_{p_{12}(0)} + a_{p_{13}(0)} + a_{p_{23}(0)})$ of the three plaquettes at one site, whose array over the periodic box of N sites a side is $-\sum_p a_p$, reads

$$\log Z_N(\beta + t) - \log Z_N(\beta) \geq -3\mathsf{a}(\nu)\, tN^3 - 144\,\beta N^2 - 72\,|t|N^2,$$

where Z_N is the partition function of that box, which does not depend on the spacing; divide by N^3 and let $N \to \infty$ (Lem. 13.11). (ii) Take points $\beta_k \downarrow \beta$ at which P is differentiable and $\nu_k \in \mathcal{G}^{\mathrm{per}}_{a,\beta_k}$, which exist by (M9); by (i), $-3\mathsf{a}(\nu_k) = P'(\beta_k)$; a subsequence converges weakly, its limit is a periodic Gibbs state at β by the closedness in (M9), and a_p being continuous, $-3\mathsf{a}(\nu_+) = \lim_k P'(\beta_k) = P'(\beta+)$; likewise $\beta_k \uparrow \beta$ gives ν_-. (iii) By Prop. YM3_I:106.3 (M9), at a first-passage datum the set $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ is a class at the cutoff K that obeys the bound system at g, and with ν it contains $k\nu/\nu(k)$ for every bounded $\mathcal{I}$-measurable $k \geq 0$ with $\nu(k) > 0$; with the invariance used in (i) this gives $\nu(ka_p) = \nu(ka_{p_0})$ for all plaquettes p, so one $\mathcal{I}$-measurable e serves all plaquettes, and the statements on conditional expectations given $\mathcal{I}$ (M9) place its values between the infimum and the supremum of the $\mathsf{a}(\rho)$, an interval of length $J(\beta)/3$ by (ii); linearity gives the conditional expectation of $\mathcal{O}_K(f)$, since $a_K^{-1}\sum_p f(x(p)) = 3a_K^{-4}\,\mathrm{Rie}_K(f)$. □

Proof of Thm. AM. Step 1: clause (a). The first bound is Prop. 13.23; $|D_K''|(I) \leq \mu_K(I) + C_0|I|$. For the convex $\Phi = D_K + \frac{1}{2}C_0 y^2$,

$$\Delta_s^2\Phi(b) = \int (s - |y - b|)_+ \,\mathrm{d}\mu_K(y) = \int_0^s \mu_K((b-u, b+u))\,\mathrm{d}u \leq C_\flat(s^2 + \delta s),$$

and $\Delta_s^2 D_K = \Delta_s^2\Phi - C_0 s^2 \geq -C_0 s^2$ because $\mu_K \geq 0$. Since K_1 does not increase, the bounds hold for every $\delta > \delta_0(K)$, hence at $\delta_0(K)$, and $\delta_0(K) \leq \delta$ once $K \geq K_1(\delta)$. For the tent $\phi_{\delta'}$, $\mathsf{p}''(b) = \int \phi_{\delta'}(b - y)\,\mathrm{d}\mu_K(y) - C_0$, and the layer-cake formula with the bound on the open intervals

$(b-\delta'(1-\delta' t), b+\delta'(1-\delta' t))$ gives $\int \phi_{\delta'}(b-y)\,\mathrm{d}\mu_K(y) \le C_\flat(1+\delta_0/\delta')$; and $\mathsf{p}(b) - D_K(b) = \int_0^{\delta'} \phi_{\delta'}(t)\Delta_t^2 D_K(b)\,\mathrm{d}t$ with $\int_0^{\delta'} \phi_{\delta'}(t)t^2\,\mathrm{d}t = \delta'^2/12$ and $\int_0^{\delta'} \phi_{\delta'}(t)t\,\mathrm{d}t = \delta'/6$.

Step 2: clause (b). $p_{K,\ell'}$ is real-analytic, so the atom of $\mu_K = (D_K + \frac12 C_0 y^2)''$ at b is the jump of $p'_{K,\infty}$ there, and $p_{K,\infty}(b) = L^{3K}P(L^K b)$ by Lem. 13.11:

$$\mu_K(\{b\}) = p'_{K,\infty}(b+) - p'_{K,\infty}(b-) = L^{4K}J(L^K b) \le C_\flat \delta.$$

For the limit of $\beta^4 J(\beta)$: by Lem. 13.24 (ii) every point of $[1,L]$ has a dilate $L^s b_\infty$ in the interior of a window $W^{(\iota)} \subset \operatorname{int} I(g^{(\iota)})$; finitely many of the sets $L^{-s_\iota}\operatorname{int} W^{(\iota)}$ cover $[1,L]$, each window with its Setting. Every $\beta \ge L^{1+\max s_\iota}$ is $L^{K'}y$ with $y \in W^{(\iota)}$ for some ι and an integer K' that tends to ∞ with β; so $\beta^4 J(\beta) = y^4 \mu^{(\iota)}_{K'}(\{y\}) \le (\max W^{(\iota)})^4 C_\flat^{(\iota)}\delta$ once $K' \ge K_1^{(\iota)}(\delta)$, for every δ.

Step 3: clause (c). Lem. 13.29 (i), (ii) at $a = a_K$, $\beta = L^K b$. With $e(\nu) = 3a_K^{-4}\mathsf{a}(\nu)$: $\nu(\mathcal{O}(f)) = a_K^{-1}\mathsf{a}(\nu)\sum_p f(x(p)) = \mathrm{Rie}_K(f)e(\nu)$, and the tangent inequality in the label reads

$$p_{K,\infty}(b') - p_{K,\infty}(b) = L^{3K}\big[P(L^K b') - P(L^K b)\big] \ge -3\mathsf{a}(\nu)L^{4K}(b'-b) = -(b'-b)e(\nu),$$

since $p'_{K,\infty}(b\pm) = L^{4K}P'(L^K b\pm)$; the end points are taken by $\nu_\pm$.

Step 4: clause (d). By (c), $\nu(x_c) - \nu'(x_c) = q^{-3}\,\mathrm{Rie}_K(f)(e(\nu)-e(\nu'))$, and for $0 \le f \le 1$ vanishing off a cube of side $\le q - a_K$, each of the three orientations contributes at most $(q/a_K)^3$ plaquettes, so $0 \le q^{-3}\,\mathrm{Rie}_K(f) \le 1$:

$$|\nu(x_c) - \nu'(x_c)| \le |e(\nu) - e(\nu')| \le J_K(b) = \mu_K(\{b\}).$$

Uniformity in $b \in W \subset W'$ follows from (b) at every δ.

Step 5: clause (e). Prop. YM3_I:100.1, applied to ν and the tuples (f), (h), bounds the difference between $S_2^T[\nu](f,\tau_v h)$ and $\nu(E_\nu[\mathcal{O}(f) - \nu(\mathcal{O}(f))\,|\,\mathcal{I}]\,E_\nu[\mathcal{O}(h) - \nu(\mathcal{O}(h))\,|\,\mathcal{I}])$ by the right member; by Lem. 13.29 (iii) the latter is $9a_K^{-8}\,\mathrm{Rie}_K(f)\,\mathrm{Rie}_K(h)\,\mathrm{Var}_\nu(\mathsf{e})$, and a random variable with values in an interval of length $J(\beta)/3$ has variance $\le \frac14(J(\beta)/3)^2$, so $9a_K^{-8}\,\mathrm{Var}_\nu(\mathsf{e}) \le \frac14(a_K^{-4}J(\beta))^2 = \frac14 J_K(b)^2$. As $-3\mathsf{a}(\nu_\pm) = P'(\beta\pm)$ and $P'(\beta+) \ge P'(\beta-)$, $\mathsf{a}(\nu_+) = \inf_\rho \mathsf{a}(\rho)$ and $\mathsf{a}(\nu_-) = \sup_\rho \mathsf{a}(\rho)$; the mean of e in a state ν is $\nu(a_p) = \mathsf{a}(\nu)$. So at ν_+ the function e is $\ge$ its mean, and at ν_- it is $\le$ its mean; in both cases the mean is an end of the interval, so e is almost surely constant. An ergodic ν makes the $\mathcal{I}$-measurable e constant. At $\nu = \frac12(\nu_+ + \nu_-)$, a periodic Gibbs state,

$$\mathrm{Cov}_\nu(F,F') = \tfrac12\,\mathrm{Cov}_{\nu_+}(F,F') + \tfrac12\,\mathrm{Cov}_{\nu_-}(F,F') + \tfrac14(\nu_+(F) - \nu_-(F))(\nu_+(F') - \nu_-(F'));$$

at $F = \mathcal{O}(f)$, $F' = \mathcal{O}(\tau_v f)$ with $\mathrm{Rie}_K(f) > 0$ the covariances in $\nu_\pm$ are $O(1/|v|)$ by the plateau at $\nu_\pm$, $\nu_+(F) - \nu_-(F) = -a_K^{-4}J(\beta)\,\mathrm{Rie}_K(f)$, and $|v| \to \infty$ gives the equality.

Step 6: clause (f). (M14) is taken with what the clause supposes, which is what (M14) asks: two frames at one radius with the block sizes L_1 and $L_2 = L_1 + 2$, Thms. YM3_I:AJ and YM3_I:AD granted at both, Thm. YM3_I:D at the choices of its constants that they determine and at a lowered choice of each, and, in each frame i, the rate of the far part, that is, numbers $\mathsf{k}_i \ge 0$ and $\mathsf{c}_{\mathrm{y},i} < \infty$, free of m, K and b, with $|\mathsf{y}^{(i)}_{m,K'}(b) - \mathsf{y}^{(i)}_{m,K}(b)| \le \mathsf{c}_{\mathrm{y},i}\,\ell^9 K^{\mathsf{p}_{2,i}} L_i^{-K/10}$ for all integers $m \ge 1$, $\ell = 2L_i^m$, all $b \in \mathcal{W}_i$ and all integers $K' \ge K \ge \mathsf{k}_i + 54m$.

The power of (M14) is derived, in §12.1 on the latent heat of the SU(2) Wilson model on $\mathbb{Z}^3$, from these data, the rate among them; so (M14) gives for every real $A > 90$ numbers C_r and β_1 with $\beta^4 J(\beta) \le \mathsf{C}_\mathrm{r}\beta^{-1/(2A)}$ for $\beta \ge \beta_1$. That proof, measured on the physical scale, compares the specific heats of the Wilson model at the block sizes L_1 and $L_2 = L_1 + 2$, whose logarithms are rationally independent (Lem. 13.5), through the exact finite-size bounds, a common near part and the far parts $\mathsf{y}^{(i)}_{m,K}$, as in Prop. 13.23 made quantitative. The rate enters at the cutoffs $K \ge \mathsf{k}_i + Am$: there $\ell^9 = 2^9 L_i^{9m} \le 2^9 L_i^{9K/A}$, so that for $K' \ge K$

$$\sup_{b\in\mathcal{W}_i} |\mathsf{y}^{(i)}_{m,K'}(b) - \mathsf{y}^{(i)}_{m,K}(b)| \le 2^9 \mathsf{c}_{\mathrm{y},i} \sup_{k\ge1} k^{\mathsf{p}_{2,i}} L_i^{-(\frac{1}{10} - \frac{9}{A})k},$$

a number free of m and K, finite as $A > 90$. With $A = 1/(2\alpha)$ for $0 < \alpha < 1/180$, and $J \le 6$ (the difference quotients of P lie in $[-6, 0]$, since $0 \le \sum_p a_p \le 6$ per site), $\mathsf{c}_*(\alpha) < \infty$; for $\alpha = 0$ use $\alpha = 1/182$. On $\mathfrak{d}(g)$ the label is $b = a_K \beta \ge g^{-2}$ by (M5), so $\beta \ge 1$ and

$$a_K^{-4} J(\beta) = b^{-4} \beta^4 J(\beta) \le b^{-4} \mathsf{c}_*(\alpha) \beta^{-\alpha} = \mathsf{c}_*(\alpha)\, b^{-4-\alpha} L^{-\alpha K} \le g^{8+2\alpha} \mathsf{c}_*(\alpha) L^{-\alpha K}.$$

By (c) at b, $3a_K^{-4} |\rho(a_p) - \nu(a_p)| = |e(\rho) - e(\nu)| \le a_K^{-4} J(\beta)$; for f with support in $[-R, R]^3$ each orientation has at most $(2R/a_K + 1)^3$ barycentres there, which gives the bound for $\mathcal{O}_K(f)$; and (d) gives the bound for the cube means. □

Proof of Thm. AN. Step 1: clause (a) and the limit at the label. By the passage, $(K, L^K \mathsf{b}) \in \mathfrak{d}(g)$ for $K \ge 1$, so $(K, L^K \mathsf{b}, \nu_K)_{K \ge \max\{K_\circ, 1\}}$ is a Gibbs sequence at g, with the constant label b. By (M13) (ii) and (v), ν_K is the one periodic Gibbs state at $(K, L^K \mathsf{b})$, the sequence converges, its limit σ is its own single component with $\ker H = \mathbb{C}\Omega$, and the spectrum of H lies in $\{0\} \cup [m_*, \infty)$; σ is invariant under $\mathbb{R}^3 \rtimes \mathrm{O}(3)$, as every limit of a convergent Gibbs sequence (M10). We use two terms. A *limit at the label* b is a limit, for $n \ge 2$ and $X \in \mathfrak{S}_n(\mathbb{R}^3)$, of $S^T_{n;K,\ell,L^K\mathsf{b}}(X^{\langle \ell \rangle})$ as K and $\ell = 2L^m$ tend to infinity, together or in either order, the bare coupling being $L^K \mathsf{b}$ at every cutoff K.

A *continuum limit along growing tori at the label* b is a continuum limit at g along growing tori (Def. 13.4 (5)) whose defining sequence (K_j, ℓ_j, β_j) has $\beta_j = L^{K_j} \mathsf{b}$, that is, all its labels equal to b, from an index on. Let $X = (f_1, \dots, f_n) \in \mathfrak{S}_n(\mathbb{R}^3)$, $n \ge 2$ (an entry 0 makes both sides 0), and take a side $\ell_1 = 2L^{m_1} \ge \ell_\circ$ with the supports in $(-\ell_1/4, \ell_1/4)^3$. By (M13) (iv), for all $K \ge \max\{K_\circ, 1\}$ and all sides $\ell \ge \ell_1$,

$$|S^T_{n;K,\ell,L^K\mathsf{b}}(X^{\langle \ell \rangle}) - S^T_n[\nu_K](X)| \le C(X)\, \theta_\circ^{\ell/(2\ell_\circ)},$$

a bound free of K, while $S^T_n[\nu_K](X) \to S^\sigma_n(X)$ as $K \to \infty$, the Gibbs sequence converging. So $S^T_{n;K,\ell,L^K\mathsf{b}}(X^{\langle \ell \rangle}) \to S^\sigma_n(X)$ when K and ℓ tend to infinity together; in the order ℓ first the inner limit is $S^T_n[\nu_K](X)$; in the order K first the inner limit exists at every fixed side $\ell \ge \ell_1$ by (M8), the supports of the periodised functions being disjoint, and it lies within $C(X)\theta_\circ^{\ell/(2\ell_\circ)}$ of $S^\sigma_n(X)$. Along a defining sequence with $\beta_j = L^{K_j}\mathsf{b}$ from an index on, $K_j \to \infty$ and $\ell_j \to \infty$, the same bound gives $S_n(X) = \lim_j S^T_{n;K_j,\ell_j,\beta_j}(X^{\langle \ell_j \rangle}) = S^\sigma_n(X)$ for $n \ge 2$; as σ_S and σ are formed from their connected functions with the one-point functions 0 ($S_1 = 0$, and $\sigma((f)) = 0$ by Def. 13.4 (2)), $\sigma_S = \sigma$.

So every limit at the label b, in any order of cutoff and volume, and every continuum limit along growing tori at the label b is σ. The class $[\varnothing]$ of the empty tuple has norm $\sigma(\varnothing)^{1/2} = 1$ and is fixed by every e^{-tH}, so it spans $\ker H$; take $\Omega = [\varnothing]$. For $X, Y \in \mathfrak{S}(\mathbb{E}_+)$, $\langle [X], \Omega \rangle = \sigma(\theta X) = \sigma(X)$ by the invariance, and P_Ω commutes with H, so

$$\sigma(\theta X \cdot \tau_{te_3} Y) - \sigma(X)\sigma(Y) = \big\langle (1 - P_\Omega)[X], e^{-tH}(1 - P_\Omega)[Y] \big\rangle,$$

and $\|e^{-tH}(1 - P_\Omega)\| \le e^{-m_* t}$ by the spectral theorem; the rotations carry this to every direction. By Thm. AL (e) 4 at g, clause (c) of that theorem applies to σ: $\sigma((f)) = 0$, the kernel $\mathbf{s} \ge 0$ does not increase, and

$$4\pi r\, \mathbf{s}(r) = \int e^{-mr}\, \mathrm{d}\rho(m), \qquad \rho \ge 0, \qquad \rho(\{0\}) = 0.$$

Step 2: the gap passes to the kernel. For real $f \in C^1_c(\mathbb{E}_+)$ the vector ψ_f is orthogonal to Ω, as $\langle \Omega, \psi_f \rangle = \sigma((f)) = 0$; so by the gap $\sigma(\theta f, \tau_{te_3} f) = \langle \psi_f, e^{-tH} \psi_f \rangle \le e^{-m_* t} \|\psi_f\|^2$. With $f_\epsilon(x) = \eta_\epsilon(x_3)\mathsf{g}(x_1, x_2)$ as in the proof of Prop. 13.25, this reads $c_\epsilon(t) \le e^{-m_* t} c_\epsilon(0)$; letting $\epsilon \to 0$ and then g tend to a point mass,

$$\mathbf{s}(2s + t) \le e^{-m_* t}\, \mathbf{s}(2s) \qquad (s > 0,\ t \ge 0),$$

that is $\mathbf{s}(r) \le e^{-m_*(r - r')} \mathbf{s}(r')$ for $0 < r' \le r$.

Step 3: no spectral mass below the gap. Fix $r' > 0$. For $0 \le m < m_*$ and $r \ge r'$, $\int e^{-m''r}\,\mathrm{d}\rho(m'') \ge e^{-mr}\rho([0,m])$, so

$$\rho([0,m]) \le e^{mr}\,4\pi r\,\mathbf{s}(r) \le 4\pi r\,e^{-(m_*-m)r}e^{m_*r'}\mathbf{s}(r') \to 0 \qquad (r\to\infty),$$

and $\rho([0,m_*)) = 0$.

Step 4: the sharper decay. As ρ is carried by $[m_*,\infty)$, for $0 < r' \le r$

$$4\pi r\,\mathbf{s}(r) = \int_{[m_*,\infty)} e^{-m''r}\,\mathrm{d}\rho(m'') \le e^{-m_*(r-r')}\int e^{-m''r'}\,\mathrm{d}\rho(m'') = e^{-m_*(r-r')}\,4\pi r'\mathbf{s}(r');$$

so $\mathbf{s}(r) = O(r^{-3})$, and $\rho([0,m]) = 0$ for $m < m_*$ gives $\rho([0,m]) = O(m^2)$.

Step 5: cube means in the limit. Profiles of distinct cells have supports at distance $\ge T/4$; as $\sigma((f)) = 0$, $S_2^\sigma = \sigma$ on pairs, and with $0 \le f_c, f_{c'} \le 1$ and $\int f_c \le q^3$,

$$\bigl|S_2^\sigma(q^{-3}f_c, q^{-3}f_{c'})\bigr| \le q^{-6}\int f_c(x)\int_{|y-x|\ge T/4}\mathbf{s}(|x-y|)\,\mathrm{d}y\,\mathrm{d}x \le q^{-3}\,4\pi\int_{T/4}^\infty r^2\mathbf{s}(r)\,\mathrm{d}r,$$

which is finite by Step 4 and the continuity of $\mathbf{s}$ on $[T/4,\infty)$. The cells $\square_c$ and $\square_{c'}$ are centred at $o+qc$ and $o+qc'$, $c,c'\in\mathbb{Z}^3$, as in clause (c); let $\mathbf{d} = c'-c$ be the difference of their indices, as in the proof of Prop. 13.25 (d). If the two cells have no common point, that is $|\mathbf{d}|_\infty \ge 2$, then the cells are at sup-distance $q(|\mathbf{d}|_\infty - 1) \ge q$, each support lies at sup-distance $\ge T/8$ from the boundary of its cell, so the supports are at distance $\ge q + T/4$, and as $\mathbf{s}$ does not increase the left member is at most $q^{-6}\|f_c\|_1\|f_{c'}\|_1\mathbf{s}(q) \le \mathbf{s}(q)$, which by Step 4 is at most $(r'/q)\,e^{-m_*(q-r')}\mathbf{s}(r')$ for $0 < r' \le q$. For the 26 cells with $|\mathbf{d}|_\infty = 1$, which share a face, an edge or a corner with the first, the bound is $\chi_\circ q^{-3}$ of the display. □

Lemma 13.30 (A derivative from the values on a diameter)**.** Let u be holomorphic on $|z| < r$ with $|\mathsf{u}| \le M$ there, and $|\mathsf{u}(x)| \le o$ for real $|x| \le r/2$, where $o > 0$. Then $|\mathsf{u}'(0)| \le (4o/r)[1 + \log_2^+(M/o)]$. ■

Sketch of proof. Write $\mathsf{u}(z) = \sum_m c_m z^m$ with $|c_m| \le Mr^{-m}$. If $o \ge M$, $|\mathsf{u}'(0)| \le M/r \le o/r$. Otherwise let $N = \lceil \log_2(M/o)\rceil \ge 1$ and $Q(x) = \sum_{m\le N} c_m x^m$; for real $|x| \le r/2$, $|\mathsf{u}(x) - Q(x)| \le M\sum_{m>N} 2^{-m} \le o$, so $|Q| \le 2o$ there. Bernstein's inequality for a real polynomial $Q_1(t) = \sum_{k\le N} c_k' t^k$ with $|Q_1| \le 1$ on $[-1,1]$ gives $|Q_1'(0)| \le N$; applied to $Q_1(t) = \mathrm{Re}[e^{i\varphi}Q(rt/2)]/(2o)$ with the phase φ that makes $e^{i\varphi}Q'(0) \ge 0$, it gives $|\mathsf{u}'(0)| = |Q'(0)| \le 4oN/r$. □

The comparison of two cutoffs stops before the terminal scale. We therefore need the vacuum constant of the steps already performed, with the stopping index retained in the two-torus estimate.

Lemma 13.31 (Vacuum constants before the terminal scale)**.** Fix the common construction choice and the real regular progression of Thm. YM3_I:Y for one admissible sequence $x_0,\dots,x_K$. Let T_{ℓ_1} and T_{ℓ_2} be admitted tori for this sequence, where $\ell_i = 2L^{m_i}$, and assume

$$LMR(x_K) \le L^{m_i}, \qquad \ell_i \ge C_L M, \qquad \kappa \ge 2\log 9216 + 5, \qquad \delta_N M \ge \kappa, \qquad E_0 \ge 2E_N.$$

Here C_L is the room constant of Prop. YM3_I:74.5, and E_N, δ_N are the normalizer constants of equations YM3_I:(1399)–(1401). At step $j+1$ put

$$\mathsf{vl}^{(j+1)}(X,z) = E^{(j+1)}(X,\mathbf{1},z), \qquad \mathsf{vn}^{(j+1)}(X,z) = \mathsf{n}^{(j+1)}(X,z).$$

The second family is the local Gaussian constant in equation YM3_I:(1399). Both families are real and bounded by $E_0 e^{-\kappa d_{j+1}(X)}$. With the extensive term $\mathsf{N}^{(j)}$ of equations YM3_I:(1402), (1405),

define

$$\begin{aligned}\mathsf{E}^{(j)}(T_\ell,T_\ell)&=\mathsf{N}^{(j)}(T_\ell,T_\ell)+\sum_X\sum_{z\in X}\bigl(\mathsf{vl}^{(j+1)}(X,z)+\mathsf{vn}^{(j+1)}(X,z)\bigr),\\ E_{(n)}[T_\ell]&=\sum_{j=0}^{n-1}\mathsf{E}^{(j)}(T_\ell,T_\ell),\qquad 0\le n\le K.\end{aligned}\tag{13.i}$$

The domains and sites in the first sum have level $j+1$. These are the constants in equations YM3_I:(1406)–(1407): on a whole torus there is no boundary layer to remove from the set of base points.

The normalizing factor of each real step satisfies

$$\log\bigl(\mathsf{pf}_j(x_j)\mathsf{Z}_{\mathbf{1}}^{(j)}\bigr)=\mathsf{N}^{(j)}(T_\ell,T_\ell)+\sum_X\sum_{z\in X}\mathsf{vn}^{(j+1)}(X,z).\tag{13.ii}$$

Consequently the constant accumulated by the zero-source chain through step n is $E_{(n)}[T_\ell]$.

Put $\ell_*=\min(\ell_1,\ell_2)$ and $C_{\mathrm{tt}}=4096\sum_{q\ge0}e^{-q}$, as in Prop. YM3_I:74.5. Then

$$\left|\frac{E_{(n)}[T_{\ell_1}]}{\ell_1^3}-\frac{E_{(n)}[T_{\ell_2}]}{\ell_2^3}\right|\le 2C_{\mathrm{tt}}E_0\sum_{s=K-n}^{K-1}L^{3s}e^{-\kappa\ell_*L^s/(5M)}.\tag{13.iii}$$

The sum is empty when $n=0$. For every integer $u\ge0$,

$$\sum_{s\ge u}L^{3s}e^{-\alpha L^s}\le 2L^{3u}e^{-\alpha L^u},\qquad \alpha=\frac{\kappa\ell_*}{5M}>4.\tag{13.iv}$$

■

Proof. We first identify the constant of one step. Write $N_j=\#\mathbb{T}^{(j)}$, $D=N_j-N_{j+1}$ and $\nu=\frac{3}{2}\log(2\pi)$. There are $3D$ positive nonpivot bonds and $3N_{j+1}$ pivot bonds. At the trivial triple the fluctuation matrix $\mathsf{A}_{\mathbf{1}}$ is positive on a real space of dimension $9D$, and the elimination map at every pivot is $L^{-3}\mathrm{id}$ on the three-dimensional colour space. The real Gaussian integral and these pivot determinants give

$$\log\mathsf{Z}_{\mathbf{1}}^{(j)}=27N_{j+1}\log L+3D\nu-\tfrac{1}{2}\log\det\mathsf{A}_{\mathbf{1}}.$$

The Haar prefactor of the step, at $t=x_j$, is

$$\log\mathsf{pf}_j(t)=-D\log\mathsf{z}(t,\varepsilon(t))+3D\bigl(\log\sigma_{\mathrm{H}}-\tfrac{3}{2}\log t\bigr)-9N_{j+1}\log L.$$

Since $D=(L^3-1)N_{j+1}$, the definition of $\mathsf{N}^{(j)}$ yields

$$\log\bigl(\mathsf{pf}_j(t)\mathsf{Z}_{\mathbf{1}}^{(j)}\bigr)=\mathsf{N}^{(j)}(T_\ell,T_\ell)+18N_{j+1}\log L-\tfrac{1}{2}\log\det\mathsf{A}_{\mathbf{1}}.$$

It remains to identify the last two terms with the local Gaussian constants. The logarithm series defining the local normalizer in the proof of Prop. YM3_I:50.2 has $T=1-c_N^{-1}\mathsf{A}_{\mathbf{1}}$, with $\|T\|<1$. Adding its diagonal terms over the nonpivot bonds gives

$$\sum_b f_b(T_\ell)=9D\log c_N-\sum_{q\ge1}\frac{1}{q}\operatorname{Tr}T^q=\log\det\mathsf{A}_{\mathbf{1}}.$$

The series is absolutely convergent. The last equality follows by applying $-\sum_{q\ge1}(1-v)^q/q=\log v$ to each positive eigenvalue v of $c_N^{-1}\mathsf{A}_{\mathbf{1}}$, which lies in $(0,2)$. Summing the local finite differences of f_b recovers $f_b(T_\ell)$, and the tents sum to one for each bond. Thus the nonpivot constants sum to $-\frac{1}{2}\log\det\mathsf{A}_{\mathbf{1}}$. Each of the $3N_{j+1}$ pivot bonds contributes $6\log L$, by equation YM3_I:(1399). This proves eq. 13.ii. The chain identity in Prop. YM3_I:114.2 adds to this logarithm the values $E^{(j+1)}(X,\mathbf{1},z)$ of the new local terms. Summing over the steps proves the assertion about its accumulated constant.

For the comparison, we retain the level in the proof of Prop. YM3_I:74.5. At level j, the localization cubes have side ML^{j-K}. A domain is common if it has a lift contained in an axis-parallel box of side less than $\ell_*/2$. The regular families have the same value on these lifts by Thm. YM3_I:Y. The Gaussian constants have the same property: the local logarithm series, pivot bond and tent are transported by the same chart, as in the proof following equation YM3_I:(1400). Thus both families define one function over the common lifts.

The sum of this function over a period of either torus, divided by its volume, has the same value. Indeed, the larger side is an integer multiple of the smaller one; a period of the larger torus contains that multiple cubed of the lifts of each site of the smaller torus. The extensive term also cancels, since

$$\ell_i^{-3}\mathsf{N}^{(j-1)}(T_{\ell_i},T_{\ell_i}) = L^{3(K-j)}\left[3(L^3-1)(\mathsf{co}^{\mathrm{b}}_{j-1}+\mathsf{co}^{\mathrm{i}}_{j-1})+\mathsf{co}^{\mathrm{s}}_{j-1}\right]$$

depends only on the common sequence. Therefore, writing $\mathsf{S}_i^{(j),\mathrm{nc}}$ for the sum of the two local families over the non-common domains on T_{ℓ_i}, we have

$$\frac{E_{(n)}[T_{\ell_1}]}{\ell_1^3}-\frac{E_{(n)}[T_{\ell_2}]}{\ell_2^3}=\sum_{j=1}^{n}\left(\ell_1^{-3}\mathsf{S}_1^{(j),\mathrm{nc}}-\ell_2^{-3}\mathsf{S}_2^{(j),\mathrm{nc}}\right).$$

The lifting argument in that proof gives $d_j(X)\ge \ell_* L^{K-j}/(2M)-2$ for each non-common domain. For a fixed site z, its domain-counting estimate is

$$\sum_{X\ni z\,;\ d_j(X)\ge D_0} e^{-\kappa d_j(X)} \le 2048\sum_{q\ge D_0} e^{-(\kappa-\log 9216)q}\le \frac{C_{\mathrm{tt}}}{2}e^{-\kappa D_0/2}.$$

The hypotheses give $t_j=\ell_* L^{K-j}/M\ge 2L^2\ge 338$, so $\frac12(t_j/2-2)\ge t_j/5$. Multiplying the last bound by $2E_0$ and by the number $\ell_i^3 L^{3(K-j)}$ of level-j sites gives

$$\ell_i^{-3}|\mathsf{S}_i^{(j),\mathrm{nc}}|\le C_{\mathrm{tt}}E_0 L^{3(K-j)}e^{-\kappa\ell_* L^{K-j}/(5M)}.$$

Sum only over $1\le j\le n$ and put $s=K-j$ to obtain eq. 13.iii. Finally the ratio of successive terms in eq. 13.iv is

$$\exp\bigl(3\log L-\alpha L^s(L-1)\bigr)\le \tfrac12,$$

because $L\ge 13$, $\alpha\ge 4$ and $\log L\le L-1$. Summing this geometric majorant from $s=u$ proves the tail bound. □

Proof of Prop. 13.7. Step 1: the identity. The translations by $a_K\mathbb{Z}^3$ and the permutations of the axes act on T_λ, preserve the weight and the Haar measure and are transitive on the plaquettes, so $\langle a_p\rangle_{K,\lambda,L^K b}=\mathsf{a}_\lambda$ is one number; there are $3(\lambda/a_K)^3$ plaquettes, and $p'_{K,\lambda}(b)=-\lambda^{-3}L^K\langle\sum_p a_p\rangle = -3a_K^{-4}\mathsf{a}_\lambda$. As each plaquette of $\mathbb{L}_K$ has exactly one image on the torus, $\sum_{p\subset T_\lambda} f^{\langle\lambda\rangle}(x(p))=\sum_{p\subset\mathbb{L}_K} f(x(p))$, so

$$\langle\mathcal{O}(f^{\langle\lambda\rangle})\rangle_{K,\lambda,L^K b}=a_K^{-1}\mathsf{a}_\lambda\sum_p f(x(p))=-\operatorname{Rie}_K(f)\,p'_{K,\lambda}(b);$$

subtract the two sides; for the tent χ_0 of half-width $1/L$ about a site, $\operatorname{Rie}_K(\chi_0)=L^{-3}$.

Step 2: the bound. By the passage both laws are at a first-passage datum, and the transfer (M7) at $n=1$, for f with support in a ball of radius 2, gives $|A_K(f;b)|\le 2\log 2\,(2\mathsf{Q}_1+2)\varrho^{-1}\|f\|_\sharp$; at $f=\chi_0$, $\|\chi_0\|_\sharp\le \mathsf{c}_\chi$. For a cube mean, $0\le q^{-3}\operatorname{Rie}_K(f)\le 1$ as in the proof of Thm. AM (d).

Step 3: the read-out at the order 0 *on one torus.* Let $\lambda\in\{\ell,\ell'\}$, $b\in\mathsf{J}$ and $K\ge K_{\mathrm{bf}}(\lambda,\mathsf{J})$. The theorem on two consecutive cutoffs (Thm. YM3_I:AR) compares on T_λ, at the label b, the run A at the cutoff $K^{\mathrm{A}}=K$ with the run B at the cutoff $K^{\mathrm{B}}=K+1$; the mark $\bullet$ stands for A or B. Its objects are taken at the order 1, the tuple being any one real Lipschitz function on T_λ that is not 0, and at the source 0, where no object reads that function (§2.3.1 and Section YM3_I:116); its conditions hold at every such K (Cor. 2.8 (a)).

The objects are: the depth $u = \lceil K/1260 \rceil$ and the level $t_1^\bullet = K^\bullet - u$ of $\bullet$ at this depth; the space of the configurations at the depth u, one space for both runs, with its Haar measure dV and its inner ball In, an open set that holds the trivial configuration $\mathbf{1}$; the chain of $\bullet$, with its exponent $\Phi^\bullet$ on In, the sum $c^\bullet(0)$ of the constants split off along it at the steps $1, \dots, t_1^\bullet$, the integral $I^\bullet = \int_{\mathsf{In}} e^{\Phi^\bullet} dV$ and the mass $T^\bullet = e^{c^\bullet(0)} I^\bullet$; and the classical start of $\bullet$ at the depth $\lceil K/9 \rceil$, with its exponent $\hat\Phi^\bullet$ on In, the sum $\hat c^\bullet(0)$ of its constants, $\hat I^\bullet = \int_{\mathsf{In}} e^{\hat\Phi^\bullet} dV$, its measure $\hat\nu^\bullet = e^{\hat c^\bullet(0)} e^{\hat\Phi^\bullet} dV$ on In and its mass $\hat T^\bullet = \hat\nu^\bullet(\mathsf{In}) = e^{\hat c^\bullet(0)} \hat I^\bullet$ (the form of the theorem, the proof of Prop. YM3_I:116.1).

The small numbers $\epsilon_{\mathrm{fg}} \le \frac{1}{8}$, $\epsilon_{\mathrm{cl}} \le \frac{1}{20}$, $\epsilon_{\mathrm{sh}} \le \frac{1}{20}$ and $\mathsf{R}_{t_1} \le \frac{1}{2}$ are those of §2.3.1 on T_λ, and $\epsilon_3 = 4\epsilon_{\mathrm{cl}} + 6\epsilon_{\mathrm{sh}}$. We write $\mathsf{R}_K[T_\lambda]$ for the number $\mathfrak{o}_K$ of (M12) on the torus T_λ (Def. 2.7 (4)); with its bound of Cor. 2.8 (a),

$$\mathsf{R}_K[T_\lambda] = \mathfrak{o}_K = 8\mathsf{R}_{t_1} + 12\epsilon_{\mathrm{fg}} + 8\epsilon_{\mathrm{cl}} + 12\epsilon_{\mathrm{sh}} \le \mathsf{C}_{\mathrm{bf}} \lambda^3 K^{\mathsf{p}_0} L^{-K/10}.$$

The links (i)–(iv) below are the four statements of Subsection 13.1.4 that the proof of clause (c) uses, each derived here from the cited construction estimates. The links (i)–(iii) give

$$\left|\log Z_{K+1,\lambda,L^{K+1}b} - \log Z_{K,\lambda,L^K b} - \mathrm{Re}(c^{\mathrm{B}} - c^{\mathrm{A}})(0)\right| \le \mathsf{R}_K[T_\lambda] \le \mathsf{C}_{\mathrm{bf}} \lambda^3 K^{\mathsf{p}_0} L^{-K/10},$$

so that $\mathsf{R}_K[T_\lambda]$ bounds the error of the read-out at the order 0 on the torus T_λ; the link (iv) identifies $c^\bullet(0)$ for Step 4.

(i) *The read-out at the order* 0, from the discarding estimate, the form and the data of the read-out at $z = 0$. With $Z^\bullet = Z_{K^\bullet,\lambda,L^{K^\bullet}b}$, the discarding estimate (Prop. YM3_I:115.2, at the level $t_1^\bullet$) gives $(1 - \mathsf{R}_{t_1}) Z^\bullet \le T^\bullet \le Z^\bullet$. The data of the read-out at $z = 0$ (Section YM3_I:116; the proofs of Props. YM3_I:116.1 and YM3_I:116.3) give $|I^\bullet/\hat I^\bullet - 1| \le 2\epsilon_{\mathrm{fg}}$, and $\hat T^{\mathrm{A}} > 0$, $\hat T^{\mathrm{B}} > 0$ with $|\hat T^{\mathrm{B}}/\hat T^{\mathrm{A}} - 1| \le \epsilon_3 \le \frac{1}{2}$. As $T^\bullet = e^{(c^\bullet - \hat c^\bullet)(0)} (I^\bullet/\hat I^\bullet) \hat T^\bullet$ with $T^\bullet > 0$ and $\hat T^\bullet > 0$, moduli and logarithms, with $-\log(1-t) \le 2t \log 2$ for $0 \le t \le \frac{1}{2}$ and $\log \frac{1+y}{1-y} \le \frac{32}{15} y$ for $0 \le y \le \frac{1}{4}$, give

$$\left|\log \frac{Z^{\mathrm{B}}}{Z^{\mathrm{A}}} - \mathrm{Re}\big[(c^{\mathrm{B}} - \hat c^{\mathrm{B}}) - (c^{\mathrm{A}} - \hat c^{\mathrm{A}})\big](0)\right| \le 1.39\,\mathsf{R}_{t_1} + 4.27\,\epsilon_{\mathrm{fg}} + 1.39\,\epsilon_3 = 1.39\,\mathsf{R}_{t_1} + 4.27\,\epsilon_{\mathrm{fg}} + 5.56\,\epsilon_{\mathrm{cl}} + 8.34\,\epsilon_{\mathrm{sh}}.$$

(ii) *The two classical starts are comparable as measures on the inner ball*, from the proof of the comparison of their masses. Let Y run through the configurations at the depth $\lceil K/9 \rceil$, with the Haar measure dY, Y_{t_1} being the block average of Y at the depth u, and let $\mathsf{k}^\bullet \ge 0$ and $\hat F^\bullet \ge 0$ be the weights and the starts of Section YM3_I:116 at the source 0, so that $\int_{\mathsf{In}} H \, \mathrm{d}\hat\nu^\bullet = \int H(Y_{t_1})\, \mathsf{k}^\bullet \hat F^\bullet \, \mathrm{d}Y$ for every bounded Borel function H on the configurations at the depth u (§2.3.2). The positive-kernel argument in the proof of Prop. YM3_I:116.1 runs with a bounded Borel function $H \ge 0$ of the block field at the depth u as a further factor, since its kernel inequalities hold against every bounded nonnegative Borel function; it gives

$$\int H(Y_{t_1})\, \mathsf{k}^\bullet \mathbf{1}\{\mathsf{k}^\circ = 0\}\, \hat F^\bullet \, \mathrm{d}Y \le \epsilon_{\mathrm{sh}} \int_{\mathsf{In}} H \, \mathrm{d}\hat\nu^\bullet \qquad (\{\bullet, \circ\} = \{\mathrm{A}, \mathrm{B}\}).$$

Where both weights are positive they are equal and

$$\hat F^{\mathrm{B}}/\hat F^{\mathrm{A}} \in [e^{-\epsilon_{\mathrm{cl}}}, e^{\epsilon_{\mathrm{cl}}}]$$

(Prop. YM3_I:116.1). Dropping from $\int_{\mathsf{In}} H \, \mathrm{d}\hat\nu^{\mathrm{B}}$ the part where $\mathsf{k}^{\mathrm{A}} = 0$, comparing the starts where both weights are positive, and using the display for $(\bullet, \circ) = (\mathrm{A}, \mathrm{B})$, and then the same with A and B exchanged, one obtains for every bounded Borel $H \ge 0$

$$e^{-\epsilon_{\mathrm{meas}}} \int_{\mathsf{In}} H \, \mathrm{d}\hat\nu^{\mathrm{A}} \le \int_{\mathsf{In}} H \, \mathrm{d}\hat\nu^{\mathrm{B}} \le e^{\epsilon_{\mathrm{meas}}} \int_{\mathsf{In}} H \, \mathrm{d}\hat\nu^{\mathrm{A}},$$
$$\epsilon_{\mathrm{meas}} = \epsilon_{\mathrm{cl}} - \log(1 - \epsilon_{\mathrm{sh}}) \le \epsilon_{\mathrm{cl}} + 1.06\,\epsilon_{\mathrm{sh}}.$$

At the real source 0 the density $\mathsf{D}^\bullet = e^{\hat c^\bullet(0)} e^{\hat\Phi^\bullet}$ of $\hat\nu^\bullet$ is positive at every point of In (the positive classical density in the proof of Prop. YM3_I:116.1), so $\log \mathsf{D}^\bullet = \operatorname{Re}\hat c^\bullet(0) + \operatorname{Re}\hat\Phi^\bullet$; taking for H the indicator functions of the sets where $\mathsf{D}^{\mathrm B} < e^{-\epsilon_{\mathrm{meas}}}\mathsf{D}^{\mathrm A}$ or $\mathsf{D}^{\mathrm A} < e^{-\epsilon_{\mathrm{meas}}}\mathsf{D}^{\mathrm B}$, the display gives

$$\big|[\operatorname{Re}\hat c^{\mathrm B}(0) + \operatorname{Re}\hat\Phi^{\mathrm B}] - [\operatorname{Re}\hat c^{\mathrm A}(0) + \operatorname{Re}\hat\Phi^{\mathrm A}]\big| \le \epsilon_{\mathrm{meas}} \qquad \text{Haar-almost everywhere on } \mathsf{In}.$$

(iii) *The constants of the two classical starts cancel*, from the value 0 and the continuity of the exponent of the classical start at the trivial configuration. At the source 0 the exponent $\hat\Phi^\bullet(V)$ of the classical start of $\bullet$ at the level $t_1^\bullet$ is the action of its bare array at a background $U(V)$ over V less the Wilson action at that background weighted by the coupling of the level (Prop. YM3_I:114.2). It vanishes at $V = \mathbf 1$ and is continuous there. Indeed the trivial configuration of the bare lattice has the action 0 and the block average $\mathbf 1$, so it lies on the minimal orbit over $\mathbf 1$, on which both terms take one value (the gauge-invariance argument preceding equation YM3_I:(2133), Props. YM3_I:114.2 and YM3_I:113.13); the weighted action vanishes at it, and so does the action of the bare array, every member of which enters less its value at the trivial pair (equation YM3_I:(874)).

The actual analytic-background argument in the proof of Prop. YM3_I:52.3 supplies a continuous representative $\Upsilon(V)$ near $\mathbf 1$. The weighted action is the finite sum $\eta_t^{-1}\sum_p c(p)a_p(\Upsilon(V))$ of continuous plaquette functions; gauge invariance identifies this sum with the action of the background. Thus it is continuous at $\mathbf 1$. The action of the array is continuous there as well: its members are local terms, holomorphic in the pair that they read (Lem. 5.26), the pair being a polynomial map of the configuration (§5.2, YM3_I:(70), YM3_I:(69)); so the action of the array is continuous at $\mathbf 1$ along the continuous choice of Thm. YM3_I:E and its real-analytic continuation. So for every $\delta > 0$ there is an open set $\mathcal N \subset \mathsf{In}$ with $\mathbf 1 \in \mathcal N$ on which $|\operatorname{Re}\hat\Phi^{\mathrm A}| \le \delta$ and $|\operatorname{Re}\hat\Phi^{\mathrm B}| \le \delta$.

The comparison (ii) holds Haar-almost everywhere, so the argument reads the open set $\mathcal N$, which has positive Haar measure: at a point of $\mathcal N$ off the null set of (ii), $|\operatorname{Re}\hat c^{\mathrm B}(0) - \operatorname{Re}\hat c^{\mathrm A}(0)| \le \epsilon_{\mathrm{meas}} + 2\delta$. As $\delta > 0$ is arbitrary, $|\operatorname{Re}(\hat c^{\mathrm B} - \hat c^{\mathrm A})(0)| \le \epsilon_{\mathrm{meas}}$, and with (i)

$$\big|\log Z^{\mathrm B} - \log Z^{\mathrm A} - \operatorname{Re}(c^{\mathrm B} - c^{\mathrm A})(0)\big| \le 1.39\,\mathsf R_{t_1} + 4.27\,\epsilon_{\mathrm{fg}} + 6.56\,\epsilon_{\mathrm{cl}} + 9.40\,\epsilon_{\mathrm{sh}} \le \mathsf R_K[T_\lambda],$$

which is the display above.

(iv) *The constants at zero source.* We claim that

$$c^\bullet(0) = E_{(n)}[T_\lambda], \qquad n = t_1^\bullet = K^\bullet - u,$$

where the right side is the vacuum constant of the first n steps of the actual progression at cutoff $K^\bullet$. The hypotheses at $K \ge \max\{K_{\mathrm{bf}}(\ell, \mathsf J), K_{\mathrm{bf}}(\ell', \mathsf J)\}$ give the admitted setting of §2.3.1 on both tori.

The couplings $x_j^\bullet = L^{K^\bullet - j} b$ have zero increments and form an admissible finite sequence. With either side $2L^m$ they give admitted data, by the proof of Lem. YM3_I:118.1. The stronger room required in Lem. 13.31 follows from $LMR(b) \le LMR(L(g^{-2} + \bar B)) \le L^{m_{\mathfrak L}(g)} \le L^m$; its proof above also gives $2L^m \ge C_L M$.

At zero source, the family of level j in the chain is the family of the canonical progression of the cut $(x_0^\bullet, \dots, x_{K^\bullet - 1}^\bullet)$, by the same proof of Lem. YM3_I:118.1. Thm. YM3_I:Y identifies these with the first $K^\bullet$ families of the progression of the full line, with their common companions on all admitted tori. They are real at the trivial configuration and satisfy the bound in Lem. 13.31; the common choice retains $\delta_N M \ge \kappa$, $E_0 \ge 2E_N$ and $\kappa \ge 2\log 9216 + 5$.

By Prop. YM3_I:114.2, step i adds the logarithm of its normalizing factor and the sum of the local values $E^{(i)}(X, \mathbf 1, z)$. Equation eq. 13.ii identifies the logarithm with $\mathsf N^{(i-1)}$ and the sum of $\mathsf{vn}^{(i)}$. Since a whole torus has no boundary, every site in each domain is a base point. Thus the constant added at step i is exactly $\mathsf E^{(i-1)}(T_\lambda, T_\lambda)$. Summation over $1 \le i \le n$ proves the claimed identity. In particular these accumulated constants are real.

Step 4: two tori. Equation eq. 13.iii gives, per volume,

$$\left|E_{(n)}[T_\ell]\ell^{-3} - E_{(n)}[T_{\ell'}]\ell'^{-3}\right| \le 2C_{\mathrm{tt}}E_0 \sum_{s=K-n}^{K-1} L^{3s}\exp[-\kappa\ell' L^s/(5M)];$$

divide Step 3 by the volumes and subtract the two tori:

$$|U_{K+1}(b) - U_K(b)| \le \ell^{-3}\mathsf{R}_K[T_\ell] + \ell'^{-3}\mathsf{R}_K[T_{\ell'}] + 2\mathsf{v}_K \le 2\mathsf{C}_{\mathrm{bf}}K^{\mathsf{p}_0}L^{-K/10} + 2\mathsf{v}_K.$$

The right member is summable in K, so U_K converges uniformly on J: clause (c).

Step 5: a holomorphic extension free of the cutoff. For $\lambda \in \{\ell, \ell'\}$, $\phi_\lambda(z) = \widetilde{Z}_\lambda(z)/\widetilde{Z}_\lambda(0)$ is holomorphic on $|z| < 2\varrho$ with $\phi_\lambda(0) = 1$ and $|\phi_\lambda| \le e^{(C_S+C_g)\lambda^3} \le e^{(C_S+C_g)\ell^3}$ by (M1); by Schwarz's lemma $|\phi_\lambda - 1| \le \frac12$, so $\log\phi_\lambda$ is holomorphic with modulus $\le \log 2$, on $|z| < \varrho/(e^{(C_S+C_g)\ell^3}+1)$. By the shift of the label, $\lambda^{-3}\log\phi_\lambda(x) = p_{K,\lambda}(b-x) - p_{K,\lambda}(b) - \lambda^{-3}\mathcal{E}_\lambda(x\cdot 1)$ for real x, the logarithm being real there by the conjugation in (M1); so, with d of Lem. 13.10,

$$\mathsf{u}_{K,b}(z) = \ell^{-3}\log\phi_\ell(z) - \ell'^{-3}\log\phi_{\ell'}(z) + d(z)$$

is holomorphic on $|z| < r_3$, bounded there by M_3, and equals $U_K(b-x) - U_K(b)$ for real $|x| < r_3$; Cauchy's estimate gives $|\partial_b^n U_K(b)| \le n! M_3 r_3^{-n}$.

Step 6: the derivative. For $b \in \mathsf{W}$ the function $\mathsf{h} = \mathsf{u}_{K+1,b} - \mathsf{u}_{K,b}$ is holomorphic on $|z| < r_3$ with $|\mathsf{h}| \le 2M_3$, and for real $|x| \le r_3/2$ both b and $b-x$ lie in J, so by Step 4, with the number $\mathsf{o}_K = 4\mathsf{C}_{\mathrm{bf}}(\mathsf{J})K^{\mathsf{p}_0}L^{-K/10} + 4\mathsf{v}_K$ of (d),

$$|\mathsf{h}(x)| = \left|(U_{K+1} - U_K)(b-x) - (U_{K+1} - U_K)(b)\right| \le \mathsf{o}_K.$$

Lem. 13.30 gives $|U'_{K+1}(b) - U'_K(b)| = |\mathsf{h}'(0)| \le (4\mathsf{o}_K/r_3)[1 + \log_2^+(2M_3/\mathsf{o}_K)]$. As o_K decays like $K^{\mathsf{p}_0}L^{-K/10}$ and its logarithm grows like K, the series $\sum_K K \sup_{\mathsf{W}}|U'_{K+1} - U'_K|$ converges; this gives the uniform convergence of U'_K and the bound of $\sup_K \sup_{\mathsf{W}}|U'_K|$.

Step 7: the contact terms. The read-out of the theorem on two consecutive cutoffs at the order 1 in the source (Thm. YM3_I:AR, equation YM3_I:(2202), at $n = 1$), on each torus with $f^{\langle\lambda\rangle}$, gives $S^T_{1;K+1}(f) - S^T_{1;K}(f)$ up to the derivative at zero source of the difference of the contact terms at the cutoffs $K+1$ and K and an error that is summable in K; subtracting the two tori, by Step 1, $A_{K+1}(f) - A_K(f) = -\operatorname{Rie}_{K+1}(f)(U'_{K+1} - U'_K)(b) - [\operatorname{Rie}_{K+1}(f) - \operatorname{Rie}_K(f)]U'_K(b)$, with $|\operatorname{Rie}_{K+1}(f) - \operatorname{Rie}_K(f)| \le \sqrt{3}a_K \operatorname{Lip}(f)(2R+2)^3$ for R the radius of a ball about the origin that holds the support of f, and $|U'_K|$ bounded; so the difference between the two tori of the derivatives of the contact terms is summable in K, uniformly on W. □

Proof of Prop. 13.8. Step 1: characteristic functions from the imaginary axis. In (a), by the passage $(K, L^K b) \in \mathfrak{d}(g)$. For real y and t the function e^{ityX_κ} of the cell κ has, by the source formula, $\|e^{ityX_\kappa}\|^N_{\mathrm{ch},\ell} = Z[ity(F_e - F_o)]/Z$, a number ≥ 0, the same for both parities of κ; with $v = y(F_e - F_o)$, $\|v\|_\sharp = |y|$, and (IM) for $\widetilde{Z}^{\mathrm{av}}$ at $u = 0$,

$$\|e^{ityX_\kappa}\|_{\mathrm{ch},\ell} \le \exp[N^{-1}\operatorname{Re}\mathcal{E}^{\mathrm{av}}_\ell(itv) + C'_I t^2 y^2 q^3].$$

The Taylor coefficients of $s \mapsto N^{-1}\mathcal{E}^{\mathrm{av}}_\ell(s(F_e - F_o))$ are real by (A1) of Lem. 13.9, and bounded uniformly in ℓ at fixed K, by the count of domains, which bounds the energies of the domains that meet one cell; so $N^{-1}\operatorname{Re}\mathcal{E}^{\mathrm{av}}_\ell(itv) = -\frac12 t^2 y^2 \mathsf{e}_\ell + O(t^4)$ uniformly in ℓ, and $\bar{\mathsf{e}}$ is finite.

Step 2: the variance from below. Choose sides $\ell_k \to \infty$ with $\mathsf{e}_{\ell_k} \to \bar{\mathsf{e}}$. The domination, for the functions $e^{ity_\kappa X_\kappa}$ of different cells, gives

$$\left|\nu(e^{it\sum_\kappa y_\kappa X_\kappa})\right| \le \exp\Big[-\tfrac12 t^2(\bar{\mathsf{e}} - 2C'_I q^3)\sum_\kappa y_\kappa^2 + O(t^4)\Big],$$

while $|\nu(e^{itY})| \ge \operatorname{Re}\nu(e^{it(Y-\nu(Y))}) \ge 1 - \frac12 t^2 \operatorname{Var}_\nu(Y)$; letting $t \to 0$ gives (a).

Step 3: covariances by variances. Apply (a) to $X_\kappa \pm X_{\kappa'}$:

$$\mathrm{Var}_\nu(X_\kappa) + \mathrm{Var}_\nu(X_{\kappa'}) \pm 2\,\mathrm{Cov}_\nu(X_\kappa, X_{\kappa'}) \ge 2(\bar{\mathsf{e}} - 2C_I' q^3),$$

which is the second assertion of (a).

Step 4: Cauchy's estimate with the chord. In (b), for $z = x + i\tau$ with $|z| = r \le \varrho/2$, the source formula gives

$$N\Lambda(z) = \mathrm{Re}\,\mathcal{E}^{\mathrm{av}}(w_z) + \log|\widetilde{Z}^{\mathrm{av}}(w_z)| - \log\widetilde{Z}(0).$$

(IM) at $u = xF$, $v = \tau(F_e - F_o)$ costs $C_I'\tau^2\ell^3$. By semi-convexity, $\mathsf{g}(t) = \log\widetilde{Z}^{\mathrm{av}}(tF)$ has $\mathsf{g}'' \ge -2C_I'\ell^3$ on $[-r, r]$, so $\mathsf{g} + C_I'\ell^3 t^2$ lies below its chord:

$$\mathsf{g}(x) \le \tfrac{1}{2}[\mathsf{g}(r) + \mathsf{g}(-r)] + C_I'\ell^3 r^2 + \mathsf{m}_r x, \qquad \mathsf{m}_r = [\mathsf{g}(r) - \mathsf{g}(-r)]/(2r).$$

The cross terms between the cells give $\mathrm{Re}\,\mathcal{E}^{\mathrm{av}}(w_z) \le N\,\mathrm{Re}\,\mathcal{E}^{\mathrm{av}}(zf) + NM_\times T^{-3}\varrho^{-2}q^3|z|^2$, as in Prop. 13.16. With $\tilde{\epsilon}(z) = \mathcal{E}^{\mathrm{av}}(zf) + N^{-1}\mathsf{m}_r z$, holomorphic with $\tilde{\epsilon}(0) = 0$ and $\tilde{\epsilon}'(0)$ real, $\Lambda(z) - \mathrm{Re}\,\tilde{\epsilon}(z) \le \frac{1}{2}\mathsf{S}_f(r) + (2C_I' + M_\times T^{-3}\varrho^{-2})q^3 r^2$ on $|z| = r$, and Lem. 13.17 (a) gives the bound, with $m = q^{-3}\tilde{\epsilon}'(0)$; Lem. 13.17 (b) gives the covariance. At $r = \hat{r}q^{-3+p/2}$ one has $(rq^3)^{-2} = \hat{r}^{-2}q^{-p}$ and $q^3r^2 = \hat{r}^2 q^{p-3} \le \hat{r}^2$.

Step 5: the cost from below. By semi-convexity at the bases $\pm tF$, $|t| \le r \le \varrho/2$,

$$\mathsf{g}(r) + \mathsf{g}(-r) - 2\mathsf{g}(0) = \int_0^r (r-t)[\mathsf{g}''(t) + \mathsf{g}''(-t)]\,dt \ge -2C_I'\ell^3 r^2,$$

and $\ell^3/N = q^3$.

Step 6: the cost from above. In (c), the walls (Prop. 13.14) at $x = \pm r$ ($r \le b$, $r \le \varrho/(256\mathsf{c})$), the identity in the proof of Prop. 13.16, the bounds $\omega(r) + \omega(-r) \le 2C_\omega r^2$ and $|d(r) + d(-r)| \le 4\mathsf{A}_1 r^2/h_1^2$ give

$$\mathsf{S}_f(r) \le q^3\Delta_r^2 D_{K,\ell}(b) + 6C_w T q^2 r + 2q^3r^2[C_2' + C_\omega + 2\mathsf{A}_1/h_1^2] \le q^3\Delta_r^2 D_{K,\ell}(b) + 6C_w T q^2 r + 2C_I q^3 r^2.$$

At $r = \hat{r}q^{-3/2}$: $q^3\Delta_r^2 D_{K,\ell}(b) \le q^3[\Delta_r^2 D_K(b) + 2\bar{\eta}(\ell)]$ by Lem. 13.11, $q^3\Delta_r^2 D_K(b) \le C_\flat \hat{r}^2 + C_\flat \hat{r} q^{3/2}\delta \le 2C_\flat$ by Thm. AM (a) with $\delta \le q^{-3/2}$ and $[b \pm r] \subset W'$, and $2q^3\bar{\eta}(\ell) = 6\ell' B' q^3/\ell \le 1$; the lower bound is Step 5.

Step 7: a power of q. In (d) the proof of Lem. 13.27 applies with $C_p q_i^{-p}$ in the place of $C_{\mathrm{LH}}(T/q_i)^2$: $C_p q_i^{-p} \ge \sigma'(\theta w_i, w_i) \ge \mathsf{s}(\sqrt{6}q_i)(1 - 3T/(4q_i))^6$, and for $\sqrt{6}q_i \le r < \sqrt{6}Lq_i$, $q_i^{-p} < (\sqrt{6}L)^p r^{-p}$; then $\rho([0, m]) \le 4\pi e\, m^{-1}\mathbf{s}(1/m)$ by Prop. 13.25.

Step 8: one frame and a growth bound. In (e), Prop. 13.19 (b) at the side ℓ_q, which lies below $6LC_{\mathrm{FS}}q^3$, gives

$$\delta \le \frac{3C_wT}{q} + C_0 x_q + x_q Y_0 \ell_q^{1/3} + \frac{6C_w C_{\mathrm{FS}} T q^2}{\ell_q} \le \frac{3C_wT}{q} + C_0 x_q + \frac{Y_0(6LC_{\mathrm{FS}})^{1/3} q}{3C_w T q^2} + \frac{C_w T}{q},$$

which is the display; the window inequality gives the three bounds. Only one frame enters.

Step 9: the variances from above. In (f), by the passage $(K, L^K b) \in \mathfrak{d}(g)$. Let ν and $0 < r \le \varrho/2$ be given, and on T_{ℓ_k}, with N_k cells, let $\tilde{\epsilon}_k(z) = \mathcal{E}^{\mathrm{av}}_{\ell_k}(zf) + N_k^{-1}\mathsf{m}_{r,k} z$, $\mathsf{m}_{r,k}$ the slope of the chord of Step 4. For $z = x + i\tau$ with $|z| = r$, the source formula, (IM) at $u = xF$, $v = \tau(F_e - F_o)$, the chord of Step 4, and the cross terms between the cells (Lem. 13.9 (A3) with $R_0 = q/2$, $\Delta = T/4$, $\rho_3 = 3\varrho/2$, so that $C_\times(2q/T) \le c_X$) give

$$\log\|e^{zX}\|_{\mathrm{ch},\ell_k} - \mathrm{Re}\,\tilde{\epsilon}_k(z) \le \tfrac{1}{2}\mathsf{S}_{f,\ell_k}(r) + 2C_I' q^3 r^2 + \tfrac{4}{9}c_X\varrho^{-2}r^2 \le \mathsf{S}_{f,\ell_k}(r) + c_3 r^2,$$

the last step by $\mathsf{S}_{f,\ell_k}(r) \ge -2C_I' q^3 r^2$ (Step 5). The functions $\tilde{\epsilon}_k$ are holomorphic on $|z| < 2\varrho$, real on the real axis and 0 at 0, and they are bounded uniformly in k, by the count of domains for the domains that meet one cell and by (M1) and the floor for the slopes; along a subsequence they converge, locally uniformly, to such a function $\tilde{\epsilon}$. The domination gives $|\nu(e^{zX})| \le \liminf_k \|e^{zX}\|_{\mathrm{ch},\ell_k}$,

so $G(z)=\nu(e^{zX})e^{-\tilde\epsilon(z)}$ is holomorphic on $|z|<2\varrho$, $G(0)=1$, and $|G|\le\exp[\limsup_k \mathsf{S}_{f,\ell_k}(r)+c_3r^2]$ on $|z|=r$. The coefficient of z^2 in G is $\frac12[\nu((X-\tilde\epsilon'(0))^2)-\tilde\epsilon''(0)]$, and Cauchy's estimate gives

$$\mathrm{Var}_\nu(X)\le\nu((X-\tilde\epsilon'(0))^2)\le\tilde\epsilon''(0)+2r^{-2}\exp\Big[\limsup_k \mathsf{S}_{f,\ell_k}(r)+c_3r^2\Big].$$

Finally $\tilde\epsilon''(0)$ is the limit of $\partial_s^2\mathcal{E}^{\mathrm{av}}_{\ell_k}(sf)|_{s=0}$ along the subsequence. By (A1) of Lem. 13.9 and the cross terms at $\rho_3=\varrho$, $N^{-1}\mathcal{E}^{\mathrm{av}}_\ell(s(F_e-F_o))=\frac12[\mathcal{E}^{\mathrm{av}}_\ell(sf)+\mathcal{E}^{\mathrm{av}}_\ell(-sf)]+N^{-1}\mathrm{cr}(s)$ with $|\mathrm{cr}(s)|\le Nc_X|s|^2/\varrho^2$ for complex $|s|<\varrho$, so Cauchy's estimate gives $|\mathrm{cr}''(0)|\le 2Nc_X/\varrho^2$, hence $\partial_s^2\mathcal{E}^{\mathrm{av}}_\ell(sf)|_{s=0}\le \mathsf{e}_\ell+2c_X\varrho^{-2}$ and $\tilde\epsilon''(0)\le\bar{\mathsf{e}}+2c_X\varrho^{-2}$. The second assertion of (f) is (a): $|\mathrm{Cov}_\nu(X_\kappa,X_{\kappa'})|\le C_{\mathrm{var}}q^{6-p}+2C_I'q^3\le(C_{\mathrm{var}}+2C_I')q^{6-p}$, as $p\le3$; divide by q^6.

Step 10: the walls and the cost. In (g), for t between 0 and x the source $xF+t(1-F)=t+(x-t)F$ has the norm $\le|x|\le\varrho/2$, and $\|1-F\|_\sharp\le1$; so semi-convexity gives $\partial_t^2\log\widetilde{Z}^{\mathrm{av}}(xF+t(1-F))\ge-2C_I'\ell^3$, and at $t=x$

$$\log\widetilde{Z}^{\mathrm{av}}(x)\ge\log\widetilde{Z}^{\mathrm{av}}(xF)+xN\upsilon_f(x)-C_I'\ell^3x^2,$$

which divided by N is the first assertion. Next, by (A1) of Lem. 13.9 at constant sources,

$$\mathsf{S}_f(r)=\Upsilon_f(r)+\Upsilon_f(-r)+N^{-1}\big[\log\widetilde{Z}(r)+\log\widetilde{Z}(-r)-2\log\widetilde{Z}(0)\big];$$

the first two terms are $\le2C_\Upsilon$; by the identity in the proof of Prop. 13.16 and the bounds of ω and of d there, the bracket is $q^3[\Delta_r^2D_{K,\ell}(b)+\omega(r)+\omega(-r)-d(r)-d(-r)]\le q^3[\Delta_r^2D_{K,\ell}(b)+2(C_\omega+2\mathsf{A}_1/h_1^2)r^2]$, as $r\le r_1\le\min\{r_2/2,h_1/2\}$; and $\Delta_r^2D_{K,\ell}(b)\le\Delta_r^2D_K(b)+2\bar\eta(\ell)$ by Lem. 13.11, while $\Delta_r^2D_K(b)\le C_\flat r^2+C_\flat\delta r$ by Thm. AM (a), since $[b-r,b+r]\subset W'$. □

14. The Haag net of the action density

Write $\phi(f)$ for the reconstructed action-density field on Minkowski space $\mathbb{R}^{1+2}$, and Ω for its vacuum vector. The polynomial energy bound gives a common invariant domain and closed smeared fields $\bar\phi(f)$. For an open region O, let $\mathcal{C}(O)$ be the bounded weak commutant of these local fields and their adjoints, and set $\mathcal{A}(O)=\mathcal{C}(O)'$. Here weak commutation is tested on their common energy domain. Write $O^\times$ for the open spacelike complement of O, and generate $\mathcal{A}(O^\times)$ from the algebras of regions contained in that complement.

Sketch M (Local algebras and wedge duality). For a volume-first limit and each of its components, $O\mapsto\mathcal{A}(O)$ is an isotonic, covariant local net. The field $\bar\phi(f)$ is affiliated with $\mathcal{A}(O)$ when $\mathrm{supp}\,f\subset O$. For a wedge W and its spacelike complement W', the vacuum is cyclic and separating, and

$$\mathcal{A}(W)'=\mathcal{A}(W'),\qquad \Delta^{it}=U(B(-2\pi t)).$$

Here U implements Poincaré transformations, $B(s)$ is the wedge-preserving boost of rapidity s, and Δ is the vacuum modular operator of $\mathcal{A}(W)$. For a double cone O, the dual completion $\mathcal{A}^d(O)=\mathcal{A}(O^\times)'$, extended to other open regions by joining the double-cone algebras, is a local Haag-dual net. Equality of $\mathcal{A}$ and $\mathcal{A}^d$ for all double cones is not asserted. ■□

Sketch of the argument. The reconstruction supplies local unbounded fields. Passing from their commutation relations to commutation of bounded operator algebras requires an additional argument: weak commutation on a common domain does not by itself identify the commutants of the closures. The lattice provides bounded operators with which to make this passage.

Use the algebra of bounded operators on the right half of the lattice time-zero plane. Reflection positivity embeds its vectors in the lattice Hilbert space. A quarter turn relates the two Euclidean quadrants. In the continuum this relation becomes the quarter power $\Delta^{1/4}$ of the boost modular operator. Convergence of the reflection-positive forms and functional calculus then carry the lattice modular translations to the boost group.

Every bounded weak limit from the lattice half-plane algebra commutes with the fields in the opposite wedge. To show that there are enough such limits, regularize the lattice insertion of a field and use the quarter-turn relation. The resulting bounded approximants reach a dense family of vectors $\phi(f)\Omega$ in the closure of the opposite-wedge commutant acting on Ω. Their error is controlled by a separated Euclidean four-point function. The modular relation

$$J\Delta^{1/2}Y\Omega = Y^*\Omega$$

identifies the two wedge algebras and yields locality and wedge duality. The weak-commutant definition then gives affiliation of the closed local fields. Intersecting the containing wedge algebras produces the stated dual completion.

The argument also passes to components. Translation-invariant weights can be approximated by distant averages, so they preserve the relevant positivity forms. The reconstructed Hilbert space and wedge algebra decompose over the component measure; their fibres agree with the component wedge algebras almost everywhere.

The periodic lattice approximants are essential to this supply of bounded operators. For a general limit of periodic Gibbs states the source establishes a supply of half-plane operators, but does not assert the whole locality theorem by this argument. Nor should wedge duality be renamed Haag duality of the original net on arbitrary bounded regions.

14.1. **Local nets and duality.** We construct the local net of the action density from the continuum limits and their components. The lattice quarter turn determines the modular data of a wedge. Passing its bounded operators to the limit gives wedge duality and the affiliation of the closed field operators with their local algebras.

We distinguish the original net from its dual completion: the latter is Haag dual, while equality with the original net is an additional assertion. Energy bounds, Gaussian examples and the lattice ring calculation clarify the scope of this distinction. The fixed-depth block-loop arguments use §10.1.

14.1.1. *Notation and hypotheses.*
Constants. We fix the radius $\varrho > 0$ and the constants (σ^-, γ) of the list $\mathcal{T}^{\mathrm{S}}[2\varrho]$ of the common sourced construction in Section YM3_I:82, among them the integers $L, M, \mathfrak{p}_0$ and the number $B_\sharp \geq 0$ that bounds the increments of the flow of the running coupling (see the objects below), and a coupling $0 < g \leq \gamma$; $\bar{B} = B_\sharp L/(L-1)$ is formed from σ^- alone, with no radius, γ or g, and $\bar{B} \leq \frac{13}{24}\gamma^{-2} < \gamma^{-2}$ (indeed, $B_\sharp\gamma^2 \leq \frac{1}{2}$ and $L \geq 13$ give $\bar{B}\gamma^2 \leq L/[2(L-1)] \leq 13/24$). C^g_{pt} is the constant of the point bound at g of equation YM3_I:(2097) among the construction results, and $\bar{\mathfrak{c}}(\mathsf{C}^g)$, with the bound system C^g at g of equation YM3_I:(2030), is the bound of the one-point constants of the components of equation YM3_I:(1993). The short-distance law of the two-point function among the construction results, the law of two points of Thm. YM3_I:AV as it holds in every component (Cor. YM3_I:127.11), holds at the couplings $0 < g \leq \gamma_{\mathrm{sd}}$, where $\gamma_{\mathrm{sd}} = \gamma_W \in (0, \gamma]$ is its ceiling equation YM3_I:(2346), a number formed from (σ^-, γ) alone. For $0 < g \leq \gamma_{\mathrm{sd}}$ we put

$$c_{\mathrm{sd}} = c_{\mathrm{sd}}(g) = 4.6 \cdot 10^{-9} L^{-2}(g^{-2} + \bar{B})^{-2}, \qquad r_{\mathrm{sd}} = r_{\mathrm{sd}}(g) = 2.2\,\mathsf{d}_0(g, \tfrac{1}{10}),$$

with $\bar{B} = B_\sharp L/(L-1)$ and the threshold d_0 of the law, equation YM3_I:(2350): two numbers that depend on g and on no sequence, limit, decomposition or ξ, and $c_{\mathrm{sd}}(g) \leq 4.6 \cdot 10^{-9} g^4$. Then for every component σ'_ξ of a limit of periodic Gibbs states at g the kernel s of its two-point function, $\sigma'_\xi((f,h)) = \iint f(x)h(y)\,\mathsf{s}(|x-y|)\,\mathrm{d}x\,\mathrm{d}y$ for f, h of disjoint supports, is ≥ 0 and non-increasing, and $\mathsf{s}(r) \geq c_{\mathrm{sd}} r^{-6}$ for $0 < r \leq r_{\mathrm{sd}}$. Indeed, with the point vectors Ψ_1 of the reconstruction recalled below and $0 < \epsilon < r/2$, $\mathsf{s}(r) = \|e^{-(r/2-\epsilon)H}\Psi_1(\epsilon e_3)\|^2$, and the truncated kernel, s minus the square of the one-point constant, is $\|e^{-(r/2-\epsilon)H}(1 - |\Omega\rangle\langle\Omega|)\Psi_1(\epsilon e_3)\|^2$, the vacuum of a component being unique; both are ≥ 0 and non-increasing in r.

For $0 < d \le \mathsf{d}_0(g, \frac{1}{10})$ let $f \ge 0$ be of class C^1, equal to 1 on the ball of radius $0.3d$ about $2de_3$ and to 0 off the ball of radius $0.9d$ about it, with $\sup f = 1$ and $\operatorname{Lip} f < 3/d$: then $(\theta f, f)$ is a test pair at the scale d with the shape parameter $\frac{1}{10}$ ([7, Lem. I.1]), its two supports lie at distances between $2.2d$ and $5.8d$, and $\int f < 3.06\, d^3$. The lower bound of Cor. YM3_I:127.11 makes the truncated two-point function at this pair at least $\frac{2}{3}\, 10^{-8} \mathsf{c}_2 \mathsf{b}^{-2}$, with $\mathsf{c}_2 = 9/16\pi^2$ YM3_I:C and the label $\mathsf{b} \le L(g^{-2} + \bar{B})$ (equation YM3_I:(1854)); as the truncated kernel is non-increasing, its value at $2.2d$ is at least $\frac{2}{3}\, 10^{-8} \mathsf{c}_2 \mathsf{b}^{-2} (3.06\, d^3)^{-2}$, and $\frac{2}{3}\, 10^{-8}\, \mathsf{c}_2\, (2.2)^6 (3.06)^{-2} > 4.6 \cdot 10^{-9}$ gives the bound at $r = 2.2d$, for the truncated kernel and so for s.

Space-time. In $\mathbb{R}^3 \ni x = (x_1, x_2, x_3) = (\vec{x}, x_3)$ the coordinate x_3 is Euclidean time and $\mathbb{E}_+ = \{x_3 > 0\}$; θ, θ_1 are the reflections at $x_3 = 0$ and at $x_1 = 0$, τ_v is the translation by v, $\mathsf{d}(x_1, x_2, x_3) = (x_3, x_2, x_1)$, and $R_\tau x = (x_1 \cos\tau - x_3 \sin\tau,\, x_2,\, x_1 \sin\tau + x_3 \cos\tau)$, also on $\mathbb{C}^3$, with $R = R_{\pi/2} = \theta_1 \mathsf{d}$; $\mathsf{I} = \{x_1 > 0, x_3 > 0\}$ and $\mathsf{II} = R\,\mathsf{I} = \{x_1 < 0, x_3 > 0\}$. An isometry r acts on functions by $\mathsf{r}h = h \circ \mathsf{r}^{-1}$ and on tuples entry by entry. In $\mathbb{R}^{1+2} \ni x = (x^0, x^1, x^2)$, $t = x^0$: $V_+ = \{x^0 > |\vec{x}|\}$, $W = \{x^1 > |x^0|\}$, $W' = -W$, $\mathsf{j}(x) = (-x^0, -x^1, x^2)$, and $B(s)(t, x^1, x^2) = (t\cosh s + x^1 \sinh s,\, t \sinh s + x^1 \cosh s,\, x^2)$ are the boosts that keep W. A wedge is a region $\mathfrak{g}W$, $\mathfrak{g}$ in the Poincaré group; a double cone is a Poincaré image of $\{|x^0| + |\vec{x}| < r\}$, upright if it is a translate of one; $O^\times$ is the union of the open sets that are spacelike separated from O; $\mathcal{F}(O) = C_c^\infty(O; \mathbb{C})$, $\mathcal{F}_\mathbb{R}(O)$ its real elements, $\mathcal{S} = \mathcal{S}(\mathbb{R}^{1+2})$, $\hat{f}$ the Fourier transform, and $f^{(a,\Lambda)} = f \circ (a, \Lambda)^{-1}$.

Data (Rec) and the reconstruction. A separated tuple is a tuple of real C_c^1 functions on $\mathbb{R}^3$ with pairwise disjoint supports; F, F' denote finite complex combinations of separated tuples, over a set if all supports lie in it, and $\bar{F}$ has the conjugate coefficients. Data (Rec) with a triple $(C_\wp, \wp, c_0)$ are the symmetric functionals σ on separated tuples, $\sigma(\varnothing) = 1$, with the following conditions (Section YM3_I:107): (Rec1) σ is invariant under the isometries of $\mathbb{R}^3$; (Rec2) σ is reflection positive at $x_3 = 0$, the classes $[F]$ of the combinations over $\mathbb{E}_+$ span the Hilbert space $\mathcal{H}$, $\Omega = [\varnothing]$, and $e^{-tH}[F] = [\tau_{te_3} F]$ with $H \ge 0$; (Rec3) the joint spectrum of H and the momenta lies in $\{E \ge |q|\}$; (Rec4) $|\sigma(\theta\bar{F} \cdot (h) \cdot F')| \le C_\wp \eta^{-\wp} \|h\|_{L^1} \|[\tau_{-\eta e_3} F]\| \|[\tau_{-\eta e_3} F']\|$ for $0 < \eta \le 1$, F, F' over $\{x_3 > \eta\}$ and $h \in C_c^1(\{|x_3| < c_0\eta\})$. (Rec5) is the uniqueness of the vacuum, $\ker H = \mathbb{C}\Omega$.

The reconstruction in Sections YM3_I:109 and YM3_I:110 gives $N = 1 + H$, $\mathcal{D}_\infty = \bigcap_k D(H^k)$, a representation U of the Poincaré group, and the field $\phi(f)$, $f \in \mathcal{S}$, on $\mathcal{D}_\infty$, with $\phi(f)^\dagger = \phi(\bar{f})$ and $[H, \phi(f)] = i\phi(\partial_t f)$ on $\mathcal{D}_\infty$, and with Lorentz covariance and locality on the span $\mathcal{D}_0$ of the vectors $\phi(f_1) \cdots \phi(f_n)\Omega$ ($\mathcal{D}_0(O)$: all $f_k \in \mathcal{F}(O)$); for $Y = \phi(f_1) \cdots \phi(f_n)$ we write $Y^\dagger = \phi(\bar{f}_n) \cdots \phi(\bar{f}_1)$. It gives the field $\mathsf{B}(s, s')$ at the point 0 between the Euclidean times $s, s' > 0$, with $e^{-tH} \mathsf{B}(s, s') e^{-t'H} = \mathsf{B}(s + t, s' + t')$, $\mathsf{B}(s, s')^* = \mathsf{B}(s', s)$ and

$$\|\mathsf{B}(s, s')\| \le C_\wp \min\{s, s', 1\}^{-\wp}, \qquad e^{-sH} \phi(f) e^{-sH} = \int f(x)\, U(x) \mathsf{B}(s, s) U(x)^{-1}\, \mathrm{d}x, \tag{14.i}$$

the insertion operators of Lem. YM3_I:107.1: for $0 < \eta \le 1$ and $h \in L^1(\{|x_3| < c_0\eta\})$, $\mathsf{B}^{\mathrm{in}}_\eta(h)$ is the bounded operator, linear in h and of norm $\le C_\wp \eta^{-\wp} \|h\|_{L^1}$, with $\langle [\tau_{-\eta e_3} F], \mathsf{B}^{\mathrm{in}}_\eta(h) [\tau_{-\eta e_3} F'] \rangle = \sigma(\theta\bar{F} \cdot (h) \cdot F')$ for F, F' over $\{x_3 > \eta\}$ when $h \in C_c^1(\{|x_3| < c_0\eta\})$, and by the insertion formula YM3_I:(2039) $\langle \psi, \mathsf{B}^{\mathrm{in}}_\eta(h) \psi' \rangle = \int h(x) \langle \psi, U(\vec{x}) \mathsf{B}(\eta + x_3, \eta - x_3) U(\vec{x})^{-1} \psi' \rangle\, \mathrm{d}x$; the form q_0 of the field at the point 0 and the time 0, $q_0(e^{-sH}\psi', e^{-s'H}\psi) = \langle \psi', \mathsf{B}(s, s')\psi \rangle$, and point vectors $\Psi_n(z)$, holomorphic on the tube $\mathsf{Tb}^{(n)} = \{z \in (\mathbb{C}^3)^n : d(z_1) > 0,\ d(z_{k+1} - z_k) > 0\}$, $d(z) = \operatorname{Re} z_3 - |\operatorname{Im}(z_1, z_2)|$, of norm $\le (C_\wp \eta^{-\wp})^n$ where these numbers are $\ge 2\eta$, $\eta \le 1$, with $[G] = \int \prod_k h_k(x_k) \Psi_n(x)\, \mathrm{d}x$ for ordered tuples $G = (h_k)$ in $\mathbb{E}_+$ whose supports lie in balls of radii below c_0 times the gaps of the heights.

With the continued translations $\mathcal{T}(\mathfrak{w})$, $\mathcal{T}(\eta e_3) = e^{-\eta H}$, and $\iota(t, \vec{x}) = (\vec{x}, -it)$, Lem. YM3_I:109.2, which links the field operators with the point vectors, reads

$$\text{(Br)} \qquad \mathcal{T}(\mathfrak{w}_1) \phi(f_1) \cdots \mathcal{T}(\mathfrak{w}_n) \phi(f_n) \Omega = \int \prod_k f_k(x_k)\, \Psi_n\big((\iota x_k + \textstyle\sum_{i \le k} \mathfrak{w}_i)_k\big)\, \mathrm{d}x \qquad (d(\mathfrak{w}_i) > 0),$$

and the products of fields are the limits of these vectors as the $\mathfrak{w}_i$ tend to 0.

The net and the conditions. $P_n = \mathbf{1}_{[n,n+1)}(H)$, $b_{mn}(f) = \|P_m\phi(f)P_n\|$, $\bar\phi(f)$ is the closure of $\phi(f)$, $\mathcal{P}$ is the joint spectral measure of the translations, $U(B(s)) = e^{isK}$ and $\Delta^s = e^{-2\pi sK}$. For open O

$$\mathcal{C}(O) = \{Q \in \mathcal{B}(\mathcal{H}) : \langle \phi(\bar f)\psi, Q\psi'\rangle = \langle \psi, Q\phi(f)\psi'\rangle \ \ \forall f \in \mathcal{F}(O),\ \psi,\psi' \in \mathcal{D}_0\}, \qquad \mathcal{A}(O) = \mathcal{C}(O)';$$

$\mathcal{L}(O) = \mathcal{A}(O)\cap\mathcal{C}(O^\times)$; $\mathcal{H}_0$ is the closure of $\mathcal{C}(W)\Omega$; for bounded open convex O, $\mathcal{A}^d(O) = \bigcap\{\mathcal{A}(\mathfrak{g}W) : \mathfrak{g}W \supset O\}$, and for other open G, $\mathcal{A}^d(G) = \bigvee\{\mathcal{A}^d(O) : O \subset G \text{ a double cone}\}$. $\mathfrak{P}$ is the unital $*$-algebra generated linearly by $\mathcal{F}(\mathbb{R}^{1+2})$, with $f^* = \bar f$ and $gh = hg$ for real spacelike separated g, h; π is its representation by the fields on $\mathcal{D}_0$ and $\mathcal{W}(f_1\cdots f_n) = \langle\Omega, \phi(f_1)\cdots\phi(f_n)\Omega\rangle$ its Wightman functional; for a subalgebra $\mathfrak{A}$, $\mathfrak{A}^+$ is the set of the finite sums $\sum Z_i^* Z_i$, $Z_i \in \mathfrak{A}$; $\mathfrak{P}(W)$ is generated by 1 and $\mathcal{F}(W)$.

(W) $\mathcal{C}(O_1)' \subset \mathcal{C}(O_2)$ for all non-empty open spacelike separated O_1, O_2.

(CP) For every upright double cone K, real $f \in \mathcal{F}(K)$ and $Y_0, \dots, Y_r$ in the algebra $\mathfrak{A}$ generated by 1, f and $\mathcal{F}(K^\times)$: if $\sum_n \lambda^n Y_n \in \mathfrak{A}^+$ for all real λ, then $\mathcal{W}(\sum_n f^n Y_n) \geq 0$.

(QCP$_f$) For real $f \in \mathcal{F}(K)$, K an upright double cone with $\bar K \subset W'$, and $\eta = \phi(f)\Omega$: if $Y_0, Y_1, Y_2 \in \mathfrak{P}(W)$ and $Y_0 + \lambda Y_1 + \lambda^2 Y_2 \in \mathfrak{P}(W)^+$ for all real λ, then $\langle\Omega, Y_0\Omega\rangle + \langle\Omega, Y_1\eta\rangle + \langle\eta, Y_2\eta\rangle \geq 0$. This is the second order of (CP) at f; its first order is the same statement with $Y_2 = 0$.

(LB$_{u,c}$) $|\langle\psi, \phi(u)\psi\rangle| \leq c\langle\psi, N\psi\rangle$ on $\mathcal{D}_0$; (LB$^\pm_{u,c}$) are its halves $\pm\langle\psi, \phi(u)\psi\rangle \leq c\langle\psi, N\psi\rangle$; (LB$_u$) holds if (LB$_{u,c}$) holds for some c, and $c_*(u) \in [0,\infty]$ is the least such c.

(QA$_f$) $\sum_n \|\phi(f)^n\Omega\|^{-1/n} = \infty$.

The objects. For a state ν on $L^{-K}\mathbb{Z}^3$ or on a torus and a tuple $X = (f_i)$ let $(X)^\nu_K = \prod_i(\mathcal{O}_K(f_i) - \nu(\mathcal{O}_K(f_i)))$, extended linearly to combinations; θ and the bar act on lattice functions by reflection and conjugation. A pair (K,β) is admitted at g if it lies in the canonical family $\mathfrak{d}(g)$ of equation YM3_I:(1843) ([7, §K.1]), that is, if it is a first-passage datum at g (Cor. YM3_I:74.2), formed at the fixed (σ^-, γ): along the flow $x_0 = \beta$, $x_k = L^{-1}x_{k-1} - b_k$ of the inverse squared running coupling, whose increments obey $|b_k| \leq B_\sharp$, K is the least index with $x_K \leq X_g = L(g^{-2} + B_\sharp)$.

Then $x_0 > x_1 > \cdots > x_K \geq \gamma^{-2}$, and the label $L^{-K}\beta$ lies in the outer interval of the labels $I^{\mathrm{lab}}(g) = [g^{-2} + B_\sharp - L^{-1}\bar B,\ L(g^{-2} + \bar B)]$ of (equation YM3_I:(1854)) (Cor. YM3_I:74.2). A periodic Gibbs state at (K,β) is a Gibbs state of the Wilson weight $e^{-\beta\sum_p a_p}$ on $L^{-K}\mathbb{Z}^3$ that is invariant under the translations of a subgroup of finite index of $L^{-K}\mathbb{Z}^3$; a torus state at (K,β) is a weak limit point, as $m \to \infty$, of the Wilson measures of the tori $L^{-K}\mathbb{Z}^3/2L^m\mathbb{Z}^3$ (Section YM3_I:91), and it is a periodic Gibbs state (Prop. YM3_I:91.2). A volume-first limit at g ([7, §K.1]) has a defining sequence (K_j, β_j, ν_j): $K_j \to \infty$, pairs (K_j,β_j) admitted at g, and torus states ν_j at (K_j,β_j) whose connected functions converge; its centred functional is $\sigma(X) = \lim_j \nu_j((X)_j)$, $(X)_j = (X)^{\nu_j}_{K_j}$.

A limit of periodic Gibbs states at g is $\sigma'(X) = \lim_j \rho_j((X)^{\rho_j}_{K_j})$ for periodic Gibbs states ρ_j of the Wilson weight at admitted pairs (K_j,β_j), $K_j \to \infty$. In general $(\cdot)_j$ is the product at K_j centred by the j-th state of the sequence at hand: $(X)_j = (X)^{\nu_j}_{K_j}$ for the centred functional, $(X)_j = (X)^{\rho_j}_{K_j}$ for a limit of periodic Gibbs states ρ_j, also where the periodic Gibbs states $\rho_{j,\xi}$ that approximate a component integrate, the ρ_j giving the centring, and, in Steps 2–7 of the proof of Thm. AP, the product on the torus Λ_j centred by the Wilson measure μ_j there. A canonical decomposition $(\mathsf{X}, \lambda, (\sigma_\xi))$ of σ has X compact metrisable, λ a probability measure of full support, $\xi \mapsto \sigma_\xi(X)$ continuous, $\xi \mapsto \sigma_\xi$ injective, and iterated cluster limits $\lim_{|v_q|\to\infty}\cdots\lim_{|v_1|\to\infty}\sigma(X^0\cdot\tau_{v_1}X^1\cdots\tau_{v_q}X^q) = \int\prod_i\sigma_\xi(X^i)\,\lambda(\mathrm{d}\xi)$; the σ_ξ are its components, and objects of σ_ξ bear the index ξ.

Lattice, norms, slabs. On $\Lambda = a\mathbb{Z}^3/\ell\mathbb{Z}^3$, ℓ/a even and ≥ 4: $G = \mathrm{SU}(2)$ with Haar measure, $t_p = \frac12\mathrm{tr}\,\partial U(p)$, $a_p = 1 - t_p$, the Wilson measure $\mu_\beta \propto e^{-\beta\sum_p a_p}dU$, and $\mathcal{O}(f) = a^{-1}\sum_p f(x(p))a_p$, $x(p)$ the barycentre of p ($\mathcal{O}_K$ at $a = L^{-K}$). $\|h\|_\sharp = \max\{\sup|h|, 40M\,\mathrm{Lip}\,h\}$ and $\|h\|_{\sharp,\mathsf{I}} = \max\{\sup|h|, 40M|\,\mathrm{Lip}\,h\}$; on $\mathbb{R}^2$, $|h|_{C^1} = \sup|h| + \sup|\nabla h|$; $\|f\|_\star = \int\mathrm{d}t\sum_{\kappa\in\mathbb{Z}^2}\sup_{\vec x\in\kappa+[-1,1]^2}(|f| +$

$|\nabla_{\vec{x}} f|)(t,\vec{x})$. A board (i,o) of $L^{-K}\mathbb{Z}^3$, $i \le K-1$ and o a site of $L^{-K}\mathbb{Z}^3$, is its partition into the cells $\square_{\mathbf{k}} = o + \mathsf{l}\mathbf{k} + [-\frac{\mathsf{l}}{2}, \frac{\mathsf{l}}{2}]^3$, $\mathbf{k} \in \mathbb{Z}^3$, cubes of side $\mathsf{l} = L^{-i}$ centred at the sites $o + \mathsf{l}\mathbf{k}$; the middle of $\square_{\mathbf{k}}$ is $\{|x - o - \mathsf{l}\mathbf{k}|_\infty \le \frac{\mathsf{l}}{4}\}$; for odd L the walls of the cells lie midway between sites and the middle planes of the cells pass through sites; and $\|\cdot\|_{\mathrm{ch}}$ is the seminorm of the chessboard estimate.

For $h \in L^1(\mathbb{R}^2)$, $\mathsf{B}_h(s,s') = \int h(\vec{x}) U(\vec{x}) \mathsf{B}(s,s') U(\vec{x})^{-1} d^2x$ is the field at time 0 smeared in space; $T_{mn}(h) = P_m e^{sH} \mathsf{B}_h(s,s') e^{s'H} P_n$ does not depend on s, s', and $\mathsf{T}(h) = \sum_{m,n} T_{mn}(h)$ is the form of the field at time 0 on the vectors of bounded energy. With $\rho_{\mathsf{l}}(\varsigma) = \frac{5}{\mathsf{l}} \cos^2 \frac{5\pi\varsigma}{2\mathsf{l}}$ on $|\varsigma| \le \frac{\mathsf{l}}{5}$ and 0 elsewhere, the plate is $\mathsf{l}_{\mathsf{l}}(h) = \int \rho_{\mathsf{l}}(\varsigma) \mathsf{B}_h(\frac{\mathsf{l}}{2} + \varsigma, \frac{\mathsf{l}}{2} - \varsigma)\, \mathrm{d}\varsigma$.

14.1.2. *From fields to bounded local operators.* There are two locality questions in this argument. The reconstructed field satisfies a commutation relation on its common invariant domain. A local net requires commuting algebras of bounded operators. Passing from the first statement to the second requires control of closures and their commutants. The energy estimates make the closures available; the lattice approximation supplies the bounded operators. Keeping these two tasks separate explains why reconstruction alone does not end the argument.

For an open region O in Minkowski space, write $\mathcal{F}(O) = C_c^\infty(O;\mathbb{C})$ and let $\mathcal{D}_0$ be the span of field polynomials applied to the vacuum. A bounded operator Q belongs to the weak commutant $\mathcal{C}(O)$ when

$$\langle \phi(\bar{f})\psi, Q\psi' \rangle = \langle \psi, Q\phi(f)\psi' \rangle, \qquad f \in \mathcal{F}(O), \quad \psi, \psi' \in \mathcal{D}_0.$$

Thus the definition involves bounded Q, although the field on either side is initially defined on $\mathcal{D}_0$. The proposed local algebra is $\mathcal{A}(O) = \mathcal{C}(O)'$. To establish locality one must show that these algebras commute for spacelike separated regions. The relation of field polynomials on $\mathcal{D}_0$ does not itself give that assertion about their bounded commutants.

The construction first uses the wedges

$$W = \{(t, x^1, x^2) : x^1 > |t|\}, \qquad W' = -W.$$

Their Euclidean antecedents are quadrants. A quarter turn exchanges the two boundaries of a quadrant, so the two Euclidean reflection forms put a finite-lattice operator on the two sides of the same Hilbert-space pairing. The finite-lattice four-square construction in proposition 14.13 is the source of the required bounded approximation. It retains the operator norm bound while keeping track of what the quarter turn does to vectors. Both pieces of information are needed: convergence of correlation functions would retain the vectors without supplying a bounded operator in the desired commutant.

In the limit, the quarter-turn operator is identified with $\Delta^{1/4}$, where the boost generator K is normalized by $\Delta^s = e^{-2\pi s K}$. This is an assertion about functional calculus, rather than a substitution of an imaginary angle into an unbounded operator. First prove convergence for the quarter turn, applies bounded functions of it, and only then reads off the corresponding bounded functions of K. This order permits the approximation to pass through the lattice Hilbert spaces.

Write $\mathsf{l} = \{x_1 > 0, x_3 > 0\}$ in Euclidean space. For real $u \in C_c^\infty(\mathsf{l})$, define the Schwartz function

$$f_u(B(s)(0,\rho,x_2)) = \int_0^{\pi/2} u(\rho\cos\alpha, x_2, \rho\sin\alpha) e^{-(s + i(\pi/4 - \alpha))^2}\, \mathrm{d}\alpha$$

on W, and set it to zero off W. Here $\rho > 0$ and s is the boost parameter. This Gaussian smoothing is the bounded functional calculus just described. Let θ_1 reflect the first Euclidean coordinate. The construction gives bounded $T_u^{(\epsilon)} \in \mathcal{C}(W')$ such that

$$\|T_u^{(\epsilon)}\Omega - \phi(f_u)\Omega\| \le \frac{\sqrt{\epsilon}}{2} \pi e^{5\pi^2/8} \|[\theta_1 u \cdot u]\|.$$

The norm on the right is determined by a separated Euclidean four-point function. It is finite before ϵ is sent to zero. Hence the field vectors $\phi(f_u)\Omega$ lie in $\overline{\mathcal{C}(W')\Omega}$. The Fourier transforms of the f_u

have no common real zero, which gives density. This is the point at which a supply of bounded lattice limits becomes sufficient to determine the wedge algebra.

Let J be the antiunitary implementing the wedge reflection in the modular construction. The modular relation on this dense family is

$$J\Delta^{1/2}Y\Omega = Y^{*}\Omega.$$

Boost invariance makes the family a core for the operator on the left. Thus a smaller algebra generated by the lattice limits and the candidate wedge algebra have the same modular conjugation. The inclusion between the algebras gives the reverse inclusion between their commutants; conjugation by J closes the inclusion back on the algebras. The detailed argument below uses this to identify the wedge algebra, prove wedge duality, and obtain

$$\mathcal{A}(W)' = \mathcal{A}(W'), \qquad \Delta^{it} = U(B(-2\pi t)).$$

Double cones require a further distinction. For a bounded convex region O, intersect the wedge algebras containing O; for general open regions, take joins over contained double cones. This defines the dual completion $\mathcal{A}^{d}$. The argument establishes its locality and duality, but does not identify it with $\mathcal{A}$ on every double cone. Wedge equality supplies the construction of $\mathcal{A}^{d}$ without settling that additional equality.

Finally, the component argument has two levels. Positivity inequalities pass to every component through the continuous decomposition of the Euclidean functional. The identification of the fibre of a decomposable wedge algebra is a direct-integral statement and holds almost everywhere. In particular, a fibre identity is not promoted to an identity at every point of the component space.

14.1.3. *Principal results.*

Theorem AO (The net of data (Rec))**.** Let σ be data (Rec) with the triple $(C_{\wp}, \wp, c_0)$; uniqueness of the vacuum is not assumed. Every statement holds also with W and W' exchanged.

(a) *Frame.* For every n the spaces $N^{n}\mathcal{D}_0$ and $N^{n}\mathcal{D}_0(O)$, $O \neq \emptyset$ open, are dense, and $\phi(f)$ on $\mathcal{D}_0$ and on $\mathcal{D}_\infty$ have one closure $\bar{\phi}(f)$. $\mathcal{C}(O) = \{Q : Q\bar{\phi}(f) \subset \bar{\phi}(f)Q,\ f \in \mathcal{F}(O)\}$ is a von Neumann algebra, $U(\mathfrak{g})\mathcal{C}(O)U(\mathfrak{g})^{-1} = \mathcal{C}(\mathfrak{g}O)$, and $\mathcal{A}$ is isotonous and covariant. For $O \neq \emptyset$, Ω is separating for $\mathcal{C}(O)$ and cyclic for $\mathcal{A}(O)$; $\bar{\phi}(f)$, $f \in \mathcal{F}(O)$, is affiliated with $\mathcal{A}(O)$, and $\mathcal{A}$ is the least net with this property; $\mathcal{C}(\bigcup_i O_i) = \bigcap_i \mathcal{C}(O_i)$ and $\mathcal{A}(\bigcup_i O_i) = \bigvee_i \mathcal{A}(O_i)$. $\mathcal{L}$ is isotonous, covariant and local; (W) holds if and only if $\mathcal{L} = \mathcal{A}$, and if and only if it holds for spacelike separated upright double cones.

(b) *The order $\wp$.* For $f \in \mathcal{S}$ and $m, n \in \mathbb{N}_0$: (b1) $b_{mn}(f) \leq eC_{\wp}\|f\|_{L^1}(m+n+2)^{\wp}$; (b2) $(|m-n|-1)^{k}b_{mn}(f) \leq b_{mn}(\partial_t^{k}f)$ if $|m-n| \geq 2$, $k \in \mathbb{N}$; (b3) for $\epsilon \in (0,1]$, $k_1 = \max\{0, \lceil(\wp-1)/\epsilon\rceil\}$ and $|m-n| > (m+n+2)^{\epsilon}$: $b_{mn}(f) \leq 3^{k_1}eC_{\wp}\|\partial_t^{k_1}f\|_{L^1}(m+n+2)$, so that the blocks are linearly bounded off every band $|m-n| \leq (m+n+2)^{\epsilon}$, $\epsilon > 0$; (b4) $b_{mn}(h*f) \leq \|h\|_{L^1}b_{mn}(f)$ for $h \in L^1(\mathbb{R}^{1+2})$ with $h*f \in \mathcal{S}$; (b5) with $\mathsf{s}_0(q) = 1+2\cdot 2^{q}+2\cdot 3^{q}$ and $\mathsf{s}(q,k) = \sum_{l\geq 1}(l+3)^{q}l^{-k}$, for real $j \geq 0$ and $\psi, \psi' \in \mathcal{D}_\infty$:

$$\|N^{j}\phi(f)\psi\| \leq 2^{j}e^{2}C_{\wp}\big[\mathsf{s}_0(\wp+j)\|f\|_{L^1} + 2\mathsf{s}(\wp+j,k_j)\|\partial_t^{k_j}f\|_{L^1}\big]\|N^{\wp+j}\psi\|, \qquad k_j = \lfloor\wp+j\rfloor+2,$$

$$|\langle\psi', \phi(f)\psi\rangle| \leq e^{2}C_{\wp}\big[\mathsf{s}_0(\tfrac{\wp}{2})\|f\|_{L^1} + 2\mathsf{s}(\tfrac{\wp}{2},k_\flat)\|\partial_t^{k_\flat}f\|_{L^1}\big]\|N^{\wp/2}\psi'\|\,\|N^{\wp/2}\psi\|, \qquad k_\flat = \lfloor\tfrac{\wp}{2}\rfloor+2,$$

and $D(H^{\wp}) \subset D(\bar{\phi}(f))$; at $\wp = 4$ the two brackets are bounded above by $195\|f\|_{L^1} + 537.82\|\partial_t^{6}f\|_{L^1}$ and $27\|f\|_{L^1} + 37.20\|\partial_t^{4}f\|_{L^1}$.

(c) *Wedge relation.* There is exactly one antiunitary J with $J\Omega = \Omega$, $J\mathcal{D}_0 = \mathcal{D}_0$ and $J\phi(f)\psi = \phi(\bar{f}\circ\mathrm{j})J\psi$ for $\psi \in \mathcal{D}_0$; $J^2 = 1$, $JU(a)J = U(\mathrm{j}a)$, $JU(B(s))J = U(B(s))$ and $J\mathcal{C}(O)J = \mathcal{C}(\mathrm{j}O)$. For $f_1, \ldots, f_n \in \mathcal{F}(W)$ and $Y = \phi(f_1)\cdots\phi(f_n)$: $Y\Omega \in D(\Delta^{1/2})$ and $J\Delta^{1/2}Y\Omega = Y^{\dagger}\Omega$; every boost-invariant dense subspace of $D(\Delta^{1/2})$ is a core of $\Delta^{1/2}$, and $\mathcal{D}_0(W)$ is one. For $Q \in \mathcal{C}(W)$: $JQ\Omega \in D(\Delta^{1/2})$ and $\Delta^{1/2}JQ\Omega = Q^{*}\Omega$. Moreover $\mathcal{C}(W') \subset \mathcal{C}(W)'$ on all of $\mathcal{H}$;

hence $O \mapsto \mathcal{C}(O^\times)$ is isotonous, covariant and local on double cones, $\mathcal{C}(O^\times) \subset \mathcal{A}^d(O)$, and (W) holds if and only if $\mathcal{C}(O^\times) = \mathcal{A}^d(O)$ for all double cones O.

(d) *What locality says.* For every $\epsilon > 0$ there is a real $f_* \in C_c^\infty(\mathbb{R}^{1+2})$, $f_* \geq 0$, $\int f_* = 1$, even in each coordinate, with support in $[-\epsilon, \epsilon]^3$ and $\hat{f}_*(p) > 0$ for all real p; it may be chosen with $\|\partial_t^k f_*\|_{L^1} \leq A_0 R_0^k (k!)^2$ for all k, $A_0 = \frac{e^3\pi}{2}$, $R_0 = \frac{3e\pi^2}{2\epsilon}$. The following are equivalent: (W); $\mathcal{C}(W)\Omega$ is dense; $\mathcal{C}(W)' = \mathcal{C}(W')$; $J\mathcal{A}(W)J \subset \mathcal{C}(W)$; (CP); $\phi(f)\Omega \in \mathcal{H}_0$ for all $f \in \mathcal{F}(W)$; $\phi(f)\Omega \in \mathcal{H}_0$ for all f in a set $\mathfrak{F} \subset \mathcal{S}$ whose Fourier transforms have no common real zero (for instance $\mathfrak{F} = \{f_s\}$, one real f_s with $\hat{f}_s$ free of real zeros); $\inf_{s\in\mathbb{R}} \operatorname{dist}(U(B(s))\phi(f_s)\Omega, \mathcal{H}_0) = 0$ for one such f_s; for one real $f_0 \geq 0$, $f_0 \neq 0$, in C_c^∞, every Lorentz transformation Λ has a translation a_Λ with $\phi(f_0^{(a_\Lambda,\Lambda)})\Omega \in \mathcal{H}_0$; ($\mathrm{QCP}_{f_s}$) for one such f_s with support in an upright double cone K, $\bar{K} \subset W'$.

The first order of (QCP_f) holds for all data (Rec), and (QCP_f) gives operators $Q_n \in \mathcal{C}(W)$ with $\|Q_n\| \leq n$ and $Q_n\Omega \to \phi(f)\Omega$. $\mathsf{D}(f) = \operatorname{dist}(\phi(f)\Omega, \mathcal{H}_0)$ is a seminorm on $\mathcal{S}$, invariant under translations and under the boosts $B(s)$, with $\mathsf{D}(h * f) \leq \|h\|_{L^1}\mathsf{D}(f)$; (W) holds if and only if $\mathsf{D} \equiv 0$. Under (W), $\mathcal{A}(W) = \mathcal{C}(W')$, Ω is cyclic and separating for it, its modular group is $\Delta^{it} = U(B(-2\pi t))$ and its modular conjugation is J. (CP) implies, for real f, g_1, g_2 with pairwise spacelike separated compact supports,

$$|\mathcal{W}(fg_1g_2)|^4 \leq \mathcal{W}(f^2g_1^2)\,\mathcal{W}(f^2g_2^2)\,\mathcal{W}(g_1^2g_2^2),$$

and $\mathcal{W}(p(f_1, f_2)) \geq 0$ for every real polynomial $p \geq 0$ on $\mathbb{R}^2$ and real spacelike separated f_1, f_2; Motzkin's polynomial $x^4y^2 + x^2y^4 - 3x^2y^2 + 1$, which is no sum of squares, is among them.

(e) *Dual net.* Assume (W), and let O, O_1, O_2 be bounded, open and convex. Then $\mathcal{A}^d(O) = \mathcal{C}(O^\times) = \mathcal{A}(O^\times)' \supset \mathcal{A}(O)$ is a von Neumann algebra; $\mathcal{A}^d$ is isotonous and covariant, $J\mathcal{A}^d(O)J = \mathcal{A}^d(\mathrm{j}O)$, Ω is cyclic and separating for $\mathcal{A}^d(O)$, and $\bar{\phi}(f)$, $f \in \mathcal{F}(O)$, is affiliated with it; $\mathcal{A}^d(O_1) \subset \mathcal{A}^d(O_2)'$ for spacelike separated O_1, O_2; $\mathcal{A}^d(O)' = \mathcal{A}(O^\times) = \mathcal{A}^d(O^\times) = \bigvee\{\mathcal{A}(\tilde{W}') : \tilde{W} \supset O \text{ a wedge}\}$; every family $O \mapsto \mathfrak{F}(O)$ of bounded operators with $\mathfrak{F}(O) \subset \mathcal{A}(O_1)'$ for all double cones $O_1 \subset O^\times$ has $\mathfrak{F}(O) \subset \mathcal{A}^d(O)$; $\mathcal{A}^d(\tilde{W}) = \mathcal{A}(\tilde{W})$ for wedges $\tilde{W}$, so $(\mathcal{A}^d)^d = \mathcal{A}^d$; and $\mathcal{A}(O) = \mathcal{A}^d(O)$ if and only if $\mathcal{C}(O^\times) \subset \mathcal{C}(O)'$. $\mathcal{A}^d$ is outer regular: for a double cone $O = (a + V_+) \cap (b - V_+)$ and $O_\epsilon = (a - \epsilon e_0 + V_+) \cap (b + \epsilon e_0 - V_+)$, $\mathcal{A}^d(O) = \bigcap_{\epsilon>0} \mathcal{A}^d(O_\epsilon)$. Without (W): $O \mapsto \bigcap_{\mathfrak{g}W\supset O} \mathcal{A}(\mathfrak{g}W)$ is local if and only if (W) holds.

(f) *Sufficient conditions.* Let $k_0 = \max\{1, \lceil \wp \rceil\}$. If f is real and $|\langle\psi, \phi(\partial_t f)\psi\rangle| \leq c\langle\psi, N\psi\rangle$ on $\mathcal{D}_0$, then $\phi(f)$ is essentially self-adjoint on $\mathcal{D}_0$; if besides $\sup_{m,n} b_{mn}(\partial_t^j f)/(m + n + 2) < \infty$ for $1 \leq j \leq k_0 + 3$, the resolvents of $\bar{\phi}(f)$ keep $D(H^k)$, $k \leq k_0$, off a strip about $\mathbb{R}$, and all of them lie in $\mathcal{C}(O)$ for every open O spacelike separated from $\operatorname{supp} f$. Each of the following implies (W) and (CP). (f1) There are real $f_i \in C_c^\infty$, $\operatorname{supp} f_i \subset \{|x| \leq \delta_i\}$, $\delta_i \to 0$, $\int f_i = 1$, $\sup_i \|f_i\|_{L^1} < \infty$, and numbers $c_i, \epsilon_i > 0$, nothing uniform in i, with $b_{mn}(\partial_t f_i) \leq c_i(m + n + 2)(1 + \log(m + n + 2))$ for $|m - n| \leq (m + n + 2)^{\epsilon_i}$. (f2) There is a real $f_0 \in C_c^\infty$, $f_0 \geq 0$, $f_0 \neq 0$, even in t, with $\langle\psi, \phi(\partial_t f_0)\psi\rangle \leq c\langle\psi, N\psi\rangle$ on $\mathcal{D}_0$. (f3) (QA_{f_0}) holds for one real $f_0 \in C_c^\infty$, $f_0 \geq 0$, $f_0 \neq 0$. (f4) $\wp \leq 1$, or $\|\mathsf{B}(s, s)\| \leq Cs^{-1}(1 + \log s^{-1})$ for $0 < s \leq 1$; then every $\phi(f)$, $f \in \mathcal{S}$ real, is essentially self-adjoint on $\mathcal{D}_0$, and the closures commute strongly at spacelike separation. (f5) There are bounded operators $\mathsf{B}_h^e(s, s')$, $s, s' > 0$, real linear in $h \in C_c^1(\mathbb{R}^2)$, with $e^{-tH}\mathsf{B}_h^e(s, s')e^{-t'H} = \mathsf{B}_h^e(s + t, s' + t')$, $\mathsf{B}_h^e(s, s')^* = \mathsf{B}_h^e(s', s)$ and $\| \int \rho_e(\varsigma/\eta)\mathsf{B}_h^e(\eta + \varsigma, \eta - \varsigma)\,\mathrm{d}\varsigma\| \leq C_e|h|_{C^1}$ for all $\eta = \frac{1}{4(k+2)}$, $k \in \mathbb{N}_0$, for one measurable $\rho_e \geq 0$ on $[-\frac{1}{2}, \frac{1}{2}]$ with $\bar{\rho}_e = \int \rho_e > 0$; and with the time average $\mathsf{e}(f) = \int q_{f_t}(e^{-itH}\cdot, e^{-itH}\cdot)\,\mathrm{d}t$ of the sharp-time forms q_h of B^e (Lem. 14.10), the boosts

$B_j(s)$ in the direction e_j, one $s>0$ and real c,c':

$$\langle\psi,\phi(f)\psi\rangle=\mathsf{c}\Big[(1+2\coth^2 s)\,\mathsf{e}(f)(\psi,\psi)-\frac{1}{2\sinh^2 s}\sum_{j=1,2}\sum_{\pm}\mathsf{e}(f\circ B_j(\pm s))(\psi_{j,\pm},\psi_{j,\pm})\Big]+\mathsf{c}'\int f\,\|\psi\|^2$$

for real $f\in C_c^\infty$ and $\psi\in\mathcal{D}_0$, $\psi_{j,\pm}=U(B_j(\pm s))^{-1}\psi$. Then (LB_f) holds for every real $f\in C_c^\infty$. If the bound of (f2) holds at $f_0=f_*$, the function of (d), then (f1) holds with the functions $f_i=f_**\mu_i$, $\mu_i\in\mathcal{S}$; this route to (W) does not use the wedge relation (c).

(g) *Mixtures and limits.* Let $(\mathsf{X},\lambda,(\sigma_\xi))$ be a canonical decomposition of σ with all σ_ξ data (Rec) with one triple. For real $u\in\mathcal{S}$ and $c\geq 0$, $(\mathrm{LB}_{u,c})$, and each of $(\mathrm{LB}^{\pm}_{u,c})$, holds for σ if and only if it holds for every σ_ξ; so do (CP) and (W); and every inequality $\mathcal{W}(a)\geq 0$, $a\in\mathfrak{P}$ fixed, passes from all σ_ξ to σ. If data (Rec) σ_j with one triple converge to σ on separated tuples, then $\mathcal{W}^{\sigma_j}(f_1\cdots f_n)\to\mathcal{W}^{\sigma}(f_1\cdots f_n)$ for all $f_k\in\mathcal{S}$: the real-time functions converge; every inequality $\mathcal{W}(a)\geq 0$, $a\in\mathfrak{P}$, passes to σ, and $(\mathrm{LB}^{\pm}_{u,c})$, (CP) and (W) pass from all σ_j to σ.

(h) *Two bounds towards a linear one.* (h1) Let $\mathsf{l},r>0$, $Q\geq 1$, $c_2\geq 0$, $\mathfrak{m}\in\mathbb{R}$, $h'\in C_c^\infty(\mathbb{R}^2)$ real, $\rho_1\geq 0$ even, of class C^1, carried by $(-\frac14,\frac14)$, $\int\rho_1=1$, and let $\hat D(x)$, $x\in[-r,r]$, be bounded operators with: (i) $x\mapsto\hat D(x)$ is the restriction of a function holomorphic in norm on a complex neighbourhood of $[-r,r]$; (ii) $\hat D(x)=\hat D(x)^*\geq 0$ and $\|\hat D(x)\|\leq Q$; (iii) $\hat D(0)=e^{-\mathsf{l}H}$; (iv) $\hat D'(0)=\mathsf{l}-\mathfrak{m}e^{-\mathsf{l}H}$ with $\mathsf{l}=\int\mathsf{l}^{-1}\rho_1(\varsigma/\mathsf{l})\mathsf{B}_{h'}(\frac{\mathsf{l}}{2}+\varsigma,\frac{\mathsf{l}}{2}-\varsigma)\,\mathrm{d}\varsigma$; (v) for every $\Lambda<\infty$ and every ψ in $\mathcal{H}_\Lambda=\mathbf{1}_{[0,\Lambda]}(H)\mathcal{H}$: $\limsup_{\epsilon\downarrow 0}\sup_{|x|\leq r}\frac{d^2}{dx^2}\langle\psi,-\log(\hat D(x)+\epsilon)\psi\rangle\leq c_2\langle\psi,N\psi\rangle$. Then for $\psi\in\mathcal{D}_\infty$

$$\big|\langle\psi,\phi(g_{\mathsf{l}}\otimes h')\psi\rangle-\mathfrak{m}\|\psi\|^2\big|\leq\frac{\mathsf{l}\langle\psi,H\psi\rangle+\log(Q+1)\|\psi\|^2}{r}+\frac{r}{2}\,c_2\langle\psi,N\psi\rangle,$$

where $\hat g_{\mathsf{l}}(E)=\mathsf{G}(\mathsf{l}E)$, $\mathsf{G}(y)=\int\rho_1(s)\cosh(sy)\,\mathrm{d}s\,\frac{y/2}{\sinh(y/2)}$. No injectivity of $\hat D(x)$ is assumed, and under (i)–(iv) the slope is the real-time average of the field: $\lim_{\epsilon\downarrow0}\frac{d}{dx}\langle\psi',\log(\hat D(x)+\epsilon)\psi\rangle|_{x=0}=\langle\psi',\phi(g_{\mathsf{l}}\otimes h')\psi\rangle-\mathfrak{m}\langle\psi',\psi\rangle$ for $\psi,\psi'\in\mathcal{H}_\Lambda$. If such bounds hold at all $\mathsf{l}=L^{-i}$ with constants $\leq C\mathsf{l}^{-N_0}$, then $(\mathrm{LB}_{k\otimes h'})$ holds for every real $k\in\mathcal{S}(\mathbb{R})$. (h2) Let $\wp<6$. For a unit timelike vector $n\in V_+$, $\mu\geq 1$, Borel sets $\Delta,\Delta'\subset\{p\in\bar V_+ : n\cdot p\leq\mu\}$ and $u\in\mathcal{S}$: $\|\mathcal{P}(\Delta')\phi(u)\mathcal{P}(\Delta)\|\leq e^2C_\wp\|u\|_{L^1}\mu^\wp$. For $\mu\geq 1$, $u\in\mathcal{S}$ and vectors ψ,ψ' of bounded energy in the spectral subspace of $\{p\cdot p\leq\mu^2\}$:

$$|\langle\psi',\phi(u)\psi\rangle|\leq C_{\mathrm{L}}(\|u\|_{L^1}+\|\mathcal{L}^4u\|_{L^1})\mu^\wp\|\psi'\|\|\psi\|,\qquad\mathcal{L}=-\partial_t^2-\partial_1^2-\partial_2^2,$$

with $C_{\mathrm{L}}=100e^2C_\wp\max\{18^\wp,6^\wp c\,\Sigma_\wp\}$, $\Sigma_\wp=\sum_{r\geq1}(r+1)r^{-8}(2r+3)^\wp$ and a numeral c, formed from $C_\wp,\wp$ alone; for the following two consequences assume additionally $\wp>0$. Then for all m,n, with $\mathcal{P}_<$ the spectral projection of $\{p\cdot p\leq(m+n+2)^{2/\wp}\}$: $\|\mathcal{P}_<P_m\phi(u)P_n\mathcal{P}_<\|\leq C_{\mathrm{L}}(\|u\|_{L^1}+\|\mathcal{L}^4u\|_{L^1})(m+n+2)$. With Π the spectral projection of $\{p:p\cdot p\leq(1+p^0)^{2/\wp}\}$, $\|\Pi e^{-sH}\phi(u)e^{-sH}\Pi\|\leq c(u)s^{-1}$ for $0<s\leq1$, $c(u)$ formed from C_{L} and the norms $\|\partial_t^ju\|_{L^1}+\|\mathcal{L}^4\partial_t^ju\|_{L^1}$, $j\leq2$. For data (Rec) of any order $\wp$ and real $f\in\mathcal{S}$, energy and invariant mass together bound the field: $\pm\phi(f)\leq c(f)(1+H+M^2)^{\wp/2}$ as forms on the vectors of bounded energy, $M^2=\int(E^2-|q|^2)\,\mathrm{d}\mathcal{P}$ the operator of the invariant mass squared, $c(f)$ formed from $C_\wp$, $\wp$ and finitely many of the norms $\|\partial_\nu^kf\|_{L^1}$, $\nu=0,1,2$.

■□

Theorem AP (Locality of the continuum limits)**.** Let $0<g\leq\gamma$, let σ be the centred functional of a volume-first limit at g, and let τ be σ or a component of a canonical decomposition of σ; every such limit, decomposition and component is admitted. On $\mathcal{H}(\tau)$:

(a) $\mathcal{C}_\tau(W')\Omega$ is dense, and (W) holds: $O\mapsto\mathcal{A}_\tau(O)$ is an isotonous, covariant and local net of von Neumann algebras, $\mathcal{L}_\tau=\mathcal{A}_\tau$, and $\mathcal{H}_0=\mathcal{H}(\tau)$.

(b) $\mathcal{A}_\tau(W) = \mathcal{C}_\tau(W') = \mathcal{C}_\tau(W)'$ and $\mathcal{A}_\tau(W)' = \mathcal{A}_\tau(W')$; Ω is cyclic and separating for $\mathcal{A}_\tau(W)$, with modular group $\Delta^{it} = U(B(-2\pi t))$ and modular conjugation J; in particular $\mathcal{A}_\tau(W) \neq \mathbb{C}1$ unless $\mathcal{H}(\tau) = \mathbb{C}\Omega$.

(c) For open O and $f \in \mathcal{F}(O)$, $\bar\phi(f)$ is affiliated with $\mathcal{A}_\tau(O)$.

(d) (CP) holds, with the two inequalities of Thm. AO(d).

(e) The dual net $\mathcal{A}^d_\tau(O) = \mathcal{C}_\tau(O^\times) = \mathcal{A}_\tau(O^\times)'$ has all the properties of Thm. AO(e): it is local, Haag dual, the largest family local relative to $\mathcal{A}_\tau$, and its own dual net.

(f) Let $\tau = \sigma$, let Λ_j be the tori of its truncated functions and $\mathcal{M}_{\Lambda_j}$ the algebras of all bounded operators on the bonds of the right halves of their time-zero planes (Prop. 14.13). Every weak limit Q, along a subsequence, of operators $Q_j \in \mathcal{M}_{\Lambda_j}$ lies in $\mathcal{C}_\sigma(W')$; the lattice modular flows converge to the boosts, $\Delta^{it}_{\Lambda_j}\psi_j \Rightarrow U(B(-2\pi t))\psi$ in the sense of Lem. 14.1; for real $u \in C^\infty_c(\mathsf{l})$ and $\epsilon > 0$ there is $T^{(\epsilon)}_u \in \mathcal{C}_\sigma(W')$, a weak limit of operators of $\mathcal{M}_{\Lambda_j}$, with $\|T^{(\epsilon)}_u\| \le (2\sqrt{\epsilon})^{-1}$ and $\|T^{(\epsilon)}_u\Omega - \phi(f_u)\Omega\| \le \frac{\sqrt{\epsilon}}{2}\pi e^{5\pi^2/8}\|[\theta_1 u \cdot u]\|$, f_u the function of Lem. 14.2(c); and $\mathcal{A}_\sigma(W)$ is the von Neumann algebra generated by the $U(\mathfrak{g})QU(\mathfrak{g})^{-1}$, $\mathfrak{g}W \subset W$, with Q or Q^* such a weak limit.

(g) For $\tau = \sigma$ and a canonical decomposition: $[G] \mapsto ([G]_\xi)_\xi$ is a unitary of $\mathcal{H}(\sigma)$ onto $\int^\oplus \mathcal{H}(\sigma_\xi)\,\lambda(\mathrm{d}\xi)$, by which we identify the two, $[G] = \int^\oplus [G]_\xi$; the semigroup, the translations, the insertions, the operators $\mathsf{B}(s,s')$ and the fields decompose; the continuous functions on X act by multiplication (the operators of Lem. 14.7(i)), generate the diagonal algebra and lie in the centre of $\mathcal{A}_\sigma(W)$; and $\mathcal{A}_\sigma(W) = \int^\oplus \mathcal{A}_{\sigma_\xi}(W)\,\lambda(\mathrm{d}\xi)$: for λ-almost every ξ, among them every ξ with $\lambda(\{\xi\}) > 0$, the fibre at ξ of the algebra of (f) is the wedge algebra of the component σ_ξ.

(h) *A supply.* Let σ' be a limit of periodic Gibbs states ρ_j at g, $c \in \bigcup_K L^{-K}\mathbb{Z}$, and B_j Borel functions, $|B_j| \le \mathsf{b}$, of finitely many bonds of $L^{-K_j}\mathbb{Z}^3$ in $\{x_3 = 0,\ x_1 \le c\}$. Then $|\rho_j(\overline{\theta(F')_j}B_j(F)_j)| \le \mathsf{b}\,\rho_j(\overline{\theta(F')_j}(F')_j)^{1/2}\rho_j(\overline{\theta(F)_j}(F)_j)^{1/2}$ for F, F' over $\mathbb{E}_+$ and large j; along a subsequence $\rho_j(\overline{\theta(F')_j}B_j(F)_j) \to \langle [F'], Q[F]\rangle$ for all such F, F', with $\|Q\| \le \mathsf{b}$; and every such Q lies in $\mathcal{C}_{\sigma'}(W + ce_1)$. The same holds for every component σ'_ξ of a canonical decomposition of σ', with ρ_j replaced, as the states that integrate, by the periodic Gibbs states $\rho_{j,\xi}$ that approximate the component (Thm. YM3_I:AP among the construction results, recalled in the results used), the ρ_j still giving the centring.

By contrast, on the space of a component an operator $e^{-sH}V_1e^{-sH}$, $s > 0$, V_1 bounded, lies in no $\mathcal{C}(O)$, $O \neq \emptyset$ open, unless it is 0 or the space is $\mathbb{C}\Omega$: weak limits of the operators $[G] \mapsto [D_j \cdot \tau_{te_3}G]$ of lattice functions D_j placed strictly between two times, in $\{s < x_3 < t - s\}$ with $s, t \in \bigcup_K L^{-K}\mathbb{Z}$, whose inner operators $[G] \mapsto [\tau_{-se_3}D_j \cdot \tau_{(t-2s)e_3}G]$ are bounded uniformly in j, on a nontrivial component space are nonlocal unless they vanish (Prop. 14.3(a)).

■□

Theorem AQ (Energy bounds of the field)**.** Let $0 < g \le \gamma$ and let σ' be a limit of periodic Gibbs states at g or a component of a canonical decomposition of one. Let $f \in \mathcal{S}$, $m, n \in \mathbb{N}_0$, $\psi, \psi' \in \mathcal{D}_\infty$, and let $c_V, \bar\Phi, c'_V, \Phi'$ be the numbers of Lem. 14.15; none of them depends on the cutoff, the sequence, σ' or ξ.

(a) *Order* 3. With $\mathsf{C}_3 = \frac{4\sqrt{8}}{\varrho}L^4\,40M(40 + 8L^2)e^{\frac{8}{3}c_V + \frac{1}{2}\bar\Phi}$ and $\mathsf{c}_k = 27L^3\mathsf{C}_3\|\partial_t^k f\|_\star$: $b_{mn}(f) \le \mathsf{c}_0(m + n + 2)^3$ for all m, n, and $b_{mn}(f) \le \mathsf{c}_k(m + n + 2)^3(|m - n| - 1)^{-k}$ if $|m - n| \ge 2$, $k \in \mathbb{N}$; $\|\phi(f)\psi\| \le 8(71\mathsf{c}_0 + 6.3\mathsf{c}_5)\|N^3\psi\|$, and $\|N^a\phi(f)\psi\| \le c_a(f)\|N^{a+3}\psi\|$ for real a, so that $D(N^3) \subset D(\bar\phi(f))$; $|\langle\psi', \phi(f)\psi\rangle| \le 8(17.1\mathsf{c}_0 + 5.7\mathsf{c}_3)\|N^{3/2}\psi'\|\|N^{3/2}\psi\|$; and $\|e^{-sH}\phi(f)e^{-sH}\| \le c(f)s^{-3}$ for $0 < s \le 1$. If f is real and $\|\partial_t^k f\|_\star \le A^{k+1}(k!)^{\mathsf{s}}$ for all $k \ge 0$, with $\mathsf{s} \ge 1$ (such $f \ge 0$, $f \neq 0$, of compact support exist for every $\mathsf{s} > 1$), then $\|\phi(f)^n\Omega\| \le C^n(n!)^{3\mathsf{s}}$ for all n, with $C = C(A, \mathsf{s}, L^3\mathsf{C}_3)$.

(b) *Order* 2. With $\mathsf{C}_2 = \frac{4(\sqrt{8}+1)}{\varrho} L^4\, 40M(40+8L^2)e^{\frac{8}{3}c'_V+\frac{1}{4}\Phi'}$: $b_{mn}(f) \le 27L^2\mathsf{C}_2\|f\|_\star(m+n+2)^2$, and $b_{mn}(f) \le 27L^2\mathsf{C}_2\|\partial_t^k f\|_\star(m+n+2)^2(|m-n|-1)^{-k}$ if $|m-n| \ge 2$; $\|\phi(f)\psi\| \le c(f)\|N^2\psi\|$, $|\langle\psi', \phi(f)\psi\rangle| \le c'(f)\|N\psi'\|\|N\psi\|$ and $\|e^{-sH}\phi(f)e^{-sH}\| \le c''(f)s^{-2}$ for $0 < s \le 1$; and for f as at the end of (a), $\|\phi(f)^n\Omega\| \le C^n(n!)^{2\mathsf{s}}$ with $C = C(A, \mathsf{s}, L^2\mathsf{C}_2)$.

(c) *At a point; at one time.* Here Thms. YM3_I:AJ and YM3_I:AD are used at every radius and for every admitted choice of the constants (σ^-, γ), at all couplings at once. With $q = \mathfrak{p}_0 + \frac{1}{6}$ and C formed from $\sigma^-, \gamma, \varrho, g$: $\|\mathsf{B}(s,s')\| \le C\mu^{-7/2}(1+\log\mu^{-1})^q$, $\mu = \min\{s, s', \frac{1}{2}\}$; so for every $\epsilon > 0$, σ' is data (Rec) with a triple $(C_\epsilon, \frac{7}{2}+\epsilon, \frac{1}{2})$, and Thm. AO applies at the order $\frac{7}{2}+\epsilon$. For real $k \in C^1_c(\mathbb{R}^2)$ carried by a square of side 2: $\|\mathsf{B}_k(s,s')\| \le C|k|_{C^1}\mu^{-5/2}(\log\frac{e}{\mu})^q$, $\mu = \min\{s, s', 1\}$; so $b_{mn}(f) \le c(f)(m+n+2)^{5/2}(1+\log(m+n+2))^q$ for all m, n.

(d) *Sharpness.* Let $0 < g \le \gamma_{\mathrm{sd}}$ and let σ' be a component. For continuous $k \ge 0$, $k \ne 0$, of compact support, $\liminf_{u\downarrow 0} u^2\|\mathsf{B}_k(u,u)\| \ge (\pi c_{\mathrm{sd}}/32)^{1/2}\|k\|_{L^2}$; so the field at one time, the form $\mathsf{T}(k)$ on the vectors of bounded energy, has no bound $\pm\mathsf{T}(k) \le cN$. $\|\mathsf{B}(s,s)\| \ge \|\mathsf{B}(s,s)\Omega\| \ge \frac{1}{8}c_{\mathrm{sd}}^{1/2}s^{-3}$ for $0 < 2s \le r_{\mathrm{sd}}$: the infimum of admissible pure-power exponents in the point bound lies in $[3, \frac{7}{2}]$. For $0 \ne h \ge 0$ in $C^1_c(\mathbb{R}^2)$, $\|\mathsf{I}_\mathsf{l}(h)\Omega\| \ge c(h)\mathsf{l}^{-2}$ for small l; so no bound $\|\mathsf{I}_\mathsf{l}(h)\| \le C\mathsf{l}^{-p}|h|_{C^1}$ holds with $p < 2$, and the exponent 2 of the plates in Lem. 14.15(c) is exact. For $0 \ne k \ge 0$ in $C^1_c(\mathbb{R}^2)$, $\epsilon \in (0,1]$ and $q' < \frac{3}{2}\epsilon$:

$$\sup\{\|T_{mn}(k)\|(m+n+2)^{-q'} :\ m,n \ge 0,\ |m-n| \le (m+n+2)^\epsilon\} = \infty;$$

in particular no bound $\|T_{mn}(k)\| \le c(m+n+2)$ holds on a band with $\epsilon > \frac{2}{3}$.

(e) *Slab families.* Let σ' be a component or the centred functional of a volume-first limit, $\mathsf{l} = L^{-i}$, $i \ge 1$, $\vec{y}^* \in L^{-i-2}\mathbb{Z}^2$, $h' \ne 0$ real and carried by $\{|\vec{x} - \vec{y}^*|_\infty \le \frac{\mathsf{l}}{4}\}$, ρ_1 as in Thm. AO(h1), and $f(x) = \mathsf{l}^{-1}\rho_1(x_3/\mathsf{l})h'(\vec{x})$. The limits $\mathbb{D}(z)$, along a subsequence, of the lattice operators $e^{z\mathcal{O}(f)-\mathcal{E}(z)}$ between reflection-positive vectors, $\mathcal{E}$ the normaliser of the one-cell bound (Prop. YM3_I:89.1), are holomorphic on $|z| < \varrho/\|f\|_\sharp$ with $\|\mathbb{D}(z)\| \le \bar{\mathsf{Q}}$, a number free of l, f, ξ, $\mathbb{D}(0) = e^{-\mathsf{l}H}$ and $\mathbb{D}'(0)$ the insertion of f plus a multiple of $e^{-\mathsf{l}H}$, and $\hat{D}(x) = \mathbb{D}(x)/\langle\Omega, \mathbb{D}(x)\Omega\rangle$, $|x| \le \varrho/(2(\bar{\mathsf{Q}}+1)\|f\|_\sharp)$, has (i)–(iv) of Thm. AO(h1) with $Q = 2\bar{\mathsf{Q}}$ and $\mathfrak{m} = q_0(\Omega,\Omega)\int h'$. ■□

Theorem AR (Examples: Wick powers of free fields; free photons)**.**

(a) *Gaussian families.* Let (M, μ) be a σ-finite measure space with $L^2(\mathsf{M}, \mu)$ separable, $\omega > 0$ and $\vec{p}$ measurable on it with $|\vec{p}| \le \omega$, κ_α, α in a finite set A, measurable, with signs ε_α, such that $\|\omega^{-1/2}\kappa_\alpha e^{-s\omega}\| \le c_\kappa s^{-\delta}$ for $0 < s \le 1$, $\delta \ge 0$, and $G = (G_{\alpha\beta})$ real and continuous on $\mathbb{R}^3 \setminus \{0\}$ with $G_{\alpha\beta}(-x) = G_{\beta\alpha}(x)$, such that the vectors $e^\alpha_x = \kappa_\alpha e^{-x_3\omega - i\vec{x}\cdot\vec{p}}$, $x \in \mathbb{E}_+$, have the inner products $\langle e^\alpha_x, e^\beta_y\rangle = \varepsilon_\alpha G_{\alpha\beta}(\theta x - y)$.

Let O(3) act on $\mathbb{R}^\mathsf{A}$ by a homomorphism ρ_A into the orthogonal matrices, with $G(\mathsf{R}x) = \rho_\mathsf{A}(\mathsf{R})G(x)\rho_\mathsf{A}(\mathsf{R})^{\mathrm{T}}$ for $\mathsf{R} \in \mathrm{O}(3)$ and $\rho_\mathsf{A}(\theta) = \mathrm{diag}(\varepsilon_\alpha)$: the reflection acts on the components by the signs (for photons A indexes the three derivatives, times the colours, and $\rho_\mathsf{A}(\mathsf{R}) = \mathsf{R}\otimes 1$; for a scalar family ρ_A is trivial). Let

$$\mathsf{P}(t) = \sum_{\underline{\alpha}} \mathsf{p}_{\underline{\alpha}} t_{\alpha_1}\cdots t_{\alpha_k} \ne 0$$

be real and k-homogeneous, $k \ge 1$, with a symmetric real tensor p, and invariant, $\mathsf{P}(\rho_\mathsf{A}(\mathsf{R})t) = \mathsf{P}(t)$ for $\mathsf{R} \in \mathrm{O}(3)$; $|\mathsf{p}|_1 = \sum|\mathsf{p}_{\underline{\alpha}}|$.

The Schwinger functions of $:\mathsf{P}(\Phi):$ for the Gaussian field $\Phi = (\Phi_\alpha)$ of the covariance G, that is, the sums over the complete pairings of n vertices with k legs each, no leg paired within its vertex, of the products of the $G_{\alpha\beta}(x_i - x_l)$ weighted by p, are data (Rec) with $C_\wp = 2^\wp 2^k|\mathsf{p}|_1(2^\delta c_\kappa)^k(k/e)^{k/2}$, $\wp = k(\delta + \frac{1}{2})$, $c_0 = \frac{1}{2}$, and a unique vacuum. A unitary V

carries $\mathcal{H}$ onto a closed subspace of the Fock space over $L^2(\mathsf{M},\mu)$ that reduces $d\Gamma(\omega)$, $d\Gamma(\vec{p})$ and the Fock operators of the field at a point, with $V\Omega=\Omega_{\mathrm{F}}$ and $VHV^{-1}=d\Gamma(\omega)$ there, and $\langle\chi,V\phi(f)\psi\rangle=\Phi_f(\chi,V\psi)$ for every $\chi\in\mathcal{D}_\infty(d\Gamma(\omega))$ and $\psi\in\mathcal{D}_\infty$, Φ_f the form of the smeared Wick polynomial.

(b) *Growth.* If $k\geq 3$ and for every $u\in C_c^\infty\setminus\{0\}$ the function

$$w_u(m_1,\dots,m_k)=\hat{u}(\sum_i\omega(m_i),\sum_i\vec{p}(m_i))\sum_{\underline{\alpha}}\mathsf{p}_{\underline{\alpha}}\prod_i\kappa_{\alpha_i}(m_i)$$

lies in $L^2(\mathsf{M}^k)\setminus\{0\}$, then $\|\phi(u)^n\Omega\|\geq c_u^n((kn)!)^{1/2}$ for real $u\in C_c^\infty\setminus\{0\}$; so (QA_u) fails, no bound $|\langle\psi,\phi(u)\psi\rangle|\leq c\langle\psi,N\psi\rangle$ holds at any real $u\neq 0$ in C_c^∞, no bound $b_{mn}(u)\leq\beta(m+n+2)^q$ with $q<\frac{k}{2}$ holds on any band $|m-n|\leq(m+n+2)^\epsilon$, and no triple of the datum has $\wp<\frac{k}{2}$.

(c) *Wick powers.* For the free field χ of mass $\mathsf{m}>0$ and $\mathsf{P}=t^k$, (a) holds with every $\delta\in(0,\frac12]$ and $c_\kappa=(8\pi\delta)^{-1/2}(2\mathsf{m})^{-\delta}$, so $:\chi^k:$ is data (Rec) of every order $\wp>\frac{k}{2}$; for $k\geq 2$, $w_u\neq 0$ as soon as $\hat{u}$ does not vanish identically on $\{E>(|\vec{k}|^2+k^2\mathsf{m}^2)^{1/2}\}$, which holds for every $u\in C_c^\infty\setminus\{0\}$, for every $u\in\mathcal{S}\setminus\{0\}$ that vanishes off $\bar{W}$, and for $u=\mathsf{g}\otimes h'$ with $\hat{\mathsf{g}}$ free of real zeros and $h'\in C_c^\infty(\mathbb{R}^2)\setminus\{0\}$; so (b) holds for $k\geq 3$. $:\chi^6:$ is data (Rec) with $(93312(\pi e)^{-3}\mathsf{m}^{-1},4,\frac12)$, a unique vacuum, $\operatorname{spec}H\subset\{0\}\cup[2\mathsf{m},\infty)$ and $S_2(x)=720(4\pi|x|)^{-6}e^{-6\mathsf{m}|x|}$. (W) and (CP) hold for every $:\chi^k:$, $k\geq 1$, $\mathsf{m}>0$; for every real homogeneous Wick polynomial in the first derivatives $\partial_\alpha\varphi^a$ of n massless free fields $\varphi=(\varphi^a)_{a\leq\mathsf{n}}$ that is invariant under $\mathrm{O}(3)$ and $\mathrm{O}(\mathsf{n})$, for instance $:(\partial\varphi\cdot\partial\varphi)^2:$ with $\wp=6$; and for the Wick cube of the generalised free field with the kernel r^{-2} ($\wp=3$, $S_2=6r^{-6}$, no gap).

So neither (LB_u) nor (QA_u), at any u, is necessary for (W) among data (Rec) with $\wp=4$, $c_0=\frac12$, a unique vacuum, a gap and $S_2\sim r^{-6}$; and for every $\wp>\frac32$ there are data (Rec) of order $\wp$ without (LB_u) at any $u\neq 0$.

(d) *Free photons.* Let φ be the free massless field on $\mathbb{R}^{1+2}$ and $\Theta=:\partial_\lambda\varphi\,\partial^\lambda\varphi:$. The Schwinger functions of $:\partial_\alpha\Phi\,\partial_\alpha\Phi:$ for the Euclidean massless free field Φ on $\mathbb{R}^3$ (the pairings of $-\partial_a\partial_b(4\pi|x|)^{-1}$) are data (Rec) with $(24/\pi e,3,\frac12)$, a unique vacuum and $S_2=12/16\pi^2r^6$; their space is the whole even Fock space, with $V\phi(f)V^{-1}=-\Theta(f)$. For real $f\in\mathcal{S}$: $\pm\Theta(f)\leq 16\pi|f|_8\,N$, $|f|_8$ the Schwartz seminorm in equation YM3_I:(1873) with $|\hat{f}(k)|\leq 16\pi^2|f|_8(1+|k|^2)^{-2}$; $\|\Theta(f)\psi\|\leq c(f)\|N\psi\|$; $\|\Theta(f_1)\cdots\Theta(f_n)\Omega\|\leq 2^n n!\prod_k c(f_k)$; and $\Theta(f)$ is essentially self-adjoint on every core of H; (W) and (CP) hold. $\|\mathsf{B}(s,s)\|\geq\|\mathsf{B}(s,s)\Omega\|=\frac{\sqrt{3}}{16\pi}s^{-3}$, so no triple has $\wp<3$, while on its own space the order is exactly 1: for real $f\neq 0$, $\alpha<1$ and every c some $\psi\in\mathcal{D}_\infty$ has $|\langle\psi,\phi(f)\psi\rangle|>c\langle\psi,N^\alpha\psi\rangle$. At one time, with $\Theta_k=\int k(\vec{x})\Theta(0,\vec{x})d^2x$, $k\in\mathcal{S}(\mathbb{R}^2)$ real, and $G=m+n+2$:

$$\|P_m\Theta_kP_n\|\leq(1+\sqrt{3})\|k\|_\infty G+2\mathsf{n}_k(|m-n|+2)\sqrt{G},\qquad \mathsf{n}_k=\tfrac{1}{2\pi}\big[10.4\|k\|_2^2+1.6\|k\|_2\|\nabla k\|_2\big]^{1/2},$$

and the second term is attained: for $k\neq 0$, every integer $\omega\geq\max\{3,7\xi_0(k)+1\}$ and every $j\in\mathbb{N}$ there are m,n with $|n-m-\omega|\leq 1$, $2j(\omega-\frac12)+\omega-1<G\leq(j+1)(2\omega+3)$ and $\|P_n\Theta_kP_m\|\geq 3\cdot10^{-3}\|k\|_2\,\omega\sqrt{G}-\kappa_k(\omega-2)G$, where $\kappa_k(R)=(2\pi)^{-2}\int_{|\xi|>R}|\tilde{k}|$ and $\xi_0(k)$ is the least R with $\int_{|\xi|\leq R}|\tilde{k}|^2\geq\frac12\|\tilde{k}\|_2^2$. So for real $k\neq 0$ in C_c^∞ the band bound $\|P_me^{-H/G}\Theta_ke^{-H/G}P_n\|\leq CG$ on $|m-n|\leq G^\epsilon$ holds if and only if $\epsilon\leq\frac12$.

(e) n *colours.* Let $\Theta=\sum_{a\leq\mathsf{n}}:\partial_\lambda\varphi^a\partial^\lambda\varphi^a:$, $\mathsf{n}\geq 1$. The closed span of Ω and the $\Theta(f_1)\cdots\Theta(f_n)\Omega$ ($f_k\in\mathcal{S}$, or $f_k\in C_c^\infty$ real) is the space of all $\mathrm{O}(\mathsf{n})$-invariant vectors, and it is the closure of the Euclidean vectors; with $\mathrm{SO}(3)$ in the place of $\mathrm{O}(3)$ at $\mathsf{n}=3$ this is false. $\|\mathsf{B}(s,s)\Omega\|=\frac{\sqrt{3\mathsf{n}}}{16\pi}s^{-3}$. For real $f\neq 0$, $\alpha<1$, every c and either sign there is $\psi\in\mathcal{D}_0$ with $\pm\langle\psi,\phi(f)\psi\rangle>c\langle\psi,N^\alpha\psi\rangle$, while $\pm\phi(f)\leq\mathsf{n}\,16\pi|f|_8N$: on the invariant space the order is exactly 1, and $\sup_{m,n}b_{mn}(f)(1+\max\{m,n\})^{-p}=\infty$ for every $p<1$.

(f) *The net of the action density of free photons.* Let $\mathsf{n} \geq 1$ and O be a double cone. For a family Φ of Wightman fields on the Fock space of n free fields $\varphi = (\varphi^a)_{a \leq \mathsf{n}}$ of a mass $\mathsf{m} \geq 0$, among them $\Theta = \sum_a :\partial_\lambda \varphi^a \partial^\lambda \varphi^a:$, $S = :\varphi \cdot \varphi: = \sum_a :\varphi^a \varphi^a:$ and the multiplet $(:\partial_\mu \varphi \cdot \partial_\nu \varphi:)_{\mu\nu}$: $\mathcal{P}_\Phi(O)$ is the algebra of the polynomials in the $\Phi(f)$, $f \in \mathcal{F}(O)$, $\mathcal{D}_0^\Phi = \mathcal{P}_\Phi(\mathbb{R}^{1+2})\Omega$ and $\mathcal{H}_\Phi$ its closure, $\mathcal{C}_\Phi(O)$ the set of the bounded Q on $\mathcal{H}_\Phi$ with $\langle \Phi(\bar{f})\psi, Q\psi' \rangle = \langle \psi, Q\Phi(f)\psi' \rangle$ for $f \in \mathcal{F}(O)$ and $\psi, \psi' \in \mathcal{D}_0^\Phi$, $\mathcal{A}_\Phi(O) = \mathcal{C}_\Phi(O)'$, and $\mathcal{A}_\Phi^d(O) = \bigcap\{\mathcal{A}_\Phi(\tilde{W}) : \tilde{W} \supset O \text{ a wedge}\}$ its dual net.

At the mass 0 the unitary V of (a), for the data of (d), (e), carries $\mathcal{D}_0$, $\mathcal{C}(O)$, $\mathcal{A}(O)$ and $\mathcal{A}^d(O)$ of the datum onto $\mathcal{D}_0^\Theta$, $\mathcal{C}_\Theta(O)$, $\mathcal{A}_\Theta(O)$ and $\mathcal{A}_\Theta^d(O)$, so that Thm. AO applies to Θ; there $\mathcal{H}_\Theta$, $\mathcal{H}_S$ and the space of the multiplet are all the space of the $\mathrm{O}(\mathsf{n})$-invariant vectors, and S is no datum (Rec), its point operators being unbounded in the infrared: its nets are those of the definition alone. For every real $f \in \mathcal{F}$, the massless square $S(f)$ is essentially self-adjoint on its polynomial cyclic domain $\mathcal{D}_0^S$. At a mass $\mathsf{m} > 0$, Θ and S are data (Rec) by (a), and the unitary V of (a) carries the nets of each onto those of the definition.

At the mass 0: $\mathcal{A}_\Theta(O)$ is the $\mathrm{O}(\mathsf{n})$-invariant part of the net of the field strengths, restricted to the invariant vectors (for $\mathsf{n} = 1$ it is generated by the $\cos \bar{\varphi}(h)$, $h \in \mathcal{F}_\mathbb{R}(O)$, $\int h = 0$, where $\bar{\varphi}(h)$ is the closure of the field smeared with the real $h \in C_c^\infty(O)$ and $\int h$ is the integral over space-time); the multiplet $:\partial_\mu \varphi \cdot \partial_\nu \varphi:$ has the same net, and for $\mathsf{n} \in \{1, 3\}$ the closures of the $T_{\mu\nu}(f)$, $T_{\mu\nu} = :\partial_\mu \varphi \cdot \partial_\nu \varphi: - \frac{1}{2}\eta_{\mu\nu}\Theta$ with $\eta = \mathrm{diag}(1, -1, -1)$, $T_{00}(f)$ among them, $f \in \mathcal{F}_\mathbb{R}(O)$, are self-adjoint and affiliated with $\mathcal{A}_\Theta(O)$; (W) holds. $\mathcal{A}_\Theta(O) \neq \mathcal{A}_\Theta^d(O)$: for real h carried by O with $\int h \neq 0$ the mean over $\mathrm{O}(\mathsf{n})$ of $e^{i\bar{\varphi}^1(h)}$ lies in $\mathcal{C}_\Theta(O^\times) = \mathcal{A}_\Theta^d(O)$ and not in $\mathcal{A}_\Theta(O)$; for $\mathsf{n} = 1$, $\cos(\mathsf{s}\bar{\pi}(g))$, $\bar{\pi}(g)$ the closure of the momentum at time 0 smeared with a real $g \in C_c^\infty(\mathbb{R}^2)$ that is 1 near the closed base of O, lies in $\mathcal{C}_\Theta(O)$ and does not commute with $\cos \bar{\varphi}(h)$ when $\mathsf{s} \int h \notin 2\pi\mathbb{Z}$.

For $\mathsf{n} = 1$ and closed subgroups $\Lambda \subset \mathbb{R}$ the even parts of the nets $\{e^{i\bar{\varphi}(f)} : f \in \mathcal{F}_\mathbb{R}(O), \int f \in \Lambda\}''$ are pairwise different local nets between $\mathcal{A}_\Theta$ and $\mathcal{A}_\Theta^d$ with the same wedge algebras. At a mass $\mathsf{m} > 0$, where $\varphi^a(h) = \varphi^a(-\mathsf{m}^{-2}\Box h)$ makes every smeared field one of a test function of integral 0 and $\Theta = \frac{1}{2}(\Box + 2\mathsf{m}^2)S$, and where, as at the mass 0 and by the same steps, which hold at every $\mathsf{m} \geq 0$, the commutators of the $\Theta(f)$, $f \in \mathcal{F}(O)$, reach the colour sums $\sum_a \varphi^a(h)\varphi^a(h')$, whose commutators have the same structure constants at every mass, for real $h, h' \in \mathcal{F}(O)$ of integral 0, hence for all real $h, h' \in \mathcal{F}(O)$ (every $Q \in \mathcal{C}_\Theta(O)$ commutes weakly with them), among them the squares $\sum_a \varphi^a(g)^2$, whose closures generate $\mathcal{A}_S(O)$, the nets of Θ and of S are one, $\mathcal{A}_\Theta = \mathcal{A}_S$ on $\mathcal{H}_\Theta = \mathcal{H}_S$, for every $\mathsf{n} \geq 1$, and the net of Θ, as that of S, is its own dual net.

The net of $S = :\varphi \cdot \varphi:$ is its own dual net at every mass; at the mass 0, where $\Theta(f) = \frac{1}{2}S(\Box f)$, $\mathcal{A}_\Theta(O) \subsetneq \mathcal{A}_S(O) = \mathcal{A}_S^d(O) = \mathcal{A}_\Theta^d(O)$, and the two nets have the same wedge algebras: on one space, with one vacuum and one family of wedge algebras, the field S is Haag dual and the field $\frac{1}{2}\Box S$ is not.

■□

Theorem AS (Lattice algebras at one time; the defect $\mathbb{Z}_2$ and its Euclidean half)**.** (a) Let G be a compact group, (Σ, B) a finite graph, $\mathcal{H}_\mathrm{ph} \subset L^2(G^\mathsf{B})$ the gauge-invariant subspace, and $\mathsf{A}(Y)$, $Y \subset \mathsf{B}$, the algebra of all gauge-invariant bounded operators of the bonds of Y on $\mathcal{H}_\mathrm{ph}$. Then $\mathsf{A}(Y)$ is a von Neumann algebra, $\mathsf{A}(Y)' = \mathsf{A}(\mathsf{B} \setminus Y)$ for every Y, $\mathsf{A}(\mathsf{B}) = \mathcal{B}(\mathcal{H}_\mathrm{ph})$, and $\mathsf{A}(Y_1 \cup Y_2) = \mathsf{A}(Y_1) \vee \mathsf{A}(Y_2)$ if Y_1, Y_2 have no common site.

In (b)–(i), $G = \mathrm{SU}(2)$. On a layer $\mathbb{Z}/N_1 \times \mathbb{Z}/N_2$, $N_i \geq 4$, let, for a set S of plaquettes, B_S be their bonds, $\mathsf{A}_\bullet(S)$ the von Neumann algebra generated by the traces M_{t_p}, $p \in S$, and the spin projections Π_b^j, $b \in \mathsf{B}_S$; $Z^1(S)$ the sign fields c on B_S with $\prod_{b \in \partial p} c_b = 1$ for $p \in S$; B^1 the coboundaries of the whole layer, $c_b = z(b_-)z(b_+)$ for a sign field z on all its sites, and $B^1(S) \subset Z^1(S)$ the coboundaries

on B_S, $c_b = z(b_-)z(b_+)$, $b \in \mathsf{B}_S$, for a sign field z on the sites of B_S; $H^1(S) = Z^1(S)/B^1(S)$; $\mathsf{V}(c)$ the unitary of $V_b \mapsto c_b V_b$, a sign field on a set of bonds being continued by 1 to the other bonds; and for a dual path γ, $\mathsf{V}_\gamma = \mathsf{V}(c)$ with $c = -1$ exactly on the bonds that γ crosses. S is admissible if it has at least three plaquettes, any two of them are joined by a chain of plaquettes of S each sharing a bond with the next, and the plaquettes of S at each site are so joined through bonds at that site.

For every closed path C' and spin j: $\mathsf{V}(c) M_{\chi_j(U(C'))} = c(C')^{2j} M_{\chi_j(U(C'))} \mathsf{V}(c)$, $\mathsf{V}(c)\mathsf{V}(c') = \mathsf{V}(cc')$; $\mathsf{V}(c) = 1$ on $\mathcal{H}_{\mathrm{ph}}$ for $c \in B^1$, $\mathsf{V}(c)$ being then the gauge transformation by the central field z; and for $c \in B^1(S)$, with z continued by 1 to the other sites, $\mathsf{V}(c) = \mathsf{V}(c^\circ)$ on $\mathcal{H}_{\mathrm{ph}}$, c° the restriction of $(z(b_-)z(b_+))_b$ to the bonds off B_S, so that $\mathsf{V}(c)|_{\mathcal{H}_{\mathrm{ph}}}$, a function of spin projections of bonds off B_S, lies in $\mathsf{A}(\mathsf{B} \setminus \mathsf{B}_S) = \mathsf{A}(\mathsf{B}_S)'$ and $\operatorname{Ad} \mathsf{V}(c)$ is trivial on $\mathsf{A}(\mathsf{B}_S)$; on a periodic layer the plaquettes on which a sign field is not closed are even in number: there is no single point operator.

(b) For admissible S: $\mathsf{A}_\bullet(S) = \mathsf{A}(\mathsf{B}_S) \cap \{\mathsf{V}(c) : c \in Z^1(S)\}'$ and $\mathsf{A}_\bullet(S)' = \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_S) \vee \{\mathsf{V}(c) : c \in Z^1(S)\}$; the $\mathsf{V}(c)$, $c \in Z^1(S)$, lie in the centre of $\mathsf{A}_\bullet(S)$; $\operatorname{Ad} \mathsf{V}(c)$ on $\mathsf{A}(\mathsf{B}_S)$ depends on the class of c in $H^1(S) = Z^1(S)/B^1(S)$ alone, so that one c of each class serves in both identities, and $\mathsf{A}(\mathsf{B}_S) = \bigoplus_\kappa \mathsf{A}(\mathsf{B}_S)_\kappa$ over the characters κ of $H^1(S)$, where $\mathsf{A}(\mathsf{B}_S)_\kappa = \{X \in \mathsf{A}(\mathsf{B}_S) : \mathsf{V}(c) X \mathsf{V}(c)^* = \kappa([c])X \text{ for all } c \in Z^1(S)\}$ is a character subspace. This is a linear grading; every summand $\neq 0$, the summand $\kappa = 1$ being $\mathsf{A}_\bullet(S)$; if every $\mathbb{Z}_2$-cycle of B_S is a sum of boundaries ∂p, $p \in S$, then $\mathsf{A}_\bullet(S) = \mathsf{A}(\mathsf{B}_S)$. $\mathsf{A}_\bullet(S)$ is the join of the $\mathsf{A}(\mathsf{B}_T)$ over the chains $T \subset S$ of three plaquettes, and also of the $\mathsf{A}(\partial p)$, $p \in S$. The traces and spin projections of one plaquette, or of two adjacent plaquettes, do not generate all gauge-invariant operators of their bonds; the traces alone generate an abelian algebra, while $\mathsf{A}(\mathsf{B}_S)$ is not abelian.

(c) *The ring.* Let $a_i < b_i < b_i' < a_i'$ be integers with $a_i' - a_i \leq N_i - 1$, $Q = [a_1, a_1'] \times [a_2, a_2']$, $D = [b_1, b_1'] \times [b_2, b_2']$, R the plaquettes of Q off D, $C = \partial D$, $W_C = M_{\frac{1}{2}\operatorname{tr} U(C)}$, $u = M_{\operatorname{sign}\operatorname{tr} U(C)}$, γ a straight dual path from a plaquette of D to a plaquette outside Q, $c_* = -1$ on the bonds of B_R that γ crosses, $v = \mathsf{V}(c_*)$, and $D_1, \dots, D_4$ the four rectangles $[a_1, a_1'] \times [a_2, b_2]$, $[b_1', a_1'] \times [a_2, a_2']$, $[a_1, a_1'] \times [b_2', a_2']$, $[a_1, b_1] \times [a_2, a_2']$. Then R is admissible, $H^1(R) = \{1, [c_*]\} \cong \mathbb{Z}_2$, and on $\mathcal{H}_{\mathrm{ph}}$

$$\begin{aligned} \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R)' = \mathsf{A}(\mathsf{B}_R) &= \mathsf{A}_\bullet(R) \vee \{W_C\} \\ &= \mathsf{A}_\bullet(R) \oplus \mathsf{A}_\bullet(R) u, \\ \mathsf{A}_\bullet(R)' = \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R) &\vee \{\mathsf{V}_\gamma\} \\ = \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R) &\oplus \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R)\mathsf{V}_\gamma, \end{aligned}$$

with the normal conditional expectation $X \mapsto \frac{1}{2}(X + vXv)$ of $\mathsf{A}(\mathsf{B}_R)$ onto $\mathsf{A}_\bullet(R)$; $vW_C v = -W_C$, $W_C \notin \mathsf{A}_\bullet(R)$, $\mathsf{V}_\gamma \notin \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R)$, $\mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R) = \mathsf{A}_\bullet(R)' \cap \{u\}'$, $u\mathsf{V}_\gamma = -\mathsf{V}_\gamma u$; each inclusion has the index two. v is central in $\mathsf{A}_\bullet(R)$, and for a closed path C' in B_R, $M_{\chi_j(U(C'))} \in \mathsf{A}_\bullet(R)$ if and only if j is an integer or $c_*(C') = 1$: the adjoint loop around the hole lies in $\mathsf{A}_\bullet(R)$, the fundamental one does not. So over a ring duality and additivity exclude each other: $Y \mapsto \mathsf{A}(Y)$ is dual and not additive at B_R, $S \mapsto \mathsf{A}_\bullet(S)$ is additive and not dual at R, and no assignment of von Neumann algebras that equals A on the closed and on the open rectangles is both.

At every spacing $\mathsf{A}_\bullet(R) = \bigvee_{i \leq 4} \mathsf{A}(\mathsf{B}_{D_i})$; if no corner $D_i \cap D_{i+1}$ is one plaquette, the multiplications by flip-invariant gauge-invariant functions of the bonds of R are generated by those of the D_i, and $M_{\chi_1(U(C))} = \sum_a M_{(T_1)_{a_4 a_1}} M_{(T_2)_{a_1 a_2}} M_{(T_3)_{a_2 a_3}} M_{(T_4)_{a_3 a_4}}$ with gauge-invariant functions $(T_i)_{ab}$ of the bonds of D_i of modulus ≤ 1.

(d) *Two discs; the whole layer.* Let D_1, D_2 be rectangles without a common site, with a row and a column of plaquettes that meet neither, S the plaquettes off $D_1 \cup D_2$, $\mathsf{B}^\circ_{D_i}$ the bonds inside D_i, γ a dual path from D_1 to D_2 whose other plaquettes lie in S off that row and column, $\mathsf{V}_1, \mathsf{V}_2$ the two fluxes of the torus, and $\mathsf{N} = \mathsf{A}(\mathsf{B}^\circ_{D_1}) \vee \mathsf{A}(\mathsf{B}^\circ_{D_2})$. Then S is admissible, $H^1(S) = \mathbb{Z}_2^3$, $\mathsf{A}(\mathsf{B}_S)' = \mathsf{N}$, $\mathsf{A}_\bullet(S)' = \mathsf{N} \vee \{\mathsf{V}_\gamma, \mathsf{V}_1, \mathsf{V}_2\}$, $\mathsf{V}_\gamma \notin \mathsf{N} \vee \{\mathsf{V}_1, \mathsf{V}_2\}$ and $\mathsf{V}_i \notin \mathsf{N} \vee \{\mathsf{V}_\gamma\}$.

On the whole layer $\mathsf{A}_\bullet(\mathsf{P})' = \mathrm{span}\{1, \mathsf{V}_1, \mathsf{V}_2, \mathsf{V}_1\mathsf{V}_2\}$, $\mathcal{H}_{\rm ph}$ is the sum of four flux sectors, none 0, and $\mathsf{A}_\bullet(\mathsf{P})$ is the sum of all bounded operators of the sectors; one disc has no defect but the fluxes: $\mathsf{A}_\bullet(\mathsf{P} \setminus D)' = \mathsf{A}(\mathsf{B}^\circ_D) \vee \{\mathsf{V}_1, \mathsf{V}_2\}$.

(e) *Three space dimensions.* On a finite part of $\mathbb{Z}^3$, or on a periodic one with sides ≥ 4, at one time: for all the plaquettes S of a box $[0, n_1] \times [0, n_2] \times [0, n_3]$, $\bigvee_{p\in S} \mathsf{A}(\partial p) = \mathsf{A}(\mathsf{B}_S)$; for the plaquettes S of the eight unit cubes of $[0, 3]^2 \times [0, 1]$ off $(1, 2)^2$, $\bigvee_{p\in S} \mathsf{A}(\partial p) = \mathsf{A}(\mathsf{B}_S) \cap \{\mathsf{V}(c_*)\}'$ and $(\bigvee_p \mathsf{A}(\partial p))' = \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_S) \vee \{\mathsf{V}(c_*)\}$, $c_* = -1$ on the four bonds of $[1, 2] \times [2, 3] \times [0, 1]$ parallel to e_1, with $\mathsf{V}(c_*)W_C = -W_C\mathsf{V}(c_*)$ for the fundamental loop C around the ring; for a dual surface $\mathfrak{s}$ that meets the ring in that section, $\mathsf{V}(c_\mathfrak{s}) = \mathsf{V}(c_*)\,\mathsf{V}(c_\mathfrak{s}|_{\mathsf{B}\setminus\mathsf{B}_S})$ with the second factor in $\mathsf{A}(\mathsf{B} \setminus \mathsf{B}_S)$: the defect $\mathbb{Z}_2$ is carried by the flip across a surface whose rim is a closed dual line in the complement that links the ring, a closed vortex line.

For a set S of plaquettes let $Z^1(S)$, $B^1(S)$ be formed as on a layer, and call $H^1(S) = Z^1(S)/B^1(S)$ its defect; call an order $p_1, \dots, p_n$ of S a shelling if for each $k \geq 2$, with X_{k-1} the set of the bonds of $p_1, \dots, p_{k-1}$, either all bonds of p_k lie in X_{k-1}, or p_k has exactly one bond in common with some p_j, $j < k$, the set $\partial p_j \cup (\partial p_k \cap X_{k-1})$ is connected, and every site of a bond of p_k off X_{k-1} that is a site of X_{k-1} lies on that set. If S has a shelling, $\bigvee_{p\in S} \mathsf{A}(\partial p) = \mathsf{A}(\mathsf{B}_S)$; a box has one. If $S = S_1 \cup S_2$, both with shellings, $I = \mathsf{B}_{S_1} \cap \mathsf{B}_{S_2}$ has r connected components, each with a cycle, and every site common to B_{S_1} and B_{S_2} is a site of I, then $\bigvee_{p\in S} \mathsf{A}(\partial p) = \mathsf{A}(\mathsf{B}_S) \cap \{\mathsf{V}(c) : c \in Z^1(S)\}'$ and the defect is $\mathbb{Z}_2^{r-1}$; the solid ring above is such a union, with $r = 2$.

(f) *The sign rule.* On every periodic lattice in three dimensions and at every β, for sign fields η on the plaquettes and s on the bonds, $(ds)_p = \prod_{b\in\partial p} s_b$, bounded Borel Φ, closed paths C_i and spins j_i:

$$Z[\eta\,\mathrm{d}s; \Phi; C_1, \dots, C_r] = \prod_i s(C_i)^{2j_i}\, Z[\eta; \Phi; C_1, \dots, C_r],$$

$$Z[\eta; \Phi; C] = \int \Phi((\eta_p t_p)_p) \prod_i \chi_{j_i}(U(C_i))\, e^{-\beta\sum_p (1-\eta_p t_p)}\, \mathrm{d}U.$$

Carrying a dual string once across a fundamental loop changes the sign, across an adjoint loop it does not; the rule holds for bounded fat odd loops F, $F(sU) = s(C)F(U)$ for s closed on the plaquettes of a tube around C; and every limit point of renormalised mixed functions of loops, pairs of monopoles and twisted action densities has it.

(g) *Block loops.* For the constants of Thm. YM3_I:D, L odd, a label $\mathsf{b} \geq \gamma^{-2}$, a torus of side $\ell = 2L^m$ with $MR(\mathsf{b}) \leq L^m$ (R the function of Thm. YM3_I:D), $\beta = L^K\mathsf{b}$ on $\Lambda_{K,\ell} = L^{-K}\mathbb{Z}^3/\ell\mathbb{Z}^3$, $\langle\cdot\rangle_K$ the Wilson measure there and $\mathcal{M}^{K-d}U$ the block field on $\Lambda_{d,\ell}$ with law μ_d^K: the block map is covariant under the centre, $\mathcal{M}^n(sU)(c) = s[c]\,\mathcal{M}^n(U)(c)$ for every U if $(ds)_p = 1$ at every plaquette with all corners in the double block of c, $s[c]$ the product of s along the straight path from c_- to c_+; for every closed path C of $\Lambda_{d,\ell}$ and bounded Borel h, $\langle h(W_C(\mathcal{M}^{K-d}U))\rangle_K$ converges along the full sequence of cutoffs, consecutive terms differing by $\leq C\ell^3 K^{\mathfrak{p}_0} L^{-K/10} \sup|h|$ for $K \geq K_{\rm bf}$ with $\lceil K/1260\rceil \geq d$, $K_{\rm bf}$ the threshold of the rate $\mathfrak{o}_K$ of Thm. YM3_I:D (Cor. 2.8 among the construction results), and so do all mixed moments of loops of block fields; the flips $\bar\xi$, $d\bar\xi = 1$, of the torus are symmetries of the limit laws μ_d, so that odd and even functions are orthogonal in $L^2(\mu_d)$; and a closed vortex line that bounds passes as a flip: for a finite union S of closed dual plaquettes of $\Lambda_{d_0,\ell}$, $d \geq d_0$, η_K the sign field that is -1 exactly on the plaquettes of $\Lambda_{K,\ell}$ pierced by $\partial\mathsf{S}$, and $\bar\xi_c = (-1)^{\#(c\cap\mathsf{S})}$: $Z_K[\eta_K; F(\mathcal{M}^{K-d}\cdot)]/Z_K[1; 1] = \int F(\bar\xi V)\,\mathrm{d}\mu_d^K(V) \to \int F(\bar\xi V)\,\mathrm{d}\mu_d(V)$ for bounded Borel F of the block bonds whose blocks miss $\partial\mathsf{S}$, the equality holding at every K and the error being $\leq \sup|F| \sum_{K'\geq K} \mathfrak{o}_{K'}$ for $K \geq K_{\rm bf}$ and $\lceil K/1260\rceil \geq d$, $\mathfrak{o}_K$ the rate of Thm. YM3_I:D; for $F = W_C F'$, with F' a bounded Borel function of such block bonds and

$F'(\bar\xi V) = F'(V)$, $W_C(\bar\xi V) = (-1)^{\#(C\cap \mathsf{S})} W_C(V)$ gives the factor $(-1)^{\#(C\cap\mathsf{S})}$, the linking number modulo 2 of C and $\partial\mathsf{S}$.

(h) *Sharp loops; flips.* In (g) let C be a lattice polygon without self-intersection that does not wind and has a neighbourhood in a cube of side $\le \ell/2$, h bounded on $[-1,1]$ and continuous off a Lebesgue null set, $\bar h = \frac{2}{\pi}\int h(w)\sqrt{1-w^2}\,\mathrm{d}w$, and F_K Borel functions, $|F_K| \le 1$, of the bonds of $\Lambda_{K,\ell}$ at a distance $\ge \mathsf{d} > 0$ from C. Then $\langle F_K h(W_C)\rangle_K - \bar h \langle F_K\rangle_K \to 0$, $W_C = \frac12 \mathrm{tr}\, U(C)$. For continuous h this is Lem. 14.9, Thm. AF of §10.1 read at one polygon (see the results used): every character of non-zero spin of the holonomy of C, times a function of modulus ≤ 1 of the bonds at a fixed distance from C, has a mean of modulus at most a constant times a positive power of $L^{-K}\log^3 L^K$ eq. 14.iii, through a rotation of the bonds of a tube of radius ρL^{-K} about C, ρ of the order of $\beta^{1/4}$ up to logarithms, so that the class of the holonomy tends to the Haar law, whose image under $\frac12\mathrm{tr}$ has the density $\frac{2}{\pi}\sqrt{1-w^2}$; a bounded h continuous off a null set lies between continuous functions $h_- \le h \le h_+$ with $\bar h_+ - \bar h_-$ as small as we please.

So, with C in a site mirror, the odd unitary u_K of (c) and every character $\chi_j(U(C))$, $j \ge \frac12$, the adjoint one included, tend weakly to 0 between the vectors of such functions in the space of reflection positivity, while $u_K^2 = 1$ and $\langle W_C^2\rangle_K \to \frac14$. For a rectangular path γ whose sides have at least 3 and at most $N-2$ bonds, $N = \ell L^K$ the number of sites on a side of the torus, γ' its bonds but the two at each corner (no two of them lie in one plaquette, which a side of $N-1$ bonds would allow), $n' = \#\gamma'$, and bounded Borel F of the other bonds: $|\langle \frac12 \mathrm{tr}\, U(\gamma) F\rangle_K| \le \mathsf{r}(4\beta)^{n'} \sup|F|$, $\mathsf{r}(x) = I_2(x)/I_1(x)$ the ratio of modified Bessel functions; the bound tends to $\exp(-3|\gamma|/8\mathsf{b})$.

The flips fade on bounded vectors: let ω_K be the state on the layer of $\Lambda_{K,\ell}$ given by the transfer operator of Section YM3_I:85 (with its trace formula, Prop. YM3_I:85.2), ζ a dual path of the layer with an end plaquette p_0, $Q_0 = M_{\mathbf{1}\{a_{p_0}<1\}}$, and X, Y bounded operators commuting with Q_0; then

$$|\omega_K(X \mathsf{V}_\zeta Y)| \le 2\|X\|\|Y\|\langle a_p\rangle_K^{1/2}, \qquad \langle a_p\rangle_K \le (21 + 3\log\beta)/\beta \quad (\beta \ge 1).$$

(i) *The Euclidean $\mathbb{Z}_2$.* In (g) let $\mathcal{R}$ be a solid ring, a union of closed cubes of side L^{-d} centred at sites of $\Lambda_{d,\ell}$, and $\bar\xi$ a flip, the sign change of the block variables crossing a dual surface S whose rim misses $\mathcal{R}$; h_D is Haar measure on the block fields at the depth D and $\mathsf{In}_{[D]} = \{V : \max_p |\partial V(p) - \mathbf{1}| < 2\varepsilon(x_{[D]})\}$, $|\cdot|$ the operator norm, the set of the small fields of Thm. YM3_I:D at the depth D, its inner ball (the common-depth definitions in Section YM3_I:116, the definitions preceding Lem. YM3_I:113.10), with $x_{[D]} = L^D\mathsf{b}$ (§2.3.2) and $\varepsilon(\mathsf{t}) = A_0 \mathsf{t}^{-1/2}(\log \mathsf{t})^{\mathfrak{p}_0}$ YM3_I:(634), A_0 a constant of the list of Thm. YM3_I:D.

There are $D > d$ and numbers $\kappa_D, \bar\kappa_D > 0$, $\mathsf{R}_{[D]} \to 0$, free of K, with $\mu_D^K(E) \ge \kappa_D\, \mathsf{h}_D(E\cap \mathsf{In}_{[D]})$ and $\mu_D^K(E) \le \bar\kappa_D\, \mathsf{h}_D(E\cap\mathsf{In}_{[D]}) + \mathsf{R}_{[D]}$ for all Borel E and $\mu_D^K(\mathsf{In}_{[D]}) \ge 1 - \mathsf{R}_{[D]}$, for all $K \ge D+1$ and for the limit law μ_D. A thick vortex is a small field: for every $D \ge d+2$ there is a block field V^1 at the depth D with values in an abelian subgroup, equal to $\bar\xi_c$ on the bonds of $\mathcal{R}$, whose plaquette angles are $\le 16\pi n_{\mathsf{S}} L^{-2(D-d)}$, n_{S} the number of dual bonds in $\partial\mathsf{S}$. Hence one positive measure θ_0 of mass $\varkappa_D > 0$, which depends on no cutoff and no surface, lies below the law of the block fields of $\mathcal{R}$ and below all its flips, for all $K \ge D+1$ and in the limit; so the law and each flip differ by at most $1-\varkappa_D$ in total variation, every odd function of the block fields of $\mathcal{R}$ with values in $[-1,1]$ has its mean in $[\varkappa_D - 1, 1-\varkappa_D]$, and for a closed path C in $\mathcal{R}$ with $\#(C\cap\mathsf{S})$ odd the fundamental block loop W_C and its sign keep an L^1-distance $\ge \mathsf{m}_C = \int |W_C|\,\mathrm{d}\theta_0 > 0$ from every flip-invariant function, at every $K \ge D+1$ and in the limit.

The limit laws have densities for Haar measure by Lem. 14.18. Conversely, let $\mathcal{R}$ be a rectangular solid ring of $\Lambda_{d,\ell}$: with $N = \ell L^d$ sites on a side of the torus, integers $a_i < b_i < b_i' < a_i'$ $(i = 1, 2)$ and $t_1 \le t_2$, in the units of $\Lambda_{d,\ell}$, $\mathcal{R}$ is the union of the four bricks $[a_1, a_1'] \times [a_2, b_2] \times [t_1, t_2]$, $[b_1', a_1'] \times [a_2, a_2'] \times [t_1, t_2]$, $[a_1, a_1'] \times [b_2', a_2'] \times [t_1, t_2]$ and $[a_1, b_1] \times [a_2, a_2'] \times [t_1, t_2]$ of sites, where $t_2 > t_1$ or no corner of two consecutive bricks is one plaquette, and where $b_i' - b_i \ge 2$, $a_i' - a_i \le N - 2$ and $t_2 - t_1 \le N - 2$, so that the hole holds a site and the four bricks hold all the bonds between the sites of $\mathcal{R}$; let α be its flip, the sign change of the block variables of the bonds $\langle x, x + e_1\rangle$ of $\mathcal{R}$ with $x_1 = b_1$ and $x_2 \le b_2$, which cross one section of the ring, $\mu = \rho \,\mathrm{d}V$ the limit law of its block fields, with a gauge-invariant density ρ, $u = \operatorname{sign} W_C$ for the loop C around the hole, $\mathfrak{L}$ the classes of all bounded gauge-invariant Borel functions of these block fields and $\mathfrak{E}$ those of the flip-invariant ones.

Then every $F \in \mathfrak{L}$ is $F = \Phi_0 + \Phi_1 u$ μ-almost everywhere with $\Phi_0, \Phi_1 \in \mathfrak{E}$ measurable for the four bricks (for the cubes of three sites if the bricks are three sites thick), $\sup|\Phi_i| \le \sup|F|$, and the same holds for the block fields of all coarser depths in $\mathcal{R}$; $\mathfrak{E} = \bigvee_i \mathfrak{L}(\mathrm{brick}_i)$ and $\mathfrak{L} = \mathfrak{E} + \mathfrak{E}u$; the adjoint loop is $\chi_1(V(C)) = \operatorname{tr}(T_1T_2T_3T_4)$ with entries $(T_i)_{ab}$ functions of the i-th brick of modulus ≤ 1. On the closed subspace $\mathfrak{h}$ of $L^2(\mu)$ of the classes of the gauge-invariant functions, on which $\mathfrak{L}$ and $\mathfrak{E}$ act by multiplication, the commutant of $\mathfrak{E}$ is $\mathfrak{L} + \mathfrak{L}v$, of type $\mathrm{I}_1 \oplus \mathrm{I}_2$, where $(vF)(V) = e(V)(\rho(\alpha V)/\rho(V))^{1/2} F(\alpha V)$ and e is the indicator of $\{\rho > 0\} \cap \alpha\{\rho > 0\}$: $v = v^*$, $v^2 = e$, $vu = -uv$, and $\int e \,\mathrm{d}\mu \ge \varkappa_D$, so that μ and its image under α are not mutually singular and $v \ne 0$: L being odd, $\mathcal{R}$ is also a union of closed cubes of side $L^{-(d+1)}$ centred at sites of $\Lambda_{d+1,\ell}$, and α agrees on the bonds of $\mathcal{R}$, up to a central gauge transformation, with the sign field into which the block map from the depth $d+1$ to the depth d turns the flip of a flat sheet S_1 of $\Lambda_{d+1,\ell}$ between two straight closed dual lines in the direction e_3 that lie over the open unit squares about a site of the hole and about a site of the same row outside $[a_1, a_1'] \times [a_2, a_2']$, so that its rim misses $\mathcal{R}$; here D and $\varkappa_D$ are those of the first half at the depth $d+1$.

And v is the limit of the self-adjoint unitaries v_K of the closed vortex line at the cutoffs: for all bounded Borel functions F, F' of the block fields of $\mathcal{R}$, with μ^K their law at the cutoff K, $|\langle F, v_K F'\rangle_{L^2(\mu^K)} - \langle F, vF'\rangle_{L^2(\mu)}| \le 2\sqrt{2} \sup|F| \sup|F'| (\sum_{K' \ge K} \mathfrak{o}_{K'})^{1/2}$ for $K \ge K_{\mathrm{bf}}$ and $\lceil K/1260\rceil \ge d$. If besides the hole is at least three bonds wide in both directions, the ring has positive height and it lies in an open ball of radius $< \ell/4$, then the law and its flip have the same null sets, $e = 1$, v is a self-adjoint unitary, and $\mathfrak{L} = \mathfrak{E} \oplus \mathfrak{E}u$ is a grading by $\mathbb{Z}_2$.

For a band $\mathbb{Z}/N \times I_2 \times I_3$ that winds around the torus, I_2, I_3 intervals of 3 to $N - 1$ sites, the defect is exactly $\mathbb{Z}_2$: $\mathfrak{L} = \mathfrak{E} \oplus \mathfrak{E}u_1$, orthogonally, $\mathfrak{E}u_1 \ne 0$, u_1 the sign of the straight loop that winds, $\mathfrak{E}$ is generated by the cubes of the band, and the flip is a unitary v with $\mathfrak{E}' = \mathfrak{L} \vee \{v\}$ on the space of the gauge-invariant classes for the law of the block fields of the band.

■□

14.1.4. *Construction estimates.* We use the reconstruction from data (Rec) of Section YM3_I:107 and Sections YM3_I:109 and YM3_I:110, as recalled in the setting: the lemma on the smeared insertion (Lem. YM3_I:107.1) with the insertion formula YM3_I:(2039), the point bound eq. 14.i (Prop. YM3_I:107.2 and Lem. YM3_I:109.2), the uniqueness of the family of point operators (Prop. YM3_I:107.2), the lemma on the energy bound (Lem. YM3_I:107.3) and the definition of the energy form q_0 (Lem. YM3_I:107.3), the field operators of Prop. YM3_I:109.1, the boundary distributions and polynomial cyclicity proved in Section YM3_I:109, and the spectrum condition in Thm. YM3_I:AO, the lemma (Br) that links the field operators with the point vectors (Lem. YM3_I:109.2), the propositions on Lorentz covariance and on locality (Props. YM3_I:110.1 and YM3_I:110.2), and the domain argument preceding Prop. YM3_I:109.1; [7, Lem. G.5], by

which a holomorphic function on a connected open subset of $\mathbb{C}^N$ that vanishes on a non-empty open set of real points vanishes; Prop. YM3_I:97.2, which turns an insertion bound for sources in small balls with a constant C_{I} into the point bound of order 4 with the constant $C_{\mathrm{pt}} = c_{\mathrm{pt}} C_{\mathrm{I}}$, $c_{\mathrm{pt}} = 320 \cdot 8^4\, \mathsf{mv}_3(40L)^4 M$ the coefficient of C_{I} in YM3_I:(1919) at $c_{\mathrm{I}} = \frac{1}{40L}$, with $\mathsf{mv}_3 = m_3$ as defined immediately after that equation, and the insertion bound for sources in small balls of one cell, Prop. YM3_I:105.1, with its proof, which the proof of Thm. AQ(c) uses up to its last step together with the insertion bound on one lattice for a pair that carries systems (Lem. YM3_I:104.4, with its normalizer YM3_I:(2017)), the systems of localised source energies of a pair and their constants ([7, Lem. N.1], with [7, eq. 288]), the window YM3_I:(1531) of the labels (Prop. YM3_I:74.1), the two source-radius conditions of Prop. YM3_I:82.2, at the same radius-independent numerical prefix, and the dilation among the construction results ([7, Lem. K.1]); and Thm. YM3_I:AP among the construction results on the limits of periodic Gibbs states and their canonical decompositions, which follows from Thm. YM3_I:D, at the constants of Prop. YM3_I:69.1 and Thms. YM3_I:AJ and YM3_I:AD.

By that theorem every limit of periodic Gibbs states at g, volume-first ones included, is data (Rec) with $(16C_{\mathrm{pt}}^g, 4, \frac{1}{2})$ and has a canonical decomposition, every component of which is data (Rec) with $(16C_{\mathrm{pt}}^g + \bar{\mathfrak{c}}(\mathsf{C}^g), 4, \frac{1}{2})$ and a unique vacuum and is approximated by lattice states: if $\sigma'(X) = \lim_j \rho_j((X)_{K_j}^{\rho_j})$, every component is $\sigma'_\xi(X) = \lim_j \rho_{j,\xi}((X)_{K_j}^{\rho_j})$ along the full sequence, with the periodic Gibbs states $\rho_{j,\xi}(F) = \rho_j(\mathbf{1}_{E_j} F)/\rho_j(E_j)$ at the same couplings, E_j events with $\rho_j(E_j) > 0$ (Thm. YM3_I:AP); these are the lattice approximants of the component; the numbers depend on no sequence, limit, decomposition or ξ; and a volume-first σ has $S_1^\sigma = 0$, its truncated functions S_n^σ, $n \geq 2$, being the limits of the joint cumulants $S_{n;j}^T$ of the $\mathcal{O}_{K_j}(f)$ under the Wilson measures μ_j on tori $\Lambda_j = L^{-K_j}\mathbb{Z}^3/\ell_j\mathbb{Z}^3$, $\ell_j = 2L^{m_j} \to \infty$.

We use besides the reflection positivity of periodic Gibbs states at the nine families of lattice mirrors (Prop. YM3_I:103.4), the chessboard estimate (Prop. YM3_I:86.3, with equation YM3_I:(1759)) and the multiple reflections (Prop. YM3_I:104.2), the one-cell bound with a source (the cell bound, Prop. YM3_I:89.1), the systems of localised source energies (Section YM3_I:88) and the one-lattice source recursion of Lem. 14.14, with its identification of the source energies and its proof on smaller cores, together with the modification of multiple reflections stated and proved in Lem. 14.15(e), and, for Thm. AQ(c), Thms. YM3_I:AJ and YM3_I:AD at every radius and every admitted choice of the constants (σ^-, γ); from Thm. YM3_I:D, that on one torus the laws of the block fields at a fixed depth converge, at a rate (Cor. 2.8), with a floor, an upper bound and a tail on the small fields, and a tail at a fraction of the threshold, read off the small-field chain (the setting of the chain, §2.3.1; Prop. YM3_I:115.2 with the definition preceding Prop. YM3_I:115.2, YM3_I:(2187) and YM3_I:(2188)–(2190); Props. YM3_I:114.2 and YM3_I:113.13; Lem. YM3_I:113.10 with the definitions preceding Lem. YM3_I:113.10, YM3_I:(2159) and YM3_I:(2161); equation YM3_I:(2180) with YM3_I:(2179); Thm. YM3_I:E and the curvature comparison in the proof of Lem. YM3_I:113.10; the condition $A_0 \geq 4B_3 C_H A_1$ in the ordered choice, written explicitly in the proof of Lem. 4.48; and YM3_I:(634), (§2.3.2), the common-depth definitions in Section YM3_I:116); the block-averaging map (equations YM3_I:(37)–(39)) with its locality, its real-analyticity on the small fields and its linear part at the unit (Prop. YM3_I:4.5, YM3_I:4.5 and YM3_I:4.6); the transfer operator of a periodic lattice in one direction and its trace formula (Section YM3_I:85, Prop. YM3_I:85.2); and the short-distance law of the two-point function recalled in the setting (Thm. YM3_I:AV, Cor. YM3_I:127.11).

The theorem on sharp loops on tori on which Thm. AS(h) rests is Thm. AF of §10.1, in its limit, as the spacing tends to 0, for polygons with a further factor G, read at one polygon with $G = F_K$, together with the bound for the characters behind that limit; it is stated with a sketch of its proof as Lem. 14.9 below: at every label $\mathsf{b} > 0$, with $\beta = L^K \mathsf{b}$ and $\langle\cdot\rangle_K$ the Wilson measure on $\Lambda_{K,\ell}$, for a lattice polygon C without self-intersection that does not wind and has a neighbourhood in

a cube of side $\le \ell/2$, and continuous h on $[-1,1]$, $\langle F_K h(\frac{1}{2}\mathrm{tr}\, U(C))\rangle_K - \bar{h}\langle F_K\rangle_K \to 0$ uniformly in Borel functions $|F_K| \le 1$ of the bonds at a fixed positive distance from C, $\bar{h} = \frac{2}{\pi}\int h(w)\sqrt{1-w^2}\,\mathrm{d}w$; its proof reads the Wilson weight on one torus, the Schwarz inequality at site mirrors and Haar measure alone.

The textbook theorems used are the spectral theorem, Stone's theorem on strongly continuous unitary groups, polar decomposition, the bicommutant theorem, von Neumann's theorem on $1+T^*T$, modular theory, reduction theory (direct integrals of separable Hilbert spaces and of von Neumann algebras over a compact metrisable space with a Borel measure: the decomposable operators are the commutant of the diagonal algebra, a von Neumann algebra between the diagonal algebra and its commutant is the direct integral of a measurable field of von Neumann algebras, whose fibres are fixed up to a null set by a countable generating family, and the commutant of a direct integral is the direct integral of the commutants), Hölder's inequality in Schatten classes, Jensen's inequality, Gronwall's lemma, the three-lines theorem, the theorems of Stone–Weierstrass, Montel, Vitali and Liouville, the extension theorem of M. Riesz, the theorems of Peter–Weyl, Clebsch–Gordan and Burnside, the theorem on quasi-analytic vectors, the duality theorem for the free field, the first fundamental theorem for orthogonal groups, Fubini's theorem, the implicit function theorem, and the fact that the zeros of a real-analytic function that does not vanish identically on a connected open set form a null set.

14.1.5. *Auxiliary estimates.*

Lemma 14.1 (Frames)**.** Let $\iota : \mathcal{V} \to \mathcal{H}$ be linear with dense range in a separable Hilbert space, and $\iota_j : \mathcal{V} \to \mathcal{H}_j$ linear, $\iota_j G$ defined for $j \ge j(G)$, with $\langle \iota_j G', \iota_j G\rangle \to \langle \iota G', \iota G\rangle$. Write $\psi_j \Rightarrow \psi$ if for every $\delta > 0$ some G has $\|\psi - \iota G\| < \delta$ and $\limsup_j \|\psi_j - \iota_j G\| < \delta$; $\psi_j \rightharpoonup \psi$ if $\sup_j \|\psi_j\| < \infty$ and $\langle \iota_j G, \psi_j\rangle \to \langle \iota G, \psi\rangle$ for all G; and, for bounded operators, $Q_j \rightharpoonup Q$ if $\sup_j \|Q_j\| < \infty$ and $\langle \iota_j G', Q_j \iota_j G\rangle \to \langle \iota G', Q \iota G\rangle$ for all G, G'. Then: (i) $\iota_j G \Rightarrow \iota G$; $\psi_j \Rightarrow \psi$ and $\psi_j' \rightharpoonup \psi'$ give $\langle \psi_j, \psi_j'\rangle \to \langle \psi, \psi'\rangle$; $\Rightarrow$ implies $\rightharpoonup$ and $\|\psi_j\| \to \|\psi\|$; and $\psi_j \rightharpoonup \psi$ gives $\|\psi\| \le \liminf_j \|\psi_j\|$. (ii) Bounded sequences of vectors and of operators have subsequences that converge in the sense of $\rightharpoonup$, the limits with $\|Q\| \le \liminf_j \|Q_j\|$. (iii) $Q_j \rightharpoonup Q$ and $\psi_j \Rightarrow \psi$ give $Q_j\psi_j \rightharpoonup Q\psi$, and $Q_j^* \rightharpoonup Q^*$. (iv) If π_j are projections with $\|(1-\pi_j)\iota_j G\| \to 0$ for all G, then $(\pi_j \mathcal{H}_j, \pi_j \iota_j)$ has the same properties, and for $\psi_j = \pi_j \psi_j$ the two meanings of $\Rightarrow$ and of $\rightharpoonup$ agree. (v) Let P_j and A be self-adjoint, $\mathcal{V}_0 \subset \mathcal{V}$ a subspace, $\mathsf{r} : \mathcal{V}_0 \to \mathcal{V}$ linear, with $\iota\mathcal{V}_0 \subset D(\mathsf{A})$, $(\mathsf{A} \pm i)\iota\mathcal{V}_0$ dense, $\mathsf{A}\iota G = \iota \mathsf{r} G$ and $\mathsf{P}_j \iota_j G = \iota_j \mathsf{r} G$ for $G \in \mathcal{V}_0$ and large j.

Then $\Phi(\mathsf{P}_j)\psi_j \Rightarrow \Phi(\mathsf{A})\psi$ for every $\Phi \in C_0(\mathbb{R})$ and all $\psi_j \Rightarrow \psi$. ■

Proof sketch. (i) The triangle inequality and the convergence of the Gram matrices, as in $|\langle \psi_j, \psi_j'\rangle - \langle \psi, \psi'\rangle| \le \|\psi_j - \iota_j G\| \sup_j \|\psi_j'\| + |\langle \iota_j G, \psi_j'\rangle - \langle \iota G, \psi'\rangle| + \|\iota G - \psi\|\|\psi'\|$. (ii) On a countable $\mathbb{Q}(i)$-subspace with dense image the forms converge along a diagonal subsequence; their limits are bounded by $\sup_j \|Q_j\|\,\|\iota G'\|\|\iota G\|$, Riesz's theorem gives Q, and the convergence of the Gram matrices passes to all of $\mathcal{V}$. (iii), (iv) The triangle inequality. (v) Call a bounded sequence (X_j) with a bounded X good if $\psi_j \Rightarrow \psi$ implies $\mathsf{X}_j\psi_j \Rightarrow \mathsf{X}\psi$; sums, products and limits uniform in j of good sequences are good. Given δ, choose $G \in \mathcal{V}_0$ with $\|\psi - (\mathsf{A} \pm i)\iota G\| < \delta$; as $(\mathsf{P}_j \pm i)\iota_j G = \iota_j(\mathsf{r}G \pm iG)$,

$$\limsup_j \|(\mathsf{P}_j \pm i)^{-1}\psi_j - \iota_j G\| \le \limsup_j \|\psi_j - \iota_j(\mathsf{r}G \pm iG)\| < \delta,$$

so the resolvents are good; the polynomials without constant term in $(x+i)^{-1}, (x-i)^{-1}$ are dense in $C_0(\mathbb{R})$, and $\|\Phi(\mathsf{P}_j)\| \le \sup|\Phi|$. □

Lemma 14.2 (Rotations and boosts of point vectors)**.** Let σ be data (Rec), $\tilde{B}(\zeta) = \iota B(\zeta)\iota^{-1}$ on $\mathbb{C}^3$, applied to each point of a $z \in (\mathbb{C}^3)^n$, so that $\tilde{B}(i\tau) = R_\tau$; for $0 < \delta \le \frac{\pi}{2}$ let $\mathsf{I}_\delta = \{x_1 > 0,\ 0 < x_3 < x_1 \tan\delta\}$, $\mathsf{I}_{\pi/2} = \mathsf{I}$, and

$$\mathsf{I}_\delta^{(n)} = \{x \in (\mathbb{R}^3)^n :\ x_1 \in \mathsf{I}_\delta,\ x_{k+1} - x_k \in \mathsf{I}_\delta\ (1 \le k < n)\},$$

an open convex cone. (a) $\tilde B(s)\mathsf{Tb}^{(n)} = \mathsf{Tb}^{(n)}$ and $U(B(s))\Psi_n(z) = \Psi_n(\tilde B(s)z)$ for real s; if $R_\tau z \in \mathsf{Tb}^{(n)}$ for $0 \le \tau \le \tau_0$, then $\Psi_n(z) \in D(e^{-\tau_0 K})$ and $e^{-\tau K}\Psi_n(z) = \Psi_n(R_\tau z)$ for $0 \le \tau \le \tau_0$; and for $x \in \mathsf{I}_\delta^{(n)}$, $R_\tau x \in \mathsf{Tb}^{(n)}$ for $0 \le \tau \le \pi - \delta$. (b) A small tuple in I_δ is a tuple $G = (h_1, \dots, h_n)$ of real $h_k \in C_c^1(\mathbb{B}(x_k^\circ, \epsilon))$, $\mathbb{B}(y, \epsilon)$ the open ball of $\mathbb{R}^3$, with centres $x^\circ = (x_k^\circ)_k$ such that $\prod_k \mathbb{B}(x_k^\circ, \epsilon) \subset \mathsf{I}_\delta^{(n)}$ and $\epsilon < c_0 \eta_{x^\circ}$, where

$$\eta_{x^\circ} = \min\Big\{1,\ \min_{0\le\tau\le\pi-\delta}(R_\tau x_1^\circ)_3,\ \tfrac12 \min_{k<n}\ \min_{0\le\tau\le\pi-\delta}\big(R_\tau(x_{k+1}^\circ - x_k^\circ)\big)_3\Big\} > 0;$$

For $n = 1$ the member involving $\min_{k<n}$ is omitted from this minimum; the small tuples in I are those in $\mathsf{I}_{\pi/2}$. For a small tuple G in I_δ and $0 \le \tau \le \pi - \delta$: $[G] \in D(e^{-\tau K})$ and $e^{-\tau K}[G] = [R_\tau G]$; in particular $\Delta^{1/4}[G] = [RG]$ for the small tuples in I. The span $\mathcal{V}_0$ of $\varnothing$ and of the small tuples in I gives a core $[\mathcal{V}_0]$ of $\Delta^{1/4}$, with $(\Delta^{1/4} \pm i)[\mathcal{V}_0]$ dense. (c) For real $u \in C_c^\infty(\mathsf{I})$ let $f_u(B(s)(0, \rho, x_2)) = \int_0^{\pi/2} u(\rho\cos\alpha, x_2, \rho\sin\alpha)\, e^{-(s+i(\pi/4-\alpha))^2} d\alpha$ on W, and $f_u = 0$ off W. Then $f_u \in \mathcal{S}$, $\operatorname{supp} f_u \subset W \cap \{r_1 \le ((x^1)^2 - (x^0)^2)^{1/2} \le r_2\}$ for some $0 < r_1 < r_2$,

$$\mathsf{F}(K)[u] = \phi(f_u)\Omega, \qquad \mathsf{F}(k) = \sqrt{\pi}\, e^{-k^2/4 + \pi k/4}, \tag{14.ii}$$

and for every $p \in \mathbb{R}^3$ some u has $\hat f_u(p) \ne 0$. ∎

Proof sketch. (a) The analytic Wightman functions are invariant under the complexified Lorentz transformations that keep the tube, and the inner products $\langle \Psi_m(w), \Psi_n(z)\rangle$ are such functions; so $\Omega \mapsto \Omega$, $\Psi_n(z) \mapsto \Psi_n(\tilde B(s)z)$ extends to an isometry. By (Br) at $\mathfrak{w}_k = \eta e_3$, then at $\mathfrak{w}_k = \eta \tilde B(s) e_3$, and $\eta \downarrow 0$, it carries $\phi(f_1)\cdots\phi(f_n)\Omega$ to $\phi(f_1 \circ B(-s))\cdots\phi(f_n \circ B(-s))\Omega = U(B(s))\phi(f_1)\cdots\phi(f_n)\Omega$; $\mathcal{D}_0$ is dense, so it is $U(B(s))$. The set of ζ with $\tilde B(\zeta)z \in \mathsf{Tb}^{(n)}$ is invariant under real shifts and holds $i[0, \tau_0]$, so it holds a strip; there $\zeta \mapsto \Psi_n(\tilde B(\zeta)z)$ is holomorphic in norm and equals $e^{i\zeta K}\Psi_n(z)$ on $\mathbb{R}$; comparison with $\int_{[-\mathsf{M},\mathsf{M}]} e^{i\zeta k} dm^K_{\chi,\Psi_n(z)}$, where $m^K_{\chi,\psi} = \langle \chi, \mathcal{P}_K(\cdot)\psi\rangle$, $\mathcal{P}_K$ the spectral measure of K, and $m^K_\psi = m^K_{\psi,\psi}$, on bounded parts of the spectrum of K at $\zeta = i\tau$ gives $\int e^{-2\tau k}\, \mathrm{d}m^K_{\Psi_n(z)} \le \|\Psi_n(R_\tau z)\|^2$ and the identity.

A point $y \in \mathsf{I}_\delta$ is $(\rho\cos\alpha, y_2, \rho\sin\alpha)$ with $\rho > 0$, $0 < \alpha < \delta$, and $(R_\tau y)_3 = \rho\sin(\alpha + \tau) > 0$ for $0 \le \tau \le \pi - \delta$; at real points $d(x) = x_3$, and R_τ is real and linear; so for $x \in \mathsf{I}_\delta^{(n)}$ the points $y = x_1$ and $y = x_{k+1} - x_k$ give $R_\tau x \in \mathsf{Tb}^{(n)}$. (b) Let G be a small tuple in I_δ with the centres x°, and $0 \le \tau \le \pi - \delta$. The balls $\mathbb{B}(R_\tau x_k^\circ, \epsilon)$ that carry the entries of $R_\tau G$ have a first height $\ge \eta_{x^\circ}$ and heights that grow by $\ge 2\eta_{x^\circ}$, and $\epsilon < c_0\eta_{x^\circ}$; so $[R_\tau G] = \int \prod_k h_k(x_k)\, \Psi_n(R_\tau x)\, \mathrm{d}x$, and on $\prod_k \mathbb{B}(x_k^\circ, \epsilon) \subset \mathsf{I}_\delta^{(n)}$, $\Psi_n(R_\tau x) = e^{-\tau K}\Psi_n(x)$ by (a); both integrands are continuous in norm and $e^{-\tau K}$ is closed, so $[G] \in D(e^{-\tau K})$ and $e^{-\tau K}[G] = [R_\tau G]$; at $\delta = \tau = \frac{\pi}{2}$ this is $\Delta^{1/4}[G] = [RG]$; and $\Delta^{1/4}\Omega = \Omega$.

Let $\chi \perp (\Delta^{1/4} + 1)[\mathcal{V}_0]$; then $\chi \perp \Omega$. Let $0 < \delta < \frac{\pi}{2}$ and G' a small tuple in I_δ with the centres x°. For $0 \le \beta \le \frac{\pi}{2} - \delta$ the tuple $R_\beta G'$ is a small tuple in I with the centres $R_\beta x^\circ$: $R_\beta \mathsf{I}_\delta \subset \mathsf{I}$, so that $R_\beta \mathsf{I}_\delta^{(n)} \subset \mathsf{I}_{\pi/2}^{(n)}$, and the heights of $R_\tau R_\beta x^\circ$, $0 \le \tau \le \frac{\pi}{2}$, are those of $R_{\beta+\tau}x^\circ$ with $\beta + \tau \in [0, \pi - \delta]$, so that $\eta_{R_\beta x^\circ} \ge \eta_{x^\circ} > \epsilon/c_0$. The function $\mathsf{f}(z) = \int e^{-zk}\, \mathrm{d}m^K_{\chi,[G']}(k)$ is continuous on $0 \le \operatorname{Re} z \le \pi - \delta$, as $\int e^{-\operatorname{Re} z\, k} d|m^K_{\chi,[G']}| \le \|\chi\|(\|[G']\| + \|e^{-(\pi-\delta)K}[G']\|)$, holomorphic inside, and $\mathsf{f}(\beta) = \langle \chi, e^{-\beta K}[G']\rangle = \langle \chi, [R_\beta G']\rangle$.

As $\chi \perp (\Delta^{1/4} + 1)[R_\beta G'] = [R_\beta G'] + [R_{\beta+\pi/2} G']$, $\mathsf{f}(\beta) + \mathsf{f}(\beta + \frac{\pi}{2}) = 0$ for $0 \le \beta \le \frac{\pi}{2} - \delta$, hence on the strip $0 < \operatorname{Re} z < \frac{\pi}{2} - \delta$; at $z = \beta_0 + it$ the finite measure $e^{-\beta_0 k}(1 + e^{-\pi k/2}) m^K_{\chi,[G']}$ has Fourier transform 0, so it vanishes, and $\chi \perp [G']$. Small tuples in I_δ, with arbitrary entries, are had about every point of the open set $\mathsf{I}_\delta^{(n)}$, and $\langle \chi, \Psi_n(\cdot)\rangle$ is continuous; so it vanishes on $\mathsf{I}_\delta^{(n)}$, an open set of real points of the tube, hence on the tube ([7, Lem. G.5]), and $\chi \perp \mathcal{D}_0$ by (Br): $\chi = 0$.

The closure of $\Delta^{1/4}$ on $[\mathcal{V}_0]$ is symmetric, ≥ 0, lies in $\Delta^{1/4}$, and the range $(\Delta^{1/4}+1)[\mathcal{V}_0]$ is dense; so it is self-adjoint, and it is $\Delta^{1/4}$. (c) f_u is smooth with all Schwartz seminorms finite because $|\partial_s^a g(s+i\tau)| \leq c_a(1+|s|)^a e^{-s^2+\tau^2}$, $g(s)=e^{-s^2}$. As $\int g(s+i\beta')e^{isk}\,\mathrm{d}s = \sqrt{\pi}e^{-k^2/4+\beta' k}$, (a) used backwards gives for $y=(0,\rho,x_2)$ and $0<\gamma\leq\alpha$

$$\mathsf{F}(K)\Psi_1(\iota B(i\alpha)y) = \int g(s+i(\gamma+\tfrac{\pi}{4}-\alpha))\,\Psi_1(\iota B(s+i\gamma)y)\,\mathrm{d}s;$$

integrate against u in polar coordinates of the (x_1,x_3)-plane, pair with $e^{-\eta H}\chi$, and let $\gamma\downarrow 0$ by dominated convergence, the norms being $\leq C_\wp \min\{1,\eta\}^{-\wp}$ because $d(\iota B(i\gamma)x+\eta e_3)\geq\eta$ on W; (Br) at $n=1$ gives $\langle e^{-\eta H}\chi, \mathsf{F}(K)[u]\rangle = \langle e^{-\eta H}\chi, \phi(f_u)\Omega\rangle$, and $e^{-\eta H}\mathcal{H}$ is dense. If $\hat{f}_u(p)=0$ for all u, the entire function $\beta\mapsto\int g(s+i\beta)e^{ip\cdot B(s)y}\,\mathrm{d}s$ would vanish, so $g * e^{ip\cdot B(\cdot)y}=0$, which is absurd as $\hat{g}$ has no zero. □

Proposition 14.3 (Operators between two Euclidean times)**.** Let σ be data (Rec) with a unique vacuum. (a) If $O\neq\emptyset$ is open, $Q\in\mathcal{C}(O)$, $s>0$ and $Q\Omega, Q^*\Omega\in D(e^{sH})$, then $Q\in\mathbb{C}1$. So a bounded operator $e^{-sH}V_1e^{-sH}$, V_1 bounded and $s>0$, lies in no $\mathcal{C}(O)$ unless it is 0 or H is bounded, that is, unless $\mathcal{H}=\mathbb{C}\Omega$; this covers $\mathsf{B}(s,s')$, every product $\mathcal{T}(\mathfrak{z})\mathsf{B}(s,s')\mathcal{T}(\mathfrak{z}')\cdots$, and every weak limit V, along a subsequence and in the sense of Lem. 14.1, of lattice operators $V_j:[G]\mapsto[D_j\cdot\tau_{te_3}G]$ on the spaces of reflection positivity at $x_3=0$ of lattice states ρ_j that converge to σ and are reflection positive at the site planes orthogonal to e_3, as periodic Gibbs states are (Prop. YM3_I:103.4), the lattice approximants of a component among them, with $s,t\in\bigcup_K L^{-K}\mathbb{Z}$ and D_j placed in $\{s<x_3<t-s\}$, whose inner operators $V_{1,j}:[G]\mapsto[\tau_{-se_3}D_j\cdot\tau_{(t-2s)e_3}G]$ are bounded uniformly in j on the spaces of reflection positivity at $x_3=0$ of the shifted states $\rho_j\circ\tau_{se_3}$, $(\rho_j\circ\tau_{se_3})(F)=\rho_j(\tau_{se_3}F)$, which are reflection positive there because ρ_j is so at the site plane $x_3=s$. (b) For every open $O\neq\emptyset$ the von Neumann algebra generated by the e^{-tH} and the $U(x)\mathsf{B}(s,s)U(x)^{-1}$, $t,s>0$, $x\in O$, is $\mathcal{B}(\mathcal{H})$. ■

Proof sketch. (a) Take $O_1\neq\emptyset$ open and $\delta>0$ with $O_1+te_0\subset O$ for $|t|<\delta$, and a product X of fields of O_1. Weak commutation gives $\langle Q^*\Omega, e^{itH}X\Omega\rangle = \langle X^\dagger\Omega, e^{-itH}Q\Omega\rangle$ for $|t|<\delta$. The left member continues to a bounded holomorphic function on $\operatorname{Im}\zeta>-s$, the right member to one on $\operatorname{Im}\zeta<s$; they agree on an interval, so they form one bounded entire function, a constant (Liouville). So the spectral measure of H for $(Q^*\Omega, X\Omega)$ sits at 0; $\mathcal{D}_0(O_1)$ is dense (Thm. AO(a)), so $Q^*\Omega\in\ker H=\mathbb{C}\Omega$, and $Q^*-\bar{c}\in\mathcal{C}(O)$ annihilates Ω, which separates $\mathcal{C}(O)$. The operators named are of the form $V=e^{-sH}V_1e^{-sH}$ with a bounded V_1: $\mathsf{B}(s,s')$ and the products by the semigroup law; for a weak limit V of the V_j, se_3 and te_3 are lattice vectors for large j, and $\tau_{se_3}\theta\tau_{se_3}=\theta$ gives, for every state ρ_j,

$$\rho_j(\theta\bar{G}'\cdot D_j\cdot\tau_{te_3}G) = (\rho_j\circ\tau_{se_3})(\theta\tau_{se_3}\bar{G}'\cdot\tau_{-se_3}D_j\cdot\tau_{(t-2s)e_3}\tau_{se_3}G),$$

that is $\langle[G'],V_j[G]\rangle = \langle[\tau_{se_3}G'],V_{1,j}[\tau_{se_3}G]\rangle$, the right member in the space of $\rho_j\circ\tau_{se_3}$; the shifted states converge to $\sigma\circ\tau_{se_3}=\sigma$, as σ is invariant under the translations, so that their spaces, with the classes $[G]$, are frames for $\mathcal{H}(\sigma)$ as in Lem. 14.1; along a further subsequence $V_{1,j}\rightharpoonup V_1$ in these frames, V_1 bounded (Lem. 14.1(ii)), and $[\tau_{se_3}G]=e^{-sH}[G]$ gives $\langle[G'],V[G]\rangle=\langle e^{-sH}[G'],V_1e^{-sH}[G]\rangle$ for all G,G'. Then $V\Omega, V^*\Omega\in e^{-sH}\mathcal{H}\subset D(e^{sH})$, and $V=c\neq 0$ would make e^{sH} bounded. If H is bounded, the joint spectrum of the translations, which is invariant under the Lorentz transformations and lies in $\{E\geq|q|\}$, is bounded, so it is $\{0\}$; then $H=0$ and $\mathcal{H}=\ker H=\mathbb{C}\Omega$. (b) An operator Y commuting with the generators commutes with $\lim_t e^{-tH}=|\Omega\rangle\langle\Omega|$, so $Y\Omega=y\Omega$ and $Y=y$ on the vectors $e^{-t_0H}\mathsf{b}(x_1)e^{-t_1H}\cdots\mathsf{b}(x_n)\Omega$, $\mathsf{b}(x)=U(x)\mathsf{B}(s,s)U(x)^{-1}$, $x_k\in O$; a vector orthogonal to all of them is, by analyticity in the differences of the points and in the times, orthogonal to all point vectors, hence to $\mathcal{D}_0$ by (Br). □

Lemma 14.4 (A second reflection)**.** Let σ be data (Rec), Q bounded and $\tilde{q}\in\mathcal{H}$ with $\langle[G_1],Q[G_2]\rangle = \langle\tilde{q},[R^{-1}(\theta G_1\cdot G_2)]\rangle$ for all tuples G_1,G_2 over II. Then $Q\in\mathcal{C}(W')$. ■

Proof sketch. For $\mathfrak{z} \in \mathbb{C}^3$ put $\mathfrak{z}^\flat = R^{-1}\mathfrak{z} = (\mathfrak{z}_3, \mathfrak{z}_2, -\mathfrak{z}_1)$ and $\mathfrak{z}^\natural = (-\overline{\mathfrak{z}_3}, \overline{\mathfrak{z}_2}, -\overline{\mathfrak{z}_1})$, so that $\mathfrak{z}^\natural = R^{-1}\theta\mathfrak{z}$ for real $\mathfrak{z}$; for a point x of II the third coordinate of $x^\flat$ and of $x^\natural$ is $-x_1 > 0$. (1) Let $w \in \mathsf{II}^m$ and $z \in \mathsf{II}^n$ be Euclidean points with growing heights, $w_{1,3} < \cdots < w_{m,3}$ and $z_{1,3} < \cdots < z_{n,3}$, whose $N = m + n$ numbers $-w_{k,1}$, $-z_{k,1}$ are pairwise different. Then the identity of the lemma holds for the kernels: $\langle \Psi_m(w), Q\Psi_n(z)\rangle = \langle \tilde{q}, \Psi_N(\mathsf{s}(w,z))\rangle$, $\mathsf{s}(w,z)$ the points $w_k^\natural$, $z_k^\flat$ in the order of their third coordinates.

Indeed, integrated against real entries g_k, h_k carried by small balls about the w_k, z_k in II, the two members are the two members of that identity at $G_1 = (g_k)$, $G_2 = (h_k)$, whose classes, and the class of $R^{-1}(\theta G_1 \cdot G_2)$ after a permutation of its entries, are integrals of point vectors; and the kernels are continuous. (2) Let $0 < r_1 < \cdots < r_N$ be these numbers, which are the heights of the points of $\mathsf{s}(w,z)$, and y_i the second coordinates, and continue in the third coordinates alone: a ket point is $(-r_i, y_i, \omega_i)$, a bra point $(-r_i, y_i, -\overline{\omega_i})$, so that $\mathsf{s}(w,z)$ has the points $\mathsf{s}_i = (\omega_i, y_i, r_i)$. The left member is holomorphic in ω on the convex open set D_1 where the real parts of the ω_i of the ket points increase from 0 in the order of the ket entries and those of the bra points decrease from 0 in the order of the bra entries; the right member is holomorphic on the convex open set $D_2 = \{|\mathrm{Im}\,\omega_1| < r_1,\ |\mathrm{Im}(\omega_{i+1} - \omega_i)| < r_{i+1} - r_i\}$, as $d(\mathsf{s}_1) = r_1 - |\mathrm{Im}\,\omega_1|$ and $d(\mathsf{s}_{i+1} - \mathsf{s}_i) = r_{i+1} - r_i - |\mathrm{Im}(\omega_{i+1} - \omega_i)|$, whatever the real parts are. $D_1 \cap \mathbb{R}^N \subset D_2$, and the two members agree there by (1); by [7, Lem. G.5] they agree on $D_1 \cap D_2$. (3) Let $p \in W'$ and $\mathsf{r} > 0$ with $\overline{\mathbb{B}(p,\mathsf{r})} \subset W'$, and $\mathbb{B}_k = \mathbb{B}(p - 8k\mathsf{r}e_1, \mathsf{r}) \subset W'$, $k \in \mathbb{N}_0$, balls of $\mathbb{R}^{1+2}$, e_1 the unit vector of the x^1-axis.

For $x \in \mathbb{B}_k$ and $x' \in \mathbb{B}_{k'}$, $k < k'$: $-x^1 > |x^0|$ and $(-x'^1) - (-x^1) \geq 6\mathsf{r} > 2\mathsf{r} \geq |x'^0 - x^0|$. Take N functions $f_i \in \mathcal{F}(\mathbb{B}_{k_i})$, one in each of N balls of pairwise different indices, and let Y be a product of some of the $\phi(f_i)$ and X a product of the others. By (Br) at $\mathfrak{w}_a = \eta e_3$ the vector X_η obtained from $X\Omega$ by putting $e^{-\eta H}$ before each field is the integral of Ψ_n at the points $\iota x + a\eta e_3 = (x^1, x^2, a\eta - ix^0)$, a counting the fields of X from the left, and the like holds for Y_η; so $r_i = -x_i^1$, $y_i = x_i^2$, $\mathrm{Im}\,\omega_i = -x_i^0$ and $\mathrm{Re}\,\omega_i = \pm a\eta$, and $\omega \in D_1 \cap D_2$ by the inequalities just written.

By (2) and as $\eta \downarrow 0$,

$$\langle Y\Omega, QX\Omega\rangle = \int \prod \bar{f} \prod f\, \mathsf{L}(x)\, \mathrm{d}x$$

with the one kernel

$$\mathsf{L}(x) = \langle \tilde{q}, \Psi_N((-ix_i^0, x_i^2, -x_i^1)_i)\rangle,$$

whichever of the fields stand in the bra Y or in the ket X. (4) Hence

$$\langle \phi(\bar{h})\psi, Q\psi'\rangle = \langle \psi, Q\phi(h)\psi'\rangle$$

for $h \in \mathcal{F}(\mathbb{B}_0)$, ψ in the span of Ω and of the products of fields, one in each of distinct balls $\mathbb{B}_k$ of odd index, applied to Ω, and ψ' in the like span with even indices $k \geq 2$. (5) These spans stay dense after every power of $1 + H$ (analyticity in the translations), and the polynomial bound of the field carries the relation to $\mathcal{D}_0$: Q commutes weakly with every $\phi(h)$, $h \in \mathcal{F}(\mathbb{B}(p,\mathsf{r}))$.

Every $p \in W'$ and every $\mathsf{r} > 0$ with $\overline{\mathbb{B}(p,\mathsf{r})} \subset W'$ are admitted, these balls cover W', and a partition of unity on the support of $h \in \mathcal{F}(W')$ gives $Q \in \mathcal{C}(W')$. □

Lemma 14.5 (Two mirrors)**.** Let σ be data (Rec) and Q a bounded operator. For combinations F, F' over $\mathsf{Q}_L = \{x_1 < 0, x_3 > 0\}$ and $h = h_+ + h_-$, $h_\pm \in C_c^1(\{x_1 > 0,\ \pm x_3 > 0\})$, put $\ell^Q_{F',F}(h) = \langle [F'], Q[F \cdot h_+]\rangle + \langle [F' \cdot \theta\bar{h}_-], Q[F]\rangle$. If for all F, F' there is $C < \infty$ with $|\ell^Q_{F',F}(h)| \leq C\,\sigma(\theta_1\bar{h} \cdot h)^{1/2}$ for all such h, then $Q \in \mathcal{C}(W)$. ■

Proof sketch. $\sigma(\theta_1\bar{h} \cdot h) = \|[\mathsf{d}h]\|^2$ is the norm of reflection positivity at the mirror $x_1 = 0$. Step 1: by Riesz, $\ell(h) = \langle \Psi, [\mathsf{d}h]\rangle$ with some $\Psi \in \mathcal{H}$; with $v(y) = \mathsf{B}(y,1)\Omega$ and $[k] = \int k(x)U(\vec{x})v(x_3)\,\mathrm{d}x$ for $k \in C_c^1(\mathbb{E}_+)$, $\ell(h) = \int h(x)\Phi(x_1, x_2; x_3)\,\mathrm{d}x$, $\Phi(x_1, x_2; \zeta) = \langle \Psi, U((\zeta, x_2))v(x_1)\rangle$, and by the spectrum condition Φ continues holomorphically to $|\mathrm{Im}\,\zeta| < x_1$: its domain holds all real ζ of both signs. Step 2: with F, F' over $\{x_3 > \tau_0\}$, $\psi = [\tau_{-\tau_0 e_3}F]$, $\psi' = [\tau_{-\tau_0 e_3}F']$ and $\mathsf{B}(\zeta, \tau_0 - \zeta) = e^{-i\mathrm{Im}\zeta H}\mathsf{B}(\mathrm{Re}\,\zeta, \tau_0 -$

$\mathrm{Re}\,\zeta)e^{i\mathrm{Im}\zeta H}$, the first mirror gives the two functions $\mathsf{G}_+(\vec{x};\zeta)=\langle Q^*[F'],U(\vec{x})\mathsf{B}(\zeta,\tau_0-\zeta)U(\vec{x})^{-1}\psi\rangle$, $0<\mathrm{Re}\,\zeta<\tau_0$, and $\mathsf{G}_-(\vec{x};\zeta)=\langle\psi',U(\vec{x})\mathsf{B}(\tau_0+\zeta,-\zeta)U(\vec{x})^{-1}Q[F]\rangle$, $-\tau_0<\mathrm{Re}\,\zeta<0$; they are Φ on real segments, so on $0<\pm\mathrm{Re}\,\zeta<\tau_0$, $|\mathrm{Im}\,\zeta|<x_1$.

Step 3: for $f\in\mathcal{F}(W)$, $|t|\le x_1-\delta$ on $\mathrm{supp}f$; by eq. 14.i,

$$\int f\,\mathsf{G}_+(\vec{x};s-it)=\langle e^{-sH}Q^*[F'],\phi(f)e^{-(\tau_0-s)H}\psi\rangle\to\langle[F'],Q\phi(f)[F]\rangle$$

and $\int f\,\mathsf{G}_-(\vec{x};-s-it)\to\langle\phi(\bar f)[F'],Q[F]\rangle$ as $s\downarrow 0$, and both tend to $\int f\,\Phi(x_1,x_2;-it)$. Step 4: the classes over Q_L are dense after every power of N (a translation along e_1 carries every class into them, and a vector orthogonal to them gives a function of the translation that vanishes on a half-line and, as $|q_1|\le E$ on the spectrum and every class is $e^{-\varepsilon H}$ of a class, continues holomorphically to a strip, hence vanishes), and the polynomial bound of the field carries the relation to $\mathcal{D}_0$. □

Lemma 14.6 (Central positivity)**.** (a) Let $\mathfrak{A}$ be a unital $*$-algebra, ω a state on it with GNS data $(\pi,\mathcal{D},\Omega)$, and $X_0=X_0^*$ central. The following are equivalent: (CP) $\omega(\sum X_0^nY_n)\ge 0$ whenever $\sum\lambda^nY_n\in\mathfrak{A}^+$ for all real λ; (SA) $\pi(X_0)$ has a self-adjoint extension on a larger Hilbert space whose bounded Borel functions F satisfy $\langle\pi(Y^*)\psi,F\psi'\rangle=\langle\psi,F\pi(Y)\psi'\rangle$ for $Y\in\mathfrak{A}$, $\psi,\psi'\in\mathcal{D}$; (CP$^\sharp$) $\omega(a)\ge 0$ whenever $(1+X_0^2)^ka\in\mathfrak{A}^+$ for some $k\ge 0$; (Ext) with $\mathscr{Q}$ the polynomials $\sum_n\lambda^nY_n$ in a real variable λ with coefficients $Y_n\in\mathfrak{A}$, $\mathrm{ev}:\mathscr{Q}\to\mathfrak{A}$ the $*$-homomorphism $\sum_n\lambda^nY_n\mapsto\sum_nX_0^nY_n$, $\mathfrak{r}=\mathbb{C}[\lambda,(1+\lambda^2)^{-1}]$ and $\mathscr{Q}_\mathfrak{r}$ the $*$-algebra of the functions $\sum_k g_k(\lambda)Y_k$, $g_k\in\mathfrak{r}$, $Y_k\in\mathfrak{A}$: $\omega\circ\mathrm{ev}$ has a linear extension $\tilde\omega$ to $\mathscr{Q}_\mathfrak{r}$ with $\tilde\omega(F^*F)\ge 0$ for all $F\in\mathscr{Q}_\mathfrak{r}$. (b) A closed symmetric operator T affiliated with a von Neumann algebra $\mathfrak{M}$ has a self-adjoint extension on $\mathcal{H}\otimes\mathbb{C}^2$ affiliated with $\mathfrak{M}\otimes M_2(\mathbb{C})$. (c) Two closed symmetric operators affiliated with commuting von Neumann algebras have strongly commuting self-adjoint extensions on $\mathcal{H}\otimes\mathbb{C}^2\otimes\mathbb{C}^2$. (d) For a linear functional L on $\mathbb{C}[x,y]$ with $L(|q|^2)\ge 0$: $L(p)\ge 0$ for all real $p\ge 0$ on $\mathbb{R}^2$ if and only if $L(a)\ge 0$ for all real a such that $(1+x^2)^ka$ is a sum of squares for some k; Motzkin's polynomial M has the level $k=1$: $(1+x^2)M=(x^2y^2-1)^2+(xy(x^2-1))^2+(x(y^2-1))^2$. ■

Proof sketch. (a) (SA)⇒(CP), (CP$^\sharp$): with E the spectral measure of the extension let $\mu_Y=\langle\Omega,E(\cdot)\pi(Y)\Omega\rangle$, $Y\in\mathfrak{A}$, a complex measure, linear in Y. By (SA), $\mu_{Z^*Z}=\langle\pi(Z)\Omega,E(\cdot)\pi(Z)\Omega\rangle\ge 0$, so $\mu_b\ge 0$ for $b\in\mathfrak{A}^+$; and $d\mu_{X_0^nY}=\lambda^nd\mu_Y$, as the extension continues $\pi(X_0)$ on $\pi(\mathfrak{A})\Omega$. If $(1+X_0^2)^ka=b\in\mathfrak{A}^+$, then $(1+\lambda^2)^kd\mu_a=d\mu_b\ge 0$, so $\mu_a\ge 0$ and $\omega(a)=\mu_a(\mathbb{R})\ge 0$: (CP$^\sharp$). If $\sum_n\lambda^nY_n\in\mathfrak{A}^+$ for all real λ, then for each real λ_0 the measure $\sum_n\lambda_0^n\mu_{Y_n}$ is positive, so the densities ρ_n of the μ_{Y_n} with respect to the sum of their variations satisfy $\sum_n\lambda_0^n\rho_n\ge 0$ for all rational λ_0 off one null set, hence $\sum_n\lambda^n\rho_n(\lambda)\ge 0$, and $\omega(\sum X_0^nY_n)=\int\sum_n\lambda^n\rho_n(\lambda)\ge 0$: (CP). (CP)⇒(Ext): the extension theorem of M. Riesz, for the real space of the hermitian $F\in\mathscr{Q}_\mathfrak{r}$, its cone K of the F with $F(\lambda)\in\mathfrak{A}^+$ for every real λ, which holds every F^*F, and the subspace of the hermitian polynomials, on whose part in K the functional $\omega\circ\mathrm{ev}$ is ≥ 0 by (CP), gives an extension $\tilde\omega\ge 0$ on K, continued complex-linearly to $\mathscr{Q}_\mathfrak{r}$, because every hermitian F is majorised by a hermitian polynomial: F is a finite sum of terms gS, $g\in\mathfrak{r}$ real, $S=S^*\in\mathfrak{A}$ (as $gY+\bar gY^*=(\mathrm{Re}\,g)(Y+Y^*)+(\mathrm{Im}\,g)\,i(Y-Y^*)$), and for a real polynomial $q\ge|g|$

$$\tfrac12q(S^2+1)-gS=\tfrac14(q-g)(S+1)^2+\tfrac14(q+g)(S-1)^2\in\mathsf{K};$$

for $|g|\le C(1+\lambda^2)^r$ one may take $q=C(1+\lambda^2)^r$, so that the extension holds at each fixed $r\in\mathbb{N}_0$, from the polynomials $\sum_{n\le 2r}\lambda^nY_n$ to the functions whose coefficients are $O((1+\lambda^2)^r)$. (CP$^\sharp$)⇒(Ext): the same theorem, with the cone of the finite sums of the F^*F, $F\in\mathscr{Q}_\mathfrak{r}$, in the place of K. If a polynomial p is such a sum $\sum_iF_i^*F_i$, then the $G_i=(1+\lambda^2)^kF_i$ are polynomials for some k, $(1+\lambda^2)^{2k}p=\sum_iG_i^*G_i$, and the $*$-homomorphism ev gives $(1+X_0^2)^{2k}\mathrm{ev}(p)\in\mathfrak{A}^+$, so $\omega(\mathrm{ev}\,p)\ge 0$ by (CP$^\sharp$). The majorisation: for $g=p_0/(1+\lambda^2)^k$, p_0 a real polynomial, $q=\frac12(1+p_0^2)\ge|g|$, and $q\mp g=n_\mp/(1+\lambda^2)^k$ with real polynomials $n_\mp\ge 0$ on $\mathbb{R}$, which are $|m_\mp|^2$ for complex polynomials $m_\mp$, a polynomial that is ≥ 0 on $\mathbb{R}$ being a sum of two squares; so $q\mp g=|u_\mp|^2$ with $u_\mp=m_\mp(\lambda-i)^k(1+\lambda^2)^{-k}\in\mathfrak{r}$, and the

identity above reads $\frac12 q(S^2+1) - gS = (\frac12 u_-(S+1))^*(\frac12 u_-(S+1)) + (\frac12 u_+(S-1))^*(\frac12 u_+(S-1))$. (Ext)⇒(SA): $\tilde\omega(F^*) = \overline{\tilde\omega(F)}$ by polarisation; in the GNS space of $\tilde\omega$, which holds that of ω isometrically, multiplication by λ is symmetric with $(\lambda \pm i)\mathscr{Q}_{\mathfrak r} = \mathscr{Q}_{\mathfrak r}$, as $(\lambda \pm i)^{-1} \in \mathfrak r$, so its closure is self-adjoint and extends $\pi(X_0)$; the multiplications by the $Y \in \mathfrak A$ commute with its resolvents, and the bounded C with $\langle \pi(Y^*)\psi, C\psi'\rangle = \langle \psi, C\pi(Y)\psi'\rangle$ for $Y \in \mathfrak A$, $\psi, \psi' \in \mathcal D$, form a weakly closed space that holds the polynomials in the two resolvents, hence, by the bicommutant theorem, all bounded Borel functions of the closure. (b) $T \oplus (-T)$ has the deficiency spaces $\mathcal N_+ \oplus \mathcal N_-$ and $\mathcal N_- \oplus \mathcal N_+$, which the flip $(x,y) \mapsto (y,x) \in 1 \otimes M_2$ exchanges isometrically; the extension that this flip defines commutes with the unitaries of $\mathfrak M' \otimes 1$. (c) Apply (b) in two tensor factors. (d) This is (CP)⇔(CP$^\sharp$) of (a) for $\mathfrak A = \mathbb C[x,y]$, $X_0 = x$ and $\omega = L/L(1)$ (if $L(1) = 0$, then $L = 0$ by the Schwarz inequality), where $\mathfrak A^+$ is the set of the sums of squares of real polynomials, as $|q|^2 = (\operatorname{Re} q)^2 + (\operatorname{Im} q)^2$: there (CP) says $L(p) \ge 0$ for every real $p \ge 0$ on $\mathbb R^2$, since $\sum_n \lambda^n Y_n \in \mathfrak A^+$ for all real λ gives $\sum_n x^n Y_n \ge 0$ on $\mathbb R^2$ (put $\lambda = x$), and a real $p = \sum_n x^n p_n(y) \ge 0$ has $\sum_n \lambda^n p_n \in \mathfrak A^+$ for every real λ, a polynomial in one variable that is ≥ 0 being a sum of two squares; and if $(1+x^2)^k a$ is a sum of squares, a is real.

The identity is checked by expanding, the three squares summing to $x^6y^2 + x^4y^4 - 2x^4y^2 + x^2y^4 - 3x^2y^2 + x^2 + 1$. □

Lemma 14.7 (Decomposition and limits)**.** Let $(\mathsf X, \lambda, (\sigma_\xi))$ be a canonical decomposition of data (Rec) σ, all σ_ξ data (Rec) with one triple, and $\mathcal W_\xi$ their Wightman functionals. (i) There is exactly one unital $*$-representation $g \mapsto M_g$ of $C(\mathsf X)$ on $\mathcal H(\sigma)$ with $\langle [F'], M_g[F]\rangle = \int g(\xi)\sigma_\xi(\theta\bar F' \cdot F)\lambda(\mathrm d\xi)$; $\|M_g\| \le \sup|g|$, and M_g commutes with e^{-tH}, the $U(x)$, the insertions and the $\mathsf B(s,s')$, keeps every $D(H^a)$, and commutes with every $\phi(f)$ on $\mathcal D_\infty$; so $M_g \in \mathcal C(O)$ for every O. (ii) For $u \in L^1(\mathbb R^{1+2})$ let $\phi_u(\psi',\psi) = \int u(x)\, q_0(U(x)^{-1}\psi', U(x)^{-1}\psi)\,\mathrm dx$ on the vectors $\psi = e^{-\eta H}\chi$, $\psi' = e^{-\eta H}\chi'$, $\eta > 0$, among them the classes $[F]$ and the $M_g[F]$, F over $\mathbb E_+$: the form of the field smeared with u, with $|\phi_u(\psi',\psi)| \le \|u\|_{L^1} C_\wp \min\{\eta,1\}^{-\wp}\|\chi'\|\|\chi\|$ and $\phi_u(\psi',\psi) = \langle \psi', \phi(u)\psi\rangle$ for $u \in \mathcal S$, by eq. 14.i; ϕ_u^ξ is the same form for σ_ξ. For combinations F, F' over $\mathbb E_+$ and $u \in L^1(\mathbb R^{1+2})$ the functions $\xi \mapsto \langle [F']_\xi, [F]_\xi\rangle$, $\langle [F']_\xi, H_\xi[F]_\xi\rangle$ and $\phi_u^\xi([F']_\xi, [F]_\xi)$ are continuous, and $\phi_u([F'], M_g[F]) = \int g\, \phi_u^\xi([F']_\xi, [F]_\xi)\,\mathrm d\lambda$; the like for 1 and for H. (iii) For $a \in \mathfrak P$, $\xi \mapsto \mathcal W_\xi(a)$ is continuous and $\langle \Omega, M_g\pi(a)\Omega\rangle = \int g\, \mathcal W_\xi(a)\,\mathrm d\lambda$; in particular $\mathcal W_\sigma = \int \mathcal W_\xi\,\mathrm d\lambda$. For $a, b_1, \dots, b_q \in \mathfrak P$ and spatial translations v_i, $\lim_{|v_q|\to\infty} \cdots \lim_{|v_1|\to\infty} \mathcal W_\sigma(a\,\tau_{v_1}b_1 \cdots \tau_{v_q}b_q) = \int \mathcal W_\xi(a) \prod_i \mathcal W_\xi(b_i)\,\lambda(\mathrm d\xi)$, each limit existing in turn. (iv) Let $a \in \mathfrak P$, $a^* = a$, and suppose that for all $b_1, \dots, b_r \in \mathfrak P$ there is R such that $\mathcal W_\sigma(b^*ab) \ge 0$ for every b obtained from a complex polynomial $\mathsf p$ in r variables by putting, at each occurrence of its i-th variable, a mean $\frac{1}{\mathsf n}\sum_{\nu=1}^{\mathsf n} \tau_{v_\nu} b_i$ over spatial translations $v_\nu = (0, \vec v_\nu)$ of length $\ge R$, all the translations that occur in b being pairwise different. Then $\mathcal W_\xi(a) \ge 0$ for every ξ. (v) If data (Rec) σ_j with one triple converge to σ on separated tuples, the forms of the fields and of the energy on the classes of combinations converge, and $\mathcal W^{\sigma_j}(f_1 \cdots f_n) \to \mathcal W^\sigma(f_1 \cdots f_n)$ for all $f_k \in \mathcal S$. ■

Proof sketch. (i) On $C(\mathsf X) \odot \{F\}$ the form $\int \bar g' g \langle [F']_\xi, [F]_\xi\rangle\,\mathrm d\lambda$ is positive; multiplication by g acts on its Hilbert space $\mathfrak K$, and $[F] \mapsto 1 \otimes F$ is an isometry $\mathsf W$ of $\mathcal H(\sigma)$ into $\mathfrak K$. For a tuple Y over $\mathbb E_+$ and far horizontal translations v the vectors $[F \cdot \tau_v Y]$ tend weakly to $\mathsf W^*\mathsf m(g_Y)\mathsf W[F]$, $g_Y(\xi) = \sigma_\xi(Y)$, and the iterated cluster limits give $\|\mathsf W^*\mathsf m(g_Y)\mathsf W[F]\| = \|\mathsf m(g_Y)\mathsf W[F]\|$, so $\mathsf m(g_Y)$ keeps the range of $\mathsf W$. The g_Y separate the points of $\mathsf X$ ($\xi \mapsto \sigma_\xi$ is injective); by Stone–Weierstrass every $\mathsf m(g)$ keeps the range, which is then all of $\mathfrak K$, and $M_g = \mathsf W^{-1}\mathsf m(g)\mathsf W$.

It commutes with e^{-tH} because each e^{-tH_ξ} is symmetric, with the translations, and with the insertions because the vector $\mathsf B^{\mathrm{in}}_\eta(h)[G]$ is a class given by one formula for σ and for all σ_ξ; hence with all $\mathsf B(s,s')$ and with the fields. (ii) For an insertion smeared in space, $\mathsf G_{\mathsf k}(z) = \int \mathsf k(\vec x)\langle e^{-(\frac\eta2 + \bar z)H}\chi', U(\vec x)\mathsf B(\frac\eta2, \frac\eta2)U(\vec x)^{-1}e^{-(\frac\eta2 - z)H}\chi\rangle d^2x$ is holomorphic on the strip $|\operatorname{Re} z| < \frac\eta2$ and

bounded by $\|\mathsf{k}\|_{L^1} C_\wp (\eta/2)^{-\wp} \|\chi'\| \|\chi\|$, the same numbers for all ξ; at real z it gives the Euclidean insertion, at $z = -it$ the real-time field.

Montel's theorem and the identity theorem carry the continuity in ξ of the Euclidean values to the imaginary axis; the same with the weight M_g. (iii) In n variables:

$$\mathcal{W}_n^{(\varepsilon)}(f_1, \dots, f_n) = \langle \Omega, \phi(f_1) e^{-\varepsilon H} \cdots e^{-\varepsilon H} \phi(f_n) \Omega \rangle$$

is a value on the imaginary axes of a function holomorphic and bounded on a polystrip, uniformly in ξ, and $|\mathcal{W}_n - \mathcal{W}_n^{(\varepsilon)}| \le \varepsilon\, \mathsf{C}(f_1, \dots, f_n)$ with C formed from the triple and the $\|\partial_t^k f_i\|_{L^1}$; Montel and the identity theorem give the continuity. For the cluster limits, the analytic Wightman functions are bounded by $(C_\wp \eta^{-\wp})^n$ uniformly in ξ and in horizontal translations, and the rate

$$\begin{aligned} &|\mathcal{W}(f_1 \cdots f_n) - \langle \Omega, e^{-\eta H} \phi(f_1) \cdots e^{-\eta H} \phi(f_n) \Omega \rangle| \\ &\qquad \le c \int_0^\eta (1 + |\log s|)\, \mathrm{d}s, \\ &\qquad c = 2^{\mathsf{m}-1} (2^\wp C_\wp)^n \max_{l \le \mathsf{m}+1} \Big\| \mathsf{D}^l \prod_k f_k \Big\|_{L^1}, \end{aligned}$$

$\mathsf{D} = \sum_k k\, \partial/\partial x_k^0$, m an integer $\ge \max\{2, n\wp\}$ (by (Br) at $\mathfrak{w}_k = \eta e_3$ the integrand is holomorphic in the $x_k^0 + ik\eta$, and $\mathsf{m}+1$ integrations by parts in the real times, with the bound of the point vectors at $\frac{\eta}{2}$, bound the derivatives in η), with c formed from the triples ($C_\wp$, $\wp$ the larger of the numbers of the triple of σ and of the one triple of the σ_ξ), n and norms that translations of the f_k do not change, so free of ξ and of the translations, exchanges the cluster limits with $\eta \downarrow 0$. (iv) Expand $b^* a b$ in the means and let the translations tend to infinity one after the other, a factor whose translation goes off commuting in $\mathfrak{P}$ with the others once it is spacelike to them.

By (iii) the terms in which no translation of b^* is one of b tend to $\int \mathcal{W}_\xi(a)$ times the product of the $\mathcal{W}_\xi(b_i)$ and $\mathcal{W}_\xi(b_i^*) = \overline{\mathcal{W}_\xi(b_i)}$ that they carry, and the others, a share $O(\frac{1}{\mathsf{n}})$ of the terms, are bounded uniformly, $\xi \mapsto \mathcal{W}_\xi(c')$ being continuous on the compact X for each $c' \in \mathfrak{P}$. So $0 \le \int \mathcal{W}_\xi(a)(|\mathsf{q}(\xi)|^2 + O(\frac{1}{\mathsf{n}}))\, \mathrm{d}\lambda$, $\mathsf{q} = \mathsf{p}(\mathcal{W}_\xi(b_1), \dots, \mathcal{W}_\xi(b_r))$, and $\int \mathcal{W}_\xi(a) |\mathsf{q}|^2\, \mathrm{d}\lambda \ge 0$. These q form a unital $*$-algebra of continuous functions that separates the points of X, as $\xi \mapsto \sigma_\xi$ is injective and $\mathcal{W}_\xi$ fixes σ_ξ (its analytic continuation to Euclidean points gives the Schwinger functions of σ_ξ); it is dense in $C(\mathsf{X})$ (Stone–Weierstrass). $\mathcal{W}_\xi(a)$ is real and continuous and λ has full support: $\mathcal{W}_\xi(a) \ge 0$ for every ξ. (v) The proofs of (ii) and (iii) use of the family only one triple and the convergence on separated tuples. □

Lemma 14.8 (Bump functions with transforms free of zeros; division)**.** (a) For $\epsilon_k > 0$ with $\sum_k \epsilon_k = \epsilon < \infty$ let u be the infinite convolution of the probability densities $\frac{e^{-x/\epsilon_k}}{\epsilon_k(1-e^{-1})}$ on $[0, \epsilon_k]$, and $w = \mathsf{u} * \mathsf{u}(-\cdot)$. Then $w \in C_c^\infty(\mathbb{R})$, $w \ge 0$, w is even, $\int w = 1$, $\operatorname{supp} w \subset [-\epsilon, \epsilon]$, $\hat{w}$ is entire, $\hat{w}(s) = |\hat{\mathsf{u}}(s)|^2 > 0$ on $\mathbb{R}$, and $\hat{w}$ has no zero in $|\operatorname{Im} \zeta| < 1/\epsilon$; $f_* = w \otimes w \otimes w$ is the function of Thm. AO(d). At $\epsilon_k = \frac{6\epsilon}{\pi^2 k^2}$: $\|w^{(k)}\|_{L^1} \le A_0 R_0^k (k!)^2$, $A_0 = \frac{e^3 \pi}{2}$, $R_0 = \frac{3e\pi^2}{2\epsilon}$. (b) For $\delta > 0$, $\gamma > 1$ and $a_k = \delta k^{-\gamma} / \sum_l l^{-\gamma}$ let $g_{\delta,\gamma}$ be the infinite convolution of the uniform probability densities on $[-\frac{a_k}{2}, \frac{a_k}{2}]$: $g_{\delta,\gamma} \in C_c^\infty(\mathbb{R})$ is ≥ 0, even, $\int g_{\delta,\gamma} = 1$, $\operatorname{supp} g_{\delta,\gamma} \subset [-\frac{\delta}{2}, \frac{\delta}{2}]$.

If $\epsilon_k = \epsilon 2^{-k}$ and $\gamma > 1$, or $\epsilon_k = 6\epsilon/\pi^2 k^2$ and $1 < \gamma < 2$, then $g_{\delta,\gamma} = w * \mu$ with a real even $\mu \in \mathcal{S}(\mathbb{R})$, whatever the sizes of δ and ϵ; so $g_{\delta,\gamma}^{\otimes 3} = f_* * \mu^{\otimes 3}$. (c) For data (Rec), real $u \in \mathcal{S}$ and real $\mu \in L^1(\mathbb{R}^{1+2})$ with $u * \mu \in \mathcal{S}$: $\langle \psi, \phi(u*\mu)\psi' \rangle = \int \mu(a) \langle U(a)^{-1}\psi, \phi(u) U(a)^{-1} \psi' \rangle\, \mathrm{d}a$ on $\mathcal{D}_\infty$, so $c_*(u * \mu) \le \|\mu\|_{L^1} c_*(u)$ and $b_{mn}(u*\mu) \le \|\mu\|_{L^1} b_{mn}(u)$; and $c_*(u \circ \Lambda^{-1}) \le ((\Lambda e_0)^0 + |\overrightarrow{\Lambda e_0}|) c_*(u)$ for Lorentz transformations Λ. (d) For $f_0 \in C_c^\infty$, $f_0 \ge 0$, $f_0 \ne 0$, every real p has a Lorentz transformation Λ with $\hat{f}_0(\Lambda^{\mathrm{T}} p) \ne 0$. ■

Proof sketch. (a) $\hat{\mathsf{u}}(\zeta) = \prod_k \frac{1 - e^{-(1 - i\zeta\epsilon_k)}}{(1-e^{-1})(1-i\zeta\epsilon_k)}$, a convergent product since $\sum \epsilon_k < \infty$; no factor vanishes for $|\operatorname{Im}\zeta| < 1/\epsilon_k$, as $|e^{-(1-i\zeta\epsilon_k)}| < 1$ there. The decay of $\hat{\mathsf{u}}$ faster than every power gives smoothness;

on $|\operatorname{Im}\zeta| \le \frac{1}{4\epsilon}$, $|\hat w(s+iq)| \ge e^{-2\epsilon|q|}\Pi(s)^{-2}$, $\Pi(s) = \prod_k (1+\frac{16}{9}s^2\epsilon_k^2)^{1/2}$, with $\Pi(s) \le e^{1/6}(1+y^2)^{(1+\log_2^+ y)/2}$, $y = \frac{4}{3}\epsilon|s|$, at the dyadic choice and $\Pi(s) \le \exp(2.6(8\epsilon|s|/\pi^2)^{1/2})$ at the other; there also $\hat w(s) \le \exp(2-2(2\epsilon|s|/e\pi^2)^{1/2})$, which gives the Gevrey bound. (b) $\hat g_{\delta,\gamma}(\zeta) = \prod_k \frac{\sin(a_k\zeta/2)}{a_k\zeta/2}$ has $|\hat g_{\delta,\gamma}(s+iq)| \le \exp(\frac{\delta}{2}|q| + 1 - (\delta|s|/2e\zeta_\gamma)^{1/\gamma})$, $\zeta_\gamma = \sum k^{-\gamma}$: it falls faster than $|\hat w|$ falls, and $\hat w$ has no zero on a strip; so $\hat g_{\delta,\gamma}/\hat w$ is holomorphic on the strip and its restriction to $\mathbb{R}$ is a Schwartz function. (c) $\phi(u*\mu) = \int \mu(a)U(a)\phi(u)U(a)^{-1}\,\mathrm{d}a$, and $U(a)$ commutes with N and with the P_m; for a Lorentz transformation $U(\Lambda)HU(\Lambda)^{-1} = \int(\Lambda^0{}_0 E + \Lambda^j{}_0 q_j)d\mathcal{P} \le ((\Lambda e_0)^0 + |\overrightarrow{\Lambda e_0}|)H$ on the spectrum $\{E \ge |q|\}$. (d) Rotate p to p' with $p'_2 = 0$; $s \mapsto \hat f_0(B(s)^{\mathrm{T}}p')$ is entire, and at $s = i\frac{\pi}{2}$ it is $\int e^{p'_1x^0 + p'_0x^1} f_0(x)\,\mathrm{d}x > 0$; so it does not vanish at some real s. □

Lemma 14.9 (Sharp loops on a torus)**.** Let $\mathsf{b} > 0$, $\ell = 2L^m$, $a_K = L^{-K}$, $\beta = L^K\mathsf{b}$, and let $\langle\cdot\rangle_K$ be the Wilson measure μ_β on $\Lambda_{K,\ell} = a_K\mathbb{Z}^3/\ell\mathbb{Z}^3$. For $j \in \frac{1}{2}\mathbb{N}$ let D^j be the irreducible representation of SU(2) of dimension $2j+1$ and χ_j its character. Let C be a closed path of $\Lambda_{K_0,\ell}$ for some K_0, read as a path of every $\Lambda_{K,\ell}$, $K \ge K_0$, that has no self-intersection, does not wind (the vectors of its bonds add up to 0 in $\mathbb{R}^3$) and has a neighbourhood in a cube of side $\le \ell/2$; $|C|$ is its length. Let $\mathsf{d} > 0$ and let F_K be Borel functions, $|F_K| \le 1$, of the bonds of $\Lambda_{K,\ell}$ at a distance $\ge \mathsf{d}$ from C. Then for every $j \ge \frac{1}{2}$ there are K_j^{sl}, formed from j, C, b, ℓ and d, and A_j^{sl}, formed from j, C, b and ℓ, with

$$\big|\langle F_K\,\chi_j(U(C))\rangle_K\big| \le A_j^{\mathrm{sl}}\,(a_K\log^3 a_K^{-1})^{j(j+1)|C|/4\pi\mathsf{b}} \qquad (K \ge K_j^{\mathrm{sl}}). \tag{14.iii}$$

Hence, for every continuous h on $[-1,1]$, $\langle F_K h(W_C)\rangle_K - \bar h\langle F_K\rangle_K \to 0$ uniformly in such F_K, where $W_C = \frac{1}{2}\operatorname{tr}U(C)$ and $\bar h = \frac{2}{\pi}\int_{-1}^1 h(w)\sqrt{1-w^2}\,\mathrm{d}w$. ■

Proof sketch. We count in units of a_K: C has $n = |C|/a_K$ steps and p_C sides; $\mathsf{c}_j = j(j+1)$, $\Lambda_C = \max\{1, |C|/\mathsf{b}\}$ and $\rho = \lfloor(\beta/\Lambda_C\log^3\beta)^{1/4}\rfloor$. *Slabs.* For a step c of C with the first point x_- and the vector e let $\Pi(c)$ be the lattice plane through x_- orthogonal to c, $\mathsf{T}_\rho(c)$, the slab of c, the set of the translates $c(y)$ of c by the vectors y of $\Pi(c) - x_-$ with $|y| < \rho$, and $\{x_- + y + s'\mathsf{e} : |y|_\infty \le \rho,\ 0 \le s' \le 1\}$ its box. Let $K \ge K_j^{\mathrm{sl}}$, where K_j^{sl} is so large that for $K \ge K_j^{\mathrm{sl}}$: $\ell/a_K \ge 10$, $\beta \ge e^3$, $\rho \ge 2$, $\frac{4}{3}\mathsf{c}_j \le 20\log\beta$, every side of C has at least $4\rho+1$ steps, two sides without a common corner are at a sup-distance $\ge 2\rho+1$, the points at a sup-distance $\le \rho+1$ from C lie in a cube of side $\le \ell/2a_K$, and $\rho a_K < \mathsf{d}$.

The steps c_i of C at least 2ρ steps away from its corners form a set I of $n_I \ge n - 4\rho\mathsf{p}_C$ steps; no step of C but c_i runs through a bond of $\mathsf{T}_\rho(c_i)$, the box of c_i holds no bond of another slab of I, and no plaquette contains bonds of two of these slabs. So, given the bonds off $B = \bigcup_{i\in I}\mathsf{T}_\rho(c_i)$, the slabs are independent under the Wilson measure, and F_K and the paths δ_i of C from c_i to the next step of I read no bond of B; with Q_i the conditional mean of $D^j(U(c_i))$, $\langle F_K\chi_j(U(C))\rangle_K = \langle F_K \operatorname{tr}\prod_i Q_i D^j(U(\delta_i))\rangle_K$, and $|\langle F_K\chi_j(U(C))\rangle_K| \le (2j+1)\langle\prod_i\|Q_i\|\rangle_K$. *One slab.* Given the outer bonds, the variables $v_y = U(c(y))$, $|y| < \rho$, of the slab of $c = c_i$ have a density $\varrho_\circ \propto \exp(-\beta\sum_e a_{p(e)})$ for Haar measure, $p(e)$ the plaquette spanned by a bond e of $\Pi(c)$ and the direction of c. Move $v_y \mapsto \exp(is\varphi_\rho(y)\,\mathsf{n}_y\cdot\frac{\sigma}{2})v_y$, with the logarithmic weight $\varphi_\rho(y) = \min\{1, \log(\rho/|y|)/\log\rho\}$, $\varphi_\rho(0) = 1$, $\varphi_\rho(y) = 0$ for $|y| \ge \rho$, and the colour $\mathsf{n}_y\cdot\sigma = P_y^{-1}(\mathsf{t}\cdot\sigma)P_y$, t a unit vector of $\mathbb{R}^3$, $P_y = U(\pi_y)$ for the path π_y of $\Pi(c)$ from x_- along the first axis and then parallel to the second axis to $x_- + y$.

These moves are left translations and keep Haar measure; integration by parts along them and the Schwarz inequality give the information inequality for the three colours at once, $\|Q_i\|^2 \le (1+\mathsf{c}_j/\bar{\mathcal{I}})^{-1}$, $\bar{\mathcal{I}}$ the largest, over t, conditional mean of $-\partial_s^2\log\varrho_\circ$ at $s=0$. Along a move, for a bond e of $\Pi(c)$ from $x_- + y$ to $x_- + y'$, $\partial_s^2 a_{p(e)} \le \frac{1}{4}(\varphi_\rho(y) - \varphi_\rho(y'))^2$, plus $\varphi_\rho(y)\varphi_\rho(y')(3a(H_e) + a_{p(e)})$, $a(g) = 1 - \frac{1}{2}\operatorname{tr}g$, if e is a rung of the comb of the π_y, a bond parallel to the first axis at a distance $t_e \ge 1$ from it; there $H_e = P_{y'}U(e)^{-1}P_y^{-1}$ is a conjugate of the holonomy around the strip S_e of the t_e plaquettes of $\Pi(c)$ between e and the first axis, and $a(H_e) \le t_e\sum_{q\in S_e} a_q$ (Stokes for a strip).

With $\sum(\varphi_\rho(y)-\varphi_\rho(y'))^2 \le \bar E_\rho = (2\pi\log\rho + 7.63)/\log^2\rho$, the sum over the bonds $\langle y,y'\rangle$ of $\mathbb{Z}^2$: $\bar{\mathcal{I}} \le \frac{\beta}{4}(\bar E_\rho + X_i)$, $X_i = \sum_e \varphi_\rho(y)\varphi_\rho(y')(12a(H_e) + 4\,\mathbb{E}_i(a_{p(e)}))$ over the rungs, $\mathbb{E}_i$ the conditional mean, and

$$\log\|Q_i\| \le -\frac{2\mathsf{c}_j}{\beta\bar E_\rho}\Big(1-\frac{2\mathsf{c}_j}{\beta\bar E_\rho}\Big) + \frac{2\mathsf{c}_j}{\beta\bar E_\rho^2}\,X_i.$$

The mean. Jensen's inequality for the conditional means, the independence of the slabs and Hölder's inequality bound $\langle\exp\sum_i \frac{2\mathsf{c}_j}{\beta\bar E_\rho^2}X_i\rangle_K$ by $\mathsf{M}_\beta^{\varpi}$, $\mathsf{M}_\beta = \sup_p\langle e^{\beta a_p}\rangle_K$, $\varpi = 40\rho^4\mathsf{c}_j n_I/(\beta\bar E_\rho)^2 \le 1$. And $\log\mathsf{M}_\beta \le 192\mathsf{b}/\ell + 4(14.15 + 3\log\beta)$: the Schwarz inequality at site mirrors, used $3\mathsf{k}$ times, gives $\mathsf{M}_\beta \le \langle\prod_q e^{\beta a_q}\rangle_K^{\mathsf{w}^{-3}}$, the product over the w^3 plaquettes q parallel to one plaquette in $[0,\mathsf{w}]^2\times\{0,2,\dots,2\mathsf{w}-2\}$, $\mathsf{w}=2^{\mathsf{k}} \in (\frac{\ell}{4a_K},\frac{\ell}{2a_K}]$; and the equation of the Wilson measure for the bonds of the box $[0,\mathsf{w}]^2\times[0,2\mathsf{w}-2]$ bounds the conditional mean of that product by $e^{2\beta|\mathsf{P}_\partial|}$, $|\mathsf{P}_\partial| \le 24\mathsf{w}^2$ the number of the plaquettes that stick out of the box, divided by the partition function of the box with free boundary, which is $\ge (1.4\cdot 10^6\beta^3)^{-(\mathsf{w}+1)^2(2\mathsf{w}-1)}$ (a spanning tree of the box, and a ball of radius $(4\beta)^{-1/2}$ about $\mathbf{1}$ for the other bonds).

As $2\mathsf{c}_j/\bar E_\rho \ge \frac{\mathsf{c}_j}{\pi}(\log\rho - 1.2144)$, $n_I/\beta \ge |C|/\mathsf{b} - 4\rho\mathsf{p}_C/\beta$, $\rho \ge \frac12(\beta/\Lambda_C\log^3\beta)^{1/4}$ and $\varpi\log\mathsf{M}_\beta$ stays bounded, the product of the $\|Q_i\|$ gives eq. 14.iii. *The law.* $\chi_{l/2}(g) = \mathsf{U}_l(\frac12\mathrm{tr}\,g)$ with the Chebyshev polynomials U_l of the second kind, $\int\chi_j(g)\,\mathrm{d}g = 0$ for $j>0$, and the law of $\frac12\mathrm{tr}\,g$ under Haar measure is $\frac{2}{\pi}\sqrt{1-w^2}\,\mathrm{d}w$. For a polynomial $h=\sum_l\hat h_l\mathsf{U}_l$, $\langle F_K h(W_C)\rangle_K - \bar h\langle F_K\rangle_K = \sum_{l\ge1}\hat h_l\langle F_K\chi_{l/2}(U(C))\rangle_K \to 0$ by eq. 14.iii, uniformly in F_K; the polynomials are dense in $C[-1,1]$, and $|W_C|\le 1$. □

Lemma 14.10 (From slabs to linear blocks)**.** Let $H\ge0$ be self-adjoint, V a real vector space with a seminorm $|\cdot|_V$, and $\mathsf{B}_h(s,s')$, $h\in V$, $s,s'>0$, bounded operators, linear in h, with $e^{-tH}\mathsf{B}_h(s,s')e^{-t'H} = \mathsf{B}_h(s+t,s'+t')$, $\mathsf{B}_h(s,s')^* = \mathsf{B}_h(s',s)$, and $\|\int\rho_e(\varsigma/\eta)\mathsf{B}_h(\eta+\varsigma,\eta-\varsigma)\,\mathrm{d}\varsigma\| \le C_e|h|_V$ for $\eta = \frac{1}{4(k+2)}$, $k\in\mathbb{N}_0$, where $\rho_e\ge0$ is measurable on $[-c_1,c_1]$, $c_1\le\frac12$, $\bar\rho_e = \int\rho_e \in(0,\infty)$. Then: (a) there is exactly one family of hermitian forms q_h on the vectors of bounded energy with $q_h(e^{-sH}\chi, e^{-s'H}\chi') = \langle\chi,\mathsf{B}_h(s,s')\chi'\rangle$; (b) $|q_h(P_m\psi,P_n\psi')| \le C_\flat|h|_V(m+n+2)\|P_m\psi\|\|P_n\psi'\|$ with $C_\flat = 8C_e/\bar\rho_e$; (c) for $t\mapsto f_t\in V$ three times continuously differentiable for $|\cdot|_V$ and of compact support, $\nu_j(f) = \int|\partial_t^j f_t|_V dt$ and $\nu^*(f) = \sum_{j\le3}\binom{3}{j}2^{3-j}\nu_j(f)$, the time average $\mathsf{e}(f)(\psi,\psi') = \int q_{f_t}(e^{-itH}\psi,e^{-itH}\psi')\,\mathrm{d}t$ has $|\mathsf{e}(f)(\psi,\psi)| \le c_e(f)\langle\psi,(1+H)\psi\rangle$, $c_e(f) = 2C_\flat[\nu_0(f) + 2(\zeta(2)+\zeta(3))\nu^*(f)]$, and extends to the form domain of H with the same bound; (d) if, for data (Rec), the forms e of such a family with $V = C^1_c(\mathbb{R}^2)$ and the field satisfy the identity of Thm. AO(f5), then $(\mathrm{LB}_{f,c(f)})$ holds with $c(f) = |\mathsf{c}|[(1+2\coth^2 s)c_e(f) + \frac{e^s}{2\sinh^2 s}\sum_{j,\pm}c_e(f\circ B_j(\pm s))] + |\mathsf{c}'|\int f|$.

For the free massless field of Thm. AR(d), with $T_{\mu\nu} = {:}\partial_\mu\varphi\,\partial_\nu\varphi{:} - \frac12\eta_{\mu\nu}{:}\partial_\lambda\varphi\,\partial^\lambda\varphi{:}$ and $\eta = \mathrm{diag}(1,-1,-1)$, one has $T_{00} = \frac12{:}(\dot\varphi^2+|\nabla\varphi|^2){:}$ and $\Theta = -2\eta^{\mu\nu}T_{\mu\nu}$, so that the field of the datum is $-\Theta = 2\eta^{\mu\nu}T_{\mu\nu}$; the identity holds with e the form of T_{00}, $\mathsf{c}=2$ and $\mathsf{c}'=0$, because every symmetric tensor satisfies $\eta^{\mu\nu}T_{\mu\nu} = (1+2\coth^2 s)T_{00} - \frac{1}{2\sinh^2 s}\sum_{j=1,2}\sum_\pm T_{uu}$, $u = \cosh s\,e_0 \pm \sinh s\,e_j$. ■

Proof sketch. (a) Two pairs of times are compared through a pair larger than both. (b) Let X be the operator $P_n\mathcal{H}\to P_m\mathcal{H}$ of q_h, and write $H = m+\delta_m$ on $P_m\mathcal{H}$, $0\le\delta_m<1$. Then $P_m\mathsf{B}_h(\eta+\varsigma,\eta-\varsigma)P_n = e^{-(\eta+\varsigma)m-(\eta-\varsigma)n}e^{-(\eta+\varsigma)\delta_m}Xe^{-(\eta-\varsigma)\delta_n}$ and $\|e^{-\alpha\delta_m}Xe^{-\alpha'\delta_n}-X\| \le (\alpha+\alpha')\|X\|$; with $\kappa = \int\rho_e(\varsigma/\eta)e^{-(\eta+\varsigma)m-(\eta-\varsigma)n}\,\mathrm{d}\varsigma$ the slab operator S_η has $\|P_m\mathsf{S}_\eta P_n - \kappa X\| \le 2\eta\kappa\|X\|$, so $\kappa(1-2\eta)\|X\| \le C_e|h|_V$; at $\eta = \frac{1}{4(m+n+2)}$, $\kappa \ge \eta\bar\rho_e e^{-3/8}$, and $\frac{16}{3}e^{3/8} < 8$. (c) The block (m,n) of $\mathsf{e}(f)$ is $\int e^{it(m-n)}e^{it\delta_m}X_{mn}(f_t)e^{-it\delta_n}\,\mathrm{d}t$; three integrations by parts for $m\ne n$ give the decay $|m-n|^{-3}\nu^*(f)$ times the bound (b); with $A+B\le2\sqrt{AB}(1+|A-B|)$, $A=m+1$, $B=n+1$, Schur's test gives the form bound. (d) In a boosted frame $U(\Lambda)HU(\Lambda)^{-1}\le e^sH$ on the spectrum, so each of the five terms is bounded by its constant times $e^s\langle\psi,N\psi\rangle$.

The free identity: $T_{uu} = \cosh^2 s\, T_{00} \pm 2\cosh s \sinh s\, T_{0j} + \sinh^2 s\, T_{jj}$; the sum over the signs and over j is $4\cosh^2 s\, T_{00} + 2\sinh^2 s(T_{11}+T_{22})$, and the combination is $T_{00} - T_{11} - T_{22} = \eta^{\mu\nu}T_{\mu\nu}$. Since $\eta^{\mu\nu}\eta_{\mu\nu} = 3$, $\eta^{\mu\nu}T_{\mu\nu} = \Theta - \frac{3}{2}\Theta = -\frac{1}{2}\Theta$, and $V\phi(f)V^{-1} = -\Theta(f)$ (Thm. AR(d)) gives $\mathsf{c} = 2$; by covariance $\mathsf{e}(f \circ B_j(\pm s))(\psi_{j,\pm}, \psi_{j,\pm})$ is the form of $T_{uu}(f)$ at ψ. □

Lemma 14.11 (Locality through a dilation)**.** (a) Let $\mathcal{R}_0$ be a $*$-algebra of bounded operators on a Hilbert space $\mathcal{F}$, Ω_{F} cyclic for it, P a projection with $P\Omega_{\mathrm{F}} = \Omega_{\mathrm{F}}$, and $\mathcal{K} = P\mathcal{F}$. Every bounded T on $\mathcal{K}$ that commutes with all $PAP|_{\mathcal{K}}$, $A \in \mathcal{R}_0$, is $PT_1P|_{\mathcal{K}}$ with $T_1 \in \mathcal{R}_0'$, $\|T_1\| \le \|T\|$. (b) For data (Rec) and non-empty open O_1, O_2 the following say the same: $\mathcal{C}(O_1)' \subset \mathcal{C}(O_2)$; there are a Hilbert space $\mathcal{F}$, an isometry $V : \mathcal{H} \to \mathcal{F}$ and a $*$-algebra $\mathcal{R}_0 \subset \mathcal{B}(\mathcal{F})$ with $\mathcal{R}_0 V\Omega$ dense, $V^*\mathcal{R}_0 V \subset \mathcal{C}(O_1)$ and $V^*\mathcal{R}_0' V \subset \mathcal{C}(O_2)$. ■

Proof sketch. (a) With T also T^*T commutes with $X = PA^*AP|_{\mathcal{K}} \ge 0$, so $\|AT\Omega_{\mathrm{F}}\|^2 = \langle T\Omega_{\mathrm{F}}, XT\Omega_{\mathrm{F}}\rangle = \langle X^{1/2}\Omega_{\mathrm{F}}, T^*TX^{1/2}\Omega_{\mathrm{F}}\rangle \le \|T\|^2\|A\Omega_{\mathrm{F}}\|^2$; so $T_1A\Omega_{\mathrm{F}} = AT\Omega_{\mathrm{F}}$ defines a bounded T_1, and $T_1 \in \mathcal{R}_0'$. For $\xi \in \mathcal{K}$ and $A_i\Omega_{\mathrm{F}} \to \xi$: $PT_1A_i\Omega_{\mathrm{F}} = (PA_iP)T\Omega_{\mathrm{F}} = T(PA_iP)\Omega_{\mathrm{F}} \to T\xi$, so $PT_1P|_{\mathcal{K}} = T$. (b) If the dilation exists and $T \in \mathcal{C}(O_1)'$, then VTV^* commutes on $V\mathcal{H}$ with the compressions of $\mathcal{R}_0$, so by (a) $T = V^*T_1V$ with $T_1 \in \mathcal{R}_0'$, and $T \in \mathcal{C}(O_2)$. Conversely take $\mathcal{F} = \mathcal{H}$, $V = 1$, $\mathcal{R}_0 = \mathcal{C}(O_1)$; it holds $\mathcal{A}(O_2) = \mathcal{C}(O_2)' \subset \mathcal{C}(O_1)$, and $\mathcal{A}(O_2)\Omega$ is dense. □

Lemma 14.12 (Weyl operators and Wick forms)**.** Let $\mathfrak{h} = L^2(\mathbb{R}^2)$, $\omega = (p^2 + \mathsf{m}^2)^{1/2}$ with $\mathsf{m} > 0$, $\mathcal{F}$ the Fock space over $\mathfrak{h}$, $H_{\mathrm{F}} = d\Gamma(\omega)$, $\kappa^x = (2\pi)^{-1}(2\omega)^{-1/2}e^{i(t\omega - \vec{x}\cdot p)}$, $u_h = \int h(x)\kappa^x dx$, $A(u) = a^*(u) + a(u)$, $\mathsf{W}[u] = e^{i\overline{A(u)}}$, $\mathcal{R}_0(O)$ the span of the $\mathsf{W}[u_h]$, $h \in \mathcal{F}_{\mathbb{R}}(O)$, and Φ_f the form of the smeared Wick power $:\chi^k:(f)$ of the free field $\chi(h) = A(u_h)$. (a) For $u \in \bigcap_r D(\omega^r)$ and every n: $\mathsf{W}[u]D(H_{\mathrm{F}}^n) = D(H_{\mathrm{F}}^n)$ and $(1+H_{\mathrm{F}})^n\mathsf{W}[u] = \mathsf{W}[u](1 + H_{\mathrm{F}} + A(i\omega u) + \langle u, \omega u\rangle)^n$ there. (b) $\Phi_f(\zeta,\zeta') = \sum_j \binom{k}{j}\int f(x)\langle a(\kappa^x)^j\zeta, a(\kappa^x)^{k-j}\zeta'\rangle\,\mathrm{d}x$ on $\mathcal{D}_\infty(H_{\mathrm{F}})$, with $|\Phi_f(\zeta,\zeta')| \le 2^k(4\pi\mathsf{m})^{-k/2}\|f\|_{L^1}\|(1+H_{\mathrm{F}})^k\zeta\|\|(1+H_{\mathrm{F}})^k\zeta'\|$; it is the form of an operator $\Phi(f)$ on $\mathcal{D}_\infty(H_{\mathrm{F}})$. (c) For real $h \in C_c^\infty$ and x spacelike to $\operatorname{supp} h$, $\langle \kappa^x, u_h\rangle = \int h(y)\frac{e^{-\mathsf{m}\mathsf{d}_{xy}}}{4\pi\mathsf{d}_{xy}}\,\mathrm{d}y$, $\mathsf{d}_{xy} = (|\vec{x}-\vec{y}|^2 - (x^0-y^0)^2)^{1/2}$: a real number; so Weyl operators of spacelike separated regions commute, and $\Phi_f(\mathsf{W}[u_h]\zeta, \mathsf{W}[u_h]\zeta') = \Phi_f(\zeta,\zeta')$ if $\operatorname{supp} f$ and $\operatorname{supp} h$ are spacelike separated. (d) For every non-empty open O the vectors $\mathsf{W}[u_h]\Omega_{\mathrm{F}}$, $h \in \mathcal{F}_{\mathbb{R}}(O)$, are total in $\mathcal{F}$. (e) For $f \in \mathcal{F}(O)$ and $B \in \mathcal{R}_0(O)'$: $\langle \Phi(\bar f)\zeta, B\zeta'\rangle = \langle \zeta, B\Phi(f)\zeta'\rangle$ for $\zeta,\zeta' \in \mathcal{D}_\infty(H_{\mathrm{F}})$. (f) *Gaussian families.* Let $(\kappa_\alpha, \varepsilon_\alpha, \mathsf{P})$, $\alpha \in \mathsf{A}$, be a Gaussian family of Thm. AR(a) on (M, μ), and now let $\mathfrak{h} = L^2(\mathsf{M}, \mu)$, $\mathcal{F}$ the Fock space over it, $H_{\mathrm{F}} = d\Gamma(\omega)$, $\kappa_\alpha^x = e^{i(t\omega - \vec{x}\cdot\vec{p})}\kappa_\alpha$, and Φ_f the form of the smeared Wick polynomial $:\mathsf{P}(\Phi):(f)$.

Put $\varsigma_\alpha = 1$ if $\varepsilon_\alpha = +1$ and $\varsigma_\alpha = i$ if $\varepsilon_\alpha = -1$, $u_h^\alpha = \varsigma_\alpha\hat h(\omega, \vec p)\kappa_\alpha$ for $h \in \mathcal{S}$, let $\mathfrak{k}(O)$ be the real span of the u_h^α, $h \in \mathcal{F}_{\mathbb{R}}(O)$, $\alpha \in \mathsf{A}$, and let $\mathcal{R}_0(O)$ now be the span of the $\mathsf{W}[u]$, $u \in \mathfrak{k}(O)$. Suppose (f1) $\mathfrak{k}(O) + i\,\mathfrak{k}(O)$ is dense in $\mathfrak{h}$ for every non-empty open O, and (f2) $\overline{\langle \kappa_\alpha^x, u\rangle} = \varepsilon_\alpha\langle \kappa_\alpha^x, u\rangle$ for $\alpha \in \mathsf{A}$, $u \in \mathfrak{k}(O)$ and x spacelike to O. Then (a)–(e) hold with the $u \in \mathfrak{k}(O)$ in the place of the u_h, $h \in \mathcal{F}_{\mathbb{R}}(O)$, and with three changes.

In (a), $u \in \bigcap_r D(\omega^r)$ with $\omega^{-1/2}u \in \mathfrak{h}$, which holds for every $u \in \mathfrak{k}(O)$. In (b), the form is

$$\Phi_f(\zeta,\zeta') = \sum_j \binom{k}{j}\sum_{\underline\alpha}\mathsf{p}_{\underline\alpha}\prod_{i>j}\varepsilon_{\alpha_i}\int f(x)\Big\langle\prod_{i\le j}a(\kappa_{\alpha_i}^x)\zeta, \prod_{i>j}a(\kappa_{\alpha_i}^x)\zeta'\Big\rangle\,\mathrm{d}x,$$

and with the weight $\varpi = \omega^{1/2}(1+\omega)^{r'}$, $r' > \delta$, the number $\mathsf{n}_* = \max_\alpha\|\varpi^{-1}\kappa_\alpha\| < \infty$ and the order $L_* = \lceil k(r' + \frac12)\rceil$ it obeys $|\Phi_f(\zeta,\zeta')| \le 2^k|\mathsf{p}|_1\mathsf{n}_*^k\|f\|_{L^1}\|(1+H_{\mathrm{F}})^{L_*}\zeta\|\|(1+H_{\mathrm{F}})^{L_*}\zeta'\|$, the operator $\Phi(f)$ being bounded by a multiple of $\|(1+H_{\mathrm{F}})^{2L_*}\cdot\|$. In (c), the invariance holds for $u \in \mathfrak{k}(O)$ and $\operatorname{supp} f$ spacelike to O, and Weyl operators of $\mathfrak{k}(O_1)$, $\mathfrak{k}(O_2)$ commute for spacelike separated O_1, O_2. The hypotheses (f1), (f2) hold for the first derivatives $\partial_\alpha\varphi^a$ of n massless free fields, the free photons with n colours of Thm. AR(c)–(e), and for the generalised free fields of Thm. AR(c), the free fields of a mixture $d\rho(\mathsf{m})$ of masses with $\varepsilon = +$, the kernel r^{-2} being the mixture $d\rho = 4\pi\,\mathsf{d}\mathsf{m}$. ■

Proof sketch. (a) On exponential vectors $\mathsf{W}[u]\mathrm{Exp}(v) = e^{-\frac{1}{2}\|u\|^2 + i\langle u,v\rangle}\mathrm{Exp}(v+iu)$, whence $H_{\mathrm{F}}\mathsf{W}[u] = \mathsf{W}[u](H_{\mathrm{F}} + A(i\omega u) + \langle u, \omega u\rangle)$ there; exponential vectors are dense in every $D(H_{\mathrm{F}}^n)$, H_{F} is closed, and an induction gives the domains. (b) Wick's theorem, with $\|a(\kappa^x)^j\zeta\| \le \|\omega^{-1}\kappa^x\|^j\|H_{\mathrm{F}}^j\zeta\|$ and $\|\omega^{-1}\kappa^x\|^2 = 1/4\pi\mathsf{m}$; the operator by Riesz after the commutator identity $\Phi_f(H_{\mathrm{F}}\zeta,\zeta') - \Phi_f(\zeta, H_{\mathrm{F}}\zeta') = i\Phi_{\partial_t f}(\zeta,\zeta')$. (c) The Euclidean two-point function $e^{-\mathsf{m}r}/4\pi r$ continued to real time is real at spacelike separation; on exponential vectors the conjugation by $\mathsf{W}[u_h]$ shifts each annihilator $a(\kappa^x)$ by $i\langle\kappa^x, u_h\rangle$ in the ket and conjugately in the bra, and the reality of the shift makes the two changes cancel. (d) $\{u_h : h \in \mathcal{F}(O)\}$, h complex, is dense in $\mathfrak{h}$ by the reflection principle and uniqueness of Fourier transforms: for $g \perp u_h$, $h \in \mathcal{F}(O)$, and η carried by a small ball, $x \mapsto \langle g, u_{\eta(\cdot - x)}\rangle = \int \bar{g}\kappa\hat{\eta}(\omega,p)e^{i(t\omega - \vec{x}\cdot p)}d^2p$ vanishes near a point and continues boundedly into every timelike direction, so it vanishes, and its Fourier transform at $t=0$ gives $\bar{g}\kappa\hat{\eta} = 0$; along an approximate unit $\hat{\eta} \to 1$, and $\kappa > 0$, so $g = 0$.

The vectors of (d) are $\mathsf{W}[u_h]\Omega_{\mathrm{F}} = e^{-\frac{1}{2}\|u_h\|^2}\mathrm{Exp}(iu_h)$ with real h. Let Ψ be orthogonal to all of them. For real $h_1,\dots,h_n \in \mathcal{F}_{\mathbb{R}}(O)$ the function $z \mapsto \langle\Psi, \mathrm{Exp}(\sum_j z_j u_{h_j})\rangle$ is entire on $\mathbb{C}^n$ and vanishes on $i\mathbb{R}^n$, since $\sum_j iy_j u_{h_j} = iu_{\sum_j y_j h_j}$; so all its Taylor coefficients at 0 vanish: $\Psi \perp \Omega_{\mathrm{F}}$ and $\Psi \perp \mathrm{Sym}(u_{h_1}\otimes\cdots\otimes u_{h_n})$. This last product is complex-linear in each u_{h_j}, and $h \mapsto u_h$ is complex-linear; so Ψ is orthogonal to it for all $h_j \in \mathcal{F}(O)$, and these symmetric tensors are total in each n-particle space by the density; hence $\Psi = 0$. (e) The Wick power of one smeared field is a real polynomial in it: $:A(u)^k: = \mathsf{h}_k(A(u))$ with $\mathsf{h}_0 = 1$, $\mathsf{h}_1 = \lambda$, $\mathsf{h}_{k+1} = \lambda\mathsf{h}_k - k\|u\|^2\mathsf{h}_{k-1}$; for $u = u_{\delta_\epsilon(\cdot - x)}$, δ_ϵ an approximate unit and $x \in \mathrm{supp} f$, the unitary group of $\overline{A(u)}$ lies in $\mathcal{R}_0(O)$, so B commutes weakly with $:A(u)^k:$ on $\mathcal{D}_\infty(H_{\mathrm{F}})$; integrate against f and let $\epsilon \downarrow 0$, the regularised Wick powers being bounded by $c(f)\|(1+H_{\mathrm{F}})^{2k}\cdot\|$ uniformly in ϵ and converging weakly. (f) The changes, clause by clause. (a): $\|a(w)\zeta\| \le \|\omega^{-1/2}w\|\|H_{\mathrm{F}}^{1/2}\zeta\|$ and $\|a^*(w)\zeta\|^2 = \|a(w)\zeta\|^2 + \|w\|^2\|\zeta\|^2$ on each n-particle space take the place of the bound by the number of particles, which uses $\mathsf{m} > 0$; so $A(u)$ is defined on $D(H_{\mathrm{F}}^{1/2})$ for $u, \omega^{-1/2}u \in \mathfrak{h}$, it is essentially self-adjoint on the vectors of finitely many particles in $\mathcal{D}_\infty(H_{\mathrm{F}})$, and the proof of (a) runs as it stands.

Every u_h^α lies in all $D(\omega^r)$ and has $\omega^{-1/2}u_h^\alpha \in \mathfrak{h}$, because $\hat{h}$ decays faster than every power and $\varpi^{-1}\kappa_\alpha \in \mathfrak{h}$. (b): in the bound of the annihilators the weight ϖ takes the place of ω,

$$\|a(g_1)\cdots a(g_m)\zeta\| \le \prod_i \|\varpi^{-1}g_i\|\|(1+H_{\mathrm{F}})^{m(r'+\frac{1}{2})}\zeta\|,$$

and

$$\|\varpi^{-1}\kappa_\alpha\|^2 = \Gamma(2r')^{-1}\int_0^\infty s^{2r'-1}e^{-s}\|\omega^{-1/2}\kappa_\alpha e^{-s\omega/2}\|^2 ds < \infty$$

by the bound $c_\kappa s^{-\delta}$ of Thm. AR(a) and $r' > \delta$; the form is hermitian, $\overline{\Phi_{\bar{f}}(\zeta',\zeta)} = \Phi_f(\zeta,\zeta')$, because $\mathsf{P}(\varepsilon t) = \mathsf{P}(t)$; the operator comes as before, from the commutator with H_{F}. (c): on exponential vectors

$$\Phi_f(\mathrm{Exp}\, v, \mathrm{Exp}\, v') = e^{\langle v, v'\rangle}\int f(x)\,\mathsf{P}\big(\bar{z}(x) + \varepsilon z'(x)\big)\,\mathrm{d}x,$$

$z_\alpha = \langle\kappa_\alpha^x, v\rangle$, $z'_\alpha = \langle\kappa_\alpha^x, v'\rangle$; under $v \to v+iu$, $v' \to v'+iu$ the argument of P changes by $-i\bar{\zeta}_\alpha + i\varepsilon_\alpha\zeta_\alpha$, $\zeta_\alpha = \langle\kappa_\alpha^x, u\rangle$, which (f2) makes 0; and $\langle u_{h'}^\alpha, u_h^\beta\rangle = \bar{\varsigma}_\alpha\int h'(x)\langle\kappa_\alpha^x, u_h^\beta\rangle\,\mathrm{d}x$ is real by (f2), so the Weyl operators commute. (d): the argument of (d) with the $u_j \in \mathfrak{k}(O)$ in the place of the u_{h_j}, and (f1) in the place of the density of the u_h. (e): with $g_\alpha = \rho_\epsilon\kappa_\alpha^x$, $\rho_\epsilon = \hat{\delta}_\epsilon(\omega,\vec{p})$, the regularised fields $\Phi_\alpha = a^*(g_\alpha) + \varepsilon_\alpha a(g_\alpha)$ are $\bar{\varsigma}_\alpha A(\varsigma_\alpha g_\alpha)$, as $\bar{\varsigma}_\alpha^2 = \varepsilon_\alpha$, and $\varsigma_\alpha g_\alpha = u^\alpha_{\delta_\epsilon(\cdot - x)} \in \mathfrak{k}(O)$ for $x \in \mathrm{supp} f$ and small ϵ, so the unitary groups of the closures of the $A(\varsigma_\alpha g_\alpha)$ lie in $\mathcal{R}_0(O)$.

The recursion

$$\Phi_{\alpha_0}{:}\Phi_{\underline{\alpha}}{:} = {:}\Phi_{(\alpha_0,\underline{\alpha})}{:} + \sum_i \varepsilon_{\alpha_0}\langle g_{\alpha_0}, g_{\alpha_i}\rangle {:}\Phi_{\underline{\alpha}\setminus i}{:}$$

makes every Wick monomial a polynomial with complex coefficients in several non-commuting $A(\varsigma_\alpha g_\alpha)$; B passes through one factor at a time, so $\langle\zeta, BZ\zeta'\rangle = \langle Z^\dagger\zeta, B\zeta'\rangle$ for such a polynomial Z and $Z^\dagger$ the polynomial with the conjugate coefficients and the products reversed; and $Z^\dagger = Z$ on $\mathcal{D}_\infty(H_\mathrm{F})$ for the regularised Wick polynomial $Z = \sum_{\underline{\alpha}} \mathsf{p}_{\underline{\alpha}}{:}\Phi_{\underline{\alpha}}{:}$, by the hermiticity of (b) at $B = 1$.

Then integrate against f and let $\epsilon \downarrow 0$ as before, with ϖ for ω in the weights and L_* for k in the norms. *The families.* For the photons, $\kappa_0 = (2\pi)^{-1}(2|p|)^{-1/2}$, $\kappa_j = -ip_j\kappa_0$, $\kappa_3 = -\omega\kappa_0$ and $\varepsilon = (+,+,-)$ (the proof of Thm. AR(d)), so $u_h^j = \hat{h}'(\omega,p)\kappa_0$ with $h' = -\partial_j h$, and $u_h^3 = \hat{h}'(\omega,p)\kappa_0$ with $h' = \partial_t h$. For real $h' \in C_c^\infty$ the function $F_{h'}(x) = \langle\kappa_0^x, \hat{h}'(\omega,p)\kappa_0\rangle$ is smooth and, by the computation of (c) at $\mathsf{m} = 0$, real on the open set of the x spacelike to $\operatorname{supp} h'$; there $\langle\kappa_j^x, \hat{h}'\kappa_0\rangle = \partial_j F_{h'}(x)$ is real and $\langle\kappa_3^x, \hat{h}'\kappa_0\rangle = -i\partial_t F_{h'}(x)$ is imaginary: (f2). $\mathfrak{k}(O) + i\,\mathfrak{k}(O)$ holds the vectors $\hat{h}(\omega,p)\,\omega\kappa_0$, $h \in \mathcal{F}(O)$, and the reflection argument of (d) with $\omega\kappa_0 > 0$ in the place of κ, $\omega - \vec{v}\cdot p \geq (1 - |\vec{v}|)|p| > 0$, gives (f1).

For n colours $\mathfrak{h}$ is the sum of n copies, $\langle\kappa_{(a,\alpha)}^x, u_h^{(b,\beta)}\rangle = \delta_{ab}\langle\kappa_\alpha^x, u_h^\beta\rangle$, and both hypotheses hold copy by copy. For a mixture of masses, $\mathsf{M} = \mathbb{R}^2 \times (0,\infty)$, $d\mu = d^2p\,\mathrm{d}\rho(\mathsf{m})$, $\omega = (p^2 + \mathsf{m}^2)^{1/2}$ and one field: $\langle\kappa^x, u_h\rangle = \int h(y) \int \frac{e^{-\mathsf{m}\mathsf{d}_{xy}}}{4\pi\mathsf{d}_{xy}}\,\mathrm{d}\rho(\mathsf{m})\,\mathrm{d}y$ is real at spacelike separation, by (c) and a ρ-integrable majorant, and the reflection argument, applied to the image of the finite measure $\bar{g}\kappa\hat{\eta}(\omega,\vec{p})\,\mu$ under $(\vec{p},\mathsf{m}) \mapsto (\omega,\vec{p})$, a homeomorphism onto $\{E > |\vec{p}|\}$, gives the density; only the weight changes. □

Proposition 14.13 (Four squares)**.** Let $\Lambda = a\mathbb{Z}^3/\ell\mathbb{Z}^3$, ℓ/a even and ≥ 4, $\beta \geq 0$, $\mu = \mu_\beta$ with the normalisation Z. The planes $\mathsf{P}_\mu = \{x_\mu \in \{0, \frac{\ell}{2}\}\}$, $\mu = 1,3$, pass through sites and cut the torus into $\mathsf{Q}_1 = \{0 < x_1, x_3 < \frac{\ell}{2}\}$, $\mathsf{Q}_2 = \theta_1\mathsf{Q}_1$, $\mathsf{Q}_3 = \theta\theta_1\mathsf{Q}_1$, $\mathsf{Q}_4 = \theta\mathsf{Q}_1$; giving each plaquette in a plane half of its weight in each of the two adjacent closed quadrants, $e^{-\beta\sum_p a_p} = w_1w_2w_3w_4$, w_i depending only on the bonds of $\bar{\mathsf{Q}}_i$. Let B_c be the bonds on the four corner lines $\mathsf{P}_1 \cap \mathsf{P}_3$; B_R the other bonds of $\mathsf{P}_3 \cap \{0 \leq x_1 \leq \frac{\ell}{2}\}$, in both planes $x_3 \in \{0, \frac{\ell}{2}\}$; $\mathsf{B}_L = \theta_1\mathsf{B}_R$; $\mathsf{B}_U = \mathsf{d}\mathsf{B}_R$, the bonds of $\mathsf{P}_1 \cap \{0 \leq x_3 \leq \frac{\ell}{2}\}$ off the corner lines; $\mathsf{X} = G^{\mathsf{B}_R}$, $\mathsf{C} = G^{\mathsf{B}_c}$, κ the map of C induced by d, $\mathcal{K} = L^2(\mathsf{X})$, and $\mathcal{H}_\Lambda = L^2(\mathsf{C}\times\mathsf{X}\times\mathsf{X})$, regarded as the space of the fields $c \mapsto \hat{\xi}_c$ of Hilbert–Schmidt operators on $\mathcal{K}$ whose left index is the θ_1-image x_L and whose right index is x_R.

For a bounded Borel function F of the bonds of $\bar{\mathsf{Q}}_1$ let A_c^F be the Hilbert–Schmidt operator on $\mathcal{K}$ with the kernel $Z^{-1/4}\int Fw_1$, integrated over the bonds of $\bar{\mathsf{Q}}_1$ in no plane, between the configuration x of B_R and the configuration x' of B_U carried to B_R by d, at the corner configuration c; $Z_c = \mathsf{A}_c^1$. For a bounded Borel F of the bonds of $\{0 \leq x_3 \leq \frac{\ell}{2}\}$ let $\iota_\Lambda F = Z^{-1/2}\int Fw_1w_2$ over the bonds of that half off $\mathsf{B}_c \cup \mathsf{B}_R \cup \mathsf{B}_L$, $\Omega_\Lambda = \iota_\Lambda 1$; and $\mathcal{M}_\Lambda = \{\hat{\xi}_c \mapsto \hat{\xi}_c q_c\}$, $\mathcal{M}'_\Lambda = \{\hat{\xi}_c \mapsto b_c\hat{\xi}_c\}$ with bounded measurable fields q_c, b_c of operators of $\mathcal{K}$.

Then:

(L1) $\langle\iota_\Lambda F', \iota_\Lambda F\rangle = \mu(\theta\bar{F}'\cdot F)$; $J_\Lambda\hat{\xi}_c = \hat{\xi}_c^*$ has $J_\Lambda\iota_\Lambda F = \iota_\Lambda(\theta_1\bar{F})$ and $J_\Lambda\mathcal{M}_\Lambda J_\Lambda = \mathcal{M}'_\Lambda$; $\mathcal{M}_\Lambda$ and $\mathcal{M}'_\Lambda$ are von Neumann algebras, each the commutant of the other.

(L2) For F_1 of $\bar{\mathsf{Q}}_1$ and F_2 of $\bar{\mathsf{Q}}_2$: $(\iota_\Lambda(F_2F_1))_c^\wedge = \mathsf{A}_{\kappa c}^{R^{-1}F_2}\mathsf{A}_c^{F_1}$; $(\mathsf{A}_c^F)^* = \mathsf{A}_{\kappa c}^{\mathsf{d}\bar{F}}$, $Z_c^* = Z_{\kappa c}$, and $\hat{\Omega}_c = Z_c^*Z_c = \rho_c^{1/2}$ with ρ_c of trace class, $\int \operatorname{Tr}\rho_c\,\mathrm{d}c = 1$.

(L3) For $Q \in \mathcal{M}_\Lambda$ there is $\tilde{q} \in \mathcal{H}_\Lambda$, $\|\tilde{q}\| \leq \|Q\|$, with $\langle\iota_\Lambda F', Q\iota_\Lambda F\rangle = \langle\tilde{q}, \iota_\Lambda(R^{-1}(\theta\bar{F}'\cdot F))\rangle$ for F, F' of the bonds of $\bar{\mathsf{Q}}_2$.

(L4) Let e_c be the projection on $(\ker Z_c)^\perp$, $e_R : \hat{\xi}_c \mapsto \hat{\xi}_c e_c$ in $\mathcal{M}_\Lambda$, $e_L = J_\Lambda e_R J_\Lambda$, $\pi = e_Le_R$, and $e^{-2\pi sK_\Lambda}\hat{\xi}_c = \rho_c^{-s}\hat{\xi}_c\rho_c^s$ on $\pi\mathcal{H}_\Lambda$: $\Delta_\Lambda = e^{-2\pi K_\Lambda}$ is the modular operator of $(\mathcal{M}_\Lambda, \Omega_\Lambda)$. There is a self-adjoint involution Σ of $\pi\mathcal{H}_\Lambda$ commuting with K_Λ such that $\mathsf{P}_\Lambda = \Sigma e^{-\frac{\pi}{2}K_\Lambda}$ is self-adjoint, $|\mathsf{P}_\Lambda| = e^{-\frac{\pi}{2}K_\Lambda}$, and $\pi\iota_\Lambda F \in D(\mathsf{P}_\Lambda)$, $\mathsf{P}_\Lambda\pi\iota_\Lambda F = \pi\iota_\Lambda(RF)$ for F of $\bar{\mathsf{Q}}_1$: the quarter turn is the fourth root of the modular operator, up to a sign.

(L5) For such F and $\xi = \mathsf{F}(K_\Lambda)\pi\iota_\Lambda F$ there is a closed, densely defined T commuting with the unitaries of $\mathcal{M}'_\Lambda$, with $T\Omega_\Lambda = \xi$ and $\|T^*T\Omega_\Lambda\| \leq \pi e^{5\pi^2/8}\|\iota_\Lambda(\theta_1\bar{F}\cdot F)\| = \pi e^{5\pi^2/8}\mu(\theta\theta_1 F\cdot$

$\theta\bar F\cdot\theta_1\bar F\cdot F)^{1/2}$; so $T^{(\epsilon)}=T(1+\epsilon T^*T)^{-1}\in\mathcal{M}_\Lambda$, $\|T^{(\epsilon)}\|\le(2\sqrt{\epsilon})^{-1}$, and $\|T^{(\epsilon)}\Omega_\Lambda-\xi\|\le\frac{\sqrt{\epsilon}}{2}\|T^*T\Omega_\Lambda\|$.

No coupling enters, and no injectivity of ρ_c is assumed: gauge invariance destroys it. ∎

Proof sketch. The maps $\theta,\theta_1,\mathsf{d}$ keep Λ, the Haar measure and the Wilson weight ($a_{rp}(\mathsf{r}U)=a_p(U)$, as $\operatorname{tr}g^{-1}=\operatorname{tr}g$ on SU(2)). (L1) $\mu(\theta\bar F'\cdot F)=Z^{-1}\int\overline{F'(\theta U)}F(U)w_1w_2(U)w_1w_2(\theta U)\,\mathrm{d}U$; at fixed (c,x_L,x_R), which θ fixes, the integrand is a function of the upper bonds times one of the lower bonds, and the two integrals are $Z^{1/2}\iota_\Lambda F$ and $Z^{1/2}\overline{\iota_\Lambda F'}$; θ_1 exchanges w_1,w_2 and x_L,x_R; $\mathcal{H}_\Lambda=L^2(\mathsf{C})\otimes\mathcal{K}\otimes\mathcal{K}$ and $\mathcal{M}_\Lambda=L^\infty(\mathsf{C})\otimes1\otimes\mathcal{B}(\mathcal{K})$. (L2) At $F=F_2F_1$ the integral over the bonds of $\mathsf{Q}_1\cup\mathsf{Q}_2$ factorises at fixed configuration of B_U; the substitution $U\to\theta_1U$ in the factor of Q_2 and the diagonal reflection d, which gives $(\mathsf{A}_c^F)^{\mathrm{T}}=\mathsf{A}_{\kappa c}^{\mathsf{d}F}$, yield the product formula; the kernel of Z_c is real. (L3) With $E=R^{-1}F$, $E'=R^{-1}F'$ of $\bar{\mathsf{Q}}_1$: $\langle\iota_\Lambda F',Q\iota_\Lambda F\rangle=\int\operatorname{Tr}((Z_cq_c^*Z_c^*)^*(\iota_\Lambda(\theta_1\bar E'\cdot E))^\wedge_{\kappa c})\,\mathrm{d}c$ and $\theta_1\bar E'=R^{-1}\theta\bar F'$; put $\hat{\bar q}_{\kappa c}=Z_cq_c^*Z_c^*$; Hölder gives $\|Z_cq_c^*Z_c^*\|_2\le\|q_c\|\|Z_c\|_4^2$, and $\int\|Z_c\|_4^4\,\mathrm{d}c=\|\Omega_\Lambda\|^2=1$. (L4) With the polar decomposition $Z_c=W_c\rho_c^{1/4}$, $W_{\kappa c}=W_c^*$; put $(\Sigma\xi)^\wedge_c=W_c^*\hat\xi_{\kappa c}W_c$, a symmetric involution of $\pi\mathcal{H}_\Lambda$ commuting with the modular flow; then $(\pi\iota_\Lambda F)^\wedge_c=\rho_c^{1/4}\mathsf{c}_c$ with $\mathsf{c}_c=W_c^*\mathsf{A}_c^Fe_c=e_c\mathsf{c}_ce_c$, while $(\Sigma\pi\iota_\Lambda RF)^\wedge_c=\mathsf{c}_c\rho_c^{1/4}=(e^{-\frac{\pi}{2}K_\Lambda}\pi\iota_\Lambda F)^\wedge_c$. (L5) The Schur multipliers $h_\gamma(\omega)=\sqrt{\pi}e^{-\omega^2/16\pi^2+\gamma\omega}$ of the eigenvalue differences ω of $\log\rho_c$ have the norm $\sqrt{\pi}e^{4\pi^2\gamma^2}$ on every Schatten class; $\mathsf{F}(\omega/2\pi)=h_{1/8}(\omega)$, so $\hat\xi_c=\rho_c^{1/4}h_{1/8}(\mathsf{c}_c)$, and with $\hat\xi_c^\sharp=\rho_c^{1/4}h_{1/8}(\mathsf{c}_c^*)$, $\hat\zeta_c=h_{3/8}(\mathsf{c}_c)h_{1/8}(\mathsf{c}_c^*)$,

$$\rho_c^{1/2}\hat\xi_c^*=\hat\xi_c^\sharp\rho_c^{1/2},\qquad\hat\xi_c\hat\xi_c^*=\hat\zeta_c\rho_c^{1/2}.$$

The first makes $Q'\Omega_\Lambda+\eta\mapsto Q'\xi$, $Q'\in\mathcal{M}'_\Lambda$, $\eta\in(1-e_R)\mathcal{H}_\Lambda$, closable, with a closure T that commutes with the unitaries of $\mathcal{M}'_\Lambda$; the second gives $T^*\xi=\zeta$ with $\|\hat\zeta_c\|_2\le\pi e^{5\pi^2/8}\|\mathsf{A}_c^F\|_4^2$ and $\|\mathsf{A}_c^F\|_4^4=\|(\mathsf{A}_c^F)^*\mathsf{A}_c^F\|_2^2=\|(\iota_\Lambda(\theta_1\bar F\cdot F))^\wedge_c\|_2^2$; (L1) turns $\|\iota_\Lambda(\theta_1\bar F\cdot F)\|$ into the four-point function. Von Neumann's theorem and the bicommutant theorem give $T^{(\epsilon)}\in\mathcal{M}_\Lambda$, and $T^{(\epsilon)}\Omega_\Lambda-\xi=-\epsilon T^{(\epsilon)}T^*T\Omega_\Lambda$. □

The one-lattice source recursion. We need more than the bound on the source energies on their original source ball: the two-cell estimate uses their continuation to larger source discs at the earlier levels. We give the induction that provides this continuation, retaining the dependence on the background radius. The local maps are those of Thm. YM3_I:W, Prop. YM3_I:51.16, Lem. YM3_I:114.1 and Prop. YM3_I:77.2.

Here is the notation used for those maps. On a domain Y at level j, P is its background triple, B is its fluctuation field, and $|B|<\mathsf{C}_{\mathcal{A}}\mathfrak{b}(x_j)$ is the field polydisc. For a family V vanishing at $B=0$, its norm at a fixed background is

$$\|V\|_{\mathrm{f},P}=\sup_{Y,B}\frac{e^{\lambda_V d_j(Y)}|V(Y;P,B)|}{1+q_Y(B)},\qquad q_Y(B)=M^{-3}\sum_b|B(b)|^2.$$

The norm $\mathsf{n}^{\mathrm{f}}(V)$ also takes the supremum over the specified background cores. For a pointed family E of level j, $\|E\|_{j,\kappa+1}$ is the supremum of $e^{(\kappa+1)d_j(X)}|E(X,P,z)|$ over its domains, background cores and marked sites. The local old-potential map is denoted by $\mathbb{V}'[\mathfrak{f}]$, the pointed integration map by $\mathbf{F}$, and the source-independent normalizer by $\mathbf{N}$.

For the admitted run $x_0>\cdots>x_K$ put $b_i=L^{-1}x_{i-1}-x_i$, so that $|b_i|\le B_\sharp$, and $v_j^w=-L^{K-j}w/x_j$. The recursion starts with the empty array and is

$$\mathfrak{f}_j[w]=((E^{(i)}[w],b_i)_{1\le i\le j},\emptyset),\qquad E^{(j+1)}[w]=\mathbf{N}^{(j+1)}+\mathbf{F}[\mathbb{V}'[\mathfrak{f}_j[w]]-\mathbb{W}^{\mathrm{src}}[v_j^w]].$$

The direct source family is equation YM3_I:2165:

$$\mathbb{W}^{\mathrm{src}}[v](Y;P,B)=x_j\mathbb{V}_v^{\mathrm{act}}(Y;P,x_j^{-1/2}B)-x_j\sum_b v(p_b)D_A\mathbb{V}_1^{\mathrm{act}}(Y;P,0)[x_j^{-1/2}B(b)\delta_b].$$

Here p_b is the fixed reference plaquette in the cube containing b_-, and $\mathbb{V}_v^{\mathrm{act}}$ is the localized weighted Wilson action. At level zero it is the finite plaquette sum with the pivot completion. All these maps use the fixed auxiliary data. Changing a core radius restricts their domains; it does not change their formulas.

This recursion gives the regular terms used by the source energies. Indeed, its negative direct source family is the one-lattice escort of Prop. YM3_I:78.6: their weighted action and reference subtraction are identical, with opposite signs. On the history with no large-field region every bond is an integration bond, so the pointed integration map is the same. Thus the regular recursion is equation YM3_I:1683, and equation YM3_I:1736 gives

$$e_i(X;w) = \sum_{z\in X\cap\mathbb{T}^{(i)}} \big[E^{(i)}[w](X,\mathbf{1},z) - E^{(i)}[0](X,\mathbf{1},z)\big].$$

In particular, the continuation constructed below continues the source energies in use, not a second choice of their factorization.

Lemma 14.14 (The source recursion on smaller cores)**.** Let (K,β) be admitted at g, with run $x_0 > \cdots > x_K \ge \gamma^{-2}$. Use the fixed one-lattice data above and the reference torus $T_{\ell'}$, where ℓ' is the least side in $\mathfrak{L}(g)$; in particular $\ell' \ge 2L^7$. Write $\bar B = B_\sharp L/(L-1)$ and $\delta(x) = x^{-1/2}\mathfrak{b}(x)$, $\mathfrak{b}(x) = A_1(\log x)^{\mathfrak{p}_0}$. The fixed core radii and size constants obey

$$\alpha_0+\alpha_1 \le a_\star^{\mathrm{S}}, \qquad c_{\mathrm{sh}} M\alpha_0 < \alpha_1, \qquad \alpha_1^2 \le c_\natural \alpha_0, \qquad s_\gamma := \frac{(1+\beta_{\mathrm{s}})\delta(\gamma^{-2})}{\alpha_0} \le s_\star^{\mathrm{S}},$$

$$E_0 \ge 2E_{\mathsf{N}}, \qquad N_\star^{\mathrm{s}} \ge \frac{L}{L-1}(E_0 + \tfrac{3}{2}\alpha_0^2 B_\sharp),$$

$$K_{\mathbf{M}}\big(\delta(\gamma^{-2}) + s_\gamma N_\star^{\mathrm{s}} + e^{-\gamma_2 \mathfrak{b}(\gamma^{-2})^2/2}\big) \le E_0/4, \qquad 8K_\Lambda s_\gamma N_\star^{\mathrm{s}} \le w_\star, \qquad 8K_P s_\gamma \le w_\star.$$

These are the absolute-core and source-recursion conditions, with their size bounds as in equations YM3_I:2169 and YM3_I:2170. Fix uniform bounds $K_{\mathrm{Q}}^\vee, K_{\mathrm{L}}^\vee$ for the quadratic and linear source coefficients, let

$$\mu_\star = \min\{w_\star/32, E_0/(16K_{\mathbf{M}}^{\mathrm{f}})\}, \qquad \sigma_{\mathrm{src}} = \mu_\star/K_{\mathrm{Q}}^\vee, \qquad K^{\mathrm{d}} \le K_{\mathbf{M}}^{\mathrm{f}},$$

and retain $4\beta_{\mathrm{s}}\kappa \ge 1$ and $\log\gamma^{-2} > 2\mathfrak{p}_0$. Then the following statements hold.

(i) On cores of radius $\bar a_0 \le 1$, at every level $j \ge 0$, the part of the direct source linear in B satisfies

$$|\mathbb{W}^{\mathrm{lin}}[v](Y;P,B)| \le \bar a_0 K_{\mathrm{L}}^\vee x_j^{1/2}\, \mathrm{sw}_j(v)(1+q_Y(B))e^{-\lambda_V d_j(Y)}.$$

The constants may be fixed uniformly over all levels and all the smaller cores in (ii). Consequently the one-step estimates apply when the oscillation allowance is divided by $\bar a_0$.

(ii) Fix $1 \le k \le K$, and put $\theta_k = \delta(x_{k-1})/\delta(\gamma^{-2})$. For every $\theta \in [\theta_k, 1]$ the recursion through level k is defined on the cores with radii $(\theta\alpha_0, \theta\alpha_1)$ for sources satisfying

$$|w|_\infty < \varrho_0^\star := \sigma_{\mathrm{src}}(x_K - \bar B), \qquad \frac{40}{\sqrt{3}} M \,\mathrm{Lip}_\infty w < \varrho_1^{[k]}(\theta) := \frac{40}{\sqrt{3}}\frac{\mu_\star}{K_{\mathrm{L}}^\vee}\frac{x_{k-1}^{1/2}}{\theta\alpha_0}.$$

It is jointly holomorphic in the source and the open background slices, local in the source, and its pointed families satisfy $\|E^{(i)}[w]\|_{i,\kappa+1} \le E_0$ for $1 \le i \le k$. At $\theta = 1$ the estimate without the factor α_0 in (i) also gives the same conclusions with second source radius $(40/\sqrt{3})(\mu_\star/K_{\mathrm{L}}^\vee)x_{k-1}^{1/2}$.

(iii) The source class at $\theta = 1$ contains the original class with radii $2\varrho, 2\varrho$. On common source classes, the families constructed at θ and at 1 agree on the smaller cores.

(iv) The pointed integration at one step is a holomorphic function of its fourth-family entry. On the admitted norm ball it is Lipschitz, with bound $K^{\mathrm{d}} e^{-\lambda_{\mathrm{out}} d}$.

(v) At the trivial background the linear source potential vanishes. The old potential vanishes to second order in the fluctuation field. If N is its weighted array size and s its absolute-core factor, then

$$|\mathbb{V}'(Y;\mathbf{1},B)| \le K_\Lambda s N e^{-\lambda_V d_j(Y)} \left(\frac{\sup|B|}{\mathsf{C}_{\mathcal{A}}\mathfrak{b}(x_j)}\right)^2.$$

For the consequence used below, let $f \neq 0$ be a real Lipschitz function vanishing off the middle of a cell of side l, put $r = \varrho/\|f\|_\sharp$, and let v'^+, v'^- be its even and odd mirror sums on $T_{\ell'}$. For $1 \le k \le K$, put $s = K - k$ and

$$\lambda_s = (1 + (s+1)\log L)^{\mathfrak{p}_0},$$
$$\epsilon_s = \max\left\{\frac{\mathsf{l}}{160M}, \min\left\{\left(\frac{24}{11}L^{-s-1}\right)^{1/2}, \frac{24\alpha_0}{11}L^{-s-1}\lambda_s\right\}\right\},$$
$$\hat\epsilon_s = \max\left\{\frac{\mathsf{l}}{160M}, \frac{24\alpha_0}{11}L^{-s-1}\right\}.$$

If X is a level-k domain that sees two cells, then $G_{k,X}(z,z') = e_k(X; zv'^+ + z'v'^-)$ continues holomorphically to a bidisc of radius $R_k \ge 4r/\epsilon_s$, with

$$|G_{k,X}(z,z')| \le 2M^3E_0e^{-\kappa d_k(X)}, \qquad |\partial_z\partial_{z'}G_{k,X}(0,0)| \le \frac{M^3E_0c_2}{32}\frac{\hat\epsilon_s^2}{r^2}e^{-\kappa d_k(X)}.$$

Here c_2 depends only on $M, \mathsf{C}_{\mathcal{A}}, A_1, \mathfrak{p}_0, \gamma$. Using the last assertion of (ii) instead gives a bidisc of radius at least $4r/\epsilon_s^\circ$, where

$$\epsilon_s^\circ = \max\left\{\frac{\mathsf{l}}{160M}, \left(\frac{24}{11}L^{-s-1}\right)^{1/2}\right\}.$$

■

Proof. We first retain the core-radius factor in the direct source calculation. At positive level write $a = \bar a_0$ and $\varepsilon = \min\{\varepsilon_\natural, c_{\mathrm{os}}a/(2c_\natural)\}$. The proof of Lem. YM3_I:114.1 gives, with $N = \#_{\pi_j}Y$,

$$e^{r'd_j(Y)}N \sup_{|A|<\varepsilon}|\mathbb{V}_v^{\mathrm{act}}(Y;P,A)| \le 672M^3a^2|v|_Y \qquad (0 \le r' \le \bar\kappa - 1).$$

For its linear Taylor term Cauchy's formula divides this bound by ε. Retain $a^2/\varepsilon = a(a/\varepsilon)$, instead of replacing it by a/ε. The bond sum obeys $\sum_b|B(b)| \le (3/2)M^3N(1+q_Y(B))$. The path from a plaquette to its reference plaquette has at most $d_j(Y)+2$ links, and $e^{-d}(d+2) \le 2$. These estimates give (i). The quotient a/ε is bounded by its value at the original radius, so the constant is uniform on smaller cores. At level zero the potential is zero unless Y is the domain of one plaquette. On the polydisc scaled by a, each shifted plaquette angle is bounded by a fixed multiple of a. Its Wilson potential is therefore bounded by a fixed multiple of a^2; Cauchy's estimate for its linear term loses one factor a. The pivot calculation in Prop. YM3_I:51.4 gives (i) at this level as well, with one fixed $K_{\mathrm{L}}^\vee$.

We next verify that the smaller cores admit every required step. The first three absolute-core conditions become

$$\theta(\alpha_0+\alpha_1) \le a_\star^{\mathrm{S}}, \qquad c_{\mathrm{sh}}M\theta\alpha_0 < \theta\alpha_1, \qquad \theta^2\alpha_1^2 \le c_\natural\theta\alpha_0.$$

They follow from $0 < \theta \le 1$. For $j < k$, monotonicity of δ gives

$$\delta(x_j) \le \delta(x_{k-1}) = \theta_k\delta(\gamma^{-2}) \le \theta\delta(\gamma^{-2}), \qquad s_j^\theta := \frac{(1+\beta_{\mathrm{s}})\delta(x_j)}{\theta\alpha_0} \le s_\gamma.$$

This is the fourth absolute-core condition. It also preserves the field-room inequalities: the allowed Taylor radius at $\theta\alpha_0$ is at least θ times the original one. The fixed auxiliary maps restrict to these cores.

For the relative weight of a source in (ii), Prop. YM3_I:73.3 gives

$$x_j \ge L^{K-j}(x_K - \bar{B}), \qquad |v_j^w| \le \frac{|w|}{x_K - \bar{B}} < \sigma_{\mathrm{src}}, \qquad \mathrm{sw}_j(v_j^w) \le x_j^{-1} M \,\mathrm{Lip}_\infty\, w.$$

Hence

$$\theta\alpha_0 x_j^{1/2}\,\mathrm{sw}_j(v_j^w) < \frac{\mu_\star}{K_{\mathrm{L}}^\vee}(x_{k-1}/x_j)^{1/2} \le \frac{\mu_\star}{K_{\mathrm{L}}^\vee}.$$

The quadratic and higher source terms cost at most $2K_{\mathrm{Q}}^\vee|v_j^w|$, by the same half-radius Taylor estimate as in Lem. YM3_I:114.1. With (i), the full source norm is therefore less than $3\mu_\star$.

We can now carry out the induction, including its counterterms. If the earlier pointed families have norm at most E_0, their weighted array has size

$$N_j = \sum_{i=1}^{j} L^{-(j-i)}(\|E^{(i)}[w]\|_{i,\kappa} + \tfrac{3}{2}\theta^2\alpha_0^2|b_i|) \le \frac{L}{L-1}(E_0 + \tfrac{3}{2}\alpha_0^2 B_\sharp) \le N_\star^{\mathrm{s}}.$$

The old-potential rule of Prop. YM3_I:51.16 thus has norm at most $K_\Lambda s_j^\theta N_j \le w_\star/8$. Adding the source gives at most $w_\star/8 + 3w_\star/32 = 7w_\star/32 < w_\star/4$. The internal potentials have total size at most $3K_P s_j^\theta \le 3w_\star/8$. Thm. YM3_I:W bounds the unsourced output by $3E_0/4$. Prop. YM3_I:77.2 bounds the added source contribution by $3K^{\mathrm{d}}\mu_\star \le 3E_0/16$. The new pointed family therefore has norm at most $15E_0/16$. At the first step the old array is empty, so the same argument starts the induction. This is the proof of Prop. YM3_I:114.2 with the displayed counterterms and relative weights retained. Holomorphy follows from the parameter version of the pointed expansion and the linearity of the old-potential rule. Locality follows inductively because a term over X reads only earlier terms and source values inside X. At $\theta = 1$, omit the factor α_0 from (i) and repeat the same argument with the smaller source radius stated in (ii).

The source-radius conditions give, for $x \ge \gamma^{-2}$,

$$4\varrho \le \sigma_{\mathrm{src}}(x - \bar{B}), \qquad 4K_{\mathrm{L}}^\vee \varrho x^{-1/2} \le 40\mu_\star/\sqrt{3}.$$

At $x = x_K$ and $x = x_{k-1}$ these imply $\varrho_0^\star \ge 4\varrho$ and $\varrho_1^{[k]}(1) \ge 4\varrho$. The recursion contains no core radius except through its domain; induction on the level therefore identifies the constructions on their common domains. This proves (iii). Assertion (iv) is the holomorphic parameter statement of Lem. YM3_I:77.1 and the entry estimate of Prop. YM3_I:77.2, at the norm reserves just checked.

For (v), the old potential is zero at $B = 0$. Its differential there is invariant under all constant gauge transformations. Averaging a traceless matrix over these transformations gives zero, so the differential vanishes. Schwarz's lemma on the field polydisc of radius $\mathsf{C}_{\mathcal{A}}\mathfrak{b}(x_j)$ gives the stated square. The same invariance makes the linear source term zero at the trivial background.

We turn to the two-variable consequence. The two mirror sums have disjoint supports in cell interiors. Thus their source belongs to the class in (ii) at θ_k whenever $|z|, |z'| < R_k$, with

$$R_k = \min\left\{ \frac{\varrho_0^\star}{\sup|f|}, \frac{\varrho_1^{[k]}(\theta_k)}{\|f\|_\sharp} \right\}.$$

The first denominator is controlled by $\sup|f| \le \mathsf{l}\,\mathrm{Lip}\, f/4 = \mathsf{l}\|f\|_\sharp/(160M)$. For the second, the run and $\bar{B} \le (13/24)\gamma^{-2}$ give

$$x_{k-1} \ge L^{s+1}(x_K - \bar{B}) \ge (11/24)L^{s+1}\gamma^{-2}, \qquad \frac{40}{\sqrt{3}}\frac{\mu_\star}{K_{\mathrm{L}}^\vee} \ge 4\varrho\gamma.$$

Using first $\theta_k\alpha_0 \le 1$ and then the formula for θ_k gives respectively

$$\frac{\gamma x_{k-1}^{1/2}}{\theta_k\alpha_0} \ge \left(\frac{11}{24}L^{s+1}\right)^{1/2}, \qquad \frac{\gamma x_{k-1}^{1/2}}{\theta_k\alpha_0} = \frac{\gamma^2 x_{k-1}}{\alpha_0}\frac{\mathfrak{b}(\gamma^{-2})}{\mathfrak{b}(x_{k-1})} \ge \frac{11}{24\alpha_0}L^{s+1}\lambda_s^{-1}.$$

For the last inequality use the decrease of $x^{-1}(\log x)^{\mathfrak{p}_0}$ and $\log((11/24)L^{s+1}\gamma^{-2})/\log\gamma^{-2} \le 1+(s+1)\log L$. These are exactly the bounds $R_k \ge 4r/\epsilon_s$. At $\theta = 1$, using the last class in (ii) gives $R_k \ge 4r/\epsilon_s^\circ$. Summing the pointed terms at the trivial background, and subtracting their values at source zero, gives $|G_{k,X}| \le 2M^3E_0e^{-\kappa d_k(X)}$.

It remains to improve the mixed coefficient. Put $x = x_{k-1}$, $b_x = \mathfrak{b}(x)$ and $R = R_k$. Use the fourth-family norm at the trivial background, with its actual denominator $1+q_Y(B)$. Assertion (v) gives

$$\|V_{\mathrm{old}}(z,z')\|_{\mathrm{f}} \le C\mathsf{A}b_x^{-2}, \qquad \|(V_{\mathrm{old}})_z\|_{\mathrm{f}} \le C\mathsf{A}b_x^{-2}R^{-1}, \qquad \|(V_{\mathrm{old}})_{zz'}\|_{\mathrm{f}} \le C\mathsf{A}b_x^{-2}R^{-2},$$

where A is the earlier-family majorant from the induction; C here depends on M and $\mathsf{C}_{\mathcal{A}}$. The direct source is linear in (z,z'), so the last entry is

$$V(z,z') = V_{\mathrm{old}}(z,z') + zW_+ + z'W_-.$$

Its part linear in B vanishes at the trivial background, and the quadratic and higher direct-source estimates give $\|W_\pm\|_{\mathrm{f}} \le C\mu_\star \mathsf{l}/r$. Here one uses $L^{s+1}/x \le (x_K-\bar B)^{-1}$, $4\varrho \le \sigma_{\mathrm{src}}(x_K-\bar B)$ and $\sup|f| \le \mathsf{l}\varrho/(160Mr)$. The direct source has zero mixed entry derivative; its first derivatives do not receive a factor b_x^{-2}.

Let $\mathfrak{F}_X$ denote the one-entry map with its source-independent terms omitted. At $V_0 = V(0,0)$ the mixed coefficient is

$$D\mathfrak{F}_X(V_0)[(V_{\mathrm{old}})_{zz'}] + D^2\mathfrak{F}_X(V_0)[(V_{\mathrm{old}})_z + W_+, (V_{\mathrm{old}})_{z'} + W_-].$$

The bounds $\mathsf{A} \le w_\star/8$, $\mu_\star \le w_\star/32$, $K^{\mathrm{d}}\mathsf{A} \le 13E_0/72$ and $K^{\mathrm{d}}\mu_\star \le E_0/16$ leave a fixed norm reserve. The Lipschitz bound in (iv) and Cauchy's estimate in that reserve bound the first and second derivatives by $CK^{\mathrm{d}}e^{-\lambda_{\mathrm{out}}d_k(X)}$ and $CK^{\mathrm{d}}w_\star^{-1}e^{-\lambda_{\mathrm{out}}d_k(X)}$. Bounding the cross products by the two squares therefore gives

$$|\partial_z\partial_{z'}G_{k,X}(0,0)| \le CM^3E_0\big[b_x^{-2}R^{-2} + \mathsf{l}^2/r^2\big]e^{-\kappa d_k(X)}.$$

Finally λ_s/b_x is bounded in terms of $A_1, \mathfrak{p}_0, \gamma$, since $x \ge (11/24)L^{s+1}\gamma^{-2}$ and $\log\gamma^{-2} > 2\mathfrak{p}_0$. Together with $R \ge 4r/\epsilon_s$, this bounds the bracket by $C\hat\epsilon_s^2/r^2$. Choosing c_2 as in the statement proves the mixed-coefficient estimate. □

Lemma 14.15 (Cell bounds)**.** Let (K,β) be admitted at g, (i,o) a board of $L^{-K}\mathbb{Z}^3$ with cells of side $\mathsf{l} = L^{-i}$, $1 \le i \le K-1$, f real Lipschitz, $f \ne 0$, vanishing off the middle of one cell, $r = \varrho/\|f\|_\sharp$ and $t = |z|/r$. Let $c_V = E_+ + C_{\mathrm{low}} + C_{\mathrm{gap}}$ be the constants of the systems of localised source energies that the pairs admitted at g carry on the sides of the family $\mathfrak{L}(g)$ of Lem. 14.14 (Thm. YM3_I:AD), and $\bar\Phi = \sup_{0<\mathsf{l}\le 1}\Phi_1(\frac{\mathsf{l}}{4},\frac{\mathsf{l}}{4};\mathsf{l})$ the supremum in l of the second-order constant of the one-cell bound, Φ_1 the function of equation YM3_I:(1786) at the constants of these systems. There are numbers c_V', formed from (σ^-,γ) and g, and Φ', $\Phi^{(1)}$, formed from (σ^-,γ), none of them depending on the cutoff, l, the cell, f or the volume, and holomorphic normalisers $\mathcal{E}^\flat, \mathcal{E}^\natural$, formed on the reference torus $T_{\ell'}$ of Lem. 14.14 from its system, the board, the cell and f, and free of the side of the torus on which the seminorm is taken, such that on every torus T_ℓ, $\ell \in \mathfrak{L}(g)$, $\ell \ge 2\mathsf{l}$:

(a) $\|e^{\mathcal{O}_K(zf)}\|_{\mathrm{ch}} \le |e^{\mathcal{E}^\flat(z)}|\exp[c_V\mathsf{l}^3 + \bar\Phi\, t^2/(1-t)]$ for $|z| < r$;

(a′) $\|e^{\mathcal{O}_K(zf)}\|_{\mathrm{ch}} \le |e^{\mathcal{E}^\natural(z)}|\exp[c_V\mathsf{l}^3 + \Phi^{(1)}\mathsf{l}\,t^2]$ for $|z| \le r$, using the last source class in Lem. 14.14(ii);

(b) $\|e^{\mathcal{O}_K(zf)}\|_{\mathrm{ch}} \le |e^{\mathcal{E}^\natural(z)}|\exp[c_V'\mathsf{l}^3 + \Phi'(|z|\|f\|_{\sharp,\mathsf{l}}/\varrho)^2]$ for $|z| \le r$;

(c) for every σ' as in Thm. AQ, $i \ge 1$ and real $h \in C^1_c(\mathbb{R}^2)$ carried by a square of side 2: $\|\mathsf{I}_{\mathsf{l}}(h)\| \le \mathsf{C}_3\mathsf{l}^{-3}|h|_{C^1}$ and $\|\mathsf{I}_{\mathsf{l}}(h)\| \le \mathsf{C}_2\mathsf{l}^{-2}|h|_{C^1}$, with the numbers of Thm. AQ(a), (b); from (a′), $\|\mathsf{I}_{\mathsf{l}}(h)\| \le C\mathsf{l}^{-5/2}|h|_{C^1}$;

(d) for σ' a limit of periodic Gibbs states ν_j, $i \ge 1$, a board (i,o) with $o \in L^{-i-2}\mathbb{Z}^3$ and $o_3 = 0$, a finite set $\mathcal{K} \subset \mathbb{Z}^2 \times \{0\}$, so that the cells $\square_{\mathbf{k}}$, $\mathbf{k} \in \mathcal{K}$, lie in the layer $\{|x_3| \le \frac{\mathsf{l}}{2}\}$, real $f_{\mathbf{k}} \in$

$C^1_c(\mathbb{R}^3)$, $f_{\mathbf{k}} \neq 0$, vanishing off the middles of the cells $\square_{\mathbf{k}}$, $r_{\mathbf{k}} = \varrho/\|f_{\mathbf{k}}\|_\sharp$, $t_{\mathbf{k}} = |z_{\mathbf{k}}|/r_{\mathbf{k}}$, and $D_j[z] = \prod_{\mathbf{k}} \exp(z_{\mathbf{k}}\mathcal{O}_{K_j}(f_{\mathbf{k}}) - \mathcal{E}^\flat_{\mathbf{k},j}(z_{\mathbf{k}}))$: along a subsequence $\nu_j \overline{(\theta(\tau_{\frac12 e_3}G')_j} D_j[z](\tau_{\frac12 e_3}G)_j) \to \langle [G'], \mathbb{D}[z][G]\rangle$ locally uniformly, with bounded operators $\mathbb{D}[z]$, $|z_{\mathbf{k}}| < r_{\mathbf{k}}$, holomorphic in norm, $\|\mathbb{D}[z]\| \le \exp[c_V \# \mathcal{K} \mathsf{l}^3 + \bar\Phi \sum_{\mathbf{k}} t^2_{\mathbf{k}}/(1-t_{\mathbf{k}})]$, $\mathbb{D}[0] = e^{-\mathsf{l}H}$ and $\partial_{z_{\mathbf{k}}}\mathbb{D}[0] = \mathsf{B}^{\mathrm{in}}_{\mathsf{l}/2}(f_{\mathbf{k}}) + c_{\mathbf{k}} e^{-\mathsf{l}H}$; for one cell and f even under the reflection at the middle plane of the cell, $\mathbb{D}(x)$ is at real x a number times a positive operator, and $\mathbb{D}(x)/\langle\Omega, \mathbb{D}(x)\Omega\rangle \ge 0$ where $\langle\Omega, \mathbb{D}(x)\Omega\rangle \neq 0$.

For a component σ'_ξ, with its lattice approximants $\rho_{j,\xi}$ (recalled in the results used) as the states that integrate, the family is had for one cell: bounded operators $\mathbb{D}(z)$ on $\mathcal{H}(\sigma'_\xi)$, limits along a subsequence, holomorphic on $|z| < \varrho/\|f\|_\sharp$, with $\|\mathbb{D}(z)\| \le \bar{\mathsf{Q}}$, $\bar{\mathsf{Q}}$ the number of Thm. AQ(e), formed from the cell bound in Prop. YM3_I:89.1, as in the proof of Thm. AQ(e), and free of l, f and ξ, with $\mathbb{D}(0) = e^{-\mathsf{l}H}$, $\mathbb{D}'(0)$ the insertion of f plus a multiple of $e^{-\mathsf{l}H}$, and with the same positivity at real x for even f; and $\|\mathsf{B}^{\mathrm{in},\xi}_\eta(h)\| \le \|\mathsf{B}^{\mathrm{in}}_\eta(h)\|$ for every insertion.

(e) *Reflections for many cells of one layer.* Let ν be a law on $L^{-K}\mathbb{Z}^3$ invariant under lattice isometries and reflection positive at every coordinate bond mirror. Fix a sequence $\ell_\bullet$ of sides tending to ∞ and $\mathfrak{C} \in \{\mathcal{F}_{\mathrm{loc}}, \mathcal{C}_{\mathrm{loc}}\}$, the bounded Borel or continuous local functions, respectively. Assume also chessboard domination: on every board and for every finite family of cell functions $F_{\mathbf{k}} \in \mathfrak{C}$,

$$\left|\nu\left(\prod_{\mathbf{k}} F_{\mathbf{k}}\right)\right| \le \liminf_{m\to\infty} \prod_{\mathbf{k}} \|F_{\mathbf{k}}\|_{\mathrm{ch},\ell_m}.$$

These are the three properties needed below, as every periodic Gibbs state of the Wilson weight has them (Prop. YM3_I:103.4, Prop. YM3_I:91.9); let $\{x_3 = c\}$ be a wall plane of the board, $\mathcal{K} \subset \mathbb{Z}^3$ finite and non-empty with $\square_{\mathbf{k}} \subset \{c \le x_3 \le c + \mathsf{l}\}$ for $\mathbf{k} \in \mathcal{K}$, $D_{\mathbf{k}} \in \mathcal{A}(\square_{\mathbf{k}}) \cap \mathfrak{C}$ functions of the bonds of the cells, and $\mathsf{N} \ge 0$ with $\prod_{\mathbf{k}\in\mathcal{K}} \|D_{\mathbf{k}}\|_{\mathrm{ch},\ell_m} \le \mathsf{N}$ for all large m. Then for all $F, F' \in \mathfrak{C}(\mathbb{H}^+_c)$, functions of the bonds of $\mathbb{H}^+_c = \{x_3 \ge c\}$,

$$\left|\nu\Big(\overline{\theta_c F'} \cdot \prod_{\mathbf{k}\in\mathcal{K}} D_{\mathbf{k}} \cdot \tau_{\mathsf{l}e_3} F\Big)\right| \le \mathsf{N}\, \|F'\|_{\nu,c} \|F\|_{\nu,c}, \qquad \|F\|^2_{\nu,c} = \nu(\overline{\theta_c F}\, F).$$

For one cell the proof of Prop. YM3_I:104.2 uses exactly these three properties and gives this estimate.

■

Proof sketch. (a) By the root identity of equation YM3_I:(1759), the sorting identity and the two counting estimates in the proof of the cell bound, Prop. YM3_I:89.1 among the construction results, the source energies that see two cells, $\mathscr{T}(z,z') = (\mathsf{l}/\ell')^3 \sum_j \sum_X e_j(X; zv'^+ + z'v'^-)$, the sum over the levels $0 \le j \le K$ and the domains X of the level j on $T_{\ell'}$ that see at least two cells, with ℓ', $T_{\ell'}$, e_j and the mirror sums $v'^\pm$ of Lem. 14.14, are holomorphic on $|z| + |z'| < 2r$ and bounded by $\bar\Phi$; their part linear in z goes into the normaliser, $\mathcal{E}^\flat(z) = \mathcal{E}^\circ(z) + \mathsf{a}z$, $\mathcal{E}^\circ$ the normaliser of the cell bound (Prop. YM3_I:89.1), which is the normaliser $\mathcal{E}$ of Thm. AQ(e), and Cauchy's estimate bounds the rest of $\mathrm{Re}\,\mathscr{T}(z,\bar z)$ by $\bar\Phi \sum_{k\ge2} t^k$. $\mathscr{T}$, $\mathcal{E}^\circ$, a and the normalisers are formed on $T_{\ell'}$ and read no side ℓ; the root identity carries the bound from $T_{\ell'}$ to every T_ℓ, $\ell \in \mathfrak{L}(g)$, $\ell \ge 2\mathsf{l}$, the difference of the two energy densities being at most C_{gap}. (a′), (b) The normaliser $\mathcal{E}^\natural = \mathcal{E}^\circ + \mathscr{T}(z,0) + \overline{\mathscr{T}(0,\bar z)}$ absorbs all unmixed terms, and only the mixed part $\mathscr{T}^{\mathrm{mx}}$ counts.

The two counting estimates for domains that see two cells, in the proof of Prop. YM3_I:89.1, give at the level $k = K - s$, with $u_s = \mathsf{l}L^s/M$, $(\mathsf{l}/\ell')^3 \sum_X e^{-\kappa d_k(X)} \le S(\kappa)\, m_s$, the sum over the domains X of the level k that see at least two cells and on which the mixed part of $G_{k,X}$ is not 0, with $m_s \le \min\{u_s^3, 27(\frac{u_s}{2}+2)^3 e^{-\delta_e[u_s/2-4]_+}\}$; here $\delta_e = \kappa - \kappa_{\mathcal{D}} > 0$ is the rate that the count of the

domains leaves (Section YM3_I:89, at $\kappa_e = \kappa$), and $\sum_s m_s \le \sigma_\Phi$, σ_Φ the number, formed from L, M and κ, that the sum of the two counting bounds in that proof gives at $R = D = \frac{\mathsf{l}}{4}$.

Cauchy's estimate on the bidiscs of Lem. 14.14, summed over the levels with these counts, gives $|\mathscr{T}^{\mathrm{mx}}(z,\bar z)| \le \Phi^{(1)}\mathsf{l}\, t^2$ from the radius $4r/\epsilon_s^\circ$, using the final assertion of Lem. 14.14(ii), whose source radius grows with the depth, and $|\mathscr{T}^{\mathrm{mx}}(z,\bar z) - \tau_{11}|z|^2| \le \Phi^{(3)}\mathsf{l}^3\lambda(\mathsf{l})^4 t^3$, $\lambda(\mathsf{l}) = (1+\log\frac{LM}{\mathsf{l}})^{\mathfrak{p}_0}$, from the radius $4r/\epsilon_s$, that is, from (i)–(iii); the coefficient $\tau_{11} = \partial_z\partial_{z'}\mathscr{T}(0,0)$ obeys $|\tau_{11}| \le \Phi^{(4)}\mathsf{l}^2 r^{-2}$ by the last bound of Lem. 14.14; and $\mathsf{l}\, t = |z|\|f\|_{\sharp,\mathsf{l}}/\varrho$ for f in the middle of a cell, where $\sup|f| \le \frac{\mathsf{l}}{4}\mathrm{Lip} f$.

So $\Phi' = \Phi^{(4)} + \Phi^{(3)}\sup_{\mathsf{l}\le 1}\mathsf{l}\lambda(\mathsf{l})^4$. (c) Tents of side l/L^2 cut the plate function $\rho_{\mathsf{l}}(x_3)h(\vec x)$ into pieces $f_{\mu,k}$, sorted into L^4 classes μ so that the pieces of one class lie in the middles of distinct cells of one layer; $\|f_{\mu,k}\|_\sharp \le F = 40M(40+8L^2)\mathsf{l}^{-2}|h|_{C^1}$ and at most $n \le 8\mathsf{l}^{-2}$ pieces in a class. For one class the reflections for many cells of one layer, (e), bound the expectation of the product of the exponentials between reflected vectors by the product of their chessboard seminorms; with (a), Cauchy's estimate at the radius $R = \varrho/2S_\mu$, $S_\mu^2 = \sum_k \|f_{\mu,k}\|_\sharp^2$, gives the insertion bound $\frac{4}{\varrho}e^{c_V n\mathsf{l}^3 + \bar\Phi/2}S_\mu$: n cells of a layer cost $\sqrt{n}$, not n; summing the classes gives $\mathsf{C}_3\mathsf{l}^{-3}$.

With (b), Cauchy at $R = \min\{\varrho/2\mathsf{l}S_\mu, \min_k \varrho/\|f_{\mu,k}\|_\sharp\}$, where the quadratic terms add up to $\le \frac14\Phi'$ and $R^{-1} \le (2\mathsf{l}S_\mu + F)/\varrho \le 2(\sqrt{8}+1)F/\varrho$, gives $\mathsf{C}_2\mathsf{l}^{-2}$. With (a′), Cauchy at $R = \min\{\varrho/(S_\mu(\Phi^{(1)}\mathsf{l})^{1/2}), \min_k \varrho/\|f_{\mu,k}\|_\sharp\}$, where the quadratic terms add up to ≤ 1 and $R^{-1} \le ((8\Phi^{(1)})^{1/2}\mathsf{l}^{-1/2} + 1)F/\varrho$, gives the exponent $\frac52$. The cutoff is removed with bounds uniform in K, through the lemma on the smeared insertion, Lem. YM3_I:107.1. (d) The reflections (e) with (a) for each cell, Montel's theorem in $\#\mathcal{K}$ variables and a diagonal choice over a countable set of combinations with dense classes give the limit; at $z = 0$ the semigroup, and the derivative is the insertion plus a number times $e^{-\mathsf{l}H}$.

For real x and f even under the reflection at the middle plane, which passes through sites, $e^{x\mathcal{O}(f)}$ is a function of the mirror bonds, positive, times a product of a function of one side and its mirror image, so reflection positivity makes the lattice operators the numbers $e^{-\mathcal{E}_j^\flat(x)}$ times positive operators; as $\mathcal{E}_j^\flat = \mathcal{E}_j^\circ + \mathsf{a}_j z$ with complex a_j, these numbers need not be positive, but along a further subsequence their phases converge, so $\mathbb{D}(x)$ is a number times a positive operator, and the division by $\langle\Omega, \mathbb{D}(x)\Omega\rangle$ removes the number. For a component the family of one cell is formed in the same way with its lattice approximants, the one-cell bound among the construction results giving the bound $\bar{\mathsf{Q}}$, free of l, f and ξ, as in the proof of Thm. AQ(e): the approximants $\rho_{j,\xi}$ are periodic Gibbs states at the couplings of the sequence (Thm. YM3_I:AP), so they have the three conditions of (e) and the bounds of the cell bound, and their moments, centred by the ρ_j, converge to those of σ'_ξ along the full sequence; so the reflections, Montel's theorem, the diagonal choice and the positivity at the middle plane run on $\mathcal{H}(\sigma'_\xi)$ as on $\mathcal{H}(\sigma')$.

For a component,

$$\|M_g \mathsf{B}_\eta^{\mathrm{in}}(h)[G]\|^2 = \int g^2 \|[X]_\xi\|^2 \,\mathrm{d}\lambda$$

with X the tuple of the insertion (Lem. 14.7(i)), and full support gives the bound at every ξ. (e) The multiple-reflection proof of Prop. YM3_I:104.2, with $D = \prod_{\mathbf{k}\in\mathcal{K}} D_{\mathbf{k}}$ in the place of its function of one cell. All the cells lie in the one layer $c \le x_3 \le c + \mathsf{l}$, and the mirror θ' at its middle plane keeps every cell; so D is a function of $\mathfrak{C}(\mathbb{H}_c^+)$,

$$D^\sharp = \overline{\theta' D} = \prod_{\mathbf{k}} \overline{\theta' D_{\mathbf{k}}},$$

and the operator $\hat D X = D \cdot \tau_{\mathsf{l}e_3} X$ has the adjoint $\widehat{D^\sharp}$ for the form of the mirror, which, with the convexity of the moments $\mathsf{s}_n = \langle F, (\widehat{D^\sharp}\hat D)^n F\rangle_{\nu,c}$, holds as in that proof.

The moment s_n is the mean of a product over $2n$ layers of the translates of the $D_{\mathbf{k}}$ and of the $D_{\mathbf{k}}^\sharp$, each with the chessboard seminorm of $D_{\mathbf{k}}$. The coarser board is chosen so that one of its

cells holds the bonds of F and all the cells $\square_{\mathbf{k}}$, $\mathbf{k} \in \mathcal{K}$; the chessboard domination on it, with the estimate from fine to coarse cells (Lem. YM3_I:104.1), the function 1 standing in the other fine cells, gives $\mathsf{s}_n \le (\sup|F|)^2 \mathsf{N}^{2n}$; the n-th root as $n \to \infty$ and the Schwarz inequality of the form give the claim. □

Lemma 14.16 (Gluing)**.** Let $G = \mathrm{SU}(2)$ and (Σ, B) a finite graph. (a) Let $Y_1, Y_2 \subset \mathsf{B}$, $I = Y_1 \cap Y_2$, $\mathsf{R}_2 = Y_2 \setminus I$, every site of Y_1 on R_2 a site of I, and every connected component of I with a cycle. Then the gauge-invariant operators of Y_1 and of Y_2 generate those gauge-invariant operators of $Y_1 \cup Y_2$ that commute with the gauge transformations of R_2 by central elements that are constant on each component of I and 1 off I; for connected I, all of them. For two plaquettes with exactly one common bond (seven bonds, six sites), the gauge-invariant operators of the two plaquettes generate all those of their union. (b) Let connected subgraphs Γ_i (boxes) cover the graph, such that at every site x any two boxes at x are joined by a chain of boxes at x in which consecutive boxes meet in a component at x with two independent cycles.

Let Z be the sign fields c with $c(C') = 1$ for every closed path C' in one box, B the coboundaries and $\mathsf{H} = Z/B$. On an open set X_0 of full Haar measure, invariant under gauge transformations and sign fields: for $V \in X_0$, a configuration V' looks as V in every box up to the gauge transformations of the box if and only if $V' = g \cdot (cV)$ with a gauge transformation g and $c \in Z$; the Borel sets invariant under gauge transformations and under Z agree on X_0 with the sets of the σ-algebra $\mathcal{S}_{\mathrm{add}}$ that the boxes generate; and for every character κ of H there is a closed path C_κ with $c(C_\kappa) = \kappa([c])$, such that every bounded gauge-invariant Borel F is $\sum_\kappa \Phi_\kappa s_\kappa$ on a set of full Haar measure in $\mathcal{S}_{\mathrm{add}}$, $s_\kappa = \operatorname{sign} \operatorname{tr} V(C_\kappa)$, Φ_κ $\mathcal{S}_{\mathrm{add}}$-measurable with $\sup|\Phi_\kappa| \le \sup|F|$. (c) For a cyclic chain of boxes $\Gamma_1, \dots, \Gamma_n$ with components $\Delta_i \subset \Gamma_i \cap \Gamma_{i+1}$ that hold two independent cycles, frames $\mathfrak{f}_i(V) \in \mathrm{SO}(3)$ formed from two holonomies of Δ_i at a site y_i, and paths C_i in Γ_i from y_{i-1} to y_i: the entries of $T_i = \mathfrak{f}_{i-1}^{\mathrm{T}} \mathrm{Ad}(V(C_i)) \mathfrak{f}_i$ are invariant under gauge transformations and under Z, measurable for Γ_i and of modulus ≤ 1, and $\chi_1(V(C_1 \cdots C_n)) = \operatorname{tr}(T_1 \cdots T_n)$ on X_0. ■

Proof sketch. (a) Write $\mathfrak{F}(Y)$ for the bounded operators of the bonds of Y, acting on $L^2(G^Y)$ and tensored with 1, $\Gamma_Y(g)$ for the gauge transformation by g on the bonds of Y alone, Γ for Γ_{B}, and $\Sigma_* = \Sigma(Y_1) \cap \Sigma(\mathsf{R}_2)$ for the common sites, which lie in $\Sigma(I)$ by hypothesis. *Fullness.* A spanning tree of a connected set J of bonds with $n \ge 1$ independent cycles fixes the gauge at all but one site, so the gauge action Γ_J on $L^2(G^J)$ is induced from the conjugation action of G on $L^2(G^n)$; by Peter–Weyl every irreducible representation of the gauge group of the sites of J with trivial total central character occurs in it, since every irreducible representation of SO(3) occurs in the conjugation action of SU(2) on $L^2(\mathrm{SU}(2))$: J is full.

Contained in. $\mathfrak{F}(Y_1)$ commutes with $\Gamma_{\mathsf{R}_2}(\zeta)$; on $\mathfrak{F}(Y_2)$ the adjoint action of $\Gamma_{\mathsf{R}_2}(\zeta)$ is that of $\Gamma(\zeta)$, because $\Gamma_I(\zeta) = 1$, both ends of a bond of I lying in one component, on which ζ is a constant central element; so the invariant operators of Y_2 commute with $\Gamma_{\mathsf{R}_2}(\zeta)$. *Contains,* in three steps. (1) The bonds of $Y_1 \cup Y_2$ are those of Y_1 and of R_2, so its operators are weak limits of sums of products AC, $A \in \mathfrak{F}(Y_1)$, $C \in \mathfrak{F}(\mathsf{R}_2)$; averaging over the gauge group and over the $\Gamma_{\mathsf{R}_2}(\zeta)$, and writing C as a weak limit of sums of members of multiplets, the right member is the weakly closed span of the sums $\sum_k A_k D_k^*$, where (A_k) in $\mathfrak{F}(Y_1)$ and (D_k) in $\mathfrak{F}(\mathsf{R}_2)$ are multiplets of one irreducible representation σ of G^{Σ_*}, invariant under the gauge transformations at the other sites, σ of trivial central character on the sites of each component of I, the mean over the ζ removing the other σ. (2) The common sites lie in $\Sigma(I)$, so the gauge group of Σ_* acts on the bonds of I (here the hypothesis on the common sites is read), and each component of I is full; so there are multiplets $\chi^{(\mathsf{r})}$ of σ in $\mathfrak{F}(I)$, invariant at the sites of I off Σ_*, with $\sum_{\mathsf{r}} \chi_k^{(\mathsf{r})*} \chi_{k'}^{(\mathsf{r})} = \delta_{kk'}$, the partial sums bounded by 1: a multiplet of σ in $\mathfrak{F}(I)$ is an intertwiner from $V_\sigma \otimes L^2(G^I)$ to $L^2(G^I)$, every type in $\sigma \otimes \Gamma_I$ has trivial central character on each component of I and so occurs in Γ_I, and partial isometries between the copies of each type give the $\chi^{(\mathsf{r})}$. (3) Then $\sum_k A_k D_k^* = \sum_{\mathsf{r}} a^{(\mathsf{r})} b^{(\mathsf{r})}$ with $a^{(\mathsf{r})} = \sum_k A_k \chi_k^{(\mathsf{r})*}$ and $b^{(\mathsf{r})} = \sum_k \chi_k^{(\mathsf{r})} D_k^*$; $a^{(\mathsf{r})}$ is

gauge invariant of Y_1, by the unitarity of σ, and $b^{(\mathsf{r})}$ is gauge invariant of Y_2, because the sites of I off Σ_* are no sites of R_2 and the sites of R_2 off Σ_* no sites of I, $\Sigma(I)$ lying in $\Sigma(Y_1)$.

For connected I every σ that occurs is trivial on the constant central elements of Σ_*, a constant central gauge transformation of all the sites of R_2 acting trivially on $\mathfrak{F}(\mathsf{R}_2)$; so the condition of commutation is void. *The two plaquettes.* On the space of the gauge-invariant vectors of a periodic layer with $N_1, N_2 \geq 4$ the invariant operators of two plaquettes with one common bond generate those of their seven bonds, by the argument with the full algebras of single plaquettes as cells that the last paragraph of the proof of Thm. AS(b), on two plaquettes with one common bond, gives on $L^2(G^7)$ itself (spin labels, Lem. 14.19, one plaquette made large, a monotone path, and the identity for the cross term at the ends of the common bond), every closed sign field of the seven bonds being a coboundary.

This is a statement on the abstract graph of the seven bonds: if the complement of a set Y of bonds of a finite graph is connected, holds a cycle and meets every site, then $X \mapsto XP$, P the projection on the gauge-invariant vectors, is injective on the invariant operators of Y, because such an operator is a sum of blocks $1 \otimes \mathcal{B}(\mathsf{K}_\tau)$ over the types τ of Γ_Y, all of trivial total central character, and the block of τ is killed by P only if the conjugate type does not occur in the gauge action on the complement, which, full and meeting every site, holds it. In a periodic layer with $N_1, N_2 \geq 4$ the complement of two plaquettes with a common bond is connected, holds cycles and meets each of their six sites. (b) Two elements of SU(2) that do not commute have the centraliser $\{\pm 1\}$: in a component with two independent cycles the two holonomies at a site fix the gauge transformation there up to the centre, and the chains carry this along; the quotients by the compact gauge groups are standard Borel spaces, and the sign of the trace of a loop that detects a character separates the classes. (c) Under a gauge transformation the frames turn by $\mathrm{Ad}(g(y_i))$, which keeps scalar and vector products; sign fields do not change Ad; and $\chi_1(V(C)) = \operatorname{tr} \mathrm{Ad}(V(C))$. □

Lemma 14.17 (Sharp loops and flips on one lattice)**.** On a torus $\Lambda_{K,\ell}$ at the coupling β, with $N = \ell L^K$ sites on a side and $\langle\cdot\rangle$ the Wilson measure: (a) for a rectangular path γ with sides of at least three and at most $N-2$ bonds, γ' its bonds but the two at each corner, $n' = \#\gamma'$, and bounded Borel F of the other bonds, $|\langle \frac{1}{2}\operatorname{tr} U(\gamma) F\rangle| \leq (I_2(4\beta)/I_1(4\beta))^{n'} \sup|F|$; (b) on a cubic torus at $\beta \geq 1$, $\langle a_p\rangle \leq (21 + 3\log\beta)/\beta$ for every plaquette; (c) for a state ω, a bounded V and a projection Q_0 with $Q_0 V Q_0 = 0$, and bounded X, Y commuting with Q_0: $|\omega(XVY)| \leq 2\|X\|\|V\|\|Y\|\omega(1-Q_0)^{1/2}$; for the flip V_ζ of a dual path ζ of a layer with an end plaquette p_0 and $Q_0 = M_{\mathbf{1}\{a_{p_0}<1\}}$, $Q_0 \mathsf{V}_\zeta Q_0 = 0$ and $\omega(1-Q_0) \leq \omega(M_{a_{p_0}})$. ■

Proof sketch. (a) Two bonds of γ' lie in no common plaquette: collinear neighbours share none; bonds of two adjacent sides could share one only at a corner, and the corners are removed; and two parallel bonds lie in a common plaquette only at the distance of one bond, while two opposite sides of γ are at least three bonds apart in the plane of γ and, as each side has at most $N-2$ bonds, at least two bonds apart also across the seam of the torus. Given all other bonds, the weight factorises over the bonds of γ', and $\frac{1}{2}\operatorname{tr} U(\gamma)$ is linear in each of them. For a bond b the sum A_b of the four staples of its plaquettes is $a\,g$ with $0 \leq a \leq 4$ and $g \in \mathrm{SU}(2)$, and by the character expansion $\int U e^{\frac{\beta}{2}\operatorname{tr}(UA_b)}\,\mathrm{d}U / \int e^{\frac{\beta}{2}\operatorname{tr}(UA_b)}\,\mathrm{d}U = \frac{I_2(\beta a)}{I_1(\beta a)} g^{-1}$, a ratio that increases with its argument; the product over γ' gives the bound. (b) $\log Z(\beta)$ is convex with $Z(0) = 1$, so $\langle \sum_p a_p\rangle_\beta \leq -\log Z(\beta)/\beta$; restricting the integral to $|U_b - 1| \leq \epsilon$ at all bonds, where $a_p = \frac{1}{2}|\partial U(p) - 1|^2 \leq 8\epsilon^2$ and the Haar measure of $\{|U-1| \leq \epsilon\}$ is $\geq 8\epsilon^3/3\pi^3$, and using that a cubic torus has as many bonds as plaquettes, $\langle a_p\rangle \leq [\log(3\pi^3/8) + 3\log\epsilon^{-1}]/\beta + 8\epsilon^2$; at $\epsilon^2 = 1/8\beta$ this is $\leq (7 + \frac{3}{2}\log\beta)/\beta$. (c) $XVY = X(1-Q_0)VY + XQ_0V(1-Q_0)Y$; the Schwarz inequality for ω, with $X(1-Q_0)X^* \leq \|X\|^2(1-Q_0)$ and $Y^*(1-Q_0)Y \leq \|Y\|^2(1-Q_0)$, bounds each term by $\|X\|\|V\|\|Y\|\omega(1-Q_0)^{1/2}$. V_ζ reverses t_{p_0}, the end plaquette having exactly one crossed bond, so it carries $\{a_{p_0} < 1\}$ onto $\{a_{p_0} > 1\}$; and $\mathbf{1}\{a_{p_0} \geq 1\} \leq a_{p_0}$. □

Lemma 14.18 (Haar-null sets under the block map)**.** Let $0 \le k < K$ and let $\mathcal{M}$ be the block-averaging map of one step from the configurations of the fine lattice $\mathbb{T}^{(k)}$ to those of the coarse lattice $\mathbb{T}^{(k+1)}$ (equations YM3_I:(37)–(39)). For every Haar-null Borel set N of coarse configurations, $\mathcal{M}^{-1}(N)$ is a Haar-null Borel set of fine configurations; so the same holds for every iterate $\mathcal{M}^n$. Hence, in the setting of Thm. AS(g), the laws μ_d^K of the block fields, $K \ge \max\{d, 1\}$, are absolutely continuous for the Haar measure h_d, and so are the limit laws μ_d. ■

Proof. The positively oriented fine bonds are the disjoint union of the bonds with both ends in one block and of the sets X_c, c a positively oriented coarse bond, where X_c is made of the L^2 fine bonds that lie over c, parallel to c, one at each offset across c: each coarse bond has L^2 fine bonds of its own. $\mathcal{M}(U)(c)$ depends only on the values of U on the bonds inside the two blocks of c and on X_c (Prop. YM3_I:4.5). At fixed values ι of the bonds inside the blocks, $\mathcal{M}(\iota, \cdot)$ is thus a product over c of Borel maps Ψ_c^ι of the bonds of X_c, and by Fubini's theorem it suffices that, for ι off a null set, every Ψ_c^ι pulls null sets back to null sets, a property that passes to products.

A contour variable is the ordered product of the U along a path; the intrinsic logarithm is the inverse of exp on $\{|X| < \pi\}$, real-analytic on $\mathrm{SU}(2) \setminus \{-\mathbf{1}\}$; $U[c]$ is the contour variable of the straight path from c_- to c_+, and $x' = x + (c_+ - c_-)$ for $x \in B(c_-)$. Equations YM3_I:(37)–(39) give $\mathcal{M}(U)(c)$ by three cases, with disjoint domains that exhaust all configurations (Prop. YM3_I:4.5). *First case:* on the open set $\mathcal{S}_c$ of the configurations whose plaquettes with all corners in the double block of c lie within $\omega_M = \frac{1}{200L^2}$ of $\mathbf{1}$ (the small fields of the double block, a condition on its bonds alone), $\mathcal{M}(U)(c) = \exp(L^{-3}\sum_{x\in B(c_-)} \log W_x) U[c]$ with $W_x = U(c_-, x)U([x, x'])U(x', c_+)U[c]^{-1}$, the $U(y, x)$ being the group means of the contour variables along the staircase paths of a block from its centre y to x, and log the series; the first case applies exactly on $\mathcal{S}_c$.

Second case: off $\mathcal{S}_c$, where none of the closed contour variables

$$Y_{x,\Gamma,\Gamma'} = U(\Gamma)U([x, x'])U(\Gamma')^{-1}U[c]^{-1},$$

$x \in B(c_-)$, Γ and Γ' staircase paths from c_- to x and from c_+ to x', equals $-\mathbf{1}$,

$$\mathcal{M}(U)(c) = \exp\big(L^{-3}\sum_x \mathrm{mean}_{\Gamma,\Gamma'} \log Y_{x,\Gamma,\Gamma'}\big) U[c]$$

with the intrinsic logarithm. *Third case:* at all other configurations $\mathcal{M}(U)(c) = U[c]$. A contour variable $Y_{x,\Gamma,\Gamma'}$ with x on the line through c_- parallel to c runs forth and back on that line and is identically $\mathbf{1}$; every other one passes once through the bond of X_c on $[x, x']$ and once through the bond of X_c on $[c_-, c_+]$, two different bonds, all its other bonds lying inside the blocks; so the set where it equals $-\mathbf{1}$ is the graph of one bond of X_c over the other bonds, a closed real-analytic submanifold of codimension 3.

The finite union $\mathcal{N}_c$ of these graphs is closed and null, and its section at every value ι of the bonds inside the blocks is null; its complement $\hat{\mathcal{O}}$ is open, connected, as the complement of finitely many closed submanifolds of codimension 3 of a connected manifold, and holds $\mathbf{1}$. The domain of the second case lies in $\hat{\mathcal{O}}$, that of the third in $\mathcal{N}_c$. In the formula of the second case every logarithm is taken of one of these closed contour variables, and no group mean occurs: products of bond variables, the intrinsic logarithm on $\mathrm{SU}(2) \setminus \{-\mathbf{1}\}$ and the exponential make it a real-analytic map F_c on $\hat{\mathcal{O}}$, jointly in the bonds inside the blocks and in X_c.

Its differential in the variables of X_c at $\mathbf{1}$: the factors $U(\Gamma)$, $U(\Gamma')$ do not read X_c, the differential of the logarithm at $\mathbf{1}$ is the identity, so $dY_{x,\Gamma,\Gamma'} = dU([x, x']) - dU[c]$ there, and the differential of F_c is $L^{-3}\sum_x dU([x, x']) = L^{-2}\sum_{b\in X_c} du_b$, each bond of X_c lying on L of the L^3 straight paths $[x, x']$: the mean over the L^2 bonds, the value that Lem. YM3_I:4.6 gives for the formula of the first case. The Gram determinant of this differential is a real-analytic function on the connected set $\hat{\mathcal{O}}$ with the value L^{-6} at $\mathbf{1}$; so its zero set $\mathcal{Z}_c$ is null, and off it F_c is a submersion in the variables of X_c.

We now prove that the first branch is a submersion at every point of $\mathcal{S}_c$. This avoids any assumption about its connected components. Let ν be the direction of c. Index the L^2 lines parallel

to c in its double block by their transverse coordinates z. Each line has one crossing bond $b_z \in X_c$; write $u_z = U(b_z)$ and let z_0 index the line through c_-. For $x \in B(c_-)$ on the line z, factor the straight transport as

$$U([x,x']) = P_x u_z Q_x, \qquad U[c] = g = P_0 u_{z_0} Q_0.$$

The factors P_x, Q_x use only bonds inside the blocks. The contour means also use only those bonds. Thus, with

$$a_x = U(c_-,x)P_x, \qquad b_x = Q_x U(c_+,x')^{-1},$$

the first-branch variables are $W_x = a_x u_z b_x g^{-1}$, and a_x, b_x stay fixed when the crossing variables vary.

Let t_z be the point of the line z whose ν-coordinate is that of c_-, so $t_{z_0} = c_-$ and $a_{c_-} = P_0$. Given $\xi \in \mathfrak{g}$, vary all crossing bonds together by

$$u_z(\epsilon) = e^{i\epsilon X_z} u_z, \qquad X_z = \mathrm{Ad}(a_{t_z})^{-1}\xi.$$

The norm of this tangent vector in the Euclidean product of the crossing variables is $L|\xi|$, and $g(\epsilon) = e^{i\epsilon\xi} g$. Moreover,

$$W_x(\epsilon) = e^{i\epsilon(\xi+\rho_x)} W_x e^{-i\epsilon\xi}, \qquad \xi + \rho_x = \mathrm{Ad}(a_x a_{t_z}^{-1})\xi.$$

The error ρ_x is small because the contour mean is close to every contour in its family. Indeed,

$$a_x a_{t_z}^{-1} = U(c_-,x)U([x,t_z])U(c_-,t_z)^{-1} = h_1 h_2.$$

To obtain the last factorization, choose the contour from c_- to x which traverses the ν direction last. It is a contour $\Gamma_\perp$ from c_- to t_z followed by $[t_z,x]$. Then

$$h_1 = U(c_-,x)U(\Gamma)^{-1}, \qquad h_2 = U(\Gamma_\perp)U(c_-,t_z)^{-1}.$$

Lem. YM3_I:4.4 gives $|h_i - \mathbf{1}| \le 1/40$ and $|W_x - \mathbf{1}| < 1/14$. Unitarity and $|\mathrm{Ad}(h)\xi - \xi| \le 2|h-\mathbf{1}||\xi|$ therefore give

$$|h_1 h_2 - \mathbf{1}| \le \frac{1}{20}, \qquad |\rho_x| \le \frac{1}{10}|\xi|.$$

Write $W_x = e^{iw_x}$ with $w_x \in \mathfrak{g}$ and $|w_x| < 0.0715$. The differential of the logarithm gives

$$\frac{d}{d\epsilon}\log W_x(\epsilon)\Big|_{\epsilon=0} = i\eta_x, \qquad |\eta_x| \le \left(1.001\cdot\frac{1}{10} + 2|w_x|\right)|\xi| < \frac{1}{4}|\xi|.$$

Here the first term bounds the logarithmic differential of the left perturbation ρ_x, and the second is the commutator from conjugating w_x by $e^{i\epsilon\xi}$.

Put $\mathsf{E} = L^{-3}\sum_x w_x$ and $\bar\eta = L^{-3}\sum_x \eta_x$. The formula for the average now yields

$$-i\frac{d}{d\epsilon}\mathcal{M}(U(\epsilon))(c)\mathcal{M}(U)(c)^{-1}\Big|_{\epsilon=0} = \mathsf{T}_U\xi := \mathsf{f}(\mathrm{ad}(i\mathsf{E}))\bar\eta + \mathrm{Ad}(e^{i\mathsf{E}})\xi, \qquad \mathsf{f}(s) = \frac{e^s-1}{s}.$$

The operator $\mathsf{f}(\mathrm{ad}(i\mathsf{E}))$ is the average, for $0 \le t \le 1$, of the isometries $\mathrm{Ad}(e^{it\mathsf{E}})$. Consequently $|\mathsf{T}_U\xi| \ge 3|\xi|/4$. It is an invertible map of $\mathfrak{g}$, and

$$\zeta \longmapsto \left(\mathrm{Ad}(a_{t_z})^{-1}\mathsf{T}_U^{-1}\zeta\right)_z$$

is a right inverse of the crossing-variable differential, of norm at most $4L/3$.

A submersion pulls Haar-null sets back to null sets, since in local coordinates it is a projection and Haar measure has a smooth positive density. The first branch is a submersion everywhere on $\mathcal{S}_c$. For the second branch, the exceptional set $\mathcal{Z}_c$ found above is null; the third branch is contained in $\mathcal{N}_c$. Fubini's theorem therefore gives, for almost every fixed internal configuration ι, the null-set property for Ψ_c^ι. There are finitely many coarse bonds, so these properties hold simultaneously outside one exceptional set of ι of measure zero. The product map has the same property, again by Fubini.

So $\mathcal{M}^{-1}(N)$ is null. Then $\mu_d^K(N) = Z_K^{-1}\int \mathbf{1}_N(\mathcal{M}^{K-d}U)\,e^{-\beta\sum_p a_p}\,\mathrm{d}U = 0$ for a null N, and $\mu_d(N) = \lim_K \mu_d^K(N) = 0$, the laws converging uniformly on all Borel sets (Cor. 2.8). □

Lemma 14.19 (One vertex)**.** Let $V_{s_1}, V_{s_2}, V_{s_3}$ be irreducible representations of SU(2), $J_i = (J_i^a)_{a\le 3}$ the spins acting in the i-th factor of $V_{s_1}\otimes V_{s_2}\otimes V_{s_3}$, $[J_i^a, J_i^b] = i\epsilon_{abc}J_i^c$, and $t_{ik} = \sum_a J_i^a J_k^a$. For every spin J the algebra generated by 1, t_{12} and t_{23} acts irreducibly on the multiplicity space $\mathsf{M}^J = \mathrm{Hom}_{\mathrm{SU}(2)}(V_J, V_{s_1}\otimes V_{s_2}\otimes V_{s_3})$: it is the algebra of all operators of M^J. ■

Proof sketch. The enveloping algebra $\mathfrak{U}$ of three copies of the Lie algebra of SU(2) maps onto the algebra of all operators of the irreducible module $V_{s_1}\otimes V_{s_2}\otimes V_{s_3}$ (Burnside), equivariantly for the diagonal adjoint action, which is locally finite; averaging over the group shows that the invariant part $\mathfrak{U}^{\mathrm{inv}}$ maps onto the commutant of the diagonal action, the sum of the algebras of all operators of the spaces M^J. By symmetrisation the graded algebra of $\mathfrak{U}^{\mathrm{inv}}$ is the algebra of the polynomial invariants of three vectors of $\mathbb{C}^3$ under SO(3), which is generated by the scalar products and the determinant (first fundamental theorem); by induction along the filtration of $\mathfrak{U}$ by the order, $\mathfrak{U}^{\mathrm{inv}}$ is generated by the J_i^2, the t_{ik} and $T_3 = \sum \epsilon_{abc}J_1^aJ_2^bJ_3^c$, and $[t_{12}, t_{23}] = \sum J_1^aJ_3^c[J_2^a, J_2^c] = -iT_3$. On the isotypic part $V_J\otimes\mathsf{M}^J$ the $J_i^2 = s_i(s_i+1)$ are numbers and, the square of the total spin being $J(J+1)$, $t_{13} = \frac12(J(J+1) - \sum_i s_i(s_i+1)) - t_{12} - t_{23}$. So 1, t_{12}, t_{23} generate the image of $\mathfrak{U}^{\mathrm{inv}}$ in the operators of M^J, which is all of them. □

14.1.6. *Proof of Thm. AO. (a)* If $\eta \perp N^n\mathcal{D}_0$, then $\langle\eta, U(x)N^n\chi\rangle = 0$ for all x and $\chi\in\mathcal{D}_0$, as $U(x)$ keeps $\mathcal{D}_0$ and commutes with N; so the spectral measure of the pair $(\eta, N^n\chi)$ vanishes and $\langle\eta,\chi\rangle = \int(1+E)^{-n}\,\mathrm{d}m_{\eta,N^n\chi} = 0$. If $\psi\perp N^n\mathcal{D}_0(O)$, the function $(a_k)_{k\le q}\mapsto\langle\psi, N^nU(a_1)\phi(f_1)\cdots U(a_q)\phi(f_q)\Omega\rangle$, $f_k\in\mathcal{F}(O_1)$, with arbitrary $q\ge 1$ with $O_1 + \{|a|<\delta\}\subset O$, vanishes for small a_k and is the boundary value of a function holomorphic in $a_k + iV_+$ (spectrum condition); by the reflection principle it vanishes for all a_k, and a partition of unity gives $\psi\perp N^n\mathcal{D}_0$.

By the bound (b5) and the core $\mathcal{D}_0$ of N^n, $n\ge\wp$, the closures from $\mathcal{D}_0$ and from $\mathcal{D}_\infty$ agree, $\langle\phi(\bar f)\chi,\eta\rangle = \langle\chi,\bar\phi(f)\eta\rangle$ for $\chi\in\mathcal{D}_\infty$, $\eta\in D(\bar\phi(f))$, and the relation that defines $\mathcal{C}(O)$ passes from $\mathcal{D}_0$ to $\mathcal{D}_\infty$. A smoothing in time turns it into $Q\bar\phi(f)\subset\bar\phi(f)Q$. Let $Q\in\mathcal{C}(O)$, $f\in\mathcal{F}(O)$, $\delta>0$ with $\mathrm{supp} f + (-\delta,\delta)e_0\subset O$, $g\in C_c^\infty((-\delta,\delta))$ and $Q(g) = \int g(t)e^{itH}Qe^{-itH}\,\mathrm{d}t$. As $e^{-itH}\phi(f) = \phi(f(\cdot + te_0))e^{-itH}$ on $\mathcal{D}_\infty$ and $f(\cdot+te_0)\in\mathcal{F}(O)$ for $|t|<\delta$, $Q(g)$ has the same relation for this f on $\mathcal{D}_\infty$; and $\langle N^n\chi, Q(g)\psi\rangle = \sum_{k\le n}\binom{n}{k}i^k\langle\chi, Q(g^{(k)})N^{n-k}\psi\rangle$ for $\chi,\psi\in\mathcal{D}_\infty$ gives $Q(g)\mathcal{D}_\infty\subset\mathcal{D}_\infty$.

With $\phi(f)^\dagger = \phi(\bar f)$: $\phi(f)Q(g) = Q(g)\phi(f)$ on $\mathcal{D}_\infty$; as $\bar\phi(f)$ is closed, $Q(g)\bar\phi(f)\subset\bar\phi(f)Q(g)$; for an approximate unit g_k, $Q(g_k)\to Q$ strongly, and closedness once more gives $Q\bar\phi(f)\subset\bar\phi(f)Q$. Conversely a bounded Q with $Q\bar\phi(f)\subset\bar\phi(f)Q$ for all $f\in\mathcal{F}(O)$ lies in $\mathcal{C}(O)$, by the relation between $\phi(\bar f)$ and $\bar\phi(f)$ above. So $\mathcal{C}(O)$ is closed under products; it holds 1, is closed under adjoints and weakly closed: a von Neumann algebra; covariance is $U(\mathfrak{g})\phi(f)U(\mathfrak{g})^{-1} = \phi(f^{\mathfrak{g}})$ on $\mathcal{D}_0$. If $Q\in\mathcal{C}(O)$ and $Q\Omega = 0$, then $\langle\psi, QX\Omega\rangle = \langle X^\dagger\psi, Q\Omega\rangle = 0$ for products X of fields of O, so $Q = 0$.

The projection onto the closure of $\mathcal{A}(O)\Omega$ lies in $\mathcal{A}(O)' = \mathcal{C}(O)$, commutes with every $\bar\phi(f)$, $f\in\mathcal{F}(O)$, and fixes Ω, so that closure holds $\mathcal{D}_0(O)$. The unitaries of $\mathcal{A}(O)' = \mathcal{C}(O)$ commute with $\bar\phi(f)$; and a net $\mathfrak{N}$ with this affiliation has $\mathfrak{N}(O)'\subset\mathcal{C}(O)$, so $\mathfrak{N}(O)\supset\mathcal{A}(O)$. A partition of unity writes $f\in\mathcal{F}(\bigcup O_i)$ as a finite sum of $f_i\in\mathcal{F}(O_i)$: $\mathcal{C}(\bigcup O_i) = \bigcap\mathcal{C}(O_i)$, and the commutants give $\mathcal{A}(\bigcup O_i) = \bigvee\mathcal{A}(O_i)$. For $O_2\subset O_1^\times$, $\mathcal{L}(O_1)\subset\mathcal{C}(O_1^\times)\subset\mathcal{C}(O_2)$ commutes with $\mathcal{A}(O_2)\supset\mathcal{L}(O_2)$. (W) says $\mathcal{A}(O_1)\subset\mathcal{C}(O_2)$ for all $O_2\subset O_1^\times$, that is $\mathcal{A}(O_1)\subset\mathcal{C}(O_1^\times)$ by additivity, that is $\mathcal{L} = \mathcal{A}$; and an open set is the union of the upright double cones in it, so additivity reduces (W) to such cones.

(b) (b1) $P_m\phi(f)P_n = P_me^{sH}(e^{-sH}\phi(f)e^{-sH})e^{sH}P_n$ with $\|P_me^{sH}\|\le e^{s(m+1)}$; at $s = (m+n+2)^{-1}$, eq. 14.i gives $b_{mn}(f)\le e\,C_\wp\|f\|_{L^1}(m+n+2)^\wp$. (b2) With $X = P_m\phi(f)P_n$ and $[H,\phi(f)] = i\phi(\partial_t f)$,

$$(m-n)X = iP_m\phi(\partial_t f)P_n - (H-m-\tfrac12)X + X(H-n-\tfrac12),\qquad \|(H-m-\tfrac12)P_m\|\le\tfrac12,$$

so $(|m-n|-1)b_{mn}(f) \le b_{mn}(\partial_t f)$; iterate. (b3) If $\wp \le 1$, then $k_1 = 0$ and the claim follows directly from (b1). Suppose $\wp > 1$, so $k_1 \ge 1$. For $|m-n| > (m+n+2)^\epsilon \ge 2^\epsilon$, $|m-n| \ge 2$ and $|m-n|-1 \ge \frac{1}{3}|m-n|$, so (b2) at k_1 and (b1) at $\partial_t^{k_1} f$ give $b_{mn}(f) \le 3^{k_1}|m-n|^{-k_1} e C_\wp \|\partial_t^{k_1} f\|_{L^1}(m+n+2)^\wp$, and $|m-n|^{k_1} > (m+n+2)^{\epsilon k_1} \ge (m+n+2)^{\wp-1}$. (b4) $\phi(h*f) = \int h(a)U(a)\phi(f)U(a)^{-1}\,\mathrm{d}a$, and $U(a)$ commutes with P_m. (b5) At $s = (1+\max\{m,n\})^{-1}$ the argument of (b1) gives $b_{mn}(f) \le e^2 C_\wp \|f\|_{L^1}(1+\max\{m,n\})^\wp$, and with $\|(H-m)P_m\| \le 1$ the argument of (b2) gives $(|m-n|-2)^k b_{mn}(f) \le b_{mn}(\partial_t^k f)$ for $|m-n| \ge 3$. With $(1+\max\{m,n\})^2 \le (1+m)(1+n)(1+|m-n|)$: $b_{mn}(f) \le e^2 C_\wp (1+m)^{\wp/2}(1+n)^{\wp/2}\gamma(m-n)$, $\gamma(d) = (1+|d|)^{\wp/2}\|f\|_{L^1}$ for $|d| \le 2$ and $\gamma(d) = (1+|d|)^{\wp/2}(|d|-2)^{-k_\flat}\|\partial_t^{k_\flat} f\|_{L^1}$ for $|d| \ge 3$; $\sum_d \gamma(d)$ is the bracket of the form bound, and Schur's test with $\sum_a (1+a)^{2q}\|P_a\psi\|^2 \le \|N^q\psi\|^2$ gives it. For the operator bound, $(2+a)^j \le 2^j(1+b)^j(1+|a-b|)^j$ and $(1+\max\{a,b\})^\wp \le (1+b)^\wp(1+|a-b|)^\wp$, and the second form of Schur's test. At $\wp = 4$: $\mathsf{s}_0(4) = 195$, $2\mathsf{s}(4,6) < 537.82$, $\mathsf{s}_0(2) = 27$, $2\mathsf{s}(2,4) < 37.20$. $\mathcal{D}_\infty$ is a core of $N^\wp$.

(c) Put $J\phi(f_1)\cdots\phi(f_n)\Omega = \phi(\bar{f}_1\circ\mathsf{j})\cdots\phi(\bar{f}_n\circ\mathsf{j})\Omega$; it is antiunitary because $\mathcal{W}(F_1\circ\mathsf{j}\otimes\cdots\otimes F_N\circ\mathsf{j}) = \mathcal{W}(F_N\otimes\cdots\otimes F_1)$, which at Euclidean points is the invariance of σ under the rotation R_π by π in the (x_1,x_3)-plane, the image of $\mathsf{j} = B(i\pi)$, which reverses the order of the Euclidean times, continued by [7, Lem. G.5]; uniqueness holds because $\mathcal{D}_0$ is dense. Call $X = \phi(f_1)\cdots\phi(f_n)$, $f_k \in C_c^\infty(\mathbb{R}^{1+2})$, an ordered product in W if $\mathrm{supp} f_1 \times\cdots\times \mathrm{supp} f_n$ lies in $W^{(n)} = \{(x_k)_k :\ x_1 \in W,\ x_{k+1}-x_k \in W\ (1\le k<n)\}$ (an order in real time along the wedge, not the order of the Euclidean heights of the setting), and put $X^{\mathsf{j}} = \phi(f_1\circ\mathsf{j})\cdots\phi(f_n\circ\mathsf{j})$, in the same order, and $X^\dagger = \phi(\bar{f}_n)\cdots\phi(\bar{f}_1)$.

As W is an open convex cone, $x_k \in W$ and $x_l - x_k \in W$ for $k<l$ on the supports: all f_k lie in $\mathcal{F}(W)$, and their supports are pairwise spacelike separated. With $\mathsf{m}(y) = y^1 - |y^0|$, so that $W = \{\mathsf{m}>0\}$ and $\mathsf{m}(B(s)y) \ge e^{-|s|}\mathsf{m}(y)$, one has $d(\iota B(s+i\tau)y) = \sin\tau\,\mathsf{m}(B(s)y) \ge \sin\tau\, e^{-|s|}\mathsf{m}(y)$ for $y\in W$, $0\le\tau\le\pi$; read at $y = x_1$ and at $y = x_{k+1}-x_k$, this puts $(\iota B(\zeta)x_k)_k$ into $\mathsf{Tb}^{(n)}$ for $0 < \mathrm{Im}\,\zeta < \pi$, and $(\iota B(\zeta)x_k + k\eta e_3)_k$ for $0 \le \mathrm{Im}\,\zeta \le \pi$, $\eta > 0$. So the function $\zeta \mapsto \langle\chi, U(B(\zeta))X\Omega\rangle$ continues through the point vectors, by (Br) and $\eta\downarrow 0$ at the edges, to a function continuous on the closed strip $0 \le \mathrm{Im}\,\zeta \le \pi$ and holomorphic inside, with the boundary value $\langle\chi, U(B(s))X^{\mathsf{j}}\Omega\rangle$ at $\mathrm{Im}\,\zeta = \pi$, $B(s+i\pi) = B(s)\mathsf{j}$; the comparison with the spectral integrals of K on its bounded spectral sets gives $X\Omega \in D(\Delta^{1/2})$ and $\Delta^{1/2}X\Omega = X^{\mathsf{j}}\Omega$.

By the definition of J ($\mathsf{j}^2 = 1$), $JX^{\mathsf{j}}\Omega = \phi(\bar{f}_1)\cdots\phi(\bar{f}_n)\Omega$, and this is $X^\dagger\Omega$, the order being reversed by locality on $\mathcal{D}_0$, as the supports are pairwise spacelike separated. A boost-invariant dense subspace $\mathcal{D}$ of $D(\Delta^{1/2})$ is a core: if $\langle\xi,\chi\rangle + \langle\Delta^{1/2}\xi, \Delta^{1/2}\chi\rangle = 0$ for $\chi\in\mathcal{D}$, then the measure $(1+e^{-2\pi k})dm^K_{\xi,\chi}$ has Fourier transform 0, and $\langle\xi,\chi\rangle = 0$ follows. The span of Ω and of the $X\Omega$, X ordered in W, is such a subspace: it is boost invariant, as $B(s)^{\times n}W^{(n)} = W^{(n)}$, and it is dense by the argument of (a): products of fields in the balls of radius $\frac{1}{4}$ about the points $(0,2k,0)$ stay ordered in W under small translations of each factor, the reflection principle carries the orthogonality to all translations, and a partition of unity to all products of fields.

Locality removes the ordering: for $Y = \phi(h_1)\cdots\phi(h_r)$, $h_k \in \mathcal{F}(W)$ in any order, and $\xi = X\Omega$, X ordered in W, the fields of $(X^{\mathsf{j}})^\dagger = \phi(\bar{f}_n\circ\mathsf{j})\cdots\phi(\bar{f}_1\circ\mathsf{j})$ lie in $\mathcal{F}(W')$, so they commute on $\mathcal{D}_0$ with those of Y, and $(X^{\mathsf{j}})^\dagger\Omega = J\xi$; hence $\langle\Delta^{1/2}\xi, Y\Omega\rangle = \langle\Omega, (X^{\mathsf{j}})^\dagger Y\Omega\rangle = \langle Y^\dagger\Omega, (X^{\mathsf{j}})^\dagger\Omega\rangle = \langle Y^\dagger\Omega, J\xi\rangle = \langle\xi, JY^\dagger\Omega\rangle$, also at $\xi = \Omega$; on the core this gives $Y\Omega \in D(\Delta^{1/2})$ and $J\Delta^{1/2}Y\Omega = Y^\dagger\Omega$ for all $h_k \in \mathcal{F}(W)$.

For $Q \in \mathcal{C}(W)$ the weak commutation, used n times, gives $\langle Q^*\Omega, X\Omega\rangle = \langle X^\dagger\Omega, Q\Omega\rangle = \langle J\Delta^{1/2}X\Omega, Q\Omega\rangle$ on the core, so $\Delta^{1/2}JQ\Omega = Q^*\Omega$. Opposite wedges: let $Q\in\mathcal{C}(W)$, $Q'\in\mathcal{C}(W')$ and $Q'_g = \int g(a)U(a)Q'U(a)^{-1}\,\mathrm{d}a$, $g\in C_c^\infty(W)$; as $W'\subset W'+a$ for $a\in W$, $Q'_g\in\mathcal{C}(W')$; it keeps $\mathcal{D}_\infty$ (smoothness in the time translation) and commutes there with the fields of W'. With $S = J\Delta^{1/2}$, J carries the graph of $\Delta^{1/2}$ onto that of $\Delta^{-1/2}$, so $\mathcal{D}_0(W')$ is a core of $S^* = J\Delta^{-1/2}$, and every R on $\mathcal{D}_\infty$ with an adjoint there that commutes with the fields of W' has $SR\Omega = R^\dagger\Omega$; and $S^*Q\Omega = Q^*\Omega$.

For products X, Y of fields of W and $R = Y^\dagger Q'_g X$:

$$\langle Y\Omega, QQ'_g X\Omega\rangle = \langle S^*Q\Omega, R\Omega\rangle = \langle SR\Omega, Q\Omega\rangle = \langle Y\Omega, Q'_g QX\Omega\rangle;$$

so $QQ'_g = Q'_g Q$, and $g \to \delta$ gives $QQ' = Q'Q$. For a wedge $\tilde W \supset O$, $\tilde W' \subset O^\times$, so $\mathcal{C}(O^\times) \subset \mathcal{C}(\tilde W') \subset \mathcal{C}(\tilde W)' = \mathcal{A}(\tilde W)$, whence $\mathcal{C}(O^\times) \subset \mathcal{A}^d(O)$; two spacelike separated double cones are parted by a wedge, which gives the locality; if $\mathcal{C}(O^\times) = \mathcal{A}^d(O)$ for all double cones, then $\mathcal{A}(O) \subset \mathcal{C}(O^\times)$ and (W) holds by (a), and the converse is (e).

(d) The function f_* is that of Lem. 14.8(a). (W) implies density: $\mathcal{A}(W') \subset \mathcal{C}(W)$ and $\mathcal{A}(W')\Omega$ is dense. Density implies wedge duality: Ω is then cyclic and separating for $\mathcal{N} = \mathcal{C}(W)$; by (c), $Q\Omega \mapsto Q^*\Omega$ is the restriction of $\Delta^{1/2} J = J\Delta^{-1/2}$ ($JKJ = -K$), and $J\mathcal{N}\Omega$ is a boost-invariant dense subspace of $D(\Delta^{1/2})$, hence a core; so modular theory gives the conjugation J, the modular operator Δ^{-1}, and $\mathcal{N}' = J\mathcal{N}J = \mathcal{C}(W')$. Wedge duality implies (W): spacelike separated upright double cones lie in some $\mathfrak{g}W'$ and $\mathfrak{g}W$, and $\mathcal{C}(K_1)' \subset \mathcal{C}(\mathfrak{g}W')' = \mathcal{C}(\mathfrak{g}W) \subset \mathcal{C}(K_2)$. $J\mathcal{A}(W)J = \mathcal{C}(W')'$, so $J\mathcal{A}(W)J \subset \mathcal{C}(W)$ is (W) for the pair (W', W), equivalent to (W) by the same chain with W and W' exchanged. (W) implies (CP): $\bar\phi(f)$, $f \in \mathcal{F}(K)$ real, is affiliated with $\mathcal{A}(K) \subset \mathcal{C}(K^\times)$; the extension of Lem. 14.6(b) has its bounded Borel functions in $\mathcal{A}(K) \otimes M_2$, which commute weakly with the fields of $K^\times$; Lem. 14.6(a) at $\mathfrak{A}$ generated by 1, f and $\mathcal{F}(K^\times)$, $X_0 = f$. (CP) implies (QCP_f), as $W \subset K^\times$ for $\bar K \subset W'$. (QCP_f) gives $\phi(f)\Omega \in \mathcal{H}_0$.

Let $\mathsf{L}(Y_0 + \lambda Y_1 + \lambda^2 Y_2) = \langle\Omega, Y_0\Omega\rangle + \langle\Omega, Y_1\eta\rangle + \langle\eta, Y_2\eta\rangle$ on the quadratic polynomials in λ with coefficients in $\mathfrak{P}(W)$; f commutes with $\mathfrak{P}(W)$ in $\mathfrak{P}$, as $\bar K$ and W are spacelike separated. On a square $(Z_0 + \lambda Z_1)^*(Z_0 + \lambda Z_1)$, L is $\|Z_0\Omega + Z_1\eta\|^2 \geq 0$ for all data; (QCP_f) is the positivity of L on the larger cone of the quadratic polynomials whose values at all real λ lie in $\mathfrak{P}(W)^+$, and the extension below is positive on this cone. Let $\mathscr{C}$ be the space of the functions $G = \sum_k g_k(\lambda) Y_k$, $Y_k \in \mathfrak{P}(W)$, g_k complex Borel with $|g_k| \leq C(1 + \lambda^2)$, and $\mathscr{C}^+$ the cone of those whose values at all real λ lie in $\mathfrak{P}(W)^+$.

Every hermitian element of $\mathscr{C}$ is a sum of terms gS, g real, $S = S^*$, and $\frac12 q(S^2+1) - gS = \frac14(q-g)(S+1)^2 + \frac14(q+g)(S-1)^2 \in \mathscr{C}^+$ for $q = C(1+\lambda^2)$, $|g| \leq q$, with $\frac12 q(S^2+1)$ a quadratic polynomial: the step of the proof of Lem. 14.6(a) at $r = 1$. The extension theorem of M. Riesz extends L to a linear $\hat\omega$ on $\mathscr{C}$ with $\hat\omega \geq 0$ on $\mathscr{C}^+$. For G, G' with $|g_k| \leq C(1 + |\lambda|)$, $G^*G \in \mathscr{C}^+$, so $\langle [G], [G']\rangle = \hat\omega(G^*G')$ gives a Hilbert space; $V\pi(Y)\Omega = [Y]$ on the dense $\mathcal{D}_0(W)$ is an isometry V of $\mathcal{H}$ into it; $\hat\pi(g)[G] = [gG]$, g bounded Borel, is bounded by $\sup|g|$, as $(\sup|g|^2 - |g|^2)G^*G \in \mathscr{C}^+$; and $Q_g = V^*\hat\pi(g)V$ commutes weakly with the fields of W on $\mathcal{D}_0(W)$, so on $\mathcal{D}_0$ by (a) and (b5): $Q_g \in \mathcal{C}(W)$, $\|Q_g\| \leq \sup|g|$.

With $g_n(\lambda) = \lambda \mathbf{1}_{|\lambda|\leq n}$, $Q_{g_n}\Omega \to \phi(f)\Omega$, because $\|[\mathbf{1}_{|\lambda|>n} Y]\|^2 \leq n^{-2}\mathcal{W}(f^2 Y^*Y)$ and $V^*[\lambda] = \phi(f)\Omega$. The first order holds for all data: if $Y_0 + \lambda Y_1 \in \mathfrak{P}(W)^+$ for all real λ, then $\langle\psi, \pi(Y_1)\psi\rangle = 0$ for $\psi \in \mathcal{D}_0$, so $\pi(Y_1) = 0$ there by polarisation, and $\mathcal{W}(Y_0 + fY_1) = \langle\Omega, Y_0\Omega\rangle \geq 0$. One vector decides: $\mathcal{H}_0$ is invariant under $\mathcal{C}(W)$, under the boosts, and under the translations $U(-a)$, $a \in \bar W$ (as $\mathcal{C}(W) \subset \mathcal{C}(W + a)$), hence, by the spectrum condition and the reflection principle along the light rays $(\pm1, 1, 0)$ and along e_2, under all translations; so its projection P_0 commutes with $\mathcal{P}$, and $\mathsf{D}(f) = \|(1 - P_0)\phi(f)\Omega\|$ is a seminorm invariant under translations and boosts, with $\mathsf{D}(h * f) \leq \|h\|_{L^1}\mathsf{D}(f)$.

For $\chi \perp \mathcal{H}_0$ the tempered distribution $\mathsf{T}(h) = \langle\chi, \phi(h)\Omega\rangle$ vanishes on all $f * k$, $f \in \mathfrak{F}$, $k \in \mathcal{S}$; if the $\hat f$, $f \in \mathfrak{F}$, have no common real zero, a partition of unity shows that T vanishes on all h with $\hat h \in C_c^\infty$, so $\phi(h)\Omega \in \mathcal{H}_0$ for all $h \in \mathcal{S}$. Then for $\chi \in \mathcal{D}_\infty \cap \mathcal{H}_0^\perp$ and $h \in \mathcal{F}(W)$, $\langle Q\Omega, \phi(\bar h)\chi\rangle = \langle\bar\phi(h)Q\Omega, \chi\rangle = \langle Q\phi(h)\Omega, \chi\rangle = 0$ for $Q \in \mathcal{C}(W)$; for $\psi \in \mathcal{H}_0 \cap \mathcal{D}_\infty$ the vector $\chi = (1 - P_0)\phi(h)\psi$ has $\|\chi\|^2 = \langle\phi(\bar h)\chi, \psi\rangle = 0$; by induction $\mathcal{D}_0(W) \subset \mathcal{H}_0$, and $\mathcal{H}_0$ is dense.

The infimum over the boosts is the distance itself, by the boost invariance of $\mathcal{H}_0$; for one $f_0 \geq 0$ the transforms of its Lorentz images have no common real zero by Lem. 14.8(d), and translations do not change zero sets. So all the conditions are equivalent, and the modular data under (W)

are those of the density step, applied to $\mathcal{A}(W) = \mathcal{C}(W')$. (CP) gives (W) also without the wedge relation: for spacelike separated upright double cones K_1, K_2, $Q \in \mathcal{C}(K_1)'$ and real $g \in \mathcal{F}(K_2)$, the extension of Lem. 14.6(a) for (K_2, g) gives bounded $Q_h = V^* h(\hat{T}) V \in \mathcal{C}(K_2^\times) \subset \mathcal{C}(K_1)$, which commute with Q, and $Q_{g_n}\psi \to \phi(g)\psi$ for ψ in the domain of the GNS representation, which holds $\mathcal{D}_0(K_2^\times)$; so Q commutes weakly with $\phi(g)$ there, and the polynomial bound carries this to $\mathcal{D}_0$.

The inequalities: under (W) the closures of $\phi(g_1), \phi(g_2)$ are affiliated with commuting algebras, and Lem. 14.6(c) gives strongly commuting self-adjoint extensions $\hat{A}_1, \hat{A}_2$; with $\psi = \phi(f)\Omega$ and their joint spectral measure,

$$\begin{aligned} |\mathcal{W}(f g_1 g_2)| &= \Big| \int \lambda_1 \lambda_2 \, \mathrm{d}\mu_{\psi,\Omega} \Big| \\ &\le \Big(\int |\lambda_1 \lambda_2| d\mu_\psi \Big)^{1/2} \Big(\int |\lambda_1 \lambda_2| d\mu_\Omega \Big)^{1/2} \\ &\le \big(\|\hat{A}_1 \psi\| \|\hat{A}_2 \psi\| \big)^{1/2} \|\hat{A}_1 \hat{A}_2 \Omega\|^{1/2}, \end{aligned}$$

and locality on $\mathcal{D}_0$ identifies the squares of the three norms with $\mathcal{W}(f^2 g_1^2)$, $\mathcal{W}(f^2 g_2^2)$, $\mathcal{W}(g_1^2 g_2^2)$; likewise $\mathcal{W}(p(f_1, f_2)) = \int p \, \mathrm{d}\mu_\Omega \ge 0$, and Motzkin's polynomial is covered by Lem. 14.6(d).

(e) Under (W), $\mathcal{A}(\tilde{W}) = \mathcal{C}(\tilde{W}')$ for every wedge by (d) and covariance. Two spacelike separated open convex sets are parted by a wedge (Hahn–Banach), so $O^\times = \bigcup\{\tilde{W}' : \tilde{W} \supset O\}$, and additivity gives $\mathcal{A}^d(O) = \bigcap \mathcal{C}(\tilde{W}') = \mathcal{C}(O^\times)$. Locality: a wedge parts O_1 from O_2. Haag duality: $\mathcal{A}^d(O)' = \mathcal{A}(O^\times) = \bigvee \mathcal{A}(O_3) \subset \bigvee \mathcal{A}^d(O_3) = \mathcal{A}^d(O^\times) \subset \mathcal{A}^d(O)'$, the joins over the double cones $O_3 \subset O^\times$; and as $O^\times = \bigcup\{\tilde{W}' : \tilde{W} \supset O\}$, (a) gives $\mathcal{A}(O^\times) = \bigvee\{\mathcal{A}(\tilde{W}') : \tilde{W} \supset O\}$.

Outer regularity: a point with a neighbourhood spacelike to O is spacelike to the compact set $\bar{O}$, hence to $\bar{O}_\epsilon$ for small ϵ, the compact sets $\bar{O}_\epsilon$ decreasing to $\bar{O}$; so $O^\times = \bigcup_{\epsilon>0} O_\epsilon^\times$, and (a) gives $\mathcal{A}^d(O) = \mathcal{C}(O^\times) = \bigcap_\epsilon \mathcal{C}(O_\epsilon^\times) = \bigcap_\epsilon \mathcal{A}^d(O_\epsilon)$. An operator commuting with all $\mathcal{A}(O_1)$, $O_1 \subset O^\times$, commutes with $\mathcal{A}(O^\times)$, so it lies in $\mathcal{A}^d(O)$. For $O \subset \tilde{W}$: $\mathcal{A}(O) \subset \mathcal{A}^d(O) \subset \mathcal{A}(\tilde{W})$, so the wedge algebras of $\mathcal{A}^d$ are those of $\mathcal{A}$, and $(\mathcal{A}^d)^d = \mathcal{A}^d$. $\mathcal{A}(O) = \mathcal{A}^d(O)$ if and only if $\mathcal{C}(O^\times) \subset \mathcal{A}(O) = \mathcal{C}(O)'$.

Covariance, J, cyclicity and affiliation follow from (a) and (c). Without (W): if the intersections are local, then $\mathcal{A}(O_1) \subset \mathcal{A}^d(O_1) \subset \mathcal{A}^d(O_2)' \subset \mathcal{A}(O_2)' = \mathcal{C}(O_2)$ for spacelike separated double cones.

(f) Let $A_j = \phi(\partial_t^j f)$ and $R_\epsilon = (1 + \epsilon N)^{-1}$. From $[H, A_0] = iA_1$: $R_\epsilon^r A_0 \chi = A_0 R_\epsilon^r \chi - i\epsilon \sum_{j=1}^r R_\epsilon^j A_1 R_\epsilon^{r-j+1} \chi$. For $A_0^* \eta = \pm i\eta$ and $\eta_\epsilon = R_\epsilon^{k_0}\eta$, the imaginary part of $\langle \eta_\epsilon, \bar{A}_0 \eta_\epsilon \rangle$ and the form bound of A_1 give $\|\eta_\epsilon\|^2 \le c \sum_j \epsilon^{1/2} \|N^{1/2} R_\epsilon^{k_0 - j + 1} \eta_\epsilon\| \, \epsilon^{1/2} \|N^{1/2} R_\epsilon^j \eta\| \to 0$, while $\eta_\epsilon \to \eta$; so $\bar{A}_0$ is self-adjoint. $N^k \phi(f) = \sum_j \binom{k}{j} i^j \phi(\partial_t^j f) N^{k-j}$ gives $\bar{A}_0 N^{-k} = N^{-k}(\bar{A}_0 + K_k)$ with K_k bounded under the block bounds, whence the resolvents keep $D(H^k)$ off a strip and, through cores, commute with $\bar{\phi}(h)$ for h spacelike to f; Neumann series reach all $z \notin \mathbb{R}$; so the resolvents lie in $\mathcal{C}(O)$. (f1) With the logarithm the self-adjointness and the commutation come from unitary groups along which the exponent of the energy falls.

Put $\mathsf{w}(x) = x(1 + \log x)$, and for real $h \in C_c^\infty$ let $v = f_i * h$. Off the band (b3) bounds the blocks of $\partial_t f_i$ linearly, so $b_{mn}(\partial_t f_i) \le c\, \mathsf{w}(m + n + 2)$ for all m, n; by (b4), $\partial_t^j v = (\partial_t f_i) * \partial_t^{j-1} h$ has the same bound with $c\|\partial_t^{j-1} h\|_{L^1}$, $j \ge 1$, and (b2) adds the decay $(|m - n| - 1)^{-3}$ for $|m - n| \ge 2$; as $\mathsf{w}(m + n + 2) \le 3.4(1 + |m - n|)\, \mathsf{w}(m + 1)^{1/2} \mathsf{w}(n + 1)^{1/2}$, Schur's test with the weights $\mathsf{w}(n + 1)$ gives $|\langle \psi', \phi(\partial_t^j v)\psi \rangle| \le c_j \|\mathsf{w}(N)^{1/2} \psi'\| \|\mathsf{w}(N)^{1/2} \psi\|$ on $\mathcal{D}_\infty$, $j \ge 1$. By $\log x = \int_0^\infty [(1 + t)^{-1} - (x + t)^{-1}] dt$, $[N + t, \phi(\partial_t^j v)] = i\phi(\partial_t^{j+1} v)$ and $\int_0^\infty \mathsf{w}(x)(x + t)^{-2} dt = 1 + \log x$, this turns into $|\langle \log N\, \psi, \phi(\partial_t^j v)\psi \rangle - \langle \phi(\partial_t^j v)\psi, \log N\, \psi \rangle| \le c_{j+1} \langle \psi, (1 + \log N)\psi \rangle$, $j \ge 0$, the left member being $|\mathfrak{l}^{(j)}(\psi, \psi)|$ for the hermitian form $\mathfrak{l}^{(j)}(\psi', \psi) = \int_0^\infty \langle (N + t)^{-1} \psi', \phi(\partial_t^{j+1} v)(N + t)^{-1} \psi \rangle dt$.

The three-lines theorem, on vectors of bounded energy, carries the bound to $\mathfrak{l}^{(j)}(N^\tau \psi', N^{-\tau}\psi)$ with the constant $\sum_{l \le \lceil |\tau| \rceil} \binom{\lceil |\tau| \rceil}{l} c_{j+1+l}$; as $\frac{d}{d\tau} \langle N^\tau \chi', A N^{-\tau} \chi \rangle = i\, \mathfrak{l}^{(0)}(N^\tau \chi', N^{-\tau} \chi)$ for $A = \phi(v)$,

and a second derivative brings in $\mathfrak{l}^{(1)}$, integration over $|\tau| \le \frac{\rho}{2}$ at $\chi' = \chi$ gives

$$|\langle N^\rho\chi, A\chi\rangle - \langle\chi, AN^\rho\chi\rangle| \le \rho\, c_1\langle\chi, N^\rho(1+\log N)\chi\rangle + \rho^2 c'(\rho)\langle\chi, N^\rho\chi\rangle \qquad (\chi\in\mathcal{D}_\infty,\ \rho\ge 0),$$

$c'(\rho) = \sum_{l\le\lceil\rho/2\rceil}\binom{\lceil\rho/2\rceil}{l} c_{l+2}$. The bounded $A_s = e^{-sH}Ae^{-sH}$ satisfy the same with the same numbers. For $\psi\in\mathcal{D}_\infty$ and $y(\lambda) = \|N^{\rho(\lambda)/2}e^{i\lambda A_s}\psi\|^2$ with the falling exponent $\rho(\lambda) = \rho_0 e^{-c_1|\lambda|}$, the term $\rho'\langle N^\rho\log N\rangle$ of y' cancels the term with the logarithm: $y' \le (\rho_0 c_1 + \rho_0^2 c'(\rho_0))\, y$ for $\lambda\ge 0$, and Gronwall's lemma bounds $\|N^{\rho(\lambda)/2}e^{i\lambda A_s}\psi\|$ uniformly in s. As $s\downarrow 0$, $\|(A_s - A_{s'})\chi\| \le c|s-s'|\|N^{q+1}\chi\|$ ($q\ge\wp$ an integer) makes the groups $e^{i\lambda A_s}$ converge strongly to a unitary group V with $V(\lambda)D(N^r)\subset D(N^{re^{-c_1|\lambda|}})$ for $r\ge 0$, so $V(\lambda)\mathcal{D}_\infty = \mathcal{D}_\infty$, and $\lambda\mapsto V(\lambda)\psi$ is continuously differentiable in every norm $\|N^a\cdot\|$ with the derivative $iAV(\lambda)\psi$; its generator is the closure of A on $\mathcal{D}_\infty$ (for η with $(A|_{\mathcal{D}_\infty})^*\eta = \pm i\eta$, $\lambda\mapsto\langle V(\lambda)\psi,\eta\rangle$ is bounded with the derivative $\pm$ itself, hence 0): $\phi(v)$ is essentially self-adjoint on $\mathcal{D}_\infty$ and on $\mathcal{D}_0$.

The group loses moments of the energy and keeps $\mathcal{D}_\infty$. For $h'\in\mathcal{S}$ whose support is spacelike separated from $\operatorname{supp} v$, $\phi(v)\phi(h') = \phi(h')\phi(v)$ on $\mathcal{D}_\infty$ (locality on $\mathcal{D}_0$, the core $\mathcal{D}_0$ of N^n, (b5)), and $\frac{d}{d\lambda}\langle\phi(\bar h')V(\lambda)\psi', V(\lambda)\psi\rangle = 0$ gives $V(\lambda)\bar\phi(h') \subset \bar\phi(h')V(\lambda)$: by (a) the bounded Borel functions of $\bar\phi(v)$ lie in $\mathcal{C}(O)$ for every open O spacelike separated from $\operatorname{supp} v$. For real $h\in\mathcal{F}(O)$ the $f_i * h$ lie in $\mathcal{F}(O)$ for large i and tend to h in $\mathcal{S}$; so $\mathcal{C}(O)$ is the commutant of the groups of the $\bar\phi(f_i*h)$ with $\operatorname{supp}(f_i*h)\subset O$ (by (a) every $Q\in\mathcal{C}(O)$ commutes with them; conversely weak commutation with the $\phi(f_i*h)$ passes to $\phi(h)$ by (b5)), and for spacelike separated O_1, O_2 the groups of O_1 lie in $\mathcal{C}(O_2)$: $\mathcal{C}(O_1)'\subset\mathcal{C}(O_2)$, which is (W) without a wedge, and (CP) by (d). (f2) The time reversal $[F]\mapsto[\bar F]$ gives the other sign for f_0 even in t; $\partial_t^j(h*f_0) = (\partial_t^{j-1}h)*\partial_t f_0$ gives the hypotheses of the first part at every $h*f_0$; by Lem. 14.8(d) the images of f_0 under rotations, boosts and translations into W' have no common real zero of their transforms, and the resolvents of the corresponding operators lie in $\mathcal{C}(W)$ and produce from Ω a total set; (d) applies. (f3) For products X of fields spacelike to f_0, $\|\phi(f_0)^n X\Omega\|^2 = \langle\phi(f_0)^{2n}\Omega, X^\dagger X\Omega\rangle \le \|\phi(f_0)^{2n}\Omega\|\|X^\dagger X\Omega\|$, so these total vectors are quasi-analytic, $\bar\phi(f_0)$ is self-adjoint with resolvents commuting with the outside fields, and the argument of (f2) applies. (f4) At $s = (m+n+2)^{-1}$ the argument of (b1) gives $b_{mn}(f)\le e\,C\|f\|_{L^1}(m+n+2)(1+\log(m+n+2))$ for every $f\in\mathcal{S}$ (for $\wp\le 1$ (b1) gives it without the logarithm), so for every $\partial_t^j f$: the hypothesis of (f1) holds for every approximate unit f_i, and the argument of (f1) runs with $v = f$ itself; so every real $\phi(f)$ is essentially self-adjoint on $\mathcal{D}_0$, and $e^{i\lambda\bar\phi(f)}\bar\phi(h) \subset \bar\phi(h)e^{i\lambda\bar\phi(f)}$ for real h whose support is spacelike separated from that of f; $\bar\phi(h)$ being self-adjoint, the group of $\bar\phi(f)$ commutes with its spectral projections: the closures commute strongly. (f5) Lem. 14.10(b)–(d) gives (LB_f) for every real $f\in C_c^\infty$, so (f2) at every f_0.

The last assertion of (f): if (f2) holds at f_*, time reversal gives the two-sided bound $c_*(\partial_t f_*) < \infty$, and with Lem. 14.8(b), (c) the functions $f_i = g^{\otimes 3}_{1/i,3/2} = f_* * \mu_i^{\otimes 3}$ have $c_*(\partial_t f_i)\le\|\mu_i\|_{L^1}^3 c_*(\partial_t f_*)$, hence $b_{mn}(\partial_t f_i)\le\|\mu_i\|_{L^1}^3 c_*(\partial_t f_*)(m+n+2)$ by polarisation; these functions have the properties that (f1) asks of the f_i.

(g) By Lem. 14.7(ii), $\phi_u([F], M_{g^2}[F]) = \int g^2\phi_u^\xi\,\mathrm{d}\lambda$ and $\langle[F], M_{g^2}N[F]\rangle = \int g^2\langle[F]_\xi, N_\xi[F]_\xi\rangle\,\mathrm{d}\lambda$; at $g=1$ the bound passes from all σ_ξ to σ; conversely, the bound for σ at the vectors $M_g[F]$ gives $\int g^2(c\,\mathsf{n}-\mathsf{a})\,\mathrm{d}\lambda\ge 0$ for all real g, with $\mathsf{a}(\xi) = \phi_u^\xi([F]_\xi,[F]_\xi)$ and $\mathsf{n}(\xi) = \langle[F]_\xi, N_\xi[F]_\xi\rangle$ continuous, and full support gives $\mathsf{a}\le c\,\mathsf{n}$ at every ξ; the classes $[F]_\xi$ form a core of N_ξ. The premise of (CP) speaks of $\mathfrak{P}$ alone, and $\mathcal{W}_\sigma = \int\mathcal{W}_\xi\,\mathrm{d}\lambda$ (Lem. 14.7(iii)): (CP) passes from all σ_ξ to σ, and so does every fixed inequality $\mathcal{W}(a)\ge 0$.

Conversely, for $a = \sum f^n Y_n$ as in (CP), which is hermitian, and b formed as in Lem. 14.7(iv), a polynomial in means of translates of elements b_i of $\mathfrak{P}$ over pairwise different spatial translations of length $\ge R$, with R so large that the translated b_i are spacelike to K, which puts b in the algebra of (CP), $\sum\lambda^n b^* Y_n b\in\mathfrak{A}^+$ and f is central, so $\mathcal{W}_\sigma(b^*ab)\ge 0$, and Lem. 14.7(iv) gives (CP) for every σ_ξ; (d) turns (CP) into (W). The limits: Lem. 14.7(v); inequalities $\mathcal{W}(a)\ge 0$ pass to the limit.

(h) (h1) Fix $\Lambda < \infty$; on $\mathcal{H}_\Lambda$ let q be the bounded operator of the form

$$\int h'(\vec{x}) q_0(U(-\vec{x})\cdot, U(-\vec{x})\cdot) d^2x;$$

then

$$\langle \psi', \mathsf{l}\psi\rangle = \int \mathsf{l}^{-1}\rho_1(\varsigma/\mathsf{l})\langle e^{-(\frac{\mathsf{l}}{2}+\varsigma)H}\psi', q\, e^{-(\frac{\mathsf{l}}{2}-\varsigma)H}\psi\rangle \,\mathrm{d}\varsigma$$

and

$$\langle \psi', \phi(g_\mathsf{l}\otimes h')\psi\rangle = \int g_\mathsf{l}(t)\langle e^{-itH}\psi', q\, e^{-itH}\psi\rangle \,\mathrm{d}t.$$

An operator integral $\int \alpha_\tau(H) q \beta_\tau(H)\,\mathrm{d}\mu(\tau)$ depends on (α,β,μ) only through its symbol

$$\int \alpha_\tau(E)\beta_\tau(E')\,\mathrm{d}\mu(\tau)$$

(approximate the functions by step functions on a fine partition of $[0,\Lambda]$). With $A(x) = \hat{D}(x) + \epsilon$ and $\log A = \int_0^\infty [(1+t)^{-1} - (t+A)^{-1}]dt$, $L_\epsilon(x) = -\log(\hat{D}(x)+\epsilon)$ is C^2 in norm and $L'_\epsilon(0) = -\int_0^\infty R_t^\epsilon \hat{D}'(0) R_t^\epsilon dt$, $R_t^\epsilon = (t+\epsilon+e^{-\mathsf{l}H})^{-1}$. As $\epsilon \downarrow 0$ on $\mathcal{H}_\Lambda$, $-L'_\epsilon(0)$ tends to the operator integral with the symbol

$$e^{-\frac{1}{2}(E+E')}\int \mathsf{l}^{-1}\rho_1(\varsigma/\mathsf{l})e^{-\varsigma(E-E')}\,\mathrm{d}\varsigma \int_0^\infty \frac{\mathrm{d}t}{(t+e^{-\mathsf{l}E})(t+e^{-\mathsf{l}E'})} = \mathsf{G}(\mathsf{l}(E-E')),$$

minus $\mathfrak{m}$; and $\phi(g_\mathsf{l}\otimes h')$ has the same symbol: this is the slope. Taylor's formula for $u_\epsilon(x) = \langle\psi, L_\epsilon(x)\psi\rangle$ at $x = \pm r$, with $L_\epsilon \geq -\log(Q+1)$ and $L_\epsilon(0) \leq \mathsf{l}H$, gives

$$r|u'_\epsilon(0)| \leq \mathsf{l}\langle H\rangle_\psi + \log(Q+1)\|\psi\|^2 + \frac{r^2}{2}\max\{\sup_{|x|\leq r} u''_\epsilon, 0\};$$

let $\epsilon \downarrow 0$ and use (v); extend from $\bigcup_\Lambda \mathcal{H}_\Lambda$ to $\mathcal{D}_\infty$ by continuity in a norm $\|N^a\cdot\|$. For every real $k \in \mathcal{S}(\mathbb{R})$, a dyadic partition χ_i of the energy transfer and the functions μ_i with $\hat{\mu}_i = \hat{k}\chi_i/\mathsf{G}(\mathsf{l}_i\cdot)$, $\|\mu_i\|_{L^1} \leq C_N \mathsf{l}_i^N$, write $k = \sum_i \mu_i * g_{\mathsf{l}_i}$ in L^1, and Lem. 14.8(c) sums the bounds. (h2) By Lorentz covariance on $\mathcal{D}_\infty$ the bound of (b1) holds in every frame for vectors with $n\cdot p \leq \mu$: boost the two spectral sets to the rest frame of n, where the argument of (b1) gives $e^2 C_\wp \|u\|_{L^1}\mu^\wp$.

The set $\{p\cdot p \leq \mu^2\}$ is cut into cells: $\{p\cdot p \leq \mu^2,\ E < 2\mu\}$ and, for $a \geq 1$, the cells $\{p\cdot p \leq \mu^2,\ 2^a\mu \leq E < 2^{a+1}\mu\}$ on which the direction of $\vec{p}$ lies in one of $\lceil 2\pi 2^a\rceil$ equal sectors, of angle about 2^{-a}. The frame of such a cell has the rapidity $a\log 2$ in the direction of the middle of its sector, and $n\cdot p < 3\mu$ on the cell. For two cells at the distance D in momentum space let n be the midpoint of the hyperbolic segment between their frames, whose length is d: then $n\cdot p \leq 3\mu e^{\mathsf{d}/2}$ on both cells, and $e^{\mathsf{d}} \leq 4(1+2D/\mu)^2$, so $n\cdot p \leq 6\mu + 12D$ there. A test function whose Fourier transform vanishes for $|k| \geq \frac{3}{4}D$ has a vanishing block between the two cells, the spectral projections being smeared translations; so, cutting $\hat{u} = \hat{\kappa}(\cdot/D)\hat{u} + (1-\hat{\kappa}(\cdot/D))\hat{u}$ with a fixed smooth $\hat{\kappa}$ equal to 1 on $\{|k| \leq \frac{1}{2}\}$ and to 0 on $\{|k| \geq \frac{3}{4}\}$, the block is that of a function of L^1 norm $\leq c\, D^{-8}\|\mathcal{L}^4 u\|_{L^1}$.

For $r\mu \leq D < (r+1)\mu$, $r \geq 1$, the block is therefore $\leq e^2 C_\wp\, c\,(r\mu)^{-8}\|\mathcal{L}^4 u\|_{L^1}(6\mu(2r+3))^\wp$, and for $D < \mu$ it is $\leq e^2 C_\wp \|u\|_{L^1}(18\mu)^\wp$. At most $100(r+1)$ cells lie at a distance $< (r+1)\mu$ from a given cell; Schur's test over the cells, with the weight $r+1$ of this count, gives the bound with the constant C_L, the sum over r converging for $\wp < 6$, with no dependence on the bound of the energy. For these two consequences let $\wp > 0$. On $P_m\mathcal{H} \cap \mathcal{P}_<\mathcal{H}$ the invariant mass is $\leq (m+n+2)^{1/\wp}$, which gives the bound of the blocks. For the last statement, $\Pi P_m = \mathcal{P}_<\Pi P_m$ and $P_n\Pi = P_n\Pi\mathcal{P}_<$ with the $\mathcal{P}_<$ of the pair (m,n), since $p\cdot p \leq (1+p^0)^{2/\wp} \leq (m+n+2)^{2/\wp}$ on both ranges; Π commutes with H, so the argument of (b2) gives $(|m-n|-1)^j\|\Pi P_m\phi(u)P_n\Pi\| \leq \|\Pi P_m\phi(\partial_t^j u)P_n\Pi\|$; the blocks of $\Pi e^{-sH}\phi(u)e^{-sH}\Pi$ are then at most $c\,e^{-s(m+n)}(m+n+2)\min\{1,(|m-n|-1)^{-2}\}$, and Schur's test with $\sup_{x\geq 0}(x+2)e^{-sx} \leq e/s$ gives the bound.

The bound by $1+H+M^2$: cut momentum space into unit cubes; for two cubes at the sup-distance d, $D=\sqrt{3}(d+1)$, and a point p_0 of energy E_0 and invariant mass m_0 in one of them, a timelike vector a with $|a|=(\sqrt{2}\mathsf{M}_1)^{-1}$, $\mathsf{M}_1^2\le 2D^2(1+E_0+m_0^2)$, has $a\cdot p\le 1$ on both cubes, and the point bound in the frame of a bounds the block of $\phi(f)$ between the two cubes by $e^2C_\wp\|f\|_{L^1}(2D)^\wp$ times the least value of $(1+p^0+p\cdot p)^{\wp/2}$ on either cube; derivatives in a coordinate along which the cubes are d apart give the decay $(d-1)^{-k}$ as in (b2), and Schur's test with the weights $(1+p^0+p\cdot p)^{\wp/2}$ of the cubes gives the form bound.

14.1.7. *Proof of Thm. AP. Step 1.* By Thm. YM3_I:AP among the construction results on the limits of periodic Gibbs states, σ is data (Rec) with $(16C_{\mathrm{pt}}^g,4,\frac12)$ and $S_1^\sigma=0$, and its truncated functions S_n^σ, $n\ge 2$, are the limits of the joint cumulants $S_{n;j}^T$ of the $\mathcal{O}_{K_j}(f)$ under $\mu_j=\mu_{\beta_j}$ on Λ_j, with $\ell_j/a_j=2L^{K_j+m_j}$ even, $a_j=L^{-K_j}$ the spacing of Λ_j; every component is data (Rec) with one triple.

Step 2 (frames). Let $\mathcal{H}_j,\iota_{\Lambda_j},\mathcal{M}_j,\pi_j,K_{\Lambda_j},\mathsf{P}_j$ be the objects of Prop. 14.13 on Λ_j at β_j, and, for tuples $G=(h_k)$ over $\mathbb{E}_+$, $(G)_j=\prod_k(\mathcal{O}_{K_j}(h_k)-\mu_j(\mathcal{O}_{K_j}(h_k)))$, the functions h_k periodised on Λ_j, and $\iota_jG=\iota_{\Lambda_j}((G)_j)$, both extended linearly. In Steps 2–7 the sign $(\cdot)_j$ is this product on the torus Λ_j, centred by μ_j; it is not the product $(X)_{K_j}^{\nu_j}$ on $L^{-K_j}\mathbb{Z}^3$ of the setting, and for a tuple G over a quadrant it is, for large j, a real function of the bonds of the closed quadrant of Λ_j, with $\mathsf{r}(G)_j=(\mathsf{r}G)_j$ for $\mathsf{r}\in\{\theta,\theta_1,R\}$ ($\mathsf{r}\mathcal{O}(f)=\mathcal{O}(\mathsf{r}f)$, μ_j being invariant).

For large j the supports lie in $(-\frac{\ell_j}{4},\frac{\ell_j}{4})^3$ and farther than $2a_j$ from the time-zero plane. For $X=\theta G'\cdot G=(f_k)$, (L1) and the invariance of the weight give $\langle\iota_jG',\iota_jG\rangle=\mu_j\prod_k(\mathcal{O}(f_k)-\mu_j\mathcal{O}(f_k))=\sum_\varpi\prod_{\mathsf{b}\in\varpi}S_{|\mathsf{b}|;j}^T(X_\mathsf{b})$, the sum over the partitions without singletons (a moment of centred variables, and a cumulant of order ≥ 2 does not see the centring); by Step 1 it tends to $\sum_\varpi\prod S_{|\mathsf{b}|}^\sigma(X_\mathsf{b})=\sigma(X)=\langle[G'],[G]\rangle$. So $(\mathcal{H}_j,\iota_j)$ is a frame for $(\mathcal{H}(\sigma),[\cdot])$, and Lem. 14.1 applies.

Step 3 (limits of half-plane operators lie in $\mathcal{C}(W')$). Let $Q_j\in\mathcal{M}_j$, $Q_j\rightharpoonup Q$ along a subsequence; take the $\tilde q_j$ of (L3), $\|\tilde q_j\|\le\sup\|Q_j\|$, and a further subsequence with $\tilde q_j\rightharpoonup\tilde q$. For tuples G_1,G_2 over II and large j, the $(G_i)_j$ are real functions of the bonds of $\bar{\mathsf{Q}}_2$, and $R^{-1}(\theta(G_1)_j\cdot(G_2)_j)=(R^{-1}(\theta G_1\cdot G_2))_j$; the limit of (L3) is the hypothesis of Lem. 14.4, so $Q\in\mathcal{C}(W')$.

Step 4 (the lattice modular flows converge to the boosts). Every subsequence of the support projections $e_{R,j}\in\mathcal{M}_j$ has a further one with $e_{R,j}\rightharpoonup\mathsf{E}$; by Step 3, $\mathsf{E}\in\mathcal{C}(W')$, and by Lem. 14.1(iii), $\mathsf{E}\Omega$ is the weak limit of $e_{R,j}\Omega_{\Lambda_j}=\Omega_{\Lambda_j}=\iota_j\varnothing$: $\mathsf{E}\Omega=\Omega$. Ω separates $\mathcal{C}(W')$, so $\mathsf{E}=1$ and $\|(1-e_{R,j})\iota_jG\|^2=\langle\iota_jG,(1-e_{R,j})\iota_jG\rangle\to 0$ along the full sequence; by (L1), $\|(1-e_{L,j})\iota_jG\|=\|(1-e_{R,j})\iota_j(\theta_1\bar G)\|\to 0$. So Lem. 14.1(iv) applies to π_j, and by (L4) (for $G\in\mathcal{V}_0$ and large j, $(G)_j$ is a function of $\bar{\mathsf{Q}}_1$ with $R(G)_j=(RG)_j$) and Lem. 14.2(b), Lem. 14.1(v) applies to P_j, $\Delta^{1/4}$, $\mathsf{r}=R$:

$$\Phi(\mathsf{P}_j)\psi_j\Rightarrow\Phi(\Delta^{1/4})\psi\quad(\Phi\in C_0(\mathbb{R})),\qquad e^{-2\pi itK_{\Lambda_j}}\psi_j\Rightarrow U(B(-2\pi t))\psi,$$

the second from the even functions $\Phi(x)=|x|^{4it}\chi_{\mathsf{M}}(x)$, χ_{M} a cut-off near 0 and ∞, since $|\mathsf{P}_j|=e^{-\frac{\pi}{2}K_{\Lambda_j}}$ and $\Delta^{1/4}$ is injective.

Step 5 (the field on the vacuum is reached). For real $u\in C_c^\infty(\mathsf{I})$ let $\Phi(x)=\mathsf{F}(-\frac{2}{\pi}\log|x|)$, $\Phi(0)=0$: an even function of $C_0(\mathbb{R})$ with $\Phi(\mathsf{P}_j)=\mathsf{F}(K_{\Lambda_j})$ and $\Phi(\Delta^{1/4})=\mathsf{F}(K)$. Step 4 and eq. 14.ii give $\xi_j=\mathsf{F}(K_{\Lambda_j})\pi_j\iota_j(u)\Rightarrow\phi(f_u)\Omega$. (L5) with $F=(u)_j$ gives $T_j^{(\epsilon)}\in\mathcal{M}_j$; along a subsequence $T_j^{(\epsilon)}\rightharpoonup T_u^{(\epsilon)}$, which lies in $\mathcal{C}(W')$ by Step 3, and by Lem. 14.1(i), (iii) and Step 2

$$\|T_u^{(\epsilon)}\Omega-\phi(f_u)\Omega\|\le\tfrac{\sqrt{\epsilon}}{2}\,\pi e^{5\pi^2/8}\|[\theta_1u\cdot u]\|,\tag{14.iv}$$

one Euclidean four-point function at separated points. With $\epsilon\to 0$, $\phi(f_u)\Omega$ lies in the closure of $\mathcal{C}(W')\Omega$.

Step 6 (density, locality). By Lem. 14.2(c) the transforms of the f_u have no common real zero; Thm. AO(d), with W and W' exchanged, gives the density of $\mathcal{C}(W')\Omega$, (W), (CP), the duality for

wedges and the modular data; Thm. AO(a), (e) give (a)–(e) for $\tau = \sigma$; $\mathcal{A}(W) = \mathbb{C}1$ would make $\mathcal{H} = \overline{\mathcal{A}(W)\Omega} = \mathbb{C}\Omega$.

Step 7 (the wedge algebra is made of lattice limits). Let $\mathfrak{M}$ be the von Neumann algebra generated by the operators of (f) and $\mathcal{N} = \mathcal{C}(W') = \mathcal{A}(W)$. For $\mathfrak{g}W \subset W$ one has $\mathfrak{g}W' \supset W'$, so $U(\mathfrak{g})\mathcal{C}(W')U(\mathfrak{g})^{-1} = \mathcal{C}(\mathfrak{g}W') \subset \mathcal{C}(W')$, and $\mathfrak{M} \subset \mathcal{N}$ by Step 3. The closure of $\mathfrak{M}\Omega$ is invariant under $\mathfrak{M}$ and under the $U(a)$, $a \in W$, hence under all translations, and it holds every $\phi(f_u)\Omega$ (Step 5); the one-vector argument of Thm. AO(d), with $\mathfrak{M}$ in the place of $\mathcal{C}(W')$, makes $\mathfrak{M}\Omega$ dense. $\mathfrak{M}\Omega$ and $\mathcal{N}\Omega$ are boost-invariant dense subspaces of $D(\Delta^{1/2})$ on which $Q\Omega \mapsto Q^*\Omega$ is the restriction of $J\Delta^{1/2}$ (Thm. AO(c) for W'), hence cores of it; so the two algebras have the one modular conjugation J, and $\mathfrak{M}' = J\mathfrak{M}J \subset J\mathcal{N}J = \mathcal{N}'$: $\mathcal{N} \subset \mathfrak{M}$.

Step 8 (components). All σ_ξ are data (Rec) with one triple. For $a = \sum f^n Y_n$ as in (CP), which is hermitian, and b formed as in Lem. 14.7(iv), a polynomial in means of translates of elements b_i of $\mathfrak{P}$ over pairwise different spatial translations of length $\geq R$, with R so large that the translated b_i are spacelike to K and so in the algebra of (CP), $\sum \lambda^n b^* Y_n b \in \mathfrak{A}^+$ for all real λ and f is central, so $\mathcal{W}_\sigma(b^*ab) \geq 0$ by Step 6; Lem. 14.7(iv) gives (CP) for every σ_ξ, and Thm. AO(d), (a), (e) give (a)–(e) for $\tau = \sigma_\xi$. The same follows from the vectors $M_g\Omega$ of Lem. 14.7(i): $M_g \in \mathcal{C}(O)$ for every O, and the extension of the proof of (W)⇒(CP) to the vector $M_g\Omega$ gives $\int g^2 \mathcal{W}_\xi(a)\,\mathrm{d}\lambda \geq 0$ for all real $g \in C(\mathsf{X})$.

Step 9 (fibres). The operators M_g of Lem. 14.7(i) commute with the semigroup, the translations and the fields, so they lie in $\mathcal{C}_\sigma(O)$ for every O, hence in $\mathcal{C}_\sigma(W') \cap \mathcal{C}_\sigma(W) = \mathcal{A}_\sigma(W) \cap \mathcal{A}_\sigma(W)'$ (Step 6). The rest is reduction theory: on a separable Hilbert space that is a direct integral over (X, λ), X compact metrisable and λ a Borel measure, the decomposable operators are the commutant $\mathcal{Z}'$ of the diagonal algebra $\mathcal{Z}$; a von Neumann algebra $\mathcal{N}$ with $\mathcal{Z} \subset \mathcal{N} \subset \mathcal{Z}'$ is $\int^\oplus \mathcal{N}_\xi\,\mathrm{d}\lambda$ for a measurable field of von Neumann algebras, whose fibres are those of a countable generating family and are fixed up to a λ-null set; and $(\int^\oplus \mathcal{N}_\xi\,\mathrm{d}\lambda)' = \int^\oplus \mathcal{N}'_\xi\,\mathrm{d}\lambda$. (1) The classes depend continuously on the entries, with constants of one triple for σ and for all ξ; so $\mathcal{H}(\sigma)$ is separable, and there are countably many combinations F_n whose classes $[F_n]_\xi$ are total in every $\mathcal{H}(\sigma_\xi)$; and $\xi \mapsto \sigma_\xi(\theta\bar{F}' \cdot F)$ is continuous: the fields $\xi \mapsto [F_n]_\xi$ are a fundamental sequence.

The Hilbert space $\mathfrak{K}$ of the proof of Lem. 14.7(i) is the direct integral $\int^\oplus \mathcal{H}(\sigma_\xi)\lambda(\,\mathrm{d}\xi)$: $g \otimes F \mapsto (g(\xi)[F]_\xi)_\xi$ is isometric by the form of $\mathfrak{K}$, and onto, as a field orthogonal to all $g[F_n]$, $g \in C(\mathsf{X})$, has $\langle [F_n]_\xi, x_\xi\rangle = 0$ for almost every ξ and all n. So $[F] \mapsto ([F]_\xi)_\xi$ is unitary, $\mathcal{H}(\sigma) = \int^\oplus \mathcal{H}(\sigma_\xi)\lambda(\,\mathrm{d}\xi)$, and M_g is the multiplication by g; the diagonal algebra $\mathcal{Z}$ is the weak closure of the M_g. (2) The semigroup, the translations and the insertions decompose, being given by one formula for σ and for all σ_ξ; so do the bounded $e^{-sH}\phi(f)e^{-sH}$, by eq. 14.i; as the e^{-sH_ξ} are injective and the polynomial bound holds with one triple, a subsequence converging fibrewise gives $(\phi(f)\psi)_\xi = \phi_\xi(f)\psi_\xi$ for almost every ξ, $\psi \in \mathcal{D}_\infty$, and by induction the fibres of $\phi(f_1)\cdots\phi(f_n)\Omega$ are $\phi_\xi(f_1)\cdots\phi_\xi(f_n)\Omega_\xi$ almost everywhere. (3) $M_g \in \mathcal{A}_\sigma(W)'$ and weak closure give $\mathcal{Z} \subset \mathcal{A}_\sigma(W)'$, so $\mathcal{A}_\sigma(W) \subset \mathcal{Z}'$ consists of decomposable operators; $M_g \in \mathcal{A}_\sigma(W)$ gives $\mathcal{Z} \subset \mathcal{A}_\sigma(W)$: $\mathcal{Z}$ lies in the centre of $\mathcal{A}_\sigma(W)$, and $\mathcal{A}_\sigma(W) = \int^\oplus \mathfrak{A}_\xi\,\mathrm{d}\lambda$ for a measurable field of von Neumann algebras $\mathfrak{A}_\xi$. (4) A decomposable bounded Q lies in $\mathcal{C}_\sigma(W')$ if and only if $Q_\xi \in \mathcal{C}_{\sigma_\xi}(W')$ for almost every ξ, and the same with W for W': "if" by integration of the relation of the fibres, by (2); "only if" because $\mathcal{C}_\sigma(W')$ is an algebra that holds the M_g, so that $\int g\,[\langle \phi_\xi(\bar{f})\psi_\xi, Q_\xi\psi'_\xi\rangle - \langle \psi_\xi, Q_\xi\phi_\xi(f)\psi'_\xi\rangle]\,\mathrm{d}\lambda = 0$ for all $g \in C(\mathsf{X})$; the bracket vanishes almost everywhere, with one null set for countably many test functions $f \in \mathcal{F}(W')$ and products of fields, and the polynomial bound with one triple carries it to all. (5) A countable family that generates $\mathcal{A}_\sigma(W) = \mathcal{C}_\sigma(W')$ has, by (4), its fibres in the von Neumann algebra $\mathcal{C}_{\sigma_\xi}(W')$ for almost every ξ, and these fibres generate $\mathfrak{A}_\xi$: $\mathfrak{A}_\xi \subset \mathcal{C}_{\sigma_\xi}(W')$ almost everywhere. (6) $\mathcal{A}_\sigma(W)' = \mathcal{C}_\sigma(W) = \int^\oplus \mathfrak{A}'_\xi\,\mathrm{d}\lambda$, and (4) for W, read on a countable generating family, gives $\mathfrak{A}'_\xi \subset \mathcal{C}_{\sigma_\xi}(W)$ almost everywhere; with wedge duality for each σ_ξ (Step 8), $\mathfrak{A}_\xi \supset \mathcal{C}_{\sigma_\xi}(W)' = \mathcal{C}_{\sigma_\xi}(W')$.

So $\mathfrak{A}_\xi = \mathcal{C}_{\sigma_\xi}(W') = \mathcal{A}_{\sigma_\xi}(W)$ for λ-almost every ξ, every ξ with $\lambda(\{\xi\}) > 0$ among them, as such a point is no null set, and $\mathcal{A}_\sigma(W) = \int^\oplus \mathcal{A}_{\sigma_\xi}(W)\,\lambda(\mathrm{d}\xi)$: for λ-almost every ξ the wedge algebra of the component σ_ξ is the fibre of the algebra of (f), the fibres of a direct integral being fixed up to null sets.

Step 10 (the supply). Here $(F)_j = (F)^{\rho_j}_{K_j}$ is the product at K_j centred by ρ_j, as in (h), also for a component, where the states $\rho_{j,\xi}$ integrate and the ρ_j give the centring. Reflection positivity of ρ_j at the site mirror $x_3 = 0$: a bounded Borel function B of bonds in the mirror is fixed by θ, and with $\mathsf{c} = (\sup|B|^2 - |B|^2)^{1/2}$, $\sup|B|^2\rho_j(\overline{\theta G}G) - \rho_j(\overline{\theta(BG)}BG) = \rho_j(\overline{\theta(\mathsf{c}G)}\mathsf{c}G) \geq 0$; the Schwarz inequality gives the first bound. $\mathcal{H}(\sigma')$ is separable; a diagonal subsequence over a countable set of combinations with dense classes, and the first bound, give the limit form and Q, $\|Q\| \leq \mathsf{b}$. For $c = 0$ and combinations F, F' over Q_L put $A_j = B_j\overline{\theta(F')_j}(F)_j$; then $\ell^Q_{F',F}(h) = \lim_j \rho_j(A_j(h)_j)$, and the Schwarz inequality for the reflection positivity at the site mirror $x_1 = 0$ gives

$$|\rho_j(A_j(h)_j)|^2 \leq \rho_j(A_j\,\theta_1\bar{A}_j)\,\rho_j(\overline{\theta_1(h)_j}(h)_j),$$

whose second factor tends to $\sigma'(\theta_1\bar{h}\cdot h)$; in the first, with $Y = F\cdot\theta_1\bar{F}$, $Y' = F'\cdot\theta_1\bar{F}'$ and the function $B_j\theta_1\bar{B}_j$ of the bonds of the mirror $x_3 = 0$, $A_j\theta_1\bar{A}_j = \overline{\theta(Y')_j}\,B_j\theta_1\bar{B}_j\,(Y)_j$, and the first mirror, used a second time, bounds it by $\mathsf{b}^2\|[Y']\|\|[Y]\|$ in the limit. So the hypothesis of Lem. 14.5 holds and $Q \in \mathcal{C}_{\sigma'}(W)$. For $c \neq 0$, ce_1 is a lattice vector for large j; the translate $\tau_{-ce_1}B_j$ is of the case $c = 0$, and by the invariance of ρ_j its limit is $U(ce_1)^{-1}QU(ce_1)$. The proof uses of the states that integrate only reflection positivity at the site mirrors $x_3 = 0$, $x_1 = 0$ and invariance under the lattice translations, which the lattice approximants of a component have; so the same holds for every component.

Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD enter only through Thm. YM3_I:AP.

14.1.8. *Proof of Thm. AQ.* Every bound below passes to the components through the last sentence of Lem. 14.15(d), which carries the norm bound of every insertion from σ' to each component; every component is data (Rec) with one triple.

(a) Step 1: the blocks at one time. Let $h \in C^1_c(\mathbb{R}^2)$ be real and carried by a square of side 2, $q = T_{mn}(h)$, $A = (H-m)P_m$, $B = (H-n)P_n$, so that $0 \leq A, B \leq 1$. As $P_m\mathsf{B}_h(s,s')P_n = e^{-sH}q\,e^{-s'H}$,

$$e^{\frac{\mathsf{l}}{2}(m+n)}P_m\mathsf{I}_\mathsf{l}(h)P_n = \int \rho_\mathsf{l}(\varsigma)e^{-\varsigma(m-n)}\,e^{-(\frac{\mathsf{l}}{2}+\varsigma)A}\,q\,e^{-(\frac{\mathsf{l}}{2}-\varsigma)B}\,\mathrm{d}\varsigma,$$

and $\|e^{-uA}qe^{-u'B} - q\| \leq (u+u')\|q\|$ with $u + u' = \mathsf{l}$; since $\int \rho_\mathsf{l}(\varsigma)e^{-\varsigma(m-n)}\,\mathrm{d}\varsigma = \int \rho_\mathsf{l}(\varsigma)\cosh(\varsigma(m-n))\,\mathrm{d}\varsigma \geq 1$, $\|q\| \leq (1-\mathsf{l})^{-1}e^{\frac{\mathsf{l}}{2}(m+n)}\|\mathsf{I}_\mathsf{l}(h)\|$. Take the least $i \geq 1$ with $L^{-i} \leq 2/(m+n+2)$: then $e^{\frac{\mathsf{l}}{2}(m+n)} \leq e$ and $\mathsf{l}^{-1} \leq L(m+n+2)$, and Lem. 14.15(c) gives

$$\|T_{mn}(h)\| \leq \tfrac{3e}{2}L^3\mathsf{C}_3(m+n+2)^3|h|_{C^1}.$$

Step 2: the field. By eq. 14.i, $\langle\chi', P_m\phi(f)P_n\chi\rangle = \int\langle e^{-itH}\chi', T_{mn}(f_t)e^{-itH}\chi\rangle\,\mathrm{d}t$ with $f_t = f(t,\cdot)$; a partition of unity on $\mathbb{R}^2$ subordinate to the squares $\kappa + [-1,1]^2$, $\kappa \in \mathbb{Z}^2$, splits the real and imaginary parts of f_t into pieces of norm $|\cdot|_{C^1} \leq 3.3\sup_{\kappa+[-1,1]^2}(|f_t| + |\nabla f_t|)$, and $2\cdot 3.3\cdot\frac{3e}{2} < 27$ gives $b_{mn}(f) \leq \mathsf{c}_0(m+n+2)^3$; Thm. AO(b2), which holds for all data, gives the decay. Step 3: the operator and form bounds. $\|P_m\phi(f)\psi\| \leq \sum_n b_{mn}(f)\|P_n\psi\|$, and $m+n+2 \leq 2(1+n)(1+|m-n|)$; so $\|\phi(f)\psi\| \leq \|\kappa\|_{\ell^1}\|N^3\psi\|$ by Young's inequality, $\kappa(d) = 8(1+|d|)^3\min\{\mathsf{c}_0, \mathsf{c}_5(|d|-1)^{-5}\}$, and $\sum_{|d|\leq 2}(1+|d|)^3 = 71$, $\sum_{|d|\geq 3}(1+|d|)^3(|d|-1)^{-5} < 6.3$; with a weight $(1+m)^a \leq (1+n)^a(1+|d|)^{|a|}$ and a larger k the same gives the bound with N^{a+3}.

For the form, $(m+n+2)^2 \leq 4(1+m)(1+n)(1+|d|)$ and Schur's test with $\kappa'(d) = 8(1+|d|)^{3/2}\min\{\mathsf{c}_0, \mathsf{c}_3(|d|-1)^{-3}\}$, whose two sums are less than 17.1 and 5.7, respectively. The blocks of $e^{-sH}\phi(f)e^{-sH}$ are $\leq e^{-s(m+n)}b_{mn}(f)$, and $\sup_{x\geq 0}(x+2)^3e^{-sx} \leq (3/s)^3e^{2s}$. Step 4: growth.

For $|m-n| = d \geq 2$, the bound of Step 2 at $\partial_t^k f$ with $\|\partial_t^k f\|_\star \leq A^{k+1}(k!)^{\mathsf{s}} \leq A^{k+1}k^{\mathsf{s}k}$, at $k = \lfloor ((d-1)/e^{\mathsf{s}}A)^{1/\mathsf{s}} \rfloor$, gives

$$b_{mn}(f) \leq \mathsf{G}\,(m+n+2)^3 e^{-\gamma'|m-n|^{1/\mathsf{s}}} \qquad \text{for all } m,n, \qquad \gamma' = \mathsf{s}e^{-1}A^{-1/\mathsf{s}},\ \mathsf{G} = 27L^3\mathsf{C}_3 A e^{\mathsf{s}+\gamma'}.$$

As $\Omega = P_0\Omega$, $\|\phi(f)^n\Omega\| \leq \sum \prod_{k\leq n} b_{m_k m_{k-1}}(f)$ over paths $0 = m_0, m_1, \dots, m_n$; with $\mathsf{D} = \sum_k |m_k - m_{k-1}| \geq \max_k m_k$, at most 2^n paths for each sequence of steps, and $\sum_k |d_k|^{1/\mathsf{s}} \geq \mathsf{D}^{1/\mathsf{s}}$,

$$\|\phi(f)^n\Omega\| \leq (2\mathsf{G})^n \sup_{\mathsf{D}\geq 0} (2\mathsf{D}+2)^{3n} e^{-\frac{\gamma'}{2}\mathsf{D}^{1/\mathsf{s}}} \Big(\sum_{d\geq 0} e^{-\frac{\gamma'}{2}d^{1/\mathsf{s}}} \Big)^n \leq C^n (n!)^{3\mathsf{s}}.$$

Such f exist: $w \otimes b$ with $b \in C_c^\infty(\mathbb{R}^2)$, $b \geq 0$, and w the infinite convolution of the uniform probability densities on $[0, i^{-\mathsf{s}}]$, $i \geq 1$, for which $\|w^{(k)}\|_{L^1} \leq \prod_{i\leq k} 2i^{\mathsf{s}} = 2^k (k!)^{\mathsf{s}}$, because the k-th derivative falls on the first k factors, each becoming a difference of two point masses divided by the width.

(b) The same four steps with the second bound of Lem. 14.15(c), $\|T_{mn}(h)\| \leq \frac{3e}{2} L^2 \mathsf{C}_2 (m+n+2)^2 |h|_{C^1}$, the exponent 2 in the place of 3, the weight $(1+n)^2$ in Step 3 for the operator bound and $[(1+m)(1+n)]^1$ for the form bound, $\sup_x (x+2)^2 e^{-sx} \leq (2/s)^2 e^{2s}$, and $(n!)^{2\mathsf{s}}$ at the end.

(c) The point. Put $\bar{B} = B_\sharp L/(L-1)$, where $B_\sharp \geq 0$ is one of the fixed constants σ^-. Thus $\bar{B}$ is independent of the source radius, γ and g, and $\bar{B} < \gamma^{-2} \leq g^{-2}$. For an integer $z \geq 2$ set

$$v_z^{-2} = L^{z-1}(g^{-2} + B_\sharp - L^{-1}\bar{B}) + L^{-1}\bar{B} - B_\sharp, \qquad G = g^{-2} + 2\bar{B}.$$

Then $L^{z-1}g^{-2} \leq v_z^{-2} \leq L^z G$. Define

$$\Lambda_z = 1 + \log(L^z G), \qquad \varrho_z = \frac{1}{2} \min\{\sigma^{\mathrm{S}}(v_z^{-2} - \bar{B}),\ \sigma_{\mathrm{L}}^{\mathrm{S}} v_z^{-1} (\log v_z^{-2})^{-\mathfrak{p}_0}\}.$$

Here σ^{S} and $\sigma_{\mathrm{L}}^{\mathrm{S}}$ depend only on σ^-. We check explicitly that this radius is allowed. The radius-independent conditions remain valid when the coupling ceiling is decreased from γ to v_z. The two source-radius conditions in Prop. YM3_I:82.2, with source radius $2\varrho_z$, are

$$2\varrho_z \leq \sigma^{\mathrm{S}}(x - \bar{B}), \qquad 2\varrho_z x^{-1/2} (\log x)^{\mathfrak{p}_0} \leq \sigma_{\mathrm{L}}^{\mathrm{S}} \quad (x \geq v_z^{-2}).$$

The first left margin is smallest at $x = v_z^{-2}$. The calibration also gives $\log \gamma^{-2} > 2\mathfrak{p}_0$. Therefore $x^{-1/2}(\log x)^{\mathfrak{p}_0}$ is decreasing for $x \geq v_z^{-2} \geq \gamma^{-2}$, and the second condition is likewise strongest at that endpoint. Both conditions hold by the definition of ϱ_z. This proves the required admissibility and also identifies ϱ_z as the largest radius permitted by these two source conditions at the fixed ceiling v_z.

Let (K, β) be admitted at g, $K > z$. Its label $\mathsf{b} = L^{-K}\beta$ lies in the window YM3_I:(1531) of Prop. YM3_I:74.1, $g^{-2} + B_\sharp \frac{L-2}{L-1} < \mathsf{b} \leq L(g^{-2} + \bar{B})$; so the running coupling g', $g'^{-2} = L^{-1}(L^z \mathsf{b} + \bar{B}) - B_\sharp$, at which the label $L^z\mathsf{b}$ of $(K-z, \beta)$ passes first, has $v_z^{-2} < g'^{-2} \leq L^z G$ and $1 + \log g'^{-2} \leq \Lambda_z$. At the radius ϱ_z and the choice (σ^-, v_z) the dilated pair $(K-z, \beta)$ is a first-passage datum at $g' \leq v_z$, and it carries systems of localised source energies with the constants $(2\varrho_z; C_{\mathrm{S}}, \kappa; C_{\mathrm{S}}, C_{g'}; C_{\mathrm{gap}})$ ([7, Lem. N.1]), where $C_{\mathrm{S}}, \kappa, C_{\mathrm{gap}}$ are the numbers of the pairs admitted at g, formed from σ^- alone, and $C_{g'} \leq C_{\mathrm{S}}(1 + \log g'^{-2}) \leq C_{\mathrm{S}}\Lambda_z$ [7, eq. 288].

As $v_z^{-2} - \bar{B} \geq \frac{12}{13} L^{z-1} g^{-2}$, $v_z \leq g L^{-(z-1)/2}$, $\log v_z^{-2} < \Lambda_z$ and $g \leq \frac{1}{2}$,

$$\frac{1}{\varrho_z} \leq 2c_\sigma\, g\, L^{-(z-1)/2} \Lambda_z^{\mathfrak{p}_0}, \qquad \Lambda_z = 1 + z\log L + \log G, \qquad G = g^{-2} + 2\bar{B},$$

$c_\sigma = \max\{\frac{13}{24\sigma^{\mathrm{S}}}, \frac{1}{\sigma_{\mathrm{L}}^{\mathrm{S}}}\}$, formed from σ^- alone: the radius of analyticity in the source grows like $1/g'$, the square root of the label g'^{-2}, up to logarithms. This is where Thms. YM3_I:AJ and YM3_I:AD are used at every radius and every admitted choice. Fix $0 < \eta \leq 1$, a real $h \in C_c^1$ carried by a ball $B(y, \frac{\eta}{40L})$ with $|y_3| \leq \frac{\eta}{8}$, and F, F' over $\{x_3 > \frac{\eta}{4}\}$. The cell: $\mathsf{l} = L^{-i}$ with $\frac{\eta}{5L} < \mathsf{l} \leq \frac{\eta}{5}$ and a centre $y^* \in L^{-i-2}\mathbb{Z}^3$ within $\frac{\mathsf{l}}{2L^2}$ of y; then the ball lies in the middle of the cell, and the cell in $\{|x_3| < \frac{\eta}{4}\}$.

The dilation by L^z among the construction results ([7, Lem. K.1]) carries periodic Gibbs states, mirrors and boards to the like and $\mathcal{O}_K(h)$ to $\mathcal{O}_{K-z}(L^z h(L^{-z}\cdot))$; it carries the cell to one of side $L^z\mathsf{l} \le L^z\eta/5$, and h to a source of norm $\max\{L^z \sup|h|, 40M \operatorname{Lip} h\}$, which is $\le \|h\|_\sharp$ as long as $L^z\eta \le 5$, since h vanishes on the boundary of its ball, so that $\sup|h| \le \frac{\eta}{40L}\operatorname{Lip} h$. The insertion bound on one lattice for a pair that carries systems, Lem. YM3_I:104.4 among the construction results, read on the dilated lattice at the radius ϱ_z, has the constant $2\mathsf{Q}_{\mathrm{mid}}(L^z\mathsf{l})/\varrho_z$ at the side of the one cell, with $\log \mathsf{Q}_{\mathrm{mid}}(\mathsf{l}') = (C_{\mathrm{S}} + C_{g'} + C_{\mathrm{gap}})\mathsf{l}'^3 + \Phi_1(\frac{\mathsf{l}'}{4}, \frac{\mathsf{l}'}{4}; \mathsf{l}')$ (equation YM3_I:(2017)) and $\Phi_1(\frac{\mathsf{l}'}{4}, \frac{\mathsf{l}'}{4}; \mathsf{l}') \le \bar\Phi$ for $0 < \mathsf{l}' \le 1$, where

$$\bar\Phi = C_{\mathrm{S}} S(\kappa)\left[\frac{27}{26} + 27\{32(C''_\Sigma + \log_L 10) + 4C'_\Sigma\}\right]$$

is the contribution of Φ_1 in equation YM3_I:(2018); so $\mathsf{Q}_{\mathrm{mid}}(L^z\mathsf{l}) \le Q_\star = e^{2C_{\mathrm{S}} + C_{\mathrm{gap}} + \bar\Phi}$, free of z, as long as $(L^z\eta/5)^3\Lambda_z \le 1$.

This is the proof of Prop. YM3_I:105.1 up to its last step, with the side of the one cell in the place of the supremum over the sides. With z fixed and the cutoff removed, $|\sigma'(\theta\bar F \cdot (h) \cdot F')| \le \frac{2Q_\star}{\varrho_z}\|h\|_\sharp \|[\tau_{-\frac{\eta}{4}e_3}F]\| \|[\tau_{-\frac{\eta}{4}e_3}F']\|$ for every $z \ge 2$ with $(L^z\eta/5)^3\Lambda_z \le 1$. As $z \mapsto (L^z\eta/5)^3\Lambda_z$ increases, such a z exists if and only if $\eta \le \eta_1 = 5L^{-2}\Lambda_2^{-1/3}$, a number $< L^{-1}$. At the largest such z: the condition gives $L^z \le 5/\eta$, so $\Lambda_{z+1} = 1 + \log L + z\log L + \log G \le \mathsf{L}(\eta)$, $\mathsf{L}(\eta) = 1 + \log(5LG) + \log\frac{1}{\eta}$; its failure at $z+1$ gives $L^{-z} < \frac{L\eta}{5}\Lambda_{z+1}^{1/3}$; hence $L^{-(z-1)/2} < \frac{L}{\sqrt{5}}\eta^{1/2}\mathsf{L}(\eta)^{1/6}$, and the insertion bound for a source in a ball of radius $\eta/40L$ holds for $\eta \le \eta_1$ with the constant $A_1\eta^{1/2}\mathsf{L}(\eta)^q$, $A_1 = \frac{4}{\sqrt{5}}LQ_\star c_\sigma g$, $q = \mathfrak{p}_0 + \frac{1}{6}$, formed from σ^- and g alone.

For $\eta_1 < \eta \le 1$ Prop. YM3_I:105.1 itself, for the undilated pairs, gives the constant $C_{\mathrm{I}} = 2\bar{\mathsf{Q}}_{\mathrm{mid}}/\varrho$, and $C_{\mathrm{I}} \le C_{\mathrm{I}}\eta_1^{-1/2}\eta^{1/2}\mathsf{L}(\eta)^q$ as $\mathsf{L}(\eta) \ge 1$. So for all $0 < \eta \le 1$ the bound holds with the running constant $\mathsf{c}_*(\eta) = A\,\eta^{1/2}\mathsf{L}(\eta)^q$, $A = \max\{A_1, C_{\mathrm{I}}\eta_1^{-1/2}\}$; the radius ϱ enters A through C_{I} alone. The data dilated by L^k, $\sigma'_k = \sigma' \circ \mathsf{D}_k^{-1}$, where $\mathsf{D}_k(f_1, \dots, f_n) = (L^k f_i(L^{-k}\cdot))_i$, extended linearly, commutes with θ, the conjugation and the product of tuples, has $\mathsf{D}_k\tau_v = \tau_{L^k v}\mathsf{D}_k$ and keeps the norms of the classes, are again data with this property, with $\mathsf{c}_*(L^{-k}\cdot)$, as $\|L^{-k}h''(L^k\cdot)\|_\sharp \le \|h''\|_\sharp$ for $k \ge 0$; and Prop. YM3_I:97.2 among the construction results, which turns an insertion bound for sources in small balls with a constant C_{I} into the point bound of order 4 with the constant $c_{\mathrm{pt}}C_{\mathrm{I}}$, $c_{\mathrm{pt}} = 320 \cdot 8^4\,\mathsf{mv}_3(40L)^4M$ the coefficient of C_{I} in equation YM3_I:(1919) at $c_{\mathrm{I}} = \frac{1}{40L}$ (with $\mathsf{mv}_3 = m_3$ in the notation following that equation), applied to σ'_k at $\eta'' = L^k\eta \in (L^{-1}, 1]$ and carried back, gives

$$\begin{aligned}&|\sigma'(\theta\bar F \cdot (h) \cdot F')| \\ &\le 16\, c_{\mathrm{pt}}\, A\, s_q\, L^{1/2}\, \eta^{-7/2}\mathsf{L}(\eta)^q\, \|h\|_{L^1}\, \|[\tau_{-\eta e_3}F]\|\, \|[\tau_{-\eta e_3}F']\|, \\ &s_q = \sup_{u \ge 0} e^{-u/2}(1+u)^q,\end{aligned}$$

for F, F' over $\{x_3 > \eta\}$ and h carried by $\{|x_3| < \frac{1}{2}\eta\}$; this is (Rec4) with $(C_\epsilon, \frac{7}{2} + \epsilon, \frac{1}{2})$, $C_\epsilon = 16c_{\mathrm{pt}}As_qL^{1/2}\sup_{\eta \le 1}\eta^\epsilon\mathsf{L}(\eta)^q$, and the point bound with the logarithm follows by taking $\epsilon = q/\mathsf{L}(\mu)$.

One time. A chain of dilates $y = 0, 1, \dots, Y$ of one pair bounds the coefficients of the mixed part of the source energies that see two cells. Let (K, β) be admitted at g, with the label $\mathsf{b} = L^{-K}\beta \ge g^{-2}$, (i, o) a board with cells of side $\mathsf{l} = L^{-i}$, $f \ne 0$ a real Lipschitz source vanishing off the middle of one cell, ℓ' the side of a reference torus, $v'^{\pm}$ the sums of the mirror copies of f in the even and in the odd cells of the board on the torus of side ℓ', and $\mathsf{N} = (\ell'/\mathsf{l})^3$. The normalised partition function $F(z, z') = Z_{K,\ell',\beta}[zv'^{+} + z'v'^{-}]/Z_{K,\ell',\beta}$, $Z_{K,\ell',\beta}[w]$ the partition function of the Wilson weight with the source w on the torus $L^{-K}\mathbb{Z}^3/\ell'\mathbb{Z}^3$, is entire with $F(0,0) = 1$, and under the dilation by L^y it is

the same number for the pair $(K-y,\beta)$ on the torus of side $L^y\ell'$ with the arrays $L^y v'^{\pm}(L^{-y}\cdot)$: one function for all $0\le y<K$.

For $y\ge 1$ put $x_y=\frac12 L^y\mathsf{b}$, $g_y=x_y^{-1/2}$, $2\varrho_y=\min\{\sigma^{\mathrm S}(x_y-\bar B),\sigma^{\mathrm S}_{\mathrm L}x_y^{1/2}(\log x_y)^{-\mathfrak p_0}\}$, $\lambda_y=2+\log x_y$, and $\varrho_0=\varrho$, $\lambda_0=2+\log g^{-2}$: at the radius ϱ_y and the choice (σ^-,g_y) the pair $(K-y,\beta)$ is a first-passage datum at g_y and carries systems of localised source energies with the constants $(2\varrho_y;C_{\mathrm S},\kappa;C_{\mathrm S},C_{g_y};C_{\mathrm{gap}})$, $C_{\mathrm S}+C_{g_y}\le C_{\mathrm S}\lambda_y$, as in (c). The system of the y-th dilate gives a factorisation $F=e^{S_y}T_y$ on the bidisc of radius $r_y=\varrho_y/\|f\|_\sharp$ (the dilated source keeps the norm $\|f\|_\sharp$ while $L^y\mathsf l\le 1$): S_y, with $S_y(0,0)=0$, is the sum of its source energies at the dilated arrays, and T_y is the remainder normalised at 0, with $|T_y|\le e^{\mathsf N V_y}$, $V_y=C_{\mathrm S}\lambda_y(L^y\mathsf l)^3$, by the stability bounds of the systems (Section YM3_I:88).

If one function on a bidisc has two factorisations $e^{S_1}T_1=e^{S_2}T_2$ with $S_\iota(0,0)=0$, $T_\iota(0,0)=1$, $|T_\iota|\le e^{B_\iota}$, Jensen's inequality on the torus of the bidisc, $\int|\log|T_\iota||\le 2B_\iota$ for each factor, bounds the Taylor coefficients of S_1-S_2 on a bidisc of radius r: $|(S_1)_{ab}-(S_2)_{ab}|\le 4(B_1+B_2)r^{-(a+b)}$ for $(a,b)\ne(0,0)$, as they are Fourier coefficients of $2\mathrm{Re}(S_1-S_2)=2(\log|T_2|-\log|T_1|)$. For the mixed parts per cell, $\mathsf m_y=\mathsf N^{-1}S_y^{\mathrm{mix}}$, $S^{\mathrm{mix}}(z,z')=S(z,z')-S(z,0)-S(0,z')+S(0,0)$ the sum of the terms $z^a z'^b$ with $a,b\ge 1$, $\mathsf m_0$ being the mixed part of the proof of Lem. 14.15, the dilates y and $y+1$ give the chain

$$|\mathsf m_{0;ab}|\le|\mathsf m_{Y;ab}|+\sum_{y=0}^{Y-1}4(V_y+V_{y+1})\min\{r_y,r_{y+1}\}^{-(a+b)}\qquad(a,b\ge1),$$

and in the top dilate Cauchy's estimate on the bidisc of radius r_Y with the two counting estimates in the proof of the cell bound (Prop. YM3_I:89.1) gives $|\mathsf m_{Y;ab}|\le\bar\Phi r_Y^{-(a+b)}$. The radii have $\varrho_y\ge\underline\varrho_y=\frac12\sigma_\wedge x_y^{1/2}(\log x_y)^{-\mathfrak p_0}$, $\sigma_\wedge=\min\{\frac{11}{13}\sigma^{\mathrm S},\sigma^{\mathrm S}_{\mathrm L}\}$, and $\underline\varrho_y/\underline\varrho_{y'}\ge L^{-(y'-y)/2}$ for $1\le y\le y'$. As $V_y\le L^{-3(Y-y)}V_Y$ while $\underline\varrho_y^{-k}\le L^{(Y-y)k/2}\underline\varrho_Y^{-k}$, the terms $1\le y<Y$ of the chain form a geometric sum, at most $5.6L^{k/2}V_Y\underline\varrho_Y^{-k}\|f\|_\sharp^k$ for $k=a+b\le5$, and the term $y=0$ is at most $8L^3C_{\mathrm S}\bar\lambda_1\mathsf l^3\varrho_\wedge^{-k}\|f\|_\sharp^k$, with $\bar\lambda_1=2+\log(\frac12L^2(g^{-2}+\bar B))$ and $\varrho_\wedge=\min\{\varrho,\frac12\sigma_\wedge(\frac12Lg^{-2})^{1/2}(\log(\frac12Lg^{-2}))^{-\mathfrak p_0}\}$.

Take $Y=i-r(\mathsf l)$, $r(\mathsf l)$ the least $r\ge1$ with $L^{3r}\ge\bar\lambda(\mathsf l)=2+\log\frac{L(g^{-2}+\bar B)}{2\mathsf l}$: in the dilate Y the cell has the side $L^{-r(\mathsf l)}$, of the order $\bar\lambda(\mathsf l)^{-1/3}$, so that $V_Y\le C_{\mathrm S}$ and $\underline\varrho_Y^{-2}\le 8L\sigma_\wedge^{-2}g^2\,\mathsf l\,\bar\lambda(\mathsf l)^{2\mathfrak p_0+1/3}$; and $\mathsf lL^2\bar\lambda(\mathsf l)^{1/3}\le1$ gives $1\le Y\le i-1$. The reference tori are those of the sides $2L^m$, $m\ge m_*$, with L^y times the side in the family of sides of the y-th dilate for all $y\le Y$, the threshold m_* being formed from σ^- and g alone. The two factorisations bound the mixed parts themselves,

$$|S_1^{\mathrm{mix}}-S_2^{\mathrm{mix}}|(z,z')\le4(B_1+B_2)\frac{tt'}{(1-t)(1-t')},$$

$t=|z|/r$, $t'=|z'|/r$, and $t\le\frac12$ on all the bidiscs of the chain for $|z|\|f\|_\sharp\le\frac12\varrho_\wedge$; this gives, for a source f in the middle of a cell of side $\mathsf l$ with $\mathsf lL^2\bar\lambda(\mathsf l)^{1/3}\le1$ and $|z|\le\frac12\varrho_\wedge/\|f\|_\sharp$,

$$\|e^{\mathcal O_K(zf)}\|_{\mathrm{ch}}\le|e^{\mathcal E_f(z)}|\exp\big[c_V\mathsf l^3+\mathsf a(\mathsf l)|z|^2\|f\|_\sharp^2\big],$$

$$\mathsf a(\mathsf l)=\big[3\bar\Phi+22.4LC_{\mathrm S}\big]\frac{8Lg^2}{\sigma_\wedge^2}\,\mathsf l\,\bar\lambda(\mathsf l)^{2\mathfrak p_0+\frac13}$$
$$+32L^3C_{\mathrm S}\bar\lambda_1\mathsf l^3\varrho_\wedge^{-2},$$

with $\mathcal E_f$ holomorphic on $|z|<\varrho/\|f\|_\sharp$, $\mathcal E_f(0)=0$, one function for all the reference sides, the normaliser of the one-cell bound that absorbs the unmixed terms. In the multiple reflections for the n cells of one layer, Lem. 14.15(e) (as in the proof of Lem. 14.15(c)), Cauchy's estimate at the radius $\min\{\mathsf a(\mathsf l)^{-1/2}\mathcal N^{-1},\varrho_\wedge/2\mathcal N_\infty\}$, $\mathcal N^2=\sum_k\|f_k\|_\sharp^2$, $\mathcal N_\infty=\max_k\|f_k\|_\sharp$, bounds the insertion of the layer by $2e^{nc_V\mathsf l^3+1}[\mathsf a(\mathsf l)^{1/2}\mathcal N+2\mathcal N_\infty/\varrho_\wedge]$; for the pieces of $\rho_{\mathsf l}\otimes k$ this is $\le C\mathsf l^{-5/2}\bar\lambda(\mathsf l)^{\mathfrak p_0+1/6}|k|_{C^1}$, and

the limit of the cutoffs gives $\|\mathsf{I}_\mathsf{l}(k)\| \le C\mathsf{l}^{-5/2}\bar\lambda(\mathsf{l})^{\mathfrak{p}_0+1/6}|k|_{C^1}$. The operator at one time has no larger norm than the plate: on the vectors of bounded energy $\mathsf{B}_k(\frac{\mathsf{l}}{2}+\varsigma,\frac{\mathsf{l}}{2}-\varsigma) = e^{-\varsigma H}\mathsf{B}_k(\frac{\mathsf{l}}{2},\frac{\mathsf{l}}{2})e^{\varsigma H}$, so the plate is the operator integral of $\mathsf{B}_k(\frac{\mathsf{l}}{2},\frac{\mathsf{l}}{2})$ with the symbol $\tilde\rho(E-E')$,

$$\tilde\rho(x) = \int \rho_\mathsf{l}(\varsigma)\cosh(\varsigma x)\,\mathrm{d}\varsigma = \frac{\sinh(ax)}{ax}\,\frac{\pi^2}{\pi^2+a^2x^2}, \qquad a=\frac{\mathsf{l}}{5}, \qquad \frac{1}{\tilde\rho(x)} = \prod_{j\ge 2}\Big(1+\frac{a^2x^2}{j^2\pi^2}\Big)^{-1},$$

and $1/\tilde\rho$ is the Fourier transform of a probability density g, a product of Fourier transforms of Laplace densities; hence $\mathsf{B}_k(\frac{\mathsf{l}}{2},\frac{\mathsf{l}}{2}) = \int \mathsf{g}(t)e^{itH}\mathsf{I}_\mathsf{l}(k)e^{-itH}\,\mathrm{d}t$ and $\|\mathsf{B}_k(\frac{\mathsf{l}}{2},\frac{\mathsf{l}}{2})\| \le \|\mathsf{I}_\mathsf{l}(k)\|$. This gives the bound at $s=\frac{\mathsf{l}}{2}$, $\mathsf{l}=L^{-i}$, for $\mathsf{l}L^2\bar\lambda(\mathsf{l})^{1/3}\le 1$, Lem. 14.15(c) gives it for the finitely many larger $\mathsf{l}=L^{-i}$, and the semigroup property gives it for all s,s'. The block bound follows from $\|P_m\phi(f)P_n\| \le e^{s(m+n+2)}\|e^{-sH}\phi(f)e^{-sH}\| \le e\int\|\mathsf{B}_{f_t}(s,s)\|\,\mathrm{d}t$ at $s=1/(m+n+2)$ and the partition of unity of (a).

(d) For a component, $\mathsf{B}_k(u,u)\Omega$ is the class of the insertion of k at the height u, and

$$\|\mathsf{B}_k(u,u)\Omega\|^2 = \iint k(\vec x)k(\vec y)\,\mathsf{s}\big((|\vec x-\vec y|^2+4u^2)^{1/2}\big)\,d^2x\,d^2y.$$

With $\mathsf{s}\ge 0$, $\mathsf{s}(r)\ge c_{\mathrm{sd}}r^{-6}$ for $r\le r_{\mathrm{sd}}$, and $u^4(|z|^2+4u^2)^{-3}$, an approximate identity on $\mathbb{R}^2$ of the mass $2\pi u^4\int_0^\infty r(r^2+4u^2)^{-3}dr = \frac{\pi}{32}$, the liminf follows for continuous $k\ge 0$ of compact support. A bound $\pm\mathsf{T}(k)\le cN$ gives $|\langle\psi',\mathsf{T}(k)\psi\rangle| \le c\|N^{1/2}\psi'\|\|N^{1/2}\psi\|$ by polarisation, so $\|\mathsf{B}_k(u,u)\| \le c\sup_E(1+E)e^{-2uE} \le c(1+(2eu)^{-1})$, against the liminf. $\|\mathsf{B}(s,s)\Omega\|^2 = \mathsf{s}(2s) \ge c_{\mathrm{sd}}(2s)^{-6}$; with the upper bound in (c), the infimum of admissible pure-power exponents lies in $[3,\frac{7}{2}]$. For the plates, $\mathsf{I}_\mathsf{l}(h)\Omega = \tilde\rho(H)\mathsf{B}_h(\frac{\mathsf{l}}{2},\frac{\mathsf{l}}{2})\Omega$ with $\tilde\rho(E)=\int\rho_\mathsf{l}(\varsigma)\cosh(\varsigma E)\,\mathrm{d}\varsigma\ge 1$, so $\|\mathsf{I}_\mathsf{l}(h)\Omega\| \ge (1-o(1))(\pi c_{\mathrm{sd}}/32)^{1/2}\|h\|_{L^2}(\mathsf{l}/2)^{-2}$.

For the blocks, put $w_D = T_{D0}(k)\Omega$, and for $0<\eta\le 1$ let $\rho=\rho^{(\eta)}\in C^1_c((-\frac{\eta}{4},\frac{\eta}{4}))$ be even and ≥ 0 with $\int\rho=1$, and $Y_\eta = \int\rho(\varsigma)\mathsf{B}_k(\eta+\varsigma,\eta-\varsigma)\,\mathrm{d}\varsigma$, the insertion of $\rho\otimes k$ at the height η (an admitted insertion, as $\rho\otimes k$ is carried by $\{|x_3|<\frac{1}{2}\eta\}$). As ρ is even, $\|P_mY_\eta P_n\| \le e^{-\eta(m+n)}\cosh(\frac{\eta}{4}|m-n|)\|T_{mn}(k)\|$, and $P_DY_\eta\Omega = e^{-\eta H}\tilde\rho_\eta(H)w_D$ with $\tilde\rho_\eta(E)=\int\rho(\varsigma)\cosh(\varsigma E)\,\mathrm{d}\varsigma\ge 1$. First, for every $D\ge 0$ and every n with $P_n\ne 0$,

$$\sum_{j=-1}^{1}\|T_{n+D+j,n}(k)\|^2 \ge \|w_D\|^2:$$

for a unit vector $\psi\in P_n\mathcal{H}$ and its far translates $\psi_t = U(te_1)\psi$, the spectral measure of H in $Y_\eta\psi_t$ tends from below on open sets to the convolution of those of $Y_\eta\Omega$ and of $e^{-2\eta H}\psi$, by clustering, which the unique vacuum of a component gives ($e^{-tH}\to|\Omega\rangle\langle\Omega|$); so the column w_D of the vacuum reappears, up to the factors $e^{-\eta(\cdot)}$, in the blocks leaving every non-empty energy shell $P_n\mathcal{H}$: on the open set $(n+D-\frac{1}{2},n+D+2)$, with the two bounds above,

$$\sum_{j=-1}^{1}e^{-2\eta(2n+D+j)}\cosh^2\big(\tfrac{\eta}{4}(D+1)\big)\|T_{n+D+j,n}(k)\|^2 \ge e^{-2\eta(D+1)-4\eta(n+1)}\|w_D\|^2,$$

and $\eta\downarrow 0$ gives the inequality, the blocks and w_D depending neither on η nor on ρ. Secondly, $\|w_D\|$ is large for infinitely many D: by the two-point law as above, the heights $2\eta+\varsigma+\varsigma'$ of the two insertions being $\le\frac{5}{2}\eta$, $\|Y_\eta\Omega\|^2 \ge (1-o(1))\,c_{\mathrm{sd}}\frac{\pi}{2}(\frac{2}{5})^4\eta^{-4}\|k\|_{L^2}^2$ as $\eta\downarrow 0$; while on $[D,D+1)$, $e^{-\eta E}\cosh(\varsigma E)\le e^{-(\eta-|\varsigma|)E}\le e^{-\frac{3}{4}\eta D}$ for $|\varsigma|\le\frac{\eta}{4}$, so that $\|Y_\eta\Omega\|^2=\sum_D\|P_DY_\eta\Omega\|^2 \le \sum_D e^{-\frac{3}{2}\eta D}\|w_D\|^2$; if $\|w_D\|^2\le(D+2)^{2q''}$ for all large D with $q''<\frac{3}{2}$, the right member would be $O(\eta^{-2q''-1})=o(\eta^{-4})$. Thirdly, every shell $P_n\mathcal{H}$ with $n\ge E_0$ is non-empty, because the support of the spectral measure of the translations holds a point (E_0,q_0) with $E_0\ge 1$ and is invariant under the boosts, along which the energy runs to infinity. Now let $q'<\frac{3}{2}\epsilon$, choose q'' with $q'/\epsilon<q''<\frac{3}{2}$, and for the infinitely many D with $\|w_D\|^2>(D+2)^{2q''}$ put $n=\lceil(D+1)^{1/\epsilon}\rceil$: the three blocks

$(n+D+j,n)$ lie in the band and $m+n+2 \le 6(D+1)^{1/\epsilon}$, so a bound $C(m+n+2)^{q'}$ in the band would give $(D+2)^{2q''} < 3C^2 6^{2q'}(D+1)^{2q'/\epsilon}$, false for large D.

(e) The multiple reflections for one cell (Prop. YM3_I:104.2) and the one-cell bound (Prop. YM3_I:89.1), with Montel's theorem and a diagonal choice over a countable set of combinations with dense classes, give $\mathbb{D}(z)$ on the whole disc with $\|\mathbb{D}(z)\| \le \bar{\mathsf{Q}}$, $\mathbb{D}(0) = e^{-\mathsf{I}H}$ and $\mathbb{D}'(0) = \mathsf{I} + c\, e^{-\mathsf{I}H}$, I the insertion of f, which is the slab operator of Thm. AO(h1)(iv) by the insertion formula YM3_I:(2039). $G(x) = \langle \Omega, \mathbb{D}(x)\Omega\rangle$ is holomorphic, $G(0) = 1$, $|G| \le \bar{\mathsf{Q}}$; by the lemma of Schwarz $|G(x) - 1| \le (1+\bar{\mathsf{Q}})|x|\|f\|_\sharp/\varrho \le \frac{1}{2}$ on the smaller disc, so $\|\hat{D}(x)\| \le 2\bar{\mathsf{Q}}$.

At real x the lattice operators are $e^{-\mathcal{E}(x)}$ times positive operators, by reflection positivity at the middle plane of the cell, a site plane, f being even under that reflection; the quotient $\hat{D}(x)$ removes the factor $e^{-\mathcal{E}(x)}$, so $\hat{D}(x) \ge 0$. Finally

$$\hat{D}'(0) = \mathbb{D}'(0) - G'(0)e^{-\mathsf{I}H} = \mathsf{I} - \langle \Omega, \mathsf{I}\Omega\rangle e^{-\mathsf{I}H}$$

, and $\langle \Omega, \mathsf{I}\Omega\rangle = q_0(\Omega,\Omega)\int h'$ because $\langle \Omega, \mathsf{B}(s,s')\Omega\rangle = q_0(\Omega,\Omega)$ for all s, s' and $\int \rho_1 = 1$. For a component the same holds with its lattice approximants, the periodic Gibbs states $\rho_{j,\xi}$ recalled in the results used.

14.1.9. *Proof of Thm. AR. (a)* Let $e^\alpha_x = \kappa_\alpha e^{-x_3\omega - i\vec{x}\cdot\vec{p}}$ for $x \in \mathbb{E}_+$, and for pairwise different points of $\mathbb{E}_+$

$$J_n(x) = \prod_i \mathsf{P}(\partial_{t_i})\Big[\exp\Big(\sum_{i<l} t_i G(x_i - x_l) t_l\Big)\mathrm{Exp}\Big(\sum_i t_i \cdot e_{x_i}\Big)\Big]_{t=0}, \qquad J(G) = \int \prod_k h_k(x_k) J_n(x)\,\mathrm{d}x,$$

$\mathrm{Exp}(u) = \sum_m u^{\otimes m}/\sqrt{m!}$. The covariance of G and the invariance of P under $\mathrm{O}(3)$ are, as in the statement, those of the orthogonal representation ρ_{A} of $\mathrm{O}(3)$ on $\mathbb{R}^{\mathsf{A}}$: $G(\mathsf{R}x) = \rho_{\mathsf{A}}(\mathsf{R})G(x)\rho_{\mathsf{A}}(\mathsf{R})^{\mathrm{T}}$, $\mathsf{P}(\rho_{\mathsf{A}}(\mathsf{R})t) = \mathsf{P}(t)$, with $\rho_{\mathsf{A}}(\theta) = \mathrm{diag}(\varepsilon_\alpha)$; and the tensor p is symmetric. So the invariance of P under θ is $\mathsf{P}(\varepsilon t) = \mathsf{P}(t)$, $(\varepsilon t)_\alpha = \varepsilon_\alpha t_\alpha$, that is $\mathsf{p}_{\underline{\alpha}}\prod_i \varepsilon_{\alpha_i} = \mathsf{p}_{\underline{\alpha}}$, and the covariance under θ is $\varepsilon G(x)\varepsilon = G(\theta x)$, $\varepsilon = \mathrm{diag}(\varepsilon_\alpha)$.

As $\langle \mathrm{Exp}\, u, \mathrm{Exp}\, v\rangle = e^{\langle u,v\rangle}$ and $\langle e^\alpha_x, e^\beta_y\rangle = \varepsilon_\alpha G_{\alpha\beta}(\theta x - y)$, the substitution $s_i = \varepsilon\tilde{s}_i$, with $\mathsf{P}(\varepsilon t) = \mathsf{P}(t)$ in the derivatives and $\varepsilon G(x_i - x_l)\varepsilon = G(\theta x_i - \theta x_l)$ in the exponent, gives $\langle J_m(x), J_n(y)\rangle = S_{m+n}(\theta x, y)$, the Schwinger function of $:\mathsf{P}(\Phi):$; so $V[G] = J(G)$ is an isometry, (Rec1) follows from the covariance of G and the invariance of P, $e^{-tH_{\mathrm{F}}}J_n(x) = J_n(x + te_3)$ gives (Rec2), $|\sum \vec{p}(m_i)| \le \sum \omega(m_i)$ gives (Rec3), and $\ker d\Gamma(\omega) = \mathbb{C}\Omega_{\mathrm{F}}$ gives (Rec5).

The point field is the Fock operator

$$\mathsf{B}_{\mathrm{F}}(s,s') = \sum_j \binom{k}{j} \sum_{\underline{\alpha}} \mathsf{p}_{\underline{\alpha}} \prod_{i>j} \varepsilon_{\alpha_i}\, a^*(e^{\alpha_1}_{se_3}) \cdots a^*(e^{\alpha_j}_{se_3})\, \Gamma(e^{-(s+s')\omega})\, a(e^{\alpha_{j+1}}_{s'e_3}) \cdots a(e^{\alpha_k}_{s'e_3}),$$

with $\mathsf{B}_{\mathrm{F}}(s,s')J_n(y) = J_{n+1}(se_3, y + (s+s')e_3)$; from

$$\|a(u_1)\cdots a(u_m)e^{-cH_{\mathrm{F}}}\| \le \prod_i \|\omega^{-1/2}u_i\|(m/2ec)^{m/2}$$

and $\|\omega^{-1/2}e^\alpha_{ue_3}\| \le c_\kappa u^{-\delta}$ one gets $\|\mathsf{B}_{\mathrm{F}}(s,s')\| \le C_{\mathsf{B}}\min\{s,s',1\}^{-\wp}$, $C_{\mathsf{B}} = 2^k|\mathsf{p}|_1(2^\delta c_\kappa)^k(k/e)^{k/2}$, hence (Rec4) with $C_\wp = 2^\wp C_{\mathsf{B}}$, $c_0 = \frac{1}{2}$. The uniqueness of the family of point operators, Prop. YM3_I:107.2, identifies $V^{-1}\mathsf{B}_{\mathrm{F}}V$ with B, and so the field with the form of the Wick polynomial, which on vectors with finitely many particles and bounded energy is

$$\Phi_f(\chi,\chi') = \sum_j \binom{k}{j} \sum_{\underline{\alpha}} \mathsf{p}_{\underline{\alpha}} \prod_{i>j} \varepsilon_{\alpha_i} \int f(x)\Big\langle \prod_{i\le j} a(\kappa^x_{\alpha_i})\chi, \prod_{i>j} a(\kappa^x_{\alpha_i})\chi'\Big\rangle\,\mathrm{d}x, \qquad \kappa^x_\alpha = e^{i(t\omega - \vec{x}\cdot\vec{p})}\kappa_\alpha;$$

the projection on $V\mathcal{H}$ commutes with H_{F}, the translations and B_{F}, which gives the identity for every $\chi \in \mathcal{D}_\infty(d\Gamma(\omega))$.

(b) The component with kn particles of $V\phi(u)^n\Omega$ is $((kn)!)^{1/2}\mathrm{Sym}(w_u^{\otimes n})$, and no component has more particles: in $\Phi_u(\chi,\chi')$ the j-th term changes the number of particles by $2j-k$, the top term $j=k$ is the creation of w_u, by Fubini and $\int u(x)\prod_i e^{i(t\omega(m_i)-\vec{x}\cdot\vec{p}(m_i))}\,\mathrm{d}x=\hat{u}(\sum\omega,\sum\vec{p})$, and w_u is a symmetric function, the tensor p being symmetric, so that $\mathrm{Sym}(w_u)=w_u$. The symmetric tensors $v^{\otimes k}$ are total among the symmetric ones, so some unit v has $c_u=|\langle v^{\otimes k},w_u\rangle|>0$, and $\|\phi(u)^n\Omega\|\geq|\langle v^{\otimes kn},\mathrm{Sym}(w_u^{\otimes n})\rangle|((kn)!)^{1/2}=c_u^n((kn)!)^{1/2}$; $m!\geq(m/e)^m$ makes $\sum_n\|\phi(u)^n\Omega\|^{-1/n}$ converge for $k\geq3$.

Let a real $u_0\neq0$ in C_c^∞ have $b_{mn}(u_0)\leq\beta(m+n+2)^q$ on a band, $q<\frac{k}{2}$; outside the band Thm. AO(b3) gives a linear bound, so $b_{mn}(u_0)\leq\beta'(m+n+2)^{\max\{q,1\}}$ for all m,n; a linear form bound at u_0 gives $b_{mn}(u_0)\leq c\sqrt{(m+2)(n+2)}$ as well. Choose $\mathsf{s}>1$ with $\max\{q,1\}\mathsf{s}<\frac{k}{2}$, and $u=h*u_0$ with $h=g\otimes g_2$, g a bump of the Gevrey class s (as in the proof of Thm. AQ(a)) and $g_2\geq0$ in $C_c^\infty(\mathbb{R}^2)$: $u\in C_c^\infty$ is real and $\neq0$, $\hat{u}=\hat{h}\hat{u}_0$ being a product of entire functions $\neq0$. Thm. AO(b2), (b4) and the choice of the number of derivatives as in the proof of Thm. AQ(a) give $b_{mn}(u)\leq Ce^{-\mathsf{a}|m-n|^{1/\mathsf{s}}}b_{mn}(u_0)$, and the sum over the paths from Ω gives $\|\phi(u)^n\Omega\|\leq C^n(n!)^{\max\{q,1\}\mathsf{s}}$, against $c_u^n(n!)^{k/2}$.

The same with a triple of order $\wp<\frac{k}{2}$ in the place of the band bound.

(c) For the free field, $\|\omega^{-1/2}\kappa e^{-s\omega}\|^2=\frac{1}{4\pi}\int_{2s\mathsf{m}}^\infty e^{-v}\frac{dv}{v}\leq\frac{1}{8\pi\delta}(2s\mathsf{m})^{-2\delta}$ for $0<\delta\leq\frac12$. For $k\geq2$, the map $(p_1,\dots,p_k)\mapsto(\sum\omega(p_i),\sum p_i)$ has an image that holds $\mathcal{V}_k=\{E>(|\vec{k}|^2+k^2\mathsf{m}^2)^{1/2}\}$, and w_u is continuous with $\kappa>0$; so $w_u\neq0$ as soon as $\hat{u}\neq0$ at a point of $\mathcal{V}_k$. For $u\in C_c^\infty\setminus\{0\}$, $\hat{u}$ is entire. For $u\in\mathcal{S}$ vanishing off $\bar{W}$ and every k_2, $(E,k_1)\mapsto\hat{u}(E,k_1,k_2)$ continues boundedly into the directions η with $\eta_1<-|\eta_0|$, since $x^0\eta_0-x^1\eta_1\geq0$ on $\bar{W}$; if it vanished on the open set where the plane meets $\mathcal{V}_k$, it would vanish on the plane.

For $u=\mathsf{g}\otimes h'$ take $\vec{k}_*$ with $\hat{h}'(\vec{k}_*)\neq0$ and E large. At $k=6$, $\delta=\frac16$: $\wp=4$ and $C_\wp=2^4\cdot2^6\cdot2c_\kappa^6\cdot(6/e)^3$ with $c_\kappa^6=(\frac{4\pi}{3})^{-3}(2\mathsf{m})^{-1}$, which is $93312(\pi e)^{-3}\mathsf{m}^{-1}$; the field changes the number of particles by an even number, so the space lies in the even Fock space, where $d\Gamma(\omega)\geq2\mathsf{m}$ off Ω_F; $S_2=6!\,G^6$, $G=e^{-\mathsf{m}r}/4\pi r$. *Locality.* Let O_1,O_2 be spacelike separated, P the projection on $V\mathcal{H}$ and $\mathcal{R}_0=\mathcal{R}_0(O_2)$. By Lem. 14.12(a), (c), for real $h\in\mathcal{F}(O_2)$ the compression $V^{-1}P\mathsf{W}[u_h]V$ commutes weakly with the fields of O_1 on $\mathcal{D}_0$, so $P\mathcal{R}_0P|_{V\mathcal{H}}\subset V\mathcal{C}(O_1)V^{-1}$; by Lem. 14.12(d) Ω_F is cyclic for $\mathcal{R}_0$, and Lem. 14.11(a) gives $(P\mathcal{R}_0P|)'=P\mathcal{R}_0'P|$; by Lem. 14.12(e), $V^{-1}P\mathcal{R}_0'PV\subset\mathcal{C}(O_2)$.

So

$$\mathcal{C}(O_1)'\subset V^{-1}(P\mathcal{R}_0P|)'V=V^{-1}P\mathcal{R}_0'PV\subset\mathcal{C}(O_2),$$

which is (W), and (CP) follows by Thm. AO(d). For the derivatives $\partial_\alpha\varphi^a$ of massless fields and for the generalised free fields the same proof runs with Lem. 14.12(f), whose hypotheses (f1), (f2) hold for them, in the place of Lem. 14.12(a)–(e): $\mathcal{R}_0(O)$ is the span of the $\mathsf{W}[u]$, $u\in\mathfrak{k}(O)$, the real span of the $u_h^\alpha=\varsigma_\alpha\hat{h}(\omega,\vec{p})\kappa_\alpha$, $h\in\mathcal{F}_\mathbb{R}(O)$, with the weight $\omega^{1/2}(1+\omega)^{r'}$, $r'>\delta$, in the place of ω in the bounds of Lem. 14.12, the reality of $\langle\kappa^x,u_h\rangle$ replaced by the reality or pure imaginarity of $\langle\kappa_\alpha^x,u\rangle$, $u\in\mathfrak{k}(O)$, according to ε_α (the derivatives in space and in time of a kernel that is real at spacelike separation), and the density of the one-particle vectors of a region by the same reflection argument with $\omega\kappa_0$ in the place of κ.

For $:(\partial\varphi\cdot\partial\varphi)^2::$ $k=4$, $\delta=1$, $\wp=6$; for the kernel r^{-2}, the mixture of masses $4\pi\,\mathrm{d}\mathsf{m}$ has $\|\omega^{-1/2}\kappa e^{-s\omega}\|^2=\frac{1}{2s}$, so $\delta=\frac12$ and $\wp=3$ at $k=3$, $S_2=3!\,r^{-6}$, and the spectrum reaches 0. The last two assertions of (c) follow from (b), applied to $:\chi^6:$ and to $:\chi^3:$.

(d) The photons are the Gaussian family of (a) with $\mathsf{M}=\mathbb{R}^2\setminus\{0\}$, $d\mu=d^2p$, $\omega=|p|$, $\mathsf{P}=\sum_\alpha t_\alpha^2$, $\kappa_0=(2\pi)^{-1}(2|p|)^{-1/2}$, $\kappa_j=-ip_j\kappa_0$, $\kappa_3=-\omega\kappa_0$, $\varepsilon=(+,+,-)$, $G_{\alpha\beta}=-\partial_\alpha\partial_\beta(4\pi|x|)^{-1}$; the form of (a) is $:|\nabla\varphi|^2-\dot{\varphi}^2:=-\Theta$. With $\|\omega^{-1/2}e^3_{ue_3}\|^2=(16\pi u^2)^{-1}$ and $\|\omega^{-1/2}e^j_{ue_3}\|^2=(32\pi u^2)^{-1}$ the three terms of B_F are bounded by $(\pi e)^{-1}\min\{s,s'\}^{-3}$ each, so $C_\wp=2^3\cdot3/\pi e=24/\pi e$; $S_2=2\sum_{\alpha\beta}G_{\alpha\beta}^2=12/16\pi^2r^6$ and $\|\mathsf{B}(s,s)\Omega\|^2=S_2(2s)$.

The span of the Θ-vectors is the even Fock space, as in (e) at $\mathsf{n}=1$. *The linear bound.* Let $\tilde f(k)=\int f(x)e^{i(k^0t-\vec k\cdot\vec x)}\,\mathrm{d}x$ (the transform $\hat f$ of the setting) and $\kappa_f(\vec k)=\frac{1}{2\pi^2}\sup_{k^0}|\tilde f(k^0,\vec k)|\le 8|f|_8(1+|\vec k|^2)^{-2}$, $\theta(f)=\int\kappa_f\,d^2k\le 8\pi|f|_8$. The field $\phi(f)=-\Theta(f)$, f real, has a part with one creator and one annihilator, with the kernel $t_f(p,q)=2\kappa_0(p)\kappa_0(q)(\vec p\cdot\vec q-\omega_p\omega_q)\tilde f(\omega_p-\omega_q,\vec p-\vec q)$, of modulus $\le\sqrt{\omega_p\omega_q}\,\kappa_f(\vec p-\vec q)$, bounded by $\theta(f)H$ by Young's inequality; and the two pair parts $\mathsf{Z}^+(w_f)+\mathsf{Z}^+(w_f)^\dagger$, $\mathsf{Z}^+(c)=\iint c(p,q)a_p^*a_q^*\,\mathrm{d}p\,\mathrm{d}q$, with the kernel $w_f(p,q)=\sum_\alpha\kappa_\alpha(p)\kappa_\alpha(q)\tilde f(\omega_p+\omega_q,\vec p+\vec q)$, of modulus $\le\frac12\sqrt{\omega_p\omega_q}\,\kappa_f(\vec p+\vec q)$, so that $w_f/\sqrt{\omega_p\omega_q}$ has the operator norm $\le\frac12\theta(f)$.

For them a square is completed: with $\epsilon>0$ and $g_p=\pm w_f(p,\cdot)/\epsilon\omega_p$, $0\le\int\omega_p\|(a_p+a^*(g_p))\psi\|^2\,\mathrm{d}p$ is $\langle\psi,H\psi\rangle$, plus the pair form at ψ divided by $\pm\epsilon$, plus $\int\omega_p\|a(g_p)\psi\|^2\,\mathrm{d}p\le\frac{\theta(f)^2}{4\epsilon^2}\langle\psi,H\psi\rangle$, plus $\int\omega_p\|g_p\|^2\,\mathrm{d}p\,\|\psi\|^2\le\frac{\mathsf{n}(f)}{4\epsilon^2}\|\psi\|^2$, $\mathsf{n}(f)=\frac{1}{4\pi^4}\iint\omega_q|\tilde f(\omega_p+\omega_q,\vec p+\vec q)|^2\le 32\pi^3|f|_8^2$; so $\pm(\mathsf{Z}^+(w_f)+\mathsf{Z}^+(w_f)^\dagger)\le(\epsilon+\frac{\theta(f)^2}{4\epsilon})H+\frac{\mathsf{n}(f)}{4\epsilon}$, and at $\epsilon=\frac{\vartheta}{2}$, $\vartheta\ge\theta(f)$: $\le\vartheta H+\mathsf{n}(f)/2\vartheta$.

At $\vartheta=8\pi|f|_8\ge\theta(f)$: $\pm\Theta(f)\le 16\pi|f|_8H+2\pi^2|f|_8\le 16\pi|f|_8N$. The same bound for $\Theta(\partial_tf)=-i[H,\Theta(f)]$ gives the operator bound and essential self-adjointness on every core of H; $\|\Theta(f)\psi\|\le c(f)(n+2)\|\psi\|$ on vectors with at most n particles gives the growth; (W) and (CP) by (c). *Order exactly one* on the even space: the squeezed vectors of (e) at $\mathsf{n}=1$. *At one time.* For $m\le n$ write $\Theta_k=\Theta^{(0)}+\Theta^{(-)}+\Theta^{(+)}$ (number-preserving, annihilating and creating parts), with $\tilde k(\xi)=\int k(\vec x)e^{-i\xi\cdot\vec x}d^2x$, $\Theta^{(0)}=\iint(4\pi^2)^{-1}\sqrt{\omega_p\omega_q}(1-\hat p\cdot\hat q)\tilde k(p-q)\,a_p^*a_q$, $\Theta^{(-)}=\iint\Xi(p,q)\,a_pa_q$, $\Xi(p,q)=-(8\pi^2)^{-1}\sqrt{\omega_p\omega_q}(1-\hat p\cdot\hat q)\tilde k(-p-q)$, and $\Theta^{(+)}=(\Theta^{(-)})^\dagger$.

The annihilation and creation densities select energies: $\langle\chi,a_pa_q\psi\rangle=0$ unless $\omega_p+\omega_q\in(n-m-1,n-m+1)$ for $\chi\in P_m\mathcal{F}$, $\psi\in P_n\mathcal{F}$. The integral operators with the kernels $\tilde k(q\mp p)$ have norms $\le(2\pi)^2\|k\|_\infty$ (Plancherel), and with the factor $1-\hat p\cdot\hat q$ at most twice that; multiplication of a kernel by a window $\eta(\omega_p+\omega_q-\omega)$, η the inverse Fourier transform of an integrable $\hat\eta$, multiplies its norm by at most $\frac{1}{2\pi}\|\hat\eta\|_{L^1}$, which is at most $\sqrt3$ (Schwarz) for $\eta_2=\mathbf{1}_{[-\frac32,\frac32]}*\mathbf{1}_{[-\frac12,\frac12]}$, a function equal to 1 on $[-1,1]$ and carried by $[-2,2]$.

With $\omega=n-m$ and $\Xi_\eta(p,q)=\Xi(p,q)\eta_2(\omega_p+\omega_q-\omega)$, $P_m\Theta^{(-)}P_n=P_m\iint\Xi_\eta a_pa_qP_n$. So $\|P_m\Theta^{(0)}P_n\|\le 2\|k\|_\infty\sqrt{(m+1)(n+1)}$ and, with $g_q=\overline{\Xi_\eta(\cdot,q)}\omega_q^{-1/2}$,

$$\mathsf{h}(\omega)^2=\iint|\Xi_\eta(p,q)|^2\omega_q^{-1}\,\mathrm{d}p\,\mathrm{d}q=\int\|g_q\|^2\,\mathrm{d}q$$

and $\|a^*(g)\chi\|^2=\|a(g)\chi\|^2+\|g\|^2\|\chi\|^2$,

$$\|P_m\Theta^{(-)}P_n\|\le\sqrt{n+1}\,[\sqrt3\|k\|_\infty\sqrt{m+1}+\mathsf{h}(\omega)],\qquad \mathsf{h}(\omega)^2\le\frac{\omega+2}{(2\pi)^4}\int|\tilde k(\xi)|^2|\mathcal{A}_\xi|\,\mathrm{d}\xi\le(\omega+2)^2\mathsf{n}_k^2,$$

$\mathcal{A}_\xi=\{p:|p|+|\xi-p|\in[\omega-2,\omega+2]\}$, of area $\le 10.4(\omega+2)$ for $|\xi|\le\frac12(\omega+2)$ (confocal ellipses) and $\le\frac{\pi}{2}(\omega+2)|\xi|$ otherwise; $\Theta^{(+)}$ vanishes between these shells for $m<n$. This is the upper bound. For the lower bound let $\Xi_1=\Xi\,\eta_1(\omega_p+\omega_q-\omega)$, $\eta_1=(\frac12-|\cdot|)_+$, $L_1=\iint\Xi_1a_pa_q$; then $[L_1,L_1^*]\ge 2\|\Xi_1\|_2^2$, and the vectors $\chi_j=(L_1^*)^j\Omega$ have $\|\chi_{j+1}\|^2\ge 2(j+1)\|\Xi_1\|_2^2\|\chi_j\|^2$; as L_1^* raises the energy by $\omega\pm\frac12$, one shell P_m in the range $[j(\omega-\frac12),j(\omega+\frac12)]$ has $\|P_nL_1^*P_m\|\ge(\frac{j+1}{3})^{1/2}\|\Xi_1\|_2$ for one of three n; the window costs the factor $\frac12$, and $\|\Xi_1\|_2^2\ge 2.3\cdot10^{-5}\|k\|_2^2\omega^3$ for $\omega\ge\max\{3,7\xi_0(k)+1\}$, because on the ring $\mathcal{A}'_\xi=\{|p|+|\xi-p|\in[\omega-\frac14,\omega+\frac14]\}$, $|\xi|\le\xi_0$, where $|p|\ge 3\xi_0$, the kernel has $|\Xi(p,\xi-p)|^2\ge\frac43|p|^2|\tilde k(\xi)|^2/64\pi^4$; the number-preserving part is at most $\kappa_k(\omega-2)G$ there, as only momentum transfers $|p-q|>\omega-2$ occur. The band statement follows by taking $j\approx(\omega+1)^{1/\epsilon}/\omega$.

(e) The cyclic space. The Θ-vectors are $\mathrm{O}(\mathsf{n})$-invariant. For the converse use the explicit bilocal calculation in steps (1)–(4) of (f) below, first as identities of Wick polynomials, before taking commutants. The timelike matrix $\mathsf{H}_{\mu\nu}=\partial_\mu\partial_\nu\Delta_{\mathrm{c}}$ is invertible. Point splitting gives its traceless symmetric contractions, and adjoining the already available trace Θ gives every local symmetric component. The two contractions $5\mathsf{X}+\mathsf{Ht}$ and $8\mathsf{t}$ in (3) then recover every bilocal component

at timelike separated supports. The commutator identity and overlapping-cone chain in (4) give the colour sums $\sum_a \varphi^a(h)\varphi^a(h')$ for arbitrary real integral-zero tests in a connected double cone, including spacelike separated supports. These are limits of polynomials in the $\Theta(f)$ on the common finite-particle energy-smooth core, with the finite-particle bounds used in those limits. This argument applies to the polynomial vectors themselves and does not use the cyclic-space identification it proves. In particular, for real g of integral zero in a double cone O, the sums of squares $\sum_a \varphi^a(g)^2$ are such limits.

One more commutator gives the squares: let $O_3 \subset O$ be a smaller double cone that holds $\operatorname{supp} g$, so that $g = \partial_\mu \mathsf{G}^\mu$ with $\mathsf{G} \in C_c^\infty(O_3)^3$, take a ball $O_1 \subset O$ spacelike to O_3, a ball $O_2 \subset O$ in the common future of O_1 and O_3, and $g_1, g_3 \in \mathcal{F}_{\mathbb{R}}(O_1)$ of integral 0 with $\Delta_{\mathrm{c}}(g_1, g_3) \neq 0$; then $\sum_a \varphi^a(g_i)\varphi^a(g)$, $i = 1, 3$, are such limits up to numbers, and, as $\Delta_{\mathrm{c}}(g, g_3) = \Delta_{\mathrm{c}}(g_1, g) = \Delta_{\mathrm{c}}(g, g) = 0$,

$$\Big[\sum_a \varphi^a(g_1)\varphi^a(g), \sum_b \varphi^b(g_3)\varphi^b(g)\Big] = i\Delta_{\mathrm{c}}(g_1, g_3) \sum_a \varphi^a(g)^2;$$

by the first fundamental theorem for $\mathrm{O}(\mathsf{n})$ the invariant polynomials in the $\varphi^a(g)$ are generated by the scalar products $\sum_a \varphi^a(g)\varphi^a(g')$, which polarisation reduces to squares; and the invariant polynomial vectors are dense in the invariant vectors. The Euclidean vectors give the same space by (a). With SO(3): $\sum \epsilon_{abc} a_a^*(u) a_b^*(v) a_c^*(w)\Omega \neq 0$ is SO(3)-invariant, has three particles and is orthogonal to the even space. $\|\mathsf{B}(s, s)\Omega\|^2 = \mathsf{n} S_2(2s)$ with the S_2 of one colour. *Order one.* For a bounded unit $u \in \mathfrak{h}$ of compact support put $A^* = \sum_a a_a^*(u)^2$; then $A(A^*)^k\Omega = 2k(2k - 2 + \mathsf{n})(A^*)^{k-1}\Omega$, and the singlet squeezed vectors $\chi_\lambda = (1 - \lambda)^{\mathsf{n}/4} \exp(\frac{\sqrt{\lambda} e^{i\beta}}{2} A^*)\Omega$, $0 \leq \lambda < 1$, are invariant unit vectors of $\mathcal{D}_\infty(H)$.

With the one-particle operator T_f of the kernel $\frac{\omega_p\omega_q - \vec{p}\cdot\vec{q}}{4\pi^2\sqrt{\omega_p\omega_q}} \tilde{f}(\omega_p - \omega_q, \vec{p} - \vec{q})$ (the kernel $-t_f$ of (d)) and $c_f(p, q) = \frac{\omega_p\omega_q - \vec{p}\cdot\vec{q}}{8\pi^2\sqrt{\omega_p\omega_q}} \tilde{f}(\omega_p + \omega_q, \vec{p} + \vec{q})$ (the w_f of (d)), the field of n colours is $\Theta(f) = d\Gamma(T_f \otimes 1) - \mathsf{Z}^+(c_f) - \mathsf{Z}^+(c_{\bar{f}})^\dagger$, $\mathsf{Z}^+(c) = \sum_a \iint c(p, q) a_a^*(p) a_a^*(q)\, \mathrm{d}p\, \mathrm{d}q$, and $\phi(f) = -\Theta(f)$; so

$$\langle \chi_\lambda, \phi(f)\chi_\lambda\rangle = \frac{\mathsf{n}}{1 - \lambda}\big[2\sqrt{\lambda}\operatorname{Re}(e^{i\beta}\mathsf{w}) - \lambda\mathsf{t}\big], \qquad \langle \chi_\lambda, H\chi_\lambda\rangle = \frac{\mathsf{n}\lambda\,\mathsf{e}}{1 - \lambda},$$

$\mathsf{t} = \langle u, T_f u\rangle$, $\mathsf{w} = \langle c_f, u \otimes u\rangle$, $\mathsf{e} = \langle u, \omega u\rangle$. Some u has $|\mathsf{t}| + 2|\mathsf{w}| > 0$: otherwise $\tilde{f}$ would vanish on $\{|k^0| \leq |\vec{k}|\}$ and on $\pm\bar{V}_+$. With Jensen's inequality $\langle \psi, N^\alpha\psi\rangle \leq \langle \psi, N\psi\rangle^\alpha$ the quotient grows like $(1 - \lambda)^{\alpha-1}$. Either sign: the one-particle operator T_f has a continuous kernel that vanishes on the diagonal, so it is 0 or indefinite; choose u with t of the sign needed, or with $\mathsf{w} \neq 0$ if $T_f = 0$, and β. These vectors are limits in every $\|N^l \cdot\|$ of vectors of $\mathcal{D}_0$. The upper bound adds the bounds of the n colours. If $b_{mn}(f) \leq \beta_p(1 + \max\{m, n\})^p$ with $p < 1$, an interpolation of this bound with the decay of Thm. AO(b2) and the crude bound $b_{mn}(f) \leq c(2 + \min\{m, n\})^\wp$, from Thm. AO(b5) at $j = 0$ and $b_{mn}(f) = b_{nm}(\bar{f})$, gives $b_{mn}(f) \leq C(2 + m)^{p'/2}(2 + n)^{p'/2}(1 + |m - n|)^{-2}$ with $p < p' < 1$, hence a form bound with $N^{p'}$, which the squeezed vectors exclude.

(f) For a field Φ among Θ, $S = {:}\varphi\cdot\varphi{:}$, the multiplet $\Theta_{\mu\nu} = {:}\partial_\mu\varphi\cdot\partial_\nu\varphi{:}$ and the $T_{\mu\nu} = \Theta_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\Theta$, all on the Fock space, $\mathcal{D}_0^\Phi$ is the span of Ω and of the $\Phi(f_1)\cdots\Phi(f_n)\Omega$, $\mathcal{H}_\Phi$ its closure, and $\mathcal{C}_\Phi(O)$, $\mathcal{A}_\Phi(O) = \mathcal{C}_\Phi(O)'$ and $\mathcal{A}_\Phi^d(O) = \bigcap\{\mathcal{A}_\Phi(\tilde{W}) : \tilde{W} \supset O\}$ are formed as in the setting, with Φ and $\mathcal{D}_0^\Phi$ in the place of ϕ and $\mathcal{D}_0$. Θ at the mass 0 (by (d), (e)), Θ at a mass $\mathsf{m} > 0$ (by (a), with $\delta = 1$, as $\|\omega^{-1/2}\kappa_\alpha e^{-s\omega}\|^2 \leq (16\pi s^2)^{-1}$) and S at $\mathsf{m} > 0$ (by (a), as in (c)) are data (Rec), and V carries their nets onto these; so Thm. AO(e) applies to them. For S at the mass 0, $\mathcal{A}_S^d$ is the net of this definition through wedges. $\mathfrak{R}(\mathsf{K})$ is the von Neumann algebra of the Weyl operators of the vectors of a closed real subspace K of the one-particle space, and K' the symplectic complement.

Let first $\mathsf{n} = 1$, $\mathsf{m} = 0$. With $\mathsf{H}_{\kappa\lambda} = \partial_\kappa\partial_\lambda\Delta_{\mathrm{c}}$ and $\mathsf{X}_{\kappa\lambda}(x, y) = {:}\partial_\kappa\varphi(x)\partial_\lambda\varphi(y){:}$, on vectors with finitely many particles and for timelike separated supports,

$$[\Theta(f_1), \Theta(f_2)] = -4i \iint f_1(x) f_2(y)\, \mathsf{H}^{\kappa\lambda}(x - y)\, \mathsf{X}_{\kappa\lambda}(x, y)\, \mathrm{d}x\, \mathrm{d}y + c\,1,$$

and $\Delta_c(z) = \mp(2\pi)^{-1}(z\cdot z)^{-1/2} \neq 0$ on $\pm V_+$, inside the cone in $1+2$ dimensions; on V_+, $-2\pi(z\cdot z)^{3/2}\mathsf{H}_{\kappa\lambda}(z) = 3z_\kappa z_\lambda/(z\cdot z) - \eta_{\kappa\lambda}$, whose mixed form has the eigenvalues $2,-1,-1$, so that H is invertible at timelike points. For Θ the following identities are used on the common energy-smooth core. First time-smooth the bounded weakly commuting operator Q as in the proof of Thm. AO(a), working in a region with closure in O; the linear energy bounds and essential self-adjointness extend the polynomial-domain identities to this core. Products and commutators are taken there, and the displayed Wick and point-splitting limits pass with their finite-particle bounds. At the end let the smoothing tend to the identity. This is a statement about these cores and limits, not an unrestricted rule for weak commutants of unbounded operators. Let $Q \in \mathcal{C}_\Theta(O)$, O open and connected. (1) The display gives the contractions $\mathsf{X}[\Psi\mathsf{H}] = \iint \Psi\,\mathsf{H}^{\kappa\lambda}\mathsf{X}_{\kappa\lambda}$, Ψ scalar in C_c^∞ of the timelike pairs of $O\times O$: one component of nine. (2) For $f\in\mathcal{F}(O)$, $\chi\in C_c^\infty(V_+)$ and small ϵ, $\Psi(x,y) = -2\pi f(y)\epsilon^{-3}\chi(\frac{x-y}{\epsilon})((x-y)\cdot(x-y))^{3/2}$ makes $\Psi\mathsf{H}$, as $3z_\kappa z_\lambda/(z\cdot z)-\eta_{\kappa\lambda}$ is homogeneous of degree 0, the function $f(y)\epsilon^{-3}\zeta(\frac{x-y}{\epsilon})$ with $\zeta(z) = \chi(z)(3z_\kappa z_\lambda/(z\cdot z)-\eta_{\kappa\lambda})$, and point splitting gives $3\mathsf{N}^{\kappa\nu}[\chi]\Theta_{\kappa\nu}(f) - (\int\chi)\Theta(f)$, $\mathsf{N}^{\kappa\nu}[\chi] = \int\chi\, z^\kappa z^\nu/(z\cdot z)$; these tensors span the traceless symmetric ones, and adjoining the trace Θ, with which Q already commutes, gives all symmetric components: Q commutes weakly with all $\Theta_{\kappa\nu}(f)$, $f\in\mathcal{F}(O)$. (3) $[\Theta_{\kappa\lambda}(f_1),\Theta_{\mu\nu}(f_2)] = -i\iint f_1 f_2\,\mathsf{Y}_{\kappa\lambda,\mu\nu} + c$, $\mathsf{Y}_{\kappa\lambda,\mu\nu} = \mathsf{H}_{\lambda\mu}\mathsf{X}_{\kappa\nu} + \mathsf{H}_{\lambda\nu}\mathsf{X}_{\kappa\mu} + \mathsf{H}_{\kappa\mu}\mathsf{X}_{\lambda\nu} + \mathsf{H}_{\kappa\nu}\mathsf{X}_{\lambda\mu}$, and with $\mathsf{t} = (\mathsf{H}^{-1})^{\lambda\mu}\mathsf{X}_{\lambda\mu}$: $(\mathsf{H}^{-1})^{\lambda\mu}\mathsf{Y}_{\kappa\lambda,\mu\nu} = 5\mathsf{X}_{\kappa\nu} + \mathsf{H}_{\kappa\nu}\mathsf{t}$ and $(\mathsf{H}^{-1})^{\kappa\nu}(\mathsf{H}^{-1})^{\lambda\mu}\mathsf{Y}_{\kappa\lambda,\mu\nu} = 8\,\mathsf{t}$; so $\iint\Psi^{\kappa\nu}\mathsf{X}_{\kappa\nu} = \iint\Upsilon^{\kappa\lambda\mu\nu}\mathsf{Y}_{\kappa\lambda,\mu\nu}$ with $\Upsilon^{\kappa\lambda\mu\nu} = \frac{1}{5}[\Psi^{\kappa\nu}(\mathsf{H}^{-1})^{\lambda\mu} - \frac{1}{8}(\Psi^{\rho\sigma}\mathsf{H}_{\rho\sigma})(\mathsf{H}^{-1})^{\kappa\nu}(\mathsf{H}^{-1})^{\lambda\mu}]$: here the invertibility of H solves for every bilocal field $\mathsf{X}[\Psi]$, Ψ a matrix in C_c^∞ of the timelike pairs. (4) For double cones K, K' in O write $K\approx K'$ if Q commutes weakly with all $\varphi(h)\varphi(h')$, $h\in\mathcal{F}_\mathbb{R}(K)$, $h'\in\mathcal{F}_\mathbb{R}(K')$ of integral 0; such h are divergences $\partial_\kappa \mathsf{k}^\kappa$, so $\varphi(h)\varphi(h')$ is a bilocal field up to a number, and (3) gives $K\approx K'$ when all pairs of $K\times K'$ are timelike.

In

$$\begin{aligned}&[\varphi(k)\varphi(k'),\varphi(l)\varphi(l')]\\ &= i\Delta_c(k',l)\varphi(k)\varphi(l')\\ &+i\Delta_c(k',l')\varphi(k)\varphi(l)\\ &+i\Delta_c(k,l)\varphi(l')\varphi(k')\\ &+i\Delta_c(k,l')\varphi(l)\varphi(k'),\end{aligned}$$

with k,k',l,l' of integral 0: $k=l$ in K and k',l' in K' give $K\approx K$ when $K\approx K'$ (the left member and the first and fourth terms are known, the third has the factor $\Delta_c(k,k)=0$, and Δ_c does not vanish identically on the pairs of functions of integral 0 in K'); k in K, k',l in K'' and l' in K' give $K\approx K'$ from $K\approx K''$, $K''\approx K''$, $K''\approx K'$ (the left member and the second, third and fourth terms are known). Every point of O has a small double cone K with $K\approx K$ (with a translate of K along the time axis), the classes of the relation so induced on O are open, and O is connected: there is one class.

With a Lebesgue number of a cover of a compact connected set that holds the supports, every $h\in\mathcal{F}_\mathbb{R}(O)$ of integral 0 is cut into pieces of integral 0 in small double cones along a chain, $h = \sum_i(\chi_i h - \vartheta_i b_i) + \sum_i \vartheta_i(b_i - b_1)$, $\vartheta_i = \int\chi_i h$, $\int b_i = 1$, χ_i a partition of unity and $b_i - b_1$ cut along the chain: Q commutes weakly with all $\varphi(h)\varphi(h')$, $h,h'\in\mathcal{F}_\mathbb{R}(O)$ of integral 0. (5) The cosine series converges on vectors with finitely many particles: Q commutes with the $\cos\bar\varphi(h)$, $h\in\mathcal{F}_\mathbb{R}(O)$, $\int h = 0$. Conversely, a bounded Q on the even space that commutes with these cosines lies in $\mathcal{C}_\Theta(O)$: $t^{-2}(1-\cos t\bar\varphi(h))\psi \to \frac{1}{2}\varphi(h)^2\psi$ on vectors with finitely many particles, polarisation gives the $\varphi(h)\varphi(h')$, with $h = \partial_\kappa\mathsf{k}^\kappa$ all bilocal fields smeared in $O\times O$, and point splitting the $\Theta_{\kappa\lambda}(f)$ and $\Theta(f)$, $f\in\mathcal{F}(O)$, whose polynomial domain lies in that of these limits.

So $\mathcal{C}_\Theta(O) = \{\cos\bar\varphi(h) : \int h = 0\}'$ on the even space, $\mathcal{A}_\Theta(O)$ is the even part of the algebra of the field strengths, $\mathfrak{R}(\mathsf{K}_0(O))$ restricted to the even space (with $\mathsf{K}_0(O)$ below), and the multiplet has the same weak commutants; the closures of the $T_{\kappa\lambda}(f)$ are self-adjoint by the linear bound and affiliated

by this identification; (W) follows from the Weyl relations. For n colours the same proof gives the net, with the $\mathrm{O}(\mathsf{n})$-invariant parts, the bounded functions of the operators $\sum_a(\sum_i \alpha_i x_{ia} + \beta_i p_{ia})^2$ on $L^2(\mathbb{R}^{n'\times\mathsf{n}})$, for every finite number n' of modes, generating the commutant of the rotations of the colours; at $\mathsf{n}=3$ it gives the affiliation of the $T_{\kappa\lambda}(f)$ as well.

The defect. For a region O let $\mathsf{K}(O)$ and $\mathsf{K}_0(O)$ be the closed real spans of the one-particle vectors of the $\varphi(h)$, $h\in\mathcal{F}_{\mathbb{R}}(O)$, respectively with $\int h = 0$; at one time, $\mathsf{K}[D]$ is the closed real span of the one-particle vectors $u_f + v_g$ of $\varphi(0,f)$, $\pi(0,g)$, $f,g\in C_c^\infty(D)$ real, and $\mathsf{K}_0[D]$ that with $\int f = 0$; $D_\rho = \{|\vec{x}|<\rho\}$ and $D_\rho^e = \{|\vec{x}|>\rho\}$. For O the double cone over the disc D_ρ the Cauchy data at the time 0 and the finite speed give $\mathsf{K}(O) = \mathsf{K}[D_\rho]$, $\mathsf{K}_0(O) = \mathsf{K}_0[D_\rho]$, $\mathsf{K}(O^\times) = \mathsf{K}[D_\rho^e]$; and outside the charges escape to infinity: for $b\in C_c^\infty(D_\rho^e)$ with $\int b = 1$, $b_n = n^{-2}b(\cdot/n)$ has $\|u_{b_n}\|^2 = n^{-1}\|u_b\|^2 \to 0$, so $u_f = \lim_n u_{f-(\int f)b_n}$ and $\mathsf{K}_0(O^\times) = \mathsf{K}(O^\times) = \mathsf{K}[D_\rho^e]$.

The symplectic complements: $\mathsf{K}[D_\rho^e]' = \overline{\mathsf{K}[D_\rho]}$, because for $w = w_1 + iw_2 \in \mathsf{K}[D_\rho^e]'$, w_1, w_2 in the real subspace of the w with $w(-\vec{k}) = \overline{w(\vec{k})}$, the distributions F, G with $\tilde{F} = w_1$, $\tilde{G} = w_2/\omega$ have $\mathrm{Im}\langle u_f + v_g, w\rangle = \frac{1}{2}(\int fG - \int gF)$, so they vanish on D_ρ^e and are carried by $\bar{D}_\rho$; a dilation by $\lambda<1$, uniformly bounded on $L^2(d^2k/\omega)$ and on $L^2(\omega\, d^2k)$ and strongly continuous as $\lambda\uparrow 1$, and a mollifier make w a limit of vectors $u_f + v_g$ with $f,g\in C_c^\infty(D_\rho)$; so $\mathsf{K}[D_\rho]' = \mathsf{K}[D_\rho^e]'' = \mathsf{K}[D_\rho^e]$; and $\mathsf{K}_0[D_\rho]' = \mathsf{K}[D_\rho^e] + \mathbb{R}v_{g_1}$, $g_1 = 1$ near $\bar{D}_\rho$, because $u_f + v_g \mapsto \int f = 2\mathrm{Im}\langle u_f + v_g, v_{g_1}\rangle$ is a continuous functional on $\mathsf{K}[D_\rho]$ with the kernel $\mathsf{K}_0[D_\rho]$.

The duality theorem for the free field, $\mathfrak{R}(\mathsf{K})' = \mathfrak{R}(\mathsf{K}')$, turns these into $\mathcal{C}_\Theta(O^\times) = \mathcal{A}_\Theta(O^\times)'$, the even part of $\mathfrak{R}(\mathsf{K}[D_\rho])$, and $\mathcal{C}_\Theta(O) = \mathcal{A}_\Theta(O)'$, the even part of $\mathfrak{R}(\mathsf{K}[D_\rho^e] + \mathbb{R}v_{g_1})$; the Weyl relation $e^{i\bar{\varphi}(h)}e^{i\mathsf{s}\bar{\pi}(g_1)} = e^{-i\mathsf{s}\int h}e^{i\mathsf{s}\bar{\pi}(g_1)}e^{i\bar{\varphi}(h)}$ gives the non-commuting pair. For $\mathsf{n}\geq 1$ colours, $h\in\mathcal{F}_{\mathbb{R}}(O)$ with $\mathsf{c} = \int h \neq 0$ and $\vec{n}$ uniform on the sphere $S^{\mathsf{n}-1}$, the mean over $\mathrm{O}(\mathsf{n})$ of $e^{i\bar{\varphi}^1(h)}$ is $X = \int e^{i\bar{\varphi}(h\vec{n})}d\vec{n}$, $\varphi(h\vec{n}) = \sum_a n_a\varphi^a(h)$, and its restriction to the invariant vectors lies in $\mathcal{C}_\Theta(O^\times)$ by the Weyl relations.

If its restriction to the invariant vectors lay in $\mathcal{A}_\Theta(O)$, then, Ω being separating for the algebra of the field of O, so that the restriction to the invariant vectors is injective on its invariant part, X would lie in the algebra of the field strengths of O, which commutes with the shifts $e^{i\vec{s}\cdot\bar{\pi}(g_1)}$, $\bar{\pi}(g_1) = (\bar{\pi}^a(g_1))_a$; but conjugation by them multiplies $e^{i\bar{\varphi}(h\vec{n})}$ by $e^{i\mathsf{c}\vec{s}\cdot\vec{n}}$, and the one-particle part of the difference on Ω, $ie^{-\|u_h\|^2/2}u_h \otimes \int(e^{i\mathsf{c}\vec{s}\cdot\vec{n}} - 1)\vec{n}\,\mathrm{d}\vec{n}$, u_h the one-particle vector of $\varphi(h)$, is $\neq 0$ for small $\vec{s}\neq 0$.

So the restriction of X lies in $\mathcal{A}_\Theta^d(O) = \mathcal{C}_\Theta(O^\times)$ and outside $\mathcal{A}_\Theta(O)$. The even parts, restricted to the even Fock space, of the nets $\mathfrak{F}_\Lambda(O) = \{e^{i\bar{\varphi}(f)} : f\in\mathcal{F}_{\mathbb{R}}(O),\ \int f\in\Lambda\}''$ are local, lie between, are injective in Λ (an h with $\int h\in\Lambda\setminus\Lambda'$ separates), and have the wedge algebras of Θ, because in a wedge the field strengths generate the field, the charges escaping to infinity inside the wedge: for $f\in\mathcal{F}_{\mathbb{R}}(W)$, $f_n = n^{-3}f(\cdot/n)\in\mathcal{F}_{\mathbb{R}}(W)$ has $\int f_n = \int f$ and $\|u_{f_n}\|^2 = n^{-1}\|u_f\|^2\to 0$, while $f - f_n$ has the integral 0. At $\mathsf{m}>0$ the cosines of all $h\in\mathcal{F}_{\mathbb{R}}(O)$ are reached, as $\varphi(h) = \varphi(\partial_\kappa(-\mathsf{m}^{-2}\partial^\kappa h))$, and the one-particle duality, which holds with $\omega = (|\vec{k}|^2 + \mathsf{m}^2)^{1/2}$, gives $\mathcal{C}_\Theta(O^\times) = \mathcal{A}_\Theta(O)$.

This is the case $\mathsf{n}=1$. For n colours at a mass $\mathsf{m}>0$ the steps (1)–(4) run with the colour sums in the place of the products of one field: the commutator of two $\Theta(f_i)$ is the colour sum of the bilocal fields, weighted by the matrix H of the second derivatives of the commutator function of the mass m, which is invertible at short timelike separations, $\mathsf{m}(z\cdot z)^{1/2} < 2$, and differs from that of the mass 0 by terms less singular at $z=0$, which the point splitting of (2) does not see; the double cones of (4) are taken small; and the commutators of the colour sums $\sum_a\varphi^a(h)\varphi^a(h')$ have the structure constants of the display of (4) at every mass.

So every $Q\in\mathcal{C}_\Theta(O)$, O open and connected, commutes weakly with the colour sums of all real $h,h'\in\mathcal{F}(O)$ of integral 0, and, by $\varphi^a(h) = \varphi^a(\partial_\kappa(-\mathsf{m}^{-2}\partial^\kappa h))$, with those of all real $h,h'\in\mathcal{F}(O)$; among them are the squares $\sum_a\varphi^a(g)^2$, whose closures generate $\mathcal{A}_S(O)$, and point splitting gives the $S(f)$, $f\in\mathcal{F}(O)$: $\mathcal{C}_\Theta(O)\subset\mathcal{C}_S(O)$, so $\mathcal{A}_S(O)\subset\mathcal{A}_\Theta(O)$ on $\mathcal{H}_\Theta = \mathcal{H}_S$. $\Theta(f) = \frac{1}{2}S((\Box + 2\mathsf{m}^2)f)$ gives the converse. So $\mathcal{A}_\Theta = \mathcal{A}_S$ on every connected open region, wedges included, and Thm. AO(e)

for Θ with the duality theorem for the free field gives, as at the mass 0 below, $\mathcal{A}^d_\Theta(O) = \mathcal{C}_\Theta(O^\times)$, the invariant part of the algebra of the field of O, which is $\mathcal{A}_S(O) = \mathcal{A}_\Theta(O)$: the net of Θ, as that of S, is its own dual net, for every $\mathsf{n} \geq 1$.

The massless square. We first establish the operator core for $S = {:}\varphi \cdot \varphi{:}$. This uses particle number, rather than the energy bound used for Θ. Work in the one-particle realization $L^2(\mathbb{R}^2, d^2p)$, with $\omega_p = |p|$. The Wick expansion of $S(f)$ has a number-preserving part and pair-creation and pair-annihilation parts. Apart from the fixed Fourier normalization and the factor 2 in the number-preserving part, their kernels are

$$K_f(p,q) = \frac{\hat{f}(\omega_p - \omega_q, p-q)}{\sqrt{\omega_p \omega_q}}, \qquad B_f(p,q) = \frac{\hat{f}(\omega_p + \omega_q, p+q)}{\sqrt{\omega_p \omega_q}}.$$

For $f \in \mathcal{S}$, the first kernel defines a bounded operator and the second is Hilbert–Schmidt. To see the first assertion, on $|p|, |q| \geq 1$ use Schur's test and $|\hat{f}(\omega_p - \omega_q, p-q)| \leq C_M (1 + |p-q|)^{-M}$, with $M > 2$. The complementary part is Hilbert–Schmidt: $|p|^{-1}$ is locally integrable in two dimensions, and the same decay controls the other variable when its radius is large. For the pair kernel, decay in the energy sum gives

$$\iint |B_f(p,q)|^2 \, d^2p \, d^2q \leq C_M \int_0^\infty \int_0^\infty (1 + r + s)^{-M} \, \mathrm{d}r \, \mathrm{d}s < \infty \qquad (M > 2).$$

The finitely many colours only change the constants.

Let $\mathcal{F}_{\leq k}$ denote the subspace of vectors with at most k particles. The creation and annihilation formulas therefore give

$$S(f)\mathcal{F}_{\leq k} \subset \mathcal{F}_{\leq k+2}, \qquad \|S(f)\psi\| \leq c_f (k+2) \|\psi\|.$$

These are particle-sector bounds; the operator $N = 1 + H$ retains its earlier meaning. For real f and $\psi \in \mathcal{F}_{\leq k}$, iteration gives

$$\|S(f)^r \psi\| \leq c_f^r \prod_{j=1}^{r} (k + 2j) \, \|\psi\|.$$

The series $\sum_{r \geq 0} t^r \|S(f)^r \psi\| / r!$ converges for $2c_f|t| < 1$. Every vector of $\mathcal{D}_0^S$ has finitely many particles, and $S(f)$ preserves $\mathcal{D}_0^S$. This domain is dense in $\mathcal{H}_S$ by its definition. The analytic-vector theorem, in the form of [6, Thm. 2, pp. 180–181], thus proves that $S(f)$ is essentially self-adjoint on $\mathcal{D}_0^S$, without using any identification of its weak commutant or its net.

The identification of $\mathcal{H}_S$ with the invariant subspace is also made before using weak commutants. The bilocal commutator construction used in (e), with S in place of Θ, gives the colour sums $\sum_a \varphi^a(g)\varphi^a(g')$ and their squares for all real compactly supported tests, with no integral-zero restriction. They are limits of polynomials in the $S(f)$ on vectors of finitely many particles, with bounds uniform on each $\mathcal{F}_{\leq k}$. The third commutator and one further commutator give the squares; polarization gives the colour sums. The first fundamental theorem for $\mathrm{O}(\mathsf{n})$ expresses every invariant field polynomial in these colour sums. Applying their polynomial approximants to the vacuum and using those finite-particle bounds shows that $\mathcal{H}_S$ contains the invariant polynomial vectors. Those vectors are dense in the invariant subspace, while every S-polynomial vector is invariant. Hence the two spaces agree.

On the common finite-particle energy-smooth domain the symmetric Wick operator extends its restriction to $\mathcal{D}_0^S$. Essential self-adjointness identifies its closure with the closure just obtained. If a bounded operator Q commutes weakly with $S(f)$ on $\mathcal{D}_0^S$, the adjoint-domain criterion first gives $Q\mathcal{D}_0^S \subset D(\overline{S(f)})$ and $\overline{S(f)}Q\psi = QS(f)\psi$ there. Approximation in the graph norm then gives $Q\overline{S(f)} \subset \overline{S(f)}Q$; the same holds for Q^*. Thus Q commutes with the bounded functions of $\overline{S(f)}$, and the Wick products and the limits described above can be taken on the common core.

For S, the squares $\sum_a \varphi^a(g)^2$ are reached for every real $g \in \mathcal{F}(O)$ by these commutators. Consequently $\mathcal{A}_S(O)$ is the invariant part of the algebra of the field of O, and $\mathcal{A}_\Theta(O) \subsetneq \mathcal{A}_S(O)$ at

$\mathsf{m} = 0$. The field strengths generate the field in every wedge, so $\mathcal{A}_S(\tilde{W}) = \mathcal{A}_\Theta(\tilde{W})$ and $\mathcal{A}_S^d = \mathcal{A}_\Theta^d$. Finally, Thm. AO(e) for Θ and the duality theorem for the free field give $\mathcal{A}_\Theta^d(O) = \mathcal{C}_\Theta(O^\times)$, the invariant part of the algebra of the field of O. This is $\mathcal{A}_S(O)$, and therefore $\mathcal{A}_S = \mathcal{A}_S^d = \mathcal{A}_\Theta^d$.

14.1.10. *Proof of Thm. AS. (a)* Let $\Gamma(g)$, $g \in G^\Sigma$, be the gauge transformations on $L^2(G^{\mathsf{B}})$, $\mathfrak{F}(Y) = \mathcal{B}(L^2(G^Y)) \otimes 1$, and $P = \int \Gamma(g)\,\mathrm{d}g$ the projection on $\mathcal{H}_{\mathrm{ph}}$. $\mathfrak{N} = \mathfrak{F}(Y) \cap \Gamma(G^\Sigma)'$ is a von Neumann algebra and $P \in \Gamma(G^\Sigma)'' \subset \mathfrak{N}'$, so $\mathsf{A}(Y) = \mathfrak{N}P$ is a von Neumann algebra with the commutant $P\mathfrak{N}'P|_{\mathcal{H}_{\mathrm{ph}}}$, and $\mathfrak{N}' = \mathfrak{F}(\mathsf{B} \setminus Y) \vee \Gamma(G^\Sigma)''$ is the weak closure of the span of the $X\Gamma(g)$, $X \in \mathfrak{F}(\mathsf{B} \setminus Y)$. As $P\Gamma(g) = P$, $PX\Gamma(g)P = PXP = PX^\natural P$, $X^\natural = \int \Gamma(g)X\Gamma(g)^{-1}\,\mathrm{d}g \in \mathfrak{F}(\mathsf{B} \setminus Y) \cap \Gamma'$; so $\mathsf{A}(Y)' = \mathsf{A}(\mathsf{B} \setminus Y)$. At $Y = \varnothing$ this is $\mathsf{A}(\mathsf{B}) = \mathcal{B}(\mathcal{H}_{\mathrm{ph}})$. Without a common site the gauge groups of Y_1 and Y_2 are independent, $(X_1 \otimes X_2)^\natural = X_1^\natural \otimes X_2^\natural$, and the products are weakly total.

The flips. $V \mapsto cV$ keeps the Haar measure; $\chi_j(-g) = (-1)^{2j}\chi_j(g)$ gives $\mathsf{V}(c)M_{\chi_j(U(C'))} = c(C')^{2j}M_{\chi_j(U(C'))}\mathsf{V}(c)$. Two groups of coboundaries are told apart. If $c_b = z(b_-)z(b_+)$ on all the bonds of the layer, z a sign field on all its sites, $\mathsf{V}(c)$ is the gauge transformation $\Gamma(z)$ by the central field z, which is 1 on $\mathcal{H}_{\mathrm{ph}}$. For a set S of plaquettes, the coboundaries $B^1(S) = \{dz|_{\mathsf{B}_S}\}$, z a sign field on the sites of B_S, form the subgroup of $Z^1(S)$ by which $H^1(S) = Z^1(S)/B^1(S)$ is formed. For $c = dz|_{\mathsf{B}_S}$ and $\tilde{z}$ the continuation of z by 1 to all sites, $\mathsf{V}(c) = \Gamma(\tilde{z})\mathsf{V}(c^\circ)$, where $c^\circ = d\tilde{z}$ on $\mathsf{B} \setminus \mathsf{B}_S$ and $c^\circ = 1$ on B_S, the sign field c° of the statement; so on $\mathcal{H}_{\mathrm{ph}}$, $\mathsf{V}(c) = \mathsf{V}(c^\circ)$, a product of the parities $\sum_j (-1)^{2j}\Pi_b^j$ of bonds $b \notin \mathsf{B}_S$, which lies in $\mathsf{A}(\mathsf{B} \setminus \mathsf{B}_S)$; it is not 1 when, say, $z = -1$ at one site x alone and a plaquette $q \notin S$ has at x one bond in B_S and one off it, for then it anticommutes with $M_{t_q} \neq 0$.

As $\Gamma(\tilde{z})$ commutes with every gauge-invariant operator and $\mathsf{V}(c^\circ)$ with every operator of the bonds of B_S, $\operatorname{Ad}\mathsf{V}(c)$ is the identity on $\mathsf{A}(\mathsf{B}_S)$ for $c \in B^1(S)$; and on $\mathcal{H}_{\mathrm{ph}}$ the flips of one class of $H^1(S)$ differ by elements of $\mathsf{A}(\mathsf{B} \setminus \mathsf{B}_S)$, so that beside $\mathsf{A}(\mathsf{B} \setminus \mathsf{B}_S)$ one flip of each class is enough. On the bonds where $c = -1$, $\mathsf{V}(c)$ acts in the spin-j sector of the bond as $(-1)^{2j}$, so it lies in the algebra of the spin projections of these bonds. On a periodic layer every bond lies in two plaquettes, so $\prod_p \prod_{b \in \partial p} c_b = \prod_b c_b^2 = 1$, and the plaquettes where c is not closed are even in number.

(b) Let Y commute with the M_{t_p}, $p \in S$, and the Π_b^j, $b \in \mathsf{B}_S$, on $L^2(G^{\mathsf{B}_S})$. Then Y keeps the polynomials in the matrix entries and commutes with the Laplacian $\Delta_b = -\sum_j j(j+1)\Pi_b^j$ of every bond; for functions F, F' with $M_F, M_{F'}$ in the algebra generated,

$$[[\Delta_b, M_F], M_{F'}] = 2M_{D_b(F,F')}, \qquad D_b(F,F') = \sum_a (J_b^a F)(J_b^a F'),$$

J_b^a the infinitesimal left translations of the bond b, so the real algebra $\mathfrak{W}$ of such functions is closed under every D_b. For two closed paths $\mathsf{M} = V_b\mathsf{a}$, $\mathsf{N} = V_b\mathsf{c}$ that run once through b, $D_b(\operatorname{tr}\mathsf{M}, \operatorname{tr}\mathsf{N}) = \frac{1}{4}[\operatorname{tr}\mathsf{M}^{-1}\mathsf{N} - \operatorname{tr}\mathsf{MN}]$ and $\operatorname{tr}\mathsf{M}\operatorname{tr}\mathsf{N} = \operatorname{tr}\mathsf{MN} + \operatorname{tr}\mathsf{M}^{-1}\mathsf{N}$: paths may be fused at a common bond. Four fusions in every chain T of three plaquettes, in a row or around a site, give the seven traces $\frac{1}{2}\mathrm{tr}$ of α_i, $\alpha_i\alpha_k$, $\alpha_1\alpha_2\alpha_3$, α_i the boundaries at one site. With $g = \mathsf{s}(g) + i\mathsf{v}(g) \cdot \sigma$,

$$\mathsf{s}(g_1g_2g_3) = \mathsf{s}_1\mathsf{s}_2\mathsf{s}_3 - \mathsf{s}_1\mathsf{v}_2 \cdot \mathsf{v}_3 - \mathsf{s}_2\mathsf{v}_1 \cdot \mathsf{v}_3 - \mathsf{s}_3\mathsf{v}_1 \cdot \mathsf{v}_2 + \det(\mathsf{v}_1, \mathsf{v}_2, \mathsf{v}_3),$$

so the seven numbers are the Gram matrix and the determinant of the $\mathsf{v}(V(\alpha_i))$, which fix the triple up to a rotation; every closed path of T is a word in the α_i, and so two configurations of T with the same seven numbers differ by a gauge transformation of T. Two chains that share two plaquettes fix the gauge transformation on their common sites up to the centre, as two non-commuting elements of SU(2) have the centraliser $\{\pm 1\}$; along the chains of S the configurations with the same functions in $\mathfrak{W}$ are those that differ by a gauge transformation and a closed sign field. Let $\mathcal{K}_S$ be the compact group of the pairs (g, c), g a gauge transformation of the sites of B_S and $c \in Z^1(S)$, acting by $((g,c)V)_b = c_b g(b_-)V_b g(b_+)^{-1}$, and λ its representation on $L^2(G^{\mathsf{B}_S})$; $\lambda(\mathcal{K}_S)$ commutes with the M_{t_p} and the Π_b^j.

Let $\delta_0 = \prod |\mathsf{v}(V(\alpha_p)) \times \mathsf{v}(V(\alpha_q))|^2$, the product over the pairs of adjacent plaquettes of S, α_p, α_q their boundaries begun at an end of the common bond; it is an invariant polynomial, $\not\equiv 0$ on the connected space G^{B_S}, so its zero set is a null set. Off it the stabiliser of V in $\mathcal{K}_S$ is $\{(z, dz)\}$, z central, which acts trivially (an element of SU(2) that commutes with two elements that do not commute is ± 1), and the seven numbers of the chains separate the orbits; so by Stone–Weierstrass on the compact sets $\{|\delta_0| \geq \epsilon\}$, M_F lies in the generated algebra for every continuous $\mathcal{K}_S$-invariant F, and Y commutes with all of them.

The polynomials are the sum of the isotypic parts Pol_π of the irreducible representations (π, V_π) of $\mathcal{K}_S$ that are trivial on the stabiliser; at a point V off the zeros of δ_0 the functionals $v \mapsto m_v(V)$ of the equivariant linear maps $m : \mathsf{V}_\pi \to \mathrm{Pol}$ fill V_π^*, the orbit of V being $\mathcal{K}_S$ modulo the stabiliser (Peter–Weyl and Stone–Weierstrass on it). With $\mathsf{n} = \dim \pi$ such maps $m^{(i)}$ whose functionals at one point V_1 form a basis of V_π^*, $\mathsf{M} = (m^{(i)}_{e_k})_{ik}$ in a basis (e_k) of V_π and $\delta_\pi = \det \mathsf{M}$, Cramer's rule writes $|\delta_\pi|^2 m_v$ as a combination of the $m_v^{(i)}$ with invariant polynomial coefficients, with which Y commutes; so $Y m_{e_k} = \sum_l m_{e_l} \tau_{lk}$ for every equivariant m on $\{\delta_\pi \neq 0\}$, with $\tau = \mathsf{M}^{-1}(Y m^{(i)}_{e_k})_{ik}$.

The vector fields $\xi_{b,F} = \sum_a (J^a_b F) J^a_b = \frac{1}{2}([\Delta_b, M_F] - M_{\Delta_b F})$, F a real invariant polynomial, commute with Y on the polynomials and with $\lambda(\mathcal{K}_S)$, and carry equivariant maps to equivariant maps; the identity for the equivariant map $\xi_{b,F} m$, set beside $\xi_{b,F}$ applied to the identity for m ($\xi_{b,F}$ is a derivation), gives $\mathsf{M}\, \xi_{b,F} \tau = 0$, so $\xi_{b,F} \tau = 0$ on $\{\delta_\pi \neq 0\}$. The $\xi_{b,F}(V)$ span the tangent space on a dense open set: at a bond b of a plaquette p of a chain $T \subset S$, with α the boundary of p begun with b and w_1, w_2 closed paths at b_- through the two bonds c_1, c_2 that a spanning tree of $\mathsf{B}_T \setminus \{b\}$ leaves out, the traces $\operatorname{tr} V(\alpha)$, $\operatorname{tr} V(\alpha w_i)$ are invariant and their derivatives $(J^a_b \cdot)_a$ are $-\mathsf{v}(V(\alpha))$, $-\mathsf{v}(V(\alpha w_i))$, whose determinant is a polynomial that is not 0 at a configuration with $\mathbf{1}$ on the tree and $V(\alpha) = i\sigma_3$, $V(\alpha w_1) = i\sigma_1$, $V(\alpha w_2) = i\sigma_2$ (the bonds b, c_1, c_2 are free there).

So $d\tau = 0$ on a dense open part of $\{\delta_\pi \neq 0\}$, τ is constant, τ^π, near V_1, and the matrix of polynomials $(Y m^{(i)}_{e_k})_{ik} - \mathsf{M}\tau^\pi$, which vanishes near V_1, vanishes identically: here analyticity enters. Hence $Y m_v = m_{\tau^\pi v}$ for all equivariant m, and with $y_\pi(\kappa) = \dim \pi \operatorname{tr}(\tau^\pi \pi(\kappa)^{-1})$, $Y P_\pi = \lambda(y_\pi) \in \lambda(\mathcal{K}_S)''$, P_π the projection on the isotypic part; summed over π, $Y \in \lambda(\mathcal{K}_S)''$. So the generated algebra is the commutant of the gauge transformations of S and of the $\mathsf{V}(c)$, $c \in Z^1(S)$; restricted to $\mathcal{H}_{\mathrm{ph}}$ and combined with (a), this is the pair of identities for $\mathsf{A}_\bullet(S)$, and it exhibits $\mathsf{A}_\bullet(S)$ as the join of the $\mathsf{A}(\mathsf{B}_T)$ over the chains.

For admissible S the join of the $\mathsf{A}(\partial p)$, $p \in S$, is $\mathsf{A}_\bullet(S)$: the generators M_{t_p} and Π^j_b, $b \in \partial p$, lie in $\mathsf{A}(\partial p)$, and $\mathsf{A}(\partial p) \subset \mathsf{A}(\mathsf{B}_T) \subset \mathsf{A}_\bullet(S)$ for a chain $T \subset S$ of three plaquettes through p, which admissibility provides. The $\mathsf{V}(c)$, $c \in Z^1(S)$, are functions of spin projections of B_S and commute with $\mathsf{A}_\bullet(S)$: they are central. $\operatorname{Ad} \mathsf{V}(c)$ is the identity on $\mathsf{A}(\mathsf{B}_S)$ for $c \in B^1(S)$ (the flips above), so $H^1(S)$ acts, and the decomposition into its characters is the spectral decomposition of this action of a finite abelian group; the summand of a character κ holds $M_{\chi_{1/2}(U(C_\kappa))} \neq 0$ for a closed path C_κ with $c(C_\kappa) = \kappa([c])$.

If every $\mathbb{Z}_2$-cycle is a sum of boundaries, every closed c is a coboundary. For one plaquette, the cyclic shift of the four bonds times the projection on the spin 1 at one corner and 0 at the others commutes with M_{t_p} and all Π^j_b and moves the charge to the next corner; for two adjacent plaquettes use reflection in the perpendicular bisector of the common bond, composed with the projection on charge 1 at one end and 0 at all other vertices. On this projected block the zero charges at the outer vertices force equal bond spins along each of the two outer three-bond paths. Reflection reverses each path and preserves each individual bond-spin projection, as well as each plaquette trace, but moves the endpoint charge; neither lies in the von Neumann algebra generated by the gauge transformations and the closed flips, which is the commutant of all invariant operators, so the traces and spin projections generate less. The traces generate multiplications, and M_{t_p} does not commute with Π^j_b, $b \in \partial p$.

Two plaquettes with one common bond: the full algebras generate. Let p, q share the bond c_0 with the ends x, y; their union has seven bonds and six sites, and at x, as at y, three bonds meet: $a \in \partial p$, c_0 and $a' \in \partial q$. Let $\mathfrak{B}$ be the algebra of all gauge-invariant bounded operators on $L^2(G^7)$, the functions of these bonds, and $\mathfrak{N} \subset \mathfrak{B}$ the von Neumann algebra generated by the gauge-invariant operators of the four bonds of p and those of the four bonds of q; we show $\mathfrak{N} = \mathfrak{B}$, which, restricted to $\mathcal{H}_{\mathrm{ph}}$ by a normal homomorphism, gives $\mathsf{A}(\partial p) \vee \mathsf{A}(\partial q) = \mathsf{A}(\partial p \cup \partial q)$.

By Peter–Weyl, $L^2(G^7) = \bigoplus_{\mathbf{j}} \mathcal{K}(\mathbf{j})$ over the spins $\mathbf{j}$ of the seven bonds, $\mathcal{K}(\mathbf{j}) = \bigotimes_z \mathcal{V}_z(\mathbf{j})$, $\mathcal{V}_z(\mathbf{j})$ the product of the spin-j_b representations of the bonds at the site z, on which the gauge group at z acts; $\mathcal{V}_z(\mathbf{j}) = \bigoplus_J V_J \otimes \mathsf{M}_z(\mathbf{j}, J)$, and a label $(\mathbf{j}, \mathbf{J})$, $\mathbf{J}$ the charges at the six sites, occurs if the polygon inequalities hold at every site. So $\mathfrak{B} = \bigoplus_{\mathbf{J}} \mathcal{B}(\mathsf{M}_{\mathbf{J}})$, $\mathsf{M}_{\mathbf{J}} = \bigoplus_{\mathbf{j}} \bigotimes_z \mathsf{M}_z(\mathbf{j}, J_z)$. (i) *Labels.* Let $Y \in \mathfrak{B}$ commute with $\mathfrak{N}$.

It commutes with the Casimir operators of the bonds, so it keeps every $\mathcal{K}(\mathbf{j})$; being gauge invariant, it keeps the charges $\mathbf{J}$; and for a corner (b_1, b_2) of p or q at z the spectral projections of the Casimir operator of the gauge group at z acting on the two bonds of that plaquette lie in $\mathfrak{N}$, so Y commutes with $t^z_{b_1 b_2} = J_{b_1,z} \cdot J_{b_2,z}$ on every $\mathcal{K}(\mathbf{j})$. At the four sites with two bonds M_z is one-dimensional; at x and at y, t_{ac_0} and $t_{c_0 a'}$ generate all operators of $\mathsf{M}_z(\mathbf{j}, J_z)$ by Lem. 14.19. So $Y = \sum y(\mathbf{j}, \mathbf{J}) P(\mathbf{j}, \mathbf{J})$, $P(\mathbf{j}, \mathbf{J})$ the projection on the label. (ii) *One plaquette made large.* If two labels with the same charges occur and agree off ∂p, y takes one value at them.

Indeed $L^2(G^{\partial p}) = \bigoplus_\pi V_\pi \otimes \mathrm{Mult}_\pi$ over the charges $\pi = (f_z)$ at the four corners of p, Mult_π with the basis $e_{\mathbf{i}}$ indexed by the spins $\mathbf{i}$ of ∂p with $f_z \in i_{b_1} \otimes i_{b_2}$ at every corner (Clebsch–Gordan is free of multiplicities); the gauge-invariant operators of ∂p are the operators $1 \otimes \mathcal{B}(\mathrm{Mult}_\pi)$, and Y, commuting with them, is the identity on every Mult_π; so y takes one value on the spins $\mathbf{i}$ of ∂p of every π that occurs with the given spins off ∂p and the given charges; and with N above all charges that occur and $e_b \in \{0, \frac{1}{2}\}$ with $e_{b_1} + e_{b_2} \equiv f_z$ modulo 1 at the four corners (they exist, the charges at the corners being fixed modulo 1 by the charges $\mathbf{J}$ and the spins off ∂p, with an integer sum), the spins $(N + e_b)_{b \in \partial p}$ belong to every such π, which joins them all. (iii) *A monotone path.* Raising the four spins of one plaquette by $\frac{1}{2}$ keeps the charges and the polygon inequalities, two spins at each corner growing by $\frac{1}{2}$.

By such raisings two labels with the same charges are brought to an integer difference, twice their difference modulo 2 being a cycle and so a sum of ∂p and ∂q; raised $2N$ times more on both plaquettes, N large, they are joined through the label that takes the larger of the two spins on the six bonds that lie on one plaquette and the smaller on c_0, changing first the bonds of p, then those of q; all labels on the way occur. With (ii) at every change, y depends on $\mathbf{J}$ alone. (iv) *The charges.* The projections E^z_J on the charge J at z lie in $\mathfrak{N}$. At the four outer sites they are spectral projections of Casimir operators of one plaquette.

At x, with $A = J_{a,x}$, $B = J_{c_0,x}$, $D = J_{a',x}$, the Casimir operator $(A+B+D)^2$ holds the cross term t_{AD} of the two outer bonds, which belongs to neither plaquette; the identity of three commuting spins

$$(B^2 + t_{AB})\, t_{AD} = A^2 t_{BD} + t_{AB} t_{BD} - [t_{AB}, t_{BD}] - [t_{AB}, [t_{AB}, t_{BD}]]$$

puts $\mathsf{F} t_{AD}$ in $\mathfrak{N}$ on every block of spins, $\mathsf{F} = B^2 + t_{AB} = \frac{1}{2}((A+B)^2 + B^2 - A^2)$, and with it $\mathsf{F}(A+B+D)^2$. As $(A+B+D)^2$ commutes with all of $\mathfrak{B}$, and F is the number $\frac{1}{2}(f(f+1) + s_c(s_c+1) - s_a(s_a+1))$ on the range where the spins of a, c_0 are s_a, s_c and $(A+B)^2 = f(f+1)$, a range given by projections of $\mathfrak{N}$, $(A+B+D)^2$ times such a range lies in $\mathfrak{N}$ where this number is not 0, and with it E^x_J times the range; the same holds where $f = 0$, as $(A+B+D)^2 = D^2$ there. The weakly closed ideal of the $n \in \mathfrak{N}$ with $nE^x_J \in \mathfrak{N}$ is $\mathfrak{N}\mathsf{p}$ with a central projection p of $\mathfrak{N}$ that holds these ranges; $1 - \mathsf{p}$ commutes with $\mathfrak{N}$, so by (i)–(iii) it is the sum of the $P(\mathbf{j}, \mathbf{J})$ over the labels of some charges, and if it were not 0, each of its ranges of the kind above would have $f > 0$ and $f(f+1) = s_a(s_a+1) - s_c(s_c+1)$; but raising p by $\frac{1}{2}$ keeps the charges and f and turns such a range into one where the number is $\frac{1}{2}(s_c - s_a) \neq 0$, a range of p: so $\mathsf{p} = 1$. (v) $\mathfrak{N}$ holds the projections on

the $\mathsf{M}_{\mathbf{J}}$, and an operator on $\mathsf{M}_{\mathbf{J}}$ that commutes with $\mathfrak{N}$ is a number by (i)–(iii); so $\mathfrak{N}$ cut down to $\mathsf{M}_{\mathbf{J}}$ is irreducible, and $\mathfrak{N} = \bigoplus_{\mathbf{J}} \mathcal{B}(\mathsf{M}_{\mathbf{J}}) = \mathfrak{B}$.

This is the two-plaquette case of Lem. 14.16(a).

(c) R deformation retracts to a circle around the hole: $H^1(R) = \mathbb{Z}_2$, generated by $[c_*]$, and $c_*(C) = -1$. $\mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R)' = \mathsf{A}(\mathsf{B}_R)$ by (a). As v represents the generator, (b) gives $\mathsf{A}(\mathsf{B}_R) = \mathsf{A}_\bullet(R) \oplus \mathsf{A}(\mathsf{B}_R)_-$, and $u = \operatorname{sign} \operatorname{tr} U(C)$ is an odd self-adjoint unitary in $\{W_C\}''$, so $\mathsf{A}(\mathsf{B}_R)_- = \mathsf{A}_\bullet(R)u$; $\frac{1}{2}(X + vXv)$ is the projection on the even part. $\mathsf{A}_\bullet(R)' = \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R) \vee \{v\}$ by (b), v being one flip of the class that is not trivial and the flips of $B^1(R)$ acting on $\mathcal{H}_{\mathrm{ph}}$ through $\mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R)$; and on $\mathcal{H}_{\mathrm{ph}}$, $v = \mathsf{V}_\gamma \mathsf{V}(c')$ with c' carried by $\mathsf{B} \setminus \mathsf{B}_R$ and $\mathsf{V}(c') \in \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R)$; $\operatorname{Ad} u$ fixes $\mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R)$ and changes the sign of V_γ ($c_\gamma(C) = -1$), which gives the second decomposition and $\mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R) = \mathsf{A}_\bullet(R)' \cap \{u\}'$.

For a closed path C', $vM_{\chi_j(U(C'))}v = c_*(C')^{2j} M_{\chi_j(U(C'))}$. If an assignment $\mathscr{N}$ equal to A on closed and open rectangles were additive and dual, then $\mathscr{N}(\mathsf{B}_R) = \bigvee_i \mathsf{A}(\mathsf{B}_{D_i}) = \mathsf{A}_\bullet(R)$ and $\mathscr{N}(\mathsf{B} \setminus \mathsf{B}_R) = \mathsf{A}(\mathsf{B}_{\mathrm{in}}) \vee \mathsf{A}(\mathsf{B}_{\mathrm{out}}) = \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R)$, the two pieces of the complement having no common site; duality would give $\mathsf{V}_\gamma \in \mathsf{A}(\mathsf{B} \setminus \mathsf{B}_R)$. The four rectangles: every chain of three plaquettes of R lies in one D_i or in the union of two adjacent ones, and two adjacent rectangles glue by Lem. 14.16(a), their overlap being a connected rectangle with a cycle; so $\mathsf{A}_\bullet(R) = \bigvee_i \mathsf{A}(\mathsf{B}_{D_i})$.

The commutative statement and the formula for the adjoint loop are Lem. 14.16(b), (c) for the four rectangles as boxes, whose overlaps hold two independent cycles when no corner is one plaquette.

(d) S deformation retracts to the torus with two holes: $H^1(S) = \mathbb{Z}_2^3$, generated by the cut γ and the two winding cuts. $\mathsf{A}(\mathsf{B}_S)' = \mathsf{A}(\mathsf{B}^\circ_{D_1} \cup \mathsf{B}^\circ_{D_2}) = \mathsf{N}$ by (a), and (b) gives $\mathsf{A}_\bullet(S)' = \mathsf{N} \vee \{\mathsf{V}_\gamma, \mathsf{V}_1, \mathsf{V}_2\}$, the products of $\mathsf{V}_\gamma, \mathsf{V}_1, \mathsf{V}_2$ meeting every class of $H^1(S)$ and the flips of $B^1(S)$ acting on $\mathcal{H}_{\mathrm{ph}}$ through N. The fundamental loop around D_1 lies in $\mathsf{A}(\mathsf{B}_S)$, commutes with N, V_1, V_2 and anticommutes with V_γ; the winding loops along the free row and column anticommute with one V_i each and commute with N and V_γ: hence the two non-memberships. On the whole layer $H^1 = \mathbb{Z}_2^2$ and $\mathsf{A}(\varnothing) = \mathbb{C}$; the four joint eigenspaces of $\mathsf{V}_1, \mathsf{V}_2$ are carried into one another by the fundamental winding loops, so none is 0. Off one disc, H^1 is generated by the two fluxes, the loop around the disc being a boundary in homology modulo 2 there.

(e) A box has a shelling, an order of its plaquettes in which each new plaquette either has all its bonds already present or is glued to an earlier one along a single bond so that the overlap with what is present is connected, and every site of a new bond of p_k that is already a site of X_{k-1} lies on the boundary ∂p_j of that earlier plaquette together with the overlap $\partial p_k \cap X_{k-1}$; Lem. 14.16(a), the two-plaquette case included, carries the generation from step to step: at a step that brings new bonds it applies to $Y_1 = X_{k-1}$ and $Y_2 = \partial p_j \cup \partial p_k$, with $I = \partial p_j \cup (\partial p_k \cap X_{k-1})$ connected and holding the cycle ∂p_j, $\mathsf{R}_2 = \partial p_k \setminus X_{k-1}$, and the site clause of the shelling as its hypothesis on the common sites, while the gauge-invariant operators of Y_2 are generated by those of ∂p_j and of ∂p_k, by the two-plaquette case.

The solid ring is the union of two shellable halves, the cubes $[i_1, i_1 + 1] \times [i_2, i_2 + 1] \times [0, 1]$ of the ring with $i_1 \in \{0, 1\}$ and those with $i_1 \in \{1, 2\}$, whose intersection has two components, the bonds of the two cubes with $i_1 = 1$, $i_2 \in \{0, 2\}$, each with a cycle; a site common to the two halves has $x_1 \in \{1, 2\}$, and the sites of the first half with $x_1 = 2$, as those of the second with $x_1 = 1$, are corners of these two cubes, so every site common to B_{S_1} and B_{S_2} is a site of I, the hypothesis of Lem. 14.16(a) on the common sites, as in the general union of (e), where a site of B_{S_1} on a bond of B_{S_2} off I is common to both; Lem. 14.16(a) gives the join as the invariant operators that commute with the central gauge transformations constant on each component, that is with $\mathsf{V}(c_*)$, and duality gives the commutant; $\mathsf{V}(c_*)$ anticommutes with W_C because $c_*(C) = -1$; for a dual surface meeting the ring in that section, $c_{\mathfrak{s}} c_*$ is carried by the complement.

With r components the central gauge transformations constant on components, modulo the global ones, form $\mathbb{Z}_2^{r-1}$.

(f) The substitution $U \mapsto sU$ keeps the product Haar measure, $t_p(sU) = (ds)_p t_p(U)$ and $\chi_j(U(C))(sU) = s(C)^{2j}\chi_j(U(C))(U)$. For a fat odd loop the same with $F(sU) = s(C)F(U)$. Both members may be multiplied by any non-zero numbers, which carries the rule to limit points.

(g) The block map is formed from parallel transports along paths in the double block of c and from averages of them, and it is covariant under gauge transformations; if $ds = 1$ on the double block, s is there the coboundary of a central field z, and $\mathcal{M}^n(sU)(c) = z(c_-)\mathcal{M}^n(U)(c)z(c_+) = s[c]\mathcal{M}^n(U)(c)$. The convergence of $\langle h(W_C(\mathcal{M}^{K-d}U))\rangle_K$, with the stated rate, is the convergence of the laws of the block fields at a fixed depth of Thm. YM3_I:D (Cor. 2.8), a bounded Borel function of the block field being tested. A flip $\bar\xi$ of the torus is the image of a closed sign field ξ of the fine lattice, $d\xi = 1$, under which the Wilson measure is invariant; by covariance the block-field laws are invariant under $\bar\xi$. For the bounding vortex line, $\eta_K = ds$ off the plaquettes pierced by $\partial\mathsf{S}$, $s = -1$ on the fine bonds that cross S; the sign rule gives $Z_K[\eta_K; F\circ\mathcal{M}^{K-d}] = Z_K[1; F\circ\mathcal{M}^{K-d}\circ s]$ and covariance gives $\mathcal{M}^{K-d}(sU) = \bar\xi\mathcal{M}^{K-d}(U)$ on the blocks that miss $\partial\mathsf{S}$; the limit and its error come from Thm. YM3_I:D; for $F = W_C F'$ the factor is $\bar\xi(C) = (-1)^{\#(C\cap\mathsf{S})}$.

(h) For continuous h the first statement is Lem. 14.9 (Thm. AF of §10.1, read at one polygon), applied to the polygon C, a closed path of some $\Lambda_{K_0,\ell}$ taken in every $\Lambda_{K,\ell}$, $K \ge K_0$, with the functions F_K of the bonds at the distance $\ge \mathsf{d}$ from C, uniformly in them; a bounded h that is continuous off a null set lies between continuous functions $h_- \le h \le h_+$ with $\overline{h_+} - \overline{h_-}$ as small as we please, and F_K is a combination of four functions with values in $[0,1]$. The law of $\frac12\mathrm{tr}\, g$ under Haar measure is $\frac{2}{\pi}\sqrt{1-w^2}\,\mathrm{d}w$; so $\bar h = 0$ for $h = \mathrm{sign}$ and for the characters χ_j, $j \ge \frac12$, written as functions of $w = \frac12\mathrm{tr}$, and $\bar h = \frac14$ for $h(w) = w^2$.

With C in a site mirror and $F_K = \overline{\theta G'_K}G_K$, $|G_K|, |G'_K| \le 1$ at a fixed distance, this is the fading between the vectors of the space of reflection positivity. The Bessel bound is Lem. 14.17(a), and $\mathsf{r}(x) = 1 - \frac{3}{2x} + O(x^{-2})$, $n' \approx |\gamma|L^K$, $4\beta = 4L^K\mathsf{b}$ give the limit $\exp(-3|\gamma|/8\mathsf{b})$. The fading of the flips is Lem. 14.17(b), (c), with $\omega_K(1-Q_0) \le \omega_K(M_{a_{p_0}}) = \langle a_{p_0}\rangle_K$, the multiplication operators of one layer having the Wilson measure as their law in the state of the transfer operator.

(i) Notation. $x_{[D]} = L^D\mathsf{b}$ (§2.3.2) is the coupling at the depth D; $\varepsilon(\mathsf{t}) = A_0\mathsf{t}^{-1/2}(\log\mathsf{t})^{\mathfrak{p}_0}$ and $\delta(\mathsf{t}) = (A_1/A_0)\varepsilon(\mathsf{t})$ YM3_I:(634), with the constants $A_0, A_1 > 0$, $\mathfrak{p}_0 \in \mathbb{N}$ of the list of Thm. YM3_I:D; $\varpi(V) = \max_p|\partial V(p) - \mathbf{1}|$, $|\cdot|$ the operator norm; and the small fields of the depth D are those of the inner ball $\mathsf{In}_{[D]} = \{V : \varpi(V) < 2\varepsilon(x_{[D]})\}$ of Thm. YM3_I:D (the common-depth definitions in Section YM3_I:116, the definitions preceding Lem. YM3_I:113.10), an open gauge-invariant set (Lem. YM3_I:113.10). $\mathsf{R}_{[D]} = 2M^{-3}(\ell L^D)^3\exp[-c'A_0^2(\log x_{[D]})^{2\mathfrak{p}_0}]\exp[(\mathsf{C}^\Sigma_{\mathrm{maj}} + 2 + C_{\mathrm{low}}(\mathsf{b}))\ell^3]$ is the number YM3_I:(2187) at the level of the depth D; it reads D and not K, and it tends to 0 as $D\to\infty$, since $c'A_0^2 > 0$ and $2\mathfrak{p}_0 \ge 2$. An element g of G with the eigenvalues $e^{\pm i\phi}$, $\phi\in[0,\pi]$, has $|g-\mathbf{1}| = 2\sin\frac{\phi}{2} \le \phi$, so $|e^{iw\sigma_3} - \mathbf{1}| \le |w|$; and replacing the four factors of $\partial(kV)(p)$ by those of $\partial k(p)$ one by one, $|\partial(kV)(p) - \partial k(p)| \le \sum_{b\in\partial p}|V_b - \mathbf{1}|$ for G-valued fields k, V and the bondwise product kV.

The floor. Let D be so large that $x_{[D]} \ge \mathsf{t}_\natural$ and $\mathsf{R}_{[D]} < 1$, let $K \ge D+1$, and let $t = K - D$ be the level of the depth D. The setting of the small-field chain of Thm. YM3_I:D asks of its top level only that it lie between 1 and $K-1$ and carry a coupling $\ge \mathsf{t}_\natural$ (§2.3.1); so it holds with the top level t, at a fixed depth. At the source 0, Prop. YM3_I:115.2 gives a bounded Borel weight $\mathfrak{w}^{\mathrm{sf}}_t \ge 0$ defined before Prop. YM3_I:115.2 such that the image of $\mathfrak{w}^{\mathrm{sf}}_t\rho_0\,\mathrm{d}U$ under $\mathcal{M}^t$ is the measure $\mathbf{1}_{\mathsf{In}_{[D]}}\nu_t$ of the chain YM3_I:(2188), $\int F\mathfrak{w}^{\mathrm{sf}}_t\rho_0\,\mathrm{d}U \le \int F\rho_0\,\mathrm{d}U$ for gauge-invariant bounded Borel $F \ge 0$ YM3_I:(2189), and $0 \le Z - \int\mathfrak{w}^{\mathrm{sf}}_t\rho_0\,\mathrm{d}U \le \mathsf{R}_{[D]}Z$ YM3_I:(2190), $Z = \int\rho_0\,\mathrm{d}U$.

So the measure of the chain lies below Z times the law on gauge-invariant functions (histories with a large field at a finer level, which may end in $\mathsf{In}_{[D]}$, are left out of it), and what it leaves out has a mass $\le \mathsf{R}_{[D]}Z$. On $\mathsf{In}_{[D]}$, which lies in the field cut (Lem. YM3_I:113.10), the measure of the chain has the density $p = e^{\mathsf{c}}e^{\Phi_t}$ for h_D, c a complex number (Prop. YM3_I:114.2), and

there $|\Phi_t| \le \mathsf{O}_D := (\ell L^D)^3[\frac{3}{2}c_\rho^2 A_0^2(\log x_{[D]})^{2\mathfrak{p}_0} + C_{\text{old}}\frac{L}{L-1}E_0]$: the bound YM3_I:(2161) of the action on a field cut, at the radius $c_\rho\varepsilon(x_{[D]})$ and the weight $x_{[D]}$, gives the first term, because $\mathsf{t}\,\varepsilon(\mathsf{t})^2 = A_0^2(\log \mathsf{t})^{2\mathfrak{p}_0}$, and the bound m_t^S YM3_I:(2179) of the added action (equation YM3_I:(2180)) gives the second; O_D reads D and not K.

A complex density of a positive measure is ≥ 0 almost everywhere; so $p = e^{\operatorname{Re}\mathsf{c}}e^{\operatorname{Re}\Phi_t}$ lies in $e^{\operatorname{Re}\mathsf{c}}[e^{-\mathsf{O}_D}, e^{\mathsf{O}_D}]$, and its mass, which lies in $[(1-\mathsf{R}_{[D]})Z, Z]$, puts $e^{\operatorname{Re}\mathsf{c}}/Z$ between $(1-\mathsf{R}_{[D]})e^{-\mathsf{O}_D}/\mathsf{h}_D(\mathsf{In}_{[D]})$ and $e^{\mathsf{O}_D}/\mathsf{h}_D(\mathsf{In}_{[D]})$: the constant c, the sum of the vacuum energies of the steps, drops out. For a gauge-invariant Borel F with $0 \le F \le 1$ on the fields of the depth D, $F \circ \mathcal{M}^t$ is gauge invariant, the block map being covariant, and this gives

$$\kappa_D \int_{\mathsf{In}_{[D]}} F\,\mathrm{d}\mathsf{h}_D \;\le\; \int F\,\mathrm{d}\mu_D^K \;\le\; \bar\kappa_D \int_{\mathsf{In}_{[D]}} F\,\mathrm{d}\mathsf{h}_D + \mathsf{R}_{[D]},$$

$$\kappa_D = \frac{(1-\mathsf{R}_{[D]})e^{-2\mathsf{O}_D}}{\mathsf{h}_D(\mathsf{In}_{[D]})},$$

$$\bar\kappa_D = \frac{e^{2\mathsf{O}_D}}{\mathsf{h}_D(\mathsf{In}_{[D]})},$$

the right inequality by YM3_I:(2189) at $1-F$ together with YM3_I:(2190); at $F = \mathbf{1}_{\mathsf{In}_{[D]}}$, $\mu_D^K(\mathsf{In}_{[D]}) \ge 1 - \mathsf{R}_{[D]}$. For a Borel set E, $\mu_D^K(E) = \int \bar{\mathbf{1}}_E\,\mathrm{d}\mu_D^K$ with the mean $\bar{\mathbf{1}}_E$ of $\mathbf{1}_E$ over the gauge group, the law being gauge invariant, while $\mathsf{In}_{[D]}$ and h_D are gauge invariant; so the floor $\mu_D^K(E) \ge \kappa_D \mathsf{h}_D(E \cap \mathsf{In}_{[D]})$ and the upper bound $\mu_D^K(E) \le \bar\kappa_D \mathsf{h}_D(E \cap \mathsf{In}_{[D]}) + \mathsf{R}_{[D]}$ hold for all Borel E, free of K, and for the limit law, the laws converging on every Borel set (Cor. 2.8).

The thick vortex. On the bonds c of the depth D let $n(c) = \#(c \cap \mathsf{S})$, let j_0 be an integer cycle carried by the dual bonds of the rim with $j_0 \equiv dn$ modulo 2 (the rim, with an even number of its dual bonds at every dual site, is a union of closed dual paths, which are oriented), and $\mathsf{r} = \lfloor L^{D-d}/4 \rfloor$. The translates $\tau_v j_0$ by the $(2\mathsf{r}+1)^3$ lattice vectors v of a box of radius $\mathsf{r}L^{-D}$ differ from j_0 by coboundaries $d\beta_v$, β_v carried within the distance $< L^{-d}$ from the rim: with the prism $(\mathsf{pr}_\lambda j)(x;\mu) = j(x;\lambda\mu)$, which takes functions on plaquettes to functions on bonds and functions on cubes to functions on plaquettes, $d\,\mathsf{pr}_\lambda j + \mathsf{pr}_\lambda dj = \tau_\lambda j - j$ and $dj_0 = 0$, summed along a path to v.

With φ the mean of the $\pi\beta_v$, ω the mean of the $\pi\tau_v j_0$, $\theta = \pi n + \varphi$ and $V_c^1 = e^{i\theta(c)\sigma_3}$: $d\theta = \omega + 2\pi\mathsf{k}$ with k integer, so $\partial V^1(p) = e^{i\omega_p\sigma_3}$, and $|\omega_p| \le \pi n_{\mathsf{S}}(2\mathsf{r}+1)^{-2} \le 16\pi n_{\mathsf{S}} L^{-2(D-d)}$, as a segment of the rim normal to p meets at most $2\mathsf{r}+1$ of the translates of p; the rim misses $\mathcal{R}$, so it keeps the distance $\ge L^{-d}$ from it (the closed cubes of side L^{-d} about sites that touch a dual bond of $\Lambda_{d,\ell}$ in the rim cover its neighbourhood of that radius, and none of them belongs to $\mathcal{R}$), and on the bonds of $\mathcal{R}$, $\varphi = 0$ and $V^1 = \bar\xi$.

Counted in $\Lambda_{d,\ell}$, n_{S} is at most the number $3(\ell L^d)^3$ of all its dual bonds; so for D so large that $48\pi(\ell L^d)^3 L^{-2(D-d)} \le \varepsilon(x_{[D]})$, which holds for large D as $\varepsilon(x_{[D]})$ decays only like $L^{-D/2}$ up to logarithms, the thick vortex of every flip has $\varpi(V^1) \le \varepsilon(x_{[D]})$: it is a small field, and one D serves all surfaces. *The common part.* Let $\eta = \frac{1}{4}\varepsilon(x_{[D]})$ and $\mathrm{vol}(\eta)$ the Haar measure of $\{g \in G : |g - \mathbf{1}| < \eta\}$. If every bond of a field W is within η of $\mathbf{1}$, then $\varpi(W) < 4\eta < 2\varepsilon(x_{[D]})$ and $\varpi(V^1 W) < \varpi(V^1) + 4\eta \le 2\varepsilon(x_{[D]})$, by the bound of a product of four: W and $V^1 W$ (bondwise product) are small fields, and $(V^1 W)|_{\mathcal{R}} = \bar\xi\, W|_{\mathcal{R}}$.

By the floor, the law $\mu_{\mathcal{R}}$ of the block fields of $\mathcal{R}$ at the depth D dominates the measure $\kappa_D \mathsf{g}(w)\,\mathrm{d}w$, $\mathsf{g}(w)$ the Haar measure of the continuations of w to small fields; on the configurations w of $\mathcal{R}$ within η of $\mathbf{1}$, $\mathsf{g}(w)$ and $\mathsf{g}(\bar\xi w)$ are $\ge c_\eta = \mathrm{vol}(\eta)^{3(\ell L^D)^3 - n_{\mathcal{R}}}$, $n_{\mathcal{R}}$ the number of the bonds of $\mathcal{R}$ at the depth D (continue w by bonds within η of $\mathbf{1}$, and $\bar\xi w$ by the same bonds times those of V^1); so $\theta_0 = \kappa_D c_\eta \mathbf{1}_{\{|w_c - \mathbf{1}| < \eta\ \forall c\}}\,\mathrm{d}w$ lies below $\mu_{\mathcal{R}}$ and below $\bar\xi_* \mu_{\mathcal{R}}$, for every flip of a surface whose rim

misses $\mathcal{R}$ and at every cutoff $K \geq D+1$, and in the limit; it reads no surface and no cutoff, and its mass $\varkappa_D = \kappa_D \mathrm{vol}(\eta)^{3(\ell L^D)^3}$ reads D alone.

The block map from the depth D to the depth d, which reads on the bonds of $\mathcal{R}$ the bonds of $\mathcal{R}$ alone and turns $\bar{\xi}$ at the depth D into $\bar{\xi}$ at the depth d by the first clause of (g), carries this down. Then $\mu_{\mathcal{R}} \wedge \bar{\xi}_* \mu_{\mathcal{R}}$ has mass $\geq \varkappa_D$, which is the bound in total variation; for odd F with values in $[-1,1]$, $\int F \, \mathrm{d}\mu_{\mathcal{R}} = \frac{1}{2} \int F \, d(\mu_{\mathcal{R}} - \bar{\xi}_* \mu_{\mathcal{R}})$ lies in $[\varkappa_D - 1, 1 - \varkappa_D]$; and for flip-invariant Φ, since $\mu \geq \frac{1}{2}(\theta_0 + \bar{\xi}_* \theta_0)$ and $W_C \circ \bar{\xi} = -W_C$,

$$\int |W_C - \Phi| \, \mathrm{d}\mu \geq \frac{1}{2} \int (|W_C - \Phi| + |W_C + \Phi|) \, \mathrm{d}\theta_0 \geq \int |W_C| \, \mathrm{d}\theta_0 = \mathsf{m}_C > 0,$$

$|W_C|$ being near 1 on the configurations near 1; the same for $\operatorname{sign} W_C$. *Densities.* If $\mathsf{h}_d(E) = 0$, the preimage of E under the block map from the depth D is Haar-null by Lem. 14.18, and the upper bound found with the floor gives $\mu_d(E) \leq \mathsf{R}_{[D]}$ for every large D, so $\mu_d(E) = 0$. *The ceiling.* On the open set X_0 of full Haar measure of Lem. 14.16(b), which has full μ-measure, the four bricks with their overlaps determine a configuration up to a gauge transformation and one flip; Lem. 14.16(b) with $\mathsf{H} = \mathbb{Z}_2$, $C_\kappa = C$ for the non-trivial character, gives $F = \Phi_0 + \Phi_1 u$ with Φ_i measurable for the bricks; for bricks three sites thick the cubes generate the bricks by the same lemma; the block field of a coarser depth in $\mathcal{R}$ is a function of that of the depth d in $\mathcal{R}$.

Lem. 14.16(c) gives the adjoint loop. *The commutant.* $\mathfrak{L}$ is maximal abelian on $\mathfrak{h}$, the space of the gauge-invariant classes of $L^2(\mu)$, the density ρ being chosen gauge invariant; $\mathfrak{E}$ is its part fixed by the flip α; on $Y = \{\rho > 0\} \cap \alpha\{\rho > 0\}$ the flip is implemented by v, a partial isometry with $v^2 = e$ and $v M_F v = M_{F \circ \alpha} e$, and on the rest of $\{\rho > 0\}$ the flip moves no mass inside; so $\mathfrak{E}' = \mathfrak{L} + \mathfrak{L} v$, isomorphic to $L^\infty(Y_1) \oplus L^\infty(Y \cap \{u = 1\}) \otimes M_2(\mathbb{C})$, $Y_1 = \{\rho > 0\} \setminus Y$; $vu = -uv$, and $v \neq 0$ because $\mu(Y) \geq \varkappa_D$: the measure $\mu \wedge \alpha_* \mu$, of the density $\min\{\rho, \rho \circ \alpha\}$, vanishes off Y and lies above θ_0.

For this last point α is realised as the flip of a surface whose rim misses $\mathcal{R}$, at the depth $d+1$. In the units of $\Lambda_{d,\ell}$ let $y_0 = (b_1 + 1, b_2 + 1)$, a site of the hole ($b'_i - b_i \geq 2$), and $y_1 = (a'_1 + 1, b_2 + 1)$, a site beyond Q ($a'_1 - a_1 \leq N - 2$); let S_1 be the flat sheet of $\Lambda_{d+1,\ell}$ in the plane $x_2 = b_2 + 1 + \frac{1}{2L}$ from the straight closed dual line of $\Lambda_{d+1,\ell}$ in the direction 3 over $y_0 + \frac{1}{2L}(1,1)$ to the one over $y_1 + \frac{1}{2L}(1,1)$, across the arm of the ring at $x_1 = b'_1$. Each line lies over the open square of side 1 about y_0 or y_1, which meets no closed cube of $\mathcal{R}$: the rim misses $\mathcal{R}$; and C crosses S_1 once. $\mathcal{R}$ is a union of closed cubes of side $L^{-(d+1)}$ about sites of $\Lambda_{d+1,\ell}$, L being odd; so the first half of (i) and (g) hold for S_1 at the depth $d+1$.

The block map from the depth $d+1$ to the depth d reads on the bonds of $\mathcal{R}$ the bonds of $\mathcal{R}$ alone, and by the first clause of (g), no plaquette in $\mathcal{R}$ being pierced by the rim, it turns the flip of S_1 into the sign field $(-1)^{\#(c \cap \mathsf{S}_1)}$ on the bonds c of $\mathcal{R}$ at the depth d; this sign field is closed on the plaquettes of the bricks and odd on C, so it differs from α by a coboundary, which acts trivially on the gauge-invariant functions. Hence θ_0, carried down from the depth $d+1$, lies below μ and below $\alpha_* \mu$, and its mass is $\varkappa_D$. This holds for every ring admitted, D and $\varkappa_D$ being those of the first half of (i) at the depth $d+1$.

If besides $b'_i - b_i \geq 3$ ($i = 1, 2$) and $a'_i - a_i \leq N - 3$ for one i, say $i = 1$, a sheet of $\Lambda_{d,\ell}$ itself serves as well, with D and $\varkappa_D$ at the depth d: in the units of $\Lambda_{d,\ell}$, the flat strip $\{x_2 = b_2 + \frac{3}{2},\ b_1 + \frac{3}{2} \leq x_1 \leq a'_1 + \frac{3}{2}\}$ between the straight closed dual lines in the direction 3 over $(b_1 + \frac{3}{2}, b_2 + \frac{3}{2})$ and over $(a'_1 + \frac{3}{2}, b_2 + \frac{3}{2})$. A closed cube of side 1 about a site meets such a line only if the site lies in one of the four columns of sites nearest to the line, and these columns hold no site of $\mathcal{R}$: about the first line they lie over $b_1 + 1, b_1 + 2$ and $b_2 + 1, b_2 + 2$, in the hole, as $b_i + 2 < b'_i$; about the second they lie over $a'_1 + 1, a'_1 + 2$ in the direction 1, beyond Q, as $a'_1 + 2 < a_1 + N$.

So the four cubes about each line, at every height, are free, and the rim of the strip misses $\mathcal{R}$; C crosses the strip once, on its side $x_1 = b'_1$, as $b_1 < b_1 + \frac{3}{2} < b'_1 < a'_1 + \frac{3}{2}$ and $b_2 < b_2 + \frac{3}{2} < b'_2$; the flip of the strip is closed on the plaquettes of the bricks and odd on C, so it differs from α by

a coboundary, and the first half of (i) at the depth d puts the measure θ_0 of that depth below μ and below $\alpha_*\mu$. *The vortex line.* At the cutoff K the flip of the block fields of $\mathcal{R}$ is the insertion of the closed vortex line $\partial\mathsf{S}_1$: (g) at the depth $d+1$, for the functions of the block fields of $\mathcal{R}$ at the depth $d+1$ that factor through the block map, together with the covariance just used, gives $\mu^K\circ\alpha = x_K\mu^K$ with $x_K = \mathbb{E}_K[X_K\,|\,V]$, X_K the ratio of the twisted to the untwisted weight, at every K; so $x_K\cdot x_K\circ\alpha = 1$, $v_KF = x_K^{1/2}F\circ\alpha$ is a self-adjoint unitary, and, with ρ_K the density of μ^K for Haar measure (Lem. 14.18), $|\langle F, v_KF'\rangle - \langle F, vF'\rangle| \le \sup|F|\sup|F'|\int|\sqrt{\rho_K\,\rho_K\circ\alpha} - \sqrt{\rho\,\rho\circ\alpha}|\,\mathsf{dh}$, which the Schwarz inequality bounds by $2\sqrt{2}\sup|F|\sup|F'|$ times the square root of the total variation between μ^K and μ, at most $\sum_{K'\ge K}\mathfrak{o}_{K'}$.

Wide rings. Let moreover the hole be at least three bonds wide in both directions, and the ring be of positive height and lie in an open ball of radius $< \ell/4$. Then a sheet of $\Lambda_{d,\ell}$ itself realises α: in the units of $\Lambda_{d,\ell}$, the flat sheet S in the plane $x_2 = b_2 + \frac{3}{2}$ from the straight closed dual line in the direction 3 over the centre $(b_1+\frac{3}{2}, b_2+\frac{3}{2})$ of four cubes of the hole (here the width three is used) to the one over the centre of four cubes beyond the ring on the same row (for which the ball leaves room) has its rim off $\mathcal{R}$ and crosses C once. Let $\xi^{(D)}$ be its flip at the depth D, π_D the limit law of the block fields of $\mathcal{R}$ at the depth D, so that $\pi_d = \mu$, and Φ_D the block map from these fields to those at the depth d; by the first clause of (g), $\Phi_D\circ\xi^{(D)} = \bar\xi\circ\Phi_D$ with the flip $\bar\xi$ of S at the depth d, which differs from α by a coboundary; as μ is invariant under gauge transformations, we take $\alpha = \bar\xi$ here.

We show that $\mu(\alpha A) = 0$ gives $\mu(A) = 0$ for a Borel set A of the block fields of $\mathcal{R}$ at the depth d.

(1) *From the measure to the fields.* Let $\mathcal{G}_D$ be the set of the $V\in\mathsf{In}_{[D]}$ whose flipped restriction $\xi^{(D)}V|_{\mathcal{R}}$ continues to a field of $\mathsf{In}_{[D]}$. If $\mu(\alpha A) = 0$, then $\mu(A)\le 1-\mu_D(\mathcal{G}_D)+\mathsf{R}_{[D]}$. Indeed, with $\tilde A = \Phi_D^{-1}A$, $\pi_D(\xi^{(D)}\tilde A) = \mu(\alpha A) = 0$; by the floor, $\int_{\xi^{(D)}\tilde A} q_D\,\mathrm{d}w = 0$, $q_D(w)$ the Haar measure of the continuations of w into $\mathsf{In}_{[D]}$, and $q_D > 0$ exactly on the projection Pr_D of the open set $\mathsf{In}_{[D]}$ to the bonds of $\mathcal{R}$; so $\tilde A\cap\xi^{(D)}\mathsf{Pr}_D$ is Haar-null. The fields of $\mathcal{G}_D$ whose restriction to $\mathcal{R}$ lies in $\tilde A$ have it in $\tilde A\cap\xi^{(D)}\mathsf{Pr}_D$; they form a Haar-null set, of μ_D-measure $\le\mathsf{R}_{[D]}$ by the upper bound; the others with the restriction in $\tilde A$ lie off $\mathcal{G}_D$.

(2) *Insertion of a straight line into a small field.* In units of L^{-D} let l_P be the straight closed dual line in the direction 3 over a dual point P of the $(1,2)$-plane, $\varrho_1\ge 1$ an integer with $2\varrho_1+4<\ell L^D$, $\mathcal{T}$ the sites whose horizontal part lies within the sup-distance $\varrho_1+\frac{3}{2}$ of P (the tube), s a sign field with $(ds)_p = -1$ exactly at the plaquettes with corners in $\mathcal{T}$ that l_P pierces, and V a field with $|\partial V(p)-\mathbf{1}|\le\delta_0$ at the plaquettes with corners in $\mathcal{T}$. Then there is a field V' with $V'_c = s_cV_c$ on every bond that has not both ends in $\mathcal{T}$, $|\partial V'(p)-\mathbf{1}|\le 6\varrho_1^{-2}+3\delta_0$ at the plaquettes with all corners in $\mathcal{T}$, and $\partial V'(p) = (ds)_p\partial V(p)$ at all the others.

Idea: take an axial gauge γ of $\mathcal{T}$, $\mathbf{1}$ on the spanning tree made of all the bonds in the direction 3 at a corner o of the unit plaquette centred at P but one, one line in the direction 1 through o in each layer, and all the bonds in the direction 2, the holonomy Υ around the torus along the bonds in the direction 3 at o being made diagonal by a constant gauge transformation; then, x measured from o, a bond in the direction 1 at x is within $|x_2|\delta_0$ of $\mathbf{1}$, and a bond in the direction 3 at x is $\mathsf{Y}_x\Upsilon_x$ with $\Upsilon_x\in\{\mathbf{1},\Upsilon\}$ and Y_x within $|x|_1\delta_0$ of $\mathbf{1}$. For positively oriented bonds put $V'_c = s_c\gamma(c_-)^{-1}e^{-i\varsigma_c\sigma_3}\gamma(c_-)V_c$ on the bonds with both ends in $\mathcal{T}$, with the profile $\varsigma_c = (1-f_c)\vartheta_c$ on the horizontal bonds, ϑ_c half the signed angle that c subtends at P, $f_c = \min\{1,(r_c^2-\frac{1}{4})/\varrho_1^2\}$, r_c the sup-distance of the midpoint of c from P, and $\varsigma_c = 0$ on the vertical bonds; define the values on reversed bonds by inversion.

Then $d\varsigma = \pi$ at the pierced plaquettes, which takes away the sign of ds there, and $|d\varsigma|\le 6\varrho_1^{-2}$ at the other horizontal ones, as $d\vartheta = 0$ there: three terms $\pm(f_{c'}-f_c)\vartheta_c$ remain, each $\le\frac{2r}{\varrho_1^2}\cdot\frac{1}{r}$, $r\ge 1$ the sup-distance of the centre of the plaquette from P; this is the source of the 6. The rotations $e^{-i\varsigma_c\sigma_3}$ commute with one another and with Υ. With $o=(0,0)$ and $P=(\frac{1}{2},\frac{1}{2})$ in a horizontal layer,

the half-angle bound is $|\vartheta_c|\,|c_- - o|_1 \le 1$ for the positively oriented horizontal bonds. It is paired with the gauge error measured from the same o. For a horizontal plaquette the direction-2 bonds are $\mathbf{1}$, leaving one commutator with a direction-1 bond. For a vertical plaquette first factor out the original plaquette product; the remaining error is one commutator with Y_x in $\mathsf{Y}_x\Upsilon_x$, since the inserted rotations commute with Υ_x. Its cost is at most $2\delta_0$ in either case. Adding the original plaquette error δ_0 gives $3\delta_0$. At the boundary $f_c = 1$, so the insertion matches the prescribed exterior field; no boundary value of the gauge at o is required.

(3) *The witness.* Let $\varrho_1 = \frac{1}{2}(L^{D-d} - 5)$. At the depth D the hole holds a square of $(2\varrho_1 + 4)^2$ sites about a dual point P_1 of the $(1,2)$-plane, and beyond the ring, on its row, there is a dual point P_2 whose square of that size misses Q (the ball leaves room); the tubes $\mathcal{T}_1, \mathcal{T}_2$ over them are disjoint and hold no site of $\mathcal{R}$. Let $s = -1$ on the bonds in the direction 2 that cross the flat strip between the vertical lines over P_1 and P_2, and $s = 1$ elsewhere: $(ds)_p = -1$ exactly at the plaquettes that the two lines pierce. On the bonds of $\mathcal{R}$ at the depth D, s and $\xi^{(D)}$ are closed on the plaquettes of the bricks and odd on the loop around the hole, so $s = \xi^{(D)} dz$ there, with signs z on the sites of $\mathcal{R}$ (Lem. 14.16(b) for the four bricks, whose group Z/B is $\mathbb{Z}_2$ and is read on that loop).

Let $\theta' < \frac{2}{3}$ and D be so large that $6\varrho_1^{-2} + 3\theta'\varepsilon(x_{[D]}) < 2\varepsilon(x_{[D]})$, as ϱ_1^{-2} is of the order of $L^{-2(D-d)}$ and $\varepsilon(x_{[D]})$ of the order of $L^{-D/2}$ up to logarithms. If $\varpi(V) < \theta'\varepsilon(x_{[D]})$, V is a small field, and the insertion in both tubes with $\delta_0 = \theta'\varepsilon(x_{[D]})$ gives V' with $V' = sV$ off the tubes, so on the bonds of $\mathcal{R}$, and $\varpi(V') < 2\varepsilon(x_{[D]})$; the gauge transform of V' by z, continued by 1, is a small field equal to $\xi^{(D)}V$ on the bonds of $\mathcal{R}$. So $\{\varpi < \theta'\varepsilon(x_{[D]})\} \subset \mathcal{G}_D$.

(4) *The tail at a fraction of the threshold.* For integers $n \ge 0$ and $D' = D - n$ with $x_{[D']} \ge \mathsf{t}_\natural$, let $\Theta_n = 4B_3 L^{-3n/2} + 2\frac{C_H A_1}{A_0}\sum_{j<n} L^{-3j/2}$, where $B_3 \ge 1$ and $C_H \ge 1$ are the constants of the list of Thm. YM3_I:D that enter the bound of the background over a small field and the contraction of the backgrounds from one level to the next (Thm. YM3_I:E and the curvature comparison in the proof of Lem. YM3_I:113.10). Then $\mu_D^K\{\varpi \ge \Theta_n\varepsilon(x_{[D]})\} \le \mathsf{R}_{[D']}$ for $K \ge D+1$, and in the limit. Run the chain with its top level at the depth D'. On the histories with $\mathfrak{w}^{\mathrm{sf}} \ne 0$ the field at the depth D' lies in the inner ball, and the field of every finer level lies in the field cut and is the field over the next coarser one moved by a fluctuation from the ball of the step; so the regularity numbers m of the backgrounds obey $\mathsf{m} < 2B_3\varepsilon(x_{[D']})$ at the top (Thm. YM3_I:E) and $\mathsf{m}_k \le L^{-2}\mathsf{m}_{k+1} + C_H\delta(x_k)$ from the level $k+1$ to the finer level k (the proof of Lem. YM3_I:113.10).

Summed down the n levels, with $\varepsilon(x_{[D-j]}) \le L^{j/2}\varepsilon(x_{[D]})$ and the same for δ, this gives $\mathsf{m} < \frac{1}{2}\Theta_n\varepsilon(x_{[D]})$ at the depth D, and a field whose background has the regularity number below m has its plaquettes below $2\mathsf{m}$ (as in the inclusions YM3_I:(2159)): $\varpi < \Theta_n\varepsilon(x_{[D]})$. The other histories have the mass $\le \mathsf{R}_{[D']}Z$ by YM3_I:(2189) at $1 - F$ and YM3_I:(2190), F the indicator of $\{\varpi(\mathcal{M}^{K-D}U) \ge \Theta_n\varepsilon(x_{[D]})\}$; the limit follows.

(5) *The end.* Let n_0 be the least integer with $L^{3n_0/2} \ge 80B_3$; it is fixed by L and B_3, and chosen so that $\Theta_{n_0} < \frac{3}{5}$: the first term of Θ_{n_0} is $\le \frac{1}{20}$, and the condition $A_0 \ge 4B_3C_HA_1$, one of the conditions on A_0 written in the proof of Lem. 4.48, with $B_3 \ge 1$ and $L \ge 13$, makes the second $\le \frac{1}{2}(1 - 13^{-3/2})^{-1} < 0.52$. For D so large that $x_{[D-n_0]} \ge \mathsf{t}_\natural$, $\mathsf{R}_{[D]} < 1$ and $6\varrho_1^{-2} < \frac{1}{5}\varepsilon(x_{[D]})$, (1) and (3) at $\theta' = \frac{3}{5}$ give $\mu(A) \le \mu_D\{\varpi \ge \frac{3}{5}\varepsilon(x_{[D]})\} + \mathsf{R}_{[D]}$, and (4) at $n = n_0$ bounds the first term by $\mathsf{R}_{[D-n_0]}$. So $\mathsf{R}_{[D-n_0]}$ bounds the mass of the histories that the chain with its top level at the depth $D - n_0$ leaves out, this chain controlling the plaquettes at the depth D, and $\mathsf{R}_{[D]}$ comes from the upper bound in (1). Both tend to 0 as $D \to \infty$, and $\mu(A)$ reads no D: $\mu(A) = 0$. With αA in the place of A the converse holds; so the law and its flip have the same null sets, $e = 1$, v is a self-adjoint unitary, and with the ceiling, whose hypothesis on the corners the positive height meets, $\mathfrak{L} = \mathfrak{E} \oplus \mathfrak{E}u$ is a grading by $\mathbb{Z}_2$.

A band that winds. The flip $\bar{\xi}_1$ of a band $\mathbb{Z}/N \times I_2 \times I_3$ is the restriction of a flip of the torus, a symmetry of the law: v is unitary, odd and even functions are orthogonal, $\mathfrak{E}u_1 \ne 0$ as $u_1^2 = 1$, and

the cubes of the band generate $\mathfrak{E}$ by Lem. 14.16(b) with the cubes as boxes; the only closed sign fields on the cubes of the band modulo coboundaries are 1 and the winding one.

14.1.11. *Scope of the conclusions.* Thm. AP(a)–(g) concern the centred functionals of volume-first limits and their components, for which the proof uses a periodic lattice; for other limits of periodic Gibbs states only (h) is asserted. The proofs of Thms. AO and AP use no energy bound beyond the polynomial one of the reconstruction, no gap, no uniqueness of the vacuum of σ and no rate of clustering. Thm. AQ(d) bounds from below the norms of the field at one instant and of one Euclidean slab. Thm. AS(g)–(i) are statements on the commutative algebras of the Euclidean block fields at a fixed depth on one torus.

15. A continuum Langevin process

Fix the physical torus and the label b, and keep the invariant thread law obtained from the compatible block-field marginals. Put $A_K = \sum_p a_p$, and write μ^K for the Wilson probability measure of [7, §2] on this fixed torus. On the cutoff lattice the stationary Wilson Langevin process has generator and Dirichlet form

$$\mathcal{L}_K = a_K^{-1} \sum_{b,\alpha} (\partial_b^\alpha)^2 - a_K^{-1} \beta\, \partial A_K \cdot \partial, \qquad \mathcal{E}_K(F) = a_K^{-1} \int |\partial F|^2 \, \mathrm{d}\mu^K.$$

The derivatives are along the three orthonormal colour directions at each link.

Sketch N (A nonfrozen reversible Markov limit). Subsequences of the stationary lattice processes converge on gauge-invariant cylinder observables to a nonconstant stationary reversible Markov process on block-field threads. Its closed Dirichlet form is the strong-and-weak limit of the lattice forms, and its semigroup mixes to the invariant thread law. The result is on a fixed torus and does not identify the limits from different subsequences. ■□

Sketch of the argument. Separate the fine field from a fixed block field using the small-field chain. On each fluctuation box the conditional Hamiltonian is strictly convex after the controlled nonlinear remainder is included. The conditional law therefore has Poincaré and logarithmic Sobolev bounds. Differentiate the conditional expectation defining one averaging step. The response of the background transports the coarse gradient, while the fluctuation covariance controls the additional term. The resulting multiplicative losses have the form $1 + C\delta(L^d b)$, and their product is finite because $\sum_d \delta(L^d b) < \infty$.

At a selected coarse depth n, retain a connected region of regular fields. Its finite-dimensional inequality supplies the remaining top-scale bound. The small-field chain omits mass $r_n \to 0$, which appears as a defect rather than a uniform spectral-gap constant:

$$\operatorname{Var}(F) \le C_n \mathcal{E}(F) + r_n \operatorname{osc}(F)^2.$$

Conditional versions control the fine field uniformly in the cutoff; the constant C_n at the top can depend on the torus.

These bounds give compactness for uniformly bounded observables of bounded energy. Along a process subsequence, they provide the lower bound for weakly convergent functions and recovery sequences converging strongly. Resolvents and semigroups then converge strongly, identifying a closed Markovian limiting form and the reversible Markov process it generates. Let Y_n be the selected coarse block field and take its plaquette observable $F = a_p(Y_n)$. Apply the coarse-gradient estimate to a recovery sequence for F. Lower semicontinuity of its Dirichlet integral on the regular coarse-field set gives $\mathcal{E}(F) > 0$. The limiting semigroup therefore moves this observable, so the process is not frozen.

For mixing, fix n in the weak variance inequality and let time increase. Its exponential term disappears, leaving the defect $r_n \operatorname{osc}(F)^2$. Now increase n to remove that defect. The analogous

entropy estimate gives relaxation of bounded probability densities in total variation on the invariant algebra.

This order of limits matters. The argument supplies constants uniform in the fine cutoff, not constants uniform in the torus volume. Stronger gauge estimates across several block levels would make the top-scale constants more explicit; the qualitative conclusion above does not require them.

15.1. **Continuum limits of the lattice Langevin dynamics.** At a fixed physical label and on a fixed torus, the Wilson Langevin processes have subsequential limits on threads of block fields. We prove that each limit is a nonconstant stationary reversible Markov process on the gauge-invariant σ-algebra. The lattice Dirichlet forms converge in the strong and weak sense. Conditional Poincaré and logarithmic Sobolev inequalities control the fine field; a connected coarse-scale region then gives weak inequalities with defects tending to zero. These estimates imply mixing and entropy relaxation. Their constants may depend on the fixed torus, and the result does not identify the processes obtained along different subsequences.

We also study observables regularised by the Wilson gradient flow. The finite-cutoff flow identities and upper bounds use no probabilistic assumption. The moment estimates use the admissibility conclusion (Adm) of Prop. 16.1, concerning conditional layer means and colour covariances. Identifying the continuum flow limits requires the approximation step described below. The nonlinear defect bounds are proved in Lem. 16.4. Lem. 16.8 proves lower propagation on a regular time interval; Prop. 16.9(c) applies it on the interval of the short-time flow theorem and extends the lower bound to every final time by current transport and the integrated curvature. These estimates do not complete the continuum-flow identification, whose remaining work is described under (Flow approximation).

The generator requires separate treatment. Some bounded flowed observables have generator images whose norms grow exponentially in the inverse square of the spacing. We therefore first insert condenser cutoffs, pass to the continuum limit, and then remove the cutoffs. The resulting generator identity is distributional.

15.1.1. *The lattice process and its limiting objects.* $G = \mathrm{SU}(2)$; $\mathfrak{g}$ is the space of hermitian traceless 2×2 matrices, $\langle X, Y\rangle = \frac{1}{2}\mathrm{tr}\, XY$, $|X| = \langle X, X\rangle^{1/2}$, so that $X^2 = |X|^2\mathbf{1}$ and $e^{iX} \in G$; d is the geodesic distance of G, the unit sphere S^3, with $\mathsf{d}(e^{iA}, \mathbf{1}) = |A|$ for $|A| \le \pi$ and $|g - h| = 2\sin\frac{1}{2}\mathsf{d}(g, h)$. The numbers L, $a_d = L^{-d}$, $\ell = 2L^m$, the torus $T_\ell = \mathbb{R}^3/\ell\mathbb{Z}^3$ of side ℓ, the lattices $\Lambda_{d,\ell} = a_d\mathbb{Z}^3/\ell\mathbb{Z}^3 \subset T_\ell$ (Section YM3_I:3) and the averaging map $\mathcal{M}$ (equations YM3_I:(37)–(39)) have the conventions just stated. The label b and setting (E) are those of §2.3.1: the constants of the construction are an admissible choice $\varsigma = (\sigma^-, \gamma)$ of the canonical list of constants and conditions (§2.3.1), subject to the conditions (Th1) and (Th2) of §2.3.1, with γ the fixed reference-coupling ceiling; the label satisfies $\mathsf{b} \ge \gamma^{-2}$ and $M\, R(\mathsf{b}) \le L^m$, with the cube size $M = L^{m'}$ and the function R of §2.3.1, $R(x)$ the least power L^j, $j \ge 0$, that is not smaller than a fixed power of $\log x$; and $\beta = L^K\mathsf{b}$, so that $\mathsf{b} = a_K\beta$.

Ω_d is the set of configurations on $\Lambda_{d,\ell}$, dV (on Ω_K: dU) its Haar probability measure, $V^u(b) = u(b_-)V(b)u(b_+)^{-1}$ the gauge action; invariant means gauge invariant. $\partial V(p)$ is the plaquette variable, $a_p = 1 - \frac{1}{2}\mathrm{tr}\,\partial V(p)$, $\varpi_p = |\partial V(p) - \mathbf{1}| = (2a_p)^{1/2}$, $\varpi(V) = \max_p \varpi_p$, $A_K = \sum_p a_p$, $A^a = a_K^{-1}A_K$, and $\mathcal{O}(f) = a_K^{-1}\sum_p f(x(p))\, a_p$ for real f on T_ℓ, $x(p)$ the barycentre of p. On Ω_K we write $N = \ell/a_K$ for the number of sites along an axis. Tangent vectors are real $\mathfrak{g}$-valued 1-forms X, with the curves $s \mapsto e^{isX}V$; $\langle X, \partial F(V)\rangle = \frac{d}{ds}F(e^{isX}V)|_{s=0}$, $\|X\|^2 = \sum_b |X(b)|^2$; $\partial_b F$ is the value of the 1-form ∂F at the bond b, and $\partial_b^\alpha F = \langle \sigma_\alpha, \partial_b F\rangle = \frac{d}{ds}F(e^{is\sigma_\alpha\delta_b}V)|_{s=0}$, with the Pauli matrices $\sigma_1, \sigma_2, \sigma_3$, a basis of $\mathfrak{g}$ that is orthonormal for $\langle\cdot,\cdot\rangle$, and δ_b the function of the bonds that is 1 at b and 0 at the others; $\Gamma_d(F) = a_d^{-1}|\partial F|^2$ on Ω_d, and $\Gamma_d(F, F')$ is its polarisation.

The vectors $\mathrm{d}_V\lambda$, $(\mathrm{d}_V\lambda)(b) = \lambda(b_-) - \mathrm{Ad}(V(b))\lambda(b_+)$ for $\mathfrak{g}$-valued site functions λ, are the gauge directions at V; a tangent vector orthogonal to them is horizontal, and the gradient of an invariant function is horizontal. $\mathrm{osc} = \sup - \inf$.

Dynamics. $\mu^K = Z_K^{-1} e^{-\beta A_K}\, dU$; $\Delta_K = a_K^{-1}\sum_{b,\alpha}(\partial_b^\alpha)^2$; $\mathcal{L}_K = \Delta_K - a_K^{-1}\beta\sum_b \langle \partial_b A_K, \partial_b \cdot\rangle$; $\mathcal{E}_K(F) = \int \Gamma_K(F)\,\mathrm{d}\mu^K$ on the closure $D(\mathcal{E}_K)$ of $C^\infty(\Omega_K)$, with $\mathcal{E}_K(F) = \infty$ off $D(\mathcal{E}_K)$; $P_t^K = e^{t\mathcal{L}_K}$, and $(U_t)_{t\in\mathbb{R}}$ is the stationary process. $J(U) = a_K^{-3}\partial A_K(U)$ is the current (YM3_I:9.2), $(J,J)_a = a_K^{-3}|\partial A_K|^2$; Φ_s is the gradient flow $\frac{d}{ds}U(b) = -i a_K J(U)(b)U(b)$, $U_s = \Phi_s U$, $v = -a_K^{-2}\sum_b \langle \partial_b A_K, \partial_b \cdot\rangle$ its vector field, and $\Lambda_\sigma(U) = \int_0^\sigma a_K^{-2}\varpi(U_s)\,\mathrm{d}s$.

Threads and limits. $\mathcal{Y} = \prod_{d\geq 0}\Omega_d$ is compact and metrisable; a thread is a point $Y = (Y_d)$ of $\mathcal{Y}$ with $Y_d = \mathcal{M}Y_{d+1}$ for all d; $\pi_{K,d} = \mathcal{M}^{K-d}$. The laws μ_d^K of $\pi_{K,d}U$ under μ^K converge in total variation to measures μ_d, with $\mu_{d+1}\circ\mathcal{M}^{-1} = \mu_d$ (Cor. 2.8), and $\mu = \mu_{\ell,\mathsf{b}}$ is the measure on threads with the marginals μ_d. A cylinder function is $\Psi(Y_d)$, Ψ bounded Borel on Ω_d, and $\Psi_K = \Psi\circ\pi_{K,d}$; L^2_{inv} is the closure in $L^2(\mu)$ of the invariant cylinder functions, which generate the invariant σ-algebra, and Π is the orthogonal projection onto L^2_{inv}.

Along $K_j\to\infty$, $H_j \in L^2(\mu^{K_j})$ converge *strongly* to $F\in L^2(\mu)$ if there are cylinder functions $\Psi_m\to F$ in $L^2(\mu)$ with $\lim_m \limsup_j \|H_j - \Psi_{m,K_j}\|_{L^2(\mu^{K_j})} = 0$, and *weakly* if $\sup_j\|H_j\| < \infty$ and $\int H_j\Psi_{K_j}\,\mathrm{d}\mu^{K_j} \to \int F\Psi\,\mathrm{d}\mu$ for every cylinder function Ψ. A *limit* is a sequence $K_j\to\infty$ along which all means $\mathbb{E}\prod_i \Psi_i(\pi_{K_j,d}U_{t_i})$ of bounded Borel Ψ_i converge, with its limit process (Y_t) on threads; $(F, Q_tF') = \mathbb{E}F(Y_0)F'(Y_t)$, $\mathcal{E}_\star(F) = \sup_{t>0} t^{-1}(F,(1-Q_t)F)$, $A_\star$ is the operator of $\mathcal{E}_\star$, and the limit is frozen if $\mathcal{E}_\star$ vanishes on L^2_{inv}, that is, if $Q_t = 1$ on L^2_{inv} for all t ($\mathcal{E}_\star$ is finite only on L^2_{inv}, Lem. 15.1(b)).

The forms converge *in the strong and weak sense* if (M1) $\mathcal{E}_\star(F) \leq \liminf_j \mathcal{E}_{K_j}(H_j)$ whenever $H_j\to F$ weakly, and (M2) every F has $H_j\to F$ strongly with $\mathcal{E}_{K_j}(H_j)\to\mathcal{E}_\star(F)$.

The chain. We recall the following objects of the small-field chain on one torus (Sections YM3_I:113, YM3_I:114, YM3_I:115), which rests on the results on one step of Prop. 6.31. $\delta(\mathsf{t}) = A_1\mathsf{t}^{-1/2}(\log\mathsf{t})^{\mathfrak{p}_0}$, $\varepsilon = \frac{A_0}{A_1}\delta$, $\mathfrak{b}(\mathsf{t}) = \mathsf{t}^{1/2}\delta(\mathsf{t})$, the threshold functions of equation YM3_I:(634), with the constants $A_0, A_1 > 0$ and $\mathfrak{p}_0\in\mathbb{N}$ of the construction; $\varepsilon_n = \varepsilon(L^n\mathsf{b})$; $\sum_d \delta(L^d\mathsf{b}) < \infty$. For an integer $n\geq 1$ with $L^n\mathsf{b}\geq \mathsf{t}_\natural$, $\mathsf{t}_\natural \geq \gamma^{-2}$ the fixed calibration of §2.3.1, a number formed from the constants alone, and for $K > n$, put $t = K - n$, so that $1\leq t\leq K-1$, as required of a top level in §2.3.1: level k is depth $d = K-k$, $V_k = \mathcal{M}^k U$, $x_k = L^d\mathsf{b}$, so that $x_k^{-1} = a_d/\mathsf{b}$.

The cuts $\mathsf{In}_k \subset \mathsf{Cut}_k \subset \Omega_d$ of the definitions preceding Lem. YM3_I:113.10 are open and invariant (Lem. YM3_I:113.10); $\mathsf{Cut}_k \subset \{\varpi < \theta_d\}$ with θ_d below; $\mathsf{I}_n = \mathsf{In}_t = \{\varpi < 2\varepsilon_n\} \subset \Omega_n$, and D_n is the connected component of $\mathbf{1}$ in I_n. R_n is the number YM3_I:(2187) at the level $t = K-n$, which depends on n, ℓ and b but not on K; it bounds the mass of the large fields (Prop. YM3_I:115.2, lost-mass bound), and by YM3_I:(2187) $\mathsf{R}_n \leq C(\ell,\mathsf{b})\,L^{3n}\exp[-c' A_0^2(\log L^n\mathsf{b})^{2\mathfrak{p}_0}]$ with the rate $c' > 0$ of §2.3.1 and $2\mathfrak{p}_0 \geq 2$, so that it tends to 0 faster than every exponential in n.

The chain measure is $\sigma^K = Z_K^{-1}\mathfrak{w}_t^{\mathrm{sf}} e^{-\beta A_K}\,dU$, with the constant Z_K of μ^K and the weight of the small-field chain of level t, the function defined before Prop. YM3_I:115.2,

$$\mathfrak{w}_t^{\mathrm{sf}}(U) = \mathbf{1}_{\mathsf{In}_t}(V_t)\prod_{k=0}^{t-1}\big[\mathbf{1}_{\mathsf{Cut}_k}\,\mathfrak{G}_k[x_k]\,\chi_{x_k}^{\mathrm{sf}}\big](V_k),$$

where $\mathfrak{G}_k[x_k]$ is the gauge-fixing factor of the axial gauge on the blocks of the level $k+1$ and $\chi_{x_k}^{\mathrm{sf}}$ the fluctuation test of the step $k+1$ ([7, §B.2]); the weight is a bounded Borel function with values in $[0,\infty)$ (Prop. YM3_I:115.2, proof and the definitions preceding Prop. YM3_I:115.2). So σ^K is a positive measure of mass in $[1-\mathsf{R}_n, 1]$ (Lem. 15.3(i)), normalised by the constant of μ^K and not by its own mass; and as its weight carries gauge-fixing factors, σ^K is not gauge invariant, and its comparisons with μ^K are made on invariant functions. σ_d^K is its image at depth d, σ_n^K the top law.

For $0 \le k < t$ and $W \in \mathsf{Cut}_{k+1}$, a background U_+ over W is any element of the minimal orbit over W with $\mathcal{M}^k U_+$ in the axial gauge, and $V_W(B)$ is the chart of the fibre $\{\mathcal{M}V = W\}$ ([7, §B.2]), defined on the convex box $\{|B(b)| < \mathfrak{b}(x_k)\}$ of the bonds that are not pivot bonds, with $V_W(B)(b) = e^{ix_k^{-1/2}B(b)}(\mathcal{M}^k U_+)(b)$ there; it carries the box into Cut_k (Lem. YM3_I:113.10). The fixed radii $0 < \alpha_0 < \alpha_1$ are two constants of the construction (Def. YM3_I:49.1 and Lem. YM3_I:82.1, Lem. YM3_I:82.1), free of n, at which the analyticity cores of the chain are taken (Def. YM3_I:49.1); Cut_{k+1} lies in the class $\mathcal{W}_{k+1}(x_k; \alpha_0, \alpha_1)$ of the sum rule (Lem. YM3_I:113.10).

One fluctuation integral p_W is set at every $W \in \mathsf{Cut}_{k+1}$ by the chart integral I of [7, eq. 191] at $\mathsf{t} = x_k$ ([7, Def. B.2] and equation [7, eq. 191]): it is the measure on the box with

$$\int h(V_W(B))\, p_W(\,\mathrm{d}B) = \frac{\mathsf{I}[hF_k](W)}{\mathsf{I}[F_k](W)} \qquad \text{for every bounded Borel } h \text{ on } \Omega_d,$$

F_k the integrand of the chain at the level k and at the source 0 (Prop. YM3_I:114.2); that is, p_W has on the box the density proportional to $\exp[-x_k G_W(B) + \mathsf{J}_W(B)]\, F_k(V_W(B))$, with the gauge-fixing form G_W and the exponent J_W of the Jacobians of the chart ([7, §B.2]). The denominator vanishes at no $W \in \mathsf{Cut}_{k+1}$: on the class of the sum rule $\mathsf{I}[F_k] = e^{\mathsf{c}_{k+1}(0)} F_{k+1} \neq 0$, $\mathsf{c}_{k+1}(0)$ the constant YM3_I:(2173) of the step (Prop. YM3_I:114.2). p_W depends on K, k and W, and not on the top level; that it is a probability measure with a positive continuous density on the box, and that its images under V_W form a regular conditional law of V_k given $V_{k+1} = W$ under σ^K, is Lem. 15.3(ii).

The tower of H is $h_0 = H$, $h_{k+1}(W) = \int h_k(V_W(B))\, p_W(\,\mathrm{d}B)$ at every $W \in \mathsf{Cut}_{k+1}$. For a finite measure ν, $\mathrm{Ent}_\nu(f) = \inf_{c>0} \int \Xi(f, c)\,\mathrm{d}\nu$, $\Xi(x, c) = x \log \frac{x}{c} - x + c \ge 0$.

Constants. $\mathsf{o}_n = [2C_{\mathrm{nl}} \frac{L}{L-1} E_0 + \frac{3}{2} c_\rho^2 A_0^2 (\log L^n \mathsf{b})^{2\mathfrak{p}_0}](\ell L^n)^3$, with the number $C_{\mathrm{nl}} = C'_{\mathrm{A}} + 2C_{\mathrm{E},0}(C_{\mathrm{lg}} + L^{4l_0})$, with C'_{A}, l_0, C_{lg} and $C_{\mathrm{E},0}$ defined exactly as in equation YM3_I:(2163) and its following sentence, using $K_{\mathrm{A}} = K^\sharp_{\mathrm{tw}}$ and $a_{\mathrm{A}} = a^\sharp$. The proof following that equation gives the bound on the non-local action, that is, the action of a bare array at the background of a field in a cut (equation YM3_I:(2180)). We also use the size $E_0 > 0$ of §15.3.1 and the constant c_ρ of the field cuts §2.3.1 (and C_H defined in [7, §P.1]); $\theta_d = 2c_\rho \varepsilon(L^d \mathsf{b})$; $\gamma_{\mathrm{r}} \in (0, 1]$ and $c_{\mathrm{r}} \ge 1$ are the numbers of Thm. YM3_I:U, entered through §15.3.1, which depend on L alone: the floor and the ceiling of a real reference form of the Gaussian integral of one step, from which the form of the step itself differs, in the ℓ^2 operator norm, by at most the defect $\frac{1}{2}\mathsf{u}_{\mathrm{G}}(LM)^{-3} \le \frac{1}{2}\gamma_{\mathrm{r}}$ (the completeness (St4) of the Gaussian structure of the step, Def. YM3_I:47.2 and Thm. YM3_I:W, with $\mathsf{u}_{\mathrm{G}} = 4C_{\mathrm{Str}}$ of Thm. YM3_I:U and the proof of Prop. YM3_I:113.4); so the form of the step lies between $\frac{1}{2}\gamma_{\mathrm{r}}$ and $c_{\mathrm{r}} + \frac{1}{2}\gamma_{\mathrm{r}}$; $\Lambda_+ = c_{\mathrm{r}} + \gamma_{\mathrm{r}}$; C_Q, c_Q, a_Q are the constants defined in Lem. 5.30, which depend only on L; for a 1-form X on $\Lambda_{K-j,\ell}$, $m_y^j(X) = \sum_{c'} e^{-\delta_0 d(y, c'_-)} |X(c')|$, $\delta_0 > 0$ the rate of decay in the bounds of the response, are the averages that enter these bounds, and B, S are the constants with $|\mathcal{H}_V(\zeta)(b)| \le \mathsf{B}\, m_y^j(\zeta)$ for b_- in the block of y and $\sum_y m_y^j(X)^2 \le \mathsf{S}^2 \|X\|^2$ (§5.3, Lem. 15.9); $\mathsf{M} = \max\{1, \mathsf{BS}\}$.

$C_{\mathrm{P}} = 4C_{\mathrm{ch}}/\gamma_{\mathrm{r}}$ (Lem. 15.3); $\epsilon_d^\flat = [1 + (C_\flat + C_{\mathrm{cov}})\delta(L^d \mathsf{b})]^2 - 1$ and $\Pi_n^\flat = \prod_{d>n}(1 + \epsilon_d^\flat)$, finite and tending to 1 (Prop. 15.5); $C^\sigma = 3\mathsf{M}^2 \Pi^\flat_{n_{\mathrm{a}}}$. Thresholds: n_2 is the least integer ≥ 1 with $L^{n_2}\mathsf{b} \ge \mathsf{t}_\natural$ such that $C_{\mathrm{Hs}}\delta(L^d \mathsf{b}) \le \frac{1}{4}\gamma_{\mathrm{r}}$ and $\theta_d \le \omega_L$ for $d \ge n_2$, with C_{Hs} the constant of Lem. 15.3(iv) and a number $\omega_L \le (3200L^3)^{-1}$ that depends only on L; $n_3 \ge n_2$ is the least integer such that for $d \ge n_3$ the number θ_d lies below the radius of §5.3 and $2\theta_d \le \varepsilon_{\mathcal{W}}\alpha_0$, with the fixed radii $\alpha_0 < \alpha_1$ of the chain and the number $\varepsilon_{\mathcal{W}}$ of Prop. YM3_I:52.3, so that θ_d is at most the threshold $\varepsilon_{\mathcal{W}} \cdot \frac{1}{2}\alpha_0$ of Prop. YM3_I:52.3 at the radii $(\frac{1}{2}\alpha_0, \frac{1}{2}\alpha_1)$; n_{a} is the least integer $\ge \max\{1, n_3\}$ such that for all $d \ge n_{\mathrm{a}}$: $L^d \mathsf{b} \ge \mathsf{t}_\natural$; $\theta_d \le \min\{a_Q, \varpi_{\mathrm{db}}\}$ (that θ_d lies below the radius of §5.3 holds already for $d \ge n_3$); $4\delta(L^d\mathsf{b}) \le r_2$; $4C_{\mathrm{db}}\delta(L^d\mathsf{b}) \le r_1$; and $C_z\delta(L^d\mathsf{b}) \le \min\{\frac{1}{2}r_{\mathrm{h}}, \frac{1}{2}, (36 + 6C_{\mathrm{pz}}^2)^{-1/2}\}$.

Here $\varpi_{\mathrm{db}} = r_{\mathrm{y}}^2/(8\pi L^3)$ and $C_{\mathrm{db}} = 8L^3$ are the numbers of [7, Def. B.2], $r_{\mathrm{h}} = \min\{r_{\mathrm{B}}, \varrho_{\mathrm{R}}^0\}$ is the radius of holomorphy of the response in §5.3, $r_1 = \min\{\frac{c_Q}{4L}, \frac{c_Q}{32C_Q L^4}\}$, $r_2 = \frac{r_1}{8C_Q L^3}$, $C_z = 4C_Q L^3$ and

$C_{\mathrm{pz}} = 24C_QL^3 + 128C_QL^4/c_Q$, so that these numbers depend only on L and the constants of §5.3; n'' is the least integer $\ge n_{\mathrm{a}}$ with $\mathsf{R}_e \le \frac12$ for all $e \ge n''$; n_1 is the least $n \ge 2$ with $L^{n-1}\mathsf{b} \ge \mathsf{t}_\natural$ and $4\theta_d \le (48000L^4)^{-1}$ for $d \ge n$.

A quantitative extension of the gauge construction. The explicit top-scale bounds below require a compatible gauge construction across several block levels. The missing estimate, referred to as *(Multiscale gauge)*, would provide finite constants $R_{\mathrm{S}}, C_{\mathrm{S}} \ge 1$, independent of the odd integer $L \ge 13$, the side ℓ, the integers n, J and the field, with the following property. If $V \in \Omega_n$, $J \ge 1$, $n - J \ge 1$, $L^J \ge R_{\mathrm{S}}$, $\varpi(\mathcal{M}^jV) < (200L^2)^{-1}$ for $0 \le j < J$, and $\frac{\pi}{2}L^J\varpi(\mathcal{M}^JV) \le 1$, there is a periodic site gauge u such that

$$\mathsf{d}(V^u(b), \mathbf{1}) \le 44\pi L^2 \sum_{j=0}^{J-1} \varpi(\mathcal{M}^jV) + C_{\mathrm{S}}L^{-J} \quad \text{at every bond } b.$$

This quantitative estimate remains to be proved. The branches (B1)(e), (B2)(d) and the explicit-top variants of (A4)(f) and (A5) below remain sketches at this estimate; the assertions using j_1, w_n and C_n^{t} retain their stated hypotheses. With $C_{\mathrm{BG}} = 2\lambda_{\mathrm{BG},3}$ of eq. 15.v, $C'' = 1 + 20\pi^2C_{\mathrm{BG}}^2$, $\kappa_0 = (8\pi C_{\mathrm{BG}})^{-2}$, $\vartheta_\infty = C_HA_1/((1 - L^{-3/2})A_0)$, which is < 1 under (E), and j_1 the least $j \ge 1$ with $L^{-3j/2}c_\rho \le \min\{\frac12(1 - \vartheta_\infty), 1/C''\}$: n_{c} is the least integer such that for $n \ge n_{\mathrm{c}}$: $n - j_1 \ge 1$, $L^{n-j_1}\mathsf{b} \ge \mathsf{t}_\natural$, $2\varepsilon_n \le C''\kappa_0$, $\ell L^n \ge \max\{\lambda_{\mathrm{BG},3}, (C''/2\varepsilon_n)^{1/2}\}$; $\mathsf{r}_n = \mathsf{R}_n + \mathsf{R}_{n-j_1}$. For the proposed variants requiring the multiscale gauge estimate, let $\alpha_n = \frac18(2\arcsin\varepsilon_n)^{1/2}$ for $n \ge 2$ (the indices 0 and 1 belong to the fixed radii), and let J_n be the least integer $J \ge 1$ with $L^J \ge \max\{R_{\mathrm{S}}, 8C_{\mathrm{S}}/\alpha_n\}$. Then $J_n \le n/4 + c$, with c independent of n and K.

For these proposed multiscale variants, n_{e} denotes the least integer ≥ 2 such that for $n \ge n_{\mathrm{e}}$: $\varepsilon_n \le \frac{1}{51}$, $\log L^n\mathsf{b} > 2\mathfrak{p}_0$, $n - J_n \ge 1$, $L^{n-J_n}\mathsf{b} \ge \mathsf{t}_\natural$, $\theta_d < (200L^2)^{-1}$ for $d \ge n - J_n$, $44\pi L^2\sum_{j<J_n}\theta_{n-j} \le \frac18\alpha_n$, and $\frac{\pi}{2}L^{J_n}\theta_{n-J_n} \le 1$. No constant of this section depends on K; $n_1, n_2, n_3, n_{\mathrm{a}}, n_{\mathrm{e}}, C^\sigma, \Pi_n^\flat$ depend only on b and the constants of the construction, and R_n, o_n, n'', n_{c} and the constants defined from them (r_n, C_n^{t}, $C_n^\flat$, C_n, C'_n) depend besides on ℓ.

Flowed observables. Fix $0 < \kappa < \frac16$. For $s > 0$ and $V \in \Omega_K$, $\mathsf{m}_s(V)$ is the least

$$\max\{s^{1/4+\kappa/2}\|A\|_\infty, s^{3/4+\kappa/2}\|\nabla A\|_\infty\}$$

over the real 1-forms A with e^{ia_KA} gauge equivalent to V (∇ forward difference quotients);

$$\mathsf{O}_K(C, \bar s) = \{U : \mathsf{m}_s(U_s) \le C \text{ for } 0 < s \le \bar s\}.$$

The *flowed observables* are the functions $H_K^\sigma = \varphi(\mathcal{O}(f_1), \dots, \mathcal{O}(f_r)) \circ \Phi_\sigma$, $\sigma > 0$, $f_i \in C^\infty(T_\ell)$ real, $\varphi \in C^\infty(\mathbb{R}^r)$ bounded with all its derivatives. For $0 < \alpha < 1$ let $\mathcal{O}^\alpha$ be the space of orbits of $\mathfrak{g}^3$-valued $C^{0,\alpha}$ connections $\mathfrak{A}$ on T_ℓ under $C^{1,\alpha}$ gauge transformations; a *flow line* is a continuous map $(0,\infty) \to \mathcal{O}^\alpha$ that near every time is the orbit of a mild solution of $\partial_s\mathfrak{A} = -\mathrm{d}_{\mathfrak{A}}^*F_{\mathfrak{A}} + \mathrm{d}_{\mathfrak{A}}(\partial_\nu\mathfrak{A}_\nu)$, the Yang–Mills gradient flow up to an infinitesimal gauge transformation; $\mathcal{O}^0(f)(\mathfrak{A}) = \frac12\int f\sum_{\mu<\nu}|F_{\mu\nu}(\mathfrak{A})|^2$.

With $P_s = e^{s\Delta_a}$ the heat semigroup of $\Lambda_{K,\ell}$, $a = a_K$, $|f|_{\eta;a} = \sup_{0<s\le1} s^{-\eta/2}\|P_sf\|_\infty$ and $X \wedge Y = \frac{1}{2i}[X, Y]$, the rough regularity quantity of a 1-form $\mathbf{B}$ is

$$\|\mathbf{B}\|_{\kappa;a} = |\mathbf{B}|_{-\frac12-\kappa;a} + \sup_{0<s\le1} s^{\frac34+\kappa}\max\big|(\tau_h\nabla_\lambda P_s\mathbf{B}_\nu) \wedge (\tau_{h'}P_s\mathbf{B}_{\nu'})\big|_{-\frac12;a},$$

the maximum over directions λ, ν, ν' and translations $\tau_h, \tau_{h'}$ in a fixed finite set containing every translation in the quadratic term of the gauge-fixed flow equation below; it bounds $s^{\frac14+\frac\kappa2}\|P_s\mathbf{B}\|_\infty$ and $\frac12 s^{\frac34+\frac\kappa2}\|\nabla P_s\mathbf{B}\|_\infty$. The rough distance of $\mathbf{B}$, $\mathbf{B}'$ adds to $|\mathbf{B} - \mathbf{B}'|_{-\frac12-\kappa;a}$ the same supremum for the products $\nabla P_s(\mathbf{B} - \mathbf{B}') \wedge P_s\mathbf{B}$ and $\nabla P_s\mathbf{B}' \wedge P_s(\mathbf{B} - \mathbf{B}')$.

15.1.2. *Why the process limit is Markov.* At a fixed physical torus, the compatible block fields give a common space on which to compare cutoff processes. An observable of one block level can be evaluated at every finer cutoff by pulling it back along the averaging map. Convergence of these observables defines the one-time thread law. Convergence of their multi-time expectations gives stationary reversible limiting processes along subsequences. There is a further issue: projecting a Markov process onto a smaller space can retain memory. The semigroup identity must therefore be proved after taking the limit.

The source makes that issue explicit. A subsequential transition family has the form $Q_t = J^* R_t J$, where J embeds the limiting L^2 space isometrically into another Hilbert space and R_t is a self-adjoint contraction semigroup there. The possible loss is measured exactly by

$$(F, Q_{2t}F) - \|Q_t F\|^2 = \|(1 - JJ^*)R_t J F\|^2.$$

Consequently, weak convergence of transition operators leaves open whether their limit is a semigroup. Strong convergence will make the two sides of the intermediate scalar-product identity agree and remove this defect. The multiscale estimates are used to obtain that strong convergence.

Fix a coarse depth n and a finer cutoff K. Successively average an invariant observable H under the conditional fluctuation probabilities. Denote the resulting conditional means by h_k. These functions form a martingale when the small-field chain is read from the coarse level to the fine level. Its variance decomposition is a sum of conditional variances. The convexity of each fluctuation integral bounds its variance by the Dirichlet energy at that step. The response estimate transports that energy to the next step with a multiplicative error whose product stays bounded. Summing the physical scale factors gives the principal approximation bound

$$\|H - (\mathbf{1}_{\mathsf{I}_n} h_{K-n}) \circ \pi_{K,n}\|^2_{L^2(\mu^K)} \le \frac{C^\sigma C_{\mathrm{P}}}{L^2 - 1} \frac{a_n^2}{b} \mathcal{E}_K(H) + 4\mathsf{R}_n \|H\|^2_\infty.$$

Here I_n is the admitted open set at depth n, $\pi_{K,n}$ is the block projection, and R_n is the mass discarded by the chain. All these objects are defined precisely below. The two errors have different origins. The factor a_n^2 comes from summing the conditional variances above depth n; R_n accounts for the part of the Wilson measure on which the chain was discarded. Both go to zero as n increases, with the physical torus held fixed.

This estimate reduces compactness to one finite-dimensional block space. For bounded observables with bounded energies, the conditional means have bounded Dirichlet integrals on compact subsets of I_n. Rellich compactness is applied on an exhaustion of that open set; no smoothness of its boundary is required. Comparison of the block law with Haar measure then transfers convergence of the conditional means to the block probability measure. Finally, the displayed approximation permits n to increase and gives strong compactness in the thread law. The order is important: finite-dimensional compactness is used at a fixed n, and the uniform approximation is used afterwards.

Apply this compactness statement to $H_K = P_t^K \Psi_K$, where Ψ is an invariant cylinder function. The lattice spectral theorem supplies

$$\mathcal{E}_K(P_t^K \Psi_K) \le \frac{\|\Psi_K\|_2^2}{2et}, \qquad t > 0.$$

Every subsequence therefore has a strongly convergent subsequence. Its limit must equal the already determined weak limit $Q_t\Psi$. Thus the entire chosen process subsequence converges strongly at this observable and time. Density extends this conclusion to the invariant L^2 space. It gives the semigroup property and the Markov property on the invariant sigma-algebra. It does not identify the limits arising from two different process subsequences.

The form domain is recovered by cylinder functions multiplied by smooth regularity cutoffs at all finer block levels. The large-field tail makes these cutoffs converge to one, and their Dirichlet costs tend to zero. This produces a common core rather than merely a collection of cylinder functions with

no energy control. The lower semicontinuity and recovery-sequence arguments then give convergence of forms in the weak and strong senses and convergence of the resolvents.

Nonconstancy is checked on the limiting form $\mathcal{E}_\star$. The conditional-mean comparison passes a positive coarse Dirichlet integral to every recovery sequence for a nonconstant plaquette observable. The coarse measure has a positive density on its regular open set, and the squared gradient of that observable is positive there away from zero curvature. Hence the limiting energy is positive. The spectral measure of the observable cannot be concentrated at zero, so its transition correlation changes with time. A positive lattice energy alone would not give this conclusion without the comparison with recovery sequences.

For mixing, the connected part of the coarse regular region gives a Poincaré inequality, while its complement contributes a defect. After transport through the chain, its useful form is

$$\operatorname{Var}_\mu(F) \le C_n \mathcal{E}_\star(F) + r_n \operatorname{osc}(F)^2, \qquad r_n \longrightarrow 0.$$

At fixed n, apply this inequality along the semigroup and let time increase; then let n increase to remove the defect. The constants C_n may depend on the physical torus. This proves the stated mixing without supplying a spectral gap uniform in volume.

15.1.3. *Markov limits, functional inequalities and flowed observables.*

Theorem AT (Nontrivial Markov limits and mixing)**.** Assume (E).

(A1) *Motion.* For every limit, every integer $n \ge n''$ and every invariant $\Phi \in C^1(\Omega_n)$, with $|\mathsf{l}_n|$ the Haar measure of l_n,

$$\mathcal{E}_\star(\Phi(Y_n)) \ge \frac{e^{-\mathsf{o}_n}}{2C^\sigma |\mathsf{l}_n|} \int_{\mathsf{l}_n} \Gamma_n(\Phi)\, \mathrm{d}V. \tag{15.i}$$

Hence $(\Psi, Q_t\Psi) < \|\Psi\|^2$ for $\Psi = a_p(Y_n)$ and every $t > 0$: no limit is frozen.

(A2) *Compactness.* Along every sequence $K_j \to \infty$, invariant $H_j \in D(\mathcal{E}_{K_j})$ with $\sup_j(\sup|H_j| + \mathcal{E}_{K_j}(H_j)) < \infty$ have a strongly convergent subsequence; its limit G has $|G| \le \sup_j \sup|H_j|$.

(A3) *The Markov property and form convergence.* For every limit, $\mathcal{E}_\star$ is a closed, densely defined Markovian form on L^2_{inv}, and $Q_t = e^{-tA_\star}$ is a strongly continuous symmetric Markov semigroup on L^2_{inv}; $P_t^{K_j} H_j \to Q_t F$ strongly for $t > 0$ whenever $H_j \to F$ strongly, in particular $P_t^{K_j}\Psi_{K_j} \to Q_t\Psi$ for every invariant cylinder function Ψ; (M1) and (M2) hold, and $(\alpha - \mathcal{L}_{K_j})^{-1} W_j \to (\alpha + A_\star)^{-1} W$ strongly for $\alpha > 0$ and invariant $W_j \to W$ strongly; (Y_t) is a stationary reversible Markov process on the invariant σ-algebra with the transition operators Q_t and the invariant measure $\mu_{\ell,\mathsf{b}}$; and $\{G \in L^2_{\mathrm{inv}} : |G| \le 1,\ \mathcal{E}_\star(G) \le 1\}$ is compact.

If invariant $H_j \in C^\infty(\Omega_{K_j})$ converge strongly to F and $\sup_j \|\mathcal{L}_{K_j} H_j\|_{L^2(\mu^{K_j})} < \infty$, then $F \in D(A_\star)$, $\mathcal{L}_{K_j} H_j \to -A_\star F$ weakly and $\mathcal{E}_{K_j}(H_j) \to \mathcal{E}_\star(F)$; under the assumptions of (C7), this hypothesis fails at the flowed observables of (C7)(b), (c), (e). If invariant $u_j \to u$ strongly with $\mathcal{E}_{K_j}(u_j) \to \mathcal{E}_\star(u)$, and invariant $v_j \to v$ weakly with $\sup_j \mathcal{E}_{K_j}(v_j) < \infty$, then $\mathcal{E}_{K_j}(u_j, v_j) \to \mathcal{E}_\star(u, v)$.

(A4) *Variance bounds and mixing.* Let $n \ge \max\{n'', n_{\mathrm{c}}\}$ and $\mathsf{C}_n = C^\sigma[\frac{C_{\mathrm{P}}}{L^2-1}\frac{a_n^2}{\mathsf{b}} + \frac{1}{2}C_n^{\mathrm{t}}]$, with C_n^{t} of (B1). For all $K > n$: (a) $\operatorname{Var}_{\mu^K}(H) \le \mathsf{C}_n \mathcal{E}_K(H) + \mathsf{r}_n \operatorname{osc}(H)^2$ for bounded invariant $H \in D(\mathcal{E}_K)$; (b) $\|P_t^K H - \mu^K(H)\|^2_{L^2(\mu^K)} \le (\mathsf{r}_n + \frac{1}{4}e^{-2t/\mathsf{C}_n})\operatorname{osc}(H)^2$ for bounded Borel invariant H and $t \ge 0$. For every limit: (c) $\operatorname{Var}_\mu(F) \le \mathsf{C}_n \mathcal{E}_\star(F) + \mathsf{r}_n \operatorname{osc}(F)^2$ for bounded $F \in L^2_{\mathrm{inv}}$, so $\mathcal{E}_\star(F) = 0$ only for constant F, and $(F, (1 - Q_t)F) > 0$ for every non-constant bounded $F \in L^2_{\mathrm{inv}}$ and $t > 0$; (d) $0 \le \mathbb{E}F(Y_0)F(Y_t) - (\int F\,\mathrm{d}\mu)^2 \le \inf_n(\mathsf{r}_n + \frac{1}{4}e^{-t/\mathsf{C}_n})\operatorname{osc}(F)^2$ for bounded $F \in L^2_{\mathrm{inv}}$, the infimum over the admitted n, which tends to 0 as $t \to \infty$; and a probability measure $\rho\,\mu$, $\rho \in L^2_{\mathrm{inv}}$, with $Q_t\rho = \rho$ for one $t > 0$ is μ.

(e) If $F_K \to F$ strongly along the full sequence of cutoffs, $\sup_K \sup|F_K| \le M_F$, F is not constant, and H_K are invariant Lipschitz functions with $\|H_K - F_K\|_{L^2(\mu^K)} \to 0$, then

$\liminf_K \mathcal{E}_K(H_K) \ge \sup_n(\mathrm{Var}_\mu F - 4\mathsf{r}_n M_F^2)/\mathsf{C}_n > 0$; and for invariant Lipschitz H_K with $\sup_K \mathrm{osc}(H_K) \le O$, $\liminf_K \mathcal{E}_K(H_K) \ge \sup_n[\liminf_K \mathrm{Var}_{\mu^K}(H_K) - \mathsf{r}_n O^2]/\mathsf{C}_n$. (f) The multiscale gauge extension remains a sketch. Completing that estimate would give all of (a)–(e) as well for $n \ge \max\{n'', n_{\mathrm{e}}\}$ with $\mathsf{R}_n < 1$, with $C_n^\flat$ of (B1)(e) in the place of C_n^{t} and $\mathsf{R}_n + \mathsf{R}_{n-J_n}$ in the place of r_n.

(A5) *Entropy relaxation.* Let $n \ge \max\{n_{\mathrm{a}}, n_{\mathrm{c}}\}$, C'_n and ς^K as in (B2). For all $K > n$: (a) if $G \in C^1(\Omega_K)$ is real and invariant, $M_G = \sup|G|$ and $m_2 = \int G^2 \,\mathrm{d}\mu^K > \mathsf{r}_n M_G^2$, then

$$\mathrm{Ent}_{\mu^K}(G^2) \le \mathsf{C}'_n \mathcal{E}_K(G) + \mathsf{r}_n M_G^2 \Big[1 + \log \frac{M_G^2}{m_2 - \mathsf{r}_n M_G^2}\Big];$$

(b) if $\psi \in C^1(\Omega_K)$ is real and invariant with $\Gamma_K(\psi) \le l^2$ and $\bar\psi = \int \psi \,\mathrm{d}\varsigma^K / \varsigma^K(\Omega_K)$, then $\mu^K(\psi - \bar\psi \ge u) \le e^{-u^2/(\mathsf{C}'_n l^2)} + \mathsf{r}_n$ for $u > 0$; (c) if $f_0 \ge 0$ is bounded Borel invariant, $m_1 = \int f_0 \,\mathrm{d}\mu^K > 0$, $\bar M = \sup f_0$ and $\mathsf{r}_n \bar M \le \frac12 m_1$, then for $\tau \ge 0$

$$\mathrm{Ent}_{\mu^K}(P_\tau^K f_0) \le e^{-4\tau/\mathsf{C}'_n} \mathrm{Ent}_{\mu^K}(f_0) + \mathsf{r}_n \bar M \Big[1 + \log \frac{2\bar M}{m_1}\Big].$$

For every limit, (a) holds for bounded $F \in D(\mathcal{E}_\star)$ with $\mu, \mathcal{E}_\star$ in the places of $\mu^K, \mathcal{E}_K$; and if $F \in L^2_{\mathrm{inv}}$, $0 \le F \le \bar M$, $m_1 = \int F \,\mathrm{d}\mu > 0$ and $\mathsf{r}_n \bar M < \frac12 m_1$, then $\mathrm{Ent}_\mu(Q_\tau F) \le e^{-4\tau/\mathsf{C}'_n} \mathrm{Ent}_\mu(F) + \mathsf{r}_n \bar M[1 + \log \frac{2\bar M}{m_1}]$; so $\mathrm{Ent}_\mu(Q_\tau F) \to 0$ as $\tau \to \infty$, and $m_1^{-1}(Q_\tau F)\,\mu \to \mu$ in total variation on the invariant σ-algebra. The corresponding extension remains a sketch at the multiscale gauge estimate; its intended conclusion is the same for $n \ge \max\{n_{\mathrm{a}}, n_{\mathrm{e}}\}$ with $\mathsf{R}_n < 1$, with the explicit top of (B1)(e).

■□

Theorem AU (Functional inequalities uniform in the cutoff)**.** Assume (E).

(B1) *The coarse-scale law.* For $n \ge n_{\mathrm{c}}$ there are an invariant Borel weight $w_n : \Omega_n \to [0,1]$ and a constant $C_n^{\mathrm{t}} < \infty$ such that for all $K > n$ and every bounded invariant locally Lipschitz ϕ on I_n: (a) $w_n \le \mathbf{1}_{\mathsf{D}_n}$ and $\sigma_n^K(w_n < 1) \le \mathsf{R}_{n-j_1}$; (b) $\mathrm{Ent}_{w_n\sigma_n^K}(\phi^2) \le C_n^{\mathrm{t}} \int \Gamma_n(\phi)\, w_n \,\mathrm{d}\sigma_n^K$; (c) $\int(\phi - c)^2 \,\mathrm{d}\sigma_n^K \le \frac12 C_n^{\mathrm{t}} \int \Gamma_n(\phi) \,\mathrm{d}\sigma_n^K + \mathsf{R}_{n-j_1} \mathrm{osc}(\phi)^2$, c the mean of ϕ under $w_n\sigma_n^K$; (d) $\mathrm{Ent}_{\sigma_n^K}(\phi^2) \le C_n^{\mathrm{t}} \int \Gamma_n(\phi) \,\mathrm{d}\sigma_n^K + \mathsf{R}_{n-j_1} \sup_{\mathsf{I}_n} \Xi(\phi^2, c')$, c' the mean of ϕ^2 under $w_n\sigma_n^K$.

Here $w_n = \min\{1, \rho/\rho_0\}$, $\rho(V) = \int \mathbf{1}_{S'}(V^g) \,\mathrm{d}g$ for a connected open $S' \subset \mathsf{D}_n$ that is a finite union of chart balls, $\rho_0 = \min_{\mathcal{K}_1} \rho > 0$ on a compact invariant set $\mathcal{K}_1 \subset \mathsf{D}_n$, and $C_n^{\mathrm{t}} = a_n\, c(S')\, e^{\mathsf{o}_n}/\rho_0$, $c(S')$ the log-Sobolev constant of Haar measure on S'. (e) (An explicit top.) The following variant remains a sketch at the multiscale gauge estimate. Let $n \ge n_{\mathrm{e}}$, $\mathsf{R}_n < 1$, $N = \ell L^n$, $w_n^{\mathrm{e}}(V) = \int \mathbf{1}\{\mathsf{d}(V^u(b), \mathbf{1}) < \alpha_n \text{ for all } b\} \,\mathrm{d}u$ (du Haar measure of the gauge group), $c_n = [(\frac12\alpha_n - \sin\frac12\alpha_n)/2\pi]^{N^3}$ and $\chi_n = \min\{1, w_n^{\mathrm{e}}/c_n\}$.

Completing that estimate would give (a)–(d) with $\chi_n \mathbf{1}_{\mathsf{I}_n}$ in the place of w_n, R_{n-J_n} in the place of R_{n-j_1}, and the constant

$$C_n^\flat = \frac{a_n e^{2\mathsf{o}_n} C_{\mathscr{D}}}{(1-\mathsf{R}_n)c_n}, \quad C_{\mathscr{D}} = 2\sqrt{e}\Big(\frac{\alpha_n}{\sin\alpha_n}\Big)^{6N^3}\Big(\frac{2\rho_{\max}}{r'}\Big)^{9N^3}\Big(\rho_{\max} + \frac{8\pi\rho_{\max}^2}{r'}\Big)^2, \quad \rho_{\max} = \alpha_n\sqrt{3N^3},\ r' = 64\alpha_n^2,$$

in the place of C_n^{t}; the set $\{\chi_n > 0\} \cap \mathsf{I}_n$ is open, invariant, path connected and contained in D_n; and for bounded invariant $\Phi \in C^1(\Omega_n)$, $\mathrm{Var}_{\mu_n^K}(\Phi) \le \frac12 C_n^\flat \int_{\mathsf{I}_n} \Gamma_n(\Phi) \,\mathrm{d}\mu_n^K + (\mathsf{R}_n + \mathsf{R}_{n-J_n})\mathrm{osc}(\Phi)^2$, and the same with μ_n in the place of μ_n^K.

(B2) *The chain with a connected coarse-scale region.* Let $n \ge \max\{n_{\mathrm{a}}, n_{\mathrm{c}}\}$, $K > n$, $\varsigma^K = w_n(V_t)\,\sigma^K$ and

$$\mathsf{C}'_n = \Big[\frac{2C_{\mathrm{P}}}{L^2 - 1}\frac{a_n^2}{\mathsf{b}} + C_n^{\mathrm{t}}\Big]\, 3\mathsf{M}^2\, \Pi_n^\flat.$$

(a) $\int f\,\mathrm{d}\varsigma^K \le \int f\,\mathrm{d}\sigma^K \le \int f\,\mathrm{d}\mu^K \le \int f\,\mathrm{d}\varsigma^K + \mathsf{r}_n \sup f$ for bounded Borel invariant $f \ge 0$; (b) $\mathrm{Ent}_{\varsigma^K}(G^2) \le \mathsf{C}'_n \int \Gamma_K(G)\,\mathrm{d}\varsigma^K$ for real invariant $G \in C^1(\Omega_K)$; (c) for $\nu \in \{\sigma^K, \mu^K\}$: $\mathrm{Ent}_\nu(G^2) \le \mathsf{C}'_n \int \Gamma_K(G)\,\mathrm{d}\nu + \mathsf{r}_n \sup_U \Xi(G(U)^2, c')$, c' the mean of G^2 under ς^K. (d) This variant remains a sketch at the multiscale gauge estimate. Its intended conclusion is: for Lebesgue-almost every label b, all m and all $n \ge \max\{n_\mathrm{a}, n_\mathrm{e}\}$ with $\mathsf{R}_n < 1$, (a)–(c) hold with $\mathbf{1}_{\mathsf{D}_n}(V_t)\,\sigma^K$ in the place of ς^K, a finite constant $C_n^\sharp$ in the place of C_n^t, and $\mathsf{R}_n + \mathsf{R}_{n-J_n}$ in the place of r_n; for every b and $\eta > 0$ they hold with the component of $\mathbf{1}$ in $\{\varpi < \tilde r\}$, for some $\tilde r \in (\varepsilon_n, 2\varepsilon_n)$ that does not depend on K, in the place of D_n, and $\mathsf{R}_n + \mathsf{R}_{n-J_n} + \eta$ in the place of r_n; with the explicit top of (B1)(e) they hold with $\chi_n(V_t)\sigma^K$, $C_n^\flat$ and $\mathsf{R}_n + \mathsf{R}_{n-J_n}$.

(e) (No weight.) For $n \ge n_\mathrm{a}$, $K > n$, real invariant $G \in C^1(\Omega_K)$, f_t the top of the tower of G^2 and $\epsilon > 0$:

$$\mathrm{Ent}_{\sigma^K}(G^2) \le \frac{6C_\mathrm{P}\mathsf{M}^2\Pi_n^\flat}{L^2-1}\frac{a_n^2}{\mathsf{b}}\int \Gamma_K(G)\,\mathrm{d}\sigma^K + \mathrm{Ent}_{\sigma_n^K}(f_t), \qquad \int \Gamma_n((f_t+\epsilon)^{1/2})\,\mathrm{d}\sigma_n^K \le 3\mathsf{M}^2\Pi_n^\flat \int \Gamma_K(G)\,\mathrm{d}\sigma^K.$$

(B3) *The fine field conditioned on a block field.* Let $n \ge n_\mathrm{a}$, $K > n$, $1 \le t' \le K-n$, $e = K - t'$, $W \in \mathsf{Cut}_{t'}$, and let P_W be the law of V_0 when $V_{t'} = W$ and $V_k = V_{V_{k+1}}(B_k)$, B_k of law $p_{V_{k+1}}$. Then P_W is a regular conditional law of U given $\mathcal{M}^{t'}U = W$ under σ^K and under ς^K, and for real invariant $G \in C^1(\Omega_K)$ with tower (g_k) and every $W \in \mathsf{Cut}_{t'}$

$$\mathrm{Ent}_{\mathsf{P}_W}(G^2) \le C_e^\mathrm{fib}\int \Gamma_K(G)\,\mathrm{d}\mathsf{P}_W, \qquad \Gamma_e(g_{t'})(W) \le 3\mathsf{M}^2\Pi_e^\flat \int \Gamma_K(G)\,\mathrm{d}\mathsf{P}_W, \qquad C_e^\mathrm{fib} = \frac{6\mathsf{M}^2 C_\mathrm{P}\Pi_e^\flat}{L^2-1}\frac{a_e^2}{\mathsf{b}},$$

a constant that depends neither on K nor on ℓ. If $\psi \in C^1(\Omega_K)$ is real and invariant and $\Gamma_K(\psi) \le l^2$ P_W-almost everywhere, then $\mathsf{P}_W(\psi - \psi_{t'}(W) \ge u) \le e^{-u^2/(C_e^\mathrm{fib} l^2)}$, $\psi_{t'}$ the member of the tower of ψ; and if $\Gamma_K(\psi) \le l^2$ on all of Ω_K, then, with $\psi_{t'} := 0$ off $\mathsf{Cut}_{t'}$, $\mu^K(|\psi - \psi_{t'}(\mathcal{M}^{t'}U)| \ge u) \le 2e^{-u^2/(C_e^\mathrm{fib} l^2)} + \mathsf{R}_n$.

■□

Admissible top depths. The assertions about flowed observables use the following condition, in the notation of Prop. 16.1.

A top depth d_0 is *admissible with constant* C_adm if $d_0 \ge n_\mathrm{a}$, $d_0 \ge d_\mathrm{num}(\ell, \mathsf{b})$, $c_\rho \varepsilon(L^{d_0}\mathsf{b}) \le C_\mathrm{BG}^{-2}$ and $4c_\mathsf{h}\delta(L^{d_0}\mathsf{b}) \le 1/4$, and, for every $K > d_0$ and $d_0 < d \le K$, (M) and (Col) hold with that same constant. Here d_num is the explicit numerical depth in the proof below. The mean and covariance-error estimates hold at every $W \in \mathsf{Cut}_{K-d+1}$; the Gaussian layer-kernel estimates hold almost surely in the chain gauge, at every pair of bare bonds. (Adm) Under (E), there are d_adm and C_adm, depending on ℓ, b and the construction constants but not on K or the chosen top depth, such that every $d_0 \ge d_\mathrm{adm}$ is admissible with C_adm. This is proved in Prop. 16.1 below.

The proof of Prop. 16.1 establishes (M) and (Col) uniformly in the cutoff and top depth whenever $d_0 \ge d_\mathrm{adm}$. Thus the flowed estimates use (Adm) as a consequence of (E).

15.1.4. *Renormalisation inputs.* We use Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD among the construction results through its results on one step (Prop. 6.31 at zero source) and on the small-field chain on one torus (Sections YM3_I:113, YM3_I:114, YM3_I:115, Section YM3_I:116): setting (E) (§2.3.1), the field cuts (the definitions preceding Lem. YM3_I:113.10, Lem. YM3_I:113.10) and the gauge action (Section YM3_I:3), the chart at the pivots ([7, §B.2]), the iteration from a level with its exponents and the bound of the non-local action (Prop. YM3_I:114.2, Prop. YM3_I:114.2, equation YM3_I:(2163), equation YM3_I:(2180)), the comparison of the chain with the bare

density through the weight $\mathfrak{w}_t^{\mathrm{sf}}$ and the bound R_n (the definitions preceding Prop. YM3_I:115.2, Prop. YM3_I:115.2), and the convergence in total variation of the laws of the block fields (Cor. 2.8).

Further construction results are cited where they are used: the constants and thresholds of the construction (equation YM3_I:(634), §15.3.1, 2.3.1, 2.3.1, 2.3.1, §2.3.1), with the old-term constants defined in equation YM3_I:(2163) and its following sentence; the current (YM3_I:9.2); the lattices and the localisation domains with their linear sizes (Section YM3_I:3, YM3_I:72 (the definitions of $\mathcal{D}_j$ and d_j at its beginning), with YM3_I:72 (the definitions of $\mathcal{D}_j$ and d_j at its beginning)); averages (equations YM3_I:(37)–(39), Prop. YM3_I:51.1 and Lem. 5.30), and of the averaging map its gauge covariance (Prop. YM3_I:4.5, YM3_I:(40)), its real-analyticity on the small fields and its measurability (Prop. YM3_I:4.5), its linear part at the identity, which is the linear average YM3_I:(42) of equation YM3_I:(42) (Lem. YM3_I:4.6), with the k-fold composites $\mathbf{Q}_k^{\mathrm{lin}}$ of the linear average, and the bound for rearranged paths (Lem. YM3_I:4.1); backgrounds over a block field, whose averages give back the block field (Defs. YM3_I:9.3 and YM3_I:9.5, §5.2); the fixed radii and the analyticity cores (Def. YM3_I:49.1 and Lem. YM3_I:82.1); the bounds of the response (Lem. 15.9, §5.3) and its Landau gauge on the kernel of the site averages (§15.3, equation YM3_I:(837)); the constant-frame row map and its response identity, with $C_{\mathcal{Q}} = 34(L^3 + 2)$ and the counting-measure kernel bounds (Lem. 5.32); the double-block constants of [7, Def. B.2]; the regularity numbers and the classes of covariance ([7, Def. B.2], [7, Lem. B.4], and the curvature comparison in the proof of Lem. YM3_I:113.10, with C_H as fixed in [7, §P.1]); backgrounds and small plaquettes (Prop. YM3_I:52.3, Prop. 15.8); the local potentials and their bounds (Prop. YM3_I:51.4), with their sums in equations YM3_I:(828)–YM3_I:(830); the outer substitution (Prop. YM3_I:51.10); the sum over polymers [7, eq. 320]; §15.3.1; Thm. YM3_I:W and equation YM3_I:(925); the Gaussian structure of a step and its numbers (Def. YM3_I:47.2, Thm. YM3_I:U and §15.3.1); the gauge-fixing factor, fluctuation test and chart integral of [7, §B.2], with the measure identity of [7, Lem. B.3]; the number $\varepsilon_{\mathcal{W}}$ Prop. YM3_I:52.3; and the consequences of the setting (§2.3.1).

These are the construction results used in this section. The admissibility conclusion (Adm) follows from (E). The multiscale-gauge and flow-approximation arguments require the separate tasks specified below. In particular, the continuum-flow identification in Prop. 16.1(b) still requires the compactness, equation identification, initial holonomies and uniqueness described under (Flow approximation).

Within the section, the proofs of (C2)(b), (C2)(c) and (C3)–(C7) start besides from the moments of the gauge-fixed field $\mathbf{B}_K$ of Prop. 16.1(a), gathered before Step 6, and from its proved conclusion (Adm) on the top depths. The moment bound under the chain measure from an admissible top depth rests on Lem. 15.3(iv), the bounds of the response and of the heat kernel, the grid lemma, and three statements on one fluctuation integral, set in the proof of Prop. 16.1(a): the bounds (M) of the conditional mean of a layer, the bounds (Col) of the part of its conditional covariance that is not scalar in colour, and the exponential bound (Q) of centred quadratic forms under the log-concave law of one integral.

A top depth d_0 is admissible when it meets the numerical and uniform-estimate conditions stated before Thm. AV; under (Adm) the chains of all the large top depths give the moments under μ^K. The statements made for all K, m and $\beta > 0$, namely Lem. 15.1(a), (C1), the short-time theory of the gauge-fixed flow in Prop. 16.1(a) (the flow on $[0, s_0(R)]$ with its four bounds, the Lipschitz bound in the rough distance and the bound of the flowed action), Lems. 16.2, 16.3 and 16.5, Lem. 16.4, Prop. 16.6(a), (b) and Prop. 16.9(a), (b) and the upper bound in (c), use no continuum-limit result. The lower bound in Prop. 16.9(c) follows from the adjoint frequency estimate and exact current transport, with the integrated curvature and current ratio displayed there. The continuum identification requires the separate (Flow approximation) step.

15.1.5. *The lattice dynamics and a common core.*

Lemma 15.1 (Lattice dynamics, subsequential limits and a common core)**.** (a) For all K, m, $\beta > 0$: $\int G\,\mathcal{L}_K F\,\mathrm{d}\mu^K = -\int \Gamma_K(G,F)\,\mathrm{d}\mu^K$ for $F, G \in C^\infty(\Omega_K)$; $\beta\,\partial_b A_K = \mathsf{b}\,a_K^2 J(U)(b)$, so $\mathcal{L}_K = \Delta_K + \mathsf{b}\,v$ is the generator of $dA = -\mathsf{b}\,J(U)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\mathcal{W}$, $dU(b) = ia_K \circ dA(b)\,U(b)$, with $\mathcal{W}$ the Brownian motion of the 1-forms for the scalar product $a_K^3 \sum_b \langle\cdot,\cdot\rangle$: no constant is tuned. P_t^K is a symmetric Markov semigroup on $L^2(\mu^K)$ that commutes with gauge transformations; (U_{-t}) has the law of (U_t); $\mathbb{E}|F(U_t) - F(U_0)|^2 = 2(F, (1 - P_t^K)F) \le 2t\,\mathcal{E}_K(F)$; for $F \in D(\mathcal{L}_K)$, $\mathbb{E}|F(U_t) - F(U_0)|^2 = 2t\,\mathcal{E}_K(F) - \mathbb{E}|\int_0^t \mathcal{L}_K F(U_s)\,\mathrm{d}s|^2 \ge 2t\,\mathcal{E}_K(F) - t^2\|\mathcal{L}_K F\|^2$; and $\mathcal{E}_K(P_t^K F) \le \|F\|^2/2et$; $v(G \circ \Phi_\sigma) = \partial_\sigma(G \circ \Phi_\sigma)$.

With Π_y the Haar average over the gauge transformations at a site y and $\Pi_S = \prod_{y\in S}\Pi_y$: $\mathcal{E}_K(F) \ge (2a_K)^{-1}\|F - \Pi_y F\|^2$ and $\|P_t^K(F - \Pi_S F)\| \le e^{-t/2a_K}\|F - \Pi_S F\|$.

(b) Assume (E). Every sequence of cutoffs has a subsequence that is a limit. For every limit: (Y_t) is stationary, reversible and positive under time reflection ($\mathbb{E}[\overline{\theta\mathsf{F}}\,\mathsf{F}] \ge 0$ for finite sums F of products of cylinder functions at times ≥ 0, θ the reflection of the times), with one-time law μ; its multi-time means are real-analytic functions of the strictly positive consecutive time differences; its finite-dimensional laws are invariant under gauge transformations chosen independently at different times; Q_t is a symmetric positivity preserving contraction of $L^2(\mu)$, $Q_t 1 = 1$, $Q_t = \Pi Q_t \Pi$, and $\|Q_t F\|^2 \le (F, Q_{2t}F)$.

If $H_j \to F$ strongly, the spectral measures of H_j for $-\mathcal{L}_{K_j}$, taken as measures on $[0,\infty]$, converge weakly to a measure ν_F of mass $\|F\|^2$ with $(F, Q_t F) = \int e^{-t\lambda} d\nu_F$ for $t > 0$; $\mathcal{E}_\star(F) = \int \lambda\,\mathrm{d}\nu_F$; $\mathcal{E}_\star$ is closed, Markovian and finite only on L^2_{inv}; (M2) holds, and $\mathcal{E}_\star(F) \le \liminf_j \mathcal{E}_{K_j}(H_j)$ whenever $H_j \to F$ strongly. The following are equivalent: $\|Q_t\Psi\|^2 = (\Psi, Q_{2t}\Psi)$ for invariant cylinder functions Ψ and $t > 0$; $Q_{t+s} = Q_t Q_s$; $P_t^{K_j}\Psi_{K_j} \to Q_t\Psi$ strongly for these Ψ; (Y_t) is a Markov process on the invariant σ-algebra with the transition operators Q_t.

They imply (M1), and $P_t^{K_j} H_j \to Q_t F$ strongly whenever $H_j \to F$ strongly. In general $Q_t = J^* R_t J$, where J is an isometry of $L^2(\mu)$ into a Hilbert space $\mathcal{K}$ and (R_t) is a semigroup of positive self-adjoint contractions of $\mathcal{K}$, and $(F, Q_{2t}F) - \|Q_t F\|^2 = \|(1 - JJ^*)R_t J F\|^2$; the four properties hold if and only if $R_t J L^2_{\mathrm{inv}} \subset J L^2_{\mathrm{inv}}$ for all t.

(c) Assume (E). For every integer $n \ge 1$ with $L^n\mathsf{b} \ge \mathsf{t}_\natural$ and every $K > n$, $\mu^K(\varpi(\pi_{K,d}U) < \theta_d$ for all $n \le d \le K) \ge 1 - \mathsf{R}_n$, and $\mu(\varpi(Y_d) < \theta_d$ for all $d \ge n) \ge 1 - \mathsf{R}_n$. Fix $\varphi_{\mathrm{cut}} \in C^\infty([0,\infty); [0,1])$ with $\varphi_{\mathrm{cut}} = 1$ on $[0,1]$, $\varphi_{\mathrm{cut}} = 0$ on $[4,\infty)$ and $|\varphi'_{\mathrm{cut}}| \le 1$, and put

$$\chi_d = \prod_p \varphi_{\mathrm{cut}}(\varpi_p^2/(2\theta_d)^2) \quad \text{on } \Omega_d, \qquad \mathsf{k}_d = 24L^{d/2}\theta_d^{-1}(\ell L^d)^3,$$

the product over the plaquettes of $\Lambda_{d,\ell}$: χ_d is smooth and invariant, takes its values in $[0,1]$, is 1 on $\{\varpi \le 2\theta_d\}$ and 0 on $\{\varpi \ge 4\theta_d\}$, and $\Gamma_d(\chi_d)^{1/2} \le \mathsf{k}_d \mathbf{1}\{\varpi \ge \theta_d\}$; the number k_d depends on ℓ and not on K, and it is at most a constant times $\ell^3 L^{4d}$. The span $\mathcal{C}$ of the functions $F = \Phi(Y_n)\prod_{d\ge n}\chi_d(Y_d)$, $n \ge n_1$, $\Phi \in C^\infty(\Omega_n)$ invariant, is dense in L^2_{inv}; the lattice forms $F_K = \Phi \circ \pi_{K,n} \prod_{n\le d\le K} \chi_d \circ \pi_{K,d}$ are smooth and invariant, $|F_K| \le \sup|\Phi|$, $F_K \to F$ strongly along the full sequence, and $\sup_K \int \Gamma_K(F_K)^p d\mu^K < \infty$ for every $p < \infty$; the cutoffs $\chi_n^K = \prod_{n\le d\le K}\chi_d \circ \pi_{K,d}$ have $\mu^K(\chi_n^K < 1) \le \mathsf{R}_n$ and $\|\Gamma_K(\chi_n^K)^{1/2}\|_{L^{2p}(\mu^K)} \to 0$ as $n \to \infty$, uniformly in K.

Consequently, for every limit, $\mathcal{E}_\star$ is densely defined, $\nu_F(\{\infty\}) = \|F - \Pi F\|^2$ for $F \in L^2(\mu)$, $(F, Q_t F) \to \|\Pi F\|^2$ and $Q_t \to 1$ strongly on L^2_{inv} as $t \to 0$, and $\frac{1}{2}\mathbb{E}|F(Y_t) - F(Y_0)|^2 \le t \sup_K \mathcal{E}_K(F_K)$ for $F \in \mathcal{C}$. ∎

Proof sketch. (a) Haar measure is invariant under left translations in each factor, so

$$\int \partial_b^\alpha (F G e^{-\beta A_K})\,\mathrm{d}U = 0,$$

which is the integration by parts; (YM3_I:9.2) and $\beta = \mathsf{b}/a_K$ give the drift. Writing a tangent vector as $X = a_K B$ with a 1-form B per unit length, $a_K\|X\|^2 = (B,B)_{a_K}$, and $\Gamma_K(F)$ is the square

of the gradient for $(\cdot,\cdot)_{a_K}$: this is the stochastic equation. The closure of $(\mathcal{E}_K, C^\infty)$ is a Dirichlet form, gauge transformations act unitarily and preserve it, so they commute with P_t^K; the two bounds are $1-e^{-t\lambda}\le t\lambda$ and $\lambda e^{-2t\lambda}\le 1/2et$ under the spectral measures of the spectral theorem, and the identity is

$$2\int(1-e^{-t\lambda})\,\mathrm{d}\nu = 2t\int\lambda\,\mathrm{d}\nu - 2\int(t\lambda-1+e^{-t\lambda})\,\mathrm{d}\nu,$$

with

$$\mathbb{E}|\int_0^t \mathcal{L}_K F(U_s)\,\mathrm{d}s|^2 = 2\int(t\lambda-1+e^{-t\lambda})\,\mathrm{d}\nu \le t^2\|\mathcal{L}_K F\|^2$$

by stationarity, ν the spectral measure of F.

The generators X_α of the rotations at y act on six bonds, $\sum_\alpha |X_\alpha F|^2 \le 6|\partial F|^2$, and the first non-zero eigenvalue 3 of the Laplacian of the unit sphere $S^3 = G$ gives $\int\sum_\alpha |X_\alpha F|^2\,\mathrm{d}\mu^K \ge 3\|F-\Pi_y F\|^2$. The commuting projections Π_y split $L^2(\mu^K)$ into joint eigenspaces invariant under P_t^K, on which $\frac{d}{dt}\|P_t^K F\|^2 = -2\mathcal{E}_K(P_t^K F)$. Since v is the field of Φ, $v(G\circ\Phi_\sigma)(U) = \frac{d}{ds}G(\Phi_\sigma\Phi_s U)|_{s=0}$.

(b) The laws of the block fields at rational times lie in a compact set of measures on the compact metrisable space $\mathcal{Y}^{\mathbb{Q}}$. The means $\langle\Psi_0, P_{s_1}^K\Psi_1\cdots P_{s_n}^K\Psi_n\rangle$ extend holomorphically and boundedly to $\operatorname{Re} s_i > 0$, so normal families carry convergence from rational to all times and give the real analyticity; convergence in total variation of the laws of the block fields carries it from continuous to Borel functions. Stationarity, reversibility and positivity under reflection hold at every cutoff, where $\mathbb{E}[\overline{\theta\mathsf{F}}\mathsf{F}]$ is the square of a norm by the Markov property, and pass to the limit.

By (a) the non-invariant parts decay in a time $O(a_K)$; this gives the independent gauges and $Q_t = \Pi Q_t \Pi$. Since 1 and the $e^{-t\lambda}$ span a dense algebra of $C([0,\infty])$, the spectral measures converge; $\frac{1}{t}(1-e^{-t\lambda})\uparrow\lambda$ gives $\mathcal{E}_\star(F) = \int\lambda\,\mathrm{d}\nu_F$, lower semicontinuity and closedness, and $(F,(1-Q_t)F) = \frac{1}{2}\mathbb{E}(F(Y_t)-F(Y_0))^2$ the Markov property of the form. $\|Q_t F\|^2 \le (F, Q_{2t}F)$ is Cauchy–Schwarz at the cutoff; recovery sequences are $\Lambda(\Lambda-\mathcal{L}_{K_j})^{-1}G_j$ with $\Lambda\to\infty$ slowly. In the equivalences, weak convergence with convergence of the norms is strong convergence, and $\|P_t^K\Psi_K\|^2 = (\Psi_K, P_{2t}^K\Psi_K)$; the strong convergence of $P_t^{K_j}\Psi_{K_j}$ gives (M1) through the quadratic forms $\langle H,(1-P_t^{K_j})H\rangle \ge 2\langle H,(1-P_t^{K_j})G_j\rangle - \langle G_j,(1-P_t^{K_j})G_j\rangle$, and the non-invariant parts are handled by (a).

For the last statement, the form $\mathbb{E}[\overline{\theta\mathsf{F}}\,\mathsf{G}]$ is hermitian and positive; $\mathcal{K}$ is its completion, $JF = [F(Y_0)]$, R_s the shift of the times by s, self-adjoint because θ reverses the shift and contractive by repeated Cauchy–Schwarz; $(F, Q_t F') = \langle JF, R_t JF'\rangle$, and Pythagoras gives the identity.

(c) The weight defined before Prop. YM3_I:115.2 of the chain with top depth n vanishes unless all $\pi_{K,d}U$, $n\le d\le K$, lie in their cuts, and by the comparison of the chain with the bare density (Prop. YM3_I:115.2, domination and lost-mass bounds) its integral against $e^{-\beta A_K}dU$ falls short of Z_K by at most $\mathsf{R}_n Z_K$; so off a μ^K-mass $\le \mathsf{R}_n$ all $\pi_{K,d}U$ lie in the cuts, and $\varpi(\pi_{K,d}U) < \theta_d$; for μ by total variation at finitely many depths (Cor. 2.8) and a monotone limit. Four statements among the construction results on the averaging map $\mathcal{M}$ enter: its gauge covariance $\mathcal{M}(U^u) = \mathcal{M}(U)^{u|}$, $u|$ the restriction of a gauge transformation u to the sites of the coarser lattice (Prop. YM3_I:4.5, YM3_I:(40)); $\mathcal{M}$ is Borel, and on the small fields, where the plaquette variables of the double block of a coarse bond lie within $(200L^2)^{-1}$ of $\mathbf{1}$, it is given at that bond by the first of the three formulas of its definition and is real-analytic (Prop. YM3_I:4.5); its linear part at $\mathbf{1}$ (Lem. YM3_I:4.6, YM3_I:(42)) is the straight average along the bonds plus the difference of the averages along the contours at the two ends of the coarse bond, which is the coarse gradient of a site function; and the bound for rearranged paths (Lem. YM3_I:4.1), which gives on the triple block of a coarse bond c, the union of the blocks of c_-, of c_+ and of the end point of the next bond c^+ in the direction of c, a gauge in which every bond variable is e^{iA_b} with $A_b\in\mathfrak{g}$, $|A_b|\le 15L^2\epsilon$, if all plaquette variables there lie within $\epsilon\le(200L^2)^{-1}$ of $\mathbf{1}$.

With them the averaging map acts on gradients of invariant functions as follows: if $\varpi(\mathcal{M}^j U) \le \epsilon'_j \le (48000L^4)^{-1}$ for $j<k$, then $\mathcal{M}^k$ is real-analytic near U and

$$\Gamma_{d+k}(\Psi\circ\mathcal{M}^k)(U) \le \prod_{j<k}(1+C_D\epsilon'_j)^2\,\Gamma_d(\Psi)(\mathcal{M}^k U), \qquad C_D = C_D(L),$$

for invariant $\Psi\in C^1(\Omega_d)$: the transpose of the differential of $\mathcal{M}$ maps horizontal vectors to horizontal vectors with norm at most $L^{-1/2}(1+C_D\epsilon)$, because the linear average is the straight average along the bonds, of norm $L^{-1/2}$ by Cauchy–Schwarz, plus the coarse gradient of the contour averages, which is a gauge direction of the block field and is orthogonal to the horizontal vectors, and the rest has norm at most $37L^2C_L\epsilon$, by the gauge covariance in the gauge of the triple block and the mean value inequality on the ball $\{|A_b|\le(3200L^2)^{-1}\}$, C_L the largest norm there of the differentials of $A\mapsto\mathcal{M}(e^{iA})(c)$ and of the rest, so that $C_D = 37L^{5/2}C_L$ depends on L alone; and $a_{d+k}^{-1}L^{-k} = a_d^{-1}$.

The cutoffs: $|\partial(\varpi_p^2)|\le 4\varpi_p$ by the computation of Lem. 16.2(a) on Ω_d, so each factor of χ_d has a gradient of norm at most $4\varpi_p/(2\theta_d)^2 \le 4/\theta_d$, which vanishes unless $2\theta_d\le\varpi_p\le 4\theta_d$, and there are $3(\ell L^d)^3$ plaquettes; this gives $\Gamma_d(\chi_d)^{1/2}\le \mathsf{k}_d\mathbf{1}\{\varpi\ge\theta_d\}$. F_K is smooth: by induction from the depth K down, the closure of the set where $\prod_{e<e'\le K}\chi_{e'}\circ\pi_{K,e'}\neq 0$ lies in $\{\varpi(\pi_{K,e'}U)\le 4\theta_{e'}$ for $e<e'\le K\}$, where $4\theta_{e'}\le(48000L^4)^{-1}$ as $e'>e\ge n\ge n_1$, so that $\pi_{K,e}$ is real-analytic near it; on that open neighbourhood the product with $\chi_e\circ\pi_{K,e}$ is smooth, it is 0 on the open complement of the closure, and the two open sets cover Ω_K; the same holds for the factor $\Phi\circ\pi_{K,n}$.

This is the role of the cutoffs: they keep every block field that enters F_K where $\mathcal{M}$ is given by its real-analytic formula. By the product rule and the averaging bound at $\epsilon'_j = 4\theta_{K-j}$, with $\prod_j(1+C_D\epsilon'_j)\le \mathsf{c}_* = \exp(4C_D\sum_{e\ge n_1}\theta_e)$,

$$\Gamma_K(F_K)^{1/2} \le \mathsf{c}_*\Big[\sup\Gamma_n(\Phi)^{1/2} + \sup|\Phi|\sum_{n\le d\le K}\mathsf{k}_d\,\mathbf{1}\{\varpi(\pi_{K,d}U)\ge\theta_d\}\Big],$$

and the event in the sum has μ^K-mass $\le\max\{\mathsf{R}_d,\mathsf{R}_{d-1}\}$, by the first statement of (c) at the depth d, or at the depth $d-1$ if $d=K$. Minkowski's inequality and the decay of R_d faster than every exponential, against $\mathsf{k}_d = O(L^{4d})$, bound every L^p norm uniformly in K, and $\|\Gamma_K(\chi_n^K)^{1/2}\|_{L^{2p}(\mu^K)}\le \mathsf{c}_*\sum_{d\ge n}\mathsf{k}_d\max\{\mathsf{R}_d,\mathsf{R}_{d-1}\}^{1/2p}\to 0$ as $n\to\infty$. Density: a bounded Borel invariant function of a small depth is μ-almost surely a Borel function of a large depth, it is approximated in $L^2(\mu_d)$ by invariant polynomials, and the cutoffs change it by $\le\mathsf{R}_d^{1/2}$.

Then $\mathcal{E}_\star(F)\le\liminf_K\mathcal{E}_K(F_K)$, $\|Q_tF-F\|^2\le 2(F,(1-Q_t)F)\le 2t\sup_K\mathcal{E}_K(F_K)$, and $\nu_F(\{\infty\}) = \|F\|^2-\lim_{t\to0}(F,Q_tF)$. Each property of (Y_t) and of $\mathcal{E}_\star$ stated in (b) and (c) is shared by the frozen process, which keeps Y_0 up to gauge transformations of the threads drawn from the Haar measure independently at different times, so that $Q_t=\Pi$ for $t>0$ and $\mathcal{E}_\star=0$ on L^2_{inv}: motion is the content of (A1). □

Proposition 15.2 (The limiting process on the common core)**.** Assume (E); let $\mathcal{C}$ and F_K be as in Lem. 15.1(c), and $T>0$.

(a) (Paths.) For $F\in\mathcal{C}$ and $1\le p<\infty$, $\mathbb{E}|F_K(U_t)-F_K(U_s)|^{2p}\le C_p(F)|t-s|^p$ uniformly in K; the processes $(F_K(U_t))_{t\le T}$ are tight in $C[0,T]$, and along every limit $F(Y_t)$ has a continuous modification, locally Hölder continuous of every exponent $<\frac12$. For $d\ge n_1$ there is $\mathsf{p}_d(T)$, which does not depend on K and tends to 0 faster than every exponential in d, such that for all $K>d$, with probability at least $1-\mathsf{p}_d(T)$, $\varpi(\pi_{K,e}U_t)<2\theta_e$ for all $d\le e\le K$ and all $t\in[0,T]$.

(b) (Energies.) For $F\in\mathcal{C}$ the limit $\mathcal{E}_\infty(F)=\lim_K\mathcal{E}_K(F_K)$ exists along the full sequence of cutoffs, and $\mathcal{E}_\infty(F)=\int\Gamma_\infty(F)\,\mathrm{d}\mu$ with

$$\Gamma_\infty(F)(Y) = \lim_{d\to\infty}a_d^{-1}\,\mathsf{t}[\tfrac23,\tfrac16;Y_d](\partial F^{[d]}(Y_d)) \quad \mu\text{-almost surely and in } L^1(\mu),$$

where $F^{[d]}$ is the lattice form of F at depth d and

$$\mathsf{t}[\alpha,\beta;V](B) = \alpha\|B\|^2 + 2\beta\sum_c \langle B(c), \mathrm{Ad}(V(c))B(c^+)\rangle,$$

c^+ the bond that follows c in its direction. $\mathcal{E}_\infty(F)$ depends only on the class of F in $L^2(\mu)$, $\mathcal{E}_\star \le \mathcal{E}_\infty$ on $\mathcal{C}$ for every limit, and for $F = \Phi(Y_d)\prod_{e\ge d}\chi_e(Y_e)$ with $\mathsf{R}_d < 1$

$$\mathcal{E}_\infty(F) \ge (\tfrac{1}{3} - \tfrac{1}{2}\bar\eta_d)(1-\mathsf{R}_d)\, e^{-\mathsf{o}_d}\, a_d^{-1}|\mathsf{l}_d|^{-1}\int_{\mathsf{l}_d}|\partial\Phi|^2 dV, \qquad \bar\eta_d \to 0.$$

(c) (The laws of the block fields.) For an integer $d \ge 1$ with $L^d\mathsf{b} \ge \mathsf{t}_\natural$ and every bounded Borel invariant $f \ge 0$ on Ω_d,

$$(1-\mathsf{R}_d)\, e^{-\mathsf{o}_d}|\mathsf{l}_d|^{-1}\int_{\mathsf{l}_d} f\,\mathrm{d}V \le \int f(Y_d)\,\mathbf{1}\{\varpi(Y_e) < \theta_e \text{ for all } e \ge d\}\,\mathrm{d}\mu \le \int f\,\mathrm{d}\mu_d$$
$$\le e^{\mathsf{o}_d}|\mathsf{l}_d|^{-1}\int_{\mathsf{l}_d} f\,\mathrm{d}V + \mathsf{R}_d \sup f,$$

and the same at every cutoff $K > d$, with μ_d^K in the place of μ_d and with the middle member $\int f(\pi_{K,d}U)\,\mathbf{1}\{\varpi(\pi_{K,e}U) < \theta_e \text{ for } d \le e \le K\}\,\mathrm{d}\mu^K$. So an invariant Haar-null set has μ_d- and μ_d^K-mass $\le \mathsf{R}_d$; and if $\mathsf{R}_d < 1$, every non-empty open subset of l_d has positive μ_d-mass and μ_d^K-mass bounded below uniformly in K.

(d) (Square field and martingales.) Let (K_j) be a limit and $F \in \mathcal{C}$. There is $\Gamma_\star(F) \in L^2_{\mathrm{inv}}$ with $(2t)^{-1}\mathbb{E}[\phi(Y_0)(F(Y_t)-F(Y_0))^2] \to \int \phi\,\Gamma_\star(F)\,\mathrm{d}\mu$ as $t \to 0$ for bounded invariant ϕ; $0 \le \Gamma_\star(F) \le \Gamma_\infty(F)$, $\int \Gamma_\star(F)\,\mathrm{d}\mu = \mathcal{E}_\star(F)$, and $\mathcal{E}_\star(\psi(F)) = \int \psi'(F)^2\Gamma_\star(F)\,\mathrm{d}\mu$ for $\psi \in C^2(\mathbb{R})$. Along a further subsequence the lattice martingales converge, and $F(Y_t) = F(Y_0) + M_t^F + D_t^F$, with a continuous martingale M^F with $\langle M^F\rangle_t = 2\int_0^t \Gamma_\infty(F)(Y_s)\,\mathrm{d}s$, and a continuous process D^F with stationary increments, $\mathbb{E}|D_t^F|^2 = 2[t\,\mathcal{E}_\infty(F) - (F,(1-Q_t)F)]$, $\mathbb{E}[D_t^F(F(Y_t)-F(Y_0))] = 0$, and $\lim_{t\to 0} t^{-1}\mathbb{E}|D_t^F|^2 = 2[\mathcal{E}_\infty(F) - \mathcal{E}_\star(F)]$; in reversed time, $F(Y_t) - F(Y_0) = \frac{1}{2}M_t^F - \frac{1}{2}(\hat M_T^F - \hat M_{T-t}^F)$ with a martingale $\hat M^F$ of the law of M^F.

The following are equivalent: $\mathcal{E}_\star(F) = \mathcal{E}_\infty(F)$; $\Gamma_\star(F) = \Gamma_\infty(F)$ μ-almost everywhere; D^F has zero energy; $\lim_{\Lambda\to\infty}\limsup_j \int_{(\Lambda,\infty)} \lambda\,\mathrm{d}\nu_F^{K_j} = 0$, ν_F^K the spectral measure of F_K. And F is frozen ($Q_tF = F$ for all t) if and only if $D^F = -M^F$.

(e) (Motion from fields with a divergence.) A vector field with a divergence at the cutoff K is a $\zeta \in L^2(\mu^K;\mathfrak{g}^{\mathrm{bonds}})$ with $\delta_K\zeta \in L^2(\mu^K)$ such that $a_K^{-1}\int\langle\zeta,\partial H\rangle\,\mathrm{d}\mu^K = -\int H\,\delta_K\zeta\,\mathrm{d}\mu^K$ for $H \in C^\infty(\Omega_K)$. If $a_{K_j}^{-1}\int|\zeta_j|^2\,\mathrm{d}\mu^{K_j} \le \mathsf{A}^2$, $\|\delta_{K_j}\zeta_j\| \le \mathsf{D}$ and $\delta_{K_j}\zeta_j \to g$ weakly, then for $F \in L^2(\mu)$ and $t > 0$

$$\mathcal{E}_\star(F) \ge \frac{(F,g)^2}{\mathsf{A}^2}, \qquad (F,(1-Q_t)F) \ge \frac{t\,(F,g)^2}{\mathsf{A}^2 + t\,\mathsf{D}^2};$$

conversely, if $\mathcal{E}_\star(F) > 0$, such fields with $(F,g) \ne 0$ exist. A field that does not depend on the configuration pairs to zero with every invariant gradient: $\int\langle\partial F, z\rangle\,\mathrm{d}\mu^K = 0$ for constant z and invariant F.

(f) (An example.) In a limit of reversible diffusions observed through a block variable whose law converges in total variation, the inequality $\mathcal{E}_\star \le \mathcal{E}_\infty$ can be strict: for the reversible diffusion on $\mathbb{R}^2$ with the weight $e^{-\frac{\kappa}{2}(x-w)^2 - \frac{1}{2}w^2}$, the form $\int(|\partial_x F|^2 + s^2|\partial_w F|^2)$ with fixed $s > 0$ and the block variable x, as $\kappa \to \infty$ the law of x converges in total variation, the square field of x is 1, the limit is an Ornstein–Uhlenbeck process, Markov and moving, and $\mathcal{E}_\star(x) = s^2/(1+s^2) < 1 = \mathcal{E}_\infty(x)$.

■

Proof sketch. (a) At every cutoff, for $H \in C^\infty(\Omega_K)$ and $0 \le t \le T$, $H(U_t) - H(U_0) = \frac{1}{2}M_t - \frac{1}{2}(\hat{M}_T - \hat{M}_{T-t})$ with forward and backward martingales of brackets $2\int \Gamma_K(H)$ (the drift drops out, and the square field alone enters); the martingale moment inequalities and stationarity give $\mathbb{E}|H(U_t) - H(U_s)|^{2p} \le c_p 2^p |t-s|^p \int \Gamma_K(H)^p d\mu^K$, and $\mathbb{P}(\sup_{t\le T}|H(U_t) - H(U_0)| \ge \lambda) \le 10T\lambda^{-2}\mathcal{E}_K(H)$. With the bounds of Lem. 15.1(c) this is uniform in K, and the criterion of Kolmogorov and Chentsov gives tightness and Hölder continuity. For the small fields apply the maximal bound to $H = \prod_{d\le e\le K} \chi'_e \circ \pi_{K,e}$, built from a second family of cutoffs, $\chi'_e = \prod_p \varphi_{\mathrm{cut}}(\varpi_p^2/\theta_e^2)$ on Ω_e with the function φ_{cut} of Lem. 15.1(c): χ'_e is 1 where $\varpi \le \theta_e$ and 0 where $\varpi \ge 2\theta_e$, and, as there, each factor has a gradient of norm at most $4\varpi_p/\theta_e^2 \le 8/\theta_e$, so that $\Gamma_e(\chi'_e)^{1/2} \le \mathsf{k}_e \mathbf{1}\{\varpi \ge \theta_e\}$ with the same numbers $\mathsf{k}_e = 24L^{e/2}\theta_e^{-1}(\ell L^e)^3$, which thus bound $\Gamma_e^{1/2}$ of both families.

As in the proof of Lem. 15.1(c), the energy of H is at most $\mathsf{c}_*^2\Sigma_d^2$, $\Sigma_d = \sum_{e\ge d} \mathsf{k}_e \max\{\mathsf{R}_e, \mathsf{R}_{e-1}\}^{1/2}$, with $\mathsf{c}_* = \exp(4C_D\sum_{e\ge n_1}\theta_e)$; $H(U_0) = 1$ off a mass $\le \mathsf{R}_d$, and $H(U_t) > 0$ forces $\varpi(\pi_{K,e}U_t) < 2\theta_e$ for $d \le e \le K$; so $\mathsf{p}_d(T) = \mathsf{R}_d + 10T\mathsf{c}_*^2\Sigma_d^2$.

(b) On the small fields $F_K = F^{[d]} \circ \pi_{K,d}$, $\partial F_K = \mathcal{D}^{(k)}(U)^* \partial F^{[d]}$ with $\mathcal{D}^{(k)}$ the differential of $\mathcal{M}^k$, and the transpose of the differential carries the quadratic forms along: $L^k \mathsf{t}[\alpha,\beta;U](\mathcal{D}^{(k)*}B) = \mathsf{t}[\mathsf{A}_L^k(\alpha,\beta);V](B) + O(\sum_j \epsilon'_j)\|B\|^2$ for horizontal B at $V = \mathcal{M}^k U$, where A_L is the linear map of the flat average, with the eigenvector $(\frac{2}{3}, \frac{1}{6})$ of eigenvalue 1 (the Gram matrix of the kernels that are a tent along the bond and a box across it) and $(\frac{1}{3}, -\frac{1}{6})$ of eigenvalue L^{-2}.

So $\Gamma_K(F_K)$ is close in $L^1(\mu^K)$, uniformly in K, to $a_d^{-1}\mathsf{t}[\frac{2}{3},\frac{1}{6};\cdot](\partial F^{[d]}) \circ \pi_{K,d}$, and total variation gives the limit along the full sequence; the same estimate between two depths along a thread gives Γ_∞. A function of $\mathcal{C}$ that vanishes μ-almost everywhere has a gradient that vanishes off an invariant Haar-null set, whose mass is controlled by (c); this gives the dependence on the class. $\mathcal{E}_\star \le \mathcal{E}_\infty$ is the lower semicontinuity of Lem. 15.1(b). The lower bound: on the threads with $\varpi(Y_e) < 2\theta_e$ for all $e \ge d$, where all cutoffs χ_e equal 1 near Y_e and $F^{[e]} = \Phi \circ \mathcal{M}^{e-d}$, the same estimate between the depths d and $e \to \infty$ gives $|\Gamma_\infty(F)(Y) - a_d^{-1}\mathsf{t}[\frac{2}{3},\frac{1}{6};Y_d](\partial\Phi(Y_d))| \le \frac{1}{2}\bar{\eta}_d a_d^{-1}|\partial\Phi(Y_d)|^2$, with $\bar{\eta}_d = 4C_{\mathrm{T}}\mathsf{c}_*^2\sum_{e>d}\theta_e$, $C_{\mathrm{T}} = C_{\mathrm{T}}(L)$ the constant of the term $O(\sum_j \epsilon'_j)$ above and c_* that of the proof of Lem. 15.1(c), so $\Gamma_\infty(F)(Y) \ge (\frac{1}{3} - \frac{1}{2}\bar{\eta}_d)a_d^{-1}|\partial\Phi(Y_d)|^2$ as $\mathsf{t}[\frac{2}{3},\frac{1}{6};V] \ge \frac{1}{3}\|\cdot\|^2$; elsewhere $\Gamma_\infty(F) \ge 0$.

The event of the middle member of (c) lies in that set of threads, and (c) at the invariant function $f = |\partial\Phi|^2$ gives the bound.

(c) At the top depth d the chain from level 0 has the density $e^{\Sigma+\Phi_t}$ with respect to Haar measure on I_d, Σ a constant (Prop. YM3_I:115.2, image identity, Prop. YM3_I:114.2), its non-classical part bounded by $C_{\mathrm{nl}}\frac{L}{L-1}E_0(\ell L^d)^3$ (equation YM3_I:(2180)) and the classical part between 0 and $\frac{3}{2}c_\rho^2 A_0^2(\log L^d \mathsf{b})^{2\mathsf{p}_0}(\ell L^d)^3$ (by YM3_I:(2161) at the weight x_t); its mass lies in $[1-\mathsf{R}_d, 1]$; the comparison of the chain with the bare density (Prop. YM3_I:115.2, domination and lost-mass bounds) gives the sandwich for μ_d^K, and total variation (Cor. 2.8) for μ_d.

The middle member: the weight of the chain with top depth d vanishes unless $\pi_{K,e}U$ lies in its cut, so in $\{\varpi < \theta_e\}$, for $d \le e \le K$; the function $f(\pi_{K,d}U)\mathbf{1}\{\varpi(\pi_{K,e}U) < \theta_e \text{ for } d \le e \le K\}$ is invariant, by the gauge covariance of $\mathcal{M}$; so, σ^K being at most μ^K on invariant functions (Lem. 15.3(i)), $\int f\,\mathrm{d}\sigma_d^K = \int f(\pi_{K,d}U)\mathbf{1}\{\cdots\}\,\mathrm{d}\sigma^K \le \int f(\pi_{K,d}U)\mathbf{1}\{\cdots\}\,\mathrm{d}\mu^K$, and $\int f\,\mathrm{d}\sigma_d^K \ge (1-\mathsf{R}_d)e^{-\mathsf{o}_d}|\mathsf{I}_d|^{-1}\int_{\mathsf{I}_d} f\,\mathrm{d}V$ by the lower bound of the density of σ_d^K (Lem. 15.3(iii)).

For μ: at a depth $D > d$ the event $\{\varpi(Y_e) < \theta_e \text{ for } d \le e \le D\}$ is an event of the block field of depth D, it contains the event at the cutoff $K > D$, and total variation at the depth D carries the bound to μ; then $D \to \infty$, a monotone limit. Positivity: for a non-empty open $O \subset \mathsf{I}_d$ and a smaller non-empty open O' with closure in O, the gauge orbit of O' is open, invariant and of positive Haar measure, and it is covered by finitely many gauge translates of O, which all have the mass of O.

(d) For bounded invariant ϕ, the forward–backward decomposition gives $(2t)^{-1}\mathbb{E}[\phi(U_0)(F_K(U_t) - F_K(U_0))^2] \le \int \phi\,\Gamma_K(F_K) + c\,\langle\phi, (1-P_t)\phi\rangle^{1/2}$ for $\phi \ge 0$; the limit in K and then in t gives

$\Gamma_\star \le \Gamma_\infty$, and, with $\mathsf{e}_t(\phi;F) = (2t)^{-1}\mathbb{E}[\phi(Y_0)(F(Y_t)-F(Y_0))^2]$, the identity $2t\,\mathsf{e}_t(\phi;F) = 2\langle \phi F, (1-Q_t)F\rangle - \langle \phi, (1-Q_t)F^2\rangle$ gives the existence. The chain rule follows from fourth moments of the increments, which are $O(t^2)$. The lattice processes $F_K(U_t)$, their martingales $M^{K,F}$ and brackets are tight in $C[0,T]$, limits of L^4-bounded martingales are martingales, and the brackets converge to $2\int \Gamma_\infty(F)(Y_s)\,\mathrm{d}s$ by (b).

At the cutoff, with the spectral measure ν of F_K, $\mathbb{E}|\int_0^t \mathcal{L}_K F_K(U_s)\,\mathrm{d}s|^2 = 2\int (t\lambda - 1 + e^{-t\lambda})\,\mathrm{d}\nu = 2[t\,\mathcal{E}_K(F_K) - (F_K,(1-P_t^K)F_K)]$ and the drift is orthogonal to the increment; these pass to the limit. The equivalences: $\mathcal{E}_\infty(F) - \mathcal{E}_\star(F) = \lim_\Lambda \lim_j \int (\lambda-\Lambda)_+\,\mathrm{d}\nu_F^{K_j}$, the energy that escapes to infinite frequency, is the energy of the drift and the integral of $\Gamma_\infty - \Gamma_\star$.

(e) Let $H_j \to F$ strongly with $\mathcal{E}_{K_j}(H_j) \to \mathcal{E}_\star(F)$; then $|\langle H_j, \delta_{K_j}\zeta_j\rangle| \le \mathsf{A}\,\mathcal{E}_{K_j}(H_j)^{1/2}$, and the left side tends to $|(F,g)|$. At a positive time, $1-e^{-x} \ge \frac{x}{1+x}$ for $x \ge 0$ (as $e^x \ge 1+x$) and $\int \frac{t\lambda}{1+t\lambda}\,\mathrm{d}\nu = \min_G\{\|H-G\|^2 + t\,\mathcal{E}_K(G)\}$, which is bounded below, for every real s, by $2s\langle H, \delta_K\zeta\rangle - s^2(\|\delta_K\zeta\|^2 + t^{-1}a_K^{-1}\int|\zeta|^2)$. Conversely, $\zeta_j = \partial(1-\mathcal{L}_{K_j})^{-1}H_j$ has $\delta_{K_j}\zeta_j = (1-\mathcal{L}_{K_j})^{-1}H_j - H_j \to (1+A_\star)^{-1}F - F$ by (A3), and $(F,(1+A_\star)^{-1}F - F) = -\int \frac{\lambda}{1+\lambda} d\nu_F \ne 0$. For constant z, $\int \partial_b F\,\mathrm{d}\mu^K$ is invariant under $\mathrm{Ad}(g)$ for all g, hence 0.

(f) The operator of the diffusion on linear functions has the trace $\kappa(1+s^2)+s^2$ and the determinant κs^2; its eigenvalues are $\lambda_+ \sim \kappa(1+s^2)$ and $\lambda_- \to s^2/(1+s^2)$, and the spectral measure of x puts a mass asymptotic to $\frac{1}{1+s^2}\lambda_+^{-1}$ at λ_+, so $\int \lambda\,\mathrm{d}\nu_x^\kappa = 1$ for all κ while $\mathcal{E}_\star(x) = s^2/(1+s^2)$. □

15.1.6. *Conditional fluctuation laws and transport estimates.*

Lemma 15.3 (Conditional laws along the renormalisation chain)**.** Assume (E), let $n \ge 1$ be an integer with $L^n\mathsf{b} \ge \mathsf{t}_\natural$, and let $K > n$. (i) $1 - \mathsf{R}_n \le \sigma^K(\Omega_K) \le 1$, and $\int f\,\mathrm{d}\sigma^K \le \int f\,\mathrm{d}\mu^K \le \int f\,\mathrm{d}\sigma^K + \mathsf{R}_n \sup f$ for bounded Borel invariant $f \ge 0$. (ii) For $0 \le k < t$, $d = K-k$, and every $W \in \mathsf{Cut}_{k+1}$, p_W is a probability measure with a positive continuous density on the box; σ_{d-1}^K is carried by Cut_{k+1}, and

$$\int \Big[\int f(V_W(B))\,p_W(\,\mathrm{d}B)\Big]\sigma_{d-1}^K(\,\mathrm{d}W) = \int f\,\mathrm{d}\sigma_d^K \qquad \text{for every Borel } f \ge 0 \text{ on } \Omega_d;$$

so the images of the p_W under the V_W form a regular conditional law of V_k given $V_{k+1} = W$ under σ^K, and under σ^K the sequence $V_t, \dots, V_0$ is a Markov chain from coarse to fine whose kernel at every $W \in \mathsf{Cut}_{k+1}$ is the image under V_W of p_W; the kernels carry bounded continuous functions on Cut_k to continuous functions on Cut_{k+1}, that is, $W \mapsto \int h(V_W(B))\,p_W(\,\mathrm{d}B)$ is continuous on Cut_{k+1} for every bounded continuous h on Cut_k; kernels and towers do not depend on the top level; $h_k(V_k)$ is the conditional mean of H given V_k; the tower of an invariant H is invariant at every point of Cut_k, with $\inf H \le h_k \le \sup H$; and a density that is a function of the first member V_t changes no kernel and no tower.

(iii) $\sigma_n^K = \mathsf{g}\,\mathrm{d}V$ on I_n, $\sup \mathsf{g} \le e^{\mathsf{o}_n} \inf \mathsf{g}$, and $(1-\mathsf{R}_n)e^{-\mathsf{o}_n} \le \mathsf{g}\,|\mathsf{I}_n| \le e^{\mathsf{o}_n}$ at every point. (iv) For $n \ge n_2$, $0 \le k < t$, $W \in \mathsf{Cut}_{k+1}$: $p_W = e^{-\Psi_W}dB$ with Ψ_W real-analytic and $\frac14\gamma_{\mathrm{r}} \le D^2\Psi_W \le \Lambda_+$; every tangent vector X of the fibre $\{\mathcal{M}V = W\}$ at $V_W(B)$ has $\|X\|^2 \le C_{\mathrm{ch}}\sum|X(b)|^2$, the sum over the bonds that are not pivot bonds; and for bounded $g \in C^1(\mathsf{Cut}_k)$

$$\mathrm{Var}_{p_W}(g\circ V_W),\quad \tfrac12\mathrm{Ent}_{p_W}\big((g\circ V_W)^2\big) \;\le\; C_{\mathrm{P}}\frac{a_d^2}{\mathsf{b}}\int \Gamma_d(g)(V_W(B))\,p_W(\,\mathrm{d}B),$$

with $C_{\mathrm{P}} = 4C_{\mathrm{ch}}/\gamma_{\mathrm{r}}$, which depends only on the constants of the construction, and not on K, k, W, ℓ or b. ■

Proof sketch. (i) is Prop. YM3_I:115.2, domination and lost-mass bounds, the comparison of the small-field chain with the bare density, with the weight $\mathfrak{w}_t^{\mathrm{sf}}$ and the bound R_n of the definitions preceding Prop. YM3_I:115.2; (ii): the weight defined before Prop. YM3_I:115.2 vanishes unless every V_j lies in its cut, so σ_{d-1}^K is carried by Cut_{k+1}. Positivity at every point, by induction over

the scales: with Σ_k the sum of the constants of the steps $1,\dots,k$ at the source 0, $e^{\Sigma_0}F_0 = F_0$ is the bare density on Cut_0 (the definition preceding Prop. YM3_I:114.2), and if $e^{\Sigma_k}F_k > 0$ on Cut_k, then $e^{\Sigma_{k+1}}F_{k+1} = \mathsf{I}[e^{\Sigma_k}F_k] > 0$ on Cut_{k+1} YM3_I:(2174), because the prefactors of the chart integral are positive, G_W and J_W are real ([7, Lem. B.3]) and the chart carries the box into Cut_k (Lem. YM3_I:113.10); so the density of p_W, a positive multiple of $e^{-x_k G_W + \mathsf{J}_W}(e^{\Sigma_k}F_k)\circ V_W$, is positive at every point of the box, and it is continuous, as $(W,B)\mapsto V_W(B), G_W(B), \mathsf{J}_W(B)$ are continuous and the chart integral of a continuous function is continuous ([7, Lem. B.3]), which makes every F_k continuous on Cut_k by the same induction.

The kernels carry bounded continuous functions to continuous functions, in the whole range $L^n\mathsf{b} \geq \mathsf{t}_\natural$: at a bounded continuous h on Cut_k the kernel at W is $\mathsf{I}[hF_k](W)/\mathsf{I}[F_k](W)$, the normalised chart integral [7, eq. 191] of the integrand of the chain; Cut_{k+1} lies in the set O^{im} (Lem. YM3_I:113.10), where the chart, its gauge-fixing form and the exponent of its Jacobians are continuous in (W,B) ([7, Lem. B.3]), and F_k is continuous on Cut_k, into which the chart carries the box; so the chart integrals of the non-negative functions $(\sup|h| \pm h)F_k$ integrate over the fixed box functions that are continuous in (W,B), Fatou's lemma on the box makes them lower semicontinuous in W, and as their sum $2\sup|h|\,\mathsf{I}[F_k] = 2\sup|h|\,e^{\mathsf{c}_{k+1}(0)}F_{k+1}$ is continuous by the induction above, both are continuous; the denominator $\mathsf{I}[F_k]$ is continuous and positive.

The identity: the factors of the weight of the levels $> k$ are functions of V_{k+1}; the measure of the chain of the level k is $\nu_k = e^{\Sigma_k}\mathbf{1}_{\mathsf{Cut}_k}F_k\,\mathrm{d}V$ the measure recursion in Prop. YM3_I:114.2, YM3_I:(2175); and by the identity of measures of the step ([7, Lem. B.3]) the image under $\mathcal{M}$ of $\mathfrak{G}_k[x_k]\chi^{\mathrm{sf}}_{x_k}\,h\,\nu_k$ has the density $e^{\Sigma_k}\mathsf{I}[h\mathbf{1}_{\mathsf{Cut}_k}F_k]$ on the open set O^{im}, which contains $\mathcal{W}_{k+1}(x_k;\alpha_0,\alpha_1)\supset \mathsf{Cut}_{k+1}$ (Lem. YM3_I:113.10), for every bounded Borel h on Ω_d, and on Cut_{k+1} this density is $e^{\Sigma_k}\mathsf{I}[hF_k]$, the chart carrying the box into Cut_k; dividing by $e^{\Sigma_k}\mathsf{I}[F_k]$ gives the identity for bounded Borel $f\geq 0$, and monotone convergence for every Borel $f\geq 0$.

As $V_{k+2},\dots,V_t$ are functions of V_{k+1}, the identity at $f = h\cdot(g\circ\mathcal{M})$ is the Markov property with the kernel p_W. The tower of an invariant H is invariant at every point by the covariance of the chain under the gauge transformations that are constant on blocks; (iii) follows from the statements on the iteration from a level: by Prop. YM3_I:115.2, image identity and Prop. YM3_I:114.2, at the top the density is $e^{\Sigma+\Phi_t}$ on I_n, Σ a constant, with a non-classical part bounded by $C_{\mathrm{nl}}\frac{L}{L-1}E_0(\ell L^n)^3$ (equation YM3_I:(2180)) and a classical part between 0 and $\frac{3}{2}c_\rho^2 A_0^2(\log L^n\mathsf{b})^{2\mathfrak{p}_0}(\ell L^n)^3$ (by YM3_I:(2161) at the weight x_t), so it oscillates by at most o_n, and its mass is that of (i).

(iv): at every $W\in\mathsf{Cut}_{k+1}$, which lies in the class $\mathcal{W}_{k+1}(x_k;\alpha_0,\alpha_1)$ of the sum rule (Lemma YM3_I:113.10), the sum identities in equations YM3_I:(828)–YM3_I:(830) and their preceding Jacobian calculation and the step (Prop. 6.31) give the density of p_W the form $e^{\mathcal{E}(P_W;B)}$, normalised, of Prop. 15.4(b). This form holds at every depth, at every W of the class of the sum rule, where the restrictions of P_W to the polymers lie in the analyticity cores and the sum rules hold, so that every term of $\mathcal{E}(P_W;\cdot)$ is formed and holomorphic in B on $\{|B(b)|<C_A\mathfrak{b}(x_k)\}$ (Thm. YM3_I:W, Prop. YM3_I:51.4 and its proof, the potential definitions in Section YM3_I:51, and §15.3.1); the conditions of n_3 enter Prop. 15.4 only through the ball of complex triples of (a) and through the response.

So the form holds for every integer $n\geq 1$ with $L^n\mathsf{b}\geq\mathsf{t}_\natural$, from n_2 on in particular, as Prop. 15.4(b) states. The density being positive by (ii), $e^{\mathcal{E}(P_W;\cdot)}$ is a constant multiple of a positive function, so the imaginary part of $\mathcal{E}(P_W;\cdot)$ is constant on the connected box, and $\Psi_W = -\mathcal{E}(P_W;\cdot) + \log\int e^{\mathcal{E}(P_W;B')}\,\mathrm{d}B'$, with the branch of the logarithm that makes it real, is real-analytic near the closed box. Its Hessian is $D^2\Psi_W = \mathsf{A}_W - D^2\sum_Y v_Y$, with the form $\mathsf{A}_W = \sum_Y \mathsf{b}_Y(P_W) + \sum_\square \hat{\mathsf{G}}_\square(P_W)$ of the Gaussian integral of one step at P_W, whose inverse is the whole covariance of the step at P_W.

It lies within $\frac{1}{2}\mathsf{u}_{\mathrm{G}}(LM)^{-3}\leq\frac{1}{2}\gamma_{\mathrm{r}}$, in the ℓ^2 operator norm, of a real reference form whose floor and ceiling are γ_{r} and c_{r} (the completeness (St4) of Def. YM3_I:47.2, Thm. YM3_I:W, the proof of Prop. YM3_I:113.4), so $\frac{1}{2}\gamma_{\mathrm{r}}\leq\mathsf{A}_W\leq c_{\mathrm{r}}+\frac{1}{2}\gamma_{\mathrm{r}}$. The potentials are bounded by a constant times

$\delta(x_k)\mathfrak{b}(x_k)^2 e^{-\lambda d(Y)}$ on the whole product of the cores and the polydisc of radius $C_A\mathfrak{b}(x_k)$, so at P_W for every W of the class of the sum rule (Thm. YM3_I:W, Prop. YM3_I:51.4 and its proof, as in the proof of Prop. 15.4(d)), and by Cauchy's estimate their Hessians are at most $C_{\mathrm{Hs}}\delta(x_k) \le \frac{1}{4}\gamma_{\mathrm{r}}$ for $n \ge n_2$; hence $\frac{1}{4}\gamma_{\mathrm{r}} \le D^2\Psi_W \le c_{\mathrm{r}} + \frac{3}{4}\gamma_{\mathrm{r}} \le \Lambda_+$.

The log-Sobolev inequality $\mathrm{Ent}_p(f^2) \le \frac{2}{\kappa}\int |\nabla f|^2 \,\mathrm{d}p$ of a probability measure $p = e^{-\Psi}dx$ on an open convex set with $D^2\Psi \ge \kappa$, at $\kappa = \frac{1}{4}\gamma_{\mathrm{r}}$ on the convex box, gives $\mathrm{Ent}_{p_W}(F^2) \le \frac{8}{\gamma_{\mathrm{r}}}\int |\nabla_B F|^2 \,\mathrm{d}p_W$, and its linearisation the Poincaré inequality with $\frac{4}{\gamma_{\mathrm{r}}}$; since $x_k^{-1} = a_d/\mathfrak{b}$, the chart gives $|\nabla_B(g\circ V_W)|^2 \le C_{\mathrm{ch}}x_k^{-1}|\partial g|^2 = C_{\mathrm{ch}}\frac{a_d^2}{\mathfrak{b}}\Gamma_d(g)\circ V_W$. □

Proposition 15.4 (Log-concavity of the fluctuation law)**.** Assume (E). For $0 \le k < K$, lengths at the level $k+1$ are measured in its units, in which the bare spacing is $\eta_{k+1} = L^{-(k+1)}$; $J^{(k+1)}(U) = \eta_{k+1}^{-3}\partial A_K(U) = L^{-3(K-k-1)}J(U)$ is the current (YM3_I:9.2) formed with this spacing; for a 1-form A and a configuration U, $(\nabla_\mu^U A_\nu)(x) = \eta_{k+1}^{-1}[\mathrm{Ad}(U(\langle x, x+\eta_{k+1}e_\mu\rangle))A_\nu(x+\eta_{k+1}e_\mu) - A_\nu(x)]$, and $|\nabla_U A|(b) = \max_\mu |(\nabla_\mu^U A_\nu)(x)|$ at the bond $b = \langle x, x+\eta_{k+1}e_\nu\rangle$. π_k^M is the partition of the torus into the cubes of side M in the units of level k (the partition π_k of Section YM3_I:72 (the definitions of $\mathcal{D}_j$ and d_j at its beginning)); $\mathcal{D}_k$ is the set of its localisation domains, the unions of cubes of π_k^M that are connected through their faces, which we call polymers; $d(Y) = d_k(Y)$ is the linear size YM3_I:72 (the definitions of $\mathcal{D}_j$ and d_j at its beginning) of a polymer Y, the least number of edges of a tree of edges of its cubes that meets all of them; $\mathrm{Bd}(Y)$ is the set of the bare bonds with an end point in Y; and for a function f of triples and a pair $(a, \mathbf{j})$ of complex 1-forms, $D_{(a,\mathbf{j})}f(P_W) = \frac{d}{dz}f(U_+, za, \mathbf{J}_+ + z\mathbf{j})|_{z=0}$.

There are $r_{\mathrm{c}} \in (0, \alpha_0/256]$, determined by the fixed radii $\alpha_0 < \alpha_1$ of the notation and the constants of Prop. YM3_I:51.10, and $C_{\mathrm{mx}} < \infty$, depending only on the constants of the construction, such that for $K > n \ge n_3$, $0 \le k < K - n$ ($k = 0$ included), $W \in \mathsf{Cut}_{k+1}$, any background U_+ over W, $\mathbf{J}_+ = J^{(k+1)}(U_+)$ and $P_W = (U_+, 0, \mathbf{J}_+)$:

(a) for every polymer $Y \in \mathcal{D}_k$ and complex 1-forms $A', \mathbf{j}$ on $\mathrm{Bd}(Y)$ with $|A'|, |\nabla_{U_+}A'|, |\mathbf{j}| < r_{\mathrm{c}}$ there, the triple $(U_+, A', \mathbf{J}_+ + \mathbf{j})$ on $\mathrm{Bd}(Y)$ lies in the analyticity core $\mathcal{U}_{k+1}^\circ(Y; \alpha_0, \alpha_1)$ (Def. YM3_I:49.1, Prop. YM3_I:51.10, first assertion);

(b) $p_W \propto e^{\mathcal{E}(P_W;B)}dB$, where

$$\mathcal{E}(P;B) = \sum_{Y\in\mathcal{D}_k}\left[-\tfrac{1}{2}(B, \mathsf{b}_Y(P)B) + v_Y(P;B)\right] - \tfrac{1}{2}\sum_{\square\in\pi_k^M}(B, \hat{\mathsf{G}}_\square(P)B),$$

$\mathsf{b}_Y = x_k D_B^2\Psi^{\mathrm{A}}(Y; P, 0)$ with Ψ^{A} of equations YM3_I:(815) and YM3_I:(816), v_Y the potentials of Thm. YM3_I:W and of the fourth family at zero source (Prop. 6.31), $\hat{\mathsf{G}}_\square$ of §15.3.1; the quadratic form, too, is a sum of local terms; and $(A', \mathbf{j}, B) \mapsto \mathcal{E}$ is holomorphic on the product of the ball of (a) with $\{|B(b)| < C_A\mathfrak{b}(x_k)\}$, $C_A \ge 3$, which contains the closed box. The formula for p_W, the locality of its quadratic form and the holomorphy of $B \mapsto \mathcal{E}(P_W; B)$ on $\{|B(b)| < C_A\mathfrak{b}(x_k)\}$ hold as well for every integer $n \ge 1$ with $L^n\mathfrak{b} \ge \mathsf{t}_\natural$, so from n_2 on, with $K > n$, $0 \le k < K - n$, at every $W \in \mathsf{Cut}_{k+1}$ and any background U_+ over W: they need of W only that it lies in the class $\mathcal{W}_{k+1}(x_k; \alpha_0, \alpha_1)$ of the sum rule (Lem. YM3_I:113.10), at which the restrictions of P_W to the polymers lie in the analyticity cores;

(c) for W at a depth $\ge n_{\mathrm{a}}$, with $\mathcal{H}^+$ the response of U_+ at level $k+1$ (§5.3), $\mathbf{U}_\xi = e^{iL^{-k-1}\mathcal{H}^+(\xi)}U_+$, $W'_\xi = \mathcal{M}^{k+1}\mathbf{U}_\xi$ and $P_\xi = (U_+, \mathcal{H}^+(\xi), J^{(k+1)}(\mathbf{U}_\xi))$, there is $r_0 > 0$, which may depend on W and K, such that for real ξ with $\sup|\xi| < r_0$: $W'_\xi \in \mathsf{Cut}_{k+1}$ is a gauge transform of $e^{i\xi}W$, and $\mathbf{U}_\xi$ lies in the minimal orbits over W'_ξ and over $\mathcal{M}^k\mathbf{U}_\xi$; there is a chart $\check{V}_\xi(B)$ of the fibre over W'_ξ, real-analytic in (ξ, B) near the closed box, $\check{V}_0 = V_W$, with values in a compact invariant

subset of Cut_k; for every bounded Borel invariant H, $h_{k+1}(e^{i\xi}W) = \int h_k(\check V_\xi(B))\,\check p_\xi(\mathrm dB)$, where $\check p_\xi$ has the density $e^{\mathcal E(P_\xi;B)}/\int e^{\mathcal E(P_\xi;B')}\,\mathrm dB'$, real-analytic in ξ uniformly on the closed box, and $\check p_0 = p_W$; the frame, $\check p_\xi$ and $\psi_\xi = \frac{d}{ds}\mathcal E(P_{s\xi};\cdot)|_{s=0}$ do not depend on the choice of the axial element;

(d) for real ζ on the box and every pair $(a,\mathbf j)$ of complex bare 1-forms, with $\mu_y(a,\mathbf j)$ the largest of $|a|$, $|\nabla_{U_+}a|$, $|\mathbf j|$ at the bare bonds with an end point in the block of $y \in \Lambda_{K-k-1,\ell}$,

$$\big|\partial_\zeta D_{(a,\mathbf j)}\mathcal E(P_W;B)\big| \le C^\sharp_{\mathrm{mx}}(x_k)\,\mathfrak b(x_k)\,\|\zeta\|\Big(\sum_y \mu_y(a,\mathbf j)^2\Big)^{1/2},$$

with $C^\sharp_{\mathrm{mx}}$ depending on x_k only through $\delta(x_k) \le 1$; and for every real ξ on $\Lambda_{K-k-1,\ell}$, $\sup_B |\partial_\zeta\partial_s\mathcal E(P_{s\xi};B)|_{s=0}| \le C_{\mathrm{mx}}\mathfrak b(x_k)\|\zeta\|\|\xi\|$, $C_{\mathrm{mx}} = \mathsf{BS}\sup_{\mathsf t\ge\mathsf t_\natural} C^\sharp_{\mathrm{mx}}(\mathsf t)$.

■

Proof sketch. (a) Prop. YM3_I:52.3 holds for free radii $b_0, b_1 \in (0,1]$, with the threshold $\varepsilon_{\mathcal W}b_0$ for $\varpi(W)$; the threshold that enters n_3 is the one at the radii $b_0 = \frac12\alpha_0$, $b_1 = \frac12\alpha_1$, that is, $2\theta_d \le \varepsilon_{\mathcal W}\alpha_0$ for $d \ge n_3$. So every $W \in \mathsf{Cut}_{k+1}$ with $K-k-1 \ge n \ge n_3$ has $\varpi(W) < \theta_{K-k-1} \le \frac12\varepsilon_{\mathcal W}\alpha_0$, and by the plain-core verification in its proof P_W lies in the core $\mathcal U^\circ_{k+1}(Y;\frac12\alpha_0,\frac12\alpha_1)$ with half the radii. The outer substitution from half the radii (Prop. YM3_I:51.10 (first assertion) at the radii α_0,α_1) carries this triple, moved by $(A',\mathbf j)$ with $|A'|, |\nabla_{U_+}A'|, |\mathbf j| < r_{\mathrm c}$, into the core with the radii α_0,α_1: the halved radii leave the room $r_{\mathrm c}$.

(b) The density in the chart comes from the sum identities in equations YM3_I:(828)–YM3_I:(830) and their preceding Jacobian calculation and the step (Prop. 6.31); the quadratic form is a sum of local terms, obtained by subtracting polynomials that agree on an open set. The formula for p_W, the locality of its quadratic form and the holomorphy of $B \mapsto \mathcal E(P_W;B)$ on $\{|B(b)| < C_A\mathfrak b(x_k)\}$ need of W only that it lies in the class $\mathcal W_{k+1}(x_k;\alpha_0,\alpha_1)$ of the sum rule (Lem. YM3_I:113.10), where the restrictions of P_W to the polymers lie in the analyticity cores and the sum rules hold, so that every term of $\mathcal E(P_W;\cdot)$ is formed and holomorphic in B there (Thm. YM3_I:W, Prop. YM3_I:51.4 and its proof); so they hold for every integer $n \ge 1$ with $L^n\mathfrak b \ge \mathsf t_\natural$, from n_2 on in particular.

Only the joint holomorphy in $(A',\mathbf j)$ on the ball of (a) uses (a), and with it the condition $2\theta_d \le \varepsilon_{\mathcal W}\alpha_0$ of n_3. (c) Moving W along the response keeps the background in the minimal orbit, so no control of the axial gauge is needed. With the residual gauge h_ξ such that the background over W'_ξ is $\mathbf U_\xi^{h_\xi}$, $\check V_\xi(B)$ is $V_{W'_\xi}(\mathrm{Ad}(h_\xi)B)$ carried back by h_ξ; its values at the pivots follow from the contraction there (the contraction leading to YM3_I:(800), with the inverse of the one-step pivot derivative bounded by $2L^2$ in YM3_I:(798)) and the implicit function theorem. Every term of $\mathcal E$ is invariant under $(P,B) \mapsto (P^u, \mathrm{Ad}(u)B)$ and depends on $P = (U,A',\mathbf J)$ only through $(e^{iL^{-k-1}A'}U,\mathbf J)$ (Thm. YM3_I:W, Def. YM3_I:52.1, Prop. YM3_I:51.4 and its proof, §15.3.1, and the pair and gauge covariance of Prop. 6.31), and $\mathrm{Ad}(h_\xi)$ keeps dB and the box: so the density depends on P_ξ alone, and no derivative of h_ξ enters.

The values lie in a compact subset because the regularity number of the backgrounds stays strictly below the cut. (d) Each local term is bounded on the whole product of the core and the polydisc (Prop. YM3_I:51.4 and its proof, Thm. YM3_I:W), for instance $|v_Y| \le 32C_V\delta(x_k)(d(Y)+1)C_A^2\mathfrak b(x_k)^2e^{-\lambda d(Y)}$ with constants C_V, λ of the step; Cauchy's estimate on a bidisc, the Cauchy–Schwarz inequality and the sum over polymers [7, eq. 320] give the first bound. At $s=0$, the derivative of $P_{s\xi}$ near y is at most $\mathsf B\,m_y(\xi)$ by Lem. 15.9, which also gives $\sum_y m_y(\xi)^2 \le \mathsf S^2\|\xi\|^2$. □

Proposition 15.5 (One-step transport of the Dirichlet energy)**.** For a level j and a $\mathfrak g$-valued function w of the bare bonds let $[w]_j^2 = \sum_y (L^j \max\{|w(b)| : b_- \in B^j(y)\})^2$, $B^j(y)$ the cube of L^{3j} bare sites centred at $y \in \Lambda_{K-j,\ell}$; then $L^{j/2}[w]_j \le L^{(j+1)/2}[w]_{j+1}$, $\|w\|^2 \le 3L^j[w]_j^2$, $[w]_0 \le \|w\|$, and $[\mathrm{Ad}(g)w]_j = [w]_j$. For $V \in \mathsf{Cut}_j$ at a depth $\ge n_{\mathrm a}$, U in the minimal orbit over V and a real 1-form

X on $\Lambda_{K-j,\ell}$, let $N_V^\flat(X) = L^{j/2}\min_\lambda[L^{-j}D\mathcal{H}_U(0)X - \mathrm{d}_U\lambda]_j$ for $j \geq 1$, $N_V^\flat(X) = \min_\lambda[X - \mathrm{d}_V\lambda]_0$ for $j = 0$, and $\Gamma_{K-j}^\flat(g)(V) = L^K \sup_{X:\, N_V^\flat(X)>0}\langle X, \partial g(V)\rangle^2/N_V^\flat(X)^2$.

$N_V^\flat$ is a seminorm that does not depend on the choice of U and vanishes exactly on the gauge directions; $N_V^\flat(X) \leq L^{j/2}\mathsf{M}\|X\|$, so $\Gamma_d \leq \mathsf{M}^2\Gamma_d^\flat$; $\Gamma_K^\flat(H) \leq 3\Gamma_K(H)$ for invariant H; and $\Gamma_d^\flat(g)$ is Borel for invariant $g \in C^1$. There are n_{a} as in the notation and $C_\flat$, depending only on L, C_Q, M and the constants of §5.3, such that, with $C_{\mathrm{cov}} = \frac{4}{\gamma_{\mathrm{r}}}C_{\mathrm{ch}}^{1/2}\mathsf{M}\sqrt{6}C_Q L^{-1/2}C_{\mathrm{mx}}$, for $K > n \geq n_{\mathrm{a}}$, $0 \leq k < K - n$, $W \in \mathsf{Cut}_{k+1}$:

(a) every real ξ' on $\Lambda_{K-k-1,\ell}$ has a representative ξ modulo gauge at W, depending on W, U_+, ξ' and not on B, with $\|\xi\| \leq \sqrt{6}C_Q L^{-(k+1)/2}N_W^\flat(\xi')$ and $N_{V_W(B)}^\flat(\check{\mathsf{d}}_k(W,B)\xi) \leq (1 + C_\flat\delta(x_k))N_W^\flat(\xi')$ for all B in the box, $\check{\mathsf{d}}_k(W,B)\xi = \frac{d}{ds}\check{V}_{s\xi}(B)|_{s=0}$; more precisely, for every bare velocity w of backgrounds at U_+ of level $k+1$ and $\xi = D\mathcal{M}^{k+1}(U_+)w$, $N_{V_W(B)}^\flat(\check{\mathsf{d}}_k(W,B)\xi) \leq (1 + C_\flat\delta(x_k))L^{(k+1)/2}[w]_{k+1}$;

(b) for real invariant $H \in C^1(\Omega_K)$ the tower has $h_k \in C^1(\mathsf{Cut}_k)$, and for $W \in \mathsf{Cut}_{k+1}$ and real ξ, with ψ_ξ of Prop. 15.4(c),

$$\langle \xi, \partial h_{k+1}(W)\rangle = \int \langle \check{\mathsf{d}}_k(W,B)\xi, \partial h_k(V_W(B))\rangle\, p_W(\,\mathrm{d}B) + \mathrm{Cov}_{p_W}(h_k \circ V_W, \psi_\xi),$$

$$\Gamma_{d-1}^\flat(h_{k+1})(W)^{1/2} \leq (1 + C_\flat\delta(x_k))\int \Gamma_d^\flat(h_k)^{1/2} \circ V_W\,\mathrm{d}p_W + C_{\mathrm{cov}}\,\delta(x_k)\Big(\int \Gamma_d^\flat(h_k) \circ V_W\,\mathrm{d}p_W\Big)^{1/2},$$

so $\Gamma_{d-1}^\flat(h_{k+1})(W) \leq (1 + \epsilon_d^\flat)\int \Gamma_d^\flat(h_k) \circ V_W\,\mathrm{d}p_W$ at every W; the same holds for $\phi_k = f_k^{1/2}$, (f_k) the tower of $G^2 + \epsilon$, G real invariant, $\epsilon > 0$;

(c) for $0 \leq k \leq K - n$,

$$\int \Gamma_{K-k}(h_k)\,\mathrm{d}\sigma_{K-k}^K \leq 3\mathsf{M}^2\Pi_n^\flat \int \Gamma_K(H)\,\mathrm{d}\sigma^K, \tag{15.ii}$$

and, pointwise, $\Gamma_e(g_{t'})(W) \leq 3\mathsf{M}^2\Pi_e^\flat \int \Gamma_K(G)\,\mathrm{d}\mathsf{P}_W$ for the law P_W of (B3) and every $W \in \mathsf{Cut}_{t'}$.

■

Proof sketch. (a) Three facts: the monotonicity of $L^{j/2}[w]_j$ in j, with constant one; $\|D\mathcal{M}^j(U)w\| \leq \sqrt{6}C_Q[w]_j$ when the plaquettes of U lie within $a_Q L^{-2j}$ of $\mathbf{1}$ (equation eq. 5.xxx); and

$$[L^{-j}D\mathcal{H}_V(\zeta)X]_j \leq \mathsf{BS}\|X\|$$

(§5.3). The key is that the gauge compensator is bounded pointwise: every bare velocity $w = L^{-j}D\mathcal{H}_U(0)X - \mathrm{d}_U\lambda$ is also $L^{-j}D\mathcal{H}_U(0)X^w + \mathrm{d}_U\nu$, $X^w = D\mathcal{M}^j(U)w$, with

$$|\nu(x)| \leq (k_6 + \tfrac{3}{2})\,\mathsf{B}\,m_y^j(X^w) + \tfrac{3}{2}L^j \max_{b_- \in B^j(y)}|w(b)| \qquad (x \in B^j(y)),$$

at the block centres by equation YM3_I:(996), with $k_6 = 4\kappa_{\mathrm{av}}$ as fixed in §5.3, and inside a block along a path, by $|\lambda(x)| \leq |\lambda(y)| + \frac{3}{2}L^j \max_{B^j(y)}|\mathrm{d}_U\lambda|$. Lift the velocity of level $k+1$ to a bare velocity w; the fluctuation B moves the background of level k by $O(\delta(x_k))$, so there is a curve of backgrounds of level k over $\check{V}_{s\xi}(B)$, up to a gauge, whose bare velocity differs from w by at most $C_\flat\delta(x_k)L^{1/2}[w]_{k+1}$ in $[\cdot]_k$; the monotonicity then costs nothing. The minimum in $N_W^\flat(\xi')$ is attained at some w; $\xi = X^w$ is the representative, and $\|\xi\| \leq \sqrt{6}C_Q[w]_{k+1}$.

The argument of (a) is deterministic. The properties of $N_V^\flat$: a curve of backgrounds whose block field moves with velocity X has a bare velocity w with $L^{j/2}[w]_j \geq N_V^\flat(X)$; $\Gamma_K^\flat \leq 3\Gamma_K$ because $\langle X, \partial H\rangle^2 = \langle X - \mathrm{d}_V\lambda, \partial H\rangle^2 \leq \|X - \mathrm{d}_V\lambda\|^2|\partial H|^2 \leq 3[X - \mathrm{d}_V\lambda]_0^2|\partial H|^2$ for invariant H and every λ; upper semicontinuity of $(V, X) \mapsto N_V^\flat(X)$ gives the Borel property.

(b) Differentiate Prop. 15.4(c) at $\xi = 0$; domination holds on the closed box, as the moving chart takes its values in a compact subset of Cut_k. The left side depends on ξ only modulo gauge: take the representative of (a); the first term is then at most $L^{-K/2}(1 + C_\flat\delta(x_k))N_W^\flat(\xi')\int\Gamma_d^\flat(h_k)^{1/2}\,\mathrm{d}p_W$. For the plain tower, the Poincaré inequality of p_W used twice gives $|\mathrm{Cov}_p(f,\psi)| \le \frac{4}{\gamma_\mathrm{r}}\|\nabla_B f\|_{L^2(p)}\sup|\nabla_B\psi|$. For the square roots, the log-Sobolev inequality of p_W with $\kappa = \frac14\gamma_\mathrm{r}$ gives $\log\int e^{s(\psi-\int\psi)}\,\mathrm{d}p \le s^2l^2/2\kappa$ when $|\nabla_B\psi| \le l$, and the variational formula of the entropy at ϕ^2 and $s(\psi - \int\psi)$, optimised in s, gives

$$|\mathrm{Cov}_p(\phi^2,\psi)| \le \frac{2l}{\kappa}\,\|\phi\|_{L^2(p)}\|\nabla_B\phi\|_{L^2(p)},$$

with equality for a Gaussian p, $\phi = e^{ax/2}$, $\psi = x$. By Prop. 15.4(d), $\sup|\nabla_B\psi_\xi| \le C_\mathrm{mx}\mathfrak{b}(x_k)\|\xi\|$; the chart factor $x_k^{-1/2}$ turns $\mathfrak{b}(x_k)$ into $\delta(x_k)$, and $L^{K/2}L^{-d/2}L^{-(k+1)/2} = L^{-1/2}$. Directions with $N_W^\flat(\xi') = 0$ contribute zero by gauge invariance; for the other directions divide by $L^{-K/2}N_W^\flat(\xi')$ (for the square roots by $2\phi_{k+1}(W) = 2\|\phi_k\circ V_W\|_{L^2(p_W)}$) and use Jensen's inequality for the probability p_W.

(c) Integrate the one-step bound against σ_{d-1}^K (Lem. 15.3(ii)) and iterate, using $\Gamma_d \le \mathsf{M}^2\Gamma_d^\flat$ at the top and $\Gamma_K^\flat \le 3\Gamma_K$ at the bottom; for the pointwise form iterate at the fixed block field along the kernels. The constant does not depend on K because $\sum_d\delta(L^d\mathfrak{b}) < \infty$. □

Proposition 15.6 (Transport in the ℓ^2 metric)**.** Assume (E). For $V \in \mathsf{Cut}_j$ with $\varpi(V)$ below the radius of §5.3, U in the minimal orbit over V and $\mathsf{Z} = L^{-j}D\mathcal{H}_U(0)$, let $N_V(\xi) = \min_\lambda\|\mathsf{Z}\xi - \mathrm{d}_U\lambda\|$ ($N_V(\xi) = \min_\lambda\|\xi - \mathrm{d}_V\lambda\|$ at $j = 0$). N_V is a seminorm that does not depend on the choice of U and vanishes exactly on the gauge directions; $N_V(\xi)^2 \le \|\mathsf{Z}\xi\|^2 \le C_\mathrm{rs}L^j\|\xi\|^2$, with C_rs depending only on L and the constants of §5.3. Every real ξ is $\xi_* + \mathrm{d}_V\phi$ with exactly one horizontal datum ξ_*, that is, with $\mathrm{d}_U^\mathsf{T}\mathsf{Z}\xi_* = 0$; on horizontal data $N_V(\xi_*) = \|\mathsf{Z}\xi_*\|$; every real ξ has $\|\xi\| \le C_\mathcal{Q}L^{-j/2}\|\mathsf{Z}\xi\|$, so $\|\xi_*\| \le C_\mathcal{Q}L^{-j/2}N_V(\xi_*)$, with $C_\mathcal{Q} = 34(L^3+2)$ as in Lem. 5.32 (a number other than the C_Q and C_Q' of Lem. 5.30); and the two metrics are equivalent: $3^{-1/2}N_V \le N_V^\flat \le \mathsf{M}C_\mathcal{Q}N_V$.

There are $C_\sharp$ and a depth $n_4 \ge n_3$, depending only on b and the constants of the construction, such that for $K - k > n_4$, $W \in \mathsf{Cut}_{k+1}$ and U_+ a background over W the following hold. All clauses of Prop. 15.4(c) hold at W, with the frame $\check V_\xi(B)$ real-analytic in (ξ, B) on $\{\sup|\xi| < r_0\}$ times the open box and with first derivatives in ξ that are bounded there. For every horizontal datum ξ at W, that is, every real ξ on $\Lambda_{K-k-1,\ell}$ with $\mathrm{d}_{U_+}^\mathsf{T}\mathsf{Z}_{k+1,U_+}\xi = 0$, $\mathsf{Z}_{k+1,U_+} = L^{-k-1}D\mathcal{H}^+(0)$, and all B in the box, $s \mapsto \check V_{s\xi}(B)$ is defined and real-analytic near 0 and

$$N_{V_W(B)}(\check{\mathsf{d}}_k(W,B)\xi) \le (1 + C_\sharp\,\delta(L^{K-k}\mathsf{b}))N_W(\xi).$$

Consequently Prop. 15.5(b), (c) hold as well in the metric N for $K > n \ge n_4$, (B3) for $K > e \ge n \ge n_4$, and (B2) in (a)–(c) for $n \ge \max\{n_4, n_\mathrm{c}\}$, in the proposed variant (d), still requiring the multiscale gauge estimate, for $n \ge \max\{n_4, n_\mathrm{e}\}$ with $\mathsf{R}_n < 1$, and in (e) for $n \ge n_4$, with $C_\mathrm{rs}\Pi_n$, $\Pi_n = \prod_{d>n}(1+\epsilon_d)$, $1 + \epsilon_d = [1 + (C_\sharp + \frac{4}{\gamma_\mathrm{r}}(C_\mathrm{ch}C_\mathrm{rs})^{1/2}C_\mathcal{Q}L^{-1/2}C_\mathrm{mx})\delta(L^d\mathfrak{b})]^2$, in the place of $3\mathsf{M}^2\Pi_n^\flat$ (in (B3): $C_\mathrm{rs}\Pi_e$ in the place of $3\mathsf{M}^2\Pi_e^\flat$). ■

Proof sketch. The range of Z. The response is in the Landau gauge on the kernel of the site averages: by equation YM3_I:(837), $\mathsf{R}^\mathrm{La}D^\mathsf{T}\mathcal{H} = 0$, $D = \mathrm{d}_U$, R^La the orthogonal projection onto $D^\mathsf{T}D\mathcal{N}$ and $\mathcal{N}$ the site functions whose site averages Q_q' vanish at the tops (§15.3); so $\mathsf{R}^\mathrm{La}\mathrm{d}_U^\mathsf{T}\mathsf{Z}\xi = 0$ for every ξ. The response is covariant, as the averaging map is, so Z carries the gauge directions $\mathrm{d}_V\psi$ of the block field to gauge directions at U; these meet $\mathrm{d}_U\mathcal{N}$ only in 0 (a $\mathrm{d}_U\mu$, $\mu \in \mathcal{N}$, in the range of Z has $\mathsf{R}^\mathrm{La}\mathrm{d}_U^\mathsf{T}\mathrm{d}_U\mu = 0$ while $\mathrm{d}_U^\mathsf{T}\mathrm{d}_U\mu$ lies in the range of R^La, so $\mathrm{d}_U\mu = 0$), and by the count of dimensions the two together span all the gauge directions at U.

A horizontal velocity of backgrounds at U, $h = \mathsf{Z}X - \mathrm{d}_U\lambda$ with $\mathrm{d}_U^{\mathsf{T}}h = 0$, is then $\mathsf{Z}X' - \mathrm{d}_U\nu$ with $\nu \in \mathcal{N}$, and $\mathsf{R}^{\mathrm{La}}\mathrm{d}_U^{\mathsf{T}}\mathrm{d}_U\nu = \mathsf{R}^{\mathrm{La}}\mathrm{d}_U^{\mathsf{T}}\mathsf{Z}X' = 0$: so $\mathrm{d}_U\nu = 0$ and $h = \mathsf{Z}X'$. The range of Z holds every horizontal velocity of backgrounds, with no compensator.

The decomposition. The part h of $\mathsf{Z}\xi$ orthogonal to the range of d_U is a horizontal velocity of backgrounds, $h = \mathsf{Z}\xi_*$; $\mathsf{Z}(\xi - \xi_*)$ is a gauge direction at U, so $\xi - \xi_*$ is a gauge direction at V; ξ_* is unique, as Z is injective (below) and a horizontal vector that is a gauge direction vanishes.

The bound. Lem. 5.32 applies in the constant frame, with bare spacing L^{-j} and terminal spacing one. It gives $\mathcal{Q}(\mathcal{H}_U(\zeta)) = \zeta$, and hence $L^jK_0\mathsf{Z}\xi = \xi$, together with $\|K_0\| \le 2L^{-3j/2} \le C_{\mathcal{Q}}L^{-3j/2}$ on counting-measure ℓ^2. Thus $\|\xi\| \le C_{\mathcal{Q}}L^{-j/2}\|\mathsf{Z}\xi\|$; on horizontal data $\|\mathsf{Z}\xi_*\| = N_V(\xi_*)$. The equivalence follows from $\|w\|^2 \le 3L^j[w]_j^2$ and $N_V^\flat(\xi) = N_V^\flat(\xi_*) \le L^{j/2}\mathsf{M}\|\xi_*\|$.

The estimate. U_+ is a background of the level k as well, and the same argument at the level k shows that the velocity of a horizontal datum ξ at W is a response one level finer, with no compensator: $\mathsf{Z}_{k+1,U_+}\xi = \mathsf{Z}_{k,U_+}\zeta$, $\zeta = D\mathcal{M}^k(U_+)\mathsf{Z}_{k+1,U_+}\xi$; the fluctuation moves the background of level k by $O(\delta(x_k))$, which costs the factor $1 + C_\sharp\delta$.

The frame from the depth n_4 on. As $n_4 \ge n_3$, θ_d lies below the radius of §5.3; U_+ is a reference of the levels $k+1$ and k; for small real ξ, W'_ξ is a gauge transform of $e^{i\xi}W$ in Cut_{k+1}, $\mathbf{U}_\xi$ lies in the minimal orbits over W'_ξ and over $\mathcal{M}^k\mathbf{U}_\xi$, and $\xi \mapsto \mathcal{M}^k\mathbf{U}_\xi$, $\xi \mapsto W'_\xi$ are real-analytic. The value of $\mathcal{M}(V)$ at a bond c of $\Lambda_{K-k-1,\ell}$ reads V on the bonds of the double block of c alone, of the pivot bonds the pivot bond of c alone, and its derivative in the value at that pivot bond is invertible, with an inverse bounded by $2L^2$, on the compact closure of the set of the restrictions of the $V_W(B)$, B in the box.

The implicit function theorem, with radii that are uniform over this compact set, gives about every base point $V_W(B)$ exactly one solution at each pivot near the value of the base point, real-analytic in the data, with bounded first derivatives. The frame $\check{V}_\xi(B)$ is $e^{ix_k^{-1/2}B(b)}(\mathcal{M}^k\mathbf{U}_\xi)(b)$ at the bonds that are not pivot bonds and this solution at the pivots; by the uniqueness at the pivots and continuation in B along τB, $\tau \in [0,1]$, from $B = 0$, it is $V_{W'_\xi}(\mathrm{Ad}(h_\xi)B)$ carried back by h_ξ. Hence, as in Prop. 15.4(c): the tower identity; the density $e^{\mathcal{E}(P_\xi;B)}/\int e^{\mathcal{E}(P_\xi;B')}\,\mathrm{d}B'$ of $\check{p}_\xi$, which reads the depth only through n_3; the values in a compact invariant subset of Cut_k; analyticity on small ξ times the open box, with ξ-derivatives bounded there, which carries the domination in the differentiation of Prop. 15.5(b); and at the depths $\ge n_{\mathrm{a}}$ the frame is that of Prop. 15.4(c).

In the differentiated step the representative is used only through $\|\xi\| \le c\,L^{-(k+1)/2}N_W(\xi)$, which the horizontal datum of the class satisfies with $c = C_{\mathcal{Q}}$; the rest of the proof of Prop. 15.5 is unchanged, with the square field Γ^N built from N as $\Gamma^\flat$ is built from $N^\flat$, $N_V(X) \le C_{\mathrm{rs}}^{1/2}L^{j/2}\|X\|$ at the top and $\Gamma_K^N \le \Gamma_K$ at the bottom. □

Lemma 15.7 (Connected regions at the coarse scale)**.**

(a) Let $\mathsf{n} \ge \lambda_{\mathrm{BG},3}$, $0 < \kappa' \le \kappa_0$, $\mathsf{n} \ge \kappa'^{-1/2}$, and V a configuration on $(\mathbb{Z}/\mathsf{n})^3$ with $\varpi(V) \le \kappa'$. Some gauge has $|V^u(b) - \mathbf{1}| \le C_{\mathrm{BG}}\sqrt{\kappa'}$ at every bond; $\{\varpi < \kappa'\}$ lies in the path component of $\mathbf{1}$ in $\{\varpi < C''\kappa'\}$; so, at $\kappa' = r/C''$, for $0 < r \le C''\kappa_0$ and $\mathsf{n} \ge \max\{\lambda_{\mathrm{BG},3}, (C''/r)^{1/2}\}$, every other component of $\{\varpi < r\}$ lies in the shell $\{r/C'' \le \varpi < r\}$.

The estimate involving J block levels remains to be proved, as described under (Multiscale gauge). The proposed estimate requires finite constants; no numerical values for them are asserted here.

(b) For a finite set E of bonds, $0 < \alpha < \pi$, $\mathsf{K} \subset \{A \in \mathfrak{g}^E : |A(b)| < \alpha\}$ open and convex and $V_0 \in G^E$, $A \mapsto e^{iA}V_0$ maps K diffeomorphically onto its image B; there Haar measure is $\prod_b(2\pi^2)^{-1}\mathsf{j}(A(b))\,\mathrm{d}A(b)$, $\mathsf{j}(X) = (\sin|X|/|X|)^2$; $-\log\prod_b\mathsf{j}$ has Hessian $\ge \frac{2}{3}$; and $\mathrm{Ent}_{dV|\mathsf{B}}(\psi^2) \le 3\int_{\mathsf{B}}|\partial\psi|^2dV$ for bounded locally Lipschitz ψ, whatever the number of bonds.

(c) If $\nu|_{S_i}$ have log-Sobolev constants c_i on open S_1, S_2 with $\nu(S_1 \cap S_2) > 0$, then $\nu|_{S_1\cup S_2}$ has $2\max_i c_i(1 + 16\nu(S_1 \cup S_2)/\nu(S_1 \cap S_2))$. If Haar measure on an open S' has the constant c,

$\rho(V) = \int \mathbf{1}_{S'}(V^g)\,\mathrm{d}g$, $\rho_0 > 0$ and $w = \min\{1, \rho/\rho_0\}$, then $\mathrm{Ent}_{w\mathsf{g}\,\mathrm{d}V}(\phi^2) \le \frac{c\,\mathsf{g}_+}{\rho_0 \mathsf{g}_-} \int |\partial\phi|^2 w\,\mathsf{g}\,\mathrm{d}V$ for invariant ϕ and Borel $0 < \mathsf{g}_- \le \mathsf{g} \le \mathsf{g}_+$, whatever the boundary of S'. For an open invariant $\mathcal{O}$, a component C of it and a compact invariant $\mathcal{K} \subset \mathsf{C}$ there is a connected finite union S' of sets $\{e^{iA}V_i : |A(b)| < \delta_i\}$ with $\mathcal{K} \subset S' \subset \mathsf{C}$.

(d) The set $\mathcal{N} \subset (0,2)$ of the $r = \varpi(V)$ at which $\sum_p c_p \partial a_p(V) = 0$ for some $c \ge 0$, $c \ne 0$, carried by the plaquettes with $\varpi_p(V) = r$, is Lebesgue-null; for $r \notin \mathcal{N}$, $\{\varpi < r\}$ has finitely many components, each a connected open set with a Lipschitz boundary. The abelian fields whose $(1,2)$ plaquettes all equal $e^{if\sigma_3}$, $f = 2\pi m/\mathsf{n}^2$, and whose other plaquettes are $\mathbf{1}$, are such critical points: $2\sin(\pi m/\mathsf{n}^2) \in \mathcal{N}$ for integers $0 < m < \mathsf{n}^2/2$.

(e) For $0 < r < 2$, every connected component of $\{\varpi < r\}$ is open, invariant, and contains a local minimiser of ϖ. For $\phi_1 \in C^2[0,c)$, $c = \frac{1}{2}r^2$, with $\phi_1' > 0$ and $\phi_1 \to \infty$ at c, the function $\Phi = \sum_p \phi_1(a_p)$ has compact sublevel sets in $\{\varpi < r\}$, every component contains a point where Φ restricted to it is least, and there $\sum_p \phi_1'(a_p)\,\partial a_p = 0$ with weights $\phi_1'(a_p) > 0$ at all plaquettes and $\nabla^2\Phi \ge 0$; $\{\varpi < r\}$ is connected if and only if every local minimiser of Φ lies in the component of $\mathbf{1}$; and $\{\varpi < r\}$ is the increasing union of the sets $\{\sum_p a_p^q < c^q\}$, $q \to \infty$.

(f) Let $0 < \alpha \le \frac{1}{32}$, $r' = 64\alpha^2$, N the number of sites along an axis, and $\mathscr{D} = \{A : |A(b)| < \alpha\ \forall b,\ |Y_p(A)| < r'\ \forall p\}$, $e^{iY_p(A)} = \partial(e^{iA})(p)$, $|Y_p| \le \pi$. Then $\mathscr{D}$ is star-shaped, $\mathscr{D} = \{s\omega : |\omega| = 1,\ 0 \le s < \varrho(\omega)\}$ with $\frac{1}{2}r' \le \varrho \le \rho_{\max} = \alpha\sqrt{3N^3}$ and $|\varrho(\omega) - \varrho(\omega')| \le \frac{4\pi\rho_{\max}^2}{r'}|\omega - \omega'|$; $A \mapsto e^{iA}$ maps $\mathscr{D}$ diffeomorphically onto $\exp\mathscr{D} = \{V : \mathsf{d}(V(b), \mathbf{1}) < \alpha,\ \mathsf{d}(\partial V(p), \mathbf{1}) < r'\}$, and $\mathrm{Ent}_{\mathbf{1}_{\exp\mathscr{D}}dV}(\phi^2) \le C_{\mathscr{D}} \int_{\exp\mathscr{D}} |\partial\phi|^2 dV$ for bounded locally Lipschitz ϕ, with $C_{\mathscr{D}}$ of (B1)(e) at $\alpha_n = \alpha$.

With $w(V) = \int \mathbf{1}\{\mathsf{d}(V^u(b), \mathbf{1}) < \alpha\ \forall b\}\,\mathrm{d}u$ and $\mathsf{In} = \{V : \mathsf{d}(\partial V(p), \mathbf{1}) < r'\ \forall p\}$: $\int_{\mathsf{In}} f\,w\,\mathrm{d}V = \int_{\exp\mathscr{D}} f\,\mathrm{d}V$ for invariant Borel $f \ge 0$; $w(V) \ge [(\frac{1}{2}\alpha - \sin\frac{1}{2}\alpha)/2\pi]^{N^3}$ if some gauge brings all bonds of V within $\frac{1}{4}\alpha$ of $\mathbf{1}$; and $\mathsf{In} \cap \{w > 0\}$ is path connected.

■

Proof sketch. (a) The gauge is built cell by cell on a periodic grid of mesh $\kappa'^{-1/2}$, along the proof of Prop. 15.8; along the ray to 0 in exponential coordinates, $||\partial e^{itA}(p) - \mathbf{1}| - |t\,\mathrm{d}A(p)|| \le 10\alpha^2$, with $dA(p) = A(b_1) + A(b_2) - A(b_3) - A(b_4)$ the plain sum of the bond function A around p, its bonds $b_1, \dots, b_4$ taken with the signs $(+,+,-,-)$ of Lem. 16.2, when $|A| \le \alpha \le \frac{1}{8}$, which keeps the ray in $\{\varpi < C''\kappa'\}$. The multilevel estimate (Multiscale gauge) is a separate task; this construction concerns one scale.

For U(1) the flux sectors are distinct components. (b) The radial function $h(\rho) = -2\log(\sin\rho/\rho)$ on $\mathbb{R}^3$ has

$$h''(\rho) = 2\sum_{m\ge1}\left[(m\pi-\rho)^{-2}+(m\pi+\rho)^{-2}\right] \ge 4\sum_{m\ge1}(m\pi)^{-2} = \tfrac{2}{3}, \qquad \frac{h'(\rho)}{\rho} = 4\sum_{m\ge1}(m^2\pi^2-\rho^2)^{-1} \ge \tfrac{2}{3};$$

$|\nabla(\psi\circ\exp)| \le |\partial\psi|$ since $\frac{d}{ds}e^{i(X+sY)} = iZe^{iX}$ with $Z = \int_0^1 \mathrm{Ad}(e^{iuX})Y\,\mathrm{d}u$, $|Z| \le |Y|$; then the log-Sobolev inequality on a convex set, and locally Lipschitz ψ by the dilations $A_0 + \vartheta(A - A_0)$, $\vartheta \to 1$. (c) The first bound by comparing means on the overlap $S_1 \cap S_2$; the second by applying the inequality on S' in each orbit through the orbit average ρ, capped at ρ_0, and the bounded change of density g; the cover of $\mathcal{K}$ by chart balls inside C is joined along paths in C. (d) Sard's theorem on the strata indexed by the set of the maximal plaquettes; off $\mathcal{N}$ Gordan's alternative gives a common descent direction of all maximal plaquettes, which makes the boundary locally a Lipschitz graph.

At the abelian flux fields $\sum_p \partial a_p$ over the $(1,2)$ plaquettes vanishes. (e) Compactness of sublevel sets in the open set; a component is closed in $\{\varpi < r\}$, so a minimising sequence in it has its limit points in it. (f) Along rays the plaquette angle grows: $\frac{d}{dt}|Y(t)| \ge |Y| - 32\alpha^2$, which gives the star shape and the Lipschitz radial function; the log-Sobolev inequality on a star-shaped domain with

Lipschitz radial function follows by mapping it to a ball, at the cost of the Jacobian and of the Lipschitz constant, which is $C_{\mathscr{D}}$. The identity is Fubini over the gauge group; the lower bound counts, at each site, the gauge transformations within $\frac14\alpha$ of the given one, a ball of normalised volume $(\frac12\alpha-\sin\frac12\alpha)/2\pi$. □

15.1.7. *Completion of the limit and flow arguments. Step 1: compactness, motion and the Markov property, (A1)–(A3).* Let $K>n\geq n''$ and let $H\in C^1(\Omega_K)$ be real and invariant, with tower (h_k). Under σ^K, $h_k(V_k)$ is a martingale from the top to the bare level (Lem. 15.3(ii)), so $\int|H-h_t(V_t)|^2\,\mathrm{d}\sigma^K=\sum_{k<t}\int\mathrm{Var}_{p_W}(h_k\circ V_W)\,\sigma^K_{d-1}(\,\mathrm{d}W)$; Lem. 15.3(iv) at $g=h_k$, eq. 15.ii, $\sigma^K\leq\mu^K$ on invariant functions, $\sum_{d>n}a_d^2=\frac{a_n^2}{L^2-1}$, and Lem. 15.3(i) for the passage to μ^K give

$$\|H-(\mathbf{1}_{\mathsf{l}_n}h_t)\circ\pi_{K,n}\|^2_{L^2(\mu^K)}\leq\frac{C^\sigma C_{\mathrm{P}}}{L^2-1}\frac{a_n^2}{\mathsf{b}}\,\mathcal{E}_K(H)+4\mathsf{R}_n\sup|H|^2,\qquad\int\Gamma_n(h_t)\,\mathrm{d}\sigma_n^K\leq C^\sigma\mathcal{E}_K(H).$$

(A2): after replacing H_j by smooth invariant functions with the same bounds (closure, a smooth contraction, the gauge average), let $c=\sup_j\sup|H_j|$, $e=\sup_j\mathcal{E}_{K_j}(H_j)$, and for $n\geq n''$ and $K_j>n$ let h_n^j be the top of the tower of H_j at $t=K_j-n$ and $g_n^j=\mathbf{1}_{\mathsf{l}_n}h_n^j$. The tops are bounded by c and, by the display and Lem. 15.3(iii), have Dirichlet integrals on l_n for Haar measure bounded uniformly in j. Rellich's theorem on compact subsets of the open set l_n, which needs no regularity of its boundary, the bound c, and diagonal sequences over an exhaustion of l_n by compact sets and over n, give one subsequence along which $h_n^j\to\phi_n$ in $L^2(\mathsf{l}_n,dV)$ for every $n\geq n''$.

The limit ϕ_n is a Haar class; as a function under μ_n we take a representative: let Ψ_n be $\mathbf{1}_{\mathsf{l}_n}$ times the gauge average of a Borel function in the class of ϕ_n with values in $[-c,c]$; it is Borel, invariant at every point, $|\Psi_n|\leq c$, and $\Psi_n=\phi_n$ Haar-almost everywhere on l_n, as ϕ_n is invariant almost everywhere. Prop. 15.2(c) for $\mu_n^{K_j}$ at $f=|\Psi_n-g_n^j|^2\leq4c^2$, which vanishes off l_n, gives $\int f\,\mathrm{d}\mu_n^{K_j}\leq e^{\mathsf{o}_n}|\mathsf{l}_n|^{-1}\int_{\mathsf{l}_n}|\phi_n-h_n^j|^2\,\mathrm{d}V+4c^2\mathsf{R}_n$, so with the display

$$\limsup_j\|H_j-\Psi_n\circ\pi_{K_j,n}\|_{L^2(\mu^{K_j})}\;\leq\;2c\,\mathsf{R}_n^{1/2}+\Big(\frac{C^\sigma C_{\mathrm{P}}}{L^2-1}\frac{a_n^2}{\mathsf{b}}\,e+4\mathsf{R}_nc^2\Big)^{1/2}=:\epsilon_n,\qquad\epsilon_n\to0.$$

Total variation enters only here: $\|\Psi_n(Y_n)-\Psi_{n'}(Y_{n'})\|_{L^2(\mu)}=\lim_j\|\Psi_n\circ\pi_{K_j,n}-\Psi_{n'}\circ\pi_{K_j,n'}\|_{L^2(\mu^{K_j})}\leq\epsilon_n+\epsilon_{n'}$, so the invariant cylinder functions $\Psi_n(Y_n)$ converge in $L^2(\mu)$ to some G, $|G|\leq c$, and $\lim_n\limsup_j\|H_j-\Psi_n\circ\pi_{K_j,n}\|_{L^2(\mu^{K_j})}=0$: $H_j\to G$ strongly. This holds along every sequence of cutoffs. (A1): let $H_j\to\Phi(Y_n)$ be a recovery sequence of Lem. 15.1(b), smooth, invariant, $\sup|H_j|\leq\sup|\Phi|$ (truncation and the gauge average do not raise the energy). The conditional mean under σ^K given V_t is an $L^2(\sigma^K)$-contraction that fixes $\Phi(V_t)$, so the tops converge to Φ in $L^2(\sigma_n^K)$, hence in $L^2(\mathsf{l}_n,dV)$ by the lower bound of the density; the bound on $\int\Gamma_n(h_t)\,\mathrm{d}\sigma_n^K$, $\inf\mathsf{g}\geq(1-\mathsf{R}_n)e^{-\mathsf{o}_n}/|\mathsf{l}_n|\geq e^{-\mathsf{o}_n}/2|\mathsf{l}_n|$ and the lower semicontinuity of the Dirichlet integral give eq. 15.i.

As $\Gamma_n(a_p)=4a_n^{-1}|\mathsf{s}_p|^2$, by the computation of Lem. 16.2(a) on Ω_n, is positive at the fields of l_n with $\varpi_p>0$, $\mathcal{E}_\star(a_p(Y_n))>0$; since $\nu_\Psi(\{\infty\})=0$ for invariant Ψ, $(\Psi,(1-Q_t)\Psi)=\int(1-e^{-t\lambda})d\nu_\Psi>0$. (A3): for an invariant cylinder function Ψ and $t>0$, $H_j=P_t^{K_j}\Psi_{K_j}$ is invariant, $|H_j|\leq\sup|\Psi|$, $\mathcal{E}_{K_j}(H_j)\leq\sup|\Psi|^2/2et$ (Lem. 15.1(a)), and $H_j\to Q_t\Psi$ weakly; by (A2) every subsequence has a strongly convergent subsequence, whose limit is the weak limit; so $H_j\to Q_t\Psi$ strongly, which is one of the equivalent properties of Lem. 15.1(b), and with it all of them, (M1), and the convergence of $P_t^{K_j}H_j$ for all $H_j\to F$.

With (M2) of Lem. 15.1(b) the forms converge in the strong and weak sense, which gives the strong convergence of the resolvents by the theorem on the convergence of resolvents under convergence of forms in the strong and weak sense; Lem. 15.1(c) gives a dense domain and strong continuity. Compactness: fix a countable dense set (Φ_i) of invariant cylinder functions; for G_m in the set choose a cylinder function Ψ_m with $\|\Psi_m-G_m\|\leq\frac1m$ and, by (M2), truncation at ±1 and the gauge average,

smooth invariant H_m on $\Omega_{K_{j(m)}}$ with $|H_m|\le 1$, $\mathcal{E}_{K_{j(m)}}(H_m)\le 1+\frac{1}{m}$ and $\|H_m-\Psi_{m,K_{j(m)}}\|\le\frac{1}{m}$, where $j(m)$ increases so fast that $\|\Psi_{m,K}-\Phi_{i,K}\|$ is within $\frac{1}{m}$ of $\|\Psi_m-\Phi_i\|$ for $i\le m$ and $K\ge K_{j(m)}$; by (A2) a subsequence of (H_m) converges strongly to some G, then $G_m\to G$ in $L^2(\mu)$ along it, and $|G|\le 1$, $\mathcal{E}_\star(G)\le 1$ by (M1).

The criterion without a cutoff: $W_j=(1-\mathcal{L}_{K_j})H_j$ is bounded in L^2, so a subsequence converges weakly to some W (a diagonal sequence over a countable dense set of cylinder functions); for invariant cylinder Ψ, $\int H_j\Psi_{K_j}\,\mathrm{d}\mu^{K_j}=\int W_j(1-\mathcal{L}_{K_j})^{-1}\Psi_{K_j}\,\mathrm{d}\mu^{K_j}\to(W,(1+A_\star)^{-1}\Psi)$, the resolvents being symmetric and strongly convergent, so $F=(1+A_\star)^{-1}W\in D(A_\star)$; the weak limit $F-W=-A_\star F$ of $\mathcal{L}_{K_j}H_j$ does not depend on the subsequence, and $\mathcal{E}_{K_j}(H_j)=\langle H_j,W_j\rangle-\|H_j\|^2\to(F,W)-\|F\|^2=\mathcal{E}_\star(F)$. The bilinear forms: with $T_j^\alpha=\alpha(\alpha-\mathcal{L}_{K_j})^{-1}$ and $T^\alpha=\alpha(\alpha+A_\star)^{-1}$, $\mathcal{E}_{K_j}(T_j^\alpha u_j,v_j)=\alpha\langle u_j-T_j^\alpha u_j,v_j\rangle\to\alpha(u-T^\alpha u,v)=\mathcal{E}_\star(T^\alpha u,v)$, as $u_j-T_j^\alpha u_j\to u-T^\alpha u$ strongly; by the spectral theorem $\mathcal{E}_{K_j}(u_j-T_j^\alpha u_j)\le\mathcal{E}_{K_j}(u_j)-\alpha\langle u_j,u_j-T_j^\alpha u_j\rangle$, whose limit $\mathcal{E}_\star(u)-\int\frac{\alpha\lambda}{\alpha+\lambda}d\nu_u$ tends to 0 as $\alpha\to\infty$; with $|\mathcal{E}_{K_j}(u_j-T_j^\alpha u_j,v_j)|\le\mathcal{E}_{K_j}(u_j-T_j^\alpha u_j)^{1/2}\mathcal{E}_{K_j}(v_j)^{1/2}$, $\mathcal{E}_\star(v)\le\liminf_j\mathcal{E}_{K_j}(v_j)$ by (M1) and $\mathcal{E}_\star(T^\alpha u,v)\to\mathcal{E}_\star(u,v)$ this gives the convergence.

Step 2: the connected coarse-scale law, (B1). The regularity number of a background U of level k is $L^{2k}\max_p|\partial U(p)-\mathbf{1}|$, the maximum over the bare plaquettes ([7, Def. B.2]), and Cut_k is the set of the block fields over which it is $<c_\rho\varepsilon(x_k)$ (the definitions preceding Lem. YM3_I:113.10). By the curvature comparison in the proof of Lem. YM3_I:113.10, these numbers contract from level to level, by the factor L^{-2} up to $C_H\delta(x_k)$: for $W\in\mathsf{Cut}_{k+1}$ and B in the box, the regularity number of every background over $V_W(B)$ is at most L^{-2} times that of a background over W plus $C_H\delta(x_k)$; starting the chain j_1 levels higher, at every U in the support of its small-field weight,

$$\varpi(V_t)\le 2\vartheta\varepsilon_n,\qquad \vartheta=\vartheta_\infty+L^{-3j_1/2}c_\rho<1.$$

The set $\mathcal{K}_0=\{\varpi\le 2\vartheta\varepsilon_n\}$ is compact and lies in I_n; as D_n and $\mathsf{I}_n\setminus\mathsf{D}_n$ are open, it is the disjoint union of the compact invariant sets $\mathcal{K}_1=\mathcal{K}_0\cap\mathsf{D}_n$ and $\mathcal{K}_2=\mathcal{K}_0\setminus\mathsf{D}_n$, which are free of K. The convexity of one box joins $V_W(B)$ to $V_W(0)$ at a fixed W only; the j_1 levels are passed with the tower of a continuous function. Let $\mu_{t+j_1}=c_\rho\varepsilon(x_{t+j_1})$ and $\mu_k=L^{-2}\mu_{k+1}+C_H\delta(x_k)$ for $t\le k<t+j_1$, so that $\mu_t\le\vartheta\varepsilon_n$, and let D'_k be the set of the $W\in\mathsf{Cut}_k$ over which the regularity number is $<\mu_k$: $\mathsf{D}'_{t+j_1}=\mathsf{Cut}_{t+j_1}$, $V_W(B)\in\mathsf{D}'_k$ for $W\in\mathsf{D}'_{k+1}$ and every B in the box (Lem. YM3_I:113.10, the curvature comparison in the proof of Lem. YM3_I:113.10), and $\mathsf{D}'_t\subset\mathcal{K}_0$ (the small-curvature averaging bound of Prop. YM3_I:14.1).

Let h be continuous on Ω_n with values in $[0,1]$, $h=1$ on $\mathcal{K}_1$ and $h=0$ on $\mathcal{K}_2$ (Urysohn), and let $h_t=h$, $h_{k+1}(W)=\int h_k(V_W(B))\,p_W(\,\mathrm{d}B)$, be its tower upward from the level t in the chain from the depth $n-j_1$. The kernels carry bounded continuous functions on Cut_k to continuous functions on Cut_{k+1} (as in the proof of Lem. 15.3(ii), where the integrands F_k are shown continuous on the cuts): the kernel at W is the normalised chart integral [7, eq. 191] of the integrand F_k of the chain at the level k; on Cut_{k+1}, which lies in the set O^{im} of the chart (Lem. YM3_I:113.10), the chart, its gauge-fixing form and the exponent of its Jacobian are continuous in (W,B), and the chart integral of a continuous function is continuous ([7, Lem. B.3]); F_k is continuous on Cut_k, into which the chart carries the box (Lem. YM3_I:113.10), and dominated convergence on the box applies.

By induction on k, h_k is continuous on Cut_k and takes on D'_k the values 0 and 1 alone: for $W\in\mathsf{D}'_{k+1}$ the function $B\mapsto h_k(V_W(B))$ is continuous on the convex box with values in $\{0,1\}$, hence constant, and as p_W is a probability, $h_{k+1}(W)=h_k(V_W(B))$ for every B in the box. Take all the fluctuations 0: for $W\in\mathsf{Cut}_{t+j_1}$ let $W^{(t+j_1)}=W$ and $W^{(k)}=V_{W^{(k+1)}}(0)=\mathcal{M}^kU_+^{(k)}$, $U_+^{(k)}$ a background over $W^{(k+1)}$. The backgrounds are nested: $U_+^{(k)}$ is a background over $W^{(k)}$ as well ([7, Lem. B.4](2)), so the regularity number over $W^{(k)}$ is exactly L^{-2} times that over $W^{(k+1)}$ (the identity $\mathsf{m}_k=L^{-2}\mathsf{m}_{k+1}$), with no $C_H\delta(x_k)$; hence $\varpi(W^{(t)})<2L^{-2j_1}c_\rho\varepsilon(x_{t+j_1})\le 2L^{-3j_1/2}c_\rho\varepsilon_n\le 2\varepsilon_n/C''$, $W^{(t)}\in\mathsf{D}_n$ by Lem. 15.7(a), so $W^{(t)}\in\mathcal{K}_1$ and $h_{t+j_1}(W)=h(W^{(t)})=1$.

On the support of the small-field weight of the chain from the depth $n-j_1$, $V_{t+j_1}\in \mathsf{Cut}_{t+j_1}$ and $V_k=V_{V_{k+1}}(B_k)\in \mathsf{D}'_k$, so $h(V_t)=h_{t+1}(V_{t+1})=\cdots=h_{t+j_1}(V_{t+j_1})=1$; as $V_t\in\mathcal{K}_0$ and $h=0$ on $\mathcal{K}_2$, $V_t\in\mathcal{K}_1$. Of the dependence on W this argument uses the continuity of the kernels alone. So $\sigma_n^K(\Omega_n\setminus\mathcal{K}_1)\le \mathsf{R}_{n-j_1}$ by Lem. 15.3(i) for the chain from depth $n-j_1$. Glue finitely many chart balls in D_n to a connected $S'\supset\mathcal{K}_1$ (Lem. 15.7(b), (c)), put $w_n=\min\{1,\rho/\rho_0\}$, $\rho_0=\min_{\mathcal{K}_1}\rho>0$, so that $w_n=1$ on $\mathcal{K}_1$ and $w_n=0$ off D_n: this is (a).

Lem. 15.7(c) with the density of σ_n^K, $\sup \mathsf{g}\le e^{\mathsf{o}_n}\inf\mathsf{g}$ (Lem. 15.3(iii)), and $\Gamma_n=a_n^{-1}|\partial\cdot|^2$ give (b) with $C_n^{\mathrm{t}}=a_nc(S')e^{\mathsf{o}_n}/\rho_0$. (c): the linearisation of (b) gives $\int(\phi-c)^2w_n\,\mathrm{d}\sigma_n^K\le\frac12 C_n^{\mathrm{t}}\int\Gamma_n(\phi)w_n\,\mathrm{d}\sigma_n^K$, and the part where $w_n<1$ costs at most $\mathrm{osc}(\phi)^2\mathsf{R}_{n-j_1}$. (d): $\mathrm{Ent}_{\sigma_n^K}(\phi^2)\le\int\Xi(\phi^2,c')\,\mathrm{d}\sigma_n^K=\mathrm{Ent}_{w_n\sigma_n^K}(\phi^2)+\int\Xi(\phi^2,c')(1-w_n)\,\mathrm{d}\sigma_n^K$. (e): the explicit-top sketch still requires the multiscale gauge estimate. The conditions of n_{e} are those of (Multiscale gauge) at $J=J_n$; a field of I_n whose block fields $\mathcal{M}^jV$, $j\le J_n$, lie in the cuts has a gauge with all bonds within $44\pi L^2\sum_{j<J_n}\theta_{n-j}+C_{\mathrm{S}}L^{-J_n}\le\frac14\alpha_n$ of $\mathbf{1}$, so $w_n^{\mathrm{e}}\ge c_n$ there by Lem. 15.7(f), $\chi_n=1$, and $\sigma_n^K(\chi_n<1)\le\mathsf{R}_{n-J_n}$ by Lem. 15.3(i) for the chain from depth $n-J_n$.

Since $r'_n=2\arcsin\varepsilon_n=64\alpha_n^2$ is the angle of the radius $2\varepsilon_n$, $\mathsf{I}_n=\{\mathsf{d}(\partial V(p),\mathbf{1})<r'_n\ \forall p\}$, and the identity of Lem. 15.7(f) carries the log-Sobolev inequality of Haar measure on $\exp\mathscr{D}$ to invariant functions under $w_n^{\mathrm{e}}dV$, then to χ_ndV at the cost c_n^{-1}, and to $\chi_n\sigma_n^K$ at the cost $e^{2\mathsf{o}_n}/(1-\mathsf{R}_n)$ of the density: this is $C_n^{\flat}$. The set $\{\chi_n>0\}\cap\mathsf{I}_n$ is the path connected set of Lem. 15.7(f); it contains $\mathbf{1}$, so it lies in D_n. For the block-field laws, $\mathrm{Var}_{\mu_n^K}(\Phi)\le\int(\Phi-c)^2\,\mathrm{d}\mu_n^K\le\int(\Phi-c)^2\,\mathrm{d}\sigma_n^K+\mathsf{R}_n\mathrm{osc}(\Phi)^2$ by Lem. 15.3(i), then (c) with χ_n and $\int\Gamma_n(\Phi)\,\mathrm{d}\sigma_n^K\le\int_{\mathsf{I}_n}\Gamma_n(\Phi)\,\mathrm{d}\mu_n^K$; for μ_n by total variation.

Step 3: weak Poincaré bounds and mixing, (A4). Let c be the mean of H under ς^K. The density $w_n(V_t)$ is a function of the first member of the chain, so it changes no kernel and no tower, and

$$\int(H-c)^2\,\mathrm{d}\varsigma^K=\int(H-h_t(V_t))^2\,\mathrm{d}\varsigma^K+\int(h_t-c)^2w_n\,\mathrm{d}\sigma_n^K\le\mathsf{C}_n\mathcal{E}_K(H),$$

the first term by the martingale sum of Step 1, the second by (B1)(b) linearised and eq. 15.ii at $k=t$; passing to μ^K costs $\mathsf{r}_n\mathrm{osc}(H)^2$ by (B2)(a). This is (a), first for smooth H and then by closure. (b): $\frac{d}{dt}\mathrm{Var}(P_t^KH)=-2\mathcal{E}_K(P_t^KH)\le-\frac{2}{\mathsf{C}_n}[\mathrm{Var}(P_t^KH)-\mathsf{r}_n\mathrm{osc}(H)^2]$, as $\mathrm{osc}(P_t^KH)\le\mathrm{osc}(H)$, and $\mathrm{Var}(H)\le\frac14\mathrm{osc}(H)^2$. (c): truncated and gauge-averaged recovery sequences and (a); if $\mathcal{E}_\star(F)=0$, then $\mathrm{Var}_\mu(F)\le\mathsf{r}_n\mathrm{osc}(F)^2$ for all admitted n. For non-constant bounded $F\in L^2_{\mathrm{inv}}$ the spectral measure ν_F is not carried by $\{0\}$, as else $\mathcal{E}_\star(F)=0$, and $\nu_F(\{\infty\})=0$ (Lem. 15.1(c)); so $(F,(1-Q_t)F)=\int(1-e^{-t\lambda})\,\mathrm{d}\nu_F>0$.

(d): $\varphi(t)=\|Q_{t/2}(F-\int F)\|^2$ has $\varphi'(t)=-\mathcal{E}_\star(Q_{t/2}(F-\int F))\le-\mathsf{C}_n^{-1}[\varphi(t)-\mathsf{r}_n\mathrm{osc}(F)^2]$ by (c) and spectral calculus, and $\varphi(0)\le\frac14\mathrm{osc}(F)^2$. If $Q_t\rho=\rho$, then $(\rho,(1-Q_t)\rho)=0$, ν_ρ sits at 0, $\mathcal{E}_\star(\rho)=0$, so $\mathcal{E}_\star(\rho\wedge M)=0$ for every M, and $\rho\wedge M$ is constant by (c): $\rho=1$. (e): apply (a) to H_K truncated at $\pm M_F$, which does not raise the energy and keeps $\|H_K-F_K\|\to0$, so that Var_{μ^K} of the truncation tends to $\mathrm{Var}_\mu F$; the second form is (a) itself.

(f): the same lines with the explicit top of (B1)(e).

Step 4: entropy along the chain and the conditional law, (B2), (B3). (a): Lem. 15.3(i) and $\int f(1-w_n(V_t))\,\mathrm{d}\sigma^K\le\sup f\,\sigma_n^K(w_n<1)$. (b): with f_k the tower of $G^2+\epsilon$ and $\phi_k=f_k^{1/2}$, since the p_W are probabilities the entropy splits along the chain,

$$\mathrm{Ent}_{\varsigma^K}(G^2+\epsilon)=\sum_{k<t}\int\mathrm{Ent}_{p_W}(f_k\circ V_W)\,\varsigma_{d-1}^K(\,\mathrm{d}W)+\mathrm{Ent}_{w_n\sigma_n^K}(f_t);$$

the fibres by Lem. 15.3(iv), $\mathrm{Ent}_{p_W}(\phi_k^2\circ V_W)\le2C_{\mathrm{P}}\frac{a_d^2}{\mathsf{b}}\int\Gamma_d(\phi_k)\circ V_W\,\mathrm{d}p_W$; the top by (B1)(b); and $\int\Gamma_d(\phi_k)\,\mathrm{d}\varsigma_d^K\le3\mathsf{M}^2\Pi_n^{\flat}\int\Gamma_K(G)\,\mathrm{d}\varsigma^K$ (Prop. 15.5(b), (c), as $\Gamma_K((G^2+\epsilon)^{1/2})\le\Gamma_K(G)$); sum over $d>n$ and let $\epsilon\to0$. (c): $\mathrm{Ent}_\nu(G^2)\le\int\Xi(G^2,c')\,\mathrm{d}\nu$ and (a). (d): for the proposed multiscale variant, the function $\mathsf{b}\mapsto2\varepsilon(L^n\mathsf{b})$ is smooth with a non-vanishing derivative where $\log L^n\mathsf{b}>2\mathfrak{p}_0$, so $2\varepsilon_n\notin\mathcal{N}$

for almost every b, for all m and n at once; by Lem. 15.7(d) D_n is then a connected Lipschitz domain, which carries a log-Sobolev inequality for the volume (the extension theorem, with the inequalities of Sobolev and Rellich), and the density bounds give $C_n^\sharp$; $\sigma_n^K(\mathsf{I}_n \setminus \mathsf{D}_n) \le \sigma_n^K(\chi_n < 1) \le \mathsf{R}_{n-J_n}$ by Step 2(e).

For every b, a radius $\tilde r \notin \mathcal{N}$ close to $2\varepsilon_n$ does the same at the cost η: the fields with $\chi_n = 1$ and $\varpi < \tilde r$ lie in the component of $\mathbf{1}$ in $\{\varpi < \tilde r\}$, since along the rays of Lem. 15.7(f) the plaquette angles stay below the larger of $32\alpha_n^2$ and their final values, and the shells $\{\tilde r \le \varpi < 2\varepsilon_n\}$ have σ_n^K-masses at most e^{o_n} times their relative Haar measure, uniformly in K. With χ_n the same lines as in (b). (e) is the splitting of (b) with the top law of σ^K itself. (B3): the splitting of (b), stopped at depth e and taken at a fixed block field W, needs no top: with $\Pi_{(e)} = \prod_{e<d\le K}(1+\epsilon_d^\flat) \le \Pi_e^\flat$,

$$\mathrm{Ent}_{\mathsf{P}_W}(G^2) \le \sum_{e<d\le K} 2C_{\mathrm{P}}\frac{a_d^2}{\mathsf{b}}\,3\mathsf{M}^2\Pi_{(e)}\int \Gamma_K(G)\,\mathrm{d}\mathsf{P}_W \le C_e^{\mathrm{fib}}\int \Gamma_K(G)\,\mathrm{d}\mathsf{P}_W,$$

and the one-step bound of Prop. 15.5(b), iterated pointwise, gives the bound on $\Gamma_e(g_{t'})(W)$; no factor depends on ℓ. P_W is a regular conditional law by Lem. 15.3(ii), also under ς^K, whose weight is a function of the top. Concentration: for real invariant ψ the log-Sobolev inequality, applied to $e^{s\psi/2}$, gives $\int e^{s(\psi-\psi_{t'}(W))}d\mathsf{P}_W \le e^{s^2C_e^{\mathrm{fib}}l^2/4}$, as $\int \psi\,\mathrm{d}\mathsf{P}_W = \psi_{t'}(W)$, then Chebyshev's inequality; for μ^K, the weight $\mathfrak{w}_t^{\mathrm{sf}}$ of σ^K vanishes unless $\mathcal{M}^{t'}U \in \mathsf{Cut}_{t'}$, and Lem. 15.3(i) bounds what μ^K adds to σ^K by R_n; both tails.

Step 5: entropy relaxation, (A5). (a) is (B2)(c) at $\nu = \mu^K$, with $\Xi(x,c) \le \bar M[1+\log(\bar M/c)]$ for $0 \le x \le \bar M$, $0 < c \le \bar M$, and $c' \ge m_2 - \mathsf{r}_n M_G^2$, as $\int G^2\,\mathrm{d}\varsigma^K \ge \int G^2\,\mathrm{d}\mu^K - \mathsf{r}_n M_G^2$ and $\varsigma^K(\Omega_K) \le 1$. (b): (B2)(b) for the normalised measure $\varsigma^K/\varsigma^K(\Omega_K)$ gives Gaussian concentration about $\bar\psi$, and (B2)(a) adds r_n. (c): $P_\tau^K f_0$ is smooth and strictly positive for $\tau > 0$, the kernel of the elliptic diffusion being positive, and

$$\tfrac{d}{d\tau}\mathrm{Ent}_{\mu^K}(P_\tau^K f_0) = -\mathcal{E}_K(P_\tau^K f_0, \log P_\tau^K f_0) \le -4\mathcal{E}_K((P_\tau^K f_0)^{1/2}) \le -\tfrac{4}{\mathsf{C}_n'}\big[\mathrm{Ent}_{\mu^K}(P_\tau^K f_0) - \mathsf{r}_n\bar M(1+\log\tfrac{2\bar M}{m_1})\big]$$

by (a) at $G = (P_\tau^K f_0)^{1/2}$, with $M_G^2 \le \bar M$, $m_2 = m_1$ and $m_1 - \mathsf{r}_n\bar M \ge \frac12 m_1$; Gronwall. For limits: (a) by recovery sequences, as integrals of continuous functions of bounded strongly convergent sequences converge; the decay first for invariant cylinder f_0 with values in $[0,\bar M]$, by (A3), and then by approximation in L^2_{inv} under the strict condition $\mathsf{r}_n\bar M < \frac12 m_1$; $\tau\to\infty$ and then $n\to\infty$ give $\mathrm{Ent}_\mu(Q_\tau F) \to 0$, and Pinsker's inequality gives the convergence in total variation.

The moments of the gauge-fixed field. Steps 6–10 use Prop. 16.1(a) through the following facts. Use the consequence (Adm) of (E) proved in Prop. 16.1, and let d_1 be a depth, free of K, such that every top depth $\ge d_1$ is admissible (the depth d_{adm} of the proof of Prop. 16.1(a)). With $\mathbb{E}^{[d]}$ the normalised chain measure from the top depth d and $\mathbf{B}_K$ its gauge-fixed field: (i) for every admissible d, $K > d$, $q < \infty$ and every index $\kappa' \in (0, \frac16)$,

$$\mathbb{E}^{[d]}\big[\|\mathbf{B}_K\|_{\kappa';a}^q\big]^{1/q} \;\le\; C(q,\kappa')\,L^{20d}, \tag{15.iii}$$

with C free of K and d: given the coarser layers, the layer of depth d' is a function of one fluctuation integral with conditional covariance at most $Ca_{d'}^2/\mathsf{b}$ and Gaussian tails (Lem. 15.3(iv)), the products of layers of two depths in the bilinear part of the rough norm are martingale differences, those of one depth are controlled by the bound (Q) of centred quadratic forms and by the cancellation in colour (Col), the conditional means by (M), all three set in the proof of Prop. 16.1(a), and the top field, of size at most a constant times L^{10d} in its regular gauge, enters to the second power; (ii) the short-time theory of the gauge-fixed flow, which uses no measure: the numbers c_1, c_2, c_3, $s_0(R)$, $a_0(R)$ at every index κ', the four bounds of A_s, ∇A_s, $A_s - P_s\mathbf{B}$, $\nabla(A_s - P_s\mathbf{B})$, monotone in R, the bound of the flowed action and the Lipschitz bound in the rough distance; (iii) for invariant Borel

$F \geq 0$ and $r > 1$,

$$\langle F \rangle_{\mu^K} \;\leq\; \mathbb{E}^{[d_1]}[F] + \sum_{d_1 \leq d \leq K-2} \mathsf{R}_d^{1-1/r}\, \mathbb{E}^{[d+1]}\big[F^r\big]^{1/r} + \mathsf{R}_{K-1} \sup F, \tag{15.iv}$$

from the chains of all the top depths $d \geq d_1$, each compared with the bare density by Lem. 15.3(i), and Hölder's inequality; and $\sum_d \mathsf{R}_d^{1/2} L^{pd}$ and $\sup_K \mathsf{R}_{K-1} L^{pK}$ are finite for every p, as R_d decays faster than every exponential in d; (iv) hence, with $\mathsf{N}' = \mathsf{N}'(U)$ of the head of Prop. 16.1 formed at the index κ', the least $\|\mathbf{B}\|_{\kappa';a}$ over the 1-forms $\mathbf{B}$ with $e^{ia_K \mathbf{B}}$ in the orbit of U, which is at most a fixed power of L^K at every field, $\mathsf{M}_q(\kappa') = \sup_K \langle \mathsf{N}'^q \rangle_{\mu^K} < \infty$, and the moments under μ^K, bounded uniformly in K, of $\sup_{0<s\leq 1} s^{\frac{3}{2}+\kappa'} A^a(U_s)$, of $(J,J)_a(U_\sigma)$ and of $a_K^{-4} \max_p a_p(U_\sigma)$ in Prop. 16.1(a), at every index $\kappa' \in (0, \frac{1}{6})$.

Step 6: flow identities and uniform estimates, (C1), (C2). (C1) is Lem. 15.1(a) with the integration by parts, and $\int \mathcal{L}_K H \, \mathrm{d}\mu^K = 0$. The formula for $P_t^K H^\sigma$: $\frac{d}{ds} P_s^K H^{\sigma + \mathsf{b}(t-s)} = P_s^K (\mathcal{L}_K - \mathsf{b}\, \partial_\sigma) H^{\sigma+\mathsf{b}(t-s)} = P_s^K \Delta_K H^{\sigma+\mathsf{b}(t-s)}$, integrated over $[0,t]$; the martingale is Itô's formula for the time-dependent function $(r, U) \mapsto H^{\sigma+\mathsf{b}(t-r)}(U)$, whose drift is $(\partial_r + \mathcal{L}_K) H^{\sigma+\mathsf{b}(t-r)} = \Delta_K H^{\sigma+\mathsf{b}(t-r)}$. (C2): on $\mathsf{O}_K(\bar{C}, \bar{s})$, for all K and $\sigma \leq \bar{s}$, $\Lambda_\sigma \leq (8\bar{C}\sigma^{\frac{1}{4}-\frac{\kappa}{2}} + 4\bar{C}^2 \sigma^{\frac{1}{2}-\kappa})/(1-2\kappa)$ (Lem. 16.2(e)) and $A^a(U_{\bar{s}/2}) \leq E_\sharp = 6\ell^3 (\bar{C}^2 (\bar{s}/2)^{-\frac{1}{2}-\kappa} + \bar{C} (\bar{s}/2)^{-\frac{3}{4}-\frac{\kappa}{2}})^2$.

For $K \geq K_0$, the least K with $a_K E_\sharp \leq e_0$, Lem. 16.2(f) at $s_1 = \bar{s}/2$ gives $a_K^{-4} \max_p a_p(U_s) \leq Z_\sharp = \frac{5}{2} E_\sharp \min\{\bar{s}/2, x_0^4 E_\sharp^{-2}\}^{-3/2}$ for $s \geq \bar{s}$, so $\Lambda_\sigma \leq \Lambda_{\bar{s}} + (2Z_\sharp)^{1/2}$ for $\bar{s} < \sigma \leq 1$. With $G = \varphi(\mathcal{O}(f_1), \dots, \mathcal{O}(f_r))$,

$$\Gamma_K(H_K^\sigma) \leq e^{24\Lambda_\sigma} \|\nabla \varphi\|_\infty^2 \sum_i \Gamma_K(\mathcal{O}(f_i))(U_\sigma), \qquad |\partial_\sigma H_K^\sigma| \leq (\Gamma_K(G)(J,J)_a)^{1/2}(U_\sigma),$$

with $\frac{\sigma}{2}(J,J)_a(U_\sigma) \leq e^{24\Lambda_\sigma} A^a(U_{\sigma/2})$, $A^a(U_{\sigma/2}) \leq 24\ell^3 \Lambda_\sigma^2 \sigma^{-2}$ for $\sigma \leq \bar{s}$ and $\leq E_\sharp$ beyond, and Lem. 16.2(d). (b): the same pointwise inequalities without a set, with the moments (iv) of Prop. 16.1(a) above, by (Adm); the Lipschitz bound in σ is the integral of ∂_σ. (c): the second flow-time derivative involves $v^2 \mathcal{O}(f)$ along the flow, which is bounded by $\mathsf{h}_s \Gamma_K(\mathcal{O}(f))^{1/2}$ plus $\|f\|_\infty$ times the square of the L^2 norm of the covariant gradient of the current and a term of the first level, $\mathsf{h}_s = a_K^{-7/2} \|\nabla^2 A_K(U_s)\, \partial A_K(U_s)\|$; a second level of the energy estimates rests on a floor of the Hessian on horizontal vectors by the action alone: Prop. 16.6(a) at $k = N-1$, where $\mathsf{e}_N^* = 2A^a$ and $a_K(N-1) \geq \frac{1}{2}\ell$, gives $a_K^{-2} \nabla^2 A_K(V)[X,X] \geq -c(A^a(V)^2 + \ell^{-2}) \|X\|^2$ for horizontal X when $\max_p a_p(V) \leq \frac{1}{2}$, with an absolute c.

With it, for every field, $0 \leq \sigma_0 < \sigma$ and $E = A^a(U_{\sigma_0})$,

$$(J,J)_a(U_\sigma) \leq e\, E \left[(\sigma - \sigma_0)^{-1} + 2c(E^2 + \ell^{-2})\right] + 3072\, \ell^3 E^6.$$

Where $a_K E \leq \frac{1}{2}$, $\max_p a_p(U_s) \leq a_K A^a(U_s) \leq \frac{1}{2}$ for $s \geq \sigma_0$, as the action does not increase along the flow, so the floor applies to the current, which is horizontal and is carried by the linearised flow: $\partial_s (J,J)_a(U_s) \leq 2c(E^2 + \ell^{-2})(J,J)_a(U_s)$, and the mean of $(J,J)_a(U_s)$ over the last interval before σ of length $\min\{\sigma - \sigma_0, (2c(E^2+\ell^{-2}))^{-1}\}$ is at most E divided by that length, as $\partial_s A^a(U_s) = -(J,J)_a(U_s)$, while $(J,J)_a(U_\sigma)$ is at most e times that mean; where $a_K E > \frac{1}{2}$, $a_K^{-1} < 2E$, and the crude bound $(J,J)_a \leq 48\ell^3 a_K^{-6}$ gives the last member. The second level needs more than this floor: with $g = \partial A_K(U_s)$, which is horizontal, and $\mathsf{Q}_s = a_K^{-5} \nabla^2 A_K(U_s)[g,g]$, $\partial_s (J,J)_a(U_s) = -2\mathsf{Q}_s$ and

$$\partial_s \mathsf{Q}_s = -2\mathsf{h}_s^2 - a_K^{-7} \nabla^3 A_K(U_s)[g,g,g],$$

and the cubic member has no sign. Plaquette by plaquette, with $|X|_p^2 = \sum_{b \in p} |X(b)|^2$ and the covariant forward differences ∇ of the field, $|\frac{d^3}{dt^3} a_p(e^{itX} V)|_{t=0}| \leq 6|X|_p^2 (|\nabla_\nu X_\mu(x)| + |\nabla_\mu X_\nu(x)|) + 25\varpi_p |X|_p^3$ for $p = p_{\mu\nu}(x)$, since the cubic term of the plaquette is a sum of triple products of the four values of X at p, carried to its base point with their signs, each of which holds the sum of

the values at two parallel bonds, a covariant difference up to $2\varpi_p|X|_p$; so $|\nabla^3 A_K(V)[X,X,X]| \le 24\sqrt{2}\,\|X\|_{\ell^4}^2 \mathsf{D}(X) + 200(2A_K(V))^{1/2}\|X\|_{\ell^6}^3$, $\mathsf{D}(X)$ the ℓ^2 norm of ∇X.

Kato's inequality, $||X_\mu(x+a_K e_\nu)| - |X_\mu(x)|| \le |\nabla_\nu X_\mu(x)|$, and the Sobolev inequality of the periodic lattice with constants free of N, $\|u\|_{\ell^6} \le N^{-1}\|u\|_{\ell^2} + \frac{4}{\sqrt{3}}\|\nabla u\|_{\ell^2}$ and $\|u\|_{\ell^4}^4 \le \|u\|_{\ell^2}\|u\|_{\ell^6}^3$, bound these norms through $\|X\|$ and $\mathsf{D}(X) + N^{-1}\|X\|$. The same tools bound the remainder of the Weitzenböck identity of Lem. 16.2(h), at most a constant times $\sum_p \varpi_p |X|_p^2$, by $\frac{1}{8}\mathsf{D}(X)^2 + c(A_K(V)^2 + N^{-2})\|X\|^2$, c taken large enough: this is the floor in the form that keeps the covariant gradient, $\nabla^2 A_K(V)[X,X] \ge \frac{1}{8}\mathsf{D}(X)^2 - c(A_K(V)^2 + N^{-2})\|X\|^2$ for horizontal X when $\max_p a_p(V) \le \frac{1}{2}$, whence at $X = g$, $\mathsf{D}(g)^2 + N^{-2}\|g\|^2 \le 8\|g\|\,\|\nabla^2 A_K(U_s)g\| + 9c(A_K(U_s)^2 + N^{-2})\|g\|^2$.

Let $\sigma_1 = \min I$, $\sigma_0 = \frac{1}{4}\sigma_1$, $\Delta = \sigma_1 - \sigma_0$ and $E = A^a(U_{\sigma_0})$. Where $a_K E \le \frac{1}{2}$, the display bounds $(J,J)_a(U_s)$ by $\bar{\mathsf{y}} = \mathrm{e}\,E[2\Delta^{-1} + 2c(E^2 + \ell^{-2})]$ for $s \ge \sigma_0 + \frac{1}{2}\Delta$, and Young's inequality absorbs the cubic member into h_s^2 ($N^{-1} = a_K/\ell$, $A_K = a_K A^a$):

$$\partial_s \mathsf{Q}_s \le -\mathsf{h}_s^2 + \mathsf{B}, \qquad \mathsf{B} \le C\Big[\bar{\mathsf{y}}^{7/3} + E^2\bar{\mathsf{y}}^3 + ((E^2+\ell^{-2})^{5/4} + (E^2+\ell^{-2})^{3/2}E^{1/2})\bar{\mathsf{y}}^{3/2}\Big],$$

C absolute. Above, Q_s is at most its mean over the interval of length $\frac{1}{2}\Delta$ before s, that is half the fall of $(J,J)_a$ over that interval divided by its length, plus $\frac{1}{2}\mathsf{B}\Delta$; below, the floor gives $\mathsf{Q}_s \ge -c(E^2+\ell^{-2})(J,J)_a(U_s)$. So for $\sigma_1 \le r_1 < r_2$, $\int_{r_1}^{r_2} \mathsf{h}_s^2 ds \le \mathsf{Q}_{r_1} - \mathsf{Q}_{r_2} + \mathsf{B}(r_2 - r_1)$, and $a_K^{-5}\mathsf{D}(g)^2 \le 8\mathsf{Q}_s + 8c(E^2+\ell^{-2})(J,J)_a(U_s)$ bounds the square of the L^2 norm of the covariant gradient of the current. With $A_f = \sum_p f(x(p))a_p$, $v^2\mathcal{O}(f) = a_K^{-5}[\langle \nabla^2 A_K\, g, \partial A_f\rangle + \nabla^2 A_f[g,g]]$, and $|v^2\mathcal{O}(f)|(U_s) \le \mathsf{h}_s \Gamma_K(\mathcal{O}(f))(U_s)^{1/2} + \|f\|_\infty[3a_K^{-5}\mathsf{D}(g)^2 + 4c(E^2+\ell^{-2})(J,J)_a(U_s)]$; integrated over s between two points of I, with the Cauchy–Schwarz inequality in s, these are polynomials in E with powers at most 11, times the square root of the length.

Where $a_K E > \frac{1}{2}$, $a_K^{-1} < 2E$, and the crude bounds $|v^2\mathcal{O}(f)| \le 1536\|f\|_\infty \ell^3 a_K^{-8}$ and $|v\mathcal{O}(f)| \le 48\|f\|_\infty \ell^3 a_K^{-6}$ become powers of E, the square of the second one the power 12. As $E \le (4/\sigma_1)^{5/3}\sup_{0<s\le 1} s^{\frac{3}{2}+\kappa}A^a(U_s)$, this gives $|\partial_\sigma H_K^{\sigma''} - \partial_\sigma H_K^{\sigma'}| \le C_I|\sigma'' - \sigma'|^{1/2}(1 + \sup_{0<s\le 1} s^{\frac{3}{2}+\kappa}A^a(U_s))^{12}$ at every field and every cutoff, and the moments of Prop. 16.1(a), by (Adm), give (c).

Step 7: convergence of flowed observables along the full sequence, (C3). Bounded strongly convergent sequences are stable under continuous functions, so $\varphi(\Theta_1,\dots,\Theta_r)$, Θ_i of Prop. 16.1(c), under the proved conclusion (Adm), would converge strongly for f_i applied to the flowed loops once the flow-approximation step is established, and $|H_K^\sigma - \varphi(\Theta_1,\dots,\Theta_r)|$ is at most $\mathrm{Lip}(\varphi)\, r\, \eta$ on $\bigcap_i \mathsf{G}_K^{(i)}$ and at most $2\sup|\varphi|$ elsewhere; let $\eta, \epsilon \to 0$. The joint convergence in law identifies the limit as G_σ, an expression independent of the cutoff; only the Lipschitz constants of the f_i and of φ were used.

By (C2)(b), $\|\mathsf{G}_\sigma - \mathsf{G}_{\sigma'}\| = \lim_K \|H_K^\sigma - H_K^{\sigma'}\| \le C|\sigma - \sigma'|$. For $\dot{\mathsf{G}}_\sigma$: by (C2)(b), (c) the functions $\sigma \mapsto \int \Psi_K \partial_\sigma H_K^\sigma \,\mathrm{d}\mu^K$ are bounded and locally Hölder continuous of exponent $\frac{1}{2}$, uniformly in K; their primitives $\int \Psi_K H_K^\sigma \,\mathrm{d}\mu^K$ converge to $\int \Psi \mathsf{G}_\sigma \,\mathrm{d}\mu$; so the derivatives converge, locally uniformly, to the derivative of the limit, which is then continuously differentiable with a Hölder continuous derivative; the bound in L^2 and the density of the cylinder functions represent the limit as $\int \Psi \dot{\mathsf{G}}_\sigma \,\mathrm{d}\mu$ with $\dot{\mathsf{G}}_\sigma \in L^2_{\mathrm{inv}}$, along the full sequence.

Step 8: condenser cutoffs and the generator limit, (C4), (C5). Let $c_3' = c_3(\kappa/2)$, $R_\flat = 2(2 + c_3')$, $\rho = R/2(2+c_3')$, $q = 4p/\kappa$; $\bar{s}(R) = 2^{-j_0(R)}$ is the largest dyadic number $\bar{s}$ with $\bar{s} \le (c_1'\rho^{-2})^8$, which lies below the time $s_0(\rho)$ of Prop. 16.1(a) at the index $\frac{\kappa}{2}$, $c_1' = c_1(\frac{\kappa}{2})$; $c_{\mathrm{st}} R^2 \bar{s}^{\frac{1}{4}-\frac{\kappa}{2}} \le 1$; and $486R^2\bar{s}^{\frac{1}{2}-\kappa} + 108R\bar{s}^{\frac{1}{4}-\frac{\kappa}{2}} \le 1$, which by Lem. 16.2(e) at $\bar{C} = 9R$ gives $\Lambda_s \le 1$ for $s \le \bar{s}$ on $\mathsf{O}_K(9R,\bar{s})$. $\bar{K}(R)$ is the least integer above the depth d_1 of the conclusion (Adm) at which $a = a_K$ has $a \le (c_2'/\rho)^3$, which lies below $a_0(\rho)$ at the index $\frac{\kappa}{2}$, $c_2' = c_2(\frac{\kappa}{2})$, $c_{\mathrm{st}} R a^{\frac{1}{2}-\kappa} \le 1$, and $4a^2 \le \bar{s}(R)$.

Here c_{st} is an absolute constant of the short-time theory of the gauge-fixed flow; all these conditions are upper bounds of $\bar s$ and of a that decrease in R. Let $g_j = \sup\{\mathsf{m}_s(U_s) : s \in (2^{-j-1},2^{-j}]\cap[a_K^2,\bar s(R)]\}$, $g_\flat = a_K^{1/2+\kappa}\min\|\mathbf{B}\|_\infty$ over $e^{ia_K\mathbf{B}}$ in the orbit of U, $\mathsf{G}^* = \max\{g_\flat, \max_j g_j\}$, and $\chi_{K,R}$ the average of $\min\{1,\max\{0,4-2\mathsf{G}^*/R\}\}$ against a product of smooth conjugation-invariant approximate identities supported on small rotations of the bond groups. On $\{\mathsf{G}^*\le 3R\}$, $\mathsf{G}^*\le c$ implies $U\in \mathsf{O}_K(3c,\bar s)$ and $\Lambda_s\le 1$ for $s\le\bar s$; there the pieces are locally Lipschitz for the L^2 metric, $\operatorname{lip} g_j \le C_L 2^{(j+1)(1-\kappa)/2}$, $\operatorname{lip} g_\flat \le 2a_K^{-1+\kappa}$, $C_L = 24e^{24}$, because the linearised flow smooths from L^2 modulo gauge: $\mathrm{Ad}(u)D\Phi_s(U)Y = \zeta + a_K^{-1}\mathrm{d}\vartheta$ with $\|\zeta\|_\infty \le 9e^{24}s^{-3/4}\|Y\|_2$ and $\|\nabla\zeta\|_\infty \le 17e^{24}s^{-5/4}\|Y\|_2$.

The rotations are chosen so small that G^* changes by at most $\frac{R}{2}$ under them; so $\chi_{K,R}=1$ on $\{\mathsf{G}^*\le R\}$, $\chi_{K,R}=0$ off $\{\mathsf{G}^*\le\frac52 R\}\subset \mathsf{O}_K(8R,\bar s(R))$, and the piece j counts in $\Gamma_K(\chi_{K,R})$ only where $g_j\ge R$. The short-time theory at the index $\frac\kappa2$, which gains $s^{\kappa/4}$, and $\mathsf{M}_q = \mathsf{M}_q(\frac\kappa2)<\infty$, the number of Prop. 16.1(a) at the index $\frac\kappa2$ under (Adm) ((iv) of the moments of the gauge-fixed field), give $\mu^K(g_j\ge R)\le \mathsf{M}_q\rho^{-q}2^{-pj}$ and $\mu^K(g_\flat\ge R)\le \mathsf{M}_q(2e^{12}/R)^q a_K^{2p}$; with $2^{(j+1)(1-\kappa)p-pj}\le 2^p 2^{-\kappa pj}$,

$$\mu^K(\chi_{K,R}<1)\le \mathsf{M}_q[2\rho^{-q}+(2e^{12}/R)^q],\qquad \int\Gamma_K(\chi_{K,R})^p d\mu^K \le (\tfrac{2}{R})^{2p}\mathsf{M}_q\Big[\frac{C_L^{2p}2^p\rho^{-q}}{1-2^{-\kappa p}}+4^p(\tfrac{2e^{12}}{R})^q\Big],$$

and we call the first of these bounds $\epsilon_p(R)$ and the second $C_\chi(p,R)$, $q=4p/\kappa$. As $\Gamma_K(\chi_{K,R})$ vanishes off the shell $\{0<\chi_{K,R}<1\}$, Hölder's inequality on the shell bounds the energy, $\mathcal{E}_K(\chi_{K,R})\le C_\chi(p,R)^{1/p}\epsilon_p(R)^{1-1/p}\le c(p)\mathsf{M}_q R^{-2-q}$, the exponents of R adding up to $-(2+\frac qp)-(q-\frac qp)$, while the mass is only $\int(1-\chi_{K,R})^2\,\mathrm{d}\mu^K\le\mu^K(\chi_{K,R}<1)\le\epsilon_p(R)\le c(p)\mathsf{M}_qR^{-q}$. The cutoff $\chi_{K,R}$ does not depend on p, so one cutoff serves all p, and

$$\bar\varkappa(R) := \min_{p\in\mathbb{N},\ p\le R}\Big[C_\chi(p,R)^{1/p}\epsilon_p(R)^{1-1/p}+\epsilon_p(R)\Big]>0$$

bounds $\mathcal{E}_K(\chi_{K,R})+\int(1-\chi_{K,R})^2\,\mathrm{d}\mu^K$ for $K\ge\bar K(R)$ and is at most $c(p)\mathsf{M}_{4p/\kappa}R^{-4p/\kappa}$ for every $p\le R$; so $R^{q'}\bar\varkappa(R)\to0$ for every q'. The numbers $\bar s(R)$, $\bar K(R)$ are monotone in R, and G^* does not increase when $\bar s$ decreases; so for $R'\ge\frac52R$, $\{\chi_{K,R}>0\}\subset\{\mathsf{G}^*_{(R)}\le\frac52R\}\subset\{\mathsf{G}^*_{(R')}\le R'\}$, on which $\chi_{K,R'}=1$. The condenser cutoff: Lem. 16.5 at $\eta=1-\chi_{K,R}$, with $\varkappa\le\mathcal{E}_{K,1}(\eta,\eta)\le\bar\varkappa(R)$. (C5): by (C1) and Lem. 16.5, $\int\chi^{\mathrm c}_{K,R}\mathcal{L}_K H\,\mathrm{d}\mu^K=\mathcal{E}_K(w_{K,R},H)$, so $|\int\chi^{\mathrm c}_{K,R}\mathcal{L}_KH^\sigma_K|\le\operatorname{osc}(\varphi)\bar\varkappa(R)$; moreover $|\int\chi^{\mathrm c}\mathsf{b}\,\partial_\sigma H-\mathsf{b}\,\partial_\sigma\langle H\rangle|=\mathsf{b}|\int w\,\partial_\sigma H|\le\mathsf{b}C_\partial\bar\varkappa(R)^{1/2}$, which gives the second line of (a).

(b) is the last part of Lem. 16.5, as $q(1-u)=1-q(u)$. (c): $\int\chi^{\mathrm c}(H'\mathcal{L}_KH-H\mathcal{L}_KH')=\mathcal{E}_K(w,HH')-2\int H\,\Gamma_K(w,H')$, and $\Gamma_K(w,\cdot)$ vanishes off the shell. (d): by (C1) and Lem. 16.5,

$$\Big|\int\chi^{\mathrm c}F\mathcal{L}_KH-\int H\mathcal{L}_KF\Big|\le\sup|H|\,[2\|F\|_\infty\varkappa+(2\mathcal{E}_K(F)^{1/2}+\|\mathcal{L}_KF\|)\varkappa^{1/2}];$$

with a test sequence, $\int H^\sigma_{K_j}\mathcal{L}_{K_j}F_j\,\mathrm{d}\mu^{K_j}\to-\int(A_\star F)\mathsf{G}_\sigma\,\mathrm{d}\mu$ by (C3); the convergence of resolvents in (A3) gives $\alpha(\alpha-\mathcal{L}_{K_j})^{-1}W_j\to\alpha(\alpha+A_\star)^{-1}W$ strongly, bounded by $\sup_j\|W_j\|_\infty$, hence test sequences; c_F uses $\bar\varkappa\le\bar\varkappa^{1/2}$. The flow-time half: $\int\chi^{\mathrm c}F_j\partial_\sigma H^\sigma_{K_j}$ differs from $\partial_\sigma\int F_jH^\sigma_{K_j}$ by at most $\sup_j\|F_j\|_\infty C_\partial\bar\varkappa(R)^{1/2}$, and $\int F_jH^\sigma_{K_j}\to(F,\mathsf{G}_\sigma)$, locally Lipschitz in σ uniformly in j; this gives the identity of distributions.

Reversibility: take $F_j=\alpha(\alpha-\mathcal{L}_{K_j})^{-1}H'_{K_j}$, a test sequence for $F=\alpha(\alpha+A_\star)^{-1}\mathsf{G}'$, and note that $A_\star\alpha(\alpha+A_\star)^{-1}$ is self-adjoint. The criteria: for $F^\Lambda=\Lambda(\Lambda+A_\star)^{-1}\mathsf{G}_\sigma$ with its test sequence, $(A_\star F^\Lambda,\mathsf{G}_\sigma)=\int\frac{\Lambda\lambda}{\Lambda+\lambda}d\nu_{\mathsf{G}_\sigma}$ increases to $\mathcal{E}_\star(\mathsf{G}_\sigma)$ as $\Lambda\to\infty$, and is at most $\sup|\varphi|\,C_1$ after $R\to\infty$; in L^2 the same bound with Cauchy–Schwarz holds on a core of $A_\star$.

Step 9: lower energy bounds, (C6). By (Adm), let $\mathbb{P}_K$ be the normalised chain measure from the least top depth d_0 with $\mathsf{R}_{d_0}\le\frac12$ such that every top depth $\ge d_0$ is admissible, as in the head of Lem. 16.7. Take R with $\mathbb{P}_K(\|\mathbf{B}_K\|_{\kappa;a}>R)\le\frac14 p_*$ (by eq. 15.iii at $q=1$ and Markov's

inequality, $R = \max\{1, 4C(1,\kappa)L^{20d_0}/p_*\}$), and d with $\theta_0 a_{d+1}^2 < \frac{3}{2}\sigma \le \theta_0 a_d^2$. On the complement $\{\|\mathbf{B}_K\|_{\kappa;a} \le R\}$ of this event and on the event of Lem. 16.7(a) at $s = \frac{3}{2}\sigma$, which meet in a $\mathbb{P}_K$-probability at least $\frac{3}{4}p_*$, Lem. 16.7(b) and the flow bounds of Prop. 16.1(a) give, for $\sigma \le \sigma_*$, where $\kappa < \frac{1}{6}$ makes the error of lower order,

$$\Gamma_K(A^a \circ \Phi_\sigma) \ge cR^{-2}\mathsf{b}^{-2}\ell^3\sigma^{-5/2+\kappa}, \qquad A^a \circ \Phi_\sigma \le S_*\sigma^{-3/2-\kappa},$$

on a closed invariant L_K with $\mathbb{P}_K(\mathsf{L}_K) \ge \frac{3}{4}p_*$, so $\mu^K(\mathsf{L}_K) \ge \frac{3}{8}p_*$ (Lem. 15.3(i), $\mathsf{R}_{d_0} \le \frac{1}{2}$); hence $\mu^K(\mathsf{L}_K \cap \{\chi_K = 1\}) \ge \frac{1}{4}p_*$, and $\Gamma_K(\varphi(A^a) \circ \Phi_\sigma) = \varphi'(A^a \circ \Phi_\sigma)^2\,\Gamma_K(A^a \circ \Phi_\sigma)$. The mean flowed action from below is the same argument at $s = \sigma$ with the last inequality of Lem. 16.7(b): $A^a \circ \Phi_\sigma \ge \frac{1}{8}k''_\Sigma \mathsf{b}^{-1}\ell^3\sigma^{-3/2}$, $k''_\Sigma = k_\Sigma\theta_0^{3/2}L^{-3}$, with μ^K-probability $\ge \frac{3}{8}p_*$, so $c_\circ = \frac{3}{64}p_*k''_\Sigma$.

From above, under μ^K and not only under a chain measure: on invariant functions the chain measure σ^K from a top depth d has a density $\bar{\mathfrak{w}} \in [0,1]$ with respect to μ^K, with $\int(1-\bar{\mathfrak{w}})\,\mathrm{d}\mu^K \le \mathsf{R}_d$ (Lem. 15.3(i)), so for invariant $F \ge 0$, by the Cauchy–Schwarz inequality, $\langle F\rangle_{\mu^K} \le \mathbb{E}^{[d]}[F] + \mathsf{R}_d^{1/2}\langle F^2\rangle_{\mu^K}^{1/2}$. Take $F = A^a \circ \Phi_\sigma$, with $\langle F^2\rangle_{\mu^K}^{1/2} \le \mathsf{C}\sigma^{-3/2-\kappa}$ by the moments (iv) of Prop. 16.1(a), under (Adm), and $d = d_\sigma$, the least depth $d \ge d_1$ with $\mathsf{R}_d \le \sigma^{2\kappa}$, an admissible top depth by (Adm); then the second member is at most $\mathsf{C}\sigma^{-3/2}$, and as R_d decays faster than every exponential in d, $L^{d_\sigma} = \sigma^{-o(1)}$ as $\sigma \to 0$.

Under $\mathbb{E}^{[d]}$, d admissible, the centred layers of all depths, with conditional covariances at most $Ca_{d'}^2/\mathsf{b}$, give $\mathbb{E}^{[d]}\|\mathrm{d}_a P_\sigma \mathbf{B}_K\|_2^2 \le c\,\mathsf{b}^{-1}\ell^3\sigma^{-3/2}$ against the heat kernel, with c free of d and K; the top field, the conditional means of the layers and the remainder of the logarithm add a lower power of σ times a number that grows at most like a power of L^d; on $\{\|\mathbf{B}_K\|_{\kappa;a} \le \sigma^{-\epsilon}\}$, $\epsilon > 0$ small, the last inequality of Lem. 16.7(b) passes from $\|\mathrm{d}_a P_\sigma \mathbf{B}_K\|_2^2$ to $2A^a \circ \Phi_\sigma$ at the cost of a lower power of σ; and off this event the bound $A^a \circ \Phi_\sigma \le c_4\ell^3(1 + \|\mathbf{B}_K\|_{\kappa;a})^{24}\sigma^{-3/2-\kappa}$ of Prop. 16.1(a) with eq. 15.iii at a large q costs at most $CL^{pd}\sigma^{-1}$.

So $\mathbb{E}^{[d]}[F] \le c\,\mathsf{b}^{-1}\ell^3\sigma^{-3/2} + CL^{pd}\sigma^{-3/2+\epsilon'}$ for large K, with some $\epsilon' > 0$, and at $d = d_\sigma$ the second member is $o(\sigma^{-3/2})$: with the bound $\mathsf{R}_{d_\sigma}^{1/2}\langle F^2\rangle_{\mu^K}^{1/2} \le \mathsf{C}\sigma^{-3/2}$ this gives $C_\circ$, for $0 < \sigma \le \sigma_{**}$. For $\vartheta < 1$ and for the probability that tends to 1: let $\psi \in C^\infty(\mathbb{R})$ take values in $[0,1]$. By Prop. 16.6(b) the square field of $\psi(\sigma^{3/2}A^a) \circ \Phi_\sigma$ is bounded on $\{\mathscr{N} \le R\}$ uniformly in K and in small σ, so by Prop. 16.6(d), that is, by the key estimate of Step 1, which rests on eq. 15.ii, the function $\chi'_K\,\psi(\sigma^{3/2}A^a \circ \Phi_\sigma)$, χ'_K the cutoff of Prop. 16.6(c), is within a distance in $L^2(\mu^K)$ that is small for large $\hat{d}_0$, uniformly in K and in small σ, of a function of the block field at depth $\hat{d}_0$; under (Adm), every $\hat{d}_0 \ge d_1$ is an admissible top depth, and $\mathsf{R}_{\hat{d}_0} \le \frac{1}{2}$ for large $\hat{d}_0$.

By the last clause of Lem. 16.7(a), at the top depth $\hat{d}_0$ in the place of d_0, off an event of probability τ_d the lower bounds hold, given the block field at depth $\hat{d}_0$, with a conditional probability that is bounded below by a positive number free of K, σ and $\hat{d}_0$; a function that is close to a function of the block field and exceeds a level with conditional probability bounded below exceeds a slightly smaller level with high probability. So the flowed action exceeds a fixed multiple of $\mathsf{b}^{-1}\ell^3\sigma^{-3/2}$ with a probability that tends to 1 as $\sigma \to 0$, uniformly in K; at the flow time $\frac{3}{2}\sigma$ this bounds $\|\mathrm{d}_a P_{3\sigma/2}\mathbf{B}_K\|_2^2$ from below, by the last inequality of Lem. 16.7(b), and the first inequality there, on $\{\|\mathbf{B}_K\|_{\kappa;a} \le R\}$ for a large R, gives the same for $\Gamma_K(A^a \circ \Phi_\sigma)$ and $\sigma^{-5/2+\kappa}$.

By Markov's inequality and the upper bound of the mean flowed action, $\mu^K(A^a \circ \Phi_\sigma > S_*\sigma^{-3/2-\kappa}) \le 2C_\circ\mathsf{b}^{-1}\ell^3\sigma^\kappa/S_*$ for $0 < \sigma \le \sigma_{**}$ and all large K, which tends to 0 with σ. Given $\vartheta < 1$, choose σ_ϑ so that the probability of these lower bounds, together with $A^a \circ \Phi_\sigma \le S_*\sigma^{-3/2-\kappa}$, exceeds $\frac{1}{2}(1+\vartheta)$ for all large K; the set $\{\chi_K = 1\}$, of mass $\ge 1-\vartheta$, meets this event in a mass $\ge \frac{1}{2}(1-\vartheta)$. The current: $\Gamma_K(A^a \circ \Phi_\sigma)(U) \le e^{24\Lambda_\sigma(U)}(J,J)_a(U_\sigma)$ by Lem. 16.2(b), since $\Gamma_K(A^a) = (J,J)_a$; by the tightness of Λ_σ (Prop. 16.1(a), under (Adm)) there is λ with $\mu^K(\Lambda_\sigma > \lambda) \le \frac{1}{8}p_*$ for large K, and on $\mathsf{L}_K \cap \{\Lambda_\sigma \le \lambda\}$, of mass $\ge \frac{1}{4}p_*$, $(J,J)_a(U_\sigma) \ge e^{-24\lambda}cR^{-2}\mathsf{b}^{-2}\ell^3\sigma^{-5/2+\kappa}$.

For small σ the same property of the current follows by a second route from the lower bound of the mean flowed action, through Prop. 16.9(d)(iii); and, by Prop. 16.9(d)(i) it gives back, conversely, the positivity in the limit of the energies $\int \chi_K \Gamma_K(\varphi(A^a)\circ\Phi_\sigma)\,\mathrm{d}\mu^K$, φ as there, for every weight with $\limsup_K \mu^K(\chi_K<1) < \frac14 p_*$.

Step 10: growth of uncut generator images, (C7). Let $U \in \mathsf{O}_K(\bar C,\bar s)$, of μ^K-mass $\ge 1-\epsilon$ (Prop. 16.1(a), by (Adm)). For large K its flow in lattice time satisfies the hypotheses of Lem. 16.4 with $T=\sigma/a_K^2$: by Lem. 16.2(e) at the flow times $s \ge a_K^{9/4}$, where $\kappa<\frac16$ gives $\varpi(U_s)\le (2\bar C+2\bar C^2)a_K^{1/8}$, and, at smaller flow times, by the bound 16 of the speed of the plaquette variables in lattice time, $\sup_{t\le T}\varpi(\Phi_{a_K^2 t}U)\le (4\bar C^2+20)a_K^{1/8}$, $\int_0^T \varpi\,\mathrm{d}t=\Lambda_\sigma$ is bounded, and the action at $T/2$ is small.

Given the other bonds, the law of the three bonds of Lem. 16.4 under μ^K has a density $\ge e^{-20\beta}$ with respect to Haar measure, as they lie in ten plaquettes, and $20\beta = 20\mathsf{b}\,a_K T/\sigma = o(T)$. So, with $\varepsilon=0.002$,

$$\int |\partial(A_K\circ\Phi_\sigma)|^q\,\mathrm{d}\mu^K \ge e^{-20\beta}(1-\epsilon)\,c\,e^{[(q-3)\lambda_1-6\lambda_2-(q+1)\varepsilon]T},$$

and the enclosures of λ_1,λ_2 give (a); at $q=4$, $\Gamma_K(A^a\circ\Phi_\sigma)^2 = a_K^{-6}|\partial(A_K\circ\Phi_\sigma)|^4$ and $\lambda_1-6\lambda_2>1.8$. (b): $G=A^a\circ\Phi_\sigma$ takes its values in $[0,6\ell^3 a_K^{-4}]$, as $A_K\le 6N^3$, and $H=\arctan G$ has $|\partial H| = a_K^{-1}|\partial(A_K\circ\Phi_\sigma)|/(1+G^2)$; so (a) at $q=4$ gives $\int|\partial H|^4\,\mathrm{d}\mu^K \ge c(\ell)\,a_K^{28}e^{1.76\,\sigma/a_K^2}$. Lem. 16.2(g) with $\sup|H|\le\frac\pi2$ gives $\frac\pi2(3\mathsf{m}+14\beta^{1/2}\mathsf{m}^{1/2}) \ge c(\ell)^{1/2}a_K^{14}e^{0.88\,\sigma/a_K^2}$; the right side tends to ∞ and $\beta^{1/2}=(\mathsf{b}/a_K)^{1/2}$, so $\mathsf{m}\ge 1$ for large K and then $\mathsf{m}\ge c\,a_K^{14.5}e^{0.88\,\sigma/a_K^2}$; and $\|\mathcal{L}_K H\| = \mathsf{m}/a_K \ge e^{0.8\,\sigma/a_K^2}$ for large K.

(c): $f=\mathfrak{f}^2$ with $\mathfrak{f}(x)=\sqrt2|\sin(\pi x_1/\ell)|$, of Lipschitz constant $\sqrt2\pi/\ell$. Let b_0 be the bond in the direction 3 at the site $a_K(k,0,0)$, $k\ge1$ an integer fixed below, $\alpha=2\pi a_K/\ell$, $\mathsf{f}_p=f(x(p))$ and $\mathsf{f}_0=1-\cos(\alpha k)\ge\frac{2}{\pi^2}\alpha^2k^2$. For p at the distance j from b_0, $|\mathsf{f}_p-\mathsf{f}_0|\le\alpha^2(j+3)(k+j+3)$, so $(\sum_p|\mathsf{f}_p-\mathsf{f}_0|^2e^{-2\theta|p|})^{1/2}\le C(\theta)\alpha^2(k+1)$; fix k with $C_\sharp C(\theta)(k+1)\le c_\sharp k^2/\pi^2$. Then Lem. 16.4(b) gives $|\partial_{b_0}(\mathcal{O}(f)\circ\Phi_\sigma)(W_\zeta)|\ge a_K^{-1}\frac{c_\sharp}{\pi^2}\alpha^2k^2e^{(\lambda_1-\varepsilon)T}$ for $\zeta\in\mathcal{S}$.

The value: with $\tilde V_t$ of Lem. 16.4 and $\mathcal{A}_f(V)=\sum_p \mathsf{f}_p\varsigma_p a_p(V)$,

$$\mathcal{O}(f)(\Phi_\sigma W_\zeta) = a_K^{-1}\Big[\mathcal{A}_f(\tilde V_T) + 2\sum_{p\ni b_0}\mathsf{f}_p\Big]\ge 0,$$

and the second term is at most $4a_K^{-1}\alpha^2(k+3)^2$, which tends to 0. For the first, a local energy inequality holds along the flow of $\tilde A$ in lattice time: with $\nu=-\partial\tilde A(\tilde V_t)$ its velocity, $f_b=f(a_Kb_-)$ and $a'=3a_K\sqrt2\pi/\ell$,

$$\tfrac{d}{dt}\mathcal{A}_f(\tilde V_t) = -\sum_b f_b|\nu(b)|^2 + \sum_b\Big\langle\nu(b),\sum_{p\ni b}(\mathsf{f}_p-f_b)\varsigma_p\,\partial_b a_p(\tilde V_t)\Big\rangle \le 64a'^2\sum_p a_p(\tilde V_t),$$

since $|\mathsf{f}_p-f_b|\le a'(2|\mathfrak{f}(a_Kb_-)|+a')$ for $b\in p$, $|\partial_b a_p|\le\varpi_p$, $|\nu(b)|\le\sum_{p\ni b}\varpi_p$, and $\sum_b(\sum_{p\ni b}\varpi_p)^2\le 32\sum_p a_p$. So $\mathcal{A}_f(\tilde V_T)\le\mathcal{A}_f(\tilde V_t)+64a'^2\int_t^T\sum_p a_p(\tilde V_u)\,\mathrm{d}u$ for $t\le T$; the mean over $t\in[\frac T2,T]$, $|\mathcal{A}_f|\le 2\sum_p a_p$, Lem. 16.4(c) and $A_K(U^\circ_{T/2})\le a_KE^{\mathrm{a}}_\sigma$ (Lem. 16.2(c), (e)) give

$$\mathcal{A}_f(\tilde V_T)\le\Big(\frac4T+64a'^2\Big)\Big(\frac32 T\,a_KE^{\mathrm{a}}_\sigma+C_\sharp\Big)\le a_K\Big(\frac4\sigma+\frac{11400}{\ell^2}\Big)\Big(\frac32\sigma E^{\mathrm{a}}_\sigma+C_\sharp a_K\Big),$$

as $64a'^2=1152\pi^2a_K^2/\ell^2\le 11400\,a_K^2/\ell^2$. So $0\le\mathcal{O}(f)\circ\Phi_\sigma\le M_\sigma$ on the fields W_ζ, $\zeta\in\mathcal{S}$, for large K, and $\int|\partial\varphi(\mathcal{O}(f)\circ\Phi_\sigma)|^4\,\mathrm{d}\mu^K\ge(\inf_{[0,M_\sigma]}|\varphi'|)^4e^{-20\beta}(1-\epsilon)\,c\,a_K^4e^{(\lambda_1-6\lambda_2-5\varepsilon)T}$; Lem. 16.2(g) as in (b). (d): the last identity of Lem. 16.2(g) gives $|\mathcal{L}_K\cos G|+|\mathcal{L}_K\sin G|\ge\Gamma_K(G)$, and (a) at $q=2p$, whose exponent $(2p-3)\lambda_1-6\lambda_2$ is positive for $p>\frac32+3\lambda_2/\lambda_1$, which is less than 1.8637; for $p>4$ the L^p norm dominates the L^4 norm.

(e): by Lem. 16.2(g) and $\nabla^2 A_K \geq -24$, $a_K^2 M^2 + 24\mathsf{b}\, M \geq \int \|\nabla^2 H\|_{\mathrm{HS}}^2 \,\mathrm{d}\mu^K$ for $M = \|\mathcal{L}_K H\|$ and $\sup|H| \leq 1$, since $\int |\partial H|^2 \,\mathrm{d}\mu^K = -a_K \int H \mathcal{L}_K H \,\mathrm{d}\mu^K$. Let $H = \sin(\mathcal{O}(f) \circ \Phi_\sigma)$; as sin is odd we may replace f by $-f$, so that $f(x_*) = \max|f| > 0$ at some x_*, and we base b_0 at a site nearest to x_*. For large K every $U \in \mathsf{O}_K(\bar{C}, \bar{s})$ satisfies the hypotheses of Lem. 16.4(d) with $\Lambda_0 = \max\{1, \sigma/\ell^2 + (8\bar{C}\sigma^{\frac{1}{4}-\frac{\kappa}{2}} + 4\bar{C}^2\sigma^{\frac{1}{2}-\kappa})/(1-2\kappa)\}$ (Lem. 16.2(e)): the supremum of ϖ along the flow tends to 0 as above, $T/N^2 = \sigma/\ell^2$, and $N^3 T^{-2} = \ell^3 a_K \sigma^{-2} \to 0$.

The fibre $g \mapsto W^{(g)}$ of the bond b_0 is totally geodesic in Ω_K, so $\mathfrak{h}(g) = H(W^{(g)})$ has $\|\nabla^2_{S^3}\mathfrak{h}\|_{\mathrm{HS}}(g) \leq \|\nabla^2 H\|_{\mathrm{HS}}(W^{(g)})$. For $(y_1, y_2) \in \mathsf{Y}$ of Lem. 16.4(d), the function $r \mapsto \mathcal{O}(f)(\Phi_\sigma W^{(g_r)}) = a_K^{-1}\mathsf{v}_f(r)$ falls by at least $f(x_*)\rho_2^2/a_K \geq 2\pi$ on $[0, r_*]$ for large K, so $\mathfrak{h}$ takes the values 1 and -1 at two points g', g'' with $\mathsf{d}(g', g'') \leq r_* \leq \frac{4}{3}\rho_2 e^{-\lambda_1' T}$; the inequalities of Sobolev and Morrey on S^3 give $|\mathfrak{h}(g') - \mathfrak{h}(g'')| \leq C\|\nabla^2_{S^3}\mathfrak{h}\|_{L^2(S^3)}\mathsf{d}(g', g'')^{1/2}$, so $\int \|\nabla^2 H\|_{\mathrm{HS}}^2(W^{(g)})\,\mathrm{d}g \geq c\, e^{\lambda_1' T}$ for the Haar probability measure dg.

The set Y has Lebesgue measure $\geq c_{\mathrm{d}} e^{-6\lambda_2'' T}$, the Haar measure in the chart $e^{iy_j} U(b_0 + \hat{e}_j)$ has a density $\geq (4\pi^2)^{-1}$ for $|y_j| \leq 1$, and given the other bonds the law of the three bonds has a density $\geq e^{-20\beta}$; so $\int \|\nabla^2 H\|_{\mathrm{HS}}^2 \,\mathrm{d}\mu^K \geq c(1-\epsilon)\exp[(\lambda_1' - 6\lambda_2'')T - 20\beta]$, with $\lambda_1' - 6\lambda_2'' > \frac{2}{5}$. This gives $a_K^2 M^2 + 24\mathsf{b} M \geq c\exp[\frac{2}{5}\sigma/a_K^2 - 20\mathsf{b}/a_K]$, hence $M \geq \exp[\frac{\sigma}{5a_K^2} - \frac{10\,\mathsf{b}}{a_K} - C]$ for sin and every $f \not\equiv 0$. For a non-constant periodic φ the same steps hold with φ in the place of sin (and $\varphi(-\,\cdot\,)$ in the place of φ if f was replaced by $-f$): for large K the fall $f(x_*)\rho_2^2/a_K$ exceeds a period of φ, so that φ takes its maximum and its minimum on the fibre within the distance r_*, $\operatorname{osc}\varphi$ replaces 2, and $\sup|\varphi|$ replaces 1 in the first inequality; this gives the exponent $\sigma/20a_K^2$ for large K.

(f): on the fields W_ζ of Lem. 16.4 the four plaquettes at b_0 stay within $\delta \leq \frac{1}{10}$ of $-\mathbf{1}$ during the flow, so $\Lambda_\sigma(W_\zeta) \geq (2-\delta)\sigma/a_K^2$, while their mass is at least $(1-\epsilon)e^{-20\beta} c_\sharp e^{-(3\lambda_1 + 6\lambda_2 + \varepsilon)\sigma/a_K^2}$; and $13(2-\delta) \geq 24.7 > 3\lambda_1 + 6\lambda_2 + \varepsilon$. □

15.1.8. *Uniformity and the order of limits.* Thm. AT holds for every limit, with constants and rates that do not depend on the subsequence. The inequalities of (A4), (A5) and (B2)(c) carry a defect, and their constants depend on ℓ; the constants of Thm. AU that are free of the volume are those of the fine field given a block field: C_e^{fib} and $3\mathsf{M}^2\Pi_e^\flat$ in (B3), in the log-Sobolev inequality of P_W, in the pointwise bound of $\Gamma_e(g_{t'})$ and in the tail under P_W, and the same two numbers at $e = n$ in (B2)(e), whose first inequality is (B3) at $t' = K - n$ integrated over the top law by the chain rule of the entropy.

The tail under μ^K in (B3) adds R_n; the first inequality of (B2)(e) keeps the entropy of the top law, which (B1) bounds at the cost C_n^{t}; and the constants, defects and thresholds of (B1) and (B2)(a)–(c), C_n^{t}, $C_n^\flat$, C_n', $\mathsf{R}_\cdot$, r_n, n_{c}, depend on ℓ. The flowed observables are taken at a fixed positive flow time; the proposed conclusion (C5) would identify the generator on them as a distribution once the flow-approximation step is completed, in the tested outer-R, inner-$\limsup_{K\to\infty}$ error sense of (C5)(d), and (C7) shows that without cutoffs the images under $\mathcal{L}_K$ of some of them are not bounded uniformly in K: for $p > 1.8637$, $\sup_K[\|\mathcal{L}_K \cos(A^a \circ \Phi_\sigma)\|_{L^p(\mu^K)} + \|\mathcal{L}_K \sin(A^a \circ \Phi_\sigma)\|_{L^p(\mu^K)}] = \infty$, so the norm is unbounded at cos or at sin, and at $\cos(\cdot + \theta)$ for all $\theta \in [0, \pi)$ but at most one, since cos and sin are linear combinations of any two of these functions.

15.2. **A uniform gauge on a box.** A comb gauge gives a bond bound proportional to the side of the box. To obtain a bound independent of that side, divide the box into cells of side comparable to the inverse square root of the plaquette bound. First fix the gauge on their edges. Extend it over each face and then each cell, keeping its already prescribed boundary values. For $SU(2)$, the required extensions can be constructed quantitatively on the unit sphere in the quaternions.

The following constants will also fix the numerical conventions in Lem. 15.7:

$$\begin{aligned} \lambda_{\mathrm{BG},1} &= \pi + 2, \qquad \rho_{\mathrm{BG},1} = 1/250, \qquad \lambda_{\mathrm{BG},2} = 2 + 12\sqrt{3}\lambda_{\mathrm{BG},1}\rho_{\mathrm{BG},1}^{-2}, \\ \lambda'_{\mathrm{BG},2} &= \lambda_{\mathrm{BG},2} + 4, \qquad \rho_{\mathrm{BG},2} = (250000\lambda'^2_{\mathrm{BG},2})^{-1}, \\ \lambda_{\mathrm{BG},3} &= 4 + 12\sqrt{3}\lambda'_{\mathrm{BG},2}\rho_{\mathrm{BG},2}^{-2}, \qquad C_{\mathrm{BG}} = 2\lambda_{\mathrm{BG},3}. \end{aligned} \tag{15.v}$$

Proposition 15.8. Let U be an $SU(2)$ configuration on the bonds of $\{0,\dots,n\}^3$, where $n \geq 0$. If $0 < \kappa \leq C_{\mathrm{BG}}^{-2}$ and $|\partial U(p) - \mathbf{1}| \leq \kappa$ at every plaquette, then some gauge u satisfies

$$|U^u(b) - \mathbf{1}| \leq C_{\mathrm{BG}}\sqrt{\kappa} \quad \text{at every bond.} \tag{15.vi}$$

The same conclusion holds on $(\mathbb{Z}/n\mathbb{Z})^3$ if $n \geq \kappa^{-1/2}$. ■

Proof. We use the operator norm, which on quaternions is their Euclidean norm, and identify $SU(2)$ with the unit sphere S^3. Unitary multiplication preserves this norm. In particular, reversing a bond does not change its distance from $\mathbf{1}$.

Avoiding a point of the sphere. If a subset $S \subset S^3$ is covered by m closed balls of radius $0 < \rho < 1/8$ and $m < (8\rho)^{-3}$, there is $q \in S^3$ such that $|z + q| > \rho$ for every $z \in S$. Indeed, put $k = \lceil (8\rho)^{-1} \rceil - 1 \geq 1$. The points

$$p_v = \frac{(1, v)}{|(1, v)|}, \qquad v \in \{-1 + 2a/k : 0 \leq a \leq k\}^3,$$

number $(k+1)^3 \geq (8\rho)^{-3} > m$. For distinct grid points v, w, the wedge product of $(1, v)$ and $(1, w)$ has norm at least $|v - w|$, whereas both vectors have norm at most 2. Consequently

$$|p_v - p_w| \geq \sin \angle((1, v), (1, w)) \geq \tfrac{1}{4}|v - w| \geq \frac{1}{2k} > 4\rho.$$

Each of the m ball centres is within distance 2ρ of at most one p_v. Choose a remaining p_v and put $q = -p_v$.

Extending a map which avoids that point. Let Y be a nonempty subset of a finite metric space (X, d). Suppose $h : Y \to S^3$ is a-Lipschitz and $|h + q| \geq \rho$. Then it extends to $H : X \to S^3$ with Lipschitz constant $4\sqrt{3}\rho^{-2}a$. To see this, use the inversion

$$I_q(z) = -q + 2\frac{z + q}{|z + q|^2}, \qquad |I_q(z) - I_q(w)| = \frac{2|z - w|}{|z + q|\,|w + q|}.$$

It is its own inverse, maps $S^3 \setminus \{-q\}$ to $q^\perp$, and maps $q^\perp$ to S^3. Its Lipschitz constants on the two sets in use are $2\rho^{-2}$ and 2, respectively. Each coordinate f of $I_q h$ in an orthonormal basis of $q^\perp$ extends by

$$F(x) = \min_{y \in Y} \{ f(y) + 2\rho^{-2} a\, d(x, y) \}.$$

The Lipschitz inequality for f shows that $F = f$ on Y; the triangle inequality shows that F has the same Lipschitz constant. The three coordinate extensions together have Lipschitz constant $2\sqrt{3}\rho^{-2}a$. Composition with I_q gives H.

The cell geometry and the comb. Put $N = \lfloor \kappa^{-1/2} \rfloor$. Then

$$N^2\kappa \leq 1, \qquad N\kappa \leq N^{-1}, \qquad N \geq \lambda_{\mathrm{BG},3}, \qquad N^{-1} \leq 2\sqrt{\kappa}.$$

For a coordinate rectangle or box with side lengths between N and $2N - 1$, any two boundary sites are joined along its boundary by a path of length at most three times their ℓ^1 distance. For sites on one face, or on intersecting faces, a monotone path has exactly that distance. For opposite faces, use either one of the two intervening faces. If their separating direction is μ and the chosen other direction is ν, the shorter detour costs at most $n_\nu + n_\mu$ in those directions; this is at most $3n_\mu$, since

$n_\nu < 2n_\mu$. The remaining coordinate contributes its ordinary difference. The rectangle proof is the same argument with two coordinates.

In a cell with lowest corner y, let $r_y(x)$ be the transport from y to x, first in direction 1, then 2, then 3. For $b = \langle x, x + e_\mu\rangle$ put $m_b = r_y(x)U(b)r_y(x + e_\mu)^{-1}$. Moving the last step through the later directions in the comb exchanges one plaquette at a time. The holonomy of a strip of k plaquettes is a product of k unitary conjugates of those plaquettes, so its distance from $\mathbf{1}$ is at most $k\kappa$. Thus, writing $x - y = (k_1, k_2, k_3)$,

$$|m_b - \mathbf{1}| \le \begin{cases} (k_2 + k_3)\kappa, & \mu = 1, \\ k_3\kappa, & \mu = 2, \\ 0, & \mu = 3. \end{cases}$$

This is the comb calculation of Lem. YM3_I:14.4, with the coordinate order reversed. It proves the estimate on an ordinary coordinate box and on the lift of a periodic cell, without any condition on the ambient torus side. On a face the bound is $2/N$, and on a cell it is $4/N$. If $n < N$ in the nonperiodic case, the same comb on the whole box gives $2n\kappa \le 4\sqrt{\kappa}$, which proves the assertion.

Suppose now $n \ge N$. Divide each coordinate interval into lengths between N and $2N - 1$: take $t_i = iN$ for $0 \le i < \lfloor n/N\rfloor$ and the last endpoint n. Their products give nodes, edges, faces and cells. Each new piece meets the union of lower-dimensional pieces exactly in its boundary; different interiors are disjoint. Every bond is contained in a cell. We can therefore define the gauge successively on nodes, edges, faces and cells.

Edges. Set $u = \mathbf{1}$ at the nodes. On an edge from y to $y' = y + n_c e_\mu$, write $U([y, y']) = e^{iY}$ with $|Y| \le \pi$, and set

$$u(y + ke_\mu) = e^{-ikY/n_c}U([y, y + ke_\mu]), \qquad 0 \le k \le n_c.$$

Both endpoint values are $\mathbf{1}$. At an edge bond the transformed field is e^{iY/n_c}, so its distance from $\mathbf{1}$ is at most π/N.

For the next two steps, write $h = ur_y^{-1}$ on the prescribed part of a piece and $w_b = U^u(b)$. Since $h(b_-)m_b h(b_+)^{-1} = w_b$, we have

$$\begin{aligned} |h(b_+) - h(b_-)| &\le |w_b - \mathbf{1}| + |m_b - \mathbf{1}|, \\ |w_b - \mathbf{1}| &\le |m_b - \mathbf{1}| + |h(b_+) - h(b_-)|. \end{aligned} \tag{15.vii}$$

Faces. On the boundary of a face, the bond differences of h are at most $\lambda_{\mathrm{BG},1}/N$; the boundary-path bound makes h $3\lambda_{\mathrm{BG},1}/N$-Lipschitz in ℓ^1 distance. The boundary cycle has at most $8N$ bonds. Sample it at intervals $s_1 = \lfloor \rho_{\mathrm{BG},1}N/\lambda_{\mathrm{BG},1}\rfloor$. The constants ensure $s_1 \ge \rho_{\mathrm{BG},1}N/(2\lambda_{\mathrm{BG},1}) \ge 1$. The image is covered by balls of radius $\rho_{\mathrm{BG},1}$ whose number is at most

$$16\lambda_{\mathrm{BG},1}/\rho_{\mathrm{BG},1} + 1 \le 17\lambda_{\mathrm{BG},1}/\rho_{\mathrm{BG},1} < 21853 < (8\rho_{\mathrm{BG},1})^{-3}.$$

The avoidance and extension arguments give an extension H over the face with bond differences bounded by $12\sqrt{3}\lambda_{\mathrm{BG},1}\rho_{\mathrm{BG},1}^{-2}/N$. Set $u = Hr_y$ there. It retains the prescribed edge values, and eq. 15.vii gives $|U^u(b) - \mathbf{1}| \le \lambda_{\mathrm{BG},2}/N$ on every face bond.

Cells. On the boundary of a cell, the bond differences of h are at most $\lambda'_{\mathrm{BG},2}/N$, and its ℓ^1 Lipschitz constant is $3\lambda'_{\mathrm{BG},2}/N$. On each face sample both coordinates at multiples of $s_2 = \lfloor \rho_{\mathrm{BG},2}N/(2\lambda'_{\mathrm{BG},2})\rfloor$. Here $s_2 \ge \rho_{\mathrm{BG},2}N/(4\lambda'_{\mathrm{BG},2}) \ge 1$. Lowering the two coordinates to these multiples uses fewer than $2s_2$ steps, so the image lies within $\rho_{\mathrm{BG},2}$ of a sampled value. The number of samples on all six faces is at most

$$m = 486\lambda'^2_{\mathrm{BG},2}\rho_{\mathrm{BG},2}^{-2}, \qquad m(8\rho_{\mathrm{BG},2})^3 \le 248832/250000 < 1.$$

The same extension gives H over the cell, with bond differences at most $12\sqrt{3}\lambda'_{\mathrm{BG},2}\rho_{\mathrm{BG},2}^{-2}/N$. Again set $u = Hr_y$. The boundary values agree, and eq. 15.vii gives

$$|U^u(b) - \mathbf{1}| \le \lambda_{\mathrm{BG},3}/N \le C_{\mathrm{BG}}\sqrt{\kappa}.$$

Every bond belongs to a cell, proving the box assertion.

The periodic lattice. Divide each coordinate circle of length n into the same integer lengths, now identifying its last endpoint with its first. Work on each piece in its coordinate lift. Put $u = \mathbf{1}$ at the periodic nodes and perform the edge, face and cell constructions just given once for each periodic piece. The edge formula agrees at both endpoints even when an edge is a full circle. At every later stage the boundary values are the values of the already defined periodic function u. The auxiliary function $h = ur_y^{-1}$ is defined on the lifted boundary; it need not have equal values on opposite lifted faces, since the comb transports there may differ. After extension, however, Hr_y equals the prescribed u at every boundary point. Thus those values agree at all identified points. No new interior point has two preimages, even when a cell spans an entire coordinate circle. The extensions therefore descend to a single gauge on the torus. Every periodic bond has a lift in one cell, with the same plaquette and comb bounds as above. This proves the periodic assertion. □

For completeness, this also gives the first component assertion of Lem. 15.7(a). Let $0 < \kappa' \le (8\pi C_{\mathrm{BG}})^{-2}$ and use the periodic gauge. Write $V^u = e^{iA}$, choosing the principal logarithm at positive bonds and extending by reversal. Then

$$|A| \le \alpha := \tfrac{\pi}{2} C_{\mathrm{BG}}\sqrt{\kappa'} \le 1/16.$$

For the product of four exponentials around a plaquette, Taylor's formula in t gives $|\partial e^{itA}(p) - \mathbf{1} - it\,\mathrm{d}A(p)| \le 8\alpha^2$: the norm of its second derivative is bounded by $(\sum_{b\in\partial p}|A(b)|)^2 \le 16\alpha^2$. Here $dA(p)$ is the unnormalised oriented sum around p. In particular, the bound $10\alpha^2$ used in Lem. 15.7 holds. Applying it at t and at 1 yields

$$|\partial e^{itA}(p) - \mathbf{1}| \le |\partial V^u(p) - \mathbf{1}| + 20\alpha^2 \le (1 + 5\pi^2 C_{\mathrm{BG}}^2)\kappa' < C''\kappa', \qquad 0 \le t \le 1.$$

The gauge group is a finite product of connected copies of $SU(2)$, so a path of gauge transformations joins V to V^u without changing its plaquette norms. Joining this path to e^{itA} places V in the component of $\mathbf{1}$ in $\{\varpi < C''\kappa'\}$. Taking $\kappa' = r/C''$ gives the shell assertion there. This single-scale construction does not supply the separate multilevel estimate (Multiscale gauge).

15.3. **Response and one-step conventions.** We record the response bound needed when differentiating the normalised measure of a step. Work on a constant torus sequence, at level $j \ge 1$, with the real regular background, geometric thresholds and datum ball of §5.3. Lengths in the following lemma are in level-j units, so the bare spacing is $\eta_j = L^{-j}$ and every terminal pillar has scale 1. Let d_j be the periodic ℓ^1 distance on the terminal lattice. In this setting it is the weighted path distance in equation YM3_I:(838). For a field X on its positive bonds put

$$m_y(X) = \sum_{c>0} e^{-\delta_0 d_j(y,c_-)}|X(c)|, \qquad S_0 = \left(\frac{1+e^{-\delta_0}}{1-e^{-\delta_0}}\right)^3, \qquad \mathsf{S} = 3S_0,$$

where $\delta_0 > 0$ is no larger than the rate in that equation.

Lemma 15.9. There is a constant B, depending only on the fixed construction constants, such that the response $\mathcal{H}_U$ obeys

$$\max\{|\mathcal{H}_U(X)(b)|, |\nabla_U\mathcal{H}_U(X)(b)|\} \le \mathsf{B}m_y(X)$$

on its datum ball, whenever b has an endpoint in the pillar of y. At zero datum, put

$$h = D\mathcal{H}_U(0)X, \qquad j_X = \frac{d}{ds}J^{(j)}(e^{i\eta_j\mathcal{H}_U(sX)}U)\Big|_{s=0}.$$

Then

$$\max\{|h(b)|, |\nabla_U h(b)|, |j_X(b)|\} \le \mathsf{B} m_y(X), \qquad \sum_y m_y(X)^2 \le \mathsf{S}^2 \sum_{c>0} |X(c)|^2. \tag{15.viii}$$

All constants are independent of j and of the torus side. ■

Proof. For one input bond, equation YM3_I:(838) bounds h, $\nabla_U h$ and $D_U^{\mathsf{T}} D_U h$ with exponential decay. Sum this bound over the input bonds. Integrating the same estimate along sX, $0 \le s \le 1$, proves the first assertion, since $\mathcal{H}_U(0) = 0$ and the datum ball is convex.

The current requires one further identity. Let $F(A)$ be the local Wilson action at $e^{i\eta_j A}U$ and set $f(z) = (e^z - 1)/z$, with its power-series value at zero. Prop. YM3_I:13.1 gives, on a complete plaquette star,

$$J^{(j)}(e^{i\eta_j A}U)(b) = f(-i\eta_j \operatorname{ad}_{A(b)})^{-1} F'(A)(b), \qquad F''(0) = D_U^{\mathsf{T}} D_U + \Delta'.$$

Since $f(z)^{-1} = 1 - z/2 + O(z^2)$, differentiation at $A = 0$ gives

$$j_X(b) = (D_U^{\mathsf{T}} D_U h)(b) + (\Delta' h)(b) + \tfrac{i\eta_j}{2}[h(b), J^{(j)}(U)(b)].$$

All stars are complete on the torus. The same proposition bounds $\Delta' h$ by $C\varepsilon$ times the supremum of h on this fixed star, where ε is the background regularity radius. Regularity also gives $|J^{(j)}(U)| \le \varepsilon$. The star meets only pillars at a fixed distance from y; translating the exponential weight over that distance costs a fixed factor. Together with $\eta_j \le 1$ and $|[h, J]| \le 2|h||J|$, the three terms therefore have the asserted bound, after increasing B.

Finally the sum of $e^{-\delta_0 d_j(y,z)}$ over a periodic lattice is at most S_0: choose least coordinate representatives and bound their sum by the sum over $\mathbb{Z}^3$. There are three positive bonds starting at each site. The nonnegative kernel defining m_y has row sums at most $3S_0$ and column sums at most S_0. Cauchy–Schwarz, first with this kernel and then after summing in y, gives an operator norm at most $\sqrt{3}S_0 \le \mathsf{S}$. □

The gauge equation used with this lemma is exactly equation YM3_I:(837). Thus the projection denoted by R^{La} in the dynamics section is the companion's R_{m}: the projection onto $D_U^{\mathsf{T}} D_U \mathcal{N}$, where $\mathcal{N}$ consists of site fields with zero prescribed site averages. This is the mean-Landau choice of Thm. YM3_I:K.

15.3.1. *The local input and its global sum.* At height k, the actual input is the one in Section YM3_I:52.4 and Thm. YM3_I:W. Keep its geometric thresholds, admissible radius system, shift hypotheses, octet condition when $k \ge 2$, numerical ceilings and array bound $N \le N_*$. In particular, $E_0 > 0$ is the fixed size of the output family there; it is chosen before the height, lattice and input array. For the density identity, keep in addition the whole-data response neighbourhood, full-torus core membership and bounded Borel integrand required in that theorem.

A family in the fourth place means the fourth potential of Def. YM3_I:52.1. Its members depend on the represented pair, are gauge invariant, vanish at $B = 0$, and are jointly holomorphic on each core slice and the field polydisc $|B| < C_{\mathcal{A}}\mathfrak{b}(\mathsf{t})$. With additional parameters, joint holomorphy is required in those parameters too. Its size is equation YM3_I:(888). For a parameter space consisting of one point this is the same definition without a parameter.

The phrase that this family *sums to S at W* has the following precise meaning. The averaged background is real, admissible, and has invertible pivots; the real cutoff chart is defined throughout its box, every background over a point of that chart exists, and P_W belongs to the full-torus core. The gauge-invariant function S is defined on both critical orbits. For every real cutoff field,

$$\sum_Y \mathbb{V}'(Y; P_W|_Y, B|_Y) = S(U_k(V_W(B))) - S(U_{k+1}(W)). \tag{15.ix}$$

These are the background and chart conditions in §5.3.1. For a bare-array potential the identity is equation YM3_I:(882). For a source it is the additional sum condition imposed at its entry into the step.

The block field of a triple is $V[P] = \mathcal{M}^k(e^{i\eta_{k+1}A'}U)$ on the level-k bonds of its domain, with the actual complex averaging domains retained. The fine-cube gauge form $\widehat{\mathsf{G}}_\square(P)$ is evaluated at this field by equation YM3_I:(823). Equivalently it is the quadratic form obtained from $\mathfrak{t}\Psi^{\mathrm{G}}(\square; P, B)$ in the proof of Prop. YM3_I:51.4. Def. YM3_I:52.1, item 2, records its pair dependence, slice holomorphy, covariance and compatibility with restrictions; its sum on the full torus is equation YM3_I:(828). At real data the inverse whole covariance differs from its real reference form, with floor γ_{r} and ceiling c_{r}, by at most $\frac{1}{2}u_{\mathrm{G}}(LM)^{-3}$. These are the actual constants of Thm. YM3_I:U, used with the choice $u_{\mathrm{G}} = 4C_{\mathrm{Str}}$ and the radius inequalities of Section YM3_I:52.4. Completeness is Def. YM3_I:47.2, condition (St4).

The expansion numbers $w_0, \mathsf{C}_{\mathrm{ex}}, c_G, c_{\mathrm{v}}, c_{\mathrm{w}}, \lambda_1, \lambda_V, w_\star$ are those defined immediately before and in equation YM3_I:(705); in particular $\mathsf{C}_{\mathrm{ex}} \geq 8e^8$. Their one-step substitution is equation YM3_I:(889), including $\lambda_{\mathrm{out}} = (1 + 4\beta_{\mathrm{s}})\kappa$ and $(LM)^3 u = 3 + c_{\mathrm{r}} + u_{\mathrm{G}}$.

16. Positive-time gradient flow

The deterministic Wilson flow Φ_σ decreases the lattice action. It is related to the Langevin generator, but its continuum identification is a separate problem. At positive time the intended limiting object is a connection modulo periodic gauge transformations, specified by its holonomies.

Sketch O (The proposed continuum flow). The sketch describes a positive-time Yang–Mills gradient flow from almost every block-field thread, with strongly convergent flowed gauge-invariant observables. Its identification requires compactness modulo gauges, passage to the limiting equation, recovery of the initial holonomies and uniqueness. The estimates below provide the preparatory bounds; these four steps remain to be established. ■□

Sketch of the argument. For a smooth lattice observable G, put $H_K^\sigma = G \circ \Phi_\sigma$ and $\Delta_K = a_K^{-1} \sum_{b,\alpha} (\partial_b^\alpha)^2$, the second-order part of the Langevin generator. The finite-dimensional identity

$$\mathcal{L}_K H_K^\sigma = (b\partial_\sigma + \Delta_K) H_K^\sigma$$

is exact. The action drift in the stochastic generator is cancelled by moving along the deterministic flow. Integrating against the stationary measure gives the corresponding identity for means; multiplying by a test observable gives its Dirichlet-form version.

The small-field chain decomposes a gauge-fixed potential into layers. Conditional estimates control the layer means and the part of their colour covariance that is not scalar. Their summation gives rough-field moments. Deterministic heat estimates then control the flow at positive times, producing uniform moments of the flowed curvature and current and high-probability regularity sets. Sufficiently deep layers satisfy the admissibility bounds needed for these estimates.

For the continuum identification, start with minimal backgrounds over converging coarse data. One must choose periodic gauges so that their positive-time potentials have convergent subsequences, locally uniformly in time. Their limit must satisfy the gauge-fixed mild equation

$$\partial_\sigma \mathfrak{A} = -\mathrm{d}_{\mathfrak{A}}^* F_{\mathfrak{A}} + \mathrm{d}_{\mathfrak{A}}(\partial_\nu \mathfrak{A}_\nu),$$

recover the prescribed smoothed holonomies as $\sigma \downarrow 0$, and be unique modulo gauge with those initial holonomies. The source explicitly leaves these linked claims as an additional approximation step. The derivative and moment estimates do not alone complete it.

There is another difficulty even after identifying the flowed observables. Positive flow time does not automatically put them in the domain of the limiting stochastic generator. Some bounded flowed observables have $\|\mathcal{L}_K H_K^\sigma\|_2$ growing exponentially in a_K^{-2}. Insert invariant regularization cutoffs $\chi_{K,R}$ whose lost mass and Dirichlet cost tend to zero as $R \to \infty$. First pass $K \to \infty$ at fixed

R; only afterwards remove R. For resolvent test functions this gives a tested generator identity, and after integrating in σ a distributional one. An unregularized operator-domain claim requires a separate uniform generator bound, which those examples show cannot be assumed for every flowed observable.

16.1. **The identification that the flow sketch requires.** Deterministic smoothing and stochastic evolution act on the same finite lattice, but their continuum limits require different arguments. Let Φ_σ be the Wilson gradient flow and let G be a smooth observable of the lattice field. Write $H^\sigma = G \circ \Phi_\sigma$. With the bond Laplacian $\Delta_K = a_K^{-1} \sum_{b,\alpha} (\partial_b^\alpha)^2$, the finite-dimensional chain rule gives

$$\mathcal{L}_K H^\sigma = (b\partial_\sigma + \Delta_K) H^\sigma.$$

Stationarity and integration by parts therefore imply

$$b\partial_\sigma \langle H^\sigma \rangle_{\mu^K} = -\langle \Delta_K H^\sigma \rangle_{\mu^K}.$$

These are finite-cutoff identities. They describe how flow time enters the observable, without yet identifying a random continuum connection or its deterministic flow.

The prospective continuum flow is most naturally specified modulo periodic gauge transformations. For a fixed admitted block depth D and block field V, choose minimal backgrounds at finer and finer cutoffs over V. The unflowed loop comparison identifies limiting holonomies $w_C^D(V)$ for fixed closed paths C. Positive-time regularity estimates then control the smoothed backgrounds in gauges. To finish the identification, one must obtain compactness modulo periodic gauges, pass the lattice equation to the continuum equation, recover these initial holonomies as flow time decreases to zero, and establish uniqueness of the resulting orbit-valued flow. The detailed estimates below provide the preparatory bounds for these four steps, which remain to be established.

The proposed limiting equation, in a local gauge representative $\mathfrak{A}_s$, is

$$\partial_s \mathfrak{A} = -\mathrm{d}_{\mathfrak{A}}^* F_{\mathfrak{A}} + \mathrm{d}_{\mathfrak{A}}(\partial_\nu \mathfrak{A}_\nu).$$

The last term is an infinitesimal gauge transformation. Thus the object to identify is a path of gauge orbits, with local representatives satisfying this equation, rather than one global choice of potentials. The initial condition is expressed by $W_C(\mathfrak{A}_s) \to w_C^D(V)$ as $s \downarrow 0$. Neither the finite-lattice chain rule nor positive-time moment bounds alone establish uniqueness for that initial condition.

A further limit issue concerns the generator. Smoothness at positive flow time does not give a bound for $\|\mathcal{L}_K H_K^\sigma\|_2$ uniform in K. The source gives bounded flowed observables for which that norm grows exponentially in a_K^{-2}. One therefore tests the generator identity with regularity cutoffs whose Dirichlet cost vanishes. First pass to the continuum with a fixed regularization; then remove that regularization. The resulting identity has the tested or distributional meaning established by this procedure. It cannot be replaced by an assertion that every bounded flowed observable lies in the domain of the limiting generator.

16.2. **The background-flow identification.**
Approximation of the fixed-depth background flow. Fix $0 < \alpha < 1$. For D sufficiently large put $\mathcal{K}_D = \{V \in \Omega_D : \varpi(V) \le \theta_D\}$, and let $U_{K,D}(V)$ be any member of the minimal orbit over V on $\Lambda_{K,\ell}$, $K > D$. The unflowed comparison in the proof of Prop. 16.1(b) defines $w_C^D(V) = \lim_{K\to\infty} W_C(U_{K,D}(V))$ for each closed lattice path C. The remaining approximation step, referred to as *(Flow approximation)*, is to prove the following compactness and identification statement. For each fixed D, every sequence $K_\nu \to \infty$, $V_\nu \to V$ in $\mathcal{K}_D$ and arbitrary minimal backgrounds U_ν over V_ν should have a subsequence whose positive-time flows converge modulo periodic gauges, locally uniformly in time in $\mathcal{O}^\alpha$, to an orbit-valued flow line. More explicitly, on each compact positive time interval, periodic gauges give real bond potentials $A_\nu(s)$ representing $\Phi_s U_\nu$ as $e^{ia_{K_\nu} A_\nu(s)}$ and

local representatives $\mathfrak{A}_s$ of the limit for which

$$\sup_{\tau\le s\le T}\max_b|A_\nu(s,b)-\mathfrak{A}_{s,\mu(b)}(x_b)|\longrightarrow 0,\qquad 0<\tau<T<\infty.$$

Here x_b is the bond midpoint and $\mu(b)$ its direction. The limit must have local mild representatives satisfying

$$\partial_s\mathfrak{A}=-\mathrm{d}^*_{\mathfrak{A}}F_{\mathfrak{A}}+\mathrm{d}_{\mathfrak{A}}(\partial_\nu\mathfrak{A}_\nu),\qquad \lim_{s\downarrow0}W_C(\mathfrak{A}_s)=w_C^D(V),$$

and one must prove uniqueness of this orbit-valued flow line from those initial holonomies. These are the compactness and identification tasks of the sketch. The positive-time derivative bounds used below are proved separately by the single-scale gauge and the short-time regularisation in the proof of Prop. 16.1(c).

Detailed sketch AV (Flowed estimates and the proposed continuum identification). Assume (E) and $0<\kappa<\frac16$; (C1) holds for all K, m, $\beta>0$, with $\mathsf{b}=a_K\beta$. (C2)(b), (C2)(c) and (C3)–(C7), which read the moments, the good sets or the field $\mathbf{B}_K$ of Prop. 16.1(a), use the consequence (Adm) of (E) proved in Prop. 16.1: every top depth from some depth on, a depth free of K, is admissible in the sense of Prop. 16.1. (C1) and (C2)(a) are free of it. The continuum identifications in (C3) and the part of (C5)(d) involving G_σ remain proposed conclusions until the approximation step above is completed. The assertions in (C7) use the nonlinear defect estimates of Lem. 16.4.

(C1) For $G\in C^\infty(\Omega_K)$, $H^\sigma=G\circ\Phi_\sigma$ and $\sigma\ge0$: $\mathcal{L}_KH^\sigma=(\mathsf{b}\,\partial_\sigma+\Delta_K)H^\sigma$. Hence $\mathsf{b}\,\partial_\sigma\langle H^\sigma\rangle_{\mu^K}=-\langle\Delta_KH^\sigma\rangle_{\mu^K}$; for $\chi\in D(\mathcal{E}_K)\cap L^\infty$, $\int\chi\,(\mathsf{b}\partial_\sigma+\Delta_K)H^\sigma\,\mathrm{d}\mu^K=-\int\Gamma_K(\chi,H^\sigma)\,\mathrm{d}\mu^K=-\int(1-\chi)\,\mathcal{L}_KH^\sigma\,\mathrm{d}\mu^K$; for $t\ge0$, $P_t^KH^\sigma=H^{\sigma+\mathsf{b}t}+\int_0^tP_s^K\Delta_KH^{\sigma+\mathsf{b}(t-s)}\,\mathrm{d}s$; and for the stationary process and $0\le r\le t$, $H^{\sigma+\mathsf{b}(t-r)}(U_r)-H^{\sigma+\mathsf{b}t}(U_0)-\int_0^r(\Delta_KH^{\sigma+\mathsf{b}(t-s)})(U_s)\,\mathrm{d}s$ is a martingale with the bracket $2\int_0^r\Gamma_K(H^{\sigma+\mathsf{b}(t-s)})(U_s)\,\mathrm{d}s$: in the frame that moves along $\sigma+\mathsf{b}t=\text{const}$ the drift of the action is gone.

(C2) (a) For $\bar C\ge1$, $0<\bar s\le1$ there is K_0 such that for every flowed observable, $0<\sigma\le1$ and $K\ge K_0$: $\Gamma_K(H_K^\sigma)\le C_\Gamma$ and $|\partial_\sigma H_K^\sigma|\le C_\partial$ on $\mathsf{O}_K(\bar C,\bar s)$, with C_Γ,C_∂ depending on $\bar C,\bar s,\ell,\kappa,\sigma$ and the norms of $\nabla\varphi$, f_i, ∇f_i, bounded for σ in compact subsets of $(0,1]$; for $\sigma\le\bar s$ this holds for every K. (b) For $0<\sigma\le1$, $1\le p<\infty$ and all K: $\|\partial_\sigma H_K^\sigma\|_{L^p(\mu^K)}\le C_p(\sigma)\sum_i\|\partial_i\varphi\|_\infty(2\|f_i\|_\infty+4\|\nabla f_i\|_\infty)$, $C_p(\sigma)$ bounded on compact subsets of $(0,1]$; so $\|H_K^\sigma-H_K^{\sigma'}\|_{L^p(\mu^K)}\le C|\sigma-\sigma'|$ locally uniformly in $\sigma,\sigma'\in(0,1]$ and in K, and, by (C1), $\sup_K\|\mathcal{L}_KH_K^\sigma\|_{L^p(\mu^K)}<\infty$ if and only if $\sup_K\|\Delta_KH_K^\sigma\|_{L^p(\mu^K)}<\infty$.

(c) For compact $I\subset(0,1]$, $1\le q<\infty$: $\sup_K\|\partial_\sigma H_K^{\sigma''}-\partial_\sigma H_K^{\sigma'}\|_{L^q(\mu^K)}\le C_I(q)|\sigma''-\sigma'|^{1/2}$ for $\sigma',\sigma''\in I$.

(C3) For every flowed observable and $\sigma>0$, $H_K^\sigma\to\mathsf{G}_\sigma$ strongly along the full sequence of cutoffs, where

$$\mathsf{G}_\sigma(Y)=\varphi(\mathcal{O}^0(f_1)(\mathfrak{F}(Y)_\sigma),\dots,\mathcal{O}^0(f_r)(\mathfrak{F}(Y)_\sigma)),$$

$\mathfrak{F}$ the map of Prop. 16.1(b); the same holds for real Lipschitz f_i and bounded Lipschitz φ, and $\sigma\mapsto\mathsf{G}_\sigma$ is locally Lipschitz from $(0,1]$ into $L^2(\mu)$. For $0<\sigma<1$ there is $\dot{\mathsf{G}}_\sigma\in L^2_{\mathrm{inv}}$ with $\int\Psi_K\,\partial_\sigma H_K^\sigma\,\mathrm{d}\mu^K\to\int\Psi\,\dot{\mathsf{G}}_\sigma\,\mathrm{d}\mu$ along the full sequence for every invariant cylinder function Ψ; $\sigma\mapsto\int\Psi\dot{\mathsf{G}}_\sigma\,\mathrm{d}\mu$ is locally Hölder continuous of exponent $\frac12$ and is the derivative of the continuously differentiable function $\sigma\mapsto\int\Psi\mathsf{G}_\sigma\,\mathrm{d}\mu$.

(C4) *Regularisation and condenser cutoffs.* There are $R_\flat\ge1$ and, for $R\ge R_\flat$, numbers $\bar s(R)\in(0,1]$, $\bar K(R)$, $\bar\varkappa(R)>0$ with $R^q\bar\varkappa(R)\to0$ for every q, and for $K\ge\bar K(R)$ invariant $\chi_{K,R}\in C^\infty(\Omega_K)$ with values in $[0,1]$ such that $\chi_{K,R}=1$ on $\{\mathsf{G}^*\le R\}$ and $\chi_{K,R}=0$ off $\{\mathsf{G}^*\le\frac52R\}\subset\mathsf{O}_K(8R,\bar s(R))$, G^* the functional of Step 8; $\mathcal{E}_K(\chi_{K,R})+\int(1-\chi_{K,R})^2\,\mathrm{d}\mu^K\le\bar\varkappa(R)$; $\sup_{K\ge\bar K(R)}\int\Gamma_K(\chi_{K,R})^p d\mu^K<\infty$ for every $p<\infty$; and $\mathbf{1}_{\{\chi_{K,R}>0\}}\le\chi_{K,R'}$ for $R'\ge\frac52R$, $K\ge\bar K(R')$.

Let $w_{K,R}$ be the potential of Lem. 16.5 for $\eta = 1 - \chi_{K,R}$, and $\chi^{\mathrm{c}}_{K,R} = 1 - w_{K,R}$, the condenser cutoff. Then $\chi^{\mathrm{c}}_{K,R} \in D(\mathcal{E}_K)$ is invariant, takes values in $[0,1]$, is 1 where $\chi_{K,R} = 1$ and 0 where $\chi_{K,R} = 0$, has $\Gamma_K(\chi^{\mathrm{c}}_{K,R}, \cdot) = 0$ off the shell $\{0 < \chi_{K,R} < 1\}$, $\mathcal{E}_K(\chi^{\mathrm{c}}_{K,R}) + \int (1 - \chi^{\mathrm{c}}_{K,R})^2 \,\mathrm{d}\mu^K \le \bar{\varkappa}(R)$, and $|\int \Gamma_K(\chi^{\mathrm{c}}_{K,R}, \psi)\,\mathrm{d}\mu^K| \le \operatorname{osc}(\psi)\,\bar{\varkappa}(R)$ for $\psi \in D(\mathcal{E}_K) \cap L^\infty$. A second family of smooth invariant cutoffs, defined without choosing a gauge, is given by Prop. 16.6.

(C5) *Static invariance and the generator.* (a) For every flowed observable, $\sigma > 0$, $R \ge R_\flat$ and $K \ge \bar{K}(R)$,

$$\Big| \int \chi^{\mathrm{c}}_{K,R}\, (\mathsf{b}\,\partial_\sigma + \Delta_K) H^\sigma_K \,\mathrm{d}\mu^K \Big| \le \operatorname{osc}(\varphi)\,\bar{\varkappa}(R);$$

for $0 < \sigma \le 1$, with $C_\partial = \sup_K \|\partial_\sigma H^\sigma_K\|_{L^2(\mu^K)}$, $|\int \chi^{\mathrm{c}}_{K,R} \Delta_K H^\sigma_K \,\mathrm{d}\mu^K + \mathsf{b}\,\partial_\sigma \langle H^\sigma_K \rangle_{\mu^K}| \le \operatorname{osc}(\varphi)\bar{\varkappa}(R) + \mathsf{b}\, C_\partial\, \bar{\varkappa}(R)^{1/2}$; and $|\int \chi^{\mathrm{c}}_{K,R} \mathcal{L}_K H \,\mathrm{d}\mu^K| \le \operatorname{osc}(H)\bar{\varkappa}(R)$ for every $H \in C^2(\Omega_K)$. (b) With $q(u) = 3u^2 - 2u^3$, $q(\chi^{\mathrm{c}}_{K,R}) \in D(\mathcal{E}_K)$ is invariant, takes values in $[0,1]$, is 1 where $\chi_{K,R} = 1$ and 0 where $\chi_{K,R} = 0$, has $\mathcal{E}_K(q(\chi^{\mathrm{c}}_{K,R})) \le \frac{9}{4}\bar{\varkappa}(R)$, and its generator is a function: $\int q(\chi^{\mathrm{c}}_{K,R})\, \mathcal{L}_K H \,\mathrm{d}\mu^K = \int H\, \mathsf{I}_{K,R} \,\mathrm{d}\mu^K$ for $H \in C^2(\Omega_K)$, $\|\mathsf{I}_{K,R}\|_{L^1(\mu^K)} \le \frac{15}{2}\bar{\varkappa}(R)$. (c) For $H, H' \in C^2(\Omega_K)$,

$$\begin{aligned} \Big| \int \chi^{\mathrm{c}}_{K,R}\, (H' \mathcal{L}_K H - H \mathcal{L}_K H') \,\mathrm{d}\mu^K \Big| &\le \operatorname{osc}(HH')\,\bar{\varkappa}(R) \\ &\quad + 2 \sup|H|\, \bar{\varkappa}(R)^{1/2} \Big(\int_{\{0 < \chi_{K,R} < 1\}} \Gamma_K(H') \,\mathrm{d}\mu^K \Big)^{1/2}. \end{aligned}$$

(d) Let (K_j) be a limit and $F \in D(A_\star)$ with a test sequence, that is, invariant $F_j \in D(\mathcal{L}_{K_j})$ with $\sup_j \|F_j\|_\infty < \infty$, $F_j \to F$ and $\mathcal{L}_{K_j} F_j \to -A_\star F$ strongly; test sequences exist for $F = \alpha(\alpha + A_\star)^{-1} W$, $\alpha > 0$, whenever W is the strong limit of uniformly bounded invariant W_j (for instance an invariant cylinder function or a G_σ). Then for $R \ge R_\flat$ with $\bar{\varkappa}(R) \le 1$

$$\limsup_{j \to \infty} \Big| \int \chi^{\mathrm{c}}_{K_j,R} F_j\, (\mathsf{b}\,\partial_\sigma + \Delta_{K_j}) H^\sigma_{K_j} \,\mathrm{d}\mu^{K_j} + \int (A_\star F)\, \mathsf{G}_\sigma \,\mathrm{d}\mu \Big| \le c_F\, \bar{\varkappa}(R)^{1/2},$$

$c_F = \sup|\varphi| \sup_j [2\|F_j\|_\infty + 2\mathcal{E}_{K_j}(F_j)^{1/2} + \|\mathcal{L}_{K_j} F_j\|_{L^2}]$: the displayed tested error tends to zero as $R \to \infty$. For every $\tau \in C^\infty_c(0,1)$ the corresponding distributional statement is

$$\begin{aligned} \lim_{R \to \infty} \limsup_{j \to \infty} \Big| \int_0^1 \tau(\sigma) \int \chi^{\mathrm{c}}_{K_j,R} F_j\, \Delta_{K_j} H^\sigma_{K_j} \,\mathrm{d}\mu^{K_j} \,\mathrm{d}\sigma \\ + \int_0^1 [\tau(\sigma)(A_\star F, \mathsf{G}_\sigma) - \mathsf{b}\,\tau'(\sigma)(F, \mathsf{G}_\sigma)] \,\mathrm{d}\sigma \Big| = 0. \end{aligned}$$

For two flowed observables H_K, H'_K at σ, σ' with limits G, G' and $\alpha > 0$,

$$\begin{aligned} \lim_R \limsup_j \Big| \int \chi^{\mathrm{c}}_{K_j,R} [\alpha(\alpha - \mathcal{L}_{K_j})^{-1} H'_{K_j}]\, \mathcal{L}_{K_j} H_{K_j} \,\mathrm{d}\mu^{K_j} \\ + (\alpha A_\star (\alpha + A_\star)^{-1} \mathsf{G}', \mathsf{G}) \Big| = 0, \end{aligned}$$

a limit that is symmetric in the two observables.

If $C_1 = \liminf_R \limsup_j \int \chi^{\mathrm{c}}_{K_j,R} |\mathcal{L}_{K_j} H^\sigma_{K_j}| \,\mathrm{d}\mu^{K_j} < \infty$, then $\mathsf{G}_\sigma \in D(\mathcal{E}_\star)$ and $\mathcal{E}_\star(\mathsf{G}_\sigma) \le \sup|\varphi|\, C_1$; if $C_2 = \liminf_R \limsup_j \|\chi^{\mathrm{c}}_{K_j,R} \mathcal{L}_{K_j} H^\sigma_{K_j}\|_{L^2(\mu^{K_j})} < \infty$, then $\mathsf{G}_\sigma \in D(A_\star)$ and $\|A_\star \mathsf{G}_\sigma\| \le C_2$. Without a cutoff, the criterion of (A3) gives $\mathsf{G}_\sigma \in D(A_\star)$, with $\mathcal{L}_{K_j} H^\sigma_{K_j} \to -A_\star \mathsf{G}_\sigma$ weakly and $\mathcal{E}_{K_j}(H^\sigma_{K_j}) \to \mathcal{E}_\star(\mathsf{G}_\sigma)$, whenever $\sup_K \|\mathcal{L}_K H^\sigma_K\|_{L^2(\mu^K)} < \infty$; this fails at the members of (C7)(b), (c), (e).

(C6) *Lower energy bounds.* There is $p_* > 0$, depending only on the constants of the construction, and there are $\sigma_*, c_* > 0$, $S_* < \infty$ such that for $0 < \sigma \le \sigma_*$, every $\varphi \in C^1(\mathbb{R})$ and every invariant Borel $\chi_K : \Omega_K \to [0,1]$ with $\mu^K(\chi_K < 1) \le p_*/8$,

$$\liminf_{K\to\infty} \int \chi_K \, \Gamma_K(\varphi(A^a) \circ \Phi_\sigma) \, \mathrm{d}\mu^K \;\ge\; c_* \, \sigma^{-\frac{5}{2}+\kappa} \inf_{[0, S_* \sigma^{-3/2-\kappa}]} \varphi'^2.$$

For every $\vartheta < 1$ there is $\sigma_\vartheta > 0$ such that the left side is positive for $0 < \sigma \le \sigma_\vartheta$, every φ with $\varphi' > 0$ and every such χ_K with $\mu^K(\chi_K < 1) \le \vartheta$; in particular $\liminf_K \mathcal{E}_K(\varphi(A^a) \circ \Phi_\sigma) > 0$. There are $\sigma_{**} > 0$ and $0 < c_\circ \le C_\circ$, $c_\circ$ depending only on the constants of the construction, with

$$c_\circ \, \mathsf{b}^{-1} \ell^3 \sigma^{-3/2} \le \liminf_K \langle A^a \circ \Phi_\sigma \rangle_{\mu^K} \le \limsup_K \langle A^a \circ \Phi_\sigma \rangle_{\mu^K} \le C_\circ \, \mathsf{b}^{-1} \ell^3 \sigma^{-3/2} \qquad (0 < \sigma \le \sigma_{**});$$

for every $\varrho \in (0,1)$ there are $c_\varrho, \sigma_\varrho > 0$ such that for $0 < \sigma \le \sigma_\varrho$ and all large K, with μ^K-probability at least $1 - \varrho$, the flowed action exceeds $c_\varrho \mathsf{b}^{-1} \ell^3 \sigma^{-3/2}$ and $\Gamma_K(A^a \circ \Phi_\sigma)$ exceeds $c_\varrho \sigma^{-5/2+\kappa}$; and for $0 < \sigma \le \sigma_*$ the flowed current does not vanish in probability: $\liminf_K \mu^K((J,J)_a(U_\sigma) \ge c_J(\sigma)) \ge \frac{1}{4} p_*$ for some $c_J(\sigma) > 0$.

(C7) *The generator without a regularisation cutoff.* Let $\bar{s}, \bar{C}$ be given by Prop. 16.1(a) for some $0 < \epsilon < 1$, and $0 < \sigma \le \bar{s}$. There is K_1 such that for $K \ge K_1$, $a = a_K$: (a) $\int |\partial(A_K \circ \Phi_\sigma)|^q \, \mathrm{d}\mu^K \ge \exp[(6.6\, q - 24.64)\, \sigma/a^2]$ for $q \in [3.74, 8]$, and $\int \Gamma_K(A^a \circ \Phi_\sigma)^2 \, \mathrm{d}\mu^K \ge e^{1.76\, \sigma/a^2}$; (b) $\|\mathcal{L}_K \arctan(A^a \circ \Phi_\sigma)\|_{L^2(\mu^K)} \ge e^{0.8\, \sigma/a^2}$; (c) for $f(x) = 1 - \cos(2\pi x_1/\ell)$, $E^{\mathrm{a}}_\sigma = \frac{3}{2}\ell^3 (2\bar{C}(\frac{\sigma}{2})^{-\frac{3}{4}-\frac{\kappa}{2}} + 8\bar{C}^2(\frac{\sigma}{2})^{-\frac{1}{2}-\kappa})^2$, $M_\sigma = (\frac{4}{\sigma} + \frac{11400}{\ell^2})(\frac{3}{2}\sigma E^{\mathrm{a}}_\sigma + 1) + 1$ and every $\varphi \in C^\infty(\mathbb{R})$, bounded with all its derivatives, with $\inf_{[0,M_\sigma]} |\varphi'| > 0$ (K_1 depending on φ): $\|\mathcal{L}_K \varphi(\mathcal{O}(f) \circ \Phi_\sigma)\|_{L^2(\mu^K)} \ge e^{0.8\, \sigma/a^2}$; (d) for $p \in [1.87, 4]$: $\|\mathcal{L}_K \cos(A^a \circ \Phi_\sigma)\|_{L^p(\mu^K)} + \|\mathcal{L}_K \sin(A^a \circ \Phi_\sigma)\|_{L^p(\mu^K)} \ge e^{0.04\, \sigma/(p a^2)}$, and for every $p > 1.8637$ the left side is unbounded in K; (e) for every real $f \in C^\infty(T_\ell)$, $f \not\equiv 0$, with K_1 and a constant C depending on f but not on K: $\|\mathcal{L}_K \sin(\mathcal{O}(f) \circ \Phi_\sigma)\|_{L^2(\mu^K)} \ge \exp[\frac{\sigma}{5a^2} - \frac{10\,\mathsf{b}}{a} - C]$, and $\|\mathcal{L}_K \varphi(\mathcal{O}(f) \circ \Phi_\sigma)\|_{L^2(\mu^K)} \ge e^{\sigma/20a^2}$ for every non-constant periodic $\varphi \in C^\infty(\mathbb{R})$, K_1 depending besides on φ; (f) $\sup_K \int e^{13\Lambda_\sigma} \, \mathrm{d}\mu^K = \infty$.

■□

16.3. **Gauge-fixed regularity and the admissibility hypothesis.**

Proposition 16.1 (Gauge-fixed fields and the gradient flow)**.** Let $0 < \kappa < \frac{1}{6}$, and let $\mathsf{N}'(U)$ be the least $\|\mathbf{B}\|_{\kappa;a}$ over the 1-forms $\mathbf{B}$ with $e^{i a_K \mathbf{B}}$ in the orbit of U. *The gauge-fixed flow* is the lattice gradient flow of the action with a gauge term: $V_s = e^{i a_K A_s}$, $A_0 = \mathbf{B}$, moves with the velocity of the Wilson flow Φ plus a gauge direction $\mathrm{d}_{V_s}\lambda_s$, λ_s the lattice form of the divergence of A_s that answers to the member $\mathrm{d}_{\mathfrak{A}}(\partial_\nu \mathfrak{A}_\nu)$ of the flow lines, fixed so that the linear part of the equation is the lattice heat operator. In the chart it reads $\frac{d}{ds} A_s(b) = a_K^{-3} \Theta^{\mathrm{ch}}(a_K A_s|_{\mathsf{N}(b)})$, $\mathsf{N}(b)$ the fifteen bonds near b, Θ^{ch} holomorphic and bounded on $\{|a_K A| \le 1\}$, and $\mathsf{M}_\Theta = \max\{1, \sup|\Theta^{\mathrm{ch}}|\}$, the supremum taken there, an absolute number; the members of Θ^{ch} of the orders one, two and at least three in A give

$$\dot{A} = \Delta_a A + \mathcal{N}(A, A) + \mathcal{R}(A), \qquad \mathcal{N}(A, A') = \sum_j c_j \, (\tau_{h_j} \nabla_{\lambda_j} A_{\nu_j}) \wedge (\tau_{h'_j} A'_{\nu'_j}),$$

a finite sum, with absolute coefficients, over the directions and the fixed finite translation set used in the rough regularity quantity.

Fields in the chain gauge. Assume (E); let $d_0 \ge n_2$ be a top depth, $K > d_0$, $t = K - d_0$, $\mathbb{E}^{[d_0]}$ the normalised chain measure from the top depth d_0, and $V_k = \mathcal{M}^k U$. On a set of full measure: U is a gauge transform of U_0, where $\mathsf{U}_t = U_{\mathrm{top}}$ is a background over the top field V_t and $\mathsf{U}_k = e^{i a_K \mathbf{h}_{K-k}} \mathsf{U}_{k+1}$ ($k < t$) is a gauge transform of the background over V_k, in a gauge that is a function of V_{k+1} (the

gauge of the chain); the layer $\mathbf{h}_d$ of the depth $d = K - k$ is a function of the fluctuation B_k of the chart at V_{k+1}, with $|\mathbf{h}_d| \le c_{\mathsf{h}} L^d \delta(L^d \mathsf{b})$, c_{h} a constant of the construction; its linear part in the fluctuation, carried by $\mathrm{Ad}(\gamma)$ below, is $\mathsf{b}^{-1/2} L^{d/2} \mathsf{J}_d B_k$, with the layer map J_d, a function of V_{k+1}, of the norms $\|\mathsf{J}_d X\|_{L^2(T_\ell)} \le c_{\mathsf{J}} L^{-3d/2} \|X\|$ and $\|\mathsf{J}_d X\|_\infty \le c_{\mathsf{J}} \max |X|$, c_{J} a constant of the construction, and at every point of the box the derivative of $\mathbf{h}_d$ in B_k is $\mathsf{b}^{-1/2} L^{d/2}$ times a map with the same two bounds; in its regular gauge γ the top is $U^\gamma_{\mathrm{top}} = e^{i a_K \mathbf{A}^{\mathrm{top}}}$ with $|\mathbf{A}^{\mathrm{top}}| \le \mathsf{a}_{\mathrm{top}} \le C L^{10 d_0}$, $\mathsf{a}_{\mathrm{top}}$ a number that reads neither K nor the field; and U is a gauge transform of $e^{i a_K \mathbf{B}_K}$,

$$\mathbf{B}_K = \tfrac{1}{i a_K} \log(e^{i a_K \mathbf{h}'_K} \cdots e^{i a_K \mathbf{h}'_{d_0+1}} e^{i a_K \mathbf{A}^{\mathrm{top}}}), \qquad \mathbf{h}'_d = \mathrm{Ad}(\gamma) \mathbf{h}_d,$$

bond by bond, the logarithm being formed by its series if $\sum_d a_K |\mathbf{h}_d| \le \frac{1}{4}$ and $a_K \mathsf{a}_{\mathrm{top}} \le \frac{1}{8}$, and otherwise by a fixed Borel $\mathfrak{su}(2)$ logarithm of norm at most π, principal off $-\mathbf{1}$, on positive bonds and extended by reversal. Admissibility and the conclusion (Adm) have the meaning stated before Thm. AV; the layer fields and their conditional estimates are specified in the proof below.

(a) There are $c_1, c_2 \in (0,1]$, $c_{\mathrm{Lip}} \in (0, c_1]$ and c_3, c_4, c_5, depending only on κ, such that for all K, m and $\beta > 0$, every real 1-form $\mathbf{B}$ on $\Lambda_{K,\ell}$ and $R \ge \max\{1, \|\mathbf{B}\|_{\kappa;a}\}$, with $s_0(R) = (c_1 R^{-2})^{4/(1-2\kappa)}$, $s_{\mathrm{Lip}}(R) = (c_{\mathrm{Lip}} R^{-2})^{4/(1-2\kappa)}$ and $a_0(R) = (c_2 R^{-1})^{2/(1-2\kappa)}$, the following hold. If $a_K \le a_0(R)$, the gauge-fixed flow A_s from $A_0 = \mathbf{B}$ exists on the closed interval $[0, s_0(R)]$, $\Phi_s(e^{i a_K \mathbf{B}})$ is gauge equivalent to $e^{i a_K A_s}$, and for $0 < s \le s_0(R)$: $\|A_s - P_s \mathbf{B}\|_\infty \le c_3 R s^{-\kappa}$, $\|\nabla(A_s - P_s \mathbf{B})\|_\infty \le c_3 R s^{-\frac{1}{2}-\kappa}$, $\|A_s\|_\infty \le (1 + c_3) R s^{-\frac{1}{4}-\frac{\kappa}{2}}$, $\|\nabla A_s\|_\infty \le (2 + c_3) R s^{-\frac{3}{4}-\frac{\kappa}{2}}$.

If besides $\mathbf{B}'$ is a real 1-form with $\|\mathbf{B}'\|_{\kappa;a} \le R$, A'_s its gauge-fixed flow and D the rough distance of $\mathbf{B}$ and $\mathbf{B}'$, then for $0 < s \le s_{\mathrm{Lip}}(R)$: $\|(A_s - A'_s) - P_s(\mathbf{B} - \mathbf{B}')\|_\infty \le c_5 \mathsf{D} s^{-\kappa}$, $\|\nabla[(A_s - A'_s) - P_s(\mathbf{B} - \mathbf{B}')]\|_\infty \le c_5 \mathsf{D} s^{-\frac{1}{2}-\kappa}$, so $\|A_s - A'_s\|_\infty \le (1 + c_5) \mathsf{D} s^{-\frac{1}{4}-\frac{\kappa}{2}}$. For every a_K: $A^a(\Phi_s e^{i a_K \mathbf{B}}) \le c_4 \ell^3 (1 + R)^{24} s^{-\frac{3}{2}-\kappa}$ for $0 < s \le 1$. These statements, the short-time theory of the gauge-fixed flow, are deterministic and rest on its equation alone; they hold on every lattice $\Lambda_{K,\ell}$, uniformly in a_K.

Assume (E): for every admissible d_0, $K > d_0$ and $q < \infty$, $\mathbb{E}^{[d_0]}[\|\mathbf{B}_K\|^q_{\kappa;a}]^{1/q} \le C(q,\kappa) L^{20 d_0}$, with $C(q,\kappa)$ free of K and d_0; and for every top depth $d_1 \ge n_2$, $K > d_1$, invariant Borel $F \ge 0$ and $r > 1$,

$$\langle F \rangle_{\mu^K} \le \mathbb{E}^{[d_1]}[F] + \sum_{d_1 \le d \le K-2} \mathsf{R}_d^{1-1/r} \, \mathbb{E}^{[d+1]}[F^r]^{1/r} + \mathsf{R}_{K-1} \sup F,$$

where $\sum_d \mathsf{R}_d^{1/2} L^{pd}$ and $\sup_K \mathsf{R}_{K-1} L^{pK}$ are finite for every p. Under (E), the conclusion (Adm) proved below gives $\mathsf{M}_q = \mathsf{M}_q(\kappa) = \sup_K \langle \mathsf{N}'^q \rangle_{\mu^K} < \infty$ for every $q < \infty$, and: for every $\epsilon > 0$ there are $\bar{s} \in (0,1]$, $\bar{C}$, $\bar{K}$ with $\mu^K(\mathsf{O}_K(\bar{C}, \bar{s})) \ge 1 - \epsilon$ for $K \ge \bar{K}$; $\sup_K \langle (\sup_{0<s\le 1} s^{\frac{3}{2}+\kappa} A^a(U_s))^q \rangle_{\mu^K} < \infty$ for every q; for $0 < \sigma \le 1$ and $1 \le p < \infty$ there is C_3, bounded for σ in compact subsets of $(0,1]$, with $\mu^K(\max_p a_p(U_\sigma) > 10^{-4}) \le C_3 a_K^p$ and $\int [(a_K^{-4} \max_p a_p(U_\sigma))^p + (J,J)_a(U_\sigma)^p] \, \mathrm{d}\mu^K \le C_3$ for all K; and for $0 < \sigma \le 1$ the laws of Λ_σ under μ^K are tight.

(b) Under (E), the flow-approximation task seeks the following conclusion. For every closed path C of bonds of some $\Lambda_{d,\ell}$ and $s > 0$, the flowed Wilson loop $W_C \circ \Phi_s$ (W_C the half trace of the holonomy) converges strongly along the full sequence of cutoffs; there are an invariant Borel set of threads of full measure and a Borel map $\mathfrak{F}$ from it into flow lines, $\mathfrak{F}(Y)_s$ represented by a $C^{1,1}$ connection for $s > 0$, such that the limit is $W_C(\mathfrak{F}(Y)_s)$. For every flowed observable and $\sigma > 0$, with $\Xi^\sigma_{K,D}(V) = H^\sigma_K(U)$ for $V \in \{\varpi < \theta_D\} \subset \Omega_D$, U any element of the minimal orbit over V in Ω_K, and $\Xi^\sigma_{K,D} = 0$ elsewhere:

$$\lim_{D\to\infty} \sup_{K>D} \|H^\sigma_K - \Xi^\sigma_{K,D} \circ \pi_{K,D}\|_{L^2(\mu^K)} = 0, \qquad \lim_{D\to\infty} \limsup_{K\to\infty} \|\Xi^\sigma_{K,D}(Y_D) - \mathsf{G}_\sigma\|_{L^2(\mu)} = 0 :$$

the flowed observable at the background of the block field of a large depth.

(c) The same flow identification is intended to give the following joint convergence under (E). For $s > 0$ and Lipschitz f the joint laws of $\mathcal{O}(f) \circ \Phi_s$ and of the $W_C \circ \Phi_s$ under μ^K converge with all moments along the full sequence to the joint law of $\mathcal{O}^0(f)(\mathfrak{F}(Y)_s)$ and the $W_C(\mathfrak{F}(Y)_s)$ under μ. Separately, under (E) alone, for $\epsilon, \eta > 0$ there are Borel sets G_K with $\mu^K(\mathsf{G}_K) \geq 1 - \epsilon$, closed paths $C_1, \dots, C_q$ and a continuous Θ on $[-1,1]^q$ with $|\mathcal{O}(f)(U_s) - \Theta((W_{C_i}(U_s))_i)| \leq \eta$ on G_K for all large K.

■

Proof sketch. (a)

The gauge-fixed flow. Write $A_s = P_s\mathbf{B} + w_s$; the mild form $w_s = \int_0^s P_{s-r}[\mathcal{N}(A_r, A_r) + \mathcal{R}(A_r)]\,\mathrm{d}r$ has a fixed point in the norm $\sup_{0<s\leq s_0(R)} \max\{s^\kappa \|w_s\|_\infty, s^{\frac{1}{2}+\kappa}\|\nabla w_s\|_\infty\}$. The first iterate $\int_0^s P_{s-r}\mathcal{N}(P_r\mathbf{B}, P_r\mathbf{B})\,\mathrm{d}r$ is bounded through the quadratic member of the rough norm, by $\|P_{s-r}f\|_\infty \leq (s-r)^{-1/4}|f|_{-\frac{1}{2};a}$ and $\int_0^s (s-r)^{-\frac{1}{4}} r^{-\frac{3}{4}-\kappa} dr \leq c\, s^{-\kappa}$; the cross terms $\mathcal{N}(P_r\mathbf{B}, w_r)$, $\mathcal{N}(w_r, P_r\mathbf{B})$, $\mathcal{N}(w_r, w_r)$ give the unweighted factor $s^{\frac{1}{4}-\frac{3}{2}\kappa}$.

This is a positive power because $\kappa < \frac{1}{6}$; multiplication by the weight s^κ in the fixed-point norm gives the contraction condition $R^2 s^{\frac{1}{4}-\frac{\kappa}{2}} \leq c_1$; the members of order at least three are bounded in the chart, $|\mathcal{R}(A)| \leq 2 \sup|\Theta^{\mathrm{ch}}|\,\|A\|_\infty^3$ for $a_K\|A\|_\infty \leq \frac{1}{2}$ by Cauchy's estimate, and the flow does not leave the chart on $[0, s_0(R)]$ for $a_K \leq a_0(R)$, by the choice of c_2, as $a_K\|\mathbf{B}\|_\infty \leq e^{12} R a_K^{\frac{1}{2}-\kappa}$. The other two bounds follow with $s^{\frac{1}{4}+\frac{\kappa}{2}}\|P_s\mathbf{B}\|_\infty \leq R$, $s^{\frac{3}{4}+\frac{\kappa}{2}}\|\nabla P_s\mathbf{B}\|_\infty \leq 2R$.

The velocities of V_s and of $\Phi_s(V_0)$ differ by a gauge direction, so the two are gauge equivalent. For two data, $(A_s - A'_s) - P_s(\mathbf{B} - \mathbf{B}')$ solves the same kind of equation, into which the quadratic member enters through $\mathcal{N}(P_r\mathbf{B}, P_r\mathbf{B}) - \mathcal{N}(P_r\mathbf{B}', P_r\mathbf{B}')$, which the rough distance bounds; the fixed point closes on $(0, s_{\mathrm{Lip}}(R)]$. The action: for $s \leq s_0(R)$ and $a_K \leq a_0(R)$, $A^a \leq \frac{3}{2}\ell^3(a_K^{-2}\varpi)^2$ and $\varpi(e^{ia_K A}) \leq 2a_K^2(\|A\|_\infty^2 + \|\nabla A\|_\infty)$ (Lem. 16.2(c), (e)) give with the four bounds $A^a(\Phi_s e^{ia_K\mathbf{B}}) \leq 6((1+c_3)^2 + 2 + c_3)^2 \ell^3 R^4 s^{-\frac{3}{2}-\kappa}$; for $s_0(R) < s \leq 1$ the action does not increase (Lem. 16.2(c)), and its value at $s_0(R)$ is at most $CR^4(R^2/c_1)^{4(\frac{3}{2}+\kappa)/(1-2\kappa)}$, a power of R below 24; for $a_K > a_0(R)$, $A^a \leq 6\ell^3 a_K^{-4} < 6\ell^3(R/c_2)^{8/(1-2\kappa)}$, a power of R below 12.

So c_4 depends only on κ.

The field of the gauge of the chain. Given $\mathcal{F}_{d-1}$, the σ-algebra of V_{K-d+1}, the law of B_{K-d} is p_W, $W = V_{K-d+1}$ (Lem. 15.3(ii)). The layers are made step by step, for $k = t-1, \dots, 0$, starting from $\mathsf{U}_t = U_{\mathrm{top}}$, a background over V_t, and ending at $k = 0$, where $\mathcal{M}^0$ is the identity, so that U is a gauge transform of U_0. At the step k, with $W = V_{k+1}$, let U_+ be a background over W whose block field $\mathcal{M}^k U_+$ is the base point $V_W(0)$ of the chart, a background of the level k as well, over $\mathcal{M}^k U_+$ (the background and chart identification in Prop. YM3_I:52.3), chosen as a Borel function of W, and let g_{k+1}, a function of V_{k+1}, be the gauge with $\mathsf{U}_{k+1}^{\mathsf{g}_{k+1}} = U_+$. The chart writes $V_k = V_W(B_k)$, and $\zeta = x_k^{-1/2} B_k$, on the bonds that are not pivot bonds, has $\sup|\zeta| < \delta(x_k) \leq \frac{1}{6}\varepsilon_{\mathrm{A}}$; by the global shift identification in Prop. YM3_I:51.5, and at $k = 0$ by the pivot chart in equation YM3_I:(800), the background over V_k is a gauge transform of $e^{iL^{-k}\mathbf{A}[U_+;\zeta]}U_+$. As $a_K L^d = L^{-k}$ and $(e^{iX}U)^g = e^{i\mathrm{Ad}(g)X}U^g$, the layer $\mathbf{h}_d = L^d \mathrm{Ad}(\mathsf{g}_{k+1}^{-1})\mathbf{A}[U_+;\zeta]$ makes $\mathsf{U}_k = e^{ia_K\mathbf{h}_d}\mathsf{U}_{k+1}$ the transform of $e^{iL^{-k}\mathbf{A}[U_+;\zeta]}U_+$ by g_{k+1}^{-1}, a function of V_{k+1}; U_k is a function of V_k, and so is the gauge g_k of the next step. The bound $|\mathbf{A}[U_+;\zeta]| \leq c_{\mathsf{h}} \sup|\zeta|$ (Prop. YM3_I:51.5 and the pivot estimate YM3_I:(800)) gives $|\mathbf{h}_d| \leq c_{\mathsf{h}} L^d \delta(L^d \mathsf{b})$. The product that defines $\mathbf{B}_K$ lies within $\frac{3}{8}$ of $\mathbf{1}$ when $\sum_d a_K|\mathbf{h}_d| \leq \frac{1}{4}$ and $a_K \mathsf{a}_{\mathrm{top}} \leq \frac{1}{8}$.

The regular gauge of the top. The regularity number of the level k of a configuration U' on the bare lattice is $L^{2k}\max_p|\partial U'(p) - \mathbf{1}|$, the maximum over the bare plaquettes p ([7, Def. P.3]); at the level 0 it is $\varpi(U')$, and at the level k it is L^{2k} times its value at the level 0 (by that definition

in the two units). For $k \geq 1$, Cut_k consists of the block fields of the level k whose backgrounds exist and have regularity numbers of the level k below $c_\rho\varepsilon(x_k)$ (the definition of the field cut). As $V_t \in \mathsf{I}_{d_0} \subset \mathsf{Cut}_t$ and $x_t = L^{d_0}\mathsf{b}$, every background U' over V_t has $L^{2t}\varpi(U') < c_\rho\varepsilon(L^{d_0}\mathsf{b})$, so $\varpi(U') < c_\rho\varepsilon(L^{d_0}\mathsf{b})L^{-2t} = c_\rho\varepsilon(L^{d_0}\mathsf{b})L^{2d_0}a_K^2$. Lem. 15.7(a) with $\mathfrak{n} = N$, at $\kappa' = \max\{c_\rho\varepsilon(L^{d_0}\mathsf{b})L^{2d_0}, \ell^{-2}\}a_K^2$, for which $N \geq \kappa'^{-1/2}$, gives a gauge γ, which a measurable selection makes depend on V_t alone in a Borel way, with $|U^\gamma_{\mathrm{top}}(b) - \mathbf{1}| \leq C_{\mathrm{BG}}\sqrt{\kappa'} \leq C_{\mathrm{BG}} \max\{(c_\rho\varepsilon(L^{d_0}\mathsf{b}))^{1/2}, \ell^{-1}\}L^{d_0}a_K \leq CL^{d_0}a_K$ at every bond; at the finitely many K with $\kappa' > \kappa_0$ or $N < \lambda_{\mathrm{BG},3}$ we take $\gamma = \mathbf{1}$, and there $\pi/a_K \leq CL^{d_0}$. So $U^\gamma_{\mathrm{top}} = e^{ia_K\mathbf{A}^{\mathrm{top}}}$, with $ia_K\mathbf{A}^{\mathrm{top}}$ a fixed Borel $\mathfrak{su}(2)$ logarithm of norm at most π, equal to the principal logarithm off $-\mathbf{1}$, at each positive bond, and extended by reversal, so that $\mathbf{A}^{\mathrm{top}}$ too depends on V_t alone in a Borel way, and $|\mathbf{A}^{\mathrm{top}}| \leq \frac{\pi}{2}a_K^{-1}|U^\gamma_{\mathrm{top}} - \mathbf{1}|$ gives $|\mathbf{A}^{\mathrm{top}}| \leq \mathsf{a}_{\mathrm{top}} = CL^{d_0} \leq CL^{10d_0}$. The bounds of J_d, and those of the derivative of the layer on the box, come from the locality of the shift. Let $\mathfrak{n}$ be the distance, in the maximum norm and in units of their side, of the cubes of π^M_{k+1} that hold β_- and b_-, for a bare bond β and a bond b of the level k, and $e^{\bar\kappa}$ the radius of the polydisc $\mathbb{D}(Y)$ of the parameters of the shift systems (Prop. YM3_I:51.5 and the hypotheses preceding it). For $\mathfrak{n} \geq 3$ the value of $\mathbf{A}[U_+;\zeta]$ at β differs from that of the shift of the cube of $(2\mathfrak{n}-1)^3$ cubes about the cube of β_-, which does not read $\zeta(b)$, by at most $2c_{\mathsf{h}} \sup|\zeta| e^{-\bar\kappa(\mathfrak{n}-2)}$: let every fine cube, that is every cube of π^M_k, in the shell of the cubes of this cube at the distance j from the cube of β_-, $2 \leq j \leq \mathfrak{n}-1$, one of its $\mathfrak{n}-2$ outer shells, carry a single variable s_j, $|s_j| < e^{\bar\kappa}$, and every fine cube outside these shells the value 1; by the holomorphy, support and restriction properties of Prop. YM3_I:51.5 the difference is then the value at $s_2 = \cdots = s_{\mathfrak{n}-1} = 1$ of a holomorphic function of these $\mathfrak{n}-2$ variables, of modulus at most $2c_{\mathsf{h}} \sup|\zeta|$ on their polydisc, and zero as soon as some s_j is zero, since that shell then separates the cubes nearer than it to the cube of β_-, among them those that hold the two ends of β, from the cubes beyond it; and Schwarz's lemma in the s_j gives the bound. So, by Cauchy's estimate, for $\sup|\zeta| \leq \frac{1}{2}\varepsilon_{\mathrm{A}}$ the derivative of $\mathbf{A}[U_+;\zeta]$ at β in the value of ζ at b is at most $4c_{\mathsf{h}}e^{-\bar\kappa(\mathfrak{n}-2)_+}$, and for $k = 0$ it vanishes unless $\mathfrak{n} \leq 1$ (the double-block dependence of the pivot chart YM3_I:(800)). As at most 125 cubes lie at a distance ≤ 2 from a given one and at most $24\mathfrak{n}^2 + 2$ at the distance $\mathfrak{n}$, Schur's test gives the two norms of J_d, and the same bounds for the derivative of the layer on the box, with $c_{\mathsf{J}} = \sqrt{27}(LM)^3 \cdot 4c_{\mathsf{h}}[125 + \sum_{\mathfrak{n}\geq 3}(24\mathfrak{n}^2+2)e^{-\bar\kappa(\mathfrak{n}-2)}]$, and $\|\mathsf{J}_d^* f\| \leq c_{\mathsf{J}} \min\{L^{-3d/2}\|f\|_2, \|f\|_1\}$.

Each layer splits as $\mathbf{h}'_d = \mathbf{n}_d + u_{(d)} + e_{(d)}$: $\mathbf{n}_d$ its conditional mean given $\mathcal{F}_{d-1}$; $u_{(d)} = \mathsf{b}^{-1/2}L^{d/2}\mathrm{Ad}(\gamma)\mathsf{J}_d\tilde B_d$, linear in the centred fluctuation $\tilde B_d = B_{K-d} - \int B\,\mathrm{d}p_W$ with a predictable coefficient; $e_{(d)}$ the centred rest, which carries besides the factor $\delta(L^d\mathsf{b})$. So $\mathbf{B}_K = u + \rho$, $u = \sum_d u_{(d)}$, ρ the sum of the $e_{(d)}$, of the $\mathbf{n}_d$, of $\mathbf{A}^{\mathrm{top}}$ and of the remainder of the ordered product beyond the sum of its factors.

The linear fluctuation. By Lem. 15.3(iv), given $\mathcal{F}_{d-1}$ every Lipschitz function of B_{K-d} has Gaussian tails, with the variance proxy $\frac{8}{\gamma_{\mathrm{r}}}$ times the square of its Lipschitz constant; for $\langle u_{(d)}, f\rangle$ this square is at most $c_{\mathsf{J}}^2\mathsf{b}^{-1}L^d \min\{\|f\|_1^2, L^{-3d}\|f\|_2^2\}$, so $(P_s u_{(d)})(x)$, $f = k_s(x,\cdot)$ the heat kernel, has the proxy $C\mathsf{b}^{-1}L^d\min\{1, L^{-3d}s^{-3/2}\}$, whose sum over d is $\leq C\mathsf{b}^{-1}s^{-1/2}$. Exponential supermartingales over the depths, with test functions that are functions of $\mathcal{F}_{d-1}$, and a union over dyadic grids of times and sites, through the grid lemma for $|\cdot|_{\eta;a}$ and the bounds of the heat kernel, give Gaussian tails of $|u|_{-\frac{1}{2}-\kappa;a}$.

The quadratic fluctuation and colour cancellation. Sort the pairs in $\mathcal{N}_s[u,u]$ by the finer index. A pair of different depths is linear in the finer layer with a predictable coefficient: a martingale difference, with Gaussian tails as before. The diagonal $\mathcal{N}_s[u_{(d)}, u_{(d)}]$, after P_r at a point, is a quadratic form $(\tilde B_d, \mathcal{Q}_d\tilde B_d)$ of the centred fluctuation, $\|\mathcal{Q}_d\|_{\mathrm{tr}} \leq C\mathsf{b}^{-1}L^d s^{-1/2}\min\{1, L^{-3d}s^{-3/2}\}$; its centred part has exponential moments by (Q), through the norm and the Hilbert–Schmidt norm of $\mathcal{Q}_d$. Its compensator, the conditional mean $\mathrm{tr}(\mathcal{Q}_d\mathrm{Cov}_W)$, would sum by the sizes to the order

$\mathsf{b}^{-1}s^{-1}$, beyond the weight $s^{\frac{3}{4}+\kappa}$; but it reads the conditional covariance of the layer on the bare lattice, in the gauge of the chain, only through its part that is antisymmetric in colour, as $X \wedge Y$ is antisymmetric.

By (Col) the scalar part drops out, and almost surely

$$|\mathrm{tr}(\mathcal{Q}_d \mathrm{Cov}_W)| \le C s^{-1/2} \min\{1, L^{-3d} s^{-3/2}\} [L^{d/2} (\log x)^{\mathfrak{p}_0+6} + \mathsf{a}_{\mathrm{top}}],$$

$x = L^d \mathsf{b}$, whose sum over d is $\le C(1+\mathsf{a}_{\mathrm{top}}) s^{-\frac{3}{4}-\frac{\kappa}{2}}$, which the weight $s^{\frac{3}{4}+\kappa}$ admits: the cancellation in colour.

Means and nonlinear remainders. The $e_{(d)}$ by the bounds of the linear part, with the factor $\delta(L^d\mathsf{b})$; the means by (M), summable in $|\cdot|_{-\kappa;a}$ through $\|\cdot\|_{L^p}$, $p = \lceil 3/\kappa \rceil$; $\mathbf{A}^{\mathrm{top}}$ by $\mathsf{a}_{\mathrm{top}}$; the ordered product, at the second order linear in the finer factor with a predictable coefficient, at the third order small; each with $|\cdot|_{-\kappa;a} \le C(1+\mathsf{a}_{\mathrm{top}})$ in every L^q. The cross members of u and ρ by the sizes, as $(-\frac{1}{2}-\kappa)+(-\kappa) \ge -\frac{1}{2}-2\kappa$. Together $\mathbb{E}^{[d_0]}[\|\mathbf{B}_K\|_{\kappa;a}^q]^{1/q} \le C(q,\kappa)(1+\mathsf{a}_{\mathrm{top}})^2$, every constant reading d_0 only through $\mathsf{a}_{\mathrm{top}}$; with $\mathsf{a}_{\mathrm{top}} \le C L^{10 d_0}$ this is the bound of the statement; for the finitely many K with $a_K \mathsf{a}_{\mathrm{top}} > \frac{1}{8}$ the principal logarithm has $\|\mathbf{B}_K\|_\infty < 8\pi \mathsf{a}_{\mathrm{top}}$.

The statements on one fluctuation integral that enter, besides Lem. 15.3(iv), the bounds of the response, the bounds of the heat kernel and the grid lemma, are, with C free of K and d_0: (M) $\|\mathbf{n}_d\|_\infty \le C\mathsf{b}^{-1}(\log L^d\mathsf{b})^{2\mathfrak{p}_0}$ and $\|\mathbf{n}_d\|_{L^2}^2 \le C\ell^3\mathsf{b}^{-3}L^{-d}(\log L^d\mathsf{b})^{8\mathfrak{p}_0+12}$; (Col) in the gauge of the chain the kernel of $\mathrm{Ad}(\gamma)\mathsf{J}_d\mathsf{A}_W^{-1}\mathsf{J}_d^*\mathrm{Ad}(\gamma)^{\mathsf{T}}$ at two bare bonds whose partition cubes are at distance N is a number times the identity in colour up to w_N, $\sum_N (24N^2+2)\mathsf{w}_N \le C[x^{-1/2}(\log x)^{\mathfrak{p}_0+6} + \mathsf{a}_{\mathrm{top}} L^{-d}]$, the number being formed at the trivial background, where the Gaussian form and the response commute with the adjoint action; and $\|\mathrm{Cov}_W - \mathsf{A}_W^{-1}\| \le C x^{-1/2}(\log x)^{\mathfrak{p}_0+3}$, A_W the Gaussian form of p_W.

(Q) There are universal constants $C_{\mathrm{HW}}, c_{\mathrm{HW}} > 0$ with the following property. Let the real symmetric $\mathcal{Q}_d$ be predictable, that is, functions of $\mathcal{F}_{d-1}$, and assume almost surely the deterministic bounds $\|\mathcal{Q}_d\| \le \mathsf{q}$ and $C_{\mathrm{HW}}\gamma_{\mathrm{r}}^{-2}\sum_d \|\mathcal{Q}_d\|_{\mathrm{HS}}^2 \le \Sigma^2$. For the actual conditional covariance set $\mathsf{X}_d = (\tilde{B}_d, \mathcal{Q}_d\tilde{B}_d) - \mathrm{tr}(\mathcal{Q}_d\mathrm{Cov}_W)$. Then $\mathbb{E}^{[d_0]} e^{\epsilon\sum_d \mathsf{X}_d} \le e^{\epsilon^2\Sigma^2}$ for $|\epsilon|\mathsf{q} \le c_{\mathrm{HW}}\gamma_{\mathrm{r}}$.

Indeed the conditional box law has Hessian at least $\gamma_{\mathrm{r}}/4$ and log-Sobolev constant $8/\gamma_{\mathrm{r}}$ (Lem. 15.3(iv)); the convex-domain criterion [4, Thm. 2.1] applies. For nonzero $\mathcal{Q}$, Adamczak's dependent-vector inequality [3, Thm. 2.3] gives the centred quadratic form Y the conditional tail $2\exp[-c\min\{\gamma_{\mathrm{r}}^2 t^2/\|\mathcal{Q}\|_{\mathrm{HS}}^2, \gamma_{\mathrm{r}} t/\|\mathcal{Q}\|\}]$. For $\epsilon > 0$, centering gives

$$\mathbb{E}e^{\epsilon Y} - 1 = \epsilon \int_0^\infty [(e^{\epsilon t}-1)\mathbb{P}(Y > t) + (1-e^{-\epsilon t})\mathbb{P}(Y < -t)]\,dt.$$

Integrating the Gaussian and exponential tails bounds the conditional logarithmic moment by $C_{\mathrm{HW}}\epsilon^2\gamma_{\mathrm{r}}^{-2}\|\mathcal{Q}\|_{\mathrm{HS}}^2$ for $|\epsilon|\|\mathcal{Q}\| \le c_{\mathrm{HW}}\gamma_{\mathrm{r}}$; apply the same argument to $-Y$. Consequently $\exp[\epsilon\sum_{j\le d}\mathsf{X}_j - C_{\mathrm{HW}}\epsilon^2\gamma_{\mathrm{r}}^{-2}\sum_{j\le d}\|\mathcal{Q}_j\|_{\mathrm{HS}}^2]$ is a conditional supermartingale. Iteration and the pathwise budget prove (Q); independence of the layers is unnecessary.

The proof of (Adm). Here $d_0 < d \le K$, $k = K-d$, $x = L^d\mathsf{b}$, $g = x^{-1/2}$, $N_k = (\ell L^d)^3$, $W \in \mathsf{Cut}_{k+1}$, U_+ is as above, $\bar{B}_W = \int B\, p_W(\,\mathrm{d}B)$, and v is the sum of the potentials, so that $\Psi_W = \frac{1}{2}(B, \mathsf{A}_W B) - v$ up to a constant (Prop. 15.4(b)). (1)

Conditions on numbers. Besides the chain, the bounds of (2) use only the following conditions on numbers. At the top depth there are six,

$$1 \le d_0 < K, \qquad L^{d_0}\mathsf{b} \ge \mathsf{t}_\natural,$$
$$\mathsf{R}_{d_0} \le \tfrac{1}{2}, \qquad C_{\mathrm{hs}}\,\delta(L^{d_0}\mathsf{b}) \le \tfrac{1}{4}\gamma_{\mathrm{r}},$$
$$4c_{\mathsf{h}}\delta(L^{d_0}\mathsf{b}) \le \tfrac{1}{4}, \qquad c_\rho \varepsilon(L^{d_0}\mathsf{b}) \le C_{\mathrm{BG}}^{-2},$$

with $C_{\mathrm{hs}} = C_{\mathrm{E},1}[338K_P + \frac{1}{2}K_\Lambda(1+\beta_{\mathrm{s}})N_\star^{\mathrm{s}}/\alpha_0]$, formed from $C_{\mathrm{E},1}$ and the numbers that make up C_V in the setting; the fifth then holds at every depth $d > d_0$ too, δ being decreasing from the reciprocal of the squared coupling floor on, a number that b does not fall below under (E) (the decreasing cutoff functions in §2.3.1). Together with them, (2) uses at the depths $d > d_0$ the bound $C_{\mathrm{Hs}}\delta(L^d\mathsf{b}) \le \frac{1}{4}\gamma_{\mathrm{r}}$, with the constant $C_{\mathrm{Hs}} = 144\,C_{\mathrm{E},1}C_V$ of the setting, which exceeds C_{hs}, as $3\cdot 144 > 338$ and $144 > \frac{1}{2}$, so that at d_0 this bound contains the fourth of the six; by the choice of n_2 it holds at every depth $d \ge n_2$, and $n_2 \le n_{\mathrm{a}} \le d_{\mathrm{adm}}$, so that it holds at d_0 and at every depth $d > d_0$ as soon as $d_0 \ge d_{\mathrm{adm}}$ and asks nothing of d_{adm} beyond its member n_{a}. At each depth $d > d_0$, with $x = L^d\mathsf{b}$ as above, $\bar\kappa \ge 1$ (Def. YM3_I:47.2) and

$$\nu_0 = \Big\lceil \frac{\log x}{2\bar\kappa} \Big\rceil, \qquad \nu_1 = \Big\lceil \frac{\lceil \log x \rceil + 2}{L} \Big\rceil,$$
$$\nu_2 = \max\Big\{\nu_1, \Big\lceil \frac{3\log x}{2\bar\kappa} \Big\rceil + 1\Big\}, \qquad \nu_3 = \max\Big\{\nu_1, 2\Big\lceil \frac{2\log x}{\bar\kappa} \Big\rceil + 19\Big\},$$

there are three conditions on cubes, at each of the three inner radii $\nu = \nu_1, \nu_2, \nu_3$:

$$2M(2\nu + 2\nu_0 + 3)^3 + 4 < L^{m+d-1},$$
$$4M(2\nu + 2\nu_0 + 3)^3\,\theta_{d-1} \le \varepsilon_{\mathcal{W}}\, c_{*,\mathrm{rg}},$$
$$4B_*M(2\nu + 2\nu_0 + 3)^3\,\theta_{d-1} \le \tfrac{1}{2}\varepsilon_{\mathcal{W}}\, a^{\mathrm{us}}(\alpha_0, \alpha_1),$$

with $\theta_{d-1} = 2c_\rho\varepsilon(x/L)$, the absolute numbers $B_* \ge 1$ and $c_{*,\mathrm{rg}} \in (0,1]$ of the local regular-gauge Thm. YM3_I:F, and the usable radius $a^{\mathrm{us}}(\alpha_0,\alpha_1)$ in equation YM3_I:(933) at the fixed radii. $(2\nu + 2\nu_0 + 3)^3$ is the number of the cubes of π^M_{k+1} at distances up to $\nu + \nu_0 + 1$ from a given one, in units of their side; they form the cube Q_ν about it, of side $(2\nu + 2\nu_0 + 3)M$ in the units of the level $k+1$, whose cubes at distances up to ν form its *inner cube*, parted from the cubes outside Q_ν by its *collar*, the ν_0 shells of the cubes at the distances $\nu + 2, \dots, \nu + \nu_0 + 1$. The *trivial triple* on a polymer Y is $P[0] = (\mathbf{1}, 0, 0)$, the value at 0 of the chart $P[\omega] = (\mathbf{1}, A, \mathbf{J})$ at the trivial pair, $\omega = (A, \mathbf{J})$, whose pair is $\mathbf{p}(P[\omega]) = (e^{iL^{-k-1}A}, \mathbf{J})$ (the triple and pair conventions of Def. YM3_I:49.1); a triple P on Y is *within r of the trivial triple in a gauge* $\mathfrak{v}$ if $P^{\mathfrak{v}}$ has the pair of $P[\omega]$ for a real ω with $\|\omega\|_Y < r$, the triple norm, namely the largest of $|A|$, $|\nabla_{\mathbf{1}}A|$ and $|\mathbf{J}|$ on $\mathrm{Bd}(Y)$. Every term of the exponent of p_W reads a triple through its pair alone (proof of Prop. 15.4(c)), and so do the structure of the step and the shift (Thm. YM3_I:U, Prop. YM3_I:51.5, and the actual-core verification in the proof of Thm. YM3_I:W). Under the three conditions Q_ν is small against the side of the torus of the level, and Thm. YM3_I:F at the level $k+1$, on Q_ν with $\mathsf{l} = (2\nu + 2\nu_0 + 3)M$ and at $\alpha = 2\theta_{d-1}/\varepsilon_{\mathcal{W}}$ (the first two conditions give its conditions on the sides and on $\mathsf{l}\alpha$), gives a gauge $\mathfrak{v}$ and a real 1-form A' with $U_+^{\mathfrak{v}} = e^{iL^{-k-1}A'}$ on the bare bonds of the concentric cube of side $\mathsf{l}+4$ and $|A'|, |\nabla_{\mathbf{1}}A'| < B_*\mathsf{l}\alpha = 2B_*(2\nu + 2\nu_0 + 3)M\theta_{d-1}/\varepsilon_{\mathcal{W}}$ on $\mathrm{Bd}(Q_\nu)$ (the potential and first-difference estimates of that theorem). Its hypotheses on U_+, at the spacing $\eta = \eta_{k+1}$, are the bounds $|\partial U_+(p) - \mathbf{1}| < \alpha\eta^2$ at the bare plaquettes p with a corner in the concentric cube of side $\mathsf{l}+4$ and $|(D^{\mathsf{T}}_{U_+}\partial U_+)(b)| < \alpha\eta^2$ at the bare bonds b with an end point in that cube; here, for a configuration U, $D^{\mathsf{T}}_U\partial U$ is the 1-form whose value at $b = \langle x, x+\eta e_\mu\rangle$ is $\eta^{-1}\sum_{\nu\ne\mu}[X_{\mu\nu}(x) - \mathrm{Ad}(U(\langle x, x-\eta e_\nu\rangle))X_{\mu\nu}(x-\eta e_\nu)]$, with $X_{\mu\nu}(y)$ the value of ∂U at the plaquette of the directions $\mu < \nu$ based at y and $X_{\nu\mu} = -X_{\mu\nu}$ (the covariant-difference convention of Section YM3_I:22), that is, the transpose, at the spacing η and at U, of the covariant exterior derivative of 1-forms into 2-forms, applied to the matrix-valued 2-form $p \mapsto \partial U(p)$. Both bounds hold at all the bare plaquettes and bonds of the torus: U_+, a background over $W \in \mathsf{Cut}_{k+1}$, has regularity number of the level $k+1$ less than $c_\rho\varepsilon(x/L) = \frac{1}{2}\theta_{d-1}$ (the definition of the field cut), so $|\partial U_+ - \mathbf{1}| < \frac{1}{2}\theta_{d-1}\eta_{k+1}^2 < \alpha\eta_{k+1}^2$; its current $\mathbf{J}_+$ is below $\theta_{d-1}/\varepsilon_{\mathcal{W}}$ at every bare bond, as $\varpi(W) < \theta_{d-1}$ admits the plain-core verification in the proof of Prop. YM3_I:52.3 at the free radii $\theta_{d-1}/\varepsilon_{\mathcal{W}}$, α_1 (as in the proof of Prop. 15.4(a)), which puts P_W in the core with these radii, and

the condition (iii) of its simple set bounds the current by the first radius (Def. YM3_I:49.1); so $|D^{\mathsf{T}}_{U_+}\partial U_+| \le \eta_{k+1}^2|\mathbf{J}_+| + \frac14\eta_{k+1}^3\theta_{d-1}^2 < \alpha\eta_{k+1}^2$ (equation YM3_I:(579)). As $|\mathrm{Ad}(\mathfrak{v})\mathbf{J}_+| = |\mathbf{J}_+|$, P_W is within $r_\nu := 4B_*M(2\nu{+}2\nu_0{+}3)^3\theta_{d-1}/\varepsilon_{\mathcal{W}}$ of the trivial triple in the gauge $\mathfrak{v}$, with $\omega = (A', \mathrm{Ad}(\mathfrak{v})\mathbf{J}_+)$, on Q_ν and on every polymer in it; r_ν is below a multiple of $x^{-1/2}(\log x)^{\mathfrak{p}_0+3}$ and below half the usable radius, so that the chart carries the ball of radius $2r_\nu$ into the analyticity cores. The cubes Q_{ν_1}, Q_{ν_2}, Q_{ν_3} serve (i), (ii), (iii) of (2) in turn; in each the collar is the room in which A_W^{-1} on the inner cube forgets the cubes outside, up to a multiple of $x^{-1/2}$ (see (i)). The six conditions at the top depth and the conditions on cubes hold for every integer $d_0 \ge d_{\mathrm{num}}(\ell,\mathsf{b})$, every $K > d_0$ and every depth $d > d_0$, where

$$
\begin{aligned}
d_{\mathrm{num}}(\ell,\mathsf{b}) := \max\Big\{1,\ &\Big\lceil\frac{\log(\mathsf{t}_{\mathrm{adm}}/\mathsf{b})}{\log L}\Big\rceil,\\
&\Big\lceil\frac{\mathsf{y}_{\mathsf{R}}(\ell,\mathsf{b}) - \log\mathsf{b}}{\log L}\Big\rceil,\\
&\Big\lceil\frac{2}{\log L}\log\frac{12\cdot10^5\,LM(\log\mathsf{b})^3}{\ell}\Big\rceil\Big\},
\end{aligned}
$$

$$
\begin{aligned}
\mathsf{t}_{\mathrm{adm}} = \max\Big\{&\mathsf{t}_\natural,\\
&\Big(\frac{4\mathfrak{p}_0}{e}\Big)^{4\mathfrak{p}_0}\max\Big\{\frac{4C_{\mathrm{hs}}A_1}{\gamma_{\mathrm{r}}},\ 16c_{\mathsf{h}}A_1,\ C_{\mathrm{BG}}^2c_\rho A_0\Big\}^4,\\
&\Big(\frac{125000\,c_\rho A_0L^{1/2}M}{\varepsilon_{\mathcal{W}}}\Big)^4\Big(\frac{4\mathfrak{p}_0+12}{e}\Big)^{4\mathfrak{p}_0+12}\\
&\qquad\max\Big\{\frac{1}{c_{*,\mathrm{rg}}},\ \frac{2B_*}{a^{\mathrm{us}}(\alpha_0,\alpha_1)}\Big\}^4\Big\},
\end{aligned}
$$

$$
\mathsf{y}_{\mathsf{R}}(\ell,\mathsf{b}) = \max\Big\{\Big(\frac{6}{c'A_0^2}\Big)^{\frac{1}{2\mathfrak{p}_0-1}},\ \Big(\frac{2\,\mathsf{E}(\ell,\mathsf{b})}{c'A_0^2}\Big)^{\frac{1}{2\mathfrak{p}_0}}\Big\},
$$

$$
\mathsf{E}(\ell,\mathsf{b}) = \log4 + 3\log\frac{\ell}{M} + \max\{\mathsf{k}(\mathsf{b}),0\}\,\ell^3,
$$

with the rate $c' > 0$ in the bound defining R_d and $\mathsf{k}(\mathsf{b}) = \mathsf{C}^{\Sigma}_{\mathrm{maj}}{+}2{+}C_{\mathrm{low}}(\mathsf{b})$, the factor of ℓ^3 in the last exponent of the definition of R_d; $\mathsf{E}(\ell,\mathsf{b}) > 0$, as $\ell = 2L^m \ge 2MR(\mathsf{b}) \ge 2M$ under (E). $\mathsf{t}_{\mathrm{adm}}$ is formed from the constants alone, and $d_{\mathrm{num}}(\ell,\mathsf{b})$ from ℓ, b and the constants of the construction; it reads neither K nor a level nor a field. For the proof: for $\mathsf{t} > 1$ and $r > 0$, $\mathsf{t}^{-1/2}(\log\mathsf{t})^r \le (4r/e)^r\mathsf{t}^{-1/4}$, as $r\log z - \frac{z}{4}$ is largest at $z = 4r$. Let $d_0 \ge d_{\mathrm{num}}(\ell,\mathsf{b})$ and $K > d_0$, which is the first of the six. Then $L^{d_0}\mathsf{b} \ge \mathsf{t}_{\mathrm{adm}}$, which gives the second and, as $\delta(\mathsf{t})$ and $\varepsilon(\mathsf{t})$ are A_1 and A_0 times $\mathsf{t}^{-1/2}(\log\mathsf{t})^{\mathfrak{p}_0}$, by that inequality at $r = \mathfrak{p}_0$ the last three; and $z_0 = \log L^{d_0}\mathsf{b} \ge \mathsf{y}_{\mathsf{R}}(\ell,\mathsf{b})$, so that half of $c'A_0^2z_0^{2\mathfrak{p}_0}$ is at least $3z_0$ and at least $\mathsf{E}(\ell,\mathsf{b})$, while the definition of R_{d_0}, at the $(\ell L^{d_0})^3$ sites of the depth d_0 and with $3d_0\log L < 3z_0$, gives $\log 2\mathsf{R}_{d_0} = \log4 + 3\log\frac{\ell}{M} + 3d_0\log L + \mathsf{k}(\mathsf{b})\ell^3 - c'A_0^2z_0^{2\mathfrak{p}_0} < \mathsf{E}(\ell,\mathsf{b}) + 3z_0 - c'A_0^2z_0^{2\mathfrak{p}_0} \le 0$: the third. At a depth $d > d_0$, $x \ge L\mathsf{t}_{\mathrm{adm}}$, and $\log x > 3$, as $\log\mathsf{b} > 2$ (the label floor of §2.3.1); as $\lceil z\rceil < z+1$, $L \ge 13$ and $\bar\kappa \ge 1$, each of ν_1,ν_2,ν_3 is less than $4\log x + 21$ and $\nu_0 < \frac12\log x + 1$, so that $(2\nu + 2\nu_0 + 3)^3 < (9\log x + 47)^3 < 15625(\log x)^3$. With $\theta_{d-1} < 2c_\rho A_0L^{1/2}x^{-1/2}(\log x)^{\mathfrak{p}_0}$ and that inequality at $r = \mathfrak{p}_0 + 3$, the left member of the second condition on cubes is less than $125000\,c_\rho A_0L^{1/2}M(\frac{4\mathfrak{p}_0+12}{e})^{\mathfrak{p}_0+3}x^{-1/4}$ and that of the third less than B_* times this number; and $x \ge \mathsf{t}_{\mathrm{adm}}$ makes these two bounds at most $\varepsilon_{\mathcal{W}}c_{*,\mathrm{rg}}$ and $\frac12\varepsilon_{\mathcal{W}}a^{\mathrm{us}}(\alpha_0,\alpha_1)$. And as $\log x \le \log\mathsf{b}\,(1 + d\log L)$ and $(1+z)^3 < 17.74\,e^{z/2}$ for $z \ge 0$, the left member of the first is less than $31251M(\log x)^3 < 6\cdot10^5M(\log\mathsf{b})^3L^{d/2}$, which is below L^{m+d-1}, as $L^{d/2} > L^{d_0/2} \ge 6\cdot10^5L^{1-m}M(\log\mathsf{b})^3$ by the last member of $d_{\mathrm{num}}(\ell,\mathsf{b})$, ℓ being $2L^m$.

A uniform second moment. For the fluctuation fields the bound holds under p_W at every W. Fix W and a bond b_0. At a bond b, the divergence theorem in the variable $B(b)$ on the ball $\{|B(b)| < \mathfrak{b}(x_k)\}$ of the box, applied to $B(b)e^{-\Psi_W}$, gives the flux $\mathfrak{b}(x_k)\int_{|B(b)|=\mathfrak{b}(x_k)} e^{-\Psi_W} \geq 0$, so $\int B(b)\cdot\nabla_b\Psi_W \,\mathrm{d}p_W \leq 3$. By Prop. 15.4(b), $\Psi_W = \frac{1}{2}(B, \mathsf{A}_W B) - v$ up to a constant, v the sum of the potentials; $\mathsf{A}_W \geq \frac{1}{2}\gamma_{\mathrm{r}}$, and Cauchy's estimate gives $|\nabla_b v| \leq C\delta(x_k)\mathfrak{b}(x_k) \leq \eta$ on the box, η a number free of x_k (the proof of Lem. 15.3(iv)). Take the weights $w_b = e^{-\mu\,\mathrm{dist}(b,b_0)}$, dist the distance in the units of the level k. The entries of A_W fall off exponentially with the distance of the bonds, as A_W is a sum of local terms with the bounds of the polymers (Prop. 15.4(b), Prop. YM3_I:51.4 and its proof), so Schur's test gives $\|w^{1/2}\mathsf{A}_W w^{-1/2} - \mathsf{A}_W\| \leq \frac{1}{4}\gamma_{\mathrm{r}}$ for a small $\mu > 0$ that reads neither K, d nor W. Then $\frac{1}{4}\gamma_{\mathrm{r}}\sum_b w_b \int |B(b)|^2 \,\mathrm{d}p_W \leq \sum_b w_b \int (3 + \eta|B(b)|)\,\mathrm{d}p_W$, and $\eta|B(b)| \leq 2\eta^2/\gamma_{\mathrm{r}} + \frac{1}{8}\gamma_{\mathrm{r}}|B(b)|^2$ gives $\int |B(b_0)|^2\,\mathrm{d}p_W \leq m_2 = \frac{8}{\gamma_{\mathrm{r}}}(3 + \frac{2\eta^2}{\gamma_{\mathrm{r}}})\sum_b w_b$, which reads neither K, d, d_0 nor W, as $\sum_b w_b \leq C\mu^{-3}$. The log-Sobolev bound of Lem. 15.3(iv), together with this second moment, gives Gaussian tails about a uniformly bounded mean. Consequently the boundary layers used below have mass smaller than every power of x. (2)

The bounds on one integral, for $d_0 \geq d_{\mathrm{adm}}$; C stands for numbers formed from $\ell/\mathfrak{b}$ and the constants of the construction. (i)

The mean of the fluctuation. Integration by parts in each variable of the box, against a cut-off that vanishes on its boundary, gives $\mathsf{A}_W\bar{B}_W$ as the mean of ∇v up to terms carried by $\{|B(b)| \geq \frac{1}{2}\mathfrak{b}(x)\}$, whose mass under p_W is at most a multiple of $e^{-c\,\mathfrak{b}(x)^2}$, $c > 0$, and so smaller than every power of x; ∇v is at most a multiple of $\delta(x)\mathfrak{b}(x) = A_1^2 x^{-1/2}(\log x)^{2\mathfrak{p}_0}$ at each bond; and the kernel of A_W^{-1} decays at the rate $\bar{\kappa}$: $\|\mathsf{A}_W^{-1}(b,b')\| \leq c_{\mathbf{K}} e^{-\bar{\kappa}(\mathfrak{n}-3)_+}$, $\mathfrak{n}$ the distance of the cubes of π_{k+1}^M that hold b_- and b'_-, with $c_{\mathbf{K}} = \frac{2}{\gamma_{\mathrm{r}}} + \frac{4}{\gamma_{\mathrm{r}}^2}(c_{\mathrm{r}} + \gamma_{\mathrm{r}})^2$, formed from γ_{r} and c_{r} alone. Indeed A_W^{-1} is the whole covariance of the complete structure of the step at P_W, a decoupled Gaussian structure with the constants $(\bar{\kappa}; \gamma_{\mathrm{r}}, c_{\mathrm{r}}, \hat{\varepsilon})$, $\hat{\varepsilon} = \frac{1}{2}\mathsf{u}_{\mathrm{G}}(LM)^{-3} \leq \frac{1}{2}\gamma_{\mathrm{r}}$ (Def. YM3_I:47.2, Thm. YM3_I:U, and the actual-core verification in Thm. YM3_I:W); on the union of the two cubes it is $\mathbf{C} + \mathbf{C}\Gamma\Gamma^{\mathsf{T}}\mathbf{C}$ at the decoupling parameters 1 (the conditioning identity in Thm. YM3_I:U), where $\mathbf{C} = \mathbf{C}(Z_\circ, s)$ and $\Gamma = \mathbf{\Gamma}(Z_\circ, s)$ are the conditioned covariance and the connecting operator, the two families of operators that the structure assigns to a region $Z_\circ$, here the union of the two cubes, and to the decoupling parameters s (Def. YM3_I:47.2); at all parameters s of the polydisc of radius $e^{\bar{\kappa}}$ this is an operator of norm at most $c_{\mathbf{K}}$, as (St3) with $\hat{\varepsilon} \leq \frac{1}{2}\gamma_{\mathrm{r}}$ gives $\|\mathbf{C}\| \leq 2/\gamma_{\mathrm{r}}$ and $\|\Gamma^{\mathsf{T}}\| = \|\Gamma\| \leq c_{\mathrm{r}} + \hat{\varepsilon} \leq c_{\mathrm{r}} + \gamma_{\mathrm{r}}$; for $\mathfrak{n} \geq 4$ its block between the two cubes is holomorphic in $\mathfrak{n} - 3$ variables, each the parameter of all the cubes of one of the shells at the distances $2, \dots, \mathfrak{n} - 2$ from the cube of b_-, the other parameters being 1 ((St1)), and vanishes as soon as one of them is 0, since that shell then parts the two cubes into different wall-components ((St2)); Schwarz's lemma in these variables gives the factor $e^{-\bar{\kappa}(\mathfrak{n}-3)}$, and for $\mathfrak{n} \leq 3$ the bound is $\|\mathsf{A}_W^{-1}\| \leq \frac{2}{\gamma_{\mathrm{r}}}$. As at most $24\mathfrak{n}^2 + 2$ cubes lie at the distance $\mathfrak{n}$ from a given one and $\bar{\kappa} \geq 1$, the rows of A_W^{-1} have bounded sums: so $\sup|\bar{B}_W| \leq Cx^{-1/2}(\log x)^{2\mathfrak{p}_0}$. The same argument, with one variable for each of the ν_0 shells of the collar of a cube Q_ν of (1), compares the parameters 1 and 0 on the cubes outside Q_ν: as $\nu_0\bar{\kappa} \geq \frac{1}{2}\log x$, on the inner cube A_W^{-1} lies within $2c_{\mathbf{K}}x^{-1/2}$ of the operator at the parameters 1 on the collar and 0 outside Q_ν, which is that of the structure on Q_ν at the restriction of P_W (restriction consistency in Thm. YM3_I:U); carried by the gauge $\mathfrak{v}$ of Q_ν it is that of the structure at $P[\omega]$ (gauge covariance and dependence on the pair), which at $P[z\omega]$ is holomorphic in z for $|z| < a^{\mathrm{us}}(\alpha_0, \alpha_1)/r_\nu$, by joint holomorphy in $(s, A', \mathbf{J})$ at a fixed real factor, taken at the real factor $\mathbf{1}$, as the chart carries the ball of radius $a^{\mathrm{us}}(\alpha_0, \alpha_1)$ into the cores (equation YM3_I:(933) and its proof), and at most $c_{\mathbf{K}}$ there, by (St3) as above, since Thm. YM3_I:U gives at every triple of the cores a structure with the same constants; so Schwarz's lemma in z puts A_W^{-1} on the inner cube, carried by $\mathfrak{v}$, within $2c_{\mathbf{K}}(x^{-1/2} + r_\nu/a^{\mathrm{us}}(\alpha_0, \alpha_1)) \leq Cx^{-1/2}(\log x)^{\mathfrak{p}_0+3}$ of the operator formed at the trivial triple on Q_ν, which commutes with the constant gauge transformations, as these fix

$P[0]$ (gauge covariance). In the ℓ^2 norm a further factor $x^{-1/2}$ comes, up to powers of $\log x$: along the laws proportional to $e^{-\frac{1}{2}(B,\mathsf{A}_W B)+\tau v}$ on the box, $0 \le \tau \le 1$, whose exponents have Hessians $\ge \frac{1}{4}\gamma_{\mathrm{r}}$, the Poincaré inequality makes the means of ∇v at $\tau = 0$ and at $\tau = 1$ differ in ℓ^2 by at most $\frac{4}{\gamma_{\mathrm{r}}} C_{\mathrm{Hs}} \delta(x) \sup \|\nabla v\|$; and at $\tau = 0$, the Gaussian law of covariance A_W^{-1} conditioned on the box, the gradient at a bond b of each potential v_Y with $b_- \in Y$ and $d(Y) < \lceil \log x \rceil$ has a mean smaller than its size by a factor $C x^{-1/2} (\log x)^{\mathfrak{p}_0+3}$: such a Y lies within $(\lceil \log x \rceil + 2)M \le \nu_1 L M$ of b_- in the units of the level k, so in the inner cube of Q_{ν_1} about the cube of b_-; in the gauge $\mathfrak{v}$ of Q_{ν_1} the triple is within r_{ν_1} of the trivial triple, at which the potentials are invariant under the constant gauge transformations; carried by $\mathfrak{v}$, the covariance lies on the inner cube within $C x^{-1/2} (\log x)^{\mathfrak{p}_0+3}$ of an operator that commutes with these (above); and $\mathfrak{g}$ holds no non-zero vector fixed by the adjoint action. The potentials with $d(Y) \ge \lceil \log x \rceil$ add at most $C\delta(x)\mathfrak{b}(x) x^{-1}$ at b, by the sum over polymers [7, eq. 320] at $p = 1$. So $\|\bar{B}_W\| \le C N_k^{1/2} x^{-1} (\log x)^{3\mathfrak{p}_0+3}$. (ii)

The mean of the shift. At a bare bond β the parts of $\mathbf{A}[U_+; gB](\beta)$ of degree two and of higher degree in B are at most $c_{\mathfrak{h}} \delta(x)^2/\varepsilon_{\mathrm{A}}$ and $2c_{\mathfrak{h}} \delta(x)^3/\varepsilon_{\mathrm{A}}^2$, by Cauchy's estimate; the mean of the part of degree two is its value at $g\bar{B}_W$, at most a multiple of $g^2 \sup |\bar{B}_W|^2$, plus its contraction with $g^2 \mathrm{Cov}_W$, and the latter is smaller than the size of that part by a factor $x^{-1/2}$, up to powers of $\log x$: by the locality of the shift at $\mathfrak{n} = \nu_2 + 1$ (above), as $\bar{\kappa}(\nu_2 - 1) \ge \frac{3}{2} \log x$, it may be replaced, at a cost below $4c_{\mathfrak{h}} \delta(x) x^{-3/2}$, by the shift of the inner cube of Q_{ν_2} about the cube of β_-, of $(2\nu_2 + 1)^3$ cubes of π_{k+1}^M; by gauge covariance in Prop. YM3_I:51.5 it may be formed in the gauge $\mathfrak{v}$ of Q_{ν_2}, in which the triple is within $r_{\nu_2} \le C x^{-1/2} (\log x)^{\mathfrak{p}_0+3}$ of the trivial one, and by pair dependence and slice holomorphy its part of degree two then differs from the one formed at the trivial triple by a multiple of r_{ν_2} times its size; at the trivial triple the part of degree two is covariant under the adjoint action of the constant gauge transformations, by covariance under constant gauges, so that its contraction with an operator that commutes with them is an element of $\mathfrak{g}^c = \mathfrak{g} \otimes \mathbb{C}$ fixed by the adjoint action, hence 0; and on the inner cube the covariance of p_W, carried by $\mathfrak{v}$, lies within $\epsilon_d^{\mathrm{cv}} + 2c_{\mathbf{K}} (x^{-1/2} + r_{\nu_2}/a^{\mathrm{us}}(\alpha_0, \alpha_1)) \le C x^{-1/2} (\log x)^{\mathfrak{p}_0+3}$ of such an operator, by (i) and (iv). Hence $|\int \mathbf{A}[U_+; gB](\beta)\, p_W(\,\mathrm{d}B) - (D\mathbf{A}[U_+; 0] g\bar{B}_W)(\beta)| \le \epsilon_d^{\mathrm{s}} \le C x^{-3/2} (\log x)^{4\mathfrak{p}_0+6}$ at every bare bond; and as $|D\mathbf{A}[U_+; 0]X| \le c_{\mathfrak{h}} \sup |X|$, and by the kernel bound of $D\mathbf{A}[U_+; 0]$ above, which reads β only through its cube, with Schur's test, $\sup_\beta |\int \mathbf{A}[U_+; gB](\beta)\, p_W(\,\mathrm{d}B)| \le \epsilon_d^{\mathrm{m}}$ and $\sum_y \max_{\beta_- \in B^k(y)} |\int \mathbf{A}[U_+; gB](\beta)\, p_W(\,\mathrm{d}B)|^2 \le \epsilon_d^{\mathrm{q}}$, the sum over the sites y of the level k, where ϵ_d^{m} is $c_{\mathfrak{h}} g$ times the bound of $\sup |\bar{B}_W|$ in (i) plus ϵ_d^{s}, and ϵ_d^{q} is $2c_{\mathfrak{J}}^2 g^2$ times the square of the bound of $\|\bar{B}_W\|$ in (i) plus $2N_k (\epsilon_d^{\mathrm{s}})^2$. (iii)

The kernel. The kernel of $D\mathbf{A}[U_+; 0] \mathsf{A}_W^{-1} D\mathbf{A}[U_+; 0]^*$ at two bare bonds β, β' is

$$\sum_{b,b'} \partial_b \mathbf{A}[U_+; 0](\beta)\, \mathsf{A}_W^{-1}(b, b')\, \partial_{b'} \mathbf{A}[U_+; 0](\beta')^{\mathsf{T}},$$

with the kernel $\mathsf{A}_W^{-1}(b, b') \in \mathrm{End}\, \mathfrak{g}$ of A_W^{-1}, and its outer factors decay at the rate $\bar{\kappa}$ in the distances of the cubes (above), and so does its middle one, by (i); so, by the triangle inequality for these distances, it is at most $C e^{-\bar{\kappa}\mathfrak{n}/2}$, $\mathfrak{n}$ the distance of the cubes of β_- and β'_-. Let $\mathfrak{n} \le n_\flat := \lceil 2 \log x/\bar{\kappa} \rceil + 9$, so that $\nu_3 = \max\{\nu_1, 2n_\flat + 1\}$, and let $\mathfrak{v}$ be the gauge of Q_{ν_3} about the cube of β_-, in which P_W is within $r_{\nu_3} \le C x^{-1/2} (\log x)^{\mathfrak{p}_0+3}$ of the trivial triple; the inner cube of Q_{ν_3} holds every cube within $n_\flat$ of the cubes of β_- and β'_-, and the bonds b, b' farther than $n_\flat$ from these add at most $C x^{-2} (\log x)^2$, as $\bar{\kappa}(n_\flat - 2) \ge 2 \log x$. There both factors lie within $C x^{-1/2} (\log x)^{\mathfrak{p}_0+3}$ of the operators formed at the trivial triple, which commute with the adjoint action of the constant gauge transformations, and a linear map of $\mathfrak{g}$ that commutes with it is a multiple of $\mathbf{1}$; so, up to that error, the factors are numbers formed at the trivial triple times $\mathrm{Ad}(\mathfrak{v}(\beta_-)^{-1}\mathfrak{v}(b_-))$ and $\mathrm{Ad}(\mathfrak{v}(b_-)^{-1}\mathfrak{v}(b'_-))$, the inner factors cancel, and the kernel is $\mathsf{s}_0(\beta, \beta')\mathrm{Ad}(\mathfrak{v}(\beta_-)^{-1}\mathfrak{v}(\beta'_-))$ up to $C x^{-1/2} (\log x)^{\mathfrak{p}_0+3}$, with a number $\mathsf{s}_0(\beta, \beta')$ formed at the trivial triple. In the gauge of the chain the kernel is multiplied

on the left by $\mathrm{Ad}((\gamma \mathsf{g}_{k+1}^{-1})(\beta_-))$ and on the right by the transpose of $\mathrm{Ad}((\gamma \mathsf{g}_{k+1}^{-1})(\beta'_-))$, which turns $\mathrm{Ad}(\mathfrak{v}(\beta_-)^{-1}\mathfrak{v}(\beta'_-))$ into $\mathrm{Ad}(\mathfrak{u}(\beta_-)^{-1}\mathfrak{u}(\beta'_-))$, $\mathfrak{u} = \mathfrak{v}\mathsf{g}_{k+1}\gamma^{-1}$. As $(\mathsf{U}_{k+1}^{\gamma})^{\mathfrak{u}} = U_+^{\mathfrak{v}} = e^{iL^{-k-1}A'}$ with $|A'| \le Cx^{-1/2}(\log x)^{\mathfrak{p}_0+3}$ on the cube, the ordered product of the values of $\mathsf{U}_{k+1}^{\gamma}$ along a path of bare bonds from β_- to β'_- of length at most $3(\mathfrak{n}+1)LML^{-d}$ lies within $3(\mathfrak{n}+1)M \sup|A'|$ of $\mathfrak{u}(\beta_-)^{-1}\mathfrak{u}(\beta'_-)$, and within $3(\mathfrak{n}+1)LM[c_{\mathsf{h}}\delta(x) + L^{-d}\mathsf{a}_{\mathrm{top}}]$ of $\mathbf{1}$, as $\mathsf{U}_{k+1}^{\gamma}$ is the ordered product of the layers of the depths below d and of the top field, which in its regular gauge has $|\mathbf{A}^{\mathrm{top}}| \le \mathsf{a}_{\mathrm{top}}$ at each bare bond. So in the gauge of the chain the kernel at β, β' is a number times $\mathbf{1}$ up to a $\mathsf{w}_{\mathfrak{n}}$ that is affine in $\mathsf{a}_{\mathrm{top}}L^{-d}$; for the pairs at distances $\mathfrak{n} > n_\flat$ the number is taken 0 and $\mathsf{w}_{\mathfrak{n}}$ is the bound $Ce^{-\bar\kappa \mathfrak{n}/2} \le Cx^{-1}e^{-\bar\kappa(\mathfrak{n}-n_\flat)/2}$ of the kernel; and $\sum_{\mathfrak{n}}(24\mathfrak{n}^2+2)\mathsf{w}_{\mathfrak{n}} \le \epsilon_d^{\Sigma}$, a number formed from x, the constants and $\mathsf{a}_{\mathrm{top}}L^{-d}$. For $k=0$, where no shift system enters, (ii) and (iii) use the local pivot construction and its holomorphy and gauge covariance in equations YM3_I:(800)–YM3_I:(803) in place of the corresponding shift properties: off the pivot bonds $\mathbf{A}[U_+;\zeta] = \zeta$ is linear in ζ and free of the triple; at the pivot bond $b_0(c)$ of a cube c the pivot value $\beta_c(P,\zeta)$ depends on the block field and on ζ through their values on P_c alone, so on the triple through its pair alone, the block field being formed from the configuration of the pair; for $\sup|\zeta| < r_P^\circ$ and a fixed real factor it is holomorphic in $(A', \mathbf{J}, \zeta)$ on the simple chart set, which contains the cores by the common-domain verification in the proof of Thm. YM3_I:W, with $|\beta_c| \le 2C_P \sup|\zeta|$; and $\beta_c(P^u, \mathrm{Ad}(u)\zeta) = \mathrm{Ad}(u(b_0(c)_-))\,\beta_c(P,\zeta)$ for every gauge transformation u of the bare sites, constant or not. Carried by the gauge $\mathfrak{v}$ of Q_ν, the pivot value is therefore the one formed at $P[\omega]$, and $z \mapsto \beta_c(P[z\omega],\zeta)$ is holomorphic and at most $2C_P\sup|\zeta|$ for $|z| < a^{\mathrm{us}}(\alpha_0,\alpha_1)/r_\nu$, where $P[z\omega]$ lies in the cores; Schwarz's lemma in z, with Cauchy's estimate in ζ, puts its parts of degree one and two in ζ within a multiple of r_ν times their size of the parts formed at the trivial triple, as pair dependence and slice holomorphy do for $k \ge 1$. (iv)

The covariance. On the convex box $\mathsf{A}_W - C_{\mathrm{Hs}}\delta(x) \le D^2\Psi_W \le \mathsf{A}_W + C_{\mathrm{Hs}}\delta(x)$, with $C_{\mathrm{Hs}}\delta(x) \le \frac{1}{4}\gamma_{\mathrm{r}}$ as $d > d_0 \ge n_2$ (the bound $C_{\mathrm{Hs}}\delta(x)$ of the Hessians of the potentials in the proof of Lem. 15.3(iv), which Cauchy's estimate draws from their bound on the whole product of the analyticity cores and the polydisc of radius $C_A\mathfrak{b}(x)$, and which so holds at the triple P_W of every $W \in \mathsf{Cut}_{k+1}$ and at every point of the box); the Brascamp–Lieb inequality gives $\mathrm{Cov}_W \le (\mathsf{A}_W - C_{\mathrm{Hs}}\delta(x))^{-1}$, and integration by parts and the Cauchy–Schwarz inequality against linear functions give $\mathrm{Cov}_W \ge (1-\iota_d)(\mathsf{A}_W + C_{\mathrm{Hs}}\delta(x))^{-1}$, with a share ι_d of the boundary of the box that is smaller than every power of x; so $\|\mathrm{Cov}_W - \mathsf{A}_W^{-1}\| \le \epsilon_d^{\mathrm{cv}} = \frac{8}{\gamma_{\mathrm{r}}^2}C_{\mathrm{Hs}}\delta(x) + \frac{2}{\gamma_{\mathrm{r}}}\iota_d$. (3)

One constant. The numbers of (2) are formed from $x = L^d\mathsf{b}$, ℓL^d and the constants of the construction, ϵ_d^{Σ} besides from $\mathsf{a}_{\mathrm{top}}L^{-d}$, in which it is affine; save for the factor $N_k = (\ell L^d)^3$ that multiplies both members of ϵ_d^{q} and stays in its bound below, $\ell L^d = (\ell/\mathsf{b})x$ enters only through counts of bonds that multiply masses of boundary layers of the box, which fall faster than every power of x. Collecting the multiples of (2) gives four numbers C_6, C_6', C_Σ, C_{cv}, formed from ℓ, b and the constants, which read neither K, d_0, d, W, a gauge nor $\mathsf{a}_{\mathrm{top}}$, with $\epsilon_d^{\mathrm{m}} \le C_6 x^{-1}(\log x)^{2\mathfrak{p}_0}$, $\epsilon_d^{\mathrm{q}} \le C_6' N_k x^{-3}(\log x)^{8\mathfrak{p}_0+12}$, $\epsilon_d^{\Sigma} \le C_\Sigma[x^{-1/2}(\log x)^{\mathfrak{p}_0+6} + \mathsf{a}_{\mathrm{top}}L^{-d}]$ and $\epsilon_d^{\mathrm{cv}} \le C_{\mathrm{cv}}x^{-1/2}(\log x)^{\mathfrak{p}_0+3}$ at $x = L^d\mathsf{b}$. From these, (M) and (Col): given $\mathcal{F}_{d-1}$ the law of B_k is p_W, $W = V_{k+1}$, and γ and g_{k+1} are functions of V_{k+1} that act at each bond by isometries, so $|\mathbf{n}_d(\beta)| = L^d|\int \mathbf{A}[U_+; gB](\beta)\, p_W(\,\mathrm{d}B)|$; the factor L^d, with $L^dx^{-1} = \mathsf{b}^{-1}$, gives $C_6\mathsf{b}^{-1}(\log L^d\mathsf{b})^{2\mathfrak{p}_0}$ as a bound of $|\mathbf{n}_d(\beta)|$ for every bare bond β; three bare bonds start at every bare site, and the norm of (M) takes at a site the euclidean norm of the three values, so $\|\mathbf{n}_d\|_\infty \le \sqrt{3}\,C_6\mathsf{b}^{-1}(\log L^d\mathsf{b})^{2\mathfrak{p}_0}$; and as $\|\mathbf{n}_d\|_{L^2}^2 = a_K^3\sum_\beta|\mathbf{n}_d(\beta)|^2$ and $B^k(y)$ holds L^{3k} bare sites, each the initial point of three bare bonds, the sum over the blocks gives $\|\mathbf{n}_d\|_{L^2}^2 \le 3a_K^3L^{3k}L^{2d}\epsilon_d^{\mathrm{q}} \le 3C_6'\ell^3\mathsf{b}^{-3}L^{-d}(\log L^d\mathsf{b})^{8\mathfrak{p}_0+12}$, as $a_K^3L^{3k} = L^{-3d}$. The kernel of $\mathrm{Ad}(\gamma)\mathsf{J}_d\mathsf{A}_W^{-1}\mathsf{J}_d^*\mathrm{Ad}(\gamma)^{\mathsf{T}}$ is that of (iii) in the gauge of the chain, in which the top field is at most $\mathsf{a}_{\mathrm{top}}$ at each bare bond; so the first two clauses of (Col) hold with C_Σ, $\mathbb{E}^{[d_0]}$-almost surely and at all pairs of bare bonds, and the third holds with C_{cv} at every W. With the four explicit

conditions, of which $d_0 \ge n_{\mathrm{a}}$ and $d_0 \ge d_{\mathrm{num}}(\ell, \mathsf{b})$ follow from $d_0 \ge d_{\mathrm{adm}} = \max\{n_{\mathrm{a}}, d_{\mathrm{num}}(\ell, \mathsf{b})\}$ and the bounds of $c_\rho \varepsilon(L^{d_0}\mathsf{b})$ and of $4c_{\mathsf{h}}\delta(L^{d_0}\mathsf{b})$ from (1), every top depth $d_0 \ge d_{\mathrm{adm}}$ is admissible with $C_{\mathrm{adm}} = \max\{\sqrt{3}\, C_6, 3C_6', C_\Sigma, C_{\mathrm{cv}}\}$: this is (Adm).

Layer estimates. At an admissible d_0, with C free of K and d_0, the chain and these statements give: (L1) for $d_0 < d \le K$ the ordered product $e^{ia_K \mathbf{h}'_d} \cdots e^{ia_K \mathbf{h}'_{d_0+1}} e^{ia_K \mathbf{A}^{\mathrm{top}}}$ of the top field and of the layers of the depths $\le d$ is $\mathsf{U}^{\gamma}_{K-d}$, a gauge transform of a background over the block field V_{K-d} at depth d; (L2) $|\mathbf{h}_d| \le c_{\mathsf{h}} L^d \delta(L^d \mathsf{b})$ at every bond, and $\sum_d a_K |\mathbf{h}_d| \le \frac{1}{4}$; (L3) for a real bond function f, every component ξ of $\langle f, \mathbf{h}'_d - \mathbf{n}_d\rangle$ has $\mathbb{E}^{[d_0]}[e^{\lambda\xi}|\mathcal{F}_{d-1}] \le e^{4\lambda^2 G_d(f)^2/\gamma_{\mathrm{r}}}$ for real λ, $G_d(f)^2 = c_{\mathsf{J}}^2 \mathsf{b}^{-1} L^d \min\{\|f\|_1^2, L^{-3d}\|f\|_2^2\}$, by Lem. 15.3(iv), as the derivative of $\mathbf{h}'_d$ in the fluctuation obeys on the box the bounds of $\mathsf{b}^{-1/2} L^{d/2} \mathsf{J}_d$; (L4) $\sup|\mathbf{n}_d| \le C\mathsf{b}^{-1}(\log L^d \mathsf{b})^{2\mathfrak{p}_0}$ by (M), a bound whose sums over $d_0 < d' \le d$, times $L^{-d/2}$, tend to 0 as $d \to \infty$, and almost surely $\|\sum_{d_0 < d \le K} \mathbf{n}_d\|_{L^{12}(T_\ell)} \le \mathsf{c}_{12}$, a number free of K and d_0, as $\|\mathbf{n}_d\|_{L^{12}(T_\ell)} \le \|\mathbf{n}_d\|_\infty^{5/6} \|\mathbf{n}_d\|_{L^2}^{1/6}$ is summable over d by (M); (L5) the moments of the fluctuation fields and of the layers: at every bond b, $\mathbb{E}^{[d_0]}|B_{K-d}(b)|^2 \le m_2$, and for $q < \infty$, $\|a_K(\sum_d |\mathbf{h}_d(b)|)^2\|_{L^q(\mathbb{E}^{[d_0]})} \le \mathsf{c}_{(q)}\mathsf{b}^{-1}$, with m_2 and $\mathsf{c}_{(q)}$ free of K, d and d_0.

The fluctuation-field second moment was proved above before the mean and covariance estimates. For the layers: (L3) at $f = a_K^{-3}\delta_b$, for which $\langle f, \cdot\rangle$ is the value at b and $G_d(f)^2 = c_{\mathsf{J}}^2 \mathsf{b}^{-1} L^d$, bounds the centred layers at b in $L^{2q}(\mathbb{E}^{[d_0]})$ by $C\sqrt{q}\, \mathsf{b}^{-1/2} L^{d/2}$, whose sum over $d \le K$ is at most $C\sqrt{q}\, \mathsf{b}^{-1/2} a_K^{-1/2}$, and the means add at most $2a_K(\sum_{d \le K} C\mathsf{b}^{-1}(\log L^d \mathsf{b})^{2\mathfrak{p}_0})^2$, a number free of K times b^{-1}.

In its regular gauge the top field is bounded by $\mathsf{a}_{\mathrm{top}} \le CL^{10 d_0}$.

Passage from the chain measure to the Wilson measure. The inequality over the top depths: on invariant functions the chain measure from the top depth d has a density $\bar{\mathfrak{w}}_d \in [0,1]$ with respect to μ^K, with $\int (1 - \bar{\mathfrak{w}}_d)\, \mathrm{d}\mu^K \le \mathsf{R}_d$ (Lem. 15.3(i)); split $1 = \bar{\mathfrak{w}}_{d_1} + \sum_{d_1 < d \le K-1} (1 - \bar{\mathfrak{w}}_{d_1}) \cdots (1 - \bar{\mathfrak{w}}_{d-1}) \bar{\mathfrak{w}}_d + (1 - \bar{\mathfrak{w}}_{d_1}) \cdots (1 - \bar{\mathfrak{w}}_{K-1})$, bound each product of the sum by its last two factors and the last product by its last factor, and use Hölder's inequality; $\sum_d \mathsf{R}_d^{1/2} L^{pd}$ and $\sup_K \mathsf{R}_{K-1} L^{pK}$ are finite, as R_d decays faster than every exponential in d.

By (Adm) let d_{adm} be a depth, free of K, such that every top depth $\ge d_{\mathrm{adm}}$ is admissible. At $d_1 = d_{\mathrm{adm}}$ the moment bound applies at every top depth $d+1$ of the inequality, and $\mathsf{N}' \le \|\mathbf{B}_K\|_{\kappa;a}$, $\mathsf{N}' \le 23 L^{3K}$: this gives $\mathsf{M}_q < \infty$. The bound of the action at $R = \max\{1, \mathsf{N}'\}$ and M_{24q} give the moments of $\sup_s s^{\frac{3}{2}+\kappa} A^a(U_s)$. The good sets: at an admissible d_0 with $\mathsf{R}_{d_0} \le \frac{1}{2}\epsilon$, on $\{\|\mathbf{B}_K\|_{\kappa;a} \le R\}$ and for $a_K \le a_0(R)$ the four bounds put U in $\mathsf{O}_K(\bar{C}, \bar{s})$, $\bar{C} = (2 + c_3)R$, $\bar{s} = s_0(R)$, and Chebyshev's inequality with Lem. 15.3(i) gives $\mu^K(\mathsf{O}_K(\bar{C}, \bar{s})^{\mathrm{c}}) \le \mathsf{R}_{d_0} + C(1,\kappa) L^{20 d_0}/R$.

With $E = A^a(U_{\sigma/2})$, Lem. 16.2(f) bounds $a_K^{-4} \max_p a_p(U_\sigma)$ and $(J,J)_a(U_\sigma)$ by polynomials in E if $a_K E \le e_0$, and otherwise by $2a_K^{-4}$ and $48\ell^3 a_K^{-6}$, against $\mu^K(a_K E > e_0) \le (a_K/e_0)^{6p} \langle E^{6p}\rangle_{\mu^K}$. Tightness of Λ_σ: on $\mathsf{O}_K(\bar{C}, \bar{s})$, which has mass $\ge 1 - \epsilon$, Λ_σ is bounded by a number free of K (Step 6).

(b) Choose D_* so that for $D \ge D_*$

$$L^{m+D} \ge 3R_1 \mathfrak{M}, \qquad L^{m+D} > 6\rho_*, \qquad 2\theta_D \le \bar{\varepsilon}, \qquad 2B_3\theta_D \le \alpha_*,$$
$$2B_3\theta_D L^{-4} < \min\{\omega_M, (\kappa_2^{\mathrm{SO}})^{-1}\},$$

where α_*, ρ_* are from Prop. 16.10. All these constants are fixed, so the conditions eventually hold because $\theta_D \to 0$. For $V \in \mathcal{K}_D$, set $\eta = L^{-(K-D)}$ and $N_D = (2L^{m+D})^3$. Take the minimal backgrounds $U^{\mathrm{A}} = U_{K,D}(V)$ and $U^{\mathrm{B}} = U_{K+1,D}(V)$. Their regularity on the larger open ball $\{\varpi < 2\theta_D\}$ and the proposition give

$$(U^{\mathrm{A}})^g = e^{i\eta A} \mathcal{M}(U^{\mathrm{B}}), \qquad \sup|A| \le 2CB_3\theta_D N_D^{1/5} \eta^{1/5}, \qquad \varpi(U^{\mathrm{B}}) \le 2B_3\theta_D (\eta/L)^2.$$

The one-step contour estimate, Lem. YM3_I:14.6(iii), gives $\mathcal{M}(U^{\mathrm{B}})(c) = e^{iY_c}U^{\mathrm{B}}[c]$, $|Y_c| \le \kappa_3\varpi(U^{\mathrm{B}})$, where $[c]$ subdivides c into its L fine bonds. A path C of physical length at most R uses at most RL^D/η bonds of the first lattice. Telescoping the products therefore gives

$$|W_C(U^{\mathrm{A}}) - W_C(U^{\mathrm{B}})| \le 2RL^D B_3\theta_D \left(CN_D^{1/5}\eta^{1/5} + \frac{\kappa_3}{L^2}\eta\right).$$

For each fixed D this is summable in K, uniformly in $V \in \mathcal{K}_D$ and in the closed paths of length at most R defined at that cutoff. Its constant may depend on D through N_D; no uniformity in D is used in this step. Thus the unflowed loops have limits $w_C^D(V)$. The positive-time assertion remains a sketch at (Flow approximation). Its proposed approximation step, together with compactness of $\mathcal{K}_D$, would give a convergent subsequence of any sequence of top data and background flows; its limit solves the specified equation with the specified initial holonomies. Uniqueness identifies the limiting orbit curve. Uniform convergence of the potential representatives gives convergence of flowed holonomies: the difference of their path products is bounded by the path length times their supremum discrepancy. The sequential assertion and compactness of $\mathcal{K}_D$ make this convergence uniform over V and bounded-length paths on compact positive time intervals. The cutoff-uniform comparison between the bare field and the background is proved separately below.

The bare field and the background over $\pi_{K,D}U$. By Lem. 15.3(i) at an admissible $d_0 < D$ with R_{d_0} small, which the proved conclusion (Adm) provides, the comparison is made under $\mathbb{E}^{[d_0]}$. There U is a gauge transform of $e^{ia_K\mathbf{B}_K}$; and $e^{ia_K\mathbf{B}_K^{\le D}}$, with $\mathbf{B}_K^{\le D}$ the logarithm of the ordered product of the top and of the layers of the depths $\le D$, the field cut at the depth D, a function of $\pi_{K,D}U$, is a gauge transform of U_{K-D}, that is of the background over $\pi_{K,D}U$. (i) $\mathbf{B}_K - \mathbf{B}_K^{\le D}$ is small in the rough distance, in every L^q uniformly in K, at the rate $L^{-\kappa D/8}$: its linear part is the sum of the martingale differences of the depths beyond D, its bilinear members are those of (a) with a factor of a depth beyond D, and its means and the remainders of the product are small in L^p.

(ii) The rough norm of the cut field has moments free of K and D: its martingale part $u_{\le D} = \sum_{d\le D} u_{(d)}$ is the conditional mean $\mathbb{E}[u|\mathcal{F}_D]$ of that of the whole field, so $|u_{\le D}|_{-\frac12-\kappa;a} \le \mathbb{E}[|u|_{-\frac12-\kappa;a}|\mathcal{F}_D]$; for a bilinear member Υ of the rough norm (one tuple of directions and translations, one colour, one time, one point) the pairs of different depths whose finer depth exceeds D have vanishing conditional means given $\mathcal{F}_D$, so $\mathbb{E}[\Upsilon(u,u)|\mathcal{F}_D] = \Upsilon(u_{\le D}, u_{\le D}) + \sum_{d>D}\mathbb{E}[\mathsf{c}_d|\mathcal{F}_D]$, $\mathsf{c}_d = \mathbb{E}[\Upsilon(u_{(d)},u_{(d)})|\mathcal{F}_{d-1}]$ the compensators of (a), whose sum is bounded almost surely by $C(1+\mathsf{a}_{\mathrm{top}})s^{-\frac34-\frac{\kappa}{2}}$; with the rest as in (a) and Jensen's inequality, $\mathbb{E}[\|\mathbf{B}_K^{\le D}\|_{\kappa;a}^q]^{1/q} \le C(q,\kappa)(1+\mathsf{a}_{\mathrm{top}})^2$.

So for a large R and a large D, off a probability that is small uniformly in K, the two fields have rough norms $\le R$ and the rough distance $\le \eta$: they are the two data of the Lipschitz bound of (a).

From the short time to the time s. Let $s_* = \min\{s_{\mathrm{Lip}}(R), \frac12 s\}$ and $a_K \le a_0(R)$. At s_* the two flowed fields are gauge transforms of e^{ia_KA}, $e^{ia_KA'}$ with the four bounds and $\|A - A'\|_\infty \le (1+c_5)\eta s_*^{-\frac14-\frac{\kappa}{2}}$; by Lem. 16.2(c), (e) the actions of the flowed fields at $\frac12 s_*$, and those of the fields $e^{ia_K((1-r)A+rA')}$, $0 \le r \le 1$, are at most $E = 6(2+c_3)^4\ell^3R^4(\frac12 s_*)^{-\frac32-\kappa}$. For $a_KE \le e_0$, Lem. 16.2(f), started at $\frac12 s_*$, bounds the curvature $a_K^{-2}\varpi$ of the flow of the bare field on $[s_*, s]$ by a number F_1, and, started at the time 0 of the fields of the segment, bounds the number Λ_{s-s_*} of their flows by a number Λ_1; both read E and s alone.

For a flowed observable H of the flow time s: by Lem. 16.2(b), (d), the square field of $H\circ\Phi_{s-s_*}$ is at most L^2, L free of K, at the fields of the segment, whose ℓ^2 length is at most $\sqrt{3}\ell^{3/2}a_K^{-1/2}\|A'-A\|_\infty$; so the two values of H differ by at most $\sqrt{3}\ell^{3/2}\mathsf{L}\|A'-A\|_\infty$. With (i), (ii) this gives the first display: the mean squared difference is at most $3\ell^3\mathsf{L}^2(1+c_5)^2\eta^2 s_*^{-\frac12-\kappa}$ plus $4\sup|\varphi|^2$ times a small probability, uniformly in K. For the loops, whose square field is of the order of the length times a_K^{-2}, the comparison is made in the supremum distance modulo gauge, in the units of a_K, which is

at most $\|A-A'\|_\infty$ at s_*: by Lem. 15.7(a) at $\kappa' = F_1 a_K^2$ the flowed bare field has at every time of $[s_*, s]$ a gauge with a potential bounded by a number $C_0(F_1, \ell)$; the short-time theory of (a), restarted from these bounded data in steps of a length $\sigma_1(F_1, \ell, \kappa)$, multiplies the distance by at most a factor C_1 in each step, and asks nothing of the second field; after $m = \lceil (s-s_*)/\sigma_1 \rceil$ steps the factor is C_1^m, free of K; and $|W_C(V) - W_C(V')|$ is at most the length of C times the distance.

Hence $\lim_D \limsup_K \|W_C \circ \Phi_s - \Psi_D \circ \pi_{K,D}\|_{L^1(\mu^K)} = 0$ with Ψ_D free of K, which is strong convergence. Two configurations with equal loops are gauge equivalent (characters of SU(2)), and the orbits with $\|\mathfrak{A}\|_\infty + \operatorname{Lip}\mathfrak{A} \le C$ form a compact set on which the loops give the topology: this gives $\mathfrak{F}$. The second display follows from the first, from (C3), and from total variation at the depth D.

(c) The good sets at the time s: by the moments of the action in (a) and Lem. 16.2(f), started at $\frac14 s$, for $\epsilon > 0$ there is F_1, free of K, such that the closed invariant set $\mathsf{G}_K = \{a_K^{-2}\varpi(\Phi_r U) \le F_1 \text{ for } \frac12 s \le r \le s\}$ has $\mu^K(\mathsf{G}_K) \ge 1-\epsilon$ for all large K. On G_K, Lem. 15.7(a) at $\kappa' = F_1 a_K^2$ gives at the time $s - \sigma_1$, $\sigma_1 = \min\{s_0''(R_1), \frac12 s\}$, where $s_0''(R) = (c_1'' R^{-2})^{4/(1-2\kappa)} \le s_0(R)$ and $c_1'' \in (0, c_1]$ depends only on κ, a gauge with a potential bounded by $C_0(F_1, \ell)$, of rough norm at most $R_1 = \max\{1, C_0 + C_0^2\}$, as $\|\mathbf{B}\|_{\kappa;a} \le \|\mathbf{B}\|_\infty + \|\mathbf{B}\|_\infty^2$; the short-time theory of (a) restarted from it gives at the time s a gauge transform $e^{ia_K A}$ of the flowed field with $\|A\|_\infty + \|\nabla A\|_\infty + \|\nabla\nabla A\|_\infty \le C$, a number $C \ge 1$ free of K: the first two members by the bounds of (a) at the time $\sigma_1 \le s_0(R_1)$ after the restart, the second differences by the same fixed point with one more derivative, which closes up to the time $s_0''(R_1)$ and gives $\|\nabla\nabla A_\sigma\|_\infty \le c_8 R_1 \sigma^{-\frac54 - \frac\kappa2}$ for $0 < \sigma \le s_0''(R_1)$, σ the time after the restart, c_8 depending only on κ; here $\sigma = \sigma_1 \le s_0''(R_1)$.

For $a_K C \le \frac{1}{24}$ and Lipschitz f, $|\mathcal{O}(f)(e^{ia_K A}) - \mathcal{O}^0(f)(\mathfrak{A})| \le c\,\ell^3 C^5 (\|f\|_\infty + \operatorname{Lip} f)\, a_K$ for the interpolated connection $\mathfrak{A}$; $\mathcal{O}^0(f)$ is continuous on the compact set of these orbits, so the theorems of Stone and Weierstrass and of Tietze give Θ; the moments come from (a). □

16.4. **Deterministic estimates along the Wilson flow.**

Lemma 16.2 (Deterministic bounds)**.** For all K, m, $\beta > 0$, every U and real X; $\partial U(p) = c_p \mathbf{1} + i\mathsf{s}_p$, $Y_1, \dots, Y_4$ the values of X at the four bonds of p carried to its base point by the parallel transport of U, with the signs $\epsilon = (+,+,-,-)$, and $Z_p(X) = \sum_i \epsilon_i Y_i$:

(a) $\langle X, \partial a_p\rangle = \langle Z_p(X), \mathsf{s}_p\rangle$, so $|\partial_b a_p| = |\mathsf{s}_p|$ for $b \in p$;

$$\nabla^2 a_p[X,X] = c_p |Z_p(X)|^2 + \sum_{i<j} \epsilon_i \epsilon_j \langle i[Y_i, Y_j], \mathsf{s}_p\rangle,$$

$$-3\varpi_p \sum_{b\in p} |X(b)|^2 \le \nabla^2 a_p[X,X] \le (4 + 3\varpi_p) \sum_{b\in p} |X(b)|^2,$$

and

$$-12\varpi(U)\|X\|^2 \le \nabla^2 A_K(U)[X,X] \le 40\|X\|^2$$

(second derivatives along $e^{isX}U$);

(b) with $w_s = D\Phi_s(U)X$: $\frac{d}{ds}\|w_s\|^2 = -2a_K^{-2}\nabla^2 A_K(U_s)[w_s, w_s]$; so $\|D\Phi_\sigma(U)\| \le e^{12\Lambda_\sigma(U)}$ and $\Gamma_K(G\circ\Phi_\sigma)(U) \le e^{24\Lambda_\sigma(U)}\Gamma_K(G)(U_\sigma)$ for $G \in C^1(\Omega_K)$; $D\Phi_s(U)\mathrm{d}_U\lambda = \mathrm{d}_{U_s}\lambda$ and $D\Phi_s(U)J(U) = J(U_s)$, and $D\Phi_\sigma(U)^{\mathsf{T}}$ maps horizontal vectors at U_σ to horizontal vectors at U;

(c) $\partial_\sigma A^a(U_\sigma) = -(J,J)_a(U_\sigma)$; $(J,J)_a(U_\sigma) \le e^{24(\Lambda_\sigma - \Lambda_s)(U)}(J,J)_a(U_s)$ for $s \le \sigma$; $A^a(V) \le \frac32 \ell^3 (a_K^{-2}\varpi(V))^2$; $A^a(U_{\sigma/2}) \le 24\ell^3 \Lambda_\sigma(U)^2 \sigma^{-2}$; and $(J,J)_a(U_\sigma) \le \frac{2}{\sigma} e^{24\Lambda_\sigma(U)} A^a(U_{\sigma/2})$;

(d) $\Gamma_K(\mathcal{O}(f)) \le 2\|f\|_\infty^2 (J,J)_a + 256\|\nabla f\|_\infty^2 A^a$ and $|v\,\mathcal{O}(f)| \le ((J,J)_a \Gamma_K(\mathcal{O}(f)))^{1/2}$;

(e) on $\mathsf{O}_K(\bar C, \bar s)$, $a_K^{-2}\varpi(U_s) \le 2\bar C s^{-\frac34 - \frac\kappa2} + 2\bar C^2 s^{-\frac12 - \kappa}$ for $0 < s \le \bar s$, and so $\Lambda_\sigma \le (8\bar C \sigma^{\frac14 - \frac\kappa2} + 4\bar C^2 \sigma^{\frac12 - \kappa})/(1-2\kappa)$ for $\sigma \le \bar s$;

(f) let $e_0 = 10^{-4}$, $c_{10} = 5.1 \cdot 10^4$, $x_0 = (1440 c_{10})^{-1}$, $E = A^a(U_{s_1})$ and $a_K E \le e_0$; at $E = 0$ set $E^{-2} = +\infty$, the flat field being stationary. Then for $s \ge s_1$, $a_p(U_s) \le e_0$, and $z_p = a_K^{-4} a_p(U_s)$ satisfies $\partial_s z_p \le \Delta_a z_p + c_{10} \bar z_p z_p^{1/2}$ (Δ_a the Laplacian on parallel plaquettes, $\bar z_p$ the largest z on a box of plaquettes about p); $a_K^{-4} \max_p a_p(U_s) \le \frac{5}{2} E \min\{s - s_1, x_0^4 E^{-2}, 1\}^{-3/2}$ for $s > s_1$; and at $s_1 = \frac{\sigma}{2} \le \frac{1}{2}$, $(J,J)_a(U_\sigma) \le e^{48}[\frac{4E}{\sigma} + 7E^{3/2}\sigma^{-3/4} + 3x_0^{-3}E^3]$;

(g) (Bochner.) For $H \in C^\infty(\Omega_K)$, with ∇^2 the Hessian of the bi-invariant metric of Ω_K, a product of unit spheres S^3 of Ricci curvature 2,

$$\|\mathcal{L}_K H\|^2_{L^2(\mu^K)} = a_K^{-2} \int \|\nabla^2 H\|^2_{\mathrm{HS}} \,\mathrm{d}\mu^K + 2a_K^{-1}\mathcal{E}_K(H) + a_K^{-2}\beta \int \nabla^2 A_K(\partial H, \partial H)\,\mathrm{d}\mu^K;$$

consequently $(\int |\partial H|^4 \,\mathrm{d}\mu^K)^{1/2} \le \sup|H|\,(3\mathsf{m} + 2(28\beta \sup|H|\,\mathsf{m})^{1/2})$, $\mathsf{m} = a_K \|\mathcal{L}_K H\|_{L^2(\mu^K)}$, and $H_K \in C^\infty(\Omega_K)$ with $\sup_K \|\mathcal{L}_K H_K\|_{L^2(\mu^K)} < \infty$ and $\liminf_K \mathcal{E}_K(H_K) > 0$ have $a_K^{-2}\beta \int \nabla^2 A_K(\partial H_K, \partial H_K)\,\mathrm{d}\mu^K \to -\infty$; and $(\mathcal{L}_K \cos G)^2 + (\mathcal{L}_K \sin G)^2 = (\mathcal{L}_K G)^2 + \Gamma_K(G)^2$ for real $G \in C^\infty(\Omega_K)$;

(h) (Weitzenböck and Kato.) $\nabla^2 A_K(U)[z,z] = \langle z, (\nabla_U^\mathsf{T}\nabla_U + R_U)z\rangle$ for horizontal z at U, where $(\nabla_U^\mathsf{T}\nabla_U z)(b) = \sum_{b' \sim b}(z(b) - \tau_{bb'} z(b'))$, the sum over the six parallel neighbours with the parallel transports $\tau_{bb'}$ of U, and $\|R_U(b,b')\| \le 4\varpi(U)\mathsf{n}(b,b')$, $\mathsf{n}(b,b') = 1$ if $b = b'$ or b, b' lie in a common plaquette and 0 otherwise. For invariant $G \in C^1(\Omega_K)$ and every bond b,

$$|\partial_b(G \circ \Phi_\sigma)(U)| \le \big[e^{4\Lambda_\sigma(U)\mathsf{n}}\, e^{-\sigma a_K^{-2}\Delta_0}\, |\partial G|(U_\sigma)\big](b),$$

Δ_0 the free Laplacian on the bonds (six parallel neighbours) and $|\partial G|(V)$ the function $b \mapsto |\partial_b G(V)|$; so $\|\partial(G \circ \Phi_\sigma)(U)\|_{\ell^q} \le e^{52\Lambda_\sigma(U)} \|\partial G(U_\sigma)\|_{\ell^q}$ for $1 \le q \le \infty$, and $\sup_b |J(U_\sigma)(b)| \le e^{52(\Lambda_\sigma - \Lambda_s)(U)} \sup_b |J(U_s)(b)|$ for $s \le \sigma$;

(i) (One-bond fibres.) For $N \ge 3$, $p \ni b_0$, $s \ge 0$, the map $g \mapsto \partial(\Phi_s U^{(g)})(p)$ of G to itself, $U^{(g)}$ the field U with the value g at b_0, is onto; so $\sup_U \varpi(U_s) = 2$, and $\sup_U \mathcal{O}(f)(U_s) \ge 2f(x(p))/a_K$ for $f \ge 0$, at every flow time. At a field with values in the centre $\{\pm\mathbf{1}\}$, $\partial A_K = 0$ and $\nabla^2 A_K[X,X] = \sum_p c_p |Z_p(X)|^2$, $c_p = \pm 1$; at the field that is $-\mathbf{1}$ at one bond and $\mathbf{1}$ elsewhere, $\|D\Phi_\sigma\| \ge e^{4\sigma/a_K^2}$.

■

Proof sketch. (a) Expand $\partial(e^{itX}U)(p)$ to second order in t, with the values of X carried to the base point of p; the bounds follow from $|\mathsf{s}_p| \le \varpi_p$, $|c_p| \le 1$, $|Z_p|^2 \le 4\sum_{b \in p} |X(b)|^2$, and four plaquettes at each bond. (b) $\frac{d}{ds}\|D\Phi_s X\|^2 = -2a_K^{-2}\nabla^2 A_K(U_s)[D\Phi_s X, D\Phi_s X]$ and Gronwall; the flow commutes with gauge transformations and moves along the current. (c) The current solves the linearised flow; A^a is non-increasing along the flow, ϖ bounds a_p, and averaging the bound of the current over $[\frac{\sigma}{2}, \sigma]$ gives the last inequality. (d) $\partial_b \mathcal{O}(f) = a_K^{-1} f(x_b)\partial_b A_K + a_K^{-1}\sum_{p \ni b}(f(x(p)) - f(x_b))\partial_b a_p$, x_b a point of b, so that $|x(p) - x_b| \le 2a_K$ for $p \ni b$, and $\sum_b (\sum_{p \ni b} \varpi_p)^2 \le 32 A_K$.

(e) $\varpi(e^{iaA}) \le 2a^2(\|A\|_\infty^2 + \|\nabla A\|_\infty)$, integrated in s. (f) The lattice Bianchi identity in a gauge near p gives the differential inequality where $a_K^4 \bar z_p \le e_0$; Duhamel's formula with $\|P_t z\|_\infty \le t^{-3/2}E$ bounds $\sup_t t^{3/2} \max_p z_p$ as long as $t^{1/4}E^{1/2} \le x_0$, and the action does not increase; then (c) on an interval before σ on which Λ grows by at most 1. (g) Bochner's formula with the drift; the limit follows from the formula, as its first member on the right is ≥ 0 and the second, $2a_K^{-1}\mathcal{E}_K(H_K)$, tends to ∞; the bound follows from $\int \langle \partial H, \partial F\rangle \,\mathrm{d}\mu^K = -a_K \int F\, \mathcal{L}_K H \,\mathrm{d}\mu^K$ at $F = H|\partial H|^2$ and using $\nabla^2 A_K \ge -28$; the identity is the chain rule $\mathcal{L}_K \phi(G) = \phi'(G)\mathcal{L}_K G + \phi''(G)\Gamma_K(G)$.

(h) Expanding $\sum_p |Z_p(X)|^2 + \sum_x |(\mathrm{d}_U^\mathsf{T} X)(x)|^2$, the products of parallel bonds give the covariant Laplacian and those of the corners of a plaquette cancel but for $2\sum_p \langle X_1, (1 - \mathrm{Ad}\,\partial U(p))X_4\rangle$; with (a) this is the identity. The adjoint of the linearised flow keeps horizontal vectors horizontal, and $|z(b)|$ is then a subsolution of the parabolic system with the off-diagonal entries ≥ 0; the comparison principle and the commutation of Δ_0 with n (Δ_0 is the same scalar parallel-neighbour convolution

in every bond direction) give the bound; $e^{-t\Delta_0}$ is doubly stochastic and n has row sums 13. The current is a forward horizontal solution, whence its maximum principle.

(i) $s \mapsto (g \mapsto \partial(\Phi_s U^{(g)})(p))$ is a homotopy from a translation of S^3, or a translation composed with $g \mapsto g^{-1}$, and a map of S^3 to itself that misses a point is homotopic to a constant; at the centre fields $\mathsf{s}_p = 0$, and $X = e\,\delta_{b_0}$ gives $\langle X, \nabla^2 A_K X\rangle = -4|e|^2$ at the flipped field. □

Lemma 16.3 (Exact identities for the Laplacian along the flow)**.** For all K, m, $\beta > 0$:

(a) for $f \in C^\infty(\Omega_K)$, $[\Delta_K, v]f = -8a_K^{-1} v f - 2a_K^{-3}\langle \nabla^2 A_K, \nabla^2 f\rangle_{\mathrm{HS}}$, and the diagonal blocks of $\nabla^2 A_K$ are $\delta_{\alpha\beta}\sum_{p\ni b}(1 - a_p)$;

(b) Φ_σ is a diffeomorphism of Ω_K, and $\log\det D\Phi_\sigma(U) = -12a_K^{-2}\int_0^\sigma (3N^3 - A_K(U_s))\,\mathrm{d}s$;

(c) at every U at which d_U is injective on the site functions (all U off a closed null set), with $\mathsf{G}_U = (\mathrm{d}_U^*\mathrm{d}_U)^{-1}$, $\mathsf{I}(U) = \frac{1}{2}\log\det \mathrm{d}_U^*\mathrm{d}_U$ and $\mathsf{j}^{\mathrm{gh}}_U(b) = i\sum_{\alpha,\gamma}\mathsf{G}_U((b_-,\alpha),(b_+,\gamma))[\sigma_\alpha, \mathrm{Ad}(U(b))\sigma_\gamma]$: $\mathsf{j}^{\mathrm{gh}}_U = \partial\mathsf{I}(U)$, and for invariant $f \in C^2(\Omega_K)$

$$\Delta_K f(U) = a_K^{-1}\sum_e \frac{d^2}{dt^2} f(e^{ite}U)\Big|_{t=0} + a_K^{-1}\sum_b \langle \mathsf{j}^{\mathrm{gh}}_U(b), \partial_b f(U)\rangle = \Delta_K^{\mathrm{hor}} f + \Gamma_K(\mathsf{I}, f),$$

(e) an orthonormal basis of the horizontal vectors at U; for a site function λ,

$$\frac{d^2}{dt^2} f(e^{it\,\mathrm{d}_U\lambda}U)|_0 = \sum_b \langle i[\lambda(b_-), \mathrm{Ad}(U(b))\lambda(b_+)], \partial_b f(U)\rangle;$$

and at $f = G\circ\Phi_\sigma$ the second member is $a_K^{-1}\langle D\Phi_\sigma(U)\mathsf{j}^{\mathrm{gh}}_U, \partial G(U_\sigma)\rangle$: the gauge term is evaluated once, at the bare field, and then carried by the linearised flow.

■

Proof sketch. (a) On a Riemannian manifold, $[\Delta, \langle\nabla\phi, \nabla\cdot\rangle]f = \langle\nabla\Delta\phi, \nabla f\rangle + 2\langle\nabla^2\phi, \nabla^2 f\rangle + 2\,\mathrm{Ric}(\nabla\phi, \nabla f)$; here $\mathrm{Ric} = 2$, and $a_K\Delta_K A_K = \sum_{b,\alpha}(\partial_b^\alpha)^2 A_K = 12(3N^3 - A_K)$, because each bond variable enters a plaquette through a matrix coefficient of the fundamental representation, an eigenfunction of the Laplacian of S^3 of eigenvalue -3, and four bonds enter each of the $3N^3$ plaquettes. (b) Liouville's formula $\frac{d}{ds}\log\det D\Phi_s = (\mathrm{div}\, v)\circ\Phi_s = -a_K^{-1}(\Delta_K A_K)\circ\Phi_s$. (c) The trace of the Hessian over an orthonormal basis splits into the horizontal part and the part along the gauge directions; along $\mathrm{d}_U\lambda$ the curve $e^{it\mathrm{d}_U\lambda}U$ differs from the gauge orbit at second order, which gives the formula for one site function by invariance of f; summing over a basis orthonormal for $\mathrm{d}_U^*\mathrm{d}_U$ gives j^{gh}, and differentiating $\log\det$ identifies it with $\partial\mathsf{I}$.

The last statement is the chain rule with (b) of Lem. 16.2. □

16.5. **The lattice defect in generator estimates.**

Lemma 16.4 (A lattice defect)**.** On $(\mathbb{Z}/N)^3$ let $\varsigma_p = -1$ on the four plaquettes that contain a bond b_0 and $+1$ elsewhere. For $N \ge N_0$ the operator $\mathrm{d}^{\mathsf{T}}\varsigma\,\mathrm{d}$ on real 1-forms, per colour the Hessian of A_K at the field that is $-\mathbf{1}$ on b_0 and $\mathbf{1}$ elsewhere, has exactly three negative eigenvalues, $-\lambda_1$ simple and $-\lambda_2$ double, $6.6 < \lambda_1 < 6.62$, $0.79 < \lambda_2 < 0.8$, with orthonormal eigenvectors e_1 (of $-\lambda_1$) and e_2, e_3 (of $-\lambda_2$) that decay exponentially away from b_0. Let $\hat e_1, \hat e_2$ be the lattice unit vectors of the two axes orthogonal to b_0, so that $b_0 + \hat e_1$, $b_0 + \hat e_2$ are the two translates of b_0 parallel to it that share a plaquette with it, and the three bonds b_0, $b_0 + \hat e_1$, $b_0 + \hat e_2$ lie in ten plaquettes; the matrix $[e_i(b_0 + \hat e_j)]_{i=1,2,3;\, j=0,1,2}$, $\hat e_0 := 0$, is upper triangular and invertible, with an inverse bounded by an absolute constant. And $\langle X, \mathrm{d}^{\mathsf{T}}\varsigma\,\mathrm{d}X\rangle \ge c_1|\mathrm{d}X|^2 - 7\sum_i\langle e_i, X\rangle^2$ with an absolute $c_1 > 0$. For $\varepsilon \in (0,1]$ and $\Lambda_0 \ge 1$ there are $N_*, T_* < \infty$ and $\delta_*, \rho_0, \theta, c_\sharp > 0$, $0 < \delta \le \frac{1}{10}$, $C_\sharp < \infty$, depending only on ε and Λ_0, such that for $N = \ell/a_K \ge N_*$, $T \ge T_*$, every bond b_0 and every U whose flow in lattice time, $U_t^\circ = \Phi_{a_K^2 t}U$, has $\sup_{t\le T}\varpi(U_t^\circ) \le \delta_*$, $\int_0^T \varpi(U_t^\circ)\,\mathrm{d}t \le \Lambda_0$, $T \le \Lambda_0 N^2$ and $A_K(U^\circ_{T/2}) \le \delta_*^2$, there is a Borel set $\mathcal{S}$ of $\zeta = (\zeta_0, \zeta_1, \zeta_2) \in \mathfrak{g}^3$, $|\zeta| \le \rho_0 \le 1$, of Lebesgue measure $\ge c_\sharp e^{-(3\lambda_1 + 6\lambda_2 + \varepsilon)T}$, such that

the field W_ζ, equal to U but for $W_\zeta(b_0) = -e^{i\zeta_0}U(b_0)$ and $W_\zeta(b_0+\hat e_j) = e^{i\zeta_j}U(b_0+\hat e_j)$ $(j=1,2)$, and the fields $\tilde V_t$, equal to $\Phi_{a_K^2 t}W_\zeta$ but for the sign of their value at b_0, have the properties (a)–(c) below. Here $a_p(\Phi_{a_K^2 t}W_\zeta) = \varsigma_p a_p(\tilde V_t) + 1 - \varsigma_p$ for every plaquette p, so that $A_K(\Phi_{a_K^2 t}W_\zeta) = 8 + \sum_p \varsigma_p a_p(\tilde V_t)$, and $t \mapsto \tilde V_t$ is the gradient flow, in lattice time, of $\tilde A = \sum_p \varsigma_p a_p + 8$. (a) $\varpi(\tilde V_t) \le \delta$ for $0 \le t \le T$: the four plaquettes of $\Phi_{a_K^2 t}W_\zeta$ at b_0 lie within δ of $-\mathbf{1}$ and all the others within δ of $\mathbf{1}$; (b) for every real function f of the plaquettes and real f_0,

$$\big|\partial_{b_0}\big(\textstyle\sum_p \mathsf{f}_p\, a_p \circ \Phi_{a_K^2 T}\big)(W_\zeta)\big| \ge \Big[c_\sharp|\mathsf{f}_0| - C_\sharp\Big(\textstyle\sum_p |\mathsf{f}_p - \mathsf{f}_0|^2 e^{-2\theta|p|}\Big)^{1/2}\Big]e^{(\lambda_1-\varepsilon)T},$$

$|p|$ the distance of p from b_0; in particular $|\partial_{b_0}(A_K \circ \Phi_{a_K^2 T})(W_\zeta)| \ge c_\sharp e^{(\lambda_1-\varepsilon)T}$; (c) $\int_{T/2}^T \sum_p a_p(\tilde V_t)\,\mathrm{d}t \le \frac{3}{2}T\,A_K(U^\circ_{T/2}) + C_\sharp$.

(d) (Traversing the fibre of one bond.) Let $\lambda_1' = \lambda_1 - 0.027$ and $\lambda_2'' = \lambda_2 + \frac{1}{700}(\lambda_1 - \lambda_2) + 10^{-3}$; then $\lambda_1' - 6\lambda_2'' > \frac{2}{5}$. For $\Lambda_0 \ge 1$ there are $\epsilon_{\mathrm{d}}, \rho_2, c_{\mathrm{d}} > 0$ and $N_{\mathrm{d}} < \infty$, depending only on Λ_0, such that the following holds for $N \ge N_{\mathrm{d}}$, $T \ge N_{\mathrm{d}}$ with $N^3 \le c_{\mathrm{d}}T^2$, every bond b_0 and every U with $\int_0^T \varpi(U_t^\circ)\,\mathrm{d}t + T/N^2 \le \Lambda_0$ and $\sup_{t\le T}\varpi(U_t^\circ) \le \epsilon_{\mathrm{d}}$. There is a compact set $\mathsf{Y} \subset \mathfrak{g}^2$ of Lebesgue measure $\ge c_{\mathrm{d}}e^{-6\lambda_2'' T}$ such that for every $(y_1,y_2) \in \mathsf{Y}$, with W the field equal to U but for $W(b_0+\hat e_j) = e^{iy_j}U(b_0+\hat e_j)$ $(j=1,2)$ and $W^{(g)}$ the field W with the value g at b_0, and for every unit vector $e \in \mathfrak{g}$, there are $x_0 \in \mathfrak{g}$, with $|(x_0,y_1,y_2)| \le 30C_\triangle\rho_2 \le 1$, where $C_\triangle = 4c_\triangle$ and $c_\triangle$ is the absolute constant that bounds the inverse of the matrix $[e_i(b_0+\hat e_j)]$, and $r_* \in (0, \frac{4}{3}\rho_2 e^{-\lambda_1' T}]$ for which $\mathsf{v}(r) = A_K(\Phi_{a_K^2 T}W^{(g_r)})$, $g_r = -e^{ire}e^{ix_0}U(b_0)$, has $\mathsf{v}(0) \in [7,9]$ and $\mathsf{v}(0) - \mathsf{v}(r_*) \ge 2\rho_2^2$. If $f \in C^\infty(T_\ell)$ is real, $f(x_*) = \max|f| > 0$ and b_0 is based at a site nearest to x_*, the same holds, with constants that depend besides on f and for $a_K \le c_{\mathrm{d}}(f)$, for $\mathsf{v}_f(r) = \sum_p f(x(p))\,a_p(\Phi_{a_K^2 T}W^{(g_r)})$ in the place of v, and with the single condition $\mathsf{v}_f(0) - \mathsf{v}_f(r_*) \ge f(x_*)\rho_2^2$ in the place of the two conditions on v. ∎

Sketch. Flipping the sign of $U(b_0)$ conjugates the flow of A_K to that of $\tilde A = \sum_p \varsigma_p a_p + 8$, whose Hessian at small fields is close to $\mathrm{d}^{\mathsf{T}}\varsigma\,\mathrm{d}$. The number of negative eigenvalues is found exactly. For $E > 0$, with $S = (\mathrm{d}^{\mathsf{T}}\mathrm{d} + E)^{1/2}$ and Π_4 the projection onto the four plaquettes, $\mathrm{d}^{\mathsf{T}}\varsigma\,\mathrm{d} + E = S(1 - 2R^{\mathsf{T}}R)S$, $R = \Pi_4 \mathrm{d}S^{-1}$; by Sylvester's law of inertia the number of eigenvalues below $-E$ is the number of eigenvalues > 1 of $2RR^{\mathsf{T}} = K(E) = 2\Pi_4[\frac{-\Delta}{-\Delta+E} - \mathrm{d}_2^{\mathsf{T}}G_E\mathrm{d}_2]\Pi_4$, d_2 the coboundary of 2-forms and $G_E = (-\Delta + E)^{-1}$, which decreases in E. On $\mathbb{Z}^3$, $K(E)$ is diagonal in three patterns of signs on the four plaquettes, taken relative to the signs of $\mathrm{d}\delta_{b_0}$: $(1,1,1,1)$, the doublet $(1,0,-1,0)$, $(0,1,0,-1)$, and $(1,-1,1,-1)$, with the entries

$$K_1 = 2 - \tfrac{E}{3}(1 - EG_E(0)), \qquad K_2 = 2 - F_2, \qquad F_2 = 2E\big(G_E(0) + G_E(e_1)\big) + 4\big(G_E(0) - G_E(e_1+e_2)\big),$$
$$K_3 = 2\big[1 - E\big(G_E(0) - G_E(e_1)\big) - 4G_E(0) + 8G_E(e_1) - 4G_E(e_1+e_2)\big],$$

where $G_E(x) = (2\pi)^{-3}\int_{[-\pi,\pi]^3} e^{ik\cdot x}\big(E + \sum_\mu 2(1-\cos k_\mu)\big)^{-1}dk$ is the Green function of $-\Delta + E$ on the sites of $\mathbb{Z}^3$. Its Neumann series $G_E(x) = \sum_{m\ge 0} c_m(x)(6+E)^{-m-1}$, with the number $c_m(x) = \sum_n \binom{m}{n}\binom{n}{(n+x_1+x_2)/2}\binom{n}{(n+x_1-x_2)/2}\binom{m-n}{(m-n+x_3)/2}$ of the walks of m steps from 0 to x (a binomial coefficient whose lower entry is not an integer between 0 and its upper entry read as 0), encloses $G_E(x)$ between its partial sum up to $m = M$ and that sum plus $E^{-1}(\frac{6}{6+E})^{M+1}$; at $M = 240$ the width is below 10^{-12} at the four energies 6.6, 6.62, 0.79, 0.8. This gives, to the digits shown, $E(1 - EG_E(0)) = 2.997958$ at $E = 6.6$ and 3.002351 at $E = 6.62$, and $F_2 = 0.99944880$ at $E = 0.79$ and 1.00144415 at $E = 0.8$; so $K_1(6.6) > 1 > K_1(6.62)$ and $K_2(0.79) > 1 > K_2(0.8)$. As $E \to 0$, $K_1 \to 2$, $K_2 \to 2 - 4(g_0 - g_2)$ and $K_3 \to 8(g_0 - g_2) - \frac{2}{3}$, $g_0 - g_2 = G_0(0) - G_0(e_1+e_2)$; since $0 \le 1 - \cos k_1\cos k_2 \le (1-\cos k_1) + (1-\cos k_2)$, the symmetry of the axes gives $0 \le g_0 - g_2 - [G_E(0) - G_E(e_1+e_2)] \le \frac{E}{3}G_E(0)$, and the series at $E = \frac{1}{10}$

with $M = 700$, of width $\le 9.3 \cdot 10^{-5}$, give $0.19195 < g_0 - g_2 < 0.19970$; so $K_2(0+) > 1.2$ and $K_3(0+) < 0.94$. As the entries decrease in E, the third pattern gives no negative eigenvalue, the first gives the simple $-\lambda_1 \in (-6.62, -6.6)$, the doublet gives the double $-\lambda_2 \in (-0.8, -0.79)$, and there is no other. On $(\mathbb{Z}/N)^3$ the Green function is $\sum_{k\in\mathbb{Z}^3} G_E(x+Nk)$, which differs from $G_E(x)$ by at most $E^{-1}\sum_{k\neq 0}(\frac{6}{6+E})^{|x+Nk|_1}$, uniformly small for $E \ge 0.7$ and $|x|_1 \le 2$, and the entries at $E \to 0$ are Riemann sums of those on $\mathbb{Z}^3$: the strict inequalities persist for $N \ge N_0$. The eigenvectors: $e_i \propto \mathrm{d}^{\mathsf{T}} G_{\lambda_i}\eta_i$, η_i the pattern of the mode on the four plaquettes, so they decay at the rate of $G_E(x) \le E^{-1}(\frac{6}{6+E})^{|x|_1}$: as $\frac{6}{6+\lambda_i} \le \frac{6}{6.79} < e^{-0.12}$, $|e_i(b)| \le C_e e^{-c_e|b|}$ with $c_e = 0.12$, an absolute number C_e and $|b|$ the distance of the bond b from b_0; $e_1(b_0)^2 \ge 2/\lambda_1$; the doublet patterns are odd under the reflections across the two planes through b_0 that contain $\hat e_2$, resp. $\hat e_1$, which fix b_0 and $b_0 + \hat e_2$, resp. b_0 and $b_0 + \hat e_1$, so $e_2(b_0) = e_2(b_0+\hat e_2) = 0 = e_3(b_0) = e_3(b_0+\hat e_1)$, while $|e_2(b_0+\hat e_1)| = |e_3(b_0+\hat e_2)| \ge 0.049$: the matrix is triangular. *The coercivity.* On the range of d the form $\langle \eta, \varsigma\eta'\rangle$ is that of $B = 1 - 2\mathsf{P}\Pi_4\mathsf{P}$, P the projection onto the range of d; the non-zero eigenvalues of $\mathsf{P}\Pi_4\mathsf{P}$ are those of $\frac12 K(0+)$, so the spectrum of B lies in $\{-1, 1-K_2(0+), 1-K_3(0+), 1\}$: $\|B\| \le 1$, $|B| \ge c_B = 0.06$ for $N \ge N_0$, and B has three negative eigenvalues. On the span of the $\mathrm{d}e_i$, three-dimensional, $B \le -\frac{\lambda_2}{12}$, as $\langle \mathrm{d}e_i, \varsigma\,\mathrm{d}e_j\rangle = -\lambda_j\delta_{ij}$ and $|\mathrm{d}Y|^2 \le 12|Y|^2$. A self-adjoint B with $\|B\| \le 1$, $|B| \ge c_B$ and an n-dimensional negative spectral subspace is $\ge c_- c_B$ on the B-orthogonal complement of every n-dimensional subspace on which $B \le -c_-$ (compare the two spectral parts in the norm of $|B|$). For $X = \sum_i a_i e_i + X'$, $a_i = \langle e_i, X\rangle$, one has $\langle \mathrm{d}X', \varsigma\,\mathrm{d}e_i\rangle = -\lambda_i\langle X', e_i\rangle = 0$, so $\langle X, \mathrm{d}^{\mathsf{T}}\varsigma\,\mathrm{d}X\rangle = -\sum_i \lambda_i a_i^2 + \langle \mathrm{d}X', \varsigma\,\mathrm{d}X'\rangle \ge -\lambda_1\sum_i a_i^2 + 0.0039(\frac12|\mathrm{d}X|^2 - 12\sum_i a_i^2)$: the bound with $c_1 = 0.0019$, as $\lambda_1 + 0.05 \le 7$. *The unstable coordinates.* Let $\tilde\Phi_t$ be the flow of $\tilde A$ in lattice time, $\tilde V_t = \tilde\Phi_t \tilde V_0$, $\tilde V_0$ the field W_ζ with the sign of its value at b_0 reversed. The nine coordinates $\xi_i(V) = \sum_p \omega_i(p)\,\mathrm{Ad}(T_p(V))\,\mathsf{s}_p(V) \in \mathfrak{g}$, $i = 1,2,3$, are the curvature paired with $\omega_i = -\lambda_i^{-1}\varsigma\,\mathrm{d}e_i$ (so $\mathrm{d}^{\mathsf{T}}\omega_i = e_i$, and $|\omega_i(p)| \le C_e e^{-c_e|p|}$ once C_e is enlarged by an absolute factor), cut off at the distance $\frac14 N$ from b_0 and carried to one root along a comb of paths: $T_x(V)$ is the ordered product of the values of V along the path of the comb from the root to the site x, and $T_p = T_x$, $T_b = T_x$ for a plaquette p and a bond b based at x; along a flow line of $\tilde A$, $\dot\xi_i = \lambda_i\xi_i + r_i$ with $|r_i| \le C(q(\tilde V_t) + N^2 e^{-c_e N/4})$, $q(V) = \sum_p e^{-c|p|}\varpi_p(V)^2$ with $c = \frac12 c_e$, a weighted sum of the plaquettes, at most $C_q\varpi(V)^2$ with an absolute C_q. The other directions are controlled by the coercivity bound in the rough field, through a lattice Weitzenböck identity and Kato's inequality, as long as the nine coordinates stay in the box $\mathfrak{B} = \{\max_i|\xi_i| \le \delta_u\}$, with the number $\delta_u > 0$ fixed below, say on $[0, t_1]$, $t_1 \le T$; the bound does not grow with T, as the unstable coordinates are integrated backwards in time. Let $D(t)$ be the ℓ^2 distance of the gauge orbits of $\tilde V_t$ and U_t°, an absolutely continuous function with $D(0) \le |\zeta| \le \rho_0$; at a minimising gauge $\tilde V_t = e^{iY_t}U_t^\circ$, Y_t horizontal at U_t°; let $\mathcal{D}_t = \sum_p |Z_p(Y_t)|^2$ be the energy of the difference and $\varpi_t^\circ = \varpi(U_t^\circ)$. (i) While $D \le \bar D$, a radius $\le \frac12$, half the derivative of D^2 is at most $-\sum_p \varsigma_p|Z_p(Y_t)|^2$ plus the source $2\sum_{p\ni b_0}\langle Z_p(Y_t), \mathsf{s}_p(U_t^\circ)\rangle$ of the four plaquettes at b_0, a member $28\varpi_t^\circ D^2$ and members of the orders three and four in Y_t; the coercivity bound applies to the first member, the three $\mathfrak{g}$-valued linear coordinates of Y_t that it leaves free equal the $\xi_i(\tilde V_t)$ up to errors bounded by $C\varpi_t^\circ$ and by $C(\varpi_t^\circ + D)(\mathcal{D}_t + (\varpi_t^\circ + N^{-2})D^2)^{1/2}$, and the Weitzenböck identity with Kato's inequality bounds $\sum_b |Y_t(b)|^4$ by $CD^2(\mathcal{D}_t + (\varpi_t^\circ + N^{-2})D^2)$; for $\bar D$ and δ_* small this gives, with absolute numbers $c' > 0$ and C, $\frac12\frac{d}{dt}D^2 \le -c'\mathcal{D}_t + C|\xi(\tilde V_t)|^2 + C(\varpi_t^\circ + N^{-2})D^2 + C(\varpi_t^\circ)^2$. (ii) Here the hypotheses $\int_0^T \varpi_t^\circ\,\mathrm{d}t \le \Lambda_0$ and $T \le \Lambda_0 N^2$ enter: they give $\int_0^T(\varpi_t^\circ + N^{-2})\,\mathrm{d}t \le 2\Lambda_0$, and with $\varpi_t^\circ \le \delta_*$ also $\int_0^T(\varpi_t^\circ)^2\,\mathrm{d}t \le \delta_*\Lambda_0$; so Gronwall's lemma gives, with $t_2 \le t_1$ the largest time with $D \le \bar D$ on $[0, t_2]$, $D(t)^2 + 2c'\int_0^t \mathcal{D}_s\,\mathrm{d}s \le e^{4C\Lambda_0}[\rho_0^2 + 2C\int_0^{t_2}|\xi(\tilde V_s)|^2 ds + 2C\delta_*\Lambda_0]$ for $t \le t_2$. (iii) The equation $\dot\xi_i = \lambda_i\xi_i + r_i$ is integrated backwards from t_2: $\xi_i(t) = e^{-\lambda_i(t_2-t)}\xi_i(t_2) - \int_t^{t_2} e^{-\lambda_i(u-t)}r_i(u)\,\mathrm{d}u$ with $|\xi_i(t_2)| \le \delta_u$, and Young's inequality gives $\int_0^{t_2}|\xi_i|^2 \le \delta_u^2/\lambda_i + 2\lambda_i^{-2}\int_0^{t_2}|r_i|^2$, a bound that does not grow with t_2. (iv) By the bound of r_i, $\int_0^{t_2}|r_i|^2 \le C[\sup_{[0,t_2]} q\int_0^{t_2} q + N^4 e^{-c_e N/2}T]$ with $q = q(\tilde V_t)$,

where $q(\tilde V_t)\le C(\mathcal D_t+(\varpi_t^\circ)^2+(\varpi_t^\circ+N^{-2})D^2)$, $\sup q\le C(\delta_*+\bar D)^2$ and $N^4e^{-c_eN/2}T\le\Lambda_0N^6e^{-c_eN/2}$. (v) Inserting (iii) and (iv) into (ii), $\mathcal J=\sup_{[0,t_2]}D^2+2c'\int_0^{t_2}\mathcal D_t\,\mathrm dt$ obeys $\mathcal J\le e^{C\Lambda_0}[\rho_0^2+\delta_u^2+\delta_*+N^6e^{-c_eN/2}]+e^{C\Lambda_0}(\delta_*+\bar D)^2\mathcal J$; for $\bar D$ and δ_* small with Λ_0, and N large with δ_*, this gives $\mathcal J\le\frac14\rho^2$ with $\rho^2=C_*(\rho_0^2+\delta_u^2+\delta_*)$, C_* depending only on Λ_0, and if $\rho\le\bar D$ the continuity of D gives $t_2=t_1$. So on $[0,t_1]$ the distance of the two flows modulo gauge stays below ρ, small with ρ_0, δ_u and δ_*, and the energy of the difference is integrable in time, $\int_0^{t_1}\mathcal D_t\,\mathrm dt\le\rho^2/c'$. *The numbers.* Given ε and Λ_0, the numbers are fixed in this order. First C_* and $\bar D$, which read Λ_0 alone, and $C_{**}=(C_*(64C_{\mathsf E}^2+2))^{1/2}$, with $C_{\mathsf E}$ the absolute bound of the inverse of the triangular matrix. Next θ and a ceiling $\bar\delta\le\frac{1}{10}$ for δ, small with ε and Λ_0, such that along flows with $\varpi(\tilde V_t)\le\bar\delta$, for N and T large, $\|\Lambda^9D\tilde\Phi_T\|\le e^{(3\lambda_1+6\lambda_2+\varepsilon)T}$, and the cone condition of (b) holds with the loss ε in the rate and leaves outside the strongest mode a part of w_T that is small against $|x_T|/(1+C_{**})$. Next δ_u, small with all these. Then $\delta_*=\delta_u^2$ and $\rho_0=8C_{\mathsf E}\delta_u$, so that the radius of (v) is $\rho=C_{**}\delta_u\le\bar D$ and $\delta=\delta_*+4\rho\le(1+4C_{**})\delta_u\le\bar\delta$. Last N_* and T_*, large with all these, so that for $N\ge N_*$ the number $N^2e^{-c_eN/4}$ is small against δ_u and $N^6e^{-c_eN/2}\le\delta_*$. These choices serve three steps below: on the faces of $\mathfrak B$, $|r_i|\le C(C_q\delta^2+N^2e^{-c_eN/4})\le\frac12\lambda_2\delta_u$, so that the exit is strict; at $t=0$ the map $\zeta\mapsto\xi(\tilde V_0)$ differs from its linear part, on the ball $|\zeta|\le\rho_0$, by a map with a Lipschitz constant at most $1/(2C_{\mathsf E})$, so that its image holds the ball of radius $\rho_0/(2C_{\mathsf E})=4\delta_u$ about $\xi(U)$, which holds $\mathfrak B$ as $|\xi(U)|\le C\delta_*\le\delta_u$; and at the end $|\partial\tilde A(\tilde V_T)|\le C(\delta_*+\rho)\le C(1+C_{**})\delta_u$, so that the errors in (b) are small multiples of $\delta_u|x_T|$. *The set.* Let $\mathcal K_T$ be the set of the ζ, $|\zeta|\le\rho_0=8C_{\mathsf E}\delta_u$, $C_{\mathsf E}$ the bound of the inverse of the triangular matrix, whose nine coordinates stay in $\mathfrak B$ up to the time T. At $t=0$ the map $\zeta\mapsto\xi(\tilde V_0)$ is close to the linear map $\zeta\mapsto(\sum_j e_i(b_0+\hat e_j)\mathrm{Ad}(T_{b_0+\hat e_j}(U))\zeta_j)_i$, invertible by the triangular matrix; on the faces of $\mathfrak B$ the exit is strict, as $\frac{d}{dt}\frac12|\xi_i|^2\ge\delta_u(\lambda_2\delta_u-|r_i|)>0$ there; so by a degree argument the end map $\Xi_T(\zeta)=\xi(\tilde V_T)$ carries $\mathcal K_T$ onto the interior of $\mathfrak B$. Put $\mathcal S=\mathcal K_T\cap\Xi_T^{-1}(\mathsf S)$, $\mathsf S=\{\frac12\delta_u<|\xi_1|<\delta_u,\ |\xi_2|,|\xi_3|<\delta_u\}$: its image under Ξ_T is the shell $\mathsf S$, of the volume $c\,\delta_u^9$, and the area formula with $\|\Lambda^9D\tilde\Phi_T\|\le e^{(3\lambda_1+6\lambda_2+\varepsilon)T}$ bounds the measure of $\mathcal S$ from below. (a) holds on $\mathcal K_T$, with $\delta=\delta_*+4\rho$. *(b) on $\mathcal S$.* With $w_T=D\tilde\Phi_T(\tilde V_0)[e\,\delta_{b_0}]$, $e\in\mathfrak g$,

$$\langle e,\partial_{b_0}(\textstyle\sum_p\mathsf f_pa_p\circ\Phi_{a_K^2T})(W_\zeta)\rangle=\mathsf f_0\langle\partial\tilde A(\tilde V_T),w_T\rangle+\sum_p(\mathsf f_p-\mathsf f_0)\varsigma_p\langle Z_p(w_T),\mathsf s_p(\tilde V_T)\rangle.$$

By a cone condition, also with the weight $e^{\theta|b|}$, w_T follows the strongest mode: $|w_T|^2\le3.4|x_T|^2$, x_T its coordinate along e_1 carried to the root, the rest of w_T small against $|x_T|$, and $|x_T|\ge\frac12|e|e^{(\lambda_1-\varepsilon)T}$. At the end the gradient is small, $|\partial\tilde A(\tilde V_T)|\le C(\delta_*+\rho)$, as $A_K(U^\circ_{T/2})\le\delta_*^2$, and, since $\varsigma\,\mathrm de_1=-\lambda_1\omega_1$, $\langle\partial\tilde A(\tilde V_T),w_T\rangle=-\lambda_1\langle\xi_1(\tilde V_T),x_T\rangle$ up to a small multiple of $\delta_u|x_T|$. The map $e\mapsto x_T(e)$ is onto; for e with $x_T(e)$ a positive multiple of $\xi_1(\tilde V_T)$, of modulus $>\frac12\delta_u$ on $\mathcal S$: $|\langle\partial\tilde A(\tilde V_T),w_T\rangle|\ge\frac14\lambda_1\delta_u|x_T|$, while the second sum is at most $21\,\delta(\sum_p|\mathsf f_p-\mathsf f_0|^2e^{-2\theta|p|})^{1/2}|x_T|$; divide by $|e|$. (c): on $\mathcal S$, up to a gauge, $\tilde V_t=e^{iY_t}U_t^\circ$ with Y_t horizontal at U_t° and small, and the energy $\mathcal D_t=\sum_p|Z_p(Y_t)|^2$ of the difference has $\int_0^T\mathcal D_t\,\mathrm dt$ bounded by a number that depends only on ε and Λ_0; as $\sum_pa_p(e^{iY}U)\le3A_K(U)+\frac32\sum_p|Z_p(Y)|^2+6\sum_p(\sum_{b\in p}|Y(b)|^2)^2$, with a quartic member that is integrable in t by the same bound and the inequality of Sobolev on the discrete torus, and as $A_K(U_t^\circ)$ does not increase, the integral over $[\frac T2,T]$ gives (c). (d): here the unstable part is read through nine coordinates of its own, taken from the velocity of the flow and cut off at a fixed radius. Let R be a radius that reads ε_c and Λ_0 alone, χ_R a cutoff of the bonds with values in $[0,1]$, equal to 1 within the distance $\frac R2$ of b_0 and to 0 beyond R, which varies by at most π/R across a plaquette, and $\mathsf t_x(V)$ the transport from the root to the site x along the comb of the coordinates ξ_i; with (σ_α) an orthonormal basis of $\mathfrak g$, $\lambda_3=\lambda_2$ and $i,\alpha=1,2,3$, let $e_{i\alpha}(V)(b)=\chi_R(b)\,e_i(b)\,\mathrm{Ad}(\mathsf t_{b_-}(V))^{-1}\sigma_\alpha$ and $u_{i\alpha}(V)=\lambda_i^{-1}\langle e_{i\alpha}(V),-\partial\tilde A(V)\rangle$. These nine numbers read the velocity of the flow along the cut eigenvectors, and they are the nine unstable coordinates of (d): $u_1=(u_{1\alpha})_\alpha\in\mathbb R^3$ is the coordinate in the strongest mode, and $(u_2,u_3)\in\mathbb R^6$ are the six slow coordinates. Along a flow line of $\tilde A$ the velocity solves the linearised

flow exactly, as the current does in Lem. 16.2(c), so that $\frac{d}{ds}u_{i\alpha}(\tilde V_s) = \lambda_i u_{i\alpha}(\tilde V_s) + \mathrm{err}_{i\alpha}(\tilde V_s)$ with $|\mathrm{err}_{i\alpha}(V)| \le C_R q_R(V) + 4\lambda_2^{-1}\epsilon_R q_R(V)^{1/2}$; here $q_R(V)$ is the sum of $\varpi_p(V)^2$ over the plaquettes within the distance $R+1$ of b_0, $\epsilon_R = \max_i \|(\mathrm{d}^\mathsf{T}\varsigma\,\mathrm{d} + \lambda_i)\chi_R e_i\| + \max_i \|(1-\chi_R)e_i\|_{\ell^{6/5}} \le CR^{5/2}(\frac{6}{6+\lambda_2})^{R/2}$ is the defect of the cut eigenvectors, and the numbers C_R read R and Λ_0 alone. For $V' = e^{i\eta}V$ with $\varpi(V)$ and $\|\eta\|$ small with R, $|\mathrm{err}_{i\alpha}(V') - \mathrm{err}_{i\alpha}(V)| \le [C_R(\varpi(V) + \|\eta\|) + 44\lambda_2^{-1}\epsilon_R]\|\eta\|$, and $|u_{i\alpha}(V') - u_{i\alpha}(V) - \langle e_{i\alpha}(V), \eta\rangle| \le \frac{\varepsilon_c}{100}\|\eta\|$ if $\varpi(V), \|\eta\| \le \epsilon_\sharp$, a number that reads R and ε_c. Let $\tilde V_s$ and $\tilde V_s' = e^{i\eta_s}\tilde V_s$ be two flow lines of $\tilde A$ in lattice time along which $\varpi \le \epsilon_\sharp$ and $n_s = \|\eta_s\| \le \epsilon_\sharp$, and $p_s = |u_1(\tilde V_s') - u_1(\tilde V_s)|$ the difference of their coordinates in the strongest mode. They obey the cone inequalities $\frac{d}{ds}\frac12 n^2 \le \lambda_2 n^2 + (\lambda_1 - \lambda_2)p^2 + \varepsilon_c n^2$ and $\frac{d}{ds}\frac12 p^2 \ge \lambda_1 p^2 - \varepsilon_c pn$, $\varepsilon_c \le 10^{-3}$; so where $n^2 \le 700p^2$ the difference p grows at the rate $\ge \lambda_1 - \varepsilon_c\sqrt{700} \ge \lambda_1'$, and this cone is never left, while where $n^2 > 700p^2$ or $p = 0$ the distance n grows at the rate $\le \lambda_2 + \frac{1}{700}(\lambda_1 - \lambda_2) + \varepsilon_c \le \lambda_2''$. As long as the nine unstable coordinates stay in the ball $\{|u| \le \rho_1\}$, the flow of $\tilde A$ from the three changed bonds stays near the flow of A_K from U, modulo gauge and in ℓ^2 of all the bonds: by an energy inequality, which holds while the distance $D(s)$ of the two flows modulo gauge is at most a number D_0, $D(s) \le C'(\Lambda_0)\rho_1 + C_R\epsilon_{\mathrm d}^{1/2}$, with $C'(\Lambda_0)$ free of ε_c and of the radii; so $\varpi(\tilde V_s) \le \epsilon_{\mathrm d} + 4D(s) \le 5C'(\Lambda_0)\rho_1 \le \epsilon_\sharp$ and $|\mathrm{err}(\tilde V_s)| \le \frac{1}{12}\lambda_2\rho_1$ in $\mathbb{R}^9$ there. The values $(x, y_1, y_2) \in \mathfrak{g}^3$ at b_0 and at the two neighbouring bonds parametrise the nine unstable coordinates at the time 0 by a map close to an invertible linear one, triangular through the matrix $[e_i(b_0 + \hat e_j)]$, whose inverse has the norm at most $c_\triangle$. The data are taken in the ball $\{|(x, y_1, y_2)| \le 30C_\triangle\rho_2\}$, of radius $C_\triangle\rho_1 = 4c_\triangle\rho_1$: there ϖ and the changes stay below $\epsilon_\sharp$, and the bound on $u_{i\alpha}(V') - u_{i\alpha}(V)$ above, with the change of the $e_{i\alpha}(V)$ over the ball, small with ρ_1, leaves to the difference of the map and its linearisation at the centre a Lipschitz constant at most $\frac{1}{2c_\triangle}$ on the ball; so the map carries the ball homeomorphically onto its image, the images of the points of the boundary sphere lie at the distance $\ge \frac{C_\triangle\rho_1}{2c_\triangle} = 2\rho_1$ from $u(U)$, the image of the centre, and as $|u(U)| \le C_R\epsilon_{\mathrm d} \le \rho_1$ the image holds the ball $\{|u| \le \rho_1\}$, $\rho_1 = 30\rho_2$; on the sphere $\{|u| = \rho_1\}$ the exit is strict, as $\frac{d}{ds}\frac12|u|^2 \ge \lambda_2\rho_1^2 - \frac{1}{12}\lambda_2\rho_1^2 > 0$ there; so by a shooting argument with Brouwer's theorem, for every $\mathsf w \in \mathbb{R}^6$ with $|\mathsf w| \le \frac14\rho_1$ there are data in the ball $\{|(x, y_1, y_2)| \le 30C_\triangle\rho_2\}$ whose nine coordinates stay in the ball $\{|u| \le \rho_1\}$ up to T, whose coordinate in the strongest mode vanishes at T, and whose six slow coordinates at T equal $\mathsf w$. Here $e_1(b_0)^2 \ge 4/\lambda_1 > 0.604$: as $e_2(b_0) = e_3(b_0) = 0$, $\delta_{b_0} = e_1(b_0)e_1 + z$ with z in the orthogonal complement of e_1, e_2, e_3, on which the form of $\mathrm{d}^\mathsf{T}\varsigma\,\mathrm{d}$ is ≥ 0, as it has no other negative eigenvalue, and the cross term vanishes, e_1 being an eigenvector; so $-4 = \langle\delta_{b_0}, \mathrm{d}^\mathsf{T}\varsigma\,\mathrm{d}\,\delta_{b_0}\rangle \ge -\lambda_1 e_1(b_0)^2$, the value -4 coming from the four plaquettes at b_0; thus $|e_1(b_0)| > 0.777$. For two such data let $s_1 \le T$ be the last time up to which $n \le \epsilon_\sharp$. On $[0, s_1]$ the pair stays out of the cone: inside it p would grow from then on, against $p_T = 0$ if $s_1 = T$, as the coordinates in the strongest mode both vanish at T, and against $p_{s_1} \ge \epsilon_\sharp/\sqrt{700} > 2\rho_1 \ge p_{s_1}$ if $s_1 < T$. Hence $n_{s_1} \le e^{\lambda_2'' s_1} n_0$, and the slow coordinates at T differ by at most $3e^{\lambda_2'' T} n_0$, by $\sqrt6 \cdot 1.01\, n_T$ if $s_1 = T$ and by $2\rho_1 < \epsilon_\sharp = n_{s_1}$ if $s_1 < T$. At the time 0 the difference η_0 lives on the three changed bonds; $p_0 \le n_0/\sqrt{700}$, $n_0 \le |\eta_0(b_0)| + |\eta_0(b_0+\hat e_1)| + |\eta_0(b_0+\hat e_2)|$ and $p_0 \ge |e_1(b_0)||\eta_0(b_0)| - |\eta_0(b_0+\hat e_1)| - |\eta_0(b_0+\hat e_2)| - \varepsilon_c n_0$, so $0.73|\eta_0(b_0)| \le 1.04(|\eta_0(b_0+\hat e_1)| + |\eta_0(b_0+\hat e_2)|)$ and $n_0 \le 2.5(|\eta_0(b_0+\hat e_1)| + |\eta_0(b_0+\hat e_2)|) \le 4|y - y'|$, y, y' the values of the data at the two neighbouring bonds: the slow coordinates at T differ by at most $12e^{\lambda_2'' T}|y - y'|$; a Lipschitz map changes Lebesgue measure at most by the sixth power of its constant, which gives the measure of $\mathsf Y$. On the fibre of b_0 through such a datum, along the geodesic g_r, the two flows differ at the time 0 by $\eta_0 = re\,\delta_{b_0}$, so $p_0 \ge (|e_1(b_0)| - \varepsilon_c)r \ge 0.75\,r \ge 0.75\,n_0$: the difference in the strongest mode starts at $\ge 0.75\,r$, inside the cone, and grows at the rate $\ge \lambda_1'$, with $n_s \le 1.35p_s$ by the same two inequalities; let r_* be the first r at which it reaches ρ_2 at the time T (meanwhile $n_s \le 1.35\rho_2 < \epsilon_\sharp$, and $|u| \le \frac13\rho_1 + 3.1n_s < \frac12\rho_1$, where $\frac13\rho_1$ bounds $|u|$ along the datum by the equation of u read backwards from T), so that $r_* \le \frac43\rho_2 e^{-\lambda_1' T}$. At the time T

the velocity of the flow is orthogonal to the strongest mode, $\langle \partial\tilde{A}(\tilde{V}_T), e_{1\alpha}(\tilde{V}_T)\rangle = -\lambda_1 u_{1\alpha}(\tilde{V}_T) = 0$, as the coordinate of the datum in that mode vanishes there, and it is of the order ρ_2 with a constant of Λ_0 alone: $\|\partial\tilde{A}(\tilde{V}_T)\| \le \|\partial A_K(U_T^\circ)\| + 16\epsilon_{\mathrm{d}} + 44D(T)$, $D(T)$ the distance of the two flows modulo gauge, where $\|\partial A_K(U_T^\circ)\|^2 \le 48e^{24\Lambda_0}\Lambda_0^2 N^3 T^{-3}$ (the condition $N^3 \le c_{\mathrm{d}}T^2$ enters here as well) and $D(T) \le C'(\Lambda_0)\rho_1 + C_R\epsilon_{\mathrm{d}}^{1/2}$; so $\|\partial\tilde{A}(\tilde{V}_T)\| \le 45\,C'(\Lambda_0)\rho_1 = 1350\,C'(\Lambda_0)\rho_2$, $\rho_2 = \rho_1/30$, by the last condition on ϵ_{d} below. The numbers are chosen in this order: $C'(\Lambda_0)$; $\varepsilon_c = \min\{10^{-3}, (4000\,C'(\Lambda_0))^{-2}\}$; the radius R, which reads ε_c and Λ_0; $\epsilon_\sharp$, which bounds ϖ and n wherever the cone inequalities are used, and D_0, the distance modulo gauge up to which the energy inequality holds, both depending on R and ε_c; ρ_1, small with R, with $5C'(\Lambda_0)\rho_1 \le \epsilon_\sharp$, $2\sqrt{700}\,\rho_1 < \epsilon_\sharp$ and $5C_\triangle\rho_1 \le \min\{1, \epsilon_\sharp\}$, so that $30C_\triangle\rho_2 \le 1$ and, as $\epsilon_{\mathrm{d}} \le \frac{1}{5}\epsilon_\sharp$ by the conditions on ϵ_{d} below, ϖ and the changes stay below $\epsilon_\sharp$ on the ball of the data; $\rho_2 = \rho_1/30$; and last ϵ_{d}, N_{d} and c_{d}, which depend on everything chosen before: small enough for the energy inequality, and such that $C_R\epsilon_{\mathrm{d}} \le \rho_1$, $\epsilon_{\mathrm{d}} + 4C_R\epsilon_{\mathrm{d}}^{1/2} \le C'(\Lambda_0)\rho_1$, $\|\partial A_K(U_T^\circ)\| + 16\epsilon_{\mathrm{d}} + 44C_R\epsilon_{\mathrm{d}}^{1/2} \le C'(\Lambda_0)\rho_1$ and $0.8\,e^{-11N_{\mathrm{d}}} \le \varepsilon_c$. At $r = r_*$ the ratio $\theta_s = n_s^2/p_s^2$ is defined on $[0,T]$, as $p_s > 0$ there, and the cone inequalities give $\theta' \le -2(\lambda_1 - \lambda_2)(\theta - 1) + 2\varepsilon_c\theta(1 + \theta^{1/2})$; as $\lambda_1 - \lambda_2 > 5.8$, the right side is at most $-11(\theta - 1) + 10\varepsilon_c$ where $1 \le \theta \le 2$, and $\theta_0 \le 0.75^{-2} < 1.8$, so that θ stays below 1.8 and $\theta_T - 1 \le 0.8\,e^{-11T} + \varepsilon_c \le 2\varepsilon_c$, by $T \ge N_{\mathrm{d}}$ and the last condition on N_{d}. Let $\hat{e}_\alpha = e_{1\alpha}(\tilde{V}_T)/\|\chi_R e_1\|$, three orthonormal vectors, and $\eta_T = \sum_\alpha c_\alpha\hat{e}_\alpha + \eta_T^\perp$ with $\eta_T^\perp$ orthogonal to them; the bound on $u_{i\alpha}(V') - u_{i\alpha}(V) - \langle e_{i\alpha}(V), \eta\rangle$ above, with $\|\chi_R e_1\| \le 1$, gives $|c| \ge p_T - \varepsilon_c n_T$, whence $\|\eta_T^\perp\|^2 = n_T^2 - |c|^2 \le (\theta_T - 1)p_T^2 + 2\varepsilon_c p_T n_T \le 5\varepsilon_c p_T^2$, and $p_T = \rho_2$ at $r = r_*$. Thus the displacement η_T at $r = r_*$ lies along the strongest mode up to the part $\eta_T^\perp$, of norm $\le (5\varepsilon_c)^{1/2}\rho_2$, so the member of the first order of the Taylor expansion of $\tilde{A}$ at the end of the flow, which sees only that part, is at most $1350\,C'(\Lambda_0)\rho_2\,(5\varepsilon_c)^{1/2}\rho_2 \le 0.76\,\rho_2^2$; the Hessian of $\tilde{A}$ on that mode is close to $-\lambda_1$, and its quadratic form at every X is at most $40\|X\|^2$, by the bounds of Lem. 16.2(a) at each plaquette, so that $\eta_T^\perp$ moves the member of the second order by at most $0.2\,\rho_2^2$, and this member is at most $-3\rho_2^2$: the fall is at least $2\rho_2^2$. The value $\mathsf{v}(0)$ is 8 plus a number of modulus at most $\sum_p a_p$ of the field with the reversed sign at b_0, and this sum is ≤ 1 by the bound used for (c), as the two flows are close and $A_K(U_T^\circ) \le 24\Lambda_0^2 N^3 T^{-2} \le 24\Lambda_0^2 c_{\mathrm{d}}$, the action being non-increasing and $\varpi(U_s^\circ) \le 4\Lambda_0/T$ at some $s \in [\frac{T}{4}, \frac{T}{2}]$. For a general f the same steps hold, with 8000 in the place of 4000 in ε_c: the plaquettes within a fixed distance of b_0 change $f(x(p))$ by $O(a_K)$ from $f(x_*)$, and the others see only the part of η_T outside the strongest mode, which is small; this gives the fall $f(x_*)\rho_2^2$. □

16.6. **Regularisation cutoffs and lower energy bounds.**

Lemma 16.5 (Condenser cutoffs)**.** For all K, m, $\beta > 0$, let $\eta \in D(\mathcal{E}_K)$, $0 \le \eta \le 1$, $\mathsf{A} = \{\eta = 0\}$, $\mathsf{B} = \{\eta = 1\}$, $D_0 = \{\phi \in D(\mathcal{E}_K) : \phi = 0 \text{ a.e. on } \mathsf{A}\cup\mathsf{B}\}$, $\mathcal{E}_{K,1}(\phi,\psi) = \mathcal{E}_K(\phi,\psi) + \int \phi\psi\,\mathrm{d}\mu^K$. There is exactly one $w \in \eta + D_0$ with $\mathcal{E}_{K,1}(w,\phi) = 0$ for $\phi \in D_0$; with $\varkappa = \mathcal{E}_{K,1}(w,w)$: $0 \le w \le 1$, $w = 0$ on A, $w = 1$ on B, $\Gamma_K(w,\cdot) = 0$ a.e. on $\mathsf{A}\cup\mathsf{B}$, $\varkappa \le \mathcal{E}_{K,1}(\eta,\eta)$, $\int w\,\mathrm{d}\mu^K \le \varkappa$, and $|\mathcal{E}_K(w,\phi)| \le \operatorname{osc}(\phi)\,\varkappa$ for every $\phi \in D(\mathcal{E}_K) \cap L^\infty$; w is invariant if η is.

With $q(u) = 3u^2 - 2u^3$: $q(w) \in D(\mathcal{E}_K)$, $q(w) = 0$ on A, $q(w) = 1$ on B, $\mathcal{E}_K(q(w)) \le \frac{9}{4}\varkappa$, and $\mathcal{E}_K(q(w),\phi) = -\int \mathsf{I}\,\phi\,\mathrm{d}\mu^K$ for $\phi \in D(\mathcal{E}_K) \cap L^\infty$, with $\mathsf{I} = 6w^2(1-w) + 6(1-2w)\Gamma_K(w)$, $\|\mathsf{I}\|_{L^1(\mu^K)} \le \frac{15}{2}\varkappa$. ■

Proof sketch. w is the element of least norm of the closed convex set $\{v \ge 1 \text{ on } \mathsf{B},\ v \le 0 \text{ on } \mathsf{A}\}$; truncation does not raise the norm, so $0 \le w \le 1$. The variational inequality $\mathcal{E}_{K,1}(w,\psi) \ge 0$ for $\psi \ge 0$ on B, $\psi \le 0$ on A, applied to $\psi = \phi\eta$, $(M-\phi)\eta$, $-\phi(1-\eta)$, $-(M-\phi)(1-\eta)$ for $0 \le \phi \le M = \operatorname{osc}\phi$, gives

$$M\Big(\int w\,\mathrm{d}\mu^K - \varkappa\Big) \le \mathcal{E}_{K,1}(w,\phi) \le M\varkappa, \qquad 0 \le \int w\phi\,\mathrm{d}\mu^K \le M\int w\,\mathrm{d}\mu^K,$$

so $|\mathcal{E}_K(w,\phi)| = |\mathcal{E}_{K,1}(w,\phi) - \int w\phi\,\mathrm{d}\mu^K| \le M\varkappa$. Gauge transformations preserve μ^K, Γ_K and the constraint set, so uniqueness gives invariance. For $q(w)$, apply the equation of w to $6w(1-w)\phi \in D_0$: $\Gamma_K(w, 6w(1-w)\phi) = 6w(1-w)\Gamma_K(w,\phi) + 6(1-2w)\phi\,\Gamma_K(w)$; and $6w^2(1-w) \le \frac{3}{2}w$, $\int \Gamma_K(w) \le \varkappa$. □

Proposition 16.6 (Gauge-invariant regularisation cutoffs)**.** For all K, m: let $0 < \bar{s} \le 1$ with $64\bar{s}^{7/16} \le \ell$ and $4a_K^2 \le \bar{s}$. For a site y and an integer $k \ge 1$ let $Q(y,k)$ be the cube of k^3 sites with lowest corner y, $\mathsf{e}_k(V;y) = a_K^{-1}\sum \varpi_p(V)^2$ over the plaquettes with base point in $Q(y,k)$, $\mathsf{e}_k^* = \max_y \mathsf{e}_k(\cdot;y)$, $\mathfrak{s}_k(V) = a_K^3 \sum_y \mathsf{e}_k(V;y)^8$, $k(r) = \lceil r^{7/16}/a_K \rceil$, and

$$\mathscr{N}(U) = \int_{a_K^2}^{\bar{s}} r\,\mathfrak{s}_{16k(r)}(U_r)\,\mathrm{d}r,$$

a smooth invariant function defined without choosing a gauge.

(a) If $\varpi(V) \le 1$, $1 \le k < N$ and X is horizontal at V, then $\nabla^2 A_K(V)[X,X] \ge \frac{1}{4}\sum_p |Z_p(X)|^2 - \hat{\lambda}_k(V)\|X\|^2$ with $a_K^{-2}\hat{\lambda}_k(V) = 16[16(144C)^3 \mathsf{e}_{k+1}^*(V)^2 + (144(a_K k)^2)^{-1}]$, C absolute: on horizontal vectors the Hessian is bounded below by the square of the energy in windows of side $a_K k$, with no supremum of the curvature and no gauge.

(b) Let $R > 0$ and K so large that $e^{64}(C_1 R)^{1/8} a_K^{11/64} \le \frac{1}{144}$, C_1 absolute. If $\mathscr{N}(U) \le R$, then $\varpi(U_r) \le \frac{1}{12}$ for $0 \le r \le \bar{s}$; $\Gamma_K(G\circ\Phi_\sigma)(U) \le e^{2\bar{\Lambda}(R)}\Gamma_K(G)(U_\sigma)$ for invariant $G \in C^2(\Omega_K)$ and $\sigma \le \bar{s}$, $\bar{\Lambda}(R) = 1 + c_\Lambda \bar{s}^{11/64} R^{1/4} + \bar{s}^{1/8}$, c_Λ absolute; for $4a_K^2 \le \sigma \le \bar{s}$, $A^a(U_{\sigma/2}) \le \mathsf{A}_\sigma(R) = \frac{1}{2}\cdot 16^{-3}(\frac{\sigma}{4})^{-25/16}\ell^{21/8}R^{1/8}$ and $(J,J)_a(U_\sigma) \le \frac{2}{\sigma}e^{2\bar{\Lambda}(R)}\mathsf{A}_\sigma(R)$; so $\Gamma_K(H_K^\sigma)$ and $|\partial_\sigma H_K^\sigma|$ are bounded on $\{\mathscr{N} \le R\}$ by numbers free of K for $4a_K^2 \le \sigma \le \bar{s}$, and for all $\sigma \ge 4a_K^2$ if besides $2a_K \mathsf{A}_{\bar{s}}(R) \le 1$; and $\Gamma_K(\mathscr{N}) \le 512\, e^{2\bar{\Lambda}(R)}\mathscr{N}'$, with $\mathscr{N}'$ the same integral with $r^2(16\rho(r))^3 a_K^3 \sum_y \mathsf{e}_{16k(r)}(\cdot;y)^{15}$, $\rho(r) = a_K k(r)$, in the place of $r\,\mathfrak{s}_{16k(r)}$.

(c) Assume (E), so that (Adm) of Prop. 16.1 holds, and let $\epsilon > 0$. There are $\bar{s}$, R, $\bar{K}$, and for every $q < \infty$ a number C_q, none depending on K, such that for $K \ge \bar{K}$ the function $\chi_K = \chi_0(\mathscr{N}/R)$, $\chi_0 \in C^\infty(\mathbb{R})$, $\chi_0 = 1$ on $(-\infty, \frac{1}{2}]$, $\chi_0 = 0$ on $[1,\infty)$, $0 \le -\chi_0' \le 3$, is smooth and invariant, vanishes off $\{\mathscr{N} \le R\}$, is 1 on the set $\mathsf{O}_K(\bar{C},\bar{s})$ of Prop. 16.1(a) formed at $\kappa = \frac{1}{64}$ and this ϵ, so that $\mu^K(\chi_K < 1) \le \epsilon$, and has $\int \Gamma_K(\chi_K)^q d\mu^K \le \mathsf{C}_q$.

(d) In the setting of (c), for a flowed observable $H = H_K^\sigma$ at a fixed $\sigma > 0$, $K \ge \bar{K}(\epsilon,\sigma)$, $\mathsf{h} = \sup_{\{\mathscr{N}\le R\}}\Gamma_K(H)$, $\mathsf{h}' = \sup_{\{\mathscr{N}\le R\}}|\partial_\sigma H|$: the vector field $\zeta_K = \chi_K \partial H$ has $a_K^{-1}\int|\zeta_K|^2\,\mathrm{d}\mu^K \le \mathsf{h}$ and the divergence $\chi_K\Delta_K H + \mathsf{b}\,\chi_K\partial_\sigma H + \Gamma_K(\chi_K, H)$; for $F \in C^1(\Omega_K)$, $|\int F\chi_K\Delta_K H\,\mathrm{d}\mu^K| \le \mathsf{h}^{1/2}\mathcal{E}_K(F)^{1/2} + [(\mathsf{hC}_1)^{1/2} + \mathsf{bh}']\|F\|_{L^2(\mu^K)}$; $|\int \chi_K(\mathsf{b}\partial_\sigma + \Delta_K)H\,\mathrm{d}\mu^K| \le (\mathsf{hC}_1)^{1/2}\mu^K(\frac{1}{2}R < \mathscr{N} < R)^{1/2}$; every sequence of cutoffs has a subsequence along which $\chi_K H_K^\sigma$ converges strongly to an element of $D(\mathcal{E}_\star)$; and $\|\chi_K H - g_n^K \circ \pi_{K,n}\|^2_{L^2(\mu^K)} \le C\,a_n^2/\mathsf{b} + 4\sup|\varphi|^2\mathsf{R}_n$ uniformly in K, g_n^K the top of the tower of $\chi_K H$, $n \ge n''$.

■

Proof sketch. (a) As $\varpi(V) \le 1$, $c_p \ge \frac{1}{2}$, and Lem. 16.2(a) gives $\nabla^2 A_K(V)[X,X] \ge \frac{1}{2}\sum_p |Z_p(X)|^2 - 3\sum_p |\mathsf{s}_p|\,|X|_p^2$, $|X|_p^2 = \sum_{b\in p}|X(b)|^2$. With $f(x) = (\sum_\mu |X_\mu(x)|^2)^{1/2}$ and $t(x)$ the largest $|\mathsf{s}_p|$ over the plaquettes with the corner x, the remainder is at most $12\mathcal{T}$, $\mathcal{T} = \sum_x t(x)f(x)^2$. For horizontal X the lattice Weitzenböck identity of Lem. 16.2(h), its remainder bounded plaquette by plaquette, gives $\|\nabla X\|^2 \le 2\sum_p |Z_p(X)|^2 + 36\mathcal{T}$, and Kato's inequality $\|\nabla f\| \le \|\nabla X\|$. In the windows $Q(y,k)$, each site lying in k^3 of them: Hölder's inequality and the inequality of Sobolev on a discrete cube, $\|f\|^2_{\ell^6(Q)} \le C_{\mathrm{Sob}}(\|\nabla f\|^2_{2,Q} + k^{-2}\|f\|^2_{\ell^2(Q)})$, $C_{\mathrm{Sob}} = 864$, give $\mathcal{T} \le E^{1/2}\|f\|^{1/2}\mathsf{B}^{3/4}$, $E = 4a_K\mathsf{e}_{k+1}^*(V)$, $\mathsf{B} = C_{\mathrm{Sob}}(\|\nabla f\|^2 + k^{-2}\|f\|^2) \le C_{\mathrm{Sob}}(2\sum_p|Z_p(X)|^2 + 36\mathcal{T} + k^{-2}\|X\|^2)$; the ℓ^2 member of the inequality of Sobolev is the source of the term $(144(a_K k)^2)^{-1}$.

Young's inequality at $(144C_{\mathrm{Sob}})^{-1}$ gives $\frac34\mathcal{T}\le\frac{1}{72}\sum_p|Z_p(X)|^2+[(144C_{\mathrm{Sob}})^3E^2+(144k^2)^{-1}]\|X\|^2$, so $12\mathcal{T}\le\frac29\sum_p|Z_p(X)|^2+\hat\lambda_k(V)\|X\|^2$, and $\frac12-\frac29>\frac14$; the absolute C of the statement is C_{Sob}. (b) Let $\mathscr{N}(U)\le R$.

The window energies. At every field a local energy inequality holds, from the derivative of a cut-off window energy along the flow: $\mathsf{e}_k(U_r;y)\le\mathsf{e}_{k+2h}(U_{r'};y-h)+64(a_Kh)^{-2}\int_{r'}^r\mathsf{e}_{k+2h}(U_\tau;y-h)\,\mathrm{d}\tau$ for $r'<r$, $y-h=y-a_Kh(1,1,1)$. At $k=16k(r)$, $h=8k(r)$, the mean over $r'\in[\frac r2,r]$, the bound $\mathsf{e}^*_k(V)^8\le8^9(a_Kk)^{-3}\mathfrak{s}_k(V)$ (a window is covered by eight windows of the same side whose corners run over a sub-cube of side $\frac k2$) and Jensen's inequality give $\mathsf{e}^*_{16k(r)}(U_r)^8\le C_1r^{-53/16}R$ for $2a_K^2\le r\le\bar s$, directly from $\mathscr{N}(U)\le R$. So $\varpi(U_r)^2\le a_K\mathsf{e}^*_{16k(r)}(U_r)\le(C_1R)^{1/8}a_K^{11/64}$ there, and for $r<2a_K^2$, as $\varpi(U_{r'})^2\le e^{32|r-r'|/a_K^2}\varpi(U_r)^2$, with the factor e^{64}: by the condition on K, $\varpi(U_r)\le\frac{1}{12}$ on $[0,\bar s]$.

The floor in two regimes. For horizontal X at U_r, $a_K^{-2}\nabla^2A_K(U_r)[X,X]\ge\frac14a_K^{-2}\sum_p|Z_p(X)|^2-\lambda(r)\|X\|^2$ with $\lambda(r)=12a_K^{-2}\varpi(U_r)$ for $r<a_K^2$ (Lem. 16.2(a), with $c_p\ge\frac12$; its integral over $[0,a_K^2]$ is ≤1) and $\lambda(r)=a_K^{-2}\hat\lambda_{k(r)}(U_r)$ for $a_K^2\le r\le\bar s$ (by (a)); Hölder's inequality with the exponents $4,\frac43$ in r bounds the integral of the window member by $c_\Lambda\bar s^{11/64}R^{1/4}$, and the member $(144\rho(r)^2)^{-1}$, $\rho(r)=a_Kk(r)\ge r^{7/16}$, contributes $\frac89\bar s^{1/8}\le\bar s^{1/8}$: $\int_0^{\bar s}\lambda(r)\,\mathrm{d}r\le\bar\Lambda(R)$.

The bounds. Lem. 16.2(b)–(d), read on horizontal vectors with λ in the place of $12a_K^{-2}\varpi$, give the bounds of $\Gamma_K(G\circ\Phi_\sigma)$, of $(J,J)_a$, of $\Gamma_K(H_K^\sigma)$ and of $\partial_\sigma H_K^\sigma$; $\mathsf{A}_\sigma(R)$ comes from $A^a=\frac12(16\rho)^{-3}a_K^3\sum_y\mathsf{e}_{16k}$, Hölder's inequality in y, the monotonicity of A^a along the flow and the mean over $[\frac\sigma4,\frac\sigma2]$. For continuation at later fixed times use the action-only horizontal Hessian bound from Prop. 16.6(a), as in Step 6 below: $a_KE\le\frac12$ gives $\max_p a_p\le\frac12$ along the later flow and a lower Hessian bound by $-c(E^2+\ell^{-2})$. This is distinct from the smaller small-action threshold in Lem. 16.2(f).

The square field of $\mathscr{N}$. With $M_r(V)=\int_r^{\bar s}w(\tau)\,\mathfrak{s}_{16k(\tau)}(\Phi_{\tau-r}V)\,\mathrm{d}\tau$, $w(\tau)=\tau\mathbf{1}_{\tau\ge a_K^2}$, and $\Xi_r=\partial M_r(U_r)$, horizontal, $\Xi_0=\partial\mathscr{N}(U)$, $\Xi_{\bar s}=0$: $\frac{d}{dr}\|\Xi_r\|^2=2a_K^{-2}\nabla^2A_K(U_r)[\Xi_r,\Xi_r]-2w(r)\langle\Xi_r,\partial\mathfrak{s}_{16k(r)}(U_r)\rangle$. The gradient of $\mathfrak{s}_{\hat k}$ is a covariant divergence: $\langle\Xi,\partial\mathfrak{s}_{\hat k}\rangle=\sum_p\mathsf{f}(p)\langle\mathsf{s}_p,Z_p(\Xi)\rangle\le\mathsf{G}_r^{1/2}(\sum_p|Z_p(\Xi)|^2)^{1/2}$, $\mathsf{f}(p)=16a_K^2\sum_{y:\,\mathrm{base}(p)\in Q(y,\hat k)}\mathsf{e}_{\hat k}(\cdot;y)^7$, $\mathsf{G}_r\le256a_K^{-1}(16\rho)^3a_K^3\sum_y\mathsf{e}_{\hat k}(\cdot;y)^{15}$; it is paid by the member $\frac14\sum_p|Z_p|^2$ of the floor, $2w\mathsf{G}_r^{1/2}(\sum_p|Z_p(\Xi)|^2)^{1/2}\le\frac12a_K^{-2}\sum_p|Z_p(\Xi)|^2+2a_K^2w^2\mathsf{G}_r$.

Gronwall's inequality from $\bar s$ back to 0 gives

$$\|\partial\mathscr{N}(U)\|^2\le2a_K^2e^{2\bar\Lambda(R)}\int w^2\mathsf{G}_r\,\mathrm{d}r\le512\,a_Ke^{2\bar\Lambda(R)}\mathscr{N}'(U),$$

that is $\Gamma_K(\mathscr{N})\le512\,e^{2\bar\Lambda(R)}\mathscr{N}'$. (c) $\{\mathscr{N}\le R\}$ contains O_K, on which the window energies are bounded by the sizes of the flowed field; at each dyadic scale r_j the square field of the functional is bounded on the good set of that scale by a positive power of r_j, and off it crudely by r_j^{-1} (by a_K^{-12} at the last scale) against a mass $\le t_Mr_j^M$ for every M, from the moments of Prop. 16.1(a) and the super-exponential decay of R_d; the sum over the dyadic scales converges in every L^q.

(d) The divergence is the product rule and (C1) of Thm. AV; the bounds are Cauchy–Schwarz; the convergence is (A2) applied to χ_KH, whose energy is at most $2\mathsf{h}+2\sup|\varphi|^2\mathsf{C}_1$; the last bound is the key estimate of Step 1 applied to χ_KH. □

Lemma 16.7 (Lower bounds for the flowed action)**.** Assume (E), so that (Adm) of Prop. 16.1 holds; let $\mathbb{P}_K$ be the normalised chain measure from the least top depth d_0 with $\mathsf{R}_{d_0}\le\frac12$ such that every top depth $\ge d_0$ is admissible, $\mathbb{E}_K$ its mean, $\mathbf{B}_K$ and $\mathsf{a}_{\mathrm{top}}$ as in Prop. 16.1, and $\|\cdot\|_2$ the norm of $L^2(T_\ell)$. For a real function f and a $\mathfrak{g}$-valued function h of the bare bonds, $\langle f,h\rangle=a_K^3\sum_b f(b)h(b)\in\mathfrak{g}$, $\|f\|_1=a_K^3\sum_b|f(b)|$, $\|f\|_2^2=a_K^3\sum_b|f(b)|^2$. For a function Y of the bonds, $\mathrm{d}Y(p)=Y(b_1)+Y(b_2)-Y(b_3)-Y(b_4)$ is the plain sum around the plaquette p, with the signs $(+,+,-,-)$ of Lem. 16.2 and no factor of length; for a 1-form A per unit length, $\mathrm{d}_aA=a_K^{-2}\mathrm{d}(a_KA)$

is its lattice curl, $(\mathrm{d}_a A)(p) = \partial_\mu^+ A_\nu(x) - \partial_\nu^+ A_\mu(x)$ for $p = (x;\mu\nu)$, $\|\mathrm{d}_a A\|_2^2 = a_K^3 \sum_p |\mathrm{d}_a A(p)|^2$, and $\mathrm{d}_a P_s = P_s \mathrm{d}_a$.

(a) There are $k_4 = 100\, c_{\mathsf{J}} \gamma_{\mathrm{r}}^{-1/2}$, c_{J} the constant of the layer maps of Prop. 16.1, $p_* = 9/(64\Lambda_+^2 k_4^4)$, $k_\Sigma = 15/(16\Lambda_+)$ and $\theta_0 > 0$, depending only on the constants of the construction, and d_L, free of K, with $\mathbb{P}_K(\|\mathrm{d}_a P_s \mathbf{B}_K\|_2^2 \ge k_\Sigma \mathsf{b}^{-1} \ell^3 a_d^{-3}) \ge p_*$ for $d \ge d_L$, $K > d$ with $a_K \mathsf{a}_{\mathrm{top}} \le \frac{1}{8}$, and $0 \le s \le \theta_0 a_d^2$. Let $k = K - d$ and $\mathbf{Q}_k^{\mathrm{lin}}$ the k-fold linear average at $\mathbf{1}$ from the bare bonds to the bonds of depth d, the composite of k maps YM3_I:(42).

For a plaquette P of depth d none of whose bonds is a pivot bond,

$$X_P = \mathrm{d}\, \mathbf{Q}_k^{\mathrm{lin}}(a_K P_s \mathbf{B}_K)(P) \in \mathfrak{g},$$

the plain sum around P, with no factor of length, is $\langle P_s f_P, \mathbf{B}_K\rangle$ with the real bond function $f_P = a_K^{-2} q_P$, q_P the kernel of $\mathsf{a} \mapsto \mathrm{d}\, \mathbf{Q}_k^{\mathrm{lin}} \mathsf{a}(P)$, $\|f_P\|_1 \le 4a_d$, $\|f_P\|_2^2 \le 4a_d^{-1}$; and X_P has $\mathbb{E}_K|X_P|^2 \ge \frac{3}{4\Lambda_+} x_k^{-1}$ and $\mathbb{E}_K|X_P|^4 \le k_4^4 x_k^{-2}$; at least $3(1-4L^{-3})(\ell L^d)^3$ of the plaquettes of depth d have no pivot bond. Conditionally: let d_0 be any admissible top depth with $\mathsf{R}_{d_0} \le \frac{1}{2}$, $\mathbb{P}_K$ the normalised chain measure from d_0, and $\mathbb{E}_K^0$ its conditional mean given the block field at depth d_0.

There are $d_L' > d_0$ and numbers $\tau_d \to 0$, which depend on d_0 but not on K, such that for $d \ge d_L'$, $K > d$ with $a_K \mathsf{a}_{\mathrm{top}} \le \frac{1}{8}$ and $0 \le s \le \theta_0 a_d^2$ there is an event $\mathcal{T}_d$ of the block field at depth d_0 with $\mathbb{P}_K(\mathcal{T}_d) \ge 1 - \tau_d$ on which, N_P being the number of the plaquettes of depth d without pivot bond and the sums running over them, $N_P^{-1} \sum_P \mathbb{E}_K^0 |X_P|^2 \ge \frac{3}{4\Lambda_+} x_k^{-1}$, $N_P^{-1} \sum_P \mathbb{E}_K^0 |X_P|^4 \le k_4^4 x_k^{-2}$, and

$$\mathbb{P}_K(\|\mathrm{d}_a P_s \mathbf{B}_K\|_2^2 \ge k_\Sigma \mathsf{b}^{-1} \ell^3 a_d^{-3} \,|\, \text{block field at depth } d_0) \ge p_*.$$

(b) Let $R \ge \max\{1, \|\mathbf{B}\|_{\kappa;a}\}$, $\bar C = (2 + c_3)R$, $a_K \le a_1(R)$, $0 < \sigma \le \min\{s_0(R), s_{\mathrm{lin}}(R)\}$, $a_K \bar C \sigma^{-1/4-\kappa/2} \le \frac{1}{8}$, $P_\sigma \mathbf{B} \ne 0$, $U = e^{i a_K \mathbf{B}}$, where c_3, $s_0(R)$, $a_0(R)$ are those of Prop. 16.1(a) and M_Θ that of its head, $s_{\mathrm{lin}}(R) = (2c_6 \bar C^2)^{-4/(1-2\kappa)}$, $a_1(R) = \min\{a_0(R), (8e^{12}R)^{-2/(1-2\kappa)}, (8(4\mathsf{M}_\Theta)^{1/6} c_3 R)^{-1/(1-2\kappa)}\}$, a positive number that reads R and κ alone, and c_6, c', c_g are absolute. For $a_K \le a_1(R)$ the gauge-fixed flow A_s from $\mathbf{B}$ has $a_K \|A_s\|_\infty \le \frac{1}{2}$ on the closed interval $[0, s_0(R)]$, so that the chart and the linearised equation serve from $s = 0$ on: $a_K \|\mathbf{B}\|_\infty \le e^{12} R a_K^{\frac{1}{2}-\kappa} \le \frac{1}{8}$, as $\|f\|_\infty \le e^{12} \|P_{a_K^2} f\|_\infty$; on $[0, a_K^2/(4\mathsf{M}_\Theta)]$ the chart variable $a_K A_s$ moves at most at the speed $a_K^{-2}\mathsf{M}_\Theta$, so that $a_K\|A_s\|_\infty \le \frac{1}{8} + \frac{1}{4}$ there; later the first bound of Prop. 16.1(a) gives

$$\begin{aligned} a_K \|A_s - P_s \mathbf{B}\|_\infty &\le (4\mathsf{M}_\Theta)^\kappa c_3 R a_K^{1-2\kappa} \\ &\le (4\mathsf{M}_\Theta)^{1/6} c_3 R a_K^{1-2\kappa} \le \frac{1}{8}. \end{aligned}$$

, and with $\|P_s \mathbf{B}\|_\infty \le \|\mathbf{B}\|_\infty$, $a_K \|A_s\|_\infty \le \frac{1}{4}$.

Then

$$\Gamma_K(A^a \circ \Phi_\sigma)(U)^{1/2} \ge \frac{\|\mathrm{d}_a P_{3\sigma/2} \mathbf{B}\|_2^2}{\|P_\sigma \mathbf{B}\|_2} - c' \ell^{3/2} \bar C^3 \sigma^{-1-\kappa}, \qquad \|P_\sigma \mathbf{B}\|_2 \le \sqrt{3} \ell^{3/2} R \sigma^{-\frac{1}{4}-\frac{\kappa}{2}},$$

$\Gamma_K(A^a \circ \Phi_\sigma)(U)^{1/2} \le \frac{8}{7}[\sqrt{2} c_g \sigma^{-1/2} \|\mathrm{d}_a P_\sigma \mathbf{B}\|_2 + c' \ell^{3/2} \bar C^3 \sigma^{-1-\kappa}]$, and $|(2A^a(U_\sigma))^{1/2} - \|\mathrm{d}_a P_\sigma \mathbf{B}\|_2| \le \sqrt{3} \ell^{3/2} (2\bar C + 8\bar C^2) \sigma^{-\frac{1}{2}-\kappa}$.

(c) Let $\mathfrak{b} > 0$, $p = e^{-\Psi} dB$ a probability measure on the box $\prod_b \{|B(b)| < \mathfrak{b}\}$ with $\Psi \in C^2$ and $0 \le D^2\Psi \le \Lambda_+$, ∂_e the derivative in one coordinate of one bond b_1, and $\eta(B) = \phi(|B(b_1)|/\mathfrak{b})$ with $\phi = 1$ on $[0, \frac{1}{2}]$ and $\phi(t) = \cos^2(\pi(t - \frac{1}{2}))$ on $[\frac{1}{2}, 1]$. Then $\mathrm{Var}_p(u) \ge (\int \eta\, \partial_e u\, \mathrm{d}p)^2/(\Lambda_+ + \pi^2 \mathfrak{b}^{-2})$ for every bounded Lipschitz u.

■

Proof sketch. (c) Integrate by parts against η, which vanishes at the faces in the direction e, so there is no boundary term: $\int \eta\,\partial_e u\,\mathrm{d}p = \int (u-\bar u)(\eta\partial_e\Psi - \partial_e\eta)\,\mathrm{d}p$, and bound the second factor in $L^2(p)$ through $D^2\Psi \le \Lambda_+$ (one more integration by parts) and $|\partial_e\eta| \le \pi/\mathfrak{b}$. (a)

Signs. They are those of the head of the lemma: the pairing $\langle f, \mathbf{h}\rangle = a_K^3 \sum_b f(b)\mathbf{h}(b) \in \mathfrak{g}$ of a real bond function f with a $\mathfrak{g}$-valued $\mathbf{h}$ on $\Lambda_{K,\ell}$; for a bond function Y without a factor of length, its plain sum $\mathrm{d}Y(p) = Y(b_1)+Y(b_2)-Y(b_3)-Y(b_4)$ around the plaquette p, with the signs of Lem. 16.2, the bonds taken as in $\partial U(p)$; and for a 1-form A per unit length, its lattice curl $\mathrm{d}_a A = a_K^{-2}\mathrm{d}(a_K A)$, with $\|\mathrm{d}_a A\|_2^2 = a_K^3 \sum_p |\mathrm{d}_a A(p)|^2$ and $\mathrm{d}_a P_s = P_s \mathrm{d}_a$. Besides, $\|f\|_r^r = a_K^3 \sum_b |f(b)|^r$ for $1 \le r < \infty$.

The roman d and d_a are these sums around a plaquette, and the italic d is a depth. For a plaquette P of depth d, $k = K-d$, X_P and f_P are those of (a): $X_P = \mathrm{d}\,\mathbf{Q}_k^{\mathrm{lin}}(a_K P_s \mathbf{B}_K)(P)$ is the sum around P of the k-fold linear average $\mathbf{Q}_k^{\mathrm{lin}}$ of equation YM3_I:(42), YM3_I:(42), applied to the bond function $a_K P_s \mathbf{B}_K$, which carries no factor of length, and f_P is the real bond function with $\mathrm{d}\,\mathbf{Q}_k^{\mathrm{lin}}(a_K A)(P) = \langle f_P, A\rangle$ for every 1-form A, so that $X_P = \langle P_s f_P, \mathbf{B}_K\rangle$. The numbers k_4, p_*, k_Σ of (a), and the number θ_0 with k_1, are

$$\theta_0 = \frac{3}{16\Lambda_+ k_1^2}, \qquad k_4 = 100\, c_{\mathsf{J}} \gamma_{\mathrm{r}}^{-1/2}, \qquad p_* = \frac{9}{64\Lambda_+^2 k_4^4}, \qquad k_\Sigma = \frac{15}{16\Lambda_+}, \qquad k_1 = 48\, c_\nabla \Big(\frac{24\, c_{\mathsf{J}}^2}{\gamma_{\mathrm{r}}(1-L^{-1})}\Big)^{1/2},$$

with the absolute constant c_∇ of the bound $\|\partial_\nu^\pm P_s f\|_1 \le c_\nabla s^{-1/2}\|f\|_1$ of the heat kernel and the constant c_{J} of the layer maps of Prop. 16.1, which enters (L3) below; they depend only on the constants of the construction.

The linear average. In the sum of YM3_I:(42) around a coarse plaquette the contour averages cancel, so one step maps the curl of a fine bond function to L^{-3} times the sum, over the sites x of the block, of its curls at the L^2 fine plaquettes of the square of side L with the corner x: a matrix with the row sums L^2 and the column sums L^{-1}. By the Cauchy–Schwarz inequality, $\sum_P |\mathrm{d}\,\mathbf{Q}_k^{\mathrm{lin}}\mathsf{a}(P)|^2 \le L^k \sum_p |\mathrm{d}\mathsf{a}(p)|^2$, and at $\mathsf{a} = a_K A$, as $\sum_p |\mathrm{d}(a_K A)(p)|^2 = a_K \|\mathrm{d}_a A\|_2^2$ and $L^k a_K = a_d$,

$$\|\mathrm{d}_a A\|_2^2 \;\ge\; a_d^{-1} \sum_P \big|\mathrm{d}\,\mathbf{Q}_k^{\mathrm{lin}}(a_K A)(P)\big|^2 : \tag{16.i}$$

the curls of the linear averages bound $\|\mathrm{d}_a A\|_2^2$ from below, with the constant one between the L^2 norms at the spacings a_K and a_d. Iterated, $\mathrm{d}\,\mathbf{Q}_k^{\mathrm{lin}}\mathsf{a}(P)$ is the mean, over the L^{3k} sites u of the block, of the sum of a around the square of side L^k with the corner u; so $a_K^2 f_P$ is a signed sum of four tents along the bonds of P, $\|f_P\|_1 \le 4a_d$, $\|f_P\|_2^2 \le 4a_d^{-1}$ and $\sum_\nu \|\partial_\nu^+ f_P\|_1 \le 24$; these norms do not grow under P_s, $\|P_s f_P\|_{12/11} \le \|P_s f_P\|_1^{5/6}\|P_s f_P\|_2^{1/6} \le 4a_d^{3/4}$, and $(1-P_s)f = -\int_0^s \sum_\nu \partial_\nu^- P_r \partial_\nu^+ f\,\mathrm{d}r$ gives $\|(1-P_s)f_P\|_1 \le 48 c_\nabla \sqrt{s}$.

The plaquette of a block field, by (c). Let $W \in \mathsf{Cut}_{k+1}$ and let P be a plaquette of level k none of whose bonds is a pivot bond, b_1 its first bond. Its four bonds are coordinates of the chart, so $\partial V_W(B)(P) = e^{i x_k^{-1/2} B(b_1) C}$ with C free of $B(b_1)$. Write $\partial V_W(B)(P) = c_P \mathbf{1} + i\mathsf{s}_P$, let (t_α) be an orthonormal basis of $\mathfrak{g}$ and $\mathsf{E} = \int_0^1 \mathrm{Ad}(e^{iu x_k^{-1/2} B(b_1)})\,\mathrm{d}u$; the derivative of $\langle \mathsf{s}_P, t_\alpha\rangle$ in the coordinate $B(b_1)^\alpha$ is $x_k^{-1/2}\,\frac12 \mathrm{tr}(t_\alpha(\mathsf{E}t_\alpha)\,\partial V_W(B)(P)) \ge x_k^{-1/2}(1-\delta(x_k)-\theta_d)$, as $|\mathsf{E}t_\alpha - t_\alpha| \le x_k^{-1/2}|B(b_1)| < \delta(x_k)$ on the box and $|\partial V_W(B)(P) - \mathbf{1}| < \theta_d$ on the cut: in the variable B of p_W, in which $D^2\Psi_W \le \Lambda_+$ holds (Lem. 15.3(iv)), the derivative carries the chart factor $x_k^{-1/2}$.

(c) at $p = p_W$, $\mathfrak{b} = \mathfrak{b}(x_k)$ and $u = \langle \mathsf{s}_P, t_\alpha\rangle$, with $\int \eta\,\mathrm{d}p_W \ge p_W(|B(b_1)| \le \frac12 \mathfrak{b}(x_k))$, summed over the three α and integrated over $W = V_{k+1}$ with Jensen's inequality, and $\mathbb{P}_K(|B_k(b_1)| > \frac12\mathfrak{b}(x_k)) \le 4m_2\mathfrak{b}(x_k)^{-2}$, m_2 the bound, free of k, of the second moments of one fluctuation field in (L5) of the proof of Prop. 16.1(a), recalled below (its Gaussian tails, Lem. 15.3(iv)), give, as

$|\partial V_W(B)(P)-\mathbf{1}| \geq |\mathsf{s}_P|$,

$$\mathbb{E}_K|\partial V_k(P)-\mathbf{1}|^2 \;\geq\; \frac{3x_k^{-1}\varsigma_d^2}{\Lambda_+ + \pi^2\mathfrak{b}(x_k)^{-2}}, \qquad \varsigma_d = \big(1-\delta(x_k)-\theta_d\big)_+\big(1-4m_2\mathfrak{b}(x_k)^{-2}\big)_+ \to 1 \quad (d\to\infty):$$

the upper bound of the Hessian of one fluctuation integral keeps the plaquette of the block field from degenerating. There are $3(\ell L^{d-1})^3$ pivot bonds, each in four plaquettes, so at most $12L^{-3}(\ell L^d)^3$ of the $3(\ell L^d)^3$ plaquettes of level k hold one.

The layers. By (Adm), as in the head of the lemma, d_0 and every larger top depth are admissible. By Prop. 16.1 (the field of the gauge of the chain), $\mathbf{B}_K = \frac{1}{ia_K}\log(e^{ia_K\mathbf{h}'_K}\cdots e^{ia_K\mathbf{h}'_{d_0+1}}e^{ia_K\mathbf{A}^{\mathrm{top}}})$, the logarithm of the ordered product of the top field, $|\mathbf{A}^{\mathrm{top}}| \leq \mathsf{a}_{\mathrm{top}}$, and of the layers $\mathbf{h}'_{d'} = \mathrm{Ad}(\gamma)\mathbf{h}_{d'}$, $d_0 < d' \leq K$, with the regular gauge γ of the top, a function of the top field, and $\mathbf{h}_{d'}$ a function of the block field at depth d'; let $\mathcal{F}_{d'}$ be the σ-algebra of that block field, which increases with d', and $\mathbf{n}_{d'} = \mathbb{E}_K[\mathbf{h}'_{d'}|\mathcal{F}_{d'-1}]$.

The facts on the layers in the proof of Prop. 16.1(a) give: (L1) the ordered product of the top field and of the layers of the depths $\leq d$ is a gauge transform of a background over the block field V_k, $k = K-d$, and the k-fold average of a background over V_k is V_k (Defs. YM3_I:9.3 and YM3_I:9.5, §5.2); (L2) $|\mathbf{h}'_{d'}| = |\mathbf{h}_{d'}| \leq c_{\mathsf{h}}L^{d'}\delta(L^{d'}\mathsf{b})$ at every bond, with the constant c_{h} of the construction, and $\sum_{d'} a_K|\mathbf{h}'_{d'}| \leq \frac{1}{4}$, as d_0 is admissible; (L3) for a real bond function f, every component ξ of $\langle f, \mathbf{h}'_{d'}-\mathbf{n}_{d'}\rangle$ has $\mathbb{E}_K[e^{\lambda\xi}|\mathcal{F}_{d'-1}] \leq e^{4\lambda^2 G_{d'}(f)^2/\gamma_{\mathrm{r}}}$ for real λ, $G_{d'}(f)^2 = c_{\mathsf{J}}^2\mathsf{b}^{-1}L^{d'}\min\{\|f\|_1^2, L^{-3d'}\|f\|_2^2\}$: the log-Sobolev inequality of one fluctuation integral (Lem. 15.3(iv)), carried by the map from the fluctuation field to the layer, whose derivative on the box obeys the bounds of the layer map with the constant c_{J}; on $L^2(T_\ell)$ this is the covariance bound $Ca_{d'}^2/\mathsf{b}$; (L4) $\sup|\mathbf{n}_{d'}| \leq \mathsf{n}_{d'} = C\mathsf{b}^{-1}(\log L^{d'}\mathsf{b})^{2\mathfrak{p}_0}$, by the bounds (M), with $L^{-d/2}\mathsf{s}_d \to 0$, $\mathsf{s}_d = \sum_{d_0<d'\leq d}\mathsf{n}_{d'}$, and almost surely $\|\sum_{d_0<d'\leq K}\mathbf{n}_{d'}\|_{L^{12}(T_\ell)} \leq \mathsf{c}_{12}$, a number free of K; (L5) at every bond b, $\mathbb{E}_K|B_{K-d'}(b)|^2 \leq m_2$, and the moments of the layers are bounded so that $\|a_K(\sum_{d'}|\mathbf{h}'_{d'}(b)|)^2\|_{L^q(\mathbb{P}_K)} \leq \mathsf{c}_{(q)}\mathsf{b}^{-1}$, with m_2 and $\mathsf{c}_{(q)}$ free of K and d'.

For $\sum_j|x_j| = s' \leq \frac{1}{2}$ the product $e^{ix_1}\cdots e^{ix_m}$ of exponentials of elements of $\mathfrak{g}$ is e^{iy} with $|y| \leq 1.02s'$ and $|y-\sum_j x_j| \leq 1.02s'^2$; so, with $a_K\mathsf{a}_{\mathrm{top}} \leq \frac{1}{8}$ and (L2), the remainder of the product formula, $\mathbf{B}_K - \mathbf{A}^{\mathrm{top}} - \sum_{d'}\mathbf{h}'_{d'}$, is at most $10a_K[(\sum_{d'}|\mathbf{h}'_{d'}|)^2 + \mathsf{a}_{\mathrm{top}}^2]$ at every bond.

The field of the depths $\leq d$. Let $d > d_0$ and let $\mathbf{B}^{\leq d}$ be the logarithm of the ordered product of the top field and of the layers of the depths $\leq d$. At every bare bond the arguments of the exponentials have the sum $s' = a_K\mathsf{a}_{\mathrm{top}} + a_K\sum_{d'\leq d}|\mathbf{h}'_{d'}| \leq L^{-k}\bar{y}_d$, $\bar{y}_d = a_d\mathsf{a}_{\mathrm{top}} + c_{\mathsf{h}}\delta(L^d\mathsf{b})/(1-L^{-1/2})$, by (L2) and $L^{d'}\delta(L^{d'}\mathsf{b}) \leq L^{-(d-d')/2}L^d\delta(L^d\mathsf{b})$ for $d' \leq d$; so $y = L^k\sup a_K|\mathbf{B}^{\leq d}| \leq 2\bar{y}_d$, and the remainder $\mathbf{B}^{\leq d} - \mathbf{A}^{\mathrm{top}} - \sum_{d'\leq d}\mathbf{h}'_{d'}$ is at most $1.02a_d^{-1}\bar{y}_d^2$ at every bond.

By (L1) and the gauge covariance of the average, $\mathcal{M}(U^u) = \mathcal{M}(U)^{u|}$ (Prop. YM3_I:4.5, YM3_I:(40)), $\mathcal{M}^k(e^{ia_K\mathbf{B}^{\leq d}})$ is a gauge transform of V_k, so its plaquette at P has the modulus $|\partial V_k(P)-\mathbf{1}|$. For $L^k\sup|\mathsf{a}| < c_Q$, $\mathcal{M}^k(e^{i\mathsf{a}})(c) = e^{iQ_k(\mathsf{a})(c)}$ with $|Q_k(\mathsf{a})(c)| \leq C_QL^k\sup|\mathsf{a}|$ and $|Q_k(\mathsf{a})(c) - \mathbf{Q}_k^{\mathrm{lin}}\mathsf{a}(c)| \leq 4C_Qc_Q^{-1}(L^k\sup|\mathsf{a}|)^2$: the linearisation of the average, with constants that depend on L alone (Prop. YM3_I:51.1 and Lem. 5.30).

At $\mathsf{a} = a_K\mathbf{B}^{\leq d}$, with the product formula for the four factors around P,

$$\big|\langle f_P, \mathbf{B}^{\leq d}\rangle\big| \;\geq\; |\partial V_k(P)-\mathbf{1}| - 4\big(8C_Q^2 + 16C_Qc_Q^{-1}\big)\,\bar{y}_d^2,$$

a bound of the modulus alone, which is the quantity that (c) bounds from below; the error tends to 0 as $d\to\infty$, uniformly in K.

The heat kernel at the depths $\leq d$. Let $f^1 = (1-P_s)f_P$, $\|f^1\|_1 \leq 48c_\nabla\sqrt{s}$. Of the parts of $\mathbf{B}^{\leq d}$, the top field, the conditional means $\mathbf{n}_{d'}$ and the remainder are bounded at every bond by $\mathsf{a}_{\mathrm{top}}$, s_d and $1.02a_d^{-1}\bar{y}_d^2$, and pair with f^1 to at most $\|f^1\|_1$ times these; the centred layers $\mathbf{h}'_{d'} - \mathbf{n}_{d'}$, $d' \leq d$, are martingale differences for $(\mathcal{F}_{d'})$, so orthogonal, and by (L3) at $f = f^1$ their pairing with f^1 has

the $L^2(\mathbb{P}_K)$ norm at most

$$\Big(3\cdot\frac{8}{\gamma_{\mathrm r}}\,c_{\mathsf J}^2\mathsf b^{-1}\|f^1\|_1^2\sum_{d'\le d}L^{d'}\Big)^{1/2}\;\le\;k_1\Big(\frac{s}{a_d^2}\Big)^{1/2}x_k^{-1/2},$$

as $\mathsf b^{-1}L^d=x_k^{-1}a_d^{-2}$. This cost does not vanish as d grows: at $s=\theta_0a_d^2$ it is $\frac14(3/\Lambda_+)^{1/2}x_k^{-1/2}$, and the range $0\le s\le\theta_0a_d^2$ of (a) is made for it.

The finer layers. Let $f=P_sf_P$. $\mathbf B_K-\mathbf B^{\le d}$ is the sum of the centred layers of the depths $d'>d$, of their conditional means and of the difference of the two remainders. $Y_P=\langle f,\mathbf B^{\le d}\rangle$ is $\mathcal F_d$-measurable and bounded, and $Z_P=\sum_{d'>d}\langle f,\mathbf h'_{d'}-\mathbf n_{d'}\rangle$ has $\mathbb E_K[Z_P|\mathcal F_d]=0$, so $\mathbb E_K|Y_P+Z_P|^2=\mathbb E_K|Y_P|^2+\mathbb E_K|Z_P|^2\ge\mathbb E_K|Y_P|^2$: in the second moment the finer layers can only add. The conditional means of the depths $>d$ pair with f to at most $4a_d^{3/4}\mathsf c_{12}+4a_d\mathsf s_d$, by Hölder's inequality and (L4), and the remainders, by (L5), to at most $40a_d(\mathsf c_{(4)}\mathsf b^{-1}+a_K\mathsf a_{\mathrm{top}}^2)+4.08\,\bar y_d^2$ in $L^4(\mathbb P_K)$.

(a). All these errors are at most $\mathsf z_dx_k^{-1/2}$ with $\mathsf z_d\to0$ uniformly in K, as $\bar y_d^2x_k^{1/2}\to0$, $a_dx_k^{1/2}=(a_d\mathsf b)^{1/2}\to0$, $a_dx_k^{1/2}\mathsf s_d\to0$ by (L4) and $a_d^{3/4}x_k^{1/2}\to0$; d_L is the least depth $>d_0$ beyond which $\bar y_d\le\min\{\frac12,\frac{1}{16C_Q},\frac14c_Q\}$, $[3/(\Lambda_++\pi^2\mathfrak b(L^d\mathsf b)^{-2})]^{1/2}\varsigma_d-\mathsf z_d\ge\frac34(3/\Lambda_+)^{1/2}$ and $\mathsf z_d\le15c_{\mathsf J}\gamma_{\mathrm r}^{-1/2}$, a depth free of K. For P without pivot bond, $d\ge d_L$ and $0\le s\le\theta_0a_d^2$,

$$\|X_P\|_{L^2(\mathbb P_K)}\;\ge\;x_k^{-1/2}\Big\{\Big[\frac{3}{\Lambda_++\pi^2\mathfrak b(x_k)^{-2}}\Big]^{1/2}\varsigma_d-k_1\Big(\frac{s}{a_d^2}\Big)^{1/2}-\mathsf z_d\Big\}\;\ge\;x_k^{-1/2}\Big(\frac34-\frac14\Big)\Big(\frac{3}{\Lambda_+}\Big)^{1/2},$$

as $k_1\theta_0^{1/2}=\frac14(3/\Lambda_+)^{1/2}$; since $(\frac34-\frac14)^2\frac{3}{\Lambda_+}=\frac{3}{4\Lambda_+}$, this is $\mathbb E_K|X_P|^2\ge\frac{3}{4\Lambda_+}x_k^{-1}$. For the fourth moment, the centred layers of all depths form a martingale whose components have, by (L3) at $f=P_sf_P$ with $\|f\|_1\le4a_d$ at the depths $d'\le d$ and $\|f\|_2^2\le4a_d^{-1}$ at the depths $d'>d$, the total variance proxy $v=\frac{4}{\gamma_{\mathrm r}}\sum_{d'}G_{d'}(f)^2\le\frac{4c_{\mathsf J}^2}{\gamma_{\mathrm r}}x_k^{-1}[\frac{16}{1-L^{-1}}+\frac{4}{L^2-1}]\le\frac{100\,c_{\mathsf J}^2}{\gamma_{\mathrm r}}x_k^{-1}$ $(L\ge3)$; so each component M^α has $\mathbb E_K|M^\alpha|^4\le64v^2$, the martingale has the $L^4(\mathbb P_K)$ norm at most $60\sqrt2\,c_{\mathsf J}\gamma_{\mathrm r}^{-1/2}x_k^{-1/2}$, the other members at most $\mathsf z_dx_k^{-1/2}$, and $\mathbb E_K|X_P|^4\le k_4^4x_k^{-2}$, for every P.

Paley and Zygmund. Let $\Sigma=a_d^{-1}\sum_P|X_P|^2$, the sum over the $N_P\ge3(1-4L^{-3})(\ell L^d)^3\ge\frac52(\ell/a_d)^3$ plaquettes without pivot bond $(L\ge3)$. By eq. 16.i at $A=P_s\mathbf B_K$, $\|\mathrm d_aP_s\mathbf B_K\|_2^2\ge\Sigma$; $\mathbb E_K\Sigma\ge a_d^{-1}N_P\frac{3}{4\Lambda_+}x_k^{-1}$, and by the Cauchy–Schwarz inequality for the pairs of plaquettes $\mathbb E_K\Sigma^2\le a_d^{-2}N_P^2k_4^4x_k^{-2}$; so $\mathbb P_K(\Sigma\ge\frac12\mathbb E_K\Sigma)\ge(\mathbb E_K\Sigma)^2/4\mathbb E_K\Sigma^2\ge\frac14(3/4\Lambda_+k_4^2)^2=p_*$, and $\frac12\mathbb E_K\Sigma\ge\frac12\cdot\frac52\cdot\frac{3}{4\Lambda_+}\,\ell^3a_d^{-3}\,x_k^{-1}a_d^{-1}=k_\Sigma\mathsf b^{-1}\ell^3a_d^{-3}$, as $x_k^{-1}a_d^{-1}=\mathsf b^{-1}$.

This is (a). The conditional form: the bounds that hold at every bond or almost surely, and the martingale structure of the layers, are conditional on the block field at depth d_0 as they stand, so the argument goes through under $\mathbb E_K^0$ with the norms $(N_P^{-1}\sum_P\mathbb E_K^0|\cdot|^r)^{1/r}$, $r=2,4$, but for the two places where unconditional moments of one layer enter, the bound m_2 and (L5); there Markov's inequality bounds their conditional means off an event of probability at most τ_d, and the inequality of Paley and Zygmund under $\mathbb E_K^0$, with the Cauchy–Schwarz inequality over P for the second moment of Σ, gives the last bound.

(b) By Duhamel's formula the linearised gauge-fixed flow from an L^2 datum z is a small perturbation of the heat flow, $\|w_s-P_sz\|_2\le2\epsilon(s)\|z\|_2$ with $\epsilon(s)=c_6\bar C^2s^{\frac14-\frac\kappa2}$; differentiating $A^a\circ\Phi_\sigma$ along $e^{ia_K(\mathbf B+rz)}$ at $z=P_\sigma\mathbf B$ and using $(\mathrm d_aP_\sigma\mathbf B,\mathrm d_aP_\sigma P_\sigma\mathbf B)=\|\mathrm d_aP_{3\sigma/2}\mathbf B\|_2^2$ and $\Gamma_K(F)^{1/2}\ge|\frac{d}{dr}F(e^{ia_K(\mathbf B+rz)})|/\|z\|_2$ gives the lower bound; the upper bound and the last inequality are the same expansion. □

The lower bound requires control of the frequency of the terminal gradient. For the action this frequency is controlled by the change in the squared current between two positive times. The following estimate makes this control uniform in the lattice spacing.

Lemma 16.8 (Lower propagation on a regular time interval)**.** Let $a = a_K$, $U_s = \Phi_s U$, $0 < \kappa < \tfrac{1}{2}$, and put

$$r = \frac{3}{4} + \frac{\kappa}{2}, \qquad \theta = 1 - r = \frac{1}{4} - \frac{\kappa}{2}.$$

Suppose that $C_0 > 0$, $\sigma > 0$, and

$$a^{-2}\varpi(U_s) \le C_0 s^{-r} \quad (0 < s \le \sigma). \tag{16.ii}$$

Write $g(s) = (J, J)_a(U_s)$ and assume $g(\sigma) > 0$. For $x \ge 0$ and $j > 0$ define

$$c_\kappa(x, j) = e^{8x/\theta}\left(\log \max\{1, j\} + \frac{64x}{\theta} + 1072x^2\right) + 32x,$$

$$E_\kappa(x, c) = 64x(1 + \theta^{-1}) + 16e^{8x/\theta}x(c + 32x) + \frac{672}{\theta}x^2 + \frac{7168}{\theta}e^{8x/\theta}x^3.$$

If $j \ge g(\sigma/2)/g(\sigma)$ and $x = C_0\sigma^\theta$, then

$$\Gamma_K(A^a \circ \Phi_\sigma)(U) \ge g(\sigma) \exp\left[-2c_\kappa(x, j) - E_\kappa(x, c_\kappa(x, j))\right]. \tag{16.iii}$$

The factor on the right is positive, decreases as j increases, and depends only on κ, C_0, σ, j. ∎

Proof. All the operators in the proof act on the finite periodic lattice at spacing a. Use the scalar product $(X, Y)_a = a^3 \sum_b \langle X(b), Y(b)\rangle$ on positive bonds. Thus $\mathrm{grad}_a F = a^{-2}\partial F$ satisfies $\|\mathrm{grad}_a F\|_a^2 = \Gamma_K(F)$, and $\mathrm{grad}_a A^a = J$. At a field V, let

$$T_\mu h(x) = \mathrm{Ad}(V_\mu(x))h(x + ae_\mu), \qquad \nabla_\mu = a^{-1}(T_\mu - 1), \qquad (DX)_{\mu\nu} = \nabla_\mu X_\nu - \nabla_\nu X_\mu.$$

Here $D_0^* X = \sum_\mu \nabla_\mu^* X_\mu$. In this proof $\Delta = \sum_\mu \nabla_\mu^* \nabla_\mu$ is positive and acts on each component; it is distinct from the heat operator Δ_a of Prop. 16.1. Set

$$\phi(V) = a^{-2}\varpi(V), \qquad \mathcal{F}(p) = \frac{\partial V(p) - \partial V(p)^*}{2ia^2}.$$

The orientations and base points of $\mathcal{F}$ agree with those of D. Differentiating the plaquette action gives $J = D^*\mathcal{F}$ and $a|J|_\infty \le 4\phi$.

We first compare the action Hessian with Δ. Define the symmetric operator H by

$$(X, HX)_a = \frac{d^2}{dt^2} A^a(e^{iatX}V)\Big|_{t=0}.$$

For $p = p_{\mu\nu}(x)$ put $\alpha_p = (-T_\nu X_\mu, -X_\nu, X_\mu, T_\mu X_\nu)$ and $S_p = \sum_{i<j}[\alpha_{p,i}, \alpha_{p,j}]$. The four perturbed factors in the plaquette can be transported to x; their sum is $a(DX)(p)$. For $P_t = \prod_i e^{it\alpha_{p,i}}$,

$$P_0'' = -\left(\sum_i \alpha_{p,i}\right)^2 - S_p.$$

Every traceless Hermitian 2×2 matrix Z has $Z^2 = |Z|^2\mathbf{1}$. Since $\partial V(p) = (1 - a_p)\mathbf{1} + ia^2\mathcal{F}(p)$, taking the trace in the second variation gives

$$(X, HX)_a = a^3 \sum_p \left((1 - a_p)|(DX)(p)|^2 + \langle \mathcal{F}(p), iS_p\rangle\right). \tag{16.iv}$$

Expansion of the squares also gives

$$\|DX\|_a^2 + \|D_0^* X\|_a^2 = (X, \Delta X)_a + w(X), \qquad w(X) = \sum_{\mu\neq\nu} (X_\mu, [\nabla_\mu, \nabla_\nu^*]X_\nu)_a.$$

The commutator compares the two transports around a plaquette, so its norm is at most 2ϕ. Thus $|w(X)| \le 4\phi\|X\|_a^2$. Moreover, $|\mathcal{F}(p)| \le \phi$, $|S_p| \le 3\sum_i |\alpha_{p,i}|^2$, and every bond belongs to four

plaquettes. Consequently the second term of eq. 16.iv has modulus at most $12\phi\|X\|_a^2$. Finally, $a_p = \frac{1}{2}|\partial V(p) - \mathbf{1}|^2$, $|\partial V(p) - \mathbf{1}| \le 2$, and $|DX(p)|^2 \le 4a^{-2}\sum_{b\in p}|X(b)|^2$ imply

$$0 \le a^3 \sum_p a_p |DX(p)|^2 \le 16\phi \|X\|_a^2.$$

It follows that

$$H + D_0 D_0^* = \Delta + \mathcal{U}, \qquad \|\mathcal{U}\| \le 32\phi. \tag{16.v}$$

In particular, $H = \Delta + \mathcal{U}$ on divergence-free forms.

We next record the two evolution equations used below. If Y_s is the differential of the flow, in the frame $\delta U_s = iaY_s U_s$, then

$$Y_s' = -H_s Y_s - \frac{a}{2} i[J_s, Y_s].$$

Indeed, for $R_X F = \frac{d}{dt} F(e^{iatX} V)\Big|_0$ one has $[R_X, R_Y] = -aR_{i[X,Y]}$. The symmetric part of the second derivative is the Hessian, and hence $R_Y J = HY - \frac{a}{2} i[J, Y]$. Differentiating $\dot U = -iaJU$ now gives the displayed equation. For a smooth gauge-invariant G with $\Gamma_K(G)(U_\sigma) > 0$, put

$$\xi(s) = \operatorname{grad}_a (G \circ \Phi_{\sigma - s})(U_s), \qquad n(s) = \frac{(\xi, H_s \xi)_a}{\|\xi\|_a^2}.$$

The pairing $(\xi(s), Y_s)_a$ is constant. Since $D\Phi_s$ is invertible, this gives

$$\xi' = H_s \xi - \frac{a}{2} i[J_s, \xi], \qquad \Gamma_K(G \circ \Phi_\sigma)(U) = \Gamma_K(G)(U_\sigma) \exp\left[-2\int_0^\sigma n(s)\, \mathrm{d}s\right]. \tag{16.vi}$$

Here ξ never vanishes, and $D_0^* \xi = 0$ because the flow is gauge covariant. The current $j(s) = J(U_s)$ satisfies

$$j' = -H_s j, \qquad D_0^* j = 0, \qquad g' = -2(j, H_s j)_a. \tag{16.vii}$$

Thus g vanishes at one time only if it vanishes identically.

The derivative of Δ has a useful cancellation. For fixed X, $\partial_s \nabla_\mu X_\nu = -i[J_\mu, T_\mu X_\nu]$. The part containing $a\nabla_\mu X_\nu$ pairs to zero with $\nabla_\mu X_\nu$. Setting $Y_\mu = \sum_\nu i[X_\nu, \nabla_\mu X_\nu]$ therefore gives

$$(X, \Delta' X)_a = -2(J, Y)_a = -2(\mathcal{F}, DY)_a.$$

The discrete product rule $\nabla_\mu [A, B] = [\nabla_\mu A, T_\mu B] + [A, \nabla_\mu B]$ shows explicitly that

$$(DY)_{\mu\lambda} = \sum_\nu i([\nabla_\mu X_\nu, T_\mu \nabla_\lambda X_\nu] - [\nabla_\lambda X_\nu, T_\lambda \nabla_\mu X_\nu] + [X_\nu, [\nabla_\mu, \nabla_\lambda] X_\nu]).$$

The first two terms contain only first differences. Translations are isometries, $|[A, B]| \le 2|A||B|$, and the sum over the three pairs of directions is bounded by twice the sum of the squared first differences. The last term uses $\|[\nabla_\mu, \nabla_\lambda]\| \le 2\phi$. Multiplying by $2|\mathcal{F}|$ and summing gives

$$|(X, \Delta' X)_a| \le 8\phi (X, \Delta X)_a + 24\phi^2 \|X\|_a^2. \tag{16.viii}$$

For either $\eta = \xi$ or $\eta = j$, let $h = (\eta, \Delta\eta)_a / \|\eta\|_a^2$ and $n_\eta = (\eta, H\eta)_a / \|\eta\|_a^2$. Then $h \ge 0$, $|h - n_\eta| \le 32\phi$, and

$$h' = \frac{2(\eta', (\Delta - h)\eta)_a + (\eta, \Delta'\eta)_a}{\|\eta\|_a^2}.$$

For ξ the leading term is $2\|(\Delta - h)\xi\|_a^2$, whereas for j it is its negative. The remaining part of ξ' has norm at most $(32 + 4)\phi\|\xi\|_a$, using $a|J|_\infty \le 4\phi$. The inequality $72\phi\|\xi\|_a \|(\Delta - h)\xi\|_a \le 2\|(\Delta - h)\xi\|_a^2 + 648\phi^2\|\xi\|_a^2$ and the corresponding bound with 64 and 512 for j give

$$h' \ge -8\phi h - 672\phi^2 \quad (\eta = \xi), \qquad h' \le 8\phi h + 536\phi^2 \quad (\eta = j). \tag{16.ix}$$

Put $\psi(s) = C_0 s^{-r}$, so $\int_0^\sigma \psi = x/\theta$. For the adjoint solution, backward integration of eq. 16.ix yields

$$h(s) \le e^{8x/\theta}\left(h(\sigma) + 1344 C_0^2 s^{1-2r}\right), \qquad 0 < s \le \sigma,$$

because $2r - 1 \ge \frac{1}{2}$. The square ψ^2 is not integrable at zero, but $s\psi^2$ is. Accordingly, integrate against the weight s:

$$\int_0^\sigma h = \sigma h(\sigma) - \int_0^\sigma s h' \le \sigma h(\sigma) + \int_0^\sigma s(8\psi h + 672\psi^2).$$

There is no boundary term at zero: for each fixed lattice the flow, the operators, and the nonzero adjoint solution are smooth there. If $c \ge \sigma n(\sigma)$, then $0 \le \sigma h(\sigma) \le c + 32x$. Combining the last two inequalities with $|n - h| \le 32\psi$ gives

$$2\int_0^\sigma n(s)\,\mathrm{d}s \le 2\sigma n(\sigma) + E_\kappa(x, c). \tag{16.x}$$

Indeed, the endpoint and $n - h$ terms give $64x(1 + \theta^{-1})$; the constant part of $h(s)$ gives at most $16e^{8x/\theta}x(c + 32x)$, since $2 - r \ge 1$; and the two remaining integrals are

$$1344 C_0^2 \int_0^\sigma s^{1-2r}\,\mathrm{d}s = \frac{672}{\theta}x^2, \qquad 21504 C_0^3 e^{8x/\theta}\int_0^\sigma s^{2-3r}\,\mathrm{d}s = \frac{7168}{\theta}e^{8x/\theta}x^3.$$

It remains to control the terminal frequency when $G = A^a$. Apply the second inequality of eq. 16.ix to the current and integrate forward from s to σ. Averaging over $\sigma/2 \le s \le \sigma$ gives

$$\frac{\sigma}{2}h(\sigma) \le e^{8x/\theta}\left(\int_{\sigma/2}^\sigma h(s)\,\mathrm{d}s + \frac{\sigma}{2}\,536\int_{\sigma/2}^\sigma \psi(s)^2\,\mathrm{d}s\right).$$

By eq. 16.vii,

$$\int_{\sigma/2}^\sigma h(s)\,\mathrm{d}s \le \frac{1}{2}\log\frac{g(\sigma/2)}{g(\sigma)} + 32\int_{\sigma/2}^\sigma \psi(s)\,\mathrm{d}s, \qquad \int_{\sigma/2}^\sigma \psi(s)^2\,\mathrm{d}s \le 2C_0^2\sigma^{1-2r}.$$

Consequently

$$0 \le \sigma h(\sigma) \le e^{8x/\theta}\left(\log\frac{g(\sigma/2)}{g(\sigma)} + \frac{64x}{\theta} + 1072x^2\right).$$

The terminal adjoint vector for $G = A^a$ is the current itself. Thus its terminal frequency satisfies $\sigma n(\sigma) \le c_\kappa(x, j)$. Equations eq. 16.vi and eq. 16.x prove eq. 16.iii. □

Proposition 16.9 (Lower energy bounds for flowed observables)**.** For all K, m, $\beta > 0$, with $a = a_K$, the lattice time $\tau = s/a^2$, $T = \sigma/a^2$ and $U_\tau^\circ = \Phi_{a^2\tau}U$: let $|X|_\infty = \max_b |X(b)|$, let $\nabla^3 A_K(V)$ be the symmetric trilinear form with $\nabla^3 A_K(V)[X, X, X] = \frac{d^3}{dt^3}A_K(e^{itX}V)|_{t=0}$, and let Z_p be as in Lem. 16.2, taken at the field V.

(a) (Log-convexity with a defect.) Let $G \in C^2(\Omega_K)$ be invariant, $U \in \Omega_K$, $\sigma > 0$ and $\partial G(U_\sigma) \ne 0$; for $0 \le \tau \le T$ let $\gamma_\tau = \partial(G \circ \Phi_{a^2(T-\tau)})(U_\tau^\circ)$, $N(\tau) = \|\gamma_\tau\|^2$, $\mathsf{q}(\tau) = \nabla^2 A_K(U_\tau^\circ)[\gamma_\tau, \gamma_\tau]$ and $\mathsf{f}_\tau = \partial A_K(U_\tau^\circ)$. Then $N > 0$, $\dot N = 2\mathsf{q}$, $(\log N)'' \ge -\eta$ with $\eta(\tau) = 2|\nabla^3 A_K(U_\tau^\circ)[\mathsf{f}_\tau, \gamma_\tau, \gamma_\tau]|/N(\tau)$, and

$$\Gamma_K(G \circ \Phi_\sigma)(U) \;\ge\; \Gamma_K(G)(U_\sigma)\,\exp\Big[-2\nu - \int_0^T \tau\,\eta(\tau)\,\mathrm{d}\tau\Big], \qquad \nu = \frac{T\,\mathsf{q}(T)}{N(T)},$$

ν the frequency of the gradient at the flowed field. For all V and real β, γ:

$$|\nabla^3 A_K(V)[\beta, \gamma, \gamma]| \le |\gamma|_\infty\{4(\sum_p |Z_p(\beta)|^2)^{1/2}\|\gamma\| + 8(\sum_p |Z_p(\gamma)|^2)^{1/2}\|\beta\| + 64\,\varpi(V)\|\gamma\|\|\beta\|\}.$$

(b) (An effective form.) In (a) let besides $0<\kappa<\frac16$, $C_F\ge1$, $0<\sigma\le1$ and $a^{-2}\varpi(U_s)\le C_F s^{-\frac34-\frac{\kappa}{2}}$ for $0<s\le\sigma$. Put $\lambda_F=6C_F$, $\mathsf{e}_1=(1+56\lambda_F e^{56\lambda_F})^{1/2}$,

$$\mathsf{r}=(\ell/a)^{3/2}|\partial G(U_\sigma)|_\infty/\|\partial G(U_\sigma)\|,$$

$\nu_+=\max\{\nu,0\}$, $\varrho_F=3.5\,e^{80\lambda_F}C_F\mathsf{r}$ and

$$\mathsf{k}_A=28e^{56\lambda_F}C_F+e^{28\lambda_F}\varrho_F(7\mathsf{e}_1+256C_F).$$

If

$$4e^{2\nu_++1}\varrho_F\big[48\mathsf{e}_1+384C_F+12\nu_+^{1/2}+32\mathsf{k}_A^{1/2}+768\varrho_F\big]\sigma^{\frac14-\frac{\kappa}{2}}\le1,$$

then $\Gamma_K(G\circ\Phi_\sigma)(U)\ge e^{-2\nu_+-\frac12}\,\Gamma_K(G)(U_\sigma)$. At $G=A^a$: $\Gamma_K(A^a)(U_\sigma)=(J,J)_a(U_\sigma)$ and $\nu=-\frac{\sigma}{2}\,\partial_\sigma\log(J,J)_a(U_\sigma)$.

(c) (Lower propagation at every time.) Let $0<\kappa<\frac16$, $\mathbf{B}$ a real 1-form, $R\ge\max\{1,\|\mathbf{B}\|_{\kappa;a}\}$ and

$$a\le a_2(R):=\min\big\{a_1(R),\ (16e^{12}R)^{-2/(1-2\kappa)},\ \big(8(16\mathsf{M}_\Theta)^{1/6}c_3R\big)^{-1/(1-2\kappa)}\big\},$$

a positive number that depends on R and κ alone, with c_3, $a_0(R)$, $s_0(R)$ of the short-time theory of the gauge-fixed flow in Prop. 16.1(a), $a_1(R)\le a_0(R)$ the threshold of Lem. 16.7(b), and $\mathsf{M}_\Theta=\max\{1,\sup|\Theta^{\mathrm{ch}}|\}$, the absolute number of the head of Prop. 16.1, with the chart equation $\frac{d}{ds}A_s(b)=a^{-3}\Theta^{\mathrm{ch}}(aA_s|_{\mathsf{N}(b)})$ of the gauge-fixed flow set there and the supremum taken where $|aA|\le1$; so $a_2(R)\le a_1(R)$, and $a\le a_2(R)$ serves both Lem. 16.7(b) and (c); let U be gauge equivalent to $e^{ia\mathbf{B}}$, $\sigma>0$, $\sigma'=\min\{\sigma,s_0(R)\}$, $(J,J)_a(U_\sigma)>0$, $\Lambda\ge\Lambda_\sigma(U)$ and $\mathsf{j}_\sharp\ge(J,J)_a(U_{\sigma'/2})/(J,J)_a(U_\sigma)$.

Put $r=3/4+\kappa/2$, $\theta=1-r$ and $C_0(R)=4((2+c_3)R)^2$. For $0<\sigma\le s_0(R)$ set $x=C_0(R)\sigma^\theta$ and

$$\mathsf{G}_\kappa^{\mathrm{short}}(R,\sigma,\mathsf{j}_\sharp)=\exp\left[-2c_\kappa(x,\mathsf{j}_\sharp)-E_\kappa(x,c_\kappa(x,\mathsf{j}_\sharp))\right],$$

where c_κ and E_κ are defined in Lem. 16.8. Then

$$\Gamma_K(A^a\circ\Phi_\sigma)(U)\ge(J,J)_a(U_\sigma)\mathsf{G}_\kappa^{\mathrm{short}}(R,\sigma,\mathsf{j}_\sharp).\tag{16.xi}$$

This positive factor is independent of K, ℓ, and b, and decreases as $\mathsf{j}_\sharp$ increases.

For arbitrary $\sigma>0$, set $s=\sigma'$, $\tau=s/2$, $x=C_0(R)s^\theta$ and $j=\max\{1,\mathsf{j}_\sharp\}$. Put

$$c_*=e^{8x/\theta}(\log j+64\Lambda+672x^2)+32x,\qquad \mathsf{G}_\kappa(R,s,\Lambda,\mathsf{j}_\sharp)=j^{-1}\exp[-2c_*-E_\kappa(x,c_*)].$$

Then

$$\Gamma_K(A^a\circ\Phi_\sigma)(U)\ge(J,J)_a(U_\sigma)\mathsf{G}_\kappa(R,\sigma',\Lambda,\mathsf{j}_\sharp).$$

The factor is positive, decreases in Λ and $\mathsf{j}_\sharp$, and is independent of the cutoff, side and label.

The upper bound $\Gamma_K(A^a\circ\Phi_\sigma)(U)\le e^{24\Lambda_\sigma(U)}(J,J)_a(U_\sigma)$ holds as well.

(d) (Positive energies and the flowed current.) Assume (E) and $0<\kappa<\frac16$; let $\varphi\in C^\infty(\mathbb{R})$ be bounded with all its derivatives and $\varphi'>0$ on $[0,\infty)$, and $0<\sigma<\sigma_2\le1$. Say that (J_σ) holds if there is $c_0>0$ with $p_0:=\liminf_K\mu^K((J,J)_a(U_\sigma)\ge c_0)>0$, and $(\mathrm{D}_{\sigma,\sigma_2})$ if $\liminf_K\langle A^a\circ\Phi_\sigma-A^a\circ\Phi_{\sigma_2}\rangle_{\mu^K}>0$. In (i) and in the first two implications of (iii) use the consequence (Adm) of (E) proved in Prop. 16.1. (i) If (J_σ) holds, then $\liminf_K\int\chi_K\Gamma_K(\varphi(A^a)\circ\Phi_\sigma)\,\mathrm{d}\mu^K>0$ for every sequence of Borel $\chi_K:\Omega_K\to[0,1]$ with $\limsup_K\mu^K(\chi_K<1)<p_0$.

(ii) Conversely, if $R\ge R_\flat$, the Borel $\chi_K:\Omega_K\to[0,1]$ vanish off $\mathsf{O}_K(8R,\bar s(R))$ and $\liminf_K\int\chi_K\Gamma_K(\varphi(A^a)\circ\Phi_\sigma)\,\mathrm{d}\mu^K>0$, then (J_σ) holds. (iii) (J_{σ_2}) implies $(\mathrm{D}_{\sigma,\sigma_2})$, and $(\mathrm{D}_{\sigma,\sigma_2})$ implies (J_σ); and $\liminf_K\langle A^a\circ\Phi_\sigma\rangle_{\mu^K}>\sup_K\langle A^a\circ\Phi_1\rangle_{\mu^K}$ implies $(\mathrm{D}_{\sigma,1})$.

■

Proof sketch. (a) $N > 0$ because $D\Phi$ is invertible. As G is invariant, γ_τ is horizontal, and along the trajectory $\dot\gamma_\tau = \nabla^2 A_K(U_\tau^\circ)\gamma_\tau - \frac{i}{2}[\mathsf{f}_\tau, \gamma_\tau]$, the Hessian taken as a symmetric operator; since $\langle \gamma, i[\mathsf{f},\gamma]\rangle = 0$, $\dot N = 2\mathsf{q}$, and a second differentiation, in which the Hessians cancel, gives $\dot{\mathsf{q}} = 2\|\nabla^2 A_K\gamma\|^2 - \nabla^3 A_K[\mathsf{f},\gamma,\gamma] + \frac{1}{6}\|i[\mathsf{f},\gamma]\|^2$. So

$$(\log N)'' = \frac{4}{N^2}\big(\|\nabla^2 A_K\gamma\|^2\|\gamma\|^2 - \langle\gamma, \nabla^2 A_K\gamma\rangle^2\big) + \frac{2}{N}\Big(\tfrac{1}{6}\|i[\mathsf{f},\gamma]\|^2 - \nabla^3 A_K[\mathsf{f},\gamma,\gamma]\Big) \geq -\eta$$

by the Cauchy–Schwarz inequality. Integrating twice from T, $\log\frac{N(\tau_1)}{N(T)} \geq -2(T-\tau_1)\frac{\mathsf{q}(T)}{N(T)} - \int_{\tau_1}^T(\tau - \tau_1)\eta(\tau)\,\mathrm{d}\tau$ for $0 \leq \tau_1 \leq T$; at $\tau_1 = 0$, $N(0) = a\Gamma_K(G\circ\Phi_\sigma)(U)$ and $N(T) = a\Gamma_K(G)(U_\sigma)$. The weight τ under the integral keeps the defect finite at rough fields. The bound of $\nabla^3 A_K$ follows from the expansion of the third derivative plaquette by plaquette, as in Lem. 16.2(a), and the Cauchy–Schwarz inequality over the plaquettes. (b) The hypothesis gives $\Lambda_\sigma \leq \lambda_F$; by Lem. 16.2(b), (c), (h), $(J,J)_a(U_s)^{1/2} \leq \mathsf{j}_0 s^{-\frac{5}{4}-\frac{\kappa}{2}}$ with $\mathsf{j}_0 = 3.5\,e^{28\lambda_F}\ell^{3/2}C_F$, and $|\gamma_\tau|_\infty \leq e^{52\lambda_F}|\gamma_T|_\infty$, $\|\gamma_\tau\| \leq e^{28\lambda_F}\|\gamma_T\|$.

Lem. 16.2(a) gives $\sum_p |Z_p(X)|^2 \leq \nabla^2 A_K(V)[X,X] + 28\varpi(V)\|X\|^2$ for all X; with $\frac{d}{d\tau}\|\mathsf{f}_\tau\|^2 = -2\nabla^2 A_K[\mathsf{f}_\tau,\mathsf{f}_\tau]$ this makes the energy $\sum_p|Z_p(\mathsf{f}_\tau)|^2$ of the current integrable in τ, $\int_0^T\sum_p|Z_p(\mathsf{f}_\tau)|^2 d\tau \leq \frac{1}{2}\|\mathsf{f}_0\|^2 + 28\lambda_F\sup_\tau\|\mathsf{f}_\tau\|^2$, and with (a) it bounds the energy $\sum_p|Z_p(\gamma_\tau)|^2 \leq \mathsf{q} + 28\varpi N$ of γ_τ. On the largest interval $[\tau_1, T]$ on which $N \geq e^{-2\nu_+ - 1}N(T)$, the bound of $\nabla^3 A_K$ and these estimates give $\int_{\tau_1}^T(\tau-\tau_1)\eta \leq \frac{1}{2}$ under the displayed condition; so (a) gives $\log\frac{N(\tau_1)}{N(T)} \geq -2\nu_+ - \frac{1}{2}$, and $\tau_1 = 0$ by continuity.

At $G = A^a$, $\partial A^a = a^{-1}\mathsf{f}$, and $\frac{d}{d\tau}\|\mathsf{f}_\tau\|^2 = -2\langle\mathsf{f}_\tau, \nabla^2 A_K\mathsf{f}_\tau\rangle$. (c) First suppose $0 < \sigma \leq s_0(R)$. Since $a \leq a_2(R) \leq a_0(R)$, Prop. 16.1(a) gives a gauge-fixed representative e^{iaA_s} of U_s for $0 \leq s \leq s_0(R)$, with

$$\|A_s\|_\infty \leq (1+c_3)Rs^{-1/4-\kappa/2}, \qquad \|\nabla A_s\|_\infty \leq (2+c_3)Rs^{-3/4-\kappa/2}.$$

For a plaquette, split its holonomy into the two differences between opposite bond potentials and the group commutator of two adjacent bond exponentials. The exponential is Lipschitz on Hermitian matrices, and the group commutator has norm at most twice the product of their norms. This gives

$$a^{-2}\varpi(e^{iaA_s}) \leq 2\left(\|\nabla A_s\|_\infty + \|A_s\|_\infty^2\right),$$

as in Lem. 16.2(e). Put $\bar C = (2+c_3)R \geq 1$. Since $s_0(R) \leq 1$ and $1/2 + \kappa \leq 3/4 + \kappa/2$, the right side is at most $4\bar C^2 s^{-3/4-\kappa/2}$. Curvature is gauge invariant, so eq. 16.ii holds with $C_0 = C_0(R)$. Here $\sigma' = \sigma$, and the ratio assumed in (c) is exactly the ratio required in Lem. 16.8. That lemma proves eq. 16.xi. Now let $\sigma > 0$ be arbitrary and put $s = \sigma'$ and $\tau = s/2$. Use the adjoint solution

$$\xi(t) = \mathrm{grad}_a(A^a \circ \Phi_{\sigma-t})(U_t), \qquad g(t) = \|J(U_t)\|_a^2.$$

It is horizontal and nonzero. The adjoint identity eq. 16.vi and $D\Phi_{\sigma-t}(U_t)J(U_t) = J(U_\sigma)$ give

$$(\xi(t), J(U_t))_a = g(\sigma), \qquad \|\xi(\tau)\|_a^2 \geq \frac{g(\sigma)^2}{g(\tau)} \geq \frac{g(\sigma)}{j}.$$

For this same solution put $h(t) = (\xi(t), \Delta_t\xi(t))_a/\|\xi(t)\|_a^2$ and $n(t) = (\xi(t), H_t\xi(t))_a/\|\xi(t)\|_a^2$. Since $h \geq 0$ and $|n-h| \leq 32\phi$,

$$\|\xi(s)\|_a^2 \leq e^{64(\Lambda_\sigma - \Lambda_s)}g(\sigma), \qquad \int_\tau^s h(t)\,\mathrm{d}t \leq \tfrac{1}{2}\log j + 32\Lambda.$$

Indeed, integrate $(\log\|\xi\|_a^2)' = 2n$ first on $[s,\sigma]$ and then on $[\tau,s]$, using the lower bound at τ. Only the integrated curvature up to σ enters here.

On $[0,s]$ the short-time estimate gives $\phi(t) \leq C_0(R)t^{-r}$. Backward integration of the adjoint inequality in eq. 16.ix, followed by averaging over $[\tau,s]$, yields

$$\tau h(\tau) \leq e^{8x/\theta}\left(\int_\tau^s h(t)\,\mathrm{d}t + 672\tau\int_\tau^s C_0(R)^2 t^{-2r}\,\mathrm{d}t\right).$$

Here $\tau\int_\tau^{2\tau} C_0(R)^2 t^{-2r}\,\mathrm{d}t \le C_0(R)^2\tau^{2\theta} \le x^2$. Consequently $\tau n(\tau) \le c_*$. The integrated adjoint estimate eq. 16.x, applied on $[0,\tau]$ to this solution, gives

$$2\int_0^\tau n(t)\,\mathrm{d}t \le 2c_* + E_\kappa(x, c_*).$$

We used $C_0(R)\tau^\theta \le x$ and monotonicity of E_κ in its nonnegative arguments. Finally,

$$\Gamma_K(A^a\circ\Phi_\sigma)(U) = \|\xi(0)\|_a^2 = \|\xi(\tau)\|_a^2 \exp\left[-2\int_0^\tau n(t)\,\mathrm{d}t\right] \ge \frac{g(\sigma)}{j} e^{-2c_* - E_\kappa(x,c_*)}.$$

Thus no power-law curvature estimate beyond the regular interval $[0,s]$ is needed. The upper bound follows as before from Lem. 16.2(b).

(d) (i): fix R, R' and s_+ such that, for large K, the regularised fields $\mathsf{O}_K(8R,\bar s(R))$ of (C4), the event that the gauge orbit holds a 1-form of rough norm $\le R'$ at the index $\frac{\kappa}{2}$, and $\{A^a\circ\Phi_\sigma \le s_+\}$ all have probability close to 1 (Prop. 16.1(a), by (Adm), and Chebyshev's inequality).

On their intersection with $\{\chi_K = 1\}\cap\{(J,J)_a(U_\sigma)\ge c_0\}$, whose mass has lower limit > 0, the proof of (C2)(a) bounds Λ_σ and $(J,J)_a(U_{\sigma'/2})$, the proved lower bound of (c) at the index $\frac{\kappa}{2}$, for $a_K \le a_2(R')$, bounds $\Gamma_K(A^a\circ\Phi_\sigma)$ below by c_0 times a positive number free of K, and $\Gamma_K(\varphi(A^a)\circ\Phi_\sigma) = \varphi'(A^a\circ\Phi_\sigma)^2\Gamma_K(A^a\circ\Phi_\sigma) \ge (\min_{[0,s_+]}\varphi')^2\Gamma_K(A^a\circ\Phi_\sigma)$. (ii): on $\mathsf{O}_K(8R,\bar s(R))$, for large K, the proof of (C2)(a) gives $\Lambda_\sigma \le \lambda_\sharp$ and $(J,J)_a(U_\sigma)\le \mathsf{j}_\sigma$ with numbers free of K, and $\Gamma_K(\varphi(A^a)\circ\Phi_\sigma) \le \|\varphi'\|_\infty^2 e^{24\lambda_\sharp}(J,J)_a(U_\sigma)$ (Lem. 16.2(b)); splitting at a level c, $\int\chi_K\Gamma_K \le \|\varphi'\|_\infty^2 e^{24\lambda_\sharp}[c + \mathsf{j}_\sigma\mu^K((J,J)_a(U_\sigma)\ge c)]$.

(iii): $A^a\circ\Phi_\sigma - A^a\circ\Phi_{\sigma_2} = \int_\sigma^{\sigma_2}(J,J)_a(U_s)\,\mathrm{d}s$ (Lem. 16.2(c)), and $(J,J)_a(U_{s'}) \le e^{24\lambda_\sharp}(J,J)_a(U_s)$ for $s\le s'$ on $\mathsf{O}_K(8R,\bar s(R))$, whose complement has small mass for large R by (Adm) (Step 8); in the second arrow the moments of $(J,J)_a(U_s)$ of Prop. 16.1(a), under (Adm), and the Cauchy–Schwarz inequality give $\langle(J,J)_a(U_\sigma)\rangle_{\mu^K} \ge 2c_0$, hence $\mu^K((J,J)_a(U_\sigma)\ge c_0) \ge c_0^2/\sup_K\langle(J,J)_a(U_\sigma)^2\rangle_{\mu^K}$. The last arrow is $\liminf_K(x_K - y_K) \ge \liminf_K x_K - \limsup_K y_K$. □

16.7. **Comparison of backgrounds at consecutive cutoffs.** The convergence argument in Prop. 16.1(b) first fixes the coarse lattice and then removes the ultraviolet cutoff. For this step a comparison whose constant depends on the coarse volume is sufficient. We prove that comparison here. The separate estimate between a rough field and its background, used subsequently in that proposition, remains uniform in the cutoff.

We measure lengths in units of the fixed coarse lattice. Its torus has side $2L^n$ and volume $N = (2L^n)^3$. Let $\eta = L^{-j}$, $j\ge 1$, and $\eta' = \eta/L$. The two fine lattices have spacings η and η'. For a configuration U at spacing ξ, write

$$F_U(p) = \frac{\partial U(p) - \partial U(p)^{-1}}{2i\xi^2}, \qquad J(U) = D_U^* F_U, \qquad \mathcal{A}^\xi(U) = \xi^{-1}\sum_p \left(1 - \tfrac{1}{2}\operatorname{tr}\partial U(p)\right).$$

For quaternionic matrices we use the real scalar product $\operatorname{Re}\operatorname{tr}(X^*Y)/2$; its norm agrees with the operator norm and restricts to the stated norm on the Lie algebra. All sums over bonds and plaquettes below use positive orientations. The pairing and L^p norms at spacing ξ include the measure ξ^3. The notation D_U denotes the covariant exterior derivative. We call U regular at radius α when

$$\sup_p|\partial U(p) - 1| < \alpha\xi^2, \qquad \sup_b|J(U)(b)| < \alpha.$$

These are the constant-scale regularity conditions used in the background construction. All constants in this subsection depend only on the fixed blocking factor L. We use the Hölder exponent $1/2$.

Proposition 16.10 (Comparison at a fixed coarse lattice)**.** There are $\alpha_* > 0$, $C < \infty$ and a fixed room radius ρ_* with the following property. Suppose the constant-torus background setting is admitted, $L^n \ge 3L\mathfrak{M}$ and $L^n > 6\rho_*$, where $\mathfrak{M}$ is its fixed analytic block size. Suppose that U^A and

U^B are regular real critical backgrounds, at radius $0 < \alpha \le \alpha_*$, on the lattices of spacings η and η', respectively, and that they have the same coarse datum:

$$\mathcal{M}^j U^A = \mathcal{M}^{j+1} U^B.$$

Then there are a gauge transformation g and a real one-form A on the η lattice such that

$$(U^A)^g = e^{i\eta A}\mathcal{M}(U^B), \qquad \|A\|_\infty \le C\alpha N^{1/5}\eta^{1/5}. \tag{16.xii}$$

The constants α_*, C, ρ_* are independent of n, j and of the coarse datum. The factor $N^{1/5}$ is retained in the estimate. ■

The proof compares the first variations of the actions on the two lattices. The average of a critical field need not be critical. Its failure of criticality is small when tested against a coarse variation and a compatible fine variation. We first quantify this fact.

Lemma 16.11 (One-step regularity and the action defect)**.** Suppose U is regular at radius α on the η' lattice, with α sufficiently small and $L^n > 6\rho_*$. Put $W = \mathcal{M}(U)$. Then W is regular at radius $C\alpha$ on the η lattice. For every real fine one-form h,

$$\left| \frac{d}{dt}\left[\mathcal{A}^{\eta'}(e^{it\eta' h}U) - \mathcal{A}^{\eta}(\mathcal{M}(e^{it\eta' h}U)) \right]_{t=0} \right| \le C\alpha\eta^{1/2}\|D_U h\|_1 + C\alpha\eta'\|h\|_{1,\sup}. \tag{16.xiii}$$

Here $\|h\|_{1,\sup} = \eta^3 \sum_{p'} \sup_{b\subset Y(p')} |h(b)|$; $Y(p')$ is the union of the four fine blocks at the corners of the coarse plaquette p'. ■

Proof. Choose ρ_* to be a fixed multiple of the radius in [7, Thm. BT], large enough to contain each averaging stencil with a collar of width four. At every coarse site that theorem, with constant level, supplies a gauge $U = e^{i\eta' P}$ on this cube with

$$|P| + |\nabla P| + |D_1^* D_1 P| + |\Delta P| \le C\alpha, \qquad |\nabla P(x) - \nabla P(y)| \le C\alpha |x-y|^{1/2}. \tag{16.xiv}$$

The enlarged cube fits in the torus by $L^n > 6\rho_*$. Its radius is fixed, so the smallness condition in that theorem follows by reducing α_*. In particular the constants in eq. 16.xiv do not depend on η or N.

We give the local estimates needed for averaging. On the fixed stencil of a coarse bond write

$$\Phi(a) = \frac{1}{i}\log \mathcal{M}(e^{ia}), \qquad \Phi(0) = 0, \qquad \Phi'(0) = Q_{\mathrm{lin}}.$$

The exponential and logarithmic formulas defining $\mathcal{M}$ are holomorphic on a ball of fixed radius, since each path contains a bounded number, depending only on L, of factors. Their first three derivatives are bounded on a smaller fixed ball by Cauchy's formula. The differential Q_{lin} is the block mean of straight path sums. Summing the oriented boundary of each path surface gives

$$dQ_{\mathrm{lin}} a(p') = L^2 \langle da\rangle_{p'}, \qquad \langle f\rangle_{p'} = L^{-5} \sum_{x\in B(p'_-)} \sum_{p\in S_x(p')} f(p). \tag{16.xv}$$

The interior edges cancel in each surface $S_x(p')$; its remaining four edges are precisely the four path sums in the left member. The mean has total mass one. Every fine plaquette occurs with total weight L^{-3} when p' is summed, and hence

$$\eta^3 \sum_{p'} \langle f\rangle_{p'} = \eta'^3 \sum_{p} f(p). \tag{16.xvi}$$

For later use we also record the elementary nonlinear curvature bounds. If $V = e^{i\xi P}$, $|P| \le a \le 1$ and $\xi \le 1/13$, then

$$F_V = D_1 P + \mathsf{n}(P), \qquad |\mathsf{n}(P)| \le 16a^2.$$

Indeed, expand the four exponentials of a plaquette to first order; the remainder is at most the square of the sum of the four angles. Cauchy's formula applied to this quadratic remainder gives

$|\mathsf{n}(P') - \mathsf{n}(P)| \le 64a \max|P' - P|$. In the current, the difference of the two nonlinear remainders on neighboring plaquettes contributes at most $64a^2$ per direction when $|\nabla P| \le a$. The parallel transport in the other term differs from the identity by at most $2\xi a$, while $|F_V| \le 2a + 16a^2$. Thus

$$|J(V) - D_1^* D_1 P| \le 200a^2. \tag{16.xvii}$$

Let $P^M = \eta^{-1}\Phi(\eta' P)$ in the gauge eq. 16.xiv. The derivative bounds for Φ give $|P^M| + |\nabla P^M| \le C\alpha$. Split it into $P^{\mathrm{lin}} = \eta^{-1} Q_{\mathrm{lin}}(\eta' P)$ and P^{nl}. For $\Psi = \Phi - \Phi'(0)$, $|\Psi'(a)| \le C|a|$ and $|\Psi''(a)| \le C$. A mixed second difference of $\Psi(\eta' P)$ is the double integral of

$$\Psi''(a(s,t))[\partial_s a, \partial_t a] + \Psi'(a(s,t))\partial_s \partial_t a,$$

where $a(s,t)$ is the bilinear interpolation of its four translated arguments. Here $|a| \le C\eta'\alpha$, $|\partial_s a| + |\partial_t a| \le C\eta'\eta\alpha$ and $|\partial_s \partial_t a| \le C\eta'\eta\alpha$. After division by η^3 this bounds $|D_1^* D_1 P^{\mathrm{nl}}|$ by $C\alpha^2$.

There is a cancellation needed to bound the linear term without a negative power of η. Fix distinct directions μ, ν, λ, put $F_{\mu\sigma} = \nabla'_\mu P_\sigma - \nabla'_\sigma P_\mu$ and $g_\sigma(w) = \eta'^{-1}[F_{\mu\sigma}(w) - F_{\mu\sigma}(w - \eta' e_\sigma)]$. The linear Stokes identity gives

$$(D_1^* D_1 P^{\mathrm{lin}})_\mu = K_\lambda(g_\nu + g_\lambda) + (K_\nu - K_\lambda) g_\nu,$$

where

$$K_\sigma f(x) = L^{-6} \sum_{z \in B(x)} \sum_{0 \le i, i', r < L} f(z + i\eta' e_\mu + (i' - r)\eta' e_\sigma).$$

The first term is bounded by $C\alpha$ because $g_\nu + g_\lambda = (D_1^* D_1 P)_\mu$ and K_λ is a positive average. To treat the second, multiply P by a product cutoff equal to one on the stencil, supported in its collar, whose first and second differences are bounded independently of the mesh, and extend it by zero. The product rule and eq. 16.xiv give $\|\Delta' \widetilde{P}\|_\infty \le C\alpha$. The two Fourier multipliers expressing $(K_\nu - K_\lambda) g_\nu$ in terms of $\Delta' \widetilde{P}_\nu, \Delta' \widetilde{P}_\mu$ are

$$-\kappa(\vartheta) \frac{\overline{d}_\nu d_\mu}{\sum_\sigma |d_\sigma|^2}, \qquad \kappa(\vartheta) \frac{|d_\nu|^2}{\sum_\sigma |d_\sigma|^2},$$

where $d_\sigma = e^{i\vartheta_\sigma} - 1$, $s(t) = L^{-1} \sum_{|q| \le (L-1)/2} e^{iqt}$, $e_L(t) = L^{-1} \sum_{q=0}^{L-1} e^{iqt}$ and

$$\kappa = s(\vartheta_\mu) e_L(\vartheta_\mu) s(\vartheta_\nu) s(\vartheta_\lambda) [s(\vartheta_\nu)^2 - s(\vartheta_\lambda)^2].$$

The last bracket vanishes to second order at zero. Each multiplier m therefore satisfies $|\partial^r m(\vartheta)| \le C|\vartheta|^{2-r}$ for $0 \le r \le 2$ near zero and is smooth elsewhere. These are also weak derivative bounds: the boundary terms on a sphere of radius δ tend to zero as $\delta \downarrow 0$. Twice integrating by parts and using Bessel's inequality gives

$$\sum_{z \in \mathbb{Z}^3} |\widehat{m}(z)| \le \left(\sum_z (1 + |z|^2)^{-2} \right)^{1/2} \|m - \Delta_\vartheta m\|_{L^2} < \infty.$$

The convolution kernels thus have ℓ^1 norms bounded only in terms of L. This bounds the second term by $C\alpha$. Together with eq. 16.xvii it proves $|J(W)| \le C\alpha$.

We prove the curvature bound and the action estimate together. In $Y(p')$ take the comb gauge based at p'_-. A comb comparison with the elementary plaquettes gives $U = e^{ia_0}$ in this gauge with $|a_0| \le C\varpi$, where $\varpi = \sup|\partial U - 1| \le \alpha\eta'^2$. Put $G_p = \partial U(p) - 1$ in this fixed gauge and $\bar{G} = \langle G \rangle_{p'}$. By eq. 16.xv, the map

$$R(a) = \partial \mathcal{M}(e^{ia})(p') - 1 - L^2 \langle \partial e^{ia} - 1 \rangle_{p'}$$

has zero value and derivative at zero. The holomorphic bounds above imply

$$|R(a_0)| \le C\varpi^2, \qquad |R'(a_0)\dot{a}| \le C\varpi|\dot{a}|. \tag{16.xviii}$$

Consequently $|\partial W - 1| \le L^2\varpi + C\varpi^2 \le C\alpha\eta^2$, completing regularity of W.

In the gauge eq. 16.xiv, curvature varies across a stencil by $C\alpha\eta^{1/2}$ in physical units: this follows from the Hölder bound for ∇P and the Lipschitz bound for $\mathsf{n}(P)$. Passing to the comb gauge changes the comparison of two curvatures by at most $C\alpha^2\eta$ in those units. Indeed, each comb has length $C\eta$, each bond angle is at most $C\eta'\alpha$, and $|\operatorname{Ad}(v)-1| \le 2|v-1|$. Thus

$$|G_p - \bar G| \le C\alpha\eta'^2\eta^{1/2}. \tag{16.xix}$$

The identity $1-\frac{1}{2}\operatorname{tr} v = |v-1|^2/2$ for $v \in SU(2)$ and eq. 16.xvi give the exact defect

$$\mathcal{A}^{\eta'}(U) - \mathcal{A}^{\eta}(W) = \eta^{-1}\sum_{p'} \delta_{p'}, \qquad \delta_{p'} = \frac{L^4}{2}\langle |G-\bar G|^2\rangle - L^2(\bar G, R) - \frac{1}{2}|R|^2.$$

Differentiate at $U_t = e^{it\eta' h}U$, retaining the comb gauge chosen at $t=0$. If $s_{p'} = \sup_{Y(p')}|h|$, the product rule on a plaquette and the derivative of the local logarithm give

$$|\dot G_p| \le \eta'^2 |D_U h(p)| + 4\varpi\eta' s_{p'}, \qquad |\dot a| \le 2\eta' s_{p'}, \qquad |\dot R| \le C\varpi\eta' s_{p'}.$$

In the derivative of the variance, the term containing $\dot{\bar G}$ vanishes because $\langle G - \bar G\rangle = 0$. Therefore

$$\dot\delta_{p'} = L^4\langle (G-\bar G, \dot G)\rangle - L^2(\bar G, \dot R) - L^2(R, \langle \dot G\rangle) - (R, \dot R).$$

Substitute eq. 16.xviii and eq. 16.xix, use $\eta' = \eta/L$ and $\alpha \le 1$, and divide by η. This yields

$$\eta^{-1}|\dot\delta_{p'}| \le C\eta^3\big[\alpha\eta^{1/2}\langle |D_U h|\rangle_{p'} + \alpha\eta' s_{p'}\big].$$

Summing and applying eq. 16.xvi proves eq. 16.xiii. □

Proof of Prop. 16.10. Put $W = \mathcal{M}(U^B)$. The preceding lemma makes W regular at radius $C\alpha$, and it has the same reduced datum as U^A. We check the setting of the companion's chart results. Take $\mathfrak{M} = M_{\mathrm{ps}}$ and choose once and for all $R_1 \ge L$ with $R_1\mathfrak{M} \ge w_S$. At depths $0,\dots,j$ let every region be the whole torus, and let the next region be empty. The required $\mathfrak{M}$-cubes exist since $\mathfrak{M} \le L^n$; the separation distances are infinite. Every layer below j is empty, so both its full and its reduced determining rows are empty. At level j all bonds are interior determining rows. There are no inward or outward rows, seam nodes or edge rows. Thus both determining maps are exactly $\mathcal{M}^j$, and both row spaces are the one-forms on the unit lattice. The local scale is one everywhere, so every weighted norm is the corresponding supremum. The same conclusions hold at the finer cutoff with depth $j+1$.

The companion's chart uses the matrix divergence regularity class. The current bound implies that class with radius $r + r^2$ when the curvature radius is r: the traceless part of the plaquette divergence is the current, and its scalar part is a difference of $1-\frac{1}{2}|\partial U - 1|^2$ at neighboring plaquettes. This is the real $SU(2)$ conversion of Lem. YM3_I:9.4. It costs only a fixed multiple of α here. The terminal torus has $2L^n \ge 6LM_{\mathrm{ps}}$ sites per direction. All geometric and regularity hypotheses of the common-chart argument in the proof of Prop. YM3_I:37.3 are therefore satisfied. After reducing α_* it gives

$$(U^A)^g = e^{i\eta A}W, \qquad RD^*_W A = 0, \qquad q(A) = 0, \qquad \|A\|_X \le C\alpha, \tag{16.xx}$$

where $\|A\|_X$ controls both $\sup|A|$ and the covariant first differences. Write $F(a) = \mathcal{A}^{\eta}(e^{i\eta a}W)$ and $F_B(b) = \mathcal{A}^{\eta'}(e^{i\eta' b}U^B)$. Here q is the nonlinear coarse constraint and $K_a = q'(a)$. Lem. YM3_I:30.8 converts criticality on the actual averaging fiber into the multiplier equation. Applying Prop. YM3_I:30.9 with zero constraint value bounds its multiplier, first at the reference W and then at U^B. Its gradient at the reference is bounded by $C\alpha$ in both cases, by regularity. It follows that on the common unit lattice

$$F'(A) = K_A^*\omega_A, \qquad F'_B(0) = (K_0^B)^*\omega_B, \qquad \|\omega_A\|_\infty + \|\omega_B\|_\infty \le C\alpha. \tag{16.xxi}$$

Here the two constraints are the actual relative constraints, before any orthogonal change of row frame. Their averaging recursions agree after the first fine step, as checked below.

We recall the precise linear and nonlinear estimates needed from Props. YM3_I:29.1 and YM3_I:30.5. With Q the corner linear constraint and T its fixed orthogonal row transport, $T^{-1}q(a) = Qa + \mathsf{C}(a)$, $\mathsf{C}(0) = 0$, and

$$\|\mathsf{C}'(a)\|_{2\to 2} \le C\alpha, \|\mathsf{C}'(a)^*\chi - \mathsf{C}'(b)^*\chi\|_\infty \le C\|a-b\|_\infty\|\chi\|_\infty \tag{16.xxii}$$

on a fixed multiple of the α ball containing the segment $[0, A]$. The first inequality follows also directly from the stated row and column bounds by applying them to the symmetric operator $\mathsf{C}'(a)^*\mathsf{C}'(a)$. Put $\Delta_R = F''(0) + D_W R D_W^*$ and $G = (\Delta_R + Q^*Q)^{-1}$, $M = QGQ^*$, $H = GQ^*M^{-1}$ and $\widetilde{G} = G - GQ^*M^{-1}QG$. These operators obey

$$QH = 1, \quad \widetilde{G}\Delta_R = 1 - HQ, \quad \Delta_R H = Q^*(M^{-1} - 1), \quad \|H\|_{2\to 2} + \|\widetilde{G}\|_{2\to 2} \le C. \tag{16.xxiii}$$

Thm. YM3_I:I gives positivity of G in this real regular setting, and Prop. YM3_I:29.1 gives positivity of M. Thus both M and G are positive definite, and $\widetilde{G}$ is positive semidefinite. To see the last assertion directly, complete the square in $(f, Gf) - (QGf, M^{-1}QGf)$. The L^2 bounds follow from the sup bounds for $\widetilde{G}, H, H^*$: a symmetric operator bounded by C in any norm has all eigenvalues bounded by C, and one applies this also to H^*H.

These facts imply coercivity along A. For $Qa = 0$, $a = \widetilde{G}\Delta_R a$ and the spectral inequality $\widetilde{G}^2 \le C\widetilde{G}$ give $(a, \Delta_R a) \ge c\|a\|_2^2$. For arbitrary h, set $a = h - HQh$. The mixed term in its quadratic form vanishes by eq. 16.xxiii, whereas the remaining HQh term is at least $-\|Qh\|_2^2$. Hence, whenever $C\|Qh\|_2 \le \|h\|_2$,

$$(h, \Delta_R h) \ge c(\|h\|_2 - C\|Qh\|_2)^2 - \|Qh\|_2^2.$$

The constraint in eq. 16.xx gives $QA = -\mathsf{C}(A)$, so eq. 16.xxii implies $\|QA\|_2 \le C\alpha\|A\|_2$. The gauge term vanishes on A. Finally $F''(0) = D_W^* D_W + \Delta'$ with $\|\Delta'\|_{2\to 2} \le C\alpha$ by the real Hessian bounds in the same results. Combining the two resulting lower bounds and reducing α_* proves

$$F''(0)[A, A] \ge cx^2, \qquad x^2 = \|D_W A\|_2^2 + \|A\|_2^2. \tag{16.xxiv}$$

We next construct the fine test form used to compare first variations. For a fine bond $b = (x, \mu)$, let Y be the lower corner of its coarse cell and $r = x - Y$, $0 \le r_\sigma < \eta$. Interpolate the four parallel coarse bonds c_s, with initial points $Y + \eta s$, $s \in \{0,1\}^{\{\nu:\nu\neq\mu\}}$, using the weights

$$w_s(x) = \prod_{\nu\neq\mu} (r_\nu/\eta)^{s_\nu}(1 - r_\nu/\eta)^{1-s_\nu}.$$

Transport their values to x along the coordinate combs in U^B and denote their weighted mean by $h^\uparrow(b)$. There is no length factor in this definition: h and $h^\uparrow$ are potentials in the same physical units. Define $h^\downarrow$ by

$$\eta h^\downarrow = Q_{\mathsf{m}}(U^B)(\eta' h^\uparrow),$$

where Q_{m} is the derivative of the relative one-step average, including its straightening gauge. The elementary interpolation estimates are

$$\begin{aligned} \|D_{U^B} h^\uparrow\|_1 &\le \|D_W h\|_1 + C\alpha\eta N\|h\|_\infty, \\ \|h^\uparrow\|_{1,\sup} &\le 3N\|h\|_\infty, \\ \|h - h^\downarrow\|_1 &\le C\eta N\|h\|_X. \end{aligned} \tag{16.xxv}$$

Here is the finite-stencil verification. In a comb gauge on the box containing the relevant neighboring cells, the fine angles are bounded by $C\alpha\eta\eta'$. Every comparison path has at most CL bonds. Replacing its transport by the identity changes the interpolated value by $C\alpha\eta^2 \sup|h|$ and its covariant curl by $C\alpha\eta \sup|h|$. Without these errors, subtracting the two linear interpolants in a plaquette gives the linear interpolation of $D_W h$ in the one transverse direction. Its weights are nonnegative; their total fine mass over each coarse plaquette is L^3. This proves the first estimate on multiplying by η'^3 and summing. The second follows from $|h^\uparrow| \le \|h\|_\infty$ and the $3N\eta^{-3}$ coarse

plaquettes. At zero fine angles the relative derivative Q_{m} is straight path averaging; applied to the interpolant it is a convex mean of neighboring values of h. At the given fine angles its derivative differs by $C\alpha\eta^2 \sup|h|$ after division by η, by the bounded second derivative of the relative average. Neighbors differ by at most $3\eta\|\nabla^W h\|_\infty + C\alpha\eta^2\|h\|_\infty$. This gives the last inequality in eq. 16.xxv.

Let $a_1(t)$ be the relative logarithm of the first average of $e^{it\eta' h^\uparrow}U^B$. It has derivative $a_1'(0) = \eta h^\downarrow$. After this first step, the reference configurations in the two averaging recursions are identical. Their subsequent relative logarithms therefore agree, by induction over the remaining j steps. Differentiating the final constraint gives

$$K_0^B h^\uparrow = K_0 h^\downarrow. \tag{16.xxvi}$$

Gauge invariance of the action removes the straightening gauge at the first step. Thus its action defect is exactly

$$E(h) = F_B'(0)[h^\uparrow] - F'(0)[h^\downarrow].$$

Put $d = F'(0) - K_0^*\omega_B$. From eq. 16.xxi and eq. 16.xxvi, $(d, h^\downarrow) = -E(h)$, and hence

$$F'(0)[h] = (d, h - h^\downarrow) + (\omega_B, K_0 h) - E(h). \tag{16.xxvii}$$

Regularity of W, the transpose row bound for K_0 and eq. 16.xxi give $\|d\|_\infty \le C\alpha$. Apply eq. 16.xiii and eq. 16.xxv with $h = A$, using $\|A\|_X \le C\alpha$ and $\|D_W A\|_1 \le (3N)^{1/2}\|D_W A\|_2$. We obtain

$$|(d, A - A^\downarrow)| + |E(A)| \le C\alpha\eta^{1/2}N^{1/2}x + C\alpha^2\eta N. \tag{16.xxviii}$$

The constraint multipliers contribute only quadratically. To see this, for fixed χ the Hessian of $(\chi, \mathsf{C}(a))$ is symmetric and has sup operator norm at most $C\|\chi\|_\infty$ by eq. 16.xxii. Its L^2 operator norm has the same bound, by the eigenvalue argument used above. Since $q(0) = q(A) = 0$, the fundamental theorem of calculus gives

$$T^{-1}K_0 A = -\int_0^1 [\mathsf{C}'(tA) - \mathsf{C}'(0)]A\,\mathrm{d}t, \qquad T^{-1}K_A A = \int_0^1 [\mathsf{C}'(A) - \mathsf{C}'(tA)]A\,\mathrm{d}t.$$

Pair these identities with $T^{-1}\omega_B$ and $T^{-1}\omega_A$, whose suprema are unchanged by the orthogonal row transport. Consequently

$$|(\omega_B, K_0 A)| + |F'(A)[A]| \le C\alpha\|A\|_2^2. \tag{16.xxix}$$

For completeness we bound the third derivative of the action; a bound on four separate bond derivatives would lose powers of the mesh. Put $\phi(t) = F(tA)$. At a plaquette with reference holonomy P, its varied holonomy has the form $Z(t) = e^{tX_1}\cdots e^{tX_4}P$, where $|X_i| \le \eta a_p$ and a_p is the largest $|A|$ on its boundary. Write $Z' = \Sigma Z$. Differentiating the four partial products gives

$$|\Sigma'| \le 12\eta^2 a_p^2, \qquad |\Sigma''| \le 72\eta^3 a_p^3, \qquad \Sigma(0) = i\eta^2 Y, \quad |Y - D_W A(p)| \le C\alpha\eta a_p.$$

The last error is precisely the transport around P rather than the identity; $|P - 1| \le C\alpha\eta^2$. Set $y_p = |Y| + 12a_p^2$. Then $|\Sigma(t)| \le \eta^2 y_p$ and $|Z(t) - 1| \le \eta^2(C\alpha + y_p)$ for $0 \le t \le 1$. Both Σ'' and Σ^3 are traceless, while $\Sigma^2 = -|\Sigma|^2 1$, since $\Sigma \in \mathfrak{su}(2)$. In

$$\frac{d^3}{dt^3}\left(1 - \tfrac{1}{2}\operatorname{tr} Z\right) = -\tfrac{1}{2}\operatorname{tr}[\Sigma'' + 2\Sigma'\Sigma + \Sigma\Sigma' + \Sigma^3]Z$$

one may therefore replace Z by $Z - 1$ in the first and last terms. With $a_p + |D_W A(p)| \le C\alpha$ the displayed bounds give

$$\left|\frac{d^3}{dt^3}\left(1 - \tfrac{1}{2}\operatorname{tr} Z\right)\right| \le C\eta^4\alpha(|D_W A(p)|^2 + a_p^2).$$

Each bond belongs to four plaquettes. Multiplication by η^{-1} and summation therefore prove $|\phi'''(t)| \le C\alpha x^2$.

Taylor's formula, eq. 16.xxiv, eq. 16.xxvii, eq. 16.xxviii and eq. 16.xxix now imply

$$cx^2 \le C\alpha x^2 + C\alpha\eta^{1/2}N^{1/2}x + C\alpha^2\eta N.$$

Choose α_* so the first term is absorbed. Young's inequality absorbs half of the remaining left member and yields

$$\|A\|_2^2 + \|D_W A\|_2^2 \le C\alpha^2 \eta N. \tag{16.xxx}$$

Finally let $s = \|A\|_\infty$ and choose $r = C\alpha$ so that $\|A\|_X \le r/2$. The norm of a component changes by at most $\eta r/2$ across an adjacent site: parallel transport is an isometry. If $s/r < \eta$, then $s < r\eta$. Otherwise set $R = \lfloor s/(r\eta) \rfloor \ge s/(2r\eta)$. On the lattice ℓ^1 ball of radius R about a maximizing site, that component has norm at least $s/2$. Its physical radius is at most $1/2$, so its sites are distinct on the torus. This ball has more than $4R^3/3$ sites, giving

$$\|A\|_2^2 \ge \eta^3 \frac{4R^3}{3} \frac{s^2}{4} \ge \frac{s^5}{24 r^3}.$$

Thus $s \le \max\{r\eta, (24 r^3 \|A\|_2^2)^{1/5}\}$. Substitution of eq. 16.xxx, with $N \ge 1$ and $\eta \le 1$, proves eq. 16.xii. □

17. A finite-size criterion for confinement

Let $Z(n_1, n_2, n_3; \beta)$ be the Wilson partition function on the periodic lattice box with the indicated integer sides. Its free-energy defect on a half box is

$$\mathsf{e}(l) = 2 \log Z(l, 2l, 2l; \beta) - \log Z(2l, 2l, 2l; \beta).$$

The following implication isolates the remaining quantitative confinement problem in one explicit inequality.

Sketch P (A uniform finite-size criterion). Suppose that for some physical side ℓ_0, label $b > 0$, and $0 < x_0 \le 0.069$,

$$\mathsf{e}(l_K) \le x_0, \qquad l_K = \frac{\ell_0}{2 a_K}, \qquad \beta = L^K b,$$

at every sufficiently large cutoff, with $l_K \in \mathbb{N}$. Put $\theta = 12.9 x_0 < 1$. Then the periodic Gibbs limit and its continuum limit are unique, the vacuum is unique, and the reconstructed scalar action-density theory has a positive gap. The characteristic gap and area constants are

$$m_* = \frac{2 \log(1/\theta)}{\ell_0}, \qquad \sigma_* = \frac{2 \log(1/\theta) + 1.67}{\ell_0^2},$$

and the cutoff states obey $0 \le \nu_K(W_{R \times T}) \le e^{-\sigma_* R T}$ for coordinate rectangles with physical sides $R, T \in a_K \mathbb{N}$. The gap and continuum assertions are understood in the construction's admitted label range. ■□

Sketch of the argument. On one lattice, trace the transfer operator over a fixed section. The defect compares the trace at one height with the trace at twice that height, and hence controls the thermal eigenvalue weights above the largest eigenvalue. Change the time direction to compare the necessary transverse sections. Successive doublings give

$$\mathsf{e}(2l) \le 12.9\, \mathsf{e}(l)^2, \qquad \mathsf{e}(2^j l) \le \theta^{2^j}/12.9.$$

This is the decisive improvement: a small initial defect becomes exponentially small in the linear size.

The cube defect is a positive sum of the three one-direction defects. Summing its dyadic relation to the pressure bounds the finite-volume excess. The resulting thermal estimates give a simple transfer top and a lower bound for the excited levels. They also compare the local laws on successive tori in total variation, producing the unique periodic Gibbs state and exponential clustering.

The vortex-tube recursion bounds the small electric-flux weight. Resolving the transfer space into flux sectors then gives an asymptotically linear lower bound for electric-flux energies. Cutting a rectangular Wilson loop into layers and applying the transfer estimates gives the area bound. The constants survive the physical scaling because the assumed criterion uses one ℓ_0 and one x_0 for

every large cutoff. Uniform volume comparison then permits the cutoff and volume limits to be interchanged, and carries the gap to the reconstructed continuum theory.

This is a genuine conditional implication. Establishing the displayed box inequality at a physical scale is still the outstanding confinement step. The source's small-torus estimates can exclude it at small sides. A test at one cutoff proves a uniform criterion only when its explicit cutoff-comparison error fits inside the margin below the threshold. Finally, the cutoff area law does not construct a nonzero random sharp-loop observable in the continuum; that is distinct from the expectation renormalization in the Wilson-loop sketch.

17.1. **A finite-lattice criterion for confinement.** Let $Z(n_1,n_2,n_3)$ be the periodic partition function at coupling $\beta>0$. We prove a fixed-lattice implication from

$$2\log Z(l,2l,2l)-\log Z(2l,2l,2l)\le x_0,\qquad l\ge 3,\qquad 0<x_0\le 0.069.$$

The transfer spectrum controls the doubling defect. A vortex-tube recursion then gives area laws, electric-flux bounds and a gap over finite sections; excess-pressure estimates give uniqueness among periodic Gibbs states and convergence of periodic boxes of arbitrary shape.

All constants and parity restrictions are stated below. Physical conclusions uniform in the cutoff require the criterion at a fixed physical side for every sufficiently large cutoff. The stronger numerical tension and the simpler tension of Prop. 17.8 are kept separate.

17.1.1. *Notation and hypotheses. Boxes.* A *box* is a triple $\underline{n}=(n_1,n_2,n_3)$ of integers ≥ 3, $|\underline{n}|=n_1n_2n_3$; the cube of side n is (n,n,n). On the periodic lattice of a box a configuration U gives each bond b an element $U(b)$ of $G=\mathrm{SU}(2)$, with $U(\bar b)=U(b)^{-1}$ for the reversed bond; dU is the product of the Haar probability measures, $\partial U(p)$ the ordered product along the boundary of a plaquette p, and $a_p=1-\frac12\operatorname{tr}\partial U(p)\in[0,2]$. For $\beta\ge 0$ we write $Z(\underline{n})=\int e^{-\beta\sum_p a_p}\,\mathrm{d}U$, which lies in $[e^{-6\beta|\underline{n}|},1]$, $F=\log Z$, and $\langle\cdot\rangle_{\underline{n}}$ for the mean under $Z^{-1}e^{-\beta\sum_p a_p}dU$. The pressure is $P=\inf_{\underline{n}}F(\underline{n})/|\underline{n}|\in[-6\beta,0]$, the *excess* of a box is $\delta(\underline{n})=F(\underline{n})-|\underline{n}|P\ge 0$, and $\delta_{\mathrm c}(n)=\delta(n,n,n)$. For integers $l\ge 3$ we put

$$\mathsf{e}(l)=2F(l,2l,2l)-F(2l,2l,2l),\quad d_1(l)=2F(l,l,l)-F(2l,l,l),\quad d_2(l)=2F(l,2l,l)$$
$$-F(2l,2l,l),\quad X(l)=8F(l,l,l)-F(2l,2l,2l).$$

Twists. For directions $\mu<\nu$ with third direction ρ, a *column* of the pair $\mu\nu$ is a set of plaquettes $p_{\mu\nu}(x+je_\rho)$, $0\le j<n_\rho$. For $z=(z_{12},z_{13},z_{23})\in\{0,1\}^3$, Z^z is Z with the sign of $\operatorname{tr}\partial U(p)$ reversed on one column of each pair $\mu\nu$ with $z_{\mu\nu}=1$, and the *twisted box law* is the probability measure with this weight; $|z|=z_{12}+z_{13}+z_{23}$. We write $Z^-=Z^{(0,1,0)}$, $a_\pm=\frac12(Z\pm Z^-)$ and $z_\pm=a_\pm/Z$; z_- is the *flux weight*. On a cube, r_j is the quotient Z^z/Z for $|z|=j$ (Lem. 17.10(c)); so $r_1=Z^-/Z=1-2z_-$.

Sections. A direction μ_0 taken as time has the side $t=n_{\mu_0}$; the two others, $\nu_1<\nu_2$, span the *section* $\Sigma=(N_1,N_2)=(n_{\nu_1},n_{\nu_2})$, and $Z(t,\Sigma)=Z(t,N_1,N_2)$, likewise for F and δ; by Lem. 17.10 these numbers do not depend on the order of the sides. Over Σ, $\lambda_0>\lambda_1\ge\lambda_2\ge\cdots\ge 0$ are the eigenvalues of the transfer operator of Lem. 17.10, with $Z(t,\Sigma)=\sum_i\lambda_i^t$ for $t\ge 3$; λ_0 is the *vacuum*, $\mathsf{r}_t(\Sigma)=\sum_{i\ge1}(\lambda_i/\lambda_0)^t$ the *thermal number*, and a number $y>0$ *lies at least η above the vacuum* if $\log(\lambda_0/y)\ge\eta$.

The eigenvalues fall into four flux sectors $e\in\{0,1\}^2$ (Lem. 17.10); Λ_e is the top of the sector e, the sectors $e\ne 00$ are those with electric flux, and over a section (N_2,N), with the first direction as time, so that $\nu_1=2$ runs along the side N_2 and $\nu_2=3$ along the side N, $\Lambda_-=\max\{\Lambda_{01},\Lambda_{11}\}$ is the top of the sectors that are odd under J_2, that is, under the *flip of the side N*: the map ι_2 of Lem. 17.10, which changes the sign of the bonds along the side N that start on one line along the side N_2. These sectors carry an electric flux along the side N, a flux line of the length N. With $\lambda_-(t,N)=\lim_s Z^-(t,s,N)^{1/s}$, the number $\varepsilon_{\mathrm v}(t,N)=\log(\lambda_0(t,N)/\lambda_-(t,N))\ge 0$ is the free

energy per unit length of the vortex line that runs along the second direction of the box (t, s, N); $\varepsilon_{\mathrm{v}}(l') = \varepsilon_{\mathrm{v}}(l', l')$.

Functions. $\varpi(x) = x/(1+\sqrt{1-x^2})$ on $[0,1]$; $\omega(D) = 2\varpi(e^D-1)^2$ on $[0, \log 2]$; $Y_1(x) = 2x + \omega(x)$, $Y_2 = 2Y_1 + \omega(Y_1)$, $\Psi = \omega(Y_2)$ and $\Phi = 4\omega + 2\omega(Y_1) + \omega(Y_2)$, where the arguments lie in $[0, \log 2]$.

Cells. For an integer $h \geq 2$, $N = 2h$, and a point $c \in (\mathbb{Z} + \frac{1}{2})^3$, the *cell* $\mathsf{C}_h(c)$ is the cube of the h layers below c, formed by the sites x with $c_\mu - h < x_\mu < c_\mu$, $\mu = 1, 2, 3$; a *function of the cell* is a bounded Borel function of the bonds with both ends in it. On the cube (N, N, N) the reflections $\mathfrak{r}_\mu \colon x_\mu \mapsto 2c_\mu - x_\mu$ map the lattice onto itself, each fixing the two planes $x_\mu = c_\mu$ and $x_\mu = c_\mu + h$; for a function A of the cell and $\epsilon \in \{0,1\}^3$ the *copy* $A^{[\epsilon]}$ is A composed with the $\mathfrak{r}_\mu$ with $\epsilon_\mu = 1$ and complex conjugated if $|\epsilon|$ is odd, and

$$\|\!| A |\!\|_N = \Big\langle \prod_{\epsilon \in \{0,1\}^3} A^{[\epsilon]} \Big\rangle^{1/8}_{(N,N,N)},$$

the mean being a number ≥ 0 by Lem. 17.21.

The infinite lattice. $\Omega = G^{\mathbb{B}}$, $\mathbb{B}$ the bonds of $\mathbb{Z}^3$; local functions are bounded Borel functions of finitely many bonds. A *cube of h layers* is $Q(x_0, h) = \{x_0 + (j_1, j_2, j_3) : 1 \leq j_\mu \leq h\}$; a function of the bonds with both ends in Q is a function on the configurations of every box with sides $\geq h+1$ through the projection of $\mathbb{Z}^3$, which is injective on Q, and box laws are laws on Ω by periodic extension. A *Gibbs state* is a probability measure on Ω under which, given the bonds outside a finite set $\mathfrak{A}$, those of $\mathfrak{A}$ have the density proportional to $e^{-\beta \sum a_p}$, the sum over the plaquettes that meet $\mathfrak{A}$; it is *periodic* if it is invariant under a subgroup of finite index of the translations.

The *symmetries of the lattice* are the maps $x \mapsto Ox + v$, O a signed permutation and $v \in \mathbb{Z}^3$. A *lattice mirror* is a plane $\{x_\mu = c\}$, $c \in \frac{1}{2}\mathbb{Z}$ (a *site mirror* if $c \in \mathbb{Z}$, a *bond mirror* otherwise), or a plane $\{x_\mu \pm x_\nu = c\}$, $\mu \neq \nu$, $c \in \mathbb{Z}$ (a *diagonal mirror*); a state s is *reflection positive* at it if $s(\overline{\mathfrak{r}A}\, A) \geq 0$ for every local A of the bonds in one closed side of the mirror, $\mathfrak{r}$ the reflection in it (bonds in a site mirror or a diagonal mirror are admitted; at a bond mirror the bonds that cross it lie in neither closed side). For a closed lattice path C, $W_C = \frac{1}{2}\operatorname{tr} U(C)$, $U(C)$ the ordered product along C; W_C is real, $|W_C| \leq 1$, and it depends neither on the starting point nor on the sense; $|C|$ is the number of bonds of C, and W_p is the loop along the boundary of one plaquette.

A *rectangle* $R \times T$ of the (μ, ν)-plane runs R steps along e_μ, T along e_ν, and back.

Hypotheses. The *criterion at the side l* is the condition: $\beta > 0$, $l \geq 3$ is an integer of either parity, $0 < x_0 \leq 0.069$, and $\mathsf{e}(l) \leq x_0$. Its numbers are $\theta = 12.9\, x_0 \leq 0.8901$, $\Lambda = \log \frac{1}{\theta} \geq 0.11642$, $\varsigma = 2\Lambda + 1.67$, $\mathsf{q} = \varsigma/2\Lambda = 1 + 0.835/\Lambda \leq 8.173$, and

$$\sigma^+ = \frac{\varsigma}{4l^2} = \frac{\log \frac{1}{\theta} + 0.835}{2l^2}, \qquad \sigma^+ l^2 = \frac{1}{2} \log \frac{e^{0.835}}{12.9 x_0} \geq 0.4757. \tag{17.i}$$

The *cube condition at the side l_0* is $X(l_0) \leq \frac{1}{20}$. For the physical scale let $L \geq 2$ be an integer and $a_K = L^{-K}$. The *uniform criterion* holds if there are $\mathsf{b} > 0$, $l_* > 0$, an integer K_0 and one $x_0 \in (0, 0.069]$ such that for every integer $K \geq K_0$ the number $l_* L^K$ is an integer ≥ 3 and the criterion holds at the side $l_* L^K$ and the coupling $L^K \mathsf{b}$ with this x_0. In the frame L is the blocking factor, a_K the spacing at the cutoff K, $\mathsf{b} = a_K \beta$ the label, and $\ell_0 = 2l_*$ the physical side of the cube of the criterion. In Lem. 17.24, in the steps of proofs that use it, and in Prop. 17.6, L is the blocking factor, an odd integer ≥ 13; the L of the uniform criterion is any integer ≥ 2, independent of it. We write x/yz for $x/(yz)$.

17.1.2. *Why doubling turns the box test into a gap.* The box defect cancels the extensive part of the free energy. At a fixed transverse section Σ, write the transfer eigenvalues as $\lambda_0 \geq \lambda_1 \geq \cdots \geq 0$ and put

$$\mathsf{r}_l(\Sigma) = \sum_{i \geq 1} (\lambda_i/\lambda_0)^l.$$

This includes any multiplicity of the largest eigenvalue. Factoring λ_0^l out of the transfer trace isolates the thermal correction $1 + \mathsf{r}_l$. The difference between twice the log trace at height l and the log trace at height $2l$ is therefore a function of these corrections, with the largest-eigenvalue contribution cancelled. The source proves

$$\log(1 + \mathsf{r}_l(\Sigma)) \le d(l; \Sigma) \le 2\log(1 + \mathsf{r}_l(\Sigma)).$$

Small defect thus controls all excited thermal weights together, including a possible top multiplicity. This is stronger than a comparison with one selected eigenvector.

Doubling time squares the eigenvalue ratios. To make a statement about larger cubes, however, one must also enlarge the transverse section. The proof alternates these two operations and changes which coordinate is time. It tracks the three sections (l, l), $(2l, l)$ and $(2l, 2l)$, bounds the intermediate defects, and closes the resulting scalar recursion. The numerical estimate that matters is

$$\mathsf{e}(2l) \le 12.9\mathsf{e}(l)^2, \qquad 0 < \mathsf{e}(l) \le 0.069.$$

With $\theta = 12.9x_0 < 1$, induction gives $\mathsf{e}(2^j l) \le \theta^{2^j}/12.9$. The exponent grows like the linear size, although the argument proceeds by doubling. This is the source of the exponential estimates that follow from a successful finite-size test.

The cube excess is controlled by a dyadic identity. If $\delta_{\mathrm{c}}(n)$ is the cube's finite-volume excess and $X(n)$ is the sum of its three directional defects, the source has

$$X(n) = 8\delta_{\mathrm{c}}(n) - \delta_{\mathrm{c}}(2n), \qquad \delta_{\mathrm{c}}(n) = \sum_{j \ge 0} 8^{-j-1} X(2^j n).$$

The pressure term cancels in the first relation. The recursion makes the second sum converge with an exponential bound. The transfer comparison then yields a simple top, a quantitative gap and rapid comparison of local laws on successive tori. In parallel, the vortex-tube estimates control the small electric-flux weight. Resolving the transfer trace into its flux sectors turns that bound into an energy cost for a long flux tube and then a rectangular Wilson area law. The gap and area law are two uses of the same small finite-volume excess, through different transfer observables.

For the continuum implication the tested box must have one physical side. Its lattice side is $l_K = \ell_0/(2a_K)$ and its coupling is $\beta_K = L^K b$. The criterion requires the same bound with the same x_0 for every sufficiently large K. This is exactly the uniformity that carries the finite-lattice gap and area constants to physical units. It also makes the volume-comparison error uniform while the cutoff is removed, allowing the volume and cutoff limits in this implication to commute.

Testing one cutoff is a different assertion. It can certify the uniform criterion only if the proved cutoff-comparison error fits within the available margin below the threshold. The detailed statement retains that margin. Likewise, small-torus asymptotics can rule out the criterion at small physical sides; a statement that the test is finite does not verify it at some larger side. Establishing it on the desired continuum trajectory remains the uncompleted confinement step.

17.1.3. *Principal results.*

Theorem AW (One lattice)**.** Assume the criterion at the side l.

(a) *The state.* For $n = 2^k l$, $k \ge 1$, and every bounded Borel function A of the bonds of a cube of h layers, $1 \le h \le n/2 - 1$,

$$|\langle A\rangle_{(n,n,n)} - \langle A\rangle_{(2n,2n,2n)}| \le 1.29 \sup|A|\, \theta^{n/2l}.$$

So $\nu(A) = \lim_k \langle A\rangle_{(2^k l, 2^k l, 2^k l)}$ exists for every local A; and for $n = 2^k l$, $k \ge 1$, and A a bounded Borel function of the bonds of a cube of h layers with $1 \le h \le n/2 - 1$, the n and A of the first bound, $|\nu(A) - \langle A\rangle_{(n,n,n)}| \le 1.29 \sup|A|\, \theta^{n/2l}/(1 - \theta^{n/2l})$. The measure ν is a Gibbs state, invariant under the symmetries of the lattice, reflection positive at every lattice mirror and ergodic under the translations; it is the only periodic Gibbs state, and the laws of the periodic boxes converge weakly to ν as their three sides tend to infinity.

(b) *Rectangles.* For integers $R, T \geq 1$ and every rectangle C of sides $R \times T$ in a coordinate plane,

$$0 \leq \nu(W_C) \leq e^{-\sigma^+ RT}; \tag{17.ii}$$

on the cube of every integer side $n \geq 2\mathsf{q}l$, for $R \leq n/2$ and $T \leq n-1$,

$$0 \leq \langle W_C \rangle_{(n,n,n)} \leq 8^{T/n} \exp\Big[-\sigma^+ RT + \frac{2\mathsf{q}RT}{ln}\Big(\frac{1}{2}\log\frac{2\mathsf{q}n}{l} + 0.86\Big)\Big];$$

and, if $R, T \geq 2$, also $\nu(W_C) \leq (\frac{8\beta}{8\beta+3})^{2(R+T)-4}$, a bound that holds in every Gibbs state.

(c) *Closed paths.* Let C be a closed lattice path that runs a bond of the direction ρ, and $\rho' \neq \rho$; let $\mathsf{A}_{\rho'\rho}(C)$ be the number of unit squares of the (ρ', ρ)-plane about which the projection of C to that plane winds an odd number of times, and $c_\rho(C)$ the number of changes from "down" to "up" in the cyclic sequence of the senses in which C runs its bonds of the direction ρ. Then $|\nu(W_C)| \leq 2^{c_\rho(C)-1} e^{-\sigma^+ \mathsf{A}_{\rho'\rho}(C)}$. More generally, for a gap τ between neighbouring layers of the direction ρ let $\ell_\tau^{\rho'}(C)$ be the number of integers α such that an odd number of the bonds of the direction ρ that C runs in that gap have the ρ'-coordinate $\leq \alpha$, so that $\sum_\tau \ell_\tau^{\rho'}(C) = \mathsf{A}_{\rho'\rho}(C)$; let $\mathsf{A}_\rho(C) = \sum_\tau \max_{\rho' \neq \rho} \ell_\tau^{\rho'}(C)$, the larger of the two counts taken gap by gap, so that $\mathsf{A}_\rho(C) \geq \max_{\rho' \neq \rho} \mathsf{A}_{\rho'\rho}(C)$; and let D be a continuous unitary representation of G on $\mathbb{C}^d$ with $D(-\mathbf{1}) = -\mathbf{1}$, and $W_C^D = d^{-1} \operatorname{tr} D(U(C))$.

Then $|\nu(W_C^D)| \leq d^{c_\rho(C)-1} e^{-\sigma^+ \mathsf{A}_\rho(C)}$; at $D(g) = g$, $d = 2$, this is the first bound with $\mathsf{A}_\rho(C)$ in the place of $\mathsf{A}_{\rho'\rho}(C)$. In particular a closed path that in each such gap runs either no bond of the direction ρ or exactly two, at the sites o_τ, r_τ of the section, has $c_\rho(C) = 1$, as its height in the direction ρ has one maximum, and $|\nu(W_C)| \leq \exp[-\sigma^+ \sum_\tau |r_\tau - o_\tau|_\infty]$. So $|\nu(W_C)| \leq e^{-\sigma^+ \sum_j |r_j - o_j|}$ for every *staircase loop*, a closed path in a coordinate plane that, in each gap between neighbouring layers of one direction ρ of the plane, runs either no bond of the direction ρ or exactly two, in opposite senses, at the ρ'-coordinates o_j and r_j; for a simple loop $\sum_j |r_j - o_j|$ is the enclosed area.

On every periodic box, of any parity, the staircase loops of a $(1,3)$-plane described in Lem. 17.12(f), with $\rho = 3$ and their bonds of the direction 3 in at most $n_3 - 2$ gaps, have the bound stated there. For a simple closed curve in a coordinate plane $4c_\rho(C) \leq |C|$, so that $|\nu(W_C)| \leq \frac{1}{2}\exp[-\sigma^+ \mathsf{A}_{\rho'\rho}(C) + 0.174\,|C|]$.

(d) *Twisted cubes.* For every integer $n \geq 2\mathsf{q}l$ and the seven $z \neq 0$, on (n, n, n):

$$z_- \leq \exp\Big[-\sigma^+ n^2 + \frac{2\mathsf{q}n}{l}\Big(\frac{1}{2}\log\frac{2\mathsf{q}n}{l} + 0.86\Big)\Big], \qquad 0 \leq 1 - Z^z/Z \leq 6z_-.$$

(e) *Flux sectors and static charges.* For the same n, over the section $n \times n$ the top of every sector with electric flux lies at least $\sigma^+ n - \frac{2\mathsf{q}}{l}(\frac{1}{2}\log\frac{2\mathsf{q}n}{l} + 0.86) - \frac{g_n}{n}$ above the vacuum, with $g_n = 0.0017$ if $n = 2^i l$ and $g_n = 0.105$ for every other n; and for $1 \leq R \leq n/2$ the transfer operator $\hat{\mathsf{T}}_{o,r}$ with two static charges at the distance R (Lem. 17.12) has

$$\|\hat{\mathsf{T}}_{o,r}\| \leq \lambda_0(n,n)\,\big[8(1 + \mathsf{r}_n(n,n))\big]^{1/n} \exp\Big[-\sigma^+ R + \frac{2\mathsf{q}R}{ln}\Big(\frac{1}{2}\log\frac{2\mathsf{q}n}{l} + 0.86\Big)\Big].$$

The same bound holds for any two sites $o \neq r$ of the section, with $R = |r - o|_\infty$, the larger of the distances of their two coordinates on the circle of n sites (Lem. 17.12(e)). So for any two sites $o \neq r$ of $\mathbb{Z}^2$, projected to the sections $n \times n$: $\limsup_{i \to \infty} \|\hat{\mathsf{T}}_{o,r}\|/\lambda_0(n,n) \leq e^{-\sigma^+ |r-o|_\infty}$ along $n = 2^i l$.

(f) *Gap.* Over every section $N_1 \times N_2$ with integers $N_1, N_2 \geq 2l$ the vacuum is simple, its eigenvector lies in the sector 00, and every λ_i, $i \geq 1$, the tops of the sectors with electric flux among them, lies at least $\frac{\Lambda}{l} + \frac{1.021}{n_0}$ above it, n_0 the largest $2^k l \leq \min\{N_1, N_2\}$. Over $N_2 \times n$, $n = 2^k l$, $k \geq 1$, and every integer $N_2 \geq n$ the same holds with $\frac{\Lambda}{l} + \frac{3.5065}{n}$, and over

$n \times n$ with $\frac{\Lambda}{l} + \frac{6.42}{n}$. For integers $N_1, N_2 \geq n = 2^k l$, $k \geq 1$, with $0.01655\, N_1 N_2\, \theta^{n/l} \leq n^2 \log 2$: $\mathsf{r}_n(N_1, N_2) \leq \frac{N_1 N_2}{41.8\, n^2}\, \theta^{n/l}$.

(g) *Flux lines.* Over a section (N_2, N), with the first direction as time, $\Lambda_- = \max\{\Lambda_{01}, \Lambda_{11}\}$ is the top of the sectors that are odd under the flip of the side N, the map ι_2 of Lem. 17.10 that changes the sign of the bonds along the side N starting on one line along the side N_2; these sectors carry a flux line of the length N.

(g1) Let $N = 2wl$, $w \geq 2$ a power of two, $\chi = 0.03\, \theta^{N/l}$, $\hat{N} = \lceil 2\sigma^+ N^2 / \log \frac{1}{\chi} \rceil$ and $\mathsf{g} = \log \hat{N} + 1.71$. Then $4 \leq \hat{N} \leq \mathsf{q}N/l + 1$. For every integer $N_2 \geq \hat{N}N/2$, Λ_- lies at least $\sigma^+ N - (2 \log \hat{N} + 1.86)/N$ above the vacuum, and the lower limit of $\log(\lambda_0/\Lambda_-)$ as $N_2 \to \infty$ is at least $\sigma^+ N - \mathsf{g}/N$. For every integer N_2 with $\max\{N, 2\mathsf{q}l\} \leq N_2 < \hat{N}N/2$, Λ_- lies at least $\sigma^+ N - \frac{2\mathsf{q}}{l}(\log \frac{2\mathsf{q}N}{l} + 0.86)$ above the vacuum. For integers N' with $N \leq N' < 2N$ and $N_2 \geq \hat{N}N/2$, over (N_2, N') the top of the sectors odd under the flip of the side N' lies at least $\sigma^+ N' - (4 \log \hat{N} + 3.72)/N - (2N_2/N^2)\, 0.01655\, \theta^{N/l}$ above the vacuum; the last term is at most $0.0331/N$ for $N_2 \leq N\theta^{-N/l}$.

(g2) Let $N \geq 4l$ be any integer, $n_0 = 2wl$, $w \geq 2$ a power of two, the dyadic multiple of l with $n_0 \leq N < 2n_0$, $\kappa = N/n_0$, $\tau = \theta^{n_0/l}$, $\chi_0 = 0.3602\, \tau$, $\mathsf{E}_0 = \sigma^+ n_0 N$, $\hat{N}'$ the least even integer $\geq 2\mathsf{E}_0 / \log \frac{1}{\chi_0}$, $\mathsf{h} = 0.01655\, \hat{N}' \tau$ and $\mathsf{g}' = \kappa \log \hat{N}' + 3.22 + (3\kappa + 0.68)\mathsf{h}$. Then $6 \leq \hat{N}' < \mathsf{q}N/l + 2$; for every integer $N_2 \geq \hat{N}' n_0/2$, Λ_- lies at least $\sigma^+ N - \kappa \mathsf{g}'/N \geq \sigma^+ N - (4 \log \hat{N}' + 18.2)/N$ above the vacuum; and for every integer N_2 with $\max\{N, 2\mathsf{q}l\} \leq N_2 < \hat{N}' n_0/2$ it lies at least $\sigma^+ N - \frac{2\mathsf{q}}{l}(\log \frac{2\mathsf{q}N}{l} + 0.86)$ above it. Consequently, with

$$C(\theta) = \max\Big\{\frac{\mathsf{q}}{2} + \frac{1}{4},\ 4 \log\Big(\mathsf{q} + \frac{1}{2}\Big) + 18.2\Big\} \leq 27,$$

for all integers $N \geq 4l$ and $N_2 \geq C(\theta) N^2/l$ the top Λ_- lies at least $\sigma^+ N - C(\theta)(1 + \log \frac{N}{l})/N$ above the vacuum. If moreover $w \geq 4$ and $\tau \leq 0.087$, then with $\hat{N}'' = \lceil 4\sigma^+ n_0^2 / \log \frac{1}{0.2728\, \tau} \rceil$ it lies at least $\sigma^+ N - (4 \log \hat{N}'' + 10.8)/n_0$ above the vacuum for every integer $N_2 \geq \hat{N}'' n_0/2$.

(g3) For every integer $N \geq 4l$, every integer $N_2 \geq \hat{N}' n_0/2$ and every rectangle C of sides $R \times T$, $T \leq N - 1$, in a plane spanned by the first and the third direction,

$$\limsup_{L_1 \to \infty} \langle W_C \rangle_{(L_1, N_2, N)} \leq 4^{T/N} \exp\Big[-\sigma^+ RT + \frac{RT}{N^2}(4 \log \hat{N}' + 18.2)\Big].$$

(h) *Boxes of every shape, cylinders and twisted boxes.* For every even $N \geq 2l$, every bounded real Borel function A of the bonds of a cube of $N/2$ layers and every periodic box with all sides $\geq N$,

$$|\langle A \rangle_{\underline{n}} - \nu(A)| \leq 5.42\, \theta^{N/4l} \sup |A|;$$

for the events of these bonds the bound is $2.71\, \theta^{N/4l}$. In particular $|\langle A \rangle_{(m,m,m)} - \nu(A)| \leq 5.42\, \theta^{\lfloor m/2 \rfloor / 2l} \sup |A|$ for every integer $m \geq 2l$ and every such A of a cube of $\lfloor m/2 \rfloor$ layers. Over every section Σ the laws of the boxes (t, Σ) converge as $t \to \infty$, on the bounded Borel functions of finitely many layers, to the law of the infinite cylinder over Σ; for every even $N' \geq 2l$, every $N \geq N'$ and every A as above of a cube of $N'/2$ layers, the cylinder over $N \times N$ satisfies the bound with N', so that these laws converge to ν as $N \to \infty$; and on the cylinder over $n \times n$, $n \geq 2l$, every rectangle with the side $R \leq n/2$ across the axis and the side T along it has a mean of modulus at most $[(1 + \mathsf{r}_n(n,n))\, z_-(n,n,n)]^{RT/n^2}$. The laws of twisted boxes whose three sides tend to infinity and whose columns recede, in the sense that every finite set of plaquettes meets the columns of only finitely many members, converge weakly to ν, and so do the mixtures, with any weights, of finitely many such sequences; if a

column passes through a fixed plaquette p_0 in every member, the law of $\operatorname{tr}\partial U(p_0)$ tends to the law of $-\operatorname{tr}\partial U(p_0)$ under ν, which is another law.

(i) *The pressure.* There is an open interval U containing β on which the criterion holds at the side $2l$ with $x_0 = 0.069$; on U the pressure is continuously differentiable, with $P'(\beta') = -3\,\nu_{\beta'}(a_p)$, $\nu_{\beta'}$ the state of (a) at β'; and $\nu(a_p) \le 1 - \frac{\beta}{4(\beta+1)} < 1$.

All constants are absolute; σ^+ and q depend only on l and x_0. Decimal upper and lower bounds used in the finite recursions are understood in their indicated directions. The interval bounds recorded in the proof of Prop. 17.17 verify the numerical part of the stronger tension estimate. ■□

Theorem AX (The criterion and the cube condition)**.** Let $\beta \ge 0$ and let $l \ge 3$ be an integer of either parity.

(a) $X(l) = \mathsf{e}(l) + 2d_2(l) + 4d_1(l)$, and $\log(1 + \mathsf{r}_l(\Sigma_j)) \le d_j(l) \le 2\log(1 + \mathsf{r}_l(\Sigma_j))$ for $\Sigma_1 = (l, l)$, $\Sigma_2 = (2l, l)$, $\Sigma_3 = (2l, 2l)$ and $(d_1, d_2, d_3) = (d_1, d_2, \mathsf{e})$; so the three terms are ≥ 0, and the cube condition at l_0 is the inequality $\mathsf{e}(l_0) \le \frac{1}{20} - 2d_2(l_0) - 4d_1(l_0)$. If $\beta > 0$, the cube condition at l_0 gives the criterion at the same side l_0, with every $x_0 \in [X(l_0), \frac{1}{20}]$, $x_0 > 0$, and $\theta \le 0.645$.

(b) Under the criterion at l: $\mathsf{e}(2^k l) \le \theta^{2^k}/12.9$ for all $k \ge 0$, and $X(2^k l) \le 0.10643\,\theta^{2^k} \le 0.04194$ for all $k \ge 3$; so the cube condition holds at every side $2^k l$ with $k \ge 3$, with $k \ge 2$ if $x_0 \le 0.0650$, and with $k \ge 1$ if $x_0 \le 0.0527$.

(c) Under the criterion at l, along $n = 2^k l$ the rates $\mathsf{m}_\mathsf{e}(n) = \frac{1}{n}\log\frac{1}{12.9\,\mathsf{e}(n)}$ $(k \ge 0)$ and $\mathsf{m}_X(n) = \frac{1}{n}\log\frac{1}{18X(n)}$ $(k \ge 3)$, with values in $(0, \infty]$, do not decrease; for $k \ge 3$, $\mathsf{m}_\mathsf{e}(n/2) - \frac{0.6501}{n} \le \mathsf{m}_X(n) \le \mathsf{m}_\mathsf{e}(n) - \frac{0.3331}{n}$; so the two rates have one common limit, which is $\ge \Lambda/l$.

(d) If $0 < \mathsf{e}(l) < 0.0740538$ at one l, then $\mathsf{e}(2^k l) \to 0$ and $X(2^k l) \to 0$ as $k \to \infty$; so the criterion holds at some side $2^k l$ with any prescribed $x_0 \in (0, 0.069]$, and the cube condition holds at $2^k l$ for all large k.

■□

Theorem AY (Physical scale, uniformly in the cutoff)**.** Assume the uniform criterion, and let ν_K be the state of Thm. AW at $l = l_* L^K$ and $\beta = L^K \mathsf{b}$, read on the lattice $a_K\mathbb{Z}^3$. Put

$$\sigma_* = \frac{\frac{1}{2}\log\frac{1}{\theta} + 0.4175}{l_*^2} \;\ge\; \frac{0.4757}{l_*^2} = \frac{1.9028}{\ell_0^2}.$$

For every $K \ge K_0$, with physical lengths in $a_K\mathbb{N}$ and energies $a_K^{-1}\log(\lambda_0/\cdot)$:

(a) $0 \le \nu_K(W_C) \le e^{-\sigma_* RT}$ for every rectangle C of a coordinate plane with physical sides R, T; and if $R, T \ge 2a_K$, also $\nu_K(W_C) \le \exp[-\frac{1}{2}\sigma_* RT - \frac{3(|C|_{\mathrm{ph}} - 4a_K)}{2(8\mathsf{b}+3a_K)}]$, $|C|_{\mathrm{ph}} = 2(R+T)$ the physical length of C;

(b) on the periodic cube of physical side $\ell \ge 2\mathsf{q}l_*$, $0 \le 1 - Z^z/Z \le 6\exp[-\sigma_*\ell^2 + \frac{2\mathsf{q}\ell}{l_*}(\frac{1}{2}\log\frac{2\mathsf{q}\ell}{l_*} + 0.86)]$ for the seven $z \ne 0$, and every sector with electric flux over $\ell \times \ell$ lies at least $\sigma_*\ell - \frac{2\mathsf{q}}{l_*}(\frac{1}{2}\log\frac{2\mathsf{q}\ell}{l_*} + 0.86) - \frac{0.105}{\ell}$ above the vacuum, with $\frac{0.0017}{\ell}$ in the place of $\frac{0.105}{\ell}$ if ℓ/l_* is a power of two;

(c) over every section whose physical sides are $\ge 2l_*$ the gap is at least $\frac{1}{l_*}\log\frac{1}{\theta}$;

(d) a flux line of physical length $\ell = 2wl_*$, $w \ge 2$ a power of two, over a section $\ell_2 \times \ell$ with $\ell_2 \ge \hat{N}\ell/2$ lies at least $\sigma_*\ell - (2\log(\mathsf{q}\ell/l_* + 1) + 1.86)/\ell$ above the vacuum, and one of every physical length $\ell \ge 4l_*$ over $\ell_2 \times \ell$ with $\ell_2 \ge C(\theta)\ell^2/l_*$ lies at least $\sigma_*\ell - C(\theta)(1 + \log\frac{\ell}{l_*})/\ell$ above it, with the number $C(\theta) \le 27$ of Thm. AW(g2);

(e) the cube condition holds at the side $8l_*L^K$, with $18X \le 1.9157\,\theta^8$; conversely, the cube condition at the sides l_*L^K, $\beta = L^K\mathsf{b}$, for all $K \ge K_0$ gives the uniform criterion with $x_0 = \frac{1}{20}$.

The numbers σ_*, q and $C(\theta)$ depend on l_* and x_0 alone, and $\hat{N}$ on ℓ/l_* and x_0 alone; none of them depends on K, b or L. In the first bound of (a) the factor in front of $e^{-\sigma_* RT}$ is 1 and there is no perimeter term. ■□

Corollary 17.1 (The static potential)**.** Assume the criterion at the side l, and let $m(R,T) = \nu(W_C)$ for a rectangle C of sides $R \times T$ in a coordinate plane, $m(R,0) = 1$.

(a) $m(R,T)$ depends on R,T alone, $m(R,T) = m(T,R)$, and $w^{RT} \le m(R,T) \le e^{-\sigma^+ RT}$, where $w = m(1,1) \ge \beta/(4(\beta+1))$.

(b) For every $n \ge 0$ and real $c_0,\dots,c_n$: $\sum_{h,h'} c_h c_{h'}\, m(R,h+h') \ge 0$ and $\sum_{h,h'} c_h c_{h'}\, m(R,h+h'+1) \ge 0$. Hence $m(R,T)^2 \le m(R,T-1)\, m(R,T+1)$ for $T \ge 1$, and there is exactly one Borel probability measure ϱ_R on $[0,1]$ with $m(R,T) = \int \lambda^T \,\mathrm{d}\varrho_R(\lambda)$ for all $T \ge 0$.

(c) $d(R;T) = \log(m(R,T)/m(R,T+1))$ does not increase in $T \ge 0$; the static potential

$$V(R) = \lim_{T\to\infty} d(R;T) = \inf_T d(R;T) = \lim_{T\to\infty} -\frac{1}{T}\log m(R,T)$$

exists in $[0,\infty)$, $\max \operatorname{supp} \varrho_R = e^{-V(R)}$, and $m(R,T_2) \le e^{-(T_2-T_1)V(R)} m(R,T_1)$ for $T_2 \ge T_1 \ge 0$, $V(R)$ being the largest number with this property. For $0 \le T_1 < T_2 \le T_3 < T_4$: $[m(R,T_2)/m(R,T_1)]^{1/(T_2-T_1)} \le [m(R,T_4)/m(R,T_3)]^{1/(T_4-T_3)}$. For $T \ge 1$, $m(R,T) = \int_{[V(R),\infty)} e^{-TE} d\mu_R(E)$, μ_R the image of the restriction of ϱ_R to $(0,1]$ under $\lambda \mapsto -\log\lambda$, a measure of mass in $(0,1]$.

(d) $\sigma^+ R \le V(R) \le R \log \frac{4(\beta+1)}{\beta}$.

(e) Under the uniform criterion, for every $K \ge K_0$ and physical R: the potential $V_K(R) = a_K^{-1} V(R/a_K)$ of ν_K in physical units is $\ge \sigma_* R$; and for every real κ the means $M_K(R,T) = e^{\kappa |C|_{\mathrm{ph}}} \nu_K(W_C)$ satisfy $M_K(R,T_2) \le e^{-(T_2-T_1)(V_K(R)-2\kappa)} M_K(R,T_1)$ for physical $T_2 \ge T_1 \ge 0$, while in the inequality of four rectangles of (c) the factors $e^{\kappa|C|_{\mathrm{ph}}}$ cancel. Along a sequence of cutoffs: let $K_j \to \infty$, $K_j \ge K_0$, let κ_j be arbitrary real numbers, and for R,T in $\bigcup_i a_i\mathbb{N}$ and all large j put $M_j(R,T) = e^{2\kappa_j(R+T)} \nu_{K_j}(W_C)$, C a rectangle of the physical sides $R \times T$, and $v_j(R) = V_{K_j}(R) - 2\kappa_j$, so that $v_j(R) \ge \sigma_* R - 2\kappa_j$. If $T_1 < T_2$ and $M_j(R,T_i) \to \mathsf{W}(R,T_i) \in (0,\infty)$ for $i = 1,2$, then

$$\limsup_j v_j(R) \le \frac{1}{T_2 - T_1} \log \frac{\mathsf{W}(R,T_1)}{\mathsf{W}(R,T_2)},$$

and if $v_j(R) \ge B$ for infinitely many j, then $\mathsf{W}(R,T_2) \le e^{-(T_2-T_1)B}\, \mathsf{W}(R,T_1)$; in particular $\mathsf{W}(R,T_2) \le e^{-(T_2-T_1)(\sigma_* R - 2\kappa)}\, \mathsf{W}(R,T_1)$ if $\kappa_j \le \kappa$ for infinitely many j. If $M_j(R,T) \to \mathsf{W}(R,T) \in (0,\infty)$ for all R,T, then $\mathsf{W}(R,T) = \mathsf{W}(T,R)$, $\mathsf{W}(R,T_2)^{T_3-T_1} \le \mathsf{W}(R,T_1)^{T_3-T_2}\, \mathsf{W}(R,T_3)^{T_2-T_1}$ for $T_1 < T_2 < T_3$, the inequality of four rectangles of (c) holds for W, and $V_{\mathrm{ren}}(R) = \lim_{T\to\infty} -\frac{1}{T}\log \mathsf{W}(R,T)$ exists in $[-\infty,\infty)$ and is the infimum of the quotients $\frac{1}{T_2-T_1}\log\frac{\mathsf{W}(R,T_1)}{\mathsf{W}(R,T_2)}$, $T_1 < T_2$. The convergence of the M_j is a hypothesis of these statements.

(f) With no criterion, in every periodic Gibbs state s at $\beta \ge 0$, $s(W_C) \ge 0$ for every rectangle C of a coordinate plane with one even side and for every square.

(g) With no criterion: let $\beta > 0$ and let s be a periodic Gibbs state that is positive at the bond mirrors on the functions of two crossing bonds, in the sense that

$$s\Big(\sum_{i,j,k,l} \overline{(\mathsf{z}_p)_{ji}}\, (\mathsf{z}_q)_{kl}\, F_{jk}\, \overline{\mathfrak{r} F_{il}}\Big) \ge 0$$

for every bond mirror, $\mathfrak{r}$ the reflection in it, z_p, z_q the values of U at the bonds from two sites x_p, x_q next to the mirror, equal or not, to their mirror images, and all continuous functions F_{jk}, $j,k \in \{1,2\}$, of finitely many bonds with both ends in the closed side of the mirror that

holds x_p and x_q, as in Lem. 17.21(c). Let $m_s(R,T) = s(W_C)$ for a rectangle C of sides $R \times T$ in a coordinate plane, $m_s(R,0) = 1$. Then $m_s(R,T)$ depends on R, T alone, $m_s(R,T) = m_s(T,R)$, and $[\beta/4(\beta+1)]^{RT} \le m_s(R,T) \le 1$ for all integers $R, T \ge 1$; both Hankel forms of (b) hold for m_s, and with them the log-convexity in T and exactly one Borel probability measure ϱ^s_R on $[0,1]$ with $m_s(R,T) = \int \lambda^T \, \mathrm{d}\varrho^s_R(\lambda)$; $d_s(R;T) = \log(m_s(R,T)/m_s(R,T+1))$ does not increase in T, $V_s(R) = \lim_T d_s(R;T) = \inf_T d_s(R;T) = \lim_T -\frac{1}{T}\log m_s(R,T)$ exists in $[0, R\log\frac{4(\beta+1)}{\beta}]$, $\max \operatorname{supp} \varrho^s_R = e^{-V_s(R)}$, $m_s(R,T_2) \le e^{-(T_2-T_1)V_s(R)} m_s(R,T_1)$ for $T_2 \ge T_1 \ge 0$, and the inequality of four rectangles of (c) holds for m_s.

On site mirrors alone, with no hypothesis at the bond mirrors: if $m_s(R,2) > 0$, as is the case for every even R, then $V_{\mathrm{e}}(R) = \lim_h -\frac{1}{2h}\log m_s(R,2h)$ exists in $[0,\infty)$, and $m_s(R,T_2) \le e^{-(T_2-T_1)V_{\mathrm{e}}(R)} m_s(R,T_1)$ for every even $T_1 \ge 0$ and every integer $T_2 \ge T_1$. The state ν of Thm. AW is positive at the bond mirrors in this sense, by Lem. 17.21(c) on the cubes of sides $2^k l$ and Thm. AW(a), as in Step 2 of the proof; so (g) contains (a)–(c), the upper bound $e^{-\sigma^+ RT}$ excepted. ■□

Proposition 17.2 (Where the criterion holds)**.**

(a) For every integer $l \ge 3$, every $x_0 \in (0, 0.069]$ and every $\beta \in (0, x_0/48l^3]$ the criterion holds at l.

(b) Let $0 < \beta < \frac{1}{6}$ and $l \ge 4$ even. Then $X(l) \le 64\, l^2 (6\beta)^{l/2}$; so the cube condition, and the criterion with every $x_0 \in [X(l), \frac{1}{20}]$, $x_0 > 0$, hold at every even l with $64\, l^2(6\beta)^{l/2} \le \frac{1}{20}$, for instance at all even $l \ge 10$ if $\beta = 0.01$, $l \ge 24$ if $\beta = 0.05$, $l \ge 62$ if $\beta = 0.1$, $l \ge 360$ if $\beta = 0.15$.

(c) For $0 < \beta < \frac{1}{6}$ there is exactly one Gibbs state, periodic or not.

■

Proposition 17.3 (What the criterion says)**.** Let $\beta \ge 0$ and let $l \ge 3$ be an integer.

(a) With $\mathsf{r} = \mathsf{r}_l(2l, 2l)$: $\log(1+\mathsf{r}) \le \mathsf{e}(l) \le 2\log(1+\mathsf{r})$. So $\mathsf{r} \le 0.0351$ implies $\mathsf{e}(l) \le 0.069$, and $\mathsf{e}(l) \le 0.069$ implies $\mathsf{r} \le 0.0357638$, in which case every λ_i, $i \ge 1$, over $2l \times 2l$, the tops of the sectors with electric flux among them, lies at least $3.33/l$ above the vacuum. Besides, $\mathsf{e}(l) \ge \log(1 + \sum_{e \ne 00}(\Lambda_e/\lambda_0)^l)$ over $2l \times 2l$; so $\mathsf{e}(l) \ge \log(1+3y)$ if each of the three sectors with electric flux has $(\Lambda_e/\lambda_0)^l \ge y$, a floor that tends to $\log 4$ as the three sectors approach the vacuum.

(b) Let $r_1 = Z^-/Z$ on the cube of side $2l$, and $\mathsf{f}(x) = 2\log(\sqrt{1-x} + \sqrt{2x})$ for $0 \le x \le 1$. Then $0 < r_1 \le 1$ and $\mathsf{e}(l) \ge \mathsf{f}(1-r_1)$. Hence $\mathsf{e}(l) \le 0.069$ forces $r_1 > 0.99937$, and $1 - r_1 \ge 0.0039$ forces $\mathsf{e}(l) \ge 0.16567$.

(c) If $\mathsf{e}(l) \le 0.069$, then $1 - Z^-/Z < 0.0039$ on the cube of every integer side $N \ge 16l$, and $\mathsf{e}(l') < 0.0234$ for every integer $l' \ge 16l$. So, if $1 - Z^-/Z \ge 0.0039$ on one cube of side N, then $\mathsf{e}(l) > 0.069$ for every integer $3 \le l \le N/16$.

(d) Under the criterion at l, the loop along the boundary of one plaquette has $l^2 \ge 0.4757/\log(1/\nu(W_p))$.

■

Remark 17.4 (Free fields). For comparison, take three free massless fields in three dimensions, one for each generator of SU(2), each with one mode for every momentum, and leave out the modes of momentum 0. Over the section $2l \times 2l$ the energies of the modes are $\pi|m|/l$, $m \in \mathbb{Z}^2 \setminus \{0\}$, so that the excited levels at the time l give

$$1 + \mathsf{r} = \prod_{m \in \mathbb{Z}^2 \setminus \{0\}} (1 - e^{-\pi|m|})^{-3}, \qquad 1 + \mathsf{r}' = \prod_{m \in \mathbb{Z}^2 \setminus \{0\}} (1 - e^{-2\pi|m|})^{-3},$$

numbers free of l, with $\log(1+\mathsf{r}) \approx 0.7201$ and $\log(1+\mathsf{r}') \approx 0.0242$. The number $D = 2\log(1+\mathsf{r}) - \log(1+\mathsf{r}')$ that stands for $\mathsf{e}(l)$ (Lem. 17.11) is approximately 1.416 at every scale, more than twenty times the bound 0.069 of the criterion. This is a computation for free fields.

Lemma 17.5 (A lower bound sufficient for anticoncentration)**.** There are absolute constants $c_{\mathrm{T}} > 0$ and $C_{\mathrm{T}} \geq 1$ such that, for every integer $N \geq 4$ and $\beta \geq C_{\mathrm{T}} N^{68}$,

$$Z(N,N,N) \geq c_{\mathrm{T}} \frac{N^2}{\beta^2} \mathsf{X}_0, \qquad \mathsf{X}_0 = (2\pi^2)^{-(2N^3-2)} \left(\frac{2\pi}{\beta}\right)^{3N^3-3} (\mathfrak{d}')^{-3/2}, \qquad \mathfrak{d}' = e^{S_0(N^{\times 3})}.$$

■

Proof. The periodic estimate in Prop. 17.47(a), proved in Prop. 11.14(g), gives

$$Z(N,N,N) \geq \tfrac{1}{2} \beta^{1-3N^3} \mathcal{Z}(N)$$

whenever $c_3 (N^{66} \log\beta/\beta)^{1/8} \leq x_*$, where $x_* \leq 1/2$. This condition holds throughout $\beta \geq C_{\mathrm{T}} N^{68}$ after an absolute increase of C_{T}. Indeed, $\log\beta/\beta$ decreases for $\beta \geq e$, and, for $C \geq e$ and $N \geq 4$,

$$\frac{N^{66}\log\beta}{\beta} \leq \frac{\log C + 68\log N}{CN^2} \leq \frac{\log C + 68\log 4}{16C} \quad (\beta \geq CN^{68}).$$

The last member tends to zero as $C \to \infty$.

Put

$$\mathcal{J}_N = N^{-2} \int_{[-\pi,\pi]^3} e^{S_0(N^{\times 3}) - S_\varphi(N^{\times 3})} \, \mathrm{d}\varphi.$$

The determinant estimate of Prop. 11.14(c) gives

$$\mathcal{J}_N \geq J_- := e^{-2C_S} \int_{[-\pi,\pi]^3} |\varphi|^{-2} d\varphi > 0.$$

The singularity is integrable in three dimensions. The normalizations in Prop. 17.47(a) give exactly

$$\frac{\beta^{1-3N^3} \mathcal{Z}(N)}{N^2 \beta^{-2} \mathsf{X}_0} = \frac{\mathcal{J}_N}{4\pi^3}.$$

Thus $c_{\mathrm{T}} = J_-/(8\pi^3)$ suffices. □

Proposition 17.6 (Failure on small tori, given Thm. YM3_I:D)**.** Given Thm. YM3_I:D among the construction results, there are numbers $t_\circ > 0$ and $\beta_\flat$, formed from the constants of Thm. YM3_I:D and of the setting (Th) of §2.3.1, the fixed calibration $\mathsf{t}_\natural$ of §2.3.1 among them, with the following properties.

(a) For every real $t \in (0, t_\circ]$ there is an integer m_t such that on the periodic cube of $2\mathsf{L}^j$ sites a side at the coupling $\beta = 2\mathsf{L}^j/t$ the quotient $r_1 = Z^-/Z$ satisfies $r_1 \leq 0.99596$ for every $j \geq m_t + 2$; there $\mathsf{e}(\mathsf{L}^j) \geq 0.16567$, and $\mathsf{e}(l) > 0.069$ for every integer $3 \leq l \leq \mathsf{L}^j/8$.

(b) For the sides $\ell_0 = 2\mathsf{L}^{m_0}$ and every label b with $\ell_0/\mathsf{b} \leq t_\circ$: $\mathsf{e}(\ell_0/2a_K) \geq 0.16567$ at $\beta = \mathsf{L}^K \mathsf{b}$ for all large K; so the uniform criterion with $L = \mathsf{L}$ and $l_* = \mathsf{L}^{m_0}$ fails at this label for every K_0 and every $x_0 \leq 0.069$.

(c) For every real $\beta \geq \beta_\flat$: $\mathsf{e}(l) > 0.069$ for every integer $3 \leq l \leq \frac{t_\circ}{16\mathsf{L}}\beta$. So the uniform criterion, for any integer $L \geq 2$, at any label $\mathsf{b} > 0$, forces $\ell_0 > \frac{t_\circ}{8\mathsf{L}}\mathsf{b}$.

(d) There are numbers $t_\star > 0$, $\mathsf{C}_\star \geq 1$ and $\mathsf{p}_\star \geq 1$, formed from the same constants, such that for every integer $j \geq 1$ and every real $\beta > 0$ with $t = 2\mathsf{L}^j/\beta \leq t_\star$, on the periodic cube of $N = 2\mathsf{L}^j$ sites a side,

$$\frac{\lambda_0(N,N)^N}{Z(N,N,N)} \leq \mathsf{C}_\star \, t \left(\log\frac{1}{t}\right)^{\mathsf{p}_\star}, \qquad \text{so that} \qquad \mathsf{e}(\mathsf{L}^j) \geq \log\frac{1}{t} - \mathsf{p}_\star \log\log\frac{1}{t} - \log\mathsf{C}_\star.$$

For a side $\ell_0 = 2\mathsf{L}^{m_0}$, $m_0 \in \mathbb{Z}$, and a label b with $\ell_0/\mathsf{b} \leq t_\star$ this says that at every cutoff $K \geq 1 - m_0$, at $\beta = \mathsf{L}^K \mathsf{b}$ and with $a_K = \mathsf{L}^{-K}$, $\mathsf{e}(\ell_0/2a_K) \geq \log\frac{\mathsf{b}}{\ell_0} - \mathsf{p}_\star \log\log\frac{\mathsf{b}}{\ell_0} - \log\mathsf{C}_\star$.

The proof of (d) rests, besides Thm. YM3_I:D, on Lem. 17.5 and the two estimates eq. 17.iii and eq. 17.iv stated in the next subsection: through them it bounds $\lambda_0(N,N)^N/Z(N,N,N)$ from above, and the floor of e follows from that ceiling.

■

Proposition 17.7 (What the trace identities alone allow)**.** Call *data* every family of numbers $Z^z(\underline{n}) > 0$, $\underline{n}$ a box and $z \in \{0,1\}^3$, with the properties (iii), (T1), (T2) and (T3) of Lem. 17.10, and form F, e, d_1, d_2, X, z_- from them as above. Let $0 < q < 1$.

(a) $Z^z(\underline{n}) = \prod_\mu (1+q^{n_\mu})^{|\underline{n}|/n_\mu}$, the same for all z, are data, with $d_1 : d_2 : \mathsf{e} = 1 : 2 : 4$ and $X = 3\mathsf{e}$ at every l.

(b) The *sheets* $Z^z(\underline{n}) = \prod_\rho (1+(-1)^{z_{\mu\nu}} q^{n_\mu n_\nu})^{n_\rho}$, $\{\mu,\nu,\rho\} = \{1,2,3\}$, are data. With $s = q^{l^2}$ they have $\mathsf{e}(l) = 8l\log(1+s^2) - 4l\log(1+s^4)$ and $X(l) = 24l\log(1+s) - 6l\log(1+s^4)$, and $X/\mathsf{e} \geq 2.95/s$ for $s \leq 0.03$ (at $l=4$, $s=0.03$: $\mathsf{e} \approx 0.028774$, $X \approx 2.837626$). So for no number C does $X(l) \leq C\,\mathsf{e}(l)$ hold for all data: at the same side, the criterion does not give the cube condition from these properties alone.

(c) For the sheets, $\frac{1}{2}q^{n^2} \leq z_-(n,n,n) \leq nq^{n^2}$, so that $-n^{-2}\log z_-(n,n,n) \to \log\frac{1}{q}$, while with $x_0 = \mathsf{e}(l)$ one has $\sigma^+ = \log\frac{1}{q} - \frac{\log(103.2e^{-0.835}l)}{2l^2} + o(1)$ and $\Lambda = 2l^2\log\frac{1}{q} - \log(103.2\,l) + o(1)$ as $q \to 0$. So an inequality $\liminf_n -n^{-2}\log z_-(n,n,n) \geq c'\Lambda/l^2 - C(l)$ with a number $c' > \frac{1}{2}$ is false for the sheets: from the properties of data alone, the coefficient $\frac{1}{2}$ of Λ/l^2 in σ^+ cannot be raised.

■

Proposition 17.8 (An area law without the numerical recursion of Prop. 17.17)**.** Assume the criterion at the side l, and let $l_j = 2^{j+1}l$. Then $l_j\varepsilon_{\mathrm{v}}(l_j) \leq 0.0165\cdot 2^{-j}\,\theta^{2\cdot 4^j}$ for all $j \geq 0$; $z_-(n,n,n) \leq \max\{1, 0.0096\sqrt{n/l}\}^{n/l}\,\theta^{n^2/2l^2}$ for $n = 2^i l$, $i \geq 1$; and $0 \leq \nu(W_C) \leq \theta^{RT/2l^2}$ for every rectangle C of sides $R \times T$ in a coordinate plane. The tension $\log\frac{1}{\theta}/2l^2 \geq 0.0582/l^2$ so obtained does not use the values $E_1,\dots,E_6$ of Prop. 17.17; the difference $0.4175/l^2$ to σ^+ is what that proposition adds. ■

17.1.4. *Construction estimates.* We use: (i) the periodic lattice of a box, its configurations and Haar measures, and Wilson's action; (ii) for a direction taken as time, the layers and the functions A_Σ and $A_\times(V,S,V') = \sum_b(1 - \frac{1}{2}\operatorname{tr} V_b S_{b_+} V_b'^{-1} S_{b_-}^{-1})$ (Section YM3_I:85, Section YM3_I:85): the map $U \mapsto (V^\tau, S^\tau)_{\tau\in\mathbb{Z}/t}$, with $V^\tau \in \mathcal{X} = G^{\mathsf{B}(\Sigma)}$ the bonds of the layer τ and $S^\tau \in G^\Sigma$ those from the layer τ to the layer $\tau+1$, carries dU to the product of the Haar measures, $\sum_p a_p = \sum_\tau[A_\Sigma(V^\tau) + A_\times(V^\tau, S^\tau, V^{\tau+1})]$ with continuous summands in $[0,2]$, and $A_\times(V',S^{-1},V) = A_\times(V,S,V')$; (iii) the crossing operator (Section YM3_I:85, Prop. YM3_I:85.1): the kernel $\mathsf{K}(V',V) = \int e^{-\beta A_\times(V,S,V')}\,\mathrm{d}S$ is symmetric, continuous and has values in $(0,1]$, and the operator K on $\mathcal{H} = L^2(\mathcal{X}, dV)$ is Hilbert–Schmidt and ≥ 0; (iv) for Lem. 17.13 and Prop. 17.6, Hölder's inequality for traces of products of an even number of Hilbert–Schmidt operators (Lem. YM3_I:86.1) and $\|A\|_{2k}^{2k} = \operatorname{Tr}(A^*A)^k$ (the norm definition preceding Lem. YM3_I:86.1); (v) for the uniqueness, the boxes and the pressure, the chessboard estimate on the tori of side $2\mathsf{L}^j$ (Prop. YM3_I:86.3), its form for two tori on one board (Cor. YM3_I:86.6), and its bound on every periodic Gibbs state (Prop. YM3_I:91.9); (vi) the invariance of every periodic Gibbs state under the symmetries of the lattice, its reflection positivity at every lattice mirror, the diagonal ones included, and its decomposition into ergodic components that are periodic Gibbs states (Props. YM3_I:103.4 and YM3_I:106.2); (vii) for Prop. 17.6 alone, Thm. YM3_I:D and the statements among the construction results through which it bears on the block fields at a fixed depth on tori that are small against the label: the setting (Th), what it gives, the weight of the small-field chain and the chain with its bound on the discarded mass (§2.3.1, Lem. YM3_I:118.1, the definitions preceding Prop. YM3_I:115.2, Prop. YM3_I:115.2); the fixed calibration and the inner balls (Defs. 2.3.1, YM3_I:113.10, Lem. YM3_I:113.10); the setting (E)

and the laws of the block fields (Defs. 2.3.1, 2.7); the numerical conditions at the chosen constants (Prop. YM3_I:69.1); the constants and the thresholds (§2.3.1 and equation YM3_I:(634)); the block-averaging map with its locality, gauge covariance, measurability and covariance under the lattice isometries (equations YM3_I:(37)–(39), Prop. YM3_I:4.5, YM3_I:4.5, [7, Lem. G.1]); the bound on the holonomies along rearranged paths (Lem. YM3_I:4.1); and, for its clause (d), the averages on small fields (Lem. YM3_I:4.4, Lem. 5.28) and the comparison of two consecutive cutoffs (§2.3.1, Section YM3_I:116, Props. YM3_I:116.1 and YM3_I:116.3 with their proofs, and §2.3.2), on which Cor. 2.8 rests, read from one cutoff to the next and summed into the estimate eq. 17.iv below, together with the signs of the constants of that comparison, among them $\kappa_{\mathrm{b}} > 0$ and $c' > 0$ (Prop. YM3_I:114.4 and §2.3.1).

From the textbooks we use the spectral theorem for compact self-adjoint operators, Jentzsch's theorem, the min–max principle, the inequalities of Young, Jensen and Schwarz, Kolmogorov's extension theorem, the weak compactness of the probability measures on a compact metrisable space, and Hausdorff's moment theorem.

Clause (d) of Prop. 17.6 rests, besides Thm. YM3_I:D and the statements of (vii), on two estimates of its own, which its proof takes up (Lem. 17.26, and Lem. 17.27 with the chain from a low cutoff). For a configuration U of a periodic cube and a site ξ let $\mathsf{a}(U,\xi) \in \mathbb{R}^5$ be the five invariants of the pair of the holonomies of U along the closed straight paths through ξ in the directions 1 and 2: their two class angles, $\alpha(g) = \arccos \frac{1}{2} \operatorname{tr} g \in [0,\pi]$ for $g \in G$, and the three entries of $(\mathsf{v}^{\mathrm{t}}\mathsf{v})^{1/2}$, v the real 3×2 matrix of their vector parts, numbers that do not change under a simultaneous conjugation of the pair. The first estimate concerns one lattice: there are absolute numbers C_{a}, c_{a}, c'_{a}, c''_{a} such that on the periodic cube of $N' \geq 4$ sites a side, at every coupling $\beta' \geq c'_{\mathrm{a}} N'^{c''_{\mathrm{a}}}$, for every closed cube Q of the half side w' in $\mathbb{R}^5$ with $(N'^5/2\beta')^{1/2} \leq w' \leq \frac{1}{20}$ and every $\omega > 0$ below a fixed negative power of N',

$$\mu_{N',\beta'}\Big(\mathsf{a}(U,\xi) \in \mathsf{Q},\ \max_p |\partial U(p) - \mathbf{1}| < \omega\Big) \leq C_{\mathrm{a}} N'^{c_{\mathrm{a}}}\, \beta'^{1/2}\, w'^3, \tag{17.iii}$$

$\mu_{N',\beta'}$ the law of that cube. Its proof combines a Gaussian ceiling with Lem. 17.5 for Wilson's measure on the cube in the gauge of a maximal tree, with constants that are powers of N': the ceiling integrates out all bonds but the three winding holonomies of ξ against the weight $\exp[-2\beta' N'^{-5} \sum_{\mu<\nu} |\mathsf{v}_\mu \times \mathsf{v}_\nu|^2]$, v_μ the vector part of the holonomy in the direction μ; the proved lower bound for $Z(N',N',N')$ supplies the normalizing denominator; and the integral of that weight over the triples whose first two members have their five invariants in Q is at most a fixed multiple of $w'^3(\beta'/N'^5)^{-3/2}$. The second estimate is drawn from the lemmas of (vii) on two consecutive cutoffs.

Let $\mathsf{b} = \ell/t$ be a label and $\ell = 2\mathsf{L}^m$ a side at which the setting (E) of §2.3.1 holds for it, with ℓ at most a fixed power of $y = \log \mathsf{b}$, and let μ_0^K be the law of the block field of the depth 0 at the cutoff K (Def. 2.7(2)). For every y above a number formed from the constants there is a lowest cutoff K_1, at which the bare cube has at most a fixed power of y sites a side, such that for every integer $K \geq K_1$ and every Borel set E of block fields of the depth 0 that the gauge transformations carry onto itself (Def. 2.7(3))

$$\mu_0^K(E) \leq e^{\mathsf{S}}(\mu_0^{K_1}(E) + \mathsf{R}_\Sigma), \tag{17.iv}$$

with S below an absolute number and $\mathsf{R}_\Sigma \leq e^{-3y^2}$. Cor. 2.8 compares two consecutive cutoffs additively; here, from one cutoff to the next, the errors ϵ_{fg} and ϵ_{cl} of §2.3.1, which decrease like powers of L^{-K} up to factors that are powers of K, y and ℓ, enter as factors on the probability, and the errors ϵ_{sh} and R_{t_1}, which are exponentially small in a power of y, as additive remainders and through the factors $(1-\epsilon_{\mathrm{sh}})^{-1}$ and $(1-\mathsf{R}_{t_1})^{-1}$ (Lem. 17.27); S and R_Σ are the sums over $K \geq K_1$ of the exponents of the factors and of the remainders. So an event whose probability at the cutoff K_1 is at most of the order of t has a probability at most of the order of t at every cutoff $K \geq K_1$, since $y \geq \log \frac{1}{t}$ makes e^{-3y^2} small against t.

Beyond (i)–(vii) the proofs use no result among the construction results; no statement of the section other than Prop. 17.6 rests on Thm. YM3_I:D or on Thms. YM3_I:AJ and YM3_I:AD. Its clause (d) combines the finite-cube lower bound of Lem. 17.5 with the estimates eq. 17.iii and eq. 17.iv.

17.1.5. *Estimates and proofs.*

Lemma 17.9 (Two identities for the crossing weight)**.** Let $(V^S)_b = S_{b_-} V_b S_{b_+}^{-1}$ and $\mathsf{K}_0(V,W) = \prod_b \exp[-\beta(1 - \frac{1}{2}\operatorname{tr} V_b W_b^{-1})]$ for $V, W \in \mathcal{X}$ and $S \in G^\Sigma$.

(a) $e^{-\beta A_\times(V,S,V')} = \mathsf{K}_0(V^g, V'^{gS})$ for all $V, V' \in \mathcal{X}$ and all $g, S \in G^\Sigma$.

(b) $\mathsf{K}_0(V,W) = \sum_\alpha c_\alpha \phi_\alpha(V)\overline{\phi_\alpha(W)}$, with α in a countable set, $c_\alpha \geq 0$, $\sum_\alpha c_\alpha = e^{\beta \#\mathsf{B}(\Sigma)}$, ϕ_α continuous and $|\phi_\alpha| \leq 1$.

■

Proof sketch. (a) $V_b((V'^S)_b)^{-1} = V_b S_{b_+} V_b'^{-1} S_{b_-}^{-1}$ is the matrix of the bond b in $A_\times$, so $e^{-\beta A_\times(V,S,V')} = \mathsf{K}_0(V, V'^S)$; $(V^g)_b((W^g)_b)^{-1} = g_{b_-} V_b W_b^{-1} g_{b_-}^{-1}$, so $\mathsf{K}_0(V^g, W^g) = \mathsf{K}_0(V,W)$ by the cyclicity of the trace; and $(V^S)^g = V^{gS}$. (b) For $v, w \in G$, $\operatorname{tr}(vw^{-1}) = \sum_{i,j} v_{ij}\overline{w_{ij}}$, with entries of modulus ≤ 1. Expand each factor $e^{-\beta}\exp[\frac{\beta}{2}\operatorname{tr} V_b W_b^{-1}]$ in powers, write out the powers and multiply over b: $\alpha = (n_b, (i_{b,s}, j_{b,s})_{s \leq n_b})_b$, $c_\alpha = e^{-\beta \#\mathsf{B}(\Sigma)} \prod_b (\beta/2)^{n_b}/n_b!$, $\phi_\alpha(V) = \prod_b \prod_s (V_b)_{i_{b,s} j_{b,s}}$; as there are 4^{n_b} index families for each n_b, $\sum_\alpha c_\alpha = e^{-\beta \#\mathsf{B}(\Sigma)} \prod_b e^{2\beta}$, and this bounds the series of the moduli. □

Lemma 17.10 (Transfer operator and flux sectors)**.** Fix a box and a direction μ_0 as time, with the other directions $\nu_1 < \nu_2$. For $i = 1, 2$ let ι_i change the sign of V_b on the bonds of the direction ν_i that start on a fixed line of the other direction of the section, $(\mathsf{J}_i \psi)(V) = \psi(\iota_i V)$, $\mathsf{J}^k = \mathsf{J}_1^{k_1}\mathsf{J}_2^{k_2}$ and $P_e = \frac{1}{4}(1 + (-1)^{e_1}\mathsf{J}_1)(1 + (-1)^{e_2}\mathsf{J}_2)$ for $e \in \{0,1\}^2$. Let $A_\Sigma^{(0)} = A_\Sigma$, let $A_\Sigma^{(1)}$ be A_Σ with the summand $1 - \frac{1}{2}\operatorname{tr}$ of one plaquette replaced by $1 + \frac{1}{2}\operatorname{tr}$, M_m the multiplication by $e^{-\beta A_\Sigma^{(m)}/2}$ and $\mathsf{T}_m = M_m \mathsf{K} M_m$; write $z \leftrightarrow (k; m)$ with $k = (z_{\mu_0\nu_1}, z_{\mu_0\nu_2})$, each pair written in its order, and $m = z_{\nu_1\nu_2}$. Then:

(i) $\mathsf{T}_m \geq 0$ is Hilbert–Schmidt with a continuous symmetric kernel with values in $(0,1]$; J_1, J_2 are commuting self-adjoint unitary involutions that commute with T_m; none of these objects depends on t.

(ii) $Z^{(k;m)} = \operatorname{Tr}(\mathsf{T}_m^t \mathsf{J}^k)$ for every integer $t \geq 3$.

(iii) Z^z does not depend on the places of the columns, and $Z^{\pi z}(\pi \underline{n}) = Z^z(\underline{n})$ for every permutation π of the directions.

Consequently: (T1) Z is symmetric in the sides; (T2) $Z(t, \Sigma) = \sum_i \lambda_i^t$ for $t \geq 3$, the $\lambda_i \geq 0$ being the eigenvalues of T_0, $\sum_i \lambda_i^2 < \infty$, and λ_0 is simple with an eigenvector in the range of P_{00}; (T3) with the eigenvalues $\mu_i(e,m) \geq 0$ of T_m on the range of P_e, which depend neither on t nor on k,

$$Z^{(k;m)} = \sum_e (-1)^{e\cdot k} W_t(e;m), \qquad W_t(e;m) = \operatorname{Tr}(\mathsf{T}_m^t P_e) = \sum_i \mu_i(e,m)^t \geq 0;$$

$\Lambda_e = \sup_i \mu_i(e,0)$ is the top of the sector e. Moreover:

(a) $W_t(00;0) \geq \lambda_0^t$;

(b) $0 \leq 1 - Z^{(k;0)}/Z \leq 2\mathsf{r}_t/(1+\mathsf{r}_t)$, so that $z_- \leq \mathsf{r}_t/(1+\mathsf{r}_t)$ when the time is the direction 1 or 3;

(c) on a cube Z^z/Z depends on $|z|$ alone, $r_2 \leq r_1$, $r_3 \leq r_1$, and $0 \leq 1 - Z^z/Z \leq |z|(1-r_1) \leq 6z_-$; with any direction as time, the weights $w_e = W_n(e;0)/Z$ of the cube of side n are $w_{00} = \frac{1}{4}(1 + 2r_1 + r_2)$, $w_{10} = w_{01} = \frac{1}{4}(1 - r_2)$, $w_{11} = \frac{1}{4}(1 - 2r_1 + r_2)$, and $1 - r_1 \leq \frac{4}{3}\mathsf{r}_n(n,n)/(1+\mathsf{r}_n(n,n))$;

(d) $a_- = \sum_i \mu_i^{n_1} = \sum_i \mu_i'^{n_3}$ and $Z^- = \sum_i \mu_i''^{n_2}$, with numbers $\mu_i, \mu_i', \mu_i'' \geq 0$ that do not depend on n_1, on n_3 and on n_2 respectively, and $\sup_i \mu_i'' = \lambda_-(n_1, n_3) \leq \lambda_0(n_1, n_3)$.

■

Proof sketch. Every plaquette is a summand of $A_\Sigma(V^\tau)$ or of $A_\times(V^\tau, S^\tau, V^{\tau+1})$. Put the columns at the origin of the section and at the crossing $\tau = 0$. A column of the pair $\nu_1\nu_2$ is one summand in every layer and turns A_Σ into $A_\Sigma^{(1)}$; a column of the pair $\mu_0\nu_i$ is the set of the crossing summands of the bonds flipped by ι_i, and since $1 - \frac{1}{2}\operatorname{tr}(-g) = 1 + \frac{1}{2}\operatorname{tr} g$, reversing it is the substitution $V^0 \mapsto \iota_i V^0$ in that crossing. Integrating each S^τ first,

$$Z^{(k;m)} = \int \mathsf{T}_m(V^1, \iota^k V^0) \prod_{\tau\neq 0} \mathsf{T}_m(V^{\tau+1}, V^\tau) \prod_\tau \mathrm{d}V^\tau = \operatorname{Tr}(\mathsf{T}_m^t \mathsf{J}^k),$$

the trace of a chain of $t \geq 3$ bounded kernels: (ii). The functions $A_\Sigma^{(m)}$ and $A_\times$ are invariant under ι_i, because the two bonds of the direction ν_i of a plaquette are both flipped or neither and $-\mathbf{1}$ is central; so J_i commutes with T_m, and $\langle\psi, \mathsf{T}_m\psi\rangle = \langle M_m\psi, \mathsf{K}M_m\psi\rangle \geq 0$: (i). (iii): changing $U(b)$ into $-U(b)$ on a set $\mathfrak{F}$ of bonds keeps dU and multiplies $\operatorname{tr}\partial U(p)$ by $(-1)^{\#(p\cap\mathfrak{F})}$; for $\mathfrak{F}$ the bonds of the direction ν that start on the line of the direction ρ through $x + e_\mu$ this sign is -1 exactly on the columns of the pair $\mu\nu$ through x and through $x + e_\mu$ (the sides are ≥ 3), so a column moves by one step; a permutation of the axes renumbers the bonds and carries a plaquette to a plaquette or to its inverse, which has the same trace. (T2) and (T3) follow from the spectral theorem on the four ranges of the P_e.

By Jentzsch's theorem λ_0 is simple with an eigenvector $\varphi > 0$, and $P_{00}\varphi = \frac{1}{4}\sum_k \varphi\circ\iota^k \geq \frac{1}{4}\varphi \neq 0$: (a). (b): $Z - Z^{(k;0)} = 2\sum_{e\cdot k \text{ odd}} W_t(e;0) \leq 2(Z - W_t(00;0)) \leq 2(Z - \lambda_0^t)$. (c): on a cube (iii) makes Z^z depend on $|z|$. With two twisted pairs, the time a direction of one of them but not of the other, $(k;m) = (10;1)$ and $r_2 = \sum_e (-1)^{e_1} W(e;1)/Z \leq \sum_e W(e;1)/Z = r^{(00;1)} = r_1$; in the same way $r_3 = r^{(11;1)} \leq r_1$; and with the time common to two twisted pairs, $(k;m) = (11;0)$, $4W(11;0)/Z = 1 - 2r_1 + r_2 \geq 0$ gives $1 - r_2 \leq 2(1 - r_1)$. With all three pairs twisted, $(k;m) = (11;1)$ for every direction as time, and (T3) at $m = 1$ gives $4W(11;1)/Z = r^{(00;1)} - r^{(10;1)} - r^{(01;1)} + r^{(11;1)} = r_1 - 2r_2 + r_3 \geq 0$; added to $1 - 2r_1 + r_2 \geq 0$ this is $1 - r_3 \leq (1 - r_1) + (1 - r_2)$, and so $1 - r_3 \leq 3(1 - r_1)$.

The formula for the weights is obtained by inverting $Z^{(k;0)}/Z = \sum_e (-1)^{e\cdot k} w_e$, with $Z^{(10;0)} = Z^{(01;0)} = Zr_1$ and $Z^{(11;0)} = Zr_2$. With $u = w_{10}$, $v = w_{11}$ one has $1 - r_1 = 2(u + v)$, $1 - r_2 = 4u$, so $r_2 \leq r_1$ means $v \leq u$, which gives $2(u + v) \leq \frac{4}{3}(2u + v) = \frac{4}{3}(1 - w_{00})$, and $1 - w_{00} \leq 1 - \lambda_0^n/Z = \mathsf{r}_n/(1 + \mathsf{r}_n)$ by (a). (d) is (T3) with the time 1, 3 and 2; $\lambda_- \leq \lambda_0$ because $Z^- \leq Z$. □

Lemma 17.11 (One positive sequence; the excess)**.** Let $\kappa_0 \geq \kappa_1 \geq \cdots \geq 0$, $\kappa_0 > 0$, $\sum \kappa_i < \infty$; $p_M = \sum_i \kappa_i^M$ for real $M \geq 1$, $\eta = \sum_{i\geq 1} \kappa_i/\kappa_0$, $s_M = \sum_{i\geq 1} (\kappa_i/\kappa_0)^M$, $D = 2\log p_1 - \log p_2$ and $D' = 2\log p_2 - \log p_4$. Then

(a) $\log(1 + \eta) \leq D \leq 2\log(1 + \eta)$;
(b) if $D < \log 2$, then $\eta \leq \varpi(e^D - 1)$;
(c) $D' \leq 2\log(1 + s_2) \leq 2s_2 \leq 2\eta^2$, so that $D' \leq \omega(D)$ if $D < \log 2$;
(d) $p_{MM'} \leq p_M^{M'}$ for real $M, M' \geq 1$;
(e) $s_M \leq s_{M_0}^{M/M_0}$ for real $M \geq M_0 \geq 1$.

For $\kappa_i = \lambda_i(\Sigma)^t$ one has $p_M = Z(Mt, \Sigma)$, $\eta = \mathsf{r}_t(\Sigma)$ and $s_M = \mathsf{r}_{Mt}(\Sigma)$ whenever Mt is an integer ≥ 3. Hence:

(f) $F/|\underline{n}|$ and $\delta/|\underline{n}|$ do not increase in any side, $F/|\underline{n}| \to P$ as the three sides tend to infinity, $\log(1 + \mathsf{r}_t(\Sigma)) \leq \delta(t, \Sigma)$, and $\log\lambda_0(N_1, N_2) \geq N_1 N_2 P$;
(g) for integers $A, B \geq N \geq 3$, the Casimir term $\varepsilon(A, B) = \log\lambda_0(A, B) - ABP$ satisfies $0 \leq \varepsilon(A, B) \leq (A + B)\,\delta_{\mathrm{c}}(N)/N^2$.

■

Proof sketch. $p_1 = \kappa_0(1 + \eta)$ and $p_M = \kappa_0^M(1 + s_M)$, so $e^D = (1 + \eta)^2/(1 + s_2)$. For $u_i \geq 0$ with the sum u and real $m \geq 1$, $\sum u_i^m \leq u^m$; this gives (d), (e) and $s_2 \leq \min\{\eta, \eta^2\}$, hence (a), and (c) as $D' = 2\log(1 + s_2) - \log(1 + s_4)$. (b): $e^D \geq (1 + \eta)^2/(1 + \eta^2) = 1 + 2\eta/(1 + \eta^2)$, and $\eta < 1$ by (a);

solve the quadratic inequality. (f): (d) with $M' = t'/t$ gives $Z(t',\Sigma) \le Z(t,\Sigma)^{t'/t}$ for $t' \ge t$, and (T1) the other sides; $\delta(t,\Sigma) = t(\log\lambda_0 - N_1N_2P) + \log(1+\mathsf{r}_t)$, with $\log\lambda_0 = \lim_s F(s,\Sigma)/s \ge N_1N_2P$. (g): put $\varkappa = \delta_{\mathrm{c}}(N)/N^2$. For $t, A, B \ge N$, $\delta(t,A,B) = t\,\varepsilon(A,B) + \log(1+\mathsf{r}_t(A,B))$, and by (f) $\log(1+\mathsf{r}_t(A,B)) \le \log(1+\mathsf{r}_N(A,B)) \le \delta(N,A,B) \le AB\varkappa$. By (f), $\varepsilon(A,B)/AB \le \delta(t,A,B)/tAB$ for every t, so $\varepsilon(A,B)/AB$ does not increase in A or B and tends to 0 as $A,B\to\infty$. With B as time, $\delta(s,t,B) = B\varepsilon(s,t) + \log(1+\mathsf{r}_B(s,t)) \le B\varepsilon(s,t) + st\varkappa$; divide by s and let $s\to\infty$: $\varepsilon(t,B)/t \le B\lim_s \varepsilon(s,t)/st + \varkappa$, and as $t\to\infty$, $\limsup_t \varepsilon(t,B)/t \le \varkappa$. With A as time, $\delta(t,A,B) \le A\varepsilon(t,B) + tB\varkappa$; divide by t and let $t\to\infty$: $\varepsilon(A,B) \le (A+B)\varkappa$. □

Lemma 17.12 (Two static charges)**.** Fix a box, the time 3, $o\in\Sigma$ and $r = o + Re_1$, $1 \le R \le n_1 - 1$; let γ be the straight path of R bonds from o to r in the section, $V(\gamma)$ its ordered product, $\hat{\mathcal{H}} = L^2(\mathcal{X}; M_2(\mathbb{C}))$ with $\langle\Psi,\Psi'\rangle = \int \operatorname{tr}(\Psi^*\Psi')\,\mathrm{d}V$,

$$(\hat{\mathsf{K}}\Psi)(V) = \iint e^{-\beta A_\times(V,S,V')}\, S_o\Psi(V')S_r^{-1}\,\mathrm{d}S\,\mathrm{d}V', \qquad \hat{\mathsf{T}} = M_0\hat{\mathsf{K}}M_0, \qquad (\Upsilon\psi)(V) = \psi(V)V(\gamma);$$

let C be the rectangle $R\times T$, $1 \le T \le n_3 - 1$, with the side γ in the layer 0, and p_ξ half the trace of the line that winds once along the direction 3 through $\xi\in\Sigma$. Then:

(a) $\hat{\mathsf{K}}$ and $\hat{\mathsf{T}}$ are Hilbert–Schmidt, self-adjoint and ≥ 0, they do not depend on n_3, and $\Upsilon^*\Upsilon = 2$;
(b) $2Z\langle W_C\rangle = \operatorname{Tr}(\mathsf{T}_0^{n_3-T}\Upsilon^*\hat{\mathsf{T}}^T\Upsilon)$ and $\operatorname{Tr}\hat{\mathsf{T}}^{n_3} = 4Z\langle p_o p_r\rangle$;
(c) $0 \le \langle W_C\rangle \le (4\langle p_o p_r\rangle)^{T/n_3}$ and $\|\hat{\mathsf{T}}\|^{n_3} \le 4Z\langle p_o p_r\rangle$;
(d) $0 \le \langle p_o p_r\rangle \le z_+^{1-\varrho}z_-^{\varrho} + z_-^{1-\varrho}z_+^{\varrho}$, $\varrho = R/n_1$, and for $R \le n_1/2$

$$0 \le \langle W_C\rangle \le \big(8z_-^{R/n_1}\big)^{T/n_3}, \qquad \frac{\|\hat{\mathsf{T}}\|}{\lambda_0(n_1,n_2)} \le \Big[8\big(1+\mathsf{r}_{n_3}(n_1,n_2)\big)z_-^{R/n_1}\Big]^{1/n_3}.$$

(e) *Any two sites.* For any two sites $o \ne r$ of the section, $\hat{\mathsf{K}}$ and $\hat{\mathsf{T}} = \hat{\mathsf{T}}_{o,r}$ are defined by the same formulas, (a) holds, $\operatorname{Tr}\hat{\mathsf{T}}^{n_3} = 4Z\langle p_o p_r\rangle$ and $\|\hat{\mathsf{T}}\|^{n_3} \le 4Z\langle p_o p_r\rangle$, and (d) holds for $\langle p_o p_r\rangle$ and $\|\hat{\mathsf{T}}\|$ with R the distance of o_1 and r_1 on the circle of n_1 sites, $1 \le R \le n_1/2$; on a cube it holds as well with R the distance of o_2 and r_2 on the circle of n_2 sites.
(f) *Staircase loops in a box.* Let $1 \le T \le n_3 - 2$, let o be a site of the section, α_j and $R_j \ge 1$ integers, $o_j = o + \alpha_j e_1$ and $r_j = o + (\alpha_j + R_j)e_1$ for $0 \le j < T$, and let C be the closed path that runs in the layer 0 straight along the line of the direction 1 from o_0 to r_0, then for $j = 0,\dots,T-1$ in turn one bond up at r_j and, if $j < T-1$, straight from r_j to r_{j+1}, in the layer T straight from r_{T-1} to o_{T-1}, and then for $j = T-1,\dots,0$ in turn one bond down at o_j and, if $j > 0$, straight from o_j to o_{j-1} ("straight" from the site of the integer α to that of α': along the $|\alpha'-\alpha|$ bonds between them). Then, with the thermal numbers over (n_1,n_2),

$$|\langle W_C\rangle| \le \big(1+\mathsf{r}_{n_3-T}\big)\prod_{j=0}^{T-1}\frac{\|\hat{\mathsf{T}}_{o_j,r_j}\|}{\lambda_0(n_1,n_2)}, \qquad \text{and if all } R_j \le \tfrac{n_1}{2}: \quad |\langle W_C\rangle| \le \big(1 + \mathsf{r}_{n_3-T}\big)\big[8(1+\mathsf{r}_{n_3})\big]^{T/n_3} z_-^{\sum_j R_j/n_1 n_3}.$$

No side need be even. ■

Proof sketch. The inequality that replaces Hölder's inequality for traces is this. Let $A, C \ge 0$ be compact with eigenvalues a_i, c_j and eigenbases (e_i), (f_j), B bounded, $p,q \ge 1$ integers and $s = p+q$. Then $\langle e_i, A^pB^*C^qBe_i\rangle = \sum_j w_{ij}a_i^pc_j^q$ with $w_{ij} = |\langle f_j, Be_i\rangle|^2 \ge 0$, the row and column sums of (w_{ij}) are $\le \|B\|^2$. A zero normaliser below makes the weighted product sum vanish, and infinite normalisers give only a trivial bound. For finite positive normalisers, Young's inequality $a^pc^q \le \mathsf{a}^{p/s}\mathsf{c}^{q/s}\big(\frac{p}{s}\frac{a^s}{\mathsf{a}} + \frac{q}{s}\frac{c^s}{\mathsf{c}}\big)$, with $\mathsf{a} = \sum w_{ij}a_i^s$ and $\mathsf{c} = \sum w_{ij}c_j^s$, gives

$$0 \le \sum_i \langle e_i, A^pB^*C^qBe_i\rangle \le \|B\|^2\Big(\sum_i a_i^s\Big)^{p/s}\Big(\sum_j c_j^s\Big)^{q/s}. \tag{17.v}$$

Its form with two operators bounds the modulus. For bounded B_1, B_2 the same eigenbases give $\sum_i \langle e_i, A^p B_1^* C^q B_2 e_i\rangle = \sum_{i,j} a_i^p c_j^q \overline{\langle f_j, B_1 e_i\rangle}\,\langle f_j, B_2 e_i\rangle$, and $\sum_{i,j} a_i^p c_j^q |\langle f_j, B_k e_i\rangle|^2 = \sum_i \langle e_i, A^p B_k^* C^q B_k e_i\rangle$ for $k = 1, 2$; Schwarz's inequality with the weights $a_i^p c_j^q \ge 0$, and then eq. 17.v for B_1 and for B_2, give

$$\Big|\sum_i \langle e_i, A^p B_1^* C^q B_2 e_i\rangle\Big| \le \|B_1\|\,\|B_2\| \Big(\sum_i a_i^s\Big)^{p/s} \Big(\sum_j c_j^s\Big)^{q/s}. \tag{17.vi}$$

(a) Symmetry follows from $A_\times(V', S^{-1}, V) = A_\times(V, S, V')$. For positivity substitute $S = g^{-1}Q$, use $e^{-\beta A_\times(V, g^{-1}Q, V')} = \mathsf{K}_0(V^g, V'^Q)$ (Lem. 17.9(a)), average over $g \in G^\Sigma$ (the left side does not depend on g) and expand K_0 by Lem. 17.9(b):

$$\langle \Psi, \hat{\mathsf{K}}\Psi\rangle = \sum_\alpha c_\alpha \operatorname{tr}(X_\alpha^* X_\alpha) \ge 0, \qquad X_\alpha = \iint \overline{\phi_\alpha(V^Q)}\, Q_o \Psi(V) Q_r^{-1}\, \mathrm{d}Q\, \mathrm{d}V.$$

The argument uses of o and r only that they are sites of the section. (b) With $\mathsf{P}_\xi = S_\xi^0 \cdots S_\xi^{T-1}$, $2W_C = \operatorname{tr}[V^0(\gamma)^* \mathsf{P}_o V^T(\gamma) \mathsf{P}_r^{-1}]$, and the layers write $2Z\langle W_C\rangle$ as the trace of the chain of the kernels of $\mathsf{T}_0^{n_3-T}$ and of $\Upsilon^* \hat{\mathsf{T}}^T \Upsilon$; at $T = n_3$ the trace over $M_2(\mathbb{C})$ gives $\operatorname{tr}\mathsf{P}_o \operatorname{tr}\mathsf{P}_r^{-1}$. (c) is eq. 17.v with $A = \mathsf{T}_0$, $C = \hat{\mathsf{T}}$, $B = \Upsilon$, $p = n_3 - T$, $q = T$ and $\sum a_i^{n_3} = Z$. (d) Take the time 1. The two winding lines become the multiplication $\hat{\mathsf{p}}$ by one function of the variables of a layer, taken at the layers o_1 and $o_1 + R$; each line holds exactly one bond flipped by ι_2, so $\hat{\mathsf{p}}$ exchanges the ranges of $P_\pm = \frac{1}{2}(1 \pm \mathsf{J}_2)$, on which $\operatorname{Tr}(\mathsf{T}_0^{n_1} P_\pm) = a_\pm$ by Lem. 17.10(ii). By eq. 17.v

$$Z\langle p_o p_r\rangle = \sum_\pm \operatorname{Tr}\big[(\mathsf{T}_0 P_\pm)^{n_1 - R}\, \hat{\mathsf{p}}\, (\mathsf{T}_0 P_\mp)^R\, \hat{\mathsf{p}}\big] \le \sum_\pm a_\pm^{1-\varrho} a_\mp^\varrho,$$

and for $\varrho \le \frac{1}{2}$ and $z_- \le z_+ \le 1$ each term is $\le z_-^\varrho Z$; finally $Z = \lambda_0^{n_3}(1 + \mathsf{r}_{n_3})$. (e) The proof of (a) uses of o and r only that they are sites of the section, and the trace of $\hat{\mathsf{T}}^{n_3}$ in (b) and the second bound of (c) do not use γ; so, for any two sites, $\hat{\mathsf{T}} \ge 0$, $4Z\langle p_o p_r\rangle = \operatorname{Tr}\hat{\mathsf{T}}^{n_3} \ge 0$ and $\|\hat{\mathsf{T}}\|^{n_3} \le 4Z\langle p_o p_r\rangle$: the sign of $\langle p_o p_r\rangle$ comes from the time 3. For the bound take the time 1. The line through $\xi \in \Sigma$ is then the function p_{ξ_2} of the variables V^{ξ_1} of the layer ξ_1, half the trace of the ordered product of the variables of its bonds of the direction 3 with the second coordinate ξ_2; whatever ξ_2 is, the line holds exactly one bond flipped by ι_2. So the multiplications $\hat{\mathsf{p}}_{o_2}$ and $\hat{\mathsf{p}}_{r_2}$, two different ones if $o_2 \ne r_2$, are real, of norm ≤ 1, and each exchanges the ranges of $P_\pm$. With R the distance of o_1 and r_1 on the circle, $1 \le R \le n_1/2$, the trace has $n_1 - R$ factors T_0 on one side of the two lines and R on the other, and, as in (d), with the names o and r exchanged if need be,

$$Z\langle p_o p_r\rangle = \sum_\pm \operatorname{Tr}\big[(\mathsf{T}_0 P_\pm)^{n_1 - R}\, \hat{\mathsf{p}}_{o_2}\, (\mathsf{T}_0 P_\mp)^R\, \hat{\mathsf{p}}_{r_2}\big];$$

eq. 17.vi with $A = \mathsf{T}_0 P_\pm$, $C = \mathsf{T}_0 P_\mp$, $B_1 = \hat{\mathsf{p}}_{o_2}$, $B_2 = \hat{\mathsf{p}}_{r_2}$, $p = n_1 - R$, $q = R$ bounds the term of $\pm$ in modulus by $a_\pm^{1-\varrho} a_\mp^\varrho$. With the sign from the time 3 this is the first statement of (d) for o and r, at every parity of n_1, n_2, n_3, and the bound of $\|\hat{\mathsf{T}}\|$ in (d) follows from it and from $\|\hat{\mathsf{T}}\|^{n_3} \le 4Z\langle p_o p_r\rangle$ as there. On a cube the time 2 does the same with o_2, r_2, the twisted pair being then the pair 23, whose quotient on the cube equals that of the pair 13 (Lem. 17.10(c)). (f) With the time 3, $U(C)$ is the ordered product of the runs in the layers and of the bonds $S_{r_j}^j$, $(S_{o_j}^j)^{-1}$; put $(\Upsilon_j \psi)(V) = \psi(V) V(o_j \to r_j)$, with $V(o_j \to r_j)$ the product along the straight run from o_j to r_j, $(\Gamma_j \Psi)(V) = V(o_{j-1} \to o_j) \Psi(V) V(r_j \to r_{j-1})$, a unitary operator of $\hat{\mathcal{H}}$ that commutes with M_0, and $\mathsf{Q} = \Upsilon_0^* \hat{\mathsf{T}}_{o_0, r_0} \Gamma_1 \hat{\mathsf{T}}_{o_1, r_1} \Gamma_2 \cdots \Gamma_{T-1} \hat{\mathsf{T}}_{o_{T-1}, r_{T-1}} \Upsilon_{T-1}$.

As in (b), $2Z\langle W_C\rangle = \operatorname{Tr}(\mathsf{T}_0^{n_3-T}\mathsf{Q})$; in an eigenbasis of T_0, $|\operatorname{Tr}(\mathsf{T}_0^{n_3-T}\mathsf{Q})| \le \|\mathsf{Q}\| \lambda_0^{n_3-T}(1 + \mathsf{r}_{n_3-T})$, with $n_3 - T \ge 2$; and $\|\mathsf{Q}\| \le 2 \prod_j \|\hat{\mathsf{T}}_{o_j, r_j}\|$, $Z \ge \lambda_0^{n_3}$. The second bound is (d) for each j. □

Lemma 17.13 (Several static charges; closed paths)**.** Fix a box with n_1 even and the time 3.

(a) For a finite set $\mathcal{V}$ of runs, each with a site $u(v) \in \Sigma$ and a sense, let $D(S) = \bigotimes_v D_v(S_{u(v)})$ on $(\mathbb{C}^2)^{\otimes\mathcal{V}}$, with $D_v(g) = g$ for the sense up and $D_v(g) = \bar g$ for the sense down, $(\hat{\mathsf{K}}_D\Psi)(V) = \iint e^{-\beta A_\times(V,S,V')} D(S)\Psi(V')\,\mathrm{d}S\,\mathrm{d}V'$ and $\hat{\mathsf{T}}_D = M_0\hat{\mathsf{K}}_D M_0$. Then $\hat{\mathsf{T}}_D \ge 0$ and $\|\hat{\mathsf{T}}_D\|^{n_3} \le \operatorname{Tr}\hat{\mathsf{T}}_D^{n_3} = 2^{\#\mathcal{V}} Z\langle\prod_v p_{u(v)}\rangle$.

(b) Let $\#\mathcal{V}$ be even, m_τ the number of the $u(v)$ in the layer $\tau \in \{1,\dots,n_1\}$ of the direction 1, and ℓ the number of τ with $m_1 + \dots + m_\tau$ odd. If $\ell \le n_1/2$, then $|\langle\prod_v p_{u(v)}\rangle| \le 2z_-^{\ell/n_1}$.

(c) Let C be a closed lattice path in a cube of h layers, $h+1 \le \min_\mu n_\mu$, that runs bonds of the direction 3 in T gaps, $n_3 \ge T+2$; in the gap j let v_j be the number of these bonds and ℓ_j the number of integers α such that an odd number of them have first coordinate $\le \alpha$, and assume $\ell_j \le n_1/2$. Then, with r taken over the section (n_1, n_2),

$$|\langle W_C\rangle| \le 2^{c_3(C)-1}(1+\mathsf{r}_{n_3-T}) \prod_j \Big[2^{\mathsf{v}_j+1}(1+\mathsf{r}_{n_3}) z_-^{\ell_j/n_1}\Big]^{1/n_3}.$$

(d) Let D be a continuous unitary representation of G on $\mathbb{C}^d$ with $D(-\mathbf{1}) = -\mathbf{1}$, and $W_C^D = d^{-1}\operatorname{tr} D(U(C))$. Then (a)–(c) hold with $D_v = D$ for the sense up and $D_v = \overline{D}$ for the sense down, with $d^{-1}\operatorname{tr} D$ of the line that winds once along the direction 3 through ξ in the place of p_ξ, $d^{\#\mathcal{V}}$ in the place of $2^{\#\mathcal{V}}$, and W_C^D, $d^{c_3(C)-1}$, $2d^{\mathsf{v}_j}$ in the place of W_C, $2^{c_3(C)-1}$, 2^{v_j+1}. ■

Proof sketch. (a) As in Lem. 17.12(a), with $D(g^{-1}Q) = D(g)^{-1}D(Q)$; the trace of the n_3-th power gives $\prod_v \operatorname{tr} D_v(\mathsf{P}_{u(v)})$, and $\operatorname{tr}\bar g = \operatorname{tr} g$. (b) With the time 1, $Z\langle\prod_v p_{u(v)}\rangle = \operatorname{Tr}\prod_\tau(\mathsf{T}_0\hat{\mathsf{p}}^\tau)$, $\hat{\mathsf{p}}^\tau$ the multiplication by the product of the line functions of the layer τ. Each line holds one flipped bond, so $\hat{\mathsf{p}}^\tau$ maps the range of P_s to that of $P_{s(-1)^{m_\tau}}$; splitting the trace over $s = \pm$, the factor of the layer τ becomes $\mathsf{T}_0 P_{s_\tau}\hat{\mathsf{p}}^\tau$, with $s_\tau = -s$ for exactly ℓ layers. Hölder's inequality for the trace of the n_1 factors (Lem. YM3_I:86.1), and $\|\mathsf{C}\hat{\mathsf{p}}\|_{n_1}^{n_1} = \operatorname{Tr}(\mathsf{C}\hat{\mathsf{p}}^2\mathsf{C})^{n_1/2} \le \operatorname{Tr}\mathsf{C}^{n_1} = a_\pm$ for $\mathsf{C} = \mathsf{T}_0 P_\pm$ (by min–max, as $0 \le \mathsf{C}\hat{\mathsf{p}}^2\mathsf{C} \le \mathsf{C}^2$), bound the term of s by $a_s^{1-\ell/n_1} a_{-s}^{\ell/n_1}$. (c) Cut C into pieces in single layers and runs of the direction 3: $2W_C = \Gamma_0 D_0(S^0)\Gamma_1 \cdots D_{T-1}(S^{T-1})\Gamma_T$, where D_j is the tensor product over the runs of the gap j and Γ_j, a function of V^j, is the tensor product over the pieces of the layer j of unitary 2×2 matrices, of cups (linear forms on $\mathbb{C}^2\otimes\mathbb{C}^2$ with the entries of a matrix of G) and of caps (vectors with such entries).

Cups and caps have the norm $\sqrt{2}$ and occur at the changes from down to up and from up to down, $c_3(C)$ times each, so $\prod_j \sup_V \|\Gamma_j(V)\| = 2^{c_3(C)}$. As in Lem. 17.12(b), $2Z\langle W_C\rangle = \operatorname{Tr}(\mathsf{T}_0^{n_3-T}\mathsf{Q})$ with $\mathsf{Q} = \Gamma_0\hat{\mathsf{T}}_{D_0}\Gamma_1\cdots\hat{\mathsf{T}}_{D_{T-1}}\Gamma_T$; in an eigenbasis of T_0, $|\operatorname{Tr}(\mathsf{T}_0^{n_3-T}\mathsf{Q})| \le \|\mathsf{Q}\|\lambda_0^{n_3-T}(1+\mathsf{r}_{n_3-T})$, and (a), (b) bound $\|\hat{\mathsf{T}}_{D_j}\|^{n_3} \le 2^{\mathsf{v}_j+1}\lambda_0^{n_3}(1+\mathsf{r}_{n_3}) z_-^{\ell_j/n_1}$; numbering the layers so that C lies in the layers $1,\dots,h$ makes the ℓ of (b) equal to ℓ_j. (d) The proofs of (a)–(c) use of the representation $g \mapsto g$ only that it is continuous and unitary, which gives the positivity of $\hat{\mathsf{T}}_D$ and pointwise unitary pieces in the layers, and that it changes its sign at $-\mathbf{1}$, which makes every line function odd under the flip and of modulus ≤ 1; a cup or a cap of D, a linear form or a vector with the entries of a matrix of $D(G)$, has the norm $\sqrt{d}$, and the trace over $(\mathbb{C}^d)^{\otimes\mathcal{V}}$ of the n_3-th power gives $d^{\#\mathcal{V}}Z$ times the mean of the product of the line functions. □

Lemma 17.14 (A floor at one length bounds the loops)**.** Let $\beta \ge 0$ and $L_1, L_2, L_3 \ge 3$ integers; take the first direction as time over the section (L_2, L_3), with its λ_0, r_s and Λ_-, and put $\phi = \log(\lambda_0/\Lambda_-) \in [0,\infty]$. Let C be a rectangle $R\times T$ of a $(1,3)$-plane with $1 \le R \le L_1 - 3$ and $1 \le T \le L_3 - 1$, and p_o, p_r the lines of Lem. 17.12 through its corners in one layer of the direction 3. Then

$$0 \le \langle W_C\rangle \le (4\langle p_o p_r\rangle)^{T/L_3}, \qquad \langle p_o p_r\rangle \le \frac{1+\mathsf{r}_{L_1-R}}{1+\mathsf{r}_{L_1}}\Big[e^{-\phi R} + z_-(L_1-R, L_2, L_3)\Big];$$

so, if $\phi > 0$, $\limsup_{L_1\to\infty}\langle W_C\rangle_{(L_1,L_2,L_3)} \le 4^{T/L_3}e^{-\phi RT/L_3}$. ■

Proof sketch. The first inequality is Lem. 17.12(c). With the time 1, as in the proof of Lem. 17.12(d), $Z\langle p_o p_r\rangle = \sum_\pm \operatorname{Tr}(\mathsf{T}_\pm^{L_1-R}\hat{\mathsf{p}}\,\mathsf{T}_\mp^{R}\hat{\mathsf{p}})$ with $\mathsf{T}_\pm = \mathsf{T}_0 P_\pm$ and $\|\hat{\mathsf{p}}\| \le 1$. The nonzero eigenvalues of T_- are those of the sectors 01 and 11, so $\|\mathsf{T}_-\| = \Lambda_-$, and $\|\mathsf{T}_+\| \le \lambda_0$. For $B \ge 0$ of trace class and $Y \ge 0$ bounded, $\operatorname{Tr}(BY) = \operatorname{Tr}(B^{1/2}YB^{1/2}) \le \|Y\|\operatorname{Tr} B$. With $B = \mathsf{T}_+^{L_1-R}$ and $Y = \hat{\mathsf{p}}\mathsf{T}_-^R\hat{\mathsf{p}}$ the term of $+$ is $\le \Lambda_-^R\lambda_0^{L_1-R}(1+\mathsf{r}_{L_1-R})$; with $B = \mathsf{T}_-^{L_1-R}$ and $Y = \hat{\mathsf{p}}\mathsf{T}_+^R\hat{\mathsf{p}}$ the term of $-$ is $\le \lambda_0^R \operatorname{Tr}\mathsf{T}_-^{L_1-R} = \lambda_0^R a_-(L_1-R,L_2,L_3) = \lambda_0^{L_1}(1+\mathsf{r}_{L_1-R})z_-(L_1-R,L_2,L_3)$. Divide by $Z = \lambda_0^{L_1}(1+\mathsf{r}_{L_1})$. As $L_1\to\infty$ the quotient of the thermal factors tends to 1, λ_0 being simple, and $z_-(s,L_2,L_3) \le \lambda_0^{-s}\operatorname{Tr}\mathsf{T}_-^s \le (\Lambda_-/\lambda_0)^{s-2}\lambda_0^{-2}\operatorname{Tr}\mathsf{T}_-^2 \to 0$ if $\phi > 0$. □

Lemma 17.15 (Flux weight and vortex line)**.**

(a) For integers $n_1' \ge n_1 \ge 3$, $n_3' \ge n_3 \ge 3$, $n_2 \ge 3$ and $\kappa = n_1'n_3'/(n_1n_3)$, a real number,

$$a_-(n_1',n_2,n_3') \le a_-(\underline{n})^\kappa, \qquad z_-(n_1',n_2,n_3') \le \bigl[z_-(\underline{n})\,e^{\delta(\underline{n})}\bigr]^\kappa.$$

(b) For all integers $t,N,n \ge 3$, with $\varepsilon = \varepsilon_{\mathrm{v}}(t,N)$,

$$2z_-(t,n,N) \le n\varepsilon + \mathsf{r}_n(t,N), \qquad n\varepsilon \le \mathsf{r}_n(t,N) - \log(1-2z_-(t,2n,N)).$$

■

Proof sketch. (a) By Lem. 17.10(d), $a_- = \sum_i \mu_i^{n_1}$ with μ_i independent of n_1, so Lem. 17.11(d) with a real exponent gives $a_-(n_1',n_2,n_3) \le a_-(n_1,n_2,n_3)^{n_1'/n_1}$; the same holds in the third side, and $Z(n_1',n_2,n_3') \ge e^{\kappa|\underline{n}|P}$. (b) With the time along the column, $Z(t,n,N) = \lambda_0^n(1+\mathsf{r}_n)$ and $Z^-(t,n,N) = \sum_i \mu_i''^{\,n}$ with $\sup_i \mu_i'' = \lambda_0 e^{-\varepsilon}$; so $1-2z_- = Z^-/Z \ge e^{-n\varepsilon}/(1+\mathsf{r}_n) \ge 1-n\varepsilon-\mathsf{r}_n$. For the second inequality, which uses that the twisted spectrum is ≥ 0, $\sum \mu_i''^{\,2n} \le \lambda_-^n \sum \mu_i''^{\,n}$ gives $\lambda_-^n \ge Z^-(t,2n,N)/Z^-(t,n,N) \ge (1-2z_-(t,2n,N))\lambda_0^{2n}/(\lambda_0^n(1+\mathsf{r}_n))$. □

Proposition 17.16 (Recursion; excess and thermal numbers)**.** Let $\beta \ge 0$ and let $l \ge 3$ be an integer.

(a) With $\Sigma_1 = (l,l)$, $\Sigma_2 = (2l,l)$, $\Sigma_3 = (2l,2l)$ and $(d_1,d_2,d_3) = (d_1,d_2,\mathsf{e})$: $\log(1+\mathsf{r}_l(\Sigma_j)) \le d_j(l) \le 2\log(1+\mathsf{r}_l(\Sigma_j))$, and $X = 4d_1 + 2d_2 + \mathsf{e}$.

(b) If $Y_2(\mathsf{e}(l)) < \log 2$, then $d_1(2l) \le \omega(\mathsf{e}(l))$, $d_2(2l) \le \omega(Y_1(\mathsf{e}(l)))$, $\mathsf{e}(2l) \le \Psi(\mathsf{e}(l))$ and $X(2l) \le \Phi(\mathsf{e}(l))$.

(c) $\Psi(x) \le 12.9x^2$ and $\Phi(x) \le 19.89x^2$ for $0 < x \le 0.069$; $\Psi(x) \le 11.03x^2$ and $\Phi(x) \le 17.71x^2$ for $0 < x \le \frac{1}{20}$; $\Psi(x) < x$ for $0 < x < 0.0740538$.

(d) Under the criterion, for $n = 2^k l$, $k \ge 1$, and $x = \mathsf{e}(n/2) \le \theta^{n/2l}/12.9$:

$$\mathsf{r}_{n/2}(n,n) \le 0.51832\,x, \quad \mathsf{r}_{n/2}(2n,n) \le 1.09990\,x, \quad \mathsf{r}_{n/2}(2n,2n) \le 2.53905\,x;$$

$$\mathsf{r}_n(n,n) \le 0.001615\,\theta^{n/l},$$

$$\mathsf{r}_n(2n,n) \le 0.00732\,\theta^{n/l},$$

$\delta_{\mathrm{c}}(n) \le 0.01655\,\theta^{n/l}$, $d_1(n) \le 0.0032288\,\theta^{n/l}$, $d_2(n) \le 0.0145396\,\theta^{n/l}$; and $\delta_{\mathrm{c}}(N) < 0.1324\,\theta^{N/2l}$ for every integer $N \ge 2l$.

(e) Let $\tilde\omega(D) = 2\log(1+\varpi(e^D-1)^2) \le \omega(D)$, and let $\tilde Y_1, \tilde Y_2, \tilde\Psi, \tilde\Phi$ be Y_1, Y_2, Ψ, Φ with $\tilde\omega$ in the place of ω. Then (b) holds with $\tilde\omega$, $\tilde Y_1$, $\tilde\Psi$, $\tilde\Phi$. With $X_0 = 0.069$ and $X_{k+1} = \tilde\Psi(X_k)$, the iterates $X_1,\dots,X_5$ are 0.0604446, 0.0432902, 0.0195722, 0.0034309, 0.0000960 to the digits shown, the X_k decrease to 0, and $\bar\delta_j = \sum_{s\ge0} 8^{-s-1}\tilde\Phi(X_{j-1+s})$ decreases in j, with $\bar\delta_4 \le 0.00073051$. If $\mathsf{e}(l) \le 0.069$, with no other hypothesis, then $\mathsf{e}(2^k l) \le X_k$ and $\delta_{\mathrm{c}}(2^j l) \le \bar\delta_j$ for all $k \ge 0$ and $j \ge 1$.

■

Proof sketch. (a) Double the sides of the cube one after the other: by (T1), X is the sum of $4[2F(l,l,l)-F(2l,l,l)]$, $2[2F(2l,l,l)-F(2l,2l,l)]$ and $2F(2l,2l,l)-F(2l,2l,2l)$, and each bracket is the D of Lem. 17.11 for $\kappa_i=\lambda_i(\Sigma_j)^l$. (b) Over $(2l,2l)$ with $\kappa_i=\lambda_i^l$: $D=\mathsf{e}(l)$ and $D'=d_1(2l)\le\omega(\mathsf{e})$. Doubling one side of the section at most doubles a defect, up to defects already bounded: by Lem. 17.11(d) with the time in the second direction, $F(l,4l,2l)\le 2F(l,2l,2l)$, so

$$2F(l,4l,2l)-F(2l,4l,2l)\le 2\mathsf{e}(l)+d_1(2l)\le Y_1(\mathsf{e}(l)),$$

and Lem. 17.11(c) over $(4l,2l)$ gives $d_2(2l)\le\omega(Y_1)$; in the same way $2F(l,4l,4l)-F(2l,4l,4l)\le Y_2$, and over $(4l,4l)$ one doubling of the time squares the defect: $\mathsf{e}(2l)\le\omega(Y_2)=\Psi$. Then (a) at $2l$. (c) $\varpi(x)/x$ and $(e^D-1)/D$ increase, so $\Psi(x)/x^2$ and $\Phi(x)/x^2$ do not decrease on $(0,0.0741]$; their values at 0.069 are 12.893528 and 19.881793, at $\frac{1}{20}$ they are 11.029686 and 17.709146, to the digits shown; $\Psi(x)/x$ increases; its positive fixed point lies in $(0.0740538,0.0740539)$, and $Y_2<\log 2$ throughout this interval. (d) Induction with (b), (c): $12.9\,\mathsf{e}(2^{k+1}l)\le(12.9\,\mathsf{e}(2^kl))^2$, so $\mathsf{e}(2^kl)\le\theta^{2^k}/12.9\le 0.069$ and $X(2^{k+1}l)\le\frac{19.89}{166.41}\theta^{2^{k+1}}\le 0.11953\,\theta^{2^{k+1}}$. At $l'=n/2$ the sections $(2l',2l')$, $(4l',2l')$, $(4l',4l')$ have at the time l' the defects x, $\le Y_1(x)$, $\le Y_2(x)$, and Lem. 17.11(b) with the monotone quotients at 0.069 gives the first three bounds. Then $\mathsf{r}_n=s_2\le\eta^2$ gives $\mathsf{r}_n(n,n)\le\frac12\omega(x)\le 0.268652\,x^2\le 0.001615\,\theta^{n/l}$, likewise $\mathsf{r}_n(2n,n)$, and $d_1(n)\le\omega(x)$, $d_2(n)\le\omega(Y_1(x))$. For the excess, the pressure cancels in $X(n)=8\delta_{\mathrm{c}}(n)-\delta_{\mathrm{c}}(2n)$, and Lem. 17.11(f) gives

$$\delta_{\mathrm{c}}(n)=\sum_{i\ge0}8^{-i-1}X(2^in)\le 0.11953\,\theta^{n/l}\sum_{i\ge0}8^{-i-1}\theta^{(2^i-1)n/l}\le 0.01655\,\theta^{n/l}.$$

For an integer N with $2^jl\le N<2^{j+1}l$, $j\ge1$, Lem. 17.11(f) gives $\delta_{\mathrm{c}}(N)\le 8\,\delta_{\mathrm{c}}(2^jl)\le 0.1324\,\theta^{2^j}<0.1324\,\theta^{N/2l}$. (e) By Lem. 17.11(b), (c), $D'\le 2\log(1+\eta^2)\le\tilde\omega(D)$, and the proof of (b) uses of ω only that it increases. The numbers $X_1,\dots,X_5$ and $\bar\delta_4$ are the values of the recursion $X_{k+1}=\tilde\Psi(X_k)$ from $X_0=0.069$ and of the series that defines $\bar\delta_4$; the X_k decrease to 0 because $\tilde\Psi\le\Psi$ and $\Psi(x)<x$ on $(0,0.0740538)$, and $\bar\delta_j$ decreases because $\tilde\Phi$ increases. If $\mathsf{e}(l)\le X_0$, the induction gives $\mathsf{e}(2^kl)\le X_k$, and $\delta_{\mathrm{c}}(2^jl)=\sum_{s\ge0}8^{-s-1}X(2^{j+s}l)\le\sum_s 8^{-s-1}\tilde\Phi(X_{j+s-1})=\bar\delta_j$. □

Proposition 17.17 (Vortex lines on tubes)**.** Assume the criterion; put $l_j=2^{j+1}l$ and $w=2^j$.

(a) $l'\varepsilon_{\mathrm{v}}(l')\le 0.01643\,\theta^{l'/l}$ for $l'=2^kl$, $k\ge1$.

(b) Let $l'=2^kl$, $k\ge1$, $l'\varepsilon_{\mathrm{v}}(l')\le e^{-E}$, $\hat N\ge2$ an integer, $y=\hat Ne^{-E}$, $\mathsf{t}=\theta^{l'/l}$, $\bar z_1=y+\frac12(0.001615\,\mathsf{t})^{2\hat N}$ and $\bar z_2=\bar z_1^4e^{8\hat N\cdot0.01655\,\mathsf{t}}<\frac12$. Then

$$2l'\varepsilon_{\mathrm{v}}(2l')\le\frac{2}{\hat N}\Big\{(0.001615\,\mathsf{t}^2)^{\hat N/2}+\frac{2\bar z_2}{1-2\bar z_2}\Big\}.$$

(c) $l_j\varepsilon_{\mathrm{v}}(l_j)\le e^{-\sigma^+l_j^2}$ for all $j\ge0$.

(d) For real $M\ge w\mathsf{q}$ with Ml_j an integer: $z_-(l_j,Ml_j,l_j)\le Me^{-\varsigma w^2}$ and $\delta(l_j,Ml_j,l_j)\le 0.01655\,M\theta^{l_j/l}$.

(e) Let $l'=l_j$ and let $N_3\ge l'$ and $n_2\ge\mathsf{q}l'^2/2l$ be integers. Over the section (n_2,N_3), with the first direction as time, the top Λ_- of the sectors odd under the flip of the side N_3 lies at least $\sigma^+N_3-\frac{N_3}{l'^2}\big[\log\frac{n_2}{l'}+0.01655\,\frac{n_2}{l'}\theta^{l'/l}\big]$ above the vacuum.

■

Proof sketch. (a) The second inequality of Lem. 17.15(b) at $t=N=n=l'$, with $z_-(l',2l',l')\le\mathsf{r}_{l'}(2l',l')\le 0.00732\,\mathsf{t}$ (Lem. 17.10(b), the time 1) and $\mathsf{r}_{l'}(l',l')\le 0.001615\,\mathsf{t}$ (Prop. 17.16(d)). (b) Three moves. The first inequality of Lem. 17.15(b) on the box $(l',2\hat Nl',l')$, with $\mathsf{r}_{2\hat Nl'}(l',l')\le\mathsf{r}_{l'}(l',l')^{2\hat N}$ (Lem. 17.11(e)), gives $z_-\le\bar z_1$. Lem. 17.15(a) with $\kappa=4$ passes to the box $(2l',2\hat Nl',2l')$, a fourth power at the cost $e^{4\delta}$, with $\delta(l',2\hat Nl',l')\le 2\hat N\delta_{\mathrm{c}}(l')$. The second inequality of Lem. 17.15(b) for the tube $2l'\times2l'$ at the time $\hat Nl'$, with $\mathsf{r}_{\hat Nl'}(2l',2l')\le\mathsf{r}_{2l'}(2l',2l')^{\hat N/2}$, returns to the vortex line.

(c) Take $\hat N_j = 4\cdot 3^j$ and write $\mathsf{t}_j = \theta^{l_j/l} = e^{-2^j\cdot 2\Lambda}$. For real E and $\mathsf{t}\in(0,1]$ let $\mathsf{H}_j(E,\mathsf{t})$ be the right member of (b) at $l'=l_j$ and $\hat N=\hat N_j$, formed with $y=\hat N_j e^{-E}$, $\bar z_1 = y + \frac12(0.001615\,\mathsf{t})^{2\hat N_j}$ and $\bar z_2 = \bar z_1^4 e^{8\hat N_j\cdot 0.01655\,\mathsf{t}}$, where $\bar z_2<\frac12$. Identically $\mathsf{H}_j = \frac{2}{\hat N_j}y^4 e^{8\hat N_j\cdot 0.01655\,\mathsf{t}}\mathsf{B}_j$, that is,

$$-\log\mathsf{H}_j(E,\mathsf{t}) = 4E - 3\log\hat N_j - 8\hat N_j\cdot 0.01655\,\mathsf{t} - \log(2\mathsf{B}_j),\qquad \mathsf{B}_j = \frac{(0.001615\,\mathsf{t}^2)^{\hat N_j/2}}{y^4\,e^{8\hat N_j\cdot 0.01655\,\mathsf{t}}} + \frac{2(\bar z_1/y)^4}{1-2\bar z_2}.$$

One value of θ serves all. Let $\Delta = 2\Lambda - 0.23284$, which is ≥ 0 as $2\log\frac{1}{0.8901}\ge 0.23284$, and $\mathring{\mathsf{t}}_j = e^{-2^j\cdot 0.23284}$, the value of t_j at $2\Lambda=0.23284$, so that $\mathsf{t}_j = \mathring{\mathsf{t}}_j e^{-2^j\Delta}$. Let $E_0=4.3414$, and let $E_1,E_2,\dots$ be numbers with $E_{j+1}\le -\log\mathsf{H}_j(E_j,\mathring{\mathsf{t}}_j)$, $\bar z_2<\frac12$ at $(E_j,\mathring{\mathsf{t}}_j)$. Then $l_j\varepsilon_{\mathrm v}(l_j)\le e^{-E_j-4^j\Delta}$ for all $j\ge 0$ and every $\theta\le 0.8901$. At $j=0$ this is (a) at $l'=2l$, as $0.01643\,\theta^2 = \exp[-\log\frac{1}{0.01643} - 0.23284 - \Delta]$ and $\log\frac{1}{0.01643}+0.23284\ge 4.3414$. From j to $j+1$ apply (b) with $E=E_j+4^j\Delta$ and $\mathsf{t}=\mathsf{t}_j$, and compare each number with its value at $(E_j,\mathring{\mathsf{t}}_j)$: y is that value times $e^{-4^j\Delta}$; $(0.001615\,\mathsf{t})^{2\hat N_j}$ is it times $e^{-2^{j+1}\hat N_j\Delta}\le e^{-4^j\Delta}$; so $\bar z_1$ is at most it times $e^{-4^j\Delta}$; as $e^{8\hat N_j\cdot 0.01655\,\mathsf{t}}$ increases in t, $\bar z_2$ is at most it times $e^{-4^{j+1}\Delta}$, and so is $\frac{2\bar z_2}{1-2\bar z_2}$, as $u\mapsto 1/(1-2u)$ increases; and $(0.001615\,\mathsf{t}^2)^{\hat N_j/2}$ is it times $e^{-2^j\hat N_j\Delta}\le e^{-4^{j+1}\Delta}$, as $\hat N_j\ge 4\cdot 2^j$.

So each of the two terms of the right member of (b) is at most its value at $(E_j,\mathring{\mathsf{t}}_j)$ times $e^{-4^{j+1}\Delta}$, and

$$l_{j+1}\varepsilon_{\mathrm v}(l_{j+1})\le \mathsf{H}_j(E_j+4^j\Delta,\mathsf{t}_j)\le \mathsf{H}_j(E_j,\mathring{\mathsf{t}}_j)\,e^{-4^{j+1}\Delta}\le e^{-E_{j+1}-4^{j+1}\Delta}.$$

Numerical bounds. The functions in the recursion are explicit and monotone on the intervals used above. Outward interval evaluation gives

x	$\Psi(x)/x^2$	$\Phi(x)/x^2$
0.069	(12.89352831, 12.89352832)	(19.88179283, 19.88179284)
0.05	(11.02968611, 11.02968612)	(17.70914610, 17.70914611).

For the successive lower bounds $E_0,\dots,E_6$ below, the differences $-\log\mathsf{H}_j(E_j,\mathring{\mathsf{t}}_j)-E_{j+1}$ exceed, respectively,

$$9.7\cdot 10^{-5},\quad 5.1\cdot 10^{-5},\quad 3.8\cdot 10^{-6},\quad 6.4\cdot 10^{-5},\quad 3.5\cdot 10^{-5},\quad 7.6\cdot 10^{-5}.$$

In each evaluation $\bar z_2<1/2$. The geometric tail displayed below gives a lower bound greater than 1.909266. The sharper defect recursion likewise gives $\bar\delta_4<0.00073051$. These bounds justify the directions of rounding used in the argument.

The numbers at $2\Lambda=0.23284$. There B_j is a number close to 2, and one may take $E_1,\dots,E_6 =$ 11.3558, 35.5848, 128.3243, 495.6445, 1962.8155, 7829.1628, each at most $-\log\mathsf{H}_j(E_j,\mathring{\mathsf{t}}_j)$ at the one before. For $j\ge 6$ the losses $3\log\hat N_j + 8\hat N_j\cdot 0.01655\,\mathring{\mathsf{t}}_j + \log(2\mathsf{B}_j)$ are at most $3\log\hat N_j+\log 4.01$, so that $E_{j+1}=4E_j-3\log\hat N_j-\log 4.01$ may be taken, and this closed recursion gives

$$4^{-j}E_j\ge 4^{-6}E_6 - \sum_{i\ge 6}4^{-i-1}(3\log 4+3i\log 3+\log 4.01)\ge 1.909266.$$

So $4^{-j}E_j\ge 0.23284+1.67$ for every $j\ge 0$, hence $4^{-j}(E_j+4^j\Delta)\ge 2\Lambda+1.67$, and $l_j\varepsilon_{\mathrm v}(l_j)\le\exp[-4^j(2\Lambda+1.67)] = e^{-\sigma^+ l_j^2}$. (d) The first inequality of Lem. 17.15(b) with $n=Ml_j$, (c), and $\mathsf{r}_{Ml_j}(l_j,l_j)\le(0.001615\,\theta^{2w})^M\le e^{-2\Lambda Mw}\le e^{-\varsigma w^2}$, as $2\Lambda\mathsf{q}=\varsigma$, give $2z_-\le(M+1)e^{-\varsigma w^2}$; Lem. 17.11(f) gives $\delta\le M\delta_{\mathrm c}(l_j)$. (e) For every integer $t\ge l'$, by Lem. 17.10(d) and Lem. 17.15(a) at the real factors t/l' and N_3/l', $\Lambda_-^t\le a_-(t,n_2,N_3)\le a_-(l',n_2,l')^{tN_3/l'^2}$; so $\log\Lambda_-\le\frac{N_3}{l'^2}[F+\log z_-](l',n_2,l')$, while $\log\lambda_0(n_2,N_3)\ge n_2N_3P$. Subtracting, $\log(\lambda_0/\Lambda_-)\ge\frac{N_3}{l'^2}[-\log z_--\delta](l',n_2,l')$, and (d) with $M=n_2/l'\ge w\mathsf{q}$ bounds the bracket below by $\varsigma w^2-\log M-0.01655\,M\theta^{l'/l}$, with $\varsigma w^2=\sigma^+l'^2$. □

Proposition 17.18 (Slabs)**.** For integers $N, a, b, t \geq 3$ let $\mathsf{e}^{(N)}(t) = 2F(t, 2t, N) - F(2t, 2t, N)$, $d^{(N)}(t) = 2F(t, t, N) - F(2t, t, N)$, $P^{(N)} = \inf_{a,b} F(a, b, N)/ab$ and $\delta^{(N)}(a, b) = F(a, b, N) - abP^{(N)}$.

(a) For every $\beta \geq 0$: $0 \leq \delta^{(N)}(a, b) \leq \delta(a, b, N)$, $\delta^{(N)}(a, b)/ab$ does not increase in a or b, $\delta^{(N)}(a, a)/a^2 \to 0$, and $\log(1 + \mathsf{r}_a(b, N)) \leq \delta^{(N)}(a, b)$; if $Y_1(\mathsf{e}^{(N)}(t)) < \log 2$, then $d^{(N)}(2t) \leq \omega(\mathsf{e}^{(N)}(t))$ and $\mathsf{e}^{(N)}(2t) \leq \omega(Y_1(\mathsf{e}^{(N)}(t)))$.

Assume now the criterion.

(b) Let $n_0 = 2^k l \leq N < 2n_0$, $k \geq 1$, $\chi' = 0.3602\,\theta^{n_0/l}$ and $t_i = 2^i n_0$. Then $2.72\,\mathsf{e}^{(N)}(t_i) \leq \chi'^{2^i}$, $\delta^{(N)}(t_i, t_i) \leq 0.133\,\chi'^{2^i}$, $\delta^{(N)}(n_0, n_0) \leq 0.01655\,\frac{N}{n_0}\theta^{n_0/l}$, $\mathsf{r}_{t_i}(t_i, N) \leq 0.14\,\chi'^{2^i}$, and $\mathsf{r}_{t_i}(b, N) \leq 0.28\,\chi'^{2^i}$ for $t_i \leq b < 2t_i$.

(c) Let $N = 2^k l$, $k \geq 1$, $\chi = 0.03\,\theta^{N/l}$ and $t_i = 2^i N$. Then $2.06\,\mathsf{e}^{(N)}(t_i) \leq \chi^{2^i}$, $d^{(N)}(t_i) \leq 0.1202\,\chi^{2^i}$, $\mathsf{r}_{t_i}(t_i, N) \leq 0.0613\,\chi^{2^i}$, $\mathsf{r}_{t_i}(2t_i, N) \leq \chi^{2^i}$ and $\delta^{(N)}(t_i, t_i) \leq 0.1826\,\chi^{2^i}$. If moreover $N = 2wl$ with $w \geq 2$, and $\hat{N}$, g are as in Thm. AW(g1), then $t_i \varepsilon_{\mathrm{v}}(t_i, N) \leq \exp[-2^i(\sigma^+ N^2 - \mathsf{g})]$ for all $i \geq 0$.

■

Proof sketch. (a) The slab of thickness N is a datum in two dimensions: $(a, b) \mapsto F(a, b, N)$ is symmetric and has the trace formula in each variable, so Lem. 17.11 and the proof of Prop. 17.16(b) go over with one side of the section frozen; $P^{(N)} \geq NP$ gives $\delta^{(N)} \leq \delta(\cdot, \cdot, N)$. (b) The pressures cancel at the start: $\mathsf{e}^{(N)}(n_0) \leq 2\delta(n_0, 2n_0, N) \leq 4\frac{N}{n_0}\delta_{\mathrm{c}}(n_0) < 0.1324\,\theta^{n_0/l} \leq 0.1049$. On $(0, 0.1049]$, $\omega(Y_1(x)) \leq 2.72x^2$ and $\omega(x) \leq 0.5593x^2$, so $2.72\,\mathsf{e}^{(N)}(t_{i+1}) \leq (2.72\,\mathsf{e}^{(N)}(t_i))^2$ and $2.72\,\mathsf{e}^{(N)}(n_0) \leq \chi'$. Since $4F(t, t, N) - F(2t, 2t, N) = 2d^{(N)}(t) + \mathsf{e}^{(N)}(t) = 4\delta^{(N)}(t, t) - \delta^{(N)}(2t, 2t)$, and for $i \geq 1$, $2d^{(N)}(t_i) + \mathsf{e}^{(N)}(t_i) \leq (2 \cdot 0.5593 + 2.72)\,\mathsf{e}^{(N)}(t_{i-1})^2 \leq 0.518842\,\chi'^{2^i}$, summation with the weights 4^{-j-1} gives $\delta^{(N)}(t_i, t_i) \leq 0.13236\,\chi'^{2^i}$ for $i \geq 1$; at $i = 0$, $\delta^{(N)}(n_0, n_0) \leq \delta(n_0, n_0, N) \leq \frac{N}{n_0}\delta_{\mathrm{c}}(n_0) < 0.0331\,\theta^{n_0/l} \leq 0.0919\,\chi'$.

Finally $\log(1 + \mathsf{r}_{t_i}(b, N)) \leq \delta^{(N)}(t_i, b) \leq \frac{b}{t_i}\delta^{(N)}(t_i, t_i)$. (c) At the dyadic thickness the start is $\mathsf{e}^{(N)}(N) = d_2(N) \leq 0.0145396\,\theta^{N/l} \leq 0.01152$ and $d^{(N)}(N) = d_1(N) \leq 0.0032288\,\theta^{N/l}$; for $x \leq 0.0116$, $\omega(Y_1(x)) \leq 2.06x^2$, $\omega(x) \leq 0.5059x^2$ and $e^x - 1 \leq 1.006x$, and for $D \leq 0.00286$, $\varpi(e^D - 1) \leq 0.5008D$. So $2.06\,\mathsf{e}^{(N)}(t_i) \leq \chi^{2^i}$; $d^{(N)}(t_i) \leq 0.5059\,\mathsf{e}^{(N)}(t_{i-1})^2 \leq 0.1202\,\chi^{2^i}$; Lem. 17.11(b) over (t_i, N) gives $\mathsf{r}_{t_i}(t_i, N) \leq 0.5008\,d^{(N)}(t_i)$, and Lem. 17.11(a) over $(2t_i, N)$ gives $\mathsf{r}_{t_i}(2t_i, N) \leq e^{\mathsf{e}^{(N)}(t_i)} - 1$; the sum with the weights 4^{-j-1} gives the bound on $\delta^{(N)}$.

In the flat tubes $t \times N$ one doubling of the width squares $t\varepsilon_{\mathrm{v}}(t, N)$. At $i = 0$, Prop. 17.17(c) gives $N\varepsilon_{\mathrm{v}}(N, N) \leq e^{-E_0}$, $E_0 = \sigma^+ N^2$. From $t = t_i$ to $2t$, with $n = \hat{N}t$ and $y_i = \hat{N}e^{-E_i}$: the first inequality of Lem. 17.15(b) at the time $2n$, with $\mathsf{r}_{2n}(t, N) \leq \mathsf{r}_t(t, N)^{2\hat{N}} \leq e^{-4E_i}$, gives $z_-(t, 2n, N) \leq (1 + 10^{-7})y_i$; Lem. 17.15(a) with $\kappa = 2$ in the width squares it at the cost $2F(t, 2n, N) - F(2t, 2n, N) \leq 4\hat{N}\delta^{(N)}(t, t) \leq \xi_i := 0.732\,\hat{N}\chi^{2^i}$; and the second inequality of Lem. 17.15(b) for the tube $2t \times N$ at the time n gives

$$2t\,\varepsilon_{\mathrm{v}}(2t, N) \leq 4.01\,\hat{N}e^{-2E_i + \xi_i} =: e^{-E_{i+1}}.$$

As $\hat{N}\chi \leq 0.7936$, $\sum_i 2^{-i-1}\xi_i \leq 0.294$, and $2^{-i}E_i \geq \sigma^+ N^2 - \log\hat{N} - \log 4.01 - 0.294 \geq \sigma^+ N^2 - \mathsf{g}$. □

Proposition 17.19 (Flux lines of every integer length)**.** Assume the criterion; let $N \geq 4l$ be an integer, n_0, κ, τ, χ_0, E_0, $\hat{N}'$, h and g' as in Thm. AW(g2), $\mathsf{E} = \sigma^+ n_0^2$ and $t_i = 2^i n_0$. Then $6 \leq \hat{N}' < \mathsf{q}N/l + 2$, $\hat{N}' \leq 1.3453\,\mathsf{E}_0 + 2$, $\mathsf{h} \leq 0.4274\,\kappa + 0.0208$, and $t_i \varepsilon_{\mathrm{v}}(t_i, N) \leq \exp[-2^i(\mathsf{E}_0 - \mathsf{g}')]$ for all $i \geq 0$. ■

Proof sketch. Numbers. $\log\frac{1}{\chi_0} = -\log 0.3602 + 2w\Lambda \geq 1.48677$; the bounds on $\hat{N}'$ follow from $2\mathsf{E}_0/\log\frac{1}{\chi_0} < 2\kappa\mathsf{q}w = \mathsf{q}N/l$, from the fact that the quotient exceeds 4 at $w = 2$ and increases in w, and from $2/1.48677 \leq 1.3453$. As $\tau = \theta^{2w} \leq 0.8901^4$, the number $c_\tau = 0.01655\,\tau$ satisfies

$c_\tau \le 0.010389$, and $\mathsf{h} = c_\tau \hat N'$. With $4\mathsf{q}\cdot 0.01655\, we^{-2w\Lambda} \le 0.8548$ (as in the proof of Thm. AW(d)), $\mathsf{h} < 0.01655(2\kappa\mathsf{q}w+2)\tau \le 0.4274\kappa + 0.0208$. By the choice of $\hat N'$, $\chi_0^{\hat N'/2} \le e^{-\mathsf{E}_0}$. Here $\mathsf{E}_0 = \kappa\mathsf{E}$ and $\mathsf{E} = \sigma^+ n_0^2 = \varsigma w^2 \ge 4\cdot 1.90284 = 7.61136$. For $\kappa\in[1,2]$, $E \ge 7.6$ and $0 < c \le 0.052$ the functions

$$\gamma_c(\kappa, E) = \kappa\big[\log(1.3453\,\kappa E + 2) + c\,(1.3453\,\kappa E + 2) - E\big]$$

increase in c, are convex in κ and decrease in E. As $\log\hat N' \le \log(1.3453\,\kappa\mathsf{E}+2)$ and $\mathsf{h} \le c_\tau(1.3453\,\kappa\mathsf{E}+2)$,

$$\log\big[\hat N' e^{-\mathsf{E}+\mathsf{h}}\big]^\kappa \le \gamma_{0.010389}(\kappa,\mathsf{E}), \qquad \log\big[\hat N' e^{-\mathsf{E}+5\mathsf{h}}\big]^\kappa \le \gamma_{0.051945}(\kappa,\mathsf{E}), \qquad 0.051945 = 5\cdot 0.010389.$$

On $\kappa\in[1,2]$, $E \ge 7.61136$ each of the two functions is largest at $E = 7.61136$ and at $\kappa = 1$ or $\kappa = 2$, where $\gamma_{0.010389}$ has approximate values -4.97953 and -8.53047, and $\gamma_{0.051945}$ approximate values -4.47090 and -6.66219. So $\bar z_0 := [\hat N' e^{-\mathsf{E}+\mathsf{h}}]^\kappa \le e^{-4.97952} \le 0.00688$ and $6.1\,[\hat N' e^{-\mathsf{E}+5\mathsf{h}}]^\kappa \le 6.1\,e^{-4.47090} \le 0.0698$. *The start.* The first inequality of Lem. 17.15(b) for the tube $n_0\times n_0$ at the time $\hat N' n_0$, Prop. 17.17(c) and $\mathsf{r}_{\hat N' n_0}(n_0,n_0) \le \chi_0^{\hat N'} \le e^{-\mathsf{E}}$ give $z_-(n_0, \hat N' n_0, n_0) \le \hat N' e^{-\mathsf{E}}$; with $\delta(n_0, \hat N' n_0, n_0) \le \hat N' \delta_{\mathrm c}(n_0) \le \mathsf{h}$, Lem. 17.15(a) at the real factor κ in the third side gives $z_-(n_0, \hat N' n_0, N) \le \bar z_0$; and the second inequality of Lem. 17.15(b) for the tube $n_0\times N$ at the time $\hat N' n_0/2$, with $\mathsf{r}_{\hat N' n_0/2}(n_0,N) \le (0.14\chi_0)^{\hat N'/2} \le \bar z_0$ (Prop. 17.18(b)), gives $n_0\varepsilon_{\mathrm v}(n_0,N) \le \frac{6.1}{\hat N'}\bar z_0 = e^{-E_0'}$, $E_0' = \mathsf{E}_0 - (\kappa-1)\log\hat N' - \kappa\mathsf{h} - \log 6.1$.

The recursion. As in Prop. 17.18(c), with the slab of the integer thickness N of Prop. 17.18(b), one doubling of the width squares $t\varepsilon_{\mathrm v}(t,N)$: $E_{i+1}' = 2E_i' - \log\hat N' - \log 4.1 - \xi_i'$, with $\xi_0' = 4\kappa\mathsf{h}$ and $\xi_i' = 0.532\,\hat N'\chi_0^{2^i}$ for $i\ge 1$; the bound $6.1[\hat N' e^{-\mathsf{E}+5\mathsf{h}}]^\kappa \le 0.0698$ keeps $y_i = \hat N' e^{-E_i'}$ so small that the step closes at every i, and $\sum_i 2^{-i-1}\xi_i' \le (2\kappa+0.68)\mathsf{h}$. Hence $2^{-i}E_i' \ge E_0' - \log\hat N' - \log 4.1 - (2\kappa + 0.68)\mathsf{h} \ge \mathsf{E}_0 - \mathsf{g}'$, as $\log 6.1 + \log 4.1 \le 3.22$. More precisely, the unused positive tail of this same recurrence gives

$$\begin{aligned} E_i' &= 2^i\Big(E_0' - \log\hat N' - \log 4.1 - \sum_{j\ge 0} 2^{-j-1}\xi_j'\Big) + \log\hat N' + \log 4.1 + \sum_{j\ge i} 2^{i-j-1}\xi_j' \\ &\ge 2^i(\mathsf{E}_0 - \mathsf{g}') + \log\hat N' + \log 4.1 + \tfrac12\xi_i'. \end{aligned}$$

Thus subtracting $\log\hat N' + \xi_i'/4$ still leaves the floor $2^i(\mathsf{E}_0 - \mathsf{g}')$, also at $i = 0$. □

Proposition 17.20 (A local function in a long box)**.** Let $\beta \ge 0$, Σ a section, $h \ge 1$ and $t \ge h+2$ integers, $s = t - h - 1$, and A a bounded Borel function of the variables of h consecutive layers and of the crossings between them. There is a number $a_\infty(\Sigma, A)$, $|a_\infty| \le \sup|A|$, that depends neither on t nor on the position of the layers, with $|\langle A\rangle_{(t,\Sigma)} - a_\infty(\Sigma,A)| \le 2\sup|A|\,\mathsf{r}_s(\Sigma)$. As $\mathsf{r}_s(\Sigma)\to 0$, the laws of the boxes (t,Σ) converge, as $t\to\infty$, on the bounded Borel functions of finitely many layers: to the law of the infinite cylinder over Σ. ■

Proof sketch. Integrating the weights of the h layers and the $h+1$ crossings with A inserted gives a kernel $\hat A$ dominated by $\sup|A|$ times that of T_0^{h+1}, so $\|\hat A\| \le \sup|A|\lambda_0^{h+1}$, and $Z\langle A\rangle = \mathrm{Tr}(\mathsf{T}_0^s\hat A) = \sum_i \lambda_i^s\langle e_i, \hat A e_i\rangle$ in an eigenbasis. With $a_\infty = \lambda_0^{-h-1}\langle e_0, \hat A e_0\rangle$, which is determined because λ_0 is simple: $Z\langle A\rangle = \lambda_0^t(a_\infty + \vartheta\sup|A|\,\mathsf{r}_s)$ with $|\vartheta|\le 1$, and $Z = \lambda_0^t(1+\mathsf{r}_t)$. The thermal number tends to 0 by dominated convergence, λ_0 being simple and $\sum\lambda_i^2 < \infty$; the limits are consistent, and Kolmogorov's theorem gives the law of the cylinder. □

Lemma 17.21 (Mirrors of a box)**.** Fix a box and a direction μ as time; let $c\in\mathbb{Z}+\frac12$, and $\mathfrak{r}$ the reflection $x_\mu \mapsto 2c - x_\mu$ of the box.

(a) Let $\vartheta \ge 2$ be an integer with $2\vartheta \le n_\mu$, and A a bounded Borel function of the bonds with both ends in the ϑ layers below c. Then $\langle A\,\overline{\mathfrak{r}A}\rangle \ge 0$ and $|\langle A\rangle|^2 \le \langle A\,\overline{\mathfrak{r}A}\rangle$; n_μ may be odd. If $n_\mu = 2\vartheta$ and A' is another such function, $|\langle A\,\overline{\mathfrak{r}A'}\rangle|^2 \le \langle A\,\overline{\mathfrak{r}A}\rangle\langle A'\,\overline{\mathfrak{r}A'}\rangle$.

(b) Let $N = 2h \ge 4$, $c \in (\mathbb{Z} + \frac{1}{2})^3$, and H_ϵ, $\epsilon \in \{0,1\}^3$, functions of the cell $\mathsf{C}_h(c)$. On the cube (N,N,N), $|\langle \prod_\epsilon H_\epsilon^{[\epsilon]} \rangle| \le \prod_\epsilon |\!|\!| H_\epsilon |\!|\!|_N$; so $|\!|\!| \cdot |\!|\!|_N$ is a seminorm, $|\langle A \rangle_{(N,N,N)}| \le |\!|\!| A |\!|\!|_N \le \sup |A|$, the mean of the product of the copies of A over $\epsilon \in E$ is at most $|\!|\!| A |\!|\!|_N^{\#E}$ in modulus for every $E \subset \{0,1\}^3$, and $|\!|\!| A |\!|\!|_N^8 \le \langle |A|^8 \rangle_{(N,N,N)}$.

(c) Let n_μ be even and the mirror lie between the layers 0 and 1 of the direction μ; let x_p, x_q be two sites of the layer 1, equal or not, z_p, z_q the values of U at the bonds from x_p to $\mathfrak{r} x_p$ and from x_q to $\mathfrak{r} x_q$, and A_{jk}, $j,k \in \{1,2\}$, bounded Borel functions of the bonds with both ends in the layers $1, \dots, n_\mu/2$. Then

$$\Big\langle \sum_{i,j,k,l} \overline{(\mathsf{z}_p)_{ji}}\, (\mathsf{z}_q)_{kl}\, A_{jk}\, \overline{\mathfrak{r} A_{il}} \Big\rangle \ge 0.$$

■

Proof sketch. Let $\mathsf{T}' = \mathsf{K}^{1/2} M_0^2 \mathsf{K}^{1/2}$, which has the eigenvalues of T_0. For a function A of ϑ consecutive layers and the $\vartheta - 1$ crossings between them let S_A be the operator whose kernel, a function of the first and the last of these layers, is the integral of A against the weights of the ϑ layers and of the inner crossings over all other variables, and $\mathsf{Y}_A = \mathsf{K}^{1/2} \mathsf{S}_A \mathsf{K}^{1/2}$, a Hilbert–Schmidt operator; $\mathsf{Y}_1 = \mathsf{T}'^{\vartheta}$. Integrating the layers in turn as in Lem. 17.10, the mean of a product of such functions on consecutive slabs, with free layers between them, is the trace of the product of their operators and of powers of T'; and, the weights being invariant under $\mathfrak{r}$ because $A_\times(V', S^{-1}, V) = A_\times(V, S, V')$, the operator of $\overline{\mathfrak{r} A}$ on the mirrored slab is Y_A^*.

So $Z\langle A \rangle = \mathrm{Tr}(\mathsf{T}'^{n_\mu - \vartheta} \mathsf{Y}_A)$ and $Z\langle A\, \overline{\mathfrak{r} A'} \rangle = \mathrm{Tr}(\mathsf{T}'^{n_\mu - 2\vartheta} \mathsf{Y}_{A'}^* \mathsf{Y}_A)$. (a) Write $\mathrm{Tr}(\mathsf{T}'^{n_\mu - \vartheta} \mathsf{Y}_A) = \mathrm{Tr}(\mathsf{T}'^{n_\mu/2} \cdot \mathsf{Y}_A \mathsf{T}'^{v})$ with $v = \frac{1}{2}(n_\mu - 2\vartheta) \ge 0$, a real power of a positive operator; Schwarz's inequality for Hilbert–Schmidt operators gives $|Z\langle A \rangle|^2 \le \mathrm{Tr}\, \mathsf{T}'^{n_\mu} \, \|\mathsf{Y}_A \mathsf{T}'^{v}\|_2^2 = Z \cdot Z\langle A\, \overline{\mathfrak{r} A} \rangle$, with $\|\mathsf{Y}_A \mathsf{T}'^{v}\|_2^2 = \mathrm{Tr}(\mathsf{T}'^{2v} \mathsf{Y}_A^* \mathsf{Y}_A) \ge 0$; at $n_\mu = 2\vartheta$, Schwarz's inequality for $\mathrm{Tr}(\mathsf{Y}_{A'}^* \mathsf{Y}_A)$ is the second statement. (b) The reflection $\mathfrak{r}_\mu$ of the cube fixes the two bond mirrors $x_\mu = c_\mu$ and $x_\mu = c_\mu + h$, and the half below the first of them is the slab of the cell.

Group the eight factors into the four with $\epsilon_\mu = 0$ and their four mirror images: the second statement of (a) gives $|\Lambda(H)|^2 \le \Lambda(H^0)\Lambda(H^1)$ for $\Lambda(H) = \langle \prod_\epsilon H_\epsilon^{[\epsilon]} \rangle$, where H^j is the family with the index ϵ_μ set to j; three directions give the first bound, and the mean of the eight copies of one function is ≥ 0. Expanding $|\!|\!| H + H' |\!|\!|_N^8$ into 2^8 such means gives the triangle inequality. With $H_\epsilon = A$ for $\epsilon \in E$ and $H_\epsilon = 1$ otherwise one gets the bound for subsets; Hölder's inequality with eight exponents 8 and the invariance of the cube law under the $\mathfrak{r}_\mu$ give the last bound. (c) Take μ as time; the mirror also fixes the plane between the layers $n_\mu/2$ and $n_\mu/2 + 1$.

Integrating in each half the layers and crossings between its two outer layers produces from A_{jk} a function Φ_{jk} of $(V^1, V^{n_\mu/2})$, and from $\overline{\mathfrak{r} A_{il}}$, by the symmetry of the weights, $\overline{\Phi_{il}}$ at the mirrored layers 0 and $n_\mu/2 + 1$. The crossing between the layers 0 and 1 carries the two cut bonds, which enter exactly as in the operator $\hat{\mathsf{K}}$ of Lem. 17.12 with the charges at the sites of x_p and x_q: the function under the mean is $\mathrm{tr}[(\overline{\mathfrak{r} A})^{\mathrm{t}} \mathsf{z}_p^{-1} A\, \mathsf{z}_q]$ for the matrices $A = (A_{jk})$, $\overline{\mathfrak{r} A} = (\overline{\mathfrak{r} A_{il}})$. The crossing between $n_\mu/2$ and $n_\mu/2 + 1$ enters through K.

Hence Z times the mean equals $\langle \Phi, (\hat{\mathsf{K}} \otimes \mathsf{K}) \Phi \rangle \ge 0$, by Lem. 17.12(a), (e), whose proof uses of the two sites only that they lie in the section, and the positivity of K. □

Lemma 17.22 (Comparison of a functional with its dominant state)**.** Let $\mathcal{A}$ be a real vector space with an element 1, N a seminorm and g a linear functional on $\mathcal{A}$ such that, with $\mathsf{R}(x) = (1+x)^8 - 1 - 8x$,

$$\mathsf{N}(1 + tG)^8 \le 1 + 8t\, \mathsf{g}(G) + \mathsf{R}(|t| \mathsf{N}(G)) \quad \text{and} \quad |\mathsf{g}(G)| \le \mathsf{N}(G)$$

for all $G \in \mathcal{A}$ and all real t. Let $\mathfrak{s}$ be a linear functional with $\mathfrak{s}(1) = 1$ and $\mathfrak{s}(G) \le Q\, \mathsf{N}(G)$ for all G, where $1 \le Q \le 1.11$. Then $|\mathfrak{s}(G) - \mathsf{g}(G)| \le 4.84\, [Q(Q-1)]^{1/2}\, \mathsf{N}(G)$. ■

Proof sketch. From $(1+u)^{1/8} \le 1+u/8$, $\mathsf{N}(1+tG) \le 1 + t\mathsf{g}(G) + \frac{1}{8}\mathsf{R}(x)$, $x = |t|\mathsf{N}(G)$; so $1+t\mathfrak{s}(G) = \mathfrak{s}(1+tG) \le Q[1+t\mathsf{g}(G)+\frac{1}{8}\mathsf{R}(x)]$, that is $t[\mathfrak{s}(G)-\mathsf{g}(G)] \le (Q-1)(1+x) + \frac{Q}{8}\mathsf{R}(x)$. If $\mathsf{N}(G)=0$ the bracket vanishes, and if $Q=1$ it vanishes as $x\to 0$. Otherwise take $t = \pm x/\mathsf{N}(G)$ with $x^2 = (Q-1)/4.74Q$, so that $x \le 0.1446$ and $\mathsf{R}(x)/8x^2 \le 4.718$:

$$|\mathfrak{s}(G)-\mathsf{g}(G)| \le \mathsf{N}(G)\Big[\frac{Q-1}{x} + (Q-1) + 4.74\,Qx\Big] = \mathsf{N}(G)\,[Q(Q-1)]^{1/2}\Big[2\sqrt{4.74} + \Big(\frac{Q-1}{Q}\Big)^{1/2}\Big],$$

and $2\sqrt{4.74} + (0.11/1.11)^{1/2} \le 4.67$. □

Proposition 17.23 (Boxes of every shape)**.** Let $\beta \ge 0$, $N = 2h \ge 4$, $c \in (\mathbb{Z}+\frac{1}{2})^3$, $\underline{n}$ a box with all sides $\ge N$, and $M_\mu = 2^{j_\mu}h$, $j_\mu \ge 1$, the largest number of this form that is $\le n_\mu$. For every function A of the cell $\mathsf{C}_h(c)$:

(a) $|\langle A\rangle_{\underline{n}}| \le \exp\{\frac{\delta_{\mathrm{c}}(N)}{8}[1+3N(\frac{1}{M_1}+\frac{1}{M_2}+\frac{1}{M_3})]\}\,\|\!|A|\!\|_N \le e^{\frac{5}{4}\delta_{\mathrm{c}}(N)}\,\|\!|A|\!\|_N$;

(b) if A is real and $\delta_{\mathrm{c}}(N) \le \frac{1}{2}$, then $|\langle A\rangle_{\underline{n}} - \langle A\rangle_{(N,N,N)}| \le 7.31\,\delta_{\mathrm{c}}(N)^{1/2}\,\|\!|A|\!\|_N$.

Under the criterion at the side l and for $N \ge 2l$: $|\langle A\rangle_{\underline{n}} - \langle A\rangle_{(N,N,N)}| \le 2.71\,\theta^{N/4l}\,\|\!|A|\!\|_N$ for every real A. None of these constants depends on the shape of the box. ■

Proof sketch. (a) *Doubling on the one box.* In the direction 1 put $A_0 = A$ and $A_{i+1} = A_i\,\overline{\mathfrak{r}^{(i)}A_i}$, $\mathfrak{r}^{(i)}$ the reflection in the plane $x_1 = c_1 + (2^i-1)h$; A_i is a function of the bonds of the 2^ih layers below that plane, and by Lem. 17.21(a), as long as $2^{i+1}h \le n_1$, $|\langle A_i\rangle_{\underline{n}}|^2 \le \langle A_{i+1}\rangle_{\underline{n}}$. So $|\langle A\rangle_{\underline{n}}| \le \langle A_{j_1}\rangle_{\underline{n}}^{2^{-j_1}}$, and A_{j_1} is a row of M_1/h alternating copies of the cell, which lies below c_2 and c_3; the same in the directions 2 and 3, always on the box $\underline{n}$, gives $|\langle A\rangle_{\underline{n}}| \le \langle \mathfrak{B}A\rangle_{\underline{n}}^{1/\mathcal{N}}$, with $\mathfrak{B}A$ an array of $\mathcal{N} = M_1M_2M_3/h^3$ copies on a block of $M_1\times M_2\times M_3$ sites.

The free layers cost their top eigenvalue. With the time 1 and Y the operator of the slab of h layers of $\mathfrak{B}_3\mathfrak{B}_2A$, $Z(\underline{m})\langle\mathfrak{B}A\rangle_{\underline{m}} = \mathrm{Tr}(\mathsf{T}'^{m_1-M_1}(\mathsf{Y}^*\mathsf{Y})^{M_1/N})$, and $\mathrm{Tr}(\mathsf{O}_1\mathsf{O}_2) \le \|\mathsf{O}_1\|\,\mathrm{Tr}\,\mathsf{O}_2$ for positive operators $\mathsf{O}_1,\mathsf{O}_2$, the second of trace class, gives

$$Z(\underline{m})\langle\mathfrak{B}A\rangle_{\underline{m}} \le \lambda_0(m_2,m_3)^{m_1-M_1}\,Z(M_1,m_2,m_3)\langle\mathfrak{B}A\rangle_{(M_1,m_2,m_3)};$$

with the times 2 and 3 in turn,

$$Z(\underline{n})\langle\mathfrak{B}A\rangle_{\underline{n}} \le \lambda_0(n_2,n_3)^{n_1-M_1}\lambda_0(M_1,n_3)^{n_2-M_2}\lambda_0(M_1,M_2)^{n_3-M_3}\,Z(\underline{M})\langle\mathfrak{B}A\rangle_{\underline{M}}.$$

On the block $\underline{M}$, whose sides lie in $N\mathbb{N}$, $\mathrm{Tr}(\mathsf{Y}^*\mathsf{Y})^q \le (\mathrm{Tr}\,\mathsf{Y}^*\mathsf{Y})^q$ in each direction gives $Z(\underline{M})\langle\mathfrak{B}A\rangle_{\underline{M}} \le [Z(N,N,N)\,\|\!|A|\!\|_N^8]^{M_1M_2M_3/N^3}$. *The count.* Write $\log\lambda_0(\mathsf{A},\mathsf{B}) = \mathsf{AB}P + \varepsilon(\mathsf{A},\mathsf{B})$ and $F = |\cdot|P+\delta$: the terms with P cancel, and with $\varepsilon(\mathsf{A},\mathsf{B}) \le (\mathsf{A}+\mathsf{B})\delta_{\mathrm{c}}(N)/N^2$ (Lem. 17.11(g)), $n_\mu < 2M_\mu$ and $\delta(\underline{n}) \ge 0$, the remaining exponent is at most $3\frac{\delta_{\mathrm{c}}(N)}{N^2}(M_1M_2+M_1M_3+M_2M_3) + \frac{M_1M_2M_3}{N^3}\delta_{\mathrm{c}}(N)$; divided by $\mathcal{N} = 8M_1M_2M_3/N^3$ this is the exponent of (a), and $M_\mu \ge N$. (b) Lem. 17.22 with $\mathcal{A}$ the real functions of the cell, $\mathsf{N} = \|\!|\cdot|\!\|_N$, $\mathsf{g} = \langle\cdot\rangle_{(N,N,N)}$, $\mathfrak{s} = \langle\cdot\rangle_{\underline{n}}$ and $Q = e^{\frac{5}{4}\delta_{\mathrm{c}}(N)}$: by Lem. 17.21(b), $\|\!|1+tG|\!\|_N^8$ is the sum over $E\subset\{0,1\}^3$ of $t^{\#E}$ times the mean of the copies of G over E, which is 1 for $E=\emptyset$, $\langle G\rangle_{(N,N,N)}$ for the eight one-element E, and at most $\|\!|G|\!\|_N^{\#E}$ else; and $\mathfrak{s}(G) \le Q\mathsf{N}(G)$ is (a).

So, if $\delta_{\mathrm{c}}(N) \le 0.08348$, which suffices for $Q \le 1.11$, the difference is at most $4.84\,[Q(Q-1)]^{1/2}\|\!|A|\!\|_N \le 5.86\,\delta_{\mathrm{c}}(N)^{1/2}\|\!|A|\!\|_N$. For $0.08348 < \delta_{\mathrm{c}}(N) \le \frac{1}{2}$, (a) and Lem. 17.21(b) bound it by $(e^{\frac{5}{4}\delta_{\mathrm{c}}(N)}+1)\|\!|A|\!\|_N$, and $(e^{5\delta/4}+1)\delta^{-1/2}$ decreases on $(0,\frac{1}{2}]$ and is ≤ 7.31 at 0.08348. Under the criterion $\delta_{\mathrm{c}}(N) < 0.1324\,\theta^{N/2l} \le \frac{1}{2}$ (Prop. 17.16(d)), and $7.31\cdot 0.1324^{1/2} \le 2.71$. □

Lemma 17.24 (The excess controls every periodic state)**.** Let $\beta > 0$, let $N = 2\mathsf{L}^j$ be the side of one of the tori of the chessboard estimate (Prop. YM3_I:86.3), and let B be the set of the bonds with both ends in one of the eight closed cubes of side $N/2$, with faces through midpoints of bonds,

that tile the torus (N,N,N). For every periodic Gibbs state s the law of s on B has a density h with respect to the law of the torus (N,N,N) on B, with $\int h^{8/7} \le e^{\delta_c(N)/7}$, and

$$\sup_A |s(A) - \langle \mathbf{1}_A \rangle_{(N,N,N)}| \le \tfrac{1}{2}(\sqrt{\epsilon} + \epsilon), \qquad \epsilon = 22.3\big(e^{\delta_c(N)/7} - 1\big),$$

the supremum over the Borel events A of the bonds of B. ■

Proof sketch. One board for all sides. Let $N' = 2\mathsf{L}^{j'}$, $j' \ge j$, be a side of the family. The cubes of side $N/2$, with faces through midpoints of bonds, that are the translates of the closed cube of B by the multiples of $N/2$ in the three directions tile the torus (N',N',N'), $2N'/N$ of them a side, an even number, as N'/N is a power of the odd number L: these *cells* form one board, and the planes between neighbouring cells are its walls. Number the cells by $k \in \{0,\dots,2N'/N-1\}^3$, the cell of B being $k=0$. For a bounded Borel function f of the bonds of B the copy $f^{[k]}$ is f composed, for $\mu = 1,2,3$, with the reflections in the k_μ walls of the direction μ that carry the cell 0 step by step to the cell k, and complex conjugated if $k_1+k_2+k_3$ is odd; it is a function of the bonds of the cell k. Put

$$\|f\|_{N'} = \Big\langle \prod_k f^{[k]} \Big\rangle_{(N',N',N')}^{(N/2N')^3},$$

the product over all $(2N'/N)^3$ cells. At $N' = N$ the cell of B is a cell $\mathsf{C}_{N/2}(c)$, the eight copies are the copies $f^{[\epsilon]}$, $\epsilon \in \{0,1\}^3$, by which $|\!|\!| \cdot |\!|\!|_N$ is defined, and $\|f\|_N = |\!|\!| f |\!|\!|_N$. With the time 1 the torus (N',N',N') is cut by the walls into $2N'/N$ slabs of $N/2$ layers, and, as in the proof of Prop. 17.23, with Y the operator of the first slab of the product (that of the cells with $k_1 = 0$), the operator of the next slab is Y^*, and $Z(N',N',N')\langle \prod_k f^{[k]} \rangle_{(N',N',N')} = \mathrm{Tr}(\mathsf{Y}^*\mathsf{Y})^{N'/N}$, a number ≥ 0 and at most $(\mathrm{Tr}\,\mathsf{Y}^*\mathsf{Y})^{N'/N}$, where $\mathrm{Tr}\,\mathsf{Y}^*\mathsf{Y}$ is the same expression on the box (N,N',N'). The same in the directions 2 and 3 gives the chessboard estimate on this board for the two tori (Prop. YM3_I:86.3, in the form of Cor. YM3_I:86.6):

$$Z(N',N',N')\,\|f\|_{N'}^{(2N'/N)^3} \le \big[Z(N,N,N)\,\|f\|_N^8\big]^{(N'/N)^3}.$$

Its root of the order $(2N'/N)^3$ is

$$\|f\|_{N'} \le \|f\|_N \exp\Big[\Big(\frac{N}{2}\Big)^3\Big(\frac{F(N,N,N)}{N^3} - \frac{F(N',N',N')}{N'^3}\Big)\Big] \le \|f\|_N\, e^{\delta_c(N)/8},$$

since $(N/2N')^3(N'/N)^3 = \frac{1}{8}$, $(N/2N')^3 = (N/2)^3/N'^3$ and $F(N',N',N')/N'^3 \ge P$. The domination of periodic Gibbs states (Prop. YM3_I:91.9), read with this same board and seminorm, gives $|s(f)| \le \liminf_{j'\to\infty} \|f\|_{N'}$. Hölder's inequality and the reflection invariance of the torus law give $\|f\|_N \le \langle |f|^8 \rangle_{(N,N,N)}^{1/8}$. So $|s(f)| \le e^{\delta_c(N)/8}\|f\|_{L^8}$, and by duality s has on B a density h with $\|h\|_{8/7} \le e^{\delta_c(N)/8}$. With $p = \frac{8}{7}$, $u^p - 1 - p(u-1) \ge 0.045\min\{(u-1)^2, |u-1|^p\}$ for $u \ge 0$; integrating at $u = h$ gives $\int_{|h-1|\le 1}(h-1)^2 + \int_{|h-1|>1}|h-1|^p \le \epsilon$, so $\int |h-1| \le \sqrt{\epsilon} + \epsilon$, and the total variation is half of it. □

Lemma 17.25 (One bond; a bound by the perimeter)**.**

(a) For $\kappa \ge 0$ let $\langle\cdot\rangle_\kappa$ be the probability measure on G with the density proportional to $e^{\kappa x_0(V)}$ with respect to the Haar measure, $x_0(V) = \frac{1}{2}\,\mathrm{tr}\,V$, and $r(\kappa) = \langle x_0 \rangle_\kappa$. Then $\langle V \rangle_\kappa = r(\kappa)\mathbf{1}$ and

$$\frac{\kappa}{\kappa+4} \le r(\kappa) \le \frac{2\kappa}{2\kappa+3}.$$

(b) Let $\beta \ge 0$, s a Gibbs state, periodic or not, C a closed lattice path and B a finite set of bonds which C runs through exactly once each, as b or as $\bar{b}$, no plaquette containing two of them. Then $|s(W_C)| \le (\frac{8\beta}{8\beta+3})^{\#B} \le \exp[-3\#B/(8\beta+3)]$. For a rectangle with $R,T \ge 2$ one may take $\#B = 2(R+T)-4$.

■

Proof sketch. (a) Conjugation invariance makes $\langle V\rangle_\kappa$ commute with G, so it is a multiple of $\mathbf{1}$, and the multiple is $r(\kappa)$. Write $V = x_0\mathbf{1} + \mathrm{i}\sum_j x_j\sigma_j$ with $\sum_{j=0}^3 x_j^2 = 1$. By the invariance of the Haar measure under $V \mapsto e^{\mathrm{i}s\sigma_j}V$, the derivatives at $s=0$ of $\int (x_j e^{\kappa x_0})(e^{\mathrm{i}s\sigma_j}V)\,\mathrm{d}V$ and of $\int (x_j x_0 e^{\kappa x_0})(e^{\mathrm{i}s\sigma_j}V)\,\mathrm{d}V$ vanish; as the derivatives of x_0 and x_j along this path are $-x_j$ and x_0, summing over $j=1,2,3$ gives

$$3\,r(\kappa) = \kappa\langle 1 - x_0^2\rangle_\kappa, \qquad 4\langle x_0^2\rangle_\kappa - 1 = \kappa\langle x_0(1-x_0^2)\rangle_\kappa.$$

The right member of the second identity lies in $[0, \kappa\langle 1-x_0^2\rangle_\kappa]$, its sign by $V\mapsto -V$; so $(4+\kappa)\langle x_0^2\rangle_\kappa \le 1+\kappa$, and the first identity gives $r(\kappa) \ge \frac{\kappa}{3}\cdot\frac{3}{4+\kappa}$; and $3r = \kappa\langle (1-x_0)(1+x_0)\rangle \le 2\kappa(1-r)$. (b) As no plaquette contains two bonds of B, given the other bonds the weight of the bonds of B factorises: $V(b)$ has the density proportional to $\exp[\beta \mathsf{k}_b \frac12 \operatorname{tr}(V(b)g_b)]$, where the sum of the four staples of b is $\mathsf{k}_b g_b$, $g_b \in G$, $0 \le \mathsf{k}_b \le 4$, as a sum of four unit quaternions. W_C is a sum of products that hold one entry of each $V(b)^{\pm1}$, $b\in B$; by (a) and $V\mapsto Vg^{-1}$ the conditional mean of $V(b)^{\pm1}$ is $r(\beta\mathsf{k}_b)g_b^{\mp1}$, so the conditional mean of W_C is $\prod_b r(\beta\mathsf{k}_b)$ times half the trace of an element of G, of modulus at most $(\frac{8\beta}{8\beta+3})^{\#B}$; the Gibbs equation of B carries this to s. In a rectangle with $R,T\ge 2$ take all bonds but the first of each side: two bonds of one plaquette are either perpendicular with a common end, which happens on C only at a corner, or parallel at the distance 1, which does not happen for opposite sides at the distance ≥ 2. Finally $\log(1+y) \ge y/(1+y)$ at $y = 3/8\beta$. □

17.1.6. *Proofs of the results.*

Proof of Thm. AW. Step 1 (the cubes converge). In a box (t,Σ) with $t\in\{n,2n\}$ a function A of a cube of $h \le n/2-1$ layers is a function of h consecutive layers, and Prop. 17.20 applies with $s \ge n/2$ and one and the same $a_\infty(\Sigma, A)$ for both t. Doubling the three sides one after the other, each time with the time in the doubled direction,

$$|\langle A\rangle_{(n,n,n)} - \langle A\rangle_{(2n,2n,2n)}| \le 4\sup|A|\,[\mathsf{r}_{n/2}(n,n) + \mathsf{r}_{n/2}(2n,n) + \mathsf{r}_{n/2}(2n,2n)] \le 4 \cdot 4.15727\sup|A|\,\mathsf{e}(n/2) \le 1.29\sup|A|\,\theta^{n/2l}$$

by Prop. 17.16(d). Summing over $n, 2n, 4n, \dots$ with $\theta^{2^i n/2l} \le \theta^{n/2l}\theta^{in/2l}$ gives the second bound of (a). On each finite set of bonds the cube laws form a Cauchy sequence in total variation; the limits are consistent, and Kolmogorov's theorem gives ν.

Step 2 (Gibbs state, symmetries). For a finite set $\mathfrak{A}$ of bonds and a large cube, the action is the sum over the plaquettes that meet $\mathfrak{A}$ plus a remainder that does not depend on the bonds of $\mathfrak{A}$; integrating these bonds first shows that every cube law, and so ν, satisfies the Gibbs equation of $\mathfrak{A}$. A symmetry of the lattice induces a map of each cube of side $2^k l$ that keeps dU and the action, hence each cube law and ν.

Step 3 (one periodic state). By Step 2, ν is periodic. Let s be a periodic Gibbs state and A a local event. For all large $N = 2\mathsf{L}^j$, A is an event of the bonds inside one cell of side $N/2$, and Lem. 17.24, applied to s and to ν, gives $|s(A) - \nu(A)| \le \sqrt{\epsilon} + \epsilon$ with $\epsilon = 22.3(e^{\delta_{\mathrm{c}}(N)/7} - 1)$, which tends to 0 as $\delta_{\mathrm{c}}(N) < 0.1324\,\theta^{N/2l}$ (Prop. 17.16(d)). So $s = \nu$. Prop. YM3_I:103.4 gives reflection positivity at every lattice mirror. By Prop. YM3_I:106.2, the ergodic components of ν are periodic Gibbs states; uniqueness makes them equal to ν, so ν is ergodic.

Step 4 (all boxes). Let the three sides of boxes $\underline{n}^{(k)}$ tend to infinity. Their laws on Ω lie in a compact set; every limit point satisfies the Gibbs equations, as in Step 2, and is invariant under all translations, as every box law is; so it is a periodic Gibbs state, that is ν, and the whole sequence converges.

Step 5 (twisted cubes). Let $n \ge 2\mathsf{q}l$ be an integer, j the largest integer with $4^j \le n/2\mathsf{q}l$, $w = 2^j$, $l' = l_j = 2wl$ and $M = n/l'$, a real number with $w\mathsf{q} \le M < 4w\mathsf{q}$ and $M^2 < 2\mathsf{q}n/l$. Lem. 17.15(a)

from the box (l',n,l') to the cube, with $\kappa = M^2$, and Prop. 17.17(d) give

$$z_-(n,n,n) \le \big[M e^{-\varsigma w^2 + 0.01655\, M\theta^{2w}}\big]^{M^2}, \qquad M^2\varsigma w^2 = \sigma^+ n^2,$$

and $0.01655\, M\theta^{2w} < 4\mathsf{q}\cdot 0.01655\, we^{-2w\Lambda} \le \frac{4\cdot 0.01655}{e}(\frac{1}{2\Lambda} + \frac{1.67}{4\Lambda^2}) \le 0.855$ and $\log M < \frac12 \log\frac{2\mathsf{q}n}{l}$: the first bound of (d). The bound for all seven $z \neq 0$ then follows from Lem. 17.10(c).

Step 6 (flux sectors, static charges). With the time 1 over $n\times n$ and $e \in \{01, 11\}$, $\Lambda_e^n \le W_n(01;0)+W_n(11;0) = a_-(n,n,n) = z_-\lambda_0^n(1+\mathsf{r}_n(n,n))$; for $e = 10$ the same holds because $Z^{(1,0,0)} = Z^{(0,1,0)}$ on the cube (Lem. 17.10(iii)). So $\log(\lambda_0/\Lambda_e) \ge \frac1n[-\log z_-(n,n,n) - \log(1+\mathsf{r}_n(n,n))]$, and $\log(1+\mathsf{r}_n(n,n))$ is at most $\mathsf{r}_n(n,n) \le 0.001615\,\theta^{n/l} \le 0.0017$ at $n = 2^i l$, and in general, with $n_0 = 2^i l \le n < 2n_0$, at most $\delta_{\mathrm{c}}(n) \le 8\delta_{\mathrm{c}}(n_0) \le 8\cdot 0.01655\cdot 0.792279 \le 0.105$ (Lem. 17.11(f), Prop. 17.16(d)). With Step 5 this is the first statement of (e); the second is Lem. 17.12(d) on the cube with $n_3 = n$, and Step 5. For any two sites $o \neq r$ it is Lem. 17.12(e) on the cube, with the time 1 or 2, whichever direction of the section realises $|r-o|_\infty$, and Step 5; for fixed o, r and $n = 2^i l$, $i\to\infty$, the factor $[8(1+\mathsf{r}_n(n,n))]^{1/n}$ tends to 1 and the correction $\frac{2\mathsf{q}R}{ln}(\frac12\log\frac{2\mathsf{q}n}{l} + 0.86)$ to 0, while $R = |r-o|_\infty$ once $n > 2|r-o|_\infty$.

Step 7 (rectangles). By the symmetries of ν and of the cube laws it suffices to take C in a $(1,3)$-plane with the side R along e_1. On the cube of an integer side $n \ge 2\mathsf{q}l$ with $R \le n/2$, $T \le n-1$, Lem. 17.12(d) gives $0 \le \langle W_C\rangle_{(n,n,n)} \le 8^{T/n} z_-(n,n,n)^{RT/n^2}$, and Step 5 gives the bound on the cube. Along $n = 2^k l$ the left side tends to $\nu(W_C)$ by Step 1, while $8^{T/n}\to 1$ and $(\log n)/n \to 0$: this is eq. 17.ii. The bound by the perimeter is Lem. 17.25(b).

Step 8 (closed paths). Let C and (ρ',ρ) be as in (c); by symmetry $(\rho',\rho) = (1,3)$. On the cubes $n = 2^k l$, which are even, Lem. 17.13(c) applies for large k: the sum of the ℓ_j is $\mathsf{A}_{13}(C)$, $\sum_j(\mathsf{v}_j+1)/n \to 0$, the thermal factors tend to 1, and $z_-(n,n,n)^{\mathsf{A}/n^2} \le e^{-\sigma^+\mathsf{A}+o(1)}$ by Step 5. For the general form let $\rho = 3$: by the symmetry of the cube that exchanges the directions 1 and 2, Lem. 17.13(b) holds with the direction 2 in the place of the direction 1, the two twisted pairs having one quotient on the cube (Lem. 17.10(c)), and in the proof of Lem. 17.13(c) each norm $\|\hat{\mathsf{T}}_{D_j}\|$ is bounded by (b) on its own; so in each gap ℓ_j may be taken as the larger of the two counts $\ell_j^1(C)$, $\ell_j^2(C)$, whose sum over the gaps is $\mathsf{A}_3(C)$, and Lem. 17.13(d) carries all this to D, with $d^{c_3(C)-1}$ in front and $\sum_j \mathsf{v}_j \log d/n \to 0$.

For a staircase loop the senses of its runs of the direction ρ are up in T consecutive gaps and then down in the same gaps, so $c_\rho = 1$, and $\mathsf{A}_{\rho'\rho}(C) = \sum_j |r_j - o_j|$. For a simple closed curve of a coordinate plane each passage from down to up and each from up to down holds at least one bond of the other direction, and between two such passages there is at least one bond of the direction ρ; so $|C| \ge 4c_\rho(C)$ and $2^{c_\rho-1} \le \frac12 e^{(\log 2/4)|C|}$.

Step 9 (gap). Let $N = N_1 \le N_2$ (by (T1)), $n_0 \le N < 2n_0$, and i with $t_i \le N_2 < 2t_i$ in Prop. 17.18(b): for $a \ge 1$, $(\lambda_a/\lambda_0)^{t_i} \le \mathsf{r}_{t_i}(N_2,N) \le 0.28\,\chi'^{2^i} \le \chi'^{t_i/n_0}$, so $\log(\lambda_0/\lambda_a) \ge \frac{1}{n_0}\log\frac{1}{\chi'} \ge \frac{\Lambda}{l} + \frac{1.021}{n_0}$. The vacuum is simple and its eigenvector lies in the sector 00 (Lem. 17.10); the tops of the sectors with electric flux are among the λ_a, $a \ge 1$. For $N = n = 2^k l$ and an integer $N_2 \ge n$, with $t_i = 2^i n \le N_2 < 2t_i$, Prop. 17.18(a), (c) give $\log(1+\mathsf{r}_{t_i}(N_2,n)) \le \frac{N_2}{t_i}\delta^{(n)}(t_i,t_i) \le 0.3652\,\chi^{2^i}$, so $(\lambda_a/\lambda_0)^{t_i} \le 0.367\,\chi^{2^i} \le \chi^{t_i/n}$ and $\log(\lambda_0/\lambda_a) \ge \frac1n\log\frac{1}{\chi} \ge \frac{\Lambda}{l} + \frac{3.5065}{n}$. Over $n\times n$, $(\lambda_a/\lambda_0)^n \le \mathsf{r}_n(n,n) \le 0.001615\,\theta^{n/l}$, and $\log\frac{1}{0.001615} \ge 6.42$. Finally, for $N_1, N_2 \ge n$, Lem. 17.11(f) gives $\log(1+\mathsf{r}_n(N_1,N_2)) \le \delta(n,N_1,N_2) \le \frac{N_1N_2}{n^2}\delta_{\mathrm{c}}(n) \le 0.01655\,\frac{N_1N_2}{n^2}\theta^{n/l} =: u \le \log 2$, and $e^u - 1 \le u/\log 2$, $\log 2/0.01655 \ge 41.8$.

Step 10 (flux lines of dyadic length). $2\sigma^+N^2/\log\frac{1}{\chi} = 2\varsigma w^2/(-\log 0.03 + 2w\Lambda)$ increases in w, is ≥ 3.83 at $w = 2$ and $< 2\mathsf{q}w = \mathsf{q}N/l$: the bounds on $\hat N$. With the time 1 over (N_2,N), Lem. 17.10(d) gives $\Lambda_-^t \le a_-(t,N_2,N) = z_-(t,N_2,N)\lambda_0^t(1+\mathsf{r}_t(N_2,N))$ for every t. Take $t = t_i = 2^i N$ with i the largest integer such that $\mathsf{M} = N_2/t \ge \hat N/2$, so that $\hat N/2 \le \mathsf{M} < \hat N$. The first

inequality of Lem. 17.15(b) at the time N_2, Prop. 17.18(c), $\mathsf{r}_{N_2}(t,N) \le \mathsf{r}_t(t,N)^{\mathsf{M}} \le e^{-2^i\sigma^+N^2}$ and $\log(1+\mathsf{r}_t(N_2,N)) \le \mathsf{M}\delta^{(N)}(t,t) \le 0.1826\,\mathsf{M}\chi^{2^i}$ give

$$\log\frac{\lambda_0}{\Lambda_-} \ge \frac{1}{t}\Big[2^i(\sigma^+N^2-\mathsf{g}) - \log\mathsf{M} - 0.1826\,\mathsf{M}\chi^{2^i}\Big] \ge \sigma^+N - \frac{2\log\hat N + 1.86}{N},$$

as $\hat N\chi \le 0.7936$. As $N_2\to\infty$, $i\to\infty$, and the bounded last two terms are divided by 2^iN: the lower limit. For $\max\{N,2\mathsf{q}l\} \le N_2 < \hat N N/2$ let w' be the largest power of two with $w' \le w$ and $2\mathsf{q}w'^2l \le N_2$, and $l' = 2w'l \le N$, so that $N_2 \ge \mathsf{q}l'^2/2l$; Prop. 17.17(e) with $N_3 = N$, $n_2 = N_2$ and $M = N_2/l'$ applies. If $w' < w$, then $8\mathsf{q}w'^2l > N_2$, so $N/l'^2 < 2\mathsf{q}N/N_2l \le 2\mathsf{q}/l$ and $M < 4\mathsf{q}w'$; if $w' = w$, then $N/l'^2 = 1/N \le 2\mathsf{q}/l$ and $M < \hat N/2 < 4\mathsf{q}w$. In both cases $\log M < \log\frac{2\mathsf{q}N}{l}$ and $0.01655\,M\theta^{l'/l} < 4\mathsf{q}\cdot 0.01655\,w'e^{-2w'\Lambda} \le 0.855$, which gives the second floor. For $N \le N' < 2N$: with the time 1, Lem. 17.15(a) in the third side gives $\Lambda_-(N_2,N')^t \le a_-(t,N_2,N') \le a_-(t,N_2,N)^{N'/N}$, and $a_-(t,N_2,N)^{1/t} \to \Lambda_-(N_2,N)$; so $\log\Lambda_-(N_2,N') \le \frac{N'}{N}\log\Lambda_-(N_2,N)$. Also $\log\lambda_0(N_2,N') \ge N_2N'P$ and $\log\lambda_0(N_2,N) \le \frac{1}{N}F(N,N_2,N) \le N_2NP + \frac{N_2}{N^2}\delta_{\mathrm{c}}(N)$ (Lem. 17.11(f)). Hence $\log\frac{\lambda_0}{\Lambda_-}(N_2,N') \ge \frac{N'}{N}[\log\frac{\lambda_0}{\Lambda_-}(N_2,N) - \frac{N_2}{N^2}\delta_{\mathrm{c}}(N)]$, and the first floor, $\delta_{\mathrm{c}}(N) \le 0.01655\,\theta^{N/l}$ and $N'/N < 2$ give the third.

Step 11 (flux lines of every integer length; the slabs). Let $N \ge 4l$. Prop. 17.19 gives $t_i\varepsilon_{\mathrm{v}}(t_i,N) \le e^{-E_i'}$ with $2^{-i}E_i' \ge \mathsf{E}_0 - \mathsf{g}'$. As in Step 10, with i the largest integer such that $\mathsf{M} = N_2/t_i \ge \hat N'/2$, the first inequality of Lem. 17.15(b) at the time N_2, with $\mathsf{r}_{N_2}(t_i,N) \le \chi_0^{2^i\hat N'/2} \le e^{-2^i\mathsf{E}_0}$, gives $z_-(t_i,N_2,N) \le \hat N'e^{-E_i'}$, and Prop. 17.18(b) gives $\log(1+\mathsf{r}_{t_i}(N_2,N)) \le \mathsf{M}\delta^{(N)}(t_i,t_i) \le \frac14\xi_i'$; so $\log(\lambda_0/\Lambda_-) \ge \frac{1}{t_i}[E_i' - \log\hat N' - \frac14\xi_i'] \ge (\mathsf{E}_0-\mathsf{g}')/n_0 = \sigma^+N - \kappa\mathsf{g}'/N$. The penultimate inequality uses the unused positive recurrence tail displayed in the proof of Prop. 17.19.

As $\kappa\mathsf{g}' = \kappa^2\log\hat N' + \kappa[3.22 + (3\kappa+0.68)\mathsf{h}]$ and $\mathsf{h} \le 0.4274\kappa + 0.0208$, $\kappa\mathsf{g}'$ increases in κ, and at $\kappa = 2$ it is at most $4\log\hat N' + 18.14$. The window $\max\{N,2\mathsf{q}l\} \le N_2 < \hat N'n_0/2$ is treated as in Step 10, with $l' = 2w'l \le n_0$. For the closed form, $\hat N'n_0/2 < (\mathsf{q}N/l+2)N/2 \le (\frac{\mathsf{q}}{2}+\frac14)N^2/l$ and $\hat N' < (\mathsf{q}+\frac12)N/l$, as $N \ge 4l$; so $C(\theta)N^2/l \ge \hat N'n_0/2$ and $4\log\hat N' + 18.2 \le 4\log\frac{N}{l} + 4\log(\mathsf{q}+\frac12) + 18.2 \le C(\theta)(1+\log\frac{N}{l})$; and $4\log 8.673 + 18.2 \le 26.85$ at $\mathsf{q} \le 8.173$. If $w \ge 4$ and $\tau \le 0.087$, the same road, with the slab of the thickness N started from $\mathsf{e}^{(N)}(n_0) \le 2\delta(n_0,2n_0,N) \le 8\cdot 0.01655\,\tau \le 0.0116$, where $\omega(Y_1(x)) \le 2.06x^2$, so that $2.06\,\mathsf{e}^{(N)}(t_i) \le (0.2728\,\tau)^{2^i}$, and with the start from Prop. 17.17(d) at $l' = n_0$ carried to the thickness N by Lem. 17.15(a) at the real factor N/n_0, gives the floor with $\hat N''$ and 10.8. (g3) is Lem. 17.14 with $L_2 = N_2$, $L_3 = N$ and $\phi \ge \sigma^+N - (4\log\hat N' + 18.2)/N$.

Step 12 (boxes of every shape, cylinders, twisted boxes). A cube of $N/2$ layers is a cell $\mathsf{C}_{N/2}(c)$, and all box laws are invariant under translations. For a real function A of the cell, Prop. 17.23 gives $|\langle A\rangle_{\underline{n}} - \langle A\rangle_{(N,N,N)}| \le 2.71\,\theta^{N/4l}\|\!|A|\!\|_N$ for every box with sides $\ge N$. If A is continuous, let the sides of a second box tend to infinity: by Step 4, $|\nu(A) - \langle A\rangle_{(N,N,N)}| \le 2.71\,\theta^{N/4l}\|\!|A|\!\|_N$, so $|\langle A\rangle_{\underline{n}} - \nu(A)| \le 5.42\,\theta^{N/4l}\|\!|A|\!\|_N \le 5.42\,\theta^{N/4l}\sup|A|$ by Lem. 17.21(b). For a bounded real Borel A, take continuous A_k with $|A_k| \le \sup|A|$ and $A_k \to A$ in L^8 of the mean of the images of $\langle\cdot\rangle_{(N,N,N)}$, $\langle\cdot\rangle_{\underline{n}}$ and ν on the bonds of the cell; both sides of the bound pass to the limit, the seminorm because $\|\!|A_k - A|\!\|_N \le \langle|A_k-A|^8\rangle_{(N,N,N)}^{1/8}$ (Lem. 17.21(b)).

For an event take $A = \mathbf{1}_E - \frac12$. For the cube of side $m \ge 2l$ take $N = 2\lfloor m/2\rfloor \le m$. The cylinder over Σ is the limit of Prop. 17.20, and the bound passes to it; with $\Sigma = (n,n)$ and $M\to\infty$, Lem. 17.15(a) in the third side gives $z_-(n,n,Mn) \le [z_-(n,n,n)\lambda_0^n(1+\mathsf{r}_n(n,n))]^M\lambda_0^{-Mn} = [(1+\mathsf{r}_n(n,n))z_-(n,n,n)]^M$, and Lem. 17.12(d) on the box (n,n,Mn) gives the area bound on the cylinder, as $8^{T/Mn}\to 1$. Twisted boxes: let the columns recede. For a finite set $\mathfrak{A}$ of bonds the plaquettes that meet $\mathfrak{A}$ carry no twist in all but finitely many members, so every limit point satisfies the Gibbs equations, as in Step 2.

The translate by e_μ of a twisted box law is the twisted law with the columns moved by one step, and by the proof of Lem. 17.10(iii) this is the image of the law under $U(b) \mapsto -U(b)$ on a set of bonds next to the columns; a fixed local function depends on none of these bonds in all but finitely many members. So every limit point is a periodic Gibbs state, that is ν; the two equations are linear in the law, so they hold for finite mixtures. If a column passes through p_0 for all members, one flip moves it off p_0 and changes the sign of $\operatorname{tr} \partial U(p_0)$, and the column so moved recedes after further flips that do not touch p_0; so the law of $\operatorname{tr} \partial U(p_0)$ tends to that of $-\operatorname{tr} \partial U(p_0)$ under ν, which differs because $\nu(\operatorname{tr} \partial U(p_0)) = 2\nu(W_p) > 0$ by Cor. 17.1(a).

Step 13 (the pressure). By Thm. AX(b), $\mathsf{e}(2l) \le \theta^2/12.9 \le 0.0615 < 0.069$ at β. Each $F(\underline{n}; \beta')$ is a smooth function of β', so there is an open interval U containing β on which $\mathsf{e}(2l; \beta') \le 0.069$: the criterion holds on U at the side $2l$ with $x_0 = 0.069$, so that what is proved above holds at each $\beta' \in \mathsf{U}$, with the state $\nu_{\beta'}$ and with $\delta_{\mathrm{c}}(N; \beta') < 0.1324 \cdot 0.8901^{N/4l}$ for $N \ge 4l$, uniformly on U. Let $N_m = 2\mathsf{L}^m$ and $f_m(\beta') = F(N_m, N_m, N_m; \beta')/N_m^3$: these are smooth, tend to P pointwise, and $f_m'(\beta') = -3\langle a_p \rangle_{(N_m,N_m,N_m)}$ by the invariance of the cube under translations and permutations of the axes. As $0 \le a_p \le 2$ is a function of one cell of the torus, Lem. 17.24 gives $|\langle a_p \rangle_{(N_m,N_m,N_m)} - \nu_{\beta'}(a_p)| \le \sqrt{\epsilon_m} + \epsilon_m$, with $\epsilon_m \to 0$ uniformly on U. So the f_m' converge uniformly on U to $-3\nu_{\beta'}(a_p)$, which is therefore continuous, and $P(\beta_2) - P(\beta_1) = \lim_m \int_{\beta_1}^{\beta_2} f_m' = \int_{\beta_1}^{\beta_2} (-3\nu_{\beta'}(a_p))\, \mathrm{d}\beta'$. Finally $\nu(a_p) = 1 - \nu(W_p) \le 1 - \frac{\beta}{4(\beta+1)}$ by Cor. 17.1(a). □

Proof of Thm. AX. (a) is Prop. 17.16(a); if $\beta > 0$, $\mathsf{e}(l_0) \le X(l_0) \le \frac{1}{20} \le 0.069$, and $12.9/20 = 0.645$. (b) By Prop. 17.16(b), (c), $12.9\,\mathsf{e}(2n) \le (12.9\,\mathsf{e}(n))^2$ as long as $\mathsf{e}(n) \le 0.069$, and the induction keeps $\mathsf{e}(2^k l) \le \theta^{2^k}/12.9$. For $k \ge 3$, $\mathsf{e}(2^{k-1} l) \le \theta^4/12.9 \le 0.04866 \le \frac{1}{20}$, so

$$X(2^k l) \le 17.71\,\mathsf{e}(2^{k-1} l)^2 \le \frac{17.71}{166.41}\,\theta^{2^k} \le 0.10643\,\theta^{2^k} \le 0.10643 \cdot 0.394013.$$

$\Phi(\Psi(x)) = \frac{1}{20}$ at $x = 0.065062\ldots$ and $\Phi(x) = \frac{1}{20}$ at $x = 0.052727\ldots$; both functions increase, $X(2l) \le \Phi(x_0)$, $X(4l) \le \Phi(\Psi(x_0))$, and $\Phi(\Psi^k(x_0))$ does not increase in k because $\Psi(x) \le x$ there. (c) $\mathsf{m}_{\mathsf{e}}(2n) \ge \mathsf{m}_{\mathsf{e}}(n)$ is $12.9\,\mathsf{e}(2n) \le (12.9\,\mathsf{e}(n))^2$. For $k \ge 3$, $\mathsf{e}(n) \le X(n) \le \frac{1}{20}$ gives $X(2n) \le 17.71\,\mathsf{e}(n)^2 \le 18X(n)^2$, so m_X does not decrease. $X(n) \ge \mathsf{e}(n)$ gives $\mathsf{m}_X(n) \le \mathsf{m}_{\mathsf{e}}(n) - \frac{1}{n}\log\frac{18}{12.9}$, with $\log\frac{18}{12.9} \ge 0.3331$; $X(n) \le 17.71\,\mathsf{e}(n/2)^2$ gives $18X(n) \le 1.91563\,(12.9\,\mathsf{e}(n/2))^2$, so $\mathsf{m}_X(n) \ge \mathsf{m}_{\mathsf{e}}(n/2) - \frac{1}{n}\log 1.91563$, with $\log 1.91563 \le 0.6501$. Both rates being monotone and squeezed, they have one limit, and $\mathsf{m}_{\mathsf{e}}(l) \ge \Lambda/l$ is the criterion. (d) $Y_2(0.0740538) \approx 0.3156 < \log 2$; for $0 < x < 0.0740538$ the iterates $\Psi^k(x)$ decrease, and to 0, since a positive limit would be a fixed point below 0.0740538; so $\mathsf{e}(2^k l) \le \Psi^k(\mathsf{e}(l)) \to 0$ and $X(2^{k+1} l) \le \Phi(\Psi^k(\mathsf{e}(l))) \to 0$; and $\beta > 0$ because $\mathsf{e}(l) > 0$. □

Proof of Thm. AY. Apply Thms. AW and AX at $l = l_* L^K$ and $\beta = L^K \mathsf{b}$, with the one x_0 of the uniform criterion; then θ, Λ, q, $C(\theta)$ do not depend on K. By eq. 17.i, $\sigma^+ l^2 = \frac{1}{2}\log\frac{1}{\theta} + 0.4175$ depends on θ alone. A rectangle of physical sides R, T has $RL^K \times TL^K$ bonds, and $\sigma^+ \cdot RL^K \cdot TL^K = \frac{\sigma^+ l^2}{l_*^2} RT = \sigma_* RT$. For $R, T \ge 2a_K$, Lem. 17.25(b) with $\#B = 2(R+T)/a_K - 4$ and $\beta = \mathsf{b}/a_K$ gives $\nu_K(W_C) \le \exp[-3(|C|_{\mathrm{ph}} - 4a_K)/(8\mathsf{b} + 3a_K)]$, and the smaller of two bounds is at most their geometric mean: (a). For $n = \ell/a_K$: $\sigma^+ n^2 = \sigma_* \ell^2$, $n/l = \ell/l_*$, $a_K^{-1}\sigma^+ n = \sigma_* \ell$, $a_K^{-1}/l = 1/l_*$ and $a_K^{-1}/n = 1/\ell$; this turns Thm. AW(d), (e), (f) and (g) into (b), (c) and (d), with $\hat{N} \le \mathsf{q}\ell/l_* + 1$ and $C(\theta) \le 27$. (e) is Thm. AX(b) at $k = 3$, with $18 \cdot \frac{17.71}{166.41} \le 1.91563 \le 1.9157$, and Thm. AX(a); finally $4 \cdot 0.4757 = 1.9028$. □

Proof of Cor. 17.1. Step 1 (symmetry). The symmetries of ν carry a rectangle to every other one with the same sides, or with the sides exchanged and run backwards.

Step 2 (two Hankel forms). At a site mirror orthogonal to e_μ that contains a site x, let γ_h run from x up h steps along e_μ, R steps along e_λ, and down to $x + Re_\lambda$ (γ_0 lies in the mirror).

Then $\sum_{j,k} U(\gamma_{h'})_{jk}\overline{U(\mathfrak{r}\gamma_h)_{jk}} = 2W$ of a rectangle $R \times (h+h')$, and the reflection positivity of ν at this mirror (Prop. YM3_I:103.4; ν is a periodic Gibbs state), applied to $A_{jk} = \sum_h c_h U(\gamma_h)_{jk}$, gives $\sum_{h,h'} c_h c_{h'} m(R, h+h') \geq 0$. At a bond mirror, take the paths γ'_h from a site x_p next to the mirror up h steps, along R steps and down to $x_q = x_p + Re_\lambda$; then $\sum \overline{(\mathsf{z}_p)_{ji}}(\mathsf{z}_q)_{kl} A_{jk}\overline{\mathfrak{r}A_{il}}$ with $A_{jk} = \sum_h c_h U(\gamma'_h)_{jk}$ is $2\sum_{h,h'} c_h c_{h'} W$ of rectangles $R \times (h+h'+1)$. Its mean is ≥ 0 on the cubes of sides $2^k l$, which are even, by Lem. 17.21(c) after a translation that brings the mirror between the layers 0 and 1, and it passes to ν by Thm. AW(a): $\sum_{h,h'} c_h c_{h'} m(R, h+h'+1) \geq 0$.

Step 3 (log-convexity). The 2×2 minors of the two forms give $m(R,T)^2 \leq m(R,T-1)m(R,T+1)$ for all $T \geq 1$.

Step 4 ($w > 0$). Given the other bonds, $U(b)$ has the density proportional to $\exp[\beta \mathsf{k} \frac{1}{2}\operatorname{tr}(U(b)g)]$, $\mathsf{k}g$ the sum of the four staples of b, $0 \leq \mathsf{k} \leq 4$; so, by the Gibbs equation and Lem. 17.25(a), $4w = \nu(\mathsf{k}\, r(\beta\mathsf{k})) \geq \nu(\beta\mathsf{k}^2/(\beta\mathsf{k}+4)) \geq \beta\nu(\mathsf{k}^2)/(4\beta+4)$. Two different staples of b are mirror images at a mirror that contains b, a site mirror if they lie in one plane and a diagonal mirror if they lie in two; so $\nu(\mathsf{k}^2) = 4 + \sum_{s \neq s'} \nu(\frac{1}{2}\operatorname{tr} s s'^{-1}) \geq 4$ by reflection positivity, and $w \geq \beta/(4(\beta+1))$.

Step 5 (positivity). From $m(1,0) = 1$, $m(1,1) = w > 0$ and Step 3, $m(1,T) > 0$ for all T; then $m(R,1) = m(1,R) > 0$ and $m(R,T) > 0$. As $T \mapsto -\log m(R,T)$ is concave and vanishes at 0, $m(R,T) \geq m(R,1)^T \geq w^{RT}$; the upper bound of (a) is Thm. AW(b).

Step 6 (the potential). The increments of a concave sequence do not increase; their limit is their infimum and also the limit of $-\frac{1}{T}\log m(R,T)$; summing the increments from T_1 to T_2 gives $m(R,T_2) \leq e^{-(T_2-T_1)V(R)} m(R,T_1)$; if a number v has this property for infinitely many T_2, dividing by T_2 gives $v \leq V(R)$; and the increasing slopes of the convex function $\log m(R,\cdot)$ give the inequality of four rectangles.

Step 7 (the measure). Let Λ be the linear functional on the real polynomials with $\Lambda(\lambda^T) = m(R,T)$. Step 2 says $\Lambda(p^2) \geq 0$ and $\Lambda(\lambda p^2) \geq 0$. With $s_r = \Lambda(\lambda^r p^2) \geq 0$, Schwarz's inequality gives $s_r^2 \leq s_0 s_{2r}$, and, $|m| \leq 1$ bounding every s_r, iteration gives $s_2 \leq s_0$ and then $s_1 \leq s_0$: $\Lambda((1-\lambda)p^2) \geq 0$; and $\Lambda(\lambda(1-\lambda)p^2) = \Lambda((1-\lambda)(\lambda p)^2) + \Lambda(\lambda((1-\lambda)p)^2) \geq 0$. Writing $\lambda^T(1-\lambda)^k$ as $\lambda^\epsilon(1-\lambda)^{\epsilon'} q^2$ with $\epsilon, \epsilon' \in \{0,1\}$ gives $\sum_i (-1)^i \binom{k}{i} m(R, T+i) \geq 0$ for all $k, T \geq 0$: the sequence is completely monotone, and Hausdorff's moment theorem gives the one measure ϱ_R. Since $m(R,T) \leq (\max\operatorname{supp}\varrho_R)^T$ and $m(R,T) \geq (\lambda_* - \varepsilon)^T \varrho_R([\lambda_* - \varepsilon, 1])$ for $\lambda_* = \max\operatorname{supp}\varrho_R > 0$, $m(R,T)^{1/T} \to \lambda_*$, so $\lambda_* = e^{-V(R)}$; for $T \geq 1$ the point 0 gives nothing, and the substitution $\lambda = e^{-E}$ gives the representation, with the mass $\varrho_R((0,1]) \geq m(R,1) > 0$.

Step 8. (d) follows from (a): $V(R) \geq \sigma^+ R$ from the upper bound, $V(R) \leq -R\log w$ from the lower one. (e) At $l = l_* L^K$, $\sigma^+ = \sigma_* a_K^2$, so $a_K^{-1} V(R/a_K) \geq a_K^{-1}\sigma^+ R/a_K = \sigma_* R$; and $M_K(R,T_2)/M_K(R,T_1) = e^{2\kappa(T_2-T_1)} m(R/a_K, T_2/a_K)/m(R/a_K, T_1/a_K)$, to which (c) applies; in the inequality of four rectangles the factors $e^{2\kappa(T'-T)}$, raised to the powers $1/(T'-T)$, give one and the same factor on both sides. Along a sequence, $M_j(R,T_2) \leq e^{-(T_2-T_1)v_j(R)} M_j(R,T_1)$ at each j, as just shown with $\kappa = \kappa_j$; so $v_j(R) \leq \frac{1}{T_2-T_1}\log\frac{M_j(R,T_1)}{M_j(R,T_2)}$, whose right member converges, and $v_j(R) \geq B$ for infinitely many j gives $B \leq \limsup_j v_j(R)$; and $V_{K_j}(R) \geq \sigma_* R$ gives $v_j(R) \geq \sigma_* R - 2\kappa_j$.

The symmetry, the inequality of three rectangles, which is the log-convexity of (b) in the form of chords, and that of four rectangles hold for each M_j, the factors $e^{2\kappa_j(R+T)}$ cancelling in both inequalities, and pass to the limits. With $\mathsf{F}(T) = -\log \mathsf{W}(R,T)$, the inequality of three rectangles says that the slope $(\mathsf{F}(T') - \mathsf{F}(T_1))/(T'-T_1)$ does not increase in $T' > T_1$; so it has a limit in $[-\infty, \infty)$ as $T' \to \infty$, $\mathsf{F}(T')/T'$ tends to this limit for every T_1, and $V_{\mathrm{ren}}(R)$ is the infimum of all these slopes. (f) A rectangle with an even side $2h$ is $\gamma(\mathfrak{r}\gamma)^{-1}$ for the site mirror halfway along that side, and a square is $\gamma(\mathfrak{r}\gamma)^{-1}$ for a diagonal mirror through two of its corners; as in Step 2, Prop. YM3_I:103.4 gives $s(W_C) \geq 0$.

Step 9 (no criterion). (g) Steps 1–7 use of ν only that it is a periodic Gibbs state, invariant under the symmetries of the lattice and reflection positive at every lattice mirror (Prop. YM3_I:103.4), and,

for the second Hankel form, that it is positive at the bond mirrors on the functions of two crossing bonds. With the hypothesis of (g) they hold for s, but for the upper bound of (a), in whose place $m_s \le 1$ stands; and $m_s(R,T) \ge w^{RT}$ gives $V_s(R) \le -R \log w \le R \log \frac{4(\beta+1)}{\beta}$. On site mirrors alone the first Hankel form, with all c but two equal to 0, gives $m_s(R, h+h')^2 \le m_s(R,2h)\, m_s(R,2h')$. At $(h-1, h+1)$, and from $m_s(R,0) = 1$ and $m_s(R,2) > 0$, induction gives $m_s(R,2h) > 0$ for all h, and $\psi(h) = -\log m_s(R,2h)$ is concave, ≥ 0, with $\psi(0) = 0$; so $\psi(h)/h$ tends to $2V_{\mathrm{e}}(R) = \inf_h(\psi(h+1) - \psi(h)) \ge 0$, and $\psi(h_2) - \psi(h_1) \ge 2(h_2 - h_1)V_{\mathrm{e}}(R)$ is the inequality for even T_1 and T_2. For an odd $T_2 = 2h_2+1 > T_1$, the pair (h_2, h_2+1) gives $m_s(R, 2h_2+1) \le [m_s(R,2h_2)\, m_s(R,2h_2+2)]^{1/2}$, and the even case twice gives the inequality. For even R: the pair $(0,1)$ gives $m_s(1,2) \ge m_s(1,1)^2 = w^2 > 0$, by Step 4, and $m_s(2,2) \ge m_s(2,1)^2 = m_s(1,2)^2 > 0$; the induction at the width 2 gives $m_s(2,2h) > 0$, and $m_s(2h,2) = m_s(2,2h)$. □

Proof of Prop. 17.2. (a) $F(l,2l,2l) \le 0$ and $-F(2l,2l,2l) \le 6\beta(2l)^3$, so $0 \le \mathsf{e}(l) \le 48\beta l^3 \le x_0$. (b) For $\mathsf{k} \in \mathbb{R}^4$ let π_{k} be the law on $G \simeq S^3$ with the density proportional to $e^{\langle U, \mathsf{k}\rangle}$. Every linear component $\langle U, \mathsf{e}\rangle$, $|\mathsf{e}| = 1$, has variance $\le \frac14$ under π_{k}: with $\kappa = |\mathsf{k}|$ and $x = \langle U, \mathsf{k}\rangle/\kappa$, the mean $A(\kappa)$ of x satisfies $A' = \operatorname{Var} x$ and, by an integration by parts, $\langle 1 - x^2\rangle = 3A/\kappa$, so $A' = 1 - 3A/\kappa - A^2$; a comparison with the positive root $\mathsf{B}(\kappa)$ of $1 - 3a/\kappa - a^2 = \frac14$, whose derivative is $< \frac14$, together with $A \le \kappa/4$, which follows from $3A = \kappa\langle 1 - x^2\rangle$ and $\langle x^2\rangle \ge \frac14$, the two identities in the proof of Lem. 17.25(a), x having the law of x_0 under $\langle\cdot\rangle_\kappa$, gives $A \ge \mathsf{B}$ and so $\operatorname{Var} x \le \frac14$, while the components orthogonal to k have the variance $A/\kappa \le \frac14$.

So two such laws differ in total variation by at most $\frac14|\mathsf{k} - \mathsf{k}'|$. The conditional law of one bond given the others is $\pi_{\beta\mathsf{h}}$, h the sum of the four unit vectors of its staples, and changing one of its 12 neighbouring bonds moves h by at most 2. For a bounded Borel function f of finitely many bonds let $\delta_b(f)$ be its oscillation in the single bond b, and $\Delta(f) = \sum_b \delta_b(f)$. The conditional expectation E_b in the bond b gives $\delta_b(E_b f) = 0$, $\delta_{b'}(E_b f) \le \delta_{b'}(f) + \frac{\beta}{2}\delta_b(f)$ for the 12 neighbours b' of b, and $\delta_{b'}(E_b f) \le \delta_{b'}(f)$ for the other bonds; so after any sequence of updates the oscillations are, bond by bond, at most the entries of the vector v that arises from $(\delta_b(f))_b$ by the linear rules $v_b \to 0$, $v_{b'} \to v_{b'} + \frac{\beta}{2}v_b$ for the neighbours b' of the updated bond b.

Let S_0 be the set of the bonds of f, $S \supset S_0$ a finite set of bonds, $d(b)$ the distance of b from S_0 in the graph of neighbouring bonds, $r = \min\{d(b) : b \notin S\}$, $\alpha = 6\beta < 1$, and f_k the function that k sweeps of updates through S, each bond of S once in a sweep, make of f. An update at b takes $v_b\alpha^{-\min\{d(b),r\}}$ from the weighted sum $\sum_{b'} v_{b'}\alpha^{-\min\{d(b'),r\}}$ and adds at most $\frac{\beta}{2}\cdot 12\, v_b\, \alpha^{-\min\{d(b),r\}-1} = v_b\, \alpha^{-\min\{d(b),r\}}$, as $d(b') \le d(b)+1$; so the weighted sum does not increase, it equals $\Delta(f)$ at the start, when v lives on S_0, where $d = 0$, and the bonds outside S carry the weight α^{-r}: $\sum_{b\notin S} v_b \le \alpha^r \Delta(f)$ at all times.

An update at $b \in S$ changes $\sum_{b'\in S} v_{b'}$ by at most $-(1-\alpha)v_b$, and in a sweep the entries of the bonds of S at the times of their updates add up to at least the sum at the beginning of the sweep; so $\sum_{b\in S} v_b \le \alpha^k \Delta(f)$ after k sweeps. As f_k is a function of the bonds of S and of their neighbours, and two configurations of these bonds are joined by changing one bond at a time, $\sup f_k - \inf f_k \le (\alpha^k + \alpha^r)\Delta(f)$. Now let $l \ge 4$ be even, S the set of the bonds with both ends in the cube $\{0, \dots, l-3\}^3$, and $f = a_p$ for the plaquette $p = p_{12}(c_0, c_0, c_0)$, $c_0 = \frac{l-4}{2}$. The ends of the bonds at distance $\le j$ from the four bonds of p lie in $[c_0 - j, c_0 + 1 + j]^3$, which lies in $[0, l-3]^3$ for $j \le c_0$; so every bond not in S has $d \ge c_0 + 1 = \frac{l}{2} - 1$, that is $r \ge \frac{l}{2} - 1$; and $\Delta(a_p) \le 8$.

The bonds of S and their neighbours lie in a cube of l sites a side, which the projections modulo l and modulo $2l$ map one to one into the tori of sides l and $2l$, the plaquettes at the bonds of S onto plaquettes; so on both tori the conditional laws of the bonds of S are those used above, the functions f_k are the same, and both means of a_p equal means of f_k, which lie in $[\inf f_k, \sup f_k]$. Letting $k \to \infty$: $|\bar a_{2l}(b) - \bar a_l(b)| \le 8(6b)^{l/2-1}$, $\bar a_n(b)$ the mean of a_p on the cube of side n at the

coupling b. As $\frac{d}{d\beta}F(n,n,n) = -3n^3\bar{a}_n(\beta)$ and $X = 0$ at $\beta = 0$,

$$X(l) = 24\,l^3 \int_0^\beta (\bar{a}_{2l}(b) - \bar{a}_l(b))\,\mathrm{d}b \le 64\,l^2(6\beta)^{l/2}.$$

At each of the four pairs of a coupling and a side named in (b), $64\,l^2(6\beta)^{l/2} \le \frac{1}{20}$; and $l \mapsto l^2(6\beta)^{l/2}$ decreases for $l > 4/\log\frac{1}{6\beta}$, which lies below each of the four sides. Thm. AX(a) gives the criterion. (c) The same contraction, with S the set of the bonds at the distance $< r$ from the bonds of f, applied to two Gibbs states s, s', for which $s(E_b F) = s(F)$ by the Gibbs equation of one bond, gives $|s(f) - s'(f)| = |s(f_k) - s'(f_k)| \le (\alpha^k + \alpha^r)\Delta(f)$; let $k \to \infty$ and then $r \to \infty$. □

Proof of Prop. 17.3. (a) Lem. 17.11(a), (b) with $\kappa_i = \lambda_i(2l,2l)^l$, so that $D = \mathsf{e}(l)$ and $\eta = \mathsf{r}$: $2\log 1.0351 \le 0.069$, and $\varpi(e^{0.069} - 1) \le 0.0357638$. The tops Λ_e, $e \ne 00$, are among the λ_i, $i \ge 1$ (Lem. 17.10), so $(\Lambda_e/\lambda_0)^l \le \mathsf{r}$, and $\log\frac{1}{0.0357638} \ge 3.33$; and $\mathsf{r} \ge \sum_{e\ne 00}(\Lambda_e/\lambda_0)^l$ gives the last statement. (b) With the time 1 over $(2l,2l)$ let $w_f = W_{2l}(f;0)/Z$ on the cube of side $2l$. Inside each sector $(\sum_i \mu_i^l)^2 \ge \sum_i \mu_i^{2l}$, so

$$\mathsf{e}(l) = 2\log Z(l,2l,2l) - \log Z(2l,2l,2l) \ge 2\log \sum_f \sqrt{w_f}.$$

By Lem. 17.10(c), $w_{00} = \frac{1}{4}(1+2r_1+r_2)$, $w_{10} = w_{01} = \frac{1}{4}(1-r_2)$, $w_{11} = \frac{1}{4}(1-2r_1+r_2)$, and $w_{11} \ge 0$, $r_2 \le r_1$ give $2r_1 - 1 \le r_2 \le r_1$; here $r_1 \le 1$ by Lem. 17.10(c), and $r_1 > 0$ because, with the time the direction that is not in the twisted pair, $Z^- = \operatorname{Tr}\mathsf{T}_1^{2l}$ is the integral over the diagonal of a kernel with values > 0 (Lem. 17.10(i), (ii)). The sum of the square roots is concave in r_2, so on this interval it is at least the smaller of its two end values; with $x = 1 - r_1 \in [0,1)$ these are $\sqrt{1-x} + \sqrt{2x}$ at $r_2 = 2r_1 - 1$ and $\frac{1}{2}\sqrt{4-3x} + \frac{3}{2}\sqrt{x}$ at $r_2 = r_1$, and the first is the smaller, as $\frac{1}{4}(4-3x) \ge 1 - x$ and $\frac{3}{2} > \sqrt{2}$.

The function f increases on $[0,\frac{2}{3}]$ and decreases on $[\frac{2}{3},1]$, with $\mathsf{f}(1) = \log 2$; and $\mathsf{f}(0.00063) > 0.069$, $\mathsf{f}(0.0039) \ge 0.16567$. (c) Let $\mathsf{e}(l) \le 0.069$ and $N \ge 16l$, and let $\bar{N} = 2^j l \le N < 2\bar{N}$, $j \ge 4$. By Lem. 17.11(f), $\log(1 + \mathsf{r}_N(N,N)) \le \log(1 + \mathsf{r}_{\bar{N}}(N,N)) \le \delta(\bar{N},N,N) \le 4\delta_{\mathrm{c}}(\bar{N}) \le 4\bar{\delta}_4 \le 0.0029221$ (Prop. 17.16(e)); and by Lem. 17.10(c), $1 - Z^-/Z = 1 - r_1 \le \frac{4}{3}(1 - e^{-0.0029221}) \le 0.0038905 < 0.0039$. For an integer $l' \ge 16l$ with $\bar{N} = 2^j l \le l' < 2\bar{N}$, $j \ge 4$: $\mathsf{e}(l') \le 2\log(1+\mathsf{r}_{l'}(2l',2l')) \le 2\delta(\bar{N},2l',2l') \le 2(2l'/\bar{N})^2\delta_{\mathrm{c}}(\bar{N}) < 32\,\bar{\delta}_4 \le 0.02338$.

The last sentence is the contrapositive of the first, at $l \le N/16$. (d) Thm. AW(b) at $R = T = 1$ gives $\sigma^+ \le \log(1/\nu(W_p))$, and $\sigma^+ l^2 \ge 0.4757$. □

The numbers of Remark 17.4. Over a torus of side $2l$ the momenta are $\frac{\pi}{l}m$, $m \in \mathbb{Z}^2$; three fields with one mode for each momentum $m \ne 0$ give the Fock levels whose sum at the time l is the product displayed, and the same at the time $2l$; in the notation of Lem. 17.11, $1 + \eta = 1 + \mathsf{r}$ and $1 + s_2 = 1 + \mathsf{r}'$. Summing $-3\log(1 - e^{-\pi|m|})$ over the shells $|m|^2 = 1, 2, 4, 5, 8, 9, 10, 13, \dots$ gives approximately 0.72010, and $-3\log(1 - e^{-2\pi|m|})$ gives approximately 0.02415; so $D \approx 1.416$.

Proof of Prop. 17.6. The exact part. On every periodic cube of an even side N, at every β, take as time a direction of the twisted pair, let ν be the other direction of that pair, let $\vartheta \ge 2$ and $k \ge 2$ be integers with $2k\vartheta = N$, and cut the cube into $2k$ slabs of ϑ layers. The transfer operator over ϑ layers, $\mathsf{Y} = \mathsf{T}'^{\vartheta}$ in the notation of the proof of Lem. 17.21, commutes with the flip J of one layer, the operator J_i of Lem. 17.10 that changes the sign of the bonds of the direction ν that cross one plane $\{x_\nu = c\}$, $c \in \mathbb{Z} + \frac{1}{2}$; $Z = \operatorname{Tr}\mathsf{Y}^{2k}$ and $Z^- = \operatorname{Tr}(\mathsf{J}\mathsf{Y}^{2k})$ (Lem. 17.10(ii), (c)); so $P_\pm = \frac{1}{2}(1 \pm \mathsf{J})$ have $\operatorname{Tr}(P_\pm \mathsf{Y}^{2k}) = \frac{1}{2}(1 \pm r_1)Z$: the weights of the two sectors of the flip are $\frac{1}{2}(1 \pm r_1)$.

A function of the bonds is *odd under the flip* if it changes its sign when the bonds of the direction ν that cross the plane $\{x_\nu = c\}$ change their signs in all layers. Let D, D' be bounded Borel functions of the bonds with both ends in one and the same slab of ϑ layers, D' odd under the

flip; of D no parity is asked. For $1 \le j \le 2k-1$ let t_j be the translation by $j\vartheta$ along the time, and $\mathsf{t} = \mathsf{t}_k$ the translation by $N/2$. With the slab operators Y_D, $\mathsf{Y}_{D'}$ of the proof of Lem. 17.21, $Z\langle D \cdot \mathsf{t}_j D'\rangle = \mathrm{Tr}(\mathsf{Y}^{2k-j-1}\mathsf{Y}_{D'}\mathsf{Y}^{j-1}\mathsf{Y}_D)$.

The weights of the layers and of the crossings do not change when all layers are flipped (proof of Lem. 17.10); so $P_\sigma \mathsf{Y} = \mathsf{Y} P_\sigma$ and, D' being odd, $\mathsf{J}\mathsf{Y}_{D'}\mathsf{J} = -\mathsf{Y}_{D'}$, that is $P_\sigma \mathsf{Y}_{D'} = \mathsf{Y}_{D'} P_{-\sigma}$. Splitting the trace over the two sectors, the sector changes between D and D':

$$Z\langle D \cdot \mathsf{t}_j D'\rangle = \sum_{\sigma=\pm 1} \mathrm{Tr}\big((P_\sigma \mathsf{Y})^{2k-j-1}\, \mathsf{Y}_{D'}\, (P_{-\sigma}\mathsf{Y})^{j-1}\, \mathsf{Y}_D\big),$$

a trace of $2k$ Hilbert–Schmidt operators. Hölder's inequality for it (Lem. YM3_I:86.1), with $\|P_\sigma \mathsf{Y}\|_{2k}^{2k} = \mathrm{Tr}(P_\sigma \mathsf{Y}^{2k})$ and $\|\mathsf{Y}_D\|_{2k}^{2k} = \mathrm{Tr}(\mathsf{Y}_D^* \mathsf{Y}_D)^k \le \sup|D|^{2k} Z$ (the norm definition preceding Lem. YM3_I:86.1; the trace is Z times the mean of a product of $2k$ copies of D and of its mirror image), gives

$$|\langle D \cdot \mathsf{t}_j D'\rangle| \le \sup|D|\, \sup|D'| \sum_{\sigma=\pm 1} \Big(\frac{1+\sigma r_1}{2}\Big)^{\frac{2k-j-1}{2k}} \Big(\frac{1-\sigma r_1}{2}\Big)^{\frac{j-1}{2k}}$$

(with $0^0 = 1$), and at $j = k$, where the two summands are equal,

$$|\langle D \cdot \mathsf{t} D'\rangle| \le 2 \sup|D|\, \sup|D'| \Big[\frac{1-r_1^2}{4}\Big]^{\frac12 - \frac{1}{2k}}.$$

The calibrated construction. The constants are chosen as in (Th1), (Th2) of §2.3.1, the setting in which Thm. YM3_I:D is used; among them are the cube size $M = \mathsf{L}^{m'}$, with $m' \ge 5$ (the setting preceding equation YM3_I:(634) has $M \ge L^4 M_1$ and $M_1 \ge L$), the exponents $r \ge 2$ and $\mathfrak{p}_0$, the constant $A_0 > 0$ of the thresholds (equation YM3_I:(634)), $\gamma \in (0, \frac12]$, and the numbers $c' > 0$, $\mathsf{C}^\Sigma_{\mathrm{maj}}$, c_{low} of (Th2). The setting (E) of §2.3.1 gives besides a label $\mathsf{b} \ge \gamma^{-2}$ and an integer $m \ge 1$ with $M\, R(\mathsf{b}) \le \mathsf{L}^m$, $R(x) = \min\{\mathsf{L}^i : i \in \mathbb{Z}_{\ge 0},\ \mathsf{L}^i \ge (\log x)^r\} \ge 1$, and $\ell = 2\mathsf{L}^m$. At the cutoff K the coupling is $\beta = \mathsf{L}^K \mathsf{b}$, and the cube has $N = \ell \mathsf{L}^K = 2\mathsf{L}^{m+K}$ sites a side. For an integer $0 \le d < K$ the block field of the depth d is $V = \mathcal{M}^{K-d} U$, $\mathcal{M}$ the block-averaging map of equations YM3_I:(37)–(39); it lives on a periodic cube of

$$\mathsf{n} = \ell \mathsf{L}^d = 2\mathsf{L}^{m+d}$$

blocks a side, each block having $\vartheta = \mathsf{L}^{K-d}$ sites a side, so that $N = \mathsf{n}\vartheta$; its law is μ_d^K (Def. 2.7). As $R \ge 1$, $\mathsf{L}^m \ge M = \mathsf{L}^{m'} \ge \mathsf{L}^5$, and with $d \ge 1$ and $\mathsf{L} \ge 13$:

$$\mathsf{n} \ge 2 \cdot 13^6.$$

Let $|\partial V - \mathbf{1}|_\infty$ be the largest $|\partial V(p) - \mathbf{1}|$ over the plaquettes of the block lattice, $x_d = \mathsf{L}^d \mathsf{b}$, $\varepsilon = A_0 x_d^{-1/2} (\log x_d)^{\mathfrak{p}_0}$ the threshold of equation YM3_I:(634) at the coupling x_d, $\delta = \exp[7\mathsf{n}^2\varepsilon] - 1$, and R_d the number R_t of the definitions preceding Prop. YM3_I:115.2 at the level $t = K - d$; as there $N_t = \ell^3 \mathsf{L}^{3d}$ and $x_t = x_d$, R_d does not depend on K. The hypotheses (H) are: $d \ge 1$; $x_d \ge \mathsf{t}_\natural$, the fixed calibration of §2.3.1; $\delta \le \frac{1}{20}$; and $\mathsf{R}_d \le \frac{1}{40}$.

Step 1: under (H) the block field is nearly flat with probability at least $\frac{39}{40}$, at every cutoff $K > d$. The data K, b, m and the top level $t = K - d$ make the setting (Th) of §2.3.1, as $1 \le t \le K-1$ and $x_t = x_d \ge \mathsf{t}_\natural$; Lem. YM3_I:118.1 gives what Prop. YM3_I:115.2 asks of it. Let $F = \mathbf{1}_{\mathsf{In}_t}(\mathcal{M}^t U)$, $\mathsf{In}_t = \{|\partial V - \mathbf{1}|_\infty < 2\varepsilon\}$ the inner ball of the definitions preceding Lem. YM3_I:113.10: a bounded gauge-invariant Borel function, by Prop. YM3_I:4.5, YM3_I:4.5 and YM3_I:113.10. The weight $\mathfrak{w}_t^{\mathrm{sf}}$ of the definitions preceding Prop. YM3_I:115.2 carries the factor $\mathbf{1}_{\mathsf{In}_t}(\mathcal{M}^t U)$, so $\mathfrak{w}_t^{\mathrm{sf}} = F\, \mathfrak{w}_t^{\mathrm{sf}}$, and the right inequality of Prop. YM3_I:115.2, domination and lost-mass bounds gives $Z - \int F \rho_0\, \mathrm{d}U \le Z - \int \mathfrak{w}_t^{\mathrm{sf}} \rho_0\, \mathrm{d}U \le \mathsf{R}_d Z$, $\rho_0 = e^{-\beta A}$ at $\beta = \mathsf{L}^K \mathsf{b}$. That is

$$\mu_d^K\big(|\partial V - \mathbf{1}|_\infty \ge 2\varepsilon\big) \le \mathsf{R}_d \le \tfrac{1}{40}.$$

Thm. YM3_I:D enters here, through Prop. YM3_I:115.2.

Step 2: a winding loop of the block field, its parity and its translate. Fix a block site ξ; let Γ_1, Γ_2, Γ_{12} be the closed paths of the block lattice from ξ with the step sequences (e_1^{n}), (e_2^{n}), $(e_1^{\mathsf{n}}, e_2^{\mathsf{n}})$, $P_i = V(\Gamma_i)$, and for each of them $D_\Gamma(U) = W_\Gamma(V) = \frac{1}{2}\operatorname{tr} V(\Gamma)$. Take the direction 3 as time, the flip of the bonds of the direction 1 that cross the plane $\{x_1 = c\}$, $c = \xi_1 + \frac{1}{2}\vartheta$, and t the translation by $N/2$ along e_3. Then: (i) D_Γ is a real Borel function with $|D_\Gamma| \le 1$ (Prop. YM3_I:4.5); (ii) it is a function of the bonds with both ends in one slab of ϑ layers, the slab of the blocks of the sites of Γ, by the locality of $\mathcal{M}^{K-d}$ (Prop. YM3_I:4.5); (iii) D_{Γ_1} and $D_{\Gamma_{12}}$ are odd under the flip: as L is odd, the plane $\{x_1 = c\}$ separates blocks at every level from d to K, on each double block the flip is the gauge transformation that is $-\mathbf{1}$ on one side of the plane, so by Prop. YM3_I:4.5 the averaging map carries the flip of the bonds that cross the plane at one level to the flip at the next level; and Γ_1, Γ_{12} run exactly one block bond that crosses the plane, Γ_2 none; (iv) $\mathsf{t}D_\Gamma = W_{\mathsf{t}\Gamma}(V)$, as t is the translation by $\frac{\mathsf{n}}{2}$ blocks and $\mathcal{M}^{K-d}$ commutes with it ([7, Lem. G.1]).

And $\mathsf{m}_\Gamma := \langle D_\Gamma \cdot \mathsf{t}D_\Gamma\rangle = \int W_\Gamma W_{\mathsf{t}\Gamma}\,\mathrm{d}\mu_d^K$ has $\mathsf{m}_{\Gamma_2} = \mathsf{m}_{\Gamma_1}$: the exchange of e_1 and e_2 about ξ keeps the law of the cube, commutes with t and with $\mathcal{M}^{K-d}$ ([7, Lem. G.1]), and carries Γ_1 to Γ_2.

On $\mathsf{B} = \{|\partial V - \mathbf{1}|_\infty < 2\varepsilon\}$ the two winding holonomies nearly commute: the step sequences $(e_1^{\mathsf{n}}, e_2^{\mathsf{n}})$ and $(e_2^{\mathsf{n}}, e_1^{\mathsf{n}})$ are rearrangements of one another, of the length $2\mathsf{n}$, and Lem. YM3_I:4.1 gives

$$|P_2P_1P_2^{-1}P_1^{-1} - \mathbf{1}| \le (1 + |\partial V - \mathbf{1}|_\infty)^{\mathsf{n}(2\mathsf{n}-1)} - 1 < e^{4\mathsf{n}^2\varepsilon} - 1 \le \delta.$$

As $|g - \mathbf{1}|^2 = 2 - \operatorname{tr} g$ for $g \in G$, $\frac{1}{2}\operatorname{tr}(P_2P_1P_2^{-1}P_1^{-1}) \ge 1 - \frac{1}{2}\delta^2$. The trace identity of Fricke, $\operatorname{tr}(ABA^{-1}B^{-1}) = (\operatorname{tr} A)^2 + (\operatorname{tr} B)^2 + (\operatorname{tr} AB)^2 - \operatorname{tr} A \operatorname{tr} B \operatorname{tr} AB - 2$ for $A, B \in \mathrm{SL}(2,\mathbb{C})$, at $A = P_2$, $B = P_1$, gives with $x_i = W_{\Gamma_i}(V)$ and $x_{12} = W_{\Gamma_{12}}(V)$

$$x_1^2 + x_2^2 + x_{12}^2 - 2x_1x_2x_{12} \ge 1 - \tfrac{1}{4}\delta^2, \qquad \text{so} \qquad 2x_1^2 + 2x_2^2 + x_{12}^2 \ge 1 - \tfrac{1}{4}\delta^2,$$

as $|x_{12}| \le 1$ gives $-2x_1x_2x_{12} \le x_1^2 + x_2^2$. On B a winding loop is also close to its translate: with ϵ the path of $\frac{\mathsf{n}}{2}$ steps e_3 from ξ to $\mathsf{t}\xi$ and $g = V(\epsilon)$, the step sequences of Γ followed by ϵ and of ϵ followed by $\mathsf{t}\Gamma$ are rearrangements of one another, of the length $\le \frac{5}{2}\mathsf{n}$, and Lem. YM3_I:4.1 gives $|gV(\mathsf{t}\Gamma)g^{-1} - V(\Gamma)| \le (1 + |\partial V - \mathbf{1}|_\infty)^{\frac{25}{8}\mathsf{n}^2} - 1 < e^{6.25\,\mathsf{n}^2\varepsilon} - 1 \le \delta$, so that $|W_{\mathsf{t}\Gamma} - W_\Gamma| \le \delta$. As $W_\Gamma W_{\mathsf{t}\Gamma} \ge W_\Gamma^2 - |W_\Gamma - W_{\mathsf{t}\Gamma}|$ and $|W_\Gamma - W_{\mathsf{t}\Gamma}| \le 2$ off B, Step 1 gives $\mathsf{m}_\Gamma \ge \mu_d^K(W_\Gamma^2) - \delta - 2\mathsf{R}_d$; with the weights $2, 2, 1$ and $\mathsf{m}_{\Gamma_2} = \mathsf{m}_{\Gamma_1}$,

$$4\mathsf{m}_{\Gamma_1} + \mathsf{m}_{\Gamma_{12}} \ge (1 - \tfrac{1}{4}\delta^2)(1 - \mathsf{R}_d) - 5(\delta + 2\mathsf{R}_d) \ge \tfrac{1599}{1600} \cdot \tfrac{39}{40} - \tfrac{1}{2} = \tfrac{30361}{64000} > 0.4743,$$

the middle member decreasing in δ and in R_d on $[0,1]$. So $\mathsf{m}_\Gamma \ge 0.0948 > 0.09$ for $\Gamma = \Gamma_1$ or for $\Gamma = \Gamma_{12}$; both loops give real functions of modulus ≤ 1 that are odd under the flip and are functions of one slab of the thickness of a block.

Step 3: the ceiling. The exact part at $D = D' = D_\Gamma$, with that Γ, $\vartheta = \mathsf{L}^{K-d} \ge 2$ and $2k = \mathsf{n}$, so that $2k\vartheta = N$ and $\frac{1}{2k} = \frac{1}{\mathsf{n}} = \frac{a_d}{\ell}$, the ratio of the block size $a_d = \mathsf{L}^{-d}$ at the depth d to the side ℓ, gives $0.09 \le \mathsf{m}_\Gamma \le 2[\frac{1}{4}(1 - r_1^2)]^{\frac{1}{2}(1-2/\mathsf{n})}$, that is, with $\mathsf{x} = (0.045)^{2/(1-2/\mathsf{n})}$,

$$r_1^2 \le 1 - 4\mathsf{x}, \qquad r_1 \le 1 - 2\mathsf{x} = 1 - 2\,(0.045)^{2/(1-2/\mathsf{n})},$$

the second because $(1 - 2\mathsf{x})^2 \ge 1 - 4\mathsf{x}$, $1 - 2\mathsf{x} > 0$ and $r_1 > 0$. The right member of the second inequality decreases in n to the limit 0.99595, and it is ≤ 0.99596 if and only if $\mathsf{n} \ge 5020$; at $\mathsf{n} = 2 \cdot 13^6$ it is ≤ 0.99595002. So under (H), $r_1 \le 0.99596$ on the cube of $2\mathsf{L}^{m+K}$ sites a side at $\beta = \mathsf{L}^K\mathsf{b}$, for every $K > d$.

Step 4: the range of (H); the numbers $t_\circ$ and m_t; (a). Let $\mathsf{C}_+ = \max\{0, \mathsf{C}_{\mathrm{maj}}^\Sigma + 2 + c_{\mathrm{low}}\}$, and let $y_\circ \ge 2$ be the least real number such that for all $y \ge y_\circ$

$$28\mathsf{L}^{11/2}M^2A_0\,y^{2r}(y + \log\mathsf{L})^{\mathfrak{p}_0}e^{-y/2} \le \log\tfrac{21}{20}, \qquad \log(16\mathsf{L}^9y^{3r}) + (\mathsf{C}_+ + \tfrac{9}{2}y)(2\mathsf{L}^2M)^3y^{3r} - c'A_0^2y^{2\mathfrak{p}_0} \le -\log 40;$$

it exists, as the first left member tends to 0 and, in the second, $2\mathfrak{p}_0 > 3r+2$ at the chosen constants (the choice $\mathfrak{p}_0 \geq 6r+1$ in Prop. YM3_I:69.1) and $c' A_0^2 > 0$. Put $t_\circ = \min\{2\mathsf{L}\gamma^2,\ 2\mathsf{L}^2/\mathsf{t}_\natural,\ 2\mathsf{L}e^{-y_\circ}\}$. For $0 < t \leq t_\circ$ let m_t be the least integer $m \geq 1$ with $M\,R(2\mathsf{L}^m/t) \leq \mathsf{L}^m$, which exists as $R(x) \leq \max\{1, \mathsf{L}(\log x)^r\}$; let $\ell_t = 2\mathsf{L}^{m_t}$, $\mathsf{b}_t = \ell_t/t$ and $y = \log \mathsf{b}_t$. Then (E) holds for the label b_t and the integer m_t, as $\mathsf{b}_t \geq 2\mathsf{L}/t_\circ \geq \gamma^{-2}$; $y \geq \log(2\mathsf{L}/t_\circ) \geq y_\circ$; and $\ell_t \leq 2\mathsf{L}^2 M y^r$, as $m_t = 1$ or $\mathsf{L}^{m_t-1} < M\,R(\mathsf{b}_t/\mathsf{L}) \leq M\mathsf{L}y^r$.

At $d=1$: $x_1 = \mathsf{L}\mathsf{b}_t \geq 2\mathsf{L}^2/t_\circ \geq \mathsf{t}_\natural$; $\mathsf{n} = \mathsf{L}\ell_t \leq 2\mathsf{L}^3 M y^r$ and $x_1 = \mathsf{L}e^y$ give $7\mathsf{n}^2\varepsilon \leq \log\frac{21}{20}$ by the first inequality, that is $\delta \leq \frac{1}{20}$; and $\log x_1 \geq y$, $2M^{-3}\ell_t^3\mathsf{L}^3 \leq 16\mathsf{L}^9 y^{3r}$ and $\ell_t^3 \leq (2\mathsf{L}^2 M)^3 y^{3r}$ give $\mathsf{R}_1 \leq \frac{1}{40}$ by the second. So (H) holds for (b_t, m_t) at $d=1$. At the cutoff $K \geq 2$ the cube has $N = 2\mathsf{L}^{m_t+K}$ sites a side and $\beta = \mathsf{L}^K\mathsf{b}_t = N/t$; by Step 3, $r_1 \leq 0.99596$ on the cube of $2\mathsf{L}^j$ sites a side at $\beta = 2\mathsf{L}^j/t$, for every $j \geq m_t+2$. Then $1-r_1 \geq 0.00404 \geq 0.0039$, and Prop. 17.3(b), (c) give $\mathsf{e}(\mathsf{L}^j) \geq \mathsf{f}(0.0039) \geq 0.16567$ and $\mathsf{e}(l) > 0.069$ for $3 \leq l \leq \mathsf{L}^j/8$: (a).

(b) The cube of the criterion at the cutoff K has $2l_*\mathsf{L}^K = 2\mathsf{L}^{m_0+K}$ sites a side and $\beta = \mathsf{L}^K\mathsf{b} = 2\mathsf{L}^{m_0+K}/t$ with $t = \ell_0/\mathsf{b} \leq t_\circ$; apply (a) with $j = m_0+K$, for all $K \geq m_t+2-m_0$. (c) Let $\bar y = \frac{4}{3}[\log(2\mathsf{L}^2M) + r\log(4r) + \log(\mathsf{L}/t_\circ)]$ and $\beta_\flat = 2\mathsf{L}^5 M\bar y^r/t_\circ$. For $\beta \geq \beta_\flat$ choose j with $2\mathsf{L}^j \leq t_\circ\beta < 2\mathsf{L}^{j+1}$ and put $t = 2\mathsf{L}^j/\beta \in (t_\circ/\mathsf{L}, t_\circ]$. With the numbers of Step 4 at this t, $y \leq \log\ell_t + \log(\mathsf{L}/t_\circ)$ and $\ell_t \leq 2\mathsf{L}^2 M y^r$; with $r\log y \leq \frac{y}{4} + r\log(4r) - r$ this gives $y \leq \bar y$, so $2\mathsf{L}^{m_t+2} = \mathsf{L}^2\ell_t \leq 2\mathsf{L}^4 M\bar y^r = \frac{t_\circ}{\mathsf{L}}\beta_\flat \leq \frac{t_\circ}{\mathsf{L}}\beta < t\beta = 2\mathsf{L}^j$, that is $j \geq m_t+2$; and (a) gives $\mathsf{e}(l) > 0.069$ for $3 \leq l \leq \mathsf{L}^j/8$, while $\mathsf{L}^j/8 > t_\circ\beta/16\mathsf{L}$. For the uniform criterion at a label b take $K \geq K_0$ with $L^K\mathsf{b} \geq \beta_\flat$: the criterion at $l = l_*L^K$ forces $l_*L^K > \frac{t_\circ}{16\mathsf{L}}L^K\mathsf{b}$.

(d) Put $\mathsf{G}_N = \log(Z(N,N,N)/\lambda_0(N,N)^N) = \log(1+\mathsf{r}_N(N,N))$. By Prop. 17.3(a), and as r_t decreases in t, $\mathsf{e}(N/2) \geq \log(1+\mathsf{r}_{N/2}(N,N)) \geq \mathsf{G}_N$; and for every integer $q \geq 2$, $Z(N,N,qN) \geq \lambda_0(N,N)^{qN}$ gives $\mathsf{G}_N \geq \mathsf{d}_q(N)/q$, $\mathsf{d}_q(N) = qF(N,N,N) - F(N,N,qN)$. Three facts bound $e^{-\mathsf{d}_q(N)} = Z(N,N,qN)/Z(N,N,N)^q$ from above: a swap on the long box, which bounds it through the largest probability on the cube (N,N,N) that the five invariants of the pair of the winding holonomies at one site lie in one small cell; a bound of that probability on a bare cube (Lem. 17.26); and a comparison of the laws of the block fields at two consecutive cutoffs in which the errors that bound quotients of densities enter as factors (Lem. 17.27), which carries the bound from one low cutoff K_1 to all higher ones. Besides Thm. YM3_I:D and the statements among the construction results named for this clause, (d) rests on Lem. 17.5 and these two lemmas, which give the estimate eq. 17.iii and, with the chain from a low cutoff, the estimate eq. 17.iv, and are stated below with their proofs.

A swap on the long box. For $g \in G$ write $g = \cos\alpha(g)\,\mathbf{1} + i\sum_j \mathsf{v}_j(g)\tau_j$, τ_1,τ_2,τ_3 the Pauli matrices, with the class angle $\alpha(g) = \arccos\frac{1}{2}\operatorname{tr} g \in [0,\pi]$ and the vector part $\mathsf{v}(g) \in \mathbb{R}^3$, $|\mathsf{v}(g)| = \sin\alpha(g)$. To a pair (g_1,g_2) of elements of G attach the real 3×2 matrix v with the columns $\mathsf{v}(g_1)$, $\mathsf{v}(g_2)$, the matrix $\mathsf{s} = (\mathsf{v}^{\mathsf{t}}\mathsf{v})^{1/2}$, and the five invariants $\mathsf{a}(g_1,g_2) = (\alpha(g_1),\alpha(g_2),\mathsf{s}_{11},\mathsf{s}_{12},\mathsf{s}_{22}) \in \mathbb{R}^5$: they do not change under a simultaneous conjugation of the pair, and if $|g_i - hg_i'h^{-1}| \leq \delta \leq 1$ for $i=1,2$ and one $h \in G$, then $|\mathsf{a}(g_1,g_2) - \mathsf{a}(g_1',g_2')|_\infty \leq 2\delta$, as $|\mathsf{v}(g) - \mathsf{v}(g')| \leq |g-g'|$, $|g-g'| \geq 2\sin\frac{1}{2}|\alpha(g)-\alpha(g')|$, and $\mathsf{v} \mapsto (\mathsf{v}^{\mathsf{t}}\mathsf{v})^{1/2}$ has the Lipschitz constant $\sqrt{2}$ for the Hilbert–Schmidt norm.

Let V be either the bare field, with $\mathsf{n} = N$, or the block field of the depth 0, on a lattice of $\mathsf{n} = \ell$ sites a side in the section, formed on the long box by the same local formulas; let $\omega \in (0,1]$ with $\delta_q = (1+\omega)^{q\mathsf{n}^2} - 1 \leq 1$. To a site ξ attach $\mathsf{a}(V,\xi) = \mathsf{a}(P_1(\xi),P_2(\xi))$, the five invariants of the pair of its winding holonomies in the directions 1 and 2. Cut $\mathbb{R}^5$ into half-open cubes c_k of a side $w \geq 2\delta_q$, and let c_k^+ be the half-open cube of side $3w$ with the same centre and the same choice of included faces, so that every point lies in exactly 3^5 of them. On the event that every plaquette of V on the box (N,N,qN) is within ω of $\mathbf{1}$, let $\xi' = \xi + ke_3$, $0 \leq k < q\mathsf{n}$, and t the transport of V from ξ to ξ' along the line of the direction 3: for $\mu = 1,2$ the transports $\mathsf{t}P_\mu(\xi')$ and $P_\mu(\xi)\mathsf{t}$ run along the step sequences $(e_3^k, e_\mu^{\mathsf{n}})$ and $(e_\mu^{\mathsf{n}}, e_3^k)$, which pass into one another by $k\mathsf{n} \leq q\mathsf{n}^2$ exchanges of adjacent

entries, and each exchange multiplies the transport by a conjugate of $\partial V(p)^{\pm 1}$ for one plaquette p, as in the proof of Lem. YM3_I:4.1; so $|\mathsf{t} P_\mu(\xi')\mathsf{t}^{-1} - P_\mu(\xi)| \le \delta_q$, with one t for both μ, and $\mathsf{a}(V,\cdot)$ stays within $2\delta_q$ of $\mathsf{a}(V,\xi)$ at all the sites of that line.

Cut the box by walls into q slabs of N layers, with the mirrors θ_i in their middle planes and sites ξ_i^- of one line of the direction 3 just below these planes; then the indicator of the event is at most $\sum_k \prod_i g_i^k\,(g_i^k \circ \theta_i)$, $g_i^k = \mathbf{1}\{\mathsf{a}(V,\xi_i^-) \in c_k^+\}$, a function of the bonds of the lower half of the slab i (for the block field, whose walls of the blocks contain these planes, as $N = \mathsf{n}\mathsf{L}^K$ with n even, by the locality of the averaging map and its covariance under the lattice isometries, Prop. YM3_I:4.5, [7, Lem. G.1]). The slab operator of $g_i^k\,(g_i^k \circ \theta_i)$ is $\mathsf{Y}^*\mathsf{Y} \ge 0$, Y that of the lower half with g_i^k, as in the proof of Lem. 17.21; and $|\operatorname{Tr}(\mathsf{O}_1 \cdots \mathsf{O}_q)| \le \prod_i \operatorname{Tr} \mathsf{O}_i$ for positive operators of trace class.

So every slab closes into a cube: $Z(N,N,qN)\langle \prod_i g_i^k\,(g_i^k \circ \theta_i)\rangle \le \prod_i Z(N,N,N)\langle g_i^k\rangle_{(N,N,N)}$. The plaquettes off the event are bounded by Schwarz's inequality at a mirror (Lem. 17.21(a)) and the same closing. With P_* the largest probability on the cube (N,N,N) that $\mathsf{a}(V,\xi)$ lies in one c_k^+, and ρ the largest probability on it that one plaquette of V is not within ω of $\mathbf{1}$, this gives $x \le 243\, P_*^{q-1} + 3q\mathsf{n}^3(\rho x)^{1/2}$ for $x = e^{-\mathsf{d}_q(N)}$, whence

$$e^{-\mathsf{d}_q(N)} \le 486\, P_*^{q-1} + 9q^2\mathsf{n}^6\rho.$$

Anticoncentration on a bare cube. For an integer $N' \ge 4$ and a coupling $\beta' > 0$ let $\mu_{N',\beta'}$ be the law $Z^{-1}e^{-\beta'\sum_p a_p}dU$ of the periodic cube of N' sites a side, $Z = Z(N',N',N')$, and $|\partial U - \mathbf{1}|_\infty$ the largest $|\partial U(p) - \mathbf{1}|$ over its plaquettes.

Lemma 17.26 (Anticoncentration on a bare cube)**.** Let $N' \ge 4$ be an integer, $\beta' \ge C_{\mathrm{T}} N'^{68}$ and $0 < \omega \le 1.6 \cdot 10^{-7} N'^{-13}$, with C_{T} of Lem. 17.5, enlarged to be at least $1.1 \cdot 10^9$. For every site ξ, every w' with $(N'^5/2\beta')^{1/2} \le w' \le \frac{1}{20}$ and every closed cube Q of the half side w' in $\mathbb{R}^5$,

$$\mu_{N',\beta'}\big(\mathsf{a}(U,\xi) \in \mathsf{Q},\ \max_p |\partial U(p) - \mathbf{1}| < \omega\big) \le C_{\mathrm{a}}\, N'^{29.5}\, \beta'^{1/2}\, w'^3. \qquad \blacksquare$$

Here C_{a} is an absolute constant. Thus the bound $C_{\mathrm{a}} N'^{c_{\mathrm{a}}} \beta'^{1/2} w'^3$ of eq. 17.iii holds with $c_{\mathrm{a}} = 29.5$, $c'_{\mathrm{a}} = C_{\mathrm{T}}$ and $c''_{\mathrm{a}} = 68$.

Sketch of the proof of Lem. 17.26. Write $\mathsf{V} = N'^3$, take $\xi = 0$, let $|\cdot|_2$ be the Euclidean norm of a family indexed by bonds or by plaquettes, and $Z[F] = \int F\, e^{-\beta'\sum_p a_p}\, \mathrm{d}U$. (i) *The gauge of a maximal tree.* Let T be the tree of the bonds $\langle x, x+e_3\rangle$ with $x_3 \le N'-2$, $\langle x, x+e_2\rangle$ with $x_3 = 0$, $x_2 \le N'-2$, and $\langle x, x+e_1\rangle$ with $x_2 = x_3 = 0$, $x_1 \le N'-2$. A bounded Borel function that the gauge transformations do not change has the same integral against dU as against the Haar measures of the bonds off T at $U = \mathbf{1}$ on T; there the winding holonomies at 0 are the values P_μ of U at the three bonds b_μ^* from $(N'-1)e_\mu$ to 0.

On the set S_0 of the other $2\mathsf{V} - 2$ bonds write $U(b) = V^P(b)\, e^{a(b)}$, $a(b) \in \mathfrak{su}(2)$, $|a(b)| < \pi$, where $V^P(b) = P_\mu$ if b is a bond of the direction μ that starts in the plane $x_\mu = N'-1$, and $V^P(b) = \mathbf{1}$ else; in $a(b)$ the Haar measure has a density at most $(2\pi^2)^{-1}$. (ii) *Small fields.* Along the tree every bond $b \in S_0$ is rebuilt from at most $2N'^2$ plaquettes, none of them a corner plaquette $p_{\mu\nu}(x)$ with $x_\mu = x_\nu = N'-1$: $|U(b) - V^P(b)| \le 2N'^2 |\partial U - \mathbf{1}|_\infty$. The same rebuilding, linearised, gives $|a|_2 \le 2N'^{5/2} |d_P a|_2$ for every triple P and every family a on S_0, d_P being the linearisation at V^P of $a \mapsto (\partial U(p))_p$, a covariant lattice curl.

So on $\{|\partial U - \mathbf{1}|_\infty < \omega\}$ every $|a(b)| \le \pi N'^2 \omega$, and at the corner plaquettes, where $\partial U(p)$ is the commutator $C_{\mu\nu} = P_\mu P_\nu P_\mu^{-1} P_\nu^{-1}$ up to four factors $e^{\pm a(b)}$, $|C_{\mu\nu} - \mathbf{1}| \le 9N'^2\omega$. Stokes' formula over the N'^2 plaquettes $p_{\mu\nu}(x)$ with $x_\rho = 0$, ρ the third direction, gives $2|\mathsf{v}(P_\mu) \times \mathsf{v}(P_\nu)| = |C_{\mu\nu} - \mathbf{1}| \le \sum_p |\partial U(p) - \mathbf{1}|$, and Schwarz's inequality then $\sum_p a_p \ge 2N'^{-2} \sum_{\mu<\nu} |\mathsf{v}(P_\mu) \times \mathsf{v}(P_\nu)|^2$. (iii) *The ceiling at a fixed triple.* Let κ be the family of the logarithms of the $C_{\mu\nu}$, placed at the corner plaquettes of the planes $\mu\nu$.

With $|e^{X_1}\cdots e^{X_k}-e^{X_1+\cdots+X_k}|\le\frac12(\sum_i|X_i|)^2$ for $X_i\in\mathfrak{su}(2)$, (ii) gives on $\{|\partial U-\mathbf{1}|_\infty<\omega\}$ that $2\sum_p a_p\ge(1-\eta_1)^2|d_Pa+\kappa|_2^2$, $\eta_1=39\pi N'^{9/2}\omega\le 2\cdot10^{-5}N'^{-8.5}$; keeping the fraction N'^{-3} of the action for the holonomies,

$$\sum_p a_p\ge\tfrac12(1-N'^{-3})(1-\eta_1)^2|d_Pa+\kappa|_2^2+\frac{2}{N'^5}\sum_{\mu<\nu}|\mathsf{v}(P_\mu)\times\mathsf{v}(P_\nu)|^2.$$

At a fixed triple the Gaussian integral over the $6\mathsf{V}-6$ real coordinates of a is at most $(2\pi/\beta'')^{3\mathsf{V}-3}D(P)^{-1/2}$, with $\beta''=\beta'(1-N'^{-3})(1-\eta_1)^2$ and $D(P)$ the determinant of $d_P^*d_P$ on the families on S_0, the shift by κ only lowering it; and $(\beta'/\beta'')^{3\mathsf{V}-3}\le(1-N'^{-3})^{-3N'^3}(1-\eta_1)^{-6N'^3}\le e^{3.2}$. (iv) *Determinants.* A triple whose three commutators lie within ϵ of $\mathbf{1}$ lies within $(2\epsilon)^{1/2}$ of a triple P' with one axis, $P'_\mu=\cos\alpha_\mu\,\mathbf{1}+i\sin\alpha_\mu\sum_j\hat n_j\tau_j$, and $D(P)\ge(1-48N'^{5/2}\delta)^{12\mathsf{V}}D(P')$ for $\max_\mu|P_\mu-P'_\mu|\le\delta$; at $\epsilon=9N'^2\omega$, $\omega\le1.6\cdot10^{-7}N'^{-13}$, this gives $D(P)^{-1/2}\le e^{0.5}D(P')^{-1/2}$.

At P' the families split into the colour along $\hat n$, on which $d_{P'}$ is the lattice curl, and the complex colour across $\hat n$, on which it is the curl twisted by the phases $e^{-2i\alpha_\mu}$ at the bonds where $V^{P'}=P'_\mu$: $D(P')=D(0)\,D(2\alpha)^2$, where $D(\theta)$, for $\theta\in\mathbb{R}^3$, is the determinant of the curl twisted by the phases $e^{i\theta_\mu}$ on the families on S_0, an even function, 2π-periodic in each θ_μ. For every momentum $k\in(\frac{2\pi}{N'}\mathbb{Z}/2\pi\mathbb{Z})^3$ the twisted curl has the eigenvalue $\sum_\mu(2-2\cos(k_\mu+\theta_\mu/N'))$ twice, besides the gradients; let $\mathfrak{d}(\theta)$ be the product of these numbers over all k, and $\mathfrak{d}'$ their product over $k\ne0$ at $\theta=0$.

Then $D(0)=\mathfrak{d}'$, the change of coordinates along the tree being unimodular, and $D(\theta)\ge\mathfrak{d}'/(1.5\cdot10^{10}N'^{24})$ for every θ: by 2π-periodicity first take $\theta\in[-\pi,\pi]^3$; for $|\theta|_\infty\le\frac{1}{64}N'^{-11/2}$ by a comparison of the forms that uses the bound $|a|_2\le2N'^{5/2}|d_Pa|_2$ of (ii), and else by Cauchy's interlacing from the determinant $|1-e^{i\theta_\mu}|^2\mathfrak{d}(\theta)$ on the families that vanish on T and at b_μ^* alone, $|\theta_\mu|=|\theta|_\infty$, together with $\mathfrak{d}(\theta)\ge4\sin^2\frac{\theta_\mu}{2}\,N'^{-2}\mathfrak{d}'$, which the product formula $\prod_{j=0}^{n-1}(2\cosh m-2\cos\frac{2\pi j+\vartheta}{n})=2\cosh nm-2\cos\vartheta$ gives. This covers the triples near $P_\mu=\pm\mathbf{1}$ as well.

Together, for every Borel set $\mathcal{P}$ of triples that the simultaneous conjugations carry onto itself,

$$Z\big[\mathbf{1}_{\mathcal{P}}(P)\,\mathbf{1}\{|\partial U-\mathbf{1}|_\infty<\omega\}\big]\le10^{12}N'^{24}\,\mathsf{X}_0\int_{\mathcal{P}}\exp\Big[-\frac{2\beta'}{N'^5}\sum_{\mu<\nu}|\mathsf{v}(P_\mu)\times\mathsf{v}(P_\nu)|^2\Big]dP,\qquad \mathsf{X}_0=(2\pi^2)^{-(2\mathsf{V}-2)}\Big(\frac{2\pi}{\beta'}\Big)^{3\mathsf{V}-3}\mathfrak{d}'^{-3/2},$$

dP is Haar measure on G^3. (v) *The normalizing lower bound.* Lem. 17.5 gives $Z\ge c_{\mathrm{T}}N'^2\mathsf{X}_0/\beta'^2$ throughout the stated range, by the quantitative periodic Laplace estimate.

(vi) *The toron integral.* Let $B=2\beta'/N'^5$, so that $B^{-1/2}\le w'\le\frac{1}{20}$. The triples with $\mathsf{a}(P_1,P_2)\in\mathsf{Q}$ carry at most $261\,w'^3B^{-3/2}$ of the integral of $\exp[-B\sum_{\mu<\nu}|\mathsf{v}(P_\mu)\times\mathsf{v}(P_\nu)|^2]$. Integrate P_3 first: with the singular values $s_1\ge s_2$ of the matrix v of the pair (P_1,P_2), in the eigenframe of $\mathsf{v}\mathsf{v}^{\mathrm t}$, $\sum_{i=1,2}|\mathsf{v}(P_i)\times\mathsf{v}(P_3)|^2\ge s_2^2x_1^2+s_1^2(x_2^2+x_3^2)$, $x=\mathsf{v}(P_3)$, so that this integral is at most $\min\{1,(Bs_1^2)^{-1}\}\min\{1,3.81(Bs_2^2)^{-1/2}\}$; and $|\mathsf{v}(P_1)\times\mathsf{v}(P_2)|=s_1s_2$.

If $|\mathsf{v}(P_i)|<0.9$ for $i=1,2$ on the whole set, the Haar measure of the pair has in the coordinates s a density at most $\frac{2}{0.19\pi^2}\,4s_1s_2(s_1+s_2)$, the product of $4s_1s_2(s_1+s_2)$ with the bound of the P_3-integral is at most $30.5\,B^{-3/2}$, and the projection of Q onto the three coordinates $(\mathsf{s}_{11},\mathsf{s}_{12},\mathsf{s}_{22})$ has volume $8w'^3$. Else $\sin\alpha(P_i)\ge0.8$ on the set for one i, and the two angles alone give $12.6\,w'^2B^{-2}\le12.6\,w'^3B^{-3/2}$: in general the triples whose two angles lie in a set $\mathcal{A}$ carry at most $2.52\,B^{-2}\int_{\mathcal{A}}\min\{B^{1/2}\mathsf{m},\mathsf{m}^{-1}\}\,\mathrm{d}\alpha_1d\alpha_2$, $\mathsf{m}=\max_i\sin\alpha(P_i)$. (vii) By (iv) and Lem. 17.5,

$$\mu_{N',\beta'}(P\in\mathcal{P},\ |\partial U-\mathbf{1}|_\infty<\omega)\le\frac{10^{12}}{c_{\mathrm{T}}}N'^{22}\beta'^2\int_{\mathcal{P}}e^{-B\sum_{\mu<\nu}|\mathsf{v}(P_\mu)\times\mathsf{v}(P_\nu)|^2}\,\mathrm{d}P.$$

Using (vi) proves the lemma with any absolute $C_{\mathrm a}\ge261\cdot10^{12}/(2^{3/2}c_{\mathrm T})$, enlarged to be at least 1.

From one cutoff to the next. In the setting (E) of §2.3.1, with the side $\ell = 2\mathsf{L}^m$ and the label b, take at every integer $K \geq 2$ the numbers of §2.3.1 at $n = 1$: the depths $u = \lceil K/1260 \rceil$ and $s = \lceil K/9 \rceil$, $t_1 = K - u$, and, with $N_{[d]} = \ell^3 \mathsf{L}^{3d}$ and $\mathsf{y}_{[d]} = (\log(\mathsf{L}^d \mathsf{b}))^{2\mathfrak{p}_0}$, the four errors

$$\epsilon_{\mathrm{fg}} = C_{\mathrm{old}} C_{\mathrm{fg}} \tfrac{\mathsf{L}}{\mathsf{L}-1} E_0 \, N_{[u]} \mathsf{L}^{-(s-u)},$$
$$\epsilon_{\mathrm{cl}} = 4\mathsf{K}_{\mathrm{w}} c_\rho^2 A_0^2 (1 + \tfrac{2\varrho}{\mathsf{b}}) \mathsf{y}_{[K]} N_{[s]} K^{1/2} \mathsf{L}^{-(K-s)/2},$$
$$\epsilon_{\mathrm{sh}} = 3(s-u) N_{[s]} \exp\bigl[C_{\mathrm{sh}} - \tfrac{1}{4}\kappa_{\mathrm{b}} A_1^2 \mathsf{y}_{[u]}\bigr],$$
$$\mathsf{R}_{t_1} = 2M^{-3} N_{[u]} \exp\bigl[-c' A_0^2 \mathsf{y}_{[u]} + (\mathsf{C}^{\Sigma}_{\mathrm{maj}} + 2 + c_{\mathrm{low}} + \tfrac{9}{2}\log \mathsf{b}) \ell^3\bigr],$$

with the constants named there. The first two fall like powers of L^{-K}, up to factors that are powers of K, ℓ and $\log \mathsf{b}$; the last two are exponentially small in $(\log(\mathsf{L}^u \mathsf{b}))^{2\mathfrak{p}_0}$. Cor. 2.8(a) bounds the difference of the means of a gauge-invariant function at two consecutive cutoffs by $\mathfrak{o}_K = 8\mathsf{R}_{t_1} + 12\epsilon_{\mathrm{fg}} + 8\epsilon_{\mathrm{cl}} + 12\epsilon_{\mathrm{sh}}$ times its supremum; here its comparison is run with the errors that bound quotients of densities kept as factors, so that an event of a probability of the order of t keeps this order.

Lemma 17.27 (One step between consecutive cutoffs)**.** In the setting (E), let $K \geq 2$ be an integer at which the conditions (k0)–(k4) and (k6) of §2.3.1 hold at $n = 1$, and put

$$\epsilon_K^{\times} = 4\epsilon_{\mathrm{fg}} + 2\epsilon_{\mathrm{cl}} + 1.06\,\epsilon_{\mathrm{sh}} + 1.39\,\mathsf{R}_{t_1}, \qquad \epsilon_K^{+} = \mathsf{R}_{t_1} + 1.29\,\epsilon_{\mathrm{sh}}.$$

Then for every integer $0 \leq d \leq u$ and every Borel set E of block fields of the depth d that the gauge transformations carry onto itself (Def. 2.7(3)),

$$\mu_d^{K+1}(E) \leq e^{\epsilon_K^{\times}} \mu_d^K(E) + \epsilon_K^{+}, \qquad \mu_d^K(E) \leq e^{\epsilon_K^{\times}} \mu_d^{K+1}(E) + \epsilon_K^{+}.$$

■

Sketch of the proof of Lem. 17.27. It suffices to take $d = u$: for $d < u$ replace E by its preimage under the averaging maps from the depth u to the depth d, which are the same on the two lattices (Section YM3_I:116) and keep it Borel and gauge invariant (Prop. YM3_I:4.5, YM3_I:4.5). Take the lattices $\mathrm{A} = (\ell, K)$ and $\mathrm{B} = (\ell, K+1)$ with the top levels t_1 and $t_1 + 1$, the objects of Section YM3_I:116 with $n = 1$ and $f_1 = 1$, at the source 0, and $\mathsf{R} = \mathsf{R}_{t_1}$, which is the same number on both (§2.3.1). For a bounded Borel function $F \geq 0$ of the block fields of the depth u that the gauge transformations do not change, and for each lattice $\bullet$, of the cutoff $K^\bullet$ and the top level $t^\bullet$, let $\langle F \rangle_0^\bullet$ be the mean of $F(\mathcal{M}^{K^\bullet - u} U)$ at the cutoff $K^\bullet$, $\langle F \rangle_1^\bullet$ the normalised integral of F against the measure of the chain on the inner ball $\mathsf{In}_{[u]}$, and $\langle F \rangle_2^\bullet$ that against the classical start on it. (1) *The discarded mass, a remainder.* The measure of the chain is the image of $\mathfrak{w}_{t^\bullet}^{\mathrm{sf}} \rho_0 \, \mathrm{d}U$ under the averaging, with $\mathfrak{w}_{t^\bullet}^{\mathrm{sf}} \geq 0$ (Prop. YM3_I:115.2, image identity); the right inequality of Prop. YM3_I:115.2, domination bound, for F and for $\sup F - F$ composed with the averaging, which are gauge invariant (Prop. YM3_I:4.5), and Prop. YM3_I:115.2, lost-mass bound, by which the discarded mass is at most R times the partition function, give $\langle F \rangle_0^\bullet \leq \langle F \rangle_1^\bullet + \mathsf{R} \sup F$ and $\langle F \rangle_1^\bullet \leq (1 - \mathsf{R})^{-1} \langle F \rangle_0^\bullet$. (2) *Forgetting, a factor.* On $\mathsf{In}_{[u]}$ the two measures have the densities $e^{c^\bullet(0)} e^{\Phi^{\bullet,0}}$ and $e^{\hat{c}^\bullet(0)} e^{\hat{\Phi}^{\bullet,0}}$ (the proof of Prop. YM3_I:116.1), both positive almost everywhere, as both measures are positive (Prop. YM3_I:115.2, image identity, Prop. YM3_I:116.1); and the real parts of the two exponents differ by at most ϵ_{fg} at every point of $\mathsf{In}_{[u]}$ (equation YM3_I:(2180)).

So the quotient of the densities lies in $\varkappa^\bullet [e^{-\epsilon_{\mathrm{fg}}}, e^{\epsilon_{\mathrm{fg}}}]$ with one number $\varkappa^\bullet > 0$, which cancels between the numerator and the normaliser: $e^{-2\epsilon_{\mathrm{fg}}} \langle F \rangle_2^\bullet \leq \langle F \rangle_1^\bullet \leq e^{2\epsilon_{\mathrm{fg}}} \langle F \rangle_2^\bullet$. (3) *The two classical starts, a factor and a remainder.* Both are integrals of $F(\mathcal{M}^{s-u} W)$ over the block fields W of the depth s against $\mathsf{k}^\bullet(W) \hat{F}_0^\bullet(W) \, \mathrm{d}W$, $\mathsf{k}^\bullet \geq 0$ (§2.3.2); where k^{A} and k^{B} are both positive they are equal and $\hat{F}_0^{\mathrm{B}} = \hat{F}_0^{\mathrm{A}} e^{\mathsf{q}_0}$, $|\mathsf{q}_0| \leq \epsilon_{\mathrm{cl}}$ (Prop. YM3_I:116.1, density comparison in the proof); and for $(\bullet, \circ) = (\mathrm{A}, \mathrm{B})$ and (B, A) the fields at which k° vanishes carry at most the fraction ϵ_{sh} of the mass of the start of $\bullet$ (§2.3.2).

So $\langle F\rangle_2^\bullet \le e^{2\epsilon_{\rm cl}}(1-\epsilon_{\rm sh})^{-1}\langle F\rangle_2^\circ + \epsilon_{\rm sh}\sup F$. Chaining (1), (2), (3), (2), (1),

$$\langle F\rangle_0^\bullet \le \frac{e^{4\epsilon_{\rm fg}+2\epsilon_{\rm cl}}}{(1-\epsilon_{\rm sh})(1-\mathsf{R})}\langle F\rangle_0^\circ + \big(\mathsf{R}+e^{2\epsilon_{\rm fg}}\epsilon_{\rm sh}\big)\sup F,$$

and (k2), (k3), (k6) give $e^{2\epsilon_{\rm fg}} \le e^{1/4} < 1.29$, $-\log(1-\epsilon_{\rm sh}) \le 1.06\,\epsilon_{\rm sh}$ and $-\log(1-\mathsf{R}) \le 2\log 2\cdot\mathsf{R} \le 1.39\,\mathsf{R}$. At $F=\mathbf{1}_E$ this is the lemma.

The chain from a low cutoff. Let $p = 2\mathfrak{p}_0$, $y = \log \mathsf{b}$, $\bar\ell(y) = 2\mathsf{L}^2 M y^r$, $\mathsf{k}(y) = \bar\ell(y)^3 y^{p+1}$, $\mathsf{N}(y) = \mathsf{L}\,\bar\ell(y)^{31} y^{10(p+1)}$, and let $K_1 = K_1(y)$ be the least integer $K \ge 10$ with $\mathsf{L}^{K/10} \ge \mathsf{k}(y)$. With the constants of §2.3.1 and the number C_+ of Step 4 put

$$a_1 = C_{\rm old}C_{\rm fg}\tfrac{\mathsf{L}}{\mathsf{L}-1}E_0\mathsf{L}^4,$$
$$a_2 = 4\mathsf{K}_{\rm w}c_\rho^2 A_0^2(1+2\varrho)\mathsf{L}^{7/2},$$
$$a_3 = \sum_{K\ge1}(1+K\log\mathsf{L})^p K^{1/2}\mathsf{L}^{-K/90},$$
$$a_4 = 529200\,e^{C_{\rm sh}}\sum_{i\ge1} i\,\mathsf{L}^{420i}e^{-\frac14\kappa_{\rm b}A_1^2(i\log\mathsf{L})^p},$$
$$a_5 = 2520\,M^{-3}\sum_{i\ge1}\mathsf{L}^{3i}e^{-c'A_0^2(i\log\mathsf{L})^p},$$
$$a_6 = \frac{256\,\mathsf{K}_u c_\rho^2 A_0^2\mathsf{L}^4}{\gamma_{\rm r}A_1^2},$$
$$\mathsf{R}(y) = 1.29\,\bar\ell(y)^3\Big[a_4 e^{-\frac14\kappa_{\rm b}A_1^2 y^p} + a_5 e^{(\mathsf{C}_+ + \frac92 y)\bar\ell(y)^3 - c'A_0^2 y^p}\Big];$$

these are formed from the constants alone, and the three series converge, as $p\ge2$, $\kappa_{\rm b}, c' > 0$ (Prop. YM3_I:114.4 and §2.3.1) and $A_0, A_1 > 0$ (equation YM3_I:(634)). Suppose that (E) holds, $\ell \le \bar\ell(y)$, $\mathsf{b} \ge \Theta_\star$, the number of the condition (k1), $y \ge \max\{8a_1, 20a_2a_3, a_6\}$ and $\mathsf{R}(y) \le e^{-3y^2}$. Then (k0)–(k4) and (k6) hold at every integer $K \ge K_1$, and the sums $\mathsf{S} = \sum_{K\ge K_1}\epsilon_K^\times$ and $\mathsf{R}_\Sigma = \sum_{K\ge K_1}\epsilon_K^+$ satisfy

$$\mathsf{S} \le \frac{17.72\,a_1 + 2a_2a_3}{y} + 1.39\,\mathsf{R}(y), \qquad \mathsf{R}_\Sigma \le \mathsf{R}(y) \le e^{-3y^2}, \qquad \ell\mathsf{L}^{K_1} < \mathsf{N}(y).$$

Indeed, for $K \ge K_1$ one has $u < \frac{K}{1260}+1$ and $\frac{K}{9} \le s < \frac{K}{9}+1$, so $4u - s \le 4 - \frac{K}{10}$, $\frac{7s-K}{2} \le \frac72 - \frac{K}{10} - \frac{K}{90}$ and $4s - K \le 4 - \frac{K}{10}$; and $\ell^3\mathsf{L}^{-K/10} \le \ell^3/\mathsf{k}(y) \le y^{-p-1}$. Hence $\epsilon_{\rm fg} \le a_1\ell^3\mathsf{L}^{-K/10}$, $\epsilon_{\rm cl} \le a_2 y^p(1+K\log\mathsf{L})^p K^{1/2}\mathsf{L}^{-K/90}\ell^3\mathsf{L}^{-K/10}$, and the left member of (k4) is at most $a_6\gamma_{\rm r}A_1^2\ell^3\mathsf{L}^{-K/10}$. Summed over $K \ge K_1$, as $(1-13^{-1/10})^{-1} \le 4.43$, these give $\sum_{K\ge K_1}(4\epsilon_{\rm fg}+2\epsilon_{\rm cl}) \le (17.72\,a_1 + 2a_2a_3)/y$. (k0) holds as $K \ge 10$, (k1) as $\mathsf{b} \ge \Theta_\star$, and $y \ge \max\{8a_1, 20a_2a_3, a_6\}$ gives (k2), (k4) and the first half of (k3).

For the two other errors, $(y+u\log\mathsf{L})^p \ge y^p + (u\log\mathsf{L})^p$, $s-u \le 140u$, $N_{[s]} \le \ell^3\mathsf{L}^{420u}$, and at most 1260 cutoffs have the same u: $\sum_{K\ge K_1}\epsilon_{\rm sh} \le a_4\ell^3 e^{-\frac14\kappa_{\rm b}A_1^2y^p}$ and $\sum_{K\ge K_1}\mathsf{R}_{t_1} \le a_5\ell^3 e^{(\mathsf{C}_+ + \frac92 y)\ell^3 - c'A_0^2y^p}$. So $\mathsf{R}_\Sigma \le \mathsf{R}(y)$, which gives the second half of (k3) and (k6), and $\sum_{K\ge K_1}(1.06\,\epsilon_{\rm sh} + 1.39\,\mathsf{R}_{t_1}) \le 1.39\,\mathsf{R}(y)$. Finally $\mathsf{k}(y) > \mathsf{L}$, so $K_1 \ge 11$ and $\mathsf{L}^{(K_1-1)/10} < \mathsf{k}(y)$: $\mathsf{L}^{K_1} < \mathsf{L}\,\mathsf{k}(y)^{10}$ and $\ell\mathsf{L}^{K_1} < \mathsf{N}(y)$. Lem. 17.27 at $d=0$ and an induction on K give $\mu_0^K(E) \le e^{\mathsf{S}_K}(\mu_0^{K_1}(E) + \mathsf{R}_K)$, with S_K and R_K the sums of the $\epsilon_{K'}^\times$ and of the $\epsilon_{K'}^+$ over $K_1 \le K' < K$; so eq. 17.iv holds: for every integer $K \ge K_1$ and every Borel set E of block fields of the depth 0 that the gauge transformations carry onto itself

$$\mu_0^K(E) \le e^{\mathsf{S}}(\mu_0^{K_1}(E) + \mathsf{R}_\Sigma).$$

The parameters. Let $\mathsf{y}_\star$ be the least real number

$$y_0 \ge \max\{122, \log\gamma^{-2}, \log\Theta_\star, 8a_1, 20a_2a_3, a_6, 2(17.72\,a_1 + 2a_2a_3)\}$$

such that at every $y \ge y_0$

$$\text{(y1)}\ \mathsf{R}(y) \le e^{-3y^2};$$
$$\text{(y2)}\ \log\mathsf{L} + 30\log\bar\ell(y) + 10(p+1)\log y \le y;$$
$$\text{(y3)}\ 26\,\mathsf{N}(y)^{28}\,\bar\ell(y)\,y^4e^{-y} \le 2.5\cdot10^{-14};$$
$$\text{(y4)}\ 26\,\mathsf{N}(y)^{6}\,\bar\ell(y)\,y^2e^{-y} \le (c_2^{\mathrm{SO}})^2;$$
$$\text{(y5)}\ 31\,(y+1)^2\,\mathsf{N}(y)^6 \le e^{2y^2};$$
$$\text{(y6)}\ C_{\mathrm{T}}\bar\ell(y)\mathsf{N}(y)^{67}e^{-y} \le 1,$$

c_2^{SO} the number of Lem. 5.28. It exists: $\bar\ell$ and N are powers of y, so the left member of (y2) is $O(\log y)$, those of (y3) and (y4) tend to 0, that of (y5) is a power of y, that of (y6) tends to zero, and the two exponents in $\mathsf{R}(y)e^{3y^2}$ are $-\frac14\kappa_{\mathrm{b}}A_1^2y^p + 3y^2 + O(\log y)$ and

$$-c'A_0^2y^p + (\mathsf{C}_+ + \frac{9}{2}y)\bar\ell(y)^3 + 3y^2 + O(\log y),$$

which tend to $-\infty$, as $p = 2\mathfrak{p}_0 > 3r+2$ at the chosen constants (the choice $\mathfrak{p}_0 \ge 6r+1$ in Prop. YM3_I:69.1) and $r \ge 2$; each condition compares continuous functions, so the numbers y_0 at which all hold on $[y_0,\infty)$ form a closed half line. Put $t_\star = 2\mathsf{L}e^{-\mathsf{y}_\star}$. Now let $j \ge 1$ and $\beta > 0$ with $t = 2\mathsf{L}^j/\beta \le t_\star$, and $N = 2\mathsf{L}^j$; let m be the least integer $m \ge 1$ with $M\,R(2\mathsf{L}^m/t) \le \mathsf{L}^m$, $\ell = 2\mathsf{L}^m$, $\mathsf{b} = \ell/t$, $y = \log\mathsf{b}$ and $K = j - m \in \mathbb{Z}$, so that $N = \ell\mathsf{L}^K$ and $\beta = \mathsf{L}^K\mathsf{b}$; and $q = \lceil y\rceil$.

Then $y \ge \log(2\mathsf{L}/t) \ge \mathsf{y}_\star$; as in Step 4, (E) holds for (b, m) and $\ell \le \bar\ell(y)$; by (y2), $\log\ell \le y/30$, so that $\log\frac1t \le y \le 2\log\frac1t$, and $y \le q < y+1$, $q \ge 122$; $e^{-y} \le t = \ell e^{-y} \le \bar\ell(y)e^{-y}$; the chain holds from $K_1 = K_1(y)$, with $e^{\mathsf{S}} \le e^{0.53} < 1.7$ and $\mathsf{R}_\Sigma \le e^{-3y^2}$; and $\beta_1 = \mathsf{L}^{K_1}\mathsf{b} = \ell\mathsf{L}^{K_1}/t$ has $\log\beta_1 = y + K_1\log\mathsf{L} < y + \log\mathsf{L} + 10\log\mathsf{k}(y) \le 2y$, by (y2).

Bare cubes. Let (N',β') be $(\ell\mathsf{L}^{K_1},\beta_1)$, the bare cube at the cutoff K_1, or (N,β) if $N < \ell\mathsf{L}^{K_1}$. Then N' is of the form $2\mathsf{L}^{j'}$, $j' \ge 1$, so even, $26 \le N' < \mathsf{N}(y)$, $\beta' = N'/t$, and $\log\beta' \le 2y$. Put $\Lambda' = 3N'^3\log(8e^8\beta'^{3/2}) + \log(3N'^3) + 3y^2$, $\omega_0 = (2\Lambda'/\beta')^{1/2}$ and $\upsilon = N'^2\omega_0$. The Haar measure of $\{g \in G : |g-\mathbf{1}| \le \beta'^{-1/2}\}$ is at least $\frac18\beta'^{-3/2}$, and $\beta'\sum_p a_p \le 24N'^3$ when all bonds lie in this set; so $Z(N',N',N') \ge (8e^8\beta'^{3/2})^{-3N'^3}$, every plaquette has $|\partial U(p) - \mathbf{1}| \ge \omega_0$ with a probability at most $Z^{-1}e^{-\beta'\omega_0^2/2}$, and

$$\mu_{N',\beta'}(|\partial U - \mathbf{1}|_\infty \ge \omega_0) \le 3N'^3\,(8e^8\beta'^{3/2})^{3N'^3}e^{-\Lambda'} = e^{-3y^2}.$$

Further $N'^3 \le \Lambda' \le 13N'^3y^2$, so $\upsilon^2 \le 26N'^6y^2t \le 26\,\mathsf{N}(y)^6\bar\ell(y)y^2e^{-y}$; and (y3), (y4) give $\omega_0 \le 1.6\cdot10^{-7}N'^{-13}$, $q\upsilon \le 1.7\cdot10^{-7}$, $\beta' \ge N'e^y/\bar\ell(y) \ge C_{\mathrm{T}}N'^{68}$ by (y6) and $\upsilon \le c_2^{\mathrm{SO}}$. With $w = 2.03\,q\upsilon$ and $w' = \frac32 w + 1.03\,\upsilon$ one has $(N'^5/2\beta')^{1/2} \le 3\upsilon \le w' \le 4q\upsilon \le \frac{1}{20}$, and Lem. 17.26 at $\omega = \omega_0$ gives, for every site ξ' and every closed cube Q of the half side w',

$$\mu_{N',\beta'}(\mathsf{a}(U,\xi') \in \mathsf{Q},\ |\partial U - \mathbf{1}|_\infty < \omega_0) \le C_{\mathrm{a}}N'^{29.5}\beta'^{1/2}(4q\upsilon)^3 \le P^{\mathrm{b}} := C_{\mathrm{b}}\,q^3y^3\,\mathsf{N}(y)^{39}\,t,$$

where $C_{\mathrm{b}} = 8500C_{\mathrm{a}}$, since $(2\Lambda')^{3/2} \le 132.6\,N'^{4.5}y^3$ and $N'^{40}/\beta' = N'^{39}t$.

The cutoffs $K \ge K_1$. Take the swap with the block field $V = \mathcal{M}^KU$ of the depth 0, $\mathsf{n} = \ell$, and $\omega = \theta_{\mathsf{L}}\mathsf{L}^{2K_1}\omega_0$, with ω_0 and υ those of $(\ell\mathsf{L}^{K_1},\beta_1)$ and the number $\theta_{\mathsf{L}} = \exp\frac{1}{2(\mathsf{L}^2-1)} < 1.003$ of Lem. 5.28: $\mathsf{n}^2\omega = \theta_{\mathsf{L}}\upsilon$, so $\delta_q \le e^{q\theta_{\mathsf{L}}\upsilon} - 1 \le 1.015\,q\upsilon$, $2\delta_q \le w$ and $\omega \le 1$. The sets $E_p = \{|\partial V(p) - \mathbf{1}| \ge \omega\}$ and $\{\mathsf{a}(V,\xi') \in c_k^+\}$ are Borel sets of block fields of the depth 0 that the gauge transformations carry onto themselves, as these conjugate $\partial V(p)$, and the two winding holonomies at ξ' by one element; so the chain gives $\mu_0^K(E) \le 1.7\,(\mu_0^{K_1}(E) + e^{-3y^2})$ for each of them.

At the cutoff K_1, where $|\partial U-\mathbf{1}|_\infty<\omega_0$ on the bare cube: $\mathsf{L}^{2K_1}\omega_0\le\upsilon\le c_2^{\mathrm{SO}}$, so $|\partial V-\mathbf{1}|_\infty<\omega$, and the winding holonomies of $V=\mathcal{M}^{K_1}U$ differ from the bare ones along the same lines by at most $13.3\,\ell\mathsf{L}^{2K_1}\omega_0=13.3\,\upsilon/\ell\le0.512\,\upsilon$ (Lem. 5.29), so that $|\mathsf{a}(V,\xi')-\mathsf{a}(U,\xi')|_\infty\le1.03\,\upsilon$; and c_k^+ lies in a closed cube of the half side $\frac32 w$. With the bounds on the bare cube $(\ell\mathsf{L}^{K_1},\beta_1)$ this gives, for the swap, $\rho\le3.4\,e^{-3y^2}$ and

$$P_*\le1.7(P^{\mathrm{b}}+2e^{-3y^2})\le\Pi(y)\,t,\qquad\Pi(y)=C_\Pi y^6\,\mathsf{N}(y)^{39},\qquad C_\Pi=2C_{\mathrm{b}}+4,$$

as $q^3\le1.025\,y^3$ and $e^{-3y^2}\le e^{-y}\le t$; and $9q^2\mathsf{n}^6\rho\le30.6\,(y+1)^2\bar\ell(y)^6e^{-3y^2}\le e^{-y^2}$, by (y5).

The cutoffs $K<K_1$, among them those with $K\le0$. Take the swap with the bare field on the cube of $N<\ell\mathsf{L}^{K_1}$ sites a side, $\mathsf{n}=N$, and $\omega=\omega_0$ of (N,β): $\mathsf{n}^2\omega=\upsilon$, $2\delta_q\le w$, $\rho\le e^{-3y^2}$, $P_*\le P^{\mathrm{b}}+e^{-3y^2}$, and $9q^2N^6\rho\le e^{-y^2}$; the same bounds hold.

The end. As $\Pi(y)\ge1$, $t\ge e^{-y}$ and $q-1\le y$, $(\Pi(y)t)^{q-1}\ge e^{-y^2}$; so in all cases $e^{-\mathsf{d}_q(N)}\le486\,P_*^{q-1}+9q^2\mathsf{n}^6\rho\le487\,(\Pi(y)t)^{q-1}$. Where $\Pi(y)t<1$, as $q\ge y\ge\log\frac1t$ and $q\ge122$,

$$\mathsf{G}_N\ge\frac{\mathsf{d}_q(N)}{q}\ge\frac{q-1}{q}\Big(\log\frac1t-\log\Pi(y)\Big)-\frac{\log487}{q}\ge\log\frac1t-\log\Pi(y)-1.06,$$

that is $\lambda_0(N,N)^N/Z(N,N,N)=e^{-\mathsf{G}_N}\le2.89\,\Pi(y)\,t$. As $\mathsf{N}(y)^{39}=\mathsf{L}^{39}(2\mathsf{L}^2M)^{1209}y^{1209r+390p+390}$ and $y\le2\log\frac1t$, this is the bound of (d), with

$$\mathsf{p}_\star=1209\,r+780\,\mathfrak{p}_0+396,\qquad\mathsf{C}_\star=3C_\Pi\,\mathsf{L}^{39}(2\mathsf{L}^2M)^{1209}\,2^{\mathsf{p}_\star},$$

formed from L, M, r and $\mathfrak{p}_0$, and with $t_\star$, formed from the constants. Where $\Pi(y)t\ge1$ the bound holds as well, as $\mathsf{C}_\star(\log\frac1t)^{\mathsf{p}_\star}\ge2.89\,\Pi(y)$ and $\lambda_0(N,N)^N\le Z(N,N,N)$. With $\mathsf{e}(\mathsf{L}^j)=\mathsf{e}(N/2)\ge\mathsf{G}_N$ this gives the floor of e. The last sentence of (d) is the first at $j=m_0+K$, as $N=\ell_0/a_K$ and $t=\ell_0/\mathsf{b}$. □

Proof of Prop. 17.7. (a) With the time μ_0 and the section (N_1,N_2), $Z(t,N_1,N_2)=(1+q^t)^{N_1N_2}\mathsf{c}^t=\sum_k\binom{N_1N_2}{k}(\mathsf{c}q^k)^t$, $\mathsf{c}=(1+q^{N_1})^{N_2}(1+q^{N_2})^{N_1}$: (T2), and (T3) with all the weight in the sector 00; (T1) and (iii) are read off. With $f(u)=\log(1+q^u)$: $F(l,l,l)=3l^2f(l)$, $F(2l,l,l)=l^2f(2l)+4l^2f(l)$, $F(2l,2l,l)=4l^2f(2l)+4l^2f(l)$, $F(2l,2l,2l)=12l^2f(2l)$; so $d_1=l^2\mathsf{h}$, $d_2=2l^2\mathsf{h}$, $\mathsf{e}=4l^2\mathsf{h}$ with $\mathsf{h}=2f(l)-f(2l)$, and $X=12l^2\mathsf{h}$. (b) Expanding the two products of the factors that contain t,

$$Z^{(k;m)}=\sum_{\mathsf{a},\mathsf{b}}\binom{N_2}{\mathsf{a}}\binom{N_1}{\mathsf{b}}(-1)^{k_1\mathsf{a}+k_2\mathsf{b}}\big(\mathsf{c}_mq^{N_1\mathsf{a}+N_2\mathsf{b}}\big)^t,\qquad\mathsf{c}_m=1+(-1)^mq^{N_1N_2}>0,$$

which is (T3) with the sector $e=(\mathsf{a},\mathsf{b})$ mod 2; (T1), (T2) and (iii) are read off. The formulas for e and X follow from $F(\underline n)=\sum_\rho n_\rho\log(1+q^{n_\mu n_\nu})$; then $X\ge24l(s-\frac{s^2}{2})-6ls^4$ and $\mathsf{e}\le8ls^2$, so $X/\mathsf{e}\ge\frac3s(1-\frac s2-\frac{s^3}{4})\ge2.95/s$ for $s\le0.03$. (c) On the cube, $Z^-/Z=(\frac{1-u}{1+u})^n$ with $u=q^{n^2}$, so $\frac12q^{n^2}\le z_-(n,n,n)\le nq^{n^2}$, while $x_0=\mathsf{e}(l)=8ls^2(1+O(s^2))$ gives $\theta=103.2\,ls^2(1+O(s^2))$ and the two expansions, with the exact constant $103.2e^{-0.835}$. If the inequality held with $c'>\frac12$, then $\log\frac1q\ge2c'\log\frac1q-C'(l)$ for all small q, which is false. □

Proof of Prop. 17.8. Put $u_j=l_j\varepsilon_{\mathrm{v}}(l_j)$ and $\mathsf{t}_j=\theta^{2^{j+1}}$. By Prop. 17.17(a), $u_0\le0.01643\,\theta^2$. Let $u_j\le0.0165\cdot2^{-j}\theta^{2\cdot4^j}$, and take $\hat N=2^{j+2}$ in Prop. 17.17(b). Then the powers of θ match: $y=\hat Nu_j\le0.066\,\theta^{2\cdot4^j}$, $(0.001615\,\mathsf{t}_j)^{2\hat N}\le0.001615^8\theta^{2\cdot4^j}$, $(0.001615\,\mathsf{t}_j^2)^{\hat N/2}\le0.001615^2\theta^{2\cdot4^{j+1}}$, and $8\hat N\cdot0.01655\,\mathsf{t}_j=16\cdot0.01655\,\mathsf{m}\theta^{\mathsf{m}}\le0.835$ with $\mathsf{m}=2^{j+1}$, as $\mathsf{m}\,0.8901^{\mathsf{m}}\le3.153$. So $\bar z_1\le0.06601\,\theta^{2\cdot4^j}$, $\bar z_2\le4.38\cdot10^{-5}\theta^{2\cdot4^{j+1}}$, and $u_{j+1}\le2^{-j-1}\cdot9.04\cdot10^{-5}\,\theta^{2\cdot4^{j+1}}$, which closes the induction.

For the cubes let $n=2^il$, $i\ge1$, $j=\lfloor(i-1)/2\rfloor$ and $M=n/l_j$, an integer with $M^2\le n/l\le2M^2$. The first inequality of Lem. 17.15(b) on (l_j,n,l_j), with $\mathsf{r}_n(l_j,l_j)\le(0.001615\,\mathsf{t}_j)^M\le0.001615\,\theta^{l_j^2/2l^2}$, gives $z_-(l_j,n,l_j)\le0.00906\,M\theta^{l_j^2/2l^2}$; and $\delta(l_j,n,l_j)\le M\delta_{\mathrm{c}}(l_j)\le0.01655\,M\theta^{l_j/l}\le$

$0.01655\, y\theta^y \le 0.0523$ with $y = \sqrt{n/l}$, as $M \le y \le l_j/l$ and $y\theta^y \le 1/e\Lambda$. Lem. 17.15(a) with $\kappa = M^2$ gives $z_-(n,n,n) \le (0.0096\, M)^{M^2\theta^{n^2/2l^2}}$, and the first factor is ≤ 1 if $0.0096\, M \le 1$, else at most $(0.0096\sqrt{n/l})^{n/l}$. Lem. 17.12(d) on these cubes and Thm. AW(a) give the area law, as $\frac{RT}{nl}\log(0.0096\sqrt{n/l}) \to 0$; and $\log\frac{1}{0.8901}/2 \ge 0.0582$. □

17.1.7. *Scope of the conclusions.* Every statement that uses the criterion is an implication from it on one lattice. Prop. 17.2 gives the criterion in two ranges of the coupling, both below $\frac16$: at every integer side $l \ge 3$ for $0 < \beta \le x_0/48l^3$ (its clause (a)), and at the even sides $l \ge 4$ with $64\, l^2(6\beta)^{l/2} \le \frac{1}{20}$ for $0 < \beta < \frac16$ (its clause (b)); from a side at which it holds the criterion passes, at the same coupling, to every integer side $l' \ge 16l$, of either parity (Prop. 17.3(c)), and to the sides $2^k l$ (Thm. AX(b)). With $\beta = L^K\mathsf{b}$ and $l = l_*L^K$ the first range reads $\mathsf{b} \le x_0 L^{-4K}/48l_*^3$ and the second $\mathsf{b} < L^{-K}/6$, while $L^K\mathsf{b} \ge \frac16$ for all large K at every label $\mathsf{b} > 0$; so neither range, nor its passage to other sides at the same coupling, contains a fixed label as K grows.

The uniform criterion is the hypothesis of Thm. AY and of Cor. 17.1(e). The number σ_* is a floor valid at each cutoff. On the centre twists of the cubes of sides $n \ge 2\mathsf{q}l$ the floors are those of the pure numbers $-\log z_-(n,n,n) \ge \sigma^+ n^2 - \frac{2\mathsf{q}n}{l}(\frac12\log\frac{2\mathsf{q}n}{l} + 0.86)$ and $-\log(1 - Z^z/Z)$, which is at least the same number less $\log 6$ (Thm. AW(d); in physical units Thm. AY(b)); the free energy $-\log(Z^z/Z)$ of a twist itself lies between 0 and $-\log(1-6z_-)$ where $6z_- < 1$, and the free energy ε_{v} of a vortex line per unit length is bounded from above (Props. 17.17, 17.8).

The energies of the flux sectors and of the flux lines, $\log(\lambda_0/\Lambda_e)$ and $\log(\lambda_0/\Lambda_-)$, have the floors of Thm. AW(e), (g) and Thm. AY(b), (d), and the gap has those of Thm. AW(f) and Thm. AY(c). For a path that lies in no coordinate plane, the area in Thm. AW(c) is that of a projection counted modulo 2, which may vanish although every surface bounded by the path is large: two parallel squares of side s at the distance s, joined by a segment run twice, form a path C with $\mathsf{A}_{\rho'\rho}(C) = 0$ for all $\rho' \ne \rho$, while every set of plaquettes whose boundary modulo 2 is the path has at least $2s^2$ members.

For two static charges at sites o, r of a section that lie on no common line of the lattice, Thm. AW(e) gives the floor $\sigma^+|r-o|_\infty$, with the maximum norm, which is at least $2^{-1/2}\sigma^+|r-o|$.

17.2. **From the box criterion to continuum confinement.** A box inequality at one fixed physical size, imposed at every sufficiently large cutoff along a fixed label, controls the infrared limit. Under this hypothesis the periodic Gibbs state is unique at each such cutoff, the continuum limit is independent of the order of the volume and cutoff limits, and its reconstructed vacuum is unique with a positive Hamiltonian gap.

The fixed-lattice transfer and tube arguments are given in §17.1. Here we prove the comparisons uniform in the cutoff, passage of the gap to the continuum, transport between labels and the single-cutoff certificates with their required thresholds and margins. The box inequality itself is an additional input; the elementary small-coupling examples do not verify it along a fixed positive label.

17.2.1. *Notation and hypotheses. One lattice.* Let $\beta \ge 0$. A *box* $\underline{n} = (n_1, n_2, n_3)$ is the periodic lattice with $n_\mu \ge 3$ sites along the side μ; $|\underline{n}| = n_1n_2n_3$, and N^3 is the cube. Every bond carries the Haar probability measure of SU(2), every plaquette p the variable $a_p = 1 - \frac12\operatorname{tr}\partial U(p) \in [0,2]$, and the box the weight $\exp(-\beta\sum_p a_p)$; $Z(\underline{n})$ is its partition function, $\mu_{\underline{n}}$ its measure, $F = \log Z \le 0$, $P = \inf_{\underline{n}} F(\underline{n})/|\underline{n}| \in [-6\beta, 0]$, and

$$\delta(\underline{n}) = F(\underline{n}) - |\underline{n}|P \ge 0, \qquad D_q(N) = q^3F(N^3) - F((qN)^3), \qquad \mathsf{e}(l) = 2F(l,2l,2l) - F((2l)^3).$$

We call δ the *excess*. For a box and a direction i, $\mathsf{d}^{(i)}(\underline{n}) = 2F(\underline{n}) - F(\underline{n}')$, $\underline{n}'$ the box with the side i doubled. None of these numbers depends on a spacing. For a bounded function Ψ of the bonds of $a\mathbb{Z}^3$ and a box, $\langle\Psi\rangle_{\underline{n}}$ is the mean of Ψ under the measure of the box extended periodically to $a\mathbb{Z}^3$.

The transfer operator. Fix a coordinate direction μ_0 as the time, the two others $\nu_1 < \nu_2$, and a cross-section $\Sigma = (A, B)$, $A, B \geq 3$; the box with t layers over Σ is (t, Σ). By Section YM3_I:85 and Prop. YM3_I:85.1 the configurations of a layer form a compact space $\mathcal{X}$ with the product Haar measure dV, a layer carries the continuous action $A_\Sigma \geq 0$ of its plaquettes, and the crossing operator K, which integrates the time-like bonds between two layers, is positive, Hilbert–Schmidt, with a continuous kernel with values in $(0, 1]$. With $\mathsf{m} = e^{-\beta A_\Sigma}$ (the function, and the operator of multiplication) we put

$$\mathsf{D} = \mathsf{D}_\Sigma = \mathsf{m}^{1/2}\mathsf{K}\,\mathsf{m}^{1/2}, \qquad \mathsf{T}' = \mathsf{K}^{1/2}\mathsf{m}\,\mathsf{K}^{1/2}.$$

D is positive, Hilbert–Schmidt, with the continuous kernel $\mathsf{m}(V')^{1/2}\mathsf{K}(V', V)\mathsf{m}(V)^{1/2} \in (0, 1]$, and $Z(t, \Sigma) = \operatorname{Tr} \mathsf{D}^t$ for every integer $t \geq 3$ (Prop. 17.31). T', which stands with the slab operators of the trace formula (M1) below, is positive as well, as $\mathsf{K} \geq 0$ and $\mathsf{m} > 0$, and has the same nonzero eigenvalues $\lambda_0 \geq \lambda_1 \geq \cdots > 0$, with multiplicities; they depend neither on t nor on the spacing. We put $\mathsf{r}_t(\Sigma) = \sum_{i \geq 1} (\lambda_i/\lambda_0)^t$, $\mathsf{G}(t; \Sigma) = \log(1 + \mathsf{r}_t)$, $\mathsf{R}_t = 1 - e^{-\mathsf{G}}$, $\varepsilon(\Sigma) = \log \lambda_0 - |\Sigma| P$, and on a cube $\mathsf{r}_N = \mathsf{r}_N(N, N)$, $\mathsf{G}_N = \mathsf{G}(N; N, N)$.

The transfer operator of a state. A periodic Gibbs state ν on $a\mathbb{Z}^3$ is invariant under $a\mathbb{Z}^3$ and reflection positive at every lattice mirror (M3), so it has the picture $(e_{\mu_0}, \frac{a}{2}, a e_{\mu_0})$ defined immediately before Lem. YM3_I:99.1. On the bounded Borel functions F, F' of finitely many bonds with both end points in $\{x_{\mu_0} \geq a\}$ the form $\langle F, F' \rangle = \nu(\overline{\Theta_0 F}\, F')$, Θ_0 the reflection $x_{\mu_0} \mapsto a - x_{\mu_0}$, is positive; $\mathcal{H}(\nu)$ is the completion of its quotient by the null vectors, $[F]$ the class of F, $\Omega = [1]$, and P_Ω the projection on $\mathbb{C}\Omega$. The *transfer operator* of ν is the operator T_ν with $T_\nu[F] = [\tau_{a e_{\mu_0}} F]$ of Lem. YM3_I:99.1: it is self-adjoint, $0 \leq T_\nu \leq 1$, $T_\nu \Omega = \Omega$, and $\langle [F], T_\nu^n [F'] \rangle = \nu(\overline{\Theta_0 F}\, \tau_{n a e_{\mu_0}} F')$, which is the number $h_{AB}(n)$ of Prop. 17.38(f) for $A = \overline{\Theta_0 F'}$ and $B = \overline{\Theta_0 F}$, functions of the slab of the layers $x_{\mu_0} \in \{(1 - \vartheta)a, \dots, 0\}$ when F, F' are functions of the bonds with both end points in the layers $a, \dots, \vartheta a$, $\vartheta \geq 2$; there $h_{AA}(0) - |\nu(A)|^2 = \|(1 - P_\Omega)[F']\|^2$.

So, if $\mathsf{e}(l) \leq x_0 \leq 0.069$ at β for one $l \geq 3$, that proposition at $n_0 = 0$ gives, with $\theta = 12.9\, x_0$, $|\langle [F], T_\nu^n [F'] \rangle - \langle [F], \Omega \rangle \langle \Omega, [F'] \rangle| \leq \theta^{n/l} \|(1 - P_\Omega)[F]\|\, \|(1 - P_\Omega)[F']\|$; the classes being dense, $\|T_\nu^n - P_\Omega\| \leq \theta^{n/l}$, hence $\ker(1 - T_\nu) = \mathbb{C}\Omega$ and $\|T_\nu\| \leq \theta^{1/l}$ on $\Omega^\perp$.

Twists and loops. A *column* of the plane $(\mu\nu)$ is the set of the plaquettes $p_{\mu\nu}(y + j e_\rho)$, $j \in \mathbb{Z}$, ρ the third direction, in lattice units. For $z \in \{0, 1\}^3$, indexed by the three coordinate planes, Z^z is the partition function with the sign of $\operatorname{tr} \partial U(p)$ reversed on one column of each plane with $z = 1$; it does not depend on the columns chosen, and $r^z = Z^z / Z$. On a cube r^z depends only on the number of twisted planes; we write r_1, r_2, r_3. With a time μ_0 we write $z = (k; m)$, $k \in \{0, 1\}^2$ recording the planes $(\mu_0 \nu_i)$ and m the plane $(\nu_1 \nu_2)$; the weights of the four classes of electric flux $f \in \{0, 1\}^2$ are $w_f = \frac{1}{4} \sum_k (-1)^{f \cdot k} r^{(k;0)}$.

For a closed lattice path C, $W_C = \frac{1}{2} \operatorname{tr} U(C)$, $U(C)$ the ordered product along C; $W_{R \times T}$ is the loop of a coordinate rectangle with $R \times T$ bonds. A *staircase loop* of the plane (13) is given by a site o, an integer $T \geq 1$ and integers α_j and $R_j \geq 1$, $0 \leq j < T$: with $o_j^k = o + \alpha_j a e_1 + k a e_3$ and $r_j^k = o + (\alpha_j + R_j) a e_1 + k a e_3$, the loop $C(\alpha, R)$ runs from o_0^0 straight to r_0^0; then, for $j = 0, \dots, T - 1$ in turn, one step up from r_j^j to r_j^{j+1} and, if $j < T - 1$, straight to r_{j+1}^{j+1}; from r_{T-1}^T straight to o_{T-1}^T; and then, for $j = T - 1, \dots, 0$ in turn, one step down from o_j^{j+1} to o_j^j and, if $j > 0$, straight to o_{j-1}^j.

It has one upward and one downward bond between any two consecutive layers that it meets, and consecutive rows need not overlap. Its physical area is $|A(C)| = a^2 \sum_j R_j$; it has $c_3(C) = 1$ and $A_{13}(C) = |A(C)|$ for the numbers c_3, A_{13} of Thm. AZ(4). A staircase loop of another coordinate plane is the image of such a loop under a lattice isometry.

The construction. Fix the common construction constants of Prop. YM3_I:69.1 and Section YM3_I:82, and use the first-passage data of Prop. YM3_I:74.1. The notation is: the blocking factor L, an odd integer; the spacings $a_K = L^{-K}$; the sides $\mathfrak{L} = \{2L^m : m \in \mathbb{N}\}$; the constants γ and $\gamma_W \leq \gamma$; the reference couplings $0 < g \leq \gamma$, the sides $\mathfrak{L}(g) \subset \mathfrak{L}$ at g and the first-passage data

at g; the numbers M, $r \ge 2$, $\mathfrak{p}_0 \in \mathbb{N}$ of the constants, $R(x) = \min\{L^j : j \in \mathbb{Z}_{\ge 0},\ L^j \ge (\log x)^r\}$, and $\mathfrak{p}_0 = 2\mathfrak{p}_0 + \frac{1}{2}$; and the interval of labels $I(g) = (g^{-2} + \bar{B},\ X_g - \bar{B}]$, $X_g = L(g^{-2} + B_\sharp)$, $\bar{B} = LB_\sharp/(L-1)$, such that all pairs $(K, L^K \mathsf{b})$ with $\mathsf{b} \in I(g)$ are first-passage data at g.

A *label* is a real $\mathsf{b} > 0$; its *line* is the set of the pairs $(K, L^K \mathsf{b})$. The torus ℓ at the cutoff K is the cube $(\ell/a_K)^3$ of $a_K\mathbb{Z}^3$; $\mu_{K,\ell,\beta}$ is its measure, extended periodically to $a_K\mathbb{Z}^3$, and $Z_{K,\ell}$ its partition function at $\beta = L^K\mathsf{b}$; $p_\ell = \ell^{-3} \log Z_{K,\ell}$, $p_\infty = a_K^{-3} P$. On $a\mathbb{Z}^3$, for a real Lipschitz function f of compact support, $\mathcal{O}_a(f) = a^{-1} \sum_p f(x(p)) a_p$, $x(p)$ the centre of p, and $\mathcal{O}_K = \mathcal{O}_{a_K}$. For $X = (f_1, \dots, f_n)$, $S^T_{n;K,\ell,\beta}(X)$ and $S^T_n[\nu](X)$ are the joint cumulants of $\mathcal{O}_K(f_1), \dots, \mathcal{O}_K(f_n)$ on the torus and in a state ν, and a periodic Gibbs state is a Gibbs state on $a_K\mathbb{Z}^3$ invariant under a subgroup of finite index of $a_K\mathbb{Z}^3$ (equation YM3_I:(1819) and the definition following it).

Gibbs sequences and their components are defined in Def. 13.4(2), and the volume-first and growing-torus limits in [7, §K.1]. We write S^σ_n for the connected functions, as in Prop. YM3_I:92.1, and use the reflection-space notation $\mathcal{H}(\sigma)$, H_σ, Ω_σ introduced before and in Prop. YM3_I:94.1. On the line of a label we write, at $\beta = L^K\mathsf{b}$,

$$\delta_K(\ell) = \delta((\ell/a_K)^3), \qquad \mathsf{e}_K(\mathsf{b}, \ell_0) = \mathsf{e}(\ell_0/2a_K), \qquad X_K(\mathsf{b}, \ell) = D_L(\ell/La_K) = L^3 \log Z_{K,\ell/L} - \log Z_{K,\ell},$$

and call X_K the *cube number*; then $\ell^3(p_\ell - p_\infty) = \delta_K(\ell)$. A *number of the constants* is formed from this fixed common construction assignment; it depends on no cutoff, coupling, side or label.

The chessboard seminorm. For $\ell'' \in \mathfrak{L}$ with $\ell''/2a_K \ge 3$, a *board* of side $\ell''/2$ is a board of [7, §B.1] of the side $\ell''/2$ with an offset $o \in a_K\mathbb{Z}^3$: its cells are the closed cubes $o + \frac{\ell''}{2}\mathbf{k} + [-\frac{\ell''}{4}, \frac{\ell''}{4}]^3$, $\mathbf{k} \in \mathbb{Z}^3$, and its walls are the planes $x_\mu = o_\mu + \frac{\ell''}{2}(t + \frac{1}{2})$, $t \in \mathbb{Z}$, which lie in $a_K(\mathbb{Z} + \frac{1}{2})$, as $\ell''/2a_K$ is odd: planes of lattice reflections through no site. The *bonds of a cell*, its *inner bonds*, are the bonds with both end points in the closed cube. A function F of the bonds of one cell has a mirror copy in every cell, its image under the reflections in the walls between, complex conjugated at an odd number of reflections.

For a torus $\ell \in \mathfrak{L}$ with $\ell \ge \ell''$, a period of the torus holds $(2\ell/\ell'')^3$ cells of the board, and $\|F\|^{(\ell'')}_{\mathrm{ch},\ell}$, the upper index in parentheses naming the side ℓ'' of the board, is the root of order $(2\ell/\ell'')^3$ of the mean under $\mu_{K,\ell,\beta}$ of the product of the copies of F in these cells: the chessboard seminorm of F at the side ℓ, for the board of F ([7, §B.1], equation [7, eq. 189]). At $\ell = \ell''$ there are eight copies, and we write $\|F\|_{\mathrm{ch},\ell''} = \|F\|^{(\ell'')}_{\mathrm{ch},\ell''}$. So $\|\cdot\|_{\mathrm{ch},\ell}$ without an upper index is always the seminorm of eight copies on the torus ℓ, for a board of side $\ell/2$: a function of a cell of side $\ell''/2 < \ell/2$ is then read under the board of side $\ell/2$ whose cell is centred at its own cell, whose walls are walls of its own board, ℓ/ℓ'' being a power of the odd L, and whose eight copies of it are among its copies under its own board; (M2) below compares the two seminorms.

On a box: for an even $N = 2h$ and a corner $c \in a(\mathbb{Z} + \frac{1}{2})^3$, the cell below c is $\mathsf{C}_h(c) = c + [-ha, 0]^3$, and for a function Ψ of its bonds $\|\Psi\|_{\mathrm{ch};N}$ is the eighth root of the mean on the cube N^3 of the product of its eight mirror copies in the planes $x_\mu = c_\mu$; at $N = \ell/a_K$, $\ell \in \mathfrak{L}$, it is the seminorm $\|\Psi\|_{\mathrm{ch},\ell}$ of the torus ℓ for the board of side $\ell/2$ whose cell is $\mathsf{C}_h(c)$.

Throughout, $0 < x_0 \le 0.069$, $\theta = 12.9\, x_0 \le 0.8901$ and $\Lambda = \log(1/\theta) \ge 0.1164$.

Definition 17.28 (The box inequality BI)**.** Let $\mathsf{b} > 0$ be real, $\ell_0 = 2cL^{m_0}$ with $c \in \mathbb{N}$ and $m_0 \in \mathbb{Z}$, and let K_0 be an integer with $m_0 + K_0 \ge 0$ and $cL^{m_0+K_0} \ge 3$. BI$[\mathsf{b}, \ell_0, K_0; x_0]$ is the statement: for every integer $K \ge K_0$, with $l_K = \ell_0/2a_K = cL^{m_0+K}$,

$$\mathsf{e}(l_K) = 2 \log Z(l_K, 2l_K, 2l_K) - \log Z(2l_K, 2l_K, 2l_K) \le x_0 \qquad \text{at } \beta = L^K\mathsf{b}.$$

Its rate is $m_* = 2\log(1/\theta)/\ell_0 \ge 0.2328/\ell_0$; then $\theta^{1/l_K} = e^{-m_* a_K}$. For every $z \in \mathbb{Z}$ the data $(L^z\mathsf{b}, L^z\ell_0, K_0 - z)$ give the same statement, since they name the same pairs $(l_K, L^K\mathsf{b})$. In the proofs below only the fact that l_K is an integer ≥ 3 is used, except where $\ell_0 = 2L^{m_0}$ is assumed. ■

Remark 17.29 (The spectral reading). At every $\beta \ge 0$ and $l \ge 3$, with $\mathsf{r} = \mathsf{r}_l(2l, 2l)$ over the cross-section $(2l, 2l)$ of the box $(l, 2l, 2l)$, Prop. 17.31(b) at $q = 2$ gives $\log(1+\mathsf{r}) \le \mathsf{e}(l) \le 2\log(1+\mathsf{r})$; so $\mathsf{r} \le e^{x_0/2} - 1$ implies $\mathsf{e}(l) \le x_0$, and $\mathsf{e}(l) \le x_0$ implies $\mathsf{r} \le \varphi(e^{x_0} - 1)$, $\varphi(\eta) = \eta/(1 + \sqrt{1-\eta^2})$; at $x_0 = 0.069$ the two numbers are 0.035102 and 0.035764. Up to these constants BI is the statement that the thermal sum $\sum_{i\ge1}(\lambda_i/\lambda_0)^{l_K}$ over the cross-section $\ell_0 \times \ell_0$ is small uniformly in the cutoff.

17.2.2. *Principal results.*

Theorem AZ (One label: one state, one limit, one vacuum, a gap, an area law)**.** Assume BI$[\mathsf{b}, \ell_0, K_0; x_0]$. Let $K \ge \max\{K_0, 1\}$ and $\beta = L^K\mathsf{b}$. No constant below depends on K.

(1) *The lattice.* (a) For every $\ell \ge \ell_0$ with $\ell/a_K \in \mathbb{N}$: $\delta_K(\ell) < 0.13736\,\theta^{\ell/\ell_0}$ and $\delta_K(\ell) \le 0.5892\,\theta^{2\ell/\ell_0} = 0.5892\,e^{-m_*\ell}$; at $\ell = 2^j\ell_0$, $\delta_K(\ell) \le 0.017170\,\theta^{2\ell/\ell_0}$; if ℓ/a_K is even, $\mathsf{e}(\ell/2a_K) < 0.3702\,\theta^{\ell/2\ell_0}$ and $\mathsf{e}(\ell/2a_K) \le 0.4079\,\theta^{\ell/\ell_0}$; on the cube ℓ/a_K all seven $0 \le 1 - r^z < 0.4121\,\theta^{\ell/\ell_0}$; for $\ell \ge L\ell_0$, $X_K(\mathsf{b}, \ell) < 0.13736\,L^3\theta^{\ell/L\ell_0}$; and for every box whose least physical side ℓ_- is $\ge \ell_0$, $\delta(\underline{n}) < |\underline{n}|(a_K/\ell_-)^3\,0.13736\,\theta^{\ell_-/\ell_0}$. (b) The lattice $a_K\mathbb{Z}^3$ has exactly one periodic Gibbs state ν_K at β; it is ergodic under $a_K\mathbb{Z}^3$, invariant under all isometries of the lattice, reflection positive at every lattice mirror, trivial on the tail σ-field, and an extreme point of the set of all Gibbs states at β; the pressure P is continuously differentiable near β with $P'(\beta) = -3\nu_K(a_p)$. (c) For every $\ell \in \mathfrak{L}$ with $\ell \ge \ell_0$, on the inner bonds of a cell of side $\ell/2$ the laws of ν_K and of the torus ℓ are at a distance in total variation at most $0.541\,\theta^{\ell/2\ell_0}$, and $|\nu_K(G) - \mu_{K,\ell',\beta}(G)| \le 0.66\,\theta^{\ell'/2\ell_0}\|G\|_{\mathrm{ch},\ell''}$ for real G of a cell of side $\ell''/2$, $\ell_0 \le \ell'' \le \ell'$ in $\mathfrak{L}$. For every $\ell \ge \ell_0$ with $N = \ell/a_K$ even, every corner c, every bounded real Borel function Ψ of the bonds of $\mathsf{C}_{N/2}(c)$ and every periodic box $\underline{n}$ with $n_1, n_2, n_3 \ge N$, of any integer sides:

$$|\langle\Psi\rangle_{\underline{n}} - \langle\Psi\rangle_{N^3}| \le 2.71\,\theta^{\ell/2\ell_0}\|\Psi\|_{\mathrm{ch};N}, \qquad |\langle\Psi\rangle_{\underline{n}} - \nu_K(\Psi)| \le 5.42\,\theta^{\ell/2\ell_0}\|\Psi\|_{\mathrm{ch};N},$$

and $|\mu_{\underline{n}}(A) - \nu_K(A)| \le 2.71\,\theta^{\ell/2\ell_0}$ for the events A of the bonds of the cell. Periodic boxes with receding closed twist sets, and finite mixtures of them, converge to ν_K as in Prop. 17.38(d). (d) Over every cross-section with physical sides $l_1 \le l_2$, $l_1 \ge \ell_0$, $l_i/a_K \in \mathbb{N}$, the top eigenvalue is simple, its eigenvector carries no electric flux, and

$$-a_K^{-1}\log(\lambda_1/\lambda_0) > m_* + \frac{1.42}{l_1}; \qquad \text{at } l_1 = \ell_0: \ -a_K^{-1}\log(\lambda_1/\lambda_0) > \frac{2}{\ell_0}\log\frac{1}{2.42\,x_0} \ge \frac{3.579}{\ell_0};$$

so the lowest level carrying electric flux, around either side, lies more than $m_* + 1.42/l_1$ above the top; by contraposition, if over a cross-section with $l_1 = l_h$ that level lies at most $1.42/l_h$ above the top at infinitely many cutoffs, then BI fails at the label b at every side $\ell_0 \le l_h$. Over the square (ℓ, ℓ), $\ell \ge \ell_0$, with $s_\ell = 2^{k-1}\ell_0 \in (\frac{\ell}{4}, \frac{\ell}{2}]$, $2^k\ell_0 \le \ell < 2^{k+1}\ell_0$, the levels $E_i = -a_K^{-1}\log(\lambda_i/\lambda_0)$ satisfy $E_1 \ge m_* + 1.5898/s_\ell \ge m_* + 3.179/\ell$ and $\#\{i \ge 1 : E_i \le E\} \le 0.20395\,e^{(E-m_*)s_\ell}$. For $t \in a_K\mathbb{N}$:

$$\sum_{i\ge1}(\lambda_i/\lambda_0)^{t/a_K} \le \exp\Big[0.37\Big\langle\frac{l_1}{2t}\Big\rangle\Big\langle\frac{l_2}{2t}\Big\rangle e^{-m_*t}\Big] - 1 \le \exp\Big[\frac{0.37}{\ell_0^2}\,l_1l_2\,e^{-m_*t}\Big] - 1 \quad (t \ge \tfrac{\ell_0}{2}),$$

$\langle u\rangle = \max\{1, u\}$, and the sum is $\le 0.2314\,e^{-m_*t}$ for $t \ge l_2/2$. No such floor holds for magnetic flux: over the square (ℓ, ℓ) the free energy of a vortex line, $E_{\mathrm{v}} = a_K^{-1}\log(\lambda_0/\|\mathsf{T}^{(1)}\|)$, satisfies $\ell E_{\mathrm{v}} < 1.0292\,\theta^{\ell/\ell_0}$, and $\ell E_{\mathrm{v}} \le e^{-\sigma_*\ell^2}$ at $\ell = 2^j\ell_0$, $j \ge 0$, σ_* as in (4). (e) If A, B are bounded functions of the bonds of two slabs between which lie n layers, then $|\nu_K(AB) - \nu_K(A)\nu_K(B)| \le e^{-m_*na_K}\sup|A|\sup|B|$. For F, G of two cells of side $\ell''/2$ of a board, $\ell'' \in \mathfrak{L}$, $\ell'' \ge \ell_0$, $j \ge 1$ cells apart in a direction, $|\nu_K(FG) - \nu_K(F)\nu_K(G)| \le 1.0350\,e^{-m_*(j-1)\ell''/2}\|F\|_{\mathrm{ch},\ell''}\|G\|_{\mathrm{ch},\ell''}$. For every n with $na_K \ge \frac{3}{4}\ell_0 + \frac{1}{2}a_K$ and every bounded real Borel F of the bonds of the cube $\{|x|_\infty \le \lfloor n/3\rfloor a_K\}$, $\|\nu_K(F\,|\,\text{bonds outside } \{|x_3| < na_K\}) - \nu_K(F)\|_{L^2(\nu_K)} \le 0.7057\,\mathrm{osc}(F)\,e^{-m_*(2n-1)a_K/6}$. (f) For $\ell \in \mathfrak{L}$, $\ell \ge \ell_0$, the free energy

$E = \log Z_\Gamma - \log Z_{K,\ell}$ of the torus sheared by half a period, $\Gamma = \mathbb{Z}\ell(e_1-e_3)\oplus\mathbb{Z}\ell e_2\oplus\mathbb{Z}\frac{\ell}{2}(e_1+e_3)$, satisfies $0 \le E \le \mathsf{e}(\ell/2a_K) \le 0.4079\,\theta^{\ell/\ell_0}$; the gap alone gives $E \ge -1.24\,\theta^{\ell/\ell_0}$.

(2) *One limit.* Let $n \ge 1$, and let $X = (f_1, \dots, f_n)$ consist of real Lipschitz functions, none 0, with pairwise disjoint compact supports in $(-\ell''/4, \ell''/4)^3$ for some $\ell'' \in \mathfrak{L}$, $\ell'' \ge \ell_0$. Put $X_i = \mathcal{O}_K(f_i) - \mu_{K,\ell'',\beta}(\mathcal{O}_K(f_i))$, $G_I = \prod_{i\in I} X_i$, $M_I = \sup_{K\in\mathbb{N}} \|G_I\|_{\mathrm{ch},\ell''}$. Then $M_I < \infty$, and with $C(X) = 0.65\sum_\pi |\pi|! \prod_{I\in\pi}(1.0174\,M_I)$, π running over the partitions of $\{1,\dots,n\}$, for every $\ell' \in \mathfrak{L}$ with $\ell' \ge \ell''$:

$$\sup_{K\ge\max\{K_0,1\}} |S_n^T[\nu_K](X) - S^T_{n;K,\ell',L^K\mathsf{b}}(X)| \le C(X)\,\theta^{\ell'/2\ell_0},$$

$$\sup_{K\ge\max\{K_0,1\}} \sup_{\min_\mu n_\mu \ge \ell'/a_K} |S_n^T[\mu_{\underline{n}}](X) - S_n^T[\nu_K](X)| \le C^{\mathrm{sh}}(X)\,\theta^{\ell'/2\ell_0},$$

$C^{\mathrm{sh}}(X) = 3.36\sum_\pi |\pi|! \prod_{I\in\pi}(1.208\,M_I)$. Hence, for $n \ge 2$, $S_n(X) = \lim_K S_n^T[\nu_K](X)$ exists along all cutoffs; it equals $\lim_{\ell'}\lim_K S^T_{n;K,\ell',L^K\mathsf{b}}(X)$, the limit of the torus cumulants along every pair of sequences $K_j \to \infty$, $\mathfrak{L} \ni \ell_j \to \infty$, and the limit of the cumulants of periodic boxes $\underline{n}^{(j)}$ at cutoffs $K_j \to \infty$ along every sequence of boxes of any shape with $a_{K_j}\min_\mu n_\mu^{(j)} \to \infty$. If $\mathsf{b} \in I(g)$, $0 < g \le \gamma$, and the supports lie in one closed ball of radius R at pairwise distances $\ge \mathsf{d}$, then for $\ell' \in \mathfrak{L}(g)$, $\ell' \ge \max\{\ell_0, 128n(R+1)\}$, the first bound holds with $C^\flat(X) = 0.65\sum_\pi |\pi|! \prod_{I\in\pi}(1.0174\,\mathsf{m}_I^\flat)$, $\mathsf{m}_I^\flat = \mathsf{M}_{8|I|}(\mathsf{d}';R)^{1/8}\prod_{i\in I}\|f_i\|_\sharp$, which depends only on g, the constants (with the radius fixed in Thms. YM3_I:AJ and YM3_I:AD), n, $\mathsf{d}' = \min\{\mathsf{d},1\}$, R and the norms $\|f_i\|_\sharp = \max\{\sup|f_i|, 40M\,\mathrm{Lip}\,f_i\}$; here $\mathsf{M}_k(\mathsf{d}';R)$ is the sum over the partitions of $\{1,\dots,k\}$ into blocks of at least two members of the products of the bounds $C^g_{|B|}(\mathsf{d}';R)$ of (M5).

With the same $C^\flat(X)$ it holds with a torus $\ell^+ \ge \ell'$ of $\mathfrak{L}$ in the place of ν_K, and with 0.69 for 0.65 with the box (q_1N, q_2N, q_3N), $N = \ell'/a_K$, $q \in \mathbb{N}^3$, in the place of ν_K; for every periodic box $\underline{n}$ with all sides $\ge \ell'/a_K$, $|S_n^T[\mu_{\underline{n}}](X) - S_n^T[\nu_K](X)| \le 3.36\sum_\pi |\pi|! \prod_{I\in\pi}(1.19\,\mathsf{m}_I^\flat)\,\theta^{\ell'/2\ell_0}$; and for $n \ge 2$, $|S_n(X)| \le C_n^g(\mathsf{d};R)\prod_i \|f_i\|_\sharp$.

(2′) *One far field.* Let $n \ge 1$, $\rho \ge 1$, $\mathsf{d} > 0$, $f_1, \dots, f_n$ real Lipschitz, none 0, with supports in $[-\rho,\rho]^3$ at pairwise sup distances $\ge \mathsf{d}$, h real Lipschitz, not 0, with support in $v + [-\rho,\rho]^3$, $\Pi = \|h\|_\sharp \prod_i \|f_i\|_\sharp$, and $\ell'' \in \mathfrak{L}$, $\ell'' \ge \max\{\ell_0, 8\rho+4\mathsf{d}+4\}$. There is C_n, formed from $n, \rho, \mathsf{d}, \ell''$, b and the constants, such that $|S^T_{n+1}[\nu_K](h,X)| \le \mathsf{C}_n\, e^{-m_*(|v|_\infty - \ell'')}\,\Pi$ for $|v|_\infty \ge \ell''$, and on every torus $\ell \ge L\ell''$ of $\mathfrak{L}$, $|S^T_{n+1;K,\ell,\beta}(h,X)| \le 3.44\,\mathsf{C}_n\, e^{-m_*(|v|_{\infty,\ell} - \ell'')}\,\Pi$ for $|v|_{\infty,\ell} \ge \frac{3}{2}\ell''$, $|\cdot|_{\infty,\ell}$ the distance on the torus.

The same two bounds hold with h replaced by any bounded real Borel function W of the bonds with both ends in $v + [-\rho,\rho]^3$, with $\mathsf{C}'_n \sup_\square \|W - \mu_{K,\ell'',\beta}(W)\|_{\mathrm{ch},\ell''}\prod_i\|f_i\|_\sharp$ in the place of $\mathsf{C}_n\Pi$, C'_n formed as C_n and the supremum over the cells $\square$ of side $\ell''/2$ of all boards that hold $v + [-\rho,\rho]^3$ at a sup distance $\ge \mathsf{d}/2$ from their walls; so the decay of a far insertion W_K, for instance a local density smeared at a fixed scale, is uniform in the cutoff whenever these seminorms are bounded in K. In particular, let u be a real bounded Lipschitz function on $\mathbb{R}^3$ that vanishes on $\{|x|_\infty < r\}$, $r \ge \ell'' + \rho$, and let $\chi_v = \chi_0(\cdot - v)$, $v \in \rho\mathbb{Z}^3$, be the translates of the tent $\chi_0(x) = \prod_\mu \max\{0, 1 - |x_\mu|/\rho\}$: a partition of unity by Lipschitz functions with $\operatorname{supp}\chi_v \subset v + [-\rho,\rho]^3$.

Then $u\chi_v = 0$ unless $|v|_\infty \ge r - \rho \ge \ell''$, $\sup_v \|u\chi_v\|_\sharp < \infty$, and for every $K \ge \max\{K_0, 1\}$

$$\sum_{v\in\rho\mathbb{Z}^3} |S^T_{n+1}[\nu_K](u\chi_v, X)| \le \mathsf{C}_n \sup_v \|u\chi_v\|_\sharp \prod_i \|f_i\|_\sharp \sum_{v\in\rho\mathbb{Z}^3,\ |v|_\infty \ge r-\rho} e^{-m_*(|v|_\infty - \ell'')},$$

a bound free of K that tends to 0 as $r \to \infty$ at a fixed bound of $\sup_v \|u\chi_v\|_\sharp$. Thus the connected-cumulant series $\sum_v S^T_{n+1}[\nu_K](u\chi_v, X)$ converges absolutely, uniformly in K, and is the limit of the cumulants with finite partial sums of the far insertions; it tends

to 0 as the region of u recedes. On a torus $\ell \geq L\ell''$ of $\mathfrak{L}$, take $\rho = \ell/q \geq 1$ with q an integer and $\ell'' \geq 8\rho + 4\mathsf{d} + 4$, use the finite set $V = \rho\mathbb{Z}^3/\ell\mathbb{Z}^3$ and the periodized tents $\chi_v^{\mathrm{per}}(x) = \sum_{k\in\mathbb{Z}^3} \chi_0(x - v + \ell k)$. For a bounded Lipschitz function u on the torus vanishing where $|x|_{\infty,\ell} < r$, $r \geq \frac{3}{2}\ell'' + \rho$, the finite sum over V has the same bound with $3.44\,\mathsf{C}_n$ and periodic distances. Its tail is bounded by the displayed infinite-grid tail, so the decay as $r \to \infty$ is uniform over these tori.

(3) *One vacuum, the Euclidean group, a gap.* Let $\mathsf{b} \in I(g)$, $0 < g \leq \gamma$. Every Gibbs sequence at g with the pairs $(K_j, L^{K_j}\mathsf{b})$, $K_j \geq \max\{K_0, 1\}$, is $(\nu_{K_j})_j$; the full sequence (ν_K) converges to a functional σ with $S_n^\sigma = S_n$ for $n \geq 2$; σ is its own and only ergodic component, invariant under $\mathbb{R}^3 \rtimes \mathrm{O}(3)$ and reflection positive at every plane; $\ker H_\sigma = \mathbb{C}\Omega_\sigma$, and $\mathcal{H}(\sigma)$ carries a Wightman theory of one Hermitian scalar field on $\mathbb{R}^{1+2}$ with a unique vacuum and the Schwinger functions of σ. Every continuum limit at g along growing tori with the pairs $(K_j, L^{K_j}\mathsf{b})$, those with the spacing removed first among them, is $(S_n^\sigma)_{n\geq 2}$. Finally $\operatorname{spec} H_\sigma \subset \{0\} \cup [m_*, \infty)$: for all F, F' in the span of separated tuples in the upper half space and $t \geq 0$,

$$\left|\sigma(\Theta\bar{F} \cdot \tau_{te_3}F') - \overline{\sigma(F)}\,\sigma(F')\right| \leq e^{-m_* t}\,\|(1 - P_\Omega)[F]\|\,\|(1 - P_\Omega)[F']\|,$$

Θ the reflection $x_3 \mapsto -x_3$, P_Ω the projection on Ω_σ; the same in every direction.

(4) *An area law.* For every rectangle C of a coordinate plane with physical sides $R, T \in a_K\mathbb{N}$,

$$0 \leq \nu_K(W_C) \leq e^{-\sigma_* RT}, \qquad \sigma_* = \frac{2\log(1/\theta) + 1.67}{\ell_0^2} \geq \frac{1.9028}{\ell_0^2};$$

$|\nu_K(W_C)| \leq e^{-\sigma_*|A(C)|}$ for every staircase loop of a coordinate plane, $|A(C)|$ its physical area; and for every closed lattice path C with a bond of the direction 3, $|\nu_K(W_C)| \leq 2^{c_3(C)-1}e^{-\sigma_* A_{13}(C)}$, where $A_{13}(C)$ is the area of the union of the squares of side a_K of the plane (13) about which the projection of C winds an odd number of times, and $c_3(C)$ is the number of passages from a downward to an upward bond of the direction 3 in the cyclic order along C; the same holds for every ordered pair of directions. The static potential $V_K(r) = \lim_{T\to\infty} -T^{-1}\log \nu_K(W_{r\times T})$, $r, T \in a_K\mathbb{N}$ physical, exists in $[0, \infty]$, is concave and non-decreasing, and $V_K(r) \geq V_K(r') + \sigma_*(r - r')$ for $r \geq r'$. For every ℓ with ℓ/a_K an integer ≥ 3: $0 \leq 1 - r_1(\ell/a_K) \leq 2\exp(-8.8 \cdot 10^{-4}\,\ell^2/\ell_0^2)$, with $5.707 \cdot 10^{-3}$ in the place of $8.8 \cdot 10^{-4}$ for $\ell \geq 128\,\ell_0$; and with $\mathsf{q} = 1 + 0.835/\log(1/\theta)$, for $\ell \geq \mathsf{q}\ell_0$: $\frac{1}{2}(1 - r_1(\ell/a_K)) \leq \exp[-\sigma_*\ell^2 + (4\mathsf{q}\ell/\ell_0)(\frac{1}{2}\log(4\mathsf{q}\ell/\ell_0) + 0.86)]$. None of these bounds depends on K, b or L.

■□

Theorem BA (All labels: one theory up to dilation)**.** Assume $\mathrm{BI}[\mathsf{b}, \ell_0, K_0; x_0]$ at one label $\mathsf{b} > 0$.

(1) There are $\beta_0 > 0$ and $C_+ < \infty$ such that for every real $\beta \geq \beta_0$ there is an integer l with $3 \leq l \leq C_+\beta$ and $\delta(l^3) \leq 0.008625$, hence $\mathsf{e}(l) \leq 0.069$, at β.

(2) For every $\mathsf{b}' > 0$ and every integer m with $L^m \geq 48C_+\mathsf{b}'$, $\mathrm{BI}[\mathsf{b}', 2L^m, K_0'; 0.069]$ holds, K_0' the least integer with $L^{K_0'}\mathsf{b}' \geq \beta_0$ and $K_0' + m \geq 1$; on every compact set of labels one side and one threshold serve. Hence Thm. AZ holds at every label, its clause (3) at every label of every $I(g)$; and the set of the labels at which BI holds at some side and some threshold is empty or all of $(0, \infty)$.

(2′) *Near the label, with the same threshold.* Let $\ell_1 \in \mathfrak{L}$, $\ell_1 \geq 75\,\ell_0$, $f(x) = \max\{0, 1 - 8|x|_\infty/\ell_1\}$ and $\mathsf{M} = \sup\{\|\mathcal{O}_K(f) - \mu_{K,\ell_1,L^K\mathsf{b}'}(\mathcal{O}_K(f))\|_{\mathrm{ch},\ell_1} : K \in \mathbb{N},\ \mathsf{b}' \in [\frac{\mathsf{b}}{2}, 2\mathsf{b}]\}$. Then $\mathsf{M} < \infty$, and with $\eta = \min\{\frac{\mathsf{b}}{2},\ 0.0086/(0.17516 \cdot 1458\,\mathsf{M})\}$: for all $|\mathsf{b}' - \mathsf{b}| \leq \eta$ and all $K \geq \max\{K_0, 1\}$, $\delta_K(\ell_1) \leq 0.008625$ at $\beta = L^K\mathsf{b}'$, so $\mathrm{BI}[\mathsf{b}', 2\ell_1, \max\{K_0, 1\}; 0.069]$; $\mathsf{b}' \mapsto S_n^{(\mathsf{b}')}(X)$ is continuous there, and $S_n^T[\nu_{K_j,\mathsf{b}_j}](X) \to S_n^{(\mathsf{b})}(X)$ along all $K_j \to \infty$, $\mathsf{b}_j \to \mathsf{b}$.

Moreover, let $\ell_0 = 2L^{m_0}$, let $\ell_1^\flat = 2L^{m_1}$ with m_1 the least $m \in \mathbb{N}$ with $2L^m \geq \ell_0$ and $L^3\, 0.13736\, \theta^{2L^m/\ell_0} \leq 0.03$, let $f^\flat$, $\mathsf{M}^\flat$ be formed as f, M with $\ell_1^\flat$ for ℓ_1, and $r = \min\{\frac{\mathsf{b}}{2}, 0.03/(16L^3\mathsf{M}^\flat)\}$. Then for all $|\mathsf{b}' - \mathsf{b}| \leq r$ and all $K \geq \max\{K_0, 1\}$, at $\beta = L^K\mathsf{b}'$: $0 \leq D_L(\ell_1^\flat/a_K) < 0.06$, $\mathsf{e}(\ell_1^\flat/a_K) < 2.2 \cdot 10^{-4}$, and $\mathrm{BI}[\mathsf{b}', L^z\ell_1^\flat, \max\{K_0, 1\}; 6.1 \cdot 10^{-9}]$ for every integer $z \geq 1$. Neither η nor r depends on the cutoff.

(3) For every $K \in \mathbb{N}$ and every real $\beta \geq \beta_0$ the lattice $a_K\mathbb{Z}^3$ has exactly one periodic Gibbs state ν at β, which is also the limit of the tori; $|\nu(AB) - \nu(A)\nu(B)| \leq e^{-c_* n/\beta} \sup|A| \sup|B|$ for bounded functions of two slabs between which lie n layers, $c_* = \log(1/0.8901)/C_+$: the correlation length in lattice units is at most β/c_*, and the transfer operator T_ν of ν (the setting, in every coordinate direction as the time) has $\ker(1 - T_\nu) = \mathbb{C}\Omega$ and norm at most $e^{-c_*/\beta}$ on $\Omega^\perp$. Moreover $0 \leq 1 - r_1(N) \leq 2\exp[-2.2 \cdot 10^{-4}(N/C_+\beta)^2]$ at β for every integer $N \geq 3$, and in lattice units every slope of the static potential of ν is at least $0.4757\,(C_+\beta)^{-2}$. For $0 < g \leq \gamma$ and $\ell_0^g = 2C_+L(g^{-2} + \bar{B})$: $|S^T_{n;K,\ell,\beta}(X) - S^T_{n;K,\ell',\beta}(X)| \leq C_g(X)\, 0.8901^{\ell/2\ell_0^g}$ for $\ell \leq \ell'$ in $\mathfrak{L}(g)$ above a side that X fixes, uniformly in the first-passage data (K, β) at g with $\beta \geq \beta_0$, with $C_g(X)$ free of K, β, ℓ, ℓ'; the same with ν for the torus ℓ'.

(4) For every $c > 0$ there is one functional σ_c such that for all real spacings $a_j \to 0$ and couplings β_j with $a_j\beta_j \to c$, and every choice of periodic Gibbs states on $a_j\mathbb{Z}^3$ at β_j, the sequence converges to σ_c. With $X_s = (f_1(\cdot/s), \dots, f_n(\cdot/s))$,

$$S_n^{\sigma_{tc}}(X) = t^{-n}\, S_n^{\sigma_c}(X_{1/t}) \qquad (t, c > 0,\ n \geq 2),$$

and σ_c is the same for every admitted blocking factor. In lattice units, $H_n(Y) = \lim_{\beta\to\infty} \kappa_n(\beta; Y)$ exists along the real β, where $\kappa_n(\beta; Y)$ is the joint cumulant of the $\mathcal{O}_{1/\beta}(y_i)$ in the periodic Gibbs state on $\beta^{-1}\mathbb{Z}^3$ at β, and $S_n^{\sigma_c}(X) = c^{-n}H_n(X_{1/c})$. For $c \in I(g)$, σ_c is the functional σ of Thm. AZ(3) at the label c; its gap scales like $1/c$.

(5) *The rotational Ward identity.* Let $\mathsf{b} \in I(g)$, $0 < g \leq \gamma$, $n \geq 2$, $\mathsf{d} > 0$, $R \geq 1$, $0 < \kappa < \frac{1}{16}$. There are C and p', depending on g, n, d, R, κ, the data of BI and the constants, but on no side and on no place, such that for all large K, all $\ell \in \mathfrak{L}(g)$ and $v \in a_K\mathbb{Z}^3$ with $|v|_\infty + R + 2 \leq \ell/4$, all $\omega \in \mathfrak{so}(3)$, and all real $\chi, f_1, \dots, f_n \in C^{1,1}(\mathbb{R}^3)$ that vanish outside $v + [-R, R]^3$, the f_i at pairwise sup distances $\geq \mathsf{d}$, χ equal to 1 at the centres and at the bonds of the plaquettes at which some f_i does not vanish, the bulk term $\mathfrak{U}^{(K,\ell,L^K\mathsf{b})}_{\omega,\chi}(f_1, \dots, f_n)$ of the rotational Ward identity YM3_I:(2108) on the torus ℓ (the joint cumulant of the $\mathcal{O}_K(f_i)$ with one insertion of the local density of the rotation ω weighted by χ, together with the like cumulants in which the insertion is weighted by one f_i) satisfies

$$\big|\mathfrak{U}^{(K,\ell,L^K\mathsf{b})}_{\omega,\chi}(f_1, \dots, f_n)\big| \leq (1 + 2|v|_1)\, C\, |\omega|\, K^{p'} L^{-\kappa K}\, \|\chi\|_{C^{1,1}} \prod_i \|f_i\|_{C^{1,1}},$$

and the same bound holds for the bulk term in the state ν_K. The bound rests, besides on BI and on Thm. YM3_I:D, on the bound (M7) of the member $\mathfrak{U}^\sharp_{\omega,\chi}$ of the bulk term of third order in the curvature.

■□

Theorem BB (One cutoff decides; a form without cutoff)**.** Let $\mathsf{b} > 0$ and $\ell = 2L^m \in 2L^{\mathbb{Z}}$. Let $\varsigma = \sum_{j\geq 0}(1 + j)^{\mathsf{p}_0} L^{-j/10}$, and let $z = z(\mathsf{b}, \ell)$ be the least integer $z \geq 0$ with $L^z\mathsf{b} \geq \gamma^{-2}$, $m + z \geq 2$ and $LM\, R(L^z\mathsf{b}) \leq L^{m+z-1}$.

(1) There are an explicit threshold $K_\star(\mathsf{b}, \ell)$, formed from the constants, b and ℓ, with $K_\star(\mathsf{b}, \ell) - z$ at or above the thresholds K_{bf} of (M4)(ii) on the two tori $L^{z-1}\ell$ and $L^z\ell$ at the label $L^z\mathsf{b}$, to which the dilation of Prop. 17.44(c) by $z = z(\mathsf{b}, \ell)$ brings the data, and the margins

$$\eta_K(\mathsf{b}, \ell) = 3\varsigma\, \mathsf{C}_{\mathrm{bf}}(L^z\mathsf{b})\, (L^z\ell)^3 (K - z)^{\mathsf{p}_0} L^{-(K-z)/10} \leq C(\mathsf{b}, \ell)\, K^{\mathsf{p}_0} L^{-K/10},$$

$\mathsf{C}_{\mathrm{bf}}(\mathsf{b}')$ a number of the constants and of b', bounded on compact sets of labels, so that $C(\cdot,\ell)$ is bounded on compact sets of labels (where $z=0$, $\eta_K = 3\varsigma\,\mathsf{C}_{\mathrm{bf}}(\mathsf{b})\,\ell^3 K^{\mathsf{p}_0} L^{-K/10}$), such that $|X_{K'}(\mathsf{b},\ell) - X_K(\mathsf{b},\ell)| \le \eta_K(\mathsf{b},\ell)$ for $K' \ge K \ge K_\star(\mathsf{b},\ell)$. So $X_\infty(\mathsf{b},\ell) = \lim_K X_K(\mathsf{b},\ell) \ge 0$ exists, uniformly on compact sets of labels. For every $y > 0$: if $X_{K_1} \le y - \eta_{K_1}$ at one $K_1 \ge K_\star$, then $X_K \le y$ for all $K \ge K_1$; $X_\infty < y$ if and only if $X_{K_1} < y - \eta_{K_1}$ at one $K_1 \ge K_\star$; $X_\infty > y$ if and only if $X_{K_1} > y + \eta_{K_1}$ at one $K_1 \ge K_\star$, and then $X_K > y$ for all $K \ge K_1$.

(2) *The cube criterion.* Let $2^k \le L/2 < 2^{k+1}$, $\Phi_L(x) = (L/2^{k+1})^3[\epsilon + \frac{20}{7}\epsilon^2]$ with $\epsilon = (12.9x)^{2^k}/12.9$, $x_1 = \max\{x \in (0, 0.069] : \Phi_L(x) \le x_0\}$, $y_1 = \frac{L^3}{8}x_1$ and $y_0 = \frac{L^3}{8}x_0$. If $X_{K_1}(\mathsf{b},\ell) \le y_1 - \eta_{K_1}(\mathsf{b},\ell)$ at one $K_1 \ge K_\star(\mathsf{b},\ell)$, then $\mathrm{BI}[\mathsf{b},\ell,K_1;x_0]$ holds; if $X_{K_1}(\mathsf{b},\ell) \le y_0 - \eta_{K_1}(\mathsf{b},\ell)$ there, then $\mathrm{BI}[\mathsf{b},2\ell/L,K_1;x_0]$ holds (with $c=2$), and with it Thm. AZ at the side $2\ell/L$, of rate $L\log(1/\theta)/\ell$. Conversely $\mathrm{BI}[\mathsf{b},2\ell/L,K_0;x_0]$ gives $X_K(\mathsf{b},L\ell) \le L^3\bar\delta_{x_0}(L)$ for all $K \ge K_0$, with $\bar\delta_x(\nu) = (\nu/2^j)^3\mathsf{c}_0(12.9x)^{2^j}$ for $2^j \le \nu < 2^{j+1}$, $\mathsf{c}_0 = \frac{20}{7\cdot 12.9^2} \approx 0.0171693$; so $X_{K_1}(\mathsf{b},L\ell) > L^3\bar\delta_{x_0}(L) + \eta_{K_1}(\mathsf{b},L\ell)$ at one $K_1 \ge K_\star(\mathsf{b},L\ell)$ refutes it. For every K with $K+m \ge 2$,

$$\min\Big\{0.069, \Big[\tfrac{7}{20}L^{-3}X_K(\mathsf{b},L\ell)\Big]^{1/2}\Big\} \le \mathsf{e}_K(\mathsf{b},\ell) \le \min\Big\{\tfrac{2}{L}X_K(\mathsf{b},\ell),\ \Phi_L\big(\tfrac{8}{L^3}X_K(\mathsf{b},\ell)\big)\Big\},$$

the last member where its argument is ≤ 0.069; so $\mathrm{BI}[\mathsf{b},\ell,K_0;x_0]$ gives $X_K(\mathsf{b},L\ell) \le \frac{20}{7}L^3x_0^2$ for $K \ge K_0$, and if $X_{K_1}(\mathsf{b},L\ell) > \frac{20}{7}L^3x_0^2 + \eta_{K_1}(\mathsf{b},L\ell)$ at one $K_1 \ge K_\star(\mathsf{b},L\ell)$, then $\mathsf{e}_K(\mathsf{b},\ell) > x_0$ for every $K \ge K_1$. The same bracket holds for $\liminf_K \mathsf{e}_K$ and $\limsup_K \mathsf{e}_K$ with X_∞.

(3) *The inequality of BI itself at one cutoff.* Let $\ell_0 = 2L^{m_0}$, $i \ge 1$ and $\ell = L^i\ell_0$. If $\mathsf{e}_{K_1}(\mathsf{b},\ell_0) \le x_0$ at one $K_1 \ge K_\star(\mathsf{b},\ell)$ with $L^3\bar\delta_{x_0}(2L^{i-1}) + \eta_{K_1}(\mathsf{b},\ell) \le y_1$, then $X_K(\mathsf{b},\ell) \le y_1$ for all $K \ge K_1$ and $\mathrm{BI}[\mathsf{b},L^i\ell_0,K_1;x_0]$ holds, of rate $L^{-i}m_*$; with y_0 in the place of y_1 it gives $\mathrm{BI}[\mathsf{b},2L^{i-1}\ell_0,K_1;x_0]$. The margin condition reads $\bar\delta_{x_0}(2L^{i-1}) \le \frac{1}{8}x_1 - L^{-3}\eta_{K_1}$; at $x_0 = 0.069$ and $i=2$ it holds, once η_{K_1} is small enough, at every odd $L \ge 33$, so at every admitted blocking factor, and at $i=1$ it never holds. So, wherever $L^3\bar\delta_{x_0}(2L) < y_1$, with $K_{\star\star}(\mathsf{b},\ell_0)$ the least $K \ge K_\star(\mathsf{b},L^2\ell_0)$ such that $\eta_{K'}(\mathsf{b},L^2\ell_0) \le y_1 - L^3\bar\delta_{x_0}(2L)$ for all $K' \ge K$, an explicit expression, the inequality $\mathsf{e}_{K_1}(\mathsf{b},\ell_0) \le x_0$ at one $K_1 \ge K_{\star\star}(\mathsf{b},\ell_0)$ gives $\mathrm{BI}[\mathsf{b},L^2\ell_0,K_1;x_0]$.

(4) The following are equivalent: (i) $\mathrm{BI}[\mathsf{b},\ell_0,K_0;x_0]$ holds for some $\ell_0 \in 2\mathbb{N}L^{\mathbb{Z}}$ and some K_0; (ii) $X_\infty(\mathsf{b},\ell) < y_0$ for some $\ell \in 2L^{\mathbb{Z}}$; (iii) $X_{K_1}(\mathsf{b},\ell) < y_0 - \eta_{K_1}(\mathsf{b},\ell)$ for some $\ell \in 2L^{\mathbb{Z}}$ and one $K_1 \ge K_\star(\mathsf{b},\ell)$; (iv) $X_\infty(\mathsf{b},\ell) \to 0$ as $2L^{\mathbb{Z}} \ni \ell \to \infty$. More precisely, (iii) at (ℓ,K_1) gives (i) at $\ell_0 = 2\ell/L$, $K_0 = K_1$; and (i) at (ℓ_0,K_0) gives $X_K(\mathsf{b},\ell) \le L^3\bar\delta_{x_0}(2\ell/L\ell_0) < 0.13736\,L^3\theta^{\ell/L\ell_0}$ for all $K \ge K_0$ and all $\ell \in 2L^{\mathbb{Z}}$ with $\ell \ge L\ell_0$.

(5) *The half box.* This branch remains a sketch at the full interacting Hessian estimate described with Thm. AH. Let $\ell_0 = 2L^{m_0}$ and J a compact set of labels. There are C and K_* with $|\mathsf{e}_{K+1}(\mathsf{b}',\ell_0) - \mathsf{e}_K(\mathsf{b}',\ell_0)| \le C\,L^{-K/10}$ for $\mathsf{b}' \in J$ and $K \ge K_*$. Hence $\mathsf{e}_\infty = \lim_K \mathsf{e}_K(\mathsf{b},\ell_0)$ exists, and with $Q_K = \sum_{k\ge K} CL^{-k/10}$, $|\mathsf{e}_K - \mathsf{e}_\infty| \le Q_K$: if $\mathsf{e}_{K_1}(\mathsf{b},\ell_0) \le x_0 - Q_{K_1}$ at one $K_1 \ge K_*$, then $\mathrm{BI}[\mathsf{b},\ell_0,K_1;x_0]$ holds, at the same side; $\mathrm{BI}[\mathsf{b},\ell_0,K_0;x_0']$ with $x_0' < x_0$ gives $\mathsf{e}_{K_1} \le x_0 - Q_{K_1}$ at every large K_1; $\mathsf{e}_\infty < x_0$ if and only if $\mathsf{e}_{K_1}(\mathsf{b},\ell_0) < x_0 - Q_{K_1}$ at one $K_1 \ge K_*$; and if $\mathsf{e}_{K_1}(\mathsf{b},\ell_0) > x_0 + Q_{K_1}$ at one $K_1 \ge K_*$, then $\mathrm{BI}[\mathsf{b},\ell_0,K_0;x_0]$ fails for every K_0.

(6) *A form without cutoff.* For every $\ell \in 2L^{\mathbb{Z}}$ and every real $\mathsf{b} > 0$ the limit $D_\infty(t) = \lim_K D_L(\ell/a_K)|_{\beta=L^K\mathsf{b}}$ exists, depends on $t = \ell/\mathsf{b}$ alone, is continuous on $(0,\infty)$, and $\sup_{t\in[t_-,t_+]}|D_L(\ell/a_K) - D_\infty(t)| \le C\,K^{\mathsf{p}_0}L^{-K/10}$ on compact sets for large K. The function $\delta_\infty(t) = \sum_{r\ge 0} L^{-3(r+1)}D_\infty(L^r t)$ is continuous and positive, $\delta_K(\ell) \to \delta_\infty(\ell/\mathsf{b})$ uniformly on compact sets of labels, $D_\infty(t) = L^3\delta_\infty(t) - \delta_\infty(Lt)$ and $(L-1)^3\delta_\infty \le D_\infty \le L^3\delta_\infty$. δ_∞ is the same for every admitted blocking factor; $t^{-3}\delta_\infty(t)$ does not increase, and $\delta_\infty(\rho t) \le$

$(3\rho^2-3\rho+1)\delta_\infty(t)\le 3\rho^2\delta_\infty(t)$ for real $\rho\ge 1$. Either $\delta_\infty\ge 0.008625$ on $(0,\infty)$, or $\delta_\infty(t_0)<0.008625$ at some t_0, and then $\delta_\infty(t)\le 0.1374\,(103.2\,\delta_\infty(t_0))^{t/2t_0}$ for all real $t\ge 2t_0$. The second case holds if and only if BI holds at some side of some label, and it holds as soon as $D_{q,\infty}(t_0)<(q-1)^3x_0/8$ at one t_0 for one admitted q, $D_{q,\infty}(t)=q^3\delta_\infty(t)-\delta_\infty(qt)$.

(7) *The size of the thresholds.* Read as inequalities, the conditions of the list of constants admit exactly the odd blocking factors $L>48\cdot 4096^2e^{24}=2.1331\ldots\cdot 10^{19}$, the least being 21 331 858 733 892 671 083. At every admissible choice of the constants, at a cutoff $K_1\ge K_\star(\mathsf{b},\ell)$ the small cube of the criterion of (2) has more than $10^{1.42\cdot 10^{11}}$ sites a side, and at a cutoff at which the criterion of (3) with $i=2$ applies each of the two boxes of BI has more than $10^{1.89\cdot 10^{12}}$ sites. Independently of all constants but L, $\eta_K(\mathsf{b},\ell)\ge 48\,\varsigma_\flat(L)\,L^6(K-z)^{5/2}L^{-(K-z)/10}$ with $\varsigma_\flat(L)=\sum_{j\ge 0}(1+j)^{5/2}L^{-j/10}$; as $X_K\ge 0$, the margin alone forces more than 10^{152} sites a side on the small cube of any inequality $X_{K_1}\le y-\eta_{K_1}$ with $y\le y_0$.

■□

To complete clause (5)(c) below and the indicated part of Prop. 17.48(b), one still needs a uniform probability estimate for tubes around holonomy classes. We call the target the **Heaviest-class estimate**. For every integer $j\ge 1$, put $N=2L^j$. For every $\beta>0$ and every $\omega>0$ with $\Xi=\beta\omega^2\ge 1$ and $\Xi N^5\omega\le 3/640$, every w with $N^2\omega\le w\le 1/10$, every pair $Q\in G^2$ and every site y of the periodic cube, its Wilson probability law satisfies

$$\mu_{N^3,\beta}\big(\mathsf{D}((P_1,P_2)(y),Q)\le w,\ \max_p|\partial U(p)-\mathbf{1}|<\omega\big)\le 10^{11}\Xi^5N^7w^6\omega^{-4}.$$

Here P_1,P_2 are the straight winding holonomies through y, and $\mathsf{D}(P,Q)=\min_{g\in G}\max_{i=1,2}|P_i-gQ_ig^{-1}|$ is the distance of pairs modulo simultaneous conjugation. This probability bound remains to be proved. The pointwise density maximum below does not control the tube measure, especially near degenerate or central pairs.

Theorem BC (Small tori without the box criterion)**.**

(1) *Comparison.* For $\beta\ge 0$, let $N,l\ge 3$ and $q\ge 2$ be integers. Then $0\le\mathsf{e}(N)\le D_2(N)\le 8\delta(N^3)$ and $D_q(N)\ge\frac{q^3}{8}D_2(N)$; $(q-1)^3\delta(N^3)\le D_q(N)\le q^3\delta(N^3)$, and $\delta(M^3)\le(3\rho^2-3\rho+1)\delta(N^3)\le 3\rho^2\delta(N^3)$ for $M=\rho N\ge N$; $\delta(\underline{n})\le(n_1n_2+n_1n_3+n_2n_3)N^{-2}\delta(N^3)$ for every box with all sides $\ge N$, so $\mathsf{e}(l)\le 16(l/N)^2\delta(N^3)$ for $l\ge N$; $\mathsf{e}(l)\le\frac{2}{L}D_L(2l/L)$ if $l=Ll'$, $l'\ge 2$; if $D_L(N)\le\frac{L^3}{8}x$ with $x\le 0.069$, then $D_2(N)\le x$, $\mathsf{e}(N)\le x$ and, for even N, $\mathsf{e}(LN/2)\le\Phi_L(x)$; $\mathsf{e}(l)\le x\le 0.069$ gives $D_L(2l)\le\frac{20}{7}L^3x^2$; $\delta(N^3)\le x_0/8$ gives $\mathsf{e}(N)\le x_0$; $\mathsf{e}(l)\le x_0$ gives $\delta(N'^3)\le 0.008625$ for $N'\ge 48l$; and $\mathsf{e}(l)\le 24\beta l^3$, so $\mathsf{e}(l)\le x_0$ whenever $0<\beta\le x_0/24l^3$.

(2) *Dichotomy.* At each $\beta\ge 0$ the following are equivalent: $\mathsf{e}(l)\le 0.069$ for one $l\ge 3$; $\mathsf{e}(l)\to 0$; $\delta(N^3)\to 0$; $\delta(N^3)\le 0.008625$ for one $N\ge 3$. Then $\delta((2N)^3)\le 183\,\delta(N^3)^2$ and $\delta(N'^3)\le 0.13736\,(103.2\,\delta(N^3))^{N'/2N}$ for $N'\ge 2N$. The numbers 0.07489 and 0.00936 may stand for 0.069 and 0.008625 in this equivalence. In the other case $\mathsf{e}(l)>0.069$ and $\delta(N^3)>0.008625$ for all $l,N\ge 3$, and indeed $\mathsf{e}(2^kl)>0.07489$ for all l and k; this case holds whenever there are two periodic Gibbs states at β, and then $\liminf_m\delta((2L^{m+K})^3)\ge\log 2$. Every phase is reached by the tori: if $\liminf_m\delta((2L^{m+K})^3)\le D<\infty$ at β, there are sides along which the tori converge to a periodic Gibbs state μ^* with $\mu^*\ge e^{-D}\nu$ for every ergodic periodic Gibbs state ν, and every periodic Gibbs state is at most $e^D\mu^*$.

(3) *A floor.* Let $0<g\le\gamma_W$, let $\mathsf{d}_0(g,\vartheta)$ and $s(\mathsf{d})$ be the threshold and the depth of the law of the two-point function at short distance of (M5), $\vartheta_1=3\sqrt{3}/8$, let $0<\mathsf{d}_\circ\le\mathsf{d}_0(g,\vartheta_1)$, let (K,β) be a first-passage datum at g with $K>s(\mathsf{d}_\circ)$, and $N_4=\lfloor 4\mathsf{d}_\circ/a_K\rfloor$. Then for every

integer $l \geq 3$

$$\mathsf{e}(l) \geq \min\{0.069,\ \tfrac{1}{12.9}\,31250^{-l/N_4}\} \geq \min\{0.069,\ 0.0775\,e^{-2.6\,l a_K/\mathsf{d}_\circ}\};$$

in particular $\mathsf{e}(\ell/2a_K) \geq \min\{0.069, 0.0775\,e^{-1.3\ell/\mathsf{d}_0(g,\vartheta_1)}\}$ whenever $\ell/a_K \geq 6$ is even, the same floor lies under D_2, 8δ and $\frac{8}{L^3}D_L$ at the side $\ell/2$, and $\mathsf{e}(l) \leq 0.069$ fails for every l with $la_K < 0.0449\,\mathsf{d}_\circ$.

(4) *What BI forces.* If $\mathrm{BI}[\mathsf{b}, \ell_0, K_0; x_0]$ holds at some label, then $\ell_0 \geq 0.0895\,(\mathsf{d}_0^\star/\kappa_0)\,\mathsf{b}$, where $\kappa_0 = L(\gamma_W^{-2} + \bar{B})$, $\mathsf{d}_0^\star = \min\{\mathsf{d}_0(g_1,\vartheta_1), \mathsf{d}_0(g_2,\vartheta_1)\}$, $g_1 = \gamma_W$ and $g_2 = (\gamma_W^{-2} + 2\bar{B}/L)^{-1/2}$. On one lattice, every box and $t \geq 3$: $2F(t,\Sigma) - F(2t,\Sigma) \geq 2\log\sum_f w_f^{1/2}$, the w_f the weights of the box $(2t,\Sigma)$; on the cube $2l$, $\mathsf{e}(l) \geq 2\log(\sqrt{1-x} + \sqrt{2x})$ for $x = 1 - r_1 \leq \frac{1}{2}$, $\mathsf{e}(l) \geq \log\frac{4}{1+3r_1}$, and $\mathsf{e}(l) < \log 2$ forces $r_1 > \frac{8}{9}$; so $\mathsf{e}(l) \leq 0.069$ forces $r_1 \geq 0.99937$. At every $K \geq K_0$, on the cube $2l_K$ at $\beta = L^K\mathsf{b}$, with $x_0 = 0.069$: $1 - r_1 \leq 0.000628$, $1 - r_2 \leq 0.001255$, $1 - r_3 \leq 0.001882$, $\delta_K(\ell_0) \leq 0.013603$, and $\sum_{i\geq 1}(\lambda_i/\lambda_0)^{l_K} \leq 0.035764$ over the cross-section $(2l_K, 2l_K)$, so $\log(\lambda_0/\lambda_i) \geq 6.66\,a_K/\ell_0$ for $i \geq 1$. $\mathrm{BI}[\cdot; 0.069]$ is false as soon as $r_1 < 0.99937$ on the cube ℓ_0 at one cutoff $K \geq K_0$.

(5) *Small tori.* Let $\vartheta(\tau,u) = \sum_{k\in\mathbb{Z}} e^{-\tau(2\pi k+u)^2}$,

$$C_\Delta = \int_0^\infty \Big[1 - e^{-\tau} - \vartheta(\tau,0)^3 + (4\pi\tau)^{-3/2}\Big]\frac{d\tau}{\tau},$$

$$\Upsilon(u) = \exp\int_0^\infty \Big[e^{-\tau} - 1 + \vartheta(\tau,0)^3 - \prod_{\mu=1}^3 \vartheta(\tau,u_\mu)\Big]\frac{d\tau}{\tau},$$

$\mathscr{J} = \int_{[-\pi,\pi]^3} du/\Upsilon(u)$ and $c' = \log(\mathscr{J}/8\pi^2) - \frac{3}{2}C_\Delta$; numerically $c' \approx 2.0919$. There are numbers $t_1, t_2, t_4 > 0$ and $C, C_\flat$ of the constants, with $\mathsf{a} = \frac{9}{2}r + \mathfrak{p}_0$ and $\mathsf{q} = 1116\,r + 720\,\mathfrak{p}_0 + 366$, such that for every side $\ell = 2L^m$, $m \in \mathbb{Z}$, every real $\mathsf{b} > 0$ and every cutoff $K \geq 1 - m$, with $t = \ell/\mathsf{b}$: (a) if $t \leq t_1$, then $|\delta_K(\ell) - \log(\mathsf{b}/\ell) - c'| \leq C\,t^{1/1500} + C\,(a_K/\ell)^2\log^{5/2}(\ell/a_K)$, and $|\delta_\infty(t) - \log(1/t) - c'| \leq C\,t^{1/1500}$; (b) if $t \leq t_2$, then $r_1(\ell/a_K) \leq \eta_\flat(t) = C_\flat(\log\frac{1}{t})^{\mathsf{a}}t^{1/500}$ at $\beta = L^K\mathsf{b}$, the weights of the four classes of electric flux on the cube satisfy $|w_f - \frac{1}{4}| \leq \frac{3}{4}\eta_\flat(t)$, and $\delta_K(\ell)$ and $\mathsf{e}(\ell/2a_K)$ are both $\geq \mathsf{G}_{\ell/a_K} \geq \log\frac{4}{1+3\eta_\flat(t)}$; so $\liminf_{t\to 0}\inf_K \delta_K(\ell) \geq \log 4$, the same for $\mathsf{e}(\ell/2a_K)$, and if $\ell_0 = 2L^{m_0}$, $\ell_0/\mathsf{b} \leq t_2$ and $\eta_\flat(\ell_0/\mathsf{b}) < 0.99937$, the inequality of $\mathrm{BI}[\mathsf{b}, \ell_0, K_0; x_0]$ fails at every cutoff $K \geq 1 - m_0$, for every $x_0 \leq 0.069$; (c) The sharper remainder in this clause requires the **(Heaviest class)** probability estimate: if $t \leq t_4$, the intended conclusion is that over the cross-section $(\ell/a_K, \ell/a_K)$ the top level carries the weight $\lambda_0^N/Z(N^3) \leq C\,t\,(\log\frac{1}{t})^{\mathsf{q}}$, $N = \ell/a_K$; so $\delta_K(\ell)$ and $\mathsf{e}(\ell/2a_K)$ are $\geq \mathsf{G}_N \geq \log\frac{\mathsf{b}}{\ell} - \mathsf{q}\log\log\frac{\mathsf{b}}{\ell} - \log C$: the coefficient of $\log(\mathsf{b}/\ell)$ in (a) is reached from below at every cutoff; (d) there are numbers $t_\circ > 0$ and $\beta_\flat$ of the constants such that for every real $\beta \geq \beta_\flat$: $\mathsf{e}(l) > 0.069$ for every integer $3 \leq l \leq t_\circ\beta$; so BI at any label b forces $\ell_0 > 2t_\circ\mathsf{b}$. On one lattice, exactly: if $1 - r_1(N) \geq 0.0039$ on one cube, then $\mathsf{e}(l) > 0.069$ for every integer $3 \leq l \leq N/16$; and $\mathsf{e}(l) \leq 0.069$ gives $\mathsf{e}(l') < 0.0234$ for every integer $l' \geq 16l$; (e) the pressure of one lattice: $|P(\beta) + 3\log\beta - z_\infty| \leq C\beta^{-1/1500}$ for all large real β, with $z_\infty = \log\frac{2}{\pi} - \frac{3}{16\pi^3}\int_{[-\pi,\pi]^3}\log\sum_\mu(2 - 2\cos p_\mu)\,dp \approx -2.96167$.

The leading logarithmic lower bound in clause (c) also holds without (Heaviest class), with q replaced by $\mathsf{p}_\star = 1209r + 780\mathfrak{p}_0 + 396$ and with possibly different constants t_4, C. Indeed, Prop. 17.6(d), with $\mathsf{L} = L$ and $j = m + K \geq 1$, applies to $N = \ell/a_K = 2L^j$ and $\beta = N/t$ and gives $\lambda_0(N,N)^N/Z(N,N,N) \leq \mathsf{C}_\star t(\log(1/t))^{\mathsf{p}_\star}$. Since $\mathsf{G}_N \leq \delta_K(\ell)$ and $\mathsf{G}_N \leq \mathsf{e}(N/2)$, taking negative logarithms gives both lower bounds.

(6) *An area from above.* If $t = \ell/\mathsf{b} > t_1$ and j is the least integer with $L^{-j}t \le t_1$, then $\delta_K(\ell) \le 3L^2(t/t_1)^2\,[\log(L/t_1) + c' + C]$ at every cutoff $K \ge 1 - m + j$. So $\sup_K \delta_K(\ell) < \infty$ on every torus at every label, and $\mathsf{e}(\ell/a_K)$, $D_L(\ell/a_K)$ are $O((\ell/\mathsf{b})^2)$ uniformly in the cutoff. ■□

17.2.3. *Construction estimates.* Beyond the exact statements on one lattice proved below, the argument uses the following statements, (M1)–(M6) and (M8), one statement of the section on plane fields and the stress tensor (§5.1), Lem. 5.16, entered as (M7), and the comparison of two cutoffs with a factor, Prop. 17.30, which is derived at the end of this subsection from the statements named in (M4)(ii).

(M1) The layers and the crossing operator of one direction (Section YM3_I:85, Prop. YM3_I:85.1): the objects of the setting; a bounded Borel function of the bonds with both ends in a slab of $\vartheta \ge 2$ consecutive layers is Φ_{τ_0} for a bounded Borel function Φ on the configurations $\mathcal{Y}_\vartheta$ of a slab, which depends neither on the length of the box in the time nor on τ_0; its slab operator $\mathfrak{Y}[\Phi] = \mathsf{K}^{1/2}\mathfrak{S}[\Phi]\mathsf{K}^{1/2}$ is Hilbert–Schmidt, $\mathfrak{Y}[1_\vartheta] = \mathsf{T}'^{\vartheta}$, $\mathfrak{Y}[\Phi^{+j}] = \mathsf{T}'^{j}\mathfrak{Y}[\Phi]$ for Φ regarded as a function Φ^{+j} on a slab thicker by j layers, read on the first ϑ of its $\vartheta + j$ layers, so that the j added layers lie above, towards the later slabs of the trace formula below, in which the first slab stands rightmost, and $\mathfrak{Y}[\Phi^\Theta] = \mathfrak{Y}[\Phi]^*$ for the conjugated mirror image Φ^Θ, which, placed on a slab of ϑ layers, is the complex conjugate of Φ read on the configuration mirrored in the plane midway between that slab and the slab of Φ; and the trace formula (Prop. YM3_I:85.2): for $r \ge 2$ functions $\Phi^1, \dots, \Phi^r$ on consecutive slabs filling the box (t, Σ),

$$Z(t,\Sigma)\,\mu_{(t,\Sigma)}\Big(\prod_s (\Phi^s)_{\tau_{s-1}}\Big) = \mathrm{Tr}(\mathfrak{Y}[\Phi^r]\cdots\mathfrak{Y}[\Phi^1]).$$

(M2) The chessboard estimate on the tori $\ell \in \mathfrak{L}$, for one board of side $\ell''/2$, $\ell'' \le \ell$ in $\mathfrak{L}$ ([7, Cor. B.1](b), (c), (e)): $\|\cdot\|^{(\ell'')}_{\mathrm{ch},\ell}$ is a seminorm on the functions of one cell, with $\|1\|^{(\ell'')}_{\mathrm{ch},\ell} = 1$ and $|\mu_{K,\ell,\beta}(F)| \le \|F\|^{(\ell'')}_{\mathrm{ch},\ell} \le \sup|F|$; in its array form $|\mu_{K,\ell,\beta}(\prod_{\mathbf{k}} F_{\mathbf{k}})| \le \prod_{\mathbf{k}} \|F_{\mathbf{k}}\|^{(\ell'')}_{\mathrm{ch},\ell}$ for functions $F_{\mathbf{k}}$ of distinct cells $\mathbf{k}$ of one period, given in all of them or in some of them only; in its fine-to-coarse form, for one board and two tori, $\|F\|^{(\ell'')}_{\mathrm{ch},\ell} \le \|F\|^{(\ell'')}_{\mathrm{ch},\ell'} \exp[(\ell''/2)^3(p_{\ell'} - p_\ell)]$ and $p_\ell \le p_{\ell'}$ for $\ell'' \le \ell' \le \ell$ in $\mathfrak{L}$; and the domination $|\nu(\prod_{\mathbf{k}} F_{\mathbf{k}})| \le \liminf_{\ell\to\infty} \prod_{\mathbf{k}} \|F_{\mathbf{k}}\|^{(\ell'')}_{\mathrm{ch},\ell}$ of a periodic Gibbs state ν, for one board and growing tori (Prop. YM3_I:91.9). Together: for a function F of a cell of side $\ell''/2$ and its board of side $\ell/2$ of the setting, whose eight copies of F are copies of F under the board of F, the array form with the constant 1 in the other cells and the fine-to-coarse form give

$$\|F\|_{\mathrm{ch},\ell} \le \|F\|^{(\ell'')}_{\mathrm{ch},\ell} \le e^{(\ell''/2)^3(p_{\ell''} - p_\ell)}\,\|F\|_{\mathrm{ch},\ell''} :$$

the seminorm of eight copies on the torus ℓ is at most that of all the copies under the board of F, which is at most a factor times the seminorm on the torus ℓ''. (M3) Limits of tori and of periodic boxes are periodic Gibbs states (Prop. YM3_I:91.2); the DLR equations and the conditional laws of a Gibbs state; all periodic Gibbs states are invariant under the isometries of the lattice and reflection positive (Prop. YM3_I:103.4); a periodic Gibbs state is an integral of ergodic ones, and it is extreme if and only if it is ergodic (Prop. YM3_I:106.1). (M4) Thm. YM3_I:D, through the following consequences: (i) on one fixed torus the cumulants $S^T_{n;K,\ell,L^K\mathsf{b}}$ of fields with disjoint supports are bounded at all cutoffs, with bounds that depend on the side, a compact set of labels and the distance of the supports (Thm. YM3_I:AS, uniform bound YM3_I:(2214)), and converge as $K \to \infty$, uniformly on compact sets of labels, to continuous functions of the label (Thm. YM3_I:AS, limit and local uniformity), with the Cauchy rate proved in Thm. YM3_I:AS: on a compact interval of labels, $|S^T_{n;K',\ell,L^{K'}\mathsf{b}}(X) - S^T_{n;K,\ell,L^K\mathsf{b}}(X)| \le C\,K^{2\mathfrak{p}_0(n+1)+\frac12}L^{-K/10}\prod_i \|f_i\|_\sharp$ for all $K' \ge K$ above a threshold, where C and the threshold depend on n, the side, the interval and a floor of the spread of the supports, and on the functions only through their norms, at every admitted choice of

the constants; (ii) for a label $\mathsf{b} \geq \gamma^{-2}$, a side $\ell = 2L^m$ with $MR(\mathsf{b}) \leq L^m$ and a cutoff $K \geq 2$ at which the conditions (k0)–(k4) and (k6) of §2.3.1 hold, with its numbers at $n = 1$ ($u = \lceil K/1260 \rceil$, $s = \lceil K/9 \rceil$, $t_1 = K - u$),

$$\big|\log Z_{K+1,\ell} - \log Z_{K,\ell} - \mathfrak{K}_K(\ell, \mathsf{b})\big| \leq 4\mathsf{R}_{t_1} + 8\epsilon_{\mathrm{fg}} + 8\epsilon_{\mathrm{cl}} + 12\epsilon_{\mathrm{sh}} \leq \mathfrak{o}_K(\ell, \mathsf{b}),$$

where $\mathfrak{K}_K$, a sum of local vacuum constants, is the real part of the difference at the source 0 of the contact terms of the two cutoffs, and $\mathfrak{o}_K = 8\mathsf{R}_{t_1} + 12\epsilon_{\mathrm{fg}} + 8\epsilon_{\mathrm{cl}} + 12\epsilon_{\mathrm{sh}}$ is the number (2.xii). This is the comparison of the two cutoffs at the source 0: the discarding of Prop. YM3_I:115.2, lost-mass bound gives $(1 - \mathsf{R}_{t_1})Z \leq T(0) \leq Z$ at each cutoff for the chain T of the top level; the form $T = e^c I$ and the contact terms of the proof of Prop. YM3_I:116.1 and Section YM3_I:116, with the proof of Prop. YM3_I:116.3, give $|I(0)/\hat{I}(0) - 1| \leq 2\epsilon_{\mathrm{fg}}$ against the classical start $\hat{T} = e^{\hat{c}}\hat{I}$; §2.3.2 and Prop. YM3_I:116.1 give $|\hat{T}_{K+1}(0)/\hat{T}_K(0) - 1| \leq 4\epsilon_{\mathrm{cl}} + 6\epsilon_{\mathrm{sh}}$ for the classical starts of the two cutoffs; and $|\log(1+\zeta)| \leq 2|\zeta|$ for $|\zeta| \leq \frac{1}{2}$.

The rate $\mathfrak{o}_K \leq \mathsf{C}_{\mathrm{bf}}\, \ell^3 K^{\mathfrak{p}_0} L^{-K/10}$ holds at every $K \geq K_{\mathrm{bf}}$, for the labels of an interval J of labels admitted on the side ℓ (Def. 2.7(5), Cor. 2.8(a), (2.xv)): $K_{\mathrm{bf}} = K_{\mathrm{bf}}(\ell, J)$ is the least cutoff from which the conditions hold and the two fast members, the large fields $8\bar{\mathsf{R}}$ and the shell term $12\bar{\epsilon}_{\mathrm{sh}}$, majorised on J, lie below half of $\mathsf{C}_{\mathrm{bf}}\ell^3 K^{\mathfrak{p}_0} L^{-K/10}$; it depends on the side, through the factor $\exp[(\mathsf{C}^{\Sigma}_{\mathrm{maj}} + 2 + C_{\mathrm{low}}(\mathsf{b}))\ell^3]$ of R_{t_1}, and on J, while C_{bf} depends on J and on no side.

The same explicit numbers give a form uniform in the label, derived in the proof of Prop. 17.47(b): there are numbers $y_\sharp$, $C_{\mathfrak{o}}$ of the constants such that at every label with $\log \mathsf{b} \geq y_\sharp$, on every side $2MR(\mathsf{b}) \leq \ell \leq 2L^3 M(\log \mathsf{b})^r$ and at every $K \geq 11(5 + 2\mathfrak{p}_0/r)\log_L \ell$, the conditions (k0)–(k4), (k6) hold and $\mathfrak{o}_K(\ell) \leq C_{\mathfrak{o}}\, \ell^{3+2\mathfrak{p}_0/r} L^{-K/11}$, by the condition $2\mathfrak{p}_0 > 3r + 2$ of the list; (iii) for a label $\mathsf{b} \geq \gamma^{-2}$, a side $\ell = 2L^m$ with $MR(\mathsf{b}) \leq L^m$, a cutoff $K \geq 2$ at which the conditions (k0)–(k4) and (k6) of §2.3.1 hold, every integer $0 \leq d \leq \lceil K/1260 \rceil$ and every bounded Borel function F, invariant under the gauge transformations, of the configurations of the lattice of the spacing L^{-d} on the torus ℓ:

$$\big|\langle F(\mathcal{M}^{K+1-d}U)\rangle_{K+1} - \langle F(\mathcal{M}^{K-d}U)\rangle_K\big| \leq \mathfrak{o}_K \sup|F|$$

(Cor. 2.8(a)), with the number $\mathfrak{o}_K$ of (ii), which is $\leq \mathsf{C}_{\mathrm{bf}}\, \ell^3 K^{\mathfrak{p}_0} L^{-K/10}$ at every $K \geq K_{\mathrm{bf}}$ by (2.xv), so that $\sum_K \mathfrak{o}_K < \infty$; the block field is taken at the fixed spacing L^{-d}, and the number $K - d$ of its averaging steps grows with the cutoff; (iv) the exact identification of the zero-source contact terms with the vacuum constants of their first $K^\bullet - u$ steps, proved in Step 3(iv) of Prop. 13.7, and their two-torus comparison in Lem. 13.31; Prop. 17.44 below retains this stopping depth and proves the resulting summable estimate; (v) the averaging map among the construction results (equations YM3_I:(37)–(39)), which sends a configuration U to its block field $V_j = \mathcal{M}^j U$ of level j, whose blocks are the cubes of L^j bare sites a side centred at the block sites, the points of $L^j\mathbb{Z}^3$, so that the block layer 1 of level q lies in the bare layers $\frac{1}{2}(L^q + 1), \ldots, \frac{1}{2}(3L^q - 1)$: it is Borel (Prop. YM3_I:4.5) and gauge covariant, it commutes with the isometries of the lattice that carry $L^j\mathbb{Z}^3$ onto itself ([7, Lem. G.1]), a block variable depends only on the bare bonds of the two blocks that its bond joins (Prop. YM3_I:4.5), on bare fields whose plaquettes all lie within ω_0 of $\mathbf{1}$, with $L^{2q}\omega_0$ below a number of L, the block plaquettes of level $j \leq q$ lie within $1.003\, L^{2j}\omega_0$ of $\mathbf{1}$ (Lem. 5.28), and on a cube of N bare sites a side the holonomies of the straight closed lines of the block field of level q lie within $13.3\, NL^q\omega_0$ of those of the bare field (Lem. 5.29); (vi) the comparison of (iii) with a factor, Prop. 17.30 at the end of this subsection, derived there from Prop. YM3_I:115.2 and the statements of Section YM3_I:116 named in (ii): in the setting of (iii), for every gauge-invariant Borel function F with values in $[0,1]$ of the block field of depth d, $\mu_d^{K+1}(F) \leq e^{m_K}\mu_d^K(F) + \epsilon_K$ and $\mu_d^K(F) \leq e^{m_K}\mu_d^{K+1}(F) + \epsilon_K$, where μ_d^K is the law of $\mathcal{M}^{K-d}U$ at the cutoff K and m_K, ϵ_K are the explicit numbers of that proposition: the errors that are powers of L^{-K} stand in the exponent m_K, the large fields of Prop. YM3_I:115.2, lost-mass bound and the shell term in the additive remainder ϵ_K.

Multiplying these inequalities: there are a number $y_\times$ and a power $\mathsf{N}(y)$ of y, both formed from the constants, such that at every label with $y = \log \mathsf{b} \ge y_\times$, on the least side ℓ at which (iii) applies, and from the cutoff $K_1 = K_1(y)$ of that proposition, at which the bare cube has $\ell L^{K_1} < \mathsf{N}(y)$ sites a side,

$$\mu_0^K(E) \le 1.7\,(\mu_0^{K_1}(E) + e^{-4y}) \qquad \text{for every } K \ge K_1$$

and every Borel set E of block fields of depth 0 that the gauge transformations carry onto itself: a factor and a remainder, not an additive error summed over the cutoffs. (M5) Thms. YM3_I:AJ and YM3_I:AD, through: (i) the bounds on cumulants at first-passage data: for $k \ge 2$, every first-passage datum (K,β) at g, every torus $\ell \in \mathfrak{L}(g)$ with $\ell \ge 16k(R+1)$ and k real Lipschitz functions with supports in $(-\frac{\ell}{4},\frac{\ell}{4})^3$, each in a ball of radius R, any two at a distance $\ge \mathsf{d}$, $|S^T_{k;K,\ell,\beta}(X)| \le C^g_k(\mathsf{d};R)\prod_i \|f_i\|_\sharp$; the bounds depend, besides on g, k, d, R and the constants, only on a radius fixed once in that theorem, and neither on the cutoff nor on the coupling nor on the side nor on the place; (ii) the theorem on the canonical decomposition, the ergodic components and the reconstruction, Thm. YM3_I:AP: each ergodic component of the limit of a convergent Gibbs sequence is a limit along the same pairs whose approximants are periodic Gibbs states at these pairs (Thm. YM3_I:AP, component approximation in the proof), and it carries the reconstruction of Thm. YM3_I:AP; (iii) the law of the two-point function at short distance (Thm. YM3_I:AV, short-distance bound and torus version), with the ceiling γ_W and the threshold $\mathsf{d}_0(g,\vartheta)$ of equations YM3_I:(2346) and YM3_I:(2349)–(2350), the depth $s(\mathsf{d})$ of equation YM3_I:(2321) and the test pairs of [7, Lem. I.1]: for $0 < g \le \gamma_W$, $\mathsf{d} \le \mathsf{d}_0(g,\vartheta)$, every first-passage datum (K,β) at g with $K > s(\mathsf{d})$, every periodic Gibbs state ν at it and every test pair (f,h) at $(\mathsf{d};\vartheta)$,

$$S_2^T[\nu](f,h) = \mathsf{c}_2\,\mathsf{b}^{-2}\,\mathcal{K}_{\text{free}}(f,h)\,(1+\varepsilon'), \qquad |\varepsilon'| \le \tfrac{1}{3}, \qquad \mathcal{K}_{\text{free}}(f,h) = \iint f(x)h(y)|x-y|^{-6}\,\mathrm{d}x\,\mathrm{d}y,$$

with $\mathsf{c}_2 > 0$ and $\mathcal{K}_{\text{free}}$ as in Thm. YM3_I:C and equation YM3_I:(2240); for $f(y) = (1 - |y - 2\mathsf{d}e_3|^2/\mathsf{d}^2)_+^2$ and $0 \le t \le 4\mathsf{d}$ the pair $(f(\Theta\,\cdot), f(\cdot - te_3))$, Θ the reflection $x_3 \mapsto -x_3$, is a test pair at $(\mathsf{d};\vartheta_1)$ ([7, Lem. I.1](c)), and $K > s(\mathsf{d})$ gives $\mathsf{d}/a_K > 4L^3M$. Besides, $I(g) = (u, Lu - 2\bar B]$ with $u = g^{-2} + \bar B$, $\bar B \ge 0$, and $g^{-2} > L^2\bar B$ for $g \le \gamma_W$ (the coupling ceiling preceding Thm. YM3_I:AV, equation YM3_I:(2346)). The labels $a_K\beta$ of all first-passage data (K,β) at g lie in the window $(g^{-2} + B_\sharp\frac{L-2}{L-1},\, L(g^{-2}+\bar B)]$ (YM3_I:(1531) in Prop. YM3_I:74.1), whose upper end is $X_g + \bar B$. (M6) The ordered choice leaves the blocking factor free upward (the choice of L in the proof of Prop. YM3_I:69.1, followed by the source choices in Section YM3_I:82): every odd integer above a threshold is the value of L at an admissible choice, and Thms. YM3_I:D, YM3_I:AJ and YM3_I:AD hold at every admissible choice. (M7) The member of third order in the curvature of the bulk term of the rotational Ward identity, on growing tori: Lem. 5.16 of the section on plane fields and the stress tensor (§5.1), which carries the bound of the cubic remainder on one torus, Lem. 5.11, to all sides at once. For $\chi, f_1, \dots, f_n$ and ω as in Thm. BA(5), this member of the bulk term of YM3_I:(2108) on the torus ℓ is

$$\mathfrak{U}^\sharp_{\omega,\chi}(f_1,\dots,f_n) = -\beta\langle \mathfrak{K}^\sharp[\chi\,\omega y]; \mathcal{O}_K(f_1); \dots; \mathcal{O}_K(f_n)\rangle_{K,\ell,\beta} + a_K^{-1}\sum_{i=1}^n \langle \mathfrak{K}^\sharp[f_i\,\omega y]; (\mathcal{O}_K(f_l))_{l\ne i}\rangle_{K,\ell,\beta},$$

the joint cumulants in which the local density of the rotation is replaced by its part $\mathfrak{K}^\sharp[\boldsymbol{\eta}] = \sum_p a_K^{-1}\boldsymbol{\eta}^{\rho(p)}(x(p))\,\mathfrak{r}^\sharp_{\rho(p)}(p)$ of [7, §H], built with the remainders $\mathfrak{r}^\sharp$ of third order in the field strength ([7, Prop. H.1]), with the weight β/a_K; here $\boldsymbol{\eta}^\rho(y) = \chi(y)(\omega y)_\rho$, resp. $f_i(y)(\omega y)_\rho$. As $\mathfrak{K}^\sharp$ depends on its field only through its values at the centres of the plaquettes, at which $\chi f_i = f_i$, this member is the cubic remainder $\mathfrak{U}^\sharp[\chi\,\omega y](f_1,\dots,f_n)$ of that section at the field $\chi\,\omega y$. Let $0 < g \le \gamma$, $n \ge 2$, $\mathsf{d} > 0$ and $\rho_0 \ge 1$. There are $C_\sharp$ and $K_\sharp$, formed from g, n, d, ρ_0 and the constants and depending on no side, such that for all first-passage data (K,β) at g with $K \ge K_\sharp$, all sides $\ell \ge 2\rho_0 + 2$ of $\mathfrak{L}(g)$, all $\omega \in \mathfrak{so}(3)$ and all real $\chi, f_1, \dots, f_n \in C^{1,1}(\mathbb{R}^3)$ that vanish outside $[-\rho_0,\rho_0]^3$, χ equal to 1

at the centres and at the bonds of the plaquettes at which some f_i does not vanish, the supports of the f_i at pairwise sup distances $\geq \mathsf{d}$,

$$|\mathfrak{U}^\sharp_{\omega,\chi}(f_1,\dots,f_n)| \leq C_\sharp\,|\omega|\,L^{-K/16}\,\|\chi\|_{C^{1,1}}\prod_i \|f_i\|_{C^{1,1}}.$$

This is Lem. 5.16 at the field $\chi\,\omega y$, for the cube of side $\frac14\ell'$ centred at 0, ℓ' the least side of $\mathfrak{L}(g)$ with $\ell' \geq 8\rho_0$, and for a rate $c \in (\frac{1}{16},\frac{1}{14})$: there $K^pL^{-cK} \leq C'L^{-K/16}$, the norm $\mathsf{H}(\eta) = \max_\rho \sup|\eta^\rho| + \max_\rho \mathrm{Lip}_\infty \eta^\rho$ of a field in that lemma, Lip_∞ the Lipschitz constant for the sup distance, is $\mathsf{H}(\chi\,\omega y) \leq C\rho_0|\omega|\,\|\chi\|_{C^{1,1}}$ with a pure number C, and $\|f_i\|_\sharp \leq \max\{1,40M\}\|f_i\|_{C^{1,1}}$. On each of the finitely many sides of $\mathfrak{L}(g)$ from $2\rho_0+2$ to ℓ' it is Lem. 5.11, whose setting holds on that side at all first-passage data at g: their labels lie in $[\gamma^{-2}, X_g+\bar B]$, and $MR(X_g+\bar B) \leq L^m$ on every side $2L^m$ of $\mathfrak{L}(g)$. The idea of the proof. On one torus the entry $\mathfrak{K}^\sharp$ enters each renormalisation step as one more local term, inside the radii of the one-step theorem with sources (Prop. 6.31). Each of its terms vanishes to the third order at $\mathbf{1}$ and is odd under the mirror of its plaquette, while the steps are covariant under the mirrors of their lattices; so only the differences of the backgrounds and their mirror images and the asymmetric part of the weight feed the families that the steps make of it, and what is fed at a level is forgotten at the rate $L^{-1/2}$ a level. The conditional mean of the entry, given the block field at a level $K - \lceil \phi K\rceil$ with a small $\phi > 0$, is therefore a number up to an oscillation of the order K^pL^{-cK} (Lem. 5.10(b)), the large fields costing less than every power of L^{-K}; the cumulant is a mixed derivative at 0 of the logarithm of the partition function with the entry and the sources, and Cauchy's estimates in the sources give $|\mathfrak{U}^\sharp| \leq CK^pL^{-cK}$ times the norms at every rate $c < \frac{1}{14}$ (Lem. 5.11), which is at most $C'L^{-K/16}$.

On all the sides at once, the carrier is put in one cell of a board of side $\frac12\ell'$, ℓ' a fixed side of $\mathfrak{L}(g)$; the chessboard estimate controls the generating function of the entry and of the sources on the torus ℓ by the mean of their mirror copies on the torus ℓ', with a factor $e^{(\ell'/2)^3(p_{\ell'}-p_\ell)}$ that is bounded uniformly at the first-passage data at g; on the torus ℓ' the bound of one torus applies to the copies, and the part of the cumulants that links different copies carries the oscillation of the conditional mean of Lem. 5.10(b); the finitely many sides between $2\rho_0+2$ and ℓ' are tori of their own. (M8) The rotational Ward identity among the construction results, which Thm. BA(5) and its proof read: the bulk term YM3_I:(2107) of equation YM3_I:(2107) and the source calculation in the proof of Prop. YM3_I:112.1, the bulk insertion of equation YM3_I:(2107) and the source calculation in the proof of Prop. YM3_I:112.1, whose mean at the sources gives the bulk term by differentiation in the source variables, and the identity YM3_I:(2108); the orbital potential $\Psi_\rho(p)$ and the remainder of third order $\mathfrak{r}^\sharp_\rho(p)$ of [7, §H], with the bound of third order [7, eq. 295], and the summation by parts of [7, Prop. H.1], which writes the bulk insertion as the sum of the spin part, a member of order 2 in the curvature and the cubic member [7, eq. 297]; the small-field density with a local insertion and its derivative as a conditional mean in Lem. 5.10(a), under its stated entry and cutoff hypotheses, obtained from the holomorphic fourth-family map of Prop. 6.31; and the two-constants inequality of Lem. YM3_I:116.2, used in the source variables.

Proposition 17.30 (Two cutoffs with a factor)**.** Let $\mathsf{b} \geq \gamma^{-2}$ and $\ell = 2L^m$ with $MR(\mathsf{b}) \leq L^m$; for a cutoff $K \geq 2$ take the numbers $\epsilon_{\mathrm{fg}}, \epsilon_{\mathrm{cl}}, \epsilon_{\mathrm{sh}}, \mathsf{R}_{t_1}$ of §2.3.1 at $n=1$ ($u = \lceil K/1260\rceil$, $s = \lceil K/9\rceil$, $t_1 = K-u$), and put

$$m_K = 4\epsilon_{\mathrm{fg}} + 2\epsilon_{\mathrm{cl}} + 1.06\,\epsilon_{\mathrm{sh}} + 1.39\,\mathsf{R}_{t_1}, \qquad \epsilon_K = \mathsf{R}_{t_1} + 1.29\,\epsilon_{\mathrm{sh}}.$$

(a) At every cutoff $K \geq 2$ at which (k0)–(k4) and (k6) hold, for every integer $0 \leq d \leq \lceil K/1260\rceil$ and every Borel function F with values in $[0,1]$ of the block fields of depth d on the torus ℓ that is invariant under their gauge transformations,

$$\mu_d^{K+1}(F) \leq e^{m_K}\mu_d^K(F) + \epsilon_K, \qquad \mu_d^K(F) \leq e^{m_K}\mu_d^{K+1}(F) + \epsilon_K,$$

μ_d^K the law of $\mathcal{M}^{K-d}U$ at the cutoff K (Def. 2.7(2)).

(b) If the conditions hold at every $K \ge K_1$, and $\mathsf{S} = \sum_{K\ge K_1} m_K$ and $\mathsf{R}_\Sigma = \sum_{K\ge K_1} \epsilon_K$ are finite, then $e^{-\mathsf{S}}\mu_d^{K_1}(E) - \mathsf{R}_\Sigma \le \mu_d^K(E) \le e^{\mathsf{S}}(\mu_d^{K_1}(E) + \mathsf{R}_\Sigma)$ for every $K \ge K_1$, every integer $0 \le d \le \lceil K_1/1260\rceil$ and every Borel set E of block fields of depth d that the gauge transformations carry onto itself.

(c) Let $p = 2\mathfrak{p}_0$, $y = \log \mathsf{b}$, $\bar\ell(y) = 2L^2 M y^r$, $\mathsf{N}(y) = L\,\bar\ell(y)^{31} y^{10(p+1)}$, and let $K_1(y)$ be the least integer $K \ge 10$ with $L^{K/10} \ge \bar\ell(y)^3 y^{p+1}$. There is a number $y_\times$, formed from the constants, such that at every label b with $y \ge y_\times$, on the least side $\ell = 2L^m$, $m \in \mathbb{N}$, with $MR(\mathsf{b}) \le L^m$, which has $\ell \le \bar\ell(y)$: the conditions (k0)–(k4) and (k6) hold at every $K \ge K_1(y)$; $\ell L^{K_1(y)} < \mathsf{N}(y)$; from $K_1 = K_1(y)$, $\mathsf{S} \le 0.53$ and $\mathsf{R}_\Sigma \le e^{-4y}$; and so $\mu_0^K(E) \le 1.7\,(\mu_0^{K_1(y)}(E) + e^{-4y})$ for every $K \ge K_1(y)$ and every Borel set E of block fields of depth 0 that the gauge transformations carry onto itself.

■

Proof. (a) Write $\langle F\rangle_K = \mu_d^K(F)$. The averaging maps from the depth u to the depth d are the same at the two cutoffs and commute with the gauge transformations, so $F \circ \mathcal{M}^{u-d}$ is a function of the same kind at the depth u, and we may take $d = u$. Three comparisons follow, each with the same function $F \ge 0$ inserted on both sides. (1) *Large fields.* By Prop. YM3_I:115.2, image identity and domination bound, the chain of the top level t_1, on its inner ball, is the image under $\mathcal{M}^{t_1}$ of $\mathfrak{w}_{t_1}^{\mathrm{sf}}\rho_0\,\mathrm{d}U$, and its integral of a gauge-invariant $F \ge 0$ is at most $\int F(\mathcal{M}^{t_1}U)\rho_0\,\mathrm{d}U$; by the discarding of Prop. YM3_I:115.2, lost-mass bound its mass is at least $(1 - \mathsf{R}_{t_1})Z$.

So the mean of F under the law exceeds its mean under the normalised chain by at most $\mathsf{R}_{t_1}\sup F$, and the latter is at most $(1-\mathsf{R}_{t_1})^{-1}$ times the former. (2) *The chain and the classical start.* On the small-field set the two measures have the positive densities $e^{c(0)}e^{\Phi}$ and $e^{\hat c(0)}e^{\hat\Phi}$ (the proof of Prop. YM3_I:116.1), and $|\Phi - \hat\Phi| \le \epsilon_{\mathrm{fg}}$ at every block field of the cut set (equation YM3_I:(2180)); so the two normalised means of F differ by a factor within $e^{\pm 2\epsilon_{\mathrm{fg}}}$, the constants $e^{c(0)}$, $e^{\hat c(0)}$ cancelling against $F = 1$. (3) *The classical starts of the two cutoffs.* They are integrals over the configurations of the depth s of the weights $\mathsf{k} \ge 0$ of the two cutoffs times densities that are positive on the cut sets (Section YM3_I:116, §2.3.2); where both weights are positive they are equal and the densities differ by a factor within $e^{\pm\epsilon_{\mathrm{cl}}}$ (Prop. YM3_I:116.1, density comparison in the proof), and where one of them vanishes the other carries at most the part ϵ_{sh} of the mass (§2.3.2); so the normalised mean at one cutoff is at most $e^{2\epsilon_{\mathrm{cl}}}(1-\epsilon_{\mathrm{sh}})^{-1}$ times that at the other plus $\epsilon_{\mathrm{sh}}\sup F$.

Chaining (1), (2), (3), (2), (1), for $\{K', K''\} = \{K, K+1\}$,

$$\langle F\rangle_{K'} \le \frac{e^{4\epsilon_{\mathrm{fg}}+2\epsilon_{\mathrm{cl}}}}{(1-\epsilon_{\mathrm{sh}})(1-\mathsf{R}_{t_1})}\,\langle F\rangle_{K''} + \big(\mathsf{R}_{t_1} + e^{2\epsilon_{\mathrm{fg}}}\epsilon_{\mathrm{sh}}\big)\sup F,$$

and (k2), (k3), (k6) give $e^{2\epsilon_{\mathrm{fg}}} \le e^{1/4} < 1.29$ and, by convexity, $-\log(1-\epsilon_{\mathrm{sh}}) \le 1.06\,\epsilon_{\mathrm{sh}}$ and $-\log(1-\mathsf{R}_{t_1}) \le 1.39\,\mathsf{R}_{t_1}$. (b) By induction on K, $\langle F\rangle_K \le e^{\sigma_K}(\langle F\rangle_{K_1} + \rho_K)$ and $\langle F\rangle_K \ge e^{-\sigma_K}\langle F\rangle_{K_1} - \rho_K$, with the sums σ_K, ρ_K of the m_k, ϵ_k over $K_1 \le k < K$; $d \le \lceil K_1/1260\rceil \le \lceil K/1260\rceil$ keeps every step in the range of (a), and $F = \mathbf{1}_E$. (c) Let $K \ge K_1(y)$; with $\ell \le \bar\ell(y)$, $\ell^3 L^{-K/10} \le y^{-p-1}$. As $u \ge K/1260$ and $s \ge K/9$, the definitions YM3_I:(2192) and YM3_I:(2193) give $\epsilon_{\mathrm{fg}} \le a_1 \ell^3 L^{-K/10}$ and $\epsilon_{\mathrm{cl}} \le a_2\, y^p(1 + K\log L)^p K^{1/2} L^{-K/90}\,\ell^3 L^{-K/10}$ with numbers a_1, a_2 of the constants, so that $\sum_{K\ge K_1(y)}(4\epsilon_{\mathrm{fg}} + 2\epsilon_{\mathrm{cl}}) \le (17.72\,a_1 + 2a_2a_3)/y$, $a_3 = \sum_{K\ge1}(1+K\log L)^p K^{1/2}L^{-K/90}$.

The numbers R_{t_1} and ϵ_{sh} carry the factors $e^{-c'A_0^2 \mathsf{y}_{[u]}}$ and $e^{-\frac14\kappa_{\mathrm{b}}A_1^2\mathsf{y}_{[u]}}$, where $\mathsf{y}_{[u]} = (y + u\log L)^p \ge y^p + (u\log L)^p$, against the growth $\ell^3 L^{3s}$, $s \le 140u$, and, in R_{t_1}, $\exp[(\mathsf{C}^{\Sigma}_{\mathrm{maj}} + 2 + c_{\mathrm{low}} + \frac92 y)\ell^3] \le \exp[O(y^{3r+1})]$; summed over the cutoffs, of which at most 1260 have the same u, and by $2\mathfrak{p}_0 > 3r+2$, which follows from the exponent choice $r \ge 2$, $\mathfrak{p}_0 \ge 6r+1$ in Thm. YM3_I:D, $\sum_{K\ge K_1(y)}(\mathsf{R}_{t_1} + 1.29\,\epsilon_{\mathrm{sh}}) \le e^{-4y}$ for all large y. The same bounds give (k0)–(k4) and (k6) at every $K \ge K_1(y)$

for large y; $y_\times$ is the least number from which all this holds, with $\mathsf{b} \ge \max\{\gamma^{-2}, \Theta_\star\}$ and $y \ge 2(17.72\,a_1 + 2a_2a_3)$.

Then $\mathsf{S} \le \frac12 + 1.39\,e^{-4y} \le 0.53$, $e^{0.53} < 1.7$, and (b) at $d = 0$ gives the last inequality. As $L < \bar\ell(y)^3$, $K_1(y) \ge 11$, and by its least choice $L^{(K_1(y)-1)/10} < \bar\ell(y)^3 y^{p+1}$, so $\ell L^{K_1(y)} < \bar\ell(y)\,L\,(\bar\ell(y)^3 y^{p+1})^{10} = \mathsf{N}(y)$. □

17.2.4. *Statements on one lattice.*

Proposition 17.31 (The exact layer)**.** Let $\beta \ge 0$, $\Sigma = (A, B)$ a cross-section with sides ≥ 3, any coordinate direction being the time, and $t, t' \ge 3$, $q \ge 2$ integers.

(a) $Z(t,\Sigma) = \operatorname{Tr} \mathsf{D}^t = \sum_i \lambda_i^t$, $\sum_i \lambda_i^2 < \infty$, λ_0 is simple and its eigenvector can be taken strictly positive. F is symmetric in the three sides, and $e^{-6\beta|\underline{n}|} \le Z \le 1$.

(b) $\mathsf{r}_{t+1} \le \mathsf{r}_t$, $\mathsf{r}_{t+t'} \le \mathsf{r}_t\mathsf{r}_{t'}$, $\mathsf{r}_{qt} \le \mathsf{r}_t^q$; $F(t,\Sigma) = t\log\lambda_0 + \mathsf{G}(t;\Sigma)$, $F(t,\Sigma)/t$ does not increase, and $F(t+t',\Sigma) \le F(t,\Sigma) + F(t',\Sigma)$;

$$qF(t,\Sigma) - F(qt,\Sigma) = q\,\mathsf{G}(t;\Sigma) - \mathsf{G}(qt;\Sigma) \in \big[(q-1)\mathsf{G}(t;\Sigma),\, q\,\mathsf{G}(t;\Sigma)\big].$$

If $2F(t,\Sigma) - F(2t,\Sigma) \le x < \log 2$, then $\mathsf{r}_t \le \varphi(e^x - 1)$ and $2F(2t,\Sigma) - F(4t,\Sigma) \le \psi(x)$, where $\varphi(\eta) = \eta/(1+\sqrt{1-\eta^2})$ and $\psi(x) = 2\log(1+\varphi(e^x-1)^2)$.

(c) $F/|\underline{n}|$ and $\delta/|\underline{n}|$ do not increase when one side grows, and $F/|\underline{n}| \to P$ along boxes all of whose sides tend to ∞; $\delta(t,\Sigma) = t\,\varepsilon(\Sigma) + \mathsf{G}(t;\Sigma)$ with $\varepsilon(\Sigma) \ge 0$, so $\mathsf{G} \le \delta$; $D_q(N) = q^3\delta(N^3) - \delta((qN)^3) \in [0, q^3\delta(N^3)]$, $\mathsf{e}(l) = 2\delta(l,2l,2l) - \delta((2l)^3)$, and

$$\delta(N^3) = \sum_{s\ge0} q^{-3(s+1)} D_q(q^s N).$$

From above, an area: for all integers $A, B, t \ge N$, $\mathsf{G}(t;A,B) \le AB\,\varkappa$ and $\varepsilon(A,B) \le (A+B-N)\varkappa$, $\varkappa = \delta(N^3)/N^2$; so for every box with all sides $\ge N$

$$\delta(\underline{n}) \le \big[n_1n_2 + n_1n_3 + n_2n_3 - N(n_1+n_2+n_3) + N^2\big]\varkappa \le (n_1n_2 + n_1n_3 + n_2n_3)\,\varkappa,$$

and $\delta(M^3) \le (3\rho^2 - 3\rho + 1)\,\delta(N^3)$ for $M = \rho N \ge N$.

(d) There are commuting unitary involutions $\mathsf{J}_1, \mathsf{J}_2$ that commute with $\mathsf{T}^{(0)} = \mathsf{D}$ and with an operator $\mathsf{T}^{(1)} \ge 0$ of the same kind (D with the sign of one plaquette of the cross-section reversed in m), such that, with $\mathsf{J}^k = \mathsf{J}_1^{k_1}\mathsf{J}_2^{k_2}$ and $\mathsf{P}_e = \frac14\sum_k (-1)^{e\cdot k}\mathsf{J}^k$,

$$Z^{(k;m)}(t,\Sigma) = \operatorname{Tr}\big(\mathsf{J}^k(\mathsf{T}^{(m)})^t\big) = \sum_e (-1)^{e\cdot k} W(e;m), \qquad W(e;m) = \operatorname{Tr}\big(\mathsf{P}_e(\mathsf{T}^{(m)})^t\big) \ge 0.$$

Each $W(e;m)$ is a sum of t-th powers of positive numbers that do not depend on t; the eigenvector of λ_0 lies in the range of P_{00}, and $W(00;0) \ge \lambda_0^t$; $Z^z > 0$ does not depend on the columns chosen. On every box: $0 \le 1 - r^{(k;0)} \le 2\mathsf{R}_t$, $0 < r^{(k;m)} \le r^{(00;m)}$, $r^{(11;m)} \ge r^{(10;m)} + r^{(01;m)} - r^{(00;m)}$, and

$$\delta(t,\Sigma) = t\,\varepsilon(\Sigma) + \log(1+\mathsf{r}_t^{00}) + \log(1/\zeta_t), \qquad \zeta_t = \frac{W(00;0)}{Z} = \frac14\Big(1 + \sum_{k\ne0} r^{(k;0)}\Big),$$

three terms ≥ 0, $1 + \mathsf{r}_t^{00} = W(00;0)/\lambda_0^t$. On a cube N^3: $r_2, r_3 \le r_1 \le 1$; all seven $0 \le 1 - r^z \le 3\mathsf{R}_N \le 3\min\{\mathsf{G}_N, \mathsf{r}_N\}$; $1 - r_1 \le \frac43\mathsf{R}_N$, $1 - r_2 \le 2(1-r_1)$ and $1 - r_3 \le 3(1-r_1)$.

(e) $2F(t,\Sigma) - F(2t,\Sigma) \ge 2\log\sum_{f\in\{0,1\}^2} w_f^{1/2}$, the w_f being the weights of the box $(2t,\Sigma)$; and $\mathsf{G}(t;\Sigma) \ge \log(1/w_{00})$, w_{00} the weight of the box (t,Σ).

(f) Let $\Gamma = \mathbb{Z}n_1e_1 \oplus \mathbb{Z}n_2e_2 \oplus \mathbb{Z}(v_1e_1 + v_2e_2 + n_3e_3)$, $v \in \mathbb{Z}^2$, and Z_Γ the partition function of the lattice with these periods. With the time e_3 over $\Sigma = (n_1, n_2)$: $Z_\Gamma = \operatorname{Tr}(\mathsf{U}_v\mathsf{D}^{n_3})$, U_v the unitary of the translation of a layer by v, and $\lambda_0^{n_3}(1 - \mathsf{r}_{n_3}) \le Z_\Gamma \le Z(\underline{n})$.

(g) *The sheared cube.* Let $N = 2h$ be even, $\Gamma^\diamond = \mathbb{Z}N(e_1 - e_3) \oplus \mathbb{Z}Ne_2 \oplus \mathbb{Z}h(e_1 + e_3)$, a period lattice of the index N^3, and $Z_{\Gamma^\diamond}$ the partition function of the lattice with these periods. Then $Z(N^3) \le Z_{\Gamma^\diamond}$.

■

Proof. (a) The weight of the box is a product over the layers of $\mathsf{m}(V^\tau)$ and of the crossing weights $e^{-\beta A_\times(V^\tau, S^\tau, V^{\tau+1})}$, each S^τ standing in one factor; integrating the S^τ (Fubini) gives

$$\int \prod_\tau \mathrm{d}V^\tau \prod_\tau \mathsf{m}(V^\tau)\mathsf{K}(V^{\tau+1}, V^\tau),$$

and splitting each $\mathsf{m}(V^\tau)$ into two roots this is

$$\int \prod_\tau \mathsf{D}(V^{\tau+1}, V^\tau)\,\mathrm{d}V^\tau = \operatorname{Tr} \mathsf{D}^t,$$

for every $t \ge 3$. With $\mathsf{A} = \mathsf{K}^{1/2}\mathsf{m}^{1/2}$, $\mathsf{A}^*\mathsf{A} = \mathsf{D}$ and $\mathsf{A}\mathsf{A}^* = \mathsf{T}'$ have the same nonzero spectrum. The kernel of D is continuous and bounded below by a positive number on a compact space; if e_0 is a real eigenvector of λ_0, so is $|e_0|$, hence so are $|e_0| \pm e_0 \ge 0$, one of which vanishes on a set of positive measure and is therefore 0.

So every eigenvector of λ_0 has a strict sign, and no two of them are orthogonal. A permutation of the coordinates carries plaquettes to plaquettes and $\operatorname{tr} \partial U$ to itself; $0 \le a_p \le 2$ on $3|\underline{n}|$ plaquettes. (b) With $x_i = (\lambda_i/\lambda_0)^t \le 1$: $x_i^{(t+1)/t} \le x_i$, $\sum x_i^q \le (\sum x_i)^q$, $\sum \lambda_i^{t+t'} \le \sum \lambda_i^t \sum \lambda_i^{t'}$; $F(t)/t = \log \|\lambda\|_{\ell^t}$ does not increase in t. The identity and its range follow from $F(t) = t \log \lambda_0 + \mathsf{G}(t)$ and $0 \le \mathsf{G}(qt) \le \mathsf{G}(t)$. If $2F(t) - F(2t) \le x$, then $\mathsf{G}(t) \le x < \log 2$, $\mathsf{r}_t < 1$, and

$$e^x \ge \frac{(1+\mathsf{r}_t)^2}{1+\mathsf{r}_{2t}} \ge \frac{(1+\mathsf{r}_t)^2}{1+\mathsf{r}_t^2} = 1 + \frac{2\mathsf{r}_t}{1+\mathsf{r}_t^2},$$

which gives $\mathsf{r}_t \le \varphi(e^x - 1)$ as $r \mapsto 2r/(1+r^2)$ increases on $[0,1]$; and $2F(2t) - F(4t) \le 2\mathsf{G}(2t) \le 2\log(1+\mathsf{r}_t^2)$. (c) is (b) in each direction, divided by the product of the two other sides; $\varepsilon(\Sigma) = \lim_t (F(t,\Sigma) - t|\Sigma|P)/t \ge 0$; in D_q and e the terms with P cancel, and the series telescopes, its tail $q^{-3S}\delta((q^S N)^3) = N^3[F/|\cdot| - P]$ tending to 0. For the area: $\mathsf{G}(t; A, B) \le \mathsf{G}(N; A, B) \le \delta(N, A, B) \le AB\varkappa$ by the monotonicity of G and of $\delta/|\underline{n}|$. With the side B as the time, $\delta(s,t,B) = B\varepsilon(s,t) + \mathsf{G}(B; s, t) \le B\varepsilon(s,t) + st\varkappa$; dividing by s and then t and letting both tend to ∞, as $\varepsilon(s,t)/st \to 0$, gives $\limsup_t \varepsilon(t,B)/t \le \varkappa$.

Since $\delta(a,b,c) = a\varepsilon(b,c) + \mathsf{G}(a; b, c) \le \delta(N,b,c) + (a-N)\varepsilon(b,c)$ for $a \ge N$, $\varepsilon(A,B) \le \varepsilon(N,B) + (A-N)\varkappa \le \varepsilon(N,N) + (A+B-2N)\varkappa$, and $\varepsilon(N,N) \le \delta(N^3)/N = N\varkappa$. The same inequality in the three sides in turn gives $\delta(\underline{n}) \le \delta(N^3) + (n_3 - N)\varepsilon(N,N) + (n_2 - N)\varepsilon(N, n_3) + (n_1 - N)\varepsilon(n_2, n_3)$, which is the bracket. (d) As -1 is central in SU(2), reversing the sign of $\operatorname{tr} \partial U(p)$ on a column that crosses one time step is the same as multiplying by -1 the bonds of one line of the next layer; this change of variables keeps the Haar measure, appears under the trace as an involution J_i of the layer variables, and commutes with m and K.

A reversed column in the plane of the cross-section lies in every layer and replaces m by m_1. Moving a column by one step is a further flip of the bonds of a line, which does not change the trace. The P_e are the isotypical projections of $k \mapsto \mathsf{J}^k$; they commute with $\mathsf{T}^{(m)}$, so every eigenspace splits along them and $W(e;m) = \sum_{i:f(i)=e} \mu_i^t$. The positive top eigenvector is invariant under the flips, so it lies in the range of P_{00}. Then $Z - Z^{(k;0)} = 2\sum_{e\cdot k \text{ odd}} W(e;0) \le 2(Z - \lambda_0^t)$, $Z^{(k;m)} \le \sum_e W(e;m) = Z^{(00;m)}$, $4W(11;m) = Z^{(00;m)} - Z^{(10;m)} - Z^{(01;m)} + Z^{(11;m)} \ge 0$, and $\delta = t\varepsilon + \log(Z/\lambda_0^t)$ with $Z/\lambda_0^t = [W(00;0)/\lambda_0^t][Z/W(00;0)]$.

On a cube, by the cubic symmetry, with $u = w_{10} = w_{01}$, $v = w_{11}$: $1 - r_1 = 2(u+v)$, $1 - r_2 = 4u$, $2u + v = 1 - w_{00} \le \mathsf{R}_N$; two planes with a common direction as the time give $r_2 \le r^{(00;1)} = r_1$, hence $v \le u$ and $1 - r_1 \le \frac{4}{3}\mathsf{R}_N$; three planes give $r_3 \ge 2r_2 - r_1$; and the sum over e with a vanishing

component gives $1-r^{(1,1,1)} \le (1-r^{z'})+(1-r^{z''})$, so $1-r_3 \le (1-r_1)+(1-r_2) \le 3(1-r_1)$. (e) Within each class $(\sum \mu_i^t)^2 \ge \sum \mu_i^{2t}$, so $Z(t,\Sigma) \ge \sum_f W_{2t}(f)^{1/2} = Z(2t,\Sigma)^{1/2} \sum_f w_f^{1/2}$; and $Z/\lambda_0^t \ge Z/W(00;0)$. (f) Integrating the layers as in (a), the last layer is glued to the first through the translation by v, which produces U_v under the trace; U_v commutes with D, so in a common eigenbasis $Z_\Gamma = \sum_i \omega_i \lambda_i^{n_3}$ with $|\omega_i| = 1$ and $\omega_0 = 1$ (the top eigenvector is positive and unique, hence invariant); Z_Γ is real and positive, so $\lambda_0^{n_3} - \sum_{i\ge 1} \lambda_i^{n_3} \le Z_\Gamma \le \sum_i \lambda_i^{n_3}$. (g) Put $\mathsf{v} = e_1 + e_3$.

The planes $x_1 + x_3 = c$, $c \in \mathbb{Z}$, are planes of lattice reflections, $(x_1, x_2, x_3) \mapsto (c - x_3, x_2, c - x_1)$; $N\mathbb{Z}^3 = \Gamma_\perp \oplus \mathbb{Z}Ne_1$ and $\Gamma^\diamond = \Gamma_\perp \oplus \mathbb{Z}h\mathsf{v}$ with $\Gamma_\perp = \mathbb{Z}N(e_1 - e_3) \oplus \mathbb{Z}Ne_2 \perp \mathsf{v}$, and both generators Ne_1, $h\mathsf{v}$ raise the height $x_1 + x_3$ by N. Cut either torus along the planes of the heights $-h, 0, h$ into two slabs of the height h, periodic under $\Gamma_\perp$. No plaquette lies in such a plane, and no bond crosses it; a plaquette cut by it has two corners z, z' in it and one on either side, and its weight is $e^{-\beta(1-\frac{1}{2}\operatorname{tr} vv'^{-1})}$, v and v' the transports from z to z' through the upper and through the lower corner.

As $\frac{1}{2}\operatorname{tr} vv'^{-1} = \frac{1}{2}\sum_{i,j} v_{ij}\overline{v'_{ij}}$ on SU(2), the exponential series writes this weight as a sum with coefficients ≥ 0 of products $\Phi(v)\overline{\Phi(v')}$ of matrix elements. With the data ϕ of a plane, its bonds and the indices of this expansion at its cut plaquettes, under the product of Haar measure and of the coefficients, the partition function of the torus $\Gamma_\perp \oplus \mathbb{Z}g$ is $\iint k_2(\tau_g\phi_0, \phi_1)\, k_1(\phi_1, \phi_0)$, where k_1, k_2 are the kernels of the two slabs between the data of their walls and τ_g, the translation by g, carries the data of the height $-h$ to those of the height h and keeps their measure.

Schwarz' inequality gives $Z(N^3)^2 \le \|k_1\|_2^2\, \|k_2\|_2^2$. The reflection in the middle plane exchanges the two sides of the cut plaquettes, conjugates the kernels, and acts on the plane of the height $-h$ as the translation by $h\mathsf{v}$; so $\|k_i\|_2^2$ is the partition function of the slab i glued to its mirror image, that is of the torus $\Gamma_\perp \oplus \mathbb{Z}h\mathsf{v} = \Gamma^\diamond$, and $Z(N^3) \le Z_{\Gamma^\diamond}$. □

Proposition 17.32 (The recursion under doubling)**.** Let $\beta \ge 0$, $l \ge 3$, and $\mathsf{e}(l) \le x_0$ at β; put $\mathsf{s}(x) = \psi(4x + 2\psi(x) + \psi(2x + \psi(x)))$.

(a) $\mathsf{e}(2l) \le \mathsf{s}(\mathsf{e}(l)) \le 12.696\, \mathsf{e}(l)^2 \le 12.9\, \mathsf{e}(l)^2$, $D_2(2l) \le 19.669\, \mathsf{e}(l)^2 \le 20\, \mathsf{e}(l)^2$, and $\mathsf{e}(2^k l) \le \theta^{2^k}/12.9$ for $k \ge 0$.

(b) $\delta((2^j l)^3) \le \frac{20}{7}\mathsf{e}(2^{j-1}l)^2 \le 0.017170\, \theta^{2^j}$, and $\delta((2^j l)^3) \le 0.01655\, \theta^{2^j}$, for $j \ge 1$; $\delta(l, 2l, 2l) \le \frac{x_0}{2} + \frac{10}{7}x_0^2$; for every integer $N \ge 2l$: $\delta(N^3) < 0.13736\, \theta^{N/2l}$; $\mathsf{e}(l') < 0.3702\, \theta^{l'/2l}$ for every integer $l' \ge l$; $D_q(N) < q^3\, 0.13736\, \theta^{N/2l}$; and $\delta(\underline{n}) < |\underline{n}|\, n_-^{-3}\, 0.13736\, \theta^{n_-/2l}$ for every box whose least side n_- is $\ge 2l$.

(c) On every cube N^3 with $N \ge 2l$: $\mathsf{G}_N \le 0.06754\, \theta^{N/2l}$, $\mathsf{r}_N < 0.1473\, \theta^{N/2l}$, all seven $0 \le 1 - r^z < 0.4121\, \theta^{N/2l}$, and $1 - r_1 \le 0.09006\, \theta^{N/2l}$, $1 - r_2 \le 0.1351\, \theta^{N/2l}$, $1 - r_3 \le 0.2252\, \theta^{N/2l}$; at $N = 2^{k+1}l$, $\mathsf{r}_N \le 0.001615\, \theta^{N/l}$.

(d) If $x_0 = 0.069$: $1 - r_1(N) < 0.0039$ for every $N \ge 16l$, and $\mathsf{e}(l') < 0.0234$ for every integer $l' \ge 16l$.

(e) *The threshold.* $\mathsf{s}(x) \le 0.99991\, x$ on $(0, 0.07489]$, and the least positive fixed point of s lies in $(0.0748941, 0.0748942)$. No threshold $\ge \log 2$ can serve for a recursion of this kind: the function $F = |\underline{n}|P + \log 2$, which in each direction is a sum of t-th powers of two equal positive numbers and has the properties of Prop. 17.31(b), (c), has $\mathsf{e} \equiv \delta \equiv \log 2$; and the positivity of the λ_i is used: $e^F = e^{|\underline{n}|P} \prod_i (1 + (-1)^{n_i})$ has $\mathsf{e} = -\infty$ at odd l and $\mathsf{e} = \log 8$ at even l.

■

Proof. Let $\epsilon = \mathsf{e}(l)$, the doubling defect at $t = l$ over the cross-section $(2l, 2l)$. We double the three sides in turn, each time with Prop. 17.31(b). First $\mathsf{a} = 2F((2l)^3) - F(2l, 2l, 4l) \le \psi(\epsilon)$. Next $y_1 = 2F(l, 2l, 4l) - F(2l, 2l, 4l) \le 2\epsilon + \mathsf{a}$, because $F(l, 2l, 4l) \le 2F(l, 2l, 2l)$, and

$\mathsf{b}' = 2F(2l,2l,4l) - F(2l,4l,4l) \le \psi(y_1)$. Last $2F(l,4l,4l) - F(2l,4l,4l) \le 4\epsilon + 2\mathsf{a} + \mathsf{b}'$, and

$$\mathsf{e}(2l) \le \psi(4\epsilon + 2\mathsf{a} + \mathsf{b}') \le \mathsf{s}(\epsilon), \qquad D_2(2l) = 4\mathsf{a} + 2\mathsf{b}' + \mathsf{e}(2l).$$

ψ and s are power series in x with coefficients ≥ 0 beginning with $\frac{1}{2}x^2$ and $8x^2$, so $\mathsf{s}(x)/x^2$ does not decrease, and $\epsilon \le 0.069$ gives $12.696\,\epsilon^2$ and $19.669\,\epsilon^2$, all arguments staying below $0.293 < \log 2$; since $12.9 \cdot 0.069 < 1$, iteration gives (a). Put $\mathsf{s}_D(x) = 4\psi(x) + 2\psi(2x + \psi(x)) + \mathsf{s}(x)$, a power series in x with coefficients ≥ 0 beginning with $14x^2$, so that $\mathsf{s}_D(x)/x^2$ does not decrease; it is at most 19.668665 at 0.069, the number 19.669 of (a) before rounding. As ψ increases, the display, with $m = 2^i l$, $i \ge 0$, in the place of l, gives $D_2(2m) \le \mathsf{s}_D(\mathsf{e}(m)) \le 19.668665\,\mathsf{e}(m)^2$, as $\mathsf{e}(m) \le \theta^{2^i}/12.9 \le 0.069$ by (a); and e decreases along these sides, as $\mathsf{e}(2m) \le 12.9 \cdot 0.069\,\mathsf{e}(m)$. The series of Prop. 17.31(c) at $q = 2$, $\delta((2^j l)^3) = \sum_{s\ge 0} 8^{-(s+1)} D_2(2^{s+j} l)$, gives for $j \ge 1$

$$\delta((2^j l)^3) \le 19.668665 \sum_{s\ge 0} 8^{-(s+1)} \mathsf{e}(2^{s+j-1} l)^2 \le \frac{19.668665}{7}\,\mathsf{e}(2^{j-1} l)^2 \le \frac{19.668665}{7 \cdot 12.9^2}\,\theta^{2^j} \le 0.016885\,\theta^{2^j},$$

which is below $\frac{20}{7}\mathsf{e}(2^{j-1}l)^2$ and below $\mathsf{c}_0\theta^{2^j}$, $\mathsf{c}_0 = \frac{20}{7\cdot 12.9^2} \approx 0.0171693$. Keeping the decay of the terms, $\mathsf{e}(2^{s+j-1}l)^2 \le \theta^{2^{s+j}}/12.9^2$, the same series gives $\delta((2^j l)^3) \le \frac{19.668665}{8\cdot 12.9^2}\,\theta^{2^j} \sum_{s\ge 0} 8^{-s}\theta^{2^{s+j}-2^j} \le 0.0147743 \cdot 1.10720\,\theta^{2^j} \le 0.01636\,\theta^{2^j}$, the sum being greatest at $j = 1$ and $\theta = 0.8901$. For $2^j l \le N < 2^{j+1} l$, $j \ge 1$, the monotone density gives $\delta(N^3) \le (N/2^j l)^3 \delta((2^j l)^3) < 8 \cdot 0.016885\,\theta^{2^j} < 0.13736\,\theta^{N/2l}$, and boxes follow against the cube of their least side. For $\mathsf{e}(l')$ with $n = 2^k l \le l' < 2n$, $k \ge 0$: $\mathsf{e}(l') \le 2\mathsf{G}(n; 2l', 2l') \le 2\delta(n, 2l', 2l') \le 2(l'/n)^2 \delta(n, 2n, 2n) < 8\delta(n, 2n, 2n) = 4[\mathsf{e}(n) + \delta((2n)^3)]$ by Prop. 17.31(b), (c), and the bound above at $j = k + 1$ with $\mathsf{e}(n) \le 0.069$ gives $4[\mathsf{e}(n) + \delta((2n)^3)] \le 4\,\mathsf{e}(n)(1 + \frac{19.668665}{7}\,0.069) \le \frac{4\cdot 1.193877}{12.9}\,\theta^{2^k} \le 0.370195\,\theta^{2^k} < 0.3702\,\theta^{l'/2l}$. (c) For $\bar{N} = 2^j l \le N < 2\bar{N}$, $j \ge 1$, with the dyadic side as the time, $\mathsf{G}_N \le \mathsf{G}(\bar{N}; N, N) \le \delta(\bar{N}, N, N) \le 4\delta(\bar{N}^3) \le \frac{4\cdot 19.668665}{7\cdot 12.9^2}\,\theta^{2^j} \le 0.0675395\,\theta^{2^j} \le 0.06754\,\theta^{N/2l}$.

Then Prop. 17.31(d): in the signs of its proof, $1 - r_1 \le \frac{4}{3}\mathsf{R}_N$, $1 - r_2 = 4u \le 2(2u + v) \le 2\mathsf{R}_N$ and $1 - r_3 \le (1 - r_1) + (1 - r_2) \le \frac{10}{3}\mathsf{R}_N$, with $\mathsf{R}_N \le \mathsf{G}_N$, which give 0.0900527, 0.135079, 0.225132 times $\theta^{N/2l}$; all seven $1 - r^z \le 3\mathsf{R}_N \le 3\mathsf{G}_N$, and $\mathsf{r}_N = e^{\mathsf{G}_N} - 1$ with $e^{cy} - 1 \le y(e^c - 1)$ for $y \le 1$. At $N = 2^{k+1} l$, with $n = 2^k l$: $\mathsf{r}_N \le \mathsf{r}_n(N, N)^2 \le \varphi(e^{\mathsf{e}(n)} - 1)^2$ by Prop. 17.31(b), and $\varphi(e^x - 1)^2/x^2$ does not decrease and is at most 0.268652 at 0.069; with $\mathsf{e}(n) \le \theta^{2^k}/12.9$ this is $\mathsf{r}_N \le 0.268652\,\theta^{N/l}/12.9^2 \le 0.001615\,\theta^{N/l}$. (d) Iterate $X_{k+1} = \mathsf{s}(X_k)$ from $X_0 = 0.069$: $X_1, \dots, X_5 \approx 0.0604446,\ 0.0432902,\ 0.0195722,\ 0.0034309,\ 0.0000960$; the series gives $\delta((2^j l)^3) \le \sum_s 8^{-(s+1)} \mathsf{s}_D(X_{j-1+s})$, with the s_D above and $\mathsf{e}(2^i l) \le X_i$ (s increases), which at $j = 4$ is at most 0.00073051; so for $N \ge 16l$, $\mathsf{G}_N \le 0.0029221$ and $1 - r_1 \le \frac{4}{3}(1 - e^{-\mathsf{G}_N}) < 0.0039$, and $\mathsf{e}(l') \le 32 \cdot 0.00073051 < 0.0234$. (e) s is star-shaped, and $\mathsf{s}(x)/x$ is approximately 0.99999914 at 0.0748941 and approximately 1.00000135 at 0.0748942.

The two examples satisfy the identities of Prop. 17.31(a)–(c) with two equal top eigenvalues, respectively with an eigenvalue $-\lambda$. □

Proposition 17.33 (Every cross-section: a simple top, a gap, the thermal law)**.** Let $\beta \ge 0$, $l \ge 3$, $\mathsf{e}(l) \le x_0$ at β, and $A \le B$ integers $\ge 2l$; let $n = 2^k l$ with $2n \le A < 4n$ and $n' = 2^j n$ with $2n' \le B < 4n'$; any direction is the time.

(a) Over (A, B) the top λ_0 is simple and lies in the sector 00, and $\lambda_1/\lambda_0 < 0.49067^{1/n}\theta^{1/l} < \theta^{1/l}$; so every level of a sector $e \ne 00$, and every level of the sector 00 but the top, lies more than $\frac{1}{l}\log\frac{1}{\theta} + \frac{1.4239}{A}$ a step above the top. If $A = 2l$: $\lambda_1/\lambda_0 < (2.42\,x_0)^{1/l}$, and $\mathsf{r}_t(2l, B) \le 0.5628\,(2.42\,x_0)^{t/l}$ for $t \ge n'$. Over the square (N, N), $N \ge 2l$, with $2n \le N < 4n$: $\mathsf{r}_n(N, N) \le 0.20395\,\theta^{n/l}$.

(b) $\mathsf{r}_t(A,B) \le 0.2314\,\theta^{t/l}$ for every integer $t \ge n'$, and for every integer $t \ge l$

$$\mathsf{G}(t;A,B) \le 0.37\,\Big\langle\frac{A}{2t}\Big\rangle\Big\langle\frac{B}{2t}\Big\rangle\,\theta^{t/l} \le 0.37\,\frac{AB}{4l^2}\,\theta^{t/l},$$

in detail $0.27263\frac{AB}{4t^2}\theta^{t/l}$ if $2t \le A$, $0.3695\frac{B}{2t}\theta^{t/l}$ if $A < 2t \le B$, and $\mathsf{r}_t(A,B) \le 0.26261\,\theta^{t/l}$ if $B < 2t$.

(c) $\mathsf{e}(l') \le 0.4079\,\theta^{l'/l}$ for every integer $l' \ge l$; $\delta(l',2l',2l') \le 0.27263\,\theta^{l'/l}$; $D_2(N) \le 4.1242\,\theta^{N/l}$ and $\delta(N^3) \le 0.5892\,\theta^{N/l}$ for every integer $N \ge 2l$.

(d) Over the square (N,N), $N \ge 2l$, the vortex free energy $\varepsilon_{\mathrm{v}}(N,N) = \log(\lambda_0/\|\mathsf{T}^{(1)}\|)$ satisfies $N\varepsilon_{\mathrm{v}}(N,N) < 1.0292\,\theta^{N/2l}$.

(e) Conversely, on one lattice: if $\lambda_1/\lambda_0 \le e^{-\mu}$ over the single cross-section $(2l,2l)$ and $3 \le s < l$, then $\mathsf{e}(l) \le 2e^{-\mu(l-s)}(e^{\delta(s,2l,2l)}-1)$; so $\mathsf{e}(l) \le x_0$ as soon as $\mu(l-s) \ge \delta(s,2l,2l) + \log(2/x_0)$. ■

Proof. Freeze the side A: the boxes (A,s,u) form a two-dimensional system, with the pressure $P_A = \inf F(A,s,u)/su$ and the excess $\delta_A(s,u) = F(A,s,u) - suP_A$, which has the properties of Prop. 17.31(b), (c) in its two sides and bounds $\mathsf{G}(t;A,u)$ from above. The first two doublings of the proof of Prop. 17.32, carried out in this system, give for its doubling defect $\eta = \mathsf{e}_A(m) = 2F(A,m,2m) - F(A,2m,2m)$: if $\eta \le 0.16521$, then $\mathsf{e}_A(2m) \le 3.4522\,\eta^2 \le 0.5704\,\eta$ and $\delta_A(2m,2m) \le 1.5514\,\eta^2$; the map $x \mapsto \psi(2x+\psi(x))$ has its least positive fixed point in $(0.2236, 0.2237)$, the threshold of this recursion. It is started at the scale $A/2$: by Prop. 17.31(b), (c) and Prop. 17.32(b),

$$\mathsf{e}_A(n) \le 2\,\mathsf{G}(n;A,2n) \le 2\,\delta(n,A,2n) \le \frac{A}{n}\,\delta(n,2n,2n) \le 0.185604\,\theta^{n/l} \le 0.16521.$$

Iterating, $\mathsf{e}_A(2^j n) \le 0.185604\,\theta^{2^j n/l}$ and $\delta_A(m,2m) \le 0.11659\,\theta^{m/l}$ along $m = 2^j n$; so for $2m \le u < 4m$, $\mathsf{r}_m(A,u) \le (e^{0.23318}-1)\theta^{m/l} \le 0.26261\,\theta^{m/l} < 1$. At $m = n'$, $u = B$ this excludes a double top and gives $(\lambda_1/\lambda_0)^{n'} < \theta^{n'/l}$, and $\mathsf{r}_t \le \mathsf{r}_{n'}^{t/n'}$; the sector is that of Prop. 17.31(d), as $\mathsf{A} = \mathsf{K}^{1/2}\mathsf{m}^{1/2}$ commutes with the flips and carries the eigenvectors of D to those of T'. *The constants of (a) and (b).* They come from a finer start and from finer constants of the same recursion. For $2n \le A < 4n$, Prop. 17.31(b) in the thickness and at the time n over the cross-section $(2n,2n)$ give $\mathsf{e}_A(n) \le \frac{A}{2n}\big[\mathsf{e}(n) + \frac{1}{2}\psi(\mathsf{e}(n))\big] \le 2.037074\,\mathsf{e}(n) \le 0.140559$, as $\mathsf{e}(n) \le 0.069$.

On $[0, 0.140559]$ the doubling of the slab, $\mathsf{e}_A(2m) \le \psi(2\mathsf{e}_A(m) + \psi(\mathsf{e}_A(m)))$, gives $\mathsf{e}_A(2m) \le 3.1072\,\mathsf{e}_A(m)^2$, the inner argument staying below $0.292638 < \log 2$. So from a start $\mathsf{e}_A(n) \le x_1 \le 0.140559$, with $\vartheta = 3.1072\,x_1 < 1$: $\mathsf{e}_A(2^j n) \le \vartheta^{2^j}/3.1072$; the series of the slab gives $\delta_A(n',2n') \le 0.19314\,\vartheta^{n'/n}$; over (B,A) with $2n' \le B < 4n'$, $\mathsf{r}_{n'} \le e^{2\delta_A(n',2n')} - 1 \le 0.4715\,\vartheta^{n'/n} < 1$, so the top is simple, $\lambda_1/\lambda_0 < \vartheta^{1/n}$, and $\mathsf{r}_t \le 0.4715\,\vartheta^{t/n}$ for every $t \ge n'$. With $x_1 = 2.037074\,\mathsf{e}(n)$ and $\mathsf{e}(n) \le \theta^{n/l}/12.9$ (Prop. 17.32(a)), $\vartheta \le 6.3296\,\mathsf{e}(n) \le 0.49067\,\theta^{n/l}$: this is the factor $0.49067^{1/n}$ in (a), and $0.4715 \cdot 0.49067 \le 0.2314$ is the number of (b).

At $A = 2l$ the hypothesis itself is the start, $\mathsf{e}_{2l}(l) = \mathsf{e}(l) \le x_0 \le 0.069$, and on $[0, 0.069]$ the same recursion has the constants 2.42 and 0.5628 in the places of 3.1072 and 0.4715: $\vartheta = 2.42\,x_0$, $\lambda_1/\lambda_0 < (2.42\,x_0)^{1/l}$ and $\mathsf{r}_t(2l,B) \le 0.5628\,(2.42\,x_0)^{t/l}$ for $t \ge n'$. For the square, the box (n,N,N) with the short side as the time gives $\mathsf{G}(n;N,N) \le \frac{N^2}{4n^2}\delta(n,2n,2n) \le 0.185604\,\theta^{n/l}$ and $\mathsf{r}_n \le (e^{0.185604}-1)\theta^{n/l}$. The thermal law (b) follows in three zones: if $2t \le A$, over the cross-section $(2t,2t)$, with $m = 2^i l \le t < 2m$, $\mathsf{r}_t(2t,2t) \le 0.20395\,\theta^{t/l}$, so $\mathsf{e}(t) \le 2\mathsf{r}_t \le 0.4079\,\theta^{t/l}$ and $\delta(t,2t,2t) = \frac{1}{2}[\mathsf{e}(t) + \delta((2t)^3)] \le 0.27263\,\theta^{t/l}$, and the monotone density carries this to (A,B); if $A < 2t \le B$, the slab gives $\delta_A(t,2t) \le 0.3695\,\theta^{t/l}$ and $\mathsf{G}(t;A,B) \le \frac{B}{2t}\delta_A(t,2t)$; if $B < 2t$, $\mathsf{r}_t \le 0.26261\,\theta^{t/l}$.

This also gives the first two clauses of (c); since $D_2(N) \le 8\mathsf{G}(N;N,N) + 4\mathsf{G}(N;N,2N) + 2\mathsf{G}(N;2N,2N)$, the three zones and the series give the last two. (d) In the box (N,N,h) with one twisted column of length h and $\zeta = \frac12(1-r)$, the second inequality of step (ii) in the proof of Prop. 17.39 below, which is exact on one lattice, gives $N\varepsilon_{\mathrm{v}}(N,N) \le \mathsf{d}(N;N,N) - \log(1 - 2\zeta(N,N;2N))$, with $\zeta(N,N;2N) \le \delta(N,N,2N) \le 2\delta(N^3)$ and $\mathsf{d}(N;N,N) \le 2\mathsf{G}_N \le 2\delta(N^3)$; with $\delta(N^3) < 0.13736\,y$, $y = \theta^{N/2l} \le 0.8901$, the first term is $< 0.27472\,y$, and $-\log(1-0.54944\,y) \le 0.75442\,y$ by convexity on $[0, 0.8901]$; the sum is $< 1.0292\,y$. (e) $\mathsf{e}(l) \le 2\mathsf{r}_l(2l,2l) \le 2e^{-\mu(l-s)}\mathsf{r}_s(2l,2l)$ and $\mathsf{r}_s = e^{\mathsf{G}(s)} - 1 \le e^{\delta(s,2l,2l)} - 1$. □

The two seminorms on a larger torus. By the setting and (M2), for $K \in \mathbb{N}$, $\beta > 0$, $\ell'' \le \ell$ in $\mathfrak{L}$ and a function F of the bonds of a cell of side $\ell''/2$, with the seminorm $\|F\|_{\mathrm{ch},\ell}^{(\ell'')}$ of all its mirror copies on the torus ℓ under its own board and the seminorm $\|F\|_{\mathrm{ch},\ell}$ of its eight mirror copies under the board of side $\ell/2$ whose cell is centred at the cell of F, the cells of the two boards being nested (Lem. YM3_I:104.1):

$$\|F\|_{\mathrm{ch},\ell} \le \|F\|_{\mathrm{ch},\ell}^{(\ell'')} \le e^{(\ell''/2)^3(p_{\ell''}-p_\ell)}\,\|F\|_{\mathrm{ch},\ell''} \le e^{\delta((\ell''/a_K)^3)/8}\,\|F\|_{\mathrm{ch},\ell''},$$

the last as $p_\ell \ge p_\infty = a_K^{-3}P$ and $(\ell''/2)^3(p_{\ell''} - p_\infty) = \frac18\delta((\ell''/a_K)^3)$.

Proposition 17.34 (The excess bounds the distance from the torus)**.** Let $K \in \mathbb{N}$, $\beta > 0$, $\ell' \in \mathfrak{L}$, $\delta' = \delta((\ell'/a_K)^3)$ and $Q' = e^{\delta'/8}$; let $\mathfrak{s}$ be either $\mu_{K,\ell,\beta}$ with $\ell \in \mathfrak{L}$, $\ell \ge \ell'$, or a periodic Gibbs state at β.

(a) For every bounded function F of the bonds of a cell of side $\ell'/2$, of any board: $|\mathfrak{s}(F)| \le Q'\|F\|_{\mathrm{ch},\ell'} \le Q'\,\mu_{K,\ell',\beta}(|F|^8)^{1/8}$.

(b) On the inner bonds of such a cell $\mathfrak{s} = h\,\mu_{K,\ell',\beta}$ with $\int h^{8/7} d\mu_{K,\ell',\beta} \le e^{\delta'/7}$, and the two laws are at a distance in total variation at most $\frac12(\sqrt{\varepsilon} + \varepsilon)$, $\varepsilon = 22.3\,(e^{\delta'/7} - 1)$.

(c) If $Q' \le 1.11$, then for every real G of the bonds of a cell of side $\ell''/2$, $\ell'' \in \mathfrak{L}$, $\ell'' \le \ell'$:

$$|\mathfrak{s}(G) - \mu_{K,\ell',\beta}(G)| \le 4.84\,[Q'(Q'-1)]^{1/2}\,\|G\|_{\mathrm{ch},\ell'} \le 4.84\,[Q'(Q'-1)]^{1/2}\,Q''\,\|G\|_{\mathrm{ch},\ell''},$$
$$Q'' = e^{\delta((\ell''/a_K)^3)/8};$$

for complex G the same with twice the constant.

(d) Let $\ell_i \in \mathfrak{L}$, $\ell_i \to \infty$, with $\limsup_i \delta((\ell_i/a_K)^3) \le D < \infty$ and $\mu_{K,\ell_i,\beta} \to \mu^*$ weakly. Then μ^* is a periodic Gibbs state; $\mu^*(A) \ge e^{-D}\mathfrak{s}(A)^8$ for every periodic Gibbs state $\mathfrak{s}$ and every Borel set A; $\mu^* \ge e^{-D}\nu$ for every ergodic periodic Gibbs state ν; and $\mathfrak{s} \le e^D\mu^*$ for every periodic Gibbs state $\mathfrak{s}$. So n mutually singular periodic Gibbs states at β have $n \le e^D$, and indeed $n \le \liminf_{\mathfrak{L}\ni\ell'\to\infty} e^{\delta'}$.

■

Proof. (a) By the fine-to-coarse form of (M2), $\|F\|_{\mathrm{ch},\ell}^{(\ell')} \le \|F\|_{\mathrm{ch},\ell'}\exp[(\ell'/2)^3(p_{\ell'} - p_\ell)]$, and since $p_\ell \ge p_\infty = a_K^{-3}P$ the exponent is at most $\frac18\ell'^3(p_{\ell'} - p_\infty) = \delta'/8$. For a torus $|\mu_{K,\ell,\beta}(F)| \le \|F\|_{\mathrm{ch},\ell}^{(\ell')}$, by the array form with the factor 1 in the other cells; and a periodic Gibbs state is dominated by the tori, $|\mathfrak{s}(F)| \le \liminf_\ell \|F\|_{\mathrm{ch},\ell}^{(\ell')}$ (M2). The second inequality is Hölder's in the array form, the mirror copies of $|F|^8$ all having the same mean. (b) By (a), $\mathfrak{s}(A) \le Q'\mu_{K,\ell',\beta}(A)^{1/8}$ on the inner bonds, so $\mathfrak{s} = h\mu$ with $\|h\|_{8/7} \le Q'$ by duality. With $p = \frac87$ and $\phi(u) = u^p - 1 - p(u-1) \ge 0.045\min\{(u-1)^2, |u-1|^p\}$ for $u \ge 0$ (Taylor's formula on $[0,2]$, monotonicity of $\phi(1+v)/v^p$ beyond), $\int\phi(h) \le e^{\delta'/7} - 1$ gives $\int|h-1| \le \sqrt{\varepsilon} + \varepsilon$. (c) We use an abstract form. Let $\mathcal{A}$ be a real vector space with a member 1, N a seminorm and g a linear functional on it with

$$\mathsf{N}(1+tG)^8 \le 1 + 8t\,\mathsf{g}(G) + R(|t|\mathsf{N}(G)), \quad R(x) = (1+x)^8 - 1 - 8x, \qquad |\mathsf{g}(G)| \le \mathsf{N}(G),$$

and $\mathfrak{s}$ a linear functional with $\mathfrak{s}(1)=1$, $\mathfrak{s}(F)\le Q\,\mathsf{N}(F)$, $1\le Q\le 1.11$. As $(1+u)^{1/8}\le 1+\frac{u}{8}$: $t[\mathfrak{s}(G)-\mathsf{g}(G)]\le (Q-1)(1+x)+\frac{Q}{8}R(x)$, $x=|t|\mathsf{N}(G)$; if $\mathsf{N}(G)=0$, the bounds on g and the linearity bound on $\mathfrak{s}$, applied to G and $-G$, give $\mathfrak{s}(G)=\mathsf{g}(G)=0$. For $Q=1$, use both signs of t and let $t\to 0$, since $R(x)=O(x^2)$. For $\mathsf{N}(G)>0$ and $Q>1$, take $t=\pm x/\mathsf{N}(G)$, $x^2=\frac{Q-1}{4.74Q}$ (so $x\le 0.1446$, where $R(x)\le 8\cdot 4.7179\,x^2$); this gives $|\mathfrak{s}(G)-\mathsf{g}(G)|\le [2\sqrt{4.74}+(0.11/1.11)^{1/2}][Q(Q-1)]^{1/2}\mathsf{N}(G)\le 4.6692\,[Q(Q-1)]^{1/2}\mathsf{N}(G)$. For $\mathsf{N}=\|\cdot\|_{\mathrm{ch},\ell'}$, $\mathsf{g}=\mu_{K,\ell',\beta}$ on the real functions of a cell, the product of the eight copies of $1+tG$ expands into $\sum_S t^{|S|}m_S$ with $m_\emptyset=1$, $m_{\{\mathbf{k}\}}=\mu(G)$ for the eight one-point sets and $|m_S|\le \mathsf{N}(G)^{|S|}$ for $|S|\ge 2$ (array form), and $Q=Q'$ by (a).

A function of a cell of side $\ell''/2$ is a function of the cell of side $\ell'/2$ of the setting, centred at its cell, as ℓ'/ℓ'' is a power of the odd L, and $\|G\|_{\mathrm{ch},\ell'}\le \|G\|_{\mathrm{ch},\ell'}^{(\ell'')}\le Q''\|G\|_{\mathrm{ch},\ell''}$ by the chain of (M2) recalled before the Proposition. (d) μ^* is a periodic Gibbs state invariant under $a_K\mathbb{Z}^3$ by (M3). For an event A of the bonds of a cell of side $\ell_i/2$, (a) with $F=1_A$ gives $\mathfrak{s}(A)\le e^{\delta_i/8}\mu_{K,\ell_i,\beta}(A)^{1/8}$, $\delta_i=\delta((\ell_i/a_K)^3)$; the conditional laws of finitely many bonds given the others are continuous in the others, so the means of local events converge, and $\mu^*(A)\ge e^{-D}\mathfrak{s}(A)^8$ for local events, hence for all Borel sets by approximation.

An ergodic ν has an invariant carrier N_ν that carries no other ergodic state, so the ergodic decomposition of μ^* gives $\mu^*(\cdot\cap N_\nu)=\mu^*(N_\nu)\,\nu$ with $\mu^*(N_\nu)\ge e^{-D}\nu(N_\nu)^8=e^{-D}$; every $\mathfrak{s}$ is an integral of ergodic states (M3), each $\le e^D\mu^*$. Mutually singular states have disjoint carriers C_j, and $\sum_j \mu^*(C_j)\le 1$ with $\mu^*(C_j)\ge e^{-D}$; for the last clause take sides along which δ' tends to its lower limit and a weakly convergent subsequence. □

Lemma 17.35 (Boxes of every shape)**.** Let $\beta\ge 0$, $N=2h\ge 4$, c a corner, $\mathsf{C}=\mathsf{C}_h(c)$, and $\underline{n}$ any box with $n_\mu\ge N$; let $M_\mu=2^{j_\mu}h$, $j_\mu\in\mathbb{N}$, be the greatest number of this form that is $\le n_\mu$, so $N\le M_\mu\le n_\mu<2M_\mu$.

(a) Let μ be a direction, $\vartheta\ge 2$ with $2\vartheta\le n_\mu$, θ' the reflection in a plane $x_\mu=c'_\mu$, $c'_\mu\in a(\mathbb{Z}+\frac12)$, and F a bounded Borel function of the bonds with both ends in the ϑ layers below that plane. Then $\langle F\,\overline{\theta' F}\rangle_{\underline{n}}\ge 0$ and $|\langle F\rangle_{\underline{n}}|^2\le\langle F\,\overline{\theta' F}\rangle_{\underline{n}}$, also for odd n_μ. On the cube N^3, $|\langle\prod_\epsilon H_\epsilon^{[\epsilon]}\rangle_{N^3}|\le\prod_\epsilon\|H_\epsilon\|_{\mathrm{ch};N}$, so $\|\cdot\|_{\mathrm{ch};N}$ is a seminorm, and $|\langle\Psi\rangle_{N^3}|\le\|\Psi\|_{\mathrm{ch};N}\le\sup|\Psi|$.

(b) For every function Ψ of the bonds of C:

$$|\langle\Psi\rangle_{\underline{n}}|\le\exp\Big\{\frac{\delta(N^3)}{8}\Big[1+3N\Big(\frac{1}{M_1}+\frac{1}{M_2}+\frac{1}{M_3}\Big)\Big]\Big\}\|\Psi\|_{\mathrm{ch};N}\le e^{\frac54\delta(N^3)}\|\Psi\|_{\mathrm{ch};N};$$

and for the boxes (q_1N,q_2N,q_3N), $q\in\mathbb{N}^3$: $|\langle\Psi\rangle_{\underline{n}}|\le\exp[\frac18(\delta(N^3)-\delta(\underline{n})/q_1q_2q_3)]\|\Psi\|_{\mathrm{ch};N}$.

(c) Let Ψ be real. If $\delta(N^3)\le\frac12$, then $|\langle\Psi\rangle_{\underline{n}}-\langle\Psi\rangle_{N^3}|\le 7.31\,\delta(N^3)^{1/2}\|\Psi\|_{\mathrm{ch};N}$; for the boxes $(q_\mu N)$ and $\delta(N^3)\le 0.8348$, the same with 1.86 for 7.31.

None of these constants depends on the shape of the box. ■

Proof. (a) With the time μ and $\mathsf{v}=\frac12(n_\mu-2\vartheta)$, the trace formula (M1) and the cyclicity of the trace give $Z(\underline{n})\langle F\rangle_{\underline{n}}=\mathrm{Tr}(\mathsf{T}'^{\,n_\mu/2}\cdot Y\mathsf{T}'^{\mathsf{v}})$, $Y=\mathfrak{P}[\Phi_F]$; Schwarz's inequality in the Hilbert–Schmidt class, with $\|\mathsf{T}'^{\,n_\mu/2}\|_2^2=Z(\underline{n})$ and $\|Y\mathsf{T}'^{\mathsf{v}}\|_2^2=\mathrm{Tr}(\mathsf{T}'^{\,2\mathsf{v}}Y^*Y)=Z(\underline{n})\langle F\overline{\theta' F}\rangle_{\underline{n}}$, is the claim. Applied in the three directions in turn this is the array bound on the cube. (b) *Doubling on a single box.* In the direction 1 let $F_0=\Psi$ and $F_{i+1}=F_i\,\overline{\theta^{(i)}F_i}$, $\theta^{(i)}$ the reflection in $x_1=c_1+(2^i-1)ha$; by (a), as long as the doubled function fits, $|\langle F_i\rangle_{\underline{n}}|^2\le\langle F_{i+1}\rangle_{\underline{n}}$; so $|\langle\Psi\rangle_{\underline{n}}|\le\langle\mathfrak{B}_1\Psi\rangle_{\underline{n}}^{2^{-j_1}}$ with an array $\mathfrak{B}_1\Psi$ of M_1/h cells.

The same in the directions 2, 3 gives an array $\mathfrak{B}\Psi$ on a block of $M_1\times M_2\times M_3$ sites with $\mathcal{N}=M_1M_2M_3/h^3$ cells and $|\langle\Psi\rangle_{\underline{n}}|\le\langle\mathfrak{B}\Psi\rangle_{\underline{n}}^{1/\mathcal{N}}$. *The free layers cost their top eigenvalue.* Let $\underline{m}$ be a box with $m_\mu\ge M_\mu$ for each μ; it is taken in turn as $\underline{n}$, (M_1,n_2,n_3) and (M_1,M_2,n_3). Take the direction 1 as the time over the cross-section (m_2,m_3), let Φ be the slab function of the cell

of $\mathfrak{B}_3\mathfrak{B}_2\Psi$, which lies in the h layers below c_1, and let $Y_\Phi = \mathfrak{Y}[\Phi]$ be its slab operator (another operator than the Y of (a)).

In the direction 1 the array $\mathfrak{B}\Psi$ is the row $\Phi, \Phi^\Theta, \Phi, \Phi^\Theta, \dots$ on M_1/h consecutive slabs of h layers, so its operators close up into $(Y_\Phi^* Y_\Phi)^{M_1/N}$, $M_1/N = 2^{j_1-1}$, and the $m_1 - M_1$ free layers carry $\mathsf{T}'^{m_1-M_1}$ (a free rest of one layer is $\mathfrak{Y}[\Phi^{+1}] = \mathsf{T}'\mathfrak{Y}[\Phi]$ of (M1)). By the trace formula (M1), $Z(\underline{m})\langle\mathfrak{B}\Psi\rangle_{\underline{m}} = \operatorname{Tr}(\mathsf{T}'^{m_1-M_1}(Y_\Phi^* Y_\Phi)^{M_1/N})$ and $\operatorname{Tr}(Y_\Phi^* Y_\Phi)^{M_1/N} = Z(M_1, m_2, m_3)\langle\mathfrak{B}\Psi\rangle_{(M_1,m_2,m_3)}$; and $\operatorname{Tr}(\mathsf{AB}) = \operatorname{Tr}(\mathsf{B}^{1/2}\mathsf{AB}^{1/2}) \le \|\mathsf{A}\| \operatorname{Tr}\mathsf{B}$ for $\mathsf{A}, \mathsf{B} \ge 0$, B of trace class.

So, direction by direction,

$$Z(\underline{n})\langle\mathfrak{B}\Psi\rangle_{\underline{n}} \le \lambda_0(n_2, n_3)^{n_1-M_1}\lambda_0(M_1, n_3)^{n_2-M_2}\lambda_0(M_1, M_2)^{n_3-M_3}\, Z(M_1, M_2, M_3)\langle\mathfrak{B}\Psi\rangle_{(M_1,M_2,M_3)}.$$

On the box (M_1, M_2, M_3), whose sides lie in $N\mathbb{N}$, the inequality $\operatorname{Tr}(Y_\Phi^* Y_\Phi)^p \le (\operatorname{Tr} Y_\Phi^* Y_\Phi)^p$ at $p = M_\mu/N$ in each direction μ, with Φ the cell of the array in that direction, reduces the array to the cube N^3, where it is the product of the eight mirror copies of Ψ: $Z(M_1, M_2, M_3)\langle\mathfrak{B}\Psi\rangle_{(M_1,M_2,M_3)} \le [Z(N^3)\|\Psi\|_{\mathrm{ch};N}^8]^{M_1M_2M_3/N^3}$. *The count.* With $\log\lambda_0(\Sigma) = |\Sigma|P + \varepsilon(\Sigma)$ and $F = |\cdot|P + \delta$ the terms with P cancel, and by the area bound $\varepsilon(\mathsf{A}, \mathsf{B}) \le (\mathsf{A} + \mathsf{B})\delta(N^3)/N^2$ for $\mathsf{A}, \mathsf{B} \ge N$ (Prop. 17.31(c)) the free layers cost at most $3\delta(N^3)N^{-2}(M_1M_2 + M_1M_3 + M_2M_3)$ against $\mathcal{N} = 8M_1M_2M_3/N^3$ cells: an area against a volume.

The boxes (q_1N, q_2N, q_3N), $q \in \mathbb{N}^3$, are treated with another array, which fills the box for every q, so that no layer is free. For a function F of the bonds in the h layers below c_μ, with slab function Φ_F and slab operator $Y_F = \mathfrak{Y}[\Phi_F]$, the time being μ, let $\mathfrak{A}_\mu F$ be the product of the $2q_\mu$ cells $\Phi_F, \Phi_F^\Theta, \Phi_F, \Phi_F^\Theta, \dots$ on all the $2q_\mu$ consecutive slabs of h layers: F and its successive conjugated mirror images in the planes $x_\mu = c_\mu + jha$. By (M1), $Z(\underline{n})\langle\mathfrak{A}_\mu F\rangle_{\underline{n}} = \operatorname{Tr}(Y_F^* Y_F)^{q_\mu} = \|Y_F\|_{2q_\mu}^{2q_\mu}$ and $Z(\underline{n}) = \operatorname{Tr}\mathsf{T}'^{2q_\mu h} = \|\mathsf{T}'^h\|_{2q_\mu}^{2q_\mu}$, the norms of the Schatten class of order $2q_\mu$; Hölder's inequality in these classes for the $2q_\mu$ factors $Y_F, \mathsf{T}'^h, \dots, \mathsf{T}'^h$ gives $Z(\underline{n})|\langle F\rangle_{\underline{n}}| = |\operatorname{Tr}(\mathsf{T}'^h \cdots \mathsf{T}'^h Y_F)| \le \|Y_F\|_{2q_\mu}\|\mathsf{T}'^h\|_{2q_\mu}^{2q_\mu-1}$, that is $|\langle F\rangle_{\underline{n}}| \le \langle\mathfrak{A}_\mu F\rangle_{\underline{n}}^{1/2q_\mu}$.

Applied on the one box $\underline{n}$ to Ψ in the direction 1, to $\mathfrak{A}_1\Psi$, which lies below c_2, in the direction 2, and to $\mathfrak{A}_2\mathfrak{A}_1\Psi$, which lies below c_3, in the direction 3, this gives $|\langle\Psi\rangle_{\underline{n}}| \le \langle\mathfrak{A}\Psi\rangle_{\underline{n}}^{1/8q_1q_2q_3}$, $\mathfrak{A} = \mathfrak{A}_3\mathfrak{A}_2\mathfrak{A}_1$, the product of the mirror copies of Ψ in all the $8q_1q_2q_3$ cells of the box. Only then is the box made smaller: with the time 1 and $F = \mathfrak{A}_3\mathfrak{A}_2\Psi$, $Z(q_1N, n_2, n_3)\langle\mathfrak{A}\Psi\rangle_{(q_1N,n_2,n_3)} = \operatorname{Tr}(Y_F^* Y_F)^{q_1} \le (\operatorname{Tr} Y_F^* Y_F)^{q_1} = [Z(N, n_2, n_3)\langle\mathfrak{A}\Psi\rangle_{(N,n_2,n_3)}]^{q_1}$, and the same in the directions 2, 3 give $Z(\underline{n})\langle\mathfrak{A}\Psi\rangle_{\underline{n}} \le [Z(N^3)\|\Psi\|_{\mathrm{ch};N}^8]^{q_1q_2q_3}$.

As $|\underline{n}| = q_1q_2q_3N^3$, the terms with P cancel, and $|\langle\Psi\rangle_{\underline{n}}| \le \exp\frac{1}{8}[\delta(N^3) - \delta(\underline{n})/q_1q_2q_3]\,\|\Psi\|_{\mathrm{ch};N}$.
(c) The abstract form of the proof of Prop. 17.34(c) with $\mathsf{N} = \|\cdot\|_{\mathrm{ch};N}$, $\mathfrak{g} = \langle\cdot\rangle_{N^3}$, $\mathfrak{s} = \langle\cdot\rangle_{\underline{n}}$: its first condition is the array bound of (a) on the cube, the eight one-point terms being $\langle G\rangle_{N^3}$ by the mirror symmetry of the cube, and $\mathfrak{s}(F) \le Q\,\mathsf{N}(F)$ is (b). With $Q = e^{\frac{5}{4}\delta(N^3)}$, which is ≤ 1.11 if and only if $\delta(N^3) \le \frac{4}{5}\log 1.11$: as $u \mapsto e^u(e^u - 1)/u$ increases, $Q(Q-1) \le \frac{1.11\cdot 0.11}{\log 1.11}\cdot\frac{5}{4}\delta(N^3)$, so $[Q(Q-1)]^{1/2} \le 1.209334\,\delta(N^3)^{1/2}$ and $4.84 \cdot 1.209334 \le 5.86 \le 7.31$; for $\frac{4}{5}\log 1.11 < \delta(N^3) \le \frac{1}{2}$, by (b) and (a) directly, $|\langle\Psi\rangle_{\underline{n}} - \langle\Psi\rangle_{N^3}| \le (e^{\frac{5}{4}\delta} + 1)\|\Psi\|_{\mathrm{ch};N}$, and $(e^{\frac{5}{4}\delta} + 1)\delta^{-1/2}$ decreases on $(0, \frac{1}{2}]$, so it is at most its value at $\frac{4}{5}\log 1.11$, which is less than 7.3025.

For the boxes $(q_\mu N)$ take $Q = e^{\delta(N^3)/8}$ by the second clause of (b): $Q \le 1.11$ if and only if $\delta(N^3) \le 8\log 1.11$, and then $Q(Q-1) \le \frac{1.11\cdot 0.11}{\log 1.11}\cdot\frac{\delta(N^3)}{8}$, $[Q(Q-1)]^{1/2} \le 0.382425\,\delta(N^3)^{1/2}$, and $4.84 \cdot 0.382425 \le 1.85094 \le 1.86$. □

Lemma 17.36 (Cells are nearly independent under every boundary condition)**.**

(a) Let $\mathcal{K}$ be a finite set, Y_κ measurable maps of a probability space (E, R) to spaces E_κ with probability measures ref_κ, and $\Delta \ge 0$ with $\log R(\exp\sum_\kappa \Phi_\kappa \circ Y_\kappa) \le \sum_\kappa[\Delta + \frac{1}{8}\log\mathrm{ref}_\kappa(e^{8\Phi_\kappa})]$

for all bounded real Φ_κ. Then for every probability $P = fR$ with $\mathsf{H} = R(f\log f) < \infty$ and all bounded real non-constant G_κ: $\sum_\kappa [P(G_\kappa \circ Y_\kappa) - \mathrm{ref}_\kappa(G_\kappa)]^2 / \operatorname{osc}(G_\kappa)^2 \le 4\mathsf{H} + 4\#\mathcal{K}\,\Delta$.

(b) Let $K \in \mathbb{N}$, $\beta > 0$, $\ell' \le \ell$ in $\mathfrak{L}$, $n = \ell/a_K$, $n' = \ell'/a_K$ and $\delta' = \delta(n'^3)$; these letters serve in (c) as well. The $(2n/n')^3 = 8(n/n')^3$ cells of a period box of a board of side $\ell'/2$ on the torus ℓ satisfy the hypothesis of (a) with $R = \mu_{K,\ell,\beta}$, ref the law of a cell under $\mu_{K,\ell',\beta}$ and $\Delta = \delta'/8$.

(c) Let D be the set of the sites of a period cube of the torus ℓ, n sites a side, that is the union of the $8(n/n')^3$ cells of side $\ell'/2$ of a period box of the board, let W be the set of the positively oriented bonds with initial point in D, which holds the bonds of these cells, and let $\gamma_W(\cdot)(\xi)$ be the law of the bonds of W given the configuration ξ outside W. For every ξ and all real non-constant $G_{\mathbf{k}}$ of the cells:

$$\sum_{\mathbf{k}} \frac{[(\gamma_W G_{\mathbf{k}})(\xi) - \mu_{K,\ell',\beta}(G_{\mathbf{k}})]^2}{\operatorname{osc}(G_{\mathbf{k}})^2} \le 144\,\beta n^2 + 4\,(n/n')^3\delta';$$

the mean over the cells is at most $\varepsilon_1(n,n')^2 = 18\beta n'^3/n + \delta'/2$.

(d) Let $\lambda \ge 3$ be odd and $\underline{n} = (\mathsf{n}_1\lambda, \mathsf{n}_2\lambda, \mathsf{n}_3\lambda)$ a periodic box with even $\mathsf{n}_1, \mathsf{n}_2, \mathsf{n}_3$, cut into the $\mathsf{n}_1\mathsf{n}_2\mathsf{n}_3$ cells of λ sites a side of a board whose walls are planes of lattice reflections. The cells satisfy the hypothesis of (a) with $R = \mu_{\underline{n}}$, ref the law of one cell of the cube $(2\lambda)^3$ under $\mu_{(2\lambda)^3}$, and $\Delta = \delta((2\lambda)^3)/8$.

■

Proof. (a) By the variational formula for the relative entropy, $P(\Phi) \le \mathsf{H} + \log R(e^\Phi)$ for $\Phi = \sum_\kappa t_\kappa(G_\kappa - \mathrm{ref}_\kappa G_\kappa)\circ Y_\kappa$; the hypothesis and Hoeffding's bound $\log \mathrm{ref}(e^{s(G-\mathrm{ref}G)}) \le s^2 \operatorname{osc}(G)^2/8$ give $\sum_\kappa t_\kappa[P(G_\kappa) - \mathrm{ref}_\kappa(G_\kappa)] \le \mathsf{H} + \#\mathcal{K}\Delta + \sum_\kappa t_\kappa^2 \operatorname{osc}(G_\kappa)^2$; optimise over the t_κ. (b) This is the proof of Prop. 17.34(a) for a product: for real $\Phi_{\mathbf{k}}$ of the cells, the array form of the chessboard estimate on the torus ℓ for the board of side $\ell'/2$ gives $\mu_{K,\ell,\beta}(\prod_{\mathbf{k}} e^{\Phi_{\mathbf{k}}}) \le \prod_{\mathbf{k}} \|e^{\Phi_{\mathbf{k}}}\|^{(\ell')}_{\mathrm{ch},\ell}$, the fine-to-coarse form gives the factor $e^{\delta'/8}$ per cell, $\|e^\Phi\|^{(\ell')}_{\mathrm{ch},\ell} \le e^{\delta'/8}\|e^\Phi\|_{\mathrm{ch},\ell'}$, and Hölder's inequality on the torus ℓ' gives $\|e^\Phi\|_{\mathrm{ch},\ell'} \le \mu_{K,\ell',\beta}(e^{8\Phi})^{1/8}$, the eight mirror copies of $e^{8\Phi}$ having the mean of one cell. (c) Identify the bonds of W with those of the torus ℓ.

The conditional law of the bonds of W and the law of the torus have the same weight but on the plaquettes that are not common to both, the $12n^2 - 3n$ plaquettes with a bond in W that meet the outside and the $6n^2 - 3n$ plaquettes of the torus that wrap around it, a surface; as $a_p \in [0,2]$, the density of the first law against the second lies between $e^{-36\beta n^2}$ and $e^{36\beta n^2}$ (Prop. YM3_I:91.8), so their relative entropy is $\mathsf{H} \le 36\beta n^2$. Then (a), with (b) on the torus ℓ and $\#\mathcal{K} = 8(n/n')^3$: the sum is at most $4\cdot 36\beta n^2 + 4\cdot 8(n/n')^3\delta'/8$, and the mean over the cells is at most $18\beta n'^3/n + \delta'/2$. (d) The reflection in a wall plane is a symmetry of the box, as every n_μ is even, and it is reflection positive by the trace formula (Lem. 17.35(a)).

So the chessboard estimate on the box holds (Prop. YM3_I:86.3): the mean of a product over the cells is at most the product of the seminorms of the factors on the box (equation YM3_I:(1756)), the $(\mathsf{n}_1\mathsf{n}_2\mathsf{n}_3)$-th roots of the means of the products of all their mirror copies. By the subadditivity in the volume (Prop. YM3_I:86.5) at $\underline{\mathsf{n}}' = (2,2,2)$, which is the trace formula with the inequality $\mathrm{Tr}(Y^*Y)^q \le (\mathrm{Tr}\,Y^*Y)^q$ in each direction as in the proof of Lem. 17.35(b), the seminorm on the box of a function G of one cell is at most $\|G\|_{\mathrm{ch};2\lambda}$ times $\exp[\frac{1}{8}F((2\lambda)^3) - \lambda^3 F(\underline{n})/|\underline{n}|] \le e^{\delta((2\lambda)^3)/8}$, as $F(\underline{n})/|\underline{n}| \ge P$.

For $G = e^\Phi$, Hölder's inequality on the cube gives $\|e^\Phi\|_{\mathrm{ch};2\lambda} \le \langle e^{8\Phi}\rangle^{1/8}_{(2\lambda)^3}$, the eight mirror copies of $e^{8\Phi}$ having the mean of one cell; this is the hypothesis of (a) with the reference law of one cell of the cube $(2\lambda)^3$. □

Lemma 17.37 (The operator of a cell; two cells cluster)**.** Let $K \in \mathbb{N}$, $\beta > 0$, a board of side $\mathsf{l} = \ell''/2$, $\ell'' \in \mathfrak{L}$, and $\ell \in \mathfrak{L}$, $\ell \geq 2\mathsf{l}$, $N = \ell/a_K = \mathsf{nw}$ with $\mathsf{w} = \mathsf{l}/a_K$ layers to a cell; a direction as the time of the cube N^3, $\Sigma = (N, N)$.

(a) For F of the bonds of one cell, Φ_F its slab function: $\|\mathfrak{Y}[\Phi_F]\| \leq Z(N^3)^{1/\mathsf{n}} \|F\|_{\mathrm{ch},\ell}^{(\ell'')} = \lambda_0^{\mathsf{w}}(1+\mathsf{r}_N)^{1/\mathsf{n}} \|F\|_{\mathrm{ch},\ell}^{(\ell'')}$, with the seminorm of the setting of all the mirror copies of F on the torus ℓ under the board of side l.

(b) Let $\mathsf{e}(l) \leq x_0$ at β, ν the periodic Gibbs state of Prop. 17.38, $\ell'' \geq 2la_K$, $Q'' = e^{\delta((\ell''/a_K)^3)/8}$. For F, G of the bonds of two cells whose indices differ by $j \geq 1$ in a direction: $|\nu(FG) - \nu(F)\nu(G)| \leq \theta^{(j-1)\mathsf{w}/l} Q''^2 \|F\|_{\mathrm{ch},\ell''} \|G\|_{\mathrm{ch},\ell''}$; on every torus $\ell \geq L\ell''$ of $\mathfrak{L}$, with $j_- = \min\{j, \mathsf{n} - j\} \geq 2$ the distance on the torus, the same with $3.44\,\theta^{(j_- - 1)\mathsf{w}/l}$.

The constants do not depend on K, β, ℓ, F, G or the offset. ■

Proof. (a) The operator norm is at most the Schatten norm of order n ($\mathsf{n} = 2\ell/\ell''$ is even), and by the trace formula (M1) $\mathrm{Tr}(\mathfrak{Y}^*\mathfrak{Y})^{\mathsf{n}/2}$ is $Z(N^3)$ times the mean of the product of the n mirror copies of F along the time, cells of the board of side l in one column, which the array form of (M2) bounds by $(\|F\|_{\mathrm{ch},\ell}^{(\ell'')})^{\mathsf{n}}$. (b) *The state ν.* A function of the second cell is the reflected image of a function of the slab of the first, so the spectral formula of Prop. 17.38(f) applies; over the cross-section (N', N'), $N' = \ell'/a_K$, $\ell' \geq \ell''$ in $\mathfrak{L}$, the covariance is $\lambda_0^{-2\mathsf{w}} \sum_{i \geq 1} (\lambda_i/\lambda_0)^n \overline{\mathsf{a}_i} \mathsf{b}_i$ with $n = (j-1)\mathsf{w}$ layers between the cells (at $n = 0$ with the term of the kernel of T' besides), and Schwarz's inequality bounds it by $(\lambda_1/\lambda_0)^n \lambda_0^{-2\mathsf{w}} \|\mathfrak{Y}[\Phi_F]\| \|\mathfrak{Y}[\Phi_G]\|$.

The gap of Prop. 17.33(a) gives $(\lambda_1/\lambda_0)^n < \theta^{n/l}$, and (a) on the cube N' with the fine-to-coarse form of (M2), $\|F\|_{\mathrm{ch},\ell'}^{(\ell'')} \leq Q'' \|F\|_{\mathrm{ch},\ell''}$ as $p_{\ell'} \geq p_\infty$, gives $\lambda_0^{-\mathsf{w}} \|\mathfrak{Y}[\Phi_F]\| \leq (1+\mathsf{r}_{N'})^{1/\mathsf{n}'} Q'' \|F\|_{\mathrm{ch},\ell''}$, $\mathsf{n}' = N'/\mathsf{w}$, the same for G; as $\ell' \to \infty$ the first factor tends to 1, and the formula passes to ν (Prop. 17.38(c), (f)). *The torus $\ell \geq L\ell''$.* The cube $N = \mathsf{nw}$ has $\mathsf{n} = 2\ell/\ell'' \geq 2L \geq 6$ cells along the time, an even number. The two cells, $j_- \geq 2$ apart the shorter way, leave $n = (j_- - 1)\mathsf{w}$ layers between them on one side and $E = (\mathsf{n} - j_- - 1)\mathsf{w} \geq n$ on the other, and $n, E \geq \mathsf{w} \geq l$, as $\ell'' \geq 2la_K$.

Write, as in the proof of Lem. 17.40, $\mathsf{T}'^m = \lambda_0^m(|e_0\rangle\langle e_0| + \mathsf{R}_{(m)})$, e_0 the top eigenvector, so that $\|\mathsf{R}_{(m)}\| = \varrho^m$, $\varrho = \lambda_1/\lambda_0$, and $\mathrm{Tr}\,|\mathsf{R}_{(m)}| = \mathsf{r}_m$, and put $y_C = \lambda_0^{-\mathsf{w}} \mathfrak{Y}[\Phi_C]$. The trace formula (M1) with the four slabs F, 1_n, G, 1_E gives $\mu_{K,\ell,\beta}(FG) = \mathrm{Tr}\big((|e_0\rangle\langle e_0| + \mathsf{R}_{(E)})\, y_G\, (|e_0\rangle\langle e_0| + \mathsf{R}_{(n)})\, y_F\big)/(1+\mathsf{r}_N)$, and the mean of one cell is $\mu_{K,\ell,\beta}(C) = [\langle e_0, y_C e_0\rangle + \mathrm{Tr}(\mathsf{R}_{(N-\mathsf{w})} y_C)]/(1+\mathsf{r}_N)$.

Expanding, the products $\langle e_0, y_F e_0\rangle\langle e_0, y_G e_0\rangle$ cancel up to the factors $(1+\mathsf{r}_N)^{-1}$ and $(1+\mathsf{r}_N)^{-2}$, and $|\mathrm{Tr}(XW)| \leq \|X\| \,\mathrm{Tr}\,|W|$ bounds the rest:

$$|\mu_{K,\ell,\beta}(FG) - \mu_{K,\ell,\beta}(F)\mu_{K,\ell,\beta}(G)| \leq \|y_F\|\,\|y_G\|\,[\mathsf{r}_N + \varrho^n + \varrho^E + \varrho^n \mathsf{r}_E + 2\mathsf{r}_{N-\mathsf{w}} + \mathsf{r}_{N-\mathsf{w}}^2],$$

the fourth term as the trace norm of $\mathsf{R}_{(E)}$ times the operator norm of $y_G \mathsf{R}_{(n)} y_F$. By (a) and the fine-to-coarse form, $\|y_C\| \leq (1+\mathsf{r}_N)^{1/\mathsf{n}} Q'' \|C\|_{\mathrm{ch},\ell''}$. Prop. 17.33 bounds $\varrho^n, \varrho^E \leq \theta^{n/l}$ by (a), and r_N, $\mathsf{r}_{N-\mathsf{w}}$, r_E by its thermal law (b) over the cross-section (N, N): in the zone $2t > N$ by $0.26261\,\theta^{t/l}$, which serves for $t = N$, $N - \mathsf{w}$ and for $t = E$ if $2E > N$, and in the zone $2t \leq N$ by $\mathsf{r}_t \leq e^{0.27263(N/2t)^2\theta^{t/l}} - 1$. Every term of the bracket, divided by $\theta^{n/l}$, and the factor $(1+\mathsf{r}_N)^{2/\mathsf{n}}$ increase with $u = \theta^{\mathsf{w}/l} \leq \theta \leq 0.8901$; at $u = 0.8901$ the product of the bracket over $\theta^{n/l}$ and $(1+\mathsf{r}_N)^{2/\mathsf{n}}$ is at most 3.33 for all even $\mathsf{n} \geq 6$ and $2 \leq j_- \leq \mathsf{n}/2$, the greatest value 3.3214 being taken at $\mathsf{n} = 6$, $j_- = 3$, where $n = E = 2\mathsf{w}$ and the bracket is at most $u^2[2 + 0.6258 + 0.26261(u^4 + 2u^3 + 0.26261\,u^8)]$. This is the torus clause, with $3.33 \leq 3.44$. □

Proposition 17.38 (One lattice under the inequality)**.** Let $K \in \mathbb{N}$, $\beta > 0$, and $\mathsf{e}(l) \leq x_0$ at β for one integer $l \geq 3$.

(a) The lattice $a_K\mathbb{Z}^3$ has exactly one periodic Gibbs state ν at β; it is ergodic under $a_K\mathbb{Z}^3$, invariant under all isometries of the lattice and reflection positive at every lattice mirror;

there is an open interval $U \ni \beta$ on which $\mathsf{e}(2l) \le 0.069$, P is continuously differentiable on U, and $P'(\beta') = -3\nu_{\beta'}(a_p)$.

(b) For every bounded real Borel F of the bonds of the cell $[-\frac{\ell'}{4}, \frac{\ell'}{4}]^3$, $\ell' \in \mathfrak{L}$, $n' = \ell'/a_K \ge 2l$, every $n = \ell/a_K$ with $\ell \ge \ell'$ in $\mathfrak{L}$, and the cube $Q_M = \{|x|_\infty \le Ma_K\}$, $M \ge n$, of $2M+1$ sites a side: $\|\nu(F \,|\, \text{bonds outside } Q_M) - \nu(F)\|_{L^2(\nu)} \le \operatorname{osc}(F)\, \varepsilon_1(n, n')$, $\varepsilon_1(n,n')^2 < 18\beta n'^3/n + 0.06868\, \theta^{n'/2l}$. So ν is trivial on the tail σ-field and an extreme point of the set of all Gibbs states at β, and for ν-almost every boundary datum the laws in growing cubes converge weakly to ν. The generator $\mathsf{H}_\nu \ge 0$ of the reversible lattice Langevin dynamics of ν, the operator of the closure of the form $\sum_b \nu(\nabla_b F \cdot \nabla_b F')$ on smooth local functions, has $\ker \mathsf{H}_\nu = \mathbb{C}1$, and $e^{-s\mathsf{H}_\nu} F \to \nu(F)$ in $L^2(\nu)$ for every F.

(b′) With a coordinate direction as the time, $\lambda = 2h+1 \ge \max\{3, l\}$, $t \ge \max\{2, l\}$ and $n = h+t$: for every bounded real Borel F of the bonds of $\{|x|_\infty \le ha_K\}$, $\|\nu(F \,|\, \text{bonds outside } \{|x_3| < na_K\}) - \nu(F)\|_{L^2(\nu)} \le \operatorname{osc}(F)\, \varepsilon_2$, $\varepsilon_2^2 = 0.27472\, (\lambda/t)^2 \theta^{t/l} + 0.06868\, \theta^{\lambda/l}$; with $h = \lfloor n/3 \rfloor$, for $n \ge \frac{1}{2}(3l+1)$, this is $\le 0.7057 \operatorname{osc}(F)\, \theta^{(2n-1)/6l}$.

(c) For every even $N \ge 2l$, every corner c, every bounded real Borel Ψ of the bonds of $\mathsf{C}_{N/2}(c)$ and every periodic box $\underline{n}$ with all sides $\ge N$: $|\langle\Psi\rangle_{\underline{n}} - \langle\Psi\rangle_{N^3}| \le 2.71\, \theta^{N/4l} \|\Psi\|_{\mathrm{ch};N}$ and $|\langle\Psi\rangle_{\underline{n}} - \nu(\Psi)| \le 5.42\, \theta^{N/4l} \|\Psi\|_{\mathrm{ch};N} \le 5.42\, \theta^{N/4l} \sup|\Psi|$; for events A of the cell, $|\mu_{\underline{n}}(A) - \nu(A)| \le 2.71\, \theta^{N/4l}$. As all sides tend to ∞ the measures of the periodic boxes converge weakly to ν, uniformly over all boxes with three sides $\ge N_0$.

(d) *Twisted boxes.* Let Γ_n be period lattices and T_n closed twist sets (Γ_n-periodic sets of plaquettes such that every elementary cube has an even number of faces in T_n; the sign of $\operatorname{tr} \partial U$ is reversed on them) that recede: the shortest period of Γ_n tends to ∞ and every finite set of plaquettes misses T_n at last. Then the laws $\mu^{\mathsf{T}_n}_{\Gamma_n,\beta}$, and all mixtures of a bounded number of such sequences with any weights (the sum over the eight sectors among them), converge weakly to ν; and if $\mathsf{T}'_n = \mathsf{T}_n \triangle d\mathsf{S}_n$ with any Γ_n-periodic set S_n of bonds, $d\mathsf{S}$ the plaquettes with an odd number of bonds in S, the joint cumulants of the twisted fields $\mathcal{O}_a^{\mathsf{T}'_n}(f) = a^{-1} \sum_p f(x(p)) a_p^{\mathsf{T}'_n}$, $a_p^{\mathsf{T}} = 1 + \frac{1}{2} \operatorname{tr} \partial U(p)$ on T and a_p elsewhere, under $\mu^{\mathsf{T}'_n}_{\Gamma_n,\beta}$ converge to those of the $\mathcal{O}_a(f)$ under ν. The word *recede* cannot be dropped: $P(\beta) > -3\beta$, so $\nu(\operatorname{tr} \partial U(p)) > 0$, and a column through a fixed plaquette p_0 gives in the limit the law of $-\operatorname{tr} \partial U(p_0)$ under ν, which differs from that of $\operatorname{tr} \partial U(p_0)$.

(e) Over every cross-section (A, B) with $A, B \ge 2l$ the top is simple, lies in the range of P_{00}, and $\lambda_1/\lambda_0 < \theta^{1/l}$ (Prop. 17.33(a)).

(f) Let Sl be a slab of $\vartheta \ge 2$ consecutive layers, Θ_n the lattice reflection such that exactly n layers lie between Sl and Θ_nSl, $A, B \in L^2(\nu)$ functions of the bonds of Sl, and $h_{AB}(n) = \nu(\overline{\Theta_n A}\, B)$. For all integers $n \ge n_0 \ge 0$,

$$|h_{AB}(n) - \overline{\nu(A)}\, \nu(B)| \le \theta^{(n-n_0)/l} \prod_{C=A,B} [h_{CC}(n_0) - |\nu(C)|^2]^{1/2},$$

and $n \mapsto h_{CC}(n) - |\nu(C)|^2$ is nonnegative and log-convex; at $n_0 = 0$, for bounded A, B: $|\operatorname{Cov}_\nu| \le \theta^{n/l} \sup|A| \sup|B|$.

■

Proof. (a) A periodic Gibbs state exists as a limit of tori (M3). If ρ and ν are two of them and A is a local event inside the central cell of side $\ell'/2$, Prop. 17.34(b) gives $|\rho(A) - \nu(A)| \le \sqrt{\varepsilon} + \varepsilon$, $\varepsilon = 22.3(e^{\delta'/7} - 1)$, and $\delta' \to 0$ as $\ell' \to \infty$ by Prop. 17.32(b); so $\rho = \nu$. The symmetries and reflection positivity come from (M3); almost every ergodic component of ν is a periodic Gibbs state, hence ν. By Prop. 17.32(a), $\mathsf{e}(2l) \le 12.9 \cdot 0.069^2 < 0.069$ at β, hence on an open interval U, the box functions being smooth in β; there the derivatives $-3\mu_{K,\ell_m,\beta'}(a_p)$ of $(\ell_m/a_K)^{-3} \log Z_{K,\ell_m}$, $\ell_m = 2L^m$, converge uniformly to $-3\nu_{\beta'}(a_p)$ by Prop. 17.34(b) and the bound of Prop. 17.32(b),

uniform on U. (b) Let D be the cube of n sites a side that is the union of the $8(n/n')^3$ cells $\square_{\mathbf{k}} = [-\frac{\ell'}{4}, \frac{\ell'}{4}]^3 + \frac{\ell'}{2}\mathbf{k}$, $0 \le \mathbf{k}_\mu < 2n/n'$, W the set of the positively oriented bonds with initial point in D, as in Lem. 17.36(c), and $F_{\mathbf{k}}$ the translate of F into $\square_{\mathbf{k}}$.

Lem. 17.36(c), integrated against ν, bounds the mean over $\mathbf{k}$ of $\|\nu(F_{\mathbf{k}}|\text{outside } W) - \mu_{K,\ell',\beta}(F)\|^2$ by $\operatorname{osc}(F)^2\varepsilon_1^2$, and by translation invariance the term of $\mathbf{k}$ is $\|\nu(F|\text{outside } W - \frac{\ell'}{2}\mathbf{k}) - \mu_{K,\ell',\beta}(F)\|^2$. Each translate $W - \frac{\ell'}{2}\mathbf{k}$ holds the cell and has its bonds in $[-\ell,\ell]^3$, so it lies in $Q_n \subset Q_M$; conditioning on the smaller field outside Q_M is a contraction in $L^2(\nu)$, so every term, and with them the mean, is at least $\|\nu(F|\text{outside } Q_M) - \mu_{K,\ell',\beta}(F)\|^2$; $\delta' < 0.13736\,\theta^{n'/2l}$; and the mean $\nu(F)$ is the best constant in $L^2(\nu)$.

Let $n \to \infty$, then $n' \to \infty$: tail triviality, which for a Gibbs state is extremality. A function in $\ker \mathsf{H}_\nu$ has $\nabla_b F = 0$ for every bond, so (the conditional law of one bond being equivalent to Haar measure) it is measurable outside every finite set of bonds, hence constant; the rest is the spectral theorem. (b′) Over the cross-section the top eigenvector e_0 is positive and continuous; in the cylinder the joint density of the two layers $V^{\pm n}$ against $\pi \otimes \pi$, $\pi = e_0^2 dV$, is $p_{2n} = \mathsf{D}^{2n}/\lambda_0^{2n} e_0 \otimes e_0$, and given them the slab of the cells has the density $\varrho = p_t(x,y)p_t(y',x')/p_{2n}(x,x')$ against its own law; the mean relative entropy is $2\iint p_t \log p_t - \iint p_{2n}\log p_{2n} \le 2\log \iint p_t^2 = 2\log(1+\mathsf{r}_{2t})$.

Lem. 17.36(a) for the n_2^2 cells of the slab (their sub-independence is Lem. 17.36(d) on the boxes $(\lambda\mathsf{n}_1, N', N')$, $N' = \lambda\mathsf{n}_2$, with even $\mathsf{n}_1, \mathsf{n}_2$, in the limit $\mathsf{n}_1 \to \infty$, the reference law being that of one cell of the cube $(2\lambda)^3$ and $\Delta = \delta((2\lambda)^3)/8)$, translation invariance of the cylinder along the cross-section, and $\log(1+\mathsf{r}_{2t}) \le \delta(2t, N', N') < 0.13736(N'/2t)^2\theta^{t/l}$ give the bound; then $N' \to \infty$ by (c) and density. (c) Lem. 17.35(c) with $\delta(N^3) < 0.13736\,\theta^{N/2l} \le \frac{1}{2}$, $7.31 \cdot 0.370622 \le 2.71$; a second box with all sides tending to ∞ gives ν; from continuous to Borel Ψ by approximation in L^8 of a measure that dominates both laws; for events take $\Psi = 1_A - \frac{1}{2}$.

The uniform convergence: limit points of box measures are periodic Gibbs states (M3), hence ν. (d) For a Γ-periodic set S of bonds, the flip $U_b \mapsto -U_b$, $b \in \mathsf{S}$, keeps the Haar measure and carries μ_Γ^{T} to $\mu_\Gamma^{\mathsf{T}\triangle d\mathsf{S}}$ (-1 is central). For a closed T and the bonds S whose plaquette swept in the direction μ lies in T, $\tau_{-ae_\mu}\mathsf{T} = \mathsf{T}\triangle d\mathsf{S}$, by counting the faces of the elementary cubes; so the translates of the twisted measure are flips of it, which a local function does not see once T_n has receded. Limit points satisfy the DLR equations away from the twist sets, hence are periodic Gibbs states, hence ν; both identities are linear in the law.

For $P(\beta) > -3\beta$: on the $\prod\lfloor n_i/3\rfloor \ge |\underline{n}|/125$ plaquettes $p_{12}(x)$, $x_i \in 3\mathbb{Z}$, which share no site, Jensen's inequality gives $F/|\underline{n}| \ge -3\beta + \frac{1}{125}\log\int e^{\frac{\beta}{2}\operatorname{tr} u}\,\mathrm{d}u > -3\beta$; P is convex, $P(0) = 0$, so $P'(\beta) > -3$, and $P' = -3\nu(a_p)$ by (a). (e) is Prop. 17.33(a). (f) In the box with $N = 2\vartheta + n + E'$ layers over (N', N'), (M1) gives $\mu(\overline{\Theta_n A}\,B) = \operatorname{Tr}(\mathsf{T}'^{E'}Y_A^*\mathsf{T}'^m Y_B)/\operatorname{Tr}\mathsf{T}'^N$, $Y_C = \mathfrak{Y}[\Phi_C]$ the slab operator of the function C. As $E' \to \infty$ the gap projects onto the top eigenvector e_0:

$$\mu(\overline{\Theta_n A}\,B) \longrightarrow \lambda_0^{-2\vartheta}\Big[\sum_{j\ge 0}(\lambda_j/\lambda_0)^n\,\overline{\mathsf{a}_j}\,\mathsf{b}_j + 1_{n=0}\,\langle \mathsf{P}_{\ker}Y_A e_0, \mathsf{P}_{\ker}Y_B e_0\rangle\Big], \qquad \mathsf{a}_j = \langle e_j, Y_A e_0\rangle,\ \mathsf{b}_j = \langle e_j, Y_B e_0\rangle,$$

e_j the eigenvectors of the nonzero eigenvalues and $\mathsf{P}_{\ker}$ the projection on $\ker\mathsf{T}'$; the term $j = 0$ is the product of the means, which converge in the same way. Schwarz's inequality and $\lambda_j/\lambda_0 < \theta^{1/l}$ for $j \ge 1$ give the bound (for $n > n_0$ the kernel term is absent on the left); $N' \to \infty$ by (c), and $L^2(\nu)$ by density. The same formula exhibits $h_{CC}(n) - |\nu(C)|^2$ as a limit of $\int x^n \varrho(\mathrm{d}x)$, $0^0 = 1$, with $\varrho = \lambda_0^{-2\vartheta}[\sum_{j\ge1}|\mathsf{c}_j|^2\delta_{\lambda_j/\lambda_0} + \|\mathsf{P}_{\ker}Y_C e_0\|^2\delta_0] \ge 0$ on $[0,1)$; such functions are ≥ 0 and log-convex. □

Proposition 17.39 (Flux and loops on one lattice)**.** Let $\beta > 0$, $l \ge 3$, $\mathsf{e}(l) \le x_0$, $\Lambda = \log(1/\theta)$, $\sigma^+ = (2\Lambda + 1.67)/4l^2$, so that $\sigma^+ l^2 = \frac{\Lambda}{2} + 0.4175 \ge 0.4757$, $\mathsf{q} = 1 + 0.835/\Lambda$, and ν as in Prop. 17.38; lengths in lattice units.

(a) $0 \le 1 - r_1(N) \le 2\exp[-2.2 \cdot 10^{-4}(N/l)^2]$ for every integer $N \ge 3$; $\frac12(1 - r_1(N)) \le \exp[-1.4268 \cdot 10^{-3}(N/l)^2]$ for $N \ge 256\,l$; and for every $N \ge 2\mathsf{q}l$,

$$\tfrac12(1 - r_1(N)) \le \exp\Big[-\sigma^+ N^2 + \frac{2\mathsf{q}N}{l}\Big(\frac12 \log\frac{2\mathsf{q}N}{l} + 0.86\Big)\Big];$$

and $1 - r^z \le 6\exp[\cdots]$ for all seven classes $z \ne 0$, with each of the three exponents.

(b) $0 \le \nu(W_C) \le e^{-\sigma^+ RT}$ for every rectangle C with $R \times T$ bonds in a coordinate plane; in the finite cube, for $n \ge 2\mathsf{q}l$, $1 \le R \le n/2$, $1 \le T \le n-1$: $0 \le \langle W_C\rangle_{(n,n,n)} \le 8^{T/n}\exp[-\sigma^+ RT + \frac{2\mathsf{q}RT}{ln}(\frac12 \log\frac{2\mathsf{q}n}{l} + 0.86)]$; and on every box, with no hypothesis on e, for $1 \le R \le n_1/2$, $1 \le T \le n_3 - 1$ and $\zeta = \frac12(1-r)$ of one twisted column of the plane (13): $0 \le \langle W_{R\times T}\rangle_{\underline{n}} \le (8\,\zeta^{R/n_1})^{T/n_3}$.

(c) $|\nu(W_C)| \le e^{-\sigma^+|A(C)|}$ for every staircase loop of a coordinate plane, and $|\nu(W_C)| \le 2^{c_3(C)-1}e^{-\sigma^+ A_{13}(C)}$ for every closed lattice path C with a bond of the direction 3, with A_{13}, c_3 as in Thm. AZ(4) at unit spacing; the same for every ordered pair of directions. Here a *staircase loop* $C = C(\alpha, R)$ of the plane (13) is the staircase loop of the setting (Twists and loops), read in lattice units: it is given by a site o, an integer $T \ge 1$ and integers α_j and $R_j \ge 1$, $0 \le j < T$; with $o_j = o + \alpha_j e_1$ and $r_j = o + (\alpha_j + R_j)e_1$ it runs at the height 0 (in the direction 3, counted from o) straight from o_0 to r_0; then, for $j = 0, \dots, T-1$ in turn, one step up above r_j from the height j to the height $j+1$ and, if $j < T-1$, straight at that height to above r_{j+1}; at the height T straight back from above r_{T-1} to above o_{T-1}; then, for $j = T-1, \dots, 0$ in turn, one step down above o_j from the height $j+1$ to the height j and, if $j > 0$, straight at that height to above o_{j-1}.

Its area is $|A(C)| = \sum_j R_j$; the staircase loops of the other coordinate planes are the images of these under the symmetries of the lattice. A staircase loop winds once about each unit square of the rows $[\alpha_j, \alpha_j + R_j] \times [j, j+1]$ and about no other, and its steps in the direction 3 are T upward ones followed by T downward ones; so $A_{13}(C) = |A(C)|$, $c_3(C) = 1$, and the first clause is a case of the second.

(d) For every weak limit point of the measures of periodic boxes all of whose sides tend to infinity, the limits of the tori among them, $w(R,T) = \langle W_{R\times T}\rangle$ satisfies $0 \le w \le 1$, $w(R,T)^2 \le w(R,T-1)w(R,T+1)$ and $w(R,T)^2 \le w(R-1,T)w(R+1,T)$, with $w(R,0) = w(0,T) = 1$; so $V(R) = \lim_T -T^{-1}\log w(R,T) = \inf_T(-T^{-1}\log w(R,T))$ exists in $[0,\infty]$, $V(0) = 0$, V is either finite everywhere or infinite at every $R \ge 1$, V is concave and non-decreasing, and, in the finite case, $V(R) \ge V(R') + \sigma_\infty(R - R')$ for $R \ge R' \ge 0$, with $\sigma_\infty = \lim_R V(R)/R \in [0,\infty)$. Under the hypothesis, in ν, which is the limit of the tori: $V(R) \ge V(R') + \sigma^+(R - R')$ for $R \ge R' \ge 0$.

(e) With $\varepsilon_{\mathrm{v}}(l') = \varepsilon_{\mathrm{v}}(l', l')$ the vortex free energy of Prop. 17.33(d) and $l_j = 2^{j+1}l$: $l_j\varepsilon_{\mathrm{v}}(l_j) \le e^{-(2\Lambda+1.67)4^j} = e^{-\sigma^+ l_j^2}$ for all $j \ge 0$.

■

Proof. For a box $(t, u; h)$, t, u the sides of a plane with one twisted column of length h, let $\zeta(t,u;h) = \frac12(1-r)$ for that twist, and let $\varepsilon_{\mathrm{v}}(t,u) = \log(\|\mathsf{T}^{(0)}\|/\|\mathsf{T}^{(1)}\|)$ over the cross-section (t,u), the time along h. (i) By Prop. 17.31(d), in each of the two sides t, u of the twisted plane ζZ and $(1-\zeta)Z$ are sums of powers of numbers ≥ 0; Hölder's inequality for such sums and the excess give

$$\zeta(t',u';h) \le [\zeta(t,u;h)\,e^{\delta(t,u,h)}]^{t'u'/tu}e^{-\delta(t',u',h)} \qquad (t' \ge t,\ u' \ge u),$$

and $\zeta(t,u;h) \le 1 - e^{-\mathsf{G}(t;u,h)} \le \delta(t,u,h)$. (ii) In the third side $Z = \mathrm{Tr}(\mathsf{T}^{(0)})^h = \lambda_0^h(1 + \mathsf{r}_h(t,u))$ and $Z^z = \mathrm{Tr}(\mathsf{T}^{(1)})^h$, with $\mathsf{T}^{(1)} \ge 0$ of norm $\lambda_0 e^{-\varepsilon_{\mathrm{v}}(t,u)}$. So $1 - 2\zeta(t,u;h) = Z^z/Z \ge e^{-h\varepsilon_{\mathrm{v}}}/(1+\mathsf{r}_h) \ge 1 - h\varepsilon_{\mathrm{v}} - \mathsf{r}_h$ if $h\varepsilon_{\mathrm{v}} \le 1$, and $2\zeta < 1$ always: the first inequality below. Further $\mathrm{Tr}(\mathsf{T}^{(1)})^{2h} \le \|\mathsf{T}^{(1)}\|^h\,\mathrm{Tr}(\mathsf{T}^{(1)})^h$, $Z^z \le Z$ at the length h (Prop. 17.31(d), with the time in the twisted plane) and

$Z \ge \lambda_0^{2h}$ at the length $2h$ give $e^{-h\varepsilon_{\rm v}} \ge (1-2\zeta(t,u;2h))/(1+\mathsf{r}_h)$, the second:

$$2\zeta(t,u;h) \le h\,\varepsilon_{\rm v}(t,u) + \mathsf{r}_h(t,u), \qquad h\,\varepsilon_{\rm v}(t,u) \le \mathsf{G}(h;t,u) - \log(1-2\zeta(t,u;2h)) \le \mathsf{d}(h;t,u) - \log(1-2\zeta(t,u;2h)),$$

where $\mathsf{d}(h;t,u) = 2F(t,u,h) - F(t,u,2h) = 2\mathsf{G}(h;t,u) - \mathsf{G}(2h;t,u) \ge \mathsf{G}(h;t,u)$ is taken for the box (t,u,h) with the side h doubled; also $\mathsf{r}_h(t,u) \ge \mathsf{G}(h;t,u)$ may stand for $\mathsf{G}(h;t,u)$. (iii) By (i) and (ii), doubling the cross-section of a long tube, on which the thermal sums lie below the fourth power, replaces $l'\varepsilon_{\rm v}(l')$ by a bounded multiple of its fourth power. Let $\mathsf{t} = \theta^{l'/l}$ at the dyadic sides $l' = l_j = 2^{j+1}l$, and let $\mathsf{k}_1 = 0.01655$ and $\mathsf{k}_2 = 0.001615$ be the constants of the bounds $\delta((2^jl)^3) \le \mathsf{k}_1\theta^{2^j}$, $j \ge 1$, of Prop. 17.32(b) and $\mathsf{r}_N \le \mathsf{k}_2\theta^{N/l}$ at $N = 2^{k+1}l$ of Prop. 17.32(c); so $\delta(l',l',h) \le (h/l')\,\mathsf{k}_1\mathsf{t}$ by the monotone density, $\mathsf{r}_{l'}(l',l') \le \mathsf{k}_2\mathsf{t}$ and $\mathsf{r}_{2l'}(2l',2l') \le \mathsf{k}_2\mathsf{t}^2$; and $\mathsf{r}_{Ms} \le \mathsf{r}_s^M$ for real $M \ge 1$ with Ms an integer, as in Prop. 17.31(b).

The start. The second inequality of (ii) at $t = u = h = 2l$, with $\mathsf{G} \le \mathsf{r}$ and $\zeta(2l,2l;4l) \le \mathsf{R}_{2l}(2l,4l) = \mathsf{r}/(1+\mathsf{r})$, $\mathsf{r} = \mathsf{r}_{2l}(2l,4l)$, by (i), gives $l_0\varepsilon_{\rm v}(l_0) \le \mathsf{r}_{2l}(2l,2l) + \log\frac{1+\mathsf{r}}{1-\mathsf{r}}$. By Prop. 17.31(b), $\mathsf{r}_{2l}(2l,2l) \le \mathsf{r}_l(2l,2l)^2 \le \varphi(e^{x_0}-1)^2$, and $\mathsf{r} \le \mathsf{r}_l(2l,4l)^2 \le \varphi(e^{y_1}-1)^2$ with $y_1 = 2F(l,2l,4l) - F(2l,2l,4l) \le 2x_0 + \psi(x_0)$, as in the proof of Prop. 17.32. Both bounds are power series in x_0 with coefficients ≥ 0 beginning with multiples of x_0^2, and so is $\log\frac{1+\mathsf{r}}{1-\mathsf{r}}$ in r; the sum divided by $\theta^2 = (12.9\,x_0)^2$ is therefore greatest at $x_0 = 0.069$, where it is at most 0.016154: $l_0\varepsilon_{\rm v}(l_0) \le 0.016154\,\theta^2$.

One doubling. Let $\hat N \ge 2$ be an integer. The first inequality of (ii) in the tube $(l',l';2\hat Nl')$, with $\mathsf{r}_{2\hat Nl'}(l',l') \le (\mathsf{k}_2\mathsf{t})^{2\hat N}$, gives $\zeta(l',l';2\hat Nl') \le \bar z_1 = \hat Nl'\varepsilon_{\rm v}(l') + \frac12(\mathsf{k}_2\mathsf{t})^{2\hat N}$; (i) with $t'u'/tu = 4$ and $\delta(l',l',2\hat Nl') \le 2\hat N\mathsf{k}_1\mathsf{t}$ gives $\zeta(2l',2l';2\hat Nl') \le \bar z_2 = \bar z_1^4 e^{8\hat N\mathsf{k}_1\mathsf{t}}$; and the second inequality of (ii) in the tube $(2l',2l';\hat Nl')$, with $\mathsf{G} \le \mathsf{r}$, $\mathsf{r}_{\hat Nl'}(2l',2l') \le \mathsf{r}_{2l'}(2l',2l')^{\hat N/2} \le (\mathsf{k}_2\mathsf{t}^2)^{\hat N/2}$ and $-\log(1-x) \le x/(1-x)$, gives, if $\bar z_2 < \frac12$,

$$2l'\varepsilon_{\rm v}(2l') \le \mathsf{U} := \frac{2}{\hat N}\Big\{(\mathsf{k}_2\mathsf{t}^2)^{\hat N/2} + \frac{2\bar z_2}{1-2\bar z_2}\Big\}.$$

The iteration. Take $\hat N_j = 4\cdot 3^j$ at $l' = l_j$, and let $\Lambda_0 = \log(1/0.8901) \approx 0.116421$. As $\hat N_j \ge 4\cdot 2^j$ and $2^{j+1}\hat N_j \ge 4^j$, every term of U at $\theta \le 0.8901$ is at most the term at $\theta = 0.8901$ times $e^{-4^{j+1}(2\Lambda-2\Lambda_0)}$, if the bounds on $l_j\varepsilon_{\rm v}(l_j)$ carry the factor $e^{-4^j(2\Lambda-2\Lambda_0)}$; so it is enough to follow numbers E_j with $l_j\varepsilon_{\rm v}(l_j) \le e^{-E_j-4^j(2\Lambda-2\Lambda_0)}$ at $\theta = 0.8901$, $\mathsf{t}_j = 0.8901^{2^{j+1}}$. The start gives such an $E_0 \ge \log(1/0.016154) + 2\Lambda_0 \ge 4.3414$; as U increases with $l'\varepsilon_{\rm v}(l')$, it is enough to take $E_0 = 4.3414$ and $E_{j+1} = -\log\mathsf{U}$ at E_j.

With $y_j = \hat N_j e^{-E_j}$ this reads $E_{j+1} = 4E_j - 3\log\hat N_j - 8\hat N_j\mathsf{k}_1\mathsf{t}_j - \log(2B_j)$, where $B_j = (\mathsf{k}_2\mathsf{t}_j^2)^{\hat N_j/2}y_j^{-4}e^{-8\hat N_j\mathsf{k}_1\mathsf{t}_j} + 2(\bar z_1/y_j)^4/(1-2\bar z_2)$; along this sequence B_j is at most 2.0920 at $j=0$, 2.0001 at $j=1$ and 2 to ten digits from $j=2$ on, $\bar z_2 \le 1.2\cdot 10^{-5}$, and the numbers $4^{-j}E_j - 2\Lambda_0$ are at least 4.1085, 2.6061, 1.9912, 1.7722, 1.7032, 1.6839, 1.6785 at $j = 0,\dots,6$. For $j \ge 6$, $8\hat N_j\mathsf{k}_1\mathsf{t}_j \le 0.000131$ and $B_j \le 2.0023$, so $E_{j+1} \ge 4E_j - 3\log\hat N_j - \log 4.01$, and summing the losses, $4^{-j}E_j \ge 4^{-6}E_6 - \sum_{i\ge6}4^{-i-1}(3\log 4 + 3i\log 3 + \log 4.01) \ge 1.909266$; so $4^{-j}E_j - 2\Lambda_0 \ge 1.6764 \ge 1.67$ for every $j \ge 0$, and $l_j\varepsilon_{\rm v}(l_j) \le e^{-(2\Lambda_0+1.67)4^j - (2\Lambda-2\Lambda_0)4^j} = e^{-(2\Lambda+1.67)4^j}$: this is (e).

From the tubes to the cubes. For an integer $N \ge 2\mathsf{q}l$ let $j \ge 0$ be the largest integer with $4^j \le N/2\mathsf{q}l$, $w = 2^j$, $l' = l_j = 2wl$ and $M = N/l'$, a real number; then $w\mathsf{q} \le M < 4w\mathsf{q}$ and $M^2 < 2\mathsf{q}N/l$. The first inequality of (ii) in the tube $(l',l';N)$, with $N\varepsilon_{\rm v}(l') = M\,l'\varepsilon_{\rm v}(l') \le Me^{-(2\Lambda+1.67)w^2}$ by (e) and $\mathsf{r}_N(l',l') \le (\mathsf{k}_2\mathsf{t})^M \le e^{-2\Lambda wM} \le e^{-(2\Lambda+1.67)w^2}$, as $M \ge w\mathsf{q}$ and $2\Lambda\mathsf{q} = 2\Lambda + 1.67$, gives $\zeta(l',l';N) \le \frac{M+1}{2}e^{-(2\Lambda+1.67)w^2} \le Me^{-(2\Lambda+1.67)w^2}$; and (i) with $t'u'/tu = M^2$ and $\delta(l',l',N) \le M\mathsf{k}_1\mathsf{t}$ gives $\frac12(1-r_1(N)) = \zeta(N,N;N) \le [Me^{-(2\Lambda+1.67)w^2}e^{M\mathsf{k}_1\mathsf{t}}]^{M^2}$.

Here $M^2(2\Lambda+1.67)w^2 = \sigma^+N^2$; $M\mathsf{k}_1\mathsf{t} < 4\mathsf{q}\mathsf{k}_1we^{-2\Lambda w} \le \frac{4\mathsf{q}\mathsf{k}_1}{2e\Lambda} = \frac{4\mathsf{k}_1}{e}\big(\frac{1}{2\Lambda} + \frac{1.67}{4\Lambda^2}\big) \le 0.85478$, the last member decreasing in $\Lambda \ge \Lambda_0$; and $0 \le \log M < \frac12\log\frac{2\mathsf{q}N}{l}$. So the exponent is at most

$-\sigma^+N^2 + M^2(\log M + 0.855) \le -\sigma^+N^2 + \frac{2\mathsf{q}N}{l}(\frac12 \log\frac{2\mathsf{q}N}{l} + 0.86)$: (a), the last display; the other two bounds of (a) are the same argument with the start at $64\,l$, where $D_2(64l) \le 0.12019\,\theta^{64}$, and Prop. 17.31(d) gives the seven classes. (iv) Two static charges at the ends o, r of the side of length R evolve by an operator $\hat{\mathsf{T}} = \mathsf{m}^{1/2}\hat{\mathsf{K}}\mathsf{m}^{1/2}$ on functions with values in the 2×2 matrices, $(\hat{\mathsf{K}}\Psi)(V) = \iint e^{-\beta A_\times(V,S,V')} S_o \Psi(V') S_r^{-1}\,\mathrm{d}S\,\mathrm{d}V'$, and

$$2Z[W_C] = \operatorname{Tr}((\mathsf{T}^{(0)})^{n_3-T}\Phi_\gamma^*\hat{\mathsf{T}}^T\Phi_\gamma) \ge 0, \qquad \operatorname{Tr}\hat{\mathsf{T}}^{n_3} = 4Z[\mathsf{PP}'],$$

$\Phi_\gamma\psi = \psi\,V(\gamma)$, P, P' the half-traces of the two straight closed lines through o and r; the positivity of $\hat{\mathsf{T}}$ is proved as that of K, with the crossing variables acting by a unitary representation. Hölder's inequality for the two spectral families gives $0 \le \langle W_C\rangle \le (4\langle\mathsf{PP}'\rangle)^{T/n_3}$, and (i) in the direction of R gives $\langle\mathsf{PP}'\rangle \le (1-\zeta)^{1-\varrho}\zeta^\varrho + \zeta^{1-\varrho}(1-\zeta)^\varrho \le 2\zeta^\varrho$, $\varrho = R/n_1 \le \frac12$: the bound on every box. With (a) on the cube n and $n\to\infty$ (Prop. 17.38(c)) the factor in front tends to 1: (b). (c) is the same argument with one charge for each passage of C through a layer: several static charges evolve by a positive operator on functions with values in $(\mathbb{C}^2)^{\otimes\mathcal{V}}$, and the projection of the path to the plane (13) is counted modulo 2 layer by layer; a passage from a downward to an upward bond costs a factor 2. (d) On every periodic box, for $T \le n_3 - 2$ and $R \le n_1 - 2$, the positivity of the operators of (iv) gives the log-convexity in T, and the reflection positivity across a plane between two charges gives it in R; these are inequalities between finitely many means of continuous local functions, and they pass to every weak limit of box measures whose sides tend to infinity.

If $w(R,T_0) = 0$ for one $T_0 \ge 1$, the log-convexity gives $w(R,T) = 0$ for all $T \ge 1$, and $V(R) = \infty$; else $T \mapsto w(R,T)^{1/T}$ does not decrease, and V is the infimum. The concavity holds in $[0,\infty]$, so V is infinite at every $R \ge 1$ as soon as it is infinite at one; with $V(R)/R \to \sigma_\infty$ it gives the slopes. Under the hypothesis $w(R,T) \le e^{-\sigma^+RT}$, so $V(R) \ge \sigma^+R$ for all R, and a concave function with $V(R) \ge \sigma^+R$ has every slope $\ge \sigma^+$. □

Lemma 17.40 (One step on a periodic box)**.** Let $\beta \ge 0$, Σ a cross-section, integers $\vartheta \ge 2$, $d \ge 3$, $N = 2d\vartheta$, $n = (d-1)\vartheta$, $\tau_s = \tau_0 + s\vartheta$; let Φ be a bounded complex Borel function on $\mathcal{Y}_\vartheta$, $\Phi^s = \Phi$ for odd s and $\Phi^s = \Phi^\Theta$ for even s, μ_N the measure of the box (N,Σ), $\varrho = \lambda_1/\lambda_0$, and

$$\mathsf{p} = \mu_N\Big(\prod_{s=1}^{2d}(\Phi^s)_{\tau_{s-1}}\Big), \qquad \mathsf{y} = [(1+\mathsf{r}_N)\,\mathsf{p}]^{1/2d}.$$

Then $\mathsf{p} \ge 0$, $\lambda_0^{-\vartheta}\|\mathfrak{Y}[\Phi]\| \le \mathsf{y}$, and

$$0 \le \mu_N(\Phi_{\tau_0}(\Phi^\Theta)_{\tau_d}) \le \big[(1+\mathsf{r}_N)\,|\mu_N(\Phi_{\tau_0})| + \mathsf{r}_{N-\vartheta}\,\mathsf{y}\big]^2 + (2+\mathsf{r}_n)\,\varrho^n\,\mathsf{y}^2.$$

■

Proof. By (M1), $Z(N,\Sigma)\mathsf{p} = \operatorname{Tr}(Y^*Y)^d = \|Y\|_{2d}^{2d} \ge \|Y\|^{2d}$, $Y = \mathfrak{Y}[\Phi]$, and $Z(N,\Sigma) = \lambda_0^N(1+\mathsf{r}_N)$: the bound on $\|Y\|$. The mean of the two cells d slabs apart is $\operatorname{Tr}(\mathsf{T}'^nY^*\mathsf{T}'^nY)/Z$; write $\mathsf{T}'^n = \lambda_0^n(|e_0\rangle\langle e_0| + \mathsf{R}_{(n)})$ with $\|\mathsf{R}_{(n)}\| = \varrho^n$ and $\operatorname{Tr}|\mathsf{R}_{(n)}| = \mathsf{r}_n$, and $\lambda_0^{-\vartheta}\langle e_0, Ye_0\rangle = (1+\mathsf{r}_N)\mu_N(\Phi_{\tau_0}) - \lambda_0^{-\vartheta}\operatorname{Tr}(\mathsf{R}_{(N-\vartheta)}Y)$ from the mean of one cell; expanding, the term $|\langle e_0, Ye_0\rangle|^2$ gives the square, and the others are bounded by ϱ^n and r_n times $\|Y\|^2$. □

Lemma 17.41 (Contraction of the seminorm)**.** Let $L' \ge 13$ be an odd integer, $k \in \mathbb{N}$, $\beta > 0$, on the lattice $L'^{-k}\mathbb{Z}^3$ with the sides $2L'^{\mathbb{N}}$. Let $\ell_f, \ell_c \in 2L'^{\mathbb{N}}$ with $d = \ell_c/\ell_f \ge L'$, $\mathsf{h} = \ell_f L'^k/2$ the number of layers of a fine cell, and let $\mathsf{e}(l) \le x_0$ at β with $24l \le \mathsf{h}$. Then for every real G of the bonds of the cell $[-\ell_f/4, \ell_f/4]^3$,

$$\big\|G - \mu_{k,\ell_c,\beta}(G)\big\|_{\mathrm{ch},\ell_c} \le 1.343\,\theta^{(d-1)\mathsf{h}/8l}\,\big\|G - \mu_{k,\ell_f,\beta}(G)\big\|_{\mathrm{ch},\ell_f},$$

where on the left G is regarded as a function of the cell $[-\ell_c/4, \ell_c/4]^3$, which holds the cell $[-\ell_f/4, \ell_f/4]^3$. The constant 1.343 depends on none of G, k, β, d, L'. ■

Proof. Let $X = G - \mu_{k,\ell_c,\beta}(G)$, $B_1 = X$, $B_{2p} = B_p \cdot \Theta_i B_p$ with Θ_i the reflection in the coarse wall plane of the direction i, and $m_p = \mu_{k,\ell_c,\beta}(B_p)$; then $m_1 = 0$, and m_8 is the eighth power of the left member. In the direction i, B_p is a function of one slab of h layers, and $\Theta_i B_p$ lies $n = (d-1)\mathsf{h}$ layers further, on the cube of $N = 2d\mathsf{h}$ sites. Lem. 17.40 with Y the slab operator of B_p, the gap of Prop. 17.33(a), and $\lambda_0^{-\mathsf{h}}\|Y\| \le \mathsf{c}\,u^p$, $\mathsf{c} = (1+\mathsf{r}_N)^{1/2d}$, $u = \|X\|_{\mathrm{ch},\ell_c}^{(\ell_f)}$, the seminorm of X on the torus ℓ_c for the fine board, of side $\ell_f/2$ (the array form of this board, applied to the pile of $2d$ copies of B_p, which are $2dp$ copies of X in cells of the fine board, with the factor 1 in the other cells, bounds the p of Lem. 17.40 by u^{2dp}), give

$$0 \le m_{2p} \le \left[(1+\mathsf{r}_N)\,m_p + \mathsf{r}_{N-\mathsf{h}}\,\mathsf{c}\,u^p\right]^2 + (2+\mathsf{r}_n)\,\theta^{n/l}\,\mathsf{c}^2 u^{2p}.$$

With the bounds of Prop. 17.32 on the r, three steps give $m_8 \le 3.68\,u^8\theta^{n/l}$. To bound u, put $G_0 = G - \mu_{k,\ell_f,\beta}(G)$ and $Q_f = e^{\delta((2\mathsf{h})^3)/8}$, $2\mathsf{h} = \ell_f L'^k$ being the number of sites of a side of the torus ℓ_f. The fine-to-coarse form gives $u \le Q_f\|X\|_{\mathrm{ch},\ell_f}$; as $X = G_0 - \mu_{k,\ell_c,\beta}(G_0)$ and $\|1\|_{\mathrm{ch},\ell_f} = 1$, $\|X\|_{\mathrm{ch},\ell_f} \le \|G_0\|_{\mathrm{ch},\ell_f} + |\mu_{k,\ell_c,\beta}(G_0)|$; and Prop. 17.34(c) at $\ell' = \ell'' = \ell_f$, with $\mathfrak{s} = \mu_{k,\ell_c,\beta}$ and $\mu_{k,\ell_f,\beta}(G_0) = 0$, gives $|\mu_{k,\ell_c,\beta}(G_0)| \le 4.84\,[Q_f(Q_f-1)]^{1/2}\|G_0\|_{\mathrm{ch},\ell_f}$. By Prop. 17.33(c) at $N = 2\mathsf{h} \ge 48l$, $\delta((2\mathsf{h})^3) \le 0.5892\,\theta^{48} \le 0.0022046$, so $Q_f \le 1.000276$ and $u \le Q_f(1 + 4.84\,[Q_f(Q_f-1)]^{1/2})\|G_0\|_{\mathrm{ch},\ell_f} \le 1.0807\,\|G_0\|_{\mathrm{ch},\ell_f}$. As m_8 is the eighth power of the left member and $n = (d-1)\mathsf{h}$, the left member is at most $3.68^{1/8}\cdot 1.0807\,\theta^{(d-1)\mathsf{h}/8l}\|G_0\|_{\mathrm{ch},\ell_f} \le 1.272\,\theta^{(d-1)\mathsf{h}/8l}\|G_0\|_{\mathrm{ch},\ell_f}$, and $1.272 \le 1.343$. □

Lemma 17.42 (A second blocking factor; dense labels)**.** (a) For every integer $L \ge 2$ there are an odd prime $L' \ge 13$ with $L' > L$ and an admissible choice of the constants with the blocking factor L'; for such L', $\alpha = \log L/\log L' \in (0,1)$ is irrational. (b) Let $\mathsf{b} > 0$, $k(K) = \lfloor K\alpha \rfloor$ and $\mathsf{c}_K = \mathsf{b}\,L'^{K\alpha - k(K)} \in [\mathsf{b}, L'\mathsf{b})$. Then $L^K\mathsf{b} = L'^{k(K)}\mathsf{c}_K$; k does not decrease, $k(K+1) - k(K) \in \{0,1\}$ and $k(K) \to \infty$, so every integer $k > k(K_1)$ is $k(K)$ for some $K > K_1$; and $\{\mathsf{c}_K : K \ge K_1\}$ is dense in $[\mathsf{b}, L'\mathsf{b}]$ for every K_1. ■

Proof. (a) is (M6) with the infinitude of primes; $L^q = L'^p$ is impossible for a prime $L' \nmid L$. (b) is arithmetic; the density is that of $\{K\alpha\}$ modulo 1 for irrational α. □

17.2.5. *Statements uniform in the cutoff.*

Proposition 17.43 (Products of action densities on one torus)**.** Let $\ell'' \in \mathfrak{L}$, $J' = [\mathsf{b}_-, \mathsf{b}_+] \subset (0,\infty)$ and $\mathsf{d} > 0$. For $k \ge 2$ let $\mathsf{B}_k(\ell'', J', \mathsf{d})$ be the bound of (M4)(i) on the k-th cumulants on the torus ℓ'' at the labels of J', for supports at pairwise sup distances $\ge \mathsf{d}$ and unit norms, and $\mathsf{M}_k = \sum_\pi \prod_{B\in\pi} \mathsf{B}_{|B|}$ over the partitions of $\{1,\dots,k\}$ into blocks of at least two members. Let $K \in \mathbb{N}$, $\mathsf{b}' \in J'$, $\beta = L^K\mathsf{b}'$, □ a cell of a board of side $\ell''/2$ with any offset, $f_1,\dots,f_r$ real Lipschitz functions, none 0, with supports in □ at sup distance $\ge \mathsf{d}/2$ from its walls and pairwise $\ge \mathsf{d}$ apart, and $X_i = \mathcal{O}_K(f_i) - \mu_{K,\ell'',\beta}(\mathcal{O}_K(f_i))$. Then for $\emptyset \ne I \subset \{1,\dots,r\}$, $X_I = \prod_{i\in I} X_i$ is a real function of the bonds of □ and

$$\|X_I\|_{\mathrm{ch},\ell''} \le \mathsf{M}_{8|I|}^{1/8} \prod_{i\in I} \|f_i\|_\sharp.$$

■

Proof. The walls of the board lie between lattice sites. Thus a plaquette whose centre lies strictly inside □ has all its corners in the closed cell; the supports lie inside the cell, so X_I depends only on its bonds. By the array form of (M2), $\|X_I\|_{\mathrm{ch},\ell''}^8$ is the mean on the torus ℓ'' of the product of the eight mirror copies of X_I; the mirror copy of $\mathcal{O}_K(f)$ is $\mathcal{O}_K(f\circ\hat{\mathsf{r}})$ for the mirror map $\hat{\mathsf{r}}$, with the same mean. So it is a moment of $8|I|$ centred fields with pairwise disjoint supports on one torus, at distances $\ge \mathsf{d}$ inside one cell and across the walls, hence a sum over the partitions into blocks of at least two members of products of joint cumulants; (M4)(i) bounds each by $\mathsf{B}_{|B|}$ times the norms. □

Proposition 17.44 (The vacuum constants of two tori; the cube number)**.**

(a) Let $\mathsf{b} \geq \gamma^{-2}$, $\ell \in \mathfrak{L}$ with $LM\,R(\mathsf{b}) \leq \ell/2$, and $K \geq 10$ with $L^{\lceil K/1260 \rceil}\mathsf{b} \geq \Theta_\star$, a number of the constants, so that the auxiliary sequence of renormalization steps is formed. At every level of the auxiliary sequence started from zero families with zero counterterms, the local numbers on the tori ℓ and $L\ell$ are equal on every region that lifts to both. Consequently

$$|L^3 \mathfrak{K}_K(\ell, \mathsf{b}) - \mathfrak{K}_K(L\ell, \mathsf{b})| \leq C\,(L\ell)^3 \exp\bigl[-\kappa_1 \ell L^{\lceil K/1260 \rceil}\bigr]$$

with numbers $C, \kappa_1 > 0$ of the constants.

(b) For every $\mathsf{b} > 0$, $\ell \in 2L^{\mathbb{Z}}$ and $K \geq K_\star(\mathsf{b}, \ell)$: $|X_{K+1}(\mathsf{b}, \ell) - X_K(\mathsf{b}, \ell)| \leq L^3 \mathfrak{o}_K(\ell/L) + \mathfrak{o}_K(\ell) + C\ell^3 e^{-\kappa_1 \ell L^{\lceil K/1260 \rceil - 1}}$ after the dilation of (c), and $\sum_{k \geq K} |X_{k+1} - X_k| \leq \eta_K(\mathsf{b}, \ell)$, with the η_K of Thm. BB(1).

(c) The dilation $(\mathsf{b}, \ell, K) \mapsto (L^z \mathsf{b}, L^z \ell, K - z)$ names the same lattices at the same couplings, so $X_K(\mathsf{b}, \ell) = X_{K-z}(L^z \mathsf{b}, L^z \ell)$, and it brings every label and side into the range of (a).

■

Proof. (a) The zero initial families and zero counterterms have a common infinite-lattice companion and satisfy the input size bound. The induction in Thm. YM3_I:Y, applied to the auxiliary sequence, therefore constructs the same family on each common lift. Its proof transports the entire local input through the chart at every step, including its domain. The floor $L^{\lceil K/1260 \rceil}\mathsf{b} \geq \Theta_\star$ is the coupling floor needed for this auxiliary run. The Gaussian numbers also agree under these charts, by the local normalizer construction in equations YM3_I:(1399)–(1400).

To estimate the contact terms, we keep the stopping depth in their vacuum constants. Put $u = \lceil K/1260 \rceil$, and mark the cutoffs $K^{\mathrm{A}} = K$ and $K^{\mathrm{B}} = K + 1$. The zero-source chain identity proved in Step 3(iv) of Prop. 13.7, or directly eq. 13.ii summed over its steps, gives

$$c_\lambda^\bullet(0) = E^\bullet_{(K^\bullet - u)}[T_\lambda], \qquad \mathfrak{K}_K(\lambda, \mathsf{b}) = c_\lambda^{\mathrm{B}}(0) - c_\lambda^{\mathrm{A}}(0).$$

Here $E^\bullet$ is built from the line $x_j^\bullet = L^{K^\bullet - j}\mathsf{b}$ on T_λ. These constants are real. The identity uses the actual chains; it does not require the error estimate comparing their masses. The line has zero increments and is admissible by the proof of Lem. YM3_I:118.1. The assumed room $LMR(\mathsf{b}) \leq \ell/2$ admits it on both tori. It also implies the side floor $\ell \geq C_L M$: the retained room condition [7, eq. 215] gives $C_L \leq 2LR(\gamma^{-2}) \leq 2LR(\mathsf{b})$.

Apply eq. 13.iii at each cutoff, with $n = K^\bullet - u$, to the sides ℓ and $L\ell$. Both lower summation indices are u. The triangle inequality gives

$$\left| \frac{\mathfrak{K}_K(\ell, \mathsf{b})}{\ell^3} - \frac{\mathfrak{K}_K(L\ell, \mathsf{b})}{(L\ell)^3} \right| \leq 4C_{\mathrm{tt}} E_0 \sum_{s \geq u} L^{3s} e^{-\kappa \ell L^s/(5M)} \leq 8C_{\mathrm{tt}} E_0 L^{3u} e^{-\kappa \ell L^u/(5M)}.$$

The last inequality is eq. 13.iv. Since $\kappa\ell/(5M) > 4$, the factor L^{3u} can be absorbed in half the exponent. For example, take

$$\kappa_1 = \frac{\kappa}{10M}, \qquad C = 8C_{\mathrm{tt}} E_0 \sup_{v \geq 1} v^3 e^{-2v} < \infty.$$

Multiplication by $(L\ell)^3$ proves (a).

(b) The zero-source read-out (M4)(ii) gives

$$|\log Z_{K+1,\ell} - \log Z_{K,\ell} - \mathfrak{K}_K(\ell, \mathsf{b})| \leq \mathfrak{o}_K(\ell, \mathsf{b}).$$

Apply this on the sides ℓ/L and ℓ, multiply the first identity by L^3, and subtract. The contact terms cancel up to the error in (a), giving the asserted bound for $X_{K+1} - X_K$. The two read-out errors have the summable rate in (M4)(ii). With $\varsigma = \sum_{j \geq 0} (1+j)^{\mathsf{p}_0} L^{-j/10}$, their tail is bounded by η_K once K is above the thresholds where (M4)(ii) and (a) apply and where the exponential term is at most $\mathsf{C}_{\mathrm{bf}} \ell^3 K^{\mathsf{p}_0} L^{-K/10}$. After the dilation in (c), these requirements are imposed on $K - z$ and

on both sides $L^{z-1}\ell$, $L^z\ell$, at label $L^z\mathsf{b}$, including their thresholds K_{bf}. This is the choice of $K_\star$ in Thm. BB(1).

(c) The dilation leaves both the bare lattice and its coupling unchanged. This proves the identity for X_K and permits the fixed dilation used in (b). □

Proposition 17.45 (Two cutoffs on boxes with mixed sides)**.** The following comparison remains a sketch at the full interacting Hessian estimate described with Thm. AH. Its intended conclusion is: for every compact set J of labels, every class $z \in \{0,1\}^3$, all sides $s_\mu \in L^{\mathbb{Z}} \cup 2L^{\mathbb{Z}}$ such that each twisted plane has a side in $L^{\mathbb{Z}}$, and every $l \in 2L^{\mathbb{Z}}$, there are C and K_* such that

$$\Big|\log Z^z_{K+1}(\underline{s};\mathsf{b}) - \log Z^z_K(\underline{s};\mathsf{b}) - \frac{|\underline{s}|}{l^3}(\log Z_{K+1,l} - \log Z_{K,l})\Big| \le C\,L^{-K/10} \qquad (\mathsf{b}\in J,\ K \ge K_*),$$

where $Z^z_K(\underline{s};\mathsf{b})$ is the partition function of the box with s_μ/a_K sites along the side μ and the twist z, at $\beta = L^K\mathsf{b}$, and $|\underline{s}| = s_1s_2s_3$. ■

Proof. The box $\underline{s}$ is mixed for z in the sense of Lem. 11.9. After a common dilation by a power of L, choose a periodic covering cube $\ell_C \in 2L^{\mathbb{Z}}$ for both this box and the reference cube l^3. The latter is mixed for class 0. Thm. AH(4), under **(Hess)**, gives the two comparisons with the same extensive term $\Delta_K[\ell_C,\mathsf{b}]$.

Subtract $|\underline{s}|/l^3$ times the comparison for l^3 from the comparison for $(\underline{s},z)$. The coefficients $|\underline{s}|/\ell_C^3$ of Δ_K cancel. The two errors are bounded by $C_1L^{-K/10}$ and $(|\underline{s}|/l^3)C_2L^{-K/10}$, uniformly on J, after increasing the common threshold. This proves the stated estimate.

The errors are $\le CL^{-K/10}$. □

Lemma 17.46 (A summable bound closes the criterion)**.** Let $(\mathsf{e}_K)_{K\ge K_*}$ be real numbers with $|\mathsf{e}_{K+1}-\mathsf{e}_K| \le \mathsf{q}_K$ and $Q_K = \sum_{k\ge K}\mathsf{q}_k < \infty$. Then (a) $\mathsf{e}_\infty = \lim \mathsf{e}_K$ exists and $|\mathsf{e}_K - \mathsf{e}_\infty| \le Q_K$; (b) if $\mathsf{e}_{K_1} \le x_0 - Q_{K_1}$ at one $K_1 \ge K_*$, then $\mathsf{e}_K \le x_0$ for all $K \ge K_1$; (c) if $\mathsf{e}_K \le x_0' < x_0$ for all large K, then $\mathsf{e}_{K_1} \le x_0 - Q_{K_1}$ at every large K_1; (d) $\mathsf{e}_\infty < x_0$ if and only if $\mathsf{e}_{K_1} < x_0 - Q_{K_1}$ at one K_1, and $\mathsf{e}_\infty > x_0$ if and only if $\mathsf{e}_{K_1} > x_0 + Q_{K_1}$ at one K_1, and then $\mathsf{e}_K > x_0$ for all $K \ge K_1$; at $\mathsf{e}_\infty = x_0$ neither holds. ■

Proof. $|\mathsf{e}_K - \mathsf{e}_{K'}| \le Q_K$ for $K' \ge K$, and $Q_K \to 0$. □

Proposition 17.47 (Small tori)**.** For an integer $N \ge 3$ let $\lambda_N(k,u) = \sum_\mu(2-2\cos\frac{2\pi k_\mu+u_\mu}{N})$ for $k \in (\mathbb{Z}/N)^3$, $u \in \mathbb{R}^3$; $\mathrm{Det}'(N) = \prod_{k\ne0}\lambda_N(k,0)$; $\Upsilon_N(u) = N^2\prod_k \lambda_N(k,u)/\mathrm{Det}'(N)$; $\mathscr{J}_N = \int_{[-\pi,\pi]^3} du/\Upsilon_N(u)$; $f_\Delta = (2\pi)^{-3}\int_{[-\pi,\pi]^3}\log\sum_\mu(2-2\cos p_\mu)\,\mathrm{d}p$; and $c'_N = \log(\mathscr{J}_N/8\pi^2) - \frac{3}{2}(\log\mathrm{Det}'(N) - f_\Delta N^3 - 2\log N)$.

(a) There are absolute numbers $c_3 > 0$ and $x_* \in (0,\frac12]$ with the following two properties. On one bare periodic cube: for every integer $N \ge 3$ and every real $\beta \ge 3$ with $\mathsf{er}_N = c_3(N^{66}\log\beta/\beta)^{1/8} \le x_*$,

$$\Big|\frac{\beta^{3N^3-1}Z(N^3)}{\mathcal{Z}(N)} - 1\Big| \le \mathsf{er}_N, \qquad \mathcal{Z}(N) = \frac{(2\pi)^{3N^3-1}\,N^2\,\mathscr{J}_N}{(2\pi^2)^{2N^3}\,4\pi\,\mathrm{Det}'(N)^{3/2}}, \qquad \log\mathcal{Z}(N) = N^3z_\infty - \log N + c'_N,$$

$Z(N^3)$ the partition function at β and z_∞ the number of (d). Consequently, for all integers $N \ge 3$, $q \ge 2$ and real $\beta \ge 3$ with $\mathsf{er} = c_3((qN)^{66}\log\beta/\beta)^{1/8} \le x_*$:

$$|D_q(N) - (q^3-1)\log(\beta/N) - q^3c'_N + c'_{qN} - \log q| \le 2(q^3+1)\,\mathsf{er}.$$

(b) There are numbers $t_0 \in (0,\frac{1}{36}]$ and C_B of the constants such that for every integer $i \ge 1$, $N = 2L^i$, and every real β with $t = N/\beta \le t_0$: $|D_L(N) - (L^3-1)\log(1/t) - L^3c'_N + c'_{LN} - \log L| \le C_Bt^{1/1500}$.

(c) With $t_1 = t_0^2$ and a number C_C of the constants: $|\delta(N^3) - \log(1/t) - c'_N| \le C_C t^{1/1500}$ for $N = 2L^i$, $t = N/\beta \le t_1$. Moreover $c'_N \to c'$ and $|c'_N - c'| \le C_A N^{-2} (\log N)^{5/2}$ with an absolute C_A.

(d) With $z_\infty = \log \frac{2}{\pi} - \frac{3}{2} f_\Delta \approx -2.96167$ there are numbers β_1, C of the constants with $|P(\beta) + 3\log\beta - z_\infty| \le C\beta^{-1/1500}$ for every real $\beta \ge \beta_1$.

■

Proof. (a) *The value.* As $\beta \to \infty$ the measure of the bare cube concentrates on the flat fields. Up to gauge these are the commuting triples of holonomies; at a generic triple the orbit is that of a maximal torus, parametrised by $u \in [0, 2\pi]^3$, and the transverse fluctuations are Gaussian with the quadratic form of the lattice Laplacian twisted by u, whose lowest eigenvalue $\lambda_N(0, u)$ is of the order $|u|^2/N^2$. The volume of the orbits, the Haar measure near them and the Gaussian integral give $\beta^{3N^3-1} Z(N^3) \to \mathcal{Z}(N)$ and $\log \mathcal{Z}(N) = (3N^3 - 1)\log 2\pi - 2N^3 \log 2\pi^2 - \log 4\pi - \frac{3}{2}\log \mathrm{Det}'(N) + 2\log N + \log \mathscr{J}_N = N^3 z_\infty - \log N + c'_N$, by $3\log 2\pi - 2\log 2\pi^2 = \log \frac{2}{\pi}$, $\log 2\pi + \log 4\pi = \log 8\pi^2$ and the definition of c'_N.

The error, with all its powers of N. (1) *The cut.* In the gauge in which the bonds of a maximal tree carry $\mathbf{1}$, which removes the gauge group exactly, with no gauge copies and no Jacobian, the field is bounded by its curvature: with $A = \sum_p a_p$, every other bond variable lies within $2NA^{1/2}$ of $\mathbf{1}$ or, at a bond that crosses the boundary of the fundamental cube, of the holonomy h_μ of its direction, and $|h_\mu h_\nu h_\mu^{-1} h_\nu^{-1} - \mathbf{1}| \le N(2A)^{1/2}$. The fields with $A \ge \alpha_\beta = 4N^3 \log\beta/\beta$ carry at most β^{-4N^3}, which is $(C/\beta)^{N^3}$ times $\mathcal{Z}(N)\beta^{1-3N^3}$: this cut is the source of $\log\beta$. (2) *The generic stratum.* On $\{A < \alpha_\beta\}$ the integral is squeezed between two Gaussian integrals: in $d_\nu \le 6N^3$ transverse variables ν with $\|\nu\|^2 \le \Lambda_\delta A$ and $|A - Q| \le 43N^2 |\nu|_\infty \|\nu\|^2$, Q the quadratic part of A and $43N^2$ the bound of the cubic remainder of the plaquettes, the integral of $e^{-\beta A}$ over $\{A < \alpha_\beta\}$ lies between $e^{\mp(d_\nu \epsilon + C\rho_\delta)}$ times the Gaussian integral of Q, $\rho_\delta = (\Lambda_\delta \alpha_\beta)^{1/2}$, $\epsilon = 43N^2 \Lambda_\delta \rho_\delta$; the Gaussian determinants need not be computed, as the limit $\mathcal{Z}(N)$ names them.

Where some holonomy has $1 - (\frac{1}{2}\operatorname{tr} h_\mu)^2 \ge \delta^2$, the base is the commuting triple, there are $6N^3 - 2$ transverse variables, and the softest of them costs of the order δ^2/N^2: $\Lambda_\delta \le 2\pi^2 N^5 + \frac{\pi^2}{4} N^2 \delta^{-2}$. (3) *The eight points* at which every h_μ is $\pm\mathbf{1}$: there the stabiliser is all of SU(2), the nine constant modes $c = (c_\mu)$ have the quartic action $N^3 |c \wedge c|^2$, $|c \wedge c|^2 = \sum_{\mu<\nu} |c_\mu \times c_\nu|^2$, and the Gaussian picture fails for $|u| \lesssim (N/\beta)^{1/4}$. An upper bound suffices there. In Landau's gauge the action is at least $\frac{1}{2}(1 - \epsilon')\|d_1 Y'\|^2 + N^3 |c \wedge c|^2$ for the non-constant part Y' of the field, d_1 the lattice exterior derivative of 1-forms, the Jacobian is at most $(\mathrm{Det}'(N)/N^3)^{3/2}(1 + 7N^2 r_\delta)^{3N^3}$ (Kirchhoff's theorem), $r_\delta^2 = 6\pi^2 N^2 \delta^2 + 24 N^5 \alpha_\beta$, and the integral over the constant modes is at most $48\pi^3 r_\delta N^{-3/2} (\beta N^3)^{-2}$; measured against $\mathcal{Z}(N)\beta^{1-3N^3}$, in which $\mathscr{J}_N = \int \mathrm{d}u / \Upsilon_N(u)$ lies between two absolute numbers, as $c_0 |u|^2 \le \Upsilon_N(u) \le c_0^{-1}$ with an absolute $c_0 > 0$ and $|u|^{-2}$ is integrable in three dimensions, the weight of these fields is at most $C\, N^{-1/2} r_\delta\, e^{C N^5 r_\delta}$. (4) *The balance.* The error of (2) is of the order of $N^8 \delta^{-3} (N^3 \log\beta/\beta)^{1/2}$, that of (3) of the order of $N^{1/2}\delta$; with $\delta = N^{-6}(N^{66} \log\beta/\beta)^{1/8}$ both, and the cut, are at most an absolute multiple of $(N^{66}\log\beta/\beta)^{1/8}$: the exponent $\frac{1}{8}$ is the balance of $\delta^{-3}\beta^{-1/2}$ against δ, all the dependence on N sits in N^{66}, and c_3, x_* are absolute.

The consequence. With $|\log(1 + x)| \le 2|x|$ for $|x| \le \frac{1}{2}$, the bound at N and at qN (the condition at qN implies the one at N) gives $\log Z(N^3) = -(3N^3 - 1)\log\beta + \log \mathcal{Z}(N) + O(2\mathsf{er})$ and the same at qN; as $-q^3(3N^3 - 1) + (3q^3 N^3 - 1) = q^3 - 1$, the terms in N^3 cancel in $D_q(N) = q^3 F(N^3) - F((qN)^3)$, and $-q^3 \log N + \log(qN) = -(q^3 - 1)\log N + \log q$. (b) Let $t \le t_0$. *The frame.* Let j_t be the least integer $j \ge 1$ with $L^j \ge LM\, R(2L^j/t)$. As $R(Lx) \le L\, R(x)$ once $\log x \ge r$, the exponent of $R(2L^j/t)$ grows by at most one when j grows by one, so equality holds at j_t: $\ell' = 2L^{j_t}$, $\mathsf{b}' = \ell'/t$ and $y' = \log \mathsf{b}'$ satisfy $\ell' = 2LM\, R(\mathsf{b}')$, which is L times the least side at which (M4)(ii), (iii) apply at the label b', and $2LMy'^r \le \ell' < 2L^2 M y'^r$, $\log\frac{1}{t} < y' < 2\log\frac{1}{t}$ for small t.

Every pair (N,β) with $N=2L^i$ and $N/\beta=t$ is the pair $(\ell'/a_K, L^K\mathsf{b}')$ of the cutoff $K=i-j_t$, of either sign, on the line of b' (Def. 17.28). Write $D_K=D_L(\ell'/a_K)$ at $\beta=L^K\mathsf{b}'$, which is $X_K(\mathsf{b}',L\ell')$. *Two cutoffs, uniformly in the label.* The number $\mathfrak{o}_K$ of (M4)(ii) is $8\mathsf{R}_{t_1}+12\epsilon_{\rm fg}+8\epsilon_{\rm cl}+12\epsilon_{\rm sh}$ ((2.xii)), with the explicit numbers of §2.3.1 at $n=1$: with $y=\log\mathsf{b}$, $u=\lceil K/1260\rceil$, $s=\lceil K/9\rceil$ and the top level $t_1=K-u$ of (M4)(ii) in R_{t_1}, $\epsilon_{\rm fg}\le c\,\ell^3L^{-34K/315}$ and $\epsilon_{\rm cl}\le c\,(y+K\log L)^{2\mathfrak{p}_0}K^{1/2}\ell^3L^{-K/9}$, while R_{t_1} carries the factor $e^{-c'A_0^2(y+u\log L)^{2\mathfrak{p}_0}}$ against $e^{(C+\frac92 y)\ell^3}$, and $\epsilon_{\rm sh}$ the factor $e^{-\frac14\kappa_{\mathfrak{b}}A_1^2(y+u\log L)^{2\mathfrak{p}_0}}$.

On the sides $2MR(\mathsf{b})\le\ell\le 2L^3My^r$ one has $\ell^3\le 8L^9M^3y^{3r}$ and $y^{2\mathfrak{p}_0}\le(\ell/2M)^{2\mathfrak{p}_0/r}$, and the condition $2\mathfrak{p}_0>3r+2$ of the list of (H3) lets $e^{-c'A_0^2y^{2\mathfrak{p}_0}}$ beat $e^{(C+\frac92 y)\ell^3}$. So there are numbers $y_\sharp$, $C_{\mathfrak{o}}$ of the constants such that at every label with $y\ge y_\sharp$, on every such side ℓ and at every $K\ge 11(5+2\mathfrak{p}_0/r)\log_L\ell$, the conditions (k0)–(k4), (k6) of §2.3.1 hold and $\mathfrak{o}_K(\ell)\le C_{\mathfrak{o}}\,\ell^{3+2\mathfrak{p}_0/r}L^{-K/11}$, uniformly in the label; this is the form needed here, as $\mathsf{b}'\to\infty$ when $t\to0$, while the constant $\mathsf{C}_{\rm bf}$ of (M4)(ii) is formed on bounded sets of labels.

The two sides ℓ', $L\ell'$ lie in this range, and (M4)(ii) on both, with Prop. 17.44(a), gives

$$|D_{K+1}-D_K|\le L^3\mathfrak{o}_K(\ell')+\mathfrak{o}_K(L\ell')+C\ell'^3L^{-K}\le C\,\ell'^{\,3+2\mathfrak{p}_0/r}L^{-K/11}\qquad(K\ge 11(5+2\mathfrak{p}_0/r)\log_L(L\ell')).$$

The base. Let $K_\circ$ be the least integer with $N_\circ=\ell'L^{K_\circ}\ge t^{-1/130}$, so that $t^{-1/130}\le N_\circ<Lt^{-1/130}$; as ℓ' is a power of $\log\frac1t$, $K_\circ$ lies above the bound on K of the display once $t\le t_0$. Let $N=2L^i$ and $\beta=N/t$. (1) If $N\le N_\circ$, then $(LN)^{66}\log\beta/\beta=L^{66}N^{65}t\log\frac{N}{t}\le L^{131}t^{1/2}(\log L+\frac{131}{130}\log\frac1t)$, and (a) at $q=L$ applies for $t\le t_0$ and bounds the left member of (b) by $2(L^3+1)c_3L^{17}t^{1/16}(\log L+\frac{131}{130}\log\frac1t)^{1/8}$. (2) If $N>N_\circ$, then $K=i-j_t>K_\circ$, and with $\mathsf{k}_M=L^3c'_M-c'_{LM}+\log L$,

$$D_L(N)-(L^3-1)\log\tfrac1t-\mathsf{k}_N=[D_K-D_{K_\circ}]+[D_L(N_\circ)-(L^3-1)\log\tfrac1t-\mathsf{k}_{N_\circ}]+[\mathsf{k}_{N_\circ}-\mathsf{k}_N]:$$

the first bracket is at most $C(1-L^{-1/11})^{-1}\ell'^{\,3+2\mathfrak{p}_0/r+1/11}t^{1/1430}$ by the display, as $L^{-K_\circ}\le\ell't^{1/130}$; the second is case (1) at $N_\circ$; the third is at most $2(L^3+1)C_AN_\circ^{-2}(\log N_\circ)^{5/2}\le 2(L^3+1)C_At^{1/65}(\log L+\frac{1}{130}\log\frac1t)^{5/2}$ by the rate of (c), whose proof does not use (b). The exponents $\frac{1}{16}$, $\frac{1}{1430}$, $\frac{1}{65}$ exceed $\frac{1}{1500}$, and the powers of $\log\frac1t$ are absorbed into C_B; t_0 is fixed by $\log\frac{1}{t_0}\ge y_\sharp$, by the place of $K_\circ$ and by the condition of (a) in case (1). (c) By Prop. 17.31(c) and $\delta(M^3)\le 3(M/N)^2\delta(N^3)$, $(1-3L^{-S-1})\delta(N^3)\le\sum_{s=0}^S L^{-3(s+1)}D_L(L^sN)\le\delta(N^3)$, and with $Y(M)=\log(\beta/M)+c'_M$ the main terms of (b) telescope:

$$\sum_{s=0}^{S}L^{-3(s+1)}\big[(L^3-1)\log(\beta/L^sN)+L^3c'_{L^sN}-c'_{L^{s+1}N}+\log L\big]=Y(N)-L^{-3(S+1)}Y(L^{S+1}N);$$

choosing L^S of the order $t^{-1/2}$ gives (c). The limit c' and the rate are those of Riemann sums for lattice determinants, written through heat kernels: $\log\mathrm{Det}'(N)$, $f_\Delta N^3$ and $2\log N$ are integrals $\int_0^\infty[\cdots]\frac{d\tau}{\tau}$ of one-dimensional heat kernels, which converge to those of $\vartheta(\tau,\cdot)$. (d) At $N=2L$ and $\beta\ge 2L/t_1$, (a) gives $F(N^3)=-(3N^3-1)\log\beta+N^3z_\infty-\log N+c'_N+O((\log\beta/\beta)^{1/8})$ and (c) gives $\delta(N^3)=\log(\beta/N)+c'_N+O(\beta^{-1/1500})$; the difference is $N^3P=-3N^3\log\beta+N^3z_\infty+O(\beta^{-1/1500})$. □

Proposition 17.48 (The flux quotient of small cubes at every cutoff)**.**

(a) There are numbers $t_2>0$, $C_{\mathfrak{b}}$ of the constants, and $\mathsf{a}=\frac92 r+\mathfrak{p}_0$, such that for every integer $j\ge1$ and every real β with $t=2L^j/\beta\le t_2$: $r_1(2L^j)\le\eta_{\mathfrak{b}}(t)=C_{\mathfrak{b}}(\log\frac1t)^{\mathsf{a}}t^{1/500}$ at β. Equivalently: for every side $\ell=2L^m$, $m\in\mathbb{Z}$, every real label b with $\ell/\mathsf{b}\le t_2$, and every cutoff $K\ge1-m$.

(b) To obtain the following sharper bound, one must establish the **(Heaviest class)** probability estimate. The intended conclusion is that there are numbers $t_4>0$, C of the constants, and $\mathsf{q}=1116\,r+720\,\mathfrak{p}_0+366$, such that for every integer $j\ge1$ and every real β with $t=2L^j/\beta\le t_4$, over the cross-section (N,N), $N=2L^j$: $\lambda_0^N/Z(N^3)\le C\,t\,(\log\frac1t)^{\mathsf{q}}$. ■

Proof. (a) *On one lattice.* (i) Multiplying by -1 the bonds of a sheet (a half-plane of bonds bounded by two columns) moves a twisted column; so Z^z does not depend on the place of its column, and reflection positivity across two layers with the column in the cut gives $0 \le Z^z[\chi\chi'] \le Z[\chi]^{1/2} Z[\chi']^{1/2}$ for $[0,1]$-valued χ, χ' of the two halves. On configurations whose block plaquettes at a level q (or the bare plaquettes) lie within ω of $\mathbf{1}$, a twist forces the holonomies P_1, P_3 of the block field around the twisted plane to anticommute up to $\varpi = (1+\omega)^{\mathsf{n}^2} - 1 \le \frac{1}{20}$, n the number of block sites a side; hence $|W_1| \le \varpi/2$ and $W_2^2 \ge 1 - 0.52\,\varpi^2$ for the half-traces $W_\mu(y)$ of the straight closed lines of the block field through a site, and

$$r_1(N) \le \max_y \mu_{N^3}(\mathcal{C}^\varpi(y)) + 3\mathsf{n}^3 \mathsf{R}^{1/2}, \qquad \mathcal{C}^\varpi(y) = \{|W_1(y)| \le \tfrac{\varpi}{2},\ W_2(y)^2 \ge 1 - 0.52\,\varpi^2\},$$

R the largest probability of a rough block plaquette. (ii) On a bare cube small against β: with $\Xi_\beta = 3N^3 \log(8e^8\beta^{3/2}) + 2\log(3N^3\beta)$ and $\omega_0 = (2\Xi_\beta/\beta)^{1/2}$, $Z \ge (8e^8\beta^{3/2})^{-3N^3}$, so a plaquette is rough, $|\partial U(p) - \mathbf{1}| \ge \omega_0$, with probability $\le (3N^3\beta)^{-2}$. Then the angle of P_1 is shifted along the seam: with $P_1(y_0) = e^{i\alpha_0 \hat n \cdot \sigma}$ at the site y_0 of the event, $0 < \alpha_0 < \pi$, σ the Pauli matrices, $\mathsf{N}(x)$ the matrix $\hat n \cdot \sigma$ carried to x along a path from y_0 that uses no seam bond $\langle x, x + e_k\rangle$, $x_k = 0$, $k = 1, 2$, and $V(b) \mapsto V(b) e^{-i s_k \mathsf{N}(b_+)}$ on the seam bonds of the direction k, this map, defined for $|s|_\infty \le \frac{1}{10}$, commutes with gauge transformations, turns $P_1(y_0)$ by $-s_1$ and $P_2(y_0)$ by $-s_2$ about $\hat n$, changes the action by at most $131 N^4 \beta \omega_0^2 |s|_\infty$ on the fields whose plaquettes all lie within ω_0 of $\mathbf{1}$, and changes the Haar measure by the factor $\sin^2(\alpha_0 - s_1)/\sin^2\alpha_0$.

Let D be the event $\mathcal{C}^\varpi(y_0)$ intersected with that of no rough plaquette. For $s \in 2\varpi\mathbb{Z}^2$ with $|s|_\infty \le s_0 = (131 N^4 \beta \omega_0^2)^{-1}$, which is $< \frac{1}{10}$, the action rises by at most 1, so the image of D has the measure $\ge e^{-1} \cos^2(0.13)\, \mu_{N^3}(D) \ge 0.36\, \mu_{N^3}(D)$; on D the angles of $P_1(y_0)$ and $P_2(y_0)$ about $\hat n$ fix s up to 1.02ϖ and 1.46ϖ, so the images of distinct points of the grid are disjoint, and there are at least $(s_0/2\varpi)^2$ of them. Hence $\mu_{N^3}(D) \le \frac{4}{0.36} \varpi^2 (131 N^4 \beta \omega_0^2)^2$, which is of the order $N^{12} \Xi_\beta^3/\beta$, as $\beta\omega_0^2 = 2\Xi_\beta$ and $\varpi \le 2.1 N^2 \omega_0$; with the rough plaquettes, $\mu_{N^3}(\mathcal{C}^\varpi) \le [6.8 \cdot 10^6 N^{12} \Xi_\beta^3 + 1]\beta^{-1}$, for the block field of a bare lattice that is not too large against β as well, by (M4)(v). (iii) *Through the cutoffs.* Write $2L^j = \ell L^K$ with $\ell = 2L^m$ the least side at which (M4) applies at the label $\mathsf{b} = \ell/t$, $y = \log \mathsf{b}$, $\ell \le 2L^2 M y^r$.

At the cutoffs $K \le K_1 = \lceil y/(25 \log L)\rceil$ the lattice is a bare cube small against β, and (ii) applies. Above K_1 we use (M4)(iii) at the depth $d = 0$: at the cutoff K the block field is $\mathcal{M}^K U$, on the unit lattice of the torus ℓ, with ℓ sites a side at every cutoff; the roughness ω and the width ϖ are chosen at K_1 and do not depend on K. The events $\mathcal{C}^\varpi(y')$ of this block field and the events that one of its plaquettes lies at a distance $\ge \omega$ from $\mathbf{1}$ are closed and gauge invariant, and the same at all cutoffs; so by (M4)(iii) their probabilities move by at most $\mathfrak{o}_K$ from one cutoff to the next, and $\sum_{K \ge K_1} \mathfrak{o}_K \le Q(y) e^{-y/250}$, Q a polynomial formed from the constants; (i) at the cutoff K then gives $r_1 \le 4\bar\ell(y)^3 Q(y)^{1/2} e^{-y/500}$, $\bar\ell(y) = 2L^2 M y^r$, which is $\le C_\flat (\log \frac{1}{t})^{\mathsf{a}} t^{1/500}$. (b) The proposed sharper estimate still requires **(Heaviest class)**; the remaining reduction uses another label of a layer and a multiplicative comparison through the cutoffs.

The top level against the heaviest cell of a label. Let Φ map the configurations on the bonds of the directions $1, 2$ of one layer into a set X, let $\mathcal{P}$ be a finite partition of X into cells whose preimages are Borel, $\sim$ a reflexive and symmetric relation on $\mathcal{P}$ with at most ν neighbours to each cell, and ι_k the cell of the label of the layer $x_3 = k$. Then over the cross-section (N, N) of an even cube $N = 2h$

$$e^{-\mathsf{G}_N} = \frac{\lambda_0^N}{Z(N^3)} \le \big(\nu^2 \sqrt{p^\sharp} + \nu\sqrt{\varepsilon_2} + \sqrt{\varepsilon_1}\big)^2 \le 2\nu^4 (p^\sharp + \max\{\varepsilon_1, \varepsilon_2\}) \quad (\nu \ge 2),$$

$p^\sharp = \max_I \mu_{N^3}(\iota_1 = I)$, $\varepsilon_1 = \mu_{N^3}(\iota_1 \not\sim \iota_{\mathsf{n}/2-1})$, $\varepsilon_2 = \mu_{N^3}(\iota_{\mathsf{n}/2-1} \not\sim \iota_{\mathsf{n}/2+1})$, in two cases: ($\alpha$) the label of a bare layer, $\mathsf{n} = N$; (β) the label of a layer of the block field $\mathcal{M}^q U$ of the bare cube, with $\mathsf{n} \ge 8$ block sites a side. Indeed, let the first slab be the bare layers $1, \dots, s$, with $s = 2$ in (α), $s = \frac{1}{2}(3L^q - 1)$ in (β), and $2s < h$, which in (β) follows from $\mathsf{n} \ge 8$. The label ι_1 is a Borel function

of the bonds with both ends in the first slab: in (α) it reads the layer $x_3 = 1$ alone, and in (β), by the locality of the averaging map (Prop. YM3_I:4.5), only the bare bonds of the blocks of the block layer 1, which, centred at the height L^q, lie in the bare layers $\frac{1}{2}(L^q+1), \dots, \frac{1}{2}(3L^q-1)$.

The reflection in the plane $x_3 = \frac{h}{2}$, the translation by he_3 and the reflection in the plane $x_3 = 0$ carry the first slab onto the layers $h-s, \dots, h-1$, the layers $h+1, \dots, h+s$ and the layers $N-s, \dots, N-1$; the layers 0 and h lie in none of these four slabs. The three maps are isometries of the bare lattice and, in (β), as $h = \frac{\mathsf{n}}{2}L^q$ is a multiple of L^q, of the block lattices, and the averaging maps commute with them ([7, Lem. G.1]); so the label ι_1, read on the configuration moved by them, is $\iota_{\mathsf{n}/2-1}$, $\iota_{\mathsf{n}/2+1}$, $\iota_{\mathsf{n}-1}$ respectively. Let χ_I be the indicator function of the label I on the first slab, $\mathfrak{Y}[\chi_I]$ its slab operator of (M1), and $Y_I = \mathfrak{Y}[\chi_I]\,\mathsf{T}'^{1/2}$, the operator $\mathsf{T}' = \mathsf{K}^{1/2}\mathsf{m}\,\mathsf{K}^{1/2}$ being ≥ 0; then $\sum_I Y_I = \mathsf{T}'^{s+1/2}$, $R = \mathsf{T}'^{h-2s-1} \geq 0$ and $\mathsf{T}'^{h} = \sum_{I,J} Y_J^* R Y_I$.

The conjugated mirror image χ_J^Θ of (M1), placed on the layers $h-s, \dots, h-1$, is χ_J read on the configuration mirrored in the plane $x_3 = \frac{h}{2}$ midway between these layers and the first slab, that is the indicator of $\iota_{\mathsf{n}/2-1} = J$; placed on the layers $N-s, \dots, N-1$, it is χ_J read on the configuration mirrored in the plane $x_3 = h$, which on the box of $N = 2h$ layers is the reflection in the plane $x_3 = 0$, that is the indicator of $\iota_{\mathsf{n}-1} = J$; and $\chi_{I'}$ placed on the layers $h+1, \dots, h+s$ is the indicator of $\iota_{\mathsf{n}/2+1} = I'$. So the trace formula of (M1), in which the first slab stands rightmost, for the functions $\chi_I^{+(h-2s-1)}$, $(\chi_J^\Theta)^{+1}$, $\chi_{I'}^{+(h-2s-1)}$, $(\chi_{J'}^\Theta)^{+1}$ on the consecutive slabs of $h-s-1$, $s+1$, $h-s-1$, $s+1$ layers from the layer 1 on, with $\mathfrak{Y}[\Phi^{+j}] = \mathsf{T}'^{j}\mathfrak{Y}[\Phi]$ (the j added layers lying above those of Φ), $\mathfrak{Y}[\chi_J^\Theta] = \mathfrak{Y}[\chi_J]^*$ and the cyclicity of the trace, gives $\mathrm{Tr}(Y_{J'}^* R Y_{I'}\, Y_J^* R Y_I) = \mathrm{Tr}(\mathsf{T}'\mathfrak{Y}[\chi_{J'}^\Theta]\, R\, \mathfrak{Y}[\chi_{I'}]\, \mathsf{T}'\mathfrak{Y}[\chi_J^\Theta]\, R\, \mathfrak{Y}[\chi_I]) = Z(N^3)\,\mu_{N^3}(\iota_1 = I, \iota_{\mathsf{n}/2-1} = J, \iota_{\mathsf{n}/2+1} = I', \iota_{\mathsf{n}-1} = J')$: probabilities of one and the same cube.

The terms with $I \not\sim J$ have together the Hilbert–Schmidt norm $\leq (Z\varepsilon_1)^{1/2}$. The terms with $I \sim J$ are $\mathbb{X}\mathbb{B}\mathbb{X}^*$, with $\mathbb{X}(\varphi_I)_I = \sum_I Y_I^* R^{1/2}\varphi_I$ and the neighbour matrix $\mathbb{B}$ of norm $\leq \nu$ (Schur's test); $\mathbb{X}^*\mathbb{X}$ is the matrix of the $R^{1/2}Y_I Y_J^* R^{1/2}$, whose Hilbert–Schmidt norms are $(Z\mu_{N^3}(\iota_1 = \iota_{\mathsf{n}/2-1} = I,\ \iota_{\mathsf{n}/2+1} = \iota_{\mathsf{n}-1} = J))^{1/2}$, so Schur's test on the neighbouring entries and the Hilbert–Schmidt norm of the others give $\|\mathbb{X}^*\mathbb{X}\| \leq \nu(Zp^\sharp)^{1/2} + (Z\varepsilon_2)^{1/2}$. Hence $\lambda_0^h = \|\mathsf{T}'^h\| \leq Z^{1/2}[\nu(\nu\sqrt{p^\sharp} + \sqrt{\varepsilon_2}) + \sqrt{\varepsilon_1}]$.

The label: the class of the pair of holonomies. For pairs $P = (P_1, P_2)$, Q in $\mathrm{SU}(2)^2$ let $\mathsf{D}(P,Q) = \min_g \max_{i=1,2} |P_i - gQ_ig^{-1}|$, a distance between the classes under simultaneous conjugation, $\mathcal{O}_w(Q) = \{P : \mathsf{D}(P,Q) \leq w\}$, and $\mathsf{v}(r)$ the volume of a ball of radius r of SU(2). For $0 < w \leq \frac{2}{35}$ a maximal family of points $Q^{(k)}$ at mutual class distances $> w$ gives a finite partition of $\mathrm{SU}(2)^2$ into Borel cells $I_k \subset \mathcal{O}_w(Q^{(k)})$, invariant under simultaneous conjugation; with $I_k \sim I_l$ if and only if $\mathsf{D}(Q^{(k)}, Q^{(l)}) \leq 3w$, two pairs at the class distance $\leq w$ lie in neighbouring cells, and $\nu \leq \mathsf{v}(\frac{7}{2}w)^2/\mathsf{v}(\frac{1}{2}w)^2 \leq 1.21 \cdot 10^5$.

The label of a layer is the cell of the pair (P_1, P_2) of the holonomies of its straight closed lines through one site, and E_k is the event that this pair lies in I_k. The label is rigid across the layers: if all plaquettes of a lattice of n sites a side lie within ϖ of $\mathbf{1}$, one conjugation carries both holonomies at $a + ke_3$ to within $k\mathsf{n}\varpi$ of those at a, so that $\mathsf{D} \leq k\mathsf{n}\varpi$. *No class is heavy on a bare even cube.* Let ρ_ω be the density, against the Haar measure of $\mathrm{SU}(2)^3$, of the three holonomies (P_1, P_2, P_3) through a site on the event that all plaquettes lie within ω of $\mathbf{1}$; it vanishes unless $|[P_\mu, P_\nu]| \leq N^2\omega$, and it is greatest at $(\mathbf{1}, \mathbf{1}, \mathbf{1})$, as follows. Choose a pointwise version of ρ_ω by solving one seam bond on each of the three straight winding loops for P_μ and integrating over all other bonds. The loops have no bonds in common, and these substitutions preserve product Haar measure. Fix μ and cut at the site planes $x_\mu = 0, h$, where $N = 2h$. Let B be their tangential bonds and X_+, X_- the private bonds in the two halves, identified by reflection. No plaquette crosses a site plane, so the cut Wilson weight factors as $w_0(B)w_+(B, X_+)w_+(B, X_-)$, with $dU = dB\,\mathrm{d}X_+\,\mathrm{d}X_-$ and nonnegative factors bounded by one. The plaquette indicators factor in the same way. At fixed B, solve one private bond on the half-path from 0 to he_μ for its transport. Its weighted Haar density k_B is bounded and hence belongs

to $L^2(G)$. If the two half-transports from 0 to he_μ are g_+, g_-, then $P_\mu = g_+ g_-^{-1}$ and its unnormalised conditional density is $q_B(p) = \int k_B(pg)k_B(g)\,\mathrm{d}g \le \int k_B(g)^2\,\mathrm{d}g = q_B(\mathbf{1})$ by Schwarz and Haar invariance. The other two loops lie in $x_\mu = 0$, so their holonomies and their solved seam bonds are functions of B. Integrating this inequality over the remaining boundary bonds with those holonomies fixed proves $\rho_\omega(P) \le \rho_\omega(P_1, \dots, P_{\mu-1}, \mathbf{1}, P_{\mu+1}, \dots, P_3)$ in the pointwise version just specified. Apply it for $\mu = 1, 2, 3$. The factorization is also the site-mirror case of the proof of Prop. YM3_I:91.4, which keeps the boundary bonds as separate Haar variables. To complete **(Heaviest class)**, one must establish the following quantitative class-probability estimate; the subsequent calculation describes its use. Take $\Xi' = \beta\omega_0^2 \ge 1$ with $\Xi' N^5 \omega_0 \le 3/640$ and $N^2\omega_0 \le w \le 1/10$.

The estimate required here is, for every pair Q and every site y,

$$\mu_{N^3}(\mathsf{D}((P_1, P_2)(y), Q) \le w, \text{ no rough plaquette}) \le 10^{11}\,\Xi'^5 N^7 w^6 \omega_0^{-4}.$$

Given this estimate, on the bare cube with ω_0 and Ξ_β of (a)(ii), $\Xi' = 2\Xi_\beta$ and $w = N^2\omega_0$, as long as $2\Xi_\beta N^5\omega_0 \le \frac{3}{640}$: where no plaquette is rough, D between the labels of the layers 1 and $\frac{N}{2} - 1$ is $< w$ and between those of the layers $\frac{N}{2} \mp 1$ it is $\le 2N\omega_0 \le w$, so $\varepsilon_1, \varepsilon_2$ are at most the probability, $\le \beta^{-2}$, that some plaquette is rough, $p^\sharp \le 10^{11}(2\Xi_\beta)^6 N^{19}\beta^{-1} + \beta^{-2}$, and, the bound increasing with ν, with $1.21 \cdot 10^5$ for ν and $2\nu^4 = 4.2872 \cdot 10^{20}$: $e^{-\mathsf{G}_N} \le 2.8 \cdot 10^{33}\Xi_\beta^6 N^{18}\,t$ on the bare cube. *Through the cutoffs.* With $y = \log \mathsf{b}$ and the side ℓ of (a)(iii), this serves at the cutoffs below the cutoff $K_1(y)$ of (M4)(vi), Prop. 17.30(c), which in (b) we call K_1: the least $K \ge 10$ with $L^{K/10} \ge \bar\ell(y)^3 y^{2\mathfrak{p}_0+1}$, so that $L^{K_1} < L\,(\bar\ell(y)^3 y^{2\mathfrak{p}_0+1})^{10}$ and the bare cube has $N_1 = \ell L^{K_1} < \mathsf{N}(y) = L\,\bar\ell(y)^{31} y^{10(2\mathfrak{p}_0+1)}$ sites a side, a fixed power of y; at the bare cubes $N \le N_1$ one has $\Xi_\beta \le 10N^3 y$, and the condition $2\Xi_\beta N^5\omega_0 \le \frac{3}{640}$ holds once y is large, as $t \le \bar\ell(y)e^{-y}$.

Above K_1 the additive transport of (a) does not serve: even summed from the higher cutoff $\lceil y/(25\log L)\rceil$ of (a)(iii), its error $\sum_K \mathfrak{o}_K \le Q(y)e^{-y/250}$ is of the order of $t^{1/250}$ up to powers of $\log\frac{1}{t}$, and swamps the first power of t; summed from K_1 it is only of the order $1/y$. We use instead the comparison with a factor of (M4)(vi), Prop. 17.30(c): from K_1 on its factors multiply to at most $e^{0.53} < 1.7$, the sums from K_1 on of the numbers of §2.3.1 being of the order $1/y$, and its remainders add up to at most e^{-4y}. At K_1 take ω_0 of (a)(ii) at $\beta_1 = N_1/t$, $w = N_1^2\omega_0$, the cells I_k of this w, and for the block field the roughness $\omega = 1.003\,L^{2K_1}\omega_0$; none of these reads K.

By (M4)(v), where all bare plaquettes lie within ω_0 of $\mathbf{1}$ (for large y, $L^{2K_1}\omega_0 \le N_1^2\omega_0$ lies below the number of L that (M4)(v) asks), the block plaquettes of $\mathcal{M}^{K_1}U$ lie within ω of $\mathbf{1}$ and its holonomies within $13.3\,N_1 L^{K_1}\omega_0 \le 0.512\,w$ of the bare ones; so the bound on the classes of the bare cube N_1, taken at $1.52\,w$, gives for each of the finitely many events E_k of the block field of depth 0 at a site $\mu_0^{K_1}(E_k) \le \beta_1^{-2} + 7.9 \cdot 10^{19} N_1^{36} y^6\,t$, and for the event $\mathfrak{R}$ of a rough block plaquette $\mu_0^{K_1}(\mathfrak{R}) \le \beta_1^{-2}$. By (M4)(vi), in the form of Prop. 17.30(c), used for each of these gauge-invariant events apart (nothing is summed over the cells: $p^\sharp$ is a maximum), at every $K \ge K_1$ each has a probability at most 1.7 times its probability at K_1 plus $1.7\,e^{-4y}$.

At the cutoff K the label (β) of the block field of depth 0, on the torus ℓ with $\mathsf{n} = \ell$ block sites a side, is rigid with ω for ϖ: between the layers 1 and $\frac{\mathsf{n}}{2} - 1$ the class distance is $< \frac{1}{2}\ell^2\omega < w$, between $\frac{\mathsf{n}}{2} \mp 1$ it is $\le 2\ell\omega < w$, so $\varepsilon_1, \varepsilon_2 \le \mu_0^K(\mathfrak{R})$. As $\beta_1^{-2} \le t$ and $e^{-4y} \le t$, the bound of the top level by the heaviest cell gives $e^{-\mathsf{G}_N} \le 6 \cdot 10^{40} N_1^{36} y^6\,t$, which is at most $C\,t\,(\log\frac{1}{t})^{\mathfrak{q}}$, as $N_1 < \mathsf{N}(y)$ and $\log\frac{1}{t} \le y \le 2\log\frac{1}{t}$; the bound $2.8 \cdot 10^{33}\Xi_\beta^6 N^{18} t \le 2.8 \cdot 10^{39}\mathsf{N}(y)^{36} y^6 t$ of the bare cubes below K_1 is smaller. □

17.2.6. *Proof of the theorems. Step 1: Thm. AZ(1).* BI is the hypothesis $\mathsf{e}(l) \le x_0$ of Props. 17.32, 17.33, 17.38, 17.39 and of Lem. 17.37 at $l = l_K$, $\beta = L^K\mathsf{b}$, with one x_0 for all K. For $N = \ell/a_K$,

$$\frac{N}{2l_K} = \frac{\ell}{\ell_0}, \qquad \theta^{1/l_K} = e^{-m_* a_K}, \qquad N \ge 2l_K \iff \ell \ge \ell_0.$$

(a) is Prop. 17.32(b), (c) and Prop. 17.33(c), with $X_K(\mathsf{b},\ell) \le L^3\delta((\ell/La_K)^3)$ (Prop. 17.31(c)). (b) is Prop. 17.38(a), (b). (c): with $y = \theta^{\ell/\ell_0} \le 0.8901$, convexity gives $\varepsilon \le 22.3(e^{0.13736/7}-1)\,y \le 0.441912\,y$ in Prop. 17.34(b), and $\frac12(\sqrt{\varepsilon}+\varepsilon) \le 0.540843\sqrt{y}$; with $y = \theta^{\ell'/\ell_0}$, $Q', Q'' \le e^{0.13736/8} \le 1.017319$ and $Q'(Q'-1) \le 0.0176182\,y$, so Prop. 17.34(c) gives $4.84 \cdot 0.132734 \cdot 1.017319 \le 0.66$. The boxes are Prop. 17.38(c) with $\theta^{N/4l_K} = \theta^{\ell/2\ell_0}$, the twisted boxes Prop. 17.38(d). (d) is Prop. 17.33(a), (b) with $A = l_1/a_K$, $B = l_2/a_K$ and t/a_K for t: $\frac{A}{2(t/a_K)} = \frac{l_1}{2t}$, $\frac{AB}{4l_K^2} = \frac{l_1 l_2}{\ell_0^2}$, $\theta^{(t/a_K)/l_K} = e^{-m_* t}$, $\frac{1.4239}{A} = a_K\frac{1.4239}{l_1}$, and $\sum_{i\ge1}(\lambda_i/\lambda_0)^{t/a_K} = e^{\mathsf{G}} - 1$; at $l_1 = \ell_0$, $(2.42x_0)^{1/l_K} = \exp[-a_K\frac{2}{\ell_0}\log\frac{1}{2.42x_0}]$, and $2\log(1/(2.42\cdot0.069)) \approx 3.5798$.

For the square, $n = s_\ell/a_K = 2^k l_K$ and $(\lambda_i/\lambda_0)^n \le i^{-1}\mathsf{r}_n \le 0.20395\,i^{-1}\theta^{n/l_K}$, which is the bound on E_i and on their number; $\log(1/0.20395) \ge 1.5898$. The vortex line: Prop. 17.33(d) and Prop. 17.39(e), at $\ell = 2^j\ell_0$, $\ell/a_K = 2^{j+1}l_K$, $\sigma^+ a_K^{-2} = \sigma_*$. (e) is Prop. 17.38(f) at $n_0 = 0$; Lem. 17.37(b) with $\theta^{(j-1)\mathsf{w}/l_K} = e^{-m_*(j-1)\ell''/2}$ and $Q''^2 \le 1.0350$; and Prop. 17.38(b′), $\theta^{(2n-1)/6l_K} = e^{-m_*(2n-1)a_K/6}$, $n \ge \frac12(3l_K+1)$ being $na_K \ge \frac34\ell_0 + \frac12 a_K$. (f) With the time e_3, the lattice Γ is the box $(\ell/2a_K; 2\ell/a_K, \ell/a_K)$ closed up with the translation by $\frac{\ell}{2}e_1$ (its periods are also $2\ell e_1$, ℓe_2, $\frac{\ell}{2}(e_1+e_3)$).

With $l' = \ell/2a_K$, Prop. 17.31(f) over the cross-section $(4l', 2l')$ gives $\lambda_0^{l'}(1-\mathsf{r}) \le Z_\Gamma \le Z(l', 4l', 2l')$, $\mathsf{r} = \mathsf{r}_{l'}(2l', 4l')$. Above: $F(l', 4l', 2l') \le 2F(l', 2l', 2l')$ (Prop. 17.31(b)), so $E \le \mathsf{e}(l')$. Below: $\log\lambda_0 = 8l'^2P + \varepsilon(2l', 4l') \ge 8l'^2P$ and $\log Z((2l')^3) = 8l'^3P + \delta_K(\ell)$, so $E \ge \log(1-\mathsf{r}) - \delta_K(\ell)$; by Prop. 17.33(b) in the zone $2t \le A$, $\mathsf{r} \le e^c - 1$ with $c = 0.54526\,y$, $y = \theta^{\ell/\ell_0}$; as $c \mapsto \log(2-e^c)$ is concave and vanishes at 0, $\log(2-e^c) \ge -2.0196\,c$ for $0 \le c \le 0.54526\cdot0.8901$; with $\delta_K(\ell) < 0.13736\,y$ the sum is $\ge -1.24\,y$.

The sign $E \ge 0$ is Prop. 17.31(g) at $N = \ell/a_K$, the lattice Γ being $\Gamma^\diamond$ in physical units.

Step 2: Thm. AZ(2). Fix $K \ge \max\{K_0, 1\}$, $\beta = L^K\mathsf{b}$, boards of the offset 0. The X_i and G_I are real functions of the cell $[-\ell''/4, \ell''/4]^3$ at every cutoff. *(i)* $M_I < \infty$: $\|G_I\|^8_{\mathrm{ch},\ell''}$ is a moment of $8|I|$ centred fields with disjoint supports on the single torus ℓ'', a polynomial in their cumulants of orders ≥ 2, which converge as $K \to \infty$ by (M4)(i) (Prop. 17.43). *(ii) One factor.* With $y = \theta^{\ell'/\ell_0}$, by Step 1, $[Q'(Q'-1)]^{1/2} \le 0.132734\sqrt{y}$ and $Q'' \le 1.0174$; Prop. 17.34(a), (c) give $|\nu_K(G_I)|, |\mu_{K,\ell',\beta}(G_I)| \le 1.0174\,M_I$ and $|\nu_K(G_I) - \mu_{K,\ell',\beta}(G_I)| \le 0.65\sqrt{y}\,1.0174\,M_I$. *(iii) The cumulant.* A cumulant of order $n \ge 2$ does not see the centring, nor, for $n = 1$, does the difference of the means of two laws; and

$$\kappa_n = \sum_\pi (-1)^{|\pi|-1}(|\pi|-1)! \prod_{I\in\pi} \mathbb{E}\Big(\prod_{i\in I} X_i\Big), \qquad \Big|\prod_I \mathsf{a}_I - \prod_I \mathsf{b}_I\Big| \le \sum_I |\mathsf{a}_I - \mathsf{b}_I| \prod_{I'\ne I} \max\{|\mathsf{a}_{I'}|, |\mathsf{b}_{I'}|\},$$

so the bound holds with $C(X)$. *(iv) Limits.* As the torus cumulants converge at each fixed ℓ' by (M4)(i), $\limsup_K - \liminf_K$ of $S_n^T[\nu_K](X)$ is at most $2C(X)\theta^{\ell'/2\ell_0}$ for every ℓ', hence 0; letting $K \to \infty$ in the bound gives the iterated limit, and along (K_j, ℓ_j) the difference is $\le C(X)\theta^{\ell_j/2\ell_0}$. *(v) Every shape.* At $N' = \ell'/a_K$ the corner $c' = \frac{\ell'}{4}(1,1,1)$ has $\mathsf{C}_{N'/2}(c') = [-\frac{\ell'}{4}, \frac{\ell'}{4}]^3$, and $\|\cdot\|_{\mathrm{ch};N'} = \|\cdot\|_{\mathrm{ch},\ell'}$ there; by Lem. 17.35(b), (c) with $\delta(N'^3) < 0.13736\,y \le \frac12$: $|\langle G_I\rangle_{\underline{n}}| \le e^{0.1717}\,1.0174\,M_I \le 1.208\,M_I$ and $|\langle G_I\rangle_{\underline{n}} - \mu_{K,\ell',\beta}(G_I)| \le 7.31\cdot0.370622\sqrt{y}\,1.0174\,M_I$; with (ii), $|\langle G_I\rangle_{\underline{n}} - \nu_K(G_I)| \le 3.36\sqrt{y}\,1.0174\,M_I$, and (iii) gives $C^{\mathrm{sh}}(X)$.

For boxes of any shape with $a_{K_j}\min_\mu n_\mu^{(j)} \to \infty$ take ℓ_j the greatest member of $\mathfrak{L}$ below that length. *(vi) A constant free of label, side and place.* For $\mathsf{b} \in I(g)$ the pairs are first-passage data at g. Put the supports by a corner of the cell, $c_\mu = \lfloor z_\mu + R\rfloor + \frac32$, z the centre of the ball: one corner serves all cutoffs because L is odd; the cell $c + [-\frac{\ell'}{2}, 0]^3$ holds the supports farther than $\frac12$ from its walls; and the $8n$ mirror images of the f_i in the planes $x_\mu = c_\mu$ have the same norms, supports each in a ball of radius R, pairwise $\ge \mathsf{d}'$ apart (two images on the two sides of a wall are more than 1 apart), all in $c + [-(2R+\frac32), 2R+\frac32]^3$, which a lattice translation carries into $(-\frac{\ell'}{4}, \frac{\ell'}{4})^3$.

So the expansion of Prop. 17.43, with the bounds (M5)(i) in the place of (M4)(i), gives $\|G_I\|_{\mathrm{ch},\ell'} \le \mathsf{m}_I^\flat$, centring by the mean on the torus ℓ' itself (the floor $128n(R+1)$ is the floor $16k(R+1)$ of (M5)(i) at $k=8n$). Then (ii), (iii) at $\ell''=\ell'$, where $4.84\cdot 0.132734 = 0.642433$ and no factor Q'' enters, give $C^\flat(X)$, for ν_K and for a torus $\ell^+ \ge \ell'$ alike, both being states $\mathfrak{s}$ of Prop. 17.34; for the boxes $(q_\mu N)$, Lem. 17.35(b), (c) give $1.86\cdot 0.370622 = 0.68936$ in the place of 0.642433; for every box, Lem. 17.35(b), (c) and (ii) give $|\langle G_I\rangle_{\underline{n}} - \nu_K(G_I)| \le (2.70925 + 0.642433)\sqrt{y}\,\mathsf{m}_I^\flat$ and $e^{\frac{5}{4}\cdot 0.13736} \le 1.19$.

For $n\ge 2$, $|S_n(X)| \le C_n^g \prod \|f_i\|_\sharp$ is the limit of (M5)(i) along the tori.

Step 3: Thm. AZ(2′). For bounded $Y, X_1, \dots, X_n$, with $X_I = \prod_{i\in I} X_i$,

$$\kappa(Y, X_1, \dots, X_n) = \sum_{\emptyset\ne I\subset[n]} \mathrm{Cov}(Y, X_I)\,\mathsf{c}([n]\setminus I), \qquad \mathsf{c}(S) = \sum_{\pi\in \mathrm{Part}(S)} (-1)^{|\pi|}|\pi|! \prod_{B\in\pi} \mathbb{E}[X_B],$$
$$\mathsf{c}(\emptyset) = 1.$$

Take a board of side $\ell''/2$ with a cell $\square$ about the tuple and a cell $\square'$ about the support of h; centring by the means on the torus ℓ'', Prop. 17.43 bounds the seminorms of the products of the centred fields in each cell, Prop. 17.34(a) bounds the means $\mathbb{E}[X_B]$, and Lem. 17.37(b) bounds $\mathrm{Cov}(Y, X_I)$ by $e^{-m_*(j-1)\ell''/2}$ times the seminorms, the indices of $\square$, $\square'$ differing by $j \ge 2(|v|_\infty - \ell'')/\ell'' + 1$ in some direction; on a torus both ways around enter. The constants depend on the label only through the compact set of Prop. 17.43. For W in the place of h the same argument runs with $Y = W$, whose centred seminorm in its cell takes the place of the bound of Prop. 17.43 for h. Summing over the plaquettes of a far region with a partition of unity by translates of a fixed bump gives the absolute convergence and the decay: with the tents χ_v, $v\in\rho\mathbb{Z}^3$, of (2′) and u vanishing on $\{|x|_\infty < r\}$, $r \ge \ell'' + \rho$, the bound for $h = u\chi_v$, which vanishes unless $|v|_\infty \ge r - \rho \ge \ell''$, summed over v, gives

$$\sum_{v\in\rho\mathbb{Z}^3} |S^T_{n+1}[\nu_K](u\chi_v, X)| \le \mathsf{C}_n \sup_v \|u\chi_v\|_\sharp \prod_i \|f_i\|_\sharp \sum_{v\in\rho\mathbb{Z}^3,\ |v|_\infty \ge r-\rho} e^{-m_*(|v|_\infty - \ell'')},$$

On the torus use V and χ_v^{per} from (2′). The conditions $\ell \ge L\ell''$ and $\ell'' \ge 8\rho + 4\mathsf{d} + 4$ give $2\rho < \ell$, so each periodized tent is a single local tent in its torus chart and $\sum_{v\in V}\chi_v^{\mathrm{per}} = 1$. Choose half-open representatives of V with minimal sup norm. They form a subset of $\rho\mathbb{Z}^3$ and satisfy $|v|_\infty = |v|_{\infty,\ell}$. The one-far-insertion bound summed over this finite set, for $r \ge \frac{3}{2}\ell'' + \rho$, is therefore bounded by the displayed infinite-grid tail, with the same $3.44\,\mathsf{C}_n$.

Step 4: Thm. AZ(3), the vacuum and the Euclidean group. For $\mathsf{b}\in I(g)$ all pairs $(K, L^K\mathsf{b})$ are first-passage data, and by Step 1 the only periodic Gibbs state there is ν_K. The moments of the centred fields under ν_K are polynomials in the cumulants $S^T_{|I|}[\nu_K]$, $|I|\ge 2$, which converge by Step 2; so (ν_K) converges, and $S_n^\sigma = S_n$. By (M5)(ii) each ergodic component of σ is a limit along the full sequence whose approximants are periodic Gibbs states at the same pairs, that is ν_K; so σ is its own component, and Thm. YM3_I:AP gives the invariance under $\mathbb{R}^3 \rtimes \mathrm{O}(3)$, reflection positivity at every plane, $\ker H_\sigma = \mathbb{C}\Omega_\sigma$ and the Wightman theory with a unique vacuum. Step 2 (iv), with its iterated limit and its bound along every pair of sequences $K_j\to\infty$, $\mathfrak{L}\ni\ell_j\to\infty$, identifies every continuum limit at g along growing tori with the pairs $(K_j, L^{K_j}\mathsf{b})$ with $(S_n^\sigma)_{n\ge 2}$, the limits with the spacing removed first among them; so each of them has the Schwinger functions of σ, and with them its invariance under $\mathbb{R}^3\rtimes\mathrm{O}(3)$, its reflection positivity, its reconstruction with a unique vacuum and, by Step 5, its gap.

Step 5: Thm. AZ(3), the gap. Let $F = \sum_i \zeta_i X_i$ be a finite combination of separated tuples of fields supported in $\{\epsilon < x_3 < \mathsf{h}'\}$, Θ the reflection $x_3\mapsto -x_3$, τ_t the translation by te_3 with $t\in a_K\mathbb{N}$, and F_K the same combination of the lattice fields $\mathcal{O}_K(f)$, centred by their means in ν_K, a function of one slab above the plane $x_3 = 0$, with n_0 layers between the slab and its mirror image. By translation invariance $\nu_K(\overline{\Theta F_K}\,\tau_t F_K)$ is $h_{F_KF_K}(n_0 + t/a_K)$, the mirror being moved to the plane

$x_3 = -t/2$; so Prop. 17.38(f) with $A = B = F_K$, $n = n_0 + t/a_K$ and $l = l_K$ gives

$$\nu_K(\overline{\Theta F_K}\,\tau_t F_K) - |\nu_K(F_K)|^2 \le e^{-m_* t}[\nu_K(\overline{\Theta F_K}\,F_K) - |\nu_K(F_K)|^2],$$

with a rate that does not depend on K and a prefactor that converges. Fix $t \in a_{K_*}\mathbb{N}$. It belongs to every finer time grid, so we may let $K \to \infty$ at this fixed time. With $\psi = (1 - P_\Omega)[F] \in \mathcal{H}(\sigma)$: $\langle \psi, e^{-tH_\sigma}\psi\rangle \le e^{-m_* t}\|\psi\|^2$ on the unbounded set $\bigcup_K a_K\mathbb{N}$ of times; so the spectral measure of ψ gives mass at most $e^{-\eta t}\|\psi\|^2$ to $[0, m_* - \eta]$ for arbitrarily large t, that is none, and such ψ are dense in $\Omega_\sigma^\perp$. The bilinear bound is the spectral theorem on $\Omega_\sigma^\perp$; the other directions follow by the invariance.

Step 6: Thm. AZ(4). Prop. 17.39 at $l = l_K$, lengths in units of a_K: $\sigma^+ a_K^{-2} = (2\Lambda + 1.67)/4l_K^2 a_K^2 = \sigma_*$, and $\sigma_* \ge (2 \cdot 0.116421 + 1.67)/\ell_0^2$ at $\theta \le 0.8901$; the areas become physical areas. For the flux quotient, $N/l_K = 2\ell/\ell_0$ turns $2.2 \cdot 10^{-4}(N/l_K)^2$ into $8.8 \cdot 10^{-4}\ell^2/\ell_0^2$, $1.4268 \cdot 10^{-3}$ into $5.707 \cdot 10^{-3}$ at $N \ge 256 l_K$, that is $\ell \ge 128\ell_0$, and $2\mathsf{q}N/l_K$ into $4\mathsf{q}\ell/\ell_0$. For the potential, $V_K(r) = a_K^{-1}V(r/a_K)$ with the V of Prop. 17.39(d), whose slopes are $\ge \sigma^+ = \sigma_* a_K^2$.

Step 7: Thm. BA(1). By Lem. 17.42 take a second admitted blocking factor $L' > L$, a prime with $\alpha = \log L/\log L'$ irrational; Thm. YM3_I:D holds in the scheme of L' (spacings L'^{-k}, sides $2L'^{\mathbb{N}}$), and $L^K\mathsf{b} = L'^{k(K)}\mathsf{c}_K$ with the labels $\mathsf{c}_K \in [\mathsf{b}, L'\mathsf{b})$ dense. In the scheme of L' let $\ell_f \in 2L'^{\mathbb{N}}$, $\ell_f \ge 24L'\ell_0$, $w_0 = \ell_f/2L'$, $f(x) = \prod_i \max\{0, 1 - |x_i|/w_0\}$ (the tent), and $\mathsf{M}_k(\ell; c) = \|\mathcal{O}_k(f) - \mu_{k,\ell,L'^k c}(\mathcal{O}_k(f))\|_{\mathrm{ch},\ell}$. (i) $\mathsf{M}_k(\ell;\cdot)^8$ is a polynomial in torus cumulants of orders 2 to 8, which by (M4)(i) at L' converges, uniformly on $[\mathsf{b}, L'\mathsf{b}]$, to a continuous function; so $\mathsf{M}_f = \sup_{k\in\mathbb{N},\ c\in[\mathsf{b},L'\mathsf{b}]} \mathsf{M}_k(\ell_f; c) < \infty$. (ii) At the labels c_K, BI gives $\mathsf{e}(l_K) \le x_0$ at $\beta = L'^{k(K)}\mathsf{c}_K$ with $l_K = \ell_0 L^K/2$, and a fine cell has $\mathsf{h} = \ell_f L'^{k(K)}/2$ layers with $\mathsf{h}/l_K \ge \ell_f/L'\ell_0 \ge 24$; Lem. 17.41 with $\ell_c = d\ell_f$, $d = L'^j$, gives

$$\mathsf{M}_{k(K)}(\ell_c; \mathsf{c}_K) \le \epsilon_\star(d) = 1.343\,\mathsf{M}_f\,\theta^{(d-1)\ell_f/8L'\ell_0}.$$

(iii) Let $x(c) = L'^{-3}D_{L'}(N)$, $N = \ell_c L'^k$, at $\beta = L'^k c$. Since $\partial_\beta \log Z(n^3)$ is $-3n^3$ times the mean plaquette variable, which is the same for all plaquettes of a torus, and the translates of the tent are a partition of unity,

$$x'(c) = -(\ell_c/w_0)^3[\mu_{k,\ell_c,\beta}(\mathcal{O}_k(f)) - \mu_{k,L'\ell_c,\beta}(\mathcal{O}_k(f))], \qquad \ell_c/w_0 = 2L'd.$$

Wherever $x \le 0.0064$, $\delta(N^3) \le (L'/(L'-1))^3 x \le 0.0086$ by Thm. BC(1), and Prop. 17.34(c) bounds the bracket by $0.16\,\mathsf{M}_k(\ell_c; c)$; so $|x'| \le 0.0058/((L'-1)\mathsf{b})$ there as soon as $\mathsf{M}_k(\ell_c; c) \le \epsilon_{\mathrm{req}}(d) = 0.036/[(2L'd)^3(L'-1)\mathsf{b}]$. The required smallness shrinks with d like d^{-3}, while $\epsilon_\star(d)/\epsilon_{\mathrm{req}}(d)$ is a constant times $d^3\theta^{(d-1)\ell_f/8L'\ell_0}$, which tends to 0 as $d = L'^j \to \infty$: fix d with $\epsilon_\star(d) \le \frac{1}{2}\epsilon_{\mathrm{req}}(d)$. By (i) the functions $\mathsf{M}_k(\ell_c;\cdot)$ converge uniformly on $[\mathsf{b}, L'\mathsf{b}]$ to a continuous limit; at the labels c_K, $K \ge K_0$, which are dense, with $k(K) \to \infty$, they are at most $\frac{1}{2}\epsilon_{\mathrm{req}}(d)$; so the limit is at most $\frac{1}{2}\epsilon_{\mathrm{req}}(d)$ everywhere, and there is $\bar{k} \ge k(K_0)$ with $\mathsf{M}_k(\ell_c; c) \le \epsilon_{\mathrm{req}}(d)$ for all $c \in [\mathsf{b}, L'\mathsf{b}]$ and all $k \ge \bar{k}$. (iv) At each $k \ge \bar{k}$ there is K with $k(K) = k$, and Prop. 17.32(b) gives $x(\mathsf{c}_K) < 0.000514$; by the mean value theorem the component of $\{x \le 0.0064\}$ that contains c_K carries $x \le 0.006314$, so it is all of $[\mathsf{b}, L'\mathsf{b}]$.

Hence $\delta(N^3) \le 0.008625$ for all c and all $k \ge \bar{k}$, and every $\beta \ge \beta_0 = L'^{\bar{k}}\mathsf{b}$ equals $L'^k c$ with $l = N \le C_+\beta$, $C_+ = \ell_c/\mathsf{b}$. Of L the argument uses only that $\log L/\log L'$ is irrational.

Step 8: Thm. BA(2), (2′), (3). At $\beta = L^K\mathsf{b}' \ge \beta_0$ there is $l \le C_+\beta$ with $\mathsf{e}(l) \le 0.069$, and Prop. 17.32(b) gives $\delta(N'^3) \le 0.13736 \cdot 0.8901^{24} < 0.008625$, so $\mathsf{e}(N') \le 0.069$, for all $N' \ge 48l$. Since $L^{m+K} \ge 48C_+\mathsf{b}'L^K \ge 48l$, the cube of BI at the side $2L^m$ qualifies; on $[\mathsf{b}_-, \mathsf{b}_+]$ take $L^m \ge 48C_+\mathsf{b}_+$ and the threshold of b_-. If BI holds at one label, it holds at all: the set of the labels is empty or $(0, \infty)$. (2′): $\mathsf{M} < \infty$ by Prop. 17.43 and (M4)(i). Fix $K \ge \max\{K_0, 1\}$ and write $\delta(s)$ for $\delta_K(\ell_1)$ at the label s. The derivative of the excess in the label is L^K times the sum over the plaquettes of the torus of the differences between the mean plaquette variable on the torus and in the periodic Gibbs state at the label s (Prop. 17.38(a)); all plaquettes having one such difference, by the symmetries of

both laws, it is a multiple R_K, $|R_K| \le 1458$, of the difference of the means of $\mathcal{O}_K(f)$ on the torus and in that state:

$$\delta(\mathsf{b}_2) - \delta(\mathsf{b}_1) = -\int_{\mathsf{b}_1}^{\mathsf{b}_2} R_K \left[\mu_{K,\ell_1,L^K s}(\mathcal{O}_K(f)) - \nu_{K,s}(\mathcal{O}_K(f))\right] \mathrm{d}s.$$

While $\delta(s) \le 0.008625$, Prop. 17.34(c) bounds the bracket by $0.17516\,\mathsf{M}$, with constants free of K. At $s = \mathsf{b}$, $\delta(\mathsf{b}) < 0.13736\,\theta^{\ell_1/\ell_0} \le 0.13736 \cdot 0.8901^{75} < 0.000025$; so δ cannot reach 0.008625 within η of b, at every cutoff. The continuity and the limits along labels follow from Step 2, made with a supremum over the labels of $[\mathsf{b} - \eta, \mathsf{b} + \eta]$ (one side $2\ell_1$, one threshold, $\theta = 0.8901$), and the continuity in the label of (M4)(i). For the second form put $N = \ell_1^\flat / a_K$. At b, Prop. 17.32(b) gives $D_L(N) < L^3\, 0.13736\, \theta^{\ell_1^\flat/\ell_0} \le 0.03$.

At a label s with $D_L(N) \le 0.069$, $\delta(N^3) \le 0.069/(L-1)^3$ by Thm. BC(1), and the derivative of $D_L(N) = L^3 F(N^3) - F((LN)^3)$ in the label is $-L^3 R_K$ times the difference of the means of $\mathcal{O}_K(f^\flat)$ on the tori $\ell_1^\flat$ and $L\ell_1^\flat$, with the number R_K of the first form, now for $f^\flat$, $|R_K| \le 1458$; Prop. 17.34(c) bounds that difference by $4.84\,[Q'(Q'-1)]^{1/2}\mathsf{M}^\flat \le 0.010813\,\mathsf{M}^\flat$. So $D_L(N)$ rises by at most 0.03 within r of b and stays below 0.06; then $\mathsf{e}(N) \le \frac{8}{L^3} D_L(N) < 0.48/L^3 < 2.2 \cdot 10^{-4}$ by Thm. BC(1), and Prop. 17.32(b) at $l = N$, with $\theta_1 = 12.9 \cdot 2.2 \cdot 10^{-4}$, gives $\mathsf{e}(L^z N/2) < 0.3702\,\theta_1^{L^z/4} \le 6.1 \cdot 10^{-9}$, which is the inequality of BI at the side $L^z \ell_1^\flat$. (3): Prop. 17.38 at this l: the rate $\theta^{n/l}$ with $\theta = 0.8901$ and $l \le C_+\beta$ is at most $e^{-c_* n/\beta}$; the transfer operator is the operator T_ν of the setting, whose matrix elements are the numbers $h_{AB}(n)$ of Prop. 17.38(f): at $\theta = 0.8901$ and $l \le C_+\beta$, $\ker(1 - T_\nu) = \mathbb{C}\Omega$ and $\|T_\nu\| \le 0.8901^{1/l} \le e^{-c_*/\beta}$ on $\Omega^\perp$; and Prop. 17.39(a) with $l \le C_+\beta$ gives the flux quotient; Prop. 17.39(d) gives the slopes, $\sigma^+ l^2 = \frac{1}{2}\log\frac{1}{0.8901} + 0.4175 \ge 0.4757$.

The labels $a_K\beta$ of all first-passage data (K, β) at g lie in the window $(g^{-2} + B_\sharp \frac{L-2}{L-1},\ L(g^{-2} + \bar{B})]$ (YM3_I:(1531) in Prop. YM3_I:74.1), so at a datum with $\beta \ge \beta_0$ the cube of the side $2l a_K \le 2C_+\beta a_K \le 2C_+ L(g^{-2} + \bar{B}) = \ell_0^g$ satisfies the inequality: Thm. AZ(1) and Step 2 hold there with ℓ_0^g for ℓ_0 and 0.8901 for θ, uniformly over these data, the seminorms of Prop. 17.43 being bounded over the window.

Step 9: Thm. BA(4). On $a\mathbb{Z}^3$ the weight does not contain a, and $\mathcal{O}_a(f(\cdot/t)) = t^{-1}\mathcal{O}_{a/t}(f)$; hence

$$\kappa_n(\beta, a/t; X) = t^n\, \kappa_n(\beta, a; X_t)$$

for the cumulants in the unique periodic Gibbs state at $\beta \ge \beta_0$. By Step 8, BI holds near every label with one side and one threshold, and Step 2, taken with a supremum over a compact set of labels and of dilations of X, gives one limit along every sequence of pairs whose labels converge, continuous in the label and the dilation. The supremum over the dilations is had through the rate of (M4)(i): by the Cauchy estimate in the proof of Thm. YM3_I:AS in Thm. YM3_I:AS, on one torus and on a compact interval of labels, $|S^T_{n;K',\ell,L^{K'}\mathsf{b}}(Y) - S^T_{n;K,\ell,L^K\mathsf{b}}(Y)| \le C\, K^{2\mathfrak{p}_0(n+1)+\frac{1}{2}} L^{-K/10} \prod_i \|y_i\|_\sharp$ for all $K' \ge K$ above a threshold, where C and the threshold depend on n, the side, the interval and a floor of the spread of the supports of $Y = (y_1, \dots, y_n)$, and on Y otherwise only through these norms.

For $Y = X_{t'}$ with t' in a compact interval $J'' \subset (0, \infty)$,

$$\|f_i(\cdot/t')\|_\sharp \le \max\{\sup|f_i|, 40M(\inf J'')^{-1} \operatorname{Lip} f_i\}$$

and the spread is at least $\inf J''$ times that of X; so the torus cumulants converge uniformly in the label and the dilation. At a fixed cutoff they are continuous in both, being finite sums with the coefficients $f_i(x(p)/t')$; so the limit is jointly continuous, and the seminorms M_I of Step 2 are bounded uniformly in $t' \in J''$ in the same way. In the scheme of L' the lattices of the line of b are the lattices of the dense labels c_K, dilated by b/c_K; so the limit of the scheme of L' at every label c is the dilation of σ_b by c/b.

The dilations by L and by L', which are trivial in their own schemes, generate a dense subgroup of $(0,\infty)$, and by continuity σ_{tc} is the dilation of σ_c for every real t; the identity above reduces arbitrary real spacings to the scheme of L, and H_n is $S_n^{\sigma_1}$, so $S_n^{\sigma_c}(X) = c^{-n} H_n(X_{1/c})$. At $c \in I(g)$, Step 4 identifies σ_c; the gap of σ_{tc} is that of σ_c divided by t.

Step 10: Thm. BA(5). For φ vanishing outside $v + [-\rho,\rho]^3$ the insertion of the rotation weighted by φ depends only on the bonds within $2a_K$ of that cube and is bounded by $3402(1 + |v|_\infty)\rho^4 L^{4K} |\omega| \sup|\varphi|$, and $\sup|\mathcal{O}_K(f)| \le 162\rho^3 L^{4K} \sup|f|$ for f vanishing outside $v + [-\rho,\rho]^3$; so in every state, for χ vanishing outside $v + [-\rho,\rho]^3$ and the f_i outside $v + [-R,R]^3$, $|\mathfrak{U}| \le \mathsf{c}_n(1+\mathsf{b})(1+|v|_\infty)\rho^4 R^{3n} L^{(4n+5)K} |\omega| \sup|\chi| \prod \sup|f_i|$, with $\mathsf{c}_n = 3402 \cdot 162^n \, \mathsf{F}_{n+1}$ and $\mathsf{F}_m = \sum_\pi |\pi|!$, the sum over the partitions π of $\{1,\dots,m\}$; in the theorem $\rho = R$. (i) *Torus against state.* The bulk term is a polynomial in means of products of local functions of one cube; by Thm. AZ(1)(c), in total variation, on all tori $\ell \ge \ell''$ and in ν_K it agrees up to $2.164\,\mathsf{c}_n(1+\mathsf{b})(1+|v|_\infty)\rho^4 R^{3n} L^{(4n+5)K} e^{-m_* \ell''/4}$ times the sup norms, which is $\le$ the same expression with L^{-K} for $L^{(4n+5)K} e^{-m_* \ell''/4}$ once $\ell'' \ge \Lambda_0 K$, $\Lambda_0 = 4(4n+6)\log L/m_*$. (ii) *The place.* For a state invariant under $a_K\mathbb{Z}^3$, translating the tuple by v adds the translation $\omega v = \sum_\lambda v_\lambda \, \omega e_\lambda$ to the generator of the insertion, and the insertion of the translation by the vector ωe_λ is the difference of the bulk terms of the tuple moved by e_λ and of the tuple, both carried by $[-R-1, R+1]^3$; summed over the components of v this gives the factor $1 + 2|v|_1$. (iii) So it suffices to bound the bulk term of tuples carried by a fixed cube about the origin on one side in $[\Lambda_0 K, L\Lambda_0 K]$ of $\mathfrak{L}(g)$, which by (i) serves for ν_K and all larger sides, and on the sides $\ell \le L\Lambda_0 K$ a power of ℓ is a power of K.

There $\mathfrak{U}$ of YM3_I:(2108) is split as follows, with the objects of (M8). A joint cumulant with the $\mathcal{O}_K(f_i)$ is the derivative in the z_i of a mean at the source $w_z = \sum_i z_i f_i$, so the bulk term of YM3_I:(2107) in equation YM3_I:(2107) and the source calculation in the proof of Prop. YM3_I:112.1 is $-\partial_z^n$ at $z = 0$ of the mean at the source w_z of the bulk insertion $(\mathfrak{S}_{\omega,\chi} + \mathfrak{K}_{\omega,\chi})[\beta - a_K^{-1} w_z]$ (equation YM3_I:(2107) and the source calculation in the proof of Prop. YM3_I:112.1). With the orbital potential $\Psi_\rho(p)$ and the remainder of third order $\mathfrak{r}_\rho^\sharp(p) = \mathfrak{r}_\rho(p) - \Psi_\rho(p) + \Psi_\rho(p_-)$ of [7, §H], the summation by parts of [7, Prop. H.1] writes this insertion as the sum of three members, affine in z: the spin part $\mathfrak{S}_\omega[\phi_z]$, $\phi_z = \beta\chi - a_K^{-1} w_z$; the member $-\sum_p \sum_\rho \tilde{c}_\rho(x(p)) \Psi_\rho(p)$, $\tilde{c}_\rho(y) = (\omega y)_\rho \, a_K^{-1}[\phi_z(y + a_K e_\rho) - \phi_z(y)]$, of order 2 in the curvature with the weight β; and the cubic member $\mathfrak{K}^\sharp[\phi_z \, \omega y]$ of [7, eq. 297], of order 3 with the weight $\beta/a_K = \mathsf{b} L^{2K}$.

Accordingly $\mathfrak{U} = \mathfrak{U}^\sharp + (\mathfrak{U} - \mathfrak{U}^\sharp)$ with

$$\mathfrak{U}^\sharp_{\omega,\chi}(f) = -\beta \langle \mathfrak{K}^\sharp[\chi \, \omega y]; \mathcal{O}_K(f_1); \dots; \mathcal{O}_K(f_n) \rangle + a_K^{-1} \sum_{i=1}^n \langle \mathfrak{K}^\sharp[f_i \, \omega y]; (\mathcal{O}_K(f_l))_{l \ne i} \rangle.$$

All but the cubic member. Use the exact lattice decomposition in Lem. 5.9(b,c,e,f), as in Step 8 of the proof of Thm. O. For $\varphi \in \{\chi, f_1, \dots, f_n\}$, antisymmetry of ω rewrites $a_K^{-1}\mathfrak{S}_\omega[\varphi]$ as an order-2 entry with weights $k^\varphi_{b,p,\rho} = a_K^{-1}(\varphi(x(p)) - \varphi(x_b)) \sum_\lambda \omega_{\rho\lambda} a_K^{-1}(x_b - x(p))_\lambda$, bounded by $|\omega| \operatorname{Lip}\varphi/4$. The bilinear orbital entry has the vector weight $\mathsf{k}^{\varphi,\rho}(y) = (\omega y)_\rho \nabla^+_\rho \varphi(y)$, whose Lipschitz norm is at most $5|\omega|(1+\ell)\|\varphi\|_{C^{1,1}}$. Lem. 5.9(c) gives $\mathfrak{J}^{\mathrm{bl}}[\mathsf{k}] = -\mathfrak{N}[\mathsf{k}]/4 + \sum_{v=1}^5 \mathfrak{J}_v[\mathsf{k}] - c_{\mathsf{k}}/4$: the five entries have orders $2,2,2,3,4$, and only the last two carry β. The null term is evaluated by the contact formulas (N3), (N4) of that lemma. They leave numbers, action densities and order-2 entries without a factor β; the numbers drop from the cumulants. Terms absorbing all the sources vanish, and merging the absorbed supports leaves spread at least $\mathsf{d} - 3a_K \ge \mathsf{d}/2$. Thus Lem. 5.15, applied in its spread form, bounds every remaining term.

For the dependence on ℓ, use the proof of Lem. 5.15 before its constants are collected. A local insertion of order m is carried to depth $u = \lceil \phi K \rceil$, $\phi = (\frac{1}{2} - c_1)/(3n+4)$. The entry contributes at most a factor ℓ^3 from its sites, and the source radii are $(2nN_t \mathsf{C}_t \|f_q\|_\sharp)^{-1}$, with $N_t = \ell^3 L^{3u}$ and $\mathsf{C}_t \le C K^{2\mathfrak{p}_0}$. Their Cauchy cost contributes ℓ^{3n}; the powers of L are $L^{(3n+4)u - \mathsf{g}_m K}$, where $\mathsf{g}_2 = 1$ and $\mathsf{g}_m = 3/2$ for $m \ge 3$. The factors β in the orders $3,4$ leave the decay $L^{-c_1 K}$. The discarded mass of

the definitions preceding Prop. YM3_I:115.2 has the form $2M^{-3}\ell^3L^{3u}\exp[-c\{\log(L^u\mathfrak{b}')\}^{2\mathfrak{p}_0}+C_g\ell^3]$. Since $\mathfrak{p}_0\ge 13$, the labels lie in a compact first-passage interval and $\ell\le L\Lambda_0K$, this mass and its fixed-order moment factors decay faster than every L^{-AK}, uniformly in these sides. The locality remainders have the same property. Collecting the weights from the preceding paragraph gives, times the norms, $|\mathfrak{U}-\mathfrak{U}^\sharp|\le C\ell^{3(n+1)}(R+5)|\omega|K^{(2n+1)\mathfrak{p}_0}L^{-c_1K}$ for every $0<c_1<\frac12$ on $2R+4\le\ell\le L\Lambda_0K$, uniformly over first-passage data at g, with C free of the side.

The cubic member obeys an estimate on all the sides at once, the bound (M7), which is Lem. 5.16 of the section on plane fields and the stress tensor (§5.1) taken at the field $\chi\,\omega y$: for $n\ge2$, $\mathsf{d}>0$ and $\rho_0\ge1$ there are $C_\sharp$ and a threshold, depending on g, n, d, ρ_0 and the constants and on no side, such that for all real $\chi, f_i\in C^{1,1}$ vanishing outside $[-\rho_0,\rho_0]^3$, χ covering the f_i, the supports of the f_i at pairwise sup distances $\ge\mathsf{d}$,

$$|\mathfrak{U}^\sharp_{\omega,\chi}(f)|\le C_\sharp|\omega|L^{-K/16}\|\chi\|_{C^{1,1}}\prod_i\|f_i\|_{C^{1,1}}$$

at all first-passage data at g with K above the threshold, on all sides $\ell\ge2\rho_0+2$ of $\mathfrak{L}(g)$. It is proved in three steps, as in the idea of the proof of (M7). On one torus, the conditional mean of the cubic insertion given the block field of the level $K-\lceil\phi'K\rceil$, $\phi'>0$ small, under the small-field chain of Thm. YM3_I:D, differs from a constant by at most a power of K times L^{-cK}, for every $c<\frac{1}{14}$ (Lem. 5.10(b)): the density is of third order, $|\mathfrak{r}^\sharp_{\rho(p)}(p)|\le C_{\mathrm{orb}}\,\bar a(p)^{3/2}$ ([7, eq. 295]), plain power counting against the weight $\mathsf{b}L^{2K}$ falls short by the factor $a_K^{-1/2}$, and the parity under the lattice mirrors, the covariance of the renormalisation step and a Hölder bound of a background against its mirror image win this factor back.

The joint cumulants of the insertion with the sources then obey the same bound, for every $c<\frac{1}{14}$, by the two-constants inequality of Lem. YM3_I:116.2 in the source variables; this is the bound of one torus, Lem. 5.11. Last, the bound on one torus serves on the finitely many sides of $\mathfrak{L}(g)$ from $2\rho_0+2$ up to a fixed side ℓ', where (M7) is Lem. 5.11, and it passes from ℓ' to every larger side (Lem. 5.16) by the chessboard estimate (M2) on the board of side $\ell'/2$, which writes the difference of a cumulant on the torus ℓ and on the torus ℓ' through cumulants on the torus ℓ' of mirror copies of the entries that join cells, every block that holds a copy of the cubic insertion carrying the small factor, at the cost of the factor $e^{(\ell'/2)^3(p_{\ell'}-p_\ell)}\le e^{\delta_K(\ell')/8}$ of the fine-to-coarse form, which Thm. BC(5)(a), (6) bound at the fixed side ℓ', uniformly in the cutoff and over the window of the labels at g.

With $\rho_0=R+1$ the sum is $\le C|\omega|K^{p'}L^{-\kappa K}$ times the norms for every $\kappa<\frac{1}{16}$. The condition $|v|_\infty+R+2\le\ell/4$ keeps the moved tuples and the insertion away from the seam of the torus, the planes $x_\mu=\pm\ell/2$ along which the cube $[-\ell/2,\ell/2]^3$ is closed up.

Step 11: Thm. BB(1)–(4). (1) is Prop. 17.44(b), (c); the statements on a height y follow from $|X_{K'}-X_K|\le\eta_K$ for $K'\ge K\ge K_\star$ and $\eta_K\to0$. (2) If $D_L(N)\le\frac{L^3}{8}x$ with $x\le0.069$, then $\mathsf{e}(N)\le D_2(N)\le\frac{8}{L^3}D_L(N)\le x$, because $D_L(N)-\frac{L^3}{8}D_2(N)=\frac{L^3}{8}\delta((2N)^3)-\delta((LN)^3)\ge0$ by the monotone density; Prop. 17.32 at $l=N$ gives $\mathsf{e}(2^kN)\le(12.9x)^{2^k}/12.9=\epsilon$, and since $2^kN\le LN/2<2^{k+1}N$,

$$\mathsf{e}(LN/2)\le2\,\delta(\tfrac{LN}{2},LN,LN)\le\Big(\frac{L}{2^{k+1}}\Big)^3[\epsilon+\tfrac{20}{7}\epsilon^2]=\Phi_L(x),\qquad D_L(2l)\le L^3\delta((2l)^3)\le\tfrac{20}{7}L^3\mathsf{e}(l)^2.$$

At $N=\ell/La_K$ we have $LN/2=\ell/2a_K$, and by (1) $X_K\le X_{K_1}+\eta_{K_1}\le y_1$ for all $K\ge K_1$; as $\Phi_L(x_1)\le x_0$, this gives BI[$\mathsf{b},\ell,K_1;x_0$]; with y_0, $\mathsf{e}(\ell/La_K)\le x_0$, which is BI at the side $2\ell/L$ with $c=2$, and Thm. AZ holds at every side $2cL^{m_0}$. Conversely BI at the side $2\ell/L$ has $l_K=\ell/La_K$, and $X_K(\mathsf{b},L\ell)=D_L(Ll_K)\le L^3\delta((Ll_K)^3)\le L^3\bar\delta_{x_0}(L)$ by the proof of Prop. 17.32(b). The bracket: the upper members are Thm. BC(1) at $l=\ell/2a_K$, $2l/L=\ell/La_K$; the lower member is the contrapositive of $D_L(2l)\le\frac{20}{7}L^3\mathsf{e}(l)^2$ for $\mathsf{e}(l)\le0.069$, at $2l=\ell/a_K$.

So BI gives $X_K(\mathsf{b}, L\ell) \le \frac{20}{7} L^3 x_0^2$, and (1) excludes this for $K \ge K_1$ when $X_{K_1}(\mathsf{b}, L\ell) > \frac{20}{7} L^3 x_0^2 + \eta_{K_1}$; the members of the bracket are continuous and monotone in X_K, and $X_K \to X_\infty$, which gives the bracket for $\liminf_K \mathsf{e}_K$ and $\limsup_K \mathsf{e}_K$. (3) At the single cutoff K_1, Prop. 17.32(b) at $l = l_{K_1}$ gives $X_{K_1}(\mathsf{b}, L^i\ell_0) = D_L(2L^{i-1} l_{K_1}) \le L^3 \bar\delta_{x_0}(2L^{i-1})$; with the margin, (1) gives $X_K \le y_1$ for all $K \ge K_1$, and (2) applies. At $i = 2$, $\bar\delta_{x_0}(2L) < 8\mathsf{c}_0 \theta^L \le 0.13736\,\theta^L$, which at $x_0 = 0.069$ and $L \ge 33$ is at most 0.0030; and for $L \ge 33$, $2^k \ge 16$, so $\Phi_L(0.05) \le 8 \cdot 1.01\,(0.645)^{16}/12.9 < 0.069$ and $x_1 \ge 0.05$, $\frac{1}{8} x_1 \ge 0.00625$: the margin condition holds once $\eta_{K_1} \le 0.003 L^3$.

At $i = 1$ it never holds at $x_0 = 0.069$, as $\bar\delta_{0.069}(2) = \frac{20}{7} \cdot 0.069^2 > \frac{1}{8} \cdot 0.069$. (4) (iii) and (1) give $X_K \le y_0$ for $K \ge K_1$, hence $\mathsf{e}(\ell/La_K) \le x_0$, which is BI at the side $\ell_0 = 2\ell/L \in 2 \cdot 2L^{\mathbb{Z}}$; (i) gives $X_K(\mathsf{b}, \ell) \le L^3 \bar\delta_{x_0}(2\ell/L\ell_0) < 0.13736 L^3 \theta^{\ell/L\ell_0}$ for $\ell \ge L\ell_0$ by Prop. 17.32(b), hence (iv); (iv) implies (ii); and (ii) implies (iii) by (1).

Step 12: Thm. BB(5), (6). (5): the reduction, requiring the interacting Hessian estimate, applies Prop. 17.45 at $\underline{s} = (\ell_0/2, \ell_0, \ell_0)$, $z = 0$, $l = \ell_0$, where $|\underline{s}|/l^3 = \frac{1}{2}$, gives

$$\mathsf{e}_{K+1} - \mathsf{e}_K = 2\Big[\log Z_{K+1}(\underline{s}; \mathsf{b}) - \log Z_K(\underline{s}; \mathsf{b}) - \tfrac{1}{2}\big(\log Z_{K+1,\ell_0} - \log Z_{K,\ell_0}\big)\Big],$$

so $|\mathsf{e}_{K+1} - \mathsf{e}_K| \le 2CL^{-K/10}$ uniformly on J, and Lem. 17.46 gives the rest. (6): $D_L(\ell/a_K) = X_K(\mathsf{b}, L\ell)$ converges uniformly on compact sets of labels by (1), with the rate of Prop. 17.44(b); the pairs $(\ell/a_K, L^K\mathsf{b})$ depend on (ℓ, b) only through ℓ/b up to the index of the cutoff, so D_∞ is a continuous function of t. The series for $\delta_K(\ell)$ of Prop. 17.31(c) at $q = L$, $\delta_K(\ell) = \sum_s L^{-3(s+1)} D_L(L^s\ell/a_K)$, has the majorant $L^{-3(s+1)} D_L(L^s\ell/a_K) \le L^{-3s}\delta_K(L^s\ell) \le C(t)\,L^{-s}$, uniformly in the large K, with $C(t)$ bounded on compact sets, by the bounds at a fixed cutoff of Thm. BC(5)(a) and (6), whose proofs do not use the present clause; dominated convergence gives $\delta_K(\ell) \to \delta_\infty(t)$, locally uniformly, and $L^3\delta_\infty(t) - \delta_\infty(Lt) = D_\infty(t)$; the inequalities $(L-1)^3\delta \le D_L \le L^3\delta$ of Thm. BC(1) pass to the limit.

One function for all block sizes. Let L' be a second admitted blocking factor with $\log L/\log L'$ irrational. At one coupling, cubes $N \le M$ with $M/N \le \rho$ have $\delta(M^3) \le \rho^3\delta(N^3)$ by the monotone density. Pass to a subsequence of the lattices (N, β) that converge to $\delta_\infty^{(L)}(t)$, and choose along it cubes $M = 2L'^j$ with $M/N \to \rho_0 \in [1, \rho]$, which the density of $\{L'^j/L^i\}$ allows; the lattices (M, β) are lattices of the scheme of L' whose labels converge, so $\delta_\infty^{(L')}(\rho_0 t) \le \rho^3 \delta_\infty^{(L)}(t)$. Letting $\rho \to 1$ and using the continuity, $\delta_\infty^{(L')} \le \delta_\infty^{(L)}$, and by symmetry the two are equal. The same comparison of nearly equal cubes, now with the lattice inequalities of Prop. 17.31(c) and Thm. BC(1),(2) between the cubes N and $M = \lfloor \rho N \rfloor$, gives that $t^{-3}\delta_\infty(t)$ does not increase, the bound $(3\rho^2 - 3\rho + 1)$ for real ρ, and the alternative for real $t \ge 2t_0$.

Positivity. $\delta_\infty \ge 0$ from the series. At every t, choosing by a dilation a label in $I(g_1) \cup I(g_2) = (\gamma_W^{-2} + \bar B, \kappa_0]$ and the side $\ell = t\mathsf{b}$, Thm. BC(1), (3) give $8\delta_K(\ell) \ge \mathsf{e}(\ell/a_K) \ge \min\{0.069, 0.0775\, e^{-2.6\ell/\mathsf{d}_\circ}\}$ at all large K, so $\delta_\infty(t) > 0$; this uses Thms. YM3_I:AJ and YM3_I:AD through (M5)(iii). *The case of BI.* If BI holds at (b, ℓ_0), then $\delta_\infty(\ell/\mathsf{b}) \le 0.13736\,\theta^{\ell/\ell_0}$ for large ℓ by Thm. AZ(1); conversely, $\delta_\infty(t_0) < 0.008625$ and $\ell/\mathsf{b} = t_0$ give $\delta_K(\ell) \le 0.008625$ for all large K, hence $\mathsf{e}(\ell/a_K) \le 0.069$: BI at the side 2ℓ.

If $D_{q,\infty}(t_0) < (q-1)^3 x_0/8$ for an admitted q, then $\delta_\infty(t_0) < x_0/8 \le 0.008625$, as $(q-1)^3\delta_\infty \le D_{q,\infty}$. In the second case $\delta_\infty < 0.008625$ on a whole interval $[t_2', Lt_2']$, and the uniform convergence on the compact set of labels $[\mathsf{b}, L\mathsf{b}]$ at one side gives, for every real β above a threshold, a cube $N \le C\beta$ with $\delta(N^3) \le 0.008625$: Thm. BA(1) a second time.

Step 13: Thm. BB(7). The conditions of the list (H3), each attached to its constant, are inequalities between numbers; the one that binds L alone, with its constant 4096, is $L > 48 \cdot 4096^2 e^{24} \approx 21\,331\,858\,733\,892\,671\,081.7\ldots$, and every odd integer above it is admitted. The list also forces floors on the other constants that enter $K_\star$ and η_K: $r \ge 2$, $\mathfrak{p}_0 \ge 13$ (so $\mathsf{p}_0 \ge 26.5$), $M \ge L^7$, $\gamma^{-2} \ge (2L)^{1602}$, $A_0 \ge 256 L^5 A_1$, and three constants: $\mathsf{c}_\mathrm{a} = \mathsf{C}_\mathrm{ex}$, the expansion constant preceding equation YM3_I:

(705), with $\mathsf{C}_{\rm ex} \geq 8e^8$; $\mathsf{c}_{\rm c} = \kappa > 0$, the constant of the list that enters $\lambda_{\rm out} = (1+4\beta_{\rm s})\kappa$ of YM3_I:(889); and $\mathsf{c}_{\rm b} = \lambda_1 = \lambda_{\rm out} + 2^3 c_{\rm w} + \mathsf{C}_{\rm ex} + \lambda_{\rm s}$ of YM3_I:(705), where $c_{\rm w} = c_{\rm v}+3$, $c_{\rm v} = 2+\mathsf{C}_{\rm ex} c_G (LM)^3 u$ with $c_G = \mathsf{C}_{\rm ex}(\gamma_{\rm r}^{-1} + c_{\rm r}^2) \geq 2\mathsf{C}_{\rm ex}$ and $(LM)^3 u = \mathsf{u}_{\rm S}^* = 3 + c_{\rm r} + \mathsf{u}_{\rm G} > 4$ by YM3_I:(889), so that $\mathsf{c}_{\rm b} > \mathsf{c}_{\rm c} + 64\,\mathsf{c}_{\rm a}^2 \geq \mathsf{c}_{\rm c} + 4096\,e^{16}$, where $4096\,e^{16} \approx 3.6397 \cdot 10^{10}$.

The threshold $K_\star$ is the least cutoff at which the conditions of (M4)(ii)–(iii) on the construction constants hold for both tori, with the thresholds $K_{\rm bf}$ of (M4)(ii) on both after the dilation by z, as in Thm. BB(1), together with the comparison of Prop. 17.44. At their floors these conditions give $(K-z)\log L > 9\,\mathsf{c}_{\rm b} > 36864\,e^{16} \approx 3.2757 \cdot 10^{11}$ for every $K \geq K_\star(\mathsf{b}, \ell)$. Indeed, at the dilated data $(L^z\mathsf{b}, L^z\ell)$, whose cutoff $K-z$ is written K in this sentence: the number $\kappa_{\rm b}$ of YM3_I:(2181) is $w_\star e^{-\lambda_V}/32M^3$ with $w_\star = w_0 e^{-4\lambda_1}/(\mathsf{C}_{\rm ex}(L+2)^3 S_{\rm s})$, $w_0 = (4S_{\rm s}L^3)^{-1}$ (equation YM3_I:(705)), so $\kappa_{\rm b}^{-1} > L^6 M^3 e^{4\mathsf{c}_{\rm b}}$; the condition $\epsilon_{\rm sh} \leq \frac{1}{20}$ of (k3) forces $\exp[\frac{1}{4}\kappa_{\rm b} A_1^2 \mathsf{y}_{[u]}] \geq 60$, so $A_1^2 \mathsf{y}_{[u]} > 16/\kappa_{\rm b}$ and $A_0^2 \mathsf{y}_{[u]} > L^{16} M^3 e^{4\mathsf{c}_{\rm b}}$; then $\epsilon_{\rm cl} \leq \frac{1}{20}$ forces $L^{(K-s)/2} \geq 80 A_0^2 \mathsf{y}_{[u]} > e^{4\mathsf{c}_{\rm b}}$, $s = \lceil K/9 \rceil$, and as $K - s \leq \frac{8}{9}K$, $K \log L > 9\,\mathsf{c}_{\rm b}$.

So the small cube of the criterion of (2) has $2L^{m+K-1} \geq 2L^{K-z+1}$ sites a side, as $m+z \geq 2$, hence more than $e^{9\mathsf{c}_{\rm b}} > 10^{1.4226 \cdot 10^{11}}$. Where the margin η_K is below the height of the criterion, as the margin condition of (3) requires of $\eta_{K_1}(\mathsf{b}, L^2\ell_0)$, the same conditions, with $\mathsf{C}_{\rm bf} \geq 64 \mathsf{K}_{\rm w} c_\rho^2 A_0^2 \mathsf{y}_{[1]} L^{7/2}$ from (2.xiii) and $\mathsf{y}_{[1]}(K-z)^{2\mathfrak{p}_0} \geq \mathsf{y}_{[u]}$, make $L^{(K-z)/10}$ larger than $e^{4\mathsf{c}_{\rm b}} \geq e^{4(\mathsf{c}_{\rm b} - \mathsf{c}_{\rm c})}$ times a power of L and of M, and $e^{4(\mathsf{c}_{\rm b}-\mathsf{c}_{\rm c})} > e^{256\,\mathsf{c}_{\rm a}^2} \geq e^{16384\,e^{16}}$; so $(K-z)\log L > 163840\,e^{16} \approx 1.4559 \cdot 10^{12}$.

At the cutoff of (3) the half side $l = \ell_0/2a_{K_1}$ of the cube of BI is then $> 10^{6.3228 \cdot 10^{11}}$, and the two boxes $(l, 2l, 2l)$ and $(2l)^3$, of $4l^3$ and $8l^3$ sites, have more than $10^{1.8968 \cdot 10^{12}}$ sites each. For the floor of η_K: $\varsigma \geq \varsigma_\flat(L)$ as $\mathfrak{p}_0 \geq \frac{5}{2}$, $\mathsf{C}_{\rm bf} \geq 2$, and $L^z\ell = 2L^{m+z} \geq 2L^2$ as $m+z \geq 2$; so $\eta_K \geq 3\varsigma_\flat \cdot 2 \cdot 8L^6 (K-z)^{5/2} L^{-(K-z)/10}$. An inequality $X_{K_1} \leq y - \eta_{K_1}$ with $X_{K_1} \geq 0$, $y \leq y_0$, forces $\eta_{K_1} \leq \frac{L^3}{8}\,0.069$, hence $K_1 - z \geq K_\flat(L)$ for the least $K_\flat$ at which the floor drops below that number, and the small cube then has $2L^{m+K_1-1} \geq 2L^{1+K_\flat}$ sites a side; at every admitted L this exceeds 10^{152} (already at $L=13$, $2L^{1+K_\flat} > 10^{152.9}$).

Step 14: Thm. BC(1)–(4). (1): $D_2(N) = 4\mathsf{d}^{(3)}(N,N,N) + 2\mathsf{d}^{(2)}(N,N,2N) + \mathsf{d}^{(1)}(N,2N,2N)$, three doubling defects ≥ 0 by Prop. 17.31(b), the last being $\mathsf{e}(N)$; $D_2 \leq 8\delta$ and $D_q - \frac{q^3}{8} D_2 = \frac{q^3}{8}\delta((2N)^3) - \delta((qN)^3) \geq 0$ by Prop. 17.31(c); $(q-1)^3 \delta(N^3) \leq D_q(N)$ is $\delta((qN)^3) \leq (3q^2 - 3q + 1)\delta(N^3)$, and the box and cube bounds are Prop. 17.31(c); $\mathsf{e}(l) \leq 2\delta(l, 2l, 2l) \leq 16(l/N)^2 \delta(N^3)$. For $l = Ll'$, $n = 2l/L$, with the time along a side of the cube $Ln = 2l$: $D_L(n) \geq LF(n, 2l, 2l) - F(2l, 2l, 2l) = L\mathsf{G}(n) - \mathsf{G}(2l) \geq L\mathsf{G}(l) - \mathsf{G}(2l) \geq \frac{L}{2}\mathsf{e}(l)$ over the cross-section $(2l, 2l)$, as $n \leq l$ and $\mathsf{e}(l) = 2\mathsf{G}(l) - \mathsf{G}(2l)$, the two other thermal defects of D_L being ≥ 0; this is $\mathsf{e}(l) \leq \frac{2}{L} D_L(2l/L)$.

The cube criterion and $D_L(2l) \leq \frac{20}{7} L^3 x^2$ are Step 11; $\mathsf{e} \leq 8\delta$; and $0.13736 \cdot 0.8901^{24} \leq 0.0084022$. Finally $F \leq 0$ and, by Jensen's inequality for the Haar measure, $F \geq -3\beta|\underline{n}|$, so $\mathsf{e}(l) \leq -F((2l)^3) \leq 24\beta l^3$. (2) is Prop. 17.32 with $x_0 = 8\delta(N^3)$, $\theta = 103.2\,\delta(N^3)$ (if $\delta(N^3) = 0$ the excess vanishes for all larger cubes), $\delta((2N)^3) \leq \frac{20}{7}\mathsf{e}(N)^2 \leq \frac{20}{7} \cdot 64\,\delta(N^3)^2$, and Prop. 17.32(e) for 0.07489. Two periodic Gibbs states exclude the first alternative by Prop. 17.38(a); by (M3) each is an integral of ergodic ones, two different ergodic ones are mutually singular, and Prop. 17.34(d) gives $2 \leq \liminf e^\delta$.

The last sentence is Prop. 17.34(d), the sides being chosen with δ tending to its lower limit and the laws converging, by the compactness of the laws on a compact space. (3): if $\mathsf{e}(l) > 0.069$ there is nothing to prove; otherwise let ν be the state of Prop. 17.38, f the bump of (M5)(iii) at radius $\mathsf{d}_\circ$ and height $2\mathsf{d}_\circ$, and $C(n)$ the covariance under ν of $\mathcal{O}_K(f \circ \Theta)$ and $\mathcal{O}_K(f(\cdot - na_K e_3))$, Θ the reflection $x_3 \mapsto -x_3$. For $0 \leq n \leq N_4$ the pair is a test pair at $(\mathsf{d}_\circ; \vartheta_1)$ and two points of the supports are between $2\mathsf{d}_\circ + na_K$ and $6\mathsf{d}_\circ + na_K$ apart; so (M5)(iii) gives $C(n) > 0$ and

$$\frac{C(0)}{C(N_4)} \leq \frac{4/3}{2/3}\Big(\frac{10\,\mathsf{d}_\circ}{2\,\mathsf{d}_\circ}\Big)^6 = 2 \cdot 5^6 = 31250.$$

By translation invariance $C(n)$ is the covariance of $\mathcal{O}_K(f)$ and its mirror image in the plane $x_3 = -\frac{n}{2}a_K$, so Prop. 17.38(f) with $A = B = \mathcal{O}_K(f)$, as in Step 5, gives $C(n) \le \theta^{n/l}C(0)$, $C \ge 0$, and C is log-convex, so all $C(n) > 0$ and the quotients $C(n+1)/C(n)$ do not decrease: $C(n) \ge C(N_4)\,31250^{-(n-N_4)/N_4}$. Taking n-th roots, $\theta^{1/l} \ge 31250^{-1/N_4}$, with $\theta = 12.9\,\mathsf{e}(l)$; and $\log 31250 < 10.35 \le 2.6\,N_4 a_K/\mathsf{d}_\circ$ as $a_K/\mathsf{d}_\circ \le \frac{1}{52}$. With $\mathsf{d}_\circ = \mathsf{d}_0$, $l = \ell/2a_K$; (1) carries the floor to D_2, 8δ, $\frac{8}{L^3}D_L$; and $\frac{1}{12.9}31250^{-l/N_4} > 0.069$ if and only if $l/N_4 < \log(1/(12.9\cdot 0.069))/\log 31250$, which $la_K < 0.0449\,\mathsf{d}_\circ$ gives, as $N_4 > 4\mathsf{d}_\circ/a_K - 1$ and $\mathsf{d}_\circ/a_K > 4L^3M$. (4): dilating by L^z, the label enters the period $(\gamma_W^{-2} + \bar{B}, \kappa_0]$, which $I(g_1)\cup I(g_2)$ covers by the form of the intervals $I(g)$ in (M5); there (3) at the side $L^z\ell_0$ gives $L^{-z}m_* \le 2.6/\mathsf{d}_0^\star$, and $L^z \le \kappa_0/\mathsf{b}$, so $m_* \le 2.6\kappa_0/(\mathsf{d}_0^\star\mathsf{b})$, which is the bound since $\log(1/\theta)/1.3 \ge 0.0895$.

On the cube $(2l)^3$, Prop. 17.31(d), (e) give $w_{00} = \frac{1}{4}(1+2r_1+r_2)$, $w_{10} = w_{01} = \frac{1}{4}(1-r_2) = u$, $w_{11} = \frac{1}{4}(1-2r_1+r_2) = v$; with $x = 1-r_1 = 2(u+v)$ the sum $\sum_f w_f^{1/2} = (1-u-\frac{x}{2})^{1/2} + 2u^{1/2} + (\frac{x}{2}-u)^{1/2}$ is concave in u on the range $\frac{x}{4} \le u \le \frac{x}{2}$ allowed by $r_2 \le r_1$ and $v \ge 0$, so least at an end of it, which is $u = \frac{x}{2}$ for $x \le \frac{1}{2}$; this gives $\mathsf{e}(l) \ge 2\log(\sqrt{1-x} + \sqrt{2x})$, which exceeds 0.069 for $x > 0.00062714$ and is at least $\log 2$ for $\frac{1}{9} \le x \le \frac{1}{2}$; for $x > \frac{1}{2}$ both end values of the sum are at least $\sqrt{2}$, so $\mathsf{e}(l) \ge \log 2$ there; and $\mathsf{e}(l) \ge \mathsf{G}_{2l} \ge \log\frac{1}{w_{00}} \ge \log\frac{4}{1+3r_1}$.

Then Prop. 17.31(d) for r_2, r_3, Prop. 17.32(b) for $\delta((2l)^3) \le \frac{20}{7}\cdot 0.069^2 \le 0.0136029$, and Prop. 17.31(b) for $\mathsf{r}_l(2l, 2l) \le \varphi(e^{0.069} - 1) \le 0.0357638$, $\log(1/0.0357638) \approx 3.33082$, $l_K a_K = \ell_0/2$.

Step 15: Thm. BC(5), (6). (5)(a) is Prop. 17.47(c) at $N = \ell/a_K$, $\beta = L^K\mathsf{b}$, where $N/\beta = \ell/\mathsf{b}$, with $|c'_N - c'| \le C_A N^{-2}(\log N)^{5/2}$; and the limit $K \to \infty$ by Thm. BB(6). (b) Prop. 17.48(a) and Prop. 17.31(d), (e): as $r_2 \le r_1$, w_{00} is the largest weight and $\frac{1}{4} \le w_{00} \le \frac{1}{4}(1+3r_1)$, $|4w_f - 1| \le 3r_1$; $\delta_K(\ell) \ge \mathsf{G}_N \ge \log\frac{1}{w_{00}} \ge \log\frac{4}{1+3r_1}$, and $\mathsf{e}(\ell/2a_K) \ge \mathsf{G}(\ell/2a_K; N, N) \ge \mathsf{G}_N$; the last assertion follows from (4): $\eta_\flat(\ell_0/\mathsf{b}) < 0.99937$ gives $1 - r_1 > 0.000628$ on the cube ℓ_0/a_K at every cutoff. (c) Completing the **(Heaviest class)** probability estimate would make the reduction in Prop. 17.48(b) give $\mathsf{G}_N = \log(Z/\lambda_0^N) \ge \log\frac{1}{t} - \mathsf{q}\log\log\frac{1}{t} - \log C$. (d) Choose $t_\circ > 0$ with $16Lt_\circ \le t_2$ and $\eta_\flat(t) \le 0.9961$ for $t \le 16Lt_\circ$, and $\beta_\flat = 1/8t_\circ$.

For $\beta \ge \beta_\flat$ there is $j \ge 1$ with $2L^j/\beta \in (16t_\circ, 16Lt_\circ]$; then $1 - r_1(2L^j) \ge 0.0039$ by (b), and Prop. 17.32(d), by contraposition, excludes $\mathsf{e}(l) \le 0.069$ for every $l \le 2L^j/16$, which includes $3 \le l \le t_\circ\beta$. For BI take K with $L^K\mathsf{b} \ge \beta_\flat$: $l_K = \ell_0 L^K/2 > t_\circ L^K\mathsf{b}$. The exact statement on one lattice is the contrapositive of Prop. 17.32(d). (e) is Prop. 17.47(d). (6): with $N = L^{-j}\ell/a_K$ at the same β, (a) at the side $L^{-j}\ell$, where $t_1/L < L^{-j}t \le t_1$, bounds $\delta(N^3)$ by $\log(L/t_1) + c' + C$, and Thm. BC(1) gives $\delta_K(\ell) \le 3L^{2j}\delta(N^3)$ with $L^j \le Lt/t_1$; then $\mathsf{e} \le 8\delta$ and $D_L \le L^3\delta$.

Remark 17.49 (Scope). The theorems do not assert BI at any label; by Thm. BC it fails at every cutoff on cubes small against the label, and it can hold only on cubes of side at least $0.0895\,(\mathsf{d}_0^\star/\kappa_0)\,\mathsf{b}$. Thm. BA(2) carries BI from one label to every other label at other sides and from other thresholds, with numbers β_0, C_+ that are not explicit. Uniqueness in Thm. AZ(1)(b) is uniqueness among the periodic Gibbs states; of Gibbs states without periods it says only that ν_K is extreme and trivial at infinity. The area law bounds the loops in the state ν_K at each cutoff. Thm. BA(5) concerns tuples carried by a cube that keeps away from the seam of the torus.

Appendix A. Index of notation and conventions

Symbol	Meaning	Defined in
$a_K, \mathsf{b}, L, T_\ell$	Lattice spacing, physical label, blocking factor and torus	§2
$V, O_K^T, \bar{O}_K^T$	Tensor labels, lattice insertion and cell-mean insertion	§2
$\mathcal{S}_{n;K}$	Truncated tensor correlations at cutoff K	§2
$\operatorname{Sym}\mathcal{E}$	Quadratic symbol of a class insertion	§2
$\mathcal{T}, \mathcal{O}$	Tensor and scalar continuum insertions	§3
$\sigma, \mathcal{H}, \Omega, H$	Clustering correlations, Hilbert space, vacuum and energy generator	§3
$\mathcal{K}_q, L_b^+, c_1, c_2$	Homogeneous kernels, two-term approximation and coefficients	§6
κ^{lat}	Coefficient of the scalar kernel of degree -5	§7
$\Psi_r, \mathcal{N}_t$	Dilated test pair and three-scale combination	§8
$G, \mathfrak{g}, C_A, C_\rho$	Compact group, Lie algebra and Casimir normalizations	§9
W_C, κ	Wilson trace and perimeter exponent	§10
$Z_{e,m}, P_e$	Electric-sector partition function and projection	§11
P, J	Pressure and its derivative jump	§12
s, ρ	Radial two-point kernel and positive spectral measure	§13
$\mathcal{A}, \mathcal{A}^d$	Action-density net and Haag-dual completion	§14
$\mathcal{L}_K, \mathcal{E}_K$	Langevin generator and Dirichlet form	§15
Φ_σ	Deterministic lattice gradient flow	§16
$\mathsf{e}, \ell_0, m_*, \sigma_*$	Box defect, physical scale, gap and area constants	§17

Derivatives of continuum tests act on $\mathbb{R}^3$; bond derivatives and covariant lattice differences act on the cutoff lattice. Stochastic time, gradient-flow time and real time in the reconstructed quantum field are distinct. Constants have the dependencies stated in each sketch. A bound uniform in the fine cutoff need not be uniform in the physical volume.

References

[1] *Scalar fields from three-dimensional lattice Yang–Mills theory*, companion manuscript YM3_I.

[2] J. Stoyanov and G. D. Lin, *Hardy's condition in the moment problem for probability distributions*, Theory Probab. Appl. **57** (2013), no. 4, 699–708. doi:10.1137/S0040585X9798631X.

[3] R. Adamczak, *A note on the Hanson–Wright inequality for random vectors with dependencies*, arXiv:1409.8457v1 (2014).

[4] A. V. Kolesnikov and E. Milman, *Riemannian metrics on convex sets with applications to Poincaré and log-Sobolev inequalities*, arXiv:1510.02971v2 (2016).

[5] G. F. Lawler and V. Limic, *Random Walk: A Modern Introduction*, Cambridge University Press, 2010. Author-hosted text, Thm. 6.3.8(a).

[6] A. E. Nussbaum, *Quasi-analytic vectors*, Ark. Mat. **6** (1965), 179–191. doi:10.1007/BF02591357.

[7] *Curvature observables, dynamics, and conditional confinement in three-dimensional lattice Yang–Mills theory*, comprehensive manuscript, 1 November 2026, 1201 pp., YM3_II.

[8] *Osterwalder–Schrader axioms for curvature bilinears*, main manuscript, revised 1 November 2026, YM3_main.